All State Engineering Services & PSUs Examinations

MECHANICAL ENGINEERING

Previous Years

Chapterwise Objective Solved Papers

Useful for : **UPSC ESE, UPSC JWM, UPPSC AE, UPRVUNL AE, UP Jal Nigam AE, UKPSC AE, UJVNL AE, HPPSC AE, BPSC AE, MPPSC AE, JPSC AE, CGPSC AE, WBPSC AE, RPSC AE, MPSC AE, OPSC AEE, GPSC AE, TSPSC AEE, APPSC AEE, TNPSC AE, KPSC AE, Nagaland PSC CTSE, Assam PSC, ISRO Scientist/Engineer, TRB Polytechnic Lect., BHEL ET, CIL MT, SJVN ET, VIZAG Steel MT, DRDO Scientist, Haryana PSC, Mizoram PSC, Oil India Ltd., NPCIL ET, Sikkim PSC, TANGEDCO AE, TSGENCO AE, Punjab PSC, RRB SSE.**

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Note : Thermodynamics, Power Plant, RAC, IC Engines, FM, HMT, Industrial Engineering Study Volume-II

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सत्यमेव जयते

UPPSC

Uttar Pradesh Public Service Commission

Government of UP, Prayagraj

Combined State Engineering Services

(General Recruitment/Special Recruitment) Examination

Examination Pattern and Syllabus

The following two objective type papers will be for the Combined State Engineering Services Examination.

PAPER-I

Subject	No. of Questions	Marks	Total Marks	Time
General Hindi	25 (Each question of 3 marks)	75	375	2.30 Hours
Main Subject (Mech. Engg.-I)	100 (Each question of 3 marks)	300		

PAPER-II

Subject	No. of Questions	Marks	Total Marks	Time
General Studies	25 (Each question of 3 marks)	75	375	2.30 Hours
Main Subject (Mech. Engg.-II)	100 (Each question of 3 marks)	300		

Personal Examination (Interview) – 100 Marks

Total – 375 + 375 + 100 = 850 Marks

SYLLABUS

General Hindi – Hindi syllabus will be made in such a way that the candidates under standing of Hindi language and efficient use of words can be checked. Its level will be of high school.

General Studies – The question paper of General Studies will include information focusing on current events and their scientific aspects of such things which come in everyday experience and which can be expected from an educated person. The question paper will also included such questions in the history, politics and geography of India, to which candidates will be able to answer without special study.

MECHANICAL ENGINEERING

PAPER-I

1. **Engineering Mechanics:** Analysis of force systems, friction, cendtroid and centre of gravity, trusses and beams, principle of virtual work, kinematics and kinetics of particle, kinematics and kinetics of rigid bodies.
2. **Mechanisms and Machines:** Velocity and acceleration of links, cams and followers gears and gear trains clutches, belt drives, brakes and dynamometers, Flywheel and governors, balancing of rotating and reciprocating masses, balancing of multi cylinder engines, Free and forced vibration, damped vibration, whirling of shafts.

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3. **Mechanics of Solids:** Stresses and strains, compound stresses strains, Torsion of circular shafts, stresses and deflections in beams unsymmetrical bending, curved beams, Thin and thick cylinders and spheres, Buckling of columns, Energy methods, helical and leaf springs.

4. **Design of Machine Elements:** Design for Static and dynamic loading, Theories of failure, fatigue principles of design of rivetted, welded and bolted joints, shafts, springs, bearings, brakes, clutches and flywheels.

5. **Engineering Materials:** Crystal systems and crystallography, crystal imperfections, Alloys and phase diagrams,

   Heat treatment, ferrous and non ferrous metals and alloys, Mechanical properties and testing.

6. **Manufacturing:** Metal casting, metal forming, metal joining, Mechanics of metal cutting, machining and machine tool operations, unconventional machining methods limits, fits and tolerances, inspection: Surface roughness, comparators, computer integrated manufacturing, Flexible manufacturing systems, jigs and fixtures.

7. **Industrial Engineering:** Production, planning and control, inventory control and operation, research, CPM and PERT.

8. **Mechatronics and Robotics:** Microprocessors and microcontrollers, Architecture, Programming, Computer interfacing Programmable logic controller, sensors and actuators, Piezoelectric accelerometers, Hall effect sensors, optical encoder, resolver, Inductosyn, Pneumatic and Hydraulic Actuators, stepper motor, control system, mathematical modeling, control signals, controllability and observability, Robotics: Robot classification, robot specification. Notation: Direct and inverse kinematics homogeneous co-ordinates and arm equation of four axix SCARA Robot.

MECHANICAL ENGINEERING

PAPER-II

1. **Thermodynamics:** Thermodynamic systems and processes, properties of pure substances, concepts and applications of zeroth, first and second law of thermodynamics, entropy, availability and irreversibility, detailed analysis of thermodynamic cycles, ideal and real gases, fuels and combustion.

2. **Fluid Mechanics:** Basic concepts and properties of fluids, manometry, fluid statics, buoyancy, equations of motion, Bernoulli's equation and applications, viscous flow of incompressible fluids, laminar and turbulent flows, flow through pipes and head losses in pipes, dimensional analysis, Forces on immersed bodies and boundary layer over a flat plate, isentropic and adiabatic flows, normal shock waves.

3. **Heat Transfer:** Modes of heat transfer, steady and unsteady heat conduction, thermocouple time constant, critical thickness of insulation, heat transfer from fins, momentum and energy equations for boundary layer flow on a flat plate. Free and forced convection, radiation heat transfer, Stefan-Boltzmann law, shape factor, black and grey body radiation heat exchange, boiling and condensation, heat exchanger analysis, LMTD and NTU – effectiveness methods.

4. **Energy conversion:** SI and CI engines, performance characteristics and testing of IC engines, combustion phenomena in SI and CI engines, carburetion and fuel injection systems, emissions and emission control. Reciprocating and rotary pumps, pelton wheel, Francis and Kaplan turbines, velocity diagrams impulse and reaction principles steam and gas turbines; Rankine and Brayton cycles with regeneration and reheat, high pressure boilers, draft, condensers. Unconventional power systems, including nuclear, MHD, biomass, wind and tidal systems, utilization of solar energy; Reciprocating and rotary compressors; theory and applications, Theory of propulsions, pulsejet and ramjet engines.

5. **Environmental control:** Vapour compression, vapour absorption, steam jet and air refrigeration systems, properties of refrigerant and their nomenclature, psychometrics properties and processes, psychrometic relations, use of pschrometic chart, load estimation, supply air conditions, sensible heat factors, air conditioning system layout, comfort chart, comfort and industrial air conditioning.

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ESE MECHANICAL ENGINEERING

Contents for syllabi of both the Papers together for Preliminary Examination/ Stage-I (objective type Paper–II) and separately for Main/ Stage-II Examination (Conventional type Paper-I and Paper – II).

PAPER–I

1. Fluid Mechanics:

Basic Concepts and Properties of Fluids, Manometry, Fluid Statics, Buoyancy, Equations of Motion, Bernoulli's equation and applications, Viscous flow of incompressible fluids, Laminar and Turbulent flows, Flow through pipes and head losses in pipes.

2. Thermodynamics and Heat transfer:

Thermodynamic systems and processes; properties of pure substance; Zeroth, First and Second Laws of Thermodynamics; Entropy, Irreversibility and availability; analysis of thermodynamic cycles related to energy conversion: Rankine, Otto, Diesel and Dual Cycles; ideal and real gases; compressibility factor; Gas mixtures.

Modes of heat transfer, Steady and unsteady heat conduction, Thermal resistance, Fins, Free and forced convection, Correlationsfor convective heat transfer, Radiative heat transfer – Radiation heat transfer co - efficient; boiling and condensation, Heat exchanger performance analysis.

3. IC Engines, Refrigeration and Air conditioning:

SI and CI Engines, Engine Systems and Components, Performance characteristics and testing of IC Engines; Fuels; Emissions and Emission Control. Vapour compression refrigeration, Refrigerants and Working cycles, Compressors, Condensers, Evaporators and Expansion devices, Other types of refrigeration systems like Vapour Absorption, Vapour jet, thermo electric and Vortex tube refrigeration. Psychometric properties and processes, Comfort chart, Comfort and industrial air conditioning, Load calculations and Heat pumps.

4. Turbo Machinery:

Reciprocating and Rotary pumps, Pelton wheel, Kaplan and Francis Turbines, velocity diagrams, Impulse and Reaction principles, Steam and Gas Turbines, Theory of Jet Propulsion – Pulse jet and Ram Jet Engines, Reciprocating and Rotary Compressors – Theory and Applications

5. Power Plant Engineering:

Rankine and Brayton cycles with regeneration and reheat, Fuels and their properties, Flue gas analysis, Boilers, steam turbines and other power plant components like condensers, air ejectors, electrostatic precipitators and cooling towers – their theory and design, types and applications;

6. Renewable Sources of Energy:

Solar Radiation, Solar Thermal Energy collection - Flat Plate and focusing collectors their materials and performance. Solar Thermal Energy Storage, Applications – heating, cooling and Power Generation; Solar Photovoltaic Conversion; Harnessing of Wind Energy, Bio-mass and Tidal Energy – Methods and Applications, Working principles of Fuel Cells.

PAPER–II

7 . Engineering Mechanics:

Analysis of System of Forces, Friction, Centroid and Centre of Gravity, Dynamics; Stresses and Strains - Compound Stresses and Strains, Bending Moment and Shear Force Diagrams, Theory of Bending Stresses - Slope and deflection-Torsion, Thin and thick Cylinders, Spheres.

8. Engineering Materials:

Basic Crystallography, Alloys and Phase diagrams, Heat Treatment, Ferrous and Non Ferrous Metals, Non metallic materials, Basics of Nano-materials, Mechanical Properties and Testing, Corrosion prevention and control

9. Mechanisms and Machines:

Types of Kinematics Pair, Mobility, Inversions, Kinematic Analysis, Velocity and Acceleration Analysis of Planar Mechanisms, CAMs with uniform acceleration and retardation, cycloidal motion, oscillating followers; Vibrations –

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Free and forced vibration of undamped and damped SDOF systems, Transmissibility Ratio, Vibration Isolation, Critical Speed of Shafts. Gears – Geometry of tooth profiles, Law of gearing, Involute profile, Interference, Helical, Spiral and Worm Gears, Gear Trains- Simple, compound and Epicyclic; Dynamic Analysis – Slider – crank mechanisms, turning moment computations, balancing of Revolving & Reciprocating masses, Gyroscopes –Effect of Gyroscopic couple on automobiles, ships and aircrafts, Governors.

10. Design of Machine Elements:

Design for static and dynamic loading; failure theories; fatigue strength and the S-N diagram; principles of the design of machine elements such as riveted, welded and bolted joints. Shafts, Spur gears, rolling and sliding contact bearings, Brakes and clutches, flywheels.

11. Manufacturing ,Industrial and Maintenance Engineering:

Metal casting-Metal forming, Metal Joining, Machining and machine tool operations, Limits, fits and tolerances, Metrology and inspection, computer Integrated manufacturing, FMS, Production planning and Control, Inventory control and operations research – CPM-PERT. Failure concepts and characteristics-Reliability, Failure analysis, Machine Vibration, Data acquisition, Fault Detection, Vibration Monitoring, Field Balancing of Rotors, Noise Monitoring, Wear and Debris Analysis, Signature Analysis, NDT Techniques in Condition Monitoring.

12. Mechatronics and Robotics:

Microprocessors and Microcontrollers: Architecture, programming, I/O, Computer interfacing, Programmable logic controller. Sensors and actuators, Piezoelectric accelerometer, Hall effect sensor, Optical Encoder, Resolver, Inductosyn, Pneumatic and Hydraulic actuators, stepper motor, Control Systems - Mathematical modeling of Physical systems, control signals, controllability and observability. Robotics, Robot Classification, Robot Specification, notation; Direct and Inverse Kinematics; Homogeneous Coordinates and Arm Equation of four Axis SCARA Robot.

GATE MECHANICAL ENGINEERING

Section 1: Engineering Mathematics

Linear Algebra: Matrix algebra, systems of linear equations, eigen values and eigenvectors.

Calculus: Functions of single variable, limit, continuity and differentiability, mean value theorems, indeterminate forms; evaluation of definite and improper integrals; double and triple integrals; partial derivatives, total derivative, Taylor series (in one and two variables), maxima and minima, Fourier series; gradient, divergence and curl, vector identities, directional derivatives, line, surface and volume integrals, applications of Gauss, Stokes and Green's theorems.

Differential equations: First order equations (linear and nonlinear); higher order linear differential equations with constant coefficients; Euler-Cauchy equation; initial and boundary value problems; Laplace transforms; solutions of heat, wave and Laplace's equations.

Complex variables: Analytic functions; Cauchy-Riemann equations; Cauchy's integral theorem and integral formula; Taylor and Laurent series.

Probability and Statistics: Definitions of probability, sampling theorems, conditional probability; mean, median, mode and standard deviation; random variables, binomial, Poisson and normal distributions.

Numerical Methods: Numerical solutions of linear and non-linear algebraic equations; integration by trapezoidal and Simpson's rules; single and multi-step methods for differential equations.

Section 2: Applied Mechanics and Design

Engineering Mechanics: Free-body diagrams and equilibrium; friction and its applications including rolling friction, belt-pulley, brakes, clutches, screw jack, wedge, vehicles, etc.; trusses and frames; virtual work; kinematics and dynamics of rigid bodies in plane motion; impulse and momentum (linear and angular) and energy formulations; Lagrange's equation.

Mechanics of Materials: Stress and strain, elastic constants, Poisson's ratio; Mohr's circle for plane stress and plane strain; thin cylinders; shear force and bending moment diagrams; bending and shear stresses; concept of shear centre; deflection of beams; torsion of circular shafts; Euler's theory of columns; energy methods; thermal stresses; strain gauges and rosettes; testing of materials with universal testing machine; testing of hardness and impact strength.

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Theory of Machines: Displacement, velocity and acceleration analysis of plane mechanisms; dynamic analysis of linkages; cams; gears and gear trains; flywheels and governors; balancing of reciprocating and rotating masses; gyroscope.

Vibrations: Free and forced vibration of single degree of freedom systems, effect of damping; vibration isolation; resonance; critical speeds of shafts.

Machine Design: Design for static and dynamic loading; failure theories; fatigue strength and the SN diagram; principles of the design of machine elements such as bolted, riveted and welded joints; shafts, gears, rolling and sliding contact bearings, brakes and clutches, springs.

Section 3: Fluid Mechanics and Thermal Sciences

Fluid Mechanics: Fluid properties; fluid statics, forces on submerged bodies, stability of floating bodies; control-volume analysis of mass, momentum and energy; fluid acceleration; differential equations of continuity and momentum; Bernoulli's equation; dimensional analysis; viscous flow of incompressible fluids, boundary layer, elementary turbulent flow, flow through pipes, head losses in pipes, bends and fittings; basics of compressible fluid flow.

Heat-Transfer: Modes of heat transfer; one dimensional heat conduction, resistance concept and electrical analogy, heat transfer through fins; unsteady heat conduction, lumped parameter system, Heisler's charts; thermal boundary layer, dimensionless parameters in free and forced convective heat transfer, heat transfer correlations for flow over flat plates and through pipes, effect of turbulence; heat exchanger performance, LMTD and NTU methods; radiative heat transfer, Stefan- Boltzmann law, Wien's displacement law, black and grey surfaces, view factors, radiation network analysis

Thermodynamics: Thermodynamic systems and processes; properties of pure substances, behavior of ideal and real gases; zeroth and first laws of thermodynamics, calculation of work and heat in various processes; second law of thermodynamics; thermodynamic property charts and tables, availability and irreversibility; thermodynamic relations.

Applications: Power Engineering: Air and gas compressors; vapour and gas power cycles, concepts of regeneration and reheat. I.C. Engines: Air-standard Otto, Diesel and dual cycles. Refrigeration and air-conditioning: Vapour and gas refrigeration and heat pump cycles; properties of moist air, psychrometric chart, basic psychrometric processes. Turbomachinery: Impulse and reaction principles, velocity diagrams, Pelton-wheel, Francis and Kaplan turbines; steam and gas turbines.

Section 4: Materials, Manufacturing and Industrial Engineering

Engineering Materials: Structure and properties of engineering materials, phase diagrams, heat treatment, stress-strain diagrams for engineering materials.

Casting, Forming and Joining Processes: Different types of castings, design of patterns, moulds and cores; solidification and cooling; riser and gating design. Plastic deformation and yield criteria; fundamentals of hot and cold working processes; load estimation for bulk (forging, rolling, extrusion, drawing) and sheet (shearing, deep drawing, bending) metal forming processes; principles of powder metallurgy. Principles of welding, brazing, soldering and adhesive bonding.

Machining and Machine Tool Operations: Mechanics of machining; basic machine tools; single and multi-point cutting tools, tool geometry and materials, tool life and wear; economics of machining; principles of non-traditional machining processes; principles of work holding, jigs and fixtures; abrasive machining processes; NC/CNC machines and CNC programming.

Metrology and Inspection: Limits, fits and tolerances; linear and angular measurements; comparators; interferometry; form and finish measurement; alignment and testing methods; tolerance analysis in manufacturing and assembly; concepts of coordinate-measuring machine (CMM).

Computer Integrated Manufacturing: Basic concepts of CAD/CAM and their integration tools; additive manufacturing.

Production Planning and Control: Forecasting models, aggregate production planning, scheduling, materials requirement planning; lean manufacturing.

Inventory Control: Deterministic models; safety stock inventory control systems.

Operations Research: Linear programming, simplex method, transportation, assignment, network flow models, simple queuing models, PERT and CPM.

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Mechanical Engineering AE & PSC Previous Exam Papers Analysis Chart

Sl No.	Exam	Proposed Year	Question Paper	Total Question
Union Public Service Commission				
1.	UPSC ESE (PRE)	18.07.2021		150
2.	UPSC ESE (PRE)	05.01.2020		150
3.	UPSC ESE (PRE)	06.01.2019		150
4.	UPSC ESE (PRE)	07.01.2018		150
5.	UPSC ESE (PRE)	08.01.2017		150
6.	UPSC ESE	1992-2016		5400
7.	UPSC CSE (PRE)	1994-2010		1200
8.	UPSC JWM	03.09.2017		100
9.	UPSC JWM Advt. No.- 50/2010	2010		100
10.	UPSC JWM Advt. No.- 52/2010	2010		100
11.	UPSC JWM Advt. No.- 16/2009	2009		100
Uttar Pradesh Public Service Commission				
12.	UPPSC AE (Paper I)	13.12.2020		100
13.	UPPSC AE (Paper II)	13.12.2020		100
14.	UPPSC AE (Paper I)	12.04.2016		100
15.	UPPSC AE (Paper II)	12.04.2016		100
Uttar Pradesh JAL NIGAM				
16.	UP JAL NIGAM AE	2016		50
Uttar Pradesh Rajya Vidyut Utpadan Nigam Limited				
17.	UPRVUNL AE	05.07.2021		150
18.	UPRVUNL AE (UPJVUNL)	07.10.2016		150
19.	UPRVUNL AE	21.08.2016		150
20.	UPRVUNL AE	2014		150
Uttarakhand Public Service Commission				
21.	UKPSC AE (Paper I)	2013		180
22.	UKPSC AE (Paper II)	2013		180
23.	UKPSC AE (Paper I)	2012		200
24.	UKPSC AE (Paper II)	2012		200
25.	UKPSC AE (Paper I)	2007		200
26.	UKPSC AE (Paper II)	2007		200
UJVNL / PTCUL/UPCL Uttarakhand				
27.	PTCUL AE	25.06.2017		126
28.	UJVNL AE	08.05.2016		126
Assam PSC				
29.	Assam PSC AE (IWT)	14.03.2021		100
30.	Assam PSC AE (PHED)	18.10.2020		100
31.	Assam PSC CCE Pre	2015		100
32.	APGCL Assistant Manager	31.01.2021		60
Andhra Pradesh PSC				
33.	APPSC Polytechnic Lect.	13.03.2020		150
34.	APPSC Polytechnic Lect. (Auto.)	12.03.2020		150
35.	APPSC AEE Screening	17.02.2019		100
36.	APPSC AEE (Mains) Mechanical	2016		150
37.	APPSC AEE Screening	2016		100
38.	APPSC AEE (Mains) Civil/Mechanical	2016		150
39.	APPSC AE Subordinate Services (Civil/Mechanical)	2016		150

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40.	APPSC Polytechnic Lect.	2012		150
41.	APPSC AEE (Civil/Mechanical)	2012		150
42.	APPSC AEE Paper - III (Mechanical)	2012		150
43.	APPSC AE(PHE)	04.12.2012		150
44.	APPSC Inspector of Factories	2009		150
45.	APPSC AE	2011		150
46.	APGENCO AE	2012		70
47.	APGENCO AE	2017		70
Airports Authority of India				
48.	AAI Junior Executive	29.11.2018		84
49.	AAI Junior Executive	26.03.2021		84
BHEL ET				
50.	BHEL ET	25.05.2019		120
Bihar Public Service Commission				
51.	BPSC AE (Paper V)	08.08.2019		50
52.	BPSC AE (Paper VI)	09.08.2019		50
53.	BPSC Poly. Tech. Lect.	24.07.2016		60
54.	BPSC Asst. Prof.	29.11.2015		60
55.	BPSC AE (Paper V)	04.12.2012		50
56.	BPSC AE (Paper VI)	05.12.2012		50
Chhattishgarh PSC				
57.	CGPSC PSC AE	15.01.2021		150
58.	CGPSC PSC AE Shift-I	26.04.2014		150
59.	CGPSC PSC AE Shift-II	26.04.2014		150
60.	CGPSC AE Shift -II	25.02.2018		150
61.	CGPSC Poly. Lect.	13.10.2017		100
62.	CGPSC AE	16.10.2016		150
63.	CGPSC Poly. Lect.	22.05.2015		100
64.	CGPSC ARTO & TSI	07.01.2017		100
65.	CGPSC Asstt. Workshop Suptd.	17.07.2016		100
Coal India Ltd.				
66.	CIL MT	27.02.2020		100
67.	CIL MT	26.03.2017		100
Defence Research and Development				
68.	DRDO Scientist	2009		100
69.	DRDO Scientist	2008		100
Graduate Aptitude Test in Engineering (GATE)				
70.	GATE	1987-2021		1000
Gujarat Public Service Commission				
71.	GPSC DEE, Class-II (GWSSB)	04.07.2021		300
72.	GPSC Engineer, Class-II Pre	19.01.2020		80
73.	GPSC AE, Class-II	23.06.2019		200
74.	GPSC EE, Class-I, Pre	28.01.2017		100
75.	GPSC ARTO/Asstt. Director of Transport, Class-II Pre	30.12.2018		200
76.	GPSC AE	26.07.2015		100
77.	GPSC Asstt. Prof.	28.08.2016		100
78.	GPSC Poly. Lect	23.10.2016		100
79.	GPSC ARTO, Class-II	01.05.2016		200
80.	GPSC Asstt. Prof. (Production)	21.08.2016		100
81.	GMB AAE	25.06.2017		100

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82.	GPSC Asstt. Director of Transport	05.03.2017		200
83.	GPSC Poly. Lect. (Automobile)	16.10.2016		100
84.	GPSC Poly. Lect	07.12.2014		100
85.	GWSSB DEE	22.05.2016		100
86.	GWSSB AAE	17.06.2018		100
87.	GWSSB AAE	27.12.2015		100
88.	Vadodara Municipal Corporation DEE	13.05.2018		80
Haryana PSC				
89.	HPSC AE (PHED) Paper -I	04.09.2020		100
90.	HPSC AE (PHED) Paper -II	05.09.2020		100
91.	HPSC Civil Services Pre	2014		100
Hindustan Petroleum Corporation Limited				
92.	HPCL Engineer Shift-I	12.08.2021		85
93.	HPCL Engineer Shift-II	12.08.2021		85
94.	HPCL (HRRL) (E1 & E2)	07.08.2021		85
Himachal Pradesh PSC				
95.	HPPS Poly. Lect.	05.07.2021		80
96.	HPPSC Workshop Superintendent	08.07.2021		80
97.	HPPSC ADF	05.03.2019		80
98.	HPPSC AE	26.07.2018		80
99.	HPPSC Asstt. Prof.	18.02.2017		80
100.	HPPSC Asstt. Prof. (TE)	20.11.2017		80
101.	HPPSC AE (PWD)	10.08.2016		80
102.	HPPSC Asstt. Prof.	18.11.2016		80
103.	HPPSC Asstt. Prof. (TE)	29.10.2016		80
104.	HPPSC Poly. Lect.	10.07.2016		80
105.	HPPSC Workshop Superintendent	12.02.2015		80
106.	HPPSC AE	21.12.2014		80
107.	HPPSC Asstt. Prof. (TE)	11.02.2015		80
108.	HPPSC ITI Principal	2015		50
ISRO Scientist / Engineer				
109.	ISRO Scientist / Engineer	12.01.2020		70
110.	ISRO Scientist / Engineer	22.04.2018		70
111.	ISRO Scientist / Engineer	27.05.2017		70
112.	ISRO Scientist / Engineer	17.12.2017		70
113.	ISRO Scientist / Engineer	03.07.2016		70
114.	ISRO Scientist / Engineer	11.10.2015		70
115.	ISRO Scientist / Engineer	24.05.2014		70
116.	ISRO Scientist / Engineer	12.05.2013		70
117.	ISRO Scientist / Engineer	2012		70
118.	ISRO Scientist / Engineer	2011		70
119.	ISRO Scientist / Engineer	2010		70
120.	ISRO Scientist / Engineer	2009		70
121.	ISRO Scientist / Engineer	2008		70
122.	ISRO Scientist / Engineer	2007		70
123.	ISRO Scientist / Engineer	2006		70
124.	ISRO Scientist / Engineer (RAC)	12.01.2020		70
125.	ISRO Scientist / Engineer (RAC)	10.03.2019		70
126.	ISRO Scientist / Engineer (RAC)	22.04.2018		70
127.	ISRO Scientist / Engineer (RAC)	07.05.2017		70
128.	ISRO Scientist / Engineer (RAC)	29.11.2015		70
129.	ISRO Scientist / Engineer (RAC)	24.05.2014		70
130.	ISRO Scientist / Engineer (RAC)	2013		70

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Jharkhand Public Service Commission				
131.	JPSC AE Paper -I	10.04.2021		100
132.	JPSC AE Paper –II	10.04.2021		100
133.	JPSC AE Pre	19.01.2010		100
134.	JPSC AE (Paper V)	2013		50
135.	JPSC AE (Paper VI)	2013		50
136.	JUVNL AEE (GTO)	05.03.2017		75
Jammu & Kashmir PSC				
137.	JK PSC Civil Services Pre	2013		120
138.	JK PSC Civil Services Pre	2010		120
139.	JK PSC Civil Services Pre	2006		120
Karnataka Public Service Commission				
140.	Karnataka PSC AE (WRD)	31.07.2021		100
141.	Karnataka PSC AE	10.09.2017		100
142.	Karnataka PSC ADF	2015		100
143.	Karnataka PSC Poly. Lect.	27.05.2017		100
144.	KPCL AE	2016		100
Kerala PSC				
145.	Kerala PSC Asstt. Manager	09.08.2021		80
146.	Kerala PSC AE	05.04.2017		80
147.	Kerala PSC AE	31.12.2015		80
148.	Kerala PSC AE	07.07.2015		80
149.	Kerala PSC AE	06.08.2015		80
150.	Kerala PSC Poly. Lect.	14.08.2016		80
151.	Kerala PSC Asstt. Prof.	27.09.2016		80
152.	Kerala PSC IOF	19.04.2016		80
153.	Kerala PSC Poly. Lect.	09.12.2014		80
154.	Kerala PSC Poly. Lect. (Automobile)	05.05.2016		80
155.	Lal Bahadur Shastri Centre For Science & Technology (Kerala) Asstt. Prof.	2014		100
Maharastra Public Service Commission				
156.	MPSC Poly. Lect. (Production)	2018		80
157.	MPSC Poly Lect.	2014		70
158.	MPSC (Deputy Engineer) (GWD)	2014		100
159.	MPSC (Deputy Engineer)	19.05.2013		80
160.	MPSC (HOD) Govt. Poly. Colleges	04.10.2014		80
Madhya Pradesh PSC				
161.	MPPSC Forest Service Exam	12.01.2016		150
162.	MPPSC AE	2016		100
163.	MPPSC AE	18.11.2015		100
Mizoram PSC				
164.	Mizoram PSC (Insp. of Factories) P-III	2019		50
165.	Mizoram PSC AE/SDO Paper-I	2014		50
166.	Mizoram PSC AE/SDO Paper-II	2014		50
167.	Mizoram PSC AE/SDO Paper-III	2014		50
Nagaland PSC				
168.	Nagaland PSC CTSE Paper-I	2018		200
169.	Nagaland PSC CTSE Paper-II	2018		200
170.	Nagaland PSC CTSE Paper-I	2017		200
171.	Nagaland PSC CTSE Paper-II	2017		200
172.	Nagaland PSC CTSE Paper-I	2016		200
173.	Nagaland PSC CTSE Paper-II	2016		200
174.	Nagaland PSC CTSE Paper-I	2015		200
175.	Nagaland PSC CTSE Paper-II	2015		200

Metallurgical & Engineering Consultant (India) Limited				
176.	MECON MT	17.03.2019		100
Nuclear Power Corporation of India Limited				
177.	NPCIL ET	2015		100
NLC India Limited				
178.	NLC GET Shift - I	17.11.2020		80
179.	NLC GET Shift - II	17.11.2020		80
Odisha Public Service Commission				
180.	OPSC AEE Paper-I	15.12.2019		180
181.	OPSC AEE Paper-II	15.12.2019		180
182.	OPSC AEE Paper-I	2015		90
183.	OPSC AEE Paper-II	2015		90
184.	OPSC Civil Services Pre	2011		120
185.	OPSC Civil Services Pre	2006		120
Oil India Limited				
186.	Senior Officer	23.12.2020		60
187.	Sr. Engineer (Drilling)	30.11.2019		60
188.	Sr. Engineer (Production)	30.11.2019		60
Punjab PSC				
189.	Punjab PSC Asstt. Municipal Engineer	15.06.2021		100
190.	Punjab PSC Poly. Lect.	06.08.2017		100
191.	Punjab PSC SDE	12.02.2017		100
192.	PSPCL AE	2012		60
Rajasthan Public Service Commission				
193.	RPSC Lecturer (Tech. Edu. Dept.) P-I	13.03.2021		150
194.	RPSC Lecturer (Tech. Edu. Dept.) P-II	13.03.2021		150
195.	RPSC ACF & FRO Gr.-I, P-III	26.02.2021		120
196.	RPSC Inspector Factory & Boilers	25.11.2020		200
197.	RPSC AE	17.12.2018		100
198.	RPSC VP ITI	06.11.2019		100
199.	RPSC Inspector Factory & Boilers	20.09.2017		100
200.	Rajasthan AE (Nagar Nigam) Shift –I	2016		84
201.	Rajasthan AE (Nagar Nigam) Shift –II	2016		84
202.	Rajasthan AE (Nagar Nigam) Shift –III	2016		84
203.	RPSC Lecturer(Tech. Edu. Dept.)	16.01.2016		100
204.	RPSC AE GWD	06.05.2016		100
205.	RPSC AE GWD	2011		100
206.	RPSC AE Drilling	05.05.2016		100
207.	RPSC AE	2013		100
208.	RPSC VP ITI	14.02.2016		100
209.	RPSC ACF	2011		200
210.	RPSC Lecturer(Tech. Edu. Dept.)	2011		100
RRB SSE				
211.	RRB SSE Shift-I	01.09.2015		16
212.	RRB SSE Shift-II	01.09.2015		15
213.	RRB SSE Shift-III	01.09.2015		15
214.	RRB SSE Shift-I	02.09.2015		15
215.	RRB SSE Shift-II	02.09.2015		15
216.	RRB SSE Shift-III	02.09.2015		15
217.	RRB SSE Shift-I	03.09.2015		16
218.	RRB SSE Shift-II	03.09.2015		15
219.	RRB SSE Shift-III	03.09.2015		15
220.	RRB SSE (Set Green)	21.12.2014		16

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221.	RRB SSE (Set Yellow)	21.12.2014		15
222.	RRB SSE (Set Red)	21.12.2014		15
SJVNL ET				
223.	SJVNL ET	28.05.2019		120
224.	SJVNL ET	06.11.2013		120
Sikkim PSC				
225.	Sikkim PSC AE	2017		75
Tamilnadu PSC				
226.	TNCSC AE	2020		25
227.	TANGEDCO AE	2015		60
228.	TNPSC ACF	2012		200
229.	TNPSC AE	2019		200
230.	TNPSC AE	2018		200
231.	TNPSC AE	2017		200
232.	TNPSC AE	2014		200
233.	TNPSC AE	2013		200
234.	TNPSC AE (Industries)	2013		200
235.	TNPSC AE	2012		200
236.	TNPSC AE	2008		200
237.	Tamilnadu TRB Poly. Tech. Lect.	2017		150
238.	Tamilnadu TRB Poly. Tech. Lect.	2012		150
239.	Tamilnadu TRB Asstt. Prof.	2012		150
Telangana PSC				
240.	TSPSC Manager (Engg.) in HMWSSB	12.11.2020		150
241.	TSPSC AEE (Civil/Mechanical)	28.08.2017		150
242.	TSPSC AEE	27.08.2017		150
243.	TSPSC AEE	18.10.2015		150
244.	TSPSC AE	2015		150
245.	TSPSC Manager (Engg.)	2015		150
TSGENCO				
246.	TSGENCO AE	14.11.2015		80
Vizag Steel MT				
247.	Vizag Steel MT Shift-I	24.01.2021		100
248.	Vizag Steel MT Shift-II	24.01.2021		100
249.	Vizag Steel MT	14.12.2020		100
250.	Vizag Steel MT	25.06.2017		100
251.	Vizag Steel MT	26.07.2015		100
252.	Vizag Steel MT (Re-Exam)	24.11.2013		100
253.	Vizag Steel MT	18.08.2013		100
254.	Vizag Steel MT	10.06.2012		100
255.	Vizag Steel MT	15.05.2011		100
West Bengal PSC				
256.	WBPSC AE	2017		200
257.	WBPSC AE	2012		100
258.	WBPSC AE	2008		100
259.	WBPSC AE	2007		100
260.	WBPSC AE	2003		100
			Total	**35125**

Note: After detailed analysis of above mentioned papers of UPSC and State Engineering Services Examination related to Mechanical Engineering **35125** (Volume-I + II) have been presented chapterwise. Questions of repeated and similar nature have been included so that the thechnique of asking question can benefit the competitors.

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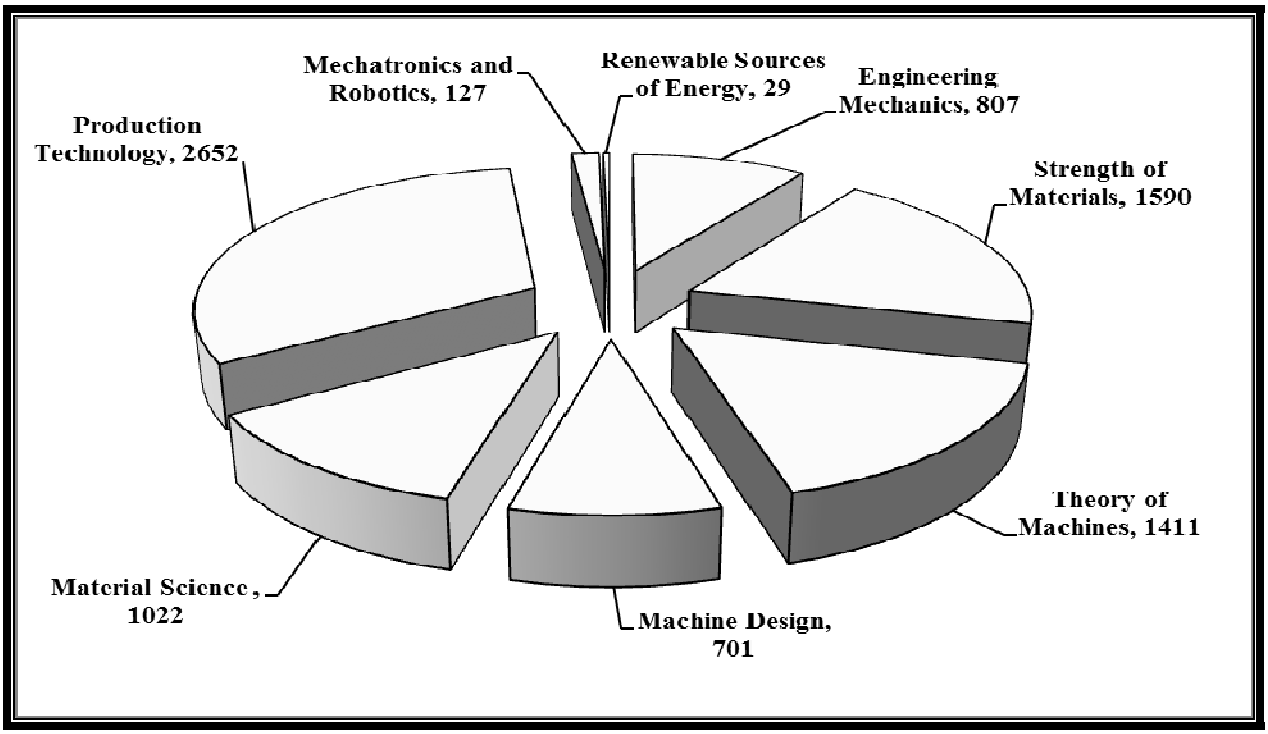
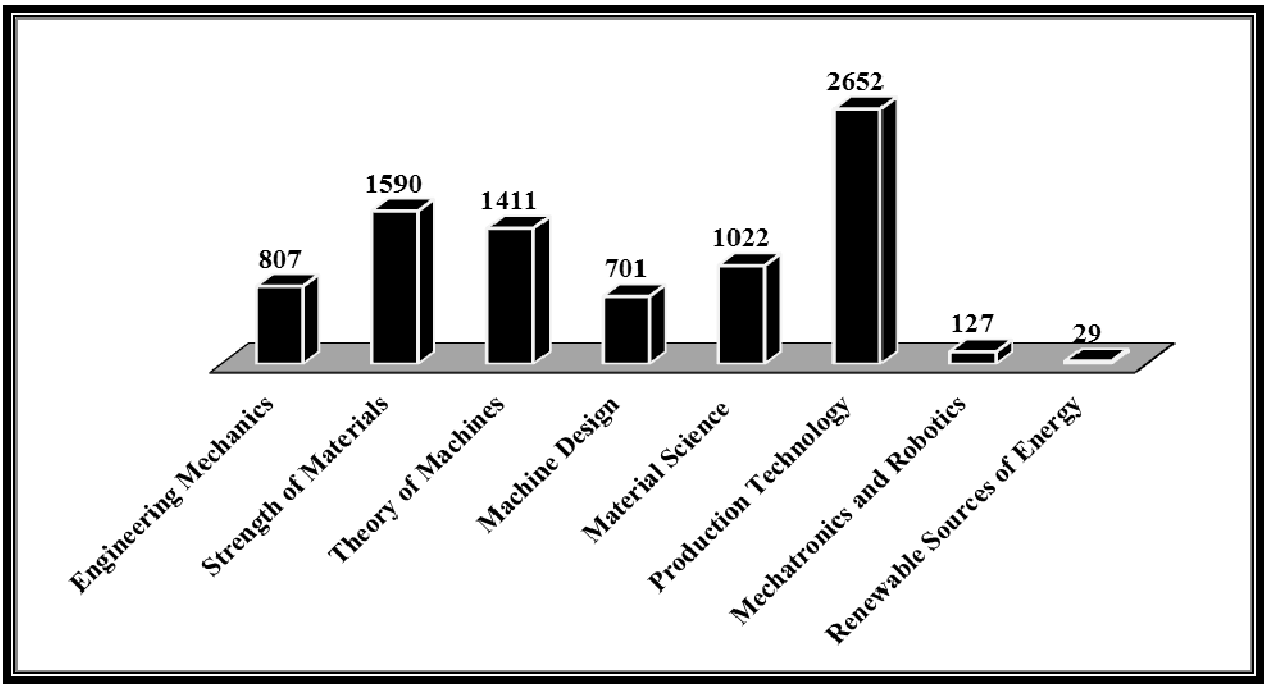
Trend Analysis of Mechanical Engineering Through Pie Chart and Bar Graph





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01.

ENGINEERING MECHANICS

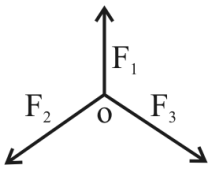
1. Forces and Force Systems

1. The forces, which meet at one point and their lines of action also lie on the same plane, are known as

(a) coplanar concurrent forces
(b) coplanar non-concurrent forces
(c) non-coplanar concurrent forces
(d) non-coplanar non-concurrent forces

Assam PSC AE (IWT) 14.03.2021
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
KPCL AE 2016
MPPSC State Forest Service Exam, 2014

Ans. (a) : Coplanar Concurrent Forces–The forces which meet at one point and their lines of action also lie on the same plane, are called coplanar concurrent forces.



Coplanar Non-Concurrent Forces–The forces which do not meet at one point but their lines action lie on the same plane are known as coplanar non-concurrent forces.

Non-Coplanar Concurrent Forces–The forces which meet at one point but their line of action do not lie on the same plane are known as non coplanar concurrent forces.

Non Coplanar Non Concurrent Forces–The forces which do not meet at one point and their line of actions also do not lie on the same plane.

2. Non coplanar concurrent forces are those where

(a) All forces do not lie in one plane and their lines of action intersect at more than one point
(b) All forces are parallel to each other and lie in single plane
(c) All forces do not lie in one plane but their lines of action pass through one point
(d) All forces lie in one plane but their lines of action pass through one point

CGPSC AE 15.01.2021
Rajasthan Nagar Nigam AE 2016, Shift-II
VIZAG STEEL MT - 2011

Ans. (c) : Non Coplanar Forces : Those forces which are not acting from a same plane.

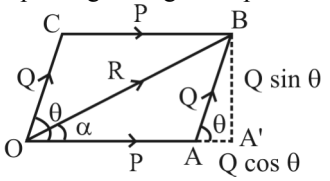
Concurrent Forces– A set of points forces is considered concurrent if all the lines of action of those forces meet come together at single point.
Coplanar Forces–Those forces which are acting in same plane.

3. If the resultant of two forces P and Q acting at an angle θ, makes an angle α with the force P, then

(a) $\tan\alpha = P\sin\theta / (P + Q\cos\theta)$
(b) $\tan\alpha = P\cos\theta / (P + Q\cos\theta)$
(c) $\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$
(d) $\tan\alpha = Q\cos\theta / (P + Q\cos\theta)$

GPSC Engineer, Class – II Pre-19.01.2020
HPPSC AE 2018
APPSC AE Subordinate Service Civil/Mech. 2016
APPSC AEE Mains (Civil Mechanical) 2016
Assam PSC CCE Pre 2015
CSE Pre-1998

Ans. (c) : Parallelogram Law of Forces– "If two forces acting at a point can be represented in magnitude and direction by the two adjacent sides of parallelogram then their resultant is represented in magnitude and direction by the two adjacent sides of parallelogram then their resultant is represented in magnitude and direction by the diagonal of the parallelogram passing through that point."



If the angle between the force is θ, then

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

The direction of resultant with P will be

$$\alpha = \tan^{-1}\left[\frac{Q\sin\theta}{P + Q\cos\theta}\right]$$

4. Which of the following states, "If a number of coplanar forces are acting simultaneously on a particle, the algebraic sum of the moments of all the forces about any point is equal to the moment of their resultant force about the same point"?

(a) Lami's theorem
(b) Varignon's principle
(c) Principle of super-position
(d) D' Alembert Principle

VIZAG MT, 14.12.2020
HPPSC AE 2018
Vizag Steel (MT) 2017
TSPSC AEE 2015
Kerala LBS Centre For Science & Tech. Asstt. Prof. 2014

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Ans. (b) : Varignon's Theorem–It states that if a number of coplanar forces are acting simultaneously on a particle the algebraic sum of the moments of all the forces about any point is equal to the moment of their resultant force about the same point.

5. According to Lami's theorem :

(a) three forces acting at a point will be in equilibrium

(b) three forces acting at a point can be represented by a triangle, each side being proportional to force.

(c) If three forces acting at a point are in equilibrium, each force is proportional to the sine of the angle between the other two

(d) If three forces acting upon a particle are represented in magnitude and direction by the side of a triangle, taken in order, they will be in equilibrium.

CGPSC AE 15.01.2021
WBPSC AE, 2017
GPSC Executive Engg. 23.12.2018

Ans. (c) : According to Lami's theorem,

If three forces acting at a point are in equilibrium each force is proportional to the sine of the angle between the other two.

6. The Lami's Theorem is applicable only for _____ force system.

(a) Non-coplanar and Non-concurrent

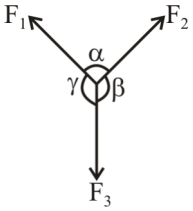
(b) Coplanar and concurrent

(c) Non-coplanar and concurrent

(d) Coplanar and non-concurrent

VIZAG Steel MT 24.01.2021
TRB Polytechnic Lecturer 2017
GMB AAE 25.06.2017

Ans. (b) : The Lami's theorem is applicable only for coplanar and concurrent.



$$\frac{F_1}{\sin\beta} = \frac{F_2}{\sin\gamma} = \frac{F_3}{\sin\alpha}$$

7. If the resultant of two equal forces has the same magnitude as either of the forces, then the angle between the two forces is :

(a) 30° (b) 60°

(c) 90° (d) 120°

HPPSC Poly. Lect. 05.07.2021
Vadodara Muncipal Corp. DEE, 2018
PTCUL AE 25.06.2017
Assam PSC CCE Pre 2015
TSPSC Managers, 2015
Kerala LBS Centre For Science & Tech. Asstt. Prof. 2014
J&K PSC Civil Services Pre, 2010

Ans. (d) : Given, Resultant of two forces has same magnitude

$$R = F$$

or $F_1 = F_2 = F$

We know that,

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$R = \sqrt{F^2 + F^2 + 2F^2\cos\theta}$$

$$F = \sqrt{2F^2 + 2F^2\cos\theta}$$

$$2F^2 + 2F^2\cos\theta = F^2$$

$$2F^2\cos\theta = F^2 - 2F^2$$

$$2F^2\cos\theta = -F^2$$

$$\cos\theta = -\frac{1}{2} = \cos 120°$$

$$\boxed{\theta = 120°}$$

8. A force is completely defined when we specify

(a) Magnitude (b) Direction

(c) Point of application (d) All of the above

CGPSC AE 15.01.2021
APPSC Poly. Lect Automobile Engg. 2020
OPSC AEE 2015 PAPER-I
UKPSC AE 2007 Paper -I

Ans. (d) : Force is a vector quantity

A force is completely defined when we specify

(i) Magnitude

(ii) Direction

(iii) Point of Application

9. A force which combines two or more forces to produce equilibrium is called

(a) resultant (b) equilibrant

(c) couple (d) moment

HPPSC Poly. Lect. 05.07.2021
Assam PSC CCE Pre 2015

Ans. (b) : A force which combines two or more forces to produce equilibrium is called equilibrant.

10. According to principle of transmissibility of forces, the effect of a force upon a body is

(a) maximum when it acts at the centre of gravity of a body

(b) different at different points in its line of action

(c) the same at every point in its line of action

(d) minimum when it acts at the C.G. of the body

Karnataka PSC AE (WRD) 31.07.2021
GPSC DEE, Class-II (GWSSB) 04.07.2021
TRB Asstt. Prof., 2012
UKPSC AE 2013 Paper-I

Ans. (c) : Principal of transmissibility–

The state of rest or of motion of a rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting anywhere on the body along the line of action of the applied forces.

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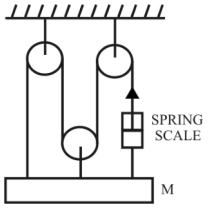
11. A spring scale indicates a tension T in the right-hand cable of the pulley system shown in the figure. Neglecting the mass of the pulleys and ignoring friction between the cable and the pulley the mass M is :

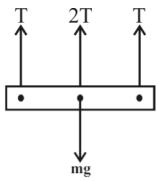


(a) $\frac{2T}{g}$ (b) $\frac{T(1+e^{4x})}{g}$

(c) $\frac{4T}{g}$ (d) None of the above

Vadodara Muncipal Corp. DEE, 2018
UPRVUNL AE 2014
OPSC Civil Services Pre. 2011
GATE 1995

Ans. (c) :



By resolving vertical forces, $\Sigma F_v=0$

$\Rightarrow T + 2T + T = mg$

$\Rightarrow 4T = mg$

$\Rightarrow \boxed{m = \frac{4T}{g}}$

12. Two equal forces equal to P inclined at θ will have their resultant with the angle with one of the force as:

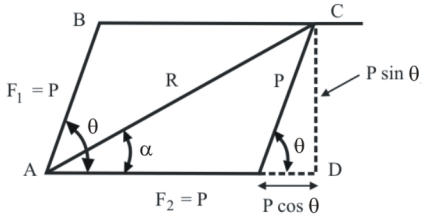
(a) 2 P cos $\frac{\theta}{2}$ and $\frac{\theta}{2}$ (b) 2 P cos $\frac{\theta}{2}$ and 2θ

(c) P cosθ and θ (d) P cos $\frac{\theta}{2}$ and $\frac{\theta}{2}$

APPSC Poly Lect. 13.03.2020
TNPSC AE 2018
JUVNL AEE (GTO) 05.03.2017
J&K PSC Civil Services Pre, 2013
VIZAG STEEL MT 2011
WBPSC AE 2003
CGPSC AE 26.04.2014 Shift-I

Ans. (a) : $F_1 = F_2 = P$

The angle between F_1 & F_2 is θ and angle between Resultant & force is α.



from $\angle ACD \rightarrow$

$(AC)^2 = (AD)^2 + (CD)^2$

$R^2 = (P + P\cos\theta)^2 + (P\sin\theta)^2$

$R^2 = [P^2 + P^2\cos^2\theta + 2P^2\cos\theta] + P^2\sin^2\theta$

$= P^2 + 2P^2\cos\theta + P^2(\sin^2\theta + \cos^2\theta)$

$= P^2 + 2P^2\cos\theta + P^2 \quad [\because \sin^2\theta + \cos^2\theta = 1]$

$= 2P^2 + 2P^2\cos\theta$

$= 2P^2(1 + \cos\theta) \quad [\because \cos\theta = 2\cos^2\theta/2 - 1]$

$= 2P^2(1 + 2\cos^2\theta/2 - 1)$

$= 4P^2\cos^2\theta/2$

$R = \sqrt{4P^2\cos^2\theta/2}$

$\boxed{R = 2P\cos\theta/2}$

$\tan\alpha = \frac{CD}{[AE + ED]} = \frac{P\sin\theta}{P + P\cos\theta} = \frac{\sin\theta}{1 + \cos\theta}$

$\tan\alpha = \frac{2\sin\theta/2\cos\theta/2}{1 + 2\cos^2\theta/2 - 1} \quad \because \sin\theta = 2\sin\theta/2\cos\theta/2$

$= \frac{\sin\theta/2}{\cos\theta/2}$

$\tan\alpha = \tan\theta/2$

$\boxed{\alpha = \theta/2}$

13. A system of masses rotating in different parallel planes is in dynamic balance if-

(a) Resultant force is zero
(b) resultant couple is zero
(c) resultant force in numerically equal to resultant couple
(d) resultant force and the resultant couple both are equal to zero

ISRO Scientist/Engineer 12.05.2013
ISRO Scientist/Engineer 2006

Ans. (d) : Dynamic Balance:-When a system of masses rotating in different parallel planes.

Its, $\Sigma F = 0, \Sigma M = 0$

Static Balance:-When a system of mass rotating in same plane.

$\Sigma F = 0$ only.

14. If a distributed force system on a beam is replaced by its statically equivalent force system, which of the following is same for both the beams?

(a) Support reactions
(b) Shear force diagram
(c) Bending moment diagram
(d) Maximum bending moment

UPPSC AE 13.12.2020, Paper-I

Ans. (a) : Support reactions.

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15. The free body diagram of a body shows the body ..... .

(a) with its surrounding and external forces acting on it.
(b) isolated from all external effects.
(c) isolated from its surrounding.
(d) isolated from its surroundings and all external actions acting upon it.

GWSSB DEE 07.07.2016

Ans. (d) : The free body diagram of a body shows the body isolated from its surroundings and all external actions acting upon it.

16. Dimensions for moment of momentum are

(a) $FL^{-1}T^{-1}$ (b) FLT
(c) FL^{-1} (d) FLT^{-1}

HPPSC Poly. Lect. 05.07.2021

Ans. (b) : Moment of Momentum = Angular momentum.
Angular momentum = mvr

$$= \text{kg-}\frac{\text{m}}{\text{sec}}\text{-m}$$

$$= \left(\frac{\text{kg - m}}{\text{sec}^2}\right)\text{m} \times \text{sec}$$

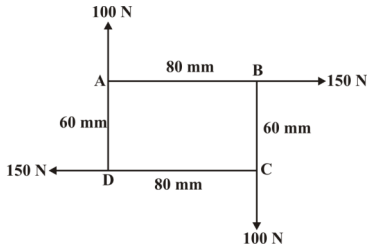
Moment of momentum = [F L T]

17. A rectangle ABCD has sides AB = CD = 80 mm and BC = DA = 60 mm. Forces of 150 N each act along AB and CD, and forces of 100 N each act along BC and DA. The resultant moment of the force system is

(a) 9000 Nmm (b) 8000 Nmm
(c) 1000 Nmm (d) 17000 Nmm

HPPSC Poly. Lect. 05.07.2021

Ans. (d) :



Resultant moment of force system,

$\Sigma M_D = 100 \times 80 + 150 \times 60$
$= 8000 + 9000$
$= 17000$ N-mm

18. Consider the following statements

1. Light year is a unit of time.
2. Momentum and impulse have the same units
3 A body falling freely under gravitational force has no weight

Which of these is / are false statement / statements?

(a) 1 only (b) 2 and 3
(c) 1 and 3 (d) 1 and 2

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : Light year is a unit of distance.

19. Given that $\vec{P}+\vec{Q}=\vec{R}$ and also P = Q = R. Then the angle between vectors $\vec{P}$ and $\vec{Q}$ is

(a) 0 (b) P
(c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{3}$

HPPSC Poly. Lect. 05.07.2021

Ans. (c) : Given,

$$P = Q = R$$

$$R = \sqrt{P^2 + Q^2 + 2PQ\ \cos\theta}$$

$$P = \sqrt{P^2 + P^2 + 2P^2\ \cos\theta}$$

$$P^2 = 2P^2\ (1 + \cos\theta)$$

$$\cos\theta = -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3} \text{ radian}$$

20. All of the following are correct statements, except

(a) Force is a vector quantity whereas displacement is a scalar quantity
(b) Space is a region which extends in all directions and contains everything in it
(c) The member of base and supplementary units in SI system are seven and two respectively
(d) The unit of amount of substance in SI system is called mole

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Ans. (a) : Force and displacement both are vector quantity, whereas distance is scalar quantity.

21. Which of the followings is the supplementary unit in SI system?

(a) ampere (b) candela
(c) radian (d) mole

HPPSC Poly. Lect. 05.07.2021

Ans. (c) : radian

22. The resultant of two forces P and Q is R. If one of the forces is reversed in direction, the resultant is S. Then for the identity $R^2 + S^2 = 2(P^2 + Q^2)$ to hold good,

(a) forces are inclined at 60° to each other
(b) forces are collinear
(c) forces act at right angles to each other
(d) forces can have any angle of inclination between them

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Ans. (d) : Given,

$$\vec{R} = \vec{P} + \vec{Q} \quad \text{.....(i)}$$

After reversing direction of one force (Q) we get

$$\vec{S} = \vec{P} - \vec{Q} \quad \text{.....(ii)}$$

Let angle between P & Q be θ
So, resultant

$$R^2 = P^2 + Q^2 + 2PQ\cos\theta \quad \text{.....(iii)}$$

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$S^2 = P^2 + (-Q)^2 - 2PQ\cos\theta$ .....(iv)

Adding equation (iii) & (iv)

$$\boxed{R^2 + S^2 = 2\left(P^2 + Q^2\right)}$$

So, the identity does not depends on angle θ.

Hence, forces can have any angle of inclination (θ) between them.

23. Identify the wrong statement about matter, particle and body

(a) A body exhibits no relative deformation between two parts when it is acted upon by a force system

(b) Matter is anything that occupies space, possesses mass and offer resistance to any external force

(c) A particle is a body of infinitely small volume and its mass is considered to be concentrated at a point

(d) A body has definite shape and consists of number of particles

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Ans. (a) : A body exhibits no relative deformation between two parts when it is acted upon by a force system

24. Newton's first law of motion gives the concept of

(a) Force (b) Inertia

(c) Energy (d) Work

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Ans. (b) : Inertia

25. All of the following are scalar quantities except

(a) Energy (b) Temperature

(c) Time (d) Displacement

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Ans. (d)

(i) Scalar quantities– Distance, speed, time, temperature etc.

(ii) Vector quantities– Displacement, force, acceleration etc.

26. For analysis purposes, a body can be modeled as rigid if:

(a) It has continuous distribution of mass.

(b) Its dimensions are small compared with the coordinates describing its motion

(c) Relative deformation between its parts is negligible

(d) It is characterized by some preferred configuration of its own

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Ans. (c) : For analysis purposes, a body can be modeled as rigid if Relative deformation between its parts is negligible.

27. Two forces of magnitude $\sqrt{2}$ N and 2 N act at origin, each making 60^0 and 30^0 respectively with positve x axis. The angle made by their resultant with positive x axis will be

(a) 12.36^0 (b) 42.36^0

(c) 72.36^0 (d) 57.36^0

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Ans. (b) : Given,

$F_1 = 2$ N

$F_2 = \sqrt{2}$ N

$F_x = F_1 \cos 30° + F_2 \cos 60°$

$= 2 \cos 30° + \sqrt{2} \cos 60°$

$F_x = 2.439$ N

$F_y = F_1 \sin 30° + F_2 \sin 60°$

$= 2 \sin 30° + \sqrt{2} \sin 60°$

$= 2.224$ N

The angle 'θ' made by resultant with positive x-axis,

$$\tan\theta = \frac{F_y}{F_x} = \frac{2.224}{2.439}$$

$\theta = \tan^{-1}(0.911)$

$\boxed{\theta = 42.36^0}$

28. A body is in equilibrium under action of three forces. The forces acting on it can be analyzed by

(a) Lami's theorem only

(b) Resolution of forces method only

(c) Either by Lami's theorem or by resolution of forces method

(d) None of the above

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Ans. (c) : A body is in equilibrium under action of three forces. The forces acting on it can be analyzed by either by Lami's theorem or by resolution of forces method.

29. A force of 2 N acts along positive y-axis and another force 2N acts along negative x-axis. The angle made by their resultant with positive x-axis will be

(a) 0^0 (b) 45^0

(c) 225^0 (d) 135^0

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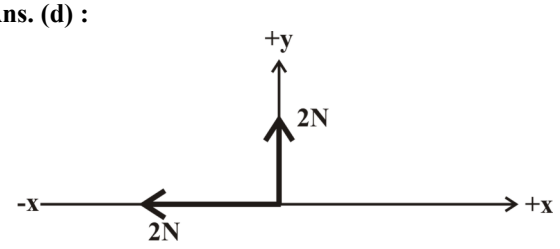
Ans. (d) :



$F_x = -2$N

$F_y = +2$N

The angle θ made by their resultant with positive x-axis will be,

$$\tan\theta = \frac{F_y}{F_x} = \frac{+2}{-2}$$

$\tan\theta = -1$

$\boxed{\theta = 135°}$

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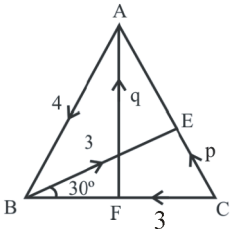
30. E and F are the mid points of sides CA and CB respectively of an equilateral triangle ABC. Forces 3, p, 3, 4, q are acting at A in the direction of the lines BE, CA, CB, AB and FA respectively.
(all forces are in Newtons)



Find the values of p and q in order that the system may be in equilibrium.

(a) p = –4.80, q = 6.12 (b) p = 4.80, q = –6.12
(c) p = 6.12, q = –4.80 (d) p = –6.12, q = 4.80

Karnataka PSC Lect., 27.05.2017

Ans (a) : $\Sigma F_v = 0$
$\Rightarrow q + p \sin60^\circ + 3 \sin30^\circ - 4 \sin60^\circ$
$\Rightarrow q + p \sin60^\circ = 1.964$
$\Rightarrow q + 0.866p = 1.964$…….(i)
$\Sigma F_H = 0$
$\Rightarrow 3 \cos30^\circ = p \cos60^\circ + 4 \cos60^\circ + 3$
$\boxed{p = -4.80}$
By using equation (i)
$q = 1.964 - 0.866(-4.80)$
$\boxed{q = 6.12}$

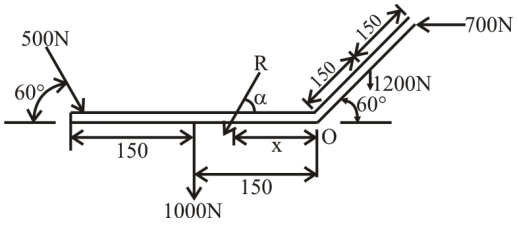
31. The unit of a physical quantity which does not depend on the unit of any other physical quantity is called

(a) independent dimension.
(b) fundamental dimension.
(c) core dimension.
(d) All of the above

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : The unit of a physical quantity which does not depend on the unit of any other physical quantity is called fundamental dimension.

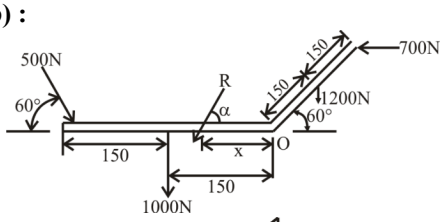
32. The system of forces acting on a bell crank is shown figure. Determine the magnitude of the resultant.

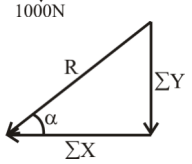


(a) 567 N (b) 2671 N
(c) 1680 N (d) 1232 N

Oil India Senior Officer 23.12.2020

Ans. (b) :





$R_X = \Sigma X = 500\cos60^\circ - 700$
$= -450$ N
$R_Y = \Sigma Y = -1200 - 1000 - 500\sin60^\circ$
$= -2633.01$
$R = \sqrt{(-450)^2 + (-2633.01)^2}$
$\boxed{R = 2671.19\,N}$

33. The resultant of two forces, one of which is double the other is 260 N. If the direction of the larger force is reversed and the other remain unaltered, the magnitude of the resultant reduces to 180 N. Determine the magnitude of the forces.

(a) 50 N, 200 N (b) 50 N, 100 N
(c) 300 N, 100 N (d) 100 N, 200 N

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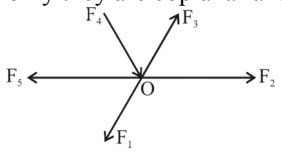
Ans. (d) : Let the forces be of magnitude P & 2P then,
$P^2 + (2P)^2 + 2P \times 2P\cos\theta = (260)^2$ .....(i)
And similarly when larger force is reversed
$P^2 + (-2P)^2 + 2(P)(-2P)\cos\theta = (180)^2$
$P^2 + 4P^2 - 4P^2\cos\theta = (180)^2$ .....(ii)
Adding equation (i) & (ii)
$2P^2 + 8P^2 = 260^2 + 180^2$
$10P^2 = 100000$
$P^2 = 10000$
$P = 100$
Larger force = 2P = 200
Hence, magnitude of the forces are 100 N & 200 N.

34. Three forces acting on a body can keep it in an equilibrium, only when they are _____.

(a) coplanar an non-concurrent
(b) non-coplanar and non-concurrent
(c) coplanar and concurrent
(d) non-coplanar and concurrent

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : Three forces acting on a body can keep it in an equilibrium, only they are coplanar and concurrent.



Coplanar concurrent

- The forces which meet at one point and their lines of action also lie on the same plane.

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35. A 100 N force acts from point A (0, 0, 0) to point B (1, 1, 1). The force vector is represented by

(a) $100\left(\bar{i}+\bar{j}-\bar{k}\right)$ (b) $\frac{100}{\sqrt{3}}\left(\bar{i}+\bar{j}-\bar{k}\right)$

(c) $\frac{100}{\sqrt{3\left(\bar{i}+\bar{j}+\bar{k}\right)}}$ (d) $\frac{100}{\sqrt{3\left(\bar{i}+\bar{j}-\bar{k}\right)}}$

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Ans. (b) : Given, F = 100 N

Point A(0, 0, 0) and B(1, 1, 1)

Unit vector AB $=\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

The force vector is given by –

$$\vec{F}=100\left(\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}\right)$$

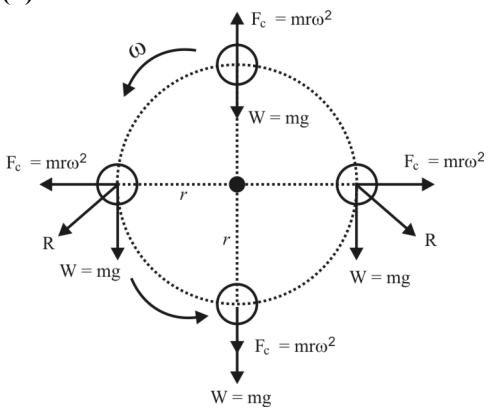
$$\boxed{\vec{F}=\frac{100}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)}$$

36. A stone of mass m at the end of a string of length *l* is whirled in a vertical circle at a constant speed. The tension in the string will be maximum when the stone is-

(a) At the top of the circle
(b) Half-way down from the top
(c) Quarter-was down from the top
(d) At the bottom of the circle

ISRO Scientist/Engineer 2006
ISRO Scientist/Engineer 2008
GPSC Executive Engg. 23.12.2018
GATE 1994

Ans. (d) :



From this fig.

At bottom the tension in string will be maximum because of both weight of stone and centrifugal force acting in downward direction

$T_{bottom} = mg + mr\omega^2$

At the top of circle the tension in string will be

$T_{Top} = mr\omega^2 - mg$ $\rightarrow$ (minimum)

37. The product of mass and acceleration of the fluid is

(a) Gravity force (b) Viscous force
(c) Inertia force (d) Elastic force

TNCSC AE 2020

Ans. (c) : Inertia force–The inertia force acting on a fluid is equal in magnitude to the mass of the element multiplied by its acceleration.

38. A force which is gradually applied and does NOT change in magnitude and direction with respect to time is called.

(a) dynamic load (b) pressing load
(c) static load (d) variable load

UPRVUNL AE 05.07.2021

Ans. (c) : A force which is gradually applied and does not change in magnitude and direction with respect to time is called static load.

39. When number of forces act simultaneously on a particle, then the resultant of these forces will have the same effect as produced by the all the forces. This is known as the principle of _____

(a) Physical independence of forces
(b) resolution of forces
(c) weight of forces
(d) transmissibility of forces

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Ans. (b) : The physical Independence of principle-It states that the effect of force on a body is not affected by the presence of other forces.

Principle of Resolution- It states, "The algebraic sum of the resolved parts of a number of forces in a given direction is equal to the resolved part of their resultant in the same direction".

40. If three _____forces are acting on a body and the body is in equilibrium, then the resultant of two forces should be equal and opposite to the third.

(a) non coplanar and non concurrent
(b) parallel and coplanar
(c) concurrent
(d) non concurrent and coplanar

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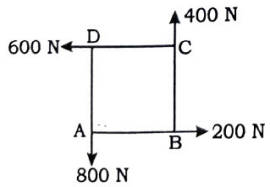
Ans. (c) : If three concurrent forces are acting on a body and the body is in equilibrium, then the resultant of two forces should be equal and opposite to the third.

41. Four forces having magnitudes of 200 N, 400 N, 600 N and 800 N respectively acting along four sides (1m each) of a square ABCD as shown in Figure. Determine the magnitude and direction of the resultant force from A along the line AB.



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(a) $400\sqrt{3}$N, 3.2m from A

(b) $400\sqrt{2}$N, 2.5m from A

(c) $300\sqrt{2}$N, 2m from A

(d) $300\sqrt{3}$N, 2.5m from A

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : $F_x = 600 - 200\ (\leftarrow)$

$F_x = 400$ N $(\leftarrow)$

$F_y = 800 - 400\ (\downarrow)$

$F_y = 400$ N $(\downarrow)$

Resultant force,

$$F_R = \sqrt{F_x^2 + F_y^2}$$

$$= \sqrt{(400)^2 + (400)^2}$$

$$\boxed{F_R = 400\sqrt{2}\ \text{N}}$$

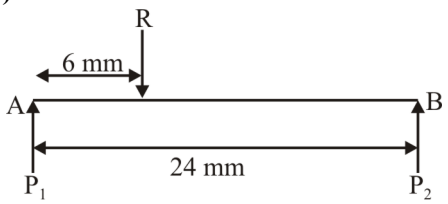
42. Two like parallel forces are acting at a distance of 24 mm apart and their resultant is 20 N. If the line of action of the resultant is 6 mm from any given force, the two forces are

(a) 15 N and 5 N (b) 20 N and 5 N

(c) 15 and 15 N (d) None of these

GPSC Executive Engineer 23.12.2018

Ans. (a) :



The resultant of P_1 & P_2 is R = 20N

Taking Moment about A,

$R \times 6 = P_2 \times 24$

$20 \times 6 = P_2 \times 24$

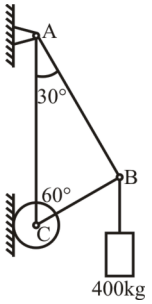
$\boxed{P_2 = 5\text{N}}$

$P_1 + P_2 = R$

$P_1 = 20 - 5$

$\boxed{P_1 = 15\ \text{N}}$

43. Considering g = 10 m/s^2, force in member AC of loaded truss is:

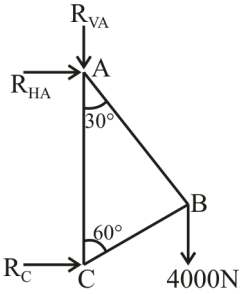


(a) 2000N compression (b) 2000N tension

(c) 1000N compression (d) 1000N tension

AAI Jr. Executive 26.03.2021

Ans. (d) :



$$\left[\text{From figure } \frac{BC}{AC} = \cos 60° = \frac{1}{2}\right]$$

$\Sigma F_x = 0 \Rightarrow R_C \times R_{HA} = 0$

$\Sigma F_y = 0 \quad R_{VA} = -4000$ N

Take moment about point A,

$\Sigma M_A = 0$

$\Rightarrow \quad R_C \times AC = 4000 \times BC\sin 60°$

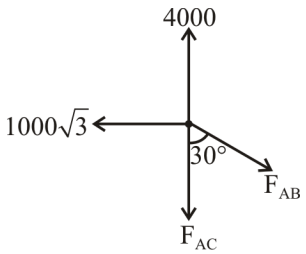
$$R_C = 4000 \times \frac{\sqrt{3}}{2} \times \frac{BC}{AC}$$

$$= 4000 \times \frac{\sqrt{3}}{2} \times \frac{1}{2}$$

$$= 1000\sqrt{3}\ \text{N}$$

So, $R_{HA} = -1000\sqrt{3}$ N

At, joint A



$\therefore \quad 1000\sqrt{3} = F_{AB}\sin 30° \Rightarrow F_{AB} = 2000\sqrt{3}$ N

So, $4000 = F_{AC} + F_{AB}\cos 30°$

$4000 = F_{AC} + 2000\sqrt{3}\cos 30°$

$\Rightarrow \quad F_{AC} = 1000$ N (Tension)

44. Which among the following is a unit of Force?

(a) Slug (b) Henry

(c) Pascal (d) Radian

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a)

Quantity	Unit
Force	Slug
Inductance	Henry
Pressure	Pascal
Angle	Radian

45. Which of the following principles states that the resultant force acting on a body together with reversed effective force or inertial force are in equilibrium?

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(a) Law of Parallelogram
(b) Newton's first Law
(c) Pascal's Law
(d) D-Alembert's Law

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : D-Alembert's Principle– D-Alembert's principle states that the resultant force acting on a body together with the reversed effective force are equilibrium consider a rigid body acted upon by a system of forces. The system may be reduced to a single resultant force (P) acting on the body whose magnitude is given by the product of the mass of the body (m) and linear acceteration (a) of the centre of mass of the body.

According to Newton's second law of motion.

$P = ma$

$P - ma = 0$

$\boxed{P + P_1 = 0}$

46. The dimensional formula $ML^2 T^{-3}$ represents

(a) Momentum (b) Work
(c) Force (d) Power

HPPSC Poly. Lect. 05.07.2021

Ans. (d) :

47. Varignon's theorem is used to find

(a) direction of the resultant force
(b) location of the resultant force
(c) magnitude of the resultant force
(d) nature of the resultant force

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Varignon's theorem is used to find location of the resultant force.

Varignon's Theorem–Moment of a force about any point is equal to the sum of the moments of the components of that force about the same point.

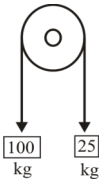
48. A rope is draped over a pulley which is freely rotating. One end of the rope is a 100 kg mass and on the other end is a 25 kg mass. Neglecting the mass of the pulley and friction, the tension in the rope is :

(a) 1000 N (b) 600 N
(c) 392 N (d) 1150 N

CGPSC AE 15.01.2021

Ans. (c) : Let a be the acceleration,

$100g - T = 100\,a$



$T = 100\,(g - a)$ ...(i)

Similarly, For 25 kg

$T = 25\,(g + a)$ ...(ii)

From (i) & (ii)

$100\,(g - a) = 25\,(g + a)$

$3g = 5a \Rightarrow a = \frac{3}{5}g$

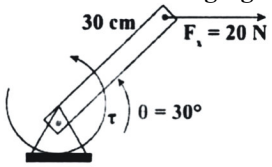
From eq. (ii)

$T = 25\,(a + g)$

$= 25\left(\frac{3}{5}g + g\right) \qquad [\because g = 9.81 m/s^2]$

$= 392.4$ N

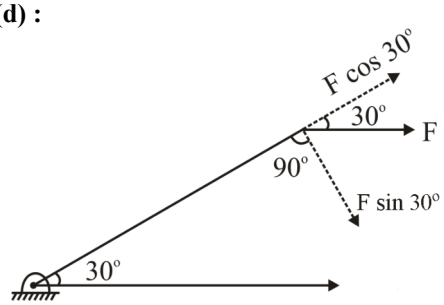
49. Calculate the resisting torque for static equilibrium in the following figure:



(a) 100 N-cm (b) 150 N-cm
(c) 200 N-cm (d) 300 N-cm

Gujarat PSC AE 2019

Ans : (d) :



$T = F \sin 30^\circ \times 30$

$= 20 \times \sin 30^\circ \times 30$

$= 300$ N-cm

50. Three forces P, Q and R are acting concurrently. The included angles are α, β and γ. According to Lami's theorem

(a) $\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$

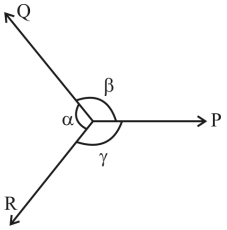
(b) $\frac{P}{\cos\alpha} = \frac{Q}{\cos\beta} = \frac{R}{\cos\gamma}$

(c) $\frac{P}{\tan\alpha} = \frac{Q}{\tan\beta} = \frac{R}{\tan\gamma}$

(d) $\frac{\sin\alpha}{P} = \frac{\sin\beta}{Q} = \frac{\sin\gamma}{R}$

HPPSC AE 2018
(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Lami's Theorem–"If a body is in equilibrium under the action of three forces, then each force is proportional to the sine of the angle between the other two forces"



$\frac{P}{\sin\alpha} = \frac{R}{\sin\beta} = \frac{Q}{\sin\gamma}$

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51. Tension in the cable supporting a lift of weight 'W' and having an acceleration of 'a' while going upward is (g is acceleration due to gravity)

(a) $W\left(1+\frac{a}{g}\right)$

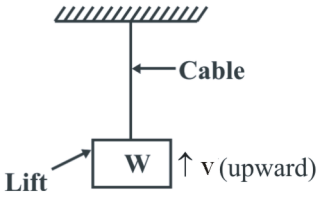
(b) $W\left(1-\frac{a}{g}\right)$

(c) $W\left(2+\frac{a}{g}\right)$

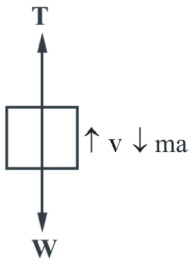
(d) $W\left(1+\frac{2a}{g}\right)$

GPSC Engg. Class-II Pre 19.01.2020
ISRO Scientist/Engineer 24.05.2014

Ans. (a) :



FBD,

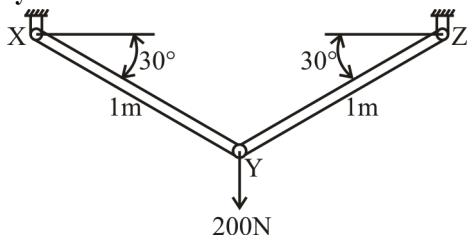


$T = W + ma \qquad \left[\because W = mg \text{ or } m = \frac{W}{g}\right]$

$= W + \frac{W}{g}a$

$T = W\left(1+\frac{a}{g}\right)$

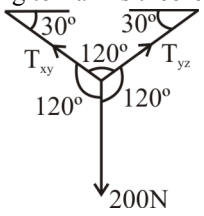
52. Two steel trusses, XY and YZ of identical size supports a load of 200 N as shown in fig. The length of the truss in 1m. The force in the truss xy in N is



(a) 100 N (b) 200 N
(c) 150 N (d) 50 N

ISRO Scientist/Engineer 12.01.2020

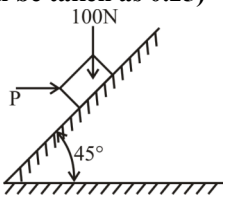
Ans. (b) : According to Lami's theorem



$\frac{T_{xy}}{\sin 120^\circ} = \frac{T_{yz}}{\sin 120^\circ} = \frac{200}{\sin 120^\circ}$

$T_{xy} = T_{yz} = 200$ N

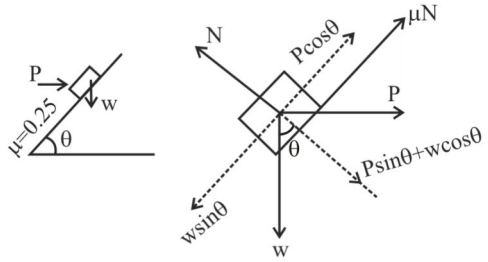
53. A block weighing 100 N is resting on a plane inclined with horizontal as shown in Fig. What horizontal force P is necessary to hold the body from sliding down the plane? (Coefficient of friction can be taken as 0.25)



(a) 30 N (b) 120 N
(c) 60 N (d) 15 N

ISRO Scientist/Engineer 12.01.2020

Ans. (c) :



F.B.D of block

$\Sigma F_y = 0$

$N = w\cos\theta + P\sin\theta$

$\Sigma F_x = 0$

$w\sin\theta = P\cos\theta + \mu N$

$w\sin\theta = P\cos\theta + \mu(w\cos\theta + P\sin\theta)$

$w\sin\theta - \mu w\cos\theta = P\cos\theta + \mu P\sin\theta$

$P = \frac{w(\sin\theta - \mu\cos\theta)}{\cos\theta + \mu\sin\theta}$

$P = \frac{100\times(\sin 45^\circ - 0.25\times\cos 45^\circ)}{\cos 45^\circ + 0.25\times\sin 45^\circ} = 60\text{N}$

54. A tension in a cable supporting a lift

(a) is less when the lift is moving downward
(b) is less when the lift is moving upward
(c) is more when the lift is moving downward
(d) remains the same whether the lift is moving upward or downward

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : A tension in a cable supporting a lift is less when the lift is moving downward.

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55. A point on a cricket bat, if hits the ball at, producing no jerk hand is called

(a) point of suspension
(b) center of percussion
(c) center of gravity
(d) mass center

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : A point on a cricket bat, if hits the ball at, producing no jerk at hand is called center of percussion.

56. The vertical door of weight W is fixed on a wall by two hinges near the top and bottom of the door. Equilibrium condition implies that

(a) top hinge tends to come out of the wall; nothing happens to the bottom hinge
(b) top hinge comes out of the wall; bottom hinge gets pushed in to the wall
(c) both the hinges tend to come out of the wall
(d) bottom hinge tends to come out of the wall; nothing happens to the top hinge

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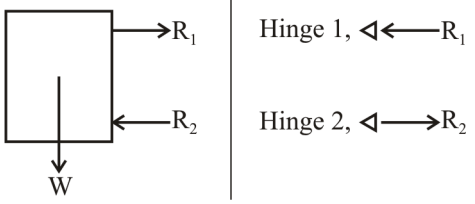
Ans. (b) : Top hinge comes out of the wall bottom hinge gets pushed into the wall.

FBD for Door | **FBD for Hinge**

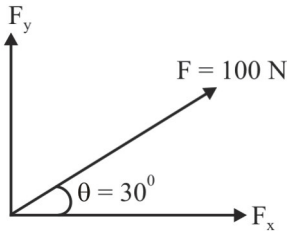


57. A force of 100 N is acting at a point making an angle of 30^0 with horizontal. Component of this force along Y-direction will be

(a) 50 N (b) 100 N
(c) 126 N (d) 86.6 N

MPSC Lect. (Prod.) 2018

Ans. (a)



$F_x = F \cos 30^0 = 100 \times \cos 30^0 = 86.60$ N

$F_y = F \sin 30^0 = 100 \times \sin 30^0 = 50$ N

58. If a body is subjected to three forces, then the body will be in equilibrium if the:

(a) resultant of any two forces is equal, collinear and acts in the same direction as that of the third force
(b) resultant of any two forces is equal, collinear and acts in the direction opposite to that of the third force
(c) vertical component of one force is equal to the sum of the horizontal components of the other two forces
(d) horizontal component of one force is equal to the sum of the vertical components of the other two forces

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : If a body is subjected to three forces, then the body will be in equilibrium if the resultant of any two forces is equal, collinear and acts in the direction opposite to that of third forces.

59. In free body diagrams, the Principle of transmissibility states that the force acting on the body is a:

(a) sliding vector (b) unit vector
(c) wedging vector (d) rolling vector

CIL MT 27.02.2020

Ans. (a) : Principle of transmissibility states that the force acting on the body is a sliding vector. That is it can be applied at any point of the body. It will give the same effect as it applied at any point other that the specific point.

60. Which of the following is not a vector quantity?

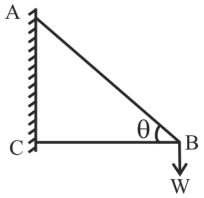
(a) Time (b) Velocity
(c) Acceleration (d) Force

APGCL AM, 2021
J&K PSC Civil Services Pre, 2010

Ans. (a) :

Vector: The physical quantity for which to define both magnitude and direction are required.
Examples: Velocity, Acceleration, force, displacement.
Scalar: The quantity for which to define only magnitude as required.
Examples: pressure, time, distance, speed etc.

61. The force induced in the string BC due to load 'W' as shown in the below figure is



(a) W sin θ (b) W cos θ
(c) W tan θ (d) W cot θ

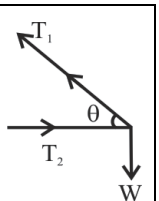
APGCL AM, 2021

Ans. (d) :

$\Sigma F_v = 0 \Rightarrow T_1 \sin\theta = W \Rightarrow T_1 = \dfrac{W}{\sin\theta}$

$\Sigma F_H = 0 \Rightarrow T_1 \cos\theta = T_2$

$\dfrac{W}{\sin\theta}.\cos\theta = T_2 \Rightarrow \boxed{T_2 = W\cot\theta}$



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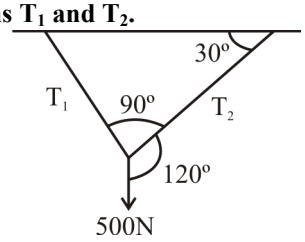
62. The resultant force acting in the couple is

(a) zero
(b) infinite
(c) twice the magnitude of single force
(d) half the magnitude of single force

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : A couple consists of two parallel forces that are equal in magnitude opposite in direction and do not chase any line of action.
It does not produce any translation, it only produce rotation.
The resultant force of a couple is zero.
But the resultant of a couple is not zero it is a pure moment.

63. A weight of 500 N is supported by two metallic ropes as shown in the figure. Find the values of tensions T_1 and T_2.



(a) $T_1 = 353.5$ N and $T_2 = 250$ N
(b) $T_1 = 250$ N and $T_2 = 433$ N
(c) $T_1 = 433$ N and $T_2 = 250$ N
(d) $T_1 = 250$ N and $T_2 = 353.5$ N

Assam Engg. College AP/Lect. 18.01.2021
Karnataka PSC Lect., 27.05.2017
GATE 2015

Ans. (c) :
Apply Lami's theorem

$$\frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 150^\circ} = \frac{500}{\sin 90^\circ}$$

$$T_1 = \frac{500 \sin 120^\circ}{\sin 90^\circ} = 433.01\text{N}$$

$$T_2 = \frac{500 \times \sin 150^\circ}{\sin 90^\circ} = 250\text{N}$$

64. A system of three forces acts on a body and keeps it in equilibrium the forces need to be :

(a) coplanar only
(b) concurrent only
(c) coplanar as well as concurrent
(d) coplanar but may or not be concurrent

HPPSC Poly. Lect. 05.07.2021
J & K PSC Screening, 2006

Ans. (c) : If three forces act on a body and keeps it in equilibrium, Forces must be coplanar as well as concurrent.

65. If the sum of all the forces acting on a body is zero, it may be concluded that the body

(a) must be in equilibrium
(b) cannot be in equilibrium
(c) may be in equilibrium provided the forces are concurrent
(d) may be in equilibrium provided the forces are parallel

VIZAG Steel MT 2015
TNPSC AE, 2008
UKPSC AE 2007 Paper -I

Ans. (c) : If the sum of all the forces acting on a body is zero, it may be concluded that the body may be in equilibrium provided the forces are concurrent.

66. D' Alembert's principle

(a) provides no special advantage on Newton's law
(b) is not applicable for kinetic problem
(c) is not dependent on inertial effect
(d) converts a kinetic problem into an equivalent static problem

TSPSC Managers, 2015

Ans. (d) : D'Alembert's principle- The vector sum of the external forces and the inertia forces acting on a moving system is zero. According to D' Alembert's principal F + (– ma) = 0 Where '–ma' is the inertia force.

67. The resultant of the two forces P and Q is R, if Q is doubled, the new resultant force is perpendicular to P, then :

(a) P = Q (b) Q = R
(c) Q = 2 R (d) P = 2 R

TSPSC Managers, 2015

Ans. (b) : Resultant

$$R^2 = P^2 + Q^2 + 2PQ\cos\theta$$

θ = Angle between P & Q

$$\tan\alpha = \frac{Q\sin\theta}{P + Q\cos\theta}$$, α is angle between P & R

$\because$ If Q is doubled, and if the angle between P & the new resultant (R) is 90° then,

$$\tan 90^\circ = \frac{(2Q)\sin\theta}{P + (2Q)\cos\theta}$$

$$(\text{infinity})\infty = \frac{2Q\sin\theta}{P + 2Q\cos\theta}$$

$$\frac{1}{0} = \frac{2Q\sin\theta}{P + 2Q\cos\theta}$$

$$P + 2Q\cos\theta = 0$$

$$P = -2Q\cos\theta$$

But $R^2 = P^2 + Q^2 + 2PQ\cos\theta$

$R^2 = P^2 + Q^2 - P^2$

$R^2 = Q^2$

$\boxed{R = Q}$

68. Two forces act at a point. The first force has X and Y components of 3N and –5N respectively. The resultant of these forces falls on the x-axis and has a magnitude of –4N. The X and Y components of the second force is:

(a) (–7, 5) (b) (–7, –5)
(c) (–7, 0) (d) (7, 0)

TSPSC Managers, 2015

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Ans. (a) : Let $\hat{i}$ and $\hat{j}$ be in the x and y directions, respectively,

$$\vec{F}_x = 3\hat{i} - 5\hat{j}$$

$$\vec{F}_y = \ell\hat{i} + y\hat{j}$$

$$\vec{F}_R = -4i$$

$$\vec{F}_x + \vec{F}_y = \vec{F}_R$$

$$+3\hat{i} - 5\hat{j} + \ell\hat{i} + y\hat{j} = -4\hat{i}$$

$$(3 + \ell)\hat{i} + (y - 5)\hat{j} = -4\hat{i}$$

$$3 + \ell = -4$$

$$\ell = -7$$

$$y - 5 = 0$$

$$y = +5$$

$$(\ell, y) = (-7, 5)$$

69. The number of independent scalar equations of equilibrium of a rigid body under concurrent force system is :

(a) 4 (b) 3

(c) 5 (d) 6

J&K PSC Civil Services Pre, 2013
TNPSC AE (Industries) 09.06.2013

Ans. (b) : If all forces meet at common point, called concurrent forces system and concurrent forces do not produce bending (try to rotate) about any these axis. Means, it should be in translatory equilibrium along all these axis. So, number of independent equation should be along three axes.

70. D'Alembert principle states that system of forces acting on a body is following type of equilibrium with the inertia force of the body :

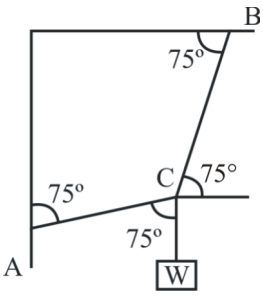
(a) static (b) dynamic

(c) both (d) none

J&K PSC Civil Services Pre, 2013

Ans. (a) : D'Alembert principle is used to convert dynamic equilibrium condition into static equilibrium with application of inertia force.

71. Two cables are tied together at C and loaded with a weight of 2000 N as shown in figure. The tension in the cable AC is

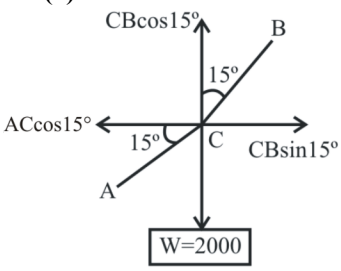


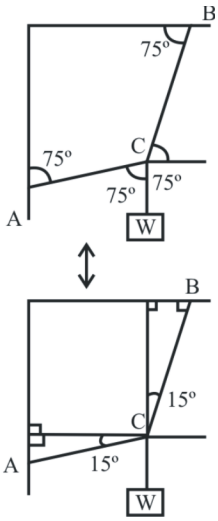
(a) 597.7 N (b) 298.35 N

(c) 1000 N (d) 1999 N

APPSC Poly. Lect. 2013

Ans. (a) :





We know that –

$\Sigma F_x = 0, \Sigma F_y = 0$

Then, $\Sigma F_x = 0$

$-CB\sin 15^\circ + AC\cos 15^\circ = 0$

$$AC = CB\frac{\sin 15^\circ}{\cos 15^\circ} = 0.2679\,CB$$

$\Sigma F_y = 0$

$ACsin15° + 2000 = CBcos15°$

$$AC(0.2588) + 2000 = 0.9659 \times \frac{AC}{0.2679}$$

$2000 = 3.6054AC - 0.2588AC$

$$AC = \frac{2000}{3.3466} = 597.62N$$

72. Vector equations of 3-dimensional system are

(a) $\Sigma F = 0, \Sigma M = 0$ (b) $\Sigma F = 0, \Sigma M \neq 0$

(c) $\Sigma F \neq 0, \Sigma M = 0$ (d) $\Sigma F \neq 0, \Sigma M \neq 0$

APPSC Poly. Lect. 2013

Ans. (a) : Vector equation of 3-dimensional system are- $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$ and $\Sigma M = 0$

73. According to Lami's theorem, the three forces

(a) must be equal

(b) must be at 120° to each other

(c) must be both of above

(d) may not be any of the two

APPSC Poly. Lect. 2013

Ans. (d) : According to Lami's theorem, the three forces must be equal, if the three forces acting at a point are in equilibrium then each force is proportional to the sine of the angle between the other two.

74. Two vectors are given by

$$\mathbf{P = 6\hat{i} + 10\hat{j} - 5\hat{k},\ Q = 5\hat{i} + 2\hat{j} + 10\hat{k}}$$

Which of the following is true for these vectors?

(a) Both the vectors are parallel

(b) Both the vectors are perpendicular to each other

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(c) Angle between the vectors is 30^0
(d) Angle between the vectors is 60^0

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given;

$$P = 6\hat{i} + 10\hat{j} - 5\hat{k}$$

$$Q = 5\hat{i} + 2\hat{j} + 10\hat{k}$$

$\because$ P.Q. = |P| |Q| cos θ

Let, θ = Angle between the two vector

$$\Rightarrow \cos\theta = \frac{P.Q}{|P||Q|} = \frac{30 + 20 - 50}{\sqrt{161} \times \sqrt{129}}$$

$$\Rightarrow \cos\theta = \frac{0}{\sqrt{161} \times \sqrt{129}} = 0 \qquad \Rightarrow \cos\theta = 0$$

$$\boxed{\theta = 90^0}$$

i.e., both the vectors are perpendicular each other.

75. An elevator has an upward acceleration of 1 m/s^2. What will be the force transmitted to the floor of the elevator by a man of weight 500 N travelling in the elevator? [g = 10 m/s^2]

(a) 550 N (b) 500 N
(c) 450 N (d) 400 N

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given; g = 10 m/s^2, a = 1 m/sec^2

mg = 500 N

$$m = \frac{500}{g} = \frac{500}{10} = 50 \text{ kg}$$

Let,

Force transmitted to the floor = F

$\Sigma F = F - mg = ma$

$F = m(g+a)$

$F = 50 \times (10+1)$

$= 50 \times 11$

$$\boxed{F = 550 \text{ N}}$$

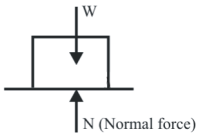
76. Normal force acts ……. to the direction of the surfaces in contact.

(a) Perpendicular (b) At and angle
(c) Tangential (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : Normal force acts perpendicular to the directions of the surface in contact.

According to Newton's 3rd law, Every action has equal and opposite reaction



W-weight of block-exerts force on surface, then Normal reaction is applied by surface on block.

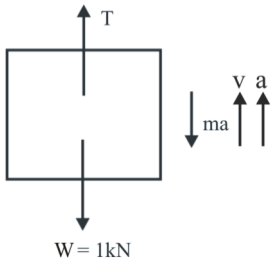
Normal reaction = N = W

77. An elevator weighing 1 kN attains an upward velocity of 4m/sec in 2 sec with uniform acceleration. The tension in the supporting cable will be:

(a) 1000 N (b) 800 N
(c) 1200 N (d) 2000 N

J & K PSC Screening, 2006

Ans. (c) : Given, W = mg = 1000



$$m = \frac{W}{g} = \frac{1000}{9.81} = 101.93 \text{kg}$$

According to Newton first law –

$v = u + at$

$4 = 0 + a \times 2$

$$\boxed{a = 2 \text{ m/s}^2}$$

Tension, $T = W + ma$

$= 1000 + 101.93 \times 2$

$$\boxed{T = 1203.8} \text{ N}$$

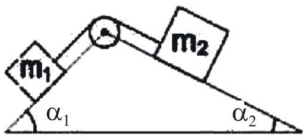
78. Two pieces of steel and brass of mass 2 kg and 1 kg respectively fall freely under the action of gravity from a tower. After a distance, the following will be identical.

(a) acceleration (b) momentum
(c) kinetic energy (d) potential energy

J & K PSC Screening, 2006

Ans. (a) : Two pieces of steel and brass of mass 2 kg and 1 kg respectively fall freely under the action of gravity from a tower. After a distance, the acceleration will be identical.

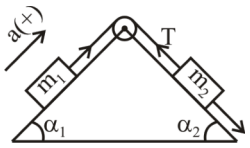
79. A frictionless double incline with angles α_1 and α_2 is shown in the figure below. It carries a set of sliding masses m_1 and m_2 that are connected with an inextensible string passing over a frictionless pulley. If $\alpha_2 = 30°$ and m_2 is twice that of mass m_1. Determine α_1 for equilibrium.



(a) 30° (b) 45°
(c) 60° (d) 75°
(e) 90°

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) :



$\Rightarrow$ $m_2 = 2m_1$

$\Rightarrow$ $\alpha_2 = 30°$

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$\alpha_1 = ?$

By balancing tension,

$m_1 \sin\alpha_1 = m_2 \sin\alpha_2$

So, $\sin\alpha_1 = \frac{1}{2} \times \frac{m_2}{m_1}$

$\sin\alpha_1 = 1$

$\alpha_1 = \sin^{-1}(1)$

$\boxed{\alpha_1 = 90°}$

80. The angle between two forces when the resultant is maximum and minimum respectively are-

(a) 0° and 180° (b) 180° and 0°

(c) 90° and 180° (d) 90° and 0°

GMB AAE 25.06.2017

Ans. (a) : The resultant force given as :

$$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

When $\theta = 0$ then the force become maximum

$$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2}$$

$$F = \sqrt{(F_1 + F_2)^2}$$

$$\boxed{F = F_1 + F_2}$$

When the $\theta = 180°$ then the force become minimum.

$$F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 180°}$$

$$F = \sqrt{F_1^2 + F_2^2 - 2F_1F_2}$$

$$F = \sqrt{(F_1 - F_2)^2}$$

$$\boxed{F = F_1 - F_2}$$

81. Two forces of 6 and 8 kg acting at right angles to each other, the resultant force will be

(a) 10 (b) 28

(c) 40 (d) 100

WBPSC AE, 2017

Ans. (a) : Given, $\theta = 90°$, $F_1 = 6$ kg, $F_2 = 8$ kg

We know that, $R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$

$$= \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 90°}$$

$$= \sqrt{6^2 + 8^2}$$

$$= \sqrt{36 + 64}$$

$$= \sqrt{100} = 10\,\text{N}$$

82. The co-ordinate of the initial and terminal points of vector are (3, 1, – 2) and (4, – 7, 10). The magnitude of vector is approximately :

(a) 50 units (b) 8.1 units

(c) 14.5 units (d) 19.0 units

J&K PSC Civil Services Pre, 2010

Ans. (c) : Given-

Magnitude of vector =

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

$$= \sqrt{(3-4)^2 + (1-(-7))^2 + [(-2-10)]^2}$$

$$= \sqrt{1 + 64 + 144}$$

$$= \sqrt{65 + 144} = \sqrt{209} = 14.456 \text{ units}$$

83. The number of independent equations available from static equations of equilibrium for plane problem are

(a) 3 (b) 6

(c) 12 (d) 9

Haryana PSC Civil Services Pre, 2014

Ans. (a) : Equations of static balance.

(i) $\sum F_x = 0$

(ii) $\sum F_y = 0$

(iii) $\sum M_z = 0$

84. According to the law of equilibrium, two forces are said to be in equilibrium only if they are

(a) equal in magnitude and opposite in direction

(b) opposite in direction

(c) collinear in action

(d) Both (A) and (C)

Haryana PSC Civil Services Pre, 2014

Ans. (d) : Two forces are said to be equilibrium if they are equal and opposite and are collinear in line of action.

$F_1 = F_2$

85. The statement – "if three forces acting at a point can be represented in magnitude and direction by the sides of a triangle taken in order, the forces are in equilibrium"– is known as

(a) Lami's theorem

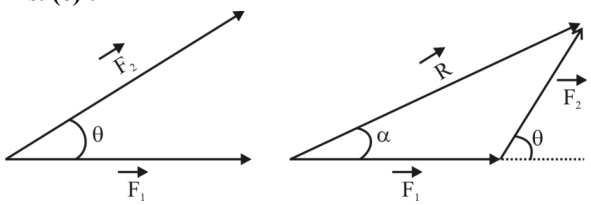
(b) Law of Polygon of forces

(c) Law of triangle of forces

(d) Newton's law of forces

Haryana PSC Civil Services Pre, 2014

Ans. (c) :



Law of triangle of forces- It states that three forces acting at a point will be in equilibrium. If it can be represented in magnitude and direction by the sides of triangle taken in order.

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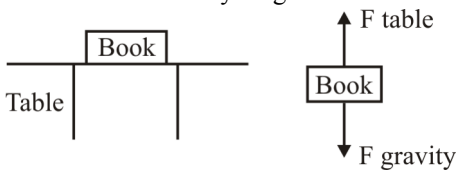
86. Free body diagram of a body in a force system is

(a) Completely isolated from the system
(b) incompletely isolated from the system
(c) completely attached to the system
(d) incompletely attached to the system

Haryana PSC Civil Services Pre, 2014

Ans. (a) : Free Body Diagram- A diagram which shows a part of system separately (completely isolated from the system) indicating all the external forces acting on it, is called as a free body diagram.

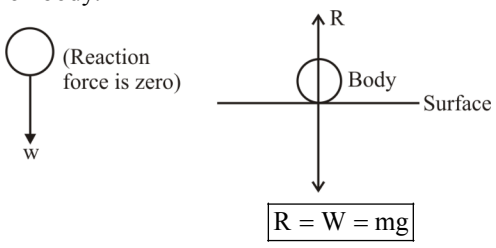


87. When a body falls freely under gravity it possesses

(a) maximum weight
(b) minimum weight
(c) no weight
(d) a weight depending upon velocity

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Body falls freely under gravity posses no weight because there is no surface which gives reaction force on body.



$R = W = mg$

88. The method of splitting a single force into two perpendicular components along the x-axis and y-axis is called as

(a) Orthogonal resolution
(b) Perpendicular resolution
(c) Both (1) and (2)
(d) None of the above

Karnataka PSC Lect., 27.05.2017

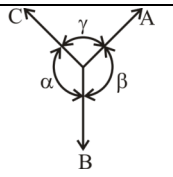
Ans (c) : The method of splitting a single force into two perpendicular components along the x-axis and y-axis is called as orthogonal resolution and perpendicular resolution.

89. A mass of 1 kg is attached to the middle of a rope, which is being pulled from both ends in the opposite directions. Taking g = 10 m/sec^2 the minimum pull required to completely straighten the rope will be :

(a) 5 N (b) 20 N
(c) 25 N (d) Infinite

MPPSC State Forest Service Exam, 2014

Ans. (d) : As we know,
From Lami's theorem,



$$\Rightarrow \quad \frac{A}{\sin\alpha} = \frac{B}{\sin\gamma} = \frac{C}{\sin\beta}$$

So, $\alpha = 90°, \beta = 90°, \gamma = 180°$

$$\frac{A}{\sin 90°} = \frac{B}{\sin 180°} = \frac{C}{\sin 90°}$$

$$\frac{A}{1} = \frac{B}{0} = \frac{C}{1}$$

$$\boxed{A = C = \infty}$$

90. The conditions of equilibrium for non-concurrent coplanar force systems are.

(a) $\Sigma F_x = 0, \Sigma F_y = 0$
(b) $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$
(c) $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma M_z = 0$
(d) $\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0, \Sigma M_x = 0, \Sigma M_y = 0, \Sigma M_z = 0$

MPPSC State Forest Service Exam, 2014
UKPSC AE 2012 Paper-I
UKPSC AE 2007 Paper -I

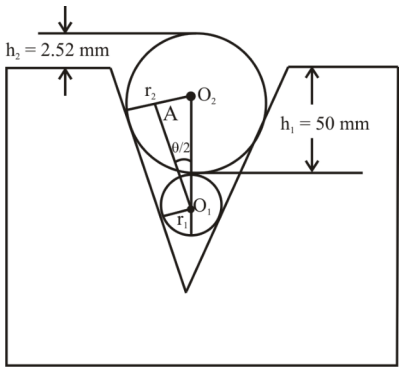
Ans. (c) : There are three equilibrium conditions that can be used for non-concurrent, non-parallel force system.
The sum of all forces, in the x-direction or horizontal is zero ($\Sigma F_x = 0$).
The sum of all forces in y-direction is zero ($\Sigma F_y = 0$).
The sum of moment at any point 'O' is zero ($\Sigma M = 0$).

91. Balls of diameter 30 mm and 15 mm were used to measure the taper of gauge. During inspection, the ball of 30 mm diameter was penetrate 2.5 mm above the top surface of the ring. This surface was located at a of 50 mm from the top of the 15 mm diameter ball. The taper as approximately

(a) 12^0 (b) 14^0
(c) 17^0 (d) 20^0

TRB Asstt. Prof., 2012

Ans. (d) :



$$\sin\frac{\theta}{2} = \frac{O_2A}{O_1O_2}$$

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$$\sin\frac{\theta}{2}=\frac{r_2-r_1}{h_1+r_1-h_2-r_2}$$

$$\sin\frac{\theta}{2}=\frac{15-7.5}{50+7.5-2.5-15}$$

$$\sin\frac{\theta}{2}=\frac{7.5}{40}$$

$$\frac{\theta}{2}=\sin^{-1}\left(\frac{7.5}{40}\right)$$

$$\boxed{\theta=21.6^\circ}$$

92. A free body diagram is a diagram :

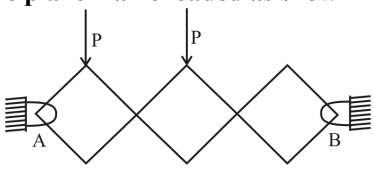
(a) drawn by free hand
(b) represents a floating body
(c) separating the body from its surroundings and replacing with force vector
(d) All of these

KPSC ADF 2015

Ans. (d) : A free body diagram is a diagram–

- Drawn by free hand
- Represents a floating body
- Separating the body from its surroundings and replacing with force vector.

93. The horizontal component of the reaction at A of the plane frame loaded as shown in figure is

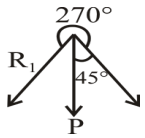


(a) 2P (b) P

(c) $\frac{P}{2}$ (d) $\frac{P}{3}$

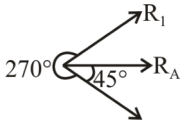
OPSC Civil Services Pre 2006

Ans. (b) :



$$\frac{R_1}{\sin 45^\circ}=\frac{P}{\sin 270^\circ}$$

$$R_1=\frac{P}{\sqrt{2}}$$

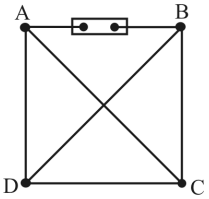


$$\frac{R_A}{\sin 270^\circ}=\frac{R_1}{\sin 45^\circ}$$

$$R_A=-\left(-\frac{P}{\sqrt{2}}\right)\times\sqrt{2}$$

$$\boxed{R_A=P}$$

94. Referring to the square frame as shown in figure, a tensile force P is produced in the bar AB by tightening a turn-buckle. The force produced in each of the members AC or BD is



(a) $\frac{P}{\sqrt{2}}$ (b) P

(c) $\sqrt{2}P$ (d) 2P

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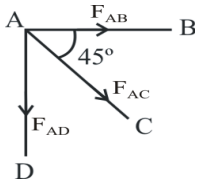
Ans. (c) :



At joint A

$F_{AC}\cos 45 = F_{AB}$

$$F_{AC}=\frac{P}{\cos 45}=\sqrt{2}P=F_{QD}$$

95. A heavy spherical ball of weight W rests on a V- shaped trough, whose sides are inclined at angles α and β to the horizontal. The contact pressure on the two sides of the trough are

(a) $\frac{W\cos\beta}{\cos(\alpha+\beta)}$ and $\frac{W\sin\alpha}{\cos(\alpha+\beta)}$

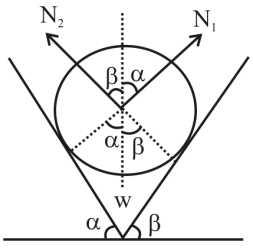
(b) $\frac{W\sin\beta}{\sin(\alpha+\beta)}$ and $\frac{W\sin\alpha}{\sin(\alpha+\beta)}$

(c) $\frac{W\cos\beta}{\cos(\alpha+\beta)}$ and $\frac{W\cos\alpha}{\cos(\alpha+\beta)}$

(d) $\frac{W\cos\beta}{\sin(\alpha+\beta)}$ and $\frac{W\cos\alpha}{\sin(\alpha+\beta)}$

OPSC Civil Services Pre 2006

Ans. (b) :



Apply Lami's theorem

$$\frac{W}{\sin(\alpha+\beta)}=\frac{N_1}{\sin(180-\beta)}=\frac{N_2}{\sin(180-\alpha)}$$

$$N_1=\frac{W\sin\beta}{\sin(\alpha+\beta)};N_2=\frac{W\sin\alpha}{\sin(\alpha+\beta)}$$

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96. What is one dyne equal to?

(a) 10^{-3} N (b) 10^{-4} N

(c) 10^{-5} N (d) 10^{-6} N

KPCL AE 2016

Ans. (c) : 1 N = 10^5 dyne

1 dyne = 10^{-5} N

97. What is the name of the force by which a body is attracted towards the earth?

(a) Center of gravity (b) Weight

(c) Mass (d) Energy

KPCL AE 2016

Ans. (b) : The force of gravity with which a body is attracted towards the centre of the earth is called the weight of the body and it is equal to the product of mass and acceleration due to gravity offered by the earth.

Weight (w) = mg

98. The ratio of load lifted to the effort applied is known as_____

(a) Lift ratio (b) Mechanical advantage

(c) Lift advantage (d) Mechanical ratio

KPCL AE 2016

Ans. (b) : The ratio of weight (load) lifted by a machine to the force (effort) applied on a machine is called mechanical advantage of the machine.

99. If a given system of concurrent forces have ∑H = 0, but ∑V # 0, then resultant of the system can be-

(a) Zero

(b) A force vertically upward or downward

(c) A force horizontally leftward or rightward

(d) Indeterminate

WBPSC AE 2003

Ans. (b) : Resultant– Resultant of a force system is a force or a couple that will have the same effect to the body both in translation and rotation, if all the forces are removed and replaced by the resultant.

R_x → The x – component of the resultant is equal to the sum of forces in the x-direction.

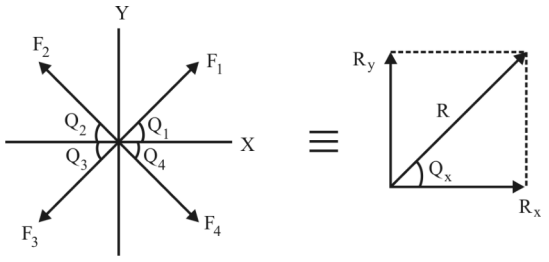
R_y → The y – component of the resultant is equal to the sum of forces in y-direction.

R_z → The z- component of the resultant is equal to the sum of forces is z-direction.

$$R_x = \Sigma F_x = F_{x_1} + F_{x_2} + F_{x_3} + ...$$

$$R_y = \sum F_y = F_{y_1} + F_{y_2} + F_{y_3} + ...$$

$$R_z = \sum F_z = F_{z_1} + F_{z_2} + F_{z_3} + ...$$



$R_x = \Sigma F_x$ $R_y = \Sigma F_4$

$$R = \sqrt{R_x^2 + R_y^2} \qquad \because \Sigma H = 0 = \Sigma F_x$$

$$= \sqrt{0 + R_y^2}$$

$$\boxed{R = R_y}$$

So, the resultant of the system can be a force vertically upward or downward.

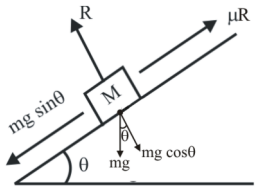
100. A block of mass M is moving down an inclined plane having coefficient of friction equal to $\frac{1}{3}\tan\theta$ where θ is the angle of inclination of the plane with the horizontal. The centre of mass of the block will have uniform acceleration equal to-

(a) $\frac{2}{3}g\cos\theta$ (b) $\frac{2}{3}g\sin\theta$

(c) $g\sin\theta$ (d) $g\cos\theta$

WBPSC AE 2003

Ans. (b) :



$$\mu = \frac{1}{3}\tan\theta$$

$R = mg\cos\theta$

F_f = friction force

$$\because \quad F_f = \frac{1}{3}\tan\theta \times mg\cos\theta$$

$$= \frac{1}{3}mg\sin\theta$$

uniform acceleration

$$F = ma = mg\sin\theta - \mu R$$

$$= mg\sin\theta - \frac{1}{3}mg\sin\theta$$

$$= mg\sin\theta\left(1 - \frac{1}{3}\right)$$

$$ma = \frac{2mg\sin\theta}{3}$$

$$\boxed{a = \frac{2}{3}g\sin\theta}$$

101. The force polygon for coplanar concurrent forces

(a) must close

(b) must not close

(c) may or may not close

(d) any of the above

WBPSC AE 2008

Ans. (a) : For equilibrium force polygon for coplanar concurrent forces must be closed.

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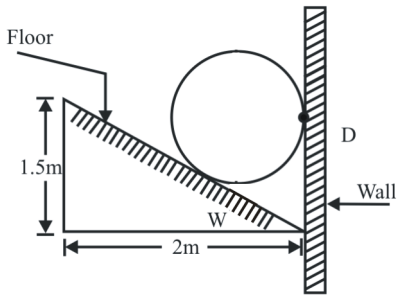
102.

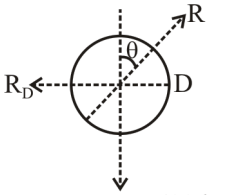


A smooth cylinder of weight 600 kN is resting on a smooth wall and floor as shown above. The reaction at D is :

(a) 300 kN (b) 450 kN
(c) 600 kN (d) 800 kN

UPSC JWM Advt. No.-52/2010

Ans. (b) :



$$\frac{600}{\sin(90+\theta)} = \frac{R_D}{\sin(180-\theta)}$$

$$\frac{\sin\theta}{\cos\theta} \times 600 = R_D$$

$$R_D = 600\tan\theta$$

$$= 600 \times \frac{1.5}{2} = 450\,\text{kN}$$

103. The forces are

1. Coplanar if the lines of action lie in a single plane.
2. Non-concurrent if the lines of action lie in different planes.
3. Concurrent if the lines of action meet at a single point.
4. Collinear if the lines of action are same.

Which of these statements are correct?

(a) 1, 2, 3 and 4 (b) 1, 3 and 4 only
(c) 2, 3 and 4 only (d) 1, 2 and 3 only

UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation :

- The forces are coplanar if the lines of action lies in a single plane.
- The forces are concurrent if the lines of action meet at a single point.
- The forces are collinear if the lines of action are same.

104. According to the D' Alembert's principle

(a) $\Sigma F - I\alpha = 0$ (b) $\Sigma F - ma = 0$
(c) $\Sigma T - I\alpha = 0$ (d) $\Sigma T - F - ma = 0$

UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation : According to D'Alembert Principle-

"For any body, the algebraic sum of externally applied forces and the forces resisting motion in any given direction is zero."

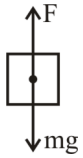
Hence, $\boxed{\Sigma F - ma = 0}$

105. A 500-kg elevator starts from rest and travels upwards with a constant acceleration of $a_c = 2$ m/s^2. What is the power output of the motor when t = 3 sec? (Take g = 10 m/s^2)

(a) 18 kW (b) 30 kW
(c) 36 kW (d) 54 kW

UPSC JWM Advt. No.-50/2010

Ans. (c) : Given, m = 500 kg, $a_c = 2$ m/s^2
t = 3 sec, g = 10 m/s^2, P = ?



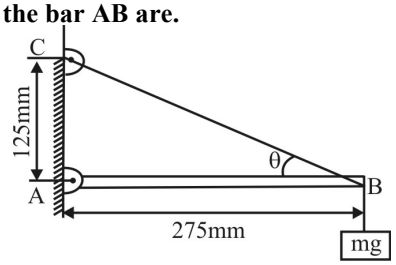
F-mg = ma
F = m(g + a)
= 500(10 + 2)
= 6000 N

Power (P) = F.v

$\Rightarrow$ v = u + at $\quad \therefore$ u = 0
= 0 + 2 × 3
v = 6 m/s

Power (P) = 6000 × 6 = 36000 W = 36 kW

106. A mass of 35 kg is suspended from a weightless bar AB which is supported by a cable CB and a pin as shown in Figure. The pin reactions at A on the bar AB are.



(a) $R_x = 343.4$ N. $R_y = 755/4$ N
(b) $R_x = 755.4$ N. $R_y = 343/4$ N
(c) $R_x = 755.4$ N. $R_y = 0$
(d) $R_x = 343.4$ N. $R_y = 0$

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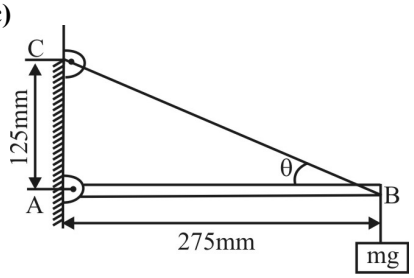
ISRO Scientist/Engineer 11.10.2015
GATE-1997

Ans. : (c)



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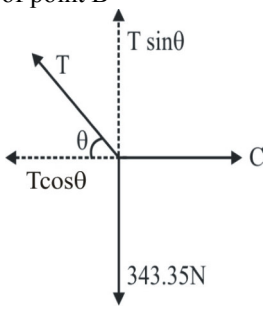
for equilibrium of point B



$T \sin\theta = 343.35$ N ..... (i)

$T \cos\theta = C$ .....(ii)

From equation (i) and (ii)

$$\tan\theta = \frac{343.35}{C}$$

$$C = \frac{343.35}{\tan\theta}$$

$$\tan\theta = \frac{125}{275}$$

$$C = \frac{343.35 \times 275}{125}$$

$C = 755.4$ N

$R_x = R_c = 755.4$ N

Reaction C and A both are same

$R_y = 0$

107. An electric lift is moving downwards with an acceleration of 3g/4. The reaction between a passenger in the lift and its floor is equal to

(a) 4/3 of the passenger's weight
(b) 1/3 of the passenger's weight
(c) 2/3 of the passenger's weight
(d) 1/4 of the passenger's weight

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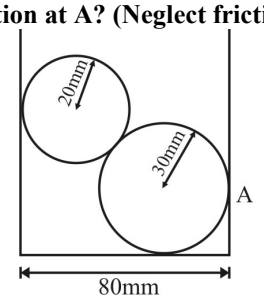
Ans. : (d) Weight of passenger W = mg N

when lift is moving downward with an acceleration $\frac{3}{4}g$ then the working gravity accleration $= g - \frac{3}{4}g = \frac{g}{4}$

Now weight of passenger $= m \times \frac{g}{4} = \frac{mg}{4}$

108. Two cylinders are kept in a container as shown in Figure. Top cylinder has a weight W and bottom cylinder has a weight of 2 W. What is the reaction at A? (Neglect friction)



(a) 3 W (b) 1.75 W
(c) 2.8 W (d) 2.5 W

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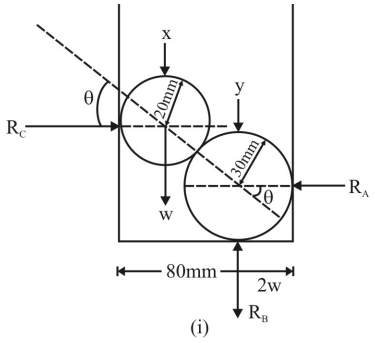
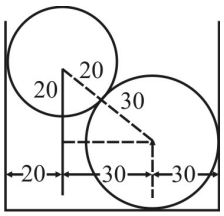
CSE Pre-2005

Ans. : (*)

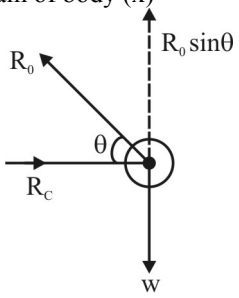


(i)



(ii)

Free body diagram of body (x)

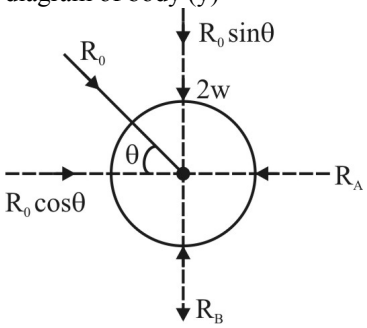


$W = R_0 \sin\theta$

$$R_0 = \frac{W}{\sin\theta} \qquad \therefore \sin\theta = \frac{4}{5}$$

$$R_0 = \frac{5W}{4}$$

Free body diagram of body (y)



$R_A = R_0 \cos\theta$

$$R_A = \frac{W}{\sin\theta} \times \cos\theta$$

$R_A = W \cot\theta$

$$R_A = \frac{3W}{4}$$

$R_A = 0.75$ W

Note : All option given is wrong.

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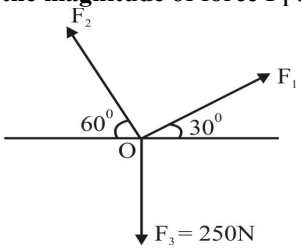
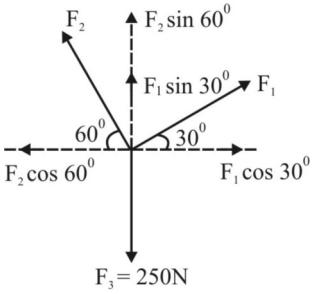
109. Three forces F_1, F_2 and F_3 are acting on a body as shown in figure and the body is in equilibrium. If the magnitude of force F_1 is 250 N. find the magnitude of force F_1 and F_2



(a) $F_1 = 85$ N and $F_2 = 125$ N
(b) $F_1 = 125$ N and $F_2 = 256.6$ N
(c) $F_1 = 125$ N and $F_2 = 215.6$ N
(d) $F_1 = 95$ N and $F_2 = 215.6$ N

UPRVUNL AE 21.08.2016

Ans. : (c)



$\Sigma F_y = 0$

$F_1 \sin 30° = F_3 - F_2 \sin 60°$

$$\frac{F_1}{2} + \frac{F_2\sqrt{3}}{2} = F_3 \quad \text{..... (i)}$$

$\Sigma F_x = 0$

$F_2 \cos 60° = F_1 \cos 30°$

$$\frac{F_2 \cos 60°}{\cos 30° \times 2} + \frac{F_2\sqrt{3}}{2} = 250$$

$F_2 = 215.6$ N

$F_1 = 125$ N

110. The free-body diagram of a satellite rotating about the earth will show the satellite isolated from its surroundings and _____

(a) Its velocity shown on it
(b) Only the force of gravity acting on it
(c) No force acting on it
(d) The force of gravity and centrifugal force acting on it

UPRVUNL AE 21.08.2016

Ans. : (d) The free-body diagram of a satellite rotating about the earth will show the satellite isolated from its surroundings and the force of gravity and centrifugal force acting on it.

111. A force F acting on a rigid body at a point P can be replaced by a force of equal magnitude and in the same direction at a point Q on the body together with a moment.

(a) Equal in magnitude to PQ times F, acting normal to the plane of F and PQ
(b) Given by F×QP
(c) Given by F×PQ
(d) Equal in magnitude to F times the distance moved in the lines of action of the force, acting in the plane of PQ and F

UPRVUNL AE 21.08.2016

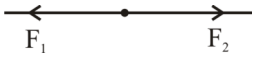
Ans. : (b) A force F acting on a rigid body at a point P can be replaced by a force of equal magnitude and in the same distance at a point Q on the body together with a moment given by F × QP.

112. Force acting on a rope in a tug of war, is an example of :

(a) Collinear force
(b) Coplanar force
(c) Coplanar concurrent force
(d) Coplanar non-concurrent force

UPRVUNL AE 05.07.2021
UPRVUNL AE 2014

Ans : (a) Force acting on a rope in a tug of war, is an example of collinear force.



113. Two equal forces are acting at a point with an angle of 60^0 between them. The magnitude of each force, if the resultant force is 36 N, is :

(a) 15.63 N (b) 25.12 N
(c) 20.78 N (d) 17.89 N

UPRVUNL AE 2014

Ans : (c) Let P is the force acting

Resultant $R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$

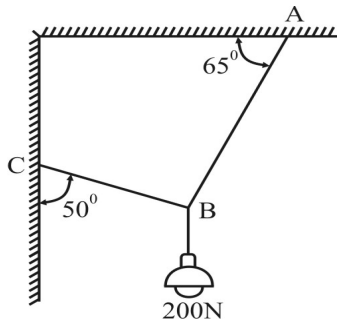
$$36 = \sqrt{P^2 + P^2 + 2P^2\cos 60^0} \qquad \{P = Q\}$$

$$36 = \sqrt{2P^2 + P^2}$$

$36 = \sqrt{3}\,P$

$P = 20.78$N

114. An electric light fixture weighing 200 N is supported (as shown in figure). Determine tensile force in wire BC.



(a) 20N (b) 87.5 N
(c)158.6 N (d) 120 N

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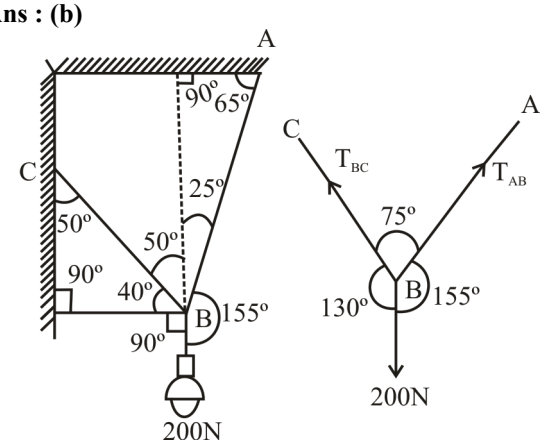
Ans : (b)



$$\frac{T_{BC}}{\sin 155^0} = \frac{W}{\sin 75^0}$$

$$T_{BC} = \frac{200}{\sin 75^o} \times \sin 155^0$$

$$\boxed{T_{BC} = 87.5N}$$

115. The principal compression member in a crane is called as

(a) boom (b) post

(c) stanchion (d) strut

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Boom – It is a principal compression member in a crane.

Post – It is a compression member in small structure with no roof.

Stanchion – A structural steel member usually larger than a strut, whose main function is to withstand axial compressive force.

Strut – It is one inclined member subjected to the axial load and supported by hinged or pin jointed support at both the ends.

116. In a system of coplanar parallel forces

(a) a resultant force exists

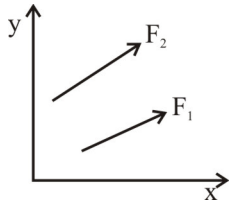
(b) a resultant force does not exist

(c) a resultant couple always exists

(d) a resultant couple can never exist

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Coplanar forces-



- Line of action of all forces lying on single plane.
- Static equilibrium equation,

$\Sigma F_x = 0, \Sigma F_y = 0, \Sigma F_z = 0$

117. A rope runs over a freely rotating frictionless pulley and two masses of 20 kg and 120 kg are hanging vertically from its two ends. The tension in the rope is

(a) less than 200N and more than 30 N

(b) More than 335 N and less than 338 N

(c) Exactly 100 kgf = 981 N

(d) Exactly 100 kgf = 1373.4 N

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Given, $m_1 = 20$ kg, $m_2 = 120$ kg

Lets T is the tension on tight side.

Balancing the force, we get.

$$T = (m_2 - m_1) \times g$$

$$T = (120 - 20) \times g = 100 \text{ kgf} = 981 \text{ N}$$

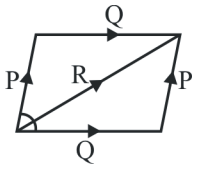
118. The resultant of two forces acting at 60° is 7 N and when acting at right angles is $\sqrt{34}$ N. The forces are

(a) 25 N and 9 N (b) 3 N and $\sqrt{5}$ N

(c) 3 N and 5 N (d) $\sqrt{3}$ N and $\sqrt{5}$ N

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : By Parallelogram Method–



$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

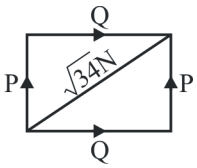
$$7 = \sqrt{P^2 + Q^2 + 2PQ\cos 60}$$

$$= \sqrt{P^2 + Q^2 + 2PQ \times \frac{1}{2}}$$

$$7 = \sqrt{P^2 + Q^2 + PQ}$$

$$49 = P^2 + Q^2 + PQ \quad .....(1)$$

When two forces acting at right angles with resultant if $\sqrt{34}$N.



$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

$$= \sqrt{P^2 + Q^2 + 2PQ\cos 90°}$$

$$\sqrt{34} = \sqrt{P^2 + Q^2}$$

$$34 = P^2 + Q^2 \quad .....(2) \quad \text{['Squaring both side']}$$

From equation (1),

$$49 = P^2 + Q^2 + PQ$$

$$49 = 34 + PQ$$

$$PQ = 49 - 34 = 15$$

$$PQ = 15$$

$$P = 3N, Q = 5N.$$

So, product of two forces is 15. Hence correct option is (c).

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119. Two equal forces are acting at a point with an angle of 60° between them. If the resultant force is equal to $60\sqrt{3}$, what is the magnitude of each force?

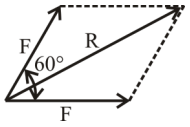
(a) 30 (b) 50
(c) 40 (d) 60

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Let two equal forces

$F_1 = F_2 = F$

By parallelogram method,



$$R = \sqrt{F^2 + F^2 + 2FF\cos 60°}$$

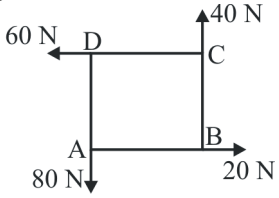
$$60\sqrt{3} = \sqrt{F^2 + F^2 + 2F^2 \times \frac{1}{2}}$$

$$60\sqrt{3} = \sqrt{3F^2}$$

$$F\sqrt{3} = 60\sqrt{3}$$

$$\boxed{F = 60}$$

120. Four forces of magnitudes 20 N, 40N, 60 N and 80 N are acting respectively along the four sides of a square ABCD as shown in figure. The magnitude of resultant is



(a) $40\sqrt{2}$ N (b) $50\sqrt{2}$ N
(c) $45\sqrt{2}$ N (d) $60\sqrt{2}$ N

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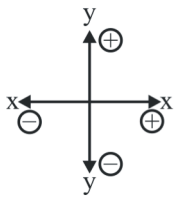
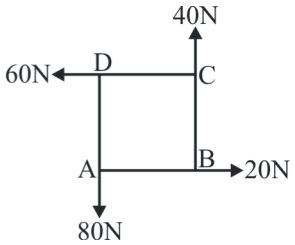
Ans. (a) :





Net horizontal force $\sum H = 0$

$F_H = -60 + 20$

$= -40N$

Net vertical force $\sum V = 0$

$F_V = 40 - 80 = -40N$

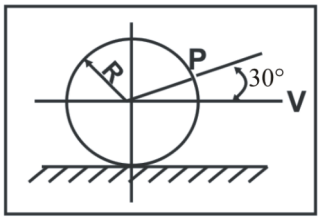
Magnitude of resultant 'F' is

$$F = \sqrt{(F_V)^2 + (F_H)^2}$$

$$= \sqrt{(-40)^2 + (-40)^2}$$

$$= 40\sqrt{2}N$$

121. A circular disc of radius R rolls without slipping at a velocity V. The magnitude of the velocity at point P shown in the figure is



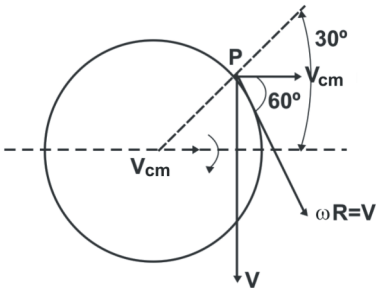
(a) $\sqrt{3}V$ (b) $\sqrt{3}/2V$
(c) $V/2$ (d) $\sqrt{2}V$

ISRO Scientist/Engineer 2012
GATE 2008

Ans. (a) : Given,

Circular disc of Radius = R

Velocity = V [∵ without slipping]



$V_{cm} = R\omega =$

$\theta = 60°$

Magnitude of velocity at point. 'P'.

$$= \sqrt{(V\sin\theta)^2 + (V + V\cos\theta)^2}$$

$$= \sqrt{(V \times \sin 60°)^2 + V^2(1 + \cos 60°)^2}$$

$$= \sqrt{\left(V \times \frac{\sqrt{3}}{2}\right)^2 + V^2\left(1 + \frac{1}{2}\right)^2}$$

$$= \sqrt{V^2 \times \frac{3}{4} + V^2\left(\frac{3}{2}\right)^2}$$

$$V_P = \sqrt{V^2\left[\frac{3}{4} + \frac{9}{4}\right]}$$

$$= V\sqrt{\frac{12}{4}}$$

$$V_P = V\sqrt{3}$$

122. The tension in the cable supporting a lift moving upwards is twice the tension when the lift moves downwards. What is the acceleration of the lift?

(a) g/4 (b) g/3
(c) g/2 (d) g

ISRO Scientist/Engineer 2011
ISRO Scientist/Engineer 2010
TNPSC AE 2014
Vizag Steel (MT) 2017

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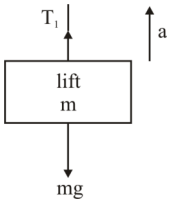
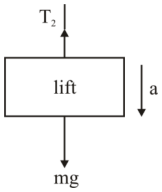
Ans. (b) : Case - I



Then, $T_1 - mg = ma$ ...(1)

Case - II



$mg - T_2 = ma$ ...(2)

Given that

$T_1 = 2\,T_2$

From equation (1) and (2)

$mg - T_2 = 2T_2 - mg$

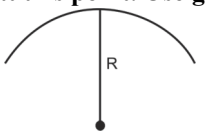
$3\,T_2 = 2\,mg$

$$\boxed{T_2 = \frac{2}{3}mg}$$

Putting the value of T_2 in equation (2)

$$mg - \frac{2}{3}mg = ma \qquad \boxed{a = \frac{g}{3}}$$

123. When a jet plane flying at 300 m/s is at the very top of its trajectory, the apparent weight of a passenger is one half of other actual weight. Find the radius of curvature R of the flight path at this point. Use g = 10 m/s^2



(a) 22 km (b) 18 km

(c) 30 km (d) 16 km

ISRO Scientist/Engineer 2011

Ans. (b) : V = 300 m/s

Let actual weight of passenger is = W

And apparent weight of passenger is = W'

So, balanced

$W' = W - M\omega^2 R$

$$= W - \frac{Mv^2}{R}$$

$$W' = \frac{W}{2} \qquad \text{(given)}$$

$$\frac{W}{2} = W - \frac{Wv^2}{Rg}$$

$$1 - \frac{v^2}{Rg} = \frac{1}{2}$$

$$\frac{v^2}{Rg} = \frac{1}{2}$$

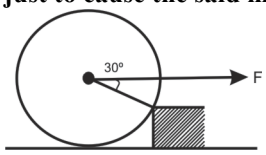
$$R = \frac{2v^2}{g}$$

$$= \frac{2\times(300)^2}{10}$$

R = 18000 m

= 18 km

124. A roller of weight W is rolled over the wooden block shown in the given figure. The pull required just to cause the said motion is



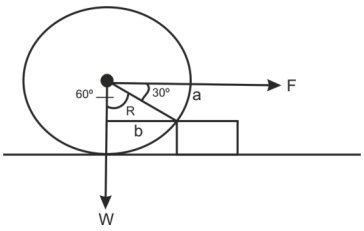
(a) W/2 (b) $\sqrt{3}$W

(c) W (d) 2W

UPPSC AE 12.04.2016 Paper-I

ISRO Scientist/Engineer 2009

Ans. (b) :



Resolving component

a = R cos 60°

b = R cos 30°

Taking moment at corner of wooden block

$\sum M = 0$

W × b = F × a

$$F = W\frac{b}{a}$$

$$= \frac{W\cos 30°}{\cos 60°}$$

$$F = W\sqrt{3}$$

125. An electric lift is moving downward with an acceleration of g/3, the vertical force between a passenger in the lift and its floor is equal to

(a) 3/4 of the passenger's weight

(b) 2/3 of the passenger's weight

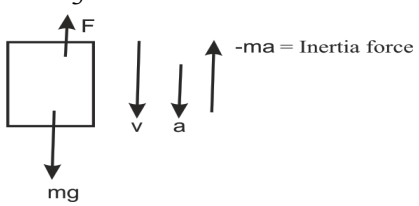
(c) Passenger's weight

(d) 4/3 of the passenger's weight

APPSC AEE 2012

ISRO Scientist/Engineer 2009

Ans. (b) : $a = \frac{g}{3}$ (downward given)



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Vertical force on the floor
= mg - ma
= m (g - a)
$= m\left(g - \frac{g}{3}\right)$
$= \frac{2mg}{3}$

126. A body of mass 10 kg moving with a velocity of 1 m/s is acted upon by a force of 50 N for two seconds. The final velocity of the body is

(a) 22 m/sec (b) 1 m/sec
(c) $\sqrt{21}$ m/sec (d) 11 m/sec

ISRO Scientist/Engineer 2009

Ans. (d) : m = 10kg, u = 1 m/s, F = 50 N, t = 2s

$a = \frac{50}{10} = 5m/s^2$
$F = ma$
$v = u + at$
$= 1 + 5\times2$
$= 11$ m/s

127. Ratio of the radii of the planets P_1 and P_2 is k and ratio of the acceleration due to gravity on them is s. Ratio of escape velocities from them will be

(a) ks (b) $\sqrt{ks}$
(c) $\sqrt{\frac{k}{s}}$ (d) $\sqrt{\frac{s}{k}}$

ISRO Scientist/Engineer 2008

Ans. (b) : The escape velocity is maximum velocity required to leave planet. It must overcome the pull of gravity. The formula of escape velocity contain a constant G which is called the "Universal gravitational constant". The unit of escape velocity is m/sec (m/s).

$$V_{esc} = \sqrt{\frac{2GM}{R}} \text{ where } g = \frac{GM}{R^2}$$

So we get $V_{esc} = \sqrt{\frac{2gR^2M}{MR}} = \sqrt{2gR}$

Ratio of escape velocity is given as

[Given, $\frac{R_1}{R_2} = k, \frac{g_1}{g_2} = s$]

$$\frac{V_1}{V_2} = \sqrt{\frac{g_1R_1}{g_2R_2}} = \sqrt{sk}$$

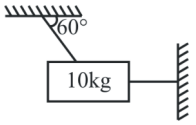
128. Which of the following is not a scalar quantity

(a) Time (b) Mass
(c) Volume (d) acceleration

ISRO Scientist/Engineer 2007
(KPSC AE 2015)

Ans. (d) : Acceleration has magnitude as well as direction, so it is a vector quantity.

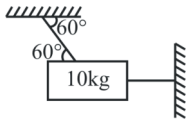
129. A 10 kg mass is hung from 2 light, inextensible strings as shown. The tension in the horizontal string is nearly



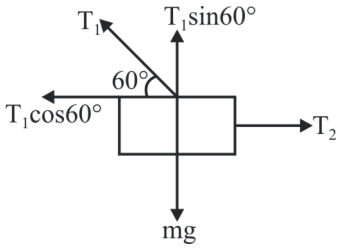
(a) 49 N (b) 57 N
(c) 100 N (d) 0 N

ISRO Scientist/Engineer 03.07.2016

Ans. (b)



FBD–



$\because$ $T_1\sin60° = mg$

$T_1 \times \frac{\sqrt{3}}{2} = 10 \times 9.81$

$T_1 = \frac{20\times9.81}{\sqrt{3}}$

Also, $T_1\cos60° = T_2$

$T_2 = \frac{20\times9.81}{\sqrt{3}} \times \frac{1}{2}$

$T_2 = \frac{10\times9.81}{\sqrt{3}}$

$T_2 = 56.63$ N

Tension in the Horizontal string $T_2 = 57$ N

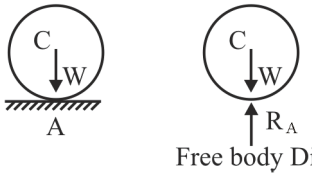
130. The free body diagram of a body is a diagram showing

(a) All the forces, including the reactions at the supports or surfaces
(b) The forces, but not showing the reactions at the supports or surfaces
(c) All the forces and the supports or surfaces
(d) Only the reactions at the supports or surfaces of the body

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : To clearly identify the various forces acting on a body in equilibrium we have to draw its free-body diagram. Only then, we can write the equilibrium equation of the body.

To draw the free body diagram of a body we remove all the supports (like wall floor, hinge, or any other body) and replace them by the reactions which these supports exert on body.



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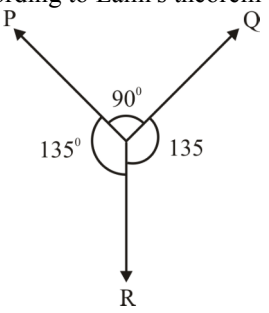
131. If three forces P, Q and R keep a body in equilibrium, the angle between P & Q is 90^0 and R is acting vertically downward, then magnitude of forces R and Q respectively in terms of P will be

(a) $\frac{P}{\sqrt{2}}, \sqrt{2}P$ (b) $\sqrt{2}P, \frac{P}{\sqrt{2}}$

(c) $P, \sqrt{2}P$ (d) $\sqrt{2}P, P$

APPSC AEE Screening Test 2016

Ans. (d) : According to Lami's theorem



$$\frac{P}{\sin 135^0} = \frac{Q}{\sin 135^0} = \frac{R}{\sin 90^0}$$

From,

$$\frac{P}{\sin 135^0} = \frac{R}{\sin 90^0}$$

$$\frac{P}{\frac{1}{\sqrt{2}}} = R \Rightarrow \boxed{R = \sqrt{2}P}$$

From,

$$\frac{Q}{\sin 135^0} = \frac{P}{\sin 135^0}$$

$$\boxed{Q = P}$$

132. A rigid body will be in equilibrium under the action of two forces only when:

(a) Forces have same magnitude, same line of action and same sense

(b) Forces have same magnitude and same line of action

(c) Forces have same magnitude, same line of action and opposite sense

(d) Forces have same magnitude

UPRVUNL AE 07.10.2016

Ans. (c) : A rigid body will be in equilibrium under the action of two forces only when forces have same magnitude, same line of action and opposite sense.

133. The resultant of two force A and B is perpendicular to A, the angle between the force A and B will be:

(a) $\theta = \cos^{-1}\left(\frac{A}{B}\right)$ (b) $\theta = \cos^{-1}\left(\frac{B}{A}\right)$

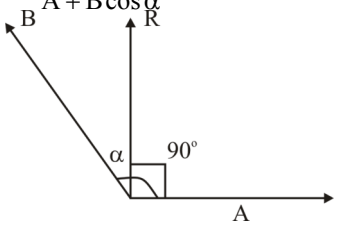
(c) $\theta = \cos^{-1}\left(-\frac{B}{A}\right)$ (d) $\theta = \cos^{-1}\left(-\frac{A}{B}\right)$

(e) $\theta = \tan^{-1}\left(\frac{B}{A}\right)$

CGPSC AE 25.02.2018

Ans. (d) : We know that-

$$\tan\theta = \frac{B\sin\alpha}{A + B\cos\alpha} \qquad \text{Given, } \theta = 90$$



$$\tan 90^o = \frac{B\sin\alpha}{A + B\cos\alpha}$$

$$\frac{1}{0} = \frac{B\sin\alpha}{A + B\cos\alpha}$$

$$A + B\cos\alpha = 0$$

$$A = -B\cos\alpha$$

$$\alpha = \cos^{-1}\left[-\frac{A}{B}\right]$$

134. Concurrent forces 2, $\sqrt{3}$, 5, $\sqrt{3}$ and 2 kN act at one of the angular points of a regular hexagon towards the remaining five angular points. Determine the magnitude and direction of the resultant force.

(a) R = 10 kN; $\alpha = 0^o$

(b) R = 12 kN; $\alpha = 180^o$

(c) R = 10 kN; $\alpha = 60^o$

(d) R = 20 kN; $\alpha = 0^o$

(e) R = 10 kN; $\alpha = 90^o$

CGPSC AE 26.04.2014 Shift-I

Ans. (c) : Resolving all forces horizontally

$$\sum H = 2\cos 0^o + \sqrt{3}\cos 30^o + 5\cos 60^o + \sqrt{3}\cos 90^o + 2\cos 120^o$$

$$= 2 + \left(\sqrt{3} \times \frac{\sqrt{3}}{2}\right) + 5\left(\frac{1}{2}\right) + \left(\sqrt{3} \times 0\right) + 2 \times \left(-\frac{1}{2}\right)$$

$$= 5 \text{ kN}$$

Resolving all forces vertically

$$\sum V = 2\sin 0^o + \sqrt{3}\sin 30^o + 5\sin 60^o + \sqrt{3}\sin 90^o + 2\sin 120^o$$

$$= 0 + \frac{\sqrt{3}}{2} + \frac{5\sqrt{3}}{2} + \sqrt{3} + \left(2 \times \frac{\sqrt{3}}{2}\right)$$

$$= 8.66 \text{ kN}$$

The magnitude of the resultant forces

$$R = \sqrt{(\sum H)^2 + (\sum V)^2}$$

$$= \sqrt{(5)^2 + (8.66)^2}$$

$$= 9.99$$

$$\simeq 10kN$$

Direction of the resultant force

$$\tan\theta = \frac{\sum V}{\sum H} = \frac{9.99}{8.66} = 1.732$$

$$\theta = 59.99 \simeq 60^o$$

135. A particle in equilibrium cannot have

(a) Linear motion with constant speed

(b) Constant velocity

(c) Zero acceleration

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(d) Zero velocity
(e) Curvilinear motion with constant speed

CGPSC AE 26.04.2014 Shift-I

Ans. (e) : A particle is said to be in equilibrium if the net force acting on the particle is zero. By Newton's second law, the acceleration of such objects will be zero.
As a result, either the body is at rest (i.e. zero velocity) or it is moving with a constant velocity (i.e. with constant speed in a straight line).

136. The force applied on a body of mass 100 kg to produce an acceleration of 5 m/s^2 is
(a) 20 N (b) 100 N
(c) 500 N (d) None of these

Vizag Steel (MT) 2017

Ans. (c) : $F = ma$
$= 100 \times 5$
$= 500$ N

137. Two parallel forces 100 kN and 75 kN act on a body and have resultant of 25 kN. Then, the two forces are
(a) Like parallel forces
(b) Unlike parallel forces
(c) Concurrent forces
(d) None of the above

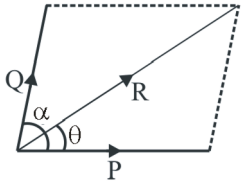
Gujarat PSC AE 2019

Ans : (b) : We know that,
$R^2 = P^2 + Q^2 + 2PQ\cos\theta$
Given, $P = 100$ kN, $Q = -75$ kN (Assume)
$R^2 = (100)^2 + (-75)^2 + 2 \times 100 \times (-75) \times \cos 0^o$
$R^2 = 625$
$R = \sqrt{625}$
$\boxed{R = 25 \text{ kN}}$
Hence, the two forces are unlike parallel forces.

138. The angle between two forces P and Q is α The resultant of these forces is
(a) $\sqrt{P^2 + Q^2 + 2PQ\sin\alpha}$
(b) $\sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$
(c) $\sqrt{P^2 + Q^2}$
(d) $\sqrt{P^2 + Q^2 - 2PQ\cos\alpha}$

UPPSC AE 12.04.2016 Paper-I
TNPSC ACF 2012

Ans : (b) Parallelogram Law of Forces:- It states that if two forces, acting simultaneously on a particle, be represented in magnitude and direction by the two adjacent side of a parallelogram, then their resultant may be represented in magnitude and direction by the diagonal of a parallelogram which passes through their points of intersection.



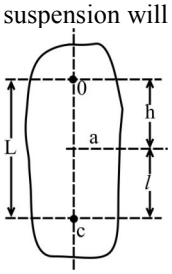
$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$

$\tan\theta = \dfrac{Q\sin\alpha}{P + Q\cos\alpha}$
$\theta \rightarrow$ Angle make by Resultant force R from P.

139. If a suspended body is struck at the centre of percussion, then the pressure on the axis passing through the point of suspension will be:
(a) Maximum (b) Minimum
(c) Zero (d) Infinity

HPPSC W.S. Poly. 2016

Ans : (c) If a suspended body is struck at the centre of percussion, then the pressure on the axis passing through the point of suspension will be zero.

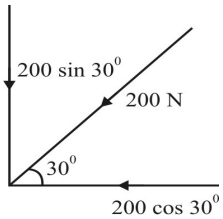


The centre of oscillation is termed as centre of percussion. It is defined as that point at which a blow may be struck on a suspended body so that the reaction at the support is zero.

140. The horizontal and vertical components of a force of 200 N acting on a body at an angle of 30° with the horizonatal is
(a) $100\sqrt{3}$ and 100 N
(b) $200\sqrt{3}$ N and 200 N
(c) $\dfrac{400}{\sqrt{3}}$ N and 400 N
(d) $300\sqrt{3}$ N and 300 N

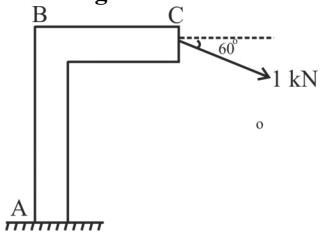
TSPSC AEE 2015

Ans : (a)



horizontal component $= 200 \cos 30°$
$F_H = 100\sqrt{3}$N
Vertical component $= 200 \sin 30°$
$F_v = 100$ N

141. What is the thrust at the point 'A' in the post shown in the figure?



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(a) 0.866 kN (b) 0.5 kN
(c) 1.388 kN (d) 1 kN

UPPSC AE 12.04.2016 Paper-I
CSE Pre-2004

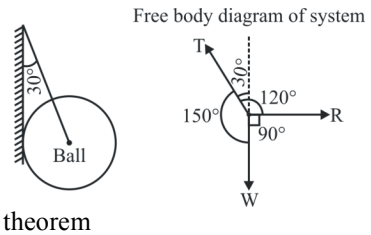
Ans : (a) Thrust at the point A= Vertical component of force 1kN
Thrust force = 1 sin 60°
Thrust force = 0.866kN

142. A ball of weight 100N is tied to a smooth wall by a cord making an angle of 30° to the wall. The tension in the cord is

(a) 200N (b) $\frac{200}{\sqrt{3}}$N
(c) 100N (d) $50\sqrt{3}$N

ESE 2017

Ans. (b) :



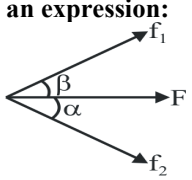
By Lami's theorem

$$\frac{T}{\sin 90°}=\frac{W}{\sin 120°}=\frac{R}{\sin 150°}$$

$$\frac{T}{\sin 90°}=\frac{W}{\sin 120°}=\frac{100}{\sqrt{3}/2}=\frac{200}{\sqrt{3}}$$

$$\boxed{T=\frac{200}{\sqrt{3}}N}$$

143. In a given figure, when angles β = α, then the components (f_1 and f_2) of force F on either side are given by an expression:



(a) $f_1=f_2=\frac{F\cos\alpha}{\sin(\alpha+\beta)}$

(b) $f_1=f_2=\frac{F\sin\alpha}{\sin(\alpha+\beta)}$

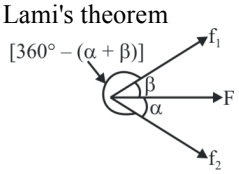
(c) $f_1=f_2=\frac{F\cos\alpha}{\cos(\alpha+\beta)}$

(d) $f_1=f_2=\frac{F\sin\alpha}{\sin(\alpha)}$

(e) $f_1=f_2=\frac{F\sin\alpha}{\sin(\beta)}$

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Using Lami's theorem



$$\frac{f_1}{\sin\alpha}=\frac{f_2}{\sin\beta}=\frac{F}{\sin\left[360°-(\alpha+\beta)\right]}$$

So, $f_1=-\frac{F\sin\alpha}{\sin(\alpha+\beta)}$

$f_2=-\frac{F\sin\beta}{\sin(\alpha+\beta)}$ $(\alpha=\beta)$

Then, $f_1=f_2=\frac{F\sin\alpha}{\sin(\alpha+\beta)}$

[–ve not considered in answer]

144. If the sum of all the forces acting on a moving object is zero, the object will

(a) continue moving with constant velocity
(b) accelerate uniformly
(c) change the direction of motion
(d) slow down and stop

UKPSC AE 2012 Paper-I

Ans. (a) : continue moving with constant velocity

145. Two equal and mutually perpendicular forces of magnitude 'P', are acting at a point. Their resultant force will be

(a) $P\sqrt{2}$, at an angle of 30° with the line of action of any one force.
(b) $P\sqrt{2}$, at an angle of 45° with the line of action of each force.
(c) $P\sqrt{2}$, at an angle of 45° with the line of action of each force.
(d) Zero

UKPSC AE 2012 Paper-I

Ans. (b) : $P\sqrt{2}$, at an angle of 45° with the line of action of each force.

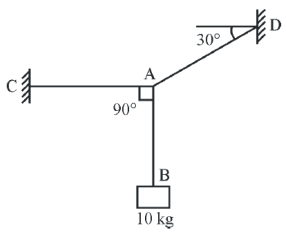
146. One end of an uniform ladder, of length L and weight W, rests against a rough vertical wall and the other end rests on rough horizontal ground. The coefficient of friction f is same at each end. The inclination of ladder when it is on the point of slipping is

(a) $\tan^{-1}\left(\frac{1-f^2}{2f}\right)$ (b) $\tan^{-1}\left(\frac{1+f^2}{2f}\right)$
(c) $\tan^{-1}\left(\frac{2f}{1+f^2}\right)$ (d) $\tan^{-1}\left(\frac{2f}{1-f^2}\right)$

UKPSC AE 2012 Paper-I

Ans. (a) : $\tan^{-1}\left(\frac{1-f^2}{2f}\right)$

147. In the following figure, the tension in the rope AC is



(a) 17.32 N (b) 56.60 N
(c) 169.90 N (d) 113.20 N

UKPSC AE 2012 Paper-I

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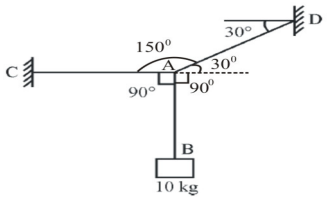
Ans. (c) :



From Lami's theorem

$$\frac{AC}{\sin 120} = \frac{AD}{\sin 90} = \frac{AB}{\sin 150}$$

$$\frac{AD}{\sin 90} = \frac{AB}{\sin 150} \Rightarrow \frac{AD}{1} = \frac{10}{0.5}$$

AD = 20 kg

$$\frac{AC}{\sin 120} = \frac{AD}{\sin 90}$$

AC = 20 × sin 120 = 20 × 0.860 kg

= 20 × 0.866 × 9.81 N

AC = 169.91 N

148. Two forces each equal to P/2 act at right angles. Their effect may be neutralized by a third force acting along their bisector in the opposite direction with a magnitude of

(a) P (b) P/2

(c) $\sqrt{2}$P (d) $\frac{P}{\sqrt{2}}$

UKPSC AE 2007 Paper -I

APPSC AEE Mains (Civil Mechanical) 2016

Ans. (d) : Given, $F_1 = F_2 = P/2$

Angle between forces is 90°

Resultant force, $F = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 90°}$

$$F = \sqrt{\left(\frac{P}{2}\right)^2 + \left(\frac{P}{2}\right)^2 + 0}$$

$$F = \sqrt{\frac{P^2}{2}}$$

$$\boxed{F = \frac{P}{\sqrt{2}}}$$

149. The resultant of forces $\vec{P} = -2\vec{i} - 3\vec{j}$ and $\vec{Q} = 3\vec{i} - 4\vec{j}$ will lie in (quadrants to be reckoned anticlockwise) quadrant

(a) first (b) second

(c) third (d) fourth

UKPSC AE 2007 Paper -I

Ans. (d) : Fourth

150. Polygon of forces is useful for computing the resultant of

(a) concurrent spatial forces

(b) coplanar parallel forces

(c) coplanar concurrent forces

(d) coplanar collinear forces

UKPSC AE 2007 Paper -I

Ans. (c) : Coplanar concurrent forces

151. The quantity whose dimensions are $M^2L^2T^{-3}$ could be the product of

(a) force and velocity (b) mass and power

(c) force and pressure (d) force and distance

UKPSC AE 2007 Paper -I

Ans. (b) : Dimension–

Product = Mass × Power

= M × power (P = F.v)

$$= M \times M \cdot \frac{L}{T^2} \cdot \frac{L}{T}$$

$$= M^2L^2T^{-3}$$

152. In case of concurrent coplanar forces, the condition of equilibrium is

(a) $\sum H = 0, \sum V = 0, \sum M = 0$

(b) $\sum H = 0, \sum V = 0$

(c) $\sum H = 0, \sum M = 0$

(d) $\sum V = 0, \sum M = 0$

UKPSC AE 2007 Paper -I

Ans. (b) : In case of concurrent coplanar forces, the condition of equilibrium is

$$\sum H = 0$$

$$\sum V = 0$$

153. If a body is in equilibrium then the following is true:

(a) There is no force acting on the body

(b) Resultant of all forces is zero but the moments of forces about any point is not zero

(c) The moments of the forces about any point is zero, but the resultant of all forces is not zero

(d) both (b) and (c)

UKPSC AE 2007 Paper -I

Ans. (d) : Both (b) and (c)

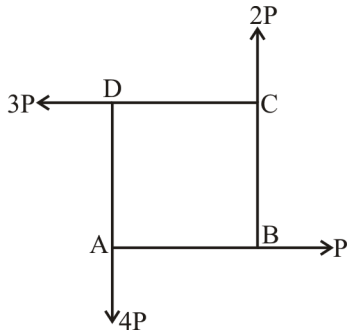
154. Four forces P, 2P, 3 P & 4P act along the sides of a square, taken in order. The resultant force is

(a) zero (b) $\sqrt{5}$P

(c) $2\sqrt{2}$P (d) 2P

UKPSC AE 2012 Paper-I

Ans. (c) :



Net horizontal force, ∑H = 0

$F_H = P - 3P = -2P$

Net vertical force, ∑V = 0

$F_V = 2P - 4P = -2P$

∵ Resultant force, $R = \sqrt{(F_H)^2 + (F_V)^2}$

$$= \sqrt{(-2P)^2 + (-2P)^2}$$

$$= \sqrt{4P^2 + 4P^2}$$

$$= \sqrt{8P^2}$$

$$= 2\sqrt{2}\,P$$

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155. According to the Newton's law of gravitation, the force of attraction, between the bodies of masses m_1 and m_2 situated at a distance 'd' apart, is given by

(a) $F = G\frac{m_1 m_2^2}{d^2}$ (b) $F = G\frac{m_1^2 m_2}{d^2}$

(c) $F = G\frac{m_1^2 m_2^2}{d^2}$ (d) $F = G\frac{m_1 m_2}{d^2}$

UKPSC AE 2012 Paper-I

Ans. (d) : According to the Newton's law of gravitation-

$$F = G\frac{m_1 m_2}{d^2}$$

156. An elevator weighing 10 kN attains an upward velocity of 4 m/s in 2 sec with uniform acceleration. The tension in the wire rope is nearly

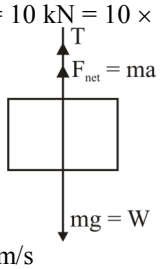
(a) 6 kN (b) 8 kN

(c) 10 kN (d) 12 kN

JWM 2017

Ans. (d) : Given,

Elevator weight W = 10 kN = 10×10^3 N



Upward force, v = 4m/s

Time, t = 2 sec

$v = u - at$

$u = 2 \times a \Rightarrow a = 2m/s^2$

$F_{net} \Rightarrow T - mg = ma$

$T = m(g + a)$

$$T = mg\left(1 + \frac{a}{g}\right)$$

$$= 10\left(1 + \frac{2}{9.81}\right)$$

Tension in wire, T = 12.03 kN

157. A force $\vec{F} = \left(10\hat{i} + 8\hat{j} - 5\hat{k}\right)$ act at a point P (2, 5,6). What will be the moment of the force about the point Q(3, 1, 4)?

(a) $-36\hat{i} + 15\hat{j} - 48\hat{k}$ (b) $18\hat{i} + 16\hat{j} - 37\hat{k}$

(c) $32\hat{i} - 31\hat{j} - 4\hat{k}$ (d) Zero

APPSC Poly Lect. 13.03.2020

Ans. (a) : We know that,

Moment = Force × perpendicular distance

Moment about Q = $\vec{F} \times \overrightarrow{PQ}$

$\overrightarrow{PQ}$ = Q – P. (vectors)

$= \left(3\hat{i} + \hat{j} + 4\hat{k}\right) - \left(2\hat{i} + 5\hat{j} + 6\hat{k}\right)$

$= \left(\hat{i} - 4\hat{j} - 2\hat{k}\right)$

Moment about Q = $\vec{F} \times \overrightarrow{PQ}$

$$\Rightarrow \vec{F} \times \overrightarrow{PQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 10 & 8 & -5 \\ 1 & -4 & -2 \end{vmatrix}$$

$$= \hat{i}(-16 - 20) - \hat{j}(-20 + 5) + \hat{k}(-40 - 8)$$

$$\vec{F} \times \overrightarrow{PQ} = -36\hat{i} + 15\hat{j} - 48\hat{k}$$

158. A moving fluid mass may be brought to a static equilibrium position by applying an imaginary inertia force of the same magnitude as that of the accelerating force but in the opposite direction. This is called

(a) Pascal's law

(b) Archimedes principle

(c) D-Alembert's principle

(d) None of the above

ISRO Scientist/Engineer 11.10.2015

Ans. (c) : D-Alembert's Principle–

- It states that under action of inertia force, and effective force, body will be in dynamic equilibrium.
- Purpose of D-Alembert's principle to convert a dynamic problem into static problem.
- Inertia force always acts opposite to direction of acceleration and not opposite to direction of motion.

159. How many equations are used in the equilibrium of particle in space ?

(a) 2 (b) 5

(c) 4 (d) 3

TNPSC AE (Industries) 09.06.2013

Ans. (d) :

160. Two forces of 10N and 20N are acting at a point in such a way that both the forces are perpendicular to each other. What is the resultant of these forces ?

(a) 22.36 (b) 23.63

(c) 26.23 (d) 21.36

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Given, P = 10, Q = 20 and θ = 90°

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta}$$

$$R = \sqrt{10^2 + 20^2 + 2 \times 10 \times 20 \times \cos 90°}$$

$$= \sqrt{500}$$

$$\boxed{R = 22.36\,N}$$

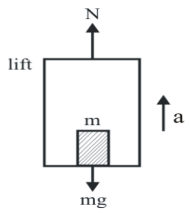
161. A body of mass 10 kg is placed in a lift moving upward with an acceleration of 2 m/s^2. What is the apparent weight of body for g = 9 m/s^2?

(a) 90 N (b) 70 N

(c) 110 N (d) 105 N

UPSC JWM Adv. No-16/2009

Ans. (c) :



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$\therefore$ Apparent Weight (N) = ma + mg
= m (a + g)
= 10 (2 + g)
= 10 × 11
= 110 N

162. If a number of forces are acting at a point, their resultant is given by

(a) $(\Sigma V)^2 + |\Sigma H|^2$

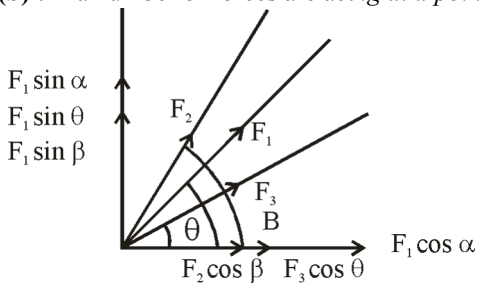
(b) $\sqrt{(\Sigma V)^2 + (\Sigma H)^2}$

(c) $(\Sigma V)^2 . (\Sigma H)^2 . 2(\Sigma V \| \Sigma H)$

(d) $\sqrt{(\Sigma V)^2 - (\Sigma H)^2 . 2(\Sigma V)(\Sigma H)}$

TNPSC ACF 2012

Ans. (b) : If a number of forces are act*ng at a poing*



Summation of all horizontal component
Vertical component
$\Sigma H = F_2 \cos\beta + F_3 \cos\theta + F_1 \cos\alpha$
$\Sigma V = F_1 \sin\alpha + F_3 \sin\theta + F_2 \sin\beta$
Resultant R = $\sqrt{(\Sigma V)^2 + (\Sigma H)^2}$

163. How can the resultant of two forces P and Q (such that P > Q) acting along the same straight line but in opposite direction, be given as?

(a) P + Q (b) P – Q
(c) P/Q (d) Q/P

VIZAG STEEL MT 2011

Ans. (b) : Solution

$R = \sqrt{P^2 + Q^2 + 2PQ\cos 180^\circ}$
$= \sqrt{P^2 + Q^2 - 2PQ}$
$= \sqrt{(P - Q)^2}$

164. One kgf/cm^2 when converted in SI units is

(a) 0.0981 MPa (b) 0.981 Pa
(c) 10^4 Pa (d) 1 P

APPSC AEE 2012

Ans. (a) : 1 kgf/cm^2 = 1 × 9.81 × 10^4
= 0.0981 × 10^6 Pa
1 kgf/cm^2 = 0.0981 MPa

165. Two bodies of different masses M_A and M_B are dropped from two different heights a and b respectively. The ratio of the time taken by the two, to drop through these distances in vacuum will be:

(a) a : b (b) b : a
(c) $\sqrt{a} : \sqrt{b}$ (d) $a^2 : b^2$

PTCUL AE 25.06.2017

Ans. (c) : During falling of bodies, a = g
v = u + gt
For initial velocity, u = 0
v = gt
$t \propto v$
$t \propto \sqrt{2gh} \propto \sqrt{h}$
Bodies dropped from height 'a' and 'b' for masses M_A and M_B respectively

$$\frac{t_A}{t_B} = \sqrt{\frac{a}{b}}$$

166. Two forces P and $P\sqrt{2}$ act on a particle in directions inclined at an angle of 135° to each other. Find the magnitude of the resultant.

(a) P (b) $P\sqrt{2}$
(c) 5P (d) None of these

PTCUL AE 25.06.2017

Ans. (a) : Resultant force,

$$R^2 = P^2 + \left(P\sqrt{2}\right)^2 + 2(P)\left(P\sqrt{2}\right)\cos 135^\circ$$
$$R^2 = P^2 + 2P^2 + 2\sqrt{2}\,P^2 \cos 135^\circ$$
$$R^2 = P^2 + 2P^2 - 2P^2$$
$$R^2 = P^2$$

Resultant, R = P

167. When two bodies of mass M and 2M are connected by light inextensible string and pass over smooth pulley with mass M on left side, then

(a) Tension in both sides of strings will be equal
(b) Tension in the left part of string will be twice than that in right part
(c) Tension in the left part of string will be half of that in right part
(d) None of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) :

168. In rigid body mechanics, equilibrium should satisfy with respect to

(a) Only forces (b) Both forces and moments
(c) Energy (d) Geometry

TANGEDCO AE 2015

Ans. (b) : In rigid bodies for equilibrium condition body should satisfy equation of forces of forces and moments both.

$\Sigma F_x = \Sigma F_y = \Sigma F_z = 0$
$\Sigma M_x = \Sigma M_y = \Sigma M_z = 0$

169. The theorem used for equilibrium of particle applied with three concurrent coplanar forces is

(a) Varignon's theorem
(b) Parallelogram theorem
(c) Lami's theorem
(d) Pythagoras theorem

TANGEDCO AE 2015

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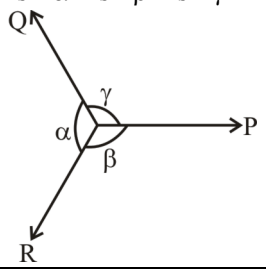
Ans. (c) : Lami's theorem– When three concurrent coplanar forces are acting and in equilibrium, then each force is proportional to the sine of the angle between the other two forces.

Mathematically, $\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$



170. Condition of static equilibrium of a planner force system is written as

(a) $\sum F = 0$ (b) $\sum M = 0$

(c) $\sum F = 0$ and $\sum M = 0$ (d) none of these

VIZAG Steel MT 2015

Ans. (c) : $\Sigma F = 0$ and $\Sigma M = 0$

171. Consider the following statements for system of forces:

1. **Two or more forces that act at the same point are called coplanar forces.**
2. **Two or more forces whose directed arrows lie in same plane are called concurrent forces.**
3. **Varignon's theorem states that the moment of several concurrent coplanar forces about any point O in their plane equals the moment of their resultant about the point O'.**
4. **Lami's theorem states that if a body is in equilibrium under the action of three forces, each force is proportional to the sine of angle between the other forces.**

Which of the above statements are correct?

(a) 1, 2 and 4 only (b) 1 and 4 only

(c) 3 and 4 only (d) 1, 2 and 3 only

ESE 18.07.2021

Ans. (c) : The correct statements are-

- Varignon's theorem states that the moment of several concurrent coplanar forces about any point O in their plane equals the moment of their resultant about the point O'.
- Lami's theorem states that if a body is in equilibrium under the action of three forces, each force is proportional to the sine of angle between the other forces.

2. Moments and Couples

172. When trying to turn a key into a lock, the following is applied :

(a) Coplanar force (b) Lever

(c) Moment (d) Couple

HPPSC W.S. Poly. 2016

Assam PSC CCE Pre 2015

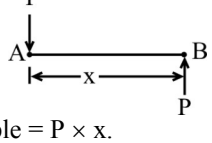
UKPSC AE-2013, Paper-I

J & K PSC Screening, 2006

GWSSB AAE, 27.12.2015

Ans : (d) When trying to turn a key into a lock, the couple is applied.

Couple:- The two equal and opposite forces, whose line of action are different, form a couple.



Moment of a couple = P × x.

173. A couple produces

(a) translatory motion

(b) rotational motion

(c) combined translatory and rotational motions

(d) none of these

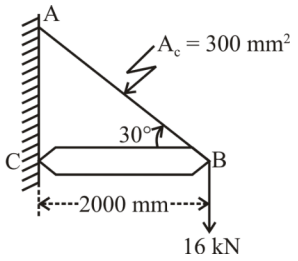
TNPSC ACF 2012

TNPSC AE, 2008

UKPSC AE 2007 Paper -I

Ans. (b) : A couple consists of two parallel forces that are equal in magnitude, opposite in sense and do not share a line of action. It does not produce any translating but only rotational motion.

174. A Jib crane consists of a rigid beam BC supported by an elastic rod AB and hinged at C. Determine the deflection of the point B when a load of 16 kN is applied. Take E = 200 GPa. Area of cross section (A_c) of elastic rod AB is 300 mm^2

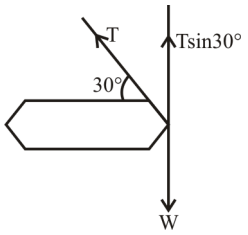


(a) 1.23 mm (b) 4.56 mm

(c) 2.46 mm (d) 5.23 mm

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) :



$$\ell_{AB} = \frac{2000}{\cos 30} = 2309.40\,\text{mm}$$

$F_y = 0$

$T\sin 30° = W$

$T = 16 \times 2 = 32$ kN

$$\delta L = \frac{T\ell_{AB}}{AE} = \frac{32\times10^3\times2309.40}{300\times200\times10^3}$$

$\delta L = 1.23$ mm

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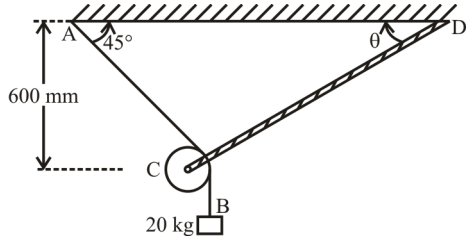
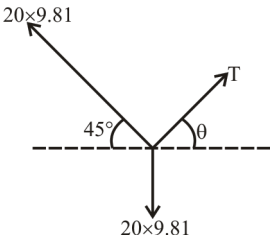
175. A load of 20 kg is supported as shown in fig. Assuming the pulley to be ideal and the equilibrium configuration as shown in figure, determine the tension in cable CD



(a) 250 N (b) 200.4 N
(c) 150.2 N (d) 100 N

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) :



$\sum F_x = 0$

$20 \times 9.81 \times \cos 45° = T\cos\theta$

$T\cos\theta = 138.73$ N .....(i)

And $\sum F_v = 0$

$T\sin\theta = 20 \times 9.81 - (20 \times 9.81 \times \sin 45°)$

$= 20 \times 9.81 (1 - \sin 45°)$

$T\sin\theta = 57.4656$ .....(ii)

Squaring and adding equations (i) and (ii) we get,

$T^2 = 138.73^2 + 57.4656^2$

$\boxed{T = 150.16\,N}$

176. For a many-particle system with particles interacting with each other, the conservation of angular momentum requires that the force of interaction between two particles should be

(a) equal and opposite
(b) equal, opposite and attractive
(c) equal, opposite and along the line joining the two particle
(d) equal, opposite along the line joining the two particles and keeps the distance between them fixed

TSPSC Manager (Engg.) HMWSSB 12.11.2020

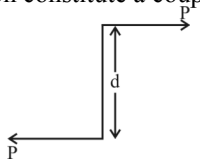
Ans. (c) : Equal opposite and along the line joining the two particle.

177. Two non-collinear parallel equal forces acting in opposite direction

(a) balance each other
(b) constitute a moment
(c) constitute a couple
(d) constitute a moment of couple

TNPSC AE 2019
CGPSC Poly. Lect. 22.05.2016

Ans. (c) : Two non-collinear parallel equal force acting in opposite direction constitute a couple.



C = P × d (clock-wise)

178. The tendency of rotation of the body along any axis is also called

(a) moment of inertia (b) moment of couple
(c) torque (d) force

Assam PSC AE (PHED), 18.10.2020

Ans. (c) : The tendency of rotation of the body along any axis also called the torque. It is the moment of the force acting perpendicular to the direction of the axis of rotation.
That is the moment is the cross product of the force and the distance between the axis and the point of action.

179. In the impact problem the following conditions must always be satisfied:

(a) the total kinetic energy remains unchanged
(b) the total linear momentum remains unchanged
(c) the magnitude of the total linear momentum is multiplied by the co-efficient of restitution
(d) none of the above

J&K PSC Civil Services Pre, 2013

Ans. (b) : During impact or collision only linear momentum remain unchanged.

180. Impulse gives a measure of the product of

(a) Force and velocity
(b) Force and acceleration
(c) Force and displacement
(d) Force and time

TSPSC AEE 2017

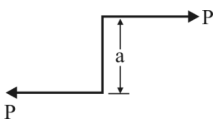
Ans. (d) : The impulse experienced by the object equals the change in momentum of the object.
Impulse = F.t
F.t = m.Δv (Product of force × time)

181. A…….. is formed by two equal magnitude and opposite direction forces displaced by perpendicular distance.

(a) Bending moment (b) Couple
(c) Shear moment (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) :



A couple is formed by two equal magnitude and opposite direction forces displaced by perpendicular distance.

$\boxed{\text{Couple} = P \times a}$ N.m

182. The unit of couple is :

(a) kN (b) kN / m
(c) kN . m (d) kN/m^2

GMB AAE 25.06.2017

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Ans. (c) : The moment of a couple is equal to the product of either force and the perpendicular distance between the line of action of both forces.
SI. unit of moment of couple is N.m

183. Dimensions for moment of momentum are :
(a) $M L^2 T^{-1}$ (b) MT^{-1}
(c) $M L^2 T^{-3}$ (d) MT^{-3}

J&K PSC Civil Services Pre, 2010

Ans. (a) : Angular momentum = $\vec{r} \times \vec{p} = I\omega$
Dimension = ML^2T^{-1}
S. I. unit = kgm^2S^{-1}

184. A zero torque acting on a system result in conservation of
(a) angular momentum (b) linear momentum
(c) angular velocity (d) energy

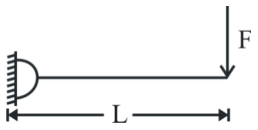
Haryana PSC Civil Services Pre, 2014

Ans. (a) : Conservation of angular momentum-
If external torque is absent in any system then angular momentum is conserved for that system.
If $T_{external} = 0$
then $\frac{d}{dt}(mvr) = 0$
Then $\boxed{\frac{d}{dt}(J) = 0}$
J = Angular Momentum (mvr)

185. What is the moment of force?
(a) moment of force is the sum of a force and its distance from an axis
(b) moment of force is the product of a force and its distance from an axis
(c) moment of force is the first derivative of force with respect to its axis
(d) none of these

KPCL AE 2016

Ans. (b) : The moment of force acting about a point or axis is found by multiplying the force (F) by the perpendicular distance from the axis (L), called the lever arm.
Moment = Perpendicular distance × Force
$M = L \times F$



186. The statement that "the moment of any force is equal to the algebric sum of the moments of the components of that force" is known as
(a) Varingon's theorem
(b) Principle of Moments
(c) Both A and B
(d) None of these

KPCL AE 2016

Ans. (c) : The principle of moments, also known as varignon's theorem, states that the moment of any force is equal to the algebric sum of the moment of the components of that force.

187. The action of a couple on a body .....if we change both the magnitudes of the forces and the arm of the couple in such a way that the moment of the couple
(a) does not change (b) always change
(c) becomes zero (d) can not be determined

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : The action of a couple on a body does not change if we change both the magnitudes of the forces and the arm of the couple in such a way that the moment of the couple remains unchanged.

188. A flat road has a curve segment with a radius of 105 m. While negotiating this curve, a vehicle slipped on its tyres as well as tried to roll over at a particular speed. This speed assuming a friction coefficient of 0.4 is :
(a) 18 m/s (b) 25 m/s
(c) 32 m/s (d) 20 m/s

UPRVUNL AE 2014

Ans : (d) Given that, $\mu = 0.4$ and curve radius = 105m
When the wheel goes under pure rolling Frictional force = Centrifugal force

$$\mu \times m \times g = \frac{mv^2}{r}$$

$$v = \sqrt{\mu gr}$$

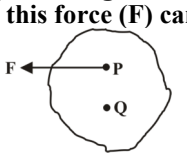
$$v = \sqrt{\mu gr} = \sqrt{0.4 \times 9.81 \times 105} = 20.29 m/s$$

189. The algebraic sum of moments of two unlike parallel forces about any point in their plane will be
(a) Zero (b) Constant
(c) Always anticlockwise (d) Always clockwise

APPSC AEE Screening Test 2016

Ans. (b) : The algebraic sum of their moments about any point in their plane is constant and it is equal to the moment of a couple.

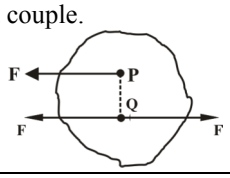
190. If a force (F) is acting on a rigid body at any point P, then this force (F) can be replaced by:

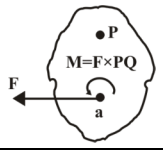


(a) An equal, opposite and parallel force (F) applied at point Q together with a couple
(b) An equal, parallel and same sense force (F) at point Q together with a couple
(c) A moment at point Q only
(d) An equal, opposite and parallel force (F) only at point Q

UPRVUNL AE 07.10.2016

Ans. (b) : If a force (F) is acting on a rigid body at any point P, then this force (F) can be replaced by an equal, parallel and same sense force (F) at point Q together with a couple.





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191. A 12 kN force produces a moment of 96 kN-m then, the moment arm is

(a) 2 m (b) 4 m
(c) 6 m (d) 8 m

TSPSC AEE 2015

Ans : (d) Given, F = 12 kN
T = 96 kN -m
We know that, $T = F \times d$

$$d = \frac{T}{F} = \frac{96}{12} = 8\,m$$

192. Which of the following statement is correct?

(a) the algebraic sum of the forces constituting the couple is zero
(b) the algebraic sum of the force, constituting the couple, about any point is same
(c) a couple cannot be balanced by a single force but can be balanced only by a couple of opposite sense
(d) all statements

TSPSC AEE 2015
UKPSC AE 2007 Paper -I

Ans. (d) : All statements are correct.

193. Varignon's theorem is related to

(a) Principle of moments
(b) Principle of momentum
(c) Principle of force
(d) Principle of inertia

UKPSC AE 2012 Paper-I

Ans. (a) : Principle of moments

194. The dimensions of angular velocity are given by

(a) $M^0L^1T^{-1}$ (b) $M^0L^2T^{-1}$
(c) $M^0L^0T^{-2}$ (d) $M^0L^0T^{-1}$

UKPSC AE 2007 Paper -I

Ans. (d) : $M^0L^0T^{-1}$

195. Opening a Limca bottle is due to

(a) moment (b) couple
(c) torque (d) parallel forces

UKPSC AE 2007 Paper -I

Ans. (a) : Moment

196. Cycle pedalling is an example of

(a) couple
(b) moment
(c) two equal and opposite forces
(d) two unequal parallel forces

UKPSC AE 2007 Paper -I

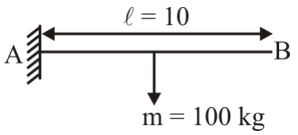
Ans. (a) : Couple

197. The length of the rod is 10 meter and its weight is 100 kg. One of its end is fixed in the wall. The moment about the fixed end of the rod will be :

(a) 1000 Nm (b) 9810 Nm
(c) 4905 Nm (d) 500 Nm

TNPSC AE (Industries) 09.06.2013

Ans. (c) :



Given,
$\ell = 10$ m
m = 100 kg
Moment about fixed end (i.e. point A),

$$M = 100 \times 9.81 \times \frac{10}{2}$$

$$\boxed{M = 4905\ N - m}$$

198. While a human walking on a smooth floor surface, what kind of reactions are produced under his foot ?

(a) Horizontal reaction
(b) Vertical reaction
(c) Horizontal and vertical reactions
(d) No reaction

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Human walking on a smooth floor surface, vertical reactions are produced under his foot.

3. Friction

199. The coefficient of friction depends on :

(a) nature of surface (b) shape of surfaces
(c) strength of surfaces (d) area of contact

Vizag Steel (MT) 2017
GWSSB DEE 07.04.2016
GPSC Poly. Lect. 07.07.2016
GWSSB AAE, 27.12.2015
UKPSC AE 2007 Paper-I
GPSC Poly. Lect. 07.12.2014
GWSSB DEE 07.07.2016

Ans. (a) : The coefficient of friction depends on nature of surface.

200. Static friction is always ____ dynamic friction

(a) Equal to (b) Less than
(c) Greater than (d) (A) and (B) both

GMB AAE 25.06.2017
TNPSC AE, 2008
UKPSC AE 2012 Paper-I
GPSC Poly. Lect. 07.12.2014

Ans. (c) : Static friction- The static friction is the frictional force develops between mating surface when subjected to external force but there is no relative motion between them.

$(\mu_s > \mu_k)$ ——— always

Dynamic or Kinetic Friction- Kinetic Friction is the force that resists the relative movement of the surface when they are in motion.

They static friction between two surface is always higher than the kinetic friction.

201. A body of weight W is placed on an inclined plane. The angle made by the inclined plane with horizontal, when the body is on the point of moving down is called

(a) angle of inclination
(b) angle of repose
(c) angle of friction
(d) angle of limiting friction

Vizag Steel (MT) 2017

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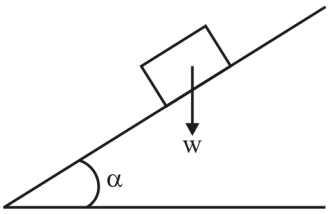
Haryana PSC Civil Services Pre, 2014
TNPSC ACF 2012

Ans. (b)



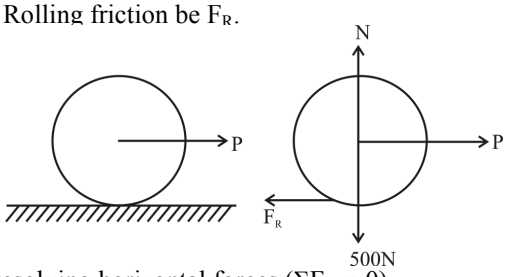
Angle of repose-
It is the angle of inclination from horizontal plane at which body starts moving down (sliding) or impending motion.

202. A steel wheel of 600 mm diameter is on a horizontal steel rail. It carries a load of 500N. The coefficient of rolling resistance is 0.3. The force in Newton, necessary to roll the wheel along the rail is

(a) 0.5 (b) 5
(c) 1.5 (d) 150

Karnataka PSC Lect., 27.05.2017
OPSC Civil Services Pre-2011
GATE 2000

Ans (a) : Diameter of wheel (d) = 600 mm
r = 300 mm
Load (W) = 500N
Coefficient of rolling resistance
(μ_R) = 0.3
Let, Rolling friction be F_R.



By resolving horizontal forces ($\Sigma F_H = 0$)
$P - F_R = 0$
$P = F_R$ ……(i)
By resolving vertical force ($\Sigma F_v = 0$)

Rolling friction, $F_R = \frac{W \times \mu_R}{r}$

$$F_R = \frac{500 \times 0.3}{300} = 0.5N$$

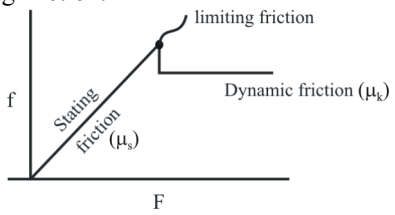
Now, from equation (i)
P = 0.5 N

203. Maximum value of frictional force acting on a body, when the body is just about to start is called :

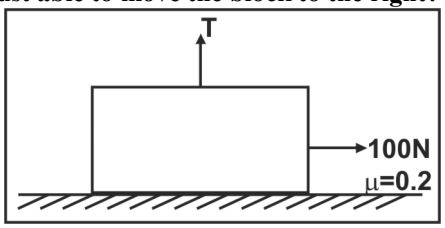
(a) limiting friction (b) static friction
(c) dynamic friction (d) none of these

MECON MT 2019
HPPSC Asstt. Prof. 29.10.2016
HPPSC Lect. (Auto) 23.04.2016
UKPSC AE 2012 Paper-I
VIZAG Steel MT 2011
UKPSC AE 2007 Paper-I

Ans. (a) : Limiting friction– The maximum friction that can be generated between two static surfaces in contact with each other. Once a force applied to the surfaces exceeds the limiting friction, motion will occur. For two dry surfaces, the limiting friction is a product of the normal reaction force and the coefficient of limiting friction.



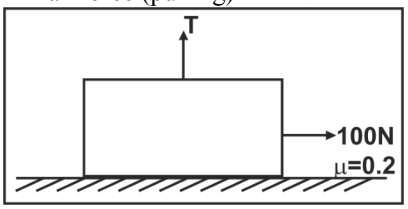
204. A block weighting 981 N is resting on Horizontal surface. The co-efficient of friction between the block and horizontal surface is μ = 0.2. A vertical cable attached to block provides partial support as shown. A man can pull horizontally with a force of 100N. What will be the tension (T) (in N) in the cable if the man is just able to move the block to the right?



(a) 176.2 (b) 196.0
(c) 481.0 (d) 981.0

ISRO Scientist/Engineer 2012
OPSC Civil Services Pre. 2011
GATE 2009

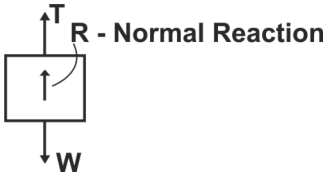
Ans. (c) : Given W = 981 N
μ = 0.2
A man force (pulling)



Let the normal Reaction at bottom of Block is R
∴ F = μR [By pulling force (man)]
100 N = μR

$$R = \frac{100}{0.2} = 500N$$

FBD, **Block**



T + R = W
T = W - R = 981 - 500
= 481 N.

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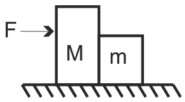
205. Two blocks with masses M and m are in contact with each other and are resting on a horizontal frictionless floor. When horizontal force F is applied to the heavier, the blocks accelerate to the right. The force between the two blocks are



(a) $\frac{mF}{(M+m)}$ (b) $\frac{MF}{m}$

(c) $\frac{mF}{M}$ (d) $(M+m)\frac{F}{m}$

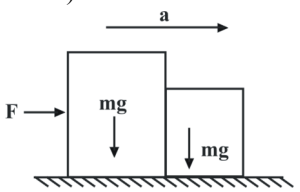
ISRO Scientist/Engineer 2011
ISRO Scientist/Engineer 2008
MPPSC AE 08.11.2015

Ans. (a) : Force between two block = ?
Let force between two blocks = F'

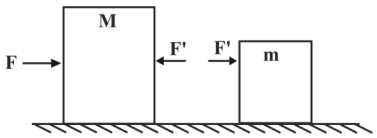
$F' = ma$ ...(1)

$F = (M + m)\, a$

$a = \frac{F}{(M+m)}$



Using FBD



Put value of a in equation (1)

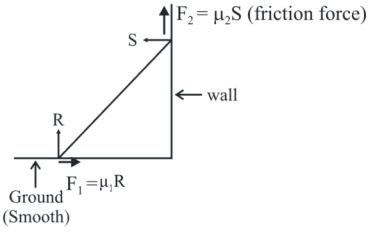
$F' = \frac{mF}{m+M}$

206. On a ladder resting on a smooth ground and leaning against rough vertical wall, the force of friction acts

(a) upward at its upper end
(b) towards the wall at the upper end
(c) towards the wall at lower end
(d) downward at its upper end

CGPSC AE 15.01.2021
BPSC Poly. Lect. 2016
TNPSC AE 2018
UKPSC AE 2007 Paper -I

Ans : (a)



for smooth ground
$\mu_1 = 0$
on a ladder resting on a smooth ground and leaning against rough vertical wall, the force of friction acts towards the wall at the upper end.

207. A rectangular block of mass 100 kg rests on a rough horizontal floor having co-efficient of friction 0.25. The minimum force required to move the block is _____.

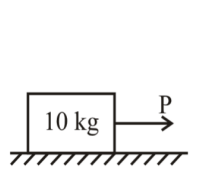
(a) 2452.5 N (b) 245.25 N
(c) 2.4525 N (d) 24.525 N

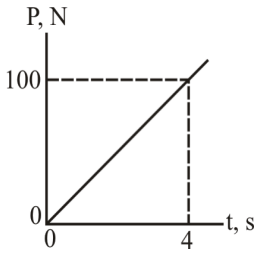
VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : Given,
m = 100 kg
$\mu = 0.25$
Minimum force required, $F = \mu R$

$F = \mu mg$
$= 0.25 \times 100 \times 9.81$
$= 245.25$ N

208. A wooden block of mass 10 kg is at rest initially as shown in the figure. A linearly time varying force is applied. The static and kinematic coefficients of friction are 0.6 and 0.4, respectively. The velocity of the block at t = 4s in m/s will be:

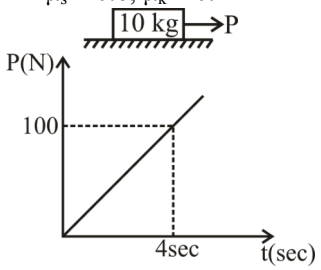




(a) 6.4 (b) 0
(c) 7.5 (d) 5

AAI Jr. Executive 26.03.2021

Ans. (a) : Given, m = 10 kg
$\mu_s = 0.6$, $\mu_k = 0.4$



Static friction force, $= \mu_s N$
$= 0.6 \times mg$
$= 0.6 \times 100 = 60$ N

Applied force at any 't' time $= \frac{100}{4} \times t = 25t$

Body will start motion when applied force is greater than 60 N i.e.,

$25t > 60$

$t > \frac{60}{25}$

$t > 2.4$ sec

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Now, from Newton's 2nd law of motion,

$$F_{net} = m\frac{dv}{dt}$$

$\Rightarrow$ $F_{net} = F_{app}$ – Kinetic friction [As body is moving after 2.4 sec]

$F_{net} = F_{app} - F_k$

$= 25t - \mu_k N$

$= 25t - 0.4 \times 100 = 25t - 40$

Thus, $25t - 40 = 10\frac{dv}{dt}$

Velocity at t = 4 sec

$$\int_{2.4}^{4}(2.5t+4)dt = \int_{0}^{v} dv$$

$$\left[2.5\frac{t^2}{2} - 4t\right]_{2.4}^{4} = v - 0$$

$v = 1.25\,(4^2 - 2.4^2) - 4 \times (4 - 2.4)$

$v = 6.4$ m/s

209. A force F, with magnitude 1N, is acting in XY-plane at a point (2 m, 4 m). It makes an angle of θ = 60° with the positive direction of x-axis. Torque of this force about the origin in Nm is:

(a) $-0.27\hat{k}$ (b) $0.27\hat{i}$

(c) $-0.27\hat{i}$ (d) $0.27\hat{k}$

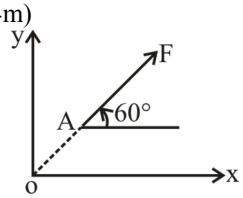
AAI Jr. Executive 26.03.2021

Ans. (a) : Given,

F = 1N (in x-y plane)

θ = 60° (wrt x-axis)

A(2m, 4m)



$\vec{F} = 1(\cos 60°\hat{i} + \sin 60°\hat{j})$

$\vec{F} = \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)$

Now, position vector of point A (2m, 4m)

$\vec{r} = 2\hat{i} + 4\hat{j}$

Torque about origin, $= \vec{r} \times \vec{F}$

$$= (2\hat{i} + 4\hat{j}) \times \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 4 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \end{vmatrix}$$

$\Rightarrow$ $\hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(\sqrt{3}-2)$

$\Rightarrow$ $(\sqrt{3}-2)\hat{k}$

$\Rightarrow$ $-0.27\hat{k}$

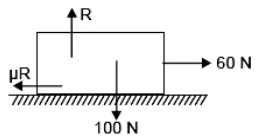
210. A body of weight 100 N is placed on a rough horizontal plane. What is the coefficient of friction if a horizontal force of 60 N just causes the body to slide over the horizontal plane?

(a) 0.4 (b) 0.5

(c) 0.6 (d) 0.9

ESE 18.07.2021

Ans. (c) :



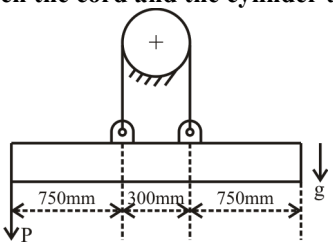
R = 100 N

Friction force = μ × R

$\Rightarrow$ $60 = \mu \times 100$

$\Rightarrow$ $\mu = 0.6$

211. A uniform beam of mass density 74 kg/m and length 1.8 m is suspended symmetrically as shown in Fig. Calculate the maximum force P that can be supported without tipping the beam. Take the coefficient of static friction between the cord and the cylinder to be 7/22



(a) 100.7 N (b) 200.9 N

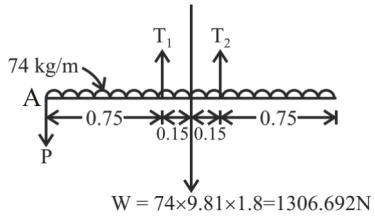
(c) 150.8 N (d) 50.2 N

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : Assume beam as UDL beam,

$$\mu = \frac{7}{22}$$

Angle of center, θ = π



For obtaining maximum force, $T_1 > T_2$

So, $\frac{T_1}{T_2} = e^{\mu\theta} = e^{\frac{7}{22}\cdot\pi} = e^1 = 2.718$

$\Sigma F_v = 0$

$T_1 + T_2 = P + W = P + 1306.692$

$P = 3.718T_2 - 1306.692$ .....(i)

$\Sigma M_A = 0$,

$-T_1 \times 0.75 - T_2 \times 1.05 + 1306.692 \times 0.9 = 0$

$\Sigma M = 0$,

$2.718T_2 \times 0.75 + T_2 \times 1.05 = 1176.0228$

$$T_2 = \frac{1176.0228}{3.0885} = 380.7747\,N$$

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$T_1 = 2.718T_2 = 1034.945$ N

From (i)

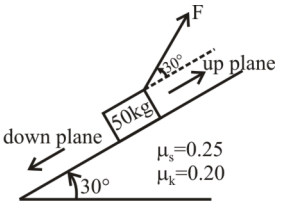
$P = 109.028$ N

If $g = 10$ m/sec^2

$\boxed{P = 111.1417\,N}$

So, option (a) $P \approx 100.7$ N

212. A force F is used to pull a 50 kg block, as shown in figure. Determine the magnitude and direction of the friction force when F = 400 N



(a) 45 N up the plane
(b) 80 N up the plane
(c) 45 N down the plane
(d) 80 N down the plane

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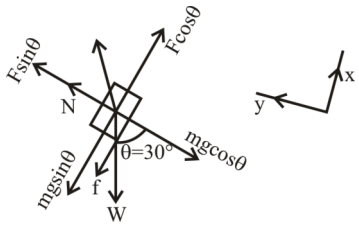
Ans. (c) :



$\sum F_y = 0$

$F\sin\theta + N = mg\cos\theta$

$N = 50 \times 10 \times \cos30° - 400 \times \sin30°$

$N = 233$ N

If (a) $F\cos\theta > mg\sin\theta + f_s$

Then the friction will be kinetic friction

(b) $F\cos\theta \le mg\sin\theta + f_s$

Then the friction will be static friction

Static friction force, $F_s = \mu_s N = 0.25 \times 233$

$= 58.25$ N

Kinetic friction force, $F_x = \mu_k N = 0.2 \times 233 = 46.6$ N

Net force along the inclined plane

$= F\cos30° - W\sin30°$

$$400\frac{\sqrt{3}}{2} - 500 \times \frac{1}{2} = 96.4\,N$$

Hence the friction will be kinetic friction and its value will be 46.6 N down the plane.

213. A sphere of mass m rolls without slipping on an inclined plane of inclination θ. Which of the following conditions should be satisfied for pure rolling?

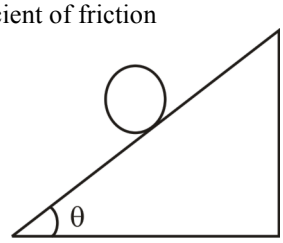
(a) $\mu < \frac{1}{7}\tan\theta$ (b) $\mu > \frac{2}{7}\tan\theta$

(c) $\mu > \frac{5}{7}\tan\theta$ (d) $\mu > \frac{3}{7}\tan\theta$

CGPSC AE 15.01.2021

Ans. (b) : Given, A sphere of mass m rolls without slipping down on inclined plane of angle of inclination θ.

Now, Coefficient of friction



$$\mu > \frac{\tan\theta}{1+\frac{R^2}{k^2}} \quad \text{... (i)}$$

where k is Radius of gyration,

Moment of Inertia of sphere.

$$I = \frac{2}{5}mR^2 = mk^2$$

$$k^2 = \frac{2}{5}R^2 \quad \text{... (ii)}$$

From (i) & (ii)

$$\mu > \frac{\tan\theta}{1+\frac{R^2}{\frac{2}{5}R^2}}$$

$$\boxed{\mu > \frac{2}{7}\tan\theta}$$

214. A box weight 1000 N is placed on the ground. The coefficient of friction between the box and the ground is 0.5. When the box is pulled by a 100 N horizontal force, the frictional force developed between the box and the ground at impending motion is

(a) 50 N (b) 75 N
(c) 100 N (d) 500

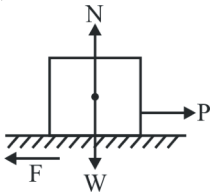
ESE 2018

Ans. (c) : Given,

$W = 1000$ N

$\mu = 0.5$

$P = 100$ N



Frictional force (F) = μN

$= 0.5 \times 1000$ ($\because$ N = W)

$= 500$ N

Since applied force (100N) is less than limiting friction, body will be at rest. As long as the body, is at rest

$F = P = 100$ N

215. A 13 m long ladder is placed against a smooth vertical wall with its lower end 5 m from the wall. What should be the coefficient of friction between the ladder and the floor so that the ladder remains in equilibrium?

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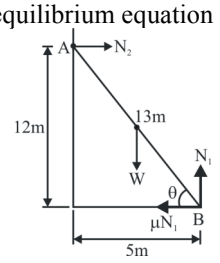
(a) 0.29 (b) 0.25
(c) 0.21 (d) 0.11

ESE 2018

Ans. (c) : Using equilibrium equation



$N_2 = \mu N_1$

$N_1 = W$ ($\because \mu_2 N_2 = 0$) Because wall is smooth

Taking moment about A

$W \times 2.5 + \mu_1 N_1 \times 12 = N_1 \times 5$

$\mu_1 = 0.208$

$\mu_1 \Rightarrow 0.21$

216. A body of weight 50 N is kept on a plane inclined at an angle of 30° to the horizontal. It is in limiting equilibrium. The coefficient of friction is equal to-

(a) $\frac{1}{\sqrt{3}}$ (b) $\sqrt{3}$

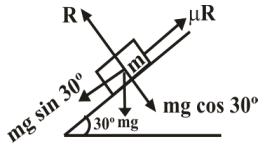
(c) $\frac{1}{50\sqrt{3}}$ (d) $\frac{\sqrt{3}}{5}$

RPSC AE 2018

Ans. (a) : Given:

$mg = 50$ N

$\theta = 30°$



For limiting equilibrium

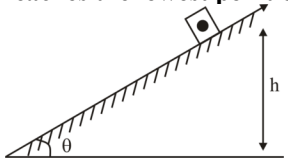
$mg \sin 30° = \mu \times R$

$mg \sin 30° = \mu \times mg \cos 30°$

$\mu = \tan 30°$

$\mu = \frac{1}{\sqrt{3}}$

217. A block of mass (m) slides down an inclined plane (coefficient of friction = μ) from the rest as in figure. What will be the velocity of block when it reaches the lowest point of plane?



(a) $2gh(1 - \mu \cot\theta)$ (b) $\sqrt{2gh(1-\mu\cot\theta)}$

(c) $\sqrt{2gh\mu(1-\cot\theta)}$ (d) $\sqrt{2gh(1-\mu\sin\theta)}$

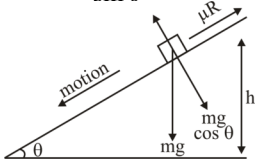
(e) $\sqrt{2gh(\mu-\tan\theta)}$

CGPSC AE 25.02.2018

Ans. (b) : Kinetic energy at lowest point on inclined plane of the block is

$\therefore \frac{1}{2}mv^2 = mgh$ – Energy loss due to friction

$= mgh - \mu R \times \frac{h}{\sin\theta}$

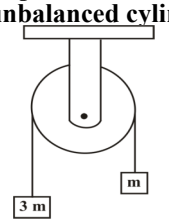


$= mgh - \mu \times mg\cos\theta \times \frac{h}{\sin\theta}$

$v = \sqrt{2gh[1-\mu\cot\theta]}$

$\frac{h}{\sin\theta}$ is distance covered by block during motion on inclined plane.

218. What is the minimum coefficient (μ) of friction between the rope and the fixed shaft which will prevent the unbalanced cylinder from moving?

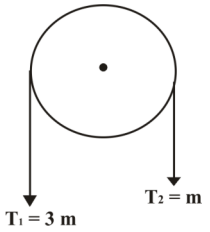


(a) $\mu = 0.333$ (b) $\mu = 0.350$

(c) $\mu = 0.253$ (d) $\mu = 0.372$

APPSC AEE SCREENING 17.02.2019

Ans. (b) :



$\frac{T_1}{T_2} = e^{\mu\theta}$

θ is angle of contact between rope and cylinder

$\theta = \pi$ radian

$\frac{3m}{m} = e^{\mu\pi}$

$3 = e^{\mu\pi}$

$1.098 = \mu\pi$

$\mu = 0.349$

≈ 0.35

219. Limiting friction depends upon

(a) Materials of the body in contact

(b) Weight of the body to be moved

(c) Roughness of surface of contact of the two bodies

(d) All of the above

Gujarat PSC AE 2019

Ans : (d) : Limiting Friction- The maximum value of frictional force, which comes into play when a body just begins to slide over the surface of the other body, is known as limiting friction. Limiting friction depends upon

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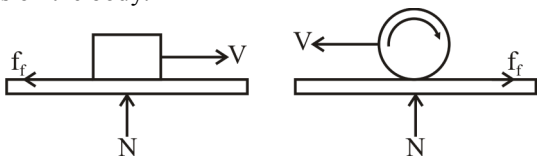
- Materials of the body in contact.
- Weight of the body to be moved.
- Roughness of surface of contact of the two bodies.

220. The energy dissipated during a full cycle of motion by dry friction damping in terms of maximum amplitude (A) and friction force (μN) is:

(a) 1μNA (b) 2μNA
(c) 4μNA (d) 3μNA

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (c) : Dry Friction Damping (or) Coulomb Damping– Dry friction damping occurs at the interface between two bodies in relative motion with contact. As indicated in fig., the friction force opposes the relative velocity direction and depends on the normal force that acts on the body.



Note– The following characteristics of a system with coulomb damping–

- The equation of motion is nonlinear with coulomb damping.
- The natural frequency of system is unchanged with the addition of coulomb damping.
- The motion is periodic with coulomb damping.
- The system comes to rest after some time with coulomb damping.
- The amplitude reduced linearly with coulomb damping.

The energy dissipated by dry friction damping in a full cycle of motion is

$$\boxed{\Delta w = 4\mu NA}$$

Where, $$A = \frac{F_o}{K}\left[\frac{1-\left(\frac{4\mu N}{\pi F_o}\right)^2}{\left(1-\left(\omega/\omega_n\right)^2\right)^2}\right]^{1/2}$$

221. For moderate speeds, the force of friction of kinetic friction remains constant. But it _____ slightly with the _____ of speed.

(a) decreases, increase
(b) decreases, decrease
(c) increases, increases
(d) increases, decrease

VIZAG Steel MT 24.01.2021

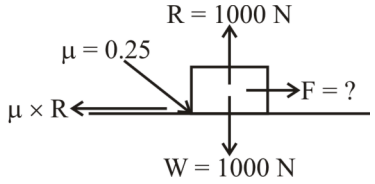
Ans. (a) : The magnitude of the kinetic friction bears a constant ratio to normal reaction between the two surfaces. But this ratio is slightly less than that in the case of limiting friction. For moderate speed, the force of friction remains constant. But it decreases slightly with the increase of speed.

222. A body of weight 1000 N is resting on a rough horizontal surface having a coefficient of friction as 0.25. The magnitude of the force required to move the body horizontally is ___.

(a) 2000 N (b) 250 N
(c) 500 N (d) 1000 N

VIZAG MT, 14.12.2020

Ans. (b) :



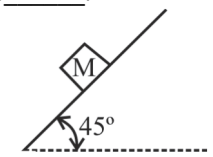
Force required to move body,

$F = \mu R$

$= 0.25 \times 1000$

$$\boxed{F = 250\,N}$$

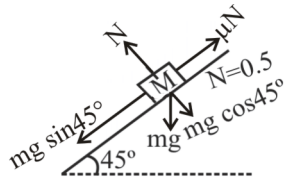
223. A body of mass (M) 10 kg is initially stationary on a 45° inclined plane as shown in figure. The coefficient of dynamic friction between the body and the plane is 0.5. The body slides down the plane and attains a velocity of 20 m/s. The distance travelled (in m) by the body along the plane is_____.



(a) 45 (b) 80
(c) 78.76 (d) 57.76

Assam Engg. College AP/Lect. 18.01.2021
UPPSC AE Paper-I 13.12.2020
GATE 2015

Ans. (d) :



Balancing forces along the inclined plane

$mg\sin 45 - \mu N = ma$

$$a = \frac{mg\sin 45 - 0.5 \times mg\cos 45}{m}$$

$$a = \frac{(9.81 - 0.5 \times 9.81) \times 1}{\sqrt{2}}$$

$$\boxed{a = 3.468\,m/s^2}$$

Distance traveled along the plane

$v^2 = u^2 + 2as$

$(20)^2 = 0 + 2 \times 3.468$

$$s = \frac{20^2}{2 \times 3.468}$$

$s = 57.67$ m

224. A block of mass 5 kg slides down from rest along a frictionless inclined plane that makes an angle of 30^0 with horizontal. What will be the speed of the block after it converse a distance of 3.6 m along the plane? [g = 10 m/s^2]

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(a) 5 m/s (b) 6 m/s
(c) 7 m/s (d) 8 m/s

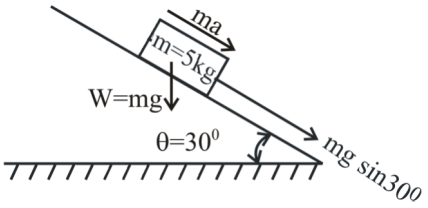
APPSC Poly Lect. 13.03.2020

Ans. (b) : Given, $g = 10\ m/s^2$

distance (s) = 3.6 m

m = 5 kg

initial velocity (u) = 0



At equilibrium condition –

$\Sigma F = ma$

$mg.\sin 30^0 - \mu R = ma$

$mg.\sin 30^0 = ma \qquad (\because \mu = 0)$

$$\Rightarrow a = g\sin 30^0 = 10 \times \frac{1}{2}$$

$$\boxed{a = 5\ m/s^2}$$

$\because v^2 = u^2 + 2as \quad = (0)^2 + 2 \times 5 \times 3.6$

$= v^2 = 2 \times 5 \times 3.6$

$= v^2 = 36$

$v = 6$ m/s

225. The maximum angle of inclination of the plane (with horizontal) at which a body remains in equilibrium under the action of friction only is termed as

(a) Angle of wedge
(b) Angle of equilibrium
(c) Angle of repose
(d) All the given answers

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : The maximum inclination of the plane at which a body can remain in equilibrium, over the plane entirely by the assistance of friction is called the Angle of Repose.

Angle of repose = Angle of limiting friction

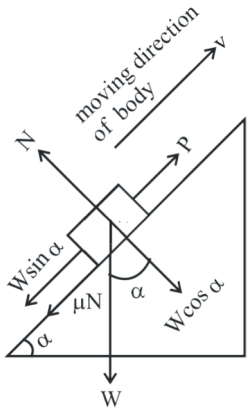
226. A body of weight W is required to move up on rough inclined plane whose angle of inclination with horizontal is α. The effort applied parallel to the plane is given by

(a) $P = W \tan \alpha$
(b) $P = W \tan (\alpha + \phi)$
(c) $P = W (\sin \alpha + \mu \cos \alpha)$
(d) $P = W (\cos \alpha + \mu \sin \alpha)$

Where $\mu = \tan \phi$ = coefficient of friction.

TNPSC AE, 2008

Ans. (c) :



N = Normal reaction

α = angle of inclination

P = Applied effort

μ = Coefficient of friction

Given,

W = Weight of the body

∴ Normal reaction on body

$N = W \cos \alpha$ ..... (i)

Now force balance along parallel to the inclined plane -

$P = W \sin \alpha + \mu N$ ..... (ii)

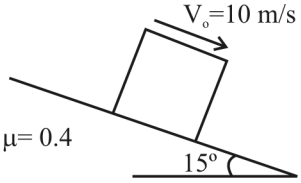
From equation (i) and (ii)

$P = W \sin \alpha + \mu W \cos \alpha$

$\Rightarrow P = W (\sin \alpha + \mu \cos \alpha)$

$P = W (\sin \alpha + \mu \cos \alpha)$

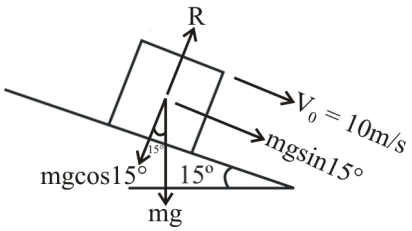
227. The 50 kg crate shown in figure down an incline with an initial velocity 10 m/s. The normal reaction of crate



(a) 233.80 N (b) 133.42 N
(c) 473.80 N (d) 500 N

APPSC Poly. Lect. 2013

Ans. (c) :



m = 50 kg, g = 9.81 m/s, μ = 0.4

Normal reaction ⇒ R = mgcos15°

= 50 × 9.81 × cos15°

= 473.78 N

228. A block weighing 100 N is resting on a surface. The coefficient of friction between the block and surface is 0.3. A horizontal force of 20N is applied on the block. What is the magnitude of force of friction acting on the block?

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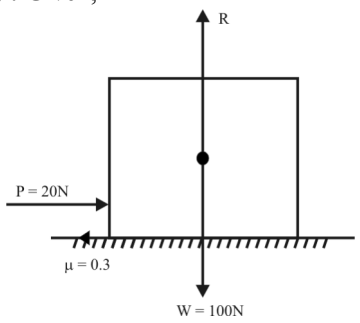
(a) 30 N (b) 100 N
(c) 50 N (d) 20 N

RPSC AEN Pre-2013

Ans. (d) : Given,



Friction force –

$F = \mu R$

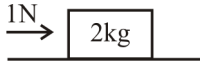
$= 0.3 \times 100 = 30$ N

Here, Applied force on block (P = 20N) less than friction force (F = 30N), So answer will be F = 20 N

229. A 2 kg block is resting on a rough surface with 0.1 coefficient of friction. A force of 1N is applied to the block as shown in figure. The friction force is-

(a) zero (b) 1N
(c) 1.96 N (d) 1.20 N

Ans. (b) :

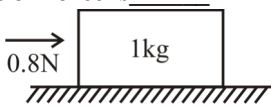


Limiting friction force between block and the surface is $2 \times 0.1 \times 9.8 = 1.96$ N

But applied force is 1N

Which is less than limiting friction, therefore the force of friction is 1N.

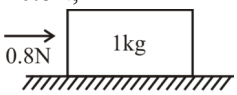
230. A 1 kg block is resting on a surface with co-efficient of friction μ = 0.1. A force 0.8 N is applied to the block as shown in the figure. The friction force is_____



(a) Zero (b) 0.8 N
(c) 0.89 N (d) 1.2 N

TSPSC AEE 2017
GATE 2011

Ans. (b) : μ = 0.1, P = 0.8N,



$F_r = \mu_s N$

$N = 9.81$N

$F_r = \mu_s N$

$= 0.1 \times 9.81$

$= 0.981$ N

$\because$ Applied force $< \mu_s N$

Then the friction force developed on the surface of block is equal to the applied force i.e equal to 0.8 N.

231. A tail stock of lathe weighs 120 kgf and the coefficient of friction as its slides is 0.12. The horizontal force (in kgf) required to slide the tail stock would be:

(a) 1000 (b) 14.4
(c) 120.12 (d) 120

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : We know that,

Frictional force (F) = μN

$F = 0.12 \times 120$ kgf

$= 14.4$ kgf

232. It the angle of friction is zero a body will experience

(a) infinite friction
(b) zero friction
(c) force of friction acting normal to the plane
(d) None of the above

WBPSC AE, 2017

Ans. (b) : Given, Angle of friction, (ϕ) = 0

So, $\tan\phi = \mu$

$\mu = \tan 0°$

$\mu = 0$

233. The slope on the road generally provided on the curve is known as

(a) Angle of repose (b) angle of friction
(c) Angle of banking (d) Angle or reaction

WBPSC AE, 2017

Ans. (c) : Angle of banking

234. A crate weighing 40 kg rests on a cart moving with an acceleration. The coefficient of friction between the cart the crate is 0.5. The maximum value of acceleration of the cart, so that the crate does not slip along the cart, is

(a) 0.25 m/sec^2 (b) 2.45 m/sec^2
(c) 4.90 m/sec^2 (d) 1.225 m/sec^2

WBPSC AE, 2017

Ans. (c) : We know that,

Force = Force due to friction

$ma = \mu mg$

$a = \mu g$

$= 0.5 \times 9.81 = 4.905$ m/s^2

235. The coefficient of friction between pulley and belt is reduced to half. If the initial ratio of tension in belt is 5, then the new value will be

(a) 2 (b) 2.236
(c) 4.47 (d) 3.64

WBPSC AE, 2017

Ans. (b) : Given, $\mu_1 = \mu$, $\mu_2 = \frac{\mu}{2}$

$$\frac{T_1}{T_2} = 5$$

We know that, $\frac{T_1}{T_2} = e^{\mu_1\theta}$

or $e^{\mu\theta} = 5$

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New value,

$$\left(\frac{T_1}{T_2}\right)_{new} = e^{\mu_2\theta}$$

$$\left(\frac{T_1}{T_2}\right)_{new} = e^{\frac{\mu}{2}\theta}$$

$$\left(\frac{T_1}{T_2}\right)^2_{new} = \left(e^{\frac{\mu}{2}\theta}\right)^2 \quad \text{[Taking both side square]}$$

$$= e^{\mu\theta}$$

$$\left(\frac{T_1}{T_2}\right)_{new} = \sqrt{5} = 2.236$$

236. A body of weight 600 N is placed on a horizontal surface which tend to move when a horizontal force of $200\sqrt{3}$ N is applied. The angle of friction is :

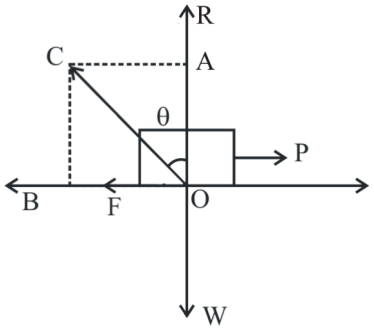
(a) 30º (b) 45º
(c) 60º (d) 90º

J&K PSC Civil Services Pre, 2010

Ans. (a) : Weight (w) = 600 N = R

Horizontal force (P) = $200\sqrt{3}$ = F (Friction force)

= (Body tends to move)



ΔAOC, $\tan\theta = \frac{AC}{OA} = \frac{P}{R}$

$$\frac{200\sqrt{3}}{600} = \frac{\sqrt{3}}{3}$$

$$= \frac{1}{\sqrt{3}}$$

$$\mu = \tan\theta = \frac{1}{\sqrt{3}}$$

$$\boxed{\theta = 30^0}$$

237. A sphere of mass m and radius R is rolling without slipping on a horizontal rough surface. If the acceleration of the mass centre of the sphere be a, the magnitude of the friction force.

(a) $\frac{1}{5}ma$ (b) $\frac{2}{5}ma$
(c) $\frac{3}{5}ma$ (d) $\frac{4}{5}ma$

WBPSC AE, 2007

Ans. (b) : We know that,

Torque (T) = $I\alpha$

$$F_f \cdot R = \frac{2}{5}mR^2 \times \left(\frac{a}{R}\right) \qquad \left[\alpha = \frac{a}{R}\right]$$

$$F_f = \frac{2}{5}ma$$

238. If the friction force or sliding velocity varies as a function of distance or time and produces oscillations, it is called a…

(a) Cold weld
(b) Adhesion
(c) Stick-slip phenomenon
(d) Ploughing of hard asperity

MPSC Dy. Engg. (GWD) 19.05.2013

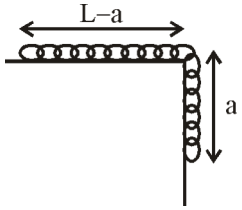
Ans. (c) : If the friction force or sliding velocity varies as a function of distance or time and produces oscillations, it is called a stick-slip phenomenon.

239. A flexible but inextensible chain of length 'L' and weight 'wL' is held on frictionless table with an initial overhang of length 'a'. The velocity 'v' with which the chain will leave the table if released is

(a) $\frac{g}{L}\sqrt{L^2 - a^2}$ (b) $\frac{g}{L}\sqrt{L^2 + a^2}$
(c) $\sqrt{\frac{g}{L}\left(L^2 - a^2\right)}$ (d) $\sqrt{\frac{g}{L}\left(L^2 + a^2\right)}$

OPSC Civil Services Pre 2006

Ans. (c) :



Initial P.E. for the above position = $W.a.\frac{a}{2}$

Final P.E. for the chain = $W.L.\frac{L}{2} = W\left(\frac{L^2}{2}\right)$

Change in K.E. of chain = change in P.E.

$$\frac{1}{2}mv^2 = W\left(\frac{L^2}{2}\right) - W\left(\frac{a^2}{2}\right)$$

$$\frac{1}{2}\frac{WL}{g}v^2 = \frac{W}{2}\left(L^2 - a^2\right)$$

$$\boxed{v = \sqrt{\frac{g}{L}\left(L^2 - a^2\right)}}$$

240. A block of mass M, supporting another block of mass m, is moved horizontally with an increasing acceleration from zero. If μ is the coefficient of static friction between the block, the acceleration a at with the upper block slips on the lower block is –

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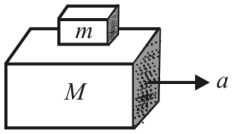
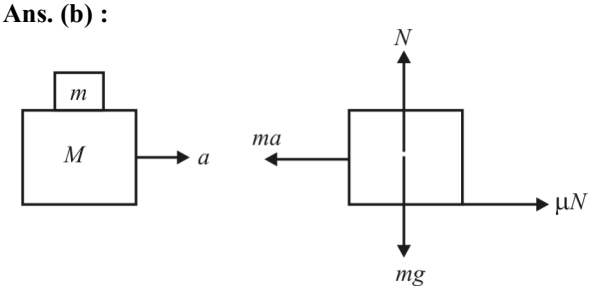


(a) μm/g (b) μg
(c) −μmM /(m + M) (d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) :



For smaller body.

N = mg

ma = μN

ma = mg μ

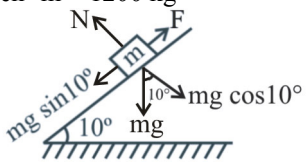
a = μg

241. A body with mass 1200 kg is located on a 10 degrees inclined plane. What is the pulling force without friction?

(a) 2044 N (b) 1200 N
(c) 120 N (d) None of these

KPCL AE 2016

Ans. (a) : Given- m = 1200 kg



F = mg sin 10° [without friction μ = 0]

= 1200 × 9.81 sin 10°

F = 2044.186 N

242. Which of the following is the friction that occurs when two objects are moving relative to each other and rub together

(a) Kinetic friction (b) Static friction
(c) Internal friction (d) None of these

KPCL AE 2016

Ans. (a) : Kinetic friction- The friction that occurs when two objects are moving relative to each other and rub together.

243. Which of the following occurs when surfaces in contact move relative to each other?

(a) the friction between the two surfaces converts kinetic energy into electrical energy
(b) the friction between the two surfaces converts kinetic energy into magnetic energy
(c) the friction between the two surfaces converts kinetic energy into thermal energy
(d) the friction between the two surfaces converts magnetic energy into electrical energy

KPCL AE 2016

Ans. (c) : When surfaces in contact move relative to each other the friction between the two surface convert kinetic energy into thermal energy.

244. What is the rolling resistance to the truck whose weight is 72,000 N and its co-efficient of rolling resistance is 0.019?

(a) 7200 N (b) 1900 N
(c) 1368 N (d) None of these

KPCL AE 2016

Ans. (c) : Given- Truck weight = 72000 N

Coefficient of rolling = 0.019

Rolling resistance = Coefficient of rolling × Weight

Rolling resistance = 0.019 × 72000

= 1368 N

245. The friction experienced by a body when it is in motion is called_____

(a) Body friction (b) Dynamic friction
(c) Angular friction (d) None of these

KPCL AE 2016

Ans. (b) : Dynamic friction is the force of friction experienced by a body when it is motion dynamic friction is the value of frictional force after slipping has occurred.

246. Which of the following statements is true?

(a) Dynamic friction is always less than static friction
(b) Dynamic friction is always greater than static friction
(c) Dynamic friction is always same as static friction
(d) Dynamic friction is double the static friction

KPCL AE 2016

Ans. (a) : Since the value of (F) in dynamic friction is little lesser than in static friction, the ratio F/N and hence, the coefficient of dynamic friction will also be a little lesser than static friction kinetic friction (also referred to a dynamic friction) is the force that resist the relative movement of the surfaces once they are in motion. The static friction between two surfaces is always higher than the kinetic friction.

$F_k < F_s$

$\mu_k < \mu_s$

247. When the two surfaces in contact have a minute thin layer of lubricant between them, it is known as .....friction.

(a) film (b) skin or greasy
(c) dry (d) solid

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : When the two surfaces in contact have a minute thin layer of lubricant between them, it is known as skin or greasy friction.

248. A car travelling at a speed of 60 km/hour is braked and comes to rest 6 seconds after the brakes are applied. The minimum coefficient of friction between the wheels and the road would be :

(a) 0.107 (b) 0.283
(c) 0.3 (d) 0.417

MPSC Poly. Lect. 2014

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Ans. (b) : Given

Initial velocity u = 60 km/hr

$= 60 \times \frac{5}{18} = 16.667 \text{m/s}$

$v = u - at$

$v = 16.667 - a \times 6$

$a = \frac{16.667}{6}$

$a = 2.777 \text{ m/s}^2$

we know that $a = \mu g$

$\mu = \frac{a}{g} = \frac{2.777}{9.81} = 0.283$

249. A body of weight 120 N rests on a rough horizontal plane. The coefficient of friction between the body and the plane is 0.3. The frictional force to be overcome for impending motion will be

(a) 30 N (b) 400 N

(c) 300 N (d) 36 N

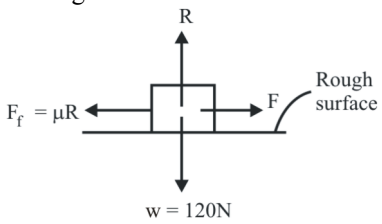
WBPSC AE 2003

Ans. (d) : F_f = friction force

F = applied effort

R = Normal force

w = Weight



$\because$ $R = w$ & $F = \mu R$ $\mu = 0.3$

$\Rightarrow$ friction force $(F_f) = \mu R = \mu w = 0.3 \times 120 = 36N$

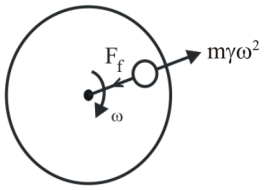
250. A small coin is placed at a distance γ from the centre of a gramophone record. The rotational speed of the record is gradually increased. If the coefficient of friction between the coin and the record is μ, the minimum angular velocity of the record for which the coin will fly off is given by

(a) $\sqrt{\frac{2\mu g}{\gamma}}$ (b) $\sqrt{\frac{\mu g}{2\gamma}}$

(c) $\sqrt{\frac{\mu g}{\gamma}}$ (d) $2\sqrt{\frac{\mu g}{\gamma}}$

WBPSC AE 2003

Ans. (c) :



Coin is at 'γ' distance from centre of gramophone $F_f = \mu mg$, F_{cf} = centrifugal force

Friction force = Centrifugal force

$F_f = F_{cf}$

$\mu mg = m\gamma\omega^2$

$\omega = \sqrt{\frac{\mu g}{\gamma}}$

251. What is Coulomb friction?

(a) it is the friction between solids and liquids

(b) it is the friction between bodies having relative motion

(c) it is the friction between two dry surfaces

(d) None of the above

WBPSC AE 2008

Ans. (c) : Coulomb law of friction :– This law states that for two dry solid surfaces sliding against one another the magnitude of kinetic friction exerted through the surface is independent of the magnitude of velocity of the slipping of the surface against each other.

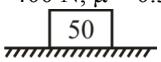
252. A 50-kg box rests on horizontal floor for which coefficient of friction is 0.3. If the box is subjected to a horizontal towing force of 400 N, what is its velocity after 5 sec from rest? (Take g = 10 m/s^2)

(a) 40 m/s (b) 30 m/s

(c) 27.5 m/s (d) 25 m/s

UPSC JWM Advt. No.-50/2010

Ans. (d) : Given, F = 400 N, μ = 0.3, time (t) = 5 sec

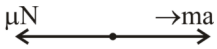


We know that,

$F_s = \mu \cdot N = 0.3 \times 50 \times 10$

$= 150 \text{ N}$

FBD of block –



$\sum F_{net} = ma$ or $F - \mu N = ma$

$250 = ma$

$a = \frac{250}{50} = 5 \text{m/s}^2$

$\Rightarrow$ $v = u + at$

$v = 0 + 5 \times 5 = 25 \text{ m/s}$

253. Which of the following is NOT one of the laws of limiting friction?

(a) The force of friction is tangential or parallel to the surfaces in contact

(b) It is dependent on the size or area of the surfaces in contact

(c) The magnitude of the limiting frictional force depends upon the nature of material of the surface in contact and on their roughness or smoothness

(d) Direction of force of friction is opposite to the direction in which the body would start moving on applying the force

JUVNL AEE (GTO) 05.03.2017

Ans. (b) : When a body just begins to slide over the surface of the other body is known as limiting friction.

Magnitude of friction force $F = \mu R_N$

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Friction force does not depend on size or area of surface contact friction depend on smoothness and roughness of the surface.

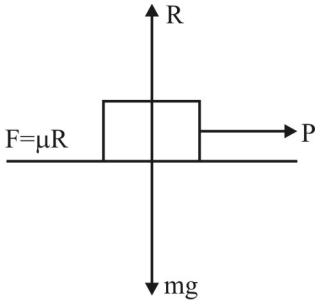
254. Which of the following statements regarding laws govern the friction between dry surfaces are correct?

1. **The friction force is dependent on the velocity of sliding.**
2. **The friction force is directly proportional to the normal force.**
3. **The friction force is dependent on the materials of the contact surfaces.**
4. **The friction force is dependent on the area of contact surfaces.**

(a) 2, 3 and 4 (b) 2 and 4
(c) 1, 2, 3 and 4 (d) 2 and 3

UPRVUNL AE 21.08.2016

Ans. : (d)



F = μR
μ = Coefficient of friction
R = Normal reaction
(i) The friction force is directly proportional to the normal force
(ii) The friction force is dependent on the materials of the contact surfaces.

255. The value of angle of repose is the same as the value of _____

(a) Limiting angle of friction
(b) Coefficient of friction
(c) Static friction
(d) normal reaction between the contact surface

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Limiting friction – The maximum value of static friction upto which body does not move and the angle which it makes with horizontal is called angle of limiting friction.

Angle of repose – It is the angle of the inclined plane with horizontal such that a body placed on it is just begins to slide.

In limiting condition –

$F = mg \sin \alpha$

$R = mg \cos \alpha$

Where, α = angle of repose

So, $\frac{F}{R} = \tan \alpha$

$\therefore \frac{F}{R} = \mu_s = \tan \theta = \tan \alpha \quad (\because \tan \theta = \mu_s)$

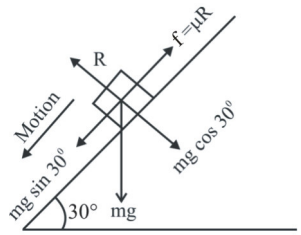
or $\boxed{\theta = \alpha}$

256. A body is resting on a plane inclined at angle of 30^0 to horizontal. What force would be required to slide it down, if the coefficient of friction between body and plane is 0.3?

(a) 5kg (b) 1 kg
(c) zero (d) 0.5 kg

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c) Force required to slide body down if



$R = mg \cdot \cos 30$

$mg \cdot \sin 30 = mg \times 0.5 = 0.5\ mg$ .....(1)

$f = \mu mg \cdot \cos 30° = 0.3 \times \frac{\sqrt{3}}{2} \times mg$

$= 0.259\ mg$ .....(2)

It means, $f < mg \cdot \sin 30°$

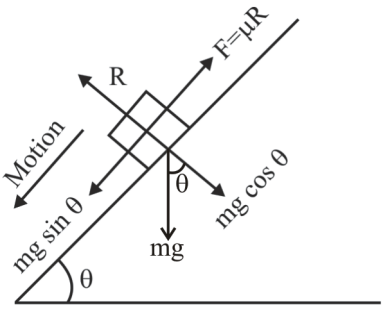
So no force will be required to slide body downward.

257. When a body slides down a surface inclined θ to horizontal, the acceleration *(f)* of the body is given by-

(a) $f = g$ (b) $f = g \cos\theta$
(c) $f = g \tan\theta$ (d) $f = g \sin\theta$

ISRO Scientist/Engineer (RAC) 29.11.2015
Haryana PSC Civil Services Pre 2014

Ans : (d)



using second law of motion

$F = (mg \sin\theta - \mu\ mg \cos\theta)$

$mf = mg\ [\sin\theta - \mu \cos\theta]$

$f = g\ [\sin\theta - \mu \cos\theta]$

but coefficient of friction μ not given and assume to be surface is smooth

so

$\mu = 0$

$f = g \sin\theta$

258. A block (of cross section 4 × 3 m size) of 20 kg mass rests on a flat horizontal surface as shown in figure below. If the coefficient of friction between the block and the surface is $\frac{1}{4}$, the force 'F' which will cause block to move is approximately given by

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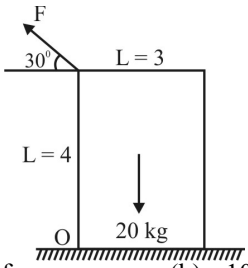
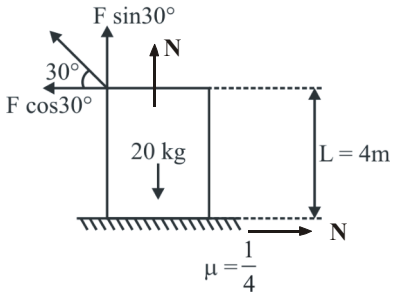


(a) 20 kgf (b) 10 kgf
(c) 5 kgf (d) 8.7 kgf

ISRO Scientist/Engineer 17.12.2017

Ans. (c) :



Given,

$m = 20$ kg $\quad \mu = \frac{1}{4}$

$N = mg - F\sin 30°$

$= 20\times 10 - F\frac{1}{2}$

$= 200 - \frac{F}{2}$

Friction force $(f) = \mu N = \frac{1}{4}\left(200 - \frac{F}{2}\right)$

Condition for block to move

$F\cos 30° > f$

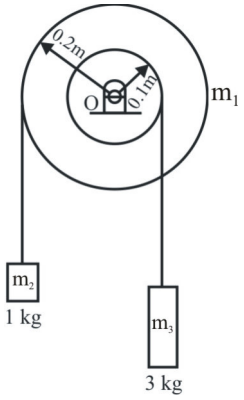
$\frac{F\sqrt{3}}{2} > 50 - \frac{F}{8}$

$F\left(\frac{\sqrt{3}}{2} + \frac{1}{8}\right) > 50$

$F > 50.45$ N

$F > 5.04$ kgf

259. The pulley system shown in figure has a mass of 1.5 kg and a radius of gyration of 0.2 m. The angular acceleration of the pulleys, when the suspended masses are released is

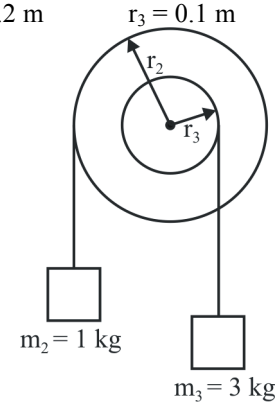


(a) 5 rad/sec^2 (b) 10/1.3 rad/sec^2
(c) 15/1.3 rad/sec^2 (d) 10 rad/ sec^2

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Given,

$m_1 = 1.5$ kg $\quad k = 0.2$ m
$m_2 = 1$ kg $\quad m_3 = 3$kg
$r_2 = 0.2$ m $\quad r_3 = 0.1$ m



$I_1 = m_1k^2 = 1.5 \times 0.2^2 = 0.06$

$I_2 = m_2r_2^2 = 1 \times 0.2^2 = 0.04$

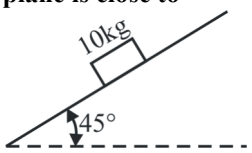
$I_3 = m_3r_3^2 = 3 \times 0.1^2 = 0.03$

$(I_1 + I_2 + I_3)\,\alpha = m_3gr_3 - m_2gr_2$

$(0.06 + 0.04 + 0.03)\,\alpha = 3 \times 10 \times 0.1 - 1 \times 10 \times 0.2$

$\alpha = \frac{1}{0.13} = \frac{10}{1.3}$ rad/sec^2

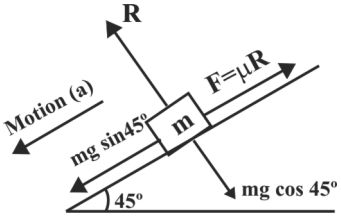
260. A body of mass 10 kg is initially stationary on a 45° inclined plane as shown in figure. The coefficient of dynamic friction between the body and the plane is 0.3. The body slides down the plane and attains a velocity of 20 m/s. The distance travelled (in meter) by the body along the plane is close to



(a) 4.8 (b) 98
(c) 41 (d) 9.8

ISRO Scientist/Engineer 07.05.2017

Ans. (c) :



From FBD for mass m is given above.

Equating the net forces along the plane, we have

$ma = mg\sin 45 - \mu(mg\cos 45°)$

$a = 9.81\times\frac{1}{\sqrt{2}}[1 - 0.3]$

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$$a = 9.81 \times \frac{1}{\sqrt{2}} \times 0.7$$

$a = 4.85$ m/s^2

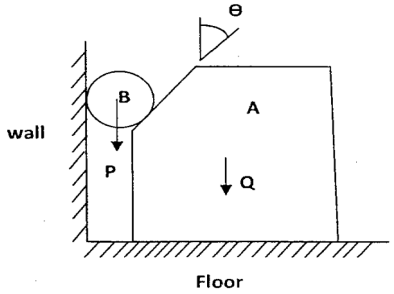
Using equation of motion

$v^2 = u^2 + 2as$

$20^2 = 0^2 + 2 \times 4.85 \times s$

$s = 41.2$ m

261. A solid block 'A' weighing 'Q' kg is resting on a flat floor. A smooth cylinder 'B' weighing 'p' kg, is placed between the solid A and the vertical wall as shown in fig. The friction between the cylinder, wall and the block A is negligible. The co-efficient of friction between the block A and floor is μ. The minimum weight P required to disturb the block A is

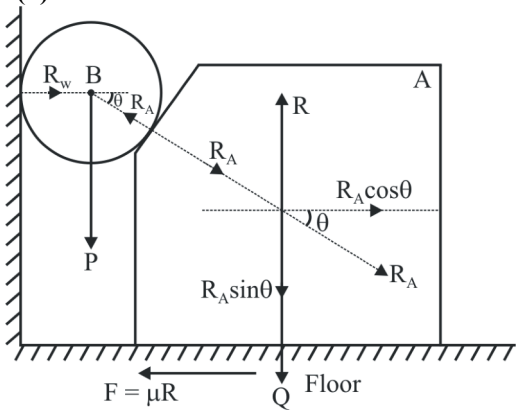


(a) $\frac{Q(1-\tan\theta)}{\mu\tan\theta}$ (b) $\frac{\mu Q\tan\theta}{(1-\mu\tan\theta)}$

(c) $\mu Q\cos\theta$ (d) $\frac{\mu Q}{\cos\theta}$

ISRO Scientist/Engineer 12.05.2013

Ans. (b) :



Using FBD –

$P = R_A \sin\theta$

$R_A = \frac{P}{\sin\theta}$

$R = R_A \sin\theta + Q$

$F = \mu R = \mu[P + Q] = \frac{P}{\tan\theta}$

$P = \mu P \tan\theta + \mu Q \tan\theta$

So arrange above equation we get

$$\boxed{P = \frac{\mu Q\tan\theta}{(1-\mu\tan\theta)}}$$

262. Angle of friction is :

(a) Angle made by resultant reaction with horizontal

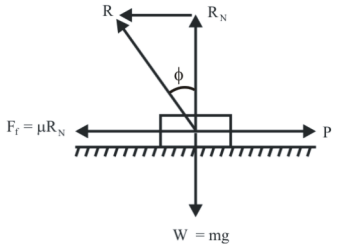
(b) Angle made by resultant reaction with normal

(c) Angle made by normal reaction with force of friction.

(d) None of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Angle of friction- Angle made by the resultant of normal reaction and limiting friction with the normal reaction is called angle of friction.

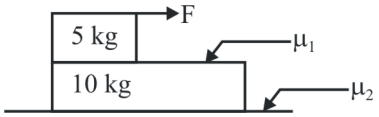


$$\tan\phi = \frac{F_f}{R_N} = \frac{\mu R_N}{R_N}$$

$\tan\phi = \mu$

Friction angle $\boxed{\phi = \tan^{-1}\mu}$

263. In the figure shown, the minimum ratio of μ_1/μ_2 so that the masses move together with the application of force F is

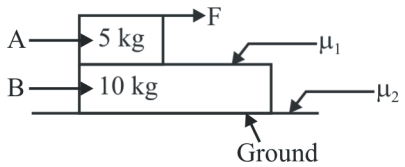


(a) 5 (b) 2

(c) 4 (d) 3

ISRO Scientist/Engineer 03.07.2016

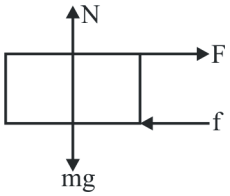
Ans. (d) :



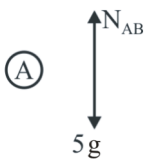
FBD (Free Body Diagram)–

For block 'A'

m = 5 kg



Frictional force– $\boxed{F_f = \mu N}$



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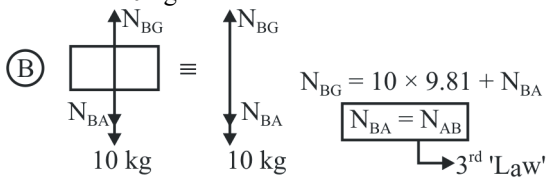
$N = mg$

$N = 5 \times 9.81$ "Newton"

$F_{fA} = \mu_1 \times 5 \times 9.81$ 'N' .....(1)

For Block 'B'–

$m = 10$ kg



$N_{BG} = 10 \times 9.81 + 5 \times 9.81$

$N_{BG} = 15 \times 9.81$

$F_{fB} = \mu_2 \times 15 \times 9.81$ .....(2)

So, applied force (F)–

Then, $\boxed{F_{fB} = F_{fA} = F}$ → Applied force on block A and B.

From equation (1) / (2)

$$\frac{F_{fA}}{F_{fB}} = \frac{\mu_1 \times 5 \times g}{\mu_2 \times 15 \times g}$$

$$\frac{F}{F} = \frac{\mu_1 \times 5}{\mu_2 \times 15}$$

$$1 = \frac{\mu_1}{3\mu_2}$$

$$\boxed{\frac{\mu_1}{\mu_2} = 3}$$

264. The maximum value of _____, which comes into play, when a body just begins to slide over the surface of other body, is known as _____.

(a) Kinetic friction, Limiting friction
(b) Dynamic friction, Limiting friction
(c) Solid friction, Limiting friction
(d) Boundary friction, Limiting friction
(e) Static friction, Limiting friction

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : The maximum value of static friction, which comes into play, when a body just begins to slide over the surface of the other body, is known as limiting friction.

Static Friction–The static friction is the frictional force that develops between mating surface when subjected to external force but there is no relative motion between them.

Dynamic Friction–The dynamic friction is the frictional force that develops between mating surface when subjected to external forces and there is relative motion between them. The dynamic friction is also known as kinematic friction.

Laws of Solid Friction [Static or Dynamic]

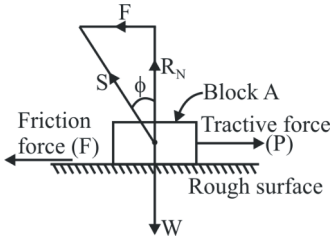
1. Friction acts tangential to the surface in contact and is in a direction opposite to that in which motion is to impend i.e. take place.
2. Friction force is maximum at the instant of impending motion. Its variation from zero to maximum value [Limiting friction] depends upon the resultant force tending to cause motion.
3. The magnitude of limiting friction bears a constant ratio to the normal reaction between the mating surface.
4. Limiting friction is independent of the area and shape of contact surface.
5. Limiting friction depends upon the nature (roughness or smoothness) of the surface in contact.
6. At low velocities between sliding surface, the friction force is practically independent of the velocity. However, slight reduction in friction occurs when the speed are high.

265. If F is the limiting friction and R_N is the normal reaction of the surfaces of contact of two bodies, then the coefficient of friction μ is expressed as:

(a) $\mu = R_N/F$ (b) $\mu = F/R_N$
(c) $\mu = 2F/R_N$ (d) $\mu = 0.5F/R_N$
(e) $\mu = 0.5R_N/F$

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) :



Where,

S = Total reaction with the normal reaction.

The ratio of friction force to normal reaction is called coefficient of friction (μ).

So, $\mu = \frac{F}{R_N}$

$\tan\phi = \mu = \frac{F}{R_N}$, $\phi \rightarrow$ Friction angle

266. The coefficient of rolling resistance is defined as the ratio between

(a) Rolling resistance to lateral load
(b) Lateral load to rolling resistance
(c) Rolling resistance to normal load
(d) Normal load to rolling resistance

TNPSC AE 2017

Ans. (c) : The coefficient of rolling resistance is defined as the ratio between Rolling resistance to normal load.

Rolling resistance coefficient (C_{rr}) = $\frac{F_R}{N}$

267. The coefficient of friction is the ratio of

(a) the normal reaction to the limiting force of friction
(b) The weight of the body to limiting force of friction
(c) the limiting force of friction to the normal reaction
(d) the weight of the body to the normal reaction

TSPSC AEE 2015
TRB Asstt. Prof., 2012

Ans : (c) The coefficient of friction is the ratio of the limiting force of friction to the normal reaction

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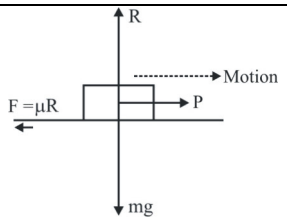


$F = \mu R$.

$$\mu = \frac{F}{R}$$

μ = Coefficient of friction.
F = Limiting force of friction
R = Normal reaction.

268. Which statement is wrong in wedge friction?
(a) To lift heavy block through small distances
(b) To lift heavy block through large force
(c) To slightly slide one end of the beam relative to another end
(d) Weight of wedge is neglected compared to weight to be lifted

TNPSC AE 2013

Ans. (b) : Wedge–
- Simple machines used to raise heavy loads.
- Force required to lift block is significantly less than block weight.
- Friction prevents wedge from sliding out.
- Minimum force D required to raise block.

269. A block resting on an inclined plane begins to slide down the plane when the angle of inclination is gradually increased to 30°. The coefficient of friction between the block and the plane is:-
(a) 0.50 (b) 0.578
(c) 0.72 (d) 0.866

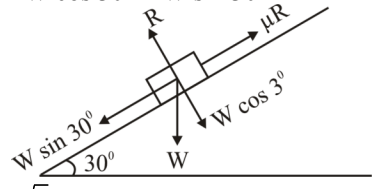
UKPSC AE-2013, Paper-I

Ans. (b) : From this figure.

$R = W \cos 30^o$

$\mu R = W \sin 30^o$

$\mu \times W \cos 30^o = W \sin 30^o$



$$\mu \times \frac{\sqrt{3}}{2} = \frac{1}{2}$$

$$\mu = 0.5773$$

270. When the applied force is less than the limiting frictional force, the body will:-
(a) Start moving (b) Remain at rest
(c) Slide backward (d) Skid

UKPSC AE-2013, Paper-I

Ans. (b) : When the applied force is less than the limiting frictional force, the body will remain at rest.

271. A 44 N block is thrust up a 30° inclined plane with an initial speed of 5 m/sec. It travels a distance of 1.5 m before it comes to rest. The frictional force acting upon it would be
(a) 18.3 N (b) 15.3 N
(c) 12.3 N (d) 9.3 N

UKPSC AE 2012 Paper-I

Ans. (b) : Given,
Wt. of block = 44 N, S = 1.5 m
initial speed, u = 5 m/s
final speed, v = 0

$v^2 = u^2 - 2as$ (Retardation)

$0 = (5)^2 - 2(a)(1.5)$

$a = 8.33 \text{ m/s}^2$

$$\text{Net force} = ma = \frac{W}{g} \times a$$

$$mg \sin 30^o + f_s = \frac{44}{9.81} \times 8.33$$

$22 + f_s = 37.37$

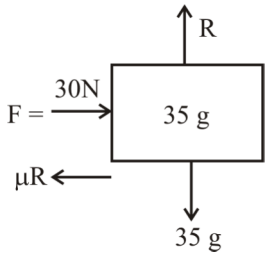
$f_s = 37.37 - 22$

friction force, $f_s = 15.37$ N

272. If a force of 30 N is required to move a mass of 35 kg on a flat surface horizontally at a constant velocity, what will be the coefficient of friction?
(a) 0.067 (b) 0.087
(c) 0.098 (d) 0.092

UKPSC AE 2012 Paper-I

Ans. (b) :



$R - 35\,g = 0$

$R = 35\,g$

$F = \mu R$

$$\mu = \frac{F}{R} = \frac{30}{35 \times 9.8} = 0.087$$

273. Two bodies of mass m and M are hung at the ends of a rope passing over a frictional pulley. The acceleration in which the heavier mass M comes down is given by the following:
(a) $\frac{g(M-m)}{M+m}$ (b) $\frac{g(M+m)}{M-m}$
(c) $\frac{gM}{M+m}$ (d) $\frac{g \times M}{M-m}$

UKPSC AE 2007 Paper -I

Ans. (a) : $\frac{g(M-m)}{M+m}$

274. Tangent of angle of friction is equal to
(a) kinetic friction
(b) limiting friction
(c) friction force
(d) coefficient of friction

UKPSC AE 2007 Paper -I

Ans. (d) : Coefficient of friction

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275. A body weight of 200 N is resting on a rough horizontal plane, and can be just moved by a force of 80 N applied horizontally. What will the value of the coefficient of friction?

(a) 0.4 (b) 0.5
(c) 0.3 (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (a) : $F = \mu R$

$$\mu = \frac{80}{200} = 0.4$$

276. Frictional force is independent of..

(a) Sliding speed
(b) area of contacting surfaces
(c) normal load
(d) both (a) & (b)

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Friction force $F_f = \mu N$

μ = coefficient of friction $F_f \propto N$

N = Normal load.

So, Friction force is independent of sliding speed & area of contacting surfaces.

277. .....provides complete restraint against transverse or longitudinal motion and against rotation.

(a) A frictionless bearing
(b) An ideal clamped support
(c) A weightless flexible string
(d) A frictionless pin joint

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : An ideal clamped support provides complete restraint against transverse or longitudinal motion and against rotation.

278. The coefficient of rolling resistance, for a steel wheel of 200 mm diameter which rolls on a horizontal steel roll, is 0.3 mm. The steel wheel carries a load of 600 N. The force necessary to roll the wheel along the rail is :

(a) 90 N (b) 180 N
(c) 90 kN (d) 270 N

UPRVUNL AE 2014

Ans : (*) Given, $\mu_r = 0.3$ mm
Diameter, d = 200 mm
Radius, r = 100 mm
Force necessary to roll the wheel along the rail,

$$F = \frac{\mu_r W}{R} = \frac{0.3 \times 600}{100}$$

$$\boxed{F = 1.8N}$$

Note-As per UPRVUNL, correct option is 'b' 180 N.

279. A heavy block of mass m is slowly placed on a conveyer belt moving with speed v. If coefficient of friction between block and the belts is μ, the block will slide on the belt through distance

(a) $\frac{v}{\mu g}$ (b) $\frac{v^2}{\sqrt{\mu g}}$
(c) $\left(\frac{v}{\mu g}\right)^2$ (d) $\frac{v^2}{2\mu g}$

ISRO Scientist/Engineer 2007

Ans. (d) : Retardation due to friction force = $-\mu g$

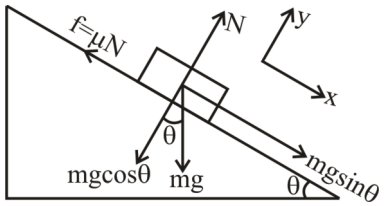
$$v^2 - u^2 = 2as$$
$$0 - u^2 = as$$
$$-u^2 = 2 \times -\mu gs$$
$$+u^2 = 2\mu gs$$
$$s = \frac{u^2}{2\mu g}$$

280. A body of weight 50 kN rests in limiting equilibrium on a rough plane, whose slope is 30°. The plane is raised to a slope of 45°. Find the value of coefficient of friction for equilibrium.

(a) 0.577 (b) 0.749
(c) 0.623 (d) 0.425

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) :



$\theta_1 = 30°$, W = 50 kN, $\theta_2 = 45°$

Consider when the slope of the plane 30°,

Sum of forces parallel to plane $\sum F_x = 0$

$$\mu N - W\sin 30° = 0 \quad .....(i)$$

Sum of forces perpendicular to plane $\sum F_y$

$$N - W\cos 30° = 0$$
$$N = W\cos 30° \quad .....(ii)$$

Putting the value of 'N' in equation (i) we get,

$$\mu W\cos 30° - W\sin 30° = 0$$
$$\mu\cos 30° - \sin 30° = 0$$

Dividing both the side by cosθ

$$\mu - \tan 30° = 0$$
$$\mu = \tan 30° = \frac{1}{\sqrt{3}} = 0.577$$

281. What is the friction experienced by a body when at rest known as?

(a) Static friction
(b) Dynamic friction
(c) Limiting friction
(d) Coefficient of friction

VIZAG STEEL MT 2011

Ans. (a) : Static friction is the friction experienced by a body when at rest.

282. A man is climbing up a ladder which is resting against a vertical wall. When he was exactly half way up, the ladder started slipping. The path traced by the man is

(a) Parabola (b) Circle
(c) Ellipse (d) Hyperbola

UPPSC AE 13.12.2020, Paper-I
ESE 2013

Ans. (b) : Circle

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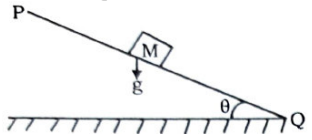
283. A block of mass M is released from point P on a rough inclined plane with angle of inclination θ as shown in figure below. The coefficient of friction is μ. If μ < tan θ, then the time taken by the block to reach point Q on the inclined plane, where PQ = S is

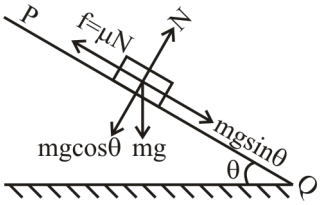


(a) $\sqrt{\frac{2S}{g\cos\theta(\tan\theta-\mu)}}$ (b) $\sqrt{\frac{2S}{g\cos\theta(\tan\theta+\mu)}}$

(c) $\sqrt{\frac{2S}{g\sin\theta(\tan\theta-\mu)}}$ (d) $\sqrt{\frac{2S}{g\sin\theta(\tan\theta+\mu)}}$

UPPSC AE 13.12.2020, Paper-I
GATE 2007

Ans. (a)



$N = mg\cos\theta$

$\mu N = mg\sin\theta$

From Newton's 2nd law

$F = ma = mg\sin\theta - \mu N$

$ma = mg\sin\theta - \mu mg\cos\theta$

$a = g\sin\theta - \mu g\cos\theta$

$S = ut + \frac{1}{2}at^2$

$S = 0 + \frac{1}{2}(g\sin\theta - \mu g\cos\theta)t^2$

$t = \sqrt{\frac{2S}{g\cos\theta(\tan\theta-\mu)}}$

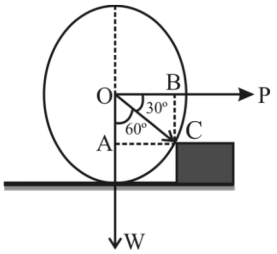
284. A roller of weight 500 N and radius 0.5 m is to be dragged over a small rectangular block by using a horizontal force P at the centre of the roller as shown in figure. The minimum magnitude of force P required for dragging the roller should be greater than

(a) 500 N (b) 400 N

(c) $500\sqrt{3}$ N (d) $500\sqrt{2}$ N

DRDO Scientists 2009

Ans. (c) :



W = weight of roller

P = horizontal force acting at the centre of roller

$OA = R\cos60^\circ$, $OB = R\cos30^\circ$

$\Sigma M_C = 0$

$\Rightarrow \quad P(OA) = W(OB)$

$P = 500\frac{\cos30^\circ}{\cos60^\circ} = 500\sqrt{3}$

285. Friction between two surfaces in contact depends on

(a) Bending moment
(b) Area of contact between the surfaces
(c) Energy possessed by the surfaces
(d) Equal and opposite normal reaction between surfaces

TANGEDCO AE 2015

Ans. (d) : Factors affecting sliding friction
1. The surface deformation of objects.
2. The roughness or smoothness of the surface of objects.
3. The original speed of either objects.
4. The size of object.
5. Amount of pressure on either object.

Friction force between two surface in contact depend on normal reaction and coefficient of friction between two surface and independent of surface area.

286. The tractive resistance during the propulsion of a wheeled vehicle depends on :

(a) Road resistance
(b) Aerodynamic resistance
(c) Gradient resistance
(d) All of these

OPSC AEE 2015 Paper-I

Ans : (d) The tractive resistance during the propulsion of a wheeled vehicle depends on:
(i) Road resistance
(ii) Aerodynamic resistance
(iii) Gradient resistance

4. Centroid, Centre of Gravity & Moment of Inertia

287. What is the unit of moment of inertia of an area.

(a) kg m^2 (b) kg m
(c) m^3 (d) m^4
(e) kg m^3

HPPSC Workshop Suptd. 08.07.2021
HPPSC AE (PWD) 10.09.2016
TNPSC ACF 2012
CGPSC AE 26.04.2014 Shift-I
AAI Jr. Executive 26.03.2021
UPRVUNL AE 05.07.2021

Ans. (d) : Units of the mass moment of inertia are kg-m^2, gram-cm^2.
Units of the area moment of inertia are m^4, mm^4.

288. The C.G. of a solid hemisphere lies on the central radius

(a) at distance 3r/2 from the plane base
(b) at distance 3r/4 from the plane base
(c) at distance 3r/5 from the plane base
(d) at distance 3r/8 from the plane base

TNPSC AE 2019
Oil India Senior Officer 23.12.2020

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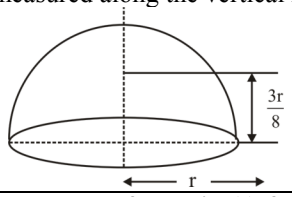
Ans. (d) : C.G. of hemisphere is at a distance of $\frac{3r}{8}$ from its base measured along the vertical radius.



289. A polar moment of Inertia (I) for hollow shaft with external diameter (D) and internal diameter (d) is given by:

(a) $\frac{32D^4}{\pi d^4}$ (b) $\frac{\pi\left(D^4-d^4\right)}{64}$

(c) $\frac{\pi\left(D^4-d^4\right)}{32}$ (d) $\frac{32\left(D^4-d^4\right)}{\pi}$

CIL MT 26.03.2017
(CGPSC Polytechnic Lecturer 2017)

Ans. (c) :

Polar moment of inertia of hollow haft = $\frac{\pi\left(D^4-d^4\right)}{32}$

290. Theorem of perpendicular axis is used in obtaining the moment of inertia of a

(a) triangular lamina (b) square lamina
(c) circular lamina (d) semicircular lamina

APPSC Poly. Lect. 2013

Ans. (a) : Theorem of perpendicular axis is used in obtaining the moment of inertia of a triangular lamina.

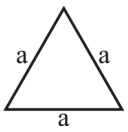
291. The centre of gravity of an equilateral triangle with each side (a) is

(a) $\frac{a\sqrt{3}}{2}$ (b) $\frac{a\sqrt{2}}{3}$

(c) $\frac{a}{2\sqrt{3}}$ (d) $\frac{a}{3\sqrt{2}}$

APPSC Poly. Lect. 2013

Ans. (c) : Centre of gravity of an equilateral triangle with each side (a) is,



Centre of Gravity of equilateral triangle = $\frac{a}{2\sqrt{3}}$

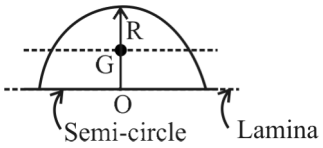
292. The distance of centroid of semi-circular lamina of radius R from the straight edge of this semi-circular lamina will be :

(a) $\frac{8R}{3\pi}$ (b) $\frac{4R}{3\pi}$

(c) $\frac{2R}{3\pi}$ (d) $\frac{R}{3\pi}$

APPSC Poly Lect. 13.03.2020
TNPSC ACF 2012
Oil India Senior Officer 23.12.2020

Ans. (b) :



Let, G = Centroid of semicircle

$\therefore$ Distance of the centroid 'G' = $\frac{4R}{3\pi}$

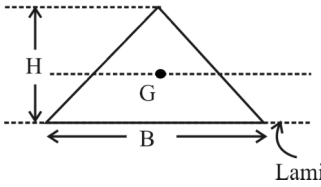
293. Moment of inertia (second moment of area) of triangular lamina about base and centroidal axis (parallel to base), respectively will be ___ and _____ . [B = length of base of triangular section, H = Height of triangular section]

(a) $\frac{BH^3}{3},\frac{BH^3}{36}$ (b) $\frac{BH^3}{36},\frac{BH^3}{12}$

(c) $\frac{BH^3}{12},\frac{BH^3}{36}$ (d) $\frac{BH^3}{3},\frac{BH^3}{12}$

APPSC Poly Lect. 13.03.2020

Ans. (c) :



Moment of inertia about base = $\frac{BH^3}{12}$(About lamina)

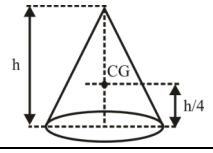
Moment of inertia about centroid axis = $\frac{BH^3}{36}$

294. The C.G. of a right circular solid cone of height h lies at _____ distance from the base.

(a) h/2 (b) h/3
(c) h/6 (d) h/4

CGPSC AE 15.01.2021

Ans. (d) : The C.G. of a right circular solid cone of height 'h' lies at h/4 distance from the base.



295. The mathematical definition of second moment of inertia with respect to x-axis of an area *dA* is:

(a) $I_x = r_s^2 A$ (b) $I_o = \int_A \rho^2 dA$

(c) $I_y = \int_A x^2 dA$ (d) $I_x = \int_A y^2 dA$

CGPSC AE 15.01.2021

Ans. (d) : Second moment of Inertia with respect to x-axis.

$$I_X = \int_A y^2 dA$$

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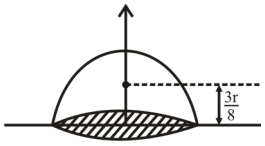
296. The center of gravity of a solid hemisphere of radius r is at _____ distance from its diametral axis.

(a) 3r/4 (b) 3r/7
(c) 3r/8 (d) 7r/8

Oil India Senior Officer 23.12.2020

Ans. (c) : The center of gravity of a solid hemisphere of radius r is at 3r/8 distance from its diametral axis.



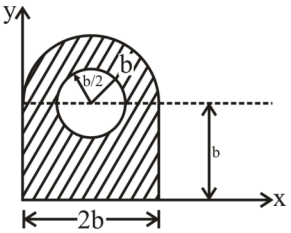
297. Unit of Area Moment of Inertia of a plane section is _____, where is "L" is the length unit.

(a) L^3 (b) L^6
(c) L^2 (d) L^4

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) :

298. Determine the centroid (x_c, y_c) of the shaded area as shown in figure with respect to given coordinate axes

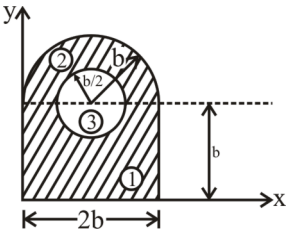


(a) $\left(b, \frac{28+3\pi}{3(8+\pi)}b\right)$ (b) $\left(\frac{b}{2}, \frac{12+2\pi}{4(2+\pi)}b\right)$

(c) $\left(b, \frac{5}{(8+\pi)}b\right)$ (d) $\left(b, \frac{1+2\pi}{2(8+\pi)}b\right)$

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (*) :



(1) → Rectangle (2b × b)

(2) → Semi-circle of radius (b)

(3) → Circle of radius $\left(\frac{b}{2}\right)$

$x_c = \frac{2b}{2} = b$ (from symmetricity)

$y_c = \frac{A_1y_1 + A_2y_2 - A_3y_3}{A_1 + A_2 - A_3}$

$\left[y_1 = \frac{b}{2}, y_2 = b + \frac{4b}{3\pi}, y_3 = b\right]$

$$= \frac{(2b \times b) \times \frac{b}{2} + \frac{\pi b^2}{2} \times \left(b + \frac{4b}{3\pi}\right) - \pi\left(\frac{b}{2}\right)^2 \times b}{2b \times b + \frac{\pi b^2}{2} - \pi\left(\frac{b}{2}\right)^2}$$

$$= \frac{b + \frac{\pi}{2}\left(b + \frac{4b}{3\pi}\right) - \frac{\pi}{4}b}{2 + \frac{\pi}{2} - \frac{\pi}{4}} = \frac{b + \frac{\pi b}{2} + \frac{\pi}{2} \times \frac{4b}{3\pi} - \frac{\pi b}{4}}{2 + \frac{\pi}{4}}$$

$$= \frac{b + \frac{\pi b}{4} + \frac{2b}{3}}{2 + \frac{\pi}{4}} = \frac{\frac{20b + 3\pi b}{12}}{\frac{8+\pi}{4}}$$

$y_c = \frac{(20+3\pi)b}{3(8+\pi)}$

Centroid of shaded region, $\left(b, \frac{(20+3\pi)b}{3(8+\pi)}\right)$

299. Polar moment of inertia of a square of side 6 cm is

(a) 108 cm^4 (b) 216 cm^4
(c) 432 cm^4 (d) 324 cm^4

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Given,

side (a) = 6 cm

Polar moment of Inertia of square,

$$J = I_x + I_y$$

$$= \frac{a^4}{12} + \frac{a^4}{12}$$

$$= \frac{(6)^4}{12} + \frac{(6)^4}{12}$$

$J = 216\ cm^4$

300. Moment of inertia of a thin spherical shell of mass M and radius R, about its diameter is

(a) MR^2 (b) $\frac{MR^2}{2}$

(c) $\frac{2}{5}MR^2$ (d) $\frac{2}{3}MR^2$

UPPSC AE 13.12.2020, Paper-I

Ans. (d)

- Moment of inertia of a thin spherical shell of mass M and radius R, about its diameter is $\frac{2}{3}MR^2$.
- Moment of inertia of a solid sphere of mass M and radius R, about its diameter is $\frac{2}{5}MR^2$.

301. Moment of inertia of a rectangle of base 12 cm and height 14 cm about its base is

(a) 672 cm^4 (b) 2016 cm^4
(c) 914.67 cm^4 (d) 2744 cm^4

GPSC DEE, Class-2 (GWSSB) 04.07.2021

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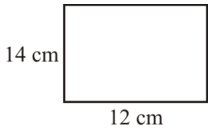
Ans. (*) :



Base = 12 cm
Height = 14 cm
Moment of Inertia of rectangle about centroidal axis,

$$I = \frac{bh^3}{12}$$

$$I = \frac{12(14)^3}{12}$$

$$\boxed{I = 2744 \text{ cm}^4}$$

Moment of inertia of rectangle about its base,

$$I_{base} = \frac{bh^3}{3} = \frac{12(14)^3}{3}$$

$$\boxed{I_{base} = 10976 \text{cm}^4}$$

Note-As per GPSC, given answer is 'b' option.

302. Polar moment of inertia (I_p), in cm^4, of a rectangular section having width, b = 2 cm and depth, d = 6 cm is :

(a) 40 (b) 20
(c) 8 (d) 80

OPSC AEE 2019 Paper-I

Ans : (a) :
Polar moment of inertia (J) = $I_x + I_y$

$$= \frac{bd^3}{12} + \frac{db^3}{12}$$

$$= \frac{2 \times 6^3}{12} + \frac{2^3 \times 6}{12}$$

$$= 40 \text{ cm}^4$$

303. Radius of gyration of a circular section with diameter D is

(a) $\frac{D}{2}$ (b) $\frac{D}{4}$
(c) $\frac{D}{3}$ (d) $\frac{D}{\sqrt{3}}$

APPSC AEE SCREENING 17.02.2019

Ans. (b) : For circular section $I = \frac{\pi}{64} D^4, A = \frac{\pi}{4} D^2$

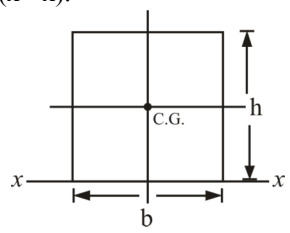
Radius of gyration $= k = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64} D^4}{\frac{\pi}{4} D^2}} = \frac{D}{4}$

304. The moment of inertia of a rectangular section of base (b), height (h) about its base will be :

(a) $\frac{hb^3}{12}$ (b) $\frac{bh^3}{3}$
(c) $\frac{bh^3}{12}$ (d) $\frac{hb^3}{3}$
(e) $\frac{bh^3}{36}$

CGPSC AE 25.02.2018

Ans. (b) : Moment of inertia of a rectangular section about its base (x - x).



$$I_{x-x} = I_{CG} + Ax^2$$

$$I_{C.G.} = \frac{bh^3}{12}, A = b \times h, \; x = \frac{h}{2}$$

$$I_{x-x} = \frac{bh^3}{12} + b \times h \times \left(\frac{h}{2}\right)^2 = \frac{bh^3}{12} + \frac{bh^3}{4} = \frac{bh^3}{3}$$

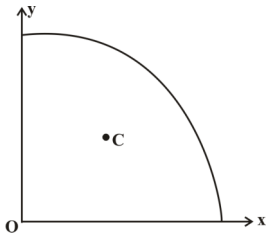
$$\boxed{I_{x-x} = \frac{bh^3}{3}}$$

305. Moment of inertia of a quarter circle (diameter = d) about its straight edge is given by :

(a) $\frac{\pi d^4}{64}$ (b) $\frac{\pi d^4}{128}$
(c) $\frac{\pi d^4}{256}$ (d) $\frac{\pi d^4}{512}$
(e) $\frac{\pi d^4}{32}$

CGPSC AE 25.02.2018

Ans. (c) :



$$I_{OX} = \frac{1}{4} \times \left[\frac{\pi d^4}{64}\right]$$

$$I_{OX} = \frac{\pi d^4}{256}$$

306. The moment of inertia of a triangular section with base b and height h about the base of the section will be

(a) $\frac{hb^3}{12}$ (b) $\frac{h^2b^2}{12}$
(c) $\frac{bh^2}{12}$ (d) $\frac{bh^3}{12}$
(e) none of the above

CGPSC AE 16.10.2016
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
Oil India Senior Officer 23.12.2020

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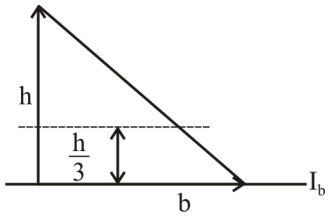
Ans. (d) : As we know,



Moment of inertia when axis passes through centroid, is

$$= \frac{bh^3}{36}$$

& Moment of inertia when axis passing through base is,

$$= \frac{bh^3}{12}$$

307. The moments of inertia of a solid circular section of radius *r* and of a hollow circular section of radii *r* and *R*, each about their diameter axes are equal then.

(a) R = 2*r* (b) R = 1.414*r*
(c) R = 1.189*r* (d) R is nearly equal to *r*

J & K PSC Screening, 2006

Ans. (c) : Moment of inertia of solid circular section

$$I_1 = \frac{\pi d^4}{64}$$

$$= \frac{\pi \times (2r)^4}{64}$$

$$I_1 = \frac{\pi \times r^4}{4}$$

Moment of inertia of Hollow circular section

$$I_2 = \frac{\pi}{4}(R^4 - r^4)$$

$$I_1 = I_2$$

$$\frac{\pi}{4} r^4 = \frac{\pi}{4}(R^4 - r^4)$$

$$r^4 = R^4 - r^4$$

$$2r^4 = R^4$$

$$R = 2^{1/4} r \quad \Rightarrow \quad \boxed{R = 1.189r}$$

308. Moment of inertia of a triangular section of base (b) and height (h) about an axis passing through its vertex and parallel to the base, is _____ than that passing through its C.G. and parallel to the base.

(a) Nine times (b) Six times
(c) Four times (d) Two times

NPCIL ET, 2015

Ans. (a) : Moment of inertia about triangular's vertex $\left(I_{vertex} = \frac{bh^3}{4}\right)$

Moment of inertia about centroid of triangular section $\left(I_{centroid} = \frac{bh^3}{36}\right)$

Moment of inertia of about base of triangular $\left(I_{base} = \frac{bh^3}{12}\right)$

Moment of Inertia,

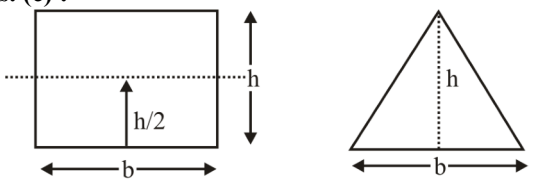
$$I_{vertex} = 9\ I_{centroid}$$

309. Ratio of moment of inertia of a rectangle and that of a triangle, having same base and height with respect to their bases will be:

(a) 2 : 1 (b) 3 : 1
(c) 4 : 1 (d) 1 : 2

Vadodara Muncipal Corp. DEE, 2018

Ans. (c) :



(I) (II)

According to Parallel Axis Theorem,

$$(I_I)_{base} = I_{CG} + Ah^2$$

$$(I_I)_{base} = \frac{bh^3}{12} + (b \times h) \times \left(\frac{h}{2}\right)^2$$

$$= \frac{bh^3}{12} + \frac{bh^3}{4}$$

$$= \frac{bh^3 + 3bh^3}{12}$$

$$= \frac{4bh^3}{12}$$

$$\boxed{(I_I)_{base} = \frac{bh^3}{3}}$$

$$(I_{II})_{about\ base} = \frac{bh^3}{12}$$

$$\frac{I_I}{I_{II}} = \frac{\frac{bh^3}{3}}{\frac{bh^3}{12}}$$

$$\boxed{\frac{(I_I)_{base}}{(I_{II})_{base}} = \frac{4}{1}}$$

310. The centre of percussion is

(a) same as centre of gravity
(b) The point in a body about which it can rotate horizontally and oscillates under the influence of gravity.
(c) the point of application of the resultant of all the forces tending to cause a body to rotate about a central axis.
(d) the metacentre of the body.

WBPSC AE, 2017

Ans. (c) : The centre of percussion is the point of application of the resultant of all the forces tending to cause a body to rotate about a central axis.

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311. The moment of inertia of a circular area of radius r with respect to a diametral axis is

(a) πr^2 (b) $\pi r^2/4$

(c) $\pi r^4/4$ (d) $\pi r^4/16$

WBPSC AE, 2017

Ans. (c) : We know that,

$$I_{\text{diametral axis}} = \frac{\pi d^4}{64} = \frac{\pi r^4}{4}$$

312. Centroid of a body is its :

(a) geometrical concentration

(b) mass concentration

(c) weight concentration

(d) density concentration

Haryana PSC Civil Services Pre, 2014

Ans. (a) :

Centroid – Geometrical concentration → Point about which whole geometry is distributed.

Centre of mass – Mass Concentration → Point at which whole mass is acting.

313. The mass moment of inertia of a thin circular disc about a diametral axis is $\frac{MR^2}{4}$, where M is the mass of circular disc & mass of a semicircular disc is m and radius R, then its mass moment of inertia about diametral axis is

(a) $\frac{mR^2}{4}$ (b) $\frac{mR^2}{8}$

(c) $\frac{mR^2}{16}$ (d) $\frac{mR^2}{32}$

WBPSC AE, 2007

Ans. (b) : MOI of circular disc

about diametrical axis = $\frac{mR^2}{4}$

MOI of semi circular plate

$$= \frac{mR^2}{4\times 2} = \frac{mR^2}{8}$$

314. A triangular sector of angle 60° is cut from a circle of radius r with its apex at the centre of the disc. The centre of gravity of the sector lies

(a) at the centre of the disc

(b) at the centre of the triangular sector

(c) somewhere in the disc

(d) Somewhere in the sector

Karnataka PSC AE, 10.09.2017

Ans. (c) : somewhere in the disc

315. A circular disc is to be made by using iron and aluminum, so that it acquires maximum moment of inertia about geometrical axis. It is possible with

(a) Iron at interior and aluminium surrounding it

(b) Aluminium at interior and iron surrounding it

(c) Using iron and aluminium layers in alternate order

(d) Sheet of iron is used at both external surface and aluminium sheet as internal layers

GPSC EE Pre, 28.01.2017

Ans. (b) : A disc is composed of rings and moment of inertia of disc can be using integral of moment of inertia of concentric elemental rings.

Moment of inertia of continuous body is found by using $I = \int R^2 dm$

For increasing the moment of inertia of a non-uniform disc, It is hence desired that the mass density is more in the interior parts of the disc.

316. Bifilar suspension method is used to find the :

(a) angular acceleration of the body

(b) moment of inertia of the body

(c) periodic time of the body

(d) frequency of vibration of the body

Punjab PSC SDE 12.02.2017

GPSC Lect. 23.10.2016

Ans. (b) : Bifilar suspension method is used to find the moment of inertia of the body.

317. Condition for stating that the centre of gravity of a body is completely determined is when:

(a) the body has (3 + 2) planes of symmetry

(b) the body has (3 + 1) planes of symmetry

(c) the body has (3 + 0) planes of symmetry

(d) the body has (3 – 2) planes of symmetry

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Condition for stating that the centre of gravity of a body is completely determined is when the body has (3 + 0) planes of symmetry.

318. .....systems are used to determine the moment of inertia of irregular bodies.

(a) Multifilar (b) Unifilar

(c) Multistar (d) Tristar

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Multifilar systems are used to determine the moment of inertia of irregular bodies.

319. The centre of gravity of a quadrant of a circle of radius R lies along its central radius at an approximate distance of–

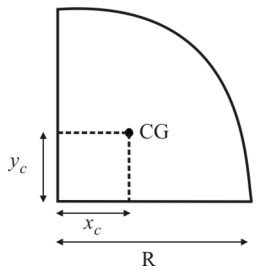
(a) 0.3 R (b) 0.44 R

(c) 0.6 R (d) 0.5 R

Karnataka PSC AE (WRD) 31.07.2021

WBPSC AE 2003

Ans. (b) :



∵ Centre of gravity (CG) of a quadrant of circle of radius R is $\Rightarrow \frac{4R}{3\pi}$.

$$y_c = x_c = \frac{4R}{3\pi}$$

$$= 0.424R$$

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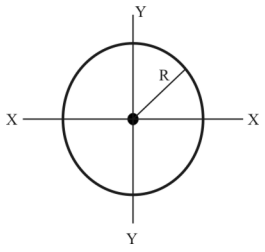
320. The moment of inertia of a uniform circular disc of mass M and radius R about any of its diameters is $\frac{1}{4}MR^2$. What is the moment of inertia of the disc about an axis passing through its centre and normal to the disc?

(a) MR^2 (b) $\frac{1}{2}MR^2$

(c) $\frac{3}{2}MR^2$ (d) $2\,MR^2$

WBPSC AE 2003

Ans. (b) :



MOI about its diameter –

$$I_{X-X} = I_{Y-Y} = \frac{MR^2}{4}$$

$\therefore$ MOI about its Z-axis or normal to the disc passing through centre.

$$I_{Z-Z} = I_{X-X} + I_{Y-Y} = \frac{MR^2}{4} + \frac{MR^2}{4}$$

$$I_{Z-Z} = \frac{2MR^2}{4}$$

$$\boxed{I_{Z-Z} = \frac{MR^2}{2}}$$

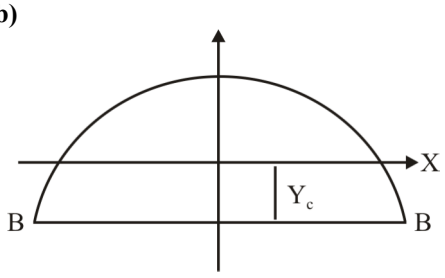
321. Moment of inertia of a semicircle about is x-axis is given by

(a) $0.22\ r^3$ (b) $0.11\ r^4$

(c) $0.14\ r^4$ (d) $0.2\ r^4$

APPSC AEE 04.12.2012
WBPSC AE 2008

Ans. (b)



Moment of inertia about BB (I_{BB}) = $\frac{\pi r^4}{8}$

Moment of inertia about x – axis (I_{x-x}) –

using parallel axis theorem

$I_{BB} = I_{x-x} + Ay_e^2$

$I_{xx} = I_{BB} - Ay_e^2$

$$= \frac{\pi r^4}{8} - \frac{\pi r^2}{2}\left(\frac{4r}{3\pi}\right)^2$$

$$= \frac{\pi r^4}{8} - \frac{8r^4}{9\pi}$$

$$= \frac{\left(9\pi^2 - 64\right)r^4}{72\pi}$$

$$\boxed{I_{xx} = 0.11\ r^4}$$

322. Regarding "'Polar moment of inertia" which of the following statement is correct?

(a) It is the moment of inertia for an area with respect to a line or axis perpendicular to the plane of the area

(b) It is the moment of inertia for an area with respect to a line or axis perpendicular to the plane of the area

(c) It is applicable to masses whereas moment of inertia is applicable to areas only

(d) It is the moment of inertia for an area with respect to a line or axis parallel to centroidal axis

WBPSC AE 2008

Ans. (b) : Polar moment of inertia it is the moment of inertia for an area with respect to a line or axis perpendicular to the plane of the area.

323. The moment of a lamina about any axis in the plane of the lamina equals the sum of moment of inertia about a parallel area of centroid axis in the plane of lamina and the product of the lamina the and square of the distance between the two axes.

The above theorem is known as

(a) theorem of parallel axes

(b) theorem of perpendicular axes

(c) three moments theorem

(d) none of the above

WBPSC AE 2008

Ans. (a) :

324. The moment of inertia about a principal axis is called

(a) mass moment of inertia

(b) second moment of inertia

(c) principal moment of inertia

(d) any of the above

WBPSC AE 2008

Ans. (c) : The moment of inertia about a principal axis is called principal moment of inertia.

325. Moment of inertia of a rectangle about its x-x-axis is given by

(a) $\frac{bd^3}{12}$

(b) $\frac{db^3}{12}$

(c) $\frac{bd^2}{6}$

(d) $\frac{bd^3}{6^3}$ where b is the breadth of a rectangle

WBPSC AE 2008

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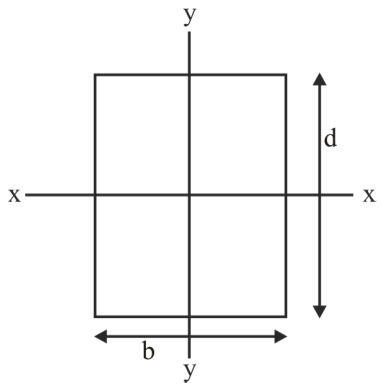
Ans. (a)



$$I_{x-x} = \frac{bd^3}{12} \text{ -----(1)}$$

$$I_{y-y} = \frac{db^3}{12} \text{ -----(2)}$$

326. .....of a body is defined as the distance from the reference axis at which the given area is assumed to be compressed and kept at thin strip, such that there is no change in its moment of inertia.

(a) Neutral axis
(b) Gyroscopic distance
(c) Second moment of area
(d) Radius of Gyration

HPPSC AE (PWD) 10.09.2016

Ans. (d) : Radius of Gyration- Radius of gyration of a body is defined as the distance from the reference axis at which the given area is assumed to be compressed that kept as thin strip such that there is no change in its moment of inertia.

$I = mk^2$

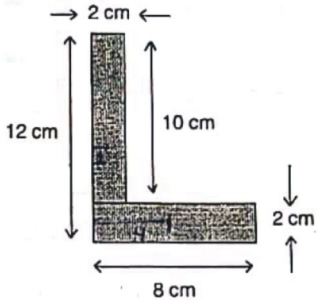
Where, I = Moment of inertia

m = mass of the body

According to the radius of gyration is

$$k = \sqrt{\frac{I}{m}}$$

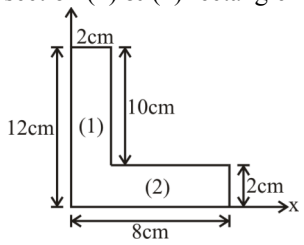
327. Find the centre of gravity of the L-section shown in figure



(a) 4.33 cm from the bottom line of the L-section and 2.33 cm from the left of the L-section
(b) 4.33 cm from the top of the L-section and 2.33 cm from the right of the L-section
(c) 3.33 cm from the bottom line of the L-section and 2.33 cm from the left of the L-section
(d) 4.33 cm from the bottom line of the L-section and 3.33 cm from the left of the L-section
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Let section (1) & (2) rectangle



For (1)

$a_1 = 10 \times 2 = 20 \text{ cm}^2$

$y_1 = 5 + 2 = 7 \text{ cm}$

$x_1 = 1 \text{ cm}$

For (2)

$a_2 = 8 \times 2 = 16 \text{ cm}^2$

$x_2 = 4 \text{ cm}, y_2 = 1 \text{ cm}$

$$\bar{x} = \frac{a_1x_1 + a_2x_2}{a_1 + a_2} = \frac{20\times1 + 4\times16}{20+16} = 2.33\text{cm}$$

$$\bar{y} = \frac{a_1y_1 + a_2y_2}{a_1 + a_2} = \frac{20\times7 + 1\times16}{20+16} = 4.33\text{cm}$$

328. One of the requirements for "dynamic equivalence" is :

(a) Mass of the model must be twice that of the original body
(b) Mass moment of inertia must equal that of the original body
(c) Centre of gravity must be displaced
(d) Mass moment of inertia must be half that of the original body.

UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation : Dynamic equivalence is the mass moment of inertia must equal to that of the original body.

329. The product of a given area and its centroidal distance from a point is:

(a) Mass moment of inertia
(b) First moment of area
(c) Area moment of inertia
(d) Centroid

JUVNL AEE (GTO) 05.03.2017

Ans. (b) : First moment of area is commonly used to determine the centroid of an area.

It is a calculated as the summation of the product of each area and its distance from the axis.

First moment of area = $\Sigma A.d$

330. The volume of the solid generated by rotating any plane figure about a non-intersecting axis in its plane is equal to the product of the area A of the figure and the distance travelled by its centroid. This is known as

(a) 2nd Pappus–Guldinus Theorem
(b) Varignon's Theorem
(c) Three Force Theorem
(d) 1st Pappus–Guldinus Theorem

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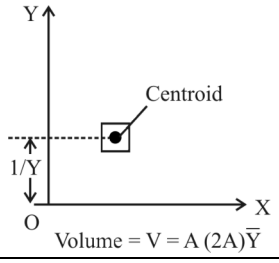
Ans. (a) : The Pappus's theorem can also be used in reverse to find the centroid of a curve or figure.

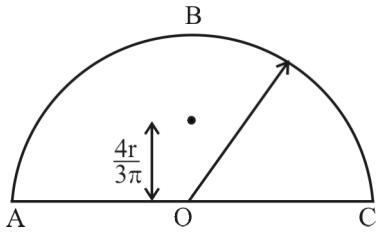


331. What is the moment of inertia of a semicircular lamina with the base horizontal about the centroidal axis parallel to x-axis, if the radius is R.

(a) $0.10976\ R^4$ (b) $0.2084\ R^4$
(c) $0.23\ R^4$ (d) $0.3142\ R^4$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Moment of inertia of a semicircular section–



We know that,
MOI of semicircular section about base AC is equal to half of the MOI of circular section about AC.

$$I_{AC} = \frac{1}{2} \times \frac{\pi}{64} \times d^4 = 0.393 r^4$$

We also know that area of semicircular section

$$a = \frac{1}{2} \times \pi r^2 = \frac{\pi r^2}{2}$$

and distance between centre of gravity of the section and the base, AC = $h = \frac{4r}{3\pi}$

MOI of the section through its C.G. and parallel to x-x axis is,

$$I_G = I_{AC} - ah^2$$

$$= \left[\frac{\pi}{8} \times r^4\right] - \left[\frac{\pi r^2}{2} \times \left(\frac{4r}{3\pi}\right)^2\right]$$

$$= \left[\frac{\pi}{8} r^4\right] - \left[\frac{8}{9\pi} r^4\right]$$

$$\boxed{I_G = 0.1097 R^4}$$

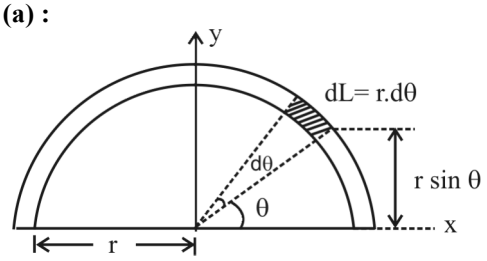
Note : The MOI about y-y axis will be the same as that about base AC i.e $0.393\ r^4$

332. The centre of gravity of a semi circular arc of radius 'r' is on the central radius from the base diameter at a distance of

(a) $2r/\pi$ (b) $3r/\pi$
(c) $4r/\pi$ (d) r/π

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



$$\bar{y} = \int \frac{y.dL}{dL}$$

$$\bar{y} = \frac{\int_0^{\pi} (r.\sin\theta)(r.d\theta)}{\int_0^{\pi} r.d\theta}$$

$$\bar{y} = \frac{r^2 \int_0^{\pi} \sin\theta.d\theta}{\int_0^{\pi} r.d\theta} = \frac{r[-\cos\theta]_0^{\pi}}{[\theta]_0^{\pi}}$$

$$\bar{y} = \frac{-r[\cos\theta]_0^{\pi}}{\pi - 0} = \frac{-r[\cos\pi - \cos\theta]}{\pi} = \frac{-r(-1-1)}{\pi}$$

$$\boxed{\bar{y} = \frac{2r}{\pi}}$$

Alternatively, we can solve it by Pappus-Guldinus equation

$$\bar{y} = \frac{\text{surface area of sphere}}{2\pi \times \text{length of arc}} = \frac{4\pi r^2}{2\pi \times \pi r} = \frac{2r}{\pi}$$

Note : Centroid of semicircular area = $\frac{4r}{3\pi}$

333. If the moments of inertia of a section about its orthogonal axes are 1×10^6 and 2×10^6 and its area is 100 units, its polar moment of inertia is

(a) 3×10^6 (b) 3.1×10^6
(c) 5×10^6 (d) 4×10^6

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Given
$I_{xx} = 1 \times 10^6$
$I_{yy} = 2 \times 10^6$
A = 100 units
Polar moment of inertia = $I_{xx} + I_{yy}$
$= 1 \times 10^6 + 2 \times 10^6$
$= 3 \times 10^6$

334. Moment of inertia (about its neutral axis) of hollow rectangular section with overall width and depth as B and D and hollow rectangular hole as b and d is

(a) $1/16\ (BD^3 - bd^3)$ (b) $1/32(BD^3 - bd^3)$
(c) $(BD^3 - bd^3)$ (d) $1/12(BD^3 - bd^3)$

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (d) : $I_{N\text{-}A} = I_1 - I_2$

$$[I_{N-A}]_1 = \frac{BD^3}{12}$$

$$[I_{N-A}]_2 = \frac{bd^3}{12}$$

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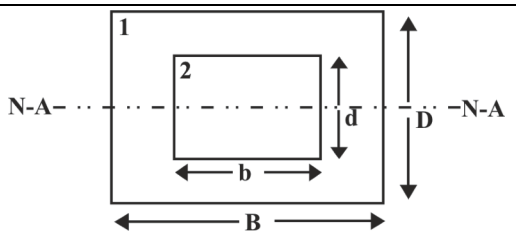


$$I_{N-A} = \frac{(BD^3 - bd^3)}{12}$$

335. According to theorem of perpendicular axes, if I_{xx} and I_{yy} be the M.I. of a lamina about *xx* and *yy* axis, then M.I. of the lamina about axis *zz*, which is perpendicular to *xx* and *yy*, equal to

(a) $I_{xx} \times I_{yy}$ (b) $I_{xx} + I_{yy}$

(c) $\frac{I_{xx}}{I_{yy}}$ (d) $\frac{I_{yy}}{I_{xx}}$

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (b) Theorem of perpendicular axis -

$I_{ZZ} = I_{XX} + I_{YY}$

336. Moment of inertia of a square of side d about the diagonal is-

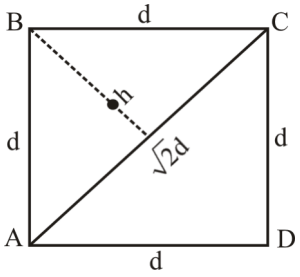
(a) $d^4/18$ (b) $d^4/24$

(c) $d^4/12$ (d) $d^4/8$

Vizag Steel (MT) 2017

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c)



Diagonal = $\sqrt{2}$d

Height (AC) of triangle = h= $\frac{AC}{\sqrt{2}} = \frac{\sqrt{2}d}{2}$

AC $\rightarrow$ just like a base of triangles of ABC & ADC

So moment of inertia of a square of side d about the diagonal AC

$$= 2 \times \left[\frac{(AC)h^3}{12}\right]$$

$$= 2 \times \left[\frac{\left(\sqrt{2}d\right).\left[\frac{\sqrt{2}d}{2}\right]^3}{12}\right]$$

$$= \frac{d^4}{12}$$

337. M.I. of a triangular section of base a and height h about an axis passing through its c.g. and parallel to base is-

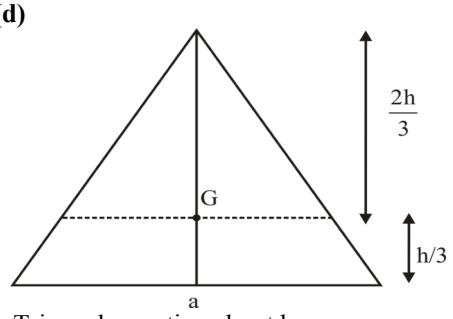
(a) $\frac{ah^3}{8}$ (b) $\frac{ah^3}{12}$

(c) $\frac{ah^3}{24}$ (d) $\frac{ah^3}{36}$

Karnataka PSC AE (WRD) 31.07.2021

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (d)



MI of a Triangular section about base

$$I_{base} = \frac{ah^3}{12}$$

MI of a Triangular section about a parallel line, passing through center of Gravity to base

$$I_{C.G.} = \frac{ah^3}{36}$$

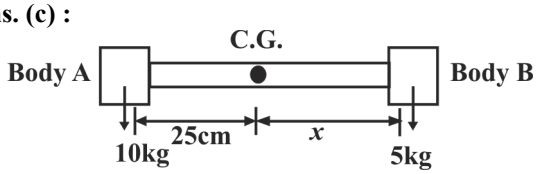
338. A body A of weight 10 kg with its C.G. 25 cm from the axis of rotation is to be balanced by another body B of weight 5 kg. The radius at which the C.G. of weight B should be placed is

(a) 2 cm (b) 12.5 cm

(c) 50 cm (d) None of these

ISRO Scientist/Engineer 24.05.2014

Ans. (c) :



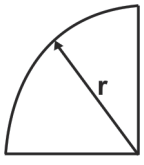
So, moment about C.G.

$$m_1 r_1 = m_2 r_2$$

$$r_2 = \frac{m_1 r_1}{m_2} = \frac{10 \times 25}{5}$$

$$r_2 = 50 \text{ cm}$$

339. Area moment of inertia for the quadrant shown below is



(a) $\pi r^4/2$ (b) $\pi r^4/4$

(c) $\pi r^4/8$ (d) $\pi r^4/16$

ISRO Scientist/Engineer 2012

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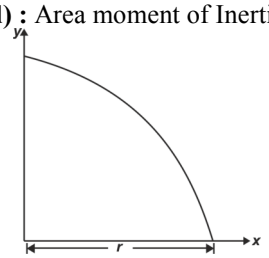
Ans. (d) : Area moment of Inertia



$$I_y = I_x = \frac{1}{4}(I_{circle})$$

$$= \frac{1}{4} \times \frac{\pi d^4}{64}$$

$$\frac{1}{4} \times \frac{\pi \times (2r)^4}{64}$$

$$= \frac{1}{4} \times \frac{\pi \times 16r^4}{64}$$

$$= \frac{\pi r^4}{16}$$

340. The mass moment of inertia of a cube with edges of length b, about an axis passing through an edge

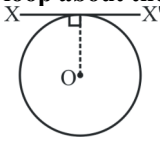
(a) $\frac{mb^2}{2}$ (b) $\frac{mb^2}{6}$

(c) $\frac{3mb^2}{2}$ (d) $\frac{2mb^2}{3}$

ISRO Scientist/Engineer 2011

Ans. (d)

341. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre O as shown in figure. The moment of inertia of the loop about the axis XX' is



(a) $\rho L^3/16\pi^2$ (b) $\rho L^3/8\pi^2$

(c) $5\rho L^3/16\pi^2$ (d) $3\rho L^3/8\pi^2$

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : $L = 2\pi R$

$$R = \frac{L}{2\pi}$$

Linear mass moment–

$$M = \rho L$$

Parallel Axis Theorem–

$$I_{XX} = I_C + MX^2$$

$$= \frac{MR^2}{2} + MR^2$$

$$I_{XX} = \frac{3}{2}MR^2$$

$$= \frac{3}{2} \times \rho L \times \left(\frac{L}{2\pi}\right)^2 = \frac{3}{2} \times \rho L \times \frac{L^2}{4\pi^2}$$

$$I_{XX} = \frac{3\rho L^3}{8\pi^2}$$

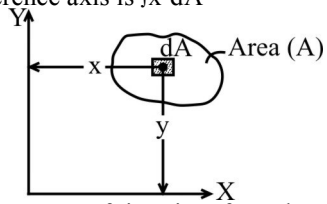
342. Moment of Inertia of an area dA at a distance x from a reference axis is:

(a) $\int x dA$ (b) $\int x^2 dA$

(c) $\int x^3 dA$ (d) $\int x^4 dA$

(HPPSC POLY. LECT. 2016)

Ans : (b) Moment of Inertia of an area dA at a distance x from a reference axis is $\int x^2 dA$



The axial moment of inertia of a plane area is the geometrical characteristic of the area defined by the integrals

$I_{xx} = \int y^2.dA$ and $I_{yy} = \int x^2.dA$

343. Moment of inertia of a body does not depend on

(a) Axis of rotation of the body

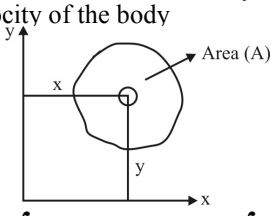
(b) Mass of the body

(c) Angular velocity of the body

(d) distribution of mass in the body

(KPSC AE 2015)

Ans : (c) Moment of Inertia of a body does not depend on angular velocity of the body



$$I_{xx} = \int y^2.dA \quad \text{and} \quad I_{yy} = \int x^2 dA$$

Where x and y the Co - ordinate of the differintial elements of area dA. Since this area dA is multiplied by the square of the distance, the moment of inertia is also called the second moment at area.

344. For principal axes, the moment of inertia will be :

(a) $I_{xy} = 0$ (b) $I_{xy} = 1$

(c) $I_{xy} = \infty$ (d) none of these

(HPPSC POLY. LECT. 2016)

Ans : (a) For principal axes, the moment of Inertia $I_{xy} = 0$

345. The radius of gyration 'k' for a solid cylinder of radius 'R' is equal to

(a) $\sqrt{2}R$ (b) $R/\sqrt{2}$

(c) 0.6324 R (d) 0.5 R

TNPSC AE 2017

Ans. (b) : The radius of gyration 'k' for a solid cylinder of radius 'R' is equal to $R/\sqrt{2}$.

346. The radius of gyration of a disc type fly wheel of diameter D is

(a) D (b) D/2

(c) D/4 (d) $\frac{3}{\sqrt{2}}D$

TNPSC AE 2018

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Ans. (c) : $I = AK^2$

$$K = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{\pi}{64}D^4}{\frac{\pi}{4}D^2}}$$

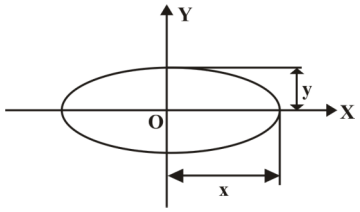
$$\boxed{K = \frac{D}{4}}$$

347. Moment of inertia of an elliptical area about the major axis is

(a) $\pi xy^3/4$ (b) $\pi xy^3/3$
(c) $\pi x^2y^3/4$ (d) πx^2y^3
(e) $\pi x^2y^3/3$

CGPSC AE 26.04.2014 Shift-I

Ans. (a) :



Moment of inertia of an elliptical area about the major axis

$$I_{xx} = \frac{\pi}{4}xy^3$$

Moment of inertia of an elliptical area about minor axis

$$I_{yy} = \frac{\pi}{4}yx^3$$

348. Statement (I) : Two circular discs of equal masses and thickness made of different materials will have same moment of inertia about their central axes of rotation.
Statement (II) : Moment of inertia depends upon the distribution of mass within the body.

ESE 2017

Ans. (d) : Mass moment of inertia for a uniform disc about its axis of rotation

$$I = \frac{MR^2}{2}$$

$$I = \frac{\rho AtR^2}{2}$$

$\because$ Mass & thickness is same,

$$\rho_1 A_1 t = \rho_2 A_2 t$$

$$I_1 = \frac{\rho_1 A_1 t R_1^2}{2} = \frac{\pi}{8}\rho_1 R_1^4 t$$

$$I_2 = \frac{\rho_2 A_2 t R_2^2}{2} = \frac{\pi}{8}\rho_2 R_2^4 t$$

If $\rho_1 > \rho_2$, $R_1 < R_2$, $I_1 > I_2$
Hence statement-I is wrong.

349. Moment of inertia of an area always least with respect to

(a) Bottom-most axis (b) Radius of gyration
(c) Central axis (d) Centroidal axis

Vizag Steel (MT) 2017
WBPSC AE, 2017

Ans. (d) : The moment of inertia about any axis passing through centroid is zero so Moment of inertia of an area always least with respect to centroidal axis.

350. Moment of Inertia about the centroidal axis of elliptical quadrant of base 'a' and height 'b' is

(a) $\frac{\pi ab}{24}(a^2+b^2)$ (b) $\frac{\pi ab}{24}(a^3+b^3)$
(c) $\frac{\pi ab}{16}(a^2+b^2)$ (d) $\frac{\pi ab}{24}(a^3+b^3)$

TNPSC AE 2013

Ans. (c) : $I = \frac{\pi ab}{16}\left(a^2+b^2\right)$

351. Length to radius ratio $\frac{l}{r}$ of a solid cylinder is such that the moments of inertia about the longitudinal and transverse axes are equal is

(a) 1 (b) $\sqrt{3}$
(c) $\sqrt{5}$ (d) 2

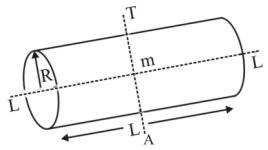
TNPSC AE 2014

Ans. (b) : MOI of a uniform solid cylinder about longitudinal axis

$$I_{L-L} = \frac{mR^2}{2}$$

MOI of a uniform solid cylinder about transfer axis

$$I_{T-A} = \frac{1}{4}mR^2 + \frac{1}{12}mL^2$$



Given-

$$I_{L-L} = I_{T-A}$$

$$\frac{mR^2}{2} = \frac{mR^2}{4} + \frac{1}{12}mL^2$$

$$\left[\frac{L}{R}\right]^2 = 3$$

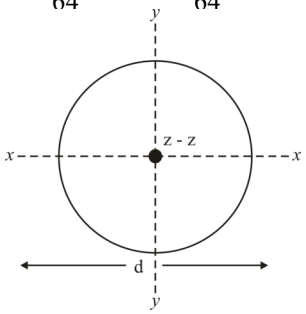
$$\left[L/R\right] = \sqrt{3}$$

352. Ratio of moment of Inertia of a circular body about its *x* axis to that about *y* axis is

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

NPCIL ET, 2015
TNPSC AE 2014

Ans. (b) : $I_{x-x} = \frac{\pi d^4}{64}$, $I_{y-y} = \frac{\pi d^4}{64}$



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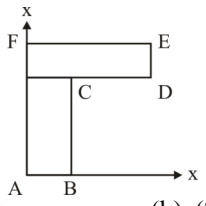
$$I_{z-z} = I_{xx} + I_{yy} = \frac{\pi d^4}{32}$$

then $$\frac{I_{x-x}}{I_{y-y}} = \frac{\frac{\pi d^4}{64}}{\frac{\pi d^4}{64}} = 1$$

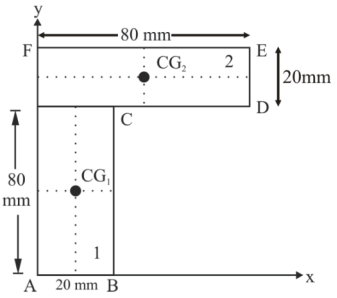
353. The coordinates of centroid of given geometry ABCDEFA [DE = 20 mm, EF = 80 mm, FA = 100 mm, AB = 20mm] will be given as : (A is origin)



(a) (60, 20) (b) (25, 60)
(c) (25, 65) (d) (65, 25)
(e) (25, 40)

CGPSC AE 25.02.2018

Ans. (c) : The coordinates of centroid is $(\bar{x}, \bar{y})$



$$\bar{x} = \frac{a_1x_1 + a_2x_2}{a_1 + a_2}$$

$$= \frac{80\times20\times10 + 80\times20\times40}{80\times20 + 80\times20}$$

$$\bar{x} = 25\text{mm}$$

$$\bar{y} = \frac{a_1y_1 + a_2y_2}{a_1 + a_2} = \frac{80\times20\times40 + 80\times20\times90}{80\times20 + 80\times20} = 65\text{mm}$$

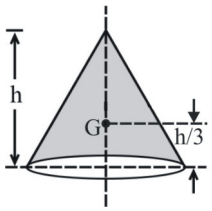
So coordinates of centroid (25, 65)

354. Centre of gravity of a thin hollow cone lies on the axis at a height of :

(a) one-fourth of the total height above base
(b) one-third of the total height above base
(c) one-half of the total height above base
(d) three-eighth of the total height above the base

HPPSC W.S. Poly. 2016

Ans : (b)



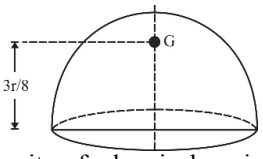
Centre of gravity of a thin hollow cone lies on the axis at a height of one-third of the total height above base.

355. The centre of gravity of a hemisphere of diameter 80 mm from its base diameter is

(a) 15 mm (b) 40 mm
(c) 20 mm (d) 10 mm

TSPSC AEE 2015

Ans : (a)



The centre of gravity of a hemisphere is at a distance of 3r/8 from base

C.G = 3r/8 (2r = d= 80mm)

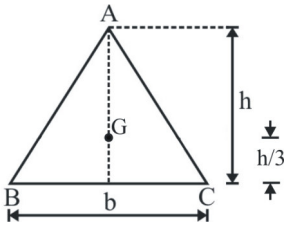
C.G = 15 mm.

356. The base of triangle is 60 mm and height is 50 mm. The moment of inertia about its base 'BC' is

(a) 2,08, 333.33mm^4 (b) 6,25,000 mm^4
(c) 9,00,000 mm^4 (d) 3,00, 000mm^4

TSPSC AEE 2015

Ans : (b)



$$I_{GG} = \frac{bh^3}{36}$$

$$I_{BC} = \frac{bh^3}{12}$$

$$I_{BC} = \frac{60\times(50)^3}{12}$$

$I_{BC} = 625000$ mm^4.

357. The centre of gravity of a plane lamina is not at its geometrical centre, if it is a

(a) circle
(b) square
(c) rectangle
(d) right angled triangle

UKPSC AE 2007 Paper -I

Ans. (d) : Right angled triangle

358. Which of the following represents the state of neutral equilibrium?

(a) Cube resting on one edge
(b) A smooth cylinder lying on a curved surface
(c) A smooth cylinder lying on a convex surface
(d) None of the above

UKPSC AE 2007 Paper -I

Ans. (d) : None of the above

359. The coordinate of centroid (x, y) of quarter circular lamina of radius (R), whose straight edges coincide with the coordinate axis in the first quadrant, are given as:

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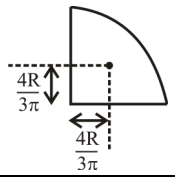
(a) $(0, 4R/3\pi)$ (b) $(4R/3\pi, 0)$

(c) $(4R/3\pi, 8R/3\pi)$ (d) $(4R/3\pi, 4R/3\pi)$

(e) $(8R/3\pi, 8R/3\pi)$

CGPSC AE 25.02.2018

Ans. (d) :



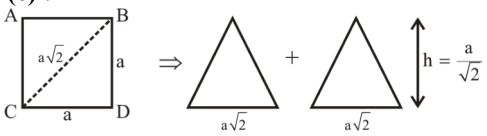
360. The area moment of inertia of a square of size 1 unit about its diagonal is

(a) $\frac{1}{3}$ (b) $\frac{1}{4}$

(c) $\frac{1}{12}$ (d) $\frac{1}{6}$

VIZAG STEEL MT 10.06.2012
VIZAG STEEL MT 18.08.2013

Ans. (c) :



Given a square with side a = 1

Diagonal of square = Base of the triangle (b) = $a\sqrt{2}$

MOI of the square about the diagonal

$$(I_{square})_{diagonal} = 2 \times (I_{triangle})_{base}$$

$$= 2 \times \frac{bh^3}{12}$$

$$= 2 \times \frac{(a\sqrt{2}) \times \left(\frac{a}{\sqrt{2}}\right)^3}{12}$$

$$(I_{square})_{diagonal} = \frac{a^4}{12} \Rightarrow \frac{1}{12} \quad [\because a = 1]$$

361. The ratio of the moment of inertia of a rectangle about its centroidal axis to that about its base, is:

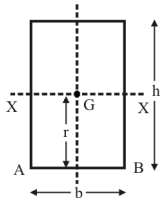
(a) 1/3 (b) 1/4

(c) 3/4 (d) 4

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : Case (I)–MOI of a rectangle about its centroidal axis (X-X)

$$I_{XX} = \frac{bh^3}{12}$$



Case (II)–MOI of a reactangle about its base (AB)

$$I_{AB} = I_{XX} + Ar^2$$

$$= \frac{bh^3}{12} + bh \times \frac{h^2}{4}$$

$$= \frac{bh^3}{12} + \frac{bh^3}{4}$$

$$= \frac{4bh^3}{12}$$

$$= \frac{bh^3}{3}$$

$$\frac{I_{XX}}{I_{AB}} = \frac{bh^3}{12 \times bh^3} \times 3$$

$$= \frac{1}{4}$$

362. What is the point through which the whole weight of body acts, irrespective of its position, known as

(a) Moment of inertia (b) Centre of gravity

(c) Centre of percussion (d) Centre of mass

VIZAG STEEL MT 2011

Ans. (b) : The point through which the whole weight of body acts, irrespective of its position, known as centre of gravity.

363. The product of moment of inertia and angular velocity is known as :

(a) angular torque (b) kinetic energy

(c) angular momentum (d) none of the above

Kerala PSC IOF 19.04.2016

Ans. (c) The product of moment of inertia and angular velocity is known as angular momentum.

Angular momentum,

= moment of inertia × angular velocity

$= I \times \omega$

$= mk^2 \times \frac{v}{k}$

$\boxed{L = mkv}$

k = Radius of gyration.

364. The formula to calculate mass moment of inertia (I) of the circular rim is:

(a) $I = m^2 \times r$ (b) $I = (m^2 \times r)/2$

(c) $I = m \times r/2$ (d) $I = m \times r^2$

UPRVUNL AE 05.07.2021

Ans. (d) : $I = m \times r^2$

365. Section modulus of a circular section about an axis through its centre of gravity is

(a) $\frac{\pi}{32}d^3$ (b) $\frac{\pi}{16}d^3$

(c) $\frac{\pi}{8}d^3$ (d) $\frac{\pi}{64}d^3$

APPSC AEE 2012

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Ans : (a) Section modulus (z) = I/y

I = Moment of Inertia $= \frac{\pi d^4}{64}$

$y = \frac{d}{2}$

$\boxed{z = \frac{\pi d^3}{32}}$

366. A man weighs 980 N on earth. What will be his weight in moon and sun where gravitation force of moon $g_m = 1.7$ m/s^2 and that of the sun $g_s = 270$ m/s^2 :

(a) $W_m = 170$ N and $W_s = 27000$ N

(b) $W_m = 1666$ N and $W_s = 264600$ N

(c) $W_m = 17$ N and $W_s = 2700$ N

(d) $W_m = 16.66$ N and $W_s = 2646$ N

TNPSC AE (Industries) 09.06.2013

Ans. (a) :

367. Polar moment of inertia of a circular area is

(a) $\frac{\pi}{32}d^4$ (b) $\frac{\pi}{64}d^4$

(c) $\frac{\pi}{4}d^4$ (d) $\frac{\pi}{4}d^3$

APPSC AEE 2012

Ans. (a) : Polar moment of inertia of circle-

J = Area moment of inertia about x-axis + Area moment of inertia about y-axis

$J = I_x + I_y = \frac{\pi}{64}d^4 + \frac{\pi}{64}d^4$

$J = \frac{\pi}{32}d^4$

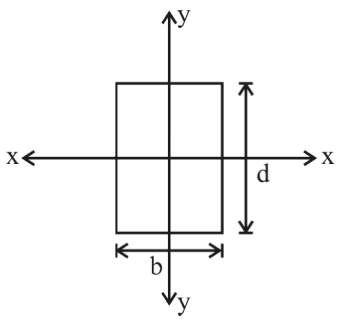
368. Radius of gyration for the rectangular section, with respect to horizontal axis. Take rectangle of width 'b' and depth 'd'.

(a) $\frac{d}{2\sqrt{3}}$ (b) $\frac{4}{\sqrt{3}}$

(c) $\frac{d}{4\sqrt{3}}$ (d) $\frac{d}{3}$

TNCSC AE 2020

Ans. (a) :



For rectangular–

Moment of inertia about horizontal axis 'x-x'

$I_{xx} = \frac{bd^3}{12}$

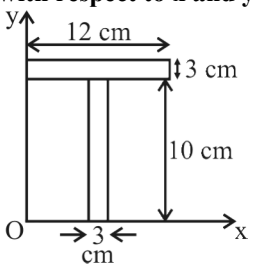
Area, A = bd

Radius of gyration (k) $= \sqrt{\frac{I}{A}}$

$\sqrt{\frac{\frac{bd^3}{12}}{bd}} = \sqrt{\frac{bd^3}{12} \times \frac{1}{bd}}$

$= \boxed{k = \frac{d}{2\sqrt{3}}}$

369. Find the centre of gravity of 'T' section given in figure with respect to x and y axis:

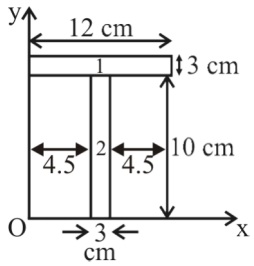


(a) 6.30 cm (b) 8.54 cm

(c) 11.25 cm (d) None of these

PTCUL AE 25.06.2017

Ans. (b) :



$A_1 = 12 \times 3 = 36$ cm^2

$A_2 = 10 \times 3 = 30$ cm^2

$x_1 = 6$ cm ; $y_1 = 11.5$ cm

$x_2 = 6$ cm ; $y_2 = 5$ cm

$x = \frac{A_1x_1 + A_2x_2}{A_1 + A_2}$

$= \frac{(36 \times 6) + (30 \times 6)}{36 + 30} = 6$

$y = \frac{A_1y_1 + A_2y_2}{y_1 + y_2} = \frac{(36 \times 11.5) + (30 \times 5)}{36 + 30}$

$= 8.545$ cm

5. Trusses and Beams

370. A framed structure of triangular shape is

(a) perfect (b) imperfect

(c) deficient (d) redundant

TSPSC Manager (Engg.) HMWSSB 12.11.2020

UPRVUNL AE 05.07.2021

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Ans. (a) : A framed structure of triangular shape is perfect.

371. If the number of members in a truss are more than that of a perfect truss. It is known as

(a) Redundant truss (b) Deficient truss
(c) Full truss (d) General type truss

APPSC AE Subordinate Service Civil/Mech. 2016
UKPSC AE, Paper-I 2013

Ans. (a) : If the number of members in a truss (m) is equal to 2j-3 then it is called perfect truss.

Stability and determinacy of truss

For 2-D truss

$m = 2j - 3 \rightarrow$ statically determinate (perfect truss)

$m < 2j-3 \rightarrow$ statically determinate but unstable (deficient truss)

$m > 2j - 3 \rightarrow$ statically indeterminate (redundant truss)

372. A perfect frame is that which satisfies the equation (Assume n = No. of members and j = No. of joints)

(a) $n = 3j - 2$ (b) $n > 3(j - 1)$
(c) $n < 3(j - 1)$ (d) $n = (2j - 3)$

TSPSC AEE 2015
ISRO Scientist/Engineer (RAC) 29.11.2015
ISRO Scientist/Engineer 24.05.2014
APPSC AE 2012
UKPSC AE Paper-I 2012
ISRO Scientist/Engineer 2009

Ans. (d) : In general let a frame have j joints and n members.

(i) if $n = 2j - 3$, then the frame is perfect frame.
(ii) if $n < 2j - 3$, then the frame is deficient (Imperfect) frame.
(iii) if $n > 2j - 3$, then the frame is redundant frame.

373. In a simple truss, if n is the total number of joints, the total number of members is equal to

(a) $2n + 3$ (b) $2n - 3$
(c) $n + 3$ (d) $n - 3$

Gujarat PSC AE 2019
OPSC Civil Services Pre. 2011
GATE 2014

Ans : (b) :

$$\boxed{m = 2n - 3}$$

Where, m = member of joints
n = Total number of joints

374. Ties are load carrying members which have many

(a) torsional loads
(b) axial compressive loads
(c) axial tension loads
(d) transverse loads

TNPSC AE, 2008

Ans. (c) : Ties are load carrying members which have many axial tension loads.

375. Method of sections in truss analysis is generally found useful to determine.

(a) Forces in all members
(b) Forces in selective members
(c) Weights of the members
(d) Bending of members

TSPSC Managers, 2015

Ans. (b) : The method of sections is a process used to solve for the unknown forces acting on members of a truss.

376. In a statically determinate problem, the number of non-trivial equilibrium equations :

(a) equals the numbers of unknowns
(b) is less than number of unknowns
(c) is more than number of unknowns
(d) cannot be determined

J&K PSC Civil Services Pre, 2013

Ans. (c) : For determinate beam number of non-trivial equilibrium equation should be more than number of unknowns.

377. If a truss consists of 8 joints, 10 members and 4 reaction components, then it is a

(a) cantilever truss (b) deficient truss
(c) redundant truss (d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Condition for redundant $\Rightarrow h > 2J - R$
Condition for deficient $\Rightarrow h < 2J - R$
Where, h – Member of truss
J – Joint
R – Reaction

$h = 10$ $h < 2J - R$
$J = 8, R = 4$ $10 < 2 \times 8 - 4$
$\boxed{10 < 12}$

So, it is deficient truss.

378. Which of the following statement(s) is/are true for truss design?

(i) All loads are applied at the joints.
(ii) The members are joined together by smooth pin.
(iii) Each truss member acts as single force member.

(a) Only (i) (b) Only (i) and (ii)
(c) Only (i) and (iii) (d) Only (iii)
(e) All (i), (ii) and (iii)

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : A truss is a structure that consists of members organised into connected triangles so that the overall assembly behaves as a single object. A truss is made up of a web of triangles joined together of enables the even distribution of weight and the handling of changing tension and compression without bending or shearing. All loads are applied at the points of member are joined together by smooth pin.

Each truss member acts as double force member as equal and opposite force (i.e., either tension or compression) at both ends of the member.

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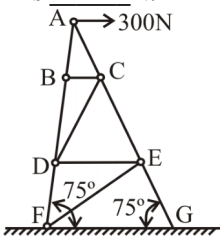
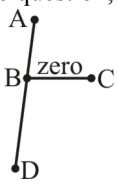
379. The figure shows an idealized plane truss. If a horizontal force of 300N is applied at point A, then the magnitude of the force produced in member CD is _____N.



(a) 300 (b) 0
(c) 116 (d) 276.5

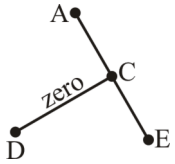
Assam Engg. College AP/Lect. 18.01.2021

Ans. (b) : According to question,



Since, members are collinear, therefore there will be no force in member BC,

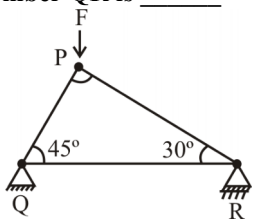
$F_{BC} = 0$



Again, $F_{CD} = 0$

Hence, the magnitude of the force produced in member CD is 0.

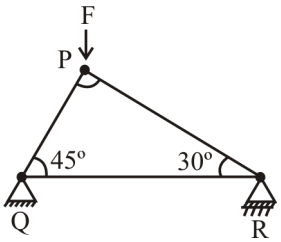
380. Consider a truss PQR loaded at P with a force F as shown in the following figure. The tension in the member QR is _____



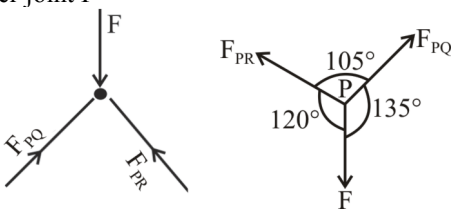
(a) 0.5 F (b) 0.63 F
(c) 0.73 F (d) 0.87 F

Assam Engg. College AP/Lect. 18.01.2021
GATE 2008
TSPSC AEE 2017

Ans. (b) : $\angle QPR = 180° - (45 + 30)$
$= 105°$



Consider joint P



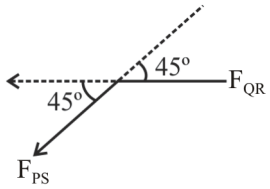
From Lami's Theorem

$$\frac{F_{PR}}{\sin 135°} = \frac{F_{PQ}}{\sin 120°} = \frac{F}{\sin 105}$$

$$F_{PQ} = \frac{F \times \sin 120°}{\sin 105°}$$

$F_{PQ} = 0.896\ F$

Consider point Q



$F_{QR} = F_{PQ} \cos 45°$
$= 0.896\ F \cos 45°$
$R_{QR} = 0.63397\ F$

381. Truss is said to be deficient, when number of members in truss are ____ than that required for perfect truss.

(a) more (b) equal to
(c) less (d) No relation

Oil India Senior Officer 23.12.2020

Ans. (c) : If number of members in truss are less than the required then it is called deficient or unperfect truss.
$[m < 2j - 3]$
where,
m = no. of members
n = number of joints
Deficient truss collapse when load is applied.

382. A mechanism converts into structure when the number of degrees of freedom is equal

(a) 0 (b) 1
(c) –1 (d) 2

UPRVUNL AE 05.07.2021

Ans. (a) : 0

383. Consider the following statements regarding types of supports and beams:

1. **When both supports of beams are roller supports, the beam is known as simply supported beam.**
2. **A beam with one end fixed and the other end free is known as fixed beam.**
3. **A beam with both ends fixed is known as cantilever beam.**
4. **A beam with one end fixed and the other simply supported is known as propped cantilever.**

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Which of the above statements is/are correct?

(a) 1 only (b) 1 and 4 only
(c) 1, 3 and 4 only (d) 2, 3 and 4 only

ESE 18.07.2021

Ans. (b) : The correct statements are-

- When both supports of beams are roller supports, the beam is known as simply supported beam.
- A beam with one end fixed and the other simply supported is known as propped cantilever.

Statements 2 and 3 are wrong because-

- A beam with one end fixed and other is free known as cantilever beam not fixed beam.
- Beam with both end fixed is known as fixed beam, not cantilever beam.

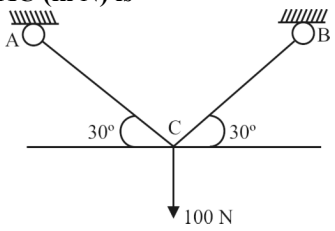
384. Mobility of a statically indeterminate structure is

(a) ≤ -1 (b) 0
(c) 1 (d) ≥ -2

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : Mobility of a statically indeterminate structure is ≤ 1

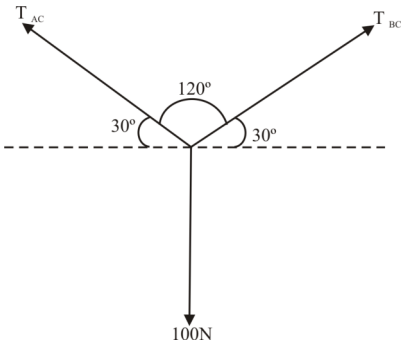
385. Two identical trusses supported a load of 100 N as shown in the figure. The length of each truss is 0.1 m; cross- sectional area is 200 mm^2, Young's modulus E=200 GPa. The force in the truss AC (in N) is



(a) 100N (b) 200N
(c) 50 N (d) 150N

PPSC Asstt. Municipal Engineer 15.06.2021
RPSC IOF & Boiler 2016
GATE 2015
RPSC IOF & Boiler 20.09.2017

Ans. (a) : By Lami's Theorem



$$\frac{T_{AC}}{\sin 120^\circ} = \frac{T_{BC}}{\sin 120^\circ} = \frac{100}{\sin 120^\circ}$$

$$T_{AC} = T_{BC} = 100N$$

386. In a truss it was found that $2j < m + 3$, where j is number of joints and m is number of member, then the truss is said to be

(a) statistically indeterminate
(b) unstable
(c) stable
(d) no such conclusion can be drawn for a truss

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) :

387. Pick the correct statement regarding analysis of forces in a truss

(a) Method of joints is used (when) as the forces at a joint are collinear
(b) Method of sections is preferred when the forces at imaginary cutting plane are non collinear
(c) Both are correct
(d) Both are wrong

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Force analysis in a truss-

- Method of joints is used (when) as the forces at a joint are collinear.
- Method of sections is preferred when the forces at imaginary cutting plane are non collinear.

388. Which of the following is an analytical method for finding out forces in a frame?

(a) Method of pins (b) Method of frames
(c) Method of joints (d) Method of hinges

UPRVUNL AE 05.07.2021

Ans. (c) : Two analytical method for finding out forces in a frame-
(i) Method of joints
(ii) Method of section

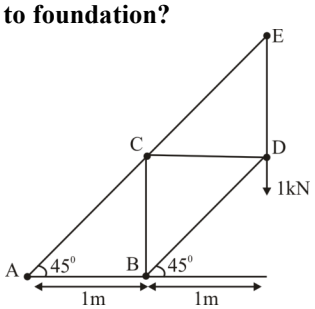
389. If $n < (2j - 3)$, where n is number of members and used in a frame structure and j is the number of joints used in the structure, then the frame is called

(a) Perfect frame (b) Deficient frame
(c) Redundant frame (d) None of the above

Gujarat PSC AE 2019
APPSC AEE Screening Test 2016

Ans : (b) : $n < (2j - 3)$ (Deficient frame)
Where, n = number of members
j = number of joints

390. What will be the axial force in the member EC, ED and DC of the plane truss (ABCDE) as shown in figure with end 'A' and 'B' is hinged to foundation?



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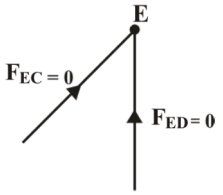
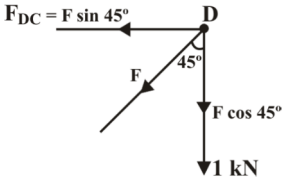
(a) $F_{EC} = F_{ED} = F_{DC} = 0$
(b) $F_{EC} = F_{ED} = 0$; $F_{DC} = 1$ kN (C)
(c) $F_{EC} = F_{ED} = 0$; $F_{DC} = 1$ kN (T)
(d) $F_{EC} = F_{ED} = F_{DC} = 1$ kN(T)
(e) $F_{EC} = F_{ED} = F_{DC} = 1$ kN (C)

CGPSC AE 25.02.2018

Ans. (c) : Point E-

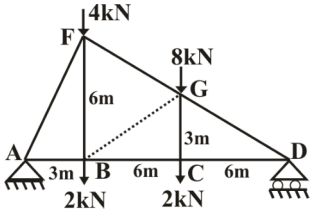


Point D-



F cos 45° = 1
F_{DC} = F sin 45°
= F cos 45°
= 1 kN (T)

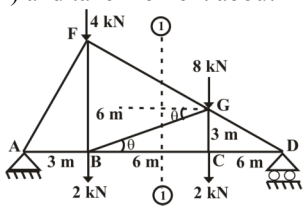
391. Determine the force in the member BG in the given truss

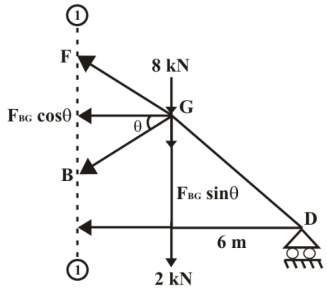


(a) 11.18 kN Tension
(b) 14.4 kN tension
(c) 11.18 kN Compression
(d) 14.40 kN compression

APPSC AEE SCREENING 17.02.2019

Ans. (c) : Take section 1 - 1 as shown in figure.
To find the force in BG, consider the right part of section (1) - (1) and take moment about 'D'.





$$\cos\theta = \frac{6}{3\sqrt{5}} = \frac{2}{\sqrt{5}}$$

$$\sin\theta = \frac{3}{3\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$$\sum M_D = 0$$

$-F_{BG}\cos\theta \times 3 - F_{BG}\sin\theta \times 6 - 2 \times 6 - 8 \times 6 = 0$

$$-F_{BG} \times \frac{6}{\sqrt{5}} \times F_{BG} \times \frac{6}{\sqrt{5}} - 60 = 0$$

$$-2\left[F_{BG} \times \frac{6}{\sqrt{5}}\right] = 60$$

$$F_{BG} = -5\sqrt{5} \text{ kN (compressive)}$$

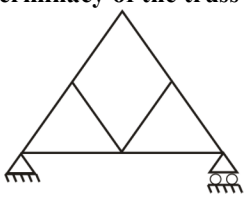
$$F_{BG} = 11.18 \text{ kN (compressive)}$$

392. A member in a truss can take
(a) axial force and bending moment
(b) only axial force
(c) only bending moment
(d) bending moment and shear force

APPSC AEE SCREENING 17.02.2019

Ans. (b) : Truss members take only axial forces (tension or compression). They cannot take shear force, bending moment and torsion.

393. Choose the correct combination of the stability and indeterminacy of the truss given.



(a) Statically indeterminate and stable
(b) Statically determinate and stable
(c) Unstable
(d) Statically determinate

APPSC AEE SCREENING 17.02.2019

Ans. (c&d) : No. of member m = 8
No. of joints, j = 6
m < 2j - 3
∴ The truss is internally unstable but determinate
∴ Both the options (c) and (d) are correct
Note- APPSC is given by answer (c) is correct.

394. Regarding the ability of a truss, the condition m + r > 2j is
(a) Necessary
(b) Sufficient
(c) Necessary and Sufficient
(d) Sufficient but not necessary

APPSC AEE SCREENING 17.02.2019

Ans. (d) : Given condition
m + r > 2j
m > 2j - r
∴ The member is structurally stable but not necessarily have m > 2j - r. It is just sufficient to have
m = 2j – r
Note- APPSC is given by answer (a) is correct.

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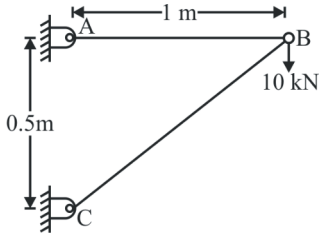
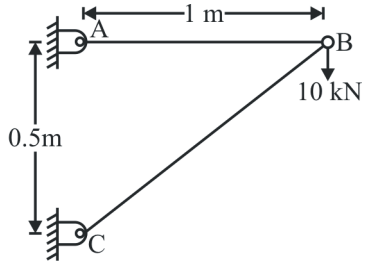
395. A two member truss ABC is configured as shown in figure. The force in the member AB is



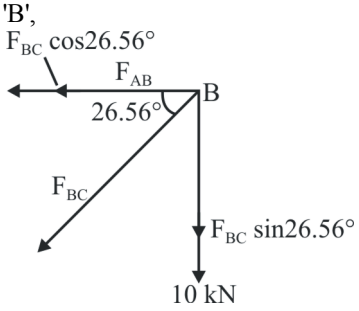
(a) 15 kN (b) 30 kN
(c) 20 kN (d) 5 kN

ISRO Scientist/Engineer 03.07.2016
UPPSC AE, Paper-I 13.12.2020

Ans. (c) :



At joint 'B',



From figure,

$$\tan\theta = \frac{0.5}{1}$$

$$\tan\theta = \frac{5}{10}$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\right)$$

$$= \tan^{-1}(0.5)$$

$$\theta = 26.56°$$

$\Sigma V = 0$,

$$F_{BC} \sin 26.56 + 10 \text{ kW} = 0$$

$$F_{BC} \times 0.447 = -10$$

$$F_{BC} = -\frac{10}{0.447}$$

$F_{BC} = 22.37$ kW (Compression)

$\Sigma H = 0$,

$$F_{AB} + F_{BC} \cos(26.56) = 0$$

$$F_{AB} - 22.37 \times 0.894 = 0$$

$$F_{AB} = +22.37 \times 0.894$$

$$= +19.90 \text{ kN}$$

$F_{AB} = \approx 20$ kN (Tension)

396. For a statistically determinate plane truss, the number of unknown support reactions is

(a) two (b) three
(c) four (d) one

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : For a statistically determinate plane truss, the number of unknown support reactions is three.

397. A Plane Truss has five joints and eight members. The truss is

(a) perfect (b) Redundant
(c) Deficient (d) None of the above

Karnataka PSC Lect., 27.05.2017

Ans (b) : Number of member (m) = 8
Number of joint (j) = 5
We know that,
For Redundant truss,

$$m > 2j - 3$$

$$8 > 2 \times 5 - 3$$

$$8 > 7$$

Note : $m = 2j - 3$, Perfect truss
$m > 2j - 3$, Redundant truss
$m < 2j - 3$, Deficient truss

398. If for a plane pin-jointed truss m = 2j-3 with the usual notion:

(a) It must be a just-rigid truss
(b) It may or may not be a just-rigid truss
(c) It cannot be over-rigid over any part of the truss
(d) It cannot be non-rigid over any part of the truss

UPRVUNL AE 21.08.2016

Ans. : (b) If for a plane pin-jointed truss m = 2j–3 with the usual notion it may or may not be a just rigid truss

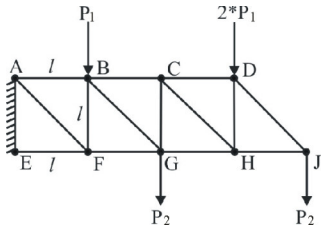
399. The maximum number of reaction components possible for a hinged support for all kinds of loading is

(a) 2 (b) 0
(c) 3 (d) 1

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Reaction components possible for hinged support = 2
Reaction components possible for fixed support = 3
Reaction components possible for roller support = 1

400. The force in member BC of steel truss structure shown in figure below is (Cross section of each member is A, Length of each member is 'ℓ')



(a) $P_1 + 2P_2$ (b) $2P_1 + P_2$
(c) $2\times(P_1 + P_2)$ (d) $(P_1 + P_2)/2$

ISRO Scientist/Engineer 17.12.2017

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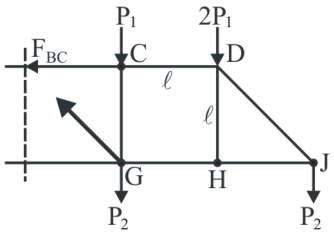
Ans. (c) :



Consider the right hand side part of the section.

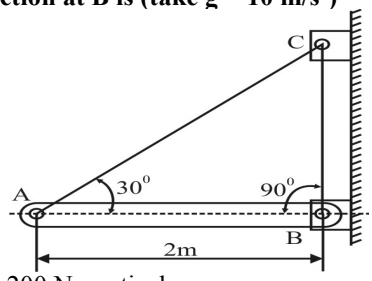
Taking moment about point G

$$F_{BC} \times \ell = 2P_1 \times \ell + P_2 \times 2\ell$$

$$F_{BC} = 2P_1 + 2P_2$$

$$F_{BC} = 2(P_1 + P_2)$$

401. The bar AB weighs 10 kg per meter and is supported by cable AC and a pin at B. Reaction at B is (take g = 10 m/s^2)



(a) 200 N, vertical
(b) 200 N, at 30^0 to horizontal
(c) 100 N vertical
(d) 100 N at 60^0 to horizontal

ISRO Scientist/Engineer 17.12.2017

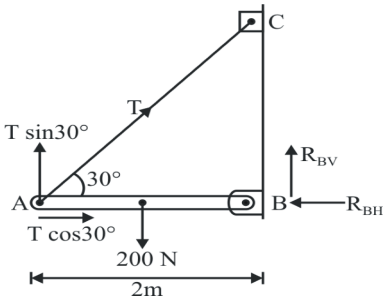
Ans. (b) : Weight of the rod

= 10 kg/m

= 10 × 10 N/m

= 100 × 2 N

= 200 N



Taking moment about point B

$$T\sin30 \times 2 - 200 \times 1 = 0$$

$$T = 200 \text{ N}$$

$\sum V = 0$

$$T\sin30° + R_{BV} - 200 = 0$$

$$200 \times \frac{1}{2} + R_{BV} - 200 = 0$$

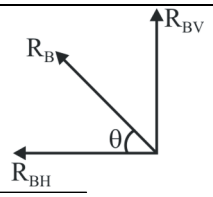
$$R_{BV} = 100 \text{ N}$$

$\sum H = 0$

$$T\cos30 - R_{BH} = 0$$

$$R_{BH} = 200 \times \frac{\sqrt{3}}{2} = 173.2 \text{ N}$$

Resultant reaction at B



$$R_B = \sqrt{R_{BV}^2 + R_{BH}^2}$$

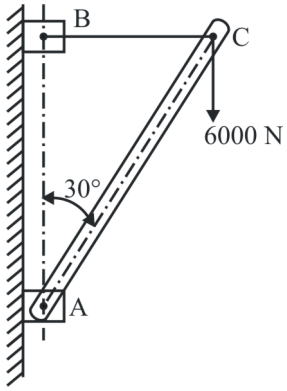
$$R_B = \sqrt{(100)^2 + (173.2)^2} = 200 \text{ N}$$

$$\theta = \tan^{-1}\left(\frac{R_{BV}}{R_{BH}}\right)$$

$$= \tan^{-1}\left(\frac{100}{173.2}\right)$$

$$= 30°$$

402. The bar AC, 10m long supports a load of 6000 N as shown in figure. The cable BC is horizontal and 5m long. Forces in the cable and the bar are



(a) 6928 N and 3464 N (b) 3464 N and 6928 N
(c) 3464 N and 3464 N (d) 6928 N and 6928 N

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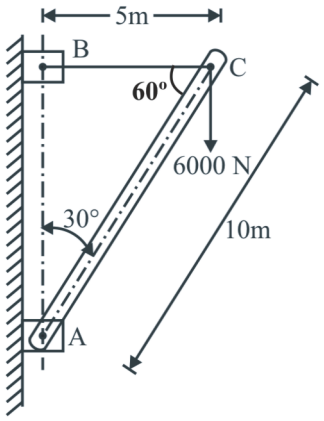
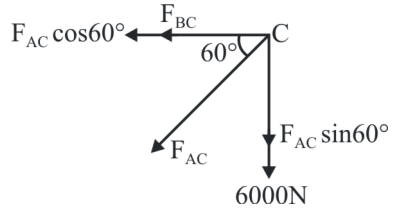
Ans. (b) :



At joint 'C'–



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$\Sigma H = 0,$

$F_{AC}\cos 60° + F_{BC} = 0$

$F_{AC} \times \frac{1}{2} = -F_{BC}$

$F_{BC} = -\frac{F_{AC}}{2}$

$F_{AC}\sin 60° + 6000 = 0$

$F_{AC} \times \frac{\sqrt{3}}{2} + 6000 = 0$

$F_{AC} \times \frac{\sqrt{3}}{2} = -6000$

$F_{AC} = \frac{-6000 \times 2}{\sqrt{3}}$

$= -4000 \times \sqrt{3}$

$F_{AC} = -6928$ N

Bar, → $\boxed{F_{AC} = 6928 \text{ N (compression)}}$

Force is cable

$F_{BC} = -\frac{F_{AC}}{2}$

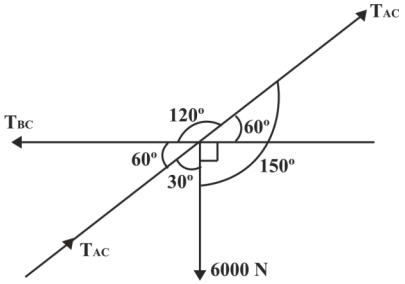
$= -\frac{(-6928)}{2}$

$= \frac{6928}{2}$

$F_{BC} = 3464$ (Tensile)

Using FBD,

Using Lami's theorem



$\frac{T_{BC}}{\sin 150} = \frac{6000}{\sin 120}$

$= \frac{T_{AC}}{\sin 90}$

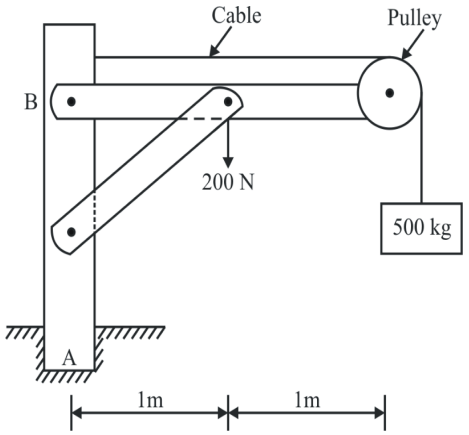
So,

$T_{AC} = \frac{6000 \times 2}{\sqrt{3}} = 6928.2N$ (compression)

$\boxed{T_{AC} = 6928.2 \text{ N (Compression)}}$

$\boxed{T_{BC} = 3464 \text{ N (Tension)}}$

403. What is the approximate reaction moment at the fixed support A for the loaded frame shown below? (Take the diameter of the pulley as 250 mm)



(a) 10010 Nm (b) 10624 Nm

(c) 9810 Nm (d) 1263 Nm

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Ans. (b) : Moment due to 500 kg weight is given as

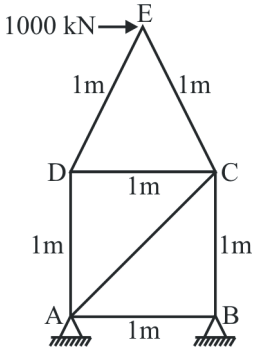
$M_1 = Mg\ (2 + 0.125)$

$M_1 = 500 \times 9.81 \times 2.125 = 10423.125$ Nm

Moment at fixed support A is given as

$M_A = M_1 + M_2 = 10423.125 + 200 = 10623.125$ Nm

404. A simple structure ABCDE is supported on a hinge at A and on rollers at B while it carries a horizontal force of 1000 kN at E as shown in the figure. Determine the force in member AC using the method of joints.

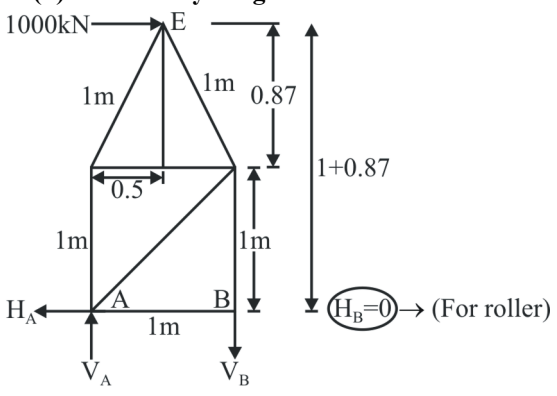


(a) 1500 kN (b) 1400 kN

(c) 1314 kN (d) 1414 kN

ISRO Scientist/Engineer 11.10.2015

Ans. (d) : Free Body Diagram–



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Finding reactions at A and B

$H_A = 1000$ kN

[Resolving forces horizontally]

Taking moment of all forces about "A".

$(\circlearrowleft)V_B \times 1 = 1000 \times 1.87\ (\circlearrowright)$

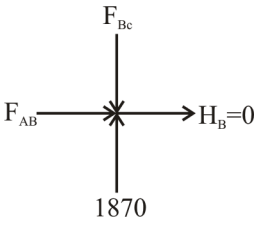
(A.C.W.) (C.W.)

$V_B = 1870$ kN

Since $V_A = V_B$

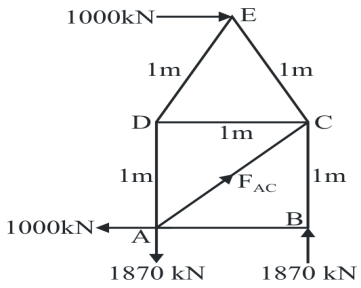
$V_A = -1870$ kN

At joint 'B'

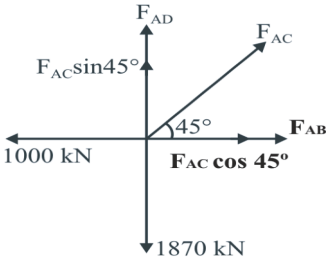


For roller support at B, $H_B = 0$

So, $F_{AB} = H_B = 0$



Considering Joint 'A'–



Resolving Forces Vertically–

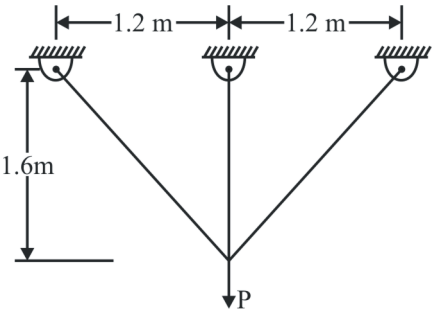
$F_{AC} \sin 45° + F_{AD} = 1870$ .....(i)

Resolving forces horizontally

$F_{AB} + F_{AC} \cos 45° = 1000$ .....(ii)

$$F_{AC} = \frac{1000}{\cos 45°} = 1414 \text{ kN}$$

405. A three bar truss to resist a service load P = 100 kN is shown in the figure. Use a load factor of 2 and consider all three members to have equal cross-sectional areas. The yield stress of the material of which the members are fabricated is 400 MPa. The area of cross section of each truss member is

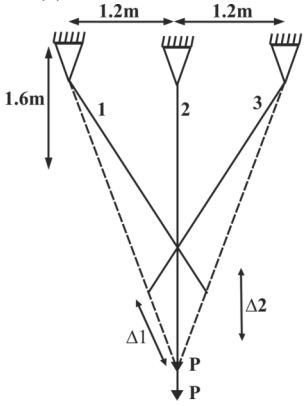


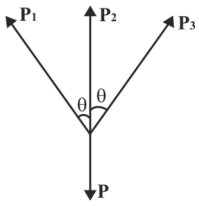
(a) $\frac{25}{26} cm^2$ (b) $\frac{26}{25} cm^2$

(c) $\frac{25}{13} cm^2$ (d) $\frac{13}{25} cm^2$

ISRO Scientist/Engineer 24.05.2014

Ans. (c) :





Equilibrium Equation

$2P_1 \cos\theta + P_2 = P$ ...(1)

Length of truss members 1 & 3

$= \sqrt{(1.2)^2 + (1.6)^2}$

$= 2$ m

$$\cos\theta = \frac{1.6}{2} = \frac{4}{5} = 0.8$$

Compatibility equation-

$\Delta_2 = \Delta_1 \cos\theta$

or $$\frac{P_2 L_2}{A_2 E_2} = \frac{P_1 L_1}{A_1 E_1} \cos\theta$$

or $$\frac{P_2}{P_1} = \frac{A_2 E_2}{A_1 E_1} \times \frac{L_1}{L_2} \cos\theta$$

$$\frac{P_2}{P_1} = \frac{1.6}{2} \times \frac{4}{5} = 0.64 \qquad [\because A_1 = A_2, E_1 = E_2]$$

From equation (1)

$2 \times P_1 \times 0.8 + P_2 = 100$

or $1.6P_1 + 0.64 \times P_1 = 100$

or $P_1 = 44.64$ kN

or $P_2 = 0.64 \times P_1 = 28.57$ kN

$$\text{Allowable stress} = \frac{\text{Permissible Stress}}{\text{FOS}}$$

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$= \frac{400}{2} = 200\,\text{MPa}$

Cross section area $A_1 = \frac{28.57 \times 10^3}{200}$

$= 142.85\ \text{mm}^2$

$= 1.42\ \text{cm}^2$

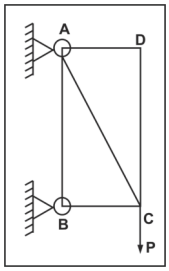
Cross-section area $A_2 = \frac{44.64 \times 10^3}{200}$

$= 223.2\ \text{mm}^2$

$= 2.23\ \text{cm}^2$

Nearest answer is $\frac{25}{13}\text{cm}^2$.

406. The stress in the rod AB in the pin jointed truss ABCD is



(a) $2P/\pi d^2$ (b) $4P/\pi d^2$

(c) 0 (d) $P/2\pi d^2$

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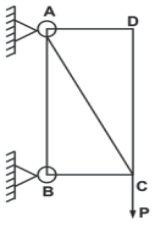
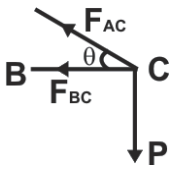
Ans. (b) :



"Force in section AD and DC = 0. According to the given conditions. Since no external force acts on joint at 'D'."

Joint 'C'



$\therefore$ $F_{AC} \sin\theta = P$

At a point A

$F_{AB} = -F_{AC} \sin\theta$

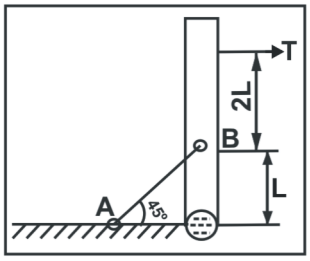
$F_{AB} = -P$

$\therefore$ $Stress = \frac{-P}{\frac{\pi d^2}{4}} = \frac{-4P}{\pi d^2}$

$= \frac{-4P}{\pi d^2}$

[$\because$ Negative sign shows compressible stress]

407. A plate is held by a chain AB through hinge joints. The plate is pulled with a force T as shown in figure. The tension in the chain is given by



(a) $(\sqrt{3}/2)T$ (b) $(\sqrt{2}/3)T$

(c) T/2 (d) $3\sqrt{2}\,T$

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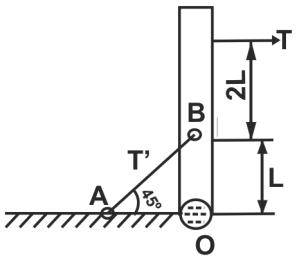
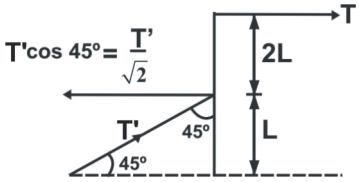
Ans. (d) :



Taking moment about O



$\frac{T'}{\sqrt{2}} \times L = T \times 3L$

$\therefore$ $T' = 3\sqrt{2}T$

408. Which one of the following is the wrong assumption:

(a) Members of the truss are pin-connected to each other

(b) Members of the truss are rigid

(c) Members of the truss are subjected to bending moments

(d) Members are of uniform cross-section

TNPSC AE 2013

Ans. (c) : Members of the truss are not subjected to bending moments.

409. Determine the nature of force in member AB and AC respectively of the truss shown in figure. A is hinge support and B is roller support. The direction of reaction at supports (R_A and R_B) is also shown. Load 50N is acting at joint C.

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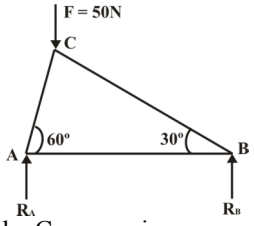
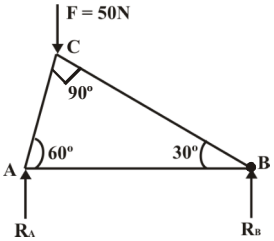


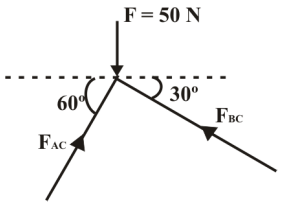
(a) Tensile, Compression
(b) Compression, Compression
(c) Tensile, Tensile
(d) Compression, Tensile

UPRVUNL AE 07.10.2016

Ans. (a) :



Joint-C



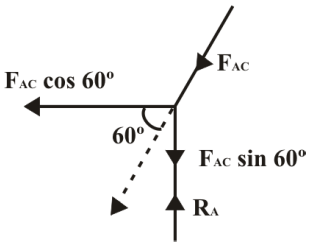
From Lami's theorem

$$\frac{F}{\sin 90} = \frac{F_{AC}}{\sin 120} = \frac{F_{BC}}{\sin 150}$$

$$F_{AC} = 50 \times \sin 120$$

$$\Rightarrow \boxed{F_{AC} = 43.3\ N}\ (Compressive)$$

Joint-A



Then,

$F_{AB} = F_{AC} \cos 60$

$= 43.3 \times \cos 60$

$F_{AB} = 21.65$ N (Tensile)

$R_A = 37.498$ N

410. In the joint method of plane truss analysis, value of forces in the member of truss can be found only when joint has:

(a) Only four unknown force members
(b) Only three unknown force members
(c) Not more than two unknown force members
(d) Any number of unknown force members

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Ans. (c) : In the joint method of plane truss analysis, value of forces in the member of truss can be found out when joint has not more than two unknown force members.

411. Which method is not there to analysing the trusses?

(a) Graphical Method (b) Analytical Method
(c) Method of Joints (d) Method of Sections

TNPSC AE 2013

Ans. (b) : Analytical method is not to analysing the trusses where as graphical method, joints method and method of sections are used to analysing the trusses.

412. For a perfect frame having 13 members, the number of joints must be :

(a) 6 (b) 8
(c) 10 (d) 13

OPSC Civil Services Pre. 2011

Ans. (b) : Given, $m = 13$

$m = 2J - 3$

$13 = 2J - 3 \Rightarrow \boxed{J = 8}$

413. In method of section, the section must pass through not more than members.

(a) 3 (b) 4
(c) 5 (d) 2

OPSC Civil Services Pre. 2011

Ans. (a) : In method of section, the section must pass through not more than 3 members.

414. Which of the following equilibrium equation should be satisfied by the joints in truss:-

(a) $\sum H = 0, \sum M = 0$
(b) $\sum H = 0, \sum V = 0$
(c) $\sum V = 0, \sum M = 0$
(d) $\sum H = 0, \sum V = 0$ and $\sum M = 0$

UKPSC AE-2013, Paper-I

Ans. (b) : No bending moment acting on the joints in truss so equilibrium equation should be satisfied by the joints in truss will be,

$\Sigma H = 0, \Sigma V = 0$

415. Choose the correct relationship between the given statements of Assertion (A) and Reason (R).

Assertion (A) : Only axial forces act in members of roof trusses.

Reason (R) : Truss members are welded together.

Code :

(a) Both (A) & (R) are correct. (R) is the correct explanation of (A).
(b) Both (A) & (R) are correct. (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

UKPSC AE 2012 Paper-I

Ans. (c) : (A) is true, but (R) is false.

416. In the analysis of truss, the force system acting at each pin

(a) is concurrent but not coplanar.
(b) is coplanar and concurrent.
(c) is coplanar and non-concurrent.
(d) does not satisfy rotational equilibrium.

UKPSC AE 2012 Paper-I

Ans. (b) : is coplanar and concurrent.

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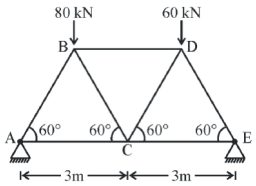
417. For truss as shown below, the forces in the member AB and AC are



(a) Tensile in each
(b) Compressive in each
(c) Compressive and Tensile respectively
(d) Tensile and Compressive respectively

UKPSC AE 2012 Paper-I

Ans. (c) : Compressive and Tensile respectively

418. The possible loading in various members framed structure are

(a) Buckling or shear
(b) Compression or tension
(c) Shear or tension
(d) Bending

UPPSC AE 12.04.2016 Paper-I

Ans : (b) The possible loading in various members framed structure are compression or tension.

419. A truss hinged at one end, supported on rollers at the other, is subjected to horizontal load only. Its reaction at the hinged end will be:

(a) Horizontal
(b) Vertical
(c) Resultant of horizontal and vertical
(d) Difference between horizontal and vertical

TRB Polytechnic Lecturer 2017

Ans. (c) : Its reaction at the hinged end will be resultant of horizontal and vertical.

$$R = \sqrt{(\Delta H)^2 + (\Delta V)^2}$$

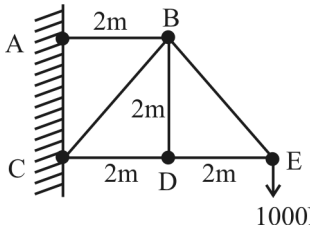
420. The force which is not considered in the analysis of the truss is

(a) External applied loads
(b) Horizontal forces on joints
(c) Vertical forces joints
(d) Support reactions
(e) Weight of the members

CGPSC AE 26.04.2014 Shift-I

Ans. (e) : Assumption for a perfect truss
(i) All the members of truss are straight and connected to each other at their ends by frictionless pins.
(ii) All external forces are acting only at pins.
(iii) All the members are assumed to be weightless.
(iv) All the members of truss and external forces acting at pins lies in same plane.
(v) Static equilibrium condition is applicable for analysis of perfect truss.

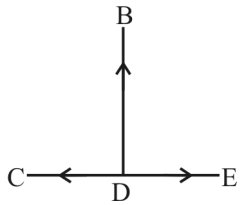
421. The force in the bar BD of the truss shown in figure is



(a) 1000 N
(b) – 1000 N
(c) 500 N
(d) 0 N

OPSC Civil Services Pre 2006

Ans. (d) :



Apply method of joint At D
$F_{DE} = F_{DC}$ Horizontal force (collinear force)
$F_{BD} = 0$ vertical force
Note- When three forces passing through a point and two of them collinear then in third member force is zero.

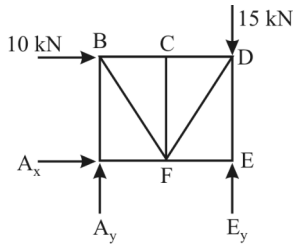
422. A truss (with j = number of - joints, m-number of members) is said to be deficient if.

(a) m – 2 j – 3
(b) m > 2j – 3
(c) m < 2j – 3
(d) j – 2m – 3

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : for truss
m = 2j – 3 – for stable
m < 2j – 3 – for unstable or deficient
m > 2j – 3 – for redundant

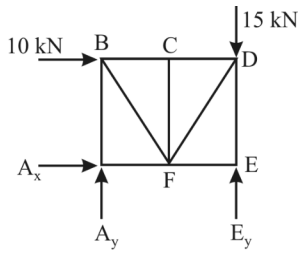
423. In the truss shown in figure, the zero force member (s) is/are



(a) CF only
(b) CF and EF
(c) AB only
(d) BF and EF

DRDO Scientists 2009

Ans. (b) :



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$A_y+E_y = 15$

$A_x+10 = 0$

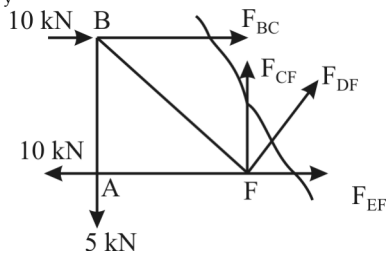
$A_x = -10kN$

$\Sigma M_A = 0$

$-10\times1-15\times2+E_y\times2 = 0$

$E_y = 20kN$

$A_y = -5kN$



$\Sigma F_x = 0,\ F_{BC}+F_{EF}+\frac{F_{DF}}{\sqrt{2}} = 0$

$\Sigma M_B = 0,$

$-10\times1+F_{DF}\times\sqrt{2}+F_{CF}\times1+F_{EF}\times1 = 0$

$\Sigma F_y = 0,\ F_{CF}+\frac{F_{DF}}{\sqrt{2}} = 5$

$\Sigma M_A = 0,$

$-10\times1-F_{BC}\times1+5\times1 = 0$

$F_{BC} = -5kN$

Now, we have

$$F_{CF}+\frac{F_{DF}}{\sqrt{2}} = 5$$

$$F_{CF}+\frac{F_{DF}}{\sqrt{2}} = 5$$

From these two equation,

$F_{CF} = F_{EF}$

CF is zero force member, so CF and EF is zero force member.

424. A plane truss has five joints and six members. The truss is

(a) Perfect (b) Redundant

(c) Deficient (d) None of these

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Given,

$m = 6,\quad j = 5$

$m = 2j - 3$

$6 < 10 - 3$

$6 < 7$

$m < 2j - 3 \rightarrow$ Deficient truss

425. A member under tension is called

(a) strut (b) tie

(c) strut-tie (d) column

APPSC AEE 2012

Ans. (b) : Tension members are also known as tie members or simply a tie. Tension members are designed as structural members that are subjected to tensile force in direction which is parallel to its longitudinal axis.

426. Hinge support is not capable of resisting

(a) horizontal force

(b) moment

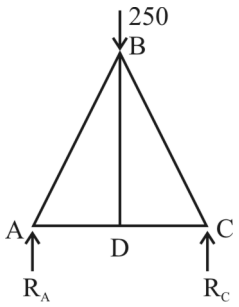
(c) vertical force

(d) both horizontal and vertical forces

DRDO Scientists 2008

Ans. (b) : Hinge support is not capable of resisting moment.

427. The axial force in the member BD of a steel truss shown in figure is



(a) 250 kN

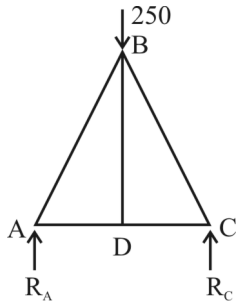
(b) –250 kN

(c) 0

(d) 125 kN

DRDO Scientists 2008

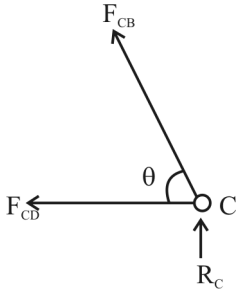
Ans. (c) :



$R_A + R_C = 250$

$R_A = R_C\ (\Sigma M_D = 0)$

$R_A = R_C = 125$ kN



$F_{CB}\sin\theta = -R_C$

$$F_{CB} = -\frac{125}{2}\times\sqrt{20}$$

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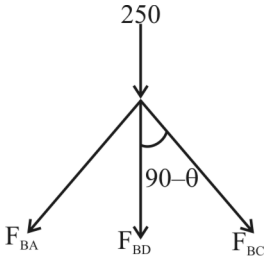
$$\sin\theta = \frac{2}{\sqrt{4^2+2^2}} = \frac{2}{\sqrt{20}}$$

$$F_{AB} = -\frac{125\sqrt{20}}{2}$$



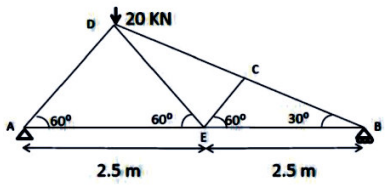
$2\,F_{BC}\sin\theta + F_{BD} = 250$

$$2\times\frac{125\sqrt{20}}{2}\times\frac{2}{\sqrt{20}} + F_{BC} = 250$$

$F_{BD} = 0$

BD is zero force member.

428. The axial forces in the members CE and DE of the plane truss shown in the figure below are equal to

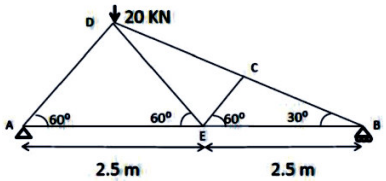


(a) 10, 10 in KN (b) 20, 20 in KN

(c) 8.7, 8.7 KN (d) 0, 0

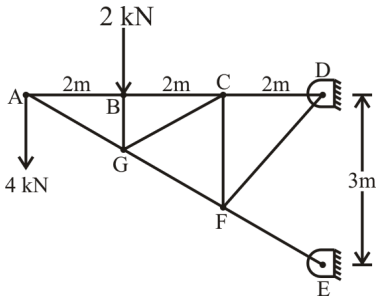
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : As we know, forced is transferred to A & B



So, there is no, Axial force on CE & DE.

429. Calculate the forces in the members CG and CF for the truss shown.



(a) $F_{CG} = \sqrt{5}\,kN, F_{CF} = 1\,kN$

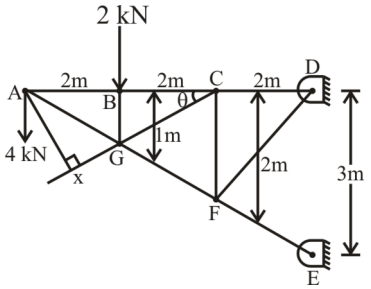
(b) $F_{CG} = \sqrt{3}\,kN, F_{CF} = 2\,kN$

(c) $F_{CG} = 2\,kN, F_{CF} = \sqrt{3}\,kN$

(d) $F_{CG} = \sqrt{5}\,kN, F_{CF} = 2\,kN$

ISRO Scientist/Engineer (RAC) 12.01.2020

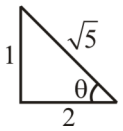
Ans. (a) :

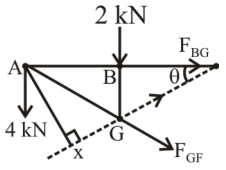


$\sum M_A = 0$

$2 \times 2 = F_{CG} \times AX$

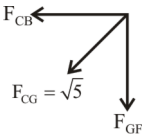
In ΔBGC $\tan\theta = \frac{1}{2}$





So $\sin\theta = \frac{1}{\sqrt{5}}$

In ΔAXC, $AX = AC\sin\theta = 4\times\frac{4}{\sqrt{5}}$



$$F_{CG} = \frac{2\times 2}{AX}$$

$F_{GF} = F_{CG}\sin\theta$

$= \sqrt{5}\times\frac{1}{\sqrt{5}} = 1\,kN = 1kN$

430. A truss having 5 joints and 7 members is termed as ..... truss.

(a) perfect (b) imperfect or deficient

(c) redundant (d) none of these

(HPPSC POLY. LECT. 2016)

Ans : (a) Perfect frame or truss:-

M = 2J - 3

M = No of member

J = No of Joints.

Given data

M = 7

J = 5

M = 2J - 3

7 = 2 × 5 - 3

LHS = RHS

A truss having 5 Joints and 7 member is termed as perfect frame.

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6. Kinematics and Kinetics of Particles & Rigid Body

431. A car moving with uniform acceleration covers 450 meter in first 5 second interval and covers 700 meters in next 5 second interval. The acceleration of the car is–

(a) 7 m/sec^2 (b) 50 m/sec^2
(c) 25 m/sec^2 (d) 10 m/sec^2

OPSC Civil Services Pre. 2011
TSPSC Managers, 2015
ISRO Scientist/Engineer 2007
HPPSC LECT. 2016
TSPSC AEE 2017
GATE 1998
ISRO Scientist/Engineer 2010

Ans. (d) : Let, the car travels with the initial velocity u and acceleration a.
Considering case-1
When displacement (s) = 450 m and time taken
t = 5 seconds
The displacement

$$s = ut + \frac{1}{2}at^2$$

$$450 = u \times 5 + \frac{1}{2}at^2 .....(i)$$

Considering case-2
When displacement (S) = 450+700 = 1150m
t = 10 seconds

$$1150 = u \times 10 + \frac{1}{2}a \times 10^2 ....(ii)$$

By solving equation (i) and (ii) we get,

$$\boxed{a = 10\,m/sec^2}$$

432. Center of percussion is a point at which

(a) total weight of the body acts
(b) center of mass acts
(c) resultant force of buoyancy acts
(d) if a blow is given, no reaction is felt at the point of suspension of the body

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : if a blow is given, no reaction is felt at the point of suspension of the body

433. The statement "The Impressed forces acting on anybody are in dynamic equilibrium with the inertia forces of the particles of the body" is related to

(a) Impulse momentum principle
(b) theorem of transmissibility of forces
(c) Lami's theorem
(d) D'Alembert's principle

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : The Impressed forces acting on any body are in dynamic equilibrium with the inertia forces of the particles of the body is related to D'Alembert's principle.

434. The acceleration of a body can be found from

(a) Area under velocity time graph
(b) slope of velocity time graph
(c) area under distance time graph
(d) slope of distance time graph

GPSC DEE, Class-2 (GWSSB) 04.07.2021

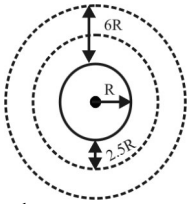
Ans. (b) : $a = \frac{dv}{dt}$
a= slope of velocity time graph

435. A geostationary satellite is orbiting the earth at a height of 6R above the surface of the earth; R being the radius of the earth. What will be the time period of another satellite at a height of 2.5R from the surface of the earth?

(a) $\frac{6}{\sqrt{2}}$ hr (b) $6\sqrt{3}$ hr
(c) $6\sqrt{3}$ hr (d) $6\sqrt{2}$ hr

WBPSC AE 2003

Ans. (d) :



From the Kepler's 3rd law.

$$T^2 \propto r^3$$

$$T_1^2 \propto r_1^3; \quad T_2^2 \propto r_2^3$$

$r_1 = 6R + R = 7R$ $\quad r_2 = 2.5R + R = 3.5R$

$$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{r_1}{r_2}\right)^3$$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2}$$

$$\frac{T_1}{T_2} = \left(\frac{7R}{3.5R}\right)^{3/2} = (2)^{3/2} = 2\sqrt{2}\text{ hours}$$

T_1 = 24 hours (Geostationary satellite time period)
= earth time period

$$T_2 = \frac{T_1}{2\sqrt{2}} = \frac{24 \times \sqrt{2}}{2\sqrt{2} \times \sqrt{2}} = \frac{24 \times \sqrt{2}}{4}$$

$$\boxed{T_2 = 6\sqrt{2}hr}$$

436. If a body is moving with a uniform acceleration (a), then the final velocity (v) of the body after time (t) is equal to

(a) $ut + \frac{1}{2}at^2$ (b) $u + at$
(c) $u^2 + 2as$ (d) none of these

Where u = initial velocity
s = distance travelled in t seconds

TNPSC AE, 2008

Ans. (b) : Final velocity (v) = u + at
Where, a = uniform acceleration
t = time

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437. The displacement of a particle undergoing rectilinear motion along the x-axis is given by $t^3 - 21t^2 + 60t$ meters. The acceleration of the particle, when its velocity is zero will be :

(a) $-18\ m/s^2$ (b) $36\ m/s^2$
(c) $9\ m/s^2$ (d) $-9\ m/s^2$

J&K PSC Civil Services Pre, 2013

Ans. (*) : $x = t^3 - 21t^2 + 60t$

velocity $\frac{dx}{dt} = 3t^2 - 42t + 60 = 0$

$t_1 = 12.38$ and $t_2 = 1.614$ for zero velocity

Acceleration $\left(\frac{d^2x}{dt^2}\right) = 6t - 42$

Acceleration at t = 12.38, $a = 32.28\ m/s^2$

t = 1.614, $a = -32.28\ m/s^2$

$a = \pm 32.28\ m/s^2$

438. When a particle moves with a uniform velocity along a circular path, then the particle has

(a) tangential acceleration only
(b) centripetal acceleration only
(c) both (a) and (b)
(d) None of the above

APPSC Poly. Lect. 2013
TNPSC AE 2013

Ans. (b) : When a particle moves, with a uniform velocity along a circular path, then there will be no tangential acceleration, but it will have only centripetal acceleration (a_c).

$$a_c = \frac{V^2}{r} = \omega^2 . r$$

439. Which of the following disciplines provide study of inertia forces arising from the combined effect of the mass and the motion of the parts?

(a) Applied mechanics (b) Kinematics
(c) Kinetics (d) Theory of machines

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) Kinetics- It is the discipline of Engineering Mechanics which provide study of inertia forces arising from the combined effect of the mass and the motion of the parts.

440. A particle moving with an acceleration of $6m/s^2$ covers 40 m during 5th second of its journey. What was the initial velocity of particle?

(a) 10 m/s (b) 11 m/s
(c) 12 m/s (d) 13 m/s

APPSC Poly Lect. 13.03.2020

Ans. (d) : Given data : Acceleration (a) = $6\ m/s^2$
distance (s) = 40 m
n = 5
We know that, distance covered in n^{th} second

$\because\ s = u + \frac{1}{2}a(2n-1)$ Let, u = initial velocity

$\Rightarrow 40 = u + \frac{1}{2} \times 6 \times (2 \times 5 - 1)$

$\Rightarrow 40 = u + 3 \times 9$

$u = 40 - 27$

$\boxed{u = 13\ m/sec}$

441. A motorist is driving at 25 m/s on the curved road of radius 500 m. He suddenly applies the brakes and that causes the speed to decrease to 15 m/s at the constant rate in 10 seconds. What will be the tangential component (a_t) and the normal component (a_n) of acceleration immediately after the application of brakes?

(a) $a_t = -1 m/s^2,\ a_n = 1 m/s^2$
(b) $a_t = -1 m/s^2,\ a_n = 1.25 m/s^2$
(c) $a_t = -1.25\ m/s^2,\ a_n = 1 m/s^2$
(d) $a_t = -1.25\ m/s^2,\ a_n = 2\ m/s^2$

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data :
Velocity (v) = 25 m/s
Radius of curved road (r) = 500 m

Normal component of acceleration $(a_n) = \frac{v^2}{r}$

$\Rightarrow a_n = \frac{(25)^2}{500} = \frac{625}{500} = \frac{25}{20}$

$\Rightarrow a_n = \frac{25}{20} = \frac{5}{4} = 1.25 m/s^2$

$\boxed{a_n = 1.25\ m/s^2}$

And, Tangential component of acceleration $(a^t) = \frac{\Delta v}{\Delta t}$

Given, $v_2 = 15$ m/s
$v_1 = 25$ m/s
t = 10 sec.

$a_t = \frac{v_2 - v_1}{\Delta t}$

$= \frac{15 - 25}{10} = \frac{-10}{10} = -1\ m/s^2$

$\Rightarrow \boxed{a_t = -1\ m/s^2}$

442. The time variation of the position of a particle in rectilinear motion is given by $x = 2t^3 + t^2 + 2t$. If 'V' is the velocity and 'a' acceleration of the particle in consistent units the motion started with

(a) V = 0, a = 0 (b) V = 0, a = 2
(c) V = 2, a = 0 (d) V = 2, a = 2

TSPSC AEE 2017
GATE 2005

Ans. (d) : $x = 2t^3 + t^2 + 2t$

$v = \frac{dx}{dt} = 6t^2 + 2t + 2$

$v_{t=0} = 2 m/s, \qquad a = \frac{dv}{dt} = 12t + 2$

$a_{t=0} = 2 m/s^2$

443. Area under the acceleration - time graph provides _____.

(a) the net change in displacement
(b) the net change in acceleration
(c) the net change in velocity
(d) the net work done

VIZAG Steel MT 24.01.2021, Shift-I

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Ans. (c) : The area under acceleration time graph gives the change in velocity.

Deceleration, $a = \frac{\Delta V}{\Delta t}$

Area between acceleration-time graph gives,

$$a \times \Delta t = \frac{\Delta V}{\Delta t} \times \Delta t = \Delta V$$

444. A body is moving with a velocity of 2 m/s. After 4 seconds, the velocity of the body becomes 5 m/s. The acceleration of the body is

(a) 0.55 m/s^2 (b) 0.65 m/s^2
(c) 0.75 m/s^2 (d) 0.45 m/s^2

ESE 18.07.2021

Ans. (c) :

$V_i = 2$ m/s
$V_f = 5$ m/s
$\Delta t = 4$ sec

$$\text{Acceleration} = \frac{\Delta V}{\Delta t} = \frac{5-2}{4} = \frac{3}{4} = 0.75\,\text{m/s}^2$$

445. A particle starts with an initial velocity of 200 cm/s and moves with a uniform retardation of 10 cm/s2. If it describes 1500 cm in time t, what is/are the possible value(s) of t?

(a) 10 sec only
(b) 10 sec and 30 sec
(c) 30 sec only
(d) 5 sec and 10 sec

ESE 18.07.2021

Ans. (b) : Given, $u = 200$ cm/s
$g = -10$ cm/s^2
$S = 1500$ cm

from second equation of motion-

$$S = ut + \frac{1}{2}gt^2$$

$\Rightarrow$ $t^2 - 40t + 300 = 0$
$t^2 - 30t - 10t + 300 = 0$
$t(t-30) - 10(t-30) = 0$
$(t-10)(t-30) = 0$
$t = 10$ sec. & $t = 30$ sec.

446. _____ is a scalar quantity

(a) velocity (b) force
(c) speed (d) Acceleration

UPRVUNL AE 05.07.2021

Ans. (c) : Speed is a scalar quantity.

447. A disc of radius 30 cm is rolling without slip with angular velocity of 10 rad/s on a horizontal surface. Which of the following statements is NOT true?

(a) Linear velocity of all the points is different
(b) Speed of all the points is different
(c) Acceleration of all the points is different
(d) Linear velocity of the point touching the horizontal surface is zero

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Speed of all the points is different.

448. A 60 kg man is weighed by a balance as 54 kg in a lift which is accelerated downwards. The acceleration of the lift is

(a) 1.26 m/s^2 (b) 1.98 m/s^2
(c) 0.98 m/s^2 (d) 1.76 m/s^2

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : $M_{eff}\, g = M(g - a)$

$$\frac{M_{eff}}{M} = \frac{g-a}{g}$$

$$\frac{54}{60} = 1 - \frac{a}{g}$$

$g = 10\,a$

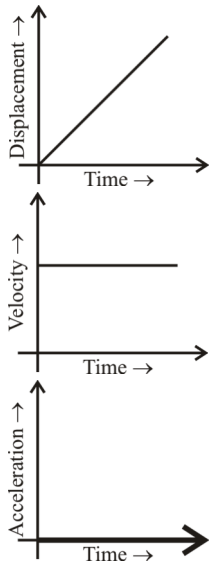
$$a = \frac{g}{10} = 0.98\,\text{m/s}^2$$

449. Which of the following statements is true, if a body moves with uniform velocity?

(a) Body covers equal distance in equal interval of velocity
(b) Body covers equal distance in equal interval of time
(c) Body covers equal distance in equal interval of speed
(d) Body covers equal distance in equal interval of acceleration

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : When the motion of a body is in such a way that it has equal displacement in equal intervals of time. The body is said to be in uniform velocity and the velocity of the particle is simply the ratio of the displacement in the given time interval to time interval.



450. If a slab A having linear velocity v rotates with an angular velocity ω about an instantaneous centre C, then what will be the distance (r) of the slab A from the centre C?

(a) $\omega = v_A + r$ (b) $v_A = r/\omega$
(c) $r = \omega/v_A$ (d) $r = v_A/\omega$

APPSC Poly Lect. Automobile Engg., 2020

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Ans. (d) : Linear velocity (v) with an angular velocity (ω) then we know that

$v = r\omega$

$v_A = r\omega$

$\boxed{r = v_A / \omega}$

451. An object of 30 kg is moved with a velocity of 2 m/s on horizontal smooth surface. What is the velocity of block for 4 sec, if force of 40 N is applied on it in the direction of force ?

(a) 2 m/s (b) 4.6 m/s
(c) 7.33 m/s (d) 9.33 m/s

APGCL AM, 2021

Ans. (c) :

Given:-

$m = 30$ kg
$V_1 = 2$m/sec
$t = 4$sec
$F = 40$N
$V_2 = ?$

Force (F) = ma

$$F = \frac{mdv}{dt}$$

$$40 = 30 \times \frac{(V_2 - V_1)}{t - 0}$$

$$40 = 30 \times \frac{(V_2 - 2)}{4}$$

$$V_2 - 2 = \frac{160}{30}$$

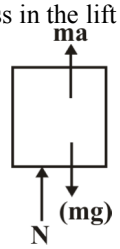
$V_2 = 5.33 + 2 = 7.33$ m/sec

452. A body exerts a force of 1 kN on the floor of the lift which moves upward with a retardation of 1.5 m/s^2. What is the mass of the body, in kg, carried in the lift?

(a) 120.33 (b) 101.94
(c) 88.42 (d) 77.32

APPSC AEE SCREENING 17.02.2019

Ans. (a) : FBD of mass in the lift



N = normal reaction between lift and mass

$\sum F_{net} = 0$

(N - mg) = m(-a) (-a = retardation)

$1\times10^3 = m(g - a)$

$1\times10^3 = m(9.81 - 1.5)$

$$m = \frac{1\times10^3}{(9.81 - 1.5)} = 120.33\ kg$$

453. A particle move along a straight line such that distance (x) traversal in t seconds is given by x = t^2 (t - 4), the acceleration of the particle will be given by the equation

(a) $3t^3 - 2t$ (b) $3t^2 + 2t$
(c) 6t - 8 (d) 6t - 4

TNPSC AE 2018

Ans. (c) $x = t^2 (t - 4)$

$$\frac{dx}{dt} = \vec{V} = 3t^2 - 8t$$

$$\frac{d^2x}{dt^2} = \frac{dV}{dt} = \vec{a} = 6t - 8$$

454. A motor car moving at a certain speed takes a left turn in a curved path. If the engine rotates in the same direction as that of wheels, then due to the centrifugal forces

(a) the reaction on the inner wheels increases and on the outer wheels decreases
(b) the reaction on the outer wheels increases and on the inner wheels decreases
(c) the reaction on the front wheels increases and on the rear wheels decreases
(d) the reaction on the rear wheels increases and on the front wheels decreases

RPSC Vice Principal ITI 2018

Ans. (b) : The reaction on the outer wheels increases and on the inner wheels decreases.

455. The motion of the particle is defined as x = t^3 + t –5, where 'x' is the displacement in meter and 't' is time in sec. Find the velocity and acceleration at 5 sec.

(a) 47 m/s, 30 m/s^2 (b) 25 m/s, 50 m/s^2
(c) 70 m/s, 60 m/s^2 (d) 76 m/s, 30 m/s^2

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Ans. (d) : Given,

Motion of particle (x) = $t^3 + t - 5$

Where x → displacement

T → time

$$\therefore \frac{dx}{dt} = 3t^2 + 1$$

Velocity (V) $\frac{dx}{dt} = 3t^2 + 1 \Rightarrow$ V/t =5sec.=$3\times5^2+1$=76 m/s

Or $\frac{d^2x}{dt^2} = 6t$

Acceleration $(a) = \frac{d^2x}{dt^2} = 6t$

Acceleration at t = 5 sec

$= 6 \times 5 = 30 m/s^2$

456. The free fall of a body is referred as:

(a) Uniform motion
(b) Uniformly accelerated motion
(c) Curvilinear motion
(d) Non- uniformly accelerated motion

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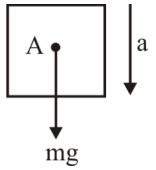
Ans. (b) :



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Free fall of a body under gravity is referred as uniformly accelerated motion.
Assume body A of weight w = mg
Free failing under gravity
$\therefore \Sigma F = mg = ma$
$a = g = 9.81\ m/s^2$
So the uniforms acceleration is $9.81\ m/s^2$

457. A particle of mass m is moving in a horizontal circle of radius r under the action of centripetal force expressed as $-C/r^2$, where C is constant. Total energy of the particle is:
(a) $-C/2r$ (b) $C/2r$
(c) $2C/r$ (d) $C/2$
AAI Jr. Executive 29.11.2018

Ans. (a) : Centripetal force $F_c = -\frac{C}{r^2} = \frac{mv^2}{r}$

$KE = \frac{1}{2}mv^2 = \frac{F_c r}{2}$

Kinetic energy K is related to centripetal force as

$K = \frac{|F_c|}{2} r = \frac{C}{2r} \quad [\text{as } K > 0]$

$K = \frac{C}{2r}$ .....(i)

If potential energy of the system is U then force can be written as

$F_c = \frac{dU}{dr}$

$-\frac{dU}{dr} = -\frac{C}{r^2}$

$\int dU = \int \frac{C}{r^2} dr$

$U = -\frac{C}{r}$ ..... (ii)

Total energy = PE + KE

$= -\frac{C}{r} + \frac{C}{2r} = \frac{-C}{2r}$

458. The relation between time and distance is $t = \alpha x^2 + \beta x$, where α and β are constants. The retardation is given by:
(a) $2\beta v^3$ (b) $2\alpha v^2$ (c) $\alpha\beta v^3$ (d) $2\alpha\beta v^3$
AAI Jr. Executive 29.11.2018

Ans. (d) : $t = \alpha x^2 + \beta x \quad \{\because d = \alpha\}$

Differentiate w.r.t. x

$\frac{dt}{dx} = 2\alpha x + \beta$

$\frac{dx}{dt} = v$

$= \frac{1}{(2\alpha x + \beta)}$ ..... (i)

Again differentiate equation (i) with respect to x

$\frac{d^2x}{dt^2} = a = \frac{-2\alpha\beta v}{(2\alpha x + \beta)^2}$

$\frac{d^2x}{dt^2} = a = \text{Acceleration}$

$a = -2\alpha\beta v^3$

$a = 2\alpha\beta v^3$ (Retardation)

459. Two points of a metal rod are moving with velocities 3u and u perpendicular to the rod and in same direction separated by a distance 'r'. Angular velocity of the rod will be:
(a) 2u/5r (b) 2u/r
(c) u/r (d) 2u/3r
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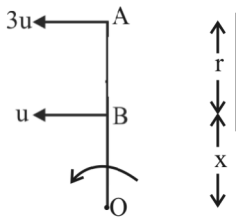
Ans. (b) :



Velocity at a point A, $3u = (r + x)\omega$ .....(i)
Velocity at a Point B, $u = x\omega$ .....(ii)
From equation (i) & (ii)

$3u - u = (r + x)\omega - x\omega$

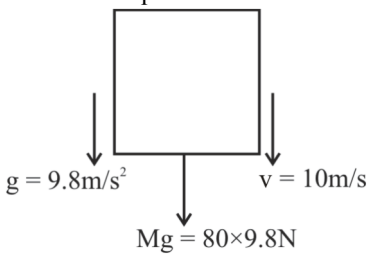
$2u = r\omega + x\omega - x\omega$

$2u = r\omega$

$\boxed{\omega = \frac{2u}{r}}$

460. A lift originally moving downwards at 10 m/s is brought to rest with a constant acceleration in a distance of 25 m. The force with which the feet of a passenger of mass 80 kg ($g = 9.8\ m/s^2$) press downwards on the floor of the lift is
(a) 160 N (b) 784 N
(c) 944 N (d) 1000 N
TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : From third equation of motion



$v^2 = u^2 + 2as$

$0 = 10^2 + 2a \times 25$

$a = \frac{-10 \times 10}{2 \times 25} = -2\ m/sec^2$

$F_{net} = M(g - a)$

$= 80 \times [9.8 - (-2)]$

$= 80 \times 11.8$

$= 944N$

461. The position of a particle which is confined to move along a straight line is $S = 4t^3 - 32t + 100$, where S is measured in meters and t in seconds. The time required for the particle to reach a velocity of 76 m/s from its initial condition at t = 0 is:
(a) 9 sec (b) 12 sec
(c) 4 sec (d) 3 sec
CIL MT 27.02.2020

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Ans. (d) : The position of the particle

$S = 4t^3 - 32t + 100$ (m)

$V = 76$ m/sec

Velocity equation $\frac{ds}{dt} = \frac{d}{dt}\left(4t^3 - 32t + 100\right)$

$V = 12t^2 - 32$

$76 = 12t^2 - 32$

$\boxed{t = 3\text{sec}}$

462. A body moves from rest with a constant acceleration of $5m/s^2$. The distance covered in 5s is most nearly to :

(a) 38 m (b) 62.5 m

(c) 96 m (d) 124 m

ESE 2018

J & K PSC Screening, 2006

Ans. (b) : Given,

Initial velocity = $u = 0$

Acceleration (a) = $5m/s^2$

Distance (s) = ?

Time $t = 5$ sec.

$s = ut + \frac{1}{2}at^2$

$= 0 \times 5 + \frac{1}{2} \times 5 \times 5^2$

$\boxed{S = 62.5}$ m

463. If a body is having motion of translation as well as motion of rotation, then the total kinetic energy is given by :

(a) $\frac{1}{2}(m + I)v^2$ (b) $\frac{1}{2}(m + I)\omega^2$

(c) $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ (d) $\frac{1}{2}m(v^2 + \omega^2)$

J & K PSC Screening, 2006

Ans. (c) : $(KE)_{Total} = (KE)_{Translation} + (KE)_{rotational}$

$(KE)_{Total} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

464. A bullet leaves the muzzle of a gun with a velocity of 400 m/sec. Assuming constant acceleration from starting point to muzzle, what is the time taken by the bullet to travel a distance of 1m through the gun barrel.

(a) 0.2 second (b) 0.5 second

(c) 0.005 second (d) None of these

J & K PSC Screening, 2006

Ans. (c) : Initial velocity,

$u = 0$

$v^2 = u^2 + 2as$

$(400)^2 = 0 + 2a \times 1$

$a = \frac{(400)^2}{2}$

$a = 80 \times 10^3$ m/s^2

$v = u + at$

$400 = 0 + (80 \times 10^3)t$

$t = \frac{400}{80 \times 10^3}$

$t = 0.005$ sec

465. A man falling from a height 'h' starts rotating midway of his fall. The vertical velocity with which the man will touch the ground will be:

(a) $\sqrt{2gh}$

(b) less than $\sqrt{2gh}$

(c) greater than $\sqrt{2gh}$

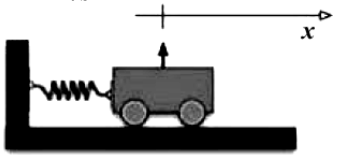
(d) less or more but never equal to $\sqrt{2gh}$

J & K PSC Screening, 2006

Ans. (a) : $V = \sqrt{2gh}$

466. A cart attached to a spring moves with an acceleration proportional to its position but of opposite sign.

$a(x) = -2x$ m/s^2



If the cart has a velocity $v = 5$ m/s when $x = 0$, determine the velocity of the cart when $x = 3$.

(a) 2.45 m/s (b) 2.5 m/s

(c) 2.65 m/s (d) 2.78 m/s

(e) 2.85 m/s

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : Given,

At $x = 0$, $v = 5$ m/s

At $x = 3$, $v = ?$

$a(x) = -2x$

$\frac{vdv}{dx} = -2x$

$vdv = -2x\,dx$

Integrating both sides,

$\int_5^v vdv = -\int_0^3 2x\,dx$

$\left[\frac{v^2}{2}\right]_5^v = -2\left[\frac{x^2}{2}\right]_0^3$

$\frac{1}{2}(v^2 - 25) = -1(9)$

$v^2 = 25 - 18$

$v = 2.65$ m/s

467. The rotation of a flywheel is given mathematically as $a = 10t - t^2$, where a, the angular acceleration, is given in radians per square second and t is in seconds. The flywheel starts from rest, makes a certain number of revolutions, then stops for a while and then reverses its direction. How many revolutions does the flywheel make during this time, prior to reversing direction?

(a) 200 (b) 225

(c) 250 (d) 275

(e) 300

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : Ans. (b) : Given,

$a = 10t - t^2$

$a = \frac{d\omega}{dt}$

$\omega = \int a\,dt$

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$= \int (10t - t^2)dt$

$\omega = \frac{10t^2}{2} = \frac{t^3}{3}$

When flywheel reverse direction, $\omega = 0$

$0 = \frac{10t^2}{2} - \frac{t^3}{3}$

$\boxed{t = 15 \text{ second}}$

At t = 15 sec, speed will reverse its direction,

$a = \frac{d^2\theta}{dt^2}$

$\frac{d\theta}{dt} = \int a\,dt = \int (10t^2 - t^2)dt$

$\frac{d\theta}{dt} = 5t^2 - \frac{t^3}{3}$

$\theta = \left[\frac{5}{3}t^3 - \frac{t^4}{12}\right]_0^{15}$

$\theta = \frac{5}{3}(15)^3 - \frac{1}{12}(15^4 - 0)$

$\theta = 1406.25$ radian,

No. of revolution = $\frac{\theta}{2\pi} = \frac{1406.25}{2\pi}$

No. of revolution = 223.8 rpm ≃ 225 rpm

468. The angular velocity of hour hand of a watch is

(a) 0.0001455 sec^{-1} (b) 0.001755 sec^{-1}

(c) 0.1047 sec^{-1} (d) 1.047 sec^{-1}

WBPSC AE, 2017

Ans. (a) : 1 hour = 2π

$\frac{2\pi}{12 \times 60 \times 60} = 0.0001453$

469. The rotation of a rigid body is defined by $\phi = 2 \sin \pi t/4$. The angular velocity and acceleration after 2 seconds will be

(a) 0,0 (b) $0, \pi^2/8$

(c) $1, \pi^2/8$ (d) $0, -\pi^2/8$

WBPSC AE, 2017

Ans. (d) : Rotation of rigid body $\phi = 2 \sin \frac{\pi t}{4}$

angular velocity $\omega = \frac{d\phi}{dt}$

$= \frac{d}{dt}\left(2 \sin \frac{\pi t}{4}\right)$

$= 2 \times \frac{\pi}{4} \cos \frac{\pi t}{4}$

$t = 2$

$\frac{\pi}{2} \times \cos \frac{\pi \times 2}{4}$

$\frac{\pi}{2} \cos \frac{\pi}{2} = 0$

angular acceleration = $\frac{\pi}{2} \times \frac{\pi}{4}\left(-\sin \frac{\pi t}{4}\right)$

$-\frac{\pi}{2}\frac{\pi}{4} \sin \frac{\pi t}{4}$

$\frac{\pi^2}{8} \sin \frac{\pi \times 2}{4}$

$\frac{-\pi^2}{8} \sin \frac{\pi}{2}$

$-\frac{\pi^2}{8} \times 1$

$= -\frac{\pi^2}{8}$

470. A particle starts from rest and moves in a straight line whose equation of motion is given by $S = 2t^3 - t^2 - 1$. The acceleration of the particle after one second will be

(a) 4 m/sec^2 (b) 6 m/sec^2

(c) 8 m/sec^2 (d) 10 m/sec^2

WBPSC AE, 2017

Ans. (d) : $S = 2t^3 - t^2 - 1$

$v = \frac{ds}{dt} = 6t^2 - 2t$

$a = \frac{dv}{dt} = 12t - 2$

t = 1 sec

$a = 12 \times 1 - 2$

$a = 10$ m/sec^2

471. When a train is rounding a curve, to prevent side thrust on the wheel flange of cars

(a) rail section is made thick on curvature.

(b) thrust eliminators are provided in the train.

(c) outside rails are raised.

(d) guide rail section are provided to take care of thrust.

WBPSC AE, 2017

Ans. (c) : When a train is rounding a curve, to prevent side thrust on the wheel flange of cars outside rails are raised.

472. The time required to stop a car moving with a velocity 20m/sec within a distance of 40m is equal to

(a) 2 sec (b) 4 sec

(c) 8 sec (d) 10 sec

WBPSC AE, 2017

Ans. (b) : u = 20 m/s

v = 0 m/s

$v^2 - u^2 = 2as$

$0^2 - 20^2 = 2 \times a \times 40$

$-400 = 80a$

$a = -5$ m/s^2

$v = u + at$

$v = 0$ $u = 20$ m/s $a = -5$ m/s^2

$0 = 20 - 5 \times t$

$5t = 20$

$t = 4$ sec

473. Time acceleration graph of a body moving with uniform acceleration is a straight line :

(a) Passing through the origin

(b) Inclined to time axis

(c) Parallel to time axis

(d) Parallel to acceleration axis

J&K PSC Civil Services Pre, 2010

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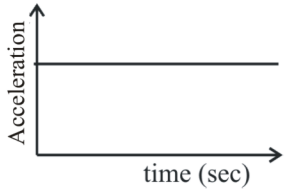
Ans. (c) : For uniformly accelerated motion, acceleration is constant with time, hence, acceleration time graph will be a straight line parallel to the time axis. For a straight line parallel to the time axis, slope equals zero.



474. An iron ball and a cotton ball of same mass are thrown from a height h in air. Then
(a) the iron ball will reach the ground first
(b) the cotton ball will reach the ground first
(c) both balls reach the ground at the same time
(d) None of the above

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Mass of both ball is m
So, weight in both cases will be mg. Hence acceleration would be 'g' by Newton's second law of motion.
Now

$$h = ut + \frac{1}{2}gt^2$$

Both balls are dropped from same height (h) & initial velocity of both balls (u = 0)

$$h = \frac{1}{2}gt^2$$

$$t = \sqrt{\frac{2h}{g}}$$

∵ height & acceleration due to gravity of balls are equals that's why both balls reach at the same time on the ground.

475. Acceleration due to gravity is
(a) same at all places
(b) more at the equator of the earth than at the poles
(c) decreasing with altitude
(d) All are correct

Haryana PSC Civil Services Pre, 2014

Ans. (c) : Acceleration due to gravity is location dependent. It is not same at all places. As we move upward or downward from earth surface then it's value decreases. Value of 'g' will be maximum at poles and minimum at equator.

476. A man moving in an accelerated train on horizontal tracks feels that his weight has
(a) decrease (b) increase
(c) zero (d) same

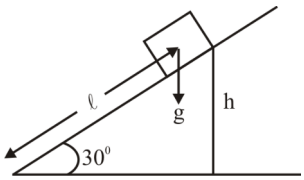
Haryana PSC Civil Services Pre, 2014

Ans. (d) : Man moving in an accelerated train on horizontal tracks feels that his weight will be same.
Man weight (apparent) changes when motion is in vertical direction means towards earth or away from earth or motion against or in direction of gravity.

477. A block slides down a smooth inclined plane at 30° in time from the top. If an identical block is released from the same point, falling freely to the ground, then it will reach in time
(a) t/2 (b) t/3
(c) t/4 (d) t

Haryana PSC Civil Services Pre, 2014

Ans. (a) :



$l \sin 30^0 = h$

$$l = \frac{h}{\sin 30^0}$$

$$l = \frac{h}{1/2}$$

$$l = 2h$$

Equation of motion along inclined plane

$$l = ut + \frac{1}{2}g\sin 30^0 t^2 \quad (u = 0)$$

$$\ell = \frac{1}{2}g \times \frac{1}{2}t^2$$

$$2h = \frac{1}{4}gt^2$$

$$t = \sqrt{\frac{8h}{g}} \quad \text{_____(i)}$$

time taken by block to reach at ground along inclined plane = t
Let, time taken by freely falling block = t_1

$$h = ut + \frac{1}{2}gt_1^2 \qquad (u = 0)$$

$$t_1 = \sqrt{\frac{2h}{g}} \quad \text{_____(ii)}$$

equation (i) divide by (ii)

$$\frac{t}{t_1} = \sqrt{\frac{8h}{g}} \times \sqrt{\frac{g}{2h}} = \sqrt{4} = 2$$

$$\boxed{t_1 = t/2}$$

478. A railway train of 200 m long passes over a bridge of 600 m long. If the train is moving at 10 m/s, the time taken to cross the bridge
(a) 60s (b) 90s
(c) 80s (d) 100s

Haryana PSC Civil Services Pre, 2014

Ans. (c)

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{200 + 600}{10} = \frac{800}{10}$$

$$\text{Time} = 80\,\text{sec}$$

479. If a body is moving in a straight line, the motion of the body is called
(a) rectilinear (b) rotational
(c) curvilinear (d) helical

Haryana PSC Civil Services Pre, 2014

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Ans. (a) :
Rectilinear Motion– Motion along straight line path.
Curvilinear Motion– Motion along curved path.
Rotational Motion– Every particle in body moves in a circle about axis of rotation.
Helical Motion– It is the superposition of straight line and circular motion.

480. The apparent weight of a man in moving lift is less than his real weight when it is going down with
(a) uniform speed
(b) an acceleration
(c) some linear acceleration
(d) retardation

Haryana PSC Civil Services Pre, 2014

Ans. (b) :

When lift is moving downward then apparent weight

$\boxed{W = m(g - a)}$

When lift moves upward then

$\boxed{W = m(g + a)}$

- Apparent weight of man in moving lift is less than his real weight (mg) when it is going down with an acceleration.

481. A body of mass 10 kg moving with a velocity of 1 m/s is acted upon by a force of 50 N for two seconds. The final velocity of the body will be –
(a) 22 m/s (b) 10 m/s
(c) $\sqrt{21}$ m/s (d) 11 m/s

TRB Asstt. Prof., 2012

Ans. (d) : Given, Force (F) = 50N
Mass (m) = 10 kg
Initial velocity (u) = 1 m/sec
Final velocity (v) = ?
Time (t) = 2 sec.
We know F = ma

$50 = 10 \times a$

$a = \frac{50}{10} = 5 m/sec^2$

From the first equation of motion.

$v = u + at$

$v = 1 + 5 \times 2$

$v = 11$ m/sec.

482. Mathematical formulation of Newton's law in Engineering Mechanics :
(a) $F = \frac{d}{dt}(mV)$ (b) $F = \frac{d}{dt}(P)$
(c) $F = \dot{P}$ (d) All of the above

KPSC ADF 2015

Ans. (d)

- Force (F) = $\frac{d}{dt}(mv)$

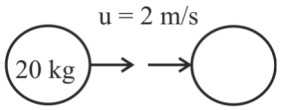
[linear momentum (P) = mv]
Now, Force = Rate of change of momentum

- $F = \frac{d}{dt}(P)$ (Rate of change of momentum)
- $F = \dot{P}$ (Rate of change of momentum)

483. A body of 20 kg mass moving with a velocity of 2 m/s is acted upon by a force of 100 N for four seconds. The final velocity will be :
(a) 22 m/sec (b) 10 m/sec
(c) $\sqrt{21}$m/sec (d) 11 m/sec

KPSC ADF 2015

Ans. (a) : m = 20 kg (moving body mass)
u = 2 m/s
$\because$ F = 100 N
t = 4 second



We know that, F = ma

$a = \frac{F}{m} = \frac{100}{20} = 5 m/s^2$

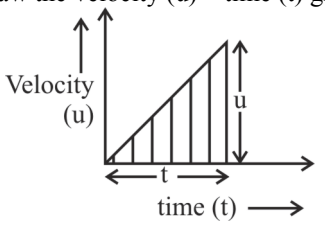
$\because$ V = u + at
= 2 + 5 × 4 = 2 + 20

$\boxed{V = 22 m/s}$

484. The velocity-time graph of a body is a straight line passing through the origin. If the slope of the graph is 'm', the distance travelled by the body in time 't' would be :
(a) $2mu^3$ (b) $\frac{mu^2}{2t^2}$
(c) $\frac{u^2}{2m}$ (d) $2mu^2$

KPSC ADF 2015

Ans. (c) : Draw the velocity (u) – time (t) graph–



Distance(s) = Area under the velocity time curve

$= \frac{1}{2} \times base \times height = \frac{1}{2} \times t \times u$

$s = \frac{ut}{2}$

Given, slope (m) = $\frac{u}{t} \Rightarrow u = mt$

or $t = \frac{u}{m}$

$s = \frac{u^2}{2m}$

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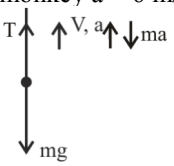
485. A monkey of mass 40 kg climbs on a rope which can withstand maximum tension of 600 N. In which of the following cases when the rope break? (Take g = 10 m/s^2 & ignore the mass of the rope)

(a) Climbs up with an acceleration of 6 m/s^2
(b) Climbs down with an acceleration of 4 m/s^2
(c) Climbs up with uniform speed of 5 m/s^2
(d) None of the mentioned

GPSC EE Pre, 28.01.2017

Ans. (a) : Given-
Monkey's mass (m) = 40 kg
g = 10 m/s
Maximum tension in rope, T_{max} = 600 N
Acceleration due to monkey a = 6 m/s^2



According to second law of motion-
T = mg + ma
T = m[g + a]
T = 40[10 + 6]
T = 640 N
Since, T > T_{max} or (640 > 600) the rope will break in this case.

486. A cylinder of mass m rolls down an inclined plane from a height h above a horizontal level. If the velocity of the axis of the cylinder is v when it reaches the lower level, then :

(a) $v = gh$ (b) $v = mgh$
(c) $v = \sqrt{(2gh)}$ (d) $v < \sqrt{(2gh)}$

BPSC Asstt. Prof. 29.11.2015

Ans. (d)

487. When a Bofors 40 mm gun is fired, the recoil can be explained by which of the following laws?

(a) Newton's first law
(b) Newton's second law
(c) Newton's third law
(d) Newton's fourth law

KPCL AE 2016

Ans. (c) : Bofors 40 mm gun after referred to simply as the before gun, is an anti-aircraft automation. When a bofors 40 mm gun is fired the, recoil by Newton's third law of motion.
∴ Action = Reaction

488. The motion of an artificial satellite orbiting the Earth at constant height is an example of

(a) Linear motion (b) Space motion
(c) Circular motion (d) Superficial motion

KPCL AE 2016

Ans. (c) : A satellite of mass m of orbiting at radius r from the center of earth. The gravitational force supplies the centripetal acceleration.

$$V_{orbit} = \sqrt{\frac{GM_{earth}}{r}}$$

The earth's gravity is the only force acting from Newton's law of gravitation.

$$F = \frac{G\,mM_e}{r^2}$$

489. What is the weight of woman standing inside a lift which is moving down with an acceleration of 9.8 m/s^2?

(a) Half of her original weight
(b) Zero
(c) One-tenth of her original weight
(d) None of these

KPCL AE 2016

Ans. (b) : Given- Mass of the woman = (m)
as it is moving down acceleration (a) = $9.8\frac{m}{s^2}$
apparent weight = m(g – a)
= m (9.8 – 9.8)
= 0 $\left[\because g = 9.8 m/sec^2\right]$

490. The force which acts along the radius of a circle and directed towards its centre is known as _____

(a) Centrifugal force (b) Radial force
(c) Centripetal force (d) None of these

KPCL AE 2016

Ans. (c) : Force which acts on a body performing circular motion along its radius and directed towards the center is known as centripetal force

$$F_c = \frac{mv^2}{r}$$

r = radius
m = mass of body
v = speed of body

491. In case of relative equilibrium of a particle with respect to a moving body, the algebraic sum of the projections, on any axis, of all real forces acting on the particle together with the ..... due to base acceleration must be zero.

(a) Frictional force (b) Inertia force
(c) Tangential force (d) Shear force

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : D'Alembert's principle of Inertial forces-
In case of relative equilibrium of a particle with respect to a moving body, the algebraic sum of the projections, on any axis, of all real forces acting on the particle together with the 'Inertia' due to base acceleration must be zero.

$$F_I - \sum_{i=1}^{n} F_i = 0$$

F_I = Inertial force.
F_i = A real force.

492. The absolute acceleration while dealing with relative motion may be considered as the geometric of ..... .

(a) base acceleration and relative acceleration
(b) relative acceleration and supplementary acceleration
(c) supplementary acceleration and based acceleration
(d) base acceleration, supplementary acceleration and relative acceleration

HPPSC Asstt. Prof. 29.10.2016

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Ans. (d) : The absolute acceleration while dealing with relative motion may be considered as the geometry sum of base acceleration, supplementary acceleration and relative acceleration.

493. A motorist is travelling at 90 kmph, when he observes a traffic light 250 m ahead of him turns red. The traffic light is timed to stay for 12 secs. If the motorist wishes to pass the light without stopping, just as it turns green, then the speed of the motor as it passes the traffic light will be .....

(a) 30 kmph (b) 45 kmph
(c) 60 kmph (d) zero

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Given that u = 90 kmph

$$= 90 \times \frac{5}{18} = 25 \text{m/sec}$$

t = 12 sec s = 250

Applying Newton's law of motion

$s = ut + 1/2\ at^2$

$$250 = 25 \times 12 + \frac{1}{2}(a) \times (12)^2$$

$a = -0.694\ \text{m/sec}^2$

Now Appling Newton's law of motion equation

$v = u + at$

$v = 25 + (-0.694)\ (12)$

$v = 16.67$ m/sec

$$v = 16.67 \times \frac{18}{5} \text{kmph}$$

= 60 kmph

494. A car travels on a horizontal circular track of radius 9 m, starting from rest at a constant tangential acceleration of 3 m/s^2. What is the resultant acceleration of the car, 2 sec after starting ?

(a) 3 m /sec^2 (b) 4 m/sec^2
(c) 5 m/sec^2 (d) 7 m/sec^2

UPSC JWM Advt. No.-50/2010

Ans. (c) : Given, $a_t = 3\ \text{m/s}^2$, R = 9 m, t = 2 sec, $\vec{a} = ?$

According to Newton's first law, v = u + at

$v = 0 + 3 \times 2 = 6$ m/s

$$\vec{a} = \sqrt{a_r^2 + a_t^2}$$

Radial acceleration, $a_r = \frac{v^2}{R} = \frac{6^2}{9} = 4 \text{m/s}^2$

Resultant acceleration, $\vec{a} = \sqrt{4^2 + 3^2} = 5 \text{m/s}^2$

495. Water drops from a tap at the rate of four droplets per second. Determine the vertical separation between two consecutive drops after the lower drop attained a velocity of 4 m/s. (Take g = 10 m/s^2)

(a) 0.49 m (b) 0.31 m
(c) 0.50 m (d) 0.30 m

UPSC JWM Advt. No.-50/2010

Ans. (*) : Time taken by lower drop to 4 m/sec

$$\frac{4}{10} = 0.4 \text{sec}$$

So distance of lower drop from origin,

$$= \frac{1}{2} gt^2$$

$$\frac{10}{2} \times (0.4)^2 = 0.8 \text{m} \quad \text{.....(i)}$$

Time after lower drop 2nd drop released,

$$= \frac{1}{4} \text{sec} = 0.25 \text{sec}$$

So, 2nd drop falling time = 0.4 – 0.25 = 0.15 sec

So, distance travelled by second drop,

$$\frac{1}{2} gt^2 = 5(0.15)^2$$

= 0.1125 m .....(ii)

So, separation = 0.8 – 0.1125 = 0.6575 m

496. A car traveling at a speed of 20 m/s is subjected to acceleration $a = 10 - \frac{t}{2}$. Determine the maximum speed of the car.

(a) 200 m/s (b) 120 m/s
(c) 100 m/s (d) 80 m/s

UPSC JWM Advt. No.-50/2010

Ans. (b) : Given, U = 20 m/s, $a = 10 - \frac{t}{2}$

The maximum velocity, when a = 0

t = 20 sec.

Acceleration, $a = \frac{dv}{dt}$

$$\Rightarrow \quad \int a dt = \int dv$$

$$\int \left(10 - \frac{t}{2}\right) dt = \int dv$$

$$10t - \frac{t^2}{4} = v - 20$$

$200 - 100 = v - 20$

$v = 120$ m/s

497. A cylinder of mass "M" and radius "R" rolls down an inclined plane of inclination "θ". What is the linear acceleration of axis of cylinder?

(Where 'g' is acceleration due to gravity)

(a) g sin θ (b) 1/3 g sin θ
(c) 2/3 g sin θ (d) 3 g sin θ

UPRVUNL AE 2014

Ans : (c) If rolling object is a solid cylinder then

$$\text{Acceleration (a)} = \frac{g \sin\theta}{1 + \left(\frac{K^2}{R^2}\right)}$$

$$I = MK^2 = \frac{1}{2} MR^2$$

$$\frac{K^2}{R^2} = \frac{1}{2} \qquad a = \frac{g \sin\theta}{1 + \left(\frac{K^2}{R^2}\right)} = \frac{g \sin\theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin\theta$$

$$\boxed{a = \frac{2}{3} g \sin\theta}$$

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498. A particle is moving under the influence of a force $F(x) = -k/(2x^2)$ along the positive x-axis. At time t = 0, it is at x = 1 m and its velocity is v = 0. What would be its velocity, what it reaches x = 0.5 m?
(Where $k = 10^{-2}$ N-m^2 and mass of particle is 10^{-2} kg)
(a) 1 m/s (b) 0
(c) 2 m/s (d) 0.5 m/s
UPRVUNL AE 2014

Ans : (d) : From Newton's Second law of motion the inertia force acting on a body is given by, F = m × 1

$$\text{Given } F(x) = -\frac{k}{2x^2} = m \times a_x$$

$$a_x = -\frac{k}{m \times 2x^2} = -\frac{1}{2x^2}$$

$$\Rightarrow a_x = \frac{d^2x}{dt^2} = -\frac{1}{2x^2}$$

$$\Rightarrow v(x) = \frac{dx}{dt} = \frac{1}{2x} + c$$

Using the initial conditions
x = 1 m, v = 0

$$\Rightarrow 0 = \frac{1}{2} + c$$

$$\Rightarrow c = -\frac{1}{2}$$

$$\therefore \frac{dx}{dt} = \frac{1}{2x} - \frac{1}{2}$$

When x = 0.5m

$$v = \frac{1}{2 \times 0.5} - \frac{1}{2} = 0.5 \text{m/s}$$

Note : – The Commission has given answer option (a).

499. A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is
(a) 36 (b) 30
(c) 24 (d) 18
GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Let total distance be 'D'
Average speed = Total distance/Total time
Hence, avg speed,

$$= \frac{D}{\frac{D}{2\times60} + \frac{D}{4\times30} + \frac{D}{4\times10}} = \frac{D}{\frac{D}{120} + \frac{D}{120} + \frac{D}{40}}$$

$$= \frac{D}{D\left(\frac{1}{120} + \frac{1}{120} + \frac{1}{40}\right)} = \frac{1}{\left(\frac{1+1+3}{120}\right)} = \frac{1}{\left(\frac{5}{120}\right)}$$

$$= \frac{120}{5} = 24 \text{km/h}$$

500. A body moves, from rest with a constant acceleration of 5 m per sec^2. The distance covered in 10 sec is most nearly.
(a) 200 m (b) 300 m
(c) 250 m (d) 500 m
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c) Using equation of motion, we have

$$S = ut + \frac{1}{2}at^2$$

u = 0, a = 5m/sec^2, t = 10sec

$$S = \frac{1}{2} \times 5 \times 10 \times 10$$

$$\boxed{S = 250 \text{ m}}$$

501. Two cars are 10 km apart and moving in the same direction at speed of 40 km/hr. A car moving in opposite direction meets these cars at interval of 10 minutes. At what speed the other car is moving?
(a) 40 km/hr (b) 25 km/hr
(c) 30km/hr (d) 20km/hr
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (d) Let us assume the two cars to be stationary, then the opposite car move with velocity (V + 40)km/hr where V is its original speed and we have added 40 since we considered two cars 10 km apart to be stationary. Speed of car relative to stationary car will be (40+V) km/hr.

$$(V + 40) = \frac{10}{\left(\frac{10}{60}\right)}$$

$$\boxed{V = 20 \text{ km/hr}}$$

502. Suppose that a particle moves on a coordinate line so that its velocity at time *t* is $v(t)=t^2-2t$ m/s. The displacement of the particle during time interval $0 \le t \le 3$ is-
(a) 4 m (b) 0m
(c) 8/3m (d) 3/8 m
ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (b) We know that

$$V = \frac{dx}{dt}$$

$$dx = Vdt$$

$$\int_0^L dx = \int_0^3 \left(t^2 - 2t\right)dt$$

$$L = \left[\frac{t^3}{3} - 2 \times \frac{t^2}{2}\right]_0^3$$

L = 9-9=0
L = zero

503. A point P moves along the path $y = x^2 - 4$. What is the displacement when the point moves from x = 1 to x = 3
(a) 8.24, @ $\tan^{-1}(4)^0$ (b) 2, @ $\tan^{-1}(1)^0$
(c) 11.3, @ $\tan^{-1}(4)^0$ (d) 11.3, @ $\tan^{-1}(1)^0$
ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Given,

$$y = x^2 - 4$$

At $x_1 = 1$, $x_2 = 3$
$y_1 = -3$, $y_2 = 5$

$$\text{Displacement} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3-1)^2 + (5+3)^2}$$

$$= \sqrt{4 + 64}$$

$$= 8.24$$

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and $\theta = \tan^{-1}\left(\frac{y_2 - y_1}{x_2 - x_1}\right)$

$= \tan^{-1}\left(\frac{5+3}{3-1}\right)$

$= \tan^{-1}(4)$

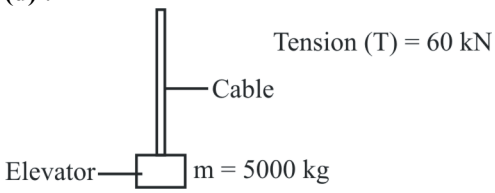
504. An elevator has a mass of 5000 kg. When the tension in the supporting cable is 60 kN, the acceleration of the elevator is nearly

(a) 8 m/s^2 (b) 12 m/s^2

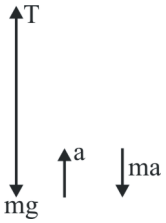
(c) –2 m/s^2 (d) 2 m/s^2

ISRO Scientist/Engineer 03.07.2016

Ans. (d) :



FBD–



$T - mg = ma$

$60000 - 5000 \times 9.81 = 5000\ a$

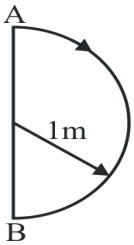
$60000 - 49050 = 5000\ a$

$10950 = 5000\ a$

$a = \frac{10950}{5000} = 2.19\ m/s^2$

$a = 2\ m/s^2$

505. In 1 second, a particle goes from point A to point B, moving in a semicircle (as shown in figure). The magnitude of the average velocity is



(a) 3.14 m/s (b) 1 m/s

(c) 2 m/s (d) Zero

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Concept of Average Velocity–

Avg. velocity $= \frac{\text{Total distance}}{\text{Time}}$

$V_{avg.} = \frac{\text{Minimum total distance}}{\text{Time}}$

$V_{avg} = \frac{2}{1} = 2\ m/s$

506. A block is made to slide down an inclined plane (30° with horizontal) which is smooth. It starts sliding from rest and takes a time 't' to reach the bottom of the plane. An identical body is freely dropped from the same point. The time the body takes to reach the bottom is

(a) t (b) $\frac{t}{2}$

(c) $\frac{t}{3}$ (d) $\frac{t}{4}$

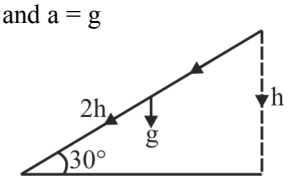
ISRO Scientist/Engineer 03.07.2016

Ans. (b) : Consider body freely drop from height 'h'.

$S = ut + \frac{1}{2}at^2$

At initial condition–

$u = 0$ and $a = g$



$2h = \frac{1}{2}\frac{g}{2}t^2$

$t = \sqrt{\frac{8h}{g}} = 2\sqrt{\frac{2h}{g}}$

If freely fall,

$h = \frac{1}{2}gt_2^2$

$t_2 = \sqrt{\frac{2h}{g}}$

$\therefore \quad \boxed{t_2 = \frac{t}{2}}$

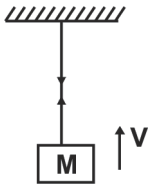
507. An elevator weighing 1000 Kg attains an upward velocity of 4m/sec in two seconds with uniform acceleration. The tension (in N) in supporting cable will be (g = 9.8 m/sec^2)

(a) 1204 N (b) 9800

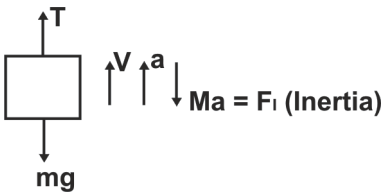
(c) 2000 N (d) 11800 N

ISRO Scientist/Engineer 2012

Ans. (d)



FBD,



M = 1000 kg

V = 4m/s

g = 9.8 m/s^2

Acceleration of lift = a

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V = u + at
4 = 0 + a × 2 [t = 2 sec]
4 = a × 2[Initial velocity u = 0]
∴ a = 2 m/s^2
∴ Tension in cable 'T'
T = mg + ma
= m (g + a)
= 1000 (9.8 + 2)
T = 11800 N

508. Two car are moving in the same direction with a speed of 45 km/hr and a distance of 10 km separates them. If a car coming from the opposite direction meets these two cars at an interval of 6 minutes, its speed would be

(a) 45 km/hr (b) 55 km/hr
(c) 65 km/hr (d) 75 km/hr

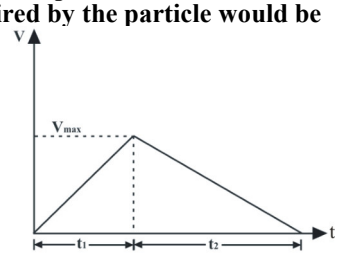
ISRO Scientist/Engineer 2011

Ans. (b) : Let speed of car moving in opposite direction is = V m/s
From relative velocity approach.
Due to same direction,
Relative velocity = V + 45

So, $V + 45 = \dfrac{10}{\left(\frac{6}{60}\right)}$

100 = V + 45
V = 100-45
V = 55 km/hr

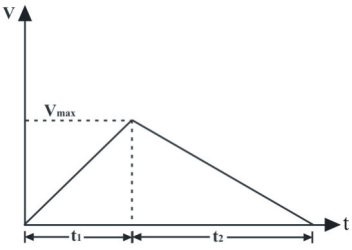
509. A particle starts from rest with a constant acceleration α m/sec^2 and after some time it decelerates at a uniform rate of β m/sec^2 till it comes to rest. If the total time taken between two rests positions is t, the maximum velocity acquired by the particle would be



(a) $\dfrac{\alpha+\beta}{2}t$ (b) $\dfrac{\alpha-\beta}{2}t$
(c) $\left(\dfrac{\alpha\beta}{\alpha+\beta}\right)t$ (d) $\left(\dfrac{\alpha-\beta}{\alpha+\beta}\right)t$

ISRO Scientist/Engineer 2010
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Let A particle starts from rest i.e. $u_1 = 0$
Acceleration $a_1 = \alpha$ m/s^2 and time = t_1
After t_2 time decelerates $a_2 = -\beta$ m/s^2
V = u + at



$V_{max} = 0 + \alpha t_1$ ...(1)
$0 = V_{max} - \beta t_2$ ...(2)
So,
$\alpha t_1 = \beta t_2$
$t_2 = \dfrac{\alpha}{\beta}t_1$
$t = t_1 + t_2$ ...(3)
$t_1 = \dfrac{\beta}{(\alpha+\beta)}.t$
So, from equation (1)
$V_{max} = \dfrac{\alpha.\beta}{(\alpha+\beta)}t$

510. A heavy block of mass m is slowly placed on a conveyer belt moving with speed v. If coefficient of friction between block and the belt is μ, the block will slide on the belt through distance

(a) $\dfrac{v}{\mu g}$ (b) $\dfrac{v^2}{\sqrt{\mu g}}$
(c) $\left(\dfrac{v}{\mu g}\right)^2$ (d) $\dfrac{v^2}{2\mu g}$

ISRO Scientist/Engineer 2008
ISRO Scientist/Engineer 2007

Ans. (d) : Mass of block = m
Speed = m
Coefficient of friction = μ
Reterdation due to friction = –μg
$u^2 - v^2 = 2$ as
$0 - v^2 = -2$ μgs
$s = \dfrac{v^2}{2\mu g}$

511. A car moving with speed u can be stopped in minimum distance x when brakes are applied. If the speed becomes n times, the minimum distance over which the car can be stopped would take the value

(a) $\dfrac{x}{n}$ (b) nx
(c) $\dfrac{x}{n^2}$ (d) n^2x

ISRO Scientist/Engineer 2008

Ans. (d) : A car moving speed = u
distance = x
speed becomes n times

So, $x = \dfrac{u^2}{2a}$ $[\because v^2 - u^2 = 2as]$

$s = \dfrac{(nu)^2}{2a}$

$s = \dfrac{n^2u^2}{2a}$ $\left(\because x = \dfrac{u^2}{2a}\right)$

$s = n^2x$

512. A particle starts with a velocity 2m/sec and moves on a straight-line track with retardation 0.1 m/sec^2. The time at which the particle is 15 from the starting point would be

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(a) 10 sec (b) 20 sec
(c) 50 sec (d) 40 sec

ISRO Scientist/Engineer 2007

Ans. (a) : u = 2m/s, a = 0.1 m/s^2

$$s = ut + \frac{1}{2}at^2$$

$$15 = 2t + \frac{1}{2}(-0.1)t^2 \quad [\text{Reterdation a} = -0.1 \text{ m/sec}^2]$$

$$2t - \frac{0.1}{2}t^2 = 15$$

$$2t - \frac{1}{20}t^2 = 15$$

$$40t - t^2 = 300$$

$$t^2 - 40t + 300 = 0$$

$$(t-10)(t-30) = 0$$

$$t = 10,\ t = 30$$

t = 10 sec

513. A stone is released from a certain height and it takes 5 seconds to reach the ground. If the stone is stopped after 3 seconds and released, the total time the stone will be in the air is :

(a) 14s (b) 4s
(c) 7s (d) 5s

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : Given, u = 0, t = 5, t′ = 3

$$S = ut + \frac{1}{2}gt^2$$

$$S_1 = \frac{1}{2}g \times 5^2$$

$$S_1 = \frac{25}{2}g$$

After 3 sec. stopped

$$S_2 = ut + \frac{1}{2}gt^2$$

$$S_2 = \frac{1}{2}g \times 3^2$$

$$S_2 = \frac{9}{2}g$$

$$S_1 - S_2 = \frac{25}{2}g - \frac{9}{2}g$$

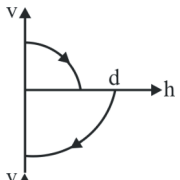
$$\Delta S = \frac{16g}{2}$$

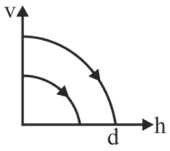
$$\frac{16g}{2} = \frac{1}{2}gt^2$$

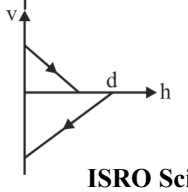
$$\boxed{t = 4}\ \text{sec}.$$

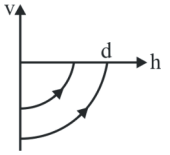
$\boxed{\text{Total time} = 4 + 3 = 7 \text{ sec.}}$

514. A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up to a height d/2. Neglecting the subsequent motion and air resistance, its velocity (v) varies with height (h) above the ground as

(a) 

(b) 

(c) 

(d) 

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : At height h = d, u = 0

At ground, velocity is maximum and downward consider velocity as negative (–ve) and upward positive (+ve).

When it return to height h velocity becomes zero.

$v^2 = u^2 + 2as$

u = 0

$\boxed{v^2 = 2as}$ → Variation parabolic

515. The motion of a body moving on a curved path is given by a equation $x = 4 \sin 3t$ and $y = 4 \cos 3t$. The resultant velocity of the car is

(a) 30 m/sec (b) 24 m/sec
(c) 12 m/sec (d) 40 m/sec

TNPSC AE 2014

Ans. (c) : $x = 4 \sin 3t$

$y = 4 \cos 3t$

velocity component in x & y direction respectively then

$$u_x = \frac{dx}{dt} = 4\cos 3t \times 3$$

$$= 12 \cos 3t$$

$$v_y = \frac{dy}{dt} = -4 \sin 3t \times 3$$

$$v_y = -12 \sin 3t$$

Resultant velocity -

$$v = \sqrt{u_x^2 + v_y^2}$$

$$= \sqrt{(12\cos 3t)^2 + (-12\sin 3t)^2}$$

$$= \sqrt{(144)\left[\sin^2 3t + \cos^2 3t\right]}$$

$$= \sqrt{144 \times 1}$$

$$\boxed{v = 12 \text{ m/s}}$$

516. The linear velocity of a body rotating at ω rad/s along a circular path of radius r is

(a) ω .r (b) ω/r
(c) $\omega^2 . r$ (d) ω^2 / r

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : We know that, linear velocity,

V = ω .r

517. The functional reaction between time t and distance x, in m, is $t = 2x^2 + 5x$.. The acceleration in m/s^2 at t = 12 s is :

(a) $\frac{-1}{121}$ units (b) $\frac{-4}{1331}$ units
(c) $\frac{-4}{121}$ units (d) $\frac{4}{1331}$ units

BHEL ET 2019

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Ans. (b) : Given-

$t = 2x^2 + 5x$

Acceleration a (m/s^2) at t = 12 sec.

$$t = 2x^2 + 5x$$

$$\frac{dt}{dx} = 4x + 5$$

or $$\frac{dx}{dt} = \frac{1}{4x+5}$$

$$v = \frac{dx}{dt} = \frac{1}{(4x+5)}$$

$$a = v\frac{dv}{dx} = \frac{1}{4x+5}\left[\frac{(4x+5)\times 0 - 1\times 4}{(4x+5)^2}\right]$$

$$a = \frac{-4}{(4x+5)^3} \quad ....(1)$$

$\therefore$ $t = 12$

$t = 2x^2 + 5x$

$2x^2 + 5x = 12$

$2x^2 + 5x - 12 = 0$

$(x + 4)(2x - 3) = 0$

$$x = \frac{3}{2}, -4$$

$x = \frac{3}{2}$ value is substitute in equation (1)

$$\left[a = \frac{-4}{\left[4\times\left(\frac{3}{2}\right)+5\right]^3} = \frac{-4}{(11)^3}\right]$$

$$= \frac{-4}{1331}\text{unit}$$

518. Time variation of the position of the particle in rectilinear motion is given by $x = 3t^3 + 2t^2 + 4t$. If v is the velocity and a is the acceleration of the particle in consistent units, the motion started with

(a) v = 0, a = 0 (b) v = 0, a = 4

(c) v = 4, a = 4 (d) v = 2, a = 4

(e) v = 4, a = 2

CGPSC AE 26.04.2014 Shift-I

Ans. (c) : Given,

$x = 3t^3 + 2t^2 + 4t$

$$\text{velocity } (v) = \frac{dx}{dt} = 9t^2 + 4t + 4 \quad ...(1)$$

$$\text{Acceleration } (a) = \frac{dv}{dt} = 18t + 4 \quad ...(2)$$

$\because$ Initially t = 0 so from equation (1) and (2)

$v = 9t^2 + 4t + 4$

at t = 0, v = 4

$a = 18t + 4$

at t =0, a = 4

519. A motorist travelling at a speed of 18 km/hr, suddenly applies the brakes and comes to rest after skidding 75 m. The time required for the car to stop is

(a) t = 30.25 sec (b) t = 29.94 sec

(c) t = 28.84 sec (d) t = 26.22 sec

TNPSC AE 2014

Ans. (b) : u = 18 km/hr

= 5 m/s

d = 75 m

v = 0

We know that after applies the brake-

$v^2 = u^2 - 2ad$

$(5)^2 = 2 \times a \times 75$

$$\boxed{a = \frac{1}{6}\,m/s^2}$$

then $v = u - a\,t$

$u = a \times t$

$$5 = \frac{1}{6} \times t$$

$$\boxed{t = 30\ sec}$$

520. The angular motion of a disc is defined by the relation ($\theta = 3t + t^3$), where θ is in radians and t is in seconds. What will be the angular position after 2 seconds?

(a) 14 rad (b) 12 rad

(c) 18 rad (d) 16 rad

CIL MT 26.03.2017

Ans. (a) : Given angular motion equation $\theta = 3t + t^3$

Motion covered in 2 seconds = $3 \times 2 + 2^3 = 14$ rad. So the angular position after 2 seconds will be 14 rad.

521. The position of a particle in rectilinear motion is given by the equation ($x = t^3 - 2t^2 + 10t - 4$), where x is in meters and t is in seconds. What will the velocity of the particle at 3s?

(a) 20 m/s (b) 25 m/s

(c) 15 m/s (d) 30 m/s

CIL MT 26.03.2017

Ans. (b) : Position $(x) = t^3 - 2t^2 + 10t - 4$

$$\text{Velocity } (v) = \frac{dx}{dt} = 3t^2 - 4t + 10$$

At t = 3s

$v = 3 \times 3^2 - 4 \times 3 + 10 = 25$ m/s

522. An object falls from the top of a tower. If comes down half the height in 2 seconds. Time taken by the object to reach the ground is:-

(a) 2.8 s (b) 3.2 s

(c) 4.0 s (d) 4.5 s

UKPSC AE-2013, Paper-I

Ans. (a) : We assume that the height of tower is h,

$$\frac{h}{2} = ut + \frac{1}{2}gt^2$$

$u = 0$, $t = 2s$

$$\boxed{h = 4g} \quad ...(i)$$

then,

$v^2 = u^2 + 2gh$

$$v = \sqrt{2gh} = \sqrt{2g \times 4g}$$

$$v = g\sqrt{8} \quad ...(ii)$$

then,

$v = u + gt$

$v = g \times t$

$$t = \frac{v}{g} = \frac{g\sqrt{8}}{g}$$

$$\boxed{t = 2.83\ s}$$

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523. Two cars 'A' and 'B' move at 15m/s in the same direction. Car 'B' is 300m ahead of car 'A'. If car 'A' accelerate at 6m/s^2 while car 'B' continues to move with the same velocity, car 'A' will overtake car 'B' after:-

(a) 7.5 s (b) 10 s
(c) 12 s (d) 15 s

UKPSC AE-2013, Paper-I

Ans. (b) : Two can A & B move initial at 15 m/s

So initial velocity of car A with respect to car B is zero.

So, car A will over take distance 300 m when accelerate at 6 m/s^2

$$s = ut + \frac{1}{2}at^2$$

$$300 = 0 \times t + \frac{1}{2} \times 6 \times t^2$$

$$\boxed{t = 10\ s}$$

524. Two balls are dropped from a common point after an interval of 1 second. If acceleration due to gravity is 10m/s^2, separation distance 3 second after the release of the first ball will be:-

(a) 5 m (b) 15 m
(c) 25 m (d) 30 m

UKPSC AE-2013, Paper-I

Ans. (c) : The first ball has traveled for 3 seconds under gravity wit acceleration g and initial velocity u = 0

$$S_1 = 0 \times 3 + \frac{1}{2} \times 10 \times (3)^2$$

S_1=45 m

Then second ball has traveled for 2 seconds

Then,

$$S_2 = 0 \times 2 + \frac{1}{2} \times 10 \times (2)^2$$

$S_2 = 20$ m

Then, distance between them will be

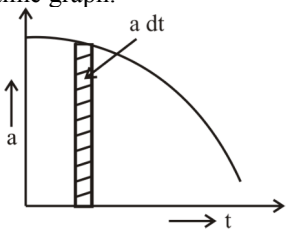
$S_1 - S_2 = 45 - 20 = 25$ m

525. ..... represents the area under acceleration - time graph.

(a) Acceleration (b) Displacement
(c) Motion (d) Change in velocity

(HPPSC POLY. LECT. 2016)

Ans : (d) Change in velocity represent the area under acceleration - time graph.



$$a = \frac{dv}{dt}$$

$$dv = a\ dt$$

$$\int dv = \int a\ dt$$

$$v_2 - v_1 = \int a\ dt$$

526. The speed of a particle moving in circular path is 600 rpm. Then, the angular velocity of the particle is

(a) 20π rad/ sec (b) 10 π rad/sec
(c) 20 /π rad/sec (d) 10/πrad / sec

TSPSC AEE 2015

Ans : (a) Speed. fo a particle = 600 rpm.

$$\text{Angular velocity} = \frac{2\pi N}{60}\ \text{rad/sec.}$$

$$= \frac{2\pi \times 600}{60}\ \text{rad/sec.}$$

Angular velocity = 20π rad/sec.

527. The tension in the cable supporting a lift is more when the lift is

(a) Moving downwards with uniform velocity.
(b) Moving upwards with uniform velocity.
(c) Moving upwards with acceleration.
(d) moving downwards with acceleration.

UPPSC AE 12.04.2016 Paper-I

Ans : (c) The tension in the cable supporting a lift is more when the lift is moving upwards with acceleration.

528. An object having 10 Kg mass and weights as 9.81 N on a spring balance. The value of "g" at that place is

(a) 9.81 m/s^2 (b) 10.m/s^2
(c) 0.981m/s^2 (d) 98.1 m/s^2

UJVNL AE 2016

Ans : (c) Given, Object mass = 10 kg
Spring weight = 9.81 N
W = mg
9.81 = 10 × g
g = 0.981 m/sec^2

529. A rigid body can be replaced by two masses placed at fixed distance apart. The two masses form an equivalent dynamic system, if (select the most appropriate answer).

(a) The sum of the two masses is equal to the total mass of the body.
(b) The centre of gravity of two masses coincide with that of the body
(c) The sum of the mass moment of inertia of the masses about their centre of gravity is equal to the mass moment inertia of the body.
(d) All of the above.

UPPSC AE 12.04.2016 Paper-I

Ans : (d) (i) The centre of gravity of two masses coincide with that of the body
(ii) The sum of the two masses is equal to the total mass of the body
(iii) The sum of the mass moment of inertia of the masses about their centre of gravity is equal to the mass moment inertia of the body

530. If the body falls freely under gravity, then the gravitational acceleration is taken as

(a) + 8.91 m/s^2 (b) – 8.91m/s^2
(c) + 9.81 m/s^2 (d) – 9.81 m/s^2

TSPSC AEE 2015

Ans : (c) If the body falls freely under gravity, then the gravitational acceleration is taken as + 9.81 m/sec^2

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531. How much force will be exerted by the floor of the lift on a passenger of 80 kg mass when lift is accelerating downward at 0.81 m/s^2?

(a) 740 N (b) 700 N
(c) 720 N (d) 680 N

CIL MT 26.03.2017

Ans. (c) : F_{net} = mg-ma = m(g-a)
= 80 (9.81-0.81) = 80×9 = 720 N

532. If velocity of a body change from 50 m/s to 200 m/s in 20 seconds, then the acceleration of the body is:

(a) 5.0 m/s^2 (b) 6.5 m/s^2
(c) 6.0 m/s^2 (d) 7.0 m/s^2
(e) 7.5 m/s^2

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : v_1 = 50 m/s, v_2 = 200 m/s, t = 20 s
Then acceleration (a) is given as
a = rate change in velocity

$$a = \frac{dv}{dt} = \frac{200-50}{20} = \frac{150}{20}$$

$$\boxed{a = 7.5\ m/sec^2}$$

533. A particle of mass 1 kg moves in a straight line under the influence of a force which increases linearly with time at the rate of 60N/ s, it being 40 N initially. The position of the particle after a lapse of 5s, if it started from rest at the origin, will be

(a) 1250 m (b) 1500 m
(c) 1750 m (d) 2000 m

ESE 2019

Ans. (c) : Given,

$$m = 1\ kg,\ \frac{dF}{dt} = 60\ N/s$$

$$dF = 60\ dt$$

$$\int dF = \int 60\ dt$$

$$F = 60\ t + c \quad .....(1)$$

at t = 0 F = 40

$$c = 40 \quad [from\ (1)]$$

$$\therefore \quad F = 60t + 40 \quad .....(2)$$

Using Newton's second law –

$$F = ma = 1 \times a = a$$

From equation (2)

$$a = 60t + 40$$

$$\frac{dv}{dt} = 60t + 40$$

$$dv = 60t \times dt + 40dt$$

Integrating it

$$\int dv = \int 60t \times dt + \int 40\ dt$$

$$v = 60\frac{t^2}{2} + 40t + c_1 \quad .....(3)$$

at t = 0, v = 0, $\therefore\ c_1 = 0$

Then $V = \frac{60t^2}{2} + 40t$

$$\frac{ds}{dt} = \frac{60t^2}{2} + 40t$$

Again integrating

$$\int ds = \int \frac{60t^2}{2} dt + \int 40t\ dt$$

$$s = \frac{60t^3}{6} + \frac{40t^2}{2} + c_2 \quad .....(4)$$

at t = 0, s = 0, $\therefore\ c_2 = 0$

then $s = \frac{60t^3}{6} + \frac{40t^2}{2}$

at t = 5 s

$$s = \frac{60\times5^3}{6} + \frac{40\times5^2}{2}, \quad s = 1750\ m$$

534. An elevator has a downward acceleration of 0.1 g m/s^2. What force will be transmitted to the floor of elevator by a man of weight 'W' travelling in the elevator?

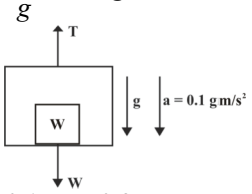
(a) W (b) W/10
(c) 11W/10 (d) 9W/10

UPRVUNL AE 07.10.2016

Ans. (d) : Newton's Second Law of motion

$$F_{net} = W - T = ma$$

$$W - T = \frac{W}{g} \times 0.1 \times g = 0.1 \times W$$



T = W - 0.1 W = 0.9 W

$$\boxed{T = \frac{9}{10}W}$$

535. Which of the following statements of D'Alembert's principle are correct?

1. **The net external force F actually acting on the body and the inertia force F_t together keep the body in a state of fictitious equilibrium**
2. **The equation of motion may be written as F + (– ma) = 0 and the fictitious force (– ma) is called an inertia force**
3. **It tends to give solution of a static problem an appearance akin to that of a dynamic problem.**

(a) 1 and 3 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2020

Ans. (d) : 1, 2 and 3

536. A 2000 kg of automobile is driven down a 5 degree inclined plane at a 100 km/h, when the brakes are applied causing a constant total braking force (applied by the road on the tires) of 7 kN. Determine the distance traveled by the automobile as it comes to a stop.

(a) 146 m (b) 152 m
(c) 135.86 m (d) 122.44 m
(e) 149.5 m

CGPSC AE 26.04.2014 Shift-I

Ans. (a) : Given,
mass (m) = 2000 kg

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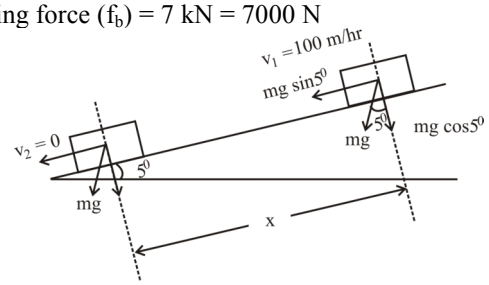
inclination (θ) = 5°

velocity (v_1) = 100 km/hr $= 100 \times \frac{5}{18} = 27.78$ m/s

braking force (f_b) = 7 kN = 7000 N



Apply work-energy principle

$$W_{1\text{-}2} = (K.E)_2 - (K.E)_1$$

$$(2000\ g \sin 5°)x - 7000\ x = 0 - \frac{1}{2}mv_1^2$$

$$-5290\ x = -771728.4$$

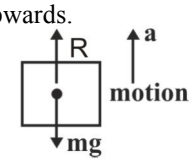
$$x = 145.88 \simeq 146 \text{ m}$$

537. Determine the apparent weight of a 70 kg man in a elevator when the acceleration of elevator is 2 m/s^2 upwards.

(a) 700 N (b) 686.7 N
(c) 826.7 N (d) 560 N
(e) 4.67 N

CGPSC AE 26.04.2014 Shift-I

Ans. (c) : As the elevator is moving upwards i.e. in the direction of reaction force, therefore, the net force would be R-mg upwards.



Following equation obtained from Newton's second law

$F_{net} = R - mg$

$R - mg = ma$

$R = mg + ma = m(g + a)$

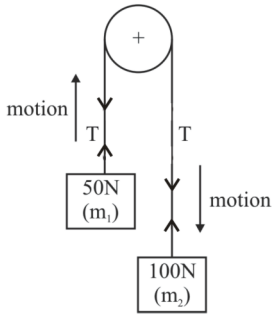
$R = 70\ (9.81 + 2) = 826.7$ N

538. Two blocks of 50 N and 100 N are connected by a light cord passing over a smooth frictionless pulley. The acceleration in blocks and tension in rope are respectively given by: [g = acceleration due to gravity]

(a) g/2, 40 N (b) g/3, 66.67 N
(c) g, 200 N (d) g/5, 40 N
(e) g/4, 50 N

CGPSC AE 25.02.2018

Ans. (b) : Using Newton's second law of motion



In equilibrium

$F_{net} = T - 50 = m_1 a$ .....(i)

$100 - T = m_2 a$ .....(ii)

from equation (i)

$$T - 50 = \frac{50}{g} \times a$$

$$T = \frac{50}{g} \times a + 50 \quad \text{.....(iii)}$$

from equation (ii)

$$100 - T = \frac{100}{g} a$$

$$T = 100 - \frac{100}{g} a \quad \text{.....(iv)}$$

equating equation (iii) & (iv)

$$\frac{100}{g} a + 50 = 100 - \frac{100}{g} a$$

$$\frac{150}{g} a = 50$$

$$\boxed{a = \frac{g}{3}}$$

summation of equation (iii) & (iv)

$$2T = 150 - \frac{50a}{g} = 150 - \frac{50}{3}$$

$$T = 66.67$$

539. A car of mass 150 kg is traveling on a horizontal track at 36 Km/hr. The time needed to stop the car is _____ (take $\mu = 0.45$).

(a) t = 2.26 sec (b) t = 3.20 sec
(c) t = 3.8 sec (d) t = 4.2 sec

TNPSC AE 2014

Ans. (a) : Given,

m = 150 kg, V = 36 km/hr = 10 m/s, $\mu = 0.45$

We know that,

Friction force = $\mu \times mg$

$F = 0.45 \times 150 \times 9.81$

$F = 662.175$ N

Let's suppose that the car stops after traveling a distance d. During this time the entire kinetic energy will be exhausted into the work done by the friction force $F \times d = \frac{1}{2} mv^2$

$$662.175 \times d = \frac{1}{2} \times 150 \times (10)^2$$

$$\boxed{d = 11.326 \text{ m}}$$

Then $v^2 = u^2 - 2ad$

$v = 0$

$u^2 = 2ad$

$(10)^2 = 2a \times 11.362$

$$\boxed{a = 4.414 \text{ m/s}^2}$$

Then $v = u - at$

$10 = 4.414 \times t$

$$\boxed{t = 2.2652 \text{ sec.}}$$

540. A car starting from rest attains a maximum speed of 100 kmph in 20 seconds. What will be its acceleration assuming it is uniform?

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(a) 1.0 m/s^2 (b) 1.4 m/s^2
(c) 1.8 m/s^2 (d) 2.0 m/s^2

CIL MT 26.03.2017

Ans. (b) : Initial velocity (u) = 0
Final velocity (v) = 100 kmph

$$= 100 \times \frac{5}{18} = \frac{500}{18} m/s$$

Time (t) = 20 sec.
v = u + at

$$\Rightarrow a = \frac{v-u}{t} = \frac{\frac{500}{18}}{20} = \frac{500}{18 \times 20} = 1.38\ m/s$$

541. ……deals with relationship between forces and the resulting motion of bodies on which they act.
(a) Dynamics of elastic bodies
(b) Statics
(c) Kinetics
(d) Theory of plasticity

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Kinetics- kinetics deals with motion of body under the action of force under consideration
According to Newton's 2nd law.

$$\boxed{\vec{F} = m\vec{a}}$$

Kinematics – kinematics deals with motion of body without consideration of force of action.
In kinematics we deals with displacement, velocity and acceleration of body.

542. Which of the following equations is an equation of motion?
(a) $v = u + at^2$ (b) $v^2 - u^2 = 2at$
(c) $s = ut + \frac{1}{2}at^2$ (d) $u = s + at$

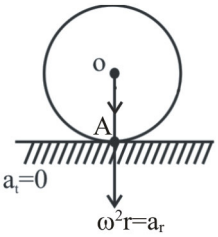
HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : According to Newton's law of motion equation,
(i) $v = u + at$
(ii) $s = ut + \frac{1}{2}at^2$
(iii) $v^2 = u^2 + 2as$
Where, u = initial velocity
v = final velocity
a = acceleration
t = time duration
s = distance

543. A solid disc of radius r rolls without slipping on the horizontal floor with angular velocity ω and angular acceleration α, the magnitude of acceleration of the point of contact on the disc is:
(a) Zero (b) rα
(c) $\sqrt{(r\alpha)^2 + (r\omega^2)^2}$ (d) rω^2

NPCIL ET, 2015, BEL PE 2015, GATE 2012

Ans. (d) :



Acceleration at point 'A'

$$a_A = \sqrt{a_T + a_r}$$

$$\overrightarrow{a_c} = 0 + \omega^2 r = \omega^2 r$$

544. Consider an attractive force which is central but inversely proportional to the distance from the centre. If a particle is in a circular orbit under such a force, its period is
(a) independent of the radius of the orbit
(b) proportional to the radius of the orbit
(c) proportional to the square of the radius of the orbit
(d) independent to the square of the radius of the orbit

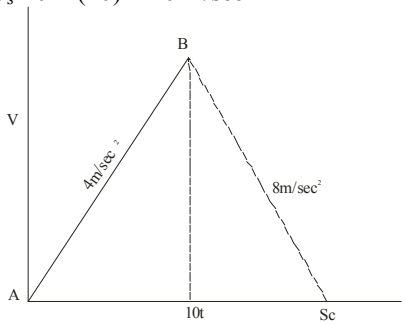
WBPSC AE, 2007

Ans. (c) : Consider an attractive force which is central but inversely proportional to the distance from the centre. If a particle is in a circular orbit under such a force, its period is proportional to the square of the radius of the orbit.

545. A motorbike starts from rest and accelerates at a rate of 4 m/sec^2 for 10 seconds and then decelerates at 8 m/sec^2 until it stops. The total distance travelled is
(a) 100 m (b) 200 m
(c) 300 m (d) 500 m

TRB Poly. Lect., 2012

Ans. (c) : For A-B
u= 0, a=4 m/sec^2, t = 10sec.
v= u+at
v_s= 0+4(10) = 40 m/sec



For B-C
v = u+ at
v_b=o+4(10)= 40 m/Sec.
= v = u + at,
u_c= 40m/sec., v_D =o
v_b= u_c+ at a= –8 m/sec^2
0 = 40+(-8)t
8t=40

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$$t = \frac{40}{8} = 5\,\text{sec}$$

Area under velocity time graph gives displacement =

$$\text{Area} = \text{Distance} = \frac{1}{2}\times 10 \times \frac{20}{40} + \frac{1}{2}\times 5 \times 20 \times 2$$

$$= 200+100$$

$$=300\ \text{m}$$

546. The linear acceleration (a) of a body of radius (r) with an angular acceleration of α rad/s², is:

(a) $a = \alpha / r$ (b) $a = \alpha . r$

(c) $a = r / \alpha$ (d) $a = \alpha^2 r$

TSPSC Managers, 2015

Ans. (b) : linear acceleration (a) = rα

Radial acceleration (a_r) = $\omega^2 r$

ω = angular velocity

r = radius

547. A particle is of weight 8 kg. When it is allowed to fall under the force of gravity its momentum at the time t = 2 sec would be ____ kg sec.

(a) 2 (b) 8

(c) 16 (d) 32

MPSC Poly. Lect. 2014

Ans. (c) : m = 8 kg

t = 2 sec

v = u + at

v = at = 9.81 × 2 = 19.62 m/s

$$\text{Momentum } P = mv = 8 \times 19.62 \left(\frac{\text{kgm}}{\text{s}}\right)$$

Momentum in kg-sec

$$P = 8 \times 19.62 \left(\frac{\text{kgm}}{\text{s}}\right) \times \frac{1}{9.81}\left(\frac{\text{s}^2}{\text{m}}\right)$$

$$\boxed{P = 16\ \text{kg-sec}}$$

548. A stone of mass 1 kg is tied to a string of length 1m and whirled in a horizontal circle at a constant angular speed 5 rad/sec. The tension in the string is :

(a) 30 N (b) 25 N

(c) 15 N (d) 10 N

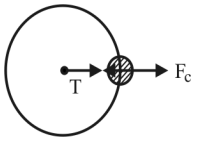
Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : $T = F_C$

$F_C = mr\omega^2$

Where, T = Tension

F_C = Centrifugal force



$$= 1\text{kg} \times 1\text{m} \times \left(\frac{5\text{rad}}{\text{sec}}\right)^2$$

$F_C = T = 25$ N

549. What is the motion of a particle rounded a fixed axis called?

(a) Translatory

(b) Rotary

(c) Circular

(d) Translatory as well as rotory

VIZAG STEEL MT 2011

TNPSC ACF 2012

Ans. (c) : The motion of particle round a fined axis is circular.

550. A body of mass 1 kg when subjected to a force results in a displacement y (measured in meters) given by $y = 5t^2 + 2t$, where t is in secs. The force acting on the body is equal to

(a) 2 N (b) 5 N

(c) 7 N (d) 10 N

DRDO Scientists 2009

Ans. (d) : Displacement velocity acceleration force

$y = 5t^2+2t$

$\dot{y} = 10t+2$

$\ddot{y} = 10$

$F = m\ddot{y} = 10\text{N}$

551. The velocity of a particle moving in a straight line is given by, $V=2t^3-t^2 + 10$. Its acceleration at t =6 sec will be

(a) 406 ms^{-2} (b) 204ms^{-2}

(c) 308ms^{-2} (d) 23 ms^{-2}

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : $V=2t^3-t^2 + 10$.

$$a = \frac{dv}{dt} = 6t^2 - 2t$$

at time (t) =6 sec

$a=6\,(6)^2-2\,(6) = 204\ \text{m/s}^2$

552. The component of acceleration directed towards center of curvature when a body moves around a curve is

(a) linear acceleration

(b) normal acceleration

(c) tangential acceleration

(d) total acceleration

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : normal acceleration

553. A particle moves along a straight line so that its displacement in metres from a fixed point to $s = 2t^3+4t^2-6t+8$. The acceleration after 4 seconds is :

(a) 8 m/s^2 (b) 176 m/s^2

(c) 122 m/s^2 (d) 56 m/s^2

TNPSC AE (Industries) 09.06.2013

Ans. (d) : $s = 2t^3+4t^2-6t+8$

$$\frac{ds}{dt} = 6t^2 + 8t - 6$$

$$\frac{d^2s}{dt^2} = 12t + 8$$

$$\text{acceleration, } a = \frac{d^2s}{dt^2}$$

acceleration, (at t=4 sec) =12(4)+8 =56 m/s^2

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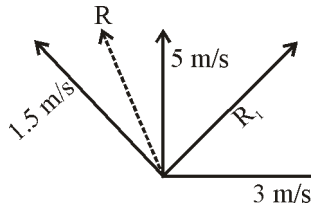
554. A ship is sailing north at a velocity of 5 m/s. The current is taking its east at the rate of 3 m/s, and a sailor is climbing a vertical mass at the rate of 1.5 m/s. What is the absolute velocity the sailor ?

(a) 5.02 m/s (b) 6.02 m/s
(c) 9.5 m/s (d) 6.5 m/s

TNPSC AE (Industries) 09.06.2013

Ans. (b) :



$R = \sqrt{R_1^2 + 1.5^2}$ ——(i)

$R_1 = \sqrt{5^2 + 3^2}$

$= \sqrt{34}$

From equation (i)

$= \sqrt{34 + 2.25}$

$= \sqrt{36.25}$

R = 6.02 m/s

555. The motion of a particle is defined by the relation $x = \frac{5}{2}t^2 - 20t + 10$, where x and t are expressed in meters and seconds respectively. The position x of the particle at zero velocity is

(a) 30 m (b) –30 m
(c) –20 m (d) 50 m

DRDO Scientists 2008

Ans. (b) : Displacement velocity

$x = \frac{5t^2}{2} - 20t + 10$

$\dot{x} = 5t - 20$

$0 = 5t - 20$

t = 4 sec.

x = ? at t = 4

So displacement

$x = \frac{5}{2} \times 16 - 20(4) + 10 = -30m$

556. A body is moving with uniform acceleration. In 4th second of its travel it covers 20 m and 30 m in 8th second. The distance traveled at the 10th seconds is (in m):

(a) 24 (b) 35
(c) 43 (d) 52

HPPSC ADF 05.03.2019

Ans. (b) : $S_n = u + \frac{a}{2}(2n-1)$

$20 = u + \frac{a}{2}(7)$

$30 = u + \frac{a}{2}(15)$

4a = 10

a = 2.5

$u = \frac{45}{4}$

$\therefore S_w = u + \frac{a}{2}(2n-1)$

$= \frac{45}{4} + \frac{5}{4} \times \frac{1}{2}(19)$

$= \frac{140}{4} = 35m$

557. A 75 Kg (736 N) man stands on a spring scale in an elevator. During the first 3 seconds of motion from rest, the tension T in the hoisting cable is 8110 N. The total mass elevator, man and scale is 750 kg (7360 N). The reading R of the scale in newton's is _____.

(a) 750 N (b) 1250 N
(c) 811 N (d) 830 N

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : Given–

t = 3 sec, u = 0

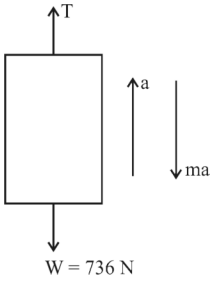
T = 8110 N

$\Sigma F = ma$

$T - W = 750 \times a$

$8110 - 7360 = 750 \times a$

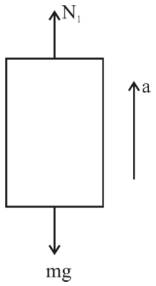
$a = 1\ m/s^2$



Now, $v = u + at$

$v = 0 + (1 \times 3)$

$v = 3\ m/s^2$



For upward (elevator) motion–

$N_1 - mg = ma$

$N_1 = ma + mg$

$= m(a + g)$

$N_1 = 75(1 + 9.81)$

$= 75 \times 10.81$

$N_1 = 810.75$ N

$N_1 \approx 811$ N

558. A particle while sliding down a smooth plane of $18.36\sqrt{2}$ m length acquires a velocity of 19.86 m/sec. The inclination of plane is

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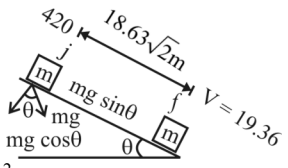
(a) 30º (b) 60º
(c) 45º (d) 75º

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) :



$v^2 = u^2 + 2as$

$(19.36)^2 = 0 + 2a \times 18.36\sqrt{2}$

$a = \dfrac{(19.86)^2}{2\times 18.36\sqrt{2}}$

$a = 7.59\ m/s^2$

$F = ma$

$mg \sin\theta = m.a$

$\sin\theta = \dfrac{7.59}{9.81}$

$\boxed{\theta = 50.73}$

Nearest option is option (c) i.e. 45°

7. Work, Power and Energy

559. If a rigid body is in equilibrium, the total virtual work of the external forces acting on the rigid body due to virtual displacement of the body is _____

(a) $-\infty$ (b) 1
(c) 0 (d) $+\infty$

Oil India Limited Sr. Engineer (Production) 30.11.2019
Karnataka PSC Lect., 27.05.2017
HPPSC AE (PWD) 10.09.2016

Ans. (c) : We know that if a system of forces acting on a body or a system of bodies be in equilibrium and the system be imagined to undergo a small displacement with the geometric conditions, then algebraic sum of the virtual work done by all the forces of the system is zero.
For rigid body is in equilibrium the total work of the external forces acting on the rigid body due to virtual displacement of the body is 0.

560. Which of the following types of energy is possessed by a tightly wound watch spring?

(a) Radial energy (b) Kinetic energy
(c) Strain energy (d) Circular energy

KPCL AE 2016
MECON MT 2019

Ans. (c) : A wound spring has potential energy. The spring does not possesses kinetic energy until it is released and starts to move. Hence energy in the form of potential energy is stored in the spring and when it is unwound it transfers its to energy to the moving parts in the form of kinetic energy.
Strain energy is a form of potential energy stored in the material.

561. When a body of mass moment of inertia about its axis (I) is rotated about the axis with an angular velocity (ω), then the kinetic energy of rotation is:

(a) $0.5\ (I^2\omega)$ (b) $2\ (I\omega^2)$
(c) $I\omega^2$ (d) $0.5\ (I\omega^2)$

CIL MT 27.02.2020
TNPSC AE 2013
TNPSC AE 2018

Ans. (d) : Kinetic energy $K = \dfrac{1}{2}mV^2$

Rotational kinetic energy due to the rotation of an object and is part of its total kinetic energy. Looking at rotational energy separately around an object's axis of rotation, the following dependence on the object moment of inertia is observed.

$(KE)_{rotational} = \dfrac{1}{2}I\omega^2$

$(KE)_{rotational} = 0.5\ (I\omega^2)$

562. Two particles with masses in the ratio 1:4 are moving with equal kinetic energies. The magnitude of their linear momentums will conform to the ratio

(a) 1:8 (b) 1:2
(c) $\sqrt{2}:1$ (d) $\sqrt{2}$

ISRO Scientist/Engineer 2007
GPSC DEE Class-2 (GWSSB) 04.07.2021

Ans. (b) :

$$\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2$$

$$\frac{m_1}{m_2} = \left(\frac{v_2}{v_1}\right)^2$$

$$\frac{v_2}{v_1} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$\frac{(momentum)_1}{(momentum)_2} = \frac{P_1}{P_2} = \frac{m_1v_1}{m_2v_2}$$

$$= \frac{1}{4}\times\frac{2}{1}$$

$$\frac{P_1}{P_2} = \frac{1}{2}$$

563. If two bodies one light and other heavy have equal kinetic energies, which one has a greater momentum

(a) Heavy body
(b) Light body
(c) Both have equal momentum
(d) It depends on the actual velocities

TRB Asstt. Prof., 2012
ISRO Scientist/Engineer 2007

Ans. (a) : Kinetic energy are same $\dfrac{1}{2}m_lv_l^2 = \dfrac{1}{2}m_hv_h^2$

$$\frac{m_l}{m_h} = \left(\frac{V_h}{V_l}\right)^2$$

the ratio of momentum

$$\frac{m_hV_h}{m_lV_l} = \left(\frac{m_h}{m_l}\right)\left(\frac{V_h}{V_l}\right)$$

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$$= \left(\frac{V_l}{V_h}\right)^2 \times \frac{V_h}{V_l} = \frac{V_l}{V_h} = \sqrt{\frac{m_h}{m_{ll}}} > 1$$

so heavy body have greater momentum

564. The wheels of a moving car posses

(a) potential energy only
(b) kinetic energy of translation only
(c) kinetic energy of rotation only
(d) kinetic energy of translation and rotation both

TNPSC AE 2019
UKPSC AE 2007 Paper -I

Ans. (d) : The wheels of a moving car posses kinetic energy of translation and rotation both.

565. A boy walks up a stalled escalator in 90 seconds. When the same escalator moves, he is carried up in 60 seconds. How much time would it take him to walk up the moving escalator?

(a) 48 seconds (b) 36 seconds
(c) 30 seconds (d) 24 seconds

ESE 2017
ISRO Scientist/Engineer 2006

Ans. (b) : Let 'ℓ' be length of escalator,

Velocity of boy $v_b = \frac{\ell}{90}$

Velocity of escalator $v_e = \frac{\ell}{60}$

If both start moving,

$$\text{Time} = \frac{\ell}{\left(\frac{\ell}{90}\right)+\left(\frac{\ell}{60}\right)} = 36 \text{ sec.}$$

566. The escape velocity on the surface of the earth is

(a) 11.2 km/s (b) 8.2 km/s
(c) 3.2 km/s (d) 1.2 km/s

UKPSC AE 2012 Paper-I
UKPSC AE 2007 Paper -I
TNPSC AE (Industries) 09.06.2013

Ans. (a) : 11.2 km/s

567. Condition for stable equilibrium of a conservative force system in terms of potential energy U is

(a) $\delta U = 0$ and $\delta^2 U = 0$
(b) $\delta U = 0$ and $\delta^2 U > 0$
(c) $\delta U = 0$ and $\delta^2 U < 0$
(d) $\delta U > 0$ and $\delta^2 U = 0$

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Considering a potential energy U and a displacement of x, the force is given by

$$F = -\frac{\partial U}{\partial x}$$

For stable equilibrium

$$F = -\frac{\partial U}{\partial x} = 0$$

i.e. $\partial U = 0$ and $\frac{\partial^2 U}{\partial^2 x} > 0$

568. A truck hauls a trailer at 40 km/hour when exerting a steady pull of 600N. If the work is done for 40 minutes, the power required is

(a) 6 kW (b) 4 kW
(c) 2.4 kW (d) 10 kW

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : Velocity is given as

$$V = 60 \times \frac{5}{18} = \frac{50}{3} \text{ m/sec}$$

We know that Power

$$P = F \times V = 600 \times \frac{50}{3}$$

P = 10,000 W
P = 10 kW

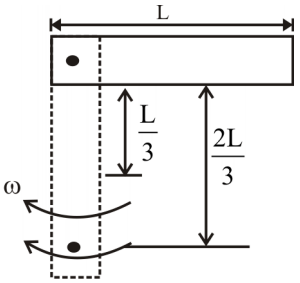
Note:- Power is work done per unit time so it does not matter for how long your hauls it.

569. A thin uniform rod of mass M and length L is hinged at its upper end, and released from rest from a horizontal position. The tension at a point located at a distance L/3 from the hinge point, when the rod becomes vertical, will be

(a) 22 Mg/27 (b) 11 Mg/13
(c) 6 Mg/11 (d) 2 Mg

GPSC EE Pre, 28.01.2017

Ans. (d) :



Taking energy equation, we get

$$\frac{1}{2} I\omega^2 = Mg \times \frac{L}{2}$$

$$\omega^2 \times \frac{1}{2} \times \frac{ML^2}{3} = Mg \times \frac{L}{2}$$

$$\omega = \sqrt{\frac{3g}{L}}$$

Let T be tension at hinge point and T be tension at $\frac{L}{3}$ position.

From FBD-

$$T = Mg + \frac{Mv^2}{r}$$

$$= M[g + r\omega^2]$$

$$= M\left[g + \frac{2L}{3} \times \frac{3g}{L}\right]$$

$$= \frac{2M}{3}[g + 2g] = \frac{2M}{3} \times 3g$$

$$T = 2Mg$$

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570. If D is the distance moved by effort and d is the distance moved by load then the velocity ratio is given by:

(a) d/D (b) D/d
(c) d×D (d) 1/(d×D)

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : $\text{Velocity Ratio} = \frac{\text{Distance moved by effort}}{\text{Distance moved by load}}$

$\text{VR} = \frac{D}{d}$

571. Capacity to exert force through a distance, is called

(a) Entropy (b) Energy
(c) Power (d) Work

Oil India Senior Officer 23.12.2020

Ans. (d) : Capacity to exert force through a distance, is called work.

572. An electric motor drives a punching press. A flywheel fitted to the press has a radius of gyration of 0.5 m and runs at 250 rpm. The press is capable of punching 800 holes per hour with each punching operation taking 1.5 seconds and requiring 12,000 N-m of work. The energy delivered by the motor during punching operation is

(a) 2000 N-m (b) 3000 N-m
(c) 4000 N-m (d) 5000 N-m

ESE 18.07.2021

Ans. (c) : Given, Actual punching time, $t_a = 1.5$ sec
Cycle time or time required to complete one ratio, $T_c = 4.5$ sec.

$\left[\because 800\,\text{hole/hr} \quad 1\,\text{hole}/4.5\,\text{sec.}\right]$

Energy required-

$E_{req} = 12000$ N-m or Joule

$\therefore\ P_{motor} = \frac{E_{req}}{T_c} = \frac{12000}{4.5}$ Joule/sec

$\therefore$ Energy delivered by motor,

$E = P_{motor} \times t_a$

$= \frac{12000}{4.5} \times 1.5$

$= 4000$ Joule

$\boxed{E = 4000\ \text{N} - \text{m}}$

573. For a conservative force, the work done is independent of

(a) path (b) time
(c) distance (d) All of the above

APPSC AEE SCREENING 17.02.2019

Ans. (d) : Conservative force are electrical force, gravitational force and elastic force etc.

574. The energy possessed by a body, for doing work by virtue of its position, is called-

(a) Potential energy (b) kinetic energy
(c) electrical energy (d) chemical energy

TNPSC AE 2018
TNPSC ACF 2012

Ans. (a) : The energy possessed by a body, for doing work by virtue of its position, is called Potential energy.

P. E. = mgh

K. E. = $\frac{1}{2}mv^2$

E. E. = i^2Rt

575. A block A is dropped down along a smooth inclined plane, while another block B is released for free fall from the same height

(a) Both will hit the ground simultaneously
(b) Block A will have higher velocity than block B while hitting the ground
(c) Block A will hit the ground earlier
(d) Block B will hit the ground earlier

Gujarat PSC AE 2019

Ans : (a) : Initially both block have same potential energy and at the lowest point both block have same kinetic energy in absence of friction.
So both block have same velocity and both will hit the ground simultaneously.

576. Wheel has mass 100 kg and radius of gyration of 0.2 m. The additional amount of energy stored in flywheel, if its speed increases from 30 rad/s to 35 rad/s, will be:

(a) 65 J (b) 650 kJ
(c) 650 J (d) 65 kJ
(e) 65 MJ

CGPSC AE 25.02.2018

Ans. (c) : Data given :

m = 100kg k = 0.2
$\omega_1 = 30$ rad/s $\omega_2 = 35$ rad/s

Then additional amount of energy & forced in flywheel,

$\Delta E = E_2 - E_1$

$\Delta E = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$

$= \frac{1}{2}\left[mk^2\right]\left[\omega_2^2 - \omega_1^2\right]$

$= \frac{1}{2} \times 100 \times (0.2)^2 [35-30][35+30]$

$\boxed{\Delta E = 650\ \text{J}}$

577. If the direction of force component and displacement is opposite, then the work done is-

(a) Positive (b) Negative
(c) Zero (d) Both (b) and (c)

MECON MT 2019

Ans. (b) : If the direction of force component and displacement is opposite, then the work done is negative.

578. If a satellite is orbiting around earth and its K.E. is doubled, then its:

(a) Orbits at higher
(b) Orbits in elliptical
(c) Falls back on earth
(d) Escape from earth's pull path

HPPSC ADF 05.03.2019

Ans. (d) : Kinetic energy of satellite = k

$\therefore\ k = kE = \frac{1}{2}mv_0^2;\quad m \rightarrow \text{mass of satelite}$

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$$v_0 = \sqrt{\frac{2k}{m}}$$

We know, for a satellite-

$$\frac{GMm}{r^2} = \frac{mv_0^2}{r} \Rightarrow v_0 = \sqrt{\frac{GM}{r}}$$

Also, escape velocity $= \sqrt{\frac{2GM}{r}} = \sqrt{2}v_0$

So, when kinetic energy is doubled k = 2k

$$v' = \sqrt{\frac{2k'}{m}} = \sqrt{\frac{2(2k)}{m}} = \sqrt{2}\sqrt{\frac{2k}{m}}$$

$$= \sqrt{2}\, v_0$$

Here, we can see on doubling the k.E., the velocity of satellite becomes equal to escape velocity.

579. The dimensional formula $[ML^2T^{-3}]$ represents

(a) work (b) power
(c) force (d) momentum

HPPSC Poly. Lect. 05.07.2021
Assam PSC CCE Pre 2015
UKPSC AE-2013, Paper-I

Ans. (b) : Dimension–

Power $= F.v$

$$= M \cdot \frac{L}{T^2} \cdot \frac{L}{T}$$

$$= ML^2T^{-3}$$

- Work is dimensionally represented as $[M^1L^2T^{-2}]$
- Force is dimensionally represented as $[MLT^{-2}]$
- Momentum is dimensionally is represented $[MLT^{-1}]$

580. What power is required to move a car along a level road at 100 km/hr against a horizontal resisting force of 2 kN along the line of motion if the overall efficiency is 85%?

(a) 100 W (b) 90 W
(c) 942 J (d) 65.3 W

CGPSC AE 15.01.2021

Ans. (*) : Given, Resisting force = 2kN

Velocity (V) = 100 km/hr

$$= 100 \times \frac{5}{18} = 27.77 \text{m/sec}$$

Overall efficiency, $\eta_o = 0.85$

$$\eta_o = \frac{F \times v}{\text{Power required}}$$

$$\text{Power required, (P)} = \frac{2 \times 27.77}{0.85}$$

$$= 65.35 \text{ kW}$$

Note- There is no option given in kW.

581. A train of weight 1200 kN is driven by an engine on a level track at a constant speed of 72 kmph. The resistance due to friction is 2% of the weight of the train. The power of the engine is_____ .

(a) 24 kW (b) 480 kW
(c) 72 kW (d) 96 kW

VIZAG MT, 14.12.2020

Ans. (b) : Given,

W = 1200 kN

v = 72 km/h = 20 m/s

Resistance due to friction = 2%

Power of engine = ?

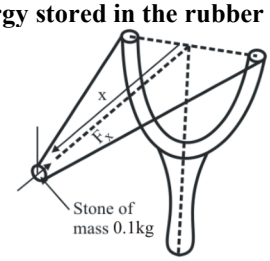
$$P = F_t \times v$$

$$= \mu \times N \times v$$

$$= 0.02 \times 1200 \times 20$$

$$\boxed{P = 480 \text{ kW}}$$

582. A stone with mass of 0.1 kg is catapulted as shown in the figure. The total force F_x (in N) exerted by the rubber band as a function of distance x (in m) is given by $F_x = 300x^2$ is the stone is displaced by 0.1 m from the un-stretched position (x = 0) of the rubber band, the energy stored in the rubber band is :



(a) 0.01 J (b) 0.1 J
(c) 1 J (d) 10 J

Vadodara Muncipal Corp. DEE, 2018
GATE 2011

Ans. (b) : Energy stored in rubber band = Workdone to place the stone

$$U = \int F_x d_x$$

$$= \int_0^{0.1} 300x^2 dx$$

$$= 300\left[\frac{x^3}{3}\right]_0^{0.1}$$

$$= 300 \times \left[\frac{0.1^3}{3} - 0\right]$$

$$\boxed{U = 0.1 \text{ J}}$$

583. If a solid cylinder and a hollow cylinder of the same mass are allowed to roll down an inclined plane simultaneously then.

(a) Hollow cylinder reach the ground first
(b) Solid cylinder reach the ground first
(c) Both reaches the ground at the same time
(d) Unpredictable

TSPSC Managers, 2015

Ans. (b) : Time taken to reach the bottom of the inclination plane.

$$t = \sqrt{\frac{2l\left(1 + \frac{k}{R^2}\right)}{g \sin\theta}}$$

Here, l is the length of inclination plane for solid cylinder $= k^2 = \frac{R^2}{2}$

For hollow cylinder $= k^2 = R^2$

Hence, solid cylinder will reach the bottom first.

584. The law of conservation of mechanical energy is valid :

(a) irrespective of the nature of force field
(b) only in gravitational field

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(c) only if the total work done by the particle in moving around in a closed trajectory
(d) only if the forces is tangential to the trajectory

J&K PSC Civil Services Pre, 2013

Ans. (a) : Law of conservation of mechanical energy is valid irrespective of the nature of force field.

585. Ten watt is equal to
(a) 10 Nm/hr (b) 1 N/mt
(c) 15 KNm/hr (d) 10 Nm/s

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) : We know that,
1 watt = 1 J/sec = 1 Nm/s [$\because$ 1J = 1 N-m]
$\therefore$ 10 watt = 10 Nm/s

586. If a system is in equilibrium and the position of the system depends upon many independent variables, the principle of virtual work states that the partial derivatives of its total potential energy with respect to each of the independent variable must be
(a) –1.0 (b) 0
(c) 1.0 (d) ∞

TSPSC AEE 2017
GATE 2006

Ans. (b) : Total potential energy = (Independent variable)
Hence for a system in equilibrium
Total P.E. = Constant
Thus P.D. of its total P.E. w.r.t. each independent variable must be zero.

587. Which of the following physical quantities is not a vector?
(a) Mass (b) Momentum
(c) Impulse (d) Acceleration

Assam PSC CCE Pre 2015

Ans. (a) : Vector Quantities– Displacement, Weight, Force, Momentum, Acceleration, Velocity
Scalar Quantities– Time, Distance mass, Area, Density, work, Temperature, Speed, Energy, Power

588. The rate of doing work is known as -
(a) Potential energy (b) Power
(c) Kinetic energy (d) Heat energy

GMB AAE 25.06.2017

Ans. (b) : Power : Rate of doing work known as power
It is amount of energy consumed per unit time.

$$P = \frac{dw}{dt} \text{ (w-work)}$$

$$P = \frac{d}{dt}(F.dx)$$

$$P = \left(\frac{dx}{dt}\right).F \qquad \left(\therefore v = \frac{dx}{dt}\right)$$

$$\boxed{P = F.v}$$

The unit of power is J/sec or watt.

589. The expression ($mv^2/2$), where m is the mass and v is the velocity; denotes
(a) centrifugal force (b) kinetic energy
(c) potential energy (d) centripetal force

Haryana PSC Civil Services Pre, 2014

Ans. (b) :

Centrifugal force = $\frac{mv^2}{r} = mr\omega^2$

Kinetic energy = $\frac{1}{2}mv^2$

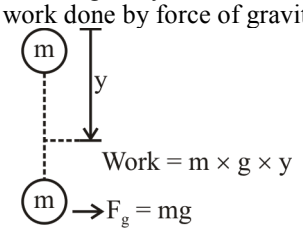
Potential energy = mgh

Centripetal force = equal and opposite of centrifugal force = $\frac{mv^2}{r}$ or $mr\omega^2$

590. When a body is freely falling, the work done by force of gravity is :
(a) Positive (b) Negative
(c) Zero (d) Infinite

MPPSC State Forest Service Exam, 2014

Ans. (a) : As the gravity force acts in downward direction, the work done by force of gravity is positive.

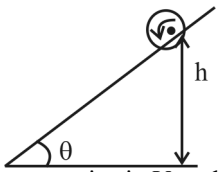


So, the work is +ve.

591. If a sphere, a solid cylinder and a hollow cylinder (all of identical masses) roll down the same inclined plane without sliding, which one will reach the bottom first?
(a) solid cylinder
(b) sphere
(c) hollow cylinder
(d) all of them simultaneously

OPSC Civil Services Pre 2006

Ans. (b) :



Let velocity at bottom point is V and Angular velocity ω
From energy conservation,

$$mgh = \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

$$mgh = \frac{1}{2}\frac{Iv^2}{r^2} + \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{mgh}{\frac{I}{2r^2} + \frac{1}{2}m}}$$

For same, radius and mass, the body which have max mass moment of Inertia 'I', will have least velocity and reach the bottom last.

Mass MOI of solid cylinder or disc = $\frac{1}{2}mr^2$

Mass MOI of hollow cylinder or ring = mr^2

Mass MOI of solid sphere = $\frac{2}{5}mr^2$

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Mass MOI of hollow sphere $= \frac{2}{3}mr^2$

Sphere have least Mass MOI so it will reach bottom first.

592. A particle moves along a straight line with a starting velocity of V_0. If the retarding force on the particle is K times the distance travelled, the distance travelled by the particle before coming to rest is

(a) $\frac{2V_0}{K}$ (b) $\frac{2V_0}{\sqrt{K}}$

(c) $\frac{V_0}{K}$ (d) $\frac{V_0}{\sqrt{K}}$

OPSC Civil Services Pre 2006

Ans. (d) : F = – kx (Restoring force)

Kinetic energy = work done by force

$$-\frac{1}{2}mv_0^2 = -\int_0^x kxdx$$

$$\frac{1}{2}mv_0^2 = \frac{kx^2}{2}$$

$$x = v_0\sqrt{\frac{m}{k}}$$

$$x = \frac{v_0}{\sqrt{k}} \quad (\text{for } m = 1kg)$$

593. When is work said to be done?

(a) When a force acts on a body and the body undergoes displacement

(b) When a force acts on a body and the body undergoes no displacement

(c) When there is no force but the body undergoes a displacement

(d) None of these

KPCL AE 2016

Ans. (a) : Work is said to be done when a force acts on a body and the body undergoes a displacement.

Work = Force × displacement

W = F × d

594. The total energy possessed by a system of moving bodies is_____

(a) Varies according to the direction of movement

(b) Constant at any point in time

(c) Hyperbolically decreasing over time

(d) None of these

KPCL AE 2016

Ans. (b) : The total energy possessed by a system of moving bodies is constant any point in time. Total mechanical energy is the energy possessed by object due to either its motion or its stored energy of position. The total amount of mechanical energy is merely the sum of these two form of energy.

Note-Commission is right answer option (b or d), but actual correct answer is option (b).

595. A planet is moving around the sun in a circular orbit of circumference 'c', The work done on the planet by the gravitational force (F) of the sun in:

(a) $\frac{F}{C}$ (b) zero

(c) $\frac{FC}{2}$ (d) FC

WBPSC AE 2008

Ans. (b) : Planet is moving in circular orbit hence. Its displacement is tangential. The force acting is in radial direction. So force (F) and displacement ds are perpendicular to each other work done W = F.ds cos θ but $\theta = 90^0$

$\boxed{W = 0}$

596. An annular disc has a mass m, inner radius R and outer radius 2R. The disc rolls on a flat surface without slipping. If the velocity of the centre of mass is v, the kinetic energy of the disc is

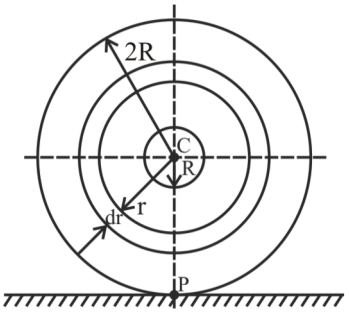
(a) $\frac{9}{6}mv^2$ (b) $\frac{11}{16}mv^2$

(c) $\frac{13}{16}mv^2$ (d) $\frac{15}{16}mv^2$

GPSC Asstt. Director of Transport 05.03.2017

GATE 2014

Ans. (c) :



$$I_C = \frac{1}{2}m(2R)^2 + \frac{1}{2}mR^2$$

$$= \frac{5}{2}mR^2$$

$$I_P = I_C + m(2R)^2$$

$$= \frac{5}{2}mR^2 + 4mR^2$$

$$= \frac{13}{2}mR^2$$

Point P can be considered as the instantaneous centre of rotation with angular velocity $\left(\frac{V}{2R}\right)$.

Kinetic energy $= \frac{1}{2}I_p\omega^2$

$$= \frac{1}{2} \times \frac{13}{2}mR^2 \times \left(\frac{V}{2R}\right)^2$$

$$= \frac{13}{16}mv^2$$

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597. Two metallic blocks having masses in the ratio 2:3 are made to slide down a friction less inclined plane starting initially from rest position. When these blocks reach the bottom of the inclined plane, they will have their kinetic energies in the ratio

(a) 2:3 (b) 3:5
(c) 3:2 (d) 7:4

ISRO Scientist/Engineer 2010
MPSC Poly. Lect. 2014
J&K PSC Civil Services Pre, 2013

Ans. (a) : Let two metallic block having mass $m_1 : m_2 = 2 : 3$

$$\frac{m_1}{m_2} = \frac{2}{3}$$

For mass m_1-
Kinetic energy

$$E_1 = \frac{1}{2} m_1 v^2$$

For mass m_2-
Kinetic energy

$$E_2 = \frac{1}{2} m_2 v^2$$

$$\frac{E_1}{E_2} = \frac{\frac{1}{2} m_1 v^2}{\frac{1}{2} m_2 v^2}$$

$$= \frac{m_1}{m_2} = \frac{2}{3}$$

598. A block whose mass is 650 gm is fastened to a spring of spring constant K equals 65 N/m whose other end is fixed. The block is pulled a distance x = 11cm from its equilibrium position at x = 0 on a smooth surface, and released from rest at t = 0. The maximum speed 'S' of the oscillating block is

(a) 11 cm/sec (b) 11 m/sec
(c) 11 mm/sec (d) 1.1 m/sec

ISRO Scientist/Engineer 2007

Ans. (d) : Maximum speed of spring will be at $x = 0$ when total spring energy will be converted into kinetic energy of mass

$$\frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$\frac{1}{2} \times 65 \times (0.11)^2 = \frac{1}{2} \times (0.650) V^2$$

V = 1.1 m/sec.

599. The units of energy in SI units

(a) Joule (b) Watt
(c) Joule/sec. (d) Watt/sec.

UJVNL AE 2016
UKPSC AE 2012 Paper-I

Ans : (a) The units of energy in SI unit Joule
Energy:-
It may be defined as the capacity to do work. The energy exists in many forms e.g. mechanical, electrical, chemical, heat, light, etc. But we are mainly concerned with mechanical energy.

600. A circular disc rolls down without slipping on an inclined plane. The ratio of its rotational kinetic energy to the total kinetic energy is.

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

UPPSC AE 12.04.2016 Paper-I

Ans : (c) Total kinetic Energy = $\frac{1}{2} I\omega^2 + \frac{1}{2} mv^2$

Rotational kinetic Energy = $\frac{1}{2} I\omega^2$

Moment of Inertia of Circular disc = $\frac{1}{2} mR^2$

Total kinetic Energy = $\frac{1}{2} \times \frac{1}{2} mR^2 \times \omega^2 + \frac{1}{2} m(R.\omega)^2$

Total kinetic Energy = $\frac{1}{4} mR^2\omega^2 + \frac{1}{2} mR^2\omega^2$

Total kinetic Energy = $\frac{3}{4} mR^2\omega^2$

Rotational kinetic Energy = $\frac{1}{2}\left(\frac{mR^2}{2}\right)\omega^2$

Rotational kinetic Energy = $\frac{1}{4} m^2R^2\omega^2$

Ratio of rotational kinetic Energy of the Total Kinetic Energy = $\frac{\frac{1}{4} m^2R^2\omega^2}{\frac{1}{4} mR^2\omega^2 + \frac{1}{2} mR^2\omega^2} = \frac{1}{3}$

601. If a particle is in static equilibrium, then the work done by the system of force acting on that particle is:

(a) Negative (b) Infinity
(c) Zero (d) Positive

CIL MT 26.03.2017

Ans. (c) : Static equilibrium is a state in which the net force and net torque acted upon the system is zero. In other words, both linear momentum and angular momentum of the system are conserved.

602. Which conversion is incorrect?

(a) 1 kWh = 3.6×10^6 Nm
(b) 1 Nm = 0.238×10^{-3} kcal
(c) 1 HP hr = 0.746 kWh
(d) 1kcal = 4.1868 Nm

UPPSC AE 12.04.2016 Paper-I

Ans : (d) (i) 1 kWh = 3.6×10^6 Nm
(ii) 1 Nm = 0.238×10^{-3} kcal
(iii) 1 HP hr = 0.746 kWh
(iv) 1cal = 4.1868 Nm

603. A body of mass 20 kg is lifted up through a height of 4 m. How much work is done? (take g = 9.81 m/s^2)

(a) 648 J (b) 684 J
(c) 748 J (d) 784 J
(e) 848 J

(CGPSC Polytechnic Lecturer 2017)

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Ans. (d) : Work done to left a mass of 20 kg upto 4 m

W = Force × distance

= m × g × 4

= 20 × 9.81 × 4

= 784.80 N-m

= 784 J

604. A man weighing 900 N climbs a staircase of 15 m height in 30 seconds. How much power is consumed?

(a) 150 watt (b) 250 watt

(c) 350 watt (d) 450 watt

(e) 2500 watt

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : W = 900 N, h = 15 m, t = 30 sec.

Power (P) = ?

We know that,

$$P = \frac{\text{Work done}}{\text{time}}$$

$$P = \frac{900 \times 15}{30}$$

P = 450 Watt

605. Torque acting on a body of moment of Inertia (I) and angular acceleration (α) is:

(a) $2\ I\alpha$ (b) $2^2\ I\alpha$

(c) $2^3\ I\alpha$ (d) $I\alpha$

TRB Polytechnic Lecturer 2017

Ans. (d) : $T = I\alpha$

606. A block of 500N is to be moved upward for a distance of 1.6 m on an inclined plane of 45º with horizontal. Work done will be (μ = 0.25):

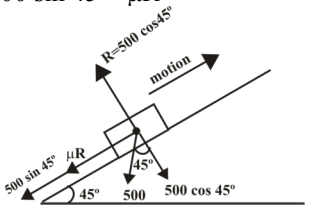
(a) 1000 Nm (b) 100 Nm

(c) $500\sqrt{2}$ Nm (d) 500 Nm

UPRVUNL AE 07.10.2016

Ans. (c) : Total force acting in upward direction

F = 500 sin 45 + μR



$$F = 500 \times \frac{1}{\sqrt{2}} + 0.25 \times 500 \times \frac{1}{\sqrt{2}}$$

$$F = 500 \times \frac{1}{\sqrt{2}} \times 1.25$$

Then to be moved upward for a distance 1.6 m on an inclined plane.

$$W = F \times d = 500 \times \frac{1}{\sqrt{2}} \times 1.25 \times 1.6$$

$$\boxed{W = 500\sqrt{2}\ N-m}$$

607. If m is the mass of the body and g is the acceleration due to gravity then the gravitational force is given by:

(a) $m \times g^3$ (b) $m \times g^2$

(c) m/g (d) m × g

CIL MT 26.03.2017

Ans. (d) : Gravitational force = mass × Acceleration due to gravity = m × g

608. A ball is thrown up. The sum of kinetic and potential energies will be maximum at

(a) the ground (b) the highest point

(c) the centre (d) all the points

TNPSC AE 2017

Ans. (d) : A ball is thrown up. The sum of kinetic and potential energies will be maximum at all the points.

$$\frac{1}{2}\ mv^2 + mgh = \text{constant.}$$

Then we can say that summation of energy will remain constant at each point.

609. The potential energy an elevator losses in coming down from the top of a building to stop at the ground floor is

(a) lost to the driving motors

(b) converted into heat

(c) lost in friction of the moving surfaces

(d) used up in lifting the counter poise weight

TNPSC AE 2017

Ans. (d) : The potential energy an elevator losses in coming down from the top of a building to stop at the ground floor is used up in lifting the counter poise weight.

610. A truck weighing 150 kN and travelling at 2 m/sec impacts with a buffer spring which compresses 1.25 cm per 10 kN. The maximum compression of the spring will be :

(a) 26.6 cm (b) 27.6 cm

(c) 28.6 cm (d) 30.6 cm

OPSC Civil Services Pre. 2011

Ans. (b) : By using conservation of mechanical energy,

$$\frac{1}{2}mv^2 = -\frac{1}{2}kx^2 \text{.....(i)}$$

(–Ve → compression of spring)

$$\frac{150 \times 1000 \times 4}{9.81} = \frac{10 \times 1000 \times x^2}{1.25 \times 10^{-2}}$$

x = 27.65 cm

611. A body is pulled through a distance of 15 m along a level track. The force applied is 400 N, acting at an angle of 60° to the direction of motion. Then the work done is

(a) 13.33 N-m (b) 3000 N-m

(c) 5196.15 N-m (d) 26.66 N-m

TSPSC AEE 2015

Ans : (b) Given,

Distance = 15 m

Applied force = 400 N.

Acting angle = 60°

work done = Fd cos 60

W = 400 × 15 cos 60

$\boxed{W = 3000N.m}$

612. A body is moving with a velocity 1 m/s and a force F is needed to stop it within a certain distance. If the speed of the body becomes three times, the force needed to stop it within the same distance would be

(a) 1.5 F (b) 3.0 F

(c) 6.0 F (d) 9.0 F

UKPSC AE 2012 Paper-I

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Ans. (d) : $\frac{1}{2}mv^2 = Fx_1 \Rightarrow \frac{m}{2} = F_1x$ .....(i)

$\frac{1}{2}m \cdot 3^2 = F_2x$ .....(ii)

From (i) & (ii)

$\boxed{F_2 = 9.0F_1}$

613. When a body is thrown up at an angle of 45° with a velocity of 100 m/sec, it describes a parabola. Its velocity on point of return down will be

(a) zero (b) 50 m/sec

(c) $\frac{100}{\sqrt{2}}$ m/s (d) $100\sqrt{2}$ m/sec

UKPSC AE 2012 Paper-I

Ans. (c) : The velocity on point of return will be the velocity at the maximum height but at the highest point only the constant horizontal velocity (4 cosθ)

$= 100 \cos 45^0$

$= \frac{100}{\sqrt{2}}$ m/s

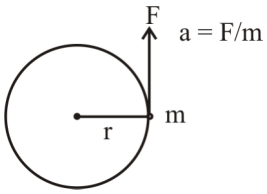
614. A thin circular ring of mass 100 kg and radius 2 m resting on a smooth surface is subjected to a sudden application of a tangential force of 300 N at a point on its periphery. The angular acceleration of the ring will be

(a) 1.0 rad/sec^2 (b) 1.5 rad/sec^2

(c) 2.0 rad/sec^2 (d) 2.5 rad/sec^2

UKPSC AE 2012 Paper-I

Ans. (b) :



Given F = 300 N, m = 100 kg, r = 2 m

Angular Acceleration (a) = rα

By the Newton's second law

F = ma

F = mrα

300 = 100 × 2 × α

$\boxed{\alpha = 1.5 \text{rad/sec}^2}$

615. A train crosses a tunnel in 30 seconds time. The speed of the train at entry and at exit from the tunnel are 36 and 54 km/hour respectively. If acceleration remains constant, the length of the tunnel is

(a) 350 m (b) 360 m

(c) 375 m (d) 400 m

UKPSC AE 2012 Paper-I

Ans. (c) : Given, time t = 30 sec

Distance at entry of tunnel

$s_1 = 36 \times \frac{5}{18} \times 30$

$s_1 = 300$ m

Distance at exit of tunnel –

$s_2 = 54 \times \frac{5}{18} \times 30$

$= 15 \times 30$

$s_2 = 450$ m

Total distance covered by train = (Entry + Exit) of tunnel

$= 450 + 300 = 750$ m

So, here length of train and length of tunnel will be equal,

∵ Train + tunnel = 750

$2s = 750$ $(s = s_1 = s_2)$

$s = 750/2 = 375$ m

616. If T_1 and T_2 are the initial and final tensions of an elastic string and x_1 and x_2 are the corresponding extensions, then the work done is

(a) $(T_2 + T_1)(x_2 - x_1)$ (b) $(T_2 - T_1)(x_2 + x_1)$

(c) $\frac{(T_2 - T_1)(x_2 + x_1)}{2}$ (d) $\frac{(T_2 + T_1)(x_2 - x_1)}{2}$

UKPSC AE 2012 Paper-I

Ans. (d) : $\frac{(T_2 + T_1)(x_2 - x_1)}{2}$

617. A motor boat whose speed in still water is 15 km/hr goes 30 km downstream and comes back in a total time of four and half hours. The stream has a speed of

(a) 3 km/hr (b) 4 km/hr

(c) 5 km/hr (d) 6 km/hr

UKPSC AE 2012 Paper-I

Ans. (c) : Data given

Speed of boat = 15 km/hr

Total time taken will 4.5 hr

Let x be the speed (T) of steam

$(T) = t_1 + t_2$

$T = \frac{30}{S_D} + \frac{30}{S_U}$ $\left\{\text{Where } \begin{matrix} SD = \text{speed of down stream} \\ SU = \text{Speed of upstream} \end{matrix}\right\}$

$T = \frac{30}{S_B + S_S} + \frac{30}{S_B - S_S}$

$4.5 = \frac{30}{15 + x} + \frac{30}{15 - x}$

$450 - 30x + 450 - 30x = 4.5\ (15+x)\ (15-x)$

$900 = 4.5\ (15^2 - x^2)$

$200 = 225 - x^2$

$x^2 = 25$

$x = 5$ km/hr

618. Identify the pair which has same dimensions :

(a) Force and power

(b) Energy and work

(c) Momentum and energy

(d) Impulse and momentum

UKPSC AE 2012 Paper-I

Ans. (b) : Energy and work

619. 0.01 kilowatt is equal to

(a) 10.01 J/s (b) 1.0 J/s

(c) 0.10 J/s (d) 0.01 J/s

UKPSC AE 2007 Paper -I

Ans. (a) : 10.01 J/s

620. The total energy possessed by moving bodies

(a) remain constant at every instant

(b) varies from time to time

(c) is maximum at the start

(d) is minimum before stopping

UKPSC AE 2007 Paper -I

Ans. (a) : Remain constant at every instant

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621. Inertia force of a body is expressed as

(a) product of mass of the body and the acceleration of its centre of gravity in the direction of acceleration

(b) product of mass of the body and the acceleration of its centre of gravity acting in an opposite direction of acceleration

(c) product of linear acceleration of the body and its mass moment of inertia in the direction of acceleration of its centre of gravity

(d) none of the above

UKPSC AE 2007 Paper -I

Ans. (b) : Product of mass of the body and the acceleration of its centre of gravity acting in an opposite direction of acceleration

622. Which one of the following is a scalar quantity?

(a) Force (b) Displacement

(c) Speed (d) Velocity

UKPSC AE 2012 Paper-I

Ans. (c) : Speed

623. A bullet of mass 0.03kg moving with a speed of 400 m/s penetrates 12 cm into a fixed block of wood. The average force exerted by the wood on the bullet will be

(a) 30 kN (b) 20 kN

(c) 15 kN (d) 10 kN

ESE 2017

UKPSC AE-2013, Paper-I

Ans. (b) : m = 0.03 kg

v = 400 m/s

K.E. of bullet = Work done

$$\frac{1}{2}mv^2 = \text{Force} \times \text{distance}$$

$$\frac{1}{2} \times 0.03 \times (400)^2 = \text{Force} \times 0.12$$

F = 20 kN

624. A cricket ball of mass 175 gm is moving with a velocity of 36 km/hr. What average force will be required to stop the ball in 0.2 second?

(a) –5.75 N (b) –6.75 N

(c) –7.75 N (d) –8.75 N

UKPSC AE 2007 Paper -I

Ans. (d) : Given, $m = 175 \times 10^{-3}$ kg

$$n = 36 \times \frac{5}{18} = 10\,m/sec$$

t = 0.2

We know that,

v = u + at

0 = u + at

–at = u

$$a = -\frac{u}{t} = \frac{10}{0.2}$$

$a = -50\ m/s^2$

Now, average force (F) = ma

$= 175 \times 10^{-3} \times -50$

$= -8.75$ N

625. A body moving with a velocity of 1 m/s has kinetic energy of 1.5 Joules. Mass of the body is:-

(a) 0.75 kg (b) 1.5 kg

(c) 3.0 kg (d) 30 kg

UKPSC AE-2013, Paper-I

Ans. (c) : We know that

$$K.E. = \frac{1}{2}mV^2$$

$$1.5 = \frac{1}{2} \times m \times (1)^2$$

$\boxed{m = 3.0\ kg}$

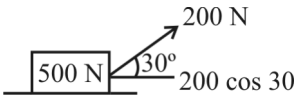
626. A block weighing 500 N resting on a horizontal floor. A force of 200 N applied at 30° with horizontal moves the block to a distance of 60 m. Work done is

(a) 10392.3 Nm (b) 12000 Nm

(c) 6000 Nm (d) 5196 Nm

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) :



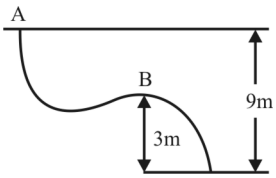
S = 60

Work done = Force × displacement

W = 200 cos 30 ×60

$\boxed{W = 10392.30\ Nm}$

627.



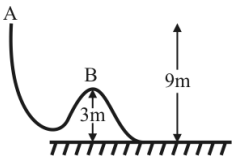
A ball starts rolling from a state of rest at point A, as shown in the figure above, Determine its velocity at point B. Assume the track is smooth.

(a) $3\sqrt{3g}$ (b) $\frac{3\sqrt{3}}{2}g$

(c) $2\sqrt{3g}$ (d) $\sqrt{3g}$

UPSC JWM Adv. No-16/2009

Ans. (c) :



Given the track is smooth

No energy loss is there,

Energy at A = Energy at B

$$mgh_1 = mgh_2 + \frac{1}{2}mv^2$$

$$mgh_1 - mgh_2 = \frac{1}{2}mv^2$$

$$g \times (9-3) = \frac{v^2}{2}$$

$$v = \sqrt{2g \times 6} = \sqrt{12g}$$

$\boxed{V = 2\sqrt{3g}}\ m/s$

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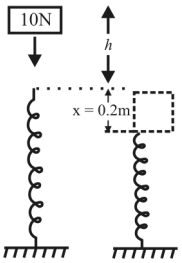
628. A block of weight 10 N falls from a height h meter on a spring of stiffness 1000 N/m, the spring is compressed by 0.2 m. What is the height?

(a) 2.2 m (b) 2.0 m
(c) 1.8 m (d) 1.6 m

UPSC JWM Adv. No-16/2009

Ans. (c) :



Applying Conservation of energy –

$$mg(h + x) = \frac{1}{2}kx^2$$

Given, mg = 10N, x = 0.2 m, k = 1000 N/m

$$\therefore \quad 10 \times (h + 0.2) = \frac{1}{2} \times 1000 \times (0.2)^2$$

$$\Rightarrow \quad h + 0.2 = \frac{500 \times 0.04}{10}$$

$$\Rightarrow \quad h + 0.2 = 2$$

$$\Rightarrow \quad \boxed{h = 1.8m}$$

629. Work done by non-conservation forces on a particle is equal to

(a) Change in kinetic energy
(b) Change in mechanical energy
(c) Change in potential energy
(d) Change in internal energy

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Work done by non-conservative forces changes the mechanical energy of system.

630. A body having weight of 1000 N is dropped from a height of 10 cm over a close coiled helical spring of stiffness 200 N/cm. The resulting deflection of spring is nearly.

(a) 5 cm (b) 16 cm
(c) 35 cm (d) 100 cm

APPSC AEE 2012

Ans. (b) : W = 1000 N, h = 10 cm, k = 200 N/cm
Resulting = ?
When the weight drops at a height of 10 cm. It has potential energy, is absorbed by the spring and undergoes a displacement 'X'

$$P.E = mg(h + x) = \frac{1}{2} \times \text{force} \times \text{displacement}$$

$$\Rightarrow W(h + x) = \frac{1}{2} \times Kx^2$$

$$1000(10 + x) = \frac{1}{2} \times 200 \times x^2$$

$$100 + 10x = x^2$$

$$x = 16 \text{ cm}$$

8. Principle of Virtual Work & simple machines

631. Virtual work refers to :

(a) Virtual work done by Virtual forces
(b) Virtual work done by Actual forces
(c) Actual work done by Actual forces
(d) Actual work done by Virtual forces

TRB Polytechnic Lecturer 2017

Ans. (b) : Virtual work is the work done by a real force acting through a virtual work done by actual forces.

632. Principle of virtual work was developed by

(a) Mohr (b) Poiseuille
(c) Euler (d) Johann Bernoulli

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Principle of virtual work was developed by Johann Bernoulli.

633. By principle of virtual work, the work done by the force is given by {dot.(.) represents dot product, cross (×) represents cross product}

(a) dU = F. dr (b) dU = F× dr
(c) dU=F(dr)tan θ (d) dU=F(dr)cot θ

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : The work done is given by the dot product of force and displacement.
dU=F.dr
dU=F dr cosθ

634. Efficiency of a machine in terms of M.A. (Mechanical advantage) and velocity Ratio (V.R) is given as _____.

(a) V.R./M.A. 100 (b) M.A. × V.R.
(c) V.R/M.A (d) M.A/V.R

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Efficiency of a machine,

$$\eta = \frac{MA}{VR}$$

MA= Mechanical advantage
VR= Velocity Ratio

635. If l_1 is the effort arm and l_2 is the load arm of the lever, the '*leverage*' is defined as

(a) $\frac{l_2}{l_1}$ (b) $\frac{l_1}{l_2}$
(c) $\frac{l_1^2}{l_2^2}$ (d) $\sqrt{\frac{l_1}{l_2}}$

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : In levers , leverage is the ratio of

$$= \frac{\text{Effort Arm}}{\text{Load Arm}} = \frac{\ell_1}{\ell_2}$$

636. In virtual work principle, the work done by the frictional force acting on wheel it rolls without slip is

(a) Zero (b) Positive
(c) Negative (d) All of these

APGCL AM, 2021
HPPSC W.S. Poly. 2016

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Ans. (a) : In virtual work principle, the work done by the frictional force acting on wheel it rolls without slip is zero.

637. The mechanical advantage of a lifting machine is the ratio of

(a) distance moved by effort to the distance moved by load
(b) load lifted to the effort applied
(c) output to input
(d) All of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Mechanical advantage of a lifting machine is the ratio of load lifted to the effort applied.

638. A man is pulling a bucket of water up to the roof of a building of 6 m height. The total weight of the rope is 20 N and the weight of the bucket with water is 100 N. The work done by the man is

(a) 300 N-m (b) 660 N-m
(c) 600 N-m (d) 360 N-m

VIZAG Steel MT 24.01.2021

Ans. (b) : Given,
Depth (h) = 6 m
Rope weight (mg) = 20 N
Weight of bucket of water (Mg) = 100 N
Work done = Change in P.E. of bucket and rope

$$\left[\because \text{Centre of mass of rope} = \frac{h}{2} = \frac{6}{2} = 3\right]$$

$$W = Mgh + mg\frac{h}{2}$$

$$= \left(M + \frac{m}{2}\right)gh$$

$$= \left(Mg + \frac{mg}{2}\right)h$$

$$= \left(100 + \frac{20}{2}\right) \times 6$$

W = 660 N-m

639. According to the Principle of 'Virtual Work', which quantity remains zero during the analysis?

(a) Virtual displacement due to external force
(b) External force due to virtual work
(c) External work due real displacement
(d) Work due to virtual displacement

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Principle of virtual work : It states if a system of force acting on a body or a system of bodies be in equilibrium and if the system be imagined to undergo a small displacement consistent with the geometrical conditions, then the algebraic sum of the virtual work done by all the forces of the system is zero.

640. Virtual work means

1. work done by real forces due to virtual displacement
2. work done by virtual forces during real displacement

(a) Both 1 and 2 are correct
(b) Both 1 and 2 are wrong
(c) 1 is correct and 2 is wrong
(d) 2 is correct and 1 is wrong

APPSC AEE SCREENING 17.02.2019

Ans. (a) :
Virtual work = Virtual force × Real displacement (or)
Virtual work = Virtual displacement × Real force

641. A lifting machine will be reversible, if the efficiency

(a) > 50% (b) equal to 100%
(c) >75% (d) <50%

GPSC Engineer, Class-II Pre- 19.01.2020
APPSC AE Subordinate Service Civil/Mech. 2016

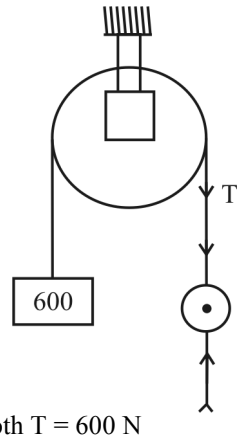
Ans. (a) : Reversible machine – If a machine is capable of doing some work in the reverse direction, after the effort is removed, then the machine is known as reversible machine. The condition for a machine to be reversible, its efficiency should be more than 50%.

642. A pulley is mounted on a vertical pillar above the head of a man of weight 800 N. The man holds one end of a rope that passes over this pulley and to the other end of which is attached a block of weight 600 N. The force with which the man's feet press against the floor is

(a) 200 N (b) 400 N
(c) 600 N (d) 800 N

OPSC Civil Services Pre 2006

Ans. (a)



If pulley is smooth T = 600 N
Force on the earth by feet = Man weight - T
= 800 – 600
= 200 N

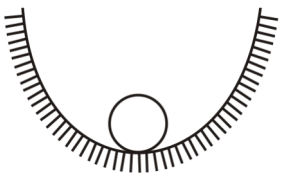
643. In the case of an unstable position of equilibrium of the body, its potential energy .....on any virtual displacement.

(a) decreases (b) increases
(c) is always zero (d) remains constant

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : There three case for position of equilibrium of body.

1. Stable condition– +



Body has minimum potential energy (U)

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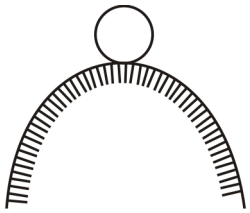
$$\boxed{\frac{d^2U}{dx^2} > 0}$$

For any virtual displacement, U increases.

2. Unstable condition–

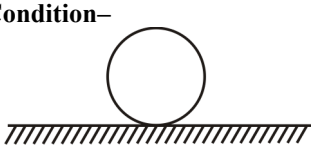


Body has maximum potential energy (U)

$$\boxed{\frac{d^2U}{dx^2} < 0}$$

• For any virtual displacement (U) decreases.

3. Neutral Condition–



Body has constant potential energy (U).

$$\boxed{\frac{d^2U}{dx^2} = 0}$$

for any virtual displacement, (U) unchanged.

644. In track type machines, the purpose of front idler and yoke is -

(a) To align track chain
(b) To replace track chain
(c) To loose track chain
(d) To tight track chain

MPSC Dy. Engineer 2014

Ans. (a) : In track type machines the purpose of front idler and yoke is to align track chain.

645. In dozer, track chains serve mounting pad for-

(a) Track frame (b) Sprocket
(c) Track shoes (d) Front idler

MPSC Dy. Engineer 2014

Ans. (c) : In dozer, track chains serve mounting pad for track shoes.
Dozer track chains are oil filled and designed with more wear material to withstand the constant movement. The NTP excavator track chain is designed with long bushings, reaching into the link to obtain a stable and wear resistant solution.

646. In a lock-gate, the reaction between two gates is
(Where P = Resultant pressure on the lock gate, and α = Inclination of the gate with normal to the side of the lock)

(a) $\frac{p}{2\sin\alpha}$ (b) $\frac{p}{\sin\alpha}$
(c) $\frac{2p}{\sin\alpha}$ (d) $\frac{2p}{\sin(\alpha/2)}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : $\frac{p}{2\sin\alpha}$

647. The law of machine can be expressed mathematically in the form of

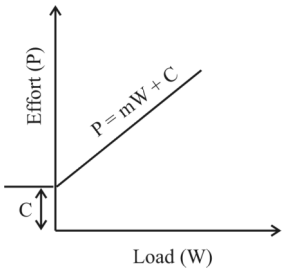
(a) linear (b) exponential
(c) parabolic (d) hyperbolic

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Law of machines – Machines which are used to lift a load are governed by the **'Law of Machine'** which states that the effort to be applied on the machine (P) is related to the weight (w) which it can lift as –

$\boxed{P = mW + C}$ → [Equation of straight line]

Where,
m and C are positive constants which are characteristics of the machine



m = slope of line
C = Effort required to start the machine

648. In lifting machines, as load increases the mechanical advantage

(a) increases
(b) decreases
(c) remains same
(d) initially increases and then decreases

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Lifting machines are those machines that are used for lifting loads. The force (or effect) is applied at one point of one machine and weight (or load) is lifted at the other point of the machine as the load increase the mechanical advantage increases.

649. What is the velocity ratio of a lifting machine that is working at an efficiency of 60% in raising a load of 7700 N by an effort of 150N?

(a) 85.55 (b) 51.33
(c) 61.72 (d) 55

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : $\text{Efficiency} = \frac{\text{Force ratio}}{\text{Velocity ratio}}$

$\text{Velocity Ratio} = \frac{\text{Force ratio}}{\text{Efficiency}}$

$VR = \frac{7700 \times 100}{150 \times 60}$

$VR = 85.55$

650. The velocity ratio as per law of machine is equal to
(Where m = slope of line and is equal to coefficient of friction, W = load lifted and c= amount of effort required by the m/c to overcome the friction)

(a) $m + \frac{c}{W}$ (b) $mW + c$
(c) $\frac{m}{W} + c$ (d) $mc + W$

APPSC AE Subordinate Service Civil/Mech. 2016

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Ans. (a) : Velocity Ratio as per law of machine –

$$VR = m + \frac{c}{W}$$

651. In general, in first order pulley system velocity ratio is given by (Where n is the number of movable pulleys.)

(a) 2^n (b) 2n

(c) n^2 (d) 2n–1

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Velocity ratio of first order pulley system = 2^n.

Where, n = number of movable pulley.

Velocity ratio of second order pulley system = n

Velocity ratio of third order pulley system = n – 1

652. Velocity ratio in wheel and differential axle is given by

(Where D= diameter of effort A, d_2 = diameter of the axle B and d_1 = diameter of the axle C.)

(a) $\frac{2D}{d_2 - d_1}$ (b) $\frac{2D}{d_2 + d_1}$

(c) $\frac{d_2 - d_1}{2D}$ (d) $\frac{d_2 + d_1}{2D}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Differential wheel and Axle

$$\text{Velocity ratio} = \frac{2D}{d_1 - d_2}$$

$$\text{Mechanical advantage} = \frac{W}{P}$$

$$\text{Efficiency} = \frac{M.A.}{V.R.}$$

653. In a first order system of pulleys, there are 3 movable pulleys. What is the mechanical advantage of the system if the efficiency of the system is 80%.

(a) 6.4 (b) 4.6

(c) 6.0 (d) 8.0

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : $V.R = 2^n = 2^3$

$$\text{Efficiency, } \eta = \frac{MA}{VR}$$

$$0.80 = \frac{MA}{2^3}$$

$MA = 0.80 \times 8 = 6.4$

654. If the screw in a screw jack is double threaded then lead of the screw is

(a) 2 × pitch (b) equal to pitch

(c) pitch / 2 (d) $(\text{pitch})^2$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : If the screw in a screw jack is double threaded then load of the screw is equal to 2 times of pitch.

655. In actual machines mechanical advantage is

(a) unity

(b) less than unity

(c) less than velocity ratio

(d) greater than velocity ratio

UKPSC AE 2007 Paper -I

Ans. (c) : An actual machine is that in which there is a loss of energy in some manner. In an actual machine, the work output is less than the work input, i.e. efficiency of an actual machine is always less than 100%.

656. Which one of the following is not an example of plane motion ?

(a) Motion of a duster on a black board.

(b) Motion of ball point of pen on the paper.

(c) Motion of a cursor on the computer screen.

(d) Motion of a nut on a threaded bolt.

UKPSC AE 2012 Paper-I

Ans. (d) : Motion of a nut on a threaded bolt.

657. The velocity ratio of a lifting machine is '8', which lifts a load 900 N by an effort of 150 N. Then, the efficiency of the machine is

(a) 75% (b) 70%

(c) 65% (d) 60%

TSPSC AEE 2015

Ans : (a) Velocity ratio (V.R) = 8

lifted load (w) = 900N

Effort (p) = 150 N.

$$\text{Efficiency } (\eta) = \frac{\text{mechanical advantage}}{\text{velocity ratio.}}$$

$$M.A = \frac{w}{P} = \frac{900}{15} = 6$$

$$\eta = \frac{6}{8}$$

$\eta = 0.75$ or 75%

658. The velocity ratio of a lifting machine is 20 and an effort of 200 N is necessary to lift a load of 3000 N. The frictional load is

(a) 7000N (b) 1000N

(c) 50 N (d) 350N

TSPSC AEE 2015

Ans : (b) Velocity ratio (V.R) = 20

Effort (P) = 200 N.

Lift a load (w) = 3000 N.

Frictional load = (P × VR - lifted load)

frictional load = (20× 200 - 3000)

Frictional load = 1000 N.

659. In an ideal machine, the output as compared to input is

(a) Less

(b) More

(c) Equal

(d) May be less more depending of efficiency

Vizag Steel (MT) 2017

Ans. (c) : In an ideal machine, the output as compared to input is equal.

Ideal machine efficiency is 100%.

660. If the algebraic sum of the virtual work for every displacement is ..... the system is in equilibrium.

(a) zero (b) one

(c) infinity (d) none of these

(HPPSC POLY. LECT. 2016)

Ans : (a) Principle of virtual work:- If a particle is in equilibrium, the total virtual work of forces acting on the particle is zero for any virtual displacement since work done by internal forces [equal, opposite and collinear) cancels each other.

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661. In the third order pulley system of the pulleys, the velocity ratio is given by

(a) $(n^2 - 1)$ (b) $(2^n - 1)$
(c) n^n (d) 2^n
(e) 2n

CGPSC AE 26.04.2014 Shift-I

Ans. (b) Velocity ratio of first order pulley system = 2^n
Velocity ratio of second order pulley system = n
Velocity ratio of third order pulley system = $2^n - 1$

662. In a lifting machine, an effort of 500 N is to be moved by a distance of 20 m to raise a load of 10,000 N by a distance of 0.8m. Determine the efficiency of the machine.

(a) 70% (b) 75%
(c) 80% (d) 85%
(e) 90%

CGPSC AE 26.04.2014 Shift-I

Ans. (c) : Given,
load (W) = 10,000 N
effort (P) = 500 N
Distance moved by the effort (D) = 20 m
Distance moved by the load (d) = 0.8 m

Mechanical Advantage (MA) = $\frac{W}{P} = \frac{10000}{500} = 20$

Velocity Ratio (VR) = $\frac{D}{d} = \frac{20}{0.8} = 25$

Efficiency = $\frac{MA}{VR} = \frac{20}{25} = 0.8 = 80\%$

663. In a lifting machine, an effort of 200 N is applied to raise a load of 800 N. what will be the velocity ratio, if efficiency is 50%.

(a) 8 (b) 6
(c) 7 (d) 9

RPSC IOF & BOILER 20.09.2017

Ans : (a) Given Effort,
P = 200 N
W = 800 N
Mechanical Advantage = $\eta \times VR$

$$\frac{W}{P} = \frac{800}{200} = 0.5 \times VR$$

V.R. = 8

664. In virtual work equation some forces are neglected. Select the most appropriate answer from the following:

(a) Reaction of a rough surface on a body which rolls on it without slipping.
(b) Reaction of any smooth surface with which the body is in contact.
(c) Reaction at a point or on an axis, fixed in space, around which a body is constrained to turn
(d) All of the above.

UPPSC AE 12.04.2016 Paper-I

Ans : (d) All of the above

665. In virtual work principle the work done by self weight of body is taken into consideration when

(a) centre of gravity moves vertically
(b) centre of gravity moves horizontally
(c) shear centre moves horizontally
(d) shear centre moves vertically

TSPSC AEE 2015

Ans. (a) : In virtual work principle the work done by self weight of body is taken into consideration when centre of gravity moves vertically.

666. The maximum mechanical advantage of a lifting machine is

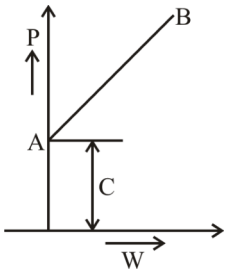
(a) 1/m
(b) m
(c) 1-m
(d) 1+m (Law of machine: $\rightarrow P - mw \div c$)

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Law of Machine–It is the relationship between the load lifted (W) and the effort applied (P). It is given by the equation,

$$P = m.W + C$$

Where, m = A constant (called coefficient of friction) which is equal to the slope of the line AB as shown in fig. 1.23, and
C = Another constant, which represent the machine friction.



Maximum Mechanical Advantage–

Mechanical advantage (MA) $= \frac{W}{P}$

$$MA = \frac{W}{mW + C} = \frac{1}{m + \frac{C}{W}}$$

For MA $\rightarrow$ max, $\frac{C}{W} \rightarrow 0$

$$\boxed{(MA)_{max} = \frac{1}{m}}$$

9. Impulse, Momentum and Collision

667. The coefficient of restitution for inelastic bodies is

(a) zero
(b) between zero and one
(c) one
(d) more than one

APPSC Poly. Lect. 2013
APPSC AEE MAINS 2016, PAPER-III
OPSC Civil Services Pre. 2011
GPSC Executive Engineer 23.12.2018
UPRVUNL AE 05.07.2021
GPSC DEE, Class-II (GWSSB) 04.07.2021

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Ans. (a) : The coefficient of restitution for inelastic bodies is e = 0.
for elastic bodies, it is e = 1

668. During inelastic collision of particles, which of the following is conserved?
(a) Total linear momentum only
(b) Total kinetic energy only
(c) Both linear momentum and kinetic energy
(d) Neither linear momentum nor kinetic energy

TSPSC AEE 2017
RPSC IOF & Boiler 20.09.2017
GATE 2007

Ans. (a) : In both elastic and inelastic collision total linear momentum remains conserved. In the inelastic collision loss in K.E. occurs because the coefficient of restitution is less than one.

669. The co-efficient of restitution of a perfectly plastic impact is
(a) 0 (b) 1
(c) 2 (d) ∞

TSPSC AEE 2017
GPSC Asstt. Director of Transport 05.03.2017
CGPSC AE 15.01.2021
TSPSC AEE 2015
GATE 2011

Ans. (a) : The coefficient of restitution is measure of the bounciness of a collision b/w two objects.

$$e = \frac{\text{speed of separation}}{\text{Speed of approach}}$$

$e = 1 \rightarrow$ for perfectly elastic collision.
$e = 0 \rightarrow$ For perfectly plastic collision.

670. Two particles with masses in the ratio 1 : 4 are moving with kinetic energies. The magnitude of their linear momentum will conform to the ratio :
(a) 1 : 8 (b) 1 : 2
(c) 2 :1 (d) 4 : 1

J&K PSC Civil Services Pre, 2010
ISRO Scientist/Engineer 2009
GPSC EE Pre, 28.01.2017
TRB Poly. Lect., 2012

Ans. (b) : Two particles with masses,

$\frac{m_1}{m_2} = \frac{1}{4}$ and $(KE)_1 = (KE)_2$

$$\therefore \frac{(\text{kinetic energy})_1}{(\text{kinetic energy})_2} = \frac{\frac{1}{2}m_1v_1^2}{\frac{1}{2}m_2v_2^2}$$

$$\frac{m_1v_1^2}{m_2v_2^2} = \frac{\frac{(m_1v_1)^2}{m_1}}{\frac{(m_2v_2)^2}{m_2}}$$

[$\therefore$ Let, linear momentum $(P_1) = m_1v_1$
linear momentum $(P_2) = m_2v_2$]

$$\frac{(KE)_1}{(KE)_2} = \frac{P_1^2}{m_1} \times \frac{m_2}{P_2^2} \Rightarrow \left(\frac{P_1}{P_2}\right)^2 \times \frac{m_2}{m_1}$$

$$\frac{(KE)_1}{(KE)_2} = \left(\frac{P_1}{P_2}\right)^2 \times \frac{m_2}{m_1}$$

$$\frac{m_1}{m_2} = \left(\frac{P_1}{P_2}\right)^2 \qquad \left[\because (KE)_1 = (KE)_2\right]$$

$$\left(\frac{P_1}{P_2}\right) = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

671. In a closed system, the total momentum is constant. This statement is known as____
(a) Law of conservation of momentum
(b) Law of closed momentum
(c) Law of Constance
(d) None of these

KPCL AE 2016
VIZAG STEEL MT 2011
UPRVUNL AE 05.07.2021

Ans. (a) : Law of conservation of momentum in a closed system, the total momentum is constant.
$\because \quad m_1v_1 = m_2v_2$

672. If e = 1.0, the bodies are
(a) inelastic (b) slightly elastic
(c) perfectly elastic (d) any of the above

WBPSC AE 2008
KPCL AE 2016
Assam PSC CCE Pre 2015
J&K PSC Civil Services Pre, 2010
Gujarat PSC AE 2019

Ans. (c)
If $e = 1 \Rightarrow$ perfectly elastic body
If $e = 0 \Rightarrow$ perfectly inelastic body
If $e < 1 \Rightarrow$ perfectly elastic body

673. A mass m_1 of 100 kg travelling with a uniform velocity of 5 m/s along a line collides with a stationary mass m_2 of 1000 kg. After the collision, both the masses travel together with the same velocity. The coefficient of restitution is
(a) 0.6 (b) 0.1
(c) 0.01 (d) 0

GPSC Asstt. Director of Transport 05.03.2017
NPCIL ET, 2015
GATE 2014

Ans. (d) :

$$\text{Co-efficient of restitution} = \frac{(\text{Velocity of separation})}{(\text{Velocity of approach})}$$

$$= \frac{v_2 - v_1}{u_1 - u_2}$$

Given,
$u_1 = 0$, $v_1 = v_2 = v$, $u_1 = 5$ m/s

$$e = \frac{(v - v)}{(5 - 0)} = 0$$

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674. Which of the following statement is correct:

(a) The kinetic energy of a body during impact remains constant

(b) The kinetic energy of a body before impact is equal to the kinetic energy of a body after impact

(c) The kinetic energy of a body before impact is less than the kinetic energy of a body after impact

(d) The kinetic energy of a body before impact is more than the kinetic energy of a body after impact

CGPSC AE 16.10.2016
TNPSC AE 2013

Ans. (d) : The kinetic energy of a body before impact is more than the kinetic energy of a body after impact is correct statement.

675. If v_1 and v_2 are the initial velocities of two bodies making direct collision and if u_1 and u_2 are their respective velocities after collision then the coefficient of restitution is given by:

(a) $\frac{(u_1 - u_2)}{(v_2 - v_1)}$ (b) $\frac{(u_1 - u_2)}{(v_1 - v_2)}$

(c) $\frac{(u_1 + u_2)}{(v_1 - v_2)}$ (d) $\frac{(u_1 - v_2)}{(u_1 - u_2)}$

CIL MT 26.03.2017
TNPSC AE 2014

Ans. (a) : The energy dissipation during impact is called by the term, coefficient of restitution, a scalar quantity

$$e = \frac{\text{velocity of separation}}{\text{velocity of approach}}$$

$$e = \frac{u_1 - u_2}{v_2 - v_1}$$

676. A ball 'A' of mass 'm' falls under gravity from a height 'h' and strikes another ball 'B' of mass 'm' which is supported at rest on a spring of stiffness 'k'. The impact between the balls is perfectly elastic. Immediately after the impact:

(a) The velocity of ball A is $\frac{1}{2}\sqrt{2gh}$

(b) The velocity of ball A is zero

(c) The velocity of both the balls is $\frac{1}{2}\sqrt{2gh}$

(d) None of the above

OPSC Civil Services Pre. 2011
TSPSC AEE 2017
GATE 1996

Ans. (b) : The velocity of ball A is zero

Velocity of ball A before collision

$u_A = \sqrt{2gh}$

Velocity of ball B before collision

$u_B = 0$

For perfectly elastic impact

Velocity of Approach = velocity of separation

$u_A - u_B = v_B - v_A$

$v_B - v_A = \sqrt{2gh}$.....(i)

Using momentum conservation

$mu_A + mu_B = mv_A + mv_B$

$m\sqrt{2gh} + 0 = m(v_A + v_B)$

$v_A + v_B = \sqrt{2gh}$.....(ii)

From (i) and (ii)

$\boxed{v_A = 0}$ $\boxed{v_B = \sqrt{2gh}}$

677. In collisions, coefficient of restitution is defined as ratio of

(a) relative velocity of bodies before impact to relative velocity of bodies after impact.

(b) relative velocity of bodies after impact to relative velocity of bodies before impact

(c) difference in velocity of heavier body before and after impact to difference in velocities of lighter body before and after impact

(d) difference in velocity of lighter body before and after impact to difference in velocities of heavier body before and after impact

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Coefficient of Restitution (e)

$$e = \frac{\text{Velocity of Separation}}{\text{Velocity of approach}}$$

678. For a perfectly elastic body, the value of co-efficient of restitution is

(a) zero (b) 0.372

(c) 0.5 (d) 1.0

VIZAG STEEL MT 18.08.2013
GATE 2011

Ans. (d) : Coefficient of restitution of perfectly elastic body is "1.0".

679. If U_1 and U_2 are the velocities of approach of two moving bodies in the same direction and their corresponding velocities of separation are V_1 and V_2 then as per Newton's law of collision of elastic bodies, the coefficient of restitution e is given by

(a) $e = \frac{V_1 - V_2}{U_1 - U_2}$ (b) $e = \frac{U_2 - U_1}{V_1 - V_2}$

(c) $e = \frac{V_1 - V_2}{U_2 - U_1}$ (d) $e = \frac{V_2 - V_1}{U_1 - U_2}$

TNPSC AE, 2008
TSPSC AEE 2015
TNPSC AE 2014

Ans. (d) :

Coefficient of Restitution (e)

$$= \frac{\text{Relative velocity after collision}}{\text{Relative velocity before collision}}$$

$$\boxed{e = \frac{V_2 - V_1}{U_1 - U_2}}$$

680. The ratio of minimum exergy which must be consumed to performance the task to the actual amount of energy consumed in performing the same task is called as

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(a) first law efficiency
(b) second law efficiency
(c) exergy efficiency
(d) None of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) :

681. Propulsive coefficient ____ if the difference between jet and airplane velocity increases.

(a) becomes zero (b) decreases
(c) does not change (d) increases

UPRVUNL AE 05.07.2021

Ans. (c) : Propulsive coefficient does not change if the difference between jet and airplane velocity increases.

682. Loss of kinetic energy due to the direct impact of two bodies depends on mass of those two bodies and

(a) difference between initial and final velocities
(b) initial velocity of the two bodies
(c) final velocity of the two bodies
(d) product of initial velocities

UPRVUNL AE 05.07.2021

Ans. (c) :

683. The ratio of magnitude of linear momentum for two objects having mass 30 kg and 10 kg respectively with equal kinetic energy is

(a) $\sqrt{\frac{1}{3}}$ (b) $(3)^2$
(c) $\sqrt{3}$ (d) $\left(\frac{1}{\sqrt{3}}\right)^2$

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : Given, $m_1 = 30$ kg, $m_2 = 10$ kg

$$K = \frac{1}{2}mv^2$$

$$K = \frac{1}{2}\frac{mv^2 \times m}{m}$$

Kinetic energy $(K) = \frac{1}{2}\frac{P^2}{m}$

$$K_1 = K_2$$

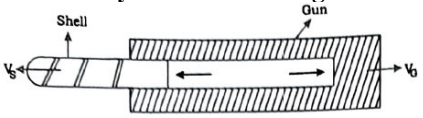
$$\frac{P_1^2}{m_1} = \frac{P_2^2}{m_2}$$

$$\frac{P_1}{P_2} = \sqrt{\frac{m_1}{m_2}}$$

$$= \sqrt{\frac{30}{10}}$$

$$= \sqrt{3}$$

684. A gun of mass 3000 kg fires horizontally a shell of mass 50 kg with a velocity of 300 m/s. What is the velocity with which the gun will recoil?



(a) – 5 m/s (b) 10 m/s
(c) 50 m/s (d) 30 m/s

UPPSC AE 13.12.2020, Paper-I

Ans. (a) : Given, $m_g = 3000$ kg, $m_b = 50$ kg
$v_g = ?$, $v_b = 300$ m/s

Before firing, both the gun and bullet are at rest, hence total momentum of bullet and gun = 0.

Applying law of conservation of momentum

Initial momentum = Final momentum

$$0 = m_g v_g + m_b v_b$$

$$v_g = -\frac{m_b v_b}{m_g}$$

$$= -\frac{50 \times 300}{3000} = -5\,m/s$$

685. Newton's second law of motion connects

(a) change of momentum and velocity
(b) momentum and acceleration
(c) momentum and rate of change of force
(d) rate of change of momentum and external force

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : Newton's second law of motion,

$$\vec{F} = m\vec{a}$$

$$F = m\frac{dv}{dt}$$

$$F = \frac{m(v_2 - v_1)}{dt}$$

$$\boxed{F = \frac{dp}{dt}}$$

External force = Rate of change of momentum.

686. A cube strikes a stationary ball exerting an average force of 50 N over at time of 10ms. The ball has mass of 0.20 kg. Its speed after the impact will be

(a) 3.5 m/s (b) 2.5 m/s
(c) 1.5 m/s (d) 0.5 m/s

ESE 2018

Ans. (b) : Given data

F = 50 N
u = 0
m = 0.2 kg
t = 10 ms = 10×10^{-3} sec

$$F_{avg} = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{\Delta t}$$

$$50 = \frac{mv}{\Delta t} = \frac{0.2 \times v}{10 \times 10^{-3}}$$

v = 2.5 m/s

687. If (F) refers to force, (m) refers to mass, (v) refers to velocity and (t) refers to time, then which of the following equation is known as momentum principle?

(a) $F = \frac{d(m^2 v)}{dt}$ (b) $F = \frac{dv}{dt}$
(c) $F = \frac{d(mv)}{dt}$ (d) $F = \frac{d(mv)}{dt^2}$

TNPSC AE 2019
UKPSC AE 2007 Paper-II

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Ans. (c) : Momentum principle- Momentum is the quantity of motion of a moving body.

From Newton's second law

$$\vec{F} = m\vec{a}$$

$$= m \times \frac{dv}{dt} \qquad m = \text{constant}$$

$$\vec{F} = \frac{d(mv)}{dt}$$

This above equation is known as momentum principle equation.

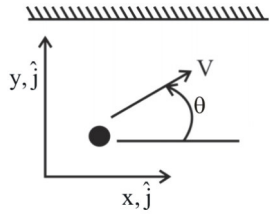
688. In elastic collisions of bodies:

(a) Only the total kinetic energy is conserved.
(b) Neither momentum of the colliding bodies nor the total kinetic energy are recoverable.
(c) Only the total momentum of the colliding objects is conserved.
(d) Both of the momentum and total kinetic energy are conserved.

CIL MT 27.02.2020

Ans. (d) : An elastic collision is an encounter between two bodies in which the total kinetic energy of the two bodies remains the same. In an ideal, perfectly elastic collision, there is no net conversion of kinetic energy into other forms such as heat, noise, or potential energy.

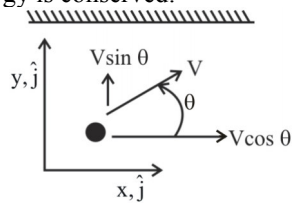
689. A point mass having mass M is moving with a velocity V at an angle θ to the wall as shown in the following figure. The mass undergoes a perfectly elastic collision with the smooth wall and rebounds. The total change in the momentum of the mass is



(a) $-2MV\cos\theta\, j$ (b) $2MV\sin\theta\, j$
(c) $-2MV\sin\theta\, j$ (d) $2MV\cos\theta\, j$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2016

Ans. (c) : Due to perfectly elastic collision momentum as well as energy is conserved.



In x- direction there is no obstruction so momentum (P_x) = constant = MV cosθ

In y direction momentum before collision $P_{y1} = MV\sin\theta\hat{j}$

In y direction momentum after collision $P_{y2} = MV\sin\theta\hat{j}$

Charge in moment $= P_{y2} - P_{y1}$

$= -MV\sin\theta\hat{j} - MV\sin\theta\hat{j}$

$= -2MV\sin\theta\hat{j}$

690. A small ball of mass 1 kg moving with a velocity of 12 m/s undergoes a direct central impact with a stationary ball of mass 2 kg. The impact is perfectly elastic. The speed (in m/s) of 2 kg mass ball after the impact will be _____.

(a) 6 (b) 8
(c) 12 (d) 16

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (b) : Given, $m_1 = 1kg$, $m_2 = 2kg$, $u_1 = 12$, $u_2 = 0$

Momentum before collision = momentum after collision

$1 \times 12 + 2 \times 0 = 1 \times V_1 + 2 \times V_2$

$V_1 + 2V_2 = 12$.....(1)

Coefficient of restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2} = 1 \qquad \text{(For perfectly elastic collision)}$$

$$e = 1$$

$$\frac{v_2 - v_1}{12 - 0} = 1$$

$v_2 - v_1 = 12$ .....(2)

Eq^n (1) + (2)

$v_2 - v_1 = 12$

$2v_2 + v_1 = 12$

$3v_2 = 24$

$v_2 = 8$ m/sec.

691. When a bullet is fired from a gun, it is coiled in the backward direction, it is due to :

(a) Impulse
(b) Inertia
(c) Conservation of momentum
(d) None of these

J&K PSC Civil Services Pre, 2010

Ans. (c) : When a bullet is fired from a gun, it is coiled in the backward direction, it is due to conservation of linear momentum.

692. Two objects moving with uniform speeds are 5m apart, after 1 second when they move towards each other and are 1m apart, when they move in same direction. The speed of the objects are:

(a) 2m/sec and 2m/sec
(b) 3m/sec and 2m/sec
(c) 3m/sec and 3m/sec
(d) 4m/sec and 6m/sec

TSPSC Managers, 2015

Ans. (b) : Given- Let two object having velocity v_1 and v_2

$$v_1 + v_2 = 5 \qquad .....(i)$$

$$v_1 - v_2 = 1 \qquad .....(ii)$$

from equation (i) and (ii)

$$2v_1 = 6$$

$$\boxed{v_1 = 3 \text{ m/sec}}$$

$v_1 = 3$ substituting in equation (i)

$$3 + v_2 = 5$$

$$v_2 = 5 - 3$$

$$\boxed{v_2 = 2\text{m/sec}}.$$

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693. A ball is dropped vertically downward from the top of a building and another one is thrown horizontally, which will strike ground first?

(a) One dropped vertically
(b) One thrown horizontally
(c) Both will strike simultaneously
(d) It will depend

TSPSC Managers, 2015

Ans. (c) : A ball is dropped vertically downward from the top of a building and another one is thrown horizontally then both will strike simultaneously. Neglect subsequent motion and air resistance.

694. A ball of mass 0.1 kg, initially at rest, is dropped from height of 1 m. Ball hits the ground and bounces off the ground. Upon impact with the ground, its velocity reduces by 20%. The height (in m) to which the ball will rise is:

(a) 0.61 (b) 0.62
(c) 0.63 (d) 0.64

NPCIL ET, 2015
GATE 2014

Ans. (d) : Given- h = 1 m

$$v = \sqrt{2gh} = \sqrt{2\times 9.81\times 1}$$

$v = 4.42944$ m/sec

$\because$ When velocity reduced 20%, so

$v_1 = 80\%$ of v

$$v_1 = \frac{80}{100}\times 4.42944$$

$$v_1 = 3.5435$$

$$h = \frac{v_1^2}{2g} = \frac{(3.5435)^2}{2\times 9.81}$$

$= 0.6399$

$h = 0.64$ m

695. Relation between coefficient of restitution (e) and velocities of bodies (v) is given by:

(a) $e = (-)\dfrac{(v_2)f-(v_1)f}{(v_2)i-(v_1)i}$

(b) $e = \dfrac{(v_2)f-(v_1)f}{(v_2)i-(v_2)i}$

(c) $e = (-)\dfrac{(v_2)i-(v_1)i}{(v_2)f-(v_1)i}$

(d) $e = \dfrac{(v_2)i-(v_1)i}{(v_2)f-(v_1)f}$

(e) $e = \dfrac{(v_2)f-(v_1)f}{(v_2)i-(v_1)f}$

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : As we know that,

Before The Collision

After the Collision

$$\text{Coefficient of restitution (e)} = \frac{\text{Velocity of separation}}{\text{Velocity of Approach}}$$

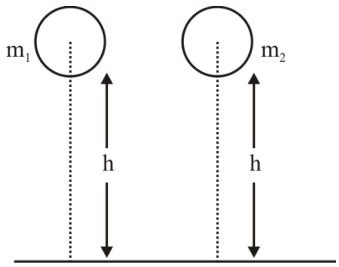
$$\boxed{e = \frac{(v_2)_f-(v_1)_f}{(v_1)_i-(v_2)_i}}$$

696. Two bodies of masses m_1 and m_2 are dropped from the height. The ratio of their momentum at the ground is

(a) m_1/m_2 (b) m_2/m_1
(c) $(m_1/m_2)^2$ (d) $(m^2/m^1)^{1/2}$

Haryana PSC Civil Services Pre, 2014

Ans. (a) :



Momentum of body (1)

$P_1 = m_1 v$

$= m_1 \times \left[\sqrt{u^2+2gh}\right]$ (u = 0)

$P_1 = m_1 \times \sqrt{2gh}$ …….. (1)

Momentum of body (2)

$P_2 = m_2 v$

$= m_2 \times \sqrt{u^2+2gh}$ (u = 0)

$P_2 = m_2 \times \sqrt{2gh}$ …….. (2)

$$\frac{P_1}{P_2} = \frac{m_1\sqrt{2gh}}{m_2\sqrt{2gh}}$$

$$\boxed{\frac{P_1}{P_2} = \frac{m_1}{m_2}}$$

697. A ball of mass 1 kg moving with a velocity of 2m/sec collides directly on a stationary ball of mass 2 kg and comes to rest after impact. The velocity of the second ball after impact will be

(a) zero (b) 0.5 m/s
(c) 1.0 m/s (d) 2.0 m/s

TRB Asstt. Prof., 2012
WBPSC AE, 2017

Ans. (c) : Before and after collision linear momentum conserved the

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$m_1 = 1$kg; $m_2 = 2$kg

u_1 and u_2 = Velocity of the body before collision.

v_1 and v_2 = Velocity of the body after collision.

$u_1 = 2$m/sec $u_2 = 0$

$v_1 = 0$ $v_2 = ?$

$1\times 2 + 0 = 0 + v_2 \times 2$

$\boxed{v_2 = 1\text{m/sec}}$

698. A ball falls vertically for 2 seconds and hits a plane inclined at 30° to the horizontal. If the co-efficient of restitution = 0.75, the time that elapses before it again hits the inclined plane is

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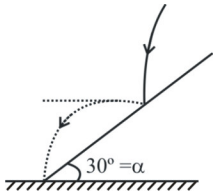
(a) 3 seconds (b) 2 seconds

(c) $\frac{3}{2}$ seconds (d) 1 second

OPSC Civil Services Pre 2006

Ans. (a) :



Velocity with which ball hits will be

$V = u + gt = g \times 2$

Velocity of ball after hitting will = $e \times V = 0.75 \times 2g = 1.5\ g$

α = Indication of plane with horizontal

β = Angle of projection with horizontal

time of flight 't' $= \frac{2u\sin(\alpha+\beta)}{g\sin\beta}$

If, projection in downward direction

$$= \frac{1.5g \times 2 \times \sin(30+0)}{g\sin 30} = 3\,\text{sec}.$$

699. A sphere impinges directly on a second identical sphere at rest. If the coefficient of restitution is 'e', the ratio of the velocity of the first sphere to that of the second sphere after the impact is

(a) $\frac{1-e}{1+e}$ (b) $\frac{1+e}{1-e}$

(c) $\frac{e}{1+e}$ (d) $\frac{e}{1-e}$

OPSC Civil Services Pre 2006

Ans. (a) : Let initial velocities are u_1 & u_2 and v_1 & v_2 are final velocity

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ $\quad m_1 = m_2,\ u_2 = 0$

$u_1 = v_1 + v_2$.....(1)

Co-efficient of restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{\text{relative velocity of seperation}}{\text{Relative velocity of approach}}$$

$$e = \frac{v_2 - v_1}{u_1} \Rightarrow eu_1 = v_2 - v_1$$

$$eu_1 = v_2\left(1 - \frac{v_1}{v_2}\right)$$

$$\frac{eu_1}{v_2} = 1 - \frac{v_1}{v_2} \text{.....(2)}$$

from eq. (1) $\frac{u_1}{v_2} = \frac{v_1}{v_2} + 1$ < putting eq. (2)

$$e\left(\frac{v_1}{v_2} + 1\right) = 1 - \frac{v_1}{v_2}$$

$$e\frac{v_1}{v_2} + e = 1 - \frac{v_1}{v_2}$$

$$\frac{v_1}{v_2} = \frac{1-e}{1+e}$$

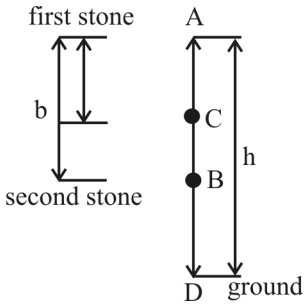
700. A stone is dropped from the top of a tower of height h. At the instant it has fallen a distance c meter, a second stone is released from rest from a point b meter below the top of the tower (b > c). The two stones strike the ground at the same time. The height of the tower is

(a) $\frac{(b+c)^2}{4c}$ (b) $\frac{(b-c)^2}{4c}$

(c) $\frac{b^2}{4c}$ (d) $\frac{4c}{(b+c)^2}$

OPSC Civil Services Pre 2006

Ans. (a) :



Velocity of first stone at 'C' $= \sqrt{2gc}$

velocity of second stone at B = 0

as both stone reach ground simultaneously

For first stone

$$h - c = \sqrt{2gc} \times t + \frac{1}{2}gt^2 \text{.....(1)}$$

For second stone

$$h - b = 0 \times t + \frac{1}{2}gt^2 \text{.....(2)}$$

or $t = \sqrt{\frac{2(h-b)}{g}}$ - putting in eq. (1)

$$h - c = \sqrt{2gc} \times \sqrt{\frac{2(h-b)}{g}} + (h-b)$$

$$b - c = \sqrt{4c(h-b)} \text{ or } \frac{(b-c)^2}{4c} + b = h$$

$$h = \frac{b^2 + c^2 - 2bc + 4bc}{4c} \text{ or } h = \frac{(b+c)^2}{4c}$$

701. A force F acts on a particle for a short time *t*. The product *F*. *t* will be equal to the change of:

(a) kinetic energy (b) impact energy

(c) momentum (d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Impact energy

702. The linear momentum of a system of rigid bodies will be conserved, if:

(a) the system is in a gravitational field

(b) no external forces act on the system

(c) kinetic energy is conserved

(d) None of the above

BPSC Asstt. Prof. 29.11.2015

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Ans. (b) : The Linear momentum of a system of rigid bodies will be conserved if no external forces act on the system.

703. A ball A of mass 0.25 kg moving on a smooth horizontal table with a velocity of 10 m/sec strikes an identical stationary ball B on the table. If the impact is perfectly plastic, the velocity of the ball B just after the impact will be :

(a) 5 m/sec (b) 10 m/sec
(c) zero (d) 2.5 m/sec

MPSC Poly. Lect. 2014

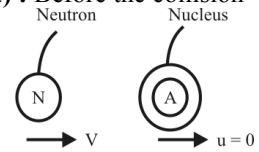
Ans. (a) : $M_A = M_B = 0.25$ kg
$u_1 = 10$ m/s.
$u_2 = 0$
$V_B = ?$
after plastic collision two bodies moves stick together coefficient of resistance is zero.
$e = 0$
conservation of momentum
$M_A \times u_1 + M_B \times u_2 = (M_A + M_B) \times V_B$
$0.25 \times 10 + 0.25 \times 0 = (0.25 + 0.25) \times V_B$
$\boxed{V_B = 5 \text{ m/s}}$

704. A neutron moving at a speed V undergoes a head on elastic collision with a nucleus of mass number A at rest. The ratio of the kinetic energies of the neutron after and before the collision is :

(a) $\left(\frac{A-1}{A+1}\right)^2$ (b) $\left(\frac{A+1}{A-1}\right)^2$
(c) $\left(\frac{A}{A+1}\right)^2$ (d) $\left(\frac{A}{A-1}\right)^2$

WBPSC AE 2003

Ans. (a) : Before the collision



after the collision

e = 1 for elastic collision

$$e = \frac{(V_2)_f - (V_1)_f}{(V_1)_i - (V_2)_i} \qquad \begin{bmatrix} f = \text{final} \\ i = \text{initial} \end{bmatrix}$$

$$1 = \frac{V_2 - V_1}{V - (0)}$$

$$[V_2 - V_1 = V] \qquad ...(i)$$

Conservation of linear momentum is applicable since, $\Sigma F_{ext} \rightarrow$ external force in the direction of collision.

$$\Sigma F_{ext} = 0$$

$$(P_{system})_i = (P_{system})_f$$

$$(m_N)V + 0 = A(m_N)V_2 + (m_N) \times V_1$$

$$V + 0 = AV_2 + V_1$$

$$[V = AV_2 + V_1] \qquad ...(ii)$$

$$\frac{(KE)_f}{(KE)_i} = \frac{(1/2\, m_N)(V_1)^2}{(1/2\, m_N)(V)^2} = \frac{(V_1)^2}{(V)} \qquad ...(iii)$$

using equation (i) & (ii)

$$AV_2 + V_1 = V$$

$$-AV_2 + AV_1 = VA$$

$$V_1 (1+A) = (1-A)V$$

$$\left[V_1 = \left(\frac{1-A}{1+A}\right)V\right] \rightarrow \text{keep is (iii)}$$

$$\frac{(KE)_f}{(KE)_i} = \left(\frac{V_1}{V}\right)^2 = \frac{\left(\frac{1-A}{1+A}\right)^2 V^2}{V^2} = \left(\frac{1-A}{1+A}\right)^2$$

$$\boxed{\frac{(KE)_f}{(KE)_i} = \left(\frac{A-1}{A+1}\right)^2}$$

705. If the momentum of a given body is doubled, its kinetic energy will

(a) get halved
(b) remain same
(c) increase by four times
(d) increase by six times

WBPSC AE 2008
UKPSC AE Paper-I, 2007

Ans. (c)

$$K.E = \frac{1}{2} mv^2 = \frac{m^2v^2}{2m}$$

$$K.E_1 = \frac{P^2}{2m} \qquad -----(1)$$

$$K.E_2 = \frac{(2P)^2}{2m} = 4\left(\frac{P^2}{2m}\right) \qquad -----(2)$$

Comparing (1) & (2)

$$\boxed{K.E_2 = 4(K.E)}$$

706. The value of coefficient of restitution lies between zero and one, when the impact is:

(a) Perfectly elastic (b) Perfectly plastic
(c) Semi-elastic (d) None of the above

HPPSC AE (PWD) 10.09.2016

Ans. (c) : The coefficient of restitution (e) is the ratio of the final to initial relative velocity between two object.

$$e = \frac{\text{Velocity of sepration}}{\text{Velocity of approach}}$$

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

If ranges from 0 to 1
e = 1 perfectly elastic
e = 0 perfectly plastic
0 < e < 1 Partially elastic collision (semi-elastic)

707. A big vehicle moving at 20km per hour hits a smaller vehicle half its mass while coasting in front of it, in neutral at 10km/hr along the same straight line on a level road. Assuming the coefficient of restitution as 0.3 estimate their speeds after collosion.

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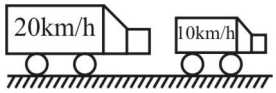


(a) $V_1 = 5.18$ m/s and $V_2 = 4.35$ m/s
(b) $V_1 = 5.35$ m/s and $V_2 = 6.18$ m/s
(c) $V_1 = 4.35$ m/s and $V_2 = 5.18$ m/s
(d) $V_1 = 2.35$ m/s and $V_2 = 3.18$ m/s

UPRVUNL AE 21.08.2016

Ans. : (c) $e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{v_2 - v_1}{20 - 10} = 0.3$

$v_2 - v_1 = 3$ km/h

Linear momentum conservation

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

$m \times 20 + \frac{m}{2} \times 10 = m \times v_1 + \frac{m}{2} \times v_2$

$25 = \frac{v_1 + v_2}{2}$ ... (ii)

for equation (i) and (ii)

$v_1 = 15.66$ Km/h = 4.3 m/sec

$v_2 = 18.6$ Km/h = 5 m/sec

708. Which of the following is NOT true?

(a) For direct impact of two bodies, coefficient of restitution is the ratio of relative velocity of approach of the two bodies to their relative velocity of separation
(b) Conservation of angular momentum implies that total angular momentum of system remains constant unless acted on by an external torque
(c) Conservation of linear momentum in a given direction implies that the surce external forces in that direction is zero
(d) The coefficient of friction is independent of area of contact

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Co-efficient of restitution

$= \frac{\text{Relative velocity of separation}}{\text{Relative velocity of approach}}$

709. For two bodies having direct impact, co-efficient of restitution is given by

(a) $\frac{\text{Velocity of approach}}{\text{Velocity of separation}}$

(b) $\frac{\text{Velocity of separation}}{\text{Velocity of approach}}$

(c) $\frac{\text{Initial velocity of the body}}{\text{Final velocity of the body}}$

(d) None of the above

ISRO Scientist/Engineer 24.05.2014

Ans. (b) : Coefficient of restitution

$e = \frac{\text{Velocity of separation}}{\text{Velocity of approach}}$

$= \frac{v_2 - v_1}{u_1 - u_2}$

where u_1 and u_2 are initial velocities of the two collide bodies and v_1, v_2 are their final velocities after collision.

710. Two objects P & Q are travelling horizontally with velocity of 8m/sec & 6m/sec from left to right. They are separated by a distance of 15m. The mass of the objects are 3 kg and 5 kg. If the coefficient of restitution is 0.7 what is the velocity (m/s) of P and Q after impact and when (seconds) and where (meters) will they impact with respect to initial positioning of Q. The corresponding answers are respectively.

(a) 7.6, 5.4, 2.1, 15 (b) 8, 6, 2.5, 7.5
(c) 7.6, 6.2, 7.5, 45 (d) None of these

ISRO Scientist/Engineer 12.05.2013

Ans. (d) : Conservation of Linear momentum

$M_P \times 8 + M_Q \times 6 = M_P V_P + M_Q V_R$

or, $3 \times 8 + 5 \times 6 = 3V_P + 5V_R$

or, $3V_P + 5V_R = 54$ ...(1)

Coefficient of Restitution

$e = \left(\frac{V_Q - V_P}{8 - 6} \right)$

$e = \frac{velocity\ of\ separation}{velocity\ of\ Approach}$

or, $V_Q - V_P = 1.4$ ...(2)

$5V_R + 3V_P = 54 \times 1$

$V_Q - V_P = 1.4 \times 3$

or, $5V_Q + 3V_Q = 54$

$3V_Q - 3V_P = 4.2$

or, $8V_Q = 58.2$

or, $8V_Q = 58.2$

$V_Q = 7.275$ m/s

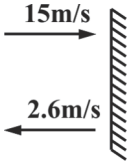
$V_P = 5.875$ m/s

711. A car crashes against a wall. The initial velocity at collision is 15m/ sec and the velocity after collision is 2.6 m/sec in the opposite direction. The mass of the car is 1500 kg, what is the average force exerted on the automobile bumper if collision lasts for 0.15 seconds.

(a) 1.76×10^5 N (b) 2.1×10^5 N
(c) 2.76×10^5 N (d) None of these

ISRO Scientist/Engineer 12.05.2013

Ans. (a) :



Change in Linear momentum $(\vec{P} = m\vec{V}cm)$

$\frac{dP}{dt} = \frac{d(mv)}{dt} = m\frac{dv}{dt} = ma = (\vec{F}R)External$

$\frac{d\vec{P}}{dt} = (\vec{F}R)Ext.$

$F = \frac{mv_1 + mv_2}{dt}$

$F = \frac{1500(15 + 2.6)}{0.15}$

$F = 1.76 \times 10^5$ N

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712. A uniform circular ring of mass M and radius r is rotating with an angular speed ω about an axis passing through its center and perpendicular to the plane of the ring. Two identical beads, each of mass m, some how get attached at two diametrically opposite points. The rotational speed of the ring will become

(a) $\frac{\omega M}{M+2m}$ (b) $\frac{\omega M}{M+m}$

(c) $\frac{2\omega M}{M+2m}$ (d) ω

ISRO Scientist/Engineer 2009

Ans. (a) : Initial angular momentum = final angular momentum

$$MR^2\omega = 2mR^2\omega' + MR^2\omega'$$

$$\omega' = \frac{\omega M}{M+2m}$$

713. Only rocket engines can be propelled to space because

(a) They can generate very high thrust
(b) They have high propulsion efficiency
(c) These engines can work on several fuels
(d) They are not air-breathing engines

ISRO Scientist/Engineer 2007

Ans. (d) : Only Rocket engines can propelled to space because in space air breathing engines will not work because no air present in space.

714. A ball is dropped from a height of 10m to a surface which rebounds to a height of 2m. The coefficient of restitution of the surface is :

(a) $\sqrt{0.2}$ (b) 0.6

(c) $\sqrt{0.5}$ (d) 08

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : Coefficient of restitution (e) $= \sqrt{\frac{h_{after}}{h_{before}}}$

$$= \sqrt{\frac{2}{10}} = \sqrt{0.2}$$

715. The angular momentum of a body of mass m rotating about an axis with angular velocity

(a) mωr (b) $m\omega r^2$

(c) mω (d) $m\omega^2 r$

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Angular momentum (L) = Iω
where $I = mr^2$
then

$\boxed{L = mr^2w}$

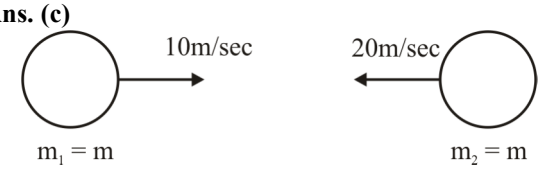
716. A perfectly elastic ball moving with a velocity 10 m/s impinges directly another perfectly elastic ball of same mass moving in opposite direction with a velocity 20m/s. The velocity of the first ball after impact will be

(a) 15 m/s (b) 20 m/s

(c) 10 m/s (d) 14.14 m/s

Kerala LBS Centre For Science & Tech. Asstt. Prof. 2014

Ans. (c)



Where u_1 & u_2 velocity first ball before and after collision and v_1 & v_2 are the velocity of second ball before and after collision respectively.

From conservation of momentum

$m_1u_1 + m_2v_1 = m_1u_2 + m_2v_2$

$u_1 + v_1 = u_2 + v_2$

$10 - 20 = u_2 + v_2$

$u_2 + v_2 = -10$ m/sec .....(i)

$$e = \frac{u_2 - v_2}{u_1 - v_1}$$

$u_1 - v_1 = u_2 - v_2$

$10 + 20 = u_2 - v_2$

$u_2 - v_2 = 30$ .....(ii)

From equation (i) & (ii)

$2u_2 = 20$

$u_2 = 10$ m/sec

And $v_2 = -20$ m/sec

717. A body of mass 20 kg falls freely under gravity. What will be its momentum after 3 seconds? (take g = 10 m/s^2)

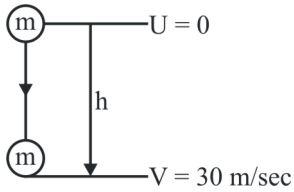
(a) 300 N sec (b) 400 N sec

(c) 500 N sec (d) 600 N sec

(e) 700 N sec

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : Given m = 20 kg, g = 10 m/sec^2
Then velocity after 3 seconds



$v = u + gt$

$v = 0 + 10 \times 3$

$\boxed{v = 30\ m/sec}$

Then momentum after 3 seconds

M = Final momentum – initial momentum

$= mv - mu$

$= 20 \times 30 - 20 \times 0$

$\boxed{M = 600\ N - sec.}$

718. A stone of mass is tied to an inextensible massless string of the length *l* and rotated in vertical circle. The minimum speed required at the top is

(a) $\sqrt{(0.5lg)}$ (b) $\sqrt{(lg)}$

(c) $\sqrt{(2lg)}$ (d) $\sqrt{(3lg)}$

(e) $\sqrt{(4lg)}$

CGPSC AE 26.04.2014 Shift-I

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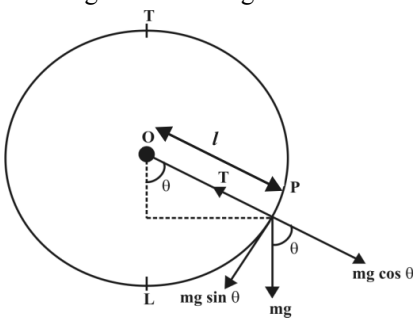
Ans. (b) : A particle of mass m is attached to a light and inextensible string. The other end of the string is fixed at O and the particle moves in a vertical circle of radius r equal to the length of the string as shown in the figure.



Consider the particle when it is at the point P and the string makes an angle θ with vertical forces acting on the particle are

T = tension in the string along its length

mg = weight of the particle vertically downwards

Hence, net radial force on the particle

$$F_R = T - mg\cos\theta$$

$$\frac{mv^2}{l} = T - mg\cos\theta$$

$$T = \frac{mv^2}{l} + mg\cos\theta$$

Since speed of the particle decrease with height, hence tension is maximum at bottom (i.e. θ = 0) and tension is minimum at top (i.e. θ = 180)

$$T_{max} = \frac{mv_L^2}{l} + mg$$

$(\cos 0^o = 1)$

$$T_{min} = \frac{mv_T^2}{l} - mg$$

$(\cos 180^o = 1)$

For v_T to be minimum, $T \simeq 0$

$$\frac{mv_T^2}{l} - mg = 0$$

$$v_T^2 = lg$$

$$v_T = \sqrt{lg}$$

719. A ball is dropped on a smooth horizontal surface from height 'h'. What will be the height of rebounce after second impact. If coefficient of restitution between ball and surface is 'e'?

(a) e^2h (b) eh

(c) e^3h (d) e^4h

UPRVUNL AE 07.10.2016

Ans. (d) : e = co-efficient of restitution

$$e = \frac{V_2}{V_1} = \frac{\sqrt{2gh_2}}{\sqrt{2gh_1}} = \sqrt{\frac{h_2}{h_1}} = \sqrt{\frac{h_3}{h_2}}$$

$h_2 = e^2h_1$

$h_3 = e^2h_2 = e^2 \times e^2\, h_1$

$\boxed{h_3 = e^4h_1}$

720. Principle of conservation of momentum is

(a) the initial momentum is greater than final momentum

(b) the initial momentum is equal to final momentum

(c) the initial momentum is smaller than final momentum

(d) the initial momentum is equal to zero

TNPSC AE 2013

Ans. (b) : The principle of conservation of momentum states that when you have an isolated system with no external forces, the initial total momentum of object before collision equals the final total momentum of the object after the collision.

721. A steel ball of weight 0.1 N falls a height 6 m and rebounds to a height of 4 m. The impulse is

(a) 0.0201 N - s (b) 0.201 N - s

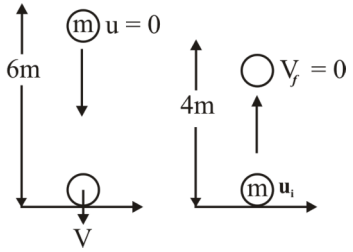
(c) 1.205 N - s (d) 12.05 N - s

TNPSC AE 2013

Ans. (b) : $v^2 = u^2 + 2 \times g \times h$

$v^2 = 2 \times 9.81 \times 6$

$\boxed{v \downarrow = 10.8498\ m/s}$



$(v_f)^2 = u_i^2 - 2gh$

$u_i^2 = 2 \times 9.81 \times 4$

$\boxed{u_i \uparrow = 8.8588\ m/s}$

then, $v_{final} = u_i \uparrow$ and $v_{initial} = V \downarrow$

Then impulse

$I = \Delta P$

$I = m\,[V_{final} - V_{initial}]$

$$= \frac{0.1}{9.81}\left[8.858 - (-10.8498)\right]$$

$I = 0.2009 = 0.201\ N - s$

722. Momentum equations are derived from

(a) First Law of Thermodynamics

(b) Newton's First Law

(c) Newton's Second Law of Motion

(d) Second Law of Thermodynamics

TNPSC AE 2013

Ans. (c) : Momentum equations are derived from Newton's Second Law of Motion.

$$F_{net} = m\vec{a} = \frac{d(mv)}{dt} = m\frac{dv}{dt}$$

$$= \frac{mv_1 - mv_2}{v_2} \quad = \frac{P_1 - P_2}{dt} = \frac{dP}{dt}$$

Equation (1) is known as momentum equation.

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723. If a constant force 'F' acts on a body of mass 'm' for time 't' and changes its velocity from u to v under an acceleration of 'a' all in the same direction, then for equilibrium of the body

(a) $F = \frac{mu}{t}$ (b) $F = \frac{mv}{t}$

(c) $F = m\left(\frac{v-u}{t}\right)$ (d) $F = m\left(\frac{v+u}{t}\right)$

TNPSC AE 2017

Ans. (c) : We know that

Newton's second law of motion-

$F_{net} = ma = m\frac{dv}{dt}$

$F \times t = m \times$ [change in velocity]

$F \times t = m\,[v_f - v_i]$ ...(1)

Then, we can say that product of force and time equal to change in linear momentum in the direction of force. acting

$$\boxed{F = m\frac{(v-u)}{t}}$$

724. A particle is dropped from a height of 3m on a horizontal floor, which has a coefficient of restitution with the ball of 1/2. The height to which the ball will rebound after striking the floor is

(a) 0.5 m (b) 0.75 m

(c) 1.0 m (d) 1.5 m

TSPSC AEE 2015

Ans. (b) : $e = \sqrt{\frac{h_2}{h_1}} = \frac{1}{2}$

$\frac{h_2}{3} = \frac{1}{4}$

$$\boxed{h_2 = 0.75\ m}$$

725. A bullet of mass 1 kg if fired with a velocity of u m/s from a gun of mass 10 kg. The ratio of kinetic energies of bullet and gun is

(a) 10 (b) 11

(c) 1.1 (d) 1.0

TSPSC AEE 2015

Ans. (a) : Total initial momentum of gun and bullet

$= m_1u_1 + m_2u_2 = 0$

Total momentum of gun and bullet after firing-

$= m_1 v_1 + m_2 v_2 = 1 \times u + 10 \times v_2$

Law of conservation of linear momentum

Total momentum after firing = Total momentum before firing

Then, $v_2 = -\frac{u}{10}$

It is recoil velocity of gun $= -\frac{u}{10}$ m/s.

Then, $(K.E.)_B = \frac{1}{2}m_1v_1^2 = \frac{1}{2}\times 1 \times u^2$

$$(K.E.)_G = \frac{1}{2}m_2v_2^2 = \frac{1}{2}\times 10 \times \left(\frac{u}{10}\right)^2 = \frac{(K.E.)_B}{10}$$

$$\boxed{\frac{(K.E.)_B}{(K.E.)_G} = 10}$$

726. Two balls of equal mass and of perfectly elastic material are lying on the floor. One of the ball with velocity 'v' is made to struck the second ball. Both the balls after impact will move with a velocity.

(a) v (b) v/2

(c) v/4 (d) v/8

TSPSC AEE 2015

Ans. (b) : Given as,

$m_1 = m_2 = m$

$u_1 = v$, $u_2 = 0$

$e = 1$

According to momentum conservation principle-

$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$

$mv + m\times 0 = mv' + mv'$

$v' \times 2m = mv$

$$\boxed{v' = \frac{v}{2}}$$

727. A lead ball with a certain velocity is made to strike a wall, it falls down, but rubber ball of same mass and with same velocity strikes the same wall. It rebounds. Select the correct reason from the following-

(a) both the balls undergo an equal change in momentum

(b) the change in momentum suffered by rubber ball is more than the lead ball

(c) the change in momentum suffered by rubber is less than the lead ball

(d) none of the above

TSPSC AEE 2015

Ans. (b) : A rubber ball which have same mass with same velocity strikes the same wall as lead ball, rubber ball rebounds because of the change in momentum suffered by rubber ball is more than the lead ball.

728. Rate of change of momentum takes place in the direction

(a) of applied force

(b) of motion

(c) opposite to the direction of applied force

(d) perpendicular to the direction of motion

(KPSC AE 2015)

Ans : (a) Rate of change of momentum takes place in the direction of applied force.

Rate of change of momentum $= \frac{d}{dt}(mv) = m\frac{dv}{dt}$

Rate of change of momentum $= m\frac{dv}{dt}$

$$\boxed{\vec{F}_{net} = m\vec{a} = m\frac{d\vec{v}}{dt}}$$

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729. Two masses 2 kg, 8 kg are moving with equal kinetic energy. The ratio of magnitude of their momentum is.

(a) 0.25 (b) 0.50
(c) 0.625 (d) 1.00

UPPSC AE 12.04.2016 Paper-I

Ans : (b) Kinetic Energy $= \frac{1}{2}mv^2$

$$(KE)_1 = \frac{p_1^2}{2m_1} \text{.....(i)}$$

$$(KE)_2 = \frac{p_2^2}{2m_2} \text{.....(ii)}$$

Given, $m_1 = 2kg, \quad m_2 = 8kg$

kinetic energy equal $\frac{p_1^2}{2m_1} = \frac{p_2^2}{2m_2}$

$$\frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2}}$$

$$\frac{p_1}{p_2} = 0.50$$

730. When two bodies collide without the presence of any other force or force fields?

(a) Their total kinetic energy must be conserved.
(b) Their total momentum must be conserved.
(c) Their collision must be direct.
(d) Both (a) and (b)

UPPSC AE 12.04.2016 Paper-I

Ans : (d) When two bodies collide without the presence of any other force or force fields.
(i) Their total kinetic energy must be conserved.
(ii) Their total momentum must be conserved.

731. If the momentum of a body is doubled, its kinetic energy will be

(a) doubled (b) quadrupled
(c) same (d) halved

UKPSC AE 2007 Paper -I

Ans. (b) : Quadrupled

We know, $K_1 = \frac{p^2}{2m}$

$$\boxed{K \propto p^2}$$

If the momentum is doubled then,

$$K_2 = 4\left(\frac{p^2}{2m}\right)$$

$$K_2 = 4K_1$$

732. The bodies which rebound after impact are called

(a) elastic (b) inelastic
(c) plastic (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (a) : Elastic

733. The total momentum of a system of moving bodies in any one direction remains constant, unless acted upon by an external force in that direction. This statement is called

(a) Principle of conservation of energy
(b) Newton's second law of motion
(c) Newton's first law of motion
(d) Principle of conservation of momentum

UKPSC AE 2007 Paper -I
GPSC Executive Engineer 23.12.2018

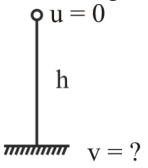
Ans. (d) : Principle of conservation of momentum

734. The velocity of a body on reaching the ground from a height 'h', is given by

(a) $v = 2gh$ (b) $v = 2gh^2$
(c) $v = \sqrt{2gh}$ (d) $v = \frac{h^2}{2g}$

APPSC Poly Lect. Automobile Engg., 2020
UKPSC AE 2007 Paper -I

Ans. (c) If a body fall from height h then



By the equation
$v^2 = u^2 + 2gh$
$v^2 = 0 + 2gh$
$v = \sqrt{2gh}$

735. Impulse is:-

(a) Minimum momentum
(b) Maximum momentum
(c) Average momentum
(d) Final momentum - Initial momentum

UKPSC AE-2013, Paper-I

Ans. (d) :
Impulse = Final momentum – Initial momentum

736. A ball of 2kg drops vertically onto the floor with a velocity of 20m/s. It rebounds with an initial velocity of 10m/s, impulse acting on the ball during contact will be:-

(a) 20 (b) 40
(c) 60 (d) 30

UKPSC AE-2013, Paper-I

Ans. (c) : We know that, impulse is equal to change in momentum
So, initial momentum
$= 2 \times 20$
$= 40$ kg - m/s
Final momentum
$= -2 \times 10$
$= -20$ kg-m/s
Then impulse
$= [-20 - 40]$
$= -60$ kg-m/s
$= 60$ kg-m/s
So, the impulse is 60 Ns acting upwards.

737. The loss of kinetic energy, during inelastic impact of two bodies having masses m_1 and m_2, which are moving with velocity v_1 and v_2 respectively, is given by

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(a) $\frac{m_1m_2}{2(m_1+m_2)}(v_1-v_2)^2$

(b) $\frac{2(m_1+m_2)}{m_1m_2}(v_1-v_2)^2$

(c) $\frac{m_1m_2}{2(m_1+m_2)}(v_1^2-v_2^2)$

(d) $\frac{2(m_1+m_2)}{m_1m_2}(v_1^2-v_2^2)$

UKPSC AE 2012 Paper-I

Ans. (a) : $\frac{m_1m_2}{2(m_1+m_2)}(v_1-v_2)^2$

738. The force applied on a body of mass 100 kg to produce an acceleration of 5 m/s^2 is

(a) 500 N (b)100 N

(c) 20 N (d)10 N

TSPSC AEE 2015

Ans : (a) Second law of Newton's :-

$F = m.\frac{dv}{dt}$

$F = ma$

$F = 500$ N.

739. A rubber ball is dropped from a height of 2 m. if there is no loss of velocity after rebounding the ball will rise to a height of

(a) 1 m (b) 2 m

(c) 3 m (d) 4 m

Vizag Steel (MT) 2017

Ans. (b) : For rebounding height attain = e^2x

then $h_1 = e^2x$

for $x = 2$ m

No loss in kinetic energy so for elastic body e = 1

$h_1 = 1 \times 2$

for second rebounding

$h_2 = e^4x$

$x = 4$ m

$h_2 = 1\times4$

$h_2 = 4$ m

then $\frac{h_1}{h_2} = \frac{2}{4}$

$\frac{h_1}{h_2} = \frac{1}{2}$

$\boxed{h_2 = 2h_1}$

740. During elastic impact, the relative velocity of the two bodies after impact is _____ the relative velocity of the two bodies before impact

(a) equal to (b) equal and opposite to

(c) less than (d) greater than

TSPSC AEE 2015

Ans. (b) : During elastic impact, the relative velocity of the two bodies after impact is equal and opposite to the relative velocity of the two bodies before impact.

741. The law of motion involved in the recoil of gun is

(a) Newton's first law of motion

(b) Newton's second law of motion

(c) Newton's third law of motion

(d) none of these

TNPSC ACF 2012

Ans. (c) : The law of motion involved in the recoil of gun is Newton's third law of motion. According to Newton's third law motion if an object A exerts a forces on object B then object B must exert a force of equal magnitude and opposite direction back on object A.

742. An impulse force $\vec{F} = 3t^2\hat{i} + 2t\hat{j} + \hat{k}$ acts at the centre of a stationary body of mass 1 kg from time t = 0 to t = 1 s. The magnitude of the velocity attained by the body is,

(a) 3 m/s (b) 2 m/s

(c) 1.8 m/s (d) 1.71 m/s

DRDO Scientists 2009

Ans. (d) : $F = 3t^2\hat{i} + 2t\hat{j} + \hat{k}$

$\frac{mdv}{dt} = F$

$dv = \int F dt$

$dv = \int (3t^2\hat{i} + 2t\hat{j} + \hat{k})dt$

$v = t^3\hat{i} + t^2\hat{j} + t\hat{k}$

$= 1\hat{i} + 1\hat{j} + 1\hat{k}$

$v = \sqrt{3} = 1.7$ m/s

743. Line of impact is :

(a) Common surface to the two colliding bodies

(b) Common tangent to the two colliding bodies

(c) Common normal to the two colliding bodies

(d) Common point to the two colliding bodies

TNPSC AE (Industries) 09.06.2013

Ans. (c) : Line of impact is common normal to the two colliding bodies.

744. A 20 kg car moving at a speed of 20 m/s to the right collides with a 35 kg car which is at rest. After the collision the 35 kg car is moving towards right at a speed of 10 m/s. The speed of the 20 kg car is

(a) 2.5 m/s (b) –2.5 m/s

(c) 50 m/s (d) 0

DRDO Scientists 2008

Ans. (a) : From momentum conservation

$m_1 v_1 = m_2 v_2 + m_1 v_f$

$20\ (20) = 35\ (10) + 20\ (v_f)$

$v_f = 2.5$ m/s

745. Two bodies collide each other and the collision is perfectly elastic. The co-efficient of restitution of these two bodies is

(a) 0 (b) 0.5

(c) 1 (d) 2

TANGEDCO AE 2015

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Ans. (c) : Coefficient of restitution,

$$e = \frac{\text{Impulse of recovery}}{\text{Impulse of deformation}} = \frac{\int F_r dt}{\int F_d dt}$$

F_r = Impulse force at recovery

F_d = Impulse force at deformation

$$e = (-)\frac{V_b' - V_a'}{V_b - V_a} = (-)\frac{\text{Velocity of separation}}{\text{Velocity of approach}}$$

1. (e = 0) = For plastic impact (no recovery)
- Only momentum is conserved.

2. (e = 1) = For elastic impact (complete recovery)
- Energy remains conserved (Initial kinetic energy = Final kinetic energy).
- Momentum is conserved.

746. A 1.0 kg ball drops vertically onto the floor with a speed of 25 m/s. It rebounds with an initial speed of 10 m/s. The impulse acting on the ball during contact will be

(a) 15 N-s (b) 25 N-s
(c) 35 N-s (d) 45 N-s

VIZAG STEEL MT 18.08.2013

Ans. (c) : Impulse (J) : It is defined as a change in the linear moment (P) with respect to time.

$J = F \times dt = \Delta P$

Given, $m = 1.0$ kg, $v_1 = 25$ m/sec., $v_2 = 10$ m/sec

$J = \Delta P = m(v_1 - v_2) = 1.0) = 1.0 \times [25 - (-10)]$

$\boxed{J = 35 \text{ N} - \text{second}}$

10. Simple Harmonic Motion and Projectile Motion

747. In simple harmonic motion the acceleration is proportional to :

(a) displacement
(b) linear velocity
(c) angular velocity
(d) rate of change of angular velocity

RPSC ACF-2011
ISRO Scientist/Engineer 2006
ISRO Scientist/Engineer 2010
WBPSC AE, 2017
CGPSC AE 26.04.2014 Shift-I
RPSC AE 2016
(HPPSC POLY. LECT. 2016)
Rajasthan Nagar Nigam AE 2016, Shift-II
ASSAM PSC CEE Pre 2015

Ans. (a) : Let the equation of SHM,

$x = A \sin(\omega t)$

$v = A\omega\cos(\omega t)$

$a = -A\omega^2 \sin(\omega t) \Rightarrow a = -\omega^2 x$

$\therefore$ $\boxed{a \propto x}$

$\because$ Acceleration is directly proportional to displacement of the particle from the mean position.

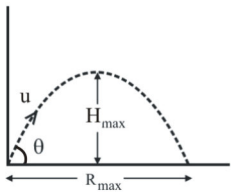
748. The range of a projectile is maximum, when the angle of projection is :

(a) 30° (b) 45°
(c) 60° (d) 90°

RPSC AE GWD, 2011
TRB Poly. Lect., 2012
UKPSC AE 2007 Paper -I
UKPSC AE 2012 Paper-I
TSPSC Managers, 2015
GPSC DEE, Class-II, (GWSSB) 04.07.2021

Ans. (b) : Range of projectile $R = \frac{u^2 \sin 2\theta}{g}$

For maximum Range $\frac{dR}{d\theta} = 0$



$$\frac{u^2}{g}\cos 2\theta \times 2 = 0$$

$$\cos 2\theta = 0$$

$$2\theta = 90°$$

$$\boxed{\theta = 45°}$$

- $H = \frac{u^2 \sin^2\theta}{2g}$
- $T = \frac{2u\sin\theta}{g}$
- R_{max} at $\theta = 45°$
- H_{max} at $\theta = 90°$

749. In order to double the period of a simple pendulum, the length of the string should be:

(a) Halved (b) Doubled
(c) Tripled (d) Quadrupled

ISRO Scientist/Engineer (RAC) 22.04.2018
UKPSC AE 2007 Paper -I
UKPSC AE 2012 Paper-I

Ans. (d) : We know that

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T \propto \sqrt{l}$$

$$T = C\sqrt{l}$$

Then, $T_2 = 2T$, $l_2 = ?$

$$\frac{T_2}{T_1} = \sqrt{\frac{l_2}{l_1}} \Rightarrow \frac{2T}{T} = \sqrt{\frac{l_2}{l_1}}$$

$$\boxed{l_2 = 4l_1}$$

750. The length of a second's pendulum is-

(a) 99.0 cm (b) 100 cm
(c) 101 cm (d) 99.4 cm

ISRO Scientist/Engineer (RAC) 29.11.2015
WBPSC AE, 2017

Ans : (d) Second's Pendulum is the term used to define the pendulum with a time period of 2 seconds. Seconds pendulums were used in vintage clocks which would advance one second every time the pendulum reaches an extreme position. Since the pendulum reaches its extreme position twice in one oscillation, the time period of a second pendulum is 2 seconds.

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$$T=2\pi\sqrt{\frac{\ell}{g}}$$

$$2=2\pi\sqrt{\frac{\ell}{9.81}}$$

$\lambda = 0.9939$ m = 99.4 cm= 994 mm

751. To take longest possible jump, an athlete should make an angle of :

(a) 60° with the ground
(b) 45° with the ground
(c) 30° with the ground
(d) None of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014
J&K PSC Civil Services Pre, 2010
BPSC Asstt. Prof. 29.11.2015

Ans. (b) : To take longest possible jump means range of projectile should be maximum.

$$R=\frac{u^2\sin 2\theta}{g}$$

For R_{max}

$\sin 2\theta = 1$

$2\theta = 90°$

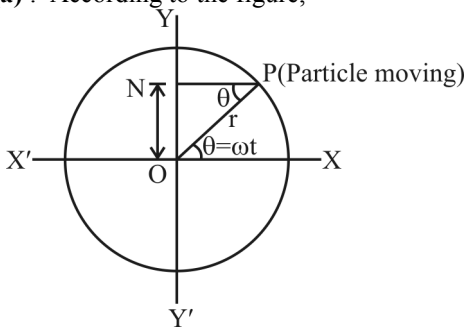
$\boxed{\theta = 45°}$

752. In S.H.M., the velocity vector w.r.t. displacement vector

(a) leads by 90° (b) lags by 100°
(c) leads by 120° (d) leads by 180°

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (a) : According to the figure,



Displacement, $\vec{y} = ON = r\sin\theta = r\sin\omega t$

$$\therefore \quad \vec{v}=\frac{d\vec{y}}{dt}=r\omega\cos\omega t$$

$= r\omega \cos \omega t$

$\vec{v} = r\omega \sin (90° + \omega t)$

$r\omega$ is constant so in SHM velocity vector w.r.t. displacement vector leads by 90°.

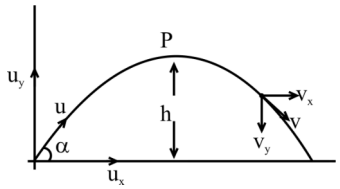
753. If the velocity of projection is u m/sec and the angle of projection is α^a, the maximum height of the projectile on a horizontal plane is :

(a) $\frac{u^2\cos^2\alpha}{2g}$ (b) $\frac{u^2\sin^2\alpha}{2g}$

(c) $\frac{u^2\tan^2\alpha}{2g}$ (d) $\frac{u^2\sin^2\alpha}{g}$

HPPSC W.S. Poly. 2016
ISRO Scientist/Engineer (RAC) 29.11.2015
MECON MT 2019

Ans : (b)



U = velocity of projection
α = angle of projection
(i) Flight - time of Projectile:-

$$T=\frac{2u\sin\alpha}{g}$$

(ii) Height of projectile:-

$$h=\frac{u^2\sin^2\alpha}{2g}$$

(iii) Range of projectile:-

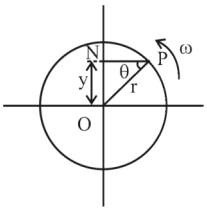
$$R=\frac{u^2\sin 2\alpha}{g}$$

754. The acceleration of a particle, moving with simple harmonic motion, at any instant is given by

(a) $\omega \cdot y$ (b) $\omega^2 \cdot y$
(c) ω^2 / y (d) ω^3 / y

GMB AAE 25.06.2017
TSPSC AEE 2015
Assam PSC CCE Pre 2015
TNPSC ACF 2012

Ans. (b) Let particle P moving with constant angular Velocity w in SHM.



In Δ OPN

$\vec{y} = r \sin \theta$

$\vec{y} = r \sin \omega t \qquad (\omega = \theta/t)$

$$\frac{d\vec{y}}{dt} = r\ \cos\omega t\ \times \omega = r\omega\cos\omega t$$

$\vec{v} = r\omega \cos \omega t$

$$\frac{d\vec{V}}{dt} = -r\omega\sin\omega t\times\omega$$

$= -r\omega^2 \sin \omega t = -\omega^2 (r \sin \omega t)$

$$\frac{d\vec{V}}{dt} = \vec{a} = -\omega^2\vec{y}$$

$\vec{a} = -\omega^2 \vec{y}$ and $\vec{a} \propto \vec{y}$ $[\omega^2 = \text{constant}]$

755. The projectile will reach the maximum height, when the vertical component of velocity of projection becomes _____.

(a) maximum
(b) independent of velocity
(c) unity
(d) zero

HPPSC Workshop Suptd. 08.07.02021

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Ans. (d) : The projectile will reach the maximum height, when the vertical component of velocity of projection becomes zero.

756. A pendulum is made of a hollow sphere of negligible mass and having a small hole in the bottom. The pendulum is made to vibrate after filling it with water.

As the water gradually flows out of the hole, the time period of the pendulum will

(a) increase
(b) decrease
(c) may increase or decrease
(d) remain unchanged

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : A pendulum is made of a hollow sphere of negligible mass and having a small hole in the bottom. The pendulum is made to vibrate after filling it with water.
As the water gradually flows out of the hole, the time period of the pendulum will increase.

757. The maximum speed of the bob of a seconds pendulum is stated to be 1 m/s. Its maximum acceleration would be

(a) $5.62\ m/s^2$ (b) $6.48\ m/s^2$
(c) $1.57\ m/s^2$ (d) $3.14\ m/s^2$

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : Time period of second pendulum

$$T = 2$$

$$T = \frac{2\pi}{\omega}$$

$$2 = \frac{2\pi}{\omega} \Rightarrow \omega = \pi$$

$$V = r\omega \Rightarrow 1 = r\omega$$

$$\boxed{r = \frac{1}{\omega} = \frac{1}{\pi}}$$

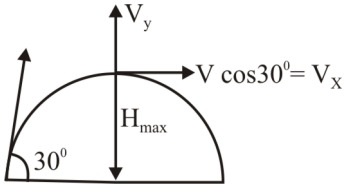
$$a = r\omega^2 = \frac{1}{\pi} \times \pi^2 = \pi = 3.14\,m/s^2$$

758. A body is projected with a velocity of 30 ms^{-1} at an angle of 30^0 to the ground. The vertical component of velocity at its maximum height will be

(a) $25.98 ms^{-1}$ (b) $0\ ms^{-1}$
(c) $15\ ms^{-1}$ (d) $30\ ms^{-1}$

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :



The vertical component of velocity at maximum height is zero, and the horizontal component is constant i.e., $v\cos\theta$

759. A body is just released from the window of a moving bus moving along a horizontal road. When observed by a person on the ground, the stone will hit the ground following a

(a) Parabolic path (b) Straight line path
(c) hyperbolic path (d) circular path

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Parabolic path.

760. The shape of the projectile is

(a) a straight line
(b) curvilinear motion
(c) a parabola
(d) a hyperbola

TANGEDCO AE 2015

Ans. (c) : The shape of the projectile is a parabola.

761. A ball is projected vertically upward with a certain velocity. It takes 40 s for its upward journey. The time (in seconds) taken for its downward journey is

(a) 10 (b) 20
(c) 30 (d) 40

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) : The time taken for its downward journey is same as its upward journey. The time taken for downward journey is 40 sec.

762. When a body moves with simple harmonic motion, the product of its periodic time and frequency is equal to

(a) zero (b) 1
(c) $\pi/2$ (d) π

Assam PSC CCE Pre 2015
UKPSC AE-2013, Paper-I

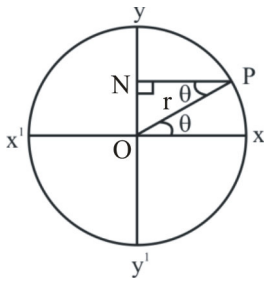
Ans. (b) : When a body moves with simple harmonic motion, the product of its periodic time and frequency is equal to 1.
Periodic time (T) × Frequency (f) = 1

763. The velocity of a particle moving with simple harmonic motion, at any instant is given by (where ω = Displacement of the particle from mean position)

(a) $\omega\sqrt{x^2 - r^2}$ (b) $\omega\sqrt{r^2 - x^2}$
(c) $\omega^2\sqrt{x^2 - r^2}$ (d) $\omega^2\sqrt{r^2 - x^2}$

Sikkim PSC (Under Secretary), 2017
APPSC Poly. Lect. 2013

Ans. (b) :



A particle moving along circumference of circle Radius r with a uniform angular velocity of ω rad/sec.

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Let
P is position of particle at some instant after t second from x.
angle turned by the particle,

$$\theta = \omega t \text{ rad}$$

Displacement at point N

$$y = ON = r\sin\theta$$

$$= r\sin\omega t \quad \text{.....(i)}$$

Differentiate with respect to time 't'

$$\frac{dy}{dt} = r\omega\cos\omega t$$

$$V = r\omega\sqrt{1-\sin^2\omega t}$$

from equation (i) we find out

$$\sin\omega t = \frac{y}{r}$$

$$V = r\omega\sqrt{\left(1-\frac{y^2}{r^2}\right)}$$

$$V = \omega\sqrt{r^2-y^2}$$

similarly

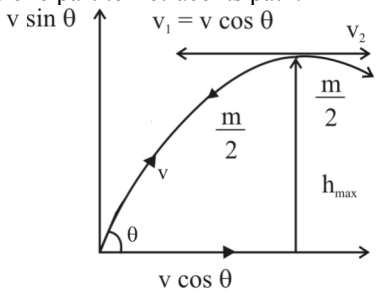
$$V = \omega\sqrt{r^2-x^2}$$

764. A shell fired from cannon with a speed 'v' at an angle 'θ' with the horizontal direction. At the highest point in its path it explodes into pieces of equal mass. One of the pieces retraces its path to the cannon. The speed of other piece immediately after explosion is:

(a) 3 v cos θ (b) 2 v cos θ
(c) 3/2 v cos θ (d) $\sqrt{3/2}$ v cos θ

ISRO Scientist/Engineer 11.10.2015
OPSC Civil Services Pre. 2011
GATE 1994

Ans. (a) : At the highest point of the trajectory the shell will have the only horizontal velocity that is –v cos θ. For its one part to retrace its path.



After explosion the piece that retraces its path is having velocity,

$$v_1 = -v\cos\theta$$

Since, there is no force acting on the shell in horizontal direction, so its linear momentum remains constant,

$$mv\cos\theta = \frac{m}{2}v_1 + \frac{m}{2}v_2$$

$$v_2 = 2v\cos\theta - v_1$$

$$v_2 = 2v\cos\theta - (-v\cos\theta)$$

$$\boxed{v_2 = 3v\cos\theta}$$

765. A harmonic motion is given by the equations x = 6 sin $\left(10t-\frac{\pi}{4}\right)$ cm where phase angle is in radian. The maximum acceleration is given by:

(a) 1200 cm/sec^2 (b) 900 cm/sec^2
(c) 60 cm/sec^2 (d) 600 cm/sec^2

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (d) : Given, For a harmonic motion

$$x = 6\sin\left(10t-\frac{\pi}{4}\right)$$

We know that for sigma
Maximum acceleration (a_{max}) = $A\omega^2$
∴ Compare given equation with

$$x = A\sin(\omega t - \phi)$$

A = 6 cm, ω = 10 rad/sec.
Maximum acceleration (a_{max}) = $A\omega^2$

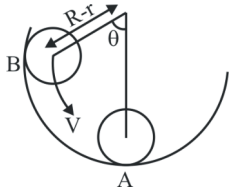
$$a_{max} = 6 \times 10^2 = 600 \text{ cm/sec}^2$$

766. The radius of arc is measured by allowing a 20 mm diameter roller to oscillate to and fro on it and the time for 25 oscillations is noted at 56.25 s. The radius of arc will be

(a) 865 mm (b) 850 mm
(c) 835 mm (d) 820 mm

ESE 2019

Ans. (b) : Given,
Radius of roller



$$r = \frac{20}{2} = 10 \text{ mm}$$

Time period of

Oscillation $T = \frac{56.25}{25} = 2.25$ sec

$$\omega_n = \sqrt{\frac{2g}{3(R-r)}}$$

$$T = 2\pi\sqrt{\frac{3(R-r)}{2g}}$$

$$2.25 = 2\pi\sqrt{\frac{3(R-0.01)}{2\times 9.81}}$$

0.8395 = R – 0.01
R = 0.849 m
R = 850 mm

767. Speed of particle executing simple harmonic motion with amplitude 'a' is half of the maximum speed. At that instant, displacement of the particle is

(a) $\frac{a}{2}$ (b) $\frac{\sqrt{3}}{2}a$
(c) $\frac{2a}{\sqrt{3}}$ (d) $3\sqrt{2}a$

ISRO Scientist/Engineer 2008
ISRO Scientist/Engineer 2006
HPPSC Poly. Lect. 05.07.2021

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Ans. (b) : x = a sin ωt

$v = a\omega \cos \omega t$

$\frac{a\omega}{2} = a\omega \cos \omega t$

$\therefore \omega t = \frac{\pi}{3}$

at $\omega t = \frac{\pi}{3}$ displacement is

$x = a \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} a$

where a is maximum displacement.

768. If the natural frequency of a spring mass system on earth is *f*, its natural frequency on moon where acceleration due to gravity is one-sixth of that on earth will be

(a) $f/6$ (b) $6f$

(c) $f/\sqrt{6}$ (d) f

ISRO Scientist/Engineer 22.04.2018

Ans. (d): We know that natural frequency is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Natural frequency is not depend on gravity so it will be remain same on earth and moon.

769. On which of the following factors does the periodic time of a simple pendulum depend?

(a) Length of the pendulum

(b) Mass of the bob

(c) Size of the bob

(d) Material of the bob

WBPSC AE 2008
ISRO Scientist/Engineer (RAC) 29.11.2015
ISRO Scientist/Engineer 2007
CIL MT 26.03.2017

Ans. (a)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Hence time period of simple pendulum depends only on length.

770. Lengths of two simple pendulums are in the ratio 16:1. They start vibrating together and are in same phase once again after the smaller pendulum has made n oscillations. The value of n is:

(a) 2/3 (b) 3/2

(c) 4/3 (d) 1/4

HPPSC ADF 05.03.2019

Ans. (d) : Let us say, in phase when shorter pendulum has mode n_1 oscillations and bigger pendulum n_2 oscillations. So total time elapsed = $T_1 \times n_1 = T_2 \times n_2$

$$\Rightarrow \frac{n_1}{n_2} = \frac{T_2}{T_1} = \sqrt{\frac{l_2}{l_1}} = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

They will again be in phase for the first time when the shorter pendulum has mode 1 oscillation and the bigger pendulum has 4 oscillation.

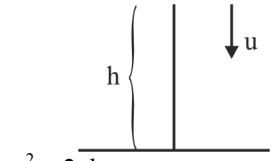
771. A monkey standing on the rooftop of a house of height h throws a ball vertically downwards and another ball horizontally with the same velocity u. The ratio of their velocities when they reach the earth's surface will be:

(a) 1 : 3 (b) 2 : 1

(c) 1 : 2 (d) 1 : 1

AAI Jr. Executive 29.11.2018

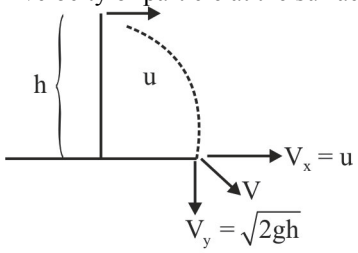
Ans. (d) : When particles thrown in vertically downward direction with velocity u then final velocity at the ground level.



$v^2 = u^2 + 2gh$

$v = \sqrt{u^2 + 2gh}$

Another particle is thrown horizontally with same velocity then velocity of particle at the surface of earth.



Horizontal component of velocity $V_x = u$

$\therefore$ Resultant velocity

$v = \sqrt{u^2 + 2gh}$

For both particles, final velocities when they reach the earth's surface are equal 1:1.

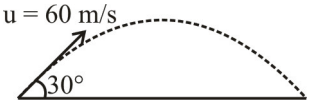
772. A projectile is fired upward at an angle 30° with a velocity of 60 m/s, The time taken by the projectile to reach the ground after the instant of firing is _____. (assume g as 10 m/s)

(a) 3 s (b) 12 s

(c) 6 s (d) 9 s

VIZAG MT, 14.12.2020

Ans. (c) :



Flight time

$$T = \frac{2u\sin\theta}{g}$$

$$= \frac{2 \times 60 \times \sin 30}{10}$$

$$\boxed{T = 6\,\text{sec}}$$

773. The radius of curvature of trajectory for a projectile is minimum, if

(a) velocity is minimum

(b) acceleration is maximum

(c) Both (a) and (b)

(d) None of the above

Assam PSC AE (IWT) 14.03.2021

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Ans. (c) : The radius of curvature is given as–

$$\rho = \frac{v^2}{a_N}$$

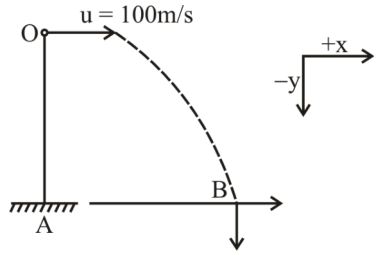
Here, v is the velocity of particle and a_N is the acceleration in normal direction. When velocity is minimum radius of curvature is minimum.
When acceleration increases radius of curvature decreases.
Hence the radius of curvature of trajectory for a projectile is minimum, if velocity is minimum and acceleration is maximum.

774. A cannonball is fired from a tower 80 m above the ground with a horizontal velocity of 100 m/s. Determine the horizontal distance at which the ball will hit the ground. (Take g = 10 m/s^2)

(a) 400 m (b) 280 m
(c) 200 m (d) 100 m

UPSC JWM Advt. No.-50/2010

Ans. (a) :



Time taken for each ground,

$$-80 = v_y \times t + \frac{1}{2} \times a_y \times t^2$$

$$-80 = 0 \times t + \frac{1}{2}(-10) \times t^2$$

$$t = 4 \text{ sec}$$

$$\text{Distance (AB)} = U_x t + \frac{1}{2} \times a_x \times t^2$$

$$= 100 \times 4 + \frac{1}{2}(0) \times 4^2$$

$$= 400 \text{ m}$$

775. Pick out the correct statement:

(a) The path traced by a projectile is trajectory
(b) The area under v-t diagram is acceleration
(c) Efficiency of simple machine is velocity ratio/mechanical advantage
(d) If efficiency is < 50% that simple machine is reversible

TNPSC AE, 2008

Ans. (a) : The path traced by a projectile that accelerates only in the vertical direction while moving at constant horizontal velocity is called a parabola. When air resistance can be neglected, the curved paths are parabolic and the paths are called trajectories.

776. A stone is dropped from the top of a tall building. If it takes 4 seconds to fall on the ground, find the height of the building. lets take g=10m/s^2.

(a) 85 m (b) 80 m
(c) 70 m (d) 75 m

VIZAG Steel MT 24.01.2021

Ans. (b) : The time of flight is,

$$T = \sqrt{\frac{2h}{g}} \qquad \left[\because h = ut + \frac{1}{2}gt^2\right]$$

When, h is the height from which ball is dropped, since, t = 4 sec.

So, $$4 = \sqrt{\frac{2 \times h}{10}} \Rightarrow 16 = \frac{2h}{10}$$

$$\Rightarrow \quad h = \frac{160}{2} = 80\,\text{m}$$

777. The maximum velocity and maximum acceleration of a particle executing simple. Harmonic motion are 2m/sec and 20 m/sec^2. The time period of motion equals :

(a) π (b) $\frac{\pi}{5}$
(c) $\frac{\pi}{10}$ (d) $\frac{\pi}{3}$

J & K PSC Screening, 2006

Ans. (b) : In SHM maximum velocity

$$y\omega = 2 \text{ m/sec} \qquad \ldots(i)$$

and maximum acceleration = $-\omega^2 y$

$$(a_{max})_{magnitude} = \omega^2 y = 20 \text{m/sec}^2 \qquad \ldots(ii)$$

divide equation (i) & (ii)

$$\frac{y\omega}{\omega^2 y} = \frac{2}{20}$$

$$\frac{1}{\omega} = \frac{1}{10}$$

$$\omega = 10$$

$$\frac{2\pi}{T} = 10$$

$$\boxed{T = \frac{2\pi}{10} = \frac{\pi}{5}}$$

778. A projectile is fired at an angle α such that its horizontal range is 500m. What is the angle α if the maximum horizontal range of the projectile is 1000m.

(a) 60° (b) 45°
(c) 22.5° (d) 15°

J & K PSC Screening, 2006

Ans. (d) : $R = \frac{u^2 \sin 2\theta}{g}$

$$500 = \frac{u^2 \sin 2\theta}{g} \qquad \ldots(i)$$

$$\text{Maximum Range} = \frac{u^2}{g}$$

$$1000 = \frac{u^2}{g}$$

In equation (i) put $\frac{u^2}{g} = 1000$

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$$500 = 1000 \sin 2\theta$$

$$\sin 2\theta = \frac{500}{1000}$$

$$2\theta = 30°$$

$$\boxed{\theta = 15°}$$

779. A fighter plane flying in the sky dives with a speed of 120 m/sec in a vertical circle of radius of 100 m. Mass of the pilot is 80 kg. Force with which the pilot presses the seat at the highest position is :

(a) 10720 N (b) 15000 N
(c) 8000 N (d) 12720 N

NPCIL ET, 2015

Ans. (a) : At the highest point of the vertical circle
Centripetal force $F_c = N + mg$……..(1)
Where N is the normal force between pilot and the seat and this is the force we have to find from equation (1) we get

$$\frac{mv^2}{r} = N + mg$$

$$N = \frac{mv^2}{r} - mg \text{ ……..(2)}$$

Given, m = 80 kg and r = 100m
v = 120 m/sec

So, $$N = \frac{80 \times (120)^2}{100} - mg$$

$$N = \frac{80 \times (120)^2}{100} - 80 \times 10$$

$$N = \frac{80 \times (120)^2}{100} - 800$$

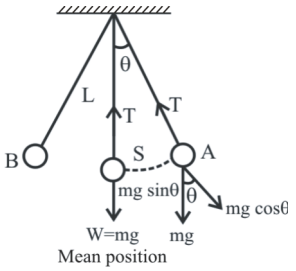
N = 10720 N

780. A simple pendulum of length of 5 m, with a bob of mass 1 kg, is in simple harmonic motion. As it passes through its mean position, the bob has a speed of 5 m/s. The net force on the bob at the mean position is:

(a) 0 (b) 2.5 N
(c) 5 N (d) 25 N

NPCIL ET, 2015
GATE 2005

Ans. (a) :



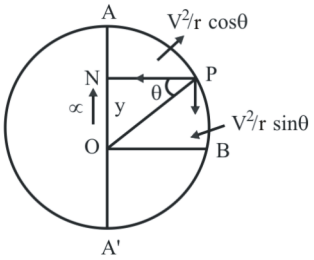
The restoring force is given by
$F = -mg\sin\theta$
$F = 0$ [At mean position, $\theta = 0$]
Here, option 'A' is correct.

781. The Maximum acceleration of particle moving with Simple Harmonic Motion is -

(a) ω (b) ω . r
(c) ω^2.r (d) ω^2/r

TNPSC ACF 2012

Ans. (c) : Let radius is *r*



From fig.

$$a = \frac{-V^2}{r}\sin\theta$$

$\because$ $v = r\omega$

and $\sin\theta = y/r$

so, $$a = \frac{-r_2\omega_2}{r} \times y/r$$

$$a = \omega^2 y$$

(i) when y = 0
a = 0
(ii) if y = r
$\boxed{a = \omega^2 r}$

782. Simple Harmonic Motion is a :

(a) Rectilinear motion with uniform acceleration
(b) Rectilinear motion with variable acceleration
(c) Curvilinear motion with variable acceleration
(d) Curvilinear motion with variable angular speed

MPPSC State Forest Service Exam, 2014

Ans. (b) : Simple harmonic motion is defined as a motion in which the restoring force is directly proportional to the displacement of the body from its mean position or we can say rectilinear motion with variable acceleration.

$$\vec{F} = -k\vec{x}$$

$$\vec{a} = -\frac{k}{m}\vec{x}$$

$\Rightarrow$ $\vec{a} = -\omega^2\vec{x}$ Where, $\omega = \sqrt{\frac{k}{m}}$

$$\boxed{\frac{d^2\vec{x}}{dt^2} = -\omega^2\vec{x}}$$

783. When a particle is executing Simple Harmonic Motion, the velocity is maximum at.

(a) The mean position
(b) Extreme position
(c) Midway between extreme and mean position
(d) One-third distance from mean position

MPPSC State Forest Service Exam, 2014

Ans. (a) : As we know,

$$v = \omega\sqrt{A^2 - y^2}$$

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At mean position.

$y = 0$

$\boxed{v_{max} = \omega A}$

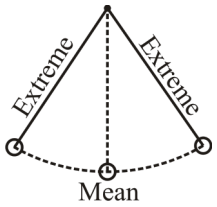
As particle executing simple harmonic motion, the velocity is maximum at mean position and zero at the extreme point.

784. A ball tied at the end of a flexible inelastic string of length 'ℓ' is executing SHM in a vertical plane. At the mean position the mass has

(a) Maximum KE and Minimum PE
(b) Maximum KE and Maximum PE
(c) Minimum KE and Maximum PE
(d) Minimum KE and Minimum PE

Karnataka PSC AE, 10.09.2017

Ans. (a) :



At mean position KE is maximum and PE is minimum.

785. In case of SHM, the period of oscillation is given by :

(a) $T = \frac{\omega}{2\pi}$ (b) $T = \frac{2\omega}{\pi}$

(c) $T = \frac{2\pi}{\omega}$ (d) $T = \frac{\pi}{2\omega}$

KPSC ADF 2015
TNPSC AE 2019

Ans. (c) : $\omega = 2\pi n$

$\omega = \frac{2\pi}{T}, \; n = \frac{1}{T}$

$\boxed{T = \frac{2\pi}{\omega}}$

786. A spring balance is attached to the ceiling of a lift. A man hangs his bas on the spring and the spring reads 49 N, when the lift is stationary. If the lift moves downwards with an acceleration of 5 m/s^2, the reading of the spring balance will be

(a) 24 N (b) 74 N
(c) 15 N (d) 49 N

GPSC EE Pre, 28.01.2017

Ans. (a) : If lift is stationary, then spring force balances weight.

F_x = mg = 49 N (k = force constant of spring, x = deflection)

$\Rightarrow \quad k = \frac{49}{x}$

and $\quad m = \frac{49}{9.8} = 5\,kg$

When lift moves with acceleration 5 m/s^2 downward we have.

Spring force = kx_2 = mg – 5 × a (pseudo force in lift)

= 49 – 5 × 5

= 49 – 25 = 24 N

787. A body of mass m is dropped from a height of h. Simultaneously another body of mass 2m is thrown up vertically with such a velocity v that they collide at height h/2. If the collision is perfectly inelastic, the velocity of combined mass at time of collision with the ground will be

(a) $\sqrt{\frac{5gh}{4}}$ (b) $\sqrt{gh}$

(c) $\sqrt{\frac{gh}{4}}$ (d) None of these

GPSC EE Pre, 28.01.2017

Ans. (d) : m_1 = 'm' mass falls a distance $\frac{h}{2}$ in time 't'

$\therefore$ Initial velocity u = 0

$$h = ut + \frac{1}{2}gt^2$$

$$\frac{h}{2} = \frac{1}{2}gt^2 \;\; \therefore t = \sqrt{\frac{h}{g}}$$

For mass $m_2 = 2m$

$h_2 = \frac{h}{2}$, finial velocity =V = initial velocity

$$\frac{h}{2} = Vt - \frac{1}{2}gt^2$$

$$= V\sqrt{\frac{h}{g}} - \frac{h}{2g}$$

$\therefore \quad V = \sqrt{gh}$

Now, let u is the velocity with which combined mass collides with ground then,

$$u = \sqrt{\left(\frac{2}{3}V\right)^2 + 2g\left(\frac{h}{3}\right)}$$

$$= \sqrt{\frac{4}{9}gh + \frac{2}{3}gh}$$

$$= \frac{\sqrt{10}}{3}\sqrt{gh} = \frac{\sqrt{10gh}}{3}$$

788. The super elevation 'e' of the order rail on a rail road curve of radius 'r' so that a car travelling at a speed 'V' around the curve will exert equal pressures on the two rails (distance between the rails being 'b') is

(a) $\frac{bV^2}{g}$ (b) $\frac{bV^2}{gr}$

(c) $\frac{gr}{bV^2}$ (d) $\frac{V^2}{gr}$

OPSC Civil Services Pre 2006

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Ans. (d) : Elevation (angle) has no unit only option (d) is unit less.

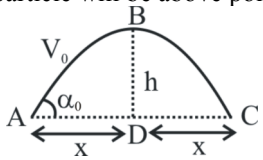
789. A projectile is fired from the level ground at an angle α_0 to the horizontal and with an initial velocity V_0. The time elapsed by the projectile to hit the ground is

(a) $\frac{2V_0\cos\alpha_0}{g}$ (b) $\frac{V_0^2\sin 2\alpha_0}{g}$

(c) $\frac{2V_0\sin\alpha_0}{g}$ (d) $\frac{V_0^2\sin^2\alpha_0}{2g}$

OPSC Civil Services Pre 2006

Ans. (c) : By vertical component of velocity V_0 find, particle will reach height point in time 't' second. and at the same time, particle will be above point D.



Vertical component of velocity at point B will be zero.

$0 = V_0\sin\alpha_0 - gt$ or $t = \frac{V_0\sin\alpha_0}{g}$

time of flight $= 2t = \frac{2V_0\sin\alpha_0}{g}$

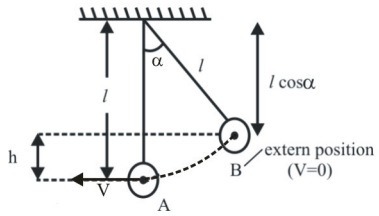
790. A pendulum has a length ℓ. Its bob is pulled aside from its equilibrium position through an angle α and then released. The speed of the bob when it passes through the equilibrium position is given by

(a) $\sqrt{2g\ell}$ (b) $\sqrt{2g\ell\cos\alpha}$

(c) $\sqrt{2g\ell(1-\cos\alpha)}$ (d) $\sqrt{2g\ell(1-\sin\alpha)}$

WBPSC AE 2003

Ans. (c) :



$h = l - l\cos\alpha$

$h = l(1-\cos\alpha)$

from energy conservation (B to A)

$KE_B + PE_B = KE_A + PE_A$

$KE_B + PE_B = KE_A + PE_A$

$0 + mgh = \frac{1}{2}mv^2 + 0$

$mgh = \frac{1}{2}mv^2$

$g[l(1-\cos\alpha)] = \frac{1}{2}v^2$

$2g\,l(1-\cos\alpha) = v^2$

$\boxed{v = \sqrt{2gl(1-\cos\alpha)}}$

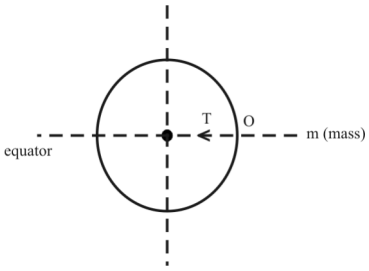
791. Consider a particle of mass m suspended vertically by a string at the equator. Let R and M denote the radius and the mass of the earth, respectively. If acceleration due to gravity at equator is $g - \omega^2R$ where ω is the angular velocity of earth's rotation about its own axis, the tension in the string is equal to

(a) $G\frac{mM}{2R^2}$ (b) $G\frac{mM}{R^2}$

(c) $G\frac{mM}{R^2} + m\omega^2R$ (d) $G\frac{mM}{R^2} - m\omega^2R$

WBPSC AE 2003

Ans. (d) :



T = tension is string.

∵ The acceleration due to gravity at a place on the surface of the earth is given by –

$g' = g - \omega^2 R\cos\phi$

where ϕ is the lattitude of the place but for equator $\phi = 0$

$\therefore\ g' = g - \omega^2R$ Which is given is question.

Therefore, the tension in the string will be –

$T = mg'$

$= m(g - \omega^2R)$

$= mg - m\omega^2R$ .....(i)

∵ mg ⇒ gravitational force

$mg \Rightarrow \frac{GmM}{R^2}$ .....(ii)

From equation (i) & (ii)

$\boxed{T = \frac{GmM}{R^2} - m\omega^2R}$

also, T = gravitational force Centrifugal force

792. The value of acceleration due to gravity at moon is g/6; where g is the value of acceleration due to gravity at earth. The value of frequency of oscillation of simple pendulum on moon as compared to earth will be-

(a) $1/\sqrt{6}$ times (b) 6 times

(c) 1/6 times (d) $\sqrt{6}$times

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) Frequency is given by on earth

$f_E = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}$ (on earth)

frequency is given by on moon

$f_m = \frac{1}{2\pi}\sqrt{\left(\frac{g/6}{\ell}\right)}$

$\frac{f_m}{f_E} = \sqrt{\frac{1}{6}}$, $\boxed{f_m = \frac{1}{\sqrt{6}}f_E}$

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793. A particle is projected at an angle θ to the horizontal and it attains a maximum height H. The time taken by the projectile to reach the highest point, of its path is

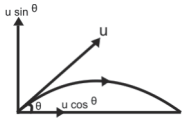
(a) $\frac{\sqrt{H}}{g}$ (b) $\sqrt{\frac{2H}{g}}$

(c) $\frac{\sqrt{2H\sin\theta}}{g}$ (d) $\frac{\sqrt{2H}}{\sin\theta}$

ISRO Scientist/Engineer 2010

Ans. (b) : $a_x = 0$, $a_y = -g$

At the highest point



$v_y = 0$

$v_y^2 - u_y^2 = -2gH$

$0 - u^2\sin^2\theta = -2gH$

$+u^2\sin^2\theta = +2gH$

$u\sin\theta = \sqrt{2gH}$

$v_y = u_y + a_y t$

$0 = \sqrt{2gH} - gt$

time taken by

$u = \frac{\sqrt{2gH}}{\sin\theta}$

$0 = \frac{\sqrt{2gH}}{\sin\theta} - gt$

$gt = \frac{\sqrt{2gH}}{\sin\theta}$

$t = \frac{\sqrt{2gH}}{\sin\theta.g} \times \frac{\sqrt{g}}{\sqrt{g}}$

$= \frac{\sqrt{2H}\times g}{\sin\theta \times g\sqrt{g}}$

$= \frac{\sqrt{2H}}{\sqrt{g}\sin\theta}$

$\theta = 90°$

$t = \sqrt{\frac{2H}{g}}$

794. For the apparent weight of a body at equator to become zero, the earth should rotate with an angular velocity of

(a) $\sqrt{\frac{2g}{r}}$ (b) $\sqrt{\frac{g}{r}}$

(c) $\sqrt{\frac{g}{2r}}$ (d) $\frac{g}{r}$

ISRO Scientist/Engineer 2010

Ans. (b) : The body will becomes weight less when acceleration toward earth of the body becomes zero.

$mg = m\omega^2 r$

$\omega = \sqrt{\frac{g}{r}}$

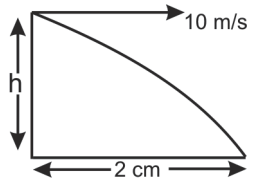
Where ω is rotation of earth, r radius of earth and m is mass of any body on earth.

795. A stone is projected horizontally from a cliff at 10 m/sec and lands on the ground below at 20 m from the base of the cliff. Find the height of the cliff. Use g=10 m/sec^2

(a) 18 m (b) 20 m

(c) 22 m (d) 24 m

ISRO Scientist/Engineer 2007



Ans. (b) :

distance = velocity × time

or 20 = 10 × t

t = 2 sec

u = 0, Initial velocity

$h = ut + \frac{1}{2}at^2$

$h = \frac{1}{2}at^2$

$\frac{1}{2}\times 10\times 4 = 20\,m$

796. Oscillation of a particle is prescribed by the equation x = 3 cos 0.25 πt. Where t is the time in seconds.

Then time taken by the particle to move from position to equilibrium to maximum displacement is

(a) 0.5 sec (b) 1.0 sec

(c) 2.0 sec (d) 3.0 sec

ISRO Scientist/Engineer 2007
ISRO Scientist/Engineer 2011

Ans. (c) :

$x = 3\cos 0.25\,\pi t$

For maximum displacement, $x_{max} = 3$

$3 = 3\cos 0.25\pi t$

$\cos 0.25\pi t = 1$

$\cos 0.25\pi t = \cos 0$

$0.25\pi t = 0$

$t = 0$

For equilibrium displacement, x = 0

$0 = 3\cos(0.25\pi t)$

$0.25\pi t = \frac{\pi}{2}$

$\frac{1}{4}\pi t = \frac{\pi}{2}$

$\boxed{t = 2\,sec}$

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797. A car is travelling on a curved road of radius 300 m at speed of 15 m/s. The normal and tangential components of acceleration respectively are given by:

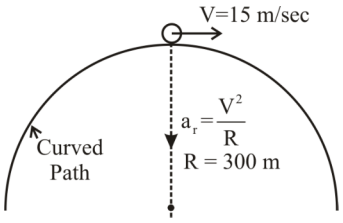
(a) 0.75 m/s^2, zero
(b) 0.75 m/s^2, 0.75 m/s^2
(c) zero, zero
(d) zero, 0.75 m/s^2

UPRVUNL AE 07.10.2016

Ans. (a) : Radial/normal components of acceleration due to change in direction continually (a_r)

$$= \frac{v^2}{r} = \frac{(15)^2}{300}$$

$a_r = 0.75$ m/s^2



In this case no tangential acceleration because of no change in angular velocity due to unifrom circular motion.

798. A particle is projected with an initial velocity of 60 m/sec at an angle of 75^o with horizontal. The maximum height attained by the particle is

(a) 171.19 m (b) 185.22 m
(c) 221.11 m (d) 198.20 m

TNPSC AE 2014

Ans. (a) : Given, u = 60 m/s, $\theta = 75^0$

The maximum height its,

$$h_{max} = \frac{u^2 \sin^2 \theta}{2g}$$

$$h_{max} = \frac{(60)^2 \times (\sin 75^o)^2}{2 \times 9.81}$$

$$\boxed{h_{max} = 171.194 \text{ m}}$$

799. The particle is projected a point 'Q' with initial velocity 'u' inclined at 'α' to x-axis, x-component of initial velocity at point 'O' is

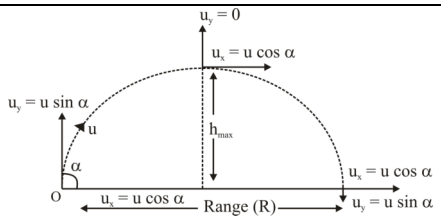
(a) $u_x = u \sin\alpha$
(b) $u_x = u \cos\alpha . \sin\alpha$
(c) $u_x = u \cos\alpha$
(d) $u_x = u \tan\alpha$

TNPSC AE 2013

Ans. (c) : $u_x = u\cos\alpha$

Time of flight (T)

$$T = 2 \times t_{peak} = \frac{2.u \sin\alpha}{g}$$



$$R = \frac{u^2 \sin 2\alpha}{g}, \quad H_{max} = \frac{u^2 \sin^2 \alpha}{2g}$$

800. A particle is projected at such an angle with the horizontal that the maximum height attained by the particle is one-fourth of the horizontal range. The angle of projection should be:-

(a) 30º (b) 45º
(c) 60º (d) 75º

UKPSC AE-2013, Paper-I

Ans. (b) : In projection motion

$$h = \frac{u^2 \sin^2 \theta}{2g}$$

and

$$\text{Range } (R) = \frac{u^2 \sin 2\theta}{g}$$

According to question

$$\frac{R}{4} = h \text{ at } \theta_{max}.$$

$$\frac{u^2 \sin 2\theta}{4 \times g} = \frac{u^2 \sin^2 \theta}{2g}$$

$$\sin\theta \cos\theta = \sin^2 \theta$$

$$\tan\theta = 1$$

$$\theta = \tan^{-1}(1)$$

$$\boxed{\theta = 45^o}$$

801. The maximum displacement of a body moving with simple harmonic motion from its mean position is called

(a) oscillation (b) amplitude
(c) beat (d) none of the above

UKPSC AE 2007 Paper -I

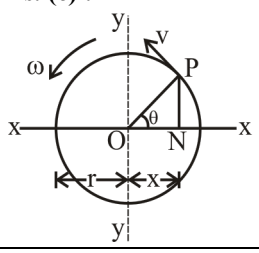
Ans. (b) : Amplitude

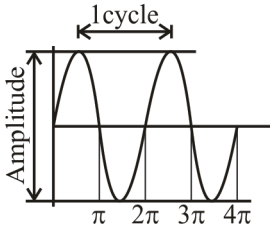
802. A body in S.H.M will have maximum velocity when its amplitude is

(a) Maximum (b) –ve maximum
(c) Zero (d) Average
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (c) :





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$\therefore \quad v_N = v\sin\theta = r\omega\sin\theta$

$v_N = \omega\sqrt{r^2 - x^2}$

The velocity is maximum when amplitude (x = 0) i.e. when N pass through O (i.e. mean position).

$\therefore \quad \boxed{v_{max} = \omega.r}$

803. The frequency of oscillation of a compound pendulum is

(a) $2\pi\sqrt{\{gh/(k_G^2 + h^2)\}}$

(b) $2\pi\sqrt{\{(k_G^2 + h^2)/gh\}}$

(c) $1/2\pi\sqrt{\{gh/(k_G^2 + h^2)\}}$

(d) $1/2\pi\sqrt{\{(k_G^2 + h^2)/gh\}}$

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : Frequency of oscillation of a compound pendulum.

We know, for simple pendulum,

$$\omega = \sqrt{\frac{g}{h}}, \quad T = 2\pi\sqrt{\frac{h}{g}}$$

For compound,

$$\boxed{f = \frac{1}{2\pi}\sqrt{\frac{gh}{k_G^2 + h^2}}}$$

804. The minimum periodic time of a compound pendulum is

(a) $1/2\pi\sqrt{(k_G/g)}$ (b) $1/2\pi\sqrt{(2k_G/g)}$

(c) $2\pi\sqrt{(k_G/g)}$ (d) $2\pi\sqrt{(2k_G/g)}$

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : The periodic time of compound pendulum is minimum when the distance between the point of suspension and the centre of gravity is equal to the radius of gyration of the body about its centre of gravity.

We know,

$$T = 2\pi\sqrt{\frac{k_G^2 + h^2}{gh}}$$

For $T_{min} \Rightarrow (h = k_G)$

$$\boxed{T_{min} = 2\pi\sqrt{\left(\frac{2k_G}{g}\right)}}$$

805. The direction of projection for the range of a projectile to be maximum on an inclined plane having angle of inclination 30° to the horizontal should be

(a) 30° with the horizontal

(b) 30° with vertical

(c) 45° with horizontal plane

(d) 45° with inclined plane

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : For the maximum range in case of the inclined plane, the angle from the inclined should be,

$$\boxed{\theta = \frac{\pi}{4} - \frac{\alpha}{2}}$$

Where, $\theta \Rightarrow$ Angle of projection from the inclined plane

$\alpha \Rightarrow$ The angle of incline from horizontal

Given, $\alpha = 30° = \frac{\pi}{6}$

For maximum range,

$$\theta = \frac{\pi}{4} - \frac{\alpha}{2}$$

$$\theta = 45° - \frac{30°}{2}$$

$$\boxed{\theta = 30°}$$

Angle of projection from the vertical = 30°.

806. In simple harmonic motion, with respect to the displacement vector, the positions of velocity vector and acceleration vector will be respectively

(a) 180 deg and 90 deg

(b) 90 deg and 180 deg

(c) 0 deg and 90 deg

(d) 90 deg and 0 deg

APPSC AEE 2012

Ans. (b) : In SHM displacement vector, velocity & acceleration vector -

Displacement Velocity	x = r cos wt ... (i) v = rw (– sin wt) v= rw cos (90+wt) ....(2)	x = r sin wt ....(i) v = rw cos wt v = rw sin (90 + wt) ... (2)
Acceleration	$a = -rw^2$ cos wt $= rw^2$ (180 +wt) ..... (3)	$a = -rw^2$ sin wt $a = rw^2$ sin (180 + wt) ..... (3)

from both table its is proved that velocity & acceleration vector is load 90º & 180 respectively from displacement vector.

807. A rigid bar of length 1.96 m hanging from a frictionless pivot exerts no horizontal component of reaction force at the pivot when a sudden blow is applied horizontally at a distance of 1 m from the point of suspension. The time period of free vibration of the bar is

(a) 2 sec

(b) 1.41 sec

(c) 1 sec

(d) 0.4 sec

DRDO Scientists 2009

Ans. (a) : Time period T is given by

$$T = 2\pi\sqrt{\frac{L}{g}} = 2\pi\sqrt{\frac{1}{9.81}} = 2s$$

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02.

STRENGTH OF MATERIALS

1. Simple Stress and Strain

1. Poisson's ratio is expressed as

(a) Lateral stress/lateral strain
(b) Longitudinal stress/longitudinal strain
(c) Lateral strain/longitudinal strain
(d) Lateral stress by longitudinal stress

VIZAG MT, 14.12.2020
TSPSC Manager (Engg.) HMWSSB 12.11.2020
RPSC Vice Principal ITI 2018
RPSC AE 2018
HPPSC AE 2018
TRB Polytechnic Lecturer 2017
CGPSC Polytechnic Lecturer 2017
HPPSC Asstt. Prof. 20.11.2017
GPSC Lect. (Auto) 16.10.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
CGPSC AE 16.10.2016
Mizoram PSC AE/SDO Paper-II 2014
TNPSC AE 2013
APPSC AEE 2012
UKPSC AE 2012 Paper-I
APPSC IOF, 2009

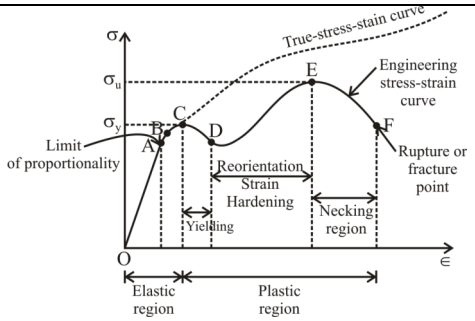
Ans. (c) : Poisson ratio (μ)-The ratio of the transverse contraction of a material to the longitudinal extension strain in the direction of the stretching force is the Poisson's Ratio for a material.
This Poisson's Ratio for most of the materials is in the range of 0 to 0.5.
When the Poisson's Ratio is 0 there is no reduction in the diameter or one can even say there is no laterally contraction happening when you are elongating the material but the density would reduce. The value 0.5 indicates the volume of the material or object will remain the same or constant during the elongation process or when the diameter decrease of material when the material is elastomeric.
Rubber (μ) = 0.5, Cork (μ) = 0.

2. Hooke's law holds good upto

(a) Yield point
(b) Limit of proportionality
(c) Ductile limit
(d) Breaking point

Haryana PSC AE (PHED) 05.09.2020, Paper-II
OPSC AEE 2019 Paper-I
Rajasthan AE (Nagar Nigam) 2016 Shift-3
Assam PSC CCE Pre 2015
UPRVUNL AE 2014
Mizoram PSC AE/SDO Paper-II 2014
SJVN ET 2013
TNPSC ACF 2012
VIZAG Steel MT 2011
UKPSC AE 2007 Paper -I

Ans. (b) : In this stress-strain diagram. Straight line O-A is called proportionality limit.



$\sigma \propto \in$; $\sigma = E\in$

So, Hooke's law holds good upto proportionality limit.
E is proportionality constant known as Young's Modulus of elasticity.
Note-E (Elastic constant) is a material property.

3. The relation between modulus of elasticity (E). modulus of rigidity (G) and bulk modulus (K) is given by:

(a) $\frac{K+G}{3K+G}$ (b) $\frac{3KG}{3K+G}$
(c) $\frac{K+G}{9K+G}$ (d) $\frac{9KG}{3K+G}$

MECON MT 2019
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
OPSC AEE 2019 Paper-I
RPSC AE 2018
TNPSC AE 2013
UJVNL AE 2016
APPSC AE 2012
UKPSC AE 2012, 2013 Paper-I
TRB Polytechnic Lecturer 2017
GPSC Engineer, Class-II Pre-19.01.2020
CSE Pre-1995
ESE 2009

Ans. (d) : Relationship between young modulus (E), Rigidity modulus (G), Bulk modulus (K), and poisson ratio (μ)-

$E = 2G(1+\mu)$, $E = 3K(1-2\mu)$

$$E = \frac{9KG}{3K+G}$$

4. In terms of Poisson's ratio (ν), the ratio of Young's modulus (E) to Shear modulus (G) of plastic material is :

(a) $2(1+\nu)$ (b) $2(1-\nu)$
(c) $(1+\nu)/2$ (d) $(1-\nu)/2$

PPSC Asstt. Municipal Engg. 15.06.2021
Oil India Limited Sr. Engineer (Production) 30.11.2019
GPSC ARTO Pre 30.12.2018
Nagaland PSC (CTSE) Paper-I 2018
GPSC Executive Engg. 23.12.2018
TSPSC AEE 2017
HPPSC Asstt. Prof. 18.11.2016
CGPSC AE 26.04.2015 Shift-I
VIZAG Steel MT 2011
GATE-2004

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Ans. (a) : The relation is,

$E = 2G(1 + \nu) = 3K(1 - 2\nu)$

$$\frac{E}{G} = 2(1+\nu)$$

5. The stress induced in a body, when suddenly loaded, is _____ the stress induced when the same load is applied gradually.

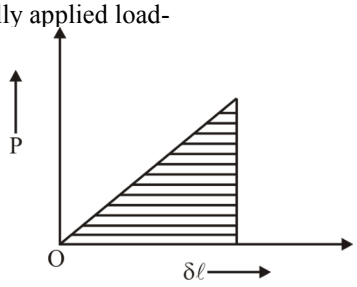
(a) equal to (b) one-half

(c) twice (d) four times

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Sikkim PSC (Under Secretary), 2017
WBPSC AE, 2017
APPSC AE Subordinate Service Civil/Mech. 2016
HPPSC Asstt. Prof. 29.10.2016
MPPSC AE 08.11.2015
Haryana PSC Civil Services Pre, 2014
APPSC AEE 2012
APPSC AE 04.12.2012

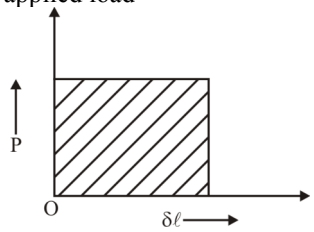
Ans. (c) :

(1) Gradually applied load-



$$\sigma_{gradual} = \frac{W}{A}$$

(2) Suddenly applied load



$$\sigma_{sudden} = \frac{2W}{A}$$

$$\sigma_{sudden} = 2\sigma_{gradual}$$

(3) Impact Load-

$$\sigma_{impact} = \frac{W}{A}\left[1 + \sqrt{1 + \frac{2h}{\delta}}\right]$$

6. A solid metal bar of uniform diameter *D* and length *L* is hung vertically. If ρ is density and *E* is the Young's Modulus, then the total elongation due to self weight is

(a) $\rho Lg/2E$ (b) $\rho L^2 g/2E$

(c) $\rho Eg/2L^2$ (d) $\rho L^2 g/E$

SJVN ET 2019
ISRO Scientist/Engineer 22.04.2018
Nagaland PSC (CTSE) Paper-I 2018
TSPSC AEE 28.08.2017 (Civil/Mech.)
GPSC Poly. Lect. 07.12.2014
UPSC JWM Adv. No.-16/2009
UKPSC AE 2007 Paper -I

Ans. (b): Elongation due to self weight

$$\delta_{self} = \frac{\rho g L^2}{2E} = \frac{\gamma L^2}{2E} \quad [\because \gamma = \rho g]$$

7. The relation between modulus of elasticity E, modulus of rigidity G, bulk modulus K and Poisson's ratio μ is

(a) $E = G(\mu + 1)$ (b) $E = 2G(\mu + 1)$

(c) $E = 4G(2\mu + 1)$ (d) $E = 2G(\mu - 1)$

Haryana PSC AE (PHED) 05.09.2020, Paper-II
WBPSC AE, 2017
ISRO Scientist/Engineer 17.12.2017
(CGPSC Polytechnic Lecturer 2017)
APPSC AEE Screening Test 2016
Mizoram PSC AE/SDO Paper-II 2014
ISRO Scientist/Engineer 2009

Ans. (b) : $E = 2G(\mu + 1)$

8. The relation between modulus of elasticity (E) & bulk modulus (k) is

(a) E= 3k(1 – 2*Poisson's ratio)

(b) E= 3k(1 + Poisson's ratio)

(c) E= 3k(1 + 2 Poisson's ratio)

(d) E= 3k(1 – Poisson's ratio)

Rajasthan Nagar Nigam AE 2016, Shift-II
Rajasthan AE (Nagar Nigam) 2016 Shift-3
J & K PSC Screening, 2006
UPRVUNL AE 21.08.2016
OPSC Civil Services Pre 2006
GATE-2002

Ans. (a) : The relation between modulus of elasticity (E) & bulk modulus (k)

$E = 3k(1 - 2\mu)$

$E = 2G(1 + \mu)$

$$E = \frac{9KG}{3K + G}$$

$$\mu = \frac{3K - 2G}{6K + 2G}$$

Where μ – Poisson's ratio

9. The value of Poisson's ratio for steel varies from:

(a) 0.20 to 0.25 (b) 0.25 to 0.35

(c) 0.35 to 0.40 (d) 0.40 to 0.55

GPSC DEE, Class-2 (GWSSB) 04.07.2021
HPPSC AE 2018
Nagaland PSC (CTSE) Paper-I 2018
Vizag Steel (MT) 2017
GMB AAE 25.06.2017
Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : Poisson's ratio (μ) – the ratio of the transverse contraction of a material to the longitudinal extension strain in the direction of stretching force. is the Poisson's ratio of material.

The value of Poisson's ratio for

Steel (μ) = 0.25 to 0.33

Rubber (μ) = 0.50

Cork (μ) = 0

Aluminium (μ) = 0.32

Concrete (μ) = 0.20

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10. The deformation of a bar under its own weight as compared to that when subjected to a direct axial load equal to its own weight will be

(a) half (b) double
(c) one fourth (d) the same

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Nagaland PSC (CTSE) Paper-I 2018
UKPSC AE-2013, Paper-I
APPSC AEE 2012, ESE 1998

Ans. (a) : Deflection due to load P

$$\delta_1 = \frac{PL}{AE}$$

Deflection due to self load

$$\delta_2 = \frac{PL}{2AE}$$

$$\frac{\delta_2}{\delta_1} = \frac{\left[\frac{PL}{2AE}\right]}{\left[\frac{PL}{AE}\right]} = \frac{1}{2}$$

11. Which of the following materials is most elastic.

(a) glass (b) brass
(c) plastic (d) steel

GWSSB AAE, 27.12.2015
APPSC AEE 2012
APPSC AE 04.12.2012
J&K PSC Civil Services Pre, 2010
CSE Pre-1995

Ans. (d) : Elasticity is the property of solid materials to return to their original shape and size after the forces deforming them have been removed.
Steel is more elastic than any other thing because steel comes back to its original shapes faster than rubber when the deforming forces are removed.

12. A steel bar of 40 mm x 40 mm square cross-section is subjected to an axial compressive load of 200 kN. If the length of the bar is 2m and E = 200 GPa, the elongation of the bar will be:

(a) 1.25 mm (b) 2.70 mm
(c) 4.05 mm (d) 5.40 mm

ISRO Scientist/Engineer (RAC) 22.04.2018
RPSC Vice Principal ITI 2018
RPSC ADE 2016
UPRVUNL AE 2014
VIZAG Steel MT 18.06.2013
VIZAG Steel MT 10.06.2012
GATE-2006

Ans. (a) : Elongation or shortenig $\delta l = \frac{Pl}{AE}$

$$= \frac{200\times10^3\times2}{(0.040\times0.040)\times200\times10^9}$$

$=1.25\times10^{-3}$ m
=1.25 mm

13. Modulus of rigidity is defined as the ratio of

(a) longitudinal stress to longitudinal strain
(b) volumetric stress to volumetric strain
(c) lateral stress to lateral strain
(d) shear stress to shear strain

Assam PSC AE (IWT) 14.03.2021
VIZAG Steel MT 24.01.2021
APPSC AEE SCREENING 17.02.2019
SJVN ET 2019
Rajasthan Nagar Nigam AE Shift-II 2016
TNPSC AE (Industries) 09.06.2013

Ans. (d) : Shear modulus or modulus of rigidity is defined as the ratio of shear stress to the shear strain.

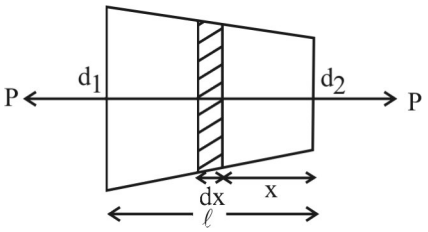
$$\text{Modulus of rigidity (G)} = \frac{\text{Shear stress}(\tau)}{\text{Shear strain}(\gamma)}$$

14. A bar uniformly tapering from a diameter d_1 at one end to a diameter d_2 at the other end, is subjected to an axial tensile load P. The length of the bar is ℓ and the young's modulus is E. The extension of the bar is

(a) $\frac{4P\ell}{\pi E d_1 d_2}$ (b) $\frac{P\ell E}{4\pi d_1 d_2}$
(c) $\frac{P\ell}{4\pi E d_1 d_2}$ (d) $\frac{4P\ell d_1 d_2}{\pi E}$

GPSC Engg. Class-2 Pre 19.01.2020
GPSC Executive Engineer 23.12.2018
APPSC AE Subordinate Service Civil/Mech. 2016
APPSC IOF, 2009
CSE Pre-1996
ESE 1995

Ans. (a) :



Consider a small strip of thickness 'dx' at a distance 'x' from smaller dia side.
Let the dia of the bar be d' at distance x

$$d' = d_1 - \left(\frac{d_1 - d_2}{\ell}\right)x$$

Let, $\left(\frac{d_1 - d_2}{\ell}\right) = K$

$\therefore d' = d_1 - Kx$

$\therefore$ cross sectional area at distance x from the smaller end $= A' = \frac{\pi}{4}d^{12} = \frac{\pi}{4}(d_1 - kx)^2$

Intensity of stress on the section = σ'

$$= \frac{P}{A'} = \frac{4P}{\pi(d_1 - kx)^2}$$

$\therefore$ Strain = $e' = \frac{\sigma'}{E} = \frac{4P}{\pi E(d_1 - kx)^2}$

$\therefore$ Extension of the length dx = e'dx

$$= \frac{4P}{\pi E(d_1 - kx)^2}.dx$$

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$\therefore$ Total extension of the bar $= \delta = \dfrac{4P}{\pi E}\int_0^{\ell}\dfrac{dx}{(d_1 - kx)^2}$

$$\delta = -\frac{4P}{\pi EK}\left[\frac{1}{d_1 - kx}\right]_0^{\ell}$$

$$\delta = \frac{4P}{\pi EK}\left[\frac{1}{d_1 - k\ell} - \frac{1}{d_1}\right]$$

put, $K = \dfrac{d_1 - d_2}{\ell}$

$$\delta = \frac{4P\ell}{\pi E(d_1 - d_2)}\left[\frac{1}{d_1 - d_1 + d_2} - \frac{1}{d_1}\right]$$

$$\delta = \frac{4P\ell}{\pi E(d_1 - d_2)}\left[\frac{1}{d_2} - \frac{1}{d_1}\right]$$

$$\delta = \frac{4P\ell}{\pi E(d_1 - d_2)} \cdot \frac{d_1 - d_2}{d_1 d_2}$$

$$\delta = \frac{4P\ell}{\pi E d_1 d_2}$$

For the particular case when the rod is of uniform dia.

$d_1 = d_2 = d$

and for this case, $\delta = \dfrac{4P\ell}{\pi E d^2}$

15. If Poisson's ratio is 0.25, then the modulus of rigidity (G) and the modulus of elasticity (E) are related as

(a) $G = 0.33\ E$ (b) $G = 0.5\ E$
(c) $G = 0.4\ E$ (d) $G = 0.25\ E$

CGPSC AE 15.01.2021
RPSC 2016
APPSC AEE Screening Test 2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
APPSC Poly. Lect. 2013
TNPSC AE, 2008

Ans. (c) Given, $\mu = 0.25$

We know that, $E = 2G(1+\mu)$

$E = 2G(1+.25)$

$E = 2G \times 1.25$

$E = 2.50G$

$E = 2G(1+\mu)$

$G = \dfrac{10}{25}E$

$G = \dfrac{2}{5}E$

$\boxed{G = 0.4E}$

16. If the value of Poisson's ratio is zero then it means that :

(a) The material is rigid
(b) The material is perfectly plastic
(c) There is no longitudinal strain in the material
(d) The longitudinal strain in the material is infinite.

OPSC AEE 2019 Paper-I
GPSC ARTO Pre 30.12.2018
Kerala PSC IOF 19.04.2016
KPSC ADF 2015
ESE 1994

Ans. (a) : If the value of Poisson's ratio is zero then it means that the material is rigid.

17. The unit of elastic modulus is the same as those of

(a) Stress, strain and pressure
(b) Strain, shear modulus and pressure
(c) Shear modulus, stress and force per unit area
(d) Strain, shear modulus and force

GPSC Executive Engg. 23.12.2018
HPPSC Asstt. Prof. 18.09.2017
APPSC AE 04.12.2012
UKPSC AE 2012 Paper-I
APPSC AEE 2012

Ans. (c) : Unit of elastic modulus (E) = MPa or Pa or N/m^2
(i) Shear modulus (G) = MPa or Pa or N/m^2
(ii) Stress (σ) = MPa or Pa or N/m^2
(iii) Force per unit area $\dfrac{F}{A} = N/m^2$

18. For a linearly elastic, isometric and homogeneous material , the number of elastic constants required to relate stress and strain are :

(a) Four (b) Two
(c) Three (d) Six

UPRVUNL AE 05.07.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
OPSC AEE 2015 Paper-I
TSPSC AEE 2015
APPSC AEE 2012
ESE 1998
CSE Pre-1990

Ans : (b) For a linearly elastic, isometric and homogeneous material, the number of elastic constant required to relate stress and strain are two.

19. If the radius of wire stretched by a load is doubled, then its Young's Modulus will be

(a) Doubled
(b) Halved
(c) Become four times
(d) Become one fourth
(e) None of the above

TNPSC AE 2017
Rajasthan Nagar Nigam AE 2016, Shift-II
CGPSC Poly. Lect. 22.05.2016
ISRO Scientist/Engineer 2006

Ans. (e) : The Young's Modulus of any materials is equal to the ratio of stress and the strain.
Its formula is given by –

$$E = \frac{\sigma}{\varepsilon}$$

or

$$E = \frac{F/A}{\Delta L/L}$$

The Young's Modulus of any material is inherited property. Its only depends on the temperature and the pressure.
So, if the radius of a wire stretched by a load is doubled then its Young's Modulus will remain unaffected.

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20. The number of elastic constants for a completely anisotropic elastic material are :

(a) 3 (b) 4
(c) 21 (d) 25

Kerala PSC Poly. Lect., 2017
APPSC AEE 2012
APPSC AE 04.12.2012
ESE 1999

Ans. (c) :

Material	Number of independent elastic constant
Homogeneous & Isotropic	2
Orthotropic	9
Anisotropic	21

21. A steel bar of 5 mm is heated from 15 °C to 40°C and it is free to expand. The bar will induce

(a) No stress (b) Shear Stress
(c) Tensile Stress (d) Compressive stress

Nagaland PSC (CTSE) Paper-I 2018
RPSC ADE 2016
APPSC AEE Mains (Civil Mechanical) 2016
Haryana PSC Civil Services Pre, 2014

Ans. (a) : A bar subjected to heating and allow to free expansion will not experience any stress. The stress produced due to oppose of expansion or contraction.

22. A solid uniform metal bar of cross-section area A and length L is hanging vertically from its upper end. If w is the total weight of bar and E the Young's modulus of elasticity, the elongation of bar due to self-weight would be

(a) WL/2AE (b) WL/4AE
(c) WL/AE (d) 2WL/AE

APPSC AEE SCREENING 17.02.2019
WBPSC AE, 2017
APPSC AEE Mains (Civil Mechanical) 2016
DRDO Scientist 2009

Ans. (a) : Specific weight, γ = Weight/Volume

$$\sigma = \frac{Ax\gamma}{A} = x\gamma$$

$$\in = \frac{\sigma}{E} = \frac{x\gamma}{E}$$

Total elongation of bar due to self weight,

$$= \int_0^L \in dx = \int_0^L \frac{x\gamma}{E} dx$$

$$= \frac{\gamma L^2}{2E} = \frac{WL}{2AE}$$

23. The material which exhibit the same elastic properties in all direction are called

(a) homogeneous (b) inelastic
(c) isotropic (d) anisotropic

APPSC AEE SCREENING 17.02.2019
Mizoram PSC AE/SDO 2014, Paper-II
APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (c) : Homogeneous Material–A material is said to be homogeneous when it exhibits same elastic properties at any point in a given direction i.e. properties are independent of point.

Example–Wood, Aluminium, Copper, Steel and Gold.

Isotropic Material–The material which exhibit the same elastic properties in all direction are called isotropic.
Example–Steel, Brass.

Homogeneous and Isotropic–A material is said to be homogeneous and isotropic when it exhibits same elastic property in any direction at any point.
Example–Steel, Copper, Gold, Aluminum.

Orthotropic Material–A material is said to be orthotropic when it exhibits different elastic property in orthogonal direction at a given point.
Example–Any layered material like plywood, graphite.

Anisotropic Material–A material which exhibits direction dependent elastic properties is known as anisotropic material.
Example–Composite material.

24. For a given material. Young's Modulus is 200 GN/m^2 and Modulus of Rigidity is 80 GN/m^2. Its Poisson Ratio will be:

(a) 0.15 (b) 0.20
(c) 0.25 (d) 0.35

SJVN ET 2013
UPSC JWM Adv. No-16/2009
UKPSC AE 2007 Paper -I
CSE Pre-2007

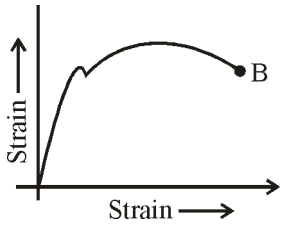
Ans. (c) : Given, E = 200 GN/m^2
G = 80 GN/m^2
$E = 2G(1 + \mu)$
$200 = 2 \times 80 (1 + \mu)$
$\boxed{\mu = 0.25}$

25. Which one of the following is rupture stress?

(a) Breaking stress
(b) Maximum load/original cross-sectional area (A)
(c) Load breaking point /Area (A)
(d) Load at breaking point/neck area

Assam PSC AE (PHED) 18.10.2020
TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-2006
TSPSC (Telangana)

Ans. (a&d) :



B = Rupture stress or Breaking stress.

26. If a material expands freely due to heating it will develop

(a) Thermal stresses (b) Tensile stress
(c) No stress (d) Bending

Assam PSC AE (PHED), 18.10.2020
(CGPSC Polytechnic Lecturer 2017)
UPRVUNL AE 2014
ISRO Scientist/Engineer 2007

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Ans. (c) : If a material expands freely due to heating it will develop no stress i.e. zero stress.

27. Young's modulus of elasticity and Poisson's ratio of a material are 1.25×10^5 MPa and 0.34 respectively. The modulus of rigidity of the material is

(a) 0.4025×10^5 MPa (b) 0.4664×10^5 MPa
(c) 0.8375×10^5 MPa (d) 0.9469×10^5 MPa

APGCL AM, 2021
APPSC AEE 2012
APPSC AE 04.12.2012
CSE Pre-1994

Ans. (b) :
Given:-

$E = 1.25 \times 10^5$ MPa
$\mu = 0.34$
$E = 2G(1 + \mu)$

$$\frac{1.25 \times 10^5}{2(1+0.34)} = G$$

$G = 46641.79$ MPa
$= 0.4664 \times 10^5$ MPa

28. The bulk modulus of a body is equal to

(a) $\frac{mE}{3(m-2)}$ (b) $\frac{mE}{3(m+2)}$
(c) $\frac{mE}{2(m-2)}$ (d) $\frac{mE}{2(m+2)}$

(where $\frac{1}{m}$ = Poisson's Ratio and E = Young's Modulus of Elasticity)

RPSC IOF, 2020
Nagaland PSC (CTSE) Paper-I 2018
TNPSC ACF 2012

Ans. (a) : The bulk modulus of a body

$$K = \frac{mE}{3(m-2)}$$

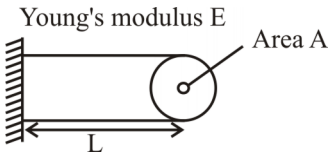
Where, K = bulk modulus
E = modulus of elasticity
$\frac{1}{m} = \mu$ = Poissons ratio

29. A circular rod of length L and area of cross-section A has a modulus of elasticity E and coefficient of thermal expansion α. One end of the rod is fixed and the other end is free. If the temperature of the rod is increased by ΔT, then

(a) Stress developed in the rod is zero and strain developed in the rod is αΔT
(b) Both stress and strain developed in the rod are zero
(c) Stress developed in the rod is EαΔT and strain developed in the rod is αΔT
(d) Stress developed in the rod is EαΔT and strain developed in the rod is zero

Assam Engg. College AP/Lect. 18.01.2021
ISRO Scientist/Engineer 07.05.2017
TSPSC AEE 2015, GATE 2014

Ans. (a) :



If temperature is increased by ΔT then
Stress developed in the rod is zero. ($\sigma_{th} = 0$) because there is no any restriction.

Strain developed in the rod $\varepsilon = \frac{\alpha \Delta T \times L}{L}$

$\varepsilon = \alpha \Delta T$

30. A steel rod 2m long is heated through a temperature of 100°C. The coefficient of linear expansion is 6.5×10^{-6}/°C and Young's Modulus is 2×10^6 N/m^2, the stress induced in the bar will be :

(a) 1000 N/m^2 (b) 1200 N/m^2
(c) 1300 N/m^2 (d) 1400 N/m^2

GPSC ARTO 01.05.2016
MPPSC AE 08.11.2015
MPPSC State Forest Service Exam, 2014

Ans. (c) : Given, $\alpha = 6.5 \times 10^{-6}$/°C, $L_0 = 2$m
$E = 2 \times 10^6$ N/m^2 $\Delta t = 100$°C
$\Delta L = L_0 \alpha \Delta T$
Thermal stress = $E \alpha \Delta T$
$= 2 \times 10^6 \times 6.5 \times 10^{-6} \times 100$
$= 1300$ N/m^2

31. Bars of copper and steel form a composite system. They are heated to a temperature of 40^{0}C. What type of stress is induced in the copper bar?

(a) Tensile
(b) Compressive
(c) Both tensile and compressive
(d) Shear.

GPSC Poly. Lect. 07.12.2014
ESE 2013
TRB Asstt. Prof., 2012

Ans. (b) : $\alpha_{copper} = 16 \times 10^{-6}$/°C
$\alpha_{steel} = 12 \times 10^{-6}$/°C
Thermal expansion of copper material is more as compare to steel hence as temperature increases the copper material elongate more than steel hence copper material experiences compression while steel material experiences tension.

32. The maximum stress produced in a bar of tapering section is at

(a) smaller end (b) larger end
(c) middle (d) anywhere

Haryana PSC Civil Services Pre, 2014
VIZAG STEEL MT 2011

Ans. (a) : The maximum stress produced in a bar of tapering section is at smaller end.

33. The Young's modulus of elasticity of a material is 2.5 times its modulus of rigidity. The poission's ratio for the material will be:

(a) 0.50 (b) 0.75
(c) 0.33 (d) 0.25

HPPSC Asstt. Prof. 18.09.2017
UPRVUNL AE 21.08.2016
CSE Pre-1997

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Ans. : (d) Relation between Young's Modulus and modulus of rigidity

$E = 2G(1 + \mu)$
$E = 2.5\ G$
$2.5\ G = 2G(1+\mu)$
$2.5 = 2 + 2\mu$
$2\mu = 0.5$
$\mu = 0.25$

34. The ratio of modulus of rigidity to modulus of elasticity for a Poisson's ratio of 0.25 would be

(a) 0.5 (b) 0.3
(c) 0.4 (d) 0.2
(e) None of the above

CGPSC AE 16.10.2016
GPSC Poly. Lect. 07.12.2014
ESE 2007

Ans. (c) : As we know,

$$G = \frac{E}{2(1+\mu)}$$

So, $\frac{E}{G} = 2(1+\mu)$ or $\frac{G}{E} = \frac{1}{2(1+\mu)}$

$$\frac{G}{E} = \frac{1}{2(1+0.25)} = 0.4$$

35. A rod of length L and diameter D is subjected to a tensile load P. Which of the following is sufficient to calculate the resulting change in diameter ?

(a) Young's modulus
(b) Shear modulus
(c) Poisson's ratio
(d) Both Young's modulus and Shear modulus

GPSC ARTO Pre 30.12.2018
BPSC Asstt. Prof. 29.11.2015
GATE-2008

Ans. (d) : We can find longitudinal elongation using Young's modulus–

$$\Delta L = \frac{PL}{AE}$$

Now to find decrease in diameter of rod, Poisson's ratio (μ) should be known –

$$\mu = -\frac{\text{Lateral strain or diametral strain}}{\text{Longitudinal strain}}$$

Hence we need both E and μ but by knowing E and G we can find μ, $E = 2G(1 + \mu)$
So, both Young's modulus and shear modulus.

36. If poisson's ratio for a material is 0.5, then the elastic modulus for the material is-

(a) 3 times its shear modulus
(b) 4 times its shear modulus
(c) Equal to its shear modulus
(d) Indeterminate

APPSC IOF, 2009
ISRO Scientist/Engineer 2006
ESE 1995

Ans. (a) : Given–
Poisson's ratio (μ) = 0.5

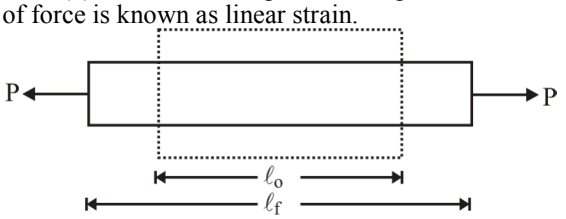
$E = 2G(1 + \mu)$
$\mu = 0.5,$
$E = 2G(1 + 0.5)$
$\frac{E}{2} = G(1.5)$
$E = 3G$

37. Deformation per unit length in the direction of force is known as

(a) Lateral strain (b) Linear strain
(c) Tensile stress (d) Linear stress

HPPSC Poly. Lect. 05.07.2021
Rajasthan Nagar Nigam AE 2016, Shift-I
UPRVUNL AE 2014

Ans. (b) : Deformation per unit length in the direction of force is known as linear strain.



Deformation ($\delta\ell$) per unit length in the direction of force/load is know as linear/longitudinal strain.

$\therefore$ linear or longitudinal strain $(\epsilon_{long}) = \frac{\delta\ell}{\ell_o} = \frac{\ell_f - \ell_o}{\ell_o}$.

38. A steel rod 10 mm in diameter and 1 m long is heated from 20 ºC to 120 ºC,E = 200 GPa and $\alpha = 12 \times 10^{-6}$ per ºC. If the rod is not free to expand, the thermal stress developed is

(a) 120 MPa (tensile) (b) 240 MPa (tensile)
(c) 120 MPa (comp. (d) 240 MPa (comp.)

RPSC Vice Principal ITI 2018
TSPSC AEE 28.08.2017 (Civil/Mechanical)
UPRVUNL AE 21.08.2016, CSE Pre 2003

Ans. (d) : Thermal strain = $e = \alpha\Delta T$
Thermal stress = $\sigma = E.e = E\ \alpha\ \Delta T$
$= (200 \times 10^9) \times (12 \times 10^{-6}) \times (120 - 20)$
$= 200 \times 10^9 \times 12 \times 10^{-6} \times 100$
= 240 MPa (compressive)
Since, body tries to expand, but it's expansion is restricted.
$\therefore$ Compressive stress will be induced.

39. In a simple tensile test, Hooke's law is valid upto the

(a) elastic limit
(b) limit of proportionality
(c) ultimate stress
(d) breaking point

TSPSC AEE 28.08.2017 (Civil/Mechanical)
TNPSC AE (Industries) 09.06.2013
APPSC AE 04.12.2012, CSE Pre 1998

Ans. (b) : In a simple tensile test, Hooke's law is valid upto the limit of proportionality.

40. The independent elastic constants for a homogeneous and isotropic material are

(a) E, G, K, ν (b) E, G, K
(c) E, G, ν (d) E, G

APPSC AEE 2012
CSE Pre-1995

Ans. (d) : Isotropic materials have the same properties in all directions. The number of independent elastic constants for such materials is 2. Out of E, G, K and μ if any two constants are known for any linear elastic and isotropic material then rest two can be derived.

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41. The true strain $\in_t$ and engineering strain $\in$ relationship is

(a) $\in_t = ln(1-\in)$ (b) $\in_t = ln(1+\in)$

(c) $\in_t = ln(1-2\in)$ (d) $\in_t = ln\dfrac{1}{(1+\in)}$

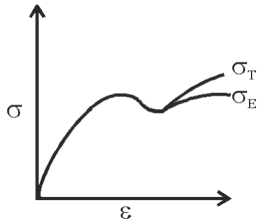
UPPSC AE 12.04.2016 Paper-I
GPSC ARTO 01.05.2016
GATE-2014

Ans : (b) True strain:-

$$\in_T = \int_{Lo}^{L_f} \frac{\delta\ell}{\ell} = \left[l\,n\right]_{Lo}^{L_f} = l\,n\left(\frac{L_f}{L_o}\right)$$

$$\in_T = l\,n\left(\frac{L_o + \Delta L}{L_o}\right)$$

$$\in_T = l\,n(1+\in)$$



$$\sigma_T = \sigma_0(1+\varepsilon)$$

42. Which of the relationship between bulk modulus (K), modulus of elasticity (E) and modulus of rigidity (G) is correct.

(a) $G = \dfrac{9KE}{K+3E}$ (b) $G = \dfrac{9KE}{E+3K}$

(c) $G = \dfrac{3KE}{E+9K}$ (d) $\dfrac{9}{E} = \dfrac{3}{G} + \dfrac{1}{K}$

ISRO Scientist/Engineer 22.04.2018
RPSC LECTURER 16.01.2016
ESE 2008

Ans. (d) :

$$E = \frac{9KG}{3K+G}$$

$$E = \frac{1}{\left[\frac{3K}{9KG}\right] + \left[\frac{G}{9KG}\right]}$$

$$E = \frac{1}{\left[\frac{1}{3G}\right] + \left[\frac{1}{9K}\right]}$$

$$\frac{1}{E} = \frac{1}{3G} + \frac{1}{9K}$$

$$\frac{9}{E} = \frac{3}{G} + \frac{1}{K}$$

43. A steel rod of diameter 1 cm and 1m long is heated from 20^0C its $\alpha = 12\times10^{-6}$/K and E = 200 GN/m^2. If the rod is free to expand, the thermal stress developed in it is–

(a) 12×10^4 N/m^2 (b) 240 KN/m^2

(c) Zero (d) Infinity

TSPSC AEE 28.08.2017 (Civil/Mechanical)
RPSC 2016
CSE Pre-2002

Ans : (c) If a material expands or contract freely due to heating or cooling. Then no stress will develop in material but if this expansion and contraction is prevented than internal resisting forces are developed in the material and because of these internal in the material.

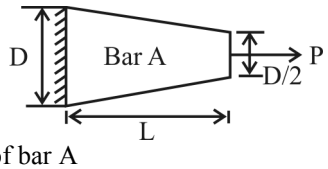
44. Two tapering bars of the material are subjected to a tensile load P. The length of both the bars are the same. The larger diameter of each of the bars is D. The diameter of the bar A at its smaller end is D/2 and that of the bar B is D/3. What is the ratio of elongation of the bar A to that of the bar B?

(a) 3 : 2 (b) 2 : 3

(c) 4 : 9 (d) 1 : 3

OPSC AEE 2019 Paper-I
TSPSC Managers, 2015
ESE 2006

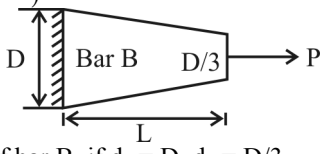
Ans : (b) :



Elongation of bar A

$$(\delta_A) = \frac{4PL}{\pi E d_1 d_2}$$

$$= \frac{4PL}{\pi E(D)(D/2)} = \frac{8PL}{\pi ED^2}$$



Elongation of bar B, if $d_1 = D$, $d_2 = D/3$

$$(\delta_B) = \frac{4PL}{\pi E d_1 d_2} = \frac{4PL}{\pi E(D)(D/3)} = \frac{12PL}{\pi ED^2}$$

$$\frac{\delta_A}{\delta_B} = \frac{8}{12} = \frac{2}{3}$$

45. For a ductile material, toughness is a measure of

(a) resistance to scratching
(b) ability to absorb energy up to fracture
(c) ability to absorb energy till elastic limit
(d) resistance to indentation

Gujarat PSC AE 2019
GPSC Executive Engineer 23.12.2018, GATE 2013

Ans. (b) : Toughness of a material is defined by energy absorbed up to fracture. It is equal to area load deformation diagram. Area under stress strain diagram up to fracture represents energy absorbed per unit volume which is known as modulus of toughness, it is measure of toughness of material.

46. If a block of material of length 25 cm, breadth 10 cm and height 5 cm undergoes a volumetric strain of 1/5000, then change in volume will be:

(a) 0.50 cm^3 (b) 0.25 cm^3

(c) 0.20 cm^3 (d) 0.75 cm^3

HPPSC Asstt. Prof. 18.09.2017
TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-1997

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Ans. (b) : Volumetric strain = $e_v = \frac{\Delta V}{V}$

Initial volume V = (25 × 10 × 5) cm^3

$e_v = \frac{1}{5000}(\text{given})$

$\Delta V = e_v \times V$

$= \frac{1}{5000} \times (25 \times 10 \times 5)$

$= 0.25\ cm^3$

47. A material has a young's modulus of 1.25×10^5 N/mm^2 and a poisson's ratio of 0.25. The bulk modulus of the material will be

(a) 83×10^5 N/mm^2 (b) 0.83×10^5 N/mm^2

(c) 8.3×10^3 N/mm^2 (d) 8.3×10^5 N/mm^2

ISRO Scientist/Engineer (RAC) 22.04.2018
Mizoram PSC AE/SDO Paper-II 2014

Ans. (b) : Given,

$E = 1.25 \times 10^5$ N/mm^2, $\mu = 0.25$

then $K = ?$

$E = 3K(1-2\mu)$

$1.25 \times 10^5 = 3 \times K \times (1 - 0.5)$

$K = \frac{2.50}{3} \times 10^5$

$K = 8.33 \times 10^4$

$K = 0.833 \times 10^5$ N/mm^2

48. A free bar of length 1 is uniformly heated from 0°C to a temperature t°C. α is the coefficient of linear expansion and E is the modulus of elasticity. The stress in the bar is

(a) α TE (b) E/αT

(c) Zero (d) None of the above

ISRO Scientist/Engineer (RAC) 07.05.2017
Karnataka PSC Lect., 27.05.2017, GATE 1995

Ans. (c) : The stress in the heated bar will be zero because the bar is free to expand so therefore no stress produced in heated bar.

49. Robert Hooke's discovered experimentally that within elastic limit

(a) Stress = strain

(b) Stress/strain = a constant

(c) Stress x strain = 1

(d) None of these

ISRO Scientist/Engineer 2011
ESE 2020

Ans. (b) : Hooke's law—According to Hook's law the stress is directly proportional to strain within elastic limit i.e. normal stress (σ) ∝ Normal strain (ε)

$\sigma = E\varepsilon$

And shear stress (τ) ∝ shearing strain (γ)

or $\tau = G\gamma$

50. The ratio of volumetric stress to the corresponding volumetric strain in a body is always constant is known as :

(a) Surface tension

(b) Bulk modulus of elasticity

(c) Compressibility

(d) Viscosity

Karnatka PSC AE (WRD) 31.07.2021
MECON MT 2019

Ans. (b) : The ratio of volumetric stress to the corresponding volumetric strain in a body is always constant is known as bulk modulus of elasticity.

51. The ratio between the change in volume and original volume of the body is called

(a) tensile strain (b) compressive strain

(c) volumetric strain (d) shear strain

UP Jal Nigam AE 2016
APPSC AEE 2012

Ans : (c)

Volumetric strain $= \frac{\text{change in volume}}{\text{original volume}}$

$$\varepsilon_v = \frac{\Delta V}{V}$$

52. A rod of copper originally 305 mm long is pulled in tension with a stress of 276 MPa. If the modulus of elasticity is 110 GPa and the deformation is entirely elastic, the resultant elongation will be nearly

(a) 1.0 mm (b) 0.8 mm

(c) 0.6 mm (d) 0.4 mm

ESE 2020
ESE 2019

Ans. (b)

Elongation,

$$\delta\ell = \frac{P\ell}{AE} = \frac{\sigma\ell}{E} = \frac{276 \times 305}{110 \times 10^3} = 0.765\,\text{mm} \simeq 0.8\,\text{mm}$$

53. Temperature stress are set up in a material when

(a) It is free to expand or contract

(b) It is first heated then cooled

(c) It is first cooled and then heated

(d) its expansion and contraction is restrained

Nagaland PSC CTSE 2017, 2016 Paper-I

Ans. (d) : Whenever there is some increase or decrease in the temperature of a body, it causes the body to expand or contract, there is no stresses are induced in the body. But, if the deformation of the body is prevented, some stresses are induced in the body, such stresses are known as thermal stresses or temperature stresses.

54. A 200 × 100 × 50 mm steel block is subjected to a hydrostatic pressure of 15 MPa. The Young's modulus and Poisson's ratio of the material are 200 GPa and 0.3 respectively. The change in the volume of the block is

(a) 100 mm^3 (b) 110 mm^3

(c) 85 mm^3 (d) 90 mm^3

(e) 80 mm^3

CGPSC AE 26.04.2015 Shift-I
GATE 2007

Ans. (d) : Given,

Volume (V) = 200 × 100 × 50 = 10^6 mm^3

Hydrostratic pressure (σ) = 15 MPa = 15 N/mm^2

Young's modulus (E) = 200 GPa = 200 × 10^3 N/mm^2

Poisson's ratio (μ) = 0.3

[∵ For Hydrostratic pressure, $\sigma_x = \sigma_y = \sigma_z = \sigma$]

Volumetric strain $(e_v) = \frac{3\sigma}{E}(1-2\mu)$

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$$\frac{\Delta V}{V} = \frac{3\sigma}{E}(1-2\mu)$$

$$\Delta V = \frac{3\sigma V}{E}(1-2\mu)$$

$$= \frac{3\times 15\times 10^6(1-2\times 0.3)}{200\times 10^3} = 90 \text{ mm}^3$$

55. _____ is the capacity of material to absorb energy when it is elastically deformed and then upon unloading, to have this energy recovered.

(a) Toughness (b) Tensile strength
(c) Plasticity (d) Resilience

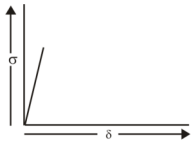
CIL MT 26.03.2017
ESE 2016

Ans. (d) : Resilience- It is energy absorbed by a member in elastic region. It denotes the capacity of material to absorb energy when it is elastically deformed and then upon unloading, to release this energy.

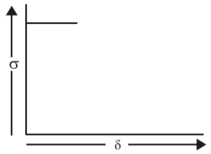
Toughness- It is energy absorbed by member just before its fracture.

56. The stress-strain curve of an ideal elastic material with strain hardening will be as-

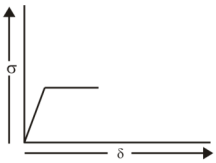
(a)



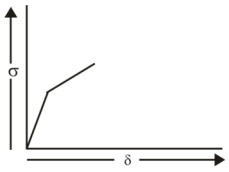
(b)



(c)

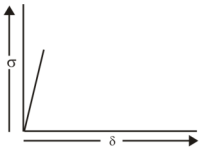


(d)

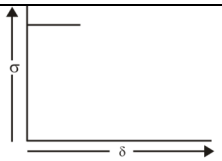


RPSC AE 2018
CSE Pre-1998

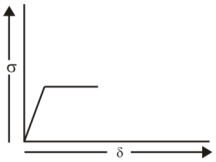
Ans. (d) : 1. The stress-strain curve for an ideal elastic material.



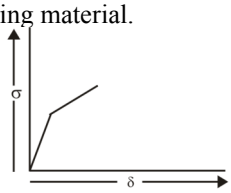
2. The stress-strain curve for rigid - Perfectly plastic material



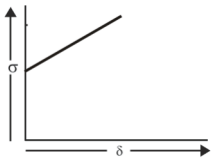
3. Stress-strain curve for elastic - Perfectly plastic material.



4. Stress-strain curve for an ideal elastic material with strain hardening material.



5. stress-strain curve for rigid - Linear hardening material

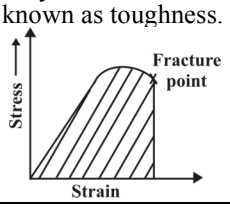


57. The ability of the material to absorb energy before fracture is known as:

(a) Toughness (b) Ductility
(c) Cold shortness (d) Hardness

UPRVUNL AE 07.10.2016
SJVN ET 2013

Ans. (a) : The ability of the material to absorb energy before fracture is known as toughness.



58. The equation for relationship between $\frac{1}{m}$, C & K is,

(a) $\frac{1}{m} = \frac{3K-2C}{6K+2C}$ (b) $\frac{1}{m} = \frac{2C-3K}{2C+6K}$

(c) $\frac{1}{m} = \frac{2K-3C}{2C+6K}$ (d) $\frac{1}{m} = \frac{3K+2C}{6K-2C}$

TNPSC AE 2013
BPSC AE 2012 Paper - VI

Ans. (a) : We know that relation between poisson ratio $\left[\mu \text{ or } \frac{1}{m}\right]$, modulus of rigidity (C) and Bulk Module (K) is given as

$$2C(1+\mu) = 3K(1-2\mu)$$

$$2C + 2C\mu = 3K - 6K\mu$$

$$\mu = \frac{1}{m} = \frac{3K-2C}{6K+2C}$$

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59. In a body, thermal stress is induced because of the existence of :
(a) Latent heat (b) Total heat
(c) Temperature gradient (d) Specific heat

OPSC AEE 2019 Paper-I
ESE 2013

Ans : (c) : Thermal stress- If the material is restrained from expanding or contracting while the temperature change, then stress builds within the part.

60. True strain for a steel bar which is doubled its length by tension is :
(a) 0.307 (b) 0.5
(c) 0.693 (d) 1.0

OPSC AEE 2019 Paper-I
ISRO Scientist/Engineer 2012

Ans : (c) :
Initial length of bar, $\ell_i = \ell$

Final length of bar, $\ell_f = 2\ell$

Engineering strain, $\in = \frac{\Delta L}{\ell} = \frac{2\ell_i - \ell_i}{\ell_i} = 1$

$True\,strain\left(\in_{true}\right) = \ln\left(1+\in\right) = \ln\left(1+1\right)$

$\in_{true} = \ln\left(2\right)$

$= 0.693$

61. Resilience of material should be considered when it is subjected to
(a) shock load (b) electroplating
(c) chemical coating (d) polishing

RPSC AE 2016
GWSSB DEE 07.07.2016
RPSC ACF-2011

Ans : (a) Resilience of material should be considered when it is subjected to shock loading.
Resilience:- It is the property of a materials to absorb energy and to resist shock and impact loads. It is measured by the amount of energy absorbed per unit volume within elastic limit this property is essential for spring materials.

62. Strain in direction at right angle to the direction of applied force is known as:-
(a) Lateral strain
(b) Shear strain
(c) Volumetric strain
(d) None of the above

UKPSC AE-2013, Paper-I
APPSC AEE 2012

Ans. (a) : Strain in direction at right angle to the direction of applied force is known as lateral strain.

63. Stress and Strain are tensor of
(a) zero-order (b) first order
(c) second order (d) None of the above

HPPSC Asstt. Prof. 2014
UKPSC AE 2012 Paper-I

Ans. (c) : Stress in second order tensor. Magnitude, direction and plane are required to describe stress that's why it is known as second order tension.

64. What does the elasticity of material enables it to do ?
(a) Regain the original shape after the removal of applied force.
(b) Draw into wires by the application of force.
(c) Resist fracture due to high impact.
(d) Retain deformation produced under load permanently.

UKPSC AE 2012 Paper-I
VIZAG STEEL MT 2011

Ans. (a) : Regain the original shape after the removal of applied force.

65. The relation between E (modulus of elasticity) and k (bulk modulus of elasticity) is
(a) $E = k\left[1-\frac{2}{m}\right]$ (b) $E = 2k\left[1-\frac{2}{m}\right]$
(c) $E = 3k\left[1-\frac{2}{m}\right]$ (d) $E = 4k\left[1-\frac{2}{m}\right]$

UPRVUNL AE 07.10.2016
UKPSC AE 2007 Paper -I

Ans. (c) : $E = 3k\left[1-\frac{2}{m}\right]$

66. Which one of the following is connect respect of poisson's ratio (μ) limits for an isotropic elastic solid:
(a) $0 < \mu < 1$ (b) $-1 < \mu < 1$
(c) $-1 < \mu < 0.5$ (d) $0 < \mu < 0.5$

PTCUL AE 25.06.2017
ESE 2004

Ans. (c) : The Poisson's ratio (μ) for any material lies in $\boxed{-1 < \mu < 0.5}$

67. Whenever some external system of forces act on a body, it undergoes deformation. As the body goes under deformation, it sets up some resistance to the deformation. What is this resistance per unit area of deformation called as?
(a) Strain
(b) Stress
(c) Pressure
(d) Modulus of elasticity

HPPSC Poly. Lect. 05.07.2021
VIZAG STEEL MT 2011

Ans. (b) : Resistance force per unit area of deformation is called stress.

68. Which one of the following information cannot be obtained from the static tensile test of a mild steel specimen?
(a) Modulus of elasticity
(b) Qualitative determination of toughness
(c) Ductility
(d) Weldability

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-2006

Ans. (d) : Weldability cannot be obtained from static tensile test.

69. A straight bimetallic strip of copper and steel is heated. It is free at ends. The strip will
(a) Expand and remain straight
(b) Not expand but bend
(c) Expand and bend also
(d) First only

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre 2002

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Ans. (c) : If heated and free to expand
since $\alpha_{copper} = 18 \times 10^{-6}$ /°c
$\alpha_{steel} = 11 \times 10^{-6}$ /°c
Copper expands more than steel and hence the bimetallic strip will bend with outer fiber as copper.

70. The work done per unit volume in elongating a body by a uniaxial force is :
(a) stress/strain (b) stress × strain
(c) $\frac{1}{2}$ stress × strain (d) None of the above

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (c) :
Workdone/volume = $\frac{1}{2}$ stress × strain = $\frac{1}{2}\sigma.\varepsilon$

71. The Poisson's ratio a material which has Young's modulus of 120 GPa and shear modulus of 50 GPa, is
(a) 0.1 (b) 0.2
(c) 0.3 (d) 0.4

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
UKPSC AE-2013, Paper-I

Ans. (b) : As, we know, the relation between E, μ & G
$E = 2G[1+\mu]$
$120 = 50 \times 2[1+\mu]$
$\mu = 0.2$

72. Thermal Strain, developed in a rod of length '*l*' and temperature rise 'ΔT', is given by- (Where α = Coefficient of Thermal Expansion; E = Young's modulus)
(a) αΔT (b) EαΔT*l*
(c) EαΔT (d) Zero

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
APPSC AEE 2012

Ans. (a) : As we know,
$\Delta L = \alpha \Delta T L$ [∴ General equation]
Strain, $\frac{\Delta L}{L} = \alpha \Delta T$

73. Which one of the following properties is more sensitive to increase in strain rate?
(a) Yield strength (b) Proportional Limit
(c) Elastic Limit (d) Tensile Strength

APPSC AEE 2012
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Yield strength is more sensitive to increase in strain rate.
Note : No answer is given by commission.

74. The ratio of Bulk Modulus to Young's Modulus of a Poisson's ratio of 0.25 will be :
(a) 1/3 (b) 2/3
(c) 1 (d) 3/2

APPSC AE Subordinate Service Civil/Mech. 2016
RPSC AE GWD, 2011

Ans. (b) : Given, Poisson's ratio (μ) = 0.25
$E = 3K(1-2\mu)$
$\frac{K}{E} = \frac{1}{3(1-2\mu)}$
$\frac{K}{E} = \frac{1}{3(1-2\times 0.25)}$
$\frac{K}{E} = \frac{1}{3\left(1-\frac{2}{4}\right)}$
$= \frac{4}{3\times 2}$
$\boxed{\frac{K}{E} = 2/3}$

75. If modulus of rigidity of a material is 80 GPa and modulus of elasticity is 200 GPa, what will be the bulk modulus of material?
(a) 133.3 GPa (b) 153.3 GPa
(c) 163.3 GPa (d) 173.3 GPa

APPSC Poly Lect. 13.03.2020
ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (a) : Given data : G = 80 GPa, E = 200 GPa
K = ?
∵ $E = 2G(1+\mu) \Rightarrow 200 = 2\times 80(1+\mu)$
$\Rightarrow 1+\mu = \frac{200}{2\times 80} \Rightarrow \mu = \frac{10}{8} - 1 = \frac{1}{4} = 0.25$
Now,
$E = 3K(1-2\mu)$
Bulk modulus (K) = $\frac{E}{3(1-2\mu)} = \frac{200}{3(1-0.50)} = \frac{200}{3\times 0.50}$
$\Rightarrow \boxed{K = 133.33 \text{ GPa}}$

76. A uniform slender cylindrical rod is made of a homogeneous and isotropic material. The rod rests on a frictionless surface. The rod is heated uniformly. If the radial and longitudinal thermal stresses are represented by σ_r and σ_z respectively, then
(a) $\sigma_r = 0, \sigma_z = 0$ (b) $\sigma_r \neq 0, \sigma_z = 0$
(c) $\sigma_r = 0, \sigma_z \neq 0$ (d) $\sigma_r \neq 0, \sigma_z \neq 0$

TSPSC AEE 2017
GATE-2005

Ans. (a) : When any material is heated or cooled and allowed to expand or contract without any restriction then they will not generate any stress, this is known as free expansion or contraction.
Hence, $\sigma_r = 0$ and $\sigma_z = 0$

77. The Young's modulus of a material is 125 GPa and Poisson's ratio is 0.25. The modulus of rigidity of the material is
(a) 30 GPa (b) 50 GPa
(c) 80 GPa (d) 100 GPa

Assam PSC CCE Pre 2015
JPSC AE 2013, Paper-V

Ans. (b) : Given as,
$E = 125$ GPa
$\mu = 0.25$
$E = 2G(1+\mu)$
$125 = 2G(1+0.25)$
$125 = 2G(1.25)$
$G = 50$ GPa

78. When a wire is stretched to double its length, the longitudinal strain produced in it is
(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

Assam PSC CCE Pre 2015
UKPSC AE-2013, Paper-I

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Ans. (b) : Given, Initial length, $\ell_i = \ell$

Final length, $\ell_f = 2\ell$

$$\text{Longitudinal strain} = \frac{\text{Change in Length}(\delta l)}{\text{Original Length}(l_o)}$$

If l_o be the original length and if stretched to double of its length, change in length $(\delta l) = (\ell_f - \ell_i) = (2 \times l - l) = l$

So, strain $(\in) = \frac{\delta l}{l_o} = \frac{l}{l} = 1$

79. Two threaded bolts A and B of the same material and length are subjected to identical tensile load. If the elastic strain energy stored in bolt A is 4 times that of bolt B and the mean diameter of bolt A is 12 mm, the mean diameter of bolt B in mm is:

(a) 16 (b) 24
(c) 36 (d) 48

NPCIL ET, 2015
GATE-2013

Ans. (b) : Strain energy is given by

$$U = \frac{P^2L}{2AE}$$

Given that-

$$U_A = 4U_B$$

$$\frac{P^2L}{2A_AE} = 4\frac{P^2L}{2A_BE}$$

Since bolts are of some material, length and subjected to identical tensile load.

$$\frac{1}{A_A} = 4\frac{1}{A_B}$$

$$\frac{1}{D_A^2} = 4\frac{1}{D_B^2}$$

$$D_B = 2D_A$$

$$D_B = 2 \times 12$$

$$D_B = 24 \text{ mm}$$

80. A solid cube of side 1 m is kept at a room temperature of 32°C. The coefficient of linear thermal expansion of the cube material is 1×10^{-5}/°C and the bulk modulus is 200 GPa. If the cube is constrained all around and heated uniformly to 42°C, then the magnitude of volumetric (mean) stress (in MPa) induced due to heating is_____

(a) 180 (b) 240
(c) 30 (d) 60

Assam Engg. College AP/Lect. 18.01.2021
GATE-2019

Ans. (d) : Given,

$\alpha = 1\times10^{-5}$/°C, K = 200 GPa, $T_1 = 32$°C, $T_2 = 42$°C

Volumetric strain $\varepsilon_v = 3 \times \varepsilon$

$= 3 \times \alpha\Delta T$

$= 3 \times 1\times10^{-5} \times (42 - 32)$

$= 3 \times 10^{-4}$

$$\text{Bulk modulus } K = \frac{\sigma_v}{\varepsilon_v} = \frac{\text{Direct stress}}{\text{Volumetric strain}}$$

$\sigma_v = K \times \varepsilon_v$

$= 200 \times 10^3 \times 3 \times 10^{-4}$

Volumetric stress $\sigma_v = 60$ MPa

81. A rod of length 20 mm is stretched to a length of 40 mm. Subsequently, it is compressed to make a rod of final length 10 mm. Consider the longitudinal tensile strain as positive and compressive strain as negative. What is the total true longitudinal strain in the rod?

(a) –0.5 (b) –1.0
(c) –0.75 (d) –0.69

Assam Engg. College AP/Lect. 18.01.2021
GATE-2017

Ans. (d) : Total true longitudinal strain

$$(\varepsilon_T) = (\varepsilon_T)_1 + (\varepsilon_T)_2$$

$$= \left[l\,n\left(\frac{l_f}{l_i}\right)\right]_1 + \left[l\,n\left(\frac{l_f}{l_i}\right)\right]_2$$

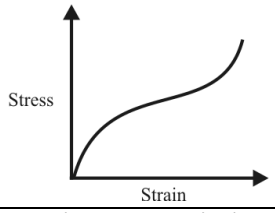
$$= l\,n\left(\frac{40}{20}\right) + l\,n\left(\frac{10}{40}\right) = -0.693$$

82. The elastic stress-strain behavior of rubber is -

(a) Linear
(b) Non-linear
(c) Plastic
(d) No fixed relationship

TNPSC AE 2018
GMB AAE 25.06.2017

Ans. (b) : The most common example of this kind of material is rubber whose stress-strain relationship can be defined as non-linearly Elastic, Isotropic, in compressible and generally Independent of strain role. Hyper elasticity provides a means of modeling the stress-strain behavior of such material.



83. For the material loaded within the elastic limit, the Young's modulus is defined as the ratio of

(a) longitudinal strain in lateral strain
(b) volumetric stress to volumetric strain
(c) longitudinal stress to longitudinal strain
(d) shear stress to shear strain

Haryana PSC AE (PHED) 05.09.2020, Paper-II
GPSC Lect. 23.10.2016
SJVN ET 2013
VIZAG Steel MT 2011

Ans. (c) :

$$\text{Young's modulus } (E) = \frac{\text{Longitudinal stress}(\sigma)}{\text{Longitudinal strain}(\varepsilon)}$$

84. When a body is subjected to three mutually perpendicular stresses of equal intensity, the ratio of direct stress to the expanding volumetric strain is known as :

(a) Young's modulus
(b) Modulus of rigidity
(c) Bulk modulus
(d) Poisson's ratio

ISRO Scientist/Engineer (RAC), 10.03.2019
Punjab PSC SDE 12.02.2017

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Ans. (c) :

$$\text{Bulk modulus (K)} = \frac{\sigma}{\left(\frac{\Delta V}{V}\right)} = \frac{\sigma}{\text{Volumetric strain}}$$

85. A copper rod 400 mm long is pulled in tension to a length of 401.2 mm by applying a tensile load of 330 MPa. If the deformation is entirely elastic, the Young's modulus of copper is

(a) 110 GPa (b) 110 MPa
(c) 11 GPa (d) 11 MPa

GPSC EE Pre, 28.01.2017
ESE 2012

Ans. (a) : Given-
$l_1 = 400$ mm
$l_2 = 401.2$ mm
Elastic of modulus = (E)

$$\text{Strain } (\varepsilon) = \frac{l_2 - l_1}{l_1}$$

$$= \frac{401.2 - 400}{400}$$

$$\text{Strain } (\varepsilon) = \frac{1.2}{400} = 0.003$$

$$\text{Modulus of elasticity (E)} = \frac{\text{stress}}{\text{strain}}$$

$$(E) = \frac{330 \times 10^6}{0.003}$$

$$= \frac{330 \times 10^6 \times 10^3}{3}$$

$$(E) = 110 \times 10^9 = 110 \text{ GPa}$$

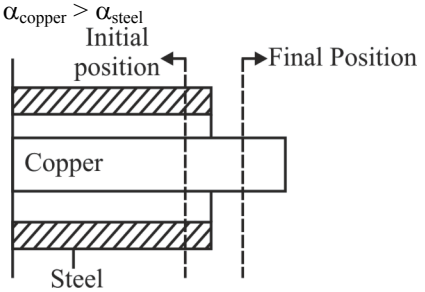
86. A composite bar made of copper and steel is heated to 120 °C from room temperature. If $\alpha_c > \alpha_s$, the stress induced in copper bar is:

(a) No stress (b) Compressive stress
(c) Shear stress (d) Tensile stress

CIL MT 27.02.2020
GPSC EE Pre, 28.01.2017

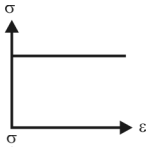
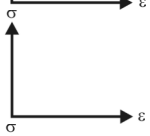
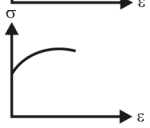
Ans. (b) : Given, T = 120 °C

$\alpha_{copper} > \alpha_{steel}$



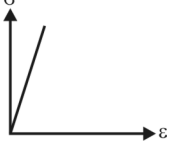
As the coefficient of expansion of copper is more as compared to that of steel.
The final expansion is less than the free expansion of copper due to temperature rise and thus compressive stress will be developed in copper bar.

87. The stress-strain relationship for the perfectly rigid material is :

(a) 
(b) 
(c) 
(d) None of the above

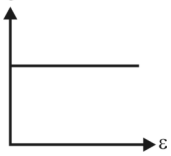
BPSC Asstt. Prof. 29.11.2015
CSE Pre-1996

Ans. (b) : (i) Linear Elastic Materials-

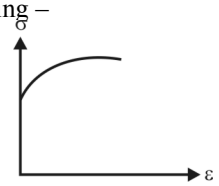


(ii) Rigid materials-

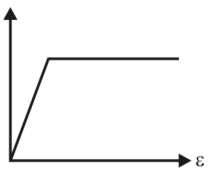
(iii) Perfectly Plastic (Non-Strain hardening)-



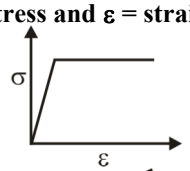
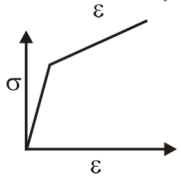
(iv) Rigid hardening –



(v) Elasto Plastic Material-



88. stress-strain relationship best describes the behavior of brittle materials such as ceramics and thermosetting plastics, (σ = stress and ε = strain)?

(a) 
(b) 

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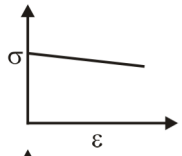
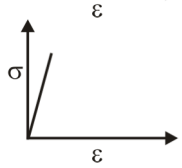
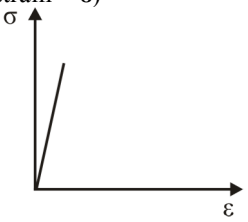
(c) 

(d) 

(e) None of the above

CGPSC Poly. Lect. 22.05.2016
GATE-2015

Ans. (d) : In stress-strain relationship best describes describes the behavior of brittle materials such as ceramics and thermosetting plastics
(σ = stress and strain = ε)



89. Normal stresses are divided into
(a) tensile and torsion
(b) compressive and thermal
(c) shear and bending
(d) tensile and compressive

UPRVUNL AE 05.07.2021

Ans. (d) : Normal stresses are divided into tensile and compressive.

90. Eccentric loading applies to _____ stress
(a) Bending (b) torsional
(c) tensile (d) thermal

UPRVUNL AE 05.07.2021

Ans. (a) : Eccentric loading applies to bending stress.

91. True stress at fracture _____ during tensile testing.
(a) becomes zero
(b) is the same as ultimate stress
(c) is higher than ultimate stress
(d) is lower than ultimate stress

UPRVUNL AE 05.07.2021

Ans. (c) : True stress at fracture is higher than ultimate stress during tensile testing.

92. Factor of safety is equal to _____
(a) $\frac{\text{Allowable stress}}{\text{Failure stress}}$
(b) $\frac{\text{Allowable load}}{\text{Failure load}}$
(c) Allowable stress × failure stress
(d) $\frac{\text{Failure stress}}{\text{Allowable stress}}$

UPRVUNL AE 05.07.2021

Ans. (d) : Factor of safety = $\frac{\text{Failure stress}}{\text{Allowable stress}}$

93. Measurements have been made for the Young's modulus of elasticity for mild steel specimen both in tension and compression. The ratio Et / Ec will be approximately equal to
(a) 0.5 (b) 0.75
(c) 1.0 (d) 1.25

HPPSC Poly. Lect. 05.07.2021

Ans. (b) : 0.75

94. The modulus of rigidity of a material whose Young's modulus is 200 Nmm^{-2} and poisson's ratio is 0.5 is
(a) 133.33 Nmm^{-2} (b) 80.67 Nmm^{-2}
(c) 66.67 Nmm^{-2} (d) 242.67 Nmm^{-2}

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Given,
$E = 200\ N/mm^2$
$\mu = 0.5$
We know,
$E = 2G(1+\mu)$
$200 = 2 \times G\ (1+0.5)$
$200 = 2G\ (1.5)$
Modulus of Rigidity, $G = 66.67\ N/mm^2$

95. Resilence is the
(a) energy stored in a body when strained within the elastic limit
(b) energy stored in a body when strained up to breaking point
(c) max strain energy stored
(d) none of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Resilence is the energy stored in a body when strained within the elastic limit.

96. In a bolt and tube assembly, due to the tightening of the nut on the bolt, _____ stress is developed in the bolt and _____ stress is developed in the tube.
(a) compressive, compressive
(b) tensile, compressive
(c) compressive, tensile
(d) tensile, tensile

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : In a bolt and tube assembly, due to the tightening of the nut on the bolt tensile stress is developed in the bolt and compressive stress is developed in the tube.

97. A copper bar of rectangular section of 20 mm × 30 mm and a length of 500 mm is subjected to a compressive load of 60 kN. The compressive stress acting on the bar is _____ N/mm^2.
(a) 150 (b) 50
(c) 200 (d) 100

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Given,
Area, A = 20 mm × 30 mm
$= 600\ mm^2$
Compressive Load, P = 60 kN, length, l = 500 mm
Compressive stress, $\sigma_c = \frac{P}{A} = \frac{60\times10^3}{600}$
$\sigma_c = 100\ N/mm^2$

98. A rod of length 2 m and diameter 50 mm is elongated by 5 mm when an axial force of 400 kN is applied. The modulus of elasticity of the material of rod will be nearly

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(a) 66 GPa (b) 72 GPa
(c) 82 GPa (d) 96 GPa

APGCL AM, 2021

Ans. (c) :

Given:-

$L = 2m,$

$P = 400kN$

$\delta L = 5mm$

$d = 50mm$

$$\text{Area} = \frac{\pi}{4}(50)^2$$

$$\sigma = \frac{\text{Load}}{\text{Area}} = \frac{400\times10^3}{\frac{\pi}{4}\times(50)^2}$$

$$\sigma = 203.82 \text{ MPa}$$

$$\text{Strain, } \varepsilon = \frac{\delta L}{L} = \frac{5}{2000} = 2.5\times10^{-3}$$

$$\text{Now stress, } \sigma = E\varepsilon$$

$$203.82 = E\times2.5\times10^{-3}$$

$$E = \frac{203.82}{2.5\times10^{-3}} = 81.5 \text{ GPa}$$

99. A tensile test was conducted on mild steel bar. The load at elastic limit was 250 kN and the diameter of steel bar was 3 cm. The value of stress is

(a) 35368×10^4 N/m^2
(b) 32463×10^4 N/m^2
(c) 35625×10^4 N/m^2
(d) 37562×10^4 N/m^2

APGCL AM, 2021

Ans. (a) : Given,

$P=250 \text{ kN} = 250 \times 10^3$ N

$d=3\text{cm}=3 \times 10^{-2}$m

$$\text{Stress}=\sigma = \frac{P}{A} = \frac{250\times10^3}{\frac{\pi}{4}\times\left(3\times10^{-2}\right)^2} = 353677.65 \times 10^3 \text{ N/m}^2$$

$$\approx 35368 \times 10^4 \text{ N/m}^2$$

100. An annealed copper strip, 9 inch (228mm) wide and 1 inch (25mm) is rolled to a thickness of 0.80 inch (20mm) in one pass.
The absolute value of true strain that the strip undergoes is :

(a) 0.20 (b) 0.25
(c) 0.223 (d) 0.225

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Given, $t_i = 25mm$, $t_f = 20mm$

$$\text{True strain, } \varepsilon_T = \ln(1+\varepsilon) = \ln\left(\frac{t_f}{t_i}\right)$$

$$= \ln\left(\frac{25}{20}\right)$$

$$\boxed{\varepsilon_T = 0.223}$$

101. Aluminium rod with original length L_o is strained. The length of the rod at any instant is given by $L(t) = L_o(1+t^2)$, where t is in minutes. The true strain at the end of 3 minutes will be:

(a) 6 (b) 9
(c) 0.6 (d) 0.9

AAI Jr. Executive 26.03.2021

Ans. (*) :

$$\text{True strain at any instant } (\varepsilon_T) = \int\frac{dL}{L} = \int\frac{dL}{L_o(1+t^2)}$$

As, $L = L_o(1 + t^2)$

$dL = L_o 2tdt$

$$\varepsilon_T = \int\frac{dL}{L_o(1+t^2)} = \int\frac{2tdt}{(1+t^2)} = \ln[1+t^2]$$

$$\left[\begin{array}{l}\text{Let}, 1+t^2 = x,\\ dx = 2tdt\end{array}\right]$$

True strain at the end of 3 minutes will be

$\varepsilon_T = \ln[1 + (3)^2] = \ln 10 = 2.30$

Note-There is no option in the given question.

102. The relationship between conventional strain, ε_0 and true strain, εT is:

(a) $\varepsilon_0 = \{e^{\varepsilon T}\}$ (b) $\varepsilon_0 = \{e^{\varepsilon T}\} - 1$
(c) $\varepsilon_0 = \{e^{-\varepsilon T}\}$ (d) $\varepsilon_0 = \{e^{\varepsilon T}\} + 1$

AAI Jr. Executive 26.03.2021

Ans. (b) : Conventional strain/Engg. strain = ε_0

True strain = εT

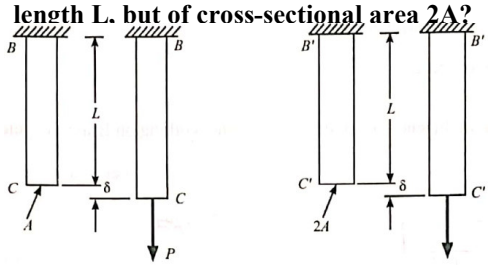
$\varepsilon T = \ln(1 + \varepsilon_0)$

$1 + \varepsilon_0 = e^{\varepsilon T}$

$\varepsilon_0 = e^{\varepsilon T} - 1$

$$\boxed{\varepsilon_0 = \{e^{\varepsilon T}\} - 1}$$

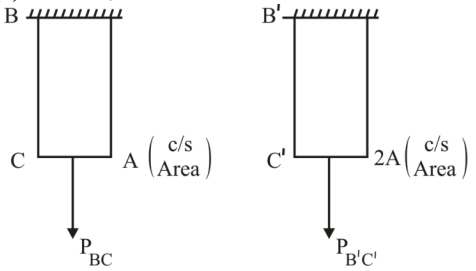
103. If a deformation δ is produced in the rod BC of cross-sectional area A of the figure below by a load P, what load is required to cause the same deformation in the rod B 'C' of the same length L, but of cross-sectional area 2A?



(a) 6P (b) 4P
(c) 2P (d) P

CGPSC AE 15.01.2021

Ans. (c) : Given,



$$\delta_{BC} = \frac{P_{BC}L}{AE}$$

$$\delta_{B'C'} = \frac{P_{B'C'}L}{2AE}$$

Acc to question, $\delta_{BC} = \delta_{B'C'}$

$$\frac{P_{BC}L}{AE} = \frac{P_{B'C'}L}{2AE}$$

$$\boxed{P_{B'C'} = 2P_{BC}}$$

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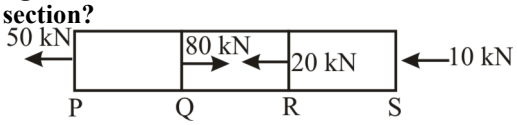
104. A steel rod of 30 mm diameter and 5 m long is connected to two grips and the rod is maintained at 95°C. Determine the stress induced in the rod when it is cooled to 30°C, if $E = 2 \times 10^5$ MN/m^2; $\alpha = 12 \times 10^{-6}$ /°C .

(a) 156 N/mm^2 (b) 65 N/mm^2
(c) 165 N/mm^2 (d) 185 N/mm^2

VIZAG Steel MT 24.01.2021

Ans. (a) : Given,
Diameter (d) = 30 mm
Length (ℓ) = 5 m
$T_1 = 95°C$, $T_2 = 30°C$
$E = 2 \times 10^5$ MN/m^2, $\alpha = 12 \times 10^{-6}$/°C
$\therefore$ Thermal stress (σ_T) = $\alpha \Delta T E$
$= 12 \times 10^{-6} \times (95 - 30) \times 2 \times 10^5 \times 10^6$
$= 12 \times 65 \times 2 \times 10^5$
$= 1560 \times 10^5$ N/m^2
$= 156$ N/mm^2

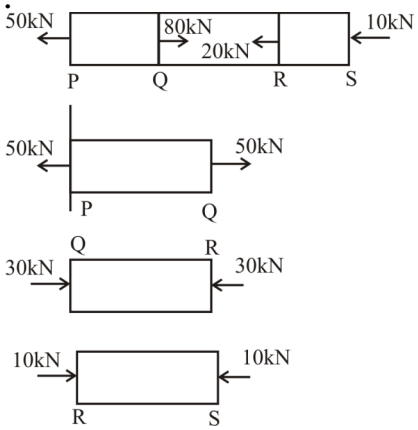
105. A bar, having 1000 mm^2 cross-sectional area, is subjected to axial loads as shown in the figure. What is the value of stress in QR section?



(a) 30 MPa Compressive
(b) 30 MPa Tensile
(c) 50 MPa Compressive
(d) 50 MPa Tensile

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) :



Stress in QR section

$$\sigma = \frac{P}{A}$$

$$= \frac{30 \times 10^3}{1000} = 30\,\text{MPa (Compressive)}$$

106. A steel rod, with all its surfaces free to deform, has young's modulus 'E', Poisson's ratio 'ν', coefficient of thermal expansion 'α'. If the rod is uniformly heated with 'ΔT' temperature rise, what is the pressure developed?

(a) $-\alpha\Delta T/(1 - 2\nu)$ (b) $\alpha\Delta T/(1 - 2\nu)$
(c) $E\alpha\Delta T/(1 - 2\nu)$ (d) 0

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I, GATE 2014

Ans. (d) : As we know, thermal stress occur only when there is resistance to its deformation. Since here its free to deform hence no thermal stress will be developed.

107. According to Hooke's law, when a material is loaded within its elastic limit, the stress is

(a) directly proportional to the strain
(b) inversely proportional to the strain
(c) proportional to the square of the strain
(d) proportional to the inverse of the square of the strain

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : According to Hooke's law, when a material is loaded within its elastic limit the stress is directly proportional to the strain.

$$\sigma \propto \varepsilon$$

$$\boxed{\sigma = \varepsilon E}$$

108. The Poisson's ratio of an elastic material is 0.3, then the ratio of its young's modulus and bulk modulus is _____.

(a) 1.2 (b) 2.1
(c) 1.4 (d) 1.6

VIZAG MT, 14.12.2020

Ans. (a) : $\mu = 0.3$

$$\frac{E}{K} = ?$$

$$E = 3K(1 - 2\mu)$$

$$\frac{E}{K} = 3(1 - 2\mu)$$

$$= 3(1 - 2 \times 0.3)$$

$$\boxed{\frac{E}{K} = 1.2}$$

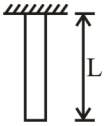
109. Maximum theoretical value of Poisson's ratio is

(a) infinity (b) 1
(c) 0.707 (d) 0.5

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : The Poisson's ratio of a stable, isotropic, linear elastic material must be between –1.0 to +0.5 because of the requirement for young's modulus, the shear modulus and bulk modulus to have positive values most material have poisson's ratio values ranging between 0.0 and 0.5.

110. A bar of steel of length L = 0.3m, fixed at one end and free at the other end as shown in the figure is uniformly heated by increasing its temperature by 40^{0}C. If the coefficient of thermal expansion is 12×10^{-6}/^{0}C and modulus of elasticity is 210 GPa, the thermal stress induced in the bar is



(a) -100.8 MPa (b) 100.8 GPa
(c) 100.8 N/m^2 (d) 0

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Since the bar is free to expand. There are no stresses developed.

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111. A 8 m long round bar made of aluminium (Modulus of elasticity 70 GPa) carries a tensile load of 720 kN. If the maximum allowable elongation is 10 mm, the minimum required diameter of the bar is

(a) 103 mm (b) 30 mm
(c) 79 mm (d) 72 mm

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Given,

$P = 720 \times 10^3$ N

$l = 8$ m

$\delta l = 10 \times 10^{-3}$ m

$E = 70$ GPa $= 70 \times 10^9$ m

Elongation, $\delta l = \frac{Pl}{AE}$

$$10 \times 10^{-3} = \frac{720 \times 10^3 \times 8}{\frac{\pi}{4} D^2 \times 70 \times 10^9}$$

$D = 0.1023$ m

$D = 102.3$ mm ≈ 103 mm

112. Failure in the case of ductile materials in static tensile loading is due to

(a) Shear stress (b) Tensile stress
(c) Compressor stress (d) Contact stress

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Failure in case of ductile materials in static tensile loading is due to shear stress.

113. The value of Poisson's ratio depends upon:

(a) Material of the test specimen
(b) Nature of load, tensile or compressive
(c) Dimensions of the test specimen
(d) Magnitude of load

CIL MT 27.02.2020

Ans. (a) : The value of Poisson's ratio depends upon material of the test specimen.

114. A square bar of side 5 mm and 200 mm long, experience an extension of 0.01 mm upon a load of 100 N. Then Young's Modulus of the material is:

(a) 800 MPa (b) 200 GPa
(c) 400 MPa (d) 80 GPa

CIL MT 27.02.2020

Ans. (d) : Given, $\ell = 200$ mm

Side = 5 mm

In square bar $(A) = (\text{Side})^2$

$A = 25$ mm^2

$\delta\ell = 0.01$ mm

$P = 100$ N

Young modulus of material,

$$E = \frac{P\ell}{A\delta\ell}$$

$$E = \frac{100 \times 200}{25 \times 0.01}$$

$E = 80$ GPa

115. For a stress-strain diagram for ductile material, the work hardening starts after the _____.

(a) elastic limit
(b) upper yield point
(c) limit of proportionality
(d) lower yield point

UPRVUNL AE 05.07.2021

Ans. (b) : For a stress-strain diagram for ductile material, the work hardening starts after the upper yield point.

116. During plastic deformation, induced stress in the material is:

(a) greater than its proportional limit but lesser than lower yield strength
(b) greater than its lower yield strength but lesser than upper yield strength
(c) greater than its ultimate strength but lesser than its fracture strength
(d) greater than its yield strength but lesser than its fracture strength

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Plastic deformation is the permanent distortion that occurs when a material is subjected to tensile, compressive bending or torsion stresses that exceed its yield strength and cause it to elongate compress buckle, bend or twist.
Induced stress is greater than its ultimate strength but lesser than its fracture strength.

117. The stress-strain curve of brittle material has:

(a) only upper yield point
(b) only lower yield point
(c) no clear yield point
(d) both lower and upper yield point

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : A stress strain curve for brittle material will be linear and no clear yield point for some materials such as concrete tensile strength is negligible compared to the compressive strength.

118. The temperature stress is a function of, 1. Coefficient of linear expansion 2. Temperature rise 3. Modulus of elasticity. The correct answer is

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Thermal stress $= \alpha\Delta TE$

Where, α = Coefficient of thermal expansion

E = Young's modulus

ΔT = Temperature rise

119. Simple stress is often called

(a) direct stress (b) transverse stress
(c) total stress (d) complete stress

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Simple stress is often called direct stress.

120. A rod of length 2 m and diameter 50 mm is elongated by 5 mm when an axial force of 400 kN is applied. The modulus of elasticity of the material of the rod will be nearly

(a) 66 GPa (b) 72 GPa
(c) 82 GPa (d) 96 GPa

ESE 2020

Ans. (c) : Elongation, $\delta = \frac{P\ell}{AE}$

$$5 = \frac{400 \times 10^3 \times 2000}{\frac{\pi}{4} \times 50^2 E}$$

$E = 81487.3$ MPa $= 81.5$ GPa ≈ 82 GPa

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121. A rod of 2m long at a temperature of 10°C is heated to 80°C. If the coefficient of thermal expansion of the rod is 0.000012/°C and E is 100 GPa, the thermal stress under the complete prevention of expansion is :

(a) 96MPa (b) 84 MPa
(c) 10 MPa (d) 12 MPa

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Given, rod length (L) = 2m

$\Delta t = (80°C - 10°C)$

$\Delta t = 70°C$

The coefficient of thermal expansion (α) = 0.000012°C

Young modulus (E) = 100 GPa

$(E) = 100 \times 10^3$ MPa

Thermal stress $(\sigma_{th}) = \alpha E \Delta t$

$= 0.000012 \times 100 \times 10^3 \times 70$

$\boxed{\sigma_{th} = 84 \text{ MPa}}$

122. During a tensile test on a specimen of 1 cm^2 cross-section, maximum load observed was 8 tonnes and area of cross-section at neck was 0.5 cm^2. Ultimate tensile strength of specimen is

(a) 4 tonnes/cm^2 (b) 8 tonnes/cm^2
(c) 16 tonnes/cm^2 (d) 22 tonnes/cm^2

RPSC IOF, 2020

Ans. (b) : Ultimate tensile strength is the maximum load upon original area.

$$\sigma = \frac{F}{A}$$

$$\sigma = \frac{8}{1}$$

$\sigma = 8$ tonnes/cm^2

123. The elongation of a freely hanging uniform steel rope. If its length is doubled will increase in the ratio of

(a) 2 : 1 (b) 1 : 1
(c) 4 : 1 (d) 8 : 1

RPSC IOF, 2020

Ans. (c) : The elongation of the freely hanging bar

$$\delta l_1 = \frac{\rho g l^2}{2E}$$

$\delta l \propto l^2$

If its length is doubled

$$\delta l_2 = \frac{\rho g (2l)^2}{2E}$$

Ratio of the elongation $\Rightarrow \frac{\delta l_1}{\delta l_2} = \frac{l^2}{(2l)^2}$

$$\frac{\delta l_1}{\delta l_2} = \frac{1}{4}$$

$$\frac{\delta l_2}{\delta l_1} = \frac{4}{1}$$

124. A specimen of an alloy is made into a cylindrical bar of 12.00 mm diameter and 200mm length. This bar is loaded in axial tension up to the proportional limit when the load is 40 kN. At this load the length was measured as 201 mm and diameter was measured as 11.98 mm. What will be the Poisson's ratio?

(a) 0.25 (b) 0.29
(c) 0.33 (d) 0.38

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given data;

$d_i = 12$ mm

$d_f = 11.98$ mm

$L_i = 200$ mm, $L_f = 201$ mm

Longitudinal strain $(\in_{long}) = \frac{L_f - L_i}{L_i}$

$$= \frac{201-200}{200} = \frac{1}{200} = 5 \times 10^{-3}$$

Lateral strain $(\in_{lat}) = \frac{d_f - d_i}{d_i}$

$$= = \frac{11.98-12}{12} = -1.666 \times 10^{-3}$$

$\therefore$ Poisson's ratio $(\mu) = \left|\frac{\text{Lateral strain}}{\text{Longitudinal strain}}\right|$

$$= \frac{1.666 \times 10^{-3}}{5 \times 10^{-3}}$$

$\Rightarrow \mu = 0.333 \quad \Rightarrow \boxed{\mu = 0.33}$

125. The stiffness of an axially loaded bar is defined as _____. [E = Modulus of elasticity, A = Area of cross-section of bar, L = Length of bar]

(a) $\frac{EA}{L}$ (b) $\frac{L}{EA}$
(c) $\frac{E}{AL}$ (d) $\frac{AL}{E}$

APPSC Poly Lect. 13.03.2020

Ans. (a) : We know that stiffness is the ratio of applied load to per unit elongation or deflection.

Stiffness for axially loaded bar (s)

$$= \frac{\text{Load}(P)}{\text{elongation}(\delta)}$$

$$= s = \frac{P}{PL/AE}$$

$$\Rightarrow \quad \boxed{\text{stiffness}(s) = \frac{AE}{L}}$$

126. A 1.25 cm diameter steel bar is subjected to a load of 2500 kg. The stress induced in the bar will be

(a) 200 MPa (b) 210 MPa
(c) 220 MPa (d) 230 MPa

ESE 2020

Ans. (a) : $\sigma = \frac{P}{A} = \frac{2500 \times 9.81}{\frac{\pi}{4}(12.5)^2} = 199.4 = 200 \text{ MPa}$

127. A copper rod 3mm in diameter when subjected to a pull of 495 N extends by 0.07 mm over a gauge length of 100 mm. The Young's Modulus for copper will be :

(a) 1×10^5 N/mm^2 (b) 1×10^6 N/mm^2
(c) 7×10^5 N/mm^2 (d) 1×10^7 N/mm^2

ISRO Scientist/Engineer (RAC), 10.03.2019

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Ans. (a) : Given,

$l = 100$ mm

$d = 3$mm

$P = 495$ N

$\delta = 0.07$ mm

Deflection, $(\delta) = \dfrac{PL}{AE}$

$$0.07 = \frac{495 \times 100}{\frac{\pi}{4} \times 3^2 \times E}$$

$$E = \frac{495 \times 100}{\frac{\pi}{4} \times 9 \times 0.07} = 100040.2499$$

$$\boxed{E = 1 \times 10^5 N/mm^2}$$

128. Engineering stress-strain curve and true stress-strain curve are equal up to :

(a) Proportional limit
(b) Elastic limit
(c) Yield point
(d) Tensile strength point

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Ans : (c) : Engineering stress-strain curve and true stress curve are equal upto yield point.

129. Strain is defined as the ratio of :

(a) Change in volume to original volume
(b) Change in length to original length
(c) Changes in cross-sectional area to original cross-sectional area
(d) Any one of these

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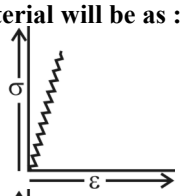
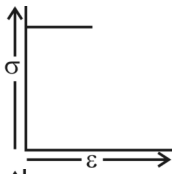
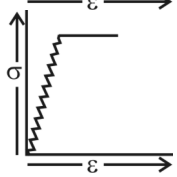
Ans : (d) : Strain is defined as the ratio of-

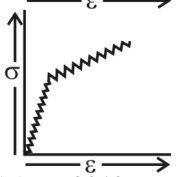
- Change in volume to original volume
- Change in length to original length.
- Change in cross-sectional area to original cross sectional area.

130. The stress-strain curve of an rigid-plastic material will be as :

(a) 

(b) 

(c) 

(d) 

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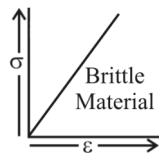
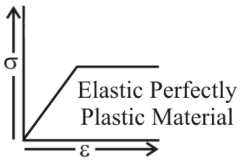
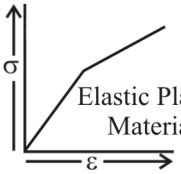
Ans : (b)







131. Failure of a material is called fatigue when it falls

(a) at the elastic limit
(b) below the elastic limit
(c) at the yield point
(d) below the yield point

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Ans : (d) : Fatigue- When a material is subjected to repeated stress, it fails at stress below the yield point stress. Such type of failure of a material is known as fatigue.

132. Poisson's ratio of perfectly linear elastic material is

(a) 0 (b) 1
(c) 0.3 (d) 0.5

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Ans : (d) : Volumetric strain for liner elastic material is given by

$$\epsilon_V = \frac{\Delta V}{V}$$

$$= \frac{(1-2\mu)}{E}\underline{(\sigma_x + \sigma_y + \sigma_z)}$$

$\Delta V = 0,$

$1 - 2\mu = 0;$

$$\boxed{\mu = 0.5}$$

133. A 10 mm diameter aluminium alloy test bar is subjected to a load of 500 N. If the diameter of the bar at this load is 8 mm, the true strain is

(a) 0.2 (b) 0.25
(c) 0.22 (d) 0.1

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Ans : (*) : True strain is given as

$$\epsilon_t = \ln\left(\frac{A_0}{A_f}\right)$$

$$= 2\ln\left(\frac{d_o}{d_f}\right)$$

Since, $d_0 = 10$ mm

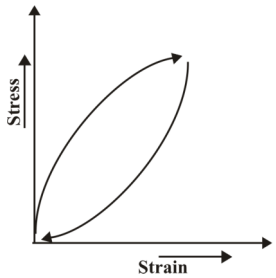
$d_f = 8$ mm

Therefore, we get,

$$\epsilon_t = 2\ln\left(\frac{10}{8}\right)$$

$$= 0.446$$

134. The following diagram is a stress-strain diagram of any material. Which kind of material is it?



(a) Plastic (b) Linear Elastic
(c) Non-linear Elastic (d) Visco-elastic

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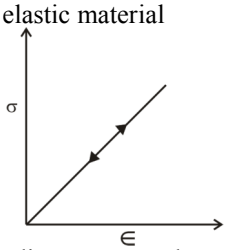
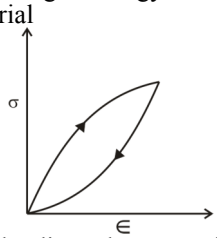
Ans. (d) : Purely elastic material



Loading and unloading to energy loss.
Viscoelastic material



Loading and unloading the area inside the curve, hysteresis loop is energy loss.

135. A metal sphere of diameter D is subjected to a suniform increase in temperature ΔT. E, ν and α are the Young's modulus, Poisson's ratio and Coefficient of thermal expansion respectively. If the ball is free to expand, the hydrostatic stress developed within the ball due to temperature change is

(a) 0 (b) $\frac{\alpha \Delta T E}{1-2\nu}$

(c) $-\frac{\alpha \Delta T E}{1-2\nu}$ (d) $\frac{\alpha \Delta T E}{3(1-2\nu)}$

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Ans. (a) : The hydrostatic stress developed within the ball due to temperature change is zero as the ball is free to expand.

136. Which mechanical property gets affected in an alloy, when it is over-aged condition :

(a) lower hardness
(b) low strain hardening rate
(c) higher yield strength
(d) higher tensile strength

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Ans. (a) :
- The ageing phenomena are associated with aluminium alloy mainly.
- Age hardening curve of the alloy basically contain of initial hardness it maximum hardness subsequently decrease hardness with ageing time which called the under ageing.

137. After which point of the Stress-Strain Diagram does metal cutting start?

(a) Proportional point (b) Ultimate point
(c) Fracture point (d) Yield point

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Ans. (c) : After fracture point of stress-strain diagram, metal cutting start.

138. A block is dimensions of upper surface 100 mm × 100 mm. The height of the block is 10 mm. A tangential force of 10 kN is applied at the centre of the upper surface. The block is displaced by 1 mm with respect to lower face. Direct shear stress in the element is :

(a) 10 MPa (b) 1 MPa
(c) 0.1 MPa (d) 100 MPa

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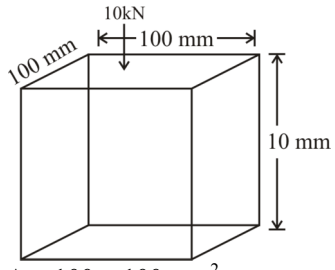
Ans. (b) : Given - Dimension = 100 × 100 × 10



$A = 100 \times 100\ mm^2$

$P = 10 \times 10^3\ N$

Direct shear stress

$$\tau = \frac{P}{A} = \frac{10 \times 10^3}{100 \times 100} = 1 N/mm^2$$

$= 1$ MPa

139. A copper rod with initial length l_o is pulled by a force. The instantaneous length of the rod is given by $l = l_o$ (1 + $2e^{4t}$), where t represents time. True strain rate at t = 0 is :

(a) $\frac{1}{3}$ (b) $\frac{8}{3}$

(c) $\frac{4}{3}$ (d) $\frac{2}{3}$

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Ans. (b) : Given - Initial length = l_o

Instantaneous length = $l = l_o (1 + 2e^{4t})$

at, t = 0

$l = l_o (1 + 2e^o)$

$l = l_o (1 + 2) = 3\ l_o$

True strain $\in_T = \int \frac{dl}{l}$

$$\in_T = \frac{\text{change in length}}{\text{Instantaneous length}}$$

Change in length $\frac{dl}{dt} = l_o \left(8e^{4t}\right)$

at, t = 0

$\frac{dl}{dt} = l_o \times 8$

True strain $\in_T = \frac{8\ l_o}{3\ l_o}$ $\boxed{\in_T = \frac{8}{3}}$

140. True stress experienced by a material is _____ than the engineering stress at a given load.

(a) lower
(b) higher
(c) equal
(d) higher or lower

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Ans. (b) : True stress experienced by a material is higher than the engineering stress at a given load.

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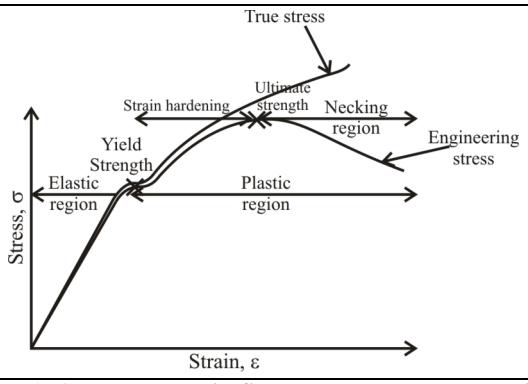
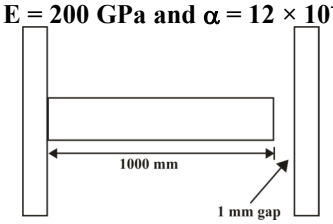


141. A 1m long rod is fixed at one end. There is a rigid wall at a distance 1 mm from the free end of the rod as depicted in the figure. What is the thermal stress generated in the rod if its temperature is increased by 100°C?
Take E = 200 GPa and $\alpha = 12 \times 10^{-6}$/°C



(a) 40 MPa (b) 80 MPa
(c) 120 MPa (d) 240 MPa

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Ans. (a) : Free expansion = $l\,\alpha\,\Delta T$

$$\ell\alpha\Delta t - \Delta = \frac{\sigma}{E}$$

$\ell\alpha\Delta t = (1000)\,(12 \times 10^{-6})\,(100) = 1.2$ mm

Expansion prevented = 1.2 - 1 = 0.2 mm

$$0.2 = \frac{Pl}{AE}$$

$$\left(\because \sigma = \frac{P}{A}\right)$$

$$\frac{2}{10} = \frac{(\sigma)(1000)}{200\times10^3}$$

$\sigma = 40$ MPa

142. If a material is heated up, its Elastic modulus
(a) decreases (b) increases
(c) remains constant (d) None of the above

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Ans. (a) : As the material is heated up, it becomes soft. It undergoes more strain for a given stress

$$\left(\because E = \frac{\sigma}{\in}\right)$$ The modulus of elasticity decreases.

143. A bar of 30 mm diameter is subjected to a pull of 60 kN. The measured extension on gauge length of 200 mm is 0.09 mm and change in diameter is 0.0039 mm. Find its Poisson's ratio.
(a) 0.309 (b) 0.299
(c) 0.289 (d) 0.279

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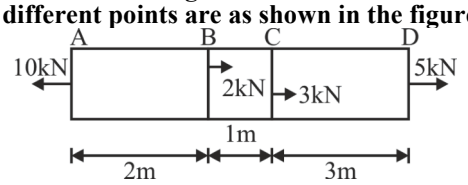
Ans. (c) : Data given-
d = 30 mm, $\Delta d = 0.0039$ mm
$l = 200$ mm, $\Delta l = 0.09$ mm
P = 60 kN
we know that
Poisson's ratio

$$\mu = \frac{\left[\frac{\Delta d}{d}\right]}{\left[\frac{\Delta l}{l}\right]} = \frac{\left[\frac{0.0039}{30}\right]}{\left[\frac{0.09}{200}\right]}$$

$$\boxed{\mu = 0.2888}$$

144. The loads acting on a 3 mm diameter bar at different points are as shown in the figure:



If E = 205 GPa, the total elongation of the bar will be nearly.
(a) 29.7 mm (b) 25.6 mm
(c) 21.5 mm (d) 17.4 mm

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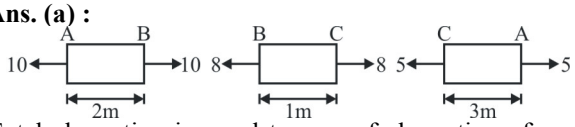
Ans. (a) :



Total elongation is equal to sum of elongation of each bar

$$\delta = \delta_{AB} + \delta_{BC} + \delta_{CA}$$

$$= \frac{p_1\ell_1}{A_1E_1} + \frac{p_2\ell_2}{A_2E_2} + \frac{p_3\ell_3}{A_3E_3}$$

$A_1E_1 = A_2E_2 = A_3E_3 = AE$

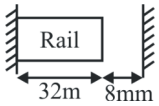
$$= \frac{10\times10^3\times2000}{AE} + \frac{8\times10^3\times1000}{AE} + \frac{5\times10^3\times3000}{AE}$$

$$= \frac{43\times10^6}{AE} = \frac{43\times10^6}{\frac{\pi}{4}\times3^2\times205\times10^3} = 29.68 \text{ mm}$$

145. Rails are laid such that there will be no stress in them at 24°C. If the rails are 32 m long with an expansion allowance of 8 mm per rail, coefficient of linear expansion $\alpha = 11\times10^{-6}$/°C and E = 205 GPa, the stress in the rails at 80°c will be nearly.
(a) 68 MPa (b) 75 MPa
(c) 83 MPa (d) 90 MPa

ESE 2019

Ans. (b) :



Given,
$\ell = 32$ m $\Delta = 8$ mm
$\alpha = 11 \times 10^{-6}$ /°C
$E = 205$ GPa $= 205 \times 10^3$ N/mm^2
$\Delta T = 80 - 24 = 56$°C
if Δ is the permissible expansion allowance then thermal

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$$\text{stress } \sigma = \frac{L\alpha\Delta t - \Delta}{L}$$

$$\ell\ \alpha\ \Delta T - \Delta = \frac{\sigma\ell}{E}$$

$$32 \times 10^3 \times 11 \times 10^{-6} \times 56 - 8 = \frac{\sigma \times 32 \times 10^3}{205 \times 10^3}$$

$$11.712 = 0.156\ \sigma$$

$$\sigma = 75.03\ \text{N/mm}^2$$

$$\sigma = 75.03\ \text{MPa}$$

146. When a load of 20 kN is gradually applied at a particular point in a beam, it produces a maximum bending stress of 20 MPa and a deflection of 10 mm. What will be the height from which a load of 5 kN should fall onto the beam at the same point if the maximum bending stress is 40 MPa?

(a) 80 mm (b) 70 mm
(c) 60 mm (d) 50 mm

ESE 2019

Ans. (c) : For 20 kN static load (P_1 = 20 kN)

δ_1 = 10 mm

$(\sigma_1)_{static}$ = 20 MPa

For 5 kN impact load (P_2 = 5 kN)

σ_{max} = 40 MPa

$(\sigma_2)_{static}$ = ? δ_2 = ?

From equation

$$\sigma = \frac{P}{A}$$

$$\sigma \propto P$$

$$\frac{(\sigma_2)_{static}}{(\sigma_1)_{static}} = \frac{P_2}{P_1}$$

$$(\sigma_2)_{static} = (\sigma_1)_{static} \times \frac{P_2}{P_1} = 20 \times \frac{5}{20} = 5\ \text{MPa}$$

deflection (δ) $\propto$ P

$$\frac{\delta_2}{\delta_1} = \frac{P_2}{P_1}$$

$$\delta_2 = \delta_1 \frac{P_2}{P_1} = 10 \times \frac{5}{20} = 2.5\ \text{mm}$$

As we know for impact load

$$\sigma_{max} = \sigma_{static}\left[1 + \sqrt{1 + \frac{2h}{\delta_{static}}}\right]$$

$$40 = 5\left[1 + \sqrt{1 + \frac{2h}{2.5}}\right]$$

h = 60 mm

147. A cylindrical specimen of steel having an original diameter of 12.8 mm is tensile tested to fracture and found to have engineering fracture strength σ_f of 460 MPa. If its cross sectional diameter at fracture is 10.7 mm, the true stress at fracture will be

(a) 660 MPa
(b) 645 MPa
(c) 630 MPa
(d) 615 MPa

ESE 2019

Ans. (a) : Given,

Initial diameter (d_i) = 12.8 mm

Final diameter (d_f) = 10.7 mm

Engineering fracture strength (σ_f) = 460 MPa

True stress at fracture

$$(\sigma_t) = \sigma_f \frac{A_i}{A_f}$$

$$= 460\left(\frac{d_i^2}{d_f^2}\right) = 460\left(\frac{12.8}{10.7}\right)^2$$

$$= 658.27\ \text{MPa} \approx 660\ \text{MPa}$$

148. If for a given material, E = 2G (E is modulus of elasticity, G is the shear modulus), then the bulk modulus K will be

(a) $\frac{E}{2}$ (b) $\frac{E}{3}$
(c) E (d) $\frac{E}{4}$

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Ans. (b) : Given, E = 2G

We have, E = 2G (1 + μ)

2G = 2G (1 + μ)

μ = 0

and E = 3K (1 - 2μ)

E = 3K [1 - 2(0)]

E = 3K

$$\Rightarrow \quad K = \frac{E}{3}$$

149. Consider the state of stress at any point as σ_{xx} = 250 MPa, σ_{zz} = 250 MPa. The Young's modulus and Poisson's ratio of the material is considered as 2 GPa and 0.18 respectively. Determine the ε_{zz} at the point.

(a) -0.125 (b) 0.103
(c) -0.103 (d) 0.125

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Ans. (b) : σ_{xx} = 250 MPa

σ_{yy} = 0

σ_{zz} = 250 MPa

E = 2 × 10^9 Pa

μ = 0.18

$$\varepsilon_{zz} = \frac{\sigma_{zz}}{E} - \mu\frac{\sigma_{yy}}{E} - \mu\frac{\sigma_{xx}}{E}$$

$$= \frac{\sigma_{zz}}{E} - \mu\frac{\sigma_{xx}}{E} = \frac{\sigma_{xx}}{E}(1-\mu)$$

$$(\because \sigma_{xx} = \sigma_{zz})$$

$$= \frac{250 \times 10^6}{2 \times 10^9}(1 - 0.18)$$

$$= 0.1025 = 0.103$$

150. Find out the Lame constants (λ and μ) for an isotropic material having modulus of elasticity (E) and Poisson's ratio (ν) as 200 GPa and 0.2, respectively.

(a) 80 GPa, 80 GPa
(b) 35.71 GPa, 166.6 GPa
(c) 55.55 GPa, 83.33 GPa
(d) 73.33 GPa, 66.66 GPa

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Ans. (c) : E = 200 GPa

Poisson's ratio ν = 0.2

$$\text{Lame constant } (\lambda) = \frac{E\nu}{(1+\nu)(1-2\nu)}$$

$$\lambda = \frac{200\times 0.2}{(1+0.2)(1-2(0.2))} = 55.55\,\text{GPa}$$

$$\mu = \frac{2E\nu}{(1+\nu)(1-\nu)} = \frac{2\times 200\times 0.2}{(1+0.2)(1-0.2)} = 83.33\ \text{GPa}$$

151. The deformation per unit length is called :

(a) Tensile stress (b) Poissons ratio

(c) Youngs modulus (d) Strain

MECON MT 2019

Ans. (d) : The deformation per unit length is called strain.

152. A bar of 20 mm diameter is subjected to pull of 50 kN. If the measured extension over the gauge length of 20 cm is 0.1 mm and the change in diameter is 0.0035 mm, Poisson's ratio and young's modulus are:

(a) 0.18, 200 GPa (b) 0.1, 108 GPa

(c) 0.25, 180 GPa (d) 0.35, 318 GPa

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Ans. (d) : Given,

Diameter of bar (d) = 20 mm

Pull load (F) = 50 kN

Gauge length (l_g) = 20 cm = 200 mm

Extension (δl) = 0.1 mm

Change in diameter (δd) = 0.0035 mm

$$\mu = \left|\frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{longitudinal}}}\right| = \frac{\delta d/d}{\delta l/l_g} = \frac{0.0035/20}{0.1/200}$$

$$\mu = 0.35$$

$$E = \frac{\sigma}{\epsilon_{\text{longitudinal}}} = \frac{F/A}{\delta\ell/\ell_g} = \frac{F\times l_g}{A\times\delta l} = \frac{50\times 10^3\times 200}{\frac{\pi}{4}(20)^2\times 0.1}$$

$$= 318.31\,\text{GPa}$$

153. The statement "Total effect of several loads applied on a body is the sum of effects of individual loads applied separately" is known as :

(a) Generalized Hooke's law

(b) St. Venant's principle

(c) Principle of superposition

(d) Uniqueness theorem

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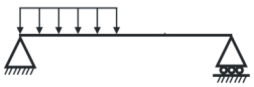
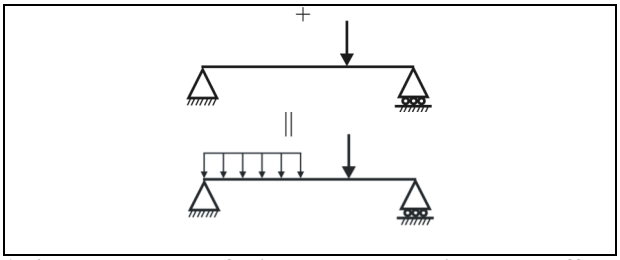
Ans. (c) : Principle of superposition- Simply states that on a linear elastic structure, the combined effect of several loads acting simultaneously is equal to the algebraic sum of the effects of each load acting individually.

In other words, for a linearly elastic member/structure the effect of several loads acting on a member is equal to the summation of the loads acting separately.





154. The value of yield stress obtained by offset method is known as :

(a) True stress (b) Elastic stress

(c) Proof stress (d) Breaking stress

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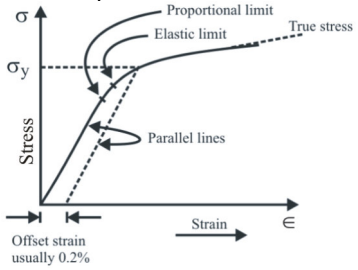
Ans. (c) : The value of yield stress obtained by offset method is known as proof stress.



155. The unit of strain is

(a) N-mm (b) N/mm

(c) mm (d) no unit

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : As we know, strain is the ratio of change in length to the total length so it has no. dimension (unit).

$$\varepsilon = \frac{\Delta L}{L}$$

156. The modulus of elasticity for mild steel; is approximately equal to

(a) 10 kN/mm^2 (b) 80 kN/mm^2

(c) 100 kN/mm^2 (d) 210 kN/mm^2

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : As we know, the modulus of elasticity for mild steel is approximately equal to 210 GPa or 210 kN/mm^2.

157. Two bars of different materials and same size are subjected to the same tensile force. If the bars have unit elongation in the ratio of 2 : 5, then the ratio of modules of elasticity of the two materials will be

(a) 2:5 (b) 5:2

(c) 4:3 (d) 3:4

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : As we know,

$$E = \frac{PL}{A\Delta L} \qquad \therefore \frac{\Delta L_1}{\Delta L_2} = \frac{2}{5}$$

So, $$E_1 \propto \frac{1}{\Delta L_1}$$

$$\frac{E_1}{E_2} = \frac{\Delta L_2}{\Delta L_1} = \frac{5}{2}$$

158. When a bar is cooled to –50°C, it will develop

(a) no stress (b) shear stress

(c) tensile stress (d) compressive stress

Nagaland PSC (CTSE) 2018, Paper-I

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Ans. (c) : When a bar is cooled to –5 temperature, it must goes to tensile stress, as the bar will resists to expand, so tensile stress will developed.

159. If the modules of elasticity for a given material is twice its modules of rigidity, then bulk modules is equal to

(a) 2C (b) 3C
(c) 2C/3 (d) 3C/2

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : We know, $E = \frac{9KC}{C+3K}$

Given, $E = 2C$

$\Rightarrow \quad 2C = \frac{9KC}{3K+C}$

$\Rightarrow \quad 2C(3K + C) = 9\,KC$

$\Rightarrow \quad 6KC + 2C^2 = 9\,KC$

$\Rightarrow \quad 2C^2 = 3\,KC$

$2C = 3\,K$

$\therefore \quad K = \frac{2C}{3}$

160. Consider a bar of length 'L', breadth 'b' and thickness 't' subjected to an axial pull or tension 'P'. The resulting volumetric strain will be equal to

(a) $\in(1-2\mu)$ (b) $2\in(1-2\mu)$
(c) $\in(1+2\mu)$ (d) 3

GPSC Executive Engineer 23.12.2018

Ans. (a) : $\in_v = \in_x + \in_y + \in_z = \left(\frac{\sigma_x + \sigma_y + \sigma_z}{E}\right)(1-2\mu)$

$\in_v = \in(1-2\mu)$

161. If Poisson's ratio of a material is 0.75, then the elastic modulus for the material is

(a) 3.5 times its shear modulus
(b) 4 times shear modulus
(c) equal to its shear modulus
(d) 5 times its shear modulus

GPSC Executive Engineer 23.12.2018

Ans. (a) : Given,

$\mu = 0.75$

$E = 2G(1+\mu)$

$= 2G\,(1+0.75)$

$= 2 \times G \times 1.75$

$= 3.50\,G$

$\boxed{E = 3.5G}$

162. Lateral strain $(\in)$ can be expressed as

(a) $\frac{\delta l}{l}$ (b) $\frac{l}{\delta l}$
(c) $\gamma \in$ (d) $-\gamma \in$

TNPSC AE 2018

Ans. (d) : Lateral strain $(\in')$

$= -\gamma \times$ longitudinal strain

$= -\gamma \times \in$

where $\gamma \rightarrow$ poisson ratio

$= \frac{\text{lateral strain}}{\text{longitudnal strain}}$

163. Allotropic metal,

(a) exists in more than one type of lattice structure depending upon temperature
(b) has equal stresses in all directions
(c) has only one lattice structure of all temperature
(d) gives equal strain in all direction

TNPSC AE 2018

Ans. (a) : Allotropic metal, exists in more than one type of lattice structure depending upon the temperature.

164. Which of the following is correct relation among elastic constants E (modulus of elasticity), G (modulus of rigidity), ν (Poisson's ratio) and K (bulk modulus)?

(a) $E = 3K(1-\nu) = 2G(1+\nu)$
(b) $E = 2G(1-\nu) = 3K(1+\nu)$
(c) $E = 3K(1-2\nu) = 2G(1+\nu)$
(d) $E = 2K(1-2\nu) = 3G(1+\nu)$
(e) $E = 3K(1+2\nu) = 2G(1-\nu)$

CGPSC AE 25.02.2018

Ans. (c) : We know that

$E = 3K\,[1 - 2\nu)$

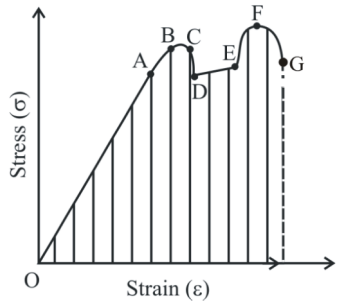
$E = 2G\,[1 + \nu)$

165. The area of under the stress-strain diagram up to the rupture point is known as

(a) Proof resilience
(b) Modulus of toughness
(c) Modulus of elasticity
(d) Modulus of resilience

HPPSC AE 2018

Ans. (b) :



Point A – Proportionality limit
Point B – Elastic limit
Point C – Upper yield Point
Point D – Lower yield Point
DE – Yielding Region
EF – Strain hardening region
F – Ultimate point
FG – Necking region
G – Breaking [Rupture point]

Modulus of Toughness [M.O.T.]–Modulus of Toughness is defined as energy observed by a component per unit volume just before its rupture.

M.O.T.–Total area of stress vs strain curve per unit volume.

166. Modulus of Rigidity is related to-

(a) Length (b) Shape
(c) Size (d) Volume

RPSC AE 2018

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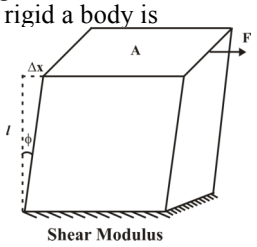
Ans. (b) : Modulus of Rigidity—The modulus of rigidity is the elastic coefficient when a shear force is applied resulting in lateral deformation. It gives us a measure of how rigid a body is



Shear Modulus

$$G = \frac{\tau_{xy}}{\gamma_{xy}} = \frac{\left(\frac{F}{A}\right)}{\left(\frac{\Delta x}{l}\right)}$$

$$= \frac{F \times l}{A\,\Delta x}$$

where

- $\tau_{xy} = \frac{F}{A}$ is shear stress.
- F is the force acting on the object.
- $\gamma_{xy} = \frac{\Delta x}{l}$ is the shear strain.
- Δx is the transverse displacement.

167. Detrimental property of a material for shock load application is-

(a) High density (b) Low toughness
(c) High strength (d) Low hardness

RPSC AE 2018

Ans. (b) : Detrimental property of a material for shock load application is low toughness.

168. A metallic rod of length 2.5 m and cross-section of 20 mm^2 is elongated by a load of 500 kN. If Poisson ratio is 0.25, then the increase in volume is: (assume E = 2 × 10^5 N/mm^2)

(a) 3000 mm^3 (b) 2125 mm^3
(c) 2000 mm^3 (d) 3125 mm^3

AAI Jr. Executive 29.11.2018

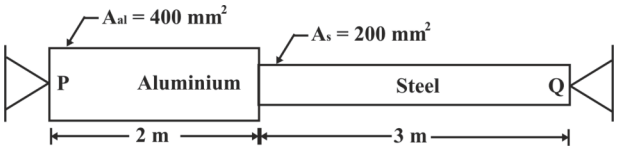
Ans. (d) : Given,
L = 2.5 m
A = 20 mm^2
P = 500 kN
μ = 0.25
E = 2 × 10^5 N/mm^2

$$\epsilon_v = \frac{(1-2\mu)(\sigma_x)}{3E} = \frac{\Delta V}{V}$$

$$\frac{(1-0.25\times 2)}{3\times 2\times 10^5} \times \frac{500\times 10^3}{20} = \frac{\Delta V}{V} = \frac{\Delta V}{20\times 2500}$$

$$\boxed{\Delta V = 3125 mm^3}$$

169. A composite bar made of aluminum and steel is fixed rigidly at supports P and Q as shown in figure. When the temperature of the bar is increased by 10°C, the reactions at the supports are (given E_{al} = 70 GPa, E_s = 210 GPa, α_{al} = 11E-6 mm/mm°C and α_s = 12 E-6 mm/mm°C)



(a) 4060 N (b) 6780 N
(c) 12240 N (d) 2412 N

ISRO Scientist/Engineer 22.04.2018

Ans. (a): After Increasing the temperature of composite bar. Total deflection due to temperature rise is given by

$$\delta_{Total} = (\alpha t l)_s + (\alpha t l)_{Al}$$

$$= 12\times 10^{-6}\times 10\times 3000 + 11\times 10^{-6}\times 10\times 2000$$

$$\delta_{Total} = 0.58 mm .....(1)$$

Another way to finding total elongation

$$\delta_{Total} = \frac{PL_s}{(AE)_s} + \frac{PL_A}{(AE)_{Al}}$$

$$\delta_{Total} = \frac{P\times 3000}{200\times 210\times 10^3} + \frac{P\times 2000}{400\times 70\times 10^3}$$

$$\delta_{Total} = \frac{P}{7000} .....(2)$$

$$P = 7000 \times \delta_{Total}$$

$$P = 7000 \times 0.5 = 4060 \text{ N}$$

170. A round uniformly tapered bar of length *L* and Young's modulus *E* has diameter of '*d*' at one end and '2*d*' at the other end. If the bar is pulled by an axial force *F*, the extension produced will be

(a) $\frac{2FL}{\pi Ed^2}$ (b) $\frac{4FL}{\pi Ed^2}$
(c) $\frac{8FL}{\pi Ed^2}$ (d) $\frac{16FL}{\pi Ed^2}$

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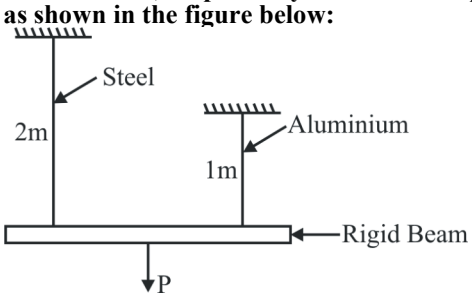
Ans. (a): Extension produced in tapered bar

$$\delta = \frac{4FL}{\pi d\times 2d\times E}$$

$$\delta = \frac{2FL}{\pi Ed^2}$$

171. A rigid beam of negligible weight is supported in a horizontal position by two rods of steel and aluminium, 2 m and 1 m long, having values of cross sectional areas 100 mm^2 and 200 mm^2, and young's modulus of 200 GPa and 100 GPa, respectively. A load P is applied as shown in the figure below:



If the rigid beam is to remain horizontal then

(a) the force P must be applied at the centre of the beam
(b) the force on the steel rod should be twice the force on the aluminium rod

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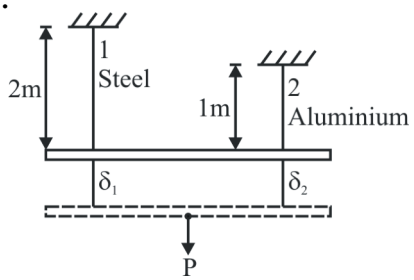
(c) the force on the aluminium rod should be twice the force on the steel rod
(d) the forces on both the rods should be equal

ESE 2018

Ans. (c) :



Let P_1 = Force in steel
P_2 = Force in aluminium

From the given condition that the rigid beam to remain horizontal.

$A_1 = 100\ mm^2$
$A_2 = 200$
$L_1 = 2\ m = 2000\ mm$
$L_2 = 1\ m = 100\ mm$
$E_1 = 200\ GP$
$E_2 = 100\ GP$
$\delta_1 = \delta_2$

$$\frac{P_1L_1}{A_1E_1} = \left(\frac{P_2L_2}{A_2E_2}\right)$$

$$\frac{P_1 \times 2000}{100 \times 200} = \frac{P_2 \times 1000}{100 \times 200}$$

$$2P_1 = P_2$$

172. The percentage elongation of a material as obtained from static tension test, depends on
(a) diameter of the test specimen
(b) gauge length of the specimen
(c) nature of end grips of the testing machines
(d) geometry of the test specimen

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : The percentage elongation of a material as obtained from static tension test, depends on gauge length of the specimen (% Elongation $= \frac{\Delta\ell}{\ell} \times 100$).

173. A heavy uniform rod of length 'L' and material density δ is hung vertically with its top end rigidly fixed. How is the total elongation of the bar under its own weight expressed?

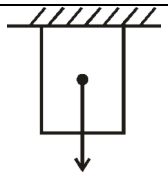
(a) $\frac{2\delta L^2 g}{E}$ (b) $\frac{\delta L^2 g}{E}$

(c) $\frac{\delta L^2 g}{\sqrt{2E}}$ (d) $\frac{\delta L^2 g}{2E}$

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Given :
Length of uniform rod = 'L'
Material density = δ
Let weight of rod be w
Total elongation of rod under its own weight

$$= \frac{WL}{2AE}$$



$W = x \times A \times L$ (x- specific weight N/m^3)
$W = (\delta h) \times A \times L$

$$\therefore \text{ Total elongation } = \frac{\delta g A \times L \times L}{2AE}$$

$$= \frac{\delta g L^2}{2E}$$

174. A round bar length *l*, elastic modulus E and Poisson's ratio is subjected to an axial pull μ, is subjected to an axial pull 'P'. What would be the change in volume of the bar?

(a) $\frac{Pl}{(1-2\mu)E}$ (b) $\frac{Pl(1-2\mu)}{E}$

(c) $\frac{Pl\mu}{E}$ (d) $\frac{Pl\mu}{(1-\mu)E}$

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Volumetric strain for cylinder

$$e_v = e_\ell + 2e_d$$

$$e_\ell = \frac{P/A}{E}$$

$$e_d = -\mu\ \frac{P/A}{E}$$

$$\therefore e_v = \frac{P}{AE}(1-2\mu)$$

$$e_v = \frac{\Delta V}{V} = \frac{P}{AE} = \frac{P}{AE}(1-2\mu)$$

$$\Rightarrow \Delta V = (A \times \ell)\ \frac{P}{AE}(1-2\mu)$$

$$\Delta V = \frac{P\ell}{E}(1-2\mu)$$

175. A 2m long rod of diameter 2 mm is subjected to an axial pull of 12 kN. The rod is extended by 0.5 cm. What is the approximate value of the modulus of elasticity of the material of the rod?
(a) 12 GPa (b) 180 GPa
(c) 125 GPa (d) 200 GPa

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (*) : Given data is wrong
Length of rod 'ℓ' = 2 m
Diameter 'd' = 2 mm = 2×10^{-3} m
Axial pull 'P' = 12 kN
Changed in length '$\Delta\ell$' = 0.5 cm = 0.5×10^{-2} M

$$\Delta\ell = \frac{P\ell}{AE}$$

$$E = \frac{P\ell}{A\Delta\ell} = \frac{\left(12\times10^3\right)\times 2}{\frac{\pi}{4}\left(2\times10^{-3}\right)^2 \times \left(0.5\times10^{-2}\right)}$$

$= 1520 \times 10^9\ N/m^2$
$= 1520\ GPa$

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176. If a material has numerically the same value for its modulus of rigidity and bulk modulus, then what is Poisson's ratio?
(a) 0.25 (b) 0.2
(c) 0.15 (d) 0.125
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Given :
Modulus of rigidity = Bulk modulus
$G = K$
Also
$2G(1 + \mu) = 3K(1 - 2\mu)$
$2 + 2\mu = 3 - 6\mu$
$8\mu = 1$
$\mu = \frac{1}{8} = 0.125$

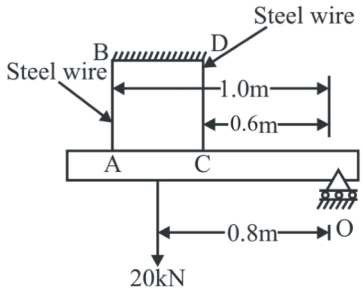
177. What is the maximum possible value of Poisson's ratio for a non-dilatants material?
(a) 0.67 (b) 0.50
(c) 0.33 (d) 0.25
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : For non dilatants material, maximum value of poisson's ratio is 0.33.

178. Modulus of elasticity and Poisson's ratio of a material are 2.1×10^5 Nmm^2 and 0.25 respectively. What is the value of modulus of rigidity of the same material in 10^5 N/mm^2?
(a) 0.84 (b) 0.70
(c) 1.40 (d) 0.50
TSPSC AEE 28.08.2017 (Civil/Mechanical)

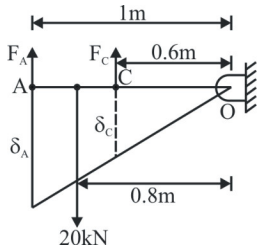
Ans. (a) : $E = 2G(1 + \mu)$
$2.1 \times 10^5 = 2G(1 + 0.25)$
$G = \frac{2.1 \times 10^5}{2 \times 1.25} = 0.84\,N/mm^2$
$G = 0.84 \times 10^5$ N/mm^2

179. A rigid bar ACO as shown is hinged at O and is held in a horizontal position by two identical vertical steel wires AB and CD. A point load of 20 kN is hung at the position shown. The tensions in wires AB and CD are



(a) 15.2 kN and 7.1 kN (b) 11.8 kN and 7.1 kN
(c) 15.2 kN and 5.0 kN (d) 11.8 kN and 5.0 kN
ESE 2017

Ans. (b) :



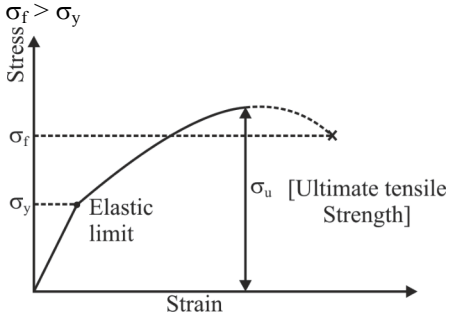
From similar triangle
$\frac{\delta_A}{\delta_C} = \frac{1}{0.6}$
$\frac{\left(\frac{F_A \cdot L}{AE}\right)}{\left(\frac{F_C \cdot L}{AE}\right)} = \frac{1}{0.6}$
$\frac{F_A}{F_C} = \frac{1}{0.6}$
$F_C = 0.6\,F_A$ .....(1)
Now $\sum M_o = 0$
$(F_A \times 1) + (F_C \times 0.6) = 20 \times 0.8$
$F_A + 0.6\,F_C = 16$ .....(2)
On solving equation (1) & (2)
$F_A = 11.76$ kN
$F_C = 7.05$ kN

180. A steel rod, 2 m long and 20 mm × 20 mm in cross section, is subjected to a tensile force of 40 kN. What will be elongation of the rod when the modulus of elasticity is 200×10^3 N/mm^2?
(a) 0.5 mm (b) 1.0 mm
(c) 1.5 mm (d) 2.0 mm
(e) 2.5 mm
(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Data given;
$L = 2\text{ m} = 2 \times 10^3$ mm
$A = 20 \times 20$ mm^2
$F = 40$ kN
$E = 200 \times 10^3$ N/mm^2
$\delta\ell = ?$
We know that,
Elongation, $\delta\ell = \frac{F \times L}{A \times E} = \frac{40 \times 10^3 \times 2 \times 10^3}{20 \times 20 \times 200 \times 10^3}$

181. For copper, the yield stress σ_y and the brittle fracture stress σ_f are related as:
(a) $\sigma_f > \sigma_y$ (b) $\sigma_y > \sigma_f$
(c) $\sigma_y = \sigma_f$ (d) $\sigma_f << \sigma_y$
TRB Polytechnic Lecturer 2017

Ans. (a) : For copper
$\sigma_f > \sigma_y$



182. Proof stress–
(a) Is the safe stress
(b) Cause a specified permanent deformation in a material usually 0.1% or less
(c) Is used in connection with acceptance tests for materials
(d) Does not exist
Nagaland PSC CTSE 2017, Paper-I

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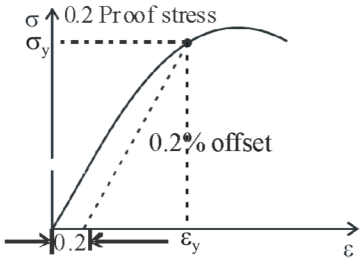
Ans. (b) : When material such as aluminium does not have an obvious yield point and yet undergoes large strain after the proportional limit is exceeded, an arbitrary yield stress may be determined by the offset of 0.1 or 0.2% of strain.

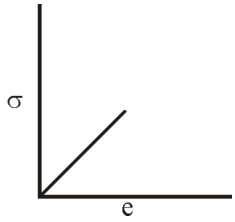


183. The stress strain curve for glass rod during tensile test would exhibit–

(a) A straight line (b) A parabola
(c) A sudden break (d) None of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Stress-strain curve for a glass rod during tensile test would exhibit, a sudden break. point Occur (due to a glass becomes a Brittle material).



184. A bar of mild steel 200 mm long and 50 mm × 50 mm in cross section is subjected to an axial load of 200 kN. If E is 200 GPa, the elongation of the bar will be

(a) 0.16 mm (b) 0.08 mm
(c) 0.04 mm (d) 0.02 mm

JWM 2017

Ans. (b) : Length of bar, L = 200 mm

Area of bar, $A = 50 \times 50\ mm = 250\ mm^2$

Axial load, $P = 200 \times 10^3\ N$

$E = 200 \times 10^3\ N/mm^2$

Elongation of bar,

$$\delta = \frac{PL}{AE} = \frac{200\times10^3\times200}{50\times50\times200\times10^3}$$

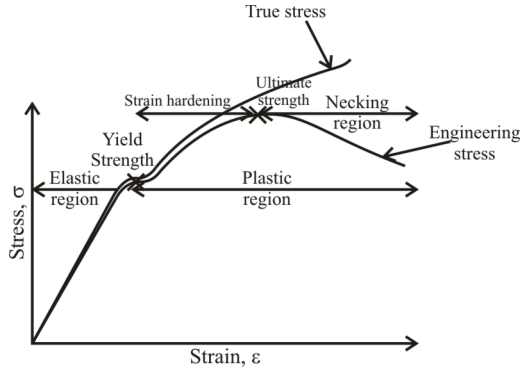
$\delta = 0.08mm$

185. The stress-strain plot for ductile materials exhibits peak at ultimate strength

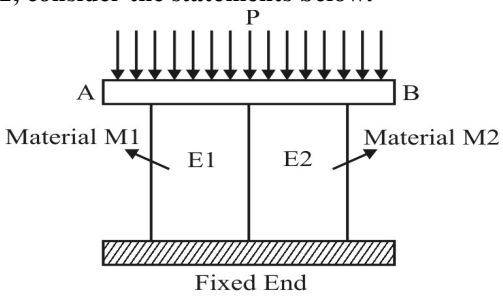
(a) because necking begins to occur where by engineering stress becomes less than the true stress
(b) because the material starts becoming weaker at microstructural level
(c) due to strain softening of the material
(d) None of the above

BPSC AE Mains 2017 Paper - VI

Ans : (a) : Because necking begins to occur, where by engineering stress becomes less than the true stress.



186. Two dimensionally equal blocks made of material M1 and M2 are placed on a flat horizontal surface as shown in figure below. Young's modulus of the materials are E1 & E2 resp. A uniform pressure 'p' is applied over the blocks through a thick plate (AB) symmetrically placed over the blocks. For E1 > E2, consider the statements below.



(a) Stress in both blocks are same
(b) Strain in both blocks are same
(c) Plate AB tilts and point B moves downward
(d) Plate AB tilts and point A moves downward

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Deflection $(\Delta) = \frac{P\ell}{AE}$

$$\Delta \propto \frac{1}{E}$$

as $E_1 > E_2$

$\Delta_1 < \Delta_2$

Hence AB tilts and B moves downward

$$\text{Stress} = \frac{P}{A} \qquad \text{(Independent of E)}$$

As two blocks are dimensionally equal having equal stresses.

187. A horizontal bar AB assumed to be rigid, is supported by two wires CE & DF, having lengths L and 2L respectively and pinned at A. If each wire has cross sectional area A, the tensile stress σ_1 and σ_2 in wires CE and DF are respectively.

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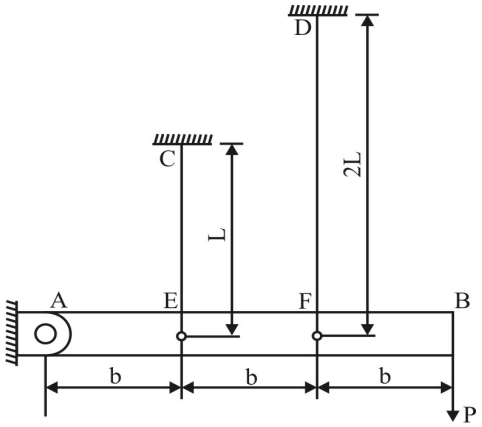
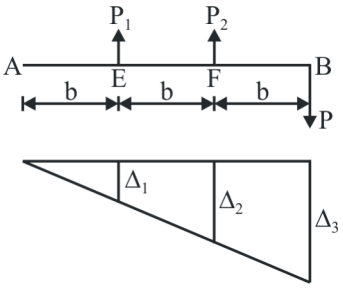


(a) $\frac{3P}{5A}$ and $\frac{6P}{5A}$ (b) $\frac{2P}{5A}$ and $\frac{4P}{5A}$

(c) $\frac{3P}{A}$ and $\frac{6P}{A}$ (d) $\frac{P}{A}$ each

ISRO Scientist/Engineer 17.12.2017

Ans. (d) :



Taking moment about point A

$P \times 3b - P_2 \times 2b - P_1 \times b = 0$

$3P = P_1 + 2P_2$ .....(1)

$\Delta_2 = 2\Delta_1$

$\frac{P_2 \times 2L}{AE} = 2\frac{P_1 \times L}{AE}$

$P_2 = P_1$ .....(2)

From equation (1) and (2)

$P = P_1 = P_2$

Tensile stresses in CE and DF

$\sigma_1 = \frac{P}{A}$

$\sigma_2 = \frac{P}{A}$

188. Two bolts A and B of same material and length are subjected to identical tensile load. If the elastic strain energy stored in bolt A is 16 times that of the bolt B and the mean diameter of bolt A is 13 mm, the mean diameter of bolt B (mm) is

(a) 52 (b) 36

(c) 24 (d) 16

ISRO Scientist/Engineer 07.05.2017

Ans. (a) : Given,

$P_A = P_B$

$U_A = 16\ U_B$

$\Rightarrow$ (Strain energy)$_A$ = 16 (Strain energy)$_B$

$SE = \frac{1}{2}P\delta \qquad \left[\because \delta = \frac{PL}{AE}\right]$

$= \frac{1}{2} \times P \times \frac{PL}{AE} = \frac{P^2L}{2AE} \qquad \left[\because A = \frac{\pi}{4}d^2\right]$

$\Rightarrow \frac{1}{2}\frac{P_A^2 L_A}{A_A E_A} = \frac{1}{2}\frac{P_B^2 L_B}{A_B E_B} \times 16$

$\frac{1}{A_A} = \frac{1}{A_B} \times 16$

$A_B = 16A_A$

$\frac{\pi}{4}d_B^2 = \frac{\pi}{4} \times 16 \times d_A^2$

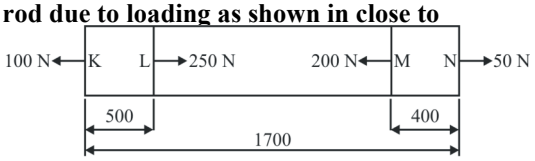
$d_B^2 = 16 d_A^2$

$d_B = \sqrt{16 \times (13)^2}$

$d_B = 4 \times 13$

$= 52$ mm

189. The figure shows an Aluminium rod of 25 mm^2 cross sectional area. It is loaded at four points, K, L, M and N. Assume E = 67 GPa for Aluminium. The total change in length of the rod due to loading as shown in close to



All dimensions are in mm

(a) 30 μm (b) −10 μm

(c) −30 μm (d) 10 μm

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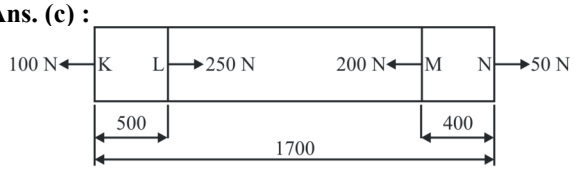
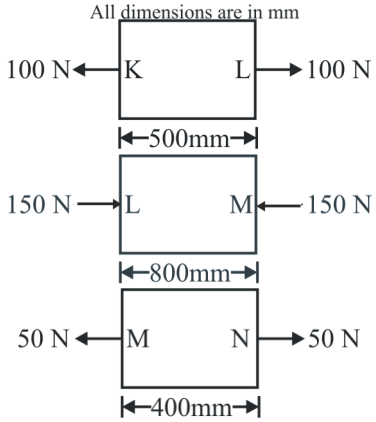
Ans. (c) :





Let, change in length of the 'rod' is ΔL

So, $\Delta L_{KL} = \left(\frac{PL}{AE}\right)_{KL}$

$= \frac{100 \times 500}{25 \times 67 \times 10^3}$

$= \frac{50}{25 \times 67}$

$(\Delta L)_{KL} = \frac{50}{1675} = 0.0298$ (Tension) ≈ 0.03

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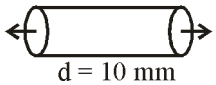
$$(\Delta L)_{LM} = \left(\frac{PL}{AE}\right)_{LM}$$
$$= \frac{150 \times 800}{25 \times 67 \times 10^3}$$
$$= \frac{120}{1675} = 0.07 \text{ (Compression)}$$
$$(\Delta L)_{MN} = \left(\frac{PL}{AE}\right)_{MN}$$
$$= \frac{50 \times 400}{25 \times 67 \times 10^3}$$
$$= \frac{20}{1675} = 0.01 \text{ (Tension)}$$
Total change in length $(\Delta L) = (\Delta L)_{KL} + (\Delta L)_{LM} + (\Delta L)_{MN}$
$= 0.03 - 0.07 + 0.01$
$= -0.03$ mm
$(\Delta L) = -30$ μm

190. A 10 mm diameter bar of mild steel of elastic modulus 200×10^9 Pa is subjected to a tensile load of 50000N, taking it just beyond its yield point. The elastic recovery of strain that would occur upon removal of tensile load will be
(a) 1.38×10^{-3} (b) 2.68×10^{-3}
(c) 3.18×10^{-3} (d) 4.62×10^{-3}

ESE 2017

Ans. (c) : Given,
d = 10 mm
$E = 200 \times 10^9$ Pa $= 200 \times 10^3$ MPa
P = 50000 N
$$\text{stress } (\sigma) = \frac{P}{A} = \frac{P}{\frac{\pi}{4}d^2} = \frac{4P}{\pi d^2}$$
The bar under loading



d = 10 mm
$$= \frac{4 \times 50000}{\pi \times 10^2}$$
$= 636.94$ N/mm^2
$= 636.94$ MPa
$\sigma = E\in$
$$\in = \frac{636.94}{200 \times 10^3}$$
$= 3.18 \times 10^{-3}$

191. A bar produces a lateral strain of magnitude 60×10^{-5} m/m when subjected to a tensile stress of magnitude 300 MPa along the axial direction. What is the elastic modulus of the material if the Poisson's ratio is 0.3?
(a) 200 GPa (b) 150 GPa
(c) 125 GPa (d) 100 GPa

ESE 2017

Ans. (b) : Given,
Lateral strain $= 60 \times 10^{-5}$
$\sigma = 300$ MPa
$\mu = 0.3$
We know that
Lateral strain = μ × longitudinal strain = $\mu \times \in_\ell$
$$\in_\ell = \frac{60 \times 10^{-5}}{0.3} = 200 \times 10^{-5}$$
$\sigma = \in E$
$$E = \frac{300}{200 \times 10^{-5}} = 150 \text{ GPa}$$

192. The modulus of rigidity of an elastic material is found to be 38.5% of the value of its Young's modulus. The Poisson's ratio μ of the material is nearly
(a) 0.28 (b) 0.30
(c) 0.33 (d) 0.35

ESE 2017

Ans. (b) : Given, G = 0.385 E
We know that $E = 2G(1 + \mu)$
$$1 + \mu = \frac{E}{2G} = \frac{1}{2 \times 0.385}$$
$1 + \mu = 1.297$
$\mu = 0.297$

193. Material Stress-strain relationship expressed by $\sigma = K\varepsilon^n$; where in K & n are :
(a) K is strength coefficient & n is yield strain
(b) K is yield stress & n is work hardening exponent
(c) K is strength coefficient & n is work hardening exponent
(d) None of the above

Punjab PSC SDE 12.02.2017

Ans. (c) : $\sigma = K\in^n$
Where,
K is strength coefficient
n is work hardening exponent

194. An aluminum bar (E = 72 GPa, ν = 0.33) of diameter 50mm cannot exceed a diameter of 50.1mm when compressed by axial force P. The maximum acceptable compressive load P is approximately.
(a) 190 kN (b) 200 kN
(c) 470 kN (d) 860 kN

HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : Given,
E = 72 GPa, Poisson ratio (ν) = 0.33, d_0 = 50mm, d_f=50.1mm
$$\in_{lat.} = \frac{\delta d}{d_0} = \frac{d_f - d_0}{d_0} = \nu\frac{\sigma}{E} = \nu\frac{P}{A.E}$$
$$\frac{d_f - d_0}{d_0} = \nu\frac{P}{\frac{\pi}{4}d_0^2.E}$$
$$P = \frac{(d_f - d_0)}{d_0} \times \frac{\pi}{4} \times \frac{d_0^2 \times E}{\nu}$$
$$P = \frac{0.1}{50} \times \frac{\pi}{4} \times \frac{(50)^2 \times 72 \times 10^3}{0.33} = 856.79\text{kN}$$
≈ 860 kN

195. A copper tube with wall thickness of 8mm must carry an axial tensile force of 175 kN. The allowable tensile stress is 90 MPa. The minimum required outer diameter is approximately.
(a) 60 mm (b) 72 mm
(c) 85 mm (d) 93 mm

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Ans. (c) :

$$\text{Stress}(\sigma) = \frac{P}{A} = \frac{P}{\frac{\pi}{4}(D_2^2 - D_1^2)} \quad \boxed{D_2 = D_1 + 2t}$$

$$D_2^2 - (D_2 - 2t)^2 = \frac{4 \times P}{\pi \times \sigma}$$

$$D_2^2 - (D_2 - 2 \times 8)^2 = \frac{4 \times 175 \times 10^3}{\pi \times 90}$$

$$\boxed{D_2 \approx 85\text{mm}}$$

196. A nylon bar (E = 2.1 GPa) with diameter 12mm, length 4.5 m, and weight 5.6 N hangs vertically under its own weight. The elongation of the bar at its free and is approximately?

(a) 0.05 mm (b) 0.07 mm
(c) 0.11 mm (d) 0.17 mm

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Ans. (a) : Given, $E = 2.1 \times 10^9$ Pa, d = 12mm, L = 4.5m, W = 5.6 N.

$$\text{Elongation due to self weight } (\delta_{\text{Self-Weight}}) = \frac{WL}{2AE}$$

$$\delta_{\text{Self-Weight}} = \frac{5.6 \times 4.5}{2 \times \frac{\pi}{4} \times (12)^2 \times 2.1 \times 10^9}$$

$$\delta_{\text{Self-Weight}} = \frac{25.2}{1899.072 \times 10^3}$$

$$\delta_{\text{Self-Weight}} = 5.305 \times 10^{-5}\ \text{m} = 0.053\text{mm}$$

197. A copper bar (d = 10 mm, E = 110 GPa) is loaded by tensile load P = 11.5 kN. The maximum shear stress in the bar is approximately.

(a) 73 MPa (b) 87 MPa
(c) 145 MPa (d) 150 MPa

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Ans. (a) : Given, P = 11.5kN, d = 10mm

$$\text{Normal stress } (\sigma_t) = \frac{P}{A} = \frac{11.5 \times 10^3}{\frac{\pi}{4} \times (10)^2}$$

$$= 146.42\text{MPa}$$

$$\text{Maximum shear stress } (\tau_{max}) = \frac{\sigma_t}{2} = \frac{146.42}{2}$$

$$= 73.21\text{MPa}$$

198. The 'Modular ratio' is the ratio of :

(a) Stress and strain
(b) Strain and stress
(c) Elasticity of two materials
(d) Change in length to original length

GMB AAE 25.06.2017

Ans. (c) : Modular ratio is defined as the ratio between modulus of elasticity of steel and modulus of elasticity of concrete it is used to finding losses in pre-stress because of the influence of creep.

199. What is the unit of Young's Modulus ?

(a) N/m^2 (b) N.m
(c) N/m^3 (d) N/m^4

GMB AAE 25.06.2017

Ans. (a) : Young's modulus (E) $= \frac{\text{Stress}}{\text{Strain}}\ (N/m^2)$

200. True stress is

(a) average load/original area
(b) average load/minimum area
(c) average load/ average area
(d) instantaneous load/instantaneous area

WBPSC AE, 2017

Ans. (d) : • True stress represents the ratio of instantaneous load to instantaneous area.

$$\sigma_T = \frac{W_i}{A_i}$$

• Engineering stress represents the ratio of average load to original cross-section of the specimen.

$$\sigma_{\text{Engineering}} = \frac{W}{A_o}$$

201. A load of 10000 kg is applied to copper cylinder 20cm long and 5cm in diameter causes the length to increase 0.4mm and diameter to decreases 0.04 mm. The Poisson ratio for copper would be

(a) 0.23 (b) 0.25
(c) 0.33 (d) 0.4

WBPSC AE, 2017

Ans. (d) : Poisson ratio $(\mu) = -\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$$\mu = \frac{0.04/50}{0.4/200} = \frac{0.04}{0.1} = 0.4$$

202. A reinforced concrete beam is considered as made of

(a) Homogeneous material
(b) Heterogeneous material
(c) Isotropic material
(d) Clad material

WBPSC AE, 2017

Ans. (b) : A reinforced concrete beam is considered as made of heterogeneous material.

203. The compressive lateral strain when a cube is subjected to mutually perpendicular tensile stresses of equal magnitude is given by

(a) $\frac{1}{\sigma} \times \frac{E}{m}$ (b) $\frac{1}{\sigma} \times \frac{m}{E}$
(c) $\frac{1}{m} \times \frac{\sigma}{E}$ (d) None of the above

Where, σ = stress, E = Young's modulus, and $\frac{1}{m}$ = Poisson's ration

Karnataka PSC AE, 10.09.2017

Ans. (c) :

$$\mu = \left| \frac{\text{lateral strain}}{\text{Longitudinal strain}} \right|$$

$$\text{Lateral strain} = \mu\left(\frac{\Delta L}{L}\right)$$

$$= \frac{\mu\sigma}{E} = \frac{\sigma}{mE}$$

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204. If the modulus of elasticity of a material is twice to its modulus of rigidity, then the Poisson's ratio of the material is equal to_____

(a) One (b) Zero
(c) 0.25 (d) 0.5

GPSC EE Pre, 28.01.2017

Ans. (b) : Given-
Modulus of elasticity (E) = 2 × modulus of rigidity (G)
$\therefore$ $E = 2G$
From, $E = 2G(1 + \mu)$
$2G = 2G(1 + \mu)$
$1 + \mu = 1$
Poisson's ratio, $\mu = 0$

205. A steel bar of 500 mm length is under tensile stress of 100 N/mm^2. If the modulus of elasticity is 2×10^{11} N/m^2, then the total elongation of the bar will be

(a) 0.25 mm (b) 2.50 mm
(c) 5.00 mm (d) 0.50 mm

APPSC AEE Screening Test 2016

Ans. (a)
Given,
length (ℓ) = 500 mm
$\sigma = 100$ N/mm^2
$E = 2 \times 10^{11}$ N/m^2 $= 2 \times 10^{11} \times 10^{-6}$ N/mm^2
$= 2 \times 10^5$ N/mm^2
We know that,

$$\text{elongation } (\delta\ell) = \frac{PL}{AE} = \frac{\sigma L}{E} = \frac{100\times500}{2\times10^5} = \frac{5}{20} = \frac{1}{4} = 0.25 \text{ mm}$$

206. An axial load P is applied on circular section of diameter D. If the same load is applied on a hollow circular shaft with inner diameter as D/2, the ratio of stress in two cases would be

(a) 4/3 (b) 9/8
(c) 1 (d) 8/3

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Axial load (P) on circular section & diameter (D) if the same load is applied on a hollow circular shaft with inner diameter as (D/2), then ratio of stress will be,
d = D/2

$$\sigma_1 = \frac{P}{A_1} \quad \sigma_1 = \frac{P}{\frac{\pi}{4}D^2}$$

$$\sigma_2 = \frac{P}{A_2} \quad \sigma_2 = \frac{P}{\frac{\pi}{4}(D^2 - d^2)}$$

$$\frac{\sigma_1}{\sigma_2} = \frac{\frac{P}{\frac{\pi}{4}(D^2)}}{\frac{P}{\frac{\pi}{4}\frac{(4D^2 - D^2)}{4}}} = \frac{D^2}{D^2 - \left(\frac{D}{2}\right)^2} = \frac{4D^2}{3D^2} = \frac{4}{3}$$

207. A compound bar having two members X and Y of length L when subjected to tensile force P would have elongation equal to (in usual notations)

(a) $\frac{PL}{A_XE_X + A_YE_Y}$ (b) $\frac{PL}{A_XE_X} + \frac{PL}{A_YE_Y}$
(c) $\frac{PL}{A_XE_X - A_YE_Y}$ (d) $\frac{PL}{A_XE_X} - \frac{PL}{A_YE_Y}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : We know,

$$\varepsilon = \frac{PL}{AE}, \quad \therefore \text{ [P \& L Remains constant]}$$

So for X & Y member

$$\varepsilon = E_X + E_Y = \frac{PL}{A_XE_X + A_YE_Y}$$

208. A vertical prismatic bar fixed at the top end loaded with P at the bottom free end, is having a unit weight of W and its length is L. If σ is the working stress, the safe cross sectional area for the when P and self weight are considered is given by

(a) $\frac{P}{\sigma + WL}$ (b) $\frac{P}{\sigma - WL}$
(c) $\frac{\sigma + WL}{P}$ (d) $\frac{\sigma - WL}{P}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : We Know, stress = $\frac{P}{A}$
So, Total stress = $\sigma - WL$

$$\sigma - WL = \frac{P}{A}$$

$$\boxed{A = \frac{P}{\sigma - WL}}$$

209. If temperature changes from t_0 to t in a simple bar with both ends constrained and having modulus of elasticity E and thermal coefficient α, the thermal stress due to the temperature rise is

(a) $E\frac{(t - t_0)}{\alpha}$ (b) $\alpha\frac{(t - t_0)}{E}$
(c) $E\alpha(t - t_0)$ (d) $\frac{E\alpha}{(t - t_0)}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : We know,
$t - t_0 = \alpha\,\Delta\,T$, '$\alpha$' thermal expansion
$t = t_0(1 + \alpha\,\Delta\,T) \Rightarrow$ thermal stress = $\frac{F}{A} = t \times \text{strain}$
So, $= E\,\alpha\,(t - t_0)$

210. Volumetric strain of a sphere of diameter increasing from d_1 to d_2 is equal to

(a) $\frac{d_2 - d_1}{d_1}$ (b) $3\frac{(d_2 - d_1)}{d_1}$
(c) $3\frac{d_2 + d_1}{2d_1}$ (d) $\frac{d_2 + d_1}{2d_1}$

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Ans. (b) : As, We know,

$$\text{Volumetric strain} = \frac{\text{Change in volume}}{\text{Original volume}}$$

$$\text{'}\varepsilon_v\text{' for sphere} = 3\times\frac{\text{Change in diameter}}{\text{Diameter}}$$

$$= 3\times\frac{(d_2 - d_1)}{d_1}$$

211. If a circular rod having Poisson's ratio μ is subjected to an axial pull and if the strain is e, then the lateral strain is equal to

(a) $-\mu e$ (b) μe

(c) $-e/\mu$ (d) e/μ

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Ans. (a) : For a circular section,

$$\text{Poisson's ratio} = -\frac{\text{Lateral strain}\,(L)}{\text{Longitudinal strain}\,(e)}$$

Lateral strain (L) = $-\mu \times e$

$$\left[\because \text{For circular section, Lateral strain} = \frac{d_f - d_0}{d_0}\right]$$

212. If a timber beam 8 cm wide and 16 cm deep is to be converted into an equivalent steel section of the same depth for analysis purpose, then the width of the equivalent section for a modular ratio of 20 will be

(a) 160 cm (b) 2.5 cm

(c) 0.4 cm (d) 12 cm

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Ans. (c) : We know, Elasticity

$$\text{Modular Ratio} = \frac{\text{Modulus of Elasticity of steel}}{\text{Modulus of Elasticity timber}}$$

$$20 = \frac{E_s}{E_t} \Rightarrow 20\ E_t = E_s$$

So, We know,

$A_s \times E_s = E_t \times A_t$

$$b_s = \frac{E_s}{20}\times\frac{16\times 8}{16\times E_s} = 0.4\text{cm}$$

213. The value of Poisson's ratio for which bulk modulus of a material will be equal to its Young's modulus

(a) 0.33 (b) 0.15

(c) 0.45 (d) 0.25

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : We know,

K = Bulk Modulus

E = Young's Modulus

$E = 3K(1-2\mu)$

So, E = K, substitute Above eq.

We get

$$\frac{1}{3} = 1 - 2\mu$$

$$\mu = \frac{1}{3} = 0.33$$

214. A brass tube has enclosed a steel bar and they equal cross-sectional area. The Young's modulus of elasticity is 200 GPa and 100 GPa for steel and brass respectively. Then the ratio of stress developed in the steel bar to that in the brass tube compression is

(a) 0.5 (b) 1

(c) 1.5 (d) 2

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : As we know,

$A_s = A_b$,

$E_s = 200$ GPa

$E_b = 100$ GPa

As we know,

$$\frac{\sigma_b A_b}{\sigma_s A_s} = \frac{E_b}{E_s} = \frac{100}{200}$$

So, $\dfrac{\sigma_s}{\sigma_b} = 2$

215. Two bars of same area and length but of different materials are subjected to same tensile force. If the bars have their axial elongation in the ratio of 4:6, then the ratio of modulus of elasticity of the two materials would be

(a) 4 : 6 (b) 6 : 4

(c) $2:\sqrt{6}$ (d) $\sqrt{6}:2$

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Ans. (b) : As we know,

$$\text{Elongation, } \delta L_1 = \frac{P_1 L_1}{A_1 E_1}$$

So for $\delta L_2 = \dfrac{P_2 L_2}{A_2 E_2}$ $\quad \therefore \begin{bmatrix} L_1 = L_2 \\ A_1 = A_2 \\ P_1 = P_2 \end{bmatrix}$

Elongation ratio

$$\frac{\delta L_1}{\delta L_2} = \frac{P_1 L_1}{A_1 E_1}\times\frac{A_2 E_2}{P_2 L_2}$$

$$\frac{\delta L_1}{\delta l_2} = \frac{E_2}{E_1}$$

$$\frac{E_1}{E_2} = \frac{\delta L_2}{\delta L_1} = \frac{6}{4}$$

$$\boxed{\frac{E_1}{E_2} = \frac{6}{4}}$$

216. The bulk modulus of elasticity of a materials is twice its modulus of rigidity. The poisson's ratio of the material is

(a) 1/7 (b) 2/7

(c) 3/7 (d) 4/7

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : We know,

$$E = 2G(1+\mu)\ \&\ E = 3K(1-2\mu)$$

$$\mu = \frac{3K - 2G}{2G + 6K} \qquad \text{Given} : \Rightarrow \boxed{K = 2G}$$

So, $\mu = \dfrac{6G - 2G}{2G + 12G} = \dfrac{4G}{14G} = \dfrac{2}{7}$

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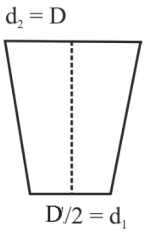
217. The diameter of a tapering rod varies from 'D' to 'D/2' in length of 'L' m. If it is subjected to an axial tension of 'P' the change in length is

(a) $4\,PL / (\pi ED^2)$
(b) $8\,PL / (\pi ED^2)$
(c) $2\,PL / (\pi ED^2)$
(d) None of the given answers

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : We know, for prismatic bar

Elongation, $\delta\ell = \dfrac{PL}{AE}$



For taper bar,

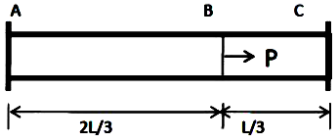
$$\delta\ell = \frac{PL}{\frac{\pi}{4}d_1 d_2 E} = \frac{PL}{\frac{\pi}{4}D \times \frac{D}{2} \times E}$$

[Given, $d_2 = D$, $d_1 = D/2$]

Elongation, $\delta\ell = \dfrac{PL}{\frac{\pi}{4}\left(\frac{D^2}{2}\right) \times E}$

$$\boxed{\delta\ell = \frac{8PL}{\pi D^2 E}}$$

218. The ratio of load shared by parts 'AB' and 'BC' of the bar shown below is

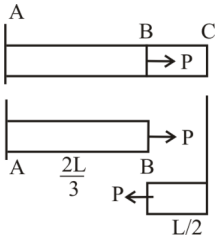


(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 1 : 2

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : We know,

$$\delta\ell = \delta\ell_1 + \delta\ell_2 = 0$$



So, $\delta\ell_1 = \dfrac{P_1 L_1}{AE}$

$\delta\ell_2 = \dfrac{P_2 L_2}{AE}$

$$\boxed{\frac{P_1}{P_2} = \frac{L_2}{L_1} = \frac{L/3}{\left(\frac{2L}{3}\right)} = \frac{1}{2}}$$

219. Let the strain produced in length and diameter of cylindrical rod be 0.02 and 0.005 respectively. Then the volumetric strain is given by

(a) 0.03 (b) 0.025
(c) 0.015 (d) 0.01

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Ans. (a) : Volumetric strain $\varepsilon_v = \varepsilon_l + 2\varepsilon_d$
$= 0.02 + 2 \times 0.005 = 0.03$

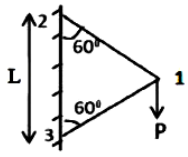
220. A material with large deformation at failure is termed

(a) Brittle (b) Elastic
(c) Ductile (d) Elasto-plastic

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : The material in which large deformation in possible before the absolute failure or rupture is termed as ductile.

221. A system of two prismatic bars of equal length (L) and equal cross-section (A) carries a vertical load P as shown in the figure given below. If their modulus of elasticity is E, the vertical displacement of the hinge 1 is



(a) $\dfrac{PL}{AE}$ (b) $\dfrac{2\,PL}{AE}$
(c) $\dfrac{3PL}{AE}$ (d) $\dfrac{2.5PL}{AE}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : As we know,

Elongation, $\delta\ell = \dfrac{PL}{AE}$

Here, P at 1 is twice the load

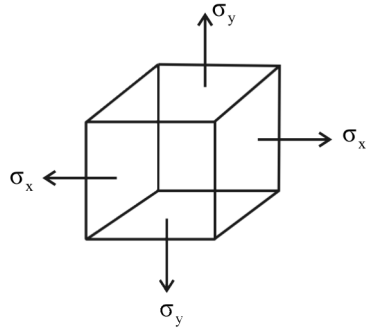
So, $\boxed{\delta\ell = 2 \times \dfrac{PL}{AE}}$

222. Which of the following is true (μ = Poisson's ratio):

(a) $0 < \mu < {}^1/_2$ (b) $1 < \mu < -1$
(c) $1 < \mu < 0$ (d) $\infty < \mu < -\infty$

UJVNL AE 2016

Ans : (a)



Total strain in x-x direction

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$$e_x = \frac{\sigma_x}{E} - \mu.\frac{\sigma_y}{E} - \mu\frac{\sigma_z}{E}$$

$$e_x = \frac{\sigma_x}{E} - \mu\left(\frac{\sigma_y}{E} + \frac{\sigma_z}{E}\right)$$

Where $\sigma_x = \sigma_y = \sigma_z = \sigma$

$$e_x = \frac{\sigma}{E} - \mu\left(\frac{\sigma}{E} + \frac{\sigma}{E}\right)$$

$$e_x = \frac{\sigma}{E} - 2\mu\left(\frac{\sigma}{E}\right)$$

$$e_x = \frac{\sigma}{E}(1 - 2\mu)$$

$$e_x + e_y + e_z = e_v$$

$$e_v = \frac{3\sigma}{E}(1 - 2\mu)$$

$$\frac{\delta v}{v} = \frac{3\sigma}{E}(1 - 2\mu)$$

This limits 2μ to maximum of 1 or the poisson's ratio lie to 0.5. No material is known to have a higher Value for poisson's ratio although μ for materials like rubber approaches this value.

Poisson's ratio

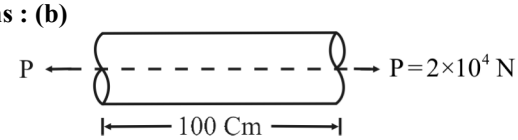
$$0 < \mu < {}^1/_2$$

223. A steel rod of 100 cm long and 1 sq cm cross sectional area has a young's modulus of elasticity 2×10^6 kgf/cm^2. It is subjected to an axial pull of 2000 kgf. The elongation of the rod will be.

(a) 0.05 cm (b) 0.1cm
(c) 0.15cm (d) 0.20cm

UJVNL AE 2016

Ans : (b)



l = 100 cm =1m, A = 1×10^{-4}m^2

P = 2×10^4 N.

E = 2×10^6kgf/cm^2 = 2×10^{11}N/m^2

$$\delta l = \frac{Pl}{AE}$$

$$\delta l = \frac{2\times10^4 \times 1}{1\times10^{-4}\times2\times10^{11}}$$

$$\delta l = 0.1cm.$$

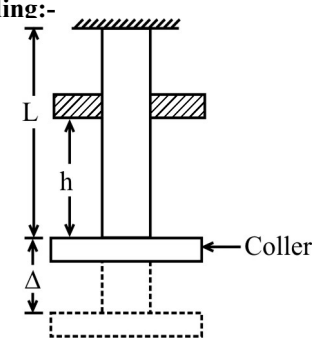
224. A test used to determine the behavior of materials when subjected to high rates of loading is known as :

(a) Hardness test (b) Impact test
(c) Fatigue test (d) Torsion test

HPPSC W.S. Poly. 2016

Ans : (b) A test used to determine the behavior of materials when subjected to high rate of loading for small time is known as impact test.

Impact loading:-



$$\sigma_{impact} = \frac{W}{A}\left[1 + \sqrt{1 + \frac{2h}{\delta}}\right]$$

225. A steel bar 10mm×10mm cross section is subjected to an axial tensile load of 20 kN. If the length of bar is 1 m and E = 200 GPa, then elongation of the bar is :

(a) 1 mm (b) 0.5 mm
(c) 0.75 mm (d) 1.5 mm

Kerala PSC IOF 19.04.2016

Ans. (a) Given,
Area = 10 × 10 = 100 mm^2
Length = 1 m = 1000 mm
Load, P = 20 kN

$$\text{Elongation } (\delta) = \frac{PL}{AE}$$

$$= \frac{20\times10^3\times1000}{100\times200\times10^3} = 1\text{mm}$$

226. The modulus of rigidity and Poisson's ratio of a material are 80GPa and 0.3 respectively, Its Young's Modulus will be:

(a) 160 GPa (b) 208 GPa
(c) 120 GPa (d) 104 GPa

Kerala PSC IOF 19.04.2016

Ans. (b) : Given, G = 80 GPa, μ = 0.3
We know, E = 2G(1 + μ)
E = 2 × 80 (1 + 0.3)
Modulus of Elasticity, E = 208 GPa

227. Yield point in fatigue loading as compared to static loading is ..... .

(a) same
(b) higher
(c) lower
(d) depends on other factors

GWSSB DEE 07.07.2016

Ans. (c) : Yield point in fatigue loading as compared to static loading is lower.

228. Residual stress in the materials ..... .

(a) acts when external load is applied
(b) becomes zero when external load is removed
(c) is independent of external
(d) is always be harmful

GWSSB DEE 07.07.2016

Ans. (c) : Residual stressess are locked in stresses within a metal object, even though the object is free of external forces.
These stresses are the result of one region of the metal being constrained by adjacenet regions from expanding, contracting, or releasing stresses can be tensile or compressive.

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229. Ability of material to resist fracture due to high impact load is called ..... .

(a) toughness (b) stiffness
(c) plasticity (d) hardness

GWSSB DEE 07.07.2016

Ans. (a) : Ability of material to resist fracture due to high impact load is called toughness.

230. Which of the following statement is correct?

(a) The stress is the pressure per unit area
(b) The strain is expressed in mm
(c) Hook's law holds good up to the breaking point
(d) Stress is directly proportional to strain within elastic limit

UP Jal Nigam AE 2016

Ans. (d) : Stress is directly proportional to strain within elastic limit.

231. Area under the stress-strain curve when load is gradually applied in tension represents the

(a) Strain energy
(b) Strain energy density
(c) Strain energy per unit weight
(d) Strain energy per unit area

RPSC LECTURER 16.01.2016

Ans. (b) : Area under the stress-strain curve when load is gradually applied in tension represents the strain energy density.

232. For ductile materials, the largest value of tensile stress that can be sustained by material before breaking is known as:

(a) Modulus of elasticity
(b) Ultimate tensile strength
(c) Yield strength
(d) Toughness

UPRVUNL AE 07.10.2016

Ans. (b) : For ductile materials, the largest value of tensile stress that can be sustained by material before breaking is known as ultimate tensile strength.

233. When the temperature of a solid metal increases-

(a) strength of the metal decreases but ductility increases
(b) both strength and ductility decrease
(c) both strength and ductility increase
(d) strength of the metal increases but ductility decreases

RPSC 2016

Ans : (a) Strength of the metal decreases but ductility increases. When the temperature of a solid metal increases, then its intermolecular bonds breaks and strength of solid metal decreases. Due to decreases its strength, the elongation of the metal increases, when we apply the load i.e. ductility increases.

234. A rod is subjected to a uniaxial load with in linear elastic limit. When the change in the stress is 200 MPa, the change in strain is 0.001. If the Poisson's ratio of the rod is 0.3, the modulus of rigidity (in GPa) is–

(a) 75.31 (b) 76.92
(c) 77.23 (d) 76.11

RPSC 2016

Ans : (b)

$$E = \frac{\text{stress}}{\text{strain}} = \frac{200\times10^6}{0.001} = 2\times10^{11}\ \text{Pa}$$

$\mu = 0.3 \Rightarrow$

$$E = 2G(1+\mu)$$

$$2\times10^{11} = 2G\,(1+0.3)$$

$$G = 7.69\times10^{10}\ \text{Pa}$$

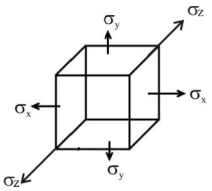
$$\boxed{G = 76.62\,\text{GPa}}$$

235. When a body is subjected to stress in all the directions, the body is said to be under..... strain.

(a) compressive (b) tensile
(c) shear (d) volumetric

(HPPSC LECT. 2016)

Ans : (d)



When a body is subjected to stress in all the direction, the body is said to be under volumetric strain.

236. Hooke's law is applicable:

(a) Plastic range, strain is proportional to stress
(b) Elastic range, strain is proportional to stress
(c) In both elastic and plastic range, strain is proportional to stress
(d) None of the above

(HPPSC LECT. 2016)
KPCL AE 2016

Ans : (b) Hooke's law is applicable up to elastic range, strain is proportional to stress.
Hooke's law:- The slope of this line is the ratio of stress to strain and in constant for a material. In this range, the material also remains elastic. When a material behaves elastically and also exhibits a linear relationship between stress and strain, it is called linearly elastic.
The slope of stress- strain curve is called the modulus of Elasticity

237. The ratio of modulus of rigidity to bulk modulus for a Poisson's ratio of 0.25 would be:

(a) 2/3 (b) 2/5
(c) 3/5 (d) 1.0

HPPSC W.S. Poly. 2016

Ans : (c) $E = 3K(1-2\mu)$ ..... (i)

$E = 2G(1+\mu)$ ..... (ii)

$3K(1-2\mu) = 2G(1+\mu)$

$$\frac{G}{K} = \frac{3(1-2\mu)}{2(1+\mu)}$$

G = Modulus of rigidity
K = Bulk modulus
E = Modulus of Elasticity
μ = Poission's ratio

$$\frac{G}{K} = \frac{3(1-2\times0.25)}{2(1+0.25)}$$

$$G/K = 3/5$$

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238. Two steel rails each of 12 m length are laid with a gap of 1.5 mm at ends at a temperature of 24°C. The thermal stress produced at a temperature of 40°C is (take E = 2 × 10^5 N/mm^2, coefficient of thermal expansion = 12 × 10^{-6} /°C)

(a) 10.5 N/mm^2 (b) 12.5 N/mm^2
(c) 13.4 N/mm^2 (d) 15.5 N/mm^2

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Given, L = 12 m = 12 × 10^3 mm
λ = gap between rails
$\Delta T = (40 - 24) = 16°C$
E = 2 × 10^5 N/mm^2
$\alpha = 12 \times 10^{-6}/°C$

Then, $\boxed{\sigma_{thermal} = \frac{(\alpha TL - \lambda)}{L} E}$

$$\sigma_{th} = \frac{(12\times10^{-6}\times16\times12\times10^3 - 1.5)}{12\times10^3}\times2\times10^5$$

σ_{th} = 13.4 N/mm^2

239. An aluminum tensile test specimen has a diameter, d_o = 25 mm and a gauge length of L_o = 250 mm. If a force of 175 kN elongates the gauge length by 1.25 mm, the modulus of elasticity of the material is nearly

(a) 71 GPa (b) 71 MPa
(c) 142 GPa (d) 142 MPa

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : Given,
L_0 = 250 mm,
P = 175 kN
= 175 × 10^3 N
$\Delta\ell$ = 1.25 mm
d_0 = 25 mm

$$A = \frac{\pi}{4}(d_0)^2 = \frac{\pi}{4}\times(25)^2$$

$$A = \frac{\pi}{4}\times(625)$$

Elongation $\boxed{\Delta\ell = \frac{PL}{AE}}$

$$1.25 = \frac{175\times10^3\times250}{\frac{\pi}{4}\times625\times E}$$

$$1.25 = \frac{4\times7\times10^3\times10}{\pi\times E}$$

$$E = \frac{28\times10^4}{1.25\times\pi}$$

E = 71301.41 N/mm^2
E = 71.301 × 10^3 N/mm^2
$\boxed{E = 71\,GPa}$

240. Hoop stress for a wooden with steel flat tyre in terms of outside diameter of the wheel D, inside diameter of the tyre d and Young's modulus E is

(a) E(D-d)/d (b) E(D/d)
(c) E(D/(D-d)) (d) E(d/D)

APPSC AEE Screening Test 2016

Ans. (a) : Given,
Inside diameter (ID) = d
Outside diameter (OD) = D

Hoop stress (σ_h)

$\boxed{(\sigma_h) = E \times strain}$

So, strain = $\left(\frac{D-d}{d}\right)$

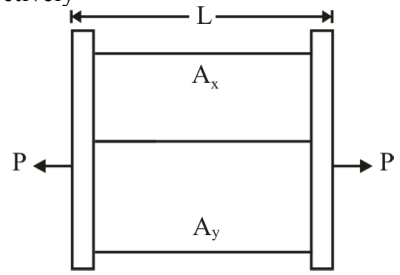
$\boxed{\sigma_h = E\times\left(\frac{D-d}{d}\right)}$

241. A tensile force of P is applied on a compound bar having two members X and Y. Then the load shared by the member X is

(a) $\frac{PA_XE_X}{A_XE_X + A_YE_Y}$ (b) $\frac{PA_YE_Y}{A_XE_X + A_YE_Y}$
(c) $\frac{P}{2}$ (d) $\frac{PE_X}{E_Y}$

APPSC AEE Screening Test 2016

Ans. (a) : Let us consider a compound bar having two member x and y and their corresponding area Ax and Ay respectively



Let the load shared by x - section be P_x and similarly for y - section be P_y.
∴ $P_x + P_y$ = total load i.e. P ---------- (1)
If the compound bars are connected in parallel, the axial deflection in both the sections x and y will be the same.
∴ Deflection in x - section = deflection in y - section

$$\frac{P_xL}{A_x.E_x} = \frac{P_yL}{A_yE_y}$$

Therefore, $P_y = \frac{P_x.A_y.E_y}{A_x.E_x}$ ---------- (2)

From (1) and (2)

$P_x + P_y = P$

$$P = P_x + \frac{P_xA_yE_y}{A_x.E_x}$$

$$P = \frac{P_xA_xE_x + P_xA_yE_y}{A_x.E_x}$$

$P_x [A_xE_x + A_yE_y] = PA_xE_x$

$$\boxed{P_x = \frac{PA_xE_x}{A_xE_x + A_yE_y}}$$

242. Stress concentration may be caused by

(a) Change in cross sectional area
(b) Change in shape

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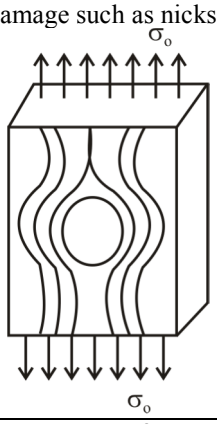
(c) Change in dimension
(d) A hole or a notch in the body

APPSC AEE Screening Test 2016

Ans. (d) : Stress concentrations occur when there are irregularities in the geometry or material of a structural component that cause on interruption to the flow of stress. This arises from such details as hole, grooves, notches and fillets. Stress concentration may also occur from accidental damage such as nicks and scratches.



243. What is the amount of stress induced in a 5 mm steel bar of length 2 m when heated from 20° C to 140° C while it is free to expand?
(a) 600 Pa (b) 1200 Pa
(c) 48 Pa (d) None of these

KPCL AE 2016

Ans. (d) : Thermal stress = $\alpha\Delta TE$
$\because$ For free to expand a thermal stress always zero.

244. Value of Poisson's ratio for ionic solids is in the range of:
(a) 0.1 (b) 0.3
(c) 0.4 (d) 0.2

UPRVUNL AE 21.08.2016

Ans. : (d) Value of poission's ratio for ionic solids is in the range of 0.2.

245. Find the modulus of elasticity for a rod which tapers uniformly from 30 mm to 15mm diameter in a length of 350mm. The rod is subjected to an axial load of 5.5 kN and extension of the rod is 0.025 mm
(a) 2.78×10^5 N/mm^2 (b) 2.18×10^5 N/mm^2
(c) 3.18×10^5 N/mm^2 (d) 3.78×10^5 N/mm^2

UPRVUNL AE 21.08.2016

Ans. : (b) Given,
$F = 5.5 \times 10^3$ N
$l = 350$ mm
$d_1 = 350$ mm
$d_2 = 15$ mm
$\Delta l = 0.025$ mm
uniformly tapers bars–

$$\Delta l = \frac{4Fl}{\pi E d_1 d_2}$$

$$E = \frac{4Fl}{\pi \times d_1 d_2 \times \Delta l}$$

$$E = \frac{4 \times 5.5 \times 10^3 \times 350}{\pi \times 350 \times 15 \times 0.025}$$

$E = 2.18 \times 10^5$ N/mm^2

246. It is desired to shrink a thin steel tyre on to a rigid wheel of 3600 mm diameter. What is the internal diameter of the tyre, if after fitting the hoop stress in the tyre is 95 N/mm^2. Take E=2×10^5 MPa
(a) 3598.29 mm (b) 3592.46 mm
(c) 3614.27 mm (d) 3618.38 mm

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Given, D = 3600 mm, σ_H = 95 N/m^2
$E = 2 \times 10^5$ MPa
According to Hooke's law

$$\sigma = E\varepsilon$$

$$\varepsilon = \frac{\sigma}{E} = \frac{95}{2 \times 10^5}$$

$$\varepsilon = 4.75 \times 10^{-4}$$

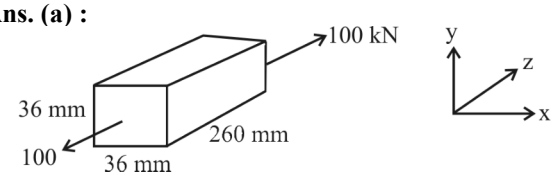
$$\frac{\delta D}{D} = \varepsilon = 4.75 \times 10^{-4}$$

$\delta D = 4.75 \times 10^{-4} \times 3600$
$\delta D = 1.71$ mm
$D - d = 1.71$ mm
$d = D - 1.71$
$d = 3600 - 1.71$
$d = 3598.29$ mm

247. A bar 36 mm × 36 mm × 260 mm long is subjected to a pull of 100 kN, in the direction of length. The decrease in each lateral dimension is 0.004 mm. The lateral strain is
(a) 1.11×10^{-4} (b) 1.25×10^{-4}
(c) 1.32×10^{-4} (d) 1.45×10^{-4}

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



Decrease in lateral dimension = 0.004 m (Given)

$$\text{Lateral strain} = \frac{\text{change in length of lateral dimension}}{\text{Original length of lateral dimension}}$$

$$\varepsilon_y = \frac{0.004}{36} = \boxed{1.11 \times 10^{-4}}$$

248. The shear strength of a metal is usually
(a) greater than its tensile strength
(b) less than its tensile strength
(c) equal to its tensile strength
(d) none of these

GPSC Lect. 23.10.2016

Ans. (b) : Shear strength is the load acting along a plane that is parallel to the direction of the force. The ratio of shear stress to the corresponding shear strain for shear stresses below the proportional limit of the material. So, shear strength of a metal is usually less than its tensile strength.

249. During tensile test on a specimen of 1 cm^2 cross-section, maximum load observed was 80 kN and area of cross-section at neck was 0.5 cm^2. U.T.S. of the specimen is:
(a) 400 MPa (b) 800 MPa
(c) 1600 MPa (d) 2200 MPa

RPSC VPITI 14.02.2016

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Ans. (b) : Given,
Maximum load = 80×10^3 N
Original cross-section area = 100 mm^2

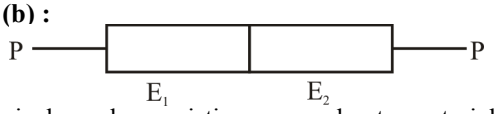
$$\text{U.T.S.} = \frac{\text{Maximum load}}{\text{Cross} - \text{section area}} = \frac{80 \times 10^3}{1 \times 100} \text{N/mm}^2$$

UTS = 800 MPa

250. Two wires of different materials but of same diameter are connected end to end and a force is applied which stretches them by 1 cm. The two wires will have the
(a) Same stress and strain
(b) Same stress but different strain
(c) Same strain but different stress
(d) Different stress and strain

RPSC VPITI 14.02.2016

Ans. (b) :



Stress is depend on resisting area and not a material.

$$\sigma_1 = \sigma_2 = \frac{P}{A}$$

$$\text{Strain} = \frac{\Delta L}{L} = \frac{\text{Stress}}{E}$$

So, strain depends on modulus of elasticity which is property of material.
So strain will be different in different wires.

251. For the bar of composite section
(a) The extension in different materials is different
(b) The total external load is equal to the total sum of the loads carried by different materials
(c) The compression in different materials is different
(d) Strain in all materials is unequal

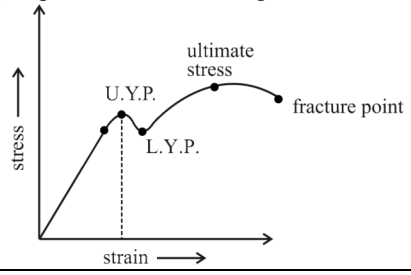
Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : For the bar of composite section the total external load is equal to the total sum of loads carried by different materials.

252. The maximum stress a material can stand before it breaks is called the
(a) Working stress
(b) Ultimate tensile stress
(c) Compressive stress
(d) Transverse stress

Rajasthan Nagar Nigam AE 2016, Shift-II

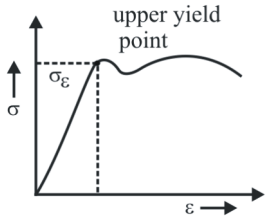
Ans. (b) : Ultimate tensile stress i.e. defined as the maximum stress that material can withstand when a force is applied. When the material are pushed beyond UTS they experience the cracking.



253. Yield strength of a material represents the stress below which the deformation is :
(a) almost elastic (b) plastic
(c) zero (d) none of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Yield strength of a material represents the stress below which the deformation is almost elastic.



254. A specimen of steel, 20 mm diameter with a gauge length of 200 mm is tested to destruction. It has an extension of 0.25 mm under a load of 80 kN and the load at elastic limit is 102 kN. The modulus of elasticity is
(a) 203718 N/mm^2 (b) 259740 N/mm^2
(c) 209740 N/mm^2 (d) 253718 N/mm^2
(e) 222718 N/mm^2

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : diameter (d) = 20 mm
length (l) = 200 mm
extension (δ) = 0.25 mm
load (P) = 80 kN = 80,000 N
load at elastic limit = 102 kN
We know that

$$\delta = \frac{Pl}{AE}$$

$$E = \frac{Pl}{A\delta} = \frac{Pl}{\frac{\pi}{4}d^2 \times \delta} = \frac{80000 \times 200}{\frac{\pi}{4} \times (20)^2 \times 0.25}$$

E = 203718.32 N/mm^2

255. A circular road of 25 mm diameter and 500 mm long is subjected to a tensile force of 60 kN. Determine modulus of rigidity and bulk modulus if Poisson's ratio = 0.3 and Young's modulus E = 2 × 10^5 N/mm^2
(a) 0.7692 × 10^5 N/mm^2 and 1.667 × 10^5 N/mm^2
(b) 0.667 × 10^5 N/mm^2 and 1.857 × 10^5 N/mm^2
(c) 0.1852× 10^5 N/mm^2 and 1.6567 × 10^5 N/mm^2
(d) 0.4692× 10^5 N/mm^2 and 1.545 × 10^5 N/mm^2
(e) 1.7562× 10^5 N/mm^2 and 1.117 × 10^5 N/mm^2

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : Data given as
d = 25 mm, l = 500 mm
F = 60 kN, E = 2 × 10^5 N/mm^2
We know that
E = 2 G (1 + μ)

$$G = \frac{2 \times 10^5}{2 \times (1 + 0.3)} = 0.7692 \times 10^5 \text{ N/mm}^2$$

$$\boxed{G = 0.7692 \times 10^5 \text{ N/mm}^2}$$

E = 3 K (1–2μ)

$$K = \frac{2 \times 10^5}{3 \times (1 - 2 \times 0.3)}$$

$$\boxed{K = 1.667 \times 10^5 \text{ N/mm}^2}$$

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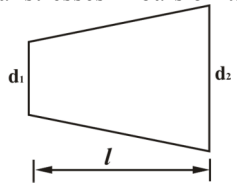
256. The steel bar AB varies linearly in diameter from 25 mm to 50 mm in a length 500 mm. It is held between two unyielding supports at room temperature. What is the stress induced in the bar, if temperature rises by 25ºC? Take E = 2 × 10^5 N/mm^2 and α = 1.667× 10^{-6}/ºC

(a) 110 N/mm^2 (b) 140 N/mm^2
(c) 120 N/mm^2 (d) 150 N/mm^2
(e) 170 N/mm^2

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Thermal stresses in bars of tapering section



Given
length of bar (l) = 500 mm
dia of smaller end of bar (d_1) = 25 mm
dia of bigger end of bar (d_2) = 50 mm
change in temperate (Δt) = 25ºC
Co-efficient of thermal expansion (α) = 12 × 10^{-6}/ºC
Young's Modulus (E) = 2 × 10^5 N/mm^2

$$\sigma = E\alpha\Delta t\frac{d_2}{d_1}$$

$$= 2\times10^5\times12\times10^{-6}\times25\times\frac{50}{25} = 120 \text{ N/mm}^2$$

257. For a circular cross section beam is subjected to a shearing force F, the maximum shear stress induced will be (where d = diameter)

(a) F/πd^2 (b) 4F/πd^2
(c) 2F/πd^2 (d) F/4d^2

TSPSC AEE 2015

Ans. (b) : $\tau_{max} = \frac{\text{shear force}}{\text{shear area}}$

$$= \frac{F}{\left[\frac{\pi}{4}\right]d^2}$$

$$\tau_{max} = \frac{4F}{\pi d^2}$$

258. Two identical circular rods made of cast iron and mild steel are subjected to same magnitude of axial force. The stress developed is within proportional limit. Which of the following observation is correct ?

(a) Both roads elongate by same amount
(b) MS rod elongates more
(c) Cl rod elongates more
(d) Both stress and strain are equal in both roads

OPSC AEE 2015 Paper-I

Ans : (c) Two identical circular rods made of cast iron and mild steel are subjected to same magnitude of axial force. The stress developed is within proportional limit. CI rod elongates more, because in proportional limit cast iron elasticity is less than mild steel elasticity.

259. The change in length due to tensile or compressive force acting on a body is given by (with usual notations)

(a) δl = AE/ Pl (b) δl = Pl/AE
(c) δl = PE/Al (d) δl= P/AlE

TSPSC AEE 2015

Ans : (b)

Hooke's low:-
Stress ∝ strain

$$\sigma = E\varepsilon$$

$$\sigma = \frac{P}{A}$$

$$e = \frac{p}{AE}$$

$$\frac{\delta\ell}{\ell} = \frac{p}{AE}$$

$$\delta l = \frac{p\ell}{AE}$$

260. A steel bar 100 mm long is subjected to a tensile stress σ. If change in length of bar is 0.05 mm what is the value of σ?
E = 200 GPa

(a) 200 MPa (b) 100 MPa
(c) 80 MPa (d) 150 MPa

TANGEDCO AE 2015

Ans. (b) : $\sigma = \in E$

$$\boxed{\sigma = \frac{\delta L}{L}\cdot E}$$

$$\sigma = \frac{0.05\text{mm}}{100\text{mm}}\times 200\,\text{GPa}$$

$= 5 \times 10^{-4} \times 200 \times 10^9$ Pa
$= 1000 \times 10^5$ Pa
σ = 100 MPa

261.The value of modulus of elasticity for steel is

(a) 70 GPa (b) 100 GPa
(c) 125 GPa (d) 200 GPa

Assam PSC CCE Pre 2015

Ans. (d)

Material	Modulus of Elasticity (E)
• Steel	200 to 220 GPa
• Wrought iron	190 to 200 GPa
• Cast iron	100 to 160 GPa
• Copper	90 to 110 GPa
• Brass	80 to 90 GPa
• Aluminium	60 to 80 GPa

262.Bulk modulus is measured in terms of

(a) N/m (b) N/m^2
(c) N-m/s (d) N-s/m^2

Assam PSC CCE Pre 2015

Ans. (b) : We know that,

$$\text{Bulk Modulus (K)} = \frac{\text{Direct stress}(\sigma)}{\text{Volumetric strain}(\in_v)}$$

Bulk modulus is measured in terms of N/m^2.

263. Which of the following materials has the least elasticity?

(a) Mild steel (b) Aluminium
(c) Rubber (d) Zinc

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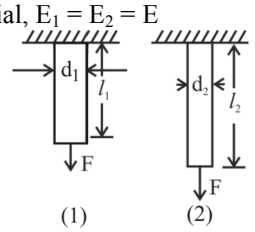
Ans. (c) : The word elastic means to regain its original position after load is removed, the highest load that can bear is steel and it is most elastic. While rubber has the least elasticity and rubber has least elastic.

264. Two wires A & B are of the same materials. Their lengths are in the ratio 1 : 2 and the diameters are in the ratio 2 : 1. If they are pulled by the same force, their increase in length will be in the ratio of :

(a) 2 : 1 (b) 1 : 4
(c) 1 : 8 (d) 8 : 1

KPSC ADF 2015

Ans. (c) :
Given data : $l_1 : l_2 = 1 : 2$
Diameter; $d_1 : d_2 = 2 : 1$
For same material, $E_1 = E_2 = E$



$\because \quad \frac{l_1}{l_2} = \frac{1}{2} \Rightarrow l_2 = 2l_1$

And $\quad \frac{d_1}{d_2} = \frac{2}{1} \Rightarrow d_1 = 2d_2$

$\therefore \quad \delta_1 = \frac{Fl_1}{A_1E}$ and $\delta_2 = \frac{Fl_2}{A_2E} \qquad \left[\because A = \frac{\pi}{4}d^2\right]$

$\therefore \quad \frac{\delta_1}{\delta_2} = \frac{Fl_1 \Big/ \frac{\pi}{4}d_1^2E}{Fl_2 \Big/ \frac{\pi}{4}d_2^2E} = \frac{l_1d_2^2}{l_2d_1^2}$

$\frac{\delta_1}{\delta_2} = \frac{l_1d_2^2}{2l_1 \times 4d_2^2} = \frac{1}{8}$

$\Rightarrow \quad \boxed{\delta_1 : \delta_2 = 1 : 8}$

265. Specific modulus of a composite is given by :
(or)
The specific stiffness is defined as a ratio of :

(a) Square root of Young's modulus/density
(b) Young's modulus/density
(c) Ultimate strength/density
(d) Square root of strength/density

APPSC Poly Lect. 13.03.2020
KPSC ADF 2015

Ans. (b) : Specific modulus is a materials property consisting of the elastic modulus per mass density. It is also known as the stiffness to weight ratio or specific stiffness.

Specific modulus $= \frac{\text{Young's Modulus}}{\text{Density}} = \frac{E}{\rho}$

266. A rigid body is very slowly dropped on another body and a deflection δ_{st} occurs in the second body. If the body be placed suddenly, the value of the impact factor will be :

(a) 0 (b) 1.0
(c) ∞ (d) 2.0

BPSC Asstt. Prof. 29.11.2015

Ans. (d) : $\sigma_{gradual} = \frac{P}{A}$, *and* *Load* $\frac{P+0}{Z}$

$\sigma_{Sudden} = \frac{2P}{A}$

$\sigma_{impact} = \frac{2P}{A}$

$= \frac{P}{A}\left[1 + \sqrt{1 + \frac{h \times AE}{Pl}}\right]$

267. Bulk modulus for a material is 200 GPa and the Poisson's ratio is 0.3. Young's modulus for that material will be

(a) 120 GPa (b) 160 GPa
(c) 210 GPa (d) 240 GPa

TSGENCO AE 14.11.2015

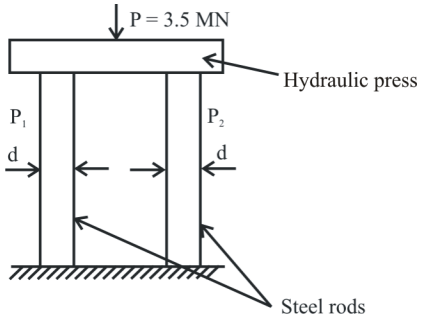
Ans. (d) : Bulk modulus (K) = 200 GPa
Poisson's ratio (μ) = 0.3
Young's modulus (E) = 3K (1–2μ)
= 3 × 200 (1–2 ×0.3)
= 600 (1–0.6)
= 600 × 0.4 = 240 GPa

268. A hydraulic press exerts a total load of 3.5 MN. This load is carried by two steel rods, supporting the upper head of the : press. The safe stress is 85 MPa and E = 210 kN/mm^2. The diameter of the rod will be

(a) 160 mm (b) 161 mm
(c) 162 mm (d) 165 mm

MPPSC AE 08.11.2015

Ans. (c) : Given data :



safe stress (σ) = 85 MPa
$= 85 \times 10^6$ N/m^2
Total load (P) = 3.5 MN = 3.5×10^6 N
Let, P_1 and P_2 load distributed equally in the steel rods

i.e., $P_1 = P_2 = \frac{P}{2} = \frac{3.5 \times 10^6 \text{ N}}{2} = 1.75 \times 10^6 \text{ N}$

Let, d = diameter of each steel rods

$\therefore \text{ stress}(\sigma) = \frac{\text{load}}{\text{cross} - \text{section area of steel rod}}$

$\Rightarrow \sigma = \frac{P_1}{\frac{\pi}{4}d^2} \Rightarrow 85 \times 10^6 \text{ N/m}^2 = \frac{1.75 \times 10^6 \text{ N}}{\pi/4\, d^2}$

$\Rightarrow d^2 = \frac{4 \times 1.75}{85\pi} m^2 = 0.0262$

$\Rightarrow d = 0.1619\text{m} \approx 0.162\text{m}$

$\Rightarrow d = 162$ mm

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269. If all the dimension of a prismatic bar be increased in the ratio of k:1, then maximum stress produced in the bar due to its own weight will increase in the following ratio

(a) 1:k (b) k^2:1
(c) k^3:1 (d) k:1

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Ans : (d) Max. stress is given as

$\sigma = \rho g x$

for max $x = \lambda$

$(\sigma_{max})_1 = \rho g \ell$

max. stress $(\sigma_{max})_2$ when $\lambda_2 = k\lambda$

$(\sigma_{max})_2 = \rho g k \ell$

$\frac{(\sigma_{max})_2}{(\sigma_{max})_1} = k : 1$

270. In a tensile testing experiment on a specimen of 1 cm^2 area, the maximum load observed was 5 tones and neck area 0.25 cm^2. The ultimate tensile strength of specimen is-

(a) 2.5 tones/cm^2 (b) 10 tones/cm^2
(c) 5 tones/cm^2 (d) 20 tones/cm^2

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Ans : (c) The ultimate tensile strength of specimen

$$\sigma = \frac{\text{max.load}}{\text{Initial Area}}$$

$$\sigma = \frac{5}{1}$$

σ =5 tones/cm^2

Always take initial area to calculate the value of stress except value of true stress.

271. For a given material assume, Young's modulus E= 300 GN/m^2 and modulus of rigidity G= 150 GN/m^2. Its bulk modulus K will be-

(a) 120 GN/m^2 (b) 100 GN/m^2
(c) 200 GN/m^2 (d) 250 GN/m^2

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Ans : (b) $E = \frac{9KG}{3K + G}$

or

$$K = \frac{EG}{3[3G - E]} = \frac{300 \times 10^9 \times 150 \times 10^9}{3[3 \times 150 - 300] \times 10^9}$$

$\boxed{K = 100 \text{ GN/m}^2}$

272. A steel wire hangs vertically under its own weight. If its density is 10000 kg/m^3 and allowable stress is 3000 kg/cm^2 then how much length it can sustain.

(a) 2500 m (b) 1250 m
(c) 3000 m (d) 5000 m

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c) Given, density (ρ) = 10000 kg/m^3 allowable stress = 3000 kg/cm^2

We know that stress in a member hangs vertically under its own weight

$$\sigma = \rho \times x$$

For length $x = \frac{3000 \times 10^4}{10000} = 3000$ m

273. The elongation of a Conical bar under its own weight is_____that of prismatic bar of the same length.

(a) Equal to (b) Half
(c) One-third (d) Two-third

TSPSC Managers, 2015

Ans. (c) : Elongation of conical bar under self weight is $\frac{1}{3}$ of the elongation of identical prismatic bar under its self weight

$$(\delta_L) = \frac{\gamma L^2}{6E} \Rightarrow \frac{1}{3} \times \frac{\gamma L^2}{2E} \qquad \left[\because \gamma = \text{Specific weight}, \ \gamma = \rho g\right]$$

Where, $\frac{\gamma L^2}{2E}$ = Prismatic elongation

274. The Poisson's ratio for cast iron varies from:

(a) 0.23 to 0.27 (b) 0.25 to 0.33
(c) 0.31 to 0.34 (d) 0.32 to 0.42

TSPSC Managers, 2015

Ans. (a) : The Poisson's ratio
For cast iron is 0.23 to 0.27
For steel is 0.25 to 0.33
For concrete is 0.2

275. The use of compound tubes subjected to internal pressure are made to :

(a) even out the stresses
(b) increases the thickness
(c) increases the diameter of the tube
(d) increase the strength

HPPSC AE 2014

Ans : (a) The use of compound tubes subjected to internal pressure are made to even out the stresses.
- So to obtain a more uniform hoop stress distributing compounding tubes are used.

276. When a bar is subjected to a change of temperature and its deformation is prevented, the stress induced in the bar is called

(a) tensile stress (b) compressive stress
(c) shear stress (d) thermal stress

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (d) : When the temperature of the bar is raised and the bar tries to expand and exerts axial pressure on wall. At the some time wall puts equal and opposite pressure on the bar which will develop compressive stress in bar. There is no mention of heating or cooling of bar. So thermal stress is correct option.

277. In a tensile test of a specimen, the ratio of maximum load to the original cross sectional area of the test piece is called

(a) ultimate stress (b) safe stress
(c) breaking stress (d) permissible stress

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : In a tensile test of a specimen, the ratio of maximum load to the original cross-sectional areas of the test piece is called ultimate stress.

278. The shear stress required to cause plastic deformation of solid metal is called

(a) proof stress (b) flow stress
(c) ultimate stress (d) none of these

Mizoram PSC AE/SDO 2014, Paper-II

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Ans. (b) : The shear stress required to cause plastic deformation of solid metal is called flow stress.

279. Consider the following statements :

Assertion (A) : An isotropic material is always homogeneous.

Reason (R) : An isotropic material is one in which all the properties are same in all the directions at every point.

of these statements,

(a) both (A) and (R) are true and (R) is the correct explanation of (A)

(b) both (A) and (R) are true but (R) is not a correct explanation of (A)

(c) (A) is true but (R) is false

(d) (A) is false but (R) is true

TNPSC AE 2014

Ans. (d) :

280. What would be the value of Young's modulus (Modulus of rigidity), for Poisson's ratio to be 0.4?

(a) 5/12 (b) 11/5

(c) 14/5 (d) 1/12

UPRVUNL AE 2014

Ans : (c) $E = 2G(1+\mu)$

$$E = 2G(1+0.4)$$

$$\frac{E}{G} = 2\times1.4 = 2.8 = \frac{28}{10} = \frac{14}{5}$$

281. A uniform rod of cross sectional area 2 mm^2 is heated from 0^0C to 40^0C. What would be the value of energy stored per unit volume if Young's modulus is 10^{11} N/m^2 and linear expansion coefficient is 12×10^{-6} per ^0C?

(a) 10250 J/m^3 (b) 12500 J/m^3

(c) 11520 J/m^3 (d) 55120 J/m^3

UPRVUNL AE 2014

Ans : (c) Given

$A = 2mm^2$ $\Delta t = 40^oC$, $E = 10^{11} N/m^2$, $\alpha = 12\times10^{-6}/^oC$

Thermal stress $((\sigma_{th}) = \alpha\,\Delta tE$

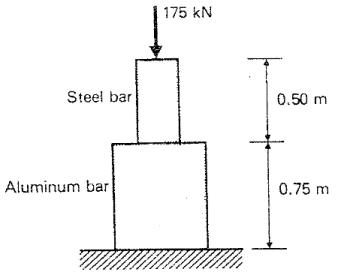
$$\sigma_{th} = 12\times10^{-6}\times40\times10^{11}$$

$$\sigma_{th} = 48\times10^6$$

Energy store per unit volume $(U) = \frac{(\sigma_{th})^2}{2E}$

$$= \frac{(48\times10^6)^2}{2\times10^{11}} = 11520\,J/m^3$$

282. A 0.75 meter aluminium bar 25×10^{-4} m^2 in cross-sectional area is attached to a 0.50 meter steel bar 15×10^{-4} m^2 in cross-sectional area, as shown in the figure. Take E (Young's modulus) value of 200 GPa for steel & 70 GPa for aluminium. Total shortening due to an axial compressive force of 175 kN is



(a) $\frac{157}{168}mm$ (b) $\frac{175}{168}mm$

(c) $\frac{175}{186}mm$ (d) $\frac{157}{186}mm$

ISRO Scientist/Engineer 24.05.2014

Ans. (b) : Length of Aluminium bar

$l_{Al} = 0.75$ m

$A_{Al} = 25 \times 10^{-4} m^2$

Length of steel bar

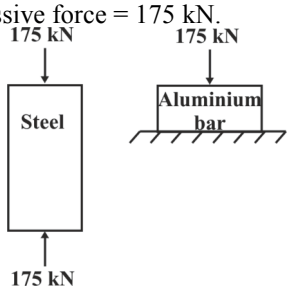
$l_{st} = 0.50$ m

$A_{st} = 15 \times 10^{-4} m^2$

$E_{steel} = 200$ GPa

$E_{Al} = 70$ GPa

Axial compressive force = 175 kN.



$$\delta_{st} = \frac{P_{st}L_{st}}{A_{st}E_{st}}$$

$$= \frac{175\times10^3\times0.5}{15\times10^{-4}\times200\times10^9}$$

$= 2.9167 \times 10^{-4}$ m (compression)

$$\delta_{Al} = \frac{P_{Al}L_{Al}}{A_{Al}E_{Al}}$$

$$= \frac{175\times10^3\times0.75}{25\times10^{-4}\times70\times10^9}$$

$= 7.5 \times 10^{-4}$ m (compression)

$\delta = \delta_{st} + \delta_{Al}$

$= (2.9167 + 7.5) \times 10^{-4}$ m

$= 10.4167 \times 10^{-4}$ m

$= 1.04167$ mm $= \frac{175}{168}$ mm

283. Member CD of the assembly shown is an aluminium bar 0.80 meter long. Cross-sectional dimension of the bar is such that, its width is three times its thickness. For functional design of the bar CD, the allowable axial stress is 70 MPa, and the total elongation is not to exceed 0.72 mm. E value for aluminium is 70 GPa. Assume the connection at C does not reduce the net area of the bar. The thickness of the bar in metre is

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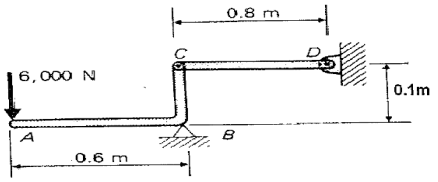

(a) $\frac{2}{\sqrt{21\times10^5}}m$ (b) $\frac{5}{\sqrt{21\times10^5}}m$

(c) $\frac{2}{\sqrt{21\times10^3}}m$ (d) $\frac{3}{\sqrt{21\times10^3}}m$

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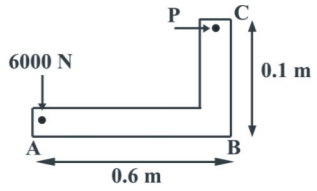
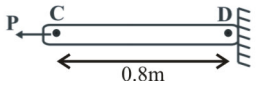
Ans. (c) :





Taking moment at B.

$\sum M_B = 0$

$6000 \times 0.6 = P \times 0.1$

$P = 36000$ N

$$\delta_{CD} = \frac{PL}{AE}$$

or $$A = \frac{PL}{E\,\delta_{CD}}$$

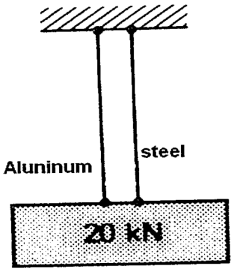
$$= \frac{36000\times0.8\times10^3}{70\times10^9\times0.72} = \frac{4}{7\times10^3}$$

$$3t^2 = \frac{4}{7\times10^3}$$

$$t^2 = \frac{4}{21\times10^3}$$

$$t = \frac{2}{\sqrt{21\times10^3}}m$$

284. A 20 kN weight is suspended by two wires as shown in the figure. The length of each wire is 2 meter. The steel wire (E value 200GPa) has an cross-sectional area of 60 × 10^{-6} m^2 and the aluminium wire (E value 70GPa) has an cross-sectional area of 120 × 10^{-6} m^2. The stress in aluminium wire is



(a) $\frac{17}{102}$ GPa (b) $\frac{7}{120}$ GPa

(c) $\frac{17}{120}$ GPa (d) $\frac{7}{102}$ GPa

ISRO Scientist/Engineer 24.05.2014

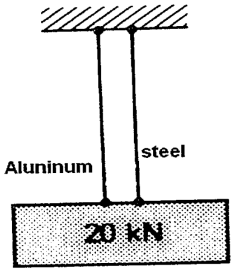
Ans. (d) : Given, P = 20 kN

L_{Al} = 2m, L_{st} = 2m

E_{Al} = 70 GPa, E_{st} = 200 GPa

$A_{Al} = 120 \times 10^{-6} m^2$

$A_{st} = 60 \times 10^{-6} m^2$



Its parallel connected-

So, $\delta_{Al} = \delta_{st}$

$$\frac{P_{AL}L_{AL}}{A_{AL}E_{AL}} = \frac{P_{st}.L_{st}}{A_{st}.E_{st}}$$

or $$\frac{P_{AL}}{P_{st}} = \frac{L_{st}}{L_{AL}}\times\frac{A_{AL}}{A_{st}}\times\frac{E_{AL}}{E_{st}}$$

$$= 1\times\frac{120\times10^{-6}}{60\times10^{-6}}\times\frac{70}{200}$$

$$= \frac{7}{10}$$

For in Parallel,

$$P_{AL} + P_{st} = P$$

$$P_{AL} + P_{st} = 20$$

or $$P_{AL} = \frac{140}{17}kN$$

$$\sigma_{AL} = \frac{P_{AL}}{A_{AL}} = \frac{140\times10^3}{17\times120\times10^{-6}}$$

$$= \frac{14}{12\times17}\times10^9\,N/m^2$$

$$\sigma_{AL} = \frac{7}{6\times17}GPa = \frac{7}{102}GPa$$

285.A bar of bronze and a bar of steel are placed between two immovable supports, A and B as shown in the figure. The physical constants are: (where L= length of bars, A = cross-sectional area, α = linear co-efficient of thermal expansion, E = Young's modulus)

Steel	Bronze
L = 0.5m	L=0.75m
A = 6 × 10^{-4} m^2	A = 12 × 10^{-4} m^2
α = 12 × 10^{-6}/ºC	α = 20 × 10^{-6}/ºC
E = 200 GPa	E = 100 GPa

The stresses in the steel & bronze, when temperature drops by 50ºC is

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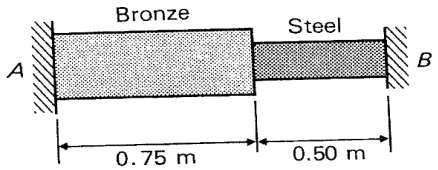

(a) Tension (b) Compression
(c) Shear (d) None of these

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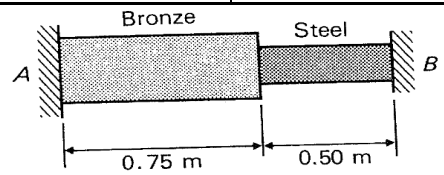
Ans. (a) :

Steel	Bronze
L = 0.5m	L = 0.75m
$A = 6 \times 10^{-4} m^2$	$A = 12 \times 10^{-4} m^2$
$\alpha = 12 \times 10^{-6}/^oC$	$\alpha = 20 \times 10^{-6}/^oC$
E = 200 GPa	E = 100 GPa

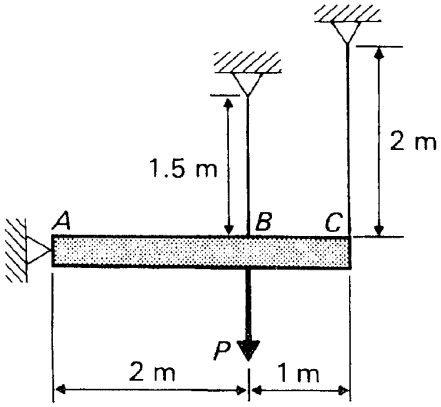


In series,

$$\Delta = \Delta_b + \Delta_s$$
$$= \alpha_b L_b \Delta t + \alpha_s L_s \Delta t$$
$$= 20 \times 10^{-6} \times 0.75 \times 50 + 12 \times 10^{-6} \times 0.5 \times 50$$
$$= 1.05\text{m (shortening)} < (0.75 + 0.5 = 1.25)$$

Note:-Due to rigid support, the shortening is prevented and tensile stresses are developed in each bars.
As temperatures the bar tends to gets shortens, to present shortening tensile stress will develop.

286. A rigid bar with wires at B & C is shown in the figure below. The cross sectional area of the wire at B is 3×10^{-4} m^2 at C is 2×10^{-4} m^2. The wires are elasto-plastic with strength 250 MPa and E value 200 GPa. The ultimate load P that can be applied to the rigid bar, as shown is



(a) 50 kN (b) 75 kN
(c) 100 kN (d) 150 kN

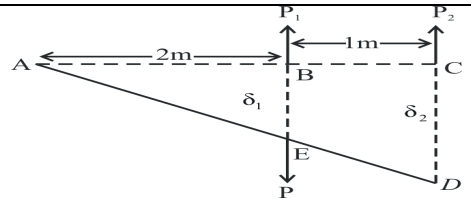
ISRO Scientist/Engineer 24.05.2014

Ans. (d) : Given, Cross-section Area of wire 'B' = $3 \times 10^{-4} m^2$

Cross-section Area of wire 'C' = $2 \times 10^{-4} m^2$

$\sigma_{ultimate} = 250\,MPa$

$E = 200\,GPa$



By Geometry,
In ΔABE and ΔACD

$$\frac{\delta_1}{2} = \frac{\delta_2}{3}$$

OR $3\delta_1 = 2\delta_2$

$$\frac{\delta_1}{\delta_2} = \frac{2}{3} \qquad \left[\because \delta_1 = \frac{P_1 L_1}{A_1 E_1}, \quad \delta_1 = \frac{P_2 L_2}{A_2 E_2}\right]$$

$$\frac{\frac{P_1 L_1}{A_1 E_1}}{\frac{P_2 L_2}{A_2 E_2}} = \frac{2}{3}$$

$$\frac{P_1}{P_2} = \frac{2}{3} \times \left(\frac{A_1}{A_2}\right)\left(\frac{E_1}{E_2}\right)\left(\frac{L_2}{L_1}\right)$$

$$\frac{P_1}{P_2} = \frac{2}{3} \times \left(\frac{3 \times 10^{-4}}{2 \times 10^{-4}}\right) \times \left(\frac{200}{200}\right) \times \left(\frac{2}{1.5}\right)$$

$$\frac{P_1}{P_2} = \frac{2}{3} \times \frac{3}{2} \times \frac{2}{1.5} = \frac{4}{3} \text{ ----------(1)}$$

Taking a moment about A. $\sum M_A = 0$

$P \times 2 = P_1 \times 2 + P_2 \times 3$
Or $2P_1 + 3P_2 = 2P$ ----------(2)
P_1 (Based on strength criterion)
$= \sigma_{ultimate} \times A_1$
$= 250 \times 10^6 \times 3 \times 10^{-4}$
$P_1 = 75$ kN
P_2 (Based on strength criterion)
$= \sigma_{ultimate} \times A_2$
$= 250 \times 10^6 \times 2 \times 10^{-4}$
$= 500 \times 10^2$
$P_2 = 50000 = 50$ kN
According to Equation (1) -

$$\frac{P_1}{P_2} = \frac{4}{3}$$

$$P_1 = \frac{4}{3} \times P_2 = \frac{4}{3} \times 50$$

= 66.67 kN
From Equation (2)-
$2P = 2P_1 + 3P_2$
$2P = 2 \times 66.67 + 3 \times 50$
$= 283.44$
P = 141.67 kN
The closest answer is option (d) 150 kN.

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287. Stress and strain are linearly related by Hooke's law in the ____ region.
(a) plastic
(b) elastic
(c) both elastic and plastic
(d) Neither elastic nor plastic

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Stress and strain are linearly related by Hooke's law in the elastic region, but Hooke's law holds good upto proportionality limit only.

288. The thermal stress in terms of coefficient of linear expansion (α), rise in temperature (ΔT) and the modulus of elasticity (E) is given by
(a) ΔT (b) $1/(E\alpha\Delta T)$
(c) $E\Delta T/\alpha$ (d) $E\alpha\Delta T$

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
APPSC Poly. Lect. 2013

Ans. (d) : Thermal stress (σ_T) = $E\alpha\Delta T$
Thermal strain (ε_T) = $\alpha\Delta T$

289. Strain energy is the
(a) energy stored in a body when strained within elastic limits
(b) energy stored in a body when strained upto the breaking of a specimen
(c) maximum strain energy which can be stored in a body
(d) proof resilience per unit volume of a material

Haryana PSC Civil Services Pre, 2014

Ans. (a) : Strain energy– When a body is subjected to external load it undergoes some deformation and stores some amount of energy. This stored energy will release when load is removed is known as strain energy.
Energy stored by body when, it is strained within elastic limit is known as strain energy.

290. For a given material, Young's modulus is 1×10^6 N/m^2 and modulus of rigidity is 0.4×10^6 N/m^2 The bulk modulus of elasticity will be.
(a) 5×10^6 N/m^2 (b) 0.67×10^6 N/m^2
(c) 67×10^6 N/m^2 (d) 6.7×10^6 N/m^2

MPPSC State Forest Service Exam, 2014

Ans. (b) : As we know,
$G = 0.4 \times 10^6$ N/m^2

$$E = \frac{9 \times KG}{3K + G}$$

$E = 1 \times 10^6$ N/m^2

$$1 \times 10^6 = \frac{9 \times K \times 0.4 \times 10^6}{3 \times K + 0.4 \times 10^6}$$

$$\boxed{K = 0.67 \times 10^6 \, N/m^2}$$

291. The no. of elastic constants in Triclinic material is:
(a) 36 (b) 9
(c) 2 (d) 21

HPPSC Asstt. Prof. 2014

Ans. (d) :

No. of elastic constant	Material
Isotropic	2
Anisotropic (Triclinic)	21
Orthotropic	9

292. The Poisson's ratio (μ) of bottle cork is:
(a) 0.5 (b) 0.3
(c) 0.25 (d) 0

HPPSC Asstt. Prof. 2014

Ans. (d) :

Material	Poisson's ratio
Cork	0
Rubber	0.5
Mild steel	0.3
Cast Iron	0.25

293. If D is the diameter of a sphere, then volumetric strain is equal to
(a) 2 times the strain of D
(b) 1.5 times the strain of D
(c) 3 times the strain of D
(d) strain of D

APPSC Poly. Lect. 2013

Ans. (c) : If D is the diameter of a sphere, then the volumetric strain is equal to three times of strain of D.

$$\text{Volumetric strain} = \frac{\delta V}{V} = \frac{3\delta D}{D} \qquad V = \frac{4}{3}\pi r^3$$

$$V = \frac{\pi}{6} D^3 \ldots\ldots(1) \quad r = \frac{D}{2}$$

$$\epsilon_v = \frac{\delta V}{V} = \frac{3\delta D}{D}$$

294. For a material with the Poisson's ratio ν, the modulus of elasticity (E) and the bulk modulus of elasticity (K) are same. Which of the following is correct?
(a) The material has $\nu = 0$
(b) The material has $\nu = 1/2$
(c) The material has $\nu = 1/3$
(d) The material has $\nu = 3/4$

RPSC AEN Pre-2013

Ans. (c) : Given, Poisson's ratio = ν
Modulus of elasticity = E
Bulk modulus = K
E = K (given)
According to,

$$E = 3K(1 - 2\nu)$$

$$E = 3E(1 - 2\nu)$$

$$1 - 2\nu = \frac{1}{3}$$

$$1 - \frac{1}{3} = 2\nu$$

$$2\nu = \frac{2}{3}$$

$$\boxed{\nu = \frac{1}{3}}$$

295. A solid cube faces similar equal normal force on all faces. Ratio of volumetric strain to linear strain on any of three axes will be:
(a) 1 (b) 2
(c) 3 (d) $\sqrt{3}$

SJVN ET 2013

Ans. (c) :

$$\epsilon_v = \frac{\delta V}{V} = \frac{3\sigma}{E}(1 - 2\mu)$$

$$\epsilon_v = 3\epsilon\,(1 - 2\mu) \qquad \left[\because \epsilon = \frac{\sigma}{E}\right]$$

• It is 3 times because cube is subjected to 3 mutually perpendicular stress.

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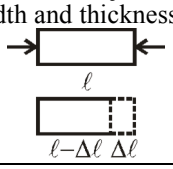
296. When a bar is subjected to a push of P, its
(a) length, width and thickness increase
(b) length, width and thickness decrease
(c) length increases, width and thickness decrease
(d) length decreases, width and thickness increase

SJVN ET 2013

Ans. (d) : When a bar is subjected to a push of P, its length decreases width and thickness increases.



297. In a tensile test, when a material is stressed beyond elastic limit, the tensile strain ___ as compared to the stress.
(a) decreases slowly
(b) increases slowly
(c) decreases more quickly
(d) increases more quickly

JPSC AE 2013, Paper-V

Ans : (d) : In a tensile test, when a material is stressed beyond elastic limit the tensile strain increases more quickly as compared to the stress.

298. A uniform taper rod of diameter 30 mm to 15 mm, length of 314 mm is subjected to 4500 N. The Young's modulus of the material is 2×10^5 N/mm^2. Extension of the bar is
(a) 0.05 mm (b) 0.5 mm
(c) 0.25 mm (d) 0.005 mm

TNPSC AE 2013

Ans. (*) : Data given-
D = 30 mm d = 15 mm
L = 314 mm P = 4500 N
$E = 2 \times 10^5$ N/mm^2
We know that extension of the taper bar.

$$\delta = \frac{4\,P.L}{\pi\, Dd.E}$$

$$\delta = \frac{4\times4500\times314}{3.14\times30\times15\times2\times10^5}$$

$$\delta = 0.02\text{mm}$$

299. A load of 4000 N has to be placed at the end of a vertical rod with stress induced is 80 N/mm^2. The diameter of the rod is :
(a) 79.79 mm (b) 7.979 mm
(c) 18.181 mm (d) 81.81 mm

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Load, P = 4000 N
Stress, $\sigma = 80$ N/mm^2

$$\sigma = \frac{P}{A} = \frac{4P}{\pi d^2}$$

$$80 = \frac{4\times4000}{\pi d^2} \Rightarrow d = 7.979\,\text{mm}$$

300. A steel rod of 1 sq. cm cross-sectional area is 1 m long and has a Young's modulus of elasticity 2×10^6 kg/cm^2. It is subjected to an axial pull of 2000 kgf. The elongation of the rod will be
(a) 0.1 cm (b) 0.05 cm
(c) 0.20 cm (d) 0.15 cm

APPSC AEE 2012

Ans. (a) : Given, A = 1 cm^2, L = 1m
E = 2×106 kg/cm^2, P = 2000 kgf

$$\text{Elongation } \delta = \frac{PL}{AE} = \frac{2000\times1}{1\times2\times10^6} = 10^{-3}$$

$\delta = 1\text{mm} = 0.1\text{cm}$

301. If a material has a modulus of elasticity of 2×10^6 kg/cm^2 and a modulus of rigidity of 0.8×10^6 kg/cm^2, then the approximate value of the Poisson's ratio of the material will be
(a) 0.26 (b) 0.31
(c) 0.45 (d) 0.2

APPSC AEE 2012

Ans. (a) : $E = 2G(1 + \mu)$

$$2\times10^6 = 2\times0.8\times10^6\,(1+\mu)$$

$$(1+\mu) = \frac{\left(2\times10^6\right)}{\left(2\times0.8\times10^6\right)} = 1.25$$

$\mu = 0.25$

302. The ratio of the deformation of a bar due to its own weight, to the deformation due to axial load equal to its weight, is :
(a) 1 (b) 1/2
(c) 2 (d) 4

APPSC AE 04.12.2012

Ans. (b) : We know that -

Deformation of a bar due to its own weight $(\Delta_1) = \frac{WL}{2AE}$

Deformation due to axial load equal to its weight $(\Delta_2) = \frac{WL}{2AE}$

$$\therefore \frac{\Delta_1}{\Delta_2} = \frac{WL/2AE}{WL/AE} = \frac{1}{2}$$

303. To measure _____ strain, strain rosettes are used :
(a) Linear (b) Shear
(c) Volumetric (d) Any of the above

APPSC AE 04.12.2012

Ans. (a) : • Strain rosettes method is used to measure Linear strain

304. A localised compressive stress at the area of contact between two members is known as :
(a) Shear (b) Crushing
(c) Bending (d) Tensile

APPSC AE 04.12.2012

Ans. (b) : A localised compressive stress at the area of contact between two member is known as crushing.

305. A square section with side 'X' of a beam is subjected to a shear force 'S'. The magnitude of shear stress at the top edge of the square is :
(a) $\frac{1.5S}{X^2}$ (b) $\frac{S}{X^2}$
(c) $\frac{0.5S}{X^2}$ (d) Zero

APPSC AE 04.12.2012

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Ans. (d) : At the top edge of the square, the magnitude of the shear stress should be zero. Because shear stress is a surface phenomena.

306. Stress-strain analysis is conducted to know which of the following properties of material?
(a) Physical properties
(b) Optical properties
(c) Mechanical properties
(d) Magnetic properties

BPSC AE 2012 Paper - VI

Ans : (c) : Mechanical properties.

307. The ratio between tensile stress and tensile strain or compressive stress and compressive strain is termed as
(a) modulus of rigidity
(b) modulus of elasticity
(c) bulk modulus
(d) modulus of subgrade reaction

APPSC AEE 2012

Ans : (b) According to Hooke's law
stress $\propto$ strain (upto proportionality limit)
$\sigma \propto \varepsilon$

$$\frac{\sigma}{\varepsilon} = E$$

E = Modulus of Elasticity.
where stress and strain both are tensile or compressive nature.

308. A prismatic bar has
(a) maximum ultimate strength
(b) maximum yield strength
(c) varying cross-section
(d) uniform cross-section

BPSC AE 2012 Paper - VI

Ans : (d) : A prismatic bar or beam is a straight structural piece that has the same cross section through its length.

309. The failure criterion for ductile materials is based on
(a) yield strength (b) ultimate strength
(c) shear strength (d) limit of proportionality

BPSC AE 2012 Paper - VI

Ans : (a) : The failure criterion for ductile materials is based on yield strength.

310. The elongation of a conical bar due to its self weight is
(a) $\frac{\gamma \ell^2}{6E}$ (b) $\frac{\gamma \ell^2}{2E}$
(c) $\frac{\gamma \ell^2}{2E}$ (d) $\frac{\gamma \ell^2}{E}$

Where γ = unit weight of the material.

APPSC AEE 2012

Ans : (a) Elongation of conical bar due to self weight.

$$\Delta = \frac{\gamma \ell^2}{6E} = \frac{1}{3} \times \text{Deflection of prismatic bar}$$

γ = Specific weight
l = length of bar
E = Modulus of Elasticity.

311. Two bars A and B are of equal length but B has an area half that of A and bar A has young's modulus double that of B. When a load 'P' is applied to the two bars, the ratio of deformation between A and B is
(a) $\frac{1}{2}$ (b) 1
(c) 2 (d) $\frac{1}{4}$

APPSC AEE 2012

Ans : (d)

Bar 'A'	Bar 'B'
$\ell_1 = \ell$	$\ell_2 = \ell$
$A_1 = A$	$A_2 = A/2$
$E_1 = 2E$	$E_2 = E$
$P_1 = P$	$P_2 = P$
$\delta\ell_1 = \frac{P_1\ell_1}{A_1E_1}$	$\delta\ell_2 = \frac{P_2\ell_2}{A_2E_2}$
$\delta\ell_1 = \frac{P\ell}{2AE}$	$\delta\ell_2 = \frac{2P\ell}{AE}$
$\frac{\delta\ell_1}{\delta\ell_2} = \frac{1}{4}$	

312. The elongation of beam of length 'l' and cross-sectional area 'A' subjected to a load 'P' is δl. If the modulus of elasticity is halved, the new elongation will be
(a) $\frac{\delta\ell}{2}$ (b) $2(\delta\ell)$
(c) $\delta\ell$ (d) $\sqrt{2}\delta\ell$

APPSC AEE 2012

Ans : (b) Case 1st :-
Elongation of beam

$$\delta\ell_1 = \frac{P\ell}{AE}$$

Case 2nd :-
Modulus of elasticity is halved $E' = \frac{E}{2}$

$$\delta\ell_2 = \frac{2P\ell}{AE}$$

$$\boxed{\delta\ell_2 = 2 \times \delta\ell_1}$$

313. A 16mm diameter central hole is bored out of a steel rod of 40mm diameter and length 1.6m. The tensile strength because of this operation
(a) increases (b) remains constant
(c) decreases (d) None of these

APPSC AEE 2012

Ans : (c) A 16mm diameter central hole is bored out of a steel rod of 40mm diameter and length 1.6m. The tensile strength because of this operation decreases.

314. The relationship between Young's modulus and shear modulus when $\frac{1}{m} = 0$, is
(a) E = 2G
(b) E = 3G
(c) E = 2G+1
(d) G = 2E

APPSC AEE 2012

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Ans : (a) We know

$$E = 2G\left(1+\frac{1}{m}\right) \quad .....(i)$$

$$E = 3K\left(1-\frac{2}{m}\right) \quad .....(ii)$$

If $\frac{1}{m} = 0$ then

$E = 2G$

$E = 3K$

315. If a rigidly connected bar of steel and copper is heated, the copper bar will be subjected to

(a) compression (b) shear

(c) tension (d) None of these

APPSC AEE 2012

Ans : (a) If a rigidly connected bar of steel and copper is heated, the copper bar will be subjected to compression.

• Because $\alpha_{copper} > \alpha_{steel}$.

316. Which of the following has no unit ?

(a) Kinematic viscosity

(b) Strain

(c) Surface Tension

(d) Bulk Modulus

UKPSC AE 2012 Paper-I

Ans. (b) : Strain = $\frac{\delta L}{L}$

317. In a static tension tests of a low carbon steel sample, the gauge length affects

(a) yield stress

(b) ultimate tensile stress

(c) percentage elongation

(d) percentage reduction in cross-sectional area

UKPSC AE 2012 Paper-I

Ans. (c) : In a static tension test of a law carbon steel sample, the gauge length affects the percentage elongation.

318. One end of a metallic rod is fixed rigidly and its temperature is raised. It will experience

(a) zero stress

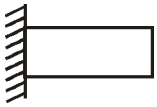
(b) tensile stress

(c) compressive stress

(d) None of the above

UKPSC AE 2012 Paper-I

Ans. (a) : zero stress



319. A metallic cube is subjected to equal pressure (P) on its all the six faces. If ϵ_v is volumetric strain produced, the ratio $\frac{P}{\epsilon_v}$ is called

(a) Elastic modulus

(b) Shear modulus

(c) Bulk modulus

(d) Strain-Energy per unit volume

UKPSC AE 2012 Paper-I

Ans. (c) : Bulk modulus

320. Select the proper sequence for the following :

1. Proportional limit
2. Elastic limit
3. Yield point
4. Fracture/failure point

(a) 1-2-3-4 (b) 2-1-3-4

(c) 1-2-4-3 (d) 2-1-4-3

UKPSC AE 2012 Paper-I

Ans. (a) : 1-2-3-4

321. A test specimen is stressed slightly beyond yield point then unloaded. Its yield strength will

(a) decreases

(b) increase

(c) remain same

(d) become equal to ultimate tensile strength

TRB Poly. Lect., 2012

Ans. (b) : A test specimen is stressed slightly beyond the yield point and then unloaded its yield strength will increase because the specimen is strain hardened due to plastic deformation.

322. A mild steel bar 1 m long, 100 mm in diameter is subjected to an axial tensile load, so that change in its length is 0.1 mm. If Poisson's ratio of mild steel is 0.3, what is the change in its diameter :

(a) 1 micron (b) 0.3 micron

(c) 0.03 micron (d) None of these

PSPCL AE, 2012

Ans. (d) : Poisson ratio 'μ' = $\frac{\frac{\delta d}{d}}{\frac{\delta l}{l}}$ or $0.3 = \frac{\delta d/100}{0.1/1000}$

or $\delta d = \frac{0.3\times100\times0.1}{1000}$

$= 3 \times 10^{-3}$ = 3 micron

323. A steel cube of volume 8×10^6 mm^3 is subjected to all round stress of 135 N/mm^2. The bulk modulus of the material is 1.35×10^5 N/mm^2. The volumetric change is :

(a) 8000 mm^3 (b) 800 mm^3

(c) 80 mm^3 (d) 8 mm^3

APGENCO AE 2012

Ans. (a) : Given, Volume (V) = $8 \times 10^6 mm^3$

$\sigma = 135$ N/mm^2

Bulk modulus of material (K) = 1.35×10^5 N/mm^2

$$\text{Bulk modulus (K)} = \frac{\sigma}{\frac{dV}{V}}$$

$$dV = \frac{\sigma V}{K}$$

$$dV = \frac{135\times8\times10^6}{1.35\times10^5}$$

$dV = 8000 mm^3$

324. A steel bar 2 m length is fixed at both ends at 20°C. The coefficient of thermal expansion is 12×10^{-6}/°C. The modulus of elasticity is $2 \times 10^5 N/mm^2$ If the temperature is reduced to 18°C, the bar will experience a stress of :

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(a) 2.4 MPa (compressive)
(b) 2.4 MPa (tensile)
(c) 4.8 MPa (tensile)
(d) 4.8 MPa (compressive)

APGENCO AE 2012

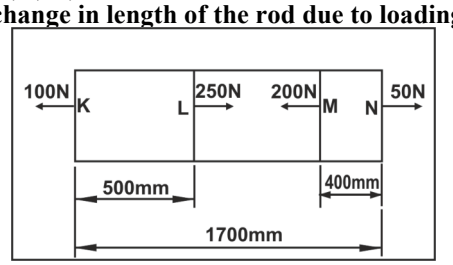
Ans. (c) : Given, length (l) = 2m
Temperature reduce (ΔT) ↓ from 20°C to 18°C
So, (ΔT)↓ – It will tensile stress
$\alpha = 12 \times 10^{-6}$ /°C, $E = 2 \times 10^5$ N/mm^2
Thermal stress (σ_T) = $\alpha \Delta T E$
$= 12 \times 10^{-6} \times (20-18) \times 2 \times 10^5$
$= 12 \times 10^{-6} \times 2 \times 2 \times 10^5$
$= \frac{48 \times 1}{10} = 4.8$ MPa (Tensile)

325. A bar of 40mm diameter is subjected to an axial load of 4kN. The extension of the bar over a gauge length of 200 mm is 0.3 mm. The decrease in diameter is 0.018mm. The Poisson's ratio is :
(a) 0.33 (b) 0.35
(c) 0.30 (d) 0.25

APGENCO AE 2012

Ans. (c) : Given, $\frac{\delta d}{d} = \frac{0.018}{40}$
$= 4.5 \times 10^{-4}$
$\frac{\delta l}{l} = \frac{0.3}{200} = 0.0015 = 15 \times 10^{-4}$
$\because$ Poisson's ratio $(\mu) = \frac{\frac{\delta d}{d}}{\frac{\delta l}{l}} \frac{4.5 \times 10^{-4}}{15 \times 10^{-4}} = 0.30$

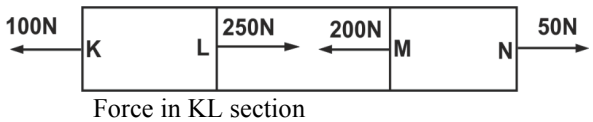
326. The figure below shows a steel rod of 25 mm^2 cross sectional area. It is loaded at four points, K,L,M,N. Assume E steel = 200 GPa. The total change in length of the rod due to loading is



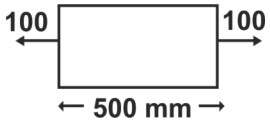
(a) 1μm (b) -10μm
(c) 10μm (d) -20μm

ISRO Scientist/Engineer 2012
GATE-2004

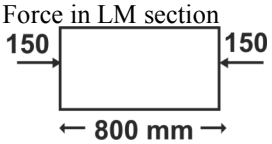
Ans. (b) : Given-
Cross-section Area of K,L,M,N section each is 25mm^2
Assume E_{steel} = 200 GPa.



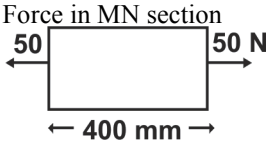
Force in KL section



Force in LM section



Force in MN section



Total change in length $= \sum \frac{PL}{AE}$
"Its applied when cross-section area of each section and 'E' is same."
$= \frac{100 \times 500 - 150 \times 800 + 50 \times 400}{25 \times 200 \times 10^3}$
$= \frac{-50000}{25 \times 200 \times 10^3}$
$= -10$ μm

327. A circular rod of 100 mm diameter and 500 mm length is subjected to a tensile force of 1000 kN. Determine the modulus of rigidity (G) if E = 2 × 10^5 N/mm^2 and Poisson's = 0.3
(a) 0.335 ×10^5 N/mm^2
(b) 0.521 ×10^5 N/mm^2
(c) 0.7692 ×10^5 N/mm^2
(d) 0.2256 ×10^5 N/mm^2

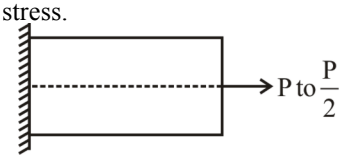
ISRO Scientist/Engineer 2011

Ans. (c) : $E = 2 \times 10^5$ N/mm^2
Poisson's ratio $\mu = 0.3$
$E = 2G(1 + \mu)$
$2 \times 10^5 = 2G(1 + 0.3)$
$G = \frac{1 \times 10^5}{1.3}$
$G = 0.7692 \times 10^5$ N/mm^2

328. What is the stress which varies from a minimum value to a maximum value of the same nature called as ?
(a) Repeated stress (b) Yield stress
(c) Fluctuating stress (d) Alternating stress

VIZAG STEEL MT 2011

Ans. (c) : The stress which varies from a minimum value to a maximum value of the same nature called as fluctuating stress.



Fluctuating fatigue load
(i.e. only magnitude of load change)

329. The highest stress that a material can withstand for a specific length of time without excessive deformation is called :
(a) Fatigue strength (b) Endurance strength
(c) Creep rupture (d) Creep strength

J&K PSC Civil Services Pre, 2010

Ans. (d) : Creep Strength – Creep strength is the maximum stress that the material can withstand for a specific length of time without excessive deformation.
Creep Rupture Strength– Creep rupture strength is the maximum stress that the material can withstand for a specific length of time without rupture.

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330. A cast iron specimen when tested in compression just fails at a compressive stress of 320 N/mm^2. The shear strength of cast iron is :

(a) 320 N/mm^2 (b) 160 N/mm^2
(c) 140 N/mm^2 (d) 80 N/mm^2

UPSC JWM Advt. No.-52/2010

Ans. (b) : Solution : Shear strength (τ)

$$= \frac{\text{Compressive Strength of cast Iron}}{2}$$

$$= \frac{(\sigma_c)_{ult}}{2}$$

$$= \frac{320}{2} = 160 N/mm^2$$

331. A bar of length L, area A and Young's modulus E is subjected to a pulling force P. The strain energy stored in the bar is

(a) $\frac{PL^2}{AE}$ (b) $\frac{PL^2}{2EI}$
(c) $\frac{P^2L}{2AE}$ (d) $\frac{P^2L}{AE}$

ISRO Scientist/Engineer 2009

Ans. (c) : Given,
Length = L
Area = A
Young modulus = E
Applied force = P

$$\text{Strain energy} = \frac{1}{2} \times P \times \delta$$

$$= \frac{1}{2} \times P \times \frac{PL}{AE} \quad \left[\because \delta = \frac{PL}{AE}\right]$$

$$= \frac{P^2L}{2AE}$$

332. A bar of 20 mm dia is tested in tension. It is observed that when a load of 38 kN is applied the extension measured over a gauge length of 200 mm is 0.12 mm and contraction in diameter is 0.0036 mm. Find the Poisson's ratio

(a) 0.2 (b) 0.3
(c) 0.25 (d) 0.33

ISRO Scientist/Engineer 2009

Ans. (b) : Given, $\Delta l = 0.12$ mm
$l = 200$ mm
$\Delta d = 0.0036$, d = 20 mm
Poisson's ratio

$$= \left(\frac{\Delta d}{d}\right) / \left(\frac{\Delta l}{l}\right) = \frac{0.0036/20}{0.12/200}$$

$$= 0.3$$

333. A steel cube of 1 m × 1 m × 1 m is subjected to hydorstatic pressure of 1 MPa. If for steel Young's modulus is 200 GPa and Poisson's ratio is 0.3, the approximate change in volume of the cube due to the hydrostatic pressure will be

(a) -6×10^{-6} m^3 (b) -2×10^{-6} m^3
(c) -1×10^{-6} m^3 (d) -0.5×10^{-6} m^3

DRDO Scientists 2009

Ans. (a) : E = 200 GPa $\quad v \cong 0.3$, Volume V = 1 m^3

$$K = \frac{E}{3(1-2v)}$$

$$= \frac{200}{3(1-0.6)} = 166.66 \text{ GPa}$$

$$K = -\frac{PV}{\Delta V}$$

$$166.66 \times 10^9 = \frac{1}{\Delta V}$$

$$\Delta V = -6 \times 10^{-6} \text{ m}^3$$

334. A steel rod of length 300 mm is held between two fixed supports so that the rod cannot elongate or contract in the axial direction. If the temperature of the rod is raised by 20°C, the axial stress induced in the rod due this rise in temperature is (for steel, E = 200 GPa and α = 11.5 × 10^{-6}/°C

(a) 46 MPa tension
(b) 46 MPa compression
(c) 23 MPa tension
(d) 23 MPa compression

DRDO Scientists 2009

Ans. (b) : $\Delta L = (\alpha \Delta T) L$

$$\frac{\Delta L}{L} = \alpha \Delta T$$

$$E = \frac{\sigma}{\alpha \Delta T}$$

$$\sigma_{th} = E \alpha \Delta T$$

$$= 200 \times 10^3 \times 11.5 \times 10^{-6} \times 20$$

= 46 MPa compression

335. Consider the following statements :
State of stress at a point when completely specified enables one to determine the :
(I) principal stresses at the point
(II) maximum shearing stress at the point
(III) stress components on any plane containing the point
Of these statements :
(a) I, II and III are correct
(b) I and III are correct
(c) II and III are correct
(d) I and II are correct

APPSC IOF, 2009, ESE-1996

Ans. (a) : The state of stress at a point when completely specified enables one to determine, principal stresses at the point, maximum shearing stress at the point and stress components on any arbitrary plane containing that point.

336. Proof resilience is the greatest energy at

(a) limit of proportionally (b)elastic limit
(c) plastic limit (d) none of these

TNPSC AE, 2008

Ans. (b) : Proof resilience is the maximum energy stored at elastic limit.

337. The ultimate tensile strength and yield strength of most of the metals, when temperature falls from 0 to -150°C will

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(a) Increase
(b) Decrease
(c) Remain same
(d) First increase and then decrease

ISRO Scientist/Engineer 2008

Ans. (a) : **Ultimate tensile stress**—It is the maximum nominal stress attained by a test specimen during a simple tensile test the ultimate strength of the materials increase with decrease in temperature.
Stainless steel has high strength and mostly preferred at Cryogenic applications.
Yield stress—It is the stress at which the strain of a material shows a rapid increase in stress, when subjected to a simple tensile test.

338. At a point in a body the normal stresses are $\sigma_x = \sigma$ and $\sigma_y = \sigma$. E is the Young's modulus and v is the Poisson's ratio of the material of the body. Assuming the material to be linearly elastic and isotropic, for plane stress condition the ratio of σ_x to $\square_x$ is

(a) $\frac{E}{1-v}$ (b) E
(c) $\frac{E}{v}$ (d) $\frac{E}{1+v}$

DRDO Scientists 2008

Ans. (a) : $\epsilon_x = \frac{\sigma_x - \nu(\sigma_y)}{E}$

$\epsilon_x = \frac{(1-\nu)\,\sigma_x}{E}$

$\frac{\sigma_x}{\epsilon_x} = \frac{E}{1-v}$

339. The linear relation between the stress and strain of a material is valid until

(a) fracture stress (b) elastic limit
(c) ultimate stress (d) proportional limit

DRDO Scientists 2008

Ans. (d) : The linear relation between the stress and strain of a material is valid until proportional limit.

340. A material has a Poisson's ratio of 0.5. If a body is made of this material and subjected to external forces (within the elastic limit) then the final volume of the body is

(a) thrice that of the initial volume of the body
(b) twice that of the initial volume of the body
(c) zero
(d) equal to the initial volume of the body

DRDO Scientists 2008

Ans. (d) : Bulk hardness of rigidity

$E = 3K(1 - 2\mu)$

$K \to \infty$ for $\mu = 0.5$ where μ is Poisson's ratio

It means material is rigid so no volume change occurs.

341. When a uniaxial tensile load is applied to a rod fixed at one end.

(a) both engineering stress and true stress in the rod depend on the type of material of the rod
(b) engineering stress in the rod depends on the type of material of the rod, but true stress in the rod does not depend on the type of material of the rod
(c) true stress in the rod depends on the type of material of the rod, but engineering stress in the rod does not depend on the type of material of the rod
(d) both engineering stress and true stress in the rod does not depend on the type of material of the rod

DRDO Scientists 2008

Ans. (c) : $\sigma_E = \frac{P}{A}$ ← Independent of material property

Engineering stress σ_E, True stress σ_T,
Engineering strain ϵ_E

$\left.\begin{aligned}\sigma &= \sigma_E(1+\epsilon_E)\\ \epsilon_E &= \frac{\sigma_E}{E}\end{aligned}\right\}$ ← Dependent on material property

342. The extension of a mild steel bar 4m long, 2000 mm^2 cross section under the action of an axial load of 20kN, if $E = 2\times10^5$ N/mm^2, is

(a) 2mm (b) 0.2mm
(c) 0.5mm (d) 0.05mm

ISRO Scientist/Engineer 2007

Ans. (b) : Given data
l = 4m = 4000 mm
A = 2000 mm^2
P = 20 kN
E = 2×10^5 N/mm^2

$\Delta L = \frac{PL}{AE}$

$= \frac{20\times10^3\times4000}{2000\times2\times10^5} = 0.2$ mm

343. The elongation produced in a bar due to its self-weight is given by

(a) $\frac{9.81\rho l^2}{E}$ (b) $\frac{9.81\rho l^2}{2E}$
(c) $\frac{9.81\rho l}{E}$ (d) $\frac{9.81\rho^2 l}{2E}$

UKPSC AE 2007 Paper -I

Ans. (b) : $\frac{9.81\rho l^2}{2E}$

344. Which of the following is not the characteristic of stress-strain curve for mild steel?

(a) The stress is proportional to the strain up to the proportional limit
(b) Percentage reduction in area may be as high as 60-70%.
(c) A neck is formed due to high stress level
(d) During plastic stage no strain hardening takes place

UKPSC AE 2007 Paper -I

Ans. (d) : During plastic stage no strain hardening takes place.

345. When a body is in a state of equilibrium under the action of any force system, the normal stress at a point within the body depends upon

(a) elementary area ΔA surrounding the point
(b) elemental force ΔF acting normal to ΔA
(c) the plane orientation containing the point
(d) all the above three

UKPSC AE 2007 Paper -I

Ans. (d) : All the above three

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346. A steel bolt passes centrally through a copper tube. At the ends nuts and washers are provided. The area of cross-section of both bolt and tube is same. If the young's modulus of steel is twice that of copper and if the assembly is tightened by rotating a nut through 60^0 on the thread of pitch 3.6 mm, then the contraction of the tube is

(a) 0.6 mm (b) 0.4 mm
(c) 0.2 mm (d) 0.1 mm

WBPSC AE, 2007

Ans. (b) : Given,

$A_s = A_c = A$

$E_s = 2E_c$ Pitch, P = 3.6 mm

The nut is rotated by 60^0

The number of turns, $n = \frac{\theta}{360^0}$

$= \frac{60}{360} = \frac{1}{6}$

Total tensile force in bolt

= Total compressive force in copper tube.

$\sigma_s A_s = \sigma_c A_c$

$\boxed{\sigma_s = \sigma_c}$

Total axial movement of nut is equals to the elongation of steel bolt + contraction of copper tube.

$$np = \frac{\sigma_s}{E_s}L + \frac{\sigma_c}{E_c}L$$

$$\frac{1}{6}\times 3.6 = \frac{\sigma_c}{\alpha E_c}L + \frac{\sigma_c}{E_c}L$$

$$0.6 = \frac{3\sigma_c L}{2E_c}$$

$$\frac{\sigma_c . L}{E_c} = \frac{2\times 0.6}{3} = 0.4\text{mm}$$

347. The area under the stress-strain curve represents:

(a) breaking strength of material
(b) toughness of material
(c) hardness of material
(d) energy required to cause failure

J & K PSC Screening, 2006

Ans. (d) : • The area under the stress strain curve represents energy required to cause failure.

• Area under stress strain curve upto fracture is known as modulus of toughness.

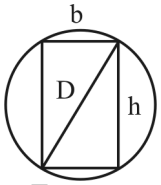
• Area under stress strain curve upto elastic limit is the modulus of resilience.

348. The width and depth of the strongest beam of rectangular section that can be cut out of a cylindrical log of wood whose diameter is 30 cm are

(a) $10\sqrt{3}$ cm and 10 cm
(b) 10 cm and $10\sqrt{2}$ cm
(c) $10\sqrt{3}$ cm and $10\sqrt{6}$ cm
(d) 10 cm and $10\sqrt{3}$ cm

OPSC Civil Services Pre 2006

Ans. (c) :



For max strength $h = \sqrt{2}\,b$

$b^2 + h^2 = D^2$

$b^2 + 2b^2 = D^2$

or $b = \frac{D}{\sqrt{3}}$

$$h = \frac{\sqrt{2}}{\sqrt{3}}\times D = \sqrt{2}b$$

$$b = \frac{30}{\sqrt{3}} = \frac{10\times\sqrt{3}\times\sqrt{3}}{\sqrt{3}} = 10\sqrt{3}$$

$$h = \sqrt{2}\times b = 10\sqrt{6}$$

349. Stiffness of a material is expressed in terms of :

(a) mass density
(b) hardness number
(c) modulus of elasticity
(d) impact strength

J & K PSC Screening, 2006

Ans. (c) : modulus of elasticity–

$$\text{stiffness}(k) = \frac{AE}{\ell}$$

Where, AE = Axial rigidity

350. If the area, length and the stress to which a bar is subjected be all doubled, then the elastic strain energy of the bar will become

(a) doubled (b) halved
(c) eight times (d) four times

WBPSC AE 2003

Ans. (d) : The strain energy formula (U)

$$U = \frac{1}{2}P\delta \qquad \left[\because\ \delta = \frac{PL}{AE}\right]$$

$$= \frac{1}{2}\,P\times\frac{PL}{AE}$$

$$= \frac{1}{2}\times\frac{P^2 L}{AE}$$

$$= \frac{1}{2}\times\frac{\sigma^2\times AL}{E}$$

$$U = \frac{\sigma^2}{2E}\times(\text{Volume})$$

Hence, If the area, length and the stress to which a bar is subjected be all doubled, then the elastic strain energy of the bar will become four times.

351. The iron bar which is fixed at both ends are heated to higher temperature. Then it is cooled slowly. During cooling, what kind of thermal stress is produced in the rod ?

(a) Tensile stress (b) Shear stress
(c) Compressive stress (d) Yield stress

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Tensile stress

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352. A steel bar of 40 mm × 40 mm square cross-section is subjected to an axial compressive load of 200 kN. If the length of the bar is 2 m and E = 200 GPa, the contraction in mm of the bar will be :

(a) 1.25 (b) 2.7
(c) 4.05 (d) 5.4

HPPSC Asstt. Prof. 18.11.2016

Ans. (a) : $\delta_L = \dfrac{PL}{AE}$

Where, P = 200 km $\Rightarrow 200 \times 10^3$ N
L = 2 m = 2000 mm
E = 200 GPa = 200×10^3 MPa
A = 40×40 mm^2

$$\delta_L = \frac{200\times10^3\times2000}{40\times40\times200\times10^3}$$

$$\boxed{\delta_L = 1.25\,\text{mm}}$$

353. Property of absorbing large amount of energy before fracture is known as:-

(a) Ductility
(b) Toughness
(c) Elasticity
(d) Hardness

UKPSC AE-2013, Paper-I

Ans. (b) : Property of absorbing large amount of energy before fracture is known as Toughness.

354. A 200 × 100 × 50 mm steel block is subjected to a hydrostatic pressure of 15 MPa. The Young's modulus and Poisson's ratio of the material are 200 GPa and 0.3 respectively. The change in the volume of the block in mm^3 is

(a) 85 (b) 90
(c) 100 (d) 110

VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012
Gate 2007

Ans. (b) : $\sigma_1 = \sigma_x = \sigma_y = \sigma_z = -15 \times 10^6$ Pa (Compression)

Now, Longitudinal strain-

$$\therefore \quad \varepsilon_x = \frac{\sigma_x}{E} - \frac{\mu\sigma_y}{E} - \frac{\mu\sigma_z}{E} \qquad (\mu = \text{Poisson's ratio})$$

$$= \frac{\sigma_1}{E}(1-2\mu)$$

$$= \frac{15\times10^6}{200\times10^9}(1-2\times0.3)$$

$$\varepsilon_x = \frac{6}{200\times10^3} = 3\times10^{-5}$$

Volumetric strain, $\varepsilon_v = \dfrac{\Delta V}{V} = 3\varepsilon_x$

$$\Delta V = 3\times3\times10^{-5}\times200\times100\times50$$

$$\boxed{\Delta V = 90\ \text{mm}^3}$$

2. Principle Stress and Strain

355. On principal plane the shear stress is.....

(a) zero
(b) unity
(c) double the value of principal stress
(d) half the value of principal stress

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
JPSC AE 10.04.2021, Paper-II
GMB AAE 25.06.2017
(HPPSC LECT. 2016)
HPPSC Asstt. Prof. 29.10.2016
RPSC ADE 2016
GPSC Lect. 23.10.2016
BPSC AE 2012 Paper - VI
APPSC AEE 2012
APPSC AEE Mains 2016 (Civil/Mech.)
APPSC AE 04.12.2012

Ans : (a) on principal plane the shear stress is zero.

Principal Stresses and Strains:-

It has been observed that at any point in a strained material, there are three planes, mutually perpendicular to each other, which carry direct stresses only, and no shear stress. A little consideration will show that out of these three direct stresses; one will be maximum, the other minimum and the third an intermediate between the two. These particular planes, which have no shear stress, are known as principal planes. The magnitude of direct stress, across a principal plane, is known as principal stress.

356. Maximum shear stress in Mohr's circle is equal to

(a) Radius of circle
(b) Diameter of circle
(c) Center of circle from y-axis
(d) Chord of circle

Gujarat PSC AE 2019
HPPSC AE 2018
APGENCO AE, 2017
GPSC Lect. 23.10.2016
APPSC AEE Screening Test 2016
ISRO Scientist/Engineer (RAC) 29.11.2015
Mizoram PSC AE/SDO 2014, Paper-II
UKPSC AE-2013, Paper-I
ISRO Scientist/Engineer 2010, 2007
ESE 2014, 2008

Ans. (a) : Maximum shear stress

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

where σ_1 and σ_2 are principal stresses.

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = \text{Radius of circle}$$

357. A Mohr's circle reduces to a point when the body is subjected to

(a) Pure shear
(b) Uniaxial stress only

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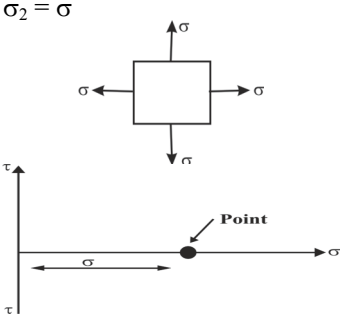
(c) Equal and opposite axial stress on two mutually perpendicular planes, the plaines being free of shear.
(d) Equal axial stress on two mutually perpendicular planes, the planes being free of shear.

MPPSC AE 08.11.2015
ISRO Scientist/Engineer 24.05.2014
ISRO Scientist/Engineer 2012
TRB Poly. Lect., 2012
ISRO Scientist/Engineer 2009

Ans. (d) : Two equal mutually perpendicular plane-
Mohr's circle—
Let two equal stress σ_1 and σ_2 having mutually perpendicular i.e.

$\sigma_1 = \sigma_2 = \sigma$



Here, $\tau = 0$ i.e. planes free of shear.
Only two equal Axial stress σ and σ

358. σ_x, σ_y and τ_{xy} are the rectangular stress components of a point. The radius of Mohr's circle is

(a) $\sqrt{\sigma_x^2 - \sigma_y^2 + \tau_{xy}^2}$ (b) $\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

(c) $\sqrt{\sigma_y^2 - \sigma_x^2 + \tau_{xy}^2}$ (d) $\sqrt{\frac{(\sigma_y - \sigma_x)^2}{2} + \tau_{xy}^2}$

CGPSC AE 25.02.2018
UPRVUNL AE 07.10.2016
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
ISRO Scientist/Engineer 2008
ESE 2006
GATE 1993

Ans. (b) : Radius of Mohr's Circle

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\frac{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}{4}}$$

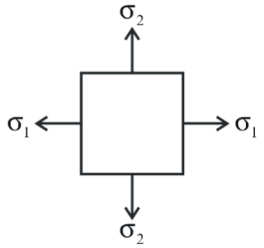
$$R = \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

359. If the principle stresses in a plane stress problem are σ_1= 100 MPa, σ_2= 40 MPa the magnitude of the maximum shear stress (in MPa) will be

(a) 60 (b) 50
(c) 30 (d) 20

GPSC Executive Engg. 23.12.2018
GPSC ARTO Pre 30.12.2018
RPSC 2016
TNPSC AE 2014
UPRVUNL AE 2014
OPSC Civil Services Pre. 2011
GATE 2009

Ans : (c) Given,
σ_1 = 100 MPa
σ_2 = 40 MPa



Maximum shear stress

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{100 - 40}{2}$$

$$\boxed{\tau_{max} = 30\text{MPa}}$$

360. For a general two dimensional stress system, what are the co-ordinates of the centre of Mohr's circle?

(a) $\frac{\sigma_x - \sigma_y}{2}, 0$ (b) $0, \frac{\sigma_x + \sigma_y}{2}$

(c) $\frac{\sigma_x + \sigma_y}{2}, 0$ (d) $0, \frac{\sigma_x - \sigma_y}{2}$

UJVNL AE 2016
UKPSC AE-2013, Paper-I
APPSC AEE 2012
ESE 2007

Ans : (c)

$$Centre\ of\ Mohr's\ circle = \left(\frac{\sigma_x + \sigma_y}{2}, 0\right)$$

$$Radius = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$$

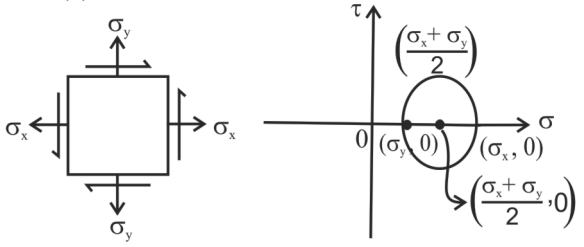
361. In Mohr's circle drawn for a general case of biaxial stresses, the distance of centre of circle from the Y-axis is

(a) $\frac{\sigma_x - \sigma_y}{2}$ (b) $\frac{\sigma_x + \sigma_y}{2}$

(c) $\sqrt{\frac{\sigma_x + \sigma_y}{2}}$ (d) $\sqrt{\frac{\sigma_x - \sigma_y}{2}}$

APGCL AM, 2021
WBPSC AE, 2017
Nagaland PSC CTSE 2016, Paper-I

Ans. (b) :



362. When two mutually perpendicular principal stresses are unequal but alike, the maximum shear stress is represented by

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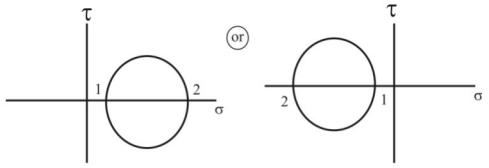
(a) the diameter or Mohr's circle
(b) half the diameter of Mohr's circle
(c) one-third the diameter of Mohr's circle
(d) one-fourth the diameter of Mohr's circle

TSPSC Managers, 2015
TRB Asstt. Prof., 2012
APPSC IOF, 2009, ESE 1994

Ans. (b) : Radius = $\tau_{maximum} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

Centre of Mohr's circle = $\left(\frac{\sigma_x + \sigma_y}{2}, 0\right)$



τ

- Maximum shear stress is represented by half the diameter of Mohr's circle.

363. If σ_1 and σ_2 be the major and minor tensile stresses, then maximum value of tangential stress is equal to-

(a) σ_2 (b) $\sigma_1 - \sigma_2$
(c) $\sigma_1 + \sigma_2$ (d) $\frac{\sigma_1 - \sigma_2}{2}$

Assam Engg. College AP/Lect. 18.01.2021
CIL MT 26.03.2017
ISRO Scientist/Engineer (RAC) 29.11.2015
GATE 2018
PPSC Asstt. Municipal Engineer 15.06.2021

Ans : (d) Maximum value of tangential stress–

$$\tau_{max.} = \frac{\sigma_1 - \sigma_2}{2}$$

364. The minimum number of strain gauges in a strain rosette is

(a) One (b) Two
(c) Three (d) Four

HPPSC AE 2018
HPPSC Asstt. Prof. 2014
MPPSC State Forest Service Exam, 2014
APPSC AEE 2012

Ans. (c) : The minimum number of strain gauge in a strain rosette is Three.
Strain Gauge Rosette–A strain gauge rosette is a term for an arrangement of two or more strain gauge that are positioned closely to measure strains along different

365. If the centre of Mohr's stress circle coincides with the origin on the σ - τ coordinates, then

(a) $\sigma_x + \sigma_y = 0$ (b) $\sigma_x - \sigma_y = 0$
(c) $\sigma_x + \sigma_y = 1/2$ (d) $\sigma_x - \sigma_y = 1/2$

Gujarat PSC AE 2019
TNPSC AE 2019
BPSC Main 2017 Paper-VI

Ans : (a) :

By centre of Mohr's circle lie at $\frac{\sigma_x + \sigma_y}{2}$

It's coincide with origin

i.e., $\frac{\sigma_x + \sigma_y}{2} = 0$ or $\sigma_x + \sigma_y = 0$

$$\boxed{\sigma_x + \sigma_y = 0}$$

Principle stresses are σ_x and $-\sigma_y$. Hence they are equal in magnitude but unlike in direction.

366. When a thick plate is subjected to external loads:

1. **State of plane stress occurs at the surface**
2. **State of plane strain occurs at the surface**
3. **State of plane stress occurs in the interior part of the plate**
4. **State of plane strain occurs in the interior part of the plate**

Which of these statements are correct?

(a) 1 and 3 (b) 2 and 4
(c) 1 and 4 (d) 2 and 3

Gujarat PSC AE 2019
ESE 2013

Ans : (c) : For a plain strain case for a given load the strain in the thickness direction is negligible because more material is available in thickness direction which will resist any deformation in that direction (due to Poisson's effect) so strain in thickness direction in thick plate is assumed to be zero.
If you thick a cracked body (part through crack) loaded in tension, the crack front in the interior will have plane strain (as the mid section is surrounded by sufficient volume of material thus making it analogous to thick section) whereas crack front at the surface will have plane stress.

367. A rectangular plate in plane stress is subjected to normal stresses σ_x = 35 MPa, σ_y = 26 MPa, and shear stress τ_{xy} = 14 MPa. The ratio of the magnitudes of the principal stresses (σ_1/σ_2) is approximately :

(a) 0.8 (b) 1.5
(c) 2.1 (d) 2.9

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : Given,

$\sigma_x = 35\,MPa$

$\sigma_y = 26\,MPa$

$\tau_{xy} = 14$ MPa

$$\sigma = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{35+26}{2} + \sqrt{\left(\frac{35-26}{2}\right)^2 + 14^2}$$

$$= \frac{61}{2} + \sqrt{\left(\frac{9}{2}\right)^2 + (14)^2}$$

$= 45.205$ MPa

$$\sigma_2 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

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$$\sigma_2 = \frac{61}{2} - \sqrt{\left(\frac{9}{2}\right)^2 + (14)^2}$$
$= 15.79$ MPa

The ratio of the magnitude of principal stress,

$$\frac{\sigma_1}{\sigma_2} = \frac{45.205}{15.790} = 2.86$$
≈ 2.9

368. For an element in pure shear $(+\tau_{xy})$, the principal stresses will be given as :

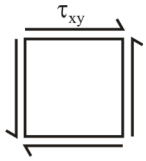
(a) $\sigma_{1,2} = \pm\tau_{xy}$ (b) $\sigma_{1,2} = \pm\tau_{xy}/\sqrt{2}$

(c) $\sigma_{1,2} = \pm\tau_{xy}\sqrt{2}$ (d) $\sigma_{1,2} = 0 + \tau_{xy}$

CIL MT 26.03.2017
UPRVUNL AE 07.10.2016

Ans. (a) : We know that

For an element in pure shear $(\pm\tau_{xy})$, then principal stresses.



$\sigma_x = \sigma_y = 0$ (Pure shear)

Then principal stresses

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$\boxed{\sigma_{1,2} = \pm\tau_{xy}}$$

369. The normal stresses at a point are σ_x = 10 MPa, σ_y = 2 MPa, and the shear stress at the at this point is 3 MPa. The maximum principal stress at this point would be

(a) 15 MPa (b) 13 MPa

(c) 11 MPa (d) 09 MPa

JWM 2017
J&K PSC Civil Services Pre, 2010
ESE 1998

Ans. (c) : Maximum principle stress,

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$\sigma_x = 10$ MPa

$\sigma_y = 2$ MPa

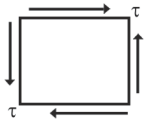
$\tau_{xy} = 3$ MPa

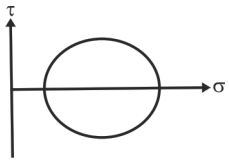
$$\sigma_1 = \frac{10+2}{2} + \sqrt{\left(\frac{10-2}{2}\right)^2 + 3^2}$$

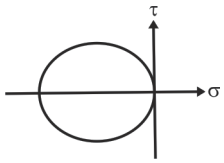
$$= 6 + \sqrt{16+9} = 6 + 5 = 11 \text{ MPa}$$

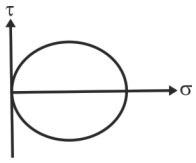
$$\boxed{\sigma_1 = 11 \text{ MPa}}$$

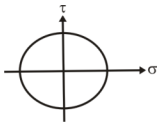
370. Which one of the following figures is the correct sketch of Mohr's circle of the given state of stress



(a)


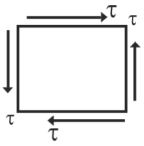
(b)


(c)


(d)


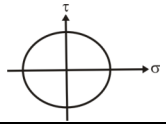
APPSC AEE SCREENING 17.02.2019
ISRO Scientist/Engineer 2011
BPSC Asstt. Prof. 29.11.2015
ESE 1996

Ans. (d) : State of stress



"Pure shear"

Mohr's circle diagram



371. If a Mohr's circle is drawn for a fluid element inside a fluid body at rest, it would be :

(a) a circle not touching the origin

(b) a circle touching the origin

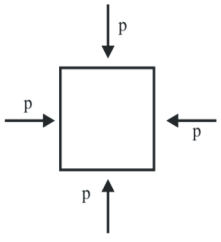
(c) a point on the normal stress axis

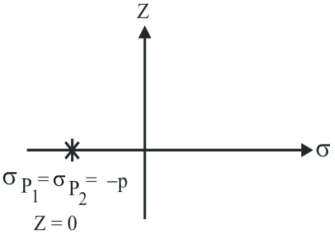
(d) a point on the shear stress axis

RPSC ACF & FRO, 26.02.2021
CGPSC AE 26.04.2015 Shift-I
TSPSC AEE 28-08-2014 (Civil/Mech.)

Ans. (c) : Mohr's circle for a fluid element inside a fluid body at rest it would be a point on the normal stress axis.

Mohr's circle reduces to a point in case of hydrostatic loading.





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372. If the principal stresses corresponding to a two-dimensional state of stress are σ_1 and σ_2. When σ_1 is greater than σ_2 and both are tensile, then which of the following would be correct criterion for failure by yielding, according to maximum shear stress theory?

(a) $\frac{\sigma_1-\sigma_2}{2}=\pm\frac{\sigma_{yp}}{2}$ (b) $\frac{\sigma_1}{2}=\pm\frac{\sigma_{yp}}{2}$

(c) $\frac{\sigma_2}{2}=\pm\frac{\sigma_{yp}}{2}$ (d) $\sigma_1=\pm2\sigma_{yp}$

RPSC IOF, 2020
Punjab PSC (Lect.) 06.08.2017

Ans. (b) : According to maximum shear stress theory for plane condition,

Absolute, τ_{max} = Larger of $\left[\frac{\sigma_1-\sigma_2}{2},\frac{\sigma_1}{2},\frac{\sigma_2}{2}\right]$

(1) If, σ_1 and σ_2 are alike in nature and $\sigma_1>\sigma_2$

Then, absolute $\tau_{max}=\frac{\sigma_1}{2}=S_{sy}$

(2) If, σ_1 and σ_2 are unlike in nature,

Then, absolute $\tau_{max}=\frac{\sigma_1-\sigma_2}{2}=S_{sy}$

According to maximum shear stress theory,

$$\frac{\sigma_{yp}}{S_{sy}}=2\Rightarrow S_{sy}=\frac{\sigma_{yp}}{2}$$

So, $\boxed{\frac{\sigma_1}{2}=\pm\frac{\sigma_{yp}}{2}}$ (Because σ_1 and σ_2 are alike in nature)

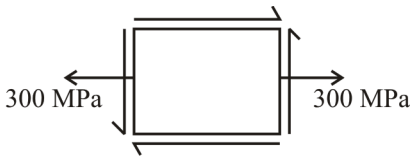
373. A body is subjected to a direct tensile stress of 300 MPa in one plane accompanied by a simple shear stress of 200 MPa. What would be the maximum normal stress?

(a) –100 MPa (b) 250 MPa

(c) 300 MPa (d) 400 MPa

GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC IOF 2020
VIZAG STEEL MT 2011
ESE 2016

Ans. (d) :



Given, $\sigma_x=300$ MPa

$\sigma_y=0$

$\tau_{xy}=200$ MPa

Maximum normal stress,

$$\sigma_1=\frac{\sigma_x+\sigma_y}{2}+\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

$$\sigma_1=\left(\frac{300+0}{2}\right)+\sqrt{\left(\frac{300-0}{2}\right)^2+(200)^2}$$

$$=150+\sqrt{(150)^2+(200)^2}$$

$$=400\text{ MPa}$$

374. Under uniaxial strain, the ratio of maximum shearing strain to uniaxial strain is–

(a) 2.0 (b) 0.5

(c) 1.0 (d) 1.5

GPSC AE (Part-B Civil) 2018
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) :

Stress, $\sigma_1/\sigma_2=\frac{\sigma_x+\sigma_y}{2}+\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$

$$\tau_{max}=\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

Strain, $\epsilon_1/\epsilon_2=\frac{\epsilon_x+\epsilon_y}{2}\pm\sqrt{\left(\frac{\epsilon_x-\epsilon_y}{2}\right)^2+\left(\frac{\phi_{xy}}{2}\right)^2}$

$$\frac{\phi_{max}}{2}=\sqrt{\left(\frac{\epsilon_x-\epsilon_y}{2}\right)^2+\left(\frac{\phi_{xy}}{2}\right)^2}$$

In question $\epsilon_x=\epsilon$, $\epsilon_y=\phi_{xy}=0$

So, $\epsilon_1=\frac{\epsilon_x}{2}$ and $\frac{\phi_{max}}{2}=\frac{\epsilon_x}{2}$ or $\phi_{max}=\epsilon_x$

ϵ_x = uniaxial strain

ϕ_{max} = max shearing strain

Ratio $=\frac{\phi_{max}}{\epsilon_x}=1$

As, $\epsilon_x=\phi_{max}$

375. A body is subjected to a tensile stress of 1200 MPa on one plane and another tensile stress of 600 MPa on a plane at right angles to the former. It is also subjected to a shear stress of 400 MPa on the same planes. What would be the maximum normal stress?

(a) 400 MPa (b) 500 MPa

(c) 900 MPa (d) 1400 MPa

TNCSC AE 2020
JPSC AE 2013, Paper- V
VIZAG STEEL MT 2011

Ans. (d) : Given, $\sigma_x=1200$ MPa, $\sigma_y=600$ MPa

$\tau_{xy}=400$ MPa

Maximum normal stress,

$$\sigma_1=\left(\frac{\sigma_x+\sigma_y}{2}\right)+\sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}^2}$$

$$=\left(\frac{1200+600}{2}\right)+\sqrt{\left(\frac{1200-600}{2}\right)^2+(400)^2}$$

$$=900+\sqrt{(300)^2+(400)^2}$$

$$\sigma_1=900+500$$

$$\boxed{\sigma_1=1400\text{ MPa}}$$

376. A plane stressed element is subjected to the state of stress given by $\sigma_x=\tau_{xy}=100$kgf/cm^2 and $\sigma=0$. Maximum shear stress in the element is equal to

(a) $50\sqrt{3}$ kgf/cm^2 (b) 100 kgf/cm^2

(c) $50\sqrt{5}$ kgf/cm^2 (d) 150 kgf/cm^2

APPSC AEE 2012
APPSC AE 04.12.2012, ESE 1997

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Ans. (c) : $\sigma_x = \tau_{xy} = 100\ \text{kgf/cm}^2,\ \sigma_y = 0$

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\frac{100}{2}\right)^2 + 100^2}$$

$$= \sqrt{50^2 + 100^2} = 50\sqrt{1+4}$$

$$\tau_{max} = 50\sqrt{5}\ \text{kgf/cm}^2$$

377. Principal stresses at a point in plane stressed element are $\sigma_x = \sigma_y = 500\ \text{N/mm}^2$. Normal stress on the plane inclined at 45° angle to x-axis will be

(a) 0 N/mm^2 (b) 700 N/mm^2
(c) 500 N/mm^2 (d) 1000 N/mm^2

UKPSC AE 2013 Paper-I
APPSC AEE 2012, ESE 1993

Ans. (c) : $\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta$

$$= \frac{500+500}{2} + \frac{500-500}{2}\cos 90$$

$$\boxed{\sigma_\theta = 500\,\text{N/mm}^2}$$

378. If a body carries two unlike principal stresses, the maximum shear stress is given by

(a) sum of the principal stresses
(b) difference of the principal stresses
(c) half the difference of the principal stresses
(d) half the sum of the principal stresses

CGPSC AE 15.01.2021
TSPSC AEE 2015
APPSC AEE 2012

Ans. (c) If a body carries two unlike principal stresses, the maximum shear stress is given by half the difference of the principal stresses.

Or

If a body carries two unlike principal stresses the maximum shear stress is given by half the sum of the principal stresses.

379. Mohr's circle can be used to determine following stress on inclined surface

(a) Normal stress (b) Principal stress
(c) Tangential stress (d) All of the above

UJVNL AE 2016
UKPSC AE-2013, Paper-I

Ans : (d) Mohr's circle can be used to determine Normal stress, principal stress, and tangential stress.

Mohr's Circle of Stresses:-

The Mohr's circle is a graphical method of finding the normal, tangential and resultant stresses on an inclined plane. It is drawn for the following two cases:-

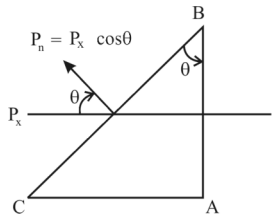
(i) When the two mutually perpendicular principal stresses are unequal and like.
(ii) When the two mutually perpendicular principal stresses are unequal and unlike.

380. Normal stress on a plane, the normal..... Which is inclined at an angle θ with the line of action of applied un axial stress σ_x is given by

(a) $\sigma_x / \sin^2\theta$ (b) $\sigma_x / \cos^2\theta$
(c) $\sigma_x \cos^2\theta$ (d) $\sigma_x \sin^2\theta$

GPSC Engg. Class-II Pre-19.01.2020
UPPSC AE 12.04.2016 Paper-I

Ans : (c)



BC –A plane which is inclined at an angle (90–θ) with the line of action of applied un-axial stress σ_x

$P_x = \sigma_x \times AB \quad [t = 1 \text{ assume}]$

$P_n = P_x \times \cos\theta = \sigma_x \times AB \times \cos\theta$

The normal stress at inclined plan BC

$$\sigma_n = \frac{P_n}{BC} = \frac{\sigma_x \times AB \times \cos\theta}{\left[\frac{AB}{\cos\theta}\right]} \quad \left[\cos\theta = \frac{AB}{BC},\ BC = \frac{AB}{\cos\theta}\right]$$

$$\boxed{\sigma_n = \sigma_x \cos^2\theta}$$

381. For a plane stress case, $\sigma_1 = 50\,\text{MPa}$, $\sigma_2 = -100\,\text{MPa}$, $\tau_{12} = 40\,\text{MPa}$, the maximum and minimum principal stresses are, respectively,

(a) 60 MPa, – 110 MPa,
(b) 50 MPa, – 110 MPa
(c) 40 MPa, – 120 MPa
(d) 70 MPa, – 130 MPa,

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : $\sigma_1 = 50\ MP_a$ $\sigma_2 = -100\ MP_a$ $\sigma_{12} = 40\ MP_a$

Max and Min principal stresses

$$\sigma_{max,\,min} = \frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau_{12}^2}$$

$$= \frac{50-100}{2} \pm \sqrt{\left(\frac{50-(-100)}{2}\right)^2 + 40^2}$$

$$= -25 \pm 85$$

$\sigma_{max,\,min} = -25+85$ & $-25-85$

$\sigma_{max,\,min} = 60, -110$

$\sigma_{max} = -110\text{MPa}$, $\sigma_{min} = 60\text{MPa}$

382. The tensile stresses at a point across two mutually perpendicular planes are 120 MPa and 60 MPa. The normal stress on a plane inclined at 30° to the axis of the minor axis is _____.

(a) 110 Mpa (b) 60 Mpa
(c) 45 Mpa (d) 105 Mpa

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Given,

$\sigma_x = 120$ MPa

$\sigma_y = 60$ MPa

$\theta = 30°$

Figure

Normal stress, $(\sigma_n)_\theta = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta$

$$= \frac{1}{2}(120+60) + \frac{1}{2}(120-60)\cos 60°$$

$$= 90 + \frac{1}{2} \times 60 \times \frac{1}{2} = 90 + 15 = 105\ \text{MPa}$$

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383. The stresses on two perpendicular planes through a point in a body are 100 MPa and 40 MPa both tensile along with a shear stress of 40 MPa. The maximum failure stress as per maximum normal stress theory is _____.

(a) 180 MPa (b) 110 MPa
(c) 70 MPa (d) 120 MPa

VIZAG Steel MT 24.01.2021

Ans. (d) : Given,

$\sigma_x = 100$ MPa
$\sigma_y = 40$ MPa
$\tau_{xy} = 40$ MPa

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{100+40}{2} \pm \sqrt{\left(\frac{100-40}{2}\right)^2 + (40)^2}$$

$$= 70 \pm \sqrt{(30)^2 + (40)^2}$$

$\sigma_{1,2} = 70 \pm 50$

Maximum (σ_1) = 70 + 50 = 120 MPa
Minimum (σ_2) = 70 – 50 = 20 MPa
Hence, the maximum failure stress as per maximum normal stresses (σ_1 = 120 MPa)

384. If a body is subjected to stresses in xy plane with stresses of 100 N/mm^2 and 20 N/mm^2 acting along x and y axes respectively. Also the shear stress acting is 30 N/mm^2. Find the maximum amount of shear stress in N/mm^2 to which the body is subjected.

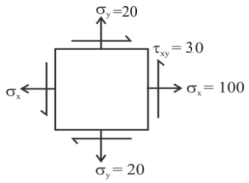
(a) $30\sqrt{5}$ (b) 50
(c) $10\sqrt{5}$ (d) Zero

APGCL AM, 2021

Ans. (b) :

Given

$\sigma_x = 100 N/mm^2$
$\sigma_y = 20 N/mm^2$
$\tau_{xy} = 30 N/mm^2$



Maximum Shear Stress e.

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{100-20}{2}\right)^2 + (30)^2}$$

$$= \sqrt{(40)^2 + (30)^2}$$

$\tau_{max} = 50 N/mm^2$

385. If a cylindrical rod of cross section A is applied with uniaxial tension P, the maximum shear stress in the rod will be :

(a) 0 (b) $\frac{1}{2}\frac{P}{A}$
(c) $\frac{P}{A}$ (d) $\frac{2P}{A}$

JPSC AE 10.04.2021, Paper-II

Ans. (b) :

Given,

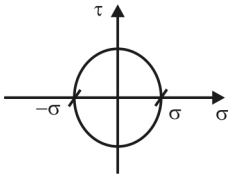
$\sigma_x = \frac{P}{A}$
$\sigma_y = 0$
$\tau_{xy} = 0$

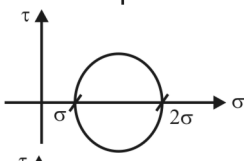
Maximum shear stress (τ_{max}) $= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

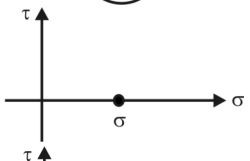
$$= \sqrt{\left(\frac{\frac{P}{A} - 0}{2}\right)^2 + 0^2}$$

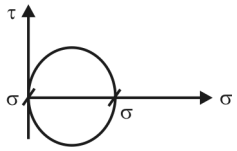
$$\boxed{\tau_{max} = \frac{P}{2A}}$$

386. Mohr's circle for membrane stresses in a thin walled spherical pressure vessel with radius r and wall thickness t, under internal pressure p will be:

(a) 

(b) 

(c) 

(d) 

JPSC AE 10.04.2021, Paper-II

Ans. (c) : For thin spherical pressure vessel

$$\boxed{\sigma_h = \sigma_L = \frac{Pd}{4t}}$$

$$\boxed{\sigma_1 = \sigma_2 = \frac{Pd}{4t}}$$

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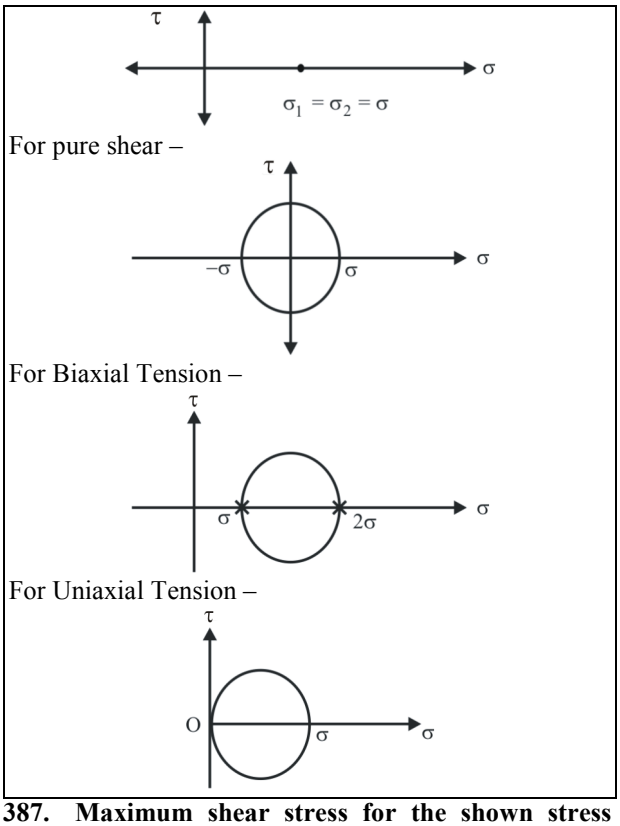
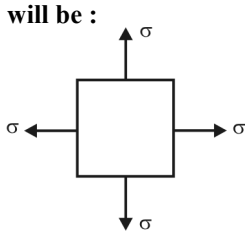


387. Maximum shear stress for the shown stress condition will be :



(a) σ/2 (b) σ
(c) 2σ (d) 0

JPSC AE 10.04.2021, Paper-II

Ans. (d) : Given,

$\sigma_x = \sigma$

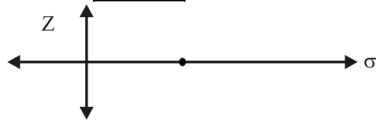
$\sigma_y = \sigma$

$\tau_{xy} = 0$

Maximum shear stress $= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \sqrt{\left(\frac{\sigma - \sigma}{2}\right)^2 + 0^2}$

$\boxed{\tau_{max} = 0}$



Mohr's circle will be point.

388. When an element is in a state of simple shear, maximum direct stresses are induced on

(a) mutually parallel planes which are at 45^0 to the planes of pure shear
(b) mutually perpendicular planes which are parallel to the planes of pure shear
(c) mutually perpendicular planes which are perpendicular to the planes of pure shear
(d) mutually perpendicular planes which are at 45^0 to the planes of pure shear

GPSC DEE, Class-2 (GWSSB) 04.07.2021

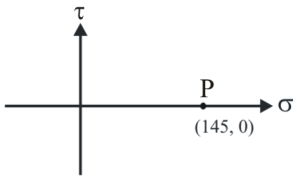
Ans. (d) : When an element is in a state of simple shear, maximum direct stresses are induced on mutually perpendicular planes which are at 45^0 to the planes of pure shear.

389. The state of stress at a point P in a two dimensional loading is such that the Mohr's circle is a point located at 145 MPa on the positive normal stress axis. The maximum and minimum principal stresses respectively from the Mohr's circle are :

(a) 0, 0
(b) 0, –145 MPa
(c) + 145 MPa, + 145 MPa
(d) + 145 MPa, – 145 MPa

RPSC ACF & FRO, 26.02.2021

Ans. (c) :



Maximum principle stress = 145 MPa
Minimum principle stress = 145 MPa
Maximum /minimum shear stress = 0 MPa

390. Principal strains occur on three planes which are

(a) inclined to one another at 45°
(b) mutually perpendicular to each other
(c) inclined to one another at 22½°
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : For the three-dimensional case it is now demonstrated that three planes of zero shear stress exist, that these planes are mutually perpendicular and that on these planes the normal stresses have maximum or minimum values.

391. A spherical container made of steel has a 5m outside diameter and a 10mm uniform wall thickness. If the internal pressure is 400 kPa, the maximum normal stress and maximum shear stress in the container are :

(a) σ_{max} = 77.4 MPa, τ_{max} = 38.7 MPa
(b) σ_{max} = 89.0 MPa, τ_{max} = 44.5 MPa
(c) σ_{max} = 49.8 MPa, τ_{max} = 2.91 MPa
(d) σ_{max} = 38.7 MPa, τ_{max} = 51.9 MPa

CGPSC AE 15.01.2021

Ans. (*) : Given, diameter of spherical container = 5m
thickness (t) = 10 mm
Internal pressure (p) = 400 kPa

Max. Normal stress, $\sigma_C = \frac{Pd}{4t}$

$= \frac{400 \times 5}{4 \times 10 \times 10^{-3}}$

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$= 50 \times 10^3$ kPa
$= 50$ MPa

Max. shear stress, $\tau_{max} = \frac{\sigma_C}{2} = \frac{50}{2}$
$\simeq 25$ MPa

Note- Official answer is (c).

392. Consider a linear elastic rectangular thin sheet of metal, subjected to uniform uniaxial tensile stress of 100 MPa along the length direction. Assume plane stress condition in the plane normal to the thickness. Young's modulus E = 200 MPa and Poisson's ratio ν = 0.3. The principal strains in the plane of sheet are_____

(a) (0.5, – 0.15) (b) (0.5, – 0.5)
(c) (0.5, 0.0) (d) (0.35, –0.15)

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) : (0.5, – 0.15)

$\sigma_x = 100$
$\sigma_y = 0$
$\nu = 0.3$

$\varepsilon_x = \frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E}$ $\varepsilon_y = \frac{\sigma_y}{E} - \nu\frac{\sigma_x}{E}$

$= \frac{100}{200} - 0.3 \times 0$ $= 0 - 0.3 \times \frac{100}{200}$

$\varepsilon_x = 0.5$ $\varepsilon_y = -0.15$

393. A strain gauge rosette is used when :

(a) direction of strain is known
(b) only magnitude of stress is known
(c) both magnitude and direction of strains are unknown
(d) magnitude is known but the direction is unknown

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : A strain gauge rosette is a term for an arrangement of two or more strain gauges that are positioned closely to measure strains along different direction of the component under evaluation. Single strain gauge can only measure strain effectively in one direction so the use of multiple strain gauges enables more measurements to be taken. Providing a more precise evaluation of strain on the surface being measured.

394. The gauge factor of a semiconductor gauge compared to a wire strain gauge is :

(a) high (b) the same
(c) infinite (d) low

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : The strain gauge should have a high value of gauge factor. As high gauge factor indicates a large change in resistance that leads to high sensitivity. The gauge resistance should be high so as to minimize the effect of undesirable variation of resistance in measurement circuit.

395. Consider a two dimensional state of stress for an element
Where, σ_x = 200 MPa
σ_y = –100 MPa
The co-ordinates of the centre of Mohr's circle are

(a) (0, 0) (b) (100, 200)
(c) (200, 100) (d) (50, 0)

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Co-ordinate of the centre of Mohr's circle is

$\left[\left(\frac{\sigma_x + \sigma_y}{2}\right), 0\right]$

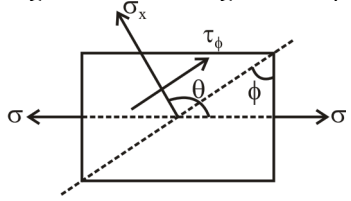
Centre $= \left[\frac{200 - 100}{2}, 0\right] = (50, 0)$

396. When a member is subjected to a direct stress (σ) in one plane, then the tangential stress on an oblique plane which is inclined at an angle Φ with the normal cross section is _____ .

(a) $\sigma(\sin^2\phi)$ (b) $\frac{\sigma(\sin 2\phi)}{2}$
(c) $\frac{\sigma(\cos 2\phi)}{2}$ (d) $\frac{\sigma(\cos^2 \phi)}{2}$

VIZAG MT, 14.12.2020

Ans. (b) : Tangential stress along inclined plane



$\tau_\phi = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$

$\tau_{xy} = 0$
$\sigma_x = \sigma$ $\theta = (90 + 90 - \phi)$
$\sigma_y = 0$ $= (180 - \phi)$

$\tau_\phi = -\left(\frac{\sigma - 0}{2}\right)\sin 2(180 - \phi)$

$\boxed{\tau_\phi = \frac{\sigma}{2}\sin 2\phi}$

397. If a element is subjected to uniform equal compressive stresses of 1 MPa in all directions so that $\sigma_x = \sigma_y = \sigma_z$. The Mohr's circle for this state of stress is a

(a) Circle with infinite radius
(b) Circle with zero radius
(c) Circle with unit radius
(d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : $\sigma_x = \sigma_y = \sigma_z$ the radius of Mohr's circle will be zero.

$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right) + (\tau_{xy})^2}$

$R = \sqrt{0 + 0} = 0$

398. For an element in plane stress, if the larger of the principal stresses is 3 times that of the smaller one, the value of the maximum shear stress is equal to

(a) the smaller principal stress
(b) the larger principal stress
(c) half of the higher principal stress
(d) sum of the two principal stress

Haryana PSC AE (PHED) 05.09.2020, Paper-II

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Ans. (c) : $\sigma_1 = 3\sigma$

$\sigma_2 = \sigma$

$\sigma_3 = 0$

Shear stress $(\tau) = \dfrac{\sigma_1 - \sigma_3}{2}$

$\tau = \dfrac{3\sigma - 0}{2} = \dfrac{1}{2}\sigma_1$

$\tau = \dfrac{1}{2}$ higher principal stress

399. For an element in plane stress, the planes of algebraically maximum shear stress occur at

(a) At 90^0 to the principal planes
(b) At the principal planes
(c) At 45^0 to the principal planes
(d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II
TSPSC AEE 28.08.2017 (Civil/Mech.)

Ans. (c) : Plane of max shear stress is at 45^0 to the principal plane.

400. A strain gauge oriented in a direction at an angle θ with x-axis, measures:

(a) $\varepsilon_x \sin^2\theta + \varepsilon_y \cos^2\theta + \tau_{xy}\cos\theta\sin\theta$
(b) $\varepsilon_x \cos^2\theta + \varepsilon_y \sin^2\theta + \tau_{xy}\cos\theta\sin\theta$
(c) $\varepsilon_x \cos^2\theta + \varepsilon_y \sin^2\theta - \tau_{xy}\cos\theta\sin\theta$
(d) $\varepsilon_x \cos^2\theta + \varepsilon_y \sin^2\theta + 2\tau_{xy}\cos\theta\sin\theta$

CIL MT 27.02.2020

Ans. (b) : A strain gauge oriented in a direction at an angle θ with x-axis measures

$= \varepsilon_x \cos^2\theta + \varepsilon_y \sin^2\theta + \tau_{xy}\cos\theta.\sin\theta$

401. A wheat stone bridge is used for electrical strain gauges, because it has:

(a) high sensitivity (b) zero sensitivity
(c) low sensitivity (d) infinite sensitivity

CIL MT 27.02.2020

Ans. (a) : A wheat stone bridge circuit has high sensivity for detecting very small variation in the values of resistance.
We can connect the strain gauge in one of the arms of a wheat stone bridge and measure the strain in term of variation in resistance.

402. If normal stress on a normal plane is σ, the maximum shear stress on a plane inclined 45 degree to the normal plane will be

(a) σ/2 (b) σ
(c) σ sin θ (d) σ cos θ

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : σ/2

403. At a point in a strained material, tensile stress of 100 MPa and compressive stress of 60 MPa are found to be principal stresses. Maximum shear stress at that point is :

(a) 20 MPa (b) 40 MPa
(c) 80 MPa (d) 60 MPa

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Given,

$\sigma_1 = +100\text{MPa}(\text{Tensile})$

$\sigma_2 = -60\text{MPa}(\text{Compressive})$

Maximum shear stress $\tau_{max} = \dfrac{\sigma_1 - \sigma_2}{2}$

$= \dfrac{100 - (-60)}{2} = 80\,\text{MPa}$

404. Mohr's circle for the state of stress defined by $\begin{bmatrix} 20 & 0 \\ 0 & 20 \end{bmatrix}$ **is a circle with :**

(a) Centre at (0, 0) and radius 20 MPa
(b) Centre at (0, 0) and radius 40 MPa
(c) Centre at (20, 0) and radius 20 MPa
(d) Centre at (20, 0) and radius zero radius

OPSC AEE 2019 Paper-I

Ans : (d) :
Data given as-

$\sigma_x = 20\,\text{MPa}$

$\sigma_y = 20\,\text{MPa}$

$\tau_{xy} = 0$ MPa

Center of Mohr's circle

$= \left(\dfrac{\sigma_x + \sigma_y}{2}, 0\right) = \left[\left(\dfrac{20+20}{2}\right), 0\right] = (20, 0)$

and we know that

$\tau_{max} = \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{\left(\dfrac{20-20}{2}\right)^2 + 0^2} = 0$

405. The state of stress at a point in a 2-D loading is such that the Mohr's circle is a point located at 175 MPa on the positive normal stress axis. The maximum and minimum principle stress, respectively, from Mohr's circle are :

(a) 0; 0 MPa
(b) + 175 MPa; + 175 MPa
(c) + 175 MPa; – 175 MPa
(d) + 175 MPa; 0 MPa

BHEL ET 2019
GATE 2003

Ans. (b) : $\sigma_x = 175\,\text{MPa}(\text{Tensile})$

$\sigma_y = 175\,\text{MPa}(\text{Tensile})$

$\sigma = \dfrac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\dfrac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$

$= \dfrac{175+175}{2} \pm \sqrt{\left(\dfrac{175-175}{2}\right) + 0}$

σ maximum = 175 MPa
σ minimum = 175 MPa

406. A point in a structural member is subjected to normal stresses (σ_x and σ_y) along two mutually perpendicular directions and shear (τ_{xy}) in xy plane. The orientation of first principal plane (θ_{p1}) is expressed as :

(a) $\tan^{-1}\left(\dfrac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$ (b) $\dfrac{1}{2}\tan^{-1}\left(\dfrac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$

(c) $2\tan^{-1}\left(\dfrac{\sigma_x - \sigma_y}{2\tau_{xy}}\right)$ (d) $\dfrac{1}{2}\tan^{-1}\left(\dfrac{\sigma_x - \sigma_y}{2\tau_{xy}}\right)$

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : Given,
By using force equilibrium along an a perpendicular to the inclined plane

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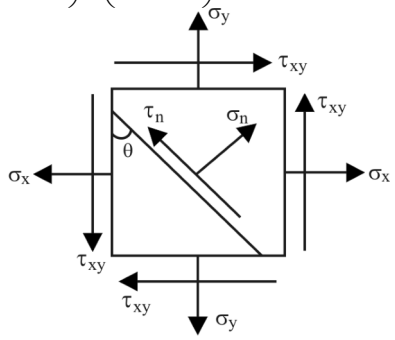
We get,

$$\sigma_n = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$



$$\tau_n = -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

We know that on the principal planes shear stress always zero.

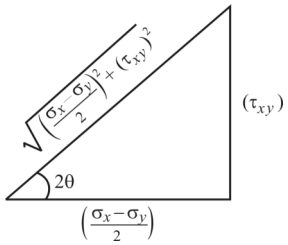
$$\therefore \quad -\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta = 0$$

$$\left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta = \tau_{xy}\cos 2\theta$$

$$\tan 2\theta = \frac{2\tau_{xy}}{(\sigma_x - \sigma_y)}$$

$$\theta = \frac{1}{2}\tan^{-1}\left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$$

So, the orientation of first principal planes (θ_{p_1}) is expresses as :



$$\theta_{p_1} = \frac{1}{2}\tan^{-1}\left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$$

407. What is the number of non-zero strain components for a plane stress problem?

(a) 6 (b) 4
(c) 3 (d) 2

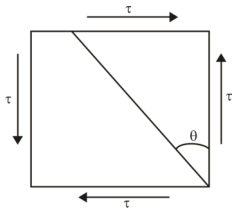
APPSC AEE SCREENING 17.02.2019

Ans. (c) : In a plane stress (2D) problem the number of non-zero strain components are $\in_x$, $\in_y$ and ϕ_{xy}

Total three number.

The 2D tensor is $\begin{bmatrix} \in_x & \phi_{xy} \\ \phi_{yx} & \in_y \end{bmatrix}$

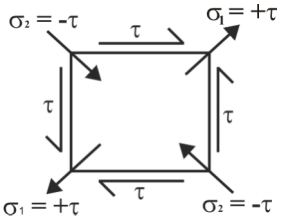
408. The state stress at a point is shown below. θ represents the principal plane corresponding to principal stress σ_1 and σ_2 ($\sigma_1 > \sigma_2$). Values of θ, σ_1, and σ_2 are



(a) 0°, 90°, τ and –τ
(b) 30°, 120°, τ and –τ
(c) 45°, 135°, τ and –τ
(d) 45°, 135°, $\frac{\tau}{2}$ and $-\frac{\tau}{2}$

APPSC AEE SCREENING 17.02.2019

Ans. (c) :



Due to pure shear diagonal tension ($\sigma_1 = +\tau$) and diagonal compression ($\sigma_2 = -\tau$) develops.
The angle between principal planes is 90°.
$\therefore \quad \theta_1 = 45°, \theta_2 = 135°, \sigma_1 = \tau, \sigma_2 = -\tau$

409. Where does principal stress occur in a component?

(a) Along the plane
(b) Perpendicular to the plane
(c) On mutually perpendicular planes
(d) Along the direction of load

TNPSC AE 2019

Ans. (c) : On mutually perpendicular planes principal stress does occur in a component.
• Shear stress τ = 0

410. A body is subjected to a direct tensile stress of 300 MPa in one plane accompanied by a simple shear stress of 200 MPa. The maximum shear stress will be

(a) 150 MPa (b) 200 MPa
(c) 250 MPa (d) 300 MPa

JPSC AE PRE 2019

Ans. (c) : Given,

$\sigma_x = 300$ MPa
$\sigma_y = 0$
$\tau_{xy} = 200$ MPa

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_2 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1 = \frac{300 + 0}{2} + \sqrt{\left(\frac{300 - 0}{2}\right)^2 + (200)^2}$$

$$= 150 + \sqrt{(150)^2 + (200)^2}$$

$$= 150 + 250$$

$$= 400 \text{ MPa}$$

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$$\sigma_2 = \frac{300+0}{2} - \sqrt{\left(\frac{300-0}{2}\right)^2 + (200)^2}$$
$$= 150 - \sqrt{(150)^2 + (200)^2}$$
$$= 150 - 250$$
$$= -100 \text{ MPa}$$
$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$
$$= \frac{400 - (-100)}{2}$$
$$= 250 \text{ MPa}$$

411. At a point in a stressed body there are normal stresses of 1N/mm^2 (tensile) on a vertical plane and 0.5 N/mm^2 (tensile) on a horizontal plane. The shearing stresses on these planes are zero. What will be the normal stress on a plane making an angle 50^o with vertical plane? [Given, (cos 50^o)^2 = 0.413]

(a) 0.6015 N/mm^2 (b) 0.4139 N/mm^2
(c) 0.5312 N/mm^2 (d) 0.7065 N/mm^2

SJVN ET 2019

Ans. (d) : Given as,

$\sigma_x = 1\text{Pa}, \sigma_y = 0.5\text{Pa}, \tau_{xy} = 0$

$\theta = 50^o$

We know that, $[\sigma_n]_{\theta=50^o} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta$

$[\sigma_n]_{\theta=50^o} = \left(\frac{1+0.5}{2}\right) + \left(\frac{1-0.5}{2}\right)\cos(2\times 50^o)$

$= 0.7065\text{Pa}$

412. Assertion (A): A plane state of stress always results in a plane state of strain.
Reason (R) : A uniaxial state of stress results in a three-dimensional state of strain.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true and R is not the correct explanation of A.
(c) A is true but R is false
(d) A is false but R is true

Gujarat PSC AE 2019
ESE 2010

Ans : (d) :
■ In a plane stress condition, the stress in the perpendicular direction to the plane is zero. But strain in that direction need not to be zero. Thus assertion is wrong.

413. Radius of Mohr's circle is represented as : [where $\sigma_{p1} > 0$ and $\sigma_{p2} > 0$ are the major and minor value of principle stresses]

(a) $\sigma_{p1} - \sigma_{p2}$ (b) $\frac{\sigma_{p1} - \sigma_{p2}}{2}$
(c) $\sigma_{p1} + \sigma_{p2}$ (d) $\frac{\sigma_{p1} + \sigma_{p2}}{2}$
(e) $\frac{\sigma_{p1} \times \sigma_{p2}}{2}$

CGPSC AE 25.02.2018

Ans. (b) : Radius of Mohr's circle is represented by

$$R = \frac{\sigma_{p1} - \sigma_{p2}}{2}$$

414. If principle stress σ_{p1} = 100 N/mm^2 (tensile) σ_{p2} = 40 N/mm^2 (compressive), then maximum shear stress will be:

(a) 70 N/mm^2 (b) 50 N/mm^2
(c) 30 N/mm^2 (d) 10 N/mm^2
(e) 5 N/mm^2

CGPSC AE 25.02.2018

Ans. (a) : Data given-

σ_{p1} = 100 N/mm^2, σ_{p2} = – 40 N/mm^2

Then max, shear stress is given as

$$\tau_{max} = \text{Radius of Mohr's circle} = \frac{\sigma_{p1} - \sigma_{p2}}{2}$$
$$\tau_{max} = \frac{100 - [-40]}{2} = 70\text{N/mm}^2$$

415. In a 3-D state of stress, the independent stress components required to define state-of-stress at a point are -

(a) 3 (b) 6
(c) 12 (d) 9

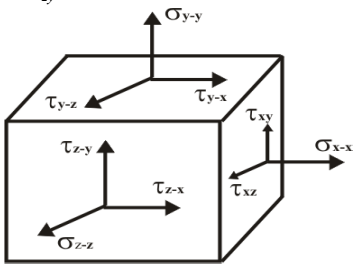
RPSC AE 2018

Ans. (b) :

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

then,

$\sigma_{x-x}, \sigma_{y-y}, \sigma_{z-z}$
$\tau_{xy} = \tau_{yx}$
$\tau_{xz} = \tau_{zx}$
$\tau_{yz} = \tau_{zy}$



In a 3-D state of stress, the independent stress components required to define state of stress at a point is six (three normal stress and three shear stress).

416. A state of plane stress consists of a uni-axial tensile stress of magnitude 8 kPa, exerted on vertical surface and of unknown shearing stresses. If the largest stress is 10 kPa, then the magnitude of the unknown shear stress will be

(a) 6.47 kPa (b) 5.47 kPa
(c) 4.47 kPa (d) 3.47 kPa

UPPSC AE 13.12.2020 Paper-I
ESE 2018

Ans. (c) : Given,

σ_x = 8 kPa, σ_1 = 10 kPa

σ_y = 0

Maximum principal stress is given by

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$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$10 = \frac{8+0}{2} + \sqrt{\left(\frac{8-0}{2}\right)^2 + \tau_{xy}^2}$$

$$10 = 4 + \sqrt{4^2 + \tau_{xy}^2}$$

$$\tau_{xy} = \sqrt{20}$$

$$\tau_{xy} = 4.47 \text{ kPa}$$

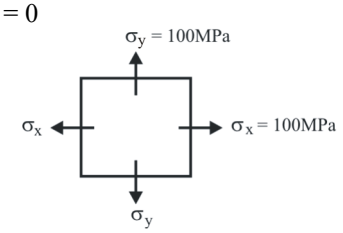
417. In a bi-axial stress problem, the stresses in x and y direction are σ_x = 100 MPa, σ_y = 100 MPa respectively. The maximum principal stress in MPa, is :

(a) 50 (b) 100
(c) 150 (d) 200

Vadodara Muncipal Corp. DEE, 2018

Ans. (b) : Given,

$\sigma_x = 100$MPa
$\sigma_y = 100$MPa
$\tau_{xy} = 0$



maximum principal stress

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{100+100}{2} + \sqrt{\left(\frac{100-100}{2}\right)^2 + 0^2}$$

$$\boxed{\sigma_1 = 100} \text{ MPa}$$

418. Consider the following statements :

1 . Two-dimensional stresses applied to a thin plate in its own plane represent the plane stress condition.

2. Under plane stress condition, the strain in the direction perpendicular to the plane is zero.

3. Normal and shear stresses may occur simultaneously on a plane

Which of the above statements is /are correct?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

GPSC ARTO Pre 30.12.2018
ESE 2009

Ans. (d) : The correct statements are –

(i) Two –dimensional stress applied to a thin plate in its own plane represents the plane stress condition.

(ii) Under plane stress condition, the strain in the direction perpendicular to the plane is not zero.

(iii) Normal and shear stress may occurs simultaneously on a plane.

419. A piece of material is subjected, to two perpendicular tensile stresses of 70 MPa and 10 MPa. The magnitude of the resultant stress on a plane in which the maximum shear stress occurs is

(a) 70 MPa (b) 60 MPa
(c) 50 MPa (d) 10 MPa

GPSC EE Pre, 28.01.2017
ESE 2012

Ans. (c) : Given-

$\sigma_x = 70$ MPa
$\sigma_y = 10$ MPa
$\tau_{xy} = 0$
$\theta = 45°$

Normal stress (σ_x)

$$\sigma_x = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_x = \frac{70+10}{2} + \frac{70-10}{2} \times \cos 90° + 0 \times \sin 90°$$

$$(\sigma_x) = \frac{80}{2} = 40 \text{ MPa}$$

Shear stress (τ) $= \frac{\sigma_x - \sigma_y}{2}\sin 2\theta - \tau_{xy}\cos 2\theta$

$$= \frac{70-10}{2}\sin(90°) - 0 \times \cos(90°)$$

$$\tau = \frac{60}{2} = 30 \text{ MPa}$$

Resultant stress (σ_r) $= \sqrt{(\sigma_x)^2 + (\tau)^2}$

$$= \sqrt{(40)^2 + (30)^2} = \sqrt{1600 + 900}$$

$= 50$ MPa

420. A horizontal beam under bending has a maximum bending stress of 100 MPa and maximum shear stress of 20 MPa. What is the maximum principal stress in the beam?

(a) 20 (b) 50
(c) $50 + \sqrt{2900}$ (d) 100

HPPSC Asstt. Prof. 18.09.2017
CSE Pre-2004

Ans. (c) : Given,

$\sigma_b = 100$ MPa
$\tau_{max} = 20$ MPa

Max. Principle stress, $\sigma = \frac{\sigma_b}{2} + \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau_{xy}^2}$

$$= \left(\frac{100}{2}\right) + \sqrt{\left(\frac{100}{2}\right)^2 + (20)^2}$$

$$= 50 + \sqrt{(50)^2 + (20)^2}$$

$$\sigma = 50 + \sqrt{2900} \text{ MPa}$$

421. What is the normal stress on a plane inclined at 45° to the axis of a square rod of side a subjected to an axial tensile force of T

(a) T/a^2 (b) $T/2a^2$
(c) $T/4a^2$ (d) $T/8a^2$

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) :

$$\sigma_\theta = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_x = \frac{T}{a^2}$$

$\sigma_y = 0$, $\tau_{xy} = 0$

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$$\therefore \sigma_\theta = \frac{\sigma_x}{2} + \frac{\sigma_x}{2}\cos 2\theta$$

$$\sigma_{45^\circ} = \frac{T}{2a^2} + \frac{T}{2a^2}\cos 90^\circ$$

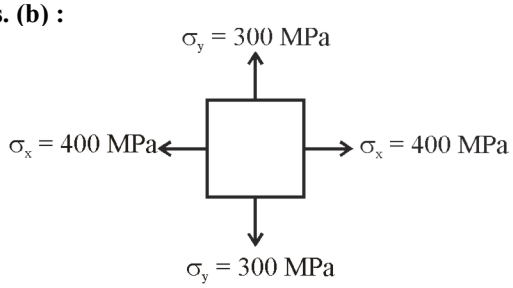
$$\sigma_{45^\circ} = \frac{T}{2a^2}$$

422. Principal stresses at a point in a stressed solid are 400 MPa and 300 MPa respectively. The normal stress on plane inclined at ± 45° to the principal planes will be

(a) 200 MPa and 5 MPa
(b) 350 MPa and both planes
(c) 100 MPa and 600 MPa
(d) 1500 Mpa and 500 MPa

TSPSC AEE 28.08.2017 (Civil/Mechanical), CSE Pre 2000

Ans. (b) :



$$\sigma_\theta = \frac{(\sigma_x + \sigma_y)}{2} + \frac{\sigma_x + \sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$\tau_{xy} = 0$

Normal stress on plane inclined at 45°

$$\sigma_\theta = \frac{400+300}{2} + \frac{400+300}{2} \times \cos 90^\circ$$

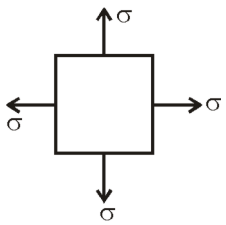
$= 350$ MPa on both planes.

423. When the two principal stresses are equal and like, the resultant stress on any plane is

(a) equal to principal stress
(b) zero
(c) one half of the principal stress
(d) one third of the principal stress

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : When two principal stresses are equal and like



Resultant stress on any plane

$$\sigma_R = \sqrt{\sigma_\theta^2 + \tau_\theta^2}$$

$\sigma_\theta = \sigma$ (irrespective of any plane)

$\tau_\theta = 0$

$\sigma_R = \sigma$ (Magnitude of principal stress)

424. The ratio of the maximum shear stress to the difference of the two principal stresses is

(a) 1/2 (b) 1/3
(c) 1/4 (d) 1.6

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : Maximum shear stress on any plane

In plane $\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$

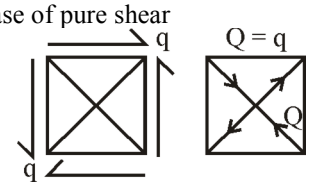
$$\frac{\tau_{max}}{\sigma_1 - \sigma_2} = \frac{\sigma_1 - \sigma_2}{2(\sigma_1 - \sigma_2)} = 1 : 2$$

425. A specimen is subjected to pure shear, the shear stress being q. Tensile and compressive stresses of intensity Q occur on planes inclined at 45° to the shear stress. What is the value of the ratio Q/q?

(a) 2 (b) 1.5
(c) 1.25 (d) 1

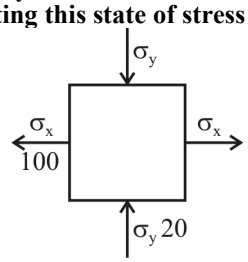
TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : Case of pure shear



$$\therefore \frac{Q}{q} = 1$$

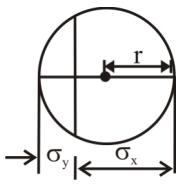
426. Following figure shows the state of stress at a certain point in a stressed body. The magnitudes of normal stresses in the x and y directions are 100 MPa and 20 MPa respectively. The radius of Mohr's stress circle representing this state of stress is



(a) 120 (b) 80
(c) 60 (d) 40

TSPSC AEE 2017

Ans. (c) :



$$r = \frac{\sigma_x - \sigma_y}{2} = \frac{100 - (-20)}{2} = 60\,\text{MPa}$$

So, Radius of Mohr's circle = 60 MPa

427. Under which loading condition Mohr's circle touches the origin of shear stress axis and having one positive principal stress?

(a) Uniaxial compression
(b) Uniaxial tension
(c) Pure torsion
(d) None of the above

Punjab PSC SDE 12.02.2017

Ans. (b) : For Uniaxial Tension Mohr's circle touches the origin of shear stress axis and having one positive principal stress.

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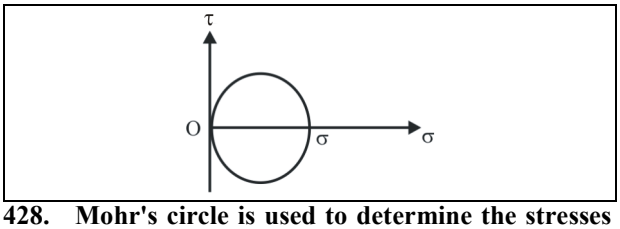


428. Mohr's circle is used to determine the stresses on an oblique section of a body subjected to :
(a) Direct tensile stress in one plane accompanied by a shear stress
(b) Direct tensile stress in two mutually perpendicular directions
(c) Direct tensile stress in two mutually perpendicular directions accompanied by a simple shear stress
(d) All of the above

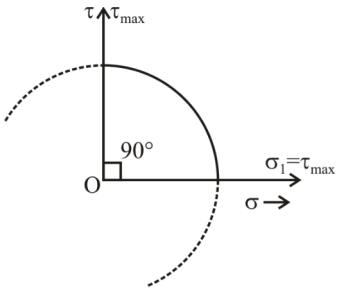
Punjab PSC SDE 12.02.2017

Ans. (d) : Mohr's circle is used to determine the stresses on an oblique section of a body subjected to.
(i) Direct tensile stress in one plane accompanied by a shear stress.
(ii) Direct tensile stress in two mutually perpendicular directions.
(iii) Direct tensile stress in two mutually perpendicular directions accompanied by a simple shear stress.

429. A shaft with a circular cross-section is subjected to pure twisting moment. The ratio of the maximum shear stress to the largest principal stress is
(a) 2.0 (b) 1.0
(c) 0.5 (d) 0

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) :



Pure twisting means pure shear, we have

$$\sigma_1 = \tau_{max}$$

Where, σ_1 is largest principle stress.
So, Ratio is equal to 1.

430. At a point in a bi-axially loaded member, the principal stresses are found to be 60 MPa and 80 MPa. If the critical stress of the material is 240 MPa, what could be the factor of safety according to the maximum shear stress theory?
(a) 2 (b) 3
(c) 4 (d) 5

JWM 2017

Ans. (b) : Principle stress, $\sigma_1 = 80$ MPa
$\sigma_2 = 60$ MPa
$\sigma_3 = 0$ MPa
Critical stress, $\sigma_y = 240$ MPa
According to maximum shear stress theory-

$$\left[\left|\frac{\sigma_1-\sigma_2}{2}\right| \text{or} \left|\frac{\sigma_2-\sigma_3}{2}\right| \text{or} \left|\frac{\sigma_3-\sigma_1}{2}\right|\right]_{max} = \frac{\sigma_y}{2\times N}$$

$$\frac{80}{2} = \frac{240}{2\times N}$$

Factor of safety N = 3

431. The principle stresses σ_1, σ_2 and σ_3 at a point respectively are 80 MPa, 30 MPa and – 40 MPa. The maximum shear stress is:
(a) 60 MPa (b) 55 MPa
(c) 35 MPa (d) 25 MPa

TRB Polytechnic Lecturer 2017

Ans. (a) : Given as,
$\sigma_1 = 80$ MPa
$\sigma_2 = 30$ MPa
$\sigma_3 = -40$ MPa
We know that maximum shear stress will be,
$\tau_{max.}$ = Maximum of

$$\left|\left(\frac{\sigma_1-\sigma_2}{2}\right),\left(\frac{\sigma_2-\sigma_3}{2}\right),\left(\frac{\sigma_3-\sigma_1}{2}\right)\right|$$

$$= \text{Max. of } \left|\left[\frac{80-30}{2}\right],\left[\frac{30-(-40)}{2}\right],\left[\frac{-40-80}{2}\right]\right|$$

$\tau_{max.} = 60$ MPa

432. Consider a plane stress case, where $\sigma_x = 3$ Pa, $\sigma_y = 1$ Pa and $\tau_{xy} = 1$ Pa. One of the principal directions w.r.t. x-axis would be
(a) 0° (b) 15°
(c) 22.5° (d) 45°

BPSC AE Mains 2017 Paper - VI

Ans : (c) : Given,
$\sigma_x = 3$ Pa
$\sigma_y = 1$ Pa
$\tau_{xy} = 1$ Pa

$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x-\sigma_y} = \frac{2\times 1}{3-1}$$

$$= 1$$

$\tan 2\theta = \tan 45°$
$\theta = 22.5°$

433. An elastic body is subjected to a tensile stress X in a particular direction and a compressive stress Y in its perpendicular direction. X and Y are unequal in magnitude. On the plane of maximum shear stress in the body there will be:
(a) No normal stress
(b) Also the maximum normal stress
(c) The minimum normal stress
(d) Both normal and shear stress

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : Maximum shear stress plane
Normal stress on maximum shear stress plane

$$= \frac{\sigma_1+\sigma_2}{2} = \frac{\sigma_x+\sigma_y}{2}$$

Shear stress on maximum shear stress plane
$(\tau_{max})_{in\ plane} = \pm$ Radius of Mohr's circle.

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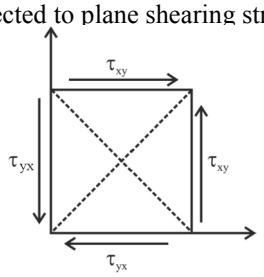
434. The maximum tangential stress acting on a plane inclined to the direction of the axial stress (P) in a tensile bar is equal to
(a) P (b) 2P
(c) P/2 (d) 3P/2
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : We know that maximum tangential stress occur at 45°, So the Axial Stress P/2→ R_{max} for 45°.

435. If an element in a body is in equilibrium under shearing stresses only, such a state of stress is called
(a) Pure rotation
(b) Pure shear
(c) Pure bending
(d) None of the given answers
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : When only two Normal stresses are present the state of stress is called Biaxial. In pure shear, the element is subjected to plane shearing stresses only.



436. An element is subjected to σ_x = 35 (Mpa (tensile) and σ_y = 20 MPa (tensile) and shear stress τ = 7.5 MPa. Then the direction of principal stresses is
(a) 45° (b) 22.5°
(c) 30° (d) 15°
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : We know that,

$$\tan 2\theta = \frac{2\tau}{\sigma_1 - \sigma_2}$$

$$\tan 2\theta = \frac{2\times 7.5}{35-20}$$

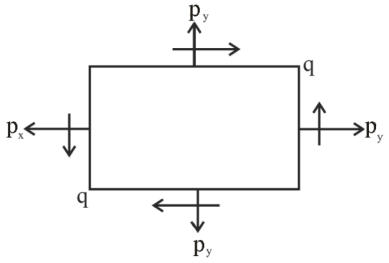
$$\tan 2\theta = 1$$

$$2\theta = 45^\circ \Rightarrow \theta = 22.5^\circ$$

437. If normal stresses of same nature p_x and p_y and shear stress q are acting on two perpendicular plans and $q = (p_x\, p_y)^{1/2}$, then the major and minor principal stresses respectively are
(a) $p_x + p_y$ and $p_x - p_y$
(b) p_x and $p_x - p_y$
(c) $0.5\,(p_x + p_y)$ and $0.5(p_x - p_y)$
(d) $p_x + p_y$ and zero
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) :

$$\left.\begin{matrix}\sigma_1\\ \sigma_2\end{matrix}\right\} = \frac{p_x + p_y}{2} \pm \sqrt{\frac{(p_x - p_y)^2}{4} + q^2}$$



$$= \frac{p_x + p_y}{2} \pm \sqrt{\frac{p_x^2 + p_y^2 - 2p_xp_y + 4p_xp_y}{4}}$$

$$= \frac{p_x + p_y}{2} \pm \sqrt{\left(\frac{p_x + p_y}{2}\right)^2}$$

$$\sigma_1 = \frac{p_x + p_y}{2} + \frac{p_x + p_y}{2} = p_x + p_y$$

$$\sigma_2 = \frac{p_x + p_y}{2} - \frac{p_x + p_y}{2} = 0$$

438. The maximum shearing stress produced by shrinkfit between two cylinders (mounted one inside the other) of $E=2\times10^6$ kg(f)/m^2 and shrinkage factor = 0.002, is equal to
(a) 2000 kg(f)/m^2 (b) 2500 kg(f)/m^2
(c) 4000 kg(f)/m^2 (d) 1000 kg(f)/m^2
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : j

439. The three dimensional state of stress at a point is given by:

$$[\sigma] = \begin{bmatrix} 60 & 20 & -20 \\ 20 & 0 & 40 \\ -20 & 40 & 0 \end{bmatrix} \text{MN/m}^2$$

What is the shear stress in the xy-plane at the same point?
(a) –20 MN/m^2 (b) –40 MN/m^2
(c) 20 MN/m^2 (d) 40 MN/m^2
(e) 300 MN/m^2
CGPSC Asstt. Workshop Supt., 17.07.2016

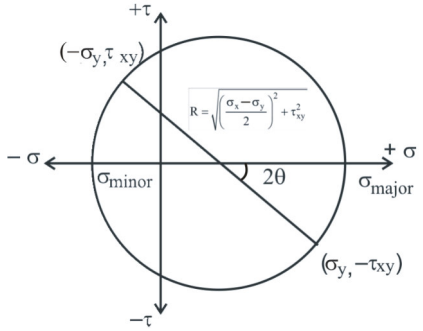
Ans. (c) : As, we know,
Three dimensional state of stress, of a point is given by–

$$\begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yx} \\ \tau_{zx} & \tau_{xy} & \sigma_z \end{bmatrix} = \begin{bmatrix} 60 & 20 & -20 \\ 20 & 0 & 40 \\ -20 & 40 & 0 \end{bmatrix}$$

So, $\tau_{yx} = \tau_{xy} = 20$ MN/m^2

440. When the principal stresses are unlike, in Mohr's circle, the normal resultant stresses will be
(a) Negative for points on the circle to the left of the vertical axis through the origin
(b) Negative for points on the circle to the right of the vertical axis through the origin
(c) Positive for points on the circle to the left of the vertical axis through the origin
(d) Either negative or positive depends on the shear stress and radius of the Mohr's circle.
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Mohr's circle for unlike principle stresses–



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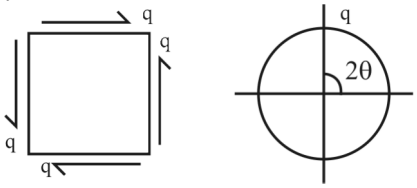
When the principal stresses are unlike in Mohr's circle, the normal and resultant stresses will be negative for points on the circle to the left of the vertical axis through the origin.

441. Complementary shear stresses of intensity q are induced at a point on a body, the plane carrying the maximum shear stress are at the following angle with the principle planes

(a) 45^0 (b) 30^0
(c) 60^0 (d) 90^0

APPSC AE Subordinate Service Civil/Mech. 2016
HPPSC Workshop Suptd. 08.07.2021
HPPSC Lect. 2016

Ans. (a) :



From mohr circle for max shear

$2\theta = 90^0$

$\theta = 45^0$

Alternatively by analytical method

$$\tau_{xy} = \frac{\sigma_1 - \sigma_2}{2} \times \sin 2\theta$$

For τ_{xy} to be max,

$\sin 2\theta = 1$ (Numerically)

$2\theta = 90^0$ or 270^0

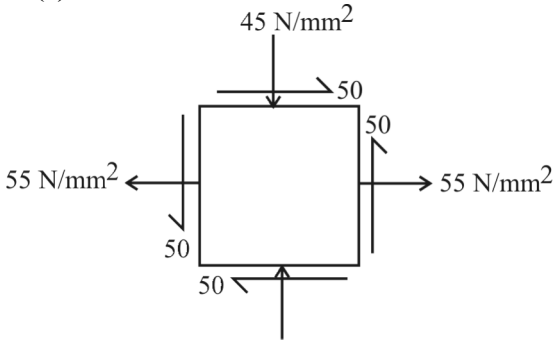
$\theta = 45^0$ or 135^0

442. The normal stresses of 55 N/mm^2 tensile and 45 N/mm^2 compressive on two mutually perpendicular planes are acting at a point in a piece of elastic material. These planes also carry shear stresses of 50 N/mm^2. The minor principal stress will be

(a) 65.71 N/mm^2 compressive
(b) 75.71 N/mm^2 tensile
(c) 65.71 N/mm^2 tensile
(d) 75.71 N/mm^2 compressive

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



Sign conversion, Tensile = + ve

Comp. = –ve

$$\sigma_{major/minor} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau^2_{xy}}$$

Considering positive sign for major principle stress and negative sign for minor principle stress.

$\sigma_x = 55$ N/mm^2, $\sigma_y = -45$ N/mm^2, $\tau_{xy} = 50$ N/mm^2

$$\sigma_{minor} = \frac{55 + (-45)}{2} - \sqrt{\left[\frac{55 - (-45)}{2}\right]^2 + 50^2}$$

$\sigma_{minor} = 5 - 70.71 = -65.71$ N/mm^2

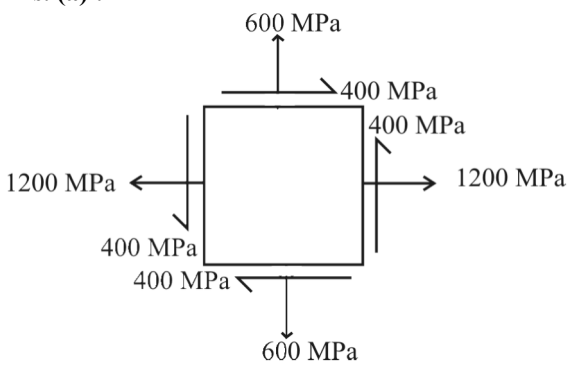
$\boxed{\sigma_{minor} = 65.71 \text{ N/mm}^2 \text{ (compressive)}}$

443. A body is subjected to tensile stress of 1200 MPa on one plane and another tensile stress of 600 MPa on a plane at right angles to the former. It is also subjected to a shear stress of 400 MPa on the same planes. The maximum shear stress will be

(a) 500 MPa (b) 1400 MPa
(c) 900 MPa (d) 400 MPa

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$$

$$\sigma_1 = \sigma_{major} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{1200 + 600}{2} + \sqrt{\left(\frac{1200 - 600}{2}\right)^2 + 400^2}$$

$= 900 + 500 = 1400$ MPa

$$\sigma_2 = \sigma_{minor} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right) - \tau^2_{xy}}$$

$= 900 - 500 = 400$ MPa

$$\tau_{max} = \frac{1400 - 400}{2} = 500 \text{ MPa}$$

444. For an inclined plane in a rectangular block subjected to two mutually perpendicular normal stresses 100 MPa and 400 MPa and shear stresses 400 MPa, the maximum normal stress will be

(a) 1200 MPa (b) 700 MPa
(c) 600 MPa (d) 200 MPa

APPSC AEE Screening Test 2016

Ans. (*) : Given,

$\sigma_x = 100$ MPa

$\sigma_y = 400$ MPa

$\tau_{xy} = 400$ MPa

Maximum normal stress $(\sigma_1)_{max}$

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$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{100 + 400}{2} \pm \sqrt{\left(\frac{(100 - 400)}{2}\right)^2 + (400)^2}$$

$$= 250 \pm \sqrt{(150)^2 + (400)^2}$$

σ_1 = 677.2 MPa (maximum normal stress)

σ_2 = 177.2 MPa (minimum normal stress)

445. The principal strains at a point in a body under biaxial state of stress are 1000×10^{-6} and -600×10^{-6}. What is the maximum shear strain at that point?

(a) 200×10^{-6} (b) 800×10^{-6}

(c) 1000×10^{-6} (d) 1600×10^{-6}

RPSC 2016
ESE 2009
GATE 1996

Ans : (d) Given,

$\in_1 = 1600 \times 10^{-6}$

$\in_2 = -600 \times 10^{-6}$

Maximum Shear strain is given by

$$\frac{\gamma_{max}}{2} = \frac{\in_1 - \in_2}{2}$$

$$\therefore \gamma_{max} = \in_1 - \in_2$$

$$= 1000 \times 10^{-6} - (-600 \times 10^{-6})$$

$$= 1600 \times 10^{-6}$$

446. Mohr's circle construction is valid for both stresses as well as the area moment of inertia, because

(a) both are tensors of first-order

(b) both are tensors of second- order

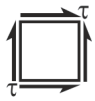
(c) both are axial vectors

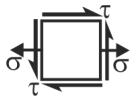
(d) both occur under plane stress condition

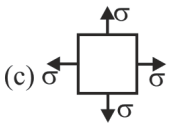
BPSC Poly. Lect. 2016

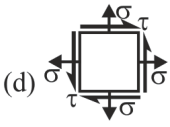
Ans : (b) Mohr's circle construction is valid for both stresses as well as the area moment of inertia because both are tensors of second order.

447. In which of the following two dimensional state of stress, Mohr's stress circle takes the shape of a point.

(a) 

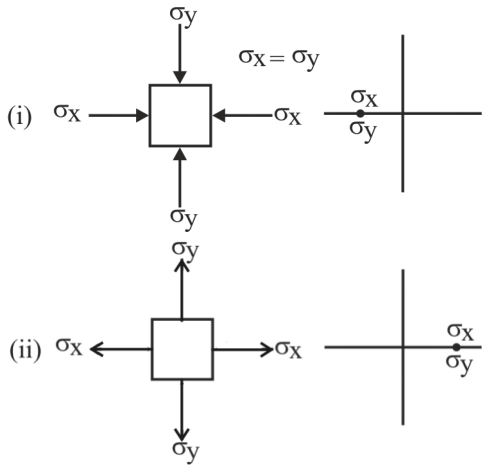
(b) 

(c) 

(d) 

UPPSC AE 12.04.2016 Paper-I
CSE Pre-1996

Ans : (c) Draw Mohr's circle like equal normal stress without shear stress :-



In both condition Mohr's stress circle takes the shape of a point.

448. The state of stress at a point under plane stress conditions is σ_{xx} = 40 MPa, σ_{yy} = 100 MPa; τ_{xy} = 40 MPa (notations have usual meaning). The radius of the Mohr's circle representing the given state of stress in MPa is

(a) 40 (b) 50

(c) 60 (d) 100

OPSC AEE 2019 Paper-I
RPSC ADE 2016
GATE 2012, ESE 2005

Ans. (b) : Radius of Mohr's circle = max shear stress

$$= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \sqrt{\left(\frac{40 - 100}{2}\right)^2 + 40^2}$$

= 50 MPa

449. On principal plane the principal stress is :

(a) Maximum/Minimum

(b) Zero

(c) Unity

(d) Depends on applied strain.

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : Principle plane is a plane on which principal stress is either maximum or minimum but shear stress on this is plane is zero. such plane is known as principal plane or plane of zero shear stress.

450. The Mohr's circle for the following loading condition has center coincide with the origin of the normal stress-shear stress co-ordinate system

(a) Uni-axial tension

(b) Uni-axial compression

(c) Bi-axial tension

(d) Pure torsion

GPSC Asstt. Prof. (Prod.) 21.08.2016

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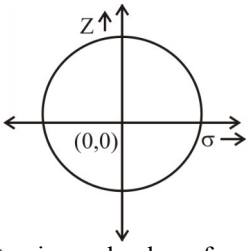
Ans. (d) :

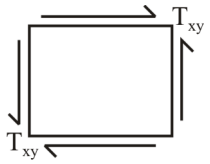


In case of pure torsion only shear force will come into play.
Then centre of circle (Mohr's) will be origin of stress shear stress co-ordinate system.

451. A cylindrical elastic body subjected to pure torsion about its axis develops :
(a) Tensile stress in a direction 45° to the axis
(b) No tensile or compressive stress
(c) Max shear stress along the axis of shaft
(d) Max shear stress at 45° to the axis

HPPSC Asstt. Prof. 18.11.2016

Ans. (a) :



$$\sigma_{1,2} = \frac{1}{2}\left[\sigma_x + \sigma_y \pm \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}\right]$$

$$\sigma_{1,2} = \frac{1}{2}\left[0 \pm \sqrt{4\tau_{xy}^2}\right]$$

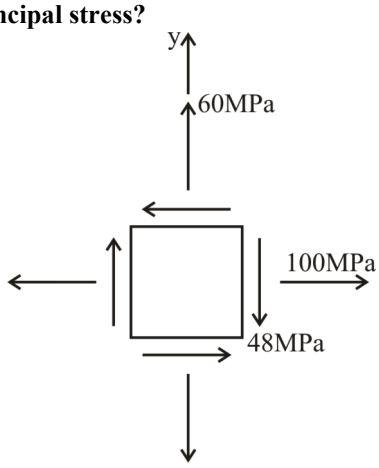
$$\sigma_{1,2} = \pm \tau_{xy}$$

$$\tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \infty$$

$$2\theta = \frac{\pi}{2}$$

$$\theta = 45°$$

452. For the state of plane surface shown, determine the principal planes and the principal stress?



(a) $\theta_p = 35.7°$ and $\sigma_{max} = +132$ MPa, $\sigma_{min} = +28$ MPa
(b) $\theta_p = 33.7°$ and $\sigma_{max} = +122$ MPa, $\sigma_{min} = +28$ MPa
(c) $\theta_p = 33.7°$ and $\sigma_{max} = +132$ MPa, $\sigma_{min} = +22$ MPa
(d) $\theta_p = 33.7°$ and $\sigma_{max} = +132$ MPa, $\sigma_{min} = +28$ MPa
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : As we know that,
Principal stresses are :

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

And $$\tan 2\theta_p = \left(\frac{2\tau_{xy}}{\sigma_x - \sigma_y}\right)$$

So, $$\theta_p = \frac{1}{2}\tan^{-1}\left(\frac{2\times 48}{100-60}\right) = 33.7°$$

$$\sigma_{1,2} = \frac{100+60}{2} \pm \sqrt{\left(\frac{100-60}{2}\right)^2 + 48^2}$$

$\sigma_{max} = 132$ MPa $\quad$ $\sigma_{min} = 28$ MPa

453. At a point in two-dimensional stress system $\sigma_x = 100$ N/mm^2, $\sigma_y = \tau_{xy} = 40$ N/mm^2. What is the radius of the Mohr circle for stress drawn with a scale of : 1 cm = 10 N/mm^2?
(a) 3 cm (b) 4 cm
(c) 5 cm (d) 6 cm
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ESE 2005

Ans. (c) : Given
$\sigma_x = 100$ N/mm^2
$\sigma_y = \tau_{xy} = 40$ N/mm^2

Radius Mohr's circle, $(R) = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

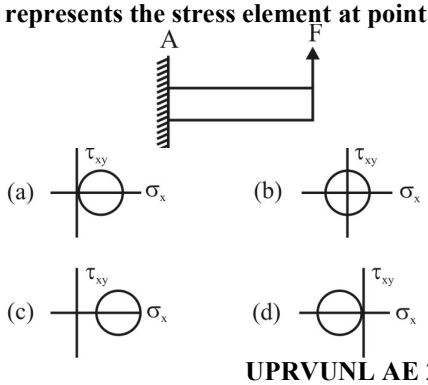
$$= \sqrt{\left(\frac{100-40}{2}\right)^2 + (40)^2}$$

$$= \sqrt{(30)^2 + (40)^2}$$

$$= \sqrt{900+1600} = \sqrt{2500}$$

$= 50$ N/mm^2
[$\because$ given, 1cm = 10N/mm^2]
$\boxed{R = 5\text{cm}}$

454. A cantilever beam is loaded as shown here. Which one of the Mohr's circle shown here represents the stress element at point A?



UPRVUNL AE 21.08.2016

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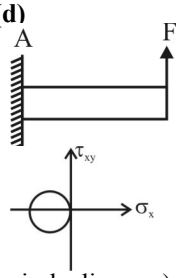
Ans. : (d)



(Mohr's circle diagram)

455. A body is subjected to a pure tensile stress of 200 units. What is the maximum shear produced in the body at some oblique plane due to the above?

(a) 0 units (b) 50 units
(c) 100 units (d) 150 units
(e) 200 units

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Maximum shear stress

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

here $\sigma_y = 0$, $\tau_{xy} = 0$, $\sigma_x = 200$ units

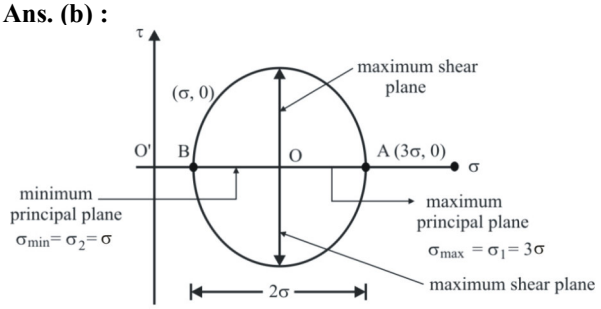
$$\tau_{max} = \frac{\sigma_x}{2} = \frac{200}{2} = 100 \text{ units}$$

456. Plane stress at a point in a body is defined by principal stresses 3σ and σ. The ratio of the normal stress to the maximum shear stresses on the plane the maximum shear stress is :

(a) 1 (b) 2
(c) 3 (d) 4

TSPSC Managers, 2015
ESE 2000

Ans. (b) :



Maximum shear stress $(\tau_{max}) = \frac{3\sigma - \sigma}{2} = \sigma$

Normal stress $(\sigma_n) = \frac{3\sigma + \sigma}{2} = 2\sigma$

(At maximum shear stress) Now,

$$\frac{\sigma_n}{\tau_{max}} = \frac{2\sigma}{\sigma} = 2$$

457. The state of stress at a point is given as

$\sigma_x = 100 N/mm^2$

$\sigma_y = 40 N/mm^2$

and $\tau_{xy} = 40 N/mm^2$

If the yield strength S_{yt} of the material is 300 MP_a, the factor of safety using maximum shear stress

(a) 3 (b) 2.5
(c) 7.5 (d) 1.25

VIZAG MT 2015

Ans. (a) : Given,

$\sigma_x = 100 N/mm^2$

$\sigma_y = 40 N/mm^2$

$\tau_{xy} = 40 N/mm^2$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{100 + 40}{2} \pm \sqrt{\left(\frac{100 - 40}{2}\right)^2 + (40)^2}$$

$$= 70 \pm \sqrt{(30)^2 + (40)^2}$$

$\sigma_{1,2} = 70 \pm 50$

$\sigma_1 = 120 N/mm^2$

$\sigma_2 = 20 N/mm^2$

$$\frac{\sigma_1 - \sigma_2}{2} \le \frac{S_{yt}}{2 FOS}$$

$$(120 - 20) \le \frac{300}{FOS}$$

$$FOS = \frac{300}{100}$$

$\boxed{FOS = 3}$

458. A machine element is subjected to the normal stress of 80 MPa and shear stress of 60 MPa. The yield strength of material is 320 MPa. The factor of safety using maximum shear stress theory will be

(a) 2.85 (b) 4
(c) 2.29 (d) none of these

VIZAG MT 2015

Ans. (c) : Given $\sigma_x = 80 MPa$

$\sigma_y = 0$

$\tau_{xy} = 60 MPa$

$$\sigma_{1,2} = \frac{80 - 0}{2} \pm \sqrt{\left(\frac{80 - 0}{2}\right)^2 + (60)^2}$$

$\sigma_1 = 112.11 N/mm^2$

$\sigma_2 = -32.11 N/mm^2$

$$\frac{\sigma_1 - \sigma_2}{2} \le \frac{S_{yt}}{2FOS}$$

$$\frac{112.11 - (-32.11)}{2} \le \frac{320}{2FOS}$$

$$FOS = \frac{320}{144.22}$$

$FOS = 2.21 \approx 2.29$

The closest answer is option (c).

459. For any state of stress it is always possible to define a new co-ordinate system, which has axis perpendicular to the planes on which the .....

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(a) maximum compressive stress act
(b) maximum shear stress act
(c) maximum normal stress act
(d) None of the above

KPSC ADF 2015

Ans. (b) : The orientations of maximum and minimum shear planes are at right angles to each other.
Maximum shear stress, it can be seen that the planes of maximum and minimum shear stress occur at an angle of 45^0 and 135^0 with respect to the plane of maximum and minimum normal stress respectively.

460. The co-ordinate of any point on Mohr's circle represent :
(a) State of stress at a point with reference to any arbitrary set of orthogonal axes passing through that point
(b) Principal stresses at a point
(c) One of the two direct stresses and shearing stress at a point
(d) Two direct stresses at a point

(HPPSC AE 2014)

Ans : (c) The coordinate of any point on Mohr's circle represent one of the two direct stresses and shearing stress at a point.

461. Ellipse of stress can be drawn only when a body is acted upon by :
(a) one normal stress
(b) two normal stresses
(c) one shear stress
(d) two normal stresses and one shear stress

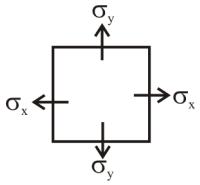
(HPPSC AE 2014)

Ans : (b) Ellipse of stress is used to find resultant stress and the angle of obliquity on any plane within a stressed body. In 2-D, it is called Ellipse of stress. In 3-D it is called Ellipsoid of stress.
The axis of ellipse are the two principle stresses.

462. State of plane stress in x-y plane is accompanied by strains along:
(a) x, y and z- axes (b) x and y- axes
(c) x and z- axes (d) y and z- axes

HPPSC Asstt. Prof. 2014

Ans. (a) :

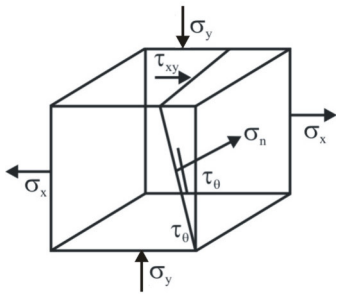


stresses
In plane stress condition, $\sigma_x \neq 0$, $\sigma_y \neq 0$, but $\sigma_z = 0$
and strains
$\varepsilon_x \neq 0$, $\varepsilon_y \neq 0$ & $\varepsilon_z \neq 0$
i.e. stain along x, y and z all exist.

463. Normal stresses of equal magnitude "p" but of opposite signs, act at a point of strained material in perpendicular direction. What is the magnitude of the resultant normal stress on plane inclined at 45^0 to the applied stresses?
(a) 2p (b) p/2
(c) p/4 (d) 0

UPRVUNL AE 2014, ESE 2005

Ans : (d)



$$(\sigma_n)_{max} = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$(\sigma_n)_{max}$ = Maximum Normal stress

τ_{xy} = Shear stress

Given $\sigma_x = \sigma_y = p$ and $\theta = 45^0$

$\sigma_x = p, \quad \sigma_y = -p$

$$(\sigma_n)_{max} = \left(\frac{p-p}{2}\right) + \left(\frac{p+p}{2}\right)\cos 90^0 - 0$$

$$\boxed{(\sigma_n)_{max} = 0}$$

464. The shear strength of every materials tends to be about ____ their tensile strength
(a) half (b) equal
(c) double (d) one third

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : Shear strength

$$\tau_{sy} = \frac{\sigma_{yt}}{2} = 0.5\sigma_{yt} \text{ (Guest \& Tresca's theory)}$$

$$\tau_{sy} = \frac{\sigma_{yt}}{\sqrt{3}} = 0.577\sigma_{yt} \text{ (Von-Mises theory)}$$

465. Strain gauge transducers are used to measure :
(a) Tensile Torsion and Compressive forces
(b) Tensile forces alone
(c) Compressive forces alone
(d) Torsional forces alone

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Strain gauge transducers are used to measure tensile torsion and compressive forces.

466. The principal stresses at a point across two perpendicular planes are 75 MN/m^2 (Tensile) and 35 MN/m^2 (Tensile). Find the tangential stress and it's obliquity on a plane at 20° with the major principal plane :
(a) 40 sin 40° (b) 20 sin 40°
(c) 40 sin 20° (d) 20 sin 20°

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Given,
$\sigma_x = 75$ MPa (tensile)
$\sigma_y = 35$ MPa (tensile)

$$\tau = \left(\frac{\sigma_x - \sigma_y}{2}\right)\sin 2\theta$$

$$= \left(\frac{75-35}{2}\right)\sin(2\times 20) = 20 \sin 40°$$

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467. The normal stresses on two perpendicular planes through a point in a stressed material are 80 MPa (tensile) and 20 MPa (tensile) respectively. A shear stress of 40 MPa is also acting on these planes. What is the value of maximum shear stress at the point?

(a) 50 MPa (b) Zero
(c) 40 MPa (d) 60 MPa

RPSC AEN Pre-2013
GATE 2015

Ans. (a) : $\sigma_x = 80$ MPa (Tensile)
$\sigma_y = 20$ MPa (Tensile), $\tau_{xy} = 40$ MPa
Maximum shear stress τ_{max} in plane–

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$$

$$= \sqrt{\left(\frac{80-20}{2}\right)^2 + (40)^2}$$

$$= \sqrt{(30)^2 + (40)^2}$$

$$= 50 \text{ MPa}$$

468. A solid circular shaft is subjected to a maximum shear stress of 140MPa. Magnitude of maximum normal stress developed in the shaft is:-

(a) 60 MPa (b) 90 MPa
(c) 110 MPa (d) 140 MPa

UKPSC AE-2013, Paper-I
CSE Pre-1995
APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (d) : Maximum normal stress,

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$\sigma_x = 0, \sigma_y = 0, \tau_{xy} = 140 \text{ MPa}$$

$$\sigma_{1,2} = 0 \pm \frac{1}{2}\sqrt{4 \times (140)^2}$$

$$\sigma_{1,2} = \pm 140 \text{ MPa}$$

Then, Maximum normal stress is 140 MPa.

469. The state of plane-stress at a point is given by σ_x = –200 MPa, σ_y = 100 MPa and τ_{xy} = 100 MPa. The maximum shear stress in MPa is

(a) 111.8 (b) 150.1
(c) 180.3 (d) 223.6

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2010

Ans. (c) : Given, $\sigma_x = -200$ MPa, $\sigma_y = 100$ MPa
$\tau_{xy} = 100$ MPa

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tau_{max} = \sqrt{\left(\frac{-200-100}{2}\right)^2 + (100)^2}$$

$$= \sqrt{150^2 + 100^2}$$

$$\tau_{max} = 180.3 \text{ MPa}$$

470. A shaft subjected to torsion experiences a pure shear stress on the surface. The maximum principle stress on the surface which is at 45° to the axis will have a value

(a) $\tau\cos45°$ (b) $2\tau\cos45°$
(c) $2\tau\sin45°\cos45°$ (d) $\tau\cos^2 45°$

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2003

Ans. (c) : It is a pure shear stress case,
$\sigma_x = \sigma_y = 0$ and $\tau_{xy} = \tau$ and $\theta = 45°$
Maximum principal stress,

$$\sigma_p = \frac{(0+0)}{2} + \left(\frac{0-0}{2}\right)\cos(2\times 45°) + \tau \times \sin(2\times 45°)$$

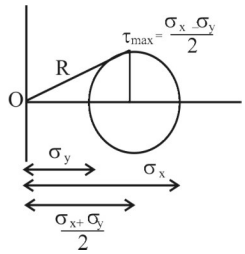
$\sigma_p = \tau.\sin2 \times 45°$
$\sigma_p = 2\tau.\sin45°.\cos45°$ $(\sin2\theta = 2.\sin\theta.\cos\theta)$
$\therefore$ The resulting normal stress (tensile or compressive) on planes inclined at 45° to the direction of the shear stresses is equal to τ.

471. If the principal stresses at a point in a strained body are σ_x and $\sigma_y (\sigma_x > \sigma_y)$, resultant stress on a plane carrying the maximum shear stress is equal to

(a) $\sqrt{\sigma_x^2 + \sigma_y^2}$ (b) $\sqrt{\sigma_x^2 - \sigma_y^2}$
(c) $\sqrt{\frac{\sigma_x^2 + \sigma_y^2}{2}}$ (d) $\frac{\sqrt{\sigma_x^2 - \sigma_y^2}}{2}$

APPSC AEE 2012

Ans : (c) If the principal stresses at a point in strained body are σ_x and σ_y then the resultant stress on a plane.
Center of Mohr's Circle



$$\left[\frac{\sigma_x + \sigma_y}{2}, 0\right]$$

Resultant stress on a plane

$$\sigma_R = \sqrt{\sigma_n^2 + \tau_{max}^2}$$

$$\sigma_R = \sqrt{\left[\frac{\sigma_x + \sigma_y}{2}\right]^2 + \left[\frac{\sigma_x - \sigma_y}{2}\right]^2}$$

$$\sigma_R = \sqrt{\left[\frac{\sigma_x^2 + \sigma_y^2 + 2\sigma_x\sigma_y + \sigma_y^2 + \sigma_y^2 - 2\sigma_x\sigma_y}{4}\right]}$$

$$\sigma_R = \sqrt{\frac{2\left[\sigma_x^2 + \sigma_y^2\right]}{4}} = \sqrt{\frac{\sigma_x^2 + \sigma_y^2}{2}}$$

472. Angle between the principal planes is

(a) 270^0 (b) 180^0
(c) 90^0 (d) 45^0

APPSC AEE 2012

Ans : (c) Angle between the principal plane is 90^0.

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473. $\sigma_x + \sigma_y = \sigma_{x'} + \sigma_{y'} = \sigma_1 + \sigma_2$
The above relation is called
(a) independency of normal stresses
(b) constancy of normal stresses
(c) first invariant of stress
(d) all the above three

UKPSC AE 2012 Paper-I

Ans. (c) : first invariant of stress

474. **When a body is subjected to direct tensile stresses (σ_x and σ_y) in two mutually perpendicular directions, accompanied by a simple shear stress τ_{xy}, then in Mohr's circle method, the circle radius is taken as**

(a) $\frac{\sigma_x - \sigma_y}{2} + \tau_{xy}$

(b) $\frac{\sigma_x + \sigma_y}{2} + \tau_{xy}$

(c) $\frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$

(d) $\frac{1}{2}\sqrt{(\sigma_x + \sigma_y)^2 + 4\tau_{xy}^2}$

UKPSC AE 2012 Paper-I

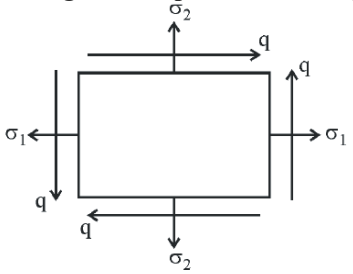
Ans. (c) : $R = \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$

475. **Choose the correct relationship in the given statements of Assertion (A) and Reason (R).**
Assertion (A) : A plane state of stress does not necessarily result into a plane state of strain.
Reason (R) : Normal stresses acting along X and Y directions will also result into strain along the Z-direction.
Code :
(a) Both (A) & (R) are correct. (R) is the correct explanation of (A).
(b) Both (A) & (R) are correct. (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

UKPSC AE 2012 Paper-I

Ans. (a) : Both (A) & (R) are correct. (R) is the correct explanation of (A).

476. **A body is subjected to two unequal like direct stresses σ_1 and σ_2 in two mutually perpendicular planes along with simple shear stress q**



Which among the following is then a wrong statement ?

(a) The principal stresses at a point are
$P_1, P_2 = \frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left[\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + q^2\right]}$

(b) The position of principal planes with the plane of stress σ_1, are
$\theta_1 = \frac{1}{2}\tan^{-1}\frac{2q}{\sigma_1 - \sigma_2}; \theta_2 = \theta_1 + 45°$

(c) Maximum shear stress is $(\sigma_t)_{max}$ = $\pm\sqrt{\left[\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + q^2\right]}$

(d) Planes of maximum shear are inclined at 45° to the principal planes.

UKPSC AE 2012 Paper-I

Ans. (b) : The position of principal planes with the plane of stress σ_1, are
$\theta_1 = \frac{1}{2}\tan^{-1}\frac{2q}{\sigma_1 - \sigma_2}; \theta_2 = \theta_1 + 45°$

477. **A tension member with a cross-sectional area of 30 mm^2 resists a load of 60 kN. What is the normal stress induced on the plane of maximum shear stress ?**
(a) 2 kN/mm^2 (b) 1 kN/mm^2
(c) 4 kN/mm^2 (d) 3 kN/mm^2

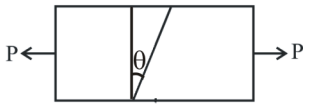
UKPSC AE 2012 Paper-I

Ans. (b) : Given, A = 30 mm^2
P = 60 kN
$\sigma_x = \frac{P}{A} = \frac{60}{30} = 2\ N/mm^2$
$\sigma_y = 0$
$\tau_{max} = \frac{\sigma_x - \sigma_y}{2} = \frac{2-0}{2} = 1 kN/mm^2$

478. **In an axially loaded bar of magnitude P and cross sectional Area A, the shear stress on an inclined plane at an angle θ is**
(a) Zero (b) P/A
(c) P/A cos (2 θ) (d) P/A sin θ cos θ

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : The shear stress



$\tau = \frac{\sigma_x - \sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta$

$\tau = \frac{P}{2A}\sin 2\theta \qquad \left\{\sigma_x = \frac{P}{A},\ \sigma_y \text{ and } \tau_{xy} = 0\right\}$

$\tau = \frac{P}{2A}(2\sin\theta.\cos\theta)$

$\tau = \frac{P}{A}\sin\theta.\cos\theta$

479. **If the Mohr's circle for a state of stress becomes a point, the state of stress is**
(a) Pure shear state of stress
(b) Uniaxial state of stress

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(c) Identical principal stresses
(d) None of the above

UKPSC AE 2012 Paper-I

Ans. (c) : Identical principal stresses

480. In a stressed field, the change in angle between two initially perpendicular lines is called
(a) Normal strain (b) Shear strain
(c) Principal strain (d) Poisson's ratio

UKPSC AE 2012 Paper-I

Ans. (b) : Shear strain

481. A body is subjected to a pure tensile stress of 100 units. What is the maximum shear produced in the body at some oblique plane due to the above?
(a) 100 units (b) 75 units
(c) 50 units (d) 0 unit

TRB Asstt. Prof., 2012
ESE 2006

Ans. (c) :

$\sigma = 100$ units

Max shear stress, $\tau_{max} = \frac{\sigma}{2} = \frac{100}{2}$

$\tau_{max} = 50$ Units

482. At a point on a plane there is normal stress + 100 MPa and a shear stress 'q', on another plane perpendicular to this plane there is normal stress + 20 MPa and a shear stress 'q'. If the maximum principal stress at the point is 110 MPa, what is maximum shear stress at the point :
(a) 50 MPa (b) 30 MPa
(c) 10 MPa (d) None of these

PSPCL AE, 2012

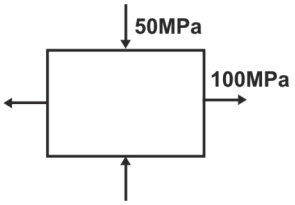
Ans. (a) : Max principal stress,

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \text{radius of mohr circle}$$

$$110 = \frac{100+20}{2} + \text{Radius}$$

Radius of mohr's circle = max. shear stress = 50 MPa

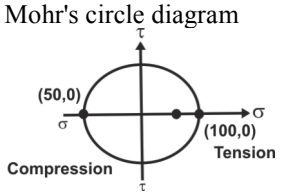
483. For the state of stress shown in the above figure, normal stress acting on the plane of maximum shear stress is



(a) 25 MPa tension
(b) 75 MPa compression
(c) 25 MPa compression
(d) 75 MPa tension

ISRO Scientist/Engineer 2011
ESE 2015

Ans. (a) : State of stress
Mohr's circle diagram



Normal stress on plane of maximum shear stress
= Normal stress at the centre of mohr circle
= Radius of Mohr circle

$$= \frac{100-50}{2} = \frac{50}{2}$$

= 25 MPa

484. A short cast iron cylindrical piece of cross-sectional area 100 mm^2 and length 25 mm is tested in compression and fails at a load of 50 kN. The shear strength of cast iron is.
(a) 500 N/mm^2 (b) 400 N/mm^2
(c) 300 N/mm^2 (d) 250 N/mm^2

UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know,
Maximum shear stress given by–

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} \quad \Rightarrow \quad \therefore \ \sigma_2 = 0$$

$$\tau_{max} = \frac{\sigma_1}{2} = \frac{50\times10^3}{2\times100} = 250\,\text{N/mm}^2 \quad [\sigma_1 = P/A]$$

485. The two-dimensional state of stress at a point is σ_x = 100 N/mm^2, σ_y = 20 N/mm^2 and τ_{zy}. If the larger principal stress at the point is 110 N/mm^2, the smaller principal stress will be
(a) 10 N/mm^2 (b) 15 N/mm^2
(c) 16 N/mm^2 (d) 18 N/mm^2

UPSC JWM Advt. No.-50/2010

Ans. (a) : Given, $\sigma_x = 100$ N/mm^2, $\sigma_y = 20$ N/mm^2
$\sigma_1 = 110$ N/mm^2, $\sigma_2 = ?$

$$\sigma_1 = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$110 = \frac{100+20}{2} + \sqrt{\left(\frac{100-20}{2}\right)^2 + \tau_{xy}^2}$$

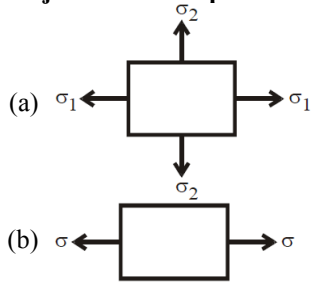
$\tau_{xy} = 30$ N/mm^2

$$\sigma_2 = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= 60 - \sqrt{40^2 + 30^2} = 60 - 50$$

$$\boxed{\sigma_2 = 10\,\text{N/mm}^2}$$

486. Which one of the following represents the state of stress on a surface element of a shaft subjected to a torque?



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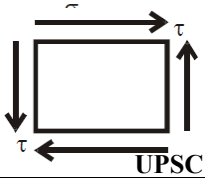
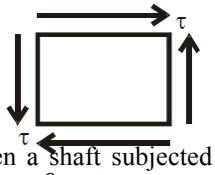
(c)

(d) 

UPSC JWM Advt. No.-50/2010

Ans. (d) : Pure shear stress only –



As we know, when a shaft subjected to torque should have only shear stress & not any normal stress.
So, the correct option is (d).

487. Which one of the following statements is true?
(a) In the principal planes, both normal and shear stresses are always zero
(b) In the principal planes, shear stress may or may not be zero and in the maximum shear planes, normal stresses may or may not be zero
(c) In the principal planes, shear stress is always zero and in the maximum shear planes, normal stresses are always zero
(d) In the principal planes, shear stresses are always zero but in the maximum shear planes, normal stresses may or may not be zero

DRDO Scientists 2009

Ans. (d) : In the principal planes, shear stresses are always zero but in the maximum shear stress planes, normal stress may or may not be zero.

488. At a point on a component loaded under plane stress condition, the stresses are σ_x = 120 MPa, σ_y = 40 MPa and τ_{xy} = 30 MPa. The possible principal stresses are
(a) 130 MPa and –30 MPa
(b) 150 MPa and –50 MPa
(c) 130 MPa and 30 MPa
(d) 150 MPa and 50 MPa

DRDO Scientists 2009

Ans. (c) : $$\sigma_1, \sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \frac{120+40}{2} \pm \sqrt{\left(\frac{120-40}{2}\right)^2 + 30^2}$$

σ_1 = 130 MPa; σ_2 = 30 MPa

489. A Rosette gauge is employed for the measurement of
(a) absolute pressure
(b) low pressure variations
(c) strain in one direction
(d) strain in more than one direction

TNPSC AE, 2008
TNPSC AE 2017

Ans. (d) : A strain gauge only measures strain in one direction in order to get principal strains it is necessary to use a strain rosette. A strain rosette is a cluster of 3 strain gauges oriented at different angles. A strain gauge rosette is an arrangement of two or more closely positioned the normal strains along different directions in the underlying surface of the test part.

490. Principal planes are the planes, on which the resultant stress is the
(a) shear stress (b) normal stress
(c) tangential stress (d) none of these

TNPSC AE, 2008

Ans. (b) : Principal planes are the planes, on which the shear stress is the zero and resultant stress is the normal stress.

491. If at a point in a body σ_x = 70 MPa, σ_y = 60 MPa and τ_{xy} = –5 MPa then the radius of the Mohr's circle is equal to
(a) $5\sqrt{5}$ MPa (b) $2\sqrt{5}$ MPa
(c) $5\sqrt{2}$ MPa (d) 25 MPa j

DRDO Scientists 2008

Ans. (c) : Radius of Mohr circle

$$= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy^2}}$$

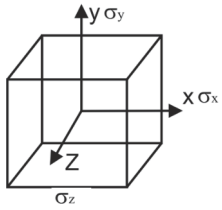
$$= \sqrt{\left(\frac{70-60}{2}\right)^2 + 25}$$

$$= 5\sqrt{2} \text{ MPa}$$

492. For a body subjected to direct stresses σ_x, σ_y and σ_z the direct strain ε_x is x direction is (where E and γ are Young's Modulus and Poisson's ratio respectively)
(a) $\varepsilon_x = \frac{1}{E}\{\sigma_x + \gamma(\sigma_y - \sigma_z)\}$
(b) $\varepsilon_x = \frac{1}{E}\{\sigma_x - \gamma(\sigma_y + \sigma_z)\}$
(c) $\varepsilon_x = \frac{1}{E}\{\sigma_x - \gamma(\sigma_y - \sigma_z)\}$
(d) $\varepsilon_x = \frac{1}{E}\{\sigma_x + \gamma(\sigma_y + \sigma_z)\}$

ISRO Scientist/Engineer 2007

Ans. (b) :



E and γ are young's modulus and Poisson ratio.
So, strain in 3-direction in a body.

$$\varepsilon_x = \frac{1}{E}[\sigma_x - \gamma(\sigma_y + \sigma_z)]$$

$$\varepsilon_y = \frac{1}{E}[\sigma_y - \gamma(\sigma_x + \sigma_z)]$$

$$\varepsilon_z = \frac{1}{E}[\sigma_z - \gamma(\sigma_x + \sigma_y)]$$

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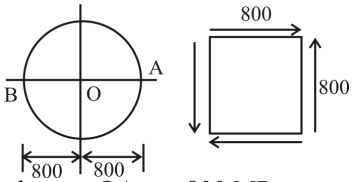
493. The state of stress at a point in a loaded member is of pure shear stresses of ± 800 MPa on two given planes at right angles to each other. The principal stresses are

(a) 800 MPa and – 800 MPa
(b) 800 MPa and 800 MPa
(c) – 800 MPa and – 800 MPa
(d) 1600 MPa and – 1600 MPa

OPSC Civil Services Pre 2006

Ans. (a) :



Max principal stress OA = + 800 MPa
Min. principal stress OB = – 800 MPa

494. If principal stress of a point in plane stressed element are $\sigma_x = \sigma_y = 5000$ N/cm^2, then the normal stress on the plane inclined at 45° to x-axis.

(a) 0 (b) 5000 N/cm^2
(c) 7070 N/cm^2 (d) 500 N/cm^2

WBPSC AE 2003
Punjab PSC (Lect.) 06.08.2017

Ans. (b) : Given, $\sigma_x = \sigma_y = 5000$ N/cm^2

Normal stress on the plane at 45°

$$(\sigma_n)_\theta = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta \quad [\text{also } \sigma_n = \sigma\cos^2\theta]$$

$$= \sigma_x \cos^2(45°) + \sigma_y \sin^2(45°)$$

$$= 5000 \times \left(\frac{1}{\sqrt{2}}\right)^2 + 5000 \times \left(\frac{1}{\sqrt{2}}\right)^2$$

$$= 5000 \text{ N/cm}^2$$

495. The maximum shear stress induced in a member which is subjected to an axial load is equal to

(a) Maximum normal stress
(b) Twice the maximum normal stress
(c) Thrice the maximum normal stress
(d) Half of maximum normal stress
(e) None of the above

CGPSC AE 16.10.2016
HPPSC AE (PWD) 10.09.2016

Ans. (d) : We know that shear stress on an oblique plane at an angle 'θ' to the cross-section of a body, which is subjected to a direct tensile stress 'σ' is equal to $\frac{\sigma}{2}\sin 2\theta$.

Thus maximum value of θ = 45°.

$$\tau_{max} = \frac{\sigma}{2}\sin 90° = \frac{\sigma}{2} = \frac{\sigma_1 - \sigma_2}{2}$$

496. The maximum shear due to an axial compression of 100 MPa is

(a) 50 MPa (b) 100 MPa
(c) 150 MPa (d) 75 MPa

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Given – Axial compression = 100 MPa

$$\tau_{max} = \frac{\sigma}{2} = \frac{100}{2} = 50 \text{ MPa}$$

3. Shear Force and Bending Moment Diagram

497. The point of contraflexure is a point, where

(a) shear force changes sign
(b) bending moment changes sign
(c) shear force is maximum
(d) bending moment is maximum

APPSC AEE SCREENING 17.02.2019
GPSC Asstt. Prof. 28.08.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
KPCL AE 2016, ESC 2005
GPSC Lect. 23.10.2016
VIZAG MT 2015
ISRO Scientist/Engineer 11.10.2015
KPSC ADF 2015
ISRO Scientist/Engineer 24.05.2014
Haryana PSC Civil Services Pre, 2014
TRB Asstt. Prof., 2012
TNPSC ACF 2012
TNPSC AE, 2008

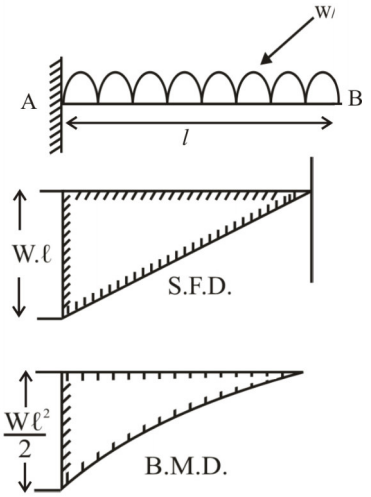
Ans. (b) : Point of contraflexure– In a beam if the bending moment changes its sign at a point, the point it self having zero bending moment, the beam changes curvature at this point of zero bending moment and this point is called point of contraflexure.

498. The bending moment diagram for a cantilever with U.D.L. over the complete span will show the curve of a (an)

(a) Ellipse (b) Rectangle
(c) Triangle (d) Parabola

Vadodara Muncipal Corp. DEE, 2018
RPSC ADE 2016
GPSC Lect. 23.10.2016
Rajasthan Nagar Nigam AE 2016, Shift-I
APPSC AEE Mains 2016 (Civil Mechanical)
Kerala PSC IOF 19.04.2016
MPPSC State Forest Service Exam, 2014
Mizoram PSC AE/SDO 2014, Paper-II
APPSC AEE 2012
J&K PSC Civil Services Pre, 2010
GATE 2001

Ans. (d)



Cantilever beam with uniformly distributed load (W/unit length).

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499. A concentrated load of P acts on a simply supported beam of span L at a distance L/3 from the left support. The bending moment at the point of application of the load is given by :

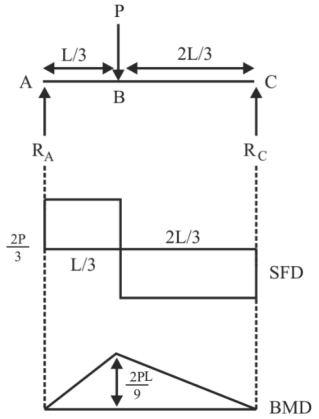
(a) PL/3 (b) 2 PL/3
(c) PL/9 (d) 2PL/9

ISRO Scientist/Engineer 22.04.2018
GPSC ARTO Pre 30.12.2018
GPSC Executive Engineer 23.12.2018
Karnataka PSC Lect. 27.05.2017
TSPSC AEE 2017
ISRO Scientist/Engineer (RAC) 07.05.2017
GPSC AE 26.07.2015
GATE 2003

Ans. (d) : $R_A = \dfrac{P \times \frac{2L}{3}}{L} = \dfrac{2P}{3}$

$$R_C = \frac{P \times \frac{L}{3}}{L} = \frac{P}{3}$$

$$M_B = \frac{2P}{3} \times \frac{L}{3} = \frac{2PL}{9}$$



500. When shear force at a point is zero, then bending moment is _____ at the point.

(a) Zero (b) Minimum
(c) Maximum (d) None of the above

Assam PSC AE (IWT) 14.03.2021
RPSC IOF, 2020
Nagaland PSC (CTSE) 2018, Paper-I
WBPSC AE, 2017
Sikkim PSC (Under Secretary), 2017
APPSC AE Subordinate Service Civil/Mech. 2016
Haryana PSC Civil Services Pre, 2014
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
APPSC Poly. Lect. 2013
UKPSC AE-2013, Paper-I
APPSC AEE 2012

Ans. (c) :
Relation between shear force (V) & bending moment (M)

$$V = \frac{dM}{dx}$$

If shear force is zero (V = 0) means first derivative of bending moment is zero means at that point bending moment will be maximum.

501. The point of contraflexure occurs in :

(a) Cantilever beam only
(b) Simply supported beam only
(c) Overhanging beam only
(d) Continuous beam only

HPPSC ADF 05.03.2019
OPSC AEE 2019 Paper-I
RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017
WBPSC AE, 2017
UPPSC AE 2016
UKPSC AE-2013, Paper-I
APPSC AEE 2012

Ans. (c) : The point of contraflexure occurs mainly in overhanging beam in which there is opposite magnitude moment acting on another side of overhanging part which makes the net moment to change its sign, by virtue of this the point of contraflexure come into picture.

But it is not always in overhanging beam it can also occur in simply supported and fixed beam and point of contraflexure always occurs in fixed beam.

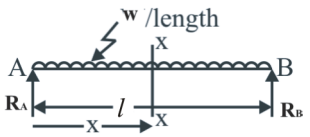
502. A simply supported beam of length *l* carries a uniformly distributed load of *w* per unit length. It will have maximum bending moment at midpoint of beam and the value will be:

(a) $\frac{wl^2}{2}$ (b) $\frac{wl^2}{4}$
(c) $\frac{wl^2}{6}$ (d) $\frac{wl^2}{8}$
(e) $\frac{wl^2}{16}$

CGPSC AE 15.01.2021
ISRO Scientist/Engineer (RAC), 10.03.2019
(CGPSC Polytechnic Lecturer 2017)
CGPSC AE 26.04.2015 Shift-I
UKPSC AE-2013, Paper-I
J & K PSC Screening, 2006

Ans. (d) : We know that

$$R_A = R_B = \frac{wl}{2}$$



Taking a section x-x from a point A at a distance x.
Then taking moment at x-x

$$M_{x\text{-}x} = R_A x - wx \cdot \frac{x}{2}$$

$$M_{x\text{-}x} = \frac{wl}{2}x - w\frac{x^2}{2}$$

For max. bending moment

$$\frac{dM_{x-x}}{dx} = 0$$

$$\frac{wl}{2} - 2w\frac{x}{2} = 0$$

$$\frac{wl}{2} = wx$$

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$$\boxed{x = \frac{l}{2}}$$

At $x = \frac{l}{2}$ [at mid] bending moment will be max. so

$$M_{x=\frac{l}{2}} = \frac{wl}{2} \times \frac{l}{2} - \frac{w \times l^2}{4 \times 2}$$

$$M_{\frac{l}{2}} = \frac{wl^2}{4} - \frac{wl^2}{8}$$

$$\boxed{M_{l/2} = \frac{wl^2}{8}}$$

503. At point of contraflexure in beam subjected to loading :

(a) Shear force is zero
(b) Bending moment is zero
(c) Bending moment is maximum
(d) Shear force is unity

Punjab PSC (Lect.) 06.08.2017
HPPSC AE (PWD) 10.09.2016
HPPSC Lect. (Auto) 23.04.2016
TNPSC AE (Industries) 09.06.2013
APPSC AE 04.12.2012
APPSC AEE 2012
TSPSC AEE 28.08.2017 (Civil/Mechanical)
DRDO Scientists 2008

Ans. (b) : A point of contraflexure is a point where the curvature of the bending moment changes sign. It is sometimes referred to as a point of inflection and will be shown later to occur at the point or points on the beam where the bending moment is zero.

504. Consider the following statements :
If at section distance from one of the ends of the beam, M represents the bending moments. V the shear force and w the intensity of loading, then

1. $\frac{dM}{dx} = V$
2. $\frac{dV}{dx} = w$
3. $\frac{dw}{dx} = y$ (deflection of the beam at the section)

Of these statements
(a) 1 and 3 are correct
(b) 1 and 2 are correct
(c) 2 and 3 are correct
(d) 1, 2 and 3 are correct
(e) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II
SJVN ET 2019, ISRO Scientist/Engineer 2011
CGPSC Poly. Lect. 22.05.2016
UKPSC AE 2012 Paper-I
APPSC AEE 2012

Ans. (b) : Bending moment = M
Shear force = V
load intensity = w

shear force (V) = $\frac{dM}{dx}$

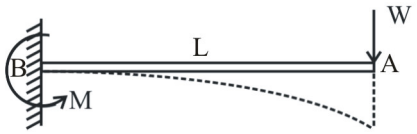
load intensity (w) = $\frac{dV}{dx}$

505. A beam is loaded as cantilever, if the load at the end is increased, then the failure will occur

(a) in the middle
(b) at the tip below the load
(c) anywhere
(d) at the support

Vizag Steel (MT) 2017
Karnataka PSC Lect., 27.05.2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
UKPSC AE-2013, Paper-I
RPSC AE GWD, 2011, CSE PRE -2005

Ans. (d) : A beam is loaded as cantilever, if the load at the end is increased then the failure will occur at the support.



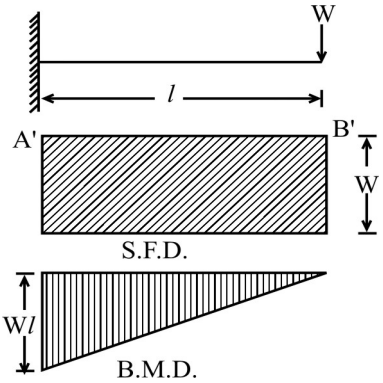
At the fixed end shear force and bending moment is maximum then failure will occur at the fixed end.

506. Maximum bending moment in a cantilever carrying a concentrated load at the free end occurs

(a) at the fixed end (b) at the free end
(c) at the mid span (d) None of these

HPPSC Workshop Suptd. 08.07.2021
GPSC Lect. 23.10.2016
HPPSC LECT. 2016
APPSC AEE 2012
WBPSC AE 2008

Ans : (a)



Maximum bending moment in a cantilever beam carrying a concentrated load at the free end occurs at the fixed end.

507. Bending moment M and torque T is applied on a solid circular shaft. If the maximum bending stress equals to maximum shear stress developed, then M is equal to :

(a) T/2 (b) T
(c) 2T (d) 4T

RPSC IOF, 2020
Vadodara Muncipal Corp. DEE, 2018
PSPCL AE, 2012

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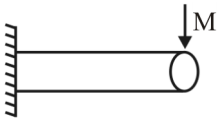
Ans. (a) : Bending equation

$$\frac{\sigma}{y}=\frac{M}{I}=\frac{E}{R}$$



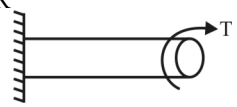
$$\sigma=\frac{M}{I}y$$

$$=\frac{M}{\frac{\pi d^4}{64}}\times\frac{d}{2}$$

$$\sigma_{max}=\frac{32M}{\pi d^3} \quad \ldots.(i)$$

Torsion equation

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{R}$$



$$\tau=\frac{TR}{J}$$

$$=\frac{T}{\frac{\pi}{32}d^4}\times\frac{d}{2}$$

$$\tau_{max}=\frac{16T}{\pi d^3} \quad \ldots.(ii)$$

According to question.

$$(\sigma)_{max}=\tau_{max} \quad (\because \text{ given, } \tau_{max}=\sigma_{max})$$

$$\frac{32M}{\pi d^3}=\frac{16T}{\pi d^3}$$

$$M=\frac{16T}{32}$$

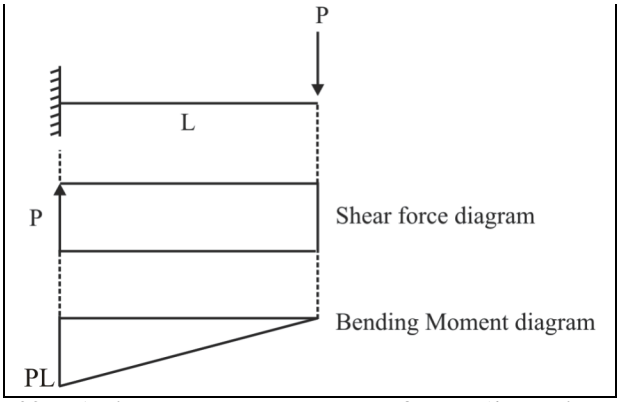
$$\boxed{M=T/2}$$

508. If the shear force acting at every section of a beam is of the same magnitude and of the same direction then it represents a :

(a) Simply supported beam with a concentrated load at the centre.
(b) Overhung beam having equal overhang at both supports and carrying equal concentrated loads acting in the same direction at the free ends.
(c) Cantilever subjected to concentrated load at the free end.
(d) Simply supported beam having concentrated loads of equal magnitude and in the same direction acting at equal distance from the supports.

GPSC ARTO Pre 30.12.2018
HPPSC Asstt. Prof. 2014
APPSC AEE 2012
APPSC IOF, 2009
ESE 1996

Ans. (c) : In a cantilever subjected to concentrated load at the free end, the shear force at every section of a beam is of same magnitude and of same direction.

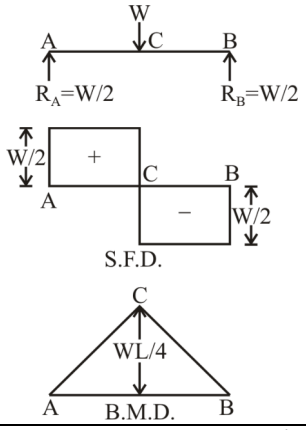


509. A simply supported beam of span (*l*) carries a point load (W) at the centre of the beam. The shear force diagram will be :

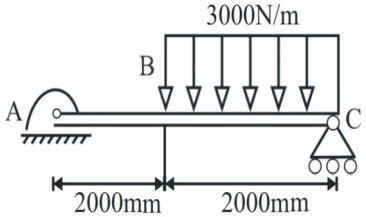
(a) a rectangle
(b) a triangle
(c) two equal and opposite rectangles
(d) two equal and opposite triangles

TSPSC Manager (Engg.) HMWSSB 12.11.2020
CGPSC AE 16.10.2016
HPPSC W.S. Poly. 2016

Ans : (c) A simply supported beam of span (*l*) carries a point load (W) at the centre of the beam. The shear force diagram will be two equal and opposite rectangles.



510. A mass less beam has a loading pattern as shown in Fig. The beam is of rectangular cross-section with a width of 30 mm and height of 100 mm



the maximum bending moment occurs at

(a) Location B
(b) 2500 mm to the right of A
(c) 2675 mm to the right of A
(d) 3225 mm to the right of A

UPPSC AE 13.12.2020, Paper-I
MPPSC AE 2016
GATE 2010

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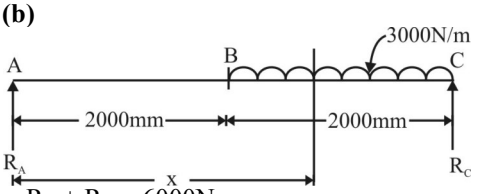
Ans : (b)



$R_A + R_C = 6000N$

Taking moment about A

$6000 \times 3 - R_C \times 4 = 0$

$R_C = 4500$ N

$R_A = 1500$ N

Taking any section from A

Shear force = $1500 - 3000\ (x - 2)$

and maximum bending moment occur where shear force = 0

$1500 - 3000\ (x - 2) = 0$

$x = 2.5$ m

or 2500 mm from A

511. The moment diagram for a cantilever beam whose free end is subjected to a bending moment

(a) Triangle (b) Rectangle

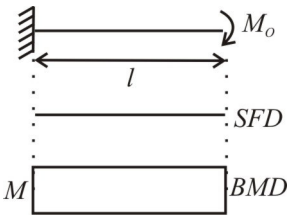
(c) Parabola (d) Cubic parabola

Vizag Steel (MT) 2017

Nagaland PSC CTSE 2016, Paper-I

WBPSC AE 2003

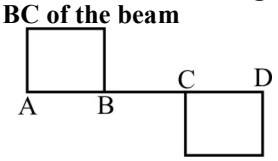
Ans. (b) :



The shear force will be zero all along the span.

The bending moment will be constant all along the span and equal to M.

512. The given figure shows the shear force diagram for the beam ABCD. Bending moment in the portion BC of the beam



(a) is zero

(b) varies linearly from B to C

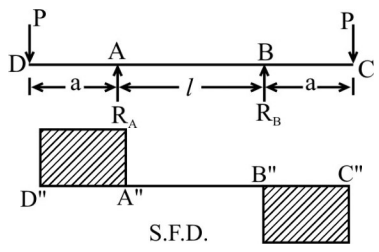
(c) parabolic variation between B and C

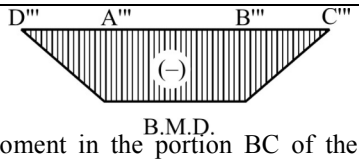
(d) is a non–zero constant

APPSC AEE 2012

ISRO Scientist/Engineer 2008

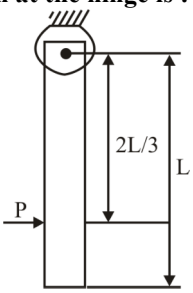
Ans : (d)





Bending moment in the portion BC of the beam is a non-zero constant.

513. A uniform rigid rod of mass 'm' and length 'L' is hinged at one end as shown in the figure. A force 'P' is applied at a distance of $\frac{2L}{3}$ from the hinge so that the rod swings to the right. The reaction at the hinge is :



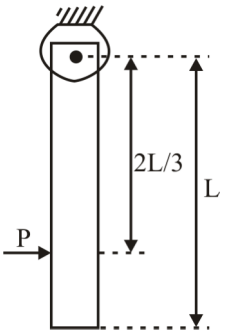
(a) – P (b) 0

(c) P/3 (d) 2P/3

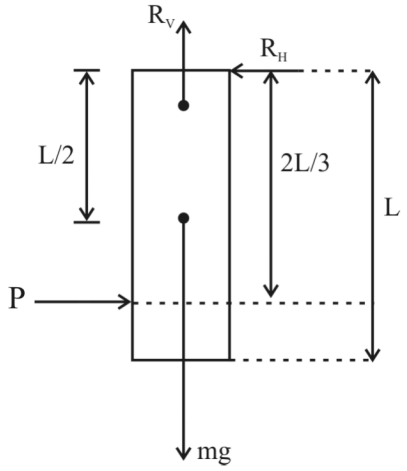
OPSC Civil Services Pre. 2011

GATE-2009

Ans. (b) :



F B D



Considering rotational moment about support

$T = I \times \alpha$

$$P \times \frac{2L}{3} = \frac{mL^2}{3} \times \alpha$$

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$$\alpha = \frac{a}{r}$$

$$\left(I_{endpoint} = \frac{mL^2}{12} + mK^2, K = \frac{L}{2}, I = \frac{mL^2}{3}\right)$$

and $r = \frac{L}{2}$ therefore, $\alpha = \frac{2a}{L}$

$$P \times \frac{2L}{3} = \frac{mL^2}{3} \times \frac{2a}{L}$$

$$a = \frac{P}{m}$$

considering horizontal equilibrium

$$\Rightarrow P - R_H = m \times a$$

$$\Rightarrow -R_H + P = m \times \frac{P}{m}$$

$$\boxed{R_H = 0}$$

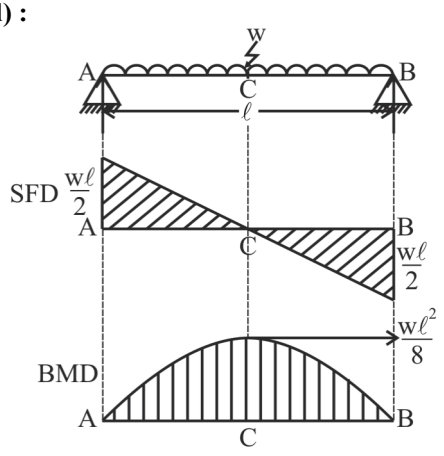
514. The maximum bending moment in a simply supported beam carrying total load W uniformly distributed on the entire span ℓ, is equal to

(a) $W\ell/2$ (b) $W\ell/3$

(c) $W\ell/4$ (d) $W\ell/8$

WBPSC AE, 2017
Mizoram PSC AE/SDO 2014, Paper-II

Ans. (d) :



$$(BM)_{max} = \frac{w\ell}{2} \times \frac{\ell}{2} - \frac{w\ell}{2} \times \frac{\ell}{4}$$

$$= \frac{w\ell^2}{2}\left(\frac{1}{2} - \frac{1}{4}\right)$$

$$= \frac{w\ell^2}{8}$$

$(\because w\ell = W)$

$$(BM)_{max} = \frac{W\ell}{8}$$

515. Which one of the following statement is correct? If the SF diagram for a beam is a right angled triangle having its base representing the length of the beam, the beam is :

(a) a cantilever having a concentrated load at its free end

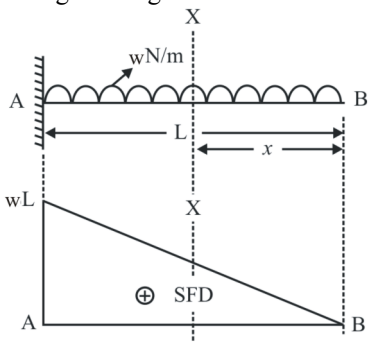
(b) a cantilever having udl over its whole length

(c) simply supported beam with a concentrated load at its mid-length

(d) simply supported beam having udl over its whole length.

TSPSC AEE 28.08.2017 (Civil/Mechanical)
UPSC JWM Adv. No-16/2009, CSE PRE 2005

Ans. (b) : For a cantilever beam having uniform distributed load (UDL) over its whole length, the shear force (SF) diagram is a right angled triangle having its base representing the length of the beam.



(Right angled Triangle)

Shear Force (F_x) = wx

at $x = 0$; $F_B = 0$

& at $x = L$; $F_A = wL$

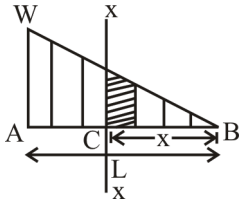
516. A simply supported beam of span *l* carries a uniformly variable load of intensity w_0x over its entire span. Maximum bending moment in the beam is

(a) $\frac{w_0 l^2}{27}$ (b) $\frac{w_0 l^2\sqrt{3}}{27}$

(c) $\frac{w_0 l^2\sqrt{2}}{9}$ (d) $\frac{w_0 l^2}{9}$

UPPSC AE 13.12.2020, Paper-I
RPSC VPITI 14.02.2016

Ans. (b) : $BM_{at\ x} = \frac{Wx^3}{6L} - \frac{WLx}{6}$



For maximum bending moment, $\frac{dM}{dx} = 0$

So, $\frac{dBM}{dx} = \frac{d}{dx}\left[\frac{Wx^3}{6L} - \frac{WLx}{6}\right]$

Shear force $= \frac{-3Wx^2}{6L} + \frac{WL}{6}$

$$-\frac{3Wx^2}{6L} + \frac{WL}{6} = 0$$

$$x = \frac{L}{\sqrt{3}}$$

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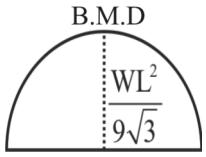
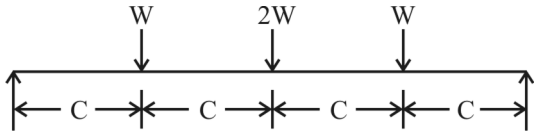
$$(BM)_{maximum} = \frac{-W\frac{L^3}{3\sqrt{3}}}{6L} + \frac{WL \times \frac{L}{\sqrt{3}}}{6}$$



$$(BM)_{max} = +\frac{WL^2}{9\sqrt{3}} \text{ or } \frac{w_0\ell^2\sqrt{3}}{27}$$

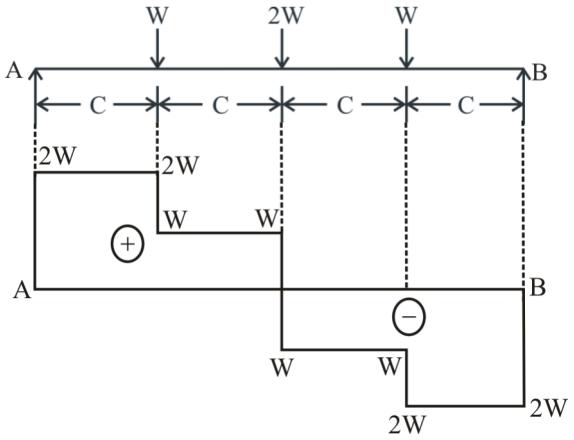
517. A simply supported beam is loaded as shown in fig. The maximum shear force in the beam will be :



(a) Zero (b) W
(c) 2W (d) 4W

APPSC AE 04.12.2012
APPSC AEE 2012, ESE 1998

Ans. (c) : A simply supported beam -



(Shear force Diagram- SFD)

From the figure -

→ Let reaction at A and B in the beam is R_A and R_B respectively.

∴ Due to symmetrical loading - $R_A = R_B$ … (i)

& $R_A + R_B = 4W$ … (ii)

∴ From equation (i) & (ii)

$R_A = R_B = 2W$

Therefore, from SFD; maximum shear force = 2W

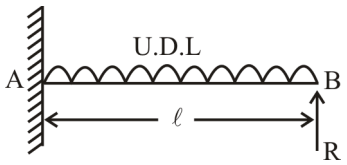
518. The reaction at the prop in a propped cantilever beam subjected to u.d.*l.* is :

(a) $\frac{Wl}{4}$ (b) $\frac{3Wl}{8}$

(c) $\frac{5Wl}{8}$ (d) $\frac{6Wl}{7}$

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (b) : Draw a propped cantilever beam subjected to U.D.L.



Let,
W = Uniform distributed load.
ℓ = Length of cantilever beam
R = Reaction at the prop
∴ Deflection at point B due to U.D.L. in cantilever beam in downward direction

$$\delta_B = \frac{W\ell^4}{8EI}(\downarrow)$$

Deflection at point B due to reaction (R) in upward direction

$$\delta_B' = \frac{R\ell^3}{3EI}(\uparrow)$$

∴ At equilibrium condition - $\delta_B' = \delta_B$

$$\Rightarrow \frac{R\ell^3}{3EI} = \frac{W\ell^4}{8EI} \Rightarrow \boxed{R = \frac{3W\ell}{8}}$$

519. The point of contraflexure is also called as :
(a) the point of inflexion
(b) a virtual hinge
(c) Either of the above
(d) None of the above

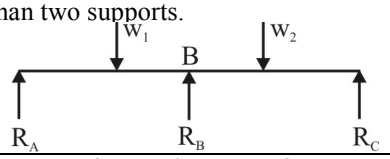
APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (c) : The point of contraflexure is also called as- the point of inflection or a virtual hinge.
Because at the point of contraflexure Bending moment changes sign and Bending moment is zero.

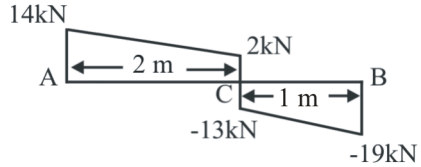
520. A continuous beam is one which is
(a) fixed at one end and free at the other end
(b) fixed at one end and free at the other end
(c) supported on more than two supports
(d) extending beyond the supports

OPSC AEE 2019 Paper-1
TSPSC AEE 2015

Ans : (c) A continuous beam is one which is supported on more than two supports.



521. The shear force diagram of a loaded beam is shown in the following figure. The maximum bending moment in the beam is



(a) 16-kN-m (b) 11-kN-m
(c) 18-kN-m (d) 18-kN-m

UPPSC AE 12.04.2016 Paper-I
CSE Pre-1997

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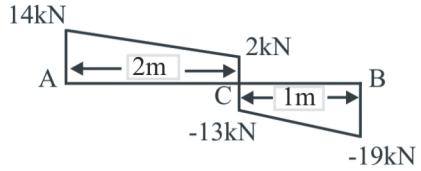
Ans : (a)



Maximum Bending Moment:-

$(BM)_{max} = 16kN-m$

We know that

$\frac{dM}{dx} = \text{shear force}(S)$

$dM = Sdx$

$\int_A^C dM = \text{Area under Shear force diagram}$

$(M_C - M_A) = 2\times2 + \frac{1}{2}\times2\times12$

$= 4 + 12 = 16$ kN-m

$M_A = 0$ [At Reaction]

So $\boxed{M_C = 16kN-m}$

and also

$M_B - M_C = -13\times1 + \left[\frac{1}{2}\times1\times(-6)\right]$

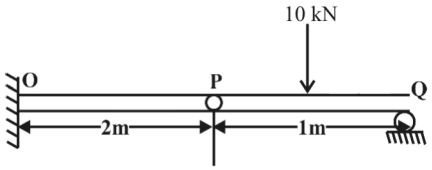
$= -13 - 3 = -16$ kN-m

$M_B = 0$ [At reaction]

$\boxed{M_C = 16\ kN-m}$

So maximum bending moment will be 16 kN-m at point C.

522. A cantilever OP is connected to another beam PQ with a pin joint as shown in figure. A load of 10 kN is applied at the mid-point of PQ. The magnitude of bending moment in (kNm) at fixed end O is–

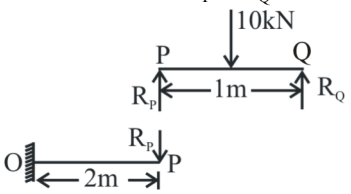


(a) 2.5 (b) 5
(c) 10 (d) 25

PPSC Asstt. Municipal Engineer 15.06.2021
RPSC 2016
GATE 2020

Ans : (c) By similarity,

$R_P = R_Q = 5kN$

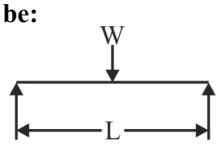


From FBD

Moment at point O

$M_o = R_P \times 2 = 5 \times 2 = 10$ kN-m

523. The BM diagram of the beam shown in the figure will be:

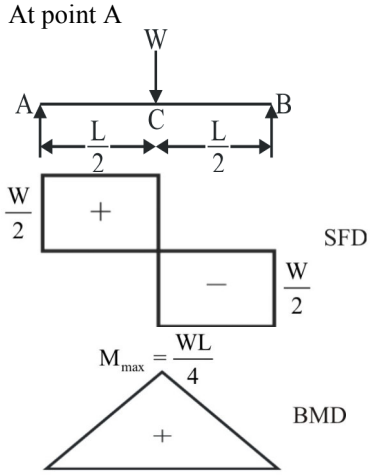


(a) A rectangle (b) A triangle
(c) A trapezium (d) A parabolic
(e) A hyperbolic

Nagaland PSC CTSE 2017, Paper-I
(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : $R_A = R_B = \frac{W}{2}$

Shear force S, At point A



$S_A = +\frac{W}{2}$

$S_C = +\frac{W}{2}$

$S_C = +\frac{W}{2} - W = -\frac{W}{2}$

$S_B = -\frac{W}{2}$

$S_B = -\frac{W}{2} + \frac{W}{2} = 0$

Bending Moment diagram

$M_A = M_B = 0$

$M_C = R_A \times \frac{L}{2} = \frac{W\times L}{4}$

524. In a beam where shear force is maximum the bending moment will be–

(a) maximum
(b) zero
(c) minimum
(d) there no such relation between the two

Nagaland PSC CTSE 2017, Paper-I
Nagaland PSC CTSE 2016, Paper-I

Ans. (d) : There no such relation between the two

525. Variation of bending moment in a cantilever carrying a load, the intensity of which varies uniformly from zero at the free end to w per unit run at the fixed end, is by :

(a) cubic law (b) parabolic law
(c) linear law (d) none of these

WBPSC AE, 2017
(HPPSC AE 2014)

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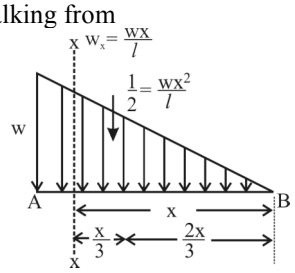
Ans : (a) Variation of bending moment in a cantilever carrying a load, the intensity of which varies uniformly from zero at the free end to w per unit run at the fixed end is by cubic law.

Section x–x talking from



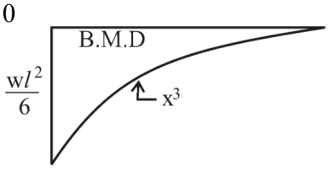
Taking moment at x–x

$$M_{x-x} = \frac{1}{2}\frac{wx^2}{l} \times \frac{x}{3}$$

$$\boxed{M_{x-x} = \frac{wx^3}{6l}}$$

Bending moment at B

$x = 0,\ M_B = 0$

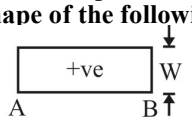


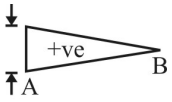
Bending Moment at A

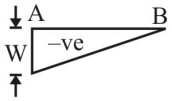
$x = l$

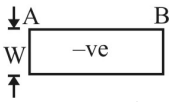
$$M_A = \frac{wl^2}{6}$$

526. The shear force of a cantilever beam of length 'L' with a point load 'W' at its free end is in the shape of the following:

(a) 

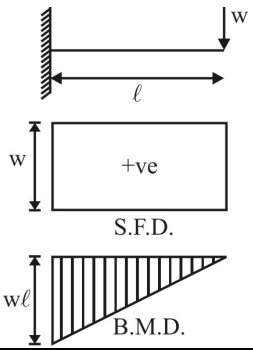
(b) 

(c) 

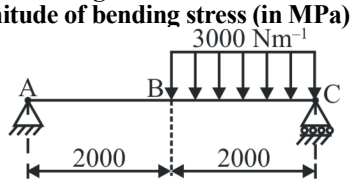
(d) 

UJVNL AE 2016
TSPSC AEE 2015

Ans : (a)



527. A massless beam has a loading pattern as shown in the figure. The beam is of rectangular cross section with a width of 30 mm and height of 100 mm. The maximum magnitude of bending stress (in MPa) is given by

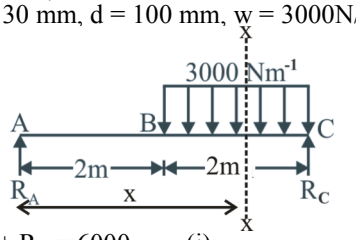


(a) 60.0 (b) 67.5
(c) 200 (d) 225

ISRO Scientist/Engineer 11.10.2015
GATE 2010

Ans. (b) : Given,

b = 30 mm, d = 100 mm, w = 3000N/m



$R_A + R_C = 6000$ .....(i)

Taking moment about point 'C'

$R_A \times 4 = 6000$

$$R_A = \frac{6000}{4} = 1500 \text{ N}$$

$R_A + R_C = 6000$

$R_C = 6000 - 1500 = 4500$ N

Taking moment about x – x

$$M_{xx} = R_A \cdot x - \frac{w(x-2)^2}{2}$$

$$\frac{dM}{dx} = R_A - \frac{2w(x-2)}{2}$$

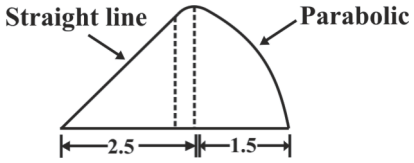
For maximum bending moment,

$$\frac{dM}{dx} = 0 \Rightarrow R_A = w(x-2)$$

$$\frac{1500}{3000} + 2 = x$$

$x = 2.5$

Maximum bending moment M at x = 2.5 m from point A.



$$M = (1500 \times 2.5) - 3000 \times \frac{(0.5)^2}{2}$$

M = 3375 N-m

Where $\boxed{Z = \frac{I}{y}}$

$$\text{Section modulus } (Z) = \frac{I}{y} = \frac{bd^3}{12 \times \frac{d}{2}}$$

$$= \frac{\frac{30}{1000} \times \left(\frac{100}{1000}\right)^3 \times 2000}{12 \times 100}$$

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$$= \frac{2.5\times10^{-6}}{100}\times 2000\times m^3 = 5\times10^{-5}\ m^3$$

Bending stress, $\sigma_{max.}$

$$\sigma_{max} = \frac{M_{max}}{Z} = \frac{3375}{5\times10^{-5}} = 67.5\times10^6\ N/m^2$$

$$= 67.5\ MPa$$

528. For a particular load distribution and support condition in a beam of length 'L', bending moment at any section 'x' (0 < x < L) is given by M(x) = Ax - Bx^2, where A and B are constants. The shear force in the beam will be zero at 'x' equal to

(a) A/2B (b) A/B
(c) 2A/B (d) A^2/B

ISRO Scientist/Engineer 12.05.2013
DRDO Scientists 2009

Ans. (a) : Bending moment

$$M = Ax - Bx^2;\quad [0 < x < L]$$

Shear force $F = \frac{dM}{dx}$

$$\frac{dM}{dx} = A - 2Bx$$

If shear force of beam (F) = 0

$$F = \frac{dM}{dx} = A - 2Bx = 0$$

$$A - 2Bx = 0$$

$$x = \frac{A}{2B}$$

529. Shear force is

(a) Rate of change of loading
(b) Sum of bending moments
(c) Rate of change of bending moment
(d) None of the above

MPSC HOD (Govt. Poly. Colleges) 04.10.2014
ISRO Scientist/Engineer 2010, 2007

Ans. (c) : Shear force is rate of change of bending moment.

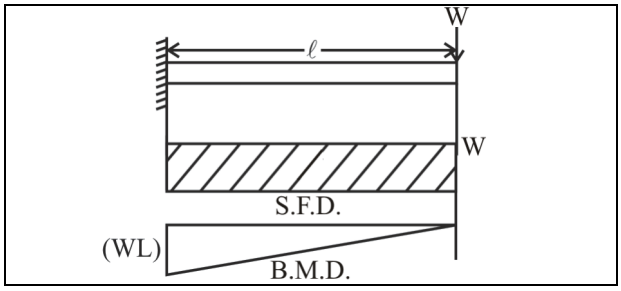
$$\text{Shear force (V)} = \frac{dM}{dx}$$

530. A cantilever of length L carries a point load W at the free end. The bending moment diagram will be a

(a) Parabola with maximum ordinate at centre of the beam
(b) Parabola with maximum ordinate at the cantilever end
(c) Triangle with maximum ordinate at the free end
(d) Triangle with maximum ordinate at the cantilever end
(e) None of the above

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (d) : The maximum bending moment of a cantilever beam of length 'L' & carrying a point load at free end, is lies at fixed end & straight in nature.



531. If for a beam $\frac{dM}{dx} = 0$ for its whole length, the beam is a cantilever.

(a) Subjected to *udl* extending from fixed end to its mid-length.
(b) Subjected to *udl* over its whole length
(c) Subjected to an end moment
(d) Having a concentrated load at its free end

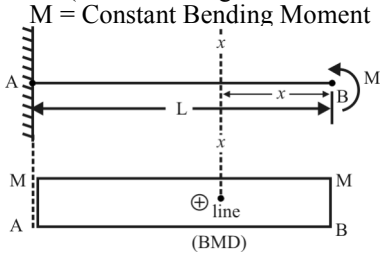
TSPSC AEE 28.08.2017 (Civil/Mechanical)
UPSC JWM Advt. No.-52/2010, CSE Pre 2007

Ans. (c) : Given,

Change in bending moment, $\frac{dM}{dx} = 0$

(for whole length of a cantilever beam)

where, M = Constant Bending Moment



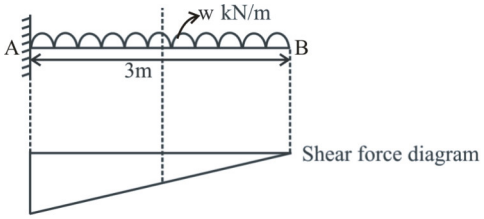
If for a beam $\frac{dM}{dx} = 0$ for it whole length, the beam is cantilever which is subjected to an end moment. Shear force is zero throughout whole length.

532. A uniformly distributed load (UDL) (in kN/m) is acting over the entire length of a 3m long cantilever beam. If the shear force at the midpoint of cantilever is 6 kN, what is the value of UDL (in kN/m)?

(a) 2 (b) 3
(c) 4 (d) 5

OPSC AEE 2019 Paper-I
GPSC ARTO Pre 30.12.2018, ESE 2009

Ans. (c) :



At mid point of cantilever beam, shear force is 6 kN

$$w\times\frac{L}{2} = 6$$

$$w\times\frac{3}{2} = 6$$

$$\boxed{w = 4\ kN/m}$$

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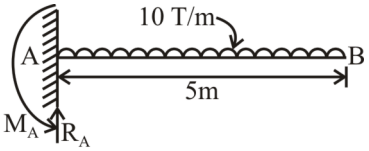
533. A cantilever beam having 5m length is so loaded that it develops a shearing force of 20 T and a bending moment of 20 T-m at a section 2m from the free end. Maximum shearing force and maximum bending moment developed in the beam under this load are respectively 50 T and 125 T-m. The load on the beam is :

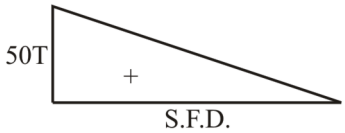
(a) 25 T concentrated load at free end
(b) 20 T concentrated load at free and
(c) 5 T concentrated load at free end and 2 T/m load over entire length
(d) 10 T/m uniformly distributed load over entire length

GPSC ARTO Pre 30.12.2018
APPSC AEE 2012
ESE 1995

Ans. (d) : By observing data, we see shear force is varying with length. Hence option (a) and (b) get eliminated for concentrated load at the ends of cantilever, shear force remains constant.

Now, considering option (d).





$R_A = 10 \times 5 = 50T$

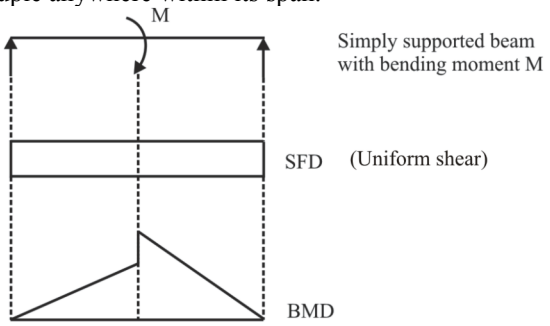
$M_A = 10 \times 5 \times \frac{5}{2} = 125T\text{-m}$

534. For the shear force to be uniform throughout the span of a simply supported beam, it should carry which one of the following loadings?

(a) A concentrated load at mid-span
(b) UDL over the entire span
(c) A couple anywhere within its span
(d) Two concentrated loads equal in magnitude and placed at equal distance from each support.

GPSC ARTO Pre 30.12.2018
TSPSC AEE 28.08.2017 (Civil/Mechanical)
MPPSC State Forest Service Exam, 2014

Ans. (c) : For the shear force to be uniform throughout the span of simply supported beam, it should carry a couple anywhere within its span.



535. A simply supported beam of length 3 m carries a concentrated load of 12 kN at a distance of 1m from left support. The maximum bending moment in the beam is.

(a) 12 kNm (b) 24 kNm
(c) 8 kNm (d) 16 kNm

ISRO Scientist/Engineer (RAC), 10.03.2019
MPPSC State Forest Service Exam, 2014

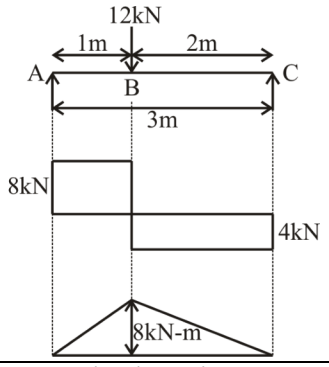
Ans. (c) :

$R_A = \frac{12 \times 2}{3} = 8kN$

$R_C = \frac{12 \times 1}{3} = 4kN$

$M_{max} = M_B = 8 \times 1$

$= 8kN\text{-m}$



536. A beam under pure bending is subjected to

(a) Same bending stress at all sections
(b) No shearing stress at any section
(c) Same shearing stress at all sections
(d) Uniformly increasing bending moment from the support to the mid-section

Karnataka PSC AE, 10.09.2017
WBPSC AE 2008

Ans. (b) : For pure bending there should be no presence of shear stress ($\tau = 0$) at any sections.

537. Bending moment at any point on a beam subjected to transverse loading will be equal to

(a) Algebraic sum of moments of forces and moments of reactions on one side of the section (point)
(b) Algebraic sum of moments of forces and moments of reaction on both sides of the section (point)
(c) Algebraic sum of moments of forces on both sides of the section (point)
(d) Algebraic sum of moments of reaction on both sides of the section (point)

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Bending moment at any point on a beam subjected to transverse loading will be equal to algebraic sum of moments of forces and moments of reactions on one side of the section (point).

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538. Which of the following are the statically determinate beams?

(a) Cantilevers (b) SSB
(c) Overhanging beams (d) All of the above

APPSC AEE 2012

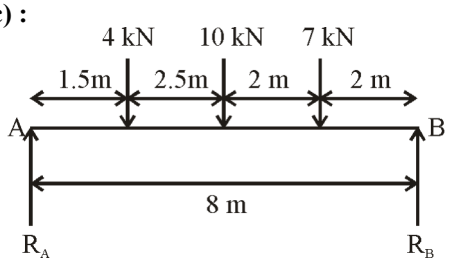
Ans. (d) : Examples of determinate structures are-
- Simply supported beams (SSB)
- Cantilever beams
- Single and double overhanging beams
- Three hinges arches etc.

539. A simply supported beam of span 8 meters carry concentrated loads of 4 kN, 10 kN and 7 kN at distances of 1.5 meters, 4 meters and 6 meters from the left support. The shear force at right support will be

(a) 4 kN (b) 10 kN
(c) 11 kN (d) 7 kN

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) :



$R_A + R_B = 21$ kN

$\sum M_A = 0$

$R_B \times 8 = 4 \times 1.5 + 10 \times 4 + 7 \times 6$

$R_B \times 8 = 6 + 40 + 42$

$R_B = 11$ kN

$R_A = 21 - R_B$

$R_A = 21 - 11 = 10$ kN

The shear force at right support will be 11 kN.

540. The differential equation $EI\,(d^4y/dx^4)$ gives

(a) Bending moment (b) Shear force
(c) Deflection (d) Rate of loading

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : EI (y) – deflection

$EI\left(\frac{dy}{dx}\right)$ – slope

$EI\frac{d^2y}{dx^2}$ – Bending moment

$EI\frac{d^3y}{dx^3}$ – shear force

$EI\frac{d^4y}{dx^4}$ – Rate of loading

541. In case of simply supported beam, if the position of point of application of bending moment is changed towards left, over the length of the beam, the value of the reactive force at the left end of the beam will

(a) remain unchanged.
(b) increase.
(c) decrease.
(d) be doubled of the previous value.

PPSC Asstt. Municipal Engineer 15.06.2021

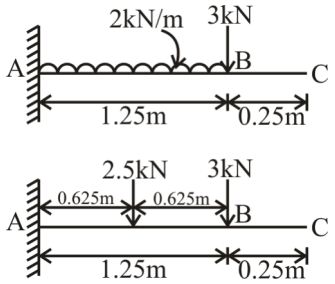
Ans. (a) : In case of simply supported beam, if the position of point of application of bending moment is changed towards left, over the length of the beam, the value of the reactive force at the left end of the beam will remain unchanged.

542. A cantilever beam 1.5 long is loaded with a uniformly distributed load of 2 kN/m run over a length of 1.25 m from the free end. It also carries a point load of 3 kN at a distance 0.25 m from the free end. The shear force acting on the fixed end is _____.

(a) 7.5 kN (b) 2 kN
(c) 3.5 kN (d) 5.5 kN

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) :



Shear force at fixed point 'A',

$(SF)_A = 3 + 2.5 = 5.5$ kN

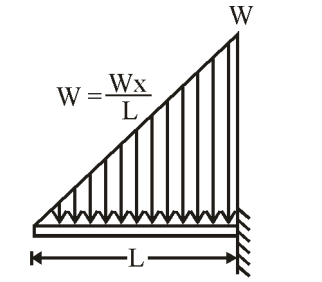
543. Maximum bending on a cantilever beam of length 'L' carrying a uniformly varying load zero at free end and 'w' per metre length

(a) $M_{max} = \frac{W.L^2}{9}$ (b) $M_{max} = \frac{W.L}{6}$

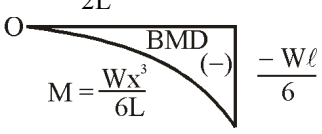
(c) $M_{max} = \frac{W.L^2}{6}$ (d) $M_{max} = \frac{W.L^2}{9\sqrt{3}}$

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) :



Shear force (SF) = $\frac{Wx^2}{2L}$



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544. A 250 mm diameter pulley is mounted on the shaft midway between two supporting bearings that are 750 mm apart. The pulley is driven by a belt, both strands pulling vertically upwards. If the tension in the tight side of the belt is 2.7 kN and the slack side is 900 N, what is maximum bending moment?

(a) 225 N-m (b) 675 N-m
(c) 300 N-m (d) 1000 N-m

CGPSC AE 15.01.2021

Ans. (b) : Given,
dia of pulley = 250 mm = 0.25 m
L = 750 mm = 0.75 m
Total load on pulley = 2.7 kN + 900 N
(W) = 2700 + 900
(W) = 3600 N

$$\text{Max. Bending Moment} = \frac{WL}{4} = \frac{3600 \times 0.75}{4}$$

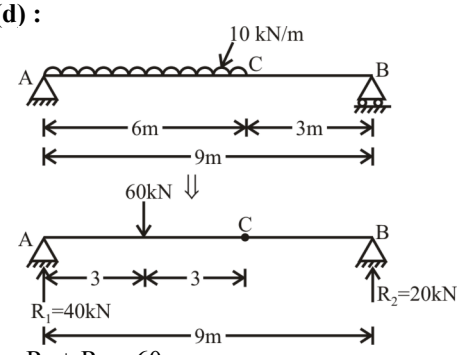
$$(BM)_{max} = 675 N - m$$

545. A simply supported beam of length 9 m carrying a uniformly distributed load of 10 kN/m for a distance of 6 m from the left end. The bending moment acting on the beam at 6 m from its left end is _____.

(a) 40 kNm (b) 80 kNm
(c) 120 kNm (d) 60 kNm

VIZAG Steel MT 24.01.2021

Ans. (d) :

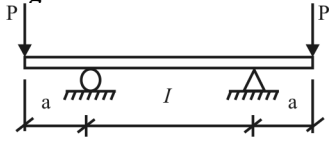


$R_1 + R_2 = 60$
$\therefore \quad \sum M_A = 0$
$60 \times 3 = R_2 \times 9$

$$R_2 = \frac{60 \times 3}{9} = 20 kN$$

$R_1 = 60 - 20 = 40$ kN
$\therefore$ The bending moment on beam at 6m from left end
$(BM) = R_1 \times 6 - 60 \times 3$
$= 40 \times 6 - 60 \times 3$
$= 240 - 180$
$= 60$ kN-m

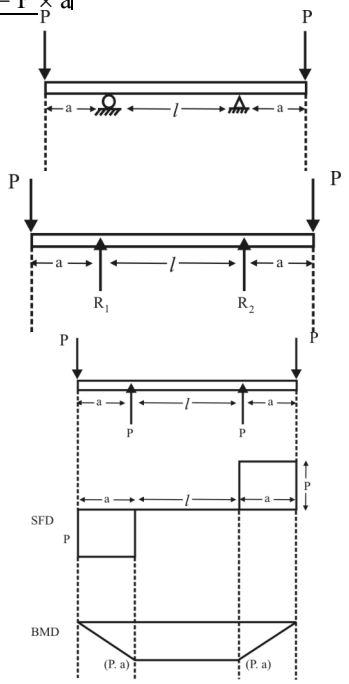
546. Maximum bending moment for the shown overhang beam with loads will be :



(a) $P\left(a + \frac{l}{2}\right)$ (b) $P.\frac{l}{2}$
(c) Pa (d) P (2a + *l*)

JPSC AE 10.04.2021, Paper-II

Ans. (c) : P. a
Beam is symmetrically loaded
So, $R_1 = R_2 = P$
Maximum Bending moment
$M = P \times a$



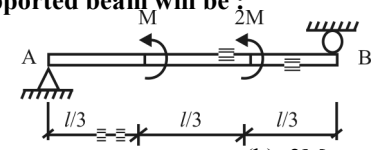
547. The part of a member is said to be in pure bending if

(a) no bending moment exists in that part
(b) no shear force and no bending moment extra in that part
(c) shear force and bending moment are maximum
(d) No shear force exists in that part

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : The part of a member is said to be in pure bending if no shear force exists.

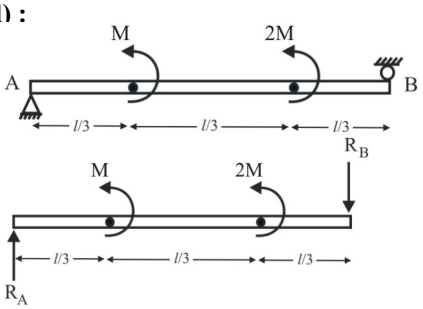
548. Reaction R_B at support B for the shown simply supported beam will be :



(a) zero (b) 3M
(c) M*l* (d) 3 M/*l*

JPSC AE 10.04.2021, Paper-II

Ans. (d) :



$\Sigma M_A = 0$
$M + 2M = R_B \times l$

$$R_B = \frac{3M}{l}$$

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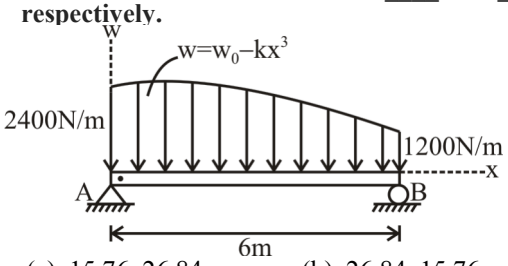
549. The beam shown here has variable load. The reactions at A and B in kN are ____ and ____, respectively.



(a) 15.76, 26.84 (b) 26.84, 15.76
(c) 6.84, 5.76 (d) 5.76, 6.84

AAI Jr. Executive 26.03.2021
BHEL ET 2019

Ans. (c) : $w = w_o - kx^3$

At $x = 0, w = w_o = 2400$

At $x = 6, w = 1200$

$1200 = 2400 - k(6)^3$

$k = 5.55$

Now, $-w_{xx} = \frac{d^2M}{dx^2}$

Shear force, SF $\frac{dM}{dx} = -\int (w_{xx})dx$

$= -\int (w_o - kx^3)dx$

$\frac{dM}{dx} = \frac{kx^4}{4} - w_o x + c_1$

$M = \frac{kx^5}{20} - \frac{w_o x^2}{2} + c_1 x + c_2$

At $x = 0, M = 0$

$c_2 = 0$

At $x = 6, M = 0$

$c_1 = 6840.036$

$\therefore \quad SF = \frac{kx^4}{4} - w_o x + 6840.036$

Reaction at A, $x = 0$

$R_A = (SF)_A = 6.84$ kN

Reaction at B, $x = 6$,

$R_B = (SF)_B = 5.76$ kN

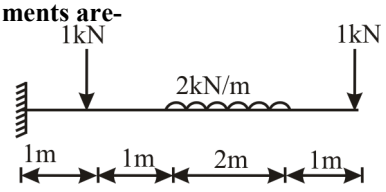
550. In a cantilever beam subjected to a point load at its free end has maximum bending moment _____.

(a) fixed end of the beam
(b) at a distance of 0.6 × length of the beam
(c) free end of the beam
(d) mid span of the beam

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : fixed end of the beam

551. For a cantilever, loaded as shown in figure, maximum shear force and maximum bending moments are-



(a) 4kN, 14 kN-m sagging
(b) 4kN, 14 kN-m hogging
(c) 6kN, 18 kN-m sagging
(d) 6kN, 18 kN-m hogging

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) :
Shear force, V = 1 + 2 × 2 + 1
= 6 kN (max)
Max. Bending moment occurs at the fixed end
M_{max} = 1 × 5 + 2 × 2 × 3 + 1 × 1
= 5 + 12 + 1
= 18 kN-m (hogging)

552. If a beam is subjected to a constant bending moment along its length then the shear force will

(a) be maximum at the center and zero at the ends
(b) zero at the center and maximum at the ends
(c) be zero at all sections along the beam
(d) has a constant value everywhere along its length

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 1997

Ans. (c) : The shear force will zero along the beam at all section, there is no shear force only it is a pure bending.

Shear force $V = \frac{dM}{dx}$

Given, M = Constant

So shear force $\boxed{V = 0}$ $\left[\because \frac{dM}{dx} = 0\right]$

553. Support is non-yielding means

(a) Slope of beam at support is zero
(b) Support is frictionless
(c) support holds the member firmly
(d) None of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) :

554. A simply supported beam PQ is loaded by a moment of 1 kN-m at the mid span of the beam of length 1m. The reaction forces R_P and R_Q at supports P and Q respectively are

(a) 1 kN downward, 1 kN upward
(b) 0.5 kN upward, 0.5 kN downward
(c) 0.5 kN downward, 1 kN upward
(d) 0.5 kN upward, 0.5 kN upward

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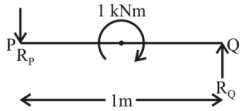
Ans. (a) :



$\sum F_v = 0$
$R_P - R_Q = 0$
$\sum M_P = 0$
$R_Q \times 1 - 1 = 0$
$R_Q = 1$ kN (↑)
$R_P = 1$ kN (↓)

555. In a beam where shear force changes its sign, the bending moment will be

(a) minimum (b) zero
(c) maximum (d) same as shear force

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Ans. (c) : Shear force $= \frac{dM}{dx}$ changes its sign, the bending moment will be maximum.

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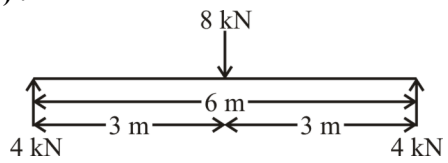
556. A simply supported beam of length 6 m carries a point load of 8 kN at the middle of its span. The bending moment at the middle of the span is ____ .

(a) 6 kNm
(b) 24 kNm
(c) 12 kNm
(d) 8 kNm

VIZAG MT, 14.12.2020

Ans. (c) :



M at mid point = 4 × 3
= 12 kN-m

557. A simply supported beam of span 8 m is carrying a uniformly distributed load of 4 kN/m over a length of 4 m from its right end. The support reactions are _____.

(a) 12 kN right support, 4 kN left support
(b) 6 kN left support, 10 kN right support
(c) 4 kN left support, 12 kN right support
(d) 8 kN left support, 8 kN right support

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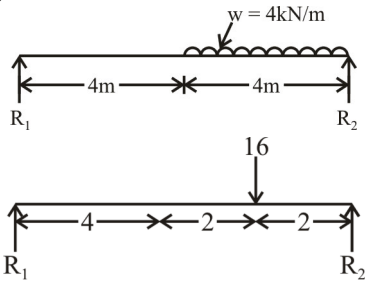
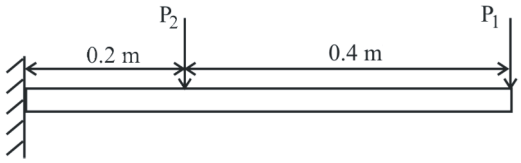
Ans. (c&a) :



$$R_1 = \frac{16 \times 2}{8} = 4\,kN$$

$$R_2 = \frac{16 \times 6}{8} = 12\,kN$$

558. For the cantilever AB shown in the following figure, if the reaction force at the fixed support A is 50 N and the area of the shear force diagram is 16 N-m, the value of the concentrated load P_2 is



(a) 16 N
(b) 64 N
(c) 15 N
(d) 35 N

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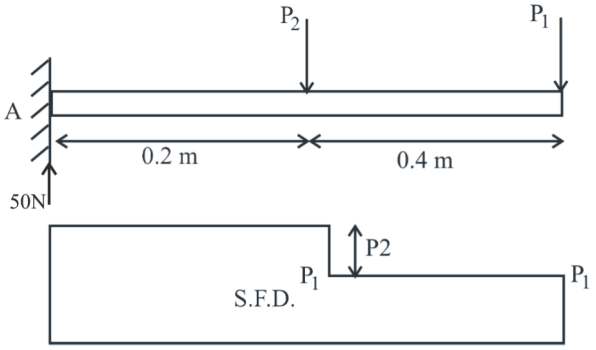
Ans. (d) :



Area of SFD = 50 × 0.2 + 0.4 × P_1
16 = 50 × 0.2 + P_1 × 0.4

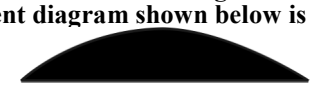
$$P_1 = \frac{6}{0.4}$$

$$\boxed{P_1 = 15}$$

$P_1 + P_2 = 50$
$15 + P_2 = 50$
$P_2 = 50 - 15$

$$\boxed{P_2 = 35\,N}$$

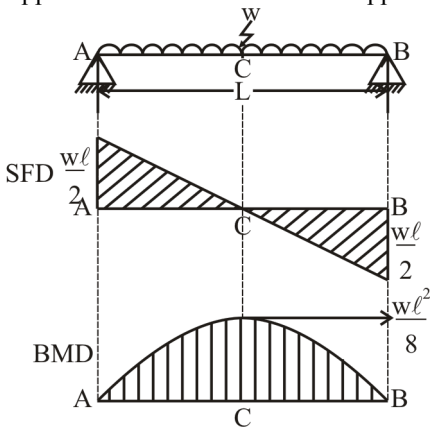
559. The beam that would generate the bending moment diagram shown below is



(a) Simply supported beam with a central point load
(b) Cantilever with uniformly distributed load
(c) Simply supported beam with uniformly decreasing load from the midspan towards the supports
(d) Simply supported beam with uniformly distributed load

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : In case of simply supported beam UDL produce parabola in BMD and straight line in SFD. Simple supported beam has zero BM at support.



560. A cantilever of span *l* is subjected to a point load W at the free end. The bending moment at the free end will be

(a) W*l* (b) zero
(c) W/*l* (d) W

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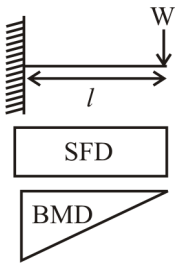
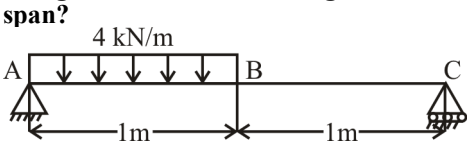
Ans. (b) :



Bending moment (M) at free end = 0
Bending moment at fixed end = W×l

561. A mass less beam has a loading pattern shown in figure. Find the bending moment at mid span?



(a) 1 kN-m (b) 3 kN-m
(c) 2 kN-m (d) 0.0 kN-m

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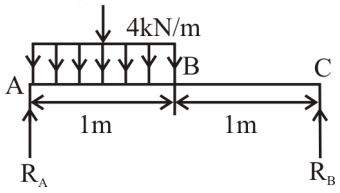
Ans. (a) :



$$R_A = \frac{4\times1.5}{2} = 3\,kN$$

$(B.M)_B = R_A \times 1 - w \times 0.5 = 3 \times 1 - 4 \times 0.5 = 1$ kN-m

562. A simply supported beam of length l, carries a load $w(x) = w_0(x)$ over the entire span. Maximum bending moment in the beam at x will be

(a) $\frac{l}{3}$ (b) $\frac{l}{\sqrt{3}}$

(c) $\frac{l\sqrt{3}}{2}$ (d) $\frac{l}{\sqrt{2}}$

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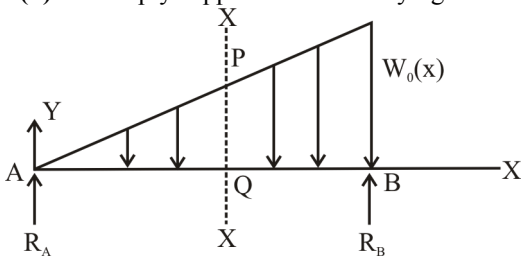
Ans. (b) : A simply supported beam carrying load



Reaction at A, $R_A = \frac{\frac{Wl}{2}\times\frac{l}{3}}{l} = \frac{Wl}{6}$

Reaction at B, $R_B = \frac{\frac{Wl}{2}\times\frac{2l}{3}}{l} = \frac{Wl}{3}$

$V_x = +R_A - \text{Area of APQ}$

$$V_x = \frac{Wl}{6} - \frac{1}{2}x.\frac{Wx}{l}$$

$$V_x = \frac{Wl}{6} - \frac{Wx^2}{2l}$$

Max. Bending moment occurs at,

$$V_x = 0$$

$$\frac{Wl}{6} - \frac{Wx^2}{2l} = 0$$

$$\frac{Wx^2}{2l} = \frac{Wl}{6}$$

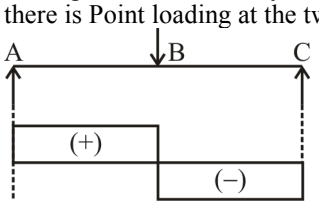
$$\boxed{x = \frac{l}{\sqrt{3}}}$$

563. When there is a sudden increase or decrease in shear force diagram between any two points, it indicates that there is

(a) No loadings between the two points
(b) Point loading at the two points
(c) Uniformly varying load between the two points
(d) Uniformly distributed load between the two points

UPPSC AE 13.12.2020, Paper-I
RPSC IOF, 2020
VIZAG Steel MT 24.01.2021

Ans. (b) : When there is a sudden increase or decrease in shear force diagram between any two points, it indicates that there is Point loading at the two points.



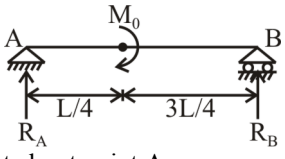
564. A simply supported beam of span L is subjected to a moment M_0 at a distance of $\frac{L}{4}$ from the left end. Magnitude of the maximum bending moment in the beam is

(a) M_0 (b) $\frac{M_0}{2}$

(c) $\frac{M_0}{4}$ (d) $\frac{3M_0}{4}$

UPPSC AE 13.12.2020, Paper-I

Ans. (d) :



Taking moment about point A

$$R_B \times L - M_0 = 0$$

$$R_B = \frac{M_0}{L}$$

$$R_A + R_B = 0$$

$$R_A = -R_B$$

$$R_A = -\frac{M_0}{L}$$

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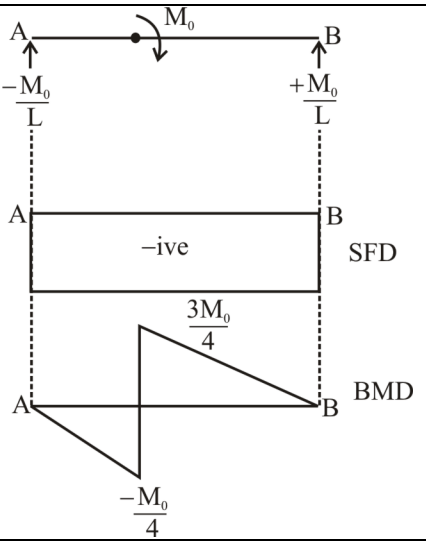


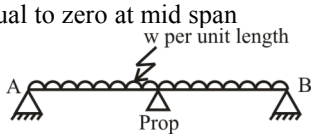
Maximum bending moment at $(x = L/4) = \frac{3M_0}{4}$

565. A simply supported beam carrying a uniformly distributed load over its whole span, is propped at the centre of the span so that the beam is held to the level of the end supports. The reaction of the prop will be

(a) half the distributed load
(b) $3/8^{th}$ the distributed load
(c) $5/8^{th}$ the distributed load
(d) 0 distributed load

RPSC IOF, 2020

Ans. (c) : The beam held to the level of end means deflection equal to zero at mid span



Deflection from distributed load $= \frac{5wl^4}{384EI}$ .....(i)

Deflection from prop load $= \frac{Pl^3}{48EI}$ .....(ii)

From equation (i) and (ii)

$$\frac{5wl^4}{384EI} = \frac{Pl^3}{48EI}$$

$$P = \frac{5}{8}wl$$

$wl \Rightarrow$ distributed load

So, The reaction (P) of the prop $= \left(\frac{5}{8}\right)^{th}$ the distributed load.

566. A simply supported beam, 6.0 m long, carries a single concentrated load of P at 1.0 m from one end. The cross-section of beam is square with the side of 100 mm. What will be the maximum value of P so that stress in material should not exceed 8 MN/m^2?

(a) 1 kN (b) 1.2 kN
(c) 1.4 kN (d) 1.6 kN

APPSC Poly Lect. 13.03.2020

Ans. (d) : Given data,
For simply supported beam length (L) = 6 m
Square cross - section side (b = d) = 100 mm = 0.1 m
Bending stress (σ_b) = 8 MN/m^2 = 8×10^6 N/m^2
Let,
P = load acting at point C from A distance = 1 m
R_A & R_B is reaction at A and B
$R_A + R_B = P$ ..... (1)
Taking moment about A, then –

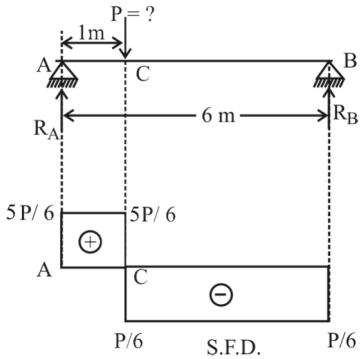
$$R_B \times 6 = P \times 1 \Rightarrow \boxed{R_B = \frac{P}{6}}$$

$$\therefore \boxed{R_A = \frac{5P}{6}}$$

Bending moment –
For section AC –
$M_x = R_A.x$ ..... (1)
For section BC –

$$M_x = R_A x - P\,(x-1) = \frac{5P.x}{6} - Px + P$$

$$M_x = -\frac{Px}{6} + P \quad .....(2)$$



From the S.F.D; shear force is zero at point C
i.e., at that point C (x=1) bending moment is maximum.

$$\therefore M_{x=1} = \frac{-P \times 1}{6} + P = \frac{5P}{6}\,N-m$$

$\because$ For square X-sectional (b=d)

$$\because I = \frac{bd^3}{12} = \frac{(0.1)\times(0.1)^3}{12}$$

$$y = \frac{d}{2} = \frac{0.1}{2}$$

Now, from bending equation –

$$\frac{M}{I} = \frac{\sigma_b}{y}$$

$$M = \frac{\sigma_b \times I}{y} = \frac{\sigma_b \times \frac{bd^3}{12}}{d/2} = \frac{\sigma_b \times d^4}{12 \times \frac{d}{2}} = \frac{8\times10^6 \times (0.1)^3}{6}$$

$$M = \frac{8\times10^6 \times 10^{-3}}{6} = \frac{8\times10^3}{6}$$

$$\Rightarrow \frac{5P}{6} = \frac{8\times10^3}{6}$$

$P = 1.6 \times 10^3$ N $\Rightarrow \boxed{P = 1.6\ kN}$

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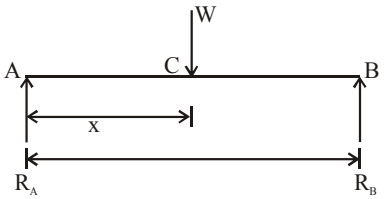
567. A simply supported beam of length *L* carrying a concentrated load W at a section which is at a distance of 'x' from one end. What will be the value of bending moment at this section?

(a) $W(x - x^2)$ (b) $W(x^2 - xL)$

(c) $W\left(x - \frac{x^2}{L}\right)$ (d) Wx

SJVN ET 2019

Ans. (c) :



$R_A + R_B = W$ .....(i)

Taking moment about A,

$R_B \times L = Wx$

$$R_B = \frac{Wx}{L}$$

$$R_A = W - R_B = W - \frac{Wx}{L}$$

Taking moment about C,

$$M_C = R_A \times x = W\left(x - \frac{x^2}{L}\right)$$

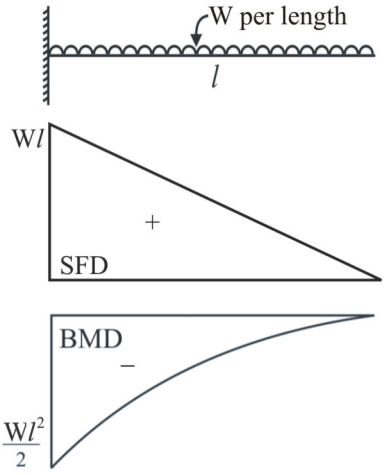
568. A cantilever beam of length *l* carry a UDL of *W* per length across the whole span. What will be the value of maximum shear stress and maximum bending moment on the beam respectively?

(a) $\frac{Wl}{2}, Wl^2$ (b) Wl, Wl^2

(c) $\frac{Wl}{2}, \frac{Wl^2}{2}$ (d) $Wl, \frac{Wl^2}{2}$

SJVN ET 2019
HPPSC AE 2018
Punjab PSC SDE 12.02.2017
GMB AAE 25.06.2017
WBPSC AE, 2017

Ans. (d) : In question they asked max shear stress that should be max shear force.

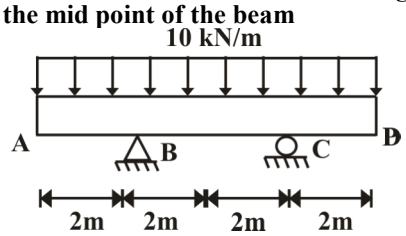


Max shear force at fixed = $W \times l$

Max value of bending moment $= \left(\frac{1}{2} W \times l\right) \times l$

$$= \frac{Wl^2}{2}$$

569. Calculate the shear force and bending moment at the mid point of the beam

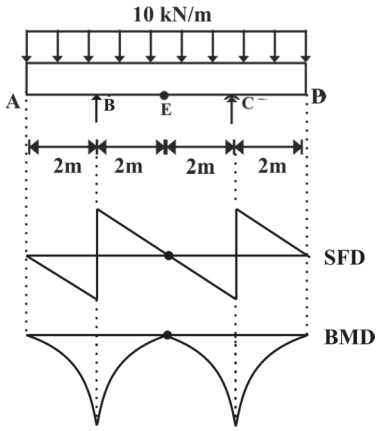


(a) 0 kN, 0 kN-m (b) -20 kN, -20 kN-m

(c) 20 kN, -20 kN-m (d) 20 kN, 0 kN-m

APPSC AEE SCREENING 17.02.2019

Ans. (a) :



Due to symmetry, R_B (or) R_C

$$= \frac{\text{Total load}}{2} = \frac{10 \times 8}{2} = 40\,\text{kN}$$

At mid span

$(SF)_E = 10 \times 4 - 40 = 0$

$(BM)_E = 40\,(2) - 10 \times 4 \times 2 = 0$

SF and BM both will be zero at mid point.

570. Consider the following:

1. **Bending moment is a moment about the longitudinal axis of a beam.**
2. **A structural component cannot have axial force and shear force together.**

(a) Only 1 is correct

(b) Only 2 is correct

(c) Both 1 and 2 are correct

(d) Both 1 and 2 are incorrect

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Ans. (d) : • Bending moment is the moment about neutral axis, but not about longitudinal axis.

• Structural components can have axial forces, shear forces together when inclined loads are acting, on a beam.

571. Out of the options given below, which one is the correct shear force diagram? B is an internal hinge

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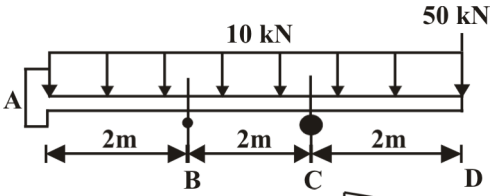
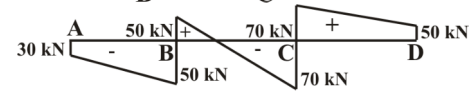
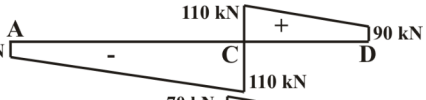
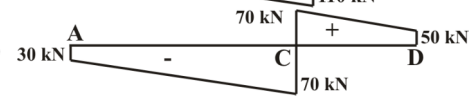


(a) 

(b) 

(c) 

(d) None of the above

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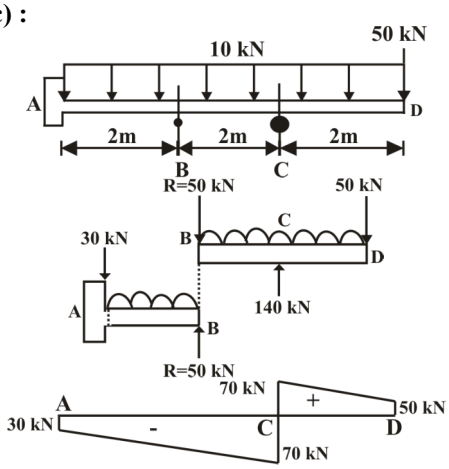
Ans. (c) :



SFD

572. Find out the Static indeterminacy of the beam in the figure below

(a) 0 (b) 2
(c) 3 (d) 6

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Ans. (b) : Degree of static indeterminacy of beam

$D_S = 3m - 3j + R_e - R_r$

m = Number of member

j = Number of joints

R_e = external reaction of support

R_r = Number of internal hinge

In the given beam,

$m = 2, j = 3, R_e = 6, R_r = 1$

$D_S = 3 \times 2 - 3 \times 3 + 6 - 1$

$= 6 - 9 + 5$

$\boxed{D_s = 2}$

573. Consider the following:

1. **In addition to equilibrium equations, compatibility equations are also required for solving indeterminate structure.**
2. **A fixed beam (two ends are fixed) is a kinematically determinate structure.**

(a) Both 1 and 2 are correct
(b) Both 1 and 2 are wrong
(c) Only 1 is correct
(d) Only 2 is correct

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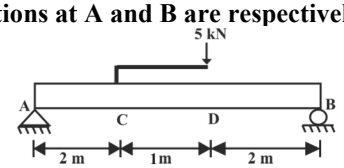
Ans. (a) : For analysing statically indeterminate structures both equilibrium and compatibility equations are required.

Fixed beam has no degree of freedom.

∴ It is kinematically determinate beam.

574. For the beam shown below, the vertical reactions at A and B are respectively



(a) 2 kN, 3 kN (b) 3 kN, 2 kN
(c) -1 kN, 1 kN (d) 1 kN, 1 kN

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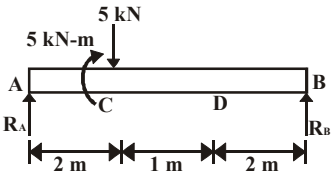
Ans. (a) :



$\sum F_y = 0$

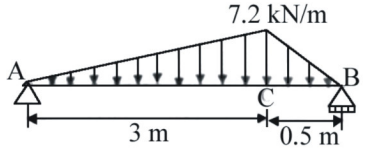
$R_A + R_B = 5$ kN

$\sum M_A = 0$

$R_B (5) = 5 \times 2 + 5 = 15/5$

∴ $R_B = 3$ kN and $R_A = 2$ kN

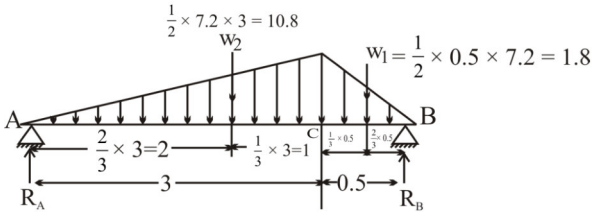
575. A simply supported beam of length 3.5 m carries a triangular load as shown in the figure below. Maximum load intensity is 7.2 N/m. The location of zero shear stress from point A is :



(a) 3 m (b) 1.5 m
(c) 2 m (d) 2.5 m

BHEL ET 2019

Ans. (c) : From section 'A C'-



$R_A + R_B = 10.8 + 1.8 = 12.6$

$\Sigma M_B = 0$

$R_A \times 3.5 - 10.8 \times 1.5 - 1.8 \times \frac{2}{3} \times 0.5 = 0$

$R_A \times 3.5 - 16.2 - 0.6 = 0$

$R_A = \frac{16.8}{3.5}$

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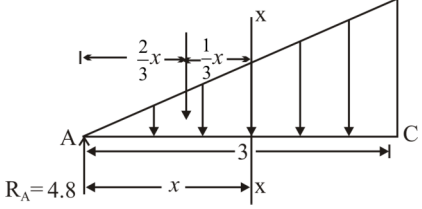
$R_A = 4.8$ kW

$R_B = 12.6 - 4.8$

$= 7.8$ kW

from section (A C) –



We know that

$L \rightarrow W$

$1 \rightarrow \frac{W}{L}$

$x \rightarrow \frac{W}{L}x$

Load at section (x - x)

$$W = \frac{1}{2} \times x \times \frac{W}{L}x$$

$$= \frac{W}{2L}x^2$$

$(SF)_A = 4.8$

$$(SF)_C = 4.8 - \frac{Wx^2}{2L}$$

zero shear force mean 'maximum bending moment'

$$4.8 - \frac{7.2x^2}{2 \times 3} = 0$$

$\boxed{x = 2m}$

576. Which of the following statements is/are correct?

1. In uniformly distributed load, the nature of shear force is linear and bending moment is parabolic.
2. In uniformly varying load, the nature of shear force is linear and bending moment is parabolic.
3. Under no loading condition, the nature of shear force is linear and bending moment is constant.

Select the correct answer using the code given below.

(a) 1 and 2 (b) 1 and 3
(c) 2 only (d) 1 only

ESE 2019

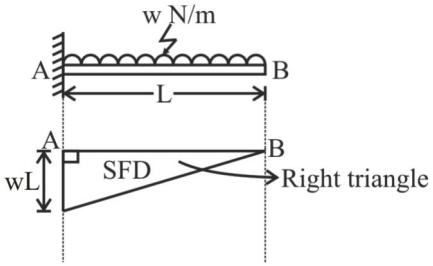
Ans. (d) : (i) In UDL, shear force is linear and bending moment is parabolic. Hence statement first is correct.
(ii) In UVL, shear force is parabolic and bending moment is cubic. Hence statement second is incorrect.
(iii) In no loading, shear force is constant and bending moment is linear. Hence statement third is also incorrect.

577. The shear force diagram of a cantilever beam of length ℓ and carrying uniformly distributed load of w per unit length will be

(a) a right angled triangle
(b) an issoscles triangle
(c) an equilateral triangle
(d) a rectangle

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) :



w = Load per unit length

At B, SF = 0

At A, $SF = w \times L = wL$

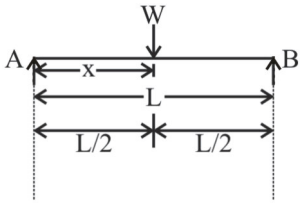
578. The maximum bending moment of a simply supported beam of span L and carrying a point load W at the centre of beam is,

(a) $\frac{WL}{4}$ (b) $\frac{WL}{2}$

(c) WL (d) $\frac{WL^2}{2}$

Nagaland PSC (CTSE) 2018, Paper-I, ESE 1996

Ans. (a) :



So from figure, $R_A = \frac{W}{2} = R_B$ $\quad \therefore R_A + R_B = W$

So for BM, at centre,

$$BM_{max} = \frac{W}{2} \times \frac{L}{2}$$

$$M_{max} = \frac{WL}{4}$$

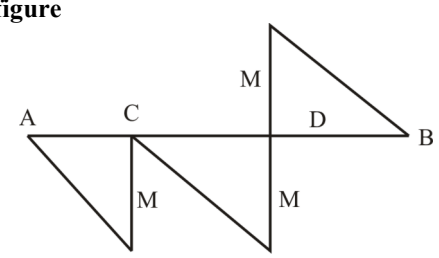
579. For a simply supported beam on two end supports the Bending Moment is maximum

(a) On the supports
(b) At mid span
(c) Where the deflection is maximum
(d) Where there is no shear force

GPSC Executive Engineer 23.12.2018

Ans. (d) : Where there is no shear force.

580. A simply supported beam AB has the bending moment diagram as shown in the following figure



The beam is possibly under the action of following loads.

(a) Couples of M at C and 2M at D
(b) Couples of 2M at C and M at D

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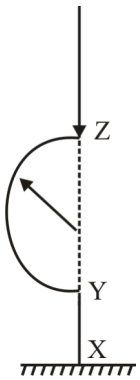
(c) Concentrated loads of M/L at C and 2M/L at D
(d) Concentrated loads of M/L at C and couple of 2M at D

GPSC ARTO Pre 30.12.2018

Ans. (a) : The beam is possibly under the action of couples of M at C and 2M at D.

581. A curved member with a straight vertical leg is carrying a vertical load at Z as shown in the figure. The stress resultants in the XY segment are:



(a) bending moment, shear force and axial force
(b) bending moment and axial force only
(c) bending moment and shear force only
(d) axial force only

GPSC ARTO Pre 30.12.2018

Ans. (d) : The stress resultants in X-Y segment is only axial force as load passes through axis of segment.
For bending moment in the section, there is some eccentricity required with the load.

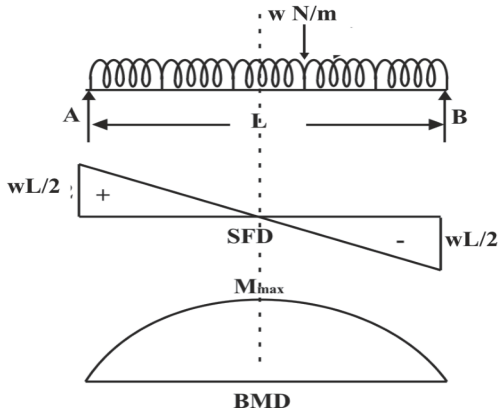
582. The shear force diagram for a simply supported beam carrying uniformly distributed load of w per unit length, consists of

(a) One right angled triangle
(b) One equilateral triangle
(c) Two right angled triangle
(d) Two rectangles

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : $R_A + R_B = wL$

$$R_A = R_B = \frac{wL}{2}$$



$BM_{max} = \frac{wL^2}{8}$ at mid point [L/2]

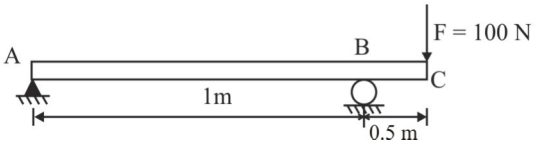
583. The reaction at the support of a beam with fixed end is referred as

(a) fixed end moment
(b) fixed end couple
(c) floating end moment
(d) floating end couple

TNPSC AE 2018

Ans. (a) : The reaction at the support of a beam with fixed end is referred as fixed end moment.

584. The beam ABC is supported at A (hinge support) and B (roller support). If a force of 100 N is applied at C as given in figure, then the reaction at the supports will be given by :

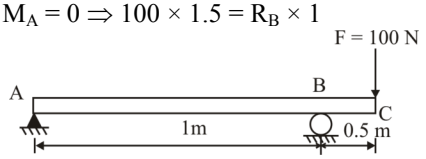


(a) $R_A = 50N(\downarrow); R_B = 150N(\uparrow)$
(b) $R_A = 50N(\uparrow); R_B = 100N(\downarrow)$
(c) $R_A = 150N(\uparrow); R_B = 50N(\downarrow)$
(d) $R_A = 150N(\downarrow); R_B = 50N(\uparrow)$
(e) $R_A = 50N(\downarrow); R_B = 150N(\downarrow)$

CGPSC AE 25.02.2018

Ans. (a) : $R_A + R_B = 100$

$M_A = 0 \Rightarrow 100 \times 1.5 = R_B \times 1$



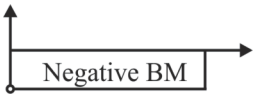
$R_B = 150N(\uparrow)$

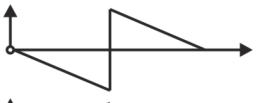
$R_A = -50N$

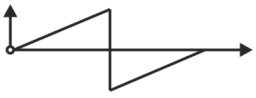
$R_A = 50N(\downarrow)$

585. Which one of the following is the correct bending moment diagram for a beam which is hinged at the ends and is subjected to a clockwise couple acting at the mid-span?

(a)

(b) 

(c) 

(d) 

ESE 2018

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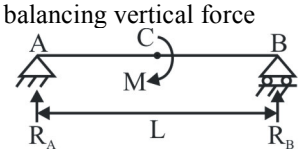
Ans. (c) : On balancing vertical force



$R_A + R_B = 0$ .....(i)

By taking moment about point A,

$R_B \times L = M$

$R_B = \frac{M}{L}$ .....(ii)

From equation (i) and (ii)

$R_A = -\frac{M}{L}$

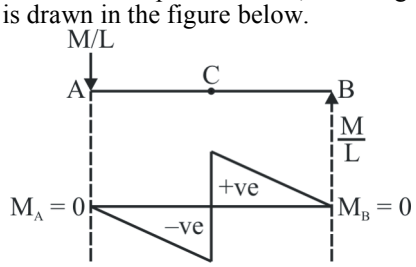
Bending moment

$M_A = M_B = 0$

$(M_c)_{Right} = R_B \times \frac{L}{2} = \frac{M}{2}$ (Sagging)

$(M_c)_{Left} = R_B \times \frac{L}{2} - M = -\frac{M}{2}$ (Hogging)

From the above computed values, bending moment diagram is drawn in the figure below.

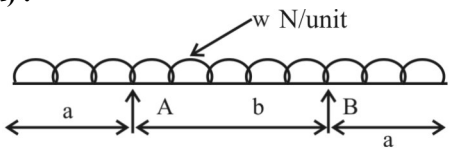


586. A beam carrying a uniformly distributed load rests on two supports 'b' apart with equal overhang 'a' at each end. The ratio b/a for zero bending moment at the mid span is

(a) 1/2 (b) 1
(c) 3/2 (d) 2

RPSC Vice Principal ITI 2018

Ans. (d) :



$R_A = R_B = \frac{(2a+b)w}{2}$

B.M. at middle of AB

$BM_{middle} = -(2a+b)\frac{w}{2} \times \frac{b}{2} + w\left(a + \frac{b}{2}\right)\left(a + \frac{b}{2}\right) \times \frac{1}{2}$

For zero B.M. at middle

$0 = -(2a+b)\frac{w}{2} \times \frac{b}{2} + \frac{w}{2}\left(a + \frac{b}{2}\right)^2$

$= -\frac{2ab - b^2}{2} + a^2 + \frac{b^2}{4} + \frac{2ab}{2}$

$= -ab - \frac{b^2}{2} + a^2 + \frac{b^2}{4} + ab$

$\boxed{\frac{b}{a} = 2}$

587. For a simply supported beam on two end supports the bending moment is maximum

(a) usually on the supports
(b) always a mid span
(c) where there is no shear force
(d) where the deflection is maximum

Karnataka PSC Lect., 27.05.2017

Ans (c) : Bending moment will be maximum if $\frac{dM}{dx} = 0$ i.e. V = 0

Bending moment will be maximum at the point at which shear force is zero (SFD changes its nature)

If M = Bending moment at the section of beam.

$\frac{dM}{dx}$ = Slope of bending moment diagram

$\frac{dM}{dx}$ = V (V = shear force at section of beam)

588. A point of contraflexure occurs in

(a) A simply supported beam
(b) A fixed beam
(c) A cantilever
(d) Fixed beam and cantilever

Karnataka PSC Lect., 27.05.2017

Ans (b) : In a beam if the bending moment changes its sign at a point, the point itself having zero bending moment, the beam changes curvature at this point of zero bending moment and this points is called point of contraflexure or point of inflection.

A point of contraflexure always occurs in a over hanging beam.

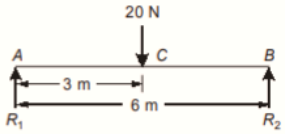
589. Calculate the bending moment at the mid-point of a 6 m long simply supported beam carrying a 20 N point load at the mid-point.

(a) 20 Nm (b) 30 Nm
(c) 45 Nm (d) 60 Nm

BPSC AE Mains 2017 Paper - VI

Ans : (b) : By symmetry,

$R_1 = R_2 = 10$ N



$(BM)_C = R_1 \times 3 = 10 \times 3$

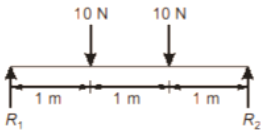
$(BM)_C = 30$ Nm

590. A 3 m long beam, simply supported at both ends, carries two equal loads of 10 N each at a distance of 1 m and 2 m from one end. The shear force at the mid-point would be

(a) 0 N (b) 5 N
(c) 10 N (d) 20 N

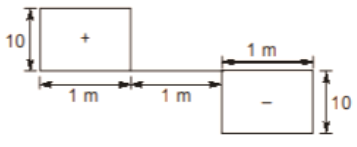
BPSC AE Mains 2017 Paper - VI

Ans : (a) :



By symmetry,

$R_1 = R_2 = 10$ N



Hence shear force at midpoint is zero.

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591. A simply supported beam of span carries over its full span a load varying linearly from zero at each end to W N/m at mid span. The maximum bending moment is

(a) $\frac{WL^2}{12}$ (b) $\frac{WL^2}{8}$

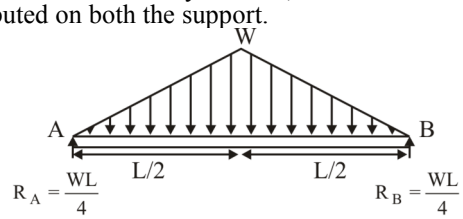
(c) $\frac{WL^2}{4}$ (d) $\frac{WL^2}{2}$

JWM 2017

Ans. (a) : Consider equilibrium of beam AB total load on beam is $2\times\left(\frac{1}{2}\times\frac{L}{2}\times W\right)$

Total load = $\frac{WL}{2}$

As the beam symmetric, the total load equally distributed on both the support.



$R_A = R_B = \frac{WL}{4}$

Bending moment (M_x)

In the region $0 < x < \frac{L}{2}$

$M_x = \frac{WL}{4}.x - \left(\frac{1}{2}\times x \times \frac{2Wx}{L}\right)\times\left(\frac{x}{3}\right)$

$= \frac{WL}{4}x - \frac{Wx^3}{3L}$

For maximum bending moment

$\frac{d(M_x)}{dx} = 0$

$\frac{WL}{4} - \frac{Wx^2}{L} = 0$

$\therefore x = \frac{L}{2}$

At. $x = \frac{L}{2}$ $\boxed{M_{max} = \frac{WL^2}{12}}$

592. The bending moment diagram for a simply supported beam is a rectangle over a larger portion of the span except near the supports. What type of load does the beam supports. What type of load does the beam carry?

(a) A uniformly distributed symmetrical load over a larger portion of the span except near the supports

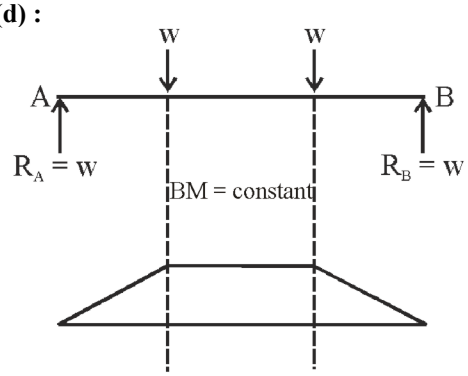
(b) A concentrated load at mid span

(c) Two identical concentrated loads equidistant from the supports and close to mid-point of the beam.

(d) Two identical concentrated loads equidistant from the mid span and close to supports

TSPSC AEE 28.08.2017 (Civil/Mechanical), CSE Pre 2007

Ans. (d) :



Two identical concentrated loads equidistant from the mid span and close to supports.

593. For a cantilever $\frac{dM}{dx}$ = constant for its whole length. What is the shape of the SF diagram for the beam?

(a) Rectangle (b) Triangle

(c) A parabola (d) A hyperbola

TSPSC AEE 28.08.2017 (Civil/Mechanical), CSE Pre 2008

Ans. (a) : $\frac{dM_x}{dx} = V_x$

For $\frac{dM_x}{dx} = \text{constant}$

Shear force,

V_x = constant

$\therefore$ Shape of SFD is rectangular.

594. SF diagram for a simply supported beam is a rectangle with its longer side equal to beam length. What type of load is acting on the beam?

(a) Concentrated load at it mid span

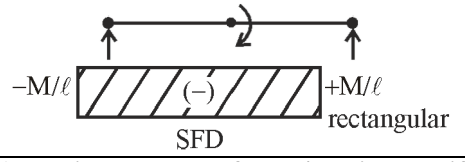
(b) UDL over its whole span

(c) Concentrated load along with a couple at a point on beam length

(d) Couple at a point on the beam length

TSPSC AEE 28.08.2017 (Civil/Mechanical), CSE Pre 2008

Ans. (d) :



595. A cantilever beam of span/carries a uniformly varying load of zero intensity at the free end and w per metre length at the fixed end. What does the integration of the ordinate of the load diagram between at the limits of free and fixed ends of the beam give?

(a) Bending moment at the fixed end

(b) Shear force at the fixed end

(c) Bending moment at the free end

(d) Shear force at the free end

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) :

$\frac{dV_x}{dx}$ = w (Rate of loading)

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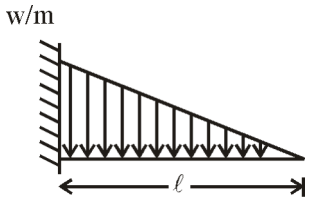


Shear force between any two points on the loaded beam is equal to total area of the leading diagram between these two points.

$$(SF)_{\text{fixed end}} = \frac{1}{2} \times \ell \times w$$

596. The diagram showing the variation of axial load along the span is called

(a) shear force diagram
(b) bending moment diagram
(c) thrust diagram
(d) influence line diagram

TSPSC AEE 28.08.2017 (Civil/Mechanical)

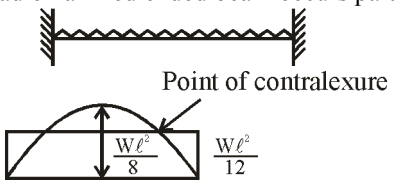
Ans. (c) : Diagram showing the variation of axial load along the span is called thrust diagram.

597. The variation of the bending moment due to the moving load on a fixed ended beam occurs

(a) linear (b) parabolic
(c) cubic (d) constant

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : The variation of the Bending moment due to a moving load on a fixed ended beam occurs parabolic.

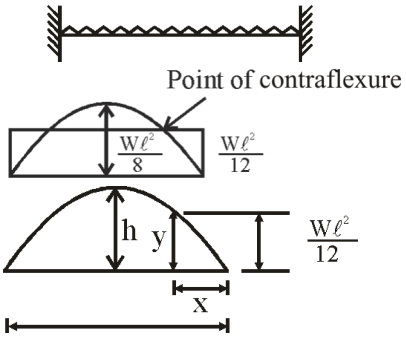


598. A prismatic beam of length L is fixed at both ends carries a uniformly distributed load. The distance of points of contra flexure from either end is

(a) 0.207 L (b) 0.211 L
(c) 0.277 L (d) 0.25 L

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) :



$$y = \frac{W\ell^2}{12}$$

$$y = \frac{4h}{\ell^2} x(\ell - x)$$

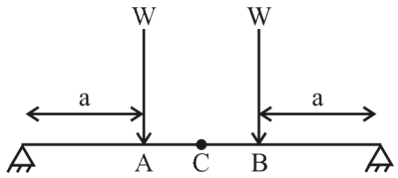
$$\frac{W\ell^2}{12} = \frac{4}{\ell^2}\left(\frac{W\ell^2}{8}\right) x(\ell - x)$$

$$\frac{\ell^2}{6} = x(\ell - x)$$

$$x^2 - \ell x + \frac{\ell^2}{6} = 0$$

$$x = 0.2111\ell$$

Directions : Q. No. 599-600 are based of the following figures:

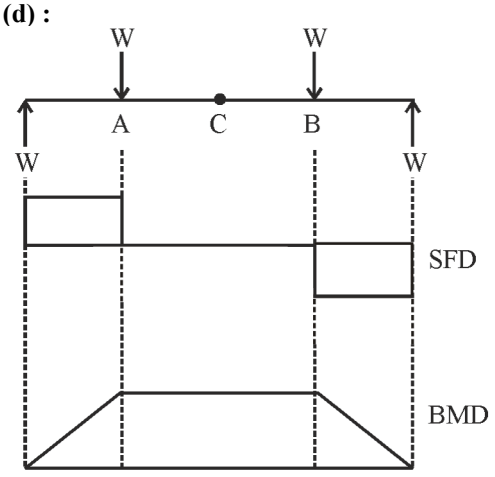


599. A beam is simply supported at the ends and loaded as shown in figure. The shearing force at the mid-point C will be:

(a) $\frac{W}{2}$ (b) W
(c) 2W (d) Zero

PTCUL AE 25.06.2017

Ans. (d) :

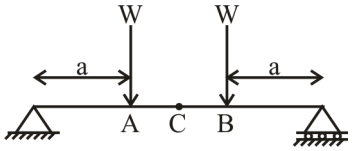


600. The bending moment between the point A and B is:

(a) Increasingly linearly
(b) Decreasingly linearly
(c) Varying parabolically
(d) Constant

PTCUL AE 25.06.2017

Ans. (d) :



By symmetric–
Reaction at supports is $R_M = W$, $R_N = W$
For SFD: (From left side)
Shear force at C, $SF_M = R_M = W$
Shear force at A (just left of A), $SF_A = W$
Shear force at A (just right of A), $SF_A = W - W = 0$
Shear force at B (just left of B), $SF_B = 0$
Shear force at B (just right of B), $SF_B = -W$

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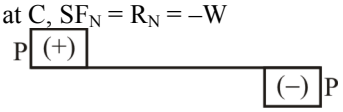
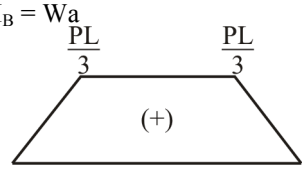
Shear force at C, $SF_N = R_N = -W$



For BMD: (From left side)

BM at A, $BM_A = Wa$

BM at B, $BM_B = Wa$



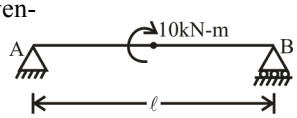
From SFD and BMD it is clear that there will be no shear force between A and B, but there will be a uniform bending moment between A and B.

601. A beam of length 2 m is pinned at both ends and is subjected to a concentrated moment 10 kNm at its center. The maximum bending moment in the beam is

(a) 10 kNm (b) 5 kNm
(c) 20 kNm (d) 15 kNm

GPSC EE Pre, 28.01.2017

Ans. (b) : Given-



Length (l) = 2 m

Concentrated moment = 10 kN-m

Maximum bending moment = $\frac{M}{l} \times \frac{l}{2}$

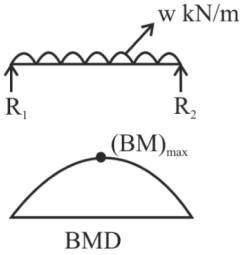
$$(BM)_{max} = \frac{10}{2} = 5\,kN-m$$

602. The number of points of contraflexure in a simple supported beam carrying uniformly distributed load, is

(a) One (b) Two
(c) Zero (d) None of these

GPSC EE Pre, 28.01.2017

Ans. (c) : Zero



In a bending moment diagram, the point of contra flexure is the point on beam where bending moment changes sign.

603. Match the following :

P.	Point of contraflexure	i.	Bending moment changes sign
Q.	Load intensity	ii.	Shear force changes sign
		iii.	Rate of change of bending moment
		iv.	Rate of change of shear force

(a) P-i, Q-iv (b) P-ii, Q-iv
(c) P-i, Q-iii (d) P-ii, Q-iii

APGENCO AE, 2017

Ans. (a) : At point of contraflexure bending moment changes its sign from positive to negative and vice-versa and at this point value of B.M. becomes zero.

Shear force $(S) = \frac{dM}{dx}$ = rate of change of bending moment.

Intensity of loading $(w) = \frac{dS}{dx}$ = rate of change of shear force.

604. The rate of change of shear force at any section represents

(a) Point of contraflexure
(b) Shear force at that section
(c) Bending moment at that section
(d) Rate of loading at that section

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Ans. (d) : Consider a simply supported beam subjected to UDL for the entire span consider a free body diagram at small portion elemental length example.

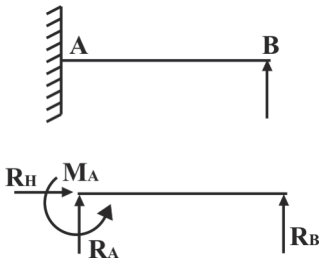
This rate of change of shear force at any section is equal to the intensity of loading.

605. A beam is fixed at one end and is vertically supported at the other end. What is the degree of statical indeterminacy?

(a) 1 (b) 2
(c) 3 (d) 4

ISRO Scientist/Engineer 07.05.2017

Ans. (a) :



- A beam is to be statically indeterminate, if the number of reaction is greater than static equilibrium condition equation.
- A beam is said to be statically determinate, if the number of reaction is equal to static equilibrium condition equation.

For propped cantilever beam—

Four reaction R_H, R_A, R_B and M_A.

And three equation of static i.e. $\sum H = 0$, $\sum V = 0$ and $\sum M = 0$

So, hence beam is statically indeterminate to one degree.

So, total No. of Reaction = 4

No. of independent equilibrium equation = 3

So, degree of static indeterminacy will be = 4 - 3 = 1.

606. A simply supported beam of length L is subjected to a varying distributed load sin(πx/L) N/m, where distance x is measured from the left support. The magnitude of the vertical reaction in N at the left support is

(a) 3L/π (b) 2L/π
(c) L/2π (d) L/π

ISRO Scientist/Engineer 07.05.2017

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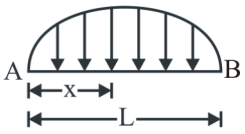
Ans. (d) :



$R_A = ?$

$$\text{Load} = \sin\left(\frac{\pi x}{L}\right) N/m$$

$$\text{Total load} = \int_0^L \sin\left(\frac{\pi x}{L}\right) dx$$

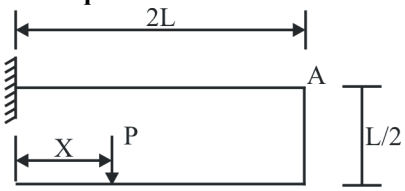
$$= -\frac{L}{\pi}\left[\cos\left(\frac{\pi x}{L}\right)\right]_0^L$$

$$\text{Total load} = -\frac{L}{\pi}[-1-1]$$

$$= \frac{2L}{\pi}$$

$$R_A = R_B = \frac{2L}{\pi} \times \frac{1}{2} = \frac{L}{\pi}$$

607. A force P is applied at a distance 'X' from the end of the beam as shown in the figure. What would be value of 'X' so that the displacement at 'A' is equal to zero?



(a) 0.5 L (b) 0.25 L
(c) 0.33 L (d) 0.66 L

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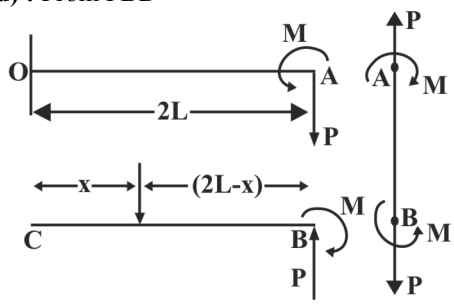
Ans. (d) : From FBD—



where,

$M = P(2L - x)$

Deflection at point (A) due to moment M.

$$\delta_M = \frac{M(2L)^2}{2EI} = \frac{4ML^2}{2EI}(\uparrow)$$

$$= \frac{4P(2L-x).L^2}{2EI}$$

Deflection at point (A) due to point load P

$$\delta_P = \frac{P(2L)^3}{3EI} = \frac{8PL^3}{3EI}(\downarrow)$$

So,

$\delta_M + \delta_P = 0$ (given)

$$\frac{4P(2L-x).L^2}{2EI} - \frac{8PL^3}{3EI} = 0$$

$$\frac{4P(2L-x).L^2}{2EI} = +\frac{8PL^3}{3EI}$$

$$2L - x = \frac{4}{3}L$$

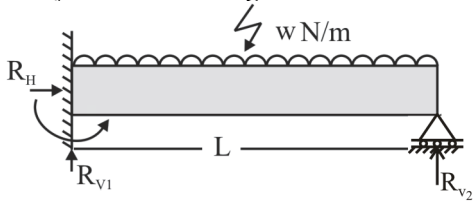
$$x = \frac{2}{3}L = 0.66L$$

608. A propped cantilever is indeterminate externally to

(a) The second degree (b) The third degree
(c) The fourth degree (d) The fifth degree

TNPSC AE 2017

Ans. (a) : A propped cantilever is indeterminate externally to the second degree.



For general loading, the total reaction components (R) are equal to (3 + 2) = 5, while the total number of conditions (r) are equal to 3. The beam is statically indeterminate externally to second degree.

Note- For vertical loading, the beam is statically determinate to single degree (In figure).

609. The difference between member of a truss and of a beam is:

(a) The members of a truss take their loads along their length whereas a beam takes loads at right angles to its length
(b) The member of the truss takes load lateral to its length whereas the beam along the length
(c) The member of the truss can be made of C.I where as the beam is of structural steel only
(d) The member of the truss can have a circular cross-section whereas the beam can have any cross-section

JWM 2017

Ans. (a) : Beams support their loads in shear and bending where as truss support loads in tension and compression.

The members of a truss take their loads along their length where as a beam takes loads at right angle to its length.

610. Shear force at any point of the beam is the algebraic sum of

(a) All vertical forces
(b) All horizontal forces
(c) Forces on either side of the point
(d) Moment of forces on either side of the point

Vizag Steel (MT) 2017

Ans. (c) :

- Shear force at any point of the beam is the algebraic sum of forces on either side of the point.
- Bending moment at any point of the beam is the algebraic sum of moment of forces on either side of the point.

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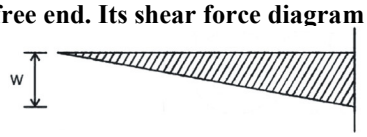
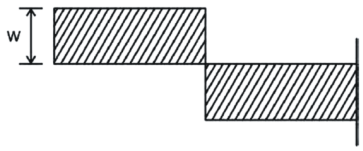
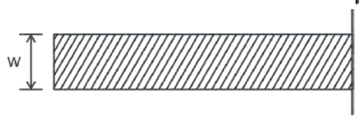
611. A cantilever carries a concentrated load (W) at its free end. Its shear force diagram will be:

(a) 

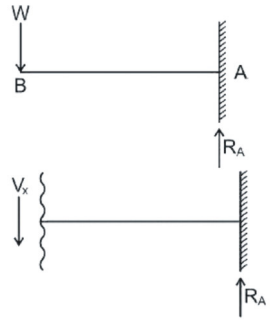
(b)

(c) 

(d) 

CIL MT 26.03.2017

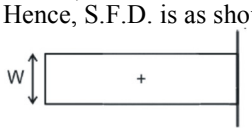
Ans. (d) : $R_A = w$



$V_x = R_A = w$

$V_A = w$, $V_B = w$

Hence, S.F.D. is as shown in figure below

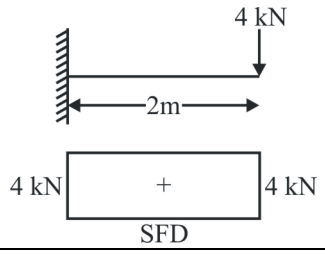


612. A cantilever beam of span 2 m with a point load of 4 kN at its free end will have a constant shear force of _____ throughout the span.

(a) 2 kN (b) 4 kN
(c) 6 kN (d) 8 kN
(e) 10 kN

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : It have constant shear force of 4 kN.



613. Basic equation of deflection (y) of the beam is represented by $EI\frac{d^2y}{dx^2} = M$, where EI flexural rigidity and M Bending moment, then $EI\frac{d^3y}{dx^3}$

(a) Shear force at the section
(b) Rate of loading
(c) Zero always
(d) Bending moment at section

UPRVUNL AE 07.10.2016
APPSC Poly. Lect. 2013

Ans. (a) :

$$EI\frac{d^2y}{dx^2} = M$$

We know that

$$\frac{dM}{dx} = \text{shear force}$$

then

$$EI\frac{d^3y}{dx^3} = \frac{dM}{dx} = \text{shear force at section}$$

614. Shear force at a section of a horizontal beam is the

(a) algebraic sum of the vertical forces to any one side of the section
(b) algebraic sum of the vertical forces on both sides of the section
(c) algebraic sum of the moments of all forces on any one side of the section
(d) algebraic sum of the moments of all forces on both sides of the section

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Shear force at a section of horizontal beam is the algebraic sum of the vertical forces to any one side of the section.

615. Bending moment is maximum at a section of a beam where

(a) Shear force changes from –ve to +ve
(b) Shear force is zero
(c) Shear force changes from +ve to –ve
(d) All the given answers

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : Bending moment at any section x

$\frac{dM_x}{dx} = V_x$ (shear force)

The bending moment is maximum at a section of a beam where shear force changes its sign or the value of shear force is zero.

616. Which of the following is giving correct relation between load(w), shear force (F) and bending moment (M)

(a) $M = \frac{dF}{dx}$ (b) $F = \frac{dM}{dx}$
(c) $w = \frac{dM}{dx}$ (d) $M = \frac{dw}{dx}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : As we know, The Rate of change of Bending moment with respect to 'x' is equal to the shearing force or the slope of the moment diagram at a given point is the shear at that point.

So shear force, $F = \frac{dM}{dx}$

617. A beam with equal overhangs is carrying UDL as shown in the following figure. The bending moment at the centre of the beam will be zero for the condition

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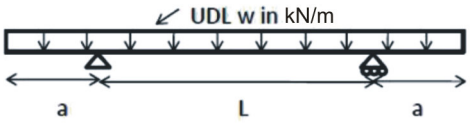
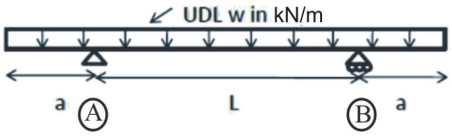


(a) a = L/3 (b) a = L/2
(c) a = 3L/2 (d) a = L

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : We know,



$$R_A = W\left(a + \frac{L}{2}\right)$$

$$M_x = R_A x - \frac{W}{2}(a+x)^2$$

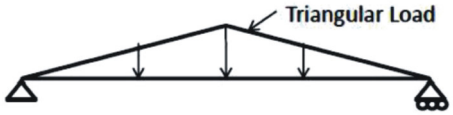
$$= W\left(a + \frac{L}{2}\right)x - \frac{W}{2}(a+x)^2$$

We know,

At $x = \frac{L}{2}$, $M = 0$,

So, We get $\frac{a}{L} = \frac{1}{2} \Rightarrow \boxed{a = \frac{L}{2}}$

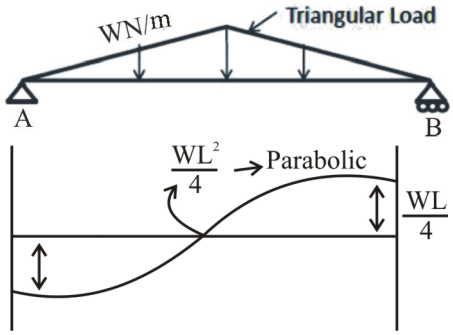
618. A simple supported beam is carrying a linearly varying load from zero at either ends to the maximum value at the mid span. Then the shape of shear force diagram (SFD) is



(a) Rectangle
(b) Triangular
(c) Second Degree Parabola
(d) cubic

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : Total load = $\frac{1}{2} \times L \times W = \frac{WL}{2}$



$$w_x = \frac{Wx}{L/2} = \frac{2Wx}{L}$$

Shear force at section x,

$$V_x = \frac{WL}{4} - \frac{1}{2} x w_x$$

$$= \frac{WL}{4} - \frac{1}{2} x \left(\frac{2Wx}{L}\right)$$

$$\boxed{V_x = \frac{WL}{4} - \frac{Wx^2}{L}} \text{ (Parabolic)}$$

$$\boxed{M_x = \frac{WL}{4} x - \frac{Wx^3}{3L}} \text{ (Cubic Parabolic)}$$

619. A simply supported beam of span L and constant width b carries a point load W at mid span. The depth of the beam required at the mid span for maximum extreme fibre stress P.

(a) $d = \frac{3WL}{2bp}$ (b) $d = \sqrt{\frac{3WL}{2bp}}$
(c) $d^3 = \frac{3WL}{2bp}$ (d) $d = \frac{3WL}{2bp^2}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : Bending moment, $M_{max} = \frac{WL}{4}$

Maximum extreme stress (Bending stress) = P

Bending stress, $\sigma_b = P = \frac{My}{I} = \frac{M\frac{d}{2}}{\left(\frac{bd^3}{12}\right)}$

$$P = \frac{\left(\frac{WL}{4}\right)\left(\frac{d}{2}\right)}{\frac{bd^3}{12}} = \frac{12}{8}\frac{WL}{bd^2}$$

$$P = \frac{3}{2}\frac{WL}{bd^2}$$

$$\boxed{d = \sqrt{\frac{3WL}{2bP}}}$$

620. The number of contra flexure points that occur in a cantilever beam subjected uniformly distributed load is

(a) 3 (b) 2
(c) 1 (d) 0

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : Point of contraflexure in a beam is a point at which bending moment changes it sign from positive to negative and vice versa. But in cantilever there is no change in sign of bending moment. So, point of contraflexure is zero.

621. A simply supported beam of span L carries a point load P at center, the slope at left end is

(a) $PL^3/(16EI)$ (b) $PL^2/(48EI)$
(c) $PL^2/(16EI)$ (d) $PL^3/(48EI)$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : For a simply supported beam, L, The general equation for this is

$$EI\frac{dy}{dx} = \frac{Px^2}{4} - \frac{PL^2}{16}$$

So, the slope of left end.

$$\boxed{\frac{dy}{dx} = \frac{PL^2}{16 \times EI}}$$

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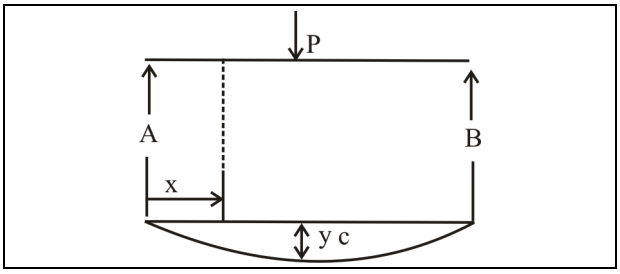

622. When equations of statics are not sufficient to determine all the reactive forces at the supports, such beams are called

(a) statically indeterminate
(b) statically determinate
(c) Imperfect
(d) Defective

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : In statics and structural mechanics a structure is statically indeterminate when the static equilibrium equations - force & moment equilibrium conditions are insufficient for determining the internal forces & reactions on that structure.

623. A simply supported beam of span length 6 m carries a uniformly distributed load of 1.5 kN/m. What is the maximum value of bending moment?

(a) 13.5 kN-m (b) 6.75 kN-m
(c) 27 kN-m (d) 9.25 kN-m

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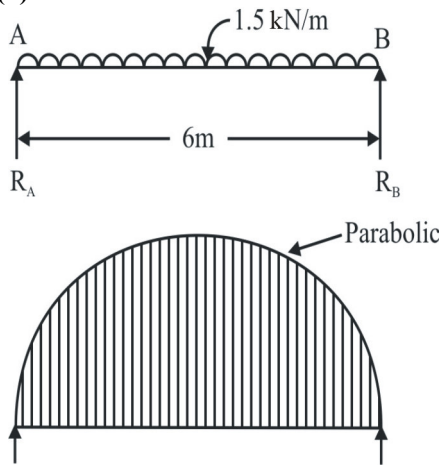
Ans. : (b)



$$(BM)_{max} = \frac{wl^2}{8}$$

$$(BM)_{max} = \frac{1.5 \times 6^2}{8}$$

$$(BM)_{max} = 6.75\,kN\text{–}m$$

624. A cantilever of length 4m is subjected to clockwise couple of 12kN-m at free end. What is the shear force at fixed end?

(a) zero (b) 12 kN
(c) 3 kN (d) 4 kN

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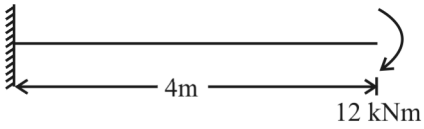
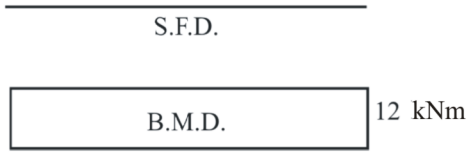
Ans. (a) :



As S.F. is due to load only and here there is no load on the beam so there will not be any S.F. in beam.



625. A simply supported beam with a gradually varying load from zero at 'B' and 'w' per unit length at 'A'. The shear force at 'B' is equal to

(a) $\frac{wl}{6}$ (b) $\frac{3wl}{2}$
(c) wl (d) $\frac{2wl}{3}$

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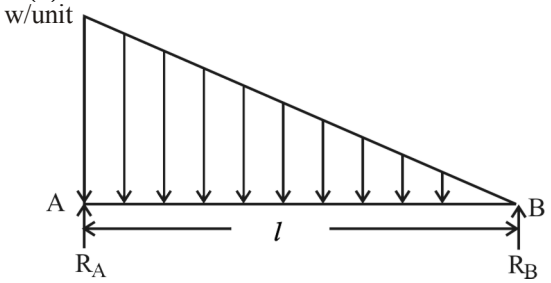
Ans. (a) :



Total load = Area of loading diagram

$$= \frac{1}{2} \times w \times l = \frac{wl}{2}$$

This load will act at the C.G. of loading diagram

i.e $\frac{l}{3}$ from support A,

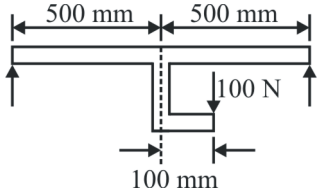
Reaction at A, $R_A = \frac{\frac{wl}{2} \times \frac{2l}{3}}{l}$

$$R_A = \frac{wl}{3}$$

Reaction at B, $R_B = \frac{\frac{wl}{2} \times \frac{l}{3}}{l}$

$$R_B = \frac{wl}{6}$$

626. In a simply supported beam loaded as shown in figure, the maximum bending moment in Nm is



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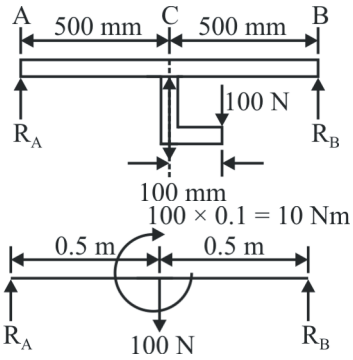
(a) 25 (b) 30
(c) 35 (d) 60

ISRO Scientist/Engineer 03.07.2016
GATE 2007

Ans. (b) :



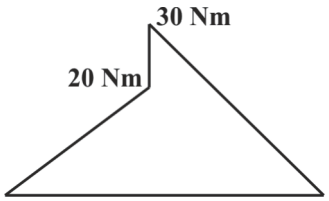
$R_A + R_B = 100$

Taking moment about point A

$R_B \times 1 = 100 \times 0.1 + 100 \times 0.5$

$R_B = 60N$

$R_A = 40N$



Moment at point (B)

$M_B = 0$

Moment at point (A) $M_A = 0$

Moment at point C = 40 × 0.5 = 20

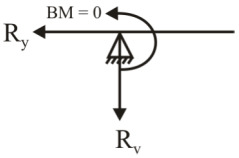
Concentrated moment (10 N-m) also act in the same direction so

Max. Bending moment at C = 20 + 10 = 30 N-m.

627. At a hinge in a beam
(a) Shear force is zero
(b) Bending moment is maximum
(c) Bending moment is zero
(d) Bending moment changes sign

APPSC AEE Screening Test 2016

Ans. (c)



Bending moment (BM) = 0

628. The bending moment for a beam with equal overhangs and carrying equal point loads each of P at the free ends, the shear force for the beam portion between the supports is
(a) Zero
(b) equal to P
(c) Linearly varying from–P to +P
(d) Linearly varying from 0 to P

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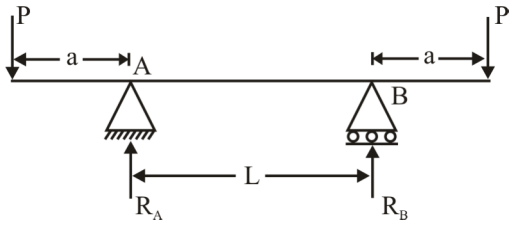
Ans. (a)



$R_A + R_B = 2P$

$R_A = P, \ R_B = P$

Hence, the shear force for the beam portion between the supports is zero.

629. The reaction at a roller support of beam will be
(a) Tangential to support
(b) Normal to support
(c) Unknown in direction
(d) Inclined to support

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Ans. (b)

Only normal/perpendicular support = R_v

$R_H = 0$

$\Sigma M = 0$

630. For a beam subjected to point loads, the shear force between point loads
(a) is constant
(b) varies linearly
(c) has parabolic variation
(d) always remain zero

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Ans. (a) : For a beam subjected to point loads, the shear force between point loads is constant.

631. Which of the following statements is true for shear force (SF) and bending moment (BM) diagram (where, w = weight per unit length)
(a) Change in BM over a small length (dM) = Area of SF diagram under that length (Vdx)
(b) Change in BM over a small length (dM) = Rate of change of SF under that length (dV/dx)
(c) Rate of change of Change in BM over a small length (dM/dx) = Rate of change of SF under that length (dV/dx)
(d) Change in SF over a small length (dV) is greater than area of loading diagram over that length (wdx)

RPSC LECTURER 16.01.2016

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Ans. (a) : We know that

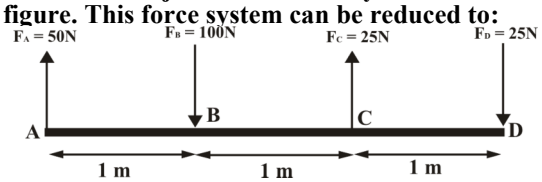
$$\frac{dM}{dx} = \text{shear force (SF)}$$

$dM = (SF) \times dx$

So,

change in BM over a small length (dM) = Area of SF diagram under that length

632. A beam is subjected to a force system shown in figure. This force system can be reduced to:



(a) A single force of 50 N (downward) at 2.5 m from A

(b) A single force of 50 N (downward) at 2.5 m from D

(c) A single force of 50 N (upward) at 2.5 m from D

(d) A single force of 50 N (upward) at 2.5 m from A

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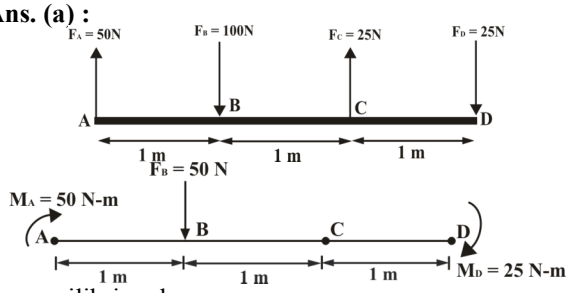
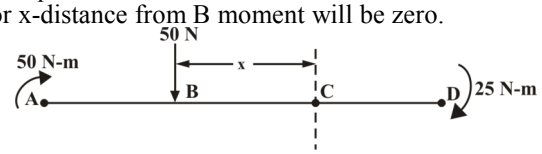
Ans. (a) :



For equilibrium beam

For x-distance from B moment will be zero.



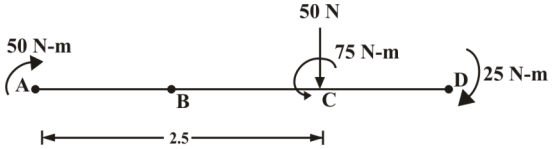
For equilibrium

$$\sum M_{CW} = \sum M_{ACW}$$

$50 + 25 = 50x$

$x = 1.5$ or

From A at a distance of 2.5



Then this force system can be reduced to single force of 50 N (downward) at 2.5 m from A.

633. Bending moment at distance L/4 from one end of a simply supported beam of length (L) with uniformly distributed load of strength w per unit length is given by

(a) $\frac{7}{32}wL^2$ (b) $\frac{5}{32}wL^2$

(c) $\frac{3}{32}wL^2$ (d) $\frac{1}{32}wL^2$

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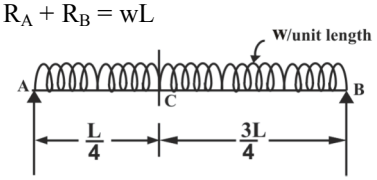
Ans. (c) : $R_A + R_B = wL$



taking moment at point A

$$R_B \times L = wL \times \frac{L}{2}$$

$$\boxed{R_B = R_A = \frac{wL}{2}}$$

taking moment at point C

$$M_c = R_A \times \frac{L}{4} - w \times \frac{L}{4} \times \frac{L}{8}$$

$$M_c = \frac{wL}{2} \times \frac{L}{4} - w\frac{L}{4} \times \frac{L}{8} = \frac{3}{32}wL^2$$

634. If the shear force diagram for a beam is triangle with length of the beam as its base, the beam is

(a) A cantilever with a point load at its free end.

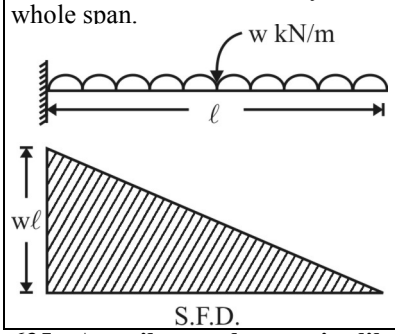
(b) A cantilever with uniformly distributed load over its whole span.

(c) A simply supported beam with a point load at its mid-point.

(d) A simply supported beam with uniformly distributed load over its whole span.

UPPSC AE 12.04.2016 Paper-I

Ans : (b) If the shear force diagram for a beam is triangle with length of the beam as its base, the beam is a cantilever with uniformly distributed load over its whole span.



635. A railway sleeper is like a beam having uniformly distributed load acting upwards (as reaction) and supports at rails with loads acting downwards due to the passing rail, such that:

(a) Greatest bending moment is at the centre of sleeper

(b) Greatest bending moment is at the end of sleeper

(c) Greatest bending moment is at the contact of rails

(d) Greatest bending moment is as small as possible

GPSC Asstt. Prof. 28.08.2016

Ans. (d) :

636. Two similar beams of same length and same load acting at the centre. First one is fixed at ends and second one is simply supported. The maximum bending moment of first with respect to second is

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(a) Double (b) Same
(c) Half (d) One fourth

GPSC Asstt. Prof. 28.08.2016

Ans. (c) : II[nd] maximum BM for simply supported beam at centre $= \frac{PL}{4}$

I[st] maximum BM for fixed end beam at centre $= \frac{PL}{8}$

$$\text{Ratio} = \frac{(BM)_I}{(BM)_{II}} = \frac{\frac{PL}{8}}{\frac{PL}{4}} = \frac{1}{2}$$

So, I[st] to II[nd] is

$$\text{Maximum } (BM)_{ss} = \frac{1}{2} \text{Maximum } (BM)_{\text{fixed}}$$

637. …….. has the unit 'N-m'.

(a) Kinetic Energy (b) Potential Energy
(c) Shear Force (d) Bending Moment

HPPSC Lect. (Auto) 23.04.2016

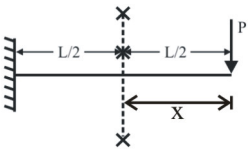
Ans. (d) : Bending moment has the unit 'N-m'
- Kinetic energy – Joule (J)
- Potential energy – Joule (J)
- Shear force – Newton (N)

638. For a cantilever beam of length 'L' having a point load 'P' at its free end, the bending moment at the mid-span of the beam is given by :

(a) PL (b) (P/2) L
(c) P/L (d) L/P

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) :



$M_{xx} = P \times x$

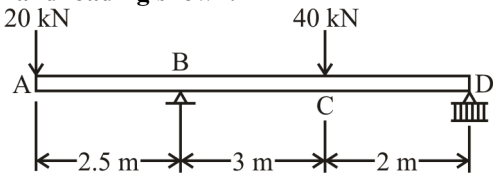
At x = L (Fixed point)

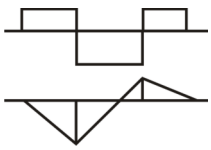
Bending moment ($M_{\text{at fixed end}} = PL$)

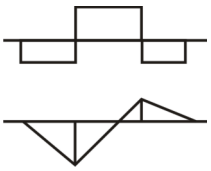
At, $x = \frac{L}{2}$ (Mid point)

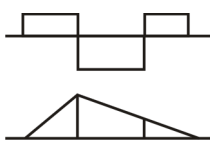
$$M_{\text{mid span}} = P.\frac{L}{2}$$

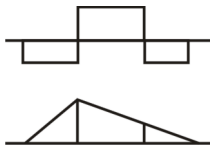
639. Draw the SFD and BMD diagram for the beam and loading shown.



(a) 

(b) 

(c) 

(d) 

(e) None of these

CGPSC AE 16.10.2016

Ans. (b) :

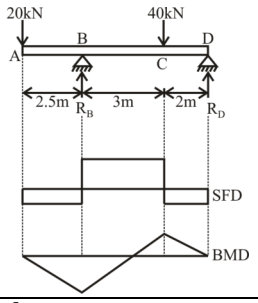
$\sum M_D = 0$

$R_B \times 5 = 40 \times 2 + 20 \times 7.5$

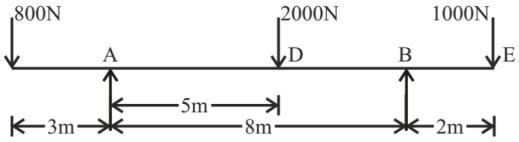
$R_B = 46$ kN

$R_D = 60 - 46$

$R_D = 14$ kN



640. A beam is loaded as shown :

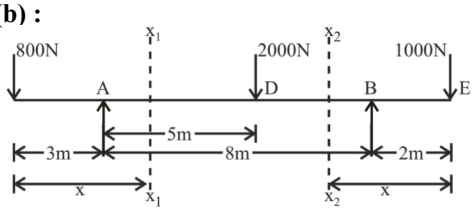


The points of contraflexure within the span AB will be

(a) 3m from A and 3 m from B
(b) 3m from A and 1.67 m from B
(c) 1.67m from A and 3 m from B
(d) 1.67m from A and 1.67 m from B
(e) None of the above

CGPSC AE 16.10.2016

Ans. (b) :



$R_A + R_B = 3800$ N

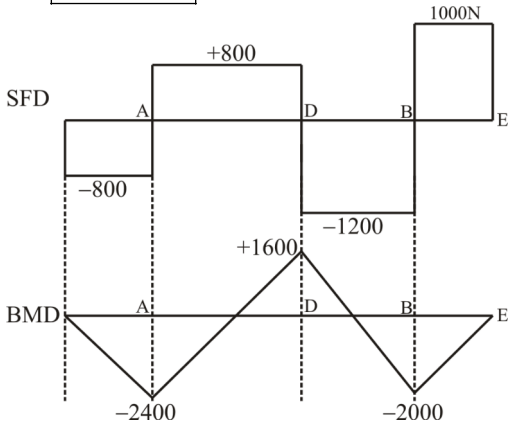
$\sum M_A = 0$

$800 \times 3 + R_B \times 8 = 2000 \times 5 + 1000 \times 10$

$\boxed{R_B = 2200\,\text{N}}$

$R_A = 3800 - 2200$

$\boxed{R_A = 1600\,\text{N}}$



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Moment in section AD (x_1-x_1)

$M = 0$

$800 \times x - 1600 \times (x - 3) = 0$

$x = 6$ m

Distance from A = 6 – 3 = 3 m

Moment in section DB (x_2-x_2)

$M = 0$

$1000x = 2200 \times (x - 2)$

$x = 3.666$

Distance from B = 3.666 – 2

= 1.667 m

641. Slope of tangent to shear force diagram gives

(a) Bending moment
(b) Couple moment
(c) Support reactions
(d) Rate of loading
(e) Rate of bending moment

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : According to relation $\frac{dF}{dx} = -W$ (constant)

So, slope of tangent to shear force diagram gives rate of loading.

642. For a simply supported beam of length L, the bending moment M is described as M = a ($x - x^3/L^2$), $0 \le x \le L$; Where a is constant. The shear force will be zero at:

(a) The supports (b) $x = \sqrt{\frac{L}{3}}$

(c) X = L (d) $X = \frac{L}{2}$

TSPSC Managers, 2015

Ans. (*) : Given – Bending moment (M)

$$M = a\left(x - \frac{x^3}{L^2}\right), \quad 0 \le x \le L$$

We know that,

$$\text{shear force } (F) = \frac{dM}{dx}$$

So, $\frac{dM}{dx} = a - \frac{3x^2}{L^2} \times a$

According to question says that shear force is zero.

$$F = 0 = \frac{dM}{dx}$$

$$a - \frac{3x^2}{L^2} \times a = 0$$

or $\frac{3x^2}{L^2} = 1$

$3x^2 = L^2$

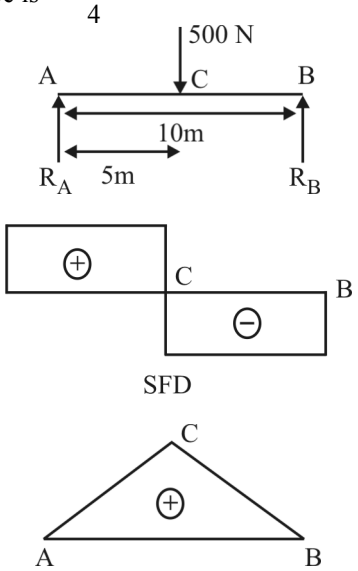
$$\boxed{x = \frac{L}{\sqrt{3}}}$$

643. A simply supported beam of span 10 m carrying a load of 500 N at the midspan will have a maximum bending moment of :

(a) 500 Nm (b) 1250 Nm
(c) 2500 Nm (d) 5000 Nm

Kerala PSC AE 06.08.2015

Ans. (b) : Formula for max bending moment for simply supported be is $= \frac{WL}{4}$



$\Sigma F_V = 0$

$R_A + R_B = 500$

take moment about 'B'

$R_A \times 10 = 500 \times 5$

$R_A = 250$ N ...(i)

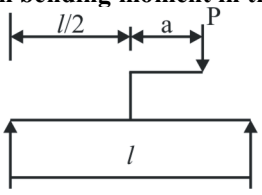
$R_B = 250$ N ...(ii)

moment about 'C' (will be max.)

$M_A = R_A \times 5 = 250 \times 5$

= 1250 Nm

644. A simply supported beam carries a load 'P' through a bracket as shown in the figure. The maximum bending moment in the beam is

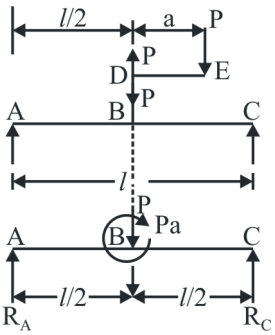


(a) $\frac{Pl}{2}$ (b) $\frac{Pl}{2} + \frac{ap}{2}$

(c) $\frac{Pl}{2} + aP$ (d) $\frac{Pl}{2} - aP$

ISRO Scientist/Engineer 11.10.2015
UPPSC AE 12.04.2016 Paper-I

Ans. (*) :



$R_A + R_C = P$ .....(i)

Take a moment about 'C'

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$$R_A \times \ell + Pa = P \times \frac{\ell}{2}$$

$$R_A \times \ell + Pa = \frac{P\ell}{2}$$

$$R_A = \frac{\frac{P\ell}{2} - Pa}{\ell} = \frac{P}{2} - \frac{Pa}{\ell}$$

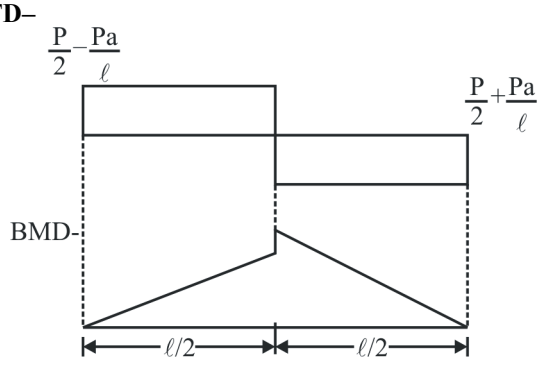
From equation (i)– $R_A + R_C = P$

$$R_A + R_C = P$$

$$R_C = P - R_A = P - \frac{P}{2} + \frac{Pa}{\ell}$$

$$R_C = \frac{P}{2} + \frac{Pa}{\ell}$$

SFD–



Maximum Bending Moment

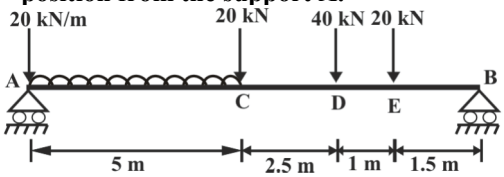
$$BM\ (M_{max}) = \left(\frac{P}{2} - \frac{Pa}{\ell}\right)\frac{\ell}{2} + Pa$$

$$= \frac{P\ell}{4} - \frac{Pa}{2} + Pa$$

$$= \frac{P\ell}{4} + \frac{Pa}{2} = \frac{1}{2}\left[\frac{P\ell}{2} + Pa\right]$$

Note-ISRO given answer (c).

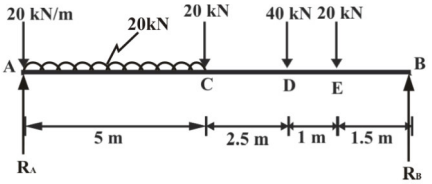
645. Find the maximum bending moment and the position from the support A.



(a) 240 Nm and 4.9 m
(b) 240.1 Nm and 4.9 m
(c) 240 Nm and 5 m
(d) 185 Nm and 7.5 m
(e) 260 Nm and 4.9 m

CGPSC AE 26.04.2015 Shift-I

Ans. (b) :



$R_A + R_B = (20 \times 5) + 20 + 40 + 20$

$R_A + R_B = 180$ ...(1)

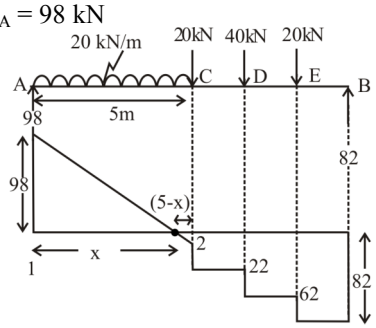
taking moment about point A

$R_B \times 10 = (20 \times 8.5) + (40 \times 7.5) + (20 \times 5) + \left(20 \times 5 \times \frac{5}{2}\right)$

$10R_B = 820$

$R_B = 82$ kN

$R_A = 98$ kN



Bending moment is maximum or minimum where shear force changes sign

$$\frac{98}{x} = \frac{2}{5-x}$$

x = 4.9 m

$$BM\big|_{x=4.9} = 98 \times 4.9 - 20 \times 4.9 \times \frac{4.9}{2}$$

= 480.2 - 240.1

= 240.1 N-m

646. The maximum bending moment of a simply supported beam of span 2m and carrying a point load 80 kN at the centre of the beam is

(a) 160 kN-m (b) 80 kN-m
(c) 320 kN-m (d) 40 kN-m

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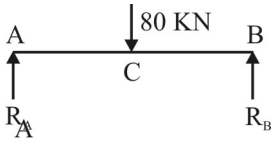
Ans : (d)



Maximum Bending moment = $\frac{wl}{4}$

$$(BM)_{max} = \frac{80 \times 2}{4}$$

$(BM)_{max} = 40$ KN-m

647. The bending moment for a certain portion of a beam is constant. For that section, shear force would be :

(a) zero (b) increasing
(c) decreasing (d) constant

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Ans. (a) : The bending moment for a certain portion of a beam is constant. For that section shear force would be zero.

648. The maximum bending moment of a simply supported beam of span *l* and carrying a point load *W* at the centre of the beam is

(a) $Wl/4$ (b) $Wl/2$
(c) Wl (d) $Wl^2/4$

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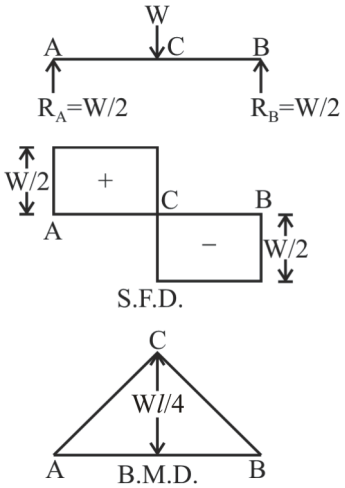
Ans. (a) For given load condition and beam.
Maximum bending moment (M_{max}) = $Wl/4$



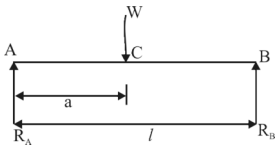
Simply supported beam carrying a point load W at the centre of the beam.

649. A simply supported beam of span 'ℓ' is subjected to a transverse point load 'W' acting at a distance of 'a' from the left support A. The reaction on the left support will be equal to

(a) W.a / ℓ (b) W. (ℓ -a)/ ℓ
(c) W/2 (d) w. ℓ /a

VIZAG MT 2015

Ans. (b) :



$$R_A = \frac{W \times BC}{AB}$$
$$= \frac{W \times (\ell - a)}{\ell}$$
$$R_B = \frac{W \times AC}{AB}$$
$$= \frac{W \times a}{\ell}$$

650. When the shear force diagram is a parabolic curve between two points, it indicates that there is

(a) a point load at two points
(b) no loading between the two points
(c) a uniformly distributed load between the two points
(d) a uniformly varying load between the two points

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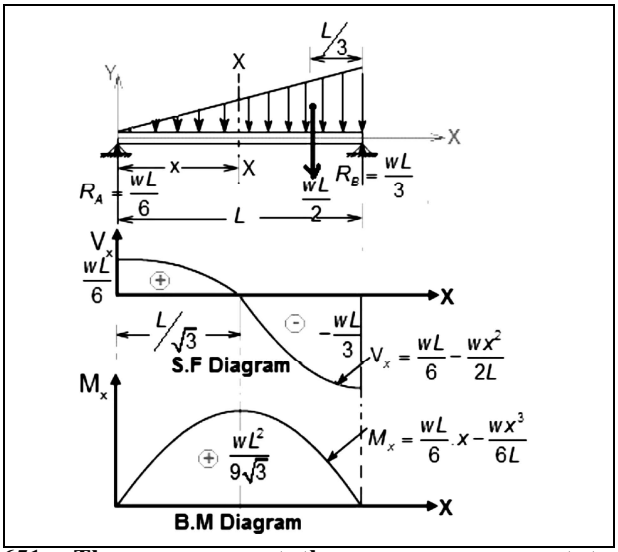
Ans. (d) : Simply supported beam with a gradually varying load, from zero, at one end to w per metre at the other end then the shear force diagram is a parabolic curve between two points.



651. The area-moment theorem …….. respect to bending of beams states that the area of the M/EI diagram between two sections of a beam gives :

(a) the difference in slopes between those two sections
(b) the difference in the maximum bending strains between those two sections
(c) the difference in deflections between those two sections
(d) the difference in strain energies between those two sections

BPSC Asstt. Prof. 29.11.2015

Ans. (a) : Moment Area method (Mohr's method)– This method is suitable for cantilevers and simply supported beams carrying symmetrical loadings and beams fixed at both ends i.e. those beams for which the area and CG area of BMD can be found easily. This means this method is not suitable for triangular loading and irregular loading. This can also be used for non-prismatic bars.
This method is suitable for :
(i) Centilevers (because slope at the fixed end is zero)
(ii) simply supported beams carrying.
(iii) Beams fixed at both ends (slope at each end is zero.

652. A simply supported beam of 2 m length is applied with uniformly distributed load of 5 kN/m through out. It is also applied with a point load of 1 kN at its centre. The maximum bending moment in the beam will be–

(a) 2 kN-m
(b) 3 kN-m
(c) 4 kN-m
(d) 5 kN-m

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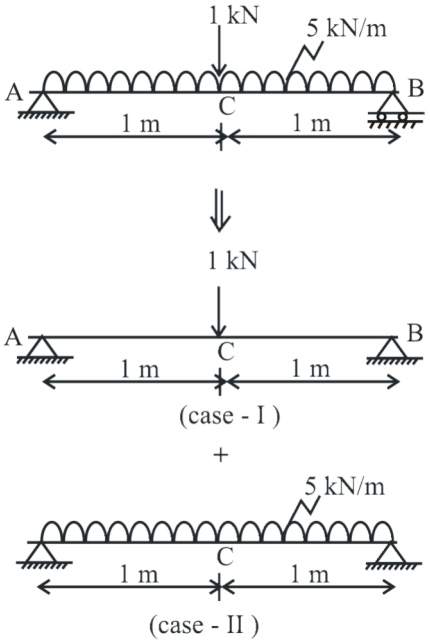
Ans. (b) : Given –
Length (l) = 2m, W = 5 kN-m



Maximum bending moment (case - I) due to point load to centre of SSB

$$(BM)_I = \frac{WL}{4} = \frac{1\times 2}{4} = 0.5 \text{ kN-m}$$

Maximum bending moment (case - II) due to uniformly load.

$$(BM)_{II} = \frac{WL^2}{8} = = \frac{5\times 2\times 2}{8} = 2.5 \text{ kN-m}$$

So, Total (case -I + case - II)
$BM_{(max)} = 2.5 + 0.5 = 3$ kN-m

653. Choose the wrong statement

(a) The shear force at any section of a beam is equal to the total sum of the forces acting on any one side of the section.
(b) The magnitude of the bending moment at any section of a beam is equal to the vector sum of the moments (about the section) due to the forces acting on the beam on any one side of the section.
(c) A diagram which shows the values of shear forces at various sections of structured member is called a shear force diagram.
(d) A simply supported beam is one which is supported on more than two supports.

MPPSC AE 08.11.2015

Ans. (d) : A simply supported beam is one which is not supported on more than two supports. fig (a) and fig (b)

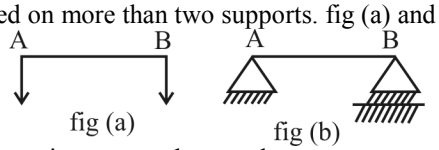


fig (a) fig (b)

If the beam is supported more than two supports called continuous beam.

654. In fixed beam of length (L) with a concentrated central load two points of contraflexure will occur, each from supports at a distance of :

(a) 1/3 (b) $1/\sqrt{3}$
(c) 1/6 (d) $1/4$

(HPPSC AE 2014)

Ans : (d)

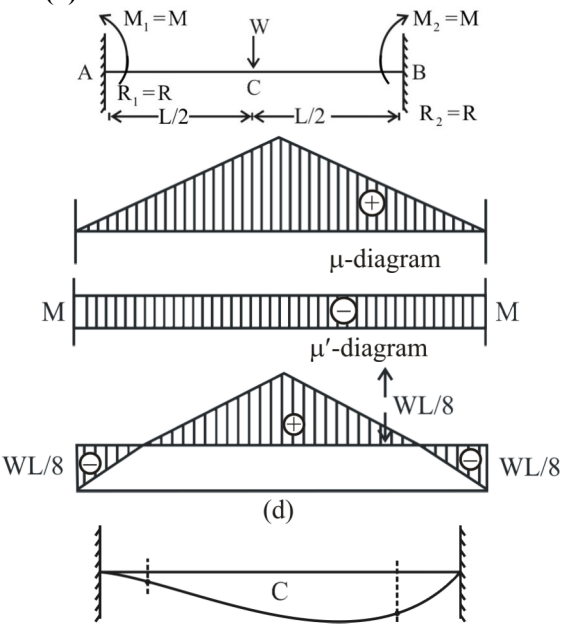


fig (d) shows the B.M diagram obtained by super-imposing and μ'diagram. At any point distance x from A

$$M_x = \mu_x + \mu'_x = \frac{Wx}{2} - \frac{WL}{8}$$

$$\text{At } x = \frac{L}{2}, M_c = \frac{W}{2}.\frac{L}{2} - \frac{WL}{8} = \frac{WL}{8}$$

Thus, the central B.M. is half of the B.M. for a freely supported beam. For point of Contraflexure

$$M_x = \frac{Wx}{2} - \frac{WL}{8} = 0$$

This gives $x = \frac{L}{4}$

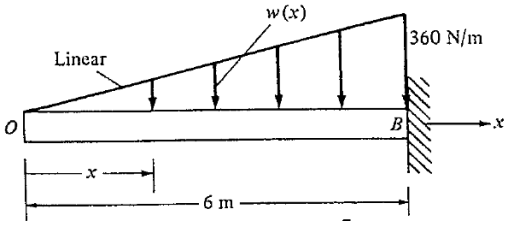
655. The three-moment for continuous beams was forwarded by :

(a) Bernoulli (b) Clapeyron
(c) Castigliano (d) Maxwell

(HPPSC AE 2014)

Ans : (b) The three moment theorem for continuous beam was forwarded by Clapeyron.

656. A cantilever beam 6 meter long as shown in figure is subjected to a linearly varying loading which has a maximum ordinate of 360 N/m at the fixed end on the right. The moment as a function of x is



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(a) $M = -3x^3$ (b) $M = -6x^3$
(c) $M = -10x^3$ (d) $M = -20x^3$

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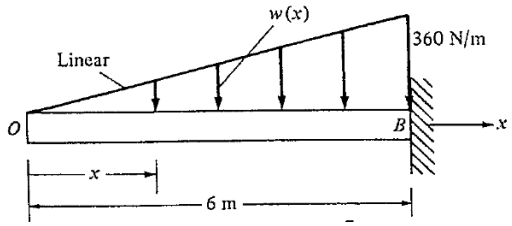
Ans. (c) :



$$W_{x-x} = \frac{-wx}{L} \quad ...(1)$$

$$(SF)_{xx} = \frac{-wx}{L} \times \frac{x}{2} \quad ...(2)$$

$$= \frac{-wx^2}{2L}$$

$$(BM)_{xx} = \frac{-wx^2}{2L} \times \frac{x}{3}$$

$$= \frac{-wx^3}{6L} \quad ...(3)$$

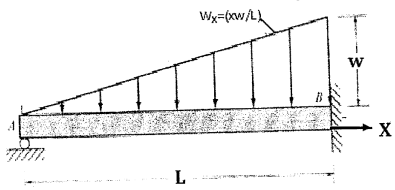
So, here, $w = 360$ N/m

$L = 6$ m

$$M = \frac{-360}{6 \times 6} \times x^3$$

$M = -10x^3$

657. Consider the beam shown in figure. The beam is simply supported at its left end A and fixed at its right end B. It carries a load that varies in intensity from zero at support A to 'w' at support B according to the relation $w_x = (x/L)w$. The reaction component at A is



(a) wL/10 (b) 2wL/5
(c) 5wL/2 (d) wL/5

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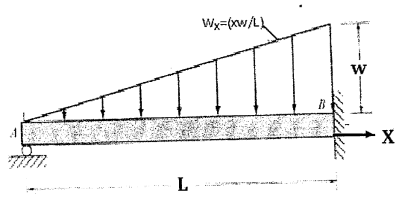
Ans. (a) : Moment at distance x from A is given as



$$M(x) = R_A x - \frac{1}{2} x \times \frac{wx}{L} \times \frac{x}{3} = R_A x - \frac{wx^3}{6L}$$

By double integration method

$$EIy'' = R_A x - \frac{wx^3}{6L}$$

Integrating once and twice we get

$$EIy' = \frac{R_A x^2}{2} - \frac{wx^4}{24L} + C_1$$

$$EIy = \frac{R_A x^3}{6} - \frac{wx^5}{120L} + C_1 x + C_2 \quad .....(1)$$

Using conditions (a) at x = 0, y = 0, thus $C_2 = 0$; and (b) at x = L, y′ = 0, we get

$$C_1 = \frac{wL^3}{24} - \frac{R_A L^2}{2}; C_2 = 0$$

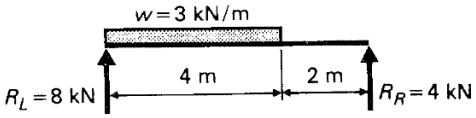
Putting above value in Eq. (1) for x = L, y = 0 we get

$$EIy = \frac{R_A x^3}{6} - \frac{wx^5}{120L} + \frac{wL^3}{24} - \frac{R_A L^2}{2}$$

$$\Rightarrow \quad 0 = \frac{-R_A L^3}{3} + \frac{wL^4}{30}$$

$$\Rightarrow \quad R_A = \frac{wL}{10}$$

658. A simply supported beam is shown in the figure below. The location is the point of maximum moment in the beam from reaction R_L is



(a) $\frac{3}{8}$ meter

(b) $\frac{8}{3}$ meter

(c) $\frac{18}{5}$ meter

(d) $\frac{21}{5}$ meter

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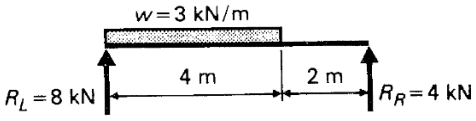
Ans. (b) :



Taking section x-x from R_L at a distance x

So at a location of max. bending moment, shear force will be zero so

$$S_{xx} = 8 - wx = 8 - 3x = 0$$

$$\boxed{x = \frac{8}{3} m}$$

659. A cantilever of length *l* carries a uniformly distributed load for a distance a (a<L) from the free end. The shear force diagram will be

(a) A rectangle
(b) A triangle
(c) A trapezium
(d) A parabola

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

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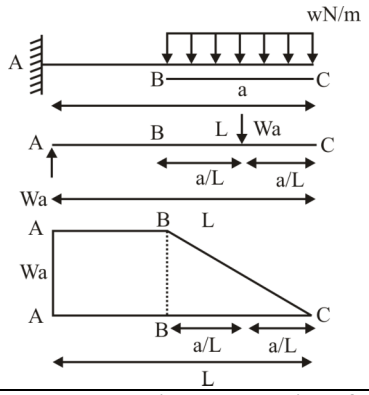
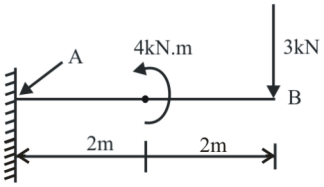
Ans. (c)



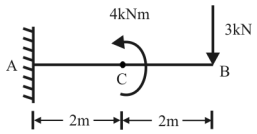
660. With the usual sign convection for bending moment, i.e. a positive bending moment causes compression in top fibre, what is sign, magnitude and location of numerically largest bending moment in the beam shown in the figure?



(a) (+) 10 kN.m at C (b) (–) 8 kN.m at A
(c) (–) 16 kN.m at A (d) (+) 4 kN.m at A

RPSC AEN Pre-2013

Ans. (b) :



Bending moment at A

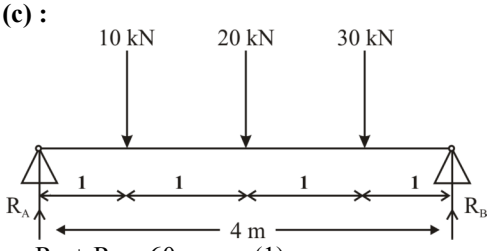
$(B.M)_A = -3 \times 4 + 4 = -8$ kN.m

661. A simply supported beam of span 4 m with hinged support at both the ends. It is carrying the point loads of 10, 20 & 30 kN at 1, 2 and 3 m from left support. The R_A & R_B are

(a) 27.5 kN, 32.5 kN (b) 15 kN, 45 kN
(c) 25 kN, 35 kN (d) 32.5 kN, 27.5 kN

TNPSC AE 2013

Ans. (c) :



$R_A + R_B = 60$ ....(1)

Taking moment about point A

$R_B \times 4 = 30 \times 3 + 20 \times 2 + 10 \times 1$

$R_B = 35$ kN

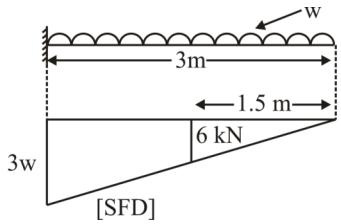
$R_A = 25$ kN

662. Uniformly distributed load 'w' act over per unit length of a cantilever beam of 3m length. If the shear force at the midpoint of beam is 6kN, what is the value of 'w':-

(a) 2 kN/m (b) 3 kN/m
(c) 4 kN/m (d) 5 kN/m

UKPSC AE-2013, Paper-I

Ans. (c) :



From shear force diagram

$$\frac{3w}{3} = \frac{6}{1.5}$$

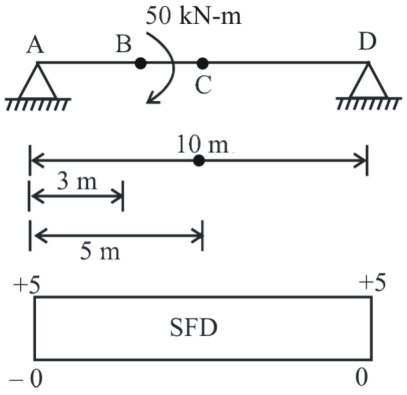
$w = 4$ kN/m

663. A beam 10 m long hinged at both the ends is subjected to a clockwise moment of 50 kNm, at a distance of 3 m from one end. What is shear force at the centre of the beam :

(a) 0 kN (b) 3 kN
(c) 5 kN (d) None of these

PSPCL AE, 2012

Ans. (c) : Shear force to constant throughout a length with value of 5 kN.

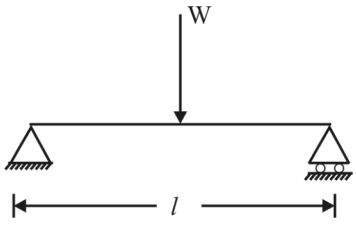


664. A beam A carries a point load at mid-span. Another identical beam B carries the same load but as uniformly distributed load over the entire span. The ratio of maximum bending moment in beam A to that in beam B will be :

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$
(c) 3 (d) 2

APGENCO AE 2012

Ans. (d) : C – 1



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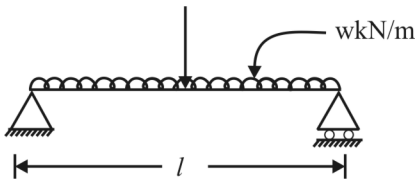
Maximum Bending moment $(BM)_{max} = \frac{Wl}{4}$

C – 2



Maximum Bending moment $(BM)_{max} = \frac{wl^2}{8}$

$(\because\ W = wL)$

$$(BM)_{max} = \frac{WL}{8}$$

$$\frac{(BM)_A}{(BM)_B} = \frac{(BM)_1}{(BM)_2} = \frac{WL}{4} \times \frac{8}{WL}$$

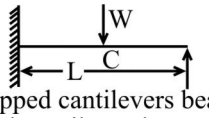
$$= 2$$

665. In a propped cantilever beam, the number of points of contra-flexure is

(a) 1 (b) 2
(c) 3 (d) 4

APPSC AEE 2012

Ans : (a)



Propped cantilevers beam

In a propped cantilever beam, the number of point of contra-flexure is one.

666. If a fixed beam is subject to a point load at mid span, total number of points of contra-flexure are

(a) 1 (b) 2
(c) 3 (d) zero

APPSC AEE 2012

Ans : (b) In a fixed beam is subjected to a point load at mid span total number of points of contra-flexure are two.

$$R_A = R_B = \frac{W}{2}$$

$$M_x = \frac{P}{8}(4x - L)$$

$$\left[x < \frac{L}{2}\right]$$

$$M_{x=0} = M_B = M_A = -\frac{PL}{8}$$

$$M_{x=L/2} = M_C = \frac{P}{8}\left[4 \times \frac{L}{2} - L\right] = \frac{PL}{8}$$

In this bending moment diagram bending moment change it's sign at two points.

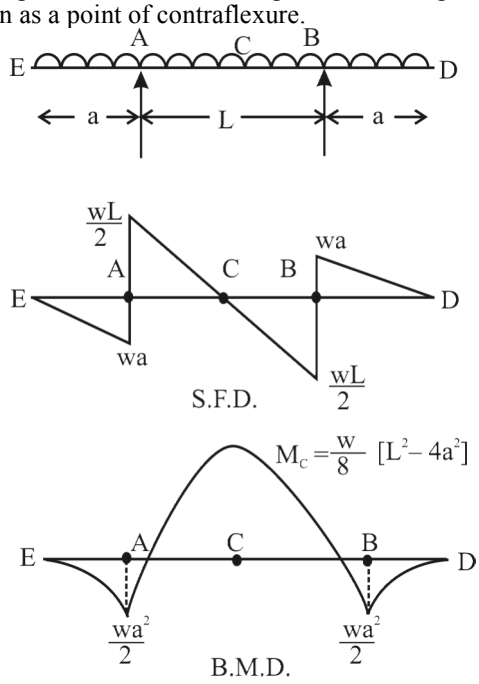
667. In a double overhanging beam carrying udl throughout its length, the number of points of contra flexure are

(a) 1 (b) 2
(c) zero (d) 3

APPSC AEE 2012

Ans : (b) In a double overhanging beam carrying UDL throughout its length, the number of points of contraflexure are two.

Points of contraflexure:– Where the bending moment will change sign from negative to positive or vice versa. Such a point, where the bending moment change sign, is known as a point of contraflexure.



668. A cantilever is subjected to udl throughout the length. If the maximum shear force is 200kN and maximum bending moment is 400kN, the span "L" of the beam in meters is

(a) 3 (b) 2
(c) 4 (d) 8

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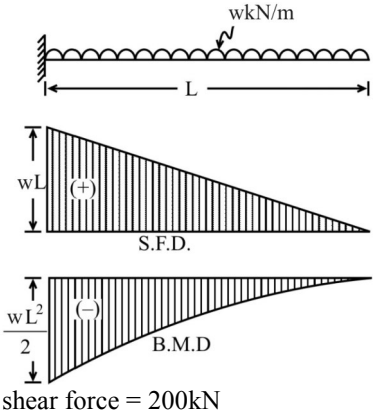
Ans : (c)



Maximum shear force = 200kN

$wL = 200kN$ ……..(i)

Maximum Bending moment

$\frac{wL^2}{2} = 400kN$ ……..(ii)

for equation (i) and (ii)

$$\frac{\frac{wL^2}{2}}{wL} = 2 \quad \boxed{L = 4m}$$

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669. A cantilever beam AB of length *l* is subjected to an anticlockwise couple of 'M' at a section C, distance 'a' from support. Then the maximum shear force is equal to

(a) M (b) $\frac{M}{2}$

(c) Zero (d) Ma

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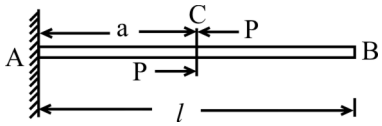
Ans : (c)



A cantilever beam AB of length *l* is subjected to an anticlockwise couple of M at a section C, distance 'a' from support, then the maximum shear force is equal to zero. Because No transverse load acting on the beam.

670. At section of a beam sudden change in BM indicates the action of

(a) point load (b) couple

(c) point load or couple (d) ud*l*

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Ans : (b) At section of a beam sudden change in BM indicates the action of couple.

671. For uniform shear force throughout the span of a simply supported beam, it should carry

(a) a concentrated load at the mid-span

(b) a couple anywhere in the sections

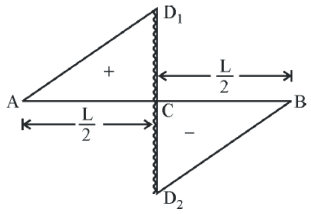
(c) ud*l* over its entire span

(d) two concentrated loads equally spaced

APPSC AEE 2012

Ans : (b) For uniform shear force throughout the span of a simply supported beam, it should carry a couple anywhere in the section.

672. The bending moment diagram for a simply supported beam AB of length 'L' is shown below :



$CD_1 = CD_2 = \frac{M}{2}$

Sagging moment : positive

Hogging moment : negative

What is the load acting on beam AB ?

(a) An upward concentrated load $\frac{M}{2}$ at C.

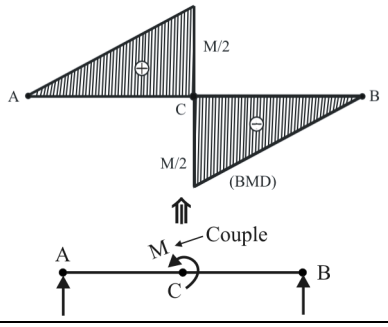
(b) A downward concentrated load $\frac{M}{2}$ at C.

(c) An anticlockwise moment 'M' at C

(d) A clockwise moment 'M' at C.

UKPSC AE 2012 Paper-I

Ans. (c) : An anticlockwise moment 'M' at C



673. Two cantilever steel beams of identical length and of rectangular section are subjected to same point load at their free end. In one beam, the longer side of section is vertical, while in the other, it is horizontal. Beams defect at free end:

(a) equally irrespective of their disposition.

(b) more in case of longer side vertical.

(c) less in case of longer side horizontal.

(d) less in case of longer side vertical.

UKPSC AE 2012 Paper-I

Ans. (d) : less in case of longer side vertical.

674. The bending moment diagram for a simply supported beam loaded in its centre is

(a) a right angled triangle

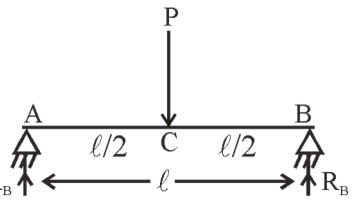
(b) an isosceles triangle

(c) an equilateral triangle

(d) a rectangle

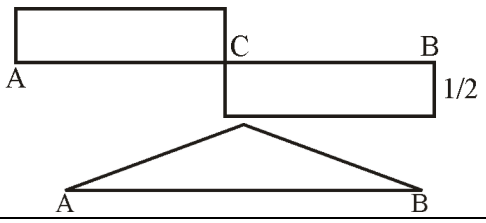
TNPSC ACF 2012

Ans. (b) :



$R_A = \frac{P \times \ell/2}{\ell}$

$R_A = P/2, \quad R_B = P/2$



675. In a beam where shear force is maximum, the bending moment will be

(a) zero

(b) minimum

(c) maximum

(d) none of these

TNPSC ACF 2012

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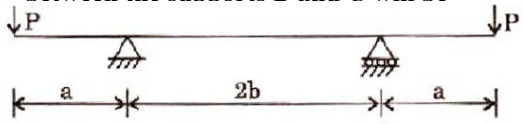
Ans. (d) : When shear force is zero, bending moment maximum.

$$F = \frac{dM}{dx}$$

In a beam, where shear force is maximum the bending moment depends upon the type of loading and type of beam.

676. For the beam shown in the fig, the elastic curve between the supports B and C will be



(a) circular (b) parabolic
(c) elliptic (d) a straight line

APPSC AEE 2012, ESE 1998

Ans. (a) : Form simple bending theory equation

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

$$\frac{1}{R} = \frac{M}{EI} = \text{constant}$$

Therefore, the elastic curve is the deflection curve which is circular between the support B and C.

677. Bending moment at any point is equal to the algebraic sum of :

(a) Moments of all vertical forces
(b) Moments of all horizontal forces
(c) Moment of forces on either side of the point
(d) All the above.

RPSC AE GWD, 2011

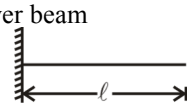
Ans. (c) : Bending moment at any point is equal to the algebraic sum of moments of forces on either side of the point.

678. What is the beam fixed at one end and free at the other end called as?

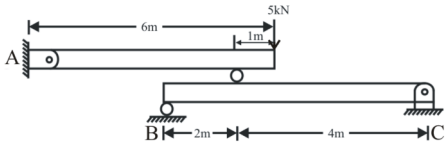
(a) Simply supported beam
(b) Fixed beam
(c) Overhanging beam
(d) Cantilever beam

VIZAG STEEL MT 2011

Ans. (d) : Cantilever beam



679.

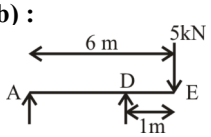


The magnitude of reaction at support B is :

(a) 2 kN (b) 4 kN
(c) 5 kN (d) 6 kN

UPSC JWM Advt. No.-52/2010

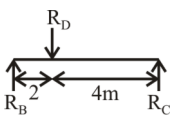
Ans. (b) :



Moment about A

$R_D \times 5 = 5 \times 6$

$R_D = 6\,kN$



Moment about C

$R_B \times 6 = 6 \times 4$

$\boxed{R_B = 4\,kN}$

680. Consider the following statements :
Shear force in beams is caused by

1. Lateral forces acting on the beam
2. Variation of bending moment along beam length
3. A couple acting on the beam

Which of the above statements is/are correct ?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know, shear force in beams is caused by the lateral forces acting on the beam and variation of bending moment along beam length.

681. Consider the following statements :
If a beam is to bend without being twisted,

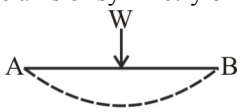
1. Plane of a load must contain one of the axes of symmetry of the beam
2. The beam cross-section must have at least an axis of symmetry

Which of the above statements is/are correct ?

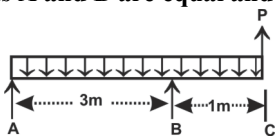
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

UPSC JWM Advt. No.-50/2010

Ans. (a) : It is the concept of shear centre that of a beam is to bend without twisting then plane of the load must contain one of the axis of symmetry of beam.



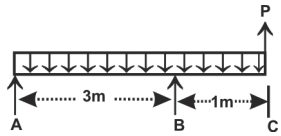
682. An overhang beam ABC of length 4m is supported at A and B 3m apart. It is loaded with UDL of 5kN/m along its entire length. Find the value of load 'P' at C such that the reactions A and B are equal and opposite.



(a) P = 10 kN (b) P = 30 kN
(c) P = 20 kN (d) P = 25 kN

ISRO Scientist/Engineer 2009

Ans. (c) : Overhang Beam ABC
length L = 4m



As mentioned reactions at A and B are equal and opposite. So, upward force P will be equal to total uniformly distributed load.

$$R_A + R_B = P \Rightarrow 5 \times 4 = 20$$

But $R_A = -R_B$

$P = 20\ kN$

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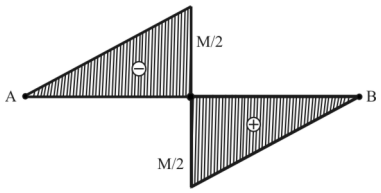
683.



The bending moment diagram for a beam AB is shown in the above figure. What is the nature of the beam?

(a) Simply supported with a concentrated load at its mid length

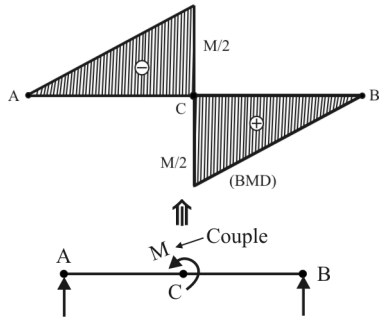
(b) Simply supported and is subjected to a couple M at its mid point

(c) Simply supported and carries a uniformly varying load from zero at the support to a maximum at its mid length

(d) A cantilever subjected to end moment M

UPSC JWM Adv. No-16/2009

Ans. (b) : The nature of the beam is simply supported and is subjected to a couple M at its mid point.



684. A beam of length '*l*' overhangs equally from its two supports. It carries three identical concentrated loads at its ends and at mid length. If the bending moment at midpoint of the beam is zero, what is the overhang length?

(a) $\frac{l}{8}$ (b) $\frac{l}{6}$

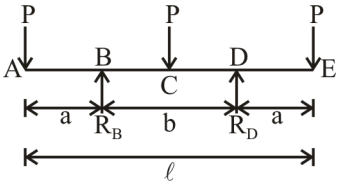
(c) $\frac{l}{4}$ (d) $\frac{L}{3}$

UPSC JWM Adv. No-16/2009

Ans. (b) : Let, a = Over hanging length

b = length of BD

l = Total length of beam



$\sum R_v = 0$

$R_B + R_D = 3P$

$\boxed{R_B = R_D = \frac{3P}{2}}$ (figure is symmetric)

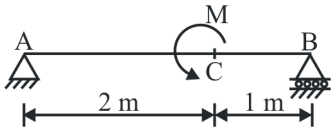
$\sum M_C = 0$

$P \times \frac{\ell}{2} - R_B \times \left(\frac{\ell}{2} - a\right) = 0$

$P \times \frac{\ell}{2} = \frac{3P}{2} \times \left(\frac{\ell}{2} - a\right)$

$\frac{\ell}{3} = \frac{\ell}{2} - a \Rightarrow \boxed{a = \frac{\ell}{6}}$

685. Couple *M* is applied at *C* on a simply supported beam *AB*. The maximum shear on *AC* will be



(a) Zero (b) M

(c) $\frac{2M}{3}$ (d) $\frac{M}{3}$

ISRO Scientist/Engineer 2008

Ans. (d) :

$R_1 + R_2 = 0$

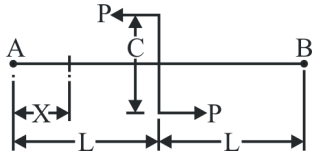
$\sum M_A = 0$

$R_2 \times 3 = -M$

$R_2 = -\frac{M}{3}$

$R_1 = \frac{M}{3}$

686. A beam AB is hinge-supported at its ends and is loaded by couple P.c. as shown in the figure. The magnitude of shearing force at a section x of the beam is



(a) 0

(b) P

(c) P/2L

(d) P.c/2L

ISRO Scientist/Engineer 2008

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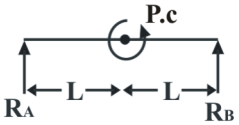
Ans. (d) : Shear force at section

$R_A + R_B = 0$

$\sum M_B = 0$

$R_A \times 2L = Pc$

$R_A = \frac{Pc}{2L}$



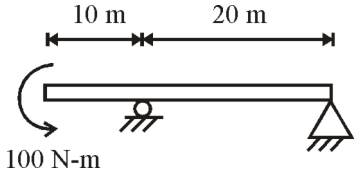
687. A discontinuity or jump in the shear force diagram occurs whenever a beam is loaded by

(a) a distributed moment
(b) a distributed force
(c) a concentrated moment
(d) a concentrated force

DRDO Scientists 2008

Ans. (d) : A discontinuity or jump in the shear force diagram occurs whenever a beam is loaded by a concentrated force.

688. For the beam loaded as shown in figure the magnitude of bending moment at the roller support is



(a) 100 Nm
(b) 1000 Nm
(c) 0
(d) 2000 Nm

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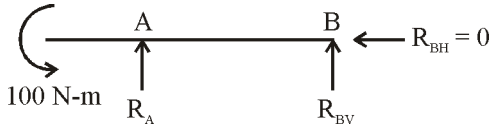
Ans. (a) :



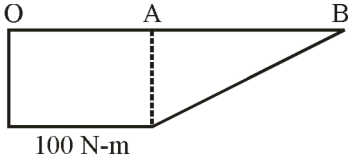
$R_A + R_{BV} = 0$

$\Sigma M_B = 0$

$-100 + R_A \times 20 = 0$

$R_A = 5$ N

$R_{BV} = -5$ N



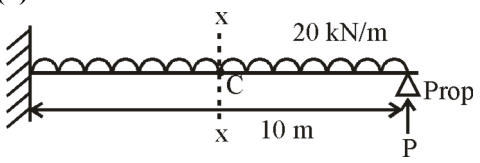
Bending moment at roller support,
= 100 N-m

689. A cantilever 10 m long carries a uniformly distributed load of 20 KN/m run throughout its length. It is propped by a force P at its free end so that the centre of the cantilever becomes the point of inflexion, then the magnitude of P is

(a) 200 KN
(b) 150 KN
(c) 100 KN
(d) none of the above

WBPSC AE, 2007

Ans. (d) :



For centre of cantilever becomes the point of inflexion,

$(BM)_{\text{at centre}} = 0$

$\Sigma M_C = 0$

$P \times 5 = 20 \times 5 \left(\frac{5}{2}\right)$

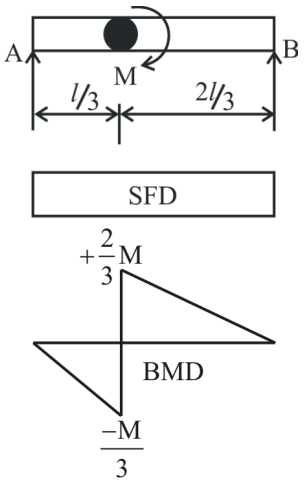
$P = 50$ kN

690. A simply supported beam of length 'L' acted upon by a couple of magnitude 'M' at a length of $\frac{2}{3}$L from one of the ends. The magnitude of maximum Bending moment in the beam is

(a) $\frac{1}{3}$M
(b) $\frac{2}{3}$M
(c) M
(d) $\frac{3}{2}$M

OPSC Civil Services Pre 2006

Ans. (b) :



$R_A + R_B = 0$

$R_B \times l + M = 0$

$R_B = \frac{M}{l}$

$R_A = -\frac{M}{l}$

from fig. magnitude of Maximum bending moment is $\frac{2}{3}$M.

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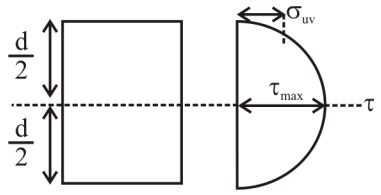
4. Bending Stresses and Shear Stresses in Beams

691. The shear stress distribution over a rectangular cross section of a beam is

(a) linear (b) cubic
(c) parabolic (d) none of these

OPSC AEE 2019 Paper-I
GPSC Executive Engineer 23.12.2018
TSPSC AEE 28.08.2017 (Civil/Mechanical)
APPSC AEE Mains 2016 (Civil Mechanical)
GPSC Lect. 23.10.2016, APPSC AEE 2012
UKPSC AE 2012 Paper-I, ESE 2004

Ans. (c) : For rectangular section beam, the shear stress distribution is parabolic and maximum shear stress is at neutral axis of the section.



692. Which one of the following statements in correct? A beam is said to be uniform strength, if :

(a) The bending moment is the same throughout the beam
(b) The shear stress is the same throughout the beam
(c) The deflection is the same throughout the beam
(d) The bending stress is the same at every section along its longitudinal axis

Assam PSC AE (IWT) 14.03.2021
RPSC IOF, 2020
OPSC AEE 2019 Paper-I
GPSC Asstt. Director of Transport 05.03.2017
TSPSC Managers, 2015
OPSC AEE 2015 Paper-I
TNPSC AE 2014, ESE 2006
J & K PSC Screening, 2006

Ans : (d) : A beam is said to be uniform strength if, the bending stress is the same at every section along its longitudinal axis.

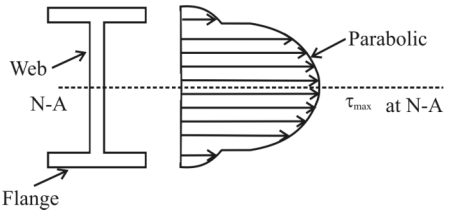
693. In an I-section of a beam subjected to transverse shear force, the maximum shear stress is developed at

(a) The bottom edges of the top flange.
(b) The top edges of the top flange.
(c) The centre of the web
(d) The upper edges of the bottom flange.

TSPSC Manager (Engg.) HMWSSB 12.11.2020
TSPSC AEE 28.08.2017 (Civil/Mechanical)
APPSC AE Subordinate Service Civil/Mech. 2016
UPPSC AE 12.04.2016 Paper-I
TNPSC AE 2014
ISRO Scientist/Engineer 2007
ESE 2008

Ans. : (c) In an I-section of a beam subjected to transverse shear force, the maximum shear stress is developed at the centre of the web.

Shear stress distribution for I-section

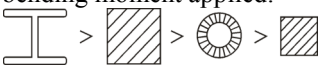


694. Which one of the following is the preferable cross-section of a beam for bending loads?

(a) Circular (b) Annular circular
(c) Rectangular (d) I-section

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TSPSC AEE 28.08.2017 (Civil/Mechanical)
TANGEDCO AE 2015, TNPSC ACF 2012
CSE Pre-2006

Ans. (d) : For a given cross-sectional area material and bending moment applied.



'Z' value decrease $\rightarrow M_R$ decreases $\rightarrow$
Bending stress $(\sigma_b)_{max}$ increases
I section is preferred for bending applications.

695. The section modulus is expressed as–

(a) I/Y (b) E/I
(c) M/1 (d) EI

OPSC AEE 2019 Paper-I
Nagaland PSC CTSE 2017, Paper-I
CGPSC AE 26.04.2015 Shift-I
APPSC AEE 2012, TNPSC ACF 2012

Ans. (a) : Section modulus is the ratio of moment of Inertia about N.A. upon the for test point of section from Neutral Axis (N.A.).

$$Z = \frac{I}{Y}$$

696. Which is the correct relation in a beam?

(a) $\frac{M}{\sigma} = \frac{I}{y} = \frac{R}{E}$ (b) $\frac{M}{I} = \frac{y}{\sigma} = \frac{E}{R}$
(c) $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$ (d) $\frac{M}{y} = \frac{E}{R} = \frac{\sigma}{I}$

OPSC AEE 2019 Paper-I
Nagaland PSC (CTSE) 2018, Paper-I
TNPSC ACF 2012, UKPSC AE 2007 Paper -I
ISRO Scientist/Engineer 2006

Ans : (c) : Bending Formula –

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

Where M = Bending moment
I = Second moment of inertia of cross section about neutral axis.
σ = Bending stress on any layer
Y = Distance of any layer from neutral layer.
E = Young's Modulus
R = Radius of curvature of the neutral

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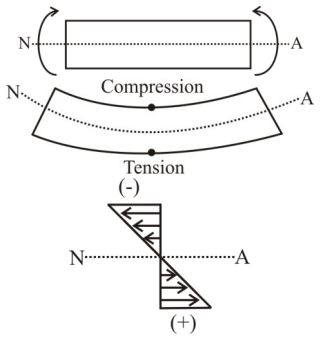
697. The neutral axis of the cross-section a beam is that axis at which the bending stress is

(a) zero (b) minimum
(c) maximum (d) infinity

UPRVUNL AE 05.07.2021
APPSC AE Subordinate Service Civil/Mech. 2016
KPCL AE 2016
Haryana PSC Civil Services Pre, 2014
APPSC AEE 2012

Ans. (a) :



N–A shows neutral axis- at neutral axis there is no bending stress.

$\sigma_{bending} = \frac{M}{I} \times y$ at neutral axis (y = 0)

$\boxed{\sigma_{bending} = 0}$

Note-Inner fibre always subjected to maximum stress.

698. Two beams of equal cross section area are subjected to equal bending moment. If one beam has square cross section and the other has circular section, then:

(a) Both the beams will be equally strong
(b) Square cross section beam will be stronger
(c) Circular section beam will be stronger
(d) Depends on the loading condition

RPSC AE (GWD) 06.05.2016
NPCIL ET, 2015, TSPSC AEE 2015
ISRO Scientist/Engineer 2011
ESE 1999

Ans. (b) : Since, cross-section area of square and circular cross-section are equal.

Let d = diameter of the cross-section are equal
a = side of square

[Given, Equal cross-section]

$$\frac{\pi}{4}d^2 = a^2$$

Now, σ_b for circular cross-section

$$\sigma_b = \frac{M.y_{max}}{I_{NA}} = \frac{M.\frac{d}{2}}{\frac{\pi d^4}{64}} = \frac{32M}{\pi d^3}$$

σ_b for square cross-section

$$\sigma_b = \frac{M.\frac{a}{2}}{\frac{a^4}{12}} = \frac{6M}{a^3}$$

$$\frac{\sigma_b \text{ for circular cross-section}}{\sigma_b \text{ for square cross-section}} = \frac{\frac{32M}{\pi d^3}}{\frac{6M}{a^3}}$$

$$= \frac{32}{\pi}\left(\frac{a^3}{d^3}\right)$$

$$= \frac{32}{\pi}\left(\sqrt{\frac{\pi}{4}}\right)^3 \cdot \left(\frac{a^3}{d^3}\right) \qquad \left[\because \frac{\pi}{4}d^2 = a^2\right]$$

$\therefore$ σ_b for circular-section > σ_b for square-section

Stress in circular beam is more so it will be less strong as compare to square beam.

699. The maximum transverse shear stress in a circular beam is (V-shear force, A-cross section area)

(a) $\frac{3V}{4A}$ (b) $\frac{4V}{3A}$
(c) $\frac{1V}{3A}$ (d) $\frac{4V}{\pi A}$

Haryana PSC AE (PHED) 05.09.2020, Paper-II
Oil India Senior Officer 23.12.2020
TSPSC AEE 28.08.2017 (Civil/Mechanical)
ESE 2006

Ans. (b) : $\tau_{max} = \frac{4}{3}\tau_{avg} = \frac{4V}{3A}$

700. The ratio of average shear stress to the maximum shear stress in a beam with a square cross section is :

(a) 2 (b) 1.5
(c) 2/3 (d) 1

OPSC AEE 2019 Paper-I
TSPSC AEE 28.08.2017 (Civil/Mechanical)
J&K PSC Civil Services Pre, 2010
GATE 1998

Ans. (c) : For square cross-section –

Maximum shear stress (τ_{max}) =

$$\frac{3}{2} \times \text{average shear stress}\left(\tau_{avg}\right)$$

$$\frac{\tau_{max}}{\tau_{avg}} = \frac{3}{2}$$

or $\frac{\tau_{avg}}{\tau_{max}} = \frac{2}{3}$

701. The rectangular beam A has a length λ and width b but depth d is double that of A. The elastic strength of beam B will be ____ as compared to beam A.

(a) same (b) double
(c) one-fourth (d) four times

ISRO Scientist/Engineer (RAC) 12.01.2020
Nagaland PSC (CTSE) 2018, Paper-I
TSPSC AEE 2015, JPSC AE 2013, Paper-V

Ans : (d) : We know that section modulus for beam A,

$$Z_A = \frac{I_A}{y_A} = \frac{bd^3/12}{d/2}$$

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$= \frac{bd^2}{6}$

And section modulus for beam B

$$Z_B = \frac{I_B}{y_B} = \frac{b(2d)^3/12}{2d/2} = \frac{2bd^2}{3}$$

(Depth is double that of A)

Since elastic strength of beam is directly proportional to their respective section modulus, therefore,

$$\frac{Z_B}{Z_A} = \frac{2bd^2}{3} \times \frac{6}{bd^2} \text{ or } Z_B = 4Z_A$$

$\boxed{Z_B = 4Z_A}$

702. If the depth is kept constant for a beam of uniform strength, then its width will vary in proportional to (where M = Bending moment)

(a) M (b) $\sqrt{M}$

(c) M^2 (d) M^3

Nagaland PSC (CTSE) 2018, Paper-I
Nagaland PSC CTSE 2016 Paper-I
TSPSC AEE 2015, ESE 1995

Ans. (a) : We know that

$$\frac{M}{I} = \frac{\sigma}{y}$$

$$I = \frac{bd^3}{12}$$

So $M \propto b$

If depth is constant then M is directly proportional to its width [b].

For a beam of uniform strength bending stress σ will be also constant

703. The ratio of maximum shear stress to average shear stress in a beam of rectangular section is

(a) 5/1 (b) 2/3

(c) 3/2 (d) 1.0

ISRO Scientist/Engineer 12.01.2020
ESE 2016, APPSC AEE 2012, GATE 1994

Ans. (c) :

Cross section	$\frac{\tau_{max}}{\tau_{avg}}$	$\frac{\tau_{NA}}{\tau_{avg}}$	$\frac{\tau_{max}}{\tau_{NA}}$
Rectangle/square	3/2	3/2	1
Circle	4/3	4/3	1
Triangle	3/2	4/3	9/8
Diamond	9/8	1	9/8

704. A wooden rectangular beam, subject to uniformly distributed load, has an average shear stress (τ_{avg}) across the section. The maximum shear stress (τ_{max}) at natural axis is:

(a) $\tau_{max} = 0.5\ \tau_{avg}$ (b) $\tau_{max} = 1.0\ \tau_{avg}$

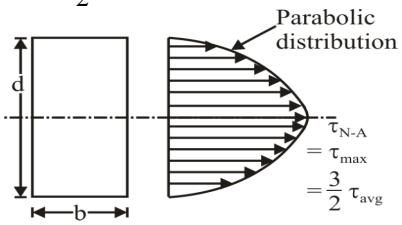
(c) $\tau_{max} = 1.5\ \tau_{avg}$ (d) $\tau_{max} = 2.0\ \tau_{avg}$

(e) $\tau_{max} = 2.5\ \tau_{avg}$

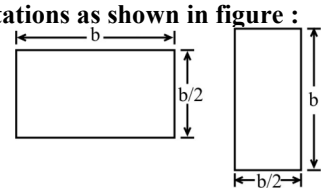
(CGPCS Polytechnic Lecturer 2017)
APPSC AEE Screening Test 2016
APPSC AEE 2012, ESE 2009

Ans. (c) Shear stress distribution across the rectangular section.

$$\tau_{max.} = \frac{3}{2}\tau_{avg} = 1.5\ \tau_{avg}$$



705. A beam cross-section is used in two different orientations as shown in figure :



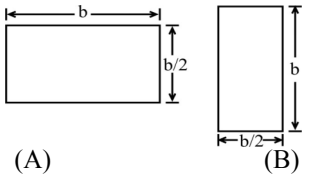
Bending moments applied in both causes are same. The maximum bending stresses induced in cases (A) and (B) are related as

(a) $\sigma_A = \sigma_B$ (b) $\sigma_A = 2\sigma_B$

(c) $\sigma_A = \frac{\sigma_B}{2}$ (d) $\sigma_A = \frac{\sigma_B}{4}$

TSPSC Manager (Engg.) HMWSSB 12.11.2020
APPSC AEE 2012, ESE 1997

Ans : (b)



(A) (B)

For first case:–

$$I_A = \frac{1}{12} \times b\left(\frac{b}{2}\right)^3 = \frac{b^4}{96}$$

$$\frac{M}{I_A} = \frac{\sigma_A}{y} \Rightarrow \frac{M \times 12 \times 8}{b^4} = \frac{\sigma_A}{b/4}$$

$$\boxed{\sigma_A = \frac{24M}{b^3}} \quad \text{..... (i)}$$

For second case

$$I_B = \left(\frac{b}{2}\right) \times (b)^3 \times \frac{1}{12}$$

$$I_B = \frac{b^4}{24}$$

$$\frac{M}{I_B} = \frac{\sigma_B}{y}$$

$$\frac{M}{\frac{b^4}{12 \times 2}} = \frac{\sigma_B}{\frac{b}{2}}$$

$$\boxed{\sigma_B = \frac{12M}{b^3}} \quad \text{..... (ii)}$$

For equation first and second $\boxed{\sigma_A = 2\sigma_B}$

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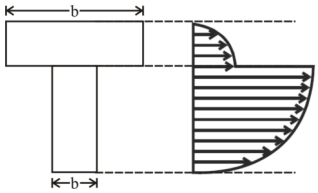
706. Which one of the following represents the shearing stress distribution over a cross section of a T-beam ?

(a)
(b)
(c)
(d)

UPSC JWM Advt. No.-50/2010
CSE Pre-2000
ESE 2000

Ans. (d) :



As we know that,

$$\tau_s = \frac{F_s A \bar{Y}}{I_{NA} b}$$

$$\tau_s \propto \frac{1}{b}$$

707. Circular beam of uniform strength can be made by varying the diameter in such way that:

(a) M/Z is constant (b) σ/Y is constant
(c) M/R is constant (d) E/R is constant

VIZAG MT 2015
UKPSC AE 2007 Paper -I

Ans. (a) : According to bending equation–

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma_b}{y_{max}}$$

$$\text{Uniform strength } (\sigma_b) = \frac{M}{Z}$$

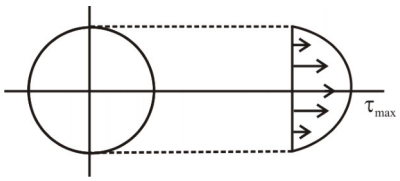
$\frac{M}{Z}$ is constant

708. In case of beam of circular cross-section subject to transverse loading, the maximum shear stress developed in the beam is greater than the average shear stress by

(a) 10% (b) 25%
(c) 50% (d) 33%

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
APGCL AM, 2021

Ans. (d) :

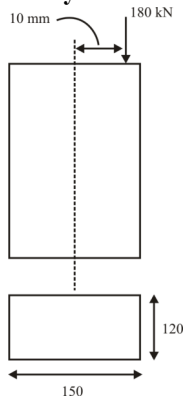


As we know,
In case of beams with circular section, the ratio of maximum shear stress to average shear stress

$$= \frac{4}{3}\tau_{avg} = 1.33\ \tau_{avg}$$

The maximum shear stress developed in the beam is 33% greater than the average shear stress.

709. A rectangular strut is 150 mm wide and 120 mm thick. It carries a load of 180 kN at an eccentricity of 10mm in a plane bisecting the thickness as shown in the figure. The maximum and minimum intensities of stress in the section will be respectively.



(a) 10 MPa and 4 MPa (b) 14 MPa and 4 MPa
(c) 10 MPa and 6 MPa (d) 14 MPa and 6 MPa

CGPSC AE 15.01.2021, 16.10.2016

Ans. (d) : Given,

P = 180 kN

Bending moment, $M = 180 \times 10^3 \times 10$

Bending stress $\Rightarrow \sigma_b = \frac{My}{I}$

$$= \frac{180 \times 10^4 \times \left(\frac{150}{2}\right) \times 12}{120 \times (150)^3}$$

$$\boxed{\sigma_b = 4\,\text{MPa}}$$

Direct stress,

$$\sigma_d = \frac{P}{A} = \frac{180 \times 10^3}{120 \times 150} = 10\text{MPa (compressive)}$$

Maximum intensity of stress $= -\sigma_b + \sigma_d$
$= -4 - 10$
$= |-14|$ MPa = 14 MPa

Minimum intensity of stress $= +\sigma_b + \sigma_d$
$= +4 - 10$
$= |-6$ MPa $| = 6$ MPa

710. When a rectangular beam is loaded transversely, the maximum compressive stress is developed on the

(a) top layer (b) bottom layer
(c) neutral axis (d) every cross-section

Sikkim PSC (Under Secretary), 2017
Haryana PSC Civil Services Pre, 2014

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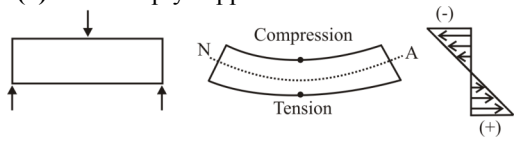
Ans. (a) : For simply supported beam–

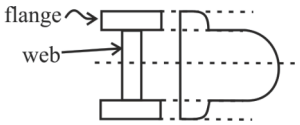


Right answer – (a)
Haryana PSC gives (b)
Note-In case of cantilever beam compressive stress is obtained at bottom layer.

711. In the case of an I-section beam maximum shear stress is at
(a) The junction of Top flange and web
(b) Middle of the web
(c) Both (A) and (B)
(d) The junction of bottom flange and web

APGCL AM, 2021, ESE 2008

Ans. (b**) :**



The maximum shear stress in I section of beam is at middle of the web.

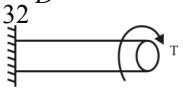
712. If a circular shaft is subjected to a torque T and a bending moment M, the ratio of the maximum shear stress to maximum bending stress is :
(a) 2 M/T (b) T/2M
(c) 2T/M (d) M/2T

ISRO Scientist/Engineer (RAC), 10.03.2019
ISRO Scientist/Engineer 12.05.2013

Ans. (b) : $\tau_{max} = \dfrac{16T}{\pi d^3}$

$$\frac{T}{J} = \frac{G\theta}{l} = \frac{\tau}{R}$$

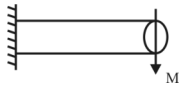
$$\tau = \frac{TR}{J} = \frac{T \times D/2}{\frac{\pi}{32} D^4}$$



$$\boxed{\tau_{max} = \frac{16T}{\pi D^3}}$$

$$\sigma_{max} = \frac{32M}{\pi d^3}$$

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$



$$\sigma = \frac{M}{I} \times y$$

$$\sigma = \frac{M}{\frac{\pi}{64} D^4} \times D/2$$

$$\boxed{\sigma_{max} = \frac{32M}{\pi D^4}}$$

$$\boxed{\frac{\tau_{max}}{\sigma_{max}} = \frac{\frac{16T}{\pi D^3}}{\frac{32M}{\pi D^3}} = \frac{T}{2M}}$$

713. Which one of the following conditions will qualify for a constant strength beam?
(a) Bending moment along the whole length of beam is constant
(b) Shear force at every section of the beam is constant
(c) The beam should be of uniform section
(d) Ratio of bending moment to its section modulus at every section must be constant.

UKPSC AE 2012 Paper-I
UPSC JWM Advt. No.-52/2010

Ans. (d) : Ratio of bending moment to its section modulus at every section must be constant for a constant strength beam

$$\text{Strength of beam} = = \frac{M}{Z}$$

Since, the strength of a beam depends upon its section modulus (Z).

714. A shaft was initially subjected to bending moment and then it was subjected to torsion. If the magnitude of bending moment is found to be the same as that of the torque, then the ratio of maximum bending stress to shear stress will be.
(a) 0.25 (b) 0.50
(c) 2 (d) 4

APPSC IOF, 2009, WBPSC AE 2003

Ans. (c) : $\left[\dfrac{M}{I} = \dfrac{\sigma}{y} = \dfrac{E}{R}\right]$

from $\dfrac{M}{I} = \dfrac{\sigma}{y} \Rightarrow M = \dfrac{\sigma I}{y}$

$$M = \frac{\sigma}{y} \times \frac{\pi d^4}{64} \qquad [y = d/2]$$

$$M = \frac{\sigma \times 2 \times \pi \times d^4}{d \times 64} = \frac{\pi \sigma d^3}{32} \qquad ...(i)$$

$\because$ $\dfrac{T}{J} = \dfrac{\tau}{R} = \dfrac{G\theta}{L}$

from $T = \dfrac{\tau . J}{R}$ $\because$ $R = d/2$ & $J = \dfrac{\pi}{32} d^4$

$$T = \frac{\tau \times 2 \times \pi d^4}{d \times 32} = \frac{\pi d^3 \times \tau}{16} \qquad ....(ii)$$

bending moment = torque

$$M = T$$

$$\frac{\pi d^3 \sigma}{32} = \frac{\pi d^3 . \tau}{16}$$

$$\frac{\sigma}{\tau} = 2$$

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715. The variation of bending stress in a curved beam is ..... in nature.

(a) Linear (b) Cubic
(c) Parabolic (d) Hyperbolic

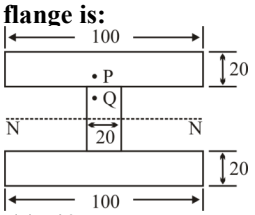
HPPSC AE 2018
ISRO Scientist/Engineer 24.05.2014

Ans. (d) : The variation of bending stress in curved beam is given as

$$\sigma_b = \frac{M\,y}{A.e.[R_n - y]}$$

Where,
M = Bending moment acting at the given section about the centroidal axis.
A = Area of cross-section
e = Distance from the centroidal axis to the neutral axis $R - R_n$
R = Radius of curvature of the centroidal axis
R_n = Radius of curvature of the neutral axis
y = Distance between the neutral axis to the considered fibre which bending stress needed to be calculated.
This equation shows that the stress distribution is Hyperbolic in nature.

716. An I-section of a beam is shown in the figure below. If the shear stress at point P which is very close to bottom of the flange is 12 MPa, the shear stress at the point Q close to the flange is:



(a) 40 MPa (b) 12 MPa
(c) Indeterminate (d) 60 MPa

BHEL ET 2019, ESE 2001

Ans. (d) : Given - shear stress of flange

$$(\tau_{flange}) = 12MPa$$

shear stress of web (τ_{web}) =?
width of flange (b_t) = 100 mm
width of web (b_w) = 20 mm

from - $\dfrac{(\tau_{flange})}{(\tau_{web})} = \dfrac{(b_{web})}{(b_{flange})}$

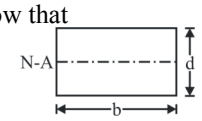
$$\frac{12}{(\tau_{web})} = \frac{20}{100} = \frac{1}{5} \qquad \boxed{(\tau_{web}) = 60MPa}$$

717. Section of the modulus (Z) for a rectangular section with width (b) and depth (d) is given by:

(a) $Z = \dfrac{bd^3}{12}$ (b) $Z = \dfrac{bd^2}{12}$
(c) $Z = \dfrac{bd^3}{6}$ (d) $Z = \dfrac{bd^2}{6}$
(e) $Z = \dfrac{bd^2}{8}$

CGPCS Polytechnic Lecturer 2017
TNPSC AE 2013

Ans. (d) : We know that



$$Z = \frac{I_{N-A}}{Y_{max.}}$$

$$Y_{max} = \frac{d}{2}$$

$$I_{N\text{-}A} = \frac{bd^3}{12}$$

$$Z = \frac{bd^3 \times 2}{12 \times d}$$

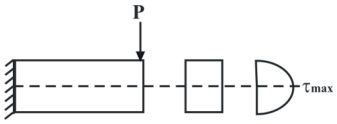
$$\boxed{Z = \frac{bd^2}{6}}$$

718. When a beam is subjected to a transverse shearing force, the shear stress in the upper fibers will be

(a) Maximum (b) Minimum
(c) Zero (d) Depends on other data

Nagaland PSC CTSE 2017, Paper-I
ISRO Scientist/Engineer 2008

Ans. (c) :



Shear stress varies parabolically. At top & bottom fibres it's value is zero & at neutral axis it's value is maximum.

719. In the case of a curved beam subjected to pure bending, which of the following is true?

(a) Neutral axis coincides with the centroidal axis
(b) Neutral axis lies between the centroidal axis and the center of curvature.
(c) Location of neutral axis depends upon the magnitude of bending moment
(d) There is no neutral axis

DRDO Scientists 2008

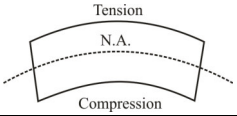
Ans. (b) : In case of curved beam subjected to pure bending, neutral axis lies between the centroidal axis and the center of curvature.
In case of normal beam, subjected to pure bending, neutral axis coincides with the centroidal axis.

720. Neutral plane of the beam

(a) Is in the middle
(b) Is one whose length remains unchanged during deformation
(c) Passes through centers of gravity
(d) Lies at the top most fiber

ISRO Scientist/Engineer (RAC) 07.05.2017 GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : At neutral plane there are no bending stress (σ_b = zero) so whose length unchanged during deformation.



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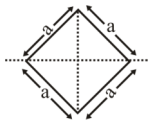
721. A beam of square section, side of square 'a' is used with its diagonal horizontal the modulus of section of beam is given by

(a) $\frac{a^4}{6\sqrt{2}}$ (b) $\frac{a^4}{12\sqrt{2}}$

(c) $\frac{a^3}{6\sqrt{2}}$ (d) $\frac{a^3}{12\sqrt{2}}$

APPSC AEE SCREENING 17.02.2019
MPSC Lect. (Prod.) 2018

Ans. (c)



$$I = \frac{a^4}{12}$$

$$Y_{max} = \frac{a}{\sqrt{2}}$$

$$\text{Section modulus } (Z) = \frac{I}{Y_{max}} = \frac{\frac{a^4}{12}}{\frac{a}{\sqrt{2}}} = \frac{a^3}{6\sqrt{2}}$$

722. If E= elasticity modulus, I = moment of inertia about the neutral axis and M = bending moment in pure bending under the symmetric loading of a beam, the radius of curvature of the beam :

(i) Increases with E (ii) Increases with M
(iii) Decreases with I (iv) Decreases with M
Which of these are correct?

(a) (i) and (ii) (b) (ii) and (iii)
(c) (iii) and (iv) (d) (i) and (iv)

OPSC AEE 2019 Paper-I, ESE 2013

Ans : (d) : In pure bending

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

$$R = \frac{E.I}{M}$$

R increase with E
R decrease with M
Statement (i) and (iv) are correct.

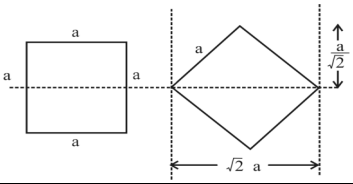
723. The ratio of flexural strength of a square section with its two sides horizontal to its diagonal horizontal is

(a) $\sqrt{2}$ (b) 2

(c) $2\sqrt{2}$ (d) $\frac{\sqrt{2}}{5}$

APPSC AEE 2012, ESE 2012

Ans : (a)



$$I_{SH} = \frac{a^4}{12}$$

$$y = \frac{a}{2}$$

$$Z_{SH} = \frac{a^3}{6}$$

$$\boxed{Z_{SH} = \frac{a^3}{6}}$$

$$I_{SD} = \frac{a^4}{12}$$

$$y = \frac{a}{\sqrt{2}}$$

$$Z_{SD} = \frac{a^4 \times \sqrt{2}}{12 \times a} = \frac{a^3}{6\sqrt{2}}$$

we know that

$$M \propto Z$$

$$\frac{M_{SH}}{M_{SD}} = \frac{Z_{SH}}{Z_{SD}} = \frac{a^3}{6} \times \frac{6\sqrt{2}}{a^3} = \sqrt{2}$$

724. For a circular section the ratio of Maximum shear stress to the average shear stress is :

(a) 1.13 (b) 1.23
(c) 1.33 (d) 1.43

GMB AAE 25.06.2017
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (c) : The ratio of maximum shear stress to average shear stress of a circular section.

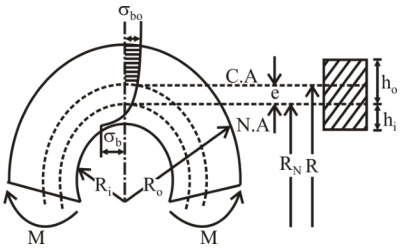
$$\boxed{\tau_{max} = \frac{4}{3}\tau_{avg}}$$

725. The maximum bending stress in a curved beam having symmetrical section always occurs at which one of the following?

(a) Inside fibre (b) Outside fibre
(c) Neutral axis (d) Centroidal axis

CGPSC AE 15.01.2021
UPSC JWM Adv. No-16/2009

Ans. (a) :



The bending stress in a straight beam is linear whereas that in a curved beam is hyperbolically varying over the cross-section.

- If the section is symmetrical, the maximum bending stress always occurs at radius r_i of the inside fiber.
- If the section is unsymmetrical, the stress may be maximum either at R_i or R_o.
- The neutral axis where the bending stress is zero is located between the centroidal axis at radius R and the neutral axis at radius R_i or R_o.

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- The distance h_o is to be considered positive for distances towards the center of curvature and negative for distances away from the center of curvature.

726. Consider the following statements :
Two beams of identical cross-section but of different materials are subjected to equal bending moments at a certain section along their length:
1. **Maximum bending stress at the section of two beams is same**
2. **Maximum shearing stress at the sections of two beam is same.**
3. **Maximum bending stress at the section of two beams will depend on the elastic modulus of the beam materials**
4. **Curvature of the beam having greater value of E will be larger.**

Which of the above statements is/are correct?
(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 4 only

UPSC JWM Adv. No-16/2009, CSE Pre 2007

Ans. (a) : According to bending equation –

$$\boxed{\frac{M}{I}=\frac{\sigma_b}{y}=\frac{E}{R}} \quad \ldots(i)$$

$$\Rightarrow \quad \boxed{\sigma_b=\frac{M.y}{I} \quad or \quad \frac{M}{Z}} \quad \ldots(ii)$$

Bending stress = σ_b

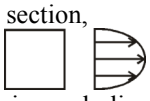
If two beams of identical cross-section but of different materials, when subjected to the equal bending moment at a certain section along their length then according to bending equation, the bending stress not depend on the elastic properties of the material i.e., Maximum bending stress of the material i.e. Maximum bending stress at the section of two beams is same.

And also Maximum shearing stress at the section of two beams is same.

727. A square section with side 'X' of a beam is subjected to a shear force 'S'. The magnitude of shear stress at the top edge of the square is
(a) $\frac{1.5S}{X^2}$ (b) $\frac{S}{X^2}$
(c) $\frac{0.5S}{X^2}$ (d) Zero

APPSC AEE 2012, WBPSC AE, 2007

Ans. (d) : For square section,



Shear stress variation is parabolic shear stress at the top edge of the square is zero.

728. A cantilever beam of rectangular cross-section is 1 m deep and 0.6 m thick. If the beam were to be 0.6 m deep and 1 m thick, then the beam would
(a) be weakend 0.5 times
(b) be weakend 0.6 times
(c) be strengthed 0.6 times
(d) have the same strength on the original beam because the cross-sectional area remains the same

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-1999

Ans. (b) : Flexural strength of beam depends upon it's section modulus.

$$Z_I=\frac{\frac{1}{12}\times 0.6\times 1^3}{0.6/2}=0.1\,m^3$$

$$Z_{II}=\frac{\frac{1}{12}\times 1\times 0.6^3}{0.6/2}=\frac{(0.6)^2}{6}=0.6\times 0.1$$

$= 0.06\ m^3$

$$Z_{11}<Z_1=\frac{0.06}{0.1}=0.6$$

Beam would be weakened by 0.6 times.

729. Three beams of 100 mm × 1200 mm section are made from RCC, aluminium and timber respectively. These three beams are loaded identically with similar supports, Maximum bending stress will occur in which beam?
(a) Timber beam
(b) Aluminium beam
(c) RCC beam
(d) The three beams will have same maximum bending stress

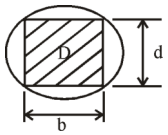
TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-2008

Ans. (d) : Bending stress does not depend upon material.

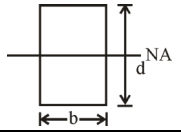
730. A beam of rectangular cross-section is to be cut from a circular beam of diameter D. What is the ratio of the depth of the beam to its width for maximum moment of resistance?
(a) $\sqrt{3}$ (b) $\sqrt{2}$
(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{3}{\sqrt{2}}$

TSPSC AEE 28.08.2017 (Civil/Mechanical)
APGENCO AE 2012

Ans. (b)



$b^2 + d^2 = D^2$
$d^2 = D^2 - b^2$
for the strongest beam in bending its 'z' should be minimum.



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$$\frac{dz}{db}=0$$

$$z=\frac{bd^2}{6}$$

$$z=\frac{b\left(D^2-b^2\right)}{6}$$

$$\frac{D^2-3b^2}{6}=0\Rightarrow b=\frac{D}{\sqrt{3}}$$

$$b^2+d^2=D^2$$

$$d^2=D^2-\frac{D^2}{3}\Rightarrow d=\sqrt{\frac{2}{3}D^2}$$

$$d=b.\sqrt{2}$$

$$\frac{\text{depth}}{\text{width}}=\sqrt{2}$$

731. If a shaft is simultaneously subjected to a torque T and bending moment M, the maximum shear stress is

(a) $\frac{16}{\pi D^3}(M+T)$

(b) $\frac{16}{\pi D^3}(M+\sqrt{M^2+T^2})$

(c) $\frac{16}{\pi D^3}(M-\sqrt{M^2+T^2}$

(d) $\frac{16}{\pi D^3}\sqrt{M^2+T^2}$

Nagaland PSC CTSE 2017, Paper-I
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : We know,

$$\sigma_b=\frac{M_b}{Z}\quad\&\quad Z=\frac{I}{Y}$$

$$\sigma_b=\frac{M}{\left(\frac{\pi d^4}{64}\right)}\times\frac{d}{2}=\frac{32M}{\pi d^3}$$

$$\tau=\frac{16T}{\pi d^3}\Rightarrow\ \therefore\ T_{equivalent}=\sqrt{M^2+T^2}$$

$$\tau_{max}=\frac{16\times T_{eq}}{\pi d^3}=\frac{16}{\pi D^3}\times\sqrt{M^2+T^2}$$

732. Section modulus (Z) of a beam depends on :

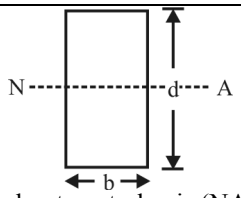
(a) The geometry of the cross-section
(b) Weight of the beam
(c) Only on length of the beam
(d) Density of the beam

HPPSC Lect. (Auto) 23.04.2016
HPPSC LECT. 2016

Ans. (a) : Section modulus is a geometrical phenomenon.

$$\text{Bending stress }(\sigma_{bending})=\frac{My_{max}}{I_{NA}}$$

$$\text{Section modulus }(Z_{N.A})=\frac{I_{NA}}{y_{max}}$$



Section modulus about neutral axis (NA)

$$Z_{NA}=\frac{\frac{bd^3}{12}}{\frac{d}{2}}=\frac{bd^2}{6}$$

$$\boxed{Z_{N.A}=\frac{bd^2}{6}}$$

733. When a shaft with diameter (d) is subjected to pure bending moment (M_b), the bending stress (σ_b) induced in the shaft is given by:

(a) $\sigma_b=\left(\frac{32M_b}{\pi d^3}\right)$ (b) $\sigma_b=\left(\frac{64M_b}{\pi d^3}\right)$

(c) $\sigma_b=\left(\frac{64M_b}{\pi d^2}\right)$ (d) $\sigma_b=\left(\frac{32M_b}{\pi d^2}\right)$

APPSC Poly Lect. 13.03.2020
CIL (MT) 2017 IInd Shift

Ans. (a) : In case of pure bending, the bending stress induced in the shaft $\sigma_b=\left(\frac{32M_b}{\pi d^3}\right)$.

734. The bending stress in a beam is _____ section modulus.

(a) inversely proportional to two times
(b) directly proportional to
(c) inversely proportional to
(d) None of the above

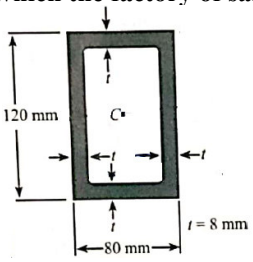
Assam PSC AE (IWT) 14.03.2021

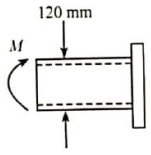
Ans. (c) : The bending moment equation

$$\frac{M}{I}=\frac{\sigma}{y_{max}}=\frac{E}{R}\qquad\sigma=\frac{M}{I/y_{max}}=\frac{M}{Z}$$

From the equation we said that bending stress inversely proportional to the section modulus
Section modulus $Z=I/y_{max}$

735. The rectangular tube shown is extruded from an aluminum alloy for which σ_y = 150 MPa, σ_U = 300 MPa and E = 70 GPa. Neglecting the effect of fillets, the bending moment M for which the factory of safety will be 3.00 is :





(a) M = 9.20 kN-m (b) M = 3.0 kN-m
(c) M = 6.1 kN-m (d) M = 15 kN-m

CGPSC AE 15.01.2021

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Ans. (a) : $\frac{M}{I}=\frac{\sigma}{y}\rightarrow M=\frac{\sigma}{y}\cdot I$

$$I=\frac{80\times120^3}{12}-\frac{(80-2\times8)(120-2\times8)^3}{12}$$

$$I=5.52\times10^{-6}\ m^4$$

$$\sigma_{allowable}=\frac{\sigma_U}{FOS}=\frac{300}{3}=100\,MPa$$

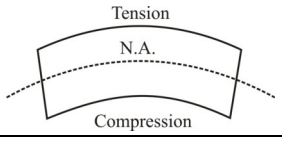
$$M=100\times10^6\times\frac{5.52\times10^{-6}}{60\times10^{-3}}=9.2\ kN\text{-}m$$

736. At the neutral axis of the beam, the layers ___.

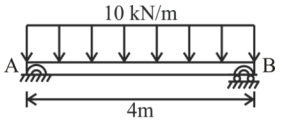
(a) do not undergo any strain
(b) are subjected to tension
(c) are subjected to compression
(d) are subjected to maximum bending stress

VIZAG Steel MT 24.01.2021

Ans. (a) : At the neutral axis of the beam, the layers do not undergo any strain and any stress. Therefore, bending stress at neutral axis is zero and no load acting at neutral axis.



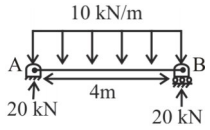
737. A simply supported beam of width 100 mm, height 200 mm and length 4 m is carrying a uniformly distributed load of intensity 10 kN/m. The maximum bending stress (in MPa) in the beam is _____ (correct to one decimal place).



(a) 30 (b) 120
(c) 90 (d) 60

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) :



Given,
b = 100 mm
D = 200 mm
$R_A + R_B = 40$
$R_A \times 4 = 10 \times 4 \times 2$
$R_A = 20$ kN
$R_B = 20$ kN

$$I=\frac{bd^3}{12}=\frac{100\times(200)^3}{12}$$

$$\frac{\sigma}{y}=\frac{M}{I}=\frac{E}{R}$$

$$\sigma=\frac{M}{I}\times y\ \left(y=\frac{D}{2}=\frac{200}{2}\right)$$

$M_{max} = 20 \times 2 - 10 \times 2 \times 1$
$= 20$ kN-m
$= 20 \times 10^6$ Nmm

$$=\frac{20\times10^6}{\frac{100\times(200)^3}{12}}\times\left(\frac{200}{2}\right)$$

$\sigma_b = 30$ MPa

738. For a cast iron simply supported beam with low strength in tension and high strength in compression, due to the transversely applied load, with beam bent concave upwards, following section will be better.

(a) Square
(b) Rectangular
(c) Circular
(d) Trapezoidal

JPSC AE 10.04.2021, Paper-II

Ans. (d) : If simply supported beam bend concave upwards then compressive stress developed at top fibre and tensile stress developed at bottom fibre.

Cast iron is low strength in tension and high strength in compression so bottom fibre has to be more area (more material) and top fibre is strong in compression so no need to add more material on top fibre so better section will be trapezoidal section.

739. Variation of bending stresses at any section on a beam is related to the bending moment by _____.

(a) St Venant's formula (b) flexural formula
(c) torsional formula (d) bending formula

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : flexural formula

740. A wooden beam AB supporting two concentrated loads P has a rectangular cross-section of width = 100 mm and height = 150 mm. The distance from each end of the beam to the nearest load is 0.5 m. If the allowable stress in bending is 11 MPa and the beam weight is negligible, the maximum permissible load will be nearly

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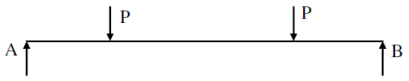
(a) 5.8 kN (b) 6.6 kN
(c) 7.4 kN (d) 8.2 kN

ESE 2020

Ans. (d) : $M_{max} = P \times 0.5 = P \times 500$ Nmm

$$\sigma_{max} = \frac{6M_{max}}{bd^2} = \frac{M_{max}}{bd^2/6}$$



$$11 = \frac{6 \times P \times 500}{100 \times 150^2}$$

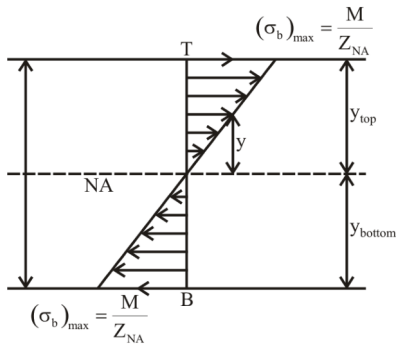
$P = 8250$ N $= 8.25$ kN

741. When a beam is subjected to bending moment alone, the bending stress at any point from the neutral axis

(a) equal
(b) varies directly proportional with respect to distance from neutral axis
(c) varies inversely proportional with respect to distance from neutral axis
(d) stress is independent of the distance

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : For pure bending condition,



Bending stress distribution

$$(\sigma_b)_{max} = \frac{M_y}{I_{NA}}$$

$$\boxed{(\sigma_b) \propto y}$$

• The bending stress at any point from the neutral axis varies directly proportional with respect to distance from neutral axis.

742. A square beam and a circular beam of same material have the same length, same allowable stress and the same bending moment. The ratio of weights of the square beam to that of circular beam is

(a) 1/2 (b) 1
(c) 1/1.12 (d) 1/3

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) : Bending stress, $\sigma = \frac{M}{Z}$

For same stress and moment,

$Z_{square} = Z_{circular}$

$$\frac{b^3}{6} = \frac{\pi}{32}d^3$$

$$b^3 = \frac{3}{16}\pi d^3$$

$$b = \left(\frac{3\pi}{16}\right)^{1/3} \cdot d$$

$$\frac{W_{square}}{W_{circular}} = \frac{A_s \times \ell}{A_{circular} \times \ell} = \frac{b^2}{\frac{\pi}{4}d^2} = \frac{4}{\pi}\left(\frac{b}{d}\right)^2 = \frac{4}{\pi}\left(\frac{3\pi}{16}\right)^{2/3}$$

$$\frac{W_{square}}{W_{circular}} = \frac{1}{1.12}$$

743. The average shear stress in a circular cross section of a beam is equal to:

(a) 0.85 times the maximum shear stress
(b) 0.55 times the maximum shear stress
(c) 0.75 times the maximum shear stress
(d) 0.25 times the maximum shear stress

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Maximum shear stress occurs at neutral axis and $y = 0$

Max shear stress $= \frac{4}{3}$ average shear stress

For circular section

Avg shear stress $= \frac{3}{4}$ Maximum shear stress

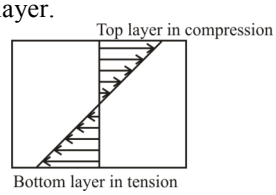
$= 0.75$ times the maximum shear stress.

744. When a rectangular beam is loaded transversely, the maximum tensile stress is developed on the

(a) top layer (b) bottom layer
(c) neutral axis (d) every cross-section

RPSC IOF, 2020

Ans. (b) : When a rectangular beam is loaded transversely, the maximum tensile stress is developed on the bottom layer.



Compressive stress developed on the top layer.
Stress is zero at the neutral axis.

745. In a triangular section, the maximum shear stress occurs at

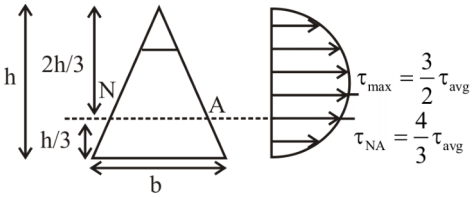
(a) Apex of the triangle (b) Mid of the height
(c) 1/3 of the height (d) base of the triangle

RPSC IOF, 2020

Ans. (b) : In a triangular section, the maximum shear stress occurs at mid of the height.

→ For rectangular section $F_{max} = h/2$

→ For triangular section, the shear stress has a parabolic variation.



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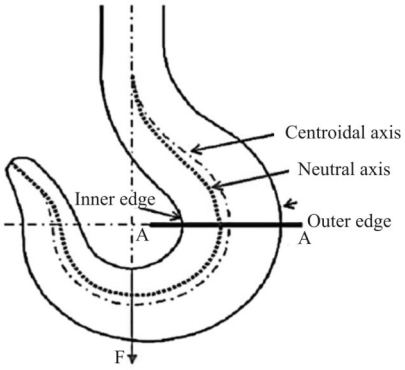
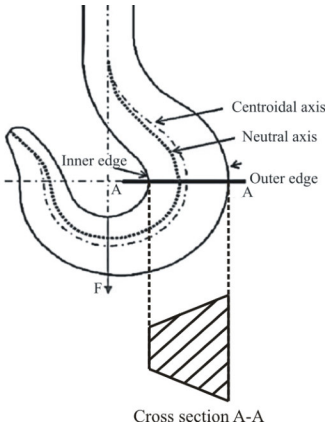
746. For a standard crane hook (as shown in figure) subjected to a constant load F, the tensile stress at a given section A-A will be maximum at___



(a) innermost edge (b) neutral axis
(c) outermost edge (d) centroidal axis

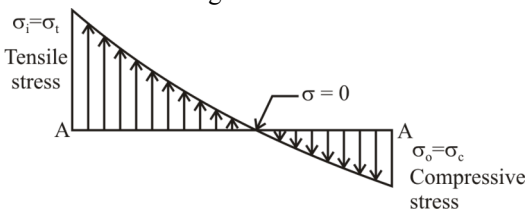
APPSC Poly Lect. 13.03.2020

Ans. (a) : For a standard crane hook (as shown in figure) subjected to a constant load F, the tensile stress at a given section A-A will be maximum at innermost edge.



Cross section A-A

Stress distribution in a given section-



Where, $\sigma_i = \sigma_t = \frac{F}{A} + \frac{MY}{A_e R}$

$\sigma_o = \sigma_c = \frac{F}{A} - \frac{MY}{A_e R}$

747. A non uniform bending refers to flexure of a beam

(a) in presence of uniformly distributed loaded in the beam
(b) in presence of uniform bending moment
(c) in presence of shear forces
(d) in absence of shear forces

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : Bending Equation–Every element of the beam is bend with the some radius of curvature

$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$ or $R = \frac{EI}{M}$

A non-uniform bending is deformation is the presence of shear force and BM change along the axis of beam.

748. The normal stresses in a cross section of a beam under lateral loading

(a) varies linearly with distance from the neutral axis
(b) varies quadratically with distance from the neutral axis
(c) does not vary with distance from the neutral axis
(d) varies inversely with distance from the neutral axis

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Ans. (a) : Lateral load produce bending. Bending stress developed in cross section is normal to cross section so called normal stress. Bending stress linearly varies with distance from the neutral stress.

749. The shear in a cross section for a rectangular beam under lateral loading

(a) does not vary with distance from the neutral axis
(b) varies linearly with distance from the neutral axis
(c) varies quadratically with distance from the neutral axis
(d) varies inversely with distance from the neutral axis

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : Shear stress in the cross section for any section beam under lateral loading varies quadratically (parabolically) with distance from neutral axis.

750. Product of allowable bending stress and sectional modulus is

(a) Moment of rigidity
(b) Radius of gyration
(c) Moment of resistance
(d) Moment of inertia

CIL MT 27.02.2020

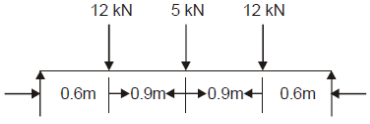
Ans. (c) : The maximum bending moment which can be carried by given section for a given maximum value of stress is known as moment of resistance (M_r)

Section modulus $(Z) = \frac{I}{y}$

$M_r = \sigma \frac{I}{y}$, $M_r = \sigma Z$

Moment of resistance=Bending stress×Section modulus

751. A beam of span 3 m and width 90 mm is loaded as shown in the figure. If the allowable bending stress is 12 MPa, the minimum depth required for the beam will be



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(a) 218 mm (b) 246 mm
(c) 318 mm (d) 346 mm

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Ans. (b) :

$R_A + R_B = 29$

$R_A \times 3 = 12 \times 2.4 + 5 \times 1.5 + 12 \times 0.6$

$R_A = 14.5$

$R_B = 14.5$

$M_{max} = 14.5 \times 1.5 - 12 \times 0.9$

$= 10.95$ kNm

$= 10.95 \times 10^6$ N mm

Bending stress $\sigma_b = \frac{MY}{I} = \frac{6\,M_{max}}{bd^2}$

$12 = \frac{6 \times 10.95 \times 10^6}{90 \times d^2}$

$d = 246.64$ mm

752. A vertical hollow aluminium tube 2.5 m high fixed at the lower end, must support a lateral load of 12 kN at its upper end. If the wall thickness is $\frac{1}{8}^{th}$ of the outer diameter and the allowable bending stress is 50 MPa, the inner diameter will be nearly

(a) 186 mm (b) 176 mm
(c) 166 mm (d) 156 mm

ESE 2020

Ans. (d) :

$t = \frac{1}{8} d_0$

$d_i = d_0 - 2t$

$= d_0 - 2 \times \frac{1}{8} d_0 = \frac{3}{4} d_0$

$\frac{d_i}{d_0} = \frac{3}{4}$

$\sigma_b = \frac{32 M_{max}}{\pi d_0^3 (1 - K^4)}$

$50 = \frac{32 \times 12 \times 10^3 \times 2500}{\pi d_0^3 \left[1 - \left(\frac{3}{4}\right)^4\right]}$

$d_0 = 207.54$

$d_i = \frac{3}{4} \times 207.54 = 155.66$ mm

753. Which of the following is NOT an example of curved beam?

(a) Chain links with tensile load
(b) Crane hooks subjected to downward load
(c) Rings with uniform tensile load
(d) Cantilever beam with point load

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Ans. (d) : Curved beams : Beam whose axis is not straight and is curved in the elevation is said to be a curved beam is the applied loads are along the y-direction and the span of the beam is along the x-direction. The axis of the beam should have a curvature in xy plane.

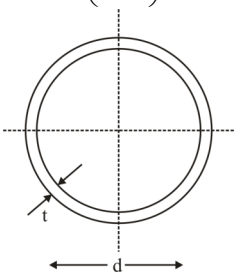
ex. : Chain links with tensile load, crane hooks subjected to downward load, ring with uniform tensile load.

754. Section modulus of hollow circle with average diameter 'd' and with thickness 't' is equal to

(a) $\frac{4}{5} td^2$ (b) $\frac{4}{5} t^2 d^2$
(c) $\frac{4}{5} t^2 d$ (d) $\frac{5}{4} td^2$

TNPSC AE 2019

Ans. (a) : $I_{x-x} = I_{y-y} = \left(\frac{\pi d^3}{8}\right) \times t$



$Ro = \frac{d}{2} + t \simeq \frac{d}{2}$

$Z_{x-x} = Z_{y-y} \left[\frac{\pi d^2}{4} \times t\right]$

$\approx \frac{4}{5} td^2$

755. Pure bending means :

(a) The bending beam shall be accompanied by twisting
(b) Shear force is zero
(c) There is no twisting
(d) None of these

OPSC AEE 2019 Paper-I

Ans : (b) : Pure bending- Pure bending is a condition of stress where a bending moment is applied to a beam without the simultaneous application of axial shear or torsional forces. Beam that is subjected to pure bending means the shear force in the particular beam is zero and no torsional or axial loads are presented. Pure bending is also the flexure (bending) of a beam that under a constant bending moment therefore pure bending only occurs therefore pure bending only occurs when the shear force in equal to zero.

756. The point within the cross-sectional plane of beam through which the resultant of the external loading on the beam has to pass through to ensure pure bending without twisting of the cross-section of the beam is called :

(a) Moment centre (b) Centroid
(c) Shear center (d) Elastic center

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Ans : (c) : Shear center is the point within the cross sectional plane of a beam through which the resultant of the external loading on the beam has to pass through to ensure pure bending without twisting of cross section of the beam.

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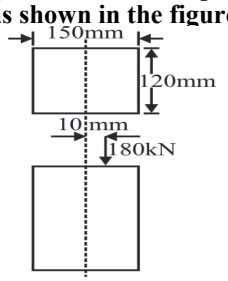
757. A rectangular strut is 150 mm wide and 120 mm thick. It carries a load of 180 kN at an eccentricity of 10 mm in a plane bisecting the thickness as shown in the figure



The maximum intensity of stress in the section will be

(a) 14 MPa (b) 12 MPa
(c) 10 MPa (d) 8 MPa

ESE 2019

Ans. (a) : Given,

P = 180 kN
b = 150 mm
d = 120 mm
e = 10 mm

Resultant normal stress is maximum at the right side fiber of the cross section, because the line of action of eccentric axial compressive load is nearer to this fiber.

Maximum intensity of stress

$$= \sigma_c + \sigma_b$$

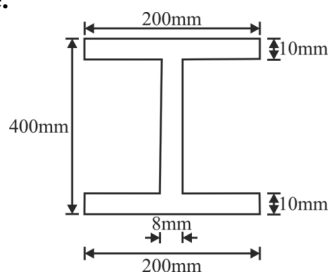
$$\sigma_{max} = \frac{-P}{A} - \frac{M}{Z}$$

$$= \frac{-P}{A} - \frac{P \times e}{\left(\frac{db^2}{6}\right)} = \frac{-P}{bd} - \frac{6P \times e}{db^2}$$

$$= \frac{-P}{bd}\left[1 + \frac{6 \times e}{b}\right] = -\frac{180 \times 10^3}{120 \times 150}\left[1 + \frac{6 \times 10}{150}\right]$$

$$= -14 \text{ MPa} = 14 \text{ MPa (compressive)}$$

758. The cross-section of the beam is as shown in the figure.



If the permissible stress is 150 N/mm^2, the bending moment M will be nearly

(a) 1.21×10^8 N mm (b) 1.42×10^8 N mm
(c) 1.64×10^8 N mm (d) 1.88×10^8 N mm

ESE 2019

Ans. (b) : Moment of inertia of I-section

$$I = I_{web} - 2\,(I_{flange})$$

$$I = \left[\frac{200 \times 400^3}{12} - 2 \times \frac{96 \times 380^3}{12}\right]$$

$$= 1.887 \times 10^8 \text{ mm}^4$$

$$\text{Section modulus } (Z) = \frac{I}{y_{max}}$$

$$= \frac{1.887 \times 10^8}{200}$$

$$= 943573.33 \text{ mm}^3$$

Permissible stress

$$\sigma_{max} = \frac{M}{Z}$$

Bending moment $M = \sigma_{max} \times Z$

$$= 150 \times 943573.33$$

$$= 1.415 \times 10^8 \text{ N-mm}$$

759. A hollow circular bar used as a beam has its outer diameter thrice the inside diameter. It is subjected to a maximum bending moment of 60 MN m. If the permissible bending stress is limited to 120 MPa, the inside diameter of the beam will be.

(a) 49.2 mm (b) 53.4 mm
(c) 57.6 mm (d) 61.8 mm

ESE 2019

Ans. (c) : Given,

$d_o = 3\,d_i$, M = 60 MN-mm = 60×10^6 N-mm

$$y_{max} = \frac{d_o}{2}$$

$$\sigma_{max} \le \sigma$$

$$\frac{M}{I} \times y_{max} \le \sigma$$

$$\frac{60 \times 10^6}{\frac{\pi}{64}\left(d_o^4 - d_i^4\right)} \cdot \frac{d_o}{2} \le 120$$

$$\frac{60 \times 10^6 \times 64}{\pi d_o^4\left[1 - \left(\frac{d_i}{d_o}\right)^4\right]} \times \frac{d_o}{2} = 120$$

$$\frac{384 \times 10^7}{2 \times 120 \times \pi\left[1 - \left(\frac{1}{3}\right)^4\right]} = d_o^3$$

d_o = 172.79 mm

$d_i = d_o/3$

d_i = 57.6 mm

Note : Bending moment value should be in MN-mm but in question it is given in MN-m.

760. In a beam of I-section, which of the following parts will take the maximum shear stress when subjected to traverse loading?

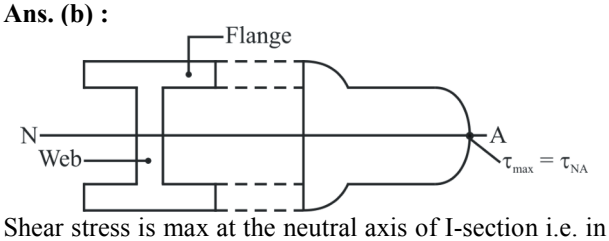
1. Flange 2. Web

Select the correct answer using code given below.

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 or 2

ESE 2019

Ans. (b) :



Shear stress is max at the neutral axis of I-section i.e. in the web portion.

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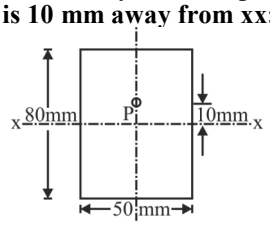
761. A pull of 100 kN acts on a bar as shown in the figure in such a way that it is parallel to the bar axis and is 10 mm away from xx:

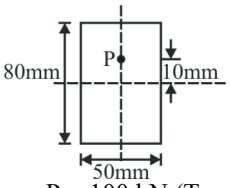


The maximum bending stress produced in the bar at xx is nearly.

(a) 20.5 N/mm^2 (b) 18.8 N/mm^2
(c) 16.3 N/mm^2 (d) 14.5 N/mm^2

ESE 2019

Ans. (b) :



P = 100 kN (Tensile)

Bending moment (M) = $100 \times 10^3 \times 10$
$= 10^6$ N-mm

$$y_{max} = \frac{80}{2} = 40 \text{ mm}$$

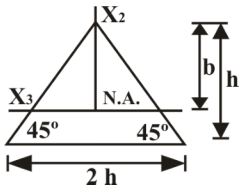
$$I = \frac{bd^3}{12} = \frac{50 \times 80^3}{12}$$

Maximum bending stress

$$\sigma_{max} = \frac{M}{I} \cdot y_{max} = \frac{10^6}{\left(\frac{50 \times 80^3}{12}\right)} \times 40$$

σ_{max} = 18.8 N/mm^2

762. The beam of triangular cross-section as shown in the figure below, is subjected to pure bending. If a plastic hinge develops at a section, determine the location of neutral axis (distance b from top) at that section. The beam material is elastic-perfectly plastic (i.e., yield stress is constant)



(a) $\frac{h}{\sqrt{3}}$ (b) $\frac{h}{2}$
(c) $\frac{h}{\sqrt{2}}$ (d) $\frac{h}{3}$

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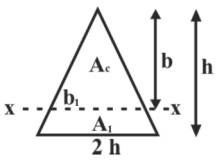
Ans. (c) :



$A_c = A_t$

$$\frac{1}{2} \times b \times b_1 = \frac{1}{2}\left[\frac{1}{2} \times 2h \times h\right]$$

$$b_1 = \frac{h^2}{b}$$

from similar triangles

$b \to b_1$

$h \to 2h$

$b \times 2h = b_1 h$

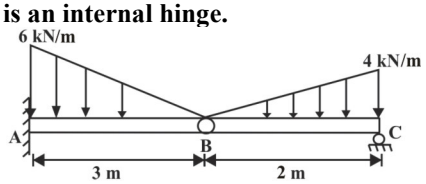
$$b = \frac{b_1}{2}$$

$$b = \frac{h^2}{b \times 2}$$

$$b^2 = \frac{h^2}{2}$$

$$b = \frac{h}{\sqrt{2}}$$

763. Calculate the shear force and bending moment at point B for the beam AB subjected to linearly varying load as shown in the figure. The value of the linearly varying load at the point is 6 kN/m and 4 kN/m, respectively. Point B is an internal hinge.



(a) 2.67 kN and 0 kN-m (b) 4 kN and 0 kN-m
(c) 4 kN and 1.33 kN-m (d) 1.33 kN and 0 kN-m

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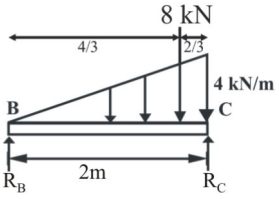
Ans. (d) :



Consider BC part

$$SF_B = R_B = \frac{\frac{1}{2} \times 4 \times 2 \times \frac{2}{3}}{2} = \frac{4}{3} = 1.33kN$$

$\boxed{R_B = 1.33 \text{ kN}}$

$BM_B = 0$ (at hinge moment is zero)

764. A beam of rectangular section 200mm × 300mm carries certain loads such that bending moment at a section A is M and at another section B it is (M + ΔM). The distance between section A and B is 1 m and there are no external loads acting between A and B. If ΔM is 20 kNm, maximum shear stress in the beam section is

(a) 0.5 MPa (b) 1.0 MPa
(c) 1.5 MPa (d) 2.0 MPa

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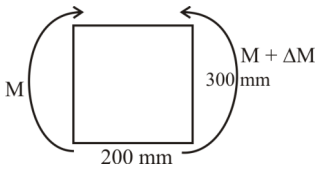
Ans. (a) :



$$\frac{dM}{dx} = F$$

$$\frac{20kN - m}{1m} = F$$

F = 20 kN

For rectangular cross-section

$$\tau_{max} = \frac{3}{2}\left[\tau_{avg}\right]$$

$$\tau_{max} = \frac{3}{2}\left[\frac{F}{bd}\right] = \frac{3}{2}\left[\frac{20\times10^3}{200\times300}\right] = 0.5 \text{ MPa}$$

765. A mild steel flat of width 100 mm and thickness 12 mm is bent into an arc of a circle of radius 10 m by applying a pure moment M. If Young's modulus E = 200 GPa, then the magnitude of M is

(a) 72 Nm (b) 144 Nm
(c) 216 Nm (d) 288 Nm

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Ans. (d) : Given, Radius of curvature (R) = 10,000 mm
Modulus of elasticity (E) = 200×10^3 MPa
Thickness (t) = 12 mm
From bending equation

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{Y}$$

$$\frac{M}{\left[\frac{100\times12^3}{12}\right]} = \frac{200\times10^3}{10,000}$$

M = 288000 N-mm
= 288 N-m

766. A rectangular beam section with depth 400 mm and width 300 mm is subjected to a bending moment of 60 kN-m. The maximum bending stress in the section is

(a) 7.50 MPa (b) 2.50 MPa
(c) 1.56 MPa (d) 0.42 MPa

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Ans. (a) : Maximum bending stress in the beam

$$f_{max} = \frac{M}{Z} = \frac{M}{\left[\frac{bd^2}{6}\right]} = \frac{60\times10^6}{\left[\frac{300\times400^2}{6}\right]} = 7.5 \text{ MPa}$$

767. In case of pure bending, the beam will bend into an arc of a/an

(a) parabola (b) hyperbola
(c) circle (d) ellipse

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Ans. (c) : If a beam is subjected to pure bending the elastic curve is a circular arc with constant radius.

768. Two beams have the same length, same allowable stresses and the same bending moment. The cross sections of the beams are a square and a rectangle with depth twice that of width. The ratio of weight of square beam to the weight of rectangular beam is:

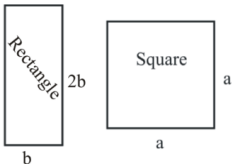
(a) 0.85 (b) 0.75
(c) 0.95 (d) 1.25

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (d) : $l_1 = l_2$

$\delta_1 = \delta_2$

$M_1 = M_2$



From bending equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{M}{I}\times y$$

$\sigma_1 = \sigma_2$

$$\frac{M_1}{I_1}\times y_1 = \frac{M_2}{I_2}\times y_2 \qquad (M_1 = M_2)$$

$$\left(\frac{y_1}{I_1}\right)_{rec.} = \left(\frac{y_2}{I_2}\right)_{sq.}$$

$$\frac{2b/2}{b\times\frac{(2b)^3}{12}} = \frac{a/2}{\frac{a^4}{12}}$$

$$\frac{2b}{8b^4} = \frac{a}{a^4}$$

$$\frac{1}{4b^3} = \frac{1}{a^3}$$

$$a^3 = 4b^3$$

$$\boxed{a = 1.587b}$$

$$\frac{\text{weight of square Beam}}{\text{weight of rectangular Beam}} = \frac{m_2 g}{m_1 g} = \frac{\rho_2 V_2 g}{\rho_1 V_1 g} = \frac{V_2}{V_1}$$

$$= \frac{a^2\times l}{2b^2\times l}$$

$$= \frac{(1.587b)^2\times l_1}{2b^2\times l_2} \qquad (l_1 = l_2)$$

= 1.25

769. A beam with rectangular cross section (10mm × 20mm) is subjected to maximum shear force of 5kN. The maximum shear stress induced is :

(a) 54.5 MPa (b) 25 MPa
(c) 37.5 MPa (d) 50 MPa

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

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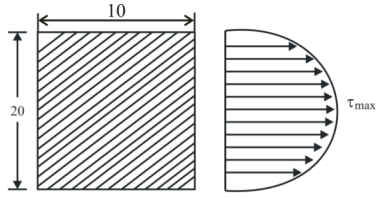
Ans. (c) : Given,

Area of rectangular cross section

$A = 10\ mm \times 20mm$

Shear force $(F_s) = 5\ kN$



$$\tau_{max} = \frac{3}{2}\tau_{avg}$$

$$= \frac{3}{2} \times \frac{F_s}{A}$$

$$= \frac{3}{2} \times \frac{5 \times 10^3}{10 \times 20}$$

$$\tau_{max} = 37.5MPa$$

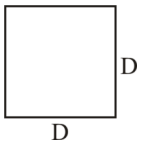
770. The ratio of moment carrying capacity of a square cross section beam of dimension D to the moment carrying capacity of a circular cross section of diameter D is :

(a) $\frac{16}{3\pi}$ (b) $\frac{16}{\pi}$

(c) $\frac{16}{5\pi}$ (d) $\frac{8}{3\pi}$

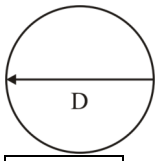
BHEL ET 2019

Ans. (a) :



Section modulus of square cross - section $Z_s = \frac{D^3}{6}$

Section modulus of circular cross - section $Z_c = \frac{\pi D^3}{32}$



$$\boxed{\sigma_b = \frac{M}{Z_{NA}}}$$

For square cross section–

$$(\sigma_b)_s = \frac{M_s}{\frac{D^3}{6}} = \frac{6M_s}{D^3}$$

For circular cross section–

$$(\sigma_b)_c = \frac{M_c}{\frac{\pi D^3}{32}} = \frac{32M_c}{\pi D^3}$$

Note- Bending stress of both beam will be same

$$(\sigma_b)_s = (\sigma_b)_c$$

$$\frac{6M_s}{D^3} = \frac{32M_c}{\pi D^3}$$

$$\boxed{\frac{M_s}{M_c} = \frac{16}{3\pi}}$$

771. If the Bending moment is increased three times, then to keep the stress in the beam same, sectional modulus shall be

(a) decreased 3 times (b) increased 3 times

(c) unchanged (d) increased 6 times

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (b) : Bending stress

$$\sigma = \frac{M}{Z}$$

For $\sigma = \frac{M}{Z} = \text{constant}$

If $M \uparrow \rightarrow 3M$

then Z also increase 3 times to keep the stress in the beam same.

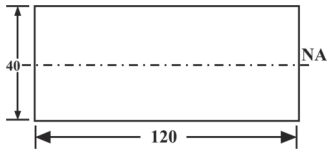
772. A beam with rectangular section of 120 mm × 40 mm is placed horizontally by mistake, (with width as 120 mm and depth as 40 mm) whereas it was designed to be placed vertically, (with width as 40 mm and depth as 120 mm). The ratio of section modulus will be

(a) 1/3 (b) 1/2

(c) 1/6 (d) 1/8

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (a) : Case-I



$$Z = \frac{I_{NA}}{Y_{max}}$$

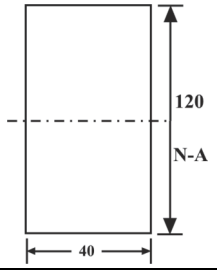
$$I_{NA} = \frac{120 \times (40)^3}{12}$$

$$Y = \frac{40}{2}$$

$$Z_1 = \frac{120 \times 40 \times 40 \times 40 \times 2}{12 \times 40}$$

$$Z_1 = 32000\ mm^3$$

Case-II



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$$I_{N-A} = \frac{40 \times (120)^3}{12}$$

$$y_{max} = \left(\frac{120}{2}\right)$$

$$Z_2 = \frac{40 \times 120 \times 120 \times 120 \times 2}{120 \times 12}$$

$$Z_2 = 96000 \text{ mm}^3$$

$$\frac{Z_1}{Z_2} = \frac{32000}{96000} = \frac{1}{3}$$

773. Neutral axis of a beam is;

(a) Layer subjected to tensile stress
(b) Layer subjected to compressive stress
(c) Layer subjected to torsion stress
(d) Layer subjected to zero stress

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (d) : Neutral axis of a beam is layer subjected to zero stress.

$$\sigma_b = \frac{M}{I} \times y \qquad \text{(at neutral axis, y = 0)}$$

$$\boxed{\sigma_b = 0}$$

774. The beams, one having square cross section and another circular cross section, are subjected to same amount of bending moment. If the cross section area and the material of both the beams are the same then

(a) Both beams will experience same deformation
(b) Maximum bending stress developed in both the beams is the same
(c) The circular beam experiences more bending stress than the square one.
(d) The square beam experiences more bending stress than the circular one

GPSC Executive Engineer 23.12.2018, GATE 2003

Ans. (c) : Area of square and circle is same,

$$\frac{\pi}{4} \times d^2 = a^2$$

Let,

a = Side of square
d = diameter of circle

$$d = \frac{2a}{\sqrt{\pi}} \text{ ……..(i)}$$

Bending stress is given as,

$$\sigma = \frac{M}{Z} \text{ ……..……(ii)}$$

Where, M = Maximum bending moment
Z = Section modulus

Section modulus

$$Z = \frac{I}{Y}$$

Greater value of section modulus stronger be the section and as per equation (ii) lower will be bending stress.

$$Z_{Square} = \left(\frac{I}{Y}\right)_{Square}$$

$$Z_{square} = \frac{9 \times a^3}{\frac{12}{9}} = \frac{a^3}{6}$$

$$\frac{9}{2}$$

$$= 0.167 \, a^3 \text{ ……..(iii)}$$

$$Z_{circle} = \left(\frac{I}{Y}\right)_{Circle} = \frac{\pi R^4}{\frac{4}{R}}$$

$$Z_{circle} = \frac{\pi}{4} R^3 = \frac{\pi}{4} \times \frac{d^3}{8}$$

$$\because R = \frac{d}{2}$$

From equation (i), we get,

$$Z_{circle} = \frac{\pi}{32} \times \left(\frac{29}{\sqrt{\pi}}\right)^3$$

$$= 0.141 \, a^3 \text{ ……..(iv)}$$

From equation (iii) and (iv),

$(Z)_{circle} < (Z)_{square}$

Hence from equation (i), we get

$\sigma_{circle} > \sigma_{square}$

775. The relationship between the radius of curvature R, bending moment M and flexural rigidity EI is given by

(a) $R = \frac{M}{EI}$ (b) $M = \frac{EI}{R}$

(c) $EI = \frac{R}{M}$ (d) $E = \frac{MI}{R}$

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : $\frac{M}{I} = \frac{E}{R}$

$\Rightarrow M = \frac{EI}{R}$

776. Due to some point load anywhere on a fixed beam, the maximum free bending moment is M. The sum of fixed end moment is

(a) M (b) 1.5 M
(c) 2.0 M (d) 3.0 M

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : Due to some point load any where on a fixed beam, the maximum free bending moment is M. The sum of fixed end moment is M.

777. A pipe of external diameter 3 cm and internal diameter 2 cm and length 4m supported at its ends. If carries a point load of 65 N at its center-section, modulus of pipe will be

(a) $\frac{65\pi}{64} \text{cm}^3$ (b) $\frac{65\pi}{32} \text{cm}^3$

(c) $\frac{65\pi}{96} \text{cm}^3$ (d) $\frac{65\pi}{124} \text{cm}^3$

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Ans. (c) : External diameter $D_0 = 3$ cm
Internal diameter $D_i = 2$ cm

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Section modulus,

$$Z = \frac{I}{y_{max}}$$

$$I = \frac{\pi}{64}\left(D_0^4 - D_i^4\right)$$

$y_{max} = D_0/2$

(distance of extreme fiber from neutral axis)

$$Z = \frac{\frac{\pi}{64}\left[3^4 - 2^4\right]}{3/2}$$

$$Z = \frac{65\pi}{96}\,cm^3$$

778. In a loaded beam under bending

(a) both the maximum normal and maximum shear stresses occur at the skin fibers.

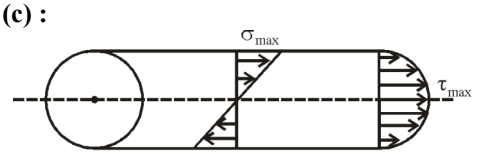
(b) both the maximum normal and maximum shear stresses occur at the neutral axis

(c) the maximum normal stress occurs at the skin fibres and the maximum shear stress occurs at the neutral axis

(d) the maximum normal stress occurs at the neutral axis while the maximum shear stress occurs at the stress fibers

TSPSC AEE 28.08.2017 (Civil/Mechanical), CSE Pre 2003

Ans. (c) :



Bending stress distribution | Shear stress distribution

Normal stress occurs maximum at extreme fiber while maximum shear stress occurs at neutral axis.

779. A beam has rectangular section 100 mm × 200 mm. If it is subjected to a maximum BM of 4 × 10^7 Nmm, then the maximum bending stress developed would be

(a) 30 N/mm^2 (b) 60 N/mm^2

(c) 90 N/mm^2 (d) 120 N/mm^2

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Rectangular section = 100 mm × 200 mm

$$I = \frac{1}{12}\times 100\times 200^3$$

$$y_{max} = \frac{200}{2} = 100$$

$$Z = \frac{I}{y_{max}} = \frac{\frac{1}{12}\times 100\times 200^3}{100} = \frac{2}{3}\times 10^6\,mm^3$$

$M_R = \sigma_{max} \times Z$

$$\sigma_{max} = \frac{M_R}{z} = \frac{4\times 10^7}{\frac{2}{3}\times 10^6}\,N\text{–}mm$$

$\sigma_{max} = 60$ N/mm^2

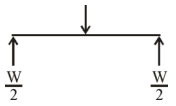
780. A timber beam is simply supported at the ends and carries a concentrated load at mid span. The maximum longitudinal stress 'f' is 12 N/mm^2 and the maximum shear stress 'q' is 1.2 N/mm^2. The ratio of span to depth would be

(a) 10 (b) 6

(c) 5 (d) 4

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c)



Maximum shearing stress $q = \frac{3}{2}q_{avg}$

(for rectangular such)

$$q = \frac{3}{2}\left(\frac{W}{2}\right)\times\frac{1}{bd}$$

$$q = \frac{3W}{4bd} \qquad \text{———(1)}$$

Maximum bending stress $f = \frac{M}{Z}$

$$f = \frac{\frac{W}{4}\ell}{bd^2/6} = \frac{3W\ell}{2bd^2} \qquad \text{———(2)}$$

$$\frac{f}{q} = \frac{2\ell}{d} = \frac{\ell}{d} = \frac{1}{2}\times\frac{12}{1.2} = 5$$

781. The moment of inertia of a given rectangular area is minimum about

(a) its longer centroidal axis

(b) its polar axis

(c) its axis along the diagonal

(d) its shorter centroidal axis

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : $I_{xx} = \frac{1}{12}bd^3$

$$I_{yy} = \frac{1}{12}db^3$$

since $b < d$

$\therefore I_{xx} > I_{yy}$

Hence MOI of a given rectangular section is minimum about longer centroidal axis.

782. A beam of uniform strength refers which the following?

(a) A beam in which extreme fibre stresses are same at all cross-section along the length of the beam

(b) A beam in which the moment of inertia about the axis of bending is a constant at all cross-section of the beam

(c) A beam in which the distribution of bending stress across the depth of cross-sections of the beam

(d) A beam in which the bending stress is uniform at the maximum bending moment cross-section

TSPSC AEE 28.08.2017 (Civil/Mechanical)

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Ans. (a) : Beams of uniform strength are those beams whose cross-section varies along length of beam, such that maximum bending stress at each cross-section reaches permissible stress and is constant through out the length of beam.

783. A mild steel plate is subjected to a moment M each at its ends such that it bends into an arc of a circle of radium 10 m. The plate has width 60 mm and thickness 10 mm. $E = 2 \times 10^5$ N/mm^2. What is the maximum bending stress produced in the plate?

(a) 100 MPa (b) 200 MPa
(c) 300 MPa (d) 400 MPa

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (a) : Thickness of plate = 10 mm
Radius of curvature = 10 m
$E = 2 \times 10^5$ N/mm^2

Flexure formula $\frac{E}{R} = \frac{\sigma}{y}$

$$\sigma = \frac{E \times y}{R}$$

$$\sigma = \frac{2 \times 10^5 \times \frac{10}{2}}{10 \times 1000} \text{ N/mm}^2$$

$\sigma = 100$ N/mm^2 = 100 MPa

784. In the design of beams for a given strength, consider that the conditions of economy of used of the material would avail as follows:

1. Rectangular cross-section is more economical than square section of the same cross-sectional area of the beam.
2. Circular section is more economical than square section.
3. I-section is more economical than a rectangular section of the same depth.

Which of the above are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

ESE 2017

Ans. (d) : For same material and same volume
$Z_I > Z_{rect} > Z_{sq} > Z_{cir}$
Square section is more economical than circular section therefore statement second is wrong.

785. Consider a simply supported beam of length, 50h, with a rectangular cross-section of depth, h, and width, 2h. The beam carries a vertical point load, P, at its mid point. The ratio of the maximum shear stress to the maximum bending stress in the beam is

(a) 0.02 (b) 0.10
(c) 0.05 (d) 0.01

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) :

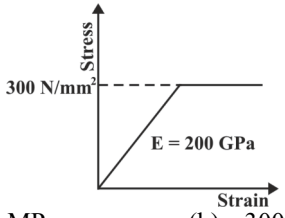
$$\sigma_{max} = \frac{M_{max}}{z} = \frac{(PL/4)}{(h^3/3)} = \frac{P \times 50h/4}{h^3/3} = 37.5\frac{P}{h^2}$$

$$\tau_{max} = \frac{3}{2}\frac{P_{max}}{A} = \frac{3P_{max}}{4h^2} = 0.75\frac{P_{max}}{h^2}$$

$$\tau_{max} = \frac{0.75P}{h^2 \times 2} \quad \therefore \quad P_{max} = \frac{P}{2}$$

$$\frac{\tau_{max}}{\sigma_{max}} = \frac{0.375}{37.5} = 0.01$$

786. The stress produced in a wire of diameter 6 mm, when it is bend around a large cylinder of diameter 2 m can be obtained as (Young's modulus of both wire and cylinder = 200 GPa, the stress strain curve of the material is given below)



(a) 600 MPa (b) 300 MPa
(c) 100 MPa (d) 200 MPa

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Given,

$\sigma = 300$ N/mm^2
$E = 200 \times 10^3$ N/mm^2

$$R = \frac{D}{2} = \frac{2}{2} = 1 \text{ m} = 1000 \text{ mm}$$

$$y_{max} = \frac{6}{2} = 3 \text{ mm}$$

From bending equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

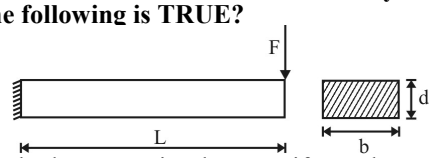
$$\sigma = y_{max}\frac{E}{R}$$

$$= 3 \times \frac{200 \times 10^3}{1000}$$

$= 600$ MPa

But $\sigma_{critical} = 300$ MPa
$\therefore$ $\sigma = 300$ MPa

787. A cantilever beam of rectangular cross section (depth 'd', width 'b' & length 'L') is subjected to a shear force 'F' at its free end as shown in figure. The beam is made of material which follown maximum shear stress theory. Which of the following is TRUE?



(a) The beam section has a uniform shear stress, $\frac{F}{bd}$
(b) Maximum shear stress in section and maximum bending stress in beam is equal when L = d/2
(c) For L > d/2, bending stress decides the failure stress
(d) Bending stress developed in the beam is independent of 'b'

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Ans. (c) (a) Shear stress varies parabolically in a rectangular cross-section.

(b) Maximum shear stress = $\frac{1.5F}{bd} \neq$ maximum bending stress.

(d) Bending stress depends on b.

788. A solid rod of 12 mm diameter was tested for tensile strength with the gauge length of 50 mm. Final length = 80 mm; Final diameter = 4 mm; Yield load = 1130 N. What is the nearest yield stress and % reduction in area respectively?

(a) 10 MPa and 10% (b) 90 MPa and 90%
(c) 10 MPa and 90% (d) 90 MPa and 10%

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Given, $d_i = 12$ mm, $L_i = 50$ mm, $L_f = 80$ mm
$d_f = 4$ mm, $P = 1130$ N, $\sigma_y =$

$$\sigma_{\text{yield stress}} = \frac{P}{A}$$

$$= \frac{1130}{\frac{\pi}{4}\times 12\times 12} = \frac{1130}{\pi\times 3\times 12} = 10 \text{ MPa}$$

Percentage reduction–

$$\% \text{ reduction area} = \frac{A_i - A_f}{A_i}\times 100$$

$$= \frac{\frac{\pi}{4}d_i^2 - \frac{\pi}{4}d_f^2}{\frac{\pi}{4}d_i^2}$$

$$= \frac{144-16}{144}\times 100 = \frac{128}{144}\times 100 = 88.88 \approx 90\%$$

789. A beam of uniform strength has constant :

(a) Shear force (b) Bending moment
(c) Cross-sectional area (d) Deflection

TRB Polytechnic Lecturer 2017

Ans. (b) : A beam of uniform strength is subjected to constant bending stress.

$$\sigma = \frac{M}{I}\times Y$$

For particular beam I & Y remains same so,

$\sigma = M =$ Constant

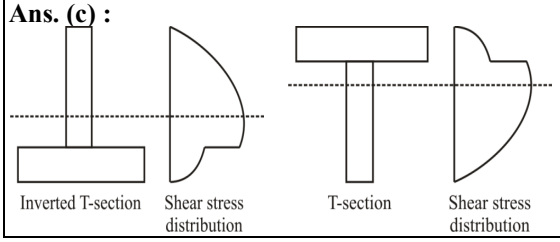
Hence, bending moment also will be constant.

790. An inverted T-section is subjected to a shear force F. The maximum shear stress will occur at:

(a) Top of the section
(b) Junction of web and flange
(c) Neutral axis of the section
(d) Bottom of the section

TRB Polytechnic Lecturer 2017

Ans. (c) :



The maximum shear stress will occur at neutral axis of the section in inverted T-section when subjected to a shear force.

791. A steel wire of 10 mm diameter is bent into a circular shape of 5 m radius. What will be the maximum stress induced in the wire, when E = 200 GPa?

(a) 50 MPa (b) 100 MPa
(c) 150 MPa (d) 200 MPa
(e) 250 MPa

(CGPCS Polytechnic Lecturer 2017)

Ans. (d) : Data given,
$d = 10$ mm
$R = 5$ m $= 5 \times 10^3$ mm
$E = 200$ GPa $= 200 \times 10^3$ N/mm^2
$\sigma = ?$

We know that pure bending equation

$$\frac{M}{I} = \frac{\sigma}{Y} = \frac{E}{R}$$

So $\sigma = \frac{E}{R}\times y_{max.}$ $\quad y_{max.} = \frac{d}{2} = \frac{10}{2} = 5$ mm

$$\sigma = \frac{200\times 10^3}{5\times 10^3}\times 5$$

$\sigma = 200$ N/mm^2

$\boxed{\sigma = 200 \text{ MPa}}$

792. A rectangular beam 100 mm wide is subjected to a maximum shear force of 50 kN. If the maximum shear stress is 3 MPa, the depth of the beam will be _____.

(a) 100 mm (b) 150 mm
(c) 200 mm (d) 250 mm
(e) 275 mm

CGPCS Polytechnic Lecturer 2017

Ans. (d) : We know that

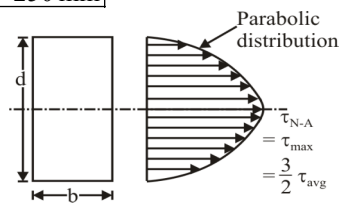
$$\tau_{max.} = \frac{3}{2}\tau_{avg.} \quad \text{[For rectangular beam]}$$

$$\tau_{avg} = \frac{\text{Shear Force}}{\text{Cross-Sectional Area}} = \frac{50\times 10^3}{100\times d}$$

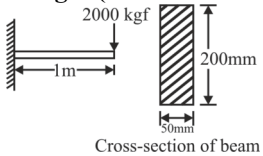
$$3 = \frac{3}{2}\times\frac{50\times 10^3}{100\times d}$$

$$d = \frac{3\times 50\times 10^3}{2\times 3\times 100}$$

$\boxed{d = 250 \text{ mm}}$



793. The maximum shearing stress induced in the beam section at any layer at any position along the beam length (shown in the figure) is equal to



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(a) 30 kgf/cm^2 (b) 40 kgf/cm^2
(c) 50 kgf/cm^2 (d) 60 kgf/cm^2

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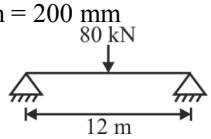
Ans. (a) : For rectangular cross section

$$\tau_{max} = \frac{3}{2}\tau_{avg}$$
$$= \frac{3}{2} \times \frac{2000}{50 \times 200} \text{kgf/mm}^2$$
$$= 30 \text{ kgf/cm}^2$$

794. A beam of rectangular section (12 cm wide × 20 cm deep) is simply supported over a span of 12 m. It is acted upon by a concentrated load of 80kN at the mid span. The maximum bending stress induced is.
(a) 400 MPa (b) 300 MPa
(c) 200 MPa (d) 100 MPa

ESE 2017

Ans. (b) : Given,
b = 12 cm = 120 mm
d = 20 cm = 200 mm



L = 12 m = 12000 mm
W = 80 kN = 80 × 10^3 N

$$M_{max} = \frac{WL}{4} = \frac{80 \times 10^3 \times 12000}{4}$$
$$= 240 \times 10^6 \text{ N–mm}$$
$$I = \frac{bd^3}{12} = \frac{120 \times 200^3}{12}$$
$$= 80 \times 10^6 \text{ mm}^4$$
$$y = \frac{d}{2} = \frac{200}{2} = 100 \text{ mm}$$

$$\text{Maximum bending stress } (\sigma) = \frac{MY}{I}$$
$$= \frac{240 \times 10^6 \times 100}{80 \times 10^6}$$
$$= 300 \text{ MPa}$$

795. In theory of simple bending an assumption is made that plane sections before bending remain plane even after bending. This assumption implies that
(a) Strain is uniform across the section
(b) Stress is uniform across the section
(c) Stress in any layer is proportional to its distance from the neutral axis
(d) Strain in any layer is directly proportional to its distance from the neutral axis

JWM 2017

Ans. (d) : The assumption made in theory of simple bending that plane sections before bending remain plane even after bending means the strain in any layer is directly proportional to its distance from the neutral axis.

796. Section modulus is defined as
(a) Moment of inertia about the neutral axis/square of the distance of neutral axis from farthest point
(b) Moment of inertia about the neutral axis/distance of the most distant point from neutral axis
(c) Bending moment/Moment of inertia
(d) None of the above

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (b) : Section Modulus—It is defined as the ratio of moment of inertia about the neutral axis to distance of the most distant point from neutral axis.
It is denoted by Z.

$$Z = \frac{I_{N-A}}{y_{max}}$$

797. Weldments in fabricated steel beams are designed for
(a) Bending stress at the flange
(b) Shear stresses in transverse plane
(c) Combination of bending and shear
(d) Bending and tensile stress

TSPSC AEE 2017

Ans. (c) : Weldments in fabricated steel beams are designed for combination of bending and shear for optimum design.

798. A beam with a rectangular section of 120 mm × 60 mm, designed to be placed vertically is placed horizontally by mistake. If the maximum stress is to be limited, the reduction in load carrying capacity would be
(a) 1/4 (b) 1/3
(c) 1/2 (d) 1/6

GPSC EE Pre, 28.01.2017

Ans. (c) : Given, depth (h) = 120 mm
width (b) = 60 mm
According to bending equation

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma_b}{y_{max}}$$
$$\sigma_b = \frac{M \times y_{max}}{I}$$

$$\text{I}^{st} \text{ case } \sigma_{b_1} = \frac{(M_{max})_1 \times \frac{h}{2}}{\frac{1}{12}bh^3}$$
$$\sigma_{b_1} = \frac{6(M_{max})_1}{bh^2}$$

$$\text{II}^{nd} \text{ case } \sigma_{b_2} = \frac{(M_{max})_2 \times \frac{b}{2}}{\frac{1}{12}hb^3}$$
$$= \frac{6(M_{max})_2}{hb^2}$$

At given- $\sigma_1 = \sigma_2$

$$\Rightarrow \quad \frac{6(M_{max})_1}{bh^2} = \frac{6(M_{max})_2}{b^2h}$$
$$\frac{(M_{max})_1}{(M_{max})_2} = \frac{bh^2}{b^2h}$$

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$$\frac{(M_{max})_1}{(M_{max})_2} = \frac{h}{b}$$

$$\frac{(M_{max})_2}{(M_{max})_1} = \frac{h}{b} = \frac{60}{120} = \frac{1}{2}$$

Note- Answer given by commission (a) but actual answer is (c).

799. For high speed engines, a rocker arm of..... should be used.

(a) rectangular section (b) I-section
(c) T-section (d) circular

GPSC EE Pre, 28.01.2017

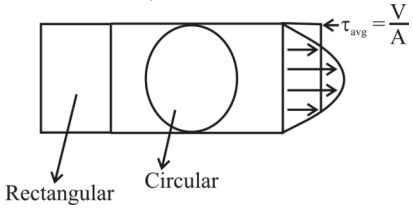
Ans. (b) : A rocker arm is an oscillating lever that conveys radial movement from the cam lobe into linear momentum at the poppet valve to open it. It may be made of cast iron, cast steel as malleable iron. In a order to reduce inertia of the rocker arm I-section is used for the high speed engines and it may be rectangular section for low speed engines.

800. If the average shear stress in a rectangular section beam is 5 N/sq. mm, then maximum shear stress for the circular section of the equal area is

(a) 7.50 N/sq.mm (b) 6.65 N/sq.mm
(c) 10 N/sq.mm (d) 2.50 N/sq.mm

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : For Circular,



For Circular,

$$\tau_{max} = \frac{4}{3} \times \frac{V}{A} = \frac{4}{3} \times \tau_{avg}$$

$$\tau_{max} = \frac{4}{3} \times 5 = 6.65 \text{ N/mm}^2$$

801. A horizontal beam of square section is placed with one diagonal (d) placed horizontally. The average shear stress occurs at

(a) d/8 from neutral axis
(b) d/4 from neutral axis
(c) neutral axis
(d) 3d/8 from neutral axis

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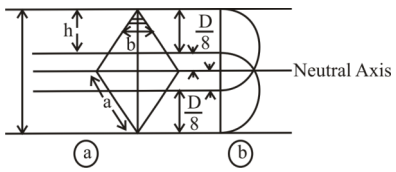
Ans. (a) :



$$h = \frac{D}{2} - y \Rightarrow \text{Base} \Rightarrow b = 2\left(\frac{D}{2} - y\right)$$

The Average shear stress occur at $\frac{D}{8}$ from Neutral Axis

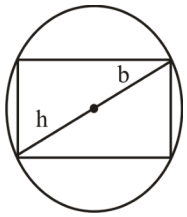
802. A long of wood is of 3 m diameter circular section. Then the width of the stronger rectangular section is bending that can be cut out from this log of wood is

(a) $\sqrt{6}$m (b) $\sqrt{3}$m
(c) $1/\sqrt{3}$m (d) $\sqrt{2}/3$m

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : As we know,
for strongest rectangle

$$b/h = \frac{1}{\sqrt{2}}$$



$$b = \frac{d}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3} \text{ m}$$

803. The bending force required for V-bending. U-bending and Edge-bending will be in the ratio of

(a) 1 : 2 : 0.5 (b) 1 : 1 : 1
(c) 2 : 1 : 0.5 (d) 1 : 2 : 1

UPRVUNL AE 21.08.2016

Ans. : (a) The bending force required for V-bending. U-bending and Edge-bending will be in the ratio of 1:2: 0.5

804. A cantilever of length 6 m is carrying a uniformly distributed load of W per unit run for a distance 4m from the free end. The value of maximum bending moment is

(a) 16W (b) 22W
(c) 18.5W (d) 21W

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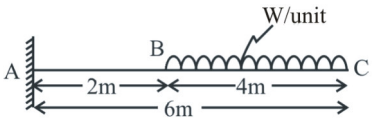
Ans. (a) :

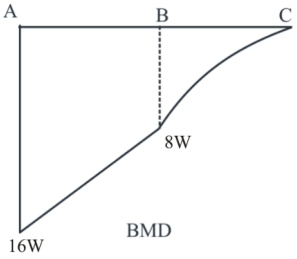


Total load = 4W and act at distance 2 m from free end for bending moment calculation.

$M_C = O$

$M_B = 4W \times 2 = 8\ W$

$M_A = 4W \times 4 = \boxed{16\ W = M_{max}}$



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805. A simply supported beam of length 7m is carrying a load whose intensity varies uniformly from zero at each end to W per unit run at the mid span. The value of maximum shear force is

(a) 1.75 W (b) 2.38 W
(c) 2.96 W (d) 3.42 W

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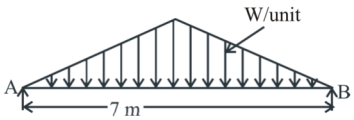
Ans. (a) :



Total load = Area of loading diagram

$$= \frac{1}{2} \times 7 \times W = 3.5W$$

Reactions at A & B will be equal due to symmetry of load, so this total load (3.5 W) is divided equal on both the supported.

$$R_A = R_B = \frac{3.5W}{2} = 1.75W$$

for uniformly varying loading, the maximum shear force (S.F.) will be at supports.

So, $\boxed{\text{Max. S.F.} = 1.75W}$

806. A cantilever of length of 2 m fails, when a load of 2300 N is applied at free end. What is the stress at failure, if the section of cantilever is 50 mm × 70 mm.

(a) 112.65 MPa (b) 121.38 MPa
(c) 132.93 MPa (d) 141.64 MPa

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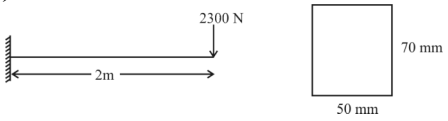
Ans. (a) :



From bending equation,

Bending stress at failure = $\sigma_f = \frac{M_{max}}{Z}$

$M_{max} = 2300 \times 2 \times 10^3$

$Z = \frac{bd^2}{6} = \frac{50 \times 70^2}{6}$

$\sigma_f = \frac{2300 \times 2 \times 10^3 \times 6}{50 \times 70^2}$

$= 112.65\,N/mm^2$

$= 112.65$ MPa

807. A beam of symmetrical section is subjected to bending moment which produces a maximum stress of 110 N/mm^2. If the beam is 260 mm deep find radius to which the longitudinal axis of beam will be bent. Take E= 2 × 10^5 MPa

(a) 236.36 m (b) 21.46 m
(c) 192.71 m (d) 28.9 m

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Ans. (a) : Given,

$\sigma_{max} = 110 N/mm^2$

d = 260 mm

$E = 2 \times 10^5$ MPa

From bending equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$R = \frac{E.y}{\sigma} = \frac{2 \times 10^5 \times 130}{110} = 236363.63 \text{ mm}$$

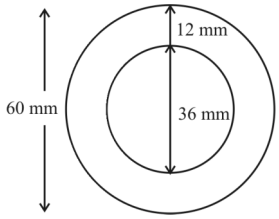
$\boxed{R = 236.36m}$

808. A hollow cast iron pipe is of external diameter 60 mm and 12 mm thick. The section modulus is

(a) 18448.13 mm^3 (b) 17017 mm^3
(c) 19362.24 mm^3 (d) 16496.75 mm^3

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



External dia of pipe (D) = 60 mm

Thickness of pipe (t) = 12 mm

Internal dia. of pipe (d) = D – 2t = 60 – 2 × 12 = 36 mm

Section modulus of hollow circular section,

$$Z = \frac{I}{y}$$

$$I = \frac{\pi}{64}\left(D^4 - d^4\right)$$

$$y = \frac{D}{2}$$

$$z = \frac{\pi}{32D}\left[D^4 - d^4\right]$$

$$Z = \frac{\pi}{32 \times 60}\left[60^4 - 36^4\right] = 18448.13 \text{ mm}^3$$

taking $\pi = 3.14$

$Z = 18448.13$ mm^3

809. For a solid circular section, the shear stress distribution is according to a

(a) parabolic law (b) constant
(c) cubic law (d) linear law

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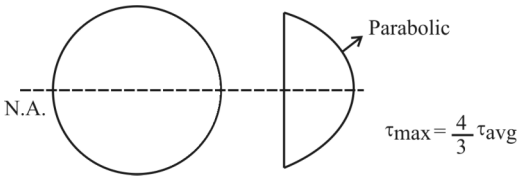
Ans. (a) : Shear stress distribution for solid circular section.



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810. A beam is triangular in section and is of base 'b' and height 'h'. If it is placed with its base horizontal, what is maximum shear stress if the shear force is S.

(a) $\frac{8S}{3bh}$ (b) $\frac{3S}{4bh^3}$

(c) $\frac{4S}{5b^2h}$ (d) $\frac{8S}{5bh}$

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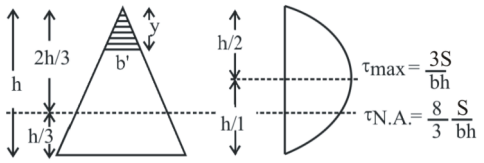
Ans. (a) :



MOI about N.A. = $I = \frac{bh^3}{36}$

Let the shear stress intensity be at a depth 'y' from the top.

Width of beam at depth 'y' from the top = $b' = \frac{b}{h}y$

$$\tau = \frac{Sa\bar{y}}{Ib} = \frac{S\left(\frac{y}{2}\frac{by}{h}\right)\left(\frac{2}{3}h - \frac{2}{3}y\right)}{\left(\frac{bh^3}{36}\right)\left(\frac{by}{h}\right)}$$

$$\tau = \left(\frac{12S}{bh^3}\right)y(h-y)$$

For τ to be max, $\frac{d\tau}{dy} = \left(\frac{12S}{bh^3}\right)(h-2y) = 0 \quad \therefore y = \frac{h}{2}$

$$\tau_{max} = \left(\frac{12S}{bh^3}\right)\frac{2}{3}h \times \frac{h}{3}$$

$$\boxed{\tau_{max} = \frac{8}{3}\frac{S}{bh}}$$

To find the shear stress at N.A, put $y = \frac{2}{3}h$

$$\tau_{N.A.} = \frac{12S}{bh^3} \times \frac{2}{3}h \times \frac{h}{3}$$

$$\boxed{\tau_{N.A.} = \frac{8S}{3bh}}$$

811. A cantilever of length ℓ is carrying a uniformly distributed load of w per unit run of a distance's from fixed end. The slope at the free end is (where EI is the flexural rigidity)

(a) $\frac{wa^3}{6EI}$ (b) $\frac{wa^4}{8EI}$

(c) $\frac{wa^4}{6EI}$ (d) $\frac{wa^3}{8EI}$

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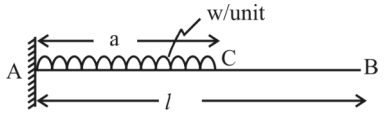
Ans. (a) :



Slope at C = slope at B

$$\boxed{\theta_C = \theta_B = \frac{wa^3}{6EI}}$$

Deflection at C,

$$y_C = \frac{wa^4}{8EI}$$

Deflection at B,

$$y_B = y_C + (\ell - a)\theta_C$$

$$y_B = \frac{wa^4}{8EI} + \frac{wa^3}{6EI} \times (l - a)$$

812. A hinged support can transmit

(a) two reactive forces

(b) one reactive forces

(c) three reactive forces

(d) four reactive forces

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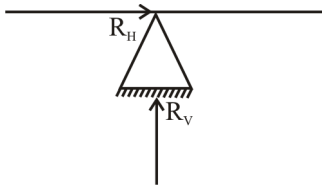
Ans. (a) : A hinged support can transmit two reactive forces.

A roller support can transmit one reactive force.

A fixed support can transmit three, reactive force.



813. The curvature of a beam at a section is given by

(a) $\frac{M}{EI}$ (b) $\frac{\sigma}{y}$

(c) $\frac{\varepsilon}{y}$ (d) $\frac{MI}{E}$

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Ans. (a) : Bending equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\frac{1}{R} = \text{curvature} = \frac{M}{EI}$$

814. A rectangular bar of width b and height d is being used as a cantilever. The loading is in a plane parallel to the side b. The section modulus is

(a) $\frac{b^2d}{6}$ (b) $\frac{bd^3}{12}$

(c) $\frac{bd^2}{6}$ (d) $\frac{bd^2}{8}$

APPSC AE Subordinate Service Civil/Mech. 2016
APPSC AEE Screening Test 2016

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Ans. (a) : Given,
Width = b
Height = d

$$\text{Section modulus} = \frac{I}{y}$$

$$I = \frac{b^3 d}{12}, \quad y = \frac{b}{2}$$

$$Z = \frac{b^3 d}{12} \times \frac{2}{b}$$

$$Z = \frac{b^2 d}{6}$$

815. Assumption made in the theory of bending is
(a) Radius of curvature is small
(b) Radius of curvature is large
(c) Transverse sections of the beam do not remain plane after bending
(d) Doesn't follow Hooke's law in bending

APPSC AEE Screening Test 2016

Ans. (b) Assumption made in the theory of bending
(i) The materials of the beam is homogeneous and isotropic.
(ii) The value of Young's Modulus of elasticity is same in tension and compression.
(iii) The transverse sections which were plane before bending remain plane after bending also.
(iv) The beam is initially straight and all longitudinal filaments bonds into circular arcs with a common centre of curvature.
(v) The radius of curvature is large compared to the dimensions of the cross-section.
(vi) Each layer of the beam is free to expend or contract, independently of the layer above or below it.

816. A beam strongest in flexural is one which has
(a) maximum bending stress
(b) maximum area of cross section
(c) maximum section modulus
(d) maximum moment of inertia

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : A beam strongest in flexural is one which has maximum section modulus. Section modulus is also known as strength of beam.

$$Z = \frac{I_{N-A}}{y_{max.}}$$

817. A component has a cross section area of 100 mm^2 and section modulus of 5×10^3 mm^3. It is subjected to an eccentric axial tensile load of 10 kN with an eccentricity of 10 mm. What will be the maximum and minimum stresses developed in the member?
(a) 120 MPa (tensile) and 80 MPa (tensile)
(b) 120 MPa (tensile) and 80 MPa (compressive)
(c) 80 MPa (tensile) and 80 MPa (compressive)
(d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (a) : Given-
$A = 100\ mm^2$
Section modulus $(Z) = 5 \times 10^3\ mm^3$
P = 10 kN
e = 10 mm

$$\sigma_{1,2} = \frac{P}{A} \pm \frac{M}{Z}$$

$$\sigma_{max} = \frac{P}{A} + \frac{M}{Z}$$

$$= \frac{10 \times 10^3}{100} + \frac{10 \times 10 \times 10^3}{5 \times 10^3}$$

= 100 + 20
= 120 MPa (Tensile)

$$\sigma_{min} = \frac{P}{A} - \frac{M}{Z}$$

= 100 – 20
= 80 MPa (Tensile)

818. Bending equation is derived by considering:
(a) Maximum bending moment on the beam
(b) The beam is simply supported
(c) The beam is a cantilever
(d) Constant pure bending moment on the beam

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Assumptions of Bending Equation–
(1) The beam should be isotropic and homogeneous.
(2) The constant pure bending moment on the beam.
(3) The section remains plane before and after bending.

819. Section modulus of a square section of side 'b' is equal to
(a) $b^3/6$ (b) $b^2/6$
(c) $b/6$ (d) $b^3/3$

TSPSC AEE 2015

Ans : (a) Square section moment of inertia $(I) = \frac{b^4}{12}$
Section modulus = I/y

$$Z_{sq} = \frac{b^4 \times 2}{12 \times b}$$

$$Z_{sq} = \frac{b^3}{6}$$

820. The maximum bending moment of a square beam of section modulus $\frac{200^3}{6}$ mm^3 is 20×10^6 N-mm. The maximum bending stress induced in the beam is
(a) 30 N/mm^2 (b) 7.5 N/mm^2
(c) 45 N/mm^2 (d) 15 N/mm^2

TSPSC AEE 2015

Ans : (d) bending moment $(M) = 20 \times 10^6$ N-mm
section modulus $(Z) = \frac{200^3}{6}\ mm^3$

$$\frac{M}{I} = \frac{\sigma_{max}}{y} = \frac{E}{R} (bending\ equation)$$

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$$\frac{M}{\sigma_{max}} = \frac{I}{y}$$

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{20 \times 10^6}{200^3 / 6}$$

$$\sigma_{max} = 15 N / mm^2$$

821. Area moment of inertia of the cross-section of the beam resist

(a) the applied normal force
(b) both the torque and bending moment
(c) the applied torque
(d) the applied bending moment of the beam

TANGEDCO AE 2015

Ans. (d) : $\frac{M}{I} = \frac{\sigma}{y}$

The beam having cross sectional area and area moment of inertia resist the applied bending moment of the beam.

822. The radius of gyration 'k' of plane figure is determined by the relation

(a) $k = (A/I)^{1/2}$ (b) $k = (I/m)^{1/2}$
(c) $C.k = (m/I)^{1/2}$ (d) $k = (I/A)^{1/2}$

VIZAG MT 2015

Ans. (d) : $I = Ak^2$

$$k = \left(\frac{I}{A}\right)^{1/2}$$

823. The stress distribution in a curved beam, when subjected to pure bending couple is :

(a) linear
(b) non-linear and hyperbolic
(c) non-linear and parabolic
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : The stress distribution in a curved beam, when subjected to pure bending couple, is non-linear and hyperbolic.

824. A beam of rectangular cross-section of breadth 10 cm and depth 20 cm is subjected to a bending moment of 20 kNm. Stress developed at a distance of 10 cm from the top face of the beam is :

(a) Zero (b) 10 kPa
(c) 20 kPa (d) 30 kPa

TSGENCO AE 14.11.2015

Ans. (a) : $\frac{\sigma}{y} = \frac{M}{I}$ (Bending equation)

$$\sigma = \left(\frac{M}{I}\right) \cdot y$$

At 10 cm from top face,

$y = 0$

$\sigma = 0$

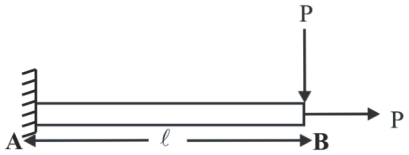
y = measured from neutral axis

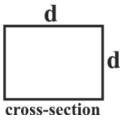
825. Two loads P act at right angles to one another at the free end of a cantilever beam having square cross-section d × d and length *l* on the vertical and horizontal faces. Maximum bending stress in the beam will be equal to-

(a) $6Pl/d^3$ (b) $24Pl/d^3$
(c) $12\ Pl/d^3$ (d) $18Pl/d^3$

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) According to this figure bending stress due to only vertical force not due to horizontal force so



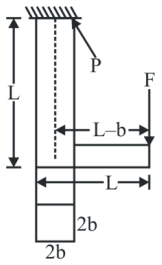


$$\frac{M}{I} = \frac{\sigma}{y_{max}}$$

$$\frac{M}{\left[\frac{d^4}{12}\right]} = \frac{\sigma}{\left[\frac{d}{2}\right]}$$

$$\boxed{\sigma = \frac{6P\ell}{d^3}}$$

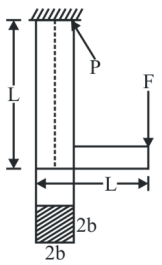
826. For the component loaded with a force F as shown in the figure, the axial stress at the corner point P is



(a) $\frac{F(3L-b)}{4b^3}$ (b) $\frac{F(3L+b)}{4b^3}$
(c) $\frac{F(3L+4b)}{4b^3}$ (d) $\frac{F(3L-2b)}{4b^3}$

ISRO Scientist/Engineer 11.10.2015, GATE 2008

Ans. (d) :



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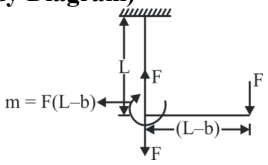
FBD (Free Body Diagram)–



$\sigma = \sigma_a + \sigma_b$

σ_a = Axial stress

$$\sigma_a = \frac{F}{2b \times 2b}$$

σ_b = (Bending Stress)

$$\sigma_b = \frac{M}{Z} = \frac{F \times (L-b)}{Z}$$

Where $$Z = \frac{I}{Y} = \frac{\frac{1}{12}(2b)(2b)^3}{\left(\frac{2b}{2}\right)}$$

$$Z = \frac{8b^3}{6} = \frac{4}{3}b^3$$

So, $\sigma = \sigma_a + \sigma_b$

$$= \frac{F}{4b^2} + \frac{F(L-b)}{\frac{4}{3}b^3}$$

$$= \frac{F}{4b^2} + \frac{3F(L-b)}{4b^3}$$

$$= \frac{Fb + 3FL - 3Fb}{4b^3} = \frac{3FL - 2Fb}{4b^3}$$

$$= \frac{F(3L-2b)}{4b^3}$$

827. A circular log of timber has diameter D. Find the dimension of the strongest rectangular section which can be cut from it.

(a) $D/\sqrt{3}$ wide and $(\sqrt{2/3})$ D deep

(b) $D^2/\sqrt{3}$ wide and $(\sqrt{2/3})$ D deep

(c) $D/\sqrt{2}$ wide and $(\sqrt{2/3})$ D deep

(d) $D/\sqrt{3}$ wide and $(\sqrt{1/3})$ D deep

(e) $\pi D/\sqrt{3}$ wide and $(\sqrt{2/3})$ D deep

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : For strongest beam

$$\text{width of beam} = \frac{D}{\sqrt{3}}$$

$$\text{depth of beam} = \sqrt{\left(\frac{2}{3}\right)}D$$

828. A rectangular beam 300 mm deep, is simply supported over a span of 4 m. Determine the uniformly load per meter, which the beam can carry, if the bending stress does not exceed 120 N/mm^2. Take moment of inertia of the beam = 8×10^6 mm^4.

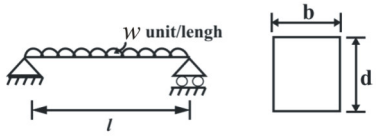
(a) 3.2 N/mm (b) 1.2 N/mm

(c) 4.2 N/mm (d) 4.5 N/mm

(e) 2.2 N/mm

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : Maximum bending stress—Maximum bending stress will occur at mid-span of beam on either top or bottom fiber of beam



$$M_{max} = \frac{wl^2}{8}$$

$$\sigma_{max} = \frac{M_{max}}{I} y$$

$$\sigma_{max} = \frac{wl^2}{8I} \times \frac{d}{2}$$

$$120 = \frac{w \times (4 \times 10^3)^2}{8 \times 8 \times 10^6} \times \left(\frac{300}{2}\right)$$

w = 3.2 N/mm

829. Bending stress in a beam cross section at a distance of 15 cm from neutral axis is 50 MPa. Determine the magnitude of bending of a distance of 10 cm from neutral axis.

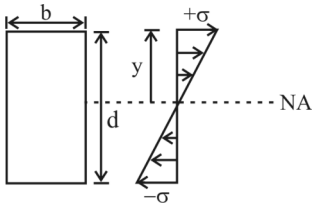
(a) 50 MPa (b) 30.43 MPa

(c) 33.33 MPa (d) 75 MPa

(e) 45.53 MPa

CGPSC AE 26.04.2015 Shift-I

Ans. (c) :



Bending stress $(\sigma) = \frac{M}{I} y$

$\sigma \propto y$

$$\frac{\sigma_2}{\sigma_1} = \frac{y_2}{y_1}$$

$$\frac{\sigma_2}{50} = \frac{10}{15}$$

σ_2 = 33.33 MPa

830. Which of the following statements is correct, in case of a curved beam?

(a) Neutral axis is nearer to centre of curvature compared to centroidal axis.

(b) Neutral axis is farther to centre of curvature compared to centroidal axis.

(c) Neutral axis matches with centroidal axis.

(d) Neutral axis shifts above or below centroidal axis depending on loading conditions.

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : In case of a curved beam neutral axis is nearer to centre of curvature compared to centroidal axis.

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831. A flat spiral mode of strip of breadth 5 mm, thickness 1 mm and length 1.5 m has been subjected to a winding couple which induces a maximum stress of 150 N/mm^2. The magnitude of winding couple is nearest to

(a) 20.8 Nmm (b) 41.6 Nmm
(c) 62.5 Nmm (d) 83.3 Nmm

TNPSC AE 2014

Ans. (c) : Data given

b = 5 mm t = 1 mm
L = 1.5 m
$\sigma_b = 150$ N/mm^2
M = ?
We know that

$$\sigma_b = \frac{12M}{bt^2}$$

$$150 = \frac{12 \times M}{5 \times (1)^2}$$

$$\boxed{M = 62.5 \text{ N} - \text{mm}}$$

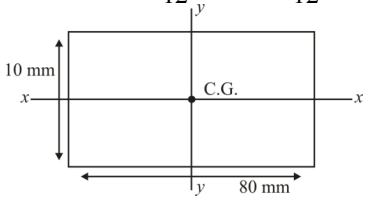
832. Moment of Inertia of the rectangle of base 80 mm and height 10 mm about its centroidal (Ixx) axis

(a) 6666.66 mm^4 = Ixx (b) 5827.21 mm^4 = Ixx
(c) 7777.22 mm^4 = Ixx (d) 6826.11 mm^4 = Ixx

TNPSC AE 2014

Ans. (a) : We know that

$$[I_{x-x}]_{CG} = \frac{80 \times [10]^3}{12} = \frac{80 \times 1000}{12}$$



$$[I_{x-x}]_{CG} = 6666.66 \text{ mm}^4$$

833. In circular plates with edges clamped and with a uniformly distributed load, the maximum radial stress occurs at :

(a) clamp edge (b) the centre
(c) the mean radius (d) none of these

(HPPSC AE 2014)

Ans : (b) In circular plates with edges clamped and with a uniformly distributed load, the maximum radial stress occurs at the centre.

834. Equivalent moment of inertia of the cross-section in terms of timber of a flitched beam made up of steel and timber is ($m = E_s/E_t$) :

(a) $(I_t + m/I_s)$ (b) $(I_t + I_s/m)$
(c) $(I_t + mI_s)$ (d) $(I_t + 2mI_t)$

(HPPSC AE 2014)

Ans : (c) Equivalent moment of inertia of the cross-section in terms of timber of a flitched beam made up-of steel and timber
is $I_t + mI_s$
$m = E_s/E_t$
E_s = Modulus of elasticity of steel.
E_t = Modulus of elasticity of timber.

835. Which one of the following statements is true?

(a) A flexure formula is used for pure bending only
(b) A flexure formula is used for bending with shear only.
(c) A flexure formula is used for bending as well as for bending with shear
(d) None of the above statements are true

HPPSC Asstt. Prof. 2014

Ans. (c) : Flexure formula or bending formula,

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

is used for bending as well as for bending with shear.

836. Which of the following assumptions is not made in curved beam?

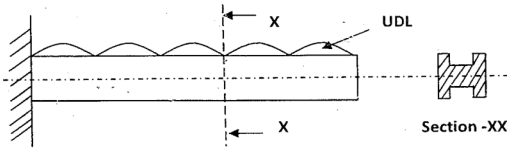
(a) Limit of proportionality is not exceeded
(b) Radial strain is not negligible
(c) Plane to transverse sections remain plane after bending
(d) Material considered isotropic and obeys Hooke's law

HPPSC Asstt. Prof. 2014

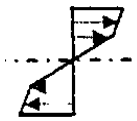
Ans. (b) : Assumptions in curved beam:

(1) Material is homogeneous and isotropic.
(2) Modulus of elasticity in tension and compression are equal.
(3) Plane sections remain plane after bending
(4) Obey Hooke's law ($\sigma_{\alpha\varepsilon}$) means limit of proportionality is not exceeded.
(5) Radial strain is negligible.

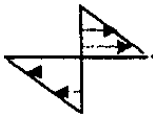
837. A cantilever beam is subjected to a UDL. The cross section of the beam is a H-Section placed as shown in Fig. The bending stress distribution across the cross section will be



(a)

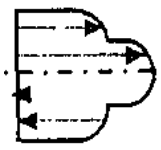


(b)



(c)

(d)



ISRO Scientist/Engineer 12.05.2013

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Ans. (c) : We know that bending stress,

$$\sigma_b = \frac{M}{I_{N-A}} \cdot y$$

Where M & I_{N-A} remain constant throughout the beam.

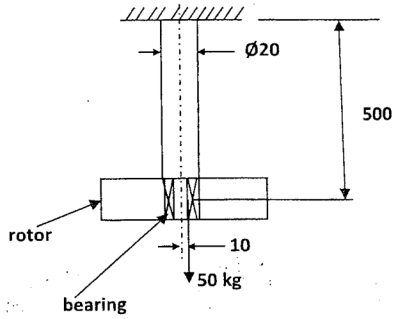
$$\sigma_b \propto y \Rightarrow \sigma_b = c \cdot y \qquad c = \frac{M}{I_{N-A}}$$

At neutral axis y = 0

$$\sigma_b = 0$$

So option (b) and (d) ruled out and varies linearly throughout the section so option (a) also ruled out (because change its shape).

838. A rod of 20 dia is fixed to the ceiling of a roof on one end. A rotor of 50 kg mass is attached to the free end with bearings. The CG of the rotor is 10 mm away from the shaft axis. The rotor is rotating at 600 rpm. The max tensile stress (in N/Sq. mm) in the rod is nearly equal to



(a) $\pi/2$ (b) 200π

(c) 300π (d) 400π

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Ans. (d) :

Given d = 20mm, e = 10mm, m = 50 kg

N = 600 rpm

$$\omega = \frac{2\pi N}{60} = \frac{2 \times 3.14 \times 600}{60}$$

$= 20\pi$

$F = m\omega^2 e$

$= 50 \times (20\pi)^2 \times 10 \times 10^{-3}$

$= 200\pi^2$ N

Let Section modulus (Z) $= \frac{\pi d^3}{32}$

$= 250\pi$ mm^3

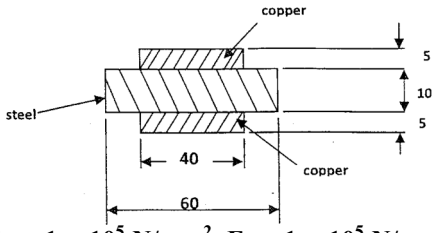
Maximum bending moment in rod = M_{max}.

$= F.L = 200\pi^2 \times 500 = 100000\ \pi^2$ N-mm.

$$\therefore \quad \sigma_{max} = \frac{M_{max}}{Z} = \frac{100000}{250\pi}\pi^2$$

$= 400\ \pi$N/mm^2

839. The cross section of a compound bar 1m long is as shown in figure. If the temperature is raised by 80°C determine the stresses (in N/mm^2) in each metal



$E_{ss} = 1 \times 10^5$ N/mm^2, $E_c = 1 \times 10^5$ N/mm2, A_{ss} = 600 mm^2, A_c = 200 mm^2

(a) $\sigma_c = 20$ (Compressive), $\sigma_{ss} = 30$ (Tensile)

(b) $\sigma_c = 30$ (Compressive), $\sigma_{ss} = 20$ (Tensile)

(c) $\sigma_c = 30$ (Tensile), $\sigma_{ss} = 20$ (Compressive)

(d) $\sigma_c = 30$ (Tensile), $\sigma_{ss} = 20$ (Tensile)

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Ans. (b) : 1. Forces on both metals are equal i.e.

$$\sigma_1 A_1 = \sigma_2 A_2$$

2. Final expansion of metal 1 is equal to final expansion of metal 2 i.e.

$$\alpha_2 t\ell_2 - \frac{\sigma_2}{E_2}\ell_2 = \alpha_1 t\ell_1 - \frac{\sigma_1}{E_1}\ell_1$$

We know that,

$\sigma_c \times 2A_c = \sigma_{ss} \times A_{ss}$

$\sigma_c \times 400 = \sigma_{ss} \times 600$

$\sigma_c : \sigma_{ss} = 3 : 2$

or $\sigma_c : \sigma_{ss} = 30 : 20$

In such question there are two equation, and nature of forces on both material will be opposite hence option (d) is ruled out so option (b) is right option.

840. Beams with four unknown reaction is

(a) In-Determinate Beams

(b) Determinate Beams

(c) Propped Beams

(d) In- Propped Beams

TNPSC AE 2013

Ans. (a) : A structure is statically indeterminate when the static equilibrium equation [Force and moment equilibrium equation condition] are insufficient for determining the internal forces and reaction on that structure.

For in-determinate beam No. of equilibrium equation < No. of Reactions.

For determinate beam No. of equilibrium equation = No. of reactions.

841. The section modulus of hollow circular section is

(a) $\frac{\pi}{16D}(D^4 - d^4)$ (b) $\frac{\pi}{32D}(D^4 - d^4)$

(c) $\frac{\pi}{32D}(D^3 - d^3)$ (d) $\frac{\pi}{16D}(D^3 - d^3)$

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Ans. (b) : $Z = \frac{I}{Y_{max}} = \frac{\pi}{64}\frac{\left[D^4 - d^4\right]}{\left[\frac{D}{2}\right]}$

$$Z = \frac{\pi\left(D^4 - d^4\right)}{32D}$$

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842. A 50 mm diameter shaft is made from carbon steel having ultimate tensile strength of 630 MPa. It is subjected to a torque which fluctuates between 2000 N- m to –800 N-m. The mean shear stress will be

(a) 24.4 N/mm^2 (b) 48.8 N/mm^2
(c) 12.2 N/mm^2 (d) 6.1 N/mm^2

APPSC Poly. Lect. 2013

Ans. (a) : d = 50mm, τ_u = 630 MPa, T_{max} = 2000 N-m, T_{min} = –800 N-m

The mean or average Torque

$$T_m = (T_{max} + T_{min})/2$$

$$= \frac{2000 + (-800)}{2}$$

$$= \frac{(2000 - 800)}{2}$$

$$= 600 \text{ N-m} = 600 \times 10^3 \text{ N-mm}$$

→ Mean or average shear stress

$$\tau_m = \frac{16.T_m}{\pi d^3}$$

$$\tau_m = \frac{\left(16 \times 600 \times 10^3\right)}{\pi \times (50)^3} = 24.4 \text{ N/mm}^2$$

843. The bending moment at a section tends to bend or deflect the beam and the internal stresses resist bending. The resistance offered by the internal stresses to the bending is called

(a) compressive stress (b) shear stress
(c) bending stress (d) elastic stress

JPSC AE 2013, Paper-V

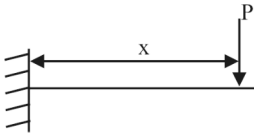
Ans : (c) : The bending moment at a section tends to bend or deflect the beam and the internal stresses resist bending. The resistance offered by the internal stresses to the bending is called bending stress.

844. For a beam of uniform strength having constant width, the depth of the beam at a distance of x from the support varies with

(a) $\sqrt{x}$ (b) x
(c) $x^{1/4}$ (d) $x^{3/4}$

APPSC Poly. Lect. 2013

Ans. (a) : Assume the beam as cantilever beam



Point load is acting at a distance x

Bending moment(BM)= Px

$$\sigma_b = \frac{My}{I}$$

$$\sigma_b = \frac{P.x.\left(\frac{d}{2}\right)}{\frac{b \times d^3}{12}}$$

$$\sigma_b = \frac{6Px}{bd^2}$$

$$d^2 \propto x$$

$$\boxed{d \propto \sqrt{x}}$$

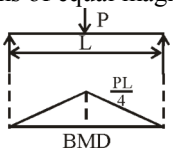
As all other things remain constant, depth of the beam varies as square root of distance x.

845. Two simply supported beams of equal lengths, cross sectional areas, and section moduli, are subjected to the same concentrated load at its mid-length. One beam is made of steel and other is made of Aluminium. The maximum bending stress induced will be in

(a) Steel beam
(b) Aluminium beam
(c) Both beams of equal magnitude
(d) The beams according to their Elastic Moduli magnitude.

UKPSC AE 2012 Paper-I

Ans. (c) : Both beams of equal magnitude



$$M_{max} = \frac{PL}{4}$$

$$\sigma_b = \frac{MY}{I}$$

• Since σ_b is depend only on geometrical shape of the beam so both beam has equal magnitude.

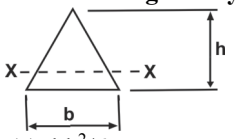
846. The flexural rigidity is the product of

(a) modulus of elasticity and mass moment of inertia
(b) modulus of rigidity and area moment of inertial
(c) modulus of rigidity and mass moment of inertia
(d) modulus of elasticity and area moment of inertia

BPSC AE 2012 Paper - VI

Ans : (d) : The flexural rigidity is the product of modulus of elasticity and area moment of inertia.

847. A beam is having a triangular cross section with its neutral axis XX. The section modulus about XX is given by



(a) $bh^2/6$ (b) $bh^2/36$
(c) $bh^2/24$ (d) $hb^2/36$

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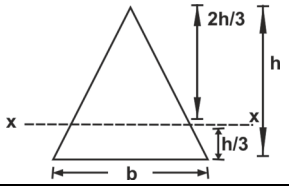
Ans. (c) :



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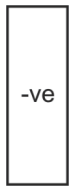
$$I_{xx} = \frac{bh^3}{36}$$

$$Z_{xx} = \frac{I_{xx}}{y_{max}} = \frac{I_{xx}}{\frac{2h}{3}}$$

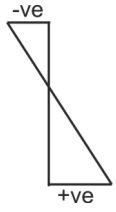
$$Z_{xx} = \frac{\frac{bh^3}{36}}{\frac{2h}{3}}$$

$$= \frac{bh^3}{36} \times \frac{3}{2h} = \frac{bh^2}{24}$$

848. A rectangular column is subjected to an eccentric load P at distance 'e' from centroidal Axis. The stress diagram at cross-section will be

(a) 

(b)

(c) 

(d)

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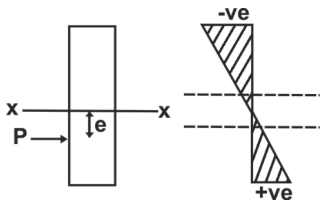
Ans. (d) :



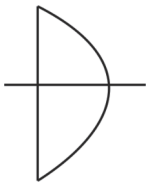
From the figure stress at any section will be

$$= \frac{-P}{A} \pm \frac{Pe}{I} \cdot y$$

Hence compressive load is taken as negative

849. Shear stress distribution for a rectangular beam subjected to transverse loading is

(a)

(b) 

(c)

(d)

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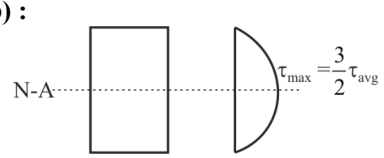
Ans. (b) :



Shear stress is acting in the parallel to the plane and Parabolically varies

At Neutral Axis, y = 0

$$\tau_{max} = \frac{3}{2}\tau_{avg} \qquad \left[\because \tau_{avg} = \frac{P}{A}\right]$$

850. A steel plate 50mm wide and 100mm thick is to be bent into a circular arc of radius 10m. If $E = 2 \times 10^5 N/mm^2$, then the maximum bending stress induced will be

(a) 200 N/mm^2 (b) 100 N/mm^2

(c) 10,000 N/mm^2 (d) 1000 N/mm^2

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Ans : (d) Given, b=50mm, d= 100mm, R = 10m

E= 2×10^5 N/mm^2

Bending equation :-

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

y = 50mm

$$\frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \times y$$

$$\sigma = \frac{2\times10^5}{10\times10^3} \times 50$$

$$\boxed{\sigma = 1000 N/mm^2}$$

851. Radius of curvature of the beam is equal to

(a) $\frac{ME}{I}$ (b) $\frac{M}{EI}$

(c) $\frac{EI}{M}$ (d) $\frac{MI}{E}$

APPSC AEE 2012

Ans : (c) Bending equation

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R.}$$

$$\frac{M}{I} = \frac{E}{R.}$$

Radius of curvature of the $(R) = \frac{EI}{M}$

852. A beam of Z-section is called a

(a) doubly symmetric section beam

(b) singly symmetric section beam

(c) a-symmetric section beam

(d) none of the above

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Ans. (c) : a-symmetric section beam

853. A uniform metal bar of weight 'W', length '*l*', cross-sectional area 'A' is hung vertically with its top end rigidly fixed. Which section of the bar will experience maximum shear stress ?

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(a) Top-section (b) Mid-section
(c) Bottom-section (d) $l/3$ from top

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Ans. (a) : Top-section

854. In theory of simple bending of beams, which one of the following assumptions is incorrect ?
(a) Elastic modulus in tension and compression are same for the beam materials.
(b) Plane sections remain plane before and after bending.
(c) Beam is initially straight.
(d) Beam material should not be brittle.

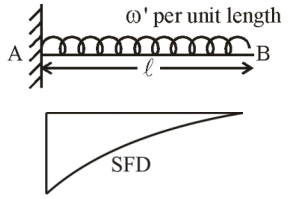
UKPSC AE 2012 Paper-I

Ans. (d) : Beam material should not be brittle.

855. The shear force of a cantilever beam of length 'ℓ' carrying a uniformly distributed load of 'ω' per unit length is …… at the free end
(a) zero (b) $\omega\ell/3$
(c) $\omega\ell/2$ (d) $\omega\ell$

TNPSC ACF 2012

Ans. (a) :



At end B the shear force is zero
$R_A = \omega\ell\, R_B = 0$

856. The strength of the beam mainly depends on
(a) bending moment (b) C.G. of the section
(c) Section modulus (d) its weight

APPSC AEE 2012

Ans. (c) : Strength of the beam mainly depends upon section modulus.
$\Rightarrow$ Strength $\propto$ Z
Section modulus Z = I/y Where Z = Section modulus
I = Area M.O.I.
y = beam width

857. In a beam of I section, the maximum shear force is carried by
(a) the upper flange (b) the web
(c) the lower flange (d) Any of these

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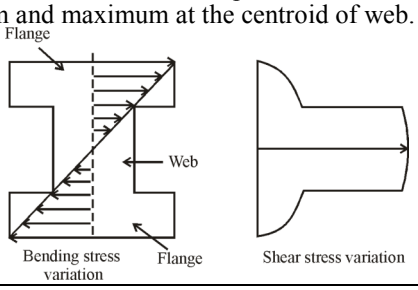
Ans. (b) : Shear force at top and bottom surface is minimum and maximum at the centroid of web.



858. The maximum shear stress induced in a shaft subjected to a shear S_a and bending stress S_b will be:
(a) $\sqrt{S_a^2 + S_b^2}$ (b) $S_a + \frac{1}{2}\sqrt{S_a^2 + S_b^2}$
(c) $\sqrt{4S_a^2 + S_b^2}$ (d) None of these

RPSC AE GWD, 2011

Ans. (d) : Maximum shear stress $\tau = \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau_s^2}$

$= \sqrt{\left(\frac{s_b - 0}{2}\right)^2 + s_a^2} = \sqrt{\left(\frac{s_b}{2}\right)^2 + s_a^2}$

$\boxed{\tau = \frac{1}{2}\sqrt{s_b^2 + 4s_a^2}}$

859. Which one of the following assumptions is *not* valid for the derivation of flexure formula?

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

(a) The beam material is linearly elastic
(b) Sections which are plane before bending remain plane after the bending
(c) The beam has large initial curvature
(d) Elastic modulus of the beam material are same in tension and compression.

UPSC JWM Advt. No.-52/2010

Ans. (c) : Explanation : The beam has large initial curvature is not a valid assumption for the derivation of flexure formula.

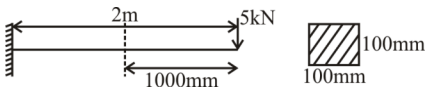
$$\boxed{\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}}$$

Assumptions made in the theory of pure Bending-
(1) The material of the beam is homogenous and isotropic.
(2) The value of Young's modulus of elasticity is same in tension and compression.
(3) The Transverse sections which were plane before bending, remain plane after bending also.
(4) The beam is initially straight and all longitudinal filaments bend into circular arcs with a common centre of curvature.
(5) The radius of curvature is large as compared to the dimensions of the cross-section.
(6) Each layer of the beam is free to expand or contract independently of the layer, above or below it.

860. A cantilever beam of square cross-section (100mm × 100mm) and length 2 m carries a concentrated load of 5 kN at its free end. What is the maximum normal bending stress at its mid-length cross-section ?
(a) 10 N/mm^2 (b) 20 N/mm^2
(c) 30 N/mm^2 (d) 40 N/mm^2

UPSC JWM Advt. No.-50/2010

Ans. (c)



$$\frac{M}{I} = \frac{\sigma_b}{y}$$

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$$\frac{5\times10^3\times1000\times12\times50}{100\times100^3}=\sigma_b$$

$\sigma_b = 30\ N/mm^2$

861. A horizontal beam with square cross-section is simply supported with sides of the square horizontal and vertical and carries a distributed loading that produces maximum bending stress σ in the beam. When the beam is placed with one of the diagonals horizontal the maximum bending stress will be :

(a) $(1/\sqrt{2})\sigma$ (b) σ

(c) $\sqrt{2}\sigma$ (d) 2σ

APPSC IOF, 2009

Ans. (c) : $\frac{M}{Z}$ = Bending stress

For rectangular beam with sides horizontal & vertical,

$$Z_1=\frac{a^3}{6}$$

For same section with diagonal, $Z_2=\frac{a^3}{6\sqrt{2}}$

Ratio of stresses $=\sqrt{2}$

862. Which one of the following conditions will cause bending of the beam without being twisted?

(a) The applied load is vertical

(b) The beam cross-section is doubly symmetric

(c) The plane of applied load contains the axis of symmetry of the beam section

(d) The beam cross-section is singly symmetric

UPSC JWM Adv. No-16/2009

Ans. (a) : The conditions which causes bending of the beams without being twisted when the applied load is vertical.

863. Match List-I with List-II and select the correct answer using the code given below the lists :

	List-I (Stress Induced)	**List-II (Component)**
(A)	Membrane stress	1. Neutral axis of beam
(B)	Torsional and direct	2. Close coiled helical shear stress spring under axial load
(C)	Double shear stress	3. Thin cylindrical shell subjected to internal fluid pressure
(D)	Maximum shear	4. Rivets in butt joint stress having two cover plates

Code :

	A	**B**	**C**	**D**
(a)	3	4	2	1
(b)	1	2	4	3
(c)	3	2	4	1
(d)	1	4	2	3

UPSC JWM Adv. No-16/2009

Ans. (c)

(A) Membrane Stress → Thin cylindrical shell subjected to internal fluid pressure.

(B) Torsional and direct shear stress → Closed coiled helical spring under axial load.

(C) Double shear stress → Rivets in butt joining having two cover plates.

(D) Maximum shear stress → Neutral axis of beam.

864. At a particular cross-section of a straight bar of rectangular cross-section, the bending moment is 10000 N-mm and the axial force acting through the centroid of the cross-section is 1000 N. If the area of cross-section is 100 mm^2 and the area moment of inertia about the centroidal axis is 1000 mm^4, the distance between the centroidal axis and the neutral axis is

(a) 0.1 mm (b) 1 mm

(c) 2 mm (d) 4 mm

DRDO Scientists 2009

Ans. (b) : $\sigma=\frac{P}{A}\pm\frac{My}{I}$

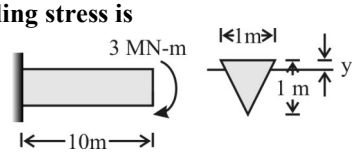
At the neutral axis, $\sigma = 0$,

$$\frac{My}{I}=\frac{P}{A}$$

or $$y=\frac{PI}{MA}=\frac{1000\times1000}{10000\times100}$$

$= 1$ mm

865. A cantilever beam has the cross-section of an isosceles triangle and is loaded as shown in figure. If the moment of inertia of the cross-section $I_{zz}=\frac{1}{36}m^4$, then the maximum tensile bending stress is

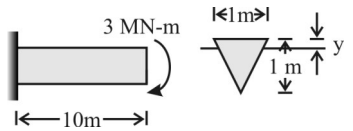


(a) 1/16 MPa (b) 72 MPa

(c) 36 MPa (d) 1/36 MPa

DRDO Scientists 2008

Ans. (c) :



$$\sigma_b=\frac{My}{I}$$

$$y=\frac{1}{3}m$$

$$I=\frac{bd^3}{36}=\frac{1\times1^3}{36}=\frac{1}{36}m^4$$

$M = 3 \times 10^6$ Nm

$$\sigma_b=\frac{3\times10^6\times1}{3\times\frac{1}{36}}=36MPa$$

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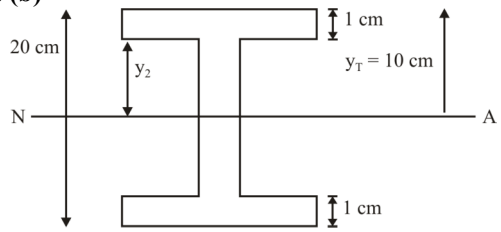
866. A beam of I section of depth 20 cm is subjected to a bending moment M. The flange thickness is 1 cm. If the minimum stress developed in the 1 section be 100 N/mm^2, the stress developed at the inner edge of the flange is

(a) 95 N/mm^2 (b) 90 N/mm^2
(c) 47.5 N/mm^2 (d) 60 N/mm^2

WBPSC AE, 2007

Ans. (b)



We know that under sagging top fibre is in compression and bottom fibre is in tension

Hence

$(\sigma_b)_{min} = -100$ N/mm^2 at top fibre

$(\sigma_b)_{max} = +100$ N/mm^2 at bottom fibre

Bending stress is given by

$$\sigma_b = \frac{m \times y}{I}$$

where y = distance from neutral axis

$\Rightarrow \sigma_b \alpha y$

Let $(\sigma_1)_2$ = Bending stress of inner fibre of top flange.

$$\Rightarrow \frac{(\sigma_b)_{Top}}{(\sigma_b)_2} = \frac{Y_T}{y_2}$$

$$\frac{100}{(\sigma_b)_2} = \frac{10 \times 10}{9 \times 10}$$

$$(\sigma_b)_2 = \frac{900}{10} = 90$$

$$\boxed{(\sigma_b)_2 = 90\,N/mm^2}$$

867. A beam of rectangular section of breadth b and depth d is tested under bending and M_{ult} is the ultimate bending moment recorded. The modulus of rupture is given by

(a) $\frac{6M_{ult}}{bd^3}$ (b) $\frac{6M_{ult}}{bd^2}$
(c) $\frac{12M_{ult}}{bd^2}$ (d) $\frac{12M_{ult}}{bd^3}$

WBPSC AE, 2007

Ans. (b) : Modulus of Rupture is given by

$$M_R = \sigma_{per} \times z$$

$$M_{ult} = \sigma_{per} \times \frac{bd^2}{6}$$

$$\boxed{\sigma_{per} = \frac{6 \times M_{ult}}{bd^2}}$$

868. Flexural rigidity (EI) is expressed in units of

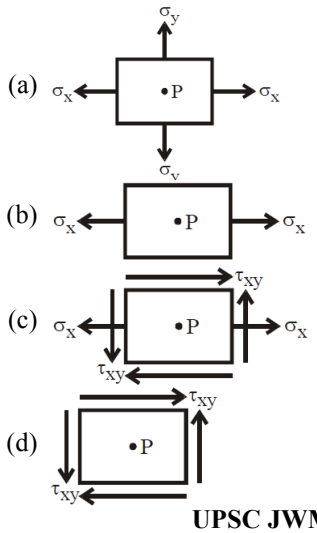
(a) kNm^2 (b) MNm
(c) kNm^{-2} (d) MNm^4

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Flexural rigidity = EI

Unit of flexural rigidity = $\frac{kN}{m^2} \times m^4 = kNm^2$

869. A cantilever beam of rectangular cross-section is subjected to a concentrated load at its free end. The state of stress at a point P on the centroidal longitudinal axis of beam is given by



UPSC JWM Advt. No.-50/2010

Ans. (d) : From question the longitudinal axis is where,

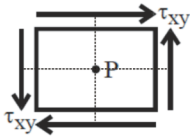
$y = 0$

So, $\frac{M}{I} = \frac{\sigma_b}{y}$

For $y = 0$

$\sigma_b = 0$

So only shear stresses are there



5. Torsion of Shafts

870. Compare the strength of solid and hollow shafts both having outside diameter D and hollow shaft having inside diameter of D/2 in torsion. The ratio of strength of solid to hollow shafts in torsion will be

(a) 0.5 (b) 0.75
(c) 15/16 (d) 1/16

VIZAG Steel MT 24.01.2021
RPSC IOF, 2020
Vadodara Municipal Corp. DEE, 2018
ISRO Scientist/Engineer (RAC) 22.04.2018, 07.05.2017
GPSC EE Pre, 28.01.2017
UPRVUNL AE 21.08.2016
Kerala PSC IOF 19.04.2016
ISRO Scientist/Engineer (RAC) 29.11.2015
MPPSC State Forest Service Exam, 2014
UKPSC AE 2012 Paper-I
TRB Asstt. Prof., 2012, ESE 2001, GATE 1993

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Ans. (c) : For solid shaft polar moment of inertia

$$J = \frac{\pi D^4}{32}$$

$$T_s = \tau \frac{\pi D^4}{32} \times \frac{2}{D} = \frac{\pi}{16} \times \tau D^3$$

For hollow shaft

$$J = \frac{\pi}{32}\left(D^4 - d^4\right) \quad \therefore\ d = \frac{D}{2}$$

$$T_H = \frac{\pi}{16}.\tau\left(\frac{D^4 - d^4}{D}\right)$$

According to question d = D/2

$$T_H = \frac{\pi}{16} \cdot \tau\left[\frac{D^4 - \frac{D^4}{16}}{D}\right]$$

$$T_H = \frac{\pi}{16} \cdot \tau\left[\frac{15}{16} D^3\right]$$

$$\frac{T_S}{T_H} = \frac{\frac{\pi}{16}\tau D^3}{\frac{\pi}{16} \cdot \tau \cdot \frac{15}{16} D^3}$$

$$\frac{T_S}{T_H} = \frac{16}{15}$$

$$\frac{T_H}{T_S} = \frac{15}{16}$$

871. Two shafts A and B are made of the same material. The diameter of shaft B is twice that of shaft A. The ratio of power which can be transmitted by shaft A to that of shaft B is

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$

(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

ISRO Scientist/Engineer (RAC), 10.03.2019
Nagaland PSC (CTSE) 2018, Paper-I
APGENCO AE, 2017
Karnataka PSC AE, 10.09.2017
Kerala PSC IOF 19.04.2016
CGPSC AE 16.10.2016
OPSC AEE 2015 Paper-I
ISRO Scientist/Engg. 11.10.2015
GPSC Lect. (Poly.) 07.12.2014
TRB Poly. Lect., 2012
ISRO Scientist/Engg. 2012
ESE 2001, GATE 1994

Ans. (c) : As we know that power P=T × ω

$$P = T \times 2\frac{\pi N}{60}$$

and $\tau = \frac{16T}{\pi d^3} \Rightarrow T = \frac{\tau \pi d^3}{16}$ (τ = shear stress)

d_A=d, d_B=2d

$$P = \frac{\tau \pi d^3}{16} \times \frac{2\pi N}{60}$$

$P \propto d^3$

$$\frac{P_A}{P_B} = \frac{d_A^3}{d_B^3} = \frac{(d)^3}{(2d)^3} = \frac{d^3}{8d^3} = \frac{1}{8}$$

$$\boxed{P_A : P_B = 1:8}$$

872. A solid circular shaft is subjected to a bending moment M and twisting moment T. What is the equivalent twisting moment T_e which will produce the same maximum shear stress as the above combination?

(a) $M^2 + T^2$ (b) M+1

(c) $\sqrt{M^2 + T^2}$ (d) M–1

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Karnataka PSC AE (WRD) 31.07.2021
RPSC ADE 2016
GPSC ARTO 01.05.2016
Kerala PSC IOF 19.04.2016
RPSC AE GWD, 2011
ISRO Scientist/Engineer 2011, ESE 2007

Ans. (c) : $\sigma = \frac{32M}{\pi d^3}$, $\tau = \frac{16T}{\pi d^3}$

Max shear stress = $\sqrt{\left(\frac{\sigma - 0}{2}\right)^2 + \tau^2}$

= radius of mohr circle

$$= \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} = \sqrt{\left(\frac{16M}{\pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2}$$

$$\tau_{max} = \frac{16}{\pi d^3}\sqrt{M^2 + T^2}$$

$$\frac{16T_{eq}}{\pi d^3} = \frac{16}{\pi d^3} \times \sqrt{M^2 + T^2}$$

T_{eq} (Equivalent torque) = $\sqrt{M^2 + T^2}$

873. Maximum shear stress developed on the surface of a circular shaft under pure torsion is 240 MPa. If the shaft diameter is doubled, then the maximum shear stress developed corresponding to the same torque will be -

(a) 120 MPa (b) 60 MPa

(c) 30 MPa (d) 15 MPa

PPSC Asstt. Municipal Engineer 15.06.2021
TSPSC AEE 2017, 2014
HPPSC Asstt. Prof. 18.11.2016
NPCIL ET, 2015
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2003

Ans. (c) : τ_1 = 240 MPa

When shaft diameter is doubled, maximum shear stress = τ_2

Let d be the original diameter

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$$\frac{\tau_1}{\tau_2}=\frac{\left(\frac{T_1}{Z_{P1}}\right)}{\left(\frac{T_2}{Z_{P2}}\right)}$$

(Since torque in both the cases is constant)

$$\frac{\tau_1}{\tau_2}=\frac{T_1}{Z_{P1}}\times\left(\frac{Z_{P2}}{T_2}\right)=\frac{(Z_{P2})}{(Z_{P1})}$$

$$\frac{\tau_1}{\tau_2}=\frac{\frac{\pi(2d)^3}{16}}{\frac{\pi d^3}{16}}=\frac{\pi}{16}\times 8d^3\times\frac{16}{\pi d^3}$$

$$\frac{\tau_1}{\tau_2}=8$$

$$\tau_2=240\times\frac{1}{8}=30\,\text{MPa}$$

$\tau_2 = 30$ MPa

874. When a shaft of diameter D is subjected to a twisting moment T and bending moment M, then equivalent bending moment M_e is given by

(a) $\sqrt{M^2+T^2}$

(b) $\sqrt{M^2-T^2}$

(c) $\frac{1}{2}\left(M+\sqrt{M^2+T^2}\right)$

(d) $\frac{1}{2}\left(M-\sqrt{M^2+T^2}\right)$

Nagaland PSC (CTSE) 2018, Paper-I
PTCUL AE 25.06.2017
MPPSC AE 2016
Kerala PSC IOF 19.04.2016
JPSC AE 2013, Paper-V
APPSC AE 04.12.2012
APPSC AEE 2012
WBPSC AE 2003, ESE 2008, 1996

Ans : (c) : Equivalent bending moment, (M_e)

$$\boxed{M_e=\frac{1}{2}\left(M+\sqrt{M^2+T^2}\right)}$$

Equivalent Twisting moment, (T_e),

$$\boxed{T_e=\sqrt{T^2+M^2}}$$

875. What is torsional rigidity?

(a) The torque required to produce a twist of one degree per unit length of the shaft

(b) The torque required to produce a twist of one radian per unit length of the shaft

(c) The torque required to produce a twist of one radian per unit area of the shaft

(d) The torque required to produce a twist of one degree per unit area of the shaft

VIZAG Steel MT 24.01.2021, Shift-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
APPSC AEE Mains 2016 (Civil Mechanical)
Rajasthan AE (Nagar Nigam) 2016 Shift-3
UPRVUNL AE 2014, WBPSC AE, 2007

Ans : (b) Torsional Rigidity– The torque required to produce a twist of one radian per unit length of the shaft

$$T=\frac{G.\theta.J}{\ell}$$

$$\text{Torsional Rigidity (GJ)} = \frac{T.\ell}{\theta}$$

876. A solid shaft can resist a bending moment of 3.0 kNm and a twisting moment of 4.0 kNm together. The maximum torque that can be applied is

(a) 7.0 kNm (b) 3.5 kNm
(c) 4.5 kNm (d) 5.0 kNm

CIL MT 27.02.2020
Vadodara Muncipal Corp. DEE, 2018
HPPSC Asstt. Prof. 18.11.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TRB Poly. Lect. 2012

Ans. (d) : M= 3.0 kNm, T= 4.0 kNm, T_e=?

$$\text{Equivalent torque} = \sqrt{M^2+T^2} = \sqrt{(3)^2+(4)^2}$$

$$T_e=\sqrt{25}=5\,\text{kNm}$$

877. A shaft can safely transmit 90 kW while rotating at a given speed. If the diameter of the shaft is doubled and speed is halved the previous shaft, the power that can be transmitted by new shaft is

(a) 360 kW (b) 90 kW
(c) 720 kW (d) 180 kW

Karnataka PSC AE (WRD) 31.07.2021
BPSC Poly. Lect. 2016
CGPSC Poly. Lect. 22.05.2016
VIZAG MT 2015
TRB Asstt. Prof., 2012, ESE 2002

Ans : (a) We know that

$$P=\frac{2\pi NT}{60}=\frac{2\pi N}{60}\times\frac{\pi d^3\tau}{16}$$

$P \propto N, d^3$

$P = kNd^3$

If diameter of the shaft is double and halved the speed.

$d_2 = 2d_1$, $N_2 = \frac{N_1}{2}$

$$\therefore \frac{P_1}{P_2}=\left(\frac{N_1}{N_2}\right)\left(\frac{d_1}{d_2}\right)^3=2\left(\frac{1}{2}\right)^3=\frac{1}{4}$$

$P_2 = 4P_1 = 4 \times 90 = 360$ kW

878. When a circular shaft is subject to torque only, the torsional shear stress is :

(a) Maximum at the axis of rotation and zero at outer surface.

(b) Maximum at outer surface and zero at axis of rotation.

(c) Uniform from axis of rotation to outer surface

(d) None of the above

JPSC AE 10.04.2021, Paper-II
OPSC AEE 2019 Paper-I
RPSC Vice Principal ITI 2018, ESE 2012

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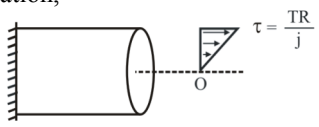
Ans. (b) : According to torsion equation,

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{R}$$

$$\tau=\frac{TR}{J}$$

$$\tau=\frac{Tr}{J}$$

At axis of rotation,



$r = 0$

$\boxed{\tau = 0}$

At outer surface

$r = R$

$$\boxed{\tau=\frac{TR}{J}}$$

879. The diameter of shaft is increased from 50 mm to 100 mm all other conditions remaining unchanged. How many times the torque carrying capacity increases?

(a) 2 times (b) 4 times

(c) 8 times (d) 16 times

GPSC ARTO 01.05.2016, ESE 2016
KPSC AE. 2015
Kerala PSC AE 06.08.2015
TANGEDCO AE 2015

Ans. (c) : Given, $d_1 = 50\text{ mm} = \frac{1}{20}\text{mm}$

$$d_2 = 100\text{ mm} = \frac{1}{10}\text{mm}$$

$\because$ equation for the shaft

$$\frac{T}{J}=\frac{\tau}{R}=\frac{G\theta}{l} \quad ...(i)$$

moment of inertia $I=\frac{\pi}{64}d^4$

$\Rightarrow$ Polar moment of inertia $[J = I_1 + I_2]$

$$J=\frac{\pi}{32}d^4$$

Now from $\frac{T}{J}=\frac{\tau}{R}$

$$T=\frac{\tau\times 2\times\pi d^4}{d\times 32}=\frac{\tau\times\pi d^3}{16}$$

$$\boxed{T\propto d^3}$$

$$T_1=\left(\frac{1}{20}\right)^3$$

$$T_2=\left(\frac{1}{10}\right)^3$$

$$\frac{T_1}{T_2}=\frac{1}{8}$$

$$\boxed{T_2 = 8T_1}$$

Thus after increasing the diameter of shaft and all other parameter remains unchanged, the torque will be 8 times of previous torque.

880. Bending moment 'M' and torque 'T' is applied on a solid circular shaft. If maximum bending stress equals to maximum shear stress developed, then M is equal to

(a) T (b) T/2

(c) 2T (d) 4T

GPSC Engg. Class-II Pre 19.01.2020
APPSC AE Subordinate Service Civil/Mech. 2016
GPSC Lect. (Poly.) 07.12.2014
ISRO Scientist/Engg. 2010

Ans. (b) : $\sigma_{b\,max}=\frac{32M}{\pi d^3}$

$$\tau_{max}=\frac{16T}{\pi d^3}$$

$$\sigma_{b_{max}}=\tau_{max}$$

$$\frac{32M}{\pi d^3}=\frac{16T}{\pi d^3}$$

$M = T/2$

881. When two shafts of same length, one of which is hollow, transmit equal torques and have equal maximum stress, then they should have equal

(a) polar modulus
(b) diameter
(c) polar moment of inertia
(d) angle of twist

ESE 2018
Rajasthan AE (Nagar Nigam) 2016 Shift-3
APPSC AEE 2012
APPSC AE 04.12.2012
CSE Pre-1994

Ans. (a) : Given, $T_1 = T_2$ & $\tau_1 = \tau_2$

Torsional formula $\frac{T}{J}=\frac{\tau}{R}$

$$T=\frac{J}{R}.\tau$$

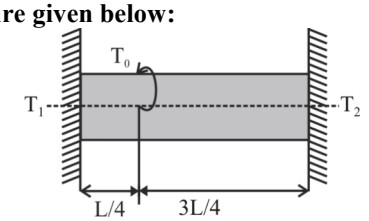
$T = \tau Z_P$ ($\therefore$ Z_P = Polar modulus)

For transmit equal torque ($T = T_1 = T_2$) and equal maximum stress ($\tau_1 = \tau_2$)

$T_1 = \tau_1 Z_{P_1}$, $T_2 = \tau_2 Z_{P_2}$

$$\boxed{Z_{P_1} = Z_{P_2}}$$

882. A solid shaft of diameter d and length L is fixed at both the ends. A Torque T_0 is applied at a distance L/4 from the left end as shown in the figure given below:



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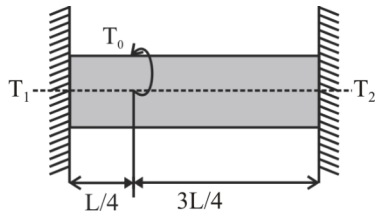
The maximum shear stress in the shaft is

(a) $(16T_0/\pi d^3)$ (b) $(12T_0/\pi d^3)$

(c) $(8T_0/\pi d^3)$ (d) $(4T_0/\pi d^3)$

NPCIL ET, 2015
OPSC Civil Services Pre. 2011
GATE 2009, ESE 2004

Ans. (b) :



$$T_1 = \frac{T_0 \times \frac{3L}{4}}{L} = \frac{3T_0}{4}$$

$$T_2 = \frac{T_0 \times \frac{L}{4}}{L} = \frac{T_0}{4}$$

$$\tau_{max} = \frac{16T}{\pi D^3}$$

$$\tau_{max} = \frac{16 \times \left(\frac{3T_0}{4}\right)}{\pi D^3}$$

$$\boxed{\tau_{max} = \frac{12T_0}{\pi D^3}}$$

883. For a round shaft subjected to pure torsion, the shear stress at the centre (axis) will be

(a) maximum
(b) minimum
(c) zero
(d) The information provided is insufficient

BPSC AE Mains 2017 Paper - VI
TSPSC AEE 28.08.2017 (Civil/Mechanical)
GPSC ARTO 01.05.2016
UKPSC AE-2013, Paper-I

Ans : (c) :

$$\frac{T}{J} = \frac{\tau}{R} = \frac{G\theta}{\ell}$$

$$\tau = \frac{T \times R}{J}$$

$$\tau_0 = \frac{T \times 0}{J} = 0$$

Hence, shear stress at centre will be zero for round shaft subjected to pure torsion.

884. The value of J in equation $T/J = \tau/r = G.\theta/L$ for a circular shaft of diameter D will be

(a) $\pi D^4/32$ (b) $\pi D^4/64$

(c) $\pi D^3/32$ (d) $\pi D^3/16$

NSPSC AE 2018, RPSC AE 2018
Punjab PSC SDE 12.02.2017
WBPSC AE, 2017

Ans. (a) : General torsion equation,

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

T = Torque or twisting moment
J = Polar moment of inertia
τ = Shear stress at outer fibre
r = Radius of shaft
G = Modulus of rigidity
θ = Angle of twist
L = Length of shaft
D = Diameter of shaft
For solid shaft,

$$J = \frac{\pi D^4}{32}$$

885. Two shafts of same length & materials are joined in series. If the ratio of their diameter is 2, then the ratio of their angle of twist will be :

(a) 2 (b) 4

(c) 8 (d) 16

ISRO Scientist/Engg. 22.04.2018
J&K PSC Civil Services Pre, 2010
WBPSC AE 2003, CSE Pre-2003, 1995

Ans. (d) : Ratio of diameter $= \frac{d_1}{d_2} = 2$

Angle of twist $\theta = \frac{TL}{GJ}$

Let two shaft have diameter d_1 and d_2, its length L_1 and L_2.

Given, $L_1 = L_2$, $\frac{d_1}{d_2} = 2$

$$\frac{\theta_1}{\theta_2} = \frac{\frac{T_1L_1}{G_1J_1}}{\frac{T_2L_2}{G_2J_2}}$$

For same material,
$G_1 = G_2$
And for series joined shaft if torque applied $T_1 = T_2$ for both in shaft

So, $\frac{\theta_1}{\theta_2} = \frac{J_2}{J_1}$

$$\frac{\theta_1}{\theta_2} = \frac{\frac{\pi}{32}d_2^4}{\frac{\pi}{32}d_1^4} = \left(\frac{d_2}{d_1}\right)^4 = \left(\frac{1}{2}\right)^4$$

$$\frac{\theta_1}{\theta_2} = \frac{1}{16}$$

$$\frac{\theta_2}{\theta_1} = 16$$

886. If the diameter of the shaft is increased two times, the torque transmitted will be

(a) Two times (b) Four times

(c) Eight times (d) Sixteen times

APGCL AM, 2021, JPSC AE Paper-II 10.04.2021
RPSC IOF, 2020

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Ans. (c)

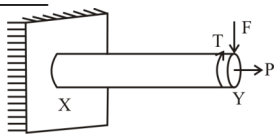
$$\frac{T}{J}=\frac{\tau}{R}$$

$$T \propto Z_p$$

$$\frac{T_1}{T_2}=\frac{\frac{\pi d_1}{16}}{\frac{\pi d_2}{16}}=\left(\frac{d_1}{d_2}\right)^3=\left(\frac{d_1}{2d_1}\right)^3=\frac{1}{8}$$

$$\boxed{T_2=8T_1}$$

887. A machine element XY, fixed at end X, is subjected to an axial load P, transverse load F, and a twisting moment T at its free end Y. The most critical point from the strength point of view is_____



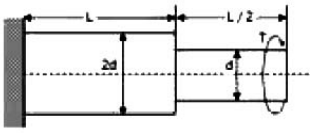
(a) a point on the circumference at location X
(b) a point at the center at location Y
(c) a point on the circumference at location Y
(d) a point at the center at location X

Assam Engg. College AP/Lect. 18.01.2021
PPSC Asstt. Municipal Engineer 15.06.2021
GATE 2016

Ans. (a) : Most critical loading point will be a point on circumference at location X

Torque (τ) and axial load (P) will be same for element XY but vertical force (F) will produce maximum stress at a point on circumference at X.

888. A torque T is applied at the free end of a stepped rod of circular cross-sections as shown in the figure. The shear modulus of the material of the rod is G. The expression for d to produce an angular twist θ at the free end is

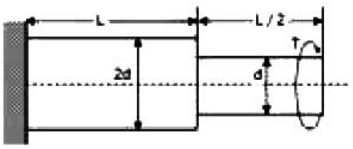


(a) $\sqrt[1/4]{\frac{32TL}{\pi\theta G}}$ (b) $\sqrt[1/4]{\frac{18TL}{\pi\theta G}}$

(c) $\sqrt[1/4]{\frac{16TL}{\pi\theta G}}$ (d) $\sqrt[1/4]{\frac{2TL}{\pi\theta G}}$

Vadodara Muncipal Corp. DEE, 2018
NPCIL ET, 2015, GATE 2011

Ans. (b) :



Given- $T_1 = T_2 = T$

$\theta = \theta_1 + \theta_2$

$$\theta=\frac{T_1L_1}{GJ_1}+\frac{T_2L_2}{GJ_2}$$

$$=\frac{T}{G}\left[\frac{L_1}{J_1}+\frac{L_2}{J_2}\right]$$

$$\theta=\frac{T}{G}\left[\frac{L_1}{\frac{\pi d_1^4}{32}}+\frac{L_2}{\frac{\pi d_2^4}{32}}\right]$$

$$\theta=\left[\frac{L}{\frac{\pi(2d)^4}{32}}+\frac{L/2}{\frac{\pi(d)^4}{32}}\right]\frac{T}{G}$$

$$=\frac{T}{G}\left[\frac{32L}{\pi\times 16d^4}+\frac{32L/2}{\pi\times d^4}\right]$$

$$=\frac{T}{G}\left[\frac{2L}{\pi d^4}+\frac{16L}{\pi d^4}\right]$$

$$\theta=\frac{T}{G}\left[\frac{18L}{\pi d^4}\right]$$

$$\theta=\frac{18TL}{G\pi d^4}\Rightarrow d^4=\frac{18TL}{\pi G\theta}$$

$$d=\sqrt[1/4]{\frac{18TL}{\pi G\theta}}$$

889. A solid circular shaft needs to be designed to transmit a torque of 50 Nm. If the allowable shear stress of the material is 140 MPa, assuming a factor of safety 2, the minimum allowable design diameter in mm is:

(a) 8 (b) 16
(c) 24 (d) 32

Assam Engg. College AP/Lect. 18.01.2021
NPCIL ET, 2015, GATE 2012

Ans. (b) : Given,

T = 50 N-m

$\tau_{allowable}$ = 140 MPa

Factor of safety (FOS) = 2

$$\tau_{safe}=\frac{\tau_{allowable}}{FOS}=\frac{140}{2}=70\,MPa$$

We know that,

$$\tau_{safe}=\frac{T}{z_p}\qquad (z_p = \text{Polar section modulus})$$

$$z_p=\frac{\pi d^3}{16}$$

$$\tau_{safe}=\frac{16T}{\pi d^3}$$

$$d^3=\frac{16T}{\pi\times\tau_{safe}}$$

$$d=\sqrt[3]{\frac{16\times 50\times 10^3}{\pi\times 70}}$$

= 15.8 mm

$d \approx 16$ mm

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890. A twisting moment 'T' and Bending moment 'M' are acting on a circular shaft. What is the ratio of maximum shear stress to maximum bending stress?

(a) 2 M/T (b) 2T/M
(c) T/M (d) T/2M

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Gujarat PSC AE 2019
UPPSC AE 12.04.2016 Paper-I

Ans. (d) : As we know, $\frac{\sigma_b}{y} = \frac{M}{I}$

$$\sigma_b = \frac{M}{I} \times y$$

$$= \frac{M}{\frac{\pi d^4}{64}} \times \frac{d}{2} = \frac{32M}{\pi d^3}$$

$$\tau_{max} = \frac{16T}{\pi d^3},$$

$$\text{So } \frac{\tau_{max}}{\sigma_b} = \frac{T}{2M}$$

891. A circular shaft subjected to torsion undergoes a twist of 1° in a length of 120 cm. If the maximum shear stress induced is limited to 1000 kg/cm^2 and if the modulus of rigidity, G = 0.8 × 10^6 kg/cm^2, then the radius of shaft in cm should be

(a) $\frac{\pi}{18}$ (b) $\frac{\pi}{27}$
(c) $\frac{18}{\pi}$ (d) $\frac{27}{\pi}$

ISRO Scientist/Engineer (RAC) 12.01.2020
TRB Poly. Lect. 2012
ISRO Scientist/Engg. 2007

Ans. (d) : From torsion equation,

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{\ell}$$

$$\frac{\tau}{r} = \frac{G\theta}{\ell} \qquad r = \frac{\tau}{\theta} \times \frac{L}{G}$$

$$\theta = 1° = \frac{\pi}{180}, L = 120cm, \tau = 1000\,kg/cm^2$$

$$G = 0.8 \times 10^6\ kg/cm^2$$

$$r = \frac{1000\,kg/cm^2}{\pi/180} \times \frac{120\,cm}{0.6\times10^6} kg/cm^2$$

$$r = \frac{27}{\pi} cm$$

892. A shaft was initially subjected to bending moment and then was subjected to torsion. If the magnitude of bending moment is found to be the same as that of the torque, then the ratio of maximum bending stress to shear stress would be

(a) 0.25 (b) 0.50
(c) 1.0 (d) 2.0

Punjab PSC (Lect.) 06.08.2017
APPSC AEE 2012, APPSC IOF 2009

Ans. (d) : Bending moment (M) = Torque (T)

Bending stress $(\sigma_b) = \frac{32M}{\pi d^3}$

Shear stress $(\tau) = \frac{16T}{\pi d^3}$ $\frac{\sigma_b}{\tau} = \frac{32M/\pi d^3}{16T/\pi d^3} = \frac{32}{16} = 2$

$(\because M = T)$

$$\boxed{\sigma_b = 2\tau}$$

893. A circular shaft can transmit a torque of 5 kN-m. If the torque is reduced to 4kN-m, then the maximum value of bending moment that can be applied to the shaft is :

(a) 1 kN-m (b) 2 kN-m
(c) 3 kN-m (d) 4 kN-m

RPSC ACF & FRO, 26.02.2021
KPSC AE 2015, APPSC IOF, 2009

Ans. (c) : Given,

T_{eq} = 5 kN-m
T = 4 kN-m
M = ?

$$T_{eq} = \sqrt{M^2 + T^2}$$

$$5 = \sqrt{M^2 + 4^2}$$

$$\boxed{M = 3 \text{ kN-m}}$$

894. A solid shaft steel of 100 mm diameter and 1.0 m long is subjected to a twisting moment T. This shaft is to be replaced by a hollow shaft having outer and inner diameters as 100 mm and 50 mm respectively. If the maximum shear stress induced in both the shafts is same, the twisting moment T transmitted by hollow shaft must be reduced by

(a) T / 4 (b) T / 8
(c) T / 16 (d) T /12

BPSC Poly. Lect. 2016, ESE 2012, ESE 1998

Ans : (c) Solid Shaft | hollow shaft
d = 100mm | d_o = 100 mm
ℓ = 1.0 m | d_i = 50 mm.
ℓ = 1.0 m.

T_s = Twisting moment for solid shaft T_h = twisting moment transmitted by hollow shaft
same material then $\tau = \tau_s = \tau_h$

for solid shaft:-

$$T_s = \frac{\pi d^3 \tau}{16}$$

for hollow shaft:-

$$T_h = \frac{\pi d_0^3 \left(1 - C^4\right)\tau}{16}$$

$$C = \frac{d_i}{d_0} = 0.5$$

$$T_h = \frac{\pi d_o^3}{16T_h} \times \frac{15}{16}$$

$$\frac{T_h}{T_s} = \frac{15}{16}$$

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$$T_h = \frac{15}{16} T_s$$

Reduce twisting moment = $T_s - T_h$

$$= T - \frac{15}{16} T$$

Reduce twisting moment = $\frac{T}{16}$

895. The polar section modulus of a solid circular shaft of diameter 'd' about an axis through its centre of gravity is

(a) $\frac{\pi}{4} d^3$ (b) $\frac{\pi}{12} d^3$

(c) $\frac{\pi}{16} d^3$ (d) $\frac{\pi}{32} d^3$

Nagaland PSC (CTSE) 2018, Paper-I
GPSC Lect. (Auto) 16.10.2016
WBPSC AE 2008

Ans. (c) : Polar section modulus $Z_P = \frac{J}{R}$

$$Z_P = \frac{\pi \times d^4}{32 \times d} \times 2$$

$$Z_P = \frac{\pi}{16} d^3$$

896. In case of a circular shaft subjected to torque the value of shear stress

(a) Is uniform through out
(b) Has maximum value at axis
(c) Has maximum value at the surface
(d) Is zero at the axis and linearly increases to a maximum value at the surface of the shaft

Nagaland PSC CTSE 2017, Paper-I
GPSC Asstt. Director of Transport 05.03.2017
Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : In case of a circular shaft subjected to torsion the value of shear stress is zero at the centre and linearly increase to a maximum value at the surface of the shaft.

897. For a circular shaft of diameter D subjected to torque T, the maximum value of the shear stress is

(a) $(64T/\pi D^3)$ (b) $(32T/\pi D^3)$
(c) $(16T/\pi D^3)$ (d) $(128T/\pi D^3)$
(e) $(8T/\pi D^3)$

RPSC 2016, CGPSC AE 26.04.2015 Shift-I
GATE 2006

Ans. (c) : From torsion equation we have

$$\frac{T}{J} = \frac{\tau}{r}$$

where T = torque
J = polar moment of inertia
τ = shear stress
r = radius

The maximum value of shear stress is given by

$$\frac{T}{\left(\frac{\pi D^4}{32}\right)} = \frac{\tau}{\left(\frac{D}{2}\right)}, \quad \tau = \frac{16T}{\pi D^3}$$

898. Angle of twist of a solid shaft of torsional rigidity GJ, length L and applied torque T will be given by:

(a) $\frac{T}{GJL}$ (b) $\frac{L}{GJT}$

(c) $\frac{TL}{GJ}$ (d) $\frac{GJ}{TL}$

SJVN ET 2019, UKPSC AE 2007 Paper -I

Ans. (c) : $\frac{\tau}{r} = \frac{T}{J} = \frac{G\theta}{\ell}$

$$\frac{G\theta}{\ell} = \frac{T}{J}$$

$$\theta = \frac{TL}{GJ}$$

899. A shaft of diameter (d) is subjected to torque (T) and bending moment (M). The value of maximum principle stress and maximum shear stress is given respectively by:

(a) $\frac{16}{\pi d^3}\left[M + \sqrt{M^2 + T^2}\right]; \frac{16}{\pi d^3}\left[\sqrt{M^2 + T^2}\right]$

(b) $\frac{16}{\pi d^4}\left[M + \sqrt{M^2 + T^2}\right]; \frac{16}{\pi d^4}\left[\sqrt{M^2 + T^2}\right]$

(c) $\frac{16}{\pi d^3}\left[\sqrt{M^2 + T^2}\right]; \frac{16}{\pi d^3}\left[M + \sqrt{M^2 + T^2}\right]$

(d) $\frac{16}{\pi d^4}\left[\sqrt{M^2 + T^2}\right]; \frac{16}{\pi d^4}\left[M + \sqrt{M^2 + T^2}\right]$

(e) $\frac{32}{\pi d^4}\left[\sqrt{M^2 + T^2}\right]; \frac{32}{\pi d^4}\left[M + \sqrt{M^2 + T^2}\right]$

CGPSC AE 25.02.2018
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Maximum principle stress σ_{max}.

$$\sigma_{max} = \frac{16}{\pi d^3}\left[M + \sqrt{M^2 + T^2}\right]$$

maximum shear stress

$$\tau_{max} = \frac{16}{\pi d^3}\left[\sqrt{M^2 + T^2}\right]$$

900. The diameter of shaft is increased from 30 mm to 60 mm, all other conditions remaining unchanged. How many times its torque carrying capacity increases?

(a) 2 (b) 4
(c) 8 (d) 16

OPSC AEE 2019 Paper-I
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : From torsion equation.

$$\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{R}$$

$$T = \frac{\tau \times J}{R}$$

$$T = \frac{\tau \times \frac{\pi}{32} d^4}{d/2}$$

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$$T = \frac{\tau d^3 \pi}{16}$$

$$T \propto d^3$$

$$\frac{T_2}{T_1} = \left(\frac{d_2}{d_1}\right)^3$$

$$\frac{T_2}{T_1} = \left(\frac{60}{30}\right)^3 = 2^3 = 8$$

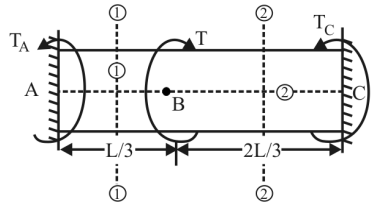
901. A hollow shaft of length *L* is fixed at its both ends. It is subjected to a torque *T* at a distance of *L*/3 from its one end. What is the reaction torque at the other end of the shaft?

(a) 2T/3 (b) T/2

(c) T/3 (d) T/4

TSPSC AEE 28.08.2017 (Civil/Mechanical)
UPSC JWM Advt. No.-52/2010

Ans. (c) : Let; *T* is the torque acting at B from the end A at a distance L/3



(Hollow shaft)

Let; T_A and T_C reaction torque on both side of fixed end of the hollow shaft due to applying torque *T*.

(1) For section – (1) (1) & section (2)- (2)

Torque ; $T_1 = T_A$; $T_2 = T_A - T$ or T_C …(i)

(2) Net torque; $T_A + T_C = T$ …(ii)

(3) $\theta_1 + \theta_2 = 0$ …(iii)

(for both end fixed)

$$\frac{T_1 L_1}{G_1 J_1} + \frac{T_2 L_2}{G_2 J_2} = 0 \qquad \begin{cases} \because GJ = \text{constant} \\ \text{i.e. } G_1 J_1 = G_2 J_2 = GJ \end{cases}$$

$$\Rightarrow T_1 L_1 + T_2 L_2 = 0$$

$$\Rightarrow T_A \times \frac{L}{3} + (T_A - T) \times \frac{2L}{3} = 0$$

$$\Rightarrow T_A + 2T_A - 2T = 0$$

$$\Rightarrow 3T_A = 2T$$

$$\boxed{T_A = \frac{2T}{3}}$$

From eqn (ii) ; $T_A + T_C = T$

$$\Rightarrow \frac{2T}{3} + T_C = T$$

$$\Rightarrow \boxed{T_C = \frac{T}{3}}$$

$T_C = T/3$;

is the torque at the other end of the shaft.

902. A uniform solid shaft of diameter D is subjected to equal magnitude of bending moment and twisting moment M. What is the maximum shear stress induced in the shaft?

(a) $\frac{32M}{\pi D^3}$ (b) $16\sqrt{2}.\frac{M}{\pi D^3}$

(c) $\frac{16M}{\pi D^3}$ (d) $\frac{16}{\sqrt{2}}.\frac{M}{\pi D^3}$

UPSC JWM Advt. No.-52/2010
UKPSC AE 2007 Paper -I

Ans. (b) : Solution : Given :

Combined effect of Bending moment and Twisting moment. Both have equal magnitude M.

$$\therefore \text{ Shear stress } (\tau) = \frac{16}{\pi D^3}\left(\sqrt{M^2 + T^2}\right)$$

$$\therefore \quad \tau_{max} = \frac{16}{\pi D^3} \times \sqrt{2M^2} \qquad (\because M = T)$$

$$\tau_{max} = 16.\sqrt{2}.\frac{M}{\pi D^3}$$

903. Torque in a solid shaft of diameter 'd' and shear stress 'τ' is given by

(a) $\frac{\pi}{8}\tau d^3$ (b) $\frac{\pi}{16}\tau d^3$

(c) $\frac{\pi}{12}\tau d^3$ (d) $\frac{\pi}{32}\tau d^3$

UPPSC AE 12.04.2016 Paper-I
WBPSC AE 2008

Ans. (b) : From Torsion equation

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

$$\frac{T}{J} = \frac{\tau}{r}$$

$$\tau = \frac{Tr}{J} = \frac{T \times d \times 32}{2 \times \pi \times d^4}$$

$$\boxed{\tau = \frac{16T}{\pi d^3}}$$

$$\boxed{T = \frac{\pi d^3}{16}\tau}$$

Where symbols have their usual meaning.

904. In case of a solid shaft the strain energy in torsion, per unit volume, is equal to

(a) $\frac{\tau^2}{2C}$ (b) $\frac{\tau^2}{4C}$

(c) $\frac{\tau^2}{6C}$ (d) $\frac{\tau^2}{8C}$

ESE 2013, WBPSC AE 2008

Ans. (b) For prismatic bar under torsion.

$$U = \frac{\tau^2}{4C} \cdot V$$

$$\frac{U}{V} = \frac{\tau^2}{4C}$$

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905. Two shafts A and B of solid circular cross-section are identical expect for their diameter d_A and d_B. The ratio of power transmitted by the shaft A to that of shaft B is

(a) $\frac{d_A}{d_B}$ (b) $\frac{(d_A)^2}{(d_B)^2}$

(c) $\frac{(d_A)^3}{(d_B)^3}$ (d) $\frac{(d_A)^4}{(d_B)^4}$

Karnataka PSC Lect., 27.05.2017
KPSC ADF 2015

Ans. (c) :

$\because$ Power (P) = T.ω $\quad \boxed{\because \text{Torque}(T) = \frac{\pi}{16}d^3\tau}$

$$P = \frac{\pi}{16}d^3.\tau.\omega$$

For shaft A; $P_A = \frac{\pi}{16}d_A^3.\tau.\omega \quad \left\{\tau_A = \tau_B = \tau,\ \omega_A = \omega_B = \omega\right\}$

For shaft B; $P_B = \frac{\pi}{16}d_B^3.\tau.\omega$

$\therefore \quad \boxed{\frac{P_A}{P_B} = \frac{(d_A)^3}{(d_B)^3}}$

906. A solid shaft transmits a torque of T. The allowable shear stress is τ. What is the diameter of the shaft ?

(a) $\sqrt[3]{\frac{16T}{\pi\tau}}$ (b) $\sqrt[3]{\frac{32T}{\pi\tau}}$

(c) $\sqrt[3]{\frac{16T}{\tau}}$ (d) $\sqrt[3]{\frac{T}{\pi}}$

OPSC AEE 2015 Paper-I
RPSC AE GWD, 2011, ESE 2008

Ans : (a) Torsion equation

$$\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{r}$$

$$\frac{32T}{\pi d^4} = \frac{2\tau}{d} = \frac{16T}{\pi d^3} = \tau$$

$$\boxed{d = \sqrt[3]{\frac{16T}{\pi\tau}}}$$

907. The section modulus of a circular section of diameter (*d*) is

(a) π/32 d^2 (b) π/32 d^3

(c) π/64 d^3 (d) π/64 d^4

Nagaland PSC (CTSE) 2018 Paper-I
GMB AAE 25.06.2017

Ans. (b) : Section Modulus – Section modulus is the ratio of moment of inertia about N.A. upon the for test point of section from Neutral axis (N.A)

$$\boxed{Z = \frac{I}{Y}}$$

for circular section

$$Z_{XX} = \frac{I_{XX}}{Y}$$

$$= \frac{\pi}{64} \times d^4 \times \frac{1}{d/2}$$

$\therefore \quad \boxed{Z_{XX} = \frac{\pi}{32}d^3 mm^3}$

908. A hollow shaft of the same cross-section area and material as that of a solid shaft, transmits:

(a) Same torque (b) Lesser torque

(c) More torque (d) None

UJVNL AE 2016, RO Scientist/Engineer 2007
ESE 2005

Ans : (c) A hollow shaft of the same cross- sectional area and materials transmits more torque than solid shaft.

given $A_s = A_h$

$$\frac{\pi}{4}d^2 = \frac{\pi}{4}\left[d_o^2 - d_i^2\right]$$

$$d^2 = d_o^2 - d_i^2 \quad \ldots\ldots\ldots\ldots \text{(i)}$$

from equation (i) we can say that

$$d_o > d$$

$$\frac{T_h}{T_s} = \frac{\frac{\pi d_o^3 \times \tau\left[1 - k^4\right]}{16}}{\left(\frac{\pi d^3}{16}\right) \times \tau}$$

$$\frac{T_h}{T_s} = \frac{d_o^3\left[1 - k^4\right]}{d^3}$$

If we assume that [hit and trail method]

$d_i = 3$ unit

$d_o = 5$ unit

$d = 4$ unit

$$\frac{T_h}{T_s} = \frac{125\left[1 - \left(\frac{3}{5}\right)^4\right]}{64} = 1.7$$

$T_h = 1.7\ T_s$

So we can say that $T_h > T_s$.

909. Angle of twist of a shaft of diameter 'd' is inversely proportional to :

(a) d (b) d^2

(c) d^3 (d) d^4

OPSC AEE 2019 Paper-I
ISRO Scientist/Engg. (RAC) 22.04.2018

Ans : (d) : By using torsion formula,

$$\theta = \frac{T\ell}{GJ} = \frac{T\ell}{G \times \frac{\pi}{32} \times d^4} \Rightarrow \theta \propto \frac{1}{d^4}$$

910. A solid circular shaft is subjected to pure torsion. The ratio of maximum shear to maximum normal stress at any point would be-

(a) 1 : 1 (b) 1 : 2

(c) 2 : 1 (d) 2 : 3

RPSC AE 2018, APPSC AEE 2012, ESE 1996

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Ans. (a) : A solid circular shaft is subjected to pure torsion, then

$$\tau_{max} = \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$\sigma_x = \sigma_y = 0$$

$$\tau_{max} = \tau_{xy}$$

$$\sigma_{max} = \frac{\sigma_x + \sigma_y}{2} + \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2}$$

$$\sigma_{max} = \tau_{xy}$$

then,

$$\frac{\tau_{max}}{\sigma_{max}} = 1:1$$

911. Two shafts, one solid and the other hollow, made of the same material, will have the same strength and stiffness, if both are of the same

(a) length as well as weight
(b) length as well as polar modulus
(c) weight as well as polar modulus
(d) length, weight as well as polar modulus

ESE 2017, Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : Solid shaft and hollow shaft are same material, so G is same.

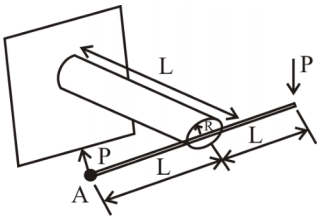
$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{\ell}$$

$$k_t = \frac{T}{\theta} = \frac{GJ}{\ell}$$

$$\tau_{max} = \frac{T}{J} \times r_{max}$$

$\therefore$ For strength and stiffness to be same, both must have same polar moment of inertia (J) and same length (ℓ).

912. A rigid horizontal rod of length 2L is fixed to a circular cylinder of radius R as shown in the figure. Vertical forces of magnitude P are applied at the two ends as shown in the figure.



The shear modulus for the cylinder is G and the Young's modulus is E.

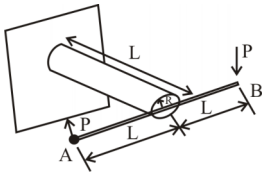
The vertical deflection at point A is

(a) $\frac{PL^3}{\pi R^4 G}$ (b) $\frac{PL^3}{\pi R^4 E}$

(c) $\frac{4PL^3}{\pi R^4 G}$ (d) $\frac{2PL^3}{\pi R^4 E}$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2016

Ans. (c) : $\frac{4PL^3}{\pi R^4 G}$



Vertical deflection at point 'A'

$(X) = \theta \times L$

$X = \theta \times L$ .....(i)

$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R}$ $\qquad T = P \times 2L$

$\theta = \frac{TL}{GJ}$

$x = \theta \times L$

$= \frac{TL^2}{GJ}$

$= \frac{P \times 2L \times L^2}{G \times \frac{\pi}{32}(D^4)}$

$= \frac{2PL^3}{G\frac{\pi}{32}(2R)^4}$

$x = \frac{4PL^3}{G\pi R^4}$

Vertical deflection at point 'A'

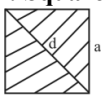
913. The polar moment of inertia of a hollow shaft of outer diameter (D) and inner diameter (d) is

(a) $\frac{\pi}{16}(D^3 - d^3)$ (b) $\frac{\pi}{16}(D^4 - d^4)$

(c) $\frac{\pi}{32}(D^4 - d^4)$ (d) $\frac{\pi}{64}(D^4 - d^4)$

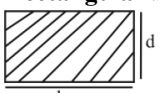
MECON MT 2019
Sikkim PSC (Under Secretary), 2017

Ans. (c) : Square section **Polar Moment of Inertia**



$J = \frac{a^4}{6} = 0.1667a^4$

Rectangular Section



$J = \frac{bd(b^2 + d^2)}{12}$

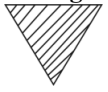
Circular section

$J = \frac{\pi D^4}{32} = 0.098D^4$

Tube section

$J = \frac{\pi}{32}(D^4 - d^4)$

Triangular section



$J = \frac{\sqrt{3}}{48}S^4 = 0.036S^4$

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914. When a shaft is subjected to twisting moment, every cross-section of the shaft will be under

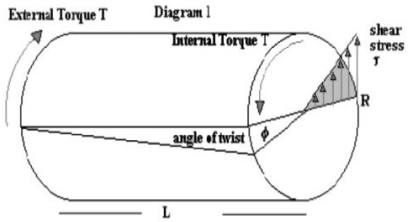
(a) tensile stress (b) compressive stress
(c) shear stress (d) bending stress

Vizag Steel (MT) 2017, Assam PSC CCE Pre 2015

Ans. (c)

Torsion equation – $\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{r}$

$$T=\frac{\pi}{16}\tau d^3$$



Shear stress in shaft, $\boxed{\tau=\frac{16T}{\pi d^3}}$

When a shaft is subjected to twisting moment, every cross-section of the shaft will be under shear stress.

915. Torsional Stiffness is _____ .

(a) quotient of shear modulus and polar moment of inertia
(b) product of shear modulus and polar moment of inertia
(c) torque produced per unit moment of inertia
(d) torque produced per unit angle of twist

VIZAG MT, 14.12.2020
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (d) : Torsional equation,

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{R}$$

$$T=\frac{GJ}{\ell}\theta$$

$$T=K_T\theta$$

$\boxed{K_T=\frac{GJ}{\ell}}$ ← Torsional stiffness

$\boxed{K_T=\frac{T}{\theta}}$ ← Torsional stiffness

Torsional stiffness is torque produced per unit angle of twist.

916. If 'P' is the Ratio of inside to outside diameter of a shaft, the ratio of torque carrying capacity of solid shaft to that of hollow shaft is given by:

(a) $(1-P^4)^{-1}$ (b) $1-P^4$
(c) P^4 (d) $\frac{1}{P^4}$

Kerala PSC Poly. Lect., 2017, ESE 2008

Ans. (a) : Maximum shear stress for hollow shaft,

$$\tau_{max}=\frac{16T_H}{\pi D^3(1-P^4)} \qquad \because\ P=\frac{D_i}{D_0}$$

Torque carrying capacity,

$$T_H=\frac{\pi D^3(1-P^4)}{16}\tau_{max} \qquad .....(1)$$

Maximum shear stress for solid shaft,

$$\tau_{max}=\frac{16T}{\pi D^3}$$

Torque carrying capacity,

$$T_S=\frac{\pi D^3}{16}\tau_{max} \qquad .....(2)$$

Equation (2) ÷ equation (1)

$$\boxed{\frac{T_S}{T_H}=\frac{1}{(1-P^4)}}$$

917. The greatest twisting moment which a shaft section can resist is equal to

(a) polar modulus × τ (b) polar modulus/τ
(c) τ/polar modulus (d) None of the above

RPSC VPITI 14.02.2016
APPSC Poly. Lect. 2013

Ans. (a) : The maximum twisting moment which a shaft can resist is product of the permissible shear stress and polar modulus.

$$\frac{T}{J}=\frac{\tau}{R}=\frac{G\theta}{\ell}$$

Where, T = Maximum Torque,
J = Polar Moment of Inertia
τ = Shear stress, R = Radius,
G = Modulus of rigidity, θ = Angle of twist

918. For the same material, length and given torque, a hollow shaft weighs _____ a solid shaft :

(a) less than (b) more than
(c) equal to (d) None of the above

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (a) : For the same material, length and given torque, a hollow shaft weighs less than a solid shaft.

919. Power transmitted by a shaft (kW) subjected to an average torque T (kNm), rotating at N rpm is equal to

(a) $\frac{2\pi NT}{3600}$ (b) $\frac{2\pi NT}{4500}$
(c) $\frac{2\pi NT}{60}$ (d) $\frac{\pi NT}{3060}$

APPSC AEE Mains 2016 (Civil Mechanical)
HPPSC Lect. 2016

Ans. (c) : As we know,
Work done per minute = force × distance
= Average torque × Angular displacement

$$=T\times\frac{2\pi N}{60}$$

$$P=\frac{2\pi NT}{60}\text{kW}$$

920. Consider the arrangement shown in the figure below where J is the combined polar mass moment of inertia of the disc and the shafts. K_1, K_2 and K_3 are the torsional stiffness of the respective shafts. The natural frequency of the torsional oscillation of the disc is given by

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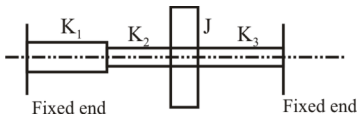

(a) $\sqrt{\left(\frac{K_1+K_2+K_3}{J}\right)}$

(b) $\sqrt{\left(\frac{K_1K_2+K_2K_3+K_3K_1}{J(K_1+K_2)}\right)}$

(c) $\sqrt{\left(\frac{K_1K_2K_3}{J(K_1K_2+K_2K_3+K_3K_1)}\right)}$

(d) $\sqrt{\left(\frac{K_1K_2+K_2K_3+K_3K_1}{J(K_2+K_3)}\right)}$

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2003

Ans. (b) : K_1 and K_2 are in series and K_3 in parallel with equivalent resistance between 1 and 2.

Equivalent stiffness between 1 and 2 (series)

$$\frac{1}{K_{12}}=\frac{1}{K_1}+\frac{1}{K_2}$$

$$K_{12}=\frac{K_1\cdot K_2}{K_1+K_2}$$

Equivalent stiffness between K_{12} and K_3 is (parallel)

$$K_{eq}=K_{12}+K_3$$

$$K_{eq}=\frac{K_1K_2}{K_1+K_2}+K_3$$

$$K_{eq}=\frac{K_1K_2+K_2K_3+K_3K_1}{(K_1+K_2)}$$

The natural frequency for torsional oscillation is given by–

$$\omega=\sqrt{\frac{K_{eq}}{J}}$$

$$\omega=\sqrt{\frac{K_1K_2+K_2K_3+K_3K_1}{J(K_1+K_2)}}$$

921. The shear stress at a point in a shaft subjected to a torque

(a) directly proportional to the polar moment of inertia and to the distance of the point from the axis

(b) directly proportional to the applied torque and inversely proportional to the polar moment of inertia

(c) directly proportional to the applied torque and the polar moment of inertia

(d) inversely proportional to the applied torque and the polar moment of inertia

GPSC Executive Engineer 23.12.2018
JPSC AE 2013, Paper-V, CSE Pre-1995

Ans. (b) : The shear stress produced by torque vary from zero at centre of cross section to maximum at surface in circumferential directions. Shear stresses in direction normal to the cross-section plane are always complimentary in nature and have equal magnitude.

By torsional formula:-

$$\frac{T}{J}=\frac{G\theta}{l}=\frac{\tau}{r}$$

Where, J = Polar moment of inertia

l = Length of shaft

r = Radius of shaft

maximum shear stresses will occur when

$$r=r_{max}=R$$

$$\tau_{max}=\frac{T}{J}\times r_{max}$$

$$\tau_{max}=\frac{T}{J}\times R\,(\text{at surface of shaft})$$

922. Maximum shear stress in a solid shaft of diameter D and length L twisted through an angle θ is τ. A hollow shaft of the same material and length having outside and inside diameters of D and $\frac{D}{2}$ respectively is also twisted through the same angle of twist θ. The value of maximum shear stress in the hollow shaft will be

(a) $\frac{16}{15}\tau$ (b) $\frac{8}{7}\tau$

(c) $\frac{4}{3}\tau$ (d) τ

UPPSC AE 13.12.2020, Paper-I, ESE 1997

Ans. (d) : $\frac{\tau_{max}}{r_{max}}=\frac{G\theta}{\ell}$

$$\tau_{max}\propto r_{max}$$

$$\frac{(\tau_{max})_H}{(\tau_{max})_S}=\frac{(r_{max})_H}{(r_{max})_S}=\frac{(D/2)}{(D/2)}$$

$$(\tau_{max})_H=(\tau_{max})_S=\tau$$

923. A line shaft rotating at 200 rpm is to transmit 20 kW. The allowable shear stress of shaft material is 42 MPa. Neglecting the bending moment, the diameter of the shaft needs to be :

(a) 35 mm (b) 50 mm

(c) 65 mm (d) 75 mm

RPSC ACF & FRO, 26.02.2021
APPSC Poly. Lect. 2013

Ans. (b) : Given, N = 200 rpm

P = 20 kW

τ = 42 MPa

d = ?

$$P=\frac{2\pi NT}{60}$$

$$20\times10^3=\frac{2\pi\times200}{60}\times T$$

$$\boxed{T=954.92\ \text{Nm}}$$

$$T=954.92\times10^3\ \text{Nmm}$$

$$\tau=\frac{16T}{\pi d^3}$$

$$d^3=\frac{16T}{\pi\tau}$$

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$$d^3 = \frac{16 \times 954.92 \times 10^3}{\pi \times 42}$$

$$d = \sqrt[3]{\frac{16 \times 954.92 \times 10^3}{\pi \times 42}}$$

$\boxed{d = 48.74}$ mm ≈ 50 mm

924. For a hollow shaft of inner diameter 'd' and outer diameter 'D' the modulus of rupture is :

(a) $8TD/\pi(D^4 - d^4)$ (b) $16TD/\pi(D^4 - d^4)$

(c) $32TD/\pi(D^4 - d^4)$ (d) $64TD/\pi(D^4 - d^4)$

SJVN ET 2019, RPSC VPITI 14.02.2016

Ans. (b) : The maximum shear stress calculated by using the torsion formula is called modulus of rupture

$$\tau_{max} = \frac{T \times D/2}{J}$$

$$\tau_{max} = \frac{T \times D/2}{\frac{\pi}{32}\left(D^4 - d^4\right)}$$

$$\tau_{max} = \frac{16TD}{\pi\left(D^4 - d^4\right)}$$

925. Two shaft A and B under pure tension are of identical length and identical weight and are made of the same material. The shaft A is solid and the shaft B is hollow. We can say that.

(a) Shaft B is better than shaft A

(b) Shaft A is better than shaft B

(c) Both the shafts are equally good

(d) Both the shafts are not good

RPSC ACF & FRO, 26.02.2021

Ans. (a) : Case- I – External diameter same

In this case solid shaft will have more strength.

Case- II – Weight is same.

In this case hollow shaft will have more strength

Case- III – Cross-sectional area same.

In this case solid shaft transmit more torque.

In question it is given that weight is identical or same so Hollow shaft (B) will have more strength, so shaft B is better than A.

926. A solid circular shaft of diameter 100 mm is subjected to an axial stress to 50 MPa. It is further subjected to a torque of 10 kNm. The maximum principal stress experienced on the shaft is closest to

(a) 41 MPa (b) 82 MPa

(c) 164 MPa (d) 204 MPa

VIZAG STEEL MT 18.08.2013

Vizag Steel MT Mechanical, 2013 Re-Exam

VIZAG STEEL MT 10.06.2012

Ans. (b) : Shear stress $(\tau) = \frac{16T}{\pi d^3}$

$$\tau = \frac{16 \times 10000}{\pi \times (0.1)^3} = 50.93 \text{ MPa}$$

Maximum principal stress-

$$\sigma_{max} = \frac{\sigma_b}{2} + \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2}$$

$$= \frac{50}{2} + \sqrt{\left(\frac{50}{2}\right)^2 + (50.93)^2}$$

$$= 25 + \sqrt{25^2 + (50.93)^2}$$

$\boxed{\sigma_{max} = 82 \text{ MPa}}$

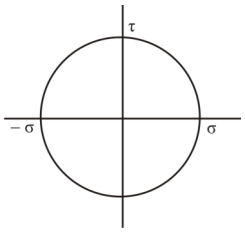
927. A circular shaft subjected to twisting moment results in maximum shear stress of 90 MPa. Then the maximum compressive stress in the material is

(a) 130 MPa (b) 100 MPa

(c) 80 MPa (d) 90 MPa

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : Mohr circle for pure shear, So, the compressive stress would be equal to applied shear stress.



$\sigma_1 = \sigma$ (Tensile)

$\sigma_2 = \sigma$ (Compressive)

$\tau = \sigma = 90$ MPa

928. A solid circular shaft, under pure torsion, develops maximum shear stress of 10 MPa on the surface. If the shaft diameter is halved, what will be the maximum shear stress developed corresponding to the same torque?

(a) 05 MPa (b) 40 MPa

(c) 20 MPa (d) 80 MPa

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : As we know,

Maximum shear stress $= \frac{16T}{\pi d^3}$

$$10 = \frac{16T}{\pi d^3} = \tau_1$$

Max. shear stress, when diameter is halved.

$$\tau_2 = \frac{16T}{\pi\left(\frac{d}{2}\right)^3}$$

$$\tau_2 = 8\left(\frac{16T}{\pi d^3}\right)$$

$$\tau_2 = 8 \times \tau_1$$

$$\tau_2 = 80 \text{ MPa}$$

929. While transmitting the same power by a shaft, if its speed is reduced by half, what should be its new diameter if the maximum shear stress induced in the shaft remains same?

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(a) $(2)^{1/2}$ of the original diameter
(b) $(1/2)^{1/2}$ of the original diameter
(c) twice of the original diameter
(d) $(2)^{1/3}$ of the original diameter

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : Given-
Power = Torque (T) × Angular speed (ω)
If P is const.

$$T \propto \frac{1}{\omega}$$

$$\text{If } \frac{T'}{T} = \frac{\omega}{\omega'} = \frac{\omega}{\left(\frac{\omega}{2}\right)} = 2$$

$$\text{Shear Stress, } \tau = \frac{16T}{\pi d^3}$$

$$T \propto d^3$$

$$\frac{T'}{T} = \left(\frac{d'}{d}\right)^3$$

$$2 = \left(\frac{d'}{d}\right)^3$$

$$\frac{d'}{d} = (2)^{1/3}$$

930. The ratio of Shear stresses of solid shaft (diameter'd') to hollow shaft (internal diameter $\frac{'d'}{3}$, external diameter 'd') When subjected to same twisting moment

(a) $\frac{81}{80}$ (b) $\frac{80}{81}$
(c) $\frac{26}{27}$ (d) $\frac{27}{26}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : d = solid shaft dia $d_o = d$ = hollow shaft external dia

$$d_i = \frac{d}{3} = \text{internal dia.}$$

$$\tau_s = \frac{16T}{\pi d^3} \text{-------(i)}$$

$$\tau_H = \frac{16Td}{\pi\left[d^4 - \left(\frac{d}{3}\right)^4\right]} \text{--------(ii)}$$

By comparing both eq.

$$\frac{\tau_s}{\tau_H} = \frac{80}{81}$$

931. Maximum shear stress developed on the surface of a solid circular shaft under pure torsion is 320 MPa. If the shaft diameter is doubled, then what is the maximum shear stress developed corresponding to the same torque?

(a) 60 MPa (b) 120 MPa
(c) 40 MPa (d) 20 MPa

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : Given-
We know,

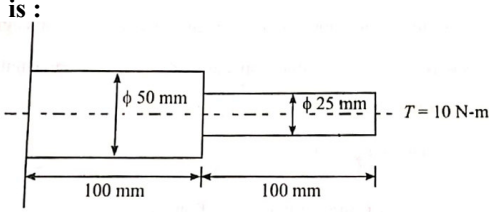
$$\text{Maximum shear stress } = \frac{16T}{\pi d^3} = 320\text{MPa}$$

So, Max. shear stress developed when diameter is doubled.

$$\tau_{new} = \frac{16T}{\pi(2d)^3} = \frac{1}{8} \times 320$$

$$\tau_{new} = 320/8 = 40\text{MPa}$$

932. A stepped steel shaft shown below is subjected to 10 N-m torque. If the modulus of rigidity is 80 GPa, the strain energy in the shaft in N-mm is :

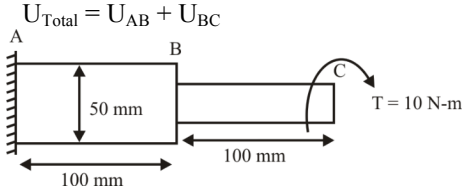


(a) 4.12 (b) 3.46
(c) 1.73 (d) 0.86

CGPSC AE 15.01.2021

Ans. (c) : Strain Energy in the shaft

$$U_{Total} = U_{AB} + U_{BC}$$



$$= \frac{T_{AB}^2 L}{2GJ} + \frac{T_{BC}^2 L}{2GJ}$$

$$= \frac{(10\times10^3)^2 \times 100}{2\times80\times10^3 \times \left[\frac{\pi}{32}(50)^4\right]} + \frac{(10\times10^3)^2 \times 100}{2\times80\times10^3 \times \left[\frac{\pi}{32}(25)^4\right]}$$

= 1.73 N-mm

933. A solid circular shaft is subjected to bending moment of 3000N and a torsion of 10,000N-m. The equivalent bending moment would be _____.

(a) 6600 N-m (b) 5920 N-m
(c) 10440 N-m (d) 6720 N-m

VIZAG Steel MT 24.01.2021

Ans. (d) : Bending moment (M) = 3000 N
Torsional moment (T) = 10,000 N
Equivalent bending moment,

$$(M_e) = \frac{1}{2}\left[M + \sqrt{M^2 + T^2}\right]$$

$$= \frac{1}{2}\left[3000 + \sqrt{(3000)^2 + (10,000)^2}\right]$$

$$= \frac{1}{2}\left[3000 + \sqrt{9000000 + 100000000}\right]$$

$$= \frac{1}{2}[3000 + 10440.3065]$$

= 6720.153 N-m

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934. The torsional rigidity of a shaft is the torque required to _____.

(a) produce maximum shear stress
(b) produce unit twist at unit length
(c) produce maximum twist
(d) produce zero twist

VIZAG Steel MT 24.01.2021

Ans. (b) : According to torsional equation–

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{R}$$

$$\frac{G\theta}{\ell}=\frac{T}{J}$$

For, $\theta = 1$ radian, length $(\ell) = 1$

$$G=\frac{T}{J}$$

$$\boxed{G \propto T}$$

935. For a hollow circular shaft under torsion T only the shear stress, τ will be expressed by :

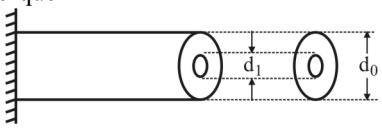
(a) $\frac{16T}{\pi(d_o^3+d_i^3)}$ (b) $\frac{32T}{\pi(d_o^3+d_i^3)}$

(c) $\frac{16\sqrt{M^2+T^2}}{\pi d_o^3}$ (d) $\frac{16Td_0}{\pi(d_o^4-d_i^4)}$

Where d_o, d_i are outer and inner diameters.

JPSC AE 10.04.2021, Paper-II

Ans. (d) : Consider,
Outside dia = d_o
Inside dia = d_i
Torque = T



then shear stress (τ) on hollow shaft.

$$\frac{T}{J}=\frac{G\theta}{\ell}=\frac{\tau}{r}$$

$$\tau=\frac{Tr}{J}$$

$$=\frac{T(d_0/2)}{\frac{\pi}{32}(d_0^4-d_i^4)}$$

$$\boxed{\tau=\frac{16Td_0}{\pi(d_0^4-d_i^4)}}$$

936. A circular shaft when subjected to equal opposite end couples whose axes coincide with axis of the shaft is said to be in

(a) pure torsion
(b) pure bending
(c) combined bending and torsion
(d) none of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : A circular shaft when subjected to equal opposite end couples whose axes coincide with axis of the shaft is said to be in pure torsion.

937. The efficiency in resisting torsional moments by hollow shaft when compared to solid shaft is

(a) less (b) more
(c) equal (d) cannot be determined

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : The efficiency in resisting torsional moments by hollow shaft when compared to solid shaft is more.

938. Two shafts will have equal strength, if

(a) diameter of both the shafts is same
(b) angle of twist of both the shafts is same
(c) material of both the shafts is same
(d) twisting moment of both the shafts is same

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Two shafts will have equal strength, if twisting moment of both the shafts is same.

939. In case of a bar under torsion if the force is applied at the outer surface of the bar and longitudinally, the shear stress is maximum at

(a) the centre of the bar.
(b) just before the outer surface.
(c) the outer surface.
(d) the mid-point of radius of the bar.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : The shear stress varries linearly zero at the centre and maximum at outer surface.

940. A solid circular shaft is subjected to a bending moment of 6000 Nm and a torque of 8000 Nm. The equivalent bending moment of the shaft is _____.

(a) 14000 Nm (b) 10000 Nm
(c) 6000 Nm (d) 8000 Nm

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Given,
Bending moment, M = 6000 N-m
Torque, T = 8000 N-m
∴ Equivalent bending moment,

$$M_e=\frac{1}{2}\left[M+\sqrt{M^2+T^2}\right]$$

$$=\frac{1}{2}\left[6000+\sqrt{(6000)^2+(8000)^2}\right]$$

$$M_e=\frac{1}{2}[6000+10000]$$

$$=\frac{16000}{2}=8000\ \text{N}-\text{m}$$

941. The polar moment of inertia of a solid circular shaft of diameter d is _____.

(a) $\frac{\pi d^3}{32}$ (b) $\frac{\pi d^4}{32}$

(c) $\frac{\pi d^4}{64}$ (d) $\frac{\pi d^3}{64}$

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : For solid circular shaft–
diameter = d

$$I_{xx}=\frac{\pi d^4}{64}$$

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$I_{yy} = \dfrac{\pi d^4}{64}$

polar moment of Inertia, $I_{zz} = I_{xx} + I_{yy}$

$= \dfrac{\pi d^4}{64} + \dfrac{\pi d^4}{64}$

$\boxed{I_{zz} = \dfrac{\pi d^4}{32}}$

942. Units of section modulus is

(a) mm (b) mm^4

(c) mm^3 (d) mm^2

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : mm^3

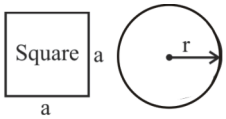
943. The cross-sections of two solid bars made of the same material are shown in the figure. The square cross-section has flexural (bending) rigidity I_1, while the circular cross section has flexural rigidity I_2, Both sections have the same cross-sectional area. The ratio $\dfrac{I_1}{I_2}$ is

(a) $\dfrac{1}{\pi}$ (b) $\dfrac{2}{\pi}$

(c) $\dfrac{\pi}{3}$ (d) $\dfrac{\pi}{6}$

Assam Engg. College AP/Lect. 18.01.2021

Ans. (c) : $\dfrac{\pi}{3}$



Area of cross section is same.

$A_1 = A_2$

$a^2 = \pi r^2$

$a = r\sqrt{\pi}$

Flexural rigidity of square $I_1 = E_1 I_{m_1}$

Flexural rigidity of circle $I_2 = E_2 I_{m_2}$

Where, $E_1 = E_2 = E$ = Young's modulus (same material)

I_{m_1} & I_{m_2} are moment of inertia of respective figures.

$$\frac{I_1}{I_2} = \frac{EI_{m_1}}{EI_{m_2}}$$

$$\frac{I_1}{I_2} = \frac{\frac{a^4}{12}}{\frac{\pi d^4}{64}} = \frac{a^4}{12} \times \frac{64}{\pi d^4} \qquad (d = 2r)$$

$$\frac{I_1}{I_2} = \frac{a^4}{12} \times \frac{64}{\pi (2r)^4}$$

$$= \frac{a^4}{12} \times \frac{64}{\pi \times 16r^4}$$

put $\left(a = r\sqrt{\pi}\right)$

$$= \frac{\left(r\sqrt{\pi}\right)^4 \times 64}{12\pi \times 16r^4}$$

$$= \frac{\pi^2 \times r^4 \times 64}{12 \times \pi \times 16r^4}$$

$$\frac{I_1}{I_2} = \frac{\pi}{3}$$

944. Two shafts of equal length and similar material in which one is hollow and other is solid are transmitting same level of torque. If the inside diameter is $\dfrac{2}{3}$ of the outside diameter of the hollow shaft, the ratio of weight of hollow shaft to weight of solid shaft is

(a) 0.642 (b) 0.358

(c) 0.732 (d) 1.444

UPPSC AE 13.12.2020, Paper-I

Ans. (a) : $T_H = T_S$

$$(\tau_{allowable} \times Z_P)_H = (\tau_{allowable} \times Z_P)_S$$

$$\frac{\pi}{16} d_o^3 (1 - k^4) = \frac{\pi}{16} d^3$$

$$\left(\frac{d_o}{d}\right)^3 = \frac{1}{1 - k^4} = \frac{1}{1 - \left(\frac{2}{3}\right)^4}$$

$$\frac{d_o}{d} = 1.076$$

Weight = ρg (Volume)

$= \rho g (A \times \ell)$

$$\frac{W_H}{W_S} = \frac{A_H}{A_S} = \frac{\frac{\pi}{4} d_o^2 (1 - k^2)}{\frac{\pi}{4} d^2}$$

$$= \left(\frac{d_o}{d}\right)^2 (1 - k^2)$$

$$= (1.076)^2 \left[1 - \left(\frac{2}{3}\right)^2\right]$$

$= 0.643$

945. What is the maximum torque transmitted by a hollow shaft of external radius 'R', internal radius 'r' and maximum allowable shear stress τ?

(a) $\dfrac{\pi}{16}(R^3 - r^3)\tau$ (b) $\dfrac{\pi}{2R}(R^4 - r^4)\tau$

(c) $\dfrac{\pi}{8R}(R^4 - r^4)\tau$ (d) $\dfrac{\pi}{32}(R^4 - r^4)\tau$

UPPSC AE 13.12.2020, Paper-I
ESE 2006

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Ans. (b) : Torque $(T) = \tau \times \frac{\pi}{16}\left[\frac{(2R)^4 - (2r)^4}{2R}\right]$

$$T = \tau \times \frac{\pi}{16} \times 16\left[\frac{R^4 - r^4}{2R}\right]$$

$$T = \tau \frac{\pi}{2R}[R^4 - r^4]$$

946. A shaft is subjected to a bending moment M = 0.75 kNm and a twisting moment T = 1 kNm. The magnitude of equivalent bending moment in shaft is

(a) 1.25 kNm (b) 1.125 kNm
(c) 1.0 kNm (d) 0.75 kNm

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : Given, M = 0.75 kNm
T = 1 kNm

Equivalent bending moment $= \frac{1}{2}\left[M + \sqrt{M^2 + T^2}\right]$

$$= \frac{1}{2}\left[0.75 + \sqrt{(0.75)^2 + 1^2}\right]$$

$$= 1 \text{ kNm}$$

947. A solid shaft of diameter 'D' carries a twisting moment that develop maximum shear stress τ. If this shaft is replaced by a hollow one of outside diameter D and inside diameter D/2, then the maximum shear stress will be

(a) 1.067 τ (b) 1.145 τ
(c) 1.335 τ (d) 2 τ

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : $k = \frac{D_i}{D_o} = \frac{1}{2}$

For solid shaft, $\tau_s = \frac{16T}{\pi D^3}$

For hollow shaft, $\tau_H = \frac{16T}{\pi D^3[1 - k^4]}$

$$\tau_H = \frac{16T}{\pi D^3\left[1 - \left(\frac{1}{2}\right)^4\right]}$$

$$\tau_H = \frac{16T}{\pi D^3} \times \frac{16}{15}$$

$$\tau_H = 1.066\ \tau$$

948. Two solid shafts 'A' and 'B' are made of the same material. The shaft 'A' is 50 mm diameter and shaft 'B' is 100 mm diameter. The ratio of strength of shaft 'B' compared to shaft 'A' is

(a) 1/2 (b) 2
(c) 4 (d) 8

ISRO Scientist/Engineer (RAC) 12.01.2020, ESE 2016

Ans. (d) : $d_A = 50$ mm
$d_B = 100$ mm

Strength of shaft is proportional to the polar section modulus.

$$\frac{Z_{P_B}}{Z_{P_A}} = \left(\frac{d_B}{d_A}\right)^3 = 8$$

949. A pump is used to transport the water. The maximum power required through the shaft is 62.8 W, when it rotates at 100 rpm. What will be the diameter of shaft (in cm) if the maximum permissible shearing stress is 100 MN/m^2? [π = 3.14]

(a) $\left(\frac{96}{314}\right)^{1/3}$ (b) $\left(\frac{196}{314}\right)^{1/3}$
(c) $\left(\frac{296}{314}\right)^{1/3}$ (d) $\left(\frac{396}{314}\right)^{1/3}$

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given data : Power (P) = 62.8 Watt
Rotational speed (N) = 100 rpm
shear stress $(\tau)_{max}$ = 100 MN/m^2
π = 3.14
Let,
d = dia of the shaft

$\because P = T\omega = \frac{2\pi NT}{60}$

$\Rightarrow T = \frac{60P}{2\pi N}$

$\tau_{max} = \frac{16T}{\pi d^3} \Rightarrow T = \frac{\pi d^3}{16} \times \tau_{max}$

$\because \frac{\pi}{16} d^3 \tau = \frac{60P}{2\pi N}$

$\Rightarrow d^3 = \frac{60P \times 16}{\pi\tau \times 2\pi N} = \frac{60 \times 62.8 \times 16}{3.14 \times 100 \times 10^6 \times 2 \times 3.14 \times 100}$

$\Rightarrow d^3 = \frac{96 \times 2}{2 \times 100 \times 3.14 \times 10^6}$

$\Rightarrow d^3 = \frac{96}{314} \times 10^{-6}$ m

$\Rightarrow d^3 = \left(\frac{96}{314}\right)^{1/3} \times 10^{-2}$ m

$\Rightarrow d = \left(\frac{96}{314}\right)^{1/3}$ cm

950. Torsional-sectional-modulus of a shaft is also called as:

(a) Sectional modulus (b) Polar modulus
(c) Torsion modulus (d) Torsional rigidity

CIL MT 27.02.2020

Ans. (b) : Polar modulus is defined as the ratio of the polar moment of inertia to the radius of shaft. It is also called as torsional section modulus (Z_P).

$$I_P = \frac{\pi}{32} D^4$$

$$Z_P = \frac{\pi}{16} D^3$$

951. A solid circular shaft needs to be designed to transmit a torque of 55 N m. If the allowable shear stress of the material is 280 N/mm^2, find the diameter (in mm) of the shaft required to transmit torque. (Assume, π = 22/7)

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(a) 5.62 (b) 10
(c) 31.62 (d) 25.0

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : For circular solid shaft,

$$\tau = \frac{16T}{\pi D^3}$$

$$D^3 = \frac{16T}{\pi\tau_{per}} = \frac{16\times 55\times 10^3}{\frac{22}{7}\times 280}$$

$D^3 = 1000$, $D = 10$ mm

952. A solid bar of circular cross-section having a diameter of 40 mm and length of 1.3 m is subjected to torque of 340 Nm. If the shear modulus of elasticity is 80 GPa, the angle of twist between the ends will be

(a) 1.26° (b) 1.32°
(c) 1.38° (d) 1.44°

ESE 2020

Ans. (a) : Angle of twist

$$\theta = \frac{T\ell}{GJ}$$

$$= \frac{340\times 10^3\times 1300}{80\times 10^3\times\frac{\pi}{32}\times 40^4}$$

$= 0.02198$ rad, $\theta = 1.26°$

953. Two shafts, one hollow and the other solid, are of same material, weight and length. If the inside diameter of the hollow shaft is half of its outer diameter, the strength of the hollow shaft to that of solid shaft will be about

(a) 1.067 (b) 1.33
(c) 1.44 (d) 1.67

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : When the hollow and solid shafts are of equal weights. In this case torsional strength is compared.

$$\frac{T_{hollow}}{T_{solid}} = \frac{1+n^2}{\sqrt{1-n^2}}$$

Given as $D_i = \frac{D_o}{2}$

$$n = \frac{D_i}{D_o}$$

$$= \frac{1}{2} = 0.5$$

$$\frac{T_{hollow}}{T_{solid}} = \frac{1+0.5^2}{\sqrt{1-0.5^2}} = 1.44$$

954. An universal coupling is used to connect two mild steel shafts transmitting a torque of 5000 N-m. Assuming that the shafts are subjected to torsion only, find the diameter of the shafts and pins. The allowable shear stresses for the shaft and pin may be taken as 60 MPa and 28 MPa respectively.

(a) 75 mm, 40 mm (b) 95 mm, 50 mm
(c) 40 mm, 85 mm (d) 75 mm, 60 mm

RPSC IOF, 2020

Ans. (a) : Given,
Allowable shear stress for shaft = 60 MPa
Transmitted torque,

$T = 5000$ N-m
$= 5000 \times 10^3$ N-mm
$= 5 \times 10^6$ N-mm

$$\tau_{shaft} = \frac{16T}{\pi d^3}$$

$$60 = \frac{16\times 5\times 10^6}{\pi d_{shaft}^3}$$

$$d_{shaft}^3 = \frac{16\times 5\times 10^6}{\pi\times 60}$$

$d_{shaft} = 75.15$ mm

For pin, allowable shear stress = 28 MPa

$$T = (2F)\times\left(\frac{d}{2}\right)$$

By convention, 2 Pin/key are used and both pin/key are under double shear.

So, $$T = 2\times(2F)\times\left(\frac{d}{2}\right)$$

$$5000 = 2\times(2\times F)\times\frac{75.15\times 10^{-3}}{2}$$

$F = 33.33 \times 10^3$ N

So, same force is applied in both shaft and pin/key

$$F = \tau_{pin}\times A_{pin} = \tau_{pin}\times\frac{\pi}{4}d_p^2$$

$$33.33\times 10^3 = 28\times 10^6\times\frac{\pi}{4}d_p^2$$

$d_p = 38.89$ mm
$d_p \approx 40$ mm

955. A solid shaft is used to transmit a power of 120 π kW at 120 rpm. The torque transmitted by the shaft is

(a) 30 kNm (b) 60 kNm
(c) 90 kNm (d) 120 kNm

Gujarat PSC AE 2019

Ans : (a) : Given-

$P = 120\ \pi$kW
$N = 120$ rpm

$$P = T\omega = \frac{T2\pi N}{60}$$

$$120\pi = \frac{T2\pi\times 120}{60}$$

$T = 30$ kNm

956. Hollow shafts are stronger than solid shafts having same weight because

(a) the stiffness of hollow shaft is less than that of solid shaft
(b) the strength of hollow shaft is more than that of solid shaft
(c) the natural frequency of hollow shaft is less than that of solid shaft
(d) in hollow shafts, material is not spread at large radius

ESE 2019

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Ans. (b) : Strength of hollow shaft is more than that of solid shaft because polar section modulus of hollow shaft is greater than that of solid shaft.

957. A propeller shaft is required to transmit 45 kW power at 500 r.p.m. It is a hollow shaft having inside diameter 0.6 times the outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm^2. The inner and outer diameters of the shaft are nearly.

(a) 21.7 mm and 39.1 mm
(b) 23.5 mm and 39.1 mm
(c) 21.7 mm and 32.2 mm
(d) 23.5 mm and 32.2 mm

ESE 2019

Ans. (b) : Given,

$P = 45$ kW
$N = 500$ rpm
$d_i = 0.6\, d_o$

$$k = \frac{d_i}{d_o} = 0.6$$

$\tau_{max} = 84$ N/mm^2

$$P = \frac{2\pi NT}{60}$$

$$45 \times 10^3 = \frac{2\pi \times 500 \times T}{60}$$

$T = 859.87$ N-m $= 859.43 \times 10^3$ N-mm

For hollow shaft

$$\tau_{max} = \frac{16T}{\pi d_o^3\left[1-\left(\frac{d_i}{d_o}\right)^4\right]}$$

$$84 = \frac{16 \times 859.43 \times 10^3}{\pi d_o^3[1-0.6^4]}$$

$$d_o^3 = 59896.95$$

$d_o = 39.12$ mm
$d_i = 23.47$ mm

958. In case of a torsional problem the assumption- "Plane sections perpendicular to longitudinal axis before deformation remain plane and perpendicular to the longitudinal axis after deformation" holds true for a shaft having

(a) circular cross-section
(b) elliptical cross-section
(c) circular and elliptical cross-section
(d) any cross-section

APPSC AEE SCREENING 17.02.2019

Ans. (a) : In case of a torsional problem the assumption-
"Plane sections perpendicular to longitudinal axis before deformation remain plane and perpendicular to the longitudinal axis after deformation" holds true for a shaft having circular cross-section only. It is not valid for other shape of cross-section.

959. A shaft turns at 200 rpm under a torque of 1800 Nm. The power transmitted is

(a) 6π kW (b) 12π kW
(c) 24π kW (d) 36π kW

APPSC AEE SCREENING 17.02.2019

Ans. (b) :

$$P = \frac{2\pi NT}{60}$$

$$= \frac{2\pi(200)(1800)}{60}$$

$= 12000\ (\pi)$
$= (12\ \pi)$ kW

960. A solid circular shaft of length 4 m is to transmit 3.5 MW at 200 rpm. If permissible shear stress is 50 MPa, the radius of the shaft is:

(a) 1.286 mm (b) 12.86 mm
(c) 0.1286 mm (d) 128.6 mm

BHEL ET 2019

Ans. (d) : Given -

Length = L = 4m
P = 3.5 MW
N = 200 rpm
$\tau = 50$ MPa, radius = r = ?
$\tau = 50 \times 10^3$ kPa

Power $P = T\omega$

$$3.5 \times 10^6 = T \times \frac{2\pi N}{60}$$

$$= T \times \frac{2\pi \times 200}{60}$$

$$3.5 \times 10^6 = 20.943\ T$$

$T = 167.120$ kN-m

Torsion equation-

$$\frac{T}{J} = \frac{T}{r}$$

$$r = \frac{\tau J}{T}$$

$$r = \frac{50 \times \frac{\pi d^4}{32}}{167.120}$$

$$\frac{d}{2} = \frac{50 \times \pi d^4}{5347.84}$$

$$\frac{5347.84}{50 \times 2 \times \pi} = d^3$$

$d = 0.25724$
$2r = 0.25724$
$r = 0.12862$ m

$\boxed{r = 128.6 \text{ mm}}$

961. The maximum shear stress developed on the surface of a solid circular shaft under pure torsion is 160 MPa. If the shaft diameter is doubled, then the maximum shear stress developed corresponding to the same torque will be:

(a) 10 MPa (b) 30 MPa
(c) 40 MPa (d) 20 MPa

BHEL ET 2019

Ans. (d) : Given

$(\tau_{max}) = 160$MPa
$d_1 = d$
$d_2 = 2d$

same torque $\boxed{T_1 = T_2 = T}$

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$$(\tau_{max})_1 = \frac{16T}{\pi d^3} \Rightarrow \quad 160 = \frac{16T}{\pi d^3} \qquad ...(1)$$

when diameter is doubled.

$$(\tau_{max})_2 = \frac{16T}{\pi(2d)^3} = \frac{16T}{8\pi d^3} \qquad ...(2)$$

from equation (1) / equation (2)

$$\frac{160}{(\tau_{max})_2} = \frac{16T}{\pi d^3} \times \frac{8\pi d^3}{16T}$$

$$\boxed{(\tau_{max})_2 = 20\,MPa}$$

962. A 50 mm diameter solid shaft is subjected to both, a bending moment and torque of 300 kN-mm & 200 kN-mm respectively. The maximum shear stress induced in the shaft is :

(a) 10.22 MPa (b) 14.69 MPa
(c) 146.9 MPa (d) 102.2 MPa

BHEL ET 2019

Ans. (b) : Given diameter of shaft d = 50 mm
Bending moment M = 300 kN-m
Torque T = 200 kN-m

$T_e = \sqrt{M^2 + T^2}$ be the equivalent torque, which acts alone producing the same maximum shearing stress

$$\tau_{max} = \frac{T_e}{\frac{\pi d^3}{16}} = \frac{16}{\pi d^3} T_e$$

$$\tau_{max} = \frac{16 \times 10^3}{\pi \times (50)^3} \sqrt{(300)^2 + (200)^2}$$

$$= \frac{16 \times 360.555 \times 10^3}{\pi \times 125000}$$

$$= \frac{5768880}{392699.081}$$

= 14.69 MPa

963. Which of the following is correct for flexible shaft?

(a) it has very low rigidity both in bending and torsion
(b) it has very high rigidity in bending and low rigidity in torsion
(c) it has low rigidity in bending and high rigidity in torsion
(d) It has very high rigidity both in bending and torsion

SJVN ET 2019

Ans. (c) : flexible shaft has low rigidity in bending and high rigidity in torsion.

964. A steel spindle transmits 4 kW at 800 r.p.m. The angular deflection should not exceed 0.25°/m length of the spindle. If the modulus of rigidity for the material of the spindle is 84 GPa, the diameter of the spindle will be

(a) 46 mm (b) 42 mm
(c) 38 mm (d) 34 mm

ESE 2019

Ans. (d) : Given, P = 4 kW
N = 800 rpm
G = 84 GPa

$$\theta = 0.25 \times \frac{\pi}{180} \times \frac{1}{1000}$$

$\theta = 4.363 \times 10^{-6}$ rad/mm

$$T = \frac{60P}{2\pi N}$$

$$= \frac{60 \times 4 \times 10^3}{2\pi \times 800} = 47.77 \text{ N-m}$$

$= 47.77 \times 10^3$ N-mm

$$\frac{\theta}{L} = \frac{T}{GJ}$$

$$\frac{4.363 \times 10^{-6}}{1} = \frac{47.77 \times 10^3}{84 \times 10^3 \times J}$$

J = 130343.90 mm^4

$$\frac{\pi}{32} d^4 = 130343.90$$

d = 33.949 mm
d = 34 mm

965. In a propeller shaft, sometimes apart from bending and twisting, end thrust will also develop stresses which would be

(a) Tensile in nature and uniform over the cross-section
(b) Compressive in nature and uniform over the cross-section
(c) Tensile in nature and non-uniform over the cross-section
(d) Compressive in nature and non-uniform over the cross-section

ESE 2019

Ans. (b) : Due to end thrust, stresses would be compressive in nature and uniform over cross section.

966. If a shaft is required to transmit twice the power at twice the speed for which it is designed, its diameter must

(a) increase two times (b) reduce two times
(c) remain same (d) increase three times

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (c) : We know that power transmitted by shaft

$P = T.\omega$

$$T = \frac{\pi D^3}{16} \tau$$

$$\frac{\pi \times D^3 \times \tau}{16} = \frac{P}{2\pi N}$$

If P' = 2P, N' = 2N
then D' = ?

$$\frac{\pi \times D'^3 \times \tau}{16} = \frac{2P}{2 \times 2\pi \times N} = \frac{P}{2\pi N} = \frac{\pi \times D^3 \times \tau}{16}$$

D' = D

Diameter must be same.

967. The power transmitted by a shaft 60 mm diameter at 180 RPM, if the permissible stress is 85 N/mm^2

(a) 68 kW (b) 650 kW
(c) 1200 kW (d) 7 kW

ISRO Scientist/Engineer (RAC) 22.04.2018

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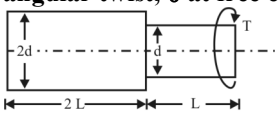
Ans. (a) : d = 60 mm, N = 180 RPM, τ = 85 MPa
We know that

$$P = T \times \omega = \frac{\pi d^3}{16} \times \tau \times \frac{2 \times \pi \times N}{60}$$

$$P = \frac{\pi \times 60^3 \times 85}{16} \times \frac{2 \times \pi \times 180}{60}$$

P = 67952226.3 N-mm/sec
P = 67.9 kW

968. Torque, *T* is applied at the free end of a stepped rod of circular cross section as shown in figure. If the shear modulus of the material is *G*, the angular twist, θ at free end will be



(a) $\frac{12TL}{G\pi d^4}$ (b) $\frac{24TL}{G\pi d^4}$
(c) $\frac{36TL}{G\pi d^4}$ (d) $\frac{48TL}{G\pi d^4}$

ISRO Scientist/Engineer 22.04.2018

Ans. (c): We know that, When Shafts are connected in series

$$\theta = \theta_1 + \theta_2$$

$$= \left(\frac{TL}{GJ}\right)_1 + \left(\frac{TL}{GJ}\right)_2$$

$$= \frac{T \times L \times 32}{G \times \pi \times d^4} + \frac{T \times 2L \times 32}{G\pi \times 2^4 \times d^4}$$

$$= \frac{T \times L \times 32}{G\pi \times d^4} + \left[1 + \frac{2}{2^4}\right]$$

$$\theta = \frac{36TL}{G\pi d^4}$$

969. The angle of twist of shaft is
(a) directly proportional to (shaft diameter)^2
(b) inversely proportional to (shaft diameter)^2
(c) directly proportional to (shaft diameter)^4
(d) inversely proportional to (shaft diameter)^4

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : As we know,

Angle of twist, $\phi = \frac{TL}{JG}$

$$J = \frac{\pi D^4}{32}$$

$$\phi \propto \frac{1}{J}$$

$$\Rightarrow \quad \phi \propto \frac{1}{D^4}$$

So, ϕ is inversely proportional to (shaft diameter)^4.(d^4)

970. Torsion bars are in parallel
(a) if same torque acts on each
(b) if they have equal angles of twist and applied torque apportioned between them
(c) are not possible
(d) if their ends are connected together

TNPSC AE 2018

Ans. (b) : Torsion bars are in parallel if they have equal angles of twist and applied torque apportioned between them.

$T = T_1 + T_2$
$\theta = \theta_1 = \theta_2$

971. Shearing stress produced on the surface of a solid shaft of diameter (d_0) is τ. The shear stress produced on the surface of hollow shaft of same material, subjected to same torque, and having the outer diameter d_0 and internal diameter d_0x is given as: [Where $x < 1$]

(a) $\frac{\tau}{1-x^4}$ (b) $\left(1-x^4\right)\tau$
(c) $\left(1-x^2\right)\tau$ (d) $\frac{\tau}{1-x^2}$
(e) $\frac{\tau}{1-2x^4}$

CGPSC AE 25.02.2018

Ans. (a) : Shear stress in solid shaft

$$\tau_s = \frac{T \times \frac{d_0}{2}}{\frac{\pi}{32} \times d_0^4} = \tau$$

Shear stress in hollow shaft is

$$\tau_H = \frac{T \times (d_o/2)}{\frac{\pi}{32} d_0^4 \left[1 - x^4\right]} \qquad [d_i = x d_o]$$

$$\tau_H = \frac{T \times (d_0/2)}{\frac{\pi}{32}\left(d_0^4\right)} \times \frac{1}{\left(1-x^4\right)}$$

$$\boxed{\tau_H = \frac{\tau}{1-x^4}}$$

972. A solid shaft is subjected to bending moment of 3.46kN-m and a torsional moment of 11.5kN-m. For this case, the equivalent bending moment and twisting moment are
(a) 7.73kN-m and 12.0kN-m
(b) 14.96kN-m and 12.0kN-m
(c) 7.73kN-m and 8.04kN-m
(d) 14.96kN-m and 8.04kN-m

ESE 2018

Ans. (a) : Given,
M = 3.46 kN-m
T = 11.5 kN-m
Equivalent bending moment is given by

$$M_e = \frac{1}{2}\left[M + \sqrt{M^2 + T^2}\right]$$

$$= \frac{1}{2}\left[3.46 + \sqrt{(3.46)^2 + (11.5)^2}\right]$$

= 7.734 kN-m
Equivalent twisting moment is given by

$$T_e = \sqrt{M^2 + T^2}$$

$$= \sqrt{(3.46)^2 + (11.5)^2} = 12 \text{ kN-m}$$

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973. A solid shaft can resist a bending moment of 6 kN m and a torque of 8 kN m applied together. The maximum torque that the shaft can resist when applied alone is

(a) 7 kN m (b) 48 kN m
(c) 10 kN m (d) 14 kN m

ISRO Scientist/Engineer 22.04.2018

Ans. (c): T = 8kN-m
M = 6kN-m
Maximum torque means equivalent torque T_{eq}.

$$T_{eq} = \sqrt{M^2 + T^2}$$
$$= \sqrt{8^2 + 6^2}$$
$$= \sqrt{100}$$
$$T_{eq} = 10 \text{ kN-m}$$

974. The diameter of shaft in any power transmission system is proportional to:

(a) speed of the shaft
(b) horse power to be transmitted
(c) torque to be transmitted
(d) allowable shear stress of the material

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : The diameter of shaft in any power transmission system is proportional to horse power to be transmitted power $P = T \times \frac{2\pi N}{60}$

and $\tau = \frac{16T}{\pi d^3}$ or $T = \frac{\tau \pi d^3}{16}$

$$P = \frac{\tau \pi d^3}{16} \times \frac{2\pi N}{60}$$
$$d^3 = \frac{480P}{\pi^2 JN}$$
$$d \propto \left(\frac{P}{N}\right)^{1/3}$$

So, diameter is directly proportional to power.

975. A 3 m long steel shaft has to transmit 7.5 kW at 3600 rpm. Required shaft diameter is given by (Take allowable shear stress = 100 N/mm^2, π^2 = 10)?

(a) 10 mm (b) 7.5 mm
(c) 25 mm (d) 12.5 mm

ISRO Scientist/Engineer 17.12.2017

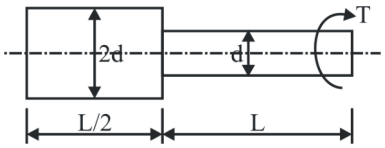
Ans. (a) : Given,
L = 3 m, P = 7.5 kW = 7500 W
N = 3600 rpm τ = 100 N/mm^2
π^2 = 10
We know that

$$P = \frac{2\pi NT}{60}$$
$$7500 = \frac{2\pi \times 3600 \times T}{60}$$
$$T = \frac{375}{6\pi} \text{N} - \text{m} = \frac{375 \times 10^3}{6\pi} \text{N} - \text{mm}$$

From torsion equation,

$$\frac{16T}{\pi d^3} = \tau$$
$$\frac{16}{\pi d^3} \times \frac{375 \times 10^3}{6\pi} = 100$$
$$d^3 = \frac{16 \times 375 \times 10^3}{\pi^2 \times 6 \times 100}$$
$$d^3 = 1013.21$$
$$d = 10.04 \text{ mm}$$

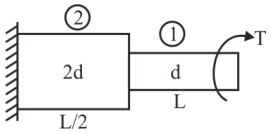
976. A torque T is applied at the free end of a stepped rod of circular cross-sections as shown in the figure. The shear modulus of the material of the rod is G. The expression for diameter 'd' to produce an angular twist θ at the free end is



(a) $\left(\frac{30TL}{\pi\theta G}\right)^{1/4}$ (b) $\left(\frac{33TL}{\pi\theta G}\right)^{1/4}$
(c) $\left(\frac{18TL}{\pi\theta G}\right)^{1/4}$ (d) $\left(\frac{24TL}{\pi\theta G}\right)^{1/4}$

ISRO Scientist/Engineer 07.05.2017

Ans. (b) :



$$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R}$$
$$\boxed{\theta = \frac{TL}{GJ}}$$
$$T = T_1 = T_2$$
$$\theta = \theta_1 + \theta_2$$
$$= \frac{T_1 L_1}{G_1 J_1} + \frac{T_2 L_2}{G_2 J_2} = \frac{TL}{G\frac{\pi d^4}{32}} + \frac{T \times \frac{L}{2}}{G \times \frac{\pi}{32}(16d^4)}$$
$$= \frac{TL}{G\pi d^4}\left[32 + \frac{1}{2} \times \frac{32}{16}\right]$$
$$\theta = \frac{TL}{G\pi d^4}[33]$$
$$d^4 = \frac{TL}{G\pi\theta}[33]$$
$$d = \left[\frac{TL}{G\pi\theta}(33)\right]^{\frac{1}{4}}$$

977. The torque (T) transmitted by sleeve coupling with D and d as outer and inner diameters, respectively, is given by:

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(a) $T = \frac{\pi}{8} \times \tau \left(\frac{D^4 - d^4}{D} \right)$

(b) $T = \frac{\pi}{12} \times \tau \left(\frac{D^4 - d^4}{D} \right)$

(c) $T = \frac{\pi}{16} \times \tau \left(\frac{D^4 - d^4}{D} \right)$

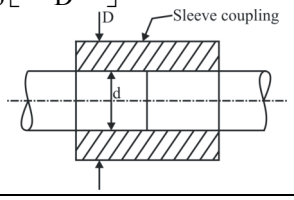
(d) $T = \frac{\pi}{32} \times \tau \left(\frac{D^4 - d^4}{D} \right)$

(e) $T = \frac{\pi}{64} \times \tau \left(\frac{D^4 - d^4}{D} \right)$

(CGPCS Polytechnic Lecturer 2017)

Ans. (c) : Torque transmitted by sleeve coupling

$$T = \frac{\pi}{16}\left[\frac{D^4 - d^4}{D}\right] \times \tau$$



978. The boring bar of a boring machine is 25 mm in diameter. During operation, the bar gets twisted though 0.01 radians and is subjected to a shear stress of 42 N/mm^2. The length of the bar is (Taking G = 0.84 × 10^5 N/mm^2)

(a) 500 mm (b) 250 mm

(c) 625 mm (d) 375 mm

GPSC EE Pre, 28.01.2017

Ans. (b) : Given-

Diameter (D) = 25 mm

$$R = \frac{25}{2} \text{ mm}$$

Twist (θ) = 0.01 radian

τ = 42 N/mm^2

G = 0.84 × 10^5 N/mm^2

From the equation of torsion

$$\frac{T}{J} = \frac{\tau_{max}}{R} = \frac{G\theta}{l}$$

$$\frac{\tau_{max}}{R} = \frac{G\theta}{l}$$

$$l = \frac{G\theta \times R}{\tau_{max}}$$

$$= \frac{0.84 \times 10^5 \times 0.01 \times \frac{25}{2}}{42}$$

l = 250 mm

979. A circular shaft of 60 mm diameter is running at 150 r.p.m. What will be the torque transmitted by the shaft if (τ = 80 MPa)?

(a) 1.1 π kN-m (b) 1.6π kN-m

(c) 2.1π kN-m (d) 2.6π kN-m

(e) 3.1π kN-m

(CGPCS Polytechnic Lecturer 2017)

Ans. (a) : d = 60 mm

N = 150 rpm

τ = 80 MPa = 80 N/mm^2

We know that

$$T = \frac{\pi d^3}{16} \times \tau = \frac{\pi \times 60 \times 60 \times 60}{16} \times 80 \text{ N} - \text{mm}$$

T = 1.08π kN-m ≃ 1.1π kN-m

980. A solid shaft is to transmit 20 kW at 200 r.p.m. The ultimate shear stress for the shaft material is 360 MPa and the factor of safety is 8. The diameter of the solid shaft shall be

(a) 42 mm (b) 45 mm

(c) 48 mm (d) 51 mm

ESE 2017

Ans. (c) : Given,

P = 20 kW = 20 × 10^3 W

N = 2000 rpm

τ_u = 360 MPa

$$T = \frac{60P}{2\pi N} = \frac{60 \times 20}{2\pi \times 200} = \frac{3}{\pi} \text{kN} - \text{m}$$

$$\tau = \frac{16T}{\pi d^3} = \frac{\tau_u}{FS} = \frac{16 \times \frac{3}{\pi}}{\pi \times d^3} = \frac{360}{8}$$

d = 47.6 mm

981. Maximum shear stress in the shaft if subjected to combine twisting moment (M_t) and bending moment (M_b)–

(a) $\frac{16}{\pi d^3}\sqrt{(M_b)^2 + (M_t)^2}$

(b) $\frac{16}{\pi d^3}\sqrt{(M_b)^2 - (M_t)^2}$

(c) $\frac{32}{\pi d^3}\sqrt{(M_b)^2 + (M_t)^2}$

(d) $\frac{32}{\pi d^3}\sqrt{(M_b)^2 - (M_t)^2}$

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Maximum, shear stress in the shaft.

$$\tau_{max} = \frac{16}{\pi d^3}\sqrt{(M_b)^2 + (M_t)^2}$$

982. A hollow shaft of 20 mm diameter and 16 mm inside diameter is subjected to a torque of 40 Nm. The shear stress at outside of the shaft will be:

(a) 53.12 N/mm^2 (b) 43.13 N/mm^2

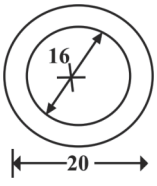
(c) 62.52 N/mm^2 (d) 34.50 N/mm^2

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (b) : D = 20 mm

d = 16 mm

Torque T = 40 N-m



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Shear stress τ

$$= \frac{16T}{\pi D^3 (1-K^4)}$$

where

$$K = \frac{d}{D} = \frac{16}{20}$$

$$\tau = \frac{16 \times 40 \times 1000}{\pi \times (20)^3 \times \left[1 - \left(\frac{16}{20}\right)^4\right]}$$

$\tau = 43.13$ N/mm^2

983. Two steel shafts, one solid of diameter D and other hollow of outside diameter D and inside diameter D/2 are twisted to the same angle of twist per unit length. Ratio of maximum shear stress in solid shaft to that in the hollow shaft is

(a) $\frac{4}{9}$ (b) $\frac{8}{7}$

(c) $\frac{16}{15}$ (d) 1

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-1998

Ans. (d) : Torsion formula

$$\frac{\tau}{r_{max}} = \frac{G\theta}{\ell}$$

for solid shaft $r_{max} = D/2$

for hollow shaft $r_{max} = D/2$

Since both are made of steel

$\therefore$ G remains same

Given angle of twist per unit length is ame

$\therefore \frac{\theta}{\ell}$ for both shafts is same

$$\frac{\delta_s}{\delta_h} = \frac{(r_{max})_s}{(r_{max})_h} = \frac{D/2}{D/2} = 1$$

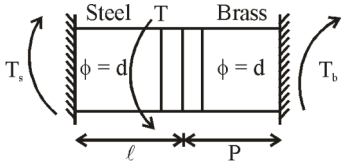
984. Steel shaft and brass shaft of same length and diameter are connected by a flange coupling. The assembly is rigidly held at its ends and is twisted by a torque through the coupling. Modulus of rigidity of steel is twice that of brass. If torque of the steel shaft is 500 Nm, then the value of the torque in brass shaft will be

(a) 250 Nm (b) 354 Nm

(c) 500 Nm (d) 708 Nm

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-2001

Ans. (a) :



$G_S = 2\ G_B$

$T_S = 500$ NM

$T_S + T_b = T$ ——— (1)

Also angle of twist at coupling is equal for both shafts

$$\frac{T_s \ell}{G_s j} = \frac{T_b \ell}{G_b J}$$

(ℓ, are equal for both shafts)

$$\frac{T_s}{T_b} = \frac{G_s}{G_b} = 2 \qquad \text{——— (2)}$$

$$T_b = \frac{500}{2} = 250\text{Nm}$$

985. When subjected to a torque, a circular shaft undergoes a twist of 1° in a length of 1200 mm, and the maximum shear stress induced is 80 N/mm^2. The modulus of rigidity of the material of the shaft is 0.8×10^5 N/mm^2. What is the radius of the shaft?

(a) $90/\pi$mm (b) $108/\pi$mm

(c) $180/\pi$mm (d) $216/\pi$mm

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : $\tau = 80$ N/mm^2

$$\frac{\theta}{\ell} = \frac{1^\circ}{1200} = \frac{1}{1200} \times \frac{\pi}{180} \qquad \text{(in radian)}$$

$G = 0.8 \times 10^5$ N/mm^2

Torsion formula

$$\frac{\tau}{r_{max}} = \frac{G\theta}{\ell}$$

$$\frac{80}{r_{max}} = \frac{0.8 \times 10^5 \times 1 \times \pi}{1200 \times 180}$$

$$r_{max} = \frac{1200 \times 180}{10^5 \pi} = \frac{216}{\pi}$$

986. Power is transmitted through a shaft, rotating at 2.5 Hz (150 rpm). The mean torque on the shaft is 20×10^3 Nm. What magnitude of power in kW is transmitted by the shaft?

(a) 50π (b) 120π

(c) 100π (d) 150π

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : N = 150 rpm

$T = 20 \times 10^3$ Nm

$$P = \frac{2\pi NT}{60 \times 1000} \text{kW}$$

$$= \frac{2\pi \times 150 \times 20 \times 10^3}{60 \times 1000}$$

$$= \frac{100\pi \times 10^3}{1000} = 100\pi \text{ kW}$$

987. A shaft of circular section is said to be in pure torsion when it is subjected to equal and opposite end couples whose axes coincides with the

(a) axis of the shaft

(b) normal line to the axis of the shaft

(c) any line passing through the section of the shaft

(d) parallel line to the axis of the shaft

APPSC AE Subordinate Service Civil/Mech. 2016

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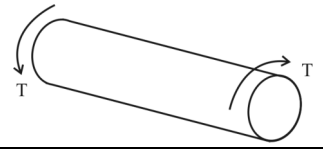
Ans. (a) : A shaft of circular section is said to be in pure torsion when it is subjected to equal and opposite end couples whose axis coincides with the axis of the shaft.



988. A tubular shaft, having an inner diameter of 30 mm and an outer diameter of 40 mm, is to be used to transmit 80 kW of power. The speed of rotation of the shaft so that the shear stress will not exceed 50 MPa is

(a) 29.6 rpm (b) 3557.4 rpm
(c) 1778.7 rpm (d) 59.2 rpm

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Inner diameter $d_i = 30$ mm
Outer diameter $d_o = 40$ mm

$P = 80$ kW

$\tau = 50$ MPa

$$T = \frac{P(kW)\times 60}{2\pi N(rpm)}\times 10^6 \quad \text{(N-mm)}$$

$$T = \frac{80\times 60}{2\pi\times N}\times 10^6$$

$$T = \frac{2400\times 10^6}{\pi\times N} = \frac{24\times 10^8}{\pi N} \quad \text{(N-mm)}$$

$$\boxed{\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R}}$$

$$\boxed{\frac{T}{J} = \frac{\tau}{R}}$$

For solid–

$$\tau_{max} = \frac{16T}{\pi d^3}$$

$$\tau_{max} = \frac{16T}{\pi d^3(1-k^4)}$$

Where $k = \frac{d_i}{d_o}$

$$50\times 10^6 = \frac{\frac{16\times 24\times 10^8}{\pi N}}{\pi d_0^3(1-k^4)}$$

$$k = \frac{d_i}{d_o} = \frac{30}{40}$$

$$k^4 = (0.75)^4$$

$$50\times 10^6 = \frac{16\times 24\times 10^8}{\pi^2\times(40)^3\times\left[1-(0.75)^4\right]\times N}$$

$$50 = \frac{16\times 24\times 10^2}{\pi^2\times 64000\times 0.68\times N}$$

$$N = \frac{16\times 24\times 100}{50\times 64000\times 0.68\times\pi^2}$$

$N = 1778.7$ rpm

989. A hollow circular shaft of outside and inside diameter 100 mm and 90 mm is subjected to a torque $T = 3\pi$ kNm. Polar moment of inertia of circular section is $\pi\times 10^{-6}$ m^4. Maximum shear stress on the shaft is given by:

(a) 175 MPa (b) 100 MPa
(c) 150 MPa (d) 125 MPa

UPRVUNL AE 07.10.2016

Ans. (c) : Data given,
$d_o = 100$ mm
$d_i = 90$ mm
$T = 3\pi$ k N-m
$J = \pi\times 10^{-6}$ m^4

$$\frac{T}{J} = \frac{\tau}{\left[\frac{d_o}{2}\right]} = \frac{G\theta}{l}$$

$$\tau = \frac{3\pi\times 10^3\times 0.05}{\pi\times 10^{-6}}$$

$\tau = 150$ MPa

990. For shaft in torsion

(a) $\frac{q}{r} = \frac{T}{J}$ (b) $\frac{C\theta}{\ell} = \frac{T}{J}$
(c) $\frac{q}{r} = \frac{C\theta}{\ell}$ (d) All the above

APPSC AEE Screening Test 2016

Ans. (d)

$$\frac{T}{J} = \frac{C\theta}{\ell} = \frac{q}{r}$$

Where,
C = Modulus of rigidity
ℓ = Length
T = Torque
q = Shear stress

991. The ratio of maximum shear stress of a solid shaft of diameter D to that of a hollow shaft having external diameter D and internal diameter 0.5D is

(a) 0.50 (b) 1.00
(c) 1.50 (d) 2.00

APPSC AEE Screening Test 2016

Ans. (*) : Given,
Inner diameter = 0.5D
Outer diameter = D, solid shaft dia = D

$$\text{Ratio} = \frac{\tau_{solid}}{\tau_{hollow}} = \frac{\frac{16T}{\pi D^3}}{\frac{16T}{\pi D^3\left[1-\left(\frac{d}{D}\right)^4\right]}}$$

$$= \frac{\pi D^3\left[1-\left(\frac{0.5D}{D}\right)^4\right]}{\pi D^3}$$

$= [1-(0.5)^4]$
$= (1-0.0625) = 0.9375 \simeq 1$

Note-Official answer given by the commission is (d).

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992. Which of the following assumption in the theory of pure torsion is false

(a) All radii get twisted due to torsion
(b) the twist is uniform along the length
(c) the shaft is uniform circular section through out
(d) cross section plane before torsion remain plane after torsion

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : Assumption in the theory of Torsion–

1. The material of shaft is uniform throughout the length.
2. The twist along the shaft is uniform.
3. The shaft is of uniform circular section throughout the length.
4. Cross-section of the shaft, which are plane before twist remain plane after twist.
5. All radii which are straight before twist remain straight after twist.

993. In a rectangular shaft is subjected to torsion, the maximum shear stress will occur

(a) Along the diagonal
(b) At the corners
(c) At the centre
(d) At the middle of the longer side

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : For a rectangular shaft is subjected to torsion the maximum shear stress will occur at the middle of the longer side.

994. Three shafts (spring constant k_1, k_2, k_3) are connected in series such that they carries the same torque (T), then spring constant (k) for composite shaft will be

(a) $k = k_1 + k_2 + k_3$

(b) $k = (k_1k_2 + k_2k_3 + k_3k_1)^{1/2}$

(c) $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$

(d) $k = \left(\frac{k_1k_2k_3}{k_1 + k_2 + k_3}\right)^{1/2}$

RPSC LECTURER 16.01.2016

Ans. (c) : In series, same torque acting on each three shafts

$T_1 = T_2 = T_3 = T$

$\theta = \theta_1 + \theta_2 + \theta_3$ ...(1)

$\theta = \frac{GJ}{TL}$

Equation No. (1) divided by T

$$\frac{\theta}{T} = \frac{\theta_1}{T} + \frac{\theta_2}{T} + \frac{\theta_3}{T}$$

$$\frac{1}{\left(\frac{T}{\theta}\right)} = \frac{1}{\left(\frac{T}{\theta_1}\right)} + \frac{1}{\left(\frac{T}{\theta_2}\right)} + \frac{1}{\left(\frac{T}{\theta_3}\right)}$$

$$\frac{T}{\theta} = K = \frac{GJ}{L}$$

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

995. The torque transmitted by a solid shaft of diameter 40 mm if the shear stress is not to exceed 400 N/cm^2, would be :

(a) $1.6 \times \pi$ N-m (b) 16π N-m
(c) $0.8 \times \pi$ N-m (d) $0.4 \times \pi$ N-m

HPPSC W.S. Poly. 2016

Ans : (b) Shaft diameter = 40mm

Shear stress (τ_{max}) = 400N/cm^2

$$\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{r}$$

$$T = \frac{\tau.J}{r}$$

$$T = \frac{\pi d^3 \tau}{16}$$

$$T = \frac{\pi \times 40^3 \times 400}{16 \times 1000 \times 100}$$

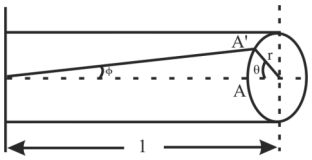
$T = 16\pi$ N-m

996. A shaft of 20 mm diameter and length 1 m is subjected to a twisting moment, due to which shear strain on the surface of the shaft is 0.001. The angular twist in the shaft is

(a) 0.1 radian (b) 0.01 radian
(c) 0.05 radian (d) 0.5 radian

UPPSC AE 12.04.2016 Paper-I

Ans : (a)



ℓ = Shaft length = 1000 mm

θ = twist angle

ϕ = shear strain = 0.001

r = shaft radius

$$\phi = \frac{AA'}{\ell} \quad .....(i)$$

$$\theta = \frac{AA'}{r} \quad .....(ii)$$

for equation (i) and (ii)

$$\phi\ell = \theta r$$

$$\theta = \frac{\phi\ell}{r}$$

$$\theta = \frac{0.001 \times 1000}{\frac{20}{2}}$$

$$\boxed{\theta = 0.1 \text{ radian}}$$

997. A shaft of J polar moment of Inertia and C modulus of rigidity is fixed at one end and subjected to torque T at the free end and the same torque at mid length in opposite direction as shown in figure, then the difference in the twist between the free end and the midpoint is equal to

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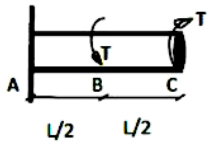
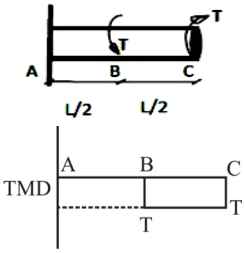


(a) $\frac{T}{2CJ}$ (b) $\frac{3T}{2CJ}$

(c) $\frac{T}{CJ}$ (d) $\frac{T}{4\,CJ}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Beam divide this Shaft into two parts AB & BC



$\theta_C - \theta_B$ is asked $\theta_B = 0$ $\therefore [T_B = T_C]$

& No torque at AB portion.

$$\therefore \Rightarrow \theta = \frac{TL}{CJ}$$

$$\theta_C - \theta_B = \frac{T \times \frac{L}{2}}{CJ} = \frac{TL}{2CJ}$$

998. Polar moment of inertia of a hollow tube of diameter 'd' and thickness 't' is

(a) $\frac{\pi d^3 t}{32}$ (b) $\frac{\pi d^3 t}{16}$

(c) $\frac{\pi d^3 t}{4}$ (d) $\frac{\pi d^4}{16}$

RPSC ADE 2016

Ans. (c) :

$I_P = \frac{\pi d^3 t}{4}$

d = diameter

t = thickness

999. The following torsional terms is analogous to bending terms except

(a) T/J corresponds to M/I

(b) G/ℓ corresponds to E/R

(c) τ/r corresponds to σ/y

(d) Gθ/ℓ corresponds to E/R

RPSC VPITI 14.02.2016

Ans. (b) : Bending moment equation $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$

Torsion equation $\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{l}$

$\frac{T}{J}$ corresponds to $\frac{M}{I}$

$\frac{\tau}{r}$ corresponds to $\frac{\sigma}{y}$

$\frac{G\theta}{l}$ corresponds to $\frac{E}{R}$

1000. The unit of torsional stress in SI units is :

(a) Kg/m^2 (b) Kg-m

(c) N/m^2 (d) N-m

HPPSC Lect. (Auto) 23.04.2016

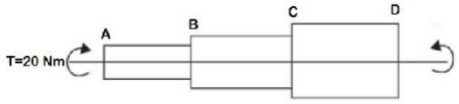
Ans. (c) : The unit of torsional stress in SI units is N/m^2

torsional shear stress

$\tau = \frac{16T}{\pi d^3}$ T = torque transmitted

d = diameter of shaft

1001. The torsional stiffness of individual sections of lengths AB, BC and CD are 40 N-m/rad, 60 N-m/rad and 120 N-m/rad respectively.



What is the angular deflection between the ends A and D of the shaft?

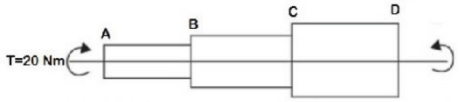
(a) 0.5 rad (b) 0.1 rad

(c) 1 rad (d) 5 rad

(e) 10 rad

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know,



$\frac{T}{\theta}$ = Torsional stiffness

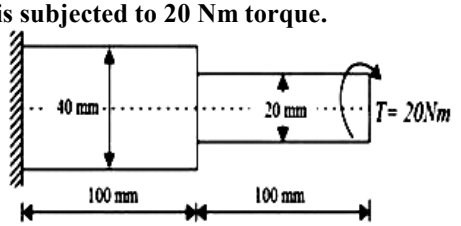
$$\theta_{AB} = \frac{1 \times 20}{40}, \quad \theta_{BC} = \frac{1 \times 20}{60}$$

$$\theta_{CD} = \frac{1 \times 20}{120}$$

So, $\theta_{AD} = \theta_{AB} + \theta_{BC} + \theta_{CD} = \frac{1 \times 20}{40} + \frac{1 \times 20}{60} + \frac{1 \times 20}{120}$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1 \text{rad}$$

1002. A stepped steel shaft shown in the figure below is subjected to 20 Nm torque.



If the modulus of rigidity is 80 GPa, then what is the strain energy, in N-mm, in the shaft?

(a) 6.4 (b) 10.8

(c) 12.7 (d) 16.9

(e) 23.7

CGPSC Asstt. Workshop Supt., 17.07.2016

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Ans. (d) : Given,
Torque (T) = 20 Nm
Modulus of rigidity (G) = 80 GPa
From figure,

$$T_1 = T_2 = T = 20 \text{ Nm}$$

$$\theta = \theta_1 + \theta_2$$

$$\theta = \frac{T_1 L_1}{J_1 G_1} + \frac{T_2 L_2}{J_2 G_2}$$

Here, $T_1 = T_2 = 20 \times 10^3$ N-mm

$L_1 = L_2 = 100$ mm

$G_1 = G_2 = 80 \times 10^3$ N/mm^2

$$\theta = \frac{TL}{G}\left[\frac{1}{J_1} + \frac{1}{J_2}\right]$$

$$\theta = \frac{20 \times 1000 \times 100}{80 \times 10^3}\left[\frac{32}{\pi \times 20^4} + \frac{32}{\pi \times 40^4}\right]$$

$$\theta = \frac{20 \times 1000 \times 100 \times 32}{\pi \times 80 \times 10^3}\left[\frac{1}{20^4} + \frac{1}{40^4}\right]$$

$$\theta = 1.69102 \times 10^{-3}$$

$$\text{Strain energy} = \frac{1}{2} T \times \theta$$

$$= \frac{1}{2} \times 20 \times 1000 \times 1.69102 \times 10^{-3}$$

Strain energy = 16.91 N-mm

1003. If the diameter of shaft is doubled its stiffness increases by a factor of:

(a) 2 (b) 6
(c) 16 (d) 24

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Torsional stiffness $(k_t) = \frac{GJ}{\ell} = \frac{G}{\ell}\frac{\pi}{32} d^4$

$$\boxed{k_t \propto d^4}$$

$$d_2 = 2d_1$$

$$\frac{(k_t)_2}{(k_t)_1} = \left(\frac{d_2}{d_1}\right)^4 = 16$$

$$\boxed{(k_t)_2 = 16(k_t)_1}$$

1004. Polar modulus is the ratio of

(a) Polar moment of inertia of the shaft to the maximum radius
(b) Moment of inertia of the shaft to the maximum radius
(c) Polar moment of inertia of the shaft to the permissible shear stress
(d) Moment of inertia of the shaft to the maximum shear stress

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Polar modulus is defined as the ratio of polar moment of inertia of the shaft to the maximum radius of the shaft. It is also called as torsional section modulus.

$$\boxed{Z_p = \frac{J}{R}}$$

1005. Torsional flexibility is the reciprocal of

(a) Torsional stiffness (b) Torsional rigidity
(c) Polar modulus (d) Twisting moment

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Torsional flexibility → It is defined as the angle of twist produced by the unit torque applied. It is reciprocal of torsional stiffness.

$$\text{Torsional flexibility} = \frac{L}{GJ} = \frac{\theta}{T}$$

$$\text{Torsional stiffness} = \frac{GJ}{L} = \frac{T}{\theta}$$

1006. A shaft is subjected to the combined bending load 'M' and tensional load 'T', the maximum shear stress at the outer surface of the shaft will be equal to

(a) $16T / \pi d^3$
(b) $32M / \pi d^3$
(c) $16(M^2 + T^2)^{1/2} / \pi d^3$
(d) $32(M^2 + T^2)^{1/2} / \pi d^3$

VIZAG MT 2015

Ans. (c) : $\tau_{max} = \frac{16}{\pi d^3}\left[\sqrt{M^2 + T^2}\right]$

1007. In a shaft with a transverse hole, as the hole to the shaft diameter ratio increases, the torsional stress concentration factor-

(a) Increases (b) Remains constant
(c) Decreases (d) Is equal to one

NPCIL ET, 2015

Ans. (c) : • As the hole to shaft diameter increase then the torsional stress concentration factor decrease.

1008. A composite shaft, consisting of two stepped portions having spring constants of K_1 and K_2, is held firmly at one end and other end is free and subjected to a torque T. Its equivalent spring constant is:

(a) $(K_1 + K_2)/2$ (b) $(K_1 K_2) / (K_1 + K_2)$
(c) $(K_1 + K_2)$ (d) $(K_1 + K_2) / (K_1 K_2)$

NPCIL ET, 2015

Ans. (b) :

$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2}$$

$$\frac{1}{K_{eq}} = \frac{K_1 + K_2}{K_1 K_2}$$

$$K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$$

1009. For two geometrically similar shafts of the same material having diameter ratio 2 : 1, if the smaller one has a torque capacity of 1 kg-m, then the larger one will carry.

(a) 2 kg-m
(b) 4 kg-m
(c) 8 kg-m
(d) 1 kg-m because of similarity

BPSC Asstt. Prof. 29.11.2015

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Ans. (c) : $T_1 = \frac{\pi}{16} \quad \tau(d_1)^3$

$$T_2 = \frac{\pi}{16}\tau(d_2)^3$$

$$\frac{T_1}{T_2} = \frac{\frac{\pi}{16}\tau(d_1)^3}{\frac{\pi}{16}\tau(d_2)^3}$$

$$\frac{T_1}{T_2} = \frac{(d_1)^3}{(d_2)^3}$$

$$\frac{T_1}{1} = \frac{(2)^3}{(1)^3}$$

$$\boxed{T_1 = 8\ kgm}$$

1010. In a hollow circular shaft of outer and inner diameters of 20 cm and 10 cm respectively, the shear stress is not to exceed 40 N/mm^2. The maximum torque which the shaft can safely transmit is

(a) 58.9 kN-m (b) 57.9 kN-m
(c) 56.9 kN-m (d) 58.7 kN-m

MPPSC AE 08.11.2015

Ans. (a) : Given, for hollow circular shaft
Outer dia. (d_o) = 20 cm = 0.2 m
Inner dia. (d_i) = 10 cm = 0.1 m
Shear stress (τ) = 40 N/mm^2 = 40 × 10^6 N/m^2
Let, T = maximum torque

$$\therefore\ T = \frac{\pi}{16} d_o^{.3}\left(1 - K^4\right).\tau\ ,\ \left(\text{Where, } K = \frac{d_i}{d_o} = \frac{0.1}{0.2} = \frac{1}{2}\right)$$

$$= \frac{\pi}{16} \times (0.2)^3 \times \left[1 - \left(\frac{1}{2}\right)^4\right] \times 40 \times 10^6$$

= 58.90 × 10^3 N-m
T = 58.90 kN-m

1011. Angle of twist allowed in case of camshaft is :

(a) Dependent on its length
(b) Restricted to ½ degree irrespective of length of the shaft
(c) Depending on the torque acting on it
(d) Dependent on the nature of the engine (i.e. 4 stroke or 2 stroke)

OPSC AEE 2015 Paper-I

Ans : (c) Angle of twist allowed in case of camshaft is Depending on the torque acting on it.

1012. A solid circular shaft of 60 mm diameter and 10 m length, transmits a torque of 2000 N-m. The value of maximum angular deflection, if the modulus of rigidity is 100 GPa is

(a) 18 degree (b) 13 degree
(c) 15 degree (d) 9 degree
(e) 5 degree

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Given, d = 60 mm
l = 10 m = 10000 mm
G = 100 GPa = 100 × 10^3 N/mm^2
T = 2000 N-m = 2000 × 10^3 N-mm

From torsion equation

$$\frac{T}{J} = \frac{G\theta}{l}$$

$$\theta = \frac{T.l}{GJ} = \frac{2000 \times 10}{100 \times 10^9 \times \frac{\pi}{32}(0.06)^4} = 0.157 \text{ radian}$$

$$\theta = 0.157 \times \frac{180}{\pi} = 9 \text{ degree}$$

1013. In case of shaft design, one of the following equation is known as stiffness equation:

(a) $\frac{T}{J} = \frac{\tau}{R}$ (b) $\frac{M}{J} = \frac{\sigma}{R}$
(c) $\frac{M}{I} = \frac{\tau}{R}$ (d) $\frac{T}{J} = \frac{G\theta}{L}$
(e) $\frac{M}{I} = \frac{G\theta}{L}$

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Equation $\left(\frac{T}{J} = \frac{G\theta}{L}\right)$ is known as stiffness equation.

1014. 100 kW is to be transmitted by each of two separate shafts 'A' and 'B'. 'A' is rotating at 250 rpm and 'B' at 300 rpm. Which shaft must have greater diameter.

(a) A
(b) B
(c) Both will have same diameter
(d) Unpredictable

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : We know that

$$N \propto \frac{1}{D}$$

then, $\frac{D_A}{D_B} = \frac{N_B}{N_A} = \frac{300}{250}$

then, $D_A > D_B$

1015. The strength of a hollow shaft for the same length, material and weight is ..... a solid shaft:

(a) Less than (b) More than
(c) Equal to (d) None of these

OPSC AEE 2015 Paper-I

Ans : (b) The strength of a hollow shaft for the same length, material and weight is more than a solid shaft.
When the shaft is subjected to pure torsional moment (T). the torsional shear stress is given by

for solid shaft : $\tau = \frac{16T}{\pi d^3}$

For hollow shaft : $\tau = \frac{16T}{\pi d_0^3\left(1 - C^4\right)}$

1016. A shaft of 10 mm diameter, whose maximum shear stress is 48 N/mm^2 can produce a maximum torque equal to

(a) 2000 π N - mm (b) 4000 π N - mm
(c) 1000 π N - mm (d) 3000 π N-mm

TSPSC AEE 2015

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Ans : (d) Shaft dia (d) = 10 mm
Max shear stress (τ_{max}) = 48 N/mm^2

$$\tau_{max} = \frac{16T}{\pi d^3}$$

$$T = \frac{\tau \times \pi d^3}{16}$$

$$T = \frac{48 \times \pi \times 10^3}{16}$$

$$T = 3000\pi \text{ N-mm.}$$

1017. The equivalent twisting moment to design a shaft subjected to the fluctuating loads will be given by

(a) $\sqrt{(K_t M)^2 + (K_m T)^2}$

(b) $\sqrt{(K_m M)^2 + (K_t T)^2}$

(c) $\sqrt{K_m M + (K_m M)^2 + (K_t T)^2}$

(d) $\frac{1}{2}\left[K_m M + \sqrt{(K_m M)^2 + (K_t T)^2}\right]$

TSPSC AEE 2015

Ans : (b) (i) Equivalent Bending moment

$$M_e = \frac{1}{2}\left[K_m M + \sqrt{(K_m M)^2 + (K_t T)^2}\right]$$

(ii) Equivalent twisting moment

$$T_e = \sqrt{(K_m M)^2 + (K_t T)^2}$$

K_m = Shock and fatigue factor for bending
k_t = shock and fatigue factor for twisting

1018. If a body is transmitting torque T kgm at angular speed of θ radians/sec, then h.p. transmitted will be-

(a) Tθ/75 (b) T/θ
(c) Tθ/50 (d) Tθ

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) We know that

$$\text{Power (P)} = T \times \omega = T \times \omega \times g\left(\frac{N-m}{sec}\right)$$

1 h.p.= 746W

$$1W = \frac{1}{746}hp$$

$$P = \frac{9.81T \times \theta}{746} \qquad \omega = \theta \text{ radian/sec}$$

$$\boxed{P = \frac{T \times \theta}{75} \text{h.p.}}$$

1019. A shaft subjected to fluctuating loads for which the normal torque (T) and bending moment (M) are 1000 N-m and 500 N-m respectively. If the combined shock and fatigue factor for bending is 1.2 and combined shock and fatigue factor for torsion is 2 then the equivalent twisting moment for the shaft is_____

(a) 2088 N-m (b) 2050 N-m
(c) 2136 N-m (d) 2188 N-m

(HPPSC AE 2014)

Ans. : (a) Equivalent twisting moment (T_e)

$$T_e = \sqrt{(K_m \times M)^2 + (K_t \times T)^2}$$

M = Bending moment = 500 N-m
T = Twisting moment = 1000 N-m
K_m = fatigue factor for bending = 1.2
K_t = fatigue factor for torsion = 2

$$T_e = \sqrt{(1.2 \times 500)^2 + (2 \times 1000)^2}$$

T_e = 2088 N-m

1020. Strain energy stored in a solid circular shaft is proportional to (GJ is torsional rigidity)

(a) GJ (b) 1/GJ
(c) 1/(GJ)^2 (d) (GJ)^2

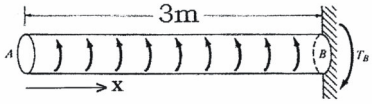
Mizoram PSC AE/SDO 2014, Paper-II

Ans. (b) : Strain energy stored in a solid circular shaft is proportional to–

$$U = \frac{1}{2}\frac{T^2 L}{GJ}$$

$$\boxed{U \propto \frac{1}{GJ}}$$

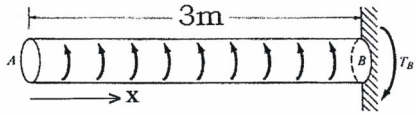
1021. Shaft AB is 3 meter long and its free at the left end and fixed torsionally at the right end. It is subjected to variable torque as shown in the figure. The intensity of the torsional loading is given by q = 2x. The internal torque T, as a function of x is



(a) $T = x^2$ (b) $T = 2x^2$
(c) $T = 4x^2$ (d) $T = 16x^2$

ISRO Scientist/Engineer 24.05.2014

Ans. (a) :



The intensity of the torsional loading q = 2x
The internal torque 'T' as the function of 'x'.

$$T = \int q\,dx$$

$$= \int 2x\,dx$$

$$= \left[\frac{2x^2}{2}\right]$$

$$T = x^2$$

1022. The variation of shear in a circular shaft subjected to torsion is :

(a) linear (b) parabolic
(c) hyperbolic (d) None of these

MPPSC State Forest Service Exam, 2014

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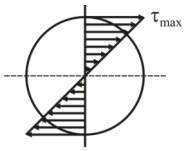
Ans. (a) : When a shaft is subjected to a torque or twisting a shearing stress is produced in the shaft. The shear stress varies from zero in the axis to a maximum at the outside surface of the shaft. The variation due to concentration loads is linear.



Solid shaft

1023. If pure torsion is applied to a piece of classroom chalk, it may crack along a 45° helical surface due to _____.
(a) maximum shear stress
(b) maximum principal stress
(c) maximum tensile stress
(d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Chalk is a brittle material hence it will fails by maximum principle stress theory.
in case of pure torsion.

$\sigma_x = 0 \quad \& \quad \sigma_y = 0$

$$\tan(2\theta_P) = \frac{\tau_{xy}}{\frac{\sigma_x - \sigma_y}{2}} = \infty$$

$2\theta_P = 90°$

$$\boxed{\theta_P = 45°}$$

1024. The torisonal stiffness of a circular shaft is
(a) Directly proportional to its length
(b) Independent of its length
(c) Inversely proportional to its length
(d) Proportional to square of its length

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c)

Torsional Stiffness $(k_t) = \frac{T}{\theta}$

$$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R} \rightarrow \frac{T}{\theta} = \frac{GJ}{L}$$

$$\boxed{k_t = \frac{GJ}{L}}$$

1025. A torque of 50 N-m applied on the wheel operating a valve. If the wheel is rotated through two revolutions, work done in Newton-metres is given by
(a) 100 (b) 25
(c) 314 (d) 628

TNPSC AE 2014

Ans. (d) : T = 50 N-m
one revolution = 360°
$\theta = 4\pi \quad (720^o)$
Then work done will be -
$W = T\theta = 50 \times 4\pi$

$$\boxed{W = 628\ N-m}$$

1026. A solid shaft is replaced by a hollow shaft with outer diameter same as the diameter of the solid shaft. The internal diameter of the hollow shaft is kept as 3/4th of its outer diameter. What is the ratio of torque transmitting capacity of hollow shaft to that of the solid shaft?
(a) 175/256 (b) 3/4
(c) 9/16 (d) 27/64

RPSC AEN Pre-2013

Ans. (a) : Given –
For Hollow shaft-
Inner diameter (ID) = D_i
Outer diameter (OD) = D_o
For solid shaft,
Diameter (D) = D_o

$$D_i = \frac{3}{4} D_o \Rightarrow \frac{D_i}{D_o} = \frac{3}{4}$$

We know that, $\tau = \frac{16T}{\pi D^3} \Rightarrow T = \frac{\pi D^3 \tau}{16}$

$\because \quad T \propto D^3$

For hollow shaft,

$$\boxed{T_{Hollow} \propto D^3\left[1-\left(\frac{D_i}{D_o}\right)^4\right]}, \quad \frac{T_{Hollow}}{T_{Solid}} = \frac{D^3\left[1-\left(\frac{D_i}{D_o}\right)^4\right]}{D^3}$$

$$\frac{T_{Hollow}}{T_{Solid}} = 1-\left(\frac{3}{4}\right)^4$$

$$= 1-\frac{81}{256}$$

$$\boxed{\frac{T_{Hollow}}{T_{Solid}} = \frac{175}{256}}$$

1027. For the two shafts connected in parallel, which of the following in each shaft is same?
(a) Torque (b) Shear stress
(c) Angle of twist (d) Torsional stiffness

UKPSC AE-2013, Paper-I

Ans. (c) : When the two shafts connected in parallel then both shafts are subjected to same angle of twist.
$\theta = \theta_1 = \theta_2$
$T = T_1 + T_2$

1028. A hollow shaft has external and internal diameters of 10cm and 5cm respectively. Torsional section modulus of shaft is:-
(a) 375 cm^3 (b) 275 cm^3
(c) 184 cm^3 (d) 84 cm^3

UKPSC AE-2013, Paper-I

Ans. (c) : We know that torsional section modulus for hollow shaft

$$Z_p = \frac{\pi}{16} D^3\left[1-K^4\right]$$

where $K = \frac{d}{D}$

putting d = 5 cm, D = 10 cm

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$$Z_p = \frac{\pi}{16} \times (10)^3 \times \left[1 - \left(\frac{5}{10}\right)^4\right]$$

$$Z_p = 184 \text{ cm}^3$$

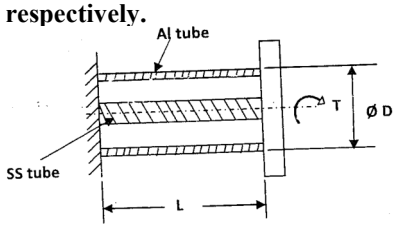
1029. Which part of the shaft cross section has maximum shear stress ?

(a) 1 mm from the centre of the surface
(b) Centre of the surface
(c) Outermost surface
(d) 2 mm from the centre of the surface

TNPSC AE (Industries) 09.06.2013

Ans. (c) : Maximum shear stress of the shaft occurs at outermost surface.

1030. A 'SS' tube is inserted into an '*Al*' tube. They are permanently fixed at one end. The other end is attached to a rigid plate. A torque 'T' is applied to the rigid plate. The circumference of the '*Al*' tube at dia 'D' at the plate and with respect to the fixed and rotates by a distance ..... mm due torque 'T'. The polar MOI & Rigidity modulus of *Al* & SS are J_{Al}, G_{Al} and J_{ss}, G_{ss} respectively.



(a) $\frac{DLT}{2(G_{Al} - G_{ss}J_{ss})}$ (b) $\frac{DLT}{2G_{Al}J_{Al}}$

(c) $\frac{DLT}{2(G_{Al}J_{Al} + G_{ss}J_{ss})}$ (d) $\frac{2DLT}{G_{ss}J_{ss}}$

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : At end, Torque = T

Circumference of the *Al* tube Diameter = D

The polar moment of Inertia (MOI) and Rigidity of *Al* and SS are J_{Al}, G_{Al} and J_{ss}, G_{ss}.

Here, *Al* and SS are connected in parallel.

So, In parallel,

$$T = T_{Al} + T_{ss} \quad ...(1)$$

Let angle of twist of Al tube is θ_{Al} and angle of twist of SS tube is θ_{ss} due to applied torque acting at cross-section.

$$\theta_{Al} = \frac{T_{Al}L}{G_{Al}J_{Al}}$$

$$\theta_{ss} = \frac{T_{ss}L}{G_{ss}J_{ss}}$$

In parallel,

$$\theta_{Al} = \theta_{ss} = \theta$$

$$\therefore \quad \frac{T_{Al}L}{G_{Al}J_{Al}} = \frac{T_{ss}L}{G_{ss}J_{ss}}$$

$$\therefore \quad T_{Al} = \frac{G_{Al}J_{Al}}{G_{ss}J_{ss}}T_{ss}$$

From Equation (1)

$$T = T_{Al} + T_{ss}$$

$$= \frac{G_{Al}J_{Al}}{G_{ss}J_{ss}}T_{ss} + T_{ss}$$

$$T = T_{ss}\left[\frac{G_{ss}J_{ss} + G_{Al}J_{Al}}{G_{ss}.J_{ss}}\right]$$

$$\therefore \quad \theta = \frac{T_{Al}L}{G_{Al}J_{Al}} = \frac{T_{SS}.L}{G_{SS}J_{SS}}$$

$$\theta_{Al} = \theta_{SS} = \frac{TL}{(G_{ss}J_{ss} + G_{Al}J_{Al})} radian$$

Distance of circumferential point.

$$d = 2R \text{ mm} \qquad \left[R = \frac{D}{2}\right]$$

$$\therefore \quad d = \frac{TDL}{2(G_{AL}J_{Al} + G_{ss}J_{ss})} mm$$

1031. Torsional rigidity of a solid cylindrical shaft of diameter 'd' is proportional to

(a) d (b) d^2

(c) d^4 (d) $\frac{1}{d^2}$

UKPSC AE 2012 Paper-I

Ans. (c) : $TR = GJ$

$$TR \propto J$$

$$TR \propto \frac{\pi}{32}d^4$$

$$TR \propto d^4$$

1032. Polar moment of inertia of an equilateral triangle of side '*x*' is given by

(a) $\frac{x^4}{16}$ (b) $\frac{x^4}{16\sqrt{3}}$

(c) $\frac{x^4}{32}$ (d) $\frac{x^4}{64}$

UKPSC AE 2012 Paper-I

Ans. (b) : $\frac{x^4}{16\sqrt{3}}$

1033. Torsional rigidity of a shaft is given by

(a) $\frac{T}{\ell}$ (b) $\frac{T}{J}$

(c) $\frac{T}{\theta}$ (d) $\frac{T}{r}$

APPSC AEE 2012

Ans : (c) Torsion Equation

$$\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{r}$$

T = Torque in N-mm

l = length of the shaft in mm

R = Radius of the circular shaft in mm

G = Modulus of rigidity of shaft material in N/mm^2

$$\frac{T}{J} = \frac{G\theta}{\ell} = \frac{\tau}{r}$$

$$GJ = \frac{T\ell}{\theta}$$

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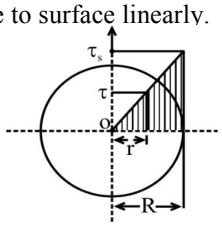
G J is known as torsional rigidity of the shaft. It is important to note that the relative stiffness of two shafts is measured by the inverse ratio of the angles of twist is equal length of shafts when subjected to equal torques.

1034. Shear stress for a circular shaft due to torque varies

(a) from surface to centre parabolically
(b) from surface to centre linearly
(c) from centre to surface parabolically
(d) from centre to surface linearly

APPSC AEE 2012

Ans : (d) Shear stress for a circular shaft due to torque varies from centre to surface linearly.



$$\frac{T}{J}=\frac{G\theta}{l}=\frac{\tau}{r}$$

The above relation states that the intensity of shear stress at any point in the cross-section of a shaft subjected to pure torsion is proportional to its distance from the centre.

1035. A circular shaft subjected to torsion undergoes a twist of 1^0 in a length of 1.2 m. If the maximum shear stress induced is 100 MPa and the rigidity modulus is 0.8×10^5 MPa, the radius of the shaft in mm should be

(a) $\frac{270}{\pi}$ (b) $\frac{\pi}{270}$
(c) $\frac{180}{\pi}$ (d) $\frac{\pi}{180}$

APPSC AEE 2012

Ans : (a) Twist angle $(\theta)=1^0$
length of shaft = 1.2 m
Maximum shear stress = 100 MPa
Modulus of rigidity $(G)=0.8\times10^5$ MPa.

Torsion equation $\frac{T}{J}=\frac{G\theta}{l}=\frac{\tau}{r}$

$$\frac{100\times10^6}{r}=\frac{0.8\times10^5\times\pi\times10^6}{180\times1.2}$$

$$\boxed{r=\frac{270}{\pi}\text{mm}}$$

1036. Two shafts are of same length and same material. The diameter and maximum shear stress of the second shaft is twice that of the first shaft. Then the ratio of power developed between the first and second shaft is

(a) 16 (b) $\frac{16}{3\sqrt{3}}$
(c) $\frac{1}{16}$ (d) $\frac{3}{16}$

APPSC AEE 2012

Ans : (c) Power develop (P) = Tω

Case 1st:-

$$T=\frac{\pi d^3\tau}{16}$$

$$P_1=\frac{\pi d^3\tau}{16}\times\omega$$

Case 2nd :-

$$P_2=\frac{\pi(2d)^3\times2\tau}{16}\times\omega$$

assume, both case angular speed same then

Ratio of power developed $\left(\frac{P_1}{P_2}\right)=\frac{1}{16}$

$$\boxed{\frac{P_1}{P_2}=\frac{1}{16}}$$

1037. The shear stress developed at a radial distance *r* is *q* . The shear stress developed at a radial stress $\frac{r}{2}$ is :

(a) 0.75 *q* (b) 0.5 *q*
(c) *q* (d) 0.25 *q*

APGENCO AE 2012

Ans. (b) : Shear stress (τ) $\propto$ radial distance (r)

$\tau \propto r \rightarrow$ In radial distance

$$\frac{\tau_1}{\tau_2}=\frac{r_1}{r_2}$$

$$\frac{q}{\tau_2}=\frac{r}{\frac{r}{2}} \Rightarrow \frac{q}{\tau_2}=\frac{2}{1}$$

$$\tau_2=\frac{q}{2}=0.5q$$

1038. The maximum shear stress occurs on the outer most fibers of a circular shaft under torsion. In a close coiled helical spring the maximum shear stress occurs on the

(a) Outermost fibers
(b) Fibers at mean diameter
(c) Innermost fiber
(d) End coils

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : Helical spring the maximum shear stress occur on the inner side of the spring for both tensile & compressive load on spring, this is because, inner shear stress is the sum of shear stress due to torsion and shear stress due to force or applied load.

1039. A shaft revolving at 'ω' radians/s transmits torque 'T' N-m. the power developed is :

(a) T.ω. Watts (b) 2πT.ω Watts
(c) 2T.ω Watts (d) 2π T/ω Watts

RPSC AE GWD, 2011

Ans. (a) :

$$\text{Power } P = T\times\omega=\frac{2\pi NT}{60}$$

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1040. A solid shaft of 100mm diameter transmits 160 HP at 200 rpm. The modulus of rigidity G = 8 × 10^5 kg/cm^2. Then the maximum angle of twist for a length of 6 meter is

(a) 5° (b) 2.5°
(c) 3.2° (d) 2°

ISRO Scientist/Engineer 2011

Ans. (b) : P = 160 HP
N = 200 rpm
G = 8 × 10^5 kg/cm^2
l = 6m
θ = ? d = 0.100 m
P = 160 × 746
P = 119360 W
= 119.360 kW

$$P = \frac{2\pi NT}{60}$$

$$119.36 = \frac{2\times 3.14\times 200\times T}{60}$$

T = 5.70 kN/m

$$\frac{T}{J} = \frac{G\theta}{l} = \frac{\tau}{r}$$

$$\frac{T}{J} = \frac{G\theta}{l}$$

$$\frac{5.70}{\frac{\pi}{32}d^4} = \frac{8\times 10^5 \times 9.81\times 10^4 \times \theta}{6\times 1000}$$

Or θ = 0.044 rad

$$= \frac{0.044\times 180}{\pi}$$

θ ≈ 2.5°

1041. Two heavy rotating masses are connected by shafts of lengths l_1, l_2 and l_3 and the corresponding diameters are d_1, d_2 and d_3. This system is reduced to a torsionally equivalent length of the shafts is

(a) $l_1 + l_2\left(\frac{d_1}{d_2}\right)^4 + l_3\left(\frac{d_1}{d_3}\right)^4$

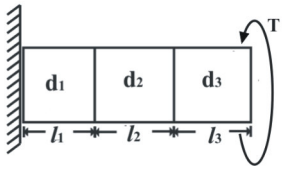
(b) $l_1 + l_2\left(\frac{d_1}{d_2}\right)^3 + l_3\left(\frac{d_1}{d_3}\right)^3$

(c) $\frac{l_1 + l_2 + l_3}{3}$

(d) $l_1 + l_2 + l_3$

ISRO Scientist/Engineer 2011

Ans. (a) :



$$\theta = \theta_1 + \theta_2 + \theta_3$$

$$\frac{Tl_e}{GJ} = \frac{Tl_1}{GJ_1} + \frac{Tl_2}{GJ_2} + \frac{Tl_3}{GJ_3}$$

$$\frac{l_e}{\frac{\pi}{32}d_e^4} = \frac{l_1}{\frac{\pi}{32}d_1^4} + \frac{l_2}{\frac{\pi}{32}d_2^4} + \frac{l_3}{\frac{\pi}{32}d_3^4}$$

$$\frac{l_e}{d_e^4} = \frac{l_1}{d_1^4} + \frac{l_2}{d_2^4} + \frac{l_3}{d_3^4}$$

$$\frac{l_e}{d_e^4} = \frac{1}{d_1^4}\left[l_1 + l_2\left(\frac{d_1}{d_2}\right)^4 + l_3\left(\frac{d_1}{d_3}\right)^4\right]$$

We assume, $d_1 = d_e$

then, we get $l_e = l_1 + l_2\left(\frac{d_1}{d_2}\right)^4 + l_3\left(\frac{d_1}{d_3}\right)^4$

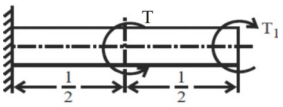
1042.

A solid circular shaft, of polar moment of inertia J and modulus of rigidity of the material G, is fixed at one end and loaded by two torques as shown in the figure. The twist at the free end of the shaft will be zero when

(a) $T_2 = 0.5\ T_1$ (b) $T_2 = T_1$
(c) $T_2 = 2T_1$ (d) $T_1 = 0$

UPSC JWM Advt. No.-50/2010

Ans. (c) :



As we know that,
According to torsion equation –

$$\frac{T}{J} = \frac{G\theta}{L}$$

$$\theta = \frac{TL}{GJ}$$

⇒ G & J are constant or same.
So, the equilibrium is zero,
∴ ΣT = 0

$$T_1 \times \ell - T_2 \times \frac{\ell}{2} = 0$$

$$\boxed{T_2 = 2T_1}$$

1043. For solid shaft subjected to a torque of 18000 Nm having a permissible shear stress of 60N/mm^2, the diameter of shaft is

(a) 115 mm (b) 121 mm
(c) 149 mm (d) 108 mm

ISRO Scientist/Engineer 2010

Ans. (a) : T = 18000 N-m
permissible shear stress τ = 60 N/mm^2

$$\tau = \frac{16T}{\pi d^3}$$

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$$60 = \frac{16 \times 18000 \times 1000}{\pi d^3}$$

$$\pi d^3 = 16 \times 3000 \times 100$$

$$d^3 = \frac{16 \times 3000 \times 100}{\pi}$$

$$d = \sqrt[3]{\frac{16 \times 3000 \times 100}{3.141}}$$

d = 115 mm

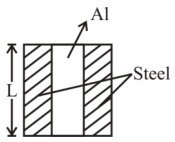
1044. A compound rod is formed by tightly inserting an aluminium rod inside a steel tube. The length of this compound rod is l and its ends are welded together so as to prevent any relative motion. This rod is subjected to a torque T, applied to its ends in opposite directions. If polar moment of inertia of the tube and rod are same and modulus of rigidity of steel is three times that of aluminium, what is the ratio of torque shared by steel tube to that of aluminium rod?

(a) 4 : 1 (b) 3 : 1

(c) 2 : 1 (d) 1 : 1

UPSC JWM Advt. No.-50/2010

Ans. (b) :



We know that,

$$\frac{T}{J} = \frac{G\theta}{L}$$

Given,

$$\ell_{A\ell} = \ell_{St} = L,\ J_{St} = J_{A\ell}$$

$$\theta = \frac{TL}{GJ} \qquad \Rightarrow \qquad G_{St} = 3\ G_{A\ell}$$

For parallel connection

$$\theta = \theta_{St} = \theta_{A\ell}$$

So, $$\frac{T_s \ell_s}{J_s G_s} = \frac{T_{A\ell} \ell_{A\ell}}{J_{A\ell} G_{A\ell}} \Rightarrow \frac{T_s}{T_{A\ell}} = \frac{G_s}{G_{A\ell}} = 3$$

$$\boxed{\frac{T_s}{T_{A\ell}} = \frac{3}{1}}$$

1045. A hollow shaft of outside diameter 40 mm and inside diameter 20 mm is to be replaced by a solid shaft of 30 mm diameter. If the maximum shear stresses induced in the two shafts are to be equal, what is the ratio of the maximum resistible torque in the hollow to that of solid shaft ?

(a) $\frac{10}{9}$ (b) $\frac{20}{9}$

(c) $\frac{30}{9}$ (d) $\frac{40}{9}$

UPSC JWM Advt. No.-50/2010

Ans. (b) : Hollow cylinder

$d_o = 40$ mm, $d_i = 20$ mm

$\tau_H = ?$

$$\frac{T}{J} = \frac{\tau}{R}$$

$$T = \tau \times \frac{J}{R} \Rightarrow k = \frac{d_i}{d_o} = \frac{1}{2}$$

$$J_H = \frac{\pi}{32} d_o^4 (1 - k^4) \Rightarrow \frac{J_H}{R} = \frac{\pi}{16} d_o^3 (1 - k^4)$$

For solid,

d = 30 mm

$\tau_s = ?$

$$J_s = \frac{\pi}{32} d^4$$

$$\therefore \qquad R = \frac{d}{2}$$

$$\frac{J}{R} = \frac{\frac{\pi}{32} d^4}{d/2}$$

$$= \frac{\pi}{16} d^3$$

$$\frac{J}{R} = \frac{\pi}{16} d^3$$

$$\frac{T_h}{T_s} = \frac{\tau_h \times \frac{\pi}{16} \times d_o^3 (1 - k^4)}{\tau_s \times \frac{\pi}{16} \times d^3} = \frac{40^3 (15/16)}{30^3} \quad \left[\because \text{Given, } \tau_H = \tau_S \right]$$

$$\boxed{\frac{T_h}{T_s} = \frac{20}{9}}$$

1046. Assertion (A) : Stress at a point is completely defined by specifying its magnitude, nature, direction and orientation of the plane containing the point.

Reason (R) : Stress is a tensor of second order.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true and R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (a) : As we know, stress is a tensor of second order therefore, it is completely defined by specifying its magnitude, nature, direction & orientation of the plane containing point.

1047. Determine the diameter of solid shaft which will transmit 90 kW at 160 rpm, if the shear stress in the shaft is limited to 60 N/mm^2

(a) 50 mm (b) 60 mm

(c) 77 mm (d) 70 mm

ISRO Scientist/Engineer 2009

Ans. (c) : Power P = 90kW

N = 160 rpm

Shear stress

$\tau = 60$ N/mm^2

$$P = \frac{2\pi NT}{60}$$

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$$90\times10^3=\frac{2\pi\times160\times T}{60}$$

Torque, T = 5371.47 N-m

Shear stress, $\tau=\frac{16T}{\pi d^3}$

$$d^3=\frac{16\times5371.47\times10^3}{\pi\times60}$$

d = 76.96 mm

$\boxed{d=77\,mm}$

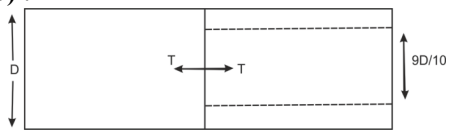
1048. One-half length of 50 mm diameter steel road is solid while the remaining half is hollow having a bore of 5 mm. The rod is subjected to equal and opposite torque at its ends. If the maximum shear stress in solid portion is τ, the maximum shear stress in the hollow portion is :

(a) $\frac{15}{16}\tau$ (b) τ

(c) $\frac{4}{3}\tau$ (d) $\frac{16}{15}\tau$

ISRO Scientist/Engineer 2009
ESE 2003, 1997

Ans. (b) :



$$\tau_{solid}=\frac{16T}{\pi D^3}$$

$$\tau_{hollow}=\frac{Tr}{J}$$

$$=\frac{T\times\frac{D}{2}}{\frac{\pi}{32}[D^4-(0.1\times D)^4]}$$

$$=\frac{16TD}{\pi D^4(1-0.1^4)}$$

$$\tau_{hollow}=\frac{16T}{\pi D^3}\times\frac{1}{0.9999}$$

$$\tau_{hollow}=\frac{\tau_{solid}}{0.9999}$$

$\Rightarrow$ $\tau_{hollow}\approx\tau_{solid}$

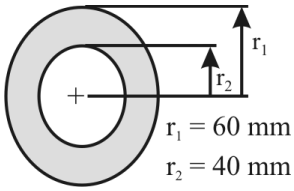
1049. A hollow cylindrical shaft is 1 m long and has inner diameter and outer diameter as 40 mm and 60 mm, respectively. When this shaft is subjected to a pure torsion of certain magnitude, maximum shear stress induced is 120 MPa. The corresponding minimum value of shear stress induced will be

(a) 20 MPa (b) 40 MPa

(c) 60 MPa (d) 80 MPa

DRDO Scientists 2009

Ans. (d) :



$$\tau\alpha\frac{1}{r}$$

$$\frac{\tau_{max}}{r_1}=\frac{\tau_{min}}{r_2}$$

$$\tau_{min}=\frac{r_2}{r_1}\tau_{max}$$

$$=\frac{40}{60}\times120=80\ MPa$$

1050. In torsion, which of the following sections will be best?

(a) triangular (b) rectangular

(c) hollow circular (d) solid cylinder

WBPSC AE 2008

Ans. (c) : $J=\frac{\pi}{32}D^4\left(1-k^4\right)$

- Hollow circular section is best in torsion as the polar moment of inertia is higher for hollow circular section (same amount of material).

1051. When a solid circular shaft is in pure tension and deforms elastically, the shearing stress in the shaft

(a) is inversely proportional to the shear modulus of elasticity

(b) varies linearly with the radial distance from the axis of the shaft

(c) varies linearly with length of the shaft

(d) is inversely proportional to the diameter of the shaft

DRDO Scientists 2008

Ans. (b) : $\frac{\tau_x}{r_x}=\frac{T}{J}=\frac{G\theta}{\ell}$

r_x – radial distance from axis of shaft.

1052. If 'T' is the torque and 'M' is the moment, then the shaft is subjected to:

(a) bending stress and direct stress

(b) shearing stress and direct stress

(c) principal stress and shearing stress

(d) principal stress and direct stress

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : If T is the torque and M is the moment, the shaft is subjected to shearing stress = $\frac{16T}{\pi d^3}$

And direct stress,

Bending stress = $\frac{32M}{\pi d^3}$

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1053. A bending moment of 400 kNm and a twisting moment of 300 kNm is applied on a circular solid shaft. If direct stress based on maximum principal stress theory is σ and shear stress based on maximum shear stress theory is τ, then $\frac{\sigma}{\tau}$ is:

(a) $\frac{5}{9}$ (b) $\frac{5}{3}$

(c) $\frac{3}{5}$ (d) $\frac{9}{5}$

AAI Jr. Executive 26.03.2021
ESE 2000

Ans. (d) : Given, Bending moment = 400 kN-m
Twisting moment = 300 kN-m
Direct stress (σ) based on max. principal stress theory

$$\sigma = \frac{16}{\pi d^3}\left[M + \sqrt{M^2 + T^2}\right]$$

$$\sigma = \frac{16}{\pi d^3}\left[400 + \sqrt{(400)^2 + (300)^2}\right]$$

$$= \frac{16}{\pi d^3}[400 + 500]$$

$$\sigma = \left(\frac{16}{\pi d^3}\right) \times (900)$$

Shear stress, $\tau = \left(\frac{16}{\pi d^3}\right)\left(\sqrt{M^2 + T^2}\right)$

$\Rightarrow$ $\left(\frac{16}{\pi d^3}\right)\left(\sqrt{(400)^2 + (300)^2}\right)$

$= \left(\frac{16}{\pi d^3}\right) \times (500)$

$\therefore$ $\boxed{\frac{\sigma}{\tau} = \frac{9}{5}}$

1054. At a certain cross-section, a shaft of 100 mm diameter is subjected to a bending moment of 4 kNm and a twisting moment of 8 kNm. Maximum principal stress induced (in N/mm^2) in the section is.

(a) 72.8 (b) 6.17

(c) 65.9 (d) 68.6

HPPSC ADF 05.03.2019

Ans. (c) : d = 100
Bending Moment, M = 4 kN-m
Twisting Moment, T = 8 kN-m
Max principal stress,

$$\sigma_1 = \frac{16}{\pi d^3}\left[M + \sqrt{M^2 + T^2}\right]$$

$$= \frac{16}{\pi (100)^3}\left[4 + \sqrt{(4^2) + (8)^2}\right]$$

$$= \frac{16}{\pi (100)^3}[4 + 8.94]$$

$$\sigma_1 = \frac{16}{\pi (100)^3} \times 12.94 \times 10^6$$

Max principal stress, $\sigma_1 = 65.9$ N/mm^2

6. Thick and Thin Cylinders and Spheres

1055. The ratio of hoop stress to longitudinal stress is___ for a thin cylinder subjected to internal pressure.

(a) 2 : 1 (b) 3 : 2

(c) 1 : 1 (d) 1 : 2

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Assam PSC AE (IWT) 14.03.2021
GPSC Executive Engineer 23.12.2018
GPSC ARTO Pre 30.12.2018
GPSC Asstt. Director of Transport 05.03.2017
GATE 2017, ESE 2017, WBPSC AE, 2017
APPSC AE Subordinate Service Civil/Mech. 2016
RPSC VPITI 14.02.2016, TSPSC AEE 2015
MPPSC State Forest Service Exam, 2014
GATE 2013, UKPSC AE 2012 Paper-I
APPSC AEE 2012, APPSC AE 04.12.2012
J&K PSC Civil Services Pre, 2010
CSE Pre-1994

Ans. (a) : We know,

$$\text{hoop stress} = \frac{pd}{2t}$$

&

$$\text{Longitudinal stress} = \frac{pd}{4t}$$

So, $= \frac{\frac{pd}{2t}}{\frac{pd}{4t}} = 2:1$

1056. Auto frettage is the method of

(a) a type of corrosion
(b) method to reduce stress concentration
(c) a type of pre-stressing
(d) a type of heat treatment

HPPSC Asstt. Prof. 18.09.2017
APGENCO AE, 2017
APPSC AEE Screening Test 2016
GPSC AE 26.07.2015, UPRVUNL AE 2014
APPSC AEE 2012, APPSC AE 04.12.2012
ESE 2005, ESE 1996

Ans. (c) : Auto frettage is the method of a type of pre-stressing of thick walled cylinders to increase its strength.

1057. Longitudinal stress in a thin cylinder is

(a) Equal to the hoop stress
(b) Half of the hoop stress
(c) Twice the hoop stress
(d) Three fourth of the hoop stress

VIZAG Steel MT 24.01.2021, Shift-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
MECON MT 2019
Rajasthan AE (Nagar Nigam) 2016 Shift-3
RPSC LECTURER 16.01.2016
APPSC AEE MAINS 2016, PAPER-III
WBPSC AE 2008
ISRO Scientist/Engineer 2006

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Ans. (b) : Longitudinal stress in a thin cylinder is half of the hoop stress.

Longitudinal stress $\sigma_L = \frac{Pd}{4t}$

Hoop stress $\sigma_H = \frac{Pd}{2t}$

$$\frac{\sigma_L}{\sigma_H} = \frac{Pd}{4t} \times \frac{2t}{Pd}$$

$$\frac{\sigma_L}{\sigma_H} = \frac{1}{2}$$

$$\boxed{\sigma_L = \frac{\sigma_H}{2}}$$

Where, $\sigma_L \rightarrow$ longitudinal stress

$\sigma_H \rightarrow$ Hoop stress

1058. A thin cylinder of inner radius 500mm and thickness 10mm is subjected to an internal pressure of 5 MPa. The hoop stress is equal to

(a) 1000 (b) 500

(c) 250 (d) 125

Vadodara Muncipal Corp. DEE, 2018
GPSC ARTO Pre 30.12.2018
GPSC Executive Engineer 23.12.2018
UPRVUNL AE 21.08.2016
HPPSC Asstt. Prof. 18.11.2016
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2011

Ans. (c) : Given

$r_i = r = 500$mm

$t = 10$mm

$P = 5$ MPa

Hoop stress $(\sigma_h) = \frac{Pd}{2t}$

$$= \frac{5 \times (2 \times 500)}{2 \times 10}$$

$$\boxed{\sigma_h = 250\,\text{MPa}}$$

1059. Thick cylinders are designed by:

(a) Lame's equation

(b) calculating radial stress which is uniform

(c) thick cylinder theory

(d) thin cylinder theory

HPPSC Asstt. Prof. 18.09.2017
RPSC AE 2016
APPSC AEE Screening Test 2016
TSPSC AEE 2015
UPRVUNL AE 2014
UKPSC AE 2012 Paper-I
RPSC ACF-2011

Ans. (a) : When the ratio of inner diameter of the cylinder to the thickness is less than 20, the cylinder is called thick walled cylinder or simply thick cylinder. Thick cylinders are designed by Lame's equation.

1060. Maximum shear stress in any point in a thin cylinder is given by_____.

(a) $\frac{pd}{4t}$ (b) $\frac{pd}{16t}$

(c) $\frac{pd}{8t}$ (d) $\frac{pd}{2t}$

Haryana PSC AE (PHED) 05.09.2020, Paper-II
VIZAG Steel MT, 14.12.2020
APPSC AE Subordinate Service Civil/Mech. 2016
Vizag Steel MT (Re-Exam) 24.11.2013
DRDO Scientists 2009, CSE Pre-1996

Ans. (c) : $\sigma_c = \frac{pd}{2t}$ $\sigma_\ell = \frac{pd}{4t}$

Maximum shear stress $= \frac{\sigma_c - \sigma_\ell}{2} = \frac{pd}{8t}$

σ_c = Circumferential stress

σ_ℓ = Longitudinal stress

p = internal pressure

d = internal diameter

t = thickness

1061. Circumferential and longitudinal strains in the cylindrical boiler under internal steam pressure are ε_1 and ε_2 respectively. Change in the volume of the boiler cylinder per unit volume will be

(a) $\varepsilon_1^2 \varepsilon_2$ (b) $\varepsilon_1 \varepsilon_2^2$

(c) $2\varepsilon_1 + \varepsilon_2$ (d) $\varepsilon_1 + 2\varepsilon_2$

ESE 2018, UPRVUNL AE 07.10.2016
APPSC AE 04.12.2012
ISRO Scientist/Engineer 2009
ISRO Scientist/Engineer 2007

Ans. (c) : Circumferential strain = ε_1

longitudinal strain = ε_2

change in the volume of the boiler cylinder per unit volume

volume of cylinder

$$V = \frac{\pi}{4} D^2 \times l$$

$$\frac{dV}{V} = 2\frac{dD}{D} + \frac{dl}{l}$$

$$e_v = 2\varepsilon_1 + \varepsilon_2$$

1062. Hoop stress in a thin cylinder of diameter 'd' and thickness 't' subjected to pressure 'p' will be

(a) $\frac{pd}{4t}$ (b) $\frac{pd}{t}$

(c) $\frac{pd}{2t}$ (d) $\frac{2pd}{t}$

MPPSC AE 2016, SJVN ET 2013
CSE Pre-2009, TNPSC AE, 2008
J & K PSC Screening, 2006

Ans. (c) : For a thin cylinder –

Hoop stress/circumferential stress $(\sigma_c) = \frac{Pd}{2t}$

Where; P = Pressure

d = dia of the cylinder

t = thickness of the cylinder

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1063. In a thick cylinder pressurized from inside, the hoop stress is maximum at

(a) The centre of the wall thickness
(b) The outer radius
(c) The inner radius
(d) Same at both inner and outer radii

GPSC AE 26.07.2015, APPSC AEE 2012
APPSC AE 04.12.2012, ESE 1998

Ans. (c) : Maximum hoop stress in thick cylinder pressurized from inside at the inner surface. Max hoop stress in thick cylinder under external pressure occurs of the outer surface.

1064. The longitudinal stress within cylindrical shell of diameter (D), length (L) and thickness (t), when subjected to an internal pressure (P) is

(a) PD/(4t) (b) PD/(2t)
(c) 2PD/t (d) 4PD/t

TNPSC AE 2019
APPSC AE Subordinate Service Civil/Mech. 2016
ISRO Scientist/Engineer (RAC) 29.11.2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a)
Safe condition

$$\sigma_L \le S_{ut}$$

Limiting condition

$$\sigma_L = S_{ut}$$

or $F_f \le F_R$ (OR) $F_f = F_R$

$$P \times \frac{\pi}{4} D^2 = \sigma_L \times \pi D t$$

$$\boxed{\sigma_L = \frac{PD}{4t}}$$

1065. A thin gas cylinder with an internal radius of 100 mm is subject to an internal pressure of 10 MPa. The maximum permissible working stress is restricted to 100 MPa. The minimum cylinder wall thickness (in mm) for safe design must be ----------

(a) 8 mm (b) 10 mm
(c) 12 mm (d) 14 mm
(e) None of the above

OPSC AEE 2019 Paper-I
CGPSC Poly. Lect. 22.05.2016
Nagaland PSC CTSE 2016 Paper-I
GATE 2014

Ans. (b)
radius r = 100 mm
d = 200 mm
pressure (P) = 10 MPa
permissible stress = 100 MPa
For safe design–

$$\frac{Pd}{2t} \le \sigma_{permissible}$$

$$\frac{10 \times 200}{2t} = 100$$

$$\boxed{t = 10\,mm}$$

1066. A thin cylinder of diameter D, wall thickness t is subjected to an internal fluid pressure p. If E is the Young's modulus and υ is the Poisson's ratio for the material of the cylinder, the expression for volumetric strain of the cylinder is

(a) $\frac{pD}{4tE}(5-4\mu)$ (b) $\frac{pD}{4tE}(4-5\mu)$
(c) $\frac{pD}{2tE}(5-4\mu)$ (d) $\frac{pD}{2tE}(4-5\mu)$

RPSC LECTURER 16.01.2016
WBPSC AE, 2007, CSE Pre-2006
ESE 2003, CSE Pre-1997

Ans. (a)

$$\varepsilon_v = \frac{\delta v}{v} = \frac{\delta \ell}{\ell} + \frac{2\delta d}{d}$$

$$\varepsilon_\ell = \frac{\delta \ell}{\ell} = \frac{\sigma_L}{E} - \frac{\mu \sigma_h}{E} = \frac{pd}{4tE} - \frac{\mu pd}{2tE}$$

$$\boxed{\varepsilon_l = \frac{pd}{4tE}(1-2\mu)}$$

$$\varepsilon_d = \frac{\delta d}{d} = \frac{\sigma_h}{E} - \frac{\sigma_L}{E} = \frac{Pd}{2tE} - \frac{Pd}{4tE}$$

$$\boxed{\varepsilon_d = \frac{Pd}{4tE}(2-\mu)}$$

$$\varepsilon_v = \frac{Pd}{4tE}(1-2\mu) + \frac{2Pd}{4tE}(2-\mu)$$

$$= \frac{Pd}{4tE}\left[1-2\mu+4-2\mu\right]$$

$$\boxed{\varepsilon_v = \frac{Pd}{4tE}(5-4\mu)}$$

1067. A thin walled spherical shell is subjected to an internal pressure. If the radius of the shell is increased by 1% and the thickness is reduced by 1%, with the internal pressure remaining the same, the percentage change in the circumferential (hoop) stress is

(a) 0 (b) 1
(c) 1.08 (d) 2.02

UPPSC AE 13.12.2020, Paper-I
TSPSC AEE 2017, GPSC ARTO 01.05.2016
GATE 2012

Ans. (d) : Hoop stress for a thin spherical shell (σ_h) $= \frac{Pr}{2t}$

By applying logarithm on both sides,

$$\log(\sigma_h) = \log\left(\frac{P}{2}\right) + \log(r) - \log(t)$$

$$\frac{d\sigma_h}{\sigma_h} = 0 + \frac{1}{r}dr - \frac{1}{t}dt$$

Percentage change $= \frac{dr}{r} \times 100 + \frac{dt}{t} \times 100$

$= 1\% + 1\% = 2\%$

Percentage increase will be 2%.
Note-Option (d) is given by commission.

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1068. What will happen in winter to a large cylindrical vessel which was sealed in summer?

(a) It buckles and collapses
(b) It becomes lighter
(c) It explodes
(d) Nothing happens to it

GPSC Asstt. Director of Transport 05.03.2017
ISRO Scientist/Engineer (RAC) 07.05.2017
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (a) : Large cylindrical vessel which was sealed in summer, is buckles and collapses in winter due pressure variation between inside and outside.

1069. A metal pipe of 1 m diameter contains a fluid having a pressure of 10 kgf/cm^2. If the permissible tensile stress in the metal is 200 kgf/cm^2, then the thickness of the metal required for making the pipe would be

(a) 5 mm (b) 10 mm
(c) 25 mm (d) 20 mm

APPSC AEE 2012 , ISRO Scientist/Engineer 2007
WBPSC AE 2003

Ans. (c) : d = 1 m
P = 10 kgf/cm^2
circumferential stress or hoop stress

$$\sigma_h = \frac{Pd}{2t} = 200$$

$$\frac{10\times1000}{2\times t} = 200$$

t = 25 mm

Note: Longitudinal stress or Axial stress

$$\sigma_l = \frac{Pd}{4t}$$

1070. A thin cylinder with both ends closed is subjected to internal pressure p. The longitudinal stress at the surface has been calculated as σ_0. Maximum shear stress at the surface will be equal to :

(a) $2\sigma_0$ (b) $1.5\sigma_0$
(c) σ_0 (d) $0.5\sigma_0$

APPSC AEE Mains 2016 (Civil Mechanical)
OPSC AEE 2015 Paper-I, ESE 1999

Ans : (d) Longitudinal stress $(\sigma_2) = \frac{Pd}{4t} = \sigma_0$

Hoop stress $(\sigma_1) = \frac{Pd}{2t} = 2\sigma_0$

Principal stresses = $2\sigma_0, \sigma_0$

Shear stress = $\frac{2\sigma_0 - \sigma_0}{2} = \frac{\sigma_0}{2} = 0.5\sigma_0$

1071. In case of thin walled cylinders the ratio of hoop strain to longitudinal strain is

(a) $\frac{2m-1}{m-2}$ (b) $\frac{2m-1}{m-1}$
(c) $\frac{m-2}{2m-1}$ (d) $\frac{m-2}{2(m-1)}$

Nagaland PSC (CTSE) 2018, Paper-I
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : In case of thin walled cylinders shell, the ratio of hoop strain to longitudinal strain is,

$$\varepsilon_c = \frac{P.d}{2tE}\left(1 - \frac{1}{2m}\right) \text{ ——(i)}$$

$$\varepsilon_\ell = \frac{P.d}{2tE}\left(\frac{1}{2} - \frac{1}{m}\right) \text{ ——(ii)}$$

from eq^n (i) & (ii)

$$\frac{\varepsilon_c}{\varepsilon_\ell} = \frac{2m-1}{m-2}$$

1072. A boiler shell of 200 cm diameter and plate thickness 1.5 cm is subjected to internal pressure of 1.5 MN/m^2 then hoop stress is:

(a) 30 MN/m^2 (b) 50 MN/m^2
(c) 100 MN/m^2 (d) 200 MN/m^2

KPSC AE 2015, SJVN ET 2013

Ans. (c) : Given,
d = 200 cm = 200 × 10^{-2} m
t = 1.5 cm = 1.5 × 10^{-2} m
P = 1.5 MN/m^2

$$\sigma = \frac{Pd}{2t}$$

$$\sigma = \frac{1.5\times200\times10^{-2}}{2\times1.5\times10^{-2}} = 100 \text{ MN/m}^2$$

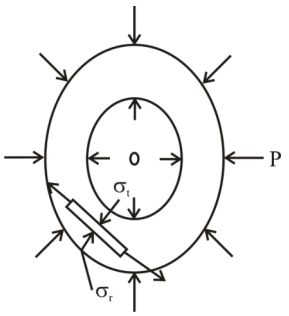
$$\boxed{\sigma = 100 \text{ MN/m}^2}$$

1073. A thin cylinder of radius r and thickness t when subjected to an internal hydrostatic pressure 'p' causes a radial displacement 'u'. Then the tangential strain caused is

(a) $\frac{du}{dr}$ (b) $\frac{1}{r}\cdot\frac{du}{dr}$
(c) $\frac{u}{r}$ (d) $\frac{2u}{r}$

APPSC AEE 2012, ESE 2002

Ans. (c) :



Data given-
Radius = r
Thickness = t
Internal hydrostatic pressure = P, σ_r = radial stress
Radial displacement = u, σ_t = tangential stress

(i) Radial strain $(\varepsilon_r) = \frac{du}{dr}$

(ii) Circumferential/Tangential strain $(\varepsilon_t) = u/r$

(iii) Axial strain $(\varepsilon_A) = \frac{\sigma_r}{E} - u\left(\frac{\sigma_r}{E} + \frac{\sigma_t}{E}\right)$

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1074. Pressure vessels are made of

(a) non-ferrous materials
(b) Sheet metal (steel)
(c) cast iron
(d) Any of the above

APPSC AEE 2012, APPSC AE 04.12.2012

Ans. (b) : Many pressure vessels are made of steel. To manufacture a cylindrical or spherical pressure vessel, railed and possibly forged parts would have to be welded together. Some mechanical properties of steel, achieved by rolling or forging, could be adversely affected by welding unless special precautions are taken.

1075. The stresses in a thick cylinder subjected to uniform pressure vary proportional to :

(a) r (b) $1/r$
(c) r^2 (d) $1/r^2$

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (d) : From the Lame's equation for thick cylinder -

$\sigma_r = \frac{b}{r2} - a$

and $\sigma_C = \frac{b}{r^2} + a$

Where;
σ_r = radial stress
σ_C = Circumfrential stress
a & b is constant

1076. As per Lame's equation, the hoop stress for thick cylinder at any points is:

(a) $(A + B/r^3)$ (b) $(A/r - B)$
(c) $(A + B/r^2)$ (d) $(A + Br)^2$

HPPSC ADF 05.03.2019
TNPSC AE, 2008

Ans. (c) : According to Lame's equation, the hoop stress for thick cylinder

Hoop stress $(\sigma_h)_{hoop} = A + \frac{B}{r^2}$

$$p = A - A + \frac{B}{r^2}$$

Radial stress $(\sigma_r) = -p = A - \frac{B}{r^2}$

A, B Lame's constant

1077. A thin walled pipe contains water at a pressure of 2N/mm^2 and discharges water into a tank. If the pipe diameter is 25 mm and wall thickness 2.5 mm, the longitudinal stress induced in the pipe is

(a) 0 (b) 2 N/mm^2
(c) 5 N/mm^2 (d) 10 N/mm^2

UPSC JWM Advt. No.-52/2010
CSE Pre-2007

Ans. (c) : Solution : Given : Pressure (P) = 2N/mm^2
Diameter of pipe (d) = 25 mm
Thickness of pipe (t) = 2.5mm

$\therefore$ Longitudinal stress $(\sigma_L) = \frac{Pd}{4t}$

$$\sigma_L = \frac{2 \times 25}{4 \times 2.5} = 5 N/mm^2$$

1078. A steel hub of 100 mm internal diameter and uniform thickness of 10 mm was heated to a temperature of 300°C to shrink-fit it on a shaft. On cooling, a crack developed parallel to the direction of the length of the hub. Consider the following factors in this regard :

1. Tensile hoop stress
2. Tensile radial stress
3. Compressive hoop stress
4. Compressive radial stress

The cause of failure is attributable to :

(a) 1 alone (b) 1 and 3
(c) 1, 2 and 4 (d) 2, 3 and 4

GPSC ARTO Pre 30.12.2018
ESE 2016

Ans. (a) : In a shrink fit, the hub is subjected to tensile or hoop stress (which is greater as compared to radial stress) and the shaft is subjected to compressive stress. Therefore, the failure (or crack on surface) is due to tensile hoop stress.

1079. A thick cylinder subjected to an internal pressure of 60MPa. If the hoop stress on the outer surface is 150 MPa, then the hoop stress on the internal surface is :

(a) 105 MPa (b) 180 MPa
(c) 210 MPa (d) 135 MPa

MPPSC AE 08.11.2015, APGENCO AE 2012

Ans. (c) : If internal pressure = P_i
External pressure = 0

Circumferential or hoop stress $(\sigma_c) = \frac{P_i r_i^2}{r_0^2 - r_i^2}\left[\frac{r_0^2}{r_i^2} + 1\right]$

At, $P_i = 60$MPa, $\sigma_c = 150$ MPa
and $r = r_0$

$$\therefore \quad 150 = 60 \ \frac{r_i^2}{r_0^2 - r_i^2}\left[\frac{r_o^2}{r_o^2} + 1\right]$$

$$150 = 2 \times 60\left[\frac{r_i^2}{r_o^2 - r_i^2}\right]$$

or $\frac{r_i^2}{r_o^2 - r_i^2} = \frac{150}{120} = \frac{5}{4} \Rightarrow \left(\frac{r_0}{r_i}\right)^2 = \frac{9}{5}$

$\because$ at $r = r_i$

$$\sigma_c = 60 \times \frac{r_i^2}{r_0^2 - r_i^2}\left[\frac{r_0^2}{r_i^2} + 1\right]$$

$$= 60 \times \frac{5}{4} \times \left[\frac{9}{5} + 1\right]$$

$\sigma_c = 210$ MPa

1080. Hoop stress and longitudinal stress in a boiler shell under internal pressure are 100 MN/m^2 and 50 MN/m^2 respectively. Young's modulus of elasticity and Poisson's ratio of the shell material are 200 GN/m^2 and 0.3 respectively. The hoop strain in the boiler shell is :

(a) 0.425×10^3 (b) 0.5×10^{-3}
(c) 0.5750^{-3} (d) 0.7×10^3

APPSC IOF, 2009, ESE 1995

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Ans. (b) : $\sigma_h = 100\ MN/m^2$, $\sigma_\ell = 50\ MN/m^2$

$E = 200\ GN/m^2$, $\mu = 0.3$

$$\varepsilon_h = \frac{1}{E}[\sigma_h - \mu\sigma_\ell] = \frac{1}{200\times10^9}[100 - 50\times0.3]\times10^6$$

$= 0.425 \times 10^{-3}$

Nearest option is (b).

1081. A thin cylindrical shell of diameter (d) and thickness (t) is subjected to an internal pressure (p). The ratio of longitudinal strain to volumetric strain is

(a) $\frac{m-1}{2m-1}$ (b) $\frac{2m-1}{m-1}$

(c) $\frac{m-2}{3m-4}$ (d) $\frac{m-2}{5m-4}$

Sikkim PSC (Under Secretary), 2017
UPPSC AE 12.04.2016 Paper-I

Ans. (d) : Longitudinal strain $(\varepsilon_\ell) = \frac{Pd}{4tE}(1-2\mu)$ .....(i)

Volumetric strain $(\varepsilon_v) = \frac{Pd}{4tE}(5-4\mu)$ .....(ii)

eq^n (i) ÷ eq^n (ii)

$$\frac{\varepsilon_\ell}{\varepsilon_v} = \frac{\frac{Pd}{4tE}(1-2\mu)}{\frac{Pd}{4tE}(5-4\mu)}$$

$$\frac{\varepsilon_\ell}{\varepsilon_v} = \frac{1-2\mu}{5-4\mu}$$

$$\because \mu = \frac{1}{m}$$

$$\therefore = \frac{1-\frac{2}{m}}{5-\frac{4}{m}}$$

$$\frac{\varepsilon_\ell}{\varepsilon_v} = \frac{m-2}{5m-4}$$

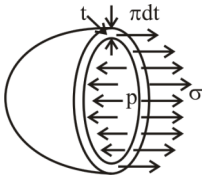
1082. A thin spherical shell of diameter (d) and thickness (t) is subjected to an internal pressure (p). The stress in the shell material is

(a) pd/t (b) pd/2t

(c) pd/4t (d) pd/8t

Sikkim PSC (Under Secretary), 2017
Karnataka PSC AE, 10.09.2017

Ans. (c) : Thin spherical shell



$$\sigma_l \times \pi dt = p \times \frac{\pi}{4}d^2$$

$$\sigma = \frac{pd}{4t}$$

1083. The typical ratio of diameter to wall thickness for a thin vessel is

(a) Less than 20 (b) More than 10

(c) Less than 2 (d) More than 20

Haryana PSC AE (PHED) 05.09.2020, Paper-II
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (d) : For thin vessel D > 20t or $\frac{D}{t} > 20$

Where, D = Diameter

t = Thickness of shell

1084. In a thick cylinder, subjected to internal and external pressures, let r_1 and r_2 be the internal and external radii respectively. Let u be the radial displacement of a material element at radius r, $r_2 \geq r \geq r_1$. Identifying the cylinder axis as z axis, the radial strain component is :

(a) u/r (b) u/θ

(c) du/dr (d) du/δθ

APPSC IOF, 2009, ESE 1996

Ans. (c) : The strains ε_r and ε_θ may be given by

$$\varepsilon_r = \frac{\partial u_r}{\partial r} = \frac{1}{E}[\sigma_r - v\sigma_\theta] \quad \text{since } \sigma_z = 0$$

$$\varepsilon_\theta = \frac{(r+u_r)\Delta\theta - r\Delta\theta}{r\Delta\theta} = \frac{u_r}{r} = \frac{1}{E}[\sigma_\theta - v\sigma_r]$$

$$\varepsilon_{rr} = \frac{du}{dr}$$

1085. A cylindrical tank with closed ends is filled with compressed air at a pressure of 500 kPa. The inner radius of the tank is 2m, and it has wall thickness of 10 mm. The magnitude of maximum in-plane shear stress (in MPa) is ___

(a) 100 (b) 50

(c) 250 (d) 25

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (d) : The maximum in plane shear stress

$$\tau_{max} = \frac{Pd}{8t}$$

$$= \frac{500\times10^3\times2\times2}{8\times10\times10^{-3}}$$

$= 25 \times 10^6\ N/m^2$

$\tau_{max} = 25\ MPa$

1086. The hoop or circumferential stress in a riveted cylindrical shell, when subjected to an internal pressure p is equal to :

(a) $\frac{pD}{4t\eta_c}$ (b) $\frac{pD}{2t\eta_c}$

(c) $\frac{pD}{2t\eta_l}$ (d) $\frac{pD}{4t\eta_l}$

Where D = internal diameter, η_l = efficiency of longitudinal joint and η_c = efficiency of circumferential joint.

CGPSC AE 15.01.2021, CGPSC AE 16.10.2016

Ans. (c) : Hoop or circumferential stress $(\sigma_c) = \frac{pD}{2t\eta_\ell}$

where, η_ℓ = Efficiency of longitudinal joint

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1087. A thin cylinder of 'D' internal diameter, is subjected to an internal pressure of 'P'. If the permissible tensile stress is σ_t' the cylinder wall thickness should be

(a) $\frac{2\sigma_t}{PD}$ (b) $\frac{PD}{\sigma_t}$

(c) $\frac{PD}{4\sigma_t}$ (d) $\frac{PD}{2\sigma_t}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
RPSC AE GWD, 2011

Ans. (d) : As we know,
Permissible tensile stress σ_t = Hoop stress in thin cylinder.

$$\text{Hoop stress} = \frac{PD}{2t}$$

$$\sigma_t = \frac{PD}{2t}$$

$$\boxed{t = \frac{PD}{2\sigma_t}}$$

1088. A thick cylinder with internal diameter d and outside diameter 2d is subjected to internal pressure P. Then the maximum hoop stress developed in the cylinder is

(a) P (b) (2/3)P

(c) (5/3)P (d) 2 P

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 2003

Ans. (c) : $d_i = d$ [internal diameter]
$d_o = 2d$ [outer diameter]
P = Pressure (internal)
In thick cylinder, maximum hoop stress,

$$\sigma_{hoop} = P \times \frac{r_2^2 + r_1^2}{r_2^2 - r_1^2} \qquad [r_1 = d, r_2 = 2d]$$

$$= P\left[\frac{(2d)^2 + (d)^2}{(2d)^2 - (d)^2}\right]$$

$$= P\left[\frac{5d^2}{3d^2}\right]$$

$$\boxed{\sigma_{hoop} = \frac{5P}{3}}$$

1089. Pick the correct statement about the hoop stress in a thick cylinder subjected to internal pressure.

(a) It varies linearly form zero at the outer surface to maximum at the inner surface.

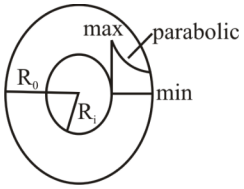
(b) It varies parabolically from a minimum stress at the outer surface (with non-zero magnitude) to a maximum value at the inner surface.

(c) It varies parabolically from zero at the outer surface to a maximum at the inner surface.

(d) It remains constant across the thickness.

HPPSC Asstt. Prof. 2014
RPSC AEN Pre-2013

Ans. (b) :



1090. The maximum shear stress in a cylinder shell of 4 m diameter and plate thickness 3 cm, subjected to an internal pressure of 3 N/mm^2 is _____.

(a) 200 N/mm^2 (b) 50 N/mm^2

(c) 100 N/mm^2 (d) 400 N/mm^2

VIZAG Steel MT 24.01.2021

Ans. (c) : Given,
Diameter (d) = 4 m = 4000 mm
Thickness (t) = 3 cm
= 30 mm
Internal pressure (P) = 3 N/mm^2

$$\text{Maximum shear tress } (\tau_{max}) = \frac{Pd}{4t}$$

$$= \frac{3 \times 4000}{4 \times 30}$$

$$\boxed{\tau_{max} = 100\,N/mm^2}$$

1091. The ratio of volumetric strain of sphere to strain in its diameter will be

(a) 2 (b) 1/3

(c) 3 (d) 1/2

GPSC DEE, Class-2 (GWSSB) 04.07.2021
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (c) :
Volumetric strain of sphere = 3 × diametric strain
Volumetric strain of cylinder = longitudinal strain + (2 × lateral strain)

1092. A 800 mm diameter pipe contains a fluid at a pressure of 3 Nmm^{-2}. If the safe stress in tension is 150 Nmm^{-2}, the minimum thickness of the pipe is

(a) 16 mm (b) 2 mm

(c) 4 mm (d) 8 mm

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) :

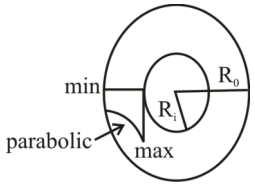
1093. In a thick cylindrical shell, the maximum radial stress at the outer surfaces of the shell is

(a) Zero (b) P

(c) –P (d) 2P

RPSC IOF, 2020

Ans. (a) : The radial stress across the thickness of a thick cylinder is maximum at the inner surface and zero at the outer surface.



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1094. A thin spherical shell of diameter 200 mm is subjected to an internal pressure of 2 MPa, If admissible tensile stress is 40 MPa, then thickness of the shell is:

(a) 2 mm (b) 5 mm
(c) 10 mm (d) 2.5 mm

CIL MT 27.02.2020

Ans. (d) : Diameter d = 200 mm
Internal pressure (P) = 2 MPa
Tensile stress σ = 40 MPa

$$\sigma = \frac{Pd}{4t}$$

$$t = \frac{Pd}{4\sigma} = \frac{2\times200}{4\times40} = 2.5 \text{ mm}$$

t = 2.5 mm

1095. A cylindrical pressure vessel has diameter 200 mm and thickness 2 mm. Find the hoop and axial stress (N/mm^2) in the cylindrical vessel, when it is subjected to an internal pressure of 5 MPa.

(a) 125, 125 (b) 125, 250
(c) 250, 250 (d) 250, 125

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : $\sigma_{hoop} = \frac{pd}{2t} = \frac{5\times200}{2\times2} = 250\,\text{MPa}$

$$\sigma_{long} \text{ or } \sigma_{axial} = \frac{1}{2}\sigma_{hoop} = 125\,\text{MPa}$$

1096. Which of the following statements regarding thin and thick cylinders, subjected to internal pressure only, is/are correct?

1. **A cylinder is considered thin when the ratio of its inner diameter to the wall thickness is less than 15.**
2. **In thick cylinders, tangential stress has highest magnitude at the inner surface of the cylinder and gradually decreases towards the outer surface**

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor

ESE 2020

Ans. (b) : If the thickness of the wall of shell is greater than 1/10 to 1/15 of its diameter.

1097. A cylindrical storage tank has an inner diameter of 600 mm and a wall thickness of 18 mm. The transverse and longitudinal strains induced are 255 × 10^{-6} mm/mm and 60 × 10^{-6} mm/mm, and if G is 77 GPa, the gauge pressure inside the tank will be

(a) 2.4 MPa (b) 2.8 MPa
(c) 3.2 MPa (d) 3.6 MPa

ESE 2020

Ans. (d) : Given,
d = 600 mm
t = 18 mm

$$\in_\ell = 60\times10^{-6}$$

$$\in_{hoop} = \in_{transverse} = 255\times10^{-6}$$

$$\in_\ell = \frac{Pd}{4tE}(1-2\mu) _____ (i)$$

$$\in_{hoop} = \frac{Pd}{4tE}(2-\mu) _____ (ii)$$

Dividing equation (i) by equation (ii),

$$\frac{\in_\ell}{\in_{hoop}} = \frac{(1-2\mu)}{2-\mu} = \frac{60\times10^{-6}}{255\times10^{-6}}$$

μ = 0.3
E = 2G(1+μ)
= 2×77×(1+0.3)
= 200.2 GPa

Putting the respective values in equation (i)

$$60\times10^{-6} = \frac{P\times600}{4\times18\times200.2}\times(1-2\times0.3)$$

P = 3.603 MPa
≈ 3.6 MPa

1098. A compressed air spherical tank having an inner diameter of 450 mm and a wall thickness of 7 mm is formed by welding. If the allowable shear stress is 40 MPa, the maximum permissible air pressure in the tank will be nearly

(a) 3 MPa (b) 5 MPa
(c) 7 MPa (d) 9 MPa

ESE 2020

Ans. (b) : $\sigma_1 = \sigma_2 = \frac{Pd}{4t}$

$$\therefore \quad \tau_{max} = \frac{Pd}{8t}$$

$$40 = \frac{P\times450}{8\times7}$$

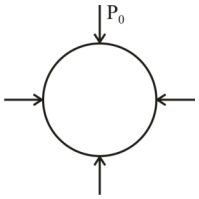
P = 4.978 MPa
P ≈ 5 MPa

1099. A thin spherical shell is subjected to an external pressure p_o. The volumetric strain of the spherical shell is
(where, d is the diameter of shell t is the thickness of the shell E is Young's modulus of elasticity of shell material
μ is Poisson's ratio of shell material)

(a) $\frac{p_o d}{4tE}(5-4\mu)$ (b) $\frac{3p_o d}{4tE}(1-\mu)$
(c) $\frac{3p_o d}{4tE}(1-2\mu)$ (d) $\frac{-3p_o d}{4tE}(1-\mu)$

UPPSC AE 13.12.2020, Paper-I

Ans. (d) :



Circumferential or hoop stress $(\sigma_c) = -\frac{P_o d}{4t}$

Volumetric strain $E_v = \frac{3\sigma}{E}(1-\mu)$

$$= -\frac{3P_o d}{4tE}(1-\mu)$$

(–ve sign is due to external pressure)

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1100. Internal and external radii of a thick cylinder are a and b. It is subjected to an internal pressure of p_i. The radial stress at a radius r in the cylinder is

(a) $\frac{a^2p_i}{(b^2-a^2)}\left(1-\frac{a^2}{r^2}\right)$ (b) $\frac{a^2p_i}{(b^2-a^2)}\left(1-\frac{b^2}{r^2}\right)$

(c) $\frac{b^2p_i}{(b^2-a^2)}\left(1-\frac{a^2}{r^2}\right)$ (d) $\frac{b^2p_i}{(b^2-a^2)}\left(1-\frac{b^2}{r^2}\right)$

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Given, Internal radius = a
External radius = b
Internal pressure = P_i
External pressure = 0
Circumferential or hoop stress is given by with internal radius (r_i) & external radius (r_0)

$$= \frac{P_i r_i^2}{r_o^2 - r_i^2}\left[1-\frac{r_o^2}{r^2}\right]$$

Similarly radial stress $(\sigma_r) = \frac{P_i a^2}{b^2-a^2}\left[1-\frac{b^2}{r^2}\right]$

1101. A thin cylinder of internal diameter D = 1 m and thickness t = 12 mm is subjected to internal pressure of 2 N/mm^2. Determine the hoop stress (N/mm^2) developed.

(a) 83 (b) 102
(c) 56 (d) 79

Oil India Senior Officer 23.12.2020

Ans. (a) : $\sigma_h = \frac{Pd}{2t} = \frac{2N}{mm^2}\times\frac{1000mm}{2\times 12mm}$
$= 83.33\ N/mm^2$

1102. A thin cylindrical shell of internal diameter D and thickness t is subjected to internal pressure p, E and μ are respectively the Elastic modulus and Poisson's ratio. The change in diameter is

(a) $\frac{pD^2}{4tE}(1-2\mu)$ (b) $\frac{pD^2}{4tE}(2-\mu)$

(c) $\frac{pt^2}{4DE}(2-\mu)$ (d) $\frac{pt^2}{4DE}(1-2\mu)$

APPSC AEE SCREENING 17.02.2019 ,CSE Pre-1998

Ans. (b) : Circumferential strain in thin cylinder

$$\epsilon_h = \frac{\sigma_h}{E}\left(1-\frac{\mu}{2}\right)$$

$$\left(\text{where } \sigma_h = \frac{PD}{2t}\right)$$

$$\therefore \quad \frac{\delta D}{D} = \frac{PD}{2tE}\left(\frac{2-\mu}{2}\right)$$

$$\therefore \quad \delta D = \frac{PD^2}{4tE}(2-\mu)$$

1103. Where does the maximum hoop stress in a thick cylinder under external pressure occur?

(a) At the outer surface
(b) At the inner surface
(c) At the mid-thickness
(d) At the 2/3rd outer radius

OPSC AEE 2019, 2015 Paper-I

Ans : (b) :

Hoop stress, $\sigma_h = \frac{A}{r^2} + B$

Hence hoop stress will be maximum at inner radius and minimum at outer radius.

1104. Thin cylindrical pressure vessel of 500 mm diameter is subjected to an internal pressure of 2 N/mm^2. If the thickness of the vessel is 20 mm, the hoop stress is

(a) 10 (b) 12.5
(c) 25 (d) 50

Gujarat PSC AE 2019

Ans : (c) :

Hoop stress $(\sigma_h) = \frac{Pd}{2t}$

P = 2 N/mm^2
d = 500 mm
t = 20 mm

$$\sigma_h = \frac{2\times 500}{2\times 20} = \frac{1000}{40}$$

$$\boxed{\sigma_h = 25\ N/mm^2}$$

1105. A spherical shell with internal diameter 320 mm and external diameter 640 mm is subjected to an internal fluid pressure of 75 N/mm^2. The hoop stress developed at the outer surface will be:

(a) 15.132 N/mm^2 (b) 16.071 N/mm^2
(c) 14.067 N/mm^2 (d) 17.173 N/mm^2

HPPSC ADF 05.03.2019

Ans. (*) : d_i = 320 mm
d_0 = 640 mm
t = $d_0 - d_i$ = 320 mm
p = 75 N/mm^2

$$\sigma_h = \frac{pd_0}{2\times t} = \frac{75\times 640}{2\times 320}$$

$$\sigma_h = 75 N/mm^2$$

1106. A spherical steel pressure vessel 400 mm in diameter with a wall thickness of 20 mm, is coated with brittle layer that cracks when strain exceeds 100 × 10^{-7}. What internal pressure will cause the layer to develop cracks? (E = 200GPa, μ = 0.3)

(a) 0.057 MPa (b) 5.7 MPa
(c) 0.57 MPa (d) 57 MPa

BHEL ET 2019

Ans. (c) : Diameter (d) = 400 mm
Thickness (t) = 20 mm
Strain (ϵ) = 100 × 10^{-7}
E = 200 GPa, μ = 0.3

$$\boxed{\epsilon = \frac{Pd}{4tE}(1-\mu)}$$

$$100\times 10^{-7} = \frac{P\times 400}{4\times 20\times 200\times 10^3}[1-0.3]$$

$$10^{-5}\times 16\times 10^6 = p\times 400\times 0.7$$

$$\frac{160}{400\times 0.7} = P \ldots \quad \boxed{P = 0.57\ MPa}$$

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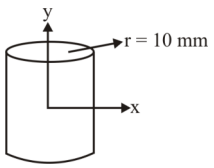
1107. A welded steel cylindrical drum made of a 10 mm thick plate has an internal diameter of 1.20 m. Find the change in diameter that would be caused by an internal pressure of 1.5 MPa. Assume that Poisson's ratio is 0.30 and E = 200 GPa (longitudinal stress, σ_y = PD/4t circumferential stress , σ_x = PD/2t).

(a) 4.590 mm (b) 0.459 mm
(c) 45.90 mm (d) 0.0459 mm

BHEL ET 2019

Ans. (b) :



t = 10 mm
P = 1.5 MPa
d = 1.20 m = 1200 mm
$\mu = 0.30$
E = 200 GPa

$$\sigma_y = \frac{Pd}{4t} = \frac{1.5\times1200}{4\times10}$$

$$\sigma_y = 45\,MPa$$

$$\sigma_x = \frac{Pd}{2t} = \frac{1.5\times1200}{2\times10} = 90\ MPa$$

$$\varepsilon_x = \frac{\sigma_x}{E} - \mu\frac{\sigma_y}{E}$$

$$= \frac{90}{20000} - 0.3\left(\frac{45}{20000}\right)$$

$$\varepsilon_x = 3.825\times10^{-4}$$

$$\varepsilon_x = \frac{\Delta D}{D}$$

$$\Delta D = \varepsilon_x \times D$$

$$= 3.825 \times 10^{-4} \times 1200 = 0.459\ mm$$

1108. Pressure vessel is said to be thin cylindrical shell, if the ratio of the wall thickness of the shell to its diameter is

(a) equal to 1/10 (b) less than 1/10
(c) more than 1/10 (d) none of these

Gujarat PSC AE 2019

Ans : (b) A thin cylindrical shell is the ratio of the wall thickness of the shell to its diameter is less than 1/10.

1109. The thickness of thin cylinder is determined on the basis of

(a) Radial stress
(b) Longitudinal stress
(c) Circumferential stress
(d) Principal shear stress

Gujarat PSC AE 2019

Ans : (c) : The thickness of thin cylinder is determined on the basis of circumferential stress.

1110. The inner diameter of a cylindrical tank for liquefied gas is 250 mm. The gas pressure is limited to 15 MPa. The tank is made of plain carbon steel with ultimate tensile strength of 340 N/mm^2, Poisson's ratio of 0.27 and the factor of safety of 5. The thickness of the cylinder wall will be.

(a) 60 mm (b) 50 mm
(c) 40 mm (d) 30 mm

ESE 2019

Ans. (d) : Maximum stress = Hoop stress in cylinder

$$\sigma_{max} = \sigma_h$$

$$\frac{\sigma_u}{FOS} = \frac{pd}{2t}$$

$$\frac{340}{5} = \frac{15\times250}{2\times t}$$

$$t = 27.57\ mm \simeq 30\ mm$$

1111. A spherical shell of 1.2 m internal diameter and 6 mm thickness is filled with water under pressure until volume is increased by 400 × 10^3 mm^3. If E = 204 GPa, Poisson's ratio ν = 0.3, neglecting radial stresses, the hoop stress developed in the shell will be nearly

(a) 43 MPa (b) 38 Mpa
(c) 33 Mpa (d) 28 Mpa

ESE 2019

Ans. (a) : Given,
d = 1.2 m = 1200 mm
r = 600 mm
t = 6 mm
$\Delta v = 400 \times 10^3\ mm^3$
$E = 204\ GPa = 204 \times 10^3\ MPa$
$\mu = 0.3$

$$\in_v = \frac{\Delta v}{v} = \frac{400\times10^3}{\frac{4}{3}\pi r^3} = \frac{400\times10^3}{\frac{4}{3}\pi(600)^3}$$

$$\in_v = 4.42 \times 10^{-4}$$

Volumetric strain ($\in_v$) = 3 × hoop strain

$$4.42 \times 10^{-4} = 3 \times \in_h$$

$$\in_h = 1.47 \times 10^{-4}$$

$$\in_h = \frac{\sigma_n}{E}(1-\mu)$$

$$1.47 \times 10^{-4} = \frac{\sigma_h}{204\times10^3}(1-0.3)$$

$$\sigma_h = 42.94 \simeq 43\ MPa$$

1112. Consider the following statements:

1. In case of a thin spherical shell of diameter d and thickness t, subjected to internal pressure p, the principal stresses at any point equal $\frac{pd}{4t}$
2. In case of thin cylinders the hoop stress is determined assuming it to be uniform across the thickness of the cylinder
3. In thick cylinders, the hoop stress is not uniform across the thickness but it varies from a maximum value at the inner circumference to a minimum value at the outer circumference.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : (i) In case of a thin spherical shell of diameter d and thickness t, subjected to internal pressure p, the principal stresses at any point is given by

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$$\sigma = \frac{pd}{4t}$$

(ii) In case of thin cylinders, the hoop stress is determined assuming it to be uniformly distributed over the thickness of the wall, provided that the thickness is small compared to radius.

(iii) For a thick cylinder hoop stress is given by

$$\sigma_h = \frac{pR^2}{r^2 - R^2}\left(\frac{r^2 + x^2}{x^2}\right) \qquad \text{(here } r \le x \le R\text{)}$$

where r and R are inner and outer radius, respectively.

σ_h is maximum when x is minimum.

σ_h is maximum at x = r

i.e. at inner surface.

Hence all the statement are correct.

1113. A thin walled cylinder (diameter = D, length = L, thickness of cylinder material = t, modulus of elasticity = E, Poission's ratio *ν*) is subjected to fluid pressure (P) inside it. The total volume of fluid that an be stored in the cylinder will be:

(a) $\frac{\pi}{4}D^2L\left[1+\frac{PD}{4tE}(5-4\nu)\right]$

(b) $\frac{\pi}{4}D^2L\left[1+\frac{PD}{4tE}(5+4\nu)\right]$

(c) $\frac{\pi}{4}D^2L$

(d) $\frac{\pi}{4}D^2L\left[\frac{PD}{4tE}(5-4\nu)\right]$

(e) $\frac{\pi}{4}D^2L\left[1+\frac{PD}{4tE}(1-\nu)\right]$

CGPSC AE 25.02.2018

Ans. (a) : We know that volumetric strain in thin walled cylinder is given as:

$$\frac{dV}{V} = \frac{Pd}{4tE}[5-4\nu]$$

$$dV = \frac{Pd}{4tE}[5-4\nu]\times V$$

Then total volume of thin walled cylinder

$$V_{Total} = V + dV$$

$$= \frac{\pi}{4}D^2 \times L + \frac{Pd}{4tE}[5-4\nu]\times\frac{\pi}{4}D^2\times L$$

$$V_{Total} = \frac{\pi}{4}D^2\times L\left[1+\frac{Pd}{4tE}(5-4\nu)\right]$$

1114. The Hoop stresses are acting across the

(a) Circumferential section
(b) Longitudinal section
(c) Radial section
(d) None of the above

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (b) :

- The Hoop stress are acting across the longitudinal section.
- The Longitudinal stress acting when two cross sectional areas of the cylinder are subjected to equal and opposite forces the stress experienced by the cylinder.

1115. A boiler shell 180cm diameter and plate thickness of 12mm is subjected to an internal pressure of 1.2MN/m^2. The hoop stress will be

(a) 45 N/mm^2 (b) 90 N/mm^2
(c) 135 N/mm^2 (d) 155 N/mm^2

WBPSC AE, 2017

Ans. (b) : $\sigma_h = \frac{pd}{2t}$

$$\sigma_h = \frac{1.2\times10^6\times180\times10^{-2}}{2\times12\times10^{-3}}$$

$$= 90\text{N/mm}^2$$

1116. In case of thin walled cylinders the ratio of hoop stress to radial stress is

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) 2 (d) None of the above

WBPSC AE, 2017

Ans. (d) : $\frac{\sigma_h}{\sigma_R} = \frac{\text{hoop stress}}{\text{radial stress}}$

$\sigma_R = 0$

- Radial stress in thin cylindrical shell and sphere radial stress is neglected and assumed zero. However its value at inner phase equal to outer pressure and outer phase it is equal to atmospheric pressure is zero but for all practical purpose its value taken zero from inner to outer surface.

1117. A cylinder is said to be thin if the thickness to diameter ratio is less than

(a) $\frac{1}{5}$ (b) $\frac{1}{10}$
(c) $\frac{1}{15}$ (d) $\frac{1}{20}$

Karnataka PSC Lect., 27.05.2017

Ans (d) : $\frac{t}{D} \le \frac{1}{20}$ —— Thin cylinder

$\frac{t}{D} > \frac{1}{20}$ —— Thick cylinder

1118. A thin cylinder of 4 m diameter has a wall thickness of 50 mm. It is subjected to an internal pressure of 2 MPa, What is the maximum shear stress induced in an element considered on the inner surface?

(a) 80 MPa (b) 41 MPa
(c) 40 MPa (d) 160 MPa

APGENCO AE, 2017

Ans. (c) : In thin cylinder inner surface is subject to three stresses

$\sigma_c = \frac{pd}{2t}$

$\sigma_{(l)} = \frac{pd}{4t}$ and radial pressure (Negligible)

Max shear stress at inner surface $= \frac{\sigma_c - 0}{2}$

$$\tau = \frac{pd}{4t} = \frac{2\times10^6\times4}{4\times50\times10^{-3}}$$

, $\tau = 40$ MPa

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1119. A 300 mm diameter basketball of 2mm wall thickness has been inflated to a pressure of 100 kPa. The value of hoop stress developed is equal to

(a) 10 MPa (b) 5 MPa
(c) 3.75 MPa (d) 2.75 MPa

Punjab PSC (Lect.) 06.08.2017

Ans. (c)

$$\text{Hoop stress} = \frac{Pd}{4t}$$
$$= \frac{100\times1000\times300}{4\times2}$$
$$= \frac{10^7\times3}{4\times2}$$
$$= 3.75 \text{ MPa}$$

1120. The purpose of longitudinal butt-joint in boiler shell is :

(a) to make cylindrical ring from steel plate
(b) to increase the length of boiler shell by connecting one ring to another.
(c) to make diameter and length of boiler shell
(d) to connect openings to shell

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : The purpose of longitudinal butt joint in boiler shell is to make cylindrical ring from steel plate.

1121. A seamless pipe of 600 mm diameter contains a fluid under a pressure of 3 N/mm^2. If the permissible tensile stress be 100 N/mm^2, the minimum thickness of the pipe is

(a) 8 mm (b) 9 mm
(c) 11 mm (d) 10 mm

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) : Given, d = 600 mm
P = 3 N/mm^2
Permissible tensile stress = 100 N/mm^2

$$\text{Hoop stress} = \frac{Pd}{2t}$$

Hoop stress = permissible tensile stress

$$\frac{Pd}{2t} = 100$$

$$t = \frac{3\times600}{2\times100} = 9 \text{ mm}$$

1122. The reason why the pressure vessels are not made of rectangular shape is

(a) These are difficult to fabricate
(b) It has been a practice to use cylindrical vessels
(c) They are not economical
(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Because gas cylinders contain high internal pressure, so we cannot use shapes like rectangle and cube because there will be many stresses associated with such shapes. They are not economical.

1123. A thin cylinder pressure vessel of 1m diameter generates steam at a pressure of 1.4 N/mm^2. What will be the wall thickness when longitudinal stress does not exceed 28 MPa?

(a) 10.5 mm (b) 12.5 mm
(c) 14.5 mm (d) 16.5 mm
(e) 18.5 mm

(CGPCS Polytechnic Lecturer 2017)

Ans. (b) : Pressure vessel diameter = 1 m = 1000 mm
Steam pressure = 1.4 N/mm^2
Longitudinal stress (σ_ℓ) = 28 MPa = 28 N/mm^2
Thickness (t) = ?
We know that

$$\sigma_\ell = \frac{Pd}{4t}$$

$$t = \frac{P\times d}{4\times\sigma_\ell} = \frac{1.4\times1000}{4\times28}$$

$$\boxed{t = 12.5 \text{ mm}}$$

1124. A thin cylindrical pressure vessel of 500 mm internal diameter is subjected to an internal pressure of 2 N/mm^2. What will be the hoop stress if the thickness of the vessel is 20 mm?

(a) 25 N/mm^2 (b) 23 N/mm^2
(c) 27 N/mm^2 (d) 29 N/mm^2

CIL (MT) 2017 IInd Shift

Ans. (a) : Hoop stress in the cylindrical pressure vessel

$$= \frac{pd}{2t} = \frac{2\times500}{2\times20} = 25 \text{ N/mm}^2.$$

1125. A 2.0 m diameter penstock pipe carries water under a pressure head of 100 m. If the wall thickness is 7.5 mm, the tensile stress in the pipe wall, in MPa, is

(a) 65.3 (b) 130.5
(c) 231.0 (d) 1305.0

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : D = 2 m
H = 100 m
t = 7.5 mm = 0.0075 m
P = ρgh (N/m^2)

$$\sigma_t = \frac{PD}{2t} = \frac{(1000\times9.81\times100)\times2}{2\times0.0075}$$

$= 130.8 \times 10^6$ N/m^2
= 130.8 MPa

1126. In design of thin cylinder, the thickness of cylinder plate is calculated from the relations:

(a) $T = \frac{PD}{ft}$ (b) $T = \frac{PD}{2ft}$
(c) $T = \frac{2PD}{ft}$ (d) None of these

Where:
D = Internal diameter of cylinder
T = Thickness of cylinder plate
ft = Tangential or hoop stress
P = Internal pressure

PTCUL AE 25.06.2017

Ans. (b) : $T = \frac{PD}{2ft}$

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1127. A cylinder of internal diameter 200 mm is subjected to internal pressure of 15 MPa. Determine the thickness of the cylinder if the material has a yield strength of 300 MPa. Take Factor of safety =2.

(a) 10 mm (b) 5 mm
(c) 7.5 mm (d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (a) : Diameter D = 200 mm
P = 15 MPa
Thickness t = ?
Yield strength (S_{ys}) = 300 MPa
Factor of safety (N) = 2

$$\sigma_{hoop} = \frac{PD}{2t}$$

Or $\sigma_{hoop} \le (\sigma_{yt})$

$$\frac{PD}{2t} = \frac{S_{ys}}{N}$$

$$\frac{15\times 200}{2\times t} = \frac{300}{2}$$

$$t = \frac{3000}{300}$$

t = 10 mm

1128. A cylinder is called a thick wall cylinder when the ratio of inner diameter of the cylinder to wall thickness is

(a) between 20 to 30 (b) less than 15
(c) between 25 to 30 (d) greater than 25

Rajasthan Nagar Nigam AE 2016, Shift-II

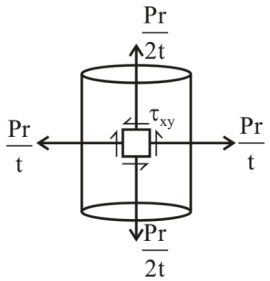
Ans. (b) : When the ratio of the inner diameter of the cylinder to the wall thickness is less than $\frac{1}{10}$ to $\frac{1}{15}$, the cylinder is called thick walled cylinder.
Example – Hydraulic cylinder
High pressure pipe
Gun barrels

1129. A thin cylinder of 100 mm internal diameter and 5 mm thickness is subjected to an internal pressure of 10 MPa and a torque of 2000 Nm. What is the magnitude of the principal stresses?

(a) 75 MPa, 25 MPa (b) 75 MPa, 50 MPa
(c) 100 MPa, 50 MPa (d) 125 MPa, 50 MPa
(e) 125 MPa, 70 MPa

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (*) : Given, d = 100 mm; r = 50 mm, t = 5 mm
P = 10 MPa
Torque, T = 2000 N-m



$$\tau_{xy} = \frac{T}{2\pi r^2 t} = \frac{2000\times 10^3}{2\pi(50)^2(5)} = 25.46\,MPa$$

$$\sigma_x = \frac{Pr}{t} = \frac{10\times 50}{5} = 100\,MPa$$

$$\sigma_y = \frac{Pr}{2t} = \frac{100}{2} = 50\,MPa$$

Principal stress,

$$\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$= \left(\frac{100+50}{2}\right) \pm \sqrt{\left(\frac{100-50}{2}\right)^2 + (25.46)^2}$$

$\sigma_{1,2} = 75 \pm 35.68$

Max. principal stress $\Rightarrow \sigma_1 = 75 + 35.68 = 110.68$ MPa
Min. principal stress $\Rightarrow \sigma_2 = 75 - 35.68 = 39.32$ MPa

1130. The volumetric strain in case of thin spherical shell when subjected to internal pressure P is

(a) $\frac{Pd}{2tE}\left(\frac{1}{2}-\frac{1}{m}\right)$ (b) $\frac{Pd}{2tE}\left(1-\frac{1}{2m}\right)$
(c) $\frac{Pd}{4tE}\left(1-\frac{1}{m}\right)$ (d) $\frac{3Pd}{4tE}\left(1-\frac{1}{2m}\right)$
(e) None of these

CGPSC AE 16.10.2016

Ans. (e) : As we know,
For thin cylinder,

$$\epsilon_v = \epsilon_\ell + 2\epsilon_h \qquad \therefore \epsilon_h = \frac{Pd}{4Et}(1-\nu)$$

$$= \frac{Pd}{4tE}(5-4\nu)$$

For spherical thin shell,

$$\epsilon_v = 3\epsilon_h \qquad \therefore \epsilon_\ell = \epsilon_h$$

$$= \frac{3Pd}{4tE}(1-\nu)$$

1131. In thick cylinder, if hoop stress is plotted w.r.t. $\left(\frac{1}{r^2}\right)$, then the curve will be

(a) Parabolic (b) Hyperbolic
(c) Linear (d) Elliptical

RPSC LECTURER 16.01.2016

Ans. (c) : In thick cylinder, if hoop stress is plotted w.r.t. $\left(\frac{1}{r^2}\right)$, then the curve will be linear.

1132. A thick cylinder, having r_o and r_i as outer and inner radii, is subjected to an internal pressure P. The maximum tangential stress at the inner surface of the cylinder is

(a) $\frac{P(r_o^2+r_i^2)}{r_o^2-r_i^2}$ (b) $\frac{P(r_o^2-r_i^2)}{r_o^2+r_i^2}$
(c) $\frac{2Pr_i^2}{(r_o^2-r_i^2)}$ (d) $\frac{P(r_o^2-r_i^2)}{r_i^2}$

UPPSC AE 12.04.2016 Paper-I

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Ans. : (a) For a thick cylinder, the hoop stress is maximum at the inner surface and is given by.

$$\sigma = \frac{\left(d_o^2 + d_i^2\right)}{\left(d_o^2 - d_i^2\right)}.p$$

$$\sigma = P.\frac{\left(r_o^2 + r_i^2\right)}{\left(r_o^2 - r_i^2\right)}$$

1133. Compound tubes are used in internal pressure cases, for following reasons

(a) For increasing the thickness.
(b) For increasing the outer diameter o the tube.
(c) The strength is more.
(d) It evens out stresses.

UPPSC AE 12.04.2016 Paper-I

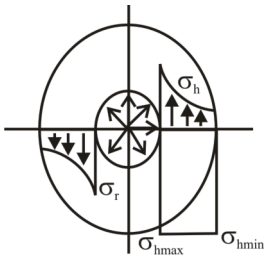
Ans. : (d) Compound tubes are used in internal pressure cases, for it evens out stresses

1134. In a thick cylinder subjected to internal pressure, the hoop stress is _____ at the outer surface and is _____ at the inner surface

(a) min, min (b) max, max
(c) max, min (d) min, max

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : The hoop stress is minimum at outer surface & its maximum at inner surface (σ_h)



1135. If a steel tyre is heated and struck on a rigid wheel, after cooling the tyre will be subjected to

(a) Bending (b) Torsion
(c) Hoop stress (d) Compression

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : If a steel tyre is heated and struck on a rigid wheel, after cooling the tyre will be subjected hoop stress, due to shrinkage from outside & expansion on inside, due to temperature variation property varies.

1136. If the hoop stress in a thin cylinder is 24 N/sq.mm, then its longitudinal stress is equal to

(a) 36 N/sq.mm (b) 24 N/sq.mm
(c) 12 N/sq.mm (d) 6 N/sq.mm

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : As we know, for cylinder

$$\sigma_H = \frac{Pd}{2t} \quad \& \quad \sigma_L = \frac{Pd}{4t}$$

So, $\sigma_H = 2\sigma_L$

$24 = 2 \times \sigma_L$

$\sigma_L = 12\,N/mm^2$

1137. A thin cylindrical shell of diameter (d), length (ℓ) and thickness (t) is subjected to an internal pressure (p). The ratio of the longitudinal strain to hoop strain is

(a) $\frac{m-2}{2m-1}$ (b) $\frac{2m-1}{m-2}$
(c) $\frac{m-2}{2m+1}$ (d) $\frac{m+2}{2m+2}$

ISRO Scientist/Engineer 11.10.2015
ESE 2004

Ans. (a) : Longitudinal strain $(\in_\ell) = \frac{1}{E}\left[\sigma_\ell - \mu\sigma_h\right]$

$$= \frac{1}{E}\left[\frac{\sigma_h}{2} - \mu\sigma_h\right]$$

$$\in_\ell = \frac{\sigma_h}{2E}[1-2\mu]$$

Hoop strain $(\in_h) = \frac{1}{E}\left[\sigma_h - \mu\sigma_\ell\right]$

$$= \frac{1}{E}\left[\sigma_h - \frac{\mu\sigma_h}{2}\right]$$

$$\in_h = \frac{\sigma_h}{2E}[2-\mu]$$

Required ratio $(\in_\ell/\in_h) = \frac{\text{Longitudinal strain}}{\text{Hoop strain}}$

$$= \frac{\frac{\sigma_h}{2E}[1-2\mu]}{\frac{\sigma_h}{2E}[2-\mu]}$$

$$= \frac{(1-2\mu)}{(2-\mu)}$$

$$= \frac{\left[1-2\times\frac{1}{m}\right]}{\left[2-\frac{1}{m}\right]} \qquad \left[\because \mu = \frac{1}{m}\right]$$

$$= \frac{m-2}{2m-1}$$

1138. A spherical vessel with an inside diameter of 2 m is made of material having an allowable stress in tension of 500 kg/cm^2. The thickness of a shell to withstand a pressure of 50 kg/cm^2 should be-

(a) 5 cm (b) 2.5 cm
(c) 10 cm (d) 1.25 cm

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) For spherical vessel

$$\sigma_h = \sigma_\ell = \frac{PD}{4t}$$

$$500 = \frac{50\times 200}{4\times t}$$

$\boxed{t = 5cm}$

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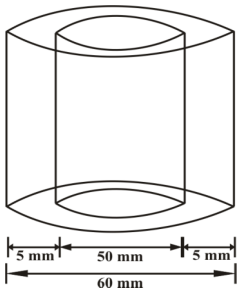
1139. The cross section area of a hollow cylinder has an internal diameter of 50 mm and a thickness of 5 mm. Moment of inertia of the cross-section about its centroidal axis is

(a) 2.848×10^5 mm^4 (b) 4.294×10^5 mm^4
(c) 1.424×10^5 mm^4 (d) 1.647×10^5 mm^4
(e) 3.294×10^5 mm^4

CGPSC AE 26.04.2015 Shift-I

Ans. (e) :



Given $d_i = 50$ mm

$d_o = 60$ mm

$$k = \frac{d_i}{d_0}$$

Moment of inertia (I) $= \frac{\pi}{64} d_0^4 \left[1 - \left(\frac{d_i}{d_o}\right)^4\right]$

$$= \frac{\pi}{64} \; 60^4 \left[1 - \left(\frac{50}{60}\right)^4\right]$$

$= 329209.37 = 3.29 \times 10^5$ mm^4

1140. A thin cylinder of internal diameter D = 1 m and thickness t = 12 mm is subjected to internal pressure of 4 N/mm^2. Determine hoop stresses developed.

(a) 83.33 N/mm^2 (b) 83.33×10^{-3} N/mm^2
(c) 166.67×10^{-3} N/mm^2 (d) 166.67 N/mm^2
(e) 166.67 N/m^2

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Given

D = 1 m = 1000 mm

t = 12 mm

P = 4 N/mm^2

In a thin cylinder pressure vessel, hoop (circumferential) stress

$$\sigma = \frac{Pd}{2t}$$

$$= \frac{4 \times 1000}{2 \times 12}$$

$= 166.67$ N/mm^2

1141. Which of the following equation is used to determine the thickness of thick cylindrical shell with closed ends and made of brittle material?

(a) Lame's equation
(b) Clavarino's equation
(c) Birnie's equation
(d) Barlow's equation

UPRVUNL AE 2014

Ans : (a) Lame's equation is used to determine cylindrical shell with closed end

$$\sigma_x = \frac{b}{r^2} + a \qquad p_x = \frac{b}{x^2} - a$$

Where p_x= Compressive Nature

σ_x= Tensile Nature

1142. A thin-walled cylinder with closed ends for which inner radius is 0.50 meter and outer radius is 0.52 meter, is subjected to internal pressure 2 MPa. The absolute maximum shearing stress on the inner surface of the cylinder shall be

(a) 37.5 MPa (b) 75 MPa
(c) 26 MPa (d) 24 MPa

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : Inner Radius r = 0.50 m

Outer Radius R = 0.52 m

Internal pressure P = 2 MPa

$$\tau_{max} = \frac{Pd}{4t}$$

So, outer diameter D = 2 × 0.52

= 1.04 m

Thickness t = R – r

= 0.52 - 0.50

t = 0.02 m

$$\tau_{max} = \frac{2 \times 1.04}{4 \times 0.02}$$

= 26 MPa.

1143. The initial hoop stress in a thick cylinder when it is wound with a wire under tension will be :

(a) zero (b) tensile
(c) compressive (d) bending

(HPPSC AE 2014)

Ans : (c) The initial hoop stress in a thick cylinder when it is wound with a wire under tension will be compressive.

Analysis of Thick Cylinders/Lame's Theorem:-

* Lame's assumption

(i) Material of shell is homogeneous

(ii) Plane section of cylinder, perpendicular to longitudinal axis remains plane under pressure.

* Subjected to internal pressure

(i) $x = R_i, \sigma_h = \frac{P\left[R_0^2 + R_i^2\right]}{R_0^2 - R_i^2}$

(ii) $x = R_0, \sigma_h = \frac{2PR_i^2}{R_0^2 - R_i^2}$

1144. Shrinking a thick cylinder over another helps:

(a) reduce the magnitude of tensile hoop stress
(b) reduce the difference between the higher and lower magnitude of tensile hoop stress
(c) remove the longitudinal stress
(d) reduce the cost

(HPPSC AE 2014)

Ans : (b) Shrinking a thick cylinder over another helps reduce the difference between the higher and lower magnitude of tensile hoop stress.

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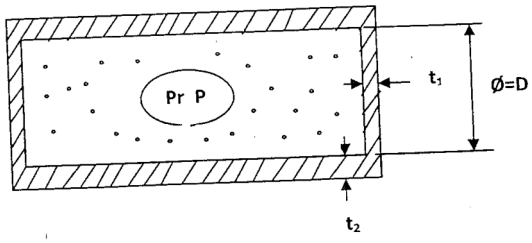
1145. A thick cylinder of inner dia 'D', wall thickness t_2 and length 'L' is sealed at its both ends with caps. The thickness of the cap is t_1. Allowable tensile yield stress = σ_y and allowable shear stress = τ_y. A gas is pumped this cylinder at pressure 'p'. The cap will yield in shear at circumference of diameter 'D' when the gas pressure applied is more than

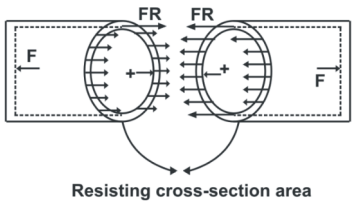


(a) $\frac{4t_1\tau_y}{D}$ (b) $\frac{8t_1\tau_y}{D}$

(c) $\frac{4t_2\sigma_y}{D}$ (d) $\frac{2t_1\tau_y}{D}$

ISRO Scientist/Engineer 12.05.2013

Ans. (a) : Busting of pressure variation (P.V.) occurs "circumferential/Axial"



$A = \pi D t_1$

Safe Condition

$F \le FR$

$F = FR$

$$P \times \frac{\pi}{4} D^2 = \tau_y \times \pi \times D \times t_1$$

or, $$P = \frac{4t_1\tau_y}{D}$$

1146. A pipe of diameter 800 mm contains fluid under a pressure of 2 N/mm^2. If the tensile strength is 100 N/mm^2, the thickness of the pipe is

(a) 16 mm (b) 4 mm

(c) 8 mm (d) 10 mm

TNPSC AE 2013

Ans. (c) : Data given -

d = 800 mm, P = 2 N/mm^2

σ_H = 100 N/mm^2

we know that

$$\sigma_H = \frac{Pd}{2t}$$

$$t = \frac{2 \times 800}{2 \times 100}$$

$$\boxed{t = 8 \text{ mm}}$$

1147. A thin spherical shell of 1.5 m diameter is 8 mm thick is filled with a liquid at the pressure of 3.2 N/mm^2, The stress induced in the shell is

(a) 75 N/mm^2 (b) 150 N/mm^2

(c) 200 N/mm^2 (d) 180 N/mm^2

TNPSC AE 2013

Ans. (b) : Data given-

d = 1.5 m = 1500 mm

t = 8 mm

P = 3.2 N/mm^2

$$\sigma_l = \sigma_H = \frac{Pd}{4t} = \frac{3.2 \times 1500}{4 \times 8}$$

$$\sigma_H = 150 \text{ N/mm}^2$$

1148. In thick cylinder, the radial stresses in the wall thickness:-

(a) is zero

(b) negligible small

(c) varies from the inner face to outer face

(d) None of the above

UKPSC AE-2013, Paper-I

Ans. (c) : In thick cylinder, the radial stresses in the wall thickness varies from inner face [maximum] to outer face [zero].

1149. A cylindrical shell of diameter 200 mm and wall thickness 5 mm is subjected to internal fluid pressure of 10 N/mm^2. Maximum shearing stress induced in the shell in N/mm^2, is

(a) 50 (b) 75

(c) 100 (d) 200

UKPSC AE 2012 Paper-I

Ans. (d) : Data Given

Diameter of shell (d) = 200 mm

thickness (t) = 5 mm

pressure (P) = 10 N/mm^2

then, maximum shear stress

$$\tau = \frac{Pd}{2t}$$

$$= \frac{10 \times 200}{2 \times 5}$$

τ = 200 N/mm^2

1150. The bursting pressure for a cold drawn seamless steel tubing of 60 mm inside diameter with 2 mm wall thickness is (The ultimate strength of steel is 380 MN/m^2)

(a) 25.33 MN/m^2 (b) 24.33 MN/m^2

(c) 26.33 MN/m^2 (d) 50.66 MN/m^2

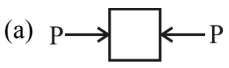
APPSC AEE 2012

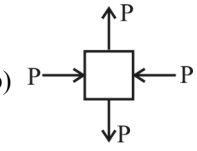
Ans. (a) : Given, d = 60 mm, t = 2m, σ_h = 380 MN/m^2

Solution- $$\sigma_h = \frac{pd}{2t} \Rightarrow P = \frac{\sigma_h \times 2t}{d} = \frac{380 \times 2 \times 2}{60}$$

P = 25.33 MN/m^2

1151. A solid thick cylinder is subjected to an external hydrostatic pressure 'P'. The state of stress in the material of the cylinder is represented as :

(a) 

(b) 

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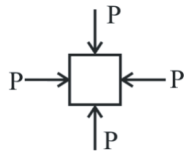
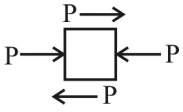
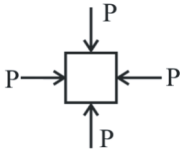
(c) 

(d) 

APPSC AE 04.12.2012

Ans. (c) : A solid thick cylinder is subjected to an external hydrostatic pressure 'P'. The state of stress represented by –



In case of hydrostatic state of stress only pure compressive stress acts on all the faces.

1152. Thin cylinders are frequently required to operate under pressures upto :

(a) 5 MN/m^2 (b) 15 MN/m^2
(c) 30 MN/m^2 (d) 250 MN/m^2

APPSC AE 04.12.2012

Ans. (c) : Thin cylinders are frequently required to operate upto 30 MN/m^2.

1153. The bursting pressure for a cold drawn seamless steel tubing of 60 mm inside diameter with 2 mm wall thickness is (The ultimate strength of steel is 380 MN/m^2)

(a) 25.33 MN/m^2 (b) 24.33 MN/m^2
(c) 26.33 MN/m^2 (d) 50.66 MN/m^2

APPSC AE 04.12.2012

Ans. (a) : For thin cylinder -

$$\sigma_c = \frac{Pd}{2t}$$

P = Bursting pressure = ?

Given,

d = Inside diameter of steel tube = 60mm

t = Wall thickness of seamless steel tube = 2mm

σ_c = Circum perential stress (or) Bursting stress (or)
= Ultimate strength of steel = 380 MN/m^2

$$\therefore P = \frac{\sigma_c \times 2t}{d} = \frac{380\,MN/m^2 \times 2 \times 2mm}{60mm}$$

$P = 25.33\ MN/m^2$

1154. For a thick-walled shell, the diameter-thickness ratio is

(a) less than 20 (b) greater than 20
(c) equal to 20 (d) equal to 20

BPSC AE 2012 Paper - VI

Ans : (a) : For a thick walled shell the diameter-thickness ratio is greater than 20

1155. In a thick cylinder, radial stress at inner surface is

(a) independent of fluid pressure
(b) more than fluid pressure
(c) less than fluid pressure
(d) equal to fluid pressure

BPSC AE 2012 Paper - VI

Ans : (d) : The radial stress for a thick walled cylinder is equal and opposite to the gauge pressure on the inside surface and zero on the outside.

1156. A thin cylindrical shell is subjected to internal pressure, such that hoop strain developed in shell is 850 microstrain. If Young's modulus is 100 GPa and Poisson's ratio is 0.3, what is hoop stress developed in shell :

(a) 200 MPa (b) 100 MPa
(c) 50 MPa (d) None of these

PSPCL AE, 2012

Ans. (b) : $\varepsilon_c = \frac{1}{E}(\sigma_c - \mu\sigma_l)$

$$850 \times 10^{-6} = \frac{1}{100 \times 10^9} \times \left[\sigma_c - 0.3 \times \frac{\sigma_c}{2}\right]$$

$$\sigma_c = 2\sigma_l = \frac{pd}{2t}$$

σ_c = 100 MPa

1157. A thick cylindrical shell is subjected to internal pressure 'p', such that maximum hoop stress developed in shell is 100 MPa. If external radius of shell is two times the internal radius, what is the magnitude of 'p'.

(a) 100 MPa (b) 60 MPa
(c) 40 MPa (d) None of these

PSPCL AE, 2012

Ans. (b) : $\sigma_h = A + \frac{B}{x^2}$

r_i = Inner radius

r_o = Outer radius

Using boundary condition –

$$A = \frac{Pr_i^2}{r_o^2 - r_i^2}$$

$$B = \frac{P \times r_i^2 \times r_o^2}{r_o^2 - r_i^2}$$

Given, $r_i = r$

$r_o = 2r$

$$A = \frac{P \times r^2}{4r^2 - r^2} = \frac{P}{3}$$

$$B = \frac{P \times 4r^2 \times r^2}{4r^2 - r^2} = \frac{4P \times r^2}{3}$$

$$\sigma_h = \frac{P}{3} + \frac{4P \times r^2}{3 \times r^2} = \frac{5P}{3} \qquad (\sigma_h = 100\ MPa)$$

$$100 = \frac{5P}{3}$$

$$\boxed{P = 60\,MPa}$$

1158. A compound cylinder with inner radius 50 mm and outer radius 70 mm is made by shrinking one cylinder to the other cylinder. The junction radius is 60 mm and the junction pressure is 11 N/mm^2. The maximum hoop stress developed in the inner cylinder would be ?

(a) 3.6 N/mm^2 compression
(b) 3.6 N/mm^2 tension
(c) 7.2 N/mm^2 compression
(d) 7.2 N/mm^2 tension

VIZAG STEEL MT 2011

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Ans. (c) : $P = 1.1\ N/mm^2$

Junction radius, $R_2 = 60$ mm

Inner radius, $R_1 = 50$ mm

Hoop stress, $\sigma_h = \dfrac{-2PR_2^2}{R_2^2 - R_1^2}$

$= \dfrac{-2 \times 1.1 \times (60)^2}{(60)^2 - (50)^2}$

$\sigma_h = 7.2 N/mm^2$ (Compressive)

1159. A thin cylinder contains fluid at a pressure of 30 kg/cm^2. The inside diameter of the shell is 60 cm and the tensile stress in the material is to be limited to 900 kg/cm^2. The shell must have minimum wall thickness of

(a) 1 mm (b) 2.7 mm

(c) 10 mm (d) 9 mm

ISRO Scientist/Engineer 2011

Ans. (c) : $P = 30\ kg/cm^2$

$d = 60$ cm

$\sigma = 900\ kg/cm^2$

$\sigma = \dfrac{Pd}{2t}$

Hoop stress (Circumferential) $\sigma = \dfrac{Pd}{2t}$

$900 = \dfrac{30 \times 60}{2 \times t}$

$t = \dfrac{30 \times 60}{900 \times 2}$

t = 1cm = 10mm

1160. A long thin cylindrical shell is fabricated by making longitudinal and circumferential weld seams. The efficiencies of longitudinal and circumferential weld seams are 80% and 75% respectively. Now this shell is subjected to internal fluid pressure. The ratio of the larger to smaller principal stresses induced in the shell material is

(a) 2 (b) $\dfrac{15}{8}$

(c) 1.75 (d) 1.50

UPSC JWM Advt. No.-50/2010

Ans. (b) : We know that, for riveted & welded,

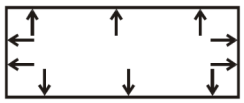
Hoop/circumferential stress,

$\sigma_h = \dfrac{pd}{2t\eta_\ell} = \dfrac{pd}{2 \times 0.8 \times t} = \dfrac{pd}{1.6t}$

Given, longitudinal stress, $\sigma_L = \dfrac{pd}{4t\eta_h} = \dfrac{pd}{4 \times 0.75 \times t} = \dfrac{pd}{3t}$

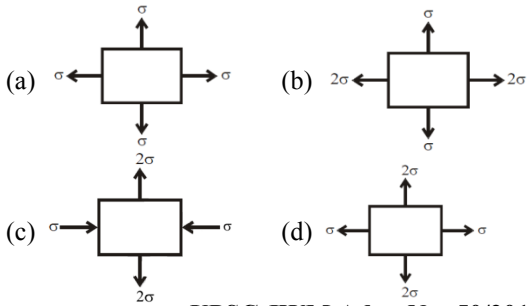
So, $\dfrac{\sigma_h}{\sigma_L} = \dfrac{pd}{1.6t} \times \dfrac{3t}{pd} = \dfrac{15}{8}$

1161.



Cylindrical shell

Principal stresses on the outside surface element of a thin cylindrical shell subjected to internal fluid pressure as shown in the figure, are represented by



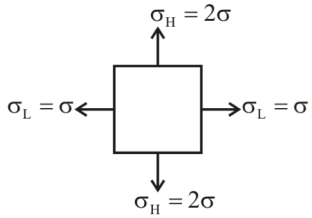
UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know that,

Hoop stress, $\sigma_H = \dfrac{PD}{2t}$

Longitudinal stress, $\sigma_L = \dfrac{PD}{4t} \Rightarrow \sigma_H > \sigma_L$

So,



1162. A compound cylinder with inner radius 5 cm and outer radius 7 cm is made by shrinking one cylinder on to the other. The junction radius is 6 cm and the junction pressure is 11 kg/cm^2. The maximum hoop stress developed in the inner cylinder is :

(a) 36 kgf/cm^2 compression

(b) 36 kgf/cm^2 tension

(c) 72 kgf/cm^2 compression

(d) 72 kgf/cm^2 tension

APPSC IOF, 2009

Ans. (c) : Hoop stress $= \dfrac{-2PR_2^2}{R_2^2 - R_1^2}$

$R_1 = 50$ mm, $R_2 = 60$ mm

$P = 11\ N/mm^2$

$= 72\ kgf/cm^2$ (Compression)

1163. A thin cylindrical shell of mean diameter D and thickness t is subjected to internal fluid pressure p_i. The ends of this shell are rigidly closed by flat circular plate. If the longitudinal stress in the shell is to be zero what external pressure p_e must be applied at the ends?

(a) $2p_i$ (b) $1.5p_i$

(c) p_i (d) $0.5p_i$

UPSC JWM Adv. No-16/2009

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Ans. (c) : If the longitudinal stress in the shell is zero i.e., In equilibrium condition, the internal fluid pressure (P_i) is equal to external (P_e).

$\boxed{P_e = P_i}$

1164. A 10 m radius thin spherical tank is to be used to store gas. If the wall thickness of the tank is 10 mm and the allowable tensile stress for the material of the tank is 125 MPa, the maximum possible gas pressure (neglecting radial stress) is

(a) 0.25 MPa (b) 0.125 MPa
(c) 0.5 MPa (d) 1 MPa

DRDO Scientists 2008

Ans. (a) : For spherical tank hoop stresses :

$$\sigma_1 = \frac{pd}{4t} \le 125$$

$$p \le 125 \times \frac{4t}{d}$$

or $$p \le 125 \times \frac{4 \times 10}{20000}$$

or $$p \le 0.25 \text{ MPa}$$

$$p_{max} = 0.25 \text{ MPa}$$

1165. A thin cylindrical pressure vessel with mean diameter 10 m and wall thickness 20 mm is subjected to an internal fluid pressure of 0.4 MPa. If the yield strength of the material of the cylinder is 200 MPa, the factor of safety according to maximum shear stress theory (neglecting radial stress) is

(a) 0.5 (b) 1
(c) 1.5 (d) 2

DRDO Scientists 2008

Ans. (d) : Maximum shear stress in thin cylindrical pressure vessel,

$$\tau_{max} = \frac{PD}{4t} = \frac{0.4 \times 10}{4 \times 20 \times 10^{-3}}$$

$\tau_{max} = 50$ MPa

According to shear stress theory,

$$\tau_{max} = \frac{\sigma_y}{2\,FOS} \Rightarrow FOS = \frac{\sigma_y}{2\tau_{max}}$$

$$FOS = \frac{200}{2 \times 50} = 2$$

1166. In thick cylinders the radial stress in the wall thickness

(a) zero
(b) negligibly small
(c) not negligible and varies from the inner surface to outer surface
(d) any of the above

WBPSC AE 2008

Ans. (c)

In thick cylinder the radial stress in the wall thickness varies from the inner surface to outer surface.

1167. Which of the following is usually considered as thin cylinders?

(a) Boiler (b) Tanks
(c) Steam pipes (d) All of the above

WBPSC AE 2008

Ans. (d) : Boilers, Tanks and steam pipes are usually considered as thin cylinder.

1168. Which of the following assumptions is made while solving problems of thick cylinders using Lame's theory

(a) the material is homogeneous and isotropic
(b) the material is stressed within the elastic limit
(c) plane sections perpendicular to the longitudinal axis of the cylinder remain plain after the application of internal pressure
(d) all of the above

WBPSC AE 2008

Ans. (d) : Assumptions in Lame's theory for thick cylinder–

1. Material is homogeneous and isotropic in which hooke's law is valid.
2. The plane section perpendicular to longitudinal axis remain plane after fluid pressure is applied. It means there will be no bending i.e. longitudinal strain is constant.
3. The material is stressed within the elastic limit and each fibre of material is free to contract and expand individually.

1169. When a thin cylinder shell is subjected to internal fluid pressure. Which of the following stress is developed in. its wall?

(a) circumferential stress (b) longitudinal stress
(c) both of (A) and (B) (d) none of the above

WBPSC AE 2008

Ans. (c)

$$\sigma_{hoop} \text{ or } \sigma_{circumferential} = \frac{Pd}{2t}$$

$$\sigma_{longitudinal} = \frac{Pd}{4t}$$

where

P = internal pressure
d = diameter of thin cylinder
t = thickness of cylinder

1170. If water in closed pipe freezes, the pipe will

(a) Rupture along a plane perpendicular to the axis of pipe
(b) Rupture of zig-zag along the length
(c) Rupture along weakest circumferential section
(d) Rupture along a line running longitudinally along the pipe

ISRO Scientist/Engineer 2007

Ans. (d) : The failure will take place along the longitudinal joint because of the circumferential/hoop stress.

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1171. In a thick cylindrical shell subjected to internal fluid pressure, the state of stress at the outer surface is

(a) three-dimensional (b) two-dimensional
(c) isotropic (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (b) : Two-dimensional

1172. The design of this cylindrical shells is based on:

(a) hoop stress
(b) longitudinal stress
(c) volumetric stress
(d) average of hoop and longitudinal

J & K PSC Screening, 2006

Ans. (a) : In thin cylindrical shell loop stress is more as compared to longitudinal stress so hoop stress is used for designing of thin cylinder.

$$\text{Hoop stress } \sigma_h = \frac{Pd}{2t}$$

$$\text{Longitudinal stress } \sigma_L = \frac{Pd}{4t}$$

1173. A thin cylindrical shell has closed hemispherical ends. The material properties are Young's modulus (E) and Poisson's Ratio (v). For no distortion to take place at the junction of cylindrical and hemispherical portions, the ratio of shell thickness of spherical portion to that to cylindrical portion is

(a) $\frac{1-v}{2+v}$ (b) $\frac{1+v}{2-v}$
(c) $\frac{1-v}{2-v}$ (d) $\frac{1+v}{2+v}$

OPSC Civil Services Pre 2006

Ans. (c) : t_1 = thickness in cylindrical portion
t_2 = thickness in spherical portion
circumferential strain in cylindrical portion

$$\in_{c_1} = \frac{Pd}{4t_1} \times \frac{1}{E}\left[2 - \frac{1}{m}\right]$$

circumferential strain in hemispherical portion

$$\in_{c_2} = \frac{1}{E}\frac{Pd}{4t_2}\left[1 - \frac{1}{m}\right]$$

For no distortion at junction $\in_{c_1} = \in_{c_2}$

$$\frac{t_2}{t_1} = \frac{m-1}{2m-1} = \frac{1-v}{2-v}$$

7. Theory of Columns

1174. A column of length '*l*' is fixed at both the ends. The equivalent length of the column is:-

(a) 2 *l* (b) 0.5 *l*
(c) 4 *l* (d) *l*

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TNPSC AE 2018
UPPSC AE 12.04.2016
UKPSC AE-2013, Paper-I
APPSC AEE 2012, APPSC AE 04.12.2012
RPSC AE GWD, 2011, TNPSC AE, 2008
WBPSC AE 2007

Ans. (b) :

Conditions	Effective length
(i) Both end fixed	$\ell/2$
(ii) Both end hinge	ℓ
(iii) One fixed, one hinge	$\ell/\sqrt{2}$
(iv) One fixed, other free	2ℓ

1175. For a long slender column of uniform cross section, the ratio of critical buckling load for the case with both ends clamped to the case with both ends hinged is :

(a) 1 (b) 2
(c) 4 (d) 16

Gujarat PSC AE 2019, BHEL ET 2019
GPSC ARTO Pre 30.12.2018
UP Jal Nigam AE 2016, RPSC 2016
TSPSC AEE 2015
BPSC Asstt. Prof. 29.11.2015
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2012

Ans. (c) :
Ratio of critical buckling

$$= \frac{\text{Critical load with both ends clamped}}{\text{Critical load with both ends hinged}}$$

$$\frac{P_{cr_1}}{P_{cr_2}} = \frac{\left(\frac{\pi^2 EI}{\ell_e^2}\right)_{\text{clamped column}}}{\left(\frac{\pi^2 EI}{\ell_e^2}\right)_{\text{hinged column}}}$$

$$\frac{P_{cr_1}}{P_{cr_2}} = \frac{\left[\frac{\pi^2 EI}{\left(\frac{\ell}{2}\right)^2}\right]_{\text{clamped column}}}{\left(\frac{\pi^2 EI}{\ell^2}\right)_{\text{hinged column}}}$$

$$= \frac{4}{1} = 4$$

1176. Slenderness ratio of a column is defined as the ratio of its length to its

(a) least radius of gyration
(b) least lateral dimension
(c) maximum lateral dimension
(d) maximum radius of gyration

Nagaland PSC (CTSE) 2018, Paper-I
HPPSC Asstt. Prof. 18.09.2017
Vizag Steel (MT) 2017, CILMT 26.03.2017
OPSC AEE 2015 Paper-I
TRB Asstt. Prof., 2012, APPSC AEE 2012
UPSC JWM Advt. No.-52/2010
ESE 2003

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Ans. (a) : Slenderness Ratio (S)

$$S = \frac{\text{Effective Length of column}}{\text{Least Radius of Gyration}}$$

$$S = \frac{L_e}{K_{min}}$$

Where $k = \sqrt{\frac{I_{min}}{A}}$

1177. If the diameter of a long column is reduced by 20 percent, the reduction in Euler buckling load in percentage is nearly:-

(a) 4 (b) 36
(c) 49 (d) 59

HPPSC W.S. Poly. 2016
TSPSC AEE 2015
UKPSC AE-2013, Paper-I
APPSC AEE 2012
APPSC AE 04.12.2012
ESE 2000

Ans. (d) : We know that

$$P_e = \frac{\pi^2 E I_{min.}}{L_e^2}$$

$$P_e \propto I_{min}$$

$$P_e \propto \frac{\pi}{64} d^4$$

$$P_e \propto d^4$$

$$P_{e_1} = kd^4$$

$$P_{e_2} = k(0.8)^4 . d^4$$

$$P_{e_2} = 0.4096\ P_{e_1}$$

Then percentage reduction in Euler buckling load

$$= \frac{P_{e_1} - P_{e_2}}{P_{e_1}} \times 100$$

$$= kd^4 \frac{[1 - 0.4096]}{kd^4} \times 100$$

$$= 59\%$$

1178. A short column of external diameter D and internal diameter d carries an external load W. The greatest ecentricity which the load can have without producing tension on the cross section of the column is-

(a) $(D+d)/8$ (b) $(D^2+d^2)/8$
(c) $(D^2+d^2)/8D$ (d) $(D^2+d^2)/8d$

GPSC Engg. Class-II Pre-19.01.2020
TSPSC AEE 28.08.2017 (Civil/Mechanical)
ISRO Scientist/Engineer 12.05.2013
ISRO Scientist/Engineer 2008
ISRO Scientist/Engineer 2006
ESE 1999

Ans. (c) : External diameter = D
Internal diameter = d
External load = W, ecentricity e = ?
Stress due to bending moment (σ_b)
= Stress due to axil load (σ_a)

$\sigma_b = \sigma_a$

$$\frac{M}{Z} = \frac{4W}{\pi(D^2 - d^2)}$$

$$\frac{64 \times W \times e \times D}{2\pi(D^4 - d^4)} = \frac{4W}{\pi(D^2 - d^2)}$$

$$e = \frac{D^2 + d^2}{8D}$$

1179. In Rankine's formula, the material constant for mild steel is

(a) $\frac{1}{9000}$ (b) $\frac{1}{5000}$
(c) $\frac{1}{1600}$ (d) $\frac{1}{7500}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Karnataka PSC AE (WRD) 31.07.2021
Karnataka PSC AE, 10.09.2017
GPSC Lect. Poly. 07.12.2014, APPSC AEE 2012

Ans : (d)

Material	Value of 'a'
wrought iron	1/9000
cast iron	1/1600
Mild Steel	1/7500
Timber	1/750

1180. For a column of length L is fixed at both the ends, corresponding Euler's critical load is

(a) $\pi^2 EI / L^2$ (b) $2\pi^2 EI / L^2$
(c) $3\pi^2 EI / L^2$ (d) $4\pi^2 EI / L^2$

ISRO Scientist/Engineer 22.04.2018
APPSC AEE Mains 2016 (Civil Mech.)
ISRO Scientist/Engineer 2011
ISRO Scientist/Engineer 2007
OPSC Civil Services Pre 2006

Ans. (d) : Euler's critical load

$$P_{cr} = \frac{\pi^2 E I_{min}}{L_e^2}$$

L_e = effective length

Both end fixed

$$L_e = \frac{1}{2} l$$

Critical load

$$P_{cr} = \frac{\pi^2 E I_{min}}{\left(\frac{1}{2} l\right)^2}$$

$$P_{cr} = \frac{4\pi^2 E I_{min}}{l^2}$$

1181. Euler's formula holds good only for

(a) short columns
(b) long columns
(c) both short and long columns
(d) weak columns

Sikkim PSC (Under Secretary), 2017
Haryana PSC Civil Services Pre, 2014

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UKPSC AE-2013, Paper-I, BPSC AE 2012 Paper-VI
APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (b) :
(1) Euler's Formula applicable only for long column which fails due to buckling.

$$P_b = \frac{\pi^2 EI_{min}}{l_e^2}$$

(2) Rankine formula is applicable for both long and short column means they consider both failure buckling as well as crushing.

$$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_b}$$

$P_R \rightarrow$ Rankine load
$P_c \rightarrow$ Crushing load
$P_b \rightarrow$ Buckling load.

1182. Match List I with List II and select the correct answer using the codes given below the lists :

	List I (End conditions of columns)		List II (Lowest critical load)
(A)	Column with both ends hinged	(I)	$(\pi^2EI)/L^2$
(B)	Column with both ends fixed	(II)	$(2\pi^2EI)/L^2$
(C)	Column with one end fixed and other end hinged	(III)	$(4\pi^2EI)/L^2$
(D)	Column with one end fixed and other end free	(IV)	$(\pi^2EI)/4L^2$

(a) A-I, B-II, C-III, D-IV
(b) A-III, B-II, C-I, D-IV
(c) A-I, B-III, C-II, D-IV
(d) A-II, B-IV, C-III, D-I

Punjab PSC (Lect.) 06.08.2017
TRB Asstt. Prof. 2012
APPSC IOF, 2009, ESE 1995

Ans. (c) : Column with both end hinged $\Rightarrow L = L_e \Rightarrow \frac{\pi^2EI}{L^2}$

Column with both ends fixed, $L_e = \frac{L}{2} \Rightarrow \frac{4\pi^2EI}{L^2}$

Column with one end fixed and other end hinged, $L_e = \frac{L}{\sqrt{2}} \Rightarrow \frac{2\pi^2EI}{L^2}$

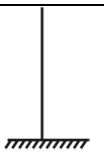
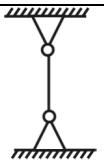
Column with one end fixed and other is free, $L_e = 2L \Rightarrow \frac{\pi^2EI}{4L^2}$

1183. When both ends of the column are hinged the crippling load at which the column just buckled is
(Where E is Young's modulus of column, I is the Moment of inertia of column and L is effective length of column)

(a) $\frac{\pi^2EI}{L^2}$ (b) $\frac{\pi^2EI}{2L^2}$
(c) $\frac{\pi EI}{L^2}$ (d) $\frac{2\pi^2EI}{L^2}$

GATE 2018
APPSC AE Subordinate Service Civil/Mech. 2016
APPSC AEE Screening Test 2016
CGPSC 26th April Shift-I
APPSC AEE 2012, GATE 2006

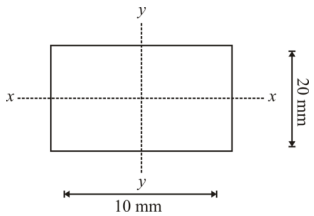
Ans. (a) :

Description	Crippling load	Eff. length	Eff. length as per IS code
	$\frac{2\pi^2EI}{L^2}$	$l_{eff} = \frac{l}{\sqrt{2}}$	$0.8\,l$
	$\frac{4\pi^2EI}{L^2}$	$leff = \frac{l}{2}$	$0.65\,l$
	$\frac{\pi^2EI}{4L^2}$	$l_{eff} = 2l$	$2\,l$
	$\frac{\pi^2EI}{L^2}$	$l_{eff} = l$	l

1184. A column has a rectangular cross-section of 10 mm × 20 mm and a length of 1 m. The slenderness ratio of the column is close to :
(a) 200 (b) 346
(c) 625 (d) 1000

Vadodara Municipal Corp. DEE 2018
GPSC Asstt. Director of Transport 05.03.2017
UP Jal Nigam AE 2016
GATE 2011

Ans. (b) : Cross sectional area (A) = 10 × 20 = 200 mm^2



$$I_{xx} = \frac{10 \times 20^3}{12} = 6666.67 \text{ mm}^4$$

$$I_{yy} = \frac{20 \times 10^3}{12} = 1666.67 \text{ mm}^4$$

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Least radius of gyration,

$$K_{min} = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{1666.67}{10\times 20}}$$

$= 2.88$ mm

Slenderness ratio (λ) is given by :

$$= \frac{\text{Effective length of column } (\ell_e)}{\text{Least radius of gyration } (K_{min})}$$

$$\lambda = \frac{1000}{2.88} = 346.409$$

1185. According to Euler's column theory, the crippling load of a column of length (l), with one end is fixed and the other end is hinged is

(a) $\pi^2 EI/l^2$ (b) $\pi^2 EI/4l^2$

(c) $2\pi^2 EI/l^2$ (d) $4\pi^2 EI/l^2$

Karnataka PSC AE (WRD) 31.07.2021
Nagaland PSC CTSE 2018 Paper-I
APPSC AEE MAINS 2016, PAPER-III
WBPSC AE, 2007

Ans. (c) :

End Condition	Euler's crippling load
Both ends hinged	$\frac{\pi^2 EI}{\ell^2}\ (l_e = l)$
Both ends fixed	$\frac{4\pi^2 EI}{\ell^2}\left(l_e = \frac{l}{2}\right)$
one end hinged and other end fixed	$\frac{2\pi^2 EI}{\ell^2}\left(l_e = \frac{l}{\sqrt{2}}\right)$
one end fixed and other end free	$\frac{\pi^2 EI}{4\ell^2}\ (l_e = 2l)$

1186. What is the expression for crippling load for a column of length L, with one end fixed and other end free?

(a) $P = \pi^2 EI/4L^2$ (b) $P = \pi^2 EI/L^2$

(c) $P = 2\pi^2 EI/L$ (d) $4\pi^2 EI/L^2$

UPRVUNL AE 21.08.2016
ESE 2012
ISRO Scientist/Engineer 2009, 2008
ESE 2006, GATE 1994

Ans. (a) : Crippling load = P_e

Column length = L

$$P = \frac{\pi^2 EI}{(L_e)^2}$$

$L_e = 2L$ (one end fixed and other end free)

$$P = \frac{\pi^2 EI}{(2L)^2} = \frac{\pi^2 EI}{4L^2}$$

1187. The buckling load for a column pinned at both ends is 10 kN. If the ends are fixed, the buckling load changes to

(a) 40 kN (b) 2.5 kN

(c) 5 kN (d) 20 kN

Karnataka PSC AE, 10.09.2017
TRB Asstt. Prof., 2012
WBPSC AE 2003, GATE 1998

Ans. (a) : We know that,

$$P_e = \frac{\pi^2 EI_{min}}{L_e^2}$$

Where, L_e = effective length

I_{min} = Moment of Inertia about centroidal axis.

$$\because \quad P_e \propto \frac{1}{L_e^2}$$

Then ratio, $\frac{(P_e)_2}{(P_e)_1} = \frac{(L_e)_1^2}{(L_e)_2^2} \Rightarrow \frac{(L)^2}{(L/2)^2}$

$(P_e)_2 = 4(P_e)_1$

$= 4 \times 10$ kN$= 40$ kN

1188. The section of a column is circular with diameter d. What is the diameter of its core :

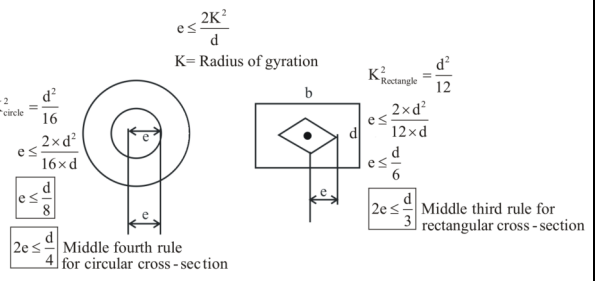
(a) d/2 (b) d/3

(c) d/4 (d) None of these

PSPCL AE, 2012
APPSC AE 04.12.2012
OPSC Civil Services Pre-2006

Ans. (c) :

Diameter of core or kernel$= \frac{2\times d}{8} = \frac{d}{4}$ for circular section



Core or kernel in column is region of cross section, when eccentric force is applied with in the region, there is no tensile force any of the fibre of column. Column are designed to bear compressive load only.

1189. Four vertical columns of the same material, height and weight have the same end conditions. The buckling load will be largest for a column having.

(a) Square cross section

(b) Hollow circular cross section

(c) Circular cross section

(d) I-section

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
MPPSC State Forest Service Exam, 2014
ESE 2009

Ans. (b) : As we know,

$$P = \frac{\pi^2 EI_{min}}{L^2}$$

So, $P \propto I_{min}$

In this inertia of the hollow circular cross section is minimum, can carry least weight, have maximum buckling.

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1190. The Euler load for a column is 1000 kN and crushing load is 1500 kN. The Rankine load is equal to:

(a) 600 kN (b) 1000 kN
(c) 1500 kN (d) 2500 kN

MPSC Poly. Lect. 2014
MPPSC State Forest Service Exam, 2014

Ans. (a) : As we know,

$$\frac{1}{P_R}=\frac{1}{P_E}+\frac{1}{P_C}$$

$$P_R=\frac{P_E\times P_C}{P_E+P_C}=\frac{1000\times1500}{1000+1500}$$

$P_R = 600$ kN

1191. Rankine Gordon formula for buckling is valid for :

(a) long column
(b) short column
(c) short and long column
(d) very long column

APPSC AEE 2019, APPSC IOF, 2009

Ans. (c) : $P=\sigma_A=\dfrac{\sigma_c A}{1+a\left(\dfrac{L}{k}\right)^2}$

σ_c = Crushing stress
A = Cross sectional area
a = Rankine's constant
L = Effective length
k = Slenderness ratio

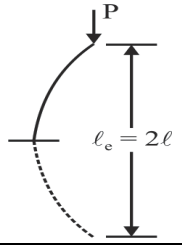
It is valid for both short and long column buckling.

1192. A column has an effective length twice the actual length, then the ends of column are:

(a) Fixed at both the ends
(b) Fixed at one end and free at the other end
(c) Hinged at both the ends
(d) Fixed at one end and hinged at the other end

CIL MT 27.02.2020
Assam PSC AE (PHED) 18.10.2020

Ans. (b) : A column has an effective length twice the actual length, then the ends of column are fixed at one end and free at the other end.



1193. If the length of a column is doubled, the critical load becomes :

(a) 1/2 of the original value
(b) 1/4 of the original value
(c) 1/8 of the original value
(d) 1/16 of the original value

HPPSC Asstt. Prof. 18.11.2016
APPSC AE Subordinate Service Civil/Mech. 2016
GATE 1998

Ans. (b) : $P_c=\dfrac{\pi^2 EI_{min}}{L_e^2}$

$$P_c\propto\frac{1}{L_e^2}$$

$$\frac{P_{c_2}}{P_{c_1}}=\frac{L_{e_1}^2}{L_{e_2}^2}=\left(\frac{1}{2}\right)^2$$

$$\boxed{P_{c_2}=\frac{P_{c_1}}{4}}$$

1194. A long column of length *l* is hinged at its ends. The equivalent length in buckling is :

(a) $2l$ (b) l
(c) $l/2$ (d) $l/4$

UKPSC AE 2012 Paper-I
UPSC JWM Advt. No.-52/2010

Ans. (b) : The equivalent Length (L_e) of a column which is hinged at both ends, if length of column is 'l'

$\therefore$ $\boxed{L_e = l}$

1195. A column that fails due to direct stress is called

(a) short column (b) long column
(c) Medium column (d) Slender column

TSPSC Manager (Engg.) HMWSSB 12.11.2020
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a)

- Short column fails due to direct stress and failure is crushing failure.
- Long column fails due to bending stresses and failure is buckling failure.
- Intermediate medium column fails due to combination of both direct stresses and bending stress and mode of failure is crushing and buckling.

1196. The buckling load will be maximum for a column, if

(a) One end of the column is fixed and other end is hinged
(b) Both ends are fixed
(c) One end is fixed and other end is free
(d) Both ends are hinged

Kerala PSC IOF 19.04.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
APPSC AEE 2012

Ans. (b)

Both end Hinged $=\dfrac{\pi^2 EI_{min}}{L^2}$

Both end fixed $=\dfrac{4\pi^2 EI_{min}}{L^2}$

One end hinged and one end fixed $=\dfrac{2\pi^2 EI_{min}}{L^2}$

One end fixed and other is free $=\dfrac{\pi^2 EI_{min}}{4L^2}$

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1197. In the theory of columns, Rankine's constant is given by:

(a) $\frac{\sigma_c}{\pi^2 E}$ (b) $\frac{\sigma_c}{\pi^2}$

(c) $\frac{\sigma_c}{E}$ (d) $\frac{\sigma_c^2}{\pi^2 E}$

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
APPSC AEE 2012

Ans. (a) : Design of intermediate column- Intermediate column fails by both crushing and buckling. Rankine's formula is used in the design of intermediate column. It is an imperical equation (Relation):

If P = Rankine's crippling load

$$\frac{1}{P}=\frac{1}{P_c}+\frac{1}{P_e}=\frac{P_c+P_e}{P_c P_e}$$

$$\Rightarrow P=\frac{P_c P_e}{P_c+P_e}=\frac{P_c}{1+\frac{P_c}{P_e}} \text{ but } P_c=\sigma_c A \ \& \ P_c=\frac{\pi^2 EI}{L_e^2}$$

$$\therefore \ P=\frac{\sigma_c A}{1+\frac{\sigma_c A}{\frac{\pi^2 EI}{L_e^2}}} \qquad \left[\because I=Ak^2 \ \& \ \lambda=\left(\frac{L_e}{k}\right)^2\right]$$

$$P=\frac{\sigma_c A}{1+\frac{\sigma_c A}{\frac{\pi^2 EAk^2}{L_e^2}}}$$

$$P=\frac{\sigma_c A}{1+\frac{\sigma_c}{\pi^2 E\lambda^2}}$$

$$P=\frac{\sigma_c A}{1+\alpha\lambda^2} \Rightarrow \alpha=\frac{\sigma_c}{\pi^2 E}$$

Where α is Rankine constant which depends on type of material of column and λ is slenderness ratio.

1198. A column is said to be a short column, when

(a) its length is very small
(b) its cross-sectional area is small
(c) the ratio of its length to the least radius of gyration is less than 80
(d) the ratio of its length to the least radius of gyration is more than 80

Vizag Steel (MT) 2017
JPSC AE 2013, Paper-V

Ans : (c) : A column is said to be short column, when the ratio of its length to the least radius of gyration is less than 80.

1199. The bucking load for a column one end fixed and other end free is 10kN. If both ends of this column is fixed, then what would be the buckling load capacity of this column ?

(a) 10 kN (b) 20 kN
(c) 80 kN (d) 160 kN

***MPPSC AE 2016,* PSPCL AE, 2012**

Ans : (d) Euler's formula for a column one end fixed and other end free

$$P_{e_1}=\frac{\pi^2 EI}{(2\ell)^2}=\frac{\pi^2 EI}{4\ell^2} \qquad (l_e=2l)$$

Euler's formula for a column both end fixed

$$P_{e_1}=\frac{4\pi^2 EI}{l^2} \qquad \left(l_e=\frac{l}{2}\right)$$

$$4\times 10=\frac{\pi^2 EI}{l^2} _____ (i)$$

$$P_{e_2}=\frac{4\pi^2 EI}{l}$$

$$(P_e)_2=4\times 40$$

$$\boxed{(P_e)_2=160\text{kN}}$$

1200. Match List I with List II and select the correct answer using the odes given below the Lists:

List-I		List II	
A	Both ends hinged	1	L
B	One end fixed and other end free	2	$L/\sqrt{2}$
C	One end fixed and other pin-joined	3	L/2
D	Both ends fixed	4	2L

Code:

	A	B	C	D
(a)	1	3	4	2
(b)	1	4	2	3
(c)	3	1	2	4
(d)	3	1	4	2

APPSC AEE 2012, ESE 2000

Ans. (b) :

Joined		**Equivalent length (l_e)**
A. Both end hinged	-	L
B. One end fixed and other end free	-	2 L
C. One end fixed and the other pin-joined	-	$L/\sqrt{2}$
D. Both end fixed	-	L/2

1201. The safe compressive load on a hollow cast iron column (one end fixed other hinged) of 150 mm external diameter, 100 mm internal diameter and 10 m length is (Use Euler's formula with a factor of safety of 5, and E = 95 GN/m^2)

(a) 74.8 kN (b) 149.6 kN
(c) 37.4 kN (d) 299.2 kN

APPSC AE 04.12.2012, APPSC AEE 2012

Ans. (a) : Given- D_o = 150 mm, D_i = 100 mm, L = 10 m
F.O.S. = 5, E = 95 GN/m^2
M.O.I.

$$I=\frac{\pi}{64}\left(D_o^4-D_i^4\right)=\frac{\pi}{64}\left(0.15^4-0.1^4\right)=1.99418\times 10^{-5}\text{ m}^4$$

Equivalent length $l_e=\frac{L}{\sqrt{2}}=\frac{10}{\sqrt{2}}=7.071\text{m}$

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Buckling load $P_e = \frac{\pi^2 EI}{l_e^2} = \frac{\pi^2 \times 95 \times 10^9 \times 1.99418 \times 10^{-5}}{(7.071)^2}$

Euler buckling load $P_e = 37.4 \times 10^4 = 374$ kN

For safe compressive load $P_c = \frac{P_e}{F.O.S.} = \frac{374}{5}$

$P_c = 74.8$ kN

1202. What is the cause of failure of a short MS strut under an axial load?

(a) Fracture stress (b) Shear stress
(c) Buckling (d) Yielding

TSPSC AEE 28.08.2017 (Civil/Mechanical)
CSE Pre-2007

Ans. (d) : The cause of failure of a short MS strut under an axial load is yielding.

1203. If one end of a hinged column is made fixed and other end free. how much is the critical load to the original value?

(a) Four times (b) One-fourth
(c) Half (d) Twice

UPRVUNL AE 07.10.2016
BPSC POLY. TEACH 2016

Ans : (b) Euler load = $\frac{\pi^2 EI}{l_e^2}$

Case– I

Initially condition [Both ends hinged]

$(P_1) = \frac{\pi^2 EI}{l_e^2}$

Case - II

One end fixed and other and free

$P_2 = \frac{\pi^2 EI}{(2l)^2} = \frac{\pi^2 EI}{4l^2} = \frac{P_1}{4}$

$P_2 = \frac{P_1}{4}$

1204. The effective length of the column of length 2.5 m, when one end is fixed and other end is free is

(a) 5 m (b) 2.5 m
(c) 1.25 m (d) 1.77 m

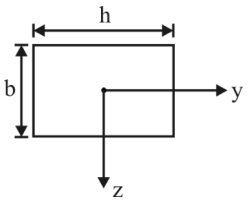
GPSC DEE, Class-2 (GWSSB) 04.07.2021

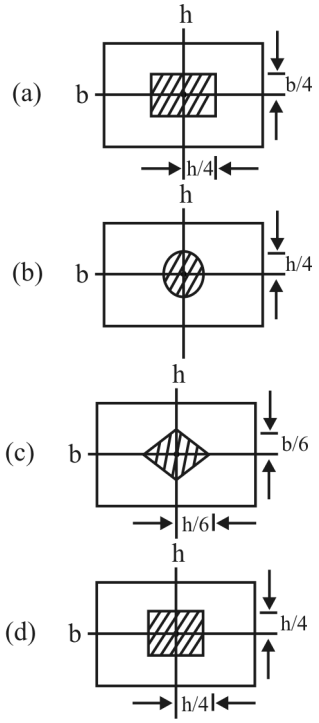
Ans. (a) : Effective length for one end fixed and other free, $L_e = 2 \times L$

$= 2 \times 2.5$

$= 5$ m

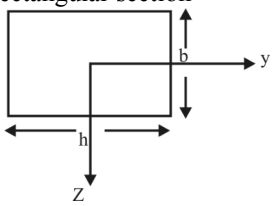
1205. The core of a section in a strut within which the load P will induce compression over the entire section for the rectangular section will be :

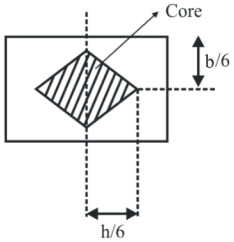




JPSC AE 10.04.2021, Paper-II

Ans. (c) : For rectangular section

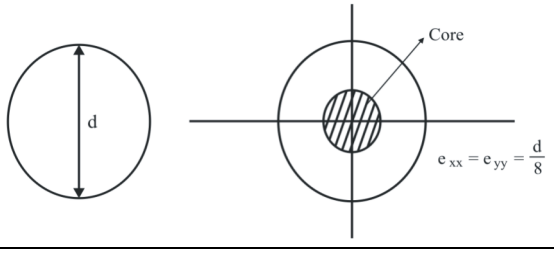




$e_{xx} = \frac{h}{6}$

$e_{yy} = \frac{b}{6}$

For Circular section –



1206. Euler's theory assumes that failure within a column is solely due to _____

(a) torsion (b) buckling
(c) sagging (d) bending

UPRVUNL AE 05.07.2021

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Ans. (b) : Euler's theory assumes that failure within a column is solely due to buckling.

1207. _____ is elastic instability which results in sudden large lateral deflection.

(a) Sagging (b) Fatigue
(c) Buckling (d) Failure

UPRVUNL AE 05.07.2021

Ans. (c) : Buckling is elastic instability which results in sudden large lateral deflection.

1208. For long column, which of the following relations holds true?

(a) Crushing load is negligible as compared to buckling load
(b) Crushing load is equal to buckling load
(c) Crushing load is more than buckling load
(d) Crushing load is less than buckling load

UPRVUNL AE 05.07.2021

Ans. (c) : Crushing load is more than buckling load

1209. The assumptions made in Euler's column theory is that

(a) the failure of column occurs due to buckling alone
(b) the length of column is very large as compared to its cross-sectional dimensions
(c) the column material obeys Hooke's law
(d) All of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : Assumptions Made in Euler's Theory of Column–

(1) The failure of column occurs due to buckling alone.
(2) The length of column is very large as compared to its cross-sectional dimensions.
(3) The column material obeys Hooke's law.

1210. The relation between equivalent length (L) and actual length (*l*) of a column whose one end is fixed and other end is hinged is given by :

(a) $L = l$ (b) $L = l/\sqrt{2}$
(c) $L = l/2$ (d) $L = 2l$

RPSC ACF & FRO, 26.02.2021

Ans. (b)

Conditions of columns	Equivalent length
Both hinged	$L_e = l$
One hinged and other fixed	$L_e = \frac{l}{\sqrt{2}}$
Both fixed	$L_e = \frac{l}{2}$
One fixed and other free	$L_e = 2l$

1211. For a long slender column of uniform cross section, the ratio of critical buckling load for the case with both ends hinged to the case with both ends clamped is

(a) 0.25 (b) 4.0
(c) 0.125 (d) 0.5

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : For both ends hinged,

$$(P_{cr})_1 = \frac{\pi^2 EI}{L^2} \quad .....(i)$$

For both ends clamped,

$$(P_{cr})_2 = \frac{4\pi^2 EI}{L^2} \quad .....(ii)$$

So, $$\frac{(P_{cr})_1}{(P_{cr})_2} = \frac{1}{4} = 0.25$$

1212. If the length of a column subjected to compressive load is increase by three times its original length, the critical buckling load becomes

(a) 1/3 of the original value
(b) 3 times the original value
(c) 1/9 of the original value
(d) 1/27 of the original value

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Critical buckling load $P_e \propto \frac{1}{L^2}$

$L_{e2} = 3L_{e1}$

$$\frac{P_{e2}}{P_{e1}} = \left(\frac{L_1}{L_2}\right)^2 = \left(\frac{1}{3}\right)^2$$

$$P_{e2} = \frac{P_{e1}}{9}$$

1213. While applying Euler's theory of columns to calculate the crippling load, equivalent length calculations are made by referencing _____.

(a) column with both ends are fixed
(b) column with one end fixed and other end hinged
(c) column with one end fixed and other free
(d) column with both ends hinged

VIZAG MT, 14.12.2020

Ans. (d) : While applying Euler's theory of columns to calculate the crippling load, equivalent length calculations are made by referencing column with both ends hinged.

1214. A structural member subjected to an axial compressive force is called

(a) beam (b) column
(c) frame (d) strut

Assam PSC AE (PHED), 18.10.2020
ESE 2008

Ans. (d) : A structural member subjected to an axial compressive force is called strut.

As per definition strut may be horizontal, inclined or even vertical.

The vertical strut is called a column.

Beam- Beam is a structural element that primarily resists loads applied laterally to the beam axis.

Frame- The frame structure is the combination of beams column and slab. Rigid structural frame refers to the frames which are capable in resisting the deformation.

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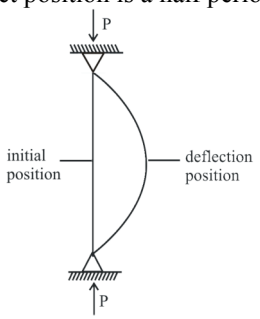
1215. The buckled shape (node shape) of a pinned-pinned end column in the first node of buckling

(a) is a half period sine wave
(b) is a full period sine wave
(c) is a quarter period sine wave
(d) none of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Deflect position is a half period sine wave.



1216. The slenderness ratio in the case of column with length L, area moment of inertia I and cross-sectional area A, is given by

(a) $L\sqrt{\frac{I}{A}}$ (b) $L\sqrt{\frac{L}{A}}$

(c) $L\sqrt{\frac{A}{I}}$ (d) $L\sqrt{\frac{LA}{I}}$

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : Slenderness ratio = $\frac{\text{Length}}{\text{Radius of Gyration}}$

and it is unitless.

Radius of gyration (k) = $\sqrt{\frac{I}{A}}$

Slenderness ratio = $L\times\sqrt{\frac{A}{I}}$

1217. The critical load in buckling of columns

(a) can be increased by increasing the strength of the column material
(b) can be increased by decreasing the strength of column material
(c) is independent of the strength of the column material
(d) none of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Critical load in buckling of column

$P_c = \frac{\pi^2 EI}{l_e^2}$

$P_c \propto E$

1218. A 2 m long pin ended column having Young's modulus (E) equal to 13 GPa can sustain 250 kN Euler's critical load for buckling. The permissible cross sectional size (I) of the column will be

(a) $7.8\times10^{-06}\ m^4$ (b) $3.9\times10^{-06}\ m^4$
(c) $1.95\times10^{-06}\ m^4$ (d) $0.975\times10^{-06}\ m^4$

Gujarat PSC AE 2019

Ans : (a) : Using Euler's formula critical buckling load is given by,

$$P_{cr} = \frac{\pi^2 EI_{min}}{\ell_e^2}$$

Given, $\ell = 2m$
$E = 13\ GPa = 13\times10^9\ Pa$
$P_{cr} = 250\ kN = 250\times10^3\ N$
$I_{min} = ?$

$$250\times10^3 = \frac{\pi^2\times13\times10^9\times I_{min}}{(2)^2}$$

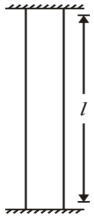
$I_{min} = 7.79\times10^{-6}\ m^4$
$= 7.8\times10^{-6}\ m^4$

1219. The slenderness ratio of a 4 m column with fixed ends having a square cross-sectional area of side 40 mm is :

(a) 173 (b) 17.3
(c) 1.73 (d) 100

BHEL ET 2019

Ans. (a) : Given A = 40 mm × 40 mm



$l = 4$ m

slenderness ratio = $\frac{L_e}{k}$

effective length of column = $L_e = \alpha L$
where α = length fixity coefficient
value 'α' for fixed end column-

$\alpha = \frac{1}{2}$

so, $L_e = \frac{1}{2}\times4 = 2m$

Slenderness ratio (s) = $\frac{L_e}{k}$

$k = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{\frac{a^4}{12}}{a^2}}$

$= \sqrt{\frac{a^2}{12}}$

$= \sqrt{\frac{40\times40\times10^{-6}}{12}}$

$= \sqrt{\frac{1600}{12\times10^6}}$

$k = 0.01154$ m

$(s) = \frac{L_e}{k} = \frac{2}{0.01154} = 173.310$

1220. Which of the following statements is correct?

(a) Euler's buckling load increases with increase in effective length.
(b) Buckling load of a column does not depend on its cross-section

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(c) If free end of a cantilever column is propped then the buckling load increases
(d) Two geometrically identical columns made of different material have same buckling load.

APPSC AEE SCREENING 17.02.2019

Ans. (c) : If the free end is supported by a propped, the effective length of column decreases. The load carrying capacity increase.

$$P_B \propto \frac{1}{L_e}$$

1221. Euler's formula for a mild steel column is not valid if the slenderness ratio is

(a) 60 (b) 90
(c) 100 (d) 120

Nagaland PSC (CTSE) 2018, Paper-I

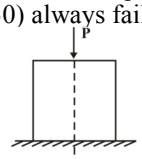
Ans. (a) : As we know, when slenderness ratio is less than 80 it is a short column from Euler's formula for mild steel column is not valid if the slenderness ratio is less than 80.

1222. What is the mode of failure of a short mild steel column (having slenderness ratio less than 10) under axial compressive load?

(a) Fracture (b) Buckling
(c) Yielding (d) Both (b) and (c)

RPSC AE 2018

Ans. (c) : Yielding is the mode of failure of short mild steel column under axial compressive load because short column (SR < 30) always fail in crushing.



In this question, given material of short column is mild steel so fracture or bucking failure not occur because fracture failure related to brittle material and buckling failure occurs in long columns (SR > 120)

1223. A Column of rectangular section ($I_{xx} > I_{yy}$) is subjected to an axial load. What is the axis about which the column will have a tending to buckle?

(a) X–X
(b) Y–Y
(c) The diagonal of the section
(d) X–X or Y–Y axis without any preference

RPSC Vice Principal ITI 2018

Ans. (b) : Column will buckle around the axis with the lowest moment of Inertia.

1224. Which one of the following statements is correct?

(a) Euler's formula holds good only for short columns
(b) A short column is one which has the ratio of its length to least radius of gyration greater than 100
(c) A column with both ends fixed has minimum equivalent length or effective length
(d) The equivalent length of column with one end fixed and other end hinged is half of its actual length

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : Euler's formula holds good for long columns only

for a short column

$$\frac{\ell}{R_{min}} < 100$$

equivalent length of column with one end fixed and other end hinged is

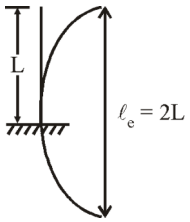
$$\ell_e = \frac{L}{\sqrt{2}}$$

1225. Which one of the following columns has effective length twice the value of actual length?

(a) Hinged-Hinged column
(b) Fixed-Fixed column
(c) Fixed-Hinged column
(d) Fixed-Free column

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : fixed-free column



1226. Critical Euler bucking load for a long column of diameter D was evaluated as P. If the diameter of the section is reduced to D/2, what is the load carrying capacity of the modified column ?

(a) P/2 (b) P/4
(c) P/8 (d) P/16

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) : $P_{cr} = \frac{\pi^2 EI}{\ell_e^2}$

$P_{cr} \propto I$

$P_{cr} \propto D^4$

$$\frac{P_{cr_1}}{P_{cr_2}} = \frac{D^4}{\left(\frac{D}{2}\right)^4}$$

$$P_{cr_2} = \frac{P}{16}$$

1227. Column C_1, has both the ends hinged while the column C_2 has one end hinged and other end fixed. What is the ratio of the critical load for C_1 to that of C_2 according to the Euler's formula?

(a) 2 (b) 1/2
(c) 4 (d) 1/4

TSPSC AEE 28.08.2017 (Civil/Mechanical)

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Ans. (b) : $P_{cr} \propto \frac{1}{\ell_e^2}$

$$\frac{(P_{cr})_{hinged-hinged}}{(P_{cr})_{hinge-fixed}} = \frac{\left(\frac{L}{\sqrt{2}}\right)^2}{(L)^2} = 1:2$$

1228. In an axially loaded compressive member with a circular cross-section of radius, r, what is the radius of the core section which is proof against tensile stress?

(a) r/2 (b) r/3
(c) r/4 (d) r/6

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : Core section which is proof against tensile stresses in case of circular reaction is of diameter $\frac{D}{4}$

$\therefore$ Core section radius will be $\frac{D}{8}$

if radius of circular section is r' then radius of core section will be

$$r' = \frac{2r}{8} = \frac{r}{4}$$

1229. A column of length 4 m, an area of cross-section 2000 mm^2. moments of inertia, I_{xx} = 720 cm^4, I_{yy} = 80 cm^4, is subjected to a buckling load. Both the ends of the column are fixed. What is the slenderness ratio of the column?

(a) 200 (b) 120
(c) 100 (d) 80

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : Length of column L = 4 m area of cross – section = A = 2000 mm^2

I_{xx} = 720 cm^4

I_{yy} = 80 cm^4

ℓ_e fixed-fixed $= \frac{L}{2} = 2m$

Slenderness ratio $\lambda = \frac{\ell_e}{k_{min}}$

$$K_{min} = \sqrt{\frac{I_{min}}{A}}$$

$$= \sqrt{\frac{80\ cm^4}{2000 \times 10^{-2}\ cm^2}}$$

$$= \sqrt{\frac{80 cm^4}{20 cm^2}} = 2 cm$$

$$\therefore \lambda = \frac{\ell_e}{k_{min}} = \frac{200 cm}{2 cm} = 100$$

1230. A pin ended column of length L modulus of elasticity E and second moment of the cross sectional area I is loaded centrally by a compressive load P. The critical buckling load (P_{cr}) is given by

(a) E/π^2L^2 (b) $\pi^2EI/3L^2$
(c) $\pi EI/L^2$ (d) π^2EI/L^2

TSPSC AEE 2017

Ans. (d) : The critical buckling load (P_{cr}) for a column having length L, modulus of elasticity E and second moment of cross sectional area I is loaded centrally both ends hinged.

$$n = 1, \quad P_{cr} = \frac{n\pi^2 EI}{L^2} = \frac{\pi^2 EI}{L^2}$$

1231. For thin cylindrical shell structures loaded in compression, the design is based on

(a) Yield strength of material
(b) Ultimate strength of material
(c) Buckling strength of the structure
(d) Shear strength of the structure

ISRO Scientist/Engineer 17.12.2017

Ans. (c)

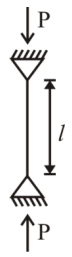
(1) Yield strength is the point where the material tends to deform. For the structures that should not deform on usage, the design is based on yield strength of material.
(2) Brittle materials generally fail at the ultimate strength of material.
(3) Buckling under axial compression is normally the controlling design consideration for thin cylindrical structures.
Thin and long cylindrical structures fail before the yield strength of the member is reached.
(4) Shear strength of a component is important for designing the dimensions and materials to be used for the manufacture or construction of the component e.g. beams, plates or bolts.

1232. In designing a connecting rod, it is considered like.....for buckling about X-axis.

(a) both ends fixed
(b) both ends hinged
(c) one end fixed and the other end hinged
(d) one end fixed and the other end free

GPSC EE Pre, 28.01.2017

Ans. (b) : In designing a connecting rod, it is considered like both end hinged for buckling about x-axis.



1233. Which of the following are correct?

P. Slenderness ratio is defined as the ratio of the length of the column to the radius of gyration.
Q. Euler's formula is applicable for short columns
R. Rankine's constant for a mild steel column is 1/7500

(a) Only P and Q are correct
(b) Only Q and R are correct
(c) Only P and R are correct
(d) P, Q and R are correct

APGENCO AE, 2017

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Ans. (c) : Euler formula is used for long column which fails due to buckling only.
Rankine formula is used for long as well as short column.

1234. If both the length and the diameter of a steel column are doubled, the elastic buckling load of the column will increase by a factor of

(a) 2 (b) 4
(c) 8 (d) 16

GPSC Asstt. Director of Transport 05.03.2017

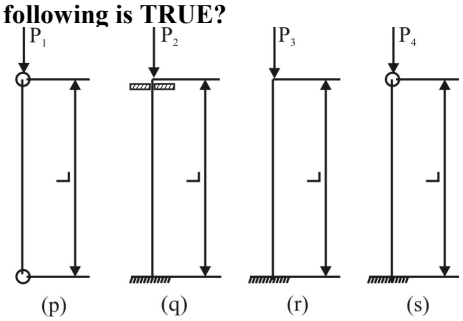
Ans. (b) : $P_{cr} = \frac{\pi^2 EI}{(L)^2}$ $L_{new} = 2$ load, $D_{new} = 2D$

$$P_{cr} \propto \frac{I}{(L)^2} \qquad I = \frac{\pi}{64} D^4$$

$$P_{cr} \propto \frac{D^4}{(L)^2} \Rightarrow P_{cr} \propto \frac{(2D)^4}{(2L)^2}$$

$P_{cr\,new} = 4\, P_{cr\,old}$

1235. Buckling loads (P_i) of 4 columns of equal length and cross section, but with different and conditions are shown below. Which of the following is TRUE?



(p) (q) (r) (s)

(a) $P_1 < P_2 < P_3 < P_4$ (b) $P_3 < P_1 < P_4 < P_2$
(c) $P_1 < P_4 < P_2 < P_3$ (d) $P_3 < P_4 < P_1 < P_2$

ISRO Scientist/Engineer 17.12.2017

Ans. (b) :

Diagram	End Condition	Effective Length	Buckling Load
p	Both ends hinged	$\ell_e = \ell$	$P_1 = \frac{\pi^2 EI}{\ell^2}$
q	Both ends fixed	$\ell_e = \frac{\ell}{2}$	$P_2 = \frac{4\pi^2 EI}{\ell^2}$
r	One end fixed and other end free	$\ell_e = 2\ell$	$P_3 = \frac{\pi^2 EI}{4\ell^2}$
s	One end fixed and other end hinged	$\ell_e = \frac{\ell}{\sqrt{2}}$	$P_4 = \frac{2\pi^2 EI}{\ell^2}$

Hence $P_3 < P_1 < P_4 < P_2$

1236. A column is known as short column if

(a) The length is more than 30 times the diameter
(b) Slenderness ratio is more than 120
(c) The length is less than 8 times the diameter
(d) The slenderness ratio is more than 32

Nagaland CTSE 2016, 2017 Ist Paper

Ans. (c) : For short column

$\frac{\ell}{d} < 8$, S.R. < 32

Medium column

$8 \geq \frac{\ell}{d} \leq 32$, $32 <$ S.R. < 120

Long column

$\frac{\ell}{d} > 32$, S.R. > 120

1237. The equivalent length of a column supported firmly at one-end and free at other end is

(or)

A compression member effectively held in position and restrained in direction at one end but not held in position or restrained in direction at the other end. If it's actual length is L, then its effective length is equal to

(a) 2L (b) 0.7 L
(c) L (d) 0.5L

Nagaland PSC CTSE 2017, Paper-I
APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Equivalent length of a column supported firmly at one-end so other end is free,

$L = 2l$

1238. Consider the following assumptions made in developing Euler's column theory:

1. The column material obeys Hooke's law
2. The failure of the column occurs due to buckling
3. The column is 'long' compared to its cross-sectional dimensions

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

JWM 2017

Ans. (d) : Euler's column theory assumption-
(i) The column is perfectly straight and of uniform cross-section.
(ii) The material is homogenous and isotropic
(iii) The material behaves elasticaly.
(iv) The load is perfectly axial and passes through the centroid of column section.
(v) The weight of the column is neglected.

1239. As the slenderness ratio of column increase, its compressive strength

(a) Increases
(b) Decreases
(c) Remains unchanged
(d) May increase or decrease depending on length

Vizag Steel (MT) 2017

Ans. (b) :

$$\text{Slenderness ratio} = \frac{\text{Effective Length}}{\text{Least Radius of gyration}}$$

So, the slenderness ratio of column increases as its compressive strength decrease.

1240. For Euler formula to be valid for mild steel struts which has yield stress of 3200 kg(f)/ sq.cm, the slenderness ratio should not be less than

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(a) 80.48 (b) 40.28
(c) 160.96 (d) 53.65

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : As we know, Euler's formula is applicable for long columns, (Long columns have slenderness ratio is greater than 80).

$$P_c = \frac{\pi^2 E I_{min}}{L_e^2}$$

$$I = AK^2$$

$$\frac{L}{K} = \text{slenderness ratio}$$

1241. Condition for no tension in a solid column of dia D the eccentricity e must be less than or equal to

(a) D/8 (b) D/6
(c) D/4 (d) D/2

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : As we know, for no tension condition

$$\sigma_b \le \sigma_a$$

$$\frac{P_e}{\left(\frac{\pi d^3}{32}\right)} \le \frac{4p}{\pi d^2}$$

$$\boxed{e \le \frac{D}{8}}$$

1242. A column that fails essentially by direct crushing at ultimate load is called

(a) Euler's column
(b) long column
(c) short column
(d) None of the given answers

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : Medium column is a column which fails due to direct stress or buckling stress. (S.R. = 32 to 120)
Shorter column (S.R is less than 32) is column which fails due to direct crushing at ultimate load.

1243. The Rankine's constant for a cast iron column with both ends hinged

(a) 1/7500 (b) 1/1600
(c) 1/9000 (d) 0.00016

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : As we know,
The Rankine's constant for cast Iron column with both ends hinged is given by.

$$a = \frac{1}{1600}, \qquad \sigma_C = 550 \text{ N/mm}^2$$

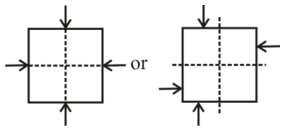
Because $\tau_{effective}$ = Length of column

1244. A short rectangular column has zero tensile stress if the eccentric, compressive load acts:

(a) In the middle three quarters of the section.
(b) In the middle half section
(c) In the middle third of the section
(d) It is not possible

GPSC Asstt. Prof. 28.08.2016

Ans. (c) :



Middle third rule states that no tension is developed in a wall or foundation if the resultant compressive force lies within the middle third of the structure.
Tensile stress is zero, if the eccentric, compressive load acts in the middle third of the section.

1245. If the length of a column is halved, the critical load _____.

(a) remains the same
(b) becomes 2 times of the original value
(c) becomes 4 times of the original value
(d) becomes 8 times of the original value
(e) becomes 16 times of the original value

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know,

$$P_e = \frac{\pi^2 EI}{L_e^2}$$

$$L_{e1} = \frac{L_e}{2}$$

$$P_{e1} = \frac{4\pi^2 EI}{L_{e1}^2}, \qquad \boxed{P_{e1} = 4P_e}$$

1246. The critical load for an ideal elastic column is often called the..... load.

(a) Gordan (b) Rankine
(c) Shanley (d) Euler

TSPSC AEE 28.08.2017 (Civil/Mechanical)
HPPSC AE (PWD) 10.09.2016

Ans. (d) : Euler's critical load is the compressive load at which a slender column will suddenly bend of buckle

$$P_{cr} = \frac{\pi^2 EI_{min}}{(L_e)^2}$$

Where,
P_{cr} = Euler critical load
E = Young's modulus
I = Minimum moment of inertia
L_e = Effective length of column

1247. The load at which the member just buckles is called

(a) Critical load (b) crushing load
(c) equilibrium load (d) ultimate load

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The critical load is the greatest load that will not cause lateral deflection (buckling). For loads greater than the critical load, the column will deflect laterally. The critical load puts the column in a state of unstable equilibrium. A load beyond the critical load causes the column to fail by buckling.
Crushing load – It is the intensity of load required to crush the material.
Ultimate local – Absolute maximum load that a structure can bear without failing.

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1248. Eulers critical load is proportional to (Where L = length of the column k = constant, k = constant depends on end conditions and EI = flexural rigidity)

(a) EI (b) L^2

(c) 1/EI (d) k^2/EI

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Crippling load = $\frac{\pi^2 EI}{L^2}$

$P \propto \frac{1}{L^2} \propto EI$

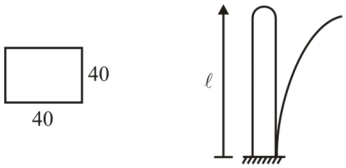
1249. The slenderness ratio of free standing column of length 4 m and 40 mm × 40 mm section is

(a) 115.2 (b) 230.4

(c) 692.8 (d) 346.8

APPSC AEE Screening Test 2016

Ans. (c)



Slenderness (λ) = $\frac{\ell_e}{k_{min}}$

effective length (ℓ_e) = 2ℓ

[$\because$ free end column]

$\ell_e = 2 \times 4$

$= 8$ m

$\ell_e = 8000$ mm

where

ℓ_e = effective length

$$k_{min} = \sqrt{\frac{I_{min}}{A}} = \sqrt{\frac{\frac{bd^3}{12}}{bd}} = \sqrt{\frac{\frac{a^4}{12}}{a^2}}$$

$$= \sqrt{\frac{a^4}{12} \times \frac{1}{a^2}}$$

$$= \sqrt{\frac{a^2}{12}} = \sqrt{\frac{40 \times 40}{12}}$$

$$\boxed{k_{min} = 11.547 mm}$$

Slenderness ratio (λ) = $\frac{8000}{11.547} = 692.82$

1250. In case of eccentrically loaded struts

(a) Hollow section is preferred

(b) Solid section is preferred

(c) Composite section is preferred

(d) Any of the above section may be used

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : A structural member subjected to an axial compressive stress is called strut. It is case of eccentrically loaded struts, a composite section is preferred. Strut may be horizontal, vertical or inclined.

1251. The factor of safety considered for Euler's formula for crippling load is

(a) 1 (b) 3

(c) 5 (d) 6

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : The factor of safety considered for Euler's formula for crippling load is 1.

1252. Which of the following is true for ideal column compressed by an axial load (P)?

(a) Column will be in stable equilibrium if P > $P_{critical}$

(b) Column will be in stable equilibrium if P < $P_{critical}$

(c) Column will be in unstable equilibrium if P < $P_{critical}$

(d) Column will buckle if P < $P_{critical}$

UPRVUNL AE 07.10.2016

Ans. (b) : If axial load P = $P_{critical}$ (Neutral)
If axial load P < $P_{critical}$ (Stable equilibrium)
If axial load P > $P_{critical}$ (Unstable equilibrium)

1253. The buckling load for a column hinged at both ends is 15kN. If the ends are fixed, the buckling load changes to :

(a) 60 kN (b) 30 kN

(c) 45 kN (d) 3.75 kN

Kerala PSC IOF 19.04.2016

Ans. (a) $P_{cr} = \frac{\pi^2 EI}{L_e^2}$

For both end hinged

$$15 = \frac{\pi^2 \times E \times I}{L^2}$$

For both end fixed

$$L_e = \frac{L}{2}$$

$$P_{cr} = \frac{4 \times \pi^2 \times E \times I}{L^2}$$

$$P_{cr} = 4 \times 15$$

$$\boxed{P_{cr} = 60 kN}$$

1254. The Euler's load for a column is 1 MN and crushing load is 1.5 MN. The Rankine load is :

(a) 1 MN (b) 0.6 MN

(c) 1.5 MN (d) 2.5 MN

Kerala PSC IOF 19.04.2016

Ans. (b) $P_R = \frac{P_c \cdot P_e}{P_c + P_e}$

$$= \frac{1.5 \times 1}{1.5 + 1} = \frac{1.5}{2.5} = \frac{3}{5} = 0.6$$

1255. Compression members tend to buckle in the direction of

(a) least radius of gyration

(b) axis of load

(c) perpendicular to axis of load

(d) minimum cross-section

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) Compression members tend to buckle in the direction of least radius of gyration.

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1256. A certain high tensile strength steel has a modulus of elasticity of 2×10^6 kg/cm^2 and a yield point stress of 6,000 kg/cm^2. Find the minimum limiting value of the slenderness ratio for which Euler's equation is valid

(a) 99 (b) 80
(c) 57 (d) 75

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c) $P_{cr.} = \dfrac{\pi^2 EI}{\ell_e^2}$ $\quad I = AK^2$

$P_{cr.} = \dfrac{\pi^2 EA}{(SR)^2}$ $\quad \left[\text{Slenderness Ratio}(SR) = \dfrac{\ell_e}{K}\right]$

So critical stress –

$$\sigma_{cr.} = \frac{\pi^2 E}{(SR)^2}$$

$$(SR)^2 = \frac{\pi^2 \times 2\times10^6}{6000}$$

$$\boxed{SR=57.357}$$

1257. A vertical column has two moments of inertia I_{xx} and I_{yy}. The column will tend to buckle in the direction of the

(a) axis of load
(b) perpendicular to the axis of load
(c) maximum moment of inertia
(d) minimum moment of inertia

ISRO Scientist/Engineer 11.10.2015

Ans. (d) : Vertical Column–
Moment of Inertia = I_{XX} and I_{YY}
We know that,
Euler's formula (P_e)

$$(P_e) = \frac{\pi^2 EI_{min}}{L_e^2} \text{ or } \frac{n\pi^2 EI_{min}}{L^2}$$

P_e = Euler's buckling load
I_{min} = min of [$I_{X\text{-}X}$ & $I_{Y\text{-}Y}$]
L = Actual length of the column
L_e = Effective length of the column
$L_e = \alpha L$
α = Length finite coefficient
$P_e \propto E$
$P_e \propto I_{min}$
$P_e \propto \dfrac{1}{L^2}$
$\boxed{P_e \propto I_{min}}$

1258. Rankine's formula is valid upto the slenderness ratio of

(a) 180 (b) 240
(c) 300 (d) 60
(e) 120

CGPSC AE 26.04.2015 Shift-I

Ans. (e) : The Rankine's formula is applicable for both long and short column.
For short column slenderness ratio is less than 32 and for long column it is more than 120.

1259. Two columns of different lengths having circular cross-sections have the same L/d ratio as 100. Both have hinged ends and the material is the same. The compressive stress P/A at the critical buckling load for the larger column will be:

(a) less
(b) equal
(c) greater than that for the smaller column
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Compressive stress $\left(\dfrac{P}{A}\right)$ at critical buckling load for both columns are equal.

$$P_{cr} = \frac{\pi^2 EI}{l_e^2}$$

$$P_{cr} = \frac{\pi^2 E\pi D_1^4}{64\ell_e^2}$$

$$\frac{P_{cr}}{\frac{\pi D_1^2}{4}} = \frac{\pi^2 ED_1^2}{16\ell_e^2}$$

$$\sigma_{C_1} = \left(\frac{P_{cr}}{\frac{\pi D_1^2}{4}}\right)_{column1} = \frac{\pi^2 E}{16}\left(\frac{1}{100}\right)^2 \qquad \because \frac{l}{D} = 100$$

Similarly,

$$\sigma_{C2} = \left(\frac{P_{cr}}{\frac{\pi D_2^2}{4}}\right)_{column2} = \frac{\pi^2 E}{16}\left(\frac{1}{100}\right)^2$$

1260. A 1.5 m long column has a circular cross-section of 5 cm diameter. One end of the column is fixed and other end is free. Taking factor of safety as 3 and E = 120 GN/m^2, safe load according to Euler's theory is

(a) 13.00 kN (b) 13.27 kN
(c) 13.47 kN (d) 13.87 kN

MPPSC AE 08.11.2015

Ans. (c) : Given, E = 120 GN/m^2, Factor of safety = 3
Length of column (L) = 1.5 m
Diameter of column (d) = 5 cm = 5×10^{-2} m
For one end fixed and other end is free for a column effective length of column = $L_e = 2L = 2 \times 1.5 = 3$ m

$$FOS = \frac{P_{max}}{P_e}$$

$$P_e = \frac{P_{max}}{3}$$

$\therefore$ Euler's load formula-

$$P_e = \frac{\pi^2 E.I_{min}}{3L_e^2} \qquad \because I_{min} = \frac{\pi}{64}d^4$$

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$$= \frac{\pi^2 \times 120 \times 10^9 \times \pi \times \left(5 \times 10^{-2}\right)^4}{3 \times 64 \times (3)^2}$$

$= 1345.75 \times 10^9 \times 10^{-8}$

$= 13457.5$ Newton

$\Rightarrow P_e = 13.457$ kN

1261. The buckling load for a given column depends on

(a) least radius of gyration (b) length of column

(c) modulus of elasticity (d) all of these

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (d) :

$$P_e = \frac{\pi^3 E I_{min}}{L_e^2}$$

P_e = Buckling load

I_{min} minimum value of I_{xx} and I_{yy}

L = actual length

$P_e \propto E$

$P_e \propto I_{min}$

$$P_e \propto \frac{1}{L_e}$$

1262. Rankine's formula for determining the load carrying capacity of a column is valid for

(a) long column only (b) short column only

(c) columns with hinged (d) all columns

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (d) : Rankine's formula for determining the load carrying capacity of column is valid for–

(i) Long column

(ii) Short column

1263. For a circular column having its ends hinged, the slenderness ratio is 160. The L/d ratio of the column is :

(a) 80 (b) 57

(c) 40 (d) 20

MPPSC State Forest Service Exam, 2014

Ans. (c) : $Ak^2 = \frac{\pi}{64} d^4$

$$A = \frac{\pi}{4} d^2$$

$$\frac{\pi}{4} d^2 \times k^2 = \frac{\pi}{64} d^4$$

$$\boxed{k = \frac{d}{4}}$$

Slenderness ratio $(\lambda) = \frac{\ell_e}{k}$

$\ell_e = \ell$ (because both ends are hinged)

$$\lambda = \frac{\ell}{d/4} = 160$$

$$\boxed{\frac{\ell}{d} = 40}$$

1264. Which of the following strength forms the basis in designing of a push rod?

(a) Tensile strength

(b) Compression strength

(c) Bending strength

(d) Buckling strength

UPRVUNL AE 2014

Ans : (d) : The push rod is designed as a column based on buckling criteria, push rods are often subjected to a compressive force which results in buckling.

1265. Short column is usually defined as one whose slenderness ratio is less than about ____

(a) 2 (b) 5

(c) 8 (d) 10

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (d) : For short column

Slenderness ratio $(\lambda) \le 32$

For medium column

$32 \le \lambda \le 100$

For long column

$\lambda > 120$

1266. Secant formula is applicable for :

(a) short columns under axial loading

(b) long columns under axial loading

(c) short columns under eccentric loading

(d) long columns under eccentric loading

(HPPSC AE 2014)

Ans : (d) Secant formula is applicable for long column under eccentric loading.

Secant formula for ν_{max} and σ_{max}

In the Euler's buckling formula we assume that the load P acts through the centroid of the cross - section. However in reality this might not always be the case : the load P might applied at an offset or the slender member might not be completely straight.

To account for this, we assume that the load P is applied at a certain distance away from the centroid. This would obviously change the way we calculate our buckling load, which is what the Secant formula is for.

1267. If the slenderness ratio for a column is 100, then it is said to be a _____ column.

(a) long (b) medium

(c) short (d) expand

Haryana PSC Civil Services Pre, 2014

Ans. (b) :

$$\text{Slenderness ratio } (\lambda) = \frac{\text{Effective length of column}}{\text{least radius of gyration}}$$

(1) Short column	–	$\lambda < 32$
(2) Medium Column	–	$32 < \lambda < 120$
(3) Long Column	–	$\lambda > 120$

1268. A loaded column fails due to

(a) stress due to direct load

(b) stress due to bending

(c) both (1) and (2)

(d) None of the above

APPSC Poly. Lect. 2013

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Ans. (c) : A loaded column fails due to both direct & bending stress.

→ The long columns, commonly fail because of buckling stress, it has direct stress very small compare to bending stress.

→ The short column primarily fails due to direct stress

1269. The ratio of the compressive critical load for a long column fixed at both the ends and a column with one end fixed and the other end being free is:-

(a) 2 : 1 (b) 4 : 1
(c) 8 : 1 (d) 16 : 1

UKPSC AE-2013, Paper-I
ESE 1997

Ans. (d) : Case I- When both ends of column are fixed

$$[P_e]_1 = \frac{4\pi^2 EI}{l^2}$$

Case II- When one end fixed and other end is free

$$[P_e]_2 = \frac{\pi^2 EI}{4l^2}$$

Then

$$\frac{[P_e]_1}{[P_e]_2} = \frac{16}{1} = 16:1$$

1270. The crippling load of a column with one end fixed and other end hinged is

(a) $\sqrt{2}$ times that of a both ends hinged colum
(b) Two times that of a both ends hinged column
(c) Four times that of a both ends hinged column
(d) Eight times that of a both ends hinged column

APPSC AEE 2012

Ans : (b)

End Conditions	Effective length
Both end hinged	L
One fixed other free	2L
Both end fixed	L/2
One end fixed and other hinged	$L/\sqrt{2}$

(i) Crippling load of a column with one end fixed and other end hinged.

$$P_1 = \frac{\pi^2 EI}{(L/\sqrt{2})^2} = \frac{2\pi^2 EI}{L^2} \quad \text{..... (i)}$$

(ii) Crippling load of a column when both end hinged

$$P_2 = \frac{\pi^2 EI}{L^2} \quad \text{.....(ii)}$$

$$\boxed{P_1 = 2 \times P_2}$$

1271. The formula given by I.S. code in calculating allowable stress for the design of eccentrically loaded columns is based on

(a) Johnson's parabolic formula
(b) Straight line formula
(c) Perry's formula
(d) Secant formula

APPSC AEE 2012

Ans : (c) The formula given by I.S. code in calculating allowable stress for the design of eccentrically loaded columns is based on Perry's formula.

1272. If the flexural rigidity of the column is doubled, then the strength of the column is increased by

(a) 16 (b) 8
(c) 2 (d) 4

APPSC AEE 2012

Ans : (c) Case 1st :-

Strength of column $(P_1) = \frac{\pi^2 EI}{L^2}$

(Flexural rigidity of column = E ×I)

Case 2st :-

Flexural rigidity of column is doubled.

strength of column $(P_2) = \frac{\pi^2 \times 2EI}{L^2}$

$$P_2 = 2 \times P_1$$

In the flexural rigidity of the column is doubled, then the strength of the column is increased by two times.

1273. The least radius of gyration for solid circular column is

(a) d (b) $\frac{d}{2}$ (c) $\frac{d}{4}$ (d) $\frac{d}{3}$

APPSC AEE 2012

Ans : (c) The least radius of gyration for solid circular column is d/4.

Radius of gyration-Radius of gyration is defined as the square root of moment of inertia divided by the area of itself.

$$r_x = \sqrt{\frac{I_{xx}}{A}}; r_y = \sqrt{\frac{I_{yy}}{A}}$$

$$= \sqrt{\frac{\pi/64 \times d^4}{\pi/4 \times d^2}} = \sqrt{\frac{d^2}{16}} = \frac{d}{4}$$

1274. In a mild steel tube 4 m long, the flexural rigidity of the tube is 1.2×10^{10} N – mm². The tube is used as a strut with both ends hinged. The crippling load in kN is given by

(a) 14.80 (b) 7.40
(c) 29.60 (d) 1.85

APPSC AEE 2012

Ans : (b) tube length = 4mm
flexural rigidity (E×I) = 1.2×10^{10} N – mm²
End condition = Both ends hinged
Effective length = *l*

Crippling load $= \frac{\pi^2 EI}{\ell^2}$

$$P_e = \frac{\pi^2 \times 1.2 \times 10^{10}}{4^2}$$

$$\boxed{P_e = 7.40 \text{ kN}}$$

1275. A member under tension is called :

(a) strut (b) tie
(c) strut-tie (d) column

APPSC AE 04.12.2012

Ans. (b) : A member under tension is called tie.

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1276. Two long columns A and B are of the same material, the same length, and the same cross-section. Column A is hinged at both ends, while column B is fixed at one end and hinged at the other. The ratio of the Euler-crippling loads for A and B is :

(a) 2 (b) 1

(c) $\sqrt{2}$ (d) $\frac{1}{2}$

VIZAG Steel MT 24.01.2021, Shift-I
APGENCO AE 2012

Ans. (d) : Column-A– both ends hinged.

$$(P_{cr})_A = \frac{\pi^2 E\, I_{min}}{l_e^2} = \frac{\pi^2 E I_{min}}{l^2}$$

Both ends hinged ($l_e = l$)

Column-B– One end hinged and other fixed

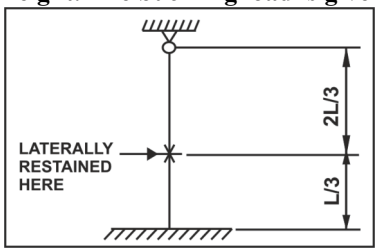
$$\left(l_e = \frac{l}{\sqrt{2}}\right)$$

$$(P_{cr})_B = \frac{\pi^2 E\, I_{min}}{\left(\frac{l}{\sqrt{2}}\right)^2} = \frac{\pi^2 E\, I_{min} \times 2}{l^2}$$

$$\text{Ratio} = \frac{(P_{cr})_A}{(P_{cr})_B} = \frac{\pi^2 E\, I_{min}}{l^2} \times \frac{l^2}{\pi^2 E\, I_{min} \times 2}$$

$$= \frac{1}{2}$$

1277. A column of length 'L' is fixed at bottom end and hinged at the other end. This column is restrained from lateral displacement at 1/3rd height. The buckling load is given by

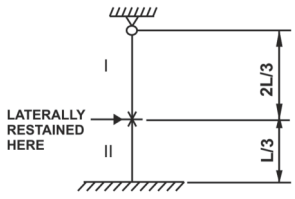


(a) $9\pi^2 EI/(4L^2)$ (b) $4\pi^2 EI/L^2$

(c) $4\pi^2 EI/(9L^2)$ (d) $\pi^2 EI/L^2$

ISRO Scientist/Engineer 2012

Ans. (a) : The column will behaves as 2 columns and restrained means it's hinged.



(i) for upper part length $= \frac{2L}{3}$

Conditions- Both ends hinged Buckling load,

$$P_{cr} = \frac{\pi^2 EI}{(Le)^2} \qquad \left[\because L_e = \frac{2L}{3}\right]$$

$$= \frac{\pi^2 EI}{\left(\frac{2L}{3}\right)^2} = \frac{9\pi^2 EI}{4L^2}$$

II for lower part length $= \frac{L}{3}$

Conditions- Up one end hinged and other end fixed

$\therefore$ Buckling load,

$$P_{cr} = \frac{\pi^2 EI}{(L_e)^2} \qquad \left[\because L_e = \frac{L}{3\sqrt{2}}\right]$$

$$= \frac{\pi^2 EI}{\left(\frac{L}{3\sqrt{2}}\right)^2} = \frac{18\pi^2 EI}{L^2}$$

Hence, we take the min. of the two buckling load, that is $\frac{9\pi^2 EI}{4L^2}$

1278. Slenderness ratio has dimension of

(a) cm (b) cm^{-1}

(c) cm^2 (d) None

UKPSC AE 2012 Paper-I

Ans. (d) : It is a dimensionless quantity.

$$S = \frac{L_e}{k}$$

$$k = \sqrt{\frac{I_{min}}{A}}$$

1279. Given that

P_E = the crippling load given by Euler

P_C = the load at failure due to direct compression

P_R = the load in accordance with the Rankine's criterion of failure

Then P_R is given by

(a) $\frac{(P_E + P_C)}{2}$ (b) $\sqrt{P_E \times P_C}$

(c) $\frac{P_E . P_C}{P_C + P_E}$ (d) None of the above

APPSC AEE Mains 2016 (Civil Mechanical)
ISRO Scientist/Engineer 2010

Ans. (c) : P_R = The load with Rankine's criterion of failure.

P_C = Load due to direct compression

P_E = Crippling load given by Euler.

$$\frac{1}{P_R} = \frac{1}{P_C} + \frac{1}{P_E}$$

$$P_R = \frac{P_C P_E}{P_C + P_E}$$

1280. Assertion (A) : A long column fails (becomes elastically unstable) due to excessive bending.

Reason (R) : Euler's buckling load is inversely proportional to the equivalent length of the column.

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Codes :

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true and R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (c) : As we know,

$$\text{Buckling load}\left(P_e\right)=\frac{\pi^2 EI}{L_e^2}$$

$$P_e \propto \frac{1}{L_e^2}$$

Euler's buckling load is inversely proportional to the square of equivalent length of the column.

1281. An aluminium column of square cross-section (10 mm × 10 mm) and length 300 mm has both ends pinned. This is replaced by a circular cross-section (of diameter 10 mm) column of the same length and made of the same material with the same end conditions. The ratio of critical stresses for these two columns corresponding to Euler's critical load, $(\sigma_{critical})_{square} : (\sigma_{critical})_{circular}$, is

(a) 1 : 4 (b) 3 : 4

(c) 4 : 3 (d) 4 : 1

DRDO Scientists 2009

Ans. (c) : Euler load P_e is given by

$$P=\frac{\pi^2 EI}{l_e^2}$$

$$\frac{P}{A}=\sigma_c=\frac{\pi^2 EI}{Al_e^2}$$

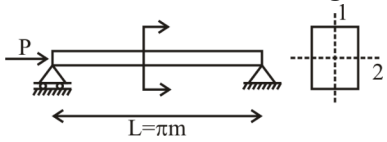
$$\sigma_c \propto \frac{I}{A}$$

$$\left(\sigma_c\right)_{circular}=\frac{\left(\pi d^4\right)(4)}{64\left(\pi d^2\right)}=\frac{d^2}{16}=\frac{(10)^2}{16}$$

$$\left(\sigma_c\right)_{square}=\frac{a^4}{12\left(a^2\right)}=\frac{a^2}{12}=\frac{(10)^2}{12}$$

$$\left(\sigma_c\right)_{square}/\left(\sigma_c\right)_{circular}=4:3$$

1282. A long slender column is pin-supported at the ends and compressed by an axial load P as shown in figure. If I_{11} = 100 cm^4, I_{22} = 200 cm^4 and Young's modulus of elasticity = 100 GPa, then the critical load for buckling is



(a) 500 kN (b) 200 kN

(c) 100 kN (d) 1000 GN

DRDO Scientists 2008

Ans. (c) : $P_{cr}=\frac{\pi^2 EI}{L^2}$ (Hinged-hinged end condition)

P_{cr} = EI (Taking I as least moment of inertia)

$= 100 \times 10^9 \times 100 \times 10^{-8}$

= 100000 N = 100 kN

1283. If the slenderness ratio of a column is more than 120 it is termed as

(a) short column (b) medium column

(c) long column (d) none of the above

WBPSC AE 2008

Ans. (c) : For long column slenderness ratio is greater than 120.

8. Strain Energy Method

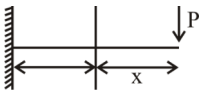
1284. A cantilever beam of length L and flexural modulus EI is subjected to a point load P at the free end. The elastic strain energy stored in the beam due to bending (neglecting transverse shear)

(a) $\frac{P^2L^3}{4EI}$ (b) $\frac{P^2L^3}{6EI}$

(c) $\frac{P^2L^3}{2EI}$ (d) $\frac{P^2L^3}{8EI}$

Assam Engg. College AP/Lect. 18.01.2021
Gujarat PSC AE 2019
GATE 2017, MPPSC AE 2016
CGPSC Poly. Lect. 22.05.2016
APPSC AEE Screening Test 2016
APPSC AEE Mains 2016 (Civil Mechanical)
(HPPSC LECT. 2016)
ISRO Scientist/Engineer 11.10.2015, 2006
OPSC Civil Services Pre. 2011, ESE 2009

Ans. (b) :



$$\text{Strain energy } U=\frac{1}{2}\times P\delta$$

$$=\frac{1}{2}\times P\times\frac{PL^3}{3EI}$$

$$\left(\delta=\frac{PL^3}{3EI}\right)$$

$$U=\frac{P^2L^3}{6EI}$$

Or

$$\text{Strain energy } U=\int_0^L\frac{M^2dx}{2EI} \qquad M=px$$

$$=\int_0^L\frac{(px)^2dx}{2EI}$$

$$=\frac{P^2}{2EI}\times\left(\frac{x^3}{3}\right)_0^L$$

$$U=\frac{P^2L^3}{6EI}$$

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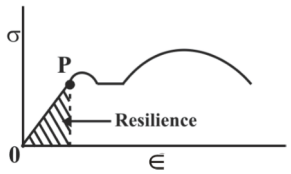
1285. The property of a material to absorb energy within elastic limits is known as

(a) Elasticity (b) Toughness
(c) Tensile strength (d) Stiffness
(e) Resilience

SJVN ET 2019
JPSC AE PRE 2019
Nagaland PSC (CTSE) 2018, Paper-I
(CGPSC Polytechnic Lecturer 2017)
RPSC LECTURER 16.01.2016
RPSC 2016
CGPSC AE 26.04.2015 Shift-I
TNPSC AE (Industries) 09.06.2013

Ans. (e) :



Resilience—The capacity of material to absorb and release strain energy within elastic limit.
Toughness—Strain energy store upto fracture point.
Modulus of toughness—Capacity of material to absorb maximum strain energy upto fracture per unit volume is called MOT.

1286. The strain energy stored in a body, when the load is gradually applied, is

(a) $\sigma E/V$ (b) $\sigma V/E$
(c) $\sigma^2 E/2V$ (d) $\sigma^2 V/2E$

where σ = stress in the material of the body, V = volume of the body and E = modulus of elasticity of the material.

RPSC IOF, 2020
Nagaland PSC (CTSE) 2018, Paper-I
Assam PSC CCE Pre 2015
TSPSC AEE 2015, TNPSC AE 2014
TRB Asstt. Prof., 2012
ESE 2006, 2002

Ans. (d) : Strain energy = $\frac{1}{2}$ stress × Strain × Vol.

The strain energy stored in a body, when the load is gradually applied is $\frac{\sigma^2 V}{2E}$.

1287. Strain energy stored in a beam due to bending is given by (Where M is bending moment, E is modulus of elasticity, I is moment of inertia, G is modulus of rigidity, L is the length of the beam, and σ is the tensile strength.)

(a) $\int \frac{M^2 dx}{2EI}$ (b) $\int \frac{\sigma^2 dx}{2EI}$
(c) $\int \frac{M^2 dx}{2GI}$ (d) $\int \frac{M^2 dx}{2EL}$

HPPSC AE 2018
UPPSC AE 12.04.2016 Paper-I
APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (a) : Strain energy stored in a beam
1. when beam is subjected to bending moment (M)

$$U_{B.M} = \int \frac{M^2.dx}{2EI}$$

2. when beam is subjected to Twisting moment (T)

$$U_{T.M} = \int \frac{T^2.dx}{2GI_P}$$

3. When beam is subjected to axial load P

$$U_{A.L} = \int \frac{P^2.dx}{2AE}$$

1288. Proof resilience may be defined as

(a) Work done in straining the material
(b) Max. strain energy that can be stored in a material per unit volume
(c) Max. strain energy that can be stored in a material under elastic condition
(d) Max. load which can be applied to a member

Nagaland CTSE 2017 Paper-I
Nagaland CTSE 2016 Paper-I
ISRO Scientist/Engineer 2009

Ans. (c) : Proof Resilience
Maximum strain energy that can be stored in a material under elastic limit.

1289. The maximum energy that can be absorbed per unit volume without creating a permanent distortion is called _____

(a) Resilience
(b) Distortion limit
(c) Modulus of resilience
(d) None of these

RPSC ACF & FRO, 26.02.2021
VIZAG Steel MT 24.01.2021
KPCL AE 2016

Ans. (c) : Modulus of Resilience- The maximum energy that can be absorbed per unit volume without creating a permanent distortion-

$$\text{Modulus of Resilience (MOR)} = \frac{\text{Strain energy}}{\text{Volume}}$$

$$(MOR) = \frac{\frac{\sigma^2}{2E} \times \text{Volume}}{\text{Volume}}$$

$$(MOR) = \frac{\sigma^2}{2E}$$

$$MOR = \frac{1}{2}\sigma_{EL}\varepsilon_{EL} \quad (\because \text{EL = Elastic limit})$$

1290. The strain energy stored in a spring, when subjected to maximum load, without suffering permanent distortion, is known as

(a) impact energy
(b) proof resilience
(c) proof stress
(d) modulus of resilience

Sikkim PSC (Under Secretary), 2017
Nagaland PSC CTSE 2017, Paper-I
Nagaland PSC CTSE 2016 Paper-I

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Ans. (b) : The strain energy stored in a spring, when subjected to maximum load, without suffering permanent distortion is known as proof resilience.

1291. Resilience of a material is considered when it is subjected to

(a) Fatigue (b) Creep

(c) Frequent heat treatment (d) Shock Loading

RPSC IOF, 2020

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) : Resilience– It is the strain energy that can be absorbed by the material without plastic deformation upto elastic limit. Toughness is the maximum strain energy that can be absorbed by the material upto fracture.

Creep– The permanent deformation of a material over a period of time at constant load and at elevated temperature is called creep.

Fatigue–The behaviour of the material under fluctuating loads is called fatigue.

1292. "Total Strain Energy" theory of failure is also known as _____.

(a) Couloumb Guest Theory

(b) Lame Rankine Theory

(c) Von - Mises Theory

(d) Beltrami-Haigh Theory

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Total Strain Energy theory of failure is also known as Beltrami-Haigh Theory.

1293. The property of material whereby it absorbs energy due to straining actions by undergoing plastic deformation is called

(a) Elasticity (b) Fatigue

(c) Resilience (d) Toughness

Oil India Senior Officer 23.12.2020

Ans. (d) : It is the property of material which absorb energy while straining by undergoing plastic deformation without fracture.

1294. If a gradual tensile load (P) is applied to an elastic bar axially, the change in the length of bar is x. What will be the strain energy stored in the material of the bar?

(a) Px (b) $\frac{1}{2}$Px

(c) 2 Px (d) Zero

APPSC Poly Lect. 13.03.2020

Ans. (b) :

Strain energy (U) = $\frac{1}{2}\times$ load $\times$ change in the length

$$\boxed{U=\frac{1}{2}P.x}$$

1295. A square bar of size 10 mm × 10 mm and length 1000 mm is subjected to 200 N axial tensile force. The bar is made of mild steel having modulus of elasticity of 200 GPa. Find the strain energy density stored in the bar under this state of loading?

(a) 10 J/m^3 (b) 20 J/m^3

(c) 2 J/m^3 (d) 5 J/m^3

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : Strain energy per unit volume = modulus of resilience

$$=\frac{\sigma_a^2}{2E}=\frac{(P/A)^2}{2E}=\frac{\left(\frac{200}{10\times10}\right)^2}{2\times200\times10^3}$$

$$=\frac{1}{10^5}N-mm/mm^3=10N-m/m^3$$

$$=10\ J/m^3$$

1296. For the state of stress of pure shear τ, the strain energy stored per unit volume in the elastic, homogeneous, isotropic material having elastic constants – Young's modulus, E and Poisson's ratio μ will be

(a) $\frac{\tau^2}{E}(1+\mu)$ (b) $\frac{\tau^2}{2E}(1+\mu)$

(c) $\frac{2\tau^2}{E}(1+\mu)$ (d) $\frac{\tau^2}{2E}(2+\mu)$

UPPSC AE 13.12.2020, Paper-I

ESE 1998

Ans. (a) : Strain energy per unit volume

$$U=\frac{\tau^2}{2G}$$

$$E=2G(1+\mu)$$

$$2G=\frac{E}{1+\mu}$$

$$U=\frac{\tau^2}{E/(1+\mu)}$$

$$\boxed{U=\frac{\tau^2(1+\mu)}{E}}$$

1297. A cantilever beam, 2 m in length is subjected to a uniformly distributed load of 10 kN/m. If E = 200 GPa and I = 1000 cm^4, the strain energy stored in the beam will be

(a) 7 Nm (b) 12 Nm

(c) 8 Nm (d) 40 Nm

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : $U=\frac{\int_0^L M_x^2dx}{2EI}=\frac{\int_0^{\ell}\left(\frac{Wx^2}{2}\right)^2dx}{2EI}$

$$=\frac{W^2}{8EI}\int_0^2x^4dx$$

$$=\frac{(10\times10^3)^2}{8\times200\times10^9\times1000\times(10^{-2})^4}\times\left(\frac{2^5}{5}\right)$$

$$=\frac{200}{5}$$

$$=40\ N\text{-}m$$

1298. Two elastic bars of equal length and same material, one is of circular cross-section of 80 mm diameter and the other of square cross-section of 80 mm side. Both absorbs same amount of strain energy under axial force. What will be the ratio of stress in circular cross-section to that of square cross-section?

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(a) 0.972 (b) 0.886
(c) 1.013 (d) 1.128

SJVN ET 2019

Ans. (d) : $U_c = U_s$

$$\frac{\sigma_c^2}{2E} \times A_c \times L = \frac{\sigma_s^2}{2E} \times A_s \times L$$

$$\frac{\sigma_c^2}{\sigma_s^2} = \frac{A_s}{A_c} = \frac{80^2}{\frac{\pi}{4} \times 80^2}$$

$$\frac{\sigma_c^2}{\sigma_s^2} = \frac{4}{\pi}$$

$$\frac{\sigma_c}{\sigma_s} = \frac{2}{\sqrt{\pi}} = \frac{2}{\sqrt{3.14}}$$

$$\frac{\sigma_c}{\sigma_s} = 1.128$$

1299. The resilience of steel can be found by integrating stress-strain curve up to the

(a) ultimate fracture point (b) upper yield point
(c) lower yield point (d) elastic point

ESE 2018

Ans. (d) : Resilience is the ability of a material to absorb energy per unit volume without permanent deformation and is equal to the area under the stress-strain curve upto the elastic limit.

1300. A circular shaft of length 'L' a uniform cross-sectional area 'A' and modulus of rigidity 'G; is subjected to a twisting moment that produces maximum shear stress 'τ' in the shaft. Strain energy in the shaft is given by the expression AL/kG, where k is equal to

(a) 2 (b) 4
(c) 8 (d) 16

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (b) : Strain energy stored in a shaft due to torque

$$U = \left(\frac{\tau^2}{4G}\right) \text{vol.}$$

$$U = \frac{\tau^2}{4G} \times A \times L$$

$$U = \frac{AL}{kG}$$

Then k = 4

1301. The strain energy corresponding to the stress at the elastic limit is known as _____

(a) Modulus of resilience (b) Toughness
(c) Proof resilience (d) Resilience

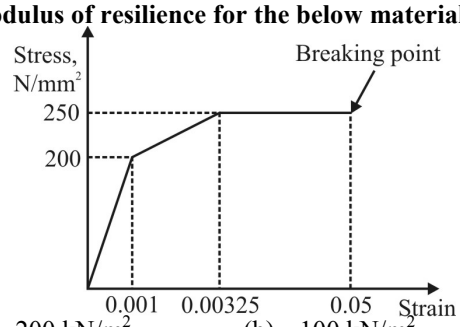
APGENCO AE, 2017

Ans. (c) : Total strain energy stored within elastic limit is known as resilience.

Total strain energy stored corresponding to elastic limit is called proof resilience.

Strain energy stored per unit volume corresponding to elastic limit is called modulus of resilience and its value remains constant and not depend upon dimension of work piece.

1302. Modulus of resilience for the below material is



(a) 200 kN/m² (b) 100 kN/m²
(c) 606.25 kN/m² (d) 12.29 MN/m²

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Modulus of resilience
= Area under stress–strain diagram up to elastic limit (EL)

$$= \frac{1}{2} \sigma_{EL} \times \in_{EL}$$

$$= \frac{1}{2} \times 200 \times 0.001 \text{ N/mm}^2$$

$= 0.1 \times 1000 \text{ kN/m}^2$

$= 100 \text{ kN/m}^2$

1303. Which one of the following statements is correct?

(a) The strain produced per unit volume is called resilience.
(b) The maximum strain produced per unit volume is called proof resilience.
(c) The least strain energy stored in a unit volume is called proof resilience.
(d) The greatest strain energy stored in a unit volume of a material without permanent deformation is called proof resilience.

ESE 2017

Ans. (d) : The greatest strain energy stored in a unit volume of a material with permanent deformation is called proof resilience.

1304. The strain energy stored in a solid circular shaft, under pure torque, per unit its volume (take max. shear stress = q and modulus of rigidity = (C) is expressed as

(a) q^2/C (b) $q/(4C)$
(c) $q^3/(4C)$ (d) $q^2/(4C)$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : We know, strain energy stored =

$$= \frac{1}{2C} \times \text{shear stress}^2 \times \frac{V}{2}$$

For per unit volume under pure torque

$$= \frac{1}{4C} \times \text{shear stress}^2$$

$$= \frac{q^2}{4C}$$

1305. A prismatic bar 1 m long and 4 sq.cm in cross sectional area is compressed by a force of 80 kN. If E = 200 kN/sq.mm, the total strain energy stored in the bar is equal to

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(a) 40 kN-mm (b) 0.05 kN-mm
(c) 400 kN-mm (d) 80 kN-mm

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : For cantilever

Strain energy (u) = $\frac{1}{2}\times\text{stress}\times\text{strain}\times\text{volume}$

Given :

F = 80 kN

L = 1m

A = 4 cm^2

$$U=\frac{F^2L}{2EA}=\frac{(80\times10^3)^2\times1000}{2\times200\times10^3\times400}$$

$$\boxed{U=40kN-mm}$$

1306. A uniform rod of length 'L', cross-section area 'A' and modulus of elasticity 'E' is held rigidly at the fixed ends and an axial load 'P' is applied at mid length of the rod. The strain energy stored is :

(a) $\frac{P^2L}{EA}$ (b) $\frac{P^2L}{4EA}$

(c) $\frac{P^2L}{8EA}$ (d) None of these

RPSC ADE 2016

Ans. (c) :

L = length of rod

A = cross-section area

E = modulus of elasticity

P = Applied load

$$U=\frac{1}{2}W\delta$$

$$\frac{1}{2}W\times\frac{WL}{AE}$$

$$W=\frac{P}{2}$$

Strain energy = $\frac{P^2L}{8EA}$

1307. A specimen 160mm^2 in cross section stretches by 0.06 mm over a 50 mm gauge length under an axial load of 35 kN. The strain energy is

(a) 1050 N-mm (b) 780 N-mm
(c) 1435 N-mm (d) 1298 N-mm

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Given,

Area of specimen A = 160 mm^2

Gauge length L = 50mm

Change in gauge length ΔL = 0.06 mm

Axial load applied P = 35 kN = 35 × 10^3 N

Now, strain energy (U) = $\frac{1}{2}\times P\times\Delta L$

$$U=\frac{1}{2}\times35\times10^3\times0.06=1050N-mm$$

1308. As compared to uniaxial tension or compression, the strain energy stored in bending is only

(a) 1/3 (b) 1/8
(c) 1/4 (d) 1/2

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Strain energy due to tension or compression

$$U_1=\frac{1}{2}\times\text{stress}\times\text{strain}\times\text{volume}$$

$$U_1=\frac{\sigma^2}{2E}\times\text{Volume} \quad \text{.... (i)}$$

Strain energy due to bending

$$U_2=\int_0^L\frac{M^2dx}{2E}=\int_0^L\frac{(Wx)^2}{2E}dx=\frac{W^2}{2E}\int_0^L x^2dx$$

$$U_2=\frac{w^2L^3}{6E} \quad \text{..... (ii)}$$

From equation (i) and (ii)

$$\frac{U_1}{U_2}=\frac{1}{3}$$

1309. What will be the strain energy stored in the metallic bar of cross sectional area of 2 cm^2 and gauge length of 10 cm if it stretches 0.002 cm under the load of 12 kN?

(a) 10 N-cm (b) 12 N-cm
(c) 14 N-cm (d) 16 N-cm

RPSC LECTURER 16.01.2016

Ans. (b) : Data given,

A = 2 cm^2, l = 10 cm

δl = 0.002 cm

P = 12 × 10^3 N

Strain energy

$$U=\frac{\sigma^2}{2E}\times volume$$

$$\sigma=\frac{P}{A}$$

$$\frac{\delta l}{l}=\frac{\sigma}{E}=e$$

then we get

$$U=\frac{1}{2}\times P\times\delta$$

$$=\frac{1}{2}\times12\times10^3\times2\times10^{-3}$$

= 12 N-cm

1310. Strain energy stored in a prismatic bar suspended from one end due to its own weight (elastic behaviour) [x = specific weight of material, A = cross-sectional area, L = length of bar]:

(a) $U=\frac{xAL^3}{6E}$ (b) $U=\frac{x^2A^2L^2}{6E}$

(c) $U=\frac{x^2AL^3}{3E}$ (d) $U=\frac{x^2AL^3}{6E}$

UPRVUNL AE 07.10.2016

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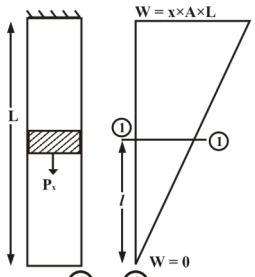
Ans. (d) :



Self weight at section ① - ①

$$P_l = \frac{W.l}{L}$$

We know that strain energy for prismatic bar suspended from one end due to its self weight

$$U = \int_o^L \frac{(P_l)^2 dl}{2(AE)_l}$$

AE = constant

$$= \int_O^L \frac{\left(\frac{Wl}{L}\right)^2 dl}{2AE}$$

$$= \frac{W^2}{2AE \times L^2}\left[\frac{l^3}{3}\right]_O^L$$

$$= \frac{(xAL)^2}{2AEL^2} \times \frac{L^3}{3}$$

$$U = \frac{x^2 AL^3}{6E}$$

1311. A square bar of side 4 cm and length 100 cm is subjected to axial load P. The same bar is then used as a cantilever beam and subjected to an end load P. The ratio of the strain energies, stored in the bar in the second case to that stored in first case, is

(a) 16 (b) 400
(c) 1000 (d) 2500

MPPSC AE 2016

Ans : (d) Square bar Area = $16cm^2$
length = 100cm.

moment of Inertia of square bar

$$= \frac{bh^3}{12} = \frac{4^4}{12} = 21.33cm^4$$

case 1^{st} : –

$$\text{Strain Energy}(U_1) = \frac{\sigma^2}{2E} \times \text{volume}$$

$$U_1 = \left(\frac{P}{16}\right)^2 \times \frac{16}{2E} \times 100 \qquad \sigma = \left(\frac{P}{16}\right)$$

$$U_1 = \frac{3.125P^2}{E}$$

case – II : –

$$U_2 = \frac{P^2L^3}{6EI}$$

$$U_2 = \frac{P^2}{E}\left[\frac{100^3 \times 12}{6 \times 4^4}\right]$$

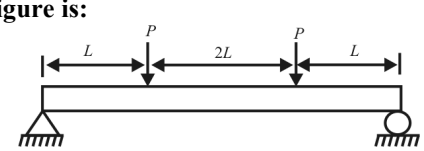
$$U_2 = \frac{7812.5P^2}{E}$$

Ratio of strain energies

$$\frac{U_2}{U_1} = \frac{7812.5P^2 \times E}{E \times 3.125P^2}$$

$$\boxed{\frac{U_2}{U_1} = 2500}$$

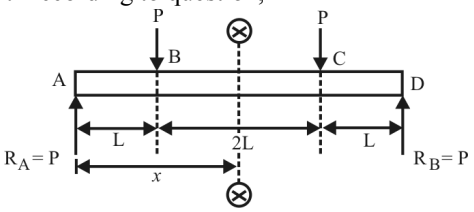
1312. The strain energy stored in the beam with flexural rigidity EI and loaded as shown in the figure is:



(a) $\frac{-P^2L^3}{3EI}$ (b) $\frac{2P^2L^3}{3EI}$
(c) $\frac{4P^2L^3}{3EI}$ (d) $\frac{8P^2L^3}{3EI}$

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : According to question,



$M_x = Px - |P(x - L)|$

For portion AB and CD,

$(M_x)_{AB} = Px = (M_x)_{CD}$

For portion BC,

$(M_x) = Px - P(x - L) = Px - Px + PL$

$(M_x)_{BC} = PL$

Strain energy in the beam due to bending is

$$U = U_{AB} + U_{BC} + U_{CD}$$

$$U = 2\int_0^L \frac{(M_x)_{AB}^2 dx}{2EI} + \int_L^{3L} \frac{(M_x^2)_{BC} dx}{2EI}$$

$$U = \frac{4P^2L^3}{3EI}$$

1313. Proof resilience is the property of the materials which indicated their capacity to bear :

(a) shocks
(b) static compressive loads
(c) static tensile loads
(d) none of these

VIZAG MT 2015

Ans. (a) : Proof resilience– The maximum energy which a body stores 'upto' elastic limit is called the 'proof resilience'. It indicates shock bearing capacity of material.

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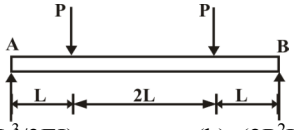
1314. Proof resilience per unit volume of a material is known as-

(a) modulus of resilience (b) resilience
(c) Proof resilience (d) toughness

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (a) Proof resilience per unit volume of a material is known as modulus of resilience.

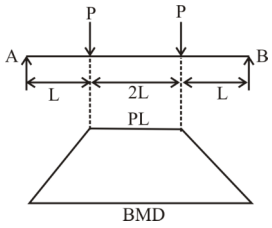
1315. Strain energy stored in beam with flexural rigidity EI and loaded as shown figure is



(a) $(P^2L^3/3EI)$ (b) $(2P^2L^3/3EI)$
(c) $(4P^2L^3/3EI)$ (d) $(6P^2L^3/3EI)$
(e) $(8P^2L^3/3EI)$

CGPSC AE 26.04.2015 Shift-I

Ans. (c) :



$R_A = R_B = P$

Total strain energy,

$$U = 2\int_0^L \frac{(Px)^2\,dx}{2EI} + \frac{(PL)^2(2L)}{2EI}$$

$$U = \frac{P^2L^3}{3EI} + \frac{P^2L^3}{EI}$$

$$\boxed{U = \frac{4}{3}\frac{P^2L^3}{EI}}$$

1316. Strain energy stored in a solid circular shaft is proportional to

(a) GJ (torsional rigidity) (b) $1/(GJ)^2$
(c) GJ^2 (d) $1/(GJ)$
(e) $1/\sqrt{GJ}$

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : For solid shaft—Let us suppose that a solid shaft is subjected to a torque which increases gradually from zero to a value T. Let θ represent the resultant angle of twist. Then, the energy stores in the shaft is equal to work done in twisting i.e.

$$U = \frac{1}{2}.T.\theta$$

$$= \frac{1}{2}.T.\left(\frac{T.l}{GJ}\right)$$

$$\left(\frac{T}{J} = \frac{G\theta}{l}\right)$$

$$= \frac{1}{2}\left(\frac{T^2l}{GJ}\right)$$

$$U \propto \frac{1}{GJ}$$

1317. Strain energy stored in a body due to a suddenly applied load compared to when applied slowly is:

(a) twice (b) four times
(c) eight times (d) half

(HPPSC AE 2014)

Ans : (b) Strain energy stored in a body due to suddenly applied load compared to when applied slowly is four times.

Stress in a body due to
gradually applied load gradually = σ
stress in a body due to suddenly applied load = 2σ
strain energy due to gradually applied load

$$U_{GAL} = \frac{\sigma^2}{2E} \times \text{Volume} \quad \text{.....(i)}$$

strain energy due to suddenly applied load

$$U_{SAL} = \frac{(2\sigma)^2}{2E} \times \text{Volume}$$

$$U_{SAL} = 4 \times \left[\frac{\sigma^2}{2E} \times \text{volume}\right]$$

$$\boxed{U_{SAL} = 4 \times U_{GAL}}$$

1318. Strain energy U, in the case of axially loaded bar is expressed as (where P is axial load, L is bar length, A is cross-section area of the bar, E is Young's Modulus of the bar, σ is axial stress, δL is elongation due to axial load)

(a) $\frac{P^2L^2}{2AE(\delta L)}$

(b) Area under stress versus strain diagram of axially loaded bar

(c) $\frac{\sigma}{2}\frac{(\delta L)}{L}$

(d) $\frac{P^2L}{2AE}$

ISRO Scientist/Engineer 24.05.2014

Ans. (d) : Strain energy = U
Axial Load = P
Length = L
Cross-Section = A
Young's modulus = E
Axial stress = σ

Elongation $\delta L = \frac{PL}{AE}$

Strain Energy U = $\frac{1}{2}$ × Force × deflection

$$= \frac{1}{2} \times P \times \frac{PL}{AE} = \frac{P^2L}{2AE}$$

1319. If the area, length and the stress to which a bar is subjected be all doubled, then the elastic strain energy of the bar will be :

(a) Doubled (b) Three times
(c) Four times (d) Sixteen times

J&K PSC Civil Services Pre, 2013

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Ans. (d) : Energy stored within elastic limit is called resilience.
Strain energy stored upto elastic limit is called proof resistance.
Energy stored upto elastic per unit volume is called modulus of resilience.

Strain energy $U_1 = \frac{\sigma^2}{2E} \times \text{volume} = \frac{\sigma^2}{2E} \times A \times l$

$$U_2 = \frac{(2\sigma)^2}{2E} \times 2A \times 2l = 16\frac{\sigma^2}{2E} \times A \times l$$

1320. The unit of strain energy in S.I. unit is
(a) watt (b) joule
(c) joule/sec (d) joule/m.
TNPSC ACF 2012

Ans. (b) : Strain Energy- It is defined as the energy stored in a body due to deformation.

$$U = \frac{1}{2}P\delta$$

$$U = \frac{1}{2} \times P \times \frac{PL}{AE} = \frac{1}{2}\frac{P^2L}{AE}$$

Unit of strain energy N-m or Joule.

1321. Principal stresses induced in a material are 60 MPa, 30 MPa and –20 MPa, E = 80 GPa, μ = 0.20. The value of total strain energy per unit volume is:
(a) 24.5 kN-m/m^3 (b) 49 kN-m/m^3
(c) 30.63 kN-m/m^3 (d) 61.25 kN-m/m^3
HPPSC ADF 05.03.2019

Ans. (c) : Given,
$\sigma_1 = 60$ MPa, $\sigma_2 = 30$ MPa, E = 80 GPa, $\mu = 0.20$
$\sigma_3 = -20$ MPa,

$$\frac{\text{Total strain energy}}{\text{Volume}} =$$

$$= \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

$$= \frac{1}{2\times80\times10^3}\left[(60)^2 + (30)^2 + (-20)^2 - 2\times0.20(60\times30 - 30\times20 - 60\times20)\right]$$

$$= \frac{1}{2\times80\times10^3}\left[3600 + 900 + 400 - 0.40(1800 - 600 - 1200)\right]$$

$$= \frac{1}{160\times10^3}\left[4900 - 0.40(0)\right]$$

$$= \frac{1}{160\times10^3} \times 4900 = \frac{4900}{160} \quad \frac{\text{kN} - \text{m}}{\text{m}^3}$$

$$= 30.625\frac{\text{kN} - \text{m}}{\text{m}^3}$$

$$\approx 30.63\frac{\text{kN} - \text{m}}{\text{m}^3}$$

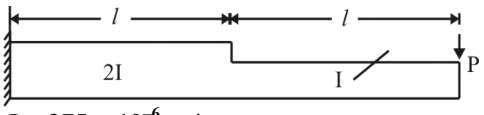
1322. If the cross-section of a member is subjected to a uniform shear stress of intensity 'q', then the strain energy stored per unit volume will be given by (C = modulus of rigidity)
(a) $2q^2/C$ (b) $2C/q^2$
(c) $q^2/2C$ (d) $C/2q^2$
APPSC AEE 2012

Ans. (c) : Shear stress intensity $\tau = q$
modules of rigidity $G = C$
Hence

$$\text{Strain energy stored per unit volume} = \frac{\tau^2}{2G}$$

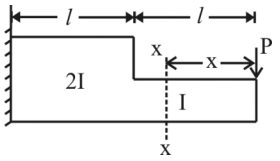
$$\boxed{\frac{\text{S.E}}{\text{Volume}} = \frac{q^2}{2c}}$$

1323.



I = 375 × 10^{-6} m4
***l* = 0.5 m**
E = 200 GPa
Determine the stiffness of the beam shown in the above fig.
(a) 12×10^{10} N/m (b) 10×10^{10} N/m
(c) 4×10^{10} N/m (d) 8×10^{10} N/m
APPSC AE 04.12.2012

Ans. (*) :



Given,
$I = 375\times10^{-6}$ m^4
$l = 0.5$ m
$E = 200$ GPa $= 200\times10^9$ Pa
Let, K = stiffness of beam
∴ Bending moment $(M_x) = P.x$

$$U = \int_0^l \frac{(Px)^2}{2EI}.dx + \int_l^{2l} \frac{(Px)}{2E(2I)}.dx$$

$$= \left[\frac{P^2x^3}{3\times2EI}\right]_0^l + \left[\frac{P^2x^3}{3\times2E\times2I}\right]_l^{2l}$$

$$= \frac{P^2l^3}{6EI} + \frac{P2(8l^3 - l^3)}{12EI} = \frac{P^2l^3}{6EI} + \frac{7\rho^2l^3}{12EI}$$

$$U = \frac{9P^2l^3}{12EI} = \frac{3P^2l^3}{4EI} \quad \text{(i)}$$

$\because$ $U = \frac{1}{2}P\times\delta = \frac{1}{2}\frac{P^2}{k}$ _____(ii) $\left\{\because \delta = \frac{p}{k}\right.$

From eq^n (1) and (2)

$$\frac{1}{2}\frac{P^2}{k} = \frac{3P^2l^3}{4EI} \Rightarrow k = \frac{2EI}{3l^3}$$

$$\Rightarrow \quad k = \frac{2\times200\times10^9\times375\times10^{-6}}{3\times(0.5)^3} = 4\times10^8 \text{ N/m}$$

No any option matching with the solution.

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1324. The strain energy per unit volume of a body subjected to a direct stress σ is given by

(a) $\frac{\sigma^2}{2E}$ (b) $\frac{\sigma}{2E}$

(c) $\frac{\sigma^2}{3E}$ (d) $\frac{\sigma}{3E}$

WBPSC AE, 2007, ESE 2016
UKPSC AE 2007 Paper-I

Ans. (a) : Strain energy per unit volume (S_{vol})

$$S_{vol} = \frac{1}{2} \times \sigma \times \varepsilon$$

$$S_{vol} = \frac{1}{2} \times \sigma \times \frac{\sigma}{E}$$

$$\boxed{S_{vol} = \frac{\sigma^2}{2E}}$$

1325. What will be the total strain energy stored in a simply supported beam of span 'L' and flexural rigidity 'EI' subjected to a concentrated load 'W' at the centre?

(a) $\frac{W^2L^3}{40\,EI}$ (b) $\frac{W^2L^3}{60\,EI}$

(c) $\frac{W^2L^3}{192\,EI}$ (d) $\frac{W^2L^3}{96\,EI}$

WBPSC AE 2003

Ans. (d) : We know that deflection $\delta = \frac{WL^3}{48EI}$ for simply supported beam in concentrated load (W) at mid span then,

$$\delta = \frac{WL^3}{48EI} = \frac{\partial U}{\partial W} \qquad (\text{Strain energy} = U)$$

Deflection, $\delta = \frac{WL^3}{48EI} = \frac{\partial U}{\partial W}$

$$\int \partial U = \int \frac{WL^3}{48EI} \partial W$$

$$\boxed{U = \frac{W^2L^3}{96EI}}$$

9. Deflection of Beams

1326. What is the maximum deflection in a cantilever beam carrying a uniformly distributed load of w per unit length? (L-length of the beam, EI-flexural rigidity of the section of the beam)

(a) $\frac{wL^4}{3EI}$ (b) $\frac{wL^3}{192EI}$

(c) $\frac{5wL^4}{384EI}$ (d) $\frac{wL^4}{8EI}$

JPSC AE 10.04.2021, Paper-II
NLCIL GET 17.11.2020, Shift-II
TNCSC AE 2020
GPSC Executive Engineer 23.12.2018
APGENCO AE, 2017, RPSC 2016
APPSC AEE Screening Test 2016
TNPSC AE (Industries) 09.06.2013
ESE 2008, GATE 2014

Ans. (d) :

Case	Max. slope	Max. deflection
• Cantilever with point load (W) at free end	$\frac{WL^2}{2EI}$	$\frac{WL^3}{3EI}$
• Cantilever UDL having load intensity (w) on whole span	$\frac{wL^3}{6EI}$	$\frac{wL^4}{8EI}$
• Simply supported point load (W) at middle of beam	$\frac{WL^2}{16EI}$	$\frac{WL^3}{48EI}$
• Simple supported UDL having load intensity (w) on whole span	$\frac{1}{24}\frac{wL^3}{EI}$	$\frac{5}{384}\frac{wL^4}{EI}$

1327. Deflection due to point load at the free end of the cantilever due to load P (with usual notation)

(a) $Pl^3/(3EI)$ (b) $Pl^2/(3EI)$

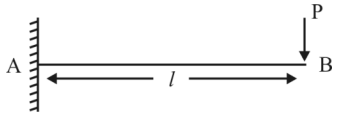
(c) $Pl^2/(2EI)$ (d) $Pl^4/(3EI)$

NLC GET 17.11.2020, Shift-II
Nagaland PSC (CTSE) 2018 Paper-I
TSPSC AEE 2015
APSC AEE 2012
APSC AE 04.12.2012
TNPSC AE, 2008

Ans. (a) : Let,

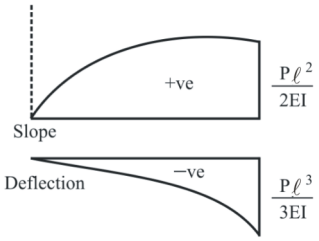
l = length of cantilever

I = moment of inertia.



Bending moment at the section = Px

or $EI\frac{d^2y}{dx^2} = -P.x$



Integrating $EI\frac{dy}{dx} = -\frac{Px^2}{2} + C_1$

at $x = l$, $\frac{dy}{dx} = 0$ $C_1 = \frac{Pl^2}{2}$

$$EI\frac{dy}{dx} = -\frac{Px^2}{2} + \frac{Pl^2}{2}$$

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Slope $\frac{dy}{dx} = \frac{P}{2EI}(l^2 - x^2)$

Integrating this equation

$$EI\ y = -\frac{Px^3}{6} + \frac{Pl^2}{2}x + C_2$$

at $x = l$, $y = 0$ $C_2 = -\frac{Pl^3}{3}$

$$EI\ \ y = -\frac{Px^3}{6} + \frac{Pl^2}{2}x - \frac{Pl^3}{3}$$

Thus deflection given by

$$y = -\frac{P}{6EI}(2l^3 - 3l^2x + x^3)$$

At the free end (x = 0), the slope and deflection are maximum

$$\boxed{\text{Deflection} = -\frac{Pl^3}{3EI}}$$

–ve sign represent deflection is downward.

1328. Maximum deflection in cantilever due to pure bending moment M at its ends is-

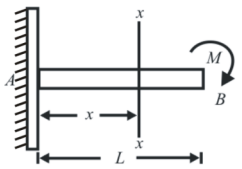
(a) $\frac{ML^2}{2EI}$ (b) $\frac{ML^2}{3EI}$

(c) $\frac{ML^2}{4EI}$ (d) $\frac{ML^2}{6EI}$

GPSC ARTO Pre 30.12.2018
TSPSC AEE 2017
UPPSC AE 12.04.2016 Paper-I
TSPSC Managers, 2015
ISRO Scientist/Engineer 2010
ISRO Scientist/Engineer 2006
GATE 2012

Ans. (a) : using double integration method, at any distance x from A, we have

$$EI\frac{d^2y}{dx^2} = -M_x$$



$$EI\frac{dy}{dx} = -Mx + C_1 \qquad \ldots(i)$$

$$EI(y) = M\frac{x^2}{2} + C_1x + C_2 \qquad \ldots(ii)$$

If

At $x = 0$, $\frac{dy}{dx} = 0$; $\boxed{C_1 = 0}$

At $x = 0$, $y = 0$; $\boxed{C_2 = 0}$

then putting the values of C_1 & C_2 in eqn. (ii)

$$EI(y) = \frac{Mx^2}{2}$$

then deflection at free end will be max . ($x = L$)

$$\boxed{y = \frac{ML^2}{2EI}}$$

and slope at free end ($x = L$)

$$\boxed{\left[\frac{dy}{dx}\right]_{x=L} = \frac{ML}{EI}}$$

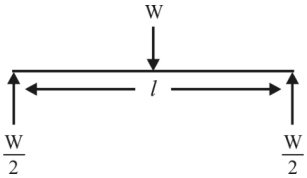
1329. A simply supported beam of span *l* carries a point load W at mid span. The downward deflection under the load will be :

(a) $\frac{Wl^3}{3EI}$ (b) $\frac{Wl^3}{8EI}$

(c) $\frac{Wl^3}{48EI}$ (d) $\frac{5}{384}\frac{Wl^3}{EI}$

SJVN ET 2019
ISRO Scientist/Engineer (RAC) 07.05.2017
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
UKPSC AE-2013, Paper-I
APPSC AE 04.12.2012
VIZAG STEEL MT 2015

Ans. (c) : Deflection, δ at mid span = $\frac{Wl^3}{48EI}$



1330. A simply supported beam of length L is subjected to uniformly distributed load w/unit length for its entire span. Taking E as Young's modulus and I as moment of inertia, the deflection at the mid span is given by:

(a) $y = \frac{-5wL^3}{384EI}$ (b) $y = \frac{-wL^3}{384EI}$

(c) $y = \frac{-wL^4}{384EI}$ (d) $y = \frac{-5wL^4}{384EI}$

MECON MT 2019
Oil India Limited Sr. Engineer (Drilling) 30.11.2019
TRB Polytechnic Lecturer 2017
CGPSC AE 26.04.2015 Shift-I
TANGEDCO AE 2015

Ans. (d) : We know that

Maximum slope (θ) $= \frac{A}{EI} = -\frac{wL^3}{24EI}$

Maximum deflection (δ) $= \frac{A\bar{X}}{EI} = \frac{\left(\frac{wL^3}{24}\right) \times \left(\frac{5}{8}.\frac{L}{2}\right)}{EI}$

$$\delta = \frac{-5}{384}\frac{wL^4}{EI}$$

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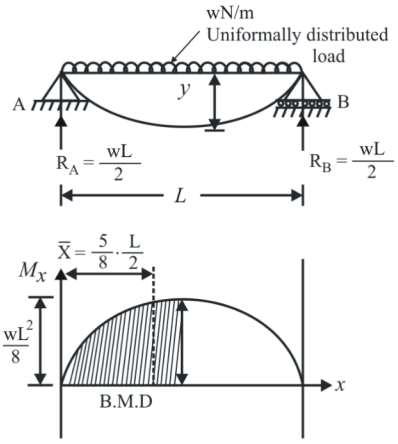
Note- Area of BMD (shaded)

$$A = \frac{2}{3} \times \left(\frac{L}{2}\right) \times \left(\frac{wL^2}{8}\right) = \frac{wL^3}{24}$$



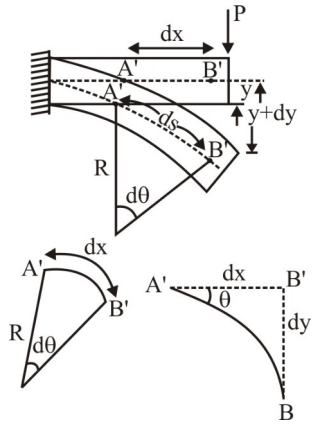
1331. The curvature equation of beam is given as:

(a) $M = EI\frac{d^3y}{dx^3}$ (b) $M = EI\frac{dy}{dx}$

(c) $M = EI\frac{d^2y}{dx^2}$ (d) $M = EI\frac{d^4y}{dx^4}$

Assam PSC AE (PHED), 18.10.2020
Oil India Ltd. Sr. Engg. (Production) 30.11.2019
APPSC AEE Screening Test 2016
APPSC AEE 2012

Ans. (c)



Curvature equation of beam-

Let ds ≅ dx

dx = Rdθ…….(i)

$$\tan\theta = \frac{dy}{dx}$$

θ is very small ⇒ tanθ = θ

$$\theta = \frac{dy}{dx} \ldots\ldots\ldots\ldots (ii)$$

From equation (i)

$$\frac{1}{R} = \frac{d\theta}{dx}$$

$$= \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

$$\therefore \frac{1}{R} = \frac{d^2y}{dx^2}$$

Now we know that equation of theory of bending

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\therefore \frac{M}{EI} = \frac{I}{R}, \text{ put the value of } \frac{1}{R}$$

$$\frac{M}{EI} = \frac{d^2y}{dx^2} \Rightarrow M = EI\frac{d^2y}{dx^2}$$

1332. The simple supported beam 'A' of length 'l' carries a central point load 'W'. Another beam 'E' is loaded with a uniformly distributed load such that the total load on the beam is 'W'. The ratio of maximum deflections between beams A and B is

(a) 5/8 (b) 8/5
(c) 5/4 (d) 4/5

TSPSC AEE 28.08.2017 (Civil/Mechanical)
TSPSC AEE 2015
VIZAG Steel MT 2011
J & K PSC Screening, 2006

Ans. (b) : $\delta_A = \frac{WL^3}{48EI}$ (simply supported beam)

$\delta_B = \frac{5wL^4}{384\,EI}$ (simply supported with uniform distributed load)

∵ Given, (W = wL)

Then $\delta_B = \frac{5\,WL^3}{384\,EI}$

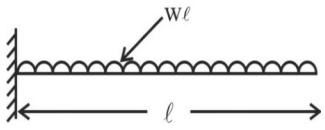
$$\frac{\delta_A}{\delta_B} = \frac{8}{5}$$

1333. A cantilever beam carries a load W uniformly distributed over its entire length. if the same load is placed at the free end of the same cantilever, then the ratio of maximum deflection in the first case to that in the second case will be is

(a) 3/8 (b) 8/3
(c) 5/8 (d) 8/5

ISRO Scientist/Engg. RAC 12.01.2020
JWM 2017
APPSC AEE 2012
WBPSC AE 2003
ESE 1996

Ans. (a) : <u>In first case –</u>



Deflection $\delta = \frac{w\ell^4}{8EI}$ $\{\because W = w\ell\}$

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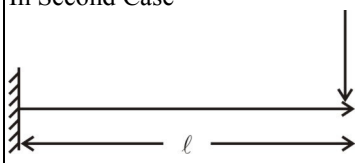
$\boxed{\delta = \frac{W\ell^3}{8EI}}$ ..... (i)

In Second Case –



Deflection $\delta = \frac{W\ell^3}{3EI}$ ..... (ii)

$$\frac{\delta_{ud\ell}}{\delta_{concentrated}} = \frac{\frac{W\ell^3}{8EI}}{\frac{W\ell^3}{3EI}}$$

$$\frac{\delta_{ud\ell}}{\delta_{concentrated}} = \frac{3}{8}$$

1334. .....theorem used based on strain energy concept and can be used for determining deflection in a beam.

(a) Euler's (b) Rankine's

(c) Gourdan's (d) Castigliano's

ISRO Scientist/Engineer 17.12.2017
HPPSC Asstt. Prof. 29.10.2016
HPPSC Lect. (Auto) 23.04.2016
APPSC AE 04.12.2012

Ans. (d) : Let, U = strain energy due to bending moment.

$$U = \int_a^b \frac{M_{x-x}^2 dx}{2(EI)_{x-x}}$$

As per this theorem,

$\boxed{\text{Deflection}(y_A) = \frac{\partial U}{\partial W_A}}$ $\boxed{\text{Slope}(\theta_A) = \frac{\partial U}{\partial M_A}}$

Where, W_A = concentrated load at A.

M_A = Concentrated moment at A.

Castigliano's theorem is used based on strain energy concept and can be used for determining deflection in a beam.

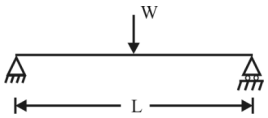
1335. If the width of a simply supported beam carrying an isolated load at its centre is doubled, the deflection of the beam at the centre is changed by:

(a) 1/2 (b) 1/8

(c) 4 (d) 2

Nagaland PSC (CTSE) 2018, Paper-I
APPSC AE Subordinate Service Civil/Mech. 2016
TSPSC Managers, 2015

Ans. (a) :



Deflection of simply supported beam (SSB)

$$(\delta) = \frac{WL^3}{48EI}$$

where, $I = \frac{bd^3}{12}$

[$\therefore$ **Given –** If width is doubled (i.e. $b_1 = 2b$)]

$$\delta_1 = \frac{WL^3}{48EI} = \frac{WL^3}{48E \times \left(\frac{2bd^3}{12}\right)} \Rightarrow \frac{WL^3}{48E(2 \times I)}$$

$\boxed{\delta_1 \Rightarrow \frac{\delta}{2}}$

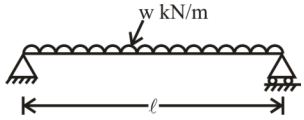
1336. The slope at the support of a simply supported beam having uniformly distributed load throughout its span is _____.

(a) $\theta_{max} = \frac{w.L^3}{24EI}$ (b) $\theta_{max} = \frac{w.L^4}{48EI}$

(c) $\theta_{max} = \frac{w.L^3}{48EI}$ (d) $\theta_{max} = \frac{w.L^4}{24EI}$

VIZAG Steel MT 24.01.2021
SJVN ET 2019
WBPSC AE, 2007

Ans. (a) :



Deflection $(\delta) = \frac{5w\ell^4}{384EI}$

Slope $(\theta) = \frac{w\ell^3}{24EI}$

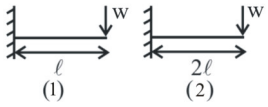
1337. In a cantilever beam, If the length is doubled while keeping the cross-section and the concentrated load acting at the free end the same, the deflection at the free and will increase by:

(a) 2.66 times (b) 3 times

(c) 6 times (d) 8 times

OPSC AEE 2019 Paper-I
ISRO Scientist/Engineer (RAC) 22.04.2018
OPSC AEE 2015 Paper-I
CSE Pre-1996

Ans : (d) : In a cantilever beam, If the length is doubled while keeping the cross-section and the concentrated load acting at the free end the same, the deflection at the free and will increase by eight times.



Cantilever beam when length = l

Deflection $(\delta_1) = \frac{w\ell^3}{3EI}$

Cantilever beam when length = $2l$

Deflection $(\delta_2) = \frac{w(2\ell)^3}{3EI}$

$\boxed{\delta_2 = 8\delta_1}$

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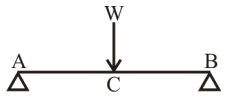
1338. Two simply supported beams 'A' and 'B' of same breadth and depth and same material carries constant central load W at the centre. The span of beam B is double that of beam A. The deflection of beam B compared to beam A will be

(a) one-half (b) one-eighth
(c) four times (d) eight times

ISRO Scientist/Engineer (RAC) 12.01.2020
CSE Pre-1996

Ans. (d) :



$$\delta = \frac{wL^3}{48EI}$$

$$\delta \propto L^3$$

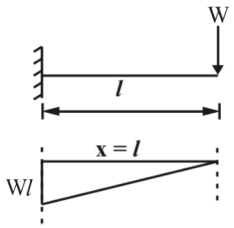
$$\frac{\delta_B}{\delta_A} = \left(\frac{L_B}{L_A}\right)^3 = 8$$

1339. In a cantilever, maximum deflection occurs where–

(a) bending moment is zero
(b) bending moment is maximum
(c) shear force is zero
(d) slope is zero

RPSC 2016
J & K PSC Screening, 2006

Ans : (a)



In the cantilever beam loaded as shown, the max bending moment is at (x = 0) but the max deflection occur at (x = l)

Maximum deflection at x = l is given by $= \frac{Wl^3}{3EI}$

1340. A cantilever beam having square cross section of side is subjected to an end load. If a increased by 19% the tip deflection decreases approximately

(a) 19% (b) 29%
(c) 41% (d) 50%

GPSC Asstt. Director of Transport 05.03.2017
RPSC 2016
GATE 2016

Ans : (d)

$$\delta = \frac{Wl^3}{3EI} = \frac{Wl^3}{3E\frac{I}{12}a^4}$$

$$\delta = \left(\frac{12Wl^3}{3E}\right)\frac{1}{a^4}$$

$$\delta \propto \frac{1}{a^4}$$

$$a_2 = a + 0.19a \Rightarrow \frac{\delta_1}{\delta_2} = \frac{a_2^4}{a_1^4}$$

$$= \frac{(a_1 + 0.19a_1)^4}{a_1^4} = \frac{a_1^4}{a_1^4}(1+0.19)^4 = 2.005$$

$$\delta_1 = 2.0053\delta_2 \Rightarrow \text{Decrease} = 1 - \frac{\delta_2}{\delta_1}$$

$$\text{Decrease} = \frac{2.0053\delta_2 - \delta_2}{2.0053\delta_2}$$

$$= 50.03\%$$

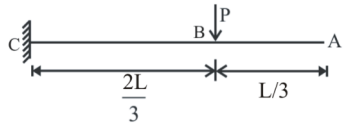
$$\approx 50\%$$

1341. A cantilever beam of length L is subjected to an on concentrated load P at a distance of L/3 from free end, what is the deflection at free end of the beam?

(a) $\frac{2PL^3}{81EI}$ (b) $\frac{3PL^3}{81EI}$
(c) $\frac{14PL^3}{81EI}$ (d) $\frac{15PL^3}{81EI}$

RPSC Vice Principal ITI 2018
ISRO Scientist/Engineer 03.07.2016
ESE 2004

Ans. (c)



Deflection at free end A is

$$\delta_A = \delta_{BC} + \theta_{BC} \times \left(\frac{L}{3}\right)$$

$$\delta_A = \frac{P\left(\frac{2L}{3}\right)^3}{3EI} + \frac{P\left(\frac{2L}{3}\right)^2}{2EI} \times \frac{L}{3}$$

$$= \frac{8PL^3}{81EI} + \frac{4PL^3}{54EI}$$

$$\boxed{\delta_A = \frac{14PL^3}{81EI}}$$

1342. The method in which a single equation is formed for all loads on the beam and the equation constructed in such a way that the integral constants apply to all the sections of the beam is

(a) Moment area method
(b) Conjugate beam method
(c) Macaulay's method
(d) method of super position

TSPSC Manager (Engg.) HMWSSB 12.11.2020
APPSC AEE Screening Test 2016

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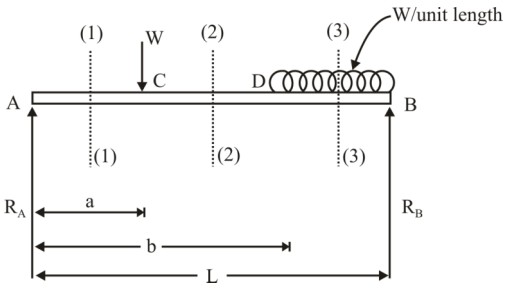
Ans. (c) : The method in which a single equations is formed for all loads on the beam and the equation constructed in such a way that the integral constants apply to all the sections of the beam is Macaulay's method.

e.g.



1343. A cantilever beam of rectangular cross section is subjected to load at free end. If the depth of the beam is double and load is halved, deflection of the free end as compared to original will be

(a) 1/2 (b) 1/8

(c) 2 (d) 1/16

Nagaland PSC CTSE 2017, Paper-I
ISRO Scientist/Engineer 2009

Ans. (d) : For cantilever beam deflection $y = \frac{WL^3}{3EI}$

$$I = \frac{bd^3}{12}$$

$$I_1 = \frac{b(2d)^3}{12}$$

$$= \frac{8bd^3}{12}$$

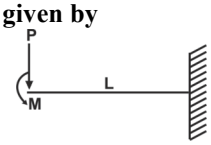
$$I_1 = 8I$$

$$y = \frac{\frac{W}{2}L^3}{3EI \times 8}$$

$$= \frac{WL^3}{6EI \times 8}$$

$$= \frac{WL^3}{48EI} = \frac{WL^3}{3EI} \times \frac{1}{16} = \frac{y}{16}$$

1344. The given figure shows a cantilever of span 'L' subjected to a concentrated load 'P' and a moment 'M' at the free end. Deflection at the free end is given by

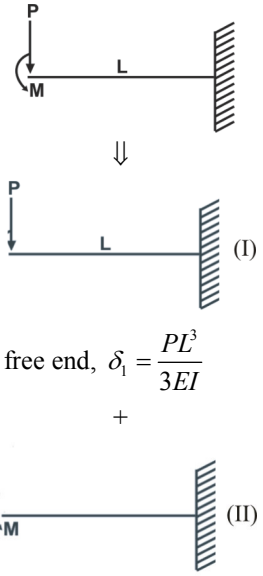


(a) $\frac{PL^2}{2EI} + \frac{ML^2}{3EI}$ (b) $\frac{ML^2}{2EI} + \frac{PL^2}{48EI}$

(c) $\frac{ML^2}{3EI} + \frac{PL^2}{2EI}$ (d) $\frac{ML^2}{2EI} + \frac{PL^3}{3EI}$

UJVNL AE 2016, ISRO Scientist/Engineer 2011
ESE 1996

Ans. (d) : For cantilever beam



Deflection at the free end, $\delta_1 = \frac{PL^3}{3EI}$

+

Deflection, $\delta_2 = \frac{ML^2}{2EI}$

So, Total Deflection (I + II)

$= \delta_1 + \delta_2$

$$= \frac{PL^3}{3EI} + \frac{ML^2}{2EI}$$

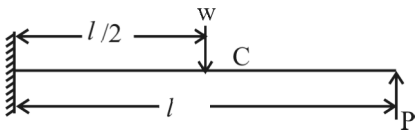
1345. A point load 'W' is acting at mid span of cantilever of length '*l*'. If the free end is supported on a rigid prop, the reaction of the prop is

(a) $\frac{5W}{13}$ (b) $\frac{5W}{11}$

(c) $\frac{5W}{16}$ (d) $\frac{7W}{9}$

APPSC AE Subordinate Service Civil/Mech. 2016
APGENCO AE 2012

Ans. (c) :



To find reaction due to prop,

Deflection at free end due to load 'W' at mid span

= Deflection at free end due to prop reaction

$$\frac{W(l/2)^3}{3EI} + \frac{W(l/2)^2}{2EI} \times \frac{l}{2} = \frac{Pl^3}{3EI}$$

$$\frac{Wl^3}{24EI} + \frac{Wl^3}{16EI} = \frac{Pl^3}{3EI}$$

$$\frac{2W + 3W}{48} = \frac{P}{3}$$

$$\boxed{P = \frac{5W}{16}}$$

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1346. A simply supported laterally loaded beam was found to deflect more than a specified value. Which of the following measures will reduce the deflection?

(a) Increase the area moment of inertia
(b) Increase the span of the beam
(c) Select a different material having lesser modulus of elasticity
(d) Magnitude of the load to be increased

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC ARTO Pre 30.12.2018

Ans. (a) : For deflection, deflection is inversely proportional to moment of inertia,

$$\text{Deflection}(\delta) = \frac{W\ell^3}{48EI}$$

$$\delta \propto \frac{1}{I}$$

Therefore, increase in moment of inertia, reduces deflection.

1347. For a fixed end of a long column, _____.

(a) The deflection and the slope is zero
(b) The deflection is zero, but the slope is maximum
(c) The deflection is maximum, but the slope is zero
(d) The deflection is zero, but the slope is minimum

VIZAG Steel MT 24.01.2021
APPSC AEE 2012

Ans. (a) : For a fixed end of a long column, the deflection and slope are zero. The fixed beam is stronger, stiffen and more stable. The slope at the supports is zero. Maximum bending moment at the centre is reduced because of fixing moments developed at supports. The deflection is zero for hinged end.

1348. The flexural rigidity for the deflection of beam is expressed as

(a) $\frac{I}{E}$ (b) $\frac{E}{I}$
(c) EI (d) $\frac{1}{EI}$

APPSC AEE Mains 2016 (Civil Mechanical)
APPSC Poly. Lect. 2013
APPSC AEE 2012

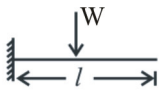
Ans. (c) : The flexural rigidity for the deflection of beam is expressed as EI.

1349. A cantilever beam of length L, moment of inertia of inertia I and Young modulus E carries a concentrated load W at the middle of its length. The slope of cantilever at the free end is:

(a) $(W\ell^2)/2EI$ (b) $(W\ell^2)/4EI$
(c) $(W\ell^2)/8EI$ (d) $(W\ell^2)/16EI$

OPSC AEE 2015 Paper-I
ESE 2001

Ans : (c)



When load acts free end then slope $= \frac{W\ell^2}{2EI}$

When load acts mid span the slope $= \frac{W(\ell/2)^2}{2EI}$

$$\text{Slope} = \frac{W\ell^2}{8EI}$$

1350. The resultant deflection of a beam under symmetrical bending is

(a) perpendicular to the neutral axis
(b) parallel to the axis of symmetry
(c) perpendicular to the axis of symmetry
(d) parallel to the neutral axis

Assam PSC AE (IWT) 14.03.2021
UKPSC AE-2013, Paper-I

Ans. (a) : The resultant deflection of a beam under symmetrical bending is perpendicular to the neutral axis.

1351. A simply supported beam is of rectangular section. It carries the uniformly distributed load over the span. The deflection at the centre is y. If the depth of the beam is doubled, the deflection at the centre is _____.

(a) 2y (b) y/8
(c) y/2 (d) 4y

VIZAG Steel MT 24.01.2021, Shift-I
WBPSC AE, 2017

Ans. (b) : Given,
Figure

Then, deflection (y) = $\frac{5wl^4}{384EI}$

$\therefore$ If depth of beam is doubled $d_1 = 2d, b_1 = b$

Now, $y_1 = \frac{5wl^4}{384 \times E \times \frac{bd^3}{12}}$

$$y_1 = \frac{5wl^4}{384 \times E \times \frac{b \times (2d)^3}{12}}$$

$$= \frac{5wl^4}{384 \times E \times \frac{bd^3}{12} \times 8} = \frac{5wl^4}{384EI8}$$

$$y_1 = \frac{1}{8} \times \frac{5wl^4}{384EI} = \frac{1}{8} \times y$$

$$\boxed{y_1 = \frac{y}{8}}$$

1352. A fixed beam of length *l* carries a point load W at the centre. The deflection at the centre is _____.

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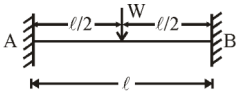
(a) one-fourth of a deflection of a simply supported beam
(b) half of the deflection of a simply supported beam
(c) double the deflection of a simply supported beam
(d) same as for a simply supported beam

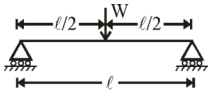
VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) :



deflection at centre,

$$(y)_1 = \frac{W\ell^3}{192EI}$$



$$(y_2) = \frac{W\ell^3}{48EI}$$

$\therefore$ So, there deflection at centre of fixed beam,

$$(y_1) = \frac{1}{4}(y_2)$$

$$= \frac{1}{4} \times \frac{W\ell^3}{48EI}$$

$$(y_1) = \frac{1}{4} \times \text{deflection of simply supported beam}$$

1353. The equation for the deflected shape of a beam carrying a uniformly distributed load and simply supported at the ends is given by :

$$y = \frac{1}{EI}\left(2x^3 - \frac{x^4}{6} - 36x\right)$$

where E is the elastic modulus of beam material and I is the second moment of beam cross-section area about neutral axis. What is the intensity of uniformly distributed load over the beam span?

(a) 12 kN/m (b) 8 kN/m
(c) 6 kN/m (d) 4 kN/m

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Equation of deflected shape of beam

$$y = \frac{1}{EI}\left(2x^3 - \frac{x^4}{6} - 36x\right)$$

$$\text{Intensity of loading (q)} = -\left(EI\frac{d^4y}{dx^4}\right)$$

$$y = \frac{1}{EI}\left(2x^3 - \frac{x^4}{6} - 36x\right)$$

$$EIy = 2x^3 - \frac{x^4}{6} - 36x$$

$$EI\frac{dy}{dx} = 2 \times 3x^2 - \frac{4x^3}{6} - 36$$

$$EI\frac{d^2y}{dx^2} = 2 \times 3 \times 2x - \frac{12x^2}{6} - 0$$

$$EI\frac{d^3y}{dx^3} = 12 - \frac{24x}{6}$$

$$EI\frac{d^4y}{dx^4} = 0 - \frac{24}{6}$$

$$EI\frac{d^4y}{dx^4} = -4$$

$$q = -EI\frac{d^4y}{dx^4}$$

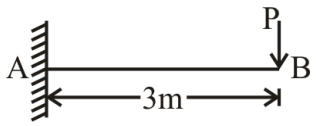
$$\boxed{q = -(-4) = 4\,kN/m}$$

1354. A cantilever beam of 3 m is carrying a point load of unknown mass and produce a slope at the free end as 0.01 rad. The deflection at the free end is _____.

(a) 20 mm (b) 5 mm
(c) 15 mm (d) 10 mm

VIZAG Steel MT 24.01.2021

Ans. (a) :



At point 'B'

$$\text{Slope } (\theta) = \frac{P\ell^2}{2EI}$$

$$\frac{P\ell^2}{2EI} = \theta$$

Given, $\theta = 0.01$ radian, $\ell = 3m$

$$0.01 = \frac{P \times (3)^2}{2EI}$$

$$P = \frac{0.01 \times 2EI}{9} = \frac{0.02EI}{9}$$

Deflection at free end 'B',

$$(\delta) = \frac{P\ell^3}{3EI}$$

$$\delta = \frac{\frac{0.02EI}{9} \times 27}{3EI}$$

$$= \frac{0.02EI \times 3}{3EI}$$

$$\delta = 0.02 \text{ m}$$

$$\delta = 20 \text{ mm}$$

1355. A force P is applied at a distance x from the end of the beam as shown in the figure. What would be the value of x so that the displacement at A is equal to zero?

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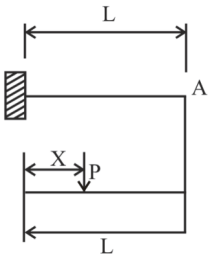
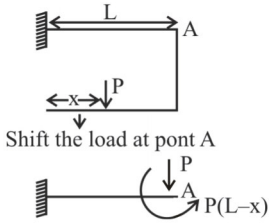


(a) 0.33L (b) 0.25L
(c) 0.5L (d) 0.66L

Assam Engg. College AP/Lect. 18.01.2021
GATE 2014

Ans. (a) : 0.33L



Deflection (displacement) at point 'A' will be zero

So, $\delta_A = \delta_1 + \delta_2 = 0$

$$-\frac{PL^3}{3EI} + \frac{P(L-x)L^2}{2EI} = 0$$

$$\frac{PL^3}{3EI} = \frac{P(L-x)L^2}{2EI}$$

$$\frac{PL^3}{3} = \frac{PL^3 - PL^2x}{2}$$

$$2PL^3 = 3PL^3 - 3PL^2x$$

$$3PL^2x = PL^3$$

$$x = L/3$$

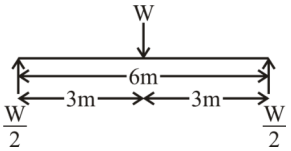
$$x = 0.33L$$

1356. A simply supported beam of length 6 m is carrying a point load W at the centre. When the slope at the ends of the beam is 0.002 radians, the deflection at the centre would be_____.

(a) 2 mm (b) 1 mm
(c) 8 mm (d) 4 mm

VIZAG MT, 14.12.2020

Ans. (d) :



Slope at ends of simply supported beam = 0.002 radians

$$\frac{W\ell^2}{16EI} = 0.002$$

Deflection at centre of beam δ_c

$$\delta_c = \frac{W\ell^3}{48EI}$$

$$\delta_c = \frac{W\ell^2}{16EI} \times \frac{\ell}{3}$$

$$\delta_c = 0.002 \times \frac{6}{3}$$

$$\delta_c = 0.004 \text{ m}$$

$$\boxed{\delta_c = 4\text{mm}}$$

1357. In deflection curve of beams, the point of contraflexure or inflection point is the point

(a) of maximum curvature
(b) of zero-curvature
(c) of zero slope
(d) none of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Point of contraflexure or inflection point occur in bending moment diagram. At this point bending moment change its sign from positive to negative and vice-versa. At this point (the point of contraflexure) bending moment becomes zero.

$$\frac{\sigma}{y} = \frac{M}{I} = \frac{E}{R}$$

R = Radius of curvature

At point of contraflexure, BM = 0

$$\frac{M}{I} = \frac{E}{R}$$

$$\frac{1}{R} = \frac{M}{EI} = 0$$

1358. A cantilever beam of length 100 mm is subjected an end load of 150 N. If Young's Modulus is 200 GPa and Moment of Inertia is 5000 mm^4, the maximum deflection of this beam is:

(a) 5 mm (b) 0.01 mm
(c) 0.5 mm (d) 0.05 mm

CIL MT 27.02.2020

Ans. (d) : Given, ℓ = 100 mm

Load W = 150 N

E = 200 GPa

I = 5000 mm^4

$$\text{Maximum deflection} = \frac{W\ell^3}{3EI}$$

$$y_{max} = \frac{150 \times (100)^3}{3 \times 200 \times 10^3 \times 5000}$$

$$y_{max} = 0.05 \text{ mm}$$

1359. Which of the following states, "The deflection between any two points is equal to the moments of its area of bending moment diagram between the two points about the last point divided by EI"?

(a) Macaulay's Method
(b) Mohr's theorem
(c) Castigliano's first theorem
(d) Double integration method

VIZAG MT, 14.12.2020

Ans. (b) : Mohr's Theorem–It states that deflection between any two points is equal to the moments of its area of bending moment diagram between the two points about the last point divided by EI.

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1360. Simply supported beam of span (l) carries UDL (w kgf/m) throughout. Deflection at center is (y). If length is doubled, deflection at center would be:

(a) 2y (b) 4y
(c) 8y (d) 16y

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) : For simply supported beam carrying UDL,

Deflection, $\delta = \frac{5}{384}\frac{w\ell^4}{EI}$

$\delta \propto \ell^4$

If length is doubled, deflection at centre,

$$\delta_f = \frac{5}{384}\frac{w(2\ell)^4}{EI}$$

$$\delta_f = (16)\frac{5}{384}\left(\frac{w\ell^4}{EI}\right)$$

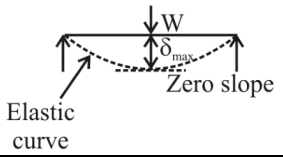
$$\boxed{\delta_{final} = 16\delta_{initial}}$$

1361. Assertion (A) : In a simply supported beam subjected to a concentrated load P at mid-span, the elastic curve slope becomes zero under the load.
Reason (R) : The deflection of the beam is maximum at mid-span.

(a) Both (A) and (R) are individually true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are individually true but (R) is NOT the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

OPSC AEE 2019 Paper-I

Ans : (a) : Elastic curve slope becomes zero at the point of maximum deflection in this case. Hence, of both assertion and reason are correct and reason is correct explanation of assertion.

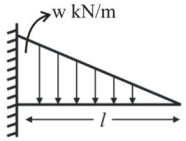


1362. The value of maximum deflection for a cantilever beam of length l and carrying uniformly varying load from zero at free end to w per unit length at fixed end is :

(a) $wl^3/8$ EI (b) $wl^3/16$ EI
(c) $wl^4/30$ EI (d) $wl^3/48$ EI

ISRO Scientist/Engineer (RAC), 10.03.2019
GATE 2019

Ans. (c) :



$$\text{Slope}(\theta) = \frac{w\ell^3}{24EI}$$

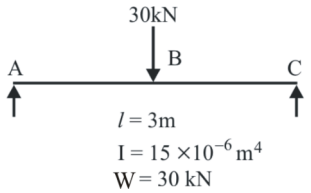
$$\text{Deflection}(\delta) = \frac{w\ell^4}{30EI}$$

1363. A girder of uniform section and constant depth is freely supported over a span of 3 metres. The point load at the midpoint is 30 kN and Moment of inertia = 15×10^{-6} m^4 and Young's Modulus = 200 GN/m^2. The deflection at centre will be :

(a) 6.6 cm (b) 8.6 cm
(c) 5.6 cm (d) 6.6 cm

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) :



Deflection of mid

$$\delta_B = \frac{Wl^3}{48EI}$$

$$= \frac{30\times10^3\times(3000)^3}{48\times200\times10^3\times15\times10^{-6}\times(10^3)^4}$$

$$\boxed{\delta_B = 5.625mm}$$

1364. A cantilever beam with rectangular cross-section is subjected to uniformly distributed load. The deflection at the tip is δ_1. If the width and depth of the beam are doubled then deflection at tip is δ_2. Then $\frac{\delta_2}{\delta_1}$ is

(a) 0.0625 (b) 16
(c) 0.5 (d) 2

APPSC AEE SCREENING 17.02.2019

Ans. (a) : Deflection $\propto \frac{1}{I}$

$$\therefore \quad \frac{\delta_2}{\delta_1} = \frac{I_1}{I_2} = \frac{\left(\frac{bd^3}{12}\right)}{\frac{(2b)(2d)^3}{12}} = \frac{1}{16} = 0.0625$$

1365. In the case of a beam simply supported at both ends, if the same load instead of being concentrated at centre is distributed uniformly throughout the length, then deflection at centre will get reduced by

(a) 1/2 times (b) 1/4 times
(c) 5/8 times (d) 3/8 times

ISRO Scientist/Engineer (RAC) 22.04.2018

Ans. (d) : Case-I
Center point load on SSB, we have deflection.

$$\delta_P = \frac{WL^3}{48EI}$$

Case-II
Deflection for UDL

$$\delta_{UDL} = \frac{5wL^4}{384EI}$$

If, W = w × L

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then

$$\delta_{UDL} = \frac{5}{384} \times \frac{WL^3}{EI} \times \frac{48}{48}$$

$$\frac{\delta_{UDL}}{\delta_P} = \frac{5}{384} \times 48 = \frac{5}{8}$$

So the reduction is given by

$$\delta_P - \delta_{UDL} = \delta_P - \frac{5}{8}\delta_P = \frac{3}{8}\delta_P$$

1366. The ratio of area under the bending moment diagram to the flexural rigidity between two points along a beam gives the change in

(a) deflection (b) slope

(c) shear force (d) bending moment

TSPSC AEE 28.08.2017 (Civil/Mechanical)

CSE Pre-1998

Ans. (b) : $\frac{d^2y}{dx^2} = \frac{M}{EI}$

Integrating it between two points A and B on elastic curve

$$\int_A^B \frac{d^2y}{dx^2} = \int_A^B \frac{M}{EI}$$

$$= \left[\frac{dy}{dx}\right]_A^B = \int_A^B \frac{Mdx}{EI}$$

$\int_A^B Mdx$ = (area of bending moment diagram between A and B)

1367. A simply supported beam AB of span L is subjected to a concentrated load W at the centre C of the span. According to Mohr's moment area method, which of the following gives the deflection under the load?

(a) Moment of the area of M/EI diagram between A and C taken about C

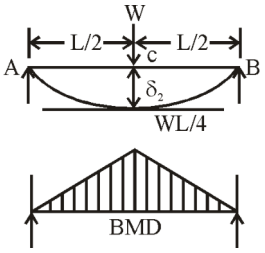
(b) Moment of the area of M/EI diagram between A and B taken about B

(c) Moment of the area of M/EI diagram between A and B taken about A

(d) Moment of the area of M/EI diagram between A and C taken about A

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (d) :



Moment of the area of M/EI diagram between A and C taken about A.

1368. The maximum deflection of a simply supported beam, 4 m long due to mid span concentrated load of 5 kN is

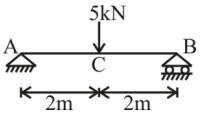
(Take E = 200 GN/m^2, I = 16 × 10^{-6} m^4)

(a) –6.8 mm (b) –3.8 mm

(c) –5.2 mm (d) –2.083 mm

Karnataka PSC AE, 10.09.2017

Ans. (d) :



$$\text{Deflection}\left(\delta_c\right) = \frac{PL^3}{48EI}$$

$$= \frac{5 \times 10^3 \times (4000)^3}{48 \times 200 \times 10^9 \times 10^{-6} \times 16 \times 10^{-6} \times 10^{12}}$$

$$= \frac{5 \times 64 \times 10^{12}}{48 \times 200 \times 16 \times 10^9}$$

= 2.083 mm

Note-(–ve) sign shows the downward deflection of beam.

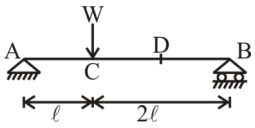
1369. A simply supported beam AB carries a point load W at a point C at a distance less than half from the left end A. The maximum deflection will be

(a) at C (b) between A and C

(c) between C and B (d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (c) :



- Deflection is maximum slope is zero in the region between point of application of load and mid span.
- Slope is maximum at a support which is nearer to the load.

1370. Governing differential equation for small deflections of elastic beams is given by

(a) $\frac{d^2y}{dx^2} = \frac{M}{EI}$ (b) $EI\frac{d^3y}{dx^3} = V(x)$

(c) $EI\frac{d^4y}{dx^4} = q(x)$ (d) All the above

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Differential equation of the elastic line is

$$EI\frac{d^2y}{dx^2} = M(x)$$

$$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$$

For shear force

$$EI\frac{d^3y}{dx^3} = V(x)$$

For shear force density (distributed load)

$$EI\frac{d^4y}{dx^4} = q(x)$$

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1371. We can find the deflection of beam carrying:
(a) Uniformly distributed load
(b) Central point load
(c) Gradually variable load
(d) All of these loads

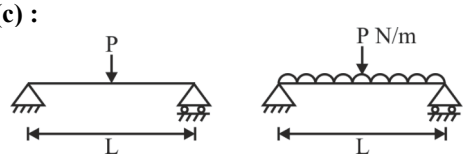
TRB Polytechnic Lecturer 2017

Ans. (d) : A subjected to UDL, central point load and also gradually variable load, deflection can find out.

1372. A uniform bar, simply supported at the ends, carries a concentrated load P at mid-span. If the same load be, alternatively, uniformly distributed over the full length of the bar the maximum deflection of the bar will decrease by.
(a) 25.5% (b) 31.5%
(c) 37.5% (d) 50.0%

ESE 2017

Ans. (c) :



$$\delta_1 = \frac{PL^3}{48EI} \qquad \delta_2 = \frac{5}{384}\frac{PL^3}{EI}$$

$$\% \text{ decrease in max deflection} = \frac{\delta_1 - \delta_2}{\delta_1} \times 100$$

$$= \frac{\frac{1}{48} - \frac{5}{384}}{\frac{1}{48}}$$

$$= 37.5\%$$

1373. A cantilever beam of length L is subjected to a point load W at free end. Then the slope at the free end is equal to
(a) $\frac{WL^3}{3EI}$ (b) $\frac{WL^2}{60EI}$
(c) $\frac{WL^3}{60EI}$ (d) $\frac{WL^2}{2EI}$

APPSC AEE Mains 2016 (Civil Mechanical)

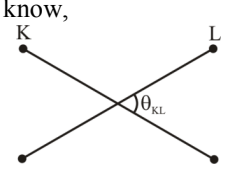
Ans. (d) : For cantilever beam,

Slope of free end (θ) = $\frac{WL^2}{2EI}$

1374. The mathematical expression for moment area theorem I for angle θ_{KL} between tangents drawn at K and L to elastic curve is
(a) $\theta_{KL} = \int_K^L \frac{MI}{E} dx$ (b) $\theta_{KL} = \int_K^L \frac{M}{EI} dx$
(c) $\theta_{KL} = \frac{M}{EI}$ (d) $\theta_{KL} = \frac{E}{MI}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (b) : As we know,



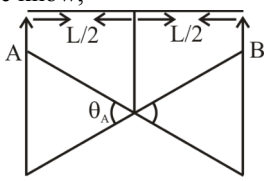
Theorem I = The change in slope between tangents draw to plastic curve at any point K & L is equal to the product of $\frac{1}{EI}$, Multiplied by area of moment.

$$\theta_{kL} = \frac{1}{EI}(\text{Area KL}) = \int_K^L \frac{M}{EI} dx$$

1375. A simply supported beam AB of L span and EI flexural rigidity is subjected to M_B moment at B. Then the rotation at A is given by
(a) $\theta_A = \frac{ML}{6EI}$ (b) $\theta_A = \frac{ML}{4EI}$
(c) $\theta_A = \frac{ML}{3EI}$ (d) $\theta_A = \frac{ML}{3EI}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : As we know,



$$\theta_A = \frac{1}{EI}(\text{Area AB})$$

$$\theta_A = \int_A^B \frac{M}{EI} dx$$

We get,

$$\theta_A = \frac{ML}{6EI}$$

1376. The width b and depth d of a cantilever beam carrying a point load at its free end are changed into 0.5b and 2d respectively. Then its maximum deflection for the same load condition reduces by
(a) one-fourth (b) half
(c) three-fourth (d) zero

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : As, we know,
Max. deflection formula for cantilever

$$\delta_{max_1} = \frac{WL^3}{3EI_1}$$

$$\therefore I_1 = \frac{bd^3}{12} \qquad \ldots(i)$$

$$I_2 = \frac{0.5b \times (2d)^3}{12} = \frac{8d^3b}{2 \times 12} = \frac{4bd^3}{12}$$

$$\therefore I_2 = 4I_1$$

$$\delta_{max_2} = \frac{WL^3}{3EI_2} = \frac{\delta_{max_1}}{4}$$

$$\text{Ratio} \Rightarrow \text{Reduction} = \delta_{max_1} - \frac{\delta_{max_1}}{4}$$

$$= \frac{3\delta_{max_1}}{4}$$

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1377. The elastic curve of a beam means

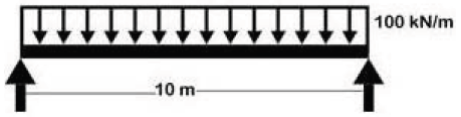
(a) BMD (b) SFD

(c) Deflection curve (d) Stress-stain curve

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : Deflection curve, is the elastic curve of a beam,

1378. A steel beam of breadth 100 mm and height 1000 mm is loaded as shown in the figure below. Assume modulus of elasticity as 200 GPa.



What is the maximum deflection in the beam?

(a) 751.25 mm (b) 781.25 mm

(c) 805.25 mm (d) 821.75 mm

(e) 851.75 mm

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (*) : Given, L = 10 m = 10000 mm

$E = 200\ GPa = 200 \times 10^9\ N/m^2$

$= 200 \times 10^3\ MPa\ or\ N/mm^2$

Breath (b) = 100 mm

Height (h) = 1000 mm

w = 100 kN/m

$W = 100 \times 10\ kN = 10^6\ N$

$$\boxed{I = \frac{bh^3}{12} = \frac{0.1\times(1)^3}{12}}$$

We know,

$$\delta = \frac{5WL^4}{384EI}$$

$$= \frac{5\times10^6\times(10000)^4\times12}{384\times200\times10^3\times(1000)^3\times100}$$

$$= 78125.0\ mm$$

Note-Official answer given by the commission (b).

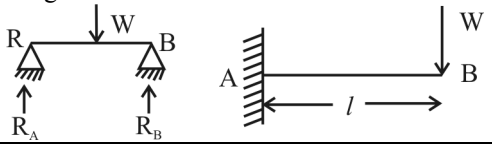
1379. Which of the following describes a single load or force that has a small contact area as to be negligible compared with the entire surface area of the supporting member?

(a) Contact load (b) Concentrated load

(c) Common load (d) None of these

KPCL AE 2016

Ans. (b) : Concentrated load describes a single load or force that has a small contact area to be negligible compared with the entire surface of area of the supporting member.



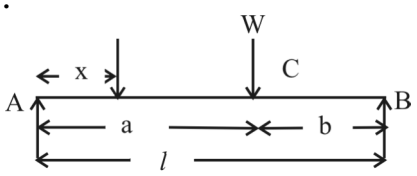
1380. A beam of length *l* simply supported at the ends carries a point load W at a distance 'a' from the left end. Also *l* -a= b. The deflection under load is given by :

(a) $\frac{Wa^2b^2}{3EIl}$ (b) $\frac{Wabl}{3EI}$

(c) $\frac{Wab}{6EIl}$ (d) $\frac{Wa^2bl}{3EI}$

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) :



Reaction at $A = \frac{Wb}{l}$

Reaction at $B = \frac{Wa}{l}$

The strain energy stored by the Beam AB (U)

$U_{AB} = U_{AC} + U_{BC}$

$$= \int_0^a \left(\frac{Wb}{l}x\right)^2 \frac{dx}{2EI} + \int_0^b \left(\frac{Wa}{l}x\right)^2 \frac{dx}{2EI}$$

$$= \frac{W^2b^2a^3}{6EIl^2} + \frac{W^2a^2b^3}{6EIl^2}$$

$$= \frac{W^2a^3b^2}{6EIl^2} + \frac{W^2a^2b^3}{6EIl^2}$$

$$= \frac{W^2a^2b^2}{6EIl^2}(a+b)$$

$\because\ a + b = l$

$$U_{AB} = \frac{W^2a^2b^2}{6EIl}$$

Deflection under load 'W' is given by –

$$\delta = \frac{\partial u}{\partial W} = \frac{(2W)a^2b^2}{6EIl}$$

$$\boxed{\delta = \frac{Wa^2b^2}{3EIl}}$$

1381. A simply supported beam of 10 m span is carrying a load of 4.8 kN at mid span. If Young's modulus of elasticity (E) is 2×10^8 kN/m^2 and moment of inertia (I) is 20 cm^4, then the maximum deflection will be

(a) 5.00 mm (b) 2.50 mm

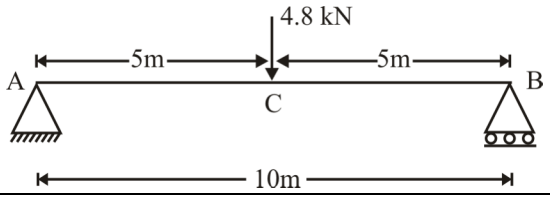
(c) 0.50 mm (d) 0.25 mm

APPSC AEE Screening Test 2016

Ans. (b) : given

$I = 20\ cm^4$

$E = 2\times10^8\ kN/m^2$



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Maximum deflection $(y_c) = \frac{W\ell^3}{48EI}$

$$= \frac{4.8\times10^3\times(10)^3}{48\times2\times10^8\times10^3\times20\times10^{-8}}$$

$$= \frac{4.8\times10^3\times10^3}{48\times2\times10^3\times20} = \frac{4.8\times10^3}{48\times2\times20}$$

$$= \frac{4800}{48\times2\times20} = \frac{100}{40} = 2.5\,\text{mm}$$

1382. A cantilever beam rectangular in cross section is subjected to a load W at its free end, causing deflection δ_1. If the load is increased to 2W, causing deflection δ_2, the value of $\frac{\delta_1}{\delta_2}$ would be

(a) 1 (b) 2
(c) 4 (d) 1/2

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : A cantilever beam rectangular in cross-section subjected, load (W), then deflection (δ_1).

$$\delta_1 = \frac{WL^3}{3EI} \quad \text{——(i)}$$

When, Load is increased to (2W), then deflection (δ_2) will be,

$$\delta_2 = \frac{2WL^3}{3EI} \quad \text{——(ii)}$$

from (i) 2 (ii)

$$\frac{\delta_1}{\delta_2} = \frac{\frac{WL^3}{3EI}}{\frac{2WL^3}{3EI}} = \frac{1}{2}$$

1383. If the depth of a rectangular beam is halved, the deflection for a beam carrying a mid point load shall be–

(a) halved (b) doubled
(c) four times (d) eight times

RPSC 2016

Ans : (d) Given, $d_1 = \frac{d}{2}$, $d_2 = d$

$$\delta = \frac{WL^3}{48EI} = \frac{WL^3}{48E\frac{bd^3}{12}}$$

$$\delta \propto \frac{1}{d^3}$$

If depth is halved $\left(d_2 = \frac{d}{2}\right)$

$$\frac{\delta_1}{\delta_2} = \frac{\left(\frac{d_1}{2}\right)^3}{d_1^3} = \frac{1}{8}$$

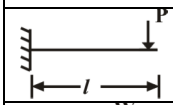
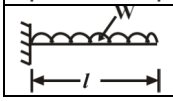
$$\boxed{\delta_2 = 8\delta_1}$$

1384. Maximum slope in case of a cantilever of length *l* carrying a load P at its end is

(a) $Pl^2/2EI$ (b) Pl^2/EI
(c) $Pl^2/4EI$ (d) $Pl^2/6EI$
(e) $Pl^2/8EI$

CGPSC AE 26.04.2015 Shift-I

Ans. (a) :

Type of load	Slope at end	Maximum deflection
	$\theta = \frac{Pl^2}{2EI}$	$\delta = \frac{Pl^3}{3EI}$
	$\theta = \frac{wl^3}{6EI}$	$\delta = \frac{wl^4}{8EI}$

1385. The ratio of central deflection due to a central load in the case of a beam freely supported at both ends to the beam fixed at both ends will be-

(a) 1/2 (b) 4
(c) 1/4 (d) 2

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (b) Central deflection for simply supported beam for central load W.

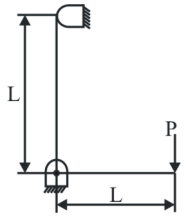
$$\delta_1 = \frac{WL^3}{48EI}$$

Central deflection for fixed beam (at both ends) for central load W.

$$\delta_2 = \frac{WL^3}{192EI}$$

Then, $\boxed{\delta_1/\delta_2 = 4}$

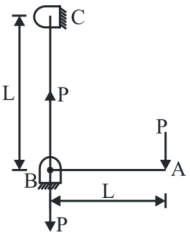
1386. A frame of two arms of equal length L is shown in the figure. The flexural rigidity of each arm of the frame is EI. The vertical deflection at the point of application of load P is



(a) $\frac{PL^3}{3EI}$ (b) $\frac{2PL^3}{3EI}$
(c) $\frac{4PL^3}{3EI}$ (d) None of the above

ISRO Scientist/Engineer 11.10.2015
GATE 2009

Ans. (c) :



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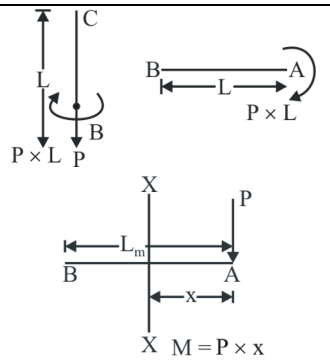


For section 'AB'

$$U_{AB} = \int \frac{M^2 dx}{2EI}$$

$$= \int_0^L \frac{M^2 dx}{2EI}$$

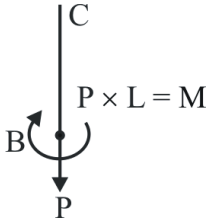
$$= \int_0^L \frac{(P \times x)^2 dx}{2EI} = \int_0^L \frac{P^2 \times x^2 dx}{2EI}$$

$$= \frac{P^2}{2EI}\int_0^L x^2 dx = \frac{P^2}{2EI}\left[\frac{x^3}{3}\right]_0^L$$

$$= \frac{P^2}{2EI}\left[\frac{L^3}{3}\right]$$

$$U_{AB} = \frac{P^2 L^3}{6EI}$$

For Section 'B'



$$U_{BC} = \int_0^L \frac{M^2 dx}{2EI}$$

$$= \int_0^L \frac{(P \times L)^2 dx}{2EI}$$

$$= \int_0^L \frac{P^2 \times L^2 dx}{2EI}$$

$$= \frac{P^2 L^2}{2EI}\int_0^L dx = \frac{P^2 L^2}{2EI}[x]_0^L$$

$$U_{BC} = \frac{P^2 L^3}{2EI}$$

So, vertical deflection,

$$= \frac{\partial U_{AB}}{\partial P} + \frac{\partial U_{BC}}{\partial P}$$

$$U_{AB} = \frac{P^2 L^3}{6EI}$$

$$\frac{\partial U_{AB}}{\partial P} = \frac{2PL^3}{6EI} = \frac{PL^3}{3EI}$$

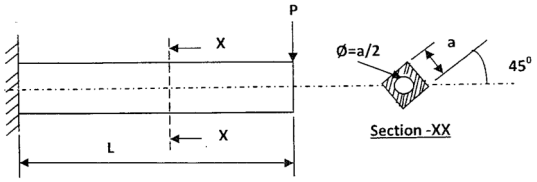
$$U_{BC} = \frac{P^2 L^3}{2EI}$$

$$\frac{\partial U_{BC}}{\partial P} = \frac{2PL^3}{2EI} = \frac{PL^3}{EI}$$

$$\text{Vertical deflection} = \frac{\partial U_{AB}}{\partial P} + \frac{\partial U_{BC}}{\partial P}$$

$$= \frac{PL^3}{3EI} + \frac{PL^3}{EI} = \frac{4}{3}\frac{PL^3}{EI}$$

1387. A concentrated load P is applied at the end of a cantilever as shown in Fig. The cross section of the beam is a square of side 'a' with a hole of dia 'a/2'. The deflection at the tip of the cantilever is given by



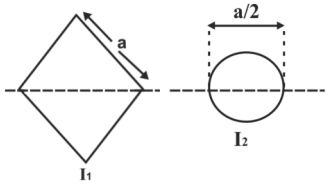
(a) $\frac{3P}{E}\frac{L^3}{a^4}$ (b) $\frac{1024P}{(256-3\pi)E}\frac{L^3}{a^4}$

(c) $\frac{1024P}{(256-\frac{\pi}{64})E}\frac{L^3}{a^4}$ (d) $\frac{256P}{(1024-3\pi)E}\frac{L^3}{a^4}$

ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Firstly find the moment of inertia of a Hollow cross-section.

Let moment of Inertia I_1 and I_2



$$I = I_1 - I_2$$

$$= \frac{a^4}{12} - \frac{\pi(a/2)^4}{64}$$

$$= \frac{a^4}{12} - \frac{\pi a^4}{16 \times 64}$$

$$= \frac{a^4}{4}\left[\frac{1}{3} - \frac{\pi}{4 \times 64}\right]$$

$$I = \frac{a^4}{4}\left[\frac{256 - 3\pi}{4 \times 3 \times 64}\right]$$

So, Deflection of Cantilever beam

$$\delta = \frac{PL^3}{3EI}$$

$$= \frac{PL^3}{3E\frac{a^4}{4}\left[\frac{256-3\pi}{4\times 3\times 64}\right]}$$

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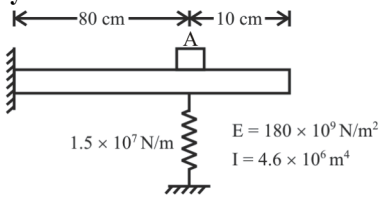
$$= \frac{4\times4\times64P\times L^3}{(256-3\pi)E\times a^4}$$

$$\therefore \text{ Deflection, } \delta = \frac{1024P}{(256-3\pi)E}\left(\frac{L^3}{a^4}\right)$$

1388. Determine the equivalent stiffness of the system shown below



(a) 6.35×10^7 N/m (b) 4×10^7 N/m
(c) 3×10^7 N/m (d) 5×10^5 N/m

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Given, $E = 180 \times 10^9$ N/m^2, $I = 4.6 \times 10^6$ m^4

$k_s = 1.5 \times 10^7$ N/m

$L_1 = 80$ cm $= 0.8$ m

$L_2 = 10$ cm $= 0.1$ m

Deflection at A, $\delta_A = \frac{PL^3}{3EI}$

Beam stiffness, $k_0 = \frac{P}{\delta_A} = \frac{3EI}{L^3}$

Equivalent stiffness, $k_{eq} = \frac{3EI}{L_1^3} + k_s$

$$= \frac{3\times180\times10^9\times4.6\times10^6}{(0.8)^3} + 1.5\times10^7$$

$$= 4851.5625 \times 10^{15} + 1.5 \times 10^7$$

$$\approx 4852 \times 10^{15} \text{ N/m}$$

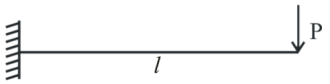
No option is matching.
But official answer is (a).

1389. If a cantilever beam of length *l*, the concentrated load *P* at the end is replaced by a bending moment *P* × *l*, the deflection will increase by :

(a) 1 time (b) 1.5 times
(c) 1.25 times (d) 2 times

J&K PSC Civil Services Pre, 2013

Ans. (b) : Case-I



Deflection at free end,

$$\delta_1 = \frac{Pl^3}{3EI} \quad \text{.....(i)}$$

Case-II

$$\delta_2 = \frac{Ml^2}{2EI} = \frac{(Pl)l^2}{2EI} \quad \text{.....(ii)}$$

From $\left(\frac{\text{ii}}{\text{i}}\right)$ –

$$\frac{\delta_2}{\delta_1} = \frac{Pl^3/2EI}{Pl^3/3EI} = \frac{3}{2} = 1.5$$

$$\boxed{\delta_2 = 1.5\delta_1}$$

1390. The slope and deflection at the centre of a simple beam carrying a central point load are

(a) zero and zero
(b) zero and maximum
(c) maximum and zero
(d) minimum and maximum

BPSC AE 2012 Paper - VI

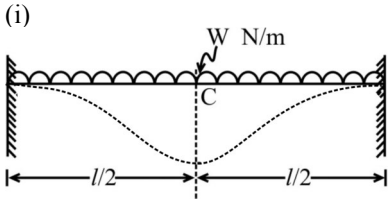
Ans : (b) : zero and maximum

1391. A beam of length, l, fixed at both ends carries a uniformly distributed load of w per unit length. If EI is the flexural rigidity, then the maximum deflection in the beam is

(a) $\frac{wl^4}{192EI}$ (b) $\frac{wl^4}{24EI}$
(c) $\frac{wl^4}{384EI}$ (d) $\frac{wl^4}{12EI}$

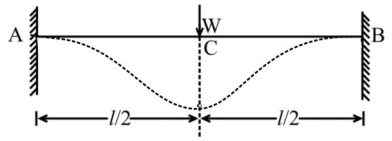
APPSC AEE 2012

Ans : (c) (i)



$$\text{Deflection } (\delta_c) = \frac{W\ell^4}{384EI}$$

(ii)



$$\text{Deflection } (\delta_c) = \frac{W\ell^3}{192EI}$$

1392. A simply supported beam span 3m, is subjected to a central point load of 5kN, then the slope at the mid span is equal to

(a) $\frac{25}{24EI}$ (b) $\frac{256}{EI}$
(c) $\frac{40}{48EI}$ (d) Zero

APPSC AEE 2012

Ans : (d) At mid span of the beam, deflection is maximum and slope is zero.

1393. A list of some theories of failure is given below:

(1) Maximum normal stress theory
(2) Maximum shear stress theory
(3) Maximum distortion theory

Which of these will be used for designing the machine components made of Cast Iron?

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(a) (2) only (b) (2) and (3)
(c) (1) an (3) (d) (1) only

RPSC AE GWD, 2011

Ans. (d) : Cast iron is a brittle material so for designing the machine components made of cast iron maximum normal stress theory is the best suited.

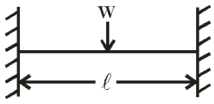
1394. The maximum deflection of a fixed beam of length 'ℓ' carrying a central point load w is

(a) $\frac{w\ell^3}{48EI}$ (b) $\frac{w\ell^3}{96EI}$

(c) $\frac{w\ell^3}{192EI}$ (d) $\frac{w\ell^3}{384EI}$

TNPSC ACF 2012

Ans. (c) :



$Y_{max} = \frac{w\ell^3}{192EI}$

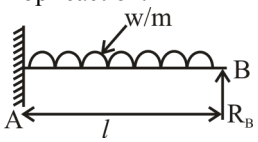
1395. The reaction at the prop in a propped cantilever beam subjected to u.d.*l*. is

(a) $\frac{Wl}{4}$ (b) $\frac{3Wl}{8}$

(c) $\frac{5Wl}{8}$ (d) $\frac{6Wl}{7}$

APPSC AEE 2012

Ans. (b) : R_B = Prop reaction.



Deflection of cantilever beam for u.d.*l*

$$\delta_B = \frac{Wl^4}{8EI} \text{.....(i)}$$

$$\Sigma\delta_B = 0$$

$$\frac{Wl^4}{8EI} - \frac{R_B.l^3}{3EI} = 0$$

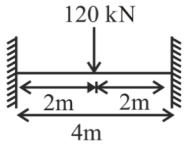
$$R_B = \frac{3}{8}Wl$$

1396. A beam length 4 m, fixed at both ends carries a point load of 120 kN at the centre. If EI for the beam is 2000 kN m^2, deflaction at the centre of beam is

(a) 1.0 mm (b) 2.0 mm
(c) 5.0 mm (d) 10.0 mm

APPSC AEE 2012

Ans. (*) :



Given,
Beam length = 4 m
P = 120 kN
EI = 2000 kN-m^2
Deflection of centre of beam

$$\delta_{max} = \frac{PL^3}{192EI} = \frac{120\times4^3}{192\times2000} = 0.02m$$

δ_{max} = 2 cm = 20 mm

1397. A simply supported beam with width 'b' and depth 'd' carries a central load W and undergoes deflection δ at the centre. If the width and depth are interchanged, the deflection at the centre of the beam would attain the value :

(a) $\frac{d}{b}\delta$ (b) $\left(\frac{d}{b}\right)^2\delta$

(c) $\left(\frac{d}{b}\right)^3\delta$ (d) $\left(\frac{d}{b}\right)^{3/2}\delta$

ISRO Scientist/Engineer 2011

Ans. (b) : In reactangle
width = b
depth = d
Deflection for S.S.B. when load acting at center of beam.

$$\delta = \frac{WL^3}{48EI}$$

$$I_1 = \frac{bd^3}{12}$$

Then Interchanged depth and width

$$I_2 = \frac{db^3}{12}$$

$$\frac{\delta_1}{\delta_2} = \frac{I_2}{I_1} = \frac{\frac{1}{12}db^3}{\frac{1}{12}bd^3}$$

$$\frac{\delta_1}{\delta_2} = \left(\frac{b}{d}\right)^2$$

$$\delta_2 = \left(\frac{d}{b}\right)^2\delta_1$$

$$\delta_1 = \delta$$

$$\delta_2 = \left(\frac{d}{b}\right)^2\delta$$

1398. If the diameter of a circular section is doubled its deflection is reduced by :

(a) 16 times (b) 8 times
(c) 4 times (d) 2 times

J&K PSC Civil Services Pre, 2010

Ans. (a) : Given, Deflection =δ
Diameter- d

$$\delta \propto \frac{1}{d^4}$$

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$$\frac{\delta_1}{\delta_2} = \left(\frac{d_2}{d_1}\right)^4 = \left(\frac{2d_1}{d_1}\right)^4$$

$$\frac{\delta_1}{\delta_2} = 16$$

$$\delta_2 = \frac{1}{16}\delta_1$$

1399. Maximum slope of a cantilever of length 6 m carrying load at a distance of 4 m from fixed end is _____.

(a) Fixed end (b) Under the load

(c) Free end (d) Can't predict

J&K PSC Civil Services Pre, 2010

Ans. (*) : Both (c) & (b) are correct.

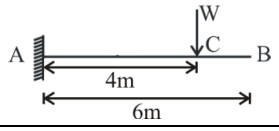
Slope at a point 'C' =

$$\theta_C = \frac{WL^2}{2EI}$$

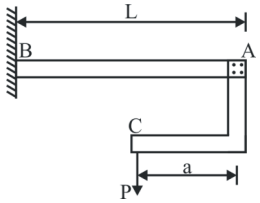
$$\theta_C = \frac{W(4)^2}{2EI} = \frac{16W}{2EI}$$

Maximum slope, $\theta_{max} = \theta_C = \theta_B = \frac{16W}{2EI}$

$\therefore$ So, hence maximum slope at free end and under the load is same.



1400. A cantilever beam *AB* of length *L* and uniform flexural rigidity *EI* has a bracket *AC* attached to its free end shown in Fig. Vertical load is applied to the free end *C* of the bracket. In order that the deflection of point *A* to be zero the ratio *a/L* should be

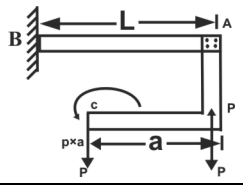


(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

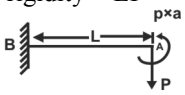
(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

ISRO Scientist/Engineer 2010

Ans. (d) :



Flexural rigidity = EI



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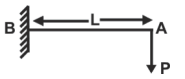
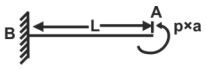


...(1)

+



...(2)

$Y_A = Y_1 + Y_2$

$$Y_1 = -\frac{PL^3}{3EI}$$

$$Y_2 = \frac{ML^2}{2EI}$$

$Y_A = 0$ [deflection at point A is zero)

$$-\frac{PL^3}{3EI} + \frac{ML^2}{2EI} = 0$$

$M = P \times a$

$$\frac{P \times a \times L^2}{2EI} = \frac{PL^3}{3EI}$$

$$\frac{a}{2} = \frac{L}{3}$$

$$\frac{a}{L} = \frac{2}{3}$$

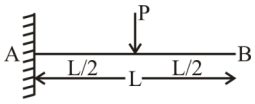
1401. A cantilever beam of length 'L' carries a concentrated load 'P' at its midpoint. What is the deflection of the free end of the beam?

(a) $\frac{PL^3}{24EI}$ (b) $\frac{PL^3}{48EI}$

(c) $\frac{PL^3}{16EI}$ (d) $\frac{5PL^3}{48EI}$

UKPSC AE 2007 Paper -I

Ans. (d) :



$$Y_{AB} = \frac{5PL^3}{48EI}$$

1402. A cantilever of length 4 m carries a uniformly distributed load w throughout its length. If the maximum bending moment in the cantilever is 80 KNm, and El is the flexural rigidity, the slope at the end of the cantilever is

(a) $\frac{640}{3EI}$ (b) $\frac{620}{3EI}$

(c) $\frac{160}{3EI}$ (d) $\frac{80}{3EI}$

WBPSC AE, 2007

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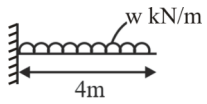
Ans. (*)



Slope $\theta = \frac{w\ell^3}{6EI}$

$\because$ Maximum bending moment $M = \frac{w\ell^2}{2}$ at fixed end in cantilever beam with UDL

$$80 = \frac{w \times (4)^2}{2}$$

$$w = \frac{160}{10} = 10 kN/m$$

$$\text{Slope } (\theta) = \frac{10 \times (4)^3}{6EI}$$

$$= \frac{64 \times 10}{6EI} = \frac{320}{3EI}$$

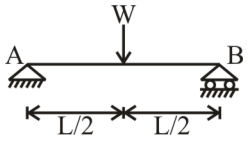
1403. A beam is simply supported at its ends over a span *l* . If the load applied at the middle of the beam is W, the minimum slope in the beam is

(a) $\frac{Wl^2}{16\ EI}$ (b) $\frac{Wl^2}{3\ EI}$

(c) $\frac{Wl^2}{2\ EI}$ (d) none of these

WBPSC AE, 2007

Ans. (a)



$$\boxed{\text{Slope}(\theta) = \frac{WL^2}{16EI}}$$

$$\boxed{\text{Deflection}(\delta) = \frac{WL^3}{48EI}}$$

1404. A simply supported beam of rectangular section 4 cm by 6 cm carries a mid-span concentrated load such that the 6 cm side lies parallel to line of action of loading and the deflection under the load is δ. If the beam is now supported with the 4 cm side parallel to line of action of loading, then the deflection under the load will be-

(a) 0.44 δ (b) 2.25 δ

(c) 1.5 δ (d) 1.75 δ

WBPSC AE 2003

Ans. (a) : Deflection at center $(\delta) = \frac{wl^3}{48EI} = \frac{wl^3}{48E\left(\frac{bd^3}{12}\right)}$

$\because$ Question says that the 6 cm side lies parallel to the line of action of loading.

$\therefore$ deflection $\delta' = \frac{wL^3}{48EI'}$ ...(i)

$$\text{Deflection, } \delta' = \frac{wL^3}{48E\frac{db^3}{12}} = \frac{wL^3 \times 12 \times d^2}{48E(db^3) \times d^2} \quad \left(I' = \frac{db^3}{12}\right)$$

$$= \frac{wL^3 \times 12 \times d^2}{48E \times (db^3) \times b^2}$$

$$= \frac{wL^3 \times d^2}{48E \times (bd^3) \times b^2}$$

$$= \left(\frac{wL^3}{48EI}\right) \times \frac{d^2}{b^2}$$

$$= \delta . \frac{d^2}{b^2} \quad \text{...(ii)}$$

$d = 4$, $b = 6$ substitute in equation (ii)

$$\delta' = \delta \times \frac{16}{36} = \delta \times \frac{4}{9} = 0.44\delta$$

$$\boxed{\delta' = 0.44\delta}$$

10. Theory of Failure

1405. Which of the following is applied to brittle materials?

(a) maximum principal stress theory
(b) maximum principal strain theory
(c) maximum strain energy theory
(d) maximum shear stress theory

JPSC AE 10.04.2021, Paper-II
APPSC Poly Lect. 13.03.2020
OPSC AEE 2019 Paper-I
Gujarat PSC AE 2019
(CGPSC Polytechnic Lecturer 2017)
RPSC AE 2016
UPRVUNL AE 07.10.2016
RPSC VPITI 14.02.2016
TSPSC AEE 2015
ISRO Scientist/Engineer 11.10.2015
TANGEDCO AE 2015
TRB Poly. Lect., 2012
RPSC ACF-2011
WBPSC AE, 2007

Ans. (a) : Maximum Principal Stress Theory (MPST) (Rankine Theory)–

Assumption–

- Neglect the shear stress
- Neglect the other principal stress (σ_2)
- It is suitable for brittle materials

(because brittle material are weak in tension) under all loading condition.

- It's not suitable for ductile material

($\because$ Shear stress is neglected)

- MPST suitable for ductile material under uni-axial state of stress and biaxial state of stress when principal stress are in same nature.

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1406. In case of crystalline and brittle materials criterion of failure, that takes place under maximum principal stress, is put forwarded by

(a) Theory of Rankine
(b) Theory due to Guest
(c) Theory of St. Venant
(d) Theory due to Beltrami and Haigh

GPSC DEE Class-2 GWSSB 04.07.2021
RPSC IOF, 2020
Kerala PSC Poly. Lect., 2017
CGPSC AE 16.10.2016
APPSC AE Subordinate Service Civil/Mech. 2016
UPRVUNL AE 21.08.2016
KPSC ADF 2015
Mizoram PSC AE/SDO 2014, Paper-II
APPSC Poly. Lect. 2013
TRB Asstt. Prof., 2012
RPSC AE GWD, 2011

Ans. (a) :

(i) Max. Principle Stress Theory/Rankine's Theory/ Lames Theory/Maximum Stress Theory

- Applicable for brittle materials (Best for brittle materials)
- Not suitable for ductile materials
- Not suitable for pure shear case.

(ii) Maximum Principal Strain Theory/Saint Venant's Theory

- Satisfactory for brittle materials
- Not suitable for pure shear case and hydrostatic loading.

(iii) Maximum Shear Stress Theory/Tresca's Theory / Guest Theory

- Applicable for ductile materials
- Most conservative theory
- Not suitable for hydrostatic loading.

(iv) Maximum Strain Energy Theory / Haigh's Theory

- Applicable for ductile material
- Not suitable for brittle material and pure shear case.

(v) Maximum Shear Strain Energy Theory / Distortion Energy Theory / Hencky Theory / Von-Mises Theory

- Most appropriate theory for ductile materials
- Perfect for pure shear case.

1407. Shear stress theory is applicable to

(a) ductile materials (b) brittle materials
(c) elastic materials (d) plastic materials

RPSC AE 2016
GWSSB DEE 07.07.2016
APPSC AEE Mains 2016 (Civil Mechanical)
TNPSC AE 2014, RPSC ACF-2011
J&K PSC Civil Services Pre, 2010
WBPSC AE 2008

Ans : (a) Shear stress theory is applicable to ductile material.
(i) Maximum principal stress theory – Brittle material
(ii) Maximum principal strain theory – Brittle material
(iii) Maximum shear stress theory – Ductile material
(iv) Maximum total strain energy theory – Ductile material.

1408. Which theory of failure will you use for aluminum components under steady loading?

(a) principal stress theory
(b) principal strain theory
(c) strain energy theory
(d) maximum shear stress theory

Karnataka PSC AE, 10.09.2017
HPPSC Asstt. Prof. 18.11.2016
GPSC ARTO 01.05.2016
VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012
GATE 1998

Ans. (d) : Aluminium is ductile material and loading is static so we will use, maximum shear stress theory only.
Maximum shear stress theory

$$\tau_{max} = \max\left\{\left|\frac{\sigma_1-\sigma_2}{2}\right|, \left|\frac{\sigma_2-\sigma_3}{2}\right|, \left|\frac{\sigma_1-\sigma_3}{2}\right|\right\}$$

$\tau_{max} > S_{sy} \rightarrow$ failed
$\tau_{max} \le S_{sy} \rightarrow$ Safe

1409. A transmission shaft subjected to bending and torsional moments should be designed on the basis of

(a) Maximum principal stress theory
(b) Maximum shear stress theory
(c) Permissible bearing pressure
(d) None of the mentioned above

GPSC EE Pre, 28.01.2017
TNPSC AE 2017
MPPSC AE 08.11.2015
TNPSC ACF 2012

Ans. (b) : Torsional moment is due to shear force, so that the design of shaft based on maximum shear stress theory, where as bending moment is due to normal force, so that the design of shaft based on maximum principal normal stress.

1410. As per maximum shear stress theory of failure. The relation between yield strength in shear (τ_y) and yield strength in tension (σ_t) is:

(a) $\tau_t = 1.2\ \sigma_t$ (b) $\tau_t = 0.7\ \sigma_t$
(c) $\tau_t = 0.3\ \sigma_t$ (d) $\tau_t = 0.5\ \sigma_t$

TNPSC AE 2019, Vizag Steel (MT) 2017
CIL MT 26.03.2017

Ans. (d) : Maximum shear stress theory (or) Guest & Tresca's Theory-According to this theory, failure of specimen subjected to any combination of load when the maximum shearing stress at any point reaches the failure value equal to that developed at the yielding in an axial tensile or compressive test of the same material.

$$\tau_y \le \frac{\sigma_t}{2}$$

Where σ_t is the yield strength in tension and τ_y is the yield strength in shear.

1411. The maximum distortion energy theory of failure is suitable to predict the failure of one of the materials is :

(a) Brittle material (b) Ductile material
(c) Plastics (d) Composite materials

UJVNL AE 2016
OPSC AEE 2015 Paper-I, ESE 2004

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Ans : (b)
Maximum Principal Stress Theory : Brittle material
Maximum Shear Stress Theory : Ductile material
Maximum Strain Energy Theory : Ductile material
Maximum Shear Strain Energy Theory : Ductile material
Maximum Principal Strain Theory : The theory can be applied for ductile and Brittle material both but result are not accurate for either case.

1412. A shaft subjected to maximum bending stress of 80N/mm^2 and maximum shearing stress equal to 30N/mm^2 at a particular section. If the yield point in tension of the material is 280 N/mm^2 and maximum shear stress theory of failure is used, then the factor of safety obtained will be :

(a) 2.5 (b) 2.8
(c) 3.0 (d) 3.5

MPSC Poly. Lect. 2014
UPRVUNL AE 2014
APPSC IOF, 2009

Ans. (b) : $\sigma_{max} = 80\ N/mm^2$
$\tau_{xy} = 30\ N/mm^2$
$\sigma_{yt} = 280\ N/mm^2$
FOS = ?

$$\text{Maximum shear stress} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tau_{max} = \sqrt{\left(\frac{80-0}{2}\right)^2 + 30^2}$$

$$\tau_{max} = 50\,MPa$$

From maximum shear stress theory

$$\tau_{max} = \frac{\sigma_{yt}}{2FOS}$$

$$50 = \frac{280}{2 \times FOS}$$

$$FOS = \frac{280}{50 \times 2}$$

$$\boxed{FOS = 2.8}$$

1413. The failure at a point in a member, when the maximum principal stress in a bi-axial stress system reaches the elastic limit of the material, in a sample tension test. This is according to

(a) Rankine's theory (b) Guest's theory
(c) Lame's theory (d) Goodman's theory

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
WBPSC AE 2003

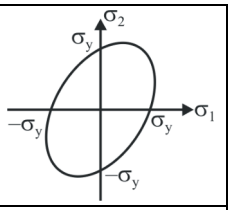
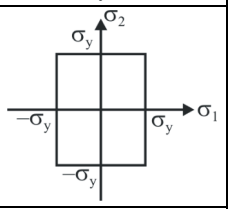
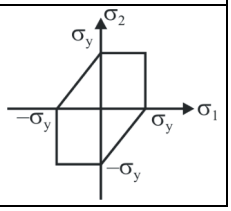
Ans. (a) : Rankine's theory– Rankine's theory assumes that failure will occur when maximum principal stress at any point reaches a value equal to the tensile stress in a simple tension specimen at failure. Rankine's theory applied for Brittle material, and not applicable for ductile material.

Guest's theory- According to this theory, the elastic failure occurs when the greatest shear stress reaches a value equal to the shear stress at elastic limit in a simple tension test. This theory is considered for Ductile material.

$$\boxed{\left(\frac{\sigma_1 - \sigma_3}{2}\right) = \left(\frac{\sigma_e}{2}\right)}$$

maximum shear stress is also known as Guest's 'or' Tresca's theory.

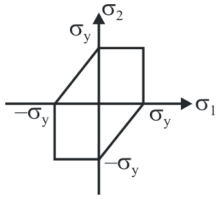
1414. Match the following criterion of material failure under biaxial stresses σ_1 and σ_2 and yield stress σ_y, with their corresponding graphic representations.

A.	Maximum shear stress criterion	L.	
B.	Maximum distortion energy criterion	M.	
C.	Maximum normal stress criterion	N.	

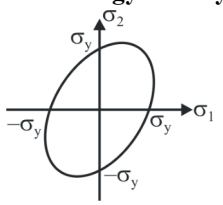
(a) A – L, B – N, C – M
(b) A – N, B – L, C – M
(c) A – M, B – N, C – L
(d) A – N, B – M, C – L

ISRO Scientist/Engineer 07.05.2017
GATE 2011, ESE 1997

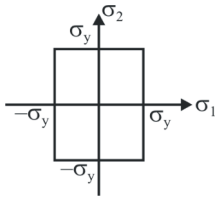
Ans. (b) : Maximum Shear Stress Theory–



Maximum Distortion Energy Theory–



Maximum Normal Stress Theory–



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1415. Guest's theory of failure is applicable for following type of materials

(a) brittle (b) ductile
(c) elastic (d) plastic
(e) tough

GPSC DEE, Class-2 (GWSSB) 04.07.2021
CGPSC AE 26.04.2015 Shift-I

Ans. (b) : Maximum shear stress theory or Guest or Tresca's theory is well justified for ductile material.

1416. Which of the following theories of failure is not suitable for ductile material

(a) Maximum shear stress theory
(b) Maximum principal strain theory
(c) Maximum total strain energy theory
(d) Maximum principal stress theory

RPSC LECTURER 16.01.2016
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

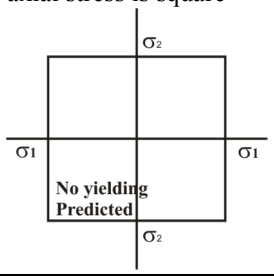
Ans. (d) : Maximum principal stress theory (Rankine's theory) is suitable for brittle material not for ductile material whereas Maximum shear stress theory (Guest's and Tresca theory), maximum principal strain theory (Saint-Venant theory) and maximum total strain energy theory (Haigh's theory) are used for ductile material.

1417. Region of safety for maximum principal stress theory under bi-axial stress is shown by:

(a) Ellipse (b) Square
(c) Pentagon (d) Hexagon

APPSC AEE SCREENING 17.02.2019
UPRVUNL AE 07.10.2016

Ans. (b) : Maximum principal stress theory— Maximum principal stress theory or normal stress theory says that, yielding occurs at a point in a body, when principle stress (maximum normal stress) in a biaxial system reaches limiting yield value of that material under simple tension test. This theory used for brittle material. This theory also known as Rankine theory. Region of safety for maximum principal stress theory under bi-axial stress is square



1418. Failure of the component occurs when the maximum shear stress in the complex system reaches the value of maximum shear stress in simple tension at the elastic limit. This is known as:

(a) Rankine theory
(b) Guest and Tresca theory
(c) Haigh theory
(d) St. Venant theory
(e) Carnot theory

CGPSC AE 25.02.2018
ISRO Scientist/Engineer 2008

Ans. (b) : Maximum shear stress theory [Guest and Tresca theory]– Failure of the component occurs when the maximum shear stress in the complex system reaches the value of maximum shear stress in simple tension at the elastic limit. This theory is used for ductile material.

1419. According to the distortion-energy theory, the yield strength in shear is

(a) 0.277 times the yield stress
(b) 0.377 times the maximum shear stress
(c) 0.477 times the yield strength in tension
(d) 0.577 times the yield strength in tension

ESE 2020
BPSC AE Mains 2017 Paper - VI

Ans. (d) : According to distortion energy theory,

$$S_{sy} = \frac{S_{yt}}{\sqrt{3}} = 0.577 S_{yt}$$

1420. Von-Mises and Tresca criteria give different yield stress for :

(a) Uni-axial stress
(b) Balanced bi-axial stress
(c) Pure shear stress
(d) All

Vadodara Muncipal Corp. DEE, 2018
GPSC EE Pre, 28.01.2017

Ans. (c) : Von-Mises and Tresca criteria gives different yield stress for pure shear stress.

1421. For designing ductile material, which of the following theories is/are used?

(a) Maximum shear stress theory
(b) Shear strain energy theory
(c) Both (A) and (B)
(d) None of the above

Assam PSC AE (PHED), 18.10.2020
APPSC AEE SCREENING 17.02.2019

Ans. (c) : For designing ductile material- Most important appropriate failure theory is maximum shear stress theory or shear strain energy theory.
For brittle material- Maximum normal stress or principal stress theory is more suitable for brittle material.

1422. Maximum distortion energy theory of failure is also called :

(a) Rankine's theory
(b) Guest's or Tresca's theory
(c) Von Mises theory
(d) St. Venant's theory

CGPSC AE 15.01.2021
UPRVUNL AE 05.07.2021

Ans. (c) : (i) Maximum Principal Stress Theory
⇒ Rankine's Theory
(ii) Maximum Shear Stress Theory
⇒ Guest's or Tresca's Theory
(iii) Maximum distortion energy Theory
⇒ Von Mises Theory
(iv) Maximum Principal Strain Theory
⇒ St. Venant's Theory

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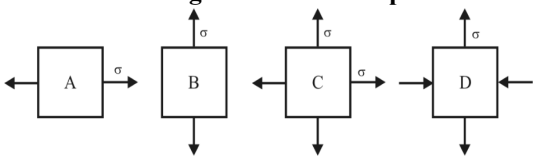
1423. Shown below are four elements a, b, c and d with different states of plane stress. Using maximum shear stress theory yield, which of the following will yield first? Assume the material is homogenous and isotropic.



(a) A (b) B
(c) C (d) D

RPSC ACF & FRO, 26.02.2021

Ans. (d) : According to max shear stress theory,

$$\tau_{max} = \max\left\{\left|\frac{\sigma_1}{2}\right|, \left|\frac{\sigma_2}{2}\right|, \left|\frac{\sigma_1-\sigma_2}{2}\right|\right\}$$

$$(\tau_{max})_A = \frac{\sigma_1}{2} = \frac{\sigma}{2}$$

$$(\tau_{max})_B = \frac{\sigma_2}{2} = \frac{\sigma}{2}$$

$$(\tau_{max})_C = \frac{\sigma_1}{2} (or) \frac{\sigma_2}{2} = \frac{\sigma}{2}$$

$$(\tau_{max})_D = \frac{\sigma_1-\sigma_2}{2} = \frac{\sigma-(-\sigma)}{2}$$

$= \sigma$ (max)

(So, element d will yield first).

1424. A tension member of diameter d is designed with F.O.S. of 3. If the load and diameter are doubled F.O.S. will be

(a) Reduced to half (b) Unchanged
(c) Tripled (d) Doubled

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : $FOS = \frac{\sigma_{allowable}}{\sigma_{max}}$

$$\frac{\sigma_{allowable}}{(P/A)} = 3$$

$$\frac{\sigma_{allowable}}{(2P/4A)} = N$$

Hence, $\frac{1}{2} = \frac{3}{N}$

$\boxed{N = 6}$

1425. A rod having cross-sectional area 100×10^{-6} m^2 subjected to a tensile load. Based on the Teresa failure criterion, if the uniaxial yield stress of the material is 200 MPa, the failure load is :

(a) 10 kN (b) 20 kN
(c) 100 kN (d) 200 kN

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : We know, Tresca's failure,

$$\tau_{max} = \frac{P}{2A} = \frac{\sigma_{yt}}{2} \text{ or } P = \sigma_{yt} \times A$$

$P = 200 \times 10^6 \times 100\times10^{-6} m^2$

$P = 20 \times 10^3 N = 20$ kN

1426. A cold rolled steel shaft is designed on the basis of maximum shear stress theory. The Principal stresses induced at its critical section are 500 MPa and -50 MPa respectively. If the shear yield stress for the shaft material is 400 MPa, the factor of safety of the design is

(a) 3 (b) 4
(c) 6 (d) 2

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : As we know,

$$\frac{\sigma_t - \sigma_c}{2} = \frac{\tau_{sy}}{FOS}$$

$$\frac{500-(-50)}{2} = \frac{400}{FOS}$$

$$\frac{550}{2} = \frac{400}{FOS}$$

$FOS = 1.45 \simeq 2$

1427. If a machine member is loaded in such a way that the three principal stresses are 600 kPa tension, 800 kPa tension and zero, the maximum shear stress at the point is :

(a) 1150 N/m^2 (b) 1000 N/m^2
(c) 700 kN/m^2 (d) 400 kN/m^2

CGPSC AE 15.01.2021

Ans. (d) : Given, rincipal stress,

$\sigma_1 = 600$ kPa, $\sigma_2 = 800$ kPa, $\sigma_3 = 0$

$$\text{Max. shear stress} = \left\{\left(\frac{\sigma_1-\sigma_2}{2}\right), \left(\frac{\sigma_2-\sigma_3}{2}\right), \left(\frac{\sigma_1-\sigma_3}{2}\right)\right\}$$

$$= \left\{\left|\frac{600-800}{2}\right|, \left|\frac{800-0}{2}\right|, \left|\frac{600-0}{2}\right|\right\}$$

$$= \{100, 400, 300\}$$

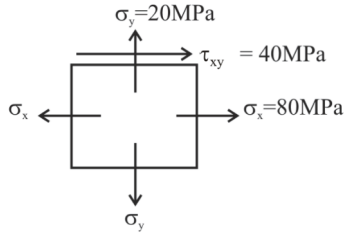
$\boxed{\tau_{max} = 400 kN/m^2}$

1428. A machine element is subjected to bi-axial state of stress : σ_x = 80 MPa; σ_y = 20 MPa; τ_{xy} = 40 MPa. If the shear strength of the material is 100 MPa, the factor of safety as per Tresca's Maximum Shear Strength theory is_____.

(a) 1.0 (b) 2.5
(c) 2.0 (d) 3.3

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (c) : 2.0



σ_x = 80 MPa
σ_y = 20 MPa
τ_{xy} = 40 MPa
τ_{xy} = 100 MPa

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Maximum shear stress $= \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \sqrt{\left(\frac{80-20}{2}\right)^2 + 40^2}$

$= 50$

Tresca's maximum shear stress theory

$$\tau_{max} = \frac{\tau_{sy}}{FOS}$$

$$50 = \frac{100}{FOS}$$

$FOS = 2$

1429. For the condition of principal stresses σ_1, σ_2 positive and $\sigma_3 = 0$, and also $\sigma_1 > \sigma_2 > \sigma_3$, the equation of boundary line of failure envelope as per maximum shear stress theory will be :

(a) $\sigma_2 - \sigma_2 = S_{yt}$ (b) $\sigma_1 = S_{sy}$

(c) $\sigma_1 = S_{yt}$ (d) $\sigma_2 = S_{yt}$

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Given, $\sigma_1 > \sigma_2 > \sigma_3 = 0$

maximum shear stress $\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$

$$\frac{\sigma_1 - \sigma_3}{2} = \tau_{max}$$

$$\frac{\sigma_1}{2} = \frac{S_{yt}}{2}$$

$$\boxed{\sigma_1 = S_{yt}}$$

1430. Failure of material is called fatigue when it fails

(a) at the elastic limit

(b) at the yield point

(c) below the elastic limit

(d) below the yield point

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Failure of material is called fatigue when it fails below the yield point.

1431. In theory of failure, maximum strain energy theory is also called _____.

(a) Tressca theory (b) Rankine theory

(c) Haigh's theory (d) Von Mises theory

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : In theory of failure, maximum strain energy theory is ab called Haigh's theory.

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + 2\mu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \le \left(\frac{S_{yt}}{N}\right)^2$$

1432. The theory of failure used in designing the ductile materials in a most accurate way is by

(1) maximum principal stress theory

(2) distortion energy theory

(3) maximum strain theory

Select the correct answer using the code given below.

(a) 1, 2 and 3 (b) 1 only

(c) 2 only (d) 3 only

ESE 2019, 2020

Ans. (c) : Since ductile materials, under static conditions, mostly fail due to shear or distortion, distortion energy theory or Von-mises theory produces most accurate results.

1433. Tresca's theory of failure is also known as,

(a) Maximum principle stress theory

(b) Maximum principle strain theory

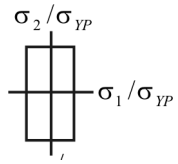
(c) Maximum shear stress theory

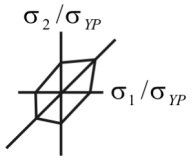
(d) Maximum distortion energy theory

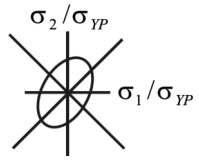
Oil India Senior Officer 23.12.2020

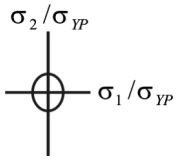
Ans. (c) : Tresca's theory of failure is also known as maximum shear stress theory.

1434. Which one of the following figures represents the maximum principal stress theory?

(a) 

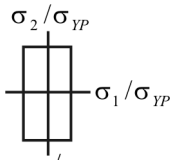
(b) 

(c) 

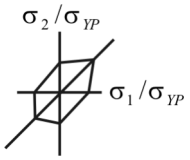
(d) 

OPSC AEE 2019 Paper-I

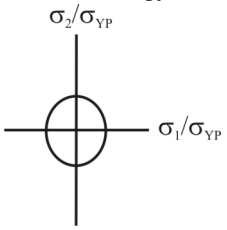
Ans : (a)



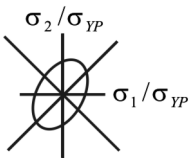
(i) Maximum shear stress theory



(ii) Maximum shear strain energy theory



(iii) Maximum strain energy theory



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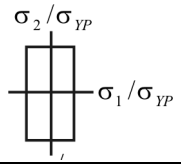
(iv) Maximum principal theory

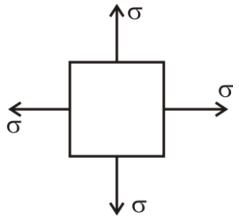


1435. Steel machine element at the critical section is in biaxial stress state with two principal stress being 300 N/mm^2 and 300 N/mm^2 (equal magnitude). Find the von Mises stress (in N/mm^2) in the member

(a) 212.1 (b) 600
(c) 424.2 (d) 300

ISRO Scientist/Engineer 12.01.2020

Ans. (d) :



$\sigma_1 = \sigma_2 = 300$

By using maximum distortion energy theory, Von-Mises Theory-

For plane stress condition,

$$\sigma_{per} = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2}$$

$\sigma_{per} = \sigma = 300$ MPa

1436. The uniaxial yield stress of a material is 300 MPa. According to von Mises criterion, the shear yield stress (in MPa) of the material is ---

(a) 173.28 (b) 150.55
(c) 124 (d) 99.87
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (a) : Given,

300 MPa = S_{yt}

According to Von - Mises criterion–

$$\frac{S_{ys}}{S_{yt}} = \frac{1}{\sqrt{3}}$$

$$\frac{S_{ys}}{300} = \frac{1}{\sqrt{3}}$$

$$S_{ys} = \frac{300}{\sqrt{3}}$$

$$\boxed{S_{ys} = 172.20 \text{ MPa}}$$

1437. Failure occurs when the maximum shear stress in the part exceeds shear stress in the tensile specimen at yield is a statement of

(a) Distortion energy theory
(b) Maximum shear stress theory
(c) Maximum strain energy theory
(d) Maximum normal stress theory

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Maximum shear stress theory

1438. The uniaxial tensile stress that would create the same distortion energy as is created by the actual combination of applied stresses is called as

(a) Effective distortion stress
(b) Impact stress
(c) Joint effective stress
(d) Von mises effective stress

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Distortion energy is associated with Von-mises.

1439. If S is the applied stress, c is the width of the crack and r the radius of curvature at the tip of the crack, Griffith's crack theory gives the concentrated stress S_c as

(a) $2S\,(c/r)^{1/2}$ (b) $2S\,(c/r)^{1/3}$
(c) $S\,(c/r)^{1/2}$ (d) $2S\,(2c/r)^{1/2}$

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : Griffith crack theory- The Griffith theory states that a crack will propagates when the reduction in potential energy that occurs due to crack growth is greater than or equal to the increase in surface energy due to the creation of new free surface. This theory is applicable to elastic materials that fracture in brittle fashion.

Griffith crack theory gives,

$$S_c = 2S\left(\frac{c}{r}\right)^{1/2}$$

1440. At a material point the principal stresses are σ_1 = 100 MPa and σ_2 = 20 MPa. If the elastic limit is 200 MPa, what is the factor of safety based on maximum shear stress theory?

(a) 1.5 (b) 2
(c) 2.5 (d) 3

APPSC AEE SCREENING 17.02.2019

Ans. (b) : $\sigma_1 = 100$ MPa

$\sigma_2 = 20$ MPa

$\sigma_3 = 0$ (Minimum principal stress)

$\sigma_y = 200$ MPa

According to Maximum Shear Stress theory

$$\tau_{max} = \frac{\sigma_y}{2FOS}$$

$$\Rightarrow \quad \frac{\sigma_1 - \sigma_3}{2} = \frac{\sigma_y}{2FOS}$$

$$\Rightarrow \quad \sigma_1 - \sigma_3 = \frac{\sigma_y}{FOS}$$

$$\Rightarrow \quad 100 - 0 = \frac{200}{FOS}$$

$$FOS = \frac{200}{100} = 2$$

1441. A point in a structural member with allowable yield strength of 300 MPa is subjected to Principal stresses σ_1 = 200 MPa, σ_2 = 50 MPa and σ_3 = –100MPa. What is the calculated value of yield stress (σ_0) and does yielding occurs?

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(a) $\sigma_0 = 150$ MPa and yielding occurs
(b) $\sigma_0 = 300$ MPa and yielding occurs
(c) $\sigma_0 = 300$ MPa and yielding does not occurs
(d) $\sigma_0 = 150$ MPa and yielding does not occurs

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : Using Tresca's criteria

$$\frac{\sigma_1 - \sigma_3}{2} \le \frac{S_{yt}}{2}$$

$$\frac{200+100}{2} \le \frac{300}{2}$$

$$150 = 150$$

Hence yielding occurs

1442. Let σ_1, σ_2 and σ_3 are the principal stresses at a material point. If the yield stress of the material is σ_y, then according to Von Mises theory yielding will not occur if

(a) $(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 < 2(\sigma_y)^2$
(b) $\max[(\sigma_1 - \sigma_2), (\sigma_2 - \sigma_3), (\sigma_3 - \sigma_1)] < \sigma_y$
(c) $(\sigma_1)^2 + (\sigma_2)^2 + (\sigma_3)^2 < (\sigma_y)^2$
(d) $(\sigma_1)^2 + (\sigma_2)^2 + (\sigma_3)^2 - 2\nu(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) < (\sigma_y)^2$

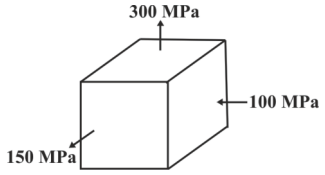
APPSC AEE SCREENING 17.02.2019

Ans. (a) : Von Mises Yielding Failure theory

$$\sigma_y \ge \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \le 2\sigma_y^2$$

1443. If an element is loaded as shown in figure below, the Von Mises stress will be close to



(a) 180 MPa (b) 350 MPa
(c) 255 MPa (d) 495 MPa

ISRO Scientist/Engineer 22.04.2018

Ans. (b) : Data given

$$\sigma_1 = -100MPa,\ \sigma_2 = 300MPa,\ \sigma_3 = 150MPa$$

We know that (according to Von Mises theory)

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2\sigma_y^2$$

$$(-100 - 300)^2 + (300 - 150)^2 + (150 - (-100))^2 = 2\sigma_y^2$$

$$160000 + 22500 + 62500 = 2 \times \sigma_y^2$$

$$\sigma_y^2 = \frac{1}{2} \times 245000 = 122500$$

$$\sigma_y = 350 \text{ MPa}$$

1444. A particular section of a shaft is subjected to a maximum shear stress equal to 30 N/mm^2 and a maximum bending moment of 80 N/mm^2. If the yield point in tension for this material is 200 N/mm^2 and maximum shear stress theory of failure for static loading is used, then the factor of safety will be:

(a) 1.75 (b) 1.5
(c) 2.0 (d) 1.2

AAI Jr. Executive 29.11.2018

Ans. (c) : Shear stress = 30 N/mm^2
Maximum bending moment = 80 N/mm^2

$$\sigma_{yield} = 200 \text{ N/mm}^2$$

$$\tau_{max} = \sqrt{\left(\frac{80-0}{2}\right)^2 + 30^2} = 50 \text{ N/mm}^2$$

According to maximum shear stress theory –

$$\tau_{max} = \frac{\sigma_{yield}}{2N}$$

$$50 = \frac{200}{2N}$$

Factor of safety (N) = 200/100 = 2

1445. For a ductile material, the limiting value of octahedral shear stress (τ_o) is related to the yield stress (S_y) as

(a) $\tau_o = S_y \frac{\sqrt{2}}{3}$ (b) $\tau_o = S_y 3\sqrt{2}$
(c) $\tau_o = S_y \frac{\sqrt{3}}{2}$ (d) None of the above

BPSC AE Mains 2017 Paper - VI

Ans : (a) : According to Von-mises (theory of failure)

$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 = 2\sigma^2_{yield}$ .....(i)

The octahedral shear stress can be given by the expression,

$$\tau_{ys} = \frac{1}{3}\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} \quad \text{.....(ii)}$$

from equation (i) and (ii)

$$\tau_{ys} = \frac{1}{3}\sqrt{2\sigma^2_{yield}}$$

$$\tau_{ys} = \frac{\sqrt{2}}{3}\sigma_{yield}$$

1446. As per Von Mises theory the shear yield stress (K) can be related with yield stress (σ_y) under state of pure shear by _____

(a) $K = \sigma_y / (3)^{0.5}$ (b) $K = \sigma_y$
(c) $K = (\sigma_y)^{0.5}/3$ (d) None of the above

Punjab PSC SDE 12.02.2017

Ans. (a) : Von Mises theory is also known as maximum shear strain energy theory or distortion energy theory this assumes that failure occur when shear strain energy (distortion energy) in the complex system is equal to that the yield point in tensile compression test.

1447. Distortion energy theory of failure is applicable to:

(a) components made of plain carbon steel
(b) components made of composites
(c) components made of cast iron
(d) components made of non-metals

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : Maximum distortion energy theory [Von-mises theory]

– best theory of failure for ductile material because it given safe and economic design.

Plain carbon steels are ductile materials.

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1448. One of the theories of failure generally used is Maximum principal strain theory. This is also known as

(a) Coulomb's Tresca and Guest's theory
(b) St. Venant's theory
(c) Haigh's theory
(d) Rankine's Theory

UPRVUNL AE 05.07.2021
Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) : The maximum principle strain in the complex stress system must be less than the elastic in simple tension if there is to be no failure.

In the limit $\varepsilon_1 = \frac{\sigma_1 - \nu\sigma_2 - \nu\sigma_3}{E}$

$$\varepsilon_1 = \frac{\sigma}{E}$$

$$\boxed{\sigma_1 - \nu\sigma_2 - \nu\sigma_3 = \sigma}$$

1449. The most applicable failure theory for metals like aluminium is

(a) Maximum principal stress theory
(b) Maximum principal strain theory
(c) Total strain energy theory
(d) Maximum distortion energy theory

ISRO Scientist/Engineer 17.12.2017

Ans. (d) The most applicable theory of failure for metals
(1) Ductile material
(a) Maximum shear stress theory
(b) Maximum distortion energy theory
(2) Brittle material
(a) Maximum principal stress theory
(b) Maximum principal strain theory
(c) Total strain energy theory

1450. A cold rolled steel shaft is designed on the basis of maximum shear stress theory. The principal stresses induced at its critical section are 60 MPa and -60 MPa respectively. If the yield stress for the shaft material is 360 MPa, the factor of safety of the design is_____

(a) 6 (b) 4
(c) 2 (d) 3

UPRVUNL AE 21.08.2016, HPPSC W.S. Poly. 2016
OPSC AEE 2015 Paper-I, ESE 2002

Ans. : (d) Maximum stress (σ_{max}) = 60 Mpa
Minimum stress (σ_{min}) = –60 Mpa

Maximum shear stress (τ_{max}) = $\frac{\sigma_{max} - \sigma_{min}}{2}$

$$\tau_{max} = \frac{60+60}{2} = 60 \text{ Mpa}$$

According to maximum shear stress theory

$$\tau_y = \text{FOS} \times \tau_{max}$$

yield point shear stress = $\frac{\text{yield stress}}{2}$

$$\tau_y = 180 \text{ MPa}$$

$$\text{FOS} = \frac{\tau_y}{\tau_{max}} = \frac{180}{60} = 3$$

FOS = 3

1451. Pick the wrong statement about various theories of failure

(a) Rankine's theory is suitable for brittle materials.
(b) Maximum shear stress theory is more suitable for ductile materials.
(c) Distortion theory is suitable for ductile materials.
(d) Distortion energy theory is easy to apply as compared to maximum shear stress theory.

RPSC AE (GWD) 06.05.2016

Ans. (d)

- Rankine's theory (maximum, principle stress theory) is suitable for brittle materials
- Maximum shear stress theory in more suitable for ductile materials.
- Distortion theory is suitable for ductile materials.

1452. Three principal stresses of 10 MPa, 20 MPa and 30 MPa (tensile in nature) are acting on a homogeneous material. Using Von Mises yield criterion, what is the value, in MPa, of the estimated shear yield stress?

(a) 16.07 (b) 17.3
(c) 18.8 (d) 19.3
(e) 20.6

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : $\sigma_x = 10$ MPa, $\sigma_y = 20$ MPa, $\sigma_z = 30$ MPa
Von-Mises, yield criterion,

$$\frac{S_{yt}}{\text{FOS}} = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)}$$

$$= \sqrt{10^2 + 20^2 + 30^2 - (10\times 20 + 30\times 20 + 30\times 10)}$$

$$= \sqrt{900 + 400 + 100 - (1100)}$$

$$\frac{S_{yt}}{F_s} = \sqrt{300} \text{ MPa} = 17.3 \text{ MPa}$$

Estimated shear stress,

$$\frac{\tau_{xy}}{F_s} = 0.577\frac{S_{yt}}{F_s} = 0.577\sqrt{300}$$

1453. What is the relationship between tensile (σ_y) and shear (K) yield stresses as per Tresca and Von mises yield criteria

(a) $K = \frac{\sigma_y}{\sqrt{2}}$ and $K = \frac{\sigma_y}{\sqrt{3}}$

(b) $K = \frac{\sigma_y}{\sqrt{2}}$ and $K = \sqrt{3}\sigma_y$

(c) $K = \frac{\sigma_y}{2}$ and $K = \frac{\sigma_y}{\sqrt{3}}$

(d) $K = \frac{\sigma_y}{2}$ and $K = \frac{\sqrt{2}}{3}\sigma_y$

GPSC Asstt. Prof. (Prod.) 21.08.2016

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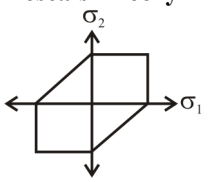
Ans. (c) : Tresca's Theory–



Hexagon

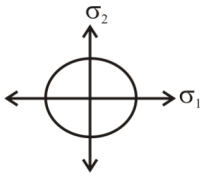
$K = \frac{\sigma_y}{2}$

K = Shear yield stress

σ_y = tensile yield stress

Tresca's theory is well used for ductile materials

Von-Mises Theory–



Ellipse

$K = \frac{\sigma_y}{\sqrt{3}}$

This is the most economical theory of failure used for ductile materials.

1454. In the case of analysis of fracture of brittle specimens and structures when all three principal stresses are compressive, ...theory seems to predict lower strengths than are actually obtained.

(a) Mcaulay's (b) Von-Tresca's
(c) Griffith's (d) Albert's

HPPSC Asstt. Prof. 29.10.2016

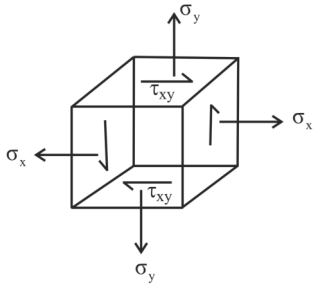
Ans. (c) : Griffith Theory of brittle fracture– Griffith proposed that a brittle material contains a population of fine small cracks and flows that have a variety of sizes, geometries and orientation which produces a stress concentration of sufficient magnitude so that the theoretical cohesive strength is reached in localized regions at a normal stress which is well below the theoretical value.

1455. The State of stress at a point is given as σ_x= 100 N / mm^2, σ_y= 40 N / mm^2 strength τ_{xy} = 40 N / mm^2. If the yield strength S_y of the material is 300 MPa, the factor of safety using maximum shear stress theory will be

(a) 3 (b) 2.5
(c) 7.5 (d) 1.25

BPSC Poly. Lect. 2016

Ans : (a)



$\sigma_x = 100$ N/mm^2, $\sigma_y = 40$ N/mm^2, $\tau_{xy} = 40$N/mm^2

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2}$$

$$\tau_{max} = \sqrt{900 + 1600}$$

$$\tau_{max} = 50 N/mm^2$$

According to maximum shear stress theory

τ_y= FOS × τ_{max}

S_y= 300 MPa

then τ_y= 150 MPa $\quad \tau_y = \frac{S_y}{2} = 0.5\ S_y$

150 = FOS × 50

$\boxed{FOS = 3}$

1456. Which theory of failure is applicable for copper components under steady load?

(a) Principal stress theory
(b) Strain energy theory
(c) Maximum shear stress theory
(d) Principal strain theory

MPPSC AE 2016

Ans : (c) Maximum shear stress theory of failure is applicable for copper components under steady load. Maximum shear stress theory or Guest and Trecas's theory is well justified for ductile materials.

1457. A transmission shaft subjected to bending loads must be designed on the basis of

(a) Maximum shear stress theory
(b) Fatigue strength
(c) Maximum normal stress and maximum shear stress theories
(d) Maximum normal stress theory

MPPSC AE 2016
ESE 1996

Ans : (b) A transmission shaft subjected to bending load must be designed on the basis of Fatigue strength
Note-As per MPPSC, official answer is (d).

1458. The factor of safety in bending for rectangular beams in terms of yield point stress (σ_{yp}) and working stress (σ_w) is

(a) $1.5\frac{\sigma_{YP}}{\sigma_W}$ (b) $1.3\frac{\sigma_{YP}}{\sigma_W}$
(c) $1.33\frac{\sigma_{YP}}{\sigma_W}$ (d) $\frac{\sigma_{YP}}{\sigma_W}$

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : As we know,

$$\text{Working stress} = \frac{\text{yield strength}}{\text{FOS}}$$

For ductile material

So, $\boxed{FOS = \frac{\sigma_{yp}}{\sigma_w}}$

1459. Consider the following

1. hard materials 2. brittle materials
3. malleable materials 4. ductile materials
5. elastic materials

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Of the above, Shear stress theory is applicable for which material

(a) 1 and 2 (b) 2 and 3
(c) 3 (d) 4
(e) 4 and 5

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Shear stress theory is applicable for ductile material.
Shear stress theory is also know as Guest & Tresca theory.

1460. Tresca theory of failure is applicable for which of the following type of materials?

(a) Elastomers (b) Plastic
(c) Ductile (d) Brittle

MPPSC AE 08.11.2015

Ans. (c) : Tresca's theory of failure is applicable for ductile materials.

1461. A solid circular shaft is subjected to a bending moment of 3000 N-m and a torque of 10000 N-m. The shafts is made of 45 C8 steel having ultimate tensile stress of 700 MPa and an ultimate shear stress of 500 MPa. Assume a factor of safety as 6. The diameter of the shaft according to the maximum shear stress theory is

(a) 84 mm (b) 85 mm
(c) 86 mm (d) 87 mm

MPPSC AE 08.11.2015

Ans. (c) : Given, Bending moment (M) = 3000 N-m
Torque(T) = 10000 N-m
σ_{ut} = 700 MPa & τ_{ut} = 500 MPa
$= 500\times10^6\ N/m^2$

Factor of safety (FOS) = 6
Let d = diameter of the shaft.

$\therefore$ Equivalent torque$(T_e) = \sqrt{M^2+T^2}$

$$\sqrt{(3000)^2+(10000)^2} = 10440.306\,N-m$$

According to the maximum shear stress theory-

$$\tau = \frac{\tau_{ut}}{FOS} = \frac{500\times10^6}{6}$$

$\therefore$ Diameter of shaft is-

$$d^3 = \frac{16T_e}{\pi\times\tau} = \frac{16\times10440.306}{\pi\times500\times\frac{10^6}{6}}$$

$$= \frac{6\times16\times10440.306\times10^{-6}}{\pi\times500}$$

$\Rightarrow$ $d^3 = 638.06\times10^{-6}$
$d = 8.609\times10^{-2}$ m
$d \approx 0.086$ m
$\Rightarrow$ d = 86 mm

1462. The slip line theory is developed to analyse:

(a) non-homogeneous plane stress deformation
(b) homogeneous plane stress deformation
(c) non-homogeneous plane strain deformation
(d) homogeneous plane strain deformation

HPPSC Asstt. Prof. 2014

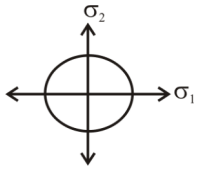
Ans. (c) : The slip line theory is developed to analyse non homogeneous plane strain deformation.

1463. The yield locus for von Mises yield criterion on π- plane is:

(a) Circle (b) Ellipse
(c) Hexagon (d) Circular cylinder

HPPSC Asstt. Prof. 2014

Ans. (b) :



Ellipse

1464. SEQA effective Von Mises stress at phase difference of 90º is minimum when the ratio of torsion stress amplitude and bending stress amplitude is near to :

(a) 0.2 (b) 0.6
(c) 1 (d) 1.6

HPPSC Asstt. Prof. 2014

Ans. (b) : According to Von-Mises

Shear yield stress $\tau_{xy} = \frac{\sigma_{yt}}{\sqrt{3}}$

$$\tau_{xy} = 0.577\,\sigma_{yt}$$

$$\frac{\tau_{xy}}{\sigma_{yt}} = 0.577 \approx 0.6$$

1465. From a uniaxial tension test, the yield strength of steel was found to be 200 N/mm^2. A steel shaft is subjected to a torque 'T', and a bending moment 'M'. The theory of failure which gives safest dimensions for the shaft and the relationship for design is

(a) Maximum Principal Strain Theory $\sigma_1 = \sigma_y$
(b) Maximum Principal Strain Theory $\frac{\sigma_1}{E} - \frac{\mu\sigma_2}{E} = \frac{\sigma_y}{E}$
(c) Maximum Shear Stress Theory $\frac{\sigma_1-\sigma_2}{2} = \frac{\sigma_y}{2}$
(d) Total Strain Energy Theory $\frac{\sigma_1^2}{2E} + \frac{\sigma_2^2}{2E} = \frac{\sigma_y^2}{2E}$

TNPSC AE 2014

Ans. (c) : Maximum shear stress theory [Guest and Tresca] of failure gives safest dimensions for the shaft and the relationship for design.

1466. If a shaft made of ductile material is subjected to combined bending and twisting moments, calculation based on which one of the following theories would give the most conservative values ?

(a) Maximum principle stress theory
(b) Maximum shear stress theory
(c) Maximum strain energy theory
(d) Maximum distortion energy theory

J&K PSC Civil Services Pre, 2013
ESE 1996, GATE 1988

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Ans. (b) :

Material	Theory of failure criterion
Brittle	max principle stress theory
ductile	max shear stress theory

1467. A rod of diameter 20 mm is subjected to a tensile load. Based on Tresca's failure criterion, if the uniaxial yield stress of the material is 300 MPa, the failure load is

(a) 20.75 kN (b) 54.00 kN
(c) 94.25 kN (d) 105.50 kN

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : Maximum shear stress theory (Guest and Tresca's theory)

$$\tau_{max} \le \frac{\sigma_y}{2} \text{(For no failure)}$$

For design max of shear stress,

$$\left[\frac{(\sigma_1-\sigma_2)}{2}, \frac{(\sigma_2-\sigma_3)}{2}, \frac{(\sigma_3-\sigma_1)}{2}\right] \le \left(\frac{\sigma_y}{2FOS}\right)$$

For uni-axial loading, ($\sigma_2 = \sigma_3 = 0$)

$$\sigma_1 \le \sigma_{yt} \quad [\because \text{FOS} = 1]$$

$d = 20$ mm, $\sigma_{yt} = \sigma_1 = 300$ MPa

$$\sigma_1 = \frac{\text{Load}}{\text{Area}}$$

$$\Rightarrow \quad \text{Load} = \sigma_1 \times \text{Area} = \sigma_1 \times \frac{\pi}{4} d^2$$

$$\text{Load} = 300 \times \frac{\pi}{4} 20^2 = 94.25\,\text{kN}$$

1468. A small element at the critical section of a component is in a bi-axial state of stress with the two principal stresses being 360 MPa and 140 MPa. The maximum working stress according to Distortion energy theory is :

(a) 220 MPa (b) 110 MPa
(c) 314 MPa (d) 330 MPa

APGENCO AE 2012

Ans. (c) : Given, $\sigma_1 = 360$ MPa

$\sigma_2 = 140$ MPa

According to maximum Distortion theory -

$$\sigma_{working} = \sqrt{\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2}$$

$$= \sqrt{(360)^2 + (140)^2 - 360 \times 140}$$

$$= 314.32 \text{ MPa}$$

1469. At a point in a strained material the principal stresses are P_1, P_2, and zero. In order that the factor of safety obtained by using the maximum shear stress theory of failure is the same as that obtained by using the distortion energy theory, the ratio of P_1, to P_2, should be

(a) -1 (b) $\frac{1}{2}$
(c) 1 (d) -2

WBPSC AE, 2007

Ans. (c) : According to maximum shear stress theory

$$\frac{\sigma_1 - \sigma_3}{2} \le \frac{\sigma_y}{N \times 2}$$

$$N = \frac{\sigma_y}{\sigma_1 - \sigma_3} = \frac{\sigma_y}{P_1 - 0} = \frac{\sigma_y}{P_1}$$

Squaring both sides

$$\boxed{N^2 = \frac{\sigma_y^2}{P_1^2}} \quad -------(1)$$

According to distortion energy theory

$$\frac{1}{2}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right] \le \frac{\sigma_y^2}{N^2}$$

$$N^2 = \frac{2\sigma_y^2}{(P_1-P_2)^2 + P_2^2 + P_1^2}$$

$$\boxed{N^2 = \frac{\sigma_y^2}{P_1^2 + P_2^2 - P_1P_2}} \quad -------(2)$$

Equating (1) & (2)

$$\frac{\sigma_y^2}{P_1^2} = \frac{\sigma_y^2}{P_1^2 + P_2^2 - P_1P_2}$$

$$P_1^2 = P_1^2 + P_2^2 - P_1P_2$$

$$P_1^2 = P_1P_2$$

$$P_2 = P_1$$

$$\boxed{\frac{P_1}{P_2} = 1}$$

11. Springs

1470. A coil is cut into two halves, the stiffness of cut coil will be:

(a) Double (b) Half
(c) Same (d) None of above

UPPSC AE 13.12.2020 Paper-I
Gujarat PSC AE 2019
BPSC AE Mains 2017 Paper - VI
GWSSB DEE 07.07.2016
GPSC Asstt. Prof. 28.08.2016
OPSC AEE 2015 Paper-I
TNPSC AE 2014, SJVN ET 2013
ESE 2002, CSE Pre-2002

Ans. (a) : $k = \frac{Gd^4}{64R^3n}$

$k \propto \frac{1}{n}$

$\frac{k_1}{k_2} = \frac{n_2}{n_1}$

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Where, k_1 and k_2 are stiffness and n_1 and n_2 are number of coils in the spring.
When a spring is cut into two equal halves.

$$\Rightarrow n_2 = \frac{n_1}{2}$$

$$\Rightarrow \frac{k_1}{k_2} = \frac{n_2}{n_1} = \frac{n_2}{2n_2}$$

$$k_2 = 2k_1$$

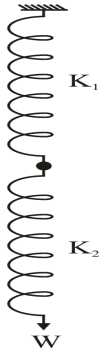
1471. Two close coiled helical springs with stiffness K_1 and K_2 respectively are connected in series. The stiffness in equivalent spring is given by

(a) $\frac{K_1K_2}{K_1 + K_2}$ (b) $\frac{K_1 - K_2}{K_1 + K_2}$

(c) $\frac{K_1 + K_2}{K_1K_2}$ (d) $\frac{K_1 - K_2}{K_1K_2}$

GPSC DEE Class-2 (GWSSB) 04.07.2021
APPSC AEE SCREENING 17.02.2019
PTCUL AE 25.06.2017
VIZAG MT 2015, TNPSC AE 2013
APPSC AEE 04.12.2012

Ans. (a) : When connected in series –
Equivalent stiffness –



$$\frac{1}{K_e} = \frac{1}{K_1} + \frac{1}{K_2}$$

$$\frac{1}{k_e} = \frac{K_1 + K_2}{K_1K_2}$$

$$\boxed{K_e = \frac{K_1K_2}{K_1 + K_2}}$$

1472. Which one of the following expresses the stress factor K used for design of closed coiled helical spring? Where C = spring index

(a) $\frac{4C-1}{4C-4}$ (b) $\frac{4C-1}{4C-4} + \frac{0.615}{C}$

(c) $\frac{4C-4}{4C-1} + \frac{0.615}{C}$ (d) $\frac{4C-4}{4C-1}$

VIZAG MT, 14.12.2020, Gujarat PSC AE 2019
CIL MT 26.03.2017, UPRVUNL AE 21.08.2016
APPSC AE 04.12.2012, ESE 2008

Ans. : (b) Wahl's stress factor for helical spring

$$K_w = \frac{4C-1}{4C-4} + \frac{0.615}{C}$$

C = Spring index (D/d)

The Wahl's stress factor (K_w) = $K_s \times K_c$
K_s = stress factor due to shear
K_c = Curvature effect factor

1473. The spring constant of a helical compression spring DOES NOT depend on :

(a) Coil diameter
(b) Material strength
(c) Number of active turns
(d) Wire diameter

Assam Engg. College AP/Lect. 18.01.2021
Gujarat PSC AE 2019
GPSC ARTO Pre 30.12.2018
GPSC Executive Engg. 23.12.2018, GATE 2016

Ans. (b) : Spring constant,

$$\boxed{k = \frac{Gd^4}{8D^3n}}$$

Stiffness depends on following –
(i) Modulus of Rigidity of spring
(ii) Wire diameter
(iii) Mean diameter
(iv) Number of active turns.

1474. Due to space constraint 5 coils are to be cut and removed from a spring with 25 active coils. Which one among the following is correct with respect to stiffness?

(a) 1.25 times that of original spring
(b) 80 percent of that of original spring
(c) The same for both springs
(d) Five percent less than that of original spring

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
APGENCO AE, 2017, ESE 2004

Ans. (a) : Deflection of spring

$$\delta = \frac{64WR^3n}{Gd^4}$$

$$\text{Stiffness} \Rightarrow \frac{W}{\delta} \propto \frac{1}{n}$$

$$\text{stiffness} \propto \frac{1}{\text{number of coil}(n)}$$

$$K_1 \propto \frac{1}{25} \qquad [n_1 = 25]$$

$$K_2 \propto \frac{1}{20} \qquad [n_2 = 20]$$

$$K_2 \propto \frac{25}{20} \times K_1$$

$$K_2 = 1.25\ K_1$$

1475. Two coiled springs each having stiffness k are placed in parallel. The stiffness of the combination will be :

(a) 4k (b) 2k

(c) $\frac{k}{2}$ (d) $\frac{k}{4}$

JPSC AE 10.04.2021, Paper-II
J & K PSC Screening, 2006
ESE 2000, CSE Pre-1998

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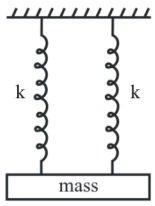
Ans. (b) : If springs are connected in parallel then



$k_{eq} = k + k$

$\boxed{k_{eq} = 2k}$

1476. A helical spring has N turns of coil diameter D and a second spring, made of same wire diameter and same material, has N/2 turns of coil of diameter 2D. If the stiffness of the first spring is k, then stiffness of the second springs will be

(a) k/4 (b) k/2

(c) 2k (d) 4k

ISRO Scientist/Engineer (RAC) 10.03.2019
TSPSC AEE 2017, ESE 1999

Ans. (a) :

$$\text{Spring stiffness } (k_1) = \frac{W}{\delta} = \frac{W}{\frac{8WD^3N}{Gd^4}} = \frac{Gd^4}{8D^3N}$$

$$k_1 = \frac{Gd^4}{8D^3N}$$

[Given, if coil diameter is doubled and no. of turns is half]

$$k_2 = \frac{Gd^4}{8\times(2D)^3\times N/2}$$

$$= \frac{Gd^4}{8\times 8D^3\times\frac{N}{2}}$$

$$= \frac{Gd^4}{8\times 4D^3\times N}$$

$$= \frac{1}{4}\left[\frac{Gd^4}{8D^3N}\right]$$

$$\boxed{k_2 = k_1/4}$$

1477. While calculating the stress induced in a closed helical spring. Wahl's factor is considered to account for

(a) the curvature and stress concentrated effect
(b) shock loading
(c) fatigue loading
(d) poor service conditions

GPSC ARTO 01.05.2016, TNPSC AE 2014
ESE 2002, ESE 1997

Ans. (a) : While calculating the stress induced in a closed helical spring. Wahl's factor is considered to account for the curvature and stress concentrated effect.

$$K = \frac{4C-1}{4C-4} + \frac{0.615}{C}$$

where $C = \frac{D}{d}$ [Spring index]

1478. An open coiled helical spring of mean diameter d is subjected to an axial force P. The wire of the spring is subjected to :

(a) Direct shear stress only
(b) Combined shear and bending only
(c) Combined shear, bending and twisting
(d) Combined shear and twisting only

GPSC DEE Class-2 (GWSSB) 04.07.2021
HPPSC Asstt. Prof. 18.09.2017
OPSC AEE 2015 Paper-I

Ans : (d) An open coiled helical spring of mean diameter d is subjected to an axial force P. The wire of the spring is subjected to combined shear and twisting only.

1479. A closed coil, helical spring is subjected to a torque about its axis. The spring wire would experience a

(a) Direct shear stress
(b) Torsional shear stress
(c) Bending stress
(d) Direct tensile stress
(e) Bending stress and shear stress both

CGPSC AE 26.04.2015 Shift-I
UPSC JWM Advt. No.-52/2010
ESE 1998

Ans. (c) : When a closed coil spring fixed at one end is subjected to twisting couple about the central axis of the spring then the bending moment will be product then spring will be experience bending stress.

1480. A spring is made of a wire of 2 mm diameter having a shear modulus of 80 GPa. The mean coil diameter is 20 mm and the number of active coils is 10. If the mean coil diameter is reduced to 10 mm, the stiffness of the spring is:

(a) increased by 16 times
(b) decreased by 8 times
(c) increased by 8 times
(d) decreased by 16 times

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
BHEL ET 2019
VIZAG Steel MT (Re-Exam) 24.11.2013
GATE 2008

Ans. (c) : $D_1 = 20$ mm

$D_2 = 10$ mm

Stiffness of spring

$$k = \frac{Gd^4}{8D^3n}$$

$$k \propto \frac{1}{D^3}$$

$$\frac{k_1}{k_2} = \left(\frac{D_2}{D_1}\right)^3$$

$$\frac{k_1}{k_2} = \left(\frac{10}{20}\right)^3$$

$$k_2 = 8k_1$$

1481. If both the mean coil diameter and wire diameter of a helical compression or tension spring be doubled, then the deflection of the spring close coiled under same applied load will:

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(a) be doubled
(b) be halved
(c) increase four times
(d) get reduced to one-fourth

GPSC ARTO Pre 30.12.2018
OPSC AEE 2019 Paper-I
ESE 2012

Ans. (b) : Deflection, $\delta_1 = \frac{8WD_1^3n}{Gd_1^4}$

Given, $D_2 = 2D_1$ $\quad \delta_2 = \frac{8WD_2^3n}{Gd_2^4}$

$d_2 = 2d_1$ $\quad \delta_2 = \frac{8W(2D_1)^3n}{G(2d_1)^4} = \frac{8}{16}\left(\frac{8WD_1^3n}{Gd_1^4}\right)$

$$\boxed{\delta_2 = \frac{\delta_1}{2}}$$

1482. In helical compression spring design formula :
$\tau = 8FCK/(\pi d^2)$
Where
F = load
τ = shear stress
C = spring index
d = wire dia.
K = Wahl factor.
The Wahl factor takes into account.
(a) the bending stress and the curvature of the spring
(b) the direct shear and the curvature of the spring
(c) the elastic constants (modulus of rigidity and Poisson's ratio) of the spring.
(d) the effect of the inactive number of coils at the ends of the spring.

NLC GET 17.11.2020, Shift-II
MPPSC AE 2016
BPSC Asstt. Prof. 29.11.2015

Ans. (b) : $K_w = K_{sh} \cdot K_c = \frac{4C-1}{4C-4} + \frac{0.615}{C}$

Where, K_w = Wahl factor
K_{sh} = Direct shear stress factor
K_c = Curvature effect factor
C = Spring index = $\frac{D}{d}$

The Wahl factor takes into account the direct shear and the curvature of the spring.

1483. The spring index in helical spring is
(a) a ratio of coil diameter to wire diameter
(b) a ratio of wire diameter to coil diameter
(c) slope of its force-deflection curve
(d) slope of its deflection force curve

Haryana PSC AE (PHED) 05.09.2020, Paper-II
Kerala PSC Poly. Lect. 09.01.2014
WBPSC AE 2008

Ans. (a) : Spring Index = $\frac{\text{Coil dia.}}{\text{Wire dia.}} = \frac{D}{d}$

1484. Stiffness of a close coiled helical spring in terms of wire diameter d, modulus of rigidity G, number of turns n and mean coil radius R is given by
(a) $\frac{Gd^4}{16nR^3}$ (b) $\frac{Gd^4}{32nR^3}$
(c) $\frac{Gd^4}{64nR^3}$ (d) $\frac{Gd^4}{96nR^3}$

APPSC IOF, 2009
WBPSC AE, 2007, ESE 1996

Ans. (c)

$$K = \frac{Gd^4}{8D^3n}$$

$$K = \frac{Gd^4}{8\times 8R^3n}$$

Stiffness, $\boxed{K = \frac{Gd^4}{64R^3n}}$

1485. Two closely coiled helical springs 'A' and 'B' are equal in all respects but the number of turns of spring 'A' is half that of spring 'B' The ratio of deflections in spring 'A' to spring 'B' is
(a) 1/8 (b) 1/4
(c) 1/2 (d) 2

Sikkim PSC (Under Secretary), 2017
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (c) : $\delta_A = \frac{64wR^3n}{Gd^4}$

Given, if $n = \frac{1}{2}$

$$\delta_A = \frac{64wR^3 1}{Gd^4 2} \quad \text{.....(i)}$$

$$\delta_B = \frac{64wR^3n}{Gd^4}$$

If, n = 1

$$\delta_B = \frac{64wR^3\times 1}{Gd^4} \quad \text{.....(ii)}$$

Ratio of deflection $\frac{\delta_A}{\delta_B}$

$$= \frac{\frac{64wR^3\times 1}{Gd^4\times 2}}{\frac{64wR^3\times 1}{Gd^4}}$$

$$\frac{\delta_A}{\delta_B} = \frac{1}{2}$$

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1486. A helical spring has spring constant k. If the wire diameter, spring diameter and the number of coils are all doubled then the spring constant of the new spring is _____

(a) k (b) 0.5 k

(c) 8 k (d) 16 k

Assam Engg. College AP/Lect. 18.01.2021
MPPSC AE 2016

Ans. (a) :

Spring Constant

$$k = \frac{W}{\delta} = \frac{W}{\frac{64WR^3n}{Gd^4}} = \frac{Gd^4}{64R^3n} = \frac{Gd^4}{64\left(\frac{D}{2}\right)^3 n}$$

$$k = \frac{Gd^4}{8D^3n}$$

[Given, if coil diameter, wire diameter and no. of coil is doubled]

$$\text{Then, } k' = \frac{G(2d)^4}{8(2D)^3 \times 2n} = \frac{16}{16}\left[\frac{Gd^4}{8D^3n}\right]$$

$$k' = k$$

1487. In a helical coil spring, if C is the spring index, then the shear correction factor is given by :

(a) $K_s = \frac{1}{2C}$ (b) $K_s = \frac{2C}{2C+1}$

(c) $K_s = \frac{2C+1}{2C}$ (d) $K_s = \frac{C+1}{2C}$

APPSC Poly Lect. 13.03.2020
CIL MT 27.02.2020

Ans. (c) :

Shear stress concentration factor $(K_s) = \frac{2C+1}{2C}$

Where; $C = \frac{D}{d}$ = Spring index for helical coil spring.

1488. The deflection of a spring with 20 active turns under a load of 1000 N is 10 mm. The spring is made into two pieces each of 10 active coils and placed in parallel under the same load. The deflection of this system is

(a) 20 mm (b) 10 mm

(c) 5 mm (d) 2.5 mm

UPPSC AE 13.12.2020 Paper-I
TRB Poly. Lect., 2012
GATE 1995

Ans. (d) : Δx= 10 mm, N= 20 turn , F= 1000N

$$K = \frac{F}{\Delta x} = \frac{1000}{10} = 100\,N/mm$$

On cutting spring in two equal parts, new 'K' will be equal 2K

Under parallel arrangement, K of the system will become 2K+2K = 4K. i.e 400 N/mm

Under same force (F), we get $\Delta x' = \frac{1000}{400} = 2.5mm.$

1489. A helical coil spring with wire diameter 'd' and mean coil diameter 'D' is subjected to axial load. A constant ratio of 'D' and 'd' has to be maintained, such that the extension of the spring is independent of 'D' and 'd' what is the ratio?

(a) D^3/d^4 (b) d^3/D^4

(c) $D^{4/3}/d^3$ (d) $d^{4/3}/D^3$

BPSC Poly. Lect. 2016
VIZAG MT 2015, ESE 2008

Ans. (a) : Deflection of spring $(\delta) = \frac{8WD^3n}{Gd^4}$

If extension of spring is independent of "D" and "d" than the ratio of "D" & "d" is –

$$\boxed{\frac{D^3}{d^4}}$$

1490. Spring driven watches and clocks utilize

(a) involute gears (b) cyloid gears

(c) epricycloid gears (d) straight rack gears

Karnataka PSC AE (WRD) 31.07.2021
RPSC ACF-2011

Ans. (b) : Spring driven watches and clock utilize cycloid gears whereas spiral spring are used in watches and toys. Torsion springs are used in places where a static position is required to be maintained against impact and load fluctuation.

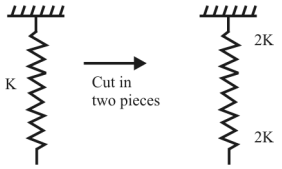
1491. A helical compression spring of stiffness K is cut into two pieces, each having equal number of turns and kept side-by-side under compression. The equivalent spring stiffness of this new arrangement is equal to :

(a) 4K (b) 2K

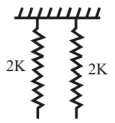
(c) K (d) 0.5K

RPSC ACF & FRO, 26.02.2021
CGPSC AE 26.04.2015 Shift-I

Ans. (a) :



Keep side by side under compression.



$$K_{eq} = 2K + 2K = 4K$$

1492. A load applied at centre of the carriage spring to straighten its leaves is known as

(a) Yield load (b) Ultimate load

(c) Proof load (d) Safe load

TSGENCO AE 14.11.2015
APPSC AEE 2012

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Ans. (c) : Proof Load– Proof load is the maximum load carrying capacity of the spring, without getting permanently distorted.

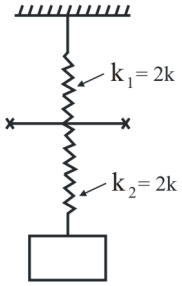
$$f_s = \frac{8WD}{\pi d^3}$$

1493. If a compression coil spring of stiffness 10 N/m is cut into two equal parts and the used in parallel the equivalent spring stiffness will be :

(a) 10 N/m (b) 20 N/m
(c) 40 N/m (d) 80 N/m

Kerala PSC AE 06.08.2015
KPSC AE 2015

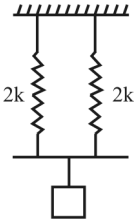
Ans. (c) : Given spring –



∴ previously stiffness of spring was K and after cutting into parts k_1 & k_2

$k_1 = 2k$ $k_2 = 2k$

Connected then in parallel



$k_{eq} = k_1 + k_2$
$= 2k + 2k$
$= 4k$ ....(i)

Putting, $k = 10$ N/m

$[k_{eq} = 4 \times 10 = 40$ N/m]

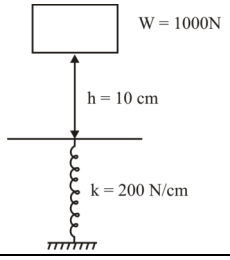
1494. A body having weight of 1000 N is dropped from a height of 10 cm over a closed coiled helical spring of stiffness 200 N/cm. The resulting deflection of spring is nearly :

(a) 5 cm (b) 16 cm
(c) 35 cm (d) 100 cm

GPSC ARTO Pre 30.12.2018
APPSC AE 04.12.2012

Ans. (b) : By conservation of energy,

Potential energy lost by body = strain energy stored in spring.



$$mg(h + x) = \frac{1}{2}kx^2$$

$$1000(10 + x) = \frac{1}{2} \times 200 \times x^2$$

$$1000 \times 10 = \frac{1}{2} \times 200 \times x^2 - 1000x$$

$$x^2 - 10x - 100 = 0$$

$$x = \frac{10 + \sqrt{(10)^2 + 400}}{2}$$

$x = 16.18$ cm

1495. A torsion bar with a spring constant 'k' is cut into 'n' equal lengths. The spring constant for each portion would be

(a) nk (b) k^n
(c) k/n (d) $k^{1/n}$

Kerala PSC Poly. Lect. 09.01.2014
ISRO Scientist/Engineer 12.05.2013

Ans. (a) : We know that

$$K \propto \frac{1}{L}$$

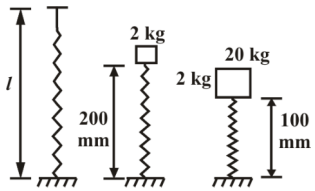
Hence, if we decrease the length into n parts the value of K increase by n times.

1496. A 2 kg pan is placed on a spring. In this condition, the length of the spring is 200 mm. When a mass of the 20 kg is placed on the pan, the length of the spring becomes 100 mm. For the spring, the un-deformed length L and the spring constant k (stiffness) are

(a) L = 220 mm, k = 1862 N/m
(b) L = 210 mm, k = 1862 N/m
(c) L = 210 mm, k = 1960 N/m
(d) L = 200 mm, k = 1960 N/m
(e) L = 200 mm, k = 2156 N/m

CGPSC AE 26.04.2015 Shift-I
ISRO Scientist/Engg. 12.05.2013
GATE 2005

Ans. (c) :



Spring force F = - kx

F = kx (compressive force)

Case (i)—When 2 kg pan is placed on spring

Force (F) = mg = 2g N

x = (*l* - 200) mm

Case (ii)—When 20 kg mass is placed on the pan

Force (F) = mg = 22g N

x = (*l* - 100) mm

From case (i) and case (ii) spring force

F = kx

2g = k (*l* - 200) ...(1)

22g = k (*l* - 100) ...(2)

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Divide equation (2) by (1)

$$\frac{22}{2}=\frac{l-100}{l-200}$$

$11l$ - 2200 = l - 100

10 l = 2100

l = 210 mm

From equation (1)

2g = k (l - 200)

2g = k (210 - 200)

$$k=\frac{2g}{10}=\frac{2\times 9.81}{10}$$

k = 1.962 N/mm

k = 1962 N/m

1497. If the wire diameter of a closed coil helical spring subjected to compressive load is increased from 1 cm to 2 cm, other parameters remaining same, then deflection will decrease by a factor of :

(a) 16 (b) 8

(c) 4 (d) 2

GPSC ARTO Pre 30.12.2018

GPSC AE 26.07.2015

Ans. (a) : Deflection $\delta_1=\frac{8WD^3n}{Gd_1^4}$

Given, d_1 = 1 cm

d_2 = 2 cm $\quad \delta_2=\frac{8WD^3n}{G(2d_1)^4}=\frac{8WD^3n}{16Gd_1^4}$

$d_2 = 2d_1$

$$\boxed{\delta_2=\frac{\delta_1}{16}}$$

1498. When a helical compression spring is subjected to an axial compressive load, then which one of the following stress will be induced?

(a) Shear (b) Bending

(c) Compressive (d) Tensile

TNPSC AE (Industries) 09.06.2013

UPSC JWM Adv. No-16/2009

Ans. (a) : When a helical compression spring is subjected to an axial compressive/tensile load, then there will be shear stress (direct + torsional) induced in the spring.

1499. Two helical tensile springs of the same material and also having identical mean coil diameter and weight, have wire diameter d and d/2. The ratio of their stiffness is

(a) 1 (b) 4

(c) 64 (d) 128

VIZAG STEEL MT 18.08.2013

VIZAG STEEL MT 10.06.2012

Ans. (c) : $\delta=\frac{64PR^3n}{Gd^4}$ or, $\left[\frac{P}{\delta}=\frac{Gd^4}{64R^3n}\right]$

Stiffness $K \propto \frac{d^4}{n}$

Since wire weight is the same,

therefore $\frac{\pi}{4}d_1^2n_1=\frac{\pi}{4}d_2^2.n_2$

$$\frac{n_1}{n_2}=\left(\frac{d_2}{d_1}\right)^2=\left(\frac{1}{2}\right)^2$$

or, $\frac{n_1}{n_2}=1/4$

$$\frac{K_1}{K_2}=\frac{d^4}{(d/2)^4}\times\frac{n_2}{n_1}$$

$$\frac{K_1}{K_2}=16\times 4$$

$$\boxed{\frac{K_1}{K_2}=64}$$

1500. A spring used to absorb shocks and vibrations is

(a) Torsion spring (b) Conical spring

(c) Leaf spring (d) Disc spring

Assam PSC AE (IWT) 14.03.2021

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : Leaf spring is used to absorb shocks and vibrations. Leaf springs are an integral part of vehicle's suspension system. They are installed to help and support the entire weight of car or truck.

1501. In design of helical springs the spring index is usually taken as

(a) 3 (b) 5

(c) 8 (d) 12

TSPSC AEE 2015

Kerala PSC Poly. Lect. 09.01.2014

Ans. (b) : In design of helical springs the spring index is usually taken as 5.

1502. A truck of mass 20 tonnes traveling a 1.6 m/s impacts with a buffer spring which compress 2 mm per kN. The maximum compression of the spring is _____.

(a) 160 mm (b) 240 mm

(c) 80 mm (d) 320 mm

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Given,

Mass of truck (m) = 20 tonne = 20 × 10^3 kg

Velocity of truck (v) = 1.6 ms

Buffer spring constant (k) = 2 mm/kN

x = maximum compression of the spring

We know that kinetic energy of truck (kE) = $\frac{1}{2}mv^2$

$$=\frac{1}{2}\times 20\times 10^3\times(1.6)^2$$

= 25600 kN-m …… (i)

Now, compression load, W = $\frac{x}{k}=\frac{x}{2}$

= 0.5x kN

∴ Work done in compressing the spring = Average compressive load × displacement

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$$= \frac{0.5x}{2} \times x = 0.25x^2 \text{ kN} - \text{m} \ldots..(ii)$$

Since, the entire kinetic energy of the truck is used to compare the spring.

$\therefore$ from equation (i) and (ii)–

$0.25x^2 = 25600$

$x^2 = \frac{25600}{0.25}$

$x = 320$ mm

1503. A closed coiled helical spring is subjected to an axial load. the stress which is not produced in the wire of the spring is___.

(a) Torsional shear stress
(b) Compressive stress
(c) Direct shear stress
(d) Bending stress

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : The stress which are produced in a closed coil helical springs are only torsional shear stress, there is not bending stress, compressive and direct shear stress are produced in wire coiled spring (helical). (tension & compressions)

1504. The longest leaf in Semi-elliptic leaf spring is known as

(a) chief leaf (b) master leaf
(c) major leaf (d) higher leaf

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : The longest leaf in semi-elliptic leaf spring is known as master leaf through which the spring from one end of bushing to other end.

1505. If a spring deflect 50 mm under a 2 kN load, the energy absorbed by the spring in one gradual application of this load is :

(a) 100 J (b) 50 J
(c) 75 J (d) 25 J

CGPSC AE 15.01.2021

Ans. (b) : Given,

deflection, $\delta = 50 \text{ mm} = 0.05 \text{ m}$

Load, $P = 2\text{kN}$

Energy absorbed by the spring,

$$E = 1/2 \times P \times \delta$$

$$= \frac{1}{2} \times 2 \times 10^3 \times 0.05$$

$E = 50$ J

1506. Two springs with stiffness 2 N/m and 3 N/m are connected in parallel, then the equivalent stiffness is _____.

(a) 3 N/m (b) 1 N/m
(c) 5 N/m (d) 2 N/m

VIZAG Steel MT 24.01.2021

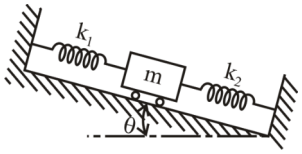
Ans. (c) : Given, $k_1 = 2$ N/m, $k_2 = 3$ N/m

Parallel connected,

$k_{eq} = k_1 + k_2$

$= 2 + 3 = 5$ N/m

1507. For the shown figure, if $k_1 = 10$ N/m, $k_2 = 20$ N/m and $\theta = 30°$, then equivalent stiffness of the system will be _____ N/m.



(a) 30 (b) 12
(c) 6.66 (d) 15

AAI Jr. Executive 26.03.2021

Ans. (a) : Stiffness of the system are in parallel,

$k_{eq} = k_1 + k_2$

$= 10 + 20$

$= 30$ N/m

1508. A helical spring is made of 12 mm diameter wire by winding it on a 120 mm diameter mandrel. If there are 10 active turns and modulus of rigidity of material is 8.2×10^4 N/mm^2 the spring constant will be

(a) 2.94 Nmm^{-1} (b) 3.64 Nmm^{-1}
(c) 9.24 Nmm^{-1} (d) 11.58 Nmm^{-1}

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (*) : Given,

Wire dia, $d = 12$ mm

Coil dia, $D = 120$ mm

No. of turn, $n = 10$

Modulus of rigidity, $G = 8.2 \times 10^4$ N/mm^2

Spring constant, $k = ?$

$$k = \frac{Gd^4}{8D^3n}$$

$$= \frac{8.2 \times 10^4 \times (12)^4}{8 \times (120)^3 \times 10}$$

$$\boxed{k = 12.3 \text{ N/mm}}$$

Note-Official answer given by commission is (c).

1509. Free length for helical compression springs having

(a) $P_n = 2d$ (b) $P_n + 3d$
(c) $2(p+d)$ (d) $P_n + 4d$

TNCSC AE 2020

Ans. (b) : Free length for helical compression springs having square ends is given as

free length (l_0) = $\boxed{P_n + 3d}$

Where, P_n = pitch

d = wire diameter

1510. A simple spring mass vibrating system has a natural frequency of N. If the spring stiffness is halved and mass is doubled, then natural frequency will become

(a) N/2 (b) 2 N
(c) 4 N (d) 8 N

RPSC IOF, 2020

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Ans. (a) : Natural frequency $f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

$$N = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \text{ .....(i)}$$

[Given, if spring stiffness is halved and mass doubled]

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k/2}{2m}}$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{4m}} \text{ .....(ii)}$$

From equation (ii) & (ii)

$$\frac{N}{f_n} = \frac{\frac{1}{2\pi}\sqrt{\frac{k}{m}}}{\frac{1}{2\pi}\sqrt{\frac{k}{4m}}}$$

$$\frac{N}{2} = f_n$$

$$f_n = \frac{N}{2}$$

So, natural frequency will be come halved

1511. The Wahl's factor is involved in the design of

(a) Gear (b) Clutch
(c) Hooke's Joint (d) Spring

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Wahl's correction factor is used in shear stress spring formula to compensate the effect of curvature and direct shear stress.

1512. For squared - ground ends in helical coil compression spring, the number of active coils is theoretically

(a) Two coils less than the total number of coils
(b) One coil less than the total number coils
(c) $\frac{1}{2}$ coil less than the total number of coils
(d) Same as the total number of coils

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) :

Spring ends	Number of effective coil
Plain end	Total coil
Plain and grounded	Total coil –1
flat/squared	total coil –2
squared and grounded	Total coil –2

1513. The angle of twist for the equivalent bar to spring is given by:

(a) $16PD^2N/Gd^4$ (b) $8PD^2N/Gd^4$
(c) $6PDN/Gd^3$ (d) $4PD^2N/Gd^4$

CIL MT 27.02.2020

Ans. (a) : Angle of twist

$$\theta = \frac{TL}{GJ}$$

$$\theta = \frac{PD/2 \times \pi DN}{G \times \pi d^4/32} \quad \left[\because M = \frac{PD}{2},\ \ell = \pi DN,\ J = \frac{\pi d^4}{32}\right]$$

$$\theta = \frac{16PD^2N}{Gd^4}$$

1514. With compression of close coiled helical spring, the wire gets subjected to

(a) tension
(b) compression
(c) shear and bending
(d) a combination of shear and tension

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : With compression of close coiled helical spring the wire gets subjected to a combination of shear and tension.

1515. A compression coil spring made of an alloy steel has mean diameter of coil as 50 mm and wire diameter as 5 mm? What will be the shear stress factor?

(a) 1.04 (b) 1.05
(c) 1.02 (d) 1.03

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Given, Coil diameter (D) = 50 mm
Wire diameter (d) = 5 mm
We know that spring index

$$C = \frac{D}{d} = \frac{50}{5} = 10$$

Shear stress factor

$$k_s = 1 + \frac{1}{2C} = 1 + \frac{1}{2\times 10}$$

$$\boxed{k_s = \frac{21}{20} = 1.05}$$

1516. When two springs are connected in parallel with stiffness 'k' and 's', the effective stiffness is:

(a) 1/k + 1/s (b) k+s
(c) k/s (d) s/k

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Spring stiffness in parallel

$k_{eq} = k_1 + k_2$

Where, $k_1 = k$

$k_2 = s$

So, $k_{equivalent} = k + s$

1517. If two identical helical coil springs are placed one over the other, total stiffness is

(a) doubled (b) halved
(c) increased four times (d) remains same

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : If two helical springs are identical then stiffness of both springs are equal.

$k_1 = k_2$

Hence connection is in series –

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

Putting $k_1 = k_2 \Rightarrow \boxed{k_{eq} = \frac{k_1}{2}}$

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1518. In which applications is leaf springs mostly used?

(a) Two-wheeler front suspensions
(b) In bicycles
(c) Airplanes
(d) In four-wheelers

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : The laminated or leaf spring (also known as flat spring or carriage spring) consists of a number of flat plates (known as leaves) of varying length held together by means of clamps and bolts. They are mostly used in automobiles.

1519. A leaf spring is to be made with seven steel plates 65 mm wide 6.5 mm thick. Calculate the length of spring to carry a central load of 2.75 kN and the bending stress is limited to 160 MPa

(a) 644.2 mm (b) 64.42 mm
(c) 74.42 mm (d) 744.2 mm

TNPSC AE 2019

Ans. (d) : Data given-

σ_b = 160 MPa b = 65 mm
t = 6.5 mm P = 2.75 kN
L = ? n = 7

We know that

$$\sigma_b = \frac{3PL}{2nbt^2}$$

$$L = \frac{2nbt^2\sigma_b}{3P} = \frac{2\times7\times65\times6.5^2\times160}{3\times2.75\times10^3}$$

$$\boxed{L = 745.64\text{mm} \simeq 744.2\text{mm}}$$

1520. Which is true statement about Belleville springs?

(a) These are used for dynamic loads
(b) These are composed of coned discs which may be stacked upto obtain variety of load-deflection characteristics
(c) These are commonly used in clocks and watches
(d) These take up torsional load

TNPSC AE 2019

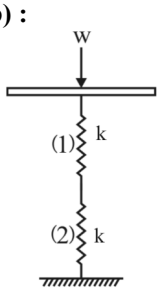
Ans. (b) : These are composed of coned discs which may be stacked upto obtain variety of load-deflection characteristics is true statement about Belleville springs.

1521. When two springs are in series (having stiffness k), the equivalent stiffness will be

(a) k (b) k/2
(c) 2k (d) k^2

Gujarat PSC AE 2019

Ans : (b) :



When two spring having stiffness k and are connected in series, then the combined stiffness of the spring ($k_{eq.}$) is given by

$$\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{k}$$

$$\frac{1}{k_{eq}} = \frac{1+1}{k}$$

$$\boxed{k_{eq} = \frac{k}{2}}$$

1522. A helical spring is subjected to an axial load W and corresponding deflection in the spring is δ. Now if the mean diameter of the spring is made half of its initial diameter keeping the material, number of turns and wire cross-section same, the deflection will be

(a) $\frac{\delta}{2}$ (b) $\frac{\delta}{8}$
(c) $\frac{\delta}{4}$ (d) 2δ

APPSC AEE SCREENING 17.02.2019

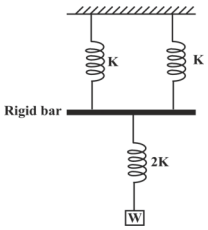
Ans. (b) : Deflection in spring $(\delta) = \frac{8WD^3n}{Gd^4} \Rightarrow \delta \alpha D^3$

$$\therefore \quad \frac{\delta_2}{\delta_1} = \left(\frac{D_2}{D_1}\right)^3$$

$$\frac{\delta_2}{\delta_1} = \left(\frac{\frac{D}{2}}{D}\right)^3$$

$$\Rightarrow \quad \delta_2 = \frac{\delta}{8}$$

1523. For the spring system given in figure, the equivalent stiffness is



(a) 5*K*/2 (b) 2 *K*/5
(c) *K* (d) 3*K*

ISRO Scientist/Engineer 22.04.2018

Ans. (c): In this system upper two springs are in parallel then equivalent spring stiffness is

$K_1 = K + K = 2K$

Then K_1 and 2K are in series then equivalent spring stiffness is K_2

$$\frac{1}{K_2} = \frac{1}{K_1} + \frac{1}{2K} = \frac{1}{2K} + \frac{1}{2K}$$

$$K_2 = K$$

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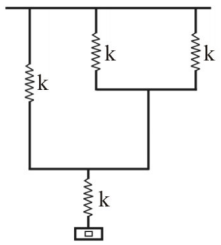
1524. Calculate equivalent stiffness of the spring for the system shown below, which has spring stiffness of 1000 N/m



(a) 7500 (b) 2000
(c) 750 (d) 4000

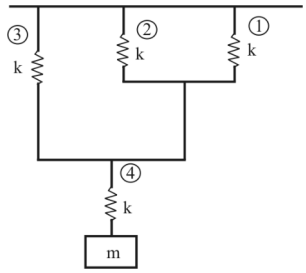
Vadodara Muncipal Corp. DEE, 2018

Ans. (c) : Given,

$k = 1000$ N/m

$k_{eq} = ?$

①&②in parallel



$k_{12} = 2k$

Now, k_{12} will be parallel with ③

$k_{13} = 2k + k = 3k$

k_{13} in series with ④

so, $k_{eq} = \frac{3k \times k}{3k + k} = \frac{3k^2}{4k} = \frac{3}{4} \times k$

$= \frac{3}{4} \times 1000$

$= 750 N/m$

1525. A closed coil helical spring has 15 coils. If five coils of this spring are removed by cutting the stiffness of the modified spring will :

(a) Increase to 2.5 times
(b) Increase to 1.5 times
(c) Reduced to 0.66 times
(d) Remain unaffected

GPSC ARTO Pre 30.12.2018

Ans. (b) : Given,

$$n = 15; \quad \text{stiffness } (k) = \frac{Gd^4}{8D^3 n} = \frac{Gd^4}{8D^3(15)}$$

$\because$ After cutting 5 coils,

$$n = 10 \quad k' = \frac{Gd^4}{8D^3(10)} \times \frac{15}{15}$$

$$k' = \frac{15}{10}\left[\frac{Gd^4}{8D^3(15)}\right]$$

$$\boxed{k' = 1.5k}$$

1526. Angle of helix in a closed coiled spring is :

(a) $< 10°$ (b) $> 10°$
(c) $= 10°$ (d) None

Kerala PSC Poly. Lect., 2017

Ans. (a)

- Helix angle of closed coil helical spring is below 10°.
- Helix angle of open coil helical spring is more than 10°.

1527. A Leaf spring in automobiles is used-

(a) To apply forces
(b) To measure forces
(c) To absorb shocks
(d) To store strain energy

GMB AAE 25.06.2017

Ans. (c) : The overall purpose a leaf spring is to provide support for a vehicle. It is also provides for a smoother ride absorbing any bumps in the road. Leaf spring are also used to locate the axle and control the height at which the vehicle rides and helps keep the tires aligned on the road.

1528. A 0.5 kg weight is attached to a light spring elongates it by 0.981 mm. The natural frequency of the system would be approximately

(a) 6 cps (b) 11 cps
(c) 16 cps (d) 21 cps

WBPSC AE, 2017

Ans. (c) : Given, $m = 0.5$ kg

Elongation, $\delta = 0.981$ mm

$$k = \frac{W}{\delta} = \frac{mg}{\delta}$$

$$= \frac{0.5 \times 9.81}{0.981 \times 10^{-3}}$$

$k = 5000$ N/m

Natural frequency, $f_n = \frac{1}{2\pi} \times \sqrt{\frac{k}{m}}$

$$\frac{1}{2\pi} \times \sqrt{\frac{5000}{0.5}} = \sqrt{\frac{50000}{5}}$$

$$f_n = \frac{1}{2\pi} \times 100 = \frac{50}{\pi}$$

$f_n = 15.916$

$f_n \approx 16$ cps or 16 cycle/sec

1529. Which of the following springs are subjected to both the torsion and bending?

P. Closed coil helical spring
Q. Closed coil helical spring subjected to axial couple
R. Open coil helical spring subjected to axial force
S. Open coil helical spring subjected to axial torque

(a) Only P and Q (b) Only Q and R
(c) Only R and S (d) P, Q and R

APGENCO AE, 2017

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Ans. (c) : Closed coil spring subjected to axial force subjected to torque only.
Closed coil spring subjected to axial couple subjected to bending only.
Open coil helical spring always experience bending and torsion both irrespective of loading either axial force or axial couple.

1530. The energy stored per unit volume in coil spring as compared to leaf spring is
(a) Equal amount (b) Double the amount
(c) Four times higher (d) Six times higher
TNPSC AE 2017

Ans. (d) : The energy stored per unit volume in coil spring as compared to leaf spring is six times higher.

1531. Due to addition of extra full length leaves the deflection of a semi-elliptic spring
(a) increases (b) decreases
(c) does not change (d) is doubled
TNPSC AE 2017

Ans. (b) : Due to addition of extra full length leaves the deflection of a semi-elliptic spring decreases.

1532. The ratio of total load on the spring to the maximum deflection is called is
(a) Spring Tension (b) Spring life
(c) Spring efficiency (d) Spring rate
TNPSC AE 2017

Ans. (d) : The ratio of total load on the spring to the maximum deflection is called is Spring rate is denoted by k. Spring rate also known as stiffness of spring.

$$K = \frac{W}{\delta_{max.}}$$

1533. For a spring mass system, the frequency of vibration is 'N' what will be the frequency when one more similar spring is added in series
(a) N/2 (b) $N/\sqrt{2}$
(c) $\sqrt{2}/N$ (d) 2N
TNPSC AE 2017

Ans. (b) : We know that,

$$N = \frac{1}{2\pi}\sqrt{\frac{s}{m}}$$

$$N \propto \sqrt{s}$$

When two similar spring is added in series then

$$s_{equivalent} = \frac{s}{2}$$

then,

$$\frac{N_2}{N_1} = \frac{\sqrt{s/2}}{\sqrt{s}} \qquad (N_1 = N)$$

$$\boxed{N_2 = \frac{N}{\sqrt{2}}}$$

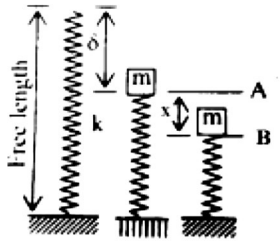
1534. The spring rate or stiffness (k) of the spring is given by:
(Where w load and δ deflection of spring)
(a) k = 2wδ (b) k = δ/w
(c) k = wδ (d) k = w/δ
CIL (MT) 2017 IInd Shift

Ans. (d) : Spring force w = kδ
So stiffness (k) = w/δ

1535. The device that permits variation in the distance between the spring eyes of leaf spring as the spring flexes is called :
(a) Spring shackle (b) Spring U bolt
(c) Spring hanger (d) Spring leaf
Kerala PSC Poly. Lect. (Auto) 05.05.2016

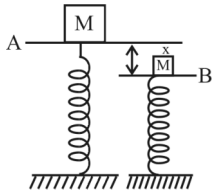
Ans. (a) The device that permits variation in the distance between the spring eyes of leaf spring as the spring flexes is called spring shackle.

1536. In the figure shown, the spring deflects by δ to position A (the equilibrium position) when a mass m is at a position B at some instant. The change in potential energy of the spring mass system from position A to position B is



(a) $\frac{1}{2}kx^2$ (b) $\frac{1}{2}kx^2 - mgx$
(c) $\frac{1}{2}k(x+\delta)^2$ (d) $\frac{1}{2}kx^2 + mgx$
RPSC ADE 2016

Ans. (b) :



$\frac{1}{2}kx^2 - mgx$

1537. The reason why a laminated spring is made up of a series of leaves is to:
(a) reduce the inter leaf friction
(b) soften the spring action and increase the maximum deflection
(c) allow the leave to slide during the bump movement
(d) overcome the weakness at the centre of a single leaf spring
RPSC VPITI 14.02.2016

Ans. (d) : As the laminated spring deflects the plates or leaves slide over each other causing inter-plate friction.
To overcome the weakness at the centre of a single leaf spring.

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1538. Shear stress in a closed coil helical spring (coil diameter 'D') of wire (diameter 'd') subjected to an axial load 'W' is

(a) $WD/\pi d^3$ (b) $(H_2 - H_1)/(H_3 - H_1)$

(c) $(H_3 + H_1)/(H_3 + H_2)$ (d) $(H_1 + H_2)/(H_1 + H_3)$

RPSC VPITI 14.02.2016

Ans. (*) : Note- No answer given by commission.

1539. The bending stress in full length leaves of a laminated spring is

(a) same as in graduated leaves

(b) 50% more than that in graduated leaves

(c) 50% less than that in graduated leaves

(d) 25% more than that in graduated leaves

RPSC AE (GWD) 06.05.2016

Ans. (b) : The bending stress in full length leaves of a laminated spring is 50% more than in graduated leaves.

1540. Which of the following types of energy is possessed by a tightly wound spring?

(a) Kinetic energy

(b) Magnetic energy

(c) Strain energy

(d) Electro-magnetic energy

KPCL AE 2016

Ans. (c) : When a tightly wound spring then energy is possessed in form strain energy. But a compressed spring has the potential energy in it due its compressed state when the compressed spring is released the potential energy changes to kinetic energy.

1541. When an open coiled helical spring (one end is fixed) in the vertical position is subjected to an axial tensile load and a couple, the strain energy stored will be due to .....

(a) bending moment only

(b) torsional moment only

(c) both bending moment and torsional moment

(d) shear force only

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : In this case both bending moment and torsional moment will apply.

■ Twisting in spring-

$$(\tau_s)_{max} = \frac{16T}{\pi d^3}$$

■ Bending in spring-

$$(\sigma_b)_{max} = \frac{32M}{\pi d^3}$$

1542. The reading of a spring balance is from 0 to 200 N and is 10 cm long. A body suspended from the spring balance is observed to oscillate vertically at 2 Hz. The mass of the body is nearly

(a) 22.5 kg (b) 12.5 kg

(c) 37 kg (d) 45 kg

ISRO Scientist/Engineer 03.07.2016

Ans. (b) : F = 200 N

Deflection = 10 cm

$(\delta) = 0.1$ m

Stiffness $k = \frac{F}{\delta}$

$$= \frac{200}{0.1} = 2000 \text{ N/m}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{2000}{m}}$$

$$4\pi = \sqrt{\frac{2000}{m}}$$

$$\frac{2000}{m} = 16\pi^2$$

$$m = \frac{2000}{16\pi^2}$$

$m = 12.66$ kg

$m \approx 12.5$ kg

1543. For a helical spring of mean diameter D. wire diameter d. number of coils n. modules of transverse elasticity C, when subjected to an axial load W the deflection would be

(a) $\frac{W_n D^3}{Cd^4}$ (b) $\frac{2Wnd^3}{CD^4}$

(c) $\frac{8Wn}{CD^4 d^3}$ (d) $\frac{8WnD^3}{Cd^4}$

Nagaland CTSE 2016-17 Ist Paper

Ans. (d) : In helical spring, when it is subjected to an axial load (W) the deflection (δ) would be,

$$\delta = \frac{8WD^3 n}{Cd^4}$$

Where,

D = mean diameter of spring

d = wire diameter

C = modules of transverse elasticity

W = axial load

n = no. of turn or coils

1544. A spring of stiffness 50 N/mm is mounted on top of second spring of stiffness 100 N/mm, what will be the deflection under the action of 500 N?

(a) 15 mm (b) 3.33 mm

(c) 5 mm (d) 10 mm

UPRVUNL AE 07.10.2016

Ans. (a) : Spring arrangement in series then,

$$\frac{1}{k_{eq}} = \frac{1}{100} + \frac{1}{50} = \frac{3}{100}$$

$$k_{eq} = \frac{100}{3} \text{ N/mm}$$

Then deflection (δ)

$$\delta = \frac{W}{k_{eq}} = \frac{500}{100} \times 3$$

$\delta = 15$ mm

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1545. When a closely coiled helical spring of mean diameter {D} is subjected to an axial load (W), the stiffness of the spring is given by

(a) Cd^4/D^3n (b) $Cd^4/2D^3n$
(c) $Cd^4/4D^3n$ (d) $Cd^4/8D^3n$

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : We know that deflection due to axial load (W) in a closely coiled helical spring is given as,

$$\delta = \frac{64WR^3n}{Cd^4}$$

$$\delta = \frac{8WD^3n}{Cd^4}$$

We know that, spring stiffness of spring.

$$K = \frac{W}{\delta}$$

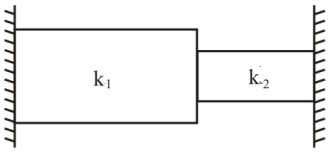
$$\boxed{K = \frac{Cd^4}{8D^3n}}$$

1546. A composite shaft consisting of two stepped portions having spring constant k_1 and k_2 is held between two rigid supports at the ends. Its equivalent spring constant is

(a) $(k_1 + k_2)/2$ (b) $(k_1 + k_2)/k_1k_2$
(c) $k_1k_2/(k_1+k_2)$ (d) $(k_1 + k_2)$

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) :



$$k_{equivalent} = k_1 + k_2$$

1547. A closed coiled helical spring is subjected to axial load (W) and absorbs 100 Nm energy at 4cm compression. The value of axial load will be:

(a) 12.5 kN (b) 5.0 kN
(c) 10.0 kN (d) 2.5 kN

UPRVUNL AE 07.10.2016

Ans. (b) : Energy absorbs in spring = 100 N-m

Average load on spring $= \frac{(W+0)}{2} = \frac{W}{2}$

Deflection in spring = 4 cm = 0.04 m

Then

$$\frac{W}{2} \times 0.04 = 100$$

$$W = \frac{200}{0.04} = 5000N$$

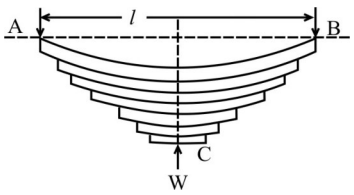
$\boxed{W = 5kN}$

1548. A laminated spring 1 m long carries a central point load of 2000 N. The spring is made of plates each 5 cm wide and 1 cm thick. The bending stress in the plates is limited to 100 N/mm^2. The number of plates required will be:

(a) 3 (b) 5
(c) 6 (d) 8

HPPSC W.S. Poly. 2016

Ans : (c)



Maximum bending stress (σ_{max}) $= \frac{3}{2}\frac{W\ell}{Nbt^2}$

l = spring length = 1m
W = central point load = 2000N
b = Width = 5 cm = 5×10^{-2}m
t = thickness = 1cm = 1×10^{-2}m
σ_{max} = maximum bending stress = 100N/mm^2

$$\sigma = \frac{3}{2}\frac{W\ell}{Nbt^2}$$

$$N = \frac{3}{2}\frac{W\ell}{\sigma bt^2}$$

$$N = \frac{3}{2} \times \frac{2000 \times 1}{100 \times 10^6 \times 5 \times 10^{-2} \times 1 \times 10^{-4}}$$

N = 6

1549. In a laminated spring the strips are provided in different lengths for

(a) Equal distribution of stress
(b) Equal distribution of strain energy
(c) Reduction in weight
(d) All are correct

MPPSC AE 2016

Ans : (a) In a laminated spring the strips are provided in distribution of bending stress.
Different length for equal distribution of bending stress therefore spring behave just like a uniform strength beam.

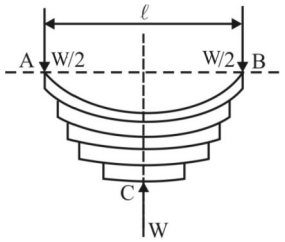
1550. In a leaf spring, the deflection at its centre is

(a) $\delta = Wl^3/8\,Enbt^3$ (b) $\delta = Wl^3/4\,Enbt^3$
(c) $\delta = 3Wl^3/8Enbt^3$ (d) $\delta = Wl^3/2Enbt^3$

TSPSC AEE 2015

(Where W = Max load on the pring
l - length of the spring
n = No. of plates
b - Width of the plates
t = thickness of the plates)

Ans : (c)



M = Maximum bending moment
I = Moment of Inerita.

Y = $\frac{t}{2}$ = half thickness of spring.

n = Number of leaf spring.

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Max bending moment = $\frac{W\ell}{4n}$

Maximum bending stress = $\frac{3}{2}\frac{W\ell}{nbt^2}$

maximum deflection = $\frac{3}{8}\frac{w\ell^3}{nEbt^3}$

1551. Type of spring used in door hinges is :

(a) Spiral spring
(b) Multi leaf spring
(c) Helical torsion spring
(d) Helical extension spring

GWSSB AAE, 27.12.2015

Ans. (c) : Helical torsion spring are particularly useful as door hinges from residential doors, garage doors, and automobile doors to large, industrial heavy duty overhead door used at loading docks and ware house.

1552. In a close-coiled helical spring subjected to an axial load and other quantities remaining the same, if the wire diameter is doubled, then the stiffness of the spring when compared to the original one will become

(a) two times (b) four times
(c) eight times (d) sixteen times

MPPSC AE 08.11.2015

Ans. (d) : Let, d = diameter of spring wire.
P = Axial load and K = stiffness of spring.
For the close - coiled helical spring, if all the quantities remaining same except diameter of spring wire.
$\therefore$ Stiffness of spring (K) $\propto d^4$.

$$\Rightarrow \frac{K_1}{K_2} = \left(\frac{d_1}{d_2}\right)^4 \qquad \because\ d_2 = 2d_1$$

$$\Rightarrow \frac{K_1}{K_2} = \left(\frac{d_1}{2d_1}\right)^4 = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

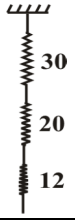
$\Rightarrow K_2 = 16\ K_1$
i.e., Sixteen times.

1553. When three springs are in series having stiffness 30 N/m, 20 N/m and 12 N/m, the equivalent stiffness will be

(a) 14 (b) 8
(c) 12 (d) 6
(e) 10

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Let stiffness of the composite spring be k_e
Given
$k_1 = 30$ N/m
$k_2 = 20$ N/m
$k_3 = 12$ N/m



Equivalent stiffness

$$\frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

$$\frac{1}{k_e} = \frac{1}{30} + \frac{1}{20} + \frac{1}{12}$$

$$\frac{1}{k_e} = \frac{2+3+5}{60}$$

$k_e = 6$ N/m

1554. A helical spring is made from a wire of 6 mm diameter and has outside diameter of 75 mm. If the permissible shear stress is 350 N/mm^2 and modulus of rigidity is 84 kN/mm^2, the axial load which the spring can carry without considering the effect of curvature.

(a) 422.5 N (b) 382.5 N
(c) 392.5 N (d) 412.5 N
(e) 402.5 N

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Given,
$d = 6$ mm, $D_0 = 75$ mm, $\tau = 350$ MPa, $G = 84$ kN/mm^2
We know that
$D = D_0 - d = 75 - 6 = 69$ mm

$\therefore$ spring index, $C = \frac{D}{d} = \frac{69}{6} = 11.5$

Neglecting the effect of curvature–
We know that shear stress factor

$K_s = 1 + \frac{1}{2C} = 1 + \frac{1}{2\times 11.5} = 1.043$ and maximum shear stress induced in the wire (τ)

$$350 = K_s \times \frac{8WD}{\pi d^3} = 1.043 \times \frac{8W \times 69}{\pi \times 6^3}$$

$$W = \frac{350}{0.848} = 412.7\ \text{N}$$

1555. Which is not a type of ends for helical compression spring

(a) Plain ends
(b) square ends
(c) half hook ends
(d) plain and ground ends
(e) square and ground ends

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : half hook ends

1556. Spring stiffness is :

(a) Load per unit deflection
(b) Load carrying capacity of the spring
(c) Ratio of mean coil diameter to wire diameter
(d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) : Spring stiffness –

$$k = \frac{W}{\delta}$$

Where, $k \rightarrow$ Spring stiffness
$W \rightarrow$ Load
$\delta \rightarrow$ Deflection

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1557. A helical spring, having N number of coils, has spring-constant 2000 N/m. If there is another helical spring of the same material and same wire diameter and having N/2 number of coils, its spring constant will be

(a) 4000 N/m (b) 2000 N/m
(c) 1000 N/m (d) 500 N/m

Haryana PSC Civil Services Pre, 2014

Ans. (a) Given,
Number of Coils = N
Spring constant (k) = 2000 N/m
If no. of coils = N/2
then k = ?

$$k = \frac{Gd^4}{8D^3N}$$

$$\boxed{k \propto \frac{1}{N}}$$

$$\Rightarrow \quad 2000 \times N = \frac{k \times N}{2}$$

$$\boxed{k = 4000\,N/m}$$

1558. The work done in stretching a spring of spring constant k by a length Δ is

(a) $k\Delta$ (b) $k\Delta^2$
(c) $k\Delta/2$ (d) $k\Delta^2/2$

MPPSC State Forest Service Exam, 2014

Ans. (d) : The work done in stretching a spring is equal to the potential energy of stretched spring.
As we know,

P.E. in a stretched spring $U = \frac{1}{2}k\Delta^2$

Where, k = Stiffness, Δ = Deflection/elongation

1559. A body weighing 500 kg falls 8 cm and strikes a 500 kg/cm spring. The deformation of the spring will be _____cm.

(a) 8 (b) 2
(c) 20 (d) 4

MPSC Poly. Lect. 2014

Ans. (d) : m = 500 kg
h = 8 cm
k = 500 kg/cm
kinetic energy of body = Energy stored in spring.

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2 \qquad [k = 500\ kg/cm = 500 \times 9.81\frac{N}{cm}]$$

$$mv^2 = kx^2$$

$$500 \times (\sqrt{2 \times 9.81 \times 8})^2 = 500 \times x^2 \times 9.81$$

$$\boxed{x = 4\,cm}$$

1560. The Levy-von Mises equation is relation between the ratios of:

(a) The components of strain and stress increments
(b) The components of strain increment and deviatoric stress and shear stresses
(c) Plastic strain increments and deviatoric stress and shear stresses
(d) The stress increments and plastic strain increments

HPPSC Asstt. Prof. 2014

Ans. (b) : Levy-von Mises flow equation

$$\varepsilon_x = \frac{\sigma_x}{E} - V\left(\frac{\sigma_y}{E} + \frac{\sigma_z}{E}\right)$$

So, Levy von Mises equation is the relation between the ratios of the component of strain increment and deviatoric stress and shear stresses.

1561. A helical spring of wire diameter 6 mm and spring index 6 is acted by an initial load of 750 N. After compressing it further by 12 mm the stress in the wire is 500 MPa. Find the number of active coils. (given : G = 84000 MPa)

(a) 10 (b) 18
(c) 24 (d) 30

UPRVUNL AE 2014

Ans : (b) Given, C = 6, d = 6 mm, τ_{max} = 500 MPa
G = 84000 MPa
Initial load = 750 N
Compression, δ = 12 mm

$$D = C \times d = 36$$

$$k_w = \frac{4 \times 6 - 1}{4 \times 6 - 4} + \frac{0.615}{6} = 1.2525$$

$$\because \quad \tau_{max} = \frac{k_w \times 8 \times F \times D}{\pi d^3}$$

$$500 = \frac{1.2525 \times 8 \times F \times 36}{\pi \times 6^3}$$

$$F = 940.6\ N$$

$$k = \frac{F}{\delta} = \frac{940.6 - 750}{12}$$

$$k = 15.83$$

No. of active coil –

$$N = \frac{Gd^4}{8D^3k}$$

$$= \frac{84000 \times 6^4}{15.83 \times 8 \times (36)^3}$$

$$N = 18.42$$

$$\boxed{N \simeq 18}$$

1562. Load p_c and p_0 respectively acting axially upon close coiled and open coiled helical springs of same wire dia, coil dia, no of coils and material to cause same deflection

(a) p_c/p_0 is 1, < 1 or > 1 depending upon α
(b) $p_c/p_0 = 1$
(c) $p_c/p_0 > 1$
(d) $p_c/p_0 < 1$

(HPPSC AE 2014)

Ans : (d) Load P_c and P_o respectively acting axially upon close coiled and open coiled helical spring of same wire dia, coil dia, no of coils and material to case same deflection

$$P_c/P_o < 1$$

1563. The load required to produce and deflection in a spring is called

(a) stiffness constant (b) strain constant
(c) rigidity constant (d) torsional constant

Mizoram PSC AE/SDO 2014, Paper-II

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Ans. (a) : Stiffness : It is defined as the load required for unit deflection in spring.

$$k = \frac{W}{\delta}$$

$k \rightarrow$ Stiffness constant

$W \rightarrow$ Load

$\delta \rightarrow$ Deflection

1564. A closed coil helical spring having mean diameter of coil 'D' and wire diameter 'd' subjected to axial force 'P'. The strain energy stored in the spring is

(a) $\frac{8PC^3n}{Gd}$ (b) $\frac{8PD^3n}{Gd^4}$

(c) $\frac{4P^2D^3n}{Gd^4}$ (d) $\frac{8PD}{\pi d^2}$

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : Strain energy $= \frac{1}{2}T.\theta$

$$= \frac{1}{2}T.\frac{TL}{GJ} = \frac{1}{2}\frac{T^2L}{GJ}$$

$$U = \frac{1}{2}\frac{\left(P\frac{D}{2}\right)^2 \times \pi Dn}{G.\frac{\pi}{32}.d^4}$$

$$\boxed{U = \frac{4P^2D^3n}{Gd^4}}$$

deflection of springs.

By castigliano's theorem

$$\delta = \frac{\partial U}{\partial P}$$

$$\boxed{\delta = \frac{8PD^3n}{Gd^4}}$$

1565. In a spring mass system if one spring of same stiffness is added in series, new frequency of vibration will be:-

(a) $\frac{\omega_n}{\sqrt{2}}$ (b) $\frac{2}{\omega_n}$

(c) $\frac{\omega_n}{2}$ (d) $\frac{\sqrt{2}}{\omega_n}$

UKPSC AE-2013, Paper-I

Ans. (a) : Case I -

$$\omega_n = \sqrt{\frac{K}{m}}$$

Case II - when spring is added in series.

$$K_{eq} = \frac{K}{2}$$

$$[\omega_n]_{II} = \sqrt{\frac{K}{2.m}}$$

$$[\omega_n]_{II} = \frac{\omega_n}{\sqrt{2}}$$

1566. Pick the correct statement about the maximum bending stress in the various leaf or laminated spring, assuming it has been designed ideally.

(a) It increases uniformly from the shortest leaf to the longest leaf.

(b) It is largest in the longest leaf.

(c) It is largest in the shortest leaf.

(d) It is same in all leafs.

RPSC AEN Pre-2013

Ans. (d) : Leaf springs (sometimes also known as laminated springs) are made up of beams of uniform strength and are normally used in automobiles to absorbs the shock and vibration produced by road bumps there by providing comfort to passengers.

1567. A closely coiled helical spring is cut into two equal parts. What will be the ratio of the deflection of any of the resulting spring to the deflection of the original spring for the same load?

(a) 2 (b) 1/2

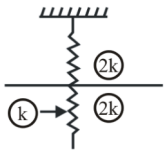
(c) 1 (d) 3/4

RPSC AEN Pre-2013

Ans. (b) : Deflection $(\delta) = \frac{8WD^3n}{Gd^4}$

If coil cut into two equal part –

Deflection $(\delta) = \frac{F}{k}$



$$\delta \propto \frac{1}{k}$$

$$\frac{\delta_1}{\delta} = \frac{\frac{F}{2k}}{F/k} \Rightarrow \boxed{\frac{\delta_1}{\delta} = \frac{1}{2}}$$

1568. A close-coiled helical spring absorbs 80 N mm of energy while extending by 4 mm. The stiffness of the spring is :

(a) 5 N/mm (b) 10 N/mm

(c) 16 N/mm (d) 20 N/mm

APPSC AE 04.12.2012

Ans. (b) : For a closed - Coiled helical spring -

Absorbe energy (U) $= \frac{1}{2}W.\delta$ … (i)

Given, δ = deflection of coil under the load W
= 4 mm

U = Absorbs energy
= 80 N-mm

$\therefore$ $W = \frac{2U}{\delta}$ [From eq. (i)]

$$W = \frac{2 \times 80N - mm}{4mm} = 40N$$

$\therefore$ Stiffness of spring (K) $= \frac{W}{\delta} = \frac{40N}{4mm} = 10N/mm$

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1569. Wahl suggested the correction in the stress factor to account for

(a) the additional transverse shear stress
(b) stress concentration
(c) fatigue stress
(d) none of these

TNPSC ACF 2012

Ans. (b) : Wahl suggested the correction in the stress factor to account for stress concentrations.

1570. A close coiled helical spring absorbs 500 Nmm of strain energy while extending by 10 mm. What is the stiffness of spring :

(a) 50 N/mm (b) 10 N/mm
(c) 5 N/mm (d) None of these

PSPCL AE, 2012

Ans. (b) : $U = \frac{1}{2}Kx^2$

$500 = \frac{1}{2} \times K \times 10^2$

K = 10 N/mm

1571. Match List – I with List – II and select the correct answer using the code given below the lists.

	List – I (Characteristic)		List – II (Member)
A.	Kernel of section	1.	Helical spring
B.	Tie and Strut	2.	Bending of beams
C.	Section modulus	3.	Eccentric loading of short column
D.	Stiffness	4.	Roof truss

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	2	1
(c)	4	1	2	3
(d)	2	3	1	4

UKPSC AE 2012 Paper-I

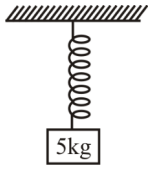
Ans. (b) :
Kernel of section - Eccentric loading of short column
Tie and Strut -Roof truss
Section modulus - Bending of beams
Stiffness - Helical spring

1572. A spring scale reads 20 N as it pulls a 5.0 kg mass across a table. what is the magnitude of the force exerted by the mass on the spring scale ?

(a) 4.0 N (b) 5.0 N
(c) 20.0 N (d) 49.0 N

UKPSC AE 2012 Paper-I

Ans. (c) :



- 5 kg mass is balance when 20 N applied then scale read 20 N.

1573. In a close-coiled helical spring

(a) plane of the coil and axis of the spring are closely attached
(b) angle of helix is large
(c) plane of the coil is normal to the axis of the spring
(d) deflection is small

BPSC AE 2012 Paper - VI

Ans : (c) : plane of the coil is normal to the axis of the spring.

1574. In a squared and ground helical spring, the effective number of turns is increased by

(a) one (b) two
(c) 15 (d) nil

TNPSC ACF 2012

Ans. (b) : In a squared and ground helical spring the effective number of turns is increased by-two.

1575. Stiffness of a spring can be increased by

(a) increasing the number of turns
(b) increasing the free length
(c) decreasing the number of turns
(d) decreasing the spring wire diameter

TNPSC ACF 2012

Ans. (c) : Stiffness of Spring $= \frac{W}{\delta} = \frac{Gd^4}{64R^3n}$

$\boxed{K \propto \frac{1}{n}}$

1576. Due to addition of extra full length leaves the deflection of a semi-elliptic spring

(a) increases (b) decreases
(c) does not change (d) is doubled

TNPSC ACF 2012

Ans. (b) : Addition of extra full length leaves the deflection of semi-elliptic spring decreases.

1577. A length of 10 mm diameter steel wire is coiled to a close coiled helical spring having 8 coils of 75 mm mean diameter and the spring has a stiffness K. If the same length of steel wire is coiled to 10 coils of 60 mm mean diameter, then the spring stiffness will be

(a) K (b) 1.95 K
(c) 1.56 K (d) 1.25 K

APPSC AEE 2012

Ans. (c) : wire diameter d = 10mm.
Mean diameter of first coil D_1 = 75mm
No. of Coils of first $N_1 = 8$
Mean diameter of second coil D_2 = 60 mm
No. of coils of second $N_2 = 10$

Then spring stiffness $k = \frac{Gd^4}{8D^3N}$

$$\frac{k_1}{k_2} = \frac{\frac{Gd^4}{8D_1^3N_1}}{\frac{Gd^4}{8D_2^3N_2}} = \frac{D_2^3N_2}{D_1^3 \times N_1} = \left(\frac{60}{75}\right)^3 \times \frac{10}{8}$$

$$\frac{k_1}{k_2} = 0.64 \Rightarrow \boxed{k_2 = 1.56k_1}$$

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1578. Cracks in helical springs used in railway carriages usually start on the inner side of the coil because of the fact that

(a) It has a lower curvature than the outer side
(b) It is subjected to higher stress than outer side
(c) It is subjected to higher cyclic loading than outer side
(d) It is more stretched than outer side during manufacturing

APPSC AEE 2012

Ans. (b) : Cracks in helical springs used in railway carriages usually start on the inner side of the coil because it is subjected to higher stress than outer side.

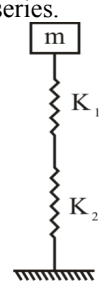
1579. If two springs with stiffness K_1 and K_2 are connected in series, then stiffness of the composite spring is given by

(a) $K_1 + K_2$ (b) $\frac{1}{K_1} + \frac{1}{K_2}$

(c) $\frac{1}{K_1} - \frac{1}{K_2}$ (d) $K_1 - K_2$

APPSC AEE 2012

Ans. (b) : If spring in series.



$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2}$$

or $$K_{eq} = \frac{K_1 . K_2}{K_1 + K_2}$$

$\Rightarrow$ If spring in parallel

$K_{eq} = K_1 + K_2$

1580. Two close-coil springs are made from a small diameter wire, one wound on 2.5 cm diameter core and the other on 1.25 cm diameter core. if each spring had 'n' coils, then the ratio of their spring constant would be

(a) 1/16 (b) 1/8
(c) 1/4 (d) 1/2

APPSC AEE 04.12.2012

Ans. (b) : $D_1 = 2.5$ cm, $D_2 = 1.25$ cm

No. of coils = n

$$\frac{K_1}{K_2} = ?$$

$$K = \frac{Gd^4}{8D^3 n}$$

$$K \propto \frac{1}{D^3}$$

$$\frac{K_1}{K_2} = \left(\frac{D_2}{D_1}\right)^3 = \left(\frac{1.25}{2.5}\right)^3 = \frac{1}{8}$$

1581. A mass is suspended at the bottom of two springs in series having stiffness 10 N/mm and 5N/mm. The equivalent spring stiffness of the two springs is nearly

(a) 0.3 N/mm (b) 3.3 N/mm
(c) 5 N/mm (d) 15 N/mm

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Given,

k_1 = stiffness of spring 1st = 10 N/mm

k_2 = stiffness of spring 2nd = 5 N/mm

For series,

k_{eq} (equivalent stiffness) $\Rightarrow \frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$

So, $k_{eq} = \frac{k_1 \times k_2}{k_1 + k_2} = \frac{10 \times 5}{15} = 3.3\,N/mm$

1582. A semi-spherical laminated leaf spring consisting of number of full length leaves say 'n_f' and number of graduated leaves say 'n_s' is subjected to a central load 'P'. The ratio of stress in full length leaves to the stress in graduated leaves will be:

(a) 1/1 (b) 2/1
(c) 3/2 (d) n_s/n_f

RPSC AE GWD, 2011

Ans. (d) : Maximum bending stress in full length leaves

$$(\sigma_b)_f = \frac{6F_f L}{n_f bt^2}$$

Maximum bending stress in Graduated leaves–

$$(\sigma_b)_s = \frac{6F_s L}{n_s bt^2}$$

Given, $F_f = F_s = F$

$\Rightarrow$ $$\boxed{\frac{(\sigma_b)_f}{(\sigma_b)_s} = \frac{n_s}{n_f}}$$

1583. Assertion (A) : In open-coiled helical springs, axial load causes normal stress and shearing stress in the spring wire.

Reason (R) : Helix angle is large in open-coiled helical springs.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : Pitch of helical springs is the axial distance between two similar points on the adjacent coils. Springs with helix angle larger than 10° is called open coil helical springs and helix angle less then 10° is called closed coil helical spring. Such spring have longer pitch.

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1584. The maximum shear stress induced in the wire of a circular section of a helical spring depends on

(a) material of the wire
(b) size of cross-section
(c) the ratio of d/D
(d) all of these

TNPSC AE, 2008

Ans. (b) : For closed coil helical spring under axial pull

$$\tau_{max} = \frac{8K_w FC}{\pi d^2}$$

where;
K_w = Wahl's factor
C = Spring index (D/d)
d = Spring wire diameter
F = Applied force
The maximum shear stress induced in the wire of a circular section of a helical spring depends on size of cross-section.

1585. Two spring of stiffness 'S_1' and 'S_2' are connected in parallel, what will be the stiffness of the composite spring?

(a) $S_2 + S_2$ (b) $S_2 \times S_2$
(c) $\frac{S_1}{S_1}$ (d) $\frac{(S_1\ S_2)}{(S_1 + S_2)}$

WBPSC AE 2008

Ans. (a) : For parallel –
equivalent stiffness (S_{eq}) = $S_1 + S_2$
For series –
equivalent stiffness $\frac{1}{S_{eq}} = \frac{1}{S_1} + \frac{1}{S_2}$

1586. Two concentric helical springs with spring constants 200 N/mm and 160 N/mm when subjected to a load of 1800 N will deflect by

(a) 5 mm (b) 10 mm
(c) 20.25 mm (d) 40.5 mm

DRDO Scientists 2008

Ans. (a) :

$$\frac{P_1}{P_2} = \frac{k_1}{k_2} = \frac{200}{160} = \frac{5}{4}$$

$$P_1 + P_2 = 1800$$

$$\left(1 + \frac{5}{4}\right) P_2 = 1800$$

$$P_2 = 800\ N$$

$$P_2 = k_2 x$$

$$800 = 160\ x$$

$$x = 5mm$$

1587. If *l* is the length of a leaf spring, t the thickness, of each leaf, then for the same stiffness, if *l* is doubled than t should be

(a) doubled (b) halved
(c) made four times (d) remain same

WBPSC AE, 2007

Ans. (a)

$$\delta_{spring}\ \alpha \frac{L^3}{t^3}$$

$$\Rightarrow K\ \alpha \frac{t^3}{L^3}$$

$$\text{or}\quad K = \frac{C \times t^3}{L^3}$$

where C = constant
for same stiffness

$$\frac{C \times t_1^3}{L_1^3} = \frac{C \times t_2^3}{L_2^3}$$

$$\frac{t_1^3}{L_1^3} = \frac{t_2^3}{(2L_1)^3}$$

$$t_1^3 = \frac{t_2^3}{8}$$

$$t_L^3 = 8t_1^3$$

$$\boxed{t_1 = 2t_1}$$

1588. In an open coiled helical spring an axial load on the spring produces which of the following stresses in the spring wire?

(a) normal (b) torsional shear
(c) direct shear (d) all of the above

UKPSC AE 2007 Paper -I

Ans. (d) : Under axial load spring will be under both bending and torsion. Thus is under complex stress therefore maximum principal stress and max. shear stress are found.

1589. When a helical coiled spring is compressed axially, it possesses

(a) potential energy (b) kinetic energy
(c) mechanical energy (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (a) : Potential energy

1590. A closed coil helical spring of stiffness 'K' is cut to $\frac{2}{3}$ of its original length. The stiffness of this spring of reduced length is

(a) $\frac{3}{2}K$ (b) K
(c) $\frac{2}{3}K$ (d) $\frac{1}{3}K$

OPSC Civil Services Pre 2006

Ans. (a) : Stiffness (K) $= \frac{W}{\delta} = \frac{Gd^4}{64R^3 n}$ or $K \propto \frac{1}{n}$

where n = no of coils

the new stiffness of spring will be $\frac{3}{2}K$

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03.

THEORY OF MACHINES

1. Mechanisms and Machines

1. The relation between the no. of pairs (p) forming a kinematic chain and the no. of links (l) is

(a) $l = 2p - 2$ (b) $l = 2p - 3$
(c) $l = 2p - 4$ (d) $l = 2p - 5$

CGPSC AE 15.01.2021
Sikkim PSC (Under Secretary), 2017
Punjab PSC (Lect.) 06.08.2017
CGPSC ARTO & TSI 07.01.2017
Rajasthan Nagar Nigam AE 2016, Shift-II
GPSC Asstt. Prof. 28.08.2016
Assam PSC CCE Pre 2015
GPSC Lect. (Poly.) 07.12.2014
APPSC Poly. Lect. 2013
TNPSC ACF 2012
TNPSC AE, 2008
CSE Pre-2000

Ans. (c) : The relation between the number of pairs(p) forming a kinematic chain and the number of links(l) is

$$l = 2p - 4$$

2. If there are L number of links in a mechanism, then the number of possible inversions would be:

(a) L (b) L/2
(c) L + 1 (d) L + 2

Nagaland PSC (CTSE) 2018 Paper-I
WBPSC AE, 2017
RPSC AE 2016
HPPSC W.S. Poly. 2016
CGPSC Poly. Lect. 22.05.2016
RPSC ACF-2011
OPSC Civil Services Pre 2006
CSE Pre-1996

Ans. (a) : If number of link = L
$\therefore$ Number of possible inversion $\leq$ L

3. Oldham's coupling is used to connect two shafts which :

(a) have lateral misalignment
(b) have two axes intersecting
(c) have lateral movements during power transmission
(d) changes axes during power transmission

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016
GPSC ARTO 01.05.2016
MPPSC AE 08.11.2015
Haryana PSC Civil Services Pre, 2014
Mizoram PSC AE/SDO 2014 Paper-II
RPSC ACF-2011

Ans. (a) : Oldham's couplings are three-piece flexible shaft coupling that are used to connect driving and driven shafts in mechanical power transmission assemblies. Flexible shaft couplings are used to counter the inevitable misalignment that occurs between connected shafts and in some cases, to absorb shock.

4. The driving and driven shafts connected by a Hook's joint will have equal speeds, if

(a) $\sin\alpha = \theta$ (b) $\tan\theta = \pm\sqrt{\cos\alpha}$
(c) $\tan\theta = \alpha$ (d) $\cot\alpha = \theta$

GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021
Nagaland PSC (CTSE) 2018 Paper-I
GPSC Asstt. Director of Transport 05.03.2017
TSPSC Managers, 2015
ESE 2012

Ans. (b) : The driving and driven shafts connected by a Hooke's joint will have equal speed if.

$$\tan\theta = \pm\sqrt{\cos\alpha}$$

5. The mechanism used in a shaping machine is :

(a) a closed 4- bar chain having 4 revolute pairs
(b) a closed 6- bar chain having 6 revolute pairs
(c) a closed 4- bar chain having 2 revolute and 2 sliding pairs
(d) an inversion of the single slider-crank chain

OPSC AEE 2019 Paper-I
GPSC ARTO Pre 30.12.2018
HPPSC Asstt. Prof. 18.11.2016
UPRVUNL AE 21.08.2016
VIZAG Steel MT 10.06.2012
PSPCL AE, 2012
GATE 2003

Ans. (d) : The mechanism used in shaping machine is quick return motion mechanism having an inversion of single slider crank chain.
The depth of cut increment by moving the workpiece and the workpiece is fed by a pawl and ratchet mechanism.

Inversions of single slider crank chain

(i) Pendulum pump or bull engine.
(ii) Oscillating cylinder engine.
(iii) Rotary internal combustion engine
(iv) Crank and slotted lever quick return motion mechanism.
(v) Whitworth quick return motion mechanism.

6. The Grubler's criterion for obtaining the degree of freedom (F) of a planar mechanism with 'n' number of links and 'j' number of binary joints, is given by :

(a) $F = 3(n-1) - j$ (b) $F = 3(n-1) - 2j$
(c) $F = 2(n-1) - j$ (d) $F = 2(n-1) - 2j$

ISRO Scientist/Engineer 07.05.2017
HPPSC Asstt. Prof. 2014
TNPSC AE (Industries) 09.06.2013
TNPSC ACF 2012
OPSC Civil Services Pre. 2011
OPSC Civil Service Pre-2011
ESE 2003

Ans. (b) : The Grubler's criterion for obtaining the degree of freedom (F) of a planar mechanism with 'n' number of links and 'j' number of binary joints, is given by : $F = 3(n-1) - 2j$

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7. Which the following is an inversion of single slider crank chain?
(a) Beam engine
(b) Watt's indicator mechanism
(c) Elliptical trammel
(d) Whitworth quick return motion mechanism

SJVN ET 2019
MPPSC State Forest Service Exam, 2014
TNPSC ACF 2012
TRB Poly. Lect., 2012
ESE 2008
OPSC Civil Services Pre 2006

Ans. (d) : Inversions of single slider crank chain
First Inversion – cylinder is fixed
Example–
(1) Reciprocating engine
Second inversion – crank is fixed
Example–
(1) Whitworth quick return mechanism.
(2) Rotary engine.
Third inversion – connecting rod is fixed
Example–
(1) Oscillating cylinder engine.
(2) Crank and slotted lever mechanism.
Fourth inversion – Slider is fixed
Example– Hand pump

8. A planar mechanism has 8 links and 10 rotary joints. The number of degrees of freedom of the mechanism, using Grubler's criterion, is :
(a) 0 (b) 1
(c) 2 (d) 3

OPSC AEE 2019 Paper-I
GPSC ARTO Pre 30.12.2018
CGPSC AE 26.04.2015 Shift-I
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2008

Ans : (b) : Degree of freedom (DOF) of a planer mechanism by Grubler's criterion,

$DOF = 3(\ell - 1) - 2j - h$

Where,
$\ell = 8$ (Number of links)
$j = 10$ (Number of binary joints)
$h = 0$ (Number of higher pairs)

$DoF = 3(8-1) - 2 \times 10 - 0$

$DOF = 1$

9. Which of the following is/are lower pair(s)?
(a) Ball and socket (b) Piston and cylinder
(c) Cam and follower (d) Both (A) and (B)

Assam PSC AE (PHED), 18.10.2020
Gujarat PSC AE 2019
WBPSC AE, 2017
UPPSC AE 12.04.2016 Paper-I
ISRO Scientist/Engineer 2008

Ans. (d) : Lower pair- A pair of links having surface or area contact between the member.
Example- Nut turning on a screw, shaft rotating in bearing, all pairs of slider crank mechanism, piston and cylinder, ball and socket etc.
Higher pair- When a pair has point or line contact between the links it is known as higher pair.
Example- Wheel rolling on a surface, cam and follower pair.

10. A ball and socket joint is example of _____ pair.
(a) Screw (b) Spherical
(c) Turning (d) Rolling

GPSC DEE, Class-2 (GWSSB) 04.07.2021
SJVN ET 2019
CIL MT 26.03.2017
HPPSC W.S. Poly. 2016
TNPSC AE (Industries) 09.06.2013

Ans. (b) : When the two elements of a pair are connected in such a way that one element (with spherical shape) turns or swivels about the other fixed element, the pair formed is called a spherical pair. The ball and socket joint, attachment of a car mirror, pen stand etc., are the examples of a spherical pair.

11. Mechanism is said to be converted to structure if the degree of freedom of mechanism reduced to :
(a) 3 (b) 1
(c) 0 (d) 2

Assam PSC AE (IWT) 14.03.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
APPSC AEE SCREENING 17.02.2019
CIL MT 26.03.2017
Nagaland PSC CTSE 2017, Paper-I
GMB AAE 25.06.2017

Ans. (c) : When the degree of freedom of a kinematic chain is zero then it is called structure. A structure with negative degree of freedom is known as super structure.

12. If the elements of a kinematic pair make surface contact when in motion, then it is called a:
(a) Lower pair (b) Closed pair
(c) Higher pair (d) Surface pair

JPSC AE 10.04.2021, Paper-II
Rajasthan AE (Nagar Nigam) 2016 Shift-3
Haryana PSC Civil Services Pre, 2014
RPSC AEN Pre-2013

Ans. (a) : Lower pair- If the element of a kinematic pair make surface contact or area contact.
Example–Nut and screw, shaft rotating in bearing Universal joint etc.
Higher Pair–If the element of a kinematic pair make point contact or line contact.
Example–Cam and follower.

13. Inversion of a mechanism means
(a) turning it upside down
(b) fixing different links in a kinematic chain
(c) changing a higher pair to lower pair
(d) changing the input and output links

GPSC Executive Engineer 23.12.2018
Assam PSC CCE Pre 2015
MPPSC State Forest Service Exam, 2014
APPSC IOF, 2009
ESE 1992

Ans. (b) : Inversion of a mechanism means fixing different links in a kinematic chain to obtain various mechanisms.

14. In a four-bar chain or quadratic cycle chain
(a) each of the four pairs is a turning pair
(b) one is a turning pair and three are sliding pairs

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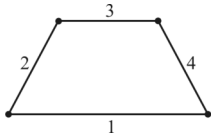
(c) two are turning pairs and two are sliding pairs
(d) three are turning pairs and one is a sliding pair

GPSC DEE, Class-2 (GWSSB) 04.07.2021
UPRVUNL AE 05.07.2021
Maharashtra PSC AE Pre 2019
JPSC AE 2013, Paper-V

Ans : (a) : A four bar chain have all the links binary and all connection a turning pair or revolute pair as shown in figure below.



Link 1, 2, 3 and 4 are connected by turning pair.

15. In a 4– bar linkage, if the lengths of shortest, longest and the other two links are denoted by s, l, p and q, then it would result in Grasshof's linkage provided that:

(a) $\ell + p < s + q$ (b) $\ell + s < p + q$
(c) $\ell + p = s + q$ (d) None of these

GPSC DEE Class-2 (GWSSB) 04.07.2021
OPSC AEE 2019 Paper-I
Nagaland PSC CTSE 2017, Paper-I
GATE 2006

Ans. (b) : According to, Grasshof's law, for continuous relative motion in four bar mechanism, it require to satisfy the equation.

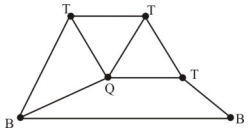
$$\ell + s < p + q$$

16. In kinematic chain, a quarternary joint is equivalent to

(a) one binary joint (b) two binary joints
(c) three binary joints (d) six binary joints

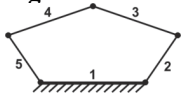
TSPSC Manager (Engg.) HMWSSB 12.11.2020
CGPSC AE 25.02.2018
UPRVUNL AE 07.10.2016
SJVN ET 2013
TNPSC ACF 2012
ESE 2005

Ans. (c) : Four links are connected at the same joint by a pin.



Q is Quaternary joint
A quaternary joint is equivalent to three binary joints.
A ternary joint is equivalent to two binary joints.

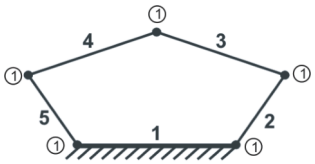
17. The number of degree of freedom of a five link place mechanism with five revolute pairs as shown in the figure is



(a) 3 (b) 4
(c) 1 (d) 2

MECON MT 2019
GPSC ARTO Pre 30.12.2018
ISRO Scientist/Engineer 2011
GATE 1993

Ans. (d) :



No. of link l = 5
No. of joint j = 5
No. of higher pair = h = 0

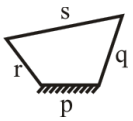
$$F = 3(l-1) - 2J - h$$
$$= 3(5-1) - 2\times5 - 0$$
$$= 3 \times 4 - 10$$
$$= 12 - 10 = \mathbf{2}$$

18. The lengths of the links of a 4 bar linkage with revolute pairs only are p, q, r and s units. Given that p<q<r<s. Which one of these links should be fixed, for obtaining a double crank mechanism?

(a) Link of length 'p' (b) Link of length 'q'
(c) Link of length 'r' (d) Link of length 's'

GPSC ARTO Pre 30.12.2018
Karnataka PSC Lect., 27.05.2017
RPSC ADE 2016
APGENCO AE 2012
GATE 2003

Ans (a) :



Shortest link is p and longest link is s, here q and r is intermediate link hence by fixing shortest link we will get double crank mechanism.

19. Which one of the following is a lower pair?

(a) cam and follower
(b) toothed gearing
(c) shaft in a bearing
(d) ball and race in bearing

RPSC AE (GWD) 06.05.2016
Kerala PSC AE 06.08.2015
Kerala PSC Poly. Lect. 09.01.2014
(KPSC AE 2015)

Ans. (c) : Kinematic Pairs According to the type of contact–
(a) Lower Pair- When the two elements of a pair have surface (area) contact while in motion the pair is called a lower pair All- sliding pair, turning pair and screw pair form lower pair.
e.g. – A nut turning on screws, a shaft rotating in a bearing, an universal joint pantograph etc.
(b) Higher Pair – When two elements have point or line contact while in motion.
e.g. – Belt rope and chain drives, gears, the cam and follower, ball and roller bearings etc.

20. Hooke's joint can be used for connecting two-

(a) Non-parallel and non-intersecting shafts
(b) Parallel and non-intersecting shafts
(c) Parallel and intersecting shafts

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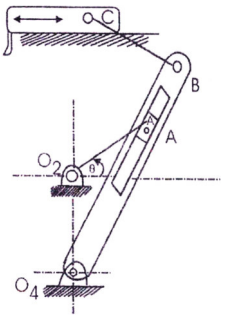
(d) Non-parallel and intersecting shafts

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
HPPSC AE (PWD) 10.09.2016
ESE 2005

Ans. (d) : Hooke joint- Hooke joint is used to connect two non-parallel but intersecting shafts. It transmits power from the gear box of the engine to the rear axle. It has two U-shaped yokes and a centre block or cross-shaped piece. The angular velocity ratio is not uniform during the cycle of operation. The amount of fluctuation depends on the angle between the two shafts. For uniform transmission of motion, a pair of universal joint should be used.

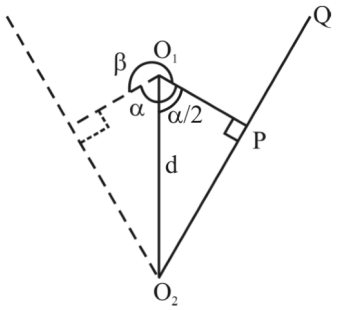
21. A simple quick return mechanism is shown in the figure. The forward to return ratio of quick return mechanism is 2 :1. If the radius of the crank O_2A is 125 mm, then the distance between the crank centre to lever centre point (O_2O_4) should be :



(a) 144.3 mm (b) 216.5 mm
(c) 240.0 mm (d) 250.0 mm

APGENCO AE, 2017
OPSC Civil Services Pre. 2011
GATE 2009

Ans. (d) :



$$\text{Quick return ratio} = \frac{\text{Time of cutting stroke}}{\text{Time of Return stroke}}$$

$$\frac{\beta}{\alpha} = \frac{2}{1}$$

$$\beta = 2\alpha$$

$$\therefore \alpha + \beta = 360^0$$

$$\alpha = 120^0$$

In triangle $\Delta O_1 O_2 P$

$$d = \frac{O_1P}{\cos\left(\frac{\alpha}{2}\right)} = \frac{125}{\cos 60^0} = 250 \text{ mm}$$

$\boxed{d = 250 \text{ mm}}$

22. A planar linkage having 8 links and 9 joints of single degree of freedom will have ..... degree of freedom.

(a) 1 (b) 2
(c) 3 (d) 4

RPSC AE 2018
AAI Jr. Executive 29.11.2018
GATE 2015

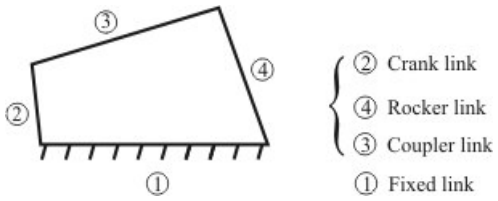
Ans. (c) : Data given:
No. of links (l) = 8
No. of joints (j) = 9
Then DOF = ?
DOF = 3(l -1) - 2j - h
DOF = 3 (8 - 1) - 2 × 9 - 0
= 3 × 7 - 18
$\boxed{\text{DOF} = 3}$

23. A mechanism is an assemblage of :

(a) three links
(b) four links or more than four links
(c) None of three mentioned here.
(d) two links

JPSC AE 10.04.2021, Paper-II
GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015

Ans. (b) : A mechanism is an assemblage of four links or more than four links.



24. In a kinematic chain the minimum number of kinematic pairs required is

(a) one (b) two
(c) three (d) four

WBPSC AE, 2017
RPSC AE 2016
RPSC ACF-2011
GATE 2002

Ans. (c) : If there is all pairs are lower pair of a kinematic chain then 4 pair required to make a kinematic chain. But, if there is atleast one higher pair involved then minimum 3 pairs required to make a kinematic chain. Cam and follower is example of a 3 pair kinematic chain.

One higher pair = Two lower pair

25. A Universal joint is an example of

(a) lower pair (b) sliding pair
(c) rolling pair (d) higher pair

Rajasthan AE (Nagar Nigam) 2016 Shift-3
GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015
UKPSC AE-2013, Paper-I

Ans. (a) : When the two element of a pair have surface contact when relatives motion take place and surface of one element slide over the surface of other the pair formed pair is known as lower pair.
Example – Turning pair, universal joint, shaft rotating in bearing automobile steering system

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26. The lead screw of a lathe with nut is a

(a) screw pair (b) sliding pair
(c) turning pair (d) rolling pair

Jharkhand Urja Vikas Nigam Ltd. AE 2017
NPCIL ET, 2015
TNPSC ACF 2012

Ans. (a) : When the nature of contact between the element of a pair is such that one element can turn about the other by screw threads, the pair is known as a screw pair. The lead screw of lathe with nut is an example of screw pair.

27. Which of the following is the inversion of double slider crank chain?

(a) Beam Engine
(b) Elliptical Trammel
(c) Watt's indicator mechanism
(d) Quick return mechanism

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC DEE, Class-2 (GWSSB) 04.07.2021
APPSC AEE 2012

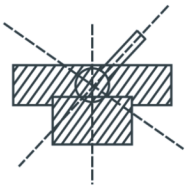
Ans. (b) : The inversion of a double slider crank chain are oldham's coupling, elliptical trammel & scotch yoke mechanism etc.

28. A spherical joint has _____ degree (s) of freedom.

(a) One (b) Two
(c) Three (d) Four

ISRO Scientist/Engineer 07.05.2017
APPSC AEE 2012
UKPSC AE 2007 Paper -I

Ans. (c) :



Linear moment x-axis or Δx not possible
Linear moment y-axis Δy, not possible
Linear moment z-axis Δz, not possible
Only three rotational motion is possible
DOF [Degree of freedom]
= 6 - Restraints
Restraints - No. of the motion which are not possible
DOF = 6 – 3 = 3

29. The following is not the inversion of slider crank mechanism

(a) Whit-worth quick return mechanism
(b) Scotch-yoke mechanism
(c) hand pump
(d) oscillating cylinder engine

GPSC Engineer, Class-II Pre-19.01.2020
GPSC ARTO 01.05.2016
Haryana PSC Civil Services Pre, 2014

Ans. (b) : Inversion single slider crank mechanism–
(i) Whit-worth quick return mechanism.
(ii) Oscillatory cylinder engine
(iii) Hand pump
Inversion of Double slider crank mechanism-
(i) Scotch yoke mechanism.
(ii) Elliptical trammel
(iii) Oldham's coupling

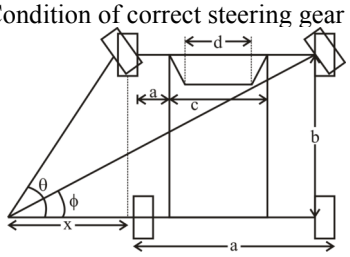
30. Condition for correct steering for a four-wheeler is

(a) $\cos\phi - \cos\theta = \frac{W}{H}$ (b) $\sin\phi - \sin\theta = \frac{W}{H}$
(c) $\tan\phi + \cot\theta = \frac{W}{H}$ (d) $\cot\phi - \cot\theta = \frac{W}{H}$
(e) none of these

Where W = Wheel base
H = Distance between the pivots of front axle, and
ϕ, θ are angles through which axis of outer and inner wheels turn, respectively.

RPSC IOF, 2020
CGPSC AE 16.10.2016
GPSC Lect. (Auto) 16.10.2016

Ans. (e) : Condition of correct steering gear –



$$\cot\phi = \frac{c+x}{b} = \frac{c}{b} + \frac{x}{b} = \frac{c}{b} + \cot\theta$$

So, $$\cot\phi - \cot\theta = \frac{c}{b} = \frac{H}{W}$$

$c \rightarrow$ Distance between the pivot centres = H
$b \rightarrow$ Wheel base of the vehicle = W

31. Automobile steering gear is an example of :

(a) Higher Pair (b) Lower Pair
(c) Sliding Pair (d) Turning Pair

Vizag Steel (MT) 2017
Assam PSC CCE Pre 2015
UPRVUNL AE 2014

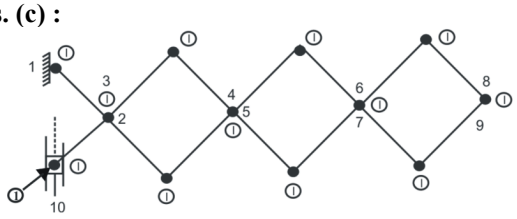
Ans. : (b) Automobile steering gear is an example of lower pair.
- Surface to surface contact is known as lower pair.
- Point to point contact is known as higher pair.

32. The kinematic planar chain shown in the given figure is a

(a) Structure
(b) Mechanism with 2 degree of freedom
(c) Mechanism with 1 degree of freedom
(d) Mechanism with more than 2 degree of freedom

OPSC Civil Services Pre. 2011
ISRO Scientist/Engineer 2009
ESE 2000

Ans. (c) :



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No. of Link = 10
No. of joint = 13
Degree of freedom
$F = 3(l-1) - 2j - h$
$= 3(10 - 1) - 2 \times 13$
$= 3 \times 9 - 26$
$= 27 - 26$
So $F = 1$

33. Scotch yoke mechanism is used to generate :
(a) Sine functions (b) Square roots
(c) Logarithms (d) Inversions

OPSC AEE 2019 Paper-I
UPRVUNL AE 21.08.2016
ESE 1992

Ans : (a) : Scotch yoke mechanism is an inversion of double slider crank mechanism. It is used to convert rotary motion into sliding motion. It generates (sin and cos) functions.

34. The number of inversions for a slider crank mechanism is :
(a) 6 (b) 5
(c) 4 (d) 3

GPSC ARTO Pre 30.12.2018
GPSC Lect. (Auto) 16.10.2016, GATE 2006

Ans. (c) : The number of inversions for a slider crank mechanism is 4.
(i) When crank is fixed – Rotary Engine, Whitworth quick return mechanism.
(ii) When connecting rod is fixed – Crank and slotted lever quick return mechanism, oscillating cylinder engine.
(iii) When slider is fixed – Pendulum pump or Bull engine.
(iv) When piston cylinder fixed – Reciprocating engine mechanism.

35. Which of the following statements is incorrect?
(a) Grashof's rule state that for a planar crank rocker four-bar mechanism, the sum of the shortest and longest link lengths cannot be less than the sum of the remaining two link lengths.
(b) Inversions of a mechanism are created by fixing different links one at a time
(c) Geneva mechanism is an intermittent motion device
(d) Gruebler's criterion assumes mobility of a planar mechanism to be one.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC ARTO Pre 30.12.2018, GATE 2010

Ans. (a) : (i) Grashof's rule states that the sum of shortest and longest link lengths less than or equal to sum of remaining two link lengths.
Grashof's law,
$S + l \le p + q$
(ii) Inversions of a mechanism are created by fixing different links one at a time.
(iii) Geneva mechanism is an intermittent motion device.
(iv) Gruebler's criterion assumes mobility of a planar mechanism to be one.

36. In slider crank mechanism, the maximum acceleration of slider is obtained when the crank is
(a) at the inner dead centre position
(b) at the outer dead centre position
(c) exactly midway position between the two dead centres
(d) slightly in advance of the midway position between the two dead centres.

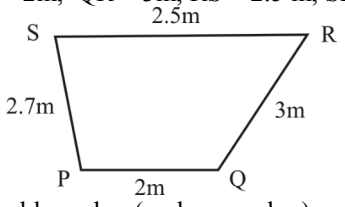
MPPSC AE 08.11.2015
ESE 2001

Ans. (a) : In slider crank mechanism, the maximum acceleration of slider is obtained when the crank is at the inner dead centre position.

37. A planar closed kinematic chain is formed with rigid links PQ = 2.0m, QR = 3.0m, RS = 2.5 m and SP = 2.7 m with all revolute joints. The links to be fixed to obtain a double rocker (rocker-rocker) mechanism is :
(a) PQ (b) QR
(c) RS (d) SP

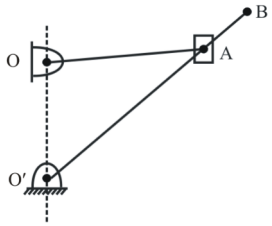
GPSC ARTO Pre 30.12.2018
GATE 2013

Ans. (c) : Given,
For a closed kinematic chain
PQ = 2m, QR = 3m, RS = 2.5 m, SP = 2.7m



To obtain double rocker (rocker - rocker) mechanism.
Opposite to shortest link should be fixed i.e. RS should be fixed.

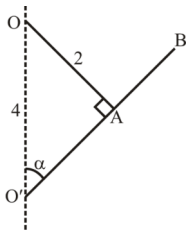
38. Figure shows a quick return mechanism. Crank OA rotates clockwise uniformly OA = 2 cm, OO' = 4 cm. The ratio of time for forward motion to that for return motion is



(a) 0.5 (b) 2.0
(c) $\sqrt{12}$ (d) 1

APGENCO AE 2012
GATE 1995

Ans. (b) : Given, $OO' = 4cm$
$OA = 2cm$



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$$\sin\alpha = \frac{2}{4}$$
$$\sin\alpha = 0.5$$
$$\boxed{\alpha = 30^\circ}$$

Ratio of time of forward motion to return motion

$$= \left(\frac{180+2\alpha}{180-2\alpha}\right) = \frac{180+60}{180-60}$$
$$= \frac{240}{120} = 2$$

39. Which one of the following is a higher pair ?

(a) Belt and Pulley (b) Turning Pair
(c) Screw Pair (d) Sliding Pair

Nagaland PSC (CTSE) 2018, Paper-I
OPSC Civil Services Pre. 2011

Ans. (a) :
■ Belt and pulley is higher pair.
■ One link is wrapped over another link.
■ A pair of two links having a rolling motion relative to each other is called rolling pair.

40. Ackerman steering gear consist of

(a) Lower pair (b) Turning pair
(c) Rolling pair (d) Cylindrical pair
(e) Sliding pair

CGPSC AE 26.04.2015 Shift-I
TNPSC AE, 2008

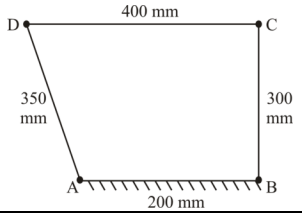
Ans. (b) : The Ackerman steering mechanism is a geometric arrangement of linkages in a steering of a vehicle to turn the inner and outer wheels at the appropriate angles. The Ackerman steering gear consists of turning pairs rather than sliding pair.

41. ABCD is a mechanism with link lengths AB = 200 mm; BC = 300 mm; CD = 400 mm and DA = 350 mm. Which one of the links should be fixed for the resulting mechanism to be a double crank mechanism?

(a) AB (b) BC
(c) CD (d) DA

TNPSC AE 2014
ESE 2004

Ans. (a) : AB = 200 mm, BC = 300 mm
CD = 400 mm, DA = 350 mm
If shortest link of the mechanism is fixed then the resulting mechanism become double crank mechanism so link AB is fixed because of AB is shortest link among them.



42. A link, which is free in space, has degree of freedom equal to

(a) 3 (b) 6
(c) 5 (d) 4

BPSC AE 2012 Paper - VI
UKPSC AE 2007 Paper -I

Ans : (b) : Degree of freedom of a pair is defined as the number of independent relative motions both translational and rotational, a pair can have Degree of freedom = 6

43. In a single slider four-bar linkage, when the slider is fixed, it forms a mechanism of :

(a) Hand pump
(b) Reciprocating engine
(c) Quick return
(d) Oscillating cylinder

OPSC AEE 2019 Paper-I
ESE 1999

Ans : (a) : The slider 4 is fixed and link-1 moves/slides through slider 4. The resulting mechanism is handpump.

44. Match the following :

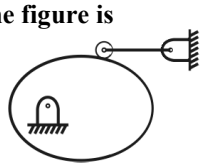
Column-I	Column-II
(P) Higher Kinematic pair	(i) Grubler's equation
(Q) Lower Kinematic pair	(ii) Line contact
(R) Quick return mechanism	(iii) Euler's equation
(S) Mobility of a linkage	(iv) Planer
	(v) Shaper
	(vi) Surfaces contact

(a) P–II, Q–VI, R–IV, S–III
(b) P–VI, Q–II, R–IV, S–I
(c) P–VI, Q–II, R–V, S–III
(d) P–II, Q–VI, R–V, S–I

OPSC AEE 2019 Paper-I
RPSC 2016

Ans : (d) :
■ Higher Kinematic Pair → Line contact
■ Lower Kinematic Pair → Surface contact
■ Quick return mechanism → Shaper
■ Mobility of a linkage → Grubler's equation

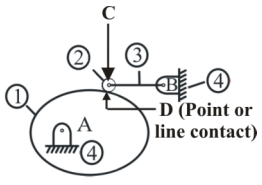
45. The degree of freedom of the mechanism shown in the figure is



(a) two (b) zero
(c) one (d) negative one

BPSC Poly. Lect. 2016
VIZAG MT 2015

Ans : (c)



link L = 4 (1, 2, 3, 4)
J = 3 (A, B, C)
h = 1 (D)
F_r = 1 (link 2 and link 3) (Redundant Motion)
DOF = 3 (L - 1) - 2J - h - F_r
DOF = 3 (4-1) - 2 × 3 - 1-1
$\boxed{\text{DOF} = 1}$

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46. The Scott Russell mechanism consist of
(a) Sliding pair only
(b) Sliding and rotary pair
(c) Turning and rotary pair
(d) Sliding and turning pair

Rajasthan Nagar Nigam AE 2016, Shift-II
RPSC AEN Pre-2013

Ans. (d) : Scott Russel mechanism has four link consists of turning pair and sliding pair.

47. The mechanism used in internal combustion engine is :
(a) crank mechanism
(b) slider mechanism
(c) slider crank mechanism
(d) circular mechanism

RPSC AE 2016
RPSC ACF-2011

Ans. (c) : Slider crank mechanism used in internal combustion engine. Internal combustion engine consists of a fixed cylinder, a reciprocating piston. The combustion gases pushes the piston which in turn rotates the crankshaft.

48. In an automobile the power is transmitted from gear box to differential through
(a) knuckle joint (b) universal joint
(c) Hooke's joint (d) bevel gears

RPSC AE 2016
RPSC ACF-2011

Ans : (b&c) In an automobile the power is transmitted from gear box to differential through universal joint.
Universal or hook's joint:- A hook's joint is used to connect two shaft which are intersecting at a small angle. The end of each shaft is forked to U-type and each fork provides to bearing for the arms of a cross. In case of automobiles, we use to hook's joint one at each end to the propeller shaft, connecting gear box on one and the differential on the other end.

49. Kinematic pairs are those which have
(a) two elements that do not permit relative motion
(b) two elements that permit relative motion
(c) elements of pair held together mechanically
(d) elements of pair not held together mechanically

RPSC VPITI 14.02.2016
TNPSC AE, 2008

Ans. (b) : Kinematic pairs are those which have two elements that permit relative motion.

50. The function of universal joint is to allow propeller shaft to :
(a) Change the length
(b) Change the inclination
(c) Transfer the torque at an angle
(d) None of these

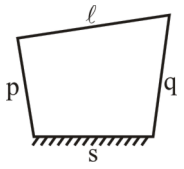
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021
GPSC Lect. (Auto) 16.10.2016

Ans. (c) : • A universal joint is a joint or coupling in a rigid rod that allows the rod to bend in any direction.
• The purpose of a universal joint is to transfer torque from one plane to another, retaining whatever rotational energy applied to the shaft.

51. The four bar mechanism in which the shortest link is fixed is termed as _____.
(a) double rocker mechanism
(b) compound mechanism
(c) Double crank mechanism
(d) crank rocker mechanism

VIZAG Steel MT 24.01.2021
Assam PSC AE (PHED), 18.10.2020

Ans. (c) :

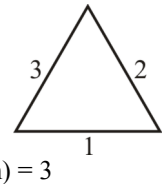


According Grashoff's Law–
$(s + \ell) \le (p + q)$
Short link (s)-fixed – Double crank mechanism

52. A framed structure of a triangular shape is _____.
(a) imperfect (b) redundant
(c) deficient (d) perfect

VIZAG Steel MT 24.01.2021 Shift-I
VIZAG MT, 14.12.2020

Ans. (d) :

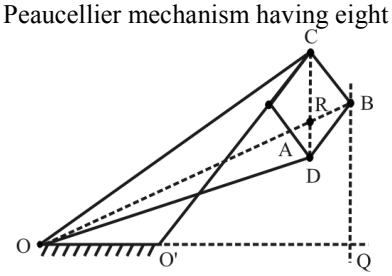


Member (m) = 3
Joint (j) = 3
For perfect frame,
$m = 2j - 3$
$3 = 2 \times 3 - 3$
$\boxed{3 = 3}$

53. Mechanism having eight links is :
(a) Porsch
(b) Backward quick return
(c) Peaucellier
(d) Short link show return

Kerala PSC AE (Irrigation) 05.04.2017
HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Peaucellier mechanism having eight links.



Peaucellier mechanism was the first true planar straight line mechanism capable of transforming rotary motion into perfect straight line motion and vice-versa.
Peaucellier mechanism has eight link and all pair is turning pair.
• Hart's mechanism has six link and all turning pair.
• Scott-Russel mechanism has four link one sliding pair and three turning pair.

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• Watt's mechanism, modified Scott-Russel mechanism, Robert mechanism, grasshopper mechanism, Tchebicheff mechanism all mechanism has four link mechanism.

54. The pair is known as a higher pair, when the relative motion between the elements of a pair is

(a) turning only
(b) sliding only
(c) rolling only
(d) partly turning and partly sliding

Nagaland PSC (CTSE) 2018 Paper-I
Sikkim PSC (Under Secretary), 2017

Ans. (d) : Higher pair- When a pair has point or line contact between the links, it is known as higher pair the motion between two element is partly turning and partly sliding.
E.g.- Cam follower, tooth gearing, ball and Roller bearing.
Lower pair- A pair of links having surface or area contact between the member is known as lower pair.
Eg.- Shaft Rotating in bearing, all pairs of slider crank mechanism (sliding pair, turning pair).

55. There is a relation between number of joints and number of links, where L = number of links and J = number of joints which constitute a kinematic chain and this is given by the expression

(a) $L = \frac{3}{2}(J+2)$ (b) $L = \frac{2}{3}(J+2)$
(c) $L = \frac{3}{2}(J+1)$ (d) $L = \frac{2}{3}(J+3)$

Nagaland PSC (CTSE) 2018 Paper-I
Karnataka PSC AE, 10.09.2017

Ans. (b) : For kinematic chain–

$$J + \frac{h}{2} = \frac{3}{2}L - 2$$

$\therefore$ If no higher pair, h = 0

$$J + 0 = \frac{3}{2}L - 2$$

$$J = \frac{3}{2}L - 2$$

$$\frac{3}{2}L = (J+2)$$

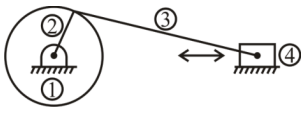
$$\boxed{L = \frac{2}{3}(J+2)}$$

56. In a single slider crank chain.

(a) each of the four pairs are a turning pair.
(b) one is turning pair and three are sliding pairs
(c) two are turning pairs and two are sliding pairs.
(d) three are turning pairs and one is a sliding pair.

Nagaland PSC (CTSE) 2018, Paper-I
MPSC Poly. Lect. 2014

Ans. (d) :



Single slider crank mechanism is 3R & 1P mechanism means.
3- Revolute or Turning pair.
1- Prismatic or sliding pair.
Double slider crank mechanism is 2R & 2P mechanism mean.
2- Revolute or Turning pair.
2 – Prismatic or sliding pair.

57. In a pantograph, all the pairs are

(a) Turning pairs (b) Sliding pairs
(c) Spherical pairs (d) Self closed pairs

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GMB AAE 25.06.2017

Ans. (a) : In a pantograph, all the pairs are turning pairs.

58. Which of the following mechanism generates intermittent rotary motion from continuous rotary motion?

(a) Scotch yoke mechanism
(b) Geneva mechanism
(c) Elliptical mechanism
(d) Whitworth mechanism

ISRO Scientist/Engineer 12.01.2020
RPSC AE 2018
ESE 2008

Ans. (b) : Geneva mechanism—A Geneva mechanism is a commonly used mechanism for producing an intermittent rotary motion from a uniform input speed.
Scotch yoke mechanism—It is one of reciprocating motion mechanism, it should be convert an rotary motion into sliding of linear motion and vice-versa.
Elliptical trammel—The Elliptical trammel (also known as the trammel of Archimedes) is a simple mechanism which can trace an exact elliptical path.
Whitworth mechanism–The Whitworth quick return mechanism converts rotary motion into reciprocating motion, but unlike the crank and slider, the forward reciprocating motion is at a different rate than backward stroke.

59. Which of the following is an inversion of single slider - crank chain?

(a) Elliptical Trammel (b) Hand Pump
(c) Scotch Yoke (d) Oldham's Coupling

RPSC AE 2018
OPSC AEE 2015 Paper-I

Ans. (b) : Hand Pump is an inversion of single slider - crank chain whereas elliptical trammel, scotch yoke and oldham's coupling are inversion of double slider crank chain.

60. Any point on a link connecting double slider crank chain will trace a

(a) Straight line (b) Circle
(c) Ellipse (d) Parabola

Gujarat PSC AE 2019
CGPSC AE 26.04.2015 Shift I
ESE 2000

Ans : (c) : At the centre the link will trace circle and it will trace ellipse on other points.

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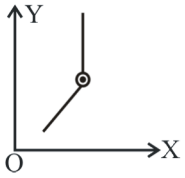
61. The two-link system, shown in the given figure, is constrained to move with planar motion. It possesses:



(a) 2- degrees of freedom
(b) 3 - degrees of freedom
(c) 4 - degrees of freedom
(d) 6 - degrees of freedom

UJVNL AE 2016
ESE 2000

Ans : (c)
ℓ = No of links = 2
j = No of Joints = 1
DOF for open chain in 2D
DOF = 3l - 2j
DOF = 3 × 2 - 2 × 1 = 4

62. Which of the following is an open pair?
(a) ball and socket joint (b) journal bearing
(c) lead screw and nut (d) cam and follower

Sikkim PSC (Under Secretary), 2017
APPSC AEE 2012

Ans. (d) : Open pair- Open pair must be held either spring action or due to gravity force.
Eg.- cam and follower in I.C. engine,
Closed pair- Two links of pair held together mechanically.

63. When two elements have point or line contact motion, the pair so formed is known as
(a) higher pair (b) lower pair
(c) closed pair (d) screw pair

Mizoram PSC AE/SDO 2014, Paper-II
OPSC Civil Services Pre 2006

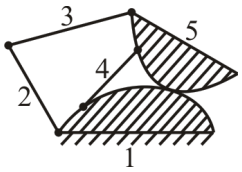
Ans. (a) : When the two elements of a pair have a line or point contact when relative motion take place between the two elements is partly turning and partly sliding, then the pair is known as higher pair. When two element have point or line contact motion, the pair is formed known as higher pair.

64. Two parallel shafts, the distance between whose axes is small and variable may be connected by:
(a) gear drive (b) universal joint
(c) knuckle joint (d) oldham's coupling

RPSC ACF-2011
CSE Pre-1998

Ans. (d) : Oldham's Coupling– It is used to connect two parallel shafts whose axes are at a small distance apart.

65. Number of degrees of freedom for the shown figure is:



(a) 0 (b) 1
(c) 2 (d) 4

AAI Jr. Executive 26.03.2021

Ans. (b) :
No. of links = 5
No. of joints = 5
No. of higher pair = 1
DOF = $3(\ell - 1) - 2j - h$
= 3(5 – 1) – 2 × 5 – 1
= 12 – 10 – 1
DOF = 1

66. The Davis steering gear fulfill the condition of correct gearing at
(a) two positions (b) three positions
(c) all positions (d) one position

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

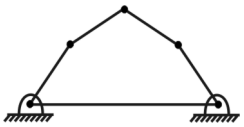
Ans. (c) : Davis steering are not commonly used, it can used in all positions due to front wheel arrangement on the other side, Ackermann steering fulfils the correct steering at only 3-positions.

67. The number of joints must be equal to _____ for a perfect frame with 5 members.
(a) 3 (b) 6
(c) 4 (d) 7

VIZAG Steel MT 24.01.2021

Ans. (c) : Member (m) = 5
For perfect frame,
m = 2j – 3
5 = 2j – 3
2j = 8
j = 4

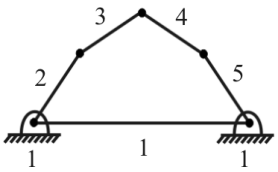
68. The degrees of freedom for the given planar mechanism will be :



(a) 1 (b) 2
(c) 3 (d) None of these

JPSC AE 10.04.2021, Paper-II

Ans. (b) :



L = 5
J = 5
Degree of Freedom
F = 3 (L – 1)– 2j – h
= 3 × (5 – 1) – 2 × 5
= 2

69. A combination of kinematic pairs, joined in such a way that the relative motion between the links is completely constrained, is called:
(a) mechanism (b) structure
(c) kinematic chain (d) inversion

UPRVUNL AE 05.07.2021

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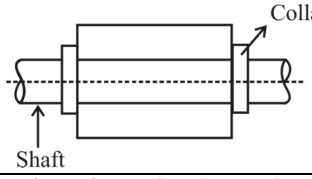
Ans. (c) : Completely Constrained Motion–When the motion between two elements of a pair is in a definite direction irrespective of the force applied, it is known as completely constrained motion.

Example – Shaft with both end collar in a circular hole.



70. Which of the following is turning pair?

(a) Piston and cylinder of a reciprocating steam engine
(b) Shaft with collars at both ends fitted in a circular hole
(c) Lead screw of a lathe with nut
(d) Ball and socket joint

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Shaft with collars at both ends fitted in a circular hole is a turning pair.

71. How may degrees of freedom a rigid body posses?

(a) 6 (b) 3
(c) 1 (d) 12

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Degree of freedom a rigid body = 6

$$= 3T + 3R \qquad \left[\because \begin{matrix} T = \text{Translational} \\ R = \text{Rotational} \end{matrix}\right]$$

72. In a four bar mechanism, the sum of the shortest and longest links should not be greater than that of the sum of the other two links. The above is called as _____.

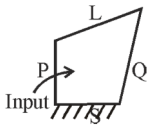
(a) Grubler Criterion (b) Hooke's Law
(c) Grashoff Law (d) Kutzbach Criterion

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : Grashof's law states that four- bar mechanism, if the sum of the length of the largest and shortest links is less than or equal the sum of length of the other two links.

Grashof's law condition satisfied-

$$\boxed{S + L \le P + Q}$$



73. A mechanism converts into structure when the number of degrees of freedom is equal

(a) 0 (b) 1
(c) –1 (d) 2

UPRVUNL AE 05.07.2021

Ans. (a) : A mechanism converts into structure when the number of degrees of freedom is equal 0.

74. A linkage has 11 links and 4 loops. What is the degree of freedom if it has only single turning pairs?

(a) 0 (b) 1
(c) 2 (d) 3

ESE 18.07.2021

Ans. (c) : Given, No. of linkages (N) = 11
No. of loop (L) = 4
Degree of freedom (F) = N – (2 × L+1)
= 11 – (2× 4 +1) = 2

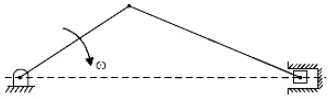
75. Which one of the following is the application of first inversion of single-slider-crank chain?

(a) Hand -pump
(b) Reciprocating engine
(c) Elliptical trammel
(d) Whitworth quick-return mechanism.

ESE 18.07.2021

Ans. (b) : First inversion of single slider crank chain

→ Reciprocating Engine
Cylinder is fixed



76. A quick-return mechanism is to be designed, where the outward stroke must consume 1.2s and the return stroke 0.8s. If the cycle time is 2.0 s/rev, what is the speed at which the mechanism should be driven?

(a) 10 rev/s (b) 30 rev/s
(c) 10 rev/min (d) 30 rev/min

ESE 18.07.2021

Ans. (d) :
Cycle time = 2 sec/rev.
So, basically, the mechanism will rotate one full relocation in 2 sec.

$$\therefore \text{speed} = \frac{60s}{2s} = 30\text{RPM}$$

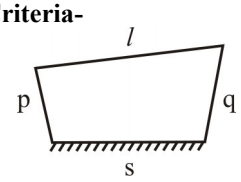
or, 30 rev/min.

77. The number of qualitatively distinct kinematic inversions possible for a Grashoff chain with four revolute pairs is_____

(a) 1 (b) 2
(c) 3 (d) 4

Assam Engg. College AP/Lect. 18.01.2021, GATE 2020

Ans. (c) : Kinematic inversions of Grashoff chain-
For Grashoff Criteria-



$s + l < p + q$
(1) Shortest link fixed- Doubled Crank
(2) Adjacent of shortest- Crank and Rocker
(3) Opposite the shortest- Double Rocker

78. In a slider-crank mechanism, there are

(a) three links (b) four links
(c) five links (d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : A slider-crank linkage is a four-link mechanism with three revolute joints and one prismatic or sliding joint. The rotation of the crank drives the linear movement the slider or the expansion of gases against a sliding piston in a cylinder can drive the rotation of the crank.

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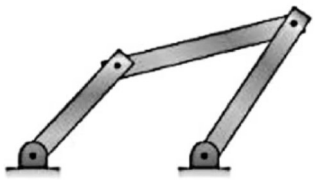
79. Which of the following is a lower pair ?
(a) Tooth gears in mesh (b) Cam and follower
(c) Roller bearing (d) All of the above
CGPSC AE 15.01.2021

Ans. (*) : All of the above are higher pair.

80. The degree of freedom for the following mechanism is :



(a) 1 (b) 0
(c) 2 (d) Infinite
APPSC Poly Lect. 13.03.2020

Ans. (a) : Number of links (l) = 4
Number of joints (j) = 4
Higher pair (h) = 0
Degree of freedom (DOF) = 3 (l–1) – 2j – h
= 3 (4–1) – 2 × 4
DOF = 9 – 8 = 1

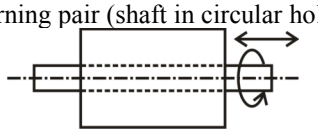
81. For a Whitworth quick return motion mechanism, $\alpha = 200^{o}$, Find the ratio of time of cutting stroke to the time of return stroke.
(a) 1.25 (b) 1.35
(c) 1.30 (d) 1.40
Assam PSC AE (PHED), 18.10.2020

Ans. (a) : Ratio of time of cutting stroke to time of return stroke for a whitworth quick return motion mechanism.

$$QRR = \frac{\alpha}{360-\alpha}$$
$$= \frac{200}{360-200}$$
$$\frac{t_C}{t_R} = 1.25$$

82. Motion of shaft in circular hole forms;
(a) Incompletely constrained motion
(b) Completely constrained motion
(c) Successfully constrained motion
(d) None of the above
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) : Incompletely Constrained Motion–When the motion between two elements of a pair in more than one direction and depends upon the direction of force applied, it is known as incompletely constrained motion.
Example–Turning pair (shaft in circular hole).



83. Ackermann steering gear mechanism is inversion of :
(a) Slider crank chain
(b) four bar chain
(c) Double slider chain
(d) Crossed slider crank chain
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : Ackermann steering gear mechanism is inversion of four bar chain.

84. Shaft with tapper key is an example
(a) incompletely constrained motion
(b) completely constrained motion
(c) successfully constrained motion
(d) All of the above
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (c) : Shaft with tapper key is an example successfully constrained motion.

85. Why is double Hooks' joint used?
(a) to obtain uniform velocity ratio
(b) to obtain variable velocity ratio
(c) Both (a) and (b)
(d) None of the above
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) : Double Hook's joint is used to obtain uniform velocity ratio.

86. Reciprocating motion of the cutting tool in shaping machines is accomplished by
(a) rack pinion mechanism
(b) crank and connecting rod mechanism
(c) cam and cam follower mechanism
(d) oscillating lever mechanism
ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : Shaping Machines–
Mechanism–Crank and slotted lever quick return motion mechanism (which is the inversion of single slider crank mechanism known as crank and connecting rod mechanism).
Note-In the shaper mechanism, tool feed is given by 'Ratched and pawl' mechanism.

87. In an IC engine, the reciprocatory motion is converted into a rotary motion by the combination of _____.
(a) rack and pinion
(b) cam and follower
(c) ratchet and pawl
(d) crankshaft and connecting rod
VIZAG MT, 14.12.2020

Ans. (d) : In an IC engine, the reciprocatory motion is converted into a rotary motion by the combination of crankshaft and connecting rod.

88. Which one of the following is not an inversion of a four bar chain?
(a) bull engine
(b) beam engine
(c) coupling rod of a locomotive
(d) watt's indicator mechanism
VIZAG MT, 14.12.2020

Ans. (a) : Bull engine is an inversion of single slider crank chain mechanism.
Beam engine, coupling rod of locomotive and watt's indicator is an example of inversion of a four bar chain.

89. A turning pair has
(a) Zero degree of freedom
(b) One degree of freedom
(c) Two degree of freedom
(d) Six degree of freedom
Haryana PSC AE (PHED) 05.09.2020, Paper-II

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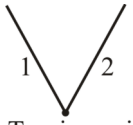
Ans. (b) : When relative motion between two elements of a pair takes place in the form of pure turning only then the pair is called turning pair.
Turning pair has one degree of freedom.



Turning pair
I_{12}

90. The degree of freedom for a planar mechanism is obtained using
(a) Kutzbach criterion
(b) Grashoff's criterion
(c) Kennedy theorem
(d) Chebyshev theorem

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Kutzbach Criterion – Degree of freedom
Grashoff's criterion – Inversion of quadratic chain
Kennedy theorem – Velocity and acceleration of machine mechanism.
Chebyshev theorem – It estimates the proportion of the measurement that lie must within one , two and three standard deviation of the mean.

91. In the process of inversion in a kinematic chain,
(a) The absolute motion of links does not change
(b) The relative motion of the links does change
(c) The link dimensions change
(d) The relative motion between the links do not change

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : In inversion, the relative motion of the links does changes.
If no. of link is ℓ
Then no. of inversion $\leq \ell$

92. When the nature of contact between the elements of a pair is such that it can only slide relative to the other, the pair is known as
(a) Screw pair (b) Spherical pair
(c) Turning pair (d) Sliding pair

RPSC IOF, 2020

Ans. (d) : When the two elements of a pair are connected in such a way that one can only slide relative to the other. The pair is known as a sliding pair.
Ex.- Piston and cylinder, cross-head and guides of reciprocating steam engine.
Ram and its guide in shaper tail stock on lathe bed.

93. Which of the mechanisms uses two identical mechanisms?
(a) Hart (b) Watt
(c) Peaucellier (d) Kempe

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Kempe mechanisms uses two identical mechanisms.

94. In a four-bar linkage if 'S' is the length of the shortest link, 'L' is the length of longest link and 'P' & 'Q' are length of other links, then the criteria for getting a triple rocker mechanism in which no links will fully rotate is
(a) $S + Q > P + L$ (b) $S + L < P + Q$
(c) $S + L = P + Q$ (d) $S + L > P + Q$

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : All three links i.e. input, output and coupler will be rocker, when Grashoff's law will not be satisfied, i.e, we will not be getting continuous relative motion in 4-bar mechanism i.e, $S + L > P + Q$

95. When two elements of a pair have surface contact while in motion and the relative motion is purely turning or sliding, then the pair is called a _____pair.
(a) Higher (b) Lower
(c) Turning (d) Sliding

MECON MT 2019

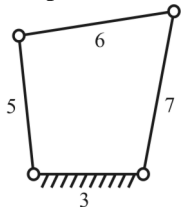
Ans. (b) : When two elements of a pair have surface contact while in motion and the relative motion is purely turning or sliding, then the pair is called a lower pair.

96. Which of the following is an inversion of four bar chain?
(a) Watts indicator mechanism
(b) Oscillating cylinder engine
(c) Whitworth quick return motion mechanism
(d) Scotch yoke mechanism

MECON MT 2019

Ans. (a) : Watts indicator mechanism

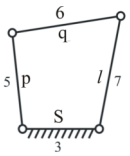
97. For the four link mechanism with the dimension of the links indicated in standard units of length, choose the type of mechanism from the given options.



(a) crank rocker (b) double crank
(c) double rocker (d) rocker

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (b) :



S + L	<	P + Q
3 + 7	<	5 + 6
10	<	11

(S + L < P + Q) means Grasshof mechanism and shortest link is fixed so it will be **Double Crank** mechanism.

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98. Match the approaches given below to perform stated kinematics/dynamics analysis of machine.

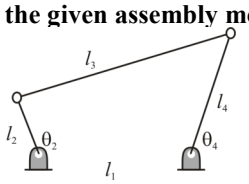
	Analysis		Approach
P.	Continuous relative rotation	1.	D' Alembert's principle
Q.	Velocity and acceleration	2.	Grubler's criterion
R.	Mobility	3.	Grashoff's law
S.	Dynamic-static analysis	4.	Kennedy's theorem

(a) P- 1, Q- 2, R- 3, S- 4;
(b) P- 3, Q- 4, R- 2, S- 1
(c) P- 2, Q- 3, R- 4, S-1;
(d) P- 4, Q- 2, R- 1, S- 3

HPPSC ADF 05.03.2019

Ans. (b) : Continuous relative rotation- Grasshoff's law
Velocity and acceleration – Kennedy's theorem
Mobility- Grubler's criterion
Dynamic-static-analysis- D' Alembert's principle

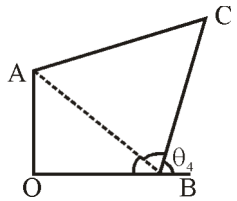
99. The figure shown below has the following details:
$l_1 = 3$ cm, $l_2 = 1.5$ cm, $l_3 = 3.5$ cm, $l_4 = 4$ cm, The value for θ_4 (in degrees) when the input link (l_2) becomes perpendicular to the fixed link for the given assembly mode will be:



(a) 45 (b) 97.44
(c) 65.34 (d) 79.44

BHEL ET 2019

Ans. (b) : When the link ℓ_2 becomes perpendicular to fixed link,



Given, OB = 3 cm, AC = 3.5 cm, BC = 4 cm, OA = 1.5 cm
In ΔAOB,

$$(AB)^2 = (OA)^2 + (OB)^2$$

$$(AB)^2 = (1.5)^2 + (3)^2$$

$$\boxed{AB = 3.354 \text{ cm}}$$

In ΔABC,

$$\cos C = \frac{(AC)^2 + (BC)^2 - (AB)^2}{2(AC)(BC)}$$

$$= \frac{(3.5)^2 + (4)^2 - (3.354)^2}{2(3.5)(4)}$$

$\cos C = 0.607$
$C = 52.615^\circ$
In ΔAOB,

$$\tan x = \frac{OA}{OB} = \frac{1.5}{3} = 0.5$$

$$\boxed{x = 26.56^\circ}$$

In ΔABC,

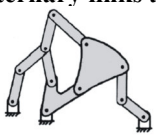
$$\cos\alpha = \frac{(AB)^2 + (BC)^2 - (AC)^2}{2(AB)(BC)}$$

$$\cos\alpha = \frac{(3.354)^2 + (4)^2 - (3.5)^2}{2(3.354)(4)}$$

$\cos\alpha = 0.559$
$\alpha = 56.012$
$\theta_4 = 180 - (x + \alpha)$
$= 180 - (26.56 + 56.012)$

$$\boxed{\theta_4 = 97.428^\circ}$$

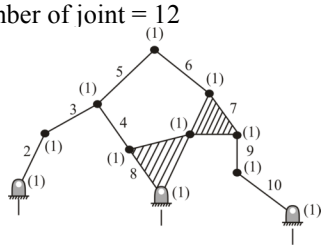
100. A mechanism is shown below. The number of ternary links and the DOF, respectively, are :



(a) 2 ; 3 (b) 2 ; 2
(c) 3 ; 2 (d) 3 ; 3

BHEL ET 2019

Ans. (d) : Number of ternary links = 3
Number of links = 10
Number of joint = 12



DOF = 3 (l – 1) – 2J – h
l = 10 , J = 12, h = 0
DOF = 3 (10 – 1) – 2 × 12
= 3 × 9 – 24 = 27 – 24

$$\boxed{DOF = 3}$$

101. DOF of an over constrained structure has _____ value.

(a) zero (b) positive or negative
(c) negative (d) positive

BHEL ET 2019

Ans. (c) : If degree of freedom is negative
i.e., F < 0, its called in determinant structure or super structure because no relative motion in between links.

102. The input parameters of inverse kinematics are

(a) Links length, wrist position
(b) Links length, joint angles
(c) Wrist position, joint angles
(d) Only links length

TNPSC AE 2019

Ans. (a) : The input parameters of inverse kinematics are links length, wrist position.

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103. Which of the following is a statically indeterminate structure?
(a) Load supported on composite member
(b) Load supported on two members
(c) Load supported on three members
(d) Thermal load supported on two members

TNPSC AE 2019

Ans. (c) : Load supported on three members is a statically indeterminate structure.

104. The number of links in a planer mechanism with only revolute joints having 21 instantaneous centers is:
(a) 5 (b) 6
(c) 7 (d) 8

HPPSC ADF 05.03.2019

Ans. (c) : $^{n}C_2 = \frac{n(n-1)}{2} = 21$

$n^2 - n - 42 = 0$

$n - 7n + 6n - 42 = 0$

$n(n-7) + 6(n-7) = 0$

$n = -6, 7$

So, no. of links = 7

105. Assertion (A) : The Ackermann steering gear is commonly used in all automobiles.
Reason (R) : It has correct inner turning angle for all positions.
(a) Both (A) and (R) are individually true and (R) is not the correct explanation of (A)
(b) Both (A) and (R) are individually true but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

OPSC AEE 2019 Paper-I

Ans : (c) : An Ackerman steering gear has only turning pairs and thus is preferred, Its drawback is that it fulfills the fundamental equation of correct gearing at the middle and the two gearing at the middle and the two extreme positions and not in all positions.

106. Which of the following statements are correct with respect to inversion of mechanisms?
1. It is a method of obtaining different mechanisms by fixing different links of the same kinematic chain.
2. It is method of obtaining different mechanisms by fixing the same links of different kinematic chains.
3. In the process of inversion, the relative motions of the links of the mechanisms produced remain unchanged.
4. In the process of inversion, the relative motions of the links of the mechanisms produced will change accordingly.

Select the correct answer using the code given below.
(a) 1 and 3 (b) 1 and 4
(c) 2 and 3 (d) 2 and 4

ESE 2019

Ans. (a) : Inversion is the process of alternately fixing each link of a kinematic chain to get different type of mechanism where the relative motion between links remain unchanged but the absolute motion between then gets inverted i.e. the sense of their motion changes from clockwise to counter clock wise rotation or from rightward movement to leftward.

107. In a mechanism having 1 degree of freedom, if one pair is higher pair, what is the minimum number of links required :
(a) 3 (b) 4
(c) 6 (d) 8

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) : In a mechanism having 1 degree of freedom, if one pair is higher pair, then the minimum number of links required 3.
For example– Cam follower mechanism.

108. Which one of the following is an exact straight-line mechanism using lower pairs?
(a) Watt's mechanism
(b) Grasshopper mechanism
(c) Robert's mechanism
(d) Peaucellier's mechanism

RPSC Vice Principal ITI 2018

Ans. (d) :
■ **Exact straight-Line motion Mechanism**
(i) Peaucellier's Mechanism
(ii) Hart's mechanism
(iii) Scott Rusell's Mechanism
■ **Approximate straight line motion Mechanism**
(i) Watt's Mechanism
(ii) Modified Scott-Russell Mechanism
(iii) Grasshopper mechanism
(iv) Techebicheff's Mechanism
(v) Roberts Mechanism

109. Which of the following does not represents the higher pair?
(a) Wheel rolling on a surface
(b) Sphere rolling on a surface
(c) Ball bearing
(d) Turning out on screw
(e) Cylinder rolling on a surface

CGPSC AE 25.02.2018

Ans. (d) : Wheel rolling on a surface, sphere rolling on a surface, ball bearing and cylinder rolling on a surface are example of higher pair where as turning nut on screw is the example of lower pair [Screw and nut pair].

110. Which of the following mechanism is made up of turning pairs?
(a) Scott-Russel mechanism
(b) Peaucellier mechanism
(c) Watt mechanism
(d) Pantograph

TNPSC AE 2018

Ans. (b) : Peaucellier mechanism- The first planar linkage capable of transforming rotary motion into perfect straight - line motion and vice versa.
Number of links is 8.

111. Which type of mechanism is used in shaper machine?
(a) Indexing mechanism
(b) Slider-crank mechanism
(c) Quick return mechanism
(d) None of the above

Vadodara Muncipal Corp. DEE, 2018

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Ans. (c) : Quick return mechanism is used in shaper machine.

112. According to linear law of Machine, when P is the effort and W is the load then

(a) P = m.W + C (b) W = P.m + C
(c) W = P.m – C (d) P = m.W – C

WBPSC AE, 2017

Ans. (a) :

113. Which of the following statements is/are true for mechanisms?

(a) A slider crank mechanism is formed if one link in slider crank chain is fixed.
(b) Three, kinematic links joined together form a mechanism
(c) Kinematic chain is a part of kinematic pair
(d) All of above

Karnataka PSC Lect., 27.05.2017

Ans (a) : If a number of bodies are assembled in such a way that the motion of one link causes constrained and predictable motions to the other links is known as a mechanism.

- Mechanism is neither a source of energy nor a producer of work but helps in proper utilization of the energy.
- A slider crank chain is a mechanism that converts reciprocating motion into rotary motion of the crank or vice-versa. But when it is used in an automobile by adding valve mechanism, cam mechanism etc. It becomes a machine.

114. Which of the following are the third inversions of slider crank mechanism?

A. Oscillating cylinder engine
B. Hand pump
C. Rotary engine
D. Crank and slotted lever
Select the code for the correct answer from the options given below:

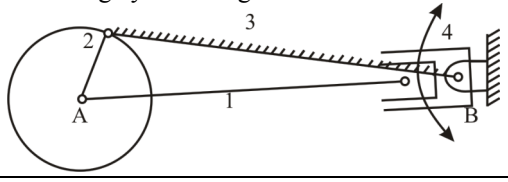
(a) A and B only (b) A and D only
(c) A, B and D only (d) All of the above

Karnataka PSC AE, 10.09.2017

Ans. (b) : 3rd inversion of single slider crank chain obtained by fixing connecting rod are

(1) Crank and slotted lever mechanism.

(2) Oscillating cylinder engine.



115. If a planner mechanism with six links and seven lower pairs, each lower pair is having one degree of freedom, then the degrees of freedom of the mechanism is

(a) 1 (b) 2
(c) 3 (d) 0

APGENCO AE, 2017

Ans. (a) : DOF = 3 (n–1) – 2j – h
3 (6–1) – 7 × 2 – 0 = 1

116. Which of the following are the inversions of a double slider crank chain?

1. Whitworth quick return motion
2. Scotch Yoke
3. Oldham's Coupling
4. Rotary Engine
Select the correct answer using the codes given below :
Codes :

(a) 1 and 2 (b) 2 and 3
(c) 1, 3 and 4 (d) 2, 3 and 4

Punjab PSC (Lect.) 06.08.2017, ESE 1993

Ans. (b) : Inversions of double slider crank chain
(i) Elliptical Trammel
(ii) Scotch Yoke
(iii) Oldham Coupling

117. Degree of freedom of a rectangular block resting on a plane table is

(a) 0 (b) 2
(c) 3 (d) 6

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : Degree of freedom of a rectangular block resting on a plane table is 3 (2T + 1R) = 3

118. Universal coupling is used to connect the shaft:

(a) Whose axes intersect as small angle
(b) Which are perfectly aligned
(c) Which are not aligned
(d) Have lateral misalignment

Punjab PSC SDE 12.02.2017

Ans. (a) : Hooke's Joint or Universal Joint– It is used to connect two non-parallel and intersecting shafts.

- It is also used for shafts with angular misalignment.
- Driving shaft runs at a uniform angular speed whereas the drivens shaft rotates at a continuously varying angular speed.
- It is used in steering mechanism and also used in transmitting power from gear box to rear axle.

119. In order to obtain the maximum mechanical advantage and consequently in order to obtain maximum mechanical efficiency we must use the machine at

(a) Minimum load (b) Maximum load
(c) No load (d) Mean load

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) : Mechanical advantage = $\frac{W}{P}$

W = load lifted
P = applied effort

For max mechanical efficiency, $\eta = \frac{W}{P} \times \frac{y}{x}$

$\frac{x}{y}$ is the velocity ratio.

For this formula $\eta = \frac{\text{Mechanical advantage}}{\text{Velocity ratio}}$

to obtain maximum M.A. in order to mechanical η_{max} used maximum load.

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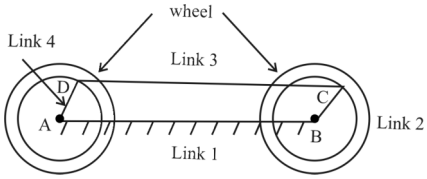
120. Inversions of four-bar chain are obtained depending upon the use of various links as lever or crank. Coupled wheel of a locomotive is an inversion of

(a) Double lever mechanism
(b) Reciprocating engine mechanism
(c) Double crank mechanism
(d) Crank and lever mechanism

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) :



The mechanism of a coupling rod of locomotive which consists 4 link.
In the mechanism the link AD and BC act as cranks and are connected to the respective wheels.
The link CD acts as coupling rod and link AB is fixed.
It is inversion of double crank mechanism.

121. In a crank and slotted lever quick return motion mechanism, the distance between the fixed centers is 160 mm and the driving crank is 80mm long. The ratio of time taken by cutting and return strokes is

(a) 0.5 (b) 1
(c) 1.5 (d) 2

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (d) : Given, OA = 160 mm, AB = 80 mm
We know that,

$$\sin(90-\alpha/2)=\frac{AB}{OA}=\frac{80}{160}$$

$$\cos\frac{\alpha}{2}=\frac{1}{2}=\cos 60°$$

Therefore, $\alpha = 120°$

$$\beta = 360 - 120° = 240°$$

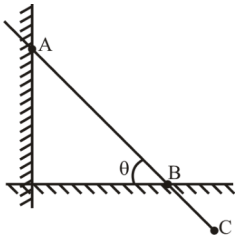
$$\text{Quick return ratio} = \frac{\beta}{\alpha}=\frac{240}{120}=2$$

122. In an elliptical trammel, the length of the link connecting the two sliders is 100mm. The tracing pen is placed on 150mm extension of this link. The major and minor axes of the ellipse traced by the mechanism would be

(a) 250 mm and 150 mm
(b) 250 mm and 100 mm
(c) 500mm and 300 mm
(d) 500 mm and 200 mm

GPSC Asstt. Director of Transport 05.03.2017, ESE 2012

Ans. (c) :



Given,

AB = 100 mm, BC = 150 mm
$x_c = BC\cos\theta = 150\cos\theta$

$$\cos\theta=\frac{x_c}{150}$$

$y_c = AC\sin\theta = 250\sin\theta$

$$\sin\theta=\frac{y_c}{250}$$

We know, ($\because \cos^2\theta + \sin^2\theta = 1$)

$$\left(\frac{x_c}{150}\right)^2+\left(\frac{y_c}{250}\right)^2=1$$

Major axis = 250 × 2 = 500 mm
Minor axis = 150 × 2 = 300 mm

123. The Ackerman steering gear mechanism is preferred to Davis steering mechanism because

(a) Whole of the mechanism in the Ackerman steering gear is on the back of the front wheels
(b) The Ackerman steering gear consists of sliding pairs
(c) The Ackerman steering gear is most economical
(d) Davis steering gear consists of turning pairs

ISRO Scientist/Engineer (RAC) 07.05.2017

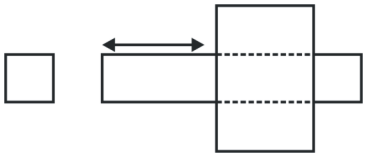
Ans. (a) : The Ackerman steering gear consists of turning pairs and Davis steering gear consist of sliding and turning pairs. The whole of the mechanism in the Ackerman steering gear is on the back of the front wheels, where as in Davis steering mechanism, it is in front of the front wheel.
The Ackerman steering gear is most economical than Davis because less wear follow exact gearing in three stage, two extreme and one is mid where as Davis steering gear satisfy exact gearing at all stage but have more wear.

124. The motion of a square bar in a square hole is example of _____.

(a) Completely constrained motion
(b) Incompletely constrained motion
(c) Successfully constrained motion
(d) Machine
(e) Structure

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Completely Constrained Motion –
Completely constrained motion is a type of constrained motion in which relative motion between the links of a kinematic pair occurs in a definite direction by itself, irrespective of external forces applied.
Example– Square bar in a square hole undergoes completely constrained motion.



Incompletely Constrained Motion– Incompletely constrained motion, the relative motion between the links depend on the direction of external forces acting

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on them. A good example of incompletely constrained motion is the motion of a shaft inside a circular hole.
Partially or Successfully Constrained Motion–A kinematic pair is said to be partially or successfully constrained if the relative motion between the links occurs in a definite direction, not by itself, but by some other means. A good example of successfully constrained motion is piston reciprocating inside a cylinder in an internal combustion engine.

125. Grubler's criterion is applicable for mechanisms with:
(a) Zero degree of freedom
(b) Single degree of freedom
(c) Two degree of freedom
(d) Three degree of freedom
(e) More than three degree of freedom

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Kutzbach Criteria–Kutzbach criteria is for determining degree of freedom of body in planar mechanism (2D)

$$DOF = 3[\ell - 1] - 2j - h \quad .....(1)$$

Here, ℓ = Number of links
j = Number of lower pair
h = Number of higher pair
If h = 0, DOF = 1 putting in equation (1)
Then we get

$$1 = 3(\ell - 1) - 2j$$

$$1 = 3\ell - 3 - 2j$$

$$\boxed{3\ell - 2j = 4}$$

This equation is known as Grubler's criteria.

126. A wheel rotating around its axle forms a _____ pair.
(a) sliding (b) turning
(c) rolling (d) screw
(e) spherical

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : A wheel rotating around its axle forms a turning pair.

127. _____permits one shaft to drive two other shafts with equal efforts at three different shafts speeds.
(a) Universal joint (b) Stub axles
(c) Differential (d) Axle housing

TNPSC AE 2017

Ans. (c) : Differential permits one shaft to drive two other shafts with equal efforts at three different shafts speeds.

128. Consider the following degrees of freedom
(i) Pitch
(ii) Roll
(iii) Yaw
DOF which is not included in half car model is
(a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (i), (ii), (iii)

TNPSC AE 2017

Ans. (a) :

129. For a Kinematic chain to be considered as mechanism
(a) Two links should be fixed
(b) One link should be fixed
(c) None of the links should be fixed
(d) There is no such criterion

Vizag Steel (MT) 2017
WBPSC AE, 2017
Mizoram PSC AE/SDO 2014, Paper-II

Ans. (b) : for a kinematic chain to be considered as mechanism one link should be fixed.

130. A planer mechanism has 10 links and 12 rotary joints. Using Grubler's criterion, the number of degrees of freedom of the mechanism is
(a) 1 (b) 3
(c) 2 (d) 4

ESE 2017

Ans. (b) : Given,
n = 10
$j_1 = 12$
$j_2 = 0$
Degree of freedom (DOF) $= 3(n - 1) - 2j_1 - j_2$
$= 3(10 - 1) - 2 \times 12 - 0$
$= 3$

131. Which of the following mechanisms are examples of forced closed kinematic pairs?
1. Cam and roller mechanism
2. Door – closing mechanism
3. Slider – crank mechanism
Select the correct answer using the code given below.
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (a) : Force closed kinematic pair– Contact between the links forming the pair is maintained by an external force/torque but not due to the geometry of surface in contact.
In general, cam and follower will be in contact either due to force of gravity or force of the spring on the follower. Door closing mechanism, has a torsion spring loaded follower in contact with door.
Note- The answer given by UPSC is option (c).

132. How many links must be there in "compound mechanism"?
(a) 1 (b) 2
(c) 3 (d) 4
(e) More than 4

CGPSC ARTO & TSI 07.01.2017

Ans. (e) : Simple and compound mechanism :-
A mechanism with four links is known as simple mechanism and mechanism with more than four link is known as compound mechanism.

133. Transmission of power from the engine to the rear axle of an automobile is by means of
(a) worm and wheel method
(b) Hooke's joint
(c) crown gear
(d) compound gears

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Transmission of power from the engine to the rear axel of an automobile is by means of Hook's joint.

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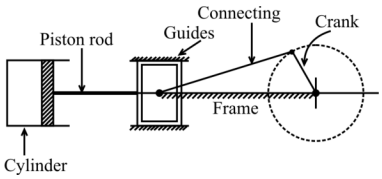
134. The no. of inversions for a slider crank mechanism is.....

(a) 4 (b) 8
(c) 6 (d) 0

(HPPSC LECT. 2016)

Ans : (a)



A single slider crank chain is a modification of the basis four bar chain. It consist of one sliding pair and three turning pair.

The method of obtaining different mechanism by fixing different links in kinematic chain, is known as inversion of the mechanism

(i) If number of links are N then no. of inversion is N

(ii) The no. of inversion for a slider crank mechanism is 4.

135. Watt mechanism is a ..... bar mechanism.

(a) 3 (b) 4
(c) 5 (d) 8

(HPPSC LECT. 2016)

Ans : (b) Watt mechanism is a four bar mechanism.

Watt mechanism:- It is a crossed four bar chain mechanism and was used by Watt for his early steam engine to guide the piston rod a cylinder to have an approximate straight line motion.

136. Mechanism having zero d.o.f. is known as:

(a) Machine (b) structure
(c) Kinematic chain (d) Link

(HPPSC LECT. 2016)

Ans : (b) Mechanism having zero degree of freedom is known as structure.

structure:- If one of the link of redundant chain is fixed. It is known as structure or a locked system.

(i) No relative motion
(ii) Capable of transmitting force only
(iii) Power energy cannot be transmitted
(iv) Degree of freedom of structure is zero
(v) Degree of freedom of super structure is less than zero.

137. Match list-I with list-II select the correct Answer using the codes given below the list:

List -I	List -II
(a) Cam and follower	1. Grubler's rule
(b) Screw pair	2. Grashof's linkage
(c) 4-bar mechanism	3. Pressure angle
(d) Degree of freedom of Planar mechanism	4. Single degree of freedom

Code

	A	B	C	D
(a)	3	4	2	1
(b)	1	2	4	3
(c)	1	4	2	3
(d)	3	2	4	1

UJVNL AE 2016, ESE 2001

Ans : (a) List -I	**List -II**
i) Cam and follower	i. Pressure angle
ii) Screw pair	ii. Single degree of freedom
iii) 4-bar mechanism	iii. Grashof's linkage
iv) Degree of freedom of Planar mechanism	iv. Grubler's rule

138. Differential gear is used in an automobile to

(a) Transmit power from the engine to driving wheels.
(b) Multiply the available engine torque.
(c) Enable the vehicle negotiate curves properly.
(d) Serves all the three functions as mentioned in (a), (b) and (c) above.

UPPSC AE 12.04.2016 Paper-I

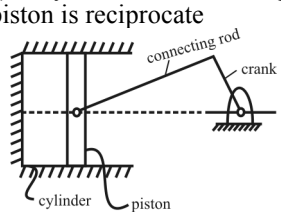
Ans : (d)
Differential gear is used in an automobile to:-
(i) Transmit power from the engine to driving wheels.
(ii) Multiply the available engine torque.
(iii) Enable the vehicle negotiate curves properly.

139. Cylinder of an I.C. engine is a :

(a) Elasto-fluid link
(b) Elasto-rigid link
(c) Translatory rotating link
(d) Fixed link

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Cylinder of an I.C. engine is a fixed lines in which piston is reciprocate



140. In order to make a mechanism with one degree of freedom, how many minimum number of links needed? The mechanisms has one higher pair

(a) 2 (b) 3
(c) 4 (d) 6

GPSC ARTO 01.05.2016

Ans. (b) : Cam and follower has 3 links where 2 are lower pair and 1 higher pair, it has DOF = 1.

141. The analysis of mechanism deals with

(a) the determination of displacement, velocity and acceleration of the links in a mechanism
(b) the determination of input and output angles of the mechanism
(c) the determination of dimensions of the link in a mechanism
(d) none of the above

GPSC ARTO 01.05.2016

Ans. (a) : The analysis of mechanism deals with the determination of displacement, velocity and acceleration of the link in a mechanism.

142. The part of an overhead valve mechanism arranged between the valve tappet and rocker arm is :

(a) Cam follower (b) Push rod
(c) Rocker shaft (d) Cam shaft

Kerala PSC Poly. Lect. (Auto) 05.05.2016

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Ans. (b) The part of an overhead valve mechanism arranged between the valve tappet and rocker arm is push rod.

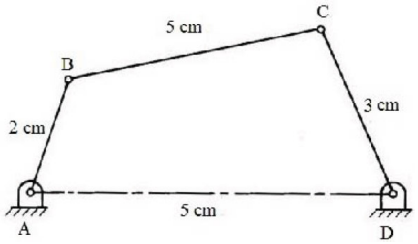
143. If the number of members (m) are more than that required by the equation m = 2j–3 (j is number of joints) such frames are called

(a) Redundant frame (b) Weak frame
(c) Perfect frame (d) Ideal frame

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : We know, In general (m = member)
→ If m = 2j – 3, then the frame is perfect
→ If m < 2j – 3, then the frame is deficient frame.
→ If m > 2j – 3, then the frame is redundant frame.

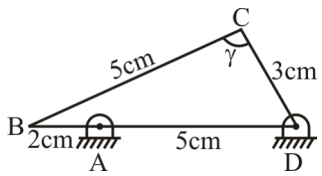
144. What will be the maximum transmission angle for the mechanism shown in the figure below?



(a) 90° (b) 110°
(c) 120° (d) 135°
(e) 150°

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) :



Transmission angle = γ

$$\cos\gamma = \frac{5^2+3^2-7^2}{2\times5\times3} = -\frac{1}{2}$$

$\gamma = 120°$

145. A Thompson indicator employs :

(a) Pantograph mechanism
(b) Grass-hopper mechanism
(c) Kemtpe's mechanism
(d) Tchebicheff mechanism

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : A Thompson indicator employs grass-hopper mechanism.
In this indicator the piston rod is hollow and serves only as a guide for a piston. The pencil mechanism is connected to the piston be a very light rod that passes through the piston rod and is attached to the piston with a swivel joint. This permits the use of a very simple and light parallel motion.

146. A wedge used to raise loads consists of.....pairs.

(a) One sliding (b) Two sliding
(c) Three sliding (d) Revolving

HPPSC AE (PWD) 10.09.2016

Ans. (c) : A wedge used to raise loads consists of three turning pairs.

147. Degrees of freedom (d.o.f.) of a pair is the number of independent relative motions, both translational and rotational and thus a kinematic pair can have:

(a) d.o.f. = 3 – no. of restraints
(b) d.o.f. = 6 – no. of restraints
(c) d.o.f. = 12 – no. of restraints
(d) d.o.f. = no. of restraints

HPPSC AE (PWD) 10.09.2016

Ans. (b) : The number of restraints can never be zero (Joint is disconnected) or six (joint becomes solid). Degree of freedom of a pair is defined as the number of independent relative motion's both translational and rotational a pair can have.

Degree of freedom = 6-no. of restraints.

148. Which one is a lower pair ?

(a) Roller bearing
(b) Tooth gears in mesh
(c) Cam and follower
(d) All of the above
(e) None of the above

CGPSC AE 16.10.2016

Ans. (e) : Lower pair, when two elements of a pair joined together with the surface contact between them, the joint is referred to as lower pair e.g. revolute, prismatic, screw, cylindrical, spherical & planar.
Roller bearing, Tooth gears in mesh, Cam and follower are higher pair.

149. A bolt and nut form

(a) Turning pair (b) Rolling pair
(c) Screw pair (d) Sliding pair
(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : The combination of sliding and turning, is an example of nuts bolt, this pair is known as screw pairs eg. Screw Nut, Screw of lathe with a nut etc.

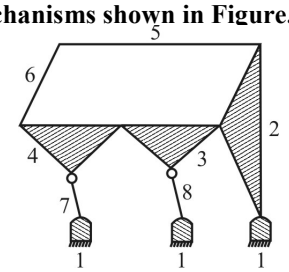
150. Consider the following pairs of parts:
1. Pair of gear in mesh
2. Belt and pulley
3. Roller Bearing
4. Cam and follower
Among these the higher pairs are:

(a) 1, 3 and 4 (b) 1, 2 and 3
(c) 2 and 4 (d) 1 and 4

UPRVUNL AE 21.08.2016

Ans. : (*) Higher pairs are
(i) Pair of gear in mesh
(ii) Belt and pulley
(iii) Roller bearing
(iv) Cam and follower.
Note-Official answer is (a).

151. Determine the number of degrees of freedom of the mechanisms shown in Figure.



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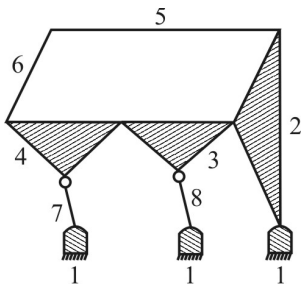
(a) 2 (b) 3
(c) 4 (d) 1

UPRVUNL AE 21.08.2016

Ans. : (d)



L = 8

Number of binary point (T) = 10

h (higher pair) = 0

DOF = 3 (l–1) –2J – h

= 3 (8–1) – 2 × 10 – 0

DOF = 1

152. Which of the following has sliding motion?

(a) crank (b) connecting rod
(c) cross head (d) cross head guide

GPSC Lect. 23.10.2016

Ans. (c) : Sliding is a type of frictional motion between two surfaces in contact e.g. cross head.

153. A slider crank chain consists of following numbers of sliding and turning pairs

(a) 1, 3 (b) 3, 1
(c) 2, 2 (d) 4, 0

GPSC Lect. 23.10.2016

Ans. (a) : Slider crank is a four bar mechanism, have one sliding pairs and three turning pairs.

154. Which of the following disciplines provides study of relative motion between the parts of a machine

(a) applied mechanics (b) mechanisms
(c) kinematics (d) kinetics

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : The study of inertia forces arising from the combined effect of the mass and motion of the parts is called kinetics while, the study of relative motion between the parts of a machine is called kinematics.

155. A circular shaft is revolving inside the bearing is an example of

(a) Sliding pair (b) Turning pair
(c) Rolling pair (d) Spherical pair

RPSC LECTURER 16.01.2016

Ans. (b) : A circular shaft is revolving inside the bearing is an example of turning pair.

156. The relation between number of links and number of lower pairs is given as n = 2p – 4. If L.H.S is less than R.H.S of this equation then the chain is

(a) constrained (b) locked
(c) structure (d) unconstrained

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : n = 2p – 4

If, L.H.S < R.H.S

Then, it become an unconstrained

157. When the motion between a pair takes place in a definite direction irrespective of direction of force applied, then it is said to be which among the following?

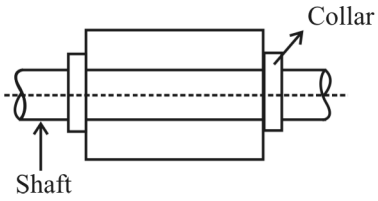
(a) Incompletely constrained motion
(b) Successfully constrained motion
(c) Completely constrained motion
(d) Partially successful constrained motion

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : There are three types of constrained motion

(i) Completely Constrained Motion–When the motion between two elements of a pair is in a definite direction irrespective of the force applied, it is known as completely constrained motion.

Example – Shaft with both end collar in a circular hole.

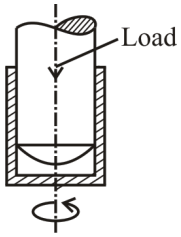


(ii) Incompletely Constrained Motion–When the motion between two elements of a pair in more than one direction and depends upon the direction of force applied, it is known as incompletely constrained motion.

Example–Turning pair (shaft in circular hole).

(iii) Successfully Constrained Motion–When the motion between two elements of a pair is possible in more than-one direction but it made to have motion only in one direction by using some external means. It is a successfully constrained motion.

Example–Foot step bearing.



In completely constrained motion, the motion between a pair takes place in a definite direction irrespective of direction of force applied.

158. A point on a link connecting double slider crank chain traces a

(a) Circle on fixed plate
(b) Ellipse on fixed plate
(c) Straight line on fixed plate
(d) Parabola on fixed plate

Rajasthan AE (Nagar Nigam) 2016 Shift-3

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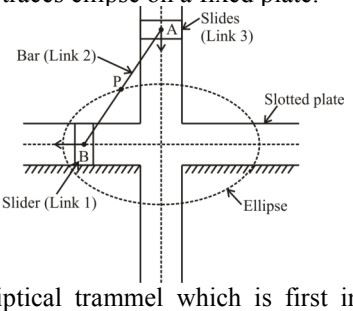
Ans. (b) : A point on a link connecting double slider crank chain traces ellipse on a fixed plate.



It is an elliptical trammel which is first inversion of double slider-crank chain. In elliptical trammel there are two turning pair & two sliding pair.

159. An inversion of four bar chain in which one link can make complete rotation while the output link oscillates is obtained when
(a) If the link adjacent to the shortest link is fixed
(b) If the shortest line is fixed
(c) If the link opposite to the shortest link is fixed
(d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (a) : An inversion of four bar chain in which one link can make complete rotation while the output link oscillation is obtained when if the link adjacent to the shortest link is fixed.
e.g. Crank-rocker

160. By fixing the connecting rod of a slider crank chain, the inversion obtained is called
(a) Whitworth quick return mechanism
(b) Rotary cylinder engine
(c) Crank and slotted lever quick return mechanism
(d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (c) : By fixing the connecting rod of a slider crank chain, the inversion is called crank and slotted lever quick return mechanism.

161. The only service that a steering linkage normally requires is :
(a) Piston-rod adjustment
(b) Lubrication
(c) Cam grinding
(d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : Lubrication is the only service that is required in steering linkage.

162. A shaft rotating in bearing forms ….. pair.
(a) Helical (b) Sliding
(c) Turning (d) Revolving

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : A shaft rotating in bearing forms turning or revolving pair.
• **Turning or revolving pair-** When one link of paired elements rotating inside other elements form turning pair.
• **Sliding pair-** When one elements slides with respect to other elements form sliding pair.
Note-Official answer is (d).

163. A truss having 4 joints and 7 members is termed as…… truss.
(a) Perfect
(b) Imperfect or deficient
(c) Redundant
(d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) :
For Perfect Truss or Statically Determinate

$$\boxed{m = 2j - 3}$$

Under Rigid or Collapsible Truss or Deficient truss

$$\boxed{m < 2j - 3}$$

Over Rigid or Redundant Truss or Statically Indeterminate

$$\boxed{m > 2j - 3}$$

$J = 4, \quad m = 7$
$m = 2j - 3$
$= 2 \times 4 - 3 = 5$
$m > 5 \Rightarrow 7 > 5$
Hence, the truss is redundant or over rigid.
Note-Option (b) given by commission.

164. The other name for Hooke's joint is ……
(a) Oldham coupling (b) Universal coupling
(c) Rigid coupling (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : The hooks joint is also known universal coupling. Hooke's joint is a Joint coupling connecting rigid rods whose axes are inclined to each other and is commonly used in shaft that transmit rotary motion.

165. The number of centers in a crank driven slider crank mechanism are
(a) 2 (b) 1
(c) 5 (d) 4

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (a) : 2 load centers is in a crank driven slider crank mechanism.

166. The degrees of freedom of a simple pendulum is given by :
(a) 2 (b) 6
(c) 0 (d) 1

TSPSC Managers, 2015

Ans. (d) : The degree of freedom of a simple pendulum = 1 because it has one input (i.e. one mass system).

167. Which one is NOT the inversion of four bar mechanism?
(a) Watt's mechanism
(b) Beam engine
(c) Pantograph
(d) Oscillating cylinder engine

NPCIL ET, 2015

Ans. (d) : Inversion's of four bar chain mechanism-
There three inversions-
(1) Beam engine or crank and lever mechanism
(2) Coupling rod of locomotive or double crank mechanism
(3) Watt's straight line mechanism or double lever mechanism
(4) Pantograph

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168. A planer mechanism has 6 links with 6 lower pairs and 2 higher pairs. Then the degree of freedom of mechanism as per Gruebler's Criterion is

(a) 1 (b) 2
(c) 3 (d) 0

TSPSC AEE 2015

Ans. (a) : Data given as,

$l = 6, \quad h = 2, \quad j = 6$

Then,

$N = 3(l - 1) - 2j - h$

$N = 3(6 - 1) - 2 \times 6 - 2$

$\boxed{N = 1}$

169. Which one of the following is not a inversion of double slider crank mechanism

(a) Elliptical trammel
(b) Oldham's coupling
(c) Scotch yoke mechanism
(d) Beam engine Mechanism

TSPSC AEE 2015

Ans. (d) : Elliptical trammel, oldham's coupling and scotch yoke mechanism are the example of double slider crank mechanism where as beam engine mechanism is the example of four bar mechanism.

170. Typewriter constitutes

(a) Structure (b) Machine
(c) Inversion (d) mechanism

(KPSC AE 2015)

Ans : (d) Typewriter constitutes mechanism.

171. Where is the Hooke's joint used in an automobile?

(a) between gearbox and propeller shaft
(b) between flywheel and clutch
(c) between differential gear and wheel
(d) between clutch and gear box

(KPSC AE 2015)

Ans : (a) Hooke's joint:- A Hooke's joint is used to connect two shaft which are Intersecting at a small angle. The end to each shaft is forked to U-type and each fork provides two bearing for the arms of a cross. The arms of the cross are perpendicular to each other. The motion is transmitted from the driving shaft to driven shaft through a cross.

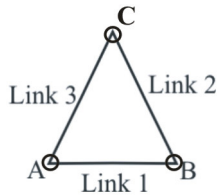
172. Locked chain satisfy equation

(i) $n = 2p - 4$, (ii) $j = \frac{3}{2}n - 2$:

(a) Only Equation (i)
(b) Only Equation (ii)
(c) Satisfy both the Equations
(d) Doesn't satisfy both the Equations

OPSC AEE 2015 Paper-I

Ans : (d)



Number of link (l) = 3

Number of pair (p) = 3

Number of Joints (J) = 3

Equation (i)

$n = 2p - 4$

$3 = 2 \times 3 - 4$

LHS > RHS

Equation (ii)

$= 3 = \frac{3}{2} \times 2 - 2$

LHS > RHS

Since the arrangement of three link, does not satisfy the equation (i) and (ii) and the left hand side is greater than the right hand side therefore it is not a kinematic chain and hence no relative motion is possible. Such type of chain is called locked chain and forms a rigid frame or structure which is used in bridges and trusses :

173. Roller bearing is a :

(a) Spherical pair (b) Higher pair
(c) Lower pair (d) Cylindrical pair

OPSC AEE 2015 Paper-I

Ans : (b) Roller bearing is a Higher pair.

Higher pair : When the two elements of a pair have a line or point contact or when relative motion takes place and the motion between the two elements is partly turning and partly sliding, then the pair is known as higher pair. A pair of friction discs, toothed gearing belt and rope drives, ball and roller bearing and cam and follower are the examples of higher pair.

174. One of the following mechanism is not exact straight line mechanism:

(a) Grasshopper Mechanism
(b) Hart Mechanism
(c) Scott Russel Mechanism
(d) Peaucellier Mechanism

OPSC AEE 2015 Paper-I

Ans : (a) Grasshopper Mechanism is not exact straight line mechanism.

Approximate straight line motion mechanism :
(1) Watt's mechanism
(2) Modified Scott-Russel mechanism
(3) Grasshopper mechanism
(4) Tchebicheff's mechanism
(5) Roberts mechanism

Exact straight line motion mechanism :
(i) Peaucellier mechanism
(ii) Hart's mechanism
(iii) Scott Russell's mechanism

175. Knee joint of a human being has ____degree(s) of freedom.

(a) One (b) Two
(c) Three (d) Zero

OPSC AEE 2015 Paper-I

Ans : (a) Knee joint of a human being has one degree of freedom.

176. The number of degrees of freedom for sphere-plane kinematic pair is :

(a) one (b) three
(c) five (d) six

MPPSC State Forest Service Exam, 2014

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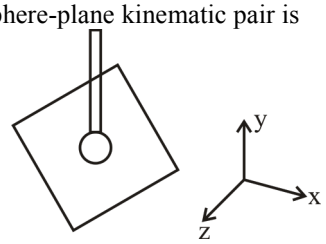
Ans. (c) : Sphere-plane kinematic pair is



Translation → x & z direct
Rotation → x, y & z direct
So have total five degree of freedom.

177. A framed structure is imperfect, if the number of member are _____ (2j – 3), where j is the number of joints.
(a) equal to (b) less than
(c) greater than (d) either (2) or (3)

MPSC Poly. Lect. 2014

Ans. (d) : For perfect truss m = 2j – 3
If m < 2j – 3 Deficient truss
m > 2j – 3 Redundant truss
Note-Official answer is (a).

178. If the number of nodes in a perfect frame is n, then the number of members is
(a) 2n-1 (b) 2n
(c) 2n-3 (d) n-1

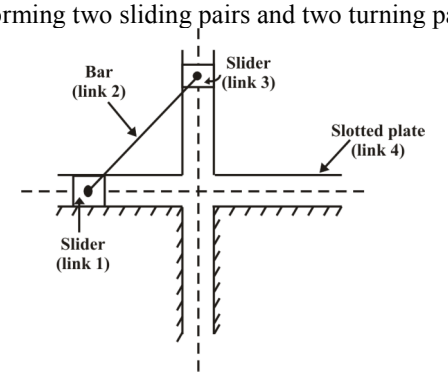
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c)
m = 2n – 3 for perfect frame
If m < 2n – 3 It means that the truss contain more members than required to be just rigid and is over rigid and statically indeterminate.
If m > 2n – 3 It means that the truss contains less members than required to be just rigid and is collapsible or under rigid.
Where,
m = Number of members in the truss
n = Number of node

179. In elliptical trammels
(a) All four pairs are sliding
(b) All four pairs are turning
(c) Two pairs are turning and other two sliding
(d) One pair turning and three sliding
(e) One pair sliding and three turning

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : • Elliptical Trammel is an inversion of double crank chain mechanism.
• A double slider crank chain consist of four links forming two sliding pairs and two turning pairs.



180. The joint in valve mechanism of reciprocating engine is
(a) knuckle joint (b) universal joint
(c) cotter joint (d) key joint
(e) pin joint

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : Applications of Knuckle joints-
(i) Joints between the tie bars in a root trusses.
(ii) Between the links of suspension bridge.
(iii)In valve mechanism of a reciprocating engine.
(iv) Fulcrum for the levers.

181. In an offset slider-crank mechanism with length of connecting rod l, crank radius r and offset e, the crank will revolve only when
(a) $l \geq (r + e)$ (b) $l \leq (r - e)$
(c) $l > (r - e)$ (d) $l < (r + e)$

TNPSC AE 2014

Ans. (a) : In an offset slider-crank mechanism with length of connecting rod l, crank radius r and offset e, the crank will revolve only when length of connecting rod l should be equal or greater than the sum of crank radius and offset e.

$$\ell \geq (r + e)$$

182. Which of the following systems has 8 links?
(a) Hart mechanism
(b) Peaucellier mechanism
(c) Whitworth Quick return mechanism
(d) Scotch yoke mechanism

TNPSC AE 2014

Ans. (b) : Mechanism	**Number of links**
Hart mechanism	6
Peaucellier mechanism	8
Whithworth quick return Mechanism	4
Scotch yoke mechanics	4

183. Theory of machine which deals with the inertia force due to combined effect of the mass and motion of the various links of the machines is known as
(a) kinetic (b) kinematics
(c) dynamics (d) statics

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : Dynamics– It is that branch of theory of machines which deal with the forces and their effects while acting upon the machine parts in motion.
• **Kinetics**–It deals with inertia forces which occur due to combined effect of mass and motion of machine element.

184. A shaft with collars both ends fitted into a circular hole is a popular example of
(a) rolling pair (b) turning pair
(c) sliding pair (d) spherical pair

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (b) : When the two element of a pair are connected in a such a way that one can only turn on revolve about a fixed axis of another links, the pair is known as turning pair. A shaft with collar at both ends fitted into a circular hole, the crankshaft in a journal bearing in an engine, lathe spindle support in head stock are the example of turning pair.

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185. If n links are connected at the same joint, the joint is equivalent to :

(a) (n – 1) binary joints
(b) (n – 2) binary joints
(c) (n – 3) binary joints
(d) (2n – 1) binary joints

J&K PSC Civil Services Pre, 2013

Ans. (a) : (n – 1) binary joints

186. If a constant velocity ratio of driving and driven shafts is required then the mechanism used is :

(a) Ackermann gearing
(b) Hooke's joint
(c) Double Hooke's joint
(d) Grass-Hopper mechanism

J&K PSC Civil Services Pre, 2013

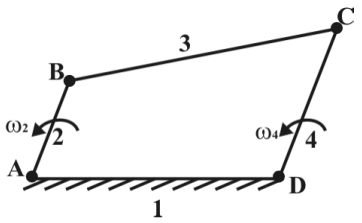
Ans. (c) : The velocity of driven shaft is not constant it varies from maximum to minimum values. In order to have constant velocity ratio of driving and driven shaft, an intermediate shaft with a hooks joint at each end is used. This is double Hooke's joint.
Ackermann gearing is used in steering and Grass hopper mechanism is used in straight line mechanism.

187. For a four-linkage in toggle position, the value of mechanical advantage is-

(a) 0.0 (b) 0.5
(c) 1.0 (d) ∞

APGENCO AE, 2017
TSPSC AEE 2017
RPSC 2016
CGPSC AE 26.04.2015 Shift-I
GATE 2006

Ans : (d)



The extreme positions of a linkage are known as "toggle position"

When, Mechanical Advantage $= \frac{\omega_2}{\omega_4} = \frac{\omega_{input}}{\omega_{output}}$

At extreme position the angular velocity of link (4) becomes 0 (i.e. $\omega_4 = 0$) and hence mechanical advantage becomes infinity.

Mechanical Advantage (M.A.) $= \frac{\omega_2}{\omega_4} = \frac{\omega_2}{0} = \infty$

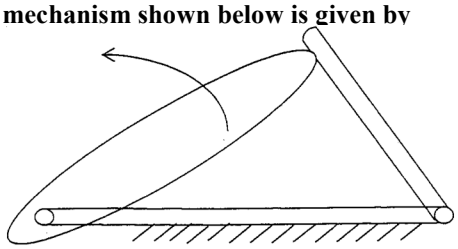
188. Capacity of accumulator indicates amount of.

(a) Work done in lifting the ram
(b) Lift of ram
(c) Work done by accumulator per second
(d) power supplied by accumulator

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Capacity of accumulator indicates amount of work done in lifting the ram.

189. The number of degrees of freedom in the 3 link mechanism shown below is given by



(a) 1 (b) 2
(c) 3 (d) 0

ISRO Scientist/Engineer 12.05.2013

Ans. (a) : Applying Kutzback equation

$$F = 3(\ell - 1) - 2J - h$$

Number of link $\ell = 3$
No. of turning pair or joint J = 2
No. of higher pair h = 1

$F = 3(3 - 1) - 2 \times 2 - 1$
$F = 3 \times 2 - 4 - 1 = 1$

190. Which of the following is an approx. straight line motion mechanism?

(a) Watt's mechanism
(b) Grasshopper mechanism
(c) Robert's mechanism
(d) All of these

SJVN ET 2013

Ans. (d) : Approximate straight line motion mechanism-
(i) Watt's mechanism.
(ii) Modified Scott-Russel mechanism
(iii) Grosshopper mechanism
(iv) Tchebicheff's mechanism
(v) Robert's mechanism

191. A completely constrained motion can be transmitted with:

(a) 1 link with pin joints
(b) 2 link with pin joints
(c) 3 link with pin joints
(d) 4 link with pin joints

SJVN ET 2013

Ans. (d) : A completely constrained motion can be transmitted with 4 link with pin joints.

192. The motion of a piston in the cylinder of a steam engine is an example of

(a) Forced constrained motion
(b) Successfully constrained motion
(c) Incompletely constrained motion
(d) Completely constrained motion

TNPSC AE 2013

Ans. (b) : Successfully constrained motion– When the motion between two elements of a pair possible in more than one direction but is made to have motion only in one direction by using some external means.
Example– Foot step bearing, A piston in a cylinder of an IC engine is made to have only reciprocating motion and no rotary motion due to constraints of the piston pin.

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193. The maximum force which acts on the connecting rod is

(a) Force due to gas pressure
(b) Force due to inertia of piston
(c) Force due to friction of connecting rod
(d) Force due to crank pin

TNPSC AE 2013

Ans. (a) : Force due to gas pressure is the maximum force acts on the connecting rod.

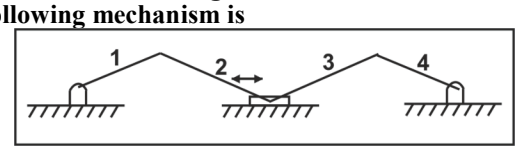
194. Which of the following mechanism is used to enlarge or reduce the size of a drawing ?

(a) Pantograph (b) Bentograph
(c) Watt mechanism (d) Four bar chain

TNPSC AE (Industries) 09.06.2013

Ans. (a) :

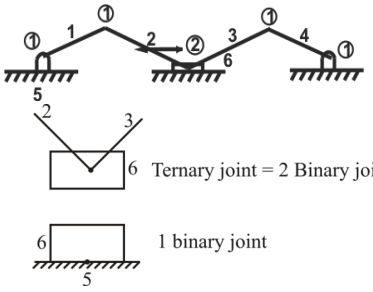
195. The number of degrees of freedom for the following mechanism is



(a) 0 (b) 1
(c) 2 (d) 4

ISRO Scientist/Engineer 2012

Ans. (b) :



No. of links $\ell = 6$

No. of joint j = 7

According to Kutzback Equations

$DOF = 3(\ell - 1) - 2j - h$

$= 3(6-1) - 2 \times 7$

$= 15 - 14 = 1$

196. In a slotted lever and crank quick return mechanism used in shapers, the beginning and end of cutting stroke occurs when

(a) cranked lever are in line with each other
(b) crank is perpendicular to lever
(c) crank is horizontal
(d) lever is horizontal

UKPSC AE 2012 Paper-I

Ans. (b) : crank is perpendicular to lever

197. Identify the wrong statement :

(a) A mechanism is an assemblage of four or more links.
(b) A slider crank chain consists of two sliding pairs and two turning pairs.
(c) A kinematic chain requires at least four links and four turning pairs.
(d) Open pairs are those whose elements are not held together mechanically.

UKPSC AE 2012 Paper-I

Ans. (b) : A slider crank chain consists of one sliding pairs and three turning pairs.

198. Which one of the following is not an example of higher pair ?

(a) Disc Cam and roller follower
(b) Spur Gear meshing teeth
(c) Ball Bearing
(d) Bush Bearing

UKPSC AE 2012 Paper-I

Ans. (d) : Bush Bearing is not an example of higher pair.

199. A framed structure is perfect, if the number of members is ..... (2j – 3), where 'j' is the number of joints.

(a) equal to (b) less than
(c) greater than (d) either (b) and (c)

TNPSC ACF 2012

Ans. (a) : m = 2J – 3 perfect
m > 2J – 3 redundant

200. Match List-I correctly with List II and select your answer using the codes given below

List I	List II
(A) Quick return mechanism	1. Lathe
(B) Apron mechanism	2. Milling machine
(C) Index mechanism	3. Shaper
(D) Regulating wheel grinding	4. Centreless

Codes

	a	b	c	d
(a)	3	2	1	4
(b)	2	3	4	1
(c)	4	2	3	1
(d)	3	1	2	4

TNPSC ACF 2012

Ans. (d) : Quick return mechanism – Shaper
Apron mechanism – Lathe
Indexing mechanism – Milling machine
Regulatory Wheel – Centre less guiding

201. When a cylinder is located in a Vee-block, the number of degrees of freedom which are arrested is

(a) 2 (b) 4
(c) 7 (d) 8

OPSC AEE 2019 Paper-I
GATE 2003

Ans. (b) :

202. Oldham's coupling and elliptical trammels are the inversion of :

(a) Crossed Slider Crank Chain
(b) Four Bar Chain
(c) Single Slider Crank Chain
(d) Double Slider Crank Chain

OPSC Civil Services Pre. 2011

Ans. (d) : Oldham's coupling scotch yoke and elliptical trammels are the inversion of double slider crank chain mechanism.

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203. A negative degree of freedom for a mechanism means :

(a) Constrained motion mechanism
(b) Unconstrained motion mechanism
(c) Any motion
(d) Statically indeterminate structure

OPSC Civil Services Pre. 2011

Ans. (d) : Statically indeterminate structure we know that mobility or degree of freedom of a statically indeterminate structure (means super structure) is always negative.

204. Which of the following is an example of sliding pair?

(a) Piston and cylinder of a reciprocating steam engine
(b) Shaft with collars at both ends fitted into a circular hole
(c) Lead screw of a lathe with nut
(d) Ball and a socket joint

VIZAG Steel MT 2011

Ans. (a) : Piston and cylinder of a reciprocating steam engine

205. What is the mechanism consisting of four links called as?

(a) Simple mechanism
(b) Complex mechanism
(c) Compound mechanism
(d) Convenient mechanism

VIZAG Steel MT 2011

Ans. (a) : Mechanism consisting of four links called as simple mechanism.

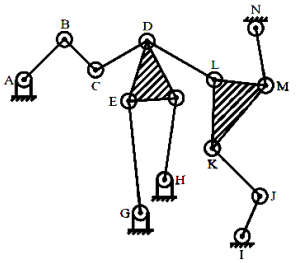
206. Unconstrained rigid link in a plane has

(a) No degree of freedom
(b) One degree of freedom
(c) Two degree of freedom
(d) Three degrees of freedom

UPSC JWM Advt. No.-52/2010

Ans. (d) : Unconstrained rigid link or body in plane has three degree of freedom. They are 2-Translational (about x, y -axis) and one-Rotational (about z-axis).

207.



The mechanism shown in the figure has

(a) One degree of freedom
(b) Two degrees of freedom
(c) Three degrees of freedom
(d) Four degrees of freedom

UPSC JWM Advt. No.-50/2010

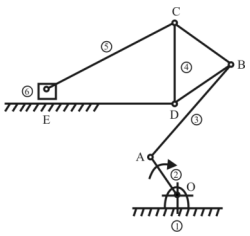
Ans. (c) : From fig. (question), $\ell = 12$, j = 15, h = 0

We know that,

Degree of freedom, $F = 3(\ell - 1) - 2j - h$

$= 3 \times 11 - 2 \times 15$ $\boxed{F = 3}$

208.

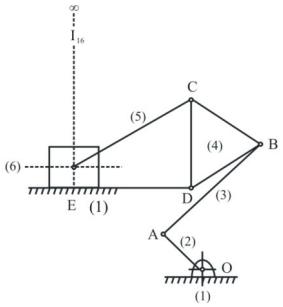


What is the location of I_{16} of the mechanism in the above figure?

(a) At E
(b) At ∞ along the line DE
(c) At ∞ along the perpendicular to DE
(d) At ∞ along the perpendicular to CE

UPSC JWM Adv. No-16/2009

Ans. (c) :



The location of I- Centre (I_{16}) of the mechanism in the figure is at ∞ perpendicular to DE when piston/object sliding on a plane surface along DE.

209. It is planned to construct a four-bar mechanism ABCD with length AB = 60 mm, BC = 100 mm, CD = 70 mm and fixed link AD = 200 mm. If at least one link is required to have a complete rotation, this mechanism is

(a) of crank-rocker type
(b) of double-crank type
(c) of double rocker type
(d) impossible to construct

DRDO Scientists 2009

Ans. (c) : S+L = 60+200 = 260 mm

P+Q = 100+70 = 170 mm

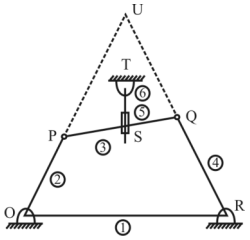
According to Grashoff's law

$S+L \le P+Q$

Here, $S + L > P + Q$ $\Downarrow$ (Double - Rocker) is not satisfied Grashoff's law

Class II mechanism, so always double rocker.

210.



A 6 –link mechanism is shown in the above figure, with a swivel trunnion at 5. What is the location of the instantaneous centre which is at ∞?

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(a) I_{35}, on a line perpendicular to TS
(b) I_{56}, on a line perpendicular to PQ
(c) I_{56}, on a line perpendicular to TS
(d) I_{15}, on a line perpendicular to TS

UPSC JWM Adv. No-16/2009

Ans. (c) : The location of the instantaneous centre I_{56} is at infinite (∞), which is on a line perpendicular to TS because slider link slide along the link TS or link 6.

211. In which of the following machine "quick return motion" incorporated?
(a) Lathe machine
(b) Broaching machine
(c) Slotting machine
(d) Milling machine

WBPSC AE 2008

Ans. (c) :

212. A pantograph is a device that reproduces a displacement
(a) On the same scale (b) On any scale
(c) On reduced scale (d) On enlarged scale

ISRO Scientist/Engineer 2008

Ans. (b) : **Pantograph**—A Pantograph is a mechanical linkage connected in a manner based on parallelograms so that the movement of one pin, in tracing an images produces identical movements in a second pin. If a line drawing is traced by the first point, an identical, enlarged or miniaturized copy with be drawn by a pen fixed to the other.

213. In ideal machines
(a) mechanical advantage is greater than velocity ratio
(b) mechanical advantage is equal to velocity ratio
(c) mechanical advantage is less than velocity ratio
(d) mechanical advantage is unity

TNPSC AE, 2008

Ans. (b) : In ideal machines, mechanical advantage is equal to velocity ratio.

$$\eta = \frac{\text{M.A.}}{\text{V.R.}}$$

Where, M.A. = Mechanical advantage
V.R. = Velocity ratio
For ideal machine $\eta = 100\%$
Then, M.A. = V.R.

214. A kinematic chain requires at least
(a) 2 links and 3 turning pairs
(b) 3 links and 4 turning pairs
(c) 4 links and 4 turning pairs
(d) 5 links and 4 turning pairs

TNPSC AE, 2008

Ans. (c) : A kinematic chain requires at least 4 links and 4 turning pairs.

215. The indicator using Watt mechanism is known as
(a) Thomson indicator (b) Richard indicator
(c) Simplex indicator (d) None of these

TNPSC AE, 2008

Ans. (b) : The indicator using Watt mechanism is called as Richard indicator.

216. A football is rolling on ground, the motion of any point on its surface is
(a) elliptical plane motion (b) helical
(c) spherical (d) plane rotation

UKPSC AE 2007 Paper -I

Ans. (b) :

217. Which of the following is not an inversion of single slider crank mechanism?
(a) Quick return mechanism
(b) Oscillating cylinder engine mechanism
(c) Pendulum pump mechanism
(d) Oldham's coupling

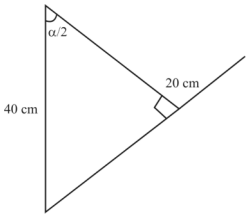
UKPSC AE 2007 Paper -I

Ans. (d) : Oldham's coupling – It is the inversion of double slider crank mechanism.

218. In a slotted lever quick return mechanism, the crank length is 20 cm while the distance between centres of crank and slotted lever rotation is 40 cm. What is the ratio of cutting stroke time to return stroke time?
(a) 2 (b) 3
(c) 4 (d) 5

UKPSC AE 2007 Paper -I

Ans. (a) :



$$\cos\frac{\alpha}{2} = \frac{20}{40} = \frac{1}{2}$$

$$\frac{\alpha}{2} = 60^\circ$$

$$\alpha = 120^\circ$$

$$\text{QRR} = \frac{360 - 120}{120} = \frac{240}{120}$$

$$\text{QRR} = 2$$

219. The number of links in a pantograph is :
(a) 3 (b) 4
(c) 5 (d) 6

J & K PSC Screening, 2006

Ans. (b) : The pantograph is a four bar mechanism used to enlarge or reduce drawings.

220. Shaft rotating inside bearing is an example of :
(a) Higher pair (b) Lower pair
(c) Unclosed pair (d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Shaft rotating inside bearing is an example of lower pair.

221. A six-link mechanism is required to have mobility or degree of freedom equal to one. All the pairs in the mechanism are single degree of freedom pairs. Pick the correct statement about the mechanism:

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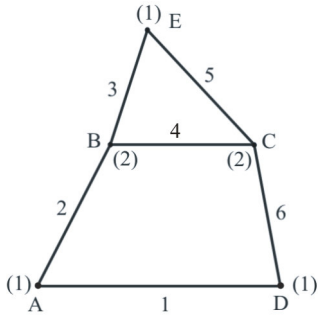
(a) It will have six pairs.
(b) It will have 7 pairs and all the links will be binary links
(c) It will have 7 pairs and two of the six links will be ternary links.
(d) It is not possible to have a mechanism with six links.

RPSC AEN Pre-2013

Ans. (c) :



link = 6, joint = 7 = (1) + (1) + (1) + (2) + (2)

Degree of freedom (F) = 3 (l – 1) – 2j – h

= 3 (6 – 1) – 2 × 7 – 0

[∵ h = higher pair = 0]

= 3 × 5 – 2 × 7

= 15 – 14

= 1

2. Velocity and Acceleration Analysis

222. The Coriolis component of acceleration is considered in

(a) Four-bar chain mechanism
(b) Quick return motion mechanism
(c) Slider-crank mechanism
(d) All of the above
(e) None of these

RPSC ACF & FRO, 26.02.2021
PPSC Asstt. Municipal Engineer 15.06.2021
CIL MT 27.02. 2020, TSPSC AEE 2017
HPPSC Asstt. Prof. 18.09.2017
CIL MT 26.03.2017
UJVNL AE 2016, RPSC AE 2016
GMB AAE 25.06.2017, CGPSC AE 16.10.2016
TANGEDCO AE 2015
TNPSC AE 2013, TSPSC AEE 2015
UKPSC AE 2012 Paper-I, HPPSC Asstt. Prof. 2014
Mizoram PSC AE/SDO 2014 Paper-II
RPSC ACF 2011

Ans. (b) : Coriolis component of acceleration exists when there is sliding motion of a slider which is sliding on a link which itself is rotating. In quick return motion mechanism and crank and slotted liver mechanism coriolis component of acceleration will exist.

Coriolis component of acceleration (a_c) = $2V\omega$.

223. If a mechanism has n links, then the number of instantaneous centers would be equal to :

(a) n (b) n(n – 1)/2
(c) n/2 (d) (n – 1)

GPSC DEE Class-2 (GWSSB) 04.07.2021
APGCL AM, 2021
ESE 2020
WBPSC AE, 2017
CGPSC Polytechnic Lecturer 2017
Nagaland PSC CTSE 2017, Paper-I
GPSC Asstt. Prof. 28.08.2016
CGPSC AE 26.04.2015 Shift-I
Assam PSC CCE Pre 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
TNPSC AE 2014
J&K PSC Civil Services Pre, 2013
APPSC Poly. Lect. 2013
ESE 2011, 1998

Ans. (b) : If number of links are 'n'

then number of instantaneous centers will be ${}^nC_2$

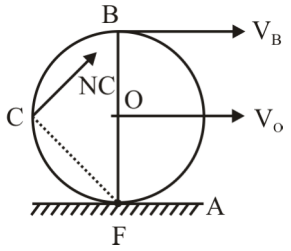
$$= \frac{n(n-1)}{2}$$

224. The instantaneous centre of a link which rolls without slipping upon another fixed link is located at

(a) The centre of the rolling link
(b) The point of contact
(c) The point on the circumference seated vertically opposite to the contact point
(d) An infinite distance on the plane surface

CIL MT 27.02.2020
UPRVUNL AE 21.08.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
UKPSC AE 2012 Paper-I
APPSC AEE 2012
APPSC IOF, 2009
ESE 2002
ESE 1995

Ans. (b)



When body rolls without slipping the point of the contact F at any instant has zero velocity. The point of contact F therefore, is the instantaneous centre of rotation of the body.

225. According to Kennedy's theorem, if three bodies are moving relative to each other, their relative instantaneous centres will lie on:

(a) Straight line (b) Ellipse
(c) Parabola (d) Circle

Oil India Senior Officer 23.12.2020
Nagaland PSC (CTSE) 2018 Paper-I
Punjab PSC (Lect.) 06.08.2017
GPSC Lect. 23.10.2016
TSPSC Managers, 2015
SJVN ET 2013

Ans. (a) : According to Kennedy's theorem if three bodies are moving relative to each other, their relative instantaneous centre will lie on straight line.

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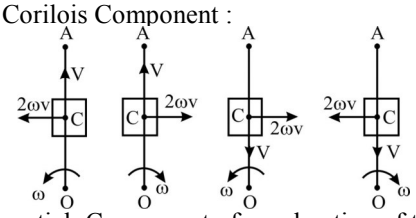
226. Coriolis component of acceleration is present exists whenever a point moves along a path that has :

(a) tangential acceleration
(b) centripetal acceleration
(c) linear motion
(d) rotational motion

MECON MT 2019
APPSC AEE Mains 2016, Paper-III
(KPSC AE 2015)
Kerala PSC AE 06.08.2015
UKPSC AE-2013, Paper-I
WBPSC AE 2008

Ans : (d) Coriolis Component of acceleration is present exists whenever a point moves along a path that has rotational motion.

Corilois Component :



This tangential Component of acceleration of the Slider B with respect to the coincident point C on the link is known as coriolis Component of acceleration and is always perpendicular to the link

∴ Coriolis Component of the acceleration of B with reference of C .

$$a_{BC}^{C} = a_{BC}^{t} = 2\omega V$$

ω = Angular velocity of the link oA .

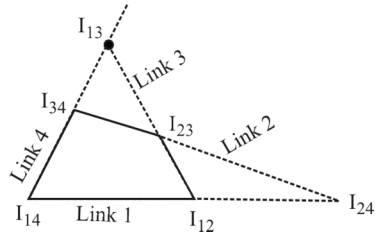
V = Velocity of Slider B with respect of Coincident Pont C .

227. The number of instantaneous centres in a four bar mechanism is

(a) 5 (b) 7
(c) 6 (d) 4

JPSC AE 10.04.2021, Paper-II
Jharkhand Urja Vikas Nigam Ltd. AE 2017
GPSC Lect. (Auto) 16.10.2016
TNPSC AE (Industries) 09.06.2013

Ans. (c) :



Number of instantaneous centres (N) in a four bar mechanism

$$N = \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

N = 6

228. The coriolis component of acceleration acts:

(a) along the sliding surface
(b) perpendicular to the sliding surface
(c) at 45° to the sliding surface
(d) parallel to the sliding surface

Punjab PSC SDE 12.02.2017
Karnataka PSC AE, 10.09.2017
TSPSC Managers, 2015
UKPSC AE-2013, Paper-I

Ans. (b) : The component 2ωV is known as the coriolis acceleration component. It comes into pircture when a body has its motion relative to a moving body, for example motion of a slider on a rotating link.

• The direction of the coriolis acceleration component is obtained by rotating the radial velocity vector 'v' through 90° in the direction of rotation of the link.

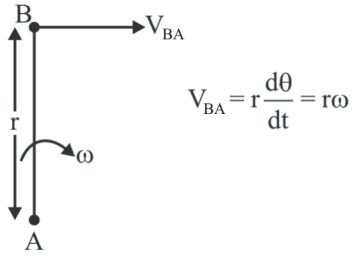
Coriolis component is always perpendicular to the sliding link and in the direction of rotation of the link.

229. The relative velocity of end B with respect to other end A in a rigid link AB is:

(a) perpendicular to AB (b) parallel to AB
(c) along AB (d) inclined to AB

CIL MT 27.02.2020
GPSC ARTO Pre 30.12.2018
HPPSC Asstt. Prof. 20.11.2017
PTCUL AE 25.06.2017

Ans. (a) : The relative velocity of any point with respect to any other point on a moving link is perpendicular to the line joining the point.

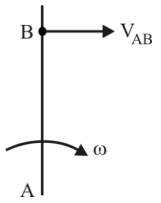


230. The linear velocity of a point B on a link rotating at an angular velocity ω relative to another point A on the same link is :

(a) $\omega^2 \times AB$ (b) $\omega \times AB$
(c) $\omega \times (AB)^2$ (d) $\frac{\omega}{AB}$

Assam PSC CCE Pre 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TNPSC ACF 2012

Ans. (b) : Relative velocity



$V_{AB} = \omega \times AB$

231. A rod of length 1 meter is sliding in a corner as shown in the figure. At an instant the velocity of point 'A' on the rod is 1m/sec when the rod makes an angle of 60^0 with the horizontal plane. The angular velocity of rod at that instant is:

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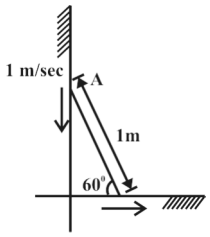
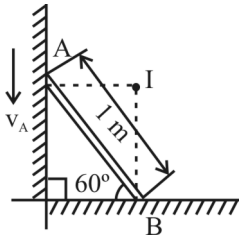


(a) 2 rad/sec (b) 1.5 rad/sec
(c) 0.5 rad/sec (d) 0.75 rad/sec

UPPSC AE 12.04.2016 Paper-I
OPSC Civil Services Pre. 2011
ESE 2009, GATE 1996

Ans. (a) :



Length of rod = 1m
velocity of point A,
$(v_A) = 1m/s$
Let the angular velocity of rod AB about I at that instant be ω

$$v_A = I_A \times \omega$$
$$v_A = (AB \cos 60^0) \times \omega$$
$$= (AB \cos 60^0) \times \omega$$
$$= 1 \times \frac{1}{2}\omega$$
$$\Rightarrow v_A = \frac{\omega}{2}$$
$$\omega = 1 \times 2 = 2 rad/s$$
$$\boxed{\omega = 2 rad/sec}$$

232. When a slider moves with a velocity 'V' on a link rotating at an angular velocity 'ω' then the magnitude of Corioli's component of acceleration is given by

(a) $\sqrt{2}\,V\omega$ (b) $V\omega$
(c) $\frac{V\omega}{2}$ (d) $2\,V\omega$

MPPSC AE 2016
TSPSC AEE 20015
UKPSC AE 2007 Paper -I

Ans. (d) : $a_c = 2\,V\omega$

233. The direction of linear velocity of any point on a link with respect to another point on the same link is

(a) Perpendicular to the link joining the points
(b) At 45^0 to the link joining the points
(c) Parallel to the link joining the points
(d) At 90^0 to the link joining the points

GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC Vice Principal ITI 2018
Rajasthan Nagar Nigam AE 2016, Shift-I
TNPSC AE Industries 09.06.2013

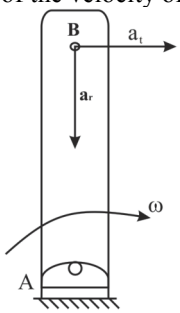
Ans. (a) : The direction of linear velocity of any point on a link with respect to another point on the same link is perpendicular to the link joining the points.

234. A point B on a rigid link AB moves with respect to A with angular velocity ω rad/s. The total acceleration of B with respect to A will be equal to

(a) vector sum of radial component and cariolis component
(b) vector sum of tangential component and cariolis component
(c) vector sum of radial component and tangential component
(d) vector difference of radial component and tangential component

Rajasthan Nagar Nigam AE 2016, Shift-II
NPCIL ET, 2015
TNPSC AE 2013

Ans. (c) : Where-
$a_t \rightarrow$ Tangential component of acceleration [due to change in angular acceleration]
$a_r \rightarrow$ Radial component of acceleration due to change in direction of the velocity of the link.



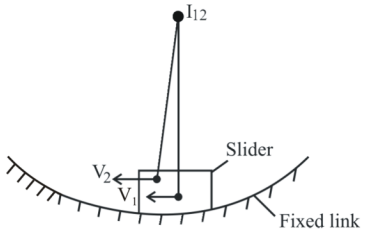
$$a = a_t + a_r$$
$$= AB \times \alpha + AB\,\omega^2$$
$$= AB\,[\alpha + \omega^2]$$

235. When a slider moves on a fixed link having curved surface, their instantaneous centre lies

(a) At the centre of circle
(b) At the pin joint
(c) On their point of contact
(d) At the centre of curvature

RPSC AE 2018
Rajasthan AE (Nagar Nigam) 2016 Shift-3
GATE 1992

Ans. (d) : When slider moves on fixed link having curved surface their instantaneous centre lies on centre of curvature of the curvilinear path in the configuration at that instant.



If n is the no. of links then,
Instantaneous centre $(I) = \frac{n(n-1)}{2}$

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236. The component of the acceleration, parallel to the velocity of the particle, at the given instant is called

(a) radial component
(b) tangential component
(c) Coriolis component
(d) axial component

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Punjab PSC SDE 12.02.2017
Rajasthan Nagar Nigam AE 2016 Shift-II

Ans. (b) : tangential component

$$\boxed{a_t = r \times \alpha}$$

237. Kliens construction can be used when

(a) crank has uniform angular velocity
(b) crank has non-uniform velocity
(c) crank has uniform angular acceleration
(d) crank has a uniform angular velocity and angular acceleration

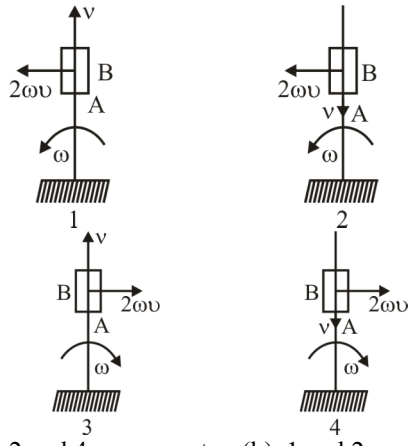
HPPSC Asstt. Prof. 18.09.2017
TNPSC AE 2014

Ans. (a) : Klein's Construction–This construction is applied only in basic single-slider crank mechanism.
When, uniform angular velocity, ω = Constant

$$\frac{d\omega}{dt} = 0$$

$$\boxed{\alpha_{crank} = 0}$$

238. The direction of Coriolis component of acceleration, 2ωv, of the slider A with respect to the coincident point B is shown in the figure below :



(a) 2 and 4 are correct (b) 1 and 2 are correct
(c) 1 and 3 are correct (d) 2 and 3 are correct

OPSC Civil Services Pre. 2011
ESE 1995

Ans. (c) : As far as direction of Coriolis acceleration is concerned,

- If velocity of Slider is away from center of rotations of link, the coriolis acceleration will be in direction of ω.
- If direction of slider velocity in toward the centre of rotation, the coriolis acceleration will be opposite to rotation ω.

So the figure 1 and 3 are correct.

239. The two links OA and OB are connected by a pin joint at O. The link OA turns with angular velocity ω_1 rad/s in the clockwise direction and the link OB turns with angular velocity ω_2 rad/s in the anticlockwise direction. The rubbing velocity at the pin joint O is given by (where r = radius of the pin at O) :

(a) $(\omega_1 + \omega_2)r$ (b) $(\omega_1 - \omega_2)r$
(c) $\omega_1 \cdot \omega_2 . r$ (d) $(\omega_1 - \omega_2)2r$

CGPSC AE 15.01.2021
ESE 1998

Ans. (a) : $\boxed{\text{Sliding Velocity } (v_s) = (\omega_1 + \omega_2)\, r}$

240. The number of Instantaneous Centers in a 6 link mechanism are

(a) 6 (b) 15
(c) 10 (d) 20

TSPSC Managers, 2015
UKPSC AE 2007, 2012 Paper-I

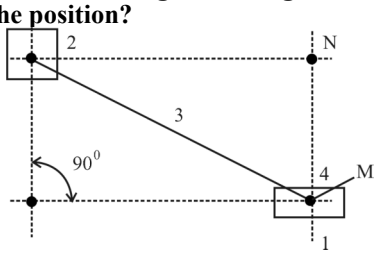
Ans. (b) : link (l) = 6

No. of Instantaneous centre $= \frac{n(n-1)}{2}$

$$= \frac{6(6-1)}{2}$$

$$= \frac{6 \times 5}{2} = 15$$

241. The figure below shows a planar mechanism with single degree of freedom. The instant centre I_{24} for the given configuration is located at the position?



(a) L (b) M
(c) N (d) ∞

RPSC ADE 2016
GATE 2004

Ans. (d) : Total instantaneous context

12 13 14
23 24
34

$I_{14} \rightarrow$ at infinite
$I_{12} \rightarrow$ at infinite
$I_{23} \rightarrow$ at L
$I_{34} \rightarrow$ at M

I_{24} will be at intersection point of line $I_{12} - I_{14}$ and I_{23}-I_{34}. These two line will intersect each other at infinity, So
I_{24} will be at infinity.
I_{13}will be at N.

242. Instantaneous center of a body rolling with sliding on a stationary curved surface lies

(a) At point of contact
(b) On the common tangent at the point of contact
(c) On the common normal at the point of contact
(d) None of these are correct

MPPSC AE 2016, GATE 1992

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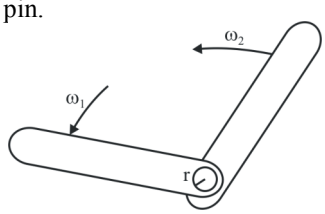
Ans : (c) Instantaneous centre of a body rolling with sliding on a stationary curved surface lies on the common normal at the point of contact.

243. If ω_1 and ω_2 are the angular velocities of the two links which are connected by a pin joint where radius of the pin is r, the rubbing velocity at the pin joint when the links are moving in same direction is given by:

(a) $(\omega_1 - \omega_2)r$ (b) $(\omega_1 + \omega_2)r$
(c) $2(\omega_1 - \omega_2)r$ (d) $2(\omega_1 + \omega_2)r$
(e) $3(\omega_1 - \omega_2)r$

CGPSC Polytechnic Lecturer 2017
APPSC AEE Mains 2016, Paper-III

Ans. (a) Rubbing Velocity– It is defined as the algebraic sum between the angular velocities of the two links which are connected by pin joints, multiplied by the radius of pin.



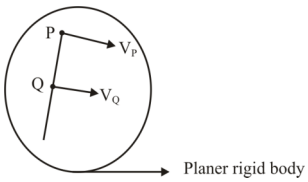
Rubbing velocity, when two links have same direction
$= (\omega_1 - \omega_2) \times r$
Rubbing velocity, when two links have opposite direction of rotation.
$= (\omega_1 + \omega_2) \times r$

244. For two points A and B located on a planar rigid body, the relative velocity between the two points:

(a) should always be perpendicular to AB
(b) should always be perpendicular to AB when the body undergoes pure translation
(c) should always be along AB
(d) can be oriented along any direction

UPRVUNL AE 2014
GATE 2010

Ans : (a)



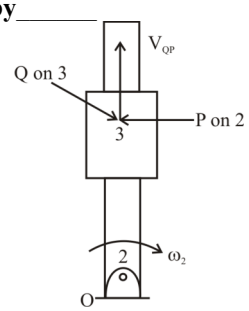
V_{PQ} = Relative velocity between P and Q
$V_{PQ} = V_p - V_Q$ Always perpendicular to PQ
Considering two points P and Q on a rigid body. The relative position of point P with respect to Q is r_{PQ}. The relative velocity of P with respect to Q is $V_{PQ} = \dot{r}_{PQ}$. If the relative distance between the P and Q does not change then V_{PQ}

$V_{PQ} = \dot{r}_{P/Q} = \omega \times \dot{r}_{P/Q}$

ω is the angular velocity of the rigid body.

Since V_{PQ} is the cross product of ω and $\dot{r}_{P/Q}$ the relative velocity of P with respect to Q will be perpendicular to PQ.
Note :– The commission has given answer option (d).

245. Link 2 rotates with constant angular velocity ω_2. Slider link 3 moves with a constant relative velocity $V_{Q/P}$, where Q is a point on slider 3 and P is a point on link 2. The magnitude and direction of Coriolis component of acceleration is given by_____



(a) $2\omega_2 V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90º in the direction ω_2
(b) $\omega_2 V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90º in the direction ω_2
(c) $2\omega_2 V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90º opposite to the direction of ω_2
(d) $\omega_2 V_{Q/P}$; direction of $V_{Q/P}$ rotated by 90º opposite to the direction ω_2

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

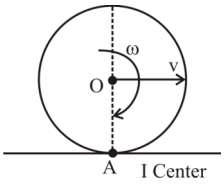
Ans. (a) : Coriolis component of acceleration
$= 2\omega_2 \times V_{Q/P}$
Direction of acceleration (Coriolis) perpendicular to slider and in direction of ω_2.

246. A circular object of radius r rolls without slipping on a horizontal level floor with the center having velocity V. The velocity at the point of contact between the object and the floor is___.

(a) V opposite to the direction of motion
(b) V in the direction of motion
(c) Zero
(d) V vertically upward from the floor

Assam Engg. College AP/Lect. 18.01.2021
GATE 2014

Ans. (c) :



Velocity of point 0 = $\omega \times OA$
$= \omega \times r$
Velocity at point of contact (A)
$= \omega \times AA$
$= \omega \times 0$
$V_A = 0$

Common data for questions 130 and 131
The circular disc shown in its plane view in the figure rotates in a plane parallel to the horizontal plane about the point O at a uniform angular velocity ω. The other points A and B are located on the line OZ at a distance r_A and r_B from O respectively.

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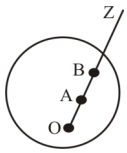


247. The velocity of point B with respect to A is a vector of magnitude

(a) 0

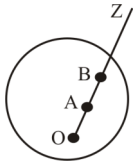
(b) $\omega(r_B - r_A)$ and direction opposite to the direction of motion of point B.

(c) $\omega(r_B - r_A)$ and direction same as the direction of motion of point B.

(d) $\omega(r_B - r_A)$ and direction being from O to Z.

TSPSC AEE 2017
GATE 2003

Ans. (c) :



$\rightarrow$ B has the linear velocity $\omega_B r_B$

$\rightarrow$ A has linear velocity $\omega_A r_A$

Obviously $\omega_B r_B > \omega_A r_A$, because, $\omega_A = \omega_B = \omega$

$r_B > r_A$

So, relative velocity = $\omega(r_B - r_A)$ in the direction of motion of point B.

248. The acceleration of point B with respect to point A is a vector of magnitude

(a) 0

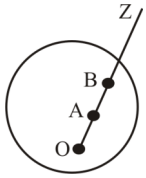
(b) $\omega(r_B - r_A)$ and direction opposite to the direction of motion of point B.

(c) $\omega(r_B - r_A)$ and direction same as the direction of motion of point B.

(d) $\omega^2(r_B - r_A)$ and direction being from Z to O.

TSPSC AEE 2017
GATE 2003

Ans. (d) :



Acceleration of point B with respect to point A

$$\alpha_{B/A} = \bar{\omega} \times v_{B/A} = \omega \times \omega(r_B - r_A) = \omega^2 (r_B - r_A)$$

and direction from Z to O

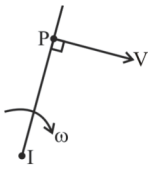
249. The velocity of any point 'P' on a body rotating with angular velocity 'ω' and with instantaneous centre of rotation 'I' is equal to:

(a) $P/\omega I$ (b) $IP\omega$

(c) $I/P\omega^2$ (d) $I^2P\omega^2/2$

HPPSC Lect. (Auto) 23.04.2016
(HPPSC LECT. 2016)

Ans. (b) :



Tangential velocity of point P with respect to I

V = radial distance × angular velocity.

$V = IP\omega$

250. The instantaneous center of rotation is a point of zero

(a) Angular velocity (b) Linear velocity

(c) Displacement (d) Acceleration

(e) Acceleration and Displacement

HPPSC Lect. (Auto) 23.04.2016
CGPSC AE 26.04.2015 Shift-I

Ans. (b) : Instantaneous center of velocity—
Any point on a rigid body or on its extension that has zero velocity is called the instantaneous center of velocity of the body.

251. A mechanism has 7 links with all binary pairs except one which is ternary pair. The number of instantaneous centres of this mechanism are:

(a) 14 (b) 21

(c) 28 (d) 42

TSGENCO AE 14.11.2015
OPSC Civil Services Pre. 2011

Ans. (c) : Number of links = 6 binary + 1 ternary

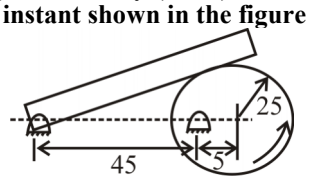
= 6 binary + 2 binary ($\because$ 2B = 1T)

= 8 binary

$$\text{Instantaneous centers of the Mechanism} = \frac{n(n-1)}{2}$$

$$= \frac{8(8-1)}{2} = 28$$

252. In the mechanism given below, if the angular velocity of the eccentric circular disc is 1 rad/s, the angular velocity (rad/s) of the follower link for the instant shown in the figure is_____



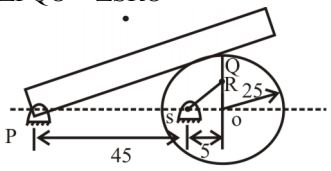
Note: All dimensions are in mm.

(a) 0.1 (b) 0.05

(c) 5.0 (d) 10.0

Assam Engg. College AP/Lect. 18.01.2021
GATE 2012

Ans. (a) : $\Delta PQO \sim \Delta SRO$



(Velocity will be some on Q & R because they are on same link)

$$\therefore \quad \frac{PQ}{RS} = \frac{PO}{OS}$$

$$PQ = \sqrt{(45+5)^2 - 25^2}$$

= 43.3 mm

Then,

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$$\frac{43.3}{RS} = \frac{50}{5}$$

$RS = 4.33$ mm

$VR = VQ = RS \times \omega$

$= 4.33 \times 1$

$VQ = 4.33$

$$\omega_{PQ} = \frac{VQ}{PQ} = \frac{4.33}{43.3} = 0.1 \text{ rad/sec.}$$

Velocity will be some on Q & R because they are on same link.

253. The component of acceleration parallel to the link is called

(a) radial (b) tangential
(c) Coriolis (d) absolute

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : The radial component also known as the normal component is parallel to the link and always exists as long as there is angular velocity.
The acceleration of a particle whose velocity changes both in magnitude and in a direction at any instant has the radial component and the tangential component.

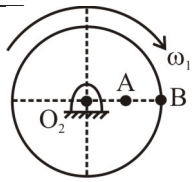
254. The velocity of an intermediate point on any of the links can be found by dividing the corresponding.....vector in the same ratio as the point divides the link.

(a) Position (b) Direction
(c) Acceleration (d) Velocity

HPPSC AE (PWD) 10.09.2016

Ans. (d) : The velocity of an intermediate point on any of the links can be found by dividing the corresponding velocity vector in the same ratio as the point divides the link.

255. Figure shows a wheel rotating about O_2. Two points A and B located along the radius of wheel have speeds of 80 m/s and 140 m/s respectively. The distance between the point A and B is 300 mm. The radius of the wheel (in mm) is_____



(a) 1400 (b) 400
(c) 700 (d) 800

Assam Engg. College AP/Lect. 18.01.2021

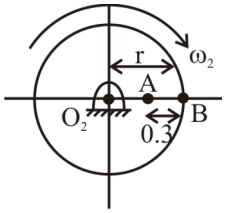
Ans. (c) : Given,

$AB = 300$ mm

$= 0.3$ m

$V_A = 80$ m/s

$V_B = 140$ m/s



Assume radius = r(m)

$$\frac{V_A}{r - 0.3} = \frac{V_B}{r}$$

$$\frac{80}{r - 0.3} = \frac{140}{r}$$

$80r = 140r - 42$

$$r = \frac{42}{60}$$

$r = 0.7$ m

or $r = 700$ mm

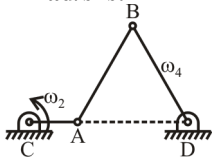
256. If three bodies have planer motion, their instantaneous centres lie on :

(a) two parallel lines
(b) two perpendicular lines
(c) two intersecting non-perpendicular lines
(d) on same straight line

JPSC AE 10.04.2021, Paper-II

Ans. (d) : According to kennedy's theorem-
If three bodies have planer motion their instantaneous centre lies on same straight line.

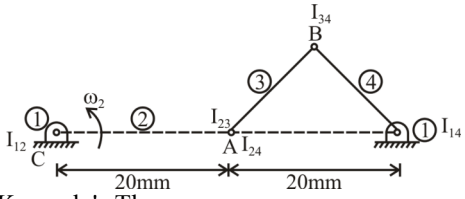
257. In the mechanism shown here, points C, A and D are in same straight line. If CA = 20 mm, CD = 40 mm and ω_2 = 4rad/s (CCW), then the value of ω_4 in rad/s is:



(a) 3.5 (CCW) (b) 4.0 (CCW)
(c) 4.0 (CW) (d) 3.5 (CW)

AAI Jr. Executive 26.03.2021

Ans. (c) : Given, $\omega_2 = 4$ rad/sec (CCW)
CA = 20 mm
CD = 40 mm



From Kennedy's Theorem,

$\omega_2(I_{12}I_{24}) = \omega_4(I_{14}I_{24})$

$4(20) = \omega_4(20)$

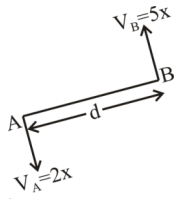
$\omega_4 = 4$rad/s (CW)

258. Two points on a link are moving with velocities of 5x and 2x perpendicular to the link in the same direction, separated by a distance d. What is the angular velocity of the link?

(a) 7x/d (b) 10x/d
(c) 3x/d (d) 2.5x/d

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) :



$V_{AB} = 5x - 2x$

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$= 3x$

We know that,

$$V_{AB} = d \times \omega_{AB}$$

$$\boxed{\omega_{AB} = \frac{3x}{d}}$$

259. Under which of the following conditions is Coriolis component encountered in the relative acceleration of two points?

1. The two points are coincident, but on different links.
2. The point on one link traces a path on the other link.
3. The link that contains the path rotates.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) : Coriolis component encountered in the relative acceleration of two points when-

- The two points are coincident, but on different links.
- The point on one link traces a path on the other link.
- The link that contains the path rotates.

260. Coriolis component of the acceleration is experienced:

(a) by a point on one link that is subjected to both dynamic and static forces.
(b) by a rotating link with varying angular velocity
(c) by a rotating link with varying angular displacement.
(d) by a point on one link that is sliding along another rotating link

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Coriolis component of the acceleration is experienced by a point on link that is sliding along another rotating link. The component $2v\omega$ is known as coriolis acceleration component.

$$a_{coriolis} = 2v\omega$$

A crank and slotted lever mechanism has coriolis component $2\omega v_{slide.}$

261. The Coriolis component of acceleration depends on

(a) velocity and acceleration of the slider
(b) angular velocity and angular acceleration of the link
(c) velocity of the slider and angular velocity of the link
(d) acceleration of slider and angular velocity of the link

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : The Coriolis component of acceleration depends on-

Coriolis acceleration, $(a_c) = 2\omega v$

v = velocity of slider

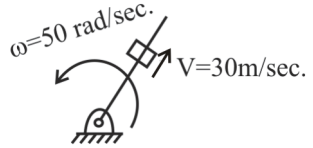
ω = Angular velocity of the link

262. If a block slides outward on a link at a uniform rate of 30 m/s, while the link is rotating at a constant angular velocity of 50 rad/s counter clockwise, the Coriolis component of acceleration is_____m/s^2.

(a) 1000 (b) 1500
(c) 3000 (d) 4500

ISRO Scientist/Engineer 12.01.2020

Ans. (c) :



Coriolis component of acceleration,

$a_c = 2\omega V = 2 \times 30 \times 50$

$a_c = 3000\ m/sec^2$

263. The magnitude of the velocity of any point on the kinematic link relative to the other point on the same kinematic link is the product of

(a) A square root of an angular velocity of the link and the distance between the two points under consideration
(b) An angular velocity of the link and the square of distance between the two points under consideration
(c) A square of an angular velocity of the link and the distance between the two points under consideration
(d) An angular velocity of the link and the distance between the two points under consideration

ESE 2020

Ans. (d) : An angular velocity of the link and the distance between the two points under consideration.

264. Where will be the instantaneous centre of rotation of a point A having linear velocity v equal to zero?

(a) Instantaneous centre of rotation will be at infinity of a point A
(b) Instantaneous centre of rotation will be at itself of a point A
(c) Instantaneous centre of rotation will be at distance r from point A
(d) Instantaneous centre of rotation will not exist for a point A

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : $v = r\omega$

$\because \quad r = 0$

$\therefore \quad v = 0$

265. Which of the graphical methods is NOT used for determining the velocity and acceleration of the reciprocating parts in an engine?

(a) Klien's construction
(b) Ritterhaus' construction
(c) Bennett's construction
(d) Mohr's circle diagram

APPSC Poly Lect. Automobile Engg., 2020

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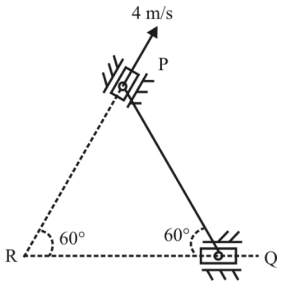
Ans. (d) : Following graphical methods is used for determining the velocity and acceleration of the reciprocating parts in an engine –
(a) Klien's construction
(b) Ritterhaus' construction
(c) Bennett's construction

266. A rigid rod PQ shown in Figure is connected to two sliders at each end through pins. The sliders can slide along PR and QR. If the velocity of the slider at P is 4m/s, the velocity of the midpoint of the rod PQ at this instant is :



(a) 4 m/s (b) 3 m/s
(c) 2 m/s (d) 1 m/s

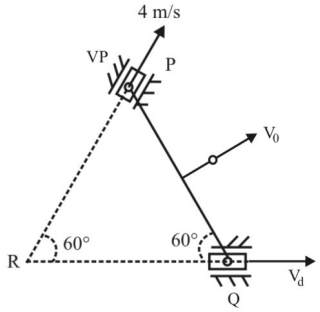
ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : $\frac{v_P}{PQ} = \frac{v_0}{OQ}$

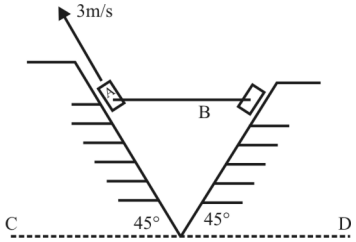
$\frac{4}{PQ} = \frac{v_0}{\frac{PQ}{2}}$

$4 = 2v_0$

$\boxed{v_0 = 2m/s}$



267. For the position of the mechanism shown in Figure, find the velocity of the slider B for the given configuration if the velocity of the slider A is 3m/s. Consider AB as a rigid link and is parallel to CD at this instant.



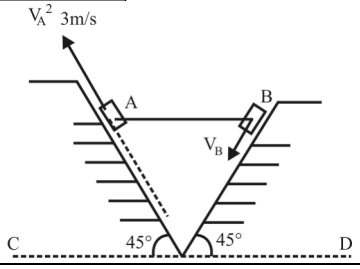
(a) 3m/s (b) $3\sqrt{2}m/s$
(c) $3/\sqrt{2}m/s$ (d) $1.5m/s$

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : $V_A \cos 45° = V_B \sin 45°$

$3 \times \frac{1}{\sqrt{2}} = V_B \times \frac{1}{\sqrt{2}}$

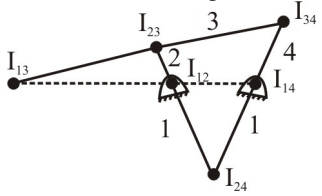
$\boxed{V_B = 3m/\sec.}$



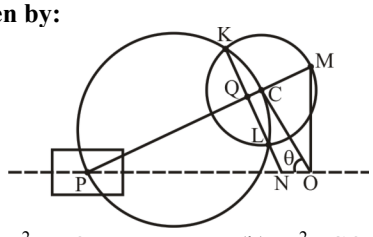
268. _____ is not associated with the vector calculus.
(a) Stokes's theorem (b) Gauess Theorem
(c) Kennedy's Theorem (d) Greens Theorem

Oil India Limited Sr. Engg. (Production) 30.11.2019

Ans. (c) : Kennedy's theorem- It is a three centre in-line theorem. Which states that if three links have relative motion with respect to each other, their relative instantaneous centre lies on straight line.



269. In the below figure, PC is the connecting rod and OC is the crank making an angle θ with the line of stroke PO and rotates with uniform angular velocity at ω rad/s. The Klein's acceleration diagram for determining the acceleration of the piston P is shown by quadrilateral CQNO. The acceleration of the piston P with respect to the crankpin C is given by:



(a) ω^2x NO (b) ω^2x CO
(c) ω^2x CN (d) ω^2x QN

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) The Klein's acceleration diagram is drawn as discussed below–
(1) First of all draw a circle with 'C' as centre as 'CM' as radius.
(2) Draw another circle with PC as diameter.
Let this circle intersect the previous circle at K and L.
(3) Join KL and produce it to intersect PO at N. Let KL intersect PC at Q.
This forms the quadrilateral CQNO, which is known as Klein's acceleration diagram.

$\boxed{\text{Hence, Acceleration of piston}\left(a_p\right) = \omega^2 \times NO}$

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270. If the rotational speed of a crank of radius 50 cm is 100 RPM and the angular acceleration is 50 rad/s^2, then the tangential acceleration of the crank will be:

(a) 50 m/s^2 (b) 25 m/s^2
(c) 0.25 m/s^2 (d) 2.5 m/s^2

AAI Jr. Executive 29.11.2018

Ans. (b) : Given, $(\alpha) = 50$ rad/sec^2

$r = 50$ cm $= 50 \times 10^{-2}$ m

Tangential acceleration $(a_t) = r\,\alpha$

$= 50 \times 10^{-2} \times 50$

$= 2500 \times 10^{-2}$

$a_t = 25$ m/sec^2

271. A slider moving at 150 mm/s on a link rotating at 60 rpm will have ..... Coriolis acceleration.

(a) 600 π mm/s^2 (b) 400 π mm/s^2
(c) 700 π mm/s^2 (d) 300 π mm/s^2

RPSC AE 2018

Ans. (a) : Data given-

$N = 60$ rpm

$V = 150$ mm/s

We know that,

Coriolis acceleration

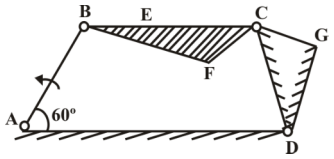
$a_c = 2\,V \times \omega$

$= \dfrac{2 \times 150 \times 2\pi \times N}{60}$

$= \dfrac{2 \times 150 \times 2\pi \times 60}{60}$

$\boxed{a_c = 600\pi \text{ mm/s}^2}$

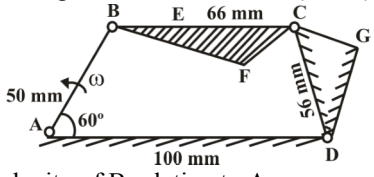
272. A four link mechanism shown in the figure has link lengths as AB = 50 mm, BC = 66 mm, CD = 56 mm and AD = 100 mm. If at an instant when $\angle DAB = 60^\circ$ and the link AB has an angular velocity of 10.5 rad/s in the counter - clockwise direction, the velocity of B relative to A (vector v_{ba}) (in m/s) of the point C is .....



(a) 0.131 (b) 0.262
(c) 0.393 (d) 0.525

RPSC AE 2018

Ans. (d) : Data given : $\omega = 10.5$ rad/s (CCW)

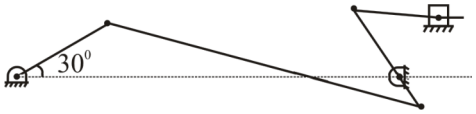


Then the velocity of B relative to A.

$V_{ba} = \omega \times AB = 10.5 \times 0.05$

$V_{ba} = 0.525$ m/s

273. Number of instantaneous centre for the mechanism will be



(a) 6 (b) 12
(c) 14 (d) 15

TNPSC AE 2018

Ans. (d) : Number of links in this mechanism (n) = 6

Number of I-centre

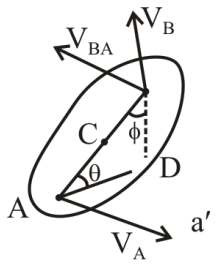
$= \dfrac{n(n-1)}{2} = \dfrac{6 \times (6-1)}{2} = 15$

274. A point B on a rigid link AB moves with respect to A with angular velocity ω rad/s. The radial component of the acceleration of B with respect to A, is

(a) $v_{BA} \times AB$ (b) v_{BA}/AB
(c) $v_{BA}^2 \times AB$ (d) v_{BA}^2 / AB

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) :



$\omega_{AB} = \dfrac{v_{BA}}{AB} = \dfrac{ab}{AB}$

Vector $a'x = a_{BA}^r$

$= \dfrac{v_{BA}^2}{AB}$

v_{BA} = Velocity of B with respect to A.

275. A wheel accelerates uniformly from rest to 1200 rpm in 20 seconds. Its angular acceleration will be _____ rad/s^2.

(a) 1π (b) 2π
(c) 3π (d) 4π
(e) 5π

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : $N_1 = 0$ $N_2 = 1200$ rpm

$\omega_1 = 0$ $\omega_2 = \dfrac{2\pi N_2}{60} = \dfrac{2 \times \pi \times 1200}{60}$

$t = 20$ sec. $\omega_2 = 40\pi$ rad/sec.

Then angular acceleration (α)

$\alpha = \dfrac{\omega_2 - \omega_1}{t} = \dfrac{40\pi - 0}{20}$

$\boxed{\alpha = 2\pi \text{ rad/sec}^2}$

276. In a slider crank mechanism if the crank rotates at uniform speed of 200 rpm and has a length of 0.2 m, its linear velocity is:

(a) 4.19 m/s (b) 20.9 m/s
(c) 5.2 m/s (d) 41.9 m/s

TRB Polytechnic Lecturer 2017

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Ans. (a) : Data given as,

$N = 200$ rpm

$\ell = 0.2$ m

$V = ?$

We know that

$$V = r\omega = r \cdot \left[\frac{2\pi N}{60}\right]$$

$$V = \frac{0.2 \times 2 \times 3.14 \times 200}{60}$$

$$V = 4.186\,m/s \simeq 4.19\ m/s$$

277. In Klein's construction for reciprocating engine mechanism, the scale of acceleration diagram will be

(a) linear scale of configuration diagram

(b) Linear scale of configuration diagram multiplied by square of angular velocity of crank

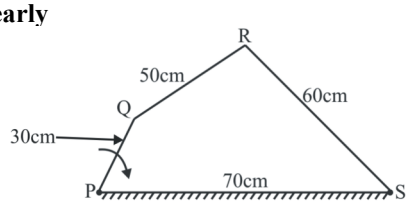
(c) square of the linear scale of configuration diagram

(d) square of the linear scale of configuration diagram multiplied by square of angular velocity of crank

BPSC AE Mains 2017 Paper - VI

Ans : (b) : Linear scale of configuration diagram multiplied by square of angular velocity of crank.

278. In the 4-bar mechanism as shown, the link PQ measures 30 cm and rotates uniformly at 100 rev/min. The velocity of point Q on link PQ is nearly



(a) 2.54 m/s (b) 3.14 m/s

(c) 4.60 m/s (d) 5.80 m/s

ESE 2017

Ans. (b) : $r_{PQ} = 30$ cm

$N = 100$ rpm

$$V_{PQ} = r_{PQ} \times \omega$$

$$= 0.3 \times \left(\frac{2\pi \times 100}{60}\right)$$

$= 3.14$ m/s

279. If two rigid bodies A and B are moving in the same plane relative to a third rigid body C, then the instantaneous centres of A relative to C, of B relative to C and B relative to A all lies.

(a) In a straight line (b) On a same point

(c) On a circle (d) None of these

PTCUL AE 25.06.2017

Ans. (a) : In a straight line

280. In the transmission system, the provision of slip joints allows a change in the :

(a) Angle of gear (b) Length of shaft

(c) Turning angle (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : In the transmission system, the provision of slip joints allow a change in the length of shaft.

281. If a point moves along a straight line which is rotating then the longitudinal component of acceleration is

(a) $\frac{V^2}{r}$ (b) $\frac{dv}{dt} - \omega^2 r$

(c) $\frac{dv}{dt}$ (d) $2v\omega + r \times \alpha$

(e) none of these

Where v = velocity of point along the straight line

ω = angular velocity of line

α = angular acceleration of the line

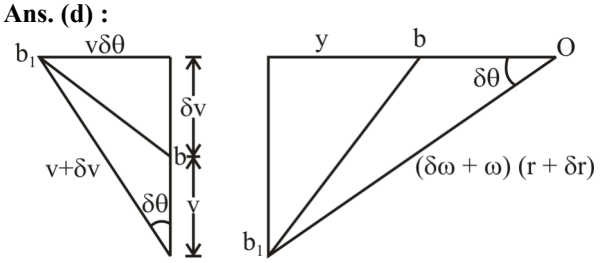
r = radius of the point

V = tangential velocity of the line

$= \omega \times r$

CGPSC AE 16.10.2016

Ans. (d) :



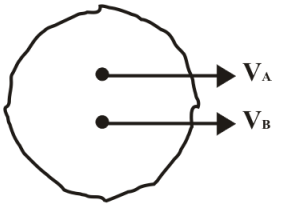
From these, we get,

$$a_{Bo} = \frac{vd\theta}{dt} + \frac{\omega dr}{dt} + \frac{rd\omega}{dt}$$

$= v \cdot \omega + \omega \cdot v + r\alpha$

$= 2v\omega + r\alpha$

282. If the direction of velocity at two different points (A and B) on a rigid body are parallel and equal in magnitude ($V_A = V_B$) as shown in figure. Then its instantaneous center will be:



(a) Outside the body but not at infinity

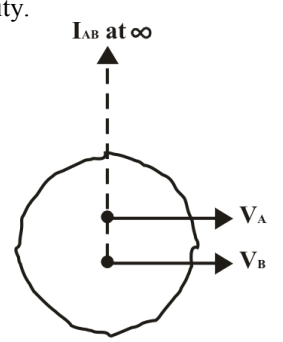
(b) At infinity

(c) On the surface of the body

(d) At centroid of the body

UPRVUNL AE 07.10.2016

Ans. (b) : If the direction of velocity at two different points (A and B) on a rigid body are parallel and equal in magnitude ($V_A = V_B$) then its instantaneous center will be at infinity.



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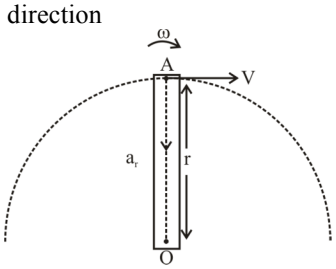
283. A link (OA) of length 'r' with centre 'O', rotates in a circular path in clockwise direction with constant angular velocity (ω). The tangential and radial acceleration of point (A) relative to center is respectively given by:

(a) $\omega^2 r$, 0 (b) $\omega^2 r$, $\omega^2 r$

(c) 0, 0 (d) 0, $\omega^2 r$

UPRVUNL AE 07.10.2016

Ans. (d) : For link OA which rotates in a circular path in clockwise direction



with constant angular velocity (ω = constant) so tangential acceleration will be zero, whereas radial or normal acceleration will be equal to $r\omega^2$

$$a_r = r\omega^2$$

284. In a general plane motion, if a body is rolling without slipping on stationary surface then the point of contact with stationary surface is the.....

(a) Axial of rotation

(b) Instantaneous centre of rotation

(c) Centroidal axis of rotation

(d) Point of inflexion

HPPSC AE (PWD) 10.09.2016

Ans. (b) : The instantaneous centre of rotation is the point fixed to a body under going planer movement that has zero velocity at a particular instant of time. At this instant the velocity vectors of the other points in the body generate a circular field around this point which is identical to what is generated by a pure rotation.

The number of instantaneous centre (N) in four bar mechanism.

$$N = \frac{n(n-1)}{2}$$

$$n = 4$$

$$N = \frac{(4-1)4}{2} = 6$$

285. Which one of the following is false for instantaneous center of rotation?

(a) At the instantaneous center of rotation, one rigid link rotates instantaneously relative to another for configuration of the mechanism concerned

(b) At the instantaneous centre of rotation, the two rigid links have no linear velocities relative to each other

(c) both (1) and (2) are true

(d) both (1) and (2) are false

TSGENCO AE 14.11.2015

Ans. (c) : Properties of instantaneous centre –

- At the instantaneous centre of rotation, one rigid link rotates instantaneously relative to another, for configuration of the mechanism concerned.
- The two rigid links have no relative velocity to each other at the instantaneous centre.
- The two rigid link have the same linear velocity relative to the third link or any other link.

286. In a link AB, the part B moves relative to A. If ω is the angular velocity and α_{AB} is the angular acceleration, the total acceleration of B relative to A will be

(a) $\omega_{AB}^2 \times AB + \alpha \times AB$

(b) $\omega_{AB} \times AB + \alpha \times AB$

(c) $\sqrt{\left(\omega_{AB}^2 \times AB\right)} + \alpha \times AB$

(d) $\sqrt{\omega_{AB}^2 \times AB + \alpha \times AB}$

TSGENCO AE 14.11.2015

Ans. (d) : ω = Angular velocity

α = Angular acceleration

link = AB

Radial Acceleration (centripetal acceleration) – Towards centre

$a_r = \omega^2 \times AB$

Tangential Acceleration – Change in magnitude of velocity

$$a_T = \alpha \times AB$$

Resultant, $a_R = \sqrt{(a_R)^2 + (a_T)^2}$

$$a_R = \sqrt{\omega^2 \times AB + \alpha \times AB}$$

287. The space Centrode of a Circular disc on a straight line path is :

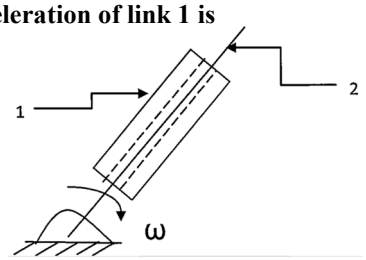
(a) A Circle (b) A parabola

(c) A straight line (d) A hyperbola

TSPSC Managers, 2015

Ans. (c) : The space centrode of a circular disc on a straight line path is straight line.

288. In the figure shown below, the relative velocity of link 1 with respect to link 2 is 15 m/s. The link 2 rotates at a constant speed of 150 rpm. The magnitude of Coriolis component of acceleration of link 1 is

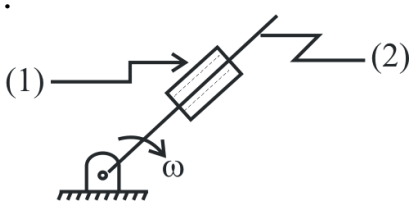


(a) 201 m/s^2 (b) 239 m/s^2

(c) 418 m/s^2 (d) 471 m/s^2

ISRO Scientist/Engineer 24.05.2014

Ans. (d) :



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$v = 15 m/s$

$\omega = \frac{2\pi n}{60}$

$N = 150 rpm$

$\omega = \frac{2\pi \times 150}{60}$

Coriolis Component of Acceleration

Coriolis Acceleration $\boxed{a_c = 2v\omega}$

$a_c = 2 \times 15 \times \frac{2\pi N}{60}$

$= 2 \times 15 \times \frac{2 \times \pi \times 150}{60}$

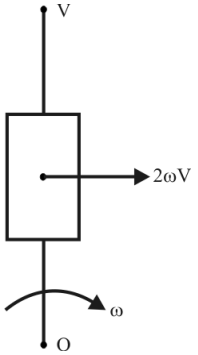
$a_c = 471 m/s^2$

289. A slider moves with uniform velocity V on a revolving link of length r with angular velocity ω. The Coriolis acceleration component of a point on the slider relative to a coincident point on the link is equal to :

(a) 2rω parallel to the link.
(b) 2ωV perpendicular to the link
(c) 2ωr perpendicular to the link
(d) 2ωV parallel to the link

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Coriolis acceleration = 2ωV



The component 2Vω is known as Coriolis acceleration component.

The direction of the Coriolis acceleration component is obtained by rotating the radial velocity vector 'V' through 90° in the direction of rotation of the link.

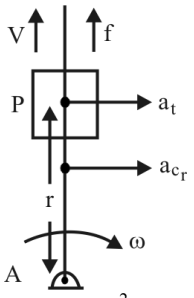
290. The component of the acceleration directed towards the centre of rotation of a revolving body is known as _____ component.

(a) tangential
(b) centripetal
(c) coriolis
(d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : The component of the acceleration directed towards the centre of rotation of a revolving body is known as centripetal component.

Tangential acceleration $(a_t) = r\alpha$



Acceleration along AP = $f - \omega^2 r$

Centripetal acceleration $a_c = \omega^2 r$

Coriolis acceleration $a_{c_r} = 2V\omega$

291. The Coriolis component of acceleration is present in:

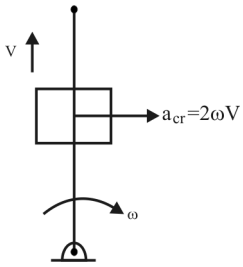
(a) four bar mechanism
(b) scotch yoke mechanism
(c) slider crank mechanism
(d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (d) : The coriolis component of acceleration is present in mechanism which have rotation & translation on same link.

It exist in –

(a) Whitworth quick return mechanism
(b) Crank & slotter lever mechanism.



292. A slider sliding at 10 cm/s on a link, which is rotating at 60 rpm. The Coriolis acceleration of the slider is

(a) 40 cm/s^2 (b) 0.4π c cm/s^2
(c) 40π cm/s^2 (d) 20 cm/s^2

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Given

Sliding Velocity (V) = 10 cm/sec.

Link rotating (N) = 60 rpm

$\omega = \frac{2\pi N}{60}$

$\omega = \frac{2 \times \pi \times 60}{60}$

$\omega = 2\pi$ rad/sec

Coriolis Acceleration $(a^c) = 2v\omega$

$= 2 \times 10 \times 2\pi$

$= 40\pi$ cm/sec^2

293. Klein's diagram is used to find

(a) Angular velocity of various parts
(b) Velocity of various parts
(c) Acceleration of various parts
(d) Displacements of the various parts
(e) Velocity and acceleration of various parts

CGPSC AE 26.04.2015 Shift-I

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Ans. (e) : Klein's diagram is used to find velocity and acceleration of various parts of single slider crank mechanism.

294. A crank of radius 12 cm is rotating at 60 rpm with an angular acceleration of 50 rad/sec^2. The tangential acceleration of the crank is

(a) 4.75 m/s^2 (b) 5.2 m/s^2
(c) 6 m/s^2 (d) 7.4 m/s^2

TNPSC AE 2014

Ans. (c) : Given, r = 12 cm, N = 60 rpm

$\alpha = 50$ rad/sec^2

$a_t = ?$

We know that the tangential acceleration of the crank is given as

$a_t = r\alpha = 0.12 \times 50$

$a_t = 6$ m/s^2

295. In which of the following mechanism, the Coriolis acceleration exists ?

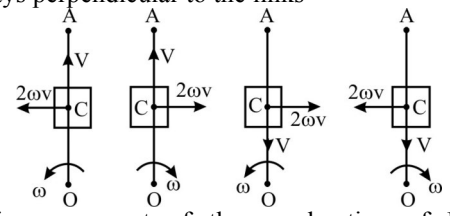
(a) Whitworth quick return mechanism
(b) Tangent cam mechanism
(c) Both (A) and (B)
(d) One of these

(HPPSC AE 2014)

Ans : (a) The Coriolis acceleration exists.

(i) Whitworth quick return mechanism

Coriolis component is on component of acceleration of the slider B with respect to the coincident point C on the link is known as coriolis component of acceleration and is always perpendicular to the links



Coriolis component of the acceleration of B with respect of C

$$a^C_{BC} = a^t_{BC} = 2\omega v$$

ω = Angular velocity of the link OA.

V = Velocity of slider respect to coincident point C.

296. The mechanical advantage for a rigid link AB, whose driving end is A whereas driven end is B (neglecting friction effect), is given by :

(a) V_A/V_B (b) V_B/V_A
(c) $V_B.V_A$ (d) None of the above

J&K PSC Civil Services Pre, 2013

Ans. (a) : Mechanical advantage is the ratio of the force produced by a machine to the force applied to it, used in a reserving the performance of a machine.

For no friction

Power input = Power output

$$\frac{F_{output}}{F_{input}} = \frac{V_{input}}{V_{output}} = \text{M.A.}$$

$$MA = \frac{V_A}{V_B} = \frac{V_{driver}}{V_{driven}}$$

297. A straight link has a straight slot along the length of the link in which a sliding block is free to slide. The slotted link is rotating at uniform angular velocity of 10 rad/s about one of its ends. The sliding block in the link is also moving outwards at a uniform speed of 5m/s relative to the slotted link. What is the magnitude of acceleration of the slider in a direction perpendicular to the link?

(a) 100 m/s^2 (b) 50 m/s^2
(c) 200 m/s^2 (d) Zero

RPSC AEN Pre-2013

Ans. (a) : Given- angular velocity (ω) = 1 rad/sec.

velocity (v) = 5 m/sec.

$\because$ magnitude of acceleration of the slider in a direction perpendicular to the link coriolis acceleration

$(a_c) = 2v\omega = 2 \times 5 \times 10$

$\boxed{a_c = 100 \text{ m/s}^2}$

298. A link is rotating about O Velocity of point P on link w.r.t. point Q on link will be perpendicular to

(a) OP
(b) OQ
(c) PQ
(d) Line in between OP and OQ

SJVN ET 2013

Ans. (c) : A link is rotating about O velocity of point P on link with respect to point Q on link will be perpendicular to PQ.

299. When a particle moves along a straight path, then the particle has

(a) normal component of acceleration only
(b) tangential acceleration only
(c) centripetal acceleration only
(d) both tangential and centripetal acceleration

TNPSC AE 2013

Ans. (b) When a particle moves along a straight path, then the particle has Radial/centripetal or normal component of $a_r = 0$

[because there is no change in the direction of the velocity (straight path)]

therefore the partial has only tangential acceleration

$a_t \neq 0$

Note : When a partial moves along a straight line or path with uniform velocity, there will be no radial as well as tangential acceleration.

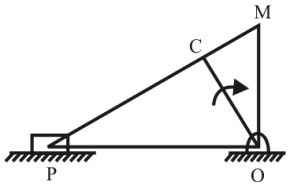
300. When two links are connected by a pin joint, their instantaneous centre lies

(a) on their point of contact
(b) at the centre of curvature
(c) at the centre of circle
(d) at the pin joint

JPSC AE 2013, Paper-V

Ans : (d) : At the pin joint.

301.



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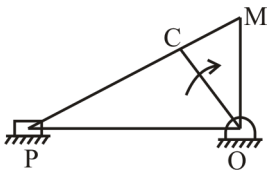
A slider crank mechanism is shown in the above figure. OC is the crank, CP is the connecting rod, PC extended cuts the vertical line from O at M. Given OC =100 mm, OM = 15 mm, CP = 220 mm, CM = 17 mm ω of cranks = 10 rad/s. Find the velocity of the piston in mm/s?

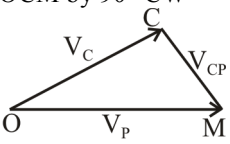
(a) 150 (b) 170
(c) 200 (d) 1000

UPSC JWM Adv. No-16/2009

Ans. (a) :



Rotating triangle OCM by 90° CW

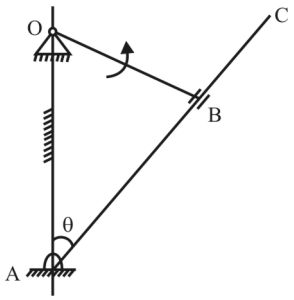


$$\frac{V_C}{OC}=\frac{V_{CP}}{CM}=\frac{V_P}{OM}=\omega$$

$$\frac{V_P}{OM}=10$$

$V_P = 10 \times 15 = 150$ mm/s

302.



A quick return mechanism is shown in the above figure. The crank OB = 100 mm running at 10 rad/s, fixed link OA = 200 mm. Which of the following statements is/are correct regarding coriolis component?

1. It is never zero
2. It is zero when θ = 0°
3. It is zero when θ = 30°
4. It is zero when $\tan\theta=\frac{1}{2}$

(a) 1 only (b) 2 and 3
(c) 2 and 4 (d) 3 and 4

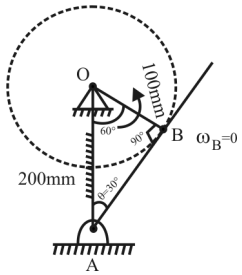
UPSC JWM Adv. No-16/2009

Ans. (b) : Coriolis Component (a_c) = 2Vω
V = Velocity of slider
ω = Angular velocity of the Crank

Case-I : When $\angle B = 90°$ (At extreme position)

i.e. $\sin\theta=\frac{OB}{OA}=\frac{100}{200}=0.5 \Rightarrow \theta=30°$

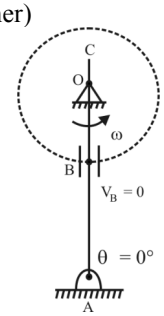
$\because \angle B = 90°$



At extreme position B:
i.e. ω = 0 (Angular Velocity)
∴ **Coriolis Component :**

$a_c = 2V\omega$
$= 2V \times 0$
$\boxed{a_c = 0}$

Case-II : When θ = 0° (i.e. fixed link and Crank Overlap on each other)



∴ $a_c = 2V\omega = 2 \times 0 \times w$
⇒ $\boxed{a_c = 0}$

303. In a single link robotic arm, the end effector slides upward along the link with a velocity of 0.5 m/s while the link rotates about revolute joint with an angular speed of 1 rad/s. When the endeffector is at a distance of 1 m from the joint, the acceleration experienced by the endeffector will be

(a) 1 m/s^2 (b) 1.41 m/s^2
(c) 1.71 m/s^2 (d) 2 m/s^2

DRDO Scientists 2009

Ans. (a) : Corolis acceleration given by
$a_c = 2\omega V = 2\times1\times0.5 = 1$ m/s^2

304. The number of links in a planar mechanism with revolute joints having 10 instantaneous centres is

(a) 3 (b) 4
(c) 5 (d) 6

DRDO Scientists 2009

Ans. (c) : Instantaneous centres $=\frac{n(n-1)}{2}=10$

$n(n-1) = 20$
$n = 5$

305. The motion of pulley (Figure) is controlled by cable C which has a constant acceleration of 0.225 m/s^2. The magnitude of angular acceleration of the pulley is

(a) 3 rad/s^2 (b) 4 rad/s^2
(c) 225 rad/s^2 (d) 6 rad/s^2

DRDO Scientists 2008

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Ans. (a) : Angular acceleration given by

$$a = r\alpha$$

$$\alpha = \frac{a}{r} = \frac{0.225}{0.075} = 3 \text{ rad/s}^2$$

306. In a reciprocating engine mechanism, the no. of links and instantaneous centres are

(a) 3, 3 (b) 4, 5
(c) 4, 6 (d) 5, 5

WBPSC AE 2008

Ans. (c) : Reciprocating engine mechanism is a single slider crank mechanism which has 4 links.

No. of instantaneous centre $= \frac{n(n-1)}{2}$

$$= \frac{4 \times 3}{2} = 6$$

307. In a mechanism, the fixed instantaneous centres are those centres which-

(a) Remain in the same place for all configurations of the mechanism
(b) Vary with the configuration of the mechanism
(c) Move as the mechanism moves, but joints are of permanent nature
(d) None of these

NPCIL ET, 2015

Ans. (a) : Fixed instantaneous centres remain in the same place for all configuration of the mechanism. The permanent instantaneous centre more when the mechanism moves, but the joints are the permanent nature.

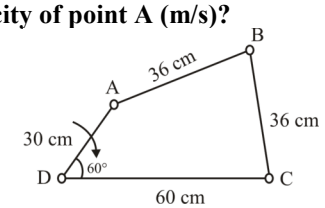
308. A body of mass 100 kg is rotating at a speed of 20 rad/s. The radius of rotation is about 20 m. The linear velocity of the body is _____ .

(a) 200 m/s (b) 800 m/s
(c) 400 m/s (d) 100 m/s

VIZAG MT, 14.12.2020

Ans. (c) : Given, M = 100 kg, ω = 20 rad/s, r = 20 m
Linear velocity (v) = $r\omega = 20 \times 20 = 400$ m/s

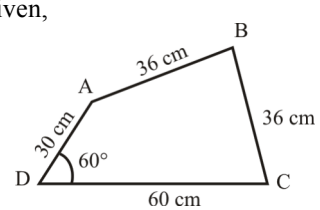
309. Consider a four bar mechanism shown in the given figure. The driving link DA is rotating uniformly at a speed of 100 rpm clockwise. Velocity of point A (m/s)?



(a) 3.00 (b) 3.14
(c) 3.25 (d) 4.00

GPSC ARTO Pre 30.12.2018, ESE 1999

Ans. (b) : Given,



N_{DA} = 100 rpm (clockwise)
r_{DA} = 30 cm = 0.3 m
$v_{DA} = r\,\omega_{DA}$

$$= r \times \frac{2\pi N_{DA}}{60}$$

$$= \frac{0.3 \times 2\pi \times 100}{60}$$

v_{DA} = 3.14 m/s

310. In a four bar mechanism the following instantaneous centre of rotation is not present:

(a) Fixed
(b) Permanent
(c) Neither fixed nor permanent
(d) Perpendicular to the line joining centre of rotations of crank and lever at infinity

UKPSC AE 2007 Paper -I

Ans. (d) : Perpendicular to the line joining centre of rotations of crank and lever at infinity

311. In a slider crank mechanism, piston velocity becomes maximum for the following configuration when crank is

(a) at inner dead centre
(b) at outer dead centre
(c) perpendicular to line of stroke
(d) perpendicular to connecting rod

UKPSC AE 2007 Paper -I

Ans. (c) : Perpendicular to line of stroke.

312. The link EF in a slider crank mechanism has a length of 0.4 m. The velocity of end E with respect to F is 4.9 m/s. The angular velocity of the link is :

(a) 0.01225 rad/s (b) 1.225 rad/s
(c) 12.25 rad/s (d) 1225.5 rad/s

TRB Polytechnic Lecturer 2017

Ans. (c) : Given,

ℓ = 0.4 m

V_{EF} = 4.9 m/s

Angular velocity $(\omega) = \frac{V_{EF}}{\ell} = \frac{4.9}{0.4}$

ω = 12.25 rad/s

313. The coriolis component of acceleration is taken into account for :

(a) Slider crank mechanism
(b) Quick return motion mechanism
(c) Four bar chain mechanism
(d) Double slider crank mechanism

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

314. The angular Velocity in rad/s of a body rotating at N rpm is

(a) π N/60 (b) 2π N/60
(c) πN/120 (d) π N/180

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : 2π N/60

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3. Cams and Follower

315. For high speed engines, the cam follower should move with

(a) Uniform velocity
(b) Simple harmonic motion
(c) Uniform acceleration and retardation
(d) Cycloidal motion

GPSC DEE Class-2 (GWSSB) 04.07.2021
JPSC AE PRE 2019
Vizag Steel (MT) 2017
UPPSC AE 12.04.2016 Paper-I
MPPSC AE 2016
Nagaland PSC CTSE 2016 Paper-I
TSPSC AEE 2015
CGPSC AE 26.04.2015 Shift-I / ESE 2002

Ans : (d) For high speed engines, the cam follower should move with cycloidal motion.
Cycloidal motion:- There are no instantaneous changes in acceleration at any point during motion. This would in turn mean that there would be no jerks and the cam follower system would show smooth operation. this makes the cam suitable for high speed application.

316. Cam size depends on

(a) Pitch circle (b) Base circle
(c) Prime circle (d) Outer circle

Karnataka PSC AE (WRD) 31.07.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TNPSC AE 2019
GMB AAE 25.06.2017
Rajasthan Nagar Nigam AE 2016, Shift-II
GWSSB DEE 07.07.2016
RPSC AE. 2016
GWSSB AAE, 27.12.2015
CGPSC AE 26.04.2015 Shift-I
RPSC ACF-2011

Ans. (b) : Base circle decides the size of cam base circle is the smallest circle tangent to the cam profile drawn from the centre of rotation of radial cam.
Pitch circle – It is the circle passing through the pitch point and concentric with the base circle.
Prime circle – It is the smallest circle drawn tangent to the pitch curve is known as prime circle.

317. Pitch point on cam is

(a) Any point on pitch curve
(b) The point on cam pitch curve having maximum pressure angle
(c) Any point on pitch circle
(d) The point on cam pitch curve having minimum pressure angle

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
CGPSC AE 15.01.2021
APPSC Poly Lect. Automobile Engg., 2020
GPSC Executive Engineer 23.12.2018
RPSC VPITI 14.02.2016
GPSC Lect. 23.10.2016
TSPSC AEE 2015
UKPSC AE 2012 Paper-I

Ans. (b) : It is the point on the pitch curve at which the pressure angle is maximum.
A high value of maximum pressure angle is not as it might jam the follower in the bearings. Maximum pressure angle
For oscillating follower = 45º
For translating follower = 30º

318. The cam and follower without a spring is an example of

(a) Sliding pair (b) Self closed pair
(c) Force closed pair (d) Lower pair

UPRVUNL AE 05.07.2021
GMB AEE 25.06.2017
CGPSC ARTO & TSI 07.01.2017
Rajasthan AE (Nagar Nigam) 2016 Shift-3
APPSC AEE Mains 2016, Paper-III

Ans. (b) : According to the type of closed kinematic pair, are of two types.
(1) Self Closed Pair–When two elements of a pair are connected together mechanically in such a way that only required kind of relative motion occurs, it is then known as self closed pair.
Note-Lower pair are self closed pair.
(2) Force-Closed Pair–When two elements of a pair are not connected mechanically but are kept in contact by the action of external forces, the pair is said to be a force-closed pair.
The cam and follower is an example of force closed pair, as it is kept in contact by the force exerted by spring and gravity.
So, cam and follower without a spring is an example of self closed pair.

319. In a cam follower mechanism, the follower needs to rise through 20 mm during 60º of cam rotation, the first 30º with a constant acceleration and then with a deceleration of the same magnitude. The initial and final speeds of the follower are zero. The cam rotates at uniform, speed of 300 rpm. The maximum speed of follower is

(a) 0.60 m/s (b) 1.20 m/s
(c) 1.68 m/s (d) 8.40 m/s

TSPSC AEE 2017
HPPSC Asstt. Prof. 18.11.2016
VIZAG Steel MT (Re-Exam) 24.11.2013
GATE 2005

Ans. (b) : Given, h = 20 mm

$$\phi = \text{Total rise} = \frac{\pi}{3} = 60°$$

$$v_{max} = \frac{2h\omega}{\phi} = \frac{2 \times 0.02 \times \left(\frac{2\pi \times 300}{60}\right)}{\pi/3}$$

v_{max} = 1.2 m/sec

320. The cam follower extensively used in air-craft engine is

(a) knife edge follower (b) flat faced follower
(c) spherical face follower (d) roller follower

APPSC AEE Mains 2016, Paper-III
TSPSC Managers, 2015
TNPS AE (Industries) 09.06.2013

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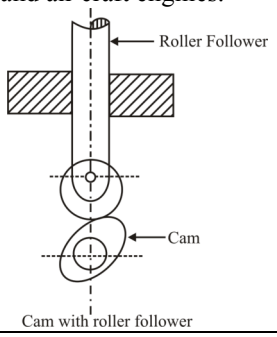
Ans. (d) : Roller follower–

- When the contacting end of the follower is a roller, it is called a roller follower.
- Since the rolling motion take place between the contacting surface therefore the rate of wear is greatly reduced.
- In roller followers also the side thurst exists between the follower and the guide.
- The roller followers are extensively used where more space is available such as in stationary gas and oil engines and air craft engines.



Cam with roller follower

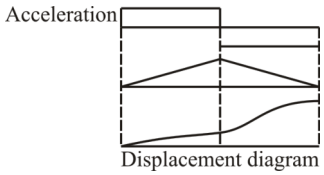
321. In the plate cam mechanism with reciprocating follower, in ……. motion the follower has a constant acceleration.

(a) Parabolic (b) MHS
(c) Sino-cycloidal (d) None of these

HPPSC Lect. (Auto) 23.04.2016
MPPSC State Forest Service Exam, 2014
ESE 2004

Ans. (a) : For constant or uniform acceleration, the velocity of the follower must change at a constant rate. The velocity diagram of the follower consists of slopping straight line. The velocity diagram represents everywhere the slope of the displacement diagram. The later must be a curve, whose slope changes at a constant rate.

Hence, the displacement diagram consists of double parabola.



Displacement diagram

322. When cam moves, the period during which the follower of cam remains at rest is known as

(a) constant period (b) fixed period
(c) dwell period (d) idle period

TSPSC AEE 2015
GPSC Lect. Poly. 07.12.2014
UKPSC AE-2013, Paper-I

Ans : (c) When cam moves, the period during which the follower of cam remains at rest is known as dwell period.

Application of dwell period:-

Cam-follower dwell mechanisms are used in pairs in sewing machines to operate the four motion feed dogs, with one cam moving the dog up and down, and the other cam moving the dog forwards and backwards. The cams in this application are usually phased 90 degrees apart allowing a pause in the up/down movement of the dog while it is being moved forwards/backwards. A separate adjustable sliding block or link is used to control the amount of forwards/ backwards movement of the dog.

323. In a cam drive, it is essential to off set the axis of follower to:

(a) Decrease the side thrust between follower and guide
(b) Decrease wear between follower and guide
(c) Take care of speed limitations
(d) Reduce cost

GPSC Asstt. Prof. 28.08.2016
MPPSC AE 08.11.2015, ESE 1998

Ans. (a) : Offsets are required in cam and followers to reduce wear and side thrust. Less stroke length means follower movement will be less. Due to less movement of follower, spring will also be compressed less. So decrease in wear between follower and guide.

324. In cycloidal motion of cam follower. The maximum acceleration of follower motion f_{mn} at $\theta = \frac{\phi}{4}$, where h = maximum follower displacement
ω = angular velocity of cam
ϕ = angle for the maximum follower displacement

(a) $\frac{h\pi\omega^2}{2\phi^2}$ (b) $\frac{3h\pi\omega^2}{2\phi^2}$
(c) $\frac{2h\pi\omega^2}{\phi^2}$ (d) $\frac{3h\pi\omega^2}{\phi^2}$

APGCL AM, 2021
ESE 2020

Ans. (c) :

For cycloidal motion, Acceleration $= \frac{2\pi\omega^2 h}{(\phi)^2} \sin\left(\frac{2\pi\phi}{\phi}\right)$

for maximum acceleration $= \theta = \phi/4$

so maximum acceleration $= \frac{2\pi\omega^2 h}{\phi^2}$

325. Let R is flank radius, r is minimum radius of the cam and θ is angle turned through by the cam. If a flat-faced follower is in contact with the flank of a circular arc cam, its lift of displacement is equal to-

(a) (R + r) (1+ cosθ) (b) (R – r) (1+ cosθ)
(c) (R – r) (1– cosθ) (d) (R + r) (1– cosθ)

TNPSC AE (Industries) 09.06.2013
WBPSC AE 2003

Ans. (c) : Displacement of flat faced follower when contact with the flank of a circular arc.

Displacement- $x = (R - r)(1 - \cos\theta)$

Velocity - $V = \omega(R - r)\sin\theta$

acceleration - $a = \omega^2(R - r)\cos\theta$

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326. A radial follower is one which-
(a) reciprocates in the guides
(b) oscillates
(c) translates along an axis passing through the cam centre of rotation
(d) None of these

OPSC AEE 2019 Paper-I
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : In radial follower, the line of motion of follower is passing through cam centre of rotation.

327. A cam mechanism imparts
(a) rotating motion
(b) oscillating motion
(c) reciprocating motion
(d) All of the above

Karnataka PSC AE (WRD) 31.07.2021
Assam PSC AE (IWT) 14.03.2021

Ans. (d) : A cam may be defined as a rotating or a reciprocating element of a mechanism which transmit reciprocating or oscillating motion to other element known as follower by direct contact.
Any complicated output motions which are otherwise difficult to produce can easily be produced with the help of cam.

328. A circle drawn with centre as the cam centre and radius equal to the distance between the cam centre and the point on the pitch curve at which the pressure angle is maximum, is called–
(a) Base circle (b) Pitch circle
(c) Prime circle (d) None of these

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : It is pitch circle, at which the pressure angle will be maximum (when this point pass through the pitch curve).
Base circle- It is the smallest circle tangent to the cam profile drawn from the centre of rotation of a radial cam.
Pressure angle- The pressure angle representing the steepness of the cam profile, is the angle between normal to the pitch curve at a point and the direction of the follower motion.
Prime circle- Smallest circle drawn tangent to the pitch curve is known as prime circle.
Pitch circle- It is a point on pitch curve at which the pressure angle is maximum.

329. In a radial cam. The follower moves:
(a) along the cam axis
(b) in a direction parallel to the cam axis
(c) in a direction perpendicular to the cam axis
(d) at an angle of 30º to the cam axis
(e) at an angle of 75º to the cam axis

CGPSC ARTO & TSI 07.01.2017
VIZAG Steel MT 2011

Ans. (c) : In radial cams, the follower reciprocates or oscillates in a direction perpendicular to the cam axis.
When the motion of the follower is along an axis passing through the centre of the cam it is know as radial follower.

330. The maximum value of the pressure angle in case of cam is kept as
(a) 20 deg (b) 25 deg
(c) 30 deg (d) 15 deg

TNPSC AE 2020
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : The maximum value of the pressure angle in case of cam is kept as 30^0.

331. The throw of a Cam is the maximum distance of the follower from
(a) Root circle (b) Base circle
(c) Pitch circle (d) Staring to finish

Gujarat PSC AE 2019
APPSC AEE Screening Test 2016

Ans. (b) : The stroke/throw, the greatest distance or angle through which the follower moves or rotates. It is the maximum distance of followers from the base circle. Base circle is a smallest circle of a cam profile drawn from the center of rotation of the cam.

332. The pressure angle and the base circle in a cam should be
(a) both as big as possible
(b) both as small as possible
(c) as big as possible and as low as possible respectively
(d) as small as possible and as big as possible respectively

BPSC Poly. Lect. 2016
VIZAG MT 2015

Ans : (d) The pressure angle and the base circle in a cam should be as small as possible and as big as possible respectively.
Base circle :- It is the smallest circle that can be drawn to the cam profile
pressure angle:- It is the angle between the direction of the follower motion and a normal to the pitch curve. This angle is very important in designing a cam profile. If the pressure angle is too large, a reciprocating follower will jam in its bearings.

333. The smallest circle that can be drawn from the centre of the cam and tangent to the pitch curve is known as :
(a) Base circle (b) Pitch circle
(c) Prime circle (d) Profile circle

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Prime circle- It is the smallest circle that can be drawn from the centre of cam and tangent to the pitch curve.
Pitch circle – It is the circle passing through the pitch and concentric with base circle.
Base circle – It is the smallest circle that can be drawn tangential to cam profiles from the centre of rotation of a radial cam.
Profile–The shape of the contoured cam surface by means of which motion is communicated to the follower.

334. In case of simple harmonic motion of a follower (rise – 90°, fall –90°, rest –90°) the maximum velocity of the follower will be :
(a) $\frac{\pi\omega s}{2\theta}$ (b) $\frac{\pi^2\omega^2 s}{2\theta}$
(c) $\frac{4\omega^2 s}{\theta}$ (d) $\frac{4\omega^2 s}{\theta^2}$

where, s – stoke of the follower
θ – cam angle
ω – angular velocity.

JPSC AE 10.04.2021, Paper-II

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Ans. (a) : In case of simple harmonic motion

$$\text{Maximum velocity } (V_{max}) = \frac{\pi}{2}\frac{\omega s}{\theta}$$

$$\text{Maximum acceleration } (a_{max}) = \frac{s}{2}\left(\frac{\pi\omega}{\theta}\right)^2$$

335. If there is constant acceleration in the first half of the follower motion and constant deceleration in the later

(a) circular (b) straight line
(c) parabolic (d) elliptical

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) :

336. Consider the following statements regarding cams:

1. **A cam in which the follower moves radially from the centre of rotation of the cam is known as a disc cam.**
2. **A globoidal cam is a double-disc cam, the two discs being keyed together and are in constant touch with the two rollers of a follower.**
3. **A conjugate cam can have two types of surfaces, convex or concave.**
4. **In a spherical cam, the follower oscillates about an axis perpendicular to the axis of rotation of the cam.**

Which of the above statements are correct?

(a) 1 and 4 only (b) 2 and 4 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

ESE 18.07.2021

Ans. (a) : Given, statements 2 and 3 are wrong because-

- Conjugate can is a double disc can, the two discs being keyed together and are in constant touch with the two rollers of followers.
- Globoidal cam have two type of surfaces, convex or concave.

337. In a radial cam translating follower mechanism, the offset is provided to

(a) Decrease the pressure angle during descent of the follower
(b) Decrease the pressure angle during ascent of the follower
(c) Increase the pressure angle during ascent of the follower
(d) Avoid any obstruction due to other machine parts

UPPSC AE 13.12.2020, Paper-I, ESE 2009

Ans. (b) : An offset provided in the radial cam translating follower mechanism, the offset is provided to decrease the pressure angle during the ascent of the because the high value of pressure angle is not desirable as it increases side thrust and performance will not be smooth.

338. Acceleration of flat faced follower, when it is in contact with flank of circular arc cam is:

(a) $\omega^2(R-r)\sin\theta$ (b) $\omega^2(R+r)\sin\theta$
(c) $\omega^2(R-r)\cos\theta$ (d) $\omega^2(R+r)\cos\theta$

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (c) :

339. When contacting end of follower is a perfectly circular flat face, it is called

(a) Mushroom follower
(b) Knife edge of follower
(c) Roller follower
(d) Spherical faced follower

Oil India Senior Officer 23.12.2020

Ans. (a) : In mushroom follower contacting end of follower is a perfectly circular flat face.

340. Path described by the trace point as referred to cams is known as

(a) Base circle (b) Prime circle
(c) Pitch circle (d) Pitch curve

RPSC IOF, 2020

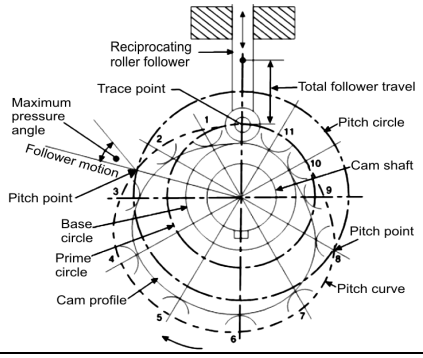
Ans. (d) : Pitch Curve– It is a path described by the trace point as referred to cams
For knife edge follower the pitch curve and the cam profile are same where as for a roller follower, they are separated by the radius of the roller.

341. The smallest circle in the cam profile is :

(a) prime circle (b) pitch circle
(c) base circle (d) pitch curve

APPSC Poly Lect. 13.03.2020

Ans. (c) : 'Base Circle' is the smallest circle in the cam profile.



342. For favorable dynamics of a high-speed cam, the acceleration curve should be necessarily made

(a) Zero (b) Continuous
(c) Positive (d) Negative

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : For favorable dynamics of a high speed cam the acceleration curve should be necessarily made continuous. Any variation in acceleration will cause jerk, which is unfavorable.

343. On the displacement diagram of a cam, the dwell portion of the cycle represents the condition of

(a) No movement of the follower as the cam continuous to rotate
(b) Movement of the follower with maximum velocity as the cam continuous to rotate
(c) Movement of the follower with maximum acceleration as the cam continuous of rotate
(d) Movement of the follower with minimum velocity as the cam continuous to rotate

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Dwell is the motion during which the follower is at rest but cam rotate continuously.

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344. The locus of the trace point if the follower is moved around the cam is known as

(a) prime circle (b) cam circle
(c) pitch (d) base circle

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : The locus of the trace point if the follower is moved around the cam is known as pitch curve.

345. The cam and follower having a line contact between them, is known as :

(a) line pair (b) higher pair
(c) lower pair (d) reciprocating pair

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Higher Pair– When the two elements of a pair have a line or point contact when relative motion takes place and the motion between the two elements is partly turning and partly sliding then the pair is known as higher pair. A pair of friction disc toothed gearing belt and rope drive, ball and roller bearing and cam and follower are the example of higher pair.

346. Which of the following is not related to the cam profile?

(a) Prime circle (b) Trace point
(c) Addendum (d) Base circle

SJVN ET 2019

Ans. (c): Addendum- Addendum is Radial distance between pitch circle and addendum circle and it is equal to 1 module.

Addendum = 1 module

347. For a given lift of the follower in a given angular motion of the cam, the acceleration /retardation of the follower will be least when the profile of the cam during the rise portion is:

(a) Such that the follower motion is simple harmonic
(b) Such that the follower motion has a constant velocity from start to end
(c) A straight line, the being a tangent cam
(d) Such that the follower velocity increases linearly for half the rise portion and then decreases linearly for the remaining half of the rise portion

OPSC AEE 2019 Paper-I, ESE 1999

Ans : (d) : Acceleration of follower in various situations

(i) SHM, $a = \left(\frac{\pi\omega}{\phi}\right)^2 \frac{h}{2}$

(ii) Constant velocity Acceleration very high at start end the rise but zero during whole motion, so overall very high

(iii) In tangent cam also, the maximum acceleration is very high.

(iv) Velocity increases linearly i.e. parabolic motion, the acceleration.

$= \frac{4\omega^2}{\phi^2}.h$

So out of given all, the least acceleration will be in which Velocity varies linearly.

348. The retardation of a flat faced follower when it has contact at the apex of the nose of a circular arm cam, is given by :

(a) $\omega^2 \times OQ$ (b) $\omega^2 \times OQ\sin\theta$
(c) $\omega^2 \times OQ\cos\theta$ (d) $\omega^2 \times OQ\tan\theta$

Where OQ = Distance between the centre of circular flank and centre of nose.

OPSC AEE 2019 Paper-I

Ans : (a) : The retardation of a flat faced follower when it has contact at the apex of the nose of a circular cam is given by $\omega^2 \times OQ$.

349. In cam and follower geometry, pitch circle is

(a) The circle passing through pitch point and concentric with the base circle
(b) The circle passing through pitch point and not-concentric with the base circle
(c) The smallest circle drawn tangent to the pitch curve
(d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (a) : In cam and follower, geometry, pitch circle is the circle passing through pitch point and concentric with the base circle.

350. Which of the following is not the shape of Cam?

(a) Spiral (b) Knife-edge
(c) Globoidal (d) Conjugate
(e) Spherical

CGPSC AE 25.02.2018

Ans. (b) : Spiral, globoidal, conjugate and spherical are shape of cam whereas knife-edge is the shape of follower in cam-follower mechanism.

351. The angle between the direction of follower motion and a normal to pitch curve is called :

(a) Pitch angle (b) Prime angle
(c) Base angle (d) Pressure angle

Kerala PSC Poly. Lect., 2017

Ans. (d) : Pressure Angle (ϕ)–It represents the steepness of cam profile, is the angle between normal to the pitch curve at a point and the direction of follower motion. A high value of maximum pressure angle is not desired. Its magnitude varies at all instant of followed motion.

352. A cam in which the follower reciprocates or oscillates in a plane parallel to the axis of the cam is known as

(a) Cylindrical Cam (b) Circular cam
(c) Reciprocating cam (d) Tangent cam

Vizag Steel (MT) 2017

Ans. (a) : Cylindrical cam– A cam in which the follower reciprocates or oscillates in a plane parallel to the axis of the cam is known as cylindrical cam.

353. A cam-and-follower system with the follower having a constant velocity is not suitable in practice because

1. The cam rotates with a uniform angular velocity
2. The follower moves with constant velocity
3. Inertial forces are large at the beginning as well as at the end of follower motion

Which of the above statements is/are correct?

(a) 1 only (b) 2 only

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(c) 3 only (d) 1, 2 and 3

UPSC JWM 2017

Ans. (c) : The follower with a constant velocity is not suitable for practical use because the inertial forces are large at the beginning as well as at the end of follower motion.

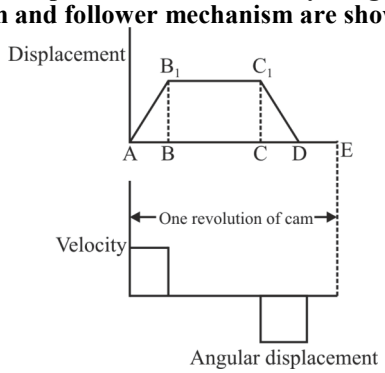
354. Statement (I): Cams used in high-speed application should have displacement, velocity and acceleration curves of the follower in continuity.

Statement (II): Abrupt changes in these curves will cause high contact stresses at the bearings and make the operation noisy.

ESE 2017

Ans. (a) : Acceleration curve with abrupt changes (such as parabolic/simple harmonic motion of follower) will exert abruptly changing contact stresses at the bearings and an the cam surface and lead to noise, surface wear and eventual failure.

355. The displacement and velocity diagrams of a cam and follower mechanism are shown:



Which of the following statements is / are correct?

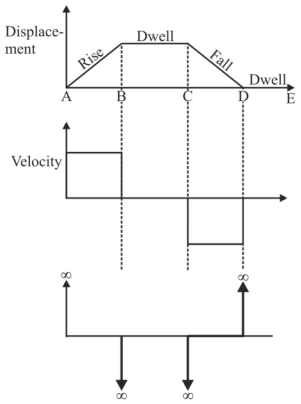
1. The acceleration of the follower at the beginning and at the end of each stroke will be zero.
2. The follower remains at rest in the dwell period.
3. During period DE, the motion of the follower is retarding.

Select the correct answer using the code given below.

(a) 1, 2 and 3 (b) 1 only
(c) 2 only (d) 3 only

ESE 2017

Ans. (c) :



Given- Profile is corresponding to uniform velocity motion of follower during rise and fall.

AB – Rise
BC – Dwell
CD – Fall
DE – Dwell

At start and end of rise and fall, velocity suddenly changes, which makes acceleration at these locations tend to ∞.

During dwell period followers does not move.

356. What is the reference point on the follower to lay the cam profile known as?

(a) Cam center (b) Pitch point
(c) Trace point (d) Prime point
(e) Flat point

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : Trace Point :- It is a reference point on the follower and is used to generate the pitch curve.

Pitch Point :- The point of contact of the pitch lines of two gears or of a rack and pinion when in mesh.

Cam Profile :- It is the actual working surface contour of the cam base circle.

It is the smallest circle drawn to the cam profile from the centre of rotation of the cam.

357. .....velocity of the following implies that the displacement of the follower is proportional to the cam displacement and the slope of the displacement curve is constant.

(a) Constant
(b) Uniformly increasing
(c) Uniformly decreasing
(d) Non-uniform

HPPSC Asstt. Prof. 29.10.2016

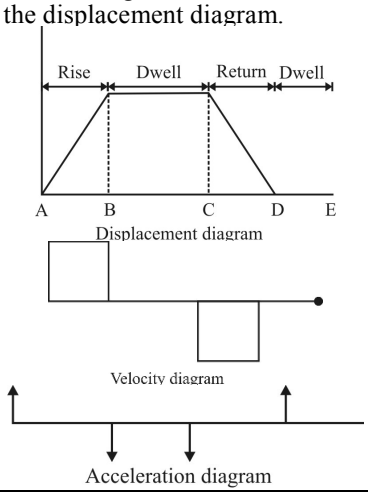
Ans. (a) : Follower motion with uniform velocity– Constant velocity of the follower implies that the displacement of the follower is proportional to the cam displacement and the slope of the displacement curve is constant.

358. What type of follower motion is represented as a straight line with arcs at the beginning and end in the displacement diagram?

(a) Harmonic (b) Uniform
(c) Combination (d) Uniformly accelerated

UPRVUNL AE 21.08.2016

Ans. : (b) Uniform tube of follower motion is represented as a straight line with arcs at the beining and end in the displacement diagram.



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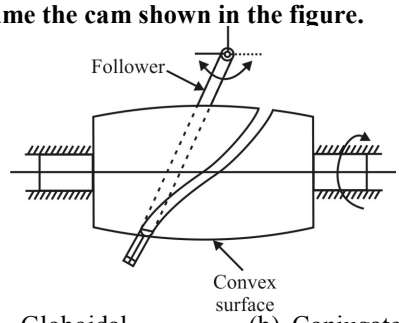
359. Name the cam shown in the figure.



(a) Globoidal (b) Conjugate
(c) Spherical (d) Cylindrical

UPRVUNL AE 21.08.2016

Ans. : (a) Globoidal cam

360. For a Roller Follower the trace point is at _____

(a) At the centre point of cam
(b) At the cam axis
(c) At the roller diameter
(d) At the centre point of roller

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : For a roller follower the trace point is at the roller centre.
Ex. Used in Aircraft engine & Gas engines.

361. The circle passing through the pitch point and concentric with the base circle of Cam profile is known as:

(a) Pitch circle (b) Prime circle
(c) Trace circle (d) Base circle

UPRVUNL AE 07.10.2016

Ans. (a) : The circle passing through the pitch point and concentric with the base circle of Cam profile is known as pitch circle.

362. The pressure angle in a cam depends on

(a) The angle of ascent
(b) The lift of the follower
(c) offset between centre lines of cam and follower
(d) All of the above

UPPSC AE 12.04.2016 Paper-I

Ans : (d) The pressure angle in a cam depends on
(i) The angle of ascent
(ii) The lift of the follower
(iii) Offset between centre lines of cam and follower.

363. If the line of movement of the follower passes through the centre of rotation of the cam, then it is called :

(a) Axial follower (b) Radial follower
(c) Offset follower (d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : If the line of movement of the follower passes through centre of rotation of the cam, then it is called Radial follower.

364. The follower of a cam has :

(a) uniform acceleration and retardation
(b) cycloidal motion
(c) simple harmonic motion
(d) any one of the above

Kerala PSC IOF 19.04.2016

Ans. (d) The follower of a cam has–
- Uniform acceleration and retardation
- Cycloidal motion
- Simple harmonic motion

365. A cam is to be designed for a knife edge follower with following data; cam lift 40 mm during 90º of cam rotation with SHM, dwell for the next 30º, during the next 60º of cam rotation the follower returns to its original position with SHM, : dwell during the remaining 180º. The radius of base circle of the cam is 40 mm. If cam rotates at 240 rpm then acceleration of the follower during its accent and descent will be

(a) 20.6, 11.8 m/s^2 (b) 20.6, 118 m/s^2
(c) 50.6, 11.8 m/s^2 (d) 50.6, 113 m/s^2

MPPSC AE 08.11.2015

Ans. (d) : Given, S = 40 mm = 0.04 m

$$\theta_0 = 90° = \frac{\pi}{2}\text{rad} = 1.571\text{rad}$$

$$\theta_R = 60° = \frac{\pi}{3}\text{rad} = 1.047\text{rad}$$

$$N = 240 \text{ rpm}$$

$$\omega = \frac{2\pi \times 240}{60} = 25.14\,\text{rad/sec}$$

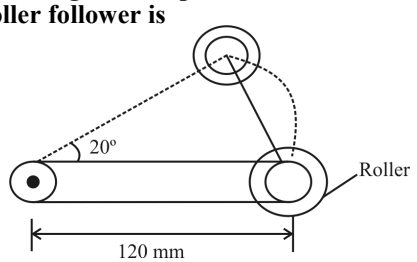
Maximum acceleration of the follower during its ascent

$$a_0 = \frac{\pi^2\omega^2 S}{2(\theta_0)^2} = \frac{\pi^2(25.14)^2 \times 0.04}{2\times(1.571)^2} = 50.6\,\text{m/s}^2$$

Maximum acceleration of follower during it's descent,

$$a_R = \frac{\pi^2\omega^2 S}{2(\theta_R)^2} = \frac{\pi^2(25.14)^2 \times 0.04}{2\times(1.047)^2} = 113.8\,\text{m/s}^2$$

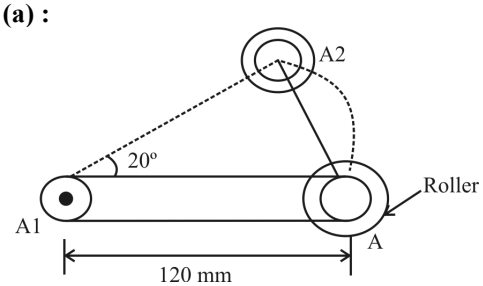
366. The angular displacement of the following roller follower is



(a) π/9 (b) 2π/9
(c) π/18 (d) 2π/18

APPSC Poly. Lect. 2013

Ans. (a) :



The angular displacement of the follower = 20º

$$= 20 \times \frac{\pi}{180} = \pi/9\,\text{rad}$$

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367. For low and moderate speed engines, the cam follower should move with
(a) simple harmonic motion
(b) uniform velocity
(c) uniform acceleration and retardation
(d) cycloidal motion

JPSC AE 2013, Paper-V

Ans : (a) : For low and moderate speed engines, the cam follower should move with simple harmonic motion.

368. Kinematic pair constituted by cam and follower mechanism is:-
(a) Higher and open type
(b) Lower and open type
(c) Lower and closed type
(d) Higher and closed type

UKPSC AE-2013, Paper-I

Ans. (a) : Kinematic pair constituted by cam and follower mechanism is higher and open type.

369. The profile of a cam in a particular zone is given by $x = \sqrt{3}\cos\theta$ and $y = \sin\theta$. The normal to the cam profile at $\theta = \frac{\pi}{4}$ is at an angle (with respect to x axis) :
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{3}$ (d) 0

APGENCO AE 2012

Ans. (c) : $\frac{\pi}{3}$

370. Angle of dwell of cam is defined as the angle
(a) during which the follower returns to its initial position
(b) of rotation of the cam for definite displacement of the follower
(c) through which the cam rotates during the period in which the follower remains in the highest position
(d) moved by the cam from the instant the follower begins to rise, till it reaches its highest position

ISRO Scientist/Engineer 2008

Ans. (c) : Angle of dwell of cam is defined as the angle through which the cam rotates during the period in which the follower remains in the highest position.

371. For a roller follower, pitch curve and cam surface are
(a) identical
(b) separated by the radius of the roller
(c) separated by the diameter of the roller
(d) separated by one and half times the radius of the roller

UKPSC AE 2007 Paper -I

Ans. (b) : Separated by the radius of the roller

372. The pressure angle of a flat footed follower in contact with a circular arc cam is
(a) 5° (b) 3°
(c) 0° (d) –1°

UKPSC AE 2007 Paper -I

Ans. (c) : 0°

373. The linear velocity of the reciprocating roller follower when it has contact with the straight flank of the tangent cam, is given by
(a) $\omega(r_1 - r_2)\sin\theta$
(b) $\omega(r_1 - r_2)\cos\theta$
(c) $\omega(r_1 + r_2)\sin\theta\sec^2\theta$
(d) $\omega(r_1 + r_2)\cos\theta\ \mathrm{cosec}^2\theta$

TNPSC AE 2013

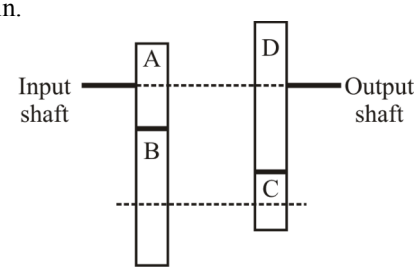
Ans. (c) : $V = \omega(r_1 + r_2)\sin\theta.\sec^2\theta$

4. Gears and Gear Trains

374. In a reverted gear train
(a) The axes of first and the last gear are parallel
(b) The axes of the first and last gear are co-axial
(c) One gear is always fixed
(d) Speed of last gear must be higher than speed of the first gear

HPPSC Workshop Suptd. 08.07.2021
APGCL AM, 2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC DEE, Class-2 (GWSSB) 04.07.2021
APPSC Poly Lect. 13.03.2020
ISRO Scientist/Engineer 12.01.2020
ESE 2020
Gujarat PSC AE 2019
TNPSC AE 2019
GPSC Executive Engineer 23.12.2018
ISRO Scientist/Engineer 07.05.2017
PTCUL AE 25.06.2017 / RPSC AE 2016
TSGENCO AE 14.11.2015
OPSC AEE 2015 Paper-I
Kerala PSC Poly. Lect. 09.01.2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
RPSC AEN Pre-2013
JPSC AE 2013, Paper-V
J&K PSC Civil Services Pre, 2010
ESE 1997

Ans : (b) When the axes of the first gear and last gear are co-axial then the gear train is known as reverted gear train.



375. The gear train usually employed in clocks is a
(a) simple gear train (b) reverted gear train
(c) sun and planet gear (d) differential gear

Assam PSC AE (IWT) 14.03.2021
Sikkim PSC (Under Secretary), 2017
Kerala PSC Poly. Lect., 2017
RPSC VPITI 14.02.2016 / KPSC ADF 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
UKPSC AE-2013, Paper-I
J&K PSC Civil Services Pre, 2013
TRB Asstt. Prof., 2012 / APPSC AEE 2012
RPSC ACF-2011 / ESE 1995

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Ans. (b) : In Reverted gear train, the first and last gear axis lie on the same axis of rotation. One of the gear is loosely mounted, when they are connected with the same shaft the application of speed reducers is applied here, the minute hand is connected with hour hand on the same axis.

376. The product of the diametral pitch and circular pitch is equal to

(a) 1 (b) $\frac{1}{\pi}$

(c) π (d) $\pi \times$ number of teeth

GPSC DEE Class-2 (GWSSB) 04.07.2021
Assam PSC AE (IWT) 14.03.2021
JPSC AE 10.04.2021, Paper-II
MECON MT 2019
Vadodara Muncipal Corp. DEE, 2018
CGPSC ARTO & TSI 07.01.2017
RPSC 2016
HPPSC W.S. Poly. 2016
MPPSC State Forest Service Exam, 2014
Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE 2007 Paper -I
OPSC Civil Services Pre 2006
GATE-1994

Ans. (c) : Circular Pitch–It is a distance measured on the circumference of the pitch circle from a point of one tooth to the corresponding point on the next tooth

$$P_c = \frac{\pi D}{T}$$

Diametral Pitch–It is the ratio of number of teeth to the pitch circle diameter in millimetres.

$$P_d = \frac{\pi}{P_c} = \frac{T}{D}$$

So, product of dimetral pitch and circular pitch is

$$P_d \times P_c = \frac{\pi D}{T} \times \frac{T}{D}$$

$$\boxed{P_d \cdot P_c = \pi}$$

377. _____ is a gear train that is useful for transmitting high velocity ratios with gears of moderate size in a comparatively lesser space.

(a) Hypocyclic gear train
(b) Reverted gear train
(c) Compact gear train
(d) Epicyclic gear train

RPSC ACF & FRO, 26.02.2021
NLCIL GET, Shift-II 17.11.2020
RPSC AE 2018
Karnataka PSC Lect., 27.05.2017
RPSC AE (GWD) 06.05.2016
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
UKPSC AE Paper-I 2012
OPSC Civil Services Pre 2006

Ans. (d) : Epicyclic gear train that is useful for transmitting high velocity ratios with gears of moderate size in a comparatively lesser space.

- More compact unit could be obtained.
- Epicyclic gears used in transmitting and computing device.
- Epicyclic gear train have two degree of freedom.

378. Which one of the following is usually preferred to specify size of a gear?

(a) Diametral pitch (b) Pressure angle
(c) Pitch circle diameter (d) Circular pitch

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Assam PSC AE (IWT) 14.03.2021
MECON MT 2019
Nagaland PSC (CTSE) 2018, Paper-I
Rajasthan Nagar Nigam AE 2016, Shift-I
RPSC AE 2016 / Assam PSC CCE Pre 2015
OPSC AEE 2015 Paper-I
UPRVUNL AE 2014

Ans. (c) Circular pitch-

- It is a distance along a pitch circle from one point on a tooth to the corresponding point on the next tooth.

$$\boxed{P_c = \frac{\pi D}{T} = \pi m}$$

m = Module
D = Pitch circle diameter
T = Number of teeth

- **Diametral Pitch–** It is the ratio of number of teeth to the pitch circle diameter but diameter should be in inches.

$$\boxed{P_d = \frac{T}{D}}$$

Relation between circular pitch (P_c) diametral pitch (P_d)

Circular pitch × Diametral Pitch = π

$$\boxed{P_c \times P_d = \pi}$$

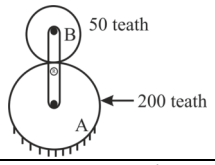
- **Pressure Angle–** It is the angle between the pressure line and the common tangent to the pitch circles. Its standard values are 14.5^0, 20^0 and 25^0.

379. A fixed gear having 200 teeth in mesh with another gear having 50 teeth. The two gears are connected by an arm. The number of turns made by the smaller gear for one revolution of arm about the centre of the bigger gear is

(a) 2 (b) 3
(c) 4 (d) 5

Nagaland PSC (CTSE) 2018 Paper-I
UPSC JWM 2017
ESE 2017, 1996 / UJVNL AE 2016
UPPSC AE 12.04.2016 Paper-I
VIZAG STEEL MT 2011
APPSC IOF, 2009
ISRO Scientist/Engineer 2006

Ans : (d)



Condition of Motion	Revolutions of elements		
Work	Arm R	Gear A	Gear B
Multiply by x	0	x	$-x \times \frac{200}{50}$
Add y	y	x+y	$-x \times \frac{200}{50} + y$

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$\because x + y = 0$ (A is fixed)

$y = 1$ (Given Data)

$x = -1$

Revolution of Gear B

$$= -x \times \frac{200}{50} + y$$

Revolution of gear B = 5

380. An epicyclic gear have .....degrees of freedom has been utilized in the differential gear of an automobile-

(a) one (b) two

(c) three (d) four

APPSC Poly Lect. 13.03.2020
Assam PSC AE (PHED), 18.10.2020
Oil India Ltd. Sr. Engineer (Production) 30.11.2019
CSE Pre-2008
HPPSC Asstt. Prof. 29.10.2016 / ESE-2011
TNPSC AE 2008

Ans. (b) : Epicyclic Gear Train– When there exists a relative motion of axes in a gear train it is called an epicyclic gear train. Thus in an epicyclic train, the axis of at least one of the gears also move relative to the frame. An epicyclic gear have two degree of freedom.

381. The helix angle for single helical gears ranges from

(a) 10 to 15° (b) 15 to 20°

(c) 20 to 35° (d) 35 to 50°

GPSC DEE Class-2 (GWSSB) 04.07.2021
GPSC EE Pre, 28.01.2017
ISRO Scientist/Engineer (RAC) 07.05.2017
TNPSC AE 2013, 2008

Ans. (c) : The helix angle for single helical gears ranges from 20° to 35°.

382. Herringbone gears are used

(a) For power transmission between shafts with axes at 90^0 to each other

(b) For stronger gear tooth

(c) For a higher velocity ratio

(d) To ensure zero axial thrust

Haryana PSC AE (PHED) 05.09.2020, Paper-II
VIZAG MT 2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TNPSC AE 2014
DRDO Scientist 2009

Ans. (d)

- Spur gear does not produce axial thrust where as for high speed and smooth application helical gears are used in car but helical gears produce axial thrust but the axial thrust can be eliminated by using axial thrust bearing of specific load capacity or by using double helical gear (Herringbone gears) which reduces noise and axial thrust.

383. In a spur gear, the circle on which the involute is generated is called

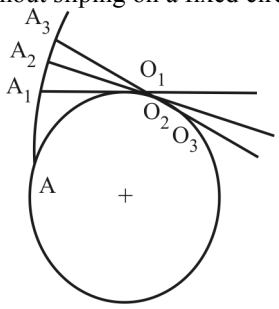
(a) pitch circle (b) clearance circle

(c) base circle (d) addendum circle

OPSC AEE 2019 Paper-I
MPPSC AE 2016
HPPSC Lect. (Auto) 23.04.2016
Mizoram PSC AE/SDO 2014 Paper-II
TRB Asstt. Prof., 2012

Ans. (c) : In a spur gear, on base circle involute profile is generated.

Involute profile– It is the locus of the point on a line, which rolls without sliping on a fixed circle.



Normal Drawn any point on the involute profile will become tangent to base circle automatically and vice-versa.

384. The train value of a gear train is

(a) equal to velocity ratio

(b) reciprocal of velocity ratio

(c) always greater than unity

(d) always equal to unity

GPSC DEE Class-II (GWSSB) 04.07.2021
GPSC Engg. Class-II Pre-19.01.2020
Nagaland PSC CTSE 2017, Paper-I
TNPSC AE 2018, 2013
J&K PSC Civil Services Pre, 2013

Ans. (b) : The train value of a gear train is reciprocal of velocity ratio

$$\text{velocity ratio} = \frac{\omega_{driver}}{\omega_{driven}} = \frac{N_1}{N_2} = \frac{T_2}{T_1}$$

$$\text{Train value} = \frac{\omega_{driven}}{\omega_{driver}} = \frac{N_2}{N_1} = \frac{T_1}{T_2}$$

$$\text{Train value} = \frac{1}{\text{velocity ratio}}$$

385. An automobile gear box has

(a) simple gear train

(b) compound gear train

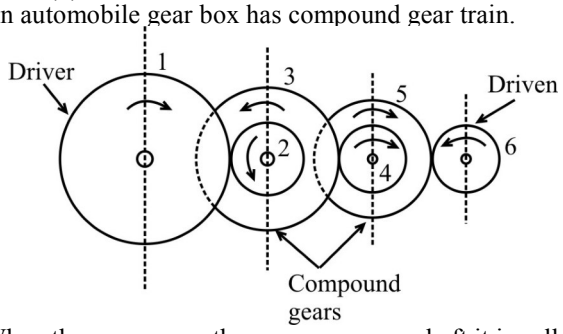
(c) epicyclic gear train

(d) compound epicyclic gear train

Nagaland PSC CTSE 2017, Paper-I
RPSC AE 2016
GWSSB DEE 07.07.2016
RPSC ACF-2011
UKPSC AE 2007 Paper-I

Ans : (b)

An automobile gear box has compound gear train.



When there are more than one gear on a shaft it is called a compound train of gear.

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386. Lewis equation in gears is used to evaluate

(a) tensile stress in bending
(b) compressive stress in bending
(c) creep stress
(d) shear stress on flank

Nagaland PSC (CTSE) 2018 Paper-I
RPSC AE 2016
RPSC VPITI 14.02.2016
UKPSC AE-2013, Paper-I
RPSC ACF-2011

Ans : (b) Lewis equation in gear is used to evaluate compressive Stress in bending.

$$W_T = \sigma_W . b . P_C . y = \sigma_W . b . \pi m . y$$

The quantity y is know as Lewis form factor or tooth form factor and W_T is called the beam strength of the tooth.

$$y = 0.124 - \frac{0.684}{T}$$

$$[\text{for} 14\frac{1}{2}^\circ \text{composite and full depth involut}]$$

$$y = 0.154 - \frac{0.912}{T} [\text{for} 20^\circ \text{ full depth involute system}]$$

387. The type of gear used for speed reduction of 50 : 1 will be

(a) Herringbone (b) Spur
(c) Bevel (d) Worm Wheel

Punjab PSC (Lect.) 06.08.2017
SJVN ET 2013
UKPSC AE-2013, Paper-I
UPPSC JWM Adv. No. 16/2009
TNPSC AE, 2008

Ans. (d) : Worm wheel type of gear used for speed reduction of 50 : 1.

388. Two mating spur gears have 40 and 120 teeth respectively. The pinion rotates at 1200 rpm and transmits a torque of 20 Nm. The torque transmitted by the gear is

(a) 10 N – m (b) 20 N – m
(c) 40 N – m (d) 60 N – m

ISRO Scientist/Engineer (RAC) 22.04.2018
TSPSC AEE 2017
APPSC AEE Mains 2016, Paper-III
Vizag Steel MT (Re-Exam) 24.11.2013
ISRO Scientist/Engineer 2012

Ans. (d) : Data given,

$Z_P = 40$ $Z_G = 120$

$N_P = 1200$ rpm $T_P = 20$ N–m

We assume that

Power transmitted by pinion = Power transmited by gear

$P_P = P_G$

$$\frac{2\pi N_P \times T_P}{60} = \frac{2\pi N_G T_G}{60}$$

$$T_G = T_P \times \frac{N_P}{N_G} \quad \left[\frac{N_P}{N_G} = \frac{Z_G}{Z_P}\right]$$

$$T_G = 20 \times \frac{120}{40}$$

$T_G = 60$ N–m

389. The motion transmitted between the teeth of gears in mesh is

(a) sliding
(b) rolling
(c) rotary
(d) partly sliding and partly rolling

UPRVUNL AE 05.07.2021
TNPSC AE 2017
GPSC Lect. 23.10.2016
Assam PSC CCE Pre 2015
Kerala PSC Poly. Lect. 09.01.2014
ESE-2007

Ans. (d) : The motion transmitted between the teeth of gears in mesh is partly sliding due to presence of sliding velocity and partly rolling due to rotation.

390. Module of spur gear is define as ratio of :

(a) the pitch circle diameter to number of teeth
(b) the number of teeth to pitch circle diameter
(c) the circular pitch to diameteral pitch.
(d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II
APPSC Poly Lect. 13.03.2020
APPSC Poly Lect. Automobile Engg., 2020
CGPSC AE 25.02.2018
GPSC Lect. (Auto) 16.10.2016
J&K PSC Civil Services Pre, 2010
WBPSC AE 2008

Ans. (a) : Module (m) = $\frac{\text{pitch circle dia (D)}}{\text{no. of teeth (T)}}$

Diameteral pitch (P_d) = $\frac{1}{m}$

391. A rack is a gear of

(a) infinite pitch
(b) infinite module
(c) infinite diameter
(d) infinite number of teeth

RPSC AE 2016
Assam PSC CCE Pre 2015
Kerala PSC Poly. Lect. 09.01.2014
RPSC ACF-2011
CSE Pre-1998

Ans : (c) A rack is a gear of infinite diameter.
We know that its base circle diameter and the profiles of the involute teeth are straight lines. since these straight profiles are tangential to the pinion profiles at the point.

392. To transmit power from one rotating shaft to another whose axes are neither parallel nor intersecting, use

(a) Spur gear (b) Spiral gear
(c) Bevel gear (d) Worm gear

VIZAG Steel MT 24.01.2021, Shift-I
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC DEE, Class-2 (GWSSB) 04.07.2021
Assam PSC AE (PHED), 18.10.2020
Jharkhand Urja Vikas Nigam Ltd. AE 2017
Vizag Steel (MT) 2017
Mizoram PSC AE/SDO 2014, Paper-II
OPSC Civil Services Pre 2006
ESE 2005, 2003
Kerala PSC AE (Irrigation) 05.04.2017
Rajasthan Nagar Nigam AE 2016, Shift-I
GWSSB DEE 07.07.2016
J&K PSC Civil Services Pre, 2010

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Ans. (b) : Worm gear– To transmit power from one rotating shaft to another whose axes are neither parallel nor intersecting known as worm gear.
Spur gear– spur gears are cylindrical and have straight teeth and are mounted on parallel shafts.
Bevel gear– Teeth are formed on a conical surface, used to transfer motion between non-parallel and intersecting shafts.
Spiral gear– Spiral gears are also known as crossed helical gears. They have high helix angle and transmit power between two non-intersecting non parallel shafts. Worm gear and hypoid gear is the type of spiral gear.

393. The number of teeth per unit length of the pitch circle diameter is called
(a) diametral pitch (b) module
(c) clearance (d) gear ratio

Assam PSC AE (PHED), 18.10.2020
GPSC Lect. 23.10.2016
KPSC ADF 2015
RPSC ACF-2011
TNPSC AE, 2008

Ans. (a) : Diametrical pitch is the number of teeth per unit length of the pitch circle diameter.
Module- It is defined as the ratio of pitch circle diameter in mm to the number of teeth.

$$M = \frac{D}{T}$$

Clearance- It is the radial distance from the top of the tooth to the bottom of the tooth, in a meshing gear. A circle passing through the top of the meshing gear is known as clearance circle.
Gear ratio- It is the ratio of number of teeth of gear to the number of teeth of pinion.

$$G = \frac{T}{t}$$

394. In a pair of spur gear with external involute profile, the interference of tooth can be minimize by:
(a) Decreasing center distance between gear pair
(b) Increasing number of gear teeth
(c) Decreasing module
(d) Decreasing pressure angle

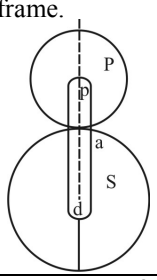
CIL MT 27.02.2020
Kernataka PSC Lect. 27.05.2017
TSPSC AEE 2017
HPPSC Asstt. Prof. 18.11.2016
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE-2000

Ans. (b) : Minimize interference in spur gear with involute profile.
(1) Increasing number of teeth
(2) Increasing center distance
(3) Increasing pressure angle

395. The gear train used in an automobile differential is
(a) simple gear train (b) compound gear train
(c) reverted gear train (d) epicyclic gear train

TSPSC AEE 2015
OPSC AEE 2015 Paper-I
(KPSC AE 2015)
TNPSC AE (Industries) 09.06.2013

Ans : (d) The gear train used in an automobile differential is epicyclic gear train.
Differential gear of an automobile: The differential gear used in the rear drive of an automobile, its functions is.
(i) To transmit motion from the engine shaft to the rear driving wheels.
(ii) To rotate the rear wheels at different speed while the automobile is taking a turn.
Epicyclic gear train:-
when there exists a relative motion of axes in a gear train, it is called an epicyclic gear train. Thus in an epicyclic train, the axis of at least one of the gears also move relative to the frame.



396. The minimum number of teeth which can be cut for standard tooth, for given pressure angle ϕ is equal to
(a) $\frac{\sin 2\phi}{2}$ (b) $2\sin^2\phi$
(c) $\frac{2}{\sin 2\phi}$ (d) $\frac{2}{\sin^2\phi}$
(e) $\frac{1}{\sin^2\phi}$

RPSC AE (GWD) 06.05.2016
CGPSC AE 26.04.2015 Shift-I
UKPSC AE 2012 Paper-I
J & K PSC Screening, 2006

Ans. (d) : Undercutting occurs if you cut deeper than the interfering point. Undercutting is a phenomenon that occurs when some part of tooth dedendum is unexpectedly cut by the edge of the generating tool. The condition for no undercutting in a standard spur gear is given by the expression.
Maximum addendum

$$m \le \frac{mZ}{2}\sin^2\phi$$

and minimum number of teeth (Z) is

$$Z = \frac{2}{\sin^2\phi}$$

397. When two bevel gears having equal teeth and equal pitch angle are used to transmit power between shafts that are intersecting at right angle, they are called
(a) Spiral bevel gears (b) Mitre gears
(c) Straight bevel gears (d) Hypoid gears

APPSC Poly Lect. Automobile Engg., 2020
VIZAG MT, 14.12.2020
GPSC Lect. 23.10.2016
TNPSC AE (Industries) 09.06.2013
ESE-2011
WBPSC AE, 2007

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Ans. (b) : Mitre gears are a special class of bevel gears. Where the shafts intersect at 90º and the gear ratio is 1:1. They are available in two types, spiral and straight tooth.

398. In involute gears the pressure angle is :

(a) Dependent on the size of teeth
(b) Dependent on the size of gears
(c) Always constant
(d) Always variable

OPSC AEE 2019 Paper-I
WBPSC AE, 2017
GPSC EE Pre, 28.01.2017
HPPSC Asstt. Prof. 18.09.2017
DRDO Scientists 2009

Ans : (c) : In involute tooth profiles, the line of action is fixed or from fundamental law of gearing, the line of action passes through a fixed point i.e. pitch point and inclined with perpendicular to the line joining centre of gears at a fixed angle. Hence pressure angle is constant throughout the teeth engagement.

399. The Minimum number of teeth in a rack and pinion for a 20° pressure angle teeth is :

(a) 20 (b) 18
(c) 22 (d) 24

VIZAG Steel MT 24.01.2021
AAI Jr. Executive 29.11.2018
TSPSC Managers, 2015
MPPSC State Forest Service Exam, 2014
UKPSC AE-2013, Paper-I
UKPSC AE 2007 Paper -I
CSE Pre-2001

Ans. (b) : $T_{min} = \frac{2A}{\sin^2 \phi}$

For minimum no. of teeth

$A = 1$

$T_{min} = \frac{2 \times 1}{(\sin^2 20)}$

$T_{min} \approx 18$

Minimum No. of teeth = 18 (If $\phi = 20°$)

400. The surface of the gear tooth below the pitch circle is called the

(a) flank (b) face
(c) undercut (d) bottom land

Assam PSC AE (IWT) 14.03.2021
Punjab PSC (Lect.) 06.08.2017
CGPSC AE 26.04.2015 Shift-I

Ans. (a) : The surface of the gear tooth above the pitch circle is called the face.
The surface of the gear tooth below the pitch circle is called the flank.

401. The tooth profile most commonly used in gear drive for power transmission is :

(a) A cycloid (b) An involutes
(c) An ellipse (d) A parabola

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TRB Poly. Lect., 2012
J&K PSC Civil Services Pre, 2010
CSE Pre-1996

Ans. (b) : For power transmission gears, the tooth form most commonly used today is the involutes profile. In involutes profile gear can be manufactured easily and the gearing has a feature that enables smooth meshing despite the misalignment of centre distance to some degree.

402. The locus of a point on the circumference of a circle which rolls without slipping on a fixed straight line is known as

(a) Cycloid (b) involute
(c) Hypocycloid (d) Epicycloid

GPSC DEE, Class-2 (GWSSB) 04.07.2021
VIZAG Steel MT 24.01.2021, Shift-I
APPSC AEE Mains 2016, Paper-III
Kerala PSC AOF 19.04.2016
GPSC Lect. (Auto), 16.10.2016

Ans. (a) : Cycloid– A cycloid is the curve traced by a point on the rim of a circle wheel as the wheel rolls along a straight line without slipping.
Involute profile– It is defined as the locus of the point on the line which roll without slipping on the fixed circle.

403. An imaginary circle on a gear which gives a rolling action, as same as the motion as the actual gear, is called:

(a) Dedendum circle (b) Pitch circle
(c) Clearance circle (d) Addendum circle

CIL MT 27.02.2020
CSE Pre-2000
UPPSC AE 12.04.2016 Paper-I
Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : Pitch Circle– It is an imaginary circle made in such a way that a pure rolling motion on this circle gives the motion which is exactly similar to the gear motion.
Dedendum Circle– A circle drawn from bottom of the teeth and concentric with the pitch circle.
Addendum Circle– A circle drawn from top of tooth and concentric to pitch circle.
Clearance Circle– A circle passing through the top of the meshing gear.

404. Law of gearing is satisfied, if

(a) two surface slides correctly
(b) common normal at the point of contact passes through the pitch point
(c) common tangent at the point of contact passes through the pitch point.
(d) Addendum > dedendum

TNPSC AE 2018
HPPSC Asstt. Prof. 20.11.2017
APPSC AEE Mains 2016, Paper-III

Ans. (b) : Law of gearing is satisfied, if common normal at the point of contact passes through the pitch point.

405. The backlash for spur gear depends upon

(a) module (b) pitch line velocity
(c) tooth profile (d) Both (a) and (b)

Punjab PSC (Lect.) 06.08.2017
Nagaland PSC CTSE 2016 Paper-I
JPSC AE 2013, Paper-V

Ans : (d) : The backlash for spur gear depends upon module and pitch line velocity.

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406. In a single reduction, a large velocity ratio is required. The best transmission is

(a) Spur gear drive (b) Helical gear drive
(c) Worm gear drive (d) Bevel gear drive

UJVNL AE 2016
RPSC Lect. (Tech. Edu. Dept.) 2011
ISRO Scientist/Engineer 2007

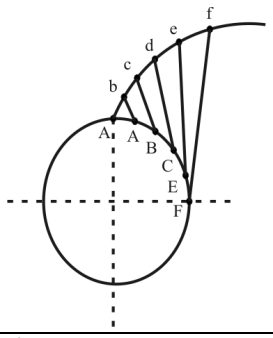
Ans. (c) : Worm gears and worm wheel have very large velocity ratio as compared to others. so worm gear drive is best in all condition.

407. When two spur gear having involute profiles on their tooth engage the line of action is tangential to :

(a) pitch circle (b) dedendum circle
(c) addendum circle (d) base circle

RPSC AE 2018
Kerala PSC AE 06.08.2015
(KPSC AE 2015
Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : Involute tooth profile is a cause made by a base circle. The definition of an involute is the spiraling curve traced by the end of an inaginary circle called the base circle



408. Gear box is used

(a) to produce torque
(b) for speed reduction
(c) to obtain variable speeds
(d) to increase efficiency of system

Vizag Steel (MT) 2017
RPSC AE 2016
RPSC ACF-2011

Ans : (c) Gear box used to obtained variable speeds.

409. If 'r' is the pitch circle radius of pinion. 'R' is pitch circle radius of driver and 'ϕ' is pressure angle, the maximum length of arc of contact for two matting gears, in order to avoid interference will be

(a) $(r + R) \sin \phi$ (b) $(r + R) \cos \phi$
(c) $(r + R) \tan \phi$ (d) $(r + R) \cot \phi$

Karnataka PSC Lect., 27.05.2017
TNPSC AE 2013
TNPSC AE, 2008

Ans. (c) : Path of contact $= (r + R) \sin \phi$

$$\text{Arc of contact} = \frac{\text{Path of contact}}{\cos\phi}$$

$$= \frac{(r+R)\sin\phi}{\cos\phi}$$

$$\boxed{\text{Arc of contact} = (r+R)\tan\phi}$$

410. Internal gears are manufactured by

(a) Hobbing
(b) Shaping with pinion cutters
(c) Shaping with rack cutter
(d) Milling

PPSC Asstt. Municipal Engg. 15.06.2021
Gujarat PSC AE 2019
GWSSB AAE, 27.12.2015
GPSC ARTO 01.05.2016

Ans : (b) : Internal gear cutting operation can be performed by shaping with pinion cutter in the case of rotating pinion type cutter indexing is not required, therefore, this type is more productive and so common.

411. The product of diametral pitch and module is equal to

(a) zero (b) infinity
(c) unity (d) two

Gujarat PSC AE 2019
Nagaland PSC (CTSE) 2018, Paper-I
Kerala PSC IOF 19.04.2016

Ans : (c) : Module $(m) = \frac{D}{T}$

$$\text{Diametral pitch } (D) = \frac{1}{\text{Module}(m)}$$

Product of diametral pitch and module is

$$= \frac{D}{T} \times \frac{T}{D} = 1$$

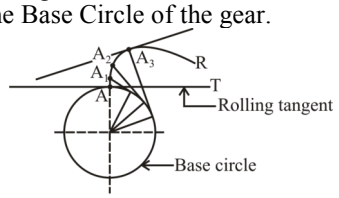
412. In a gear having involute teeth, the normal to the involute is a tangent to the

(a) pitch circle (b) base circle
(c) addendum circle (d) dedendum circle

Sikkim PSC (Under Secretary), 2017
CGPSC AE 26.04.2015 Shift-I
UPSC JWM Adv. No-16/2009

Ans. (b) : Involute Profile–Involute is a curve generated by point on a tangent which rolls on a circle without slipping. The involute profile on a gear will be generated through a generating circle and this generating circle will be known as base circle. It is a fundamental property of a gear its radius will not change in any condition for a gear.

In an involute gear, the normal to the involute is the tangent to the Base Circle of the gear.



Involute Profile

413. A fixed gear having 100 teeth meshes with another gear having 25 teeth, the centre lines of both the gears being joined by an arm so as to form an epicyclic gear train. The number of rotations made by the smaller gear for one rotation of arm is.

(a) 3 (b) 4
(c) 5 (d) 6

MPPSC AE 08.11.2015
MPPSC State Forest Service Exam, 2014
APPSC AEE 2012
CSE Pre-1995 / ESE-2009

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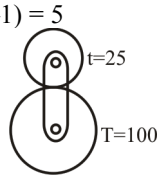
Ans. (c) :

Arm	$T_P(100)$	t(25)
0	+x	–4x
+1	1+x	1–4x

$N_P = 0,\ x = -1$

$N_t = 1 - 4(-1) = 5$



414. An involute pinion and gear are in mesh. If both have the same size of addendum, then there will be an interference between the :
(a) Tip of the gear tooth and flank of pinion
(b) Tip of the pinion and flank of gear
(c) Flanks of both gear and pinion
(d) Tip of both gear and pinion

RPSC ACF & FRO, 26.02.2021
APPSC IOF, 2009
ESE 1996

Ans. (a) : An involute pinion and gear are in mesh. If both have the same size of addendum, then there will be an interference between the tip of the gear tooth and flank of pinion.

415. For 20° pressure angle, minimum number of teeth will be:
(a) 6 (b) 12
(c) 17 (d) 20

SJVN ET 2013
J & K PSC Screening, 2006

Ans. (c) : $t \geq \dfrac{2a_r}{\sin^2\phi}$ ($a_r = 1$ for standard addendum)

$$\therefore t_{min} = \frac{2}{\sin^2 20^o} = 17.1 \text{ or } 18$$

416. The interference in cycloidal teeth
(a) is absent completely
(b) depends on number of teeth
(c) depends on condition of meshing
(d) depends on pressure angle
(e) is maximum

Nagaland PSC CTSE 2017, Paper-I
CGPSC AE 26.04.2015 Shift-I

Ans. (a) : In cycloidal gears, no interference exists provided the centre distance is properly maintained, unlike in involute gears.

417. Backlash is
(a) Sum of the clearances of pinion and gear teeth
(b) The amount by which the width of a tooth space exceeds the thickness of meshing tooth on pitch circle
(c) Difference between dedendum and addendum of gear tooth
(d) Difference between the pitch circles of meshing gears

APPSC AEE Mains 2016, Paper-III
TNPSC AE, 2008

Ans. (b) : Backlash– Backlash is defined as the amount by which a tooth space exceeds the thickness of an engaging tooth. The general purpose of backlash is to present gears from jamming together and making contact on both sides of their teeth simultaneously. Lack of backlash may cause noise.

418. The minimum number of teeth on the pinion, in order to avoid interference for $14\frac{1}{2}$ full depth system is
(a) 12 (b) 14
(c) 18 (d) 32

TNPSC AE 2013
UKPSC AE 2007 Paper-I

Ans. (d) : In the design of gears, it is required to decide the number of teeth on the pinion and gear. There is a limiting value of the minimum number of teeth on the pinion.

So, the minimum number of teeth to avoid interference is given by,

$$Z_{min} = \frac{2}{\sin^2\alpha}$$

So, for $\alpha = 14\frac{1}{2}$ (full depth system)

$Z_{min} = 32$

419. A high pressure angle for spur gear leads to
(a) minimum axial thrust
(b) greater backlash
(c) more interference
(d) wide base and stronger teeth

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : A high pressure angle for spur gear leads to wide base and stronger teeth because if pressure angle increases, strength of tooth increases.

420. Purpose of using differential gear in automobile is to
(a) control speed
(b) avoid jerks
(c) help in turning
(d) improve speed performance

HPPSC Asstt. Prof. 20.11.2017
GPSC Lect. 23.10.2016

Ans. (c) : Differential gear, in automotive mechanism, gear arrangement that permits power from the engine to be transmitted to a pair of driving wheel, as when turning a corner or transverse an uneven road.

421. The path of the point of contact between the involute teeth profile gears is :
(a) Circle (b) Straight line
(c) Complex curve (d) Parabola

HPPSC W.S. Poly. 2016
RPSC AE (GWD) 06.05.2016

Ans : (b) The path of the point of contact between the involute teeth profile gear is straight line.
In actual practice following are the two types of teeth commonly used – 1. Cycloidal teeth, 2. Involute teeth
An involute of a circle is a plane curve generated by a point on a tangent, which rolls on the circle without slipping which in unwrapped from a reel as.

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422. If t and T are the number of teeth on pinion and gear, G is gear ratio, A_p is the fraction by which the standard addendum of one module is multiplied and ϕ is the pressure angle, the expression minimum number of teeth on pinion to avoid interference when pinion and gear have equal number of teeth is given as

(a) $t = 2\,A_p / (1-3\sin^2\phi + 1)^{1/2}$
(b) $t = 2\,A_p / (1+3\sin^2\phi + 1)^{1/2}$
(c) $t = 2\,A_p / (1-3\sin^2\phi - 1)^{1/2}$
(d) $t = 2\,A_p / [(1+3\sin^2\phi)^{1/2} - 1]$

BPSC Poly. Lect. 2016
VIZAG MT 2015

Ans : (d)
Minimum number of teeth on the pinion in order to avoid interference.

$$t = \frac{2Ap}{\sqrt{\left[1+\frac{T}{t}\left(\frac{T}{t}+2\right)\sin^2\phi\right]-1}}$$

t = Number of teeth on the pinion
T = Number of teeth on the wheel
m = Module of the teeth

r = pitch circle radius of pinion = $\frac{mt}{2}$

G = Gear ratio = $\frac{T}{t} = R/r$

ϕ = pressure angle.
if the pinion and wheel have equal teeth, then G=1 therefore the above equation reduces to

$$t = \frac{2A_p}{\sqrt{(1+3\sin^2\phi)}-1}$$

423. In a simple gear train, there is odd number of idlers. The direction of rotation of the driver and the driven gears will be

(a) same
(b) opposite
(c) depends upon the number of teeth of the gears
(d) depends upon the diameter of idlers used

MPPSC AE 2016
UKPSC AE 2012 Paper-I

Ans. (a) : In a simple gear train, there is odd number of idlers. The direction of rotation of the driver and the driven gears will be same.

424. The working surface above the pitch surface of a gear tooth is termed as

(a) addendum (b) dedendum
(c) flank (d) face

GPSC Executive Engg. 23.12.2018
UKPSC AE 2007 Paper -I
ESE-1998

Ans. (d) : • **Face-** The working surface above the pitch surface of the gear tooth is termed as face.
• **Flank-** The portion of tooth profile below the pitch surface.
• **Addendum-** It is the radial height of the teeth above the pitch circle. Its standard value is one module.
• **Dedendum-** It is the radial depth of the tooth below the pitch circle. Its standard value is 1.157 module.

425. For full depth involute spur gears, minimum number of teeth on the pinion to avoid interference depends on :

(a) Pressure angle (b) Speed ratio
(c) Circular pitch (d) Pitch diameter

GPSC Executive Engineer 23.12.2018
Kerala PSC Asstt. Prof. 27.09.2016

Ans. (a) : Interference- When two gears are mesh at one instant there is a chance to mate involute portion with non-involute portion of mating gear. This phenomenon is known as interference.
• Minimum number of teeth on the pinion to avoid interference depends on pressure angle.
• When we increase the pressure angle then base radius decrease but we cannot increase the pressure angle beyond the 20°.

426. For a helical gear, if θ is helix angle, and Z is number of teeth on pinion, then what is virtual number of teeth:

(a) $Z/\cos\theta$ (b) $Z/\cos^2\theta$
(c) $Z/\cos^3\theta$ (d) None of these

TNPSC AE 2018
PSPCL AE, 2012

Ans. (c) :
Vertical number of teeth of helical gear = $\frac{z}{\cos^3\theta}$
Where θ = helix angle

427. The normal circular pitch in helical gear is given by :

(a) $p\sin\psi$ (b) $\frac{p}{\sin\Psi}$
(c) $p\cos\psi$ (d) $\frac{p}{\cos\Psi}$

KPSC ADF 2015
OPSC AEE 2015 Paper-I

Ans. (c) : The normal circular pitch (P_n) for helical gear is :

$$\boxed{P_n = P\cos\Psi}$$

Where, P_n = Normal circular pitch
P = Transfer circular pitch

428. Tumbler gears are used to:

(a) give the desired direction of movement to the lathe carriage
(b) give the desired angle to the tapper turning
(c) give a varying speed to the headstock
(d) give a varying speed to tailstock, when needed

NLCIL GET 17.11.2020, Shift-II
WBPSC AE, 2017

Ans. (a) : Tumbler gear mechanism is used to change the direction of lead screw and feed rod in lathe machines by engaging tumbler gear the carriage can be moved automatically from tailstock end to head stock end or moved from headstock end to tail stock end.
So, tumbler gear is used to give the desired direction of movement to the lathe carriage.

429. Spur gears used for connecting……..

(a) Shafts intersecting at 45 degrees
(b) Parallel shafts

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(c) Non-parallel, non intersecting shafts
(d) Shafts intersecting at 90 degrees.

HPPSC Lect. (Auto) 23.04.2016
DRDO Scientist 2008

Ans. (b) : Spur gears used for connecting.
- Parallel shafts.
- Tooth are parallel to shaft axis.
- Sudden engagement and disengagement.
- High impact stresses comes on profile and no axial thrust.

430. If T is the number of teeth on a helical gear and ϕ is the helix angle for the teeth, the formative number of teeth is written as

(a) $T \sec^3 \phi$ (b) $T \sec^2 \phi$
(c) $T/\sec^3 \phi$ (d) $T \operatorname{cosec} \phi$

APPSC Poly. Lect. 2013
TNPSC AE (Industries) 09.06.2013

Ans. (a) : T - Actual no. of teeth on a helical gear
ϕ - is the helix angle for the teeth
then, Formative number of teeth is written as, = $T\sec^3 \phi$
Formative or equivalent no. of teeth for a helical gear may be defined as the number of teeth that can be generated on the surface of a cylinder having a radius equal to the radius of curvature at the tip of the minor axis of an ellipse obtaining by taking a section of the gear in the normal plane.

431. For a speed ratio of 100, smallest gear box is obtained by using

(a) A pair of spur gears
(b) A pair of helical and pair of spur gear compounded
(c) A pair of bevel and a pair of spur gear compounded
(d) A pair of helical and a pair of worm gear compounded

RPSC IOF, 2020
APPSC Poly. Lect. 2013

Ans. (d) : For a speed ratio of 100, smallest gear box is obtained by using a pair of helical and a pair of worm gear compounded.

432. Two parallel and coplanar shafts are connected by :

(a) Spur gears (b) Bevel gears
(c) Spiral gears (d) Double helical gears

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Rajasthan Nagar Nigam AE 2016, Shift-I
TSPSC Managers, 2015

Ans. (a) : Spur Gears– Two parallel and coplanar shafts are connected by spur gears.
Bevel Gears–Bevel gears connect two shafts above axes intersect at an angle after than a right angle, uses – to change the axis of rotational motion.
Spiral Gears–It is used to connect known parallel shaft are commonly called spiral gear or helical gears or crossed axis helical gear. If the shaft angle is 90 degrees, the gears will be of the same hand and the sum of the helix angles will be equal to the shaft angle.
Double Helical Gears–A double helical gear incorporates two opposites oriented helical gears of the same helix angle together such that the two opposing thrust forces annihilates one another.

433. In spiral bevel gears, the axes are

(a) non-parallel and non-intersecting and the teeth are curved
(b) non-parallel and non-intersecting and the teeth are straight
(c) intersecting, the teeth are curved and oblique
(d) intersecting, the teeth are curved and can be ground

TNPSC AE 2019
TNPSC AE, 2008

Ans. (c) : A spiral bevel gear is a bevel gear with helical teeth formed along spiral lines an the gear face. Spiral bevel gear are most often used in vehicle differentials automotive, marine and aerospace.
Non-parallel and non-intersecting and the teeth are curved and are called skew bevel gear.

434. The axial thrust on the worm (W_A) is given by

(a) $W_A = W_T \cdot \tan \phi$ (b) $W_A = W_T / \tan \phi$
(c) $W_A = W_T \cdot \tan \lambda$ (d) $W_A = W_T / \tan \lambda$

TNPSC AE 2014
TNPSC AE, 2008
TNPSC AE (Industries) 09.06.2013

Ans. (d) : The axial thrust on the worm (W_A) is given by

$$W_A = \frac{W_T}{\tan \lambda}$$

Where; W_T = Tangential force on the worm.

435. In a 20^0 pressure angle involutes gear, how can interference be avoided?

(a) Using more than 8 teeth
(b) Cutting involutes profile accurately
(c) Using more than 20 teeth
(d) Using as small number of teeth as possible

APGENCO AE, 2017
GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC ACF 2012

Ans. (c) : Interference in gear can be avoided by –
(i) Increasing centre distance
(ii) Increasing no. of teeth on pinion
(iii) Increasing pressure angle
(iv) Stubbing or cutting some portion of addendum

436. The locus of a point on a thread unwound from a cylinder will be

(a) an involute (b) a helix
(c) a straight line (d) a cycloid

JPSC AE 10.04.2021, Paper-II
Punjab PSC (Lect.) 06.08.2017

Ans. (a) : The locus of a point on a thread unwound from a cylinder will be an involute.

437. In gears, the contact ratio is the ratio of

(a) length of path of contact to the circular pitch
(b) length of arc of contact to the circular pitch
(c) length of arc of approach to the circular pitch
(d) length of arc of recess to the circular pitch

CGPSC AE 15.01.2021
JPSC AE 2013, Paper-V

Ans : (b) :

$$\text{Contact ratio} = \frac{\text{length of arc of contact}}{\text{circular pitch}}$$

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438. Gears used for non-intersecting perpendicular shafts are :

(a) spur gears
(b) helical gears
(c) double helical gears
(d) hypoid gears

RPSC AE 2016
RPSC ACF-2011

Ans. (d) : Gears used for non-intersecting and perpendicular shafts are hypoid gears.

Spur gear is used to connect two parallel and coplanar shaft.

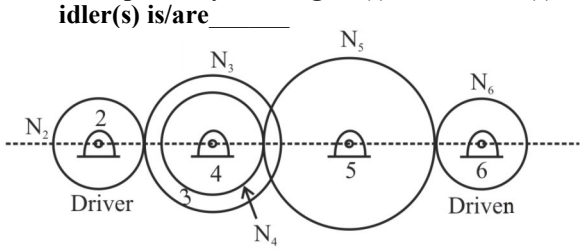
439. When both pinion and gear are made of the same material, then from the design point of view :

(a) pinion is the determining factor
(b) gear is the determining factor
(c) any one of the two may be taken as determining factor
(d) criteria is strength of any one of the teeth

RPSC AE 2016
RPSC ACF-2011

Ans. (a) : When both pinion and gear are made of the same material, then from the design point of view based on pinion. Pinion weaker than gear because a pinion tooth is loaded and unloaded more number of times than the gear tooth you can verify this by putting a mark on one pinion tooth and count how many times it meshes. Pinion has less material in the blank to sustain the same radial as well as thermal load.

440. A gear train is made up of five spur gears as shown in the figure. Gear 2 is driver and Gear 6 is driven member. N_2, N_3, N_4, N_5 and N_6 represent number of teeth on gears 2, 3, 5 and 6 respectively. The gear(s) which act(s) as idler(s) is/are_____



(a) Only 5
(b) Only 3
(c) Only 4
(d) Both 3 and 5

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (a) : $\frac{\omega_2}{\omega_6} = \frac{T_3 \times T_5 \times T_6}{T_2 \times T_4 \times T_5}$

$$\boxed{\frac{\omega_2}{\omega_6} = \frac{T_3 \times T_6}{T_2 \times T_4}}$$

So, Gear 5 act as a idler gear.

441. In which one of the following tooth profiles, does the pressure angle remain constant throughout the engagement of teeth?

(a) Cycloidal
(b) Involute
(c) Conjugate
(d) Epicycloid

ESE 2020
Haryana PSC Civil Services Pre, 2014

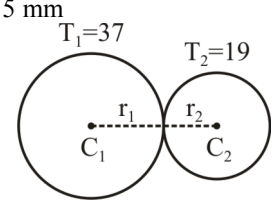
Ans. (b) : In involute profile tooth, because of the involute property to be satisfied, pressure angle does not change with point of contact position.

442. 20° full depth involute profile 19 tooth pinion and 37 teeth gear are in mesh. If the module is 5 mm, then the centre distance between the gear pair is

(a) 140 mm
(b) 150 mm
(c) 280 mm
(d) 300 mm

UPPSC AE 13.12.2020, Paper-I
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2006

Ans. (a) : m = 5 mm



Pitch circle radius of gear $(r_1) = \frac{mT_1}{2}$

Pitch circle radius of pinion $(r_2) = \frac{mT_2}{2}$

Centre distance $(C_1C_2) = r_1 + r_2$

$$= \frac{m}{2}(T_1 + T_2)$$

$$= \frac{5}{2}(37 + 19)$$

$$= 140 \text{ mm}$$

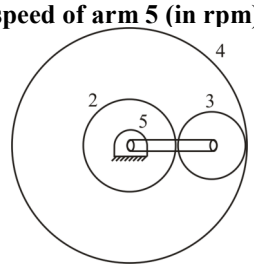
443. From the start of engagement to the end of engagement, the pressure angle of gear, having involutes profile, is :

(a) zero
(b) varies linearly
(c) varies exponentially
(d) constant

APPSC Poly Lect. 13.03.2020
ESE 2020

Ans. (d) : From the start of engagement to the end of engagement, pressure angle (ϕ) of gear of Involute profile should be constant.

444. In the gear train shown, gear 3 is carried on arm 5, Gear 3 meshes with gear 2 and gear 4. The number of teeth on gear 2, 3 and 4 are 60, 20 and 100 respectively. If gear 2 is fixed and gear 4 rotates with an angular velocity of 100 rpm in the counterclockwise direction, the angular speed of arm 5 (in rpm) is



(a) 100 rpm
(b) 62.5 rpm
(c) 125.5 rpm
(d) 500 rpm

Gujarat PSC AE 2019 / GATE 2016

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Ans : (b) : The table for calculation

Action	Arm 5	Gear 2	Gear 3	Gear 4
1. Fix Arm to Gear 2	0	x	$=-\frac{60}{20}x$ $=-3x$	$=\frac{20}{100}\times(-3x)$ $=-0.6\,x$
2. Add 'y' to each	y + 0	y + x	y – 3x	y – 0.6x

Conditions-

1. Gear - 2 is fixed

$\therefore$ x + y = 0

x = – y

2. Gear - 4 rotates with 100 rpm (ccw)

$\therefore$ y – 0.6 x = 100

y + 0.6 y = 100

$$y=\frac{100}{1.6}=62.5=62.5 \text{ rpm (ccw)}$$

445. Match list I with list II and select the correct answer from the codes given below:

List I (Gear type)	List II (Application)
A. Worm gears	1. Parallel shafts
B. Cross helical gears	2. Non parallel intersecting shafts
C. Bevel gears	3. Non-parallel, non-intersecting shafts
D. Spur gears	4. Large speed ratio

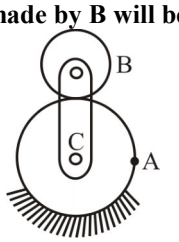
Code:

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	3	4	1	2
(d)	4	3	2	1

PTCUL AE 25.06.2017
GATE 1995

Ans. (d) :

446. In the Epicyclic gear train shown below, A is fixed, A has 100 teeth and B has 20 teeth. If the arm C makes three revolutions, the number of revolutions made by B will be



(a) 12 (b) 15
(c) 18 (d) 24

TSPSC AEE 2017
ESE 1997

Ans. (c) : From figure

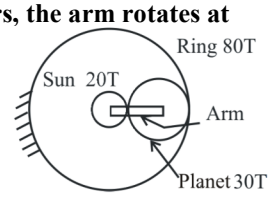
For 1 revolution of C,

$$N_B=1+\frac{T_A}{T_B}=6$$

For 3 revolution

$N_B = 6 \times 3 = 18$

447. The sun gear given in the figure is driven clockwise at 100 rpm. The ring gear is held stationary. For the number of teeth shown on the gears, the arm rotates at ____.



(a) 0 rpm (b) 20 rpm
(c) 33.33 rpm (d) 66.67 rpm

RPSC ADE 2016
GATE 2001

Ans. (b) :

Action	Rotation			
	arm	sun (T_s)	planet (T_P)	ring (T_R)
Arm to fixed sun gear 'S' rotate x revolution CCW	0	+x	$-\frac{T_s}{T_P}x$	$-\frac{T_s}{T_R}x$
Add + y' revolution in all member	+y	+y	$+y-\frac{T_s}{T_P}x$	$y-\frac{T_s}{T_R}x$
Total revolution	y	x+y	$-\frac{T_s}{T_P}x+y$	$-\frac{T_s}{T_R}x+y$

x+ y = – 100 ..... (1)

$-\frac{20}{80}x+y=0$ ..... (2) ring is fixed

x = – 80, y = –20

rotation of arm = y = – 20 rpm

i.e. = 20 rpm clockwise.

448. Which one of the following is used to convert a rotational motion into a translational motion?

(a) Bevel gears
(b) Double helical gears
(c) Worm gears
(d) Rack and pinion gears
(e) None of the above

GPSC Executive Engineer 23.12.2018
CGPSC Poly. Lect. 22.05.2016

Ans. (d) : Rack and pinion gears

- Rack and pinion gear system are used to transfer rotational motion into translational motion.
- The flat tooth part is the rack and the gear is pinion.
- The rack has teeth cut into it and they mesh with the teeth of the pinion gear.

449. The maximum efficiency for spiral gears is

(a) $\frac{\cos(\theta+\Phi)+1}{\cos(\theta-\Phi)+1}$ (b) $\frac{\cos(\theta-\Phi)+1}{\cos(\theta+\Phi)+1}$

(c) $\frac{\sin(\theta+\Phi)+1}{\cos(\theta+\Phi)+1}$ (d) $\frac{\cos(\theta-\Phi)+1}{\sin(\theta+\Phi)+1}$

Nagaland PSC CTSE 2016 Paper-I
CSE Pre-1998

Ans. (a) : The maximum efficiency of spiral gears is given by, $\eta_{max}=\frac{\cos(\theta+\Phi)+1}{\cos(\theta-\Phi)+1}$

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450. Match the item in Column I and II

Column-I	Column-II
p. Addendum	**1.** Cam
q. Instantaneous center of velocity	**2.** Beam
r. Section modulus	**3.** Linkage
s. Prime Circle	**4.** Gear

Select the correct answer using the codes given below-

(a) p-4, q-3, r-2, s-1 **(b) p-4, q-2, r-3, s-1**
(c) p-3, q-2, r-1, s-4 **(d) p-3, q-4, r-1, s-2**

RPSC 2016
GATE 2006

Ans : (a) p-4, q-3, r-2, s-1

451. An external gear with 60 teeth meshes with a pinion of 20 teeth, module being 6 mm. What is the centre distance in mm?

(a) 120 (b) 180
(c) 240 (d) 300

Assam PSC CCE Pre 2015
ESE 2009

Ans. (c) : Centre distance,

$$d = \frac{m}{2}(T_1 + T_2)$$

$$= \frac{6}{2}(60 + 20) = \frac{6}{2} \times 80$$

$$= 240 \text{ mm}$$

452. Match List I with List II :

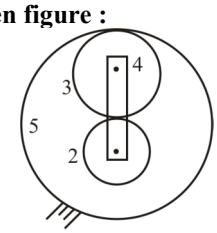
List I	List II
A. Helical gear	(i) Non-interchangeable
B. Herring bones gear	(ii) Zero axial thrust
C. Worm gears	(iii) Quiet motion
D. Hypoid gears	(iv) High speed reduction

	A	B	C	D
(a)	(i)	(ii)	(iii)	(iv)
(b)	(iii)	(ii)	(i)	(iv)
(c)	(iii)	(i)	(iv)	(ii)
(d)	(iii)	(ii)	(iv)	(i)

MPSC Poly. Lect. 2014
CSE Pre-2009

Ans. (d) : Helical gear – Quiet motion.
Herring bones gear – zero axial thrust
Worm gears – High speed reduction
Hypoid gears – Non- interchangeable.

453. An epicyclic gear train is shown schematically in the given figure :

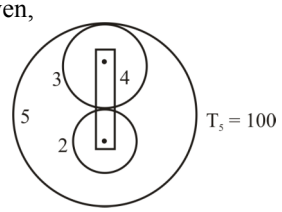


The sun gear 2 on the input shaft is a 20 teeth external gear. The planet gear 3 is a 40 teeth external gear. The ring gear 5 is a 100 teeth internal gear. The ring gear 5 is fixed and gear 2 is rotating at 60 rpm ccw. The arm 4 attached to the output shaft will rotate at :

(a) 10 rpm ccw (b) 10 rpm cw
(c) 12 rpm ccw (d) 12 rpm cw
(ccw = counter clockwise and cw = clock wise)

OPSC Civil Services Pre. 2011
GATE 2009

Ans. (a) : Given,



clockwise direction taken as positive

Condition of motion	Arm	Gear-2 $T_2 = 20$	Gear-3 $T_3 = 40$	Gear-5 $T_5 = 100$
(1) Arm is fixed	0	x	$-\frac{20x}{40}$	$-\frac{20x}{100}$
(2) Arm is free	y	x+y	$y - \frac{x}{2}$	$y - \frac{x}{5}$

$$\therefore \quad y - \frac{x}{5} = N_5 = 0$$

$$x = 5y$$

$$x + y = -60$$

$$\therefore \quad 6y = -60$$

$$y = -10 \text{ rpm}$$

$$\boxed{N_{arm} = 10 \text{ rpm}(ccw)}$$

454. The type of gear used to keep noise minimum should be

(a) Cyclical (b) Involute
(c) Bevel (d) Helical

APGCL AM, 2021

Ans. (d) : Helical gears are gradually engaged and disengaged, that's why this type of gears are used to keep noise minimum.

455. An automobile generally employs _____ gear train.

(a) simple (b) reverted
(c) compound (d) epicyclic

UPRVUNL AE 05.07.2021

Ans. (c) : An automobile generally employs compound gear train.

456. Consider the following statements regarding gears:

1. The ratio of number of teeth on the gear to that on the pinion is known as gear ratio.
2. The circle passing through the tips of teeth is called dedendum circle.
3. The circle passing through the roots of teeth is called addendum circle.
4. Backlash is the difference between the space width and the tooth thickness along the pitch circle.

Which of the above statements are correct?

(a) 1, 2 and 3 only (b) 1 and 4 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

ESE 18.07.2021

Ans. (b) : Statements 2 and 3 are wrong because-

- The circle passing through the tips of teeth is called addendum circle.
- The circle passing through the roots of teeth is called dedendum circle.

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457. The number of teeth of a spur gear is 30 and it rotates at 200 rpm. What is the pitch line velocity if it has a module of 2 mm?

(a) 341.7 mm/s (b) 497.2 mm/s
(c) 628.3 mm/s (d) 758.5 mm/s

ESE 18.07.2021
HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Given:
Number of teeth, T = 30
Module,m = 2mm
Rotational speed, N = 200 rpm
Pitch line velocity, $v = \omega \times r$

$$= \frac{2\pi N}{60} \times \frac{mT}{2}$$

$$\therefore v = \left(\frac{2\pi \times 200}{60} \times \frac{2 \times 30}{2}\right) mm/s.$$

v = 628.32 mm/s

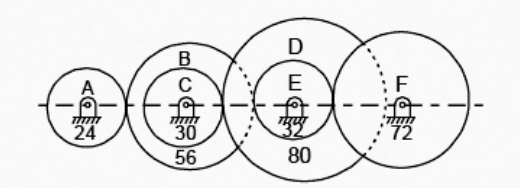
458. The difference between the tooth space and tooth the thickness as measured on the pitch circle is called.

(a) working depth (b) clearance
(c) face width (d) backlash

GPSC AE 26.07.2015
MPSC Poly. Lect. 2014

Ans. (d) : The difference between the tooth space and tooth thickness is called backlash.

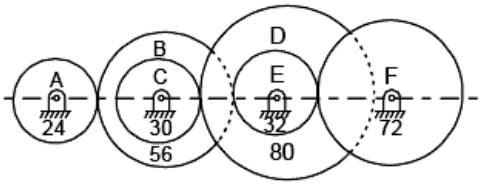
459. The compound gear train shown in the figure below consists of compound gear B-C and DE. All gears are mounted on parallel shafts. The motor shaft rotating at 800 rpm is connected to the gear A and the output shaft to the gear F. The number of teeth on gears A, B, C, D, E and F are 24, 56, 30, 80, 32 and 72 respectively. What is the speed of the gear F?



(a) 57.14 rpm (b) 32.51 rpm
(c) 74.63 rpm (d) 69.72 rpm

ESE 18.07.2021

Ans. (a) :



$$\frac{N_F}{N_A} = \frac{\text{Product of teeth on driving gear}}{\text{Product of teeth on driven gear}}$$

$$\frac{N_F}{800} = \frac{24 \times 30 \times 32}{56 \times 80 \times 72}$$

$N_F = 0.07142 \times 800$

$\boxed{N_F = 57.14 \text{ rpm}}$

460. Which of the following factors can cause misalignment of the teeth on the pinion relative to those on the gear?

1. Inaccurate gear teeth
2. Misalignment of the axes of shafts carrying gears
3. Thermal distortions during operation

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) : factors responsible for misalignment of the teeth on the pinion-

- Inaccurate gear teeth.
- Misalignment of the axes of shafts carrying gears.
- Thermal distortions during operation.

461. Two parallel shafts are connected by spur gearing. The approximate distance between the shafts is 600 mm. One shaft rotates at 120 rpm and the other rotates at 360 rpm. The module is 8 mm. Number of teeth on the wheels are _____.

(a) 114.38 (b) 120.40
(c) 132.44 (d) 108.36

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Given,
Centre distance, x = 600 mm
N_1 = 120 rpm
N_2 = 360 rpm
module, m = 8 mm
zed, D_1 = diameter of the first gear
D_2 = diameter of second gear
We know that,

$$\frac{N_1}{N_2} = \frac{D_2}{D_1} = \frac{120}{360} = \frac{1}{3}$$

$D_1 = 3D_2$ ———(I)

$$\therefore \text{ centre distance, } \quad 600 = \frac{1}{2}(D_1 + D_2)$$

$D_1 + D_2 = 1200$ ———(II)
From equation (I) and (II) –
$3D_2 + D_2 = 1200$
$4D_2 = 1200$
D_2 = 300 mm
and $D_1 = 3 \times 300 = 900$ mm
$\therefore$ Number of teeth on the first gear

$$\boxed{\text{module} = \frac{D_1}{T_1}}$$

$$T_1 = \frac{D_1}{m} = \frac{900}{8} = 112.5$$

and Number of teeth on the second gear,

$$T_2 = \frac{D_2}{m} = \frac{300}{8}$$

$T_2 = 37.5$
$T_2 = 38$

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462. In order to have smooth operation, the minimum number of teeth on the smaller sprocket, for moderate speeds, should be
(a) 15 (b) 17
(c) 21 (d) 25
GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : 17

463. Gears used in drive to the differential of an automobile are
(a) straight bevel gears.
(b) spiral bevel gears.
(c) zero bevel gears.
(d) All of the above
PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : The spiral bevel gear is a bevel gear with helical feat. The main application of this gear is in vehicle differential where the direction of drive from the driven shaft must be turned 90º.

464. A spur gear is designed to transmit the power 10 kW at a pitch line velocity of 10 m/s. Diameter of driving gear is 1 m. The tangential force between the driver and the follower and the transmitted torque, respectively, are:
(a) 1 kN and 0.5 kNm
(b) 0.5 kN and 0.5 kNm
(c) 1 kN and 1 kNm
(d) 0.5 kN and 1 kNm
AAI Jr. Executive 26.03.2021

Ans. (a) : Given, Power = 10 kW, dia = 1 m
Pitch line velocity = 10 m/s

$$P = F \times v$$

$$F = \frac{P}{v}$$

$$F = \frac{10}{10} = 1\,kN \quad \text{(Tangential force)}$$

Transmitted torque,

$$T = F \times r$$

$$T = 1 \times 0.5$$

$$T = 0.5\ kNm$$

465. Hypoid gears are approximations of _____ through they look like spiral gears.
(a) epicycloids (b) pinion
(c) cycloids (d) hyperboloids
HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Hypoid gears are approximations of hyperboloids through they look like spiral gears.

466. Spur and helical gears can be planed using _____ process.
(a) Whitworth (b) Markovian
(c) Sunderland (d) Taylor's
HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : Spur and helical gears can be planed using Sunderland process.

467. The locus of all such points on the coupler which have stationary curvature at the instant is known as _____.
(a) circling curve
(b) spherical stationary curvature
(c) centre of curvature
(d) None of the above
HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : The locus of all such points on the coupler which have stationary curvature at the instant is known as circling curve.

468. The control gear to epicylic gear unit is called
(a) sun gear (b) ring gear
(c) planet gear (d) internal gear
HPPSC Poly. Lect. 05.07.2021

Ans. (a) : sun gear

469. Addendum circle for a spur gear has a diameter equal to :
(a) pitch circle diameter
(b) pitch circle diameter – 2m
(c) pitch circle diameter + 2m
(d) None of the above
JPSC AE 10.04.2021, Paper-II

Ans. (c) : Diameter of addendum Circle = D + 2m
Diameter of dendum Circle = D – 2 × 1.157m
where,
D = Pitch circle dia of gear
m = module of gear.

470. If two gears A with m_A = 2mm and t_A = 50 and B with m_B = 4mm and t_B = 30 teeth are in mesh with each other such that A rotates with 100 rpm, then revolution of gear B is approximately near to :
(a) 60 rpm (b) 100 rpm
(c) 92 rpm (d) 83 rpm
JPSC AE 10.04.2021, Paper-II

Ans. (d) : Given,

$m_A = 2mm$

$t_A = 50$

$m_B = 4mm$

$t_B = 30$

$N_A = 100$ rpm

$N_B = ?$ $\qquad m = \frac{D}{T}$

$$\frac{N_B}{N_A} = \frac{D_A}{D_B}$$

$$\frac{N_B}{100} = \frac{50 \times 2}{30 \times 4} \Rightarrow \boxed{N_B = 83\,rpm}$$

471. A pinion and gear are in mesh with a gear ratio of 2. The moment of inertias of the pinion and gear are 3 kg m^2 and 5kg m^2 respectively. For the pinion to have an angular acceleration of 4 rad/s^2, the torque to be applied to the pinion shaft is :
(a) 5 N-m (b) 10 N-m
(c) 14 N-m (d) 17 N-m
RPSC ACF & FRO, 26.02.2021

Ans. (b) : Given,

$$\text{Gear ratio}\left(\frac{Z_2}{Z_1}\right) = 2$$

$$I_{pinion}(I_1) = 3\,kg\,m^2$$

$$I_{gear}(I_2) = 5\,kg\,m^2$$

$$\frac{Z_2}{Z_1} = \frac{D_2}{D_1} = \frac{\omega_1}{\omega_2} = 2$$

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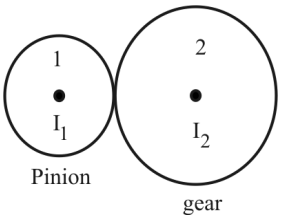


In meshing power transmission is same.

$$P_1 = P_2$$

$$T_1\omega_1 = T_2\omega_2$$

$$T_1 = T_2 \times \frac{\omega_2}{\omega_1} = I_2\alpha\frac{\omega_2}{\omega_1} = 5 \times 4 \times \frac{1}{2}$$

$$\boxed{T_1 = 10}$$

472. In comparison to reverse gear, the torque obtained and speed ratio of top gear are

(a) less, less (b) more, less
(c) equal, more (d) less, more

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (d) : less, more

473. The adjustment for backlash in a differential is provided between :

(a) Crown wheel and sun gear
(b) Crown wheel and the planet gear
(c) Sun gear and the planet gear
(d) Crown wheel and drive pinion

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

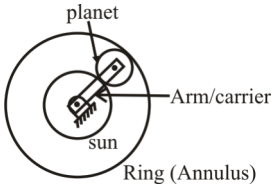
Ans. (b) : The adjustment for backlash in a differential is provided between crown wheel and the planet gear.

474. In a simple epicyclic gear-set, the output member to increase torque in reverse is always the :

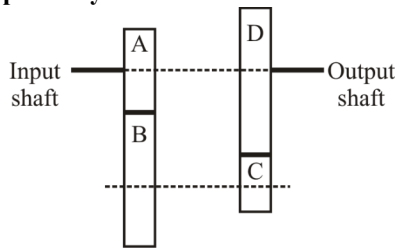
(a) Ring gear (b) Sun gear
(c) Planet carrier (d) None of these

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (a) :



475. In compound gear train as shown in fig., what will be the speed of output shaft if input shaft runs at 2700 rpm? The number of teeth on gears A, B, C, D are 16, 24, 12 & 36 respectively



(a) 5400 rpm (b) 2700 rpm
(c) 600 rpm (d) 1350 rpm

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021 Paper-II

Ans. (c) :

$$\frac{N_D}{N_A} = \frac{T_A \times T_C}{T_B \times T_D}$$

$$\frac{N_D}{2700} = \frac{16 \times 12}{24 \times 36}$$

$$\boxed{N_D = 600\text{rpm}}$$

476. A pair of Spur gears consists of a 20 teeth pinion meshing with a 120 teeth gear. The module is 4 mm, what will be the center distance?

(a) 560 mm (b) 140 mm
(c) 700 mm (d) 280 mm

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : As we know,

Centre distance $(a) = \frac{D_p + D_g}{2}$

D_p = Diameter of pinion
D_g = Diameter of gear
$D = mZ \Rightarrow m$ = module, Z = No. of teeth.

$$a = \frac{m\left[Z_p + Z_g\right]}{2} = \frac{4\left[20 + 120\right]}{2}$$

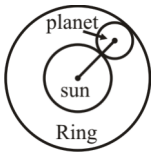
$$a = 280 \text{ mm}$$

477. If the annular wheel of an epicyclic gear train has 200 teeth and the planet wheel has 40 teeth, the number of teeth on the sun wheel is

(a) 160 (b) 120
(c) 80 (d) 40

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) :



As we know,

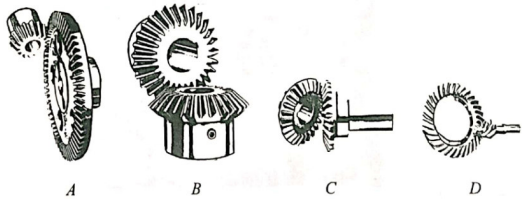
$$R_{ring} = 2\,R_{planet} + R_{sun}$$

$$R \propto T, \quad m = \frac{2R}{T} = \text{const.}$$

So, $T_R = 2 \times T_P + T_s$

$200 = 2 \times 40 + T_s \Rightarrow T_s = 120$ teeth

478. The figure below shows some of the gears used in engineering practices. Which combination of gearing is correct in order of A, B, C, D?



(a) Bevel gears (A), Miter gears (B), Angular bevel gears (C), Hypoid gears (D)
(b) Miter gears (B), Angular bevel gears (C), Hypoid gears (D), Bevel gears (A)
(c) Angular bevel gears (C), Hypoid gears (D), Bevel gear (A), Miter gear (B)
(d) Hypoid gears (D), Angular bevel gears (C), Miter gears (B), Bevel gears (A).

CGPSC AE 15.01.2021

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Ans. (a) : Bevel gears (A), Miter gears (B), Angular bevel gears (C), Hypoid gears (D).

179. The gears are termed as medium velocity gears, if their peripheral velocity is :

(a) $1-3\,m/s$ (b) $30-50\,m/s$

(c) $15-30\,m/s$ (d) $3-15\,m/s$

CGPSC AE 15.01.2021

Ans. (d) : The gears are termed as medium velocity gears, if their peripheral velocity is 3-15 m/s.

480. The following data is given, module of the gear is 4 mm, number of teeth in gear and pinion is 24 and 12 respectively, what is the centre distance?

(a) 144 mm (b) 72 mm

(c) 18 mm (d) 36 mm

VIZAG Steel MT 24.01.2021

Ans. (b) : Given,

$m = 4$ mm

$T_g = 24$

$T_p = 12$

Centre distance $(R) = R_g + R_p$

$$= \frac{mT_g}{2} + \frac{mT_p}{2}$$

$$= m\left[\frac{24}{2} + \frac{12}{2}\right]$$

$$= 4[12 + 6] = 4 \times 18$$

$$= 72 \text{ mm}$$

481. For the given statements:

I. Mating spur gear teeth is an example of higher pair.

II. A revolute joint is an example of lower pair.

Indicate the correct answer.

(a) Both I and II are false

(b) Both I and II are true

(c) I is false and II is true

(d) I is true and II is false

Assam Engg. College AP/Lect. 18.01.2021, GATE 2014

Ans. (b) : Mating gear teeth is an example of higher pair because their is a point of contact.

Revolute joint is an example of lower pair because their is surface or area contact.

482. A spur gear with 20° full depth teeth is transmitting 20 kW at 200 rad/s. The pitch circle diameter of the gear is 100 mm. The magnitude of the force applied on the gear in the radial direction is_____.

(a) 1.39 kN (b) 2.78 kN

(c) 0.73 kN (d) 0.36 kN

Assam Engg. College AP/Lect. 18.01.2021

Ans. (c) : Given,

Power (P) = 20 kW

$\omega = 200$ rad/sec

D = 100 mm

Power (P) = 20 kW

$T \times \omega = 20 \times 10^3$

$T \times 200 = 20 \times 10^3$

$T = 100$ Nm

$$F_t \times \frac{D}{2} = 100$$

$$F_t = \frac{100 \times 2}{0.1} = 2000\,N$$

Tangential force

$F_t = F_n \cos\phi = 2000$ N

$$F_n = \frac{2000}{\cos 20^\circ} = 2128.35\,N$$

Radial force

$F_r = F_n \sin 20^\circ = 2128.35 \times \sin 20^\circ = 727.94$ N

$= 0.73$ kN

483. In a gear drive, the ratio of teeth of meshing gears is:

(a) inversely proportional to the product of their pitch diameters.

(b) directly proportional to the product of their pitch diameters.

(c) inversely proportional to the ratio of their pitch diameters.

(d) directly proportional to the ratio of their pitch diameters.

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : In a gear drive, the ratio of teeth of meshing gear is directly proportional to the ratio of their pitch diameters.

484. A compound gear train has pinions of 25 and 40 teeth and gears of 75 teeth and 80 teeth. The overall gear ratio is

(a) 6 (b) 3.2

(c) 1.75 (d) 1.5

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Overall gear ratio $= \frac{80}{40} \times \frac{75}{25} = 6$

485. When the bevel gears connect two shafts whose axes intersect at an angle greater than a right angle and one of the bevel gears has a pitch angle as right angle, then it is known as _____.

(a) mitre gear (b) crown bevel gear

(c) internal bevel gear (d) angular bevel gears

VIZAG MT, 14.12.2020

Ans. (b) : Crown Bevel Gear–When the bevel gears connect two shaft whose axes intersect at an angle greater than a right angle and one of the bevel gears has a pitch angle as right angle, then it is known as crown bevel gear.

486. The radial dimension between the pitch circle and the bottom of a tooth space is known as -

(a) Dedendum (b) Diametral pitch

(c) Addendum (d) Circular pitch

MECON MT 2019

Ans. (a) : The radial dimension between the pitch circle and the bottom of a tooth space is known as dedendum.

487. Two parallel shafts at a distance D apart are to be connected by spur gears (gear and pinion). Which equation represents the correct relationship between centre distance D, diameters of gear (d_1) and pinion (d_2)?

(a) $D = d_1 + d_2$ (b) $D = 2(d_1 + d_2)$

(c) $D = (d_1 + d_2)/2$ (d) $D = d_1 - d_2$

APPSC Poly Lect. Automobile Engg., 2020

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Ans. (c) : Center Distance–It is the distance between the center of rotation of the gears in mesh,

$$C = R_1 + R_2 = \frac{D_1 + D_2}{2} = m\left(\frac{T_1 + T_2}{2}\right)$$

488. Which of the following system is able to take heavy loads?
(a) $14\frac{1}{2}°$ full depth involute system
(b) 18° stub involute system
(c) $14\frac{1}{2}°$ composite system
(d) 20° stub involute system

Oil India Senior Officer 23.12.2020

Ans. (d) :

489. The velocity of sliding _____ the distance of the point of contact from the pitch point.
(a) is directly proportional to
(b) is inversely proportional to
(c) is equal to cosθ multiplied by
(d) does not depend upon

TNCSC AE 2020

Ans. (a) : The velocity is directly proportional to the distance of the point of contact from the pitch point.

$\boxed{V_{sliding} = (\omega_1 + \omega_2)QP}$

Where, ω_1 = angular velocity of gear
ω_2 = angular velocity of pinion
QP = distance of the point of contact from the pitch point.

490. The circular pitch of spur gears is defined as the ratio of
(a) The circumference of the pitch circle to the number of teeth
(b) The circumference of the pitch circle to the diameter of the pitch circle
(c) The diameter of pitch circle to the number of teeth
(d) The number of the teeth to the diameter of pitch circle

GPSC Engineer, Class-II Pre-19.01.2020
RPSC IOF, 2020

Ans. (a) : The circular pitch of spur gears is defined as the ratio of the circumference of the pitch circle to the number of teeth.

491. In case of cycloidal tooth profile gears in mesh
(a) The pressure angle is always constant throughout the contact of mating teeth
(b) The path of contact is a straight line
(c) The variation in centre distance affects the angular velocity ratio
(d) There is interference in these gears.

RPSC IOF, 2020

Ans. (c) : A cycloid is the locus of a point on the circumference of a circle that rolls without slipping on the flat surface in case of cycloidal teeth the pressure angle varies from the maximum at the beginning of engagement to zero when the point of contact coincides with pitch point P and then again increases to maximum in the reverse direction.

492. A spur gear has a module of 3.5 mm, length of path of contact of $11\sqrt{3}$ mm and pressure angle 30^0. The contact ratio of the gear will be :
(a) 1.7 (b) 3.0
(c) 2.0 (d) 1.0

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given data : Module (m) = 3.5 mm
Length of path of contact = $11\sqrt{3}$ mm
Pressure angle (ϕ) = 30^0
Contact Ratio (C.R.) = ?

$$\text{Contact ratio (C.R.)} = \frac{\text{Arc of contact}}{\text{Circular pitch}}$$

$\because$ Circular Pitch = $\frac{\pi D}{T} = \pi m$ {Where; m = module}
$= 3.5\ \pi$

$\because$ Arc of contact = $\frac{\text{Path of contact}}{\cos\phi} = \frac{11\sqrt{3}}{\cos 30^0} = 22$ mm

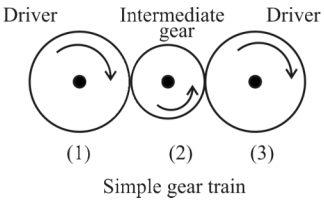
$\therefore$ C.R. = $\frac{22}{3.5\pi} = \frac{22}{3.5 \times 22/7} = \frac{7}{3.5} = \frac{10}{5} = 2$

$\Rightarrow$ $\boxed{\text{C.R.} = 2.0}$

493. In a simple gear train, 1st gear (Driver) has 40 number of teeth, 2nd gear gas has 20 number of teeth and last gear has 80 number of teeth. The train value of the gear train is
(a) 0.5 (b) 1
(c) 2 (d) 1.2

APPSC Poly Lect. 13.03.2020

Ans. (a) :



Simple gear train

Given,
Teeth, $T_1 = 40$, $T_2 = 20$, $T_3 = 80$

$\therefore$ Train value = $\frac{T_1}{T_2} \times \frac{T_2}{T_3} = \frac{T_1}{T_3} = \frac{40}{80}$

Train value = $\frac{1}{2} = 0.5$

494. In gears, angle of obliquity is the angle :
(a) between the direction of the force applied by the driving gear and the direction of the resultant reaction by driven gear.
(b) between the common normal to two gear teeth at the point of contact and the common tangent at the pitch point.
(c) between two shafts on which meshing gears are mounted.
(d) between the direction of the force applied by the driving gear and the direction of the friction force.

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : In gears, angle of obliquity is the angle between the common normal to two gear teeth at the point of contact and the common tangent at the pitch point. It is usually denoted by □ .

The standard pressure angles are $14\frac{1}{2}°$ and 20°

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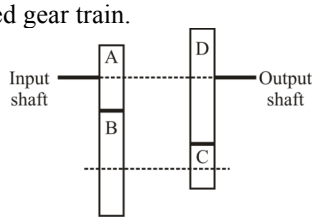
495. In lathe back gears, the following gear train is used :

(a) compound gear train
(b) reverted gear train
(c) simple gear train
(d) epicyclic gear train

NLCIL GET 17.11.2020, Shift-II

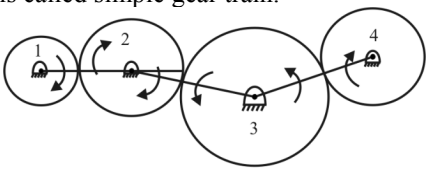
Ans. (b) : Back gear is a gear mounted at the back of the headstock that allows the chuck to rotate slowly with greatly increased turning power. By engaging back gear and so reducing rpm but increasing torque even the largest face plate mounted jobs can be turned successfully.

Reverted gear- It is such a arrangement is used in clock and in simple lathe where back gear is used to given slow speed to the chuck in this gear train if the axis of the first and last wheels of a compound gear coincide is called reverted gear train.



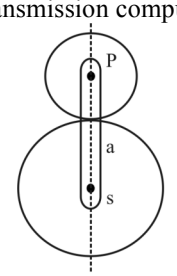
Reverted gear

Simple gear train- A series of gears capable of receiving and transmitting motion form one gear to another is called simple gear train.



Simple gear train

Epicyclic gear train- Important application of epicyclic gear train are in transmission computing devices.



Epicyclic gear train

496. A single reduction gear of 100 kW with a pinion 200 mm pitch circle diameter and angular speed 50 rad/s is supported in bearings on either side. The tangential load on the pinion is _____.

(a) 15 kN (b) 20 kN
(c) 5 kN (d) 10 kN

VIZAG MT, 14.12.2020

Ans. (b) : P = 100 kW
d = 200 mm
ω = 50 rad/sec
$P = T \times \omega$
$100 \times 10^3 = T \times 50$
$\boxed{T = 2000\,N}$
$T = F_t \times r$
$F_t \times r = 2000$
$F_t \times \left[\frac{d}{2}\right] = 2000$
$F_t \times \frac{0.2}{2} = 2000$
$F_t = 20{,}000$ N
$\boxed{F_t = 20\,kN}$

497. The number of teeth on each of the two spur gears in mesh are 40. The teeth have 20° profile and the module is 6 mm. If the arc of contact is 1.5 times the circular pitch, then its value is _____.

(a) 28.26 mm (b) 42.39 mm
(c) 56.52 mm (d) 14.13 mm

VIZAG MT, 14.12.2020

Ans. (a) : No. of teeth (T) = 40
$\phi = 20°$
m = 6 mm
$\text{Arc of contact} = \frac{\text{Path of contact}}{\cos\phi}$
Circular pitch $P_c = m\pi = 6 \times \pi = 6\pi$
Arc of contact = $1.5\,P_c = 1.5 \times 6\pi$
= 28.26 mm

498. The minimum number of teeth on a pinion in order to avoid interference for a 20 degree stub involute system is _____.

(a) 12 (b) 14
(c) 18 (d) 32

VIZAG MT, 14.12.2020

Ans. (b) : The minimum number of teeth on a pinion in order to avoid interference for a 20° stub involute system.

$$T_P = \frac{2A_P}{\left[\sqrt{1+G(G+2)\sin^2\phi} - 1\right]}$$

$$T_P = \frac{2}{\sqrt{1+1\times 3\sin^2 20° - 1}}$$

$\boxed{T_P = 12.32}$ ≈ next higher value 13

So answer will be 14.

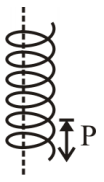
499. The distance between the corresponding points on adjacent teeth measured on the pitch circle is called

(a) helical pitch (b) normal pitch
(c) gear pitch (d) circular pitch

Assam PSC AE (PHED), 18.10.2020

Ans. (d) : Circular pitch- The distance between the corresponding point on adjacent teeth measured on the pitch circle is called circular pitch.

Helical Pitch- The pitch of a helix is the height of one complete helix turn measured parallel to the axis of helix.



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Normal pitch- It is the distance between point of intersection of the line of action of gear teeth with the working faces of two adjacent.
Gear pitch- The pitch of gear is defined as the distance between two identical points on two adjacent gear teeth.

500. The angle at which teeth of the gear are inclined to the axis of a gear is called as
(a) pitch angle (b) normal angle
(c) helix angle (d) gear angle
Assam PSC AE (PHED), 18.10.2020

Ans. (c) : Helix angle- The angle at which teeth of the gear are inclined to the axis of a gear is called as helix angle.
Pitch angle- Pitch angle is the angle between an element of a pitch cone and its axis.

501. The manufacturing of gear tooth with generating process leads to
(a) Undercutting of gear tooth
(b) Interface of gear tooth
(c) Non-conjugate tooth profiles
(d) None of the above
Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Generating process (combination of straight moment of tool and rotation of work piece by spindle) is method by gear cutting.

502. For spur gears with involute profile teeth, when the centre distance between the gear centres is increased
(a) Pressure angle decrease
(b) Pitch circle diameter remains unchanged
(c) Pressure angle increase
(d) None of the above
Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : As the centre distance increases, the radius of pitch circle increases and pressure angle increases.

503. The spiral bevel gears can be considered as
(a) Bevel counterparts of helical gear
(b) Spur counterparts of helical gear
(c) Bevel counterparts of worm gear
(d) Spur counterparts of worm
Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : The spiral bevel gears can be considered as bevel counterparts of helical gear.

504. The Lewis equation for gear design is (w_t– tangential) forces, P_c-circular pitch, p_d-diametral pitch, F-face width and Y-Lewis form factor, A - face area)
(a) $\sigma_b = \frac{FY}{w_t p_d}$ (b) $\sigma_b = \frac{w_t}{AY}$
(c) $\sigma_b = \frac{w_t P_c}{FY}$ (d) $\sigma_b = \frac{w_t P_d}{F.Y}$
Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : $w_t = F.m.Y.\sigma_b$
Where, m = module $= \frac{D}{T}$
Diametrical pitch $p_d = \frac{T}{D}$
$w_t = \frac{F.Y.\sigma_b}{p_d}$ $\boxed{\sigma_b = \frac{w_t.p_d}{F.Y}}$

505. Tooth profile of gear which can accommodate slight variation in center distance is
(a) cycloid (b) epi-cycloid
(c) involute (d) circular
TSPSC Manager (Engg.) HMWSSB 12.11.2020

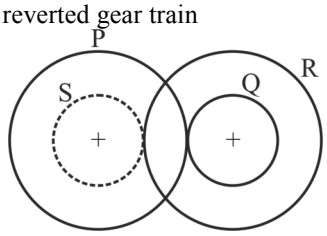
Ans. (c) : In involute profile center distance can be slightly varied but in cycloidal profile exact center distance are required.

506. A pinion rotates at 1200 rpm and transmits a torque of 20 Nm. This pinion of 40 teeth mesh with the other spur gear having 60 teeth. The torque transmitted by the gear is:
(a) 15 Nm (b) 20 Nm
(c) 40 Nm (d) 30 Nm
CIL MT 27.02.2020

Ans. (d) : In gear transmission power is always constant. If we neglect friction on teeth and bearing
Power P = Torque × Angular velocity
$T_1\omega_1 = T_2\omega_2$
$T_1N_1 = T_2N_2$ $\frac{N_2}{N_1} = \frac{\text{Teeth on pinion}}{\text{Teeth on gear}}$
$T_2 = \frac{20\times1200}{800}$ $N_2 = \frac{1200\times40}{60}$
$T_2 = 30$ Nm $N_2 = 800$ rpm

507. In a reverted gear train with four gears, two gears P and Q are meshing, Q – R is a compound gear, while R and S are meshing. The modules of P and R are 6 mm and 5 mm respectively. The numbers of teeth in P, Q and R are 24, 36 and 40 respectively. The number of teeth in gear S is:
(a) 42 (b) 28
(c) 38 (d) 32
CIL MT 27.02.2020

Ans. (d) : A reverted gear train



Formula,
$m(T_1 + T_2) = m'(T_3 + T_4)$
$6(24 + 36) = 5(40 + T_4)$
$160 = 5T_4$
$T_4 = 32$ Teeth

508. A gear train in which each shaft carries more than one gear except input and output shafts is known as:
(a) Compound (b) Reverted
(c) Planetary (d) Simple
Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : A gear train is combination of gears used to transmit motion from one shaft to another. It becomes necessary when it is required to obtain large speed reduction within a small space.
Types of gear train are :
1. Simple gear train
2. Compounded gear train

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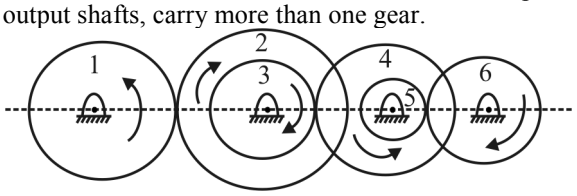
3. Reverted gear train
4. Planetary gear train.
In compounded gear train a series of gear are connected in such a way that two or more gears rotates about an axis with the same angular velocity. It also carries some of the intermediate shafts i.e. other than the input and output shafts, carry more than one gear.



$$\text{Train value}\left(\frac{N_6}{N_1}\right)=\frac{\text{Product of Number of teeth on driving gears}\,(T_1\times T_3\times T_5)}{\text{Product of Number of teeth on driven gears}\,(T_2\times T_4\times T_6)}$$

509. Which of the following is the correct relation for gear?
(a) Circular pitch = π × Module
(b) Circular pitch = 2 × Module
(c) Circular pitch = 2π × Module
(d) Circular pitch = Module

SJVN ET 2019

Ans. (a) : Circular pitch = $\frac{\pi D}{T}$

Module = $\frac{D}{T}$

Circular pitch = π × Module

510. Face of a tooth in spur gear is :
(a) The surface between pitch circle and top land
(b) The surface between top land and bottom land
(c) The surface between pitch circle and bottom land
(d) The surface of the top of the tooth

SJVN ET 2019

Ans. (a) : The surface between pitch circle and top land is known as face of a tooth in spur gear.

511. In meshing gears with involute gears teeth, the contact begins at the intersection of the :
(a) Line of action and the addendum circle of the driven gear
(b) Line of action and the pitch circle of the driven gear
(c) Dedendum circle of the driver gear and the addendum circle of the driven gear
(d) Addendum circle of the driver gear and the pitch circle of the driven gear

OPSC AEE 2019 Paper-I, ESE 2007

Ans : (a) : In meshing gear with involute gear teeth, the contact begins at the intersection of the line of action and the addendum circle of the driven gear. The contact point moves along the line of action throughout the engagement.

512. Match the following :

Type of gears	**Arrangement of Shafts**
(P) Bevel gears	(i) Non-parallel offset shafts
(Q) Worm gears	(ii) Non-parallel intersecting shafts
(R) Herringbone gears	(iii) None-parallel non-intersecting shafts
(S) Hypoid gear	(iv) offset of the shaft

(a) P–IV,Q–II, R-I, S–II
(b) P–II,Q–III, R-IV, S–I
(c) P–III,Q–II, R-I, S–IV
(d) P–I,Q–III, R-IV, S–II

OPSC AEE 2019 Paper-I, GATE 2004

Ans : (b) :
(P) Bevel gears – Non-parallel intersecting shafts
(Q) Worm gears- Non-parallel non-intersecting
(R) Herringbone- Herringbone gear are used to transmit power between two parallel shafts.
(S) Hypoid gear- Advantage of hypoid gear is that the offset of the shaft is so great that the shafts may continue past each other. Therefore, multiple power take-offs from a single shaft with several pinion is possible.

513. Match the type of gears with their most appropriate description :

Type of gear	**Description**
(P) Helical	**(i) Axes non-parallel and intersecting**
(Q) Spiral	**(ii) Axes parallel and teeth are inclined to the axis**
(R) Hypoid	**(iii) Axes parallel and teeth are parallel to the axis**
(S) Rack and pinion	**(iv) Axes are perpendicular and intersecting and teeth are inclined to the axis**
	(v) Axes are perpendicular and used for large speed reduction
	(vi) Axes parallel and one of the gears has infinite radius

(a) P– ii, Q–iv, R-i, S–vi
(b) P– ii, Q–vi, R-iv, S–ii
(c) P– i, Q–iv, R-v, S–vi
(d) P– vi, Q–iii, R-i, S–v

OPSC AEE 2019 Paper-I, GATE 2008

Ans : (a) : Helical- Axes parallel and teeth are inclined to the axis.
Spiral- Axes are perpendicular and intersecting and teeth are inclined to the axis.
Hypoid- Axes non parallel and non-intersecting.
Rack and pinion- Axes parallel and one of the gears has infinite radius.

514. Automatic transmission in automobile is based on
(a) Epicyclic gear train
(b) Compound gear train
(c) Simple gear train
(d) Inverted gear train

Gujarat PSC AE 2019

Ans : (a) : Automatic transmission in automobile is based on epicyclic gear train.

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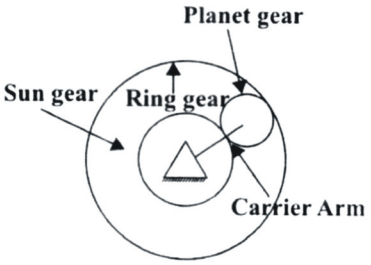
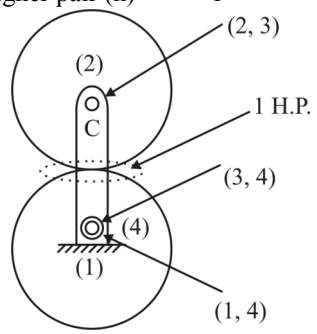
515. The number of degrees of freedom of the planetary gear train shown in the figure is



(a) -1 (b) 0
(c) 1 (d) 2

Gujarat PSC AE 2019

Ans : (d) : Number of links (l) = 4
Number of binary joints (j) = 3
Number of higher pair (h) = 1



Now, from Grubler's equation, Degree of freedom is given by

$$F = 3(l-1) - 2j - h$$
$$= 3(4-1) - 2 \times 3 - 1$$
$$= 2$$

516. Law of Gearing is about

(a) Constant torque between meshed gears
(b) Constant force between meshed gears
(c) Constant moment between meshed gears
(d) Constant velocity between meshed gears

Gujarat PSC AE 2019

Ans : (d) : Law of gearing- According to the law of gearing, the common normal at the point of contact between a pair of teeth must always pass through the pitch point.

517. Automobile gears are generally manufactured by

(a) Hobbing (b) Stamping
(c) Punching (d) Rolling

Gujarat PSC AE 2019

Ans : (a) : Gear hobbing- It is a machining process in which gear teeth are progressively generated by a series of cuts with a helical cutting tool. Gear hobbing is generative type of process in gear manufacturing.

518. For the two gears meshing in velocity ratio equal to 1/3, module = 4 mm; centre distance = 200 mm. The number of teeth on the gear wheel are

(a) 150 (b) 125
(c) 100 (d) 75

Gujarat PSC AE 2019

Ans : (d) : Velocity ratio $= \frac{1}{3} = \frac{T_1}{T_2}$

$$T_2 = 3T_1$$

Module (m) = 4 mm

$$\text{Centre distance } (C) = m\left(\frac{T_1 + T_2}{2}\right) = 200$$

$$T_2 = 3T_1$$

$$4\left(\frac{T_1 + 3T_1}{2}\right) = 200$$

$$4T_1 = 100$$

$$T_1 = 25$$

$$T_2 = 3 \times 25$$

$$\boxed{T_2 = 75}$$

519. The centre distance C between two gears, in terms of base circle radii R_1, R_2 and the pressure angle ϕ, is

(a) $\frac{\cos\phi}{R_1 + R_2}$ (b) $\frac{R_1 + R_2}{\cos\phi}$
(c) $\left(\frac{R_1}{R_2}\right)\cdot\cos\phi$ (d) $\left(\frac{R_2}{R_1}\right)\cdot\cos\phi$

ESE 2019

Ans. (b) : The centre distance between two gears
C = sum of pitch circle radii
$= r_1 + r_2$

$$= \frac{R_1}{\cos\phi} + \frac{R_2}{\cos\phi} = \frac{R_1 + R_2}{\cos\phi}$$

where r_1 and r_2 are pitch circle radius.

520. Which one of the following is not the correct statement with respect to the involute profile toothed gears in mesh?

(a) Pressure angle remains constant from the start till the end of the engagement.
(b) The base circle diameter and the pitch circle diameter of the two mating involutes are proportional.
(c) When two involutes are in mesh, the angular velocity ratio is proportional to the size of the base circles.
(d) The shape of the involute profile depends only on the dimensions of the base circle.

ESE 2019

Ans. (c) : Important points for involute profile toothed Gears in mesh.

(i) The pressure angle, from the start of the engagement of the teeth to the end of the engagement, remains constant in involute gears.
(ii) The base circle diameters and pitch diameters of two mating involutes are directly proportional.
(iii) When two involute are in mesh, the angular velocity ratio is inversely proportional to the size of the base circles.
(iv) The shape of the involute profile depends only on the dimensions of the base circle.

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521. For the follower with stroke S, following the cycloidal motion, the radius of the rolling circle will be

(a) $S \times 2\pi$ (b) $\frac{S}{2\pi}$

(c) $\frac{2\pi}{S}$ (d) $S + 2\pi$

ESE 2019

Ans. (b) : For cycloidal motion the circumference of the rolling circle is equal to stroke i.e. follower displacement.

Hence $2\pi r = s$

Radius of rolling circle

$$r = \frac{s}{2\pi}$$

522. The number of teeth of a spur gear is 30 and it rotates at 200 rpm. What will be its pitch line velocity if it has a module of (6/π) mm?

(a) 600 mm/s (b) 300 mm/s

(c) 191 mm/s (d) 164 mm/s

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (a) : Given, No of teeth (T) = 30

N = 200 rpm

Module (m) = $6/\pi$ mm $= \frac{6}{\pi}$ mm

$$m = \frac{D}{T} = \frac{6}{\pi}$$

$$D = \frac{6}{\pi} \times T = \frac{6}{\pi} \times 30 = 57.29mm$$

Pitch line velocity $v = \frac{\pi DN}{60} = \frac{\pi \times 57.20 \times 200}{60}$

= 600 mm/sec.

523. Operating pressure angle for involute gear teeth is :

(a) Maximum at engagement of teeth

(b) Minimum at engagement of teeth

(c) Same at all points of contact

(d) Vary uniformaly at all points of contact

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : Same at all points of contact.–

In involute profile gear tooth pressure angle remains constant.

524. A simple gear train consists of gear A and B, having module 2 mm and centre distance of shafts equal to 115 mm. If the pitch circle diameter of driver (gear A) is 46 mm, the train value of the gear train is:

(a) 1/2 (b) 1/3

(c) 1/4 (d) 1/2.5

HPPSC ADF 05.03.2019

Ans. (c) : Given,

m = 2

Centre of distance, $R_A + R_B = 115$ mm

$D_A = 46$

$m\,T_A = 46$

$$\boxed{T_A = \frac{46}{2} = 23}$$

Centre of distance, $R_A + R_B = 115$ mm

$$\frac{m}{2}(T_A + T_B) = 115$$

$$\frac{2}{2}(23 + T_B) = 115$$

$$\boxed{T_B = 92}$$

Train value $\frac{T_A}{T_B} = \frac{23}{92}$

$$\boxed{\text{Train value} = \frac{1}{4}}$$

525. Which of the following gears are used between parallel shafts?

(a) Straight Bevel gears (b) Spiral Bevel gears

(c) Worm & worm gears (d) Herringbone gears

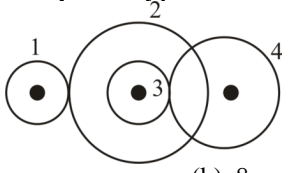
(e) Hypoid gears

CGPSC AE 25.02.2018

Ans. (d) : Herringbone Gear– Herringbone gears, also called double helical gears, are gear sets designed to transmit power through parallel shafts.

Herringbone gear are mostly used on heavy machinery. These gears overcome the problem of axial thrust presented by single helical gear, by having two sets of teeth that are set in 'V' shape. So axial thrust in case of these gear is zero. Double helical gear most difficult to manufacture.

526. A compound gear train is shown in figure. The power is transmitted from the shaft of gear 1 to the shaft of gear 4 with intermediate shaft having gear 2 and gear 3. The speed ratio will be: [Number of teeth on gears 1, 2, 3, 4 are 10, 80, 30, 60 respectively].



(a) 4 (b) 8

(c) 12 (d) 16

(e) 20

CGPSC AE 25.02.2018

Ans. (d) : Speed ratio of compound train

$$\frac{N_1}{N_4} = \frac{T_2 \times T_4}{T_1 \times T_3} = \frac{80 \times 60}{10 \times 30} = 16$$

527. Circular pitch is equal to [D = pitch diameter, T = number of teeth]

(a) Addendum + dedendum

(b) Addendum – dedendum

(c) Tooth thickness × width of space

(d) Tooth thickness – width of space

(e) None of these

CGPSC AE 25.02.2018

Ans. (e) : Circular pitch of gear- The distance between corresponding points of consecutive gear teeth measured along the pitch circle.

It is denoted by P.

$$P = \frac{\pi D}{T}$$

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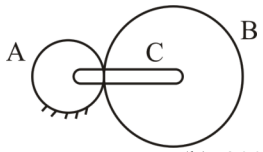
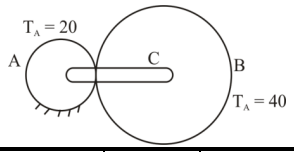
528. Ref. the figure gear B rotates at 150 rpm about its own axis. The arm C will rotate by $T_A = 20$; $T_B = 40$.



(a) 150 rpm (b) 200 rpm
(c) 100 rpm (d) 250 rpm

TNPSC AE 2018

Ans. (c) :



Motion	Arm	A	B
Arm fixed let A rotated + x rpm [Clock wise]	0	+ x	$-x \times \frac{20}{40}$
Arm free	y	y + x	$y - \frac{x}{2}$

Then
Gear A is fixed

$$y + x = 0$$
$$y = -x \quad ...(1)$$

Gear B rotates at 150 rpm

$$y - \frac{x}{2} = 150$$
$$y + \frac{y}{2} = 150$$
$$\frac{3y}{2} = 150$$
$$\boxed{y = 100 \text{ rpm}}$$

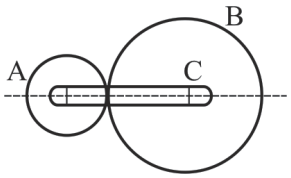
529. The interference between a given pinion tooth and a gear tooth can be avoided by using
1. smaller pressure angle
2. larger pressure angle
3. less number of teeth on the gear for a pinion with predefined number of teeth
4. more number of teeth on the gear for a pinion with predefined number of teeth

Which of the above statements are correct?
(a) 1 and 4 (b) 1 and 3
(c) 2 and 4 (d) 2 and 3

ESE 2018

Ans. (c) : (i) To avoid interference pressure angle should be large as it results in large centre distance.
(ii) During meshing interference takes place when addendum of gear digs into portion between base and root circle of pinion. To avoid it we define the minimum number of teeth on pinion. Number of teeth on gear should be more as it results in reduction of size of gear teeth.

530. A gear train is as shown in the figure below, in which gears A and B have 20 and 40 teeth, respectively. If arm C is fixed and gear A rotates at 100 rpm, the speed of gear B will be



(a) 90 rpm (b) 75 rpm
(c) 50 rpm (d) 20 rpm

ESE 2018

Ans. (c) : $\frac{N_B}{N_A} = \frac{T_A}{T_B}$

$$\frac{N_B}{100} = \frac{20}{40}$$
$$N_B = 50 \text{ rpm}$$

531. Consider the following statements:
1. In spur gears, the contact occurs abruptly on a line parallel to the axis, and the disengagement too is abrupt.
2. In helical gears, both loading and unloading are gradual, and therefore, these happen more smoothly and less noisily.
3. When two gears mesh, any arbitrary shape of the tooth can be chosen for the profile of the teeth of any one of the two gears, and the profile for the other shall be obtained by applying the law of gearing.

Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) :
(i) In spur gears, the contact takes place on a line parallel to the axis abruptly and disengagement is also abrupt.
(ii) In helical gears, the loading and unloading is gradual and therefore, the action is more smooth and less noisy.
(iii) It has been proved that any arbitrary shape of the tooth can be chosen for profile of teeth of one of the two gears in mesh and the profile for the other may be determined to satisfy the law of gearing.

532. Large speed reductions (greater than 20) in one stage of a gear train are possible through
(a) Helical gearing (b) Bevel gearing
(c) Spur gearing (d) Worm gearing

GPSC Executive Engineer 23.12.2018, GATE 2002

Ans. (d) : Worm gears- It is a special case of a spiral gear in which the larger wheel usually has a hollow or concave shape.
- The smaller of the two wheels is called the worm which also has a large spiral angle.
- The sliding velocity of a worm gear is higher as compared to other types of gears. Velocity ratio is very high (50 : 1 to 100 : 1).

533. A gear box using epicyclic gear train is used to supply 100 kW power to a machine. The input speed of supply gear shaft is 100 rad/s. The speed of output shaft of a gear box is 10 rad/s in opposite sense of supply speed. The holding torque on the fixed gear of the train is:

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(a) 5 kNm (b) 25 kNm
(c) 9 kNm (d) 18 kNm

AAI Jr. Executive 29.11.2018

Ans. (c) : Holding torque,

$$T_3 = T_1\left[\frac{\omega_1}{\omega_2} - 1\right]$$

$$P = T_1 \times \omega$$

$$100 \times 10^3 = T_1 \times 100$$

$$T_1 = 1000 \text{ N-m}$$

$$T_3 = 1000\left[\frac{100}{10} - 1\right]$$

$$= 1000 \times 9 = 9 \text{ kN-m}$$

$$T_3 = 9 \text{ kN-m}$$

534. The velocity ratio in the case of compound train of wheels is equal to-

(a) $\frac{\text{Number of teeth on first driver}}{\text{Number of teeth on last follower}}$

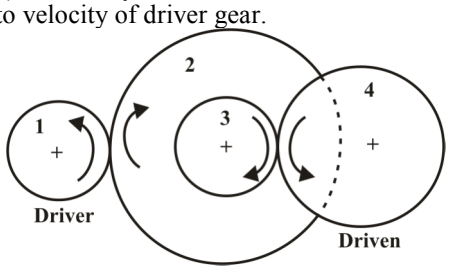
(b) $\frac{\text{Number of teeth on last follower}}{\text{Number of teeth on first driver}}$

(c) $\frac{\text{Product of teeth on the driver}}{\text{Product of teeth on the followers}}$

(d) $\frac{\text{Product of teeth on the followers}}{\text{Product of teeth on the driver}}$

RPSC AE 2018
ESE-2000

Ans. (d) : Velocity Ratio—It is the ratio of velocity of driver to velocity of driver gear.



$$\frac{N_1}{N_4} = \frac{T_2 \times T_4}{T_1 \times T_3}$$

535. The maximum efficiency of worm and worm wheel is : [Coefficient of friction = μ]

(a) $\frac{1}{1+\sin\left(\tan^{-1}\mu\right)}$ (b) $\frac{1-\sin\left(\tan^{-1}\mu\right)}{1+\sin\left(\tan^{-1}\mu\right)}$

(c) $\frac{1}{1+\tan\left(\sin^{-1}\mu\right)}$ (d) $\frac{1}{1+\mu}$

(e) $1-\sin\left(\tan^{-1}\mu\right)$

CGPSC AE 25.02.2018

Ans. (b) : Maximum efficiency of worm and worm wheel is given as

$$[\eta_{max}]_{WW} = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin\left(\tan^{-1}\mu\right)}{1+\sin\left(\tan^{-}\mu\right)}$$

$$\theta = \tan^{-1}\mu$$

536. If the number of teeth on the wheel rotating at 300 r.p.m. is 90, then the number of teeth on the mating pinion rotating at 1500 r.p.m. is

(a) 15 (b) 18
(c) 20 (d) 60

RPSC Vice Principal ITI 2018

Ans. (b) :

$$\frac{T_A}{T_B} = \frac{N_B}{N_A}$$

$$\frac{90}{T_B} = \frac{1500}{300}$$

$$\boxed{T_B = 18}$$

537. The circle passing through the top of the teeth of a gear is known as

(a) Pitch circle (b) Addendum circle
(c) Dedendum circle (d) Base circle

HPPSC AE 2018

Ans. (b) : The circle passing through the top of the teeth of a gear is known as Addendum circle.

538. In a simple gear train the wheel has 150 teeth and the pinion has 50 teeth. What is the speed of pinion if the wheel rotates at 120 rpm?

(a) 300 rpm (b) 330 rpm
(c) 360 rpm (d) 375 rpm
(e) 4 00 rpm

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : Given,

$Z_1 = 150$, $N_1 = 120$ rpm

$Z_2 = 50$ $N_2 = ?$

From Gear Ratio

$$\frac{Z_1}{Z_2} = \frac{N_2}{N_1}$$

$$\frac{\cancel{150}}{\cancel{50}} = \frac{N_2}{120}$$

$$N_2 = 360\text{rpm}$$

539. A gear has 40 teeth of module 5 mm. Find the pitch circle diameter.

(a) 40 mm (b) 100 mm
(c) 140 mm (d) 200 mm
(e) 240 mm

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Given,

Teeth $(\tau) = 40$

Module (M) = 5

$$\text{Module} = \frac{D}{T}$$

$D = 5 \times 40$

$D = 200$ mm

540. Which is the correct relationship of gears?

(a) Module = pitch circle diameter/number of teeth
(b) root circle diameter = pitch circle diameter × module
(c) pitch circle diameter = number of teeth/ module
(d) circular pitch × π = diametral pitch

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(e) Circular pitch = pitch circle diameter × number of teeth

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : $\text{Module} = \frac{D}{T}$

$$\text{Module} = \frac{\text{Pitch circle Diameter (D)}}{\text{Number of Teeth (T)}}$$

541. It is required to drive a shaft at 720 rpm by means of a belt from a parallel shaft with a pulley of 300 mm diameter on it and running at 240 rpm. What is the diameter of the pulley required on the driven shaft?

(a) 100 mm (b) 120 mm
(c) 130 mm (d) 145 mm
(e) 150 mm

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Form the Relation

$$\boxed{\frac{D_1}{D_2} = \frac{N_2}{N_1}} \text{ ____(i)}$$

Given, $N_1 = 720$ rpm ; $D_1 = ?$
$N_2 = 240$ rpm ; $D_2 = 300$ mm

From eq^{h} (i)

$$\frac{D_1}{300} = \frac{240}{720}$$

$$\boxed{D_1 = 100\text{mm}}$$

542. When the gear ratio through the transmission is 0.8 : 1, the transmission is in :

(a) overdrive (b) underdrive
(c) direct drive (d) neutral
(e) deep drive

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Overdrive :-It is a transmission gearing that lowers the engine's per minute (RPM) at select time to bring on humorous beneficial effect.
The point of overdrive is fuel efficiency. If the car can move faster while relaxing the engine. The car's efficiency improve additionally, with engine's reduction in workload, the car is consequently less noisy and ride comfort improves.
In overdrive the gearing makes it so that the input shaft is rotating slower than the output shaft and the car is over driving beyond its peak power point.

543. Steering ratio for cars is approximately:

(a) 2-5 (b) 10-20
(c) 30-40 (d) 50-60
(e) 70-80

CGPSC ARTO & TSI 07.01.2017

Ans. (b) :

544. The smallest gears inside the differential casing are:

(a) pinion gears (b) sun gears
(c) side gears (d) ring gears
(e) both, sun gears and side gears

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : The smallest gear inside the differential casing are pinion gears. Differential gear, in automotive mechanics, gear arrangement that transmit power from the engine to be transmitted to a pair of driving wheels, dividing the force equally between them but permitting them to follow path of different lengths as when turning a corner or traversing an uneven read.

545. For small cars, suitable steering gear box is:

(a) rack and pinion (b) worm and wheel
(c) cam and roller (d) worm and nut
(e) cam and worm

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Most car- based vehicles use rack and pinion steering in which the steering wheel turns a pinion gear that moves on a rack to turn the front wheel. others use recalculating ball systems in which ball bearing are mounted on a worm type steering gear.

546. The part of the vehicle by which outer wheel of a vehicle runs faster than inner wheel while taking turn is known as:

(a) Differential (b) Brake
(c) Clutch (d) Engine
(e) Chassis

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Differential Gear :- It is a system that transmits an engine's torque to the wheels. The differential takes the power from the engine and splits it, allowing the wheels to spin at different speed-turn it around a corner and you will have no issues as each wheel is able to turn independently from the other.
Brake :- A brake is a mechanical device that inhibits motion by absorbing energy from a moving system. it is used for slowing or stopping vehicle wheel axle , or to prevent its motion, most often accomplished by means of friction.
Chassis :- It is the main support structure of the vehicle which is also known as frame. It bears all the stresses on the vehicle in both static and dynamic condition.

547. How can interference in involute gears be avoided?

(a) Varying the centre distance by changing the pressure angle only
(b) Using modified involute composite system only
(c) Increasing the addendum of small wheel and reducing it for the larger wheel only
(d) Any of the above

HPPSC Asstt. Prof. 18.09.2017

Ans. (d) : Any of the above

548. The circle passing through the bottom of the teeth of gear is known as:

(a) dedendum circle (b) prime circle
(c) addendum circle (d) base circle

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : Dedendum circle–The circle, passing through the bottom of the teeth of gear.
or
It is a radial distance of a tooth from the pitch circle to the bottom of the tooth.
Addendum circle–It is the circle drawn through the top of teeth and is concentric with the pitch circle.

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Pitch circle–It is an imaginary circle which by pure rolling action, would give the same motion of the actual gear.
Prime circle–It is smallest circle which can be drawn tangent to the locus of the centerline of the follower.
Base circle–Base circle of gear is the base circle for the involute curve. An involute curve is the trace of a point at the end of taut string that unwinds from a cylinder, this cylinder is called base circle.

549. According to law gearing:
(a) dedendum should be equal to 1.157 m
(b) teeth should of cycloidal type
(c) teeth should be of involute type
(d) none of these

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Law of gearing–The law of gearings fulfilled by the goal, tooth profiles to maintain a constant angular velocity ratio between two gears.
2. For constant angular velocity ratio of the two gears the common normal at the point of contact of the two mating teeth must pass through the pitch point.
3. Velocity of sliding = $(w_1 + w_2) \times$ distance between the pitch point and point of contact.
So, answer is none of these.

550. Match list I with list II and select the correct answer from the codes given below:

List I	List II
A. Gears	1. Sprocket
B. Belts	2. Friction
C. Chains	3. Slip
D. Clutch	4. Flank

Code:

	A	B	C	D
(a)	4	3	1	2
(b)	1	2	3	4
(c)	2	3	4	1
(d)	3	4	1	2

PTCUL AE 25.06.2017

Ans. (a) :

551. What is the preferred order of sequence for the following drives from high speed to low speed?
(a) Rack and pinion, spur gears, worm and worm wheel
(b) Worm and worm wheel, spur gears, rack and pinion
(c) Worm and worm wheel, rack and pinion, spur gears
(d) Spur gears, worm and worm wheel, rack and pinion

UPSC JWM 2017

Ans. (c) : The preferred order of sequence for the high speed to low speed drives as following-
Worm and worm wheel
Rack and pinion
Spur gears

552. Design load factor for (i) a helical gear is smaller than that for (ii) a straight tooth spur gear because
(a) Load concentration factor may be same in both but dynamic load factor is smaller in (i)
(b) Both load concentration factor and dynamic load factor are smaller in (i)
(c) Load concentration factor is smaller even though dynamic load factor may be higher in (i)
(d) Load concentration factor is smaller in (i) even as the dynamic load factor is the same for both

UPSC JWM 2017

Ans. (a) :

553. The minimum recommended top land for hardened gears in terms of module m is
(a) 0 m (b) 0.4 m
(c) 1 m (d) 1.2 m

BPSC AE Mains 2017 Paper - VI

Ans : (c) : The minimum recommended top land for hardened gears in terms of module m is 1 m.

554. Pressure angles more than 25° are not used in involute profiles because
1. The gears are difficult to manufacture
2. Minimum number of teeth for under cutting increases enormously
3. Bearing radial load from normal reaction between gears will be very high
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

UPSC JWM 2017

Ans. (c) : Pressure angles more than 25^o are not used in involute profiles because bearing radial load from normal reaction between gears will be very high.

555. Gears with involute profile have become popular in comparison to those with cycloidal profiles because:
1. They are interchangeable
2. They are insensitive to centre-distance variation
3. Involute rack cutter is straight-sided
4. They occupy lesser space

Which of the above statements are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 3 and 4 only (d) 1 and 4 only

UPSC JWM 2017

Ans. (b) : Gear with involute profile have become popular because of
• They are insensitive to centre-distance variation.
• Involutes rack cutter is straight sided.

556. In case of two meshing gears both with involute tooth profiles, contact is made when the tip of the:
(a) Driven wheel touches the flank of a tooth of the driving wheel
(b) Driven wheel touches the face of a tooth of the driving wheel
(c) Driving wheel touches the face of a tooth of the driven wheel
(d) Driving wheel touches the flank of a tooth of the driven wheel

UPSC JWM 2017

Ans. (a) : In case of two meshing gears both with involute tooth profiles, contact is made when the tip of the Driven wheel touches the flank of a tooth of the driving wheel.

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557. The direction of motion of the driven gear of a simple gear train having an odd number of idler gears will be

(a) Same as that of the driving gear
(b) Opposite to that of the driving gear
(c) Dependent upon the number of teeth on the driving gear
(d) Dependent upon the number of teeth on the driven gear

UPSC JWM 2017

Ans. (a) : The direction of motion of the driven gear of a simple gear train having an odd number of idler gears will be same as that of the driving gear.
Note : Even number of gears, driver and follower turn in opposite direction.

558. A spur gear with 32 teeth and module 4 mm is rotating at 400 rpm. The circular pitch is:

(a) 7π mm (b) 6π mm
(c) 5π mm (d) 4π mm

UPSC JWM 2017

Ans. (d) : Circular pitch for spur gear,

$$P_c = \pi m = \pi \frac{D}{T}$$

Module, m = 4 mm

$$P_c = 4\pi$$

559. In a simple epicyclic gear train for transmission of torque. The following component must be held stationary

(a) Sun gear (b) Annular gear
(c) The carrier unit (d) Any one of the above

TNPSC AE 2017

Ans. (d) : In a simple epicyclic gear train for transmission of torque any one, sun gear or annular gear or the carries unit [arm] may be fixed.

560. The problem of interference in an involute pinion, which has to come in external mesh with a gear, can be avoided by

(a) increasing the pressure angle
(b) positive profile correction on pinion and corresponding negative profile correction on gear
(c) machining the pinion teeth through hobbing so that the interfering portion gets undercut
(d) All of the above

BPSC AE Mains 2017 Paper - VI

Ans : (d) : The problem of interference in an involutes pinion, which has to come in external mesh with a gear, can be avoided by,

- Increasing the pressure angle.
- Positive profile correction on pinion and corresponding negative profile correction on gear.
- Machining the pinion teeth through hobbing so that the interfering portion gets undercut.

561. Statement (I): All worm drives (worm and worm wheel) are reversible.
Statement (II): The worm and worm wheel are made of different materials.

ESE 2017

Ans. (d) : All worm drives are not reversible. Only low speed ratio worm drives are reversible while high speed ratio drives are non reversible.
So statement-I s wrong.
The worm wheel in general made from phosphor-bronze alloy, which is costly. The worm is usually made of hardened alloy steel.

562. Consider the following statements:
In the case of gears of involute profiles, increase in the centre to centre distances between the mounting shafts
1. increases the pressure angle
2. will not affect the law of gearing
3. shortens the path of contact
4. increases the contact ratio
Which of the above statements are correct?

(a) 1, 2 and 4 (b) 1, 2 and 3
(c) 1, 3 and 4 (d) 2, 3 and 4

ESE 2017

Ans. (b) : If the distance between centre is increased then pressure angle also increases, path of contact decreases and law of gearing will be unaffected.

563. If the speed of a mechanism is to be reduced in the ratio of 1000:1, which of the following systems would you choose?

(a) Bevel gear system
(b) Spur gear system
(c) Differential gear system
(d) Worm and worm wheel system

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : As, we know for a single start worm, each full turn of worm advances the gear by one tooth. So a gear with 24 teeth will provide a gear reduction of 24:1. So, for a reduction ratio of 1000:1, worm and worm wheel system is best.

564. In order to test the efficiency of a reducer gear train 1 kW input was given at the input end at a speed of 1440 rpm and at a output end the measured torque was 56.36 N.m. If the ratio of speed reduction in this unit is 10: 1, the efficiency is about

(a) 78% (b) 85%
(c) 63% (d) 96%

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Given,
Speed reduction = 10 : 1

$$\text{Output speed} = \frac{1440}{10} = 144 \text{ rpm}$$

$$\text{Output power} = T.\omega = 56.36 \times \frac{2\pi \times 144}{60}$$

$$= 849.88 \text{ W}$$

$$\text{Efficiency} = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{849.88}{1000} \times 100 = 85\%$$

565. The differential gear is fitted on rear axle of automobiles; its function is

(a) To rotate the front wheels at different speeds
(b) To rotate the back wheels at the same speed during turning
(c) To rotate the back wheel at different speeds during turning

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(d) To permit the two back wheels to rotate at the different speeds when driving in the straight path.

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : The differential gear is fitted on rear axle of automobiles; its function is to rotate the back wheel at different speeds during turning.

566. Consider the following profiles of mating gear teeth:

1. Involute profiles
2. Cycloidal profiles
3. Conjugate profiles

Which of these satisfy the law of gearing?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : All 3-profiles satisfy the law of gearing.

567. In the case of involute system of gears, if the centre distance is changed

(a) The velocity ratio changes
(b) The pressure angle changes
(c) The pitch circle remain unaffected
(d) The law of gearing fails to get satisfied

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : In the case of involute system of gears, if the centre distance is changed the pressure angle changes.

568. Which of the following is FALSE?

(a) A reverted gear train is one in which the first and last gears are on the same axis.
(b) Train value for a gear is the ratio of product of driven tooth numbers to product if driving tooth numbers.
(c) A planetary gear train is one in which the axes of some of the gears may have a motion.
(d) A compound gear train is one which has two or more gears on each axis.

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Train value $= \dfrac{\text{Product of driving tooth}}{\text{Product of driven tooth}}$

569. Velocity factor is used to take care of

(a) effect of high velocity
(b) possibility of fatigue failure
(c) possibility of high wear
(d) pitting

TNPSC AE 2017

Ans. (b) : Velocity factor is used to take care of possibility of fatigue failure. The ratio of the static load at failure to the dynamic load at failure was taken as the velocity factor (C_v). This means that an identical gear pair running at this speed would have a dynamic load equal to C_v times the static tangential load.

570. A spur gear with 22 number of teeth and a 28 mm pitch circle diameter will have a circular pitch as:

(a) 2 mm (b) 4 mm
(c) 6 mm (d) 8 mm
(e) 10 mm

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Number of teeth (T) = 22
Pitch circle diameter (D) = 28 mm
Then circular pitch (P) = ?

We know that

$$P = \frac{\pi D}{T} = \frac{22}{7} \times \frac{28}{22}$$

$$\boxed{P = 4 \text{ mm}}$$

571. In case of helical gear, the relation between the normal pitch (P_N), axial pitch (P_C) and helix angle (α), is given by:

(a) $P_C = \dfrac{P_N}{\cos\alpha}$ (b) $P_C = \dfrac{P_N}{\sin\alpha}$
(c) $P_N = \dfrac{P_C}{\cos\alpha}$ (d) $P_N = \dfrac{P_C}{\sin\alpha}$
(e) $P_N = \dfrac{P_N}{\cos 2\alpha}$

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : In case of helical gear, the relation between the normal pitch (P_N), axial pitch P_C and helix angle (α) is given by

$$P_C = \frac{P_N}{\cos\alpha}$$

572. The train value for T_1 number of teeth on driver and T_2 number of teeth on driven gear is given by:

(a) $\dfrac{T_1}{T_2}$ (b) $\dfrac{T_2}{T_1}$
(c) $\dfrac{1}{2}\left(\dfrac{T_1}{T_2}\right)$ (d) $\dfrac{1}{2}\left(\dfrac{T_2}{T_1}\right)$
(e) $(T_1 - T_2)$

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Speed Ratio–It is defined as the ratio of speed of driver gear to speed of driven gear

$$\text{Speed ratio} = \frac{\omega_{DVR}}{\omega_{DVN}}$$

$$\text{Then train value} = \frac{1}{\text{Speed Ratio}} = \frac{\omega_{DVN}}{\omega_{DVR}}$$

$$\text{Train value} = \frac{\omega_{DVN}}{\omega_{DVR}} = \frac{T_1}{T_2}$$

573. Stub teeth are provided for-

(a) High velocity ratios
(b) High power transmitting capacity
(c) Low noise
(d) Reducing number of the teeth for compact construction

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Stub teeth are provided for reducing number, of teeth for compact construction, reduced interference in involutes gears, to improve capability to with stand shock and vibrations.

574. Velocity factor for ordinary cut gears operating at velocity 6 m/s is equal to-

(a) 0.5 (b) 0.75
(c) 0.33 (d) 1

Nagaland PSC CTSE 2017, Paper-I

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Ans. (c) : Velocity factor for ordinary gears operating at (velocity upto 12.5 m/sec) is equal to,

$$C_V = \frac{3}{3+V}$$

Here V = 6m/sec

$$C_V = \frac{3}{3+6} = 0.33$$

For carefully cut gear (for velocity upto 12.5 m/sec)

$$C_V = \frac{4.5}{4.5+V}$$

For very accurately cut and ground metallic gears (for velocity upto 20 m/sec)

$$C_V = \frac{6}{6+V}$$

For precision gear cut coin high accuracy (for velocity upto 20 m/sec)

$$C_V = \frac{0.75}{0.75+\sqrt{V}}$$

575. Involute profile is widely used for gear tooth because–

(a) pressure angle remains constant
(b) face and flank form a continuous curve
(c) involute rack has straight sided teeth
(d) all the above factors

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Involute profile is widely used for gear tooth because all the above factors.

576. An epicyclic gear train has 3 shafts A, B and C. A is the input shaft running at 100 r.p.m. clockwise. B is the output shaft running at 250 r.p.m. clockwise. The torque on A is 50 kNm (clockwise), C is a fixed shaft. The torque needed to fix C is

(a) 20 kN m (anti-clockwise)
(b) 20 kN m (clockwise)
(c) 30 kN m (anti-clockwise)
(d) 30 kN m (clockwise)

ESE 2017

Ans. (c) : Given,

$N_A = 100$ rpm (CW)
$N_B = 250$ rpm (CW)
$\tau_A = 50$ kNm (CW)
$N_C = 0$
$\sum \tau N = 0$
$\tau_A N_A + \tau_B N_B + \tau_C N_C = 0$
$(50 \times 100) + (\tau_B \times 250) + 0 = 0$
$\tau_B = -20$ kNm
$\tau_B = 20$ kNm (anti clockwise)

Now $\sum \tau = 0$
$\tau_A + \tau_B + \tau_C = 0$
$\tau_C = -(\tau_A + \tau_B)$
$= -(50 - 20) = -30$ kNm
$= 30$ kNm (anti clockwise)

577. When the addendum of pinion and wheel is such that the path of approach and path of recess are half of their maximum possible values then the length of path of contact is given by,

r pitch circle radius of pinion
R pitch circle radius of wheel
'ϕ' pressure angle.

(a) $\frac{(R+r)\sin\phi}{2}$ (b) $\frac{(R+r)^2\cos\phi}{2}$
(c) $\frac{(R+r)^2\tan\phi}{2}$ (d) $\frac{(R+r)^2}{2\sin\phi}$

Kerala PSC Poly. Lect., 2017

Ans. (a) : $\frac{(R+r)\sin\phi}{2}$

578. Which is the incorrect relationship of gears?

(a) Circular pitch × Diameteral pitch = π
(b) Module = P.C.D. / No. of teeth
(c) Dedendum = 1.157 × module
(d) Addendum = 2.157 × module

GMB AAE 25.06.2017

Ans. (d) : Addendum = 2 × Module is the correct relationship of gears

579. In case of helical gears the angle between the plane containing the axes of the gears and plane passing through the pitch point is known as:

(a) Pitch angle (b) Lead angle
(c) Helix angle (d) pressure angle

WBPSC AE, 2017

Ans. (d) : In case of helical gears the angle between the plane containing the axes of the gears and plane passing through the pitch point is known as pressure angle.

580. In a speed gear box speeds are arranged in

(a) A.P. (b) G.P.
(c) H.P. (d) L.P.

WBPSC AE, 2017

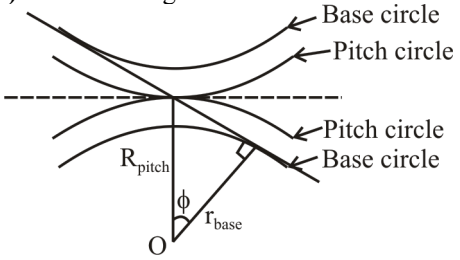
Ans. (b) : In a speed gear box speeds are arranged in G.P. (Geometric Progression).

581. For an involute gear, the ratio of base circle radius and pitch circle radius is equal to

(a) sin φ (b) cos φ
(c) sec φ (d) cosec φ

Sikkim PSC (Under Secretary), 2017

Ans. (b) : In involute gear



ϕ = Pressure angle
$r_{base} = R_{pitch}.\cos\phi$

$$\frac{r_{base}}{R_{pitch}} = \cos\phi$$

582. Which of the following statements is/are true for gear drives?

(a) Gear drives can transmit very large power
(b) Gear drives have low transmission efficiency

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(c) Gear drives require more space
(d) All of the above

Karnataka PSC Lect., 27.05.2017

Ans (a) : The gears are used to transmit power from one shaft to another when the shafts are at a small distance apart. It transmits exact velocity ratio. It has high efficiency, reliable service and compact layout. The gear drives has positive drives.

583. The common normal at the point of contact between a pair of teeth must always pass through
(a) Pitch circle (b) Pitch surface
(c) Pitch point (d) None of the above

Karnataka PSC Lect., 27.05.2017

Ans (c) : Pitch point- It is a common point of contact between two pitch circles.

584. An external gear with 60 teeth meshes with a pinion of 20 teeth, module being 8 mm. What is the centre distance in mm?
(a) 960 (b) 320
(c) 480 (d) 160

GPSC EE Pre, 28.01.2017

Ans. (b) : Given,
$T_g = 60$
$T_p = 20$
Module (m) = 8 mm

$$\text{Centre distance} = \frac{m}{2}\left(T_g + T_p\right)$$
$$= \frac{8}{2}(60+20) = \frac{8\times 80}{2}$$
$$= 4 \times 80$$
$$= 320 \text{ mm}$$

585. Suggest gearing arrangement for a stirrer to run at 36 rpm being driven by a motor running at 1440 rpm
(a) Helical gear (b) Differential gear
(c) Spur gear (d) Worm gear

APGENCO AE, 2017

Ans. (d) : When large (say more than 10) gear ratio is required. Worm gears are used. They are used in feed grass cutting machine or grass chaff cutter.

586. The details of two crossed helical gears for speed reduction is as follows. Pitch circle diameter in the plane of rotation for one gear is 80 mm and the helix angle is 30^0. For the other gear pitch circle diameter in the plane of rotation is 120 mm and the helix angle is 22.5^0. If the input speed is 1440 rpm, what is the output speed?
(a) 700 rpm (b) 800 rpm
(c) 1200 rpm (d) 900 rpm

APGENCO AE, 2017

Ans. (d) : $V_r = \frac{V_2}{V_1} = \frac{\omega_2}{\omega_1} = \frac{d_1}{d_2}\frac{\cos\phi_1}{\cos\phi_2} = \frac{N_2}{N_1}$

$\frac{N_2}{1440} = \frac{80\cos 30}{120\cos 22.5}$ or $N_2 = 900$ rpm

587. For two gears to mesh
(a) The diametral pitch of both gears must be the same
(b) Tooth thickness of both gears must be same
(c) Number of teeth on both gears must be same
(d) Whole depth of both gears must be same

Punjab PSC (Lect.) 06.08.2017

Ans. (a) : For two gears to mesh the diametral pitch of both gears must be the same.

588. In the design of spur gears, the Barth's equation is used to express
(a) Velocity Factor
(b) Geometry Factor
(c) Size Factor
(d) Surface Condition Factor

Punjab PSC (Lect.) 06.08.2017

Ans. (a) : The modified Lewis equation for bending stress is

$$P_t = K_v\, mb\sigma_b y$$

Where K_v is known as velocity factor and is given by Barth's equation below for known pitch line velocity V in m/s and is given by,

$$K_v^1 = \frac{6}{6+V}$$

589. Helical gears can be cut on
(a) universal milling machine
(b) horizontal milling machine
(c) vertical milling machine
(d) planetary milling machine

Punjab PSC (Lect.) 06.08.2017

Ans. (a) : Helical gears can be cut on universal milling machine.

590. Two mating spur gears have 25 teeth and 75 teeth. The pinion rotates at 1260 rpm. What is the rotational speed of the gear?
(a) 420 rpm (b) 840 rpm
(c) 1060 rpm (d) 1260 rpm

Punjab PSC (Lect.) 06.08.2017

Ans. (a)

$$\frac{T_1}{T_2} = \frac{N_2}{N_1} \Rightarrow \frac{75}{25} = \frac{1260}{N_1}$$
$$N_1 = \frac{1260}{3} = 420 \text{ rpm}$$

591. The minimum number of teeth to avoid interference in a two wheel set of equal size, when addendum is one module, is given by
(a) $\frac{2}{\sqrt{1+3\sin^2\phi}-1}$ (b) $2\left[\sqrt{1+3\sin^2\phi}-1\right]$
(c) $2\left[\sqrt{1+3\sin^2\phi}+1\right]$ (d) $\frac{2}{\sqrt{1+3\text{sub}^2\phi}+1}$

Punjab PSC (Lect.) 06.08.2017

Ans. (a)

$$T = \frac{2a_w}{\sqrt{1+\frac{1}{G}\left(\frac{1}{G}+2\right)\sin^2\phi}-1}$$

This gives the minimum number of teeth on the wheel for the given values of the gear ratio, the pressure angle and the addendum coefficient a_w.
For $a_w = 1$ i.e., when the addendum is equal to one module

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$$T \geq \frac{2}{\sqrt{1+\frac{1}{G}\left(\frac{1}{G}+2\right)\sin^2\phi}-1}$$

For equal number of teeth on the pinion and the wheel, G = 1 and

$$T_{min} = \frac{2}{\sqrt{1+3\sin^2\phi}-1}$$

592. Among the following, which are standard interchangeable Tooth Profiles?
(a) The 14.5° composite system
(b) The 20° full depth involutes system
(c) The 20° stub involutes system
(d) All of the above

Punjab PSC SDE 12.02.2017

Ans. (d) : All of the above

593. Hypoid gears are used for :
(a) Parallel shafts
(b) Intersecting shafts
(c) Non-parallel non-intersecting shafts
(d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : Hypoid gears are used for non-parallel non intersecting shafts. This provided more contact area and better tooth strength which allow more torque to be transmitted and high gear ratio to be used.
For parallel shaft – Spur gear, Helical gear,
Intersecting shaft – Bevel Gear.

594. Two mating spur gears have 40 and 120 teeth respectively. The pinion rotates at 12000 rpm and transmits a torque of 20 Nm. The torque transmitted by gear is:
(a) 80 Nm (b) 30 Nm
(c) 40 Nm (d) 60 Nm

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (d) : From Gears Ratio

$$GR = \frac{D_2}{D_1} = \frac{Z_2}{Z_1} = \frac{N_1}{N_2} = \frac{T_2}{T_1}$$

Where D_1 = Diameter of pinion
D_2 = Diameter of Gear
Z_1 = Teeth on pinion
Z_2 = Teeth on Gear
N_1 = Speed of Pinion
N_2 = Speed of Gear
T_2 = Torque transmitted from the Gears.
T_1 = Torque transmitted from the Pinion.

Given $Z_1 = 40$ teeth; $Z_2 = 120$ teeth
$N_1 = 12000$ rpm
and $T_1 = 20$N-m $T_2 = ?$
From the above relation.

$$\frac{Z_2}{Z_1} = \frac{T_2}{T_1}$$

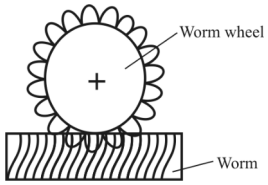
$$\frac{120}{40} = \frac{T_2}{20}$$

$$\boxed{T_2 = 60\text{N} - \text{m}}$$

595. A worm is basically a :
(a) Spur gear (b) Bevel gear
(c) Helical gear (d) Herringbone gear

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : A worm is basically a helical gear. Worm gears are constructed of a worm and a gear, with non-parallel non-intersecting shaft, oriented 90° to each other.



596. The central gear of an epicyclic gear box known as :
(a) Ring gear (b) Sun gear
(c) Planet gear (d) Internal gear

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) Sun gear

597. The ratio of the number of teeth and pitch circle diameter is called ..... .
(a) pitch (b) circular pitch
(c) diametral pitch (d) module

GWSSB DEE 07.07.2016

Ans. (c) : The ratio of the number of teeth and pitch circle diameter is called diametral pitch.

598. The torque that can be transmitted safely by the spur gear tooth at zero pitch line velocity is known as:
(a) Average torque (b) Maximum torque
(c) Minimum torque (d) Stalling torque

RPSC ADE 2016

Ans. (d) : Stalling torque is the torque produced by a mechanical device whose output rotational speed is zero. It may also means that the torque load that causes the output rotational speed of a device to become zero.
The torque that can be transmitted safely by the spur gear tooth at zero pitch line velocity is known as stalling torque.

599. The ratio of face width to transverse pitch of a helical gear with the helix angle is normally
(a) Equal to 1/tan α
(b) more than 1.15/tan α
(c) less than 1.05/tan α
(d) Equal to 2/tan α

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : The ratio of face width to transverse pitch of a helical gear with the helix angle is more than $\frac{1.15}{\tan\alpha}$

600. In cycloidal profile gears, the pressure angle is:
(a) Dependent on the size of teeth
(b) Depends on diameter of gears
(c) Constant
(d) Varies with velocity ratio

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : In cycloidal profile, pressure angle varies from maximum at the beginning of engagement, reduces to zero at the pitch point and again increases to maximum at the end of engagement resulting in improper running of gears, pressure angle depends on no. of teeth size of teeth.

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601. In a mitre bevel gear, the pitch cone angle is:

(a) Less than 90°
(b) 90°
(c) Greater than 90°
(d) Can be less or greater than 90°

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : Mitre gears are a special class of bevel gears, where the shafts intersect at 90° and the gear ratio is 1:1. So, it have 90° of pitch cone angle.
For mitre bevel gear–

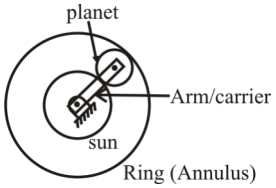
$$\left[\frac{\omega_1}{\omega_2} = 1\right]$$

602. If The annulus wheel of an epicyclical gear train has 120 teeth and planet has 20 teeth, the number of teeth on the sun gear is:

(a) 100 (b) 80
(c) 60 (d) 40

GPSC Asstt. Prof. 28.08.2016

Ans. (b) :



$R_{annulus} = 2R_{planet} + R_{sun}$
$R \propto T$ i.e. radius is proportional to teeth.

$$\text{Module } (m) = \frac{2R}{T} = \text{Constant}$$

$120 = 2 \times 20 + T_{sun}$
$T_{sun} = 80$ teeth

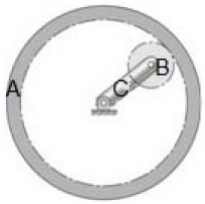
603. The pressure angle commonly used for involutes splines is

(a) 14½° (b) 20°
(c) 22½° (d) 30°

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Involute splines are available in several standard pressure angles. The 30° pressure angle spline are most commonly used pressure angle.

604. In the epicyclic gear train shown in the figure below, A is rotating at 10 rpm clockwise. A has 140 teeth and B has 35 teeth.



If the arm C makes five revolutions, then what is the ratio of speed of A to that of B?

(a) 1 : 5 (b) 2 : 5
(c) 5 : 1 (d) 5 : 2
(e) 5 : 3

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) :

Arm(c)	$(T_B = 35)B$	$A(T_A = 140)$
0	+x	$+x \cdot \frac{35}{140}$
y	y + x	$y + x \cdot \frac{35}{140}$

$N_C = y = 5$

$$N_A = 10 = y + x\left(\frac{35}{140}\right)$$

$$10 = y + x \times \frac{1}{4}$$

$$10 = 5 + \frac{x}{4}$$

$x = 20$
$N_B = y + x = 5 + 20 = 25$ rpm
$N_B = 25$ (ACW)

Therefore, $\frac{N_A}{N_B} = \frac{10}{25} = \frac{2}{5}$

605. Match the gear types in List-I with their applications with List-II.

	List-I		List-II
(a)	Worm gear	1.	Parallel shafts
(b)	Cross helical gear	2.	Non parallel, intersecting shaft gear
(c)	Bevel gear	3.	Non parallel, non-intersecting shaft
(d)	Spur gear	4.	Large speed ratio

(a) A-4, B-2, C-3, D-1 (b) A-4, B-3, C-2, D-1
(c) A-1, B-2, C-3, D-4 (d) A-1, B-3, C-2, D-4
(e) A-2, B-4, C-1, D-3

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : A Worm gear have large speed ratio of 40:1.
Cross Helical Gear–Sets are used to transmit power and motion between non-intersecting & non-parallel axes.
Bevel gear are gears where the axes of the two shafts intersect and the tooth-bearing faces of gears.

606. In order to test the efficiency of reduced gear train, 2 kW input was given at the input end at a speed of 1800 rpm and at the output end the measured torque was 60 Nm. If the ratio of speed reduction in this unit is 9 : 1, the efficiency is about:

(a) 63% (b) 72%
(c) 78% (d) 85%
(e) 96%

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : Speed reduction = 9:1

$$\text{Output speed} = \frac{N}{9} = \frac{1800}{9} = 200\,\text{rpm}$$

$$\text{Output power} = T.\omega = 60 \times \frac{2\pi \times 200}{60}$$

$$= 400\,\pi \text{ N-m}$$

$$\% \text{ efficiency} = \frac{\text{Output}}{\text{Input}} \times 100 = \frac{400\pi}{2000} \times 100 = 63\%$$

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607. A 15 teeth spur pinion has a module 2mm, runs at 20 rev/s and driver gear at a speed of 10 rev/s. Find the number of teeth on the gear and the theoretical centre to centre distance.

(a) 30 teeth, 45 mm (b) 45 teeth, 45 mm
(c) 45 teeth, 30 mm (d) 8 teeth, 30 mm

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Given,

$T_P = 15$ teeth
$m = 2$ mm
$\omega_p = 20$ rev/sec.
$\omega_D = 10$ rev/s

$$\frac{T_D}{T_P} = \frac{\omega_p}{\omega_D}$$

Teeth on driver gear $T_D = \frac{20}{10} \times 15 = 30 \text{teeth}$

$\boxed{T_D = 30 \text{ teeth}}$

Centre distance $x = r_1 + r_2$

$$= \frac{m}{2}(T_P + T_D) = \frac{2}{2} \times (15 + 30)$$

$\boxed{x = 45}$ mm

608. …….gears are used for transmitting motion/power between intersecting shafts.

(a) Spur (b) Helical
(c) Worms end worm (d) Bevel

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Bevel Gear–Bevel gears have a cone shaped appearance and are used to transmit force between two shaft, which intersect at one point (intersecting shafts). A bevel gear has a cone as its pitch surface and its teeth are cut along the cone.
Spur – parallel shaft.
Helical – parallel shaft.
Worms and worm wheel – Neither parallel nor intersecting.

609. ……provides a continuous variation of gear ratio from lowest to highest.

(a) Torque convertor (b) Gear box
(c) Both (a) and (b) (d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Torque Convertor provides a continuous variation of gear ratio from lowest to highest.

610. If the pitch circle diameter of a spur gear is 60 mm and number of teeth are 20. What would be the diametral pitch?

(a) 3 (b) 0.3
(c) 9.42 (d) 0.106

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Pitch circle Dia (D) = 60 mm
No. of teeth (T) = 20

$$P_d = \frac{T}{D_{mm}} = \frac{20}{60}$$

$P_d = 0.33$

611. Worm gear is an example of which of the following simple machines?

(a) Lever (b) Screw
(c) Wedge (d) Pulley

KPCL AE 2016

Ans. (b) : A worm is an example of screw, one of the six simple machines. One of the major advantages of worm gear drive units are that they can transfer motion in 90º degrees. The worm in the worm gear drive may have single or multiple starts.

612. When the leading edges of the teeth are not parallel to the axis of rotation, but are set at an angle so that can be meshed in parallel or crossed orientations, they are called_____

(a) Helical gears (b) Spur gears
(c) Worm gears (d) None of these

KPCL AE 2016

Ans. (a) : Where the leading edges of the teeth are not parallel to the axis of rotation, but are set at an angle so that can be meshed in parallel or crossed orientations, they are called helical gears.

613. .....is the distance travelled by a point on either pitch circle of the two wheels during the period of contact of a pair of gear teeth.

(a) Path of contact (b) Line of contact
(c) Arc of contact (d) Radius of contact

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Arc of contact- The locus of a point on the pitch circle from the beginning to the end of engagement of two mating gear is known as the arc of contact.

$$\text{Arc of contact} = \frac{\text{Path of contact}}{\cos\phi}$$

614. When a pinion and cutting rack are in mesh and due to interference the material removed from pinion is known as:

(a) undercutting (b) backlash
(c) arc cutting (d) traction

HPPSC AE (PWD) 10.09.2016

Ans. (a) : Undercutting- A portion of addendum of one gear falls inside the base circle of meshing gear. The profile of the gear inside the base circle is radial. A gear having its material removed in this manner is known as undercut and this process is known as undercutting.

615. A spur gear has a module of 3 mm, number of teeth 16, face width 36 mm, and a pressure angle of 20 degree. It is transmitting a power of 3 kW at 20 rev/s. Taking a velocity factor of 1.5 and a form factor of 0.3, the stress in the gear tooth in MPa will be :

(a) 32 (b) 46
(c) 58 (d) 70

HPPSC Asstt. Prof. 18.11.2016

Ans. (b) : $m = 3$ mm, $\phi = 20°$ $C_v = 1.5$
$t = 16$ mm, $P = 3$ kW $y = 0.3$
$w = 36$ mm, $N = 20$ rev/s

For the safe condition

$$= F_t \times C_v \times s \le bmy(\sigma_b) \quad .....(1)$$

$$P = T \times \omega = F_t \times \frac{d}{2} \times \omega$$

$$3 \times 10^3 = F_t \times \frac{48}{2} \times 2 \times \pi \times 20 \qquad \left(\because m = \frac{d}{t} \Rightarrow 3 = \frac{d}{16}\right)$$

$d = 48$ mm

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$$F_t = \frac{3\times10^3\times2}{48\times2\times\pi\times20}$$ Put the value equation (1)

$$\frac{3\times10^3\times2}{48\times2\times\pi\times20}\times1.5\times1 = 36\times3\times0.3\sigma_b$$

$$\sigma_b = \frac{3\times10^3\times60\times1.5}{24\times2\times2\times20\times36\times3\times8.5} = 46.05\,\text{MPa}$$

616. A bronze spur pinion rotating at 600 rpm drives a cast iron spur gear at a transmission ratio of 4 : 1. The allowable static stresses for the bronze pinion and cast iron gear are 84 MPa and 105 MPa respectively. The pinion has 16 standard 20° full depth involute teeth of module 8 mm. The face width of both the gears is 90 mm. The power that can be transmitted from the stand point of strength is

(a) 31.64 kW (b) 78.70 kW
(c) 40.2 kW (d) 316.4 kW
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Given, $N_p = 600$ rpm, V.R. $= T_g/T_p = 4$
$\sigma_{OP} = 84$ MPa, $\sigma_{OG} = 105$, $T_P = 16$
$m = 8$ mm, $b = 90$ mm, $C_S = 1$

$$v = \frac{\pi D_P N_P}{60} = \frac{\pi\times m\times T_P\times N_P}{60}$$

$$v = \frac{\pi\times0.128\times600}{60} = 4.02\,\text{m/s}$$

$$C_v = \frac{3}{3+v} = 0.427$$

$$W_{T(P)} = \sigma_{OP}\times C_v\times b\times\pi\times m\times Y_P$$

$$Y_P = 0.154 - \frac{0.912}{T_P} = 0.154 - \frac{0.912}{16} = 0.097$$

$$W_{T(P)} = 84\times0.427\times90\times\pi\times8\times0.097$$

$$W_{T(P)} = 7870\text{ N}$$

$$P = \frac{W_T\cdot v}{C_s} = \frac{7870\times4.02}{1} = 31637\,\text{W}$$

$$= 31.64\text{ kW}$$

617. In a simple gear train of three gear wheels, the first gear has 20 teeth, second gear 10 teeth and the third gear 40 teeth. The speed ratio of the third gear to first gear and the direction are

(a) 1/2 and opposite direction
(b) 1/2 and same direction
(c) 2 and opposite direction
(d) 2 and same direction
(e) none of these

CGPSC AE 16.10.2016

Ans. (b) : As we know, gear ratio

$$\frac{N_2}{N_1} = \frac{T_1}{T_2}$$

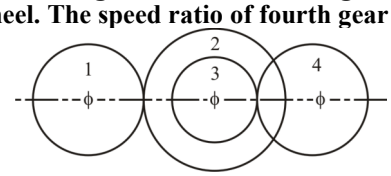
For simple gear strain

$$\frac{N_1}{N_3} = \frac{N_1}{N_2}\times\frac{N_2}{N_3} = \frac{T_2}{T_1}\times\frac{T_3}{T_2}$$

$$\frac{N_1}{N_3} = \frac{T_3}{T_1} = \frac{40}{20}$$

$$\frac{N_3}{N_1} = \frac{1}{2}$$ & same direction

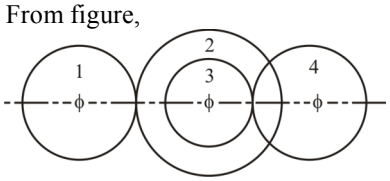
618. The following figure shows a compound gear train with gears 2 and 3 forming a compound wheel. The speed ratio of fourth gear is



(a) $\frac{N_4}{N_1} = \frac{T_4}{T_1}$ (b) $\frac{N_4}{N_1} = \frac{T_1\times T_3}{T_2\times T_4}$
(c) $\frac{N_4}{N_1} = \frac{T_1\times T_2}{T_3\times T_4}$ (d) $\frac{N_4}{N_1} = \frac{T_2\times T_4}{T_1\times T_3}$
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : From figure,



As we know, for compound gear train,

$$\frac{N_4}{N_1} = \frac{\text{Product of teeth of driver}}{\text{Product of teeth of follower}}$$

$$\boxed{\frac{N_4}{N_1} = \frac{T_1\times T_3}{T_2\times T_4}}$$

619. Which of the following is true for involute gears?

(a) Interference is inherantly absent.
(b) Variation in centre distance of shaft increases radial force.
(c) A convex flank is always in contact with conclave flank.
(d) Pressure angle is constant through the teeth engagement.
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (d) : For involute gears, the pressure angle is constant throughout the teeth engagement.

620. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I	List-II
A. Interference	1. Arc of approach, arc of recess, circular pitch
B. Dynamic load on tooth	2. Lewis equation
C. Static load	3. Minimum number of teeth on pinion
D. Contact ratio	4. Inaccuracies in tooth profile

Code :

	A	B	C	D
(a)	3	4	1	2
(b)	1	2	3	4
(c)	4	3	2	1
(d)	3	4	2	1

(e) None of the above

CGPSC Poly. Lect. 22.05.2016

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Ans. (d)

A. Interference	3. Minimum number of teeth on pinion
B. Dynamic load on tooth	4. Inaccuracies in tooth profile
C. Static load	2. Lewis equation
D. Contact ratio	1. Arc of approach, arc of recess, circular pitch

621. In a machine tool gear box, the smallest and largest spindles are 100 rpm and 1120 rpm respectively. If there are 8 speeds in all, the fourth speed will be :

(a) 400 rpm (b) 280 rpm
(c) 800 rpm (d) 535 rpm
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (b)

$N_{max} = 1120$ rpm
$N_{min} = 100$ rpm, $n = 8$

According speed selection in gear box–

$N_{max} = N_{min}\, r^{n-1}$

$1120 = 100\,(r)$

$$r^7 = \frac{1120}{100} \Rightarrow r = \left(\frac{1120}{100}\right)^{1/7}$$

$r = 1.412$

$\therefore$ Fourth speed,

$N_4 = N_{min}\,(r)^{4-1}$
$= 100\,(1.412)^3$

$\boxed{N_4 = 281.51 \text{ rpm}}$

622. The number of teeth of a spur gear is 30 and it rotates at 200 rpm. What will be its circular pitch and the pitch line velocity if it has a module of 2 mm?

(a) $p = 2\pi$ mm. $V_p = 37699$ mm/min
(b) $p = 2\pi$ mm. $V_p = 36699$ mm/min
(c) $p = 0.5\pi$ mm. $V_p = 37699$ mm/min
(d) $p = 2\pi$ mm. $V_p = 39699$ mm/min

UPRVUNL AE 21.08.2016

Ans. : (a) $m = 2$ mm

$D = mT = 60$ mm

Circular pitch $= \dfrac{\pi D}{T} = m\pi = 2\pi$

Velocity on periphery $= r\omega \quad \left(\omega = \dfrac{2\pi N}{60}\right)$

$= 30 \times \dfrac{2\pi \times 200}{60}$

$= 200\,\pi$ mm/sec

$= 200\,\pi \times 60$ mm/min

$V = 37699.1$ mm/min

623. The velocity ratio between pinion and gear in a gear drive is 2 : 3 the module of teeth is 4.0 mm and sum of number of teeth on pinion and gear is 99. What is the centre distance between pinion and the gear?

(a) 49.5 mm (b) 198 mm
(c) 99 mm (d) 148.5 mm

UPRVUNL AE 21.08.2016

Ans. : (b) Velocity Ratio $= \dfrac{\omega_{gear}}{\omega_{pinion}} = \dfrac{2}{3} = \dfrac{T_{pinion}}{T_{gear}}$

$m = 4$ mm

$T_{pinion} + T_{gear} = 99$

Distance between centre $= \dfrac{m}{2}\left(T_{pinion} + T_{gear}\right) = \dfrac{4}{2}(99)$

Distance b/w centre = 198

624. The transverse section of a helical gear is identical to

(a) bevel gear (b) spur gear
(c) worm gear (d) spiral gear

GPSC Lect. 23.10.2016

Ans. (b) : The transverse section of a helical gear is identical to spur gear.

625. In case of Worm Wheel and Worm the maximum efficiency is obtained when

(a) Either Worm or Worm Wheel is driver
(b) Worm Wheel is driver
(c) Worm is driver
(d) Depends on other factors

APPSC AEE Screening Test 2016

Ans. (a) : In case of worm wheel and worm the maximum efficiency is obtained when either worm or worm wheel is driver.

626. Gears of watches are generally manufactured by

(a) Powder Metallurgy
(b) Machining on Gear Shaper
(c) Die Casting
(d) Stamping

APPSC AEE Screening Test 2016

Ans. (d) : Gears of watches are generally manufactured by stamping.

627. Spring driven watches and clocks utilise

(a) involute gears (b) cycloid gears
(c) epicycloid gears (d) straight rack gears

RPSC AE 2016

Ans : (b) Spring driven watches and clock utilize cycloid gears teeth.
A cycloid is the curve traced by a point on the circumference of a circle which rolles without slipping an a fixed straight line.

628. For a reverted gear trains the speed ratio is given by

(a) $\dfrac{\text{Product of the no. of teeth on drives}}{\text{Product of no. of teeth on drivens}}$

(b) $\dfrac{\text{Product of the no. of teeth on drivens}}{\text{Product of no. of teeth on driver}}$

(c) $\dfrac{\text{Addition of the no. of teeth on drivens}}{\text{Addition of no. of teeth on drivers}}$

(d) $\dfrac{\text{Addition of the no. of teeth on drivens}}{\text{Product of no. of teeth on drivers}}$

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : For, a reverted gear trains, the speed ratio is given by,

$$S.R = \frac{\omega_{DVR}}{\omega_{DVN}} = \frac{T_{DVN}}{T_{DVR}}$$

$$S.R = \frac{\text{Product of the no. of teeth on drivens}}{\text{Product of no. of teeth on driver}}$$

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629. Velocity factor for ordinary cut gears operating at velocity 9m/s is equal to

(a) 0.5 (b) 0.75
(c) 0.25 (d) 1

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Velocity factor from ordinary cut gears operating at velocity 9 m/se. then

$$C_v = \frac{3}{3+v} = \frac{3}{3+9} = \frac{3}{12} = 0.25$$

630. Stub tooth are cut on gears

(a) Subjects to shock and vibration
(b) To reduce centre distance between shafts
(c) To reduce friction torque
(d) To improve transmission efficiency

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Stub tooth are cut on gears subjects to shock and vibration.

631. Surface endurance limit of gear material is dependent upon its

(a) Brinnel Hardness number
(b) Elastic Strength
(c) Toughness
(d) Yield Strength

APPSC AEE Mains 2016, Paper-III

Ans. (a) : Surface endurance limit of gear material is dependent upon its Brinnel Hardness Number.

632. "Addendum" is defined as

(a) The radius of addendum circle
(b) The radial height of the tooth below the pitch circle
(c) The radial height of the tooth above the pitch circle
(d) It is the full depth of the tooth

RPSC Lecturer 16.01.2016

Ans. (c) : Addendum—It is defined as the radial distance from pitch circle to addendum circle.

633. If a spur gear has module of 4 mm, its circular pitch will be given by

(a) 2 mm (b) 3 mm
(c) 6.28 mm (d) 12.56 mm

RPSC Lecturer 16.01.2016

Ans. (d) : m = 4

Circular pitch (P_c) = ?

$$P_c = \frac{\pi D}{T}$$

$$m = \frac{D}{T}$$

$P_c = \pi m = 3.14 \times 4$

$P_c = 12.56$ mm

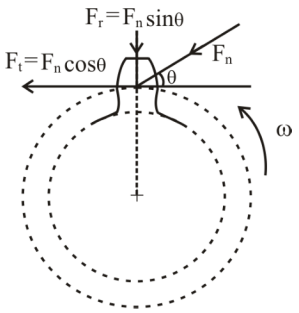
634. Power transmitted (Watts) by spur gear may be given as [Ft = tangential component of force (N), Fr = radial component of force (N), n = rotational speed (rpm), and d = pitch diameter (m)]

(a) $\frac{\pi dnF_t}{60}$ (b) $\left(\frac{\pi dn}{60}\right)^2 F_t$

(c) $\left(\frac{\pi dn}{60}\right)^2 F_r$ (d) $\frac{\pi dnF_r}{60}$

UPRVUNL AE 07.10.2016

Ans. (a) :



We know that,

$P = T \times \omega$

$$P = F_t \times \frac{d}{2} \times \frac{2\pi n}{60}$$

$$P = \frac{\pi dn}{60} \times F_t$$

635. Number of pairs of teeth in contact is given by (where, ϕ = pressure angle):

(a) (Arc of contact)/(Circular pitch)
(b) (Arc of contact × cos ϕ)/(Circular pitch)
(c) (Arc of contact)/(cos ϕ × Circular pitch)
(d) (Path of contact)/(Circular pitch)

UPRVUNL AE 07.10.2016

Ans. (a) : Contact ratio—The contact ratio is defined as the ratio of the length of arc of contact to the circular pitch or contact ratio is the average number of teeth in mesh during a contact cycle, its value always equal to or greater than one.

$$C.R. = \frac{\text{Length of arc of contact}}{\text{Circular pitch}}$$

636. Herringbone gear is a type of

(a) spiral bevel gear
(b) crossed helical gear
(c) hypoid gear
(d) double-helical gear with no groove in between the left and right inclinations

APPSC AEE Mains 2016, Paper-III

Ans. (d) : Herringbone gear is a type of double-helical gear with no groove in between the left and right inclinations.

637. Which of the following characteristics are applicable to stub 20^0 involute system-

(a) small interference
(b) strong tooth
(c) high production cost
(d) gear with small number teeth

RPSC 2016

Ans : (c) 20^0 Stub involute system require high production cost.
And minimum number of teeth on the pinion for 20^0stub involute is 14.

638. A pair of spur gears with module 5 mm and a centre distance of 450 mm is used for a speed reduction of 5 : 1 the number of teeth on pinion is-

(a) 60 (b) 90
(c) 30 (d) 50

RPSC 2016

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Ans : (c) m = 5 mm
c = 450 mm
speed reduction = 5 : 1
Speed reduction is inverse of number of teeth ratio

Speed reduction $\frac{N_P}{N_G}=\frac{T_G}{T_P}=5$ _____(1)

Center distance (c) = R + r

$$=\frac{m}{2}(T_G+T_P)$$

$$450=\frac{5}{2}(T_G+T_P)$$

$180 = T_G + T_P$
$180 = 5T_P + T_P = 6T_P$ (from 1)
$\boxed{T_P = 30}$

639. Profiles used for spur gears are:
(a) Epicycloids profiles
(b) Hypocycloid profiles
(c) Thread profiles
(d) Involutes profiles

(HPPSC LECT. 2016)

Ans : (d)
Profiles used for spur gear are Involute profiles.
An involute of a circle is a plane curve generated by a point on a tangent, which rolls on the circle without slipping or by a point on a string which in unwrapped from a reel. In connection with toothed wheels, the circle is known as base circle.

640. To accurately cut gears operating at velocities up to 20 m/s, the velocity factor is equal to :
(a) $\frac{3}{3+v}$ (b) $\frac{6}{6+v}$
(c) $\frac{9}{9+v}$ (d) $\frac{0.75}{1+v}+0.25$

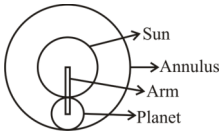
HPPSC W.S. Poly. 2016

Ans : (b) C_v = Velocity factor
(i) $C_v = \frac{3}{3+V}$, for ordinary cut gears operation at velocity up to 12.5 m/sec
(ii) $C_v = \frac{4.5}{4.5+V}$, for carefully cut gears operating at velocity upto 12.5 m/sec
(iii) $C_v = \frac{6}{6+V}$, for very accurately cut and ground metallic gear operation at velocity up to 10 m/sec.

641. In an Epicyclic gear train the number of teeth on Sun and annulus are 80 and 200 respectively. The number of teeth on planet Gear are-
(a) 60 (b) 30
(c) 90 (d) 100

TSPSC Managers, 2015

Ans. (a) :



We know that,
Epicyclic gear train
$R_{annulus} = 2R_{planet} + R_{sun}$
$\left[\because R \propto T \text{ i.e. radius is propartional to teeth}\left(m=\frac{2R}{T}=\text{constant}\right)\right]$
So, $200 = 2 \times T_{planet} + 80$
$\Rightarrow T_{planet} = \frac{120}{2} = 60$

642. In order to cut 30 T gear we use No. 4 cutter which is suitable for the range 2 T to 34 T. If helical teeth with helix angle α are to be cut, the size of cutter is given by
(a) (number of teeth) /α
(b) (number of teeth) /cos α
(c) (number of teeth) /$(\cos\alpha)^2$
(d) (number of teeth) /$(\cos\alpha)^3$

VIZAG MT 2015

Ans. (d) : Size of cutter $=\frac{(\text{number of teeth})}{(\cos\alpha)^3}$

643. Interference is inherently absent in the following types of gears :
(a) Involute (b) Cycloidal
(c) Stub (d) Hypocycloidal

KPSC ADF 2015

Ans. (b) : Interference is inherently absent in the cycloidal types of gears.

644. The pressure angle recommended by BIS for gear is :
(a) 14.5^o (b) 20^o
(c) 25^o (d) 30^o

KPSC ADF 2015 / ESE 2006

Ans. (b) : The pressure angle recommended by BIS for gear is 20^o.

645. The Lewis form factor of gear tooth depends on
(a) the elastic modulus of the tooth
(b) the module
(c) the number of teeth
(d) the pitch circle diameter

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : $y=\left(\frac{t^2}{6hm}\right)$ = Lewis form factor

$P_t = mb\sigma_b y$ the value of y independent of the size of tooth and depend only on No. of teeth on gear and system of teeth beam strength and replaced by S_b.
$S_b = mb\sigma_b y$ known as lewis equation
Note- Lewis form factor for 20° full depth involute is

$$\boxed{y=\pi\left(0.154-\frac{0.912}{T}\right)}$$

646. Undercut of a gear tooth can be avoided by :
(a) decreasing the pressure angle
(b) improving the heat treatment
(c) using smaller rake angle for the cutter
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (d) : If the addendum of the mating gear is more than the limiting value then undercut is taking place.

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647. Power transmitted by an involute gear is given by
(F_T, F_R, v are Tangential force, Radial force and velocity respectively)
(a) $P = F_T \times v$ (b) $P = F_R \times v$
(c) $P = (F_T + F_R) \times v$ (d) $P = (F_T - F_R) \times v$
TSGENCO AE 14.11.2015

Ans. (a) : Given–
F_T = Tangential force
F_R = Radial force
V = velocity
Power transmitted, P = Tangential force × velocity
$P = F_T \times V$

648. An internal gear having 60 teeth is meshing with an external gear having 20 teeth. Module is 6 mm. The centre distance of two gears is
(a) 120 mm (b) 180 mm
(c) 240 mm (d) 300 mm
MPPSC AE 08.11.2015

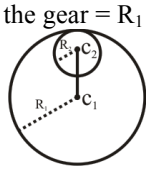
Ans. (a) : Let, T teeth of gear
Given, Teeth of gear (T_1) = 60 & $T_2 = 20$
module (m) = 6 mm $\Rightarrow m_1 = m_2 = m = 6$ mm

$\because \quad m = \dfrac{D}{T}$ where, D = diameter of gear
T = teeth of gear

$\therefore D = mT$

$\therefore$ radius of gear (R) $= \dfrac{D}{2} = \dfrac{mT}{2}$

Let, R_1 and R_2 are the radius of meshing gear.
$\therefore$ centre distance of the gear = $R_1 - R_2$



$$= \frac{mT_1}{2} - \frac{mT_2}{2} = \frac{m}{2}(T_1 - T_2)$$

$$= \frac{6}{2} \times (60 - 20) = 3 \times 40 = 120 \text{ mm}$$

Note- Option (c) is given by the commission.

649. For two wheels of equal size with 20^o Pressure angle and addendum equal to one module, the minimum number of teeth of each wheel must be
(a) 10 (b) 13
(c) 15 (d) 17
TSPSC AEE 2015

Ans. (b) : Data given as,
$G = 1, m = 1, \phi = 20^o$
then

$$t_{min} = \frac{2A_p}{\left(\sqrt{1 + G(G+2)\sin^2\phi} - 1\right)}$$

$$t_{min} = \frac{2}{\sqrt{\left(1 + 3\sin^2\phi\right)} - 1}$$

$$= \frac{2}{\sqrt{1 + 3\left(\sin 20^o\right)^2} - 1} = 12.32 \simeq 13$$

650. Gears produced by die casting are generally used for
(a) high torque transmission
(b) light duty
(c) low temperature application
(d) high axial thrust transmission
(KPSC AE 2015)

Ans : (b) Gear produced by die casting are generally used for light duty.
Methods of manufacturing gears:-
(i) Casting **(ii)** Hot rolling
(iii) Extruding **(iv)** Stamping
(v) Coining **(vi)** Powder- metallurgy

651. Which one of the following pairs is not correctly matched?
(a) Positive drive-belt drive
(b) High velocity ratio-worm gearing
(c) Non-parallel and non-intersecting-spiral gear
(d) Diminished noise and smooth operation-helical gear
OPSC AEE 2015 Paper-I

Ans : (a) (i) Negative drive: Belt drive
(ii) High Velocity ratio: worm gearing
(iii) Non-Parallel and non-intersecting: spiral gear
(iv) Diminished noise and smooth operation: Helical gear

652. A 1.5 kW motor is running at 1440 rpm. It is to be connected to stirrer running at 36 rpm. The gearing arrangement suitable for this application is
(a) differential gear (b) helical gear
(c) worm gear (d) spur gear
GPSC AE 26.07.2015
GATE 2000

Ans. (c) : worm gear

653. In a Gear Train of n wheels, the speed ratio is defined as
(a) $\dfrac{N_1}{N_n}$ (b) $\dfrac{N_n}{N_1}$
(c) $\dfrac{T_1}{T_n}$ (d) $\dfrac{T_n}{T_1}$
TANGEDCO AE 2015

Ans. (a) : Speed ratio = $\dfrac{1}{\text{Train value}}$

Train value = $\dfrac{N_n}{N_i}$

$\therefore$ Speed ratio = $\dfrac{N_i}{N_n}$

654. Minimum center distance is to be maintained between two gears with respect to
(a) Yield strength of the material
(b) Ultimate strength of the material
(c) Fatigue strength of the material
(d) Compressive strength of the material
TANGEDCO AE 2015

Ans. (d) : Compressive strength of the material.

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655. In a fully automatic transmission the
(a) torque is continuously transmitted to the driving wheels
(b) power is continuous transmitted to the driving wheels
(c) tractive effect is continuously transmitted to the driving wheels
(d) driving force is continuously transmitted to the driving wheels

TANGEDCO AE 2015

Ans. (a) : In fully automatic transmission in the torque is continuously transmitted to the driving wheels.

656. In an involute pinion and gear system, the interference can be avoided by:
(a) decreasing number of teeth in pinion
(b) lowering the pressure angle
(c) taking long and short addendum system
(d) making the gears using from milling cutter

HPPSC Asstt. Prof. 2014

Ans. (c) : Methods to remove interference:-
(1) Undercut
(2) By increasing the number of teeth
(3) By increasing the pressure angle
(4) Stub tooth
(5) Increasing the centre distance
(6) Decrease the addendum of gear and increase same for pinion.

657. Which of the following gives the value of face angle of a bevel gear?
(a) pitch angle + addendum angle
(b) pitch angle – addendum angle
(c) axial pitch
(d) diametral pitch

UPRVUNL AE 2014

Ans : (a) value of face angle of bevel gear
Face Angle= Pitch Angle + addendum Angle

658. Which of the following gives the value of root angle of a bevel gear?
(a) pitch angle + addendum angle
(b) pitch angle – addendum angle
(c) pitch angle + dedendum angle
(d) pitch angle – dedendum angle

UPRVUNL AE 2014

Ans : (d) The value of root angle of bevel gear.
Root Angle = Pitch Angle - Dedendum Angle.

659. Among the following types of gears, which gears offer a refinement over spur gears?
(a) Skew Gear
(b) Advanced spur gears
(c) External Gears
(d) Helical Gears

UPRVUNL AE 2014

Ans. : (d) Helical gears offer a refinement over spur gears.
- Helical gears have teeth that are set on as angle to the gear axis since the teeth engage more gradually, they have a smother and quieter operation than spur gear.
- Helical gear also have greater tooth strength and a higher load carrying capacity.

660. Which type of Gear meshing should be employed to transmit torque to a linear force?
(a) Epicyclical Gear (b) Helical Gear
(c) Bevel Gear (d) Rack and pinion

ISRO Scientist/Engineer 24.05.2014

Ans. (d) : Type of Gear meshing should be employed to transmit torque to a linear force rack and pinion.

661. Consider the following statements
1. Cheaper low quality gear in small numbers
2. The tooling costs are reasonable
3. Poor surface finish and dimensional accuracy
4. Due to low precision and high backlash, they are noisy
5. They are suited for non-critical applications

Out of the above which are the characteristics of sand cast gears?
(a) 1, 2, 3 and 4 (b) 1, 2, 4 and 5
(c) 2, 3, 4 and 5 (d) 1, 2, 3, 4 and 5
(e) 1, 4 and 5

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : The characteristics of sand cast gears are
(i) Cheaper low quality gear in small numbers.
(ii) The tooling costs are reasonable
(iii) Poor surface finish and dimensional accuracy.
(iv) Due to low precision and high backlash, they are noisy.
(v) They are suited for non-critical application.

662. When the pitch angle in Bevel gears is less than 90^o it is referred to as an
(a) external Bevel gear
(b) internal Bevel gear
(c) crown gear
(d) meter gear

TNPSC AE 2014

Ans. (a) : When the pitch angle in Bevel gears is less than 90^o it is referred to as an external Bevel gear.

663. Reduction ratio of a worm gear drive with a 60 teeth wheel and a double start worm with 12 threads is
(a) 60 (b) 30
(c) 12 (d) 5

TNPSC AE 2014

Ans. (b) :

Velocity ratio	Number of starts
20 and above	Single start
12 - 36	Double start
8 - 12	Triple start
6 - 12	Quadruple start
4 - 10	Sextuple start

664. Two mating spur gears have 30 and 90 teeth respectively. The pinion rotates at 1200 rpm and transmits a torque of 20 Nm. The torque transmitted by the gear is
(a) 6.6 Nm (b) 20 Nm
(c) 40 Nm (d) 60 Nm

TNPSC AE 2014

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Ans. (d) : We assume that there is a no loss in power transmission

so, $P_P = P_G$

$[T\omega]_P = [T\omega]_G$

$T_P N_P = T_G . N_G$

$$T_G = 20 \times \frac{90}{30} \qquad \left[\frac{N_P}{N_G} = \frac{Z_G}{Z_P}\right]$$

$\boxed{T_G = 60\ N-m}$

665. For a reverted your train following details are given

(i) Number of teeth on driving gear A and C are 20 and 30 respectively

(ii) Number of teeth on driven gears of B and D are 60 and 50 respectively. if the driving gear 'A' rotates with 600 rpm. the final driven gear D will rotate at

(a) 200 rpm (b) 120 rpm

(c) 300 rpm (d) 240 rpm

GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : $\frac{N_D}{N_A} = \frac{T_A \times T_C}{T_B \times T_D} = \frac{20 \times 30}{60 \times 50}$

$$N_D = \frac{600}{2} \times \frac{2}{5}$$

$$N_D = \frac{240}{2} \text{rpm}$$

$\boxed{N_D = 120 \text{rpm}}$

666. For the design of gear teeth, according to Buckingham equation

(a) $F_d > F_w$ (b) $F_w = F_d$

(c) $F_r > F_w$ (d) $F_w > F_d$

Where F_t = Tangential load

F_w = Wear load

F_d = Dynamic load

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : Tangential tooth load, $F_T = \sigma_b$ mby

Wear load, $F_w = D_p.b.Q.K.$

$$Q = \frac{2G}{G+1}$$

Dynamic load, $F_d = S.C_v. F_{leff}$

S = Service Factor

C_v = Velocity factor

F_{eff} = Effective tangential force

D_p = Dia of pinion

b = Face width of tooth

For safe design

$\boxed{F_w > F_d}$

667. In a conventional lathe, the lead screw has 4 threads per inch. It is required to cut the threads with 2 mm pitch. The train value for the given set of change gear will be

(a) 0.25 (b) 3.175

(c) 0.5 (d) 0.315

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) :

668. The circular pitch of a toothed wheel with 24 teeth and a module of 4.25 mm is :

(a) 1.35 mm (b) 4.25 mm

(c) 6.67 mm (d) 13.35 mm

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : $P_C = \pi m = \pi \times 4.25 = 13.35$ mm

669. The maximum efficiency of a worm and worm wheel system in terms of friction angle □ is :

(a) 1– cos/1 + cos □ (b) 1 – sin □/1 + cos □

(c) 1 – sin □/1 + sin □ (d) 1 – tan □/1 + tan □

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : 1 – sin □/1 + sin □

670. A worm has a lead angle of 22.5^0. This corresponds to a helix angle of :

(a) 22.5^0 (b) 45^0

(c) 55^0 (d) 67.5^0

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : $\lambda = 22.5$ (λ = Lead angle)

$\Psi = 90 - \lambda$ (Ψ = Helix angle)

$= 90 - 22.5 = 67.5$

671. Which one of the following types of gears can be used for speed reduction purpose only

(a) spur gears

(b) bevel gears

(c) helical gears

(d) worm and worm-wheel

Haryana PSC Civil Services Pre, 2014

Ans. (d) : worm and worm wheels are used for speed reduction only because worm and worm wheels have very high velocity ratio.

672. Initial contact between the driving and driven gear teeth occurs at a point where –

(a) addendum circle of driven wheel interacts the line of action

(b) dedendum circle of driven wheel interacts the line of action

(c) dedendum circle of the driver wheel interacts the line of action

(d) addendum circle of the driver wheel interacts the addendum circle of driven gear

Haryana PSC Civil Services Pre, 2014

Ans. (a) : Initial contact between the driving and driven gear teeth occurs at a point where addendum circle of driven gear interacts the line of action.

673. A pair of 20^0 full depth involute spur gears having 30 and 50 teeth respectively and module 4 mm, are in mesh. The addendum of larger gear will be:

(a) 3 mm (b) 6 mm

(c) 9 mm (d) None of these correct

MPPSC State Forest Service Exam, 2014

Ans. (d) : Addendum–It is the radial height of the teeth above the pitch circle. Its standard value is one module. So, addendum = 1 module = 4 mm

674. In hypoid gears, axes of the shaft are.

(a) parallel

(b) intersecting at any angles

(c) intersecting at 90° only

(d) neither parallel nor intersecting

MPPSC State Forest Service Exam, 2014

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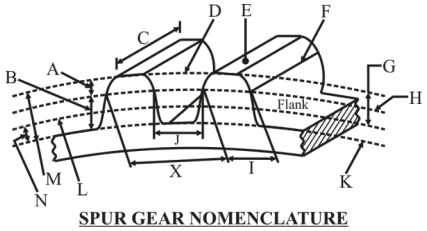
Ans. (d) : Hypoid gear are a type of spiral bevel gear with the difference that hypoid gears have axes, that are non-intersecting & non-parallel.

675. What does top Land refer to in the figure given below?

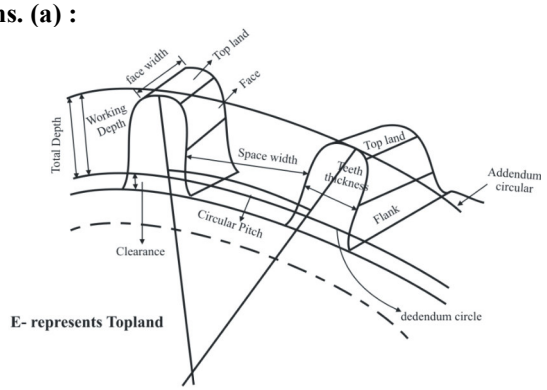


SPUR GEAR NOMENCLATURE

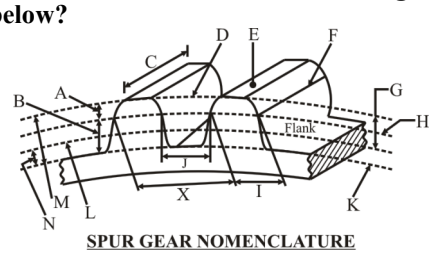
(a) E (b) Both E and F
(c) C and E (d) Only C

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) :



676. What does X refer to in the figure given below?



SPUR GEAR NOMENCLATURE

(a) Circular Pitch (b) Tooth Space
(c) Clearance (d) Face width

MPSC Dy. Engg. (GWD) 19.05.2013

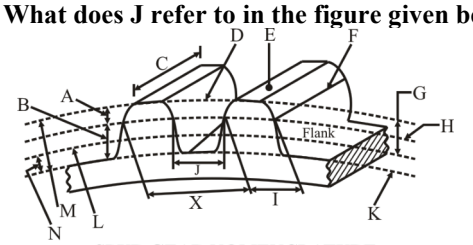
Ans. (a) : X represents circular pitch.

Circular pitch is the distance measured along the circumference of pitch circle from a point an one tooth to a corresponding point on the adjacent tooth.

$$\text{Circular Pitch}\left(P_C\right) = m\pi = \frac{\pi d}{T}$$

m = module
d = Pitch circle dia
T = number of teeth.

677. What does J refer to in the figure given below?



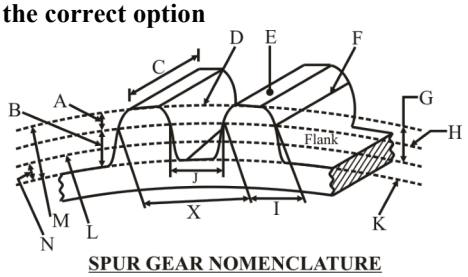
SPUR GEAR NOMENCLATURE

(a) Circular Pitch (b) Tooth Space
(c) Clearance (d) Face width

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : J refers to Tooth space. Tooth space is the space between adjacent tooth.

678. Refer to the figure given below and pick out the correct option



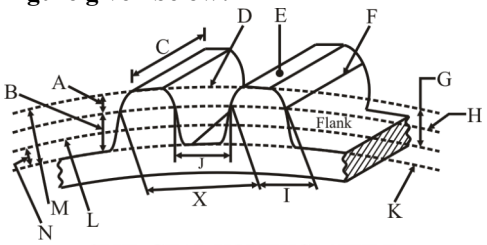
SPUR GEAR NOMENCLATURE

(a) M = A + B – G (b) M = G + N
(c) N = G – A – B (d) G = A + B + M

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : M = G + N
Total depth = working depth + clearance

679. What represents the Tooth thickness in the figure given below?



SPUR GEAR NOMENCLATURE

(a) I (b) X
(c) M (d) N

MPSC Dy. Engg. (GWD) 19.05.2013

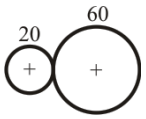
Ans. (a) : Tooth thickness is shown by I

680. Two meshed gears, where the small gear has twenty teeth and big gear has sixty teeth; for every three rotations of small gear, the big gear rotates.

(a) only once (b) three times
(c) nine times (d) none of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) :



$$\frac{N_1}{N_2} = \frac{T_2}{T_1}$$

$$N_2 = N_1 \frac{T_1}{T_2}$$

$$= \frac{20 \times 3}{60} = \frac{60}{60}$$

$$= 1$$

For three rotation of small gear big gear rotates only once.

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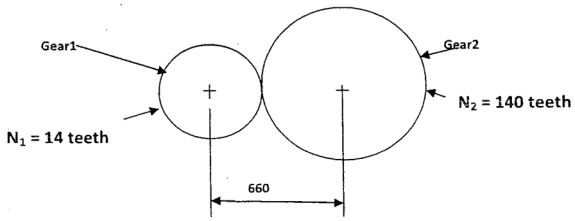
681. In the gear train of 1:10 as shown in the figure the pinion transmits 250 kw at 1800 rpm. What is the tangential load on the gear tooth



(a) 221 kN (b) 22.1 kN
(c) 25.1 kN (d) 251 kN

ISRO Scientist/Engineer 12.05.2013

Ans. (b) :

$$\text{Centre Distance } c = \frac{m}{2}(T_1 + T_2)$$

$$m = \frac{2c}{(T_1 + T_2)} = \frac{2 \times 660}{14 + 140}$$

$$m = 8.57$$

$$\text{Pinion Radius } r = \frac{mT_1}{2} = \frac{8.57 \times 14}{2}$$

$$= 60 \text{ mm}$$

$$\text{Torque} = \frac{P}{\omega} \qquad [\because P = T\omega]$$

$$\text{Torque} = \frac{250 \times 1000}{\frac{2\pi N}{60}}$$

$$T = \frac{250 \times 60 \times 1000}{2\pi N}$$

$$\text{Torque } T = \frac{250 \times 60000}{2 \times 3.14 \times 1800}$$

$$= 1326.964 \text{ N-m}$$

$F_t \times$ pinion radius = 1326.964

$$F_t = \frac{1326.964}{60}$$

$$= 22116\text{N}$$

$F_t = 22.116$kN

682. Compare to spur gears, helical gears:
(a) Run more smoothly
(b) Run with more vibrations and noise
(c) Consume more power
(d) Run exactly alike

SJVN ET 2013

Ans. (a) : Helical gear- Teeth are inclined to the axis of rotation, the angle provides more gradual engagement of the teeth during meshing, transmits motion between parallel shafts.
• The angled teeth engage more gradually than spur gear teeth, causing them to run more smoothly and quietly.

683. To design 12 speed gear box, the minimum number of intermediate shaft is
(a) 3 (b) 2
(c) 1 (d) 4

TNPSC AE 2013

Ans. (b) :

684. A pair of helical gear number of teeth in pinion is 80 and number of teeth in gear is 320. The ratio factor is
(a) 2 (b) 1.6
(c) 1 (d) 2.6

TNPSC AE 2013

Ans. (b) : Ratio factor of gear

$$= \frac{2 \times 320}{320 + 80} = \frac{2 \times 320}{400} = 1.6$$

685. What is the term used in gear arrangement to define the amount by which tooth space exceeds the thickness on an engaging tooth?
(a) Lead (b) Backlash
(c) Run out (d) Pitch Error

TNPSC AE 2013

Ans. (b) : Backlash – Backlash is the amount by which a tooth space exceeds the thickness of a gear tooth engaged in mesh.

686. A motor running at 1200 rpm drives a compressor at 800 rpm through bevel gear arrangement. The pinion has 30 teeth and the number of teeth in the gear is
(a) 18 teeth (b) 45 teeth
(c) 48 teeth (d) 24 teeth

TNPSC AE 2013

Ans. (b) : Given, $N_P = 1200$ rpm, $N_G = 800$ rpm
$Z_P = 30$ $Z_G = ?$
We know that

$$\text{speed ratio} = \frac{N_{\text{Driver}}}{N_{\text{Driver}}}$$

$$\frac{N_P}{N_G} = \frac{Z_G}{Z_P}$$

$$Z_G = \frac{1200}{800} \times 30$$

$$\boxed{Z_G = 45}$$

687. Which type of gears are used in connecting two co-planer and intersecting shafts?
(a) Spur gear
(b) Bevel gear
(c) Helical gear
(d) Worm and worm wheel

UKPSC AE-2013, Paper-I

Ans. (b) : Bevel gear are used in connecting two co-planer and intersecting shafts.

688. Which one of the following is the preferred mode of transmission of power from one shaft to another when distance between the shafts is relatively small–
(a) Gears (b) Belts
(c) Ropes (d) Chains

UKPSC AE-2013, Paper-I

Ans. (a) : Gears are the preferred mode of transmission of power from one shaft to another when distance between the shafts is relatively small.

689. Which pair of gears usually has higher frictional losses–
(a) Spur gears
(b) Helical gears

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(c) Bevel gears
(d) Worm and worm wheel

UKPSC AE-2013, Paper-I

Ans. (d) : Pair of gears usually has higher frictional losses in worm and worm wheel.

690. The outer circle of spur gear is called as:-
(a) Pitch circle (b) Addendum circle
(c) Dedendum circle (d) Base circle

UKPSC AE-2013, Paper-I

Ans. (b) : The outer circle of spur gear is called as addendum circle.

691. Axes of a pair of spur gears are 200 mm apart. The gear ratio is 3:1 and number of teeth on pinion is 20. The module of the gear is:-
(a) 4 mm (b) 5 mm
(c) 8 mm (d) 10 mm

UKPSC AE-2013, Paper-I

Ans. (b) : Given as
$G = 3 : 1$
$x = 200$ mm
$Z_p = 20$
$\frac{Z_G}{Z_p} = 3$
$Z_G = 3 \times 20 = 60$
We know tat
$r_p + r_G = x$
$\frac{m}{2}\left[Z_p + Z_G\right] = 200$
$m\,[20 + 60] = 400$
$\boxed{m = 5 \text{ mm}}$

692. A 20° full depth involute spur pinion of 4 mm module and 21 teeth is to transmit 15 kW at 960 rpm. Its face width is 25 mm.
Given that the tooth geometry factor is 0.32 and the combined effect of dynamic load and allied factors intensifying the stress is 1.5, the minimum allowable stress (in MPa) for gear material is
(a) 242.0 (b) 166.5
(c) 121.0 (d) 74.0

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : Given, $m = 4$ mm, $T_p = 21$
$P = 15$ kW $= 15 \times 10^3$W
$N = 960$ rpm, $b = 25$ mm, $y = 0.32$

$$R = \frac{mT_2}{2} = \frac{4\times10^{-3}\times21}{2} = 42\times10^{-3}\,m$$

$P = \omega T = F_T \times R \times \omega$

$$15\times10^3 = F_t \times 42\times10^{-3} \times \frac{2\pi\times960}{60}$$

$F_t = 3552.26$ N
Force transmitted

$$F_t = \frac{\sigma_b \times b \times y \times m}{f_s} \Rightarrow 3552 = \frac{\sigma_b \times 25 \times 0.32 \times 4}{1.5}$$

$\sigma_b = 166.5$ MPa

693. The cone distance and face width of the bevel gear are 200 mm and 50 mm respectively. What is the bevel factor ?
(a) 0.25 (b) 0.5
(c) 0.75 (d) 1

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

$$\text{Bevel factor} = \frac{L-B}{L}, \quad L = \text{Cone distance} = 200$$

B = Face width = 50

$$= \frac{200-50}{200}$$
$$= 0.75$$

694. When the addenda on pinion and wheel is such that the path of approach and path of recess are half of their maximum possible values, then the length of the path of contact is given by
Where r = Pitch circle radius of the pinion
R = Pitch circle radius of the driver and
ϕ = Pressure angle
(a) $\frac{(r+R)\sin\phi}{2}$ (b) $\frac{(r+R)\cos\phi}{2}$
(c) $\frac{(r+R)\tan\phi}{2}$ (d) $\frac{(r+R)\sec\phi}{2}$

TNPSC AE (Industries) 09.06.2013

Ans. (a) : $\frac{(r+R)\sin\phi}{2}$

695. Lewis equation in spur gears is applied :
(a) To weaker of the pinion or gear
(b) To the gear
(c) To the pinion
(d) to stronger of the pinion or gear

TNPSC AE (Industries) 09.06.2013

Ans. (a) : To weaker of the pinion or gear

696. In a spur gear arrangement, the diameter of the pinion is 100 mm and its number of teeth is 40. The diameter of the gear is 50 mm and its number of teeth is 20. Determine whether the above gears mesh correctly :
(a) Do not mesh
(b) Mesh correctly
(c) Transmit less power
(d) Produces the backlash

TNPSC AE (Industries) 09.06.2013

Ans. (b) : Mesh correctly it $m_1 = m_2$

$$m_1 = \frac{D_1}{T_1} = \frac{100}{40} = \frac{5}{2}$$

$$m_2 = \frac{D_2}{T_2} = \frac{50}{20} = \frac{5}{2}$$

697. An engine running at 150 r.p.m. drives a shaft with belt arrangement. If diameter of engine pulley is 55 cm and shaft pulley 33 cm, find the speed of shaft:-
(a) 100 rpm (b) 150 rpm
(c) 200 rpm (d) 250 rpm

UKPSC AE-2013, Paper-I

Ans. (d) : Given as
$N_1 = 150$ rpm
$d_1 = 55$ cm
$d_2 = 33$ cm
We know that

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$$\frac{N_1}{N_2}=\frac{d_2}{d_1}$$

$$N_2=N_1\times\frac{d_1}{d_2}$$

$$N_2=150\times\frac{55}{33}$$

$N_2 = 250$ rpm

698. The blank size of the gear of 3 mm module and 40 number of teeth is

(a) 13.33 mm (b) 120 mm
(c) 123 mm (d) 129 mm

APPSC Poly. Lect. 2013

Ans. (*) : Blank size of the gear, T = 40, m = 3 mm
$d = mT = 3 \times 40 = 120$
Blank size also known as outside diameter
Blank size = $d + 2m = m \times (T + 2)$
Blank size = $3 \times (40 + 2)$
Blank size = 126 mm

699. For a cycloidal tooth profile, pressure angle at (i) Commencement of engagement, (ii) Pitch point and (iii) End of engagement will be :

(a) constant
(b) zero, maximum, zero
(c) maximum, zero, maximum
(d) zero, zero, maximum

J&K PSC Civil Services Pre, 2013

Ans. (c) : In an involute tooth profile, pressure angle remain constant from engagement to disengagement but in a cycloidal tooth profile pressure angle changes from maximum at engagement and zero at pitch point and again become maximum at end of engagement.

700. A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gears is involute with 20º pressure angle, 12 mm module and 10 mm addendum. The radius of addendum circle of pinion is

(a) 180 mm (b) 190 mm
(c) 200 mm (d) 210 mm

APPSC Poly. Lect. 2013

Ans. (b) : t = 30, T = 80, $\phi = 20^o$, m = 12 mm, Ad = 10 mm

$$r_{pinion}=\frac{m.t}{2},\ R_{gear}=\frac{m.T}{2}=\frac{12\times80}{2}=480mm$$

$$r_{pinion}=\frac{12\times30}{2}=180mm$$

$\rightarrow$ Radius of addendum circle of pinion (r_A) = r + Addendum = (180 + 10 = 190 mm)
$\rightarrow$ Radius of addendum circle of gear (R_A) = R + Addendum = 480 + 10
$R_A = 490$ mm

701. Which of the following gears is not employed to connect non-parallel and non-intersecting shafts?

(a) Worm and worm gear
(b) Hypoid gears
(c) Double helical gears
(d) Crossed helical gears.

RPSC AEN Pre-2013

Ans. (c) : A double helical gear is equivalent to a pair of helical gear secured together, one having a right-hand helix and other having a left-hand helix. Axial thrust which occurs in case of single helical gears is eliminated in double helical gears.

702. Which of the following is not correct for involute gears?

(a) The path of contact of two gears during mesh is along a straight line.
(b) Slight change of centre distance does not affect correct tooth action.
(c) The pressure angle remains constant.
(d) Interference is inherently absent.

RPSC AEN Pre-2013

Ans. (d) : An involute is defined as the locus of a point on a straight line which rolls without slipping on the circumference of a circle.
In involute gears-
- The path of contact of two gears during mesh is along a straight line.
- Slight change of centre distance does not affect correct tooth action.
- The pressure angle remains constant.
- **Interference**–When two gears are in mesh at one instant there is a chance to mate involute portion with non-involute portion of mating gear. This phenomenon is known as "interference".

703. A 2 kW motor is running at 1440 rpm. It is to be connected to a stirrer running at 36 rpm. The gearing arrangement suitable for this application to

(a) differential gear (b) helical gear
(c) spur gear (d) worm gear

TRB Poly. Lect., 2012

Ans. (d) : Velocity reduction $=\frac{36}{1440}=\frac{1}{40}$ for, this kind of reduction worm gear arrangement is most suitable

704. The gear reduction in a worm gear set with worm gear of 50 teeth and worm of double lead thread would be

(a) 50 : 1 (b) 100 : 1
(c) 25 : 1 (d) 40 : 1

TRB Asstt. Prof., 2012

Ans. (b) : The gear reduction in a worm gear set with worm gear of 50 teeth and worm of double lead thread would be 100 : 1.

705. In spur gear drive the number of teeth in pinion is assumed between 16-20 to avoid

(a) break off due to fatigue
(b) pitting of gears
(c) interference
(d) seizure

TRB Asstt. Prof., 2012

Ans. (c) : In spur gear drive the number of teeth in pinion is assumed between 16-20 to avoid interference.

706. In a spur gear, if t = thickness of tooth at bottom, h = length of tooth as cantilever from bottom to the point of application of load, m = module of tooth, then Lewis from factor is given by :

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(a) $6\,t^2/hm$ (b) $t^2/6\,hm$
(c) $hm/6t^2$ (d) None of these

PSPCL AE, 2012

Ans. (b) : A cantilever of depth h and t thickness and width 'b'

Bending stress $\sigma_w = \frac{(w_T \times h) \times t/2}{bt^3/12}$

$\sigma_w = w_T \times \frac{h}{bt^2} \times 6$

or $w_T = \sigma_w \times \frac{bt^2}{6h}$

and according Lewis equation, $w_T = \sigma_w \times b \times m \times y$

Where, y = Lewis form factor

equating $\frac{t^2}{6h} = my$ or $y = \frac{t^2}{6hm}$

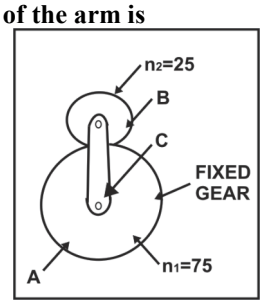
707. Surface hardness of a spur gear tooth is 250 BHN, what is its stress load factor.

(a) 1.0 (b) 0.8
(c) 0.64 (d) None of these

PSPCL AE, 2012

Ans. (a) : Stress load factor $k = 0.16\ [BHN/100]^2$
$= 0.16\ [250 \div 100]^2 = 1$

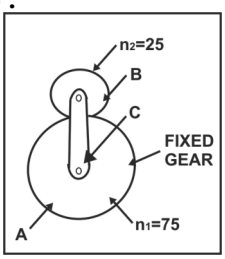
708. Planetary gear is in contact with the fixed gear as shown in the figure. The number of rotations made by the planetary gear for one rotation of the arm is



(a) 3 (b) 4
(c) 1 (d) 1/3

ISRO Scientist/Engineer 2012

Ans. (b) :



Taking $n_1 = 75$, $n_2 = 25$

Condition	Arm	A = 75	B = 25
Arm fixed	0	+x	$-\frac{75}{25}x = -3x$
Arm free	y	y + x	y - 3x

y + x = 0 and y = 1
x = -1
$\omega_B = 1 - 3x$
$= 1 - 3 \times (-1)$
$\omega_B = 4$

709. The standard value of addendum is

(a) 2 module (b) 1.157 module
(c) 1 module (d) 3 module

BPSC AE 2012 Paper - VI

Ans : (c) : The standard value of addendum is 1 module.

710. Spur gears have/are

(a) straight teeth perpendicular to the axis.
(b) curved teeth perpendicular to the axis.
(c) not subjected to axial thrust due to tooth load.
(d) subjected to axial thrust due to tooth load.

UKPSC AE 2012 Paper-I

Ans. (c) : Spur gears have/are not subjected to axial thrust due to tooth load.

711. When teeth formed on the cones are straight, the gears are known as

(a) worm gear (b) helical gear
(c) straight bevel (d) spiral bevel

UKPSC AE 2012 Paper-I

Ans. (c) : When teeth formed on the cones are straight, the gears are known as straight bevel.

712. When two gear teeth are in mesh, then pure rolling occurs at the

(a) root of tooth (b) tip of tooth
(c) pitch point (d) flank

UKPSC AE 2012 Paper-I

Ans. (c) : When two gear teeth are in mesh, then pure rolling occurs at the pitch point.

713. The module is the reciprocal of

(a) diametral pitch (b) circular pitch
(c) addendum (d) deoendum

TNPSC ACF 2012

Ans. (a) : $m = \frac{D}{T}$ $P_e = \frac{T}{D}$

$\boxed{m = \frac{1}{P_d}}$

714. The spur gears are used for gear ratios up to

(a) 6 (b) 2
(c) 10 (d) 20

TNPSC ACF 2012

Ans. (a) : Law velocity < 3 MIS
Medium Velocity 3 to 15 MIS –spur gear
High Velocity > 15 MIS worm gear

715. Herringbone gears are virtually

(a) spur gear (b) miter gear
(c) stub teeth gear (d) double helical gear

TNPSC ACF 2012

Ans. (d) : Herringbone gears : (Double Helical gear)- If there is no groove in between the gear is known as herringbone gear.

716. The bevel gears used for connecting non-intersecting shafts are

(a) Miter gears (b) Hypoid gears
(c) Spiral bevel gears (d) Zerol gears

TNPSC ACF 2012

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Ans. (b) :
Hypoid gears– (Non parallel, non intersecting)
Miter years– Intersecting gear
Gear of same size and connects shafts at 90º are known as miter gears.
Spiral bevel gear– When teeth are inclined, they are known as spiral level gear.

717. Efficiency of work gear used for hoist is less than
(a) 90% (b) 70%
(c) 60% (d) 50%

TNPSC ACF 2012

Ans. (d) : Efficiency of worm gear

$$\eta_{max} = \frac{1-\sin\theta}{1+\sin\theta} < 50\%$$

718. The outside diameter of an involute gear is equal to pitch circle diameter plus
(a) 2 addendum
(b) 2 dedendum
(c) addendum + dedendum
(d) 2.157 module

APPSC AEE 2012

Ans. (a) : Outside diameter (D_0) = D_{pitch} + 2 addendum.

719. Static design of gears uses
(a) Johnson equation
(b) Rankine equation
(c) Buckingham's equation
(d) Lewis equation

APPSC AEE 2012

Ans. (d) : Lewis equation-
Beam strength of gear tooth
$(f_t)_{max}$= b.m.y $(\sigma)_{permissible}$
(y) = Lewis form factor
Where, $\left\{ y = \frac{t^2}{6hm} \right\}$
Beam strength is define as a maximum value of tangential load that a gear tooth can bear without any bending failure.
Safe condition $F_{actual} \le$ Beam strength $(f_t)_{max}$.

720. In the formulation of Lewis equation for toothed gearing, it is assumed that tangential tooth load 'F_1' acts on the :
(a) pitch point
(b) tip of the tooth
(c) root of the tooth
(d) whole face of the tooth

APPSC AE 04.12.2012
APPSC AEE 2012

Ans. (a) : Lewis equation for tooth gearing, it is assumed that tangential tooth load 'F_1' acts on the pitch point.

$$F_1 = \frac{P}{v} \times C_s$$

Where, P = Power transmitted in watts
v = Pitch line velocity in m/s
C_s = Service factor

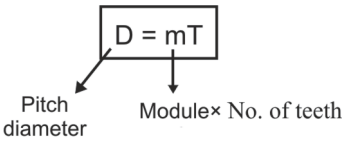
721. Pitch diameter is equal to product of
(a) Circular pitch and number of teeth
(b) Working depth and number of teeth
(c) Clearance and number of teeth
(d) Module and number of teeth

ISRO Scientist/Engineer 2011

Ans. (d) : Pitch diameter = ?

$$\text{Module (m)} = \frac{D}{T}$$

Or



722. What is the face of the gear tooth
(a) Surface of the top of tooth
(b) Surface of tooth above the each surface
(c) Width of tooth below the surface
(d) Width of tooth pitch circle

VIZAG STEEL MT 2011

Ans. (a) : Surface of the top of tooth

723. The contact ratio or engagement factor in case of gears should be :
(a) 1 (b) 1.3 to 1.5
(c) 1.8 to 2.0 (d) 2.0 to 2.5

UPSC JWM Advt. No.-52/2010

Ans. (b) : Contact ratio or engagement factor one pair is engaged in full engagement period. But 21% to 50% times of this engagement period is like that in which along with this pair one more pair, i.e., total two pairs are engaged. Therefore number of pairs engaged in one engagement period its average value comes out to be 1.3 to 1.5.

724. In spiral gears, if α_1 and α_2 are spiral angles and θ is the shaft angle efficiency is maximum when :
(a) $\alpha_1 = \theta + \phi$, $\alpha_2 = \theta - \phi$
(b) $\alpha_1 = (\theta + \phi)/4$, $\alpha_2 = (\theta - \phi)/4$
(c) $\alpha_1 = (\theta + \phi)/2$, $\alpha_2 = (\theta - \phi)/2$
(d) $\alpha_1 = \theta + \phi \tan 45°$, $\alpha_2 = \theta - \phi \tan 45°$

UPSC JWM Advt. No.-52/2010

Ans. (c) : In spiral gears, if α_1 and α_2 are spiral angles and θ is the shaft angle efficiency is maximum when
$\alpha_1 = (\theta + \phi)/2$
$\alpha_2 = (\theta - \phi)/2$

725. Match List I with List II and select the correct answer using the code given below the lists :

	List I		List II
A.	Double-helical gears connecting parallel shafts	1.	Rack and pinion
B.	Two non-parallel or intersecting but coplanar shafts	2.	Spiral gear
C.	Two non-intersecting and non- parallel shafts connected by gears	3.	Bevel gear
D.	Gear of a shaft meshing externally with a straight toothed member	4.	Herringbone gear

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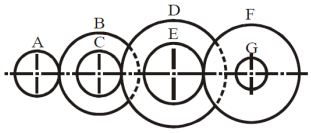
Code:

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	2	3	1
(c)	1	3	2	4	(d)	4	3	2	1

UPSC JWM Advt. No.-50/2010

Ans. (d) :
- Bevel gear are, two non-parallel or intersecting but coplanar shafts.
- Spiral gear are, two non-intersecting & non-parallel shafts connected by gears.
- Rack and pinion, are gear of a shaft meshing externally with a straight toothed member.
- Herringbone gear is a double helical gears connecting parallel shaft.

726.



The gearing arrangement used in a machine-tool device is shown in the figure. The motor shaft is connected to gear A and rotates at 750 r.p.m. The gear wheels B, C, D and E are fixed to parallel shafts. The last gear F is fixed on the output shaft. The number of teeth on each wheel is as given below :

Gear	A	B	C	D	E	F
No. of teeth	20	50	25	75	30	60

The speed of the output shaft on which the gear G is mounted is

(a) 200 r.p.m. (b) 100 r.p.m.
(c) 80 r.p.m. (d) 50 r.p.m.

UPSC JWM Advt. No.-50/2010

Ans. (d) : Given, $N_A = 750$ rpm

For (A, B) –

$$\frac{\omega_A}{\omega_B} = \frac{T_B}{T_A} \quad .....(1)$$

For (C, D) –

$$\frac{\omega_C}{\omega_D} = \frac{T_D}{T_C} \quad .....(2)$$

For (E, F) –

$$\frac{\omega_E}{\omega_F} = \frac{T_F}{T_E} \quad .....(3)$$

(1) × (2) × (3) –

$$\frac{\omega_A}{\omega_B} \times \frac{\omega_C}{\omega_D} \times \frac{\omega_E}{\omega_F} = \frac{T_B}{T_A} \times \frac{T_D}{T_C} \times \frac{T_F}{T_E}$$

$$\frac{\omega_A}{\omega_F} = \frac{T_B}{T_A} \times \frac{T_D}{T_C} \times \frac{T_F}{T_E} \quad \left[\because \omega_B = \omega_C,\ \omega_D = \omega_E, \omega_F = \omega_G\right]$$

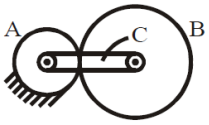
$$\frac{\omega_A}{\omega_G} = \frac{T_B}{T_A} \times \frac{T_D}{T_C} \times \frac{T_F}{T_E}$$

or $$\frac{N_A}{N_G} = \frac{T_B}{T_A} \times \frac{T_D}{T_C} \times \frac{T_F}{T_E}$$

$$= \frac{50}{20} \times \frac{75}{25} \times \frac{60}{30} = \frac{5}{2} \times 3 \times 2 = 15$$

$$N_G = \frac{750}{15} = 50\,\text{rpm} \quad \boxed{N_G = 50\,\text{rpm}}$$

727.



The figure shows the gear wheel A with the number of teeth T_A fixed to the frame. The arm C with the gear wheel B which has the number of teeth T_B revolves about the gear wheel A. The ratio of the speed of gear B to the speed of the arm is given by

(a) $\frac{N_B}{N_C} = 1 + \frac{T_B}{T_A}$ (b) $\frac{N_B}{N_C} = 1 + \frac{T_A}{T_B}$

(c) $\frac{N_B}{N_C} = 1 - \frac{T_B}{T_A}$ (d) $\frac{N_B}{N_C} = 1 - \frac{T_A}{T_B}$

UPSC JWM Advt. No.-50/2010

Ans. (b) :

Arm C	Gear A	Gear B
0	+x	$-x\frac{T_A}{T_B}$
y	y + x	$y - x\frac{T_A}{T_B}$

A is fixed so,

$y + x = 0,\ y = -x$

So gear B,

$$= y + y\frac{T_A}{T_B}$$

$$\frac{N_B}{N_C} = \frac{y\left[1 + \frac{T_A}{T_B}\right]}{y}$$

$$\boxed{\frac{N_B}{N_C} = 1 + \frac{T_A}{T_B}}$$

728. Consider the following statements :
The form factor of a spur gear tooth depends upon the :
(I) Number of teeth
(II) Pressure angle
(III) Addendum modification coefficient
(IV) Circular pitch
Of these statements :
(a) I and III are correct
(b) II and IV are correct
(c) I, II and III are correct
(d) I and IV are correct

APPSC IOF, 2009, ESE 1996

Ans. (c) : The value of Lewis form factor used in design of gears is dependent on number of teeth of a gear, geometry of tooth profile, pressure angle and doesn't depend on circular pitch.

729. Which of the following statements are correct?
(I) For constant velocity ratio transmission between two gears, the common normal at the point of contact must always pass through a fixed point on the line joining the centres of rotation of the gears.

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(II) For involute gears to pressure angle changes with change in centre distance between gears
(III) The velocity ratio of compound gear train depends up on the number of teeth of the input and output gears only
(IV) Epicyclic gear trains involve rotation of at least one gear axis about some other gear axis.

Select the correct answer using the codes given below :

(a) I, II and III (b) I, III and IV
(c) I, II and IV (d) II, III and IV

APPSC IOF, 2009, ESE 1996

Ans. (c) :

730. Ratio of pitch circle diameter in millimeters to the number of teeth, is known as

(a) Module (b) Circular pitch
(c) Diametric pitch (d) Clearance

ISRO Scientist/Engineer 2009

Ans. (a) : Module = m

$$m = \frac{D}{T}(mm)$$

where D = diameter takes always in (mm)

Module (m) is defined with ratio diameter (D) and its number of teeth."

Note: For two mating gear, $m_1 = m_2$

731. The blank size of the gear of 3 mm module and 40 number of teeth is

(a) 120 mm (b) 123 mm
(c) 126 mm (d) 129 mm

ISRO Scientist/Engineer 2009

Ans. (c) : We know that pitch circle diameter is given by

$D = m \times T$

$D = 3 \times 40 = 120$

Blank size from which the gear is cut given by

$D_B = D + 2 \times m$

$= 120 + 2 \times 3 = 126$ mm

732. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I		List-II
(A)	Involute profile	1. Is a straight line
(B)	Profile for the basic rack for involute	2. Pitch circle
(C)	Tooth thickness is measured at	3. Base circle
(D)	Tooth base pitch is measured at	4. Cannot exist inside its base circle

Code :

	A	B	C	D
(a)	3	1	2	4
(b)	4	1	2	3
(c)	3	2	1	4
(d)	4	2	1	3

UPSC JWM Adv. No-16/2009

Ans. (b) : (A) Involute profile – Cannot exist inside its base circle

(B) Profile for the basic rack – It is a straight line **for involute.**

(C) Tooth thickness is measured at – Pitch circle

(D) Tooth base pitch is measured at – Base circle

733. Which one of the following statement is correct?

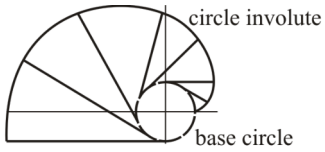
The radius of curvature of the involute profile of gear tooth is:

(a) Constant
(b) Maximum at base circle
(c) Zero at base circle
(d) Maximum at middle

UPSC JWM Adv. No-16/2009

Ans. (c) : The radius of curvature of the involute profile of gear tooth is zero at Base circle.

An involute, specifically a circle involute, is a geometric curve that can be described by the trace of unwrapping a tight string which is tangent to a circle, known as the base circle.



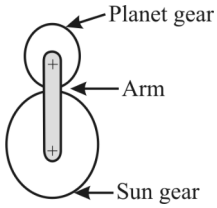
Circle involute from an unwrapped string

734. In an epicyclic gear train with a sun gear, planet gear and a moving arm, the ratio of the number of teeth of the sun gear to that of the planet gear is 3. If the planet gear is fixed and the arm has an angular velocity of 150 rpm, the sun gear will rotate with

(a) 50 rpm (b) 100 rpm
(c) 150 rpm (d) 200 rpm

DRDO Scientists 2009

Ans. (d) :



Arm	Sun gear	Planet gear
0	x	$-x\frac{T_S}{T_P}$
y	x + y	$y - x\frac{T_S}{T_P}$

Given,

$N_P = 0$, $N_A = 150$ rpm,

$y - 3x = 0$

$\Rightarrow y = 3x$, $y = 150$ rpm, $z = 50$ rpm

$N_S = x + y = 150 + 50$

$= 200$ rpm

735. A gear set consists of a 16-tooth pinion driving a 40-tooth gear. The modules is 12 mm. The addendum and dedendum are 12 mm and 15 mm, respectively and the gears are cut using 20° pressure angle. The centre distance is

(a) 36 mm (b) 136 mm
(c) 236 mm (d) 336 mm

DRDO Scientists 2009

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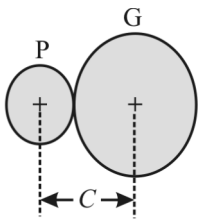
Ans. (d) :



$T_P = 16, T_G = 40, m = 12$

$r_p = \frac{mT_p}{2}, \quad r_G = \frac{mT_G}{2}$

$C = r_P + r_G$

$= \frac{m}{2}(T_P + T_G) = 6 \times 56$

$= 336$ mm

736. Two measuring gears must have same
(a) number of teeth (b) addendum
(c) dedendum (d) module

TNPSC AE, 2008

Ans. (d) : Two measuring gears must have same module and these tooth must have conjugate profile otherwise law of gearing will not satisfy.

737. Gears which connect inclined shafts, which if produced, would intersect at some angle in the same plane are known as
(a) spur gears (b) bevel gears
(c) spiral hypoid gears (d) worm wheels

TNPSC AE, 2008

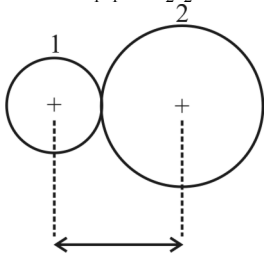
Ans. (b) : Gears which connect inclined shafts, which if produced, would intersect at same angle in the same plane are known as bevel gears.

738. Two parallel shafts whose axes are separated by a distance of 75 mm are to be connected by a spur gear set so that the output shaft rotates at 50% of the speed of the input shaft. Which of the following could be the possible pitch circle diameters of the gears?
(a) 25 mm and 50 mm
(b) 30 mm and 60 mm
(c) 50 mm and 100 mm
(d) 60 mm and 120 mm

DRDO Scientists 2008

Ans. (a) : By law of gearing

$\omega_1 r_1 = \omega_2 r_2$



$\omega_1 = 0.5\ \omega_2$

$\therefore \quad r_1 = 0.5\ r_2$

$r_1 + r_2 = 75$

$1.5\ r_2 = 75 \Rightarrow r_2 = 50$ mm

$r_1 = 25$ mm

739. Two skew shafts may be connected for power transmission through
(a) spur gearing
(b) worm gearing
(c) helical gearing
(d) miter gearing

WBPSC AE, 2007

Ans. (b) : Two skew shafts may be connected for power transmission through worm gearing.

740. Efficiency in a normal spur gear pair will be
(a) above 98%
(b) 70 to 80%
(c) less than 50%
(d) less than 30%

UKPSC AE 2007 Paper -I

Ans. (a) : Efficiency in a normal spur gear pair will be above 98%.

741. In an epicyclic gear train, the number of planets in any given row will be
(a) equal to 1
(b) equal to or more than 1
(c) more than 1
(d) more than 2

UKPSC AE 2007 Paper -I

Ans. (c) : In an epicyclic gear train, the number of planets in any given row will be more than 1.

742. In a single involute gear manufacturing the following two are predetermined:
(a) Addendum circle and pitch circle diameter
(b) Base circle and addendum circle diameter
(c) Pitch circle and dedundum circle diameter
(d) None of the above

UKPSC AE 2007 Paper -I

Ans. (b) : In a single involute gear manufacturing the following two are predetermined base circle and addendum circle diameter.

743. Normal efficiency in a single reduction worm gear pair with the velocity ratio 70 would be
(a) above 98% (b) 50 to 80%
(c) less than 30% (d) less than 10%

UKPSC AE 2007 Paper -I

Ans. (b) : Normal efficiency in a single reduction worm gear pair with the velocity ratio 70 would be 50 to 80%.

744. Calculate the tooth width of a gear having 30 number of teeth and with module 5 mm.
(a) 7.85 mm (b) 8.50 mm
(c) 6.352 mm (d) 5.55 mm

UKPSC AE 2007 Paper -I

Ans. (a) : Circular pitch = Tooth thickness + Space width

$P_c = 2 \times$ Tooth thickness

Tooth thickness $= \frac{P_c}{2} = \frac{5 \times 3.14}{2} = 7.85$ mm

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5. Belts, Ropes and Chains

745. Creep in belt is due to-

(a) Material of the pulley
(b) Larger size of the driver pulley
(c) Uneven extensions and contractions due to varying tension
(d) Expansion of belt

Assam PSC AE (PHED), 18.10.2020
Kerala PSC AE (Irrigation) 05.04.2017
GPSC Lect. 23.10.2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
NPCIL ET, 2015
Kerala PSC AE 06.08.2015
GWSSB AAE, 27.12.2015
UPSC JWM Adv. No-16/2009
TNPSC AE, 2008 / ESE 2001

Ans. (c) : Creep in belt drives when the belt passes from the slack side to the tight side a certain portion of the belt extends. Similarly when the belt passes from the tight side to slack side it contracts. This phenomenon of alternative stretching and contraction of the belt result in relative motion between the belt and the pulley surface. This relative motion is called creep.

746. The centrifugal tension in belts :

(a) increases power transmitted
(b) decreases power transmitted
(c) have no effect on the power transmitted
(d) increases power transmitted unto a certain speed and then decreases.

TNPSC AE 2018
HPPSC Asstt. Prof. 20.11.2017
GMB AAE 25.06.2017
Nagaland CTSE 2017 Paper-I
UPPSC AE 12.04.2016 Paper-I
TNPSC AE Industries 09.06.2013
ISRO Scientist/Engg. 2009
WBPSC AE 2008

Ans. (c) : The centrifugal tension in belts have no effect on the power transmitted.

$$\left[\text{But theoretically, } P = (T - T_C)V\left(1 - \frac{1}{e^{\mu\theta}}\right)\right]$$

Power transmit is decreased. But it is very less. So that we can neglect it.

747. The objective of 'crowning' of the flat pulleys of belt drive is to–

(a) prevent the belt from running off the pulley
(b) increases the power transmission capacity
(c) increase the belt velocity
(d) prevent the belt joint from damaging the belt surface

ISRO Scientist/Engg. RAC 10.03.2019
Nagaland PSC CTSE 2017, Paper-I
HPPSC W.S. Poly. 2016
APPSC AEE MAINS 2016, PAPER-III
Rajasthan Nagar Nigam AE 2016 Shift-I
(KPSC AE 2015)
Kerala PSC AE 06.08.2015
(HPPSC AE 2014)
RPSC ACF 2011

Ans. (a) : The objective of 'crowning' of the flat pulleys of belt drive is to prevent the belt from running off the pulley.

748. Which of the following is the condition of maximum power transmission for belt? [T = Maximum permissible tension in belt, T_C = Centrifugal tension, V_{max} = velocity at maximum power transmission, m = mass/length of belt]

(a) $T = 3T_c;\ V_{max} = \frac{T}{3m}$
(b) $T = 3T_c;\ V_{max} = \sqrt{\frac{T}{3m}}$
(c) $T_c = 3T;\ V_{max} = \sqrt{\frac{T}{3m}}$
(d) $T_c = 3T;\ V_{max} = \frac{T}{3m}$
(e) $T = T_c;\ V_{max} = \frac{T}{m}$

MECON MT 2019
UPPSC AE 13.12.2020 Paper-I
Gujarat PSC AE 2019
CGPSC AE 25.02.2018
GPSC EE Pre, 28.01.2017
RPSC AE (GWD) 06.05.2016
UKPSC AE 2013

Ans. (b) : condition of maximum power transmission for belt is given as

$$T = 3T_c;\ V_{max} = \sqrt{\frac{T}{3m}}$$

749. In flat belt drive the condition for maximum power transmission is given by: (where T maximum tension and T_c centrifugal tension in belt)

(a) $T = 3T_c$ (b) $T = 2T_c$
(c) $T = \pi T_c$ (d) $T = 3\pi T_c$

CIL MT 26.03.2017
Punjab PSC (Lect.) 06.08.2017
RPSC AE 2016
TAN GEDCO AE 2015
Mizoram PSC AE/SDO 2014
Kerala PSC Poly. Lect. 09.01.2014
ISRO Scientist/Engg. 2012
TNPSC ACF 2012
RPSC ACF-2011
UKPSC AE 2007 Paper-I / ESE 2008

Ans. (a) : Centrifugal Tension: Since the belt continuously runs over the pulleys, therefore some centrifugal force is caused, whose effect is to increase the tension on both the tight as well as the slack sides. The tension caused by centrifugal force is called centrifugal tension (T_c). For maximum power transmission $T = 3T_c$

750. Idler pulley is used

(a) For changing the direction of motion of the belt
(b) For applying tension
(c) For increasing velocity ratio
(d) All of the above

APPSC AEE Screening Test 2016
RPSC AE 2016
RPSC ACF-2011
ISRO Scientist/Engineer 2007

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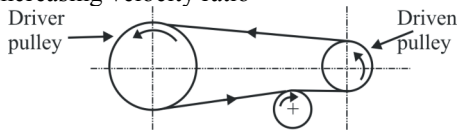
Ans. (d) : Idler pulley is used–
- For changing the direction of motion of the belt
- For applying tension
- For increasing velocity ratio



751. Creep in a belt drive is due to :
(a) improper crowning
(b) plasticity of belt material
(c) differential elongation of belt due to difference in tension on two sides of the pulley
(d) Change in the coefficient of friction due to over heating

GPSC Lect. 23.10.2016
KPSC AE 2015
Kerala PSC AE 06.08.2015
TNPSC AE 2014
UKPSC AE 2012 Paper-I

Ans : (c)
Creep in a belt is due to differential elongation of belt due to difference in tension on two sides of the pulley.

Creep of belt : $\frac{N_2}{N_1}=\frac{d_1}{d_2}\times\frac{E+\sqrt{\sigma_2}}{E+\sqrt{\sigma_1}}$

σ_1 and σ_2 = stress in the belt on the fight and slack side respectively
E = Young's modulus for the material of the belt

752. A 28 mm 6 × 7 haulage wire rope indicates :
(a) 6 regular lay and 7 long lay
(b) 7 regular lay and 6 long lay
(c) 6 number of strands and 7 wires in each strand
(d) 7 number of strands and 6 wires in each strand

GPSC Asstt. Prof. 28.08.2016
GPSC AE 26.07.2015
Kerala PSC Poly. Lect. 09.01.2014
UPSC JWM Advt. No.-52/2010

Ans. (c) : A 28 mm 6 × 7 haulage wire rope indicates 6 number of strands and 7 wires in each strand

753. The included angle in the V-belt drive ranges from
(a) 15^o to 30^o (b) 30^o to 45^o
(c) 45^o to 60^o (d) 60^o to 90^o

HPPSC AE 2018
APPSC Poly. Lect. 2013
UKPSC AE 2007 Paper -I

Ans. (b) : The included angle in the V-Belt drive ranges from 30^o to 45^o.

Ratio of tensions for V-belt $\left(\frac{T_1}{T_2}\right)=e^{(\mu\,\mathrm{cosec}\,\alpha)\theta}$

θ = Angle of contact, α = semi-angle of groove.
- They are efficient-performing with an average of 94-98% efficiency.
- Higher velocity ratio (up to 10) can be obtained.

754. In a flat belt drive, slip between the driver and belt is 1% and that between belt and follower is 3%. If the pulley diameters are same, the velocity ratio of the drive is:-
(a) 0.99 (b) 0.98
(c) 0.97 (d) 0.96

UKPSC AE-2013, Paper-I
APPSC AEE 2012
APPSC IOF 2009

Ans. (d) : Velocity ratio

$$\frac{N_2}{N_1}=\frac{d_1}{d_2}\left[1-\frac{S}{100}\right]$$

where

$$S=S_1+S_2-0.01\,S_1S_2$$
$$S=1+3-0.01\times1\times3$$
$$S=3.97$$
$$\frac{N_2}{N_1}=1\times\left[1-\frac{3.97}{100}\right]$$
$$\frac{N_2}{N_1}=0.9603$$

755. Pulley in a belt drive act as
(a) Cylindrical Pair (b) Turning Pair
(c) Rolling Pair (d) Sliding Pair

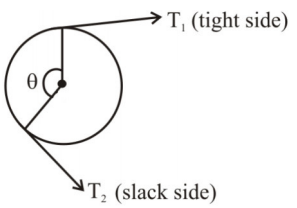
Rajasthan Nagar Nigam AE 2016, Shift-I
RPSC AE 2016
RPSC ACF-2011

Ans. (c) : Rolling Pair– When the links of a pair have a rolling motion relative to each other they form a rolling pair e.g. rolling wheel on a flat surface, ball and roller bearings etc.
- **Turning Pair–** When one link has a turning or revolving motion relative to the other, they constitute a turning or revolving pair.
  Example- in slider – crank mechanism, all pairs except the slider and guide pair are turning pairs.
- **Spherical Pair–** When one link in the form of a sphere turn inside a fixed link, it is a spherical pair.
  Example- ball in a socket joint.

756. It T_1 is the tension on tight side and T_2 is the tension on slack side of a belt over a pulley and it 'θ' is the angle of lap of the belt around the pulley, which expression holds true to relate tensions
(a) $\frac{T_2}{T_1}=e^{\mu\theta}$ (b) $T_1-T_2=e^{\mu\theta}$
(c) $T_2-T_1=e^{\mu\theta}$ (d) $\frac{T_1}{T_2}=e^{\mu\theta}$

GPSC DEE Class-II (GWSSB) 04.07.2021
HPPSC AE 2018
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) :



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T_1 - (tight side tension)
T_2 - (Slack side tension)
μ - Coefficient of friction
θ - Angle of lap of belt around the pulley.

$$\frac{T_1}{T_2} = e^{\mu\theta}$$

757. The sum of the tensions when the belt is running on the pulley is
(a) Less than the initial tension
(b) More than the initial tension
(c) Half the initial tension
(d) More than twice the initial tension

APPSC AEE Screening Test 2016
RPSC AE 2016
RPSC ACF-2011, ESE 1996

Ans. (d) : The sum of tension (T_1) and (T_2)when the belt is transmitting power is always greater than twice the initial tension (T_i)

Initial tension $T_i = \frac{T_1 + T_2}{2}$

Considering centrifugal tension,

$$T_i = \frac{T_1 + T_2 + 2T_c}{2}$$

Where, T_c = Centrifugal tension

758. The relation between the pitch of the chain (P) and pitch circle diameter of the sprocket (D) is given by
(a) $P = D\sin\left(\frac{90°}{T}\right)$ (b) $P = D\sin\left(\frac{120°}{T}\right)$
(c) $P = D\sin\left(\frac{360°}{T}\right)$ (d) $P = D\sin\left(\frac{180°}{T}\right)$

Nagaland PSC CTSE 2016 Paper-I
TNPSC AE Industries 09.06.2013
TNPSC ACF 2012

Ans. (d) : The relations between the pitch of the chain (ρ) and pitch circle diameter of the sprocket (D) is, given by:- $P = D\sin\left(\frac{180°}{T}\right)$

where, T = No, teeth on the sprocket.

759. The velocity ratio of two pulleys connected by an open belt or crossed belt is
(a) Directly proportional to their diameters
(b) Inversely proportional to their diameters
(c) Directly proportional to the square of their diameters
(d) Inversely proportional to the square of their diameters

GPSC DEE, Class-2 (GWSSB) 04.07.2021
HPPSC W.S. POLY. 2016
TNPSC AE 2014

Ans. (b) : The velocity ratio of two pulleys connected by an open belt or crossed belt is inversely proportional to their diameters.

760. The ratio of tension on the tight side to that on the slack side in a flat belt drive is:
(a) Proportional to the product of coefficient of friction and lap angle
(b) An exponential function of the product of coefficient of friction and lap angle
(c) Proportional to the lap angle
(d) Proportional to the coefficient of friction

OPSC AEE 2019 Paper-I
GPSC Executive Engineer 23.12.2018, GATE 2000

Ans : (b) : The ratio of tension on the tight side to that on the slack side in a flat belt drive is

$$\boxed{\frac{T_1}{T_2} = e^{\mu\theta}}$$

Where,
T_1 = Tension in tight side
T_2 = Tension in slack side
μ = Coefficient of friction
θ = Angle of lap (in radian)

761. The difference between tensions on the tight and slack sides of a belt drive is 3000 N. If the belt speed is 15m/s, the transmitted power in kW is:
(a) 45 (b) 22.5
(c) 90 (d) 100

Vadodara Muncipal Corp. DEE, 2018
MPSC Poly. Lect. 2014
GATE 1998

Ans. (a) Given, $T_1 - T_2 = 3000$ N & V = 15m/s.
Power $(P) = (T_1 - T_2) \times V$
$= 3000 \times 15$
$= 45000$ Watt.
$= 45$ kW

762. Which belt drive is recommended in cone pulley, if the sum of radii of the pulley on the driving and driven shaft is constant?
(a) Open belt drive only
(b) The drive is recommended depending upon the torque transmitted
(c) Cross belt drive only
(d) Both open and cross belt drive

Rajasthan AE (Nagar Nigam) 2016 Shift-3
APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : In a cross belt drive, both the pulley rotate in opposite directions. If sum of the radii of the two pulleys be constant, then length of the belt required will also remain constant provided the distance between centres of the pulleys remain unchanged.

763. The cross-section most commonly used in flat belt drive pulleys is
(a) elliptical (b) rectangular
(c) I-section (d) circular

Nagaland PSC CTSE 2017, Paper-I
TNPSC AE 2013, 14

Ans. (a) : The cross-section most commonly used in flat belt drive pulleys is elliptical.

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764. Slip in belt drive is :
(a) loss of power
(b) difference between velocities of two pulleys
(c) difference between angular velocities of two pulleys
(d) difference between linear speed of the rim of pulley and the belt on it

RPSC AE 2016
RPSC ACF-2011

Ans. (d) : Slip is defined as insufficient friction grip between pulley (driven/driver) and belt. Slip is the difference between the linear velocities of pulley (driven/driver) and belt.

765. The wire ropes make contact at
(a) bottom of groove of the pulley
(b) sides of groove of the pulley
(c) sides and bottom of groove of the pulley
(d) anywhere in the groove of the pulley

Nagaland CTSE 2016, 2017 Ist Paper
TNPSC AE, 2008

Ans. (a) : The wire ropes makes contact at bottom of the groove of the pulley so that a clearance must be there in below the rope or (Belt) and the V-groove of the pulley in order to prevent touching at the bottom, to minimize/narrower from wear.

766. Average tensions on the tight side and slack side of a flat belt drive are 700 N and 400 N respectively. If linear velocity of the belt is 5m/s, the power transmitted will be:-
(a) 1.5 kW (b) 2.5 kW
(c) 2.8 kW (d) 3.0 kW

Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE-2013, Paper-I

Ans. (a) : Given as,
$T_1 = 700$ N
$T_2 = 400$ N
$V = 5$ m/s
Power $(P) = (T_1 - T_2) V$
$P = (700 - 400) \times 5$
$P = 1.5$ kW

767. In belt drive power transmitted is given by: (Where T_t, T_s and v are tight side tension, slack side tension and linear velocity of belt respectively).
(a) $= \frac{(T_t - T_s)}{2v}$ (b) $= \frac{(T_t + T_s)}{2v}$
(c) $(T_t - T_s)\, v$ (d) $(T_t + T_s)\, v$

CIL MT 26.03.2017
UKPSC AE 2012

Ans. (c) : Power Transmitted in belt drive $= (T_t - T_s) V$ watt
Where T_t the tension on tight side (N)
T is the tension on slack side (N)
V is the velocity of the belt in m/s

768. In a multiple V-belt drive system, when a single belt is damaged, it is preferable to change the complete set in order to
(a) reduce vibration
(b) reduce slip
(c) ensure proper alignment
(d) ensure uniform loading

MECON MT 2019
ISRO Scientist/Engineer (RAC) 22.04.2018,
ESE 1993

Ans. (d) : When a single belt is damaged in a multiple V-belt drive system, it is preferable to change the complete set of V-belt for ensure uniform loading.

769. Considering centrifugal tension, the power transmitted by belt drive is maximum at velocity V equal to
Note- T is total tension on tight side and m is mass per unit length of belt.
(a) $V = \left\{\frac{T}{m}\right\}^{\frac{1}{2}}$ (b) $V = \left\{\frac{T}{3m}\right\}^{\frac{1}{2}}$
(c) $V = \left\{\frac{3T}{m}\right\}^{\frac{1}{2}}$ (d) $V = \left\{\frac{2T}{m}\right\}^{\frac{1}{2}}$

CGPSC ARTO & TSI 07.01.2017
MPPSC AE 08.11.2015
ESE 2014

Ans. (b) : $P = T_1\left(1 - \frac{1}{e^{\mu\theta}}\right) v$

$P = T_1 C v$ Where, $C = \left(1 - \frac{1}{e^{\mu\theta}}\right)$

But $T_1 = T - T_c$ and $T_c = mv^2$
$\therefore P = \left(TV - mv^3\right) C$

Differentiating w.r.t. V and equating to zero.
$\frac{dp}{dv} = \frac{d}{dv}\left(Tv - mv^3\right)$
$0 = T - 3mv^2$
$T = 3mv^2$
$v = \sqrt{\frac{T}{3m}}$

770. A belt of mass m is running with a velocity of v. The centrifugal tension in the belt is
(a) $mv^{3/2}$ (b) mv^2
(c) mv^3 (d) mv

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : Given–
Mass of belt,
$\boxed{T_c = mv^2}$

771. In block brakes, the ratio of shoe width and drum diameter is kept between
(a) 0.1 to 0.25 (b) 0.25 to 0.50
(c) 0.50 to 0.75 (d) 0.75 to 1.0

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : In a block brakes the ratio of shoe width and drum diameter is of range 0.25 to 0.50

772. Crowning on pulley helps
(a) in increasing velocity ratio
(b) in decreasing the slip of the belt
(c) in automatic adjustment of belt position so that belts run centrally

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(d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Pulley crowning is an assential design element for tracking contact of flat conveyor and power transmission belts. The purpose of crown pulley to assist with belt tracking. In most operating conditions, a belt will follow the path of highest tension. So, it makes sense to have the diameter of pulley larger in the centre than as the edges so that belts run centrally.

773. The Law of belting states that the centre line of the belt when it approaches a pulley must lie in the _____plane of that pulley.

(a) mid (b) side
(c) front (d) end

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : The Law of belting states that the centre line of the belt when it approaches a pulley must lie in the mid plane of that pulley.

774. If d_1 and d_2 are the diameters of the larger and smaller pulleys, and x is the centre distance between the pulleys, then the length of the cross belt drive is _____.

(a) $(d_1 + d_2)\pi/2 + 2x + (d_1 - d_2)^2/4x$
(b) $(d_1 + d_2)\pi/2 + 2x + (d_1 + d_2)^2/4x$
(c) $(d_1 + d_2)/2 + 2x + (d_1 + d_2)^2/4x$
(d) $(d_1 - d_2)\pi/2 + 2x + (d_1 + d_2)^2/4x$

VIZAG MT, 14.12.2020
ESE 2002

Ans. (b) : Length of cross belt drives –

$$L = \pi(r_1 + r_2) + \frac{(r_1 + r_2)^2}{x} + 2x$$

$$L = \frac{\pi}{2}(d_1 + d_2) + \frac{(d_1 + d_2)^2}{4x} + 2x$$

Length of open belt drives

$$L = \frac{\pi}{2}(d_1 + d_2) + \frac{(d_1 - d_2)^2}{4x} + 2x$$

775. A shaft runs at 80 rpm and drives another shaft at 150 rpm through belt drive. The diameter of driven pulley is 600 mm. What will be the diameter of driven pulley assuming belt thickness as 5 mm?

(a) 320.00 mm (b) 300.12 mm
(c) 317.67 mm (d) 304.85 mm

SJVN ET 2019

Ans. (c) : Velocity ratio

$$\frac{N_2}{N_1} = \left(\frac{d_1 + t}{d_2 + t}\right)\left(1 - \frac{s}{100}\right)$$

Assume slip (s) = 0

$$\frac{150}{80} = \frac{600 + 5}{d_2 + 5}$$

$$d_2 + 5 = \frac{80 \times 605}{150}$$

$$d_2 + 5 = 322.66$$

$$d_2 = 322.66 - 5$$

$$= 317.66\,\text{mm}$$

776. Sources of power loss in a chain drive are given below :

(i) Friction between chain and sprocket teeth.
(ii) Overcoming the chain stiffness
(iii) Overcoming the friction in shaft bearing
(iv) Frictional resistance to the motion of the chain in air or lubricant

The correct sequence of descending order of power loss due to these sources is:

(a) (i) (ii) (iii) (iv)
(b) (i) (ii) (iv) (iii)
(c) (ii) (i) (iii) (iv)
(d) (ii) (i) (iv) (iii)

OPSC AEE 2019 Paper-I

Ans : (a) :

- Since the maximum wear surface is at teeth. So it is chain and sprocket surface for maximum loss.
- The least friction losses are due to motion of chain in air or lubricant.

777. For constant velocity ratio positive drive large distance between drive and driven shaft, one would used

(a) Flat belt drive (b) V- belt drive
(c) Chain drive (d) Gear drive

GPSC AE 26.07.2015

Ans. (c) : Chain drive

778. The diameter of the manila and cotton ropes used for power transmission usually ranges from :

(a) 10-20 mm (b) 15-25 mm
(c) 20-25 mm (d) 38-55 mm

OPSC AEE 2019 Paper-I

Ans : (d) : The diameter of manila and cotton ropes usually ranges from 38 mm to 55 mm. The size of the rope is usually, designed by its circumference or 'girth'.

779. What will be the speed ratio of the pulley due to creep, if stress-strain curve for belt is assumed to be parabolic? [where, N_1 = speed of driving pulley, N_2 = speed of driven pulley, σ_1 & σ_2 = stress on tight and slack side of belt respectively, E = Modulus of elasticity of belt, D_1 & D_2 = diameter of driving and driven pulley respectively]

(a) $\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \left[\frac{E + \sigma_2}{E + \sigma_1}\right]$

(b) $\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \left[\frac{E + \sqrt{\sigma_2}}{E + \sqrt{\sigma_1}}\right]$

(c) $\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \left[\frac{E - \sqrt{\sigma_2}}{E - \sqrt{\sigma_1}}\right]$

(d) $\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \left[\frac{E - \sigma_2}{E - \sigma_1}\right]$

(e) $\frac{N_2}{N_1} = \frac{D_1}{D_2} \times \left[\sqrt{\frac{E - \sigma_2}{E - \sigma_1}}\right]$

CGPSC AE 25.02.2018

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Ans. (b) : speed of the pulley due to creep

$$\frac{N_2}{N_1}=\frac{D_1}{D_2}\times\left[\frac{E.+\sqrt{\sigma_2}}{E+\sqrt{\sigma_1}}\right]$$

780. It is usually preferable in chain drive to use
(a) even number of teeth of sprocket
(b) odd number of teeth on sprocket
(c) either even or odd, but certain minimum number
(d) maximum number of teeth permissible on sprocket

TNPSC AE 2018

Ans. (b) : of course, a higher number of teeth 19 or 21 gives better life expectancy to the chain with less noise during operation. It is preferable to use an odd number of teeth for the driving sprocket in combination with an even number of chain links for uniform wear and tear on the teeth and rollers.

781. Crowning of flat pulleys is generally done
(a) to reduce belt friction
(b) to prevent belt joint from damaging the belt surface
(c) to prevent belt from running off the pulley
(d) in case of cross belt drives only

RPSC AE 2016

Ans : (c) Crowning of flat pulleys is generally done to prevent belt from running off the pulley.

782. Which one of the following is a positive drive?
(a) Crossed flat belt drive (b) Rope drive
(c) V-belt drive (d) Chain drive

Nagaland PSC CTSE 2017, Paper-I
Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : Positive drive means movement without slip such as the case in the linking between the crankshaft an the camshaft in the reciprocating engine, some positive drive are chain drive, gear both come under positive drive.

783. Stretching in a belt can be controlled by
(a) decreasing belt length
(b) increasing centre distance
(c) increasing pulley diameter
(d) reducing stress in the belt

TNPSC AE 2018

Ans. (d) : Reducing stress in the belt, stretching in a belt can be controlled.

784. If 'μ' is the actual coefficient of friction in a belt moving in a grooved pulley, the groove angle being '2α', the virtual coefficient of friction will be:
(a) $\frac{\mu}{\cos\alpha}$ (b) $\frac{\mu}{\sin\alpha}$
(c) $\mu\sin\alpha$ (d) $\mu\cos\alpha$

Kerala PSC Poly. Lect., 2017

Ans. (b) : If 'μ' is the actual coefficient of friction in a belt moving in a grooved pulley, the groove angle being '2α', the virtual coefficient of friction will be $\frac{\mu}{\sin\alpha}$.

785. In a band brake the ratio of tight side band tension to the tension on the slack side is 3. If the angle of overlap of band on the drum is 180°, the coefficient of friction between drum and the band is
(a) 0.202 (b) 0.254
(c) 0.335 (d) 0.345

TSPSC AEE 2017

Ans. (d) : Given, $\frac{\text{Tension in tight side}}{\text{Tension in slack side}}=\frac{T_1}{T_2}=3$

$\frac{T_1}{T_2}=e^{\mu\theta}$, θ = Angle of overlap = 180° = π radian

$3=e^{\mu\theta}$
$\ln(3)=\ln e^{\mu\theta}$
$\ln(3)=\mu\theta$
$\mu=\frac{1}{\pi}ln(3)=0.35$

786. Consider the following drives :
A. V-belt
B. Flat belt
C. Toothed belt
D. Round belt
Which of the above is/are positive drive(s)?
Select the code for the correct answer from the options given below:
(a) A and B only (b) B only
(c) C only (d) C and D only

Karnataka PSC AE, 10.09.2017

Ans. (c) : Positive drive provide constant velocity ratio during operation. Ex.-toothed belt, gear.
Non-Positive Drive–Non- constant velocity ratio and prove to slip and creep.
Ex.- belt drives, rope drives.

787. By the law of belting
(a) Two non-parallel shafts can be run in either direction without any intermediation.
(b) A guide pulley helps two non-parallel shafts to run in either direction.
(c) The centre line of the belt when it approaches an idler pulley should be at the edge of the plane.
(d) All of the above

Karnataka PSC AE, 10.09.2017

Ans. (b) : By the law of belting a guide pulley helps two non-parallel shafts to run in either direction.

788. Which of the following wire ropes will be more wear resistant?
(a) 6 × 19 (b) 6 × 7
(c) 6 × 37 (d) All of the above

GPSC EE Pre, 28.01.2017

Ans. (b) : 6×7 type wire ropes will be more wear resistant. The 6×7 type having six strands with 7 wires in each strand.

Wire in each strand
↑
6 × 7
↓
Six strands

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789. Creep of belt can be controlled by
(a) Decreasing belt length
(b) Reducing stress in belt
(c) Increasing center distance
(d) Reducing belt velocity

GPSC EE Pre, 28.01.2017
ESE 2012

Ans. (b) : When the belt passes from slack side to tight side, a certain portion of the belt extends and it contracts again when the belt passes from tight side to slack side. Due to these changes in length, there is a relative motion between the belt and the pulley surfaces.

790. If T_1 is the tension on tight side (Newtons) & T_2 is the tension (Newtons) on slack side of the belt over a pulley, if v is the linear velocity of the belt in m/sec. What would be the magnitude of the power transmitted in H.P.?
(a) $v(T_1 - T_2)/1000$ (b) $vT_1 T_2/1000$
(c) $v(T_1 - T_2)/75$ (d) $vT_1 T_2/75$

Punjab PSC SDE 12.02.2017

Ans. (*) : Power = $(T_1 - T_2)v$ Watt

1 HP = 746 Watt

$$1 \text{ Watt} = \frac{1}{746}\text{HP}$$

$$(T_1 - T_2)v \text{ Watt} = \frac{(T_1 - T_2)v}{746}\text{HP}$$

791. The power transmitted by means of belt drives depends upon:
(a) Velocity of the belt
(b) Tension under which the belt is placed on the pulleys
(c) Arc of contact between the belt and the smaller pulley
(d) All of the above

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (d) : The power transmitted by means of belt drives depends upon velocity of belt, tension under which the belt is placed on the pulley and also arc of contact between the belt and the smaller pulley.

Power (P) = $(T_1 - T_2) \times V$

$$\frac{T_1}{T_2} = e^{\mu\theta}$$

θ = arc of contact between the belt and pulley.
μ = coefficient of friction.

792. When a belt happens to start slipping
(a) it can no longer transmit any power as kinetic friction is less than static friction
(b) it will still continue to transmit power with some slipping
(c) power transmission efficiency will increase
(d) flat belts do not slip

BPSC AE Mains 2017 Paper - VI

Ans : (b) : It will still continue to transmit power with some slipping.

793. The tight and slack sides of a belt connecting two pulleys are having tensions of 25 N and 15 N respectively, while the belt is running at 10 m/s. The power transmitted is
(a) 250 W
(b) 150 W
(c) 100 W
(d) Data provided is insufficient for calculating power

BPSC AE Mains 2017 Paper - VI

Ans : (c) : Power = $(T_1 - T_2) \times V$
$= (25\text{-}15) \times 10 = 100$ W

794. Compared to a belt drive, a chain drive is generally characterised by
(a) constant velocity ratio
(b) absence of slip
(c) heavier construction
(d) all of the above

BPSC AE Mains 2017 Paper - VI

Ans : (d) : Chain drive is positive drive. It maintains constant velocity ratio. It has pure rolling motion. In case of positive drive there is no slipping means no energy loss.

795.The objective of idler pulley in belt drive is to–
(a) decrease the tendency transmission capacity
(b) increase the power transmission capacity
(c) increase the wrap angle and belt tension
(d) all the above objectives

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) :

796. The belt slip occurs due to-
(a) heavy load
(b) loose belt
(c) driving pulley too small
(d) any one of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : The belt slip occur due to, heavy load, loose belt, driving pulley too small.

797. For same pulley diameters, centre distance, belt speed and belt and pulley materials,
(a) open belt drive transmits more power than crossed belt drive
(b) crossed belt drive transmits more power than open belt drive
(c) open and crossed belt drives transmit same power
(d) power transmission does not depend upon open and crossed types of constructions

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : For same pulley diameters, centre distance, belt speed and belt and pulley materials, crossed belt drive transmit more power than open drive.

798. In the designation 6×19 for the wire rope, 6 and 19 stand for:
(a) diameter of wire rope and number of strands respectively
(b) diameter of wire rope and number of wires respectively
(c) number of wires and diameter of wire rope respectively
(d) number of wires and number of strands respectively

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(e) number of strands and number of wires respectively

CGPSC ARTO & TSI 07.01.2017
ESE 2011, 2003, 2001, GATE 2003

Ans. (e) : In the designation 6×19 for the wire rope, 6 and 19 stand for number of strands and number of wires respectively.

799. The tensions in the sides of the belt are T_1 and T_2 respectively. If the speed of the belt is 75 m/sec. then what is the power transmitted by the belt?

(a) $75(T_1+T_2)$ (b) $75\ (T_1+T_2+1)$

(c) $\dfrac{(T_1-T_2)}{75}$ (d) $75(T_1 - T_2)$

(e) $75\left(\dfrac{T_1}{T_2}\right)$

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Power Transmission in Belt Drive

$$\boxed{P=(T_1-T_2)V}$$

The give V = 75 m/sec

$$\boxed{P=75(T_1-T_2)}$$

800. The power transmitted by a belt drive is maximum when running at a velocity equal to:

(a) $\sqrt{\dfrac{T_m}{3W}}$ (b) $\sqrt{\dfrac{T_m}{3Wg}}$

(c) $\sqrt{\dfrac{T_m g}{W}}$ (d) $\sqrt{\dfrac{T_m g}{3W}}$

PTCUL AE 25.06.2017

Ans. (d) : $\sqrt{\dfrac{T_m g}{3W}}$

801. A prime mover running at 300 r.p.m. drives a D.C. generator at 500 r.p.m. by a belt drive. Diameter of the pulley on the outside shaft of the prime mover is 600 mm. Assuming slip of 3%, what is the approximate diameter of generator pulley if the belt running over it is 6 mm thick?

(a) 346.7 mm (b) 406.6 mm

(c) 506.5 mm (d) 606.4 mm

RPSC VPITI 14.02.2016

Ans. (a) : Velocity ratio

$$\frac{N_2}{N_1}=\frac{d_1+t}{d_2+t}\left(1-\frac{S}{100}\right)$$

$$\frac{500}{300}=\frac{600+6}{d_2+6}\left(1-\frac{3}{100}\right)$$

$d_2 = 346.69$ mm

≈ 346.7 mm

802. The idler pulley is employed in belt drives to

(a) Take up the stretch in the belt

(b) To increase the angle of lap

(c) To facilitate power transmission to a number of shafts

(d) All of these

RPSC AE (GWD) 06.05.2016

Ans. (d) : The idler pulley is employed in belt drives to

- Take up the stretch in the belt
- To increase the angle of lap
- To facilitate power transmission to a number of shafts.

803. Choose most appropriate option based on the following two sentences.

(i) For flat and round belts, there is joint, but no limitation on centre distance.

(ii) For V – and timing belt, there is also no limitation on centre distance; However they are having no joints.

(a) sentence (i) and (ii) both are right

(b) sentence (i) is right and sentence (ii) is wrong

(c) sentence (ii) is right and sentence (i) is wrong

(d) sentence (i) and (ii) both are wrong

GPSC Lect. (Auto) 16.10.2016

Ans. (b) :

804. A band brake having bandwidth of 80 mm, drum diameter of 250 mm, coefficient of friction 0.25 and angle of wrap 270 degrees is required to exert a friction torque of 1000 Nm. The maximum tension in kN developed in the band is :

(a) 1.88 (b) 3.56

(c) 6.12 (d) 11.56

HPPSC Asstt. Prof. 18.11.2016

Ans. (d) : b = 80 mm $\Rightarrow$ 0.08 m

D = 250 mm = 0.250 m

$$R=\frac{D}{2}=0.125\ ,\ \mu=0.25$$

$$\theta=270^\circ=\frac{3\pi}{2}=4.712 \text{ radius}$$

$F_t = 1000$ N-m

$$\frac{T_1}{T_2}=e^{\mu\theta}$$

$$T_1=T_2e^{\mu\theta}=e^{6.25\times4.712}\times T_2$$

$T_1 = 3.25\ T_2$

$F_t = (T_1 - T_2) \times R$

$$1000=\left(T_1-\frac{T_1}{3.25}\right)\times0.125$$

$$\boxed{T_1=11.56\,kN}$$

805. A shaft running at 120 rpm drives a parallel shaft at 100 rpm. If pulley on the driving shaft is 75 cm in diameter, the diameter of the pulley on the driven shaft, neglecting belt thickness, and taking belt thickness at 15 mm will be respectively

(a) 49.7 cm and 50 cm

(b) 50 cm and 49.7 cm

(c) 50 cm and 47.8 cm

(d) 49.7 cm and 47.8 cm

(e) None of these

CGPSC AE 16.10.2016

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Ans. (e) : As we know, for belt drive.

$$\frac{N_2}{N_1}=\frac{d_1+t}{d_2+t}$$

$$\frac{100}{120}=\frac{75+t}{d_2+t} \quad [\because \text{ Neglecting thickness}]$$

$d_2 = 90$ cm

Considering belt thickness 15 mm

$$\frac{100}{120}=\frac{75+15}{d_2+15} \Rightarrow d_2 = 93\text{cm}$$

806. In multi V-belt transmission, if one of the belt is broken, we have to change the

(a) broken belt
(b) broken belt and its adjacent belt
(c) all the belts
(d) none of these

GPSC Lect. 23.10.2016

Ans. (c) : In multi V-belt transmission, if one belt is broken, we have to change all the belts.

807. Velocity ratio for pulley drive is given by (where D1 = diameter of driving pulley, D2 = diameter of driven pulley, t = thickness of belt, S = total percentage slip)

(a) $\left(\frac{D_1+t}{D_2+t}\right)\left(\frac{100+S}{S}\right)$

(b) $\left(\frac{D_1+t}{D_2+t}\right)\left(\frac{100-S}{S}\right)$

(c) $\left(\frac{D_1+t}{D_2+t}\right)\left(\frac{100-S}{100}\right)$

(d) $\left(\frac{D_1+t}{D_2+t}\right)\left(\frac{100+S}{100}\right)$

RPSC LECTURER 16.01.2016

Ans. (c) : Velocity ratio for pulley drive is given by

$$V=\left(\frac{D_1+t}{D_2+t}\right)\left(\frac{100-S}{100}\right)$$

808. Which of the following statements is not correct for the ratio of friction tension in flat belt $\left(\frac{T_1}{T_2}=e^{\mu\theta}\right)$

(a) T_1 = tension on the tight side
(b) T_2 = tension on the tight side
(c) θ = angle of lap over the pulley
(d) μ = coefficient of friction between belt and pulley

RPSC LECTURER 16.01.2016

Ans. (b) :

$$\left(\frac{T_1}{T_2}=e^{\mu\theta}\right)$$

where
T_1 = Tension on tight side
T_2 = Tension on slack side
θ = angle of lap over the pulley
μ = Coefficient of friction between belt and pulley

809. The convexity provided on the rim of the pulley is known as:

(a) Grooving (b) Caulking
(c) Forming (d) Crowning

(HPPSC LECT. 2016)

Ans : (d) The convexity provided on the rim of the pulley is known as crowning.

Crowning:- The cast iron pulleys are generally made with rounded rims. This slight convexity is known as crowning. The crowing tends to keep the belt in centre on a pulley rim while in motion. The crowning may be 9 mm for 300 mm width of pulley face.

810. The power transmitted by a belt is dependent on the centrifugal effect in the belt. The maximum power can be transmitted when the centrifugal tensions is

(a) 1/3 of the tension (T_1) on the tight side
(b) 1/3 of the total tension (T_1) on the tight side
(c) 1/3 of the tension (T_2) on the slack side
(d) 1/3 of the tension (T_1) and (T_2)

MPPSC AE 2016
ESE 2002, 1996

Ans : (b)

condition for the transmission of maximum power:-

$P = (T_1-T_2)V$

T_1 = Tension in tight side

T_2 = Tension in slack side

V = Velocity of belt in m/sec

$$\frac{T_1}{T_2}=e^{\mu\theta} \quad \text{or} \quad T_2=\frac{T_1}{e^{\mu\theta}}$$

$$P=\left(T_1-\frac{T_1}{e^{\mu\theta}}\right)V=\left(1-\frac{1}{e^{\mu\theta}}\right)V=T_1C.V$$

$$C=\left(1-\frac{1}{e^{\mu\theta}}\right)$$

$T_1 = T - T_c$

T = Maximum tension to which the belt

T_c = Centrifugal tension

$P=(T-T_C)V.C$

$P=(TV-mv^3)C$

$P=(Tv-mv^3)C$

$$\frac{dp}{dv}=0, \frac{d}{dv}(TV-mv^3)C=0$$

$T = 3T_C$

It shows that when the power transmitted is maximum 1/3 rd of the maximum tension absorbed as centrifugal tension.

811. Constant velocity ratio between two shafts can be obtained, if they are connected by

(a) V-belts and pulleys
(b) Sprocket and chains
(c) Gears
(d) Universal joint

UPPSC AE 12.04.2016 Paper-I

Ans : (c) Constant velocity ratio between two shafts can be obtained. If they are connected by gears.

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Velocity ratio:- Velocity ratio $= \frac{1}{\text{Gear ratio}}$

Gear ratio:-

$$G = \frac{T}{t}$$

Here
T= Number of teeth of gear
t = Number of teeth of pinion (small gear)

812. Idler pulley in belt drives is used when
(a) high velocity ratio is desired at a long distance
(b) high velocity ratio is desired at a short distance
(c) when long life of the belt is desirable
(d) high forces are required to be transmitted

TSGENCO AE 14.11.2015

Ans. (b) : Idler pulley is used for increasing velocity ratio, also for changing the direction of motion of the belt. Idler pulleys are engine pulley's that are responsible for guiding and tensioning the engine drive bolts. Idler pulley are used to take up slack, change the direction of transmission or provide clutching action in any industry material, material handing or any other mechanical purpose.

813. Velocity ratio of a flat belt drive with pulley radius, R_1 and R_2 belt thickness is 't' and slip factor 's' will equal

(a) $VR = \frac{2R_2 + t}{2R_1 + t}(1+s)$

(b) $VR = \frac{R_2 + t}{R_1 + t}(1+s)$

(c) $VR = \frac{R_2 + 2t}{R_1 + 2t}(1-s)$

(d) $VR = \frac{2R_2 + t}{2R_1 + t}(1-s)$

TSGENCO AE 14.11.2015

Ans. (d) : We know that –

$$\frac{N_2}{N_1} = \frac{d_1 + t}{d_2 + t}\left(1 - \frac{S}{100}\right)$$

$$\text{Velocity ratio} = \frac{d_2 + t}{d_1 + t}(1-s)$$

$$\text{velocity ratio} = \frac{2R_2 + t}{2R_1 + t}(1-s)$$

814. Wire ropes used in lifts and hoists are :
(a) Cross ply (b) Regular lay
(c) Long lay (d) Reverse laid ropes

OPSC AEE 2015 Paper-I

Ans : (c) Wire ropes used in lifts and hoists are long lay.
Wire ropes:– When a large amount of power is to be transmitted over long distance from one pulley to another then wire rope are used . The wire ropes widely used in elevators, mine hoists, cranes.

815. The splines are designated as "straight side spline 18 × 11× 112 - IS : 2610". The meaning is the straight side spline of
(a) length 18 mm, thickness 11 mm and width 112 mm
(b) width 18 mm, length 11 mm and thickness 112 mm
(c) width 18 mm, thickness 11 mm and length 112 mm
(d) thickness 18 mm, width 11 mm and length 112 mm

TSPSC AEE 2015

Ans : (c) Straight side spline 18 × 11 × 112 - IS:2610" means
width = 18mm
thickness = 11 mm
length = 112 mm

816. Jockey pulley is used to
(a) Change the direction
(b) change the velocity
(c) reduce the belt tension
(d) increase the angle of contact

TSPSC AEE 2015

Ans : (d) Jockey pulley is used to increase the angle of contact.

817. For maximum power transmission, the velocity of the belt which has the mass per meter length of the belt 1 kg and maximum tension of 1200 N is equal to
(a) 34.64 m/s (b) 24.49m/s
(c) 20m/s (d) 29.6 m/s

TSPSC AEE 2015

Ans : (c) For maximum power transmission

$3mv^2 = T_{max}$

$T = 3mv^2$

$V = \sqrt{\frac{T}{3m}}\ m/sec.$

T = 1200 N

m = 1 Kg

$V = \sqrt{\frac{1200}{3 \times 1}}$

V = 20 m/sec.

818. The average speed of chain on the sprocket is given by
(a) $v = \frac{DN}{1000}\ m/min$ (b) $v = \frac{PTN}{1000}\ m/min$
(c) $v = \frac{2\pi N}{1000}\ m/min$ (d) $v = \frac{TN}{1000}\ m/min$

(Where P = Pitch in mm
T- No of teeth
N = Speed of sprocket
D = diameter of sprocket in mm)

TSPSC AEE 2015

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Ans : (b) The average speed of chain on the sprocket

$$v = \frac{PTN}{1000}\ m/min$$

$$v = \frac{PTN}{60 \times 1000}\ m/sec$$

Relation between pitch and pitch circle diameter

$$D = P\ cosec\left(\frac{180^o}{T}\right)$$

P = pitch in mm
T = No of teeth
N = Speed of sprocket
D = Diameter of the pitch circle.

819. In order to have smooth operation, the minimum number of teeth on the small sprocket for moderate speed should be :
(a) 15 (b) 17
(c) 21 (d) 25

OPSC AEE 2015 Paper-I

Ans : (b) In order to have smooth operation, the minimum number of teeth on the small sprocket for moderate speed should be 17.

820. Which of the following causes the variation in chain speed?
(a) Chordal action (b) Creep
(c) Slip (d) Backlash

UPRVUNL AE 2014

Ans : (a) Chordal action are used for variation in chain speed.

821. In order to increase the angle of wrap it is preferable to go in for
(a) Crossed belt drive
(b) Open belt drive
(c) Horizontal open belt drive
(d) Vertical open belt drive

TNPSC AE 2014

Ans. (a) : A crossed belt drive is used to rotate driven pulley in the opposite direction of driving pulley and in order to increase value of warp angle enables more power can be transmitted than open belt drive. The angle of contact is more in crossed belt drive, so there are less chances of slipping.

822. A V-belt designated as B 4430 L_p represents
(a) A V-belt of basic length 4430 mm
(b) A V-belt of B cross-section and pitch length 4430 mm
(c) A V-belt of B cross-section and nominal inside length 4430 mm
(d) A V-belt of B cross-section and 4430 times multiple of piches

TNPSC AE 2014

Ans. (b) : A V-belt designated as B 4430 L_p represents A V-belt of B cross-section and pitch length 4430 mm.

823. For maximum power transmission in a belt drive the condition is
T_{max} = Maximum tension
T_1 = Tight side tension
T_2 = Slack side tension
(a) $T_{max} = mV^2$ (b) $T_{max} = 3mV^2$
(c) $T_{max} = \frac{T_1 + T_2}{2}$ (d) $T_{max} = 3T_1$

TNPSC AE 2014

Ans. (b) : We know that centrifugal tension (T_C) in belt drive is given as
$T_C = mV^2$
and we know that relation between maximum tension (T_{max}) and centrifugal tension (T_C) is belt drive is

$$T_C = \frac{T_{max}}{3}$$

then,

$$T_{max} = 3T_C = 3mV^2$$

where m $\rightarrow$ mass of belt per unit length

824. In a horizontal belt drive, it is preferable to have
(a) tight side on the top
(b) slack side on the top
(c) tight side tension twice slack side tension
(d) equal tensions on both sides

TNPSC AE 2014

Ans. (b) : The slack side of the belt is preferably placed an the top side because the slack side of the belt, due to its self-weight, will sag. For this reason the angle of contact between the belt and the pulleys will increase.

825. The life of an open belt in comparison to a cross belt is
(a) more (b) less
(c) equal (d) unpredictable

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : In open belt does not touch itself, so no rubbing take place. Thus belts exhibits longer life. Due to crossing between two pulley belt rubs against itself and thus undergoes grounded wear. Thus belt has shorter life.

826. In a V-belt drive, the belt touches the pulley
(a) at bottom
(b) at sides only
(c) at bottom and sides
(d) some cases bottom and some cases sides

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (b) : In case of V belt drive the belt should not touch the bottom of groove in the pulley, the belts touches the pulley at sides only.

827. 'V' belt designated by A 914 denote
(a) The belt section and its inside length in mm
(b) The belt section and its outside length in mm
(c) The belt section and its pitch length in mm
(d) The belt section and its tenside strength in N/m^2.

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : A 914 denotes belt section
914 – its inside length (mm)

828. Which type of belt is used in computer printer?
(a) Flat belt (b) 'V' belt
(c) Timing belt (d) Rope

GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : Timing belt is used in computer printer.

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829. Which of the following is not a common section of V belts?
(a) F (b) E
(c) C (d) A
Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) : F

830. Due to slip of the belt, the velocity ratio of the belt drive
(a) decreases
(b) increases
(c) first increases then decreases
(d) None of the above
APPSC Poly. Lect. 2013

Ans. (a) : Due to slip of the belt, the velocity ratio of the belt drive decreases.

831. When the speed of belt increases
(a) the coefficient of friction between the belt and pulley increases
(b) the coefficient of friction between the belt and pulley decreases
(c) the power transmitted will decrease
(d) the power transmitted will increase
JPSC AE 2013, Paper-V

Ans : (d) : $P = (T_1 - T_2)V$
So if speed increased, Power transmitted should also increases.

832. Which statement is wrong, while providing the crown in the pulley?
(a) The crown on pulley helps to hold the belt on pulley in running condition
(b) The crown on pulley helps the belt from running off the pulley
(c) The crown on pulley helps to maintain the speed
(d) The crown on pulley helps the belt to running equilibrium position near the mid plane of the pulley
TNPSC AE 2013

Ans. (c) : Function of crown in the pully-
1. The crown on pulley helps to hold the belt on pulley in running condition.
2. The crown on pulley helps the belt from running off the pulley.
3. The crown on pulley helps the belt to running equilibrium position near the mid plane of the pulley.
4. The crown on the pulley does not helps to maintain the speed.

833. Which parameter is not required while selecting the flat belt?
(a) Power to be transmitted
(b) The input and output speeds
(c) The centre distance depending on the space
(d) Semicone angle
TNPSC AE 2013

Ans. (d) : Flat Belts– Use materials for flat belts are leather, canvas, cotton and rubber. These belts are used to connects shaft upto 8 - 10 m apart with speeds as high as 22 m/s.
Parameter is required while selecting the flat belt– (1) Power to be transmitted. (2) The input and output speeds. (3) The centre distance depending on the space.

834. A belt drives a pulley of 200 mm diameter such that the ratio of tensions in the tight side and the slack side is 1 : 2, the maximum tension in the belt is not to exceed 240 kN. The speed of pulley is 60 rpm. Find the tension on tight side and slack side of the belt.
(a) 340 kN, 360 kN (b) 200 kN, 240 kN
(c) 360 kN, 120 kN (d) 240 kN, 200 kN
Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) : Given, D = 200 mm = 0.2 m

$$\frac{T_1}{T_2} = \frac{1}{2}$$

Maximum tension = 240 kN, N = 60 rpm
Since, $T_1 > T_2$
Therefore,

$$\boxed{T_1 = 240\,kN} \quad \text{(tight side)}$$

$$\frac{T_1}{T_2} = \frac{1}{2} \Rightarrow T_2 = 2 \times 240 = 480\,kN$$

$$\boxed{T_2 = 200\,kN} \quad \text{(slack slide)}$$

835. In a flat belt drive the belt can be subjected to a maximum tension of 100 N. In maximum power transmission condition, the centrifugal tension Tc is :
(a) 100 N (b) 50 N
(c) 33.33 N (d) 3.33 N
TNPSC AE (Industries) 09.06.2013

Ans. (c) : $T_{max} = 3T_c$

$$100 = 3T_c$$

$$T_c = \frac{100}{3}$$

$$\boxed{T_c = 33.33\,N}$$

836. The flat belt tension at the tight side is 2400 N. The mass of the belt per unit length is 2 kg/m. Find the velocity of the belt for the maximum power transmission :
(a) 10 m/s (b) 20 m/s
(c) 30 m/s (d) 40 m/s
TNPSC AE (Industries) 09.06.2013

Ans. (b) : For maximum power transmission
$T_{max} = 3\,mv^2$

$$\sqrt{\frac{T}{3m}} = v$$

T = 2400 N
m = 2 kg/m

$$v = \sqrt{\frac{2400}{3 \times 2}}$$

$$\boxed{v = 20\,m/sec}$$

837. In a stepped pulley drive, the speed of driven shaft is changed by :
(a) without stopping the driving shaft
(b) Stopping the driving shaft
(c) Lower the speed of driving shaft
(d) Increase the speed of driving shaft
TNPSC AE (Industries) 09.06.2013

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Ans. (a) : In a stepped pulley drive, the speed of driven shaft is changed by without stopping the driving shaft.

838. In a belt drive, maximum tension is 3 kN, mass per unit length is 0.2 kg, what should be the pitch line velocity :

(a) 31.6 m/s (b) 61.2 m/s
(c) 70.7 m/s (d) None of these

PSPCL AE, 2012

Ans. (c) : $T_c = mv^2$, where m = mass per unit / length
$T_c =$ centrifugal tension, T = max tension

For maximum power $T_c = \frac{T_{max}}{3}$

$mv^2 = \frac{3000}{3}$

$0.2v^2 = \frac{3000}{3}$ or v = 70.7 m/s

839. The cross-section of a V-belt is

(a) circular (b) trapezoidal
(c) rectangular (d) elliptical

BPSC AE 2012 Paper - VI

Ans : (b) : A V-belt is mostly used in factories and workshop, where a great amount of power is to be transmitted from one pulley to another when the two pulleys are very near to each other. The included angle for the V-belt is usually from 30^o - 40^o. The cross-section of a V-belt is trapezoidal.

840. The groove angle of the rope is usually

(a) 45^o (b) 30^o
(c) 25^o (d) 15^o

BPSC AE 2012 Paper - VI

Ans : (a) : The groove angle of the rope is usually 45^o.

841. Slip is the result of

(a) insufficient function between the belt and the pulley
(b) unequal elongation of belt due to T_1 and T_2
(c) elongation of belt due to T_1
(d) none of these

Where, T_1 – Tension in tight side, T_2 = Tension in slack side

TNPSC ACF 2012

Ans. (b) : Slip of the belt- A firm frictional grip between belt and shaft is essential but sometimes it becomes insufficient. This may cause some forward motion of the belt without caring the driven puly with it. This is called slip of the belt.

842. By increasing the center distance the ratio of tensions in an open belt drive is

(a) increased (b) decreased
(c) not affected (d) none of these

TNPSC ACF 2012

Ans. (a) : By increasing the centre distance the ratio of tension in an open belt drive is increased.

843. The pitch line velocity of the belt drive should not exceed

(a) 50 m/s (b) 20 m/s
(c) 10 m/s (d) 40 m/s

TNPSC ACF 2012

Ans. (b) : Pitch line, velocity is measured at the pitch line of the gear, which is midway along the length of the gear teeth for circular gear, the pitch circle which is an imaginary circle that rolls without slipping when aligned with the pitch circle of mating gear pitch line velocity of the belt drive should not exceed 20 m/s.

844. The crown height of a flat belt pulley varies between

(a) 0 mm to 5 mm (b) 0.3 mm to 1 mm
(c) 2 mm to 5 mm (d) none of these

TNPSC ACF 2012

Ans. (b) : The crown height of flat belt pully varies between 0.3 mm to 1 mm.

845. Wire ropes are used for applications having

(a) Low speed and low tension
(b) High speeds and high tension
(c) High speeds and low tension
(d) Low speeds and high tension

APPSC AEE 2012

Ans. (d) : Wire ropes are used for application having low speed and high tension. It is used for dynamically for lifting and hoisting in cranes and elevators, and for transmission of mechanical power.

846. Consider the following statements:
A splined shaft is used for
(i) transmitting power
(ii) holding a flywheel rigidly in position.
(iii) moving axially the gear wheels mounted on it.
(iv) mounting V-belts pulley on it.
code:

(a) ii and iii are correct
(b) i and iv are correct
(c) ii and iv are correct
(d) i and iii are correct

APPSC AEE 2012, ESE 1998

Ans. (d) : The transmission of motion between two shafts has been carried out by the use of mechanical connection's including splines are used to provide torque and speed from the shaft to the hub or vice-versa. Compared to other mechanical connections, spline is primary suitable for high torque and then when it is essential an axial transfer of the hub which should be locked during the motion.

847. The maximum power to be transmitted by the belt drive, the ratio of centrifugal tension to permissible tension is :

(a) 3 (b) 2
(c) 0.5 (d) $\frac{1}{3}$

J&K PSC Civil Services Pre, 2010

Ans. (d) : Centrifugal tension in a belt is given by–

$T_c = m \times v^2$

Where, m = Mass of belt per meter length in kg
For a maximum power transmission velocity of belt (v) is given by–

$v = \sqrt{\frac{T_{max}}{3m}}$

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$$T_c = mv^2$$
$$= m\left(\sqrt{\frac{T_{max}}{3m}}\right)^2 = m\frac{T_{max}}{3m}$$
$$T_c = \frac{T_{max}}{3}$$

848. When a belt drive is transmitting maximum power ?
(a) effective tension is equal to centrifugal tension
(b) effective tension is half of centrifugal tension
(c) driving tension on slack side is equal to the centrifugal tension
(d) driving tension on the tight side is twice the centrifugal tension

APPSC IOF, 2009
ESE 1996

Ans. (d) : When a belt drive is transmitting maximum power, driving tension on the tight side is twice the centrifugal tension.

849. The minimum nominal pitch dia. for a V-pulley is
(a) 50 mm (b) 65 mm
(c) 75 mm (d) 90 mm

TNPSC AE, 2008

Ans. (a) : The minimum nominal pitch dia. for a V-pulley is 50 mm.

850. The actual length of the belt is slightly less than the calculated
(a) to give initial tension
(b) due to creep
(c) due to slip
(d) to provide strength

TNPSC AE, 2008

Ans. (a) : The actual length of the belt is slightly less than the calculated to give initial tension.

851. A step cone pulley is provided in a lathe to
(a) reverse the spindle rotation
(b) change the spindle rotation
(c) drive the lead screw
(d) provide feed

TNPSC AE, 2008

Ans. (b) : A step cone pulley is provided in a lathe to change the spindle rotation.
Advantage of cone pulley–
- Easy to maintain
- Can take up heavy load
- Positive drive when the back gear is in engagement
- Its steeped pulley mounted on the main spindle which is free to revolve.
- The head stock having cone pulley drive is known as cone pulley headstock.

852. In a belt drive, the pair between belt and pulley is
(a) higher (b) lower
(c) spherical (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (a) : higher

853. The effective coefficient of friction for a vee belt on a pulley is roughly x times that of a corresponding flat belt. x =
(a) 3 (b) 2
(c) 0.333 (d) 0.111

UKPSC AE 2007 Paper -I

Ans. (a) : 3

854. The normal velocity ratio in belt drive is around
(a) 30 (b) 20
(c) 10 (d) 3 to 4

UKPSC AE 2007 Paper -I

Ans. (d) : 3 to 4

855. Which of the following drives is best to be suited for large velocity ratios?
(a) chain drive
(b) spur gear drive
(c) helical gear drive
(d) worm and worm wheel

J & K PSC Screening, 2006

Ans. (d) : Worm and worm wheel are used for very high velocity ratio.

6. Brake and Dynamometer

856. The brakes commonly used in railways trains is
(a) shoe brake
(b) band brake
(c) band and block brake
(d) internal expanding brake

TNPSC AE 2018
Kerala PSC AE (Irrigation) 05.04.2017
RPSC AE 2016

Ans. (a) : The brake shoe carries the brake block. The block was originally made of wood but is not usually cast iron. When the brake is applied, the shoe moves and presses the block against the tread of the wheel. As well as providing braking effort this also 'Scrubs' the wheel and keeps it clean.

857. Which of the following is a transmission type of dynamometer?
(a) Froude's hydraulic dynamometer
(b) Belt or Tatham dynamometer
(c) Prony brake dynamometer
(d) Torsion dynamometer

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (*) : Types of Dynamometers–
(1) Absorption Dynamometer-
(i) Prony Brake dynamometer
(ii) Rope brake dynamometer
(iii) Hydraulic dynamometer
(2) Transmission Dynameters-
(i) Belt transmission dynamometer
(ii) Epicyclic dynamometer
(iii) Torsion dynamometer
Note- option (b) & option (d) both are correct

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858. In a band and block brake, the ratio of tensions on the tight and slack sides of band is given by (where μ = Coefficient of friction between the blocks and the drum, θ = Semi-angle of each block subtending at the center of drum, and n = Number of blocks)

(a) $T_1/T_2 = \mu\theta \times n$

(b) $T_1/T_2 = (\mu\theta)^n$

(c) $T_1/T_2 = [(1-\mu \tan\theta)/(1+\mu \tan\theta)]^n$

(d) $T_1/T_2 = [(1+\mu \tan\theta)/(1-\mu \tan\theta)]^n$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : As we know, for a band and block brakes, balancing forces

$$\frac{(T_1 - T_2)\cos\theta}{(T_1 + T_2)\sin\theta} = \frac{\mu R_N}{R_N} \Rightarrow T_1 - T_2 = \mu \tan\theta\left[T_1 + T_2\right]$$

$$\frac{T_1}{T_2} = \left[(1+\mu\tan\theta)/(1-\mu\tan\theta)\right]^n$$

859. The heat generated in brake depends upon, Where P = intensity of normal pressure (N/mm^2)
V = rubbing speed (m/min)

(a) PV (b) P/V

(c) PV^2 (d) $1/2PV^2$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : Heat generated by breaks = H_g

$$H_g = f \times F.N \times V$$

$$= P \times \underbrace{\frac{f \times A}{\text{Constant}}} \times V \quad \left[\because P = \frac{F}{A} \Rightarrow F = P \times A\right]$$

$$H_g \propto PV$$

860. A rope brake dynamometer attached to the crank shaft of an IC engine measure a brake power of 10 kW when the speed of rotation of the shaft is 400 rad/s. The shaft torque (in N-m) sensed by the dynamometer is

(a) 25 Nm (b) 20 Nm

(c) 10 Nm (d) 15 Nm

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : $P = T\omega$

$$T = \frac{P}{\omega} = \frac{10 \times 10^3}{400} = \frac{10000}{400}$$

$$T = 25 \text{ Nm}$$

861. For block brakes θ = angle of lap. μ = co-efficient of friction; if θ > 45°, the equivalent coefficient of friction is

(a) $\mu' = \mu\left(\frac{4\sin\theta}{\sin\theta + \theta}\right)$

(b) $\mu' = \mu\left(\frac{4\sin\theta/2}{\sin\theta/2 + \theta/2}\right)$

(c) $\mu' = \mu\left(\frac{4\sin\theta/2}{\sin\theta + \theta}\right)$

(d) $\mu' = \mu\left(\frac{\sin\theta}{\sin\theta + \theta}\right)$

RPSC IOF, 2020

Ans. (c) : The equivalent co-efficient of friction

$$\mu' = \mu\left(\frac{4\sin\theta/2}{\sin\theta + \theta}\right)$$

862. Which of the following is a transmission dynamometer?

(a) Rope brake dynamometer

(b) Epicyclic gear train dynamometer

(c) Prony Brake dynamometer

(d) Heenan and Froude dynamometer

APPSC Poly Lect. 13.03.2020

Ans. (b) : Epicyclic gear-train dynamometer is a transmission dynamometer.

Types of Dynamometers–

(1) Absorption dynamometer

(i) Prony brake dynamometer

(ii) Rope brake dynamometer

(2) Transmission dynamometer

(i) Belt transmission dynamometer

(ii) Epicyclic dynamometer

(iii) Torsion dynamometer

863. For power measurement, the following dynamometer is used when the speed is high and the viscous force is small:

(a) Rope brake dynamometer

(b) Froude water vortex dynamometer

(c) Telsa fluid friction dynamometer

(d) Amsler dynamometer

CIL MT 27.02.2020

Ans. (c) : In power measurement Telsa fluid friction dynamometer is used if speed is high and viscous force is small.

864. Which of the following is an absorption type dynamometer?

(a) Prony brake dynamometer

(b) Epicyclic- train dynamometer

(c) Torsion dynamometer

(d) Belt transmission dynamometer

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : Absorption dynamometers

(1) Prony brake dynamometer

(2) Rope brake dynamometer

(3) Hydraulic dynamometer

Transmission dynamometers

(1) Belt transmission dynamometer

(2) Epi-cyclic dynamometer

(3) Torsion dynamometer

865. In brake, when angle of contact is less than 60°, which brake is preferable?

(a) Single block shoe brake

(b) Pivoted block shoe brake

(c) Double block shoe brake

(d) None of these

Oil India Senior Officer 23.12.2020

Ans. (a) : In brake, when angle of contact is less than 60°, single block shoe brake is preferable.

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866. The ratio of tension on slack side to the tension on tight side after n blocks for band and block brakes is given as: [when, μ = Coefficient of friction, 2θ = Angle subtended by each block at the centre, n = Number of blocks]

(a) $\left(\frac{1-\mu\tan\theta}{1+\mu\tan\theta}\right)^n$ (b) $\left(\frac{1+\mu\tan\theta}{1-\mu\tan\theta}\right)^n$

(c) $\left(\frac{1-\mu\tan\theta}{1+\mu\tan\theta}\right)^{n+1}$ (d) $\left(\frac{1+\mu\tan\theta}{1-\mu\tan\theta}\right)^{n+1}$

(e) $\left(\frac{1+\mu\cos\theta}{1-\mu\cos\theta}\right)^{n+1}$

CGPSC AE 25.02.2018

Ans. (a) : The ratio of tension on slack side to the tension on tight side after n blocks for band and block brakes is given as

$$\left(\frac{1-\mu\tan\theta}{1+\mu\tan\theta}\right)^n$$

867. The brake bleeding process removes ___ from the system.

(a) air (b) vacuum
(c) excess fluid (d) excess pressure
(e) dust particles

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Brake bleeding is the procedure performed on hydraulic brake systems where by the brake lines (the pipes anal noses containing the broke fluid) are purged of any air bubbles.

868. Brake fade is:

(a) loss of pedal
(b) loss of friction
(c) loss of hydraulic fluid pressure
(d) low brake fluid level
(e) poor vacuum

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Brake fade tends to happen when the generated heat due to friction does not have time to dissipate, such as when you repeatedly or continuously apply the brakes when driving fast. This excessive build up of heat causes brakes to become less responsive.

869. Rope brake dynamometer uses

(a) oil as lubricant
(b) water as lubricant
(c) grease as lubricant
(d) no lubricant

RPSC AE 2016

Ans : (d) Rope brake dynamometer uses no lubricants.

Rope Brake Dynamometer:-

It is another form of absorption type dynamometer which is most commonly used for measuring the brake power of the engine. It consists of one, two or more ropes wound around the flywheel or rim of a pulley fixed rigidly to the shaft of an engine.

$$\text{B.P.} = \frac{(W-S)\pi DN}{60} \text{ watts}$$

W = Dead load in newtons
S = Spring balance reading in newtons
D = Diameter of the wheel in metres
N = Speed of the engine shaft in r.p.m.

870. Dynamometer is a device, which is used to measure

(a) Speed of machine
(b) RPM of a machine
(c) Weight of the machine
(d) None of the above

UJVNL AE 2016

Ans : (d) A dynamometer is a brake but in addition it has a device to measure the frictional resistance. Knowing the frictional resistance, we may obtain the torque transmitted and hence the power of the engine

Dynamometer

Absorption dynamometer	Transmission dynamometer
i) Rope brake	i. Epicyclic train
ii) Prony brake	ii. Belt transmission
	iii. Torsion dynamometer

871. The commonly used type of brakes in motor cars is :

(a) Shoe brake
(b) Band brake
(c) Internal expanding brake
(d) All of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : The internal expanding type of brake is commonly used in motor car and light trucks.
When the brakes are applied to all the four wheels, the braking distance (i.e. the distance in which the vehicle is brought to rest after applying the brakes) will be the least.

872. A dynamometer is used for measuring :

(a) speed of a machine
(b) torque exerted by or on a machine
(c) kinetic energy of the system
(d) none of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : A dynamometer is used for measuring torque exerted by or on a machine.

873. Prony brake is used for testing of

(a) Small engines
(b) Large engines
(c) Engines having small flywheel
(d) slow speed engines

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Prony brake is used for testing of small engines.

874. The principal object of a brake is to

(a) balance the energy (b) give energy
(c) absorb energy (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (c) : Absorb energy

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7. Static and Dynamic Forces Analysis

875. The acceleration of the piston in a reciprocating steam engine, neglecting the weight of the connecting rod is given by -
ω = Angular velocity of the crank
r = Radius of the crank
θ = Angle turned by the crank from inner dead centre
n = Ratio of length of connecting rod to crank radius

(a) $\omega r\left(\cos\theta+\frac{\cos 2\theta}{n}\right)$

(b) $\omega^2 r\left(\cos\theta+\frac{\cos 2\theta}{n}\right)$

(c) $\omega r\left(\sin\theta+\frac{\sin 2\theta}{n}\right)$

(d) $\omega^2 r\left(\sin\theta+\frac{\sin 2\theta}{n}\right)$

APPSC Poly Lect. Automobile Engg., 2020
UPPSC AE 13.12.2020, Paper-I
CIL MT 26.03.2017
Nagaland PSC CTSE 2016 Paper-I
TNPSC AE 2013

Ans. (b) : $V_P = r\omega\left[\sin\theta+\frac{\sin 2\theta}{2n}\right]$

$$a_p = r\omega^2\left[\cos\theta+\frac{\cos 2\theta}{n}\right]$$

Angular velocity and angular acceleration of connecting rod.

$$\omega_{pc} = \frac{\omega\cos\theta}{n}$$

$$\alpha_{pc} = -\frac{\omega^2\sin\theta}{n}$$

876. When the crank is at the inner dead center, in a horizontal reciprocating engine, then the velocity of the piston will be:
(a) Zero (b) Maximum
(c) Minimum (d) Infinity

TSPSC Managers, 2015
TANGEDCO AE 2015

Ans. (a) : The velocity of the piston -

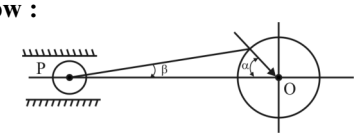
$$V_p = r\omega\left(\sin\theta+\frac{\sin 2\theta}{2n}\right)$$

Here, θ is the angle of the crank makes with respect to the horizontal when the crank is that inner dead centre, (θ = 0).

$$V = r\omega\left[\sin 0+\frac{\sin 2\times 0}{2n}\right]$$

$$\boxed{V = 0}$$

877. The cross head velocity in the slider Crank mechanism, for the position shown in Fig. below :



(a) $V_c\cos(90-\overline{\alpha+\beta})\cos\beta$

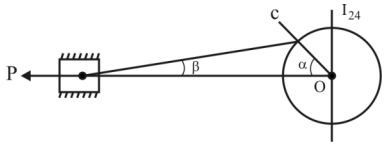
(b) $V_c\cos(90-\overline{\alpha+\beta})\sec\beta$

(c) $V_c\cos(90-\overline{\alpha-\beta})\cos\beta$

(d) $V_c\cos(90-\overline{\alpha-\beta})\sec\beta$

APGENCO AE 2012
GATE 1997

Ans. (b) :



Velocity of cross head = WOI_{24}
Consider ΔOCI_{24}

$$OI_{24} = \frac{OC\cos(90-\overline{\alpha+\beta})}{\cos\beta}$$

$V_c = OC \times W$

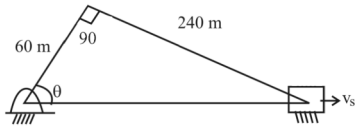
$$\Rightarrow \quad OI_{24}W = V_c\cos(90-\overline{\alpha+\beta})\sec\beta$$

878. A slider crank mechanism with crank radius 60 mm is perpendicular to connecting rod having length 240 mm. The crank is rotating with uniform angular speed of 10 rad/s, counter clockwise. For the given configuration, the speed of slider is _____m/s.
(a) 618 (b) 0.618
(c) 6.18 (d) 61.8

RPSC ADE 2016
GATE 2014

Ans. (b) :
Given,
l = 240 mm
r = 60 mm

$$n = \frac{l}{r} = 4$$



$$\tan\theta = \frac{l}{r} = \frac{240}{60} = 4$$

$\theta = 75.96^0$

$$v_s = \omega r\left[\sin\theta+\frac{\sin 2\theta}{2n}\right]$$

$$= 10\times 0.06\left[\sin 75.96+\frac{\sin 151.92}{2\times 4}\right]$$

= 0.6 × 1.0288 = 0.617

OR

Since slider is perpendicular to the crank hence, $v_s = \omega r = 10 \times 0.06 = 0.6$ m/s

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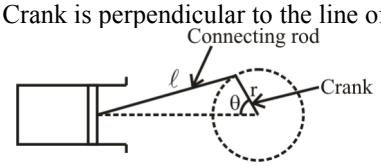
879. In a slider-crank mechanism, the velocity of the piston is maximum when

(a) crank and connecting rod are perpendicular to each other
(b) crank and connecting rod are in the same line
(c) crank is perpendicular to the line of stroke
(d) crank is at 120 degrees with the line of stroke

TSPSC Manager (Engg.) HMWSSB 12.11.2020
ESE 2003

Ans. (c) : Crank is perpendicular to the line of stroke.



Velocity of piston,

$$v_p = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2n}\right] \quad \text{Where, } n = \frac{\ell}{r}$$

For maximum velocity, $\theta = 90°$

$$\boxed{v_p = r\omega}$$

880. In a certain slider crank mechanism, lengths of crank and connecting rod are equal. If the crank rotates with a uniform angular speed of 14 rad/s and the crank length is 300 mm, maximum acceleration of the slider (in m/s^2) is–

(a) 11.76 m/s^2 (b) 117.6 m/s^2
(c) 1176.2 m/s^2 (d) 1.1762 m/s^2

RPSC 2016
GATE 2015

Ans : (b)

Acceleration of slider $\Rightarrow a = r\omega^2 (\cos\theta + \frac{\cos 2\theta}{n})$

Given, $r = l$ $n = \frac{l}{r}$

$n = 1$

Condition of Maximum Acceleration

$\Rightarrow \sin\theta + 2\sin(2\theta) = 0 \Rightarrow \frac{\partial a}{\partial \theta} = 0$, which gives $(\theta = 0)$

$a_{max} = 0.3\times 14^2 [\cos 0 + \cos(2\times 0)] = 117.6$ m/s^2

881. When the primary direct crank of a reciprocating engine positioned at 30° clockwise, the secondary reverse crank for balancing will be at

(a) 30° anticlockwise (b) 60° anticlockwise
(c) 30° clockwise (d) 60° clockwise

UPPSC AE 13.12.2020, Paper-I
ESE 1997

Ans. (b) : If the primary direct crank of a reciprocating engine is located at 30° clockwise, then secondary direct crank will be located 60° anticlockwise.

882. A slider crank mechanism has the maximum acceleration of slider when the crank is –

(a) at the inner dead centre position
(b) at the outer dead centre position
(c) exactly midway position between two dead centers
(d) none of these

RPSC 2016
MPPSC AE 08.11.2015, ESE 2001

Ans : (a) A slider crank mechanism has the maximum acceleration of slider when the crank is at the inner dead centre position.

883. For a slider crank mechanism, the velocity and acceleration of the piston at inner dead centre will be

(a) 0 and 0 (b) 0 and $\omega^2 r$
(c) 0 and $< \omega^2 r$ (d) 0 and $> \omega^2 r$

TNPSC AE 2014
Kerala LBS Centre for Sci. & Tech. Asstt. Prof. 2014

Ans. (d) : At inner dead centre $\theta = 0^0$

$$\text{then } V_p = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2n}\right]$$

at, $\theta = 0^o$

$$\boxed{V_p = 0}$$

$$a_p = r\omega^2\left[\cos\theta + \frac{\cos 2\theta}{n}\right]$$

at, $\theta = 0^0$

$$a_p = r\omega^2\left[\cos 0^o + \frac{\cos 0^o}{n}\right]$$

$$a_p = r\omega^2\left[1 + \frac{1}{n}\right]$$

so we can say that

$a_p > r\,\omega^2$

884. How many times the secondary unbalanced forces are maximum during the one revolution of the crank?

(a) 6 (b) 8
(c) 4 (d) 2

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : • The secondary unbalanced force is maximum, when $\theta = 0^o$, 90^o, 180^o, and 360^o. Thus The secondary force is maximum four times in one revolution of the crank.

• The primary unbalanced force is maximum when $\theta = 0^o$ or $\theta = 180^o$. Thus the primary force is maximum twice in one revolution of the crank.

885. In a reciprocating engine, the force along the connecting rod F_Q is

(a) $\frac{F_P}{\sqrt{n^2 - \sin^2\theta}}$ (b) $\frac{F_P}{2\sqrt{n^2 - \sin^2\theta}}$
(c) $\frac{nF_P}{2\sqrt{n^2 - \sin^2\theta}}$ (d) $\frac{nF_P}{\sqrt{n^2 - \sin^2\theta}}$

where
F_P = Force on piston
$n = \frac{L}{r}$

ESE 2020

Ans. (d) : $F_Q = \frac{F_P}{\cos\phi} = \frac{nF_P}{\sqrt{n^2 - \sin^2\theta}}$

886. When the piston reaches at the TDC and BDC, what will be the velocity of the piston while changing the direction at these dead centres?

(a) Maximum velocity at both TDC and BDC

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(b) Maximum velocity at TDC and zero velocity at BDC
(c) Zero velocity at TDC and maximum velocity at BDC
(d) Zero velocity at both TDC and BDC

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : We know velocity of piston –

$$V = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2n}\right]$$

If piston at top dead centre (TDC) (i.e. $\theta = 0°$)
Then, $V = 0$
If the piston at bottom dead centre (BDC) (i.e. $\theta = 180°$)
Then, $V = 0$
Velocity of piston at inner and outer dead centre will be zero.

887. The frequency of secondary force as compared to primary force for ratio of connecting rod length to crack radius of 4 is

(a) Half (b) Twice
(c) Four times (d) Sixteen times

TNPSC AE 2019

Ans. (b) : Twice

888. In a crank and slider mechanism if ω is angular velocity of the crank, r is radius of the crank, θ is the angle turned by the crank from inner dead centre and n is the ratio of the connecting rod to crank radius, then the velocity of the piston V_p is:

(a) $V_p = \omega r\left(\sin\theta - \frac{\sin 2\theta}{2n}\right)$

(b) $V_p = \omega r\left(\sin\theta + \frac{\sin 2\theta}{2n}\right)$

(c) $V_p = \omega r\left(\sin 2\theta - \frac{\sin\theta}{2n}\right)$

(d) $V_p = \omega r\left(\sin 2\theta + \frac{\sin\theta}{2n}\right)$

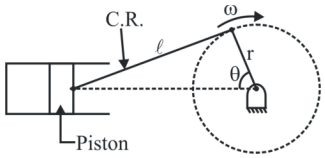
(e) $V_p = \omega^2 r\left(\sin\theta - \frac{\sin 2\theta}{2n}\right)$

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Displacement of Piston From IDC to ODC

$$x = r\left[(1-\cos\theta) + \left\{n - \sqrt{(n^2 - \sin^2\theta)}\right\}\right]$$

where $n = \frac{\ell}{r}$



Velocity of piston V_p

$$V_p = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2\theta}}\right] \qquad n^2 >>> \sin^2\theta$$

$$V_p = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2n}\right]$$

Acceleration of piston (a_p)

$$a_p = r\omega^2\left[\cos\theta + \frac{\cos 2\theta}{n}\right]$$

Angular velocity of connecting rod

$$\omega_{C.R.} = \frac{\omega\cos\theta}{\sqrt{n^2 - \sin^2\theta}} \qquad \text{If } \sin\theta \to 0$$

$$\omega_{C.R.} = \frac{\omega\cos\theta}{n}$$

Angular acceleration of connecting rod

$$\alpha_{C.R.} = -\frac{\omega^2\sin\theta}{n}$$

889. The secondary force in a crank-piston mechanism

(a) arises due to obliquity of connecting rod
(b) acts at double the frequency as that of the primary force
(c) is smaller in magnitude than the primary force
(d) All of the above

BPSC AE Mains 2017 Paper - VI

Ans : (d) : The secondary force in a crank piston mechanism is-

- Arise due to obliquity of connecting rod.
- Acts at double the frequency as that of the primary force.
- Is smaller in magnitude than the primary force.

890. A piston having mass m = 3 kg executes pure simple harmonic motion, whose displacement in meters is given by x = 0.05 sin (10t). Its inertia force at time t = 0 will be

(a) 0 N (b) –0.15 N
(c) –1.5 N (d) –15 N

BPSC AE Mains 2017 Paper - VI

Ans : (a) : m = 3 kg
$x = 0.05\sin(10t)$ meter.
$\dot{x} = 0.05\times 10\cos(10t)$
$\ddot{x} = -0.05\times 10\times 10\sin(10t)$
Inertia force,
$F = m\ddot{x}$
$= -3\times 0.05\times 100\sin(10t)$
At time, t = 0 second
F = 0 N

891. The piston of a steam engine moves with simple harmonic motion. The speed of rotation of crank is 120 rpm with a stroke of 2 m. What is the velocity of piston when it is 0.5 m from the centre?

(a) $4\pi\sqrt{3}$ (b) $\pi\sqrt{3}$
(c) $2\pi\sqrt{3}$ (d) $3\pi\sqrt{3}$

ISRO Scientist/Engineer 03.07.2016

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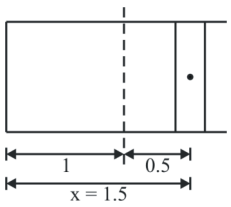
Ans. (c) : N = 120 rpm

ℓ = 2 m

x = 1.5



$\therefore \quad r = \frac{\ell}{2} = 1$ m

$x = r(1 - \cos\theta)$

$1.5 = 1(1 - \cos\theta)$

$\theta = 120°$

$v = r\omega \sin\theta$

$= 1 \times \frac{2\pi N}{60} \sin 120°$

$= \frac{2\pi \times 120}{60} \times \frac{\sqrt{3}}{2}$

$= 2\pi\sqrt{3}$ m/s

892. In a slider crank mechanism for l/r ratio of 4, the percentage of stroke converted by pistion corresponding to 90° movement of crank from top dead centre will be

(a) 0 % (b) less than 50%

(c) greater than 50% (d) 100%

APPSC AEE Mains 2016, Paper-III

Ans. (c) : displacement of piston

$$x = r\left[(1 - \cos\theta) + n - \sqrt{n^2 - \sin^2\theta}\right]$$

Given,

$$n = \frac{\ell}{r} = 4, \qquad \theta = 90°$$

$$x = \frac{L}{2}\left[1 + 4 - \sqrt{(4)^2 - 1}\right]$$

$$x = \frac{L}{2}\left[1 + 4 - \sqrt{15}\right]$$

$$= \frac{L}{2} \times 1.127$$

percentage of stroke converted, % x = greater than 50%

893. When the angle of inclination of the crank with inner dead centre of a reciprocating engine is 45°, then :

(a) Primary force is maximum

(b) Secondary force is maximum

(c) Primary force is zero

(d) Secondary force is zero

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (d) : $F = mr\omega^2\left[\cos\theta + \frac{\cos 2\theta}{n}\right]$

$$= \underset{\text{Primary Force}}{\underbrace{mr\omega^2 \cos\theta}} + \underset{\text{Secondary Force}}{\underbrace{mr\omega^2 \frac{\cos 2\theta}{n}}}$$

When $\theta = 45°$

- Secondary force will be zero.

894. In petrol engine mechanism , the piston is at its dead centre position when piston:

(a) Acceleration is zero

(b) Acceleration is maximum

(c) Velocity is zero

(d) Velocity is infinity

(a) (a) and (d) (b) (a) and (c)

(c) (b) and (c) (d) (b) and (d)

RPSC ADE 2016

Ans. (c) : Petrol engine mechanism, the piston is at its dead centre position when piston.

(b) Acceleration is maximum

(c) velocity is zero.

Velocity of piston$(V_p) = r\omega\left[\sin\theta + \frac{\sin 2\theta}{2n}\right]$

Acceleration$(a_p) = r\omega^2\left[\cos\theta + \frac{\cos 2\theta}{n}\right]$

for dead centre ($\theta = 0$)

$$v_p = r\omega\left[\sin\theta + \frac{\sin 2\times 0}{2n}\right] = 0$$

$$a_p = r\omega^2\left[\cos\theta + \frac{\cos 2\times 0}{n}\right]$$

$$a_p = r\omega^2\left[1 + \frac{1}{n}\right] \text{ is maximum}$$

895. Consider that connecting a rod and the crank of an IC engine are the two sides of a triangle. The maximum area of this triangle is when the crank is at 60°. What is the ratio of the connecting rod length to the crank radius?

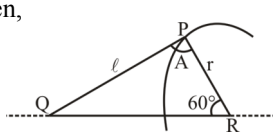
(a) $1:\sqrt{3}$ (b) $2:\sqrt{3}$

(c) $\sqrt{2}:\sqrt{3}$ (d) $\sqrt{3}:1$

(e) $\sqrt{3}:2$

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) : Given,



$$\Delta = \frac{1}{2}(PQ)(PR)\sin A$$

Area will be max for A = 90°

Ratio of connecting rod length to crank radius

$$\frac{\ell}{r} = \tan 60° \Rightarrow \boxed{\frac{\ell}{r} = \frac{\sqrt{3}}{1}}$$

896. In a reciprocating horizontal engine, the inertia forces due to reciprocating mass help the piston effort at

(a) $\theta = 30^0$ (b) $\theta = 45^0$

(c) $\theta = 120^0$ (d) $\theta = 180^0$

TSGENCO AE 14.11.2015

Ans. (d) : In a reciprocating horizontal engine, the piston effort is the net force acting on the piston or cross head pin along ($\theta = 180^0$) the line of stroke. It is denoted by F_P.

Piston effort P_F =Net load on the piston ± Inertia force

$= F_L + F_I$ (Negative frictional resistance)

$P_F = F_L \pm F_I - R_f$ (considering frictional resistance)

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897. In an I.C. engine piston executes approximately S.H.M. if:
(a) length of connecting rod is very large in comparison of the length of crank
(b) length of crank is very large in comparison of the length of connecting rod
(c) length of connecting rod is equal to the length of crank
(d) piston does not execute S.H.M. at all

HPPSC Asstt. Prof. 2014

Ans. (a) : Piston executes SHM if length of connecting rod is very large in comparison of the length of crank.

$$F_u = mr\omega^2 \cos\theta + \frac{mr\omega^2 \cos 2\theta}{n}$$

$$n = \frac{l}{r}$$

l is kept very high to reduce secondary unbalance force.
If l >> r means n is large.

So, secondary unbalance force $\left[\frac{mr\omega^2 \cos 2\theta}{n}\right]$ is neglected.
Hence $F_u = mr\omega^2\cos\theta$
Means, force is cosine which leads to SHM.

898. The piston effort for a vertical engine includes
(a) Pressure force due to gas and inertia force
(b) Pressure force due to gas, inertia force and frictional resistance
(c) Pressure force due to gas, inertia force and weight of the piston
(d) Pressure force due to gas, inertia force, weight of the piston and frictional resistance

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) Piston Effort (F) = $\left(F_{gas} - F_I\right) \pm M_g$

where
F_{gas} = Pressure force due to gas
F_I = Inertia force
M_g = Piston weight

899. The velocity of a piston in a reciprocating pump mechanism is maximum when the obliquity is
(a) zero (b) maximum
(c) average (d) minimum

APPSC Poly. Lect. 2013

Ans. (d) : The velocity of a piston in a reciprocating pump mechanism is maximum, when the obliquity is minimum.

Velocity of piston $(v_p) = r\omega\left[\sin\theta + \frac{\sin 2\theta}{n}\right]$

$v_p \uparrow \quad n \downarrow$

n = obliquity ratio = ℓ/r

900. Dynamics of machines deals with :
(a) relative motion between parts neglecting the consideration of forces
(b) forces acting on the parts of machine
(c) apparatus for applying mechanical power
(d) number of interrelated parts each having a definite motion

RPSC ACF-2011
RPSC AE 2016

Ans. (b) : Dynamics of Machines– Analysis of the forces and couples on the members of the machine due to external forces (static force analysis) also analyse the forces and couples due to accelerations of machine members.

901. For the same crank length and uniform angular velocity of the crank in an offset slider crank mechanism, if the connecting rod length is increased by 1.5 times, the velocity of piston will
(a) remain unchanged
(b) increase 1.5 times
(c) decrease by 1.5 times
(d) increase by $1.5\sqrt{2}$ times

DRDO Scientists 2009

Ans. (c) : $V_1 = \omega r\left(\sin\theta + \frac{\sin 2\theta}{2n}\right)$

$$V_2 = \omega r\left(\sin\theta + \frac{\sin 2\theta}{3n}\right)$$

from these two equation, $V_2 < V_1$
V_2 will decrease but correct quantification can not be done with available data.
Among the available options, best answer is (c).

902. In a slider-crank mechanism, the crank is rotating with an angular velocity of 20 rad/s in counterclockwise direction. At the instant when the crank is perpendicular to the direction of the piston movement, velocity of the piston is 2 m/s. Radius of the crank is
(a) 100 cm (b) 10 cm
(c) 1 cm (d) 0.1 cm

DRDO Scientists 2008

Ans. (b) : Velocity of piston

$$V_p = \omega r\left(\sin\theta + \frac{\sin 2\theta}{2n}\right)$$

In this case $\theta = 90^\circ$, $Vp = \omega r$

$$r = \frac{V_p}{\omega} = \frac{2}{20} = 0.1 \text{ m} = 10 \text{ cm}$$

903. The maximum angular acceleration of the connecting rod with crank to connecting rod ratio 1.5 and crank running at 3000 rpm is around
(a) 1.8×10^4 rad/s^2 (b) 9×10^4 rad/s^2
(c) 0.9×10^4 rad/s^2 (d) 3×10^4 rad/s^2

UKPSC AE 2007 Paper -I

Ans. (a) : $\alpha_{C.R} = \frac{-\omega^2 \sin\theta}{n}$

For maximum $\theta = 90°$

Given, $\frac{r}{\ell} = \frac{1}{5}$

$$n = \frac{\ell}{r} = 5$$

$$\alpha_{C.R} = \frac{-\left(\frac{2\pi \times 3000}{60}\right)^2}{5}$$

$$\boxed{\alpha_{C.R} = 1.9 \times 10^4 \text{ rad/s}^2}$$

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904. If W is weight of reciprocating parts, ω is angular speed of crank, r is radius of crank, θ is angle of inclination of crank with the line of stroke and n is ratio of length of connecting rod to radius of crank, then the force to accelerate the reciprocating mass in reciprocating engine is equal to-

(a) $(W/g)\,\omega^2 r\left[\cos\theta+\frac{\cos 2\theta}{n}\right]$

(b) $(W/g)\,\omega^2 r\,(\cos\theta+\cos 2\theta)$

(c) $(W/g)\,\omega^2 r\,(n\cos\theta+\cos 2\theta)$

(d) $(W/g)\,\omega^2 r\,(n\sin\theta+\sin 2\theta)$

WBPSC AE 2003

Ans. (a) : ∵ Piston displacement

$$x = r[(1-\cos\theta)+(n-\sqrt{n^2-\sin^2\theta})] \quad ...(i)$$

Velocity $V=\frac{dx}{dt}=\frac{dx}{d\theta}.\frac{d\theta}{dt}=\omega\frac{dx}{d\theta}$

$$V = r\omega\left[\sin\theta+\frac{\sin 2\theta}{2\sqrt{n^2-\sin^2\theta}}\right] \quad ...(ii)$$

acceleration $a=\frac{dV}{dt}=\omega\frac{dV}{d\theta}$

$$a = r\omega^2\left[\cos\theta+\frac{\cos 2\theta}{n}\right] \quad ...(iii)$$

force to accelerate F = ma [keep the value of a]

$$F=\frac{W}{g}.a$$

$$\boxed{F=(W/g)r\omega^2\left[\cos\theta+\frac{\cos 2\theta}{n}\right]}$$

8. Flywheel and Governors

905. The maximum fluctuation of the energy is the :

(a) sum of maximum and minimum energies

(b) difference between the maximum and minimum energies

(c) ratio of the maximum energy and minimum energy

(d) ratio of mean resisting torque to the work done per cycle

Rajasthan Nagar Nigam AE 2016, Shift-I
CGPSC Poly. Lect. 22.05.2016
OPSC AEE 2015 Paper-I
Haryana PSC Civil Services Pre, 2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
J&K PSC Civil Services Pre, 2013
UKPSC AE 2013 Paper-I
APPSC AEE 2012, ESE 1998

Ans. (b) : The maximum fluctuation of the energy is the difference between the maximum and minimum energies.

906. If the rotating mass of a rim type flywheel is distributed on another rim type flywheel whose mean radius is half the mean radius of the former, then the energy stored in the later at the same speed will be :

(a) Four times the first one

(b) Same at the first one

(c) One fourth of the first one

(d) One and half times the first one

JPSC AE 10.04.2021, Paper-II
Nagaland PSC (CTSE) 2018, Paper-I
UPRVUNL AE 21.08.2016
Kerala PSC IOF 19.04.2016
Assam PSC CCE Pre 2015
MPPSC State Forest Service Exam, 2014
TNPSC AE 2014
J&K PSC Civil Services Pre, 2013
JPSC AE 2013, Paper-V
ISRO Scientist/Engineer 2011
ISRO Scientist/Engineer 2006, ESE 2002

Ans. (c) : For rim type flywheel radius of gyration = radius of wheel.

$$KE=\frac{1}{2}I\omega^2=\frac{1}{2}mk^2\times\omega^2$$

For first wheel $(KE)_1=\frac{1}{2}mk^2\omega^2$

For second wheel $(KE)_2=\frac{1}{2}m\left(\frac{k}{2}\right)^2\omega^2$

$$(KE)_2=\frac{(KE)_1}{4}$$

907. When the speed of the engine fluctuates continuously above and below the mean speed then the governor is said to be :

(a) Hunting (b) Stable

(c) Unstable (d) Isochronous

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
VIZAG Steel MT 24.01.2021
APPSC Poly Lect. 13.03.2020
MECON MT 2019, CSE Pre 2010,
OPSC AEE 2019 Paper-I
Sikkim PSC (Under Secretary) 2017, ESE 2016, 2013
NPCIL ET, 2015 / KPSC AE 2015
Kerala PSC AE 06.08.2015
MPPSC State Forest Service Exam, 2014
J & K PSC Screening, 2006

Ans. (a) : Hunting–A governor is said to be hunt if the speed of engine fluctuates continuously above and below the mean speed. This is caused by a too sensitive governor, which changes the fuel supply by a large amount when a small change in speed of rotation take place.

908. The sensitiveness of the governor is equal to

(a) $\frac{\omega_1-\omega_2}{\omega}$ (b) $\frac{\omega}{\omega_1-\omega_2}$

(c) $\frac{\omega}{\omega_2-\omega_1}$ (d) $\frac{\omega_2-\omega_1}{\omega}$

(Where ω_1 = maximum speed
ω_2 = minimum speed
ω = mean speed)

TNPSC AE 2019 / OPSC AEE 2019 Paper-I
HPPSC AE 2018 / UJVNL AE 2016
TSPSC AEE 2015, CGPSC AE 26.04.2015 Shift-I
TNPSC AE 2014 / TNPSC AE 2013
ISRO Scientist/Engineer 2010

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Ans : (a) Sensitiveness of the governor

$$= \frac{\text{Range speed}}{\text{mean speed.}} = \frac{\omega_1 - \omega_2}{\omega}$$

ω_1 = max speed

ω_2 = min speed.

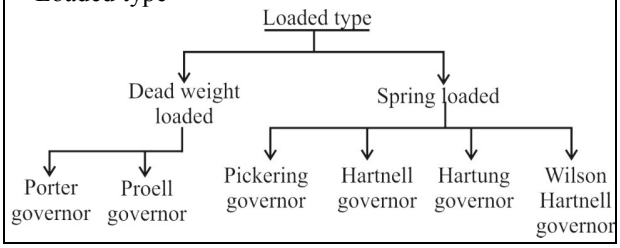
$\omega = \frac{\omega_1 + \omega_2}{2}$ (Mean speed)

909. A porter governor is a/an

(a) pendulum-type governor
(b) dead weight governor
(c) spring-loaded governor
(d) inertia governor

Haryana PSC AE (PHED) 05.09.2020, Paper-II
Assam PSC AE (PHED), 18.10.2020
HPPSC AE 2018
GPSC Lect. 23.10.2016
Mizoram PSC AE/SDO 2014, Paper-II
JPSC AE 2013, Paper-V
UKPSC AE-2013, Paper-I

Ans : (b) : Type of governors-
(i) Centrifugal governors
(ii) Inertia governors
Type of Centrifugal governor
■ Pendulum type- watt governor (simplest)
■ Loaded type



910. The ratio of the height of a porter governor (when the length of arms and links are equal) to the height of a Watt's governor is m = Mass of the ball, and M = Mass of the load on the sleeve.

(a) $\frac{m}{m+M}$ (b) $\frac{M}{m+M}$
(c) $\frac{m+M}{m}$ (d) $\frac{m+M}{M}$

VIZAG MT, 14.12.2020
Kerala PSC Poly. Lect., 2017
ISRO Scientist/Engineer 07.05.2017
TSPSC Managers, 2015
VIZAG MT 2015
TNPSC AE (Industries) 09.06.2013

Ans. (c) : Ratio of height of porter governor (when the length of arms and links are equal) to the height of a Watt's governor.

$\boxed{\frac{h_{porter}}{h_{watt}} = \frac{M+m}{m}}$ – When length of arms and links are equal.

Where, m = Mass of ball
M = Mass of the load on the sleeve.

911. In a Hartnell governor, if the stiffness of spring is increased, then the governor will be

(a) more sensitive (b) less sensitive
(c) hunting (d) insensitive

Nagaland PSC (CTSE) 2018, Paper-I
WBPSC AE, 2017
GPSC Lect. (Poly.) 07.12.2014
UKPSC AE-2013, Paper-I
OPSC Civil Services Pre. 2011
UKPSC AE 2007 Paper -I

Ans. (b) : Stiffness of Hartnell governor

$$\boxed{K = 2\left(\frac{a}{b}\right)^2 \frac{(F_1 - F_2)}{r_1 - r_2}} = 2\left(\frac{a}{b}\right)^2 \left(mr\omega^2 - mr'\omega'^2\right)$$

$\boxed{K \uparrow, r - r' \downarrow}$

r–r' Proportional to sleeve movement so that also decreases making governor less sensitive.

912. If N_1 and N_2 are maximum and minimum equilibrium speeds then sensitiveness of governor will be :

(a) $\frac{N_1 + N_2}{2N_1}$ (b) $\frac{N_1 + N_2}{2N_2}$
(c) $\frac{N_1 - N_2}{N_1 + N_2}$ (d) $\frac{2(N_1 - N_2)}{(N_1 + N_2)}$

JPSC AE 10.04.2021, Paper-II
WBPSC AE, 2017
HPPSC Asstt. Prof. 2014
J & K PSC Screening, 2006
ESE 2014, ESE 2005

Ans. (d) : Generally,
By considering performance of engine

$$\text{Sensitivity} = \frac{N_1 - N_2}{N_{mean}}$$

$$= \frac{2(N_1 - N_2)}{N_1 + N_2}$$

913. If a flywheel has mass moment of inertial I, rotating at a mean speed ω kinetic energy is E and coefficient of fluctuation of speed is K, then maximum fluctuation of energy (E) in a flywheel will be equal to

(a) $I\omega^2K$ (b) 2 KE
(c) $I\omega(\omega_1-\omega_2)$ (d) all of the above

Nagaland PSC (CTSE) 2018, Paper-I
BPSC Poly. Lect. 2016
VIZAG MT 2015
MPSC Poly. Lect. 2014
TNPSC ACF 2012
J&K PSC Civil Services Pre, 2010

Ans. (d) : The maximum fluctuation of energy is the difference between the maximum and minimum energies maximum fluctuation of energy (ΔE_{max}).

$\Delta E = I\omega^2K = 2KE$

$\Delta E = I\omega(\omega_1 - \omega_2)$

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914. A Hartnell governor is a

(a) pendulum type governor
(b) spring loaded governor
(c) dead weight governor
(d) inertia governor

GPSC DEE, Class-2 (GWSSB) 04.07.2021
CGPSC ARTO & TSI 07.01.2017
GPSC Lect. 23.10.2016
Kerala PSC IOF 19.04.2016
GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Hartnell governor is a spring controlled centrifugal governor, in which a spring controls the movement of the ball and hence the sleeve.

915. The speed of an engine varies from 210 rad/s to 190 rad/s. During cycle the change in kinetic energy is found to be 400 Nm. The inertia (I) of the flywheel in kgm^2 is–

(a) 0.10 (b) 0.20
(c) 0.30 (d) 0.40

GPSC ARTO Pre 30.12.2018
TSPSC AEE 2017
RPSC 2016, GATE 2007
Vizag Steel MT (Re-Exam) 24.11.2013

Ans : (a) Maximum speed (ω_1) = 210 rad/s
Minimum speed (ω_2) =190 rad/s
Maximum fluctuation of energy is given by relation,
$\Delta E = I\omega^2 C_s$

$$\text{Where, } C_s = \frac{\omega_1 - \omega_2}{\frac{\omega_1 + \omega_2}{2}} = 0.1$$

$$\text{and, } \omega = \frac{\omega_1 + \omega_2}{2} = 200\,\text{rad/s}$$

$$400 = I \times (200)^2 \times 0.1$$

$$I = \frac{400}{(200)^2 \times 0.1} = 0.10\,\text{kgm}^2$$

$$\boxed{I = 0.10\,\text{kgm}^2}$$

916. A flywheel has radius of gyration of 1 m and mass 2500 kg. The starting torque is 1500N-m and may be assumed constant. The angular acceleration of the flywheel will be :

(a) 0.3 rad/s^2 (b) 0.6 rad/s^2
(c) 0.9 rad/s^2 (d) 1.2 rad/s^2

RPSC ACF & FRO, 26.02.2021
Oil India Senior Officer 23.12.2020
RPSC Vice Principal ITI 2018
MPPSC State Forest Service Exam, 2014

Ans. (b) : Given, k = 1 m, m = 2500 kg, T = 1500 N-m
α = Angular acceleration
$I = mk^2 = 2500 \times 1^2 = 2500\ \text{kg} - \text{m}^2$
$T \Rightarrow 1500 = I\alpha$

$$\alpha = \frac{1500}{2500} = 0.6\,\text{rad/s}^2$$

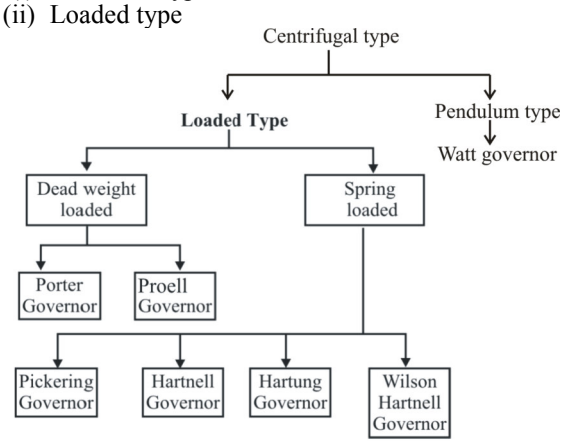
917. A Hartnell governor is a governor of the

(a) deadweight type (b) pendulum type
(c) inertia type (d) centrifugal type

Karnataka PSC Lect., 27.05.2017
Assam PSC CCE Pre 2015
Haryana PSC Civil Services Pre, 2014
TNPSC ACF 2012

Ans. (d) : Type of Governor
(i) Centrifugal Governor
(ii) Inertia Governor

Type of Centrifugal Governor
There are two type of centrifugal governor
(i) Pendulum type – Watt Governor
(ii) Loaded type



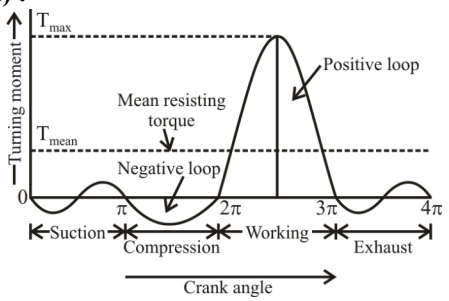
- Dead weight type is also called gravity controlled governor.

918. In a turning moment diagram, the variations of energy above and below the mean resisting torque line is called

(a) fluctuation of energy
(b) maximum fluctuation of energy
(c) coefficient of fluctuation of energy
(d) none of the above

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017
Nagaland CTSE 2017 Paper-I
JPSC AE 2013 Paper-I

Ans. (a) :



In a turning moment diagram, the variations of energy above and below the mean resisting torque line is called fluctuation of energy.

- **Fluctuation of energy :-** In a turning moment diagram, the variations of energy above and below the mean resisting torque line is called fluctuation of energy.
- **Maximum fluctuation of energy :-** The difference between the maximum and minimum energies is known as maximum fluctuation of energy.
- **Coefficient of fluctuation of speed :-** The ratio of the maximum fluctuation of speed to the mean speed is called coefficient of fluctuation of speed.
- **Coefficient of fluctuation of energy :-** Is defined as the ratio of the maximum fluctuation of energy to the work done per cycle.

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919. Sensitivity of an isochronous governor is:-

(a) Zero (b) One
(c) Two (d) Infinity

TSPSC Manager (Engg.) HMWSSB 12.11.2020
GPSC Asstt. Prof. 28.08.2016
UKPSC AE-2013, Paper-I
UKPSC AE 2007 Paper -I, ESE 1997

Ans. (d) : Sensitivity of an isochronous governor is infinity.

920. A flywheel of moment of inertia 9.8 kgm^2 fluctuates by 30 rpm for a fluctuation of energy of 1936. The mean speed of the flywheel is in rpm.

(a) 600 (b) 900
(c) 968 (d) 2940

Kerala PSC Poly. Lect., 2017
Kerala PSC IOF 19.04.2016
PSPCL AE, 2012
GATE 1998

Ans. (a) : $\Delta E = \frac{1}{2}I\omega_1^2 - \frac{1}{2}I\omega_2^2$

$= \frac{1}{2}I\left(\omega_1^2 - \omega_2^2\right)$

$= \frac{1}{2}I\left(\omega_1 - \omega_2\right)\left(\omega_1 + \omega_2\right)$

$= I(\omega_1 - \omega_2)\ \omega_{mean}$

$1936 = 9.8 \times \frac{30 \times 2\pi}{60} \times \omega_{mean}$

$\omega_{mean} = 62.88$ rad/sec

$\omega_{mean} = \frac{2\pi N}{60}$

$\frac{60.88 \times 60}{2\pi} = N$

N = 600.4830 rpm

921. The speed variations of the engine during a cycle for constant output load is reduced by a :

(a) D-slide value
(b) Governor
(c) Fly-wheel
(d) Mayers expansion value

SJVN ET 2019
NPCIL ET, 2015
UKPSC AE 2012 Paper-I
J & K PSC Screening, 2006

Ans. (c) : Flywheel reduces speed fluctuation during a cycle for a constant load, but flywheel does not control the mean speed of the engine if the load changes.
Governor controls speed fluctuations during a cycle for a constant load & governor also controls the mean speed of the engine if load changed.

922. A circular solid disc of uniform thickness 20 mm, radius 200 mm and mass 20 kg, is used as a flywheel. If it rotates at 600 rpm, the kinetic energy of the flywheel, in joules is_____

(a) 1580 (b) 790
(c) 3160 (d) 395

Assam Engg. College AP/Lect. 18.01.2021
GPSC Executive Engineer 23.12.2018
RPSC 2016
GATE 2012

Ans. (b) : Radius = 200 mm
Mass = 20 kg
N = 600 rpm

Kinetic energy of fly wheel (KE) = $\frac{1}{2}I\omega^2$

$= \frac{1}{2} \times \left[\frac{1}{2}mr^2\right] \times \left[\frac{2\pi N}{60}\right]^2$

$= \frac{1}{4} \times 20 \times 0.2^2 \times \left[\frac{2\pi \times 600}{60}\right]^2$

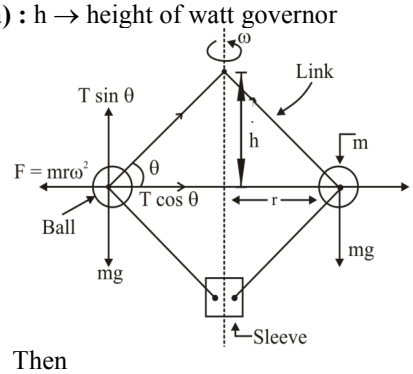
KE = 789.56 Joule

923. The height of watt governor is: [g = acceleration due to gravity, ω = angular velocity in rad/s]

(a) $\frac{g}{\omega^2}$ (b) $\frac{g}{2\omega^2}$
(c) $\frac{g}{4\omega^2}$ (d) $\frac{g}{8\omega^2}$
(e) $\frac{g}{10\omega^2}$

CGPSC AE 25.02.2018
UJVNL AE 2016
BPSC Poly. Lect. 2016

Ans. (a) : h → height of watt governor



Then

$T \sin\theta = mg$ ...(1)

$T \cos\theta = mr\omega^2$ ...(2)

(1) ÷ (2)

$\tan\theta = \frac{g}{r\omega^2}$ $\tan\theta = \frac{h}{r}$

$\frac{h}{r} = \frac{g}{r\omega^2} \Rightarrow \boxed{h = \frac{g}{\omega^2}}$ $\boxed{h = \frac{895}{N^2}}$

924. If C_f is the coefficient of speed fluctuation of a flywheel then the ratio of $\omega_{max}/\omega_{min}$ will be

(a) $\frac{1-2C_f}{1+2C_f}$ (b) $\frac{2-C_f}{2+C_f}$
(c) $\frac{1+2C_f}{1-2C_f}$ (d) $\frac{2+C_f}{2-C_f}$
(e) $\frac{1+C_f}{1-C_f}$

RPSC 2016
CGPSC AE 26.04.2015 Shift-I
GATE 2006

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Ans. (d) : Coefficient of speed fluctuation

$$C_f = \frac{(\omega_{max} - \omega_{min}) \times 2}{(\omega_{max} + \omega_{min})}$$

$$= \frac{\left(\frac{\omega_{max}}{\omega_{min}} - 1\right)}{\left(\frac{\omega_{max}}{\omega_{min}} + 1\right)} \times 2 = \frac{k-1}{k+1} \times 2$$

$$\frac{C_f}{2} = \frac{k-1}{k+1}$$

$$\left(k = \frac{\omega_{max}}{\omega_{min}}\right)$$

$C_f (k + 1) = 2 (k - 1)$

$$k = \frac{2 + C_f}{2 - C_f}$$

925. The height of Watt's governor is proportional to

(a) N (speed) (b) N^2
(c) 1/N (d) $1/N^2$
(e) $1/\sqrt{N}$

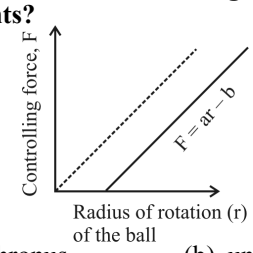
Nagaland PSC (CTSE) 2018, Paper-I
CGPSC AE 26.04.2015 Shift-I
CSE Pre-1999

Ans. (d) : If N is the speed of arm and ball about the spindle axis, then the height of the governor (h) is given by

$$h = \frac{895}{N^2}$$

From this expression, we see that the height of a Watt's governor proportional to $1/N^2$.

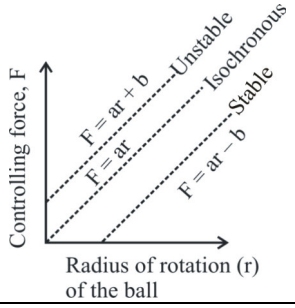
926. The controlling force curve for a spring controlled Governor is given in the figure. Which of the following conditions it represents?



(a) Isochronus (b) unstable
(c) stable (d) hunting

APPSC IOF, 2009
WBPSC AE, 2007
OPSC Civil Services Pre 2006

Ans. (c) :



927. For a flywheel operating with maximum and minimum angular speeds during the cycle as ω_1 and ω_2 respectively, the coefficient of fluctuation of speed is given by

(a) $\frac{\omega_1 - \omega_2}{\omega_1}$ (b) $\frac{\omega_2 - \omega_1}{\omega_2}$
(c) $\frac{\omega_1 - \omega_2}{2(\omega_1 - \omega_2)}$ (d) $\frac{2(\omega_1 - \omega_2)}{(\omega_1 + \omega_2)}$

Haryana PSC AE (PHED) 05.09.2020, Paper-II
UPSC JWM 2017 / DRDO Scientists 2008

Ans. (d) : Co-efficient of fluctuation of speed

$$= \frac{\text{Max. speed} - \text{Min. speed}}{\text{Average Speed}} = \frac{2(\omega_1 - \omega_2)}{(\omega_1 + \omega_2)}$$

928. The ratio of maximum fluctuation of speed to the mean speed is called.

(a) fluctuation of speed
(b) coefficient of fluctuation of speed
(c) maximum fluctuation of speed
(d) none of these

UPRVUNL AE 05.07.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE, 2008

Ans. (b) : The ratio of maximum fluctuation of speed to the mean speed is called coefficient of fluctuation of speed (C_s).

$$C_s = \frac{N_{max} - N_{min}}{N_{mean}}$$

929. Function of a governor is to :

(a) control the engine speed
(b) maintain the speed of engine constant
(c) store energy and give up whenever required
(d) adjust variation of speed by varying the input to the engine

RPSC AE 2016
GPSC Lect. (Poly.) 07.12.2014 / RPSC ACF-2011

Ans. (d) : Function of a governor is to adjust variation of speed by varying the input to the engine.

930. Which of the following governor cannot be isochronous ?

(a) Watt (b) Hartnell
(c) Proel (d) Porter

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
(HPPSC AE 2014) / BPSC AE 2012 Paper - VI
CSE Pre 2009

Ans : (d) Porter governor cannot be isochronous.
A governor is said to be isochronous when the equilibrium speed is constant (i.e. range speed is zero) for all radii of rotation of the balls within the working range, neglecting friction. The isochronism is the stage of infinite sensitivity.
Let us consider the case of a Porter governor running at speeds N_1 and N_2 r.p.m. We assume that friction force is zero (f = 0)

$$(N_1)^2 = \frac{m + \frac{M}{2}(1+q)}{m} \times \frac{895}{h_1}$$

and $$(N_2)^2 = \frac{m + \frac{M}{2}(1+q)}{m} \times \frac{895}{h_2}$$

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For isochronism, range of speed should be zero i.e. N_2-N_1=0 or N_2=N_1. Therefore from equations (i) and (ii), h_1=h_2, which is impossible in case of a Porter governor. Hence a Porter governor cannot be isochronous

931. When the sleeve of a porter governor moves upwards, the governor speed

(a) increases
(b) decreases
(c) remains unaffected
(d) first increases and then decreases

JPSC AE Pre 2019
Nagaland PSC CTSE 2017, Paper-I
TSPSC AEE 2015

Ans : (a) When the sleeve of a porter governor moves upwards, the governor speed increase. If the sleeve of a watt governor is loaded with a heavy mass, it becomes a porter governor

932. Isochronism in a governor is desirable when

(a) The engine operates at low speeds.
(b) The engine operates at high speeds.
(c) Only one speed is desired to be kept at all loads.
(d) The engine operates at variable speeds.

UPPSC AE 12.04.2016 PAPER-I
HPPSC W.S. POLY. 2016
OPSC Civil Services Pre. 2011

Ans : (c) Isochronous Governor:- A governor is said to be isochronous when the equilibrium speed is constant for all radii of rotation of the balls within the working range, neglecting friction. The isochronism is the stage of infinite sensitivity.

Let us consider the case of a porter governor running at speeds N_1 and N_2 rpm. Assume f = 0

$$N_1^2 = \frac{m+\frac{m}{2}(1+K)}{m} \times \frac{895}{h_1} \quad \text{.....(i)}$$

$$N_2^2 = \frac{m+\frac{m}{2}(1+K)}{m} \times \frac{895}{h_2} \quad \text{.....(ii)}$$

For isochronism range of speed should be zero e.g. N_2-N_1= 0 or $N_2 = N_1$. Therefore from equation (i) and (ii)

$h_1 = h_2$ which is impossible in case of a porter governor. Hence a porter governor cannot be isochronous.

933. A hunting governor is -

(a) More stable (b) Less sensitive
(c) More sensitive (d) None of these

GMB AAE 25.06.2017
RPSC VPITI 14.02.2016
J&K PSC Civil Services Pre 2010

Ans. (c) : A Governor is said to be sensitive when it readily respond to small change of speed.

$$\text{Sensitiveness} = \frac{\text{Range of Speed}}{\text{Mean speed}}$$

If a governor is too sensitive, it may fluctuate continuously when the load on the engine falls. the sleeve rises rapidly to a maximum position. The speed subsequently rises and becomes more than the average with the result that the sleeve again rises to reduce the fuel supply. This process is known as hunting. hunting depends range of speed if range of speed increase the sensitiveness become more.

934. A flywheel connected to a punching machine has to supply energy of 400 Nm while running at a mean angular speed of 20 radians/s. If the total fluctuation of speed is not to exceed ± 2%, the mass moment of inertia of the flywheel in kg – m^2 is

(a) 25 (b) 50
(c) 100 (d) 125

GPSC Asstt. Director of Transport 05.03.2017
APGENCO AE, 2017
GATE 2013

Ans. (a) : Given,

$\Delta E = 400$ Nm, $\omega_{mean} = 20$ rad/s

$C_s = \pm 2\%$

$$C_s = 2 \times \frac{2}{100} = 0.04$$

$$\Delta E = I\omega_{mean}^2 \times C_s \Rightarrow 400 \text{ Nm} = I(20)^2 \times 0.04$$

$I = 25$ kg-m^2

935. In Gramophones for adjusting the speed of the turntable, the following type of governor is commonly employed–

(a) Hartung governor
(b) Wilson Hartnell governor
(c) Pickering governor
(d) Inertia governor

Vizag Steel (MT) 2017
HPPSC Asstt. Prof. 18.09.2017
TSPSC AEE 2015, ESE 2005

Ans. (c) : Pickering governor is used to drive a Gramophone.

936. The supply of working fluid to the engine to suit the load conditions is controlled by:

(a) flywheel (b) governor
(c) throttle valve (d) all of these

HPPSC Asstt. Prof. 2014
UPSC JWM Adv. No-16/2009
ISRO Scientist/Engineer 2008

Ans. (b) : Governor is a mechanical device which maintain speed of engine constant when there is variation of load by means of controlling supply of fuel.

937. Which of the following statement is correct?

(a) Flywheel reduce speed fluctuations during a cycle for a constant load, but flywheel does not control the mean speed of the engine if the load changes
(b) Flywheel does not reduce speed fluctuations during a cycle for a constant load, but flywheel does control the mean speed of the engine if the load changes
(c) Governor control a speed fluctuations during a cycle for a constant load, but governor does not control the mean speed of the engine if the load change

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(d) Governor controls speed fluctuations during a cycle for a constant load, and governor also controls the mean speed of the engine if the load changes

OPSC AEE 2019 Paper-I
OPSC Civil Services Pre. 2011
GATE 2001

Ans : (a) : The flywheel used in machines serves as a reservoir which stores energy during the period when the supply of energy is more than the requirement and releases it during the period when the requirement of energy is more the supply.
The flywheel controls the speed variations caused by the fluctuations of the engine turning moment during each cycle of operation for constant load.

938. For a certain engine having an average speed of 20 rps, a flywheel approximated as a solid disc, is required for keeping the fluctuation of the speed within 2% about the average speed. The fluctuation of kinetic energy per cycle is found to be 2kJ. What is the least possible mass of the flywheel if its diameter is not to exceed 1 m?

(a) 40 kg (b) 62 kg
(c) 51 kg (d) 73 kg

RPSC ADE 2016
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2003

Ans. (c) : Given, N = 20 rps
$C_s = 0.02$, $\Delta E = 2000$ J

$$R = \frac{1}{2} = 0.5 \text{ m}$$

Average speed
$\omega = 2\pi N = 2\pi \times 20 = 125.66$ rad/s
We know that,

$$\Delta E = I\omega^2 C_s$$

$$\Delta E = \frac{MR^2}{2} \times \omega^2 \times C_s$$

$$M = \frac{2\Delta E}{R^2 \times \omega^2 \times C_s} = \frac{2 \times 2000}{(0.5)^2 \times (125.66)^2 \times 0.02}$$

$= 50.66 = 51$ kg.

939. The effort of a governor is the force exerted by the governor on the

(a) Balls (b) Sleeve
(c) Upper links (d) Lower links

TSPSC AEE 2017
TSPSC AEE 2015

Ans. (b) : Governor effort: It is defined as the mean force exerted, on the sleeve during a given change in speed.

940. The speed range of a Watt governor is

(a) 20 r.p.m. to 50 r.p.m.
(b) 60 r.p.m. to 80 r.p.m.
(c) 80 r.p.m. to 120 r.p.m.
(d) 120 r.p.m. to 200 r.p.m.

Assam PSC AE (PHED), 18.10.2020
Assam PSC CCE Pre 2015

Ans. (b) We know that,

$$h = \frac{g}{\omega^2} = \frac{9.81}{(2\pi N/60)^2} = \frac{895}{N^2}$$

- The speed range of watt governor is 60 r.p.m. to 80 r.p.m.

941. Which of the following machines uses flywheel?

(a) Milling machine (b) Broaching machine
(c) Drilling machine (d) Punching machine

VIZAG Steel MT 24.01.2021
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : Flywheel in Punching Press–The flywheel can also be used to perform the same function when the torque is constant and the load varies during the cycle. Such an application is found in punching press or in riveting machine. Thus flywheel has to absorb excess energy available at one stage and has to make up the deficient energy at the other stage to keep to fluctuations of speed within permissible limits.

942. The coefficient of steadiness of a flywheel running between the maximum speed N_1 rpm and minimum speed N_2 rpm is given by

(a) $\frac{N}{N_2 - N_1}$ (b) $\frac{N_1 - N_2}{N}$
(c) $\frac{N}{N_1 - N_2}$ (d) $\frac{N_2 - N_1}{N}$

Where N is the average speed $N = \frac{N_1 + N_2}{2}$

Karnataka PSC AE, 10.09.2017
RPSC 2016

Ans. (c) : Coefficient of fluctuation of speed,

$$C_s = \frac{\text{Range of speed}}{\text{Mean speed}}$$

$$\text{Coefficient of steadiness} = \frac{1}{C_s}$$

$$= \frac{\text{Mean speed}}{\text{Range of speed}}$$

$$= \frac{N_{mean}}{N_1 - N_2}$$

943. The approximate height of the Watt governor, when it rotates at 60 rpm is:

(a) 0.25 m (b) 0.76 m
(c) 0.12 m (d) 0.51 m

NLCIL GET 17.11.2020, Shift-II
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Given N = 60 rpm
Height of watt governor

$$h = \frac{895}{N^2}$$

$$h = \frac{895}{(60)^2}$$

$$h = 0.2486 \text{ m}$$

$$\boxed{h \approx 0.25 \text{m}}$$

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944. Hunting in a governor occurs due to
(a) worn-out guides of the sleeve
(b) fixed position of balls for each speed within working range
(c) friction
(d) none of these

TNPSC AE (Industries) 09.06.2013
TNPSC AE, 2008

Ans. (d) : A governor is said to be hunting if the speed of the engine fluctuates continuously above and below the mean speed. This is caused by a too sensitive governor which changes the fuel supply by a large amount when a small change in the speed of rotation takes place.

945. A governor is said to be isochronous when
(a) the equilibrium speed is constant for all radii of rotation of the balls within working range
(b) the range of speed is zero for all radii of rotation of the balls within working range
(c) any one of these
(d) none of these

APPSC Poly Lect. Automobile Engg., 2020
TNPSC AE, 2008

Ans. (c) : A governor is said to be isochronous when the equilibrium speed ($\omega_1=\omega_2=\omega$) is constant (i.e., range of speed is zero) for all radii of rotation of the balls within the working range, neglecting friction.

946. The effort of a governor is defined as the force to be applied on its sleeve for
(a) 5 percent change in speed
(b) 3 percent change in speed
(c) 1 percent change in speed
(d) no change in speed

APPSC Poly. Lect. 13.03.2020
BPSC AE 2012 Paper - VI

Ans : (c) : The effort of governor is the mean force exerted at the sleeve for a given percentage change of speed or lift of the sleeve.

947. A flywheel is a device which controls
(a) the mean speed of rotation of the engine shaft over a long period of time
(b) the speed variation caused by cyclic fluctuation of energy
(c) the fluctuation of energy over a long period
(d) the fuel supply to control the mean speed of the engine shaft

GPSC Lect. (Auto) 16.10.2016
OPSC Civil Services Pre 2006

Ans. (b) : Flywheel controls the fluctuation of speed variation caused by cyclic fluctuation of energy.

948. The radius of gyration for the solid disc flywheel is equal to
(a) $\frac{D}{2\sqrt{2}}$ (b) $\frac{D}{\sqrt{2}}$
(c) $\frac{D}{2}$ (d) $\frac{2\sqrt{2}}{D}$

TSPSC AEE 2015
UKPSC AE 2007 Paper -I

Ans : (a) Radius of gyration disc type flywheel:-

$$I = MK^2 = \frac{MR^2}{2}$$

$$K_D = \frac{R}{\sqrt{2}} = \frac{D}{2\sqrt{2}}$$

K_D = Radius of gyration disc type flywheel

949. The ratio of maximum fluctuation of energy to the work done per cycle is called as
(a) fluctuation of energy
(b) coefficient of fluctuation of speed
(c) coefficient of fluctuation of energy
(d) fluctuation of speed

APPSC Poly Lect. Automobile Engg., 2020
TSPSC AEE 2015

Ans : (c) Coefficient of fluctuation of energy is the ratio of maximum fluctuation of energy to the work done per cycle.

$$C_f = \frac{E_{max} - E_{min}}{\text{work done per cycle}}$$

950. For a governor running at constant speed, the force acting on the sleeve is
(a) Constant (b) Minimum
(c) Maximum (d) Zero

OPSC AEE 2019 Paper-I
Vizag Steel (MT) 2017

Ans. (d) : At constant speed, no sleeve movement therefore zero force.

951. In which of the following case, the turning moment diagram will have largest variations?
(a) Single cylinder two stroke engine
(b) Four stroke single cylinder engine
(c) Four cylinder four stroke engine
(d) six cylinder four stroke engine

RPSC ACF & FRO, 26.02.2021

Ans. (b) :

952. A flywheel of mass 6000 kg starts from rest, and gets up a speed of 20 rad/s in 2 minutes. The radius of gyration of the wheel is 100 cm. The average torque exerted on the wheel is _____.
(a) 2000 Nm (b) 500 Nm
(c) 1000 Nm (d) 750 Nm

VIZAG Steel MT 24.01.2021

Ans. (c) : Given, Mass (m) = 6000 kg, ω = 20 rad/s
t = 2 min = 120 sec
Radius of gyration (k) = 100 cm
We know that,

$\omega = \omega_0 + \alpha t$
$\omega_0 = 0$ (Initially)
$\omega = \alpha t$

$$20 = \alpha \times 120 \Rightarrow \alpha = \frac{20}{120}$$

$$\alpha = \frac{1}{6}\text{rad/s}^2$$

Average torque (T) = $I\alpha$

$$= mk^2 \times \alpha$$

$$= 6000 \times (100 \times 10^{-2})^2 \times \frac{1}{6}$$

$$= 1000 \text{ N-m}$$

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953. The normal acceleration of a fly wheel of diameter 1.6 m rotating at 6 rads^{-1} will be

(a) 28.8 rads^{-2} (b) 9.6 ms^{-2}
(c) 28.8ms^{-2} (d) 9.6 rads^{-2}

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Given, $\omega = 6$ rad/sec

Diameter (D) = 1.6 $\Rightarrow$ radius (r) $= \frac{1.6}{2} = .8$

Normal acceleration $(\alpha) = \frac{V^2}{r}$

$= \frac{\omega^2 r^2}{r}$ $[V = r\omega]$

$= \omega^2 \times r$

$= .8 \times (6)^2$

$= 28.8$ m/sec^2

954. The height of the watt's governor h and the speed of rotation N are related using the expression _____.

(a) $h = 895/N^2$ (b) $h = 895/N^3$
(c) $h = 985/N^2$ (d) $h = 985/N^3$

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Given, In watt's governor–
h = height of governor
N = speed rotation

$\therefore \boxed{h = \frac{895}{N^2}}$

955. In flywheels, if the turning moment is positive, then the crank shaft _____ and the work is done _____ the steam.

(a) accelerates, on (b) accelerates, by
(c) decelerates, by (d) decelerates, on

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : When the turning moment is positive, the crackshaft accelerates and the work is done by the steam.

• When the turning moment is negative, the crankshaft retards and the work is done on the steam.

956. The steam turbine can be governed by the following methods except

(a) Reaction governing
(b) Throttle governing only
(c) Nozzle control governing only
(d) Combination of throttle and nozzle control governing.

ESE 18.07.2021

Ans. (a) : Steam turbine is governed by
(i) Throttle governing
(ii) Nozzle control governing
(iii) Combination of throttle and nozzle governing.

· Reaction governing is done to increase the efficiency of turbine by the use of reaction blade.

957. Find change in vertical height of a watt governor when its speed increases from 60 rpm to 61 rpm

(a) 2.4 mm (b) 8.0 mm
(c) 8.9 mm (d) 8.2 mm

Oil India Senior Officer 23.12.2020

Ans. (b) : $\Delta h = 895\left[\frac{1}{N_1^2} - \frac{1}{N_2^2}\right]$

$\Delta h = 895\left[\frac{1}{60^2} - \frac{1}{61^2}\right]$

$\Delta h = 0.00808$ m

$\Delta h = 8$ mm

958. Which of the following equations does NOT represent the maximum fluctuation of energy of a flywheel (ΔE)?
(Consider ω, ω_1 and ω_2 mean, maximum and minimum angular speeds of the flywheel respectively, I – moment of inertia of the flywheel, K – coefficient of fluctuation of speed.)

(a) $\Delta E = \omega.I(\omega_1 - \omega_2)$ (b) $\Delta E = 2EK$
(c) $\Delta E = I\omega^2.K$ (d) $\Delta E = \frac{I\omega^2}{2}$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : The maximum fluctuation of energy

$\boxed{\Delta E = \frac{1}{2} I\omega_{max}^2 - \frac{1}{2} I\omega_{min}^2}$

$\Delta E = \frac{1}{2} I\left(\omega_{max}^2 - \omega_{min}^2\right)$

$\Delta E = I \frac{(\omega_{max} - \omega_{min})}{\frac{(\omega_{max} + \omega_{min})}{2}} \cdot \left(\frac{\omega_{max} + \omega_{min}}{2}\right)^2$

$\Delta E = I\omega^2 k$

$\Delta E = 2EK$

Where, I = Moment of inertia of flywheel

ω = Mean angular speed $\left(\frac{\omega_{max} + \omega_{min}}{2}\right)$

K = Coefficient of fluctuation of energy $\left(\frac{\omega_{max} - \omega_{min}}{2}\right)$

E = Kinetic energy at mean speed $\left(\frac{1}{2} I\omega^2\right)$

959. A solid cast iron disk, 1 m in diameter and 0.2 m thick, is used as a flywheel. It is rotating at 350 rpm. It is brought to rest in 1.5 s by means of a brake. If the mass density of cast iron is 7200 kg/m3, the torque exerted by the brake will be nearly.

(a) 3.5 kN m (b) 4.5 kN m
(c) 5.3 kN m (d) 6.3 kN m

ESE 2020

Ans. (a) : Mass of flywheel

$$m = \rho v = \rho \frac{\pi}{4} d^2 \times t$$

$$= 7200 \times \frac{\pi}{4} \times (1)^2 \times 0.2$$

$$= 1130.97 \text{ kg}$$

Moment of Inertia

$$(I) = \frac{mR^2}{2} = \frac{1130.97 \times (0.5)^2}{2} = 141.37 \text{ kg-m}^2$$

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$\omega = \omega_0 + \alpha t$

$0 = \frac{2\pi \times 350}{60} + \alpha + 1.5$

$\alpha = \frac{-2\pi \times 350}{60 \times 1.5} = -24.43 \text{ rad/s}^2$

Torque (T) = I α

= 141.37×24.43 = 3453.67 Nm

= 3.45 kNm

T ≈ 3.5 kNm

960. Centrifugal force of the rim in a big flywheel causes stress in arms which is

(a) shear (b) tensile

(c) compressive (d) bending

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : Centrifugal force of the rim in a big flywheel causes stress in arms which is tensile.

961. The maximum and minimum speeds of a flywheel are 630 rpm and 600 rpm. Then the coefficient of fluctuation of speed:

(a) 0.0487 (b) 0.0476

(c) 0.0300 (d) 0.0500

CIL MT 27.02.2020

Ans. (a) : $N_{max} = 630$ rpm

$N_{min} = 600$ rpm

$N_{mean} = \frac{630 + 600}{2} = 615$ rpm

Coefficient of fluctuation of speed $C_s = \frac{N_{max} - N_{min}}{N_{mean}}$

$C_s = \frac{630 - 600}{615}$

$= \frac{30}{615}$

$C_s = 0.0487$

962. The maximum fluctuation of energy in a flywheel is proportional to

(a) Maximum fluctuation of speed

(b) Mass moment of inertia of the flywheel

(c) Both (A) and (B)

(d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) :

Fluctuation of energy $(\Delta E) = \frac{\pi^2}{900} \times I \times N \times (N_1 - N_2)$

$\Delta E = \frac{\pi^2}{900} \times I \times N^2 \ C_s$

$\Delta E \propto I$

$\Delta E \propto C_s$

$E \propto (N_1 - N_2)$

963. If v is linear velocity of the mean radius of the flywheel rim, ρ is the density of the rim material and σ is the hoop stress due to centrifugal force, then they are related by the expression _____ .

(a) $v = (\sigma/\rho)$ (b) $v = \sqrt{(\sigma/\rho)}$

(c) $v = (\rho/\sigma)$ (d) $v = \sqrt{(\rho/\sigma)}$

VIZAG MT, 14.12.2020

Ans. (b) : Given,

Linear velocity = v

Circumferential stress = σ

Density = ρ

$\sigma = \rho v^2$

$v^2 = \frac{\sigma}{\rho}$

$\boxed{v = \sqrt{\sigma/\rho}}$

964. In a horizontal cross compound steam engine, maximum fluctuation of energy is 20 kJ/cycle, the fluctuation of speed is to be kept within ± 0.5 percent of the mean speed of 10 rad/s, the radius of gyration is 2m. The mass of the flywheel is :

(a) 2225 kg (b) 5000 kg

(c) 6324 kg (d) 1250 kg

NLC GET 17.11.2020, Shift-II

Ans. (b) : Given,

Fluctuation of energy (ΔE) = 20 kJ/cycle

Fluctuation of speed, $C_s = \pm 0.5\%$

$C_s = 1\% = 0.01$

Radius of gyration = 2 m

$\Delta E = I\omega_m^2 C_s$

$20 \times 10^3 = mk^2\omega_m^2 C_s$

$20000 = m \times (4) \ (10)^2 \times 0.01$

m = 5000 kg

965. The flywheel of a steam engine has a radius of gyration of 1 m and mass of 2500 kg. The starting torque of the engine is 2500 N-m. The kinetic energy of such a flywheel after 10 sec from rest position will be–

(a) 120 kN-m (b) 135 kN-m

(c) 130 kN-m (d) 125 kN-m

APPSC Poly Lect. 13.03.2020

Ans. (d) :Given, Mass (m) = 2500 kg

Radius of gyration (k)= 1 m

Starting torque (T) = 2500 N-m

We know that,

Moment of inertia (I)= $mk^2 = 2500 \times (1)^2 = 2500$ kg–m^2

T = I α

$\Rightarrow \alpha = \frac{T}{I} = \frac{2500 \text{ N} - \text{m}}{2500} = = 1 \text{ rad/s}^2$

Let,

ω_1 = Angular speed at rest = 0

ω_2 = Angular speed after 10 seconds

t = Time in seconds

We also know that

$\Rightarrow \omega_2 = \omega_1 + \alpha t = 0 + 1 \times 10 = 10$ rad/sec

$\Rightarrow \omega_2 = 10 \text{ rad/sec}$

Kinetic energy of flywheel (K.E.) = $\frac{1}{2} I\omega^2$

The kinetic energy of such a flywheel after 10 sec from rest position

$\text{K.E.} = \frac{1}{2} I\omega_2^2 = \frac{1}{2} \times 2500 \times (10)^2 = 1250 \times 100$

K.E. = 125 kN-m

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966. In a rim type of flywheel, the major mass is

(a) concentrated around periphery
(b) concentrated at the centre
(c) contributed due to arms
(d) balanced by centripetal forces

RPSC IOF, 2020

Ans. (a) : In a rim type of flywheel the major mass is concentrated around periphery mass equally distributed at rim of flywheel for uniform motion.

967. Mass of flywheel is 5000 kg and radius of gyration is 1.8 m. From the turning moment diagram maximum fluctuation of energy is found to be 52 kJ. If the mean speed of engine is 120 rpm, its maximum speed is:

(a) 115.6 rpm (b) 121.22 rpm
(c) 128.34 rpm (d) 132.58 rpm

HPPSC ADF 05.03.2019

Ans. (b) : Given,
Mass (m) = 5000 kg
Radius of gyration (k) = 1.8
Energy (E) = 52 kJ
ω_{mean} = 120 rpm

$$E = \frac{1}{2} I\left(\omega_{max}^2 - \omega_{min}^2\right)$$

$$E = \frac{1}{2} mk^2 \frac{\left(\omega_{max} + \omega_{min}\right)\left(\omega_{max} - \omega_{min}\right) \times 2}{2} \qquad [I = mk^2]$$

$$E = \frac{1}{2} mk^2 \omega_{mean}\left(\omega_{max} - \omega_{min}\right)$$

$52 \times 10^3 = 5000 \times (1.8)^2 \times 120\ (\omega_{max} - \omega_{min})$
$\omega_{max} - \omega_{min} = .026$ .....(i)
$\omega_{max} + \omega_{min} = 2 \times 120$ .....(ii)
From (i) & (ii)
$2\omega_{max} = 240.026$
Then,
ω_{max} = 120.01 rpm

968. If each ball of a porter governor has a mass of m, mass of sleeve is M and percentage change in speed is C then governor power for different inclinations of upper and lower arms is given by:

(a) $\left[m + \frac{M}{2}(1+k)\right] gh\left(\frac{C^2}{1+2C}\right)$

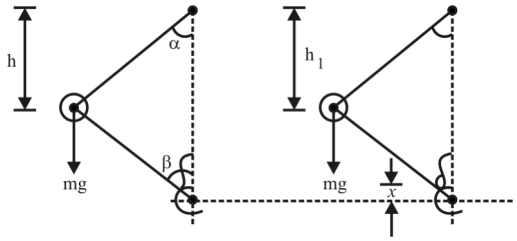
(b) $\left[m + \frac{M}{2}(1+k)\right] gh\left(\frac{4C^2}{1+2C}\right)$

(c) $\left[m + \frac{M}{2}(1+k)\right] gh\left(\frac{C^2}{1+C}\right)$

(d) $\left[m + \frac{M}{2}(1+k)\right] gh\left(\frac{4C^2}{1+C}\right)$

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : Power of porter governor
Let x = lift of governor
Governor power = P × x



If the height of the governor at speed N is h and at an increased speed (1 + C) N is h_1. Then

$x = 2(h - h_1)$

As there is no resultant force at sleeve in two equilibrium positions, therefore

$$h = \frac{m+M}{m} \frac{895}{N^2} \text{ and } h_1 = \frac{m+M}{m} \times \frac{895}{(1+c)^2 N^2}$$

$$\frac{h_1}{h} = \frac{1}{(1+c)^2} \text{ or } h_1 = \frac{h}{(1+c)^2}$$

We know that

$$x = 2(h - h_1) = 2\left[h - \frac{h}{(1+c)^2}\right] = 2h\left[1 - \frac{1}{(1+c)^2}\right]$$

$$= 2h\left[\frac{1 + C^2 + 2C - 1}{1 + C^2 + 2C}\right] = 2h\left(\frac{2C}{1+2C}\right) \quad C^2 \approx 0$$

$\therefore$ Governor power = C (m + M) g × 2h $\left(\frac{2C}{1+2C}\right)$

$$= (m+M)gh\left(\frac{4C^2}{1+2C}\right)$$

or Governor power

$$= \left[m + \frac{M}{2}(1+k)\right] gh\left(\frac{4C^2}{1+2C}\right)$$

969. A flywheel absorbs 24 kJ of energy while increasing its speed from 210 rps to 214 rps. What will be its kinetic energy at 220 rps?

(a) 825.1 kJ (b) 790.4 kJ
(c) 936.4 kJ (d) 684.9 kJ

SJVN ET 2019

Ans. (d) :

$$E = \frac{1}{2} I\left(\omega_2^2 - \omega_1^2\right)$$

$$2400 = \frac{1}{2} I\left(214^2 - 210^2\right)$$

I = 28.30 kgm^2
K.E. at 220 rps

$$E = \frac{1}{2}(28.30)(220)^2 = 684905.66 \text{ J} = 684.905 \text{ kJ}$$

970. The turning moment diagram for a single cylinder double acting steam engine consists of +ve and –ve loops above and below the average torque line. For the +ve loop, the ratio of the speeds of the flywheel at the beginning and the end is which one of the following?

(a) Less than unity (b) Equal to unity
(c) Greater than unity (d) Zero

OPSC AEE 2019 Paper-I, ESE 2007

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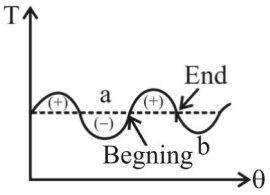
Ans : (a) : The given turning moment diagram,



∴ Ratio of flywheel speed at 'a' and 'b'

$$= \frac{\text{Speed at 'a'}}{\text{Speed at 'b'}}$$

Since the loop is positive i.e. above average torque and flywheel gains energy, so speed should be more at 'b' than 'a'.

∴ Speed ratio < 1

971. In watt governor, the weight of the ball is 50 N and the friction at the sleeve is 10 N. The coefficient of detention would be

(a) 5.0 (b) 0.5
(c) 0.2 (d) 0.1

Gujarat PSC AE 2019

Ans : (c) : Given, W = 50 N = mg = 50 N

f = 10 N

For watt governor, M = 0

$$\frac{N_{max} - N_{min}}{N_{mean}} = \frac{f}{(M+m)g}$$

So, coefficient of detention

$$= \frac{f}{mg} = \frac{10}{50} = 0.2$$

972. The change in governor height for a Watt governor when speed varies from 100 r.p.m. to 101 r.p.m. will be nearly

(a) 1.8 mm (b) 2.6 mm
(c) 3.4 mm (d) 4.2 mm

ESE 2019

Ans. (a) : Given,

$N_1 = 100$ rpm

$N_2 = 101$ rpm

Height of watt governor is given by

$$h = \frac{895000}{N^2} \text{ mm}$$

$$\Delta h = h_1 - h_2 = 895000\left[\frac{1}{N_1^2} - \frac{1}{N_2^2}\right]$$

$$= 895000\left[\frac{1}{100^2} - \frac{1}{101^2}\right]$$

= 1.763 mm = 1.8 mm

973. The coefficient of fluctuation of speed in flywheel for punching press is in the range of

(a) 0.003 – 0.06 (b) 0.03 – 0.05
(c) 0.06 – 0.2 (d) Above 0.2

TNPSC AE 2019

Ans. (c) : The coefficient of fluctuation of speed in flywheel for punching press is in the range of 0.06 – 0.2.

974. A three-cylinder single-acting engine has its cranks at 120°. The turning moment diagram for each cycle is a triangle for the power stroke with a maximum torque of 60 Nm at 60° after the dead centre of the corresponding crank. There is no torque on the return stroke. The engine runs at 400 r.p.m. the power developed will be

(a) 1745 W (b) 1885 W
(c) 1935 W (d) 1995 W

ESE 2019

Ans. (b) : Energy produced in one cycle per cylinder = Area of turning moment diagram

$$= \frac{1}{2} \times \pi \times 60$$

= 30 π Joule

Total energy produced by 3 cylinder = 90 π

$$\text{Mean torque} = \frac{\text{Total energy produced per cycle}}{\text{Angle rotated by crank per cycle}}$$

$$T_{mean} = \frac{90\pi}{2\pi} = 45 \text{ Nm}$$

$N_{mean} = 400$ rpm

Power produced = $T_{mean} \times \omega_{mean}$

$$= 45 \times \frac{2\pi \times 400}{60}$$

= 1885 W

975. A circular solid disc of uniform thickness 20mm, radius 400 mm and mass 10 kg, is used as a flywheel. If it rotates at 300 rpm, the kinetic energy of the flywheel, in Joules is

(a) 395 (b) 790
(c) 1580 (d) 3160

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : We know that kinetic energy in Flywheel

$$KE = \frac{1}{2} I\omega^2$$

$$KE = \frac{1}{2} \times \frac{mr^2}{2} \times \left(\frac{2 \times \pi \times N}{60}\right)^2$$

$$= \frac{1}{2} \times \frac{10 \times (0.4)^2}{2} \times \left(\frac{2 \times \pi \times 300}{60}\right)^2$$

$$KE = 394.784 \simeq 394.784 \text{ J}$$

976. Sensitiveness of the governor is defined as the ratio of the

(a) mean speed to the maximum equilibrium speed
(b) mean speed to the minimum equilibrium speed
(c) difference of the maximum and minimum equilibrium speeds to the mean speed
(d) sum of the maximum and minimum equilibrium speeds to the mean speed

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : Sensitiveness is the defined as the ratio of the difference between the maximum and minimum equilibrium speeds to the mean equilibrium speed,

$$\text{Sensitiveness} = \left[\frac{\omega_2 - \omega_1}{\omega}\right]$$

$$\therefore \quad \omega = \frac{\omega_1 + \omega_2}{2}$$

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977. If 'k' is the coefficient of speed fluctuation of a flywheel, then the ratio $\frac{\omega_{min}}{\omega_{max}}$ will be

(a) $\frac{1+2k}{1-2k}$ (b) $\frac{2-k}{2+k}$

(c) $\frac{2+k}{2-k}$ (d) $\frac{1-2k}{1+2k}$

Karnataka PSC AE, 10.09.2017

Ans. (b) : we know that,

$$k = \frac{\omega_{max} - \omega_{min}}{\left(\frac{\omega_{max} + \omega_{min}}{2}\right)}$$

$$\therefore \frac{k}{2} = \frac{\omega_{max} - \omega_{min}}{\omega_{max} + \omega_{min}}$$

$$\frac{k}{2} = \frac{1-(\omega_{min}/\omega_{max})}{1+(\omega_{min}/\omega_{max})}$$

$$\frac{k}{2}\left[1+\left(\frac{\omega_{min}}{\omega_{max}}\right)\right] = 1-\left(\frac{\omega_{min}}{\omega_{max}}\right)$$

$$\therefore \frac{\omega_{min}}{\omega_{max}} = \frac{2-k}{2+k}$$

978. Which of the following governors is/are not suitable for high speeds?
A. Watt Governor
B. Hartnell Governor
C. Hartung Governor
D. Wilson-Hartnell Governor
Select the code for the correct answer from the options given below :

(a) C only (b) A only

(c) Both A and C (d) Both C and D

Karnataka PSC AE, 10.09.2017

Ans. (b) : Spring loaded governor are suitable for high speeds.
- Pickering Governor
- Hartnell Governor
- Hartung Governor
- Wilson-Hartnell Governor

979. In Meyer's Expansion valve, main valve is driven by an eccentric having an angle of advance.

(a) 10° – 15° (b) 15° – 25°

(c) 25° – 30° (d) 30° – 40°

GMB AAE 25.06.2017

Ans. (c) : In Meyer's expansion valve, main valve is driven by an eccentric having angle of advance 25° – 30°.

980. The primary function of the flywheel is

(a) To limit the fluctuations of speed during each cycle

(b) To absorb energy during those periods of crank rotation when turning moment is less than the resisting moment

(c) To maintain constant speed of rotation of the crank shaft when the load on the engine increases.

(d) To maintain constant speed of rotation of the crank shaft when the load on the engine decreases

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (a) : The flywheel has 3 main features. The first is to maintain a rotating mass (inertia) to assist the engine rotation and provide a more consistent delivery of torque during running.

981. If the speed of the flywheel is to be maintained within ±1% of the mean speed, then the coefficient of the fluctuation of speed is:

(a) 0.01 (b) 0.02

(c) 0.03 (d) 0.04

(e) 0.05

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : We know that,
Coefficient of fluctuation of speed

$$C_s = \left[\frac{\omega_{max.} - \omega_{min.}}{\omega}\right]$$

$$C_s = \left[\frac{1.01\omega - 0.99\omega}{\omega}\right]$$

$$\boxed{C_s = 0.02}$$

982. A mass m is attached to a shaft at a radius r rotating at ω rad/s. The centrifugal force exerted by the mass is given by:

(a) $(m \cdot r)$ (b) $(m \cdot \omega \cdot r)$

(c) $(2 \cdot m \cdot \omega \cdot r)$ (d) $(m \cdot \omega^2 \cdot r)$

(e) $(n \cdot \omega \cdot r^2)$

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : A mass m is attached to a shaft at a radius r rotating at ω rad/s. The centrifugal force exerted by the mass is given by

$$F_C = mr\omega^2$$

983. A flywheel of a steam engine has a mass 2000 kg and a radius of gyration of 2 m. Its mass moment of inertia will be _____ kg. m^2.

(a) 6000 (b) 8000

(c) 16000 (d) 24000

(e) 32000

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Given, Mass of flywheel (m) = 2000 kg
Radius of gyration (k) = 2 m
Then moment of inertia for disc type flywheel

$$I_{Disc} = \frac{mk^2}{2} = \frac{2000 \times (2)^2}{2} = 4000 \text{ kg-m}^2$$

For Rim type flywheel

$$I_{Rim} = mk^2 = 2000 \times (2)^2 = 8000 \text{ kg-m}^2$$

In option 8000 kg-m^2 is given so answer will be 8000 kg-m^2.

984. A flywheel of 1500 mm diameter is fit to the crankshaft of an engine, which rotates at 300 rpm. What will be its angular velocity?

(a) 31.02 rad/sec (b) 31.42 rad/sec

(c) 32.02 rad/sec (d) 32.42 rad/sec

(e) 33.02 rad/sec

(CGPSC Polytechnic Lecturer 2017)

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Ans. (b) : D = 1500 mm, N = 300 rpm, ω = ?
We know that,

$$\omega = \frac{2\pi N}{60} = \frac{2\times 3.142\times 300}{60}$$

$$\boxed{\omega = 31.42 \text{ rad/sec.}}$$

985. For a Hartnell governor, the loads on the spring at the lowest and highest equilibrium speeds are 1150 N and 85 N, respectively. If the lift of the governor is 1.5 cm, then spring stiffness would be
(a) 700 N/cm (b) 710 N/cm
(c) 725 N/cm (d) 690 N/cm

UPSC JWM 2017

Ans. (b) : Load on the spring at the lowest speed = 1150N
Load on the spring at the highest speed = 85N
Lift of Governor, x = 1.5 cm

$$\text{Spring stiffness, } k = \frac{1150-85}{1.5\text{cm}} = 710\text{N/cm}$$

986. Which of the following is a correct statement?
(a) A flywheel regulates speed during one cycle while the governor regulates speed from cycle to cycle
(b) The flywheel is an essential element of every prime mover
(c) A flywheel controls the quantity or quality of the working agent
(d) A governor regulates speed by storing kinetic energy

BPSC AE Mains 2017 Paper - VI

Ans : (a) : A flywheel regulates speed during one cycle while the governor regulates speed from cycle to cycle.

987. If a flywheel having mass of 100 kg and radius of gyration 10 cm is rotating at 10 rad/s, its rotational kinetic energy will be
(a) 10 J (b) 25 J
(c) 50 J (d) 100 J

BPSC AE Mains 2017 Paper - VI

Ans : (c) : Given,
m = 10 kg
k = 10 cm = 0.1m
ω = 10 rad/s

$$KE = \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}mk^2\omega^2$$

$$KE = \frac{1}{2}\times 100\times 0.1^2\times 10^2$$

$$= 50 \text{ J}$$

988. The rim of a flywheel is subjected to
(a) direct tensile stress and bending stress
(b) torsional shear stress and bending stress
(c) direct shear stress and bending stress
(d) compressive stress and bending stress

ESE 2017

Ans. (a) : Due to rotation, rim is subjected to radially outwards centrifugal force.
Due to centrifugal force, tensile stress develops. Considering an element subtending angle dθ at centre, internal resistance in hoop/circumferential direction produces direct tensile stress and due to distance between distributed centrifugal force and supports there will be bending stresses. Portion of beam between two arms will act as fixed beam and uniformly loaded and hence, bending stress develops.

989. A flywheel fitted to a steam engine has a mass of 500 kg and radius of gyration 300mm. The starting torque is 900 N m. What is the kinetic energy after 10 s?
(a) 225 kJ (b) 450 kJ
(c) 900 kJ (d) 1800 kJ

ESE 2017

Ans. (c) : Given,
m = 500 kg
k = 300 mm = 0.3 m
T = 900 N-m
t = 10 sec
We know that

$$I = mk^2$$

$$= 500\times (0.3)^2$$

$$= 45 \text{ kg-m}^2$$

$$T = I\alpha$$

$$\alpha = \frac{T}{I} = \frac{900}{45} = 20 \text{ r/s}^2$$

$$\omega = \omega_0 + \alpha t$$

$$= 20\times 10$$

$$= 200 \text{ r/s}^2$$

$$\text{K.E.} = \frac{1}{2}I\omega^2 = \frac{1}{2}\times 45\times (200)^2 = 900 \text{ kJ}$$

990. In a Hartnell governor, the mass of each ball is 4 kg. The maximum and minimum centrifugal forces on the balls are 1800 N and 100 N at radii 25 cm and 20 cm, respectively. The lengths of vertical and horizontal arms of the bell-crank levers are the same. What is the spring stiffness?
(a) 780 N/cm (b) 740 N/cm
(c) 720 N/cm (d) 680 N/cm

ESE 2017

Ans. (d) : Given, m = 4 kg a = b
at r_1 = 20 cm, F_1 = 100 N
r_2 = 25 cm, F_2 = 1800 N

$$\text{Spring stiffness (k)} = 2\left(\frac{a}{b}\right)^2\left[\frac{F_2-F_1}{r_2-r_1}\right]$$

$$= 2\times 1\left[\frac{1800-100}{25-20}\right]$$

$$= 680 \text{ N/cm}$$

991. Consider the following statements:
1. In stationary constant speed engines, the spring-loaded governor mechanism is fitted on the cam-shaft of the engine.
2. Hunting occurs when the governor is not sensitive.
3. Isochronous governors have the same speed over a wide range of governor action.
4. A governor is said to be unstable if the radius of rotation falls as the speed increases.

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Which of the above statements are correct?

(a) 1, 2, 3 and 4 (b) 1, 2 and 4 only
(c) 1, 3 and 4 only (d) 2 and 3 only

ESE 2017

Ans. (c) : Statement second is wrong because hunting occurs when a governor is too sensitive.

992. A flywheel on a motor speeds up from rest to 1000 r.p.m. in 6 seconds. The number of revolutions made thereby is nearly.

(a) 200 (b) 100
(c) 50 (d) 20

ESE 2017

Ans. (c) : Given,

Flywheel on a motor speeds up from rest $\omega_0 = 0$ rad/sec

Time taken = 6 sec

$$\text{Final angular speed } (\omega) = \frac{2\pi N}{60}$$

$$= \frac{2\pi \times 1000}{60}$$

$$= 104.71 \text{ r/s}$$

$$\omega = \omega_0 + \alpha t$$

$$104.71 = \alpha t$$

$$\alpha = \frac{104.71}{6} = 17.4 \text{ r/s}^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$(104.71)^2 = 2 \times 17.4 \times \theta$$

$$\theta = 315.06 \text{ rad}$$

$$\text{No. of revolutions} = \frac{\theta}{2\pi} \simeq 50 \text{ approx}$$

993. A flywheel 200 cm in diameter is rotating at 100 rpm. A point on the rim of flywheel will not have:

(a) Angular velocity.
(b) Centripetal acceleration
(c) Tangential acceleration.
(d) All of the above.

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : The point rotating on the rim of flywheel, there its velocity is changing, the point must have radial acceleration. The point must rotate with constant angular speed (ω). Therefore, its linear speed is constant, by this reason, the point does not have a tangential acceleration.

994. For two governors A and B, and lift of sleeve of governor A is more than that of governor B, for a given fractional change in speed. It indicates that:

(a) governor A is more sensitive than governor B
(b) governor B is more sensitive than governor A
(c) both governors A and B are equally sensitive
(d) none of the above

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : Sensitivity of a governor– A governor is said to be sensitive when its readily response to a small change.

The lift of sleeve of governor A is more than that of governor B, then it means governor A is more sensitive than governor B.

995. A centrifugal watt governor is fitted with two balls, each of mass 2.5 kg. Find the angular velocity of the governor, when it is running at 75 rpm.

(a) 1.5π rad/sec (b) 1.75π rad/sec
(c) 2.5π rad/sec (d) 2.75π rad/sec
(e) 3π rad/sec

CGPSC ARTO & TSI 07.01.2017

Ans. (c) : Given,

m = 2.5 kg

N = 75 rpm

$$\omega = \frac{2\pi N}{60}$$

$$= \frac{2 \times \pi \times 75}{60}$$

$$= 2.5\pi \text{ rad/sec}$$

996. Governor power is defined as

(a) The work done at the sleeve for a given percentage change of speed
(b) The work done at the sleeve for a given percentage change of mass
(c) The work done at the sleeve for a given percentage change of force
(d) The work done at the sleeve for a given percentage change of volume

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (a) : Governor power is defined as the work done at the sleeve for a given percentage change of speed.

Power = Effort governor × displacement

$$\boxed{\text{Effort} = \frac{1}{2} \times s \times h}$$

Where, s = Effort of governor

h = displacement of sleeve

997. A flywheel fitted to an engine has a mass of 1000 kg and the radius of gyration is 200 mm. The starting torque of engine is 500 Nm and is assumed constant. Find the kinetic energy of the flywheel after 5 seconds.

(a) 150 kJ (b) 200 kJ
(c) 250 kJ (d) 275 kJ
(e) 300 kJ

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (*) : Given, m = 1000 kg, k = 200 mm

T = 500 Nm, t = 5 sec

K.E. = ?,

$I = m.k^2 = 1000 \times 0.2^2$

$= 40 \text{ kgm}^2$

$T = I\alpha$

$500 = 40 \times \alpha$

$\alpha = 12.5 \text{ rad/s}^2$

$\omega = \omega_0 + \alpha t$

$\omega = 12.5 \times 5 = 62.5 \text{ rad/s}$

$$KE = \frac{1}{2} \times I \times \omega^2$$

$$= \frac{1}{2} \times 40 \times 62.5^2 = 78125$$

$= 78.125$ kJ

Note-Official answer given by commission (b).

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998. A flywheel of moment of inertia 9 kg m^2 fluctuates by 40 rpm for a fluctuation in energy of 2400 Joules. The mean speed of flywheel (in rpm) is approximately:

(a) 500 (b) 525
(c) 600 (d) 675
(e) 725

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know,
In flywheel,

$$\Delta E = \frac{1}{2} I\left(\omega_{max}^2 - \omega_{min}^2\right)$$

$= I\Delta\omega\omega_m$, $\therefore$ I = 9 kgm^2

$\Rightarrow$ $\Delta\omega$ = 40 rpm, ω_m = mean speed

ΔE = 2400 Joules,

$\Rightarrow$ $\Delta\omega$ = 40 rpm

$2400 = 9 \times 40 \times \omega_m$

$$\omega_m = \frac{60}{9} = 6.7\,rad/s$$

ω_m = 6.7 rad/s $\approx$ 600 rpm

999. A governor with a range of speed .....is known as an isochronous governor.

(a) 1000 (b) infinity
(c) unity (d) zero

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : Isochronous Governor– A governor with sensitivity equal to infinity is treated as isochronous governor. For all position of sleeves, governor has same speed. The range of speed is zero.

1000. .....governor is also known as Radial-Spring Governor.

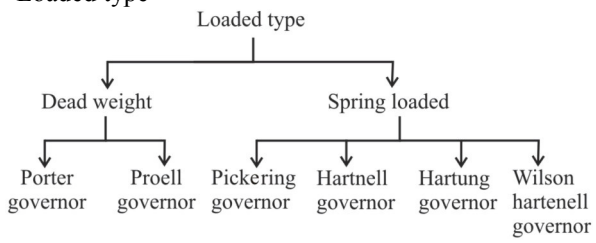
(a) Inertia (b) Wilson-Hartenell
(c) Proell (d) Watt

HPPSC AE (PWD) 10.09.2016

Ans. (b) : Types of governor
(i) Centrifugal governors
(ii) Inertia governors

Type of centrifugal governor

- Pendulum type- Watt governor
- Loaded type



1001. Change in vertical height of a Watt governor when its speed increases from 60 rpm to 61 rpm and when it increases from 150 rpm to 151 rpm will be respectively

(a) 810 mm and 0.52 mm
(b) 0.52 mm and 810 mm
(c) 14 mm and 0.3 mm
(d) 0.3 mm and 14 mm
(e) None of these

CGPSC AE 16.10.2016

Ans. (e) : We know,

$$h = \frac{895}{N^2}$$

For 60 & 61 rpm

So, $$\Delta h = \frac{895}{60^2} - \frac{895}{61^2} = 0.008084 = 8\,mm$$

For 150 & 151 rpm

$$\Delta h = \frac{895}{150^2} - \frac{895}{151^2} = 0.000525$$

Δh = 0.5 mm

1002. What is the value of radius of gyration of disc type flywheel as compared to rim type flywheel for the same diameter?

(a) $\sqrt{2}$ times (b) $1/\sqrt{2}$ times
(c) 2 times (d) 1/2 times
(e) None of the above

CGPSC Poly. Lect. 22.05.2016, ESE 2004

Ans. (b)
For solid disc

$$I = \frac{1}{2}MR^2$$

or $$\frac{1}{2}MR = mk_1^2$$

$$\left[\because I = mk^2\right]$$

$$k_1 = \frac{R}{\sqrt{2}} \quad .....(I)$$

For Rim type disc –

$MR^2 = Mk_2^2$

$k_2 = R$ ..... (II)

From equation (I and II) –

$$\boxed{k_1 = \frac{1}{\sqrt{2}} \times k_2}$$

Radius of gyration of disc type flywheel as compared to $\frac{1}{\sqrt{2}}$ rim type disc.

1003. The function of governor in turbine is

(a) keeping the speed of rotation of runner constant for varying loads
(b) keeping the load constant for varying speed of rotations
(c) keeping the speed of rotation of runner in proportion to the varying loads
(d) keeping the discharge through the runner constant for varying loads

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : A steam turbine governor is a component of turbine control system that regulates rotational speed in response to changing load conditions. The governor output - signal manipulates the position of steam inlet valve or nozzles which in turn regulates the steam flow to the turbine.

1004. Which of the following governors is not spring controlled?

(a) Hartnell governor
(b) Hartung governor

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(c) Wilson-hartnell governor
(d) Porter governor

RPSC LECTURER 16.01.2016

Ans. (d) : Hartnell governor, Hartung governor and Wilson-hartnell governor are spring controlled governors whereas Porter governor is dead weight type governor.

1005. The minimum speed (rpm) of engine at full load condition and maximum speed (rpm) of engine at no load condition is given by 245 and 255 respectively. The sensitiveness of governor will be:

(a) 1% (b) 2%
(c) 4% (d) 3%

UPRVUNL AE 07.10.2016

Ans. (c) : $N_{max} = 255$
$N_{min} = 245$
Then, sensitiveness of governor will be the ratio of range of speed to average of speed.

$$= \frac{(N_{max} - N_{min})}{\left[\frac{N_{max} + N_{min}}{2}\right]}$$

$$= \frac{2\times(255-245)}{255+245} = \frac{2\times10}{500}\times100 = 4\%$$

1006. The maximum fluctuation of energy of flywheel:

(a) is directly proportional to coefficient of fluctuation of speed
(b) is directly proportional to square of angular velocity of flywheel
(c) is directly proportional to moment of inertia of flywheel
(d) all of the above

HPPSC W.S. Poly. 2016

Ans : (d) Maximum fluctuation of energy of flywheel.

$E_f = E_{max} - E_{min}$

$$E_f = \frac{1}{2}I\omega_1^2 - \frac{1}{2}I\omega_2^2$$

$$E_f = \frac{1}{2}I(\omega_1 - \omega_2)(\omega_1 + \omega_2)$$

$E_f = I(\omega_1 - \omega_2)\omega$ ……..(i)

$E_f = \frac{I(\omega_1 - \omega_2)}{\omega}.\omega^2$ multiplying and dividing by ω

$E_f = I\omega^2 . C_s$ …… (ii)

$$C_s = \left(\frac{\omega_1 - \omega_2}{\omega}\right)$$

$$E_f = 2 \times \frac{I\omega^2 C_s}{2}$$

$E_f = 2\ EC_s$ …….. (iii)

1007. Flywheel maintains consistency of power transmission due to

(a) Its light weight
(b) Its high moment of inertia.
(c) its speed of rotation
(d) Its capability to be rotated by wind power.

UJVNL AE 2016

Ans : (b) Flywheel maintains consistency of power transmission due to its high moment of Inertia.

$E_{flc} = I\omega^2 C_s$.

I = moment of Inertia.
ω = mean angular speed
C_s = coefficient of fluctuation

1008. A flywheel is fitted to the crank shaft of an engine having W amount of indicated work per revolution. Permissible limits of coefficient of fluctuation of energy and speed are C_E and C_S respectively. The kinetic energy of the flywheel is given by

(a) $2\frac{W.C_E}{C_S}$ (b) $\frac{W.C_E}{2C_S}$
(c) $\frac{W.C_E}{C_S}$ (d) $\frac{W.C_s}{2C_E}$

UPPSC AE 12.04.2016 Paper-I, ESE 1993

Ans : (b)
Coefficient of fluctuation of energy

$$(C_E) = \frac{\text{fluctuation of energy}}{\text{work done/cycle}}$$

$$C_E = \frac{\frac{1}{2}I\omega_1^2 - \frac{1}{2}I\omega_2^2}{W}$$

$$C_E = \frac{\frac{1}{2}I(\omega_1 - \omega_2)(\omega_1 + \omega_2)}{W}$$

$$C_E = \frac{I(\omega_1 - \omega_2)\times\omega_{mean}}{W}$$

$$\omega_{mean} = \frac{C_E \times W}{I(\omega_1 - \omega_2)}$$

Kinetic energy of flywheel $(KE) = \frac{1}{2}I\omega^2_{mean}$

$$KE = \frac{1}{2}I \times \frac{C_E \times W}{I(\omega_1 - \omega_2)}\ \omega_{mean}$$

$$\boxed{KE = \frac{W.C_E}{2.C_S}}$$

1009. The cross sectional area of the flywheel is given by

(a) $\frac{m^2}{\pi D\rho}$ (b) $\frac{m}{\pi D\rho}$
(c) $\frac{m}{\pi D^2\rho}$ (d) $\frac{m}{\pi D\rho^2}$

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : Cross sectional area of the flywheel is given by.

$$A = \frac{m}{\pi D\rho}$$

Where m is mass of flywheel

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1010. Which of the following components regulates the idling speed and maximum speed of the engine by controlling the rate of fuel delivery in a diesel engine?
(a) Carburetor (b) Inlet valve
(c) Governor (d) None of these

KPCL AE 2016

Ans. (c) : Governor regulates the idling speed and maximum speed of the engine by controlling the rate of fuel delivery in a diesel engine.

1011. ……governing is employed in diesel engine.
(a) Quality (b) Quantity
(c) Hit and miss (d) none of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Quantitative Governing- In this system of governing the quality of charge (i.e. air-fuel ratio of the mixture) is kept constant but the quantity of mixture supplied to engine cylinder is varied by means of throttle valve which is regulated by the centrifugal governor through rack and pinion arrangement. This is used for SI or Petrol engine.
Qualitative governing- In this system of governing a control valve is fitted in the fuel delivering pipe which control the quantity of fuel to be mixed in the charge. The movement of control valve is regulated by the centrifugal governor through rack and pinion arrangement. This is used for CI or diesel engine.
Hit and miss governing- In this system of governing. Whenever the engine starts running at higher speed some explosions are omitted or missed. This is widely used in IC engines of smaller capacity or gas engines.

1012. In …… the governor stops the flow of fuel in the engine cylinder which will then perform the idle cycle.
(a) Run and Miss governing
(b) Quantitative/Qualitative Governing
(c) Hit and Miss Governing
(d) None of the above

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Hit and Miss Governing–In this system of governing whenever the engine starts running of higher speed. Some explosions are omitted or missed. This is done with the help of centrifugal governor. This method of governing is widely used for IC. engine of smaller capacity.

1013. A Porter governor has maximum and minimum equilibrium speeds of 200 r.p.m. and 150 r.p.m. respectively. If the effective load on the sleeve is 300 N, the governor effort would be
(a) 16.7 N (b) 58.3 N
(c) 75 N (d) 100 N

Assam PSC CCE Pre 2015

Ans. (d) : Given, $N_1 = 150$ rpm, $N_2 = 200$ rpm
Effective load on sleeve (F) = 300 N
Governor effort (P) = F × C

$$= 300 \times \left[\frac{200-150}{150}\right]$$

$$\boxed{P = 100\,N}$$

1014. The spokes used in a flywheel increases its :
(a) Angular velocity
(b) Angular momentum
(c) Moment of inertia
(d) Centripetal action

GWSSB AAE, 27.12.2015

Ans. (c) : Mass moment of Inertia (I) = mk^2
Where, m = mass, k = radius of gyration
In case of spokes max amount of mass of wheel lie at rim which increase radius of gyration case result increase moment of Inertia.

1015. The basic purpose of using a flywheel on rotating machinery is to :
(a) store and release energy as required.
(b) control rotational speed
(c) balance the rotating parts
(d) allow rapid acceleration.

GWSSB AAE, 27.12.2015

Ans. (a) : A flywheel controls the speed variations caused by the fluctuation of the engine turning moment during each cycle of operation.
A flywheel used in machines serves as a reservoir, which stores energy during the period when the supply of energy is more than the requirement, and releases it during the period when the requirement of energy is more than the supply.

1016. In a flywheel safe stress is 7×10^6 N/m^2 and density of flywheel material 700 kg/m^3. The maximum peripheral velocity will be :
(a) 100 m/s (b) 10 m/s
(c) 10^4 m/s (d) None of the above

KPSC ADF 2015

Ans. (a) : Given data;
Stress (σ) = 7×10^6 N/m^2
Density (ρ) = 700 kg/m^3
Maximum peripheral velocity of a flywheel (V)

$$= \sqrt{\frac{\text{Stress}(\sigma)}{\text{Density}(\rho)}} = \sqrt{\frac{\sigma}{\rho}}$$

$$\Rightarrow \quad V = \sqrt{\frac{7\times10^6}{700}} = \sqrt{(10)^4}$$

$$\Rightarrow \quad V = 10^2 = 100 \text{ m/s}$$

1017. The speed variation of the engine caused by the fluctuations of engine turning moment is controlled by
(a) Flywheel (b) Throttle valve
(c) Governor (d) D-slide valve

(KPSC AE 2015)

Ans : (a) A flywheel used in machines serve as reservoir, which stores energy during the period when supply of energy is more than the requirement and release it during the period when the requirement of energy is more than the supply.

1018. One of the following is not pure rotation:
(a) Windmill
(b) The minutes hand of a clock
(c) Bicycle wheel
(d) Flywheel

OPSC AEE 2015 Paper-I

Ans : (c) Bicycle wheel is not pure rotation. Bicycle wheel partially rotational and translatory motion.

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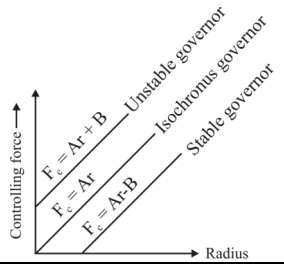
1019. Which type of governor has controlling force diagram a straight line passing through the origin.

(a) proell governor
(b) unstable governor
(c) isochronous governor
(d) porter governor

Haryana PSC Civil Services Pre, 2014

Ans. (c)



1020. Absorption of energy into a flywheel is:

(a) at constant speed
(b) accompanied by increase of speed
(c) accompanied by decrease of speed
(d) not concerned with increase/decrease of speed

HPPSC Asstt. Prof. 2014

Ans. (b) : Absorption of energy into a flywheel is accompanied by increase of speed.
Flywheel is a mechanical reservoir which stores energy during period when supply of energy is more than requirement and release when requirement is more than supply. It does not maintain constant speed. It simply reduces fluctuation of speed.

1021. Which of the following stresses acts on the spokes of the flywheel?

(a) Direct shear stress
(b) Torsional shear stress
(c) Tensile stress
(d) Compressive stress

UPRVUNL AE 2014

Ans : (c) Spokes of flywheel experience centrifugal force because of this force tensile stress developed in the spokes of flywheel.

1022. For the same lift of sleeve, range of speed of proell governor as compared to porter governor is–

(a) Less (b) More
(c) Equal (d) Half

UPRVUNL AE 2014

Ans. : (b) For the same lift of sleeve, range of speed of proell governor as compared to proter governor is more.
Since, Proell governor is more sensitive than Porter governor therefore the range of speed of the Proell governor is more.

1023. If the size of the flywheel in a punching machine is increased, then

(a) fluctuation of speed as well as fluctuation of energy will decrease
(b) fluctuation of speed will decrease but the fluctuation of energy will increase
(c) fluctuation of speed will increase and the fluctuation of energy will decrease
(d) fluctuation of speed as well as fluctuation of energy will increase

TNPSC AE 2014

Ans. (b) : If the size of the flywheel in a punching machine is increased, then fluctuation of speed will decrease but the fluctuation of energy will increase

1024. Proell Governor as compared to porter governor

(a) is more sensitive
(b) is less sensitive
(c) requires weights of smaller size
(d) is more sensitive and requires weights of smaller size

TNPSC AE 2014

Ans. (d): Proell Governor as compared to porter governor is more sensitive and requires weights of smaller size. Proell governor use to reduce mass of ball than Porter governor for same sensitivity.

$$N^2 = \frac{a}{e}.\frac{895}{h}\left[1+\frac{Mg \pm f}{2mg}(1+k)\right] \qquad (a < e)$$

1025. The collapse speed of a rotating solid disc is given by :

(a) $\omega = \frac{1}{R}\sqrt{\frac{3\sigma_y}{\rho}}$ (b) $\omega = \frac{1}{R}\sqrt{\frac{\sigma_y}{3\rho}}$

(c) $\omega = \frac{1}{R}\sqrt{\frac{2\sigma_y}{\rho}}$ (d) $\omega = \frac{1}{R}\sqrt{\frac{\sigma_y}{2\rho}}$

Where σ_y is yield stress, ρ is the density and R is the radius of the disc.

(HPPSC AE 2014)

Ans : (a) The collapse speed of rotating solid disc is given by $\omega = \frac{1}{R}\sqrt{\frac{3\sigma_y}{\rho}}$

σ_y = yield stress.
ρ = density
R = Radius of the disc.

1026. In an engine, the work done by inertia forces in cycle is

(a) positive (b) zero
(c) negative (d) None of the above

APPSC Poly. Lect. 2013

Ans. (b) :

1027. A machine requires torque increasing linearly from 0 to 100 N.m using its cycle of operation of one revolution. The mean speed of the shaft is $10\sqrt{\pi}$ rad/s. Assuming the hub and arms do not contribute to the moment of inertia, what will be the mass of a rim type flywheel of 1m mean diameter, if the total fluctuation of speed has to be limited to 1%?

(a) 25 kg (b) 50 kg
(c) 100 kg (d) 250 kg

RPSC AEN Pre-2013

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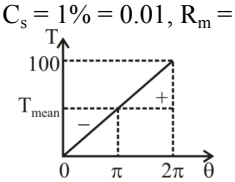
Ans. (c) : Given, $C_s = 1\% = 0.01$, $R_m = 0.5$ m



$$\frac{1}{2}\times 100\times 2\pi = T_{mean}\times 2\pi$$

$$\boxed{T_{mean} = 50\,Nm}$$

At, $0 \rightarrow E = E_0$

At, $\pi \rightarrow E = E_0 - \frac{1}{2}\times 50\times\pi = E_0 - 25\pi$

At, $2\pi \rightarrow E = E_0 - 25\pi + 25\pi = E_0$

$E_{max} = E_0$

$E_{min} = E_0 - 25\pi$

$\Delta E = E_0 - (E_0 - 25\pi)$

$$\boxed{\Delta E = 25\pi}$$

$\Delta E = I\omega^2 C_s$

$25\pi = m \times (0.5)^2 \times \left(10\sqrt{\pi}\right)^2 \times 0.01$

$25 = 0.01 \times m \times 0.25 \times 100$

$$\boxed{m = 100\,kg}$$

1028. Which one of the following does not require a flywheel?
(a) Steam engine
(b) Engine driven press
(c) CI engine
(d) Gas turbine

UKPSC AE-2013, Paper-I

Ans. (d) : Gas turbine does not require a flywheel.

1029. A spring controlled governor is found unstable. It may be made stable by:-
(a) Increasing spring stiffness
(b) Decreasing spring stiffness
(c) Increasing ball weight
(d) Decreasing ball weight

UKPSC AE-2013, Paper-I, ESE 1994

Ans. (b) : A spring controlled governor is found unstable. It may be made stable by decreasing spring stiffness.

1030. A flywheel starting from rest and accelerating uniformly performs 25 revolutions in 3 seconds. Its angular velocity in rpm after 10 seconds will be.
(a) 20 rpm (b) 25 rpm
(c) 45 rpm (d) 50 rpm

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (*) : $\omega_o = 0$

$n_1 = 25$ revolution, $t_1 = 3$ sec

$\theta_1 = 2\pi n_1 = 2\pi \times 25 = 50\pi$

$\theta = \omega_o t + \frac{1}{2}\alpha t^2$

$50\pi = (0\times t) + \frac{1}{2}\times\alpha\times 3^2$

$\alpha = \frac{100\times\pi}{9} = 34.9\,rad/sec^2$

$\omega_2 = \omega_o + \alpha t$

$t_2 = 10$ sec

$\omega_2 = (0) + (34.9 \times 10)$

$\omega_2 = 349$ rad/sec

$349 = 2 \times \pi \times N_2$

$$\boxed{N_2 = 55.54 rps = 3332.4 rpm}$$

No option is matching.
But official answer is given (d).

1031. If the speed of the engine varies between 390 and 410 rpm in a cycle of operation, the coefficient of fluctuation of speed will be
(a) 0.01 (b) 0.03
(c) 0.05 (d) 0.07

UKPSC AE 2012 Paper-I

Ans. (c) : Given $N_{max} = 410$
$N_{min} = 330$

Then coefficient of fluctuation of speed (C_S)

$$(C_S) = = \frac{N_{max} - N_{min}}{N_{mean}}$$

$$= \frac{N_{max} - N_{min}}{\frac{N_{max} + N_{min}}{2}} = \frac{410 - 390}{\frac{410 + 390}{2}}$$

$$C_S = \frac{20}{400} = 0.05$$

1032. In a governor, if the equilibrium speed is constant for all radii of rotation of balls, the governor is said to be
(a) stable governor (b) unstable governor
(c) inertia governor (d) isochronous governor

UKPSC AE 2012 Paper-I

Ans. (d) : isochronous governor

1033. A Hartnell governor has its controlling force given by F = p + qr, where r is the radius of the balls and p and q are constants. The governor will be isochronous when
(a) p = 0 and q is positive
(b) p is positive and q = 0
(c) p is negative and q is positive
(d) p is positive and q is also positive

APPSC AEE 2012, ESE 1993

Ans. (a) : Hartnell governor –
Controlling force equation
(i) for unstable governor

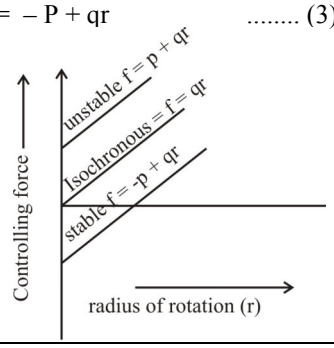
$f_c = P + qr$ ..... (i)

(ii) for Isochronous governor

$f = 0$, q = positive $f_c = qr$ ..... (2)

(iii) for stable governor

$f_c = -P + qr$ ..... (3)



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1034. The moment of inertia of Flywheel is 200 kg m^2 starting from rest it is moving with a uniform acceleration of 0.5 rad/sec^2. after 10 seconds from start, its kinetic energy will be :

(a) 250 N-m (b) 500 N-m
(c) 5000 N-m (d) 25000 N-m

J&K PSC Civil Services Pre, 2010

Ans. (*) : Given
$I = 200 \text{ kg - m}^2$
$\alpha = 0.5 \text{ rad/sec}^2$
time (t) = 10 sec.
Let, initial angular velocity (ω_0) = 0
Then, $\omega = \omega_0 + \alpha t$
$\omega = 0 + 0.5 \times 10$
$\omega = 5$ rad/sec
kinetic energy (KE) = $\frac{1}{2}I\omega^2$
$= \frac{1}{2} \times 200 \times 5 \times 5$
$= 100 \times 25 = 2500$ N-m

1035. Match List I with List II and select the correct answer using the code given below the lists :

	List I		List II
A.	Pendulum-type governor	1.	Pickering governor
B.	Dead weight governor	2.	Rites governor
C.	Spring-controlled governor	3.	Proell governor
D.	Inertia governor	4.	Watt governor

Code:

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	2	3	1	4
(c)	4	1	3	2	(d)	2	1	3	4

UPSC JWM Advt. No.-50/2010

Ans. (a) :
• **Pendulum Type Governor**–This type of governor has a mass known as governor ball, watt governor is an example of pendulum governor.
• **Dead Weight Governor**–Porter governor is a dead weight loaded type governor works on the action of gravity. e.g. Proell governor
• **Spring Controlled Governor**–A centrifugal governor is used to control the speed of rotating machinery.
e.g. Wilson Hartnell governor, Pickering governor
• **Inertia Governor**–This governor works on a different principle compared to centrifugal governors.
e.g. rites governor.

1036. If the governor balls have one particular fixed radius for each given speed in the equilibrium position, then what is the status of the governor?

(a) Sensitive (b) Insensitive
(c) Stable (d) Unstable

UPSC JWM Adv. No-16/2009

Ans. (c) : If the governor balls have **one particular fixed radius for each given speed** in the equilibrium position is called stable governor.

1037. In a reciprocating engine, the maximum fluctuation in energy is found to be 2 J and the variation in speed is limited to 2% of the mean speed. If the mean speed is 100 rad/s and the radius of gyration of the flywheel is 0.1 m, the mass of the flywheel is

(a) 3 kg (b) 2 kg
(c) 1 kg (d) 0.5 kg

DRDO Scientists 2009

Ans. (c) : $\frac{1}{2}I\left(\omega_1^2 - \omega_2^2\right) = 2$
$\frac{1}{2}mK^2\left(\omega_1 - \omega_2\right)\left(\omega_1 + \omega_2\right) = 2$
$m(0.1)^2 (0.2) (100) (100) = 2$
or m = 1 kg

1038. Match the following and select the correct answer using the codes given below :

(A)	Hunting	(I)	One radius of rotation for each speed
(B)	Isochronism	(II)	Too sensitive
(C)	Stability	(III)	Mean force exerted at the sleeve during change of speed
(D)	Effort	(IV)	Constant equilibrium speed for all radii of rotation

(a) A-II, B-IV, C-I, D-III
(b) A-III, B-I, C-IV, D-II
(c) A-II, B-I, C-IV, D-III
(d) A-I, B-II, C-III, D-IV

APPSC IOF, 2009, ESE 1996

Ans. (c) :
Hunting – Too sensitive
Isochronism – One radius of rotation for each speed
Stability – Constant equilibrium speed for all radii of rotation
Effort – Mean force exerted at the sleeve during change of speed

1039. In hit and miss governing

(a) mixture strength is maintained constant
(b) quantity of fuel is varied to suit load on engine
(c) the fuel supply is cut off completely during one or more number of cycles
(d) none of these

TNPSC AE, 2008

Ans. (c) : In a hit and miss governing the fuel supply is cut off completely, during one or more number of cycles. A hit and miss engine is a type of flywheel engine. A flywheel engine is an engine that has a large flywheel or set of flywheels connected to the crankshaft. Virtually all hit and miss engines where of the enclosed crankcase.

1040. Controlling force curve is a plot between controlling force and

(a) radius of rotation (b) speed of rotation
(c) range of speed (d) sleeve lift

TNPSC AE, 2008

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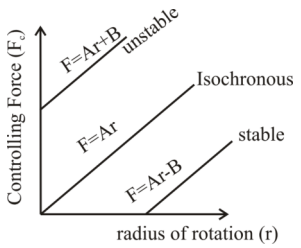
Ans. (a) : Controlling force curve is a plot between controlling force and radius of rotation.



1041. Centrifugal governors are preferred to inertia type governors because an inertia governors.
(a) has less controlling force
(b) is highly sensitive and more prone to hunting
(c) posses problems in the balancing of inertia forces
(d) has high initial and maintenance cost

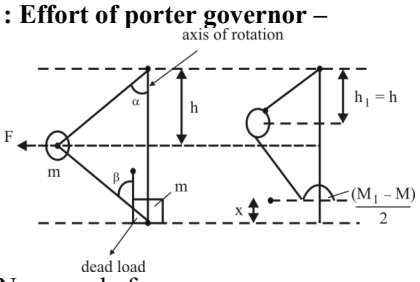
J & K PSC Screening, 2006

Ans. (c) : Centrifugal governors are preferred to inertia type governor because an inertia governors poses problem in the balancing of inertia forces.

1042. If w and W are weights of ball and sleeve respectively, C is percentage increase in speed then the effort of a Porter governor is equal to
(a) C(w+W) (b) C(w–W)
(c) C/(W–w) (d) C/(W+w)

WBPSC AE 2003

Ans. (a) : Effort of porter governor –



N = speed of governor
C = % increase in speed
∴ Increase speed = CN
final speed = N + CN = (1 + C) N
x = lifted height.

If we increase mass after increase in speed then height of ball will become equal to previous height ($h_1 = h$)

We know that,

h = height of governor

$$h = \frac{m+M}{m}\left(\frac{895}{N^2}\right) \quad ..(i) \quad [M_1 > m]$$

$$h = \frac{m+M_1}{m}\cdot\frac{895}{N^2(1+C)^2} \quad ..(ii)$$

from equation (i) & (ii)

$$\frac{m+M}{m}\cdot\frac{895}{N^2} = \frac{m+M_1}{m}\cdot\frac{895}{N^2(1+c)^2}$$

$M_1 = (m + M)(1 + c)^2 – m$

$M_1 – M = (m + M)[(1 + c)^2 – 1]$

P = mean exerted force on the slow during the increase in speed or the effort of governor.

$$P = \left(\frac{m_1 = M}{2}\right)g = \frac{(m+M)[(1+c)^2-1]}{2} = \frac{(m+M)[1+c^2+2C-1]}{2}$$

$$= \frac{(m+M)(2C)g}{2} \qquad \because [C^2 <<<< 0$$

$$= (m+M)gC$$

effort of governor = (mg + mg)C

$\boxed{P = (w + W)C}$

Power of governor = effort of g or × lift of slow

= (m + M)Cg × x

after device ⇒ $\boxed{\text{Power} = \frac{4C^2}{1+2C}(m+M)gh}$

9. Balancing

1043. Which one of the following can completely balance several masses revolving in different planes on a shaft?
(a) Two equal masses in any two planes
(b) Two masses in any two planes
(c) A single mass in one of the planes of the revolving masses
(d) A single mass in any one plane

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
UPPSC AE 13.12.2020, Paper-I
OPSC AEE 2019 Paper-I
MPPSC AE 2016, ESE 2005
UPPSC AE 12.04.2016 Paper-I
Assam PSC CCE Pre 2015 / KPSC ADF 2015
HPPSC Asstt. Prof. 2014
J&K PSC Civil Services Pre, 2010
APPSC AEE 2012

Ans. (b) : By using mass balancing or weight balancing technique, two balancing mass in different planes can completely balance several masses revolving in different planes.

1044. Maximum magnitude of the unbalanced force in a line perpendicular to the line of stroke is known as
(a) swaying couple (b) hammer blow
(c) unbalanced force (d) resultant force

VIZAG Steel MT 24.01.2021
Jharkhand Urja Vikas Nigam Ltd. AEE 2017
GPSC Asstt. Prof. 28.08.2016
APPSC AEE MAINS 2016, PAPER-III
RPSC AE. 2016
TNPSC AE 2013
RPSC ACF-2011
TNPSC AE, 2008

Ans. (b) : The maximum magnitude of unbalanced force along the perpendicular to the line of stroke is known hammer blow. The effect of hammer blow is to cause the variation in pressure between the wheel and the rail.

1045. A disturbing mass m_1 attached to a rotating shaft may be balanced by a single mass m_2 attached in the same plane of rotation as that of m_1 such that (taking r as the radial distance) using this equation.
(a) $m_1/r_1 = m_2/r_2$ (b) $m_1r_1 = m_2r_2$

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(c) $m_1m_2 = r_2r_1$ (d) $m_1/r_2 = m_2/r_1$

GPSC DEE, Class-2 (GWSSB) 04.07.2021
ESE 2020
NLCIL GET 17.11.2020, Shift-II
Nagaland PSC CTSE 2017, Paper-I
TNPSC AE (Industries) 09.06.2013
TNPSC ACF 2012

Ans. (b) : Balancing of a single rotating mass by a single mass rotating in the same plane.

$$F_{c_1} = F_{c_2}$$
$$m_1w^2r_1 = m_2w^2r_2$$
$$\boxed{m_1r_1 = m_2r_2}$$

1046. If the ratio of length of connecting rod to crank radius increases, then.

(a) Primary unbalanced force increases.
(b) Primary unbalanced force decreases.
(c) Secondary unbalanced force increases.
(d) Secondary unbalanced force decreases.

TNPSC AE 2018, 2017, 2013
UPPSC AE 12.04.2016 Paper-I
Nagaland CTSE 2017, 2016 Ist Paper
Kerala PSC AE 06.08.2015
KPSC AE 2015
WBPSC AE 2008, 2007
UKPSC AE 2007 Paper-I
ESE 2014, 2010, 1999

Ans : (d) If the ratio of length of connecting rod to crank radius increases, then secondary unbalanced force decreases.

Fp= Primary unbalanced force = $mr\omega^2\cos\theta$

Fs= Secondary unbalanced force = $\frac{mr\omega^2}{n}\cos 2\theta$

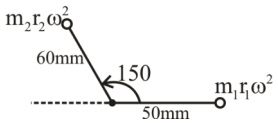
$$n = \frac{\ell}{r}$$

1047. A rotating disc 1 m diameter has two eccentric masses of 0.5 Kg each at radii of 50 mm and 60 mm at angular positions of 0° and 150°, respectively. A balancing mass of 0.1 Kg is to be used to balance the rotor. What is radial position of the balancing mass?

(a) 50.5 mm (b) 120.6 mm
(c) 150.3 mm (d) 280 mm

TSPSC AEE 2017
CGPSC AE 26.04.2015 Shift-I
VIZAG STEEL MT 18.08.2013
ISRO Scientist/Engineer 2012
VIZAG STEEL MT 10.06.2012
GATE 2005

Ans. (c) : Given, $m_1 = m_2 = 0.5$ kg



Along x- axis–

$0.5(-60\times10^{-3}.\cos30°+50\times10^{-3})\times\omega^2 = 0.1\omega^2\times x\times10^{-3}$

$x = -9.8076$ mm

Along y-axis–

$0.5(60\times10^{-3}\sin30°)\omega^2 = 0.1\omega^2y$

$y = 150$ mm

$r = \sqrt{x^2+y^2} = 150.32$mm

1048. In order to have complete balance of several revolving masses mounted in different planes and different angular positions over a shaft

(a) The resultant force must be zero
(b) The resultant couple must be zero
(c) The resultant force as well as couple must be zero
(d) Either the resultant force or the resultant couple must be zero

GPSC ARTO Pre 30.12.2018
GPSC Asstt. Director of Transport 05.03.2017
GPSC ARTO 01.05.2016
MPPSC State Forest Service Exam, 2014
OPSC Civil Services Pre 2006, ESE 2012

Ans. (c) : In order to have complete balance of several revolving masses mounted in different planes and different angular positions over a shaft, the resultant forces as well as couple must be zero.

1049. To ensure dynamic balancing of a shaft:

(a) the net couple due to dynamic forces acting on the shaft is equal to zero.
(b) the net dynamic force acting on the shaft is equal to zero.
(c) the net mass and moment is equal to zero.
(d) the net dynamic force and net couple due to dynamic forces acting on the shaft is equal to zero.

APPSC Poly Lect. Automobile Engg., 2020
NLC GET 17.11.2020, Shift-II
TANGEDCO AE 2015

Ans. (d) : To ensure dynamic balancing of a shaft the net dynamic forces and net couple due to dynamic forces acting on the shaft is equal to zero.

1050. The primary unbalanced force of a reciprocating mass m at radius r and angular velocity ω is:

(a) $mr\omega^2$ (b) $mr\omega^2\sin\theta$
(c) $mr\omega^2\cos\theta$ (d) $mr\omega^2\cos\left(\frac{2\theta}{n}\right)$

Oil India Limited Sr. Engineer (Drilling) 30.11.2019
MPPSC AE 2016 / UKPSC AE-2013, Paper-I

Ans. (c) : Primary and secondary unbalanced force of a reciprocating mass (m) at radius (r) and angular velocity (ω).

Primary unbalanced Force (F_P) = $m.r.\omega^2\cos\theta$

Secondary unbalanced Force (F_s) = $m.r.\omega^2\frac{\cos 2\theta}{n}$

1051. The resultant unbalanced force is minimum in reciprocating engines when the part of the reciprocating mass balanced by rotating masses is :

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

TNPSC AE 2019
Karnataka PSC AE, 10.09.2017
OPSC Civil Services Pre. 2011

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Ans. (b) : The resultant unbalanced force is minimum in reciprocating engines when the part of the reciprocating mass balanced by rotating masses is $\frac{1}{2}$.

1052. The secondary unbalanced force due to a reciprocating mass has:

(a) same frequency as a primary force
(b) twice the frequency as of primary force
(c) four times the frequency as of primary force
(d) none of the above

Vadodara Muncipal Corp. DEE, 2018
HPPSC Asstt. Prof. 2014
TNPSC AE 2014

Ans. (b) : Twice the frequency of primary force.
Unbalanced force due to reciprocating mass m.

$$= mr\omega^2\left[\cos\theta + \frac{\cos 2\theta}{n}\right]$$

$$= mr\omega^2\cos\theta + \frac{mr\omega^2\cos 2\theta}{n}$$

$\downarrow$ $\downarrow$

Primary force secondary force

$$\omega = \frac{d\theta}{dt} \qquad \theta = \int \omega ds$$

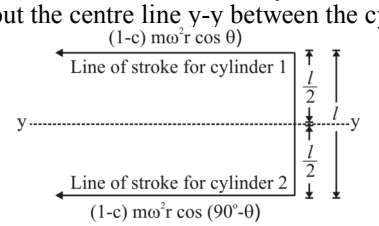
Hence frequency of secondary force is twice of frequency of primary force.

1053. The swaying couple is maximum or minimum when the angle of inclination of the crank to the link of stroke is ? =

(a) 90^o and 225^o (b) 135^o and 180^o
(c) 45^o and 225^o (d) 135^o and 315^o

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018, Paper-I
TSPSC AEE 2015

Ans. (c) : Swaying couple- The unbalanced forces along the line of stroke for the two cylinders constitute a couple about the centre line y-y between the cylinders.



This couple has swaying effect about a vertical axis, and tends to sway the engine alternately in clockwise and anticlockwise direction. Hence the couple is known as swaying couple.

$$\text{Swaying couple} = (1-c)\frac{m\omega^2 r\ell}{2}(\cos\theta + \sin\theta)$$

For maximum magnitude of couple

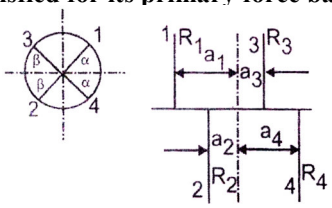
$$\frac{\partial}{\partial\theta}\left[(1-c)m\omega^2 r\frac{\ell}{2}(\cos\theta+\sin\theta)\right] = 0$$

$\theta = 45^o$ or 225^o

Then maximum swaying couple

$$= \pm\frac{1}{\sqrt{2}}(1-c)m\omega^2 r\ell$$

1054. A four cylinder symmetrical in-line engine is shown in figure. Reciprocating weights per cylinder are R_1 and R_2 and the corresponding angular disposition of the crank are α and β. Which one of the following equations should be satisfied for its primary force balance ?



(a) $a_1 \tan\alpha = a_2 \tan\beta$
(b) $\cos\alpha = 0.5 \sec\beta$
(c) $R_1 a_1 \sin 2\alpha = R_2 a_2 \sin 2\beta$
(d) $R_1 \cos\alpha = R_2 \cos\beta$

OPSC Civil Services Pre. 2011
ESE 1998

Ans. (d) : $R_1 \cos\alpha = R_2 \cos\beta$

$\Rightarrow m_1 r w^2 (\cos\theta_1 + \cos\theta_2 + \cos\theta_3 + \cos\theta_4) = 0$

$\theta_1 = \alpha_1$,

$\theta_2 = 180^o + \beta$

$\theta_3 = 180^o - \beta$

$\theta_4 = (360 - \alpha)$

For primary force balance

$$\frac{R_1}{g}rw^2\cos\alpha + \frac{R_2^2}{g}rw^2\cos(180^o+\beta)$$

$$\frac{R_2}{g}rw^2\cos(180^o-\beta) + \frac{R_1}{g}rw^2\cos(360^o-\alpha) = 0$$

$$\therefore R_1\cos\alpha - R_2\cos\beta - R_2\cos\beta + R_1\cos\alpha = 0$$

$$2R_1\cos\alpha - 2R_2\cos\beta = 0$$

$$\boxed{R_1\cos\alpha = R_2\cos\beta}$$

1055. The static balancing is satisfactory for low speed rotors but with increasing speeds, dynamic balancing becomes necessary. This is because then,

(a) unbalanced couples are caused only at higher speeds
(b) unbalanced forces are not dangerous at higher speeds
(c) effects of unbalances are proportional to the square of the speed
(d) effects of unbalances are directly proportional to the speed

Nagaland PSC (CTSE) 2018, Paper-I
APPSC AE 2012

Ans. (c) : The static balancing is statisfactory for low speed rotors but with increasing speeds, dynamics balancing becomes necessary because of effects of unbalance force. Unbalance forces are proportional to the square of the speed.

1056. In order to facilitate the starting of locomotive in any position, the crank of a locomotive with two cylinders are placed :

(a) 45° (b) 90°
(c) 120° (d) 180°

Kerala PSC AE 06.08.2015
(KPSC AE 2015)

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Ans. (b) : In order to facilitate the starting of locomotive in any position, the crank of a locomotive with two cycle are placed at 90° to each other.

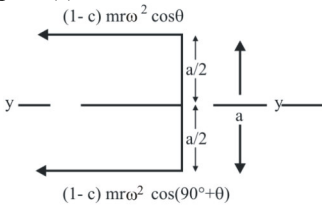
1057. Swaying couple is

(a) primary unbalanced force
(b) average of primary and secondary unbalanced force
(c) secondary unbalanced force
(d) both option A and C

GPSC Lect. (Auto) 16.10.2016
GWSSB DEE 07.07.2016

Ans. (a) : Swaying couple – Swaying Couple is due to primary unbalance force. The unbalance force along the line of stroke due to two cylinder produce a couple about centre line between the cylinders is known as swaying couple.

Swaying couples (s)–



$$= (1-c)\, mr\omega^2 \cos(90° + \theta) \times \frac{a}{2}$$

$$= (1-c) mr\omega^2 a(\cos\theta + \sin\theta)$$

for maximum magnitude $\frac{\partial s}{\partial \theta} = 0$

$\theta = 45°$ or $225°$

maximum swaying couple $= \pm \frac{a}{\sqrt{2}}(1-c) mr\omega^2$

1058. The unbalanced force due to reciprocating masses

(a) varies in magnitude but constant in direction
(b) varies in direction but constant in magnitude
(c) varies in both magnitude and direction
(d) constant in both direction and magnitude

JPSC AE 2013, Paper-V
GPSC Executive Engg. 23.12.2008

Ans : (a) : The unbalanced force due to reciprocating masses varies in magnitude but constant in direction.

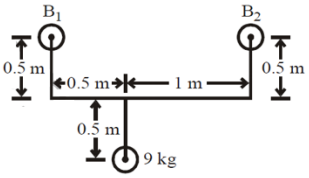
1059. The balancing of rotating the reciprocating parts of an engine is necessary when it turns at

(a) slow speed (b) medium speed
(c) high speed (d) constant speed

TNPSC AE 2019
Mizoram PSC AE/SDO 2014, Paper-II

Ans. (c) : The balancing of rotating the reciprocating parts of an engine is necessary when it turns at high speed.

1060.

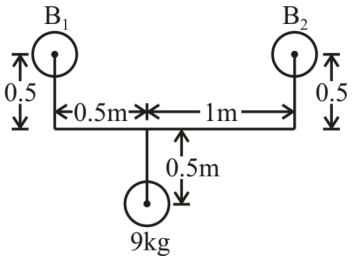


A mass of 9 kg is attached to a rotating shaft at a radius of 0.5 m. The balancing masses B_1 and B_2 are attached at a radius of 0.5 m as shown in the figure. The planes of rotation of the three masses are parallel. If the shaft rotates at 100 r.p.m., the magnitude of the balancing mass B_2 is

(a) 9 kg (b) 6 kg
(c) 4 kg (d) 3 kg

UPSC JWM Advt. No.-50/2010
OPSC Civil Services Pre 2006, ESE 2000

Ans. (d) :



$\sum F_c = 0$
$\sum M = 0$
So, $B_1 \times R_1 + B_2 \times R_2 = 9 \times R_3$
$\therefore$ R is same for all
$B_1 + B_2 = 9$ .....(i)
$\sum M_0 = 0$
$B_1 R_1 \ell_1 = B_2 R_2 \ell_2$
$B_1 \times 0.5 = B_2 \times 1m$
$B_1 = 2B_2$
So, $3B_2 = 9$
$B_2 = 3kg$ & $B_1 = 6kg$

1061. In the statement "an eccentric mass rotating a 3000 rpm will create X times more unbalanced force than 50% of the same mass rotating at 300 rpm." X stands for:

(a) 200 (b) 50
(c) 100 (d) 10

UPRVUNL AE 21.08.2016
APPSC IOF 2009

Ans. : (a) (i) when mass m is rotates at 3000 rpm

The unbalance force $(F_1) = ma \times \left(\frac{2\times\pi\times 3000}{60}\right)^2$

(ii) when mass $\frac{m}{2}$ is rotates at 300 rpm

The unbalance force $(F_2) = \frac{m}{2} \times a \times \left(\frac{2\pi\times 300}{60}\right)^2$

According to question

$F_2 \,.\, X = F_1$
$X = 200$

1062. A variation in the.....force (effort) of an engine is caused by the unbalanced portion of the primary force which acts along the line of stroke of a locomotive engine.

(a) Shear (b) Reactive
(c) Tensile (d) Tractive

VIZAG MT, 14.12.2020
HPPSC AE (PWD) 10.09.2016

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Ans. (d) : Tractive force- A variation in the tractive force (effort) of an engine is caused by the unbalanced portion of the primary force which acts along the line of stroke of a locomotive engine.

1063. A balance mass of value $\frac{2}{3}$m is placed diametrically opposite to the crank at crank radius r. The unbalance force along the line of stroke of a reciprocating engine is

(a) $\frac{2}{3}m\omega^2 r\cos\theta$ (b) $\frac{1}{3}m\omega^2 r\cos\theta$

(c) $\frac{2}{3}m\omega^2 r\sin\theta$ (d) $\frac{1}{3}m\omega^2 r\sin\theta$

(e) None of these

Where ω = crank speed in rad/sec
r = crank radius
θ = crank angle with inner dead centre

CGPSC AE 15.01.2021 / CGPSC AE 16.10.2016

Ans. (a) : Primary unbalanced force,

$F_P = m\omega^2 \times r\cos\theta$ For reciprocating crank

Secondary unbalanced force

$$F_S = m\omega^2 \times r \times \frac{\cos 2\theta}{n}$$

So the force along the line of stroke of reciprocating engine is

$$F_P = \frac{2}{3}m \times \omega^2 \times r\cos\theta \qquad \therefore m_1 = \frac{2}{3}m$$

1064. The partial balancing means

(a) best balancing of engines
(b) balancing of engines
(c) balancing partially the reciprocating masses
(d) All of the above

Assam PSC AE (IWT) 14.03.2021
GWSSB DEE 07.07.2016

Ans. (c) : To balance the reciprocating masses partially is known as partial balancing due to the partial balancing of the reciprocating parts there is an unbalanced primary force along the line of stroke and also an unbalanced primary force perpendicular to the line of stroke.

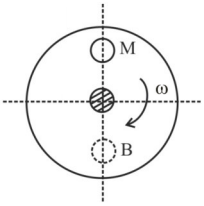
1065. The mass 'M' attached to shaft rotating at ω rad/sec at radius 'r' from axis of shaft is balanced by mass B at radius 'b' from axis of shaft, in the same plane of rotation. The necessary condition for balancing is

(a) Mωr = BMb (b) $\frac{M\omega^3}{r} = \frac{B\omega^2}{b}$

(c) Mr = Bb (d) $\frac{M}{B} = \frac{r}{b}$

RPSC IOF, 2020
TSGENCO AE 14.11.2015

Ans. (c) :



r = radius of rotation of mass M
b- radius of rotation of balancing mass B
The centrifugal force of mass M.

$F_{C_1} = M\omega^2 r$ $F_{C_1} = F_{C_2}$

$F_{C_2} = B\omega^2.b$ $M\omega^2 r = B\omega^2 b$

$Mr = Bb$

1066. Usually following fraction of reciprocating masses is balanced in case of reciprocating engines :

(a) full (b) half
(c) one-quarter (d) two –thirds

RPSC AE 2016
RPSC ACF-2011

Ans. (d) : Two third (2/3) of reciprocating masses is balanced in case of reciprocating engines. Since the shaking force and a shaking couple vary magnitude and direction during the engine cycle, therefore they causes very objectionable vibrations. Thus the purpose of balancing the reciprocating masses is to eliminate the shaking force and a shaking couple.

1067. The following data relate to a single-cylinder reciprocating engine : mass of reciprocating parts = 40 kg, mass of revolving parts = 30 kg at crank radius, speed = 150 rpm, stroke = 350 mm. If 60% of the reciprocating parts and all the revolving parts are to be balanced, what is the balance mass required at a radius of 320 mm?

(a) 15.27 kg (b) 21.43 kg
(c) 24.96 kg (d) 29.53 kg

ESE 18.07.2021
UPRVUNL AE 21.08.2016

Ans. (d) :

Stroke = 2 × crank radius (r) = 350 mm
Mass to be balanced (M) = Revolving mass + 60% of reciprocating mass.
= (30 + 0.6 × 40) kg
= (30 + 24) kg
= 54 kg

Now, $\underbrace{M\omega^2.r}_{\text{Mass to be balanced part}} = \underbrace{m\omega^2 \times R}_{\text{Balanced mass}}$

$$\Rightarrow \left(54 \times \frac{0.35}{2}\right) = (m \times 0.32)$$

or, $$m = \left(\frac{54 \times 0.35}{0.32 \times 2}\right) kg$$

= 29.53kg

1068. In a locomotive, the effect of "hammer blow" can be reduced by:

(a) Balancing whole of the reciprocating parts
(b) Uncoupling the pair of wheels
(c) Decreasing the speed and having pairs of wheels coupled together

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(d) Increasing the speed

CIL MT 27.02.2020
MPPSC AE 08.11.2015

Ans. (c) : The addition of extra weights on the wheels reduces the unbalanced reciprocating forces on the locomotive but causes it to be out of balance vertically creating hammer blow between 40% and 50% of the reciprocating weights on each side were balanced by rotating weights in the wheels.

1069. The maximum value of secondary unbalanced force as compared to the maximum value of primary, unbalanced force in reciprocating engines is-

(a) 1/n times (b) n times
(c) 2n times (d) n^2 times

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : As we know,
Max. Primary force = $m\omega^2.r$
Max secondary force = $m.\ \omega^2.\ \frac{r}{n}$
By comparing above $= \frac{1}{n}$ times

1070. Which of the following in-line engines working on four-stroke cycle is completely balanced inherently?

(a) 6 Cylinder engine (b) 3 Cylinder engine
(c) 4 Cylinder engine (d) 2 Cylinder engine

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 2008

Ans. (a) : A 6- cylinder engine in line engines working on four strokes cycle is completely balanced inherently no extra mass balance & torque balancing is require, less vibration etc.

1071. When balanced weights are introduced in a plane parallel to the plane of rotation of disturbing weight, the minimum number of balance weights, for balancing single revolving disturbing weight will be

(a) One (b) Two
(c) Three (d) Four

GPSC Engineer, Class-II Pre-19.01.2020
CSC Pre 2009

Ans. (b) : The minimum number of balance weights, for balancing single revolving disturbing weight will be two.

1072. In balancing, reference plane is a plane which is:

(a) Passing through plane of rotation of rotating weight
(b) Passing through plane of rotation of balancing weight
(c) At 45 deg to rotating weight
(d) At 45 deg to balancing weight

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : In balancing, reference plane is a plane which is passing through plane of rotation of balancing weight.

1073. The necessary condition in which two masses become dynamically equivalent to the body is represented by which equation? (Consider, L_1 and L_2 – as distance of two masses from the center of gravity of the body K_G – as radius of gyration of the body.)

(a) $L_1.\ L_2 = K_G^2$ (b) $L_1.\ L_2 = K_G$
(c) $L_1/L_2 = K_G^2$ (d) $L_2 = K_G^2.L_1$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : The necessary condition in which two masses dynamically equivalent

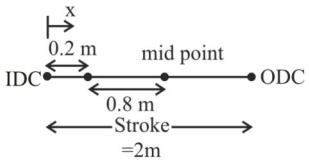
$$\boxed{L_1L_2 = K_G^2}$$

1074. The piston of an engine moves with simple harmonic motion. The crank rotates at 120 r.p.m. with a stroke of 2 meters. The velocity of the piston, when it is at a distance of 0.8 metre from the center is

(a) 4.8π (b) 2.4π
(c) 1.2π (d) 0.6π

ISRO Scientist/Engineer 12.01.2020

Ans. (b) :



Stroke = 2m = 2 × crank radius (r)
Crank radius (r) = 1m

$$x = r\left[\left(1-\cos\theta\right)+\left(n-\sqrt{n^2-\sin^2\theta}\right)\right]$$

$$n = \frac{L}{r} = \text{large}$$

So, $n^2 >>> \sin^2\theta$

$x = r[1-\cos\theta]$.....(i) (SHM)

$0.2 = 1[1-\cos\theta] \Rightarrow \cos\theta = \frac{4}{5}, \Rightarrow \sin\theta = \frac{3}{5}$

Differentiating equation (i) w.r.t. time

$$\frac{dx}{dt} = v = r\omega\sin\theta = \frac{r\times 2\pi N}{60}\times\sin\theta$$

$$v = 1\times 2\pi\times\frac{120}{60}\times\frac{3}{5}$$

$$v = \frac{12\pi}{5} = 2.4\pi \text{ m/sec.}$$

1075. For a single cylinder reciprocating engine speed is 500 rpm, stroke is 150 mm, mass of reciprocating parts is 21 kg; mass of revolving parts is 15 kg at crank radius. If two-thirds of reciprocating masses and all the revolving masses are balanced, the mass at a radius of 150 mm will be

(a) 7.5 kg (b) 10.5 kg
(c) 12.5 kg (d) 14.5 kg

ESE 2020

Ans. (d) : Given, N = 500 rpm

$$\omega = \frac{2\pi\times 500}{60} = 52.3598 \text{ rad/s}$$

$$r = \frac{150}{2} \text{ mm} = 75 \text{ mm} = 0.075 \text{ m}$$

$m_{reci} = 21$ kg
$m_{rot} = 15$ kg

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b = 150 mm = 0.150 m
B = ?

Mass to be balanced = Rotating + $\frac{2}{3}$ Reciprocating

$= \left(15 + \frac{2}{3} \times 21\right) = 15 + 14 = 29$ kg

Total mass to be balanced
m = 29 kg
$mr\omega^2 = B \cdot b \cdot \omega^2$
$29 \times 0.075 = B \times 0.150$

$B = \frac{29 \times 0.075}{0.150}$ kg

B = 14.5 kg

1076. A four cylinder four-stroke inline engine is not balanced for
(a) primary forces (b) primary couples
(c) secondary forces (d) secondary couples

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : A four cylinder four stroke inline engine is not balanced for secondary forces.

1077. In a coupled engine, the rotating and reciprocating masses must be treated:
(a) together and the balanced masses for the two systems are to be suitably combined in the wheel
(b) together and the balanced masses for the two systems are to be suitably combined in the coupling rod
(c) separately and the balanced masses for the two systems are to be suitably combined in the wheel
(d) separately and the balanced masses for the two systems are to be suitably combined in the coupling rod

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : In a coupled engine, the rotating and reciprocating masses must be treated separately and the balanced masses for the two systems are to be suitably combined in the wheel.

1078. For dynamic balancing of rotors, the minimum number of balanced planes required for balancing the unbalance in multiple planes is
(a) One
(b) Two
(c) Three
(d) Depends on the number of unbalance mass

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : For dynamic balancing minimum two mass at two different planes are required.

1079. If the rotational speed of a single cylinder reciprocating IC engine is 955 rpm, the frequency of secondary unbalance forces is approximately
(a) 1910 rad/s (b) 100 rad/s
(c) 200 rad/s (d) 15.9 rad/s

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : Given, N = 955 rpm

Secondary unbalance force $\frac{m\omega^2 r \cos 2\theta}{n}$

reaches at max/min value two times in one revolution.

So frequency of secondary unbalance force = $2 \times \frac{2\pi N}{60}$

$= \frac{2\pi \times 955}{60} \times 2 = 200$ rad/s

1080. In case of slider crank mechanism if the mass of slider = 1 kg, crank radius = 10 cm, length of the connecting rod = 40 cm, crank speed = 100 rad/s, then the inertia force at the zero crank angle (from inner dead centre) is equal to :
(a) 1250 N (b) 1050 N
(c) 1750 N (d) 950 N

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given, mass of slider (m) = 1 kg
Crank radius (r) = 10 cm
Length of the connecting rod (ℓ) = 40 cm
Crank speed (ω) = 100 rad/s

$\therefore \quad n = \frac{l}{r} = \frac{40}{10} = 4$

we know that,

$F_{inertia} = mr\omega^2 \left[\cos\theta + \frac{\cos 2\theta}{n}\right]$

$F_{inertia} = 1.0 \times 0.1 \times 100^2 \left(\cos 0° + \frac{\cos 0°}{n}\right)$

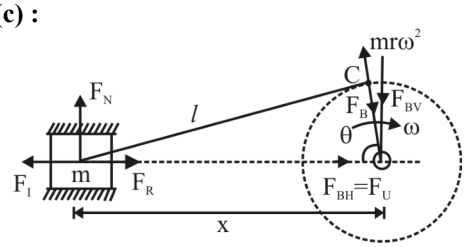
$= 1 \times 0.1 \times 10000 \left(1 + \frac{1}{4}\right) = 1250$

$F_{inertia} = 1250$ N

1081. Inertia force of reciprocating mass is given as:
(a) $F_i = m\omega^2 r \left[\cos 2\theta + \frac{\cos\theta}{2n}\right]$
(b) $F_i = m\omega^2 r \left[\cos\theta + \frac{\cos 2\theta}{2n}\right]$
(c) $F_i = m\omega^2 r \left[\cos\theta + \frac{\cos 2\theta}{n}\right]$
(d) $F_i = m\omega^2 r \left[\cos 2\theta + \frac{\cos 2\theta}{n}\right]$

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (c) :



We know that,
1. Primary unbalanced force (F_p) = $mr\omega^2 \cos\theta$

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$(F_p)_{max} \rightarrow \theta = 0^\circ$ or 180°

$(F_p)_{max} = mr\omega^2$ [It is maximum twice in one revolution]

2. Secondary unbalanced force $(F_s) = mr\omega^2 \dfrac{\cos 2\theta}{n}$

$(F_s)_{max} \rightarrow \theta = 0^\circ, 90^\circ, 180^\circ$ and 270°

$(F_s)_{max} = mr\omega^2 \times \dfrac{r}{n}$ [It is maximum four times in one revolution]

Unbalanced force

$$F_u = F_i = F_p = F_s = mr\omega^2 \cos\theta + mr\omega^2 \frac{\cos 2\theta}{n}$$

$$\therefore \text{Inertia force } (F_i) = mr\omega^2 \left[\cos\theta + \frac{\cos 2\theta}{n}\right]$$

1082. In order to balance the reciprocating masses

(a) Only primary forces and couples must be balanced
(b) Only secondary forces and couples must be balanced
(c) Both (a) and (b)
(d) None of (a), (b) and (c)

JPSC AE PRE 2019

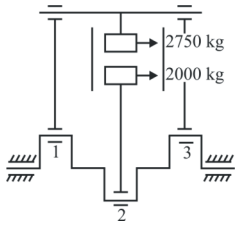
Ans. (c) : In order to balance the reciprocating masses
- Primary forces and couples must be balanced.
- Secondary forces and couples must be balanced.

1083. A vertical single-cylinder opposed piston type engine has reciprocating parts of mass 2000 kg for the lower piston and 2750 kg for the upper piston. The lower piston has a stroke of 60 cm and the engine is in primary balance. If the ratio of the length of connecting rod to crank is 4 for the lower piston and 8 for the upper piston, and when the crankshaft speed is of 135 r.p.m., the maximum secondary unbalanced force will be

(a) 48935.5 N (b) 46946.5 N
(c) 44968.5 N (d) 42989.5 N

ESE 2019

Ans. (c) :



Lower piston crank radius $(r_1) = \dfrac{60}{2} = 30$ cm

Upper piston crank radius = r_2

$m_1 = 2000$ kg

$m_2 = 2750$ kg

$$n_1 = \frac{\ell_1}{r_1} = 4$$

$$n_2 = \frac{\ell_2}{r_2} = 8$$

N = 135 rpm

$$\omega_1 = \omega_2 = \frac{2\pi \times 135}{60} = 14.137 \text{ r/s}$$

$$m_1r_1 = m_2r_2$$

$$r_2 = \frac{m_1r_1}{m_2}$$

$$= \frac{2000 \times 30}{2750}$$

$$= 21.81 \text{ m}$$

$$\text{Unbalanced force} = \frac{m_1r_1\omega_1^2}{n_1} + \frac{m_1r_2\omega_2^2}{n_2}$$

$$= \left[\frac{2000 \times 0.3 \times (14.137)^2}{4}\right] + \left[\frac{2750 \times 0.2181 \times (14.137)^2}{8}\right]$$

$$= 29978.2 + 14983.48$$

$$= 44{,}961.70 \text{ N}$$

1084. The reciprocating mass is balanced when primary force is

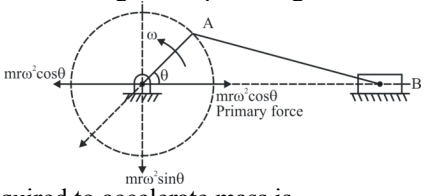
1. balanced by the mass = $cmr\omega^2 \cos\theta$
2. unbalanced by the mass = $cmr\omega^2 \cos\theta$
3. balanced by the mass = $(1-c)mr\omega^2 \cos\theta$
4. unbalanced by the mass $(1-c)mr\omega^2 \cos\theta$

Select he correct answer using the code given below.

(a) 1 and 3 (b) 2 and 3
(c) 1 and 4 (d) 2 and 4

ESE 2019

Ans. (c) : Balancing of reciprocating mass



Force required to accelerate mass is

$$F = \underbrace{mr\omega^2 \cos\theta}_{\text{Primary accelerating force}} + \underbrace{mr\omega^2 \frac{\cos\theta}{n}}_{\text{Secondary accelerating force}}$$

Primary balancing of reciprocating mass

■ If c is the fraction of the reciprocating mass, primary force balanced by the mass

$= cmr\omega^2 \cos\theta$

■ Primary force unbalance by the mass

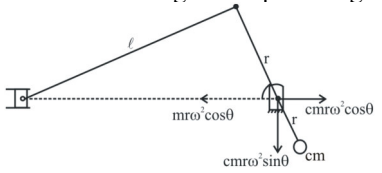
$= (1 - c)\, mr\omega^2 \cos\theta$

1085. In case of partial balancing of single cylinder reciprocating engine, what is the primary disturbing force along the line of stroke ?

(a) $cmr\omega^2 \cos\theta$ (b) $(1-c^2)\, mr\omega^2 \cos\theta$
(c) $(1-c)\, mr\omega^2 \cos\theta$ (d) $(1-c^2)\, mr\omega^2 \cos 2\theta$

OPSC AEE 2019 Paper-I, ESE 2006

Ans : (c) : Partial balancing of reciprocating engine



$\therefore$ Primary disturbing force along the line of stroke,

$= mr\omega^2 \cos\theta - cmr\omega^2 \cos\theta = (1-c)\, mr\omega^2 \cos\theta$

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1086. When the primary direct crank of a reciprocating engine makes an angle of θ with the line of stroke, then the secondary direct crank will make an angle of ..... with the line of stroke.

(a) θ/2 (b) θ
(c) 2θ (d) 4θ

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : When primary direct crank of a reciprocating engine makes an angle θ, with line of stroke, then secondary direct crank will make an angle of '2θ' with the line of stroke.

Primary unbalanced = $m\omega^2 r\cos\theta$

Secondary unbalanced $= m\omega^2 r\frac{\cos 2\theta}{n}$

1087. A single-cylinder reciprocating engine works with a stroke of 320 mm, mass of reciprocating parts as 45 kg and mass of revolving parts as 35 kg at crank radius. If 60% of the reciprocating parts and all the revolving parts are to be balanced, then the balancing mass required at a 300 mm radius is nearly

(a) 33.1 kg (b) 36.3 kg
(c) 39.5 kg (d) 42.7 kg

ESE 2018

Ans. (a) : Length of stroke = 320 mm
Mass of reciprocating parts = 45 kg
Mass of revolving parts = 35 kg
Mass of reciprocating parts to be balanced
$= 0.6 \times 45 = 27$ kg
Therefore mass to be balanced = 27 + 35 = 62 kg

Crank radius $= \frac{320}{2} = 160$ mm

$m_c r_c = mr$
$m_c \times 300 = 62 \times 160$
$m_c = 33.06$ kg

1088. Partial balancing in locomotives results in

(a) hammer blow, variation of tractive effort, swaying couple
(b) least wear
(c) most smooth operation
(d) better performance of engine

TNPSC AE 2018

Ans. (a) : Hammer blow, variation of tractive effort, swaying couple due to partial balancing in locomotives.

1089. The primary unbalanced force is maximum when the angle of crank with the line of stroke is

(a) 45° (b) 90°
(c) 135° (d) 180°

TSPSC AEE 2017

Ans. (d) : The primary unbalanced force is maximum, when θ = 0° or 180°. Thus, the primary force is maximum twice in one revolution of the crank. The maximum primary unbalanced force is given by $F_{p(max)} = m\omega^2 r$

1090. Which of the following statements is correct?

(a) If a rotor is statically balanced, it is always dynamically balanced
(b) If a rotor is dynamically balanced is may or may not be statically balanced and vice-versa
(c) If a rotor is dynamically balanced, it is always statically balanced
(d) None of the above

Karnataka PSC Lect., 27.05.2017

Ans (c) : If a rotor is dynamically balanced, it is always statically balanced.

1091. A three cylinder radial engine driven by a common crank of radius r has the cylinders spaced at 120° among each other. The mass of the reciprocating parts per cylinder is m kg. The primary unbalance force at a crank shaft speed of ωrad/s

(a) $\frac{2}{3}m\omega^2 r$ (b) $3m\omega^2 r$
(c) $\frac{1}{3}m\omega^2 r$ (d) $m\omega^2 r$

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (a) : Primary unbalanced force
$F_p = m.\omega^2.r\cos\theta$
$\theta = 120°$,
Primary unbalanced force in max at
0° and 180°
For 120°

$F_p = \frac{2}{3}m\omega^2 r$

1092. In dynamic balancing of machines, which of the following is NOT a method of measuring the magnitude and location of correction weight for a two plane correction?

(a) Pivoted cradle method
(b) Mechanical compensation method
(c) Pivoted compensation method
(d) Nodal point method

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : In dynamic balancing of machines, pivoted compensation method is NOT a method of measuring the magnitude and location of correction weight for a two plane correction

1093. Balancing of industrial rotors in a balancing machine is usually done at speeds between 400 rpm and 700 rpm, because:

1. Industrial rotors have lateral critical speeds more than 5000 rpm
2. The rotor has critical speed generally less than 100 rpm
3. Speeds lower than 400 rpm do not create centrifugal forces of magnitudes measurable in this set up
4. Balanced rotors have no critical speed

Which of the above statements are correct?

(a) 1 and 4 only (b) 1 and 3 only
(c) 2 and 3 only (d) 2 and 4 only

UPSC JWM 2017

Ans. (a) : Balancing of industrial rotors in a balancing machine is usually done at speed between 400 rpm and 700 rpm because industrial rotors have lateral critical speeds more than 500 rpm and a rotor balanced for 400 rpm to 700 rpm will still balanced for more than 5000 rpm. Balancing rotors have no critical speed.

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1094. Statement (I): There is no balancing methodology in the case of reciprocating engines.
Statement (II): Balancing of dynamic forces is achieved mostly by resorting to multi-cylinder engine concept.

ESE 2017

Ans. (c) : Single cylinder reciprocating engine can not be balanced completely by any method. But in multi cylinder engine case (six cylinder), it is possible to balance the engine in particular configuration.
Note : Answer given by UPSC is option (A).

1095. Consider the following statements:
1. Balancing of several masses rotating in the same plane can be effected by a single mass.
2. Balancing of several masses in different planes can be done by 2 masses in 2 planes on either side of the reference plane or on the same side.
3. Reciprocating masses cannot be completely balanced by rotating masses.
4. Secondary unbalanced forces will be negligible compared to primary imbalance forces.

Which of the above statements are correct?
(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only
(c) 3 and 4 only (d) 1, 2 and 4 only

ESE 2017

Ans. (a) : All statements are correct.

1096. A body of mass 10 kg with its CG 200 mm from the axis of rotation is to be completely balanced by another mass B of 5 kg placed in the same plane. The radius at which the CG of mass B should be is
(a) 500 mm (b) 400 mm
(c) 300 mm (d) 200 mm

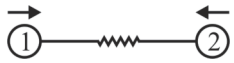
ESE 2017

Ans. (b) : Given,
$m_1 = 10$ kg $r_1 = 200$ mm
$m_2 = 5$ kg $r_2 = ?$
$m_1r_1 = m_2r_2$
$10 \times 200 = 5 \times r_2$
$r_2 = 400$ mm

1097. Two steel balls of 2 kg and 4 kg mass, respectively, are pressed on the two ends of a spring, all pre – placed on a smooth surface. When released, the smaller ball moves with an acceleration of 2 m/s^2. The simultaneous acceleration of the bigger ball will be
(a) 0.5 m/s^2 (b) 1 m/s^2
(c) 2 m/s^2 (d) 4 m/s^2

ESE 2017

Ans. (b) :



Given, $m_1 = 2$ kg $m_2 = 4$ kg
$a_1 = 2$ m/s^2 $a_2 = ?$
Spring force on each mass F = ma
$m_1a_1 = m_2a_2$
$2 \times 2 = 4 \times a_2$
$a_2 = 1$ m/s^2

1098. A system in dynamic balance implies that
(a) The system is critically damped.
(b) The system is at its critical speed.
(c) The system is also statically balanced.
(d) There will be no wear of bearings.

UPPSC AE 12.04.2016 Paper-I, ESE 1993

Ans : (c)
Static Balancing:-A system of rotating masses is said to be in static balance if the combined mass centre of the system lies on the axis of rotation.
Dynamic Balancing:- A system of rotating masses is in dynamic balancing when these does not exist any resultant centrifugal force as well as resultant couple.

1099. The tractive force is maximum or minimum when the angle of inclination of the crank to the line of stroke is equal to
(a) 90° and 225° (b) 135° and 180°
(c) 180° and 225° (d) 135° and 315°

UPPSC AE 12.04.2016 Paper-I

Ans : (d) The tractive force is maximum or minimum when the angle of inclination of the crank to the line of stroke is equal to 135° and 315°
Variation of Tractive force:- A variation in the tractive force (effort) of an engine is caused by the unbalanced portion of the primary force which acts along the line of stroke of a locomotive engine.
* Total unbalanced primary force or the variation in the Tractive force = $(1-c)mr\omega^2 (\cos\theta - \sin\theta)$. This is maximum when $\theta = 135^\circ$ or 315°
* Maximum variation in Tractive force = $\pm\sqrt{2}(1-c)mr\omega^2$

1100. Hammer blow in reciprocating engine varies with:
(a) (Rotational speed)^3 (b) (Rotational speed)^2
(c) (Rotational speed)$^{1/2}$ (d) Rotational speed

UPRVUNL AE 07.10.2016

Ans. (b) : Hammer blow in reciprocating engine varies with square of Rotational speed.
Hammer blow—Hammer blow is the maximum vertical unbalanced force caused by the mass provided to balance the reciprocating masses. Its value is $mr\omega^2$. Thus, it varies as a square of the speed.

1101. During balancing of inline engines, if a mass is transferred to the crankpin, the axial component parallel to the cylinder axis of the resulting centrifugal force represents ..... .
(a) any balanced force
(b) the tertiary balanced force
(c) the tertiary vertical force
(d) the primary unbalanced force

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : The primary unbalanced force parallel to the cylinder axis = (1–C) mrw^2 cosθ

1102. A shaft between two bearings is rotating with a disturbing mass 25 kg at a radius of 2 cm. What is the balancing mass to be kept at a radius of 5 cm?
(a) 12.5 kg (b) 5 kg
(c) 10 kg (d) 25 kg

TSPSC Managers, 2015

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Ans. (c) : For balancing mass

$m_1r_1 = m_2r_2$

given– $m_1 = 25kg$, $r_1 = 2cm$

$m_2 = ?$, $r_2 = 5cm$

Now, $m_1r_1 = m_2r_2$

$25 \times 2 = m_2 \times 5$

$\boxed{m_2 = 10\ kg}$

1103. What is the condition for dynamic balancing of a shaft-rotor system?

(a) $\Sigma M = 0$ and $\Sigma F = 0$

(b) $\Sigma M = 0$

(c) $\Sigma F = 0$

(d) $\Sigma F + \Sigma M = 0$

where ΣF is the total force and ΣM is the total torque.

Assam PSC CCE Pre 2015

Ans. (a) : For dynamic balancing of a shaft rotor system

$\Sigma M = 0;\ \ \Sigma F = 0$

Dynamic Balancing–A rotating mass is said to be dynamically balanced when it does not vibrate in its running state. To make a rotating mass dynamically balanced it must first be statically balanced.

- Dynamic balance is a balance due to the action of inertia forces

Note-(a) is correct answer, but commission take (b) as correct answer.

1104. A two-cylinder in-line engine with identical cylinders has cranks at 180° to each other. When the engine runs, the unbalanced forces are :

(a) both the primary and secondary forces

(b) only the secondary force

(c) only the primary force

(d) Both forces are balanced.

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : two cylinder in-line engine with identical cylinders has cranks at 180° to each other. When the engine runs, the unbalanced forces are only the secondary force.

1105. In one revolution of the crank, the maximum value of primary force occurs

(a) twice (b) three times

(c) four times (d) six times

Assam PSC CCE Pre 2015

Ans. (a) : The primary unbalancing force attains its maximum value twice in one revolution of crank. The hammer blow in locomotive engines occurs due to unbalanced force along the line of stroke of the piston.

1106. In dynamic analysis of planar mechanisms, the equivalent offset inertia force may be kept at a distance (h) of

(a) $\boxed{h = \frac{k^2\alpha}{f_g}}$ (b) $\boxed{h = \frac{k\,\alpha}{f_g}}$

(c) $\boxed{h = h^2\alpha}$ (d) $\boxed{h = \frac{k^2\alpha^2}{f_g}}$

TSPSC AEE 2015

Ans. (a) : Equivalent offset inertia force is given as,

$$h = \frac{k^2\alpha}{f_g}$$

1107. For static balancing

(a) dynamic forces acting on the shaft must be zero

(b) net couple acting on the shaft due to dynamic forces must be zero

(c) both (A) and (B)

(d) none of the above

HPPSC Asstt. Prof. 2014

Ans. (a) : Dynamic forces acting on the shaft must be zero.

For static balancing

$\Sigma F_c = 0$

For dynamic balancing

$\Sigma F_c = 0,\ \Sigma M = 0$

1108. The conditions for static balancing and dynamic balancing in rotating masses are

(a) $\Sigma mr^2 = 0$ and $\Sigma mr^2l = 0$ respectively

(b) $\Sigma mr^2l = 0$ and $\Sigma mr^2 = 0$ respectively

(c) $\Sigma mr = 0$ and $\Sigma mrl = 0$ respectively

(d) $\Sigma mrl = 0$ and $\Sigma mr = 0$ respectively

Haryana PSC Civil Services Pre, 2014

Ans. (c) Let F_c = Centrifugal force

M = Couple

Condition for static balancing

$\Sigma F_c = 0$

$\Sigma mr\omega^2 = 0$

$\boxed{\Sigma mr = 0}$ .....(i)

Condition for dynamic balancing

$\Sigma F_c = 0$ .....(ii)

$\Sigma M = 0$ .....(iii)

$\Rightarrow$ $\boxed{\Sigma mr\ell = 0}$

1109. If different masses are rotating about an axis in a single plane at different radii, then for their complete balance, it is sufficient to analyze the conditions of -

(a) static and dynamic balancing both

(b) static balancing only

(c) dynamic balancing only

(d) none of these

Haryana PSC Civil Services Pre, 2014

Ans. (b) : Different masses rotating in a single plane at different radii, then only static balancing is sufficient for complete balancing.

1110. Balancing of inertia forces due to reciprocating parts of a single cylinder engine is done by extending the crank to the opposite side and placing a counter-mass there. Pick the correct statement about balancing of inertia forces in this way:

(a) Only primary inertia force can be completely balanced.

(b) Only secondary inertia force can be completely balanced

(c) The primary inertia force can never be completely balanced in this way.

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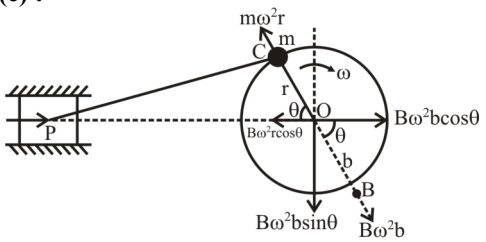
(d) If only a fraction of reciprocating part's mass is balanced, then the primary inertia force can be completely balanced.

RPSC AEN Pre-2013

Ans. (c) :



Partial Balancing of Unbalanced Primary Force in a Reciprocating Engine

The primary inertia force can never be completely balanced when balancing of inertia forces due to reciprocating parts in a single cylinder engine is done by extending the crank to the opposite side on placing of counter-mass.

$\therefore$ Inertia force due to reciprocating or force required to accelerate the reciprocating parts.

$$F_I = F_R = \text{Mass} \times \text{Acceleration}$$

$$= m\omega^2 r\left[\cos\theta + \frac{\cos 2\theta}{n}\right] \qquad \left[\because n = \frac{\ell}{r}\right]$$

1111. Which one of the following statements is correct in the context of balancing of engines?

(a) Magnitude of primary unbalancing force is less than the secondary unbalancing force

(b) The primary unbalancing force attains its maximum value twice in one revolution of the crank

(c) The unbalanced force due to reciprocating masses varies in magnitude and direction

(d) The hammer blow in the engine occurs due to unbalanced force along the line of stroke of the piston.

APPSC AEE 2012

Ans. (b) : Statement correct in the context of balancing of engines– The primary unbalancing force attains its maximum value twice in one revaluation of the crank.

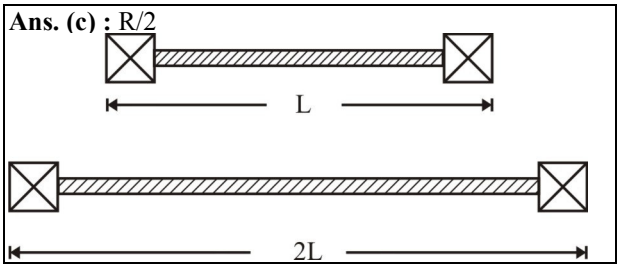
1112. A statically balanced rotor is supported on two bearings L m apart. At high speed of the rotor, dynamic reaction on left bearing is R. Now the right side bearing is shifted to a new position 2L m apart from the left bearing. What shall be the dynamic reaction on left bearing if the rotor speed is same?

(a) R (b) 2R

(c) R/2 (d) R/4

APPSC AEE 2012

Ans. (c) : R/2



1113. A thin disc is balanced at speed 300 rpm by adding a balancing mass of 4 gms at a radius 30 mm on a particular radial line. The balancing mass at speed 600 rpm on the same radius as previous, would be

(a) 4 gms (b) 16 gms

(c) 8 gms (d) 2 gms

APPSC AEE 2012

Ans. (a) : Mass will not be change by different revolution M = 4gms

1114. Any distributed mass can be replaced by two point masses to have the same dynamical properties, if

(a) The sum of the two masses is equal to the total mass.

(b) The combined centre of mass coincides with that of the rod.

(c) The moment of inertia of two point masses about perpendicular axis through their combined centre of mass is equal to that of the rod.

(d) All the above.

UKPSC AE 2012 Paper-I

Ans. (d) : Any distributed mass can be replaced by two point masses to have the same dynamical properties, if
(a) The sum of the two masses is equal to the total mass.
(b) The combined centre of mass coincides with that of the rod.
(c) The moment of inertia of two point masses about perpendicular axis through their combined centre of mass is equal to that of the rod.

1115. Static balancing involves balancing of

(a) forces (b) couples

(c) masses (d) All the above

UKPSC AE 2012 Paper-I

Ans. (a) : Static balancing involves balancing of forces.
For static balancing $\sum F = 0$
For dynamic balancing $\sum F = 0$ & $\sum M = 0$

1116. Choose the correct relationship between the given statements of Assertion (A) and Reason (R).

Assertion (A) : A dynamically balanced system of multiple rotors on a shaft can rotate smoothly at the critical speeds of the system.

Reason (R) : Dynamic balancing eliminates all the unbalanced forces and couples from the system.

Code :

(a) Both (A) and (R) are true. (R) is the correct explanation of (A).

(b) Both (A) and (R) are true. (R) is not the correct explanation of (A).

(c) (A) is true, but (R) is false.

(d) (A) is false, but (R) is true.

UKPSC AE 2012 Paper-I

Ans. (a) : Both (A) and (R) are true. (R) is the correct explanation of (A).

1117. Hammer blow in locomotives results in :

(a) Pulsating torque

(b) Tendency to lift the wheels from rail

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(c) Uneven speed
(d) Variable horizontal force

OPSC Civil Services Pre. 2011

Ans. (b) :
- Hammer blow amount has to be limited in order to avoid lifting of wheels from rails of a locomotive.
- In locomotive where primary forces are partially balanced, the maximum magnitude of the unbalanced force along the perpendicular to the line of stroke is known as hammer blow.

1118. Hammer blow occurs when the centre of gravity of the balanced weight is :
(a) Directly above the wheel centre
(b) Directly below the wheel centre
(c) Directly above or below the wheel centre
(d) Perpendicular to the wheel vertical plane

J&K PSC Civil Services Pre, 2010

Ans. (c) : Hammer blow– Hammer blow is maximum vertical unbalanced force cause by mass provided to balance the reciprocating masses. Its value is = $mr\omega^2$
m = balance mass
- It varies as square of the speed.
- At high speeds, force of the hammer blow could exceed the static load on the wheel and the wheels can be lifted vertically upwards in the direction of hammer blow.

1119. Consider the following statements :
Effect of unbalanced primary force along the line of stroke produces
1. Swaying couple
2. Variations in tractive force
3. Hammer blow

Which of these statements are correct ?
(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know that the effect of unbalanced primary force along the line of stroke produces, tractive force and perpendicular to the line of stroke produces swaying couple.

1120. In a V-twin engine the reciprocating masses can be completely balanced by a rotating mass, if the angle 2α between the cylinder is :
(a) 60° (b) 90°
(c) 120° (d) 150°

UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation – The most common V-angle for a V-twin engine is 90°, which can achieve a perfect primary balance (if the correct counter weight is used)

1121. 'Direct and reverse crank method' of balancing is useful in which one of the following?
(a) Machine tools (b) Steam engines
(c) Inline engines (d) Radial engines

UPSC JWM Adv. No-16/2009

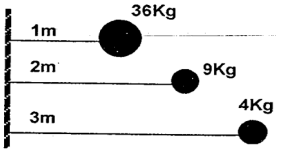
Ans. (d) : The method of direct and reverse cranks are used in balancing of radial engines or V-engines in which the connecting rods are connected to a common crank. Since the plane of rotation of various cranks (in Radial or V-engines) is same, therefore there is no unbalanced primary or secondary couple.

1122. For a multi cylinder in line engine the primary forces can be completely balanced by suitably arranging the crank angles, provided that the number of cranks is not less than
(a) three (b) four
(c) six (d) two

WBPSC AE, 2007

Ans. (c) : A four cylinder engine is majorly balanced as primary force, primary couple and secondary couple are balanced.
A six cylinder engine is completely balanced.

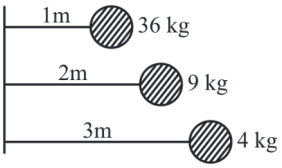
1123. The figure shows 3 small spheres that rotate about a vertical axis. The perpendicular distance between the axis and the center of each sphere is given. Mass of highest rotational inertia about that axis, is-



(a) 36 (b) 4
(c) 9 (d) All same

ISRO Scientist/Engineer 2006

Ans. (d) :



Mass of rotational inertia,
$I = \text{mass} \times (\text{radius})^2$
$I_1 = m_1r_1^2 = 36 \times (1)^2$
$= 36 \text{ kg-m}^2$
$I_2 = m_2r_2^2 = 9 \times (2)^2 = 36 \text{ kg-m}^2$
$I_3 = m_3(r_3)^2 = 4 \times (3)^2 = 36 \text{ kg-m}^2$
Since, $\boxed{I_1 = I_2 = I_3}$
∴ Since, all ball has same mass of rotational inertia.

1124. Which of the following is the correct expression for the secondary unbalanced forces of reciprocating masses
(a) $F_s = m\omega^2 r \times (\cos 2\theta)/n^2]$
(b) $F_s = m\omega^2 r \times (\cos\theta)/n^2]$
(c) $F_s = m\omega^2 r \times (\cos 2\theta)/n]$
(d) $F_s = m\omega^2 r \times (\cos\theta)/n]$

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Expression for the secondary unbalanced forces of reciprocating masses is;

$$F_s = \frac{m\omega^2 r \times (\cos 2\theta)}{n}$$

Where, $n = \frac{\ell}{r}$
ℓ = length of Connecting rod
r = radius of Crank
m = mass, of reciprocating parts
ω = Angular Velocity of crank

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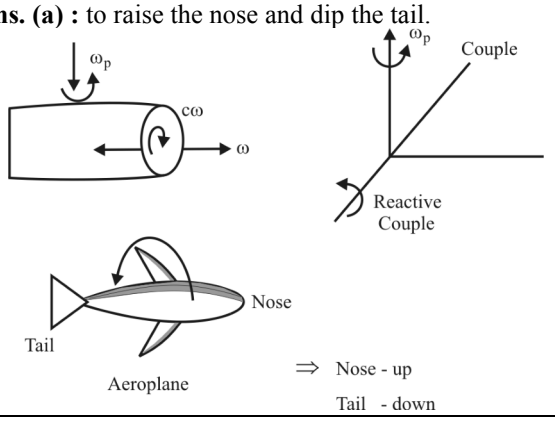
10. Gyroscope

1125. The engine of an aeroplane rotates in clockwise direction when seen from the tail end and the aeroplane takes turn to left. The effect of gyroscopic couple on the aeroplane will be :

(a) to raise the nose and dip the tail
(b) to dip the nose and raise the tail
(c) to raise the nose and tail
(d) to dip the nose and tail

CGPSC AE 25.02.2018
(KPSC AE 2015)
TSPSC Managers, 2015
Kerala PSC AE 06.08.2015
TNPSC AE 2019, 2018, 2013

Ans. (a) : to raise the nose and dip the tail.

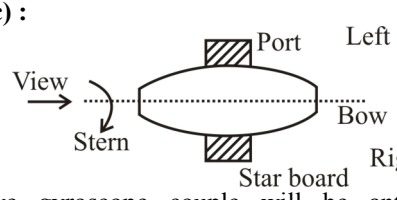


1126. The rotor of a ship in the clockwise direction when viewed from stern and the ship takes left turn. The effect of gyroscope couple acting on it will be

(a) To raise the bow and stern
(b) To lower the bow and stern
(c) To raise the bow and lower the stern
(d) To raise the stern and lower the bow

Karnataka PSC AE, 10.09.2017
HPPSC Asstt. Prof. 18.09.2017
TSPSC AEE 2015 / TNPSC AE 2013

Ans. (c) :



Reactive gyroscope couple will be anticlockwise which will lower the stern and raise bow.

1127. If the air screw of an aeroplane rotates clockwise when viewed from the rear and aeroplane takes a right turn, the gyroscopic effect will be

(a) Tends to raise the tail and depress the nose
(b) Tends to raise the nose and depress the tail
(c) Tilt the aeroplane about the spin axis
(d) None of these

RPSC ACF & FRO, 26.02.2021
RPSC Vice Principal ITI 2018
RPSC VPITI 14.02.2016
MPPSC State Forest Service Exam, 2014

Ans. (a) : The effect of gyroscopic couple on the aeroplane will be to raise the nose and dip the tail. The engine of an aeroplane rotates in clockwise direction when seen from the tail end and the aeroplane takes a turn to the right. The effect of gyroscopic couple on the aeroplane will be to dip the nose and raise the tail.

1128. Which of the following effects is more dangerous for a ship

(a) rolling (b) waving
(c) pitching (d) steering

Gujarat PSC AE 2019
RPSC AE. 2016
SJVN ET 2013
RPSC ACF-2011

Ans : (c) Pitching is more dangerous for a ship
Pitching:-The pitching is the moment of a complete ship up and down in a vertical plane about transverse axis.
Rolling:-In case of rolling of a ship, the axis of precession is always parallel to the axis of spin for all positions

1129. The effect of gyroscopic couple, acting on a ship pitch upward, will be to

(a) move the ship towards star board
(b) move the ship towards port
(c) move the ship in clockwise direction when viewed from stern
(d) none of these

TSPSC AEE 2017
TNPSC AE, 2008
WBPSC AE 2008

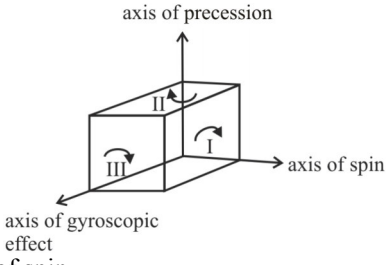
Ans. (a) : The effect of gyroscopic couple, acting on a ship pitch upward, will be to move the ship towards star board.

1130. The axis of spin, the axis of precession and the axis of gyroscopic torque are in

(a) two parallel planes
(b) two perpendicular planes
(c) three perpendicular planes
(d) three parallel planes

TSPSC Manager (Engg.) HMWSSB 12.11.2020
OPSC AEE 2015 Paper-I
ISRO Scientist/Engg. 2008

Ans. (c) : The axis of spin, the axis of precession and the axis of gyroscopic torque are in three perpendicular planes.



I – Plane of spin
II- Plane of precession
III- Plane of gyroscopic effect
Gyroscopic couple = $I\omega\omega_p$

1131. A disc with mass moment of inertia (I) and an angular velocity ω rad/s is spinning about the axis of spin. The angular velocity of precession of the axis of spin is (ω_p), the causing precession will be given by :

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(a) $\left(I\omega^2\omega_p\right)$ (b) $\left(I\omega\omega_p\right)$

(c) $\left(\frac{1}{2}I\omega^2\omega_P\right)$ (d) $\left(\frac{1}{2}\omega\omega_p\right)$

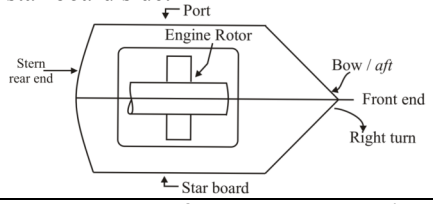
GPSC Class-2 (GWSSB) 04.07.2021
CIL MT 26.03.2017 / WBPSC AE, 2007

Ans. (b) : Mass moment of inertial = I
Angular velocity = ω rad/s
Angular velocity of precession = ω_p
Angular acceleration (α) = $\omega\omega_p$
Precession Torque = $I\omega\omega_P$

1132. When the pitching motion causes the bow to rise, the rotor rotating in clockwise sense as seen from stern) the gyroscopic effect tends to
(a) turn the ship towards port side
(b) turn the ship towards star-board side
(c) depress the stern
(d) raise the stern

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE 2014 / TNPSC ACF 2012

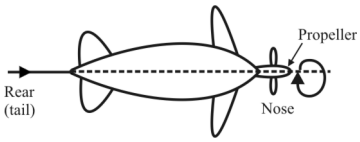
Ans. (b) : When the pitching motion causes the bow to rise, the rotor rotating in clockwise sense as seen from stern) the gyroscopic effect tends to turn the ship towards star-board side.



1133. The propeller of the aeroplane is rotating clockwise when viewed from the rear end and aeroplane takes a left turn, the gyroscopic effect will:
(a) Tend to raise the tail and depress the nose
(b) Tend to raise the tail and tilt the nose
(c) Tend to raise the nose and depress the tail
(d) Tend to raise the nose and tilt the tail.

UPPSC AE 13.12.2020 Paper-I
Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (c) : The propeller of aeroplane is rotating clockwise direction when viewed from the rear end and aeroplane takes a left turn, the gyroscopic effect will tends of raise the nose and depress the tail of the aeroplane.



When the aeroplane take the right turn and viewed from the rear end of plane, the gyroscopic effect will tends to dip the nose and raise the tail of the aeroplane.

1134. As the axes of the rolling of the ship and of the rotor are ....., there is no precession of the axis of spin and thus there is no gyroscopic effect.
(a) Perpendicular
(b) Parallel
(c) Inclined at any angle
(d) Inclined at 45 degrees

HPPSC Asstt. Prof. 29.10.2016
HPPSC AE (PWD) 10.09.2016

Ans. (b) : As the axes of the rolling of the ship and of the rotor one parallel, there is no precession of the axis of spin and thus there is no gyroscopic effect.
Rolling– In rolling spin axis become coinciding or parallel to precession axis so there will be no gyroscopic effect.

$$\text{Gyroscopic Couple } C_g = J\frac{d\theta}{dt} = Iww_p$$

1135. In an automobile, if the vehicle makes a left turn, the gyroscopic torque
(a) increase the forces on the outer wheels
(b) decreases the forces on the outer wheels
(c) does not affect the forces on the outer wheels
(d) None of these

UPRVUNL AE 05.07.2021
RPSC IOF, 2020

Ans. (a) : In an automobile, if the vehicle takes as left turn, the reactive gyroscopic couple on it is in clockwise when viewed from the rear end of the vehicle. The reaction couple is provided by equal and opposite forces on the outer and the inner wheels of the vehicle.
The gyroscopic torque increases the forces on the outer wheels.

1136. The gyroscopic effects due to rotating parts of an aircraft on a curved course depend on
(a) Flight altitude (b) Flight velocity
(c) Radius if the curve (d) Both (2) & (3)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Assam PSC AE (IWT) 14.03.2021

Ans. (d) : The gyroscopic effect due to rotating parts of a turbo jet engine of an air craft on a curved course depend on flight velocity & radius of curve.

1137. A disc spinning on its axis at 20 rad/s will undergo precession when a torque 100 N-m is applied about an axis normal to it. If the mass moment of inertia is 1 kg-m^2, then the angular velocity of precession is?
(a) 0.2 rad/s (b) 5 rad/s
(c) 10 rad/s (d) 200 rad/s

ISRO Scientist/Engineer 12.01.2020
APPSC Poly. Lect. 2013

Ans. (b) : ω = 20 rad/sec.
Gyroscopic couple, C = 100 N-m
I = 1 kg-m^2
ω_p = ?
$C = I.\omega.\omega_p$
$100 = 1 \times 20 \times \omega_p$
ω_p = 5 rad/sec.

1138. A uniform disc (Diameter d = 150 mm, m = 5 kg) is mounted centrally on a horizontal shaft as shown in Fig. which runs in bearings (A and B) which are 10 cm apart. The disc spins, in the direction shown in the Fig, with a uniform speed of 1000 rpm. The shaft processes with a uniform velocity of 60 rpm in the horizontal plane in the anti-clockwise direction when looking from top.

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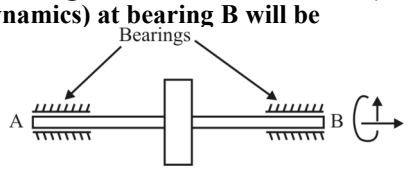
The magnitude of total reaction (i.e. static + dynamics) at bearing B will be



(a) 21.5 N (b) 67.96 N
(c) 92 N (d) 117 N

RPSC ACF & FRO, 26.02.2021

Ans. (b) : $I = \frac{1}{2}mr^2$

Couple (C) = $I\omega\omega_p$

$$= \frac{1}{2} \times 5 \times 0.075^2 \times \left[\frac{2\pi \times 1000}{60}\right] \times \left[\frac{2\pi \times 60}{60}\right]$$

C = 9.2527

The magnitude of total reaction at each bearing due to gyroscopic couple will be–

$$\text{Reaction} = \frac{C}{l} = \frac{9.2527}{0.1} = 92.52\text{N}$$

Reaction on each bearing due to weight of disc,

$$R_A = R_B = \frac{5 \times 9.81}{2} = 24.5\text{N}$$

Reaction at bearing A,

$R_{A1} = F + R_A = 92 + 24.5 = 116.5$ N (upward)

Reaction at bearing B,

$\boxed{R_{B1} = F - R_B = 92 - 24.5 = 67.5 \text{ N (downward)}}$

1139. A car is moving on a curved horizontal road of radius 100 m with a speed of 20 m/s. The rotating masses of the engine have an angular speed of 100 rad/s in clockwise direction when viewed from the front of the car. The combined moment of inertia of the rotating masses is 10 kg-m^2. The magnitude of the gyroscopic moment (in N-m) is_____.

(a) 2000 (b) 200
(c) 100 (d) 1000

Assam Engg. College AP/Lect. 18.01.2021

Ans. (b) : 200

Gyroscope couple = $I\omega\omega_p$

$$= 10 \times 100 \times \left(\frac{20}{100}\right)$$

= 200 N-m

1140. Vertically hanging curtains between the compartments in an aero plane are seen to bend forward. This indicates that the aero plane is

(a) **slowing down for landing**
(b) moving steadily at constant altitude
(c) taking a bend
(d) taking off

HPPSC Poly. Lect. 05.07.2021

Ans. (a) :

1141. Which of the following statements is/are false for active gyroscopic couple?

(a) Reactive gyroscopic couple--------- active gyroscopic couple are opposite in direction.
(b) In right hand rule, curled fingers denote direction of precession.
(c) In active gyroscopic couple spin vector and precession -------- parallel to each other.
(d) All the above statements are false.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) :

1142. If the axes of the rolling of the ship and of the stabilizing rotor are parallel, it will result in

(a) A higher bow and lower stern
(b) A lower bow and higher stern
(c) Turning towards left
(d) No gyroscopic effect

ESE 2020

Ans. (d) : No gyroscopic effect.

1143. The engine shaft of an aircraft rotates in counter - clockwise direction, when viewed from nose. If the aircraft is taking turn to the left, the gyroscopic couple will cause :

(a) lowering of the nose and raising of the tail
(b) lowering of both the nose and tail
(c) raising of both the nose and tail
(d) raising of the nose and lowering of the tail

APPSC Poly Lect. 13.03.2020

Ans. (d) : Raising of the nose and lowering of the tail.

1144. A rotating mass having moment of inertia of 30 kg m^2 rotates at 800 rpm and travelling in a curve of 170 m radius at a speed of 240 km/hr. It will experience a gyroscopic reaction of

(a) 10 m-kgf (b) 100 m-kgf
(c) 1000 m-kgf (d) None of these

RPSC IOF, 2020

Ans. (b) : Gyroscopic reaction is given as

$$= I\omega\omega_p$$

$$= I\frac{2\pi \times 800}{60} \times \frac{240 \times 10^3}{3600 \times 170}$$

$= 30 \times 83.77 \times 0.392$

= 985.529 kg m^2/sec or N-m

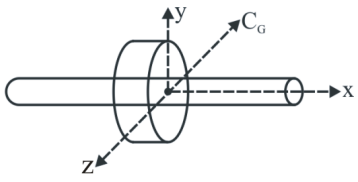
= 100.56 m-kgf

1145. The active gyroscopic torque in gyroscope about a horizontal axis represents

(a) the torque required to cause the axis of spin to process in the vertical plane
(b) the torque required to cause the axis of spin to process in the horizontal plane
(c) the force required to cause the axis of spin to process in the horizontal plane
(d) the force required to cause the axis of spin to process in the vertical plane

ESE 2019

Ans. (b) :



y – axis (axis of precession)
x – axis (axis of spin)
C_G – active gyroscopic couple (horizontal)
xz plane–Plane of precession (horizontal plane)

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1146. Consider the following statements:

1. Gyroscopic effects generate forces and couples which act on the vehicles, and these effects must be taken into account while designing their bearings.
2. Rolling motion of a ship usually occurs because of the difference in buoyancy on the two sides of the ship due to a wave.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

ESE 2018

Ans. (c) 1. Gyroscopic effect arises when there is precession of rotating rotor. Gyroscopic couple is generated which changes the bearing reactions. Hence gyroscopic effects to be considered while designing bearing.
2. Rolling motion of a ship usually occurs because of the difference in buoyancy on the two sides of the ship due to a wave.

1147. An aeroplane moving with a velocity of 300 Kmph negotiates a turn along a circular path of radius 600 m. If the angular velocity of the rotating parts of the plane is 120 rad/s and the moment of inertia of the rotating parts is 60 kg.m^2, the gyroscopic couple is

(a) 500 Nm (b) 1000 Nm
(c) 3000 Nm (d) 5000 Nm

UPSC JWM 2017

Ans. (b) : Aeroplene velocity V = 300 Kmph = $300\times\frac{5}{18}$ m/s

Radius of turn taken, R = 600 rpm
Angular velocity, ω = 120 rad/s
Moment of inertia, I = 60 kg-m^2
Gyroscopic couple, $C = I\,\omega\,\omega_p$

$$= I\omega\frac{V}{R}$$

$$= 60\times120\times\frac{300}{600}\times\frac{5}{18}$$

C = 1000 Nm

1148. An aircraft cruising at 360 kmph takes a right turn on an arc of 100 m radius. The turbines and propellers have a total mass of 500 kg with radius of gyration of 25 cm. The engine rotates at 2000 r.p.m. The magnitude of the gyroscopic couple generated is

(a) 6.55 kN m (b) 7.65 kN m
(c) 9.81 kN m (d) 13.1 kN m

ESE 2017

Ans. (a) : Given,
v = 360 kmph = 100 m/s
r = 100 m
m = 500 kg
k = 25 cm = 0.25 m
N = 2000 rpm
We know that $I = mk^2 = 500\times(0.25)^2 = 31.25$ kg m^2

$$\omega = \frac{2\pi N}{60} = \frac{2\pi\times2000}{60} = 209.44 \text{ r/s}$$

$$\omega_p = \frac{v}{r} = \frac{100}{100} = 1 \text{ rad/s}$$

Gyroscopic couple (c) = $I\omega\omega_p = 31.25\times209.44\times1$
= 6545 N-m = 6.54 kNm

1149. The effect of gyroscopic couple on rolling of ships is

(a) Very high (b) Very low
(c) No effect (d) Moderate

MPPSC AE 2016

Ans : (c) No effect of gyroscopic couple on rolling of ships.
Gyroscopic Effect on Rolling:- As the axes of the rolling of the ship and of the rotor are parallel, there is no precession of the axis of spin and thus there is no gyroscopic effect. In the same way, the effects on steering,, pitching or rolling can be observed when the plane of the spin of the rotating masses is horizontal but along the longitudinal axis of the vessel; or when the axis is vertical.

1150. The rotor of a ship with moment of inertia 200 kgm^2 and rotational speed of 200 rad/s. What will be the gyroscopic couple when ship turns to left at radius of 100 m with speed 5 m/s?

(a) 20 Nm (b) 2000 Nm
(c) 200 Nm (d) 2 Nm

UPRVUNL AE 07.10.2016

Ans. (b) : Data given,
moment of Inertia (I) = 200 kgm^2
Rotational speed ω = 200 rad/s

$$\omega_p = \frac{V}{R} = \frac{5}{100}$$

We know that Gyroscopic couple

$$C = I\ \omega\ \omega_p$$

$$C = 200\times200\times\frac{5}{100}$$

C = 2000 N-m

1151. The turbine rotor of a ship has a mass of 2.2 tonnes and rotates at 1800 rpm clockwise when viewed from left. The radius of gyration of the rotor is 320 mm. Determine the gyroscopic couple when the ship turns right at a radius of 250 m with a speed of 25 km/h.

(a) 1180 N-m (b) 3397 N-m
(c) 4246 N-m (d) 0

KPSC ADF 2015

Ans. (a) : Given data,
Mass (m) = 2.2 tones = 2200 kg
Radius of gyration (k) = 320 mm = 0.32 m, N = 1800 rpm
Radius about ship turned (R) = 250 m,
speed (V) = 25 km/hr

$$= \frac{25\times1000}{3600}\text{m/s} = 25\times\frac{5}{18}\ \text{m/sec}$$

$\therefore$ Gyroscopic couple (C) = $I\ \omega\ \omega_p$

$\because$ $I = mk^2,\ \omega_p = \frac{V}{R}$, $\omega = \frac{2\pi N}{60}$

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$$= mk^2 \times \frac{2\pi N}{60} \times \frac{V}{R}$$

$$= 2200 \times (0.32)^2 \times \frac{2\pi \times 1800}{60} \times \frac{25 \times 1000}{3600 \times 250}$$

$$= 2200 \times 0.1024 \times \frac{10\pi}{6}$$

$$\boxed{C = 1179.56\,N\text{-}m}$$

$$\boxed{C \simeq 1180\,N\text{-}m}$$

1152. Net reaction of ground on wheels due to gyroscopic couple due to wheels and the dead weight and centrifugal force of a vehicle negotiating a curve

(a) increases on inner wheels and decreases on outer wheels
(b) decreases on inner wheels and increases on outer wheels
(c) decreases on all the wheels
(d) increases on all the wheels

TSGENCO AE 14.11.2015

Ans. (b) : Net reaction of ground on wheels due to gyroscopic couple due to wheels and the dead weight and centrifugal force of a vehicle negotiating a curved decreases on inner wheels and increase on outer wheels. As a vehicle moves on curved path, a centrifugal force also acts on the vehicle in the outward direction at the centre of mass of vehicle.

Centrifugal force $(F_c) = mr\omega^2$

$$= mr\left(\frac{v}{r}\right)^2$$

$$F_c = \frac{mv^2}{r}$$

Forces on each outer wheel $= \frac{F_c}{2\omega}(\text{up wards})$

Forces on each inner wheel $= \frac{F_c}{2\omega}(\text{down wards})$

1153. The gyroscopic acceleration depends upon :

(a) Instantaneous value of axis ω
(b) The rate at which the axis of spin changes its speed
(c) Both (a) and (b)
(d) None of these

OPSC AEE 2015 Paper-I

Ans : (c) The gyroscopic acceleration depends upon
(i) Instantaneous value of ω.
(ii) The rate at which the axis of spin changes its speed.

1154. When a four wheeler moving forward at a speed above critical takes a turn to the right the wheel(s) that tends to leave the ground is

(a) outer front wheel
(b) outer rear wheel
(c) both the inner wheels
(d) both the outer wheels

TNPSC AE 2014

Ans. (c) : When a four wheeler moving forward at a speed above critical takes a turn to the right the wheel(s) that tends to leave the ground is both the inner wheels.

1155. A bicycle remains stable in running through a bend because of :

(a) Coriolis action
(b) Radius of curve path
(c) Gyroscopic and centrifugal action
(d) None of these

MPSC Poly. Lect. 2014

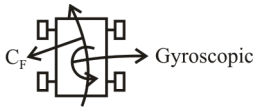
Ans. (c) : A bicycle remains stable in running through a bend or curved path because of centrifugal action & gyroscopic effect.

1156. When car takes a turn, for the stability of car :

(a) only gyroscopic effect should be considered
(b) only centrifugal effect should be considered
(c) both gyroscopic and centrifugal effects should be considered
(d) only speed of the car should be considered

MPPSC State Forest Service Exam, 2014

Ans. (c) : When a car takes turns, for the stability of car both gyroscopic & centrifugal effects should be considered.



1157. A flywheel weight $\frac{981}{\pi} kg$ and has a radius of gyration of 100 cm. It is given a spin of 100 rpm about its horizontal axis. The whole assembly is rotating about a vertical axis at 6 rad/sec. The gyroscopic couple experienced will be

(a) 2,000 kgm (b) 19,620 kgm
(c) 20,000 kgm (d) 1,962 kgm

ISRO Scientist/Engineer 2009

Ans. (a) : $m = \frac{981}{\pi}$

Radius of gyration (k) = 100 cm
N = 100 rpm
ω_p = 6 rad/sec

$$\omega = \frac{2\pi N}{60}$$

$$= \frac{2\pi \times 100}{60}$$

$$= \frac{10\pi}{3} rad/\sec$$

$I = mk^2$

$$= \frac{981}{\pi} \times (1)^2$$

$$= \frac{981}{\pi}$$

Gyroscopic couple = $I\omega\omega_p$

$$\frac{981}{\pi} \cdot \frac{10\pi}{3} .6$$

= 19620 N-m

Dividing by 9.81 m/sec^2, we get
= 2000 Kg-m

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11. Vibrations

1158. Ratio of maximum displacement of the forced vibration to the deflection due to static force is known as

(a) critical damping coefficient
(b) logarithmic decrement
(c) magnification factor
(d) damping factor

VIZAG Steel MT 24.01.2021
CIL MT 27.02.2020
TRB Poly. Lect. 2017
GPSC ARTO 01.05.2016
Kerala PSC AE 06.08.2015/ VIZAG MT 2015
KPSC ADF 2015/ KPSC AE 2015
J&K PSC Civil Services Pre, 2013
TNPSC ACF 2012 / TNPSC AE, 2008
CSE Pre-2003, 1994

Ans. (c) : Ratio of maximum displacement of the forced vibration to the deflection due to static force is known as magnification factor (M.F.).

$$MF = \frac{A}{F_0/s}$$

$$MF = \frac{1}{\sqrt{\left\{1-\left(\frac{\omega}{\omega_n}\right)^2\right\}^2 + \left(\frac{2\xi\omega}{\omega_n}\right)^2}}$$

1159. Whirling speed of a shaft coincides with the natural frequency of its

(a) Torsional vibration
(b) Transverse vibration
(c) Longitudinal vibration
(d) Coupled bending vibration

Nagaland PSC (CTSE) 2018, Paper-I
TSPSC AEE 2017
MPPSC AE 08.11.2015
TANGEDCO AE 2015
MPSC Poly. Lect. 2014
UKPSC AE-2013, Paper-I
TRB Poly. Lect., 2012
ISRO Scientist/Engineer 2011
CSE Pre-1995
Vadodara Muncipal Corp. DEE, 2018
CGPSC AE 15.01.2021
CGPSC AE 16.10.2016
OPSC AEE 2015 Paper-I

Ans. (b) : Whirling of shaft occurs, when the natural frequency of transverse vibration matches with the frequency of rotating shaft.

1160. In a spring mass system, the mass of the system is doubled and the spring stiffness is made half. The natural frequency of vibration is

(a) Halved (b) Doubled
(c) Unaffected (d) Quadrupled

ESE 2017
GPSC ARTO 01.05.2016
GPSC Lect. (Auto) 16.10.2016
TSPSC Managers, 2015
TSPSC AEE 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
APPSC AEE 2012
ISRO Scientist/Engineer 2007

Ans. (a)

$$(\omega_n)_1 = \sqrt{\frac{k}{m}}$$

Given

$$k_1 = \frac{k}{2}$$

$$m_1 = 2m$$

$$(\omega_n)_2 = \sqrt{\frac{k_1}{m_1}} = \sqrt{\frac{k}{2\times 2m}}$$

$$(\omega_n)_2 = \frac{1}{2}\sqrt{\frac{k}{m}}$$

$$(\omega_n)_2 = \frac{(\omega_n)_1}{2}$$

1161. At a certain speed, revolving shaft tends to vibrate violently in transverse directions. The speed is known as .....

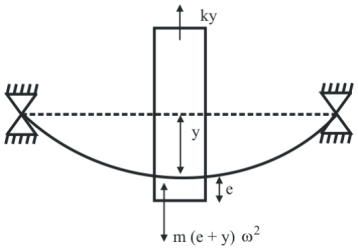
(a) whirling speed (b) critical speed
(c) whipping speed (d) all of these

TSPSC Manager (Engg.) HMWSSB 12.11.2020
TRB Polytechnic Lecturer 2017
HPPSC Asstt. Prof. 18.09.2017
Kerala PSC Asstt. Prof. 27.09.2016
Nagaland PSC CTSE 2016 Paper-I
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
UKPSC AE 2012 Paper-II

Ans. (d) : Whirling Speed- It is the speed at which the shaft tends to vibrate violently in transverse direction. This is also known as critical speed or whipping speed of shaft.

The critical speed of shaft essentially depends upon –
(1) The eccentricity of the C.G. of the rotating masses from the axis of rotation of the shaft.
(2) Diameter of the disc.
(3) Span of the shaft.
(4) Type of support connection of its ends.

$$y = \frac{e}{\left(\frac{k}{m\omega^2} - 1\right)}$$



$$y = \frac{e}{\left(\frac{\omega_C}{\omega}\right)^2 - 1}$$

$$\omega_C = \sqrt{\frac{k}{m}}$$

At $\omega = \omega_n = \omega_c$ (critical speed of shaft)

$y = \infty$

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1162. The rate of decay of oscillations is known as

(a) critical damping
(b) damping coefficient
(c) transmissibility
(d) logarithmic decrement

VIZAG Steel MT 24.01.2021, Shift-I
Assam PSC AE (PHED), 18.10.2020
Jharkhand Urja Vikas Nigam Ltd. AEE 2017
CGPSC Poly. Lect. 22.05.2016
RPSC AE 2016, RPSC ACF-2011
TNPSC AE, 2008

Ans. (d) : Logarithmic decrement- The rate of decay of oscillations is known as logarithmic decrement it is used to find the damping ratio of an under damped system in the time domain the method of logarithmic decrement becomes less and less precise as the damping ratio increases past about 0.5 it does not apply at all for a damping ratio greater than 1.0 because the system is over damped.

$$\text{logarithmic decrement, } \delta \quad \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

Critical damping- It is defined as the threshold between over damping and under damping.

Damping coefficient- Damping coefficient is a material property that indicates whether a material will bounces back or return energy to system.

Transmissibility- The state of rest or of motion of a rigid body is unaltered if a force acting on the body is replaced by another force of the same magnitude and direction but acting any where on the body along the line of action of the applied force.

1163. If the damping factor for a vibrating system is unity, then the system will be

(a) over-damped (b) under-damped
(c) critically damped (d) without vibrations

VIZAG MT, 14.12.2020
Assam PSC CCE Pre 2015
RPSC AE 2016
APPSC AEE MAINS 2016, PAPER-III
UKPSC AE 2012 Paper-I
RPSC ACF 2011, CSE Pre-1996

Ans. (c) : Damping factor–

$$\xi = \frac{C}{C_c} = \frac{\text{Actual damping coefficient}}{\text{Critical damping coefficient}}$$

$$= \frac{\sqrt{(c/2m)^2}}{\sqrt{S/m}} = \frac{C}{2\sqrt{Sm}}$$

Here, $C_c = 2\sqrt{Sm}$

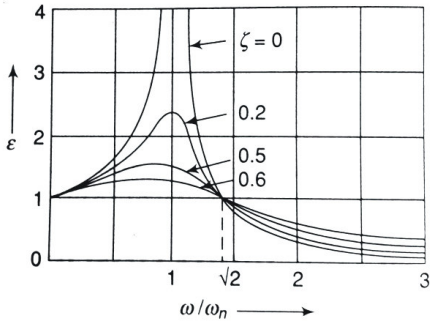
$\xi = 1$, The damping is critical
$\xi > 1$, The system is over-damped
$\xi < 1$, The system is under-damped

1164. If ratio of excitation and natural frequency of vibration $\omega/\omega_n = \sqrt{2}$; the transmissibility of vibration will be :-

(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
CGPSC AE 15.01.2021
Nagaland PSC (CTSE) 2018 Paper-I
UKPSC AE-2013, Paper-I
J&K PSC Civil Services Pre, 2010
APPSC IOF, 2009 / ESE 1995

Ans. (b) :



If $\frac{\omega}{\omega_n} = \sqrt{2}$

Then, the transmissibility (ε) of vibration will be 1.

1165. A shaft carrying two rotors at its ends will have-

(a) No node (b) One node
(c) Two nodes (d) Three nodes

NPCIL ET, 2015
TSPSC AEE 2015
OPSC AEE 2015 Paper-I
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TNPSC AE 2014 /CSE Pre-2006

Ans. (b) : Rotor (n) = 2
Number of Node = (n – 1)
= 2 – 1
= 1

1166. At the nodal point in a shaft, the amplitude for torsional vibration will be-

(a) Zero (b) Maximum
(c) Minimum (d) Infinity

CIL MT 27.02.2020
Nagaland PSC (CTSE) 2018 Paper-I
NPCIL ET, 2015
Mizoram PSC AE/SDO 2014 Paper-II
ISRO Scientist/Engineer 2010
WBPSC AE 2008

Ans. (a) : At the node point in a shaft, the amplitude for torsional vibration will be zero.

1167. For forced damped vibration system, the vibration isolation is possible only when

(a) $\frac{\omega}{\omega_n} = 1$ (b) $\frac{\omega}{\omega_n} < 1$
(c) $\frac{\omega}{\omega_n} < \sqrt{2}$ (d) $\frac{\omega}{\omega_n} > \sqrt{2}$

RPSC IOF, 2020
NLC GET 17.11.2020 Shift-II
CIL MT 27.02.2020
TNPSC AE 2019
GPSC Lect. (Auto) 16.10.2016
TSPSC AEE 2015

Ans. (d) : Transmissibility ratio-

$$\varepsilon = \frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2}$$

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When $\frac{\omega}{\omega_n} > 1, \varepsilon$ is negative. This means that there is a phase difference of 180° between the transmitted force and the disturbing force. The value of ω/ω_n must be greater than $\sqrt{2}$ if ε is to be less than 1 and it is the numerical value of ε independent of any phase difference between the forces that may exist which is important.

1168. A mass of 1 kg is attached to the end of a spring with stiffness 0.7 N/mm. The critical damping co-efficient of this system will be _____ Ns/m.

(a) 1.40 (b) 18.522
(c) 52.92 (d) 529.2

CGPSC AE 15.01.2021
UJVNL AE 2016
NPCIL ET, 2015
MPSC Poly. Lect. 2014
VIZAG STEEL MT 2011
WBPSC AE 2003

Ans. (c) : Given,

$m = 1kg$

$k = 0.7\ N/mm = 0.7 \times 1000\ \frac{N}{m} = 700\ N/m$

$C_c = ?$

Critical Damping coefficient

$(C_C) = 2\sqrt{km}$

$= 2\sqrt{700 \times 1}$

$\boxed{C_C = 52.915 \frac{Ns}{m}}$

1169. A shaft has two heavy rotors mounted on it. The transverse natural frequencies considering each of the rotor separately are 100 Hz and 200 Hz respectively. The lowest critical speed is

(a) 5367 rpm (b) 6000 rpm
(c) 9360 rpm (d) 12000 rpm

RPSC ACF & FRO, 26.02.2021
RPSC IOF, 2020
ISRO Scientist/Engineer 2012
TRB Poly. Lect., 2012
VIZAG STEEL MT 2011
ESE 2017

Ans. (a) : Dunkerley's empirical formula

$$\frac{1}{f_n^2} = \frac{1}{(f_{n1})^2} + \frac{1}{(f_{n2})^2}$$

$(f_n)_1 = 100$ Hz

$(f_n)_2 = 200$ Hz

$$\frac{1}{(f_n)^2} = \frac{1}{10000} + \frac{1}{40000}$$

$$\frac{1}{f_n^2} = \frac{5}{40000}$$

$$f_n^2 = \frac{40000}{5}$$

$$f_n = \frac{200}{\sqrt{5}}$$

Critical speed = Whirling speed

$= 200/5^{½} \times 60$

Critical speed = 5367 rpm

1170. Critical damping is a function of

(a) mass and stiffness
(b) mass and damping coefficient
(c) stiffness and natural frequency
(d) natural frequency and damping coefficient

Vadodara Muncipal Corp. DEE, 2018
Punjab PSC (Lect.) 06.08.2017
SJVN ET 2013
TRB Asstt. Prof., 2012
WBPSC AE 2003

Ans. (a) : Critical damping is the function of mass & stiffness,

Critical damping just prevent vibration or is just sufficient to allow the object to return to its rest position in the shortest period of time

$\boxed{\omega_n = \sqrt{k/m}}$ [ω_n = natural frequency

Critical damping → $\boxed{\text{Critical} = 2\sqrt{km}}$ ($\xi = 1$)

$\boxed{\xi = \frac{C}{C_{critical}}}$ ← damping ratio

1171. In free vibrations viscous damping force is proportional to the

(a) velocity of the mass
(b) displacement of the mass
(c) acceleration of the mass
(d) none of these

GPSC Asstt. Director of Transport 05.03.2017
GPSC Lect. 23.10.2016
RPSC AE 2016, APPSC AEE 2012
RPSC ACF-2011
TNPSC AE 2017, SJVN ET 2013

Ans. (a) : In viscous damping, the damping force (F) is proportional to the velocity $(\dot{x})$ of vibrating body

$F \propto \dot{x}$

$F = C\dot{x}$

Where C is called as the viscous damping coefficient.

1172. Consider a single degree-of-freedom system with viscous damping excited by a harmonic force. At resonance, the phase angle (in degree) of the displacement with respect to the exciting force is

(a) 0 (b) 45
(c) 90 (d) 135

Oil India Limited Sr. Engineer (Drilling) 30.11.2019
GPSC Asstt. Director of Transport 05.03.2017
Karnataka PSC Lect. 27.05.2017
RPSC AE 2016
UKPSC AE 2012 Paper-I
RPSC ACF-2011 / GATE 2014

Ans. (c) : Phase angle,

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$$\tan\phi = \frac{2\zeta\frac{\omega}{\omega_n}}{\left(1-\frac{\omega^2}{\omega_n^2}\right)}$$

At resonance,

$\omega = \omega_n$

$\tan\phi = \infty$

$\phi = 90°$

1173. When a body is subjected to transverse vibrations, the stress induced in a body will be:

(a) Shear stress
(b) Torsional shear stress
(c) Compressive stress
(d) Bending stress

UPRVUNL AE 05.07.2021
GMB AAE 25.06.2017
RPSC VPITI 14.02.2016
NPCIL ET, 2015
UPRVUNL AE 2014

Ans. (d) : Transverse vibrations; when the particle of the shaft or disc move approximately perpendicular to the axis of the shaft, then vibrations are known as transverse vibrations. In this case, the shaft is straight and bent horizontally and bending stresses are induced in the shaft.

1174. In vibration isolation system, if $\frac{\omega}{\omega_n} > 1$, then the phase difference between the transmitted force and the disturbing force is :

(a) 0° (b) 90°
(c) 180° (d) 270°

TSPSC Managers, 2015
KPSC ADF 2015
APPSC Poly. Lect. 2013
TNPSC AE (Industries) 09.06.2013
CSE Pre-2007

Ans. (c) : Given- $\frac{\omega}{\omega_n} > 1$

There is a phase difference of 180° between the transmitted force and the disturbing force.

1175. The factor which affects the critical speed of a shaft is-

(a) Diameter of the disc (b) Span of the shaft
(c) Eccentricity (d) All of these

NPCIL ET, 2015
UKPSC AE-2013, Paper-I
TNPSC ACF 2012
CSE Pre-2010
WBPSC AE 2008

Ans. (d) : Critical speed of shaft

$$y = \frac{e}{\left(\frac{\omega}{\omega_n}\right)^2 - 1}$$

Critical speed depends upon factor-

- diameter of disc
- span of the shaft
- eccentricity

1176. A vibrating machine is isolated from the floor using springs. If the ratio of excitation frequency of vibration of machine to the natural frequency of the isolation system is equal to 0.5, the transmissibility ratio of isolation is :

(a) 1/2 (b) 3/4
(c) 2 (d) 4/3

TSPSC AEE 2017
RPSC ADE 2016
APGENCO AE 2012
GATE 2004

Ans. (d) : Given,

$$\left(\frac{\omega}{\omega_n}\right) = 0.5$$

$$\text{Transmissibility ratio } \varepsilon = \frac{\sqrt{1+\left(\frac{2\xi\omega}{\omega_n}\right)^2}}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2+\left(\frac{2\xi\omega}{\omega_n}\right)^2}}$$

Assuming $\xi = 0$

$$\varepsilon = \frac{1}{1-\left(\frac{\omega}{\omega_n}\right)^2} = \frac{1}{1-(0.5)^2} = \frac{1}{1-\frac{1}{4}} = 4/3$$

1177. In forced vibrations the magnitude of damping force at resonance equals

(a) Inertia force (b) Impressed force
(c) Zero (d) Spring force

RPSC AE 2016
CGPSC AE 26.04.2015 Shift-I
SJVN ET 2013
RPSC ACF-2011

Ans. (b) : In force vibrations the magnitude of damping force at resonance equals impressed force.

1178. A shaft with torsional stiffness (q) has a disc of mass moment of inertia (I) attached at the end, then the natural frequency (f_n) of free torsional vibration of the shaft is given by:

(a) $f_n = 2\pi\sqrt{qI}$ (b) $f_n = \frac{1}{2\pi}\sqrt{qI}$

(c) $f_n = \frac{1}{2\pi}\sqrt{\frac{q}{I}}$ (d) $f_n = 2\pi\sqrt{\frac{q}{I}}$

CIL MT 26.03.2017
ISRO Scientist/Engg. (RAC) 07.05.2017
GPSC Lect. Poly. 07.12.2014
TNPSC AE (Industries) 09.06.2013

Ans. (c) : Mass moment at Inertia = I
Torsional stiffness = q

Frequency of torsional vibration $f_n = \frac{1}{2\pi}\sqrt{\frac{q}{I}}$

1179. The ratio of actual damping coefficient to the critical damping coefficient is called as

(a) Magnification factor

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(b) Damping factor
(c) Critical damping ratio
(d) Logarithmic decrement

UPRVUNL AE 05.07.2021
VIZAG Steel MT 24.01.2021, Shift-I
TNPSC AE 2017
(KPSC AE 2015)

Ans : (b) Damping factor (ξ) =

$$\frac{\text{Actual damping coefficient}(C)}{\text{Critical damping coefficient}(C_c)}$$

$$\xi = \sqrt{\frac{(c/2m)^2}{s/m}} = \frac{c}{2\sqrt{s\,m.}}$$

$$c_c = 2\sqrt{s\,m}$$

$\xi = 1$ critical damping
$\xi > 1$ over damping
$\xi < 1$ Under damping

1180. The natural frequency of spring mass system on earth is ω_n. The natural frequency of this system on the moon ($g_{moon} = g_{earth}/6$) is

(a) ω_n (b) $0.408\ \omega_n$
(c) $0.204\ \omega_n$ (d) $0.167\ \omega_n$

GPSC ARTO Pre 30.12.2018
TSPSC AEE 2017
Karnataka PSC AE, 10.09.2017
CGPSC AE 26.04.2015 Shift-I, GATE 2010

Ans. (a) : We know natural frequency of spring mass system is, $\omega_n = \sqrt{k/m}$

This equation, depends only on mass but not on weight. Therefore it is independent of gravity of a system, so it will remain ω_n.

1181. The equation of motion for a damped free vibration is, $3\ddot{x} + 9\dot{x} + 27x = 0$, the damping factor will be:

(a) 0.25 (b) 0.5
(c) 0.75 (d) 1.0

Vadodara Muncipal Corp. DEE, 2018
TRB Asstt. Prof., 2012
WBPSC AE 2003
ESE 2000

Ans. (b) : Given,

$$3\ddot{x} + 9\dot{x} + 27x = 0$$

Compare with given equation from.

$$m\ddot{x} + c\dot{x} + kx = 0$$

$m = 3,\ c = 9,\ k = 27$

Damping factor $(\xi) = \frac{c}{2\sqrt{km}} = \frac{9}{2\sqrt{27\times 3}} = 0.5$

1182. Critical speed of shaft depends on

(a) mass
(b) stiffness
(c) mass and stiffness
(d) mass, stiffness and eccentricity

GPSC Asstt. Director of Transport 05.03.2017
RPSC AE 2016
RPSC ACF-2011
ESE 2012, 1996

Ans : (c) Critical speed of shaft depends on mass and stiffness.

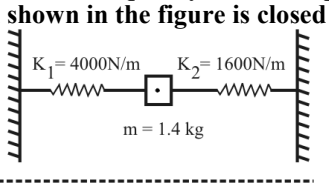
Critical or whirling speed of a shaft:-

The speed at which the shaft runs so that the additional deflection of the shaft from the axis of rotation become infinite is known as critical or whirling speed.

$$\text{critical speed } (\omega_c) = \sqrt{\frac{s}{m}} = \sqrt{\frac{g}{\delta}}\ \text{Hz}$$

Critical speed of shaft depends on mass and stiffness.

1183. The natural frequency of the spring mass system shown in the figure is closed to :



(a) 8 Hz (b) 10 Hz
(c) 12 Hz (d) 14 Hz

BPSC Asstt. Prof. 29.11.2015
VIZAG MT 2015
VIZAG STEEL MT 18.08.2013
GATE 2008

Ans. (b) : Mass of spring mass system (m) = 1.4 kg
Stiffness of spring 1 (k_1) = 4000 N/m
Stiffness of spring 2 (k_2) = 1600 N/m

Now, the equivalent spring stiffness,

$K_{eq} = k_1 + k_2 = 4000 + 1600$
$= 5600$ N/m

Natural frequency,

$$\omega_n = \sqrt{\frac{k_{eq}}{m}} = \sqrt{\frac{5600}{1.4}} = 63.245\,rad/\text{sec}.$$

$$f_n = \frac{1}{2\pi}\omega_n = \frac{1}{2\pi}\times 63.245$$

$$f_n = 10.065\,Hz$$

$$\approx 10.0\,Hz$$

1184. In free vibrations, acceleration vector leads the displacement vector by

(a) π (b) $\pi/2$
(c) $\pi/6$ (d) $2\pi/3$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
UKPSC AE 2012 Paper-I

Ans. (a) : SHM,

displacement = $x = x\sin(\omega_n t + \phi)$

Velocity = $\dot{x} = \omega_n \times \cos(\omega_n t + \phi)$

Acceleration $\ddot{x} = -\omega_n^2 \times \sin(\omega_n t + \phi)$

By comparing $x\ \&\ \ddot{x} = \omega_n^2 \times \sin\left[180^\circ + (\omega_n t + \phi)\right]$

So, acceleration is leads by $180^\circ = \pi$

1185. Natural frequency of transverse vibrations of a shaft carrying load at the centre of the span is:

(a) $f_n = \frac{5.63}{\sqrt{\delta}}\ \text{Hz}$ (b) $f_n = \sqrt{\frac{4.97}{\delta}}$
(c) $f_n = \sqrt{\frac{5.63}{\delta}}\ \text{Hz}$ (d) $f_n = \frac{0.4987}{\sqrt{\delta}}\ \text{Hz}$

RPSC 2016

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TSPSC Managers, 2015
OPSC Civil Services Pre 2006

Ans. (d) : Natural frequency of shaft.

$$\omega_n = \sqrt{\frac{g}{\delta}} = \sqrt{\frac{g}{\text{deflection}}}$$

$$\text{Time period } (t_p) = \frac{2\pi}{\omega_n}$$

$$\text{Frequency } (f_n) = \frac{1}{t_p} = \frac{1}{2\pi} \times \omega_n \Rightarrow \frac{1}{2\pi} \times \sqrt{\frac{g}{\delta}}$$

$$\Rightarrow \frac{3.1320}{2\pi} \times \frac{1}{\sqrt{\delta}} \text{ Hz}$$

$$f_n \Rightarrow \frac{0.4987}{\sqrt{\delta}} \text{ Hz}$$

1186. A shaft of mass (m_c) and stiffness (s) is fixed at one end and carries a mass (m) at the other end. The natural frequency of its longitudinal vibration is equal to

(a) $2\pi\sqrt{\{s/(m+m_c)\}}$

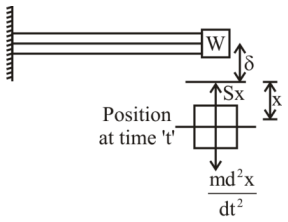
(b) $2\pi\sqrt{\left[s/\{m+(m_c/3)\}\right]}$

(c) $1/2\pi\sqrt{\{s/(m+m_c)\}}$

(d) $1/2\pi\sqrt{\left[s/\{m+(m_c/3)\}\right]}$

Oil India Senior Officer 23.12.2020
Nagaland PSC (CTSE) 2018, Paper-I
OPSC Civil Services Pre-2006

Ans. (d) :



s = Stiffness
x = xsin ω + t
x = Displacement

$$\frac{dx}{dt} = \omega \times x \sin \omega t$$

$$v = \frac{dx}{dt} = \omega \cdot x$$

$$= \frac{1}{2} \times mv^2 = \frac{1}{2} \times m\omega^2 \times x^2$$

$$= \left(\frac{s \cdot x}{2}\right) x = \frac{1}{2} s \cdot x^2$$

$$\frac{1}{2} m\omega^2 x^2 = \frac{1}{2} \times s \cdot x^2$$

$$\omega = \sqrt{\frac{s}{m}}$$

$$t_p = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{s}{m}} \Rightarrow f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{s}{m}}$$

For longitudinal vibration,

$$\left[f_n = \frac{1}{2\pi}\sqrt{\frac{s}{m + \frac{m_c}{3}}}\right]$$

1187. $\partial^2u/\partial t^2 = C^2\partial^2u/\partial x^2$ represents the equation for

(a) Vibration of a stretched string
(b) Heat flow of a thin rod
(c) Motion of a projectile in a gravitational field
(d) Oscillation of a simple pendulum

MPSC Poly. Lect. 2014
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 1999

Ans. (a) : Wave Equation–It is a second-order linear partial differential equation for the description of waves (like mechanical waves). The partial differential equation is given as,

$$A\frac{\partial^2u}{\partial x^2} + B\frac{\partial^2u}{\partial x \partial y} + C\frac{\partial^2u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} = F$$

$B^2 - 4AC < 0$	Elliptical	2 - D heat equation
$B^2 - 4AC = 0$	Parabolic	1 - D Heat equation
$B^2 - 4AC > 0$	Hyperbolic	1 - D wave equation

For one-dimensional equation,

$$\alpha^2 \frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial t^2}$$

$$\boxed{B^2 - 4AC} \quad \text{.....(i)}$$

Where, $A = \alpha^2$, $B = 0$, $C = -1$
Put all the values in equation (i)

$$0 - 4(\alpha^2)(-1)$$

$$4\alpha^2 > 0$$

So, this is a one-dimensional wave equation or vibration of stretched spring.

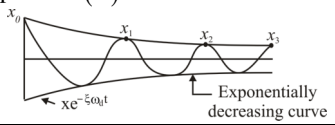
1188. In underdamped vibrating system the amplitude of vibration with reference to time

(a) increases linearly
(b) increases exponentially
(c) decreases linearly
(d) decreases exponentially

TNPSC AE 2019
RPSC Vice Principal ITI 2018
CGPSC AE 26.04.2015 Shift-I

Ans. (d) : In underdamped vibrating system the amplitude of vibration with reference to time decreases exponentially.

$$\text{Amplitude } (x) = X.e^{-\xi.\omega_d.t}$$



1189. Critical speed of the shaft is affected by

(a) diameter and eccentricity of the shaft
(b) span and eccentricity of the shaft
(c) diameter and span of the shaft
(d) span of the shaft

RPSC Vice Principal ITI 2018
RPSC Lect. (Tech. Edu. Dept.) 2011, ESE 2000

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Ans. (c) : The critical speed or whirling or whipping speed is the speed at which the shaft tends to vibrate violently in transverse direction. It depends on–

- The eccentricity of the rotating mass from the axis of rotation of the shaft.
- Diameter of the disc
- Span (length) of the shaft, and
- Types of supports connected at its ends.

For equilibrium, $y = \dfrac{e}{\left[\left(\dfrac{\omega_n}{\omega}\right)^2 - 1\right]}$

1190. Logarithmic decrement of a damped single degree of freedom system is δ. If stiffness of the spring is doubled and the mass is made half, then the logarithmic decrement of the system will be equal to :

(a) δ/4 (b) δ/2
(c) δ (d) 2 δ

ISRO Scientist/Engineer 12.05.2013
J&K PSC Civil Services Pre, 2010
CSE Pre-1997

Ans. (c) : Logarithmic decrement $(\delta) = \dfrac{2\pi\xi}{\sqrt{1-\xi^2}}$

So, logarithmic decrement depend upon damping factor.

Given–damping factor $(\xi) = \dfrac{C}{2\sqrt{km}}$

If, $k_1 = 2k$, $m_1 = \dfrac{m}{2}$

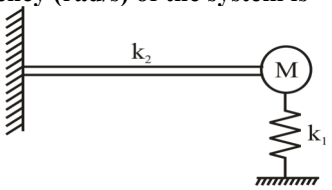
$$\xi_1 = \frac{C}{2\sqrt{k_1 m_1}} = \frac{C}{2\sqrt{2k \times \dfrac{m}{2}}}$$

$$\xi_1 = \frac{C}{2\sqrt{km}}$$

$$\boxed{\xi_1 = \xi}$$

Since, there is no change in damping factor, so there is no change in logarithmic decrement.

1191. A cantilever beam of negligible weight is carrying a mass M at its free end, and is also resting on an elastic support of stiffness k_1 as shown in the figure below. If k_2 represents the bending stiffness of the beam, the natural frequency (rad/s) of the system is-



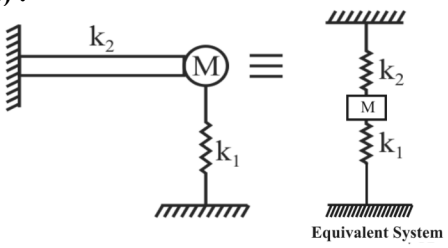
(a) $\sqrt{\dfrac{(k_1 k_2)}{M(k_1 + k_2)}}$ (b) $\sqrt{\dfrac{2(k_1 + k_2)}{M}}$

(c) $\sqrt{\dfrac{(k_1 + k_2)}{M}}$ (d) $\sqrt{\dfrac{(k_1 - k_2)}{M}}$

ESE 2013
ISRO Scientist/Engineer 2006

GATE 1993

Ans. (c) :



From equivalent system,
k_1 & k_2 are in
Parallel then
$k_{eq} = k_1 + k_2$
Natural frequency is given by

$$\omega_n = \sqrt{\frac{k_{eq}}{M}} = \sqrt{\frac{(k_1 + k_2)}{M}}$$

1192. What is the number of nodes in a shaft carrying three rotors?

(a) Zero (b) 2
(c) 3 (d) 4

OPSC AEE 2019 Paper-I
TNPSC AE 2018
Sikkim PSC (Under Secretary), 2017
TSPSC Managers, 2015
WBPSC AE, 2007, ESE 2006

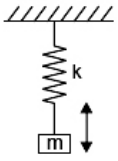
Ans : (b) : Given as, n = no. of Rotor in shaft
∴ For 3- rotor system
Nodes = 3–1 = 2 nodes

1193. A mass 'm' attached to a light spring oscillates with a period of 2 seconds. If the mass is increased by 2 kg, the period increases by 1 second. What is the value of the mass?

(a) 0.9 kg (b) 1.6 kg
(c) 2.1 kg (d) 2.7 kg

ESE 18.07.2021
MPSC Poly. Lect. 2014
ISRO Scientist/Engineer 2006
GATE 1994

Ans. (b) : Given,
T = 2 sec.
$m_1 = m$
$m_2 = m + 2$



frequency of oscillation:

$$\omega_n = \sqrt{\frac{k}{m_1}}$$

$$\text{Time period} = \frac{2\pi}{\omega_n} = 2\pi\sqrt{\frac{m_1}{k}}$$

Now, Initially, $(m_1 = m)$

$$2 = 2\pi\sqrt{\frac{m}{k}} \quad \text{....(i)}$$

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Later, $(m_2 = m + 2)$

$$3 = 2\pi\sqrt{\frac{m+2}{k}} \quad ....(ii)$$

Dividing (ii) by (i)

$$\frac{3}{2} = \sqrt{\frac{m+2}{m}}$$

or, $\quad \frac{9}{4} = \frac{m+2}{m}$

or, $\quad 9m = 4m + 8$

or, $\quad 5m = 8$

or, $\quad \boxed{m = 1.6kg}$

1194. In a forced vibration with viscous damping, maximum amplitude occurs when forced frequency is

(a) equal to natural frequency
(b) slightly less than natural frequency
(c) slightly greater than natural frequency
(d) zero

Assam PSC CCE Pre 2015
ESE 1999
CSE Pre-1999

Ans. (b) : The frequency of maximum amplitude in forced damped vibration is,

$$\frac{\omega}{\omega_n} = \sqrt{1 - 2\xi^2}$$

Magnification factor $= \frac{x}{x_{st}}$

$$= \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)\right]^2 + \left[2\xi\left(\frac{\omega}{\omega_n}\right)\right]^2}}$$

Amplitude $(A) = \frac{F_o}{\sqrt{(k - m\omega^2) + (c\omega)^2}}$

$$A = \frac{F_o / k}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\xi\frac{\omega}{\omega_n}\right]^2}}$$

1195. A vehicle suspension system consists of a leaf spring and a damper. The stiffness of the leaf spring is 3.6 kN/m and damping constant of the damper is 400 Ns/m. If the mass is 50 Kg, then the damping factor and damped natural frequency respectively are

(a) 0.471 and 1.19 Hz (b) 0.471 and 7.48 Hz
(c) 0.666 and 1.35 Hz (d) 0.666 and 8.50 Hz

UPRVUNL AE 2014
ISRO Scientist/Engineer 2012
OPSC Civil Services Pre. 2011
GATE 2009

Ans. (a) : Given

k = 3.6 kN/m
C = 400 Ns/m
m = 50 kg

$$\xi = \frac{C}{2\sqrt{km}}$$

$$\xi = \frac{400}{2\sqrt{3.6 \times 1000 \times 50}}$$

$\xi = 0.471$

Natural frequency (ω)

$$\omega_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{3.6 \times 10^3}{50}}$$

$\omega_n = 1.35$ Hz

$$\omega_d = \sqrt{1 - \xi^2}\ \omega_n$$

Frequency of Damped vibration

$$\omega_d = 1.35\sqrt{1 - (0.47)^2}$$

$\omega_d = 1.19$ Hz

1196. Under logarithmic decrement, the amplitude of successive vibrations are

(a) Constant
(b) In arithmetic progression
(c) In geometric progression
(d) In logarithmic progression

RPSC Vice Principal ITI 2018
APPSC IOF, 2009
ISRO Scientist/Engineer 2007, ESE 1992

Ans. (c) : The ratio of two successive oscillation amplitude is constant in an under damped system. Natural logarithm of the ratio is called logarithmic decrement

$$\delta = \ln\frac{x_2}{x_1} = \ln\frac{x_3}{x_2}$$

$$x_2^2 = x_1 . x_3$$

Hence, amplitude of successive vibration are in geometric progression.

1197. When shaking force is transmitted through the spring damping becomes detrimental when the ratio of its frequency to the natural frequency is greater than _____

(a) 1.00 (b) 0.50
(c) 0.25 (d) $\sqrt{2}$

UPRVUNL AE 21.08.2016
APPSC AEE 2012, ESE 1996

Ans. : (d) When shaking force is transmitted through the spring damping becomes detrimental when the ratio of its frequency to the natural frequency is greater than $\sqrt{2}$

$$\frac{\omega}{\omega_n} > \sqrt{2}$$

1198. The equation for free vibration of a system if $\ddot{x} + 36\pi^2 x = 0$. Its natural frequency is

(a) 6 Hz (b) 3π Hz
(c) 3 Hz (d) 6π Hz

OPSC AEE 2019 Paper-I
TRB Asstt. Prof., 2012, ESE 1995

Ans. (c) : $\ddot{x} + 36\pi^2 x = 0$

$$\omega_n^2 = 36\pi^2$$

$$\boxed{\omega_n = 6\pi}$$

$$\omega_n = \frac{6\pi}{2\pi} = 3\text{Hz}$$

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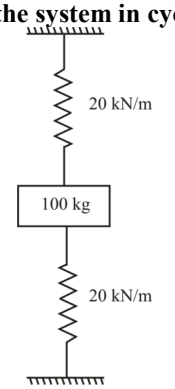
1199. As shown in the figure a mass of 100 kg is held between two springs. The natural frequency of vibration of the system in cycle is :



(a) $1/2\pi$ (b) $5/\pi$
(c) $10/\pi$ (d) $20/\pi$

GPSC ARTO Pre 30.12.2018
ISRO Scientist/Engineer 11.10.2015
ISRO Scientist/Engineer 2006

Ans. (c) : Given,
$k_1 = k_2 = 20$ kN/m
Spring are connected in parallel combination
So, $k_{eq} = k_1 + k_2 = 40$ kN/m = 40000 N/m

Natural frequency, $f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{40000}{100}}$$

$$f_n = \frac{1}{2\pi}\times 20 = \frac{10}{\pi} \text{ cycles}$$

1200. Critical damping is the :
(a) largest amount of damping for which no oscillation occurs in free vibration
(b) smallest amount of damping for which no oscillation occur in free vibration
(c) largest amount of damping for which the motion is simple harmonic in free vibration
(d) smallest amount of damping for which the motion is simple harmonic in free vibration

Assam Engg. College AP/Lect. 18.01.2021
GPSC ARTO Pre 30.12.2018
CGPSC Poly. Lect. 22.05.2016
GATE 2014

Ans. (b) : Critically damped system never executes a cycle, it approaches equilibrium with exponentially decaying displacement, because the system returns to equilibrium in shortest time without any oscillations and in critically damped free vibrations, the damping force is just sufficient to dissipate energy within one cycle of motion.
The value of damping factor (ξ) is 1.
eg. AK 47 gun.

1201. The equation of motion for a single degree of freedom system with viscous damping is $4\ddot{x} + 9\dot{x} + 16x = 0$. The damping ratio is :
(a) 9/128 (b) 9/16
(c) $9/8\sqrt{2}$ (d) 9/8

APGENCO AE 2012
ESE 1996

Ans. (b) : Given, $4\ddot{x} + 9\dot{x} + 16x = 0$

comparison by $m\ddot{x} + c\dot{x} + kx = 0$
$m = 4, \quad c = 9, \quad k = 16$

damping ratio $(\xi) = \frac{C}{2\sqrt{mk}}$

damping ratio $(\xi) = \frac{9}{2\sqrt{4\times 16}}$

$$= \frac{9}{2\times 8} = \frac{9}{16}$$

$$= \frac{9}{16}$$

1202. The maximum velocity of a body in SHM with amplitude of 150mm and a frequency of 2 vibrations /sec is equal to _____m/sec.
(a) 0.01885 (b) 0.1885
(c) 1.885 (d) 18.55

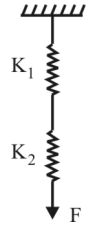
WBPSC AE, 2017
MPSC Poly. Lect. 2014

Ans. (c) : Given, Amplitude (A) = 150 mm
frequency (f) = 2 vibrations/sec.
= 2Hz
maximum velocity (v) = $A\omega$
$= 0.150 \times 2\pi f$
$= 0.150 \times 2\pi \times 2$
$\boxed{v = 1.8849 \text{ m/s}}$

1203. For two springs having same stiffness (K) in series, the equivalent stiffness would be
(a) K^2 (b) $\sqrt{K}$
(c) $\frac{K}{2}$ (d) 2K

ISRO Scientist/Engg. 2010
WBPSC AE 2003

Ans. (c) : Spring stiffness-
(i) When connected in series :



$$\frac{1}{K_{eq}} = \frac{1}{K_1} + \frac{1}{K_2}$$

If $K_1 = K_2 = K$ (given)

then $\frac{1}{Keq} = \frac{2}{K} \Rightarrow \boxed{K_{eq} = K/2}$

(ii) When spring connected in parallel :

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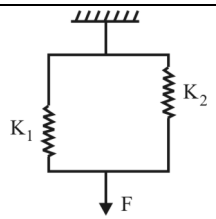
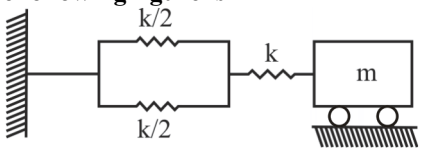


$K_{eq} = K_1 + K_2$

$\boxed{K_{eq} = 2K}$

1204. The natural frequency of the system shown in the following figure is



(a) $\sqrt{k/2m}$ (b) $\sqrt{k/m}$

(c) $\sqrt{2k/m}$ (d) $\sqrt{3k/m}$

TSPSC AEE 2017
GATE 2007

Ans. (a) : $\omega_n = \sqrt{\frac{k_{eq}}{m}}$

So, $k_{parallel} = \frac{k}{2} + \frac{k}{2} = k$

Now, $k_{parallel} = k$ and k is series

$\frac{1}{k_{eq}} = \frac{1}{k} + \frac{1}{k} = \frac{2}{k}$

$k_{eq} = \frac{k}{2}$

1205. The danger of breakage and vibration is maximum :

(a) Below critical speed
(b) Near critical speed
(c) Above critical speed
(d) None of the above

PSPCL AE, 2012, VIZAG STEEL MT 2011

Ans. (b) : Near critical speed – At critical speed, the resonance occurs at the shaft which means actual and natural frequency are equal. at resonance ($\omega=\omega_n$) vibration is maximum, there will be a danger of breakage.

1206. In steady state forced vibration, the amplitude of vibration at resonance is —— damping coefficient.

(a) Equal to
(b) Directly proportional to
(c) Inversely proportional to
(d) Independent of

APGCL AM, 2021
JPSC AE 2013, Paper-V

Ans. (c) :

$$\text{Amplitude (A)} = \frac{F_o/s}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)+\left(\frac{2\xi\omega}{\omega_n}\right)^2}}$$

at resonance condition $\frac{\omega}{\omega_n} = 1$

$$A = \frac{fo/s}{2\xi}$$

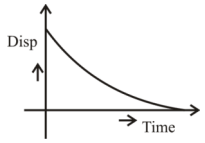
$$\boxed{A \propto \frac{1}{\xi}}$$

1207. When the mass of a critically damped single degree of freedom system is deflected from its equilibrium position and released it will :

(a) return to equilibrium position without oscillation
(b) oscillate with increasing time period
(c) oscillate with decreasing amplitude
(d) oscillate with constant amplitude

APPSC IOF, 2009, ESE 1996

Ans. (a) : When a mass of a critically damped single degree of freedom system is deflected from its equilibrium position & released then it will return to equilibrium position without oscillation.



1208. The critical damping coefficient C_e of a vibrating system of mass m and natural frequency ω_n is

(a) $m\omega_n$ (b) $\frac{\omega_n}{m}$

(c) $2m\omega_n$ (d) $\frac{2\omega_n}{m}$

GPSC Asstt. Director of Transport 05.03.2017
WBPSC AE, 2007

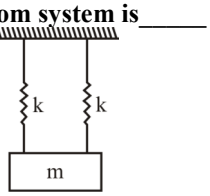
Ans. (c)

$$2\xi\omega_n = \frac{C}{m}$$

for $\xi = 1 \Rightarrow C = C_c$

$\boxed{C_c = 2\ m\omega_n}$

1209. A mass m is attached to two identical springs having spring constant k as shown in the figure. The natural frequency ω of this single degree of freedom system is_____



(a) $\sqrt{\frac{k}{m}}$ (b) $\sqrt{\frac{2k}{m}}$

(c) $\sqrt{\frac{k}{2m}}$ (d) $\sqrt{\frac{4k}{m}}$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2017

Ans. (b) : $\sqrt{\frac{2k}{m}}$

Springs are connected in parallel

then, $k_{eq} = k_1 + k_2$

$= k + k$

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$k_{eq} = 2k$

Natural frequency

$$\omega_n = \sqrt{\frac{k_{eq}}{m}} = \sqrt{\frac{2k}{m}}$$

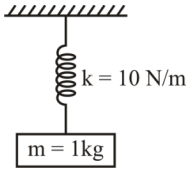
1210. A single degree of freedom system having mass 1 kg and stiffness 10 kN/m initially at rest is subjected to an impulse force of magnitude 5 kN for 10^{-4} seconds. The amplitude in mm of the resulting free vibration is_____.

(a) 0.5 (b) 1.0

(c) 5.0 (d) 10.0

Assam Engg. College AP/Lect. 18.01.2021
GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (c) :



At t = 0, mass is at rest, F = 5 kN for 10^{-4} seconds & amplitude = x

$$F = m \times \frac{dv}{dt}$$

$$F\int_{t_1}^{t_2} dt = m\int_{v_1}^{v_2} dv$$

$$F(t_2 - t_1) = m \times (v_2 - v_1)$$

$$5 \times 10^3 \times 10^{-4} = 1 \times (v_2 - 0)$$

$$v_2 = 0.5 \text{ m/s}$$

$$\frac{1}{2}mv_2^2 = \frac{1}{2}kx^2$$

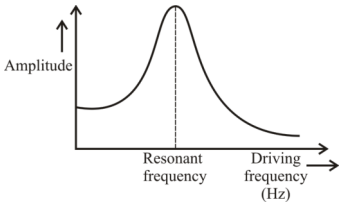
$$\Rightarrow \quad x = \sqrt{\frac{mv_2^2}{k}} = \sqrt{\frac{1\times 0.5\times 0.5}{10\times 1000}} = 5\,\text{mm}$$

1211. At resonance, the amplitude of vibration is

(a) very large
(b) small
(c) zero
(d) depends upon frequency

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Gujarat PSC AE 2019

Ans. (a) : At resonance, the amplitude of vibration is very large.



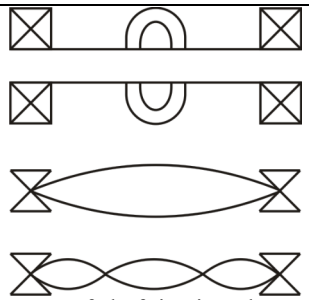
1212. If two nodes are observed at a frequency of 1800 rpm during whirling of a simply supported long slender rotating shaft, the first critical speed of the shaft in rpm is

(a) 200 (b) 450

(c) 600 (d) 900

GPSC Asstt. Director of Transport 05.03.2017
HPPSC Asstt. Prof. 18.11.2016

Ans. (a) :



Whirling frequency of shaft is given by,

$$f = \frac{\pi}{2} \times n^2 \sqrt{\frac{gEI}{wL^4}}$$

n = no. of loops

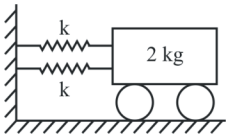
$$f_1 = \frac{\pi}{2}\sqrt{\frac{gEI}{wL^4}}$$

$$f_3 = 3^2 f_1 = 1800 \text{ rpm}$$

[2 nodes represents 3 loop]

$$f_1 = \frac{1800}{9} = 200\,\text{rpm}$$

1213. A mass of 2 kg is attached to two identical springs each with stiffness k = 40 kN/m as shown in the figure. Under frictionless condition, the natural frequency of the system in Hz is close to



(a) 32 (b) 16

(c) 24 (d) 200

ISRO Scientist/Engineer 07.05.2017
GATE 2011

Ans. (a) : $k_{eq} = k + k = 2k$

$$f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k_{eq}}{m}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{2\times 40\times 10^3}{2}} = \frac{1}{2\pi}\sqrt{4\times 10^4}$$

$$= \frac{2\times 10^2}{2\pi} = \frac{100}{\pi}$$

$$= 31.83$$

$$\simeq 32$$

1214. Damping is beneficial only when :

(a) $\frac{\omega}{\omega_n} = 1$ (b) $\frac{\omega}{\omega_n} < 1$

(c) $\frac{\omega}{\omega_n} < \sqrt{2}$ (d) $\frac{\omega}{\omega_n} > \sqrt{2}$ b

TRB Polytechnic Lecturer 2017
GATE 1992

Ans. (c) : Damping is beneficial upto a ratio of $\frac{\omega}{\omega_n} < \sqrt{2}$, ($\in$ will $\uparrow$, $\in > 1$)

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whereas no need of damping less damping. If ratio of $\frac{\omega}{\omega_n} > \sqrt{2}$, ($\in$ will $\downarrow$, $\in < 1$)

Note : No damping is best ($\in = 0$)

1215. For a vibrating system with viscous damping, the characteristics equation is given as:-

$$M\ddot{x} + c\dot{x} + kx = 0$$

If the roots of the characteristics equation are real and equal, the system is:-

(a) Over damped
(b) Critically damped
(c) Underdamped
(d) Cannot be predicted

UKPSC AE-2013, Paper-I
WBPSC AE 2008

Ans. (b) : The characteristics equation given as -

$M\ddot{x} + c\dot{x} + kx = 0$

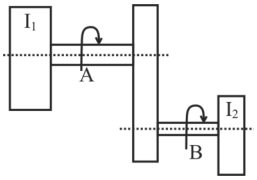
Then the roots of this equation will be

$$\alpha_1, \alpha_2 = \frac{-\frac{c}{m} \pm \sqrt{\left(\frac{c}{m}\right)^2 + 4\omega_n^2}}{2}$$

$$\alpha_1, \alpha_2 = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 + \omega_n^2}$$

If the roots of the characteristics equation are real and equal, the system will be critically damped.

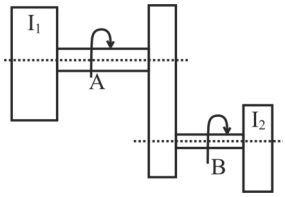
1216. A torsional system with discs of moment of inertia I_1 and I_2 are shown in figure, which are gear driven. The ratio of speed of shaft B to shaft A is G. The equivalent moment of inertia of disc on shaft B at the speed of shaft 'A' is equal to



(a) GI_2 (b) G^2I_2
(c) I_2/G (d) I_2/G^2

UPPSC AE 12.04.2016 Paper-I
ESE 1995

Ans : (b)



$$I'_B = I_B(\text{on } B) \times \left(\frac{\omega_B}{\omega_A}\right)^2$$

$$\boxed{I'_B = G^2.I_2}$$

1217. In a spring mass system, the mass is m and the spring constant is k. The critical damping coefficient of system is 0.1 kg/sec, in another spring mass system, the mass is 2m and the spring constant is 8k. the critical damping coefficient (in kg/sec) of this system is–

(a) 4 x 0.1 (b) 4.5 x 0.1
(c) 3.8 x 0.1 (d) 5.2 x 0.1

GPSC Executive Engineer 23.12.2018
RPSC 2016

Ans : (a)

Case – 1

Mass of the system (m) = m

spring constant = k

critical damping coefficient $C_{C_1} = 0.1$ kg/s

Case – 2

Mass of the system (m) = 2m

spring constant = 8 k

Critical damping coefficient $C_{C_2} = 2\sqrt{km}$

The ratio of critical damping coefficient of two system is,

$$\frac{C_{C_2}}{C_{C_1}} = \frac{2\sqrt{8k \times 2m}}{2\sqrt{km}}$$

$$\frac{C_{C_2}}{C_{C_1}} = \frac{2\sqrt{8k \times 2m}}{2\sqrt{km}} = 0.4\,\text{kg/s}$$

$$\boxed{C_{C_2} = 0.4\,\text{kg/s}}$$

1218. The equation of motion for a damped vibration is given by $6\ddot{x} + 9\dot{x} + 27x = 0$. The damping factor will be

(a) 0.25 (b) 0.5
(c) 0.35 (d) 0.75

ISRO Scientist/Engineer 12.05.2013
ESE 2000

Ans. (c) : Equation of Damped vibration

$6\ddot{x} + 9\dot{x} + 27x = 0$

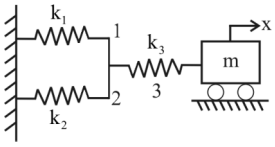
m = 6

C = 9

k = 27

$$\xi = \frac{C}{C_c} = \frac{C}{2\sqrt{km}} = \frac{9}{2\sqrt{27 \times 6}} = 0.3535$$

1219.



Which one of the following is the correct value of the natural frequency (ω_n) of the system given above ?

(a) $\left[\frac{1}{\left\{\frac{1}{k_1 + k_2} + \frac{1}{k_3}\right\} m}\right]^{\frac{1}{2}}$

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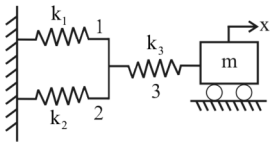
(b) $\left(\frac{3k}{m}\right)^{\frac{1}{2}}$

(c) $\left(\frac{k}{2m}\right)^{\frac{1}{2}}$

(d) $\left[\frac{k_3+\left[\frac{1}{\frac{1}{k_3}+\frac{1}{k_2}}\right]}{m}\right]^{\frac{1}{2}}$

MPPSC AE 2016, ESE 2005

Ans : (a)



$$\frac{1}{K_{eq}}=\frac{1}{K_1+K_2}+\frac{1}{K_3}$$

$$\omega=\sqrt{\frac{k_{eq}}{m}}=\sqrt{\frac{1}{\left(\frac{1}{K_1+K_2}+\frac{1}{K_3}\right)m.}}$$

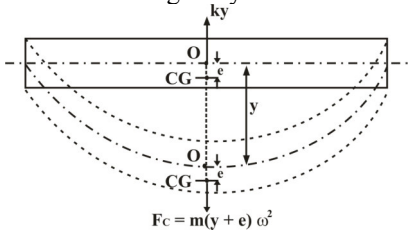
1220. A shaft carries a weight 'w' at the centre. The CG of the weight is displacement by an amount 'e' from the axis of the rotation. If 'y' is the additional displacement of the CG from the axis of rotation due to the centrifugal force, then ratio of 'y' to e (where 'ω_c' is the critical speed of shaft and ω is the angular speed of shaft) is given by

(a) $\frac{1}{\left[\frac{\omega_c}{\omega}\right]^2+1}$ (b) $\frac{1}{\left[\frac{\omega_c}{\omega}\right]^2-1}$

(c) $\left[\frac{\omega_c}{\omega}\right]^2+1$ (d) $\left[\frac{\omega_c}{\omega}\right]^2-1$

MPPSC AE 2016
ESE 2001

Ans : (b) O = Point on axis which the shaft rotates
CG = Centre of gravity



out ward centrifugal force = Resisting force

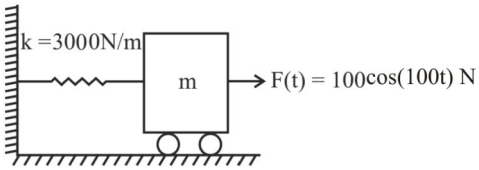
$$m(y+e)\omega^2=Ky$$

$$\frac{y}{e}=\frac{1}{\left(\frac{K}{m\omega^2}-1\right)^2}$$

$$\frac{y}{e}=\frac{1}{\left(\frac{\omega_n}{\omega}\right)^2-1}$$

ω_n = Natural frequency

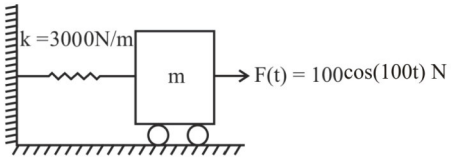
1221. A mass m attached to a spring is subjected to harmonic force as shown in figure below. The amplitude of the force motion is observed to be 50 mm. The value of m (in kg) is



(a) 0.1 (b) 1.0
(c) 0.3 (d) 0.05

TSPSC AEE 2017
GATE 2010

Ans. (a) :



Given: A = 50 mm
F(t) = 100 cos(100t)

$$m\ddot{x}+kx=100\cos(100t)$$

$\omega = 100$,

$\Rightarrow$ $x = A\cos\omega t$

$$\dot{x}=\frac{dx}{dt}=-A\omega\sin\omega t$$

$$\ddot{x}=\frac{d\dot{x}}{dt}=-A\omega^2\cos\omega t$$

$m(-A\omega^2\times\cos\omega t)+k(A\cos\omega t)=100\cos\omega t$

$-m\omega^2A+kA=100$

$k = 3000$N/m

$-m(0.05)(100)^2+3000(0.05)=100$

$$m=\frac{10}{100}=0.1\text{kg}$$

1222. Transmissibility is defined as

(a) Ratio of force applied to the force transmitted to the foundation
(b) Ratio of force transmitted to the foundation to the input force
(c) Sum of the forces applied and the force transmitted to the foundation
(d) Difference of force applied vis-a-vis the force transmitted to the foundation

UPSC JWM 2017

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CSE Pre-2006
Oil India Senior Officer 23.12.2020
APGENCO AE 2012

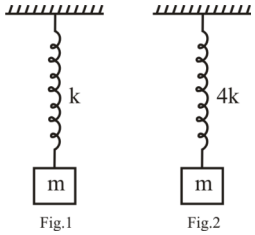
Ans. (b) : Transmissibility factor

$$(\varepsilon) = = \frac{\text{Transmitted}}{\text{Force applied}}$$

$$\boxed{\varepsilon = \frac{F_T}{F_O}}$$

$$\varepsilon = \frac{\sqrt{1+\left(2\xi\frac{\omega}{\omega_n}\right)^2}}{\sqrt{\left\{1-\left(\frac{\omega}{\omega_n}\right)^2\right\}^2+\left(2\xi\frac{\omega}{\omega_n}\right)^2}}$$

1223. Two vibratory systems are shown in the given figures. The ratio of the natural frequency of longitudinal vibration of the second system to that of the first is :



(a) 4 (b) 2
(c) 0.5 (d) 0.25

GPSC ARTO Pre 30.12.2018
CSE Pre-1998

Ans. (b) : For fig 1, $\omega_1 = \sqrt{\frac{k}{m}}$ ....(i)

For fig 2, $\omega_2 = \sqrt{\frac{4k}{m}} = 2\sqrt{\frac{k}{m}}$ ....(ii)

from (i) & (ii)

$$\boxed{\frac{\omega_2}{\omega_1} = 2}$$

1224. Large military field guns which come to initial position after firing in shortest possible time are

(a) Critically damped (b) Under damped
(c) Over damped (d) Undamped

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 2011

Ans. (a) : Critical damping is defined as the threshold between over damping & under damping, the oscillator returns to equilibrium position as quickly as possible, without oscillating & passes it once at most same thing happen with heavy guns.

1225. A system has viscous damped output. There is no steady-state lag if input is :

(a) Unit step displacement
(b) Step velocity
(c) Harmonic
(d) Step velocity with error-rate damping

OPSC AEE 2019 Paper-I, ESE 2001

Ans : (a) : Unit step displacement

1226. A mass M is attached to a spring whose upper end is fixed. The mass and stiffness of the spring are m and K resp. The natural frequency of the system would be

(a) $\frac{1}{2\pi}\sqrt{\frac{K}{M+m}}$ (b) $\frac{1}{2\pi}\sqrt{\frac{2K}{M+m}}$
(c) $\frac{1}{2\pi}\sqrt{\frac{3K}{m+3M}}$ (d) $\frac{1}{2\pi}\sqrt{\frac{2K}{m+2M}}$

VIZAG Steel MT 24.01.2021
TNPSC AE 2018

Ans. (c) : $f_n = \frac{1}{2\pi}\sqrt{\frac{K}{\left(M+\frac{m}{3}\right)}}$

$$f_n = \frac{1}{2\pi}\sqrt{\left(\frac{3K}{3M+m}\right)}$$

1227. In under damped vibrating system, if x_1 and x_2 are the successive values of the amplitude on the same side of the mean position, then the logarithmic decrement is equal to

(a) $\frac{x_1}{x_2}$ (b) $\log\left(\frac{x_1}{x_2}\right)$
(c) $\log_e\left(\frac{x_2}{x_1}\right)$ (d) $\log(x_1.x_2)$

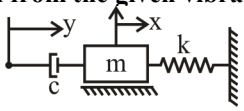
HPCL Engineer 12.08.2021 Shift-I
APPSC AEE MAINS 2016, PAPER-III
WBPSC AE, 2007

Ans. (b) : The logarithmic decrement

$$\delta = \log\frac{x_i}{x_{i+1}} = \log\frac{x_1}{x_2}$$

$$\delta = ln\left(\frac{x_1}{x_2}\right) = \xi\omega_n T_d = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

1228. What will be the differential equation developed from the given vibrating system?



(a) $m(\ddot{x}-\ddot{y})+c(\dot{x}-\dot{y})+k(x-y)=0$
(b) $m\ddot{x}+c(\dot{x}-\dot{y})+kx=0$
(c) $m\ddot{x}+c\dot{x}+k(x-y)=0$
(d) $m(\ddot{x}-\ddot{y})+c(\dot{x}-\dot{y})+kx=0$

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2006

Ans. (b) : Differential equation governing the above vibration system is given by.

$$m\left(\frac{d^2x}{dt^2}\right)+c\left(\frac{dx}{dt}-\frac{dy}{dt}\right)+kx=0$$

$$m\ddot{x}+c(\dot{x}-\dot{y})+kx=0$$

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1229. The equation of motion of a harmonic oscillator is given by $d^2x/dt^2 + 2\zeta\omega_n\, dx/dt + \omega_n^2 x = 0$ and the initial conditions at t = 0 are x(0) = x, dy/dx (0) = 0. The amplitude of x(t) after n complete cycles is

(a) $X\exp\{-2n\pi(\zeta/\sqrt{(1-\zeta^2)})\}$

(b) $X\exp\{2n\pi(\zeta/\sqrt{(1-\zeta^2)})\}$

(c) $X\exp\{-2n\pi(\sqrt{(1-\zeta^2)}/\zeta^2)\}$

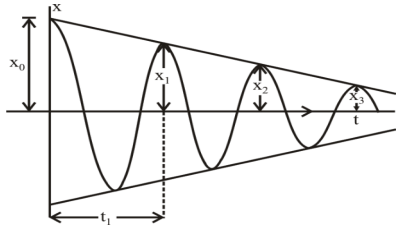
(d) X

Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2007

Ans. (a) : Harmonic oscillator

$$\frac{d^2x}{dt^2} + 2\xi\omega_n\frac{dx}{dt} + \omega_n^2 x = 0$$

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2 x = 0$$



On comparison with

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$\Rightarrow \quad \ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0$$

We get,

$$2\xi\omega_n = \frac{c}{m} \qquad \frac{k}{m} = \omega_n^2$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\xi = \frac{c}{2m\sqrt{k/m}} = \frac{c}{2\sqrt{mk}}$$

$$\text{Logarithmic decrement} = \ln\left(\frac{X_1}{X_2}\right) = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

$$\frac{X_1}{X_2} = e^{\frac{2\pi\xi}{\sqrt{1-\xi^2}}} = \text{Constant}$$

$$\text{Also, } \frac{X_0}{X_1} = \frac{X_1}{X_2} = \frac{X_2}{X_3} = ... = \frac{X_{n-1}}{X_n} = \exp\left(\frac{2\pi\xi}{\sqrt{1-\xi^2}}\right)$$

For n complete cycle in time t_1 we have,

$$\frac{X_n}{X} = \frac{X_n}{X_{n-1}}\cdot\frac{X_{n-1}}{X_{n-2}}.....\frac{X_3}{X_2}\cdot\frac{X_2}{X_1}\cdot\frac{X_1}{X}$$

$$\frac{X_n}{X} = \exp\left(-\frac{2\pi\xi}{\sqrt{1-\xi^2}}\right)$$

$$\Rightarrow \quad \boxed{X(t) = X\exp\left(-\frac{2\pi\xi n}{\sqrt{1-\xi^2}}\right)}$$

1230. For effective vibration isolation, the frequency ratio $\frac{\omega_n}{\omega}$ must be:

(a) equal to $\frac{1}{\sqrt{2}}$ (b) more than $\frac{1}{\sqrt{2}}$

(c) less than $\frac{1}{\sqrt{2}}$ (d) less than $\frac{5}{\sqrt{2}}$

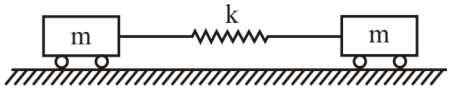
AAI Jr. Executive 26.03.2021
APPSC AEE 2012

Ans. (c) : For effective vibration isolation,

$$\frac{\omega}{\omega_n} > \sqrt{2}$$

$$\boxed{\frac{\omega_n}{\omega} < \frac{1}{\sqrt{2}}}$$

1231. Consider the system of two wagons shown in the figure. The natural frequencies of this system are



(a) $0, \frac{\sqrt{2k}}{m}$ (b) $\frac{\sqrt{k}}{m}, \frac{\sqrt{2k}}{m}$

(c) $\frac{\sqrt{k}}{m}, \frac{\sqrt{k}}{2m}$ (d) $0, \frac{\sqrt{k}}{2m}$

VIZAG STEEL MT 18.08.2013
GATE 1999

Ans. (a) :

$$0, \frac{\sqrt{2k}}{m}$$

1232. When elastic vibrates without external forces owing to inherent forces, it is called _____ vibrations

(a) forced (b) internal

(c) free (d) external

UPRVUNL AE 05.07.2021

Ans. (c) : When elastic vibrates without external forces owing to inherent forces, it is called free vibrations.

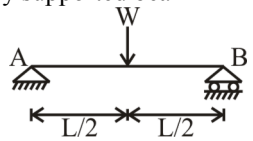
1233. The natural frequency of a simply supported beam of length l with mass M at its centre, flexural rigidity EI and negligible beam mass is

(a) $\frac{1}{2\pi}\sqrt{\frac{48EI}{Ml^3}}$ (b) $\frac{1}{2\pi}\sqrt{\frac{3EI}{Ml^3}}$

(c) $\frac{1}{2\pi}\sqrt{\frac{EI}{Ml^3}}$ (d) None of the above

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : Simply supported beam



Here static deflection

$$\Delta = \frac{W\ell^3}{48EI} = \frac{Mg\ell^3}{48EI}$$

Circular natural frequency

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$$\omega_n = \sqrt{\frac{g}{\Delta}} = \sqrt{\frac{g}{\frac{Mg\ell^3}{48EI}}}$$

$$\omega_n = \sqrt{\frac{48EI}{M\ell^3}}$$

Linear natural frequency

$$f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{48EI}{M\ell^3}}$$

1234. A rotary system having damping coefficient of 50 N/m/s rad, when twisted with an angular velocity of 2 rad/s will experience the damping torque of _____.

(a) 100 Nm (b) 25 Nm
(c) 200 Nm (d) 50 Nm

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : damping torque

$$Td = k_v.\ \frac{d\theta}{dt}$$

$= k_v.\ \omega$
$= 50 \times 2 = 100$ N-m

Where, $k_v = C$
= damping coefficient

1235. What is the critical speed of the shaft if its natural frequency of transverse vibration is 2.85 Hz?

(a) 171 rpm (b) 285 rpm
(c) 570 rpm (d) 142.5 rpm

ESE 18.07.2021

Ans. (a) :
The critical speed of the shaft is revolution per second is equal to the natural frequency of transverse vibration is Hz.

Now, f = 2.85 Hz
$\therefore$ Nc = 2.85 rps
N = 60 × Nc = (60 × 2.85) = 171 RPM

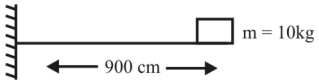
1236. A steel cantilever beam of length 900 cm has moment of inertia 0.0002m^4. A mass of 10 kg is attached to the free end of the beam. If the mass is displaced slightly and released. The natural frequency of the system. Consider E = 50 GPa.

(a) 2028.60 rad/sec (b) 256.60 rad/sec
(c) 202.86 rad/sec (d) 64.15 rad/sec

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Given, I = 0.0002m^4

E = 50 GPa = 50 × 10^{3}Pa



$$\omega_n = \sqrt{\frac{g}{\delta}}$$

$$= \sqrt{\frac{g}{\frac{mgl^3}{3EI}}}$$

$$\omega_n = \sqrt{\frac{3EI}{ml^3}}$$

$$= \sqrt{\frac{3 \times 50 \times 10^9 \times 0.0002}{10 \times 9^3}}$$

$$\omega_n = 64.15\,\text{rad/sec.}$$

1237. A cantilever shaft 50mm diameter and 300 mm long has a disc of mass 100 kg at its free end. The Young's modulus for the shaft material is 200 GN/m^2. The frequency of longitudinal vibrations of the shaft will be :

(a) 57.5 Hz (b) 575 Hz
(c) 5.75 Hz (d) 41 Hz

RPSC ACF & FRO, 26.02.2021

Ans. (b) : Given,
d = 50mm = 0.05m
l = 300mm
m = 100 kg
E = 200 GPa = 200 × 10^9 Pa

$$I = \frac{\pi}{64}d^4$$

$$= \frac{\pi}{64} \times (0.05)^4$$

$$I = 3.067 \times 10^{-7}\,m^4$$

$$A = \frac{\pi}{4} \times d^2$$

$$= \frac{\pi}{4} \times 0.05^2$$

$$= 1.96 \times 10^{-3}\,m^2$$

$$\delta = \frac{Wl}{AE} = \frac{100 \times 9.81 \times 0.3}{1.96 \times 10^{-3} \times 200 \times 10^9} = 7.49428 \times 10^{-7}\,m$$

Frequency of longitudinal vibrations.

$$f_n = \frac{0.4985}{\sqrt{\delta}} = 575.83\,Hz$$

1238. The function of the shock absorber is :

(a) to absorb the energy
(b) to release the energy
(c) to dissipate the energy
(d) to generate the energy

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (a) : A shock absorber or damper is a mechanical or hydraulic device designed to absorb damp shock impulse. It does this by converting the kinetic energy of the shock into another form of energy (typically heat) which is then dissipated. Most shock absorbers are a form of dashpot.

1239. In a vibrating system the spring has stiffness 32 N/m and the mass 2 kg. The system is having a damper whose coefficient of viscous damping is 8 N-s/m. The system is:

(a) Over damped system
(b) Under damped system
(c) Critical damped system
(d) undamped system

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) :

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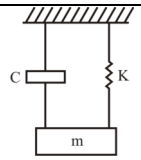


For general eq^{n}

$2\ddot{x}+8\dot{x}+32x=0$

$m\ddot{x}+c\dot{x}+kx=0$

m = 2kg, C = 8 Ns/m, K = 32 N/m

$\xi=\frac{C}{2\sqrt{mK}}=\frac{8}{2\sqrt{2\times32}}=\frac{8}{2\times8}=0.5$

$\xi<1$, So system is under damped

1240. The damping ratio for a viscously damped spring mass system, governed by the relationship

$$m\frac{d^2x}{dt^2}+C\frac{dx}{dt}+kx=F(t)$$ **is given by**

(a) $\sqrt{\frac{c}{mk}}$ (b) $\frac{c}{\sqrt{km}}$

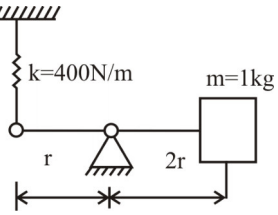
(c) $\frac{c}{2\sqrt{km}}$ (d) $\sqrt{\frac{c}{2mk}}$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2017

Ans. (c) : $2\xi\omega_n=\frac{c}{m}$

$$\boxed{\xi=\frac{c}{2\sqrt{km}}}$$

1241. Considering massless rigid rod and small oscillations, the natural frequency (in rad/s) of vibration of the system shown in the figure is

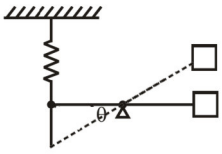


(a) $\sqrt{\frac{400}{1}}$ (b) $\sqrt{\frac{400}{4}}$

(c) $\sqrt{\frac{400}{3}}$ (d) $\sqrt{\frac{400}{2}}$

Assam Engg. College AP/Lect. 18.01.2021
GATE 2015

Ans. (b) :



$\Sigma M=0$

$k(r\theta)r+m(2r\ddot{\theta})\times2r=0$

$4m\ddot{\theta}+k\theta=0$

$\ddot{\theta}+\frac{k}{4m}\theta=0$

Compare above equation from

$\ddot{\theta}+\omega_n^{\,2}\theta=0$

$\omega_n=\sqrt{\frac{k}{4m}}=\sqrt{\frac{400}{4\times1}}=\sqrt{\frac{400}{4}}$

1242. In viscous damping system, $m\ddot{x}+c\dot{x}+sx=0$, over damping occurs when.

(a) $\frac{c^2}{2m}>\frac{s}{m}$ (b) $\frac{c^2}{2m}<\frac{s}{m}$

(c) $\frac{c^2}{4m^2}>\frac{s}{m}$ (d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : $m\ddot{x}+c\dot{x}+sx=0$

For over damping $\xi>1$

$\xi=\frac{C}{C_c}=\frac{C}{2\sqrt{sm}}$

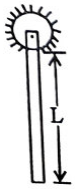
$\frac{C}{2\sqrt{sm}}>1$

Squaring both side.

$\frac{C^2}{4sm}>1$

$$\boxed{\frac{C^2}{4m^2}>\frac{s}{m}}$$

1243. A thin uniform rod of length L and mass M is free to rotate in vertical plane as shown in figure below. The time period of its oscillation in vertical plane is

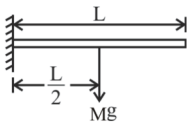


(a) $T=2\pi\sqrt{\frac{L}{2g}}$ (b) $T=2\pi\sqrt{\frac{2L}{3g}}$

(c) $T=2\pi\sqrt{\frac{L}{g}}$ (d) $T=2\pi\sqrt{\frac{3L}{4g}}$

UPPSC AE 13.12.2020, Paper-I

Ans. (b) :



Torque $(\tau)=Mg\times\frac{L}{2}$

Moment of inertia $(I)=\frac{ML^2}{3}$

Time period $(T)=2\pi\sqrt{\frac{I}{\tau}}$

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$$= 2\pi\sqrt{\frac{ML^2/3}{MgL/2}} = 2\pi\sqrt{\frac{2L}{3g}}$$

$$T = 2\pi\sqrt{\frac{2L}{3g}}$$

1244. Smallest and largest natural frequency of a n degree freedom system are ω_1 and ω_n respectively. Approximate natural frequency estimated by Rayleigh's and Dankerley's methods are ω_r and ω_d respectively. Which of the following statements is true?

(a) $\omega_r < \omega_1$ and $\omega_d < \omega_1$
(b) $\omega_r < \omega_1$ and $\omega_d > \omega_1$
(c) $\omega_r > \omega_1$ and $\omega_d > \omega_1$
(d) $\omega_r > \omega_1$ and $\omega_d < \omega_1$

UPPSC AE 13.12.2020, Paper-I

Ans. (d) :
- Dunkerley's method (ω_d) (Approximate results which are less than the actual natural frequency of the system).
- Reyleigh Method (ω_r) or Energy method (Gives accurate results that are slightly greater than the actual natural frequency of the system).

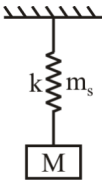
Given that the smallest natural frequency of the system is ω_1, therefore as per the definition of two methods mentioned, $\omega_r > \omega_1$ and $\omega_d < \omega_1$.

1245. The effect of the mass of spring can be considered for calculating natural frequency of a spring mass system by adding 'n' times the mass of spring to the main mass. The value of 'n' is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

UPPSC AE 13.12.2020, Paper-I

Ans. (b) :



Where, M is suspended mass m_s is spring mass
Natural frequency of system

$$\omega_n = \sqrt{\frac{k}{\left(M + \frac{m_s}{3}\right)}}$$

$$= \sqrt{\frac{k}{(M + nm_s)}}$$

Where, $n = \frac{1}{3}$

1246. How does the effect of the inertia constraints in longitudinal free vibration affect the natural frequency of the system?

(a) Inertia constraint does not have any effect on natural frequency of the system
(b) Inertia constraint increases the value of natural frequency of the system
(c) Inertia constraint decreases the value of natural frequency of the system
(d) Inertia constraint increases the value of natural frequency of the system by two times

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : The effect of the inertia constraints in longitudinal free vibration affect the natural frequency of the system inertia constraint decreases the value of natural frequency of the system.

1247. Which of following options defines resonance in the system?

(a) When natural frequency coincides with that of the operating frequency of the system
(b) When stiffness of the structure coincides with that of the stiffness of the system
(c) When mass of the structure coincides with that of the mass of the system
(d) When the frequency of the surrounding noise coincides with that of the noise of the operating system

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Ans. (a) : Resonance occurs when the natural frequency of a system coincides with any expected force vibration frequencies.

1248. The free vibration in the form of twisted and untwisted of the shaft about an axis is known as :

(a) damped free longitudinal vibration
(b) damped longitudinal vibration
(c) free torsional vibration
(d) damped torsional vibration

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Ans. (c) : Free torsional vibration :– When the particle of a shaft or disc in a circle about the axis of a shaft then the vibrations are known as torsional vibration the shaft is twisted and unit wisted alternately and torsional shear stress are induced in the shaft.
Natural frequency of free torsional vibration

$$f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{q}{I}}$$

where
q = Torsional stiffness
I = Mass moment of Inertia of disc

1249. A shaft of span 1 m and diameter 25 mm is simply supported at the ends. It carriers a 1.5 kN concentrated load at mid-span. If E is 200 GPa, its fundamental frequency will be nearly

(a) 3.5 Hz (b) 4.2 Hz
(c) 4.8 Hz (d) 5.5 Hz

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Ans. (d) : Frequency (f)$= \frac{\omega_n}{2\pi}$

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$$

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$$= \frac{1}{2\pi}\sqrt{\frac{g}{\frac{w\ell^3}{48EI}}}, \qquad \left\{\delta = \frac{w\ell^3}{48EI} \text{ for SSB}\right\}$$

$$= \frac{1}{2\pi}\sqrt{\frac{9810\times 48\times 200\times 10^3\times \pi\times 25^4}{1.5\times 10^3\times 1000^3\times 64}}$$

$\boxed{f = 5.5 \text{ Hz}}$

1250. A vibrating system consists of mass of 50 kg, a spring with a stiffness of 30 kN/m and a damper. If damping is 20% of the critical value, the natural frequency of damped vibrations will be

(a) 16 rad/s (b) 20 rad/s
(c) 24 rad/s (d) 28 rad/s

ESE 2020

Ans. (c) : Given, m = 50 kg
K = 30 kN/m = 30000 N/m
$\xi = 0.20$

$$\omega_n = \sqrt{\frac{K}{m}}$$

$$= \sqrt{\frac{30000}{50}} = 24.49 \text{ rad/sec}$$

$$\frac{\omega_d}{\omega_n} = \sqrt{1-\xi^2}$$

$$\omega_d = \sqrt{1-(0.20)^2}\times 24.49$$

$$= 24.48 \text{ rad/sec}$$

$$\approx 24 \text{ rad/sec}$$

1251. A refrigerator unit having a mass of 35 kg is to be supported on three springs, each having spring stiffness s. The unit operates at 480 rpm. If only 10% of the shaking force is allowed to transmit to the supporting structure, the value of stiffness will be nearly

(a) 2.7 N/mm (b) 3.2 N/mm
(c) 3.7 N/mm (d) 4.2 N/mm

ESE 2020

Ans. (a) : m= 35 kg, N= 480 rpm
Stiffness = k

$$\omega = \frac{2\pi N}{60} = \frac{2\pi\times 480}{60} = 50.27 \text{ rad/s}$$

Transmissibility, $\epsilon_T = 0.10 = \frac{1}{\pm(1-r^2)}$

{When $\xi = 0$ }
Consider negative sign

$10 = -1 + r^2$
$r^2 = 11$
$r = \sqrt{11}$

$$\frac{\omega}{\omega_n} = 3.317$$

$$\omega_n = \frac{50.27}{3.317}$$

$$\sqrt{\frac{k_{eq}}{m}} = 15.16$$

$k_{eq} = 15.16^2\times 35 = 8043.89$ N/m

$$k = \frac{k_{eq}}{3} = 2.68\times 10^3 \text{ N/m}$$

$\boxed{k = 2.7 \text{ N/mm}}$

1252. If the damping coefficient is 10 Ns/m, mass of the body is 5 kg and the angular velocity is 240 rad/min, then damping ratio is :

(a) 0.050 (b) 0.008
(c) 0.004 (d) 0.250

NLC GET 17.11.2020, Shift-II

Ans. (d) : Given, damping coefficient (C) = 10 Ns/m
mass of the body (m) = 5 kg
Angular velocity (ω) = 240 rad/min.

$$\text{damping ratio } (\xi) = \frac{\text{damping coefficient}}{\text{critical damping}}$$

Critical damping coefficient $\rightarrow C_C = 2m\omega_n$

$$= 2\times 5\times \frac{240}{60} = 40$$

$$\text{Damping ratio} = \frac{C}{C_C} = \frac{10}{40} = 0.25$$

$\xi = 0.25$

1253. A simple spring mass system has a natural frequency of N. If the spring stiffness becomes 1/4 and the mass is doubled, then the natural frequency becomes _____.

(a) $\frac{N}{2\sqrt{2}}$ (b) $2\sqrt{2}N$
(c) $\frac{N}{2}$ (d) 2 N

VIZAG MT, 14.12.2020

Ans. (a) : Natural frequency $N = \sqrt{\frac{k}{m}}$

$$N_2 = \sqrt{\frac{k/4}{2m}}$$

$$= \sqrt{\frac{k}{8m}}$$

$$N_2 = \frac{1}{2\sqrt{2}}\sqrt{k/m}$$

$$\boxed{N_2 = \frac{N}{2\sqrt{2}}}$$

1254. When a rotor of mass 50 kg is mounted on a shaft, a bending deflection of 0.1 mm is observed at the place where the rotor is placed. The dangerous speed of the rotor is _____.

(a) 4485 rpm (b) 997 rpm
(c) 1495 rpm (d) 2990 rpm

VIZAG MT, 14.12.2020

Ans. (d) : Given, m = 50 kg
$\delta = 0.1$ mm

$$\omega_n = \omega_c = \sqrt{k/m} = \sqrt{g/\delta}$$

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$$= \sqrt{\frac{9.81}{0.1\times10^{-3}}}$$
$$= 313.209 \text{ rad/sec}$$
$$\omega_n = \frac{2\pi N}{60}$$
$$N = \frac{\omega_n \times 60}{2\pi}$$
$$\boxed{N = 2990.92 \text{ rpm}}$$

1255. Damping capacity of a material is its ability to
(a) absorb shock
(b) absorb vibration
(c) withstand compression
(d) absorb impact load

Assam PSC AE (PHED), 18.10.2020

Ans. (b) : Damping capacity is the ability of a material to absorb vibration by internal friction, converting the mechanical energy into heat.

1256. In a vibratory system, the magnitude of damping force during an oscillation is constant in
(a) Viscous damping (b) Coulomb damping
(c) Structural damping (d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : In coulomb damping the magnitude of damping force remains constant. The difference between static and dynamic friction is usually neglected and the force is assumed to be independent of frequency and displacement.

1257. The response of a vibrator system with viscous damping becomes oscillatory when the system damping becomes.
(a) Underdamped (b) Overdamped
(c) Critically damped (d) Both (B) and (c)

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : For underdamped condition response of a vibratory system become oscillatory.

1258. The whirling of rotating shaft supported in bearings primarily involve
(a) Natural vibration of the shaft
(b) Torsional vibration of the shaft
(c) Axial vibration of the shaft
(d) Lateral vibration of the shaft

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Whirling of rotating shaft supported in bearing vibrate in direction perpendicular to axis and involve lateral vibration of shaft.

1259. Consider two single degree of freedom vibratory systems. If the static deflection of the mass in the case of first system is twice that of the second system, the time period of the undamped vibration of the first system is
(a) Twice that of the second system
(b) Half that of the second system
(c) One-fourth that of the second system
(d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : T = Time period = $2\pi\sqrt{\frac{\delta}{g}}$

If $T_1 = 2\pi\sqrt{\frac{2\delta}{g}}$ and $T_2 = 2\pi\sqrt{\frac{\delta}{g}}$

Then, $\boxed{T_1 = \sqrt{2} \times T_2}$

1260. For a single degree of freedom-mass-damper vibrator system under a constant amplitude harmonic force excitation, the largest steady state vibration response is observed at the frequency that is
(a) Same as the undamped natural frequency of the system
(b) Higher than the undamped natural frequency of the system
(c) Lower than the undamped natural frequency of the system
(d) Dependent on the amplitude of the external force

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Amplitude of damped forced vibration.

$$A = \frac{F_0}{\sqrt{\left(s - m\omega^2\right)^2 + (c\omega)^2}}$$

for undamped force vibrator

$$A = \frac{F_0}{s - m\omega^2}$$

The amplitude will be max if system is undamped and $s - m\omega^2 = 0$ $\omega_n = \omega$

1261. A body of mass 100 kg with natural frequency 10 rad/sec has a critical damping:
(a) 10 N/m/sec (b) 1000 N/m/sec
(c) 3142 N/m/sec (d) 2000 N/m/sec

CIL MT 27.02.2020

Ans. (d) : Given, Mass(m) = 100 kg
Natural frequency $\omega_n = 10$ rad/sec
Critical damping $C_c = 2m\omega_n$
$= 2 \times 100 \times 10$
$= 2000$ N/m/sec

1262. Vibrations of transient in nature occurs in:
(a) Transverse vibrations
(b) Undamped vibrations
(c) Torsional vibrations
(d) Damped vibrations

CIL MT 27.02.2020

Ans. (d) : When there is a reduction in amplitude over every cycle of vibration the motion is said to be damped vibration. This is due to fact that a certain amount of energy possessed by the vibrating system is always dissipated in overcoming frictional resistance to the motion.

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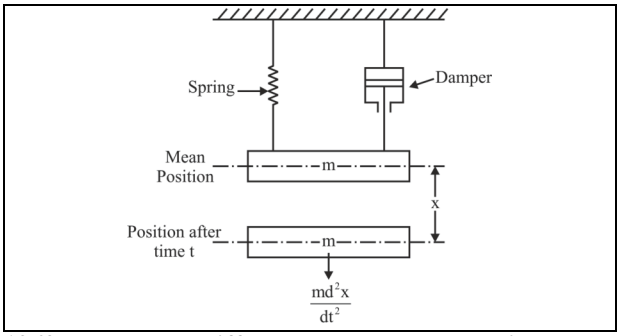


1263. The phase difference between transmitted and disturbing force for a negative isolation factor is:

(a) 90° (b) 120°
(c) 180° (d) 45°

CIL MT 27.02.2020

Ans. (c) : If the isolation factor is negative then the ratio ω/ω_n is greater than 1, in this phase difference between the disturbing force and the transmitted force is 180°.

1264. A vibrating system consists of a mass of 200 kg, a spring of stiffness 80 N/mm and damper with a damping coefficient of 800 Ns/m. Then the nearest circular frequency of damped vibration is:

(a) 10 rad/s (b) 20 rad/s
(c) 15 rad/s (d) 25 rad/s

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Given, m = 200 kg

$s = 80 \text{ N/mm} = 80 \times 10^3 \text{ N/m}$

c = 800 N/m/sec

Natural frequency $\omega_n = \sqrt{\frac{s}{m}} = \sqrt{\frac{80\times10^3}{200}}$

$\omega_n = 20$ rad/sec

And circular frequency of damped vibration

$$\omega_d = \sqrt{\omega_n^2 - a^2}$$

$$= \sqrt{\omega_n^2 - \left(\frac{c}{2m}\right)^2}$$

$$= \sqrt{20^2 - \left(\frac{800}{2\times200}\right)^2}$$

$= 19.9$ rad/sec

$\omega_d \approx 20$ rad/sec

1265. In order to limit resonance amplitude during the vibration isolation:

(a) stops need not to be provided but dampers may be provided
(b) only stops must be provided
(c) only dampers must be provided
(d) dampers need not to be provided but stops may be provided

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : As the damping is a linear function of velocity, the vibration amplitude at resonant frequency can be reduced by increasing the damping. While isolation performance over the whole frequency range could degrade greatly.

During the vibration isolation dampers need not to be provided but stop may be provided.

1266. An overdamped system

(a) does not vibrate at all
(b) vibrates with frequency more than natural frequency of system
(c) vibrates with frequency less than natural frequency of system
(d) vibrate with frequency equal to natural frequency of system

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : An overdemped system does not vibrate at all.

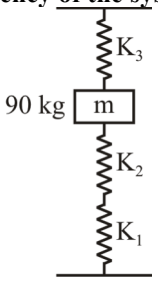
1267. Forced vibration is also known as

(a) steady-state vibration (b) transient vibration
(c) transformed vibration (d) variable vibration

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Forced vibration is also known as steady-state vibration.
When analyzing forced vibrations we always neglect the transient response of the system and calculate only the steady state behavior.

1268. A machine component of 90 kg mass needs to be held in position using three springs as shown below. The spring constants K_1, K_2 and K_3 are 4, 4 and 8 N/m respectively. Find the natural frequency of the system in rad/sec.



(a) 0.33 (b) 0.42
(c) 0.13 (d) 3.0

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : Here k_1 and k_2 are in series where as k_3 is in parallel.

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

k = 2 N/m

Now, Total $k_{eq} = (k + k_3) = 2 + 8 = 10$ N/m

So, Natural frequency,

$$\omega_n = \sqrt{\frac{k_{eq}}{M}} = \sqrt{\frac{10}{90}} = \frac{1}{3}\text{rad/sec}$$

$\omega_n = 0.33$ rad/sec.

1269. A driver of mass 100 kg is standing at the tip of a spring board of negligible mass. The natural frequency of the spring board with the driver is 1.6 Hz. What is the static deflection at the tip of the spring board when the driver is standing at the tip?

(a) 0.1 mm (b) 981 mm
(c) 98.1 mm (d) 9.81 mm

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Given, $f_n = 1.6$ Hz

By Rayleigh's method, if Δ = static deflection, then,

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$$f_n = \frac{1}{2\pi}\sqrt{\frac{g}{\Delta}}$$

$$1.6 = \frac{1}{2\pi}\sqrt{\frac{9.81}{\Delta}}$$

$\Delta = 97.06$ mm

Closest answer is 98.1 mm.

1270. A machine component of natural frequency 20 rad/s is subjected to a base motion from the machine which is harmonic in natural with amplitude 3 m/s^2 at 10 rad/s. What is the peak amplitude of relative displacement of the components if the damping is negligible?

(a) 0.1 mm (b) 1.0 mm
(c) 10.0 mm (d) 100.0 mm

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Given, $\omega_n = 20$ rad/sec

$a_{max} = 3$ m/sec^2 $= \frac{F_0}{m}$

$(F_0 = me\omega^2 = ma_{max})$

$\omega = 10$ rad/sec.

$$\frac{\omega}{\omega_n} = \frac{10}{20} = \frac{1}{2}$$

Amplitude

$$A = \frac{(F_0/s)}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}} \qquad (\xi = 0)$$

$$A = \frac{(F_0/s)}{1-\left(\frac{\omega}{\omega_n}\right)^2} = \frac{(F_0/m)/(s/m)}{1-\left(\frac{\omega}{\omega_n}\right)^2}$$

$$A = \frac{\frac{F_0/m}{\omega_n^2}}{1-\left(\frac{\omega}{\omega_n}\right)^2}$$

$$A = \frac{a_{max}/\omega_n^2}{1-\left(\frac{\omega}{\omega_n}\right)^2} = \frac{3/(20)^2}{1-\left(\frac{1}{2}\right)^2}$$

$$= \frac{3/400}{3/4} = \frac{4}{400} = 0.01\,\text{meter}$$

A = 10 mm

1271. Which one of the following relations with usual notations will hold good in a dynamic vibration absorber system under tuned conditions?

(a) $k_1k_2 = m_1m_2$ (b) $k_1m_2 = m_1k_2$
(c) $k_1m_1 = k_2m_2$ (d) $k_1 + k_2 = m_1 + m_2$

ESE 2019

Ans. (b) : For turning of dynamic vibration absorber,

$$\sqrt{\frac{k_1}{m_1}} = \sqrt{\frac{k_2}{m_2}}$$

i.e. $k_1m_2 = k_2m_1$

1272. The frequency of oscillation is the number of cycles per unit time described by the particle, given by the relation

(a) $f = \frac{\omega}{2\pi}$ (b) $\frac{1}{f} = \frac{\omega}{2\pi}$
(c) $f' = \frac{2\pi r}{T}$ (d) $f' = \frac{2\pi NT}{T\omega}$

ESE 2019

Ans. (a) : f = linear frequency
T = Time period of oscillation
ω = angular frequency

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$f = \frac{\omega}{2\pi}$$

1273. A vertical shaft of 100 mm diameter and 1 m length has its upper end fixed at the top. The other end carries a disc of 5000 N and the modulus of elasticity of the shaft material is 2 × 10^5 N/mm^2. Neglecting of the longitudinal vibrations will be nearly

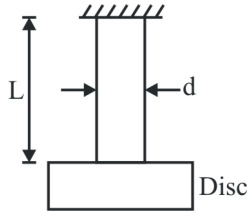
(a) 279.5 Hz (b) 266.5 Hz
(c) 253.5 Hz (d) 241.5 Hz

ESE 2019

Ans. (a) : Given,
d = 100 mm = 0.1 m
L = 1 m
mg = 5000

$$m = \frac{5000}{9.81} = 509.68\ \text{kg}$$

E = 2 × 10^5 N/mm^2



Spring stiffness $k = \frac{AE}{L}$

∴ Natural frequency of longitudinal vibration

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{AE}{mL}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{\frac{\pi}{4}d^2 \times E}{mL}} = \frac{1}{2\pi}\sqrt{\frac{\frac{\pi}{4}\times 0.1^2 \times 2\times 10^5}{509.68\times 1}}$$

= 279.4 Hz

1274. The accurate method of finding the natural frequency of transverse vibration of a system of several loads attached to some shaft is

(a) Dunkerley method
(b) energy method
(c) Stodola method
(d) Dunkerley and energy method

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Ans. (b) : Dunkerley method and stodola methods are numerical methods but approximate energy method gives accurate frequency.

1275. The speed at which the shaft runs, so that the deflection of the shaft from the axis of rotation becomes infinite, is known as

(a) whipping speed (b) damping speed
(c) resonant speed (d) gravitational speed

ESE 2019

Ans. (a) : The speed at which the shaft runs so that the additional deflection of the shaft from the axis of rotation becomes infinite, is known as critical or Whirling or Whipping speed.

1276. For a vibrating system the successive amplitudes of vibration obtained under free effects are 0.70, 0.28, 0.25, 0.23 and 0.067 respectively. The value of damping ratio of the system is given by :

(a) 9.28 (b) 0.0928
(c) 0.00928 (d) 0.928

BHEL ET 2019

Ans. (b) : $\delta = \frac{1}{n}\ln\left(\frac{x_n}{x_{n+1}}\right) = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$

$\Rightarrow \frac{1}{4}\ln\left(\frac{0.70}{0.067}\right) = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$

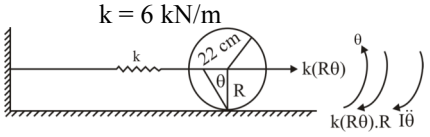
$\boxed{\xi = 0.0928}$

1277. What is the natural frequency of a cylinder having mass 7 kg and radius 22 cm that is connected to a spring of stiffness 6 kN/m at the centre of the cylinder and rolls on a rough surface?

(a) 2.4 Hz (b) 4.8 Hz
(c) 6.4 Hz (d) 3.8 Hz

BHEL ET 2019

Ans. (c) : Given - M = 7kg
R = 22 cm
k = 6 kN/m



$I\ddot{\theta} + k R^2\theta = 0$

$\left(\frac{1}{2}MR^2 + MR^2\right)\ddot{\theta} + kR^2\theta = 0$

$\frac{3}{2}MR^2\ddot{\theta} + kR^2\theta = 0$

$\ddot{\theta} + \frac{k}{\frac{3M}{2}}\theta = 0$

$\omega_n = \sqrt{\frac{2k}{3M}} = 23.9045$

In 'Hz' value

$f_n = \frac{\omega_n}{2\pi} = \frac{23.9045}{2\pi}$

$= 3.8045$ Hz

1278. Magnification factor is maximum at a frequency ratio (r):

(a) $r = \sqrt{1-\xi^2}$ (b) $r = \sqrt{1+2\xi^2}$
(c) $r = \sqrt{1-2\xi}$ (d) $r = \sqrt{1-2\xi^2}$

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Ans. (d) : Magnification-factor (M.F.)- It is defined as the ratio of the amplitude of steady state variation 'X' to the zero frequency defection X_{ST} It is also know as dynamic magnifier.

$$(M.F.) = \frac{1}{\sqrt{\left[1-\left(\frac{\omega}{\omega n}\right)^2\right]^2 + \left[2\xi\frac{\omega}{\omega n}\right]^2}}$$

For max Mf,

$\left[(1-r)^2\right]^2 + (2\xi r)^2$ Should be minimum

$(1-r^2) + (2\xi r)^2$

$1 + r^4 - 2r^2 + 4\xi^2 r^2$

$1 + r^4 + r^2(4\xi^2 - 2)$

For Differentiate max. MF w.r.t r

$4r^3 + 2r(4\xi^2 - 2) = 0$

$2r^2 + 4\xi^2 - 2 = 0$

$r^2 = 1 - 2\xi^2$

$\boxed{r = \sqrt{1-2\xi^2}}$

1279. If single degree of freedom vibratory system has mass = 3kg. stiffness = 100 N/m and damping coefficient = 3 N-s/m then logarithmic decrement is:

(a) 0.46 (b) 0.55
(c) 0.32 (d) 0.65

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Ans. (b) : Given-
For single degree of freedom vibratory system
m = 3kg., k = 100 N/m
C = 3N-s/m

$\therefore \quad \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}}$

$2 \times \xi\, \omega_n = \frac{C}{m}$

$2 \times \xi \times \frac{10}{\sqrt{3}} = \frac{3}{3} = 1$

$\xi = \frac{\sqrt{3}}{20}$

$\delta = \frac{2\pi\xi}{\sqrt{1-\xi^2}} = \frac{2\pi\sqrt{3}}{20\sqrt{1-\frac{3}{(20)^2}}}$

$\delta = 0.546$

1280. A concentrated mass m is attached at centre of a rod of length 2 L as shown in figure. The rod is kept in a horizontal equilibrium position by a spring of stiffness 4 k. For very small amplitude of vibration, neglecting the weight

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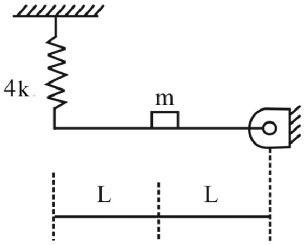
of the rod and spring, the undamped natural frequency (w_n) of the system is:



(a) $\sqrt{(2k/m)}$ (b) $\sqrt{(4k/m)}$

(c) $\sqrt{(8k/m)}$ (d) $\sqrt{(16k/m)}$

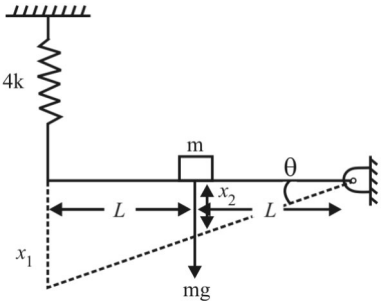
ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (d) : D, Alembert principle

$\Sigma T_{ext} + T_{inertia} = 0$

$4kx_1 \times 2L + mx_2 \times L = 0$

$mx_2 + 4kx_1 \times 2 = 0$



$I\times\alpha+4k\times2L\times2L\theta = 0$

$mL^2\ddot{\theta}+16kL^2\theta = 0$

$m\ddot{\theta} + 16\,k\theta = 0$

$\ddot{\theta}+\frac{16k}{m}\theta = 0$

$\ddot{\theta}+w_n^2\theta = 0$

$w_n^2 = \frac{16k}{m}$

$\boxed{w_n = \sqrt{\frac{16k}{m}}}$

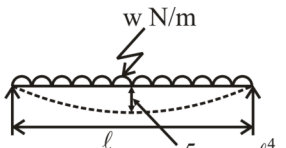
1281. The natural frequency of free transverse vibrations due to uniformly distributed load acting over a simply supported shaft is

(a) $\frac{0.4985}{\sqrt{\delta_s}}$ (b) $\frac{0.5615}{\sqrt{\delta_s}}$

(c) $\frac{0.571}{\sqrt{\delta_s}}$ (d) $\frac{0.6253}{\sqrt{\delta_s}}$

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) :



$\delta = \frac{5}{384}\times\frac{w\ell^4}{EI}$

Natural frequency due to UDL –

$f_n = \frac{\omega}{2\pi} = \frac{\pi^2}{2\pi}\sqrt{\frac{EIg}{w\ell^4}}$

$= \frac{\pi}{2}\sqrt{\frac{EIg}{w\ell^4}}$ .....(1)

Static deflection due to UDL –

$\delta_s = \frac{5w\ell^4}{384EI}$ or $\frac{EI}{w\ell^4} = \frac{5}{384\delta_s}$

So, from equation (1)

$f_n = \frac{\pi}{2}\sqrt{\frac{5g}{384\delta_s}}$

$\boxed{f_n = \frac{0.5615}{\sqrt{\delta_s}}\text{ Hz}}$

1282. The rotating shafts tend to vibrate violently at whirling speeds because

(a) the system is unbalanced

(b) bearing centre line coincides with the axis

(c) the shafts are rotating at very high speeds

(d) resonance is caused due to the heavy mass of the rotor

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Rotating shafts tends to vibrate violently at whirling speeds because resonance is caused due to the heavy weight of motor.

1283. In a vibrating system, if the actual damping coefficient is 40 N/m/s and critical damping coefficient is 420 N/m/s, then logarithmic decrement is equal to

(a) 0.2 (b) 0.4

(c) 0.6 (d) 0.8

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) :

Given, Actual damping coefficient C = 40 N/m/s

Critical damping coefficient C_c = 420 N/m/s

Logarithmic decrement $\delta = \frac{2\pi C}{\sqrt{(C_c)^2 - C^2}}$

$= \frac{2\pi\times40}{\sqrt{(420)^2-(40)^2}}$

$\boxed{\delta = 0.6}$

1284. If X_n = Amplitude at the end of n^{th} oscillation; X_0 = Amplitude at the start of oscillation. Then for underdamped system, which of the following relation is correct? [n = Number of oscillations]

(a) $\frac{X_0}{X_1} = \left(\frac{X_0}{X_n}\right)^n$ (b) $\frac{X_0}{X_1} = \frac{X_0}{X_n}$

(c) $\frac{X_0}{X_1} = \left(\frac{X_0}{X_n}\right)^{1/n}$ (d) $\frac{X_0}{X_1} = \left(\frac{X_0}{X_n}\right)^{1/n}$

(e) $\frac{X_0}{X_1} = \left(\frac{X_0}{X_n}\right)^{n-1}$

CGPSC AE 25.02.2018

Ans. (c) : We know that, for underdamked system,

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$$\frac{x_0}{x_1}=\frac{x_1}{x_2}=\frac{x_2}{x_3}=\frac{x_3}{x_4}=\ldots\ldots=\frac{x_{n-1}}{x_n}$$

then $\frac{x_0}{x_1}\times\frac{x_1}{x_2}\times\frac{x_2}{x_3}\times\frac{x_3}{x_4}\times\ldots\ldots\ldots\frac{x_{n-1}}{x_n}=\left[\frac{x_0}{x_1}\right]^n$

$$\left[\frac{x_0}{x_1}\right]^n=\left[\frac{x_0}{x_n}\right]$$

$$\boxed{\left[\frac{x_0}{x_1}\right]=\left[\frac{x_0}{x_n}\right]^{\frac{1}{n}}}$$

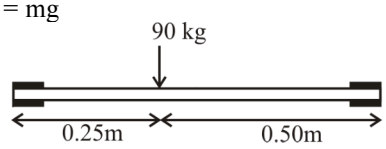
1285. A shaft of length 0.75 m, supported freely at its ends, is carrying a mass 90 kg at 0.25 m. from one end. Find the fundamental frequency of transverse vibration. Take E = 200 GPa shaft dia 0.05 m

(a) 84.95 Hz (b) 84.85 Hz
(c) 48.95 Hz (d) 49.85 Hz

TNPSC AE 2018

Ans. (d) : Data given-

m = 90 kg E = 200 GPa
d = 0.05 m L = 0.75 m
F = mg



$$\delta=\frac{(m\times g).a^2b^2}{3.E.I.L}$$

$$\delta=\frac{90\times9.81\times(0.25)^2\times(0.5)^2}{3\times200\times10^9\times\left(\frac{\pi}{64}\right)\times(0.05)^4\times0.75}$$

$$\delta=9.9974\times10^{-5}\,m$$

Then,

$$f=\frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$$

$$f=\frac{1}{2\pi}\sqrt{\frac{9.81}{9.9974\times10^{-5}}}$$

$$\boxed{f=49.88\ Hz}$$

1286. Which of the following are the basic building block elements for a mechanical system where forces and straight line displacements are involved without any rotation?

1. Spring
2. Dashpot
3. Mass
4. Moment of inertia

Select the correct answer using the code given below.

(a) 1, 2 and 4 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 3

ESE 2018

Ans. (d) : The basic building block elements for a mechanical system where forces and straight line displacements are involved, are

(i) Spring (k) [F = kx]

(ii) Dashpot (B) $\left[F=B\frac{dx}{dt}\right]$

(iii) Mass (m) $\left[F=m\frac{d^2x}{dt^2}\right]$

A translating mechanical system can be modelled as a single degree of freedom by using simple spring-mass-damper system. Mass moment of inertia does not affect a translating (straight line displacement) system.

1287. Which of the following is correctly matched?

(a) Coulomb - Energy principle
(b) Rayleigh - Dynamic equilibrium
(c) D'Alembert - Damping force
(d) Fourier - Frequency domain analysis

TNPSC AE 2018

Ans. (d) : Coulomb method - Retaining wall
Rayleigh method - Energy principle
D'Alembert method - Dynamic equilibrium
Fourier - Frequency domain analysis.

1288. If a spring mass system, the mass is 0.1 kg and the stiffness of the spring is 1 kN/m. By producing a damper, the frequency of oscillation is found to be 90% of the original value. What is the damping coefficient of the damper?

(a) 1.2 Ns/m (b) 3.4 Ns/m
(c) 8.7 Ns/m (d) 12.0 Ns/m

GPSC ARTO Pre 30.12.2018

Ans. (c) : Given,

m = 0.1 kg, $\omega_d = 0.9\,\omega_n$
k = 1 kN/m = 1000 N/m

$$\omega_d=\sqrt{1-\xi^2}\,\omega_n$$

$$0.9\omega_n=\sqrt{1-\xi^2}\,\omega_n$$

$$\xi=0.43$$

$$2\xi\sqrt{km}=C$$

Damping coefficient,

$$C=2\times0.43\ \sqrt{1000\times0.1}\ =8.7\ N-s/m$$

1289. A particle is reciprocating simple harmonic motion (SHM) with an amplitude of 0.003 m and frequency of 20 Hz. The maximum acceleration of the particle attained during the motion is:

(a) 57.3 m/s^2 (b) 50.3 m/s^2
(c) 47.3 m/s^2 (d) 55.3 m/s^2

AAI Jr. Executive 29.11.2018

Ans. (c) : Maximum acceleration (a_{max}) = $A\omega^2$
f = 20 Hz
$a = A(2\pi f)^2$
$a = 0.003\ (2\times3.14\times20)^2$
$a_{max} = 0.003\ (2\times3.14\times20)^2$
$= 47.3\ m/sec^2$

1290. The static deflection of the shaft under flywheel is 25 mm. Assuming g = 10 m/s^2, the critical speed in rad/s will be:

(a) 30 (b) 20
(c) 10 (d) 40

AAI Jr. Executive 29.11.2018

Ans. (b) : Given, $g = 10\ m/sec^2$
$\delta = 25$ mm

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Critical speed $= \sqrt{\frac{g}{\delta}}$

$= \sqrt{\frac{10}{25\times10^{-3}}} = \sqrt{\frac{10\times10^{3}}{25}}$

$= \sqrt{\frac{10000}{25}} = \sqrt{400}$

= 20 rad/sec

1291. If the length of the cantilever beam is reduced to 1/4th, the natural frequency of the mass M at the end of this cantilever beam of negligible mass will be increased by a factor of

(a) 2 (b) $\sqrt{8}$

(c) 4 (d) 8

ISRO Scientist/Engineer 22.04.2018

Ans. (d): We know that

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$$

deflection of cantilever beam with mass M at its end is given by

$$\delta = \frac{Mg\times L^3}{3EI}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{g\times 3EI}{Mg\times L^3}}$$

$$f \propto \frac{1}{L^{3/2}}$$

Then

$$\frac{f_2}{f_1} = \left(\frac{L_1}{L_2}\right)^{3/2} = \left(\frac{L}{L/4}\right)^{3/2}$$

$$f_2 = 8f_1$$

1292. A perfectly balanced rotor shaft supported between short bearings at both ends shows a deflection of 2 mm at middle. Then the critical speed of the shaft in rpm will be nearly,

(a) 335 (b) 670

(c) 2010 (d) 1340

ISRO Scientist/Engineer 22.04.2018

Ans. (b) : We know that

$$\omega_n = \sqrt{\frac{g}{\delta}}$$

$$\omega_n = \sqrt{\frac{9.81}{0.002}}$$

ω_n = 70.035 rad/sec.

Critical speed of the shaft in rpm is given by

$$N_c = \frac{60\times\omega_n}{2\pi}$$

$$N_c = \frac{60\times 70.035}{2\times 3.14}$$

N_c = 669.12 rpm

$N_c \simeq$ 670 rpm

1293. Statement (I): Resonance is a special case of forced vibration in which the natural frequency of the body is the same as the impressed frequency of the external periodic force whereby the amplitude of the forced vibration peaks sharply.

Statement (II): The amplitude of forced vibration of a body increases with increase in the frequency of the externally impressed periodic force.

ESE 2017

Ans. (c) : Resonance is a special case of forced vibrations. When the frequency of an externally applied periodic force on a body is equal to its natural frequency, the body starts vibrating with an increased amplitude.

The amplitude of forced vibrations increases when the frequency of the impressed force approaches the natural frequency of the driven body.

1294. Consider the following statements:

Artefacts to prevent harmful effects resulting from vibrations of an unbalanced machine fixed on its foundation include.

1. mounting the machine on springs thereby minimizing the transmission of forces
2. using vibration isolating materials to prevent or reduce the transmission of forces
3. moving the foundation so as to have only one degree of freedom towards reducing the transmission of forces.

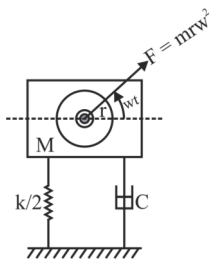
Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only

(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) :



Spring and damper are used between machine and ground/foundation. Appropriate choice of spring and damper characteristics will enable the displacement transmissibility/amplification factor to be below one.

$$A = \left(\frac{mr}{M}\right)\frac{q^2}{(1-q^2)} \text{ if } \xi = 0$$

$$q = \frac{w}{w_n}$$

As k ↑→ w_n ↑, q↓, A↓

1295. A car of mass 1450 Kg is constructed on a chassis supported by four springs. Each spring has a force constant of 40000 Nm. The combined mass of the two people occupying the car is 150 kg. What is the period of execution of two complete vibrations?

(a) 0.63 s (b) 1.59 s

(c) 4.96 s (d) 1.26 s

ESE 2017

Ans. (d) : Total mass (m) = 1450 + 150

= 1600 kg

Total force constant (k) = 4 × 40000

= 160000 N/m

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$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$= 2\pi\sqrt{\frac{1600}{160000}} = 0.628 \text{ sec}$$

Time for two complete oscillation = 2T

$= 2 \times 0.628$

$= 1.26$ sec

1296. A shaft of diameter 50 mm, length 900 mm and torsional stiffness 87 Nm/rad, carries two flywheels of mass moments of inertia 10 kgm^2 and 30 kgm^2 respectively. The distance of the node from smaller flywheel end is:

(a) 300 mm (b) 425 mm

(c) 550 mm (d) 675 mm

UPSC JWM 2017

Ans. (d) : Distance of node is inversely proportional to the mass moment at inertia

$$\ell \propto \frac{1}{I}$$

$$\ell_1 I_1 = \ell_2 I_2 = \ell_3 I_3$$

Shaft length = $\ell_1 + \ell_2 = 900$mm

First flywheel mass moments of inertia, $I_1 = 10$ kgm^2

Second flywheel mass moment of inertia, $I_2 = 30$ kgm^2

$$\ell_1 I_1 = \ell_2 I_2$$

$$\ell_1 (10) = \ell_2 (30)$$

$$\ell_1 = 3\ell_2$$

$$\ell_2 = \ell_1 / 3$$

$$\ell_1 + \frac{\ell_1}{3} = 900$$

$$\frac{4\ell_1}{3} = 900$$

$$\ell_1 = \frac{2700}{4} = 675 \text{ mm}$$

Distance of the node from smaller flywheel end

$\boxed{\ell_1 = 675 \text{ mm}}$

1297. Two rotors are mounted on a shaft. If the unbalanced force due to one rotor is equal in magnitude to the unbalanced force due to the other rotor, but positioned exactly 180° apart, then the system will be balanced:

(a) statically

(b) dynamically

(c) statically as well as dynamically

(d) neither statically nor dynamically

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) :

1298. A vibratory system consists of a mass 25 Kg, a spring of stiffness 10,000 N/m, and a dashpot with damping Co-efficient of 100 N-s/m. The value of critical damping of the system is

(a) 50 N-s/m (b) 5000 N-s/m

(c) 1000 N-s/m (d) 500 N-s/m

TSPSC AEE 2017

Ans. (c) : $C = 2\sqrt{ms} = \sqrt{4ms}$

$C = \sqrt{4 \times 25 \times 10000} = 1000 \text{N-s/m}$

1299. If the frequency of oscillation on earth is n, then the frequency of oscillation on the moon will be

(a) 6n (b) $\sqrt{6n}$

(c) $\frac{n}{6}$ (d) $\frac{n}{\sqrt{6}}$

WBPSC AE, 2017

Ans. (d) : For simple pendulum,

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}$$

$$\frac{f_e}{f_m} = \sqrt{\frac{g_e}{g_m}}$$

$$\frac{f_e}{f_m} = \sqrt{\frac{g_e}{\frac{g_e}{6}}} \quad \Rightarrow f_m = \frac{n}{\sqrt{6}}$$

Note:- For spring mass system frequency doesn't change with location.

1300. The spring stiffness of an automobile suspension system is 3.6 kN/m and the damping constant of the damper is 400 Ns/m. The mass is 50 Kg. What is the damped natural frequency?

(a) 8.48 Hz (b) 1.4 Hz

(c) 1.19 Hz (d) 7.48 Hz

APGENCO AE, 2017

Ans. (c) : Given, $K = 3.6 \times 10^3$ N/m

C = 400 N-s/m

m = 50 kg

$$\xi = \frac{C}{2\sqrt{km}} = \frac{400}{2\sqrt{3.6 \times 10^3 \times 50}} = 0.471$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{3.6 \times 10^3}{50}} = 8.48 \text{ rad/sec}$$

$$\omega_d = \omega_n\sqrt{1 - \xi^2}$$

$$= 8.48\sqrt{1 - 0.471^2} = 7.48 \text{ rad/sec}$$

Damped natural frequency in Hz

$$f_d = \frac{7.48}{2\pi} = 1.19 \text{ Hz}$$

1301. In simple harmonic motion, the acceleration is proportional to :

(a) time period (b) displacement

(c) velocity (d) momentum

Punjab PSC (Lect.) 06.08.2017

Ans. (b) : An object is undergoing SHM if the acceleration of the object is directly proportional to its displacement from its equilibrium position.

In SHM, the displacement of particle at an instant is given by

$y = r \sin \omega t$

velocity (v) = $\frac{dy}{dt} = r\omega \cos\omega t$

Acceleration (a) = $\frac{dv}{dt} = -\omega^2 r \sin\omega t$

$a = -\omega^2 y$

$\boxed{a \propto y}$

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1302. A cantilever beam of cross section area 'A', moment of Inertia 'I' and length 'L' is having natural frequency ω_1. If the beam is accidently broken into halves, the natural frequency of the remaining cantilever beam ω_2 will be such that

(a) $\omega_2 < \omega_1$

(b) $\omega_2 > \omega_1$

(c) $\omega_2 = \omega_1$

(d) Cannot be obtained from the given data

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : $\omega = \sqrt{\frac{g}{\delta}}$

$$\delta = \frac{W\ell^3}{3EI} \text{ (assumed)}$$

$$\omega = \sqrt{\frac{3EIg}{W\ell^3}}$$

$$\omega \propto \frac{1}{\ell^{3/2}}$$

as $\omega \uparrow \rightarrow L \downarrow$

$\omega_2 > \omega_1$

1303. Which of the following is TRUE for vibration of a mechanical system

(a) Damping ratio (ξ) is the ratio of the critical damping to the actual damping

(b) The damping can be obtained from the response of the system under forced vibration using logarithmic decrement method

(c) For damped mechanical systems, the amplitude of vibration tends to become infinity when excitation frequency (ω) reaches the system's natural frequency (ω_n)

(d) Amplitude ratio and transmissibility of a vibration isolator is unity when the frequency of exciting force is $\sqrt{2}$ times the natural frequency

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Amplitude ratio and transmissibility of a vibration isolator is unity when the frequency of exciting force is $\sqrt{2}$ times the natural frequency.

(1) For any damped system, the damping ratio (ξ) is defined as the ratio of the damping constant to the critical damping constant.

(2) Logarithmic decrement is used to obtain damping in free vibration.

(3) Dampers absorb the energy and do not allow the vibration amplitude to reach the infinity in resonance phase, while in conservative systems without any damper; the amplitude reaches infinity when resonance happen.

i.e. at $\omega = \omega_n$

Resonance occurs

Amplitude $\rightarrow \infty$

(4) Transmissibility of vibration

$$T_r = \frac{\sqrt{1+\left(2\xi\frac{\omega}{\omega_n}\right)^2}}{\sqrt{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(2\xi\frac{\omega}{\omega_n}\right)^2}}$$

Put $\frac{\omega}{\omega_n} = \sqrt{2}$

$$T_r = \frac{\sqrt{1+4\xi^2.2}}{\sqrt{(1-2)^2+4\xi^2.2}}$$

$$= \frac{\sqrt{1+8\xi^2}}{\sqrt{1+8\xi^2}} = 1$$

1304. Longitudinal vibrations are said to occur when the particles of a body moves–

(a) perpendicular to its axis

(b) Parallel to its axis

(c) In a circle about its axis

(d) None of these

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : When the particles of the shaft or disc moves parallel to the axis of the shaft, these vibration are known as longitudinal vibration and if it moves perpendicular to the axis of shaft, then the vibration can be known as transverse vibrations.

1305. In an underdamped vibration system, logarithmic decrement is given by:

(a) $\delta = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$ (b) $\delta = \frac{2\pi\xi^2}{\sqrt{1-\xi}}$

(c) $\delta = \frac{2\pi\xi}{\sqrt{1+\xi^2}}$ (d) $\delta = \frac{2\pi\xi^2}{\sqrt{1+\xi}}$

TRB Polytechnic Lecturer 2017

Ans. (a) : In an underdamped vibration system, logarithmic decrement is

$$\delta = \ell n\left[\frac{x_i}{x_{i+1}}\right]$$

$$\delta = \xi\omega_n.T_d = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

1306. Though vibration cannot be eliminated completely, it can be suppressed to a greater extent using :

(a) Damper (b) Accumulator

(c) Receiver (d) Reducer

CIL MT 26.03.2017

Ans. (a) : The process of diminishing the vibration is called damping. The equipment used for this purpose is called damper.

1307. Natural frequency (ω_n) of a passenger car whose weight is w Newton and whose suspension has a combined stiffness of k N/mm is given by:

(a) $\omega_n = \sqrt{\frac{1}{km}}$ (b) $\omega_n = \sqrt{km}$

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(c) $\omega_n = \sqrt{\frac{k}{m}}$ (d) $\omega_n = \sqrt{\frac{km}{2}}$

CIL MT 26.03.2017

Ans. (c) : Natural frequency $\omega_n = \sqrt{\frac{k}{m}}$

Where k is the combined stiffness and m is the mass.

1308. When a system is subjected to force vibrations, then under steady-state conditions

(a) it vibrates at its natural frequency
(b) it vibrates at the imposed frequency
(c) it vibrates at the mean of natural and imposed frequencies
(d) None of the above

BPSC AE Mains 2017 Paper - VI

Ans : (b) : It vibrates at the imposed frequency.

1309. Calculate the critical speed of a light weight, vertically mounted shaft carrying a rotor of mass 200 g at its mid-point. The stiffness of shaft at the location of rotor is 72000 N/m.

(a) 360 rad/s (b) 19 rad/s
(c) 360000 rad/s (d) 600 rad/s

BPSC AE Mains 2017 Paper - VI

Ans : (d) : Given,
$m = 200 \times 10^{-3}$kg
k = 72000 N/m

$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{72000}{200\times10^{-3}}} = \sqrt{360\times10^3}$
= 600 rad/sec.

1310. For a vibrating system if the damping factor is unity, the maximum magnification factor will occur for ω/ω_n equal to

(a) 1 (b) 0
(c) 0.707 (d) >> 1

GPSC ARTO 01.05.2016

Ans. (a) : Given, damping factor 'ζ' = 1
Max. magnification factor at (ω/ω_n) = ?

Magnification factor = $\frac{1}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2+\left[2\zeta\left(\frac{\omega}{\omega_n}\right)\right]^2}}$

Where, $\{q = \omega/\omega_n\}$

$= \frac{1}{\sqrt{\left(1-q^2\right)^2+\left(2\zeta q\right)^2}}$

Max M.F. at q = 0

$\boxed{(M.F)_{max} = 1}$

1311. The equation of motion for a damped viscous vibration is $\ddot{x}+3\dot{x}+9x=0$. The damping factor will be

(a) 0.25 (b) 0.5
(c) 0.75 (d) 1.0

GPSC ARTO 01.05.2016

Ans. (b) : $\ddot{x}+3\dot{x}+9x=0$ Compared to equation
$m\ddot{x}+c\dot{x}+kx=0$
then, m = 1, c = 3, k = 9
Damping Coefficient, $c_c = 2\sqrt{km} = 2\sqrt{9\times1} = 6$

Damping factor $\xi = \frac{c}{c_c} = \frac{3}{6} = \frac{1}{2} = 0.5$

$\boxed{\xi = 0.5}$

1312. The natural frequency of an undammed vibratory system is 100 rad/s. a damper with a damping factor of 0.8 is introduced into the system. The frequency of vibration of damped system in rad/s is

(a) 100 (b) 80
(c) 75 (d) 60

GPSC ARTO 01.05.2016

Ans. (d) : Given, frequency of undamped vibration system (ω_n) = 100 rad/sec.
Damping factor (ξ) = 0.8

frequency of damped system $(\omega_d) = \omega_n\sqrt{1-\xi^2}$

$(\omega_d) = 100\sqrt{1-0.8^2} = 100 \times 0.6$

$\therefore \boxed{(\omega_d) = 60 \text{ rad/sec}}$

1313. A double acting shock absorber usually has :

(a) Pressure acting only one side
(b) Equal pressure on both side
(c) Unequal pressure on both sides
(d) None of the above

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) : A double acting shock absorber usually has unequal pressure on both sides.

1314. Critical whirling speed of a propeller shaft is increased by :

(a) Increasing its length
(b) Decreasing its diameter
(c) Decreasing its length
(d) None of the above

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c) Critical whirling speed of a propeller shaft is increased by decreasing its length.

1315. The secondary critical speed of shaft occurs at ..... .

(a) twice the speed of primary critical speed
(b) half the speed of primary critical speed
(c) four times the speed of primary critical speed
(d) one-fourth the speed o primary critical speed

GWSSB DEE 07.07.2016

Ans. (a) : The secondary critical speed of shaft occurs at twice the speed of primary critical speed.

1316. The usefulness of critical damping is that it .....

(a) totally eliminates vibrations
(b) provide basis of determining critical damping
(c) enables measurements of damping
(d) provides measure of the relative amount of damping in system

GWSSB DEE 07.07.2016

Ans. (c) : The usefulness of critical damping is that it enables measurements of damping.

1317. An over damped system when disturbed from equilibrium position with an initial velocity will ..... .

(a) vibrate above equilibrium position
(b) immediately return to equilibrium position
(c) not cross the equilibrium position

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(d) return to equilibrium position after 1 to 2 oscillation

GWSSB DEE 07.07.2016

Ans. (c) : An over damped system when disturbed from equilibrium position with an initial velocity will not cross the equilibrium position.

1318. A single degree freedom of spring mass is subjected to a sinusoidal force of 10 N amplitude and frequency ω along the axis of the spring. The stiffness of the spring is 150 N/m, damping factor is 0.2 and undamped natural frequency is low. At steady state, the amplitude of vibration (in m) is approximately–

(a) 0.05 (b) 0.07
(c) 0.70 (d) 0.90

RPSC 2016

Ans : (b)
Sinusoidal force (F) = 10 N
Stiffness of Spring (k) = 150 N/m
Damping factor (ξ) = 0.2
Let us consider,
Undamped Natural frequency (ω_n) = 10ω

$$\Rightarrow \frac{\omega}{\omega_n} = \frac{\omega}{10\omega} = 0.1$$

$$\text{Amplitude, } A = \frac{\left(\frac{F}{k}\right)}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}}$$

$$A = \frac{\frac{10}{150}}{\sqrt{[1-(0.1)^2]^2 + [2\times 0.2\times 0.1]^2}}$$

$$A = 0.0673\,m \approx 0.07\,m$$

1319. A vibrating beam has following degrees of freedom

(a) 0 (b) 1
(c) 2 (d) 3

RPSC AE 2016

Ans : (d) A vibrating beam has three degree of freedom.

1320. In a spring dash pot, mass system if m= mass, k = spring stiffness and ω_n = natural frequency of vibration, then critical damping is equal to

(a) $2\sqrt{km}$
(b) $2m.\omega_n$
(c) Both (a) and (b)
(d) neither (a) nor (b)

UPPSC AE 12.04.2016 Paper-I

Ans : (c) The critical damping coefficient (C_c) may be obtained by substituting C_c for C in the condition for critical damping e.g.

$$\left(\frac{C_c}{2m}\right)^2 = \frac{s}{m} \text{ or } C_c = 2m\sqrt{\frac{k}{m}}$$

$$C_c = 2m.\omega_n$$

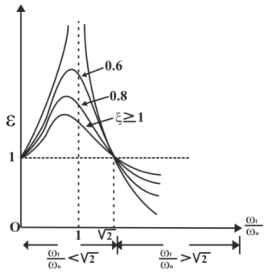
The critical damping coefficient (C_c) is the amount of damping required for a system to be critically damped.

1321. In a forced vibration system, for which value of frequency ratio $\left(\frac{\omega_f}{\omega_n}\right)$, the transmissibility is same for all the values of damping factors.

(a) 1 (b) 2
(c) $\sqrt{2}$ (d) $\frac{1}{2}$

UPPSC AE 12.04.2016 Paper-I

Ans : (c) Transmissibility:- It is defined as the ratio of the force transmitted to the force applied. It is a measure of the effectiveness of the vibration isolating materials.



(i) $\frac{\omega_f}{\omega_n} < \sqrt{2}$ Need of insulation is more (cement, concrete etc. used)

(ii) $\frac{\omega_f}{\omega_n} = \sqrt{2}$ No need of insulation

(iii) $\frac{\omega_f}{\omega_n} > \sqrt{2}$ Need of insulation is less (like spring)

1322. Which of the following is the correct expression for natural frequency of free longitudinal vibration

(a) $f_m = \frac{1}{2\pi}\sqrt{\frac{m}{s}}$ (b) $f_m = \frac{1}{2\pi}\sqrt{\frac{s}{m}}$
(c) $f_m = \frac{1}{\pi}\sqrt{\frac{m}{s}}$ (d) $f_m = \frac{1}{\pi}\sqrt{\frac{s}{m}}$

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : The natural frequency of free longitudinal and transverse vibrations is given by;

$$f_m = \frac{1}{2\pi}\sqrt{\frac{s}{m}} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}} \text{ Hz.}$$

m = mass of the body
s = Sfiffnes of the body
δ = Static deflection of the body

1323. A rotor is said to be rigid if

(a) it rotates at rotational speed below 4% of critical speed
(b) It rotates at rotational speed below 70% of critical speed
(c) It rotates at rotational speed below 60% of critical speed
(d) It rotates at rotational speed below 20% of critical speed

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : As, a general rule, rotors that operates below 70% of their critical speed are considered rigid and,

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when balanced at on speed will be balanced at any other normal operating speed below 70% of its critical speed.

1324. Damping force per unit velocity is known as
(a) Damping factor
(b) Damping coefficient
(c) Logarithmic decrement
(d) Stiffness of the spring

RPSC LECTURER 16.01.2016

Ans. (b) : Damping force (F_D)
= Damping coefficient (C) × Velocity ($\dot{x}$)
$= C \times \dot{x}$
$F_D = C \times \dot{x}$
$$C = \frac{F_D}{\dot{x}}$$

1325. The system in which the amplitude of vibration decreases with time and ratio of amplitude of successive oscillate being constant is known as:
(a) Critically damped system
(b) Underdamped system
(c) Undamped system
(d) Overdamped system

UPRVUNL AE 07.10.2016

Ans. (b) : The system in which the amplitude of vibration decreases with time and ratio of amplitude of successive oscillate being constant is known as Underdamped system.
In Underdamped system with each overshoot, some energy in the system is dissipated, and the oscillations die towards zero.

1326. The ratio of two consecutive amplitudes of an underdamped vibrating system is given by: (where δ = logarithmic decrement):
(a) $e^{4\delta}$ (b) $e^{3\delta}$
(c) e^{δ} (d) $e^{2\delta}$

UPRVUNL AE 07.10.2016

Ans. (c) : The ratio of two consecutive amplitude of an underdamped vibrating system.

$$\frac{x_i}{x_{i+1}} = e^{\delta}$$

$$\ln\left(\frac{x_i}{x_{i+1}}\right) = \delta$$

$$\ln\left(\frac{x_i}{x_{i+1}}\right) = \delta = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

where,
δ = logarithmic decrement

1327. A 0.5 kg weight attached to a light spring elongates by 0.981 mm. The natural frequency of the system should be
(a) 6 Hz (b) 16 Hz
(c) 26 Hz (d) 32 Hz

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : Given,
m = 0.5 kg, δ = 0.981 mm
We know that-

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{9.81}{0.981\times10^{-3}}}$$
$$f_n = 15.92 \simeq 16\,\text{Hz}$$

1328. A critical speed
(a) the shaft runs smoothly
(b) the shaft runs with noise due to vibration
(c) the shaft becomes dynamically unstable with negligible lateral amplitudes
(d) the shaft becomes dynamically unstable with large lateral amplitudes

RPSC 2016

Ans : (c) The shaft becomes dynamically unstable with negligible lateral amplitudes.
The critical speed essentially depends on critical or whirling or speed is the speed at which the shaft tends to vibrate violently in transverse direction.
- The eccentricity of the C.G. of the rotating masses from the axis of rotation of the shaft.
- Diameter of the disc.
- Span (length) of the shaft.
- Type of supports connections at its ends.

$$y = \frac{e}{\left(\frac{\omega_n}{\omega}\right)^2 - 1}$$

At critical speed,

$$\omega = \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{\delta}}$$

ω = Angular velocity of shaft
k = Stiffness of shaft
e = initial eccentricity of centre of mass of rotor
m = mass of rotor
y = additional of rotor due to centrifugal force.

1329. The torsional vibrational frequency of a shaft with n oscillating masses will be–
(a) n (b) n–1
(c) n+1 (d) 2n

RPSC 2016

Ans : (b) Torsional vibrational frequency of a shaft with n oscillating masses will be (n – 1).

1330. A mass of 1 kg attached to a light spring extends it by 0.981 mm. the natural frequency of vibration will be approximately–
(a) 26 Hz (b) 21 Hz
(c) 16 Hz (d)11 Hz

RPSC 2016

Ans : (c)

$$\omega_n = \sqrt{\frac{g}{\delta}} \Rightarrow \omega_n = 2\pi f$$

$$f = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}},$$

$$f = \frac{1}{2\pi}\sqrt{\frac{9.81}{9.81\times10^{-4}}} \quad (\delta = 0.981\text{mm} = 9.81\times10^{-4}\,\text{m})$$
$$= 15.91\,\text{Hz}$$
$$f \approx 16\,\text{Hz}$$

1331. Calculate logarithmic decrement if damping factor is 0.33.

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(a) 1.36 (b) 3.23
(c) 5.16 (d) 2.19

RPSC VPITI 14.02.2016

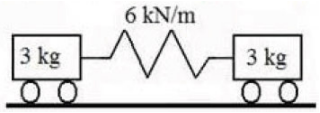
Ans. (d) : Damping factor $\xi = 0.33$

Logarithmic decrement $\delta = \dfrac{2\pi\xi}{\sqrt{1-\xi^2}}$

$$\delta = \frac{2\pi(0.33)}{\sqrt{1-(0.33)^2}}$$

$$\delta = 2.19$$

1332. Consider the system of two wagons shown in the figure below.



What are the natural frequencies of this system?

(a) 0, 2 (b) 0, 1
(c) $\sqrt{2}, 1$ (d) $\sqrt{2}, \sqrt{2}$
(e) $\sqrt{2}, 2$

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Ans. (a) : Natural frequencies will be,

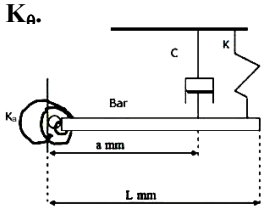
$$\omega_n = \sqrt{\frac{k_{eq}}{m}} = \sqrt{\frac{2k}{m}}$$

For $m_1 = 3$ kg, $m_2 = 3$ kg

$$\omega_{n_1} = \sqrt{\frac{0}{m}} = 0$$

$$\omega_{n_2} = \sqrt{\frac{2\times k}{m_2}} = \sqrt{\frac{2\times 6}{3}} = 2$$

1333. A uniform rigid bar of mass m kg is hinged at the left end, and suspended with the help of spring and damper arrangement as shown in the figure below. The stiffness of torsional spring is K_θ.



Ignore the hinge dimensions. What is the undamped natural frequency of oscillations of the rod about the hinge point?

(a) $\sqrt{\dfrac{KL^2+K_\theta}{3mL^2}}$ (b) $\sqrt{\dfrac{KL^2+K_\theta}{mL^3/3}}$
(c) $\sqrt{\dfrac{KL^2+K_\theta}{3mL^3}}$ (d) $\sqrt{\dfrac{KL^2+K_\theta}{mL^2/3}}$
(e) $\sqrt{\dfrac{KL^2-K_\theta}{mL^3/3}}$

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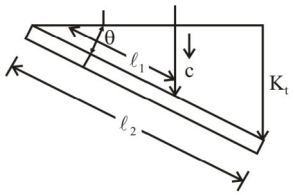
Ans. (d) : ℓ_1 = a mm, ℓ_2 = L mm

For static equilibrium,



For θ deflection equation will be,

$$I\cdot\ddot{\theta} + Ca^2\theta + (K_\theta + KL^2)\theta = 0$$

Where, $I = \dfrac{mL^2}{3}$

$$w = \sqrt{\frac{(K_\theta + KL^2)\times 3}{mL^2}}$$

1334. Which of the following represents the transverse frequency of vibration for a shaft, if its inertia is considered?

(a) $fn = \dfrac{1}{2\pi}\sqrt{\dfrac{q}{\frac{I+I_s}{3}}}$ (b) $fn = \dfrac{1}{2\pi}\sqrt{\dfrac{q}{I+\frac{I_s}{3}}}$
(c) $fn = \sqrt{\dfrac{q}{\frac{I+I_s}{3}}}$ (d) $fn = \sqrt{\dfrac{q}{I+\frac{I_s}{3}}}$
(e) $fn = \dfrac{1}{2\pi}\sqrt{\dfrac{3q}{I+I_s}}$

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Ans. (b) : For a shaft,

Transverse frequency (fn) $= \dfrac{1}{2\pi}\sqrt{\dfrac{q}{I+\frac{I_s}{3}}}$

1335. Magnification factor depends upon:

(i) The ratio of frequencies
(ii) The damping factor
(iii) Instant of force applied

(a) Only (i) (b) Only (i) and (ii)
(c) Only (i) and (iii) (d) Only (ii) and (iii)
(e) (i), (ii) and (iii)

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Ans. (b) : The dynamic magnification factor value varies depending upon the types of loading conditions. It is further compared with the rule-based equation. Thus, the ratio of amplitude of motion caused by external forces. It depends on ratio of frequencies & damping factor.

Magnification factor,

$$\in = \frac{\sqrt{1+\left(\frac{2\xi\omega}{\omega_n}\right)^2}}{\sqrt{\left[1-\left(\frac{\omega}{\omega_n}\right)^2\right]^2+\left(\frac{2\xi\omega}{\omega_n}\right)^2}}$$

1336. In a spring mass system, the static deflection caused by mass m in spring of stiffness K is δ. The natural frequency of free longitudinal vibration of the system will be

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(a) $f_n = \frac{1}{2\pi}\sqrt{\frac{K}{mg}}$ (b) $f_n = \frac{1}{2\pi}\sqrt{\frac{mg}{\delta}}$

(c) $f_n = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$ (d) $f_n = \frac{1}{2\pi}\sqrt{\frac{m}{K}}$

(e) None of these

CGPSC AE 16.10.2016

Ans. (c) : As we know,
The static deflection 'δ' leads to time period,

$$T = 2\pi\sqrt{\frac{\delta}{g}}$$

$$f_n = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g}{\delta}}$$

1337. A shaft of 50 mm diameter and 1 m length carries a disc which has mass eccentricity equal to 190 microns. The displacement of the shaft at a speed which is 90% of critical speed in microns is ___

(a) 810 (b) 800

(c) 820 (d) 900

UPRVUNL AE 21.08.2016

Ans. : (a) We know that

$$\frac{y}{c} = \frac{1}{\left(\frac{\omega_n}{\omega}\right)^2 - 1}$$

Shaft speed = 0.9 × Critical speed

$\omega = 0.9\ \omega_n$

Eccentricity = 190 micron

$$\frac{y}{c} = \frac{1}{\left(\frac{1}{0.9}\right)^2 - 1}$$

y = 810 microns

1338. The natural frequency of an undamped vibrating system is 200 rad/sec. A damper with a damping factor of 0.8 is introduced into the system. The frequency of vibration of the damped system in rad/s is_____

(a) 120 (b) 60

(c) 75 (d) 80

UPRVUNL AE 21.08.2016

Ans. : (a) frequency of vibration = 120

$$\frac{\omega_d}{\omega_n} = \sqrt{1-\xi^2}$$

$$\frac{\omega_d}{200} = \sqrt{1-0.8^2}$$

$$\frac{\omega_d}{200} = 0.6$$

$\omega_d = 120\,\text{rad/sec}$

1339. When the motion is repeated in equal intervals of time, it is known as

(a) static motion
(b) periodic motion
(c) non periodic motion
(d) none of these

GPSC Lect. 23.10.2016

Ans. (b) : Periodic motion, in physics motion repeated in equal intervals of time eg. a bouncing ball etc.

1340. Vibration of a system that occurs in the absence of external force is called?

(a) Self excited vibration
(b) Force vibration
(c) Free vibration
(d) None of these

GPSC Lect. 23.10.2016

Ans. (c) : Free vibration is used to indicate that there is no external force causing the motion, and that the motion is primarily the result of initial conditions, such as an initial displacement of mass element of the system from an equilibrium position.

1341. When the frequency of external exciting force is equal to the natural frequency of vibration of the system

(a) the amplitude of vibration is zero
(b) the amplitude of vibration is insignificantly small
(c) the amplitude of vibration is very large
(d) the amplitude of vibration may be large or small depending upon the magnitude of frequency

TSGENCO AE 14.11.2015

Ans. (c) : When the frequency of external exciting force is equal to the natural frequency $\left(\text{i.e.}\frac{\omega}{\omega_n} = 1\right)$ or vibration of the system, then the amplitude of vibration system is very large.

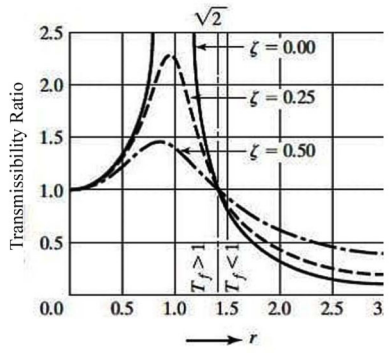
1342. When $\omega/\omega_n = 1$, transmissibility is where ω = frequency of excitation, rad/s ω_n = Natural frequency rad/s

(a) ∞ (b) 0

(c) 1 (d) $\sqrt{2}$

TSPSC AEE 2015

Ans. (a) :



1343. The amplitude ratio of two successive oscillations of a damped vibratory system is

(a) more than one (b) less than one

(c) equal to one (d) variable

TSPSC AEE 2015

Ans. (a) : The amplitude ratio of two successive oscillations of a damped vibratory system is more than one.

1344. Seismic vibrations are measured with

(a) Richter scale (b) Vibrometers

(c) Accelerometers (d) Gyroscope

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TSPSC AEE 2015

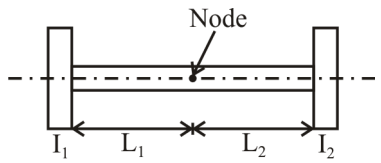
Ans.(a) Seismic vibrations are measured with Richter scale.

1345. The flywheels of moments of inertia I_1 and I_2 are fixed to two ends of a uniform shaft, and they from a torsional vibration system. If I_1 is greater than I_2, then the node point will be :

(a) nearer to I_1
(b) nearer to I_2
(c) at the mid-point of the shaft
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (a) :



$I_1L_1 = I_2L_2 = C$

So, if I_1 is greater then L_1 must be lesser.

1346. The door closers operated normally at :

(a) Over damping (b) Under damping
(c) Critical damping (d) No damping

TSPSC Managers, 2015

Ans. (a) : The door closures operated normally at over damping.

1347. In a spring mass system, 4 kg mass is suspended through a spring of stiffness 100 N/m. What is the circular frequency of the system?

(a) 15 rad/s (b) 5 rad/s
(c) 10 rad/s (d) 4 rad/s

TSPSC Managers, 2015

Ans. (b) : Given – m = 4kg
k = 100 N/m

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{4}} = \sqrt{25}$$

= 5 rad/sec

1348. Earthquakes produces the following type of vibrations.

(a) Torsional (b) Transverse
(c) Seismic (d) Longitudinal

TSPSC Managers, 2015

Ans. (c) : Earthquakes produces seismic type of vibration. Earthquakes is measured by seismometer.

1349. According to Dunkerley's empirical equation, the frequency of the transverse vibration of the system of several loads attached to the same shaft is

(a) $\frac{1}{f_n} = \frac{1}{f_{n_1}} + \frac{1}{f_{n_2}} + \frac{1}{f_{n_3}} ..... + \frac{1}{f_{n_s}}$

(b) $\frac{1}{f_n^2} = \frac{1}{f_{n_1}^2} + \frac{1}{f_{n_2}^2} + \frac{1}{f_{n_3}^2} + ... \frac{1}{f_{n_s}^2}$

(c) $f_n = f_{n_1} + f_{n_2} + f_{n_3} + ...f_{n_s}$

(d) none of these

VIZAG MT 2015

Ans. (b) : $\frac{1}{f_n^2} = \frac{1}{f_{n_1}^2} + \frac{1}{f_{n_2}^2} + \frac{1}{f_{n_3}^2} + ... \frac{1}{f_{n_s}^2}$

1350. Two springs of stiffness 100N/m each are connected in series and support mass of 2 kg. The natural frequency of the system will be

(a) $(\pi / 2.5)$Hz (b) $(2.5 / \pi)$Hz
(c) $(\sqrt{50}/2\pi)$Hz (d) $(\sqrt{50})$Hz

VIZAG MT 2015

Ans. (b) : Two springs connected in series

$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$$

$$= \frac{100 \times 100}{100 + 100} = \frac{100 \times 100}{200} = 50 \text{ N/m}$$

$$f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k_{eq}}{m}} = \frac{1}{2\pi}\sqrt{\frac{50}{2}}$$

$$= \frac{5}{2\pi}\text{Hz} = \frac{2.5}{\pi}\text{Hz}$$

1351. In a damped free vibration of a mass supported on a spring and a damper, where the damping forces is proportional to the velocity the ratio of two successive amplitudes

(a) remains constant
(b) gradually decreases and varies linearly with time
(c) gradually decreases and varies exponentially with time
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (a) : remains constant.

1352. Two rotors supported by a shaft has a natural frequency ω_n. If one of the rotors is fixed, the natural frequency

(a) increase (b) decreases
(c) remains same (d) becomes zero

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Shaft has a natural frequency ω_n . If one of the rotors is fixed the natural frequency is decreases.

1353. An elastic beam, simply supported at the ends, carries a concentrated mass at the middle of its length. If the natural frequency of the beam neglecting the weight is ω_n, considering the weight of the beam the natural frequency of the system will be :

(a) greater than ω_n
(b) less than ω_n
(c) zero
(d) equal to ω_n, as the natural frequency is independent of concentrated load.

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : $\omega_{n_1} = \sqrt{\frac{g}{\Delta_1}}$

$$\omega_{n_2} = \sqrt{\frac{g}{\Delta_1 + \Delta_2}}$$

As $\Delta_1 + \Delta_2 > \Delta_1$

So, $\omega_{n_2} < \omega_{n_1}$

Δ_1 = Deflection of simply supported beam by point load at mid span.

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$$\Delta_1 = \frac{WL^3}{48EI}$$

Δ_2 = Deflection of simply supported beam by considering weight of beam which is in the form of UDL.

$$\Delta_2 = \frac{5wL^4}{384EI}$$

1354. Resonance is a aphenomenon where frequency of exciting force is ..... the natural frequency of the system

(a) double (b) half
(c) equal to (d) thrice

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

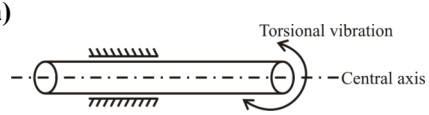
Ans. (c) : Resonance is a special case of forced vibration in which the natural frequency of vibration at the body is the same as the impressed of external periodic force.

1355. The particles of a body vibrate along a circular arc, whose centre lies in the axis of the body. The type of vibration is then

(a) Torsional vibration
(b) Longitudinal vibration
(c) Transverse vibration
(d) Lateral vibration

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a)



1356. The type of damping due to sliding of two dry surfaces is termed as

(a) Viscous damping
(b) Coulomb damping
(c) Structural damping
(d) Interfacial damping

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Sliding friction, also termed coulomb, is the damping force commonly used to model friction arising from relative motion of two surfaces

1357. Natural frequency of a system is due to

(a) free vibration (b) forced vibrations
(c) resonance (d) damping

TNPSC AE 2014

Ans. (a) : Natural frequency of a system is due to free vibration.

1358. Frequency of vibrations is usually expressed in

(a) Number of cycles per hour
(b) Number of cycles per second
(c) Number of cycles per minute
(d) Number of cycles per day

Mizoram PSC AE/SDO 2014, Paper-II

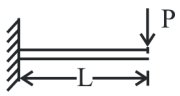
Ans. (b) : The number of cycle that a vibrating object completes in one second is called frequency. The unit of frequency is hertz (Hz). One hertz equal one cycle per second.

1359. The fundamental natural frequency of the cantilever beam with point load P acting at the free end, in rad/sec is

(a) $\sqrt{\frac{3EIg}{PL^3}}$ (b) $\sqrt{\frac{PL^3}{3EIg}}$
(c) $\frac{1}{2\pi}\sqrt{\frac{3EIg}{PL^3}}$ (d) $\frac{1}{2\pi}\sqrt{\frac{PL^3}{3EIg}}$

ISRO Scientist/Engineer 24.05.2014

Ans. (a) :



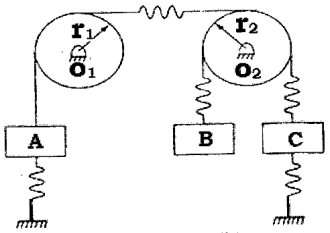
Natural frequency $\omega_n = \sqrt{\frac{g}{\Delta}}$

δ = Deflection

$$\delta = \frac{PL^3}{3EI}$$

$$\omega_n = \sqrt{\frac{g}{PL^3/3EI}} = \sqrt{\frac{3EIg}{PL^3}} \text{ rad/sec}$$

1360. Determine the number of degrees of freedom used in the analysis of the mechanical system of below mentioned figure.



(a) 3 (b) 7
(c) 5 (d) 4

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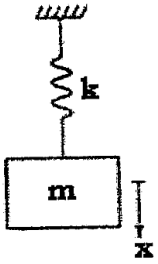
Ans. (d) : In the above mechanism, two rotation of the pulleys and two vertical motions of the masses B and C. Hence answer given in this particular case is 4.

1361. The difference in the Potential energies between two arbitrary positions downward displacement X_1 and X_2. ($X_2 > X_1$).



(a) $\frac{1}{2}k[x_1 - x_2]^2$ (b) $\frac{1}{2}k\left[(x_2)^2 - (x_1)^2\right]$
(c) $\frac{1}{2}k[(x_1) - (x_2)]$ (d) $\frac{1}{2}k[x_1 - x_2]$

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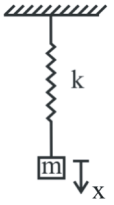
Ans. (b) :



The potential energy due to downward

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displacement x_1

$P_1 = \frac{1}{2} k x_1^2$ ---------(1)

The potential energy due to displacement 'x_2'

$P_2 = \frac{1}{2} k x_2^2$ ---------(2)

$\therefore$ Their potential difference

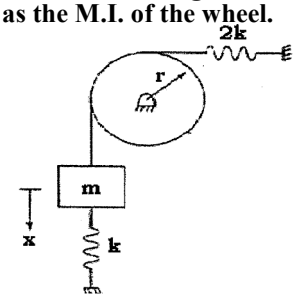
$\because\ x_2 > x_1$ so, $P_2 > P_1$

$= P_2 - P_1$

$= \frac{1}{2} k x_2^2 - \frac{1}{2} k x_1^2$

$= \frac{1}{2} k \left[(x_2)^2 - (x_1)^2 \right]$

1362. Determine m_{eq} and k_{eq} for the system in below Fig., when X, the downward displacement of a block, measured from the system's equilibrium position, is used as the generalized coordinate. Take I as the M.I. of the wheel.



(a) $K_{eq} = 3k, m_{eq} = m + \frac{I}{r^2}$

(b) $K_{eq} = 3k, m_{eq} = mr^2 + I$

(c) $K_{eq} = 3k, m_{eq} = m - \frac{I}{r^2}$

(d) $K_{eq} = k/3, m_{eq} = m + \frac{I}{r^2}$

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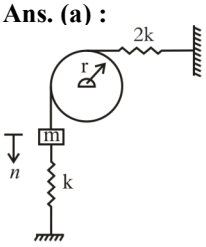
Ans. (a) :



2k, k is connected in parallel.

$k_{eq} = k + 2k$

$k_{eq} = 3k$

Moment of Inertia of wheel = I

$E = \frac{1}{2} m v^2 + \frac{1}{2} I \left(\frac{v}{r} \right)^2 \qquad [v = r\omega]$

$= \frac{1}{2} m v^2 + \frac{1}{2} I \frac{v^2}{r^2}$

$= \frac{1}{2} v^2 \left[m + \frac{I}{r^2} \right]$

So, $\boxed{m_{eq} = m + \frac{I}{r^2}}$

1363. The system described by the following governing equation of motion, will have how many natural frequencies?

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} + \begin{bmatrix} 2k & -k \\ -k & 3k \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

(a) 4 (b) 3

(c) 2 (d) 1

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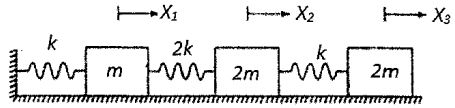
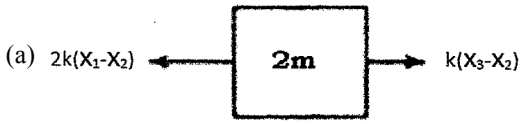
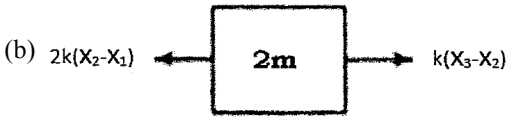
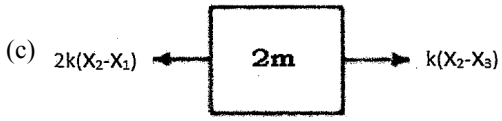
Ans. (c) : Number of natural frequencies
= Number of eigen values of stiffness matrix
"As the matrix is of the order of 2×2. So number of eigen values are two.

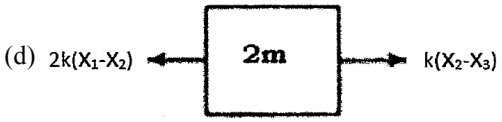
1364. Find out the correct free body diagram from the middle mass for the given spring-mass system in below figure



(a) 

(b) 

(c) 

(d) 

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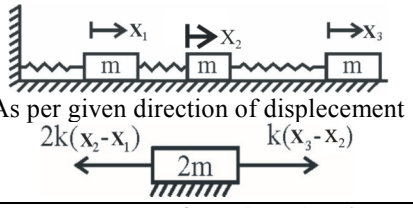
Ans. (b) :



1365. For what value of m (mass) will resonance occur for the spring mass system given below with the given spring constants?

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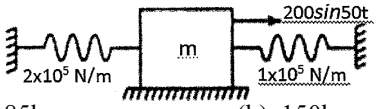

(a) 85kg (b) 150kg
(c) 100kg (d) 120kg

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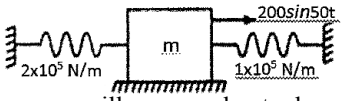
Ans. (d) :



Resonance will occur only at when

$$\boxed{\text{Natural frequency } (\omega_n) = \text{Excitation frequency } (\omega)}$$

Here $\omega = 50$ rad/sec

In Parallel

$$k_{eq} = k_1 + k_2$$
$$= 2 \times 10^5 + 1 \times 10^5$$
$$k_{eq} = 3 \times 10^5 \text{ N/m}$$
$$\omega_n = \omega = \sqrt{\frac{k}{m}}$$
$$m = \frac{k}{\omega^2} = \frac{3\times10^5}{50\times50} \qquad \boxed{m = 120\text{kg}}$$

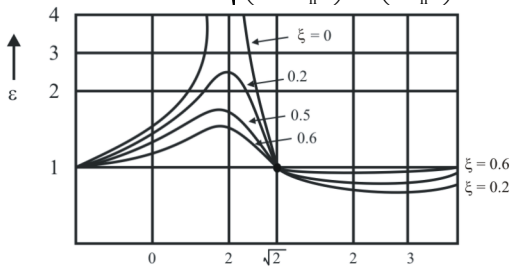
1366. If frequency ratio is more than $\sqrt{2}$ in a vibration isolation system then for all values of amplitude ratio the transmissibility is ____.

(a) less than unity (b) more than unity
(c) equal to one (d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : Transmissibility ratio (ε)

$$\varepsilon = \frac{F_t}{F_o} = \sqrt{\frac{1+\frac{2\xi\omega}{\omega_n}}{\left(1-\frac{\omega^2}{\omega_n^2}\right)^2+\left(\frac{2\xi\omega}{\omega_n}\right)^2}}$$



$$\boxed{\frac{\omega}{\omega_n} > \sqrt{2} \Rightarrow \varepsilon < 1}$$

1367. The frequency of damped vibrations is always _____ the natural frequency.

(a) equal to (b) more than
(c) less than (d) double

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : Frequency of damped vibration

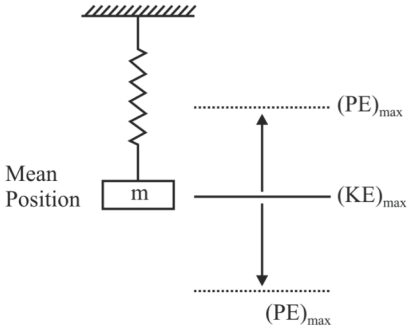
$$\omega_d = \omega_n\sqrt{1-\xi^2}$$
$$\boxed{\omega_d < \omega_n}$$

1368. Choose the correct option for the given blank space "While using _____ method to find the natural frequency of free vibrations, it is assumed that kinetic energy at mean position is equal to the potential energy at the extreme positions."

(a) energy (b) equilibrium
(c) Kennedy's (d) Rayleigh's

Haryana PSC Civil Services Pre, 2014

Ans. (d)



KE – Kinetic Energy
PE – Potential Energy

According to Rayleigh's method

$(KE)_{max} = (PE)_{max}$
$(KE)_{mean} = (PE)_{at\ extreme}$

$$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$$
$$mv^2 = kx^2$$

Maximum velocity $\boxed{v = \omega_n x}$

$$m(x\omega_n)^2 = kx^2$$
$$\boxed{\omega_n = \sqrt{k/m}}$$

1369. A particle is performing SHM. If its maximum velocity is 1.5 m/sec and its maximum acceleration is 12 m/sec^2 the frequency of motion would be ____Hz.

(a) $16/\pi$ (b) $4/\pi$
(c) $2/\pi$ (d) $1/\pi$

MPSC Poly. Lect. 2014

Ans. (b) : Maximum velocity = $x\omega = 1.5$ m/s
Maximum acceleration = $\omega^2 x = 12$ m/s^2

$$\frac{x\omega}{\omega^2 x} = \frac{1.5}{12}$$
$$\omega = \frac{12}{1.5} = 8\,\text{rad/sec.}$$
$$\omega = 2\pi f = 8$$
$$f = \frac{8}{2\pi}$$
$$\boxed{f = \frac{4}{\pi}\text{ Hz}}$$

1370. A mass of 8 kg hanging from free end of spring. If stiffness of spring is 2 N/cm, then determine the angular frequency of the system

(a) 5 rad/s (b) 25 rad/s
(c) 0.2 rad/s (d) 0.04 rad/s

UPRVUNL AE 2014

Ans. : (a) Given, k = 2N/cm = 200N/m, m = 8Kg

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Angular frequency $(\omega) = \sqrt{\frac{k}{m}}$

$\sqrt{\frac{200}{8}} = 5$ rad/s

1371. Which of the following gives the expression for the magnification factor at resonance in forced vibration system?
(Where s is stiffness of the spring, 'c' is the damping coefficient or damping force unit velocity and ω_n is the natural circular frequency)
(a) $s/(c\omega_n)$ (b) $(c\omega_n)/s$
(c) $\sqrt{(s/(c\omega_n))}$ (d) $\sqrt{(c\omega_n)/s}$

UPRVUNL AE 2014

Ans. : (a)
s = Spring stiffness
c = Damping coefficient
ω_n= Natural circular frequency

$$M.F = \frac{1}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}}$$

$$(MF)_{\omega=\omega_n} = \frac{1}{2\xi} = \frac{1}{2\times\left(\frac{c}{2\sqrt{sm}}\right)}$$

$$(MF)_{\omega=\omega_n} = \frac{\sqrt{sm}}{c}\times\frac{1}{\omega_n}\times\sqrt{\frac{s}{m}}$$

$$(MF)_{\omega=\omega_n} = s/(c\omega_n)$$

1372. A reciprocating engine running at 90 rad/s, is supported on springs and static deflection of spring is 4 mm. When engine runs, the frequency of vibration of system will be : (Take g = 10 m/s^2)
(a) 50 rad/s (b) 75 rad/s
(c) 100 rad/s (d) 160 rad/s

UPRVUNL AE 2014

Ans. : (a) $\omega_n = \sqrt{\frac{S}{m}} = \sqrt{\frac{g}{\delta}}$

$$\omega_n = \sqrt{\frac{10}{4\times10^{-3}}} = \sqrt{\frac{10^4}{4}} = \frac{100}{2} = 50\,\text{rad/sec}$$

$\omega_n = 50$ rad/sec

Where
S = Spring stiffness

1373. When a shaft is rotating at a speed which is less than critical speed, the phase difference between displacement and centrifugal force is
(a) 180^o (b) 90^o
(c) 45^o (d) 0^o

TNPSC AE 2014

Ans. (d) : The phase difference between displacement and centrifugal force, when a shaft is rotating at a speed which is less than critical speed will be 0^o.

1374. A flexible rotor shaft system comprises of a 10 kg rotor disc placed in the middle of a massless shaft of diameter 30 mm and length 500 mm between bearing (shaft is being taken massless as the equivalent mass of the shaft is included in the rotor mass) mounted at the ends. The bearings are assumed to simulate simply supported boundary conditions. The shaft is made of steel for which the value of E is 2.1 × 10^{11} Pa. What is the critical speed of rotation of the shaft?
(a) 60 Hz (b) 90 Hz
(c) 135 Hz (d) 180 Hz

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : Given, m = 10 kg = mass of rotor
d = Diameter of shaft = 30×10^{-3} m
ℓ = Length of shaft = 500×10^{-3} m = 0.5 m
E for steel = 2.1×10^{11} N/m^2
The given shaft can be considered as a simply supported beam for which the stiffness

$$k = \frac{48EI}{\ell^3}$$

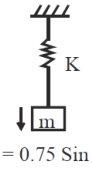
$$I = \frac{\pi d^4}{64} = \frac{\pi\times0.03^4}{64} = 3.97\times10^{-8}\,m^4$$

Natural frequency, $(\omega_n) = \sqrt{\frac{k}{m}} = \sqrt{\frac{48EI}{m\ell^3}}$

$$= \sqrt{\frac{48\times2.1\times10^{11}\times3.97\times10^{-8}}{10\times0.5^3}} = 565.81$$

$$f_n = \frac{\omega_n}{2\pi} = \frac{565.81}{2\pi} = 90\,\text{Hz}$$

1375. A spring mass system shown in Figure is actuated by a load P = 0.75 sin2t. If mass of the block is 0.25 kg and stiffness of the spring is 4 N/m, displacement of the block will be:-



(a) 0.25 (b) 0.5
(c) 1.0 (d) 2.25

UKPSC AE-2013, Paper-I

Ans. (a) :
P = 0.75 Sin 2t
P = F_0 Sinωt
F_0 = 0.75 N
ω = 2
m = 0.25 kg
k = 4 N/m

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{4}{0.25}} = 4$$

$$\xi = 0$$

$$A = \frac{F_0/k}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\xi\frac{\omega}{\omega_n}\right)^2}}$$

$$\frac{0.75/4}{\sqrt{\left(1-\left(\frac{2}{4}\right)^2\right)^2 + 0}} = \frac{0.75\times4}{4\times3} = 0.25$$

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$A = 0.25$

1376. A single degree of freedom spring mass damper system has a spring of stiffness k = 4 kN/m, mass m = 10 kg and a damper of damping constant c = 240 N-sec/m, its damped natural frequency will be :

(a) 18 rad/sec (b) 16 rad/sec
(c) 20 rad/sec (d) 22 rad/sec

J&K PSC Civil Services Pre, 2013

Ans. (b) : Frequency of damped vibration

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\zeta = \frac{C}{C_c} = \frac{C}{2\sqrt{km}} = \frac{240}{2\sqrt{4\times1000\times10}}$$

$$= \frac{240}{2\times10^2\times2} = 0.6$$

$$\omega_n = \sqrt{k/m} = \sqrt{\frac{4000}{10}} = 20 \text{ rad/sec}$$

$$\omega_d = 20\sqrt{1-0.6^2} = 16 \text{ rad/sec.}$$

1377. A rotating shaft carries a single rotor and the static deflection of the shaft at the rotor is 0.1 m. What will be transverse natural frequency of the shaft in Hertz, if the acceleration due to gravity is 10 m/s^2?

(a) 5π (b) $5/\pi$
(c) 10π (d) None of the above

RPSC AEN Pre-2013

Ans. (b) : Given-
Deflection (δ) = 0.1 m
g = 10 m/s^2

$$\text{Natural frequency } (f_N) = \frac{1}{2\pi}\times\sqrt{\frac{g}{\delta}}$$

$$= \frac{1}{2\pi}\sqrt{\frac{10}{0.1}} \Rightarrow \frac{1}{2\pi}\sqrt{\frac{100}{1}}$$

$$= \frac{1}{2\pi}\times10 \Rightarrow \frac{5}{\pi}$$

1378. An automotive engine having a mass of 135 kg is supported on 4 springs with linear characteristics. Each of the 2 front springs have stiffness of 3MN/m while the stiffness of each of 2 rear springs is 4.5 MN/m. The engine speed (rpm) at which resonance is likely to occur is

(a) $10^3/(6\pi)$ (b) $1/(6\pi)$
(c) $10^4/(\pi)$ (d) $10^3/(3)$

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : Given, Equivalent stiffness
k = 2 × 4.5 + 2 × 3 = 15 MN/m
m = 135 kg

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{15\times10^6}{135}}$$

$\omega_n = 0.3333 \times 10^3$ rad/sec

$$N = \frac{60\omega_n}{2\pi} = \frac{60\times0.3333\times10^3}{2\times\pi}$$

$$N = \frac{10000}{\pi} = \frac{10^4}{\pi}$$

1379. For a second order system with natural frequency 2 rad/sec and damping ratio ξ = 0.3, the value of peak time t_p without derivative feedback control is

(a) 2π secs (b) 5π secs
(c) 0.2π secs (d) 0.25π secs

APPSC AEE 2012

Ans. (*) : Natural frequency (ω_n) = 2rad/sec
Damping factor (ξ) = 0.3
Peak time (t_p) =?

$$t_p = \frac{\pi}{\omega_n\sqrt{1-\xi^2}} = \frac{\pi}{2\sqrt{1-0.3^2}} = \frac{\pi}{2\times0.954}$$

$$\boxed{t_p = 0.524\pi} \text{ Seconds}$$

1380. In the statement "an eccentric mass rotating at 3000 rpm will create X time more unbalanced force than 50% of the same mass rotating at 300 rpm", what is the value of X?

(a) 59 (b) 100
(c) 200 (d) 10 mew^2

APPSC AEE 2012

Ans. (c) : Unbalanced force, $F = mr\omega^2 = mr\left(\frac{2\pi N}{60}\right)^2$

$$\therefore F \propto mN^2 \qquad \left\{\omega = \frac{2\pi N}{60}\right\}$$

$$X = \frac{F_1}{F_2} = \frac{m_1N_1^2}{m_2N_2^2} = \frac{m\times(3000)^2}{\frac{m}{2}\times(300)^2} = 200$$

Then,

$$\boxed{X = \frac{F_1}{F_2} = 200}$$

1381. When a body vibrates under the influence of an external force. it is said to have

(a) Free vibrations
(b) Forced vibrations
(c) Damped vibrations
(d) Under damped vibrations

APGENCO AE 2012

Ans. (b) : When a body vibrates under the influence of an external force, its said to have forced vibrations.

1382. A shaft having one end fixed and a disc oscillating forcely at the other end. With increase in shaft stiffness, the natural frequency of vibration will :

(a) Increase (b) Decrease
(c) Remains same (d) Both (A) and (B)

APGENCO AE 2012

Ans. (a) : A shaft having one end fixed and a disc oscillating forcefully at the other end with increase in shaft stiffness, the natural frequency of vibration will increase.

$\boxed{F_{unbalance} = F_0 \sin\omega t}$

1383. A mass 'M' is suspended from a ceiling through a string of length 'L'. This mass moves in a circle of horizontal radius 'r' at constant

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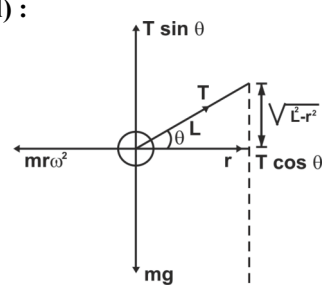
speed such that it is a conical pendulum. The tension in the string will be

(a) $mgL/(2\sqrt{L^2-r^2})$ (b) $mgL/\sqrt{L^2}$

(c) $mgL/\sqrt{r^2}$ (d) $mgL/(\sqrt{L^2-r^2})$

ISRO Scientist/Engineer 2012

Ans. (d) :



$T \sin\theta = mg$

$$\sin\theta = \frac{mg}{T}$$

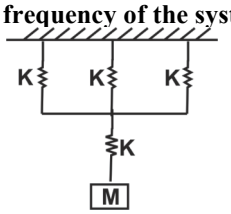
Or $$\sin\theta = \frac{\sqrt{L^2-r^2}}{L}$$

$\therefore$ $$T = \frac{mg}{\sin\theta}$$

$$T = \frac{mg}{\frac{\sqrt{L^2-r^2}}{L}}$$

$$T = \frac{mgL}{\sqrt{L^2-r^2}}$$

1384. Four linear elastic springs are connected to mass 'M' as shown in Figure. The natural frequency of the system is

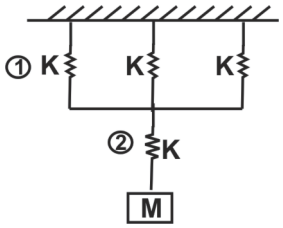


(a) $(\sqrt{4K/3m})/(2\pi)$ (b) $(\sqrt{4K/m})/(2\pi)$

(c) $(\sqrt{K/4m})/(2\pi)$ (d) $(\sqrt{3K/4m})/(2\pi)$

ISRO Scientist/Engineer 2012

Ans. (d) : Natural frequency $(f_n) = \frac{1}{2\pi}\sqrt{\frac{K_{eq}}{m}}$



$K_1 = 3K$

$K_2 = K$

$$K_{eq} = \frac{3K.K}{3K+K} = \frac{3K}{4}$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{3K}{4m}}$$

1385. In the viscous damped vibration, the logarithmic decrement value over five cycles is found to be 8.11. What is viscous damping factor of vibratory system?

(a) 20% (b) 25%

(c) 30% (d) 15%

ISRO Scientist/Engineer 2012

Ans. (b) : Logarithmic decrement value over five cycles = 8.11

Logarithmic decrement

$$\delta = \ln\frac{X_0}{X_5} = 5\zeta\omega_n T_d$$

where ζ = critical damping ratio or damping factor, δ = logarithmic decrement, ω_n = Natural frequency, T_d = Time of cycle

$$8.11 = 5 \times \zeta \times \frac{\omega_d}{\sqrt{1-\zeta^2}} \times \frac{2\pi}{\omega_d}$$

$\Rightarrow$ $\zeta = 0.25 = 25\%$

1386. Which of the following parameters has higher value during whirling of a shaft?

(a) Speed (b) Acceleration

(c) Frequency (d) Amplitude

BPSC AE 2012 Paper - VI

Ans : (d) : The speed at which the shafts runs so that the additional deflection of the shaft from the axis of rotation becomes infinite is known as critical or whirling speed. It is equal to the natural frequency of transverse vibrations but it's unit will be revolutions per second (r.p.s.)

1387. A machine mounted on a single coil spring has a period of free vibration of T. If the spring is cut into four equal parts and placed in parallel and the machine is mounted on them, then the period of free vibration of the new system will be

(a) 16 T (b) T/4

(c) 4 T (d) T/16

ISRO Scientist/Engineer 2011

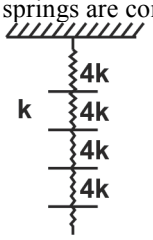
Ans. (b) : Time period

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \quad ...(1)$$

$$\omega = \sqrt{\frac{k}{m}}$$

When spring is cut 4 parts effective spring constant (k_{eq}) becomes 4k (for one spring).

When 4 springs are connected in parallel



$k_{eq} = 4k + 4k + 4k + 4k$

$k_{eq} = 16k$

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$$T' = 2\pi\sqrt{\frac{m}{16k}}$$

Hence, $T' = \frac{2\pi}{4}\sqrt{\frac{m}{k}}$ ...(2)

$$T' = \frac{T}{4}$$

1388. The rotor shaft of a large electric motor supported between short bearings at both the ends shows a deflection of 1.8 mm in the middle of the rotor. Assuming the rotor to be perfectly balanced and supported at knife edges at both the ends, the likely critical speed of the shaft is:

(a) 350 rpm (b) 705 rpm
(c) 2810 rpm (d) 4430 rpm

OPSC Civil Services Pre. 2011

Ans. (b) : Given,
Deflection in shaft $\delta = 1.8 \text{ mm} = 1.8 \times 10^{-3}$ m
Critical speed of shaft,

$$\omega_c = \omega_n = \sqrt{\frac{g}{\delta}} = \sqrt{\frac{9.81}{1.8\times10^{-3}}} = 73.824 \text{ rad/s}$$

$$N = \frac{1}{2\pi}\times\omega_c\times60 = \frac{73.824\times60}{2\pi}$$

$N = 704.967$ rpm

$\boxed{N = 705 \text{ rpm}}$

1389. A machine of mass 100 kg supported on four rubber pads at the four corners of the machine is found to have maximum amplitude of vibration in the vertical direction at a frequency of 10 rad/s. If the damping of the system is negligible, the stiffness of each rubber pad is

(a) 0.25 kN/m (b) 2.5 kN/m
(c) 10 kN/m (d) 40 kN/m

DRDO Scientists 2009

Ans. (b) : Frequency = 10 rad/s

$$\text{Frequency} = \sqrt{\frac{k_{eff}}{m}}$$

$k_{eff} = 10$ kN/m
$k_{eff} = 4k$

$$(k)_{\text{for rubber pad}} = \frac{10}{4} = 2.5 \text{kN/m}$$

1390. A precision electronic instrument of mass 10 gm is to be placed in a space vehicle whose surface vibrates with a frequency in the range of 20-40 rad/s. The vibration can be isolated if the stiffness of the instrument is

(a) less than 2 N/m (b) equal to 8 N/m
(c) equal to 2 N/m (d) more than 8 N/m

DRDO Scientists 2009

Ans. (a) : Natural frequency of precision instrument should be out of range of 20-40 rad/s
$\omega_n < 20$ rad/s or $\omega_n > 40$ rad/s

$$k < m\omega_n^2 = 10\times10^{-3}\times1600$$

or $k < 16$ N/m

1391. For a simply supported, long and slender shaft in the whirling of shaft experiment, the first critical speed is found to be at 100 rpm. When a point mass is placed at the middle of the shaft, the second critical speed will be at

(a) 100 rpm (b) 350 rpm
(c) 400 rpm (d) 420 rpm

DRDO Scientists 2009

Ans. (c) : $\omega_{cr} = n^2\sqrt{\frac{g}{\Delta}}$

n = 1 for first critical speed
n = 2 for second critical speed

$$(\omega_{cr})_2 = (\omega_{cr})_1 \times 4$$
$$= 4\times100 = 400$$

1392. In an under damped single degree of freedom vibrating system, if the amplitude ratio between the 1st and 4th cycle is 16, then the amplitude ratio between the 1st and 2nd cycle is

(a) ln 2 (b) ln 4
(c) 2 (d) 4

DRDO Scientists 2009

Ans. (c) : Amplitude ratio

$$\frac{X_1}{X_5} = \left(\frac{X_1}{X_2}\right)^4$$

$$16 = \left(\frac{X_1}{X_2}\right)^4$$

$$\Rightarrow \quad \left(\frac{X_1}{X_2}\right)^4 = 2^4$$

So

$$\therefore \quad \frac{X_1}{X_2} = 2$$

1393. For a spring-mass damper system with spring mass 0.9 kg, stiffness 100 N/m and an attached mass of 3.7 kg, the natural frequency of the system is

(a) 4.8 rad/s (b) 5 rad/s
(c) 5.2 rad/s (d) 6 rad/s

DRDO Scientists 2009

Ans. (b) : $\omega_n = \sqrt{\frac{k}{m + \frac{m_s}{3}}} = \sqrt{\frac{100}{3.7 + 0.3}} = 5$ rad/s

1394. Two springs of equal length but having stiffness of 10 N/mm and 15 N/mm support a mass of 2 ton in series. Find the frequency of vibration

(a) $\frac{1}{2\pi}\sqrt{3}Hz$

(b) $\frac{1}{2\pi}\sqrt{10}Hz$

(c) $\frac{1}{2\pi}\sqrt{15}Hz$

(d) $\frac{1}{2\pi}\sqrt{2}Hz$

ISRO Scientist/Engineer 2009

Ans. (a) : Let two springs of stiffness k_1 and k_2
Given, $k_1 = 10$ N/mm

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$k_2 = 15$ N/mm
$m = 2$ ton $= 2 \times 1000$ kg
1 ton = 1000 kg
When springs are in series

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\frac{1}{k_{eq}} = \frac{1}{10} + \frac{1}{15}$$

$k_{eq} = 6$ N/mm
Frequency of vibration

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{6\times10^3}{2000}}$$

$$f = \frac{1}{2\pi}\sqrt{3}\text{ Hz}$$

1395. If two gears have moment of inertias as I_1 and I_2 respectively and mesh with a speed ratio $\omega_2/\omega_1 = n$, then equivalent moment of inertia of both gears referred to first one is

(a) $n^2 I_1 + I_2$ (b) $I_1 + nI_2$

(c) $nI_1 + I_2$ (d) $I_1 + n^2 I_2$

ISRO Scientist/Engineer 2009

Ans. (d) : Given,

$$\frac{\omega_2}{\omega_1} = n$$

Power between two gears is given as

$$T_1\omega_1 = T_2\omega_2$$

$$T_1 = nT_2$$

If some external torque is applied on gear 1 to accelerate the system then the acceleration ratio is given by

$$\frac{\alpha_2}{\alpha_1} = n$$

The torque on 1 to accelerate only 1 is $T_1 = I_1\alpha_1$
The torque on 2 to accelerate only 2 is $T_2 = I_2\alpha_2 = I_2 n\alpha_1$
The torque on 1 to accelerate only 1 is given by

$$T_1\omega_1 = T_2\omega_2$$

$$T_1 = (I_2 n\alpha_1)(n) = I_2 n^2\alpha_1$$

So total torque on 1 to acceleration 1 and 2 is

$$I_{eq}\alpha_1 = T_1 + T_1$$

$$\Rightarrow \quad I_{eq}\alpha_1 = I_1\alpha_1 + n^2 I_2\alpha_1$$

$$\Rightarrow \quad I_{eq}\alpha_1 = (I_1 + n^2 I_2)\alpha_1$$

$$\Rightarrow \quad I_{eq} = (I_1 + n^2 I_2)$$

1396. The natural frequency of undamped vibrations in a single degree of freedom system is 10 rad/s. If the system is damped with damping constant $\xi = 2.0$, then what will be damped natural frequency of vibrations?

(a) Zero (b) Less than 10

(c) Equal to 10 (d) More than 10

UPSC JWM Adv. No-16/2009

Ans. (a) :

$\xi > 1 \Rightarrow$ Over damped system
[Over damping]
[Coulomb's damping]
$\xi = 1 \Rightarrow$ Critically damped system
[Critically damping]
$\xi < 1 \Rightarrow$ Under damped systems
[Under damping]
[Viscous damping]
Given, condition is overdamped vibration ($\xi = 2$)
Since there is no periodic vibration in overdamped system.
So damped natural frequency will be zero.

1397. Natural frequency (rad/s) of mass M on the free end of a cantilever beam of negligible mass length L and flexural rigidity EI (fig.) is

(a) $\sqrt{\frac{EI}{3ML^3}}$ (b) $\frac{1}{3}\sqrt{\frac{EI}{ML^3}}$

(c) $\sqrt{\frac{3EI}{ML^3}}$ (d) $\sqrt[3]{\frac{EI}{ML^3}}$

DRDO Scientists 2008

Ans. (c) : Natural frequency con.

$$\omega_n = \sqrt{\frac{g}{\Delta}}$$

For a cantilever beam

$$\Delta = \frac{MgL^3}{3EI}$$

$$\omega_n = \sqrt{\frac{3EI}{(ML^3)}}\text{rad/s}$$

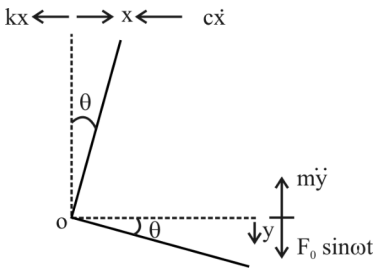
1398. For the system shown in figure, the angular displacement θ, the undamped natural frequency of the system in rad/s is

(a) $\frac{a}{b}\sqrt{\frac{k}{m}}$ (b) $\frac{b}{a}\sqrt{\frac{k}{m}}$

(c) $\sqrt{\frac{ak}{bm}}$ (d) $\sqrt{\frac{bk}{am}}$

DRDO Scientists 2008

Ans. (a) :



$$\frac{x}{a} = \frac{y}{b} = \theta$$

$$x = \frac{a}{b}y$$

$\sum M_o = 0$ (moment about pivot point)

$$\Rightarrow \quad (m\ddot{y})b + kxa + c\dot{x} \to a = F_0 b\sin\omega t$$

$$\Rightarrow \quad mb^2\ddot{\theta} + ka^2\theta + ca^2\dot{\theta} = F_0 b\sin\omega t$$

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Undamped natural frequency.

$$\omega_n = \left(\frac{ka^2}{mb^2}\right)^{1/2} = \frac{a}{b}\sqrt{\frac{k}{m}}$$

1399. For a one degree of freedom system described by the differential equation $10\ddot{x} + 200\dot{x} + 810x = 0$ (units as per SI system), which of the following is true?

(a) The system is under damped
(b) The system is over damped
(c) The system is critically damped
(d) The system has no damping

DRDO Scientists 2008

Ans. (b) : On comparing given equation with

$$\ddot{x} + (2\xi\omega_n)\dot{x} + \omega_n^2 x = 0$$

We will get

$$2\xi\omega_n = 20 \text{ and } \omega_n^2 = \frac{810}{10}$$

$$\xi\omega_n = 10 \text{ and } \omega_n = 9$$

$$\xi = \frac{10}{9} = 1.1$$

$$\xi > 1$$

(Hence, system is overdamped)

1400. The overshoot and the setting time are maximum with

(a) under damped system
(b) over damped system
(c) critically damped system
(d) damped system

TNPSC AE, 2008

Ans. (a) : The overshoot and the setting time are maximum with under damped system.

1401. In a continuous system, the number of degrees of freedom would be

(a) one (b) two
(c) three (d) four

WBPSC AE 2008

Ans. (*) : In a continuous system the number of degree of freedom would be infinite.

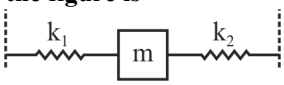
1402. Which type of vibration does the equation $\frac{d^2x}{dt^2} + \frac{K}{m}.x = 0$ represent?

(a) Forced vibration
(b) Free vibration
(c) Resonance vibration
(d) Periodically forced vibration

WBPSC AE 2008

Ans. (b) : Free vibration is used to indicate that there is no external force causing the motion, and that the motion is primarily the result of initial conditions, such as an initial displacement of mass element of the system from an equilibrium position.

1403. A block of mass 'm' is connected to two springs of stiffness K_1 and K_2 as shown in the figure f_1 and f_2 are the frequencies of the block when connected to K_1 and K_2 independently. Then the frequency of 'm' when connected to K_1 and K_2 as in the figure is



(a) $f = f_1 + f_2$ (b) $f = f_1^2 + f_2^2$
(c) $f = \sqrt{f_1^2 + f_2^2}$ (d) $f = \sqrt{f_1^2 - f_2^2}$

ISRO Scientist/Engineer 2007

Ans. (c) :

$$f_1 = \frac{1}{2\pi}\sqrt{\frac{k_1}{m}}$$

$$f_2 = \frac{1}{2\pi}\sqrt{\frac{k_2}{m}}$$

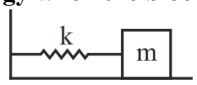
f = when spring are in parallel

$$k_{eq} = k_1 + k_2$$

$$f_{eq} = \frac{1}{2\pi}\sqrt{\frac{k_1 + k_2}{m}}$$

$$f_{eq} = \sqrt{f_1^2 + f_2^2}$$

1404. The block has a kinetic energy of 3J and the spring connecting the block has an elastic energy of 2J (shown in figure).
When the block is at x = +2.0 cm. What is the kinetic energy when the block is at x = 0.



(a) 3J (b) 2J
(c) 1J (d) 5J

ISRO Scientist/Engineer 2007

Ans. (d) : From energy conservation
[total energy at (x = 2 cm) = total energy at (x = 0)]

$$K.E_1 + P.E_1 = K.E_2 + P.E_2$$

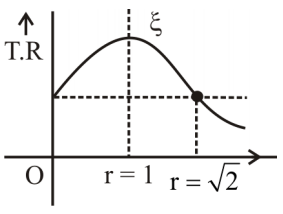
$$3 + 2 = K.E_2 \Rightarrow K.E_2 = 5J$$

1405. In a system for vibration isolation, if the frequency ratio is greater than zero and less than $\sqrt{2}$, then for all values of the damping factors, the transmissibility

(a) less than unity (b) equal to unity
(c) greater than unity (d) zero

WBPSC AE, 2007

Ans. (c) :



Curve between transmissibility ratio and r $\left(=\frac{\omega}{\omega_n}\right)$

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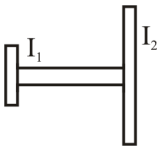
By this curve it can be inferred, when $0 < \frac{\omega}{\omega_n} < \sqrt{2}$

$\varepsilon > 1$

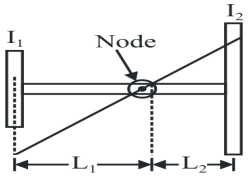
1406. In the two-rotor system shown in the figure, ($I_1 < I_2$), a node of vibration situated.



(a) Between I_1 and I_2 but nearer to I_1
(b) Between I_1 and I_2 but nearer to I_2
(c) Exactly in the middle of the shaft
(d) Nearer to I_1 but outside

ISRO Scientist/Engineer 2006

Ans. (b) :



$\ell_1 + \ell_2 = \ell$ .....(1)

$\sqrt{\frac{S_{T_1}}{I_1}} = \sqrt{\frac{S_{T_2}}{I_2}}$ .....(2)

(S_T = Torsional stiffness of shaft)

$\frac{S_{T_1}}{I_1} = \frac{S_{T_2}}{I_2}$

$\frac{GJ}{I_1 L_1} = \frac{GJ}{I_2 L_2} \Rightarrow \frac{L_1}{L_2} = \frac{I_2}{I_1}$

$L_1 I_1 = L_2 I_2$

From (1) and (2) we get,

given,

$\because \quad I_2 > I_1$

so, $L_2 < L_1$

So, the node of vibration is situated nearer to I_2 [because $I_2 > I_1$]

1407. A disc of moment of inertia I_1 is mounted on a shaft of moment of inertia I_2 . What is the natural frequency of torsional vibrations, if torsional rigidity of shaft is q.

(a) $\frac{1}{2\pi}\sqrt{\frac{q}{I_1}}$ (b) $\frac{1}{2\pi}\sqrt{\frac{q}{I_2}}$

(c) $\frac{1}{2\pi}\sqrt{\frac{3q}{3I_1 + I_2}}$ (d) $\frac{1}{2\pi}\sqrt{\frac{q}{I_1 + I_2}}$

J & K PSC Screening, 2006

Ans. (c) : $f_n = \frac{1}{2\pi}\sqrt{\frac{q}{I_1 + I_2/3}}$

$$\boxed{f_n = \frac{1}{2\pi}\sqrt{\frac{3q}{3I_1 + I_2}}}$$

1408. Which of the following relationships between the force F on a particle and the particles position x implies simple harmonic oscillation

(a) F= -7x (b) $F=-200x^2$
(c) F=10x (d) $F= 5x^2$

ISRO Scientist/Engineer 2006

Ans. (a) : Force F on a particle and the particles position x implies simple harmonic oscillation–

Force (F) = – kx

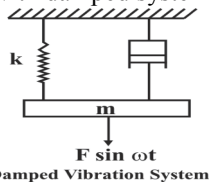
k = Spring constant

1409. With symbols having the usual meanings, the single degree of freedom system, $m\ddot{x} + c\dot{x} + kx = F\sin\omega t$ represents

(a) Free vibrations with damping
(b) Free vibrations without damping
(c) Forced vibrations with damping
(d) Forced vibrations without damping

ISRO Scientist/Engineer 2006

Ans. (c) : $m\ddot{x} + c\dot{x} + kx = F\sin\omega t$ represents Forced vibrations with damped system–



Damped Vibration System

1410. Polar moment in inertia (I_p) of a circular disc is to be determined by suspending it by a wire and noting the frequency of oscillations (f).

(a) $I_P \propto \frac{1}{f^2}$ (b) $I_P \propto f^2$

(c) $I_P \propto f$ (d) $I_P \propto \frac{1}{f}$

ISRO Scientist/Engineer 2006

Ans. (a) : Radius of gyration,

$k \propto \frac{1}{f_n}$

[$\because I = mk^2$]

$\therefore \quad \boxed{I \propto \frac{1}{f_n^2}}$

1411. For over damped system, damping factor will be-

(a) $\xi > 1$ (b) $\xi < 1$
(c) $\xi = 1$ (d) $\xi \le 1$

WBPSC AE 2003

Ans. (a) : The damping ratio is a measure describing how rapidly the oscillations decay from one bounce to the next. The damping ratio is a system parameter denoted by ξ.

for Undamped [$\xi = 0$]
Under damped [$\xi < 1$]
Over damped [$\xi > 1$],
Critically damped [$\xi = 1$]

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04.

MACHINE DESIGN

1. Design For Static and Dynamic Loading

1. The Notch sensitivity (q) is determined by the equation in terms of where K_f = fatigue stress concentration factor and K_t = theoretical stress concentration factor is:

(a) $(K_f + 1) / (K_t + 1)$ (b) $(K_f - 1) / (K_t + 1)$
(c) $(K_f + 1) / (K_t - 1)$ (d) $(K_f - 1) / (K_t - 1)$

CGPSC AE 15.01.2021
HPCL Engineer 12.08.2021
CIL MT 27.02.2020
UPPSC AE 13.12.2020, Paper-I
CGPSC AE 16.10.2016
KPSC ADF 2015
CGPSC AE 26.04.2015 Shift-I
GATE 2004

Ans. (d) : An estimate of the effect of a notch or hole of a given size and shape on the fatigue properties of a material measured by–

$$q = \frac{(k_f - 1)}{(k_t - 1)}$$

k_f – Fatigue notch factor
k_t – Stress concentration factor

2. Stress concentration factor is defined as the ratio of

(a) maximum stress to the endurance limit
(b) nominal stress to the enurance limit
(c) maximum stress to the nominal stress
(d) nominal stress to the maximum stress

RPSC IOF, 2020
Nagaland PSC (CTSE) 2018, Paper-I
GPSC EE Pre, 28.01.2017
CGPSC AE 16.10.2016
WBPSC AE 2008

Ans. (c) : As we know, the stress concentration factor is defined as the ratio of maximum stress to the nominal stress,

$$k_t = \frac{\sigma_{max}}{\sigma_{nominal}}$$

3. In static loading, stress concentration is more serious in

(a) Ductile material (b) Malleable material
(c) Elastic material (d) Brittle material

Gujarat PSC AE 2019
MPPSC State Forest Service Exam, 2014
J&K PSC Civil Services Pre, 2013
UPSC JWM Advt. No.-52/2010

Ans. (d) : The effect of stress concentration factor under static loading is not serious for ductile material because the geometry near the discontinuity is rearranged by the phenomenon of local yielding hence K_t can be neglected ($K_t = 1$).

The effect of K_t under static loading is more serious for brittle material because they does not permit any yielding hence K_t cannot be neglected for brittle materials.

Hence in static loading, stress concentration is more serious in Brittle materials.

4. Factor of safety is the ratio of

(a) lower yield and working stresses
(b) ultimate and working stresses
(c) upper yield and working stresses
(d) ultimate and lower yield stresses

APPSC AE Subordinate Service Civil/Mech. 2016
RPSC AE 2016
APPSC AE 2012
RPSC ACF-2011

Ans. (a) : Factor of safety is the ratio of lower yield stress to the working stress.

$$FOS = \frac{\text{Lower yield stress}}{\text{Working stress}}$$

* For ductile material, in factor of safety calculation yield stress is the criterion.
* For brittle material ultimate stress is the criterion in factor of safety calculation.

5. Stress concentration is caused due to :

(a) variations in load acting on a member
(b) variations in Properties of materials in a member
(c) abrupt change of cross-section
(d) all of the above

Punjab PSC SDE 12.02.2017
HPPSC Asstt. Prof. 18.09.2017
GWSSB AAE, 27.12.2015
UKPSC AE 2007 Paper -I

Ans. (c) : Stress concentration is defined as the localization of high stresses due to the irregularties present in the component and abrupt changes of the cross-section.

• **Stress concentration factor (K_t)–**

$$K_t = \frac{\text{Highest value of actual stress near discontinuity}}{\text{Nominal stress obtained by elementry equations for minimum cross section}}$$

$$K_t = \frac{\sigma_{max}}{\sigma_0} = \frac{\tau_{max}}{\tau_0}$$

• Fatigue Stress Concentration Factors (K_f)

$$K_f = \frac{\text{maximum stress in notched specimen}}{\text{stress limit in notch of free specimen}}$$

$$K_f = \frac{\text{Endurance limit in a notch free specimen}}{\text{Endurance limit of notched specimen}}$$

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6. Resistance to fatigue of a material is measured by:
(a) Young's modulus
(b) modulus of rigidity
(c) elastic limit
(d) endurance limit

RPSC AE 2016
RPSC ACF-2011
UKPSC AE 2007 Paper -I

Ans. (d) : Resistance to fatigue of a material is measured by endurance limit. The fatigue of a substance is the stress or pressure it undergoes as it is exposed to repetitive processes or cycles that lead to its structure breaking down and the material eventually fracturing.

7. Stress concentration in a machine component of ductile materials is not so harmful as it is in brittle materials because
(a) In ductile materials local yielding may distribute stress concentration
(b) Ductile materials have large Young's modulus
(c) Poisson's ratio is larger in ductile materials
(d) Modulus of rigidity is larger in ductile materials

MPPSC AE 2016
ESE 1996
GATE 1989

Ans : (a) The stress concentration effect in ductile material for static loading has no serious effect because there material undergoes local yielding and distribution the stress where maximum value is reached.

8. Endurance limit or fatigue limit is the maximum stress that a member can withstand for a infinite number of load applications without failure when subjected to
(a) Dynamic loading
(b) Static loading
(c) Combined static and dynamic loading
(d) Completely reversed loading

JPSC AE 10.04.2021, Paper-II
RPSC VPITI 14.02.2016
HPPSC Asstt. Prof. 2014
WBPSC AE 2003

Ans. (d) : Completely reverse loading represents a loading condition where an object is subjected to alternating tensile and compressive stress.
Where mean stress is zero.
In this endurance limit or fatigue limit is the maximum stress that a member can withstand for a finite number of cycles without failure when subjected to completely reversed loading.

9. Which process will improve the fatigue life of a part?
(a) shot peening (b) electroplating
(c) chemical coating (d) polishing

RPSC IOF, 2020
TNPSC AE 2019
GPSC EE Pre, 28.01.2017
RPSC AE 2016
TSPSC Managers, 2015
KPSC ADF 2015
SJVN ET 2013
TRB Poly. Lect., 2012
RPSC ACF-2011

Ans. (a) : Shot peening is a – Surface enhancement processes used to input compressive residual stresses into fatigue-prone metals this process increase fatigue strength by delaying the initiation of cracking.

10. If the size of a standard specification for fatigue testing machine is increased, then endurance limit for the material will
(a) Have same value as that of standard specification
(b) Increase
(c) Decrease
(d) None of these

UPPSC AE 13.12.2020 Paper-I
RPSC Vice Principal ITI 2018
ESE 1996

Ans. (c) : The endurance strength of non-standard sample or any other loaded component

$S_e = k_a.k_b.k_c.\ k_d.\ S'_e$

where S'_e – endurance strength of standard size specimen
k_a = size factor
k_b = surface finish factor
k_c = Reliability factor
k_d = Loading factor
Assuming all factors i.e. k_b and k_d as same as standard specimen i.e, unity. Endurance strength of a component.

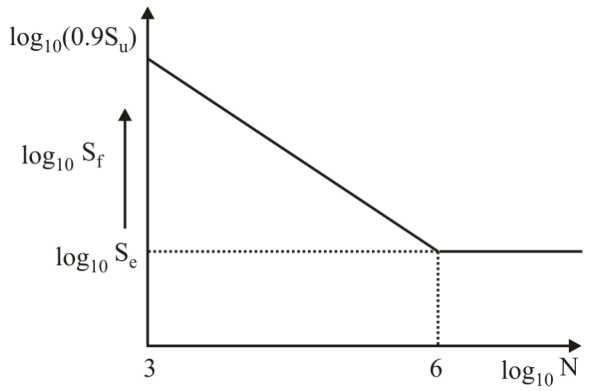
$\therefore S_e = k_a \times 1 \times 1.S'_e = k_a.S'_e$

We know that as size increases k_a reduces hence endurance limit of material decreases.

11. The S-N curve for steel becomes asymptotic nearly at
(a) 10^3 cycles (b) 10^4 cycles
(c) 10^6 cycles (d) 10^9 cycles

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Haryana PSC AE (PHED) 05.09.2020, Paper-II
Punjab PSC (Lect.) 06.08.2017
CGPSC AE 26.04.2015 Shift-I
VIZAG STEEL MT 18.08.2013
SJVN ET 2013
VIZAG STEEL MT 10.06.2012
GATE 2004

Ans. (c) : S N Diagram– By joining the 0.9 S_{ut} at 1000 cycles and S_e at 10^6 cycle by a straight line on a log s-log N graph we will get this S-N diagram for variable load.



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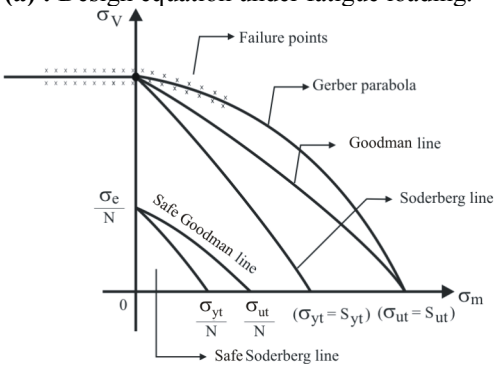
12. Which is a more conservative and commonly use failure criterion for designing parts subjected to mean plus alternating stresses?

(a) Soderberg line
(b) Modified –Goodman line
(c) Gerber parabola
(d) All of these

GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC Vice Principal ITI 2018
MPPSC AE 2016
MPPSC AE 08.11.2015
GATE 2015
MPSC HOD (Govt. Poly. Colleges) 04.10.2014
APPSC AEE 2012
ESE 1995
CGPSC Poly. Lect. 22.05.2016

Ans. (a) : Design equation under fatigue loading.



Soderberg equation

$$\frac{1}{N}=\left(\frac{\sigma_m}{\sigma_{yt}}\right)+k_f\left(\frac{\sigma_V}{\sigma_e}\right)$$

Good man line equation.

$$\frac{1}{N}=\left(\frac{\sigma_m}{\sigma_{ut}}\right)+k_f\left(\frac{\sigma_V}{\sigma_e}\right)$$

Where, σ_m = Mean stress
σ_v = Variable stress/Amplitude stress
σ_{yt} = Yield strength
σ_{ut} = Ultimate strength
σ_e = Endurance strength
N = Factor of safety (FS)

Soderberg equation is more conservative and commonly used failure criteria for designing part subjected to mean plus alternating stress.

13. Stress concentration in cyclic loading is more serious in ..... .

(a) ductile materials
(b) brittle materials
(c) equally serious in both cases
(d) depends on other factors

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GWSSB DEE 07.07.2016

Ans. (b) : Stress concentration in cyclic loading is more serious in brittle materials.

14. The failure that happens due to cyclic loading is called

(a) static failure (b) shear failure
(c) creep failure (d) fatigue failure

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : The failure that happens due to cyclic loading is called fatigue failure.

15. Which of the following is given by S-N curve for carbon steels?

(a) Ultimate Strength (b) Yield Strength
(c) Toughness (d) Fatigue limit

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : S-N curve consist of ultimate strength and life of the material, is drawn by using no. of revolution for cyclic loading on material to determine its fatigue limit which is less than ultimate strength.

16. A machine component is subjected to fluctuating stress that varies from 40 MPa to 100 MPa. The corrected endurance limit stress for the machine component is 270 MPa. The ultimate tensile strength and yield strength of material are 600 MPa and 450 MPa respectively. The factor of safety against fatigue as per Soderberg's criterion is :

(a) 3.75 (b) 4.39
(c) 4.5 (d) 5.41

RPSC ACF & FRO, 26.02.2021

Ans. (a) : Given,

$\sigma_1 = 40$ MPa
$\sigma_2 = 100$ MPa
$\sigma_e = 270$ MPa
$\sigma_{ut} = 600$ MPa
$\sigma_{yt} = 450$ MPa

$$\sigma_{mean}=\frac{100+40}{2}=70\ \text{MPa}$$

$$\sigma_{amp}=\frac{100-40}{2}=30\ \text{MPa}$$

From soderberg equation,

$$\frac{\sigma_{mean}}{\sigma_{yt}}+\frac{\sigma_{amp}}{\sigma_e}=\frac{1}{FOS}$$

$$\frac{70}{450}+\frac{30}{270}=\frac{1}{FOS}$$

$$\boxed{FOS = 3.75}$$

17. When steel is subjected to fatigue loading, the ratio of the endurance limit to the ultimate tensile strength is :

(a) 0.50 (b) 0.35
(c) 0.65 (d) 0.20

RPSC ACF & FRO, 26.02.2021

Ans. (a) : When steel is subjected to fatigue loading the ratio of the endurance limit to the ultimate tensile strength is 0.5.

$$\boxed{\sigma_e = 0.5\ \sigma_{ut}}$$

18. Allowable stress in ductile material is given by

(a) Ultimate strength/factor of safety
(b) Fracture strength/factor of safety
(c) Yield strength/factor of safety
(d) Elastic limit/factor of safety

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : For ductile material

$$\text{Working/allowable stress} = \frac{\text{Yield stress}}{FOS}$$

For brittle material

$$\text{Working/allowable stress} = \frac{\text{Ultimate stress}}{FOS}$$

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19. Which statement (s) is/are true?

I. A parabolic curve joining S_e on the ordinate to S_{ut} on the abscissa is called the Gerber line

II. A straight line joining S_e on the ordinate to S_{yt} on the abscissa is called the soderberg line

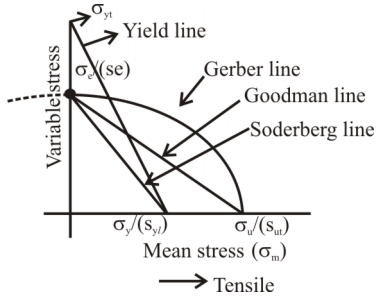
III. A straight line joining S_e on the ordinate to S_{ut} on the abscissa is called the Goodman line

IV. A straight line joining S_{yt} on the ordinate to S_{yt} on the abscissa is called the limit on first cycle of stress

(a) Only I is correct
(b) I, II and III are correct
(c) I, II and IV are correct
(d) All four are correct

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d)



Soderberg's Line : $\frac{\sigma_m}{S_{yt}}+\frac{\sigma_a}{\sigma_e}\le\frac{1}{N}$

Goodmann's Line : $\frac{\sigma_m}{S_{ut}}+\frac{\sigma_a}{\sigma_e}\le\frac{1}{N}$

Langer's Line : $\frac{\sigma_m}{S_{yt}}+\frac{\sigma_a}{S_{yt}}\le\frac{1}{N}$

ASME Ellipse : $\left(\frac{\sigma_m}{S_{yt}}\right)^2+\left(\frac{\sigma_a}{\sigma_e}\right)^2\le\frac{1}{N^2}$

Gerber Parabola : $\left(\frac{\sigma_m N}{S_{ut}}\right)^2+\left(\frac{\sigma_a N}{\sigma_e}\right)\le 1$

20. A machine part of steel with ultimate tensile strength = 500 MPa is subjected to a residual compressive stress of 50 MPa, the surface finish and sizes are such that combined fatigue strength reduction factor is 1.5. The effective fatigue strength is :

(a) 225.3 MPa (b) 183.3 MPa
(c) 150 MPa (d) 275.3 MPa

CGPSC AE 15.01.2021

Ans. (b) : Corrected endurance limit

$$\sigma_e=\frac{500+50}{2}=275\,\text{MPa}$$

The effective fatigue strength is $\sigma=\frac{275}{1.5}=183.3\,\text{MPa}$

21. Endurance limit of a beam subjected to pure bending decreases with

(a) decrease in the surface roughness and increase in the size of the beam
(b) increase in the surface roughness and decrease in the size of the beam
(c) decrease in the surface roughness and decrease in the size of the beam
(d) increase in the surface roughness and increase in the size of the beam

Assam Engg. College AP/Lect. 18.01.2021
GATE 2019
DRDO Scientists 2008

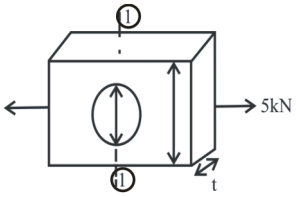
Ans. (d) : Endurance limit of a beam subjected to pure bending decreases with increase in the surface and increase in the size of the beam.

22. A flat plate 45 mm wide and "t" mm wide is subjected to a tensile force of 5 kN. The plate has a circular hole of diameter 30 mm with the centre coinciding with the diagonal intersection point of the rectangle. If stress concentration factor = 2.16, find the thickness of the plate if maximum allowable tensile stress is 80 N/mm^2 .

(a) 8 mm (b) 9 mm
(c) 10 mm (d) 12 mm

APGCL AM, 2021

Ans. (b) : Given,
K_t (stress concentration factor) = 2.16
τ_{max} = 80 N/mm^2
width = 45mm
thickness = t mm



Area of section 1-1 = (45-30)×t = 15t

Nominal stress $=\frac{5\times10^3}{(15t)}$

(induced stress)
at section 1-1

$K_t=\frac{\text{maximum stress}}{\text{Nominal stress}}$

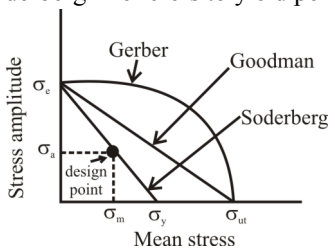
$2.16=\frac{80}{\frac{5\times10^3}{15t}}\Rightarrow t=\frac{2.16\times5\times10^3}{80\times15}=9\,\text{mm}$

23. A parabolic curve joining endurance limit on the ordinate and ultimate tensile strength on abscissa is called.

(a) Gerber Line
(b) Goodman's Line
(c) Soderbergs Line
(d) Modified Goodman's Line

APGCL AM, 2021
Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : Soderberg line refers to yield point failure.



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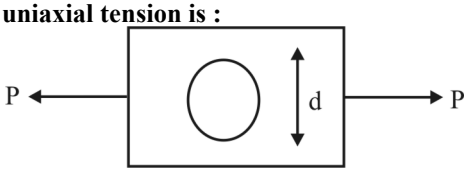
Soderberg Criterion $\frac{\sigma_a}{\sigma_e}+\frac{\sigma_m}{\sigma_y}=\frac{1}{FOS}$

Goodman Criterion $\frac{\sigma_a}{\sigma_e}+\frac{\sigma_m}{\sigma_{ut}}=\frac{1}{FOS}$

Gerber Criterion $\frac{FOS\times\sigma_a}{\sigma_e}+\left(\frac{FOS\times\sigma_m}{\sigma_{ut}}\right)^2=1$

Where,
σ_a = Stress amplitude
σ_e = Endurance limit
σ_m = Mean stress
σ_y = Yield point
σ_{ut} = Ultimate stress
FOS = Factor of safety

24. Stress concentration factor in an infinite plate with a transverse hole, loaded by in plane uniaxial tension is :



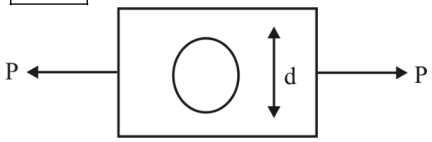
(a) 2 (b) 2.5
(c) 3 (d) 4

Kerala LBS Centre For Science & Tech. Asstt. Prof. 2014
JPSC AE 10.04.2021, Paper-II

Ans. (c) : For circular,

$K_t=1+\frac{2d}{d}$

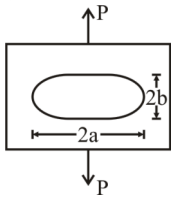
$K_t=3$



$K_t=3$ for infinite plate having transverse hole.

For ellipse,

$K_t=1+\frac{2a}{b}$



The stress concentration factor, K_t, is the ratio of maximum stress at a hole, fillet, or notch to the remote stress. For a hole in infinite plate, $K_t=3$

25. The limiting load beyond which the material no longer behaves elastically is known as
(a) Breaking Load
(b) Limiting Load
(c) Load Bearing Capacity
(d) Elastic Limit

APGCL AM, 2021

Ans. (d) : The property by virtue of which certain materials refers back to their original position offer the removal of the external force, is called elasticity.
The body will regain its previous shape and size only when the deformation caused by the external force, is within a certain limit. Thus there is limiting value of force for this and the value of stress corresponding to this limiting force is known as the elastic limit of the material.

26. The design calculations for members subject to fluctuating loads with the same factor of safety yields by using
(a) gerber (b) soderberg relation
(c) goodman relation (d) max stress relation

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : The design calculations for members subject to fluctuating loads with the same factor of safety yields by using goodman relation.

27. Consider the following statements regarding crack:
1. Crack is more likely to occur in the regions of discontinuity such as oil holes.
2. Crack is more likely to occur in the regions of irregularities in machining operations such as stamp mark.
3. Crack is more likely to occur in the internal cracks due to defects in materials like blow holes.

Which of the above statements are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) :
The crank is more likely to occur in the following regions:
(i) Regions of discontinuity, such as oil holes, keyways, screw threads etc.
(ii) Region of irregularitics in machining operation such as scratches on the surface stamp mark, inspection mark etc.
(iii) Internal crack due to defects in materials like blow holes.

28. A cast steel bar having an ultimate strength of 120 MPa is subjected to a reversed, repeated, bending load. The bar will be machined to a rectangular cross-section, 150 mm wide × 200 mm high. What is the equivalent diameter?
(a) 14 mm (b) 30 mm
(c) 140 mm (d) 300 mm

ESE 18.07.2021

Ans. (c) : For rectangular cross- section of width B and depth H, the equivalent diameter is-

$$de=\sqrt{\frac{0.05\times B\times H}{0.0766}}$$

$$=\sqrt{\frac{5\times150\times200}{7.66}}$$

$$=\boxed{de=139.93\approx140\,mm}$$

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29. The endurance limit of a material with finished surface in comparison to rough surface is

(a) more (b) less
(c) same (d) unpredictable

RPSC IOF, 2020
GWSSB DEE 07.07.2016

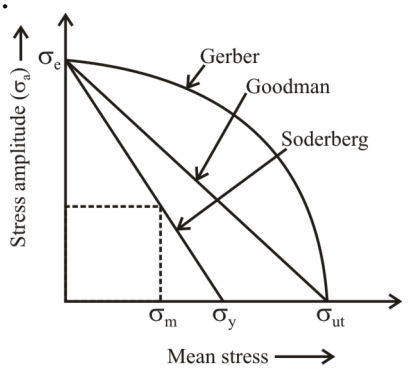
Ans. (a) : The endurance limit of a material is defined as the stress below which a material can endure and infinite number of repeated load cycles without exhibiting failure.
The endurance limit of a material with finished surface in comparison to rough surface is more.

30. A line joining σ_y (yield strength) on mean stress axis and σ_e (endurance limit) on stress amplitude axis is called as:

(a) Goodman line (b) Soderberg line
(c) Gerber line (d) Langer line

CIL MT 27.02.2020
TNCSC AE 2020

Ans. (b) :



1. Gerber Line–A parabolic curve joining σ_e on the ordinate to σ_{ut} on the abscissa is called the Gerber line.

2. Soderberg Line–A straight line joining σ_e on the ordinate to σ_y on the abscissa is called the soderber line.

$$\frac{\sigma_a}{\sigma_e}+\frac{\sigma_m}{\sigma_y}=\frac{1}{FS}$$

3. Goodman Line–A straight line joining σ_e on the ordinate to σ_{ut} on the abscissa is called the Goodman line.

$$\frac{\sigma_a}{\sigma_e}+\frac{\sigma_m}{\sigma_{ut}}=\frac{1}{FS}$$

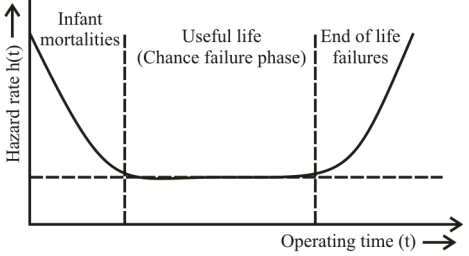
Where, σ_a = Stress amplitude σ_y = Yield point
σ_e = Endurance limit σ_{ut} = Ultimate stress
σ_m = Mean stress FOS = Factor of safety

31. In the Bath-Tub curve representing failure rate with respect to time, the useful life is represented by

(a) Debugging phase (b) Wear out phase
(c) Chance failure phase (d) All of these

RPSC IOF, 2020

Ans. (c) : The bath tub curve is generated by mapping the rate of early infant mortality failures when first introduced, the rate of random failures with constant failure rate during its "useful life" and finally the rate of "wear out" failures as the product exceeds its design life time.



Bath Tub Curve

32. Creep resistance decreases due to

(a) low stacking fault energy
(b) fine dispersed size
(c) high melting point
(d) small grain size

RPSC IOF, 2020

Ans. (d) : Creep resistant materials are used in machines and facilities operated at high temperature.
Small grain size lowers creep resistance and large grain size increases creep resistance ideal microstructure is as single crystal.

33.Fatigue strength does not increased by

(a) fine grain size
(b) shot peening
(c) grinding and lapping surface
(d) decarburization

RPSC IOF, 2020

Ans. (d) : During decarburization, the carbon diffuses from the surface at the metal, thus weakening the metal. This diffusion increases at higher temperatures the effect of decarburization not only brings down the strength, but also increases. The shear strain below the metal surface.

34. A 13 mm diameter tensile specimen has 50 mm gauge length. If the load corresponding to the 0.2% offset of 6800 kg, the yield stress will be nearly

(a) 31 kg/mm^2 (b) 43 kg/mm^2
(c) 51 kg/mm^2 (d) 63 kg/mm^2

ESE 2020

Ans. (c) : Yield stress, $\sigma=\dfrac{6800}{\frac{\pi}{4}\times(13)^2}=51.23\,kg/mm^2$

35. A component is subjected to varying tensile stresses such that maximum stress = 50 MPa and minimum stresses = 10 MPa. It is made of the material having yield stress = 300 MPa and endurance stress = 100 MPa. According to the Soderberg criteria, the factor of safety will be :

(a) 2 (b) 2.5
(c) 3.33 (d) 4

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given,
σ_{max} = 50 MPa, σ_{min} = 10 MPa
σ_y = 300 MPa, σ_e = 100 MPa
Factor of safety (FOS) = ?

$$\sigma_a=\frac{\sigma_{max}-\sigma_{min}}{2}$$
$$=\frac{50-10}{2}=20\text{ MPa}$$

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$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$$
$$= \frac{50+10}{2}$$
$$= 30 \text{ MPa}$$

Soderberg Equation –

$$\frac{\sigma_a}{\sigma_e} + \frac{\sigma_m}{\sigma_y} = \frac{1}{FOS}$$
$$\Rightarrow \frac{20}{100} + \frac{30}{300} = \frac{1}{FOS}$$
$$\Rightarrow \frac{1}{FOS} = \frac{1}{5} + \frac{1}{10} = \frac{3}{10}$$
$$\Rightarrow FOS = 10/3 = 3.3333$$
$$\Rightarrow \boxed{FOS \approx 3.33}$$

36. For the given material, if the stress concentration factor is 2 and Notch sensitivity is 0.1, then the fatigue stress concentration factor is :

(a) 1.0 (b) 1.2
(c) 1.1 (d) 0.9

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given $q = 0.1, k_t = 2$

We know that,

$$\text{Notch sensitivity } (q) = \frac{k_f - 1}{k_t - 1}$$
$$0.1 = \frac{k_f - 1}{2 - 1}$$
$$k_f = 0.1 + 1$$
$$k_f = 1.1$$

37. A 120 mm wide uniform plate is to be subjected to a tensile load that has a maximum value of 250 kN and a minimum value of 100 kN. The properties of the plate material are: endurance limit stress is 225 MPa, yield point stress is 300 MPa. If the factor of safety based on yield point is 1.5, the thickness of the plate will be nearly

(a) 12 mm (b) 14 mm
(c) 16 mm (d) 18 mm

ESE 2020

Ans. (a) : $\sigma_{max} = \frac{250 \times 10^3}{120 \times t}$, $\sigma_{min} = \frac{100 \times 10^3}{120 \times t}$

Where,

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2}$$
$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$$
$$\sigma_m = \frac{175 \times 10^3}{120 \times t}, \quad \sigma_a = \frac{75 \times 10^3}{120 \times t}$$
$$\frac{\sigma_m}{S_{yt}} + \frac{\sigma_a}{S_e} = \frac{1}{N}$$
$$\frac{175 \times 10^3}{120 \times t \times 300} + \frac{75}{120 \times t \times 225} = \frac{1}{1.5}$$
$$\frac{4.861}{t} + \frac{2.77}{t} = \frac{1}{1.5}$$
$$t = 11.45$$
$$\approx 12 \text{ mm}$$

38. For which one of the following loading conditions is the standard endurance strength multiplied by a load factor, Ke = 0.9?

(a) Reversed beam bending loads
(b) Reversed axial loads with no bending
(c) Reversed axial loads with intermediate bending
(d) Reversed torsion loads

ESE 2020

Ans. (c) : Reversed axial loads with intermediate bending.

39. A steel connecting rod having S_{ut} = 1000 MN/m^2, S_{yt} = 900 MN/m^2 is subjected to a completely reversed axial load of 50 kN. By neglecting any column action, if the values of k_e = 0.85, k_b = 0.9, k_a = 0.82, k_t = 1.5, q = 0.6 and N = 2, the diameter of the rod will be nearly

(a) 20 mm (b) 23 mm
(c) 26 mm (d) 29 mm

ESE 2020

Ans. (b) : $\frac{k_f - 1}{k_t - 1} = q$

$$\frac{k_f - 1}{1.5 - 1} = 0.6$$
$$k_f = 1.3$$
$$k_d = \frac{1}{k_f} = 0.769$$
$$\sigma_e^* = \frac{S_{ut}}{2} = 500$$
$$\sigma_e = k_a \times k_b \times k_c \times k_d \times k_e \ldots\ldots \sigma_e^*$$
$$\sigma_e = 0.82 \times 0.9 \times 0.85 \times 0.769 \times 500$$
$$= 241.19 \text{ MPa}$$

Strength criterion, Soderberg equation,

$$\sigma_{max} \le \sigma_{per}$$
$$\sigma_{max} \le \left[\frac{\sigma_e}{N} = \frac{\sigma_e^* k_a k_b k_c k_d}{N} \right]$$
$$\frac{50000}{\frac{\pi}{4} d^2} \le \frac{241.19}{2}$$
$$d \ge 22.97$$
$$d = 23 \text{ mm}$$

40. A cast iron bed plate for a pump has a crack length of 100 µm. If the Young's modulus of cast iron is 210 GN/m^2 and the specific surface energy is 10 J/m^2, the fracture strength required will be nearly

(a) 1.0×10^8 N/m^2 (b) 1.2×10^8 N/m^2
(c) 1.4×10^8 N/m^2 (d) 1.6×10^8 N/m^2

ESE 2020

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Ans. (b) : Given,
Crack length (c) = 100 μm =100×10^{-6} m
Young's modulus(E)= 210×10^{9} N/m^2
Specific surface energy(γ)= 10 J/m^2
Fracture strength for brittle material is given by

$$\sigma=\sqrt{\frac{2E\gamma}{\pi c}}=\sqrt{\frac{2\times210\times10^{9}\times10}{\pi\times100\times10^{-6}}}=1.15\times108\ \text{N/m}^2$$
$$\approx 1.2\times10^{8}\ \text{N/m}^2$$

41. A metal alloy machine component is subjected to fluctuating tensile stress from 100 N/mm^2 to 200 N/mm^2. The material has yield and endurance strength of 450 and 200 N/mm^2 respectively. Find the factor of safety of the machine component.
(a) 5/6 (b) 12/7
(c) 1/3 (d) 6/5

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : Given, σ_{max} = 200 N/mm^2, σ_{min} = 100 N/mm^2
σ_y = 450 N/mm^2, σ_e = 200 N/mm^2

$$\sigma_m=\frac{\sigma_{max}+\sigma_{min}}{2}=150\,\text{MPa}$$

$$\sigma_v=\frac{\sigma_{max}-\sigma_{min}}{2}=50\,\text{MPa}$$

By assuming Soderberg's equation,

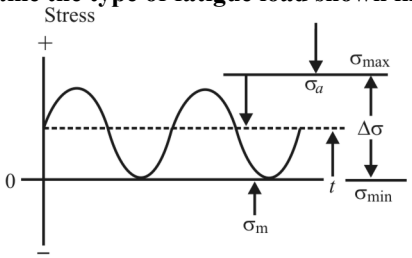
$$\frac{\sigma_m}{\sigma_y}+\frac{\sigma_v}{\sigma_e}=\frac{1}{N}$$
$$\frac{150}{450}+\frac{50}{200}=\frac{1}{N}$$
$$\Rightarrow \quad N=\frac{12}{7}$$

42. The process which does NOT improve the fatigue strength of a material is
(a) Shot peening of the surface
(b) Electroplating of the surface
(c) Polishing of the surface
(d) Cold rolling of the surface

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : The process which does not improve the fatigue strength of a material is electroplating of the surface.

43. Name the type of fatigue load shown in figure:



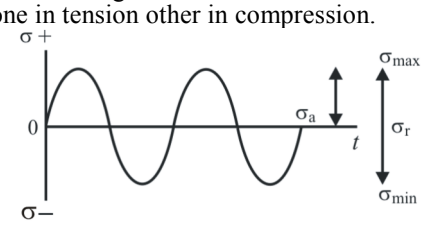
(a) Repeated (b) Reversed
(c) Fully reversed (d) Fluctuating

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

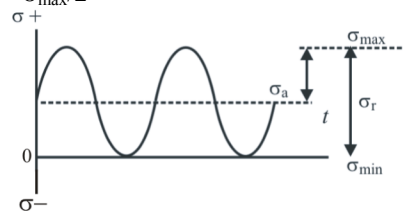
Ans. (a) : Fatigue Loading- Fatigue loading is primarily type of loading which causes cyclic variations in the applied stress or strain on a component. Thus any variable loading is basically a fatigue loading.

Types of variable (fatigue) loading /stress-

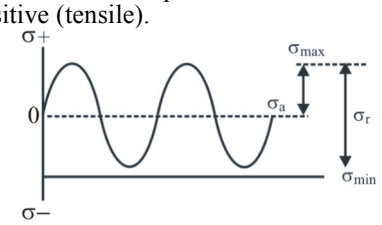
(i) Completely reversible loading/stressing- Stress variable is such that the mean stress is zero;
Same magnitude of maximum and minimum stress, one in tension other in compression.



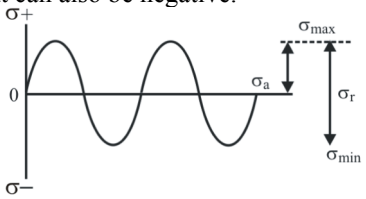
(ii) Repeated loading/stressing- Stress variation such that the minimum stress is zero. Mean and amplitude stress have the same value for repeated loading σ_{min} = 0, $\sigma = \sigma_a = \sigma_{max}/2$



(iii) Fluctuating stressing- Both minimum and maximum stresses are positive and mean stress also being positive (tensile).



(iv) Alternating stressing- Positive maximum stress and negative minimum stress, mean stress is generally positive but can also be negative.



44. The areas of fatigue failure in a part may be in the
(1) region having slow growth of crack with a fine fibrous appearance.
(2) region having faster growth of crack with a fine fibrous appearance.
(3) region of sudden fracture with a coarse granular appearance.
(4) region of gradual fracture with a coarse granular appearance.
Select the correct answer using the code given below.
(a) 2 and 3 (b) 2 and 4
(c) 1 and 4 (d) 1 and 3

ESE 2019

Ans. (d) : Fatigue failure is recognised through
(i) Crack having fine fibrous appearance associated with slow growth rate.
(ii) Crack having coarse granular appearance associated with faster growth rate.

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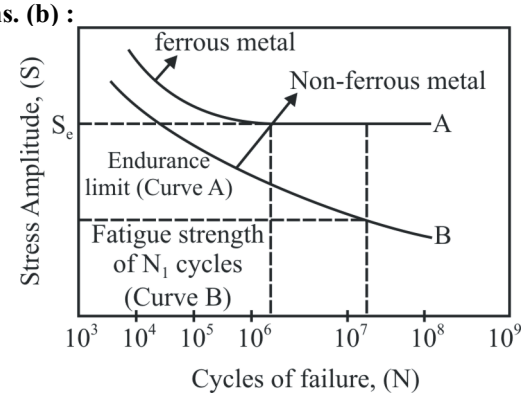
45. Fatigue strength for non-ferrous materials is usually defined at _____ stress cycles.
(a) 10^4 (b) 10^7
(c) 10^9 (d) 10^{12}

AAI Jr. Executive 29.11.2018
ISRO Scientist/Engineer 07.05.2017

Ans. (b) :



For non-ferrous materials, there is no particular limiting stress i.e. as stress reduce, number of cycle to failure keep increasing. Hence corresponding to 10^7 cycle is considered as characteristic of material and known as fatigue strength and corresponding number of cycle is called fatigue life.

46. Notch sensitivity varies between:
(a) 0 – 10 (b) 0.1 – 100
(c) 1 – 100 (d) 0 – 1
(e) 0 – 0.1

CGPSC AE 25.02.2018

Ans. (d) : Notch sensitivity Factor (q)– It is defined as the ratio of increase of actual stress over nominal stress in fatigue to the increase stress over nominal stress in static loading.

$$q = \frac{K_f - 1}{K_t - 1} \qquad \boxed{0 \le q \le 1}$$

where, K_f = actual (or) fatigue stress concentration factor

K_t = Theoretical (or) static stress concentration factor.

47. In a structure subjected to fatigue loading, the minimum and maximum stresses developed in a cycle are 200MPa and 400MPa respectively. The value of stress amplitude (in MPa) is
(a) 600 (b) 300
(c) 200 (d) 100

GPSC Executive Engineer 23.12.2018

Ans. (d) : Given:
Minimum stress (σ_{min}) = 200 MPa
Maximum stress (σ_{max}) = 400 MPa
Stress amplitude is given by,

$$\sigma_a = \frac{400 - 200}{2} = 100\,\text{MPa}$$

48. The process of shot peening increases the fatigue life of steel spring mainly because it results in
(a) Damping of vibration
(b) Surface hardening
(c) Increased stiffness of the material
(d) Residual compression at the surface

GPSC Executive Engineer 23.12.2018

Ans. (d) : The process of shot peening increases the fatigue life of steel spring mainly because it results in residual compression at the surface.

49. Which theory of failure will you use for steel components under steady loading?
(a) Principal stress theory
(b) Principal strain theory
(c) Strain energy theory
(d) Maximum shear stress theory

TSPSC AEE 2017

Ans. (d) : The maximum shear stress theory states that failure occurs when the maximum shear stress from a combination of principal stresses equal or exceeds the value obtained for the shear stress at yielding in the uniaxial tensile test or steady load.

50. If theoretical stress concentration factor is 0.4 and fatigue stress concentration factor is 0.6, then the notch sensitivity is
(a) 2/3 (b) 4/3
(c) 3/2 (d) 3/4

TSPSC AEE 2017

Ans. (a) : Notch sensitivity is given by –

$$q = \frac{k_f - 1}{k_t - 1} = \frac{0.6 - 1}{0.4 - 1} = \frac{0.4}{0.6} = \frac{2}{3}$$

51. Find the thickness of a 120 mm wide uniform plate for safe continuous operations, if the plate is to be subjected to tensile load that has a maximum value of 250 kN and minimum value 100 kN. The properties of the plate materials are Endurance limit stress = 225 MPa
Yield point stress = 300 MPa
Factor of safety = 1.5
(a) 11.46 mm (b) 5.82 mm
(c) 8.53 mm (d) 9.46 mm

Karnataka PSC Lect., 27.05.2017

Ans (a) : Given, b = 120 mm, W_{max} = 250 kN, W_{min} = 100 kN, σ_e = 225 MPa = 225 N/mm^2, σ_y = 300 MPa = 300 N/mm^2, F.O.S. = 1.5
Let t = Thickness of plate in mm.
$\therefore$ Area A = b × t = 120t mm^2
We know that mean or average load,

$$W_{mean} = \frac{W_{max} + W_{min}}{2} = \frac{250 + 100}{2}$$

$$W_{mean} = 175\text{kN} = 175 \times 10^3\,\text{N}$$

$$\therefore \text{Mean stress } (\sigma_m) = \frac{W_{mean}}{A} = \frac{175 \times 10^3}{120t}\,\text{N/mm}^2$$

$$\text{Variable load } (W_v) = \frac{W_{max} - W_{min}}{2} = \frac{250 - 100}{2}$$

$$W_v = 75\text{ kN} = 75 \times 10^3\,\text{N}$$

$$\therefore \text{Variable stress } (\sigma_v) = \frac{W_v}{A} = \frac{75 \times 10^3}{120t}\,\text{N/mm}^2$$

According to Soderberg's formula,

$$\frac{1}{\text{F.O.S.}} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v}{\sigma_e}$$

$$\frac{1}{1.5} = \frac{175 \times 10^3}{120t \times 300} + \frac{75 \times 10^3}{120t \times 225} = \frac{4.86}{t} + \frac{2.78}{t} = \frac{7.64}{t}$$

$$\frac{1}{1.5} = \frac{7.64}{t}$$

t = 7.64 × 1.5 = 11.46 mm

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52. Short time fatigue life testing is done by which of the following?

(a) Soderberg's equation
(b) Goodman's equation
(c) Gerber's equation
(d) Miner's equation

Karnataka PSC AE, 10.09.2017

Ans. (d) : Miner's equation is a way to calculate the fatigue damage of a material through a linear damage accumulation model and finding short life.

53. What is the length of lap for a flat lap brazing joint if the factor of safety is 2, the tensile strength is 600 MPa, shear strength of braze filler metal is 300 MPa and the wall thickness of thinner member is 10 mm?

(a) 20 mm (b) 30 mm
(c) 50 mm (d) 40 mm

TSPSC AEE 2017

Ans. (d) : Given,
length of lap for a flat lap brazing joint = ?
(Y) Factor of safety = 2
(T) Tensile strength = 600 MPa
(L) Shear strength of braze filler metal = 300 MPa
(W) Thickness of thinner member = 10 mm
$\because$ length of lap area

$$X = \frac{Y.T.W}{L}$$

$$X = \frac{2 \times 600 \times 10}{300}$$

$$= 2 \times 2 \times 10$$

$$= 40 \text{ mm}$$

54. Which type of residual stresses improve fatigue strength of the weld?

(a) Uniaxial tensile stress
(b) Biaxial tensile stress
(c) Triaxial tensile stress
(d) Compressive stress

Punjab PSC SDE 12.02.2017

Ans. (d) : Improve Fatigue Strength–

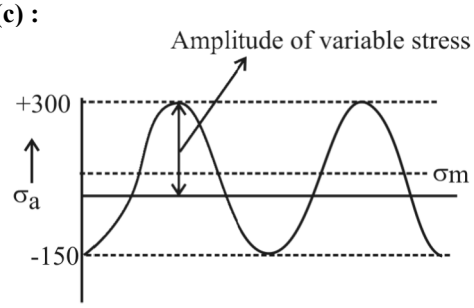
- By residual compressive stresses fatigue failure is always a tensile failure. Therefore residual compressive stresses will counter it. This is done by the process called shot peening.
- By providing fillets, we can change its microstructure thus we can basically increase its mechanical property that is fatigue strength.
- **Hammering-** The hammering process has been used to improve the fatigue resistance of the components.
- **Cold rolling-** Cold rolling gives excellent surface finish and increases material strength due to work hardening.
- **Burnishing-** In this metal is plastically deformed and convert the uneven surface of workpiece at normal temperature to smooth surface so as to change the surface structure, mechanical properties, shape and size. It improves the fatigue and wear strength of the workpiece.

55. A machine component is subjected to a flexural stress which fluctuates between +300 MN/m^2 and –150MN/m^2, then the variable stress is

(a) 250 MN/m^2 (b) 200 MN/m^2
(c) 225 MN/m^2 (d) 75 MN/m^2

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) :



σ_a = amplitude stress
σ_m = mean stress

The mean stress = $\frac{1}{2}(\sigma_{max} + \sigma_{min})$

$$\sigma_m = \frac{1}{2}(300 - 150)$$

$$\sigma_m = 75 \text{ MN/m}^2$$

The amplitude of stress over mean stress is called amplitude of variable stress and this value from graph is

$$\sigma_v = 300 - 75$$

$$\sigma_v = 225 \text{ MN/m}^2$$

II[nd] Method–

$$\sigma_v = \frac{(\sigma_1 - \sigma_2)}{2} = \frac{300 - (-150)}{2}$$

$$= \frac{450}{2} = 225 \text{ MN/m}^2$$

56. A bar is subjected to fluctuating tensile load from 20 kN to 100 kN. The material has yield strength of 240 MPa and endurance limit in reversed bending is 160 MPa. According to the Soderberg principle, the area of cross-section in mm^2 of the bar for a factor of safety of 2 is

(a) 400 (b) 600
(c) 750 (d) 1000

GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (d) : Soderberg criteria is :

$$\frac{\sigma_a}{\sigma_e} + \frac{\sigma_m}{\sigma_y} = \frac{1}{FOS}$$

$$\sigma_m = \frac{100 + 20}{2A} = \frac{60}{A}$$

$$\sigma_a = \frac{100 - 20}{2A} = \frac{80}{2A} = \frac{40}{A}$$

$$\Rightarrow \quad \left[\frac{40}{160A} + \frac{60}{240 \times A}\right] \times 1000 = \frac{1}{2}$$

$$A = 1000 \text{ mm}^2$$

57. According to the Gerber method, the variable stress is given by:
(where σ_m - mean stress, σ_u - ultimate stress and σ_e - endurance limit for reversible loading and FS - factor of safety)

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(a) $\sigma_v = \sigma_e \left[\frac{1}{FS} - \left(\frac{\sigma_m}{\sigma_u} \right)^3 FS \right]$

(b) $\sigma_v = \sigma_e \left[\frac{1}{FS} + \left(\frac{\sigma_m}{\sigma_u} \right)^3 FS \right]$

(c) $\sigma_v = \sigma_e \left[\frac{1}{FS} - \left(\frac{\sigma_m}{\sigma_u} \right)^2 FS \right]$

(d) $\sigma_v = \sigma_e \left[\frac{1}{FS} + \left(\frac{\sigma_m}{\sigma_u} \right)^2 FS \right]$

(e) $\sigma_v = \sigma_e \left[\frac{1}{FS} - \left(\frac{\sigma_m}{\sigma_u} \right)^2 \right]$

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : According to the Gerber method, the variable stress is given by

$$\sigma_v = \sigma_e \left[\frac{1}{FS} - \left(\frac{\sigma_m}{\sigma_u} \right)^2 \times FS \right]$$

58. The failure of a material under varying load, after a number of cycles of such load, is known as

(a) Ductile failure (b) Brittle failure
(c) Impact failure (d) Fatigue failure

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : The failure of a material under varying load, after a number of cycles of such load, is known as fatigue failure, material to be fracture by means brittle cracking under repeated cyclic stresses.

59. The fatigue strength of mild steel is:

(a) Lower than the yield strength
(b) More than the yield strength
(c) More than its tensile strength
(d) Lower than its tensile strength

TRB Polytechnic Lecturer 2017

Ans. (a) : The fatigue strength of mild steel is lower than the yield strength.

60. The stress-concentration factor (K) is:

(a) Ratio of maximum stress occurring near discontinuity to average stress at critical section.
(b) Ratio of minimum stress occurring near discontinuity to average stress at critical section.
(c) Ratio of average stress at critical section to minimum stress occurring near discontinuity.
(d) Ratio of average stress at critical section to maximum stress occurring near discontinuity.

CIL MT 2017 IInd shift

Ans. (a) : Stress concentration factor (K) is defined as the ratio of the maximum stress in a member (at a notch or a fillet) to the nominal or average stress at the same section based upon net area. The value of K_t depends upon the material and geometry of the part.

61. Which of the following characteristics is generally exhibited by brittle materials?

(a) High impact strength
(b) Ultimate compressive strength higher than ultimate tensile strength
(c) Good fracture toughness
(d) All of the above

BPSC AE Mains 2017 Paper - VI

Ans : (b) : Ultimate compressive strength higher than ultimate tensile strength.

62. A machine component is subjected to a flexural stress, which fluctuates between 300 MN/m^2 and -150 MN/m^2. Taking the yield strength = 0.55 of the ultimate strength, endurance strength = 0.50 of the ultimate strength and factor of safety to be 2, the value of the minimum ultimate strength according to modified Goodman relation will be

(a) 1100 MN/m^2 (b) 1075 MN/m^2
(c) 1050 MN/m^2 (d) 1025 MN/m^2

ESE 2017

Ans. (c) : Given,

$\sigma_{max} = 300$ MPa

$\sigma_{min} = -150$ MPa

$$\text{Mean stress } (\sigma_m) = \frac{\sigma_{max} + \sigma_{min}}{2} = \frac{300 - 150}{2} = 75 \text{ MPa}$$

$$\text{Alternating stress } (\sigma_a) = \frac{\sigma_{max} - \sigma_{min}}{2}$$

$\sigma_a = \sigma_v$ = Alternative (or) Variable stress

$$= \frac{300 - (-150)}{2} = 225 \text{ MPa}$$

Equation for goodman line

$$\frac{\sigma_a}{\sigma_e} + \frac{\sigma_m}{\sigma_{ut}} = \frac{1}{FS}$$

$$\frac{225}{0.5\,\sigma_{ut}} + \frac{75}{\sigma_{ut}} = \frac{1}{2}$$

$\sigma_{ut} = 1050$ MPa

63. Consider the following statements:
For a component Made of ductile material, the failure criterion will be

1. endurance limit, if the external force is fluctuating.
2. fatigue, if the external force is fluctuating.
3. yield stress, if the external force is static.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) : (1) Endurance limit is the maximum stress amplitude of a component can withstand. It is the strength of material.
(2) Failure of ductile material under fluctuating load is called fatigue.
(3) Failure of ductile material under static load occurs at yield stress.

64. Factor of safety for brittle materials is the ratio of

(a) Maximum stress to working stress
(b) Yield stress to working stress
(c) Ultimate stress to working stress
(d) All of the above

Karnataka PSC Lect., 27.05.2017

Ans (c) : Factor of safety for brittle material is the ratio of ultimate stress to working stress.

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65. The factor of safety for machine parts subjected to reversed stresses is :

(a) ratio of yield strength to maximum stress
(b) ratio of endurance limit to amplitude stress
(c) ratio of ultimate tensile strength to maximum stress
(d) ratio of endurance limit to mean stress

HPPSC Asstt. Prof. 20.11.2017
RPSC Vice Principal ITI 2018

Ans. (b) : Factor of Safety (FOS) $= \dfrac{\text{Failure stress}}{\text{Permissible stress}}$

For machine parts subjected to reversed stresses.

$$F.O.S = \frac{\text{Endurance limit}}{\text{Amplitude stress}}$$

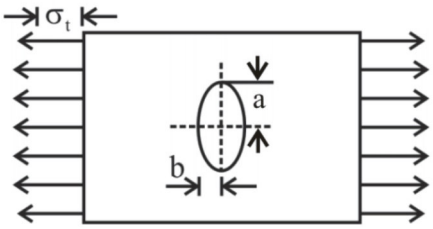
66. A large uniform plate containing the hole is subjected to a uniform axial tension 'σ'. The maximum stress in the plate is

(a) $\sigma_{max} = 5\sigma$ (b) $\sigma_{max} = 2\sigma$
(c) $\sigma_{max} = 3\sigma$ (d) $\sigma_{max} = \sigma$

RPSC VPITI 14.02.2016
APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Stress concentration factor for elliptical hole

$$K_t = 1 + 2\left(\frac{a}{b}\right)$$



For circular hole a = b

$$K_t = 1 + 2\left(\frac{a}{a}\right)$$

$$K_t = 3$$

$$K_t = \frac{\sigma_{max}}{\sigma_0}$$

$$3 = \frac{\sigma_{max}}{\sigma_0}$$

$$\sigma_{max} = 3\sigma_0$$

67. The endurance limit is found by testing specimen of standard size, usually 7.5 mm, in a standard rotating beam test. What will be the actual endurance limit of parts with size greater than used in the test?

(a) less than the value determined with standard test.
(b) more than the value determined with standard test.
(c) same as the value determined with standard test.
(d) less than the value determined with standard test upto certain size than it will increase with the increasing size.

RPSC AEN Pre-2013

Ans. (a) : Fatigue Testing Machine–In this test the rotating beam specimen is small with 7.5 mm diameter. The larger the machine part, the greater the probability that a flow exists some where in the component. The chances of fatigue failure originating at any one of these flows are more. The endurance therefore reduces with increasing size of the component.

Actual endurance strength $\sigma_e = \sigma \cdot k_a \cdot k_b \cdot k_c \cdot k_d$

k_a = Size factor

As size increases, size factor decreases.

Endurance limit of parts less than the value determined with standard test.

68. Few statements about creep in materials are given below :

(a) Creep means regaining the original shape after removal of stresses.
(b) Creep implies development of additional strains when the material is loaded for prolonged period of time.
(c) Creep is more at elevated temperatures
(d) Creep means loading the material in plastic range.

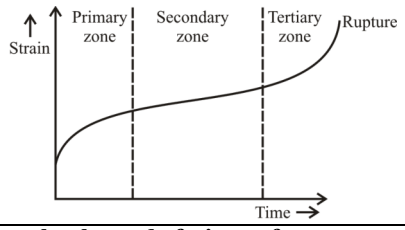
Which of the above statements are all correct?

(a) (a), (b) and (c) (b) (a) and (d)
(c) (b) and (c) (d) (b) and (d)

RPSC AEN Pre-2013

Ans. (c) : To about creep in materials –

- Creep implies development of additional strains when the material is loaded for prolonged period of time.
- Creep is more at elevated temperatures.



69. The shock and fatigue factors used in the design of shafts as per ASME for the case of gradually applied load are respectively

(a) 1.0 and 1.0 (b) 1.0 and 1.5
(c) 1.25 and 1.0 (d) 1.5 and 1.0

RPSC AE (GWD) 06.05.2016

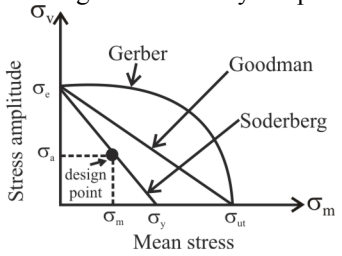
Ans. (d) : 1.5 and 1.0

70. Soderberg's line refers to :

(a) Fracture failure of material
(b) Yield point failure
(c) Elastic limit
(d) Fatigue Failure

RPSC AE GWD, 2011

Ans. (b) : Soderberg line refers to yield point failure.



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Soderberg Criterion $\boxed{\frac{\sigma_a}{\sigma_e}+\frac{\sigma_m}{\sigma_y}=\frac{1}{FOS}}$

Goodman Criterion $\boxed{\frac{\sigma_a}{\sigma_e}+\frac{\sigma_m}{\sigma_{ut}}=\frac{1}{FOS}}$

Gerber Criterion $\boxed{\frac{FOS\times\sigma_a}{\sigma_e}+\left(\frac{FOS\times\sigma_m}{\sigma_{ut}}\right)^2=1}$

Where,
σ_a = Stress amplitude
σ_e = Endurance limit
σ_m = Mean stress
σ_y = Yield point
σ_{ut} = Ultimate stress
FOS = Factor of safety

71. A plate of rectangular section is 101 mm side and 10 mm thick. The plate has 1 mm diameter circular hole in the centre of width. It is subjected to a tensile load of 10 kN along the longitudinal axis. The tensile stress across the section of hole is:

(a) 10 N/mm^2 (b) 20 N/mm^2
(c) 30 N/mm^2 (d) 40 N/mm^2

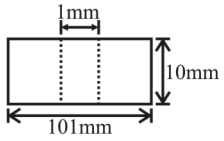
GPSC Asstt. Prof. 28.08.2016

Ans. (c) : $F = \sigma \times A$

$10 \times 10^3 = (b - d_h) \times t \times \sigma_0$

$10 \times 10^3 = (101 - 1) \times 10 \times \sigma_0$

$\boxed{\sigma_0 = 10\,N/mm^2}$



Side view of plate

And theoretical stress concentration factor,

$k_t = 1+\frac{2a}{b}$

$k_t = 1+\frac{2a}{a}$ (For circular hole)

$\boxed{k_t = 3}$

And $k_t = \frac{\sigma_{max}}{\sigma_0} \Rightarrow \sigma_{mx} = k_t \times \sigma_0$

$\boxed{\sigma_{max} = 3\times10 = 30\ N/mm^2}$

72. A ductile material having an endurance limit of 225 N/mm^2 and the yield point at 300 N/mm^2 is stressed under variable load. The maximum and minimum stresses are 150 N/mm^2 and 50 N/mm^2. The fatigue stress concentration factor is 1.5. What is the available factor of safety for this load?

(a) 1.5 (b) 1.33
(c) 2.0 (d) 3.0
(e) 4.0

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : As we know, Soderberg equation,

$\frac{\sigma_a}{S_e}+\frac{\sigma_m}{S_y}=\frac{1}{FOS}$

$\sigma_a = \frac{150-50}{2} = 50\,MPa$

$\sigma_m = \frac{150+50}{2} = 100\,MPa$

$S_e = \frac{225}{1.5} = 150\,MPa$

So, $\frac{50}{150}+\frac{100}{300}=\frac{1}{FOS}$

$\frac{1}{FOS}=\frac{2}{3}$

$\boxed{FOS = 1.5}$

73. The form stress factor K_r the actual stress concentration factor K and notch sensitivity q are related by an expression

(a) $K_r = 1 + q\,(K-1)$
(b) $K = 1 + q\,(K_r - 1)$
(c) $q = 1 + K\,(K_r - 1)$
(d) $q = 1 - K\,(K_r + 1)$

WBPSC AE, 2007

Ans. (a) Notch sensitivity is given by –

$q = \frac{K_r - 1}{K - 1}$

$q \times (K-1) = K_r - 1$

$\boxed{K_r = 1 + q(K-1)}$

74. Soderberg and Goodman equations are concerned with the determination of the factor of safety for

(a) static stresses
(b) variable stresses
(c) stress concentration effect
(d) either (a) or (c) above

MPPSC State Forest Service Exam, 2014
WBPSC AE, 2007

Ans. (b) : Soderberg and Goodman equations are used to determine factor of safety when the material is subjected to fatigue loading i.e. alternating or variable stresses.

$\frac{\sigma_m}{S_{yt}}+\frac{\sigma_a}{\sigma_e}\le\frac{1}{N}$ (Soderberg equation)

$\frac{\sigma_m}{S_{ut}}+\frac{\sigma_a}{\sigma_e}\le\frac{1}{N}$ (Goodman equation)

σ_m = mean stress
σ_a = stress amplitude
σ_e = endurance strength

75. Soderberg line is used for designing machine elements made of

(a) brittle materials
(b) ductile materials
(c) elastic materials
(d) both brittle and ductile materials

TRB Asstt. Prof., 2012

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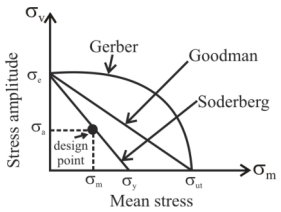
Ans. (b) :



- Soderberg line – Ductile material (Yield point fracture)
- Goodman line – Brittle material.

76. The size factor is employed to find out the endurance limit of a part from the endurance limit of the material (as determined from the standard rotating beam specimen). For sizes greater than 50 mm, this factor is

(a) 0.85 (b) 0.75
(c) 0.9 (d) 1.25

RPSC AE (GWD) 06.05.2016

Ans. (b) :

Diameter (d) mm	k_b (Value of size factor)
$D \le 07.5$	1.00
$7.5 < d \le 50$	0.85
$D > 50$	0.75

77. Which of the following prevents yielding in the material?

(a) Goodman line (b) Soderberg line
(c) Gerber parabola (d) Distortion elliptic line

WBPSC AE 2008

Ans. (b) : Soderberg line prevents yielding in the material.

78. A thin spherical pressure vessel of 200 mm diameter and 1 mm thickness is subjected to an internal pressure varying from 4 to 8 MPa. Assume that the yield, ultimate and endurance strength of material are 600, 800 and 400 MPa respectively. The factor of safety as per Goodman's relation is :

(a) 2.0 (b) 1.6
(c) 1.4 (d) 1.2

HPPSC Asstt. Prof. 18.11.2016
GATE 2007

Ans. (b) : $\sigma_1 = \sigma_2 = \dfrac{Pr}{2t}$

$$\sigma_{max} = \frac{8\times100}{2\times1} = 400\,MPa$$

$$\sigma_{min} = \frac{4\times100}{2\times1} = 200\,MPa$$

$$\sigma_v = \frac{400-200}{2} = 100$$

$$\sigma_m = \frac{400+200}{2} = 300$$

For Goodman equation

$$\frac{\sigma_v}{S_e} + \frac{\sigma_m}{S_{ut}} = \frac{1}{N}$$

$$\frac{100}{400} + \frac{300}{800} = \frac{1}{N}$$

$$\boxed{N = 1.6}$$

79. Select the correct statement.

(a) Fatigue strength improves due to presence of residual compressive stress
(b) Brittle and ductile materials follow Goodman straight line for tensile mean stress
(c) Fatigue strength reduction due to stress concentration is always proportional to stress concentration factor for all materials
(d) Fatigue strength of a material is same for all sizes
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Only the statement which is true, is fatigue strength improves due to presence of residual compressive stress because shot peening & cold rolling is done to increase the fatigue strength.

80. Which one is not useful to increase the fatigue resistance of a metal alloy?

(a) Polish the surface to remove stress amplification sites
(b) Increase the number of internal defects (pores, etc.) by means of altering processing and fabrication techniques
(c) Modify the design to eliminate notches and sudden contour changes
(d) Harden the outer surface of the structure by case hardening (carburizing, nitriding) of shot peening.

HPPSC Asstt. Prof. 2014

Ans. (b) : Internal defects reduces the fatigue strength of component.
Fatigue strength can be improved by following methods
(1) Improve surface finish
(2) Eliminate notches and sudden contour changes
(3) Case hardening
(4) Shot peening.

81. Match List I with List II and select the correct answer using the code given below the lists :

	List I		List II
A.	Von-Mises yield criterion	1.	Fatigue design
B.	Soderberg's law	2.	Failure in machine elements made of ductile materials
C.	Sommerfield number	3.	Design of journal bearings
D.	Buckingham equation	4.	Dynamic tooth load in gears in mesh

Code:

	A	B	C	D		A	B	C	D
(a)	4	1	3	2	(b)	2	1	3	4
(c)	4	3	1	2	(d)	2	3	1	4

UPSC JWM Advt. No.-50/2010

Ans. (b) :

- Von-Mises yield criterion is used to determine the stress for the failure of machine elements made up of ductile materials.
- **Soderberg's Law–**

$$\frac{\sigma_m}{S_{yt}} + \frac{\sigma_a}{S_e} = 1$$

is used to determine the fatigue strength of the material.

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• **Sommerfeld Number**–It is a dimensionless quantity used extensively in hydro dynamic lubrication analysis e.g. Bearings journal, Buckingham equation, Lewis factor equation for gear tooth calculations, used to design the gear tooth.

82. Fatigue failure results due to fluctuating stresses when the stress magnitude is
(a) more than ultimate tensile strength
(b) more than yield strength but lower than ultimate tensile strength
(c) lower than yield strength
(d) none of these

GPSC Lect. 23.10.2016

Ans. (c) : Fatigue failure results due to fluctuating stresses when the stress magnitude, which is generally lower than the ultimate tensile strength of the component and sometimes it fails at a stress below the yield strength of the material.

83. A 4 meter long solid circular cylindrical shaft transmits 1600 kW at 500 rpm. The endurance limit of the material is 210 MPa and a factor of safety of 3 is to be used with respect to fatigue failure. Use G = 77.5 GPa. The design diameter 'D' of the shaft in meter is
(a) $\sqrt[3]{\frac{69}{4375\pi^2}}$ (b) $\sqrt[3]{\frac{4375}{96\pi^2}}$
(c) $\sqrt[3]{\frac{69}{4735\pi^2}}$ (d) $\sqrt[3]{\frac{96}{4375\pi^2}}$

ISRO Scientist/Engineer 24.05.2014

Ans. (d) : Torque is given by

$$T = \frac{60 \times P}{2\pi N} = \frac{60 \times 1600 \times 10^3}{2\pi \times 500} = \frac{96000}{\pi} Nm$$

Using Soderberg equation, we have, this is used for ductile material

$$\frac{\tau_m}{\tau_y} + \frac{\tau_v}{\tau_e} = \frac{1}{N}$$

where

$$\tau_m = \frac{\tau + (-\tau)}{2} = 0 \text{ and } \tau_v = \frac{\tau - (-\tau)}{2} = \tau$$

So, we get

$$0 + \frac{\tau_v}{210} = \frac{1}{3} \Rightarrow \tau_v = \tau = 70 \text{ MPa}$$

Using Torsion equation, we get

$$d = \left(\frac{16T}{\pi \cdot \tau}\right)^{1/3} \Rightarrow d = \sqrt[3]{\frac{96}{4375\pi^2}}$$

84. Surface endurance limit of gear material depends on its
(a) Surface hardness
(b) Modulus of elasticity
(c) Elastic strength
(d) All of the above

ISRO Scientist/Engineer 2008

Ans. (a) : Surface endurance limit of gear material depends upon "surface hardness".

85. A rotating shaft with steady bending loads must be designed against____.
(a) fatigue failure (b) static failure
(c) both these (d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : A rotating shaft with steady bending loads must be designed against fatigue failure.

86. The factor of safety for fatigue loading is the ratio of
(a) Elastic limit to the working stress
(b) Young's modulus to the ultimate strength
(c) Endurance limit to the working stress
(d) Elastic limit to linear deformation

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

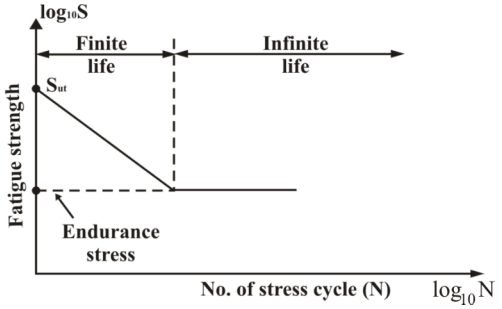
Ans. (c)

$$\text{(FOS) for fatigue loading} = \frac{\text{Endurance Limit}}{\text{Working Stress}}$$

87. S-N curve represents the:
(a) Fracture toughness (on y-axis) and numbers of fully reversed stress cycle (on x-axis)
(b) Resilience (on y-axis) and numbers of fully reversed stress cycle (on x-axis)
(c) Fatigue strength (on y-axis) and numbers of fully reversed stress cycle (on x-axis)
(d) Hardness (on y-axis) and numbers of fully reversed stress cycle (on x-axis)

UPRVUNL AE 07.10.2016

Ans. (c) : S-N curve represents the fatigue strength (on y-axis) and numbers of fully reversed stress cycle (on x-axis).



S-N curve—S-N curve, a graphical representation of a cyclic loading, is a plot of fatigue strength versus the number of cycles of the material through gets before it fails.

88. Select the correct answer out of the following alternatives about 'Cyclic Stresses'.
(a) That a material can tolerate are much greater than stresses produced under static loading.
(b) Can lead to fatigue if the stress level is above the endurance limit.
(c) Can lead to fatigue if the stress level is below the endurance limit.
(d) Are not introduced in the axle of a running train.

UKPSC AE 2012 Paper-I

Ans. (b) : Can lead to fatigue if the stress level is above the endurance limit.

89. The ratio of endurance limit in shear to endurance limit in flexure is
(a) 0.33 (b) 0.4
(c) 0.5 (d) 0.55

SJVN ET 2013

Ans. (d) : The ratio of endurance limit in shear to endurance limit in flexure is 0.55.

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90. In Concurrent Engineering, design and manufacturing are
(a) Sequentially considered
(b) Separately considered
(c) Simultaneously considered
(d) Independently considered

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : In Concurrent Engineering, design and manufacturing are simultaneously considered.

91. The effective stress in wire ropes during normal working is equal to the stress due to
(a) axial load plus stress due to bending
(b) acceleration/retardation of masses plus stress due to bending
(c) axial load plus stress due to acceleration /retardation
(d) bending plus stress due to a acceleration /retardation

JPSC AE 2013, Paper-V

Ans : (a) : The effective stress in wire ropes during normal working is equal to the stress due to axial load plus stress due to bending.

92. Select the wrong statement. Fatigue crack initiates on surface because :
(a) in most cases stress is highest on surface
(b) surface is machined
(c) surface is inherently weaker than the inside
(d) there may exist some stress concentration on surface

(HPPSC AE 2014)

Ans : (b) Fatigue crack initiates on surface because
(i) In most cases stress is highest on surface
(ii) Surface is inherently weaker than the inside
(iii) There may exist some stress concentration on surface.

93. Goodman straight line relation suggests that variable stress component in the presence of a positive mean stress :
(a) decreases
(b) increases
(c) remains unaffected
(d) increases or decreases depending upon σ_u

(HPPSC AE 2014)

Ans : (a) Goodman straight line relation suggests that variable stress component in the presence of a positive mean stress decreases.

94. The factor to be considered while designing a machine element is
(a) Selection of the materials
(b) Effect of environment
(c) Mechanism
(d) All the above

TSPSC AEE 2015

Ans : (d) The following steps are involved in the process of machine design
(i) Products Specification
(ii) Selection of mechanism
(iii) Effect of environment
(iv) Layout of Configuration
(v) Design of individual component
(vi) Preparation of drawing
(vii) Selection of materials.

95. Stress concentration factor for a machine component is dependent
(a) Only on the geometry of the component
(b) Only on the material of the component
(c) On geometry as well as material of the component
(d) On factors other than geometry and material of the component

UPRVUNL AE 2014
Punjab PSC SDE 12.02.2017

Ans. : (c) On geometry as well as material of the component

96. Which one of the following features improves the fatigue strength of a metallic material ?
(a) Increasing the temperature
(b) Scratching the surface
(c) Over stressing
(d) Under stressing

APPSC AE 04.12.2012
ESE 2000

Ans. (d) : When we applying the stress at the surface of metallic material, then the fatigue strength of material improves.

97. The Soderberg line for ductile materials gives upper limit for
(a) Any combination of mean and alternating stress.
(b) Alternating Compressive stress
(c) Alternating tensile stress
(d) Mean stress

APPSC AEE Screening Test 2016

Ans. (a) : The Soderberg line for ductile materials gives upper limit for any combination of mean and alternating stress.

$$\frac{\sigma_m}{S_y}+\frac{\sigma_a}{S_e}=\frac{1}{FOS}$$

Mean stress $(\sigma_m)=\dfrac{\sigma_1+\sigma_2}{2}$

Alternating stress $(\sigma_a)=\dfrac{\sigma_1-\sigma_2}{2}$

98. In design analysis, factor of safety is defined as the ratio between material strength and
(a) yield strength (b) ultimate strength
(c) maximum load (d) allowable strength

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (d) :

Factor of safety (FOS) $=\dfrac{\text{Ultimate load (strength)}}{\text{Allowable load (stress)}}$

As understand from the above equation the allowable stress is always less than the ultimate failure stress. Hence FOS is always greater than 1.

99. Failure of the machine component due to fatigue takes place
(a) At elastic limit (b) Below yield point
(c) Below elastic limit (d) At ultimate point

Mizoram PSC AE/SDO 2014, Paper-II
GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : Failure of the machine component due fatigue takes place below yield point.

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Fatigue strength–
It is the highest stress that a material can with stand for a given no. of cycles without breaking.
Fatigue strength is affected by environmental such as corrosion.

100. Increase in the radius of fillets
(a) improves the appearance of machine part
(b) reduces the strength of the part
(c) reduces stress concentration factor
(d) increases the notch sensitivity index

TNPSC ACF 2012

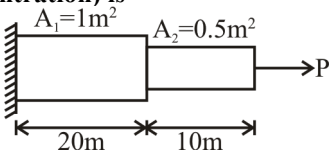
Ans. (c) : Increase the radius of fillets reduces stress concentration factor.

101. As per IS: 4326 - 1993 the crushing strength of the masonry unit should not be less than
(a) 35 MPa (b) 3.5 MPa
(c) 10 MPa (d) 5.0 MPa

GPSC AE 26.07.2015

Ans. (b) :

102. A rod of variable cross-section (areas A_1 and A_2) fixed at one end is subjected to an axial force P as shown in figure. If the allowable normal stress in tension is 100 MPa then the maximum allowable load P (neglecting stress concentration) is



(a) 100 MN (b) 200 MN
(c) 50 MN (d) 50 N

DRDO Scientists 2008

Ans. (c) : $\sigma_{max} = \dfrac{P}{A_{min}}$

$\sigma_{max} \le \sigma_{allowable}$

$\dfrac{P}{A_{min}} \le \sigma_{allowable}$

$\dfrac{P}{0.5} \le 100 \times 10^6$

$P \le 50$ MN

103. The phenomenon of the decreased resistance of a material to repeated stresses is called
(a) fatigue (b) creep
(c) freezing (d) melting

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : Fatigue is a failure mechanism that involves the racking of materials & structural components due to cyclic stress. This phenomenon of the decreased resistance of a material to repeated stresses.

104. Endurance limit is related to
(a) fatigue (b) creep
(c) impact (d) yielding

TANGEDCO AE 2015

Ans. (a) : Endurance Limit–The stress below which a material can endure and infinite number of repeated load cycles without fatigue failure.

105. What is the stress concentration factor for a reinforced butt weld :
(a) 2.0 (b) 1.5
(c) 1.2 (d) None of these

PSPCL AE, 2012

Ans. (c) : Stress concentration factor for a reinforced butt weld generally 1.1 to 1.3.

106. A static load is mounted at the centre of a shaft rotating at uniform angular velocity. This shaft will be designed for
(a) The maximum compressive (static)
(b) Fatigue loading
(c) The maximum tensile stress (static)
(d) The maximum bending moment (static)

Vizag Steel MT (Re-Exam) 24.11.2013
Karnataka PSC Lect. 27.05.2017

Ans. (b) : Fatigue Loading–Fatigue loading is primarily the type of loading which causes cyclic variation in the applied stress or strain on a component. Thus any variable loading is basically a fatigue loading.
Example–Rotating shaft with a static point load at the centre.

107. The ratio of the ultimate stress to the design stress is known as :
(a) elastic limit (b) strain
(c) factor of safety (d) bulk modulus

RPSC ACF & FRO, 26.02.2021
MPSC Poly. Lect. 2014

Ans. (c) : $(FOS)_{Brittle\ material} = \dfrac{Ultimate\ stress}{Design\ stress}$

$(FOS)_{Ductile\ material} = \dfrac{Yield\ stress}{Design\ stress}$

108. A thin spherical pressure vessel with 500 mm inner diameter is welded from steel plates having yield strength, σ_y = 200 MPa and endurance limit, σ_{en} = 100 MPa. The vessel is subjected to internal pressure which varies from 2 MPa to 6 MPa. Using Soderberg's equation for fatigue failure and considering a factor of safety of 3, the thickness of the vessel is
(a) 5 mm (b) 10 mm
(c) 12 mm (d) 15 mm

DRDO Scientists 2009

Ans. (d) : $\sigma_1 = \sigma$,
$\sigma_2 = 3\sigma$,

$$\sigma = \frac{Pd}{4t} = \frac{2 \times 500}{4 \times t} = \frac{250}{t} \text{MPa}$$

As per soderberg's equation,

$$\frac{1}{N} = \frac{\sigma_m}{\sigma_y} + \frac{\sigma_v}{\sigma_e}$$

$$\sigma_m = \frac{\sigma_1 + \sigma_2}{2} = 2\sigma$$

$$\sigma_v = \frac{\sigma_2 - \sigma_1}{2} = \sigma$$

σ_2 = 200 Mpa, σ_3 = 100 MPa

So, $\dfrac{1}{3} = \sigma\left[\dfrac{2}{200} + \dfrac{1}{100}\right]$

$\therefore \quad \sigma = \dfrac{100}{6}$ MPa

$\therefore \quad \dfrac{250}{t} = \dfrac{100}{6}$

Or t = 15 mm

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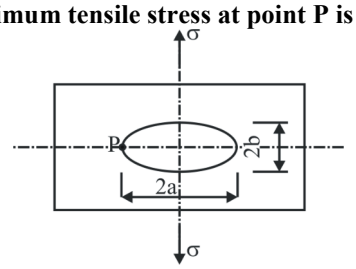
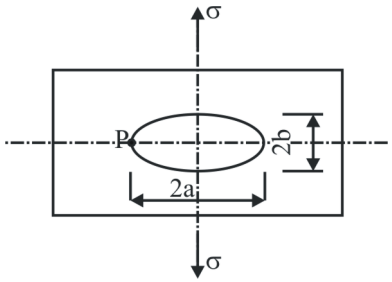
109. For a plate with an elliptical hole subjected to tensile stress, σ as shown in the figure, the maximum tensile stress at point P is



(a) σ (b) 2σ
(c) 3σ (d) more than 3σ

ISRO Scientist/Engineer 07.05.2017

Ans. (d) :



Theoretical stress concentration factor

k_t = Theoretical stress concentration factor

$$= \frac{\text{Maximum stress}}{\text{Nominal stress}} = \frac{\sigma_{max}}{\sigma}$$

$$k_t = 1 + 2 \times \frac{2a}{2b} = 1 + 2 \times \frac{a}{b}$$

$\frac{a}{b} > 1$ [From figure]

$$k_t = 1 + 2\left(\frac{a}{b} > 1\right)$$

$k_t > 3$

$$\frac{\sigma_{max}}{\sigma} > 3 \Rightarrow \sigma_{max} > 3\sigma$$

110. A static load mounted at the centre of a shaft rotating at uniform angular velocity. This shaft will be designed for

(a) the maximum compressive stress (static)
(b) the maximum tensile stress (static)
(c) the maximum bending moment (static)
(d) fatigue loading

Karnataka PSC Lect., 27.05.2017

Ans (d) : As the shaft is rotating in nature, each fiber of the shaft will under go tensile and compressive loads so the shaft is under the action of cyclic load. So the design should be for fatigue loading.

111. Stress concentration occurs when-

(a) a body is subjected to excessive stress
(b) a body is subjected to unidirectional stress
(c) a body is subjected to fluctuating stress
(d) a body is subjected to non-uniform stress distribution

RPSC 2016

Ans : (d) Stress concentration occurs when these is sudden change in the geometry of the body due to cracks sharp corners, holes and decrease in the cross section area.

112. The ability of a material to withstand varying loads is called ___

(a) tensile strength (b) impact strength
(c) fatigue strength (d) creep strength

UPRVUNL AE 05.07.2021

Ans. (c) : The ability of a material to withstand varying loads is called fatigue strength.

113. Failure of a materials is called fatigue when it fails

(a) at the elastic limit
(b) below the elastic limit
(c) at the yield point
(d) below the yield point

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : Failure of a materials is called fatigue when it fails below the yield point.

114. A steel specimen is subjected to the following principal stresses: 120 MPa tensile, 60 MPa tensile and 30 MPa compressive. If the proportionality limit for the steel specimen is 250 MPa; the factor of safety as per maximum shear stress theory will be nearly

(a) 1.3 (b) 1.7
(c) 2.3 (d) 2.7

ESE 2020

Ans. (b) : $\sigma_1 = 120$ MPa

$\sigma_2 = 60$ MPa

$\sigma_3 = -30$ MPa

$S_{yt} = 250$ MPa

For maximum shear stress theory

$$\frac{S_{yt}}{FOS} = \sigma_1 - \sigma_3$$

$$FOS = \frac{S_{yt}}{\sigma_1 - \sigma_3}$$

$$FOS = \frac{250}{120 + 30}$$

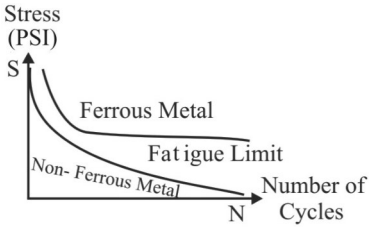
FOS = 1.7

115. S-N curves are connected with

(a) Toughness (b) Hardness
(c) Creep (d) Fatigue

UPPSC AE 12.04.2016 Paper-I

Ans : (d)



S-N curve reveals a substantial difference in fatigue characteristics between ferrous metals (like iron, steel and titanium) and non-ferrous metals (like aluminum, magnesium and copper). Ferrous metals exhibit a "fatigue limit" stress below which they can endure an infinite number of repetitive stress cycles without failing. Non-ferrous metals have no fatigue limit, and will always fail eventually if subjected to enough stress cycles.

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2. Design of Shafts, Keys and Couplings

116. Normally in the assembly of pulley, key and shaft, the

(a) pulley is made the weakest
(b) key is made the weakest
(c) key is made the strongest
(d) shaft is designed as weakest

VIZAG Steel MT 24.01.2021
OPSC AEE 2019 Paper-I
Vadodara Muncipal Corp. DEE, 2018
RPSC AE 2018
RPSC Asstt. Engg. 2018
WBPSC AE, 2017
UJVNL AE. 2016
RPSC AE 2016
MPPSC State Forest Service Exam, 2014
ESE 2013
APPSC AEE 2012
RPSC ACF-2011
ESE 1998

Ans. (b) : As we know, the failure is mainly at key because it is easy to replace, so that it cost less in a assembly of pulley, key & shaft.

117. The piston rod and the cross head in a steam engine are usually connected by means. of:

(a) Cotter joint (b) Knuckle joint
(c) Ball joint (d) Universal joint

Assam PSC AE (IWT) 14.03.2021
AGujarat PSC AE 2019
Vizag Steel (MT) 2017
UJVNL AE. 2016
RPSC 2016
Kerala PSC AE 06.08.2015
KPSC AE. 2015
TNPSC AE 2013
JPSC AE 2013, Paper-V
ESE 1998

Ans : (a) The piston rod and the cross head in a steam engines are usually connected by means. of cotter joint.

Cotter Joint:-

A cotter joint is used to connect two rod either subjected to tensile force or compressive axial force with no relative motion between them.

Application of Cotter Joint:-

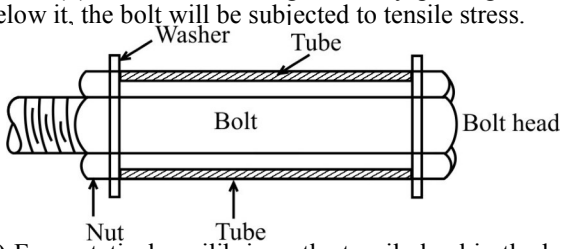
* Joint between the piston rod and the cross head of a steam engine.
*Joint between the slide spindle and the fork of the valve mechanism.
*Joint between the piston rod and the tail or pump rod.
* Foundation bolt.

118. When a nut is tightness by placing washer it, the bolt will be subjected to following type of loads

(a) compression (b) tension
(c) shear (d) combined load

RPSC IOF, 2020
Nagaland PSC CTSE 2017, Paper-I
RPSC AE 2016
CGPSC Poly. Lect. 22.05.2016
UJVNL AE 2016
TSPSC AEE 2015
RPSC ACF-2011
ESE 1998

Ans : (b) When a nut is tightened by placing washer below it, the bolt will be subjected to tensile stress.



(i) From statical equilibrium, the tensile load in the bolt will be equal to the compressive load in the tube.
(ii) $P_b A_b = P_c A_c$
(iii) $\Delta_b + \Delta_c$ = movement of the nut

119. A cotter joint is used to connect two rods which are in :

(a) Compression only
(b) Tension and Compression only
(c) Tension only
(d) Shear only

UPRVUNL AE 05.07.2021
Nagaland PSC (CTSE) 2018 Paper-I
GPSC Asstt. Director of Transport 05.03.2017
GPSC ARTO 01.05.2016
GPSC Lect. 23.10.2016
KPSC AE. 2015
CGPSC AE 26.04.2015 Shift-I
MPPSC State Forest Service Exam, 2014
Mizoram PSC AE/SDO 2014, Paper-II
WBPSC AE 2008ESE 2002

Ans : (b) Connect two Co-axial rods, which are subjected to either axial tensile force or axial compressive force.

Application of cotter Joint:-

(i) Foundation bolt
(ii) Joint between the piston rod and the tail or pump rod.
(iii) Joint between the slide spindle and the fork of the valve mechanism
(iv) Joint between the piston and the cross head of a steam Engine.

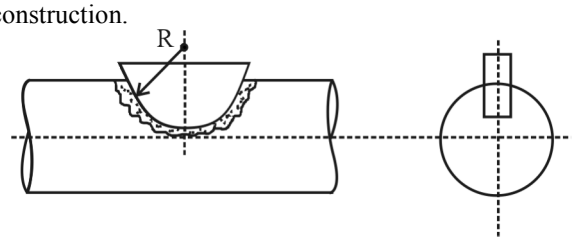
120. Which of the following is an easily adjustable key which is a piece from a cylindrical disc having segmental cross-section in front view?

(a) Feather key (b) Gib head key
(c) Flat saddle key (d) Woodruff key

Assam PSC AE (IWT) 14.03.2021
Rajasthan AE (Nagar Nigam) 2016 Shift-3
MPPSC AE 2016

Ans. (d) : The woodruff key is an easily adjustable key. It is a piece from a cylindrical disc having segmented cross section.

This key is largely used in machine tool and automobile construction.



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121. When the shafts are slightly misaligned then most suitable coupling to connect them is:

(a) Rigid coupling
(b) Flexible coupling
(c) Oldham-hook coupling
(d) None of these

HPPSC Workshop Suptd. 08.07.2021
RPSC AE (GWD) 06.05.2016
HPPSC Asstt. Prof. 29.10.2016
HPPSC LECT. 2016

Ans : (b)

(i) When the shafts are slightly misaligned then most suitable coupling to connect them is flexible coupling.
(ii) When the shafts are not slightly misaligned the most suitable coupling to connect them is rigid coupling.

Classification of coupling

Rigid coupling	**Flexible Coupling**
(i) Muff coupling	(i) Bushed pin coupling
(ii) Split muff coupling	(ii) Oldham Coupling
(iii) Flange rigid coupling	(iii) Universal coupling

122. A bolt is designed with a large factor of safety, on the basis of

(a) Direct tensile stress
(b) Direct compressive stress
(c) Direct bending stress
(d) Direct shear stress

RPSC AE 2016
RPSC ACF-2011
ISRO Scientist/Engineer 2010

Ans. (a) : A bolt is designed with a large factor of safety, on the basis of direct tensile stress. Basis of direct tensile stress, bolt designed with the large factor of safety.

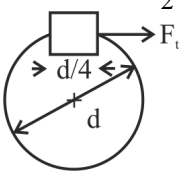
123. Square key of side "d/4" each and length *l* is used to transmit torque "T" from the shaft of diameter "d' to the hub of a pulley. Assuming the length of the key to be equal to the thickness of the pulley, the average shear stress developed in the key is given by

(a) $4T/\ell d$ (b) $16T/\ell d^2$
(c) $8T/\ell d^2$ (d) $16T/\pi d^3$

OPSC AEE 2019 Paper-I
APPSC AEE MAINS 2016, PAPER-III
GATE 2003

Ans. (c) : Shear area of the key $= \frac{d}{4}\times l$

Torque transmit by key = $T = F_t \times \frac{d}{2}$



$$T = \tau \times \frac{d}{4}\times \ell \times \frac{d}{2}$$

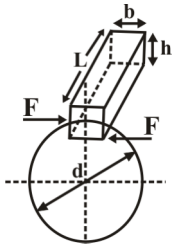
$$\tau = \frac{8T}{d^2\ell}$$

124. The shearing area of a Key of length 'L' breadth 'b' depth 'h' is equal to

(a) b × h (b) L × h
(c) L × b (d) $L\times\frac{h}{2}$

MPPSC AE 2016
UKPSC AE-2013, Paper-I
ESE 1998

Ans : (c)



T= Torque transmitted by the shaft
b = Width of key
L = length of key
h = depth of key
Torque transmitted by the shaft

$$T = F\times\frac{d}{2}$$

$$T = L\times b\times\frac{d}{2}$$

According to this above figure shear plain area = L × b

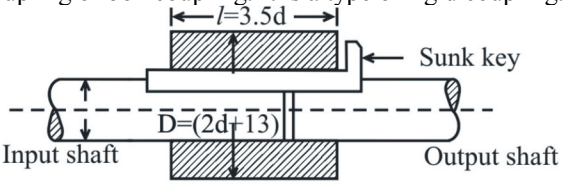
Crushing or Bearing area $= L\times\frac{h}{2}$

125. Sleeve of muff coupling is designed as a

(a) thin vessel (b) thick vessel
(c) solid shaft (d) hollow shaft

RPSC AE 2016
Nagaland CTSE 2016 Ist Paper
TNPSC ACF 2012
RPSC ACF-2011

Ans : (d) Sleeve of muff coupling is designed as a hollow shaft. Muff Coupling is also called sleeve coupling or box coupling. It is a type of rigid coupling.



Muff coupling

126. In case of clamp coupling, power is transmitted by means of :

(a) friction force (b) shear resistance
(c) crushing resistance (d) none of the above

Nagaland PSC CTSE 2017, Paper-I
HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : In a clamp or compression coupling, the power is transmitted from one shaft to the other by means of key and the friction between the muff and shaft.
In case of clamp coupling, power is transmitted by means of friction force.

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127. Allen bolts are :

(a) Self locking bolts
(b) Self screwing bolts
(c) Provided with hexagonal depression in head
(d) Counter shunk

Nagaland PSC CTSE 2017, Paper-I
GWSSB DEE 07.07.2016
UPRVUNL AE 2014

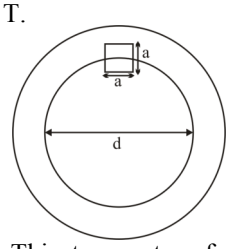
Ans. : (c) Allen bolts are provided with hexagonal depression in head.
• Self locking bolts are developed to uphold unstability in the bolts and also prevent the possibility of disassembly for the pre load.

128. A square key is used to transmit the torque (T) from the shaft to the hub. The relationship between induced compressive stress (σ_c) and shear stress (τ) is given by:

(a) $\sigma = \tau$ (b) $\sigma = 2\tau$
(c) $\sigma = \tau/2$ (d) $\sigma = \tau/4$
(e) $\sigma = \tau/6$

JPSC AE 10.04.2021, Paper-II
CGPSC AE 25.02.2018

Ans. (b) : Let us assume that torque transmitted by shaft is T.



This torque transfer to hub with the help of square key [*l, a, a*]

Then tangential force acting on the key

$$F = \frac{2T}{d}$$

shear stress in square key

$$\tau = \frac{\text{shear force}}{\text{shear area of the key}} = \frac{\frac{2T}{d}}{[a \times \ell]} = \frac{2T}{d \times a \times \ell}$$

Compressive stress in square key

$$\sigma_{comp.} = \frac{\text{compressive force}}{\text{compression area of the key}} = \frac{\left[\frac{2T}{d}\right]}{\left[\frac{a}{2} \times \ell\right]}$$

$$\sigma_{comp} = \left[\frac{2T}{d \times a \times \ell}\right] \times 2 = 2\tau$$

$$\boxed{\sigma_{comp} = 2\tau}$$

129. Match List –1 (Type of keys) with List–II (Characteristics) and select the correct answer using the codes given below the lists :

List–I	**List–II**
(A) Woodruff Key	**(i) Loose fitting, Light Duty**
(B) Kennedy Key	**(ii) Heavy duty**
(C) Feather Key	**(iii) Self-aligning**
(D) Flat Key	**(iv) Normal industrial use**

	(A)	**(B)**	**(C)**	**(D)**
(a)	(ii)	(iii)	(i)	(iv)
(b)	(iii)	(ii)	(i)	(iv)
(c)	(ii)	(iii)	(iv)	(i)
(d)	(iii)	(ii)	(iv)	(i)

OPSC AEE 2019 Paper-I
ESE 1997

Ans : (b) :
(A) Woodruff Key- It is semicircular key having circular portion in the shaft which can self align according to taper in the hub and shaft.
(B) Kennedy Key- There are two keys set aligned 90º. This key strong in shear and crushing and used in heavy duty applications.
(C) Feather Key- This is similar to square key and fixed in one member of joint (shaft or hub) in such a way that other member can move axially.
(D) Flat Key- Most of industrial keys are industrial.

130. Consider the following statements in respect of flexible couplings :

(i) The flanges of flexible coupling are usually made of grey cast iron FG200
(ii) In the analysis of flexible coupling, it is assumed that the power is transmitted by the shear resistance of the pins
(iii) Rubber bushes with brass lining are provided to absorb misalignment between the two shafts

Which of the statements given above are correct?

(a) (i), (ii) and (iii) (b) Only (i) and (ii)
(c) Only (ii) and (iii) (d) Only (i) and (iii)

OPSC AEE 2019 Paper-I
ESE 2006

Ans : (d) :
(i) The flanges of flexible couplings are made by grey cast iron (FG 200) because of shock/vibration absorbing capacity.
(ii) The power is transmitted from driving flange to driven flange by shear resistance of pins connecting these flanges. Bending of pins is also checked.
(iii) Since rubber can absorb some misalignment of shafts, so it is used between surfaces of pin and flange hole.

131. When large quantities of bolts are to be purchased the quantity is usually specified in terms of

(a) number (b) volume
(c) weight (d) packets

RPSC AE 2016
RPSC ACF-2011

Ans : (c) When large quantities of bolt are to be purchased the quantity is usually specified in terms of weight.

132. Which of the following steel key is usually strong in failure by shear and crushing?

(a) rectangular (b) square
(c) circular (d) flat

RPSC AE 2016
RPSC ACF-2011

Ans : (b) Square key is usually strong in failure by shear and crushing.

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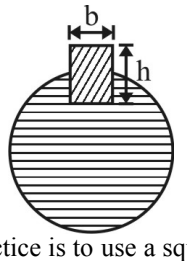

"The industrial practice is to use a square key with sides equal to one-quarter of the shaft diameter and length at least 1.5 times the shaft diameter

$b = h = d/4$ and $l = 1.5\ d$

b = width of key (mm)

h = height of key (mm)

ℓ = length of key (mm)

d = diameter of shaft . (mm)

133. Which key is preferred for the condition where a large amount of impact torque is to be transmitted in both direction of rotation?

(a) Woodruff key (b) Feather key

(c) Gib head key (d) Tangent key

UJVNL AE 2016
PSPCL AE, 2012

Ans : (d) The tangent keys are fitted in pair at right angles. Each key is to withstand torsion in both direction. These are used in large heavy duty shafts.

134. Which one of the following loadings is considered for the design of axles?

(a) Bending moment only
(b) Twisting moment only
(c) Combined bending moment and torsion
(d) Combined action of bending moment and axial thrust

APPSC AEE 2012
ESE 1995

Ans. (a) : bending moment only
Because axles are only meant for support, they do not transmit torque. So only designed for bending.

135. A screw is specified by its

(a) major diameter (b) minor diameter

(c) pitch diameter (d) Pitch

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Nagaland PSC (CTSE) 2018, Paper-I
MPSC Poly. Lect. 2014
JPSC AE 2013, Paper-V

Ans. (a) : As we know, a screw have 3-diameter major, minor & pitch diameter, the screw is specified by its outer or major diameter.

- To hold parts together-act as fastners (eg: V-threads)
- To transmit motion & power (Square, Acme threads)

136. A flange coupling is used

(a) for intersecting shafts
(b) for collinear shafts
(c) for small shaft rotating at slow
(d) for parallel shafts

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : Flange couplings is a type of connector between turning flanges that have two arrangements of flanges, flanges are fitted at the end of shafts. They are only used for collinear shafts.

137. If there is no place to accommodate the nut, then one would choose the

(a) through bolts (b) tap bolts

(c) studs (d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Tap bolts are directly threaded into the clamped parts and does not require any nut.

138. _____couplings are usually used to transmit torque from one shaft to another when the two shafts are slightly misaligned.

(a) Rigid (b) Flexible

(c) Stack (d) Pin

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) :

139. Which is the false statement in relation to propeller shaft?

(a) a long propeller shaft tends to sag and whirl
(b) shaft length may be decreased by using a divided propeller shaft supported by intermediate bearings
(c) it transmits the drive from the gear box to the rear axle in front engine, rear drive vehicles
(d) decreasing the shaft length to half would decrease the critical speed two times

HPPSC Poly. Lect. 05.07.2021

Ans. (d) :

140. The knuckle pin in a two rod knuckle joint under axial load will be subjected to shear stress. The failure area of the pin under shear will be :

(a) $\frac{\pi}{4}d^2$ (b) πd

(c) $\frac{\pi}{2}d^2$ (d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : The knuckle pin a two rod knuckle joint under axial load.
Then failure of the pin under shear stress.

$$= 2 \times \frac{\pi}{4}d^2$$

$$= \frac{\pi}{2}d^2$$

141. Types of stresses in sunk key are

(a) shear
(b) shear and compressive
(c) shear and bending
(d) compressive and tensile

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : A sunk key is a key in which half the thickness of the key fits into the keyway on the shaft and other on the hub. Stresses in sunk key are shear and compressive.

142. Consider a square key mounted in the keyway of a shaft engaging a hub of a pulley. The key has cross section of 6 mm × 6 mm and a length of 40 mm. The shaft is of diameter 20 mm transmits a torque of 48 Nm. The average bearing stress in the key is -

(a) 20 MPa (b) 28.5 MPa

(c) 40 MPa (d) 200 MPa

Haryana PSC AE (PHED) 05.09.2020, Paper-II

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Ans. (c) : For key
w = 6 mm
t = 6 mm
l = 40 mm
Shaft, D = 20 mm
Torque = 48 N-m
Torque = force × radius

$$T = \sigma_b \times \left(\frac{t}{2} \times l\right) \times \frac{d}{2}$$

$$1000 \times 48 = \sigma_b \times \frac{6}{2} \times 40 \times 10$$

$$\sigma_b = 40 \text{ N/mm}^2$$

σ_b = 40 MPa

143. In an unprotected type of flange coupling, the outer diameter of the hub is _____.
(a) 3 times the diameter of the inner hub
(b) twice the diameter of the inner hub
(c) 1.5 times the diameter of the inner hub
(d) 1.75 times the diameter of the inner hub

VIZAG MT, 14.12.2020

Ans. (b) : In an unprotected type flange coupling, the outer diameter of the hub is twice the diameter of the inner hub.

144. The failure of a key is due to :
(a) Shearing (b) Tension
(c) bending (d) twisting

APPSC Poly Lect. 13.03.2020

Ans. (a) : Failure of a key is due to shearing only.

145. The _____ coupling requires accurate alignment of input and output shafts.
(a) flexible disk (b) rigid
(c) helical (d) bush pin

APPSC Poly Lect. 13.03.2020

Ans. (b) The rigid coupling requires accurate alignment of input and output shafts.

146. In designing a sleeve and cotter joint, the outside diameter of the sleeve is taken as
(a) 1.5 d (b) 2.5 d
(c) 3 d (d) 4 d

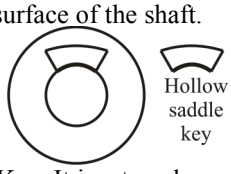
RPSC IOF, 2020

Ans. (b) : The various proportions for the sleeve and cotter joint in terms of the diameter of rod (d)
Out side diameter of sleeve d_1 = 2.5 d
Diameter of enlarged end of rod d_2
Inside diameter of sleeve = 1.25 d
Length of sleeve, L = 8d
Thickness of cotter t = 0.31 d
Width of cotter b = 1.25 d

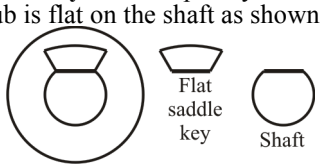
147. A saddle key
(a) is provided in the hub only and hollowed to fit the shaft
(b) has flat surface and the shaft is planned off to accommodate the key
(c) is fitted such that each withstand torsion is in one direction only
(d) None of these

RPSC IOF, 2020

Ans. (a) : A saddle key is a key which fits in the keyway of the hub only. In this case, there is no key way on the shaft. It is likely to slip round the shaft under load. Therefore it is used for comparatively light loads.
Two types of saddle key is there–
(i) Hollow Saddle Key–It is a taper key which fits in a keyway in the hub and the bottom of the key is shaped to fit the curved surface of the shaft.



(ii) Flat Saddle Key–It is a tape key which fits in a key way in the hub is flat on the shaft as shown in fig.

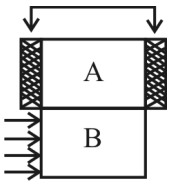


Note-No answer is given by commission.

148. Which of the following stresses are associated with the design of pins in bushed pin-type flexible coupling?
(i) Bearing stress
(ii) Bending stress
(iii) Axial tensile stress
(iv) Transverse shear stress
Select the correct answer using the codes given below :
(a) (i), (ii) and (iv) (b) (ii), (iii) and (iv)
(c) (i), (ii), and (iii) (d) (i), (ii) and (iv)

OPSC AEE 2019 Paper-I
APPSC AEE 2012
ESE 2000, 1998

Ans : (d) : The free body diagram of pin in flexible coupling.



Portion 'A' is in driving flange and 'B' in driven flanges so the pin is subjected to following stresses.
(i) Bearing stress on cylindrical surfaces A and B.
(ii) Bending stress because the forces from driving and driven flanges are not collinear.
(iii) The transverse shear at common plane of surface A and B.
(iv) No axial stress at all.

149. Splines are used when :
(a) Power to be transmitted is low
(b) High rotational speeds are involved
(c) High torque is to be transmitted
(d) There is need for axial relative motion between the shaft and hub

OPSC AEE 2019 Paper-I

Ans : (d) : Splines are used to provide toqrue and speed from the shaft to the hub or vice-versa.

150. The coupling used to connect two shafts with large angular misalignment is :
(a) A Flange coupling
(b) An Oldham's coupling
(c) A Flexible bush coupling
(d) A Hooke's joint

OPSC AEE 2019 Paper-I

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Ans : (d) : Hooke's joint is used to connect two intersecting shaft and used in automobile to connect propeller shaft.
A Hooke's joint is used to connect two shafts, which are intersecting at a small angle.

151. The holes in the flange coupling for coupling the two changes together by bolts are reamed because it permits
(a) equal sharing of load by bolts
(b) avoidance of stress concentration
(c) avoidance of any injury during dismantling
(d) less wear, tear and vibration

TNPSC AE 2018

Ans. (a) : The holes in the flange coupling for coupling the two changes together by bolts are reamed because it permits equal sharing of load by bolts.

152. The type of joint used in cycle chain is
(a) Cotter joint (b) Knuckle joint
(c) Gib and cotter joint (d) Riveted joint

TNPSC AE 2018

Ans. (b) : Knuckle joint - Knuckle joint is a type of mechanical joint used in structures, to connect to intersecting cylinderical, whose axes lie on the same plane. It permits some angular movement between the cylindrical rods. It is specially designed to withstand tensile loads.

153. The type of coupling used for high torque and low speed is
(a) Muff coupling
(b) Bushed pin flexible coupling
(c) Disc coupling
(d) Oldham's coupling

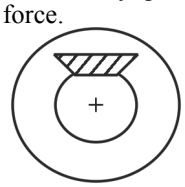
TNPSC AE 2018

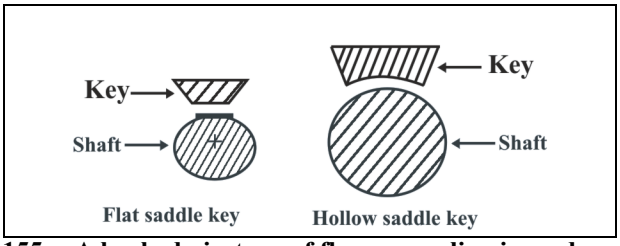
Ans. (d) : Oldham's Coupling- An Oldham coupling has three discs, one coupled to the input, one coupled to the output and a middle disc that is joined to the first two by tongue and groove. The tongue and groove on one side is perpendicular to the tongue and groove on the other. The middle disc rotates around its centre at the same speed as the input and output shafts. Its centre traces a circular orbit, twice per rotation, around the midpoint between input and output shafts. Often spring are used to reduce backlash of the mechanism.

154. In case of saddle key, power is transmitted by means of
(a) Friction force
(b) Shear resistance of key
(c) Torsional resistance of key
(d) Tensile force

ISRO Scientist/Engineer (RAC) 22.04.2018
APPSC AEE Screening Test 2016
TSPSC AEE 2017
RPSC AE GWD, 2011
JWM 2017
Nagaland CTSE 2016 Ist Paper
GPSC Lect. 23.10.2016

Ans. (a) : In case of saddle key, power is transmitted by means of frictional force.





155. A bushed pin-type of flange coupling is used
(a) When the shafts are not in exact alignment
(b) For intersecting shafts
(c) For small shafts rotating at slow speeds
(d) For parallel shaft

GPSC EE Pre, 28.01.2017

Ans. (a) : A bushed pin-type of flange coupling is used when the shafts are not in exact alignment. Bush pin flange coupling is used to connect of shaft which having small parallel misalignment, angular misalignment or axial misalignment.

156. The purpose of rubber bushing in flexible coupling is to
(a) allow proper fixation of pin
(b) avoid wear of pin
(c) avoid failure of C.I. flanges
(d) allow small misalignment of the shaft

APGENCO AE, 2017

Ans. (d) : The purpose of rubber bushing in flexible coupling is to allow small mis-alignment of the shaft. It is used in small capacity generator to couple alternator and engine also used in transmission shaft of automobile.

157. Which type of key is preferred for mounting shifting gears in a gear box?
(a) an involute (b) a helix
(c) a straight line (d) a cycloid

Punjab PSC (Lect.) 06.08.2017

Ans. (d) : Splines are ridges or teeth on a drive shaft that mesh with grooves in a mating piece and transfer torque to it, maintaining the angular correspondence between them.

158. Which of the following keys is rectangular or square in cross section and uniform in width and thickness throughout its length?
(a) Parallel sunk key (b) Feather key
(c) Gib head key (d) Taper sunk key

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : Parallel sunk key is rectangular or square in cross section and uniform in width and the thickness throughout its length.
- In this power is transmitted by shear resistance at the key.
- It is suitable for heavy duty application.
- Rectangular (flat) key is more stable as compare with square key.

159. The usual proportion for the width of the key used between shaft (of diameter d) and hub of pulley for transmitting power is
(a) d/8 (b) d/6
(c) d/2 (d) d/4

ISRO Scientist/Engineer (RAC) 07.05.2017

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Ans. (d) : Generally width of the sunk key (rectangular) is taken as $\frac{d}{4}$ and thickness of key is taken as $\frac{d}{6}$.
For square sunk key both width and thickness are equal and equal to $\frac{d}{4}$.

160. Which of the following key transmits power in a taper shaft?
(a) Barth key (b) Woodruff key
(c) Kennedy key (d) Tangent key

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Woodruff keys are used for tapered shaft.

161. During crushing failure of key, The area under crushing is–
(a) width
(b) Length
(c) both (1) and (2)
(d) length and half of thickness

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : When the key failure due to crushing, its failure area under crushing is length and half of thickness (due to given compressive force)

162. The distance between two parallel shafts is 18 mm and they are connected by an oldhams coupling, the driving shaft revolves at 160 rpm. The maximum speed of sliding of the tongue is:
(a) 0.302 m/s (b) 0.6 m/s
(c) 3.2 m/s (d) 6 m/s

TRB Polytechnic Lecturer 2017

Ans. (a) : $V_{sliding} = r.\omega$
$= 0.018 \times \frac{2 \times \pi \times 160}{60} = 0.3014$ m/s

163. _____permits the motion to be transmitted from the gear box output shaft to the pinion shaft of the differential, irrespective of the inclination of the drive shaft.
(a) Riveted joints (b) Welded joints
(c) Slip joints (d) Universal joints

TNPSC AE 2017

Ans. (d) : Universal joints permits the motion to be transmitted from the gear box output shaft to the pinion shaft of the differential, irrespective of the inclination of the drive shaft.

164. Taper on the cotter and slot is provided–
(a) On both the sides (b) On one side only
(c) On none of the sides (d) None of the above

Vizag Steel (MT) 2017

Ans. (b) : Taper on the cotter and slot is provided on one side only.

165. Which of the following is a wrong statement?
(a) Key is generally used for transmitting rotational motion
(b) Cotter is generally employed for transmitting axial motion
(c) The keys are loaded in shear
(d) The cotter have to withstand tensile or compressive loads

BPSC AE Mains 2017 Paper - VI

Ans : (d) : The cotter have to withstand tensile or compressive loads

166. The empirical formula for the design of shaft is given as:
(a) $d = \sqrt{\frac{R.P.M.}{C \times H.P.}}$ cm (b) $d = \sqrt{\frac{H.P. \times C}{R.P.M.}}$ cm
(c) $d = \sqrt[3]{\frac{H.P. \times C}{R.P.M.}}$ (d) None of these

Where:
H.P. = Horse Power
R.P.M. = Revolution Per Minute
C = Constant

PTCUL AE 25.06.2017

Ans. (c) :

167. The thickness of cotter is generally taken equal to (where. d is the diameter of two rods which are connected by cotter joint)
(a) 0.15 d (b) 0.3 d
(c) 0.4 d (d) 0.5 d

Nagaland CTSE 2016 Ist Paper

Ans. (b) : Cotter is a flat wedge shaped piece of rectangular cross-section and its width tapered. The taper varies from 1 in 48 to 1 in 24. A cotter joint is a temporary fastening & is used to connect rigidly two Co-axial rods or bars which are subjected to axial tensile or compressive forces. The thickness of the cotter is generally taken equal to 0.31d.

168. Two parallel shafts, the distance between whose axes is small and variable may be connected by
(a) gear drive (b) universal joint
(c) knuckle joint (d) Oldham's coupling

RPSC AE 2016
TSPSC AEE 2015

Ans : (d) Two parallel shaft, the distance between whose axes is small and variable may be connected by Oldham's coupling
Oldham's Coupling:- An Oldham's coupling is used for connecting two parallel shafts whose axes are at a small distance apart. The shafts are coupled in such a way that if one shaft rotates, the other shaft also rotates at the same speed.

169. Oldham's coupling is used to connect two shafts which
(a) have lateral misalignment
(b) have two axes intersecting
(c) have lateral movements during power transmission
(d) changes axes during power transmission

RPSC AE 2016
Nagaland PSC (CTSE) 2018 Paper-I

Ans : (a) Shaft coupling are divided into two main group as follows
1. Rigid coupling:- It is used to connect two shafts which are perfectly aligned
(a) sleeve or muff coupling
(b) split - muff coupling
(c) Flange coupling
2. Flexible Coupling:- It is used to connect two shafts having both lateral and angular misalignments
(a) Bushed pin type coupling
(b) Universal coupling
(c) Oldham coupling.

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170. ..... is a cylindrical rod threaded at both the ends and left plain in the middle.

(a) Shaft (b) Spindle
(c) stud (d) Bolt

HPPSC LECT. 2016
CGPSC AE 26.04.2015 Shift-I
APPSC Poly Lect. 13.03.2020
GWSSB DEE 07.07.2016
Kerala PSC IOF 19.04.2016
GWSSB AE 27.12.2015

Ans : (c) Stud is a cylindrical rod threaded at both the ends left plain in the middle.
Studs:- A stud is a round bar threaded at both ends. One end of the stud is screwed into a tapped hole of the parts to be fastened, while the other end receives a nut on it. Studs are chiefly used instead of tap bolts for securing various of covers e.g. covers of engine and pump cylinders.

171. In a butt or half-lap coupling, the sleeve is normally fitted to the shafts by means of _____.

(a) Saddle key (b) Feather key
(c) Woodruff key (d) Sunk key

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : In a butt or half-lap coupling, the sleeve is normally fitted to the shafts by means of Saddle key.

172. The product of the tangential force acting on the shaft and its distance from the axis of the shaft (i.e. radius of shaft) is known as :

(a) Bending moment (b) Twisting moment
(c) Torsional rigidity (d) None of the above

UP Jal Nigam AE 2016

Ans. (b) : The product of the tangential force acting on the shaft and its distance from the axis of the shaft (i.e., radius) is known as twisting moment.

173. During shear failure of key. The area under shear is

(a) width (b) Length
(c) both (1) and (2) (d) length and thickness

Nagaland CTSE 2016 Ist Paper

Ans. (c) : A key failure, under shearing, the tangential shearing force acting at the circumference of the shaft.
F = Area resisting shearing × shear stress
ℓ = Length of key
w = width of key

$$F = \ell \times w \times \tau$$

$$\text{Torque, } T = F.\frac{d}{2}$$

$$T = \ell \times w.\tau.\frac{d}{2}$$

$\Rightarrow$ During shearing, key fails under area of both length & width, as we see above.

174. A shaft subjected to fluctuating loads for which the normal torque (T) and bending moment (M) are 1000 N-m and 500 N-m respectively. If the combined shock and fatigue factor for bending is 1.2 and combined shock and fatigue factor for torsion is 2 then the equivalent twisting moment for the shaft is _____

(a) 2088 N-m (b) 2050 N-m
(c) 2136 N-m (d) 2188 N-m

UPRVUNL AE 21.08.2016

Ans. : (a) Equivalent twisting moment (T_e)

$$T_e = \sqrt{(K_m \times M)^2 + (K_t \times T)^2}$$

M = Bending moment = 500 N-m
T = Twisting moment = 1000 N-m
K_m = fatigue factor for bending = 1.2
K_t = fatigue factor for torsion = 2

$$T_e = \sqrt{(1.2 \times 500)^2 + (2 \times 1000)^2}$$

T_e = 2088 N-m

175. Shaft is generally designed for :

(a) Bending and torsional deflection
(b) Slope at bearing and shaft supported elements.
(c) Shear deflection due to transverse loading of shaft
(d) All of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Shaft is generally designed for –
(A) Bending and torsional deflection
(B) Slope at bearings and shaft support elements.
(C) Shear deflection due to transverse loading of shaft.

176. The strength of a link chain is the:

(a) Strength of the strongest link
(b) Strength of the weakest link
(c) Average strength of all the links
(d) Means of strongest and weakest links

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : The strength of a link is the strength of the weakest link in a kinematic chain.

177. The key will fail in which of the following manner?

(a) Shearing only
(b) Crushing only
(c) Both shearing and crushing
(d) None of these

RPSC VPITI 14.02.2016

Ans. (c) : The force imposed due to the rotation of the shaft causes the failure of the key.
These are two types of failure.
(i) Shear failure of keys
(ii) Crushing failure of key
(i) During rotation the shaft and the machine element, say hub, each element exerts equal and opposite forces introduce shear stress along the radius of the shaft.
The key then starts deformation and after some time it fails.
(ii) During rotation the shaft and the hub impose compressive force causes its deformation the key is then permanently deformed under this force and finally crushing occurs.

178. A knuckle joint with a fork of thickness of each eye as 10 mm and the outside diameter as 20 mm is subjected to a load of 15 kN. The diameter of the pin is 10 mm. What will be the magnitude of tensile stress in the fork?

(a) 100 MPa (b) 75 MPa
(c) 50 MPa (d) None of these

RPSC AE (GWD) 06.05.2016

Ans. (b) : Given,
P = 15 kN = 15×10^3 N
b = 10 mm, d = 10 mm
d_o = 20 mm

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Magnitude of tensile stress in fork or eye

$$\sigma_t = \frac{P/2}{b(d_o - d)}$$

$$= \frac{15\times10^3}{10(20-10)\times2}$$

$$\sigma_t = \frac{150}{2} = 75 \text{ MPa}$$

Where,
P = load
d_o = Outside diameter of eye or fork (mm)
d = diameter of knuckle pin (mm)
b = thickness of eye end of rod.

179. A list of couplings is given below :
(1) Rigid flexible coupling
(2) Bush type flange coupling
(3) Split muff coupling
(4) Universal (Hook's) coupling
Which of these in used for perfectly aligned shafts?
(a) (1) and (2) (b) (1) and (3)
(c) (2) and (3) (d) (2) and (4)

RPSC AE GWD, 2011

Ans. (c) : Bush type flange coupling and split muff coupling is used for perfectly aligned shaft.
Note-Rigid coupling are used where shaft are coaxial and perfectly aligned. Here rigid flexible coupling is given.

180. The shafts are designed for :
(a) strength (b) rigidity
(c) both (a) and (b) (d) none of the above

J & K PSC Screening, 2006

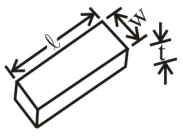
Ans. (c) : Shafts are designed for both strength and rigidity.

181. The design of the key in pulley-shaft assembly is done by considering.
(a) shearing only
(b) crushing only
(c) both crushing and shearing
(d) principle stress and strain

MPPSC State Forest Service Exam, 2014

Ans. (c) : As we know, when we design a key, both crushing & shearing is considered,

$$T = \ell \times w \times \tau \times \frac{d}{2} \quad \text{for shearing}$$



$$T = \ell \times \frac{t}{2} \times \sigma_c \times \frac{d}{2} \text{ for crushing}$$

182. Shaft carrying transverse load and transmitting power is subjected to.
(a) bending stress only
(b) torsional stress only
(c) both bending and torsion stress
(d) axial stresses only

MPPSC State Forest Service Exam, 2014

Ans. (c) : Transmission shaft are rotating members & transmit power & torque, used to support transmission shaft are generally subjected to two kinds of loads, torsion to transmitted torque & bending from transverse loads from gears.

183. The function of the propeller shaft is…
(a) to connect the engine to the drive wheels
(b) to permit variation in speed
(c) to permit sharp turns smoothly
(d) a driving shaft for connecting the transmission of the main shaft to the differential at the rear axles.

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Propeller shaft is a kind of driving shaft use for connecting the transmission of the main shaft to the differential at the rear axles.

184. Splined shaft means a shaft with
(a) Woodruff key
(b) feather key
(c) key integral with the shaft
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Splined shaft means a shaft with key integral with the shaft.

185. The type of stresses developed in key is/are :
(a) Shearing, bearing and bending stresses
(b) Bearing stress only
(c) Shear stress only
(d) Both shear and bearing stresses

UPSC JWM Advt. No.-52/2010

Ans. (d) : Type of stress developed in key are both shear and bearing (compressive/crushing) stress.
• Design on the basis of failure due to shear stress.

$$\tau = \frac{2T}{Db\ell}$$

• Design on the basis of failure due to compressive stress.

$$\sigma_c = \frac{4T}{Db\ell}$$

186. A cotter and key can be compared by which of the following statement
(a) Cotter is used to connect a rod which is subjected to axial loading, whereas keys are used in for twisting moment
(b) Cotter are used to connect shafts transmitting twisting moments whereas keys are used for axial loading
(c) Both cotters and keys are used for axial loading
(d) Both cotters and keys are used for twisting moment

TSGENCO AE 14.11.2015

Ans. (a) : Cotter– The cotter is usually made of mild steel or wrought iron. A cotter joint is a temporary fastening and is used to connect rigidity to co-axial shafts which are subjected to axial tensile or compressive forces.
Key– A key is a piece of mild steel inserted between the shaft and hub of the pulley to connect these together in order to prevent relative motion between them. Keys are used as temporary fastening and subjected to crushing and shear stresses. Keys are used in for twisting moment of shaft.

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187. Which one of the following holds true for coupling and clutch

(a) a coupling cannot be engaged / disengaged frequently
(b) a clutch can be engaged / disengaged frequently
(c) both (1) and (2) are true
(d) both (1) and (2) are false

TSGENCO AE 14.11.2015

Ans. (c) : Couplings – Coupling are used to connect two shaft for torque transmission in varied applications. It may be connect two shaft units such as a motor and generator or it may be to form a long line shaft by connecting shafts of standard lengths say 6-8 m by couplings.

Clutch – A Clutch is a machine member used to connect the driving shaft to a driven shaft, so that the driven shaft may be started or stopped at will, without stopping the driving shaft.

A clutch can be engaged/ disengaged frequently.

188. A solid shaft is to transmit 20 kW at 200 rpm. The ultimate shear stress for the steel may be taken as 360 MPa and a factor of safety as 8. The diameter of solid shaft is

(a) 45 mm (b) 46 mm
(c) 48 mm (d) 50 mm

MPPSC AE 08.11.2015

Ans. (c) : Let, T = transmit torque

Given, Power (P) = 20 kW, Speed (N) = 200 rpm

$= 20 \times 10^3 W$

$\because \quad P = T\omega = \dfrac{T \times 2\pi N}{60}$

$\Rightarrow \quad T = \dfrac{60P}{2\pi N} = \dfrac{60 \times 20 \times 10^3}{2\pi \times 200} = 954.92\ Nm$

Let, D = diameter of shaft = ?

$\because \quad T = \dfrac{\pi}{16} D^3 \tau$

where, τ = Allowable shear stress

$= \dfrac{\text{Ultimate shear stress}}{\text{Facter of safety}}$

$\tau = \dfrac{360MP}{8} = 45\ MPa$

$= 45 \times 10^6\ N/m^2$

$\therefore \quad D^3 = \dfrac{16T}{\pi\tau}$;

$= \dfrac{16 \times 954.92}{\pi \times 45 \times 10^6}$;

$\Rightarrow \quad D^3 = 108.0759 \times 10^{-6}\ m^3$

$\Rightarrow \quad D = 4.763 \times 10^{-2} m$

$\Rightarrow \quad = 0.0476\ m$

$\Rightarrow \quad D = 47.6\ mm$

$\simeq 48\ mm$

189. A solid steel shaft transmits 40kW of power at a speed of $\dfrac{75}{\pi} Hz$. The internal torque needed in the shaft is

(a) $\dfrac{812}{3} Nm$ (b) $\dfrac{800}{3} Nm$
(c) $\dfrac{541}{2} Nm$ (d) $\dfrac{400}{3} Nm$

ISRO Scientist/Engineer 24.05.2014

Ans. (b) : P = 40 kw

$f = \dfrac{75}{\pi} Hz$

$\omega = 2\pi f = 2\pi \times \dfrac{75}{\pi}$

$\omega = 150$ rad/s

$P = 40 \times 10^3\ W = T \times \omega$

or $T = \dfrac{P}{\omega} = \dfrac{40 \times 10^3}{150} = \dfrac{800}{3} = 266.67\ Nm$

190. A transmission shaft includes:

(a) Counter Shaft (b) Line Shaft
(c) Over Head Shaft (d) All of these

SJVN ET 2013

Ans. (d) : A transmission shaft includes
- Counter shaft
- Line shaft
- Overhead shaft

191. A square key is to be used to fix a gear to a 35 mm diameter shaft. The hub length of the gear is 60 mm. Both the shaft and key are to be made of the same material, having the allowable shear stress of 55 N/mm^2. If the torque to be transmitted is 395 N.m, the minimum dimensions of key cross section in mm is

(a) 4.85×4.85 (b) 5.2×5.2
(c) 7.35×7.35 (d) 8.43×8.43
(e) 6.84×6.84

CGPSC AE 26.04.2015 Shift-I

Ans. (e) : Given as,

$l = 60$ mm, d = 35 mm

$\tau = 55\ N/mm^2$

T= 395 N-m

depth of key (t) = width of key (w)

then,

T = shear stress × shear area $\times \dfrac{d}{2}$

$395 \times 1000 = 55 \times w \times 60 \times \dfrac{35}{2}$

w = 6.8398 mm

then cross - section of key = 6.839 × 6.839

192. A pulley is connected to a power transmission shaft of diameter 'd' by means of a rectangular sunk key of width length '*l*'. The width of the key is taken as d/4. For full power transmission, the shearing strength of the key is equal to the torsional shearing strength of the shaft. The ratio of the length of the key to the diameter of the shaft (*l*/d)is

(a) $\dfrac{\pi}{4}$ (b) $\dfrac{\pi}{2}$
(c) $\dfrac{3\pi}{4}$ (d) $\dfrac{\pi}{\sqrt{2}}$
(e) π

CGPSC AE 26.04.2015 Shift-I

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Ans. (b) : Shearing strength of key

$$F = \tau\left(\frac{d}{4}.l\right)$$

Torque (T) = $F.\frac{d}{2}$

$$= \tau\left(\frac{d}{4}.l\right).\frac{d}{2}$$

Torsional shearing

$$\frac{T}{J} = \frac{\tau}{r}$$

$$\frac{T}{\left(\frac{\pi d^4}{32}\right)} = \frac{\tau}{\left(\frac{d}{2}\right)}$$

$$T = \pi d^3 \times \frac{\tau}{16}$$

For same strength

$$\tau\left(\frac{d}{4}.l\right).\frac{d}{2} = \pi d^3 \times \frac{\tau}{16}$$

$$\boxed{\frac{l}{d} = \frac{\pi}{2}}$$

193. Woodruff key is used for :

(a) Heavy torque at high speeds
(b) Low torque at high speeds
(c) Heavy torque at low speeds
(d) Low torque at low speeds

OPSC AEE 2015 Paper-I

Ans : (b) Woodruff key is used for low torque at high speed.

The advantages of woodruff key are as follows:

(i) The woodruff key can be used on tapered shaft because it can align by slight rotations in the seat.

(ii) The extra depth of key in the shaft prevents its tendency to slip over the shaft.

194. The initial tension of M30 bolt used for a fluid tight joint is

(a) 852 N (b) 852000 N
(c) 8520 N (d) 85200 N

TSPSC AEE 2015

Ans : (d) The initial tension of M30 bolt used for a fluid tight joint is 85200 N.

195. To design the bolts of a flange coupling, the crushing stress induced in the bolts is equal to

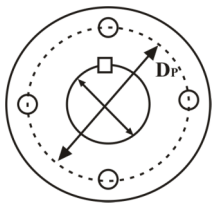
(a) $\frac{2T}{d_1 t_f n D_p}$ (b) $\frac{2T}{d_1 t_f D_p}$

(c) $\frac{2T}{d_1 n D_p}$ (d) $\frac{4T}{d_1 t_f n D_p}$

(Where d_1 = Nominal diameter of bolt;
D_p = Pitch circle diameter ;
n = no. of bolts;
t_f = thickness of flange

TSPSC AEE 2015

Ans : (a)



Let, n = number of bolts
Total force

$$F = n F_1$$

$$F_1 = \frac{F}{n}$$

$$F = \frac{2T}{D_P} \Rightarrow F_1 = \frac{2T}{n.D_P}$$

Crushing stress $\sigma_c = \frac{F_1}{d_1 \times t_F} = \frac{2T}{d_1 \times t_F \times n \times D_P}$

F_1 = Force acting on the bolt
t_F = Thickness of Flange

196. Consider the following

1. Bearing pressure **2. Bending moment**
3. Torsional moment **4. Axial tensile force**
5. Axial compressive force

Of the above, which is used for design of fulcrum pin of lever

(a) 1 (b) 2
(c) 1 and 2 (d) 3 and 4
(e) 2, 3 and 5

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : Bearing pressure is used for design of fulcrum pin of lever.

197. A shaft is subjected to fluctuating loads for which the normal torque (T) and bending moment (M) are 1000 N-m and 500 N-m respectively. If the combined shock and fatigue factor for bending is 1.5 and combined shock and fatigue factor for torsion is 2, then the equivalent twisting moment for the shaft is:

(a) 2000 N-m (b) 2050 N-m
(c) 2100 N-m (d) 2136 N-m

NPCIL ET, 2015

Ans. (d) : Normal torque or torsion (T) = 1000 N-m
Bending moment (M) = 500 N-m
Fatigue factor for torsion $k_T = 2$
Fatigue factor for bending $k_m = 1.5$
Equivalent twisting moment is given by –

$$T_e = \sqrt{(k_m \times M)^2 + (k_t T)^2}$$

$$= \sqrt{(1.5 \times 500)^2 + (2 \times 1000)^2}$$

$$= 2136.00093 \text{ N-m}$$

198. During shearing failure of the key, the area under shear is

(a) both length and thickness
(b) both width and thickness
(c) both length and width
(d) only length

Mizoram PSC AE/SDO 2014, Paper-II

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Ans. (c) : Shear Design of Key–

$$(T_{max})_{shear} = \frac{1}{2} \times db\ell\tau_{permissible}$$

Where, $(T_{max})_{shear}$ = Shear strength of key in terms of torque

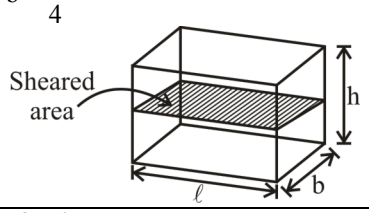
d = Diameter of shaft,

ℓ = Length of key

b = Width of key

$\tau_{permissible}$ = Permissible shear stress,

Generally, $b = \frac{d}{4}$



199. A shaft directly coupled to a power source is called

(a) line shaft (b) counter shaft

(c) flexible shaft (d) jack shaft

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : A shaft directly coupled to a power source is called line shaft.

200. A round key is used for

(a) light duty (b) medium duty

(c) heavy duty (d) extra heavy duty

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : A round key is used for light duty.

201. In a rigid type of flange coupling, the coupling halves are joined by four bolts at 112 mm pcd. The coupling transmits 11 kw power at 1500 rpm. The shear force on each bolt will be

(a) 312.5 N (b) 500 N

(c) 1250 N (d) 750 N

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Power 'P' $= \frac{2\pi NT}{60}$

$$T = \frac{60P}{2\pi N} = \frac{60 \times 11 \times 10^3}{2\pi \times 1500}$$

T = 70 Nm

Shear force

$$F_t = \frac{2T_f}{D \times n}$$

$$= \frac{2 \times 70 \times 10^3}{112 \times 4}$$

$$\boxed{F_t = 312.5N}$$

202. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I	**List-II**
(A) Rigid coupling	1. Permits greater torque capacity for a given size
(B) Oldham coupling	2. Permits substantial angular misalignment of intersecting axis
(C) Universal joint	3. Permits substantial lateral offset and little angular alignment
(D) Gear coupling	4. Permits almost no misalignment

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	2	3	1
(c)	1	3	2	4
(d)	4	3	2	1

UPSC JWM Adv. No-16/2009

Ans. (d) :

(A) Rigid Coupling --(4)-- Permits almost 'no misalignment'

(B)Oldham Coupling --(3)—Permits substantial lateral offset and little angular alignment

(C) Universal Joint --(2)-- Permits substantial angular misalignment of intersecting axes.

(D) Gear Coupling --(1)-- Permits greater torque capacity for a given size.

203. A small value of permissible stress is used for the bolts of a flange coupling due to

(a) combined loading

(b) initial lightening

(c) stress concentration

(d) none of these

TNPSC ACF 2012

Ans. (b) : A small value of permissible stress is used for the bolts of a flange coupling due to initial lightening.

204. Number of bolt in flange coupling should not be less than

(a) 3 (b) 2

(c) 4 (d) 6

GPSC AE 26.07.2015

Ans. (c) : 4

205. A key connecting a flange coupling to a shaft is likely to fail in

(a) Shear (b) Tension

(c) Torsion (d) Bending

GPSC AE 26.07.2015

Ans. (a) : A key connecting a flange coupling to a shaft is likely to fail in shear.

206. In fluid coupling, the ratio of output shaft torque to input shaft torque is :

(a) One (b) Less than one

(c) Greater than one (d) Zero

TNPSC AE (Industries) 09.06.2013

Ans. (a) : One

207. Length of cotter, in sleeve and cotter joint is taken as : (where d is the diameter of the rod)

(a) 1.5 d (b) 2.5 d

(c) 3 d (d) 4 d

TNPSC AE (Industries) 09.06.2013

Ans. (d) : 4 d

208. In shafts with keyways, the allowable stresses are usually _____ of the value given

(a) 25% (b) 50%

(c) 75% (d) 95%

APPSC AEE 2012

Ans. (c) : In shaft with keyways the allowable stresses are usually 75% of the value given.

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209. A square key is used to key a gear to a 40 mm diameter shaft. The hub length of the gear is 50 mm. Both shaft and the key are made of same material having allowable shear strength of 50 MPa. If a torque of 500 Nm is to be transmitted, what is the minimum value of the dimension of the sides of the square key?

(a) 20 mm (b) 10 mm
(c) 5 mm (d) 1 mm

DRDO Scientists 2008

Ans. (b) : Shearing area = hub length (or key length) × width of key (or side of square key)

$= l \times d$

$$d = \frac{2T}{\tau \ell d} = \frac{2 \times 500 \times 1000}{50 \times 50 \times 40} = 10 \text{ mm}$$

210. A Hooke's joint is used to connect two

(a) Coplanar and Nonparallel shafts
(b) Non coplanar and Nonparallel shafts
(c) Coplanar and Parallel shafts
(d) All the above option

ISRO Scientist/Engineer 2009

Ans. (a) : Hooke's Joint
Its also called universal joint (Universal coupling, U-joint, carbon joint, spicer as hardy spicer joint as Hook's joint) is a joint as coupling connected rigid rods whose axes are inclined to each other, and is commonly used in shaft that transmit rotary motion.

211. Which of the following stresses are associated with the tightening of a nut on a stud?

1. **Tensile stresses due to stretching of stud.**
2. **Bending stresses of stud.**
3. **Transverse shear stresses across threads.**
4. **Torsional shear stress in threads due to frictional resistance.**

Select the correct answer using the codes given below:

Codes:

(a) 1, 2 and 3 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 4

ISRO Scientist/Engineer 2007

Ans. (b) : The tightening of a nut on a stud the following stress are associated–
1. Tensile stresses due to stretching of stud.
2. Transverse shear stresses across threads
3. Torsional shear stresses in threads due to frictional resistance.

3. Design of Threaded Joints and Power Screws

212. The maximum efficiency of a screw jack is .....

(a) $\frac{1-\sin\phi}{1+\sin\phi}$ (b) $\frac{1+\sin\phi}{1-\sin\phi}$
(c) $\frac{1-\tan\phi}{1+\tan\phi}$ (d) $\frac{1+\tan\phi}{1-\tan\phi}$

GPSC DEE Class-2 (GWSSB) 04.07.2021
MECON MT 2019
RPSC AE 2018
Nagaland PSC CTSE 2018, 2017 Paper-I
GPSC Lect. (Auto) 16.10.2016
CGPSC Poly. Lect. 22.05.2016
TNPSC AE 2013
TNPSC AE (Industries) 09.06.2013
ESE 2007

Ans. (a) : Efficiency of screw jack

$$\eta = \frac{\tan\alpha}{\tan(\alpha+\phi)}$$

$$\eta = \frac{(\sin\alpha/\cos\alpha)}{\left[\frac{(\tan\alpha+\tan\phi)}{1-\tan\alpha.\tan\phi}\right]}$$

$$= \frac{\frac{\sin\alpha}{\cos\alpha}\left[1-\tan\alpha.\tan\phi\right]}{\left[\tan\alpha+\tan\phi\right]}$$

$$= \frac{\frac{\sin\alpha}{\cos\alpha}\left[1-\frac{\sin\alpha}{\cos\alpha}\times\frac{\sin\phi}{\cos\phi}\right]}{\left[\frac{\sin\alpha}{\cos\alpha}+\frac{\sin\phi}{\cos\phi}\right]}$$

$$= \frac{\frac{\sin\alpha}{\cos\alpha}\left[\cos\alpha.\cos\phi-\sin\alpha.\sin\phi\right]}{\left[\sin\alpha.\cos\phi+\sin\phi.\cos\alpha\right]}$$

$$= \frac{\frac{\sin\alpha}{\cos\alpha}\left[\frac{\cos\alpha.\cos\phi-\sin\alpha.\sin\phi}{\sin\alpha.\cos\phi+\sin\phi.\cos\alpha}\right]}{}$$

$$= \frac{2\sin\alpha\times\cos(\alpha+\phi)}{2\cos\alpha\times\sin(\alpha+\phi)}$$

$$= \frac{\sin(2\alpha+\phi)-\sin\phi}{\sin(2\alpha+\phi)-\sin\phi} \quad ...(1)$$

For max. efficiency

$$\sin(2\alpha+\phi)=1$$

$$2\alpha+\phi=\sin^{-1}(1)=\frac{\pi}{2}$$

$$2\alpha=\frac{\pi}{2}-\phi$$

$$\alpha=\frac{\pi}{4}-\frac{\phi}{2}$$

Putting this value in equation (1) and we get-

$$\eta_{max}=\frac{1-\sin\phi}{1+\sin\phi}$$

213. The efficiency of a screw jack is maximum, when

(a) $\alpha = 45^\circ + \frac{\phi}{2}$ (b) $\alpha = 45^\circ - \frac{\phi}{2}$
(c) $\alpha = 90^\circ + \phi$ (d) $\alpha = 90^\circ - \phi$

Where α = Helix angle and ϕ = Angle of friction.

OPSC AEE 2019 Paper-I
SJVN ET 2019
WBPSC AE, 2017
APPSC AE Subordinate Service Civil/Mech. 2016
APPSC AEE MAINS 2016, PAPER-III
UPPSC AE 12.04.2016 Paper-I
APPSC Poly. Lect. 2013
ESE 2014, 2004
GATE 1997

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Ans : (b) Efficiency of screw jack

$$(\eta) = \frac{\tan\alpha}{\tan(\alpha+\phi)}$$

$$\eta = \frac{\sin\alpha.\cos(\alpha+\phi)}{\cos\alpha.\sin(\alpha+\phi)}$$

$$1-\eta = 1-\frac{\sin\alpha.\cos(\alpha+\phi)}{\cos\alpha.\sin(\alpha+\phi)}$$

$$1-\eta = \frac{\sin\phi}{\cos\alpha.\sin(\alpha+\phi)}$$

Multiplying and dividing by 2

$$1-\eta = \frac{2\sin\phi}{2\cos\alpha.\sin(\alpha+\phi)}$$

$$= \frac{2\sin\phi}{\sin(2\alpha+\phi)+\sin\phi}$$

Now for efficiency to be greatest, $1-\eta$ should be least i.e $\sin(2\alpha+\phi)$ should be maximum

$\therefore \quad \sin(2\alpha+\phi) = 1$

$2\alpha+\phi = 90^\circ$

$\boxed{\alpha = 45-\phi/2}$

214. In a screw jack, the effort required to lift the load W is given by (where α = Helix angle, and ϕ Angle of friction) :

(a) P = W tan (α – ϕ) (b) P = W cos (α – ϕ)
(c) P = W tan (α + ϕ) (d) P = W cos (α + ϕ)

ISRO Scientist/Engineer 12.01.2020
TNPSC AE 2019
RPSC Vice Principal ITI 2018
CGPSC AE 25.02.2018
HPPSC Asstt. Prof. 20.11.2017
GPSC Lect. (Poly.) 7.12.2014
TNPSC AE 2013

Ans. (c) : Given,
P = Effort, W = load, α = Helix angle/lead angle
ϕ = friction angle.

- Effort required to raise the load
  P = W tan (α + ϕ)
- Effort required to lower the load
  P = W tan (ϕ – α)

215. The shock-absorbing capacity (resilience) of bolts can be increased by

(a) increasing the shank diameter above the core diameter of threads
(b) reducing the shank diameter to the core diameter of threads
(c) decreasing the length of shank portion of the bolt
(d) pre-heating of the shank portion of the bolt

ESE 2019
OPSC AEE 2019 Paper-I
HPPSC Asstt. Prof. 20.11.2017
MPPSC AE. 2016
TPSC AE. 2015
JPSC AE 2013, Paper-V
TNPSC AE (Industries) 09.06.2013
TRB Poly. Lect., 2012
ESE 2007

Ans. (b) : The shock absorbing capacity of bolts can be increased by reducing the shank diameter of the bolt to the core diameter of threads.
This can be done either by reducing diameter of unthread part or making hole in thread part along axis. This concept is used in bolt of uniform strength.

216. What is the efficiency of a self-locking power screw?

(a) 70% (b) 60%
(c) 55% (d) <50%

Assam PSC AE (IWT) 14.03.2021
OPSC AEE 2019 Paper-I
TSPSC Managers, 2015
TSPSC AEE 2015
GPSC AE 26.07.2015
TNPSC AE, 2008
ESE 2002
ESE 1997

Ans : (d) :
The efficiency of screw jack

$$\eta = \frac{\tan\alpha}{\tan(\phi+\alpha)}$$

For self locking

$\phi \geq \alpha$

$$\eta \leq \frac{\tan\phi}{\tan^2\phi}$$

$$\leq \frac{\tan\alpha}{\left[\frac{2\tan\phi}{1-\tan 2\phi}\right]} \leq \frac{1-\tan^2\phi}{2}$$

$$\leq \frac{1}{2}-\frac{1}{2}\tan^2\phi$$

in ideal case i.e. zero friction

$\phi = 0$

$$\eta \leq \frac{1}{2}-0$$

$$\eta_{max} = 0.5 = 50\%$$

217. One of the following power screws transmit power in one direction only :

(a) Buttress thread (b) Acme tread
(c) Square thread (d) Trapezoidal thread

TNPSC AE 2018
GPSC EE Pre, 28.01.2017
GPSC Asstt. Prof. 28.08.2016
OPSC AEE 2015 Paper-I
UPRVUNL AE 2014
ISRO Scientist/Engineer 2011

Ans : (a) Buttress thread power screws transmit power in one direction only.
Buttress thread:– It is used in vices. It combines the advantage of square and trapezoidal threads.
It is used where a heavy axial force acts along the screw axis in one direction only.
Advantage:–
(i) It has higher efficiency compared with trapezoidal thread.
(ii) Thread milling machine is used to manufacture it.
(iii) Stronger than square and trapezoidal thread tooth.
Buttress threads also used for power screws but they can transmit power only in one direction, used in screw jack, vices etc.

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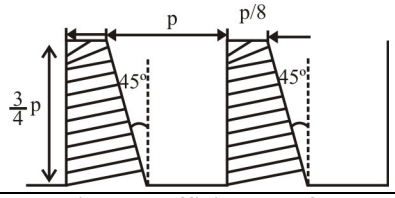

218. The maximum efficiency of a screw jack provided with square threads and angle of friction 30° will be:

(a) 25% (b) 33%
(c) 40% (d) 50%

HPPSC ADF 05.03.2019
RPSC AE 2016
HPPSC W.S. Poly. 2016
UKPSC AE 2012 Paper-I
RPSC ACF-2011

Ans. (b) : Given, $\phi = 30°$

For screw jack maximum efficiency is given by–

$$\eta_{max} = \frac{1-\sin\phi}{1+\sin\phi} = \frac{1-\sin 30°}{1+\sin 30°}$$

$$= \frac{1-\frac{1}{2}}{1+\frac{1}{2}} = \frac{\frac{2-1}{2}}{\frac{2+1}{2}}$$

$$= \frac{1}{3}\times 100$$

$$= 33\%$$

219. The function of a washer is to :

(a) provide cushioning effect
(b) provide bearing area
(c) absorb shocks and vibrations
(d) provide smooth surface in place of rough surface

Vizag Steel (MT) 2017
RPSC AE 2016
MPSC Poly. Lect. 2014
RPSC ACF-2011
ISRO Scientist/Engineer 2008

Ans. (b) : Washer, machine component that is used in conjunction with a screw fastener such as a bolt and nut and that usually serves either to keep the screw from loosening or to distribute. The load from the nut or bolt head over a larger area. For load distribution, thin flat rings of shaft steel are used.

220. The efficiency of screw jack is

(a) $\frac{\tan(\alpha+\phi)}{\tan\alpha}$ (b) $\frac{\tan\alpha}{\tan(\alpha+\phi)}$

(c) $\frac{\tan(\alpha-\phi)}{\tan\alpha}$ (d) $\frac{\tan\alpha}{\tan(\alpha-\phi)}$

TNPSC AE 2019
Nagaland PSC (CTSE) 2018 Paper-I
APPSC AEE MAINS 2016, PAPER-III
RPSC AE (GWD) 06.05.2016
TNPSC AE 2013

Ans. (b) : The efficiency of screw jack is given as

$$\eta_{S.J.} = \frac{\tan\alpha}{\tan(\alpha+\phi)} = \frac{\text{Ideal effort}}{\text{Actual effort}}$$

A screw will be self locking if the friction angle is greater than helix angle or coefficient of friction is greater than of helix tangent of helix angle.

$$\mu(\text{or})\tan\phi > \tan\alpha$$

221. The resilience of a bolt may be increased by :

(a) Decreasing its length
(b) Increasing its length
(c) Decreasing its shank diameter
(d) Increasing its shank diameter

TNPSC AE 2013
APPSC Poly. Lect. 2013
RPSC ACF-2011
UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation : Resilience is defined as, Ability of a material to absorb energy in the elastic region when it is strained.

$$\text{Resilience} = \frac{\sigma^2}{2E}\times Volume$$

{∵ Volume = Area × Length = A × L}

$$\text{Resilience} = \left(\frac{\sigma^2.A}{2E}\right)\times L$$

$\Rightarrow$ Resilience $\propto$ L

Hence, The resilience of a bolt may be increased by increasing its length.

Resilience of a bolt may be increased by-

(1) Reducing the shank diameter equal to or less than core diameter.

(2) Increasing the shank length.

222. A bolt of M 24 × 2 means that–

(a) the pitch of the thread is 24 mm and depth is 2 mm
(b) the cross-sectional area of the threads is 24 mm 2
(c) the nominal diameter of bolt is 24 mm and the pitch is 2 mm.
(d) the effective diameter of the bolt is 24 mm and there are two threads per cm

CGPSC AE 16.10.2016
Nagaland CTSE 2017, 2016 Ist Paper
WBPSC AE 2008

Ans. (c) : A, bolt M 24×2 means, where 'M' stand for metric and number 24 is for nominal diameter or major diameter of bolt in (mm) & '2' is the pitch of the threads in (mm).

223. On a triple start, thread screw

(a) Lead = pitch (b) Lead = 3× pitch
(c) Lead = (1/2) × pitch (d) Lead = 9× pitch

APPSC Poly Lect. 13.03.2020
MPPSC AE 2016
UJVNL AE 2016

Ans : (b) Lead = n × pitch
where n = number of start
on a triple start, thread Screw
Lead = 3×pitch
on a double start, thread Screw
Lead = 2×pitch

224. For self locking of the screw, which of the following is correct?

(a) Coefficient of friction > tangent of helix angle

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(b) Coefficient of friction < tangent of helix angle
(c) These is no relation between coefficient of friction and tangent of helix angle for the screw to be self locking
(d) Efficiency of self locking square thread power screw is always greater than 50%
(e) Tangent of helix angle is always zero in self locking screw

APPSC Poly Lect. 13.03.2020
CGPSC AE 25.02.2018
CIL MT 26.03.2017

Ans. (a) : Self locking screw- The torque required to lower the load can be obtained by

$$T = \frac{Wd_m}{2}\tan(\phi - \alpha)$$

$\phi \rightarrow$ friction angle

$\alpha \rightarrow$ Load angle

If $\phi < \alpha$, the torque required to lower the load is negative. It indicates a condition that no force is required to lower the load. The load it self will being to turn the screw and descend down, unless a restraining torque is applied. This condition is also called overhauling of the screw.

If $\phi \geq \alpha$, then a positive torque is required to lower the load under this condition, the load will not turn the screw and will not descend on its own unless an effort p is applied. In this case, the screw is said to be 'self-locking'.

"A screw will be self-locking if the coefficient of friction is equal to or greater than the tangent of the helix angle".

225. The bolts in a rigid flanged coupling connecting two shafts transmitting power are subjected to
(a) shear force and bending moment
(b) axial force
(c) torsion
(d) torsion and bending moment

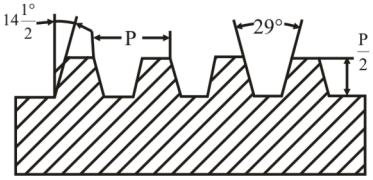
GPSC Executive Engineer 23.12.2018
MPPSC State Forest Service Exam, 2014
TRB Poly. Lect., 2012
ESE 2002
GATE 1996

Ans. (a) : The bolts in a rigid flanged coupling connecting two shaft transmitting power are subjected to shear & bending stresses while transmitting torque.

226. The thread angle of ACME thread is :
(a) 10^0 (b) 19^0
(c) 29^0 (d) 39^0

APPSC Poly Lect. 13.03.2020
APPSC AEE MAINS 2016, PAPER-III
CGPSC AE 26.04.2015 Shift-I
OPSC AEE 2015 Paper-I
GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : Acme Threads–The acme threads, shown in figure, have thread angle equal to 29°.
Advantage of Acme Threads–
(i) Acme threads permit the use of split nut which can compensate the wear.
(ii) Acme threads are stronger than the square threads in shear because of the larger cross section at the root.
(iii) Acme threads can transmit power in either directions.



Thread	Angle
Square thread	0^0
Buttress thread	45^0
British Association	47.5^0
Acme thread	29^0
Trapezoidal thread	30^0

227. Turn-buckle has
(a) right-hand threads on both the ends
(b) left-hand threads on both the ends
(c) left-hand threads on one end and right-hand threads on the other
(d) left-hand threads on one hand

ISRO Scientist/Engineer 2012
Nagaland PSC CTSE 2017, Paper-I
WBPSC AE, 2007

Ans. (c) : Turnbuckle mechanical device that connects the threaded ends of two rods and permits them to be adjusted for length or, tension.
The turnbuckle has two collinear holes, one of which has right-hand threads and the other left-hand threads that mate will the right and left handed threads on the ends of the rods.

228. To ensure self locking in a screw jack it is essential that helix angle is
(a) larger than friction angle
(b) smaller than friction angle
(c) equal to friction angle
(d) such as to give maximum efficiency in lifting

ISRO Scientist/Engineer 12.05.2013
ISRO Scientist/Engineer 2011
ESE 1996
TNCSC AE 2020
UPSC JWM Adv. No-16/2009

Ans. (b) : Helix angle is smaller than friction angle. We know that a screw will be self-locking if and only if its efficiency is below 50%.

229. Multiple threads are used to secure
(a) low efficiency
(b) high efficiency
(c) high load lifting capacity
(d) high mechanical advantage

Nagaland PSC (CTSE) 2018, Paper-I
HPPSC Asstt. Prof. 18.09.2017
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : When we donot get enough efficiency or power from a single screw thread we uses multiple threads to secure high efficiency & max power output.

230. The motion of a nut on a threaded bolt is
(a) Helical (b) Plane
(c) Spherical (d) None of the above

UKPSC AE 2012 Paper-I
UKPSC AE 2007 Paper -I

Ans. (a) : Helical

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231. In a metric thread designated by M12×1.5 specify that the thread has :

(a) cross sectional area 12 mm^2 and depth 1.5
(b) nominal diameter 12mm and pitch 1.5
(c) nominal diameter 12 mm and number of threads per mm is 1.5
(d) Pitch 1.5 and depth 12 mm

Kerala PSC AE 06.08.2015
KPSC AE. 2015

Ans : (b) In a metric thread designated by M12×1.5 specify that the thread has
(i) Nominal diameter =12mm
(ii) Pitch = 1.5

232. Eyebolts are used–

(a) to prevent relative motion between two parts
(b) to absorb shocks and vibrations
(c) to lift and transport machines and heavy objects on shop floor
(d) to prevent loosening of threads in bolted assembly

Nagaland PSC CTSE 2017, Paper-I
APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Eyebolts are used to lift and transport machines and heavy objects on shop floor.

233. The frictional torque for trapezoidal thread at mean radius while raising load is given by (W = load, d_m = mean diameter, ϕ = angle of friction, α = Helix angle)

(a) $\frac{Wd_m}{2}\cdot\frac{(\mu\sec\theta+\tan\alpha)}{(1-\mu\sec\theta.\tan\alpha)}$

(b) $\frac{Wd_m}{2}\cdot\frac{(\mu+\tan\alpha)}{(1-\tan\alpha)}$

(c) $\frac{Wd_m}{2}\cdot\frac{(\mu\sec\theta-\tan\alpha)}{(1+\mu\sec\theta.\tan\alpha)}$

(d) $\frac{Wd_m}{2}\cdot\frac{(\mu-\tan\alpha)}{(1+\tan\alpha)}$

Nagaland PSC CTSE 2017, Paper-I
Nagaland CTSE 2016 Ist Paper

Ans. (a) : Expression, to find the frictional torque for trapezoidal thread at mean radius, while raising the load is given by,
W = Load
d_m = mean diameter
ϕ = angle of friction
α = Helix angle

$$T=\frac{Wd_m}{2}\cdot\frac{(\mu\sec\theta+\tan\alpha)}{(1-\mu\sec\theta.\tan\alpha)}$$

234. The most important dimension in the design of nut is :

(a) inside diameter (b) height
(c) pitch diameter (d) thread size

RPSC AE 2016
RPSC ACF-2011

Ans. (b) : The most important dimension in the design of nut is height.

235. Diameter of washer is generally taken :

(a) equal to nut size
(b) less than nut size
(c) bigger than nut size
(d) any size irrespective of nut size

RPSC AE 2016
RPSC ACF-2011

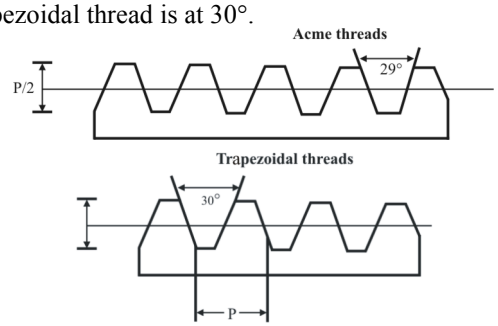
Ans. (c) : Flanged nut is hexagonal nut with a washer that is a flat circular disc attached to it. It is thus provided with a larger bearing surface. A bolt can be used in a comparatively large size hole with help of this nut. It is widely used in automobiles.

236. Which of the following is Trapezoidal thread.

(a) ACME (b) Square
(c) Buttress (d) Metric

TSPSC Managers, 2015
ISRO Scientist/Engineer 11.10.2015

Ans. (a) : Trapezoidal threads are similar to ACME threads, except the thread angle. The American ACME form has an included flank angle 29°. The metric Trapezoidal thread is at 30°.



237. Power screws are used to convert

(a) Linear motion to rotary motion
(b) Rotary motion into translatory motion
(c) Both (a) & (b)
(d) None of these

APGENCO AE 2012
ISRO Scientist/Engineer 2009

Ans. (b) : Power screws are used to convert rotary motion into translatory motion.

238. A knuckle joint is used to transmit

(a) axial compressive force only
(b) axial tensile force only
(c) bending moment
(d) torsional moment

GPSC Lect. 23.10.2016
UKPSC AE 2007 Paper -I

Ans. (b) : A knuckle joint is a form of pin point that is used to transmits tension loads while allowing rotation in one plane.

239. Which of the following stresses are associated with the tightening of a nut on a stud?

(a) Transverse shear stresses across threads
(b) Tensile stresses due to stretching of stud
(c) Torsional shear stresses in threads due to frictional resistance
(d) Bending stresses of stud

UPRVUNL AE 21.08.2016
J&K PSC Civil Services Pre, 2010

Ans. : (b) Tensile stresses are induced due to tightening action on the nut. Then compressive stresses will get induced on the threads of screw. But imagine if two ends of the screw are fixed then on tightening the nut over it one more stress that is shear stress will also get introduced in the screw.

240. Bolts in the flanged end of a pressure vessel are usually pre-tensioned.
Indicate which of the following statements is not true

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(a) Pre-tensioning helps to seal the pressure vessel.
(b) Pre-tensioning reduces the maximum tensile stress in the bolts.
(c) Pre-tensioning increases the fatigue life of the bolts.
(d) Pre-tensioning helps to reduce the effect of pressure pulsations in the pressure vessel

Karnataka PSC AE, 10.09.2017
GATE 1998

Ans. (b) : Pre-tensioning of bolts are done to get following in the flanged end of a pressure vessel.
- To increase fatigue life of the bolts
- To reduce the effect of pressure pulsations in the pressure vessel.
- Pre-tensioning helps to seal the pressure vessel.

241. The square threaded power screw is self-locking when (Coefficient of friction - μ) Lead angle - λ)
(a) $\mu = \tan(\lambda) / \lambda$ (b) $\tan(\lambda) > \mu$
(c) $\lambda \geq \tan\mu$ (d) $\mu \geq \tan(\lambda)$

Haryana PSC AE (PHED) 05.09.2020, Paper-II
Gujarat PSC AE 2019

Ans. (d) : For self locking, $\phi \geq \alpha$
Where ϕ = friction angle = $\tan^{-1}\mu$
$\lambda = \alpha$ = Helix angle or lead angle
$\tan^{-1}\mu \geq \lambda$ or $\mu \geq \tan\lambda$

242. In a bolted joint two members are connected with an axial tightening force of 2200 N. If the bolt used has metric threads of 4 mm pitch, the torque required for achieving the tightening force is
(a) 0.7 Nm (b) 1.4 Nm
(c) 2.8 Nm (d) 1.0 Nm

TSPSC AEE 2017
UPRVUNL AE 2014
GATE 2004

Ans. (b) : Given,
Tightening force = 2200 N
P = 4 mm = 0.004m

$$\text{Torque} = \text{tighting force} \times \frac{0.004}{2\pi}$$

$$= 2200 \times \frac{0.004}{6.28} = 1.4\text{N} - \text{m}$$

243. If a fastener is threaded into a tapped hole, then the fastener is likely to be called as
(a) screw (b) bolt
(c) washer (d) screw or bolt

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Screw–If a fastener is threaded into a tapped hole, then the fastener is likely to be called as screw.
Bolt–A bolt is a mechanical fastener with a threaded shaft bolts are closely related to screw.
Washer–A washer is a flat ring of metal put under the head of a bolt or nut to spread the load when the bolt or nut is tightened. Put washers under the bolts to prevent damage to the surface being fixed.

244. Which type of joints is better when the product is subjected to large vibrations : welded or threaded?
(a) Welded
(b) Threaded
(c) Both have same results
(d) Depends on the magnitude of the vibrational force

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Welded joints is better when the product is subjected to large vibrations because threaded joints loosen when subjected to vibrations.

245. A power screw is turned at a constant rate by application of steady torque of 150 N-m. How much work is being spent per revolution.
(a) 942 J (b) 4000 J
(c) 650 J (d) 740 J

CGPSC AE 15.01.2021

Ans. (a) : Given, Torque (T) = 150 N-m
$\theta = 2\pi$ (for per revolution)
Work spent (W) = $T \times \theta$
$W = 150 \times 2\pi$
$W = 942.47$ J

246. The diameter of the hole, that must be drilled in the head of the bolt to make the bolt of uniform strength, is equal to ——. Where D_o is the outside diameter of thread and D_c is the core diameter of thread.
(a) $D_o - D_C$ (b) $(D_O + D_C)/2$
(c) $\sqrt{(D_O^2 - D_C^2)}$ (d) $(D_O - D_C)/2$

APGCL AM, 2021

Ans. (c) : For uniform strength of bolt, an axial hole is drilled through the Head as for as the thread portion. Such that area of Shank = Root area of the thread.
if D = Diameter of the Hole.
Do = Outer diameter of thread.
Dc = Root or core diameter of thread.

Then $\frac{\pi}{4}D^2 = \frac{\pi}{4}\left((D_O)^2 - (D_C)^2\right)$

$$\boxed{D = \sqrt{D_O^2 - D_C^2}}$$

247. In case of a bolted joint, if the gasket is too soft, the resultant load, P_b on bolt will be :
(a) initial tension, P_i only
(b) external load, P, only
(c) sum of initial tension and external load, $(P_i + P)$
(d) difference of external load and initial tension, $(P - P_i)$

JPSC AE 10.04.2021, Paper-II

Ans. (c) : In case of bolted joint. If gasket is too soft, the resultant load P_b on bolt will be sum of initial tension (P_i) and external load (P)

$$\boxed{P_b = P_i + P}$$

248. A screw thread with M12 × 1.25 designation indicates :
(a) A coarse thread, with 12 mm major diameter
(b) A coarse thread, with 12 mm minor diameter
(c) A fine thread with 12 mm major dia and 1.25 mm pitch.
(d) A fine thread with 12 mm minor dia and 1.25 mm pitch.

JPSC AE 10.04.2021, Paper-II

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Ans. (c) : M12 × 1.25 indicates a fine thread with 12 mm major dia and 1.25 mm pitch.

249. Transverse shear stress at the root of the nut τ_n is expressed by :

(a) $\frac{W}{\pi d_c tZ}$ (b) $\frac{W}{\pi dtZ}$

(c) $\frac{4W}{\pi d_c^2}$ (d) $\frac{16Wt}{\pi d_c^3}$

Where W – load, d – major dia,
d_c – core dia
t – thread thickness at the root of the nut and
Z – number of threads in engagement with the nut.

JPSC AE 10.04.2021, Paper-II

Ans. (b) : Transverse shear stress at the root of the nut τ_n

$$\tau_n = \frac{W}{\pi d_c tZ}$$

Transverse shear stress in thread of screw τ_s

$$\tau_s = \frac{W}{\pi dtZ}$$

250. Which aspect is not true regarding a universal joint?

(a) the speed of driven shaft undergoes linear variation
(b) the constant velocity universal joints are much costlier and complicated in construction
(c) the magnitude of variation in the speed of driven shaft becomes considerable for large angle of inclination
(d) the speed of the driven shaft does not remain uniform

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : the speed of driven shaft undergoes linear variation

251. A universal joint allows the propeller shaft to

(a) bend side ways
(b) change length
(c) transfer torque at an angle
(d) change inclination

HPPSC Poly. Lect. 05.07.2021

Ans. (c) : A universal joint allows the propeller shaft to transfer torque at an angle.

252. The maximum efficiency of a screw jack is a function of

(a) helix angle (b) angle of friction
(c) load lifted (d) effort

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : The maximum efficiency of a screw jack is a function of angle of friction.

253. Pre-tensioning of a bolted joint is used to

(a) strain harden the bolt head.
(b) decrease stiffness of the bolted joint.
(c) increase stiffness of the bolted joint.
(d) prevent yielding of the thread root.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : Pre-tensioning of a bolted joint is used to increase stiffness of the bolted joint.

254. The yield strength of bolt material is 300 MPa and factor of safety is 2.5. What is the maximum principal stress using maximum principal stress theory?

(a) 750 MPa (b) 120 MPa
(c) 27.38 MPa (d) 10.95 MPa

ESE 18.07.2021

Ans. (b) : Given, σ_{yt} = 300 MPa
Factor of safety, FOS = 2.5
Using maximum principal stress theory,

Maximum principal stress, $\sigma_1 = \frac{\sigma_{yt}}{FOS}$

$$\sigma_{max} = \frac{300}{2.5} = 120 MPa$$

255. Consider the following statements regarding typical analysis of bolt failure:

1. 15% failure of bolt occur at the fillet under the head.
2. 50% failure of bolt occur at the end of threads on the shank.
3. 80% failure of bolt occur in the threads that are in contact with the nut.

Which of the above statements is/are correct?

(a) 1 only (b) 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (a) :
A typical analysis of bolt failures indicate that
- 15% failures of bolt occur at the fillet under the head.
- 20% failures of bolt occur at the end of threads on the shank.
- 65% failures of bolt occur in the threads that are in contact with the nut.

256. A circular solid rod of diameter 'd' welded to a rigid flat plate by a circular fillet weld of throat thickness 't' is subjected to a twisting moment 'T'. The maximum shear stress induced in the weld is

(a) $\frac{T}{\pi td^2}$ (b) $\frac{2T}{\pi td^2}$

(c) $\frac{4T}{\pi td^2}$ (d) $\frac{8T}{\pi td^2}$

UPPSC AE 13.12.2020, Paper-I
ESE 2003

Ans. (b) : The maximum shear stress occurs in the throat area. Thus, for a given torque (T) the maximum shear stress in the weld is

$$\tau_{max} = \frac{T\left(\frac{d}{2} + t\right)}{I_P}$$

Where, I_P = Polar moment of area at the throat section

$$= \frac{\pi}{32}\left[(d+2t)^4 - d^4\right] = \frac{\pi}{4}td^3$$

When, $t << d$

$$\tau_{max} = \frac{T \times \frac{d}{2}}{\frac{\pi}{4}td^3} = \frac{2T}{\pi td^2}$$

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257. Which among the following is NOT a standard locking device used to prevent loosening of the nut or bolt?

(a) Lock nut (b) Sawn nut
(c) Castle nut (d) Screwed crow nut

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Standard locking device used to prevent loosening of the nut or bolt.
(I) lock nut, (II) sawn nut, (III) castle-nut.

258. Ring nut is used where

(a) the nut is to be locked by a screw through a ring
(b) the nut to be used without washer
(c) the nut is to be put above the main nut
(d) the nut is non-removable

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Ring nut is used where the nut is to be locked by a screw through a ring.

259. If 'n' number of bolts (made of material with shear stress 's') are subjected to a shear load of 'S', then the diameter of the bolt is given as:

(a) $\sqrt{\frac{4S}{\pi sn}}$ (b) $\sqrt{\frac{sn}{4\pi sn}}$
(c) $\sqrt{\frac{4\pi S}{sn}}$ (d) $\sqrt{\frac{4s}{\pi Sn}}$

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : $\frac{S/n}{\frac{\pi}{4}d_c^2} = s$

Diameter of bolts $= \sqrt{\frac{4S}{\pi sn}}$

260. The washer is generally specified by its-

(a) Outer diameter (b) Hole diameter
(c) Thickness (d) Mean diameter

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : A small flat metal, rubber or Plastic ring fixed between two joining surface or between nut and a bolt to spread the pressure or act as a spacer or seal. It is generally specified by its hole diameter.

261. The power screw efficiency is given by (P= Load, T = Torque, L - Lead of thread)

(a) $\frac{PL}{2\pi T}$ (b) $\frac{4\pi T}{PL}$
(c) $\frac{T}{2PL}$ (d) $\frac{PL}{T}$

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Efficiency of power screw

$$\eta = \frac{P.L}{2\pi T}$$

Where, P = Load (N)
T = Torque (Nm)
L = Lead or pitch distance (mm)

262. A stud is a

(a) Square headed screw
(b) Headless fastener
(c) Hexagonal headed bolt
(d) Round headed screw

Haryana PSC AE (PHED) 05.09.2020, Paper-II

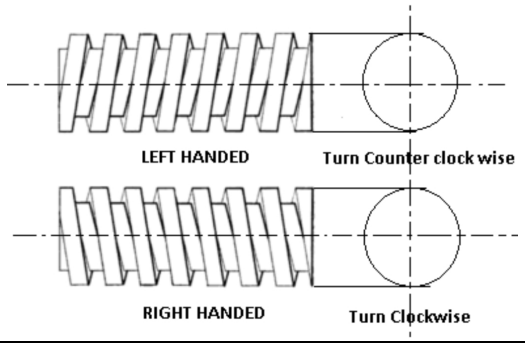
Ans. (b) : Stud is an externally threaded headless fastener.

263. If a nut, turned in clockwise direction screws on bolt, then the threads are called as

(a) Right - handed threads
(b) Left - handed threads
(c) Left - right hand threads
(d) Right - left hand threads

APPSC Poly Lect. 13.03.2020

Ans. (a) : If a nut, turned in clockwise direction screws on bolt, then the threads are called as right-handed threads.

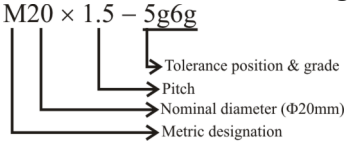


264. In the M20 metric screw threads, number 20 represents :

(a) nominal diameter (b) minor diameter
(c) root diameter (d) effective diameter

APPSC Poly Lect. 13.03.2020

Ans. (a) : ISO Metric Screw Thread Designation–



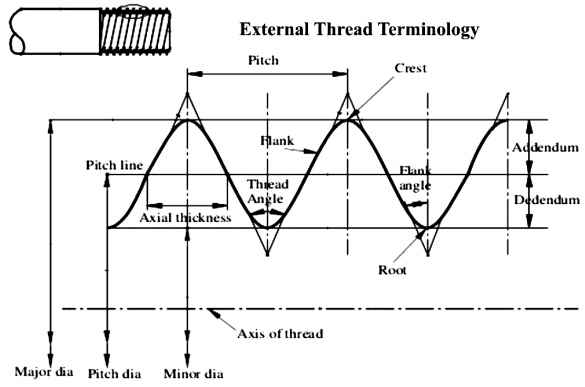
Nominal size (basic major diameter) and pitch expressed in millimeters.

265. The diameter of the screw at the crests is called:

(a) major diameter (b) minor diameter
(c) effective diameter (d) root diameter

APPSC Poly Lect. 13.03.2020
CGPSC Poly. Lect. 22.05.2016

Ans. (a) : The diameter of the screw at the crest is called major diameter



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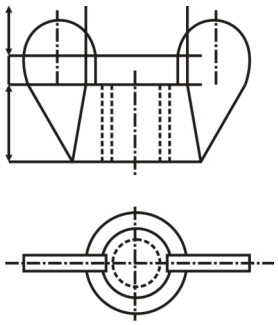
266.



The figure given shows the _____ type of nut.
(a) ring (b) wing
(c) capstan (d) dome

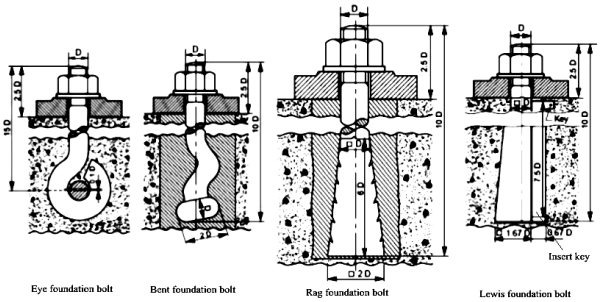
APPSC Poly Lect. 13.03.2020

Ans. (b) : The figure given shows as wing type of nut.

267. Which one of the foundation bolts is fixed in position by inserting a key on the face opposite to the tapered face?
(a) Eye foundation bolt
(b) Rang foundation bolt
(c) Lewis foundation bolt
(d) Cotter foundation bolt

APPSC Poly Lect. 13.03.2020

Ans. (c) : Lewis foundation bolts is fixed in a position by inserting a key on the face opposite to the tapered face.
Different type of foundation bolt–



268. A _____ is used for turning a nut or for holding bolt head while the nut is being screwed on or off the bolt.
(a) washer (b) spanner
(c) screw (d) bolt

APPSC Poly Lect. 13.03.2020

Ans. (b) : A spanner is used for turning a nut or for holding bolt head while the nut is being screwed on or off the bolt.

269. In screw threads drawing, SQ 40 × 3 indicates :
(a) metric threads with major diameter 40 mm and pitch 3 mm
(b) square threads with minor diameter 40 mm and width 3mm
(c) square threads with nominal diameter 40 mm and pitch 3 mm
(d) square threads with minor diameter 40 mm and pitch 3 mm

APPSC Poly Lect. 13.03.2020

Ans. (c) : In screw thread drawing, SQ 40 × 3, indicates Square threads with nominal diameter 40 mm and pitch 3 mm.

270. A rod is enclosed centrally in a tube and the assembly is tightened by rigid washers. If the assembly is subjected to compressive load, then
(a) rod is under compression
(b) tube is under compression
(c) both rod and tube are under compression
(d) tube is under tension, rod under compression

RPSC IOF, 2020
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : A rod is enclosed centrally in a tube and the assembly is tightened by rigid washers, if the assembly is subjected to compressive load, then both rod tube are under compression.

271. What is the safe static tensile load for a M36 × 4C bolt of mild steel having yield stress of 280 MPa and factor of safety 1.5?
(a) 285 kN (b) 190 kN
(c) 142.5 kN (d) 95 kN

RPSC IOF, 2020

Ans. (b) : $\text{Stress} = \frac{\text{load}}{\text{area}} = \frac{W}{\pi \times \frac{d^2}{4}}$

$\text{Safe static tensile load } w_s = \frac{W}{\text{F.O.S.}}$

$= \frac{\text{Stress} \times \pi \times d^2}{4 \times \text{FOS}}$

$= \frac{280 \times \pi \times 36^2}{4 \times 1.5}$

$w_s = 190 \text{ kN}$

272. The torque required to tighten the bolt comprises of the
(a) torque required in overcoming thread friction only
(b) torque required in inducing the pre-load only
(c) torque required in overcoming circumferential hoop stress
(d) torque required in overcoming thread friction and inducing the pre-load and also the torque required to overcome collar friction between the nut and the washer

ESE 2019

Ans. (d) : The torque required to tighten a bolt comprises of
■ The part required to overcome thread friction.
■ The part required to induce preload or pretension in the bolt.
■ The part required to overcome collar friction between the nut and the washer.

273. The railway carriage couplings have
(a) square threads (b) acme threads
(c) knuckle threads (d) buttress threads

Nagaland PSC (CTSE) 2018, Paper-I
MECON MT 2019

Ans. (c) : The railway carriage couplings have knuckle threads.

274. A locking device in which the bottom cylindrical portion is recessed to receive the tip of the locking set screw, is called
(a) castle nut (b) jam nut
(c) ring nut (d) sawn nut

Nagaland PSC (CTSE) 2018, Paper-I

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Ans. (c) : A locking device in which the bottom cylindrical portion (Bolt) is recessed to receive the tip of the locking set screw is called a ring nut worked as a washer.

275. A locking device extensively used in automobile industry is a
(a) jam nut (b) castle nut
(c) sawn nut (d) ring nut

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : A castle nut has notches that allow if to pinned to prevent rotation, so they are most widely used in automobile industry. It can stand high vibrations.

276. The taper on cotter varies from
(a) 1 in 15 to 1 in 10 (b) 1 in 24 to 1 in 20
(c) 1 in 32 to 1 in 24 (d) 1 in 48 to 1 in 24

Nagaland PSC (CTSE) 2018, Paper-I
Nagaland PSC CTSE 2017, Paper-I
RPSC IOF, 2020
Punjab PSC (Lect.) 06.08.2017

Ans. (d) : As we know, the taper on the cotter varies from,
1 in 48 to 1 in 24
According to the load & size.

277. In a steam engine, the valve rod is connected to an eccentric rod by means of
(a) cotter joint (b) knuckle joint
(c) universal joint (d) flange coupling

Nagaland PSC (CTSE) 2018, Paper-I
HPPSC Asstt. Prof. 18.09.2017

Ans. (b) : Knuckle joint is a mechanical joint used to connect two rods which are under a tensile load, so in a steam engine, the valve rod is connected to an eccentric rod by means of knuckle joint because if with stand tensile load.

278. The mean dia of screw jack having pitch 10 mm is 50 mm. A load of 20 kN is lifted to the dist. of 170 mm. Find the work done in lifting the load when the load rotates with the screwing. The external and internal dia. of the bearing surface of the loose head are 60 mm and 10 mm. resp. Take μ = 0.08.
(a) 72.25 N-m (b) 31.85 N-m
(c) 7718 N-m (d) 10025 N-m

TNPSC AE 2018

Ans. (c) : p = 10 mm W = 20 kN = 20×10^3 N
d = 50 mm D_2 = 60 mm or R_2 = 30 mm
$\tan\phi = \mu_1 = \mu = 0.08$ D_1 = 10 mm or R_1 = 5 mm
We know–

$$\tan\alpha = \frac{p}{\pi d} = \frac{10}{\pi \times 50} = 0.0637$$

∴ force required at the circumference of the screw to lift the load,

$$P = W \tan(\alpha + \phi) = W\left[\frac{\tan\alpha + \tan\alpha}{1 - \tan\alpha\tan\phi}\right]$$

$$= 20\times10^3\left[\frac{0.0637 + 0.08}{1 - 0.0637\times0.08}\right]$$

= 2890 N
and torque required to overcome friction at the screw,

$$T = P \times d/2 = 2890 \times \frac{50}{2} = 72250\ \text{N-mm} = 72.25\ \text{N-m}$$

∵ load is lifted to the distance of 170 mm vertically, and the distance moved by the screw is one rotation is 10 mm (equal to pitch = 10 mm)
So, Number of rotations made by the screw

$$N = \frac{170}{10} = 17 \text{ rotates}$$

1) When the load rotates with the screw work done in lifting the load
$T = 2\pi N = 72.25 \times 2\pi \times 17 = 7718$ N-m

279. In case of a knuckle joint the pin is likely to fail in
(a) Compression (b) Crushing
(c) Tension (d) Double shear

WBPSC AE, 2017

Ans. (d) : Knuckle joint
Knuckle joint is used to connect two rods which are under a tensile load,
Knuckle joint assembly consists of the-
- Single eye
- double eye or fork
- Knuckle pin

The mode of failure are
- Shear failure of pin (single shear)
- Crushing of pin against rod
- Tensile failure of flat end bar.

280. The lead screw of a lathe has_____ threads.
(a) single start (b) double start
(c) multi-start (d) any one of these

Sikkim PSC (Under Secretary), 2017

Ans. (a) : A single start thread has one continuous thread running along the body of the screw. A trapezoidal thread is used in the lead screw of lathe machine. It has a 30° included angle. Stronger at the root. Easy to manufacture compare to square thread.

281. A screw thread is obtained
(a) When the hypotenuse of a right-angled triangle is wrapped around the circumference of a cylinder.
(b) When a point moves on the circumference of a cylinder at a ratio of diameter to the pitch of thread.
(c) When a line moves on the circumference of a cylinder with an angle parallel to its axis.
(d) When one of the sides of an equilateral triangle is wrapped around the circumference of a cylinder.

Karnataka PSC AE, 10.09.2017

Ans. (a) : Screw thread is obtained when the hypotenuse to a right angled triangle is wrapped around the circumference of a cylinder.

282. In design of power screws, Acme threads are preferred over the square threads because Acme threads
(a) offer better efficiency than square threads
(b) have less wear and therefore more life
(c) are easier to machine and permit use of a split nut
(d) require lower torque to raise the load

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : Square Thread–The square threads, because of their high efficiency, are widely used for transmission of power in either direction.

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- The square threads are not so strong as V-threads but they offer less frictional resistance to motion than whitworth threads.
  Application of square threads–Feed mechanism in machine tools, valve and spindle, screw jack etc.
- Acme Thread/Trapezoidal Thread–It is a modification of square thread. It is much stronger than square thread and can be easily produced.
- Since having high root, so high load carrying capacity.
- V-groove so used for frequent engaged and disengaged used in lathe machines because lathe tool continuously engaged / disengaged.
- Application–High power transmission devices of induction machine.
- Thread angle = 29^0

283. The thickness of the lock nut is about one-half or two-third thickness of the standard nut. This statement holds true for
(a) Castle nut (b) Grooved nut
(c) Jam nut (d) Sawn nut
Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Jam nut or lock nut- this is about one half or the third thickness of standard nut.

284. The threaded bolts A and B of same material and length are subjected to identical tensile load. If the elastic strain energy stored in bolt A is 4 times that of the bolt B and the mean diameter of bolt A is 12mm, the mean diameter of bolt B in mm is
(a) 16 (b) 24
(c) 36 (d) 48
GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Strain energy is given by

$$U = \frac{F^2L}{2AE}$$

Given, $U_A = U_B, \frac{F^2L}{2A_AE} = \frac{4F^2L}{2A_BE}$

Same material, length and subjected to identical tensile load.

So, $\frac{1}{A_A} = \frac{4\times1}{A_B} \Rightarrow \frac{1}{D_A^2} = 4\times\frac{1}{D_B^2}$

$D_B = 2D_A$
$D_B = 2 \times 12$
$D_B = 24$ mm

285. The designation M 33×2 of a bolt means
(a) Metric thread of 33 nos in 2 cm
(b) Metric thread with cross section area of 33 mm^2
(c) Metric thread of 33 mm outside diameter and 2mm pitch
(d) Bolt of 33 mm nominal diameter having 2 threads per cm.
ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (c) :

M (↓ Metric Thread) 33 (↓ Outside Diameter (mm)) × 2 (↓ Thread Pitch (mm))

286. Which of the following is FALSE for bolted joints?
(a) Bolted joints are preloaded to abode vibration loosening
(b) Threads made by thread rolling process are preferred over machined threads in bolts
(c) When external load is applied to a preloaded bolt joint, a bigger percentage of that load relieves compression of flanges and remaining percentage increases tension in bolts
(d) Bolted joints are designed such that bolt stiffness is higher than flange stiffness
ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Bolts are made of mild steel having lower stiffness while flanges are made of cast iron having higher stiffness.

287. A cup is provided in screw Jack–
(a) to reduce the friction
(b) to prevent rotation of load
(c) to increases load capacity
(d) to increase efficiency
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : A cup is provided in screw Jack to prevent rotation of load.

288. While designing screw threads, adequate length of engaged threads between the screw and nut is provided so as to prevent failure of threads due to–
(a) direct shear stress
(b) torsional shear stress
(c) tensile stress
(d) compressive stress
Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : While designing screw threads, adequate length of engaged threads between the screw and nut is provided so as to prevent failure of threads due to direct shear stress.

289. Setscrews are :
(a) Small screws
(b) Headless screws
(c) Used to prevent relative motion between parts
(d) Self-threading
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Setscrews is used to prevent relative motion between two parts. (eg. between hub and the shaft in small power transmission)- it is headless and subjected to only compressive force only.

290. An eye bolt is to be used lifting a load of 70 kN and the tensile stress is not to exceed 100 MPa. The core diameter of the bolt will be given by:
(a) $\left(\frac{2100}{\pi}\right)^{\frac{1}{2}}$ (b) $\left(\frac{2800}{\pi}\right)^{\frac{1}{2}}$
(c) $\left(\frac{210}{\pi}\right)^{\frac{1}{2}}$ (d) $\left(\frac{280}{\pi}\right)^{\frac{1}{2}}$
CIL MT 26.03.2017

Ans. (b) : Given load P = 70 kN = 70×10^3 N
Tensile stress σ_t = 100 MPa
Load on the bolt

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$$P = \frac{\pi}{4} \times (d_c)^2 \times \sigma_t$$

Where d_c is the core diameter

$$70 \times 10^3 = \frac{\pi}{4} \times d_c^2 \times 100$$

$$\Rightarrow d_c = \sqrt{\frac{2800}{\pi}}$$

291. Which of the following screw threads does not have a symmetric section?
(a) Buttress (b) Square
(c) Whitworth (d) Acme

BPSC AE Mains 2017 Paper - VI

Ans : (a) : Buttress threads does not have symmetric section.

292. Ball bearing type screws are found in following application
(a) Screw jack
(b) Aero plane engines
(c) Crane
(d) Steering mechanism

Vizag Steel (MT) 2017

Ans. (d) : Ball bearing type screws used in aircraft and missiles to move control surfaces, especially for electric fly by wire, and in automobile power steering to translate rotary motion from an electric motor to axial motion of the steering rack.

293. Which of the following screw thread is used for jacks, vices and clamps?
(a) square threads
(b) trapezoidal threads
(c) buttress threads
(d) acme threads

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Square thread is used for Jacks, vices and clamps are square threads.

294. The object of caulking in a riveted joint is to make the joint:
(a) free from corrosion (b) stronger in tension
(c) free from stress (d) leak proof

HPPSC Asstt. Prof. 18.09.2017

Ans. (d) :

295. An allen bolt is ..... .
(a) provided with hexagonal depression in head
(b) self locking bolt
(c) used in high speed components
(d) provided with countersunk head

GWSSB DEE 07.07.2016

Ans. (a) : An allen bolt is provided with hexagonal depression in head.

296. Following type of thread is used in cold drinks plastic bottle cap.
(a) Single start (b) multi start
(c) acme (d) buttress

GWSSB DEE 07.07.2016

Ans. (b) :

297. A M24 × 3 represents a bolt having:
(a) Outer bolt diameter 24 mm and threads of pitch 3 mm
(b) Nominal bolt diameter 24 mm and threads of pitch 3 mm
(c) Inner bolt diameter 24 mm and threads of pitch 3 mm
(d) None of above

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : M24 × 3
M – Metric
24 – Represent Nominal bolt diameter.
3 – Pitch of threads.
A bolt is a form of threaded fastener with an external male thread represents a matching preformed female thread such as a nut.

298. A Lewis bolt is a/an:
(a) foundation bolt (b) stud bolt
(c) eye bolt (d) tap bolt

(HPPSC LECT. 2016)

Ans : (a) A Lewis bolt is an foundation bolt

299. A lead screw with half nut mechanism in a lathe, free to rotate in both directions, has
(a) V-threads
(b) Whitworth threads
(c) Acme threads
(d) British standard threads

UJVNL AE 2016

Ans : (c) A lead screw with half nut mechanism in a lathe free to rotate in both direction, has Acme thread.

Thread

v-thread	**Square thread**
i) BSW → 55°	Square thread - 0°
ii) BA → $47^1/_2$°	Knuckle thread - No angle
iii) Metric thread → 60°	Acme thread - 29°
iv) pipe thread → 55°	Butters thread - 45°

300. Which one of the following threads is having smallest included angle?
(a) Acme thread (b) BSW thread
(c) Buttress thread (d) Unified thread

HPPSC W.S. Poly. 2016

Ans : (a)

Thread	**Angle**
Acme thread	29°
BSW thread	55°
Butters thread	45°
Square thread	0°
BA thread	$47\frac{1}{2}$°

301. The square threads are usually found in–
(a) spindle of bench vices
(b) railway carriage coupline
(c) feed mechanism of machine tools
(d) screw cutting lathes

RPSC 2016

Ans : (c) Square thread– They have normally very high efficiency and hence they are widely used for transmission of power in either direction. Such type of threads are usually found on the feed mechanism of machine tools.

302. The fulcrum pin of lever is designed on the basis of
(a) torsional moment
(b) axial tensile force
(c) bending moment
(d) bearing pressure

GPSC Lect. 23.10.2016

Search on Telegram @apna_pdf

Ans. (d) : Fulcrum of lever is a pin joint and is designed based on bearing and bending

Bearing pressure ($P_{bearing}$) $= \dfrac{\text{Load}}{\text{Projected Area}}$

303. Which of the following has a self locking property?
(a) Nut cracker (b) Scissors
(c) Tongs (d) Screw

KPCL AE 2016

Ans. (d) : Screw has self-locking property. The self-locking principle states that, A screw will be self-locking if the coefficient of friction is equal to or greater than the tangent of the helix angle.
For self-locking screw

$$\tan\theta \geq \tan\alpha$$

$$\mu \geq \frac{\ell}{\pi dm}$$

304. .....is used for "Two Wire Method" to measure effective diameter of a screw thread.
(a) Diameter measuring machine
(b) Micro-diameter gauge
(c) Macro-diameter gauge
(d) Collimator

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Diameter measuring machine is used for (Two-wire) method to measure effective diameter of a screw thread.

305. Which among the following flat wedge shaped piece of rectangular cross-section and its width is tapered (either on one side or both sides) from one end to another for an easy adjustment?
(a) Cotter (b) Piston rod
(c) Spigot (d) Sprocket

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (a) A cotter is a flat wedge-shaped piece of steel as shown in figure below. This is used to connect rigidly two rods which transmit motion in the axial direction, without rotation. These joints may be subjected to tensile or compressive forces along the axes of the rods.
Application- Its usually used in connecting a piston rod to the cross-head of a reciprocating steam engine, a piston rod and its extension as a fail as pump rod, strap end of connecting rod etc.

306. Why is it desirable to provide hexagonal headed bolts and nuts in various assemblies in comparison with square headed bolts and nuts?
(a) These are cheaper
(b) They have higher strength
(c) These are easier to grip
(d) These can be tightened in narrow spaces

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : The use of a hexagonal shape makes fasteners easier to turn by reducing the required turning space.

307. To avoid reversibility of a screw lifting machine, the leads angle is restricted to:
(a) 5º (b) 7º
(c) 10º (d) 15º

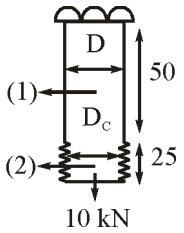
GPSC Asstt. Prof. 28.08.2016

Ans. (b) : The lead angle is the helix angle of thread based on the lead surface, for preventing/avoiding reversibility of a screw lifting machine is up to 7º.

308. M 20 has a shank of 50 mm length and a thread of 25 mm length. It is subjected to a tensile load of 10 kN. Young's modulus of bolt material is 205000 N/mm^2. D_C = 0.8d. The strain energy stored in the bolt, of the shank of the bolt is turned down to root of the thread, is:
(a) 67 N.mm (b) 84 N.mm
(c) 114 N.mm (d) 50 N.mm

RPSC ADE 2016

Ans. (*) : Given,
Shank
$L_1 = 50$ mm
$D_1 = D = 20$ mm
$P = 1$ kN
Thread-
$L_2 = 25$ mm
$D_2 = 0.8 \times D = 16$ mm
$E = 205 \times 10^3$ N/mm^2
Total strain energy (U)



$$U = U_1 + U_2 = \frac{P^2 L_1}{2A_1 E} + \frac{P^2 L_2}{2A_1 E}$$

$$= \frac{P^2}{2E}\left[\frac{L_1}{A_1} + \frac{L_2}{A_2}\right]$$

$$= \frac{10^8}{2\times 205\times 10^3}\left[\frac{50\times 4}{\pi\times 20^2} + \frac{25\times 4}{\pi\times 16^2}\right]$$

$$U = 69.14 \text{ N-mm}$$

Note- But official answer is option (b)

309. Which thread is more suited in power screw to take load on both directions?
(a) Acme thread (b) Square thread
(c) Buttress thread (d) None of these

ISRO Scientist/Engineer 03.07.2016
Nagaland PSC CTSE 2018, 2017 Paper-I
ESE 2012, 1999

Ans. (a&b) : Both acme and square thread can take load in both direction but manufacturing of acme thread is easy than square thread.

310. The angle between the flanks of British Standard Whitworth (BSW) thread is
(a) 1250° (b) 47.5°
(c) 55° (d) 69°

(KPSC AE 2015)
Nagaland PSC (CTSE) 2018, Paper-I
Kerala PSC Poly. Lect. 09.01.2014
Punjab PSC SDE 12.02.2017

Ans : (c) The angle between the flanks of British Standard whitworth thread is 55°

311. Self locking in power screw is better achieved by
(a) Using multistrart threads
(b) Increasing helix angle
(c) decreasing helix angle and increasing the co-efficient of friction
(d) decreasing co-efficient of friction

GPSC AE 26.07.2015

Ans. (c) :

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312. If t, τ and P are thickness, permissible shear stress and applied axial load respectively on a cotter joint, the mean width of the collar is equal to

(a) $P/2t\tau$ (b) $2P/t\tau$
(c) $P/t\tau$ (d) none of these

VIZAG MT 2015

Ans. (a) : Mean width of the collar, $w_{mean} = \frac{P}{2t\tau}$

313. A screw is said to be an overhauling screw, if its efficiency is:

(a) Less than 50% (b) More than 50%
(c) Equal to 50% (d) None of these

NPCIL ET, 2015

Ans. (b) : A screw is said to be an overhauling screw if its efficiency is more than 50%

$$\eta = \frac{\tan\alpha}{\tan(\alpha+\phi)}$$

- Overhauling screw, ($\phi < \alpha$)
- Self locking screw, ($\phi \geq \alpha$)
- Efficiency of overhauling screw is more than 50% ($\eta > 50\%$)
- Efficiency of self locking screw is less than 50% ($\eta < 50\%$)

314. For power screw applications, a square thread can be preferred to trapezoidal thread for :

(a) easier manufacture
(b) better strength
(c) better efficiency
(d) more available standards

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Square thread- The square threads because of their high efficiency, are widely used for transmission of power in either direction.

- The square threads are not so strong as V-threads but they offer less frictional resistance to motion.

Application of square threads- Feed mechanism in machine tools, value and spindle, screw jack etc.

- **Acme Thread/Trapezoidal Thread-**
- It is a modification of square thread. It is much stronger than square thread and can be easily produced.
- Since having high root, so high load carrying capacity.

Application- devices of induction machine

- thread angle = 29°

315. The force required to move an object of F Newton downwards with the help of a power screw with ϕ angle of friction and α helix angle of the screw will be

(a) F tan (ϕ – α) (b) F tan (ϕ + α)
(c) F tan (ϕ × α) (d) F tan (ϕ / α)

TSGENCO AE 14.11.2015

Ans. (a) : Force required to move an object of F (Newton) downwards with the help of a power screw.

$$P = F \tan(\phi - \alpha)$$

Where, ϕ = friction angle
α = helix angle

316. According to Indian standard specifications the total number of designated grades of fundamental tolerances are :

(a) 18 (b) 21
(c) 24 (d) 28

Kerala PSC AE 06.08.2015

Ans. (a) : According to the Indian Standard Specification the total No. of designated grade of fundamental tolerances are 18.

317. Which of the following threads is used for transmission of power?

I. Square thread
II. Buttress thread
III. Acme thread

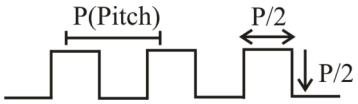
Choose the correct answer from the options given below.

(a) I and II (b) II and III
(c) I and III (d) I, II and III

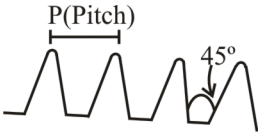
UPRVUNL AE 2014

Ans : (d) All of the above mention are used in transmission of power

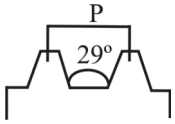
(i) Square thread- It is used in srcrew jack, lead screw etc.



(ii) Buttress Thread- Power transmitted in only one direction.



(iii)Acme thread- It is used in Lathe machine.



318. The largest diameter of an external or internal screw thread is known as

(a) major diameter (b) minor diameter
(c) roof diameter (d) pitch diameter

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : Major Diameter–The longest diameter of the thread of the screw or nut. The term major diameter replaces the term outside diameter as applied to the thread of a screw.

319. Self-locking is not possible in the case of

(a) simple band brake
(b) differential band brake
(c) simple block
(d) internal expanding shoe brake

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : A band brake consist of a flexible band of leather. One or more ropes or a steel line with friction materials which improves friction of the drum. In simple band break one end of the band is attached to the fulcrum of the lever arm hence self locking is not possible.

320. A bolt having grade of 12.9, has nominal tensile strength of

(a) 1200 MPa (b) 129 MPa
(c) 900 MPa (d) 12 Kg/ mm^2

GPSC Lect. (Poly.) 07.12.2014

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Ans. (a) :

Class	Nominal Size Range (mm)	Mechanical Properties		
		Proof Load (MPa)	Min. Yield Strength (MPa)	Min. Tensile Strength (MPa)
Class 12.9	1.6 mm - 100 mm	970	1100	1220

321. The equivalent coefficient of friction for *V* threads is
(a) equal to actual coefficient of friction
(b) less than actual coefficient of friction
(c) greater than actual coefficient of friction
(d) not related to the actual coefficient of friction

TNPSC AE 2014

Ans. (c) : The equivalent coefficient of friction for *V* threads is greater than actual coefficient of friction.

322. Recirculating ball screws are used because
(a) they are easy to manufacture
(b) power required for driving them is small due to small friction
(c) frictional resistance is more compared to ACME threads
(d) variable friction present due to the recirculating balls

TNPSC AE 2014

Ans. (b) : Recirculating ball screws are used because power required for driving them is small due to small friction.

323. While designing a screw in a screw jack against buckling failure, the end conditions for a screw are taken as
(a) both ends fixed
(b) both ends hinged
(c) one end fixed and other end hinged
(d) one end fixed and other end free

TNPSC AE 2014
ESE 1995

Ans. (d) : While designing a screw in a screw jack against buckling failure, the end conditions for a *screw* are taken as one end fixed and other end free.

324. In order to reduce thermal stress, fastening bolts must be made of materials possessing
(a) High coefficient of linear expansion
(b) Low coefficient of linear expansion
(c) Zero coefficient of linear expansion
(d) Negative coefficient of linear expansion

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) In order to reduce thermal stress, fastening bolts must be made of materials having low coefficient of linear expansion.

325. The normal lead, in a worm having multiple start threads, is given by
(a) $l_N = l/\cos\lambda$ (b) $l_N = l.\cos\lambda$
(c) $l_N = l$ (d) $l_N = l\tan\lambda$

TNPSC AE 2013

Ans. (b) : The term normal pitch is used for a warm having single start threads. In case of a warm having multiple starts threads, the term normal lead is used.

$l_n = l\cos\lambda$

$\lambda \rightarrow$ lead angle

326. Taper parallel method and Rollers method are conveniently used to measure
(a) Major diameter of External Thread
(b) Minor diameter of External Thread
(c) Major diameter of Internal Thread
(d) Minor diameter of Internal Thread

TNPSC AE 2013

Ans. (d) : Taper parallel method and Rollers method are conveniently used to measure minor diameter of internal thread.

327. A screw jack has pitch of 12 mm with a mean radius of thread equal to 22.5 mm. A lever 600 mm long is used to raise a load of 1800 kg, the coefficient of friction is 0.10. Find the helix angle and friction angle.
(a) 6.2°, 4.5° (b) 4.85°, 5.7°
(c) 6.37°, 4.42° (d) 4.2°, 5.5°

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : p = 12 mm, r = 22.5 mm, $\mu = 0.10$
Length of lever = 600 mm
Load = 1800 kg

Helix angle α, $\tan\alpha = \frac{p}{2\pi r} \Rightarrow \alpha = \tan^{-1}\left(\frac{P}{2\pi r}\right)$

$$\alpha = \tan^{-1}\left(\frac{12}{2\times\pi\times 22.5}\right)$$

$\alpha = 4.85°$

The friction angle $\phi = \tan^{-1}(\mu)$

$\phi = \tan^{-1}(0.10)$

$\phi = 5.71°$

328. Which one of the following is NOT true about pitch error in screw thread ?
(a) It increases the effective diameter of a bolt
(b) It increases the effective diameter of a nut
(c) It occurs due to incorrect tool work velocity ratio
(d) If there is some error in pitch, then the total length of the tread engaged will be either too great or small

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

329. In order to obtain bolt of uniform strength
(a) Increase the shank diameter
(b) Increase its length
(c) Drill an axial hole in thread portion so that shank area is twice the root area.
(d) Drill an axial hole through head up to the threaded portion so that shank area is equal to root area.

RPSC AE GWD, 2011

Ans. (d) : In order to obtain bolt of uniform strength drill an axial hole through head up to the threaded portion to that shank area is equal to root area.

330. The effective diameter of an external or internal screw thread, is known as
(a) Minor diameter (b) Major diameter
(c) Pitch diameter (d) None of these

ISRO Scientist/Engineer 2011

Ans. (c) : The effective diameter of an external or internal screw thread, is known as pitch diameter.

Search on Telegram @apna_pdf

331. A screw having coefficient of friction equal to or greater than the tangent of the load angle is known as

(a) Machine screw
(b) Self locking screw
(c) Set-screw
(d) Ball-bearing type screws

ISRO Scientist/Engineer 2010

Ans. (b) : A screw having coefficient of friction equal to or greater than the tangent of the load angle is known as self locking screw.

332. Assertion (A) : A gib and cotter joint is usually used in small end of a connecting rod of a reciprocating engine mechanism.

Reason (R) : The gib provides a larger bearing surface for the cotter to slide on and thus the tendency of cotter to slacken back owing to friction is considerably decreased.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (a) : A gib and cotter joint is usually used in small end of a connecting rod of a reciprocating engine mechanism. Gib provides larger bearing surface for the cotter to slide on due to the increased holding power-this tendency of cotter to slacken back owing to friction is decreased.

333. Match List I with List II and select the correct answer using the code given below the List :

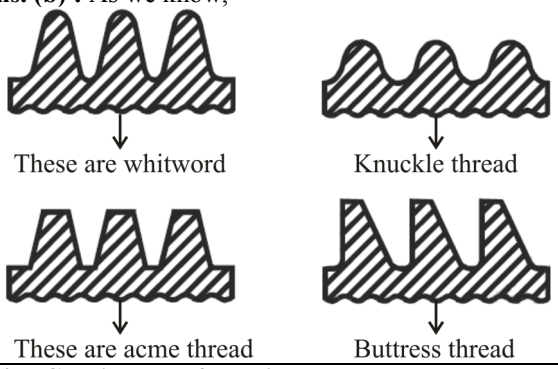
	List I (Screw thread designation)		List II (Form of screw thread)
A.	Knuckle	1.	
B.	Acme	2.	
C.	Buttress	3.	
D.	Whitworth	4.	

Code:

	A	B	C	D		A	B	C	D
(a)	1	2	4	3	(b)	3	2	4	1
(c)	1	4	2	3	(d)	3	4	2	1

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know,



334. Consider the following statements :
The coefficient of friction in power screw thread is :
1. Independent of axial load
2. Practically independent of speed
3. Increases with heavier lubricants
4. Varying in the range of 0.25 to 0.35
Which of the above statements is/are correct?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 3 and 4

UPSC JWM Adv. No-16/2009

Ans. (b) : The coefficient of friction in power screw thread is :
(1) Independent of axial load.
(2) Practically independent of speed

335. The circumferential joint of a boiler of diameter 1 m is made of lap joint with rivet diameter of 20 mm. If the allowable shear stress for the material of the rivet is 50 MPa and the boiler has to withstand an internal pressure of 2.5 MPa, then the minimum number of rivets required to prevent the failure of the joint is

(a) 150 (b) 125
(c) 100 (d) 75

DRDO Scientists 2009

Ans. (b) : $n \times \frac{\pi}{4} d^2 \times t = \frac{\pi}{4} D^2 \times P$

$$\Rightarrow \quad n = \frac{D^2 P}{d^2 \times \tau}$$

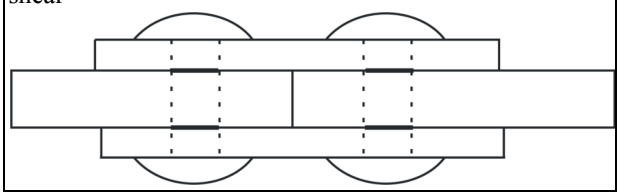
$$= \frac{1^2 \times 2.5 \times 10^6}{0.02^2 \times 500 \times 10^6} = 125$$

336. A double strap butt joint is always in

(a) Single shear
(b) Double shear
(c) Either single or double shear
(d) None of the above

ISRO Scientist/Engineer 2008

Ans. (b) : Double strap butt joint always in double shear



Search on Telegram @apna_pdf

337. A screw jack used for lifting the load is
(a) a reversible machine
(b) a non-reversible machine
(c) an ideal machine
(d) none of these

TNPSC AE, 2008

Ans. (b) : A screw jack used for lifting the load is a non-reversible machine or self locking machine.
Non-reversible Machine if a machine is not capable of doing some work in the reversed direction, after the effort is removed, then the machine is known as non-reversible or self locking machine. The condition for a machine to be non-reversible or self locking is that its efficiency should be less than 50%.

338. Lag bolt is generally used in
(a) wooden construction
(b) electrical equipment
(c) fastening castings
(d) all of these

TNPSC AE, 2008

Ans. (a) : A lag bolt is a mechanical fastening with a threaded body that is driven through two adjacent surfaces for the purpose of adjoining them securely. This piece is most often used to link wooden elements with other wooden or concrete foundations.

339. The pin in a knuckle joint is subjected to
(a) crushing and double shear
(b) crushing and single shear
(c) single shear only
(d) double shear only

WBPSC AE, 2007

Ans. (a) : Knuckle joint is used to connect two rods under tensile loading. This tensile loading causes crushing and shear at two sections i.e. double shear in the pin of joint.

340. If the lead angle of a worm is 22½°, then the helix angle will be
(a) 22½° (b) 30°
(c) 45° (d) 67½°

UKPSC AE 2007 Paper -I

Ans. (d) : 67½°

341. Consider the following statements regarding the ends of the pressure vessels flanged by pre-tensioned bolts:
1. Pre-tensioning helps to seal the pressure vessel.
2. Pre-tensioning reduces the maximum tensile stress in the bolts.
3. Pre-tensioning countermands the fatigue life of the bolts.
4. Pre-tensioning helps to reduce the deleterious effect of pressure pulsations in the pressure vessel.

Which of the above statements are correct?
(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 2 and 4 only (d) 1, 2, 3 and 4

ESE 2017

Ans. (b) : Statement-2 is wrong because pre-tensioning does not reduce the maximum tensile stress in the bolts.

4. Design of Riveted and Welded Joints

342. The transverse fillet welded joints are designed for
(a) Tensile strength
(b) Bending strength
(c) Shear strength
(d) Compressive strength

GPSC DEE Class-2 (GWSSB) 04.07.2021
Karnataka PSC AE (WRD) 31.07.2021
RPSC IOF, 2020
Oil India Senior Officer 23.12.2020
TNPSC AE 2019
Rajasthan Nagar Nigam AE 2016, Shift-I
UPRVUNL AE 2014
TNPSC AE 2013
WBPSC AE 2008

Ans. (a) : Transverse Fillet welded joints– The weld is subjected to tensile stress and failure will occur at minimum cross-section of weld at throat
∴ Permissible tensile stress,

$$\sigma_t = \frac{P}{t\,\ell} = \frac{P}{0.828h\,\ell}$$

If there are weld of equal length

$$\sigma_t = \frac{P}{0.828\,h\,\ell}$$

Allowable load for a transverse fillet weld is 1.17 times more than that of a parallel fillet weld.
Parallel fillet welded joint is designed for shear strength of weldment throat.

343. If the ratio of rivet hole to the pitch of rivet is 0.25, then the tearing efficiency of the joint is
(a) 0.50 (b) 0.75
(c) 0.25 (d) 0.87

Karnataka PSC Lect., 27.05.2017
ISRO Scientist/Engineer 11.10.2015
MPPSC AE 08.11.2015
MPPSC State Forest Service Exam, 2014
TRB Asstt. Prof., 2012
GATE 1996

Ans (b) : Given, $\frac{d}{p} = 0.25$
Tearing efficiency of the joint –

$$\eta_{tearing} = \frac{p-d}{p} = 1 - \frac{d}{p} = 1 - 0.25$$

$$\eta_{tearing} = 0.75$$

344. Which of the following welded joints is designed for shear strength?
(a) transverse fillet welded joint
(b) parallel fillet welded joint
(c) butt welded joint
(d) transverse butt weld joint

CGPSC AE 15.01.2021
Punjab PSC SDE 12.02.2017
CGPSC AE 16.10.2016
APPSC AEE MAINS 2016, PAPER-III
SJVN ET 2013
TNPSC AE (Industries) 09.06.2013

Search on Telegram @apna_pdf

Ans. (b) : Parallel fillet welded joints are designed for shear strength.

345. The diameter of rivets in mm for a plate of thickness t mm is taken as

(a) t (b) 2t

(c) 1.41 $\sqrt{t}$ (d) 6.05 $\sqrt{t}$

TSPSC AEE 2017
APPSC AE Subordinate Service Civil/Mech. 2016
CGPSC AE 26.04.2015 Shift-I
UPRVUNL AE 2014
TNPSC AE 2008

Ans. (d) : Diameter of rivets using unwin's empirical formula –

$d = 6\sqrt{t}$

where,

t = thickness of plate

346. If the tearing efficiency of a riveted joint is 60% then the ratio of pitch of diameter of rivet is

(a) 0.20 (b) 0.33

(c) 0.40 (d) 0.50

HPPSC W.S. Poly. 2016
RPSC AE 2016
KPSC AE 2015
GPSC Lect. (Poly.) 07.12.2014
RPSC ACF-2011

Ans : (c) Tearing efficiency $(\eta_t) = \frac{(p-d).t.\sigma_t}{p.t.\sigma_t}$

Tearing efficiency = $\frac{p-d}{p}$

$$0.60 = 1-\frac{d}{p}$$

$$\frac{d}{p} = 0.40$$

d = diameter of screw. (mm)

p = pitch of rivet (mm)

347. If the tearing efficiency of a rivetted joint is 50%, then the ratio of diameter of rivet to the pitch is :

(a) 0.20 (b) 0.30

(c) 0.50 (d) 0.70

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016
KPSC ADF 2015
APPSC Poly. Lect. 2013

Ans. (c) :

$\because$ Tearing efficiency of rivited joint $(\eta) = \left(1-\frac{d}{p}\right)$

Where; d = diameter of rivet; p = pitch of rivet.

Given; efficiency $(\eta) = 50\% = 0.5$

$$\therefore \quad 0.5 = 1-\frac{d}{p}$$

$$\Rightarrow \quad \frac{d}{p} = 1-0.50$$

$$\Rightarrow \quad \boxed{\frac{d}{p} = 0.50}$$

348. A rivet is specified by

(a) length of the shank
(b) diameter of the shank
(c) both length and diameter of the shank
(d) type of head

TSPSC Manager (Engg.) HMWSSB 12.11.2020
TNPSC AE 2017
SJVN ET 2013
WBPSC AE 2008

Ans. (b) : The size of rivet is specified by its shank diameter.

349. Rivets are made of

(a) Soft material (b) Ductile material

(c) Hard material (d) Brittle material

Nagaland PSC CTSE 2017, Paper-I
CGPSC AE 26.04.2015 Shift-I
ISRO Scientist/Engineer 2009
ISRO Scientist/Engineer 2008

Ans. (b) : Rivets

Rivets are made by ductile material, i.e. means,

(1) Aluminium
(2) Aluminium with added magnesium
(3) High carbon steel
(4) Stainless steel

Stainless steel rivets are strong hard and can withstand wear.

Stainless steel rivets are used for aircraft, automotive, locomotive and power generation industries.

They are used for their strength and ability to resist corrosion.

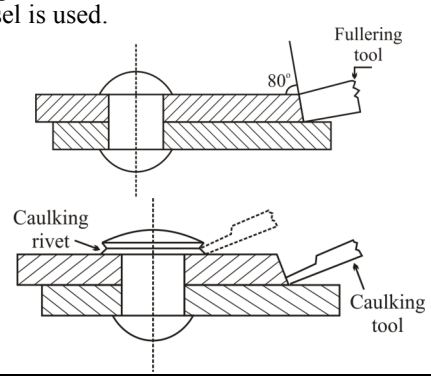
(5) Copper
(6) Copper Nickel
(7) Brass

350. Caulking and fullering processes as used:

(a) to lubricate the bearing
(b) to reduce the friction between matting parts of gears
(c) to obtain self locking of screw
(d) to obtain leak proof riveted joint
(e) to cool the bearing

TNPSC AE 2019
RPSC Vice Principal ITI 2018
CGPSC AE 25.02.2018
TNPSC AE 2013

Ans. (d) : Caulking and fullering- Caulking and fullering are used to make tight contact between metal-parts. Both are operations in riveting to ensure complete tightness of parts. Caulking is done with the help of caulking tool which is a blunt chisel and the fullering flat chisel is used.



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351. The permissible stress in a fillet weld is 100 N/mm^2. The fillet weld has equal leg lengths of 15 mm each. The allowable shearing load on weldment per cm length of the weld is:

(a) 22.5 kN (b) 15.0 kN
(c) 10.6 kN (d) 7.5 kN

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 18.09.2017
MPPSC AE 2016
ESE 1995

Ans. (c) : Given, $\tau = 100\ N/mm^2$, $t = 15$ mm

$\ell = 100$ mm

$$F_s = 0.707 \times t \times \ell \times \tau_s$$

$$\frac{F_s}{\ell} = 0.707 \times 15 \times 100 \times 10 = 10.605 \frac{kN}{cm}$$

352. A bench vice has the following type of thread

(a) metric thread (b) acme thread
(c) buttress thread (d) square thread

Nagaland PSC CTSE 2017, Paper-I
GWSSB AAE, 27.12.2015
KPSC AE 2015

Ans : (c) A bench vice has buttress thread.

Buttress thread-

(i) It is used in vice. It combines the advantage of square and trapezoidal thread.
(ii) It is used where a heavy axial force acts along the screw axis in one direction only.

Buttress thread Advantage:-

(i) It has higher efficiency compared with trapezoidal thread.
(ii) Stronger than square and trapezoidal thread both.

353. Which of the following is a permanent fastening?

(a) bolts (b) keys
(c) screws (d) rivets

RPSC AE 2016
RPSC ACF-2011
UKPSC AE 2007 Paper -I

Ans : (d) Rivets is a permanent fastening.

Permanent fastenings:- The permanent fastenings are those fastenings which can not be disassembled without destroying the connecting components. The examples of permanent fastenings in order of strength are soldered, brazed, welded and riveted joints.

354. In a butt welded joint, the size of weld is the throat of the weld

(a) 0.5 time (b) equal to
(c) $\sqrt{2}$ times (d) double

Nagaland PSC (CTSE) 2018, Paper-I
MPPSC AE 08.11.2015
JPSC AE 2013, Paper-V

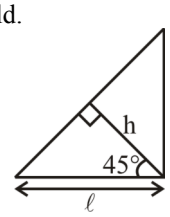
Ans : (b) : In case of butt joint, the length of leg or size of weld is equal to throat thickness which is equal to thickness of plates.

355. In the case of transverse fillet welded joint, the size of weld is given as :

(a) 0.25 × throat of the weld
(b) 0.50 × throat of the weld
(c) 2 × throat of the weld
(d) $\sqrt{2}$ × throat of the weld

NLC GET 17.11.2020, Shift-II
GPSC EE Pre, 28.01.2017

Ans. (d) : Transverse Fillet Weld Joint– A fillet weld is called transverse if the direction of the weld is perpendicular to the direction of the force acting on the joint. Double transverse Fillet weld is preferred to avoid warp out of shape size of weld is given as $\sqrt{2}$ times of the throat of the weld.



ℓ = Size of weld

h = Throat of weld

$$\cos 45° = \frac{h}{\ell} \qquad \left(\because \cos 45° = \frac{1}{\sqrt{2}}\right)$$

$$\boxed{\ell = \sqrt{2}h}$$

$$\boxed{\text{Size of weld} = \sqrt{2} \times \text{throat of weld}}$$

356. Socket joint is mostly used for pipes which

(a) carry steam at high pressure
(b) carry water at low pressure
(c) are buried in the earth
(d) carry fluid at high pressure

GWSSB AAE, 27.12.2015
TNPSC AE, 2008

Ans. (b) : Socket joint is mostly used for pipes which carry water at low pressure.

357. According to IBR, safety factor of rivet should be less than :

(a) 2 (b) 3
(c) 4 (d) 8

TSPSC Managers, 2015
TNPSC AE, 2008

Ans. (c) : According to IBR, safety factor of rivet should be less than 4.

358. A 60 mm long and 6 mm thick fillet weld carries a steady load of 15 kN along the weld. The shear strength of the weld material is equal to 200 MPa. The factor of safety is :

(a) 2.4 (b) 3.4
(c) 4.8 (d) 6.8

HPPSC Asstt. Prof. 18.11.2016
GATE 2006

Ans. (b) : $F = 0.707 \times t \times \ell \times \tau_s$

Where, $F = 15$ kN

$\ell = 60$ mm

$t = 6$ mm

$$\tau_s = \frac{15 \times 10^3}{0.707 \times 6 \times 60}$$

$\tau_s = 58.94$ MPa

Given shear strength = 200 MPa

$$FOS = \frac{200}{58.94} = 3.4$$

359. Two metals plates of thickness '*t*' and width '*w*' are joined by a fillet weld of 45° as shown in fig. When subjected to a pulling force '*F*' the stress induced in the weld will be

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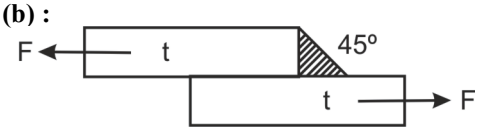
(a) $\frac{F}{wt}$ (b) $\frac{F}{wt\sin 45}$

(c) $\frac{F.\sin 45}{wt}$ (d) $\frac{2F}{wt}$

ISRO Scientist/Engineer 2009
ESE 1998

Ans. (b) :



$$Stress\ \tau = \frac{F}{A}$$

$$\tau = \frac{F}{w \times t\sin 45^\circ}$$

$$\tau = \frac{F}{wt\sin 45^\circ}$$

360. If the tearing efficiency of a riveted joint is 75%, then the ratio of diameter of rivet to the pitch is equal to:

(a) 0.2 (b) 0.25

(c) 0.50 (d) 0.6

RPSC VPITI 14.02.2016
SJVN ET 2013

Ans. (b) : Tearing efficiency of riveted joints is the ratio of minimum strength of the joint against failure by tension shear to that of member without a joint.

$$\eta_{tearing} = \frac{(p-d)t\sigma_t}{pt\sigma_t} = \frac{p-d}{p} = 1-\frac{d}{p}$$

$$1-\frac{d}{p} = 0.75$$

$$\frac{d}{p} = 1-0.75 = 0.25$$

Ratio of rivet diameter and pitch of rivet $\left(\frac{d}{p}\right) = 0.25$

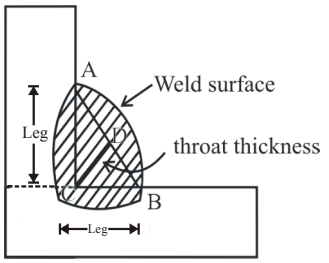
361. In a fillet welded joint, the weakest area of the weld is :

(a) toe (b) throat

(c) root (d) face

MPPSC AE 2016
ESE 2001

Ans : (b)



$$\text{Throat thickness}(t) = \frac{S}{\sqrt{2}}$$

S = leg length

In a fillet welded joint, the weakest area of the weld is throat.

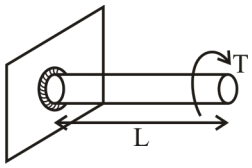
362. A circular solid rod of diameter d welded to a right flat plate by a circular fillet weld of throat thickness t is subject to a twisting moment T. The maximum shear stress induced in the weld is :

(a) $T/\pi td^2$ (b) $2T/\pi td^2$

(c) $4T/\pi td^2$ (d) $2T/\pi td^3$

Kerala PSC Poly. Lect. 09.01.2014
ESE 2003

Ans. (b) :



We know,

$$\frac{\tau}{R} = \frac{T}{J}$$

$$\tau = \frac{TR}{J}$$

$$\tau = \frac{T\cdot d/2}{\frac{\pi}{4}\cdot d^3\cdot t} \qquad \left[J = \frac{\pi d^3 t}{4},\ R = \frac{d}{2}\right]$$

$$\tau = \frac{2T}{\pi d^2 t}$$

Where, T = Twisting moment

d = Diameter of rod

t = Throat thickness

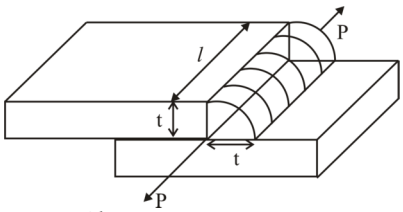
363. In welded joint the throat of weld as compared to size of weld is

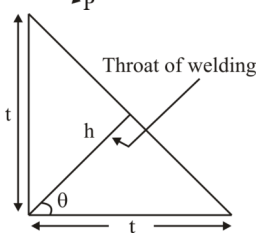
(a) about same size (b) about 0.7 times

(c) about 0.5 times (d) about 0.25 times

TNPSC AE 2019
RPSC Vice Principal ITI 2018

Ans. (b) : $h = \frac{t}{\sin\theta + \cos\theta}$





$$\tau = \frac{\text{shear load}}{\text{shear area}}$$

$$= \frac{P}{l\times h} \quad t \rightarrow \text{size of weld}$$

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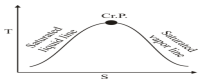


P, l & t are constant, $\tau = f(\theta)$

For τ_{max}, $\frac{d\tau}{d\theta} = 0$

$\cos\theta - \sin\theta = 0 \Rightarrow \boxed{\theta = 45^o}$

$h = \frac{t}{\sin 45^o + \cos 45^o} = \frac{t}{\sqrt{2}}$

$\boxed{h = 0.707\ t}$

364. In riveted joints, due to tearing of the plate at an edge and the rivet holes, a joint may fail. This can be avoided by keeping the margin m equal to (where d = diameter of the rivet hole)

(a) the margin m = 4.5d
(b) the margin m = 1.5d
(c) the margin m = 3.5d
(d) the margin m = 2.5d

JPSC AE 10.04.2021, Paper-II
Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (b) : Riveted joint, mainly three type of failure are considered.
(i) Shear failure of rivet
(ii) Tensile failure of plate between rivets and
(iii) Crushing failure of the plate
The margin is the distance between the edge of the plate to the centerline of the rivets in the nearest row usually

$\boxed{m = 1.5d}$

Where, m = margin
d = dia of rivets

365. The plane of maximum shear stress in the conventional 45° fillet weld subjected to parallel loading is :

(a) $67\frac{1}{2}°$ (b) $27\frac{1}{2}°$
(c) 45° (d) 90°

CGPSC AE 15.01.2021
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Shear stress in parallel weld,

$\tau = \frac{P(\sin\theta + \cos\theta)}{hl}$

The plane with maximum shear stress,

$\frac{\partial\tau}{\partial\theta} = \frac{P}{hl}(\sin\theta + \cos\theta) = 0$

$\cos\theta - \sin\theta = 0$

$\cos\theta = \sin\theta$

$\tan\theta = 1$

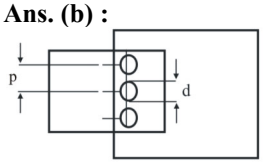
therefore, $\boxed{\theta = 45°}$

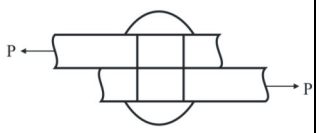
366. When two plates are joined by rivets, the pull required to tear off the plate across a row of rivets per pitch length is equal to-

(a) ptf_t (b) $(p–d)tf_t$
(c) $(p+d)\ tf_t$ (d) $(p+\sqrt{d}\ )tf_t$

NLC GET 17.11.2020, Shift-II
WBPSC AE 2003

Ans. (b) :





Single riveted lap joint
we known that tearing area per pitch length
$A = (p - d)\ t$
Tearing resistance or pull required to tear off the plate per pitch length [$P_t = A_t\ f_t$]
$P_t = (p - d)t\ f_t$
where, d = diameter of rivet
p = pitch of rivets
t = thickness of plate
f_t = Permissible tensile stress

367. A rivet is specified as a 20 mm rivet. What does it mean?

(a) Head dia 20 mm
(b) Shank dia 20 mm
(c) Both head dia and shank dia 20 mm
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Rivet is specified by shank diameter. So, rivet specified as a 20 mm rivet means its shank dia 20 mm.

368. When two plates are butt together and riveted with cover plates with two rows of rivets, the joint is known as

(a) lap joint
(b) butt joint
(c) single-riveted single-cover butt joint
(d) double-riveted double-cover butt joint

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : When two plates are butt together and riveted with cover plates with two rows of rivet, the joint is known as double-riveted double-cover butt joint.

369. A/An ____ is a joint in which the rivets are arranged in such a way that rivets in different rows are located opposite to each other.

(a) zig zag joint (b) lap joint
(c) elevated joint (d) chain riveted joint

UPRVUNL AE 05.07.2021

Ans. (d) : A/An chain riveted joint is a joint in which the rivets are arranged in such a way that rivets in different rows are located opposite to each other.

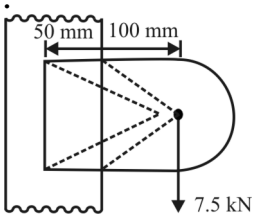
370. _____ rivet is a type of rivet.

(a) Blind (b) Sync
(c) Sledge (d) None of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) :

371. Magnitude of primary shear stress in case of eccentrically loaded welded connection with throat t of the weld as shown in figure will be in N/mm^2.



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(a) $\frac{7.5}{t}$ (b) $\frac{75}{t}$

(c) $\frac{750}{t}$ (d) $\frac{7500}{t}$

JPSC AE 10.04.2021, Paper-II

Ans. (b) : Primary shear stress

$$\tau = \frac{\text{load}}{\text{Area}}$$

$$= \frac{7.5\times10^3}{2\times50\times t}$$

$$\boxed{\tau = \frac{75}{t}\ \frac{N}{mm^2}}$$

372. For a parallel fillet weld, the ratio between throat, t and leg, h is :

(a) $\sqrt{2}$ (b) $2\sqrt{2}$

(c) $\frac{1}{2\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}$

JPSC AE 10.04.2021, Paper-II

Ans. (d) : For a parallel fillet weld the ratio between throat, t and leg h is :

$$t = \frac{h}{\sqrt{2}}$$

$$\boxed{\frac{t}{h} = \frac{1}{\sqrt{2}}}$$

373. If tearing efficiency of a riveted joint is 80%, then the ratio of diameter d of rivet to the pitch p will be :

(a) 0.2 (b) 0.25

(c) 0.3 (d) 0.4

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Tearing efficiency,

$$\eta_{tearing} = \left(1-\frac{d}{P}\right)$$

$$1-\frac{d}{P} = 0.80$$

$$\frac{d}{P} = 1-0.80$$

$$\boxed{\frac{d}{P} = 0.20}$$

374. Consider the following statements regarding welded and riveted joints:

1.Welded assemblies are tight and leak proof as compared with riveted assemblies.

2. Single-welded V-joint is less reliable than square butt joint.

3. Welding results in a thermal distortion of the parts, thereby inducing residual stresses.

Which of the above statements is/are correct?

(a) 1 only (b) 1 and 3 only

(c) 2 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (b) :

- Welded joints are subjected to residual stresses due to non-uniform heating of the parts being joined. There is always a possibility that localised thermal stresses may result from uneven heating and cooling during fusion and subsequent cooling. This also results in distortion.
- Single welded V-joint is more reliable than square butt joint. Single welded V-joint with backing strip is more reliable than single welded V-joint without backing strip. Double welded V-joint is more reliable than single welded V-joint with backing strip. The cost also increases with the reliability of the joint.

375. Which type of welding joint is used while joining the two pipes of equal diameters?

(a) Corner joint (b) Lap joint

(c) Butt joint (d) Edge joint

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Butt joint : A joint is a technique in which two pieces of material are joined by simply placing their ends together with out any special shaping the name of 'butt' joint comes from the way the material is joined together. It is used white joining the two pipes of equal diameters.

376. A small weld used to temporarily hold two pieces together during actual welding joint is known as :

(a) tack weld (b) fillet weld

(c) puddle (d) weld bead

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : Tack welding involves welding two or more metal pieces together by merely appling pressure and heat to the area to be weld. Tack welding joins the two pieces of metal by using electrodes to send electrical current through the work pieces. The part are locally heated.

377. The welded joint designed for shear strength is

(a) parallel fillet (b) transverse fillet

(c) circumferential joint (d) butt joint

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Parallel fillet welded joints are weak in shear so, they are designed for shear strength.

378. In a hexagonal nut, width across flats by approximate standard method is _____ (where D = nominal diameter).

(a) 1.0*D + 3 mm (b) 2*D mm

(c) 1.5*D + 3 mm (d) 4*D mm

APPSC Poly Lect. 13.03.2020

Ans. (c) : Let;

b = width across flats for a hexagonal nut.

$\therefore\ b = 1.5\times D + 3mm$ Where; D = nominal diameter

379. Which one of the following is NOT a rivet head?

(a) snap head (b) pan head

(c) low head (d) high button head

APPSC Poly Lect. 13.03.2020

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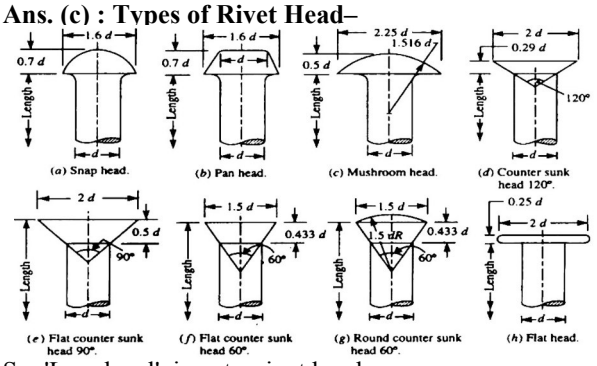
Ans. (c) : Types of Rivet Head–



So, 'Low head' is not a rivet head.

380. During crushing or bearing failure of riveted joints

(a) The holes in the plates become oval shaped and joints become loose
(b) There is tearing of the plate at an edge
(c) The plates will crack in radial directions and joints fail
(d) The rivet heads will shear out by applied stress

ESE 2020

Ans. (a) : Crushing strength of plate/rivet crushing failure of the plate/rivet. This type of failure occurs when the compressive stress between the shank of the rivet and the plate exceeds the yield stress in compression. The failure results in elongating the rivet hole in the plate and loosening of the joint.

381. If 's' is the size of the weld, 'ℓ' is the length of the weld and 't' is the allowable tensile stress, then the tensile strength of the joint for double transverse fillet weld is:

(a) $0.707\ s\ell t$ (b) $1.414\ s\ell t$

(c) $0.707\ \pi s\ell t$ (d) $1.414\ \pi s\ell t$

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : The tensile strength of the welded joint for

single fillet $= \dfrac{s\ell t}{\sqrt{2}}$

For double fillet $= \dfrac{2s\ell t}{\sqrt{2}}$

$= \sqrt{2}\, s\ell t$

Allowable tensile strength = $1.414\ s\ell t$

382. Two plates 16mm thick are joined by a double riveted lap joint. The rivets are 25 mm in diameter. If the maximum tensile load on the joint is 48 kN, the crushing stress induced between the plates and the rivet is :

(a) 15 MPa (b) 60 MPa
(c) 30 MPa (d) 90 MPa

NLC GET 17.11.2020, Shift-II

Ans. (b) : Crushing of rivets– Some times, the rivets do not actually shear off under the tensile stress, but are crushed. Due to this the rivet hole becomes of an oval shape and hence the joint becomes loose. The Failure of rivets in such a manner is also known as bearing Failure

$P_c = nd\ t\ \sigma_c$

where P_c = Pull (Tensile) is required to crush rivet
n = no. of rivet per pitch length
σ_c = safe permissible crushing stress
d = diameter of rivet

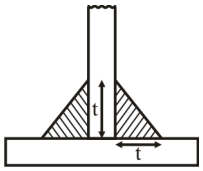
Given, $P_c = 48\ kN = 48 \times 10^3 N$
n = 2
d = 25 mm
t = 16mm
$\sigma_c = ndt\ \sigma_c$
$48 \times 10^3 = 2 \times 16 \times 25 \times \sigma_c$
$\sigma_c = 60\ MPa$

383. The T joint of steel plates which is subjected to 350 kN load is developed using intermitted 4 double fillet welds of length each of 40 mm at an interval of 100 mm. Allowable shear strength of the weld metal is 100 MPa. Determine the leg length of the weld

(a) 12.45 mm (b) 15.45 mm
(c) 17.25 mm (d) 20.25 mm

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) :



$\because$ 4 double filler welds
t = leg length/leg size
$p = 4 \times 2 \times 0.707\ t\ell\tau_{per}$
$350 \times 10^3 = 4 \times 2 \times 0.707 \times t \times 40 \times 100$
t = 15.45 mm

384. According to Unwin's formula, the relation between diameter of rivet hole (d) and thickness of plate (t) is given by

(a) $d = \sqrt{3}\ t$, where $t < 8$ mm
(b) $d = 5t$ Where $t > 8$ mm
(c) $d = \sqrt{7}\ t$, where $t < 8$ mm
(d) $d = 6\sqrt{t}$, where $t > 8$ mm

Oil India Senior Officer 23.12.2020

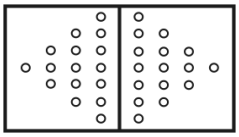
Ans. (d) : $d = 6\sqrt{t}$, where $t > 8$ mm

385. In a riveted joint, when the number of rivets decrease from the innermost row to outermost row, the joint is said to be :

(a) Chain riveted (b) Zig-zag riveted
(c) Diamond riveted (d) None of these

ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) :



386. The distance between the edge of the plate to the center line of the rivets in the nearest row is known as:

(a) Pitch of riveted joint
(b) Margin of riveted joint
(c) Transverse pitch
(d) Diagonal pitch
(e) Circular pitch

CGPSC AE 25.02.2018
CGPSC ARTO & TSI 07.01.2017

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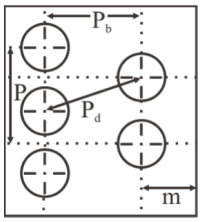
Ans. (b) : The distance between the edge of the plate to the centre line of the rivets in the nearest row is known as margin of riveted joint. A joint may fail due to tearing of the plate at an edge. This can be avoided by keeping the margin (m = 1.5 d), where d is the diameter of the rivet hole.



387. Tensile strength of the plates for Butt joint is : [σ = Permissible tensile stress, *t* = Thickness of plate, *l* = Length of weld, η = Efficiency of joint]

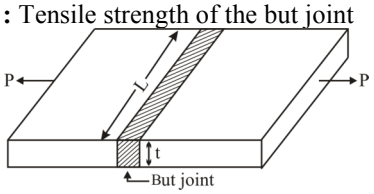
(a) $\frac{\sigma tl}{\eta}$ (b) $\sigma tl\eta$

(c) $\frac{\sigma tl\eta}{2}$ (d) $\frac{\sigma tl}{2\eta}$

(e) $\frac{\sigma t^2 l}{2\eta}$

CGPSC AE 25.02.2018

Ans. (b) : Tensile strength of the but joint



$P_{theoritical} = \sigma \times t \times L$

$\eta = \frac{P_{actual}}{P_{theoretical}}$

$P_{actual} = P_{theoretical} \times \eta = \sigma \times t \times L \times \eta$

388. A riveted joint may fail by

1. tearing of the plate at an edge
2. tearing of the plate across a row of rivets
3. shearing of rivets

Which of the above are correct?

(a) 1 and 2 only (b) 1 and 3 only

(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : A riveted joint may fail in the following ways.
(i) Tearing of the plate at an edge.
(ii) Tearing of the plate across a row of rivets.
(iii) Shearing of the rivets.
(iv) Crushing of the rivets.

389. If weld throat thickness for the fillet weld is 4.2 mm, the weld size is approximately :

(a) 5mm (b) 7.1 mm

(c) 8.2 mm (d) 6 mm

Vadodara Muncipal Corp. DEE, 2018

Ans. (d) : Given, Weld throat thickness (h) = 4.2 mm

weld size (t) = $\sqrt{2}$ × throat thickness

$= \sqrt{2} \times 4.2$

= 5.94

= 6 mm

390. The factor of safety provided for the cotter is less than that for the spigot and socket end because

1. There is stress concentration in the cotter
2. There is no stress concentration in the cotter
3. The cotter is not as weaker as the spigot and socket end
4. The cotter has been made from materials different from that of the spigot and socket end

Which of the above statements are correct?

(a) 1 and 3 only (b) 1 and 4 only

(c) 2 and 3 only (d) 2 and 4 only

JWM 2017

Ans. (c) : Spigot is the male part of the joint, it has a rectangular slot for passing the cotter through it. Spigot has a collar which rests against the socket end.

Socket is the female part of the joint, it also has a rectangular slot for passing the cotter through it.

Cotter is a wedge shaped piece of metal which actually connects two part which are not rotating.

The factor of safety for cotter is less than the spigot and socket because, there is no stress concentration in the cotter and resulting high strength than spigot and socket end.

391. In welding joint, it is preferable to design groove joint in such a way that it would be subject to

(a) Tension only
(b) Compression only
(c) Tension or Compression only
(d) Tension and compression simultaneously

GPSC EE Pre, 28.01.2017

Ans. (c) : In welding joint, it is preferable to design groove joint in such a way that it would be subject to tension or compression only.

392. Root opening of a Single Vee groove weld joint of plate thickness "t" is

(a) $\frac{t}{3}$ (b) $\frac{t}{4}$

(c) $\frac{t}{5}$ (d) $\frac{t}{6}$

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : As per welding codal provision the root opening of single Vee groove weld joint is t/4.

393. Compressive residual stresses in weld joints

(a) Decreases tensile strength
(b) Increases cracking tendency
(c) Increases fatigue strength
(d) Coarsen the grain structure

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Compressive residual stresses in weld joints increases fatigue strength.
Compressive residual stresses the fatigue performance of the weld joints is based on simple concept of lowering the effective applied tensile stresses by inducing compressive residual stress. Compressive residual stress to some extant cancels the magnitude of externally applied tensile stress. Improvement in fatigue performance of the weld joint can vary from 20-30%. There are many methods namely shot peening, overloading, spot heating and past-weld heat treatment, which can be used to induce compressive residual stress.

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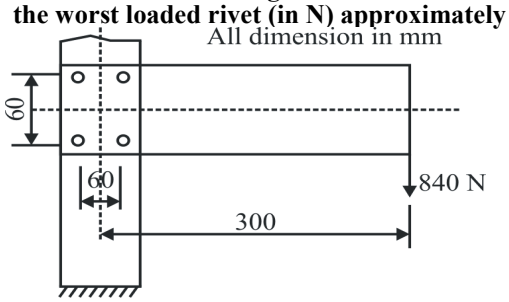
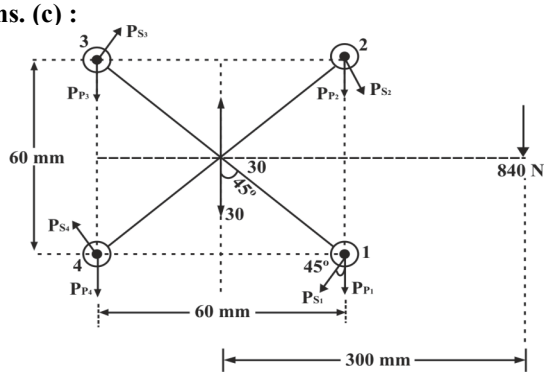
394. A rectangular steel plate is joined to a vertical post using four identical rivets arranged as shown below in the figure. The shear load on the worst loaded rivet (in N) approximately is



(a) 1700 (b) 1500
(c) 1650 (d) 1750

ISRO Scientist/Engineer 07.05.2017

Ans. (c) :



Primary shear force,

$$P_{P_1} = P_{P_2} = P_{P_3} = P_{P_4}$$

$$= \frac{P}{4} = \frac{840}{4} = 210 \text{ N}$$

For secondary shear force,

$$\frac{P_{S_1}}{r_1}(r_1^2 + r_2^2 + r_3^3 + r_4^2) = P \times e$$

where, $r_1 = r_2 = r_3 = r_4 = 30\sqrt{2}$ mm

$$\Rightarrow \quad P_{S_1} = \frac{(P \times e) r_1}{(r_1^2 + r_2^2 + r_3^3 + r_4^2)}$$

$$= \frac{840 \times 300 \times 30\sqrt{2}}{7200} = 1485 \text{ N}$$

$$R_{max} = \sqrt{P_{P_1} + P_{S_1} + 2P_{P_1}P_{S_1} \cos 45^\circ}$$

$$= \sqrt{210^2 + 1485^2 + 2 \times 210 \times 1485 \cos 45^\circ}$$

$$= 1640.23 \text{ N}$$

So, the shear load on the worst loaded rivet (1 or 2) is 1640 N.

395. Among the following which is not part of solid rivets?

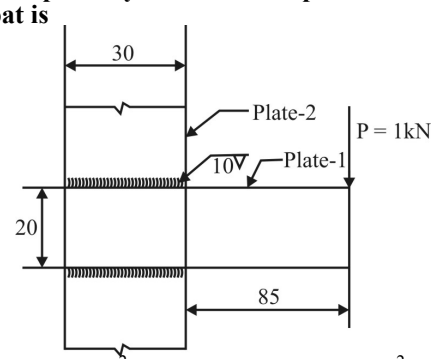
(a) Bifurcated rivet (b) Pan head rivet
(c) Flat countersunk rivet (d) Snap head rivet

Punjab PSC SDE 12.02.2017

Ans. (a) : Bifurcated rivet is not part of solid rivets.

396. A 10 mm thick plate-1 is welded to another plate-2 (10 mm thick) using fillet weld of thickness 10 mm, on two sides of plate-1 as shown in figure. Plate-1 projects outside plate-2 by 85 mm. An end load of 1 kN acts on plate-

1. The primary shear stress produced in the throat is



(a) 173 N/mm^2 (b) 2.4 N/mm^2
(c) 85 N/mm^2 (d) None of these

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Primary shear stress $= \frac{P\sqrt{2}}{2b \times s}$

$$= \frac{1000 \times \sqrt{2}}{2 \times 30 \times 10}$$

$$= 2.35 \text{ N/mm}^2$$

$$\simeq 2.4 \text{ N/mm}^2$$

397. The strength of a properly welded joint as compared to base metal would be

(a) Same
(b) More
(c) Less
(d) Two can't be compared
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (b) : The strength of a properly welded joint as compared to base metal would be more.

398. The permissible stress in a filled weld is 200 N/mm^2. The filled weld has equal leg lengths of 15mm each. The allowable shearing load on weldment per cm length of the weld is _____

(a) 15.0 kN (b) 22.5 kN
(c) 7.5 kN (d) 21.2 kN

UPRVUNL AE 21.08.2016

Ans. : (d) Shear stress in fillet weld = 200 N/mm^2
leg length (s) = 15 mm

Allowable shearing load per cm length = $\frac{T \times S}{\sqrt{2}}$

load/cm length is = $\frac{200 \times 15}{\sqrt{2}}$ N/mm

load/cm length = 21.2 kN/mm.

399. What is the tearing strength of a plate of a riveted joint whose pitch is 50 mm and rivet diameter is 20mm? The plate thickness is 10mm and permissible tensile stress is 120N/mm^2?

(a) 36 kN (b) 24 kN
(c) 12 kN (d) none of the above

Kerala PSC IOF 19.04.2016

Ans. (a) $F_s = (P - d)t \times \tau_s$

$$= (50 - 20) \times 10 \times 120 \text{N}$$

$$F_s = 36 \text{kN}$$

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400. For a single v butt weld joint the effective throat thickness is 10 mm and length of weld is 100mm. If the safe stress is 120 N/mm^2, the permissible load is equal to :

(a) 100 kN (b) 120 kN
(c) 60 kN (d) none of the above

Kerala PSC IOF 19.04.2016

Ans. (b) $F_s = t \times \ell \times \tau_s$
$= 10 \times 100 \times 120N$
$= 120kN$

401. Aircraft body is usually fabricated by ..... .

(a) welding (b) precasting
(c) rivetting (d) casting

GWSSB DEE 07.07.2016

Ans. (c) : Aircraft body is usually fabricated by rivetting.

402. The stress along the contact surface of a rivet and the plate is

(a) bearing stress (b) shearing stress
(c) temperature stress (d) hoop stress

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Bearing stress– It is a contact pressure between two separate bodies. In this case it is the stress between the contact surface of a rivet and the plate.
Shearing stress – The stress which acts parallel to a plane on which a force has been applied. Shear stresses tends to promote sliding along a plane.
Temperature stress – Stresses due to change in temperature.
Hoop stress – This stress is directed along the tangent to the circumference of the cylinder. This stress is tensile in nature. It is also called circumferential stress.

403. Rivets in lap joints are subjected to

(a) axial and shear forces
(b) axial forces only
(c) double shear
(d) single shear alone

APPSC AE Subordinate Service Civil/Mech. 2016
TSPSC AEE 2015

Ans. (a) : Rivets in lap joint are subjected to axial and shear force.
It fails in single shear or due to bending stresses.
Double cover butt joint is free from bending stress and is subjected to double shear.

404. .....is an example for foundation bolt.

(a) Plow bolt (b) Lag bolt
(c) Rag bolt (d) Stove bolt

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Rag bolt is type of dowel screws also known as lay screws are heavy duty screws designed for metal to wood applications. They have a square or hexagonal head that can be used with the standard wrench and pliers.

405. To secure a leakless or fluid tight joint, a process known as .....is employed during riveting.

(a) Clocking (b) Caulking
(c) Crowning (d) Clinch forming

HPPSC AE (PWD) 10.09.2016

Ans. (b) : To secure a leakless or fluid tight joint a process known as caulking is employed during riveting.

406. A joint where material is removed from each of the members so that the resulting joint is the thickness of the thickest member is known as_____

(a) Lap joint (b) Half lap joint
(c) Material joint (d) Elbow joint

KPCL AE 2016

Ans. (b) : In a half lap joint or halving joint, material is removed from both of the members so that the resulting joint has the thickness of the thickness member. Most commonly in half lap joints the members are of the same thickness and half the thickness of each is removed.

407. Stress relaxation is the phenomenon

(a) In which stress remains constant on increasing load
(b) Stress reduced on increasing load
(c) In which deformation tends to loosen the joint and produces a reduced stress
(d) In which parts are not loaded

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : In stress relaxation phenomenon deformation tends to loosen the joint and produces a reduced stress.
Stress relaxation is a gradual reduction in stress with time at constant strain.
stress relaxation occurs in polymers when they are held in strained state for long period of time.

408. Single 'V' and single 'U' butt welds are used for sheets of thickness

(a) 5-15mm (b) upto 5 mm
(c) 20-25mm (d) 15-20mm

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : Single and single butt welds are used for sheets of thickness 5 to 15 mm.

409. Which of the following joints have more corrosion resistance?

(a) Welded joint (b) Riveted joints
(c) Soldering joints (d) Threaded joint

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : The joint welded with activating flux have good resistance and have better resistance to gain boundary corrosion.
Example – Soldering joint

410. A cylindrical section having no joint is known as

(a) Homogeneous section
(b) Heterogeneous section
(c) Jointless section
(d) Seamless section

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (d) : A cylindrical section having no joint is known as seamless section.

411. A riveted joint in which every rivet of a row is opposite to the other rivet of the outer row, is known as

(a) Diamond riveted joint
(b) Zig-Zag riveted joint
(c) Gold riveted joint
(d) Chain riveted joint

Rajasthan Nagar Nigam AE 2016, Shift-I

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Ans. (d) : A riveted joint in which every rivet of a row is opposite to other rivet of the outer row, is known as chain riveted joint.

412. A welded material is 50 mm long and 5 mm thick. The factor of safety is 3.54. If it carries a steady load of 20 KN along the weld, what is the shear strength of the weld material?

(a) 200 MPa (b) 275 MPa
(c) 325 MPa (d) 350 MPa
(e) 400 MPa

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : As we know,
T = 20 kN, d = 50 mm
FOS = 3.54

$$P = 0.707t\ell\left(\frac{\tau_{per}}{FOS}\right)$$

$$20\times10^3 = 0.707\times5\times50\times\frac{\tau}{3.54}$$

Shear strength of weld material,

$$\tau = 400.56\ N/mm^2 \simeq 400\ MPa$$

413. Two plates are joined by a parallel fillet weld. The plate is subjected to a tensile load of 340 kN. The working stress in shear for weld material is 80 MPa. The length of 15 mm weld received to resist load is:

(a) 400 mm length of weld adjusted on both sides
(b) 200 mm length of weld adjusted on both sides
(c) 400 mm length on each side
(d) 100 mm length of weld on each side

RPSC ADE 2016

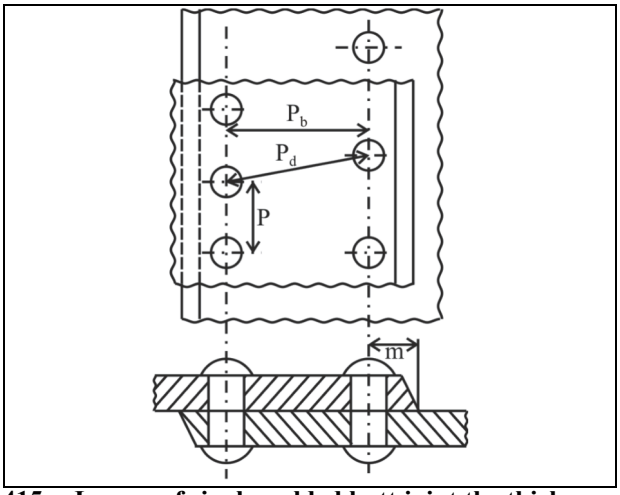
Ans. (a) : As load is acting in longitudinal direction, the joint will fail in shear.
l = total length of weld
force = stress × area
$340 \times 10^3 = 80 \times 10^6 \times 0.015 \times l$
l = 0.2833 m
and this total length should be adjusted on both side to avoid eccentric loading.
For safe design total length of weld should more than 283.3 mm or say l = 400 mm.

414. The distance between the centers of the rivets in adjacent rows of zig zag riveted joint is known as

(a) Back Pitch (b) Lap
(c) Diagonal Pitch (d) Pitch

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : **Pitch (P)–**It is the distance from the centre of one rivet to the centre of next measured parallel to the seam.
Back Pitch (P_b)– It is the perpendicular distance between the centre line of the successive rows.
Marginal Pitch (M)–It is the distance between the centre of rivet hole to the nearest edge of the plate.
Diagonal Pitch (P_d)–The smallest distance between centres of two rivets holes in adjacent row of a zig-zag riveted joint is called diagonal pitch.



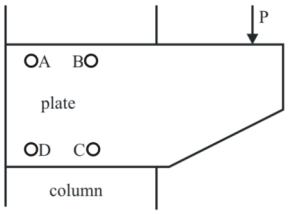
415. In case of single welded butt joint the thickness of plate must be :

(a) 10 – 12 mm (b) 1 – 6 mm
(c) 8 – 10 mm (d) 12 – 15 mm

TSPSC Managers, 2015

Ans. (b) : In case of single welded butt–joint the thickness of plate must be 1mm to 6mm.

416. A plate, connected to a column by 4 rivets, carries a load P. If the centres of the rivets form a square and if they are designed on the basic of shear stresses, then the load carried by:



(a) the rivets A and B is maximum
(b) the rivets A and D is maximum
(c) the rivets B and C is maximum
(d) the rivets C and D is maximum

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : The rivets which are nearer to the applied load will experience maximum load.

417. A 50 mm diameter solid shaft is welded to a flat plate by 10 mm fillet weld. The maximum torque that the welded joint can sustain if the maximum shear stress intensity in the weld material is not to exceed 80 MPa is

(a) 2 kN-m (b) 2.1 kN-m
(c) 2.22 kN-m (d) 2.35 kN-m

MPPSC AE 08.11.2015

Ans. (c) : Given data : d = 50 mm = 0.05m, S = size of weld = 10 mm = 0.01m
Let, T = maximum torque on shaft

$$\because (\tau_s)_{max} = \frac{2T}{\pi\times0.707\times S\times d^2}$$

$$\Rightarrow T = \frac{\pi\times0.707\times S\times d^2\times(\tau_s)_{max}}{2}$$

$$= \frac{\pi\times0.707\times0.01\times(0.05)^2\times80\times10^6}{2}$$

$$\Rightarrow T = 2.22\times10^{-3}\times10^6 \Rightarrow T = 2.22\ kN\text{-}m$$

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418. Tearing efficiency of rivet is given by (P = pitch of rivets, d = diameter of river hold)

(a) (2p - d)/P (b) (p - d)/4P
(c) (p - 2d)/P (d) (p - d)/P

TSPSC AEE 2015

Ans. (d) : Tearing efficiency of riveted joints is the ratio of minimum strength of the joint against failure by tension shear to that of member without a joint.

$$\eta_{tearing} = \frac{(p-d)t\sigma_t}{pt\sigma_t} = \frac{p-d}{p} = 1-\frac{d}{p}$$

419. In a butt weld joint the throat of weld as compared to size of the weld is in the ratio :

(a) 2 : 1 (b) 1 : 2
(c) 1 : 1 (d) 1 : 3

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : 1 : 1

420. Which of the following statements are correct?

1. As per Indian Boiler Regulations the thickness of the boiler shell should not be more than 7 mm.
2. **Minimum distance between the rows of the rivets in chain riveting joints should be equal to the diameter of rivet hole.**
3. **For longitudinal joint in boilers a butt joint with two cover plates should be used.**
4. **A butt joint with double strap is in double shear.**

Choose the correct answer from the options given below.

(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

UPRVUNL AE 2014

Ans : (c)
(3) For longitudinal joint in boilers a butt joint with two cover plates should be used.
(4) A butt joint with double strap is in double shear

421. Which of the following is true regarding T joint weld?

(a) It is used where severe loading is encountered and the upper surface of both pieces must be in the same plane
(b) It is used to join two pieces of metal in the manner as rivet joint meals
(c) It is used to join two pieces perpendicularly
(d) It is used where longitudinal shear is present

UPRVUNL AE 2014

Ans : (c) A T-joint is used when two components are perpendicular to each other in the form of T. The end of one face is welded to other side of a component.
When severe loading is encountered and the upper surface of both pieces are in the same plane then Joggled weld joint is used. Welded Joints are mostly of two types

- Butt Joint : It is used when the components are lying in the same plane. It connects the end of the two plates.
- Fillet Joint : It is used between two overlapping components. When longitudinal shear stress is present Parallel fillet welds are used.

Note :– The commission has given answer option (d).

422. A single riveted lap joint is used to join the two 6 mm plates. If the tensile load acting on the plates in 50 kN, determine the diameter of the rivet

(a) 14 mm (b) 17 mm
(c) 22 mm (d) 19 mm
(e) 20 mm

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : We know that,
Unwin's formula is given as

$d = 6\sqrt{t}$

$d = 6 \times \sqrt{6} = 14.696$ mm

From option 14.696 mm is greater than 14 mm and less than 17 mm. So answer should be 17 mm.

423. Locked-in-stresses of welded joints are negligible in-

(a) Fusion welding method
(b) Torque welding method
(c) Low temperature joining method
(d) Induction brazing method

MPSC Dy. Engineer 2014

Ans. (c) : Locked-in-stresses welded joints are negligible in low temperature joining method.
Locked in stress (Residual stresses) were defined as the internal stresses that exist in a body or part of body in the absence of any externally applied load. It exist in high temperature fabrication processes.

424. In a gib and cotter joint, the gib and cotter are subjected to :

(a) Single shear
(b) Double shear
(c) Single shear and crushing
(d) Double shear and crushing

UPRVUNL AE 2014
ESE 2006

Ans. : (d) In a gib and cotter joint the gib and cotter are subjected to double shear and crushing.

425. A 50 mm long and 5 mm thick fillet weld carries a steady load of 20 kN along the weld. The shear strength of the weld material is equal to 220 MPa. The factor of safety is

(a) 2.95 (b) 3.4
(c) 2.75 (d) 4.2
(e) 3.8

CGPSC AE 26.04.2015 Shift-I

Ans. (e) : Given, l = 50mm, t = 5 mm
P = 20 kN = 20,000 N
For parallel fillet weld

$$\tau_s = \frac{P}{A} = \frac{P}{0.707\,t.\ell} = \frac{20\times10^3}{0.707\times5\times50}$$

$\tau_s = 113.154$ MPa

$$FOS = \frac{220}{\tau_s} = \frac{220}{113.154}$$

FOS = 1.94
for double fillet weld = 1.94 × 2 = 3.88

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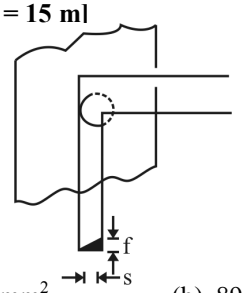
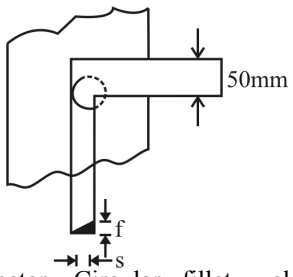
426. The throat area of the circular shown in figure is, [take s = 15 m]



(a) 1666 mm^2 (b) 899 mm^2
(c) 3333 mm^2 (d) 1444 mm^2

APPSC Poly. Lect. 2013

Ans. (a) :



50 mm diameter, Circular fillet weld t = throat thickness, S = size of weld,

t = S.sin 45° = 15 × 0.707, [Where, S = 15 mm]

Throat area for a circular fillet weld (A) = t × π D

= 15 × 0.707 × π × 50

= 1666 mm^2

427. A backing ring is used inside the pipe joint when making a:

(a) Butt Weld (b) Fillet Weld
(c) Sleeve Weld (d) Socket Weld

SJVN ET 2013

Ans. (a) : A backing ring is used inside the pipe joint when making a butt weld.

428. A plate 100 mm wide and 10 mm thick is to be welded to another plate by means of double parallel fillets. The plates are subjected to a static load of 77 kN. Find the length of weld if the permissible shear in the weld does not exceed 55 MPa.

(a) 100 mm (b) 150 mm
(c) 200 mm (d) 250 mm

TNPSC AE 2013

Ans. (a) : Data given as-

b = 100 mm, h = 10 mm

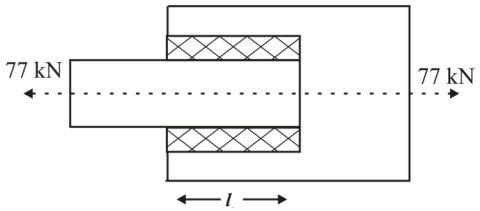
P = 77 kN τ = 55 MPa

We know that, when there are two welds of equal length on two sides then,

$$P = 2(0.707\, h\ell\tau)$$

$$\ell = \frac{77\times1000}{2\times0.707\times10\times55}$$

$$\ell = 99 \approx 100 \text{ mm}$$



429. Single riveted lap joint for 8 mm thick plates with 8.5 mm diameter rivets at a pitch of 50 mm. The stress in the plate is 110 N/mm^2. The tearing strength per pitch is

(a) 36.52 × 10^3 N (b) 7.48 × 10^3 N
(c) 37.84 × 10^3 N (d) 35.2 × 10^3 N

TNPSC AE 2013

Ans. (a) : Data given as-

t =8 mm, d_r = 8.5 mm

p = 50 mm, σ_t = 110 N/mm^2

Tearing strength per pitch

$$P_t = (p-d)\times t\times\sigma_t$$

= (50 – 8.5) × 8× 110 = 36.52 × 10^3 N

430. What is the diameter of butt joint if the thickness of the plate is 25 mm ?

(a) 30 mm (b) 25 mm
(c) 20 mm (d) 15 mm

TNPSC AE (Industries) 09.06.2013

Ans. (a) : Given, Plat thickness = 25 mm

We know,

Diameter of Rivet (d) = $6\sqrt{t}$

$= 6\sqrt{25}$

= 30 mm

431. A 60 mm long and 6 mm thick fillet weld carries a steady load of 15 kN along the weld. The shear strength of the weld material is equal to 200 MPa. The factor of safety is

(a) 2.4 (b) 3.4
(c) 4.8 (d) 6.8

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : $P_{PFW} = 0.707 \times t \times L \times \tau_s$

(Where PFW = Parallel fitted weld)

Given, L = 60 mm, t = 6 mm, P_{PFW} = 15 kN

τ_s = 200 MPa

$$\tau_s = \frac{15\times10^3}{0.707\times6\times60}$$

τ_s = 58.9344 MPa

Given the shear strength of the material = 200 MPa.

$$\text{Factor of safety} = \frac{200}{58.9344} = 3.4$$

432. When one gib is used in a gib and cotter joint, then the width of gib should be taken as

(a) 0.45 B (b) 0.55 B
(c) 0.65 B (d) 0.75 B

Where B = Total width of gib and cotter.

APPSC Poly. Lect. 2013

Ans. (b) : In the joint, as one gib is used the proportion of which are width of gib, b_1 = 0.55B. & width of cotter, b = 0.45B

B = total width of gib and cotter.

433. _____ riveting is used in structural units.

(a) Chain (b) Zig-zag
(c) Diamond (d) Non of the above

APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (c) : Chain riveted joint- In this type the accommodated rows of rivets may lie exactly opposite to each other over straight lines.

Zig-Zag riveted joint- Unlike the above type, here the rows may appear staggered and not complementing each other.

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434. Two strips of equal lengths and widths are joined together by two rivets, one at each end. One strip is of copper and the other of steel. Now, the temperature of this assembly is lowered, the rivets will undergo.

(a) Bending (b) Single shear
(c) Double shear (d) Both (a) & (b) above

UKPSC AE 2012 Paper-I

Ans. (b) : Single shear

435. A boiler plate thickness is 20mm. The rivet diameter will be :

(a) 10 mm (b) 20 mm
(c) 30mm (d) 40mm

RPSC ACF-2011
RPSC AE 2016

Ans. (c) : given- boiler plate thickness (t) = 20mm

Unwin's formula – $d = 6\sqrt{t}$

$d = 6\sqrt{20}$

$= 6 \times 4.472$

$d = 27\,mm \approx 30$ mm will be most suitable answer.

436. Two plates, subjected to a direct load, are connected by means of the following types of riveted joints:

(1) Lap joint
(2) Butt joint with single cover plate
(3) Butt joint with double equal cover Plates
(4) Butt joint with double unequal cover plates

Select the joint/joints in which all the rivets fail in double shear using the codes below:

(a) (1) only (b) (2), (3) and (4)
(c) (3) and (4) (d) (3) only

RPSC AE GWD, 2011

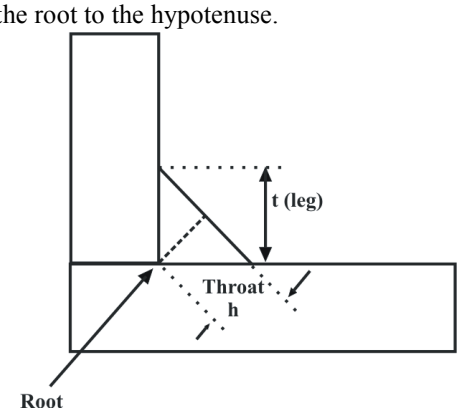
Ans. (c) : Butt joint with double equal or unequal cover plates when subjected to directed load will fail in double shear.

437. In fillet weld throat is

(a) Perpendicular distance from the root to the hypotenuse
(b) The smaller size of triangle of the fillet
(c) The larger size of triangle of the fillet
(d) None of the above

ISRO Scientist/Engineer 2010

Ans. (a) : In fillet weld throat is perpendicular distance from the root to the hypotenuse.



438. Efficiency of riveted joint is the ratio of

(a) shearing strength of rivet to strength of unriveted plate
(b) crushing strength of rivet to strength of unrivetd plate
(c) tearing strength of plate to strength of unriveted plate
(d) strength of riveted joint to strength of unriveted plate

TNPSC AE, 2008

Ans. (d) : Efficiency of riveted joint is the ratio of strength of riveted joint to strength of unriveted plate.

Efficiency of rivet =

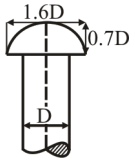
$$\frac{\text{Least value of [strength of tearng, shearing, crushing]}}{\text{strength of unriveted plate}}$$

439. Snap head rivets are used in

(a) aircraft body (b) ship building
(c) structural work (d) all of these

TNPSC AE, 2008

Ans. (d) : Body plates such as in structural work, ship building, bridges, tanks, aircraft body then the steel rivets are used snap head.



440. Tearing efficiency of a double riveted lap joint having a pitch of 60 mm and rivet hole diameter of 18 mm is

(a) 30% (b) 35%
(c) 60% (d) 70%

DRDO Scientists 2008

Ans. (d) : Tearing efficiency $= \frac{p-d}{p} \times 100$

$= \frac{60-18}{60} \times 100 = 70\%$

441. The centre-to-centre distance between two consecutive rivets in a row is called

(a) margin (b) pitch
(c) back pitch (d) diagonal pitch

Assam PSC AE (IWT) 14.03.2021
TNPSC AE (Industries) 09.06.2013

Ans. (b) : Pitch–It is the distance from the centre of one rivet to the centre of the next rivet measured parallel to the seam.
Margin–It is the distance between the centre of rivet hole to the nearest edge of the plate.
Back Pitch–It is the perpendicular distance between the centre lines of the successive rows.
Diagonal Pitch–It is the distance between the centres of the rivets in adjacent rows of zig-zag riveted joint.

442. The double riveted joint with two cover plates for boiler shell is 1.5 m in diameter subjected to steam pressure of 1 MPa. If the joint efficiency is 75%, allowable tensile stress in the plate is 83 MN/m^2, compressive stress is 138 MN/m^2 and shear stress in the rivet is 55 MN/m^2, the diameter of rivet hole will be nearly

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(a) 8 mm (b) 22 mm
(c) 36 mm (d) 52 mm

ESE 2020

Ans. (b) : Thickness, $t = \frac{PD}{2\sigma_{per}\eta} + CA$

$$t = \frac{1\times1.5}{2\times83\times0.75} + 2 = 14\text{mm} \geq 8\text{mm}$$

Hence by Unwin's formula

$$d = 6\times\sqrt{t} = 21 \text{ mm}$$

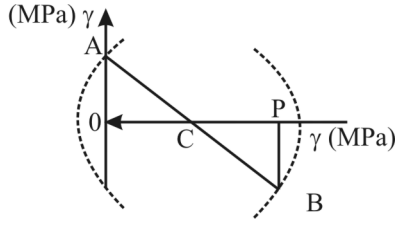
$$d_h = 22 \text{ mm}$$

443. A steel bolt is subjected to a direct tensile load of 24 kN and a shear load of 9 kN. If the area of cross-section resisting tension and shear is 300 mm^2 and the yield strength in tension for the bolt material is 200 MPa, the operating factor of safety according to maximum shear stress theory is

(a) 4 (b) 3
(c) 2 (d) 1

DRDO Scientists 2009

Ans. (c) :



$$\sigma_{xx} = \frac{24\times10^3}{300} = \frac{240}{3} = 80\,\text{MPa}$$

$$\tau_{xy} = \frac{9\times10^3}{300} = 30\text{MPa}$$

PB = 30
OP = 80
CP = OC = 40

$$BC = \sqrt{30^2+40^2} = 50$$

Maximum shear stress developed = 50 MPa (By maximum shear stress theory)

$$\tau_y = \frac{\sigma_y}{2} = 100\text{MPa}$$

$$FOS = \frac{100}{50} = 2$$

444. A 6 mm fillet weld is 50 mm long and carries a steady load of 12000 N along the weld as shown in figure. The weld metal has yield strength of 360 MPa. The value of factor of safety is

(a) 1.59 (b) 3.18
(c) 4.18 (d) 6.36

DRDO Scientists 2008

Ans. (b) : Shear stress

$$\tau = \frac{P}{0.707hl} = \frac{12000}{0.707\times6\times50} = 56.57 \text{ N/mm}^2$$

$$FOS = \frac{\text{yield strength}/2}{\text{shear stress induced}} = \frac{180\times10^6}{56.57\times10^6} = 3.18$$

445. A 60 mm long and 6 mm thick fillet weld carries a steady load of 15 kN along the weld. Te shear strength of the weld material is 200 MPa. The factor of safety is :

(a) 2.4 (b) 3.4
(c) 4.8 (d) 6.8

Kerala PSC Poly. Lect. 09.01.2014

Ans. (b) : $F_s = 0.707\times t\times \ell\times\tau_s$

$$= 0.707\times6\times60\times200\text{N}$$

$$= 50.904\,\text{kN}$$

$$FOS = \frac{F_s}{F}$$

$$= \frac{50.904}{15}$$

$$= 3.39$$

5. Design of Bearings, Brakes and Clutches

446. In a multiple disc clutch, n_1 is the number of discs on the driving shaft, and n_2 is the number of discs on the driven shaft. The number of pairs of contact surfaces is given by:

(a) $n_1 + n_2$ (b) $n_1 + n_2 + 1$
(c) $n_1 + n_2 - 1$ (d) $\frac{n_1 - n_2}{2}$
(e) $\frac{n_1 + n_2}{2}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
UPPSC AE 13.12.2020, Paper-I
OPSC AEE 2019 Paper-I
CGPSC ARTO & TSI 07.01.2017
Karnataka PSC Lect., 27.05.2017
Kerala PSC IOF 19.04.2016
Nagaland CTSE 2016 Ist Paper
Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE-2013, Paper-I
ESE 2001, 2008, 2011

Ans. (c) : In multi plate clutch, number of pairs of contact surface (n)

$$\boxed{n = n_1 + n_2 - 1}$$

Due to the high number of pairs of contact surfaces, the torque transmitted higher n is always a whole number.

447. If the load on a ball bearing is halved, its life

(a) Remains unchanged
(b) Increases two times
(c) Increase four times
(d) Increases eight times

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Assam PSC AE (PHED), 18.10.2020
UPPSC AE 13.12.2020 Paper-I
GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017
TSPSC AEE 2015
ESE 2005
GATE 1988

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Ans. (d) : As we know,
L = Life of bearing
P = Load
for ball bearing

$$L = \left(\frac{C}{P}\right)^3$$

$$L = \frac{C^3}{P^3}$$

$$L_1 P_1^3 = L_2 \left[\frac{P_1}{2}\right]^3 \qquad \therefore P_2 = \frac{P_1}{2}$$

$L_2 = 8\ L_1$
Increases by 8- times.

448. Two identical ball bearing P and Q are operating at loads 30 kN and 45 kN respectively. The ratio of the life of bearing P to the life of bearing Q is–
(a) 81/76 (b) 27/8
(c) 9/4 (d) 3/2

GPSC Asstt. Director of Transport 05.03.2017
BPSC Asstt. Prof. 29.11.2015
VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2011

Ans : (b) Life of ball bearing

$$L = \frac{C}{P_e}^3$$

Where, L = Life of ball bearing
P_e = Equivalent load
C = Basic dynamic capacity

$$\frac{L_P}{L_Q} = \left[\frac{(P_e)_Q}{(P_e)_P}\right]^3$$

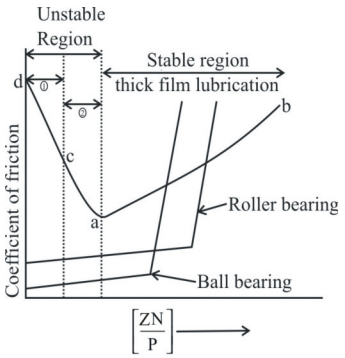
$$= \frac{45}{30}^3 = \frac{3}{2}^3 = \frac{27}{8}$$

$$\boxed{\frac{L_P}{L_Q} = \frac{27}{8}}$$

449. In case of a sliding contact bearing, the bearing characteristic number is given by (where, Z = Viscosity, N = Speed, P= Pressure):
(a) $\frac{ZP}{N}$ (b) $\frac{ZN}{P}$
(c) $\frac{Z}{NP}$ (d) $\frac{P}{ZN}$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
JPSC AE 10.04.2021, Paper-II
APPSC Poly Lect. 13.03.2020
APPSC Poly Lect. Automobile Engg., 2020
CGPSC AE 25.02.2018
TNPSC AE 2013
TNPSC AE (Industries) 09.06.2013
CSE Pre 1997

Ans. (b) :



(1) Thin film lubrication
(2) Semi (or) mixed lubrication.

Bearing characteristics Number– $\frac{ZN}{P}$ is termed as bearing characteristics number, the part of curve 'ab' represents the region of thick film lubrication between a & b, viscosity or the speed are so low or the pressure "P" is so great. So air combination will reduced the film thickness so that the partial metal to metal contact take place.
In 'd-c' region there are thin film, it is unstable region due to increase in load, temperature increase and over all value of ZN/P is decrease as a result increase in coefficient of friction.
At a point 'a' minimum amount of friction occurs. At this point the value of ZN/P is known as bearing modulus.

450. The bearing characteristic number of a hydrodynamic journal bearing depends on :
(a) length width and speed
(b) viscosity load and speed
(c) viscosity, speed and bearing pressure
(d) None of the above

BPSC Poly. Lect. 2016
VIZAG MT 2015
BPSC Asstt. Prof. 29.11.2015
ESE 2013, 1995
APPSC IOF, 2009

Ans. (c) : The bearing characteristic Number of a hydrodynamic journal bearing depends on viscosity, speed and bearing pressure.
The bearing characteristic number in journal bearing is defined as–

$$= \frac{\mu N_s}{P} = \frac{\text{Viscosity} \times \text{speed}}{\text{Pressure}}$$

451. The life of a ball bearing at a load of 10 kN is 8000 hours. If the load is increased to 20 kN, keeping all other conditions the same, then its life in hours is
(a) 4000 (b) 2000
(c) 1000 (d) 500

BHEL ET 2019
Vadodara Muncipal Corp. DEE, 2018
GPSC ARTO 01.05.2016
ISRO Scientist/Engineer 11.10.2015
GPSC AE 26.07.2015
Kerala LBS Centre for Sci. & Tech. Asstt. Prof. 2014
OPSC Civil Services Pre. 2011
GATE 2007

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Ans. (c) : Given, $p_1 = 10$ kN, $L_1 = 8000$ hrs

$p_2 = 20$ kN

$$L_1 = \left(\frac{c}{p_1}\right)^n$$

Here, c = Dynamic load capacity

p_1 = Equivalent dynamic load

$\frac{c}{p_1}$ = Loading rate

Note : Point remember

n = 3 for ball bearing

$n = \frac{10}{3}$ for roller bearing

$$\therefore \quad L_1 = \left(\frac{c}{p_1}\right)^3$$

$$\therefore \quad \frac{L_1}{L_2} = \left(\frac{p_2}{p_1}\right)^3$$

$L_2 = 1000$ hrs.

452. The frictional torque transmitted by a cone clutch is same as that of :

(a) Flat pivot bearing
(b) Conical pivot bearing
(c) Flat collar bearing
(d) Trapezoidal pivot bearing

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017
Rajasthan Nagar Nigam AE 2016, Shift-I
TNPSC ACF 2012

Ans. (d) :

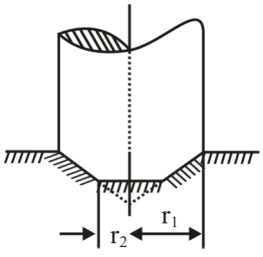
	Friction Torque	
Trapezoidal or Truncated conical pivot Bearing	Uniform pressure	Uniform wear
	$\frac{2}{3}\mu.W\left(\frac{R_2^3 - R_1^3}{R_2^2 - R_1^2}\right)\text{cosec}\,\alpha$	$\frac{1}{2}\mu W \quad (R_1 + R_2)\ \text{cosec}\alpha$
Cone clutch	$\frac{2}{3}\mu W\left(\frac{R_2^3 - R_1^3}{R_2^2 - R_1^2}\right)\text{cosec}\,\alpha$	$\frac{1}{2}\mu W \quad (R_1+R_2)\ \text{cosec}\alpha$

Trapezoidal pivot bearing – Area of the bearing surface,

$A = \pi [(r_1)^2 - (r_2)^2]$

∴ Intensity of uniform pressure,

$$P_n = \frac{W}{A} = \frac{W}{\pi\left[(r_1)^2 - (r_2)^2\right]}$$



453. Starting friction is low in :

(a) hydrostatic lubrication
(b) hydrodynamic lubrication
(c) mixed (or semi fluid) lubrication
(d) boundary lubrication

ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC ARTO Pre 30.12.2018
TSPSC Managers, 2015
GPSC Executive Engineer 23.12.2018
GATE 1992

Ans. (a) : Starting friction is low in hydrostatic lubrication.
In hydrostatic lubrication, the pump is used, this will create a thin film between journal and bearing therefore starting friction is low.

454. The bearing that is used when heavy loads are to be carried with an oscillatory motion (e.g. piston pin bearings in heavy diesel engines), is:

(a) Tapered roller bearing
(b) Cylindrical roller bearing
(c) Needle roller bearing
(d) Spherical roller bearing

RPSC IOF, 2020
NLC GET 17.11.2020, Shift-II
OPSC AEE 2019 Paper-I
GPSC EE Pre, 28.01.2017
ESE 2012g

Ans. (c) : Needle roller bearing– It is used when heavy loads are to be carried with an oscillatory motion.
Example– Piston pin bearing in heavy duty diesel engine.
Tapered roller bearing– It is rolling element bearing that can support axial forces as well as radial forces.
Cylindrical roller bearing– Cylindrical roller bearing are also available in sealed or split designs. In sealed bearing the rollers are protected from contaminants water.
Spherical roller bearing- A spherical roller bearing is a rolling element bearing that permits rotation with low friction and permit angular misalignment.

455. Spherical roller bearings are normally used :

(a) For increased radial load & to compensate for angular misalignment
(b) For increased thrust load
(c) When there is less radial space
(d) None of these

OPSC AEE 2019 Paper-I
GPSC ARTO Pre 30.12.2018
Vadodara Muncipal Corp. DEE, 2018
GATE 1992

Ans. (a) : Spherical roller bearings are normally used for increased radial load & to compensate for angular misalignment.

456. The frictional torque transmitted in a flat pivot bearing considering uniform pressure is :

(a) $\frac{1}{2}\mu WR$ (b) $\frac{2}{3}\mu WR$

(c) $\frac{3}{4}\mu WR$ (d) μWR

CGPSC AE 15.01.2021
JPSC AE 10.04.2021, Paper-II
Kerala PSC IOF 19.04.2016
TNPSC AE 2014

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Ans. (b) : The frictional Torque transmitted in Flat pivot bearing.

(1) Uniform Pressure Theory

$$T = \frac{2}{3}\mu WR$$

(2) Uniform Wear Theory

$$T = \frac{1}{2}\mu WR$$

457. A ball-bearing is characterized by basic static capacity = 11,000 N and dynamic capacity = 18,000 N. This bearing is subjected to equivalent static load = 5500 N. The bearing loading ratio and life in million revolutions respectively are :

(a) 3.27 and 52.0 (b) 3.27 and 35.0

(c) 2.00 and 10.1 (d) 1.60 and 4.1

OPSC AEE 2019 Paper-I
UPRVUNL AE 07.10.2016
UPRVUNL AE 2014
ESE 2001

Ans : (b) :

C = 18000 N,

n = 3 for ball bearings

P = 5500 N

$$L_{10} = \left(\frac{C}{P}\right)^n$$

L_{10} = rated bearing life in million revolutions

$$\therefore L_{10} = \left(\frac{18000}{5500}\right)^3$$

$= (3.27)^3 = 35$ million revolutions

$$\text{Loading ratio} = \frac{C}{P} = \frac{18000}{5500} = 3.27$$

458. Which one of the following is not a friction clutch?

(a) Disc or plate clutch

(b) Cone clutch

(c) Centrifugal clutch

(d) Jaw clutch

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Kerala PSC Poly. Lect. 09.01.2014
APPSC Poly. Lect. 2013
ISRO Scientist/Engineer 2006
ESE 2002

Ans. (d) : Friction clutches are gradually engaging clutches. Driving shaft may be rotating at full speed while the driven shaft either stationary or rotating at much lower speed is brought into connection with the former friction clutches depends on the frictional force between two bodies which are pressed together.

■ Friction clutches obtained following clutches.

(1) Disc or Plate Clutch–

■ **Single Plate Friction Clutches–**Generally these clutches are bigger in size. The large size is required to dissipate heat generated. So these are used in trucks.

■ **Multi-Plate Friction Clutches–**These clutches are small in size and cooling oil is used for heat removal. So they are used in scooter, bike etc.

(2) Centrifugal Clutches–In centrifugal clutches applications there is no need of gear box and it used in mopeds, textile machinery etc.

(3) Cone Clutch–Cone clutches are friction clutches. They are simple in construction and are easy to disengage. However, the driving and driven shaft must be perfectly co-axial for efficient functioning of the clutch. A cone clutches consists of two working surface i.e. inner and outer cone.

(4) Jaw Clutches–It is positive drive clutch i.e. no slipping and used in low speed machinery e.g. rolling milling.

459. When a shaft rotates in anticlockwise direction at slow speed in a bearing, then it will

(a) move towards upward of the bearing making metal to metal contact

(b) move towards right of the bearing making metal to metal contact

(c) move towards left of the bearing making metal to metal contact

(d) move towards downward of the bearing making metal to metal contact

CGPSC AE 15.01.2021
ESE 2014
TNPSC AE 2013

Ans. (c) : When a shaft rotates in anticlockwise direction at slow speed in a bearing, then it will move towards left of the bearing making metal to metal contact.

460. The frictional torque transmitted by a disc or plate clutch is same as that of

(a) flat pivot bearing

(b) flat collar bearing

(c) conical pivot bearing

(d) trapezoidal pivot bearing

TNPSC AE 2019
OPSC AEE 2015 Paper-I
TNPSC AE 2013

Ans. (b) : The frictional torque transmitted by a disc or plate clutch is same as that of flat collar bearing.

$$T_f = \frac{2}{3}\mu W\left[\frac{R_0^3 - R_0^3}{R_0^2 - R_1^2}\right] \quad [\text{UPT}]$$

$$T_f = \frac{1}{2}\mu W(R_0 + R_i) \quad (\text{UWT})$$

461. The criterion in the design of hydrodynamic journal bearing is :

(a) Revolving factor

(b) Sommerfeld number

(c) Bosch number

(d) Prandtl number

GPSC ARTO Pre 30.12.2018
TSPSC AEE 2017
HPPSC Lect. (Auto) 23.04.2016
VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012
GATE 2005

Ans. (b) : Sommerfeld number is the criteria for design of hydrodynamic journal bearings.

$$\text{Sommerfeld number} = S = \frac{Zn}{P}\left(\frac{D}{C}\right)^2$$

Z = Absolute viscosity of the lubricant

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n = Speed of rotating shaft in rev/sec.
P = Load per unit projected area
D = Journal diameter
C = Diameteral clearance.
Sommerfeld number is very important in lubricant analytic because it contains all the variables normally specified by the designs.
It remains same for a bearing.

462. Considering the safe design, the friction clutch should be designed
(a) Assuming uniform pressure
(b) Assuming uniform wear
(c) Assuming any criteria, either uniform pressure or uniform wear
(d) Assuming uniform pressure for high torque and uniform wear for low torque

RPSC IOF, 2020
TRB Polytechnic Lecturer 2017
UKPSC AE 2012 Paper-I

Ans. (b) : A friction clutch is very useful in the transmission of the power of shafts and machines which must be started and stopped frequently.
Assuming uniform wear for friction clutch for safe design. In this, wear is uniformly distributed over the entire surface the friction disk. This theory is used by worm-out clutches.

463. If n total number of plates on driving and driven shaft in the multi plate clutch, then number of pairs of contact surfaces will be.....
(a) n + 1 (b) n + 2
(c) n – 1 (d) n – 2

GPSC Lect. 23.10.2016
TNPSC AE, 2008
WBPSC AE, 2007

Ans. (c) : n – 1

464. A clutch has outer and inner diameter of 100 mm and 40 mm respectively. Assuming a uniform pressure of 2 MPa and coefficient of friction of linear material 0.4; the torque carrying capacity of the clutch is
(a) 196 Nm (b) 372 Nm
(c) 148 Nm (d) 275 Nm
(e) 490 Nm

TSPSC AEE 2017
CGPSC AE 26.04.2015 Shift-I
VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2008

Ans. (a) : Using uniform pressure theory torque carrying capacity of clutch is given by

$$T = \frac{2}{3}\mu W\left(\frac{R^3 - r^3}{R^2 - r^2}\right)$$

$$= \frac{2}{3}\pi\mu(R^3 - r^3)\left[\frac{W}{\pi(R^2 - r^2)}\right]$$

$$= \frac{2}{3}\pi\mu P(R^3 - r^3)$$

$$= \frac{2}{3}\times 3.14\times 0.4\times 2\left[\left(\frac{100}{2}\right)^3 - \left(\frac{40}{2}\right)^3\right]$$

= 195936 Nmm = 195.93 N-m
≃ 196 N-m

465. Rated life of a ball bearing in relation to load (P) varies as
(a) 1/P (b) P^2
(c) $1/P^2$ (d) P^3
(e) $1/P^3$

TNPSC AE 2018
CGPSC AE 26.04.2015 Shift-I
TRB Poly. Lect., 2012

Ans. (e) : The equation for the basic rating life for dynamically loaded ball bearing is as follows

$$L_{10} = \left(\frac{Cr}{P}\right)^3 \times 10^6$$

L_{10} = Basic Rating life
C_r = Basic Dynamic Load Rating
P = Equivalent load

$$\boxed{L_{10} \propto \frac{1}{P^3}}$$

466. Deep groove ball bearings are used for
(a) heavy thrust load
(b) small angular displacement of shafts
(c) radial load at high speed
(d) combined thrust and radial load at high speed

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 18.09.2017
RPSC LECTURER 16.01.2016
GPSC AE 26.07.2015
TRB Asstt. Prof., 2012

Ans. (d) : These bearing can take very good radial as well as axial road. The radial load is continuous and steady, but axial load can very due to impact etc. steady axial load of very high magnitude is there, these bearings are not suitable.

467. In a ball bearing, a ball is subjected to
(a) compressive stress
(b) tensile stress
(c) shear stress
(d) cyclic stress or fatigue

WBPSC AE, 2017
RPSC AE 2016
RPSC ACF-2011

Ans : (d) In a ball bearing, a ball is subjected to cycle stress or fatigue loading.
The rolling contact bearings are of the following two types:
(a) Ball bearing (b) roller bearing
The ball and roller bearings consist of an inner race which is mounted on the shaft or journal and an outer race which is carried by the housing or casing. In between the inner and outer race, there are balls or rollers. A number of balls or rollers are used and these are held at proper distance by which are assembled after the balls have been properly spaced. The ball bearings are used for light loads and the roller bearings are used for heavier loads.

468. In an oil lubricated journal bearing, the coefficient of friction
(a) Is constant at all speeds
(b) Is maximum at almost zero speed and decreases linearly with increasing speed
(c) Is minimum at optimum speed and increases with increasing speed
(d) Is minimum at almost zero speed and increases with increase of speed

GPSC Asstt. Prof. 28.08.2016
APPSC IOF, 2009
ESE 1995

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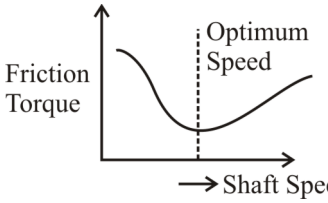
Ans. (c) :



From graph, the coefficient of friction is minimum at optimum speed and increases with increasing speed.

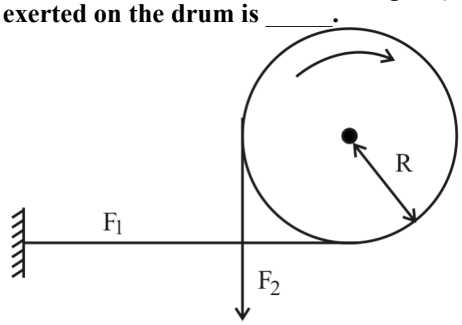
469. The radius of friction circle for a radius r rotating inside a bearing is

(a) r sin ϕ (b) r cos ϕ
(c) r tan ϕ (d) r cot ϕ

APPSC Poly. Lect. 2013
WBPSC AE, 2007
J & K PSC Screening, 2006

Ans. (a) : The radius of a friction circle for a shaft rotating inside a bearing is (where, r = radius of shaft and tan ϕ = coefficient of friction b/w the shaft and bearing)
Radius of friction circle = r sin ϕ

470. The forces F_1 and F_2 in a brake band and the direction of rotation of the drum are as shown in the figure. The coefficient of friction is 0.25. The angle of wrap is 3π/2 radii. It is given that R =1 m and F_2 = 1 N. The torque (in N-m) exerted on the drum is _____.



(a) 2.248 (b) 4.248
(c) 0.693 (d) 1.308

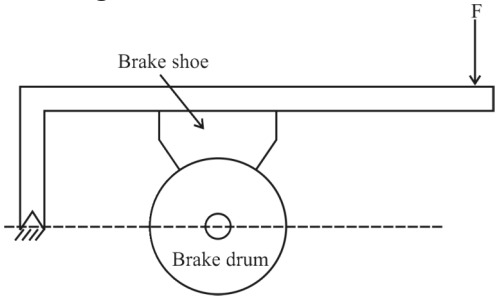
RPSC ADE 2016
GATE 2016

Ans. (a) : As rotation is clock wise and pulley is driver
So, $F_1 > F_2$

$$\frac{F_1}{F_2} = \frac{F_1}{1} = e^{\left(0.25\times\frac{3\pi}{2}\right)} \qquad (\because F_2 = 1N, R = 1m)$$

$F_1 = 3.2481$
Torque = $(F_1 - F_2) \times R$ = 2.248 N-m

471. For the brake shown in figure, which of the following is true ?



(a) Self energizing for clockwise rotation of the drum
(b) Self energizing for anticlockwise rotation of the drum
(c) Self energizing for rotation in either direction of the drum
(d) Not of the self energizing type

RPSC ADE 2016
GATE 2016

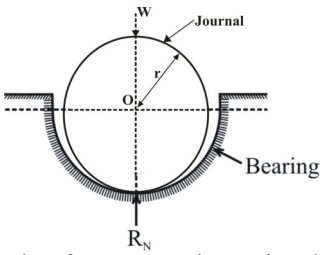
Ans. (a) : For self energizing the direction of friction should be such that it help in application of force i.e. moment of friction on the block should be clockwise which is possible if drum rotate clockwise and friction force towards right.

472. For supporting the pressure perpendicular to the axis of the shaft, the bearing preferred is:

(a) Journal bearing (b) Pivot beating
(c) Thrust beating (d) Footstep bearing

(HPPSC LECT. 2016)
HPPSC Asstt. Prof. 29.10.2016

Ans : (a) For supporting the Pressure perpendicular to the axis of the shaft the bearing preferred is Journal bearing.



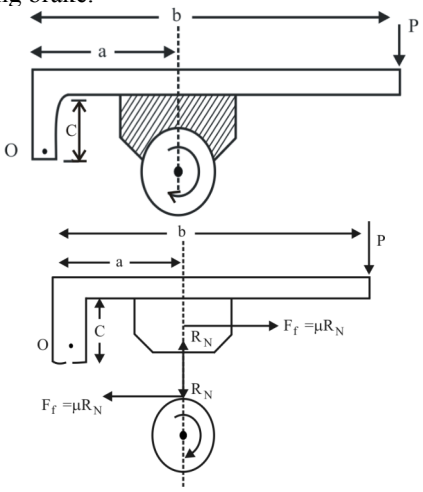
A Journal bearing forms a turning Pair. The fixed outer element of a turning pair is called a bearing and the portion of the inner element which fits in the bearing is a called a journal. The journal is slightly less in diameter than the bearing, in order to permit the free movement of the journal in a bearing.

473. When the friction force helps the applied force in applying the brake is called as _____.

(a) self-locking (b) automatic
(c) self-energizing (d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014
WBPSC AE, 2007

Ans. (c) : When the friction force helps the applied force in applying the brake, the brake is called as self energizing brake.



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$$P \times b + \mu R_N \times C - R_N \times a = 0$$

$$P = \frac{R_N}{b}(0 - \mu C)$$

(i) When a > μC; (P = + ve) in this case the friction force μN help to reduce the magnitude of braking force or support to applied force is known as self energised breaking system.

(ii) When a = μC; P = 0; which implies that force needed to apply the brake is zero i.e. self locking brake.

474. Rope brake dynamometer uses

(a) oil as lubricant (b) water as lubricant
(c) grease as lubricant (d) no lubricant

MPPSC AE 08.11.2015
RPSC ACF-2011

Ans. (d) : There is no lubricant used in the Rope brake dynamometer

475. The most suitable bearing for carrying very heavy loads with slow speed is :

(a) Hydrodynamic bearing
(b) Ball bearing
(c) Roller bearing
(d) Hydrostatic bearing

BPSC Asstt. Prof. 29.11.2015
MPPSC State Forest Service Exam, 2014

Ans. (d) : Hydrostatic bearings can support large loads without journal rotation and provide large (accurate & controllable) direct stiffness as well as damping (energy dissipation) Co-efficients.

476. Frictional torque transmitted in a conical pivot bearing considering uniform wear is

(a) $\frac{1}{2}\mu WR \operatorname{cosec}\alpha$ (b) $\frac{2}{3}\mu WR \operatorname{cosec}\alpha$

(c) $\frac{3}{4}\mu WR \operatorname{cosec}\alpha$ (d) $\mu WR \operatorname{cosec}\alpha$

HPPSC Asstt. Prof. 2014
TNPSC AE, 2008

Ans. (a) : According to theory of uniform wear for a conical pivot bearing –

Frictional torque (T)= $\frac{1}{2}\frac{\mu W}{\sin\alpha}R$

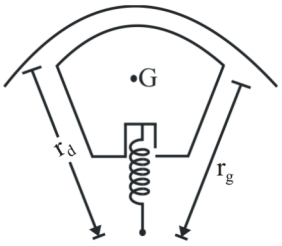
Frictional torque (T) = $\frac{1}{2}\mu W\ R \operatorname{cosec}\alpha$

477. In a centrifugal clutch, the force with which the shoe presses the driven member is the

(a) Sum of centrifugal and spring force
(b) Difference of centrifugal and spring force
(c) Centrifugal force
(d) Spring force

Nagaland PSC CTSE 2017, Paper-I
GPSC Lect. (Poly.) 07.12.2014

Ans. (b) :



Normal force on drum by shoes

$N = F_{c_2} - F_{c_1}$

$= mr_g\omega_R^2 - k.x$

At minimum speed

$F_{c_1} = b_s = F_{c_1} = mrg\omega_l^2$

$N = mr_g\omega_R^2 - mr_g\omega_l^2$

$N = mr_g(\omega_R^2 - \omega_l^2)$

frictional torque

$T_f = F_f \times r_d$

$T_f = \mu mr_g(\omega_R^2 - \omega_l^2)$

n = no. of shoes

total torque

$(T_f) = n.\mu r_g r_d(\omega_R^2 - \omega_l^2)$

ω_R – rated speed

478. In thick film hydrodynamic journal bearings, the coefficient of friction :

(a) Decreases with increases in load
(b) Increases with increase in load
(c) Is independent of load
(d) None of the above

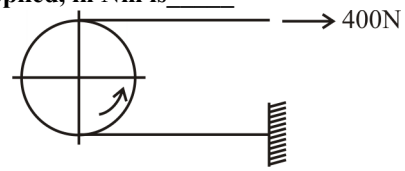
GPSC Executive Engg. 23.12.2018
Kerala PSC Asstt. Prof. 27.09.2016

Ans. (a) : Hydrodynamic journal bearing is a bearing operating with hydrodynamic lubrication, in which the bearing surface is separated from the journal surface by the lubricant film generated by the journal location. Most of engine bearings are hydrodynamic journal bearing. Used in–turbine, compressor, generators gear boxes & pumps.

$$f \propto \frac{\mu N}{P}$$

Thus, in thick film hydrodynamic journal bearings the coefficient of friction decreases with increases in load.

479. A force of 400N is applied to the brake drum of 0.5 m diameter in a band brake system as shown in the figure, where the wrapping angle is 180°. If the coefficient of friction between the drum and the band is 0.25, the braking torque applied, in Nm is_____



(a) 100.6 (b) 15.7
(c) 22.1 (d) 54.4

Assam Engg. College AP/Lect. 18.01.2021
GATE 2012

Ans. (d) : 54.4

Given

$T_1 = 400$ N

$\theta = 180^\circ = \pi$

Coefficient of friction (μ) = 0.25

$\frac{T_1}{T_2} = e^{\mu\theta}$

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$\frac{400}{T_2} = e^{0.25\times\pi}$

$T_2 = 182.37N$

Braking Torque $(M_t) = (T_1 - T_2) \times R$

$= (400 - 182.37) \times \frac{0.5}{2}$

$M_t = 54.4$ Nm

480. Given a self-aligning ball bearing with basic dynamic load rating (C_{10}' for 10^6 revolutions) of 35 kN. Find the expected life (in 10^6 revolutions), if the equivalent radial load on the bearing is 45 kN.

(a) 0.8 to 1.0 (b) 0.5 to 0.8
(c) below 0.5 (d) above 1.0

Assam Engg. College AP/Lect. 18.01.2021
GATE 2018

Ans. (c) : C = 35 kN
P = 45 kN

$L = \left(\frac{C}{P}\right)^3$ for ball bearing

$L = \left(\frac{35}{45}\right)^3$

L = 0.4705 Millions of revolution

Means life is below 0.5×10^6

481. In the multiple disc clutch, if there are 6 discs on the driving shaft and 5 discs on the driven on the driven shaft, then the number of pairs of contact surfaces will be equal to:

(a) 11 (b) 12
(c) 22 (d) 10

HPPSC Asstt. Prof. 18.09.2017
ESE 1997

Ans. (d) : $n = n_1 + n_2 - 1$
$= 5 + 6 - 1$
$= 10$

482. When the intensity of pressure between the block and brake drum is uniform, the angle of contact between the block and brake drum is less than

(a) 45^0 (b) 90^0
(c) 60^0 (d) 80^0

APGCL AM, 2021
TNPSC AE 2017

Ans. (a) : For uniform pressure between block and brake drum, the angle of contact between the block and brake drum. Should be less than 45^0

If $2\theta \le 45^0$, The pressure distribution between block and Drum is uniform.

If $2\theta > 45^0$, The pressure distribution between block and Drum is parabolic.

483. Which of the bearings given below SHOULD NOT be subjected to a thrust load ?

(a) Deep groove ball bearing
(b) Angular contact ball bearing
(c) Cylindrical (straight) roller bearing
(d) Single row tapered roller bearing

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
APGCL AM, 2021

Ans. (c) : Cylindrical or straight roller bearings are subjected to radial load.

Name of bearing	Advantage	Dis Advantage
1. Deep groove ball bearing	• Load carrying capacity is High • It can take load in axial as well as radial direction	• Self alignment is not possible • poor rigidity due to point contact
2. Angular contact ball bearing	• Load carrying capacity is greater than deep groove ball bearing • Both radical and axial load.	• Two bearing are required to take thrust in both direction • Assembled with specific magnitude of pre load.
3. Cylindrical (straight) roller bearing	• Radial load carrying capacity in high frictional loss is less	• Can not take thrust load. • Self alignment is not possible.
4. Single row tapered roller bearing	• Easily assembled'. • Take heavy radial and thrust load.	• Pre load is required. can not tolerate mis-alignment.

484. The effect of increasing the stiffness springs of centrifugal clutch is

(a) The decrease of engagement speed
(b) The increase of engagement speed
(c) The increase of frictional force at maximum speed
(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017
ESE 2012

Ans. (b) : By increasing or decreasing the stiffness of spring of centrifugal clutch there is increase or decrease of engagement speed respectively, due to increase or decrease in restoring force.

485. The expected life of a ball bearing subjected to a load of 9800 N and operating at 1000 RPM is 3000 hrs. What is the expected life of the same bearing for similar load of 4900 N and speed of 2000 RPM

(a) 12000 hrs. (b) Remain same
(c) 6000 hrs. (d) 1500 hrs.

ISRO Scientist/Engineer (RAC) 22.04.2018
GPSC Executive Engineer 23.12.2018
GATE 1987

Ans. (a) : We know that

$$L_{90} = \left(\frac{K}{C}\right)^n$$

$h = \frac{10}{3}$, for roller bearing

$h = 3$, for ball bearing

$$L_{90} = \left(\frac{K}{C}\right)^3 \times 10^6 \; rev$$

$$L = 60NL_h \; rev$$

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where,

L = life in million revolutions

L_h = life in hours

N = RPM

K = dynamic load capacity

C = equivalent dynamic load

So,

$$60 \times N \times L_h = \left(\frac{K}{C}\right)^3 \times 10^6$$

$$60 \times 1000 \times 3000 = \left(\frac{K}{9800}\right)^3 \times 10^6 \quad ...(1)$$

and

$$60 \times 2000 \times L_h = \left(\frac{K}{4900}\right)^3 \times 10^6 \quad ...(2)$$

From equation $(2) \div (1)$

$$\frac{2000 \times L_h}{1000 \times 3000} = \left(\frac{9800}{4900}\right)^3 = 8$$

$$\boxed{L_h = 12000\, hrs}$$

486. Petroff's equation is used to find out :

(a) load carrying capacity of the bearing
(b) frictional losses in the bearing
(c) unit bearing pressure on the bearing
(d) pressure distribution around the periphery of the journal

RPSC Vice Principal ITI 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : Petroff's equation is used to find frictional losses in the bearing.

Petroff's equation.

$$\boxed{\mu = 2\pi^2\left(\frac{Zn}{P}\right)\left(\frac{d}{c}\right)}$$

Where, μ = Coefficient of friction

Z = Absolute viscosity of lubricant in kg/m-s

n = Speed on the journal in rpm

P = Bearing pressure on the projected bearing area in N/mm^2

d = Diameter of the journal

c = Diametral clearance

487. The co-efficient of friction for the clutch facing is approximately

(a) 0.1 (b) 0.4
(c) 0.8 (d) 1.2

TNPSC AE 2017
Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (b) : The co-efficient of friction for the clutch facing is approximately 0.4.

488. Consider the following statements :
Radius of friction circle for a journal bearing depends upon:
(i) Coefficient of friction
(ii) Radius of the journal
(iii) Angular speed of rotation of the shaft
Which of the statements given above are correct?

(a) (i), (ii) and (iii) (b) Only (i) and (ii)
(c) Only (ii) and (iii) (d) Only (i) and (iii)

OPSC AEE 2019 Paper-I
ESE 2006

Ans : (b) :
Radius of friction circle = $\mu \times r$
Where, r = radius of journal
μ= Coefficient of friction

489. A hydrodynamic slider bearing develops load bearing capacity mainly because of :

(a) Slider velocity
(b) Wedge shaped oil film
(c) Oil compressibility
(d) Oil viscosity

OPSC AEE 2019 Paper-I
ESE 2008

Ans : (b) : Due to converging and diverging portions of journal bearing, the thick film lubrication will be generated.

490. In case of single plate new clutches and brakes, the friction radius is equal to ____, where D and d are outer and inner diameters of friction lining respectively.

(a) $\frac{1}{3}\left(\frac{D^3 - d^3}{D^2 - d^2}\right)$ (b) $\frac{1}{2}\left(\frac{D^3 - d^3}{D^2 - d^2}\right)$

(c) $\frac{1}{4}\left(\frac{D^3 - d^3}{D^2 - d^2}\right)$ (d) $\frac{1}{4}(D + d)$

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
HPPSC AE 2018

Ans. (a) : $T = \frac{2}{3}\mu W\left[\frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}\right]$

$T = \mu W R_m$

$$R_m = \frac{2}{3}\left[\frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}\right]$$

$$R_m = \frac{2}{3}\left[\frac{(D/2)^3 - (d/2)^3}{(D/2)^2 - (d/2)^2}\right]$$

$$R_m = \frac{1}{3}\left[\frac{D_o^3 - D_i^3}{D_o^2 - D_i^2}\right]$$

491. A sliding bearing which can support steady loads without any relative motion between the journal and bearing is known as

(a) zero film bearing
(b) boundary lubricated bearing
(c) hydrostatic lubricated bearing
(d) hydrodynamic lubricated bearing

TSPSC Manager (Engg.) HMWSSB 12.11.2020
GPSC Asstt. Prof. 28.08.2016

Ans. (c) : Hydrostatic lubricated bearing- Hydrostatic lubrication is defined as a system of lubrication in which the load-supporting fluid film, separating the two surfaces are created by an external source, like a pump, supplying sufficient fluid under pressure.

492. A ball-bearing is characterized by basic static capacity = 11000 N and dynamic capacity = 18000 N. This bearing is subjected to equivalent static load = 6000 N. The bearing loading ratio and life inmillion revolutions respectively are:

(a) 3 and 27 (b) 1.60 and 4.1
(c) 3 and 35 (d) 2 and 27

UPRVUNL AE 21.08.2016
ESE 2001

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Ans. : (a) Loading ratio = $\frac{C}{P}=\frac{18000}{6000}=3$

Life (million revolutions)

$$\left(\frac{C}{P}\right)^3=\left(\frac{18000}{6000}\right)^3=27$$

493. A journal bearing has a shaft diameter of 40 mm and a length of 40 mm. The shaft is rotating at 20 rad/s and the viscosity of the lubricant is 20 mPa.S. The clearance is 0.020 mm. The loss of torque due to the viscosity of the lubricant is approximately

(a) 0.400 Nm (b) 0.252 Nm
(c) 0.040 Nm (d) 0.652 Nm
(e) 0.425 Nm

CGPSC AE 26.04.2015 Shift-I
GATE 2008

Ans. (c) : Given,

$d = 40 \text{ mm} = 0.04 \text{ m}$
$l = 40 \text{ mm} = 0.04 \text{ m}$
$\omega = 20 \text{ rad/s}$
$\mu = 20 \text{ m Pa-s} = 20 \times 10^{-3} \text{ Pa.s}$
$C = 0.020 \text{ mm} = 0.02 \times 10^{-3} \text{ m}$
$V = r\,\omega = 0.02 \times 20 = 0.4$

We know that,

$$\tau = \frac{\mu V}{C} = \frac{20\times10^{-3}\times0.4}{0.02\times10^{-3}}$$

$\tau = 400 \text{ N/m}^2$

Force (F) = $\tau A = \tau . \pi\, d l$
$= 400\times3.14\times0.04\times0.04$
$= 2.0096 \text{ N}$

Torque = F × r
$= 2.0096 \times 0.02$
$= 0.0401$

494. Which of the following clutches is of positive type

(a) Hydraulic (b) Cone
(c) Disc (d) Centrifugal
(e) Jaw

Nagaland PSC CTSE 2017, Paper-I
CGPSC AE 26.04.2015 Shift-I

Ans. (e) : The positive clutches are used when positive drive is required. The simplest type of a positive clutch is a jaw or claw clutch. The jaw clutch permits one shaft to drive another through a direct contact of interlocking jaws.

495. The load carrying capacity of a hydrodynamically lubricated journal bearing depends on

(a) Speed only
(b) Viscosity only
(c) Radial clearance only
(d) Speed, Radial clearance and Viscosity

Gujarat PSC AE 2019
Punjab PSC (Lect.) 06.08.2017

Ans : (d) : The load carrying capacity of a hydrodynamically lubricated journal bearing depends on speed, radial clearance and viscosity.

496. Tapered roller bearing can take

(a) Radial load only
(b) axial load only
(c) both radial and axial loads and the ratio of these being less than unity
(d) both radial and axial loads and the ratio of these being greater then unity

APPSC AEE 2012
ESE 1996

Ans. (b) : Tapered roller bearing are bearing that can take both axial loads (Sideways or thrust forces) as well as radial loads (downwards force). the ratio of fr/ra<1

497. Ball bearing are, usually, made from

(a) Low carbon steel
(b) High carbon steel
(c) Medium carbon steel
(d) Chrome steel

GPSC Engineer, Class-II Pre-19.01.2020
VIZAG Steel MT 2011
Gujarat PSC AE 2019

Ans : (d) : Ball bearings are usually made from chrome steel.

498. Match List-I with List-II and select the correct answer using the codes given below the lists :

	List -I	List-II
(A)	Ball bearing	(1) Carrying both radial and thrust loads
(B)	Tapered roller Bearing	(2) Self aligning property
(C)	Spherical roller Bearing	(3) Light Loads
(D)	Needle roller Bearing	(4) Heavy loads with oscillatory motion

Codes :

	A	B	C	D
(a)	1	2	3	4
(b)	3	1	2	4
(c)	2	4	1	3
(d)	4	3	2	1

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 1997

Ans. (b) : • Pairs of tapered roller bearing are used in car & vehicle, they cope up with both radial and axial forces.
• **Ball Bearing** is a type of antifriction bearing, and used for light axial loads.
• **Spherical roller** bearing is a bearing having rolling element as roller allows rotation with low friction, they have self alignment property.
• **Needle roller** bearing, with a cage to separate and guide the rollers. It is used in heavy duty engine. Needle roller bearing can withstand high speed & misalignment.

499. For a block brake, the equivalent coefficient of friction is (Where 2θ is the angle of contact and μ is coefficient of friction.)

(a) $\frac{2\sin2\theta}{2\theta+\sin2\theta}\mu$ (b) $\frac{4\sin\theta}{2\theta+\sin2\theta}\mu$
(c) $\frac{4\sin2\theta}{2\theta+\sin2\theta}\mu$ (d) $\frac{2\sin\theta}{2\theta+\sin2\theta}\mu$

HAL Design Trainee 13.08.2021 Shift- II
HPPSC AE 2018

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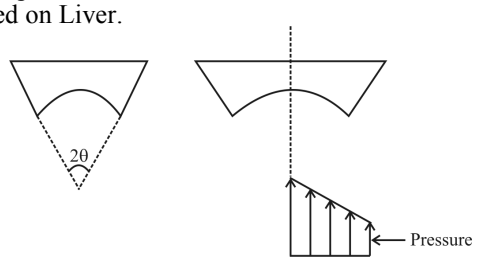
Ans. (b) : When break shoe is long ($2\theta > 45^o$) pressure is non-uniformly distributed and wear will be more where pressure is more hence to avoid this shoe is pivoted on Liver.



For less wear

$$\mu_{eqv.} = \frac{4\mu \sin\theta}{2\theta + \sin 2\theta}$$

500. To restore stable operating condition in a hydrodynamic journal bearing, when it encounters higher magnitude loads.
(a) Oil viscosity is to be decreased
(b) Oil viscosity is to be increased
(c) Oil viscosity index is to be increased
(d) Oil viscosity index is to be decreased

Kerala PSC Poly. Lect., 2017
GATE 1997

Ans. (b) : To restore stable operating condition in a hydrodynamic journal bearing, when it encounters higher magnitude loads oil viscosity is to be increased.

501. The performance of a hydrodynamic journal bearing is a function of
(a) Bearing modulus
(b) Speed
(c) Sommerfeld number
(d) Reynolds number

UPRVUNL AE 21.08.2016
CSE Pre 2005

Ans. : (a) The performance of a hydrodynamic journal bearing is the function of Bearing modulus

502. In designing a plate clutch, assumption of uniform wear conditions is made because :
(a) It is closer to real life situation
(b) It leads to safer design
(c) It leads to cost effective design
(d) No other assumption is possible

Mizoram PSC AE/SDO 2014, Paper-II
CGPSC Poly. Lect. 22.05.2016
UPRVUNL AE 2014

Ans. : (a) In designing a plate clutch assumption of uniform wear conditions is made because. It is closer to real life situation.

$$W = \frac{1}{2}\mu\,\omega\,R$$

503. Anti-friction bearings are :
(a) sleeve bearings
(b) gas lubricated bearings
(c) ball and roller bearings
(d) journal bearings

TSPSC Managers, 2015
TNPSC AE 2018
OPSC Civil Services Pre 2006
GPSC ARTO Pre 30.12.2018
DRDO Scientists 2008

Ans. (c) : Ball and Roller bearings are Anti-friction bearing.

504. A bearing material should not possess the characteristic of high _____.
(a) coefficient of friction
(b) hardness
(c) melting point
(d) thermal conductivity

Karnataka PSC AE (WRD) 31.07.2021
APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : A bearing material should possess the characteristics of high hardness, high melting point and high thermal conductivity but low coefficient of friction.

505. An axle is subjected to
(a) Bending only (b) Direct load only
(c) Torsion only (d) Bending and torsion

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
ESE 1995

Ans. (a) : Axle is designed and manufactured in such a way that it can only take bending load. It is not used for torque transmission.

506. If the stiffness of spring of centrifugal clutch is increased, it causes:
(a) increase the friction torque at maximum speed
(b) no change in engagement speed
(c) increase of engagement speed
(d) decrease of engagement speed

UPRVUNL AE 07.10.2016
ESE 2012

Ans. (c) : If the stiffness of spring of centrifugal clutch is increased, it causes increase of engagement speed.

507. Which of the following are the functions of lubrication in a bearing unit?
1. To protect the bearing components from corrosion.
2. To absorb heat from the bearing unit.
3. To carry heat away from the bearing unit.
Select the correct answer using the code given below:
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) : The functions of lubrication in a bearing unit-
- To protect the bearing components from corrosion.
- To absorb heat from the bearing unit.
- To carry heat away from the bearing unit.

508. Consider the following statements regarding clutches:
1. Dry clutch has higher coefficient of friction compared to wet clutch.
2. The torque capacity of wet clutch is high compared to dry clutch.
3. The engagement in a dry clutch is smoother than in case of wet clutch.
Which of the above statements is/are correct?
(a) 1 only (b) 1 and 3 only
(c) 2 only (d) 1, 2 and 3

ESE 18.07.2021

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Ans. (a) :

Dry Clutch	Wet Clutch
1. As lubricant is not present, coefficient of friction is relatively high.	1. Presence of lubricant decreases friction.
2. Because of higher friction, torque carrying capacity is higher.	2. As friction is less, torque carrying capacity is lower.
3. High friction leads to sudden engagement of clutch	3. Lesser friction result in smooth engagement of clutch.

509. Which of the bearings given below should not be subjected to a thrust load?
(a) Deep groove ball bearing.
(b) Angular contact ball bearing.
(c) Cylindrical (straight) roller bearing.
(d) Single row tapered roller bearing.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : Cylindrical (straight) roller bearing

510. Which is true in respect of clutch plate and engine flywheel in an automobile ?
(a) the clutch plate gets connected to the flywheel when the driver presses the clutch pedal
(b) the clutch plate gets disconnected when the clutch pedal is pressed
(c) clutch plate is always connected to the flywheel
(d) the clutch plate always remains disconnected from the flywheel

HPPSC Poly. Lect. 05.07.2021

Ans. (b) : the clutch plate gets disconnected when the clutch pedal is pressed

511. The brake bleeding system serves to free the system from
(a) vacuum (b) excess pressure
(c) excess fluid (d) air

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : air

512. The coefficient of friction for the clutch facings is approximately
(a) 0.4 (b) 0.1
(c) 1.0 (d) 0.75

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : 0.4

513. Which is not true in the context of the free wheel clutch ?
(a) appreciable saving in fuel on long downward slope
(b) **effective use of brakes needed to stop the vehicle on a downward slope**
(c) simplified gear changing at low speeds
(d) transmission subjected to greater wear and tear

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : transmission subjected to greater wear and tear

514. In a _____bearing, the inner ring has two raceways and the outer ring has a single spherical raceway with its center
(a) rigid (b) self-aligning
(c) pseudo-aligning (d) elastic

HPPSC Workshop Suptd. 08.07.2021

Ans. (b) : self-aligning

515. In a journal bearing, the load acting on the shaft is
(a) axial
(b) co-axial
(c) transverse to its axis
(d) inclined to the axis

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : In a journal bearing, the load is acting on the shaft in transverse to its axis.
Journal bearing support a load acting in the direction normal to the rotating shaft. It is the most common bearing among sliding bearing.

516. In bearing design, the term 'thin film' is where:
(a) lubricant does not prevent completely the contact between journal and bearing
(b) there is a perfect lubrication
(c) all the conditions of thick film lubrication are met
(d) the lubricant prevents contacts with journal and bearing completely

CGPSC AE 15.01.2021

Ans. (a) : In bearing design the term 'thin film' is where lubricant does not prevent completely the contact between journal and bearing.

517. In a Journal bearing, the attitude is the ratio of _____ for hydrodynamic lubrication condition.
(a) Eccentricity and radial clearance
(b) Radial clearance and minimum film thickness
(c) Eccentricity and diametral clearance
(d) diametral clearance and minimum film thickness

VIZAG Steel MT 24.01.2021

Ans. (a) : In a journal bearing, the attitude defining the static performance of the journal bearing as functions of operating speed, lubricant viscosity and applied load. In journal bearing the attitude is ratio of eccentricity and radial clearance for hydrodynamic lubrication condition.

518. In a Journal bearing, the diameter of the journal is 0.2 m, its speed is 1000 rpm and the load on the bearing is 53 kN. Considering coefficient of friction is 0.0072, the heat generated will be nearly:
(a) 4 kW (b) 3 kW
(c) 2 kW (d) 1 kW

VIZAG Steel MT 24.01.2021

Ans. (a) : Given, Diameter (D) = 0.2 m
N = 1000 rpm
Bearing load (W) = 53 kN
$\mu = 0.0072$

Heat generated $(Q_g) = \mu WV = \mu W\left(\frac{\pi DN}{60}\right)$

$= 0.0072 \times 53 \times \pi \times \frac{0.2 \times 1000}{60}$

= 3.996 kW
= 4 kW

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519. The frictional torque transmitted for uniform pressure, in case of a conical pivot bearing having semi-angle of the cone as θ, is equal to

(a) $\frac{\mu WR}{\sin\theta}$ (b) $\frac{\mu WR}{2\sin\theta}$

(c) $\frac{2\mu WR}{\sin\theta}$ (d) $\frac{2\mu WR}{3\sin\theta}$

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : $\frac{2\mu WR}{3\sin\theta}$

520. A 100 mm long journal bearing supports a load of 4000 N on a 50 mm diameter shaft. The pressure acting on the bearing is _____.

(a) 0.8 N/mm^2 (b) 0.4 N/mm^2

(c) 1.2 N/mm^2 (d) 1.6 N/mm^2

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Given,
length, l = 100 mm
load, w = 4000 N
diameter, d = 50 mm

bearing pressure, $P = \frac{w}{A}$

$= \frac{w}{ld}$

$= \frac{4000}{100\times50}$

$= \frac{4}{5}N/mm^2$

$= 0.8\ N/mm^2$

521. Which is/are not the characteristics of wet clutch is comparison of dry clutch?

(1) Has more life
(2) High torque transmitting capacity
(3) High cost
(4) High coefficient of friction

(a) (1), (3) (b) (2), (3)

(c) (2), (4) (d) (4)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (d) :

Dry Clutch	Wet Clutch
• A dry clutch is a clutch which does not have oil present between the clutch plates.	• Oil is used between clutch plates to lubricate and cool the components.
• The sound in dry clutch is more and keeps increasing as the clutch continue to undergo wear & tear.	• The sound in this is less. Oil keep the surface clean and provides smooth performance and longer life.

522. The common clutch used between the engine and synchromesh gear box is :

(a) Cone (b) Single plate (dry)

(c) Multiplate wet (d) Electro-magnetic

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (d) : Electro-Magnetic Clutch–Electro-magnetic clutches are most suitable for remote operation since no mechanical linkages are required to control their engagement, providing fast, smooth operation.

523. If the brake wheels gets locked before the vehicle stops, the vehicle is said to be :

(a) Slipping (b) Skidding

(c) Sliding (d) Rolling

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (b) : If the brake wheels gets locked before the vehicle stops, the vehicle is said to be skidding.

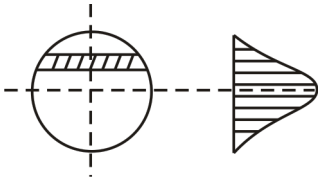
524. When partial journal bearing has no clearance, then the bearing is called

(a) Sleeve bearing (b) Slipper bearing

(c) Guide bearing (d) Fitted bearing

Oil India Senior Officer 23.12.2020

Ans. (c) : The circular section, shear stress distribution is parabolic. The max shear stress is at neutral axis and end of the reaction has zero stress.



Max shear stress

$Z_{max} = \frac{4}{3}Z_{avg}$

$\frac{Z_{avg}}{Z_{max}} = \frac{3}{4}$

525. Find the dynamic load carrying capacity of a roller bearing if the shaft rotates at 1500 rpm, radial load acting on the bearing is 6 kN and expected life for 90% life of the bearing is 8100 hours.

(a) 6 kN (b) 54 kN

(c) 54000 kN (d) 60000 kN

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Given, W = 6 kN

L_{10h} = 8100 hr

h = 1500 rpm

Life of bearing in million revolution is given by

$L_{10} = \frac{60N(L_{10h})}{10^6} = \frac{60\times1500\times8100}{10^6}$

L_{10} = 729 mr (million revolution)

The maximum radial load can be given by equation

$L_{10} = \left(\frac{C}{W}\right)^k$ (Where, k = 3 for ball bearing)

$729 = \left(\frac{C}{6}\right)^3$

C = 54 kN

526. If 'w' is the load on a cylindrical journal of diameter (d) and length (l), then bearing pressure is

(a) $\frac{2w}{\pi d^2}$ (b) $\frac{4w}{\pi d^2 l}$

(c) $\frac{w}{\pi d l}$ (d) $\frac{w}{d l}$

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Bearing pressure $= \frac{w}{dl}$

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527. In single row angular contact ball bearings, the basic static load relates to the:
(a) radial component of the load
(b) full axial and partial radial component of the load
(c) axial component of the load
(d) full radial and partial axial component of the load

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : In single row angular contact ball bearings, the basic static load relates to the full radial and partial axial component of the load.

528. The highest permissible temperatures recommended for automobile brakes with asbestos block lining range from:
(a) 200 – 325 degree C (b) 180 – 225 degree C
(c) 100 – 155 degree C (d) 80 – 125 degree C

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : The highest permissible temperatures recommended for automobile brakes with asbestos block will be above to 150ºC, So the temperature range will be 118ºC-225ºC.

529. For a clutch, the lower torque capacity is obtained using an assumption of
(a) Uniform wear assumption
(b) Uniform pressure assumption
(c) Uniform speed assumption
(d) Uniform torque assumption

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Torque = μWR

where $R = \frac{r_1 + r_2}{2}$ for uniform wear

$R = \frac{2}{3}\left(\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2}\right)$ for uniform pressure

530. In the case of drum brakes, for the brake to be self-energizing, the friction moment acts
(a) Opposite to the actuating moment
(b) Orthogonal to the actuating moment
(c) To add to the actuating moment (moment due to applied load)
(d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : A self energizing brake is one in which effort due to friction acts in same direction as that of effort due to applied load i.e moment gets add up.

531. The bearings that generate simply supported end conditions more closely than others are
(a) Cylindrical Roller bearings
(b) Self-aligning ball bearings
(c) Double -row deep groove ball bearings
(d) Taper roller bearings

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Self aligning ball bearing generate simply supported and condition the rest other bearing behave like fixed support.

532. The bearing that is not ideally suited for taking up axial load is
(a) Spherical roller bearing
(b) Needle roller bearings
(c) Angular contact bearing
(d) Taper roller bearings

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Needle roller bearing is not suitable for axial load. Axial load can cause the rollers to run against the filling slot which leads to early failure.

533. The clearance ratio in the case of hydrodynamic journal bearings is typically
(a) 0.1 – 0.2
(b) 0.001 – 0.002
(c) 0.01 – 0.02
(d) 0.00001 – 0.00002

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Clearance to radius ratio for hydrodynamic journal bearing is 0.001 to 0.002.

534. The Sommerfeld number in a hydrodynamic bearing is a function of
(a) Fluid viscosity (b) Load on the bearing
(c) Radial clearance (d) All of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Sommerfeld number = $\frac{ZN}{P} \times \left(\frac{D}{C}\right)^2$

D =diameter of journal bearing
C = diameter clearance
Z = viscosity kg/m-sec
P = (MPa)
N = rps

535. In rolling element bearings, the life in number of revolutions, which 90% of a random sample of bearings of given size reach or exceed at their design load is typically referred to as
(a) L90 life
(b) L10 life
(c) Bearing residual life
(d) L50 life

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : L90 life → shows reliability
L10 life → shows failure

536. The basic dynamics load rating for a rolling element bearing is a load that will give a life of
(a) One million revolution of the inner race
(b) 0.1 million revolutions of the inner race
(c) Ten thousand revolution of the inner race
(d) One hundred million revolution of the inner race

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : The basic dynamic load rating for a rolling element bearing is a load that will gives a life of 10^6 revolution.

537. A clutch is operating under uniform wear, axial load of 500N, friction coefficient of 0.3 and the mean radius of the contact surfaces 0.1m. If the torque transmitted is 60 Nm, then the number of pairs of contact surfaces is :
(a) 6 (b) 4
(c) 2 (d) 8

NLC GET 17.11.2020, Shift-II

Ans. (b) : Given, Axial load (W) = 500 N
Friction coefficient (μ) = 0.3

Mean radius of contact surface(R) $= 0.1m = \frac{r_1 + r_2}{2}$

Torque transmitted (T) = 60 N.m
number of pair (n) = ?
Torque transmitted of clutch $T = n\mu WR$
$60 = n \times 0.3 \times 500 \times 0.1$
$n = 4$

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538. A bearing supports a radial load (F_r) of 7000 N and a thrust load (F_a) of 2100 N. The desired life of the ball bearing is 160×10^6 revolution at 300 rpm. If the load is uniform and steady, service factor is 1, radial factor (X) is 0.65, thrust factor (Y) is 3.5, k = 3 and rotational factor is 1, the basic dynamic load rating of a bearing will be nearly.

(a) 65 kN (b) 75 kN
(c) 85 kN (d) 95 kN

ESE 2020

Ans. (a) : Equivalent Radial Load $P_e = XVF_r + yF_a$

$P_e = 0.65 \times 1 \times 7000 + 3.5 \times 2100$

$= 11900$ N

$$L_{90} = \left(\frac{C}{P_e}\right)^k$$

Dynamic Capacity Load

$C = P_e \times (L_{90})^{1/k}$

$C = 11900 \times (160)^{1/3}$

$C = 64603.14$ N

$C = 64.6$ kN

$C \approx 65$ kN

539. The torque transmitting capacity of friction clutches can be increased by:

(a) Use of friction material with a lower coefficient of friction
(b) Decreasing the mean radius of the friction disk
(c) Increasing the mean radius of the friction disk
(d) Decreasing the plate pressure

ESE 2020

Ans. (c) : The torque transmitting capacity of friction clutches can be increased by increasing the mean radius of the friction disk.

$$T = \frac{2}{3}\mu w\left(\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2}\right) = \mu w R_m$$

$T \propto R_m$

540. What is the correct formula to represent the average pressure (p_{av}) on contact surface of the clutch? (Consider W – axial thrust, r_1 – radius of external face of the friction plate, r_2 – radius of internal face of the friction surface.)

(a) $p_{av} = \dfrac{W}{\pi\left[(r_1)^2 - (r_2)^2\right]}$

(b) $p_{av} = \dfrac{2W}{\pi\left[(r_1)^2 - (r_2)^2\right]}$

(c) $p_{av} = \dfrac{2W(r_1 - r_2)}{\pi\left[(r_1)^2 - (r_2)^2\right]}$

(d) $p_{av} = \dfrac{2W(r_1 - r_2)}{\pi\left[(r_1)^2 - (r_2)^2\right]}$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : The average pressure (P_{av}) on the friction or contact surface is given

$$P_{av} = \frac{\text{Total force on friction surface}}{\text{cross-sectional area of friction surface}}$$

$$\boxed{P_{av} = \frac{W}{\pi\left[(r_1)^2 - (r_2)^2\right]}}$$

541. Which of the following equations represents the total frictional torque (T) induced in a multiple disc clutch, assuming it as uniform wear? (Consider n – number of pairs of contact surfaces, μ – coefficient of friction, r_1– radius external face of the friction plate, r_2 – radius of internal face of the friction surface.)

(a) $T = \mu W \dfrac{2(r_1)^3 - (r_2)^3}{3(r_1)^2 - (r_2)^2}$

(b) $T = \eta\ \mu W \dfrac{2(r_1)^3 - (r_2)^3}{3\left[(r_1)^2 - (r_2)^2\right]}$

(c) $T = \eta\ \mu W \dfrac{r_1 + r_2}{2}$

(d) $T = \eta\ \mu W \dfrac{2(r_1)^3 + (r_2)^3}{3r_2}$

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Frictional torque acting on the friction surfaces or on the clutch.

$T = \eta\mu WR$

where

R = Mean radius of friction surfaces.

$$\boxed{T = \eta\mu W \frac{r_1 + r_2}{2}}$$

542. The part of the machine whose function is to engage or disengage the driver shaft with driven shaft at will, without stopping the engine is known as :

(a) governor (b) brake
(c) steering (d) clutch

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Clutch : A clutch is a machine member used to connect a driving shaft to a driven shaft so that the driven shaft may be started or stopped without stopping the driving shaft. The use of a clutch is mostly found in automobiles.

543. In a single block brake, when the line of action of the tangential braking Force (F_t) passes through the fulcrum O of the lever of length (L) subjected to external force (P) at end of lever in downward direction, what will be the equation of Braking torque (T_B) of this brake? (Consider μ as coefficient of friction and x is distance of reaction force from fulcrum O and drum of radius (r).)

(a) $T_B = \dfrac{\mu.P.L.r}{x}$ (b) $T_B = \dfrac{P.L.r}{(x - \mu)}$

(c) $T_B = \dfrac{\mu.P.L}{x.r}$ (d) $T_B = \dfrac{\mu.P.x.r}{L - 1}$

APPSC Poly Lect. Automobile Engg., 2020

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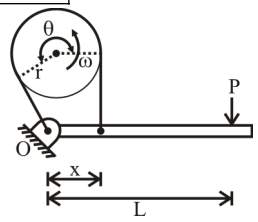
Ans. (a) : Braking torque

$$T_B = \frac{\mu.P.L.r}{x}$$



544. While calculating the torque capacity of the old clutch, which analysis theory is more accurate and why?
(a) Uniform pressure theory is more accurate as it assumes uniform pressure in old clutch plates.
(b) Uniform pressure theory is more accurate as it gives non-uniform pressure in old clutch plates.
(c) Uniform wear theory is more accurate as it assumes uniform wear in old clutch plates.
(d) Uniform wear theory is more accurate as it assumes non-uniform wear in clutch plates.

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : The uniform pressure theory is applicable only when the friction lining is new and for old clutches UWT is applicable.
So, for the safe designing of the clutch it is better to assume UWT as a design criteria.

545. In a ventilated disc brake
(a) a duct directs air towards the caliper for cooling while the vehicle is moving
(b) Caliper is covered with cooling fins
(c) disc contains many small holes for optimum cooling performance
(d) disc contains radial vanes between its rubbing surface for optimum cooling performance

TNCSC AE 2020

Ans. (d) : In a ventilated disc brake disc contains radial vanes between its rubbing surface for optimum cooling performance ventilated brake discs are used in modern vehicles such as passenger and racing cars as they require high speed. Ventilated discs are designed to run coder than solid discs is normal operating conditions.

546. A cage in a roller bearing keeps the
(a) outer and inner race together
(b) balls and outer race together
(c) balls and inner race together
(d) balls together at required distance

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : A cage in a roller bearing keeps the balls together at required distance.

547. Time for engagement of a clutch depends on
(a) driver shaft speed
(b) driven shaft speed
(c) both driver and driven shaft speeds
(d) does not depend on speed

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Clutch is defined as the machine element which is used to engage and disengage the driver and driven shaft speeds without stopping the prime mover.
Clutches are used to avoid frequent start or stopping of prime mover but driven machinery can be stop and start frequently.

548. A self locking brake is
(a) when the brake is applied without any effort
(b) drum gets locked in case of emergency
(c) both the shoes are self-energizing
(d) brake is always jammed

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : A brake is said to be self energised brake if the direction of moment due to effort and frictional force are in the same direction. It will reduce effort applied to the brake.
• If $b = \mu c \Rightarrow P = 0 \Rightarrow$ Self locking condition
• If $b < \mu c \Rightarrow P = -ve \Rightarrow$ uncontrollable breaking action
Self locking means that the effort needed to applied on the brake is zero or that once contact is made between the block and the drum, the brake is applied itself.

549. If the ball bearing indentified as SKF 2015, 3115 and 4215 :
(a) Bore is common but width is increasing
(b) Outer diameter is common but bore is increasing
(c) Width is common but outer diameter is decreasing
(d) Bore is common but outer diameter is decreasing

OPSC AEE 2019 Paper-I

Ans : (a) : According to SKF, bearing is designed by three numbers. Last two digits multiplied by 5 gives shaft diameter.
100- Extra light load
200- Light load
300- Medium load
400- Heavy load

550. Antifriction bearings are
(a) Journal bearing (b) Pedestal bearing
(c) Collar bearing (d) Needle bearing

Gujarat PSC AE 2019

Ans : (d) : Antifriction bearing are needle bearing.

551. In automobiles, Hooke's joint is used between which of the following?
(a) Clutch and gear box
(b) Gear box and differential
(c) Differential and wheels
(d) Flywheel and clutch

OPSC AEE 2019 Paper-I

Ans : (b) : Hooke's joint is used between gear box and differential. It is provided on propeller shaft.
Hooke's joint is used to connect two nonparallel intersecting shafts.
It is also used for shaft with angular misalignment.

552. To avoid self-engagement in cone clutch, its semi-cone angle is always kept
(a) smaller than the angle of static friction
(b) equal to the angle of static friction
(c) greater than the angle of static friction
(d) half of the angle of static friction

ESE 2019

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Ans. (c) : To avoid self engagement in cone clutch, the semi cone angle should always be kept greater than the angle of static friction.

553. A bicycle and rider travelling at 12 km/hr on a level road have a mass of 105 kg. A brake is applied to a rear wheel having 800 mm diameter. The pressure on the brake is 80 N and the coefficient of friction is 0.06. The number of turns of the wheel before coming to rest will be

(a) 48.3 revolutions (b) 42.6 revolutions
(c) 38.3 revolutions (d) 32.6 revolutions

ESE 2019

Ans. (a) : Given,

$V = 12 \text{ km/hr} = 12 \times \frac{5}{18} = 3.33 \text{ m/s}$

$m = 105 \text{ kg}$

$d = 800 \text{ mm} = 0.8 \text{ m}$

$r = 400 \text{ mm} = 0.4 \text{ m}$

$F = 80 \text{ N}$

$\mu = 0.06$

Assuming the bicycle and rider as a ring, the moment of inertia

$I = mr^2 = 105 \times 0.4^2 = 16.8 \text{ kg-m}^2$

Initial angular velocity of disc

$\omega_i = \frac{V}{r} = \frac{3.33}{0.4} = 8.325 \text{ r/s}$

Friction force on whirl (f) = $\mu F = 0.06 \times 80$

$= 4.8 \text{ N}$

Torque due to brake

$\tau = r \cdot f$

$= 0.4 \times 4.8$

$= 1.92 \text{ N-m}$

Angular acceleration $\alpha = \frac{\tau}{I}$

$= 0.114 \text{ rad/s}$

$-\alpha$ because retardation take place due to break apply

From equation of motion

$\omega_f^2 = \omega_i^2 - 2\alpha\theta$

$0 = \omega_i^2 - 2\alpha\theta$

Angular displacement

$\theta = \frac{\omega_i^2}{2\alpha} = \frac{(8.325)^2}{2 \times 0.114}$

$= 303.97 \text{ radius}$

Numbers of revolutions during braking

$n = \frac{\theta}{2\pi} = \frac{303.97}{2 \times 3.14}$

$n = 48.4 \text{ revolutions}$

554. A taper roller bearing has a dynamic load capacity of 26 kN. The desired life for 90% of the bearings is 8000 hr and the speed is 300 r.p.m. The equivalent radial load that the bearing can carry will be nearly

(a) 5854 N (b) 5645 N
(c) 5436 N (d) 5227 N

ESE 2019

Ans. (a) : Given,

$L_{90} = 8000 \times 60 \times 300$

$= 144 \times 10^6 \text{ revolutions}$

$= 144 \text{ MR (Million Revolution)}$

Equivalent radial load

$L_{90} = \left(\frac{C}{P}\right)^{10/3}$ k = 10/3 for taper roller bearing

$P = \frac{C}{(L_{90})^{0.3}} = \frac{26000}{(144)^{0.3}}$

$= 5854.16 \text{ N}$

555. What tapers in a tapered roller bearing?

(a) Inner race (b) Outer race
(c) Roller (d) Cage

OPSC AEE 2019 Paper-I

Ans : (c) : The tapered rollers are guided in tapered raceways. The roller set is retained by the cage, to the inner rings (cone) between guide flange and locating flange. The outer ring (cup) is loose. Tapered roller bearings equally support radial and axial loads.

556. Total torque transmitted by a single plate clutch (both sides are effective) with axial spring load of 1 kN, inner radii 10 cm and outer radii 15 cm will be [Consider coefficient of friction as 0.5, assuming uniform wear]?

(a) 75 N-m (b) 150 N-m
(c) 100 N-m (d) 125 N-m

SJVN ET 2019

Ans. (d) : By using uniform wear theory

$\text{Torque} = \frac{1}{2}\mu W(R_1 + R_2)n$

$= \frac{1}{2} \times 0.50 \times 1000 \frac{(10+15) \times 2}{100}$

$= 125 \text{ Nm}$

557. Self lubricating bearings can be manufactured by_____

(a) Extrusion (b) Drawing
(c) Forming (d) Powder Metallurgy

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (d) : Self lubricating bearings can be manufactured by powder metallurgy and it is also called sintered metal self lubricating bearings. For such type of bearings there is no need to supplying continuous lubricating oils.
When bush heated due to friction it release oil stored in interconnected pores.
Oil forms hydrodynamic layer due to wedge action.

558. Failure of a bearing depends on

(a) Number of revolution
(b) Radial load applied
(c) Axial load
(d) All of the above

Gujarat PSC AE 2019

Ans : (d) : Failure of bearing depends on
- Number of revolution
- Radial load applied
- Axial load

559. A radial load of 6.5 kN is applied on a cylindrical roller bearing. The desired life of the bearing with 90% reliability is 18000 hours. With the application factor 1.7, if the shaft rotates at 1600 rpm, then the required basic dynamic load rating for the bearing is given by :

(a) 103.4 MN (b) 103.4 kN
(c) 1.042 kN (d) 10.34 kN

BHEL ET 2019

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Ans. (b) ∵ load life L = 60 × L_H × N

Where

L_H = life in hour

L = life in cycle

N = RPM

Given, K = 1.7, N = 1600 rpm, L_H = 18000 hr

F_r = 6.5 kN

$L = (C/Pe)^{10/3}$ ⇒ For roller bearing

⇒ $L.(Pe)^{10/3} = C$

i.e. $\left(\frac{P}{k}\right)^{10/3} \times L_{90} = (F_r)^{10/3} \times L_r$

$\left(\frac{P}{1.7}\right)^{10/3} \times 10^6 = (6.5)^{10/3} \times 60 \times 18000 \times 1600$

$\left(\frac{P}{1.7}\right)^{10/3} = (6.5)^{10/3} \times 96 \times 18$

$P = (6.5 \times 1.7)^{10/3} \times 96 \times 18$

P = 103.4 kN

560. In a plate clutch, the axial force is 4 kN. The inside radius of contact surface is 50 mm and the outside radius is 100 mm. For uniform pressure, the mean radius of friction surface will be:

(a) 78 mm (b) 60 mm

(c) 75 mm (d) 80 mm

HPPSC ADF 05.03.2019

Ans. (a) : Mean radius of friction surface for uniform pressure

$R_m = \frac{2}{3}\left(\frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}\right)$

$R_m = \frac{2}{3}\left(\frac{100^3 - 50^3}{100^2 - 50^2}\right)$ [where R_0 = 100, R_i = 50

$= \frac{2}{3} \times \frac{1750}{15}$

=77.77

$\boxed{R_m = 78\,m}$

561. The journal diameter of a full journal bearing is 60 mm. The diameter of the bush bore is 60.60 mm and bush length is 30 mm. If the journal rotates at 1500 rpm and the average viscosity of lubricant is 0.03 Ns/m², the power loss will be:

(a) 60.2 W (b) 90.7 W

(c) 102.8 W (d) 12.55 W

HPPSC ADF 05.03.2019

Ans. (d) : Given,

D = 60 mm = 60 × 10^{-3} m

D_c = 60.60 = 60.60 × 10^{-3} m

C = D–D_c

= 0.6 × 10^{-3} m

ℓ = 0.03 m, n = 1500 rpm × $\frac{1}{60}$ = 25 rps

z = μ = 0.03 N-s/m² or Pa-s.

Power loss ⇒ P_{loss} = μ WV

$\because \mu = 2\pi^2 \frac{z_n}{P} \times \frac{D}{C}$

$= 2\pi^2 \times \frac{z_n}{W} \times \ell D \times \frac{D}{C}$

$= 2\pi^2 \times \frac{0.03 \times 25}{W} \times 0.03 \times \frac{60 \times 60 \times 10^3}{10^6 \times 0.6}$

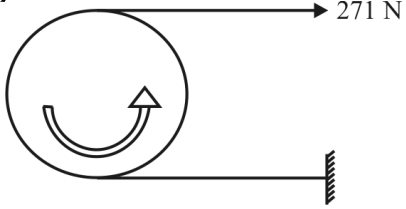
$= \frac{2.6647}{W}$

$P_{loss} = \frac{2.6647}{W} \times W \times \pi Dn$ $(V = \pi Dn)$

$= 2.6647 \times \pi \times \frac{60}{10^3} \times 25$

= 12.55 W

562. A force of 271 N is applied to the break drum of 2m diameter in band brake system as shown in figure, where the wrapping angle is 180°. If the coefficient of friction between the drum and the band is 7/22 the braking torque applied in Nm is :



(a) 271 (b) 100

(c) 171 (d) 371

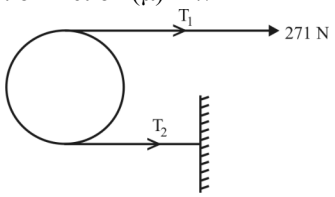
ISRO Scientist/Engineer (RAC), 10.03.2019

Ans. (c) : Given, T_1 = 271 N

θ = π (180°)

D = 2*m*

Coff. of friction (μ) = 7/22



$\frac{T_1}{T_2} = e^{\mu\theta}$

$\frac{271}{T_2} = e^{\left[\frac{7}{22}\right] \times \pi}$

$T_2 = \frac{271}{e^{\frac{7}{22} \times \pi}} = 99.73$

Braking Torque = $(T_1 - T_2) \times R$

$(M_t) = (271 - 99.73)\frac{2}{2}$ $(D = 2m)$ $(R = D/2)$

$\boxed{M_t = 171.26N - m}$

563. Which of the following is not true for Rolling contact bearing?

(a) Good damping ability

(b) Can support combined load

(c) Low starting friction

(d) Compact in axial direction

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

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Ans. (a) : Rolling contact Bearing- These bearings are also called anti-friction bearing due to its low friction characteristics.
Characterstics of Rolling contact bearing.
1. These can be used for radial load, thrust load or combination of thrust and radial load.
2. Almost maintenance free & low starting friction.
3. Low starting and running friction except at very high speed.
4. Accuracy of shaft alignment.
5. Ability to with stand momentary shock loads.

Disadvantage
1. More noisy at very high speed
2. Low resistance to shock loading
3. More initial cost
4. Design of bearing housing complicated.

564. Which one of the following is not a desirable characteristics of friction clutches.
(a) The moving parts should be weight enough
(b) should have good heat conductivity
(c) should have high coeff. of friction
(d) should have high wear resistance

TNPSC AE 2018

Ans. (a) : The moving parts should not be weight enough also a desirable characteristics of friction clutches.

565. Which type of bearing (s) provides a bearing surface for forces acting along axis to the shaft?
(a) Thrust bearing (b) Journal bearing
(c) Linear bearing (d) None of the above

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) : A thrust bearing supports load acting along the axis of shaft journal bearing supports load perpendicular to the axis of shaft.

566. In a journal bearing, the diameter of the journal is 0.15 m, its speed is 900 rpm and the load on the bearing is 40 k N. Considering μ = 0.0072, the heat generated will be nearly
(a) 1 kW (b) 2 kW
(c) 3 kW (d) 4 kW

ESE 2018

Ans. (b) : Given,
Diameter of bearing (d) = 0.15 m
Radius of bearing (r) = 0.075 m
Speed (N) = 900 rpm
Load (W) = 40 kN
$\mu = 0.0072$
Heat generated = $\mu W r\omega$

$$= \mu W r \frac{2\pi N}{60}$$

$$= 0.0072 \times 40 \times 0.075 \times \frac{2\pi \times 900}{60}$$

$$= 2.036 \text{ kW}$$

567. Starting friction is low in
(a) Boundary lubrication
(b) Mixed (or semi fluid) lubrication
(c) Hydrostatic lubrication
(d) Hydrodynamic lubrication

GPSC Executive Engineer 23.12.2018

Ans. (c) : Hydrostatic lubrication is defined as a system of lubrication in which the load supporting fluid film, separating the two surfaces is created by on external source, like a pump, supplying sufficient fluid under pressure.
- Hydrostatic bearings have no starting friction and no rubbing at any operating speed or load.

568. Life of a ball bearing at a load of 10 kN is 24000 hours. If the load is increased to 20 kN (all other conditions are same), its life in hour is
(a) 1000 h (b) 3000 h
(c) 6000 h (d) 12000 h

ISRO Scientist/Engineer 22.04.2018

Ans. (b) : We know that

$$L = \left[\frac{K}{C}\right]^n$$

where, n = 3 for ball bearing

$n = \frac{10}{3}$ or 3.33 for roller bearing

K = Dynamic load capacity, $C_1 \rightarrow$ equivalent dynamic load

$$L \propto \frac{1}{C^3} \text{ for ball bearing}$$

$$\frac{L_2}{L_1} = \left[\frac{C_1}{C_2}\right]^3 = \left(\frac{10}{20}\right)^3 \Rightarrow L_2 = 24000 \times \frac{1}{8}$$

$$\boxed{L_2 = 3000\, hr}$$

569. Which of the following is true for bearings?
(a) In hydrostatic lubricated bearings, externally pressurized lubricant is fed into the bearing
(b) Hydrostatic lubricated bearings require the motion of surface to generate the lubricating film
(c) Hydrostatic lubricated bearing can operate at very high speed only
(d) In hydrodynamic lubricated bearings, externally pressurized lubricant is fed into the bearing
(e) Hydrodynamic lubricated bearing can operate at very low speed only

CGPSC AE 25.02.2018

Ans. (a) : Hydrostatic Bearing– Hydrostatic bearing are externally pressurized fluid bearings, where the fluid is usually oil, and the pressurization is done by a pump. Hydrodynamic bearings rely on the high speed of the journal to pressurize the fluid in a wedge between faces.

570. For the cone clutches, face width of friction linings is [where, α = semi cone angle, D = outer diameter of cone, d = inner diameter of cone]
(a) $\frac{D}{\sin\alpha}$ (b) $\frac{D-d}{\sin\alpha}$
(c) $\frac{D-d}{2}\sin\alpha$ (d) $\frac{D-d}{2\sin\alpha}$
(e) $\frac{D}{2}\sin\alpha$

CGPSC AE 25.02.2018

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Ans. (d) : For the cone clutches, fact width of friction linings is given by $= \frac{D-d}{2\sin\alpha}$

571. The relationship between the dynamic load carrying capacity (C), equivalent dynamic load (P) and bearing life (L) is given by : [where p = constant and depends upon the type of bearing]

(a) $L = \left(\frac{C}{P}\right)^{1/p}$ in million revolution

(b) $L = \left(\frac{P}{C}\right)^{1/p}$ in million revolution

(c) $L = \left(\frac{C}{P}\right)^{p}$ in million revolution

(d) $L = \left(\frac{P}{C}\right)^{p}$ in million revolution

(e) $L = \left(\frac{C}{P}\right)^{2p}$ in million revolution

CGPSC AE 25.02.2018

Ans. (c) : The relationship between the dynamic load carrying capacity (C) equivalent dynamic load (P) and bearing life (L) is given as:

$L = \left[\frac{C}{P}\right]^{p}$ in million revolution where

p = 3 for ball bearing

$p = \frac{10}{3}$ (3.33 for roller bearing)

572. The frictional force and frictional torque acting on the drum of centrifugal clutch by each shoe is given as: [ω_1 = engagement speed (rad/s), ω_2 = running speed (rad/s), m = mass of each shoe, R = Radius of drum, r = Radius of centre of gravity of shoe]

(a) $\mu mR\left(\omega_2^2-\omega_1^2\right); \mu mRr\left(\omega_2^2-\omega_1^2\right)$

(b) $\mu m\frac{R^2}{r}\left(\omega_2^2-\omega_1^2\right); \mu mR^2\left(\omega_2^2-\omega_1^2\right)$

(c) $\mu mr\left(\omega_2^2-\omega_1^2\right); \mu mRr\left(\omega_2^2-\omega_1^2\right)$

(d) $\mu mr\left(\omega_2^2-\omega_1^2\right); \mu mr^2\left(\omega_2^2-\omega_1^2\right)$

(e) $\mu mr\left(\omega_2^2-\omega_1^2\right); \mu mR^2\left(\omega_2^2-\omega_1^2\right)$

CGPSC AE 25.02.2018

Ans. (c) : The frictional force and frictional torque acting on the drum of centrifugal clutch by each shoe is given by $\mu mr\left(\omega_2^2-\omega_1^2\right), \mu mRr\left(\omega_2^2-\omega_1^2\right)$ respectively.

573. The angle through which the brake drum rotates during the braking period is 20 radians and the energy absorbed by the brakes is 100 kJ. The braking torque capacity is :

(a) 2×10^6 Nm (b) 5×10^3 N/m

(c) 5×10^3 Nm (d) 2×10^6 N/m

(e) 50 kN.m

CGPSC AE 25.02.2018

Ans. (c) : T = braking torque capacity

$\omega = 20$ radian

Energy absorbed by the brakes = Torque capacity × angle rotates during the braking period

$= T \times \omega$

$T \times \omega = 100$ kJ

$T = \frac{100}{20} = 5\times10^3 \text{ N-m}$

574. If a bearing is designated by number 208, it refers to:

(a) light series bearing with 8 mm bore

(b) light series bearing with 40 mm bore

(c) medium series bearing with 20 mm bore

(d) medium series bearing with 40 mm bore

(e) heavy series bearing with 20 mm bore

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : If the bearing is desighe by number 208. It refers to light series bearing with 40 mm bore.

575. Which of the following is correct for clutch?

(a) Clutch is engaged while changing of gears.

(b) Clutch is disengaged while starting of engine.

(c) Clutch should be engaged suddenly to avoid jerk.

(d) Clutch should have low coefficient of friction.

(e) Clutch is disengaged while the vehicle is moving.

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Function of clutch is transmitting the torque from the engine to the drive train. Smoothly deliver the power form the engine to enable smooth vehicle movement. Perform quietly and to reduce drive related vibration. Protect the drive train when given the inappropriate use.

Clutch is disengaged while starting of engine.

576. Gear box is required for:

(a) smooth running of the car

(b) controlling fuel supply

(c) acceleration

(d) regulation of torque and speed

(e) retardation

CGPSC ARTO & TSI 07.01.2017

Ans. (d) : Gearbox used to increase torque which reducing the speed of a prime mover output shaft (e.g a moter crankshaft). This means that the output shaft of a gearbox rotates at a slower rate than input shaft and this reduction in speed produces a mechanical advantage.

577. Which type of rear axle is used on a heavy duty truck?

(a) Semi floating (b) Fully floating

(c) Quarter floating (d) Non floating

(e) Drive shaft

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : There are three type of Rear Axle

1. Semi Floating Axle
2. Full-Floating Axle
3. Three Quarter Floating Axle.

Full Floating Axle :-

True to its name this axle type effectively floats in place and maintains its position thanks to two bearing. It is designed to transmit driving torque only. Full floating axles tend to work best for larger vehicles, such as

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heavy- duty trucks. Some midsize trucks that have targer towing capacities or those that use four wheel drive extensively can also benefit from full floating axles.

578. Excessive back and forth movement of the crankshaft is prevented by:

(a) thrust bearing (b) sleeve bearing
(c) shaft journal (d) harmonic balance

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : A crank shaft is a part in a reciprocating engine that rotates due to the reciprocating motion of the pistons. However in a pump, the crankshafts motion causes the piston to move back and forth.
Excessive back and fourth movement of crank shaft is prevented by thrust bearings.

579. The parking brakes employed in vehicles are operated

(a) Mechanically (b) Hydraulically
(c) Pneumatically (d) Electronically

TNPSC AE 2017

Ans. (a) : The parking brakes employed in vehicles are operated mechanically.

580. Cushioning springs in clutch plate are meant to reduce

(a) Torsional vibrations (b) Vehicle speed
(c) Jerky starts (d) None of the above

TNPSC AE 2017

Ans. (c) : Cushioning springs in clutch plate are meant to reduce Jerky starts.

581. Two controlling devices in the automatic transmission operated by hydraulic pressure are the bonds and

(a) pistons (b) gears
(c) planetary gear sets (d) clutches

TNPSC AE 2017

Ans. (d) : Two controlling devices in the automatic transmission operated by hydraulic pressure are the bonds and clutches.

582. In a full journal bearing the angle of contact of the bearing with journal is -

(a) 120° (b) 180°
(c) 270° (d) 360°

GMB AAE 25.06.2017

Ans. (d) : When the angle of contact of the bearing with the journal is 360°. Then bearing is called a full journal bearing. This type of bearing is commonly used in industrial machinery to accommodate bearing loads in any radial direction.

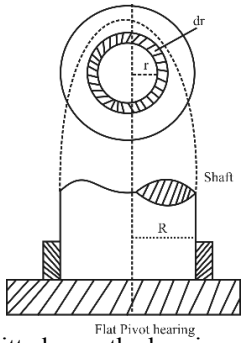
583. The frictional torque transmitted in a flat pivot bearing, considering uniform wear, is (where μ Coefficient of friction, W = Load over the bearing, and R = Radius of bearing surface)

(a) $\frac{1}{2}\mu WR$ (b) $\frac{2}{3}\mu WR$

(c) $\frac{3}{4}\mu WR$ (d) μWR

Sikkim PSC (Under Secretary), 2017

Ans. (a) :



W = load transmitted over the bearing surface
R = Radius of bearing surface
φ = Intensity of pressure
μ = Coefficient of friction

Frictional Torque Transmitted in a Flat Pivot Bearing–

For uniform wear theory,

$$T_{f(UWT)} = \frac{1}{2}\mu WR$$

For uniform pressure theory,

$$T_{f(UPT)} = \frac{2}{3}\mu WR$$

$$\frac{T_{f(UPT)}}{T_{f(UWT)}} = \frac{4}{3} = 1.33$$

So, $T_{f(UPT)}$ is 33% greater than frictional torque by $T_{f(UWT)}$ in case of flat pivot bearing.
There is always friction torque by UPT is greater than frictional torque by UWT. This is true for all cases.

584. Why is the load carrying capacity low in ball bearings?

a. Due to line contact between inner race and the ball.
b. Due to greater axial dimensions.
c. Due to less radial dimensions
d. Due to point contact between inner race and the ball

Select the code from correct answer for the options given below:

(a) Only a and c (b) Only a
(c) Only d (d) Only b and d

Karnataka PSC Lect., 27.05.2017

Ans (c) : Bearing is defined as a machine element whose function is to support a rotating element like shaft and prevents its motion in the direction of applied load, In fluid type bearing all the load is beared by some fluid (lubricant).
Note- In ball bearing- point contact
In roller bearing- Line contact

585. Why is a dog clutch used?

(a) Used to connect and disconnect machine parts
(b) Used to control torque
(c) It helps in rotating the shafts at different speeds
(d) All of the above

Karnataka PSC Lect., 27.05.2017

Ans (a) : A dog clutch is a type of clutch that couples two rotating shafts or other rotating components not by friction but by interference or clearance fit.
The two parts of the clutch are designed such that one will push the other, causing both to rotate at the same speed and will never slip.

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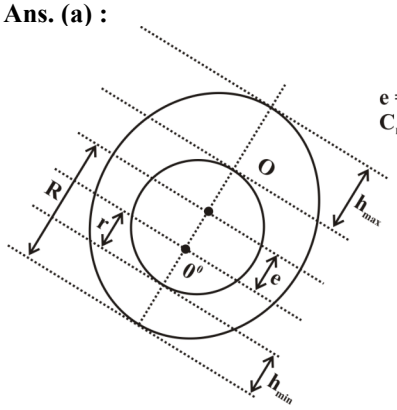
586. For a hydrodynamic journal bearing, if 'h' is the film thickness and 'C' is the diametral clearance, then the eccentricity ratio 'ξ' is given by

(a) $\xi = \frac{1-2h_{min}}{C}$ (b) $\xi = \frac{1-h_{min}}{C}$

(c) $\xi = \frac{1+2h_{min}}{C}$ (d) $\xi = \frac{1+h_{min}}{C}$

Karnataka PSC AE, 10.09.2017

Ans. (a) :



h = film thickness

C = diametral clearance

ξ = eccentricity

$C = D - d$

$\frac{C}{2} = R - r = C_r$

C_r = radial clearance

$\xi = \frac{e}{C_r}$

$h_{min} + r + e = R$

$h_{min} = C_r - \xi C_r$

$h_{min} = \frac{C}{2}(1-\xi)$

$1 - \xi = \frac{2h_{min}}{C}$

$\boxed{\xi = \frac{1-2h_{min}}{C}}$

587. The axial force 'W_e' for engaging cone clutch is

(a) $W_n (\sin\alpha + 0.25 \cos\alpha)$

(b) $W_n (\cos\alpha + 0.25 \sin\alpha)$

(c) $W_n (\sin\alpha + 0.25\ \mu \cos\alpha)$

(d) $W_n (\cos\alpha + 0.25\ \mu \sin\alpha)$

Where, W_n = normal force,

α = face angle of cone,

μ = coefficient of friction.

Karnataka PSC AE, 10.09.2017

Ans. (c) : $W_e = W_n (\sin\alpha + 0.25\mu \cos\alpha)$

588. For a radial ball bearing, the desired rated life is 10000 hrs for a speed of 360 rpm and radial load of 7 kN. The basic load rating for the bearing will be

(a) 42 kN (b) 45 kN

(c) 35 kN (d) 40 kN

GPSC EE Pre, 28.01.2017

Ans. (a) : Given,

t = 10000 hrs

P_e = 7 kN

C = ?

N = 360 rpm

We know that, $L_{10} = \left(\frac{C}{P_e}\right)^k$

Life = L_{10} = 10000 × 60 × 360

= 216 × 10^6

= 216 Million Revolution

= 216 MR

$216 = \left(\frac{C}{7}\right)^3$ [∵ k = 3, ball bearing]

$6 = \frac{C}{7}$

C = 42 kN

589. A journal bearing of diameter 40 mm and length of 20 mm carries a load of 4 kN. The average bearing pressure is

(a) 0.005 N/cm^2

(b) 500 Pa

(c) 500 N/cm^2

(d) None of the mentioned above

GPSC EE Pre, 28.01.2017

Ans. (c) : Given-

Diameter (d) = 40 mm

Length (*l*) = 20 mm

Load (W) = 4 kN = 4× 10^3 N

W = pressure on the bearing in N/mm^2 × projected area of the bearing (*l* × d)

$W = p \times (l \times d)$

$4\times10^3 = p \times 20 \times 40$

$p = \frac{4\times10^3}{20\times40}\frac{N}{mm^2}$

$= \frac{4\times10^3}{8}\frac{N}{cm^2}$

$= \frac{4000}{8} N/cm^2$

= 500 N/cm^2

590. Ball bearing type screws are found in following application

(a) Screw Jack

(b) Aeroplane engines

(c) Crane

(d) Steering Mechanism

GPSC EE Pre, 28.01.2017

Ans. (d) : Ball bearing types screws are found is steering mechanism.

591. A bearing is designated by the number 405. It means that a bearing is of

(a) light series with bore of 5 mm

(b) medium series with bore of 15 mm

(c) heavy series with bore of 25 mm

(d) light series with width of 20 mm

GPSC EE Pre, 28.01.2017

Ans. (c) : A bearing is designated by the number 405. It means that a bearing is of heavy series with bore of 25 mm.

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592. Self locking is not possible in case of
(a) simple block brake
(b) simple band brake
(c) differential band brake
(d) internal expanding shoe brake

APGENCO AE, 2017

Ans. (b) : As one end of band is attached to the fixed pin or fulcrum of lever in simple band brake so self locking is not possible.

593. The dynamic capacity of 6205 bearing is 10.8 kN. Then the dynamic capacity of 6305 bearing is
(a) 10 kN (b) 8 kN
(c) 5 kN (d) 16.2 kN

APGENCO AE, 2017

Ans. (d) : The tooth position in bearing nomenclature represents bearing rating or dynamic capacity the bearing is designated by only the number. Thousand place number or any other letter provide additional information.
1 – extra light
2 – light
3 – medium
4 – heavy
The last two digit from 04 onward when multiplied by gives the both dia in multi meter.

594. The rated life of a ball bearing is :
(a) proportional to the load
(b) inversely proportional to the load
(c) inversely proportional to the square of the load
(d) inversely proportional to the cube of the load

Punjab PSC (Lect.) 06.08.2017

Ans. (d) : The rated life of Ball Bearing

$$= \left(\frac{\text{Dynamic Capacity (c)}}{\text{Dynamic Equivalent Load } (P_e)} \right)^n$$

$$L = \left(\frac{C}{P_e} \right)^3$$

n = 3, Ball Bearing

$n = \frac{10}{3}$, for Roller Bearing

595. In a machinery, a shaft is supported on one end by a taper roller bearing. At the other end, one may use
(a) a ball bearing
(b) another taper roller bearing
(c) a self – aligning ball bearing
(d) no other bearing

Punjab PSC (Lect.) 06.08.2017

Ans. (b) : Taper Roller Bearing–They are arranged in such a way that the axes of individual rolling elements intersects in a common apex point on the axis of the bearing. This is essential requirement of pure rolling motion.
- When it is subjected to pure radial load it induces a thrust component and vice-versa.
- Preferred in impact and fatigue loading.

Advantage
- It can take heavy radial and thrust load.
- More rigid.
- It can be easily disassembled / assembled.
- Maximum load bearing capacity.

Application–Car, trucks, differential, propeller shaft, rail road axle-box, large size bearing in rolling mills.

596. The design of hydrostatic bearings commonly involve more complex lubrication systems, and requires specialized design and application practices. One of the advantages of hydrostatic bearing is
(a) low stiffness
(b) being best for heavy load and extremely high speed
(c) complete elimination of wear
(d) being best for light load and extremely high speed

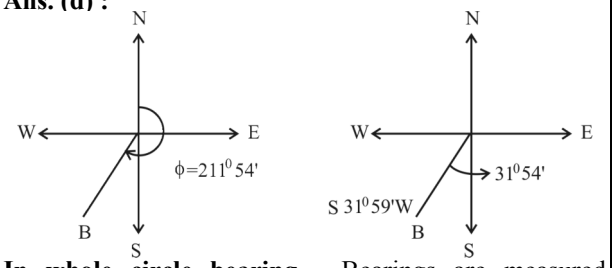
Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Advantages of hydrostatic bearings–
1. They can take heavy loads even at exceptionally low speed.
2. They have very low frictional loss even at starting.
3. They have high positional accuracy.
4. The lead carrying capacity of hydrostatic bearing is independent at the speed.
5. In this bearing complete elimination at wear.

597. If the whole circle bearing is 211ϕ 54' , then its reduced bearing will be
(a) N 31ϕ 54' E (b) S 31ϕ 54' E
(c) N 31ϕ 54' W (d) S 31ϕ 54' W

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) :



In whole circle bearing - Bearings are measured clockwise from north of reference meridian.
In reduced bearing - bearing are measured either clockwise or anti clockwise from north/south whichever is close to line.

598. A flat end foot step bearing supports a vertical shaft of 150 mm diameter rotating at 10rad/s. The shaft carriers a vertical load of 20 kN. Assuming uniform pressure distribution and coefficient of friction equal to 0.05, the power lost in friction is
(a) 500W (b) 750W
(c) 1000W (d) 1125W

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : Given,
D = 150 mm or R = 75 mm = 0.075 m, μ = 0.05
N = 100 rpm
ω = 2πN/60 = 10.47 rad/s
We know,
W = 20 kN = 2 × 10³ N

$$T = \frac{2}{3} \times \mu W.R = \frac{2}{3} \times 0.05 \times 20 \times 10^3 = 50 \text{N-m}$$

P = T.ω = 50 × 10.47 = 523.5 ≈ 500 W

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599. The frictional torque transmitted in a conical pivot bearing considering uniform pressure is : (R: radius of shaft, α: semi angle of cone, μ: Coefficient of friction, W: load on bearing)

(a) $(1/2)\mu WR\,Cosec\alpha$ (b) $(3/4)\mu WR\,Cos\alpha$
(c) $(2/3)\mu WR\,Cos\alpha$ (d) $(2/3)\mu WR\,Cosec\,\alpha$

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : Frictional torque transmitted conical pivot bearing (uniform pressure)

$$=\left(\frac{2}{3}\right)\mu WR\,\text{cosec}\,\alpha$$

600. A ball bearing designated by the number 408 is a:

(a) Extra light bearing (b) Light bearing
(c) Medium bearing (d) Heavy bearing
(e) Extra heavy bearing

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : The most common ball bearing are available in four series as follows–
1. Extra light (100)
2. Light (200)
3. Medium (300)
4. Heavy (400)

If a ball bearing designated by the number 408, it means that bearing is of heavy series whose bore is 08 × 5 i.e. 40 mm.

601. In the case of a shoe brake, equivalent coefficient of friction is considered when the angle of contact is greater than _____.

(a) 30° (b) 45°
(c) 60° (d) 90°
(e) 120°

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : In the case of a shoe brake, equivalent coefficient of friction is considered when the angle of contact is greater than 60°.

602. A journal bearing of diameter 30 mm supported in sliding bearing has a maximum load of 1.8 kN, the allowable bearing pressure is 6 N/mm^2. What will be the length of sliding bearing?

(a) 50 mm (b) 40 mm
(c) 30 mm (d) 20 mm
(e) 10 mm

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Bearing diameter (d) = 30 mm
Max. load (F) = 1.8×10^3 N
Bearing pressure (P) = 6 N/mm^2
Length of bearing = ℓ = ?
We know that

$$\text{Bearing pressure (P)} = \frac{\text{Max. Load}}{\text{Projected area of sliding bearing}}$$

$$6 = \frac{F}{\ell \times d} = \frac{1.8\times10^3}{\ell \times 30}$$

$$\ell = \frac{1.8\times10^3}{6\times30}$$

$$\boxed{\ell = 10\text{ mm}}$$

603. Which is not a type of rolling contact bearing :

(a) Deep groove ball bearing
(b) Cylindrical roller bearing
(c) Journal bearings
(d) Spherical roller bearing

TRB Polytechnic Lecturer 2017

Ans. (c) : Journal bearings are sliding contact type bearings whereas deep groove ball bearing, cylindrical roller bearing and spherical roller bearing are type of rolling contact bearings.

604. In applications like power presses, and rolling mills, where synchronous operation is required, the most preferred type of clutch is:

(a) Electromagnetic clutch
(b) Positive contact clutch
(c) Friction clutch
(d) Fluid clutch

TRB Polytechnic Lecturer 2017

Ans. (b) : The positive clutches are used when positive (no slip) drives are required. These clutches transmit power from the drive shaft to the driven shaft by the interlocking of jaws or teeth. They are rarely used as compared to friction clutch.

605. The clutch used in trucks is–

(a) centrifugal clutch
(b) cone clutch
(c) multi-plate clutch
(d) single plate clutch

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : The clutch used in trucks is single plate clutch.

606. The clutch used in scooters is–

(a) multi-plate clutch (b) single plate clutch
(c) centrifugal clutch (d) cone clutch

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : The clutch used in scooters is multi-plate clutch.

607. The cone clutches have become obsolete because–

(a) strict requirement of coaxiality of two shafts
(b) difficult to disengage
(c) difficult construction
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : The cone clutches have become obsolete because strict requirement of coaxiality of two shafts.

608. In the beginning of engagement of a centrifugal clutch,

(a) the centrifugal force on shoe is equal to spring force
(b) the centrifugal force on shoe is less than spring force
(c) the centrifugal force on shoe is slightly more than spring force
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : In the beginning of engagement of a centrifugal clutch, the centrifugal force on shoe is slightly more than spring force.

609. In case of multi-disk clutches, oil is used,

(a) to reduce the heat
(b) to carry away the heat
(c) to lubricate the contacting surfaces
(d) for all above functions

Nagaland PSC CTSE 2017, Paper-I

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Ans. (b) : In case of multi-disk clutches, oil is used, to carry away the heat.

610. For a single row angular contact ball bearing, the basic dynamic capacity is 55 kN and the dynamic equivalent load is 3950 N. The rated life of the bearing is:
(a) 2700 Million revolutions
(b) 2200 Million revolutions
(c) 1700 Million revolutions
(d) 1200 Million revolutions

JWM 2017

Ans. (a) : Basic dynamic capacity = 55 kN
Dynamic equivalent load = 3950 N
For ball bearing, n = 3

$$\text{Life} = \left(\frac{\text{Dynamic capacity}}{\text{Dynamic equivalent load}}\right)^n$$

$$= \left(\frac{55\times10^3}{3950}\right)^3$$

Life = 2699.58 ≃ 2700 Million revolution

611. Single plate clutch is used in
(a) four wheelers
(b) two wheelers
(c) mopeds
(d) applications where initial torque is high

TNPSC AE 2017

Ans. (a) : Application of single plate clutch—Single plate clutches are used in trucks, buses and cars etc.
- Single plate clutches used where large radial space is available.
- As sufficient surface area is available for the heat dissipation in single plates clutches, no cooling oil is required. Therefore, single plate clutches are dry type.

612. _____are welded to the rear wheel house panel, the floor panel and the rear of the rocker panel in a car.
(a) Rear doors (b) Rear windows
(c) Rear quarter panels (d) Trunk lid

TNPSC AE 2017

Ans. (c) : Rear quarter panels are welded to the rear wheel house panel, the floor panel and the rear of the rocker panel in a car.

613. There are three types of Disc Brake
(a) Fixed Caliper, Tab-Action and Two-Piston
(b) Fixed Caliper, Sliding Caliper and Floating Caliper
(c) Floating Caliper, Swinging Caliper and Proportioning Caliper
(d) Fixed caliper, floating caliper and Swinging caliper

TNPSC AE 2017

Ans. (b) : There are three types of Disc Brake fixed caliper, sliding caliper and floating caliper.

614. The sliding dog clutch in the constant mesh gear box is to transmit the power from the
(a) Primary shaft to lay shaft
(b) Lay shaft to output shaft
(c) Primary shaft to output shaft
(d) None of the above

TNPSC AE 2017

Ans. (c) : The sliding dog clutch in the constant mesh gear box is to transmit the power from the Primary shaft to output shaft.

615. In a fully automated centrifugal clutch the reaction plate is installed in between
(a) The pressure plate and cover pressing
(b) The pressure plate and fly wheel
(c) The pressure plate and the driven plate
(d) Cover processing and bob weight

TNPSC AE 2017

Ans. (a) : In a fully automated centrifugal clutch the reaction plate is installed in between The pressure plate and cover pressing.

616. In hydrostatic bearing, pressure to lubricant is supplied by
(a) External sources
(b) Partially external and partially from rotation of journal
(c) Not Supplied by external sources
(d) Shaft driven pump

Vizag Steel (MT) 2017

Ans. (a): Hydrostatic bearings are externally pressurized fluid bearings, where the fluid is usually oil, water. The pressurization is done by a pump.

617. Which of the following bearings are most suitable for supporting high axial thrust?
(a) Radial ball bearings
(b) Needle bearings
(c) Cylindrical roller bearings
(d) Tapered roller bearings

BPSC AE Mains 2017 Paper - VI

Ans : (d) : Tapered roller bearings.

618. The brake pipe in hydraulic brakes is made up of :
(a) PMC (b) Steel
(c) Carbon (d) Plastic

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : The brake pipe in hydraulic brakes is made up of steel.

619. Oil in journal bearing should be applied at the point where load is ..... .
(a) nil or lightest
(b) maximum
(c) average
(d) None of three mentioned here

GWSSB DEE 07.07.2016

Ans. (a) :

620. The parts of the clutch cover assembly that hold the pressure against the clutch plate are the :
(a) Release levers (b) Thrust bearings
(c) Struts (d) Springs

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d)

621. Braking is produced by the frictional effect between the brake drum and the :
(a) Wheel Studs
(b) Wheel rim
(c) Brake shoes
(d) Wheel cylinder pistons

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c)

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622. In a ventilated disc brake :
(a) A duct directs air towards the caliper for cooling while the vehicle is moving
(b) Caliper is covered with cooling fins
(c) Disc contains many small holes for optimum cooling performance
(d) Disc contains radial vanes between its rubbing surfaces for optimum cooling performance

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (d)

623. In the case of flat pivot bearing the rubbing velocity is :
(a) maximum at the centre of the contact area
(b) zero at the centre and maximum at the outer radius
(c) uniform throughout the contact area
(d) zero at the outer radius

Kerala PSC IOF 19.04.2016

Ans. (b)

624. Which of the following is an example of friction clutch?
(a) disc (b) cone
(c) plate (d) all of the above

Kerala PSC IOF 19.04.2016

Ans. (d) all of the above

625. For new clutches, it is generally assumed that :
(a) Wear is uniform
(b) Pressure is uniform
(c) Both pressure and wear are uniform
(d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (b) : For new clutches, uniform pressure theory is used. When the mating component in clutch, bearing are new, then the contact between surface may be good over the whole surface. It means that the pressure over the rubbing surfaces is uniformly distributed.

626. Which of the following dynamometer is used for measuring large power along the propeller shaft of turbine
(a) Rope brake dynamometer
(b) Belt transmission dynamometer
(c) Electric dynamometer
(d) Torsion dynamometer

RPSC VPITI 14.02.2016

Ans. (d) : Torsion dynamometer is used for measuring large power along the propeller shaft of the turbine.
Types of dynamometer–
1. Absorption dynamometer
   - Prony brake dynamometer
   - Rope brake dynamometer
2. Transmission dynamometer
   - Belt transmission dynamometer
   - Epicyclic dynamometer
   - Torsion dynamometer

627. Which among the following is advantage of a positive clutch?
(a) They can be engaged at high speed
(b) When both driven and drive shafts are at rest, some relative motion may be required to engage
(c) Due to no slip between contact surfaces no heat is generated
(d) Shock accompanies engagement at any speed

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : In positive clutch power transmission is achieved by means of interlocking of jaws or teeth.
Example – Jaw clutch, Toothed clutches
- It can transmit large torque with no slip and no heat generated during engagement/ disengagement.

628. The reliability of ball bearing selected from manufacture's catalogue is:
(a) 90% (b) 99%
(c) 50% (d) more than 90%

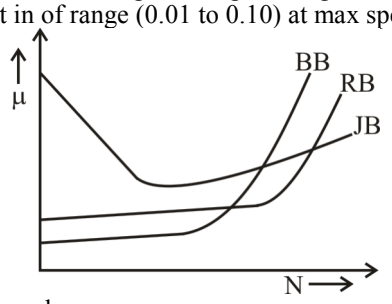
GPSC Asstt. Prof. 28.08.2016

Ans. (a) : The basic rating life L_{10} is a service life of 90% reliability, when used under normal usage conditions for bearings of high manufacturing quality where the inside of the bearing is of a standard design made from bearing steel materials.

629. Which bearing has a lower coefficient of friction at high speed:
(a) Ball bearing
(b) Roller bearing
(c) Spherical roller bearing
(d) Sliding bearing

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : A sliding bearing having a friction of coefficient in of range (0.01 to 0.10) at max speed.



For high speed

$$\mu_{BB} > \mu_{RB} > \mu_{JB}$$

JB = Journal bearing (sliding bearing)
RB = Roller bearing
BB = Ball bearing

630. The life of a ball bearing at a load of 20 KN is 27000 hours. If the load is increased to 30 KN, keeping all other conditions same, what is its life (in hours)?
(a) 4000 (b) 6000
(c) 8000 (d) 10000
(e) 12000

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know,

$$L \propto \left[\frac{1}{P}\right]^3$$

So, $$L_2 = L_1\left[\frac{P_1}{P_2}\right]^3$$

$$= 27000\left[\frac{20000}{30000}\right]^3 = 27000 \times 0.296$$

$$\boxed{L_2 = 8000\,\text{hr}}$$

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631. Axial operation jaw clutches having self-locking tooth profile:
(a) can work only with load
(b) can be engaged only when unloaded
(c) can be disengaged only when unloaded
(d) can be disengaged at any speed
(e) can either work only with load or can be disengaged only when unloaded

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) : Claw clutch is a mechanical clutch in which jaws or claws interlock when pushed together-called also positive clutch. Axial operations, having self locking tooth profile can be disengaged at any speed.

632. For good bearing material :
1. the material should have a satisfactory compressive and fatigue strength
2. The material should be soft, with a low modulus of elasticity and a low melting point.
Choose most appropriate option.
(a) Statement 1 is right and 2 is wrong
(b) Statement 1 is wrong and 2 is right
(c) Both statements 1 and 2 is right
(d) Both statements 1 and 2 are wrong

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : For a good bearing material.
1. The material should have a satisfactory compressive and Fatigue strength.
2. The material should be soft with a low modulus of elasticity and a low melting point.
Hence both statements are correct.

633. Which of the following materials is not suitable for bearings or brushes used in automotives?
(a) Gun metal
(b) Admiralty brass
(c) Bronze and phosphorous bronze
(d) Both option (a) and (c)

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Gun metal – Steam and hydraulic castings, gears, valves, pump and steam fittings, bearing and housing.
Admiralty brass – Heat exchanger tubes, condenser tubes, pump barrel.
Phosphorous bronze- Springs, bolts, bushing and bearings, electrical switches with moving or sliding parts, dental bridges etc.

634. This is not a friction material used in brakes and clutches.
(a) Waven-cotton lining
(b) Woven asbestos lining
(c) Glass-wool cotton lining
(d) Sintered or cermet metal pad

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Friction material used in brakes and clutches of –
(1) Sintered or cermet metal pad.
(2) Woven asbestos lining
(3) Waven –cotton lining
Glass-wool cotton lining is not used in brakes and clutches.

635. …… is used to engage and disengage the engine from transmission.
(a) Gear box (b) Propeller shaft
(c) Differential (d) Clutch

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Clutch is used to engage and disengage the engine from transmission.

636. An engine developing 45 kW at 1000 rpm is fitted with a cone clutch built inside the flywheel. The cone has a face angle of 12.5° and a maximum mean diameter of 500 mm. The coefficient of friction is 0.2. The normal pressure on the clutch face is not to exceed 0.1 N/mm^2. The face width required, and the axial spring force necessary to engage the clutch will be respectively
(a) 54.7 mm and 2277.5 N
(b) 109.4 mm and 2290 N
(c) 54.7 mm and 1145 N
(d) 109.4 mm and 1145 N
(e) None of these

CGPSC AE 16.10.2016

Ans. (*) : Given, P = 45 kW, N = 1000 rpm, $\alpha = 12.5°$
D = 2R, R = 250 mm, $\mu = 0.2$, $P_0 = 0.1$

$$T = \frac{P \times 60}{2\pi N} = \frac{45 \times 10^3 \times 60}{2\pi \times 1000} = 430\,\text{Nm}$$

$T = 2\pi\mu P_0 . R^2 b$
$430 \times 10^3 = 2\pi \times 0.2 \times 0.1 \times 250^2 \times b$
$b = 54.7$ mm
$W_n = P_0 \times 2\pi R.b = 0.1 \times 2\pi \times 250 \times 54.7$
$W_n = 8640$ N
$W_A = W_n(\sin\alpha + 0.25\mu \cos\alpha)$
$= 8640(\sin 12.5 + 0.25 \times 0.2 \times \cos 12.5)$
$= 2290$ N
Note-This question has been deleted by CGPSC.

637. A simple band brake is applied to a rotating drum of 750 mm diameter. The angle of lap of the band on the drum is 270°. One end of the band is attached to the fulcrum and the other end to a pin at a distance of 90 mm from the fulcrum. If the coefficient of friction is 0.25 and a braking force of 54 N is applied at a distance of 900 mm from the fulcrum, the braking torque when the drum rotates clockwise and when it rotates anti clockwise, will be respectively
(a) 140.25 Nm and 456.2 Nm
(b) 456.2 Nm and 140.25 Nm
(c) 304.15 Nm and 93.5 Nm
(d) 93.5 Nm and 304.15 Nm
(e) None of these

CGPSC AE 16.10.2016

Ans. (a) : Given, P = 54 N

$$\theta = 270° = 270 \times \frac{\pi}{180} = 4.713\,\text{rad}$$

Drum diameter, d = 750 mm $\Rightarrow$ r = 375 mm
For clockwise rotation,
Taking moment about A, $\sum M_A = 0$
$T_1 \times 90 = P \times 900$
$T_1 \times 90 = 54 \times 900$
$\boxed{T_1 = 540\,\text{N}}$

From, $\frac{T_1}{T_2} = e^{\mu\theta} = e^{(0.25 \times 4.713)}$

$\boxed{T_2 = 166.22\,\text{N}}$

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Braking torque, $T = (T_1 - T_2) \times r$

$= (540 - 166.22) \times 0.375$

$T = 140.16$ N-m

For anticlock wise, $T_2 = 540$ N

$$\frac{T_1}{T_2} = e^{(0.25\times4.713)} \Rightarrow T_1 = 1754.28\,N$$

Braking torque, $T = (T_1 - T_2) \times r = 455.358$ N-m

638. A slender shaft supported on two bearing at its ends carries a disc with an eccentricity e from the axis of rotation. The critical speed of the shaft is N. If the disc is replaced by a second one of the same weight but mounted with an eccentricity 2e. critical speed of the shaft in the second case is ___

(a) 2 N (b) $\frac{1}{2}$ N

(c) $\frac{1}{\sqrt{2}}$ N (d) N

UPRVUNL AE 21.08.2016

Ans. : (d) Critical speed of shaft is independent of the eccentricity.

639. The load-carrying surface of the bearing are separated by a relatively thick film of lubricant, so as to prevent metal-to-metal contact, such lubrication is called

(a) hydrostatic (b) hydrodynamic

(c) boundary (d) none of these

GPSC Lect. 23.10.2016

Ans. (b) : Hydrodynamic lubrication is a term that defines a situation in which two rubbing surfaces are separated by a thick film of a lubricant. This situation is often beneficial and lubrication is used to reduce friction and/or wear of rubbings solids.

640. In a ball bearing, if the number of balls is reduced to half and diameter of the ball is increased to 4 times, then static load capacity of the ball bearing is

(a) reduced 4 times (b) reduced 8 times

(c) increased 4 times (d) increased 8 times

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Stribeck's equation gives the static load capacity of bearing.

$$C = \frac{kd^2z}{5}$$

Given that– $Z_2 = \frac{Z_1}{2}$ and $d_2 = 4d_1$

$$\frac{C_2}{C_1} = \left(\frac{d_2}{d_1}\right)^2\left(\frac{Z_2}{Z_1}\right)$$

$$= \left(\frac{1}{2}\right)(4)^2$$

$$C_2 = 8\,C_1$$

641. According to uniform wear theory the torque developed by a disc clutch is given by

(a) $T = 0.25\,\mu.W.R$ (b) $T = 0.5\,\mu.W.R$

(c) $T = 0.75\,\mu.W.R$ (d) $T = \mu.W.R$

Where W = Axial force with which the friction surfaces are held together;

μ = Coefficient of friction; and

R = Mean radius of friction surfaces

Nagaland CTSE 2016 Ist Paper

Ans. (d) : Torque for uniform wear rate, where

W = Axial force with which friction surfaces are held together

μ = Coefficient of friction

R = Mean radius of friction surfaces

$$R = \frac{r_1 + r_2}{2}$$

$T = \mu.W.R.$

642. The ratio of torque transmitting capacity of single cone clutch to the single plate clutch is equal to

(a) 1 (b) 2.5

(c) 3.62 (d) 4.62

Nagaland CTSE 2016 Ist Paper

Ans. (a) : Ratio of torque transmitting capacity of single cone clutch to the single plate clutch is equal to; 1

1) For, Single cone clutch, $T = \mu WR$

(Torque)

R = mean radius of clutch

2) For, Single plate clutch, $T = n\mu WR$

n = 2

643. Braking torque on the drum of shoe brake is given by

(a) (Normal reaction on the block) × (radius of drum)

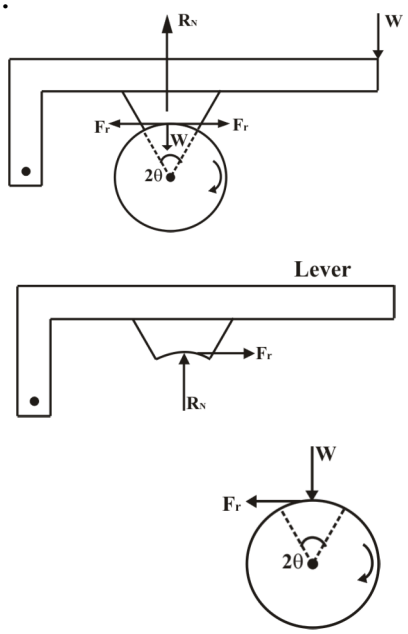
(b) (Frictional force on the block) × (radius of drum)

(c) (Force applied at the level end) × (radius of drum)

(d) 2 × (Force applied at the level end) × (radius of drum)

RPSC LECTURER 16.01.2016

Ans. (b) :



So, braking torque on the drum of shoe brake is given as

T = Frictional force on the block × radius of drum

$T = F_r \times R$

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644. The total normal force (F) acting on the area of contact of disc clutch (Assuming uniform distribution of interface pressure) is: (where P is the Intensity of pressure, r_o and r_i outer and inner diameters of disc respectively)

(a) $F = 2\pi \mathrm{Pr}_o(r_o - r_i)$ (b) $F = 2\pi \mathrm{Pr}_i(r_o - r_i)$

(c) $F = \pi \mathrm{P}(r_0^2 - r_i^2)$ (d) $F = \pi \mathrm{P}(r_o^4 - r_i^4)^{1/2}$

UPRVUNL AE 07.10.2016

Ans. (c) : The total normal force (F) acting on the area of contact of disc clutch is

$$F = \pi P(r_o^2 - r_i^2)$$

645. Journal bearing works satisfactory with L/D ratio in the range of: (where L = length of bearing, D = diameter of journal)

(a) 10-20 (b) 0.01-0.1

(c) 1-2 (d) 0.1-0.2

UPRVUNL AE 07.10.2016

Ans. (c) : Journal bearing works satisfactory with L/D ratio in the range of 1-2.

646. The machine members which are used to gradually transmit motion of a driving member to another member at rest, without shock, are termed as

(a) bearings (b) couplings

(c) clutches (d) cotters

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Clutches are used to gradually transmit motion of a driving member to another member at rest, without shock.

647. As per the uniform wear theory, brakes and clutches friction radius is equal to :

(a) $\frac{R + r}{2}$ (b) $\frac{R^2 + r^2}{2}$

(c) $\frac{R^2 + r^2}{2Rr}$ (d) R + r

TSPSC Managers, 2015

Ans. (a) : As per the uniform wear theory, brakes and clutches friction radius is equal to

$$R_f = \frac{R + r}{2}$$

As per the uniform pressure theory or a new clutch–

$$R_f = \frac{2(R^3 - r^3)}{3(R^2 - r^2)}$$

648. In a single plate clutch $\mu F, r_0$ and r_i are coefficient of friction, axial force, outer radius and inner radius respectively. The frictional torque using uniform wear theory is equal to

(a) $(1/2)\mu F(r_o + r_i)$ (b) $(2/3)\mu F(r_o + r_i)$

(c) $\mu F(r_o + r_i)$ (d) $2\mu F(r_i + r_i)$

VIZAG MT 2015

Ans. (a) : The frictional torque using uniform wear theory is equal to

$$T_f = \frac{1}{2}\mu F(r_o + r_i)$$

649. The thrust of propeller shaft is absorbed by 6 collars. The rubbing surfaces of these collars have outer diameter 300 mm and inner diameter 200 mm. If the bearing pressure amounts to 0.4 N/mm², the axial load acting on the bearing is (in Newton):

(a) 84260 (b) 86420

(c) 94260 (d) 96240

NPCIL ET, 2015

Ans. (c) : Given- No. of collars (n) = 6

Outer diameter (D) = 300 mm

Inner diameter (d) = 200 mm

R = 150 mm, r = 100 mm

Bearing pressure (P) = 0.4 N/mm²

$$P = \frac{\text{Load(W)}}{n\pi(R^2 - r^2)}$$

$$0.4 = \frac{W}{6 \times \pi\left[(150)^2 - (100)^2\right]}$$

$W = 0.4 \times 6 \times \pi\,[(150)^2 - (100)^2]$

$W = 94247.77N$

650. Which of the following bearing has a low starting friction?

(a) Ball bearing

(b) Roller bearing

(c) Journal bearing

(d) Taper roller bearing

KPSC ADF 2015

Ans. (a) : Ball bearing has a low starting friction.

651. The ratio of tight and slack side tensions in a V-belt or rope is :

(a) $e^{\mu\theta \sin\alpha}$ (b) $\frac{\mu\theta}{e^{\cos\alpha}}$

(c) $e^{\mu\theta \cos\alpha}$ (d) $\frac{\mu\theta}{e^{\sin\alpha}}$

KPSC ADF 2015

Ans. (d) : Let, T_1 = tension in the V-belt/rope tight side

T_2 = tension in the V-belt/rope slack side

$$\therefore \quad \boxed{\frac{T_1}{T_2} = (e)^{\frac{\mu\theta}{\sin\alpha}}}$$

652. The uniform wear theory for disc clutches states that :

(a) frictional force per unit area is uniform all over the disc surface

(b) pressure is proportional to the velocity of sliding

(c) pressure is inversely proportional to radius

(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Pressure distribution (P) for uniform wear theory

$$P = \frac{W}{2\pi R_i(R_o - R_i)}$$

$$P \propto \frac{1}{R_i}$$

$W \rightarrow$ Axial Load

$R_o \rightarrow$ Outer radius

$R_i \rightarrow$ Inner radius

So, Pressure distribution is inversely proportional to radius in uniform wear theory.

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653. In a flat pivot bearing, the total moment of friction force

(a) for uniform wear is greater than that for uniform pressure
(b) for uniform wear is lesser than that for uniform pressure
(c) for uniform wear is equal to that of uniform pressure
(d) for uniform wear may be more or less and cannot be predicted

TSGENCO AE 14.11.2015

Ans. (b) : Flat pivot bearing – Moment of friction force is a torque on bearing.
For uniform pressure – total frictional torque on bearing

$$T_p = \frac{2}{3}\mu WR$$

For uniform wear – Total friction torque on bearing

$$T_w = \frac{1}{2}\mu WR$$

$$\frac{T_p}{T_w} = \frac{4}{3} = 1.33$$

Hence torque transmitted under uniform pressure condition is 33% more than uniform wear condition.

654. For a band brake, the width of the band for a drum diameter greater than 1m, should not exceed

(a) 150 mm (b) 200 mm
(c) 250 mm (d) 300 mm

MPPSC AE 08.11.2015

Ans. (a) : For a band brake, the width of the band for a drum diameter greater than 1m, should not exceed 150 mm.

655. Which of the following is not a correct statement?

(a) The balls in a ball bearing remain in position due to centrifugal force
(b) cage is made hardest of all parts in ball bearings
(c) Spherical roller bearings have the capacity for self-alignment
(d) Roller bearing have higher load capacity than that of ball bearings for a given overall size

(KPSC AE 2015)

Ans : (a) Following correct statement are:-
(i) Cage is made hardest have the capacity for self-alignment
(ii) Spherical roller bearing have the capacity for self - alignment
(iii) Roller bearing have higher load capacity than that of ball bearing for a given overall size.

656. To increase the capacity of ball bearing ..... is provided to increase the number of balls in the bearing :

(a) Grease (b) Collar
(c) Filling notch (d) Less gap

OPSC AEE 2015 Paper-I

Ans : (c) To increase the capacity of ball bearing filling notch is provided to increase the number of balls in the bearing.

657. A 150 mm diameter shaft supporting a load of 10 kN has a speed of 1500 rpm. The shaft runs in a bearing whose length is 1.5 times the shaft diameter. The coefficient of friction is 0.02. The bearing pressure is _____ N/mm^2

(a) 0.24 (b) 0.296
(c) 0.34 (d) 0.46

MPSC Poly. Lect. 2014

Ans. (b) : Given, d = 150 mm
W = 10 kN
N = 1500 rpm
L = 1.5 d
$\mu = 0.02$

$$\text{Bearing Pressure (P)} = \frac{W}{Lxd} = \frac{10\times10^3}{1.5\times150}$$

$$= 0.29629 \ \frac{N}{mm^2}$$

658. Which statement is true from the following statements for full journal bearing?

(a) The temperature rise due to fluid shear increases on increase of Sommerfield number and it is more for longer bearing
(b) The temperature rise due to fluid shear increases on increase of Sommerfield number and it is less for longer bearing
(c) The temperature rise due to fluid shear decreases on increase of Sommerfield number and it is more for longer bearing
(d) The temperature rise due to fluid shear decreases on increase of Sommerfield number and it is less for longer bearing

HPPSC Asstt. Prof. 2014

Ans. (b) : The temperature rise due to fluid shear increases on increase of Sommerfield Number and it is less for longer bearing.

659. In a tilted thrust bearing operating under hydrodynamic lubrication, the maximum pressure lies:

(a) at the exit point
(b) at the entry point
(c) in between the entry and exit point
(d) at entry or exit point depending on the direction of rotation of the shaft

HPPSC Asstt. Prof. 2014

Ans. (c) : In tilted hydrodynamic bearing the maximum pressure lies in between the entry and exit point.

660. What assumption regarding the lubricant film is made in Petroff's equation?

(a) It is converging
(b) It is diverging
(c) It is uniform
(d) It is converging or diverging

UPRVUNL AE 2014

Ans : (c) In Petroff's equation, the lubricant film distributed uniformally.
Petroff's equation is defined as the group of dimensionless parameters that govern the frictional properties of the journal bearing. This equation is based on the following assumptions.
(1) The shaft is concentric with the bearing.
(2) The bearing is subjected to light load.

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661. In an internal expanding shoe brake, more than 50% of the total braking torque is supplied by ____.
(a) leading shoe
(b) tailing shoe
(c) leading as well as trailing shoe
(d) none of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : In an internal expanding shoe brake more than 50% of the total braking torque is supplied by leading shoe.

662. A multi-disc clutch has three discs on the driving shaft and two discs on the driven shaft. The number of pairs of contact surfaces is
(a) 2 (b) 3
(c) 4 (d) 5

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c)

$$\boxed{n = n_1 + n_2 - 1}$$

n = No. of pair contact surface
n_1 = No. of pair on driven.
n_2 = No. of pair on driver
Given
$n_1 = 3$
$n_2 = 2$
$n = 3 + 2 - 1$
$n = 4$

663. A journal bearing with hydrodynamic lubrication is running steadily with a certain amount of minimum film thickness. When the load and speed are doubled, how does the minimum film thickness vary?
(a) remains unchanged
(b) gets doubled
(c) gets reduced to one fourth of original value
(d) gets reduced to half of original value

TNPSC AE 2014

Ans. (a) : remains unchanged

664. A 15 cm shaft turns 900 rpm in a journal bearing of length 20 cm. If the load on the bearing is 3×10^4 N, the bearing pressure will be approximately
(a) 75 N/cm^2 (b) 100 N/cm^2
(c) 170 N/cm^2 (d) 32 N/cm^2

TNPSC AE 2014

Ans. (b) : Data Given-
d = 15 cm, N = 900 rpm
L = 20 cm, W = 3×10^4 N
We know that bearing pressure is given as

$$P = \frac{W}{D \times L} = \frac{3 \times 10000}{15 \times 20}$$

$$\boxed{P = 100\,N/cm^2}$$

665. Which of the following does not belong to the category of sliding contact bearing?
(a) pivot bearing (b) ball bearing
(c) bush bearing (d) foot step bearing

TNPSC AE 2014

Ans. (b) : Pivot bearing, bush bearing and foot-step bearing are belong to the category of sliding contact bearing where as ball bearing and roller bearing are belong to the category of rolling type bearing.

666. A journal bearing running at 900 rev/min has a bearing pressure of 100 N/cm^2. If the lubricant used has absolute viscosity of 15 centipoise, the bearing characteristic number will be equal to
(a) 29.8 (b) 13.5
(c) 9.94 (d) 2.85

TNPSC AE 2014

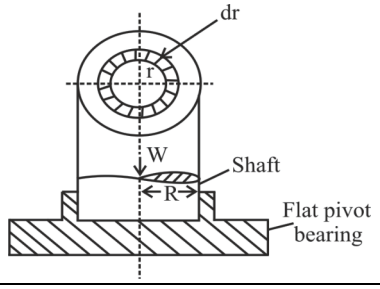
Ans. (b) : Bearing characteristics number $= \frac{ZN}{P}$
Z = Absolute viscosity of lubricant (kg/m.s)
N = Speed of the journals (rpm)
P = bearing pressure (N/mm^2)
It is non-dimensional number and used in design of journal bearing.
Z = 15 centi poise
1CP = 0.001 kg/ms
N = 900 rev/min
P = 100 N/cm^2 $\frac{100}{100}$ N/mm^2
P = 1N/mm^2

$$BCN = \frac{ZN}{P} = \frac{0.015 \times 900}{1} = 13.5$$

667. In case of pivot bearing, the wear is :
(a) maximum at the centre
(b) zero at the centre
(c) uniform throughout the contact are
(d) minimum at maximum radius

(HPPSC AE 2014)

Ans : (b) In case of pivot bearing, the wear is zero at the centre.
When a vertical shaft rotates in a flat pivot bearing (known as foot step bearing), the sliding friction will be along the surface of contact between the shaft and the bearing.
Let
W = Load transmitted over the bearing surface
R = Radius of bearing surface
p = Intensity of pressure per unit area of bearing surface between rubbing surfaces
μ = Coefficient of friction



668. The rated life of a bearing under a given load (L) is inversely proportional to the
(a) cube of load (b) cubic root of load
(c) square of load (d) square root of load

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (a) : Life of bearing–The approximate rating of serve life of a ball or roller bearing is based on the given fundamental equation.

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$$L = \left(\frac{C}{P}\right)^k \times 10^6 \text{ revolution} \begin{cases} \text{Where L is rating....} \\ C = S \text{ is a basic dynamic load} \\ P = \text{Equivalent dynamic load} \end{cases}$$

For a ball bearing

$$L \propto \frac{1}{P^k} \text{ or life} \propto \frac{1}{(\text{Load})^k} \begin{cases} k = 3 \text{ ball bearing} \\ k = \frac{10}{3} \text{ roller bearing} \end{cases}$$

669. Which one of the following bearings is most suitable for carrying very heavy loads with slow speed?
(a) hydrodynamic bearing
(b) ball bearing
(c) roller bearing
(d) hydrostatic bearing

Mizoram PSC AE/SDO 2014, Paper-II

Ans. (d) : Hydrostatic bearings can support large load without journal rotation and provide large (accurate and controllable) direct stiffness as well as damping (energy dissipation) coefficients.

670. The type of brakes commonly used in automobiles is :
(a) Shoe brake
(b) Bank brake
(c) Band and Block brake
(d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) :

671. Which type of spring is usually used for an automobile clutch?
(a) Spiral spring
(b) Leaf spring
(c) Closed coil helical spring
(d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) :

672. For a band brake, the width of drum diameter greater than 1 exceed
(a) 150 mm (b) 200 mm
(c) 250 mm (d) 300 mm

APPSC Poly. Lect. 2013

Ans. (a) : For a band brake, the width of the band for a drum diameter greater than 1, should not exceed 150 mm.

673. A bearing is designated by the number X410. It means that it is a bearing of.
(a) light series with bore diameter of 10mm
(b) heavy series with bore diameter of 50 mm
(c) medium series with bore diameter of 50 mm.
(d) light series with bore diameter of 50mm

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Bearing number X410 – heavy series with bore dia of 50 mm.

674. Brake shoe applied on rotating train wheel is an example of :
(a) dry friction (b) boundary friction
(c) fluid friction (d) mixed friction

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Brake shoe applied on rotating train wheel is an example of dry friction.

675. In hydrodynamic bearings usual clearance provided per mm of diameter of shaft is:
(a) 0.01 micron (b) 0.1 micron
(c) 1 micron (d) 10 micron

SJVN ET 2013

Ans. (c) : In hydro dynamic bearings usual clearance provided per mm of diameter of shaft is 1 micron.

676. In block brakes, the ratio of shoe width and wheel diameter is kept between
(a) 0.1 and 0.25 (b) 0.25 and 0.50
(c) 0.50 and 0.75 (d) 0.75 and 1.0

SJVN ET 2013

Ans. (b) : In block brakes, the ratio of shoe width and wheel diameter is kept between 0.25 and 0.50.

677. The total frictional torque for a thrust bearing is given by _____ .
Where μ is coefficient of friction, W – load transmitted, R – outer radius of collar and r – inner radius of coller.

(a) $\frac{1}{3}\mu W\left[\frac{R^3 - r^4}{R^2 - r^2}\right]$ (b) $\mu W\left[\frac{R^3 - r^2}{R^2 - r^2}\right]$

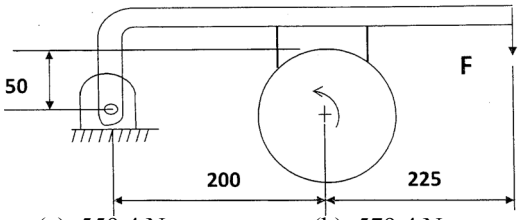
(c) $2\mu W\left[\frac{R^3 - r^3}{R^2 - r^2}\right]$ (d) $\frac{2}{3}\mu W\left[\frac{R^3 - r^3}{R^2 - r^2}\right]$

TNPSC AE 2013

Ans. (d) : $T_f = \frac{2}{3}\mu W\left[\frac{R^3 - r^3}{R^2 - r^2}\right]$ .... (UPT)

$T_f = \mu W\left[\frac{R + r}{2}\right]$ .... (UWT)

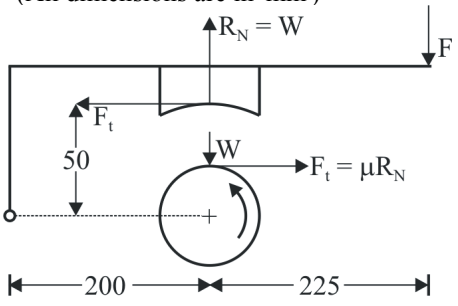
678. A block brake with 400 mm diameter is used to brake a torque of 100 Nm as shown in the figure. If the coefficient friction is 0.25 at the brake surface what is the value of force F to be applied at the end of the lever.



(a) 559.4 N (b) 579.4 N
(c) 439.4 N (d) 1000 N

ISRO Scientist/Engineer 12.05.2013

Ans. (d) : Given,
Block Brake
Diameter d = 400 mm
Torque T = 100 Nm
Coefficient of friction μ = 0.25
(All dimensions are in 'mm')



Torque, T = 100 Nm

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$$r = \frac{d}{2} = \frac{400}{2} = 200 \text{ mm} = 0.2 \text{ m}$$

$$\therefore \quad F_t = \frac{T}{r} = \frac{100}{0.2} = 500\text{N}$$

$$F_t = \mu R_N$$

$$R_N = \frac{F_t}{\mu} = \frac{500}{0.25} = 2000\text{N}$$

$$\sum M_P = 0$$

$$F(200+225) = R_N \times 200 + F_t \times 50$$

or $$F \times 425 = 2000 \times 200 + 500 \times 50$$

$$F = 1000\text{N}$$

679. Complete the following with the correct alternative : Rolling contact bearing as compared to sliding contact bearing…
(a) has lower starting torque
(b) requires considerable axial space
(c) generate less noise
(d) are costly

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Rolling contact bearing as compared to sliding contact bearing has lower starting torque.
- Low starting and low running friction except at very high speed.
- Accuracy of soft alignment.
- Small overall dimension.

680. The rolling element bearings are
(a) hydrostatic bearings
(b) squeeze film bearings
(c) antifriction bearings
(d) grease lubrication bearings

TRB Asstt. Prof., 2012
GPSC DEE Class-2 (GWSSB) 04.07.2021

Ans. (c) : An antifriction bearing, also known as a rolling contact bearing is justified over a journal or fluid film bearing when very little friction is needed for low differential surface speeds of course, the extra mechanical complexity of antifriction bearing drives up their cast when compared to similar journal bearing.

681. Babbit lining is used on brass bearings to
(a) increase bearing resistance
(b) increase compressive strength
(c) provide any friction properties
(d) increase wear resistance

TRB Asstt. Prof., 2012

Ans. (d) : Babbit lining is used on brass bearings to increase wear resistance.

682. Which bearing offers lowest friction
(a) hydrostatic Bearing
(b) Roller Bearing
(c) Acro-static Bearing
(d) Hydrodynamic Bearing

ISRO Scientist/Engineer 2012

Ans. (c) : Acro-static bearing offers lowest friction.

683. Which of the following brakes is commonly used in motor cars ?
(a) Band Brake
(b) Shoe Brake
(c) Internal expanding Shoe Brake
(d) All the above

UKPSC AE 2012 Paper-I

Ans. (c) : Internal expanding Shoe Brake

684. The frictional torque transmitted in a conical pivot bearing considering uniform wear is
(a) $\frac{1}{2} \times \mu WR \operatorname{cosec}\alpha$
(b) $\frac{2}{3} \times \mu WR \operatorname{cosec}\alpha$
(c) $\frac{3}{4} \times \mu WR \operatorname{cosec}\alpha$
(d) $\mu WR \operatorname{cosec}\alpha$

Where,
μ = co efficient of friction
W = load over the bearing
R = radius of the shaft

TNPSC ACF 2012

Ans. (a) : flat color bearing to conical pivot bearing μ_{cone}- μ_{flate}
According to uniform wear is

$$T_f = \frac{1}{2}\frac{r}{\sin\alpha} R_{eff} W$$

$$\boxed{T_F = \frac{1}{2}\mu W R_{eff} \operatorname{cosec}\alpha}$$

685. Ball bearing threads are used in a
(a) screw jack
(b) machine tool feed mechanism
(c) screw press
(d) shaft straightener

TNPSC ACF 2012

Ans. (b) : Ball bearing threads are used in a machine tool feed mechanism.

686. A localised compressive stress at the area of contact between two members is known as
(a) Shear (b) Crushing
(c) Bending (d) Tensile

APPSC AEE 2012

Ans. (b) : Crushing stress- Crushing stress (bearing stress) is a special type of localized compressive stress which occurs at the surface of contact of two members that are relatively at rest. It is taken into account in the design of riveted joints, cotter joints, knuckle joints.

687. The brake commonly used on train boggies is :
(a) internal expanding
(b) band brake
(c) band and block brake
(d) shoe brake

RPSC ACF-2011

Ans. (d) : The brake commonly used in locomotive boggies is shoe brake.

688. Which of the following is an example of friction clutch?
(a) disc clutch (b) cone clutch
(c) centrifugal clutch (d) all of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : all of the above

689. Match List I with List II and select the correct answer using the code given below the Lists :

List I	List II
A. Single row deep groove ball bearing	1. Under heavy radial load without misalignment,
B. Spherical roller bearing	2. Under high radial load but limited space

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C.	Straight roller bearing	3. Under some radial as well as some Thrust load
D.	Needle bearing	4. Under heavy load with misalignment
		5. Under combined load but little misalignment

Code :	A	B	C	D
(a)	3	1	4	2
(b)	5	4	1	3
(c)	3	4	1	2
(d)	5	1	4	3

UPSC JWM Advt. No.-52/2010

Ans. (c) :

(A) Single row deep groove ball bearing →	Under some radial as well as some thrust
(B) Spherical roller bearing →	Under heavy load with misalignment
(C) Straight roller bearing →	Under heavy radial load without mis-alignment, sufficient space.
(D) Needle bearing →	Under high radial load but limited radial space.

690. Match List I with List II and select the correct answer using the code given below the Lists :

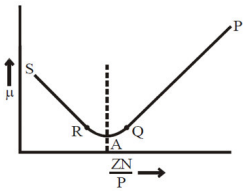
	List I	List II
A	Mixed film lubrication	1. Elasto hydrodynamic
B.	Full film lubrication	2. Wear rate virtually zero
C.	Boundary lubrication	3. Some asperity contact between the surfaces
D.	Hydrostatic lubrication	4. There is always some metal to metal contact in the surface

Code :	A	B	C	D
(a)	3	1	4	2
(b)	2	1	4	3
(c)	3	4	1	2
(d)	2	4	1	3

UPSC JWM Advt. No.-52/2010

Ans. (a) : A-3, B-1, C-4, D-2

691.



The variation of coefficient of friction (μ) with bearing characteristic number $\left(\frac{ZN}{P}\right)$ is shown in the figure. Match List I with List II and select the correct answer using the code given below the Lists :

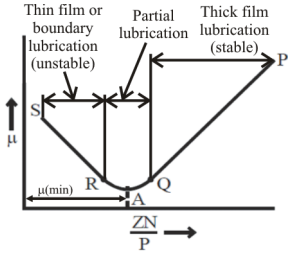
	List I		List II
A.	S–R	1.	Thin film or boundary lubrication (unstable)
B.	R–Q	2.	Partial lubrication
C.	A	3.	Limiting condition of bearing modulus
D.	Q–P	4.	Thick film lubrication (stable)

Code:

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	2	3	1
(c)	1	3	2	4	(d)	4	3	2	1

UPSC JWM Advt. No.-50/2010

Ans. (a) :



As we know,
- SR → is shows for thin film for unstable lubrication.
- From R to Q, it is for partial lubrication of bearing (journal bearing).
- 'A' shows the limiting condition of bearing modulus.

692. Which one of the following statement is correct?
In case of clutches, the uniform pressure theory:
(a) gives a higher friction torque than uniform wear theory
(b) gives a lower friction torque than uniform wear theory
(c) is applicable for old clutches
(d) is not applicable for multiplate clutches which are old

UPSC JWM Adv. No-16/2009

Ans. (a) : In case of clutches, the uniform pressure theory gives 33.33% higher friction torque than uniform wear theory.

693. Consider the following statements with respect to the square jaw clutch:
1. It has low slip.
2. Engagement at any speed is accompanied by shock.
3. Has low heat generation
4. Cannot be engaged at high speed.
Which of the above statements is/are correct?
(a) 1 only (b) 1 and 2
(c) 2 and 4 (d) 3 and 4

UPSC JWM Adv. No-16/2009

Ans. (c) : Positive Clutch–
(i) These are positive drives hence no slip occurs.
(ii) Sudden engagement hence impact and jerk.
(iii) Only limited to low speed application.
(iv) Clutch can not be engage when load.
(v) Power transmission capacity more.

694. Consider the following statement :
The fatigue life 'L' of rolling bearing is:
1. Inversely proportional to the third power of the load for ball bearing.
2. Inveserly proportional to the fourth power of the load for ball bearing
3. Inversely proportional to the 10/3 power for roller bearing
4. Inversely proportional to the third power of the load for roller bearing
Which of the above statements is/are correct?

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(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

UPSC JWM Adv. No-16/2009

Ans. (d) :

$\because$ Fatigue life $(L) = \left(\frac{C}{P_e}\right)^n$ million revolution

Where, n = 3 ; for ball bearing

$n = \frac{10}{3}$; for roller bearing

$\because$ C = Dynamic load capacity

P_e = Equivalent dynamic load

Therefore,

$\Rightarrow$ The fatigue life 'L' of rolling bearing is Inversely proportional to third power of the load for ball bearing.

$\Rightarrow$ And Inversely proportional to the 10/3 power for roller bearing.

695. A multiple disc clutch has 4 steel and 3 bronze discs and each surface has a mean radius of 60 mm. The coefficient of friction is 0.25. Assuming uniform wear, the power capacity of the clutch corresponding to a rotational speed of 400 rpm and an axial force of 300 N is

(a) 360 π watt (b) 270 π watt
(c) 180 π watt (d) 90 π watt

DRDO Scientists 2009

Ans. (a) : Power = $n\mu Fr\omega$

$$= \frac{6 \times 0.25 \times 300 \times 60 \times 10^{-3} \times 2\pi \times 400}{60}$$

= 360 π watt

696. Radial pressure in a hydrodynamic bearing is

(a) uniform
(b) independent of load
(c) maximum where film thickness is minimum
(d) maximum where film thickness is maximum

DRDO Scientists 2009

Ans. (c) : Radial pressure in a hydrodynamic bearing is maximum where film thickness is minimum.

697. Which of the following is a friction clutch?

(a) Cone clutch (b) Band clutch
(c) Disc clutch (d) All of these

TNPSC AE, 2008

Ans. (d) : In the friction clutches, power transmission is achieved by means of friction between contacting surface example. cone clutch, disc clutch, band clutch etc.

698. The purpose of providing multiple collars on a flat collar pivot bearing is to

(a) Increase frictional resistance
(b) To establish self-sustaining bearing conditions
(c) Distribute the frictional load because of limiting friction
(d) Distribute the axial load because of limiting bearing pressure on a collar

ISRO Scientist/Engineer 2008

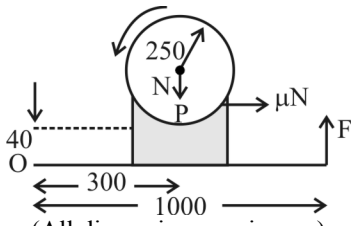
Ans. (d) : The purpose of providing multiple collars on a flat collar pivot bearing is to distribute the axial load because of limiting bearing pressure on a collar.

699. A 250 mm radius brake drum contacts a single shoe as shown in figure and sustains a torque of 300 Nm at 500 rpm. Coefficient of friction is 0.3. Force F required for braking in the case of counterclockwise rotation of the brake drum is

(a) 1152 N (b) 1248 N
(c) 4000 N (d) 2000 N

DRDO Scientists 2008

Ans. (b) :



(All dimensions are in mm)

Taking moment about 0.

$\Sigma M_o = 0$

$\Rightarrow$ F(1000) – N(300) – μN(40) = 0

$\Rightarrow$ F(1000) = 300 N + 12 N

$\Rightarrow$ F = 0.312 N

Resisting torque,

$T = 300 \times 10^3$ Nmm

$T^1 = (\mu N)R$

Normal reaction

$\therefore$ $N = \frac{300 \times 10^3}{0.3 \times 250} = 4000$ N

$F = 0.312 \times 4000 = 1248$ N

700. Chain drive is used for

(a) short distance
(b) large distance
(c) medium diameter of sprockets
(d) diameter is not considered

UKPSC AE 2007 Paper -I

Ans. (b) : Large distance

701. A thin circular disc is rolling with a uniform linear speed, along a straight path on a plane surface. Consider the following statements in this regard-

1. **All points of the disc have the same velocity**
2. **The center of the disc has zero acceleration.**
3. **The center of the disc has centrifugal acceleration.**
4. **The point on the disc making contact with the plane surface has zero acceleration**

Of these statements

(a) 1 and 4 are correct (b) 3 and 4 are correct
(c) 3 alone is correct (d) 2 alone is correct

ISRO Scientist/Engineer 2006

Ans. (d) : ■ Linear velocity is same for all points but angular velocity is different at every point hence resultant velocity is different for all points.

■ The center of the disc moves with the average velocity of the points of the top and the bottom. Centre point is stationary and does not change with direction. So, there is no acceleration.

■ As centre point has no acceleration then it has no centrifugal acceleration.

■ The point on the disc making contact with the ground has zero velocity for pure rolling. Therefore, it has zero acceleration.

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05.

MATERIAL SCIENCE

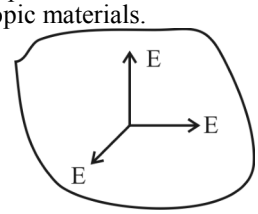
1. Material Properties

1. The materials which exhibit the same elastic properties in all directions are called–

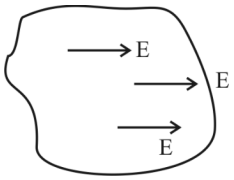
(a) Homogeneous (b) Inelastic
(c) Isotropic (d) Isentropic

HPPSC Poly. Lect. 05.07.2021
JPSC AE 10.04.2021 Paper II
TSPSC Manager (Engg.) HMWSSB 12.11.2020
Gujarat PSC AE 2019
ISRO Scientist/Engineer (RAC) 07.05.2017, 22.04.2018
Vizag Steel (MT) 2017
Nagaland PSC CTSE 2016, 2017 Paper-I
KPCL AE 2016
Rajasthan Nagar Nigam AE 2016 Shift-I, III
KPCL AE 2016
UKPSC AE 2012, 2007 Paper-I

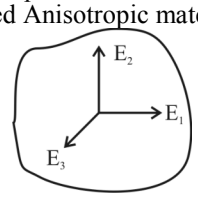
Ans. (c) :
Isotropic material -The materials which exhibit the same elastic properties in all directions at a given point are called isotropic materials.



Homogeneous materials– Those materials which have same elastic properties at all the points in any particular direction.



Anisotropic materials– Those materials which exhibit different elastic properties in different direction at a given point is called Anisotropic materials.



2. In the tensile test, the phenomenon of slow extension of the material, i.e., stress increasing with the time at a constant load is called

(a) creeping (b) yielding
(c) breaking (d) plasticity

TNPSC AE 2018
HPPSC LECT 2016
TSPSC AEE 2015
UKPSC AE 2013 Paper -I
ISRO Scientist/ Engineer 2009
UKPSC AE 2007 Paper –I
ESE 2007

Ans. (a) : Creeping, stress increasing with time at a constant load is the phenomenon of slow extension of the material in the tensile test.

3. The property of a material which enables it to resist fracture due to high impact loads is known as

(a) Brittleness (b) Ductility
(c) Toughness (d) Hardness

APPSC Poly Lect. 13.03.2020
Maharashtra PSC AE Pre 2019
Nagaland PSC CTSE 2018 Paper-I
APPSC AEE MAINS 2016, PAPER-III
GPSC ARTO 01.05.2016

Ans. (c) : Toughness – Toughness is the ability of material to absorb energy and plastically deform without fracture.

4. Slow plastic deformation of a material under constant stress is:

(a) Creep (b) Proof deformation
(c) Gradual deformation (d) Fatigue

JPSC AE 10.04.2021 Paper-II
Assam PSC AE (IWT) 14.03.2021
CIL MT 27.02.2020
APPSC Poly. Lect. Automobile Engg. 2020
UKPSC AE-2013, Paper-I

Ans. (a) : Creep is a tendency of a solid material to move slowly or deformed permanently under the influence of material.

5. Ductility of a material can be defined as :

(a) Ability to undergo large permanent deformations in compression
(b) Ability to recover its original
(c) Ability to undergo large permanent deformations in tension
(d) All of these

OPSC AEE 2019 Paper-I
Gujarat PSC AE 2019
TNPSC AE 2018
UKPSC AE 2007 Paper -I

Ans. (c) : Ductility– It is the property of a material by virtue of which it can be drawn into wires under the action of tensile force. A ductile material must have a high degree of plasticity and strength so that large deformation can take place without failure or rupture of the material.

6. Which one of the following materials is highly elastic?

(a) Rubber (b) Brass
(c) Steel (d) Glass

Gujarat PSC AE 2019
GPSC ARTO Pre 30.12.2018

Search on Telegram @apna_pdf

Rajasthan AE Nagar Nigam 2016 Shift-3
UKPSC AE-2013, Paper-I

Ans. (c) : Steel is highly elastic material
Steel has larger value of Young's Modulus of elasticity and hence steel has more elasticity than rubber.

7. The unique property of cast iron is its high
(a) malleability (b) ductility
(c) toughness (d) damping characteristics

JPSC AE 10.04.2021
ESE 2018
GWSSB AEE 07.07.2016
GWSSB AAE 27.12.2015

Ans. (d) : In grey cast iron, the graphite exist in the form of flakes. The flakes act as stress raises, thus reducing ductility.
The graphite flakes gives the material the capacity to damper vibration by the internal friction caused by these flakes.
Hence grey cast iron is mostly used material where vibration damping is important.

8. For a ductile material, toughness is a measure of:
(a)Resistance to scratching
(b)Ability to absorb energy upto fracture
(c)Ability to absorb energy till elastic limit
(d)Resistance to indentation

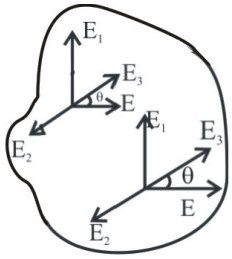
GPSC ARTO Pre. 30.12.2018
GPSC ASST director of transport 5.03.2017
UP Jal Nigam AE 2016

Ans. (b) : Toughness of a material is defined by energy absorbed up to fracture. It is equal to area under load deformation graph. Area under stress strain diagram up to fracture represents energy absorbed per unit volume, which is known as modulus of toughness. It is measure of toughness of material.

9. Anisotropic materials are having :
(a) Same properties at a point in all directions
(b) Different properties at different locations
(c) Different properties at a point in all directions
(d) Same properties at different locations

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC ARTO Pre 30.12.2018
KPSC ADF 2015

Ans. (c) : Anisotropic materials are having different properties at a point in all directions.



10. The property of a material which allows it to be drawn into a smaller section is
(a) plasticity (b) ductility
(c) elasticity (d) malleability

Assam PSC AE (PHED), 18.10.2020
Kerala PSC IOF 19.04.2016
VIZAG STEEL MT 2011

Ans. (b) : Ductility-It is property of a materials enabling it to be drawn into wire with the application of tensile force the ductile materials such as Mild steel, Copper, Aluminum, Nickel, Zinc, Tin and Lead.
Plasticity- It is property of a material which retains the deformation produced under load permanently.
Elasticity- It is the property of materials to regain its original shape after deformation when the external force are removed.
Malleability- It is a special case of ductility which permits material to be rolled or hammered into thin sheets. A malleable material should be plastic but it is not essential to be so strong.

11. The fatigue life of a part can be improved by :
(a) Polishing (b) Coating
(c) Shot peening (d) Electroplating

TSPSC manager engineering HMWSSB 12.11.2020
Vadodara Muncipal Corp. DEE, 2018
MPPSC state forest service exam 2014

Ans. (c) : Fatigue life of a part can be improved by shot peening process. Shot peening is a surface enhancement process used to impact compressive residual stress into fatigue-prone metals. This process increases fatigue strength by delaying the initiation of cracking.

12. The impact strength of a material is an indication of
(a) toughness of the material
(b) tensile strength
(c) capability of being hot worked
(d) resilience

Gujarat PSC AE 2019
ISRO Scientist/Engineer 22.04.2018
Rajasthan Nagar Nigam AE 2016 Shift- I

Ans : (a) : Toughness of a material is defined by energy absorbed upto fracture. It is equal to area under load deformation diagram. Area under stress strain diagram upto fracture represents energy absorbed per unit volume which is known as modulus of toughness, it is measure of toughness of material.

13. The ability of a material to withstand bending without fracture is known as :
(a) Mechanical strength (b) Stiffness
(c) Toughness (d) Ductility

Gujarat PSC AE 2019
HPPSC W.S. Poly. 2016
VIZAG Steel MT 2011

Ans : (c) The ability of a material to with stand bending without fracture is known as toughness.
Toughness:- It is the property of a material to resist fracture due to high impact loads like hammer blows. The toughness of the material decreases when it is heated. It is measured by the amount of a unit volume of the material has absorbed after being stress up to the point of fracture.

14. The property of a material necessary for forgings, in stamping image on coins and in ornamental work is
(a) elasticity (b) plasticity
(c) ductility (d) malleability

ASSAM PSC CCE Pre. 2015
TANGEDO AE 2015
JPSC AE 2013, Paper-V

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Ans : (b) : Plasticity- It is a property of a material which retains the deformation produced under load permanently. This property of material is necessary for forgings in stamping images on coins, and in ornamental works.

15. Failure due to repeated or reversal stresses is called

(a) hardness (b) Creep
(c) brittleness (d) Fatigue

TSPSC Manager (Engg.) HMWSSB 12.11.2020
TSPSC AEE 28.08.2017 (Civil/Mechanical)
TSPSC AEE 2015

Ans : (d) Failure due to repeated or reversal stresses is called fatigue.
Fatigue:- When a material is subjected to repeated stresses, it fails at stresses below the yield point stresses. Such type of failure of material is known as fatigue. This property is considered in designing shafts, connecting rods, spring, gears etc.

16. When the temperature of a solid metal increases :

(a) strength of the metal decreases but ductility increases
(b) both strength and ductility of the metal decreases
(c) both strength and ductility of the metal increases
(d) strength of the metal increases but ductility decreases

Mizoram PSC (Insp. of Factories) 2019 Paper-III
HPPSC Asstt. Prof. 18.11.2016
GATE 2005

Ans. (a) : When the temperature of a solid metal increases strength of the metal decreases but ductility increases.

17. The highest stress that a material can with stand for a specified length of time without excessive deformation is called.

(a) fatigue strength
(b) endurance strength
(c) creep strength
(d) creep rupture strength

Karnataka PSC lect. 27.05.2017
Kerala PSC Poly. Lect., 2017

Ans. (c) : The highest stress that a material can with stand for a specified length of time without excessive deformation is called creep strength.

18. Which property is most important for material selection of gas turbine blade?

(a) Creep (b) Hardness
(c) Toughness (d) Corrosion resistance

Rajasthan AE (Nagar Nigam) 2016 Shift-3
UPPSC AE 12.04.2016 Paper - I

Ans. (a) : Creep is most important property for the material selection of gas turbine.
Creep is the progressive time dependent inelastic deformation under mechanical load and high temperature.

19. The property of a material by virtue of which it can be beaten or rolled into plates is called

(a) malleability (b) ductility
(c) plasticity (d) elasticity

Assam PSC AE (IWT) 14.03.2021
HPPSC Poly. Lect. 05.07.2021

Ans. (a) : Malleability–The property of a material by virtue of which it can be beaten or rolled into plates is called malleability.
Ductility–It is property of a material enabling it to be drawn into wire with the application of a tensile force.
Plasticity–It is property of a material which retains the deformation produced under load permanently.
Elasticity–It is the property of a material to regain its original shape after deformation when the external force are removed.

20. Which of the following metals has the lowest melting point :

(a) Aluminium (b) Silver
(c) Zinc (d) Tin

GPSC ARTO Pre. 30.12.2018
J & K PSC Screening, 2006

Ans. (d) :

Metals		Melting Point
Aluminium	→	660.3°C
Silver	→	961.8°C
Zinc	→	419.5°C
Tin	→	231.9°C

21. Lathe bed is made up of :

(a) Mild steel (b) Cast steel
(c) Close grain Cast Iron (d) Pig Iron

WBPSC AE 2017
J & K PSC Screening, 2006

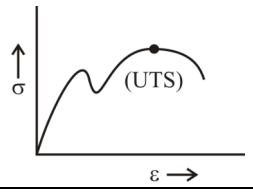
Ans. (c) : Lathe bed is made up of close grain cast iron.

22. In the engineering stress-strain curve for mild steel, the Ultimate Tensile Strength (UTS) refers to_____

(a) Fracture stress (b) Proportional limit
(c) Yield stress (d) Maximum stress

APGCL AM 2021
Assam Engg. College AP/Lect. 18.01.2021

Ans. (d) : In the engineering stress strain curve for mild steel, the ultimate tensile strength (UTS) refers to maximum stress.



23. The ability of a material to absorb energy in the plastic range is called

(a) resilience (b) creep
(c) fatigue strength (d) toughness

Nagaland PSC (CTSE) 2018, Paper - I
Sikkim PSC (Under Secretary), 2017

Ans. (d) : The ability of a material to absorb energy in plastic deformation till the point of fracture is known as toughness.
Fatigue- The behavior of the material under fluctuating loads is called fatigue.

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Creep- The permanent deformation of a material over a period of time at constant load and elevated temperature is called creep.
Resilience- It is the strain energy that can be absorbed by the material without plastic deformation up to elastic limit.

24. On loading a ductile material in excess of a certain value, gradual increase in elongation occurs with time. Such phenomenon is known as
(a) Creep
(b) Fatigue
(c) Stress concentration
(d) None of the above

ISRO Scientist/Engineer 2012, 2008

Ans. (a) : On loading a ductile material in excess of a certain value gradual increase in elongation occurs with time such phenomenon is known as **creep**.

25. What is the property of a material which enables it to be drawn into wires with the application of tensile force called as?
(a) Plasticity (b) Elasticity
(c) Ductility (d) Malleability

Kerala PSC IOF 19.04.2016
VIZAG STEEL MT 2011

Ans. (c) : Wire is a virtue of ductility

26. A material is known as allotropic or polymorphic if it :
(a) Has a fixed structure under all conditions
(b) Exists in several crystal forms at different temperature
(c) Responds to heat treatment
(d) Has its atoms distributed in random pattern

OPSC AEE 2019 Paper-I
APPSC AEE MAINS 2016, PAPER-III

Ans : (b) : A material is known as allotropic or polymorphic if it exists in several crystal forms at different temperatures.

27. Malleability is
(a) maximum energy that can be stored in a body due to external loading
(b) flattened into thin sheets
(c) energy stored in a body when strained
(d) ability to deform

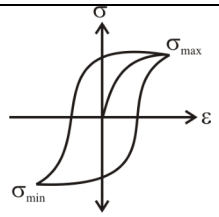
Gujarat PSC AE 2019
UJVNL AE 2016

Ans : (b) : Malleability- It is a special case of ductility which permits materials to be rolled or hammered into thin sheets. A malleable material should be so strong. The malleable materials commonly used in engineering practice (in order of diminishing malleability) are lead soft steel wrought iron, copper and aluminium.

28. The loss of strength in compression due to overloading is known as:
(a) hysteresis (b) relaxation
(c) creep (d) Bauschinger effect

APPSC AEE Screening Test 2016
HPPSC Asstt. Prof. 2014

Ans. (d) : Bauschinger effect is a phenomenon by which plastic deformation of a metal increases the yield strength in the direction of plastic flow and decreases the yield strength in the opposite direction.



Loss of strength in compression due to overloading is known as Bauschinger effect.

29. The materials which show direction dependent properties are called
(a) Homogeneous materials
(b) Viscoelastic materials
(c) Isotropic materials
(d) Anisotropic materials

UPPSC AE 13.12.2020, Paper-I
ESE 2007

Ans. (d) : Homogeneous Materials–Those materials which have same properties at all the points.
Isotropic Material–Those material which have same elastic properties in all the direction.
Anisotropic Material–Those materials which show direction dependent properties are called anisotropic material.

30. Materials exhibiting time bound behaviour are known as
(a) Visco elastic (b) Anelastic
(c) Isentropic (d) Resilient

TNPSC AE 2019
ISRO Scientist/Engineer 2007

Ans. (a) : Viscoelasticity is the property of material that exhibit both viscous and elastic characteristic when undergoing deformation. Viscous materials, like water, resist shear flow and strain linearly with time when a stress is applied.

31. Which of the following material has maximum ductility–
(a) grey cast iron (b) mild Steel
(c) alloy steel (d) high carbon steel

GMB AAE 25.06.2017
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Mild steel

32. The following equation is related to corrosion rate
(a) Nernst equation (b) Faraday's equation
(c) Newton's equation (d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Faraday's equation

33. All of the following statements are correct, except
(a) The limit of proportionality and the point of elastic limit coincide for ductile materials
(b) An elastic material recovers fully after unloading but not instantaneously
(c) A zero value of the Young's modulus of elasticity implies that the material is highly elastic
(d) A material that undergoes no deformation till its yield point, but thereafter flows at a constant stress is known as rigid plastic

HPPSC Poly. Lect. 05.07.2021

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Ans. (c) : A zero value of the Young's modulus of elasticity implies that the material is highly elastic

34. Pick the odd one out

(a) elastic strength (b) stiffness
(c) resilience (d) endurance limit

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : Endurance limit

35. To improve the machinability, the alloying element to steel is:

(a) cobalt (b) vanadium
(c) sulphur (d) manganese

AAI Jr. Executive 26.03.2021

Ans. (d) : To improve the machinability, the alloying element to steel is manganese.

36. _____ cast iron has little or no elongation

(a) Nodular (b) Grey
(c) White (d) malleable

UPRVUNL AE 05.07.2021

Ans. (c) : White cast iron has little or no elongation.

37. Work hardening in ductile material increases its

(a) Ductility
(b) Plasticity
(c) Electrical Conductivity
(d) Strength

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : Work hardening is a metallurgical process which increases in hardness of a metal induced, deliberately or accidently, by hammering etc. It is done below the recrystallization temperature, due to which it's strength get increased because pre-deformation.

38. Hardness is

(a) the property of a material due to which it is able to resist abrasion, indentation and scratching
(b) the susceptibility of a material to get hardened
(c) the same as hardenability
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Hardness is the property of a material due to which it is able to resist abrasion, indentation and scratching.

39. Hardenability of steel is a measure of

(a) the ability to retain its hardness when it is heated to elevated temperatures
(b) the ability to harden when it is cold worked
(c) the maximum hardness that can be obtained when it is austenitized and then quenched
(d) the depth to which required hardening is obtained when it is austenitized and then quenched

Assam Engg. College AP/Lect. 18.01.2021

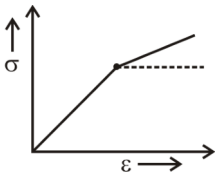
Ans. (d) : Hardenability is a measure of the depth upto which required hardening is obtained when it is austenitized and then quenched. Jomminy's test is performed to measure hardenability.

40. In a linearly hardening plastic material, the true stress beyond initial yielding

(a) increases linearly with the true strain
(b) decreases linearly with the true strain
(c) first increases linearly and then decreases linearly with the true strain
(d) remain constant

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) :



Elasto plastic material with strain hardening.

41. Which of the following structure of steel is magnetic in nature?

(a) δ (b) v
(c) α (d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : The alpha (α) form of iron is magnetic and stable at all temperatures below 910°C.

42. If a material fails below its yield point failure would be due to :

(a) Straining (b) Fatigue
(c) Creep (d) Sudden loading

JPSC AE 10.04.2021, Paper-II

Ans. (b) : Fatigue failure is the formation and propagation of cracks due to a repetitive or cyclic load. Most fatigue failure are caused by cyclic loads significantly below the loads that would result in yielding of material.

43. Which portion of the creep curve provides the information on steady state creep rate?

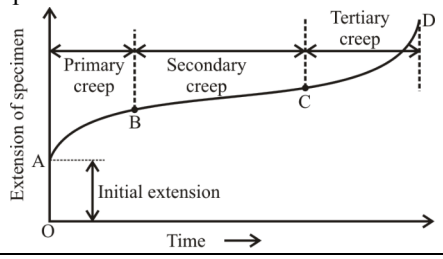
(a) Primary stage (b) Secondary stage
(c) Tertiary stage (d) All the above

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : → **Primary Region-** Characterized by transient creep with creep rate.

→ **Secondary Region-** Is characterized by steady state creep. (Creep strain rate)

→ **Tertiary Region-** Is characterized by increasing creep strain rate.



44. Strain ageing leads to:

(a) increase in yield strength
(b) decrease in hardness
(c) decrease in yield strength
(d) increase in ductility

NLCIL GET 17.11.2020, Shift-II

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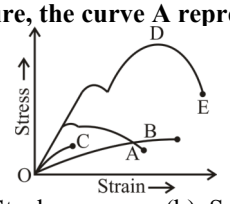
Ans. (a) : In general, strain ageing increases strength to some extent but results in a loss of some ductility. It is known as strain ageing is accompanied by a decrease in fracture toughness, which is usually reflected by an increase in the brittle transition temperature.

45. In a stress-strain diagram as shown in the below figure, the curve A represents



(a) Mild Steel (b) Soft Brass
(c) Low Carbon steel (d) Cold rolled steel

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) :

46. Oxides, nitrides, carbides and silicates of metal are called as:

(a) Organic materials
(b) Ceramic materials
(c) Ferrous materials
(d) Non-ferrous materials

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : Oxides, nitrides, carbides and silicates of metal are called as ceramic materials.

47. Which of the following is not improved by grain size reduction?

(a) Hardness (b) Toughness
(c) Elasticity (d) Strength

RPSC IOF, 2020

Ans. (c) : Grain size reduction increases the hardness, toughness and the strength of the material. Elasticity is not increased by grain size reduction.

48. _____ in steel is mainly responsible for the occurrence of hair line crack.

(a) Sulphur (b) Hydrogen
(c) Nitrogen (d) Manganese

RPSC IOF, 2020

Ans. (b) : There are three factors which combines to cause cracking.
(1) Hydrogen generated by the welding process.
(2) A hard brittle structure which is susceptible to cracking.
(3) Tensile stresses acting on the welded joint.

49. Coaxing is a process of

(a) improving the fatigue properties, attained by under-stressing and then raising the stress in small increments
(b) Decreasing the hardness by full annealing
(c) Increasing the uniaxial tensile strength by heating above recrystallization temperature and quenching in oil media
(d) Maintaining the ductility of the material by chemical treatment

ESE 2020

Ans. (a) : Improving the fatigue properties, attained by under-stressing and then raising the stress in small increments.

50. The mechanical properties of forged components compared to cast components are

(a) Inferior (b) Superior
(c) Similar (d) Unpredictable

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : The mechanical properties of forged components compared to cast components are superior.

51. Which one of the following is the process to refine the grains of metal after it has been distorted by hammering or cold working?

(a) Annealing (b) Softening
(c) Recrystallizing (d) Normalizing

RPSC IOF, 2020

Ans. (c) : Recrystallization is a process by which deformed grains are replaced by a new set of defects-free grains that nucleate and grow until the original grains have been entirely consumed.

52. Fully recoverable but time-dependent deformation is known as :

(a) elastic deformation
(b) anelastic deformation
(c) viscoelastic deformation
(d) plastic deformation

APPSC Poly Lect. 13.03.2020

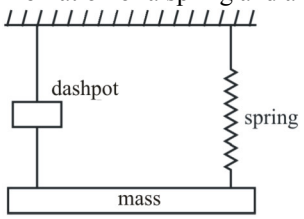
Ans. (b) : Fully recoverable but time-dependent deformation is known as anelastic deformation.

53. Which of the following is true for Voigt-Kelvin model of viscoelastic materials?

(a) It is a parallel combination of a spring and a dashpot
(b) It is a series combination of a spring and a dashpot
(c) It is a parallel combination of two springs
(d) It is a parallel combination of two dashpots

APPSC Poly Lect. 13.03.2020

Ans. (a) : Voigt-Kelvin Model of viscoelastic materials is a parallel combination of a spring and a dashpot.



In Maxwell Model of viscoelastic material it is a series combination of a spring and a dashpot.

54. In the absence of clear yielding, criteria for yielding corresponds to :

(a) 0.5 percent strain (b) 0.8 percent strain
(c) 0.2 percent strain (d) 0.1 percent strain

NLC GET 17.11.2020, Shift-II

Ans. (c) : In the absence of clear yielding criteria for yielding corresponds to 0.2% strain. In the absence of a distinct yield point, a 0.2% offset is used to obtain an approximate yield point. The proportion limit and the yield point are close to one another on a stress-strain curve but they each have a different representation and thus do not corresponds to the same point on the curve.

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55. A bar subjected to static loading within elastic limit develops additional strain over long periods of time due to the phenomenon of

(a) Fatigue (b) Fracture
(c) Creep (d) Plasticity

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : When a bar is subjected to constant load for a long period of time is termed as creep.

56. Lowering of the yield stress when deformation in one direction is followed by deformation in the opposite direction is known as:

(a) Ludwig effect (b) Boldtzman effect
(c) Bauschinger effect (d) Maudsley effect

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Bauschinger Effect–Lowering of the yield stress when deformation in one direction is followed by deformation in the opposite direction.
The Bauschinger effect refers to a property of materials where the materials stress-strain characteristics change as a result of the microscopic stress distribution of the material.

57. Decreasing grain size in a polycrystalline material :

(a) Increases yield strength and corrosion resistance
(b) Decreases yield strength and corrosion resistance
(c) Decreases yield strength but increases corrosion resistance
(d) Increases yield strength but decreases corrosion resistance

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : Polycrystalline material – These are composed of individual grains oriented in a random manner to the loading axis when no texture is present in the material.
Decreasing grain size in polycrystalline material increases yield strength and corrosion resistance.

58. Which of the following statement is true about brittle fracture?

(a) High temperature and low strain rates favour brittle fracture
(b) Many metal with HCP crystal structure commonly show brittle fracture
(c) Brittle fracture is always preceded by noise
(d) Cup and cone formation is characteristic for brittle materials

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : Brittle fracture is the sudden, very rapid cracking of equipment under stress where the material exhibited little or no evidence of ductility or plastic degradation before the fracture occurs.
Metals with hexagonal close packed (H.C.P.) crystal structure commonly show brittle fracture.

59. Magnesium is extruded and not rolled because :

(a) It has low melting point
(b) It has a low density
(c) Its reactivity with roll material is high
(d) It has a close-packed hexagonal structure

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : Magnesium is extruded and not rolled because it has a low melting point.

60. With the increase of percentage of carbon in the steel, which one of the following properties does increase?

(a) Modulus of elasticity (b) Ductility
(c) Toughness (d) Hardness

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) : As carbon % increases

1. Strength ↑	(a) Modulus of toughness ↓
2. Modulus of Resilience ↑	(b) Ductility ↓
3. Wear resistance ↑	(c) Malleability ↓
4. Surface hardness ↑	(d) Formability ↓
5. Cast ↑	(e) Machininability ↓

- Modulus of elasticity remains constant for all carbon steels.

61. The element with the lowest density and smallest atomic weight is which one of the following?

(a) aluminum (b) argon
(c) helium (d) hydrogen

GPSC ARTO Pre 30.12.2018

Ans. (d) : The first chemical element with the lowest density and smallest atomic weight is hydrogen and the highest density is osmium.

62. Which one of the following materials is closest to diamond in hardness?

(a) aluminum oxide (b) carbon dioxide
(c) cubic boron nitridc (d) silicon dioxide

GPSC ARTO Pre 30.12.2018

Ans. (c) : CBN (Cubic Boron Nitride) is closest to diamond in hardness.

63. In most brittle materials, the ultimate strength in compression is larger than the ultimate strength in tension due to

(a) Necking in tension
(b) Presence of flaws and microscopic cracks or cavities
(c) Non-linearity of stress-strain diagram
(d) Severity of tensile stress as compared to compressive stress

ISRO Scientist/Engineer 22.04.2018

Ans. (b): In brittle materials, the ultimate strength in compression is larger than the ultimate strength in tension due to presence of flows and microscopic cracks or cavities.

64. Permanent deformation of solid material under the influence of long - term exposure of high level of mechanical stresses that are still below the yield strength along with subjected to heat is .....

(a) Elasticity (b) Isotropy
(c) Hardness (d) Creep

RPSC AE 2018

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Ans. (d) : Creep—Permanent deformation of solid material under the influence of long - term exposure of high level of mechanical stresses that are still below the yield strength along with subjected to heat is known as creep.
Elasticity—If the material is elastic, the object will return to its initial shape and size when these force are removed.
Isotropy—Any material are considered to be isotropic if the properties are not dependent on the direction.
Hardness—Material hardness is the property of the material which enables it to resist plastic deformation, usually by penetration or by indentation.

65. The resistance of a material of elastic deformation is called:
(a) Ductility (b) Toughness
(c) Hardness (d) Stiffness
(e) Hot hardness

CGPSC AE 25.02.2018

Ans. (d) : The resistance of a material to elastic deformation is called stiffness.

66. Which one of the following properties is more sensitive to increase in strain rate:
(a) Yield strength (b) Proportional limit
(c) Elastic limit (d) None or these

PTCUL AE 25.06.2017

Ans. (a) : more sensitive to increase yield strength.

67. As soon as the external forces causing deformation in a perfectly elastic body are withdrawn, the elastic deformation disappears
(a) Only partially
(b) completely over a prolonged period of time
(c) completely and instantaneously
(d) completely after an initial period of rest

TSPSC AEE 28.08.2017 (Civil/Mechanical)

Ans. (c) : If a body regains its original configuration immediately and completely after removal of deformation force it is called as perfectly elastic body.

68. Which of the following is a general physical property of metals?
(a) Creep (b) Density
(c) Magnetic flux (d) Brittleness
(e) Thermal stability

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Physical properties can be observed or measured without changing the composition of matter. Physical properties are used to observe and describe matter. Physical properties include: appearance, texture, color, odor, melting point, boiling point, density, solubility, polarity, and many others.

69. Which of the following properties will be the meaningful indicator/indicators of uniform rate of elongation of a test piece of a structural material before necking happens in the test piece?
1. Ductility 2. Toughness 3. Hardness
Select the correct answer using the code given below.
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

ESE 2017

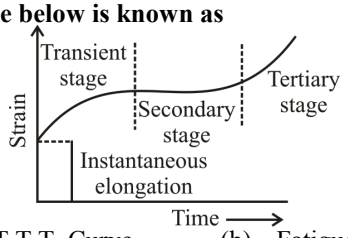
Ans. (a) : Ductility is the meaningful indicator of uniform rate of elongation in any material along with ductility there is strain rate sensitivity index, that tells us about the rate of elongation since strain rate is not given, ductility will be the answer.

70. The phenomenon under which the strain of a material varies at constant stress is known as
(a) Hysteresis (b) Creep
(c) Viscoelasticity (d) Strain hardening

WBPSC AE, 2017

Ans. (b) : The phenomenon under which the strain of a material varies at constant stress is known as creep.

71. Curve below is known as



(a) T.T.T. Curve (b) Fatigue Curve
(c) Creep Curve (d) Resilience Curve

Karnataka PSC Lect., 27.05.2017

Ans (c) : Creep Curve

72. For carbon-carbon composites, the following statement(s) is/are true:
(a) C-C composites are structures in which both matrix and the reinforcement are carbon.
(b) C-C composites have high temperature strength and toughness.
(c) The high cost of C-C composites is a major issue.
(d) All of the above

Karnataka PSC AE, 10.09.2017

Ans. (d) : C–C composite (carbon fibre reinforced carbon).
- Carbon fibre reinforced in a matrix of graphite.
- They have high toughness and temperature strength.
- Highly composites.

73. Match Material property (List-I) with Test to determine material property (List-II)

List-I (Material property)		List-II (Test to determine material property)
A. Ductility	I.	Impact Test
B. Endurance limit	II.	Fatigue Test
C. Toughness	III.	Tension Test
D. Resistance to penetration	IV.	Hardness Test

Select the code for the correct answer from the options given below :

	A	B	C	D
(a)	I	II	IV	III
(b)	II	III	I	IV
(c)	III	II	I	IV
(d)	III	II	IV	I

Karnataka PSC AE, 10.09.2017

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Ans. (c) : Tension test is done to determine ductility. Fatigue test is done for endurance limit.
Impact test is done for toughness.
Hardness test is done to determine penetration resistance.

74. Babbit lining is used on brass or bronze bearings to
(a) Increase bearing resistance
(b) Increase compressive strength
(c) Provide any friction properties
(d) Increase wear resistance

Karnataka PSC AE, 10.09.2017

Ans. (d) : Babbit lining is used on brass or bronze to increase wear resistance.

75. Match the Materials (List-I) with Applications (List-II)

List-I (Materials)		List-II (Applications)
A. Engineering ceramics	I.	Bearings
B. Fibre-reinforced plastics	II.	Control rods in nuclear reactors
C. Synthetic carbon	III.	Aerospace industry
D. Boron	IV	Electrical insulator

Select the code for the correct answer from the options given below:

	A	B	C	D
(a)	III	III	IV	I
(b)	IV	III	II	I
(c)	II	III	I	IV
(d)	IV	III	I	II

Karnataka PSC AE, 10.09.2017

Ans. (d) : Engineering ceramics are used as electrical insulator. Fibre-reinforced plastics are used in Aerospace industry. Synthetic carbon are used in bearings.
Boron is used in control rods in nuclear reactors.

76. Brittle fracture is more dangerous than ductile fracture because
(a) No warning sign
(b) Crack propagates at very high speeds
(c) No need for extra stress during crack propagation
(d) All of the mentioned above

GPSC EE Pre, 28.01.2017

Ans. (d) : Brittle fracture is more dangerous than ductile fracture because
- No warning sign
- Crack propagates at very high speeds
- No need for extra stress during crack propagation.

77. Addition of which one of the following elements, shifts the lower critical temperature line in iron-iron carbide diagram towards the higher side?
(a) Chromium (b) Nickel
(c) Molybdenum (d) Aluminium

GPSC EE Pre, 28.01.2017

Ans. (c) : Addition of Molybdenum shifts the lower critical temperature line in iron-Iron carbide diagram towards the higher side.

78. Statement (I): Directionally solidified materials have good creep resistance.
Statement (II): Directionally solidified materials may be so loaded that there is no shearing stress along, or tensile stress across, the grain boundaries.

ESE 2017

Ans. (a) : The sharp temperature gradient used in directional solidification forces the grains to grow continuously from one end of the casting to the other. The final casting has a columnar grain structure with few or no transverse grain boundaries, thus providing high creep resistance.

79. The principle of super position is valid only for
(a) Any elastic materials
(b) All metals
(c) Linear elastic materials
(d) Homogeneous material

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : Super position principle, also known as super position property, states that for all linear systems, the Net response caused by two or more stimuli is the sum of the responses that would have been caused by each stimuli individually. So, it is valid for Linear elastic materials.

80. Creep of a material is
(a) Continued deformation with time under sustained loading
(b) Disappearance of deformation on removal of load
(c) Not being ductile
(d) To become brittle

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (a) : In material science, creep is the tendency of a solid material to move slowly or deform permanently under the constant stress. It is continuous deformation with time under sustained loading.

81. Plastic response of a material to compressive force is known as
(a) Elasticity (b) Ductility
(c) Plasticity (d) Malleability

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (d) : Malleability describes the property of a metal's ability to be distored below compression. It is a physical property of metals by which they can be hammered, shaped & rolled into a very this sheets without rupturing.

82. What is the most widely used reinforcement in modern day FRP tennis racket?
(a) Magnesium (b) Aluminium
(c) Carbon (d) Glass
(e) Iron

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : Carbon fibre is one of highest and have good strength, as compared to others. It is 2.7 times stronger than steel.

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83. The property by virtue of which the material absorb maximum energy before the fracture takes place is called

(a) Hardness (b) Toughness
(c) Creep (d) Fatigue

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Toughness– It is the property of material by virtue of which it absorbs maximum energy up to the fracture limit.

84. The stress at which a material begins to deform plastically is known as_____

(a) Elastic limit (b) Deform point
(c) Irreversible plasticity (d) All the above

KPCL AE 2016

Ans. (a) : The stress at which a material begins to deform plastically is known as elastic limit.

85. Phosphorus in small amount.....the strength and hardness of steels.

(a) Increases (b) Decreases
(c) Makes no change in (d) None of these

HPPSC AE (PWD) 10.09.2016

Ans. (a) : Up to 0.10% of phosphorus in low alloy high strength steels will increase the strength as well as improve the steel's resistance against corrosion. The possibility of being brittle increases when the content in hardened steel is to high.

86. Orthotropic materials have

(a) different properties in orthogonal directions
(b) different properties in different directions
(c) different strength in different directions
(d) same properties in different direction

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Orthotropic materials have different properties in orthogonal directions.

- Isotropic materials have identical properties in all direction at a point.
- Anisotropic materials have different properties in different direction at a point.
- Homogeneous materials have same properties at every point throughout the volume.

87. Under torsion, brittle materials generally fail

(a) along a plane perpendicular to it's longitudinal axis
(b) in the direction of minimum tension
(c) along surfaces forming a 45 degree with the longitudinal axis
(d) none of these

GPSC Lect. 23.10.2016

Ans. (c) : Brittle materials fail in tension and therefore, a shaft made of a brittle material and under torsion will break on a surface that is in the shape of 45° helix.

88. When a material is strain hardened?

(a) its yield strength reduces and ductility increases
(b) its yield strength increases and ductility reduces
(c) both yield strength and ductility increases
(d) both yield strength and ductility reduces

ISRO Scientist/Engineer 03.07.2016

Ans. (b) : Yield strength increases because of dislocation movements and dislocation generation within the crystal structure.

89. Match the items in column-1 and column-2 and choose the correct combination

	Column-1		Column-2
A.	Resilience	1.	Ability of the material to return to the original shape when load is removed.
B.	Fatigue	2.	Capacity of a material to absorb energy elastically
C.	Stiffness	3.	Ability of the material to absorb certain amount of energy without failure.
D.	Elasticity	4.	Resistance of the material to elastic deformation
		5.	Failure of material at loads less than that at normal conditions

(a) A – 2, B – 5, C – 3, D – 1
(b) A – 3, B – 2, C – 1, D – 4
(c) A – 3, B – 5, C – 4, D – 2
(d) A – 2 , B – 5, C – 4, D – 1

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Resilience → Capacity of material to absorb energy elastically.

Fatigue → Failure of material at loads less than that at normal condition.

Stiffness → Resistance of material to elastic deformation.

Elasticity → Ability of material to return the original shape when load is removed.

90. The ratio of the largest load in a test to the original cross-sectional area of the test piece is called:

(a) Elastic limit (b) Yield stress
(c) Ultimate stress (d) None of the above

UP Jal Nigam AE 2016

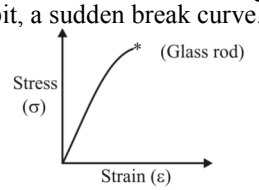
Ans. (c) : The ratio of the largest load in a test to the original cross-sectional area of the test piece is called ultimate stress.

91. The stress strain curve for a glass rod during tensile test would exhibit

(a) A straight line (b) A parabola
(c) A sudden break (d) None of the above

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : The stress-strain curve for a glass rod during test would exhibit, a sudden break curve.



92. An allotropic material has

(a) Atoms distributed in random patterns
(b) Atoms distributed in fixed pattern
(c) Different crystal structure at different temperatures
(d) fixed structure at all temperatures

APPSC AEE MAINS 2016, PAPER-III

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Ans. (c) : An allotropic material has different crystal structure at different temperatures.

93. The Erichsen cupping number of a metal sheet indicates its

(a) Ductility (b) Hardenability
(c) Toughness (d) Drawing ability

UPPSC AE 12.04.2016 Paper-I

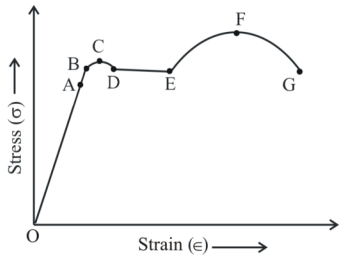
Ans : (d) The Erichsen cupping number of a metal sheet indicates its drawing ability. So this test can be classified as stretch forming.

94. Strain hardening of a material is

(a) increases in stress with strain beyond yield stress
(b) increase in strain with stress beyond ultimate stress
(c) increase in hardness as strain in increased
(d) increase in hardness under repeated loading

APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : Stress strain curve for mid steel –



OA = Linear curve
A = Proportional limit
B = Elastic limit
C = Upper yield point
D = Lower yield point
DE = Plastic region
EF = Strain hardening region
F = Ultimate stress point
FG = Necking region
G = Fracture point
Strain hardening in a material increase beyond yield stress.

95. The ability of material to absorb energy when it is deformed elastically and release energy upon unloading is called

(a) Ductility (b) Hardness
(c) Resilience (d) Toughness

APPSC AEE Mains 2016 (Civil Mechanical)

Ans. (c) : Resilience is defined as the ability of the material to absorb energy when deformed elastically and to release this energy when unloaded. A resilient material absorb energy within elastic range without any permanent deformation.

96. Creep is essentially a-

(a) High temperature phenomenon
(b) Medium temperature phenomenon
(c) Low temperature phenomenon
(d) Phenomenon which is independent of temperature

NPCIL ET, 2015

Ans. (a) At low temperatures and low stress, creep is essentially non existent and all strain is elastic. At low temperature and high stress materials experience plastic deformation rather than creep. At high temperature and low stress, diffusional creep tends to be dominant, while at high temperature and high stress dislocation creep tends to be dominant.

97. Ductile fracture mode experiences :

(a) Nil deformation
(b) Faster crack propagation
(c) Low energy absorption with unstable cracks
(d) Major plastic deformation before crack propagation

KPSC ADF 2015

Ans. (d) : Ductile fracture mode experiences major plastic deformation before crack propagation.

98. The ability of a metal to withstand elongation or bending is known as:

(a) Ductility (b) Malleability
(c) Stiffness (d) Brittleness

OPSC AEE 2015 Paper-I

Ans : (c) The ability of a metal to withstand elongation or bending is known as stiffness.
Stiffness : It is the ability of a materials to resist deformation under stress. The modulus of elasticity is the measure of stiffness.

99. The toughness of a material, when it is heated

(a) decreases (b) increases
(c) remains same (d) stabilise

TSPSC AEE 2015

Ans : (a) The toughness or more specifically fracture toughness increases with increase in temperature.

100. Elasticity is the ability regarding its deformation to _____ its original shape.

(a) retain (b) regain
(c) resist (d) uniform

TSPSC AEE 2015

Ans : (b) Elasticity :- It is the property of a materials to regain its original shape after deformation when the external forces are removed. This property is desirable for materials used in tools and machines. It may be noted that steel is more elastic than rubber.

101. Work hardening of a steel component reduces:

(a) Malleability (b) Hardness
(c) Ductility (d) Toughness

OPSC AEE 2015 Paper-I

Ans : (c) During yielding more stress is required for the same strain to be produced in ductile materials. The phenomenon is known as work hardening. Each strain strengthen the material for future strains.

102. Hardness of the steel greatly improves with

(a) annealing (b) normalizing
(c) stress releaving (d) tempering

GPSC AE 26.07.2015

Ans. (b) : Hardness of the steel greatly improve normalizing.

103. Percentage elongation during tensile test is indicative of

(a) Ductile (b) Malleability
(c) Creep (d) Fatigue strength

GPSC AE 26.07.2015

Ans. (a) : Percentage elongation during tensile test is indicative of ductile.

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104. Puddling is the process employed for converting :

(a) Iron ore into pig iron
(b) Pig iron into wrought iron
(c) Pig iron into cast iron
(d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (b) : Puddling is the process employed for converting pig iron into wrought iron.

105. The melting point of mild steel in degree centigrade is about :

(a) 850 (b) 1000
(c) 1250 (d) 1500

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : Melting point of mild steel in degree centigrade is 723.

106. The property of the material due to which it can cut another material is known as

(a) ductility (b) malleability
(c) hardness (d) none of these

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (c) : The property of the material due to which it can cut another material is known as hardness.

107. Which one of the following micro constituents has maximum hardness?

(a) Austenite (b) Pearlite
(c) Sorbite (d) Cementite

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : Cementite

108. Type of alloyed plain carbon steel used for drawing dies, punches and drills is

(a) high carbon steel
(b) medium carbon steel
(c) high speed steel
(d) none of these

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : The high carbon steel is used as a standard material for dies and cutting elements.

109. Increasing temperature has the following effect on stress-strain curves

(a) It raises ductility
(b) It lowers toughness
(c) It raises the yield stress
(d) The sensitivity of strength to strain rate decreases

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : Increasing temperature raises ductility.

110. Percent elongation is an indicator of which mechanical property?

(a) UTS (b) Yield strength
(c) Ductility (d) Resilience

HPPSC Asstt. Prof. 2014

Ans. (c) : As percentage elongation increases then material will be more ductile.

111. Which of the following are the strongest materials available and are used for applications where high strength is required at relatively low cost and where weight is not of primary importance?

(a) Ferrous metals (b) Non-ferrous metals
(c) Diamonds (d) Carbon

UPRVUNL AE 2014

Ans : (a) Ferrous metal are used for low cost and high strength.

112. Match List-I with List-II and select the correct answer:

List-I (Process)	List-II (Property)
(P) Cold Rolling	(1) Indentation
(Q) Hardness	(2) Ductility
(R) Impact loads	(3) Malleability
(S) Wire drawing	(4) Resilience

(a) P-4, Q-2, R-3, S-1 (b) P-3, Q-1, R-4, S-2
(c) P-1, Q-3, R-4, S-2 (d) P-3, Q-1, R-2, S-4

ISRO Scientist/Engineer 24.05.2014

Ans. (b) : Cold Rolling - Malleability
Hardness - Indentation
Impact Load - Resilience
Wire drawing - Ductility

113. A property often used to characterize strength and hardness at elevated temperature is

(a) creep (b) fatigue
(c) hot hardness (d) residual hardness
(e) Ultimate Red Strength

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : A property often used to characterize strength and hardness at elevated temperature is hot hardness.

114. Which one of the following is not a strengthening mechanism?

(a) Alloying (b) Heat treatment
(c) Strain hardening (d) Metallizing

TRB Poly. Lect., 2012

Ans. (d) : Alloying, heat treatment & strain hardening are responsible for strengthening the products in strengthening mechanism appropriate heating re-crystallization temperature (cold working) while metallizing is coating of metal on surface of objects.

115. The ability of material of resist elastic or plastic deformation under applied load is called

(a) elasticity (b) plasticity
(c) malleability (d) rigidity

TRB Asstt. Prof., 2012

Ans. (d) • Elasticity is the ability of a body to resist against external force and to return to its original size and shape when external force is removed.

- Plasticity is defined as the ability of a solid material to undergo, permanent deformation against external force.
- Malleability and ductility are relatable. A malleable material is one in which a thin sheet can be easily formed by hammering or rolling. In other words the material has the ability to deform under compressive stress.
- In contrast, ductility is the ability of a solid material to deform under tensile stress.
- When an external force is applied to the solid material, there would not be any change in the shape. This shows that the particles are closely packed and the attraction between these particle are very strong.

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116. Creep depends on :

(a) Pressure (b) Temperature
(c) Load applied (d) Stiffness

APGENCO AE 2012

Ans. (b) : Creep is dependent on temperature and stress. Increase temperature for a constant stress can decrease the time of the second stage and hence accelerate failure.

117. Which one of the following is a wrong statement?

(a) In low carbon steels, phosphorus is added to raise its yield point
(b) Lining of open hearth furnace controls impurities in steel
(c) Blast furnace uses coke as fuel
(d) Manganese is added in blast furnace for better fluidity.

ISRO Scientist/Engineer 2012

Ans. (a) :

118. Assertion (A) : Little energy is required to break materials such as glass, polystyrene and some cast irons. Conversely, rubber and many steels absorb considerable energy in the fracture process.
Reason (R) : The service limit in many engineering products is not the yield or ultimate strength, rather may be the energy associated with fracture propagation.
Code :

(a) Both (A) and (R) are true and (R) explains (A) correctly.
(b) Both (A) and (R) are true but (R) does not explain (A) correctly.
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

UKPSC AE 2012 Paper-I

Ans. (a) : Both (A) and (R) are true and (R) explains (A) correctly.

119. Assertion (A) : In general, materials deform more readily at elevated temperature.
Reason (R) : Plastic deformation commonly arises from dislocation movements that involve a continual displacement of atoms to new neighbours at elevated temperature.
Code :

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) explains (A) correctly.
(d) Both (A) and (R) are true, but (R) does not explain (A) correctly.

UKPSC AE 2012 Paper-I

Ans. (c) : Both (A) and (R) are true and (R) explains (A) correctly.

120. The ductile-brittle transition temperature

(a) depends on size and shape of material, rate of loading, presence of notches, impurities and operating temperature
(b) depends on size but does not depend on shape of material
(c) does not depend on size of material
(d) does not depend on rate of loading but depends on presence of impurities

UKPSC AE 2012 Paper-I

Ans. (a) : It depends on size and shape of material, rate of loading, presence of notches, impurities and operating temperature.

121. Ability of a material to exhibit considerable elastic recovery on release of load, is known as:

(a) Toughness (b) Stiffness
(c) Resilience (d) Hardness

OPSC Civil Services Pre. 2011

Ans. (c) : Ability of a material to exhibit considerable elastic recovery on release of load, is known as resilience.

122. Toughness of a material means its :

(a) Strength (b) Fatigue Resistance
(c) Machinability (d) Softening

OPSC Civil Services Pre. 2011

Ans. (a) : Toughness is the ability of the material to absorb energy during plastic deformation up to fracture.

- Toughness is a good combination of strength and ductility.
- Toughness is by calculating the area under the load and deformation curve from a tensile test.

123. Consider the following properties of engineering materials :

1. Density 2. Conductivity
3. Elasticity 4. Hardness
5. Ductility

Which of these are mechanical properties of a material ?

(a) 1, 2 and 4 only (b) 1, 2, 3 and 4
(c) 3, 4 and 5 (d) 2, 4 and 5

UPSC JWM Advt. No.-50/2010

Ans. (c) : As, we know the mechanical properties of the materials are, elasticity, hardness & ductility.

124. Shock resisting steel should have

(a) Low wear resistance
(b) High wear resistance
(c) Toughness
(d) Poor hardenability

ISRO Scientist/Engineer 2010

Ans. (c) : Shock resisting steel should have toughness. Toughness is used because it absorb shock easily. Shock resisting steel should have more toughness.

125. Match List-I and List-II and select the correct answer using the code given below the lists-

List-I (Match property)	List-II (Related to)
(A) Malleability	1. Wire drawing
(B) Hardness	2. Impact loads
(C) Resilience	3. Cold rolling
(D) Isotropy	4. Indentation
	5. Direction

	(A)	(B)	(C)	(D)
(a)	4	2	1	3
(b)	3	4	2	5
(c)	5	4	2	3
(d)	3	2	1	5

ISRO Scientist/Engineer 2010

Ans. (b)

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126. High creep resistance is due to :
(a) finer grains
(b) coarser grains
(c) independent of grain size
(d) none of the above

APPSC IOF, 2009

Ans. (b) : At high temperature high creep resistance is observed with coarse grain size. Fine grain size are not desirable for creep resistance.

127. Striations on a fractured surface indicate :
(a) ductile failure (b) brittle fracture
(c) creep fracture (d) fatigue fracture

APPSC IOF, 2009

Ans. (d) : The presence of striations is used in failure analysis as an indication that a fatigue crack has been growing.

128. Phenomenon of progressive extension of material with time at constant load is called
(a) Plasticity (b) Creep
(c) Yield (d) Breaking

ISRO Scientist/Engineer 2009

Ans. (b) : Creep: Creep is defined as the phenomenon of progressive extension of material with time at constant load is called creep.

129. Which of the following materials CANNOT be heat treated ?
(a) alloy steel
(b) high carbon steel
(c) medium carbon steel
(d) low carbon steel

DRDO Scientists 2009

Ans. (d) : low carbon steel

130. Which one of the following type of stress-strain relationship best describes the behaviour of material at temperature above their respective recrystallization point ?
(a) rigid plastic
(b) perfectly elastic
(c) elastic and perfectly plastic
(d) elastic and strain hardening

DRDO Scientists 2009

Ans. (b) : perfectly elastic

131. A body which is permanently deformed is said to have undergone
(a) Elastic deformation
(b) Limit of elastic deformation
(c) Plastic deformation
(d) Uniform deformation

ISRO Scientist/Engineer 2008

Ans. (c) : A body which is permanently deformed is plastic deformation.

132. Which of the following properties of mild steel cannot be determined by a static tensile test of the sample?
(a) Ultimate tensile strength
(b) Ultimate shear strength
(c) Ductility
(d) Poisson's ratio

UKPSC AE 2007 Paper -I

Ans. (b) : Ultimate shear strength of mild steel cannot be determined by a static tensile test of the sample.

133. A material shall be called ideal plastic material if
(a) it does not strain harden
(b) it strain hardens
(c) if it is made of plastic
(d) none of the above

UKPSC AE 2007 Paper -I

Ans. (a) : An ideal plastic material is one that experiences no work hardening or does not strain hardening during plastic deformation.

134. A material in which the atoms are arranged chaotically is called
(a) amorphous (b) mesomorphous
(c) crystalline (d) none of the above

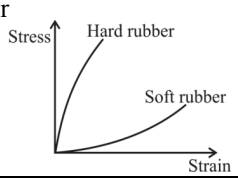
UKPSC AE 2007 Paper -I

Ans. (a) : The amorphous material is a non-crystalline solid and the glass is amorphous material produced through melt quenching.

135. Which of the following material has non-linear elastic behaviour?
(a) Mild Steel (b) Aluminium
(c) Cast iron (d) Rubber

UKPSC AE 2007 Paper -I

Ans. (d) : Rubber



136. Hardness of steel depends on
(a) Amount of carbon it contains
(b) The shape and distribution of the carbides in iron
(c) Method of fabrication
(d) Contents of alloying elements

ISRO Scientist/Engineer 2006

Ans. (a) : The term hardened steel is often used for a medium or high carbon steel that has been given heat treatment and then quenching followed by tempering. Hardness of steel depends on the amount of carbon present in metal. Only steel that is high in carbon can be hardened and tempered. If a metal does not contain the necessary quantity of carbon, then its crystalline structure cannot be broken.

2. Atomic Structures, Crystallography and Phase Diagrams

137. Coordination number for FCC crystal structure is:-
(a) 4 (b) 6
(c) 8 (d) 12

RPSC Lect. (Tech. Edu. dept.) 13.03.2021 Paper-I
RPSC ACF & FRO 26.02.2021
Mizoram PSC (Insp. of Factories) 2019 Paper-III
Oil India Limited Sr. Engineer (Prod.) 30.11.2019
Mizoram PSC AE/SDO 2014 Paper-III
UKPSC AE-2013, Paper-I, ESE 2003

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Ans. (d) : Coordination number for FCC crystal structure is 12.

138. Which of the following is an amorphous material?

(a) Lead (b) Rubber
(c) Mica (d) Glass

ISRO Scientist/Engineer 22.04.2018
GWSSB DEE 07.07.2016
Rajasthan AE Nagar Nigam 2016 Shift-III
(KPSC AE 2015)
UKPSC AE 2012 Paper-I
TRB Poly Lect. 2012
UKPSC AE 2007 Paper -I

Ans : (d) Glass is an amorphous material.
Amorphous material:- The material in which the atoms are arranged chaotically is called amorphous material.

139. What is the packing factor of FCC crystal structure?

(a) 0.64 (b) 0.68
(c) 0.74 (d) 0.78

APPSC Poly Lect. 13.03.2020
OPSC AEE 2019 Paper-I
SJVN ET 2019
RPSC AE 2018
UPPSC AE 12.04.2016 Paper- I
Kerala PSC Asstt. Prof. 27.09.2016
UKPSC AE 2013 Paper-I
TRB Poly Lect. 2012

Ans : (c) :

Type of cell		Packing factor
Simple cubic (SC)	–	0.52
Hexagonal Close packed (HCP)	–	0.74
Body centered cubic (BCC)	–	0.68
Face-centered cubic (FCC)	–	0.74

140. What is the crystallographic structure of austenite?

(a) Simple cubic
(b) Face-centered cubic (F.C.C.)
(c) Body-centered cubic (B.C.C.)
(d) Hexagonal close-packed (H.C.P.)
(e) It cannot be determined

GPSC Executive Engg. 23.12.2018
VIZAG Steel MT (Re.Exam) 24.11.2018
RPSC AE 2018
GPSC ARTO Pre 30.12.2018
CGPSC Asstt. Workshop Supt., 17.07.2016
APPSC AEE MAINS 2016, PAPER-III
CGPSC AE 16.10.2016
GPSC Asst. Prof. (Prod) 21.08.2016
UP Jal Nigam AE 2016
Rajasthan Nagar Nigam AE 2016 Shift-I
CGPSC AE 15.01.2015
TSPSC AEE 2015
VIZAG Steel MT 10.06.2012
APPSC IOF 2009
DRDO Scientist 2008

Ans. (b) : Austenite stainless steels are defined by their face-centered cubic crystal structure (FCC).

141. Pearlite is a combination of :

(a) Ferrite and cementite
(b) Cementite and gamma iron
(c) Ferrite and austenite
(d) Ferrite and iron graphite

GPSC DEE Class-2 (GWSSB) 04.07.2021
OPSC AEE 2019 Paper-I
GPSC ARTO Pre 30.12.2018
Kerala PSC AE (Irrigation) 05.04.2017
Punjab PSC Lect. 06.08.2017
RPSC ADE 2016
Kerala PSC Asstt. Prof. 27.09.2016
APPSC Poly Lect. 2013
TRB Asst. Prof. 2012
TSPSC AEE 2015

Ans : (a) : Pearlite consists of 87% ferrite (layer are thick) and 13% cementite (thin layer) arranged alternatively in a lamelle formation.

142. Which of the following constituents of steels is softest and least strong?

(a) Austenite (b) Pearlite
(c) Cementite (d) Ferrite

JPSC AE 10.04.2021 Paper- II
CIL MT 27.02.2020
Rajasthan Nagar Nigam AE 2016 Shift- I
OPSC AEE 2019 Paper-I

Ans : (d) : Ferrite– Ferrite is known as α solid solution.

- It is an interstial solid solution of a small amount of carbon dissolved in α (BCC) iron.
- Stable form of iron below 912^0C.
- The maximum solubility is 0.025%C at 723^0C and it dissolves only 0.008%C at room temperature.
- It is the softest structure that appears on the diagram.

143. What is the crystal structure of alpha iron(Ferrite)?

(a) Body-centered cubic
(b) Cubic structure
(c) Face-centered cubic
(d) Hexagonal close packed

UKPSC AE 2013, 2012 Paper I
Rajasthan AE (Nagar Nigam) 2016 Shift-3
JPSC AE 10.04.2021
WBPSC AE 2003
Kerala PSC IOF 19.04.2016

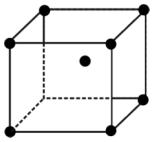
Ans. (a) :

Crystal Structure	Coordination No. (z)	Atomic Radius (r)	Atomic Packing Factor (APF)	Example
BCC	6	$r = \frac{a\sqrt{3}}{4}$	0.68	Fe (α-iron, δ-iron) Li, Na, K, Rb, Cs, Fr, Cr, Mo, Ta, W, V

Search on Telegram @apna_pdf

FCC	8	$r = \frac{a}{2\sqrt{2}}$	0.74	Fe (γ-iron), Al, Ca, Ce, Cu, Au, Pb, Ni, Pt, Ag
HCP	12	$r = \frac{a}{2}$	0.74	Zn, Ti, He, Cd, Co, Mg, Be, Zr

In α iron, the unit cell contains one atom at each of its 8 corner and 1 atom at the body centre, So α iron has structure of BCC.



144. The crystal structure of gamma iron is :

(a) Body centred cubic
(b) Face centred cubic
(c) Hexagonal close packed
(d) Cubic structure

Karnataka PSC AE 10.09.2017
Kerala LBS Center for Sci. & Tech. Asst. Prof. 2014
OPSC AEE 2019 Paper-I
Kerala PSC AE (Irrigation) 05.04.2017

Ans : (b) : The crystal structure of gamma iron is face centered cubic.

145. The metallic structure of mild steel is

(a) Hexagonal close packed
(b) Cubic structure
(c) Body-centered cubic
(d) Face-centered cubic

Rajasthan Nagar Nigam AE 2016, Shift-II
GMB AEE 25.06.2017
UKPSC AE 2013 Paper I
CGPSC Poly lect 22.05.2016

Ans. (c) : BCC structure – In this type of structure the unit cell contain one atom at each of its corners and another atom at the body centre.

Example –α iron (below 910^0C)
tungsten
vanadium

146. A FCC crystalline structure unit cell shares :

(a) 8 atoms (b) 9 atoms
(c) 13 atoms (d) 14 atoms

JPSC AE 10.04.2021, Paper-II
ISRO scientist / Engineer 22.04.2018
TSPSC manager (Engg.) HMWSSB 12.11.2020
ISRO Scientist/Engineer (RAC) 07.05.2017
Kerala PSC IOF 19.04.2016

Ans. (d) : FCC structure have 8 corner atoms and 6 atoms on each face centre.

So, A FCC crystalline structure unit cell shares
Total = 8 + 6 = 14 atoms

Atom per unit cell $= \frac{1}{8}\times 8 + \frac{1}{2}\times 6$
$= 4$ atoms per unit cell.

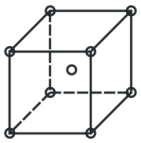
147. An atomic packing factor (APF) for the BCC unit cell of hard spheres atoms will be

(a) 0.63 (b) 0.68
(c) 0.73 (d) 0.78

TSGENCO AE 14.11.2015
ESE 2019
UKPSC AE 2007 Paper -I
CIL MT 26.03.2017
APPSE AEE 2016
UPPSC AE 13.12.2020 Paper-I
AAI Jr. Executive 26.03.2021

Ans. (b) : ■ Effective number of atom in the unit cell of BCC structure

$= 8\times\frac{1}{8} + 1 = 2$ atom



■ Relation between lattice constant and radius of atom

$4r = a\sqrt{3}$

$r = \frac{a\sqrt{3}}{4}$

$$\text{Atomic packing factor (APF)} = \frac{n\times\frac{4}{3}\pi r^3}{\text{Volume of unit cell}}$$

$$= \frac{n\times\frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3}$$

$$= \frac{2\times\frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3}$$

$= 0.68$

148. The effective number of lattice points in the unit cell of simple, body centered cubic and face centered cubic space lattices, respectively, are

(a) 1, 2, 2 (b) 1, 2, 4
(c) 2, 3, 4 (d) 2, 4, 4

GPSC Asstt. Prof. (Prod.) 21.08.2016
ISRO scientist / engineer 11.10.2015
VIZAG Steel MT 10.06.2012
VIZAG Steel MT 18.08.2013
GPSC Executive Engg. 23.12.2018
GATE 2009

Ans. (b) :

	No. of Atoms	r & a relation	APF	Coordination no.
SC	1	$2r = a$	0.52	6
BCC	2	$4r = \sqrt{3}a$	0.68	8
FCC	4	$4r = \sqrt{2}a$	0.74	12

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149. The fractional amount of volume or space occupied by atoms in a unit cell is termed as:

(a) Atomic packing density
(b) Volume density
(c) Inter-atomic space
(d) Linear atomic density
(e) Average volume of an atom

OPSC AE Mechanical Paper–I 15 Dec 2019
(CGPSC Polytechnic Lecturer 2017)
JWM 2017
HPPSC LECT. 2016

Ans. (a) : Atomic Packing Density– The ratio of volume occupied by average no. of atom to the volume of unit cell.

$$\text{Atomic Packing Factor} = \frac{\text{Volume occupied by } N_{avg}}{\text{Volume of unit cell}}$$

$$(APF) = \frac{N_{avg} \times \frac{4}{3}\pi r^3}{\text{Volume of unit cell}}$$

150. Gibb's phase rule is given by :- P = number of phases, F = number of degree of freedom; C = number of components.

(a) F = C + P (b) F = C + P - 2
(c) F = C - P - 2 (d) F = C - P + 2

UJVNL AE 2016
TNPSC AE 2018
HPPSC LECT. 2016
CGPSC Polytechnic Lecturer 2017
ESE 1999

Ans : (d) Gibb's phase rule:-
F = C - P +2
F = Degree of freedom of the system
C = No. of components of the system
P = No. of phase in equilibrium in the system
* The degree of freedom decreases as the no. of phase increases
* A two component system can not have more than 4 phase in equilibrium.

151. Effective number of atoms per unit cell of FCC structure is :

(a) 4 (b) 6
(c) 8 (d) 12

ISRO Scientist / engineer 12.01.2020, 22.04.2018
APPSC IOF, 2009, ESE 2003

Ans. (a) : FCC has a Co-ordination number of 12 and contains 4 atoms per unit cell.
The hexagonal closed packed (HCP) has a Co-ordination number of 12 and contains 6 atom per unit cells.

152. The Iron-Carbon diagram and T-T-T curves are determined respectively under

(a) Equilibrium and non-equilibrium conditions
(b) Non-equilibrium and equilibrium conditions
(c) Equilibrium conditions for both
(d) Non-equilibrium conditions for both

Karnataka PSC Lect., 27.05.2017
GWSSB AAE 27.12.2015
Mizoram PSC (Insp. of factories) 2019 Paper III

Ans (a) : • Iron- Carbon phase diagram is based on equilibrium condition where as TTT diagram is based on non equilibrium condition.

• The principal source of information on the actual process of austenite decomposition under non-equilibrium condition is the TTT diagram, which relates the transformation of austenite to the time and temperature conditions to which it is subjected.

153. The percentage of carbon in Eutectoid steel is

(a) 0.02% (b) 0.3%
(c) 0.6% (d) 0.8%

Karnataka PSC Lect., 27.05.2017
GPSC Asst. Prof. 28.08.2016
TSGENCO AE 14.11.2015

Ans (d) : Eutectoid point-
(i) Temperature = 723ºC
(ii) C-content in iron = 0.83%
Standard value = 0.76%C, Refer by WB Calister)
(iii) Eutectoid reaction

$$\text{Austenite } (\gamma\text{- iron}) \underset{\text{Heating}}{\overset{\text{Cooling}}{\rightleftharpoons}} \text{Ferrite } (\alpha\text{-iron}) + \text{Cementite } (Fe_3C)$$

(iv) This mixture of α- iron and cementite are in alternate layers called pearlite (α-iron + Fe_3C).

154. The crystalline structure of aluminum is :

(a) body centered cubic
(b) face centered cubic
(c) closed packed hexagonal
(d) body centered tetragonal

HPPSC Asst. Prof. 20.11.2017
ISRO Scientist/Engineer 24.05.2014
GPSC ARTO Pre 30.12.2018

Ans. (b) :
(i) Examples of FCC structure –
Aluminium, Copper, Nickel, γ-iron, Gold, Silver.
(ii) Examples of BCC structure –
Iron, Chromium, Tungsten, Niobium, Bainite etc.
(iii) Examples of HCP structure –
Beryllium, Cadmium, Magnesium, Titanium, Zinc and Zirconium.

155. Hardness of Martensite is about-

(a) RC 32 (b) RC 48
(c) RC 65 (d) RC 80

ISRO scientist / Engineer 2012
GMB AAE 25.06.2017
ISRO Scientist/Engineer 2007

Ans. (c) : Hardness of martensite is about RC 65

156. Eutectic reaction for iron - carbon system occurs at-

(a) 600ºC (b) 723ºC
(c) 1147ºC (d) 1490ºC

RPSC AE 2018, UJVNL AE 2016
Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : Peritectic Reaction—It take place at 1442ºC at 0.18% C

$$\delta + L \rightleftharpoons \gamma$$

Eutectic Reaction—It take place at 1147ºC at 4.3% C.

$$L \rightleftharpoons (\gamma + Fe_3C)$$

$$L \rightleftharpoons S_2 + S_3$$

$\gamma + Fe_3C \rightarrow$ Ledeburite $\rightarrow$ It is eutectic mixture of austenite and cementite.

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Eutectoid reaction— It take place at 727°C at 0.8% C.

$\gamma \rightleftharpoons \alpha + Fe_3C$

$S_2 \rightleftharpoons S_4 + S_3$

$\alpha + Fe_3C \rightarrow$ A mixture of pearlite

157. Austempering of steels results in greater,
(a) hardness (b) toughness
(c) brightness (d) ductility

TNPSC AE 2018, Gujarat PSC AE 2019
APPSC Poly. Lect. Automobile Engg. 2020

Ans. (b) : The austempered steel has higher strength and ductility compared to cast steel with increasing austempering temperature, hardness and strength decreased but the percentage of elongation increased.

158. Delta iron occurs at the temperature,
(a) above recrystallisation temperature
(b) above melting point
(c) between 1400°C and 1539°C
(d) between 910°C and 1400°C

TNPSC AE 2018
Mizoram PSC AE/SDO 2014 Paper-III
UPPSC AE 13.12.2020 Paper-I

Ans. (c) :

Name	Type	Carbon content	Temperature	Structure
Ferrite	α	0.022%	Below 725°C	BCC
Austenite	γ	2.14%	910°C-1410°C	FCC
δ-iron	δ	0.1%	1410°C-1539°C	BCC
Cementite	Fe_3C	6.61%	below 1175°C	ortho-rhombic

159. The weight percentage of carbon, in cementite is
(a) 13.00% (b) 6.67%
(c) 4.00% (d) 0.40%

BPSC AE Mains 2017 Paper - VI
PSPCL AE 2012
OPSC AEE 2015 Paper-I

Ans : (b) : Weight percent of carbon in cementite

$= \frac{12 \times 100}{3 \times 56 + 12}$

$= 6.67\%$

160. Which one of the following statements about a phase diagram is INCORRECT?
(a) Solid solubility limits are depicted by it.
(b) It gives information on transformation rates.
(c) It indicates the temperature at which different phases start to melt.
(d) Relative amount of different phases can be found under given equilibrium conditions.

Assam Engg. College AP/Lect. 18.01.2021
APGCL AM 2021

Ans. (b) : Phase diagram does not provide information on transformation rates.

161. On heating, one solid phase results in another solid phase and a liquid phase during _____ reaction.
(a) Eutectoid (b) Peritectic
(c) Eutectic (d) Peritectoid

RPSC Vice Principal ITI 2018
Assam PSC AE (IWT) 14.03.2021

Ans. (b) :
■ **Eutectic reaction :-**
Liquid → solid 1 + solid 2
■ **Eutectoid reaction :-**
Solid → Solid 2 + Solid 3
■ **Peritectic reaction :-**
Liquid + Solid 1 → Solid 2
■ **Peritectoid reaction :-**
Solid 1 + Solid 2 → Solid 3

162. How many space lattices does the Bravis lattices consist of?
(a) 3 (b) 7
(c) 9 (d) 14

UPPSC AE 12.04.2016 Paper- I
RPSC Lect. (Tech Edu. Dept.) 13.03.2021 Paper-I

Ans : (d) There are 14 possible Bravais lattices in three-dimensional space.
These 14 Bravais lattices are grouped into seven lattice system:
Triclinic, Monoclinic, Orthorhombic, Tetragonal, Rhombohedral, Hexagonal, and Cubic.

163. Eutectic product in Fe-C system is called
(a) Pearlite (b) Ledeburite
(c) Bainite (d) Spheroidite

GPSC EE Pre, 28.01.2017
BPSC AE Mains 2017 Paper - VI

Ans. (b) : Ledeburite is an eutectic mixture of austenite and cementite in the form of alternate layers the average carbon content in ledeburite is 4.3%.

164. When steel containing less than 0.85% carbon is cooled slowly below the lower critical point, it contains
(a) Ferrite mainly
(b) Pearlite mainly
(c) Ferrite and pearlite
(d) Pearlite and cementite

GPSC EE Pre, 28.01.2017

Ans. (c) : When steel containing less than 0.85% carbon is cooled slowly below the lower critical point it contains ferrite and pearlite.

165. Which of the following is a line defect found in metal crystals?
(a) Grain boundary (b) Cracks
(c) Vacancy (d) Dislocation

APGENCO AE, 2017
UKPSC AE 2012 Paper- I
ISRO Scientist/Engineer 2011
BPSC Poly. Lect. 2016
(KPSC AE. 2015)

Ans. (d) : Point defect or zero dimensional defect.
(a) Vacancy defect (b) Interstitial defect
(c) Substitutional defect (d) Frenkel defect
(e) Schottky defect
Surface defect –
(a) Grain boundary defect (b) Twin boundary defect
(c) Tilt boundary defect (d) Stacking faults
Line defect –
(a) Edge dislocation (b) Screw dislocation

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166. Stainless steels are highly corrosion resistance due to the presence of

(a) Chromium (b) Manganese
(c) Molybdenum (d) Nickel

UPPSC AE 13.12.2020, Paper-I

Ans. (a) : Stainless steel is well known for being resistant to corrosion from things that make plain steel and iron (and other material) rust away due to the presence of chromium.

167. Martensite is a super saturated solution of carbon in ..... .

(a) alpha iron (b) gamma iron
(c) beta iron (d) delta iron

GWSSB DEE 07.07.2016
RPSC IOF 2020
BPSC AE Mains 2017 Paper - VI
Mizoram PSC (Insp. of factories) 2019 Paper- III
Kerala PSC AE (Irrigation) 05.04.2017
GWSSB DEE 07.07.2016
KPSC AE. 2015, ESE 2001

Ans. (a) : Martensite is a super saturated solution of carbon in alpha iron.

168. The predominant structure of a hypereutectoid steel that has been quenched at above its upper critical temperature will be

(a) Austenite (b) Martensite
(c) Troosite (d) Surbite

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : The predominant structure of a hypereutectoid steel that has been quenched at above its upper critical temperature will be martensite.

169. Fine grains of austenite

(a) decrease hardenability
(b) increase hardenability
(c) first decrease, then increase hardenability
(d) first increase, then decrease hardenability

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : Fine grains of austenite decrease hardenability.

170. A steel with 0.8% C is called

(a) hypoeutectoid steel
(b) hyper eutectoid steel
(c) eutectoid steel
(d) None of these

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : A steel with 0.8% C is called eutectoid steel.

171. When a steel is heated to above its upper critical temp, the structure produced is one of

(a) Martensite (b) Austenite
(c) Pearlite (d) Sorbite

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Steel is heated to above its upper critical temperature, the structure produced is austenite.

172. Eutectoid reaction occurs at

(a) 600^0C (b) 723^0C
(c) 1173^0C (d) 1493^0C

GPSC DEE, Class-2 (GWSSB) 04.07.2021, ESE 1995

Ans. (b) : Eutectoid reaction occurs at 723°C.

173. In a eutectic system, two elements are completely

(a) insoluble in solids and liquids
(b) soluble in liquid state
(c) soluble in solid state
(d) insoluble in liquid state

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GPSC ARTO Pre 30.12.2018, ESE 1999

Ans. (b) : In a eutectic system, two elements are completely soluble in liquid state.

174. Upper critical temp for steel

(a) is constant
(b) depends upon the rate of heating
(c) varies according to the carbon in steel
(d) depends upon the rate of cooling

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Upper critical temp for steel varies according to the carbon in steel.

175. Carbon occurs in steel in the combined state with iron to form the component

(a) Ferrite (b) Cementite
(c) Peralite (d) Bainite

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Carbon occurs in steel in the combined state with iron to form the component cementite.

176. The number of atoms per unit cell and the number of slip systems, respectively, for a face-centred cubic (FCC) crystal are

(a) 3, 3 (b) 3, 12
(c) 4, 12 (d) 4, 48

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) :

177. Which one of the is not an allotropic form of iron?

(a) β (b) δ (c) α (d) γ

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : At atmospheric pressure three allotropic forms of iron exist alpha iron (α-Fe), gamma iron (γ-Fe) and delta iron (δ-Fe).

178. Which one of the following metallic crystals has face centred cubic crystal structure form?

(a) Titanium (b) Cadmium
(c) Silver (d) Magnesium

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : Face centred cubic crystal structure–

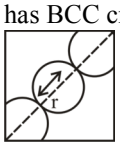
- Silver, gold, copper, nickel, gamma iron, aluminium etc.

179. Molybdenum is having _____ crystal structure.

(a) BCC (b) HCP
(c) FCC (d) SC

Oil India Senior Officer 23.12.2020

Ans. (a) Molybdenum has BCC crystal structure where



$4r = \sqrt{3}a$, No. of atoms = 2 , APF = 68%

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180. The temperature above which a material loses its ferroelectric properties is :
(a) Ferro temperature
(b) Insulation temperature
(c) Curie temperature
(d) Critical temperature

TNPSC AE (Industries) 09.06.2013

Ans. (c)

181. Eutectoid steel contains the following percentage of carbon:
(a) 0.02% (b) 0.03%
(c) 0.33% (d) none of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Eutectoid steel contain 0.78 percentage of carbon.
This steel contains about 0.78 percent carbon in pure iron carbon alloy. In alloy steel the eutectoid carbon content is usually lower steel with less than the eutectoid carbon content are known as hypoeutectoid and contain free ferrite as well as pearlite.
carbon % less than 0.78% – hypoeutectoid steel
carbon % more than 0.78% – hypereutectoid steel.

182. Hypo eutectoid steels have carbon content:
(a) equal to 0.93%
(b) less than 0.83%
(c) more than 0.93% and upto 2%
(d) more than 2%

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Hypoeutectoid steel have less than 0.83% of carbon.
- Eutectoid steel has equal to 0.83% of carbon.
- Hyper eutectoid steel is more than 0.83% of carbon.

183. Twin boundaries are which type of crystal defect?
(a) Line defect (b) Point defect
(c) Surface defect (d) None of the above

UPPSC AE 13.12.2020, Paper-I
APPSC Poly Lect. 13.03.2020

Ans. (c) : Types of surface defects are –
(i) External surface (ii) Grain boundaries
(iii) Tilt boundary (iv) Twist boundary
(v) Twin boundary
Types of point defect –
(i) Vacancy, (ii) Interstitial, (iii) Frenkel, (iv) schottky.
Types of line defect –
(i) Edge dislocation
(ii) Screw dislocation

184. A bond that can be strong in some cases and weak in some other cases is _____.
(a) Van der Walls bond (b) ionic bond
(c) metallic bond (d) covalent bond

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Covalent and ionic bonds are both typically considered strong bonds. However other kinds of more temporary bonds can also form between atoms or molecules.
Covalent bond that can be strong in some cases and weak in some other cases.

185. Iron carbon equilibrium diagram :
(a) indicates the phase changes occurring during heating and cooling
(b) correlates the microstructure and properties of steel and cast iron
(c) is made by plotting carbon percentage and temperature
(d) all of these

Kerala PSC IOF 19.04.2016

Ans. (d)
- Indicates the phase changes occurring during heating and cooling.
- Correlates the microstructure and properties of steel and cast iron.
- Is made by plotting carbon percentage and temperature.

186. The iron has a crystal structure:
(a) Simple cubic
(b) Body-centered cubic
(c) Face centered cubic
(d) Close packed hexagonal

GPSC Lect. 23.10.2016
CIL MT 27.02.2020

Ans. (b) : In BCC structure unit cell contains one atom at each of its 8 corners and another atom at center.
Examples– α iron, tungsten, vanadium.
In FCC– γ iron (910 °C to 1400 °C), copper, silver, gold, aluminium etc.

187. Many composite materials are composed of just two phase know as–
(a) Matrix and dispersed
(b) Solid and liquid
(c) Liquid and gas
(d) Solvent ad solute

Nagaland PSC CTSE 2016, 2017 Paper-I

Ans. (a) : Composite material is a material composed of two or more distinct phases (matrix phase and dispersed phase).

188. Metals which can be plastically deformed very easily have unit cell of:
(a) cubic structure (b) H.C.P. structure
(c) F.C.C. structure (d) B.C.C. structure
(e) either H.C.P. structure or B.C.C. structure

CGPSC Asstt. Workshop Supt., 17.07.2016
Karnataka PSC lect 27.05.2017

Ans. (a) : Metals having cubic structure (simple) can be plastically deformed very easily have unit cell this, because of large gap between each atoms. [less packing factor].

189. Which of the following is a point imperfection?
(a) Vacancy
(b) Interstitial imperfection
(c) Frenkel imperfection
(d) All of these

SJVN ET 2013
UKPSC AE-2013, Paper-I

Ans. (d) : Point imperfections or zero imperfections
- Vacancy
- Interstitial imperfection.
- Frenkel imperfection.

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190. How many atoms are there in an HCP crystal structure?

(a) 8 (b) 4
(c) 6 (d) None of these

PSPCL AE 2012
Gujarat PSC AE 2019

Ans : (c) : The hexagonal closed packed (HCP) has a coordination number of 12 and contains 6 atoms per unit cell.

191. Iron-carbon alloy containing 1.7 to 4.3% carbon is called

(a) Eutectoid cast Iron
(b) Hyper eutectic cast Iron
(c) Hypo-eutectic cast Iron
(d) Eutectoid steel

TNPSC AE 2017
GPSC Engg. Class-2 Pre 19.01.2020

Ans. (c) : Iron-carbon alloy containing 1.7 to 4.3% carbon is called Hypo-eutectic cast Iron.

192. Which of the following is not true for Ferrite?

(a) It has BCC structure
(b) It is soft and ductile
(c) Solubility of carbon in ferrite is very high
(d) It is alpha-iron and highly magnetic
(e) It is alpha-iron and non-magnetic

CGPSC AE 25.02.2018
UPRVUNL AE 07.10.2016

Ans. (c) : Solubility of carbon in ferrite is very high

193. Which of the following is not a structural steel shape?

(a) I (b) T
(c) H (d) V

GMB AAE 25.06.2017
Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Not structural steel shape V.

194. Which of the following is a line defect?

(a) Edge dislocation (b) Impurity
(c) Twinning (d) Vacancy

SJVN ET 2019
GPSC ARTO 01.05.2016

Ans. (a) : Line Defect-
(i) Edge dislocation defect
(ii) Screw dislocation defect

195. The crystal structure of brass is

(a) BCC (b) FCC
(c) HCP (d) Orthorhombic

Gujarat PSC AE 2019
Nagaland PSC CTSE 2017, Paper-I

Ans : (b) : Examples of FCC– Nickel, gold, copper, brass, lead, aluminium.

196. What is coordination number BCC crystal structure?

(a) 4 (b) 8
(c) 12 (d) 1

APPSC Poly Lect. 2013
OPSC AEE 2019 Paper-I

Ans : (b) :

Crystal structure		Coordination number
SC	–	6
BCC	–	8
FCC	–	12
HCP	–	12

197. Carbon content is highest in :-

(a) Mild steel
(b) Eutectoid steels
(c) Hypoeutectoid steels
(d) Hypereutectoid steels

UKPSC AE-2013, Paper-I
Punjab PSC (Lect.) 06.08.2017

Ans. (d) : Carbon content is highest in hypereutectoid steels.

198. An eutectoid steel consists of

(a) wholly pearlite
(b) wholly austenite
(c) pearlite and ferrite
(d) pearlite and cementite

ISRO Scientist/Engineer 11.10.2015
GPSC Engg. Class-2 Pre 19.01.2020
Sikkim PSC (Under Secretary) 2017
UKPSC AE-2013, Paper-I

Ans. (a) : Eutectoid Reaction– (727°C, 0.8%C)
Decomposition reaction

$$\text{Solid} \xrightarrow[-\Delta T]{} \text{Solid}$$

$$\gamma \xrightarrow{-\Delta H} \boxed{\alpha + Fe_3C} \text{ Pearlite}$$

$$S_2 \rightleftharpoons S_4 + S_3$$

Pearlite– Eutectoid mixture of ferrite and cementite.

199. Pure iron is the structure of -

(a) Ferrite (b) Pearlite
(c) Austenite (d) Cementite

GMB AAE 25.06.2017
UKPSC AE 2012 Paper-I

Ans. (a) : Ferrite, also known as alpha Ferrite or alpha iron is a material science term of pure iron with a body centred cubic (BCC) crystal structure, In pure iron, Ferrite is stable below 910°C, above this the Face centred cubic form of iron, austenite (γ-iron) is stable.

200. If a particular Fe-C alloy contains less than 0.83% carbon it is called.

(a) High speed steel
(b) Hypoeutectoid steel
(c) Hypereutectoid steel
(d) Cast iron

ESE 2018
GPSC ARTO Pre 30.12.2018

Ans. (b) : If a particular Fe-C alloy contains less than 0.83% carbon it is called Hypoeutectoid steel. If Fe-C alloy contains more than 0.83% carbon it is called Hypereutectoid steel.

201. Which one of the following cast irons consists of carbon in rosette form?

(a) White cast iron (b) Gray cast iron
(c) Malleable cast iron (d) Nodular cast iron

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Kerala PSC Poly. Lect. 09.01.2014
ESE 2009

Ans. (c) : Malleable cast irons consists of carbon in rosette form.

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202. Which one of the following related to the most stable arrangement of atoms in a crystal is not correct?

(a) Preserves electrical neutrality
(b) Maximizes strong ion-ion repulsion
(c) Satisfies discreteness of all covalent bonds
(d) Packs the atoms as closely as possible

ESE 18.07.2021

Ans. (b) : Stability of atoms in crystal depends on :

- Electrical neutrality.
- Atoms should have greater atomic packing factor.
- The ease of formation of the positive and negative ions from respective neutral atoms.
- Satisfies discreteness of all the covalent bonds.
- As many anions as possible to surround the cation for maximum reduction in electrostatic energy.

203. Iron at 20°C is BCC with atoms of atomic radius 0.124 nm. What is the lattice constant 'a' for the cube edge of the iron unit cell?

(a) 0.2864 nm (b) 0.1496 nm
(c) 0.2173 nm (d) 0.1756 nm

ESE 18.07.2021

Ans. (a) : Given atomic radius r = 0.124 nm

$$4r = \sqrt{3}a$$

$$4 \times 0.124 = \sqrt{3} \times a$$

or $a = 0.286365 \approx 0.2864$ nm

204. Copper has the FCC crystal structure and a unit cell with a lattice constant of 0.361 nm. What is the interplanar spacing 'd_{220}' ?

(a) 0.085 nm (b) 0.174 nm
(c) 0.206 nm (d) 0.128 nm

ESE 18.07.2021

Ans. (d) : Given, lattice constant, a = 0.361 nm

$$h = 2, \quad k = 2, \quad \ell = 0$$

Interplanar spacing-

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + \ell^2}}$$

or, $$d_{220} = \frac{0.361}{\sqrt{2^2 + 2^2 + 0}}$$

$$= 0.12763 \approx 0.128$$

205. Which one of the following is not a step of lever rule of determination of phase amounts?

(a) A tie line is constructed across the two-phase region at the temperature of the alloy.
(b) The overall alloy composition is located on the tie line.
(c) Perpendiculars are dropped from these intersections to the horizontal composition axis, from which the composition of each of the respective phases is read.
(d) The fraction of one-phase is computed by taking the length of tie line from the overall alloy composition to the phase boundary for the other phase and dividing by the total tie line length.

ESE 18.07.2021

Ans. (c) : The Lever Rule is used to find the fractions of liquid phase in the binary alloy in the twophase state. The rule can be obtained by using the law of conservation of mass.
The tie line must be used along with a procedure that is generally termed as lever rule (or the inverse lever rule) which is as follows (same is marked in figure).
(1) The tie line is constructed across the two phase region at the temperature of the alloy.
(2) The overall alloy composition is located on the tie line.
(3) The fraction of one phase is computed by taking the length of tie line from the overall alloy composition to the phase boundary for the other phase and dividing by the total tie line length.
(4) The fraction of the other phase is determined in the same manner.

206. The property of a metal existing in more than one crystalline form is

(a) Crystallisation (b) Solidification
(c) Polymorphism (d) Resilescence

TSPSC AEE 2015

Ans : (c) The property of a metal existing in more than one crystalline form is polymorphism.

207. The difference between Graphite and Diamond is that

(a) Diamond is transparent while Graphite is opaque.
(b) Diamond is insulator while Graphite is conductor.
(c) Diamond has all primary bonds while Graphite has three primary and one secondary bonds.
(d) All the above

UKPSC AE 2012 Paper-I

Ans. (d) :

- Diamond is transparent while Graphite is opaque.
- Diamond is insulator while Graphite is conductor.
- Diamond has all primary bonds while Graphite has three primary and one secondary bonds.

208. Graphene is

(a) A two dimensional material consisting of a single layer of carbon atoms arranged in a honeycomb or chicken wire structure.
(b) A tube-shaped material made of carbon, having a diameter measuring on the nanometre scale.
(c) Completely non-transparent material.
(d) None of the above.

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : Graphene– Graphene is a 2-D carbon allotrope with the carbon atoms arranged, in a 2-D honeycomb lattice. It is an extremely thin material alongwith being flexible and transparent.

209. Choose the correct eutectic reaction in steel being cooled from liquid state

(a) $\text{Liquid} \xrightarrow[738^0C]{4.3\%C} \text{Austenite} + \text{Cementite}$
(b) $\text{Liquid} \xrightarrow[1148^0C]{4.3\%C} \text{Austenite} + \text{Cementite}$
(c) $\text{Liquid} \xrightarrow[1148^0C]{4.9\%C} \text{Austenite} + \text{Cementite}$
(d) $\text{Liquid} \xrightarrow[738^0C]{4.9\%C} \text{Austenite} + \text{Cementite}$

VIZAG STEEL MT 2015

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Ans. (b) : The correct eutectic reaction in steel being cooled from liquid state.

Liquid $\xrightarrow[1148^0C]{4.3\%C}$ Austenite + cementite

210. Copper has an atomic radius of 0.128 nm (1.28A°), a FCC crystal structure and an atomic weight of 63.5 g/mol. Its theoretical density in g/cm^3 is :

(a) 8.94 (b) 8.89
(c) 8.50 (d) 8.40

RPSC ACF & FRO, 26.02.2021, ESE 2015

Ans. (b) : Given,
Atomic radius = 0.128 nm
In FCC effective number of atoms in unit cell (N_{av}) = 4
Atomic weight of crystal = 63.5 g/mol
In, FCC relation between edge length and atomic radius

Edge length, $a = \frac{4r}{\sqrt{2}}$

$$a = \frac{4\times0.128\times10^{-9}}{\sqrt{2}}$$

$a = 3.620 \times 10^{-10}$ m

Volume of unit cell, $a^3 = (3.620 \times 10^{-10})^3$
$= 4.745 \times 10^{-29}$ m^3

Density $(\rho) = \frac{N_{av} \times \text{Atomic weight}}{\text{Avogadro No.} \times a^3}$

$$= \frac{4\times63.5}{6.023\times10^{23}\times4.745\times10^{-29}}$$

$= 8887602.791$ gm/m^3
$= 8.88$ gm/cm^3

211. A screw dislocation.

1. lies parallel to its Burger's vector.
2. lies perpendicular to its Burger's vector
3. moves in perpendicular direction to the Burger's vector.
4. moves in an inclined direction to the Burger's vector.

Select the correct answer using the codes given below :

(a) 1 and 4 (b) 1 and 3
(c) 2 and 3 (d) 2 and 4

RPSC ACF & FRO, 26.02.2021, ESE 2003

Ans. (b) : A screw dislocation-
(1) It lies parallel to its burger's vector.
(2) It moves in a direction perpendicular. to the burger vector

Edge dislocation –
(1) It moves in direction of Burger's vector
(2) An edge dislocation may be described as an extra plane of atoms within a crystal structure.

212. Movement of block of atoms along certain crystallographic phases and directions termed as

(a) Glide (b) Twinning
(c) Slip (d) Jog

Kerala PSC Poly. Lect., 2017
ESE 2007

Ans. (c) : Movement of block of atoms along certain crystallographic phases and directions termed as slip.

213. The correct order of co-ordination number in SC, BCC, FCC and HCP unit cells is :

(a) 12, 12, 8, 6 (b) 6, 8, 12, 12
(c) 6, 6, 8, 12 (d) 12, 8, 6, 12

Kerala PSC Poly. Lect., 2017

Ans. (b) : Co-ordination number is the number of nearest neighbour to a particular atom in the crystal.

S.No.	Properties	Simple cubic	BCC	FCC
1.	No. of atoms in unit cell	1	2	4
2.	Atomic radius (r)	$\frac{a}{2}$	$\frac{a\sqrt{3}}{4}$	$\frac{a\sqrt{2}}{4}$
3.	Volume of unit cell	a^3	a^3	a^3
4.	Volume of atoms in a unit cell	$\frac{\pi a^3}{6}$	$2\times\frac{\pi\sqrt{3}a^3}{16}$	$4\times\frac{\pi\sqrt{2}a^3}{24}$
5.	Atoms packing factor	0.52	0.68	0.74
6.	Co-ordination no.	6	8	12

214. Which one of the following metals is not of face centered cubic structure?

(a) Aluminium (b) Chromium
(c) Copper (d) Nickel

TRB Poly. Lect., 2012

Ans. (b) : Among the given, only chromium which is not FCC structure, chromium lead to in BCC structure.

215. Peritectoid reaction is

(a) Liquid $\rightleftharpoons$ Solid 1 + Solid 2

(b) Liquid + Solid 1 $\rightleftharpoons$ Solid 2

(c) Solid 1 $\rightleftharpoons$ Solid 2 + Solid 3

(d) Solid 1 + Solid 2 $\rightleftharpoons$ Solid 3

APPSC Poly. Lect. 2013

Ans. (d) : Peritectoid phase diagram involve transformation of two solid phases into a different solid phase on cooling. It is given by-

$\gamma + \beta$ (Solid phase) $\underset{\text{Heating}}{\overset{\text{Cooling}}{\rightleftharpoons}}$ α (Solid phase)

or Solid 1 + Solid 2 $\rightleftharpoons$ Solid 3

216. Iron at room temperature has BCC structure and atomic radius f 1.24Å. The lattice constant of iron will be

(a) 3.864 Å (b) 2.864 Å
(c) 1.864 Å (d) 0.864 Å

VIZAG STEEL MT 2015

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Ans. (b) : $R = \frac{a\sqrt{3}}{4}$

$a = \frac{4R}{\sqrt{3}} = \frac{4 \times 1.24}{1.7}$

$= \frac{4 \times 12}{17}$

$= 2.864 Å$

217. Copper has FCC structure, its atomic radius is 1.28 Å and atomic mass is 63.5 The density of copper will be

(a) 8.9×10^3 kg/mm^3 (b) 8.9×10^3 kg/cm^3

(c) 8.9×10^3 kg/m^3 (d) 8.9×10^3 g/mm^3

VIZAG STEEL MT 2015
ESE 2015

Ans. (c)

$$\text{Density (g/m}^3) = \frac{\text{mass} \times \text{No.of atoms in a Unit cell}}{\text{Volume} \times \text{Avagadro's Number}}$$

Volume of a unit cell $= a^3 = \left(2\sqrt{2}R\right)^3$

$= 47.46 \times 10^{-30}$ m^3

$= \frac{4 \times 63.5}{47.46 \times 10^{-30} \times 6.023 \times 10^{23}}$

$= 8.885 \times 10^6$ g/m^3 $\approx 8.9 \times 10^3$ kg/m^3

218. A Plane in a unit cell is described by its miller indices a (632). The plane intersects x, y and z respectively at a point whose distances from origin are

(a) 6, 3 and 2 units

(b) 1/6, 1/3 and 1/2 units

(c) 1/3, 2/3 and 1 units

(d) 1/2, 1 and 2/3 units

VIZAG STEEL MT 2015

Ans. (c) : Take reciprocal

$\frac{1}{6}\frac{1}{3}\frac{1}{2}$

Intercepts are

$\frac{6}{6}\frac{6}{3}\frac{6}{2}$

1 2 3

$\frac{1}{3}\frac{2}{3}$ 1 Units

219. Which of the following pairs describe the correct coordination number for body centered cubic and face centered cubic unit cell structures respectively?

(a) 8 and 6 (b) 12 an 8

(c) 6 and 8 (d) 8 and 12

RPSC AEN Pre-2013

Ans. (d) :

Unit cell	Coordination Number (CN)
BCC	8
FCC	12
HCP	12

220. Which of the following unit cells are arranged in ascending order of atomic packing factor? (The abbreviations used are BCC = Body Centered Cubic, FCC = Face Centered Cubic, SC = Simple Cubic, HCP = Hexagonal Close Packing).

(a) BCC, FCC, SC (b) BCC, FCC, HCP

(c) HCP, FCC, BCC (d) FCC, HCP, BCC

RPSC AEN Pre-2013

Ans. (b) :

Unit cell	Atomic Packing Factor (APF)
SC	0.52
BCC	0.68
FCC	0.74
HCP	0.74

Ascending order – BCC < FCC < HCP.

221. Consider the following statements about edge dislocations in crystal structures.

(a) The Burger vector is perpendicular to the dislocation line.

(b) The Burger vector is parallel to the dislocation line.

(c) Only shear stress field exists.

(d) It can exhibit both climb and glide motions.

RPSC AEN Pre-2013

Ans. (a)

- The Burger vector is perpendicular to the dislocation line in edge dislocation.
- The Burger vector is parallel to dislocation line in screw dislocation.

222. As per Gibb's phase rule, if number of components is equal to 2, then the number of phases will be

(a) ≤ 2 (b) ≤ 3

(c) ≤ 4 (d) ≤ 5

RPSC IOF, 2020
ESE 2002

Ans. (c) : Gibb's phase rule

Given C = 2 P + F = C + 2

P + F = 4

Let F = 0, 2, 2, 3..... P = 4 – F

(P) number of phases ≤ 4

223. The fine grain size in austenite steel increase

(a) toughness (b) creep strength

(c) hardenability (d) rough machinability

RPSC IOF, 2020

Ans. (a) : Effect the austenitic grain size of steels. Coarse grains raise the ductile to brittle transition temperature. One of the most significant effect of fine grains is that fine grains improve the strength and toughness considerably as compared to coarse grains.

224. Match List-I (Material) with List-II (Structure) and select the correct answer using the codes given below the lists :

	List -I	List-II
(A)	Charcoal	(1) FCC
(B)	Graphite	(2) HCP
(C)	Chromium	(3) Amorphous
(D)	Copper	(4) BCC

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Codes :

	A	B	C	D
(a)	2	3	1	4
(b)	3	2	1	4
(c)	3	2	4	1
(d)	2	3	4	1

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 1998

Ans. (c) : A graphite has HCP (Hexagonal closed packed), chromium has BCC copper has FCC at room temp and charcoal have Amorphous lattice.

225. If (3 2 6) are the Miller indices of a plane, the intercepts made by the plane on the three crystallographic axes are

(a) (a, b, c) (b) (2a, 3b, c)
(c) (a, 2b, 3c) (d) (2a, b, 3c)

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : Miller indices are (3 2 6)
Hence crystallographic axis are,

$$\frac{6a}{3},\frac{6b}{2},\frac{6c}{6}=(2a,3b,c)$$

226. Screw dislocation in a material is a ___defect.

(a) Line (b) Point
(c) Plane (d) Volumetric

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : A screw dislocation is a topographical defect of a crystal lattice, if one moves around the dislocation the lattice plane shifts by one layer. The Burgers vector of a screw dislocation is parallel to the dislocation line.

227. Which one of the following is correct for Burger's vector in screw dislocation?

(a) Perpendicular to the dislocation line
(b) Inclined to the dislocation line
(c) parallel to the dislocation line
(d) Opposite to the dislocation line

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 2009

Ans. (c) : The Burgers vector of a screw dislocation is parallel to the dislocation line.
The Burger vector of a edge dislocation is perpendicular to the dislocation line.

228. What type of fracture occurs when crack propagation is along grain boundaries?

(a) Intergranular fracture
(b) Transgranular fracture
(c) Cup and cone fracture
(d) Granular helicoidal

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : Intergranular fracture occurs when a crack propagates along the grain boundaries of a material, usually when these grain boundaries are weakened. The more commonly seen transgranular fracture, occurs when the crack grows through the material grains.

229. A cation vacancy and an anion vacancy in a crystal of the type AB is called

(a) Schottky defect (b) Frenkel defect
(c) pair of vacancies (d) None of the above

Assam PSC AE (IWT) 14.03.2021

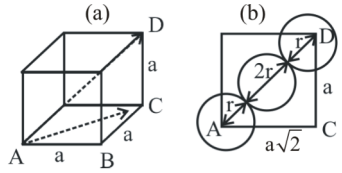
Ans. (a) : Schottky Defect–A cation vacancy and an anion vacancy in a crystal of the type AB is called schottky defect.
Frenkel Defect–The defect forms when an atom or smaller ion leaves its place in the lattice creating a vacancy and becomes an interstitial by lodging in a near by location.
Pair of Vacancies–A vacancy (or pair of vacancy in an ionic solid) is sometimes called a schottky defect. Interstitial defects are atoms that occupy a site in the crystal structure at which there is usually not an atom.

230. Atomic radius of BCC structure is given by ____.
[a = length of cube edge]

(a) $\frac{a\sqrt{3}}{4}$ (b) $\frac{4a}{\sqrt{3}}$
(c) $\frac{a}{2\sqrt{2}}$ (d) $\frac{a}{2}$

APPSC Poly Lect. 13.03.2020

Ans. (a) : Atomic radius of BCC structure –
In ΔACD



a = length of cube edge
Let,
r = atomic radius
Figure (a); AC = $\sqrt{a^2+a^2}=a\sqrt{2}$
From figure (b)

$$AD=\sqrt{\left(a\sqrt{2}\right)^2+a^2}=a\sqrt{3}$$

$$\Rightarrow 4r=a\sqrt{3}$$

$$\Rightarrow \boxed{r=\frac{a\sqrt{3}}{4}}$$

231. Which of the following is NOT a point defect?

(a) Edge dislocation
(b) Vacancy
(c) Substitutional impurity
(d) Interstitial impurity

APPSC Poly Lect. 13.03.2020
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : The defect which take place due to imperfect packing of atom during crystallisation are known as point defect.
Types of point defect
(a) Frenkel defect (b) Schottky defect
(c) Vacancy defect (d) Interstitial defect
(e) Substitutional defect

232. Gibb's phase rule is given by the equation where F = number of degrees of freedom; c = number of components, p = number of phases in, n = number of external factors

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(a) $F = c + n - p$ (b) $f = c + n + p$
(c) $F = c + 2n - p$ (d) $F = c + 2n - 2p$

TSPSC AEE 2017

Ans. (a) : Gibb's phase rule $F = c + n - p$

233. The following structures are arranged in the order of decreasing hardness:
(a) Martensite, troosite, bainite, sorbite
(b) Martensite, bainite, troosite, sorbite
(c) Martensite, sorbite, trosite, bainite
(d) Trossite, bainite, sorbite, Martensite

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : Martensite have highest hardness > troosite > bainite > sorbite

234. The temperature at which carbon steel, when being heated, transforms from pearlite to austenite, this is around 1330°F (721°C) for 0.83 percent steel, is called :
(a) Recalescence point (b) Decalescence point
(c) Tripple point (d) Null point

CGPSC AE 15.01.2021

Ans. (b) : Decalescence point- The temperature where carbon steel transforms from pearlite to Austenite; 1300°F for 0.83% carbon steel.
Recalescence Point- The temperature where carbon steel transfer from Austenite to pearlite when being slowly cooled.

235. In the state at which iron exists below the lower critical temperature, the atoms form a body-centered cube, is known as :
(a) Beta iron (b) Gamma iron
(c) Pearlite (d) Alpha iron

CGPSC AE 15.01.2021

Ans. (d) : In the state at which iron exists below the lower critical temperature, the atoms form a body-centred cube is known as Alpha-iron.

236. Example of metals that crystallize in BCC, FCC & Close-packed Hexagonal, respectively are _____.
(a) Alpha iron, Zinc & delta iron
(b) Alpha iron, delta iron & Zinc
(c) Gamma iron, Nickel & Zinc
(d) Alpha iron, gamma iron & Zinc

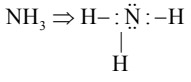
VIZAG Steel MT 24.01.2021

Ans. (d) : The crystallize in BCC, FCC and close-packed Hexagonal, respectively are α-Iron, γ-Iron and Zinc.

237. A nitrogen atom shares the electrons with the 3 hydrogen atoms to form ammonia is an example of _____ bond.
(a) Ionic (b) Vander Waals forces
(c) Covalent (d) Metallic

VIZAG Steel MT 24.01.2021

Ans. (c) : A nitrogen atom shares the electrons with 3 hydrogen atoms to form ammonia is an example of covalent bond.



238. In an eutectic reaction :
(a) A liquid solution solidifies at constant temperature to form one modified liquid phase and one or more solid phases.
(b) A solid solution melts at constant temperature to form one solid phase and two or more liquid phases.
(c) A liquid solution solidifies at constant temperature to form two or more solid phases
(d) A solid solution melts at constant temperature to form two or more liquid phases.

NLC GET 17.11.2020, Shift-II

Ans. (c) : In an eutectic reaction a liquid solution solidifies at constant temperature to form two or more solid phases.
A eutectic reaction is a three phase reaction by which on cooling a liquid transform into two solid phase at same time. It is a phase reaction, but a special one. For example liquid alloy becomes a solid mixture of alpha and Beta at a specific temperature.

239. A family of directions is represented by :
(a) [hkl] (b) (hkl)
(c) [uvw] (d) <uvw>

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Miller Indices–
- <uvw> represents a family of directions
- [hkℓ] represents a direction
- (hkℓ) represent a plane
- {hkℓ} represents a family of planes

240. The phase that does not appear in Fe-C equilibrium phase diagram is _____.
(a) Austenite (b) Cementite
(c) Ferrite (d) Martensite

VIZAG MT, 14.12.2020

Ans. (d) : Martensite does not appears in Fe-C equilibrium diagram.

241. Which of the following properties of a crystalline material depends on crystal defects?
(a) Hardness (b) Elastic modulus
(c) Density (d) Yield strength
(e) Toughness

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) : Whenever, any crystal defects is found in a crystalline material it effects materials properties, such as its yield strength & ultimate strength.

242. For the molybdenum, which has a body centered cubic lattice structure, the number of atoms per unit cell is :
(a) 1 (b) 2
(c) 4 (d) 6

Vadodara Muncipal Corp. DEE, 2018

Ans. (b) For BCC the number of atoms per unit cell

$$= \frac{1}{8} \times 8 + 1$$
$$= 2$$

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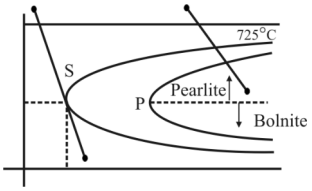
243. Critical cooling rate on Time – Temperature – Transformation (TTT) curve is that slowest rate of cooling at which all austenite is transformed into:

(a) coarse perlite (b) 100% martensite
(c) fine pearlite (d) none of the above

GPSC Lect. (Auto) 16.10.2016
APPSC Poly. Lect. 2013

Ans. (b) :



If the cooling rate is such that it just touches the nose of TTT diagram is called critical cooling rate (CCR). Any cooling rate greater than or equal to CCR will not produce pearlite and is the microstructure Carbon will freeze at its location and forms a super saturated solution of Carbon in α iron with BCT (Body Centered Tetragonal) structure. This phase of iron is called martensite.

244. Hypoeutectoid steel has carbon content :

(a) equal to 0.83%
(b) less than 0.83%
(c) more than 0.83% and up to 2%
(d) more than 2%

GMB AAE 25.06.2017

Ans. (b) : (i) Hypoeutectoid – less than 0.83%
(ii) Eutectoid steel – 0.8%
(iii) Hyper – eutectoid = greater than 0.8%
(iv) Cementite consist of 93.33% iron and 6.67 carbon
(v) Pearlite consists of 87% ferrite and 13% cementite.

245. Pearlite in steels is formed as a result of the following transformation :

(a) peritectic (b) eutectic
(c) eutectoid (d) monotectic

APPSC IOF, 2009

Ans. (c) : Pearlite is the product of the decomposition of austenite by eutectoid reaction and comprise a lamellar arrangement of ferrite and cementite.

246. For cubic crystal structures, interplanar spacing is given by :

(a) $\frac{a}{h^2+k^2+\ell^2}$ (b) $\frac{a}{(h^2+k^2+\ell^2)^2}$

(c) $\frac{a}{\sqrt{(h^2+k^2+\ell^2)}}$ (d) $a\times\sqrt{(h^2+k^2+\ell^2)}$

APPSC IOF, 2009

Ans. (c) : We know,

$$d_{hk\ell}=\frac{a}{\sqrt{h^2+k^2+\ell^2}}$$ for cubic crystal structure

247. Martensitic transformation in steels is :

(a) diffusion less
(b) equilibrium process
(c) diffusion controlled
(d) nucleation and growth process

APPSC IOF, 2009

Ans. (a) : The Martensite transformation is a diffusionless first order phase transformation in the solid state which proceeds by the Nucleation and growth of the new phase.

248. For a simple cubic structure the packing factor is

(a) 60% (b) 52%
(c) 62% (d) 50%

WBPSC AE, 2007

Ans. (b)

Structure	Side & Radius Relation	No. of atom	Atomic packing factor	Coordination No.
Simple cubic	a = 2r	1	52.4%	6
BCC	$a=\frac{4r}{\sqrt{3}}$	2	68%	8
FCC	$a=\frac{4r}{\sqrt{2}}$	4	74%	12
HCP	–	6	74%	12

249. Copper has fcc structure and the atomic radius is 0.1278 nm. The inter planer spacing for the plane (110) is

(a) 0.1205 mm (b) 0.3615 mm
(c) 0.2556 mm (d) 0.2605 mm

WBPSC AE, 2007

Ans. (b)

$$d_{hk\ell}=\frac{a}{\sqrt{h^2+k^2+\ell^2}}$$

$$2\times 0.1278=\frac{a}{\sqrt{1^1+1^2+0^2}}$$

$$\boxed{a=0.3614\ \text{mm}}$$

250. δ-iron has the following structure:

(a) bcc (b) fcc
(c) hcp (d) hcc

WBPSC AE, 2007

Ans. (a) : δ iron has bcc structure, it is formed above 1410°C.

251. The structure of eutectoid steel is

(a) ferrite (b) cementite
(c) pearlite (d) none of the above

WBPSC AE, 2007

Ans. (c) : Structure of eutectoid steel is pearlite.

252. The atomic diameter of a FCC crystal (lattice parameter is a) is :

(a) $\frac{a}{\sqrt{2}}$ (b) $\frac{a}{2}$

(c) 2a (d) $\frac{a}{2\sqrt{2}}$

MPPSC State Forest Service Exam, 2014

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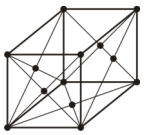
Ans. (a) : For FCC,



$$a = \frac{4r}{\sqrt{2}}$$

$$a = \frac{2.2r}{\sqrt{2}}$$

$$a = \frac{2d}{\sqrt{2}}$$

$$a = \frac{2d}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$a = \frac{2d\sqrt{2}}{2}$$

$$a = d\sqrt{2}$$

$$\boxed{d = \frac{a}{\sqrt{2}}}$$

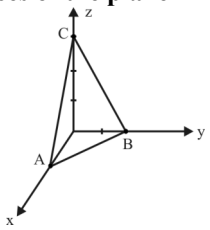
253. After rapid cooling, austenite will transform into

(a) Pearlite (b) Bainite
(c) Martensite (d) Fine pearlite

MPPSC State Forest Service Exam, 2014

Ans. (c) : Accelerated cooling is applied at the temperature of the maximum rate of the fernitic transformation. Accelerated cooling is then applied and the remaining austenite is transformed into martensite.

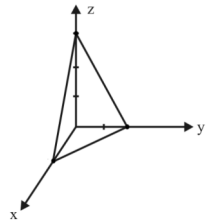
254. Miller indices of the plane ABC will be :



(a) 123 (b) 321
(c) 632 (d) 211

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Intercept cut on x axis = 1
Intercept cut on y axis = 2
Intercept cut on z axis = 3



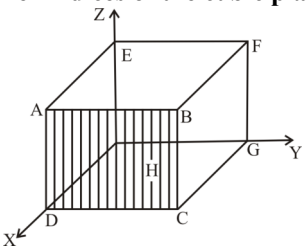
Miller Indices –

$$\left(\frac{1}{1}\frac{1}{2}\frac{1}{3}\right)$$

(6 3 2)

Note- The official answer given by JPSC is (a).

255. The Miller indices of the cubic plane ABCD is



(a) (0 1 0) (b) (0 0 1)
(c) (1 0 1) (d) (1 0 0)

Karnataka PSC AE, 10.09.2017

Ans. (d) : ABCD plane is parallel to YZ plane.
$\therefore$ Miller indices for (X Y Z) = (1 0 0)

256. Consider the following statements

A. Steels which contain 0.8% carbon are known as eutectoid steels.

B. Cementite contains 13% carbon and 87% ferrite.

Which of the above statements is/are correct?

Select the code for the correct answer from the options given below

(a) A only (b) B only
(c) Both A and B (d) Neither A nor B

Karnataka PSC AE, 10.09.2017

Ans. (a) : Steel containing 0.8% carbon are called eutectoid steel. Only option (a) is correct.
• **Note-**The official answer given by commission is (d).

257. Four types of Bravais Lattices is observed in :

(a) Rhombohedral (b) Orthorhombic
(c) Triclinic (d) Monoclinic

KPSC ADF 2015

Ans. (b) : Four types of Bravais Lattices is observed in Orthorhombic crystal system.
- Simple, Base centered, face-centered and body-centered orthorhombic cells.

258. The nearest distance between two neighbour atoms in case of Body Centered Cubic (BCC) structure is :

(a) $\frac{\sqrt{3}a}{2}$ (b) $\frac{2a}{\sqrt{3}}$

(c) $\frac{\sqrt{2}a}{2}$ (d) $\frac{2a}{\sqrt{2}}$

KPSC ADF 2015

Ans. (a) : In the BCC structure atoms has nearest neighbours at a distance

$$d = 2r = \frac{\sqrt{3}a}{2} \qquad \left[\because r = \frac{\sqrt{3}a}{4}\right]$$

259. Select the correct sequence for a pure iron solidifying from the liquid state :

(a) Liquid $\rightarrow \alpha$ ferrite $\rightarrow \delta$ ferrite $\rightarrow \gamma$ ferrite
(b) Liquid $\rightarrow \gamma$ ferrite $\rightarrow \delta$ ferrite $\rightarrow \alpha$ ferrite
(c) Liquid $\rightarrow \delta$ ferrite $\rightarrow \gamma$ ferrite $\rightarrow \alpha$ ferrite
(d) Liquid $\rightarrow \gamma$ ferrite $\rightarrow \alpha$ ferrite $\rightarrow \delta$ ferrite

KPSC ADF 2015

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Ans. (c) : Sequence for pure iron solidifying from the liquid state :

- Liquid $\rightarrow$ δ ferrite $\rightarrow$ γ ferrite $\rightarrow$ α ferrite

260. Choose the correct statements related to Martensite transformation :

A. Martensite transformation takes place due to slow diffusion of carbon atoms

B. FCC austenite transforms to body centered tetragonal crystal structure

C. Transformation is dependent on temperature and is time independent

D. Martensite transformation is unique only to iron-carbon alloy

(a) Statement A is correct and B is incorrect
(b) Statement C is incorrect and D is correct
(c) Statement A and D are correct
(d) Statement B and C are correct

KPSC ADF 2015

Ans. (d) : Martensite transformation–

- FCC austenite transforms to body centered tetragonal (BCT) crystal structure (Martensite)
- Transformation is dependent on temperature and it is independent of time.

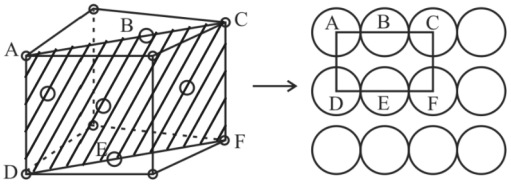
261. Planar density of an FCC unit cell in a (110) plane (assuming atomic radius R) is given by :

(a) $\frac{1}{2\sqrt{2}R^2}$ (b) $\frac{1}{2R^2}$

(c) $\frac{1}{4\sqrt{2}R^2}$ (d) $\frac{1}{4R^2}$

KPSC ADF 2015

Ans. (c) : Planar density (PD) of the (110) plane for the FCC crystal–



Plane area (A_P) = AC × AD

AC = 4R

AD = a = $2\sqrt{2}R$

Area (A_P) = $8\sqrt{2}R^2$

For FCC, Number of atoms (n) along (110) plane = 2

$\therefore$ Planar density (PD) = $\frac{n}{A_p} = \frac{2}{8\sqrt{2}R^2}$

$= \frac{1}{4\sqrt{2}R^2}$

262. The structure which has the highest packing of atoms

(a) Hexagonal close packed lattice
(b) Body centered cubic lattice
(c) Face centered cubic lattice
(d) Both (a) & (c)

ISRO Scientist/Engineer 24.05.2014
MPPSC State Forest Service Exam, 2014

Ans. (d)

Structure	Atomic packing factor
BCC	0.68
HCP	0.74
FCC	0.74
SC	0.52
Diamond Cubic	0.34

263. The surface imperfection which separates crystal of different orientation in a poly crystalline aggregate is called.

(a) Edge dislocation (b) Stacking fault
(c) Grain boundary (d) Screw dislocation

GPSC ARTO Pre 30.12.2018
ESE 2008

Ans. (c) : A grain boundary is the interface between two grains or crystalline, in a polycrystalline material. Grain boundaries are 2D defects in the crystal structure and tend to decrease the electrical and thermal conductivity of material.

264. Out of which cannot be removed from structure of a material?

(a) Point defect (b) Line defect
(c) Grain boundary (d) Dislocation

GPSC ARTO Pre 30.12.2018

Ans. (d) : Dislocation cannot be removed from structure of a material.
Dislocation is a linear crystallographic defect or irregularity within a crystal structure that contains an abrupt change in the arrangement of atoms. The movement of dislocations allow atoms to slide over each other at low stress levels and is known as glide or slip.

265. The following metal does not have HCP crystal structure :

(a) Cobalt (b) Cadmium
(c) Nickel (d) Zinc

Punjab PSC SDE 12.02.2017

Ans. (c) : Hexagonal closed packed (HCP)–
Zn, Mg, Co

- **Face centered cubic (FCC)-** Ni, Pt, Cu, Ag, Au, Al, Pb.
- **Body centered cubic (BCC)-** Li, Na, V, Cr, Mo, W, Fe.
- **Simple cubic (SC)-** Polonium (Po), Mn

266. If one solid phase splits into two solid phases on heating, the reaction is

(a) eutectic (b) peritectic
(c) eutectoid (d) peritectoid

OPSC Civil Services Pre 2006

Ans. (c) Liquid + solid (A) $\rightarrow$ Solid (B) **Peritectic**
Solid (A) + Solid (B) $\rightarrow$ Solid (C) **Peritectoid**
Solid (A) $\rightarrow$ Solid (B) + Solid (C) **Eutectoid**
Liquid $\rightarrow$ Solid (A) + Solid (B) **Eutectic**

267. During solidification of an alloy when a mixture of solid and liquid metals is present, the solid-liquid mixture is referred to as which one of the following :

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(a) eutectic composition. (b) ingot segregation
(c) liquidus (d) mushy zone

HPPSC Asstt. Prof. 20.11.2017

Ans. (d) : During solidification of an alloy when a mixture of solid and liquid metals is present, the solid liquid mixture is referred as mushy zone.

268. α-iron space lattice is

(a) bcc (b) hcp
(c) fcc (d) either bcc or fcc

WBPSC AE 2003

Ans. (a)

(i) α- iron space lattice is bcc (body centered cubic)

BCC → Coordination No. = 8
No of atoms per unit cell = 2
total No. of atoms in BCC (N) = 9
APF = 0.68

(ii) for simple cubic →

n = 1, coordination No (CN) = 6
(N) = 8
APF = 0.52

(iii) for FCC → n = 4, N = 14, CN = 12
APF = 0.74

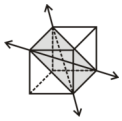
(iv) for diamond cubic → n = 8, CN = 4, APF = 0.37

269. For BCC structure the representation of slip plane is given by:

(a) (111) (b) (110)
(c) (011) (d) (101)

HPPSC AE (PWD) 10.09.2016

Ans. (b) :

Crystal structure	Slip plane	Unit of geometry
BCC	(110)	

270. Which one of the following metal possesses Hexagonal-close packed structure?

(a) Zinc (b) Copper
(c) Tungsten (d) Iron

UPSC JWM Advt. No.-52/2010

Ans. (a) : Explanation- Zinc, has the Hexagonal-closed packed structure.
Zinc – HCP
Copper – FCC
Tungsten – BCC
Iron – FCC or BCC (Depending upon existing temperature).

271. The eutectic reaction in Iron-Carbon diagram refers to, when

(a) liquid metal changes to cementite and austenite
(b) ferrite changes to cementite and austenite
(c) austenite changes to ferrite and cementite
(d) cementite and ferrite combine to form austenite

TSGENCO AE 14.11.2015

Ans. (a) : Eutectic reaction
At 4.30%C and 1147°C

$L(4.30\%C) \leftrightarrow \gamma(2.14\%C) + Fe_3C$

In eutectic reaction, the liquid solidifies as a phase mixture of austenite (containing 2.14%C) and cementite. This phase mixture is known as ledeburite.
Where, γ = Austenite
Fe_3C = Cementite

272. Read the following statements and select the correct answer

(a) Ferrite is α iron (BCC) which does not have more than 0.025% of carbon in solid form
(b) Cementite is iron carbide (Fe_3C), which has 6.67% of carbon
(c) Tempering is used for relieving the internal stresses

(a) Statement a and b are true and c is false
(b) Statement a and c are true and b is false
(c) Statements b and c are true and a is false
(d) Statement a, b and c are true

TSGENCO AE 14.11.2015

Ans. (a)

- Solubility of carbon in δ-ferrite (BCC) is 0.025%.
- Fe_3C (cementite) – It is a compound of iron and carbon having a maximum carbon solubility of 6.67% of carbon.

273. The maximum number of phases that may be present for a ternary system assuming that pressure is held constant:

(a) 4 (b) 3
(c) 2 (d) 0

HPPSC Asstt. Prof. 2014

Ans. (a) : For a ternary system C = 3
At constant pressure N = 1
From Gibb's phase rule,

P + F = C + N
P + F = 3 + 1

When F = 0, P will have its maximum value of 4, which means that the maximum number of phases present for this situation is 4.

274. In metals subjected to cold working, strain hardening effect is due to

(a) Slip mechanism
(b) Twining mechanism
(c) Dislocation mechanism
(d) Fracture mechanism
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
ESE 1997

Ans. (c) : In metals subjected to cold working, strain hardening effect is due to dislocation mechanism. The movement of dislocations allow atoms to slide and each other at low stress levels and is known as slide or slip.

275. For Rhombohedral space lattice, which one of the following is correct?

(a) $\alpha = \beta = \gamma = 90$ (b) $\alpha = \beta = \gamma \neq 90$
(c) $\alpha = \gamma = 90 \neq \beta$ (d) $\alpha \neq \beta \neq \gamma \neq 90$
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (b) : For Rhombohedral space lattice,

$\alpha = \beta = \gamma \neq 90$

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276. If the Number of components in a system are known to be 3, then according to Gibb's phase rule, the number of phases will be :

(a) ≤ 2 (b) ≤ 3

(c) ≤ 4 (d) ≤ 5

UPRVUNL AE 2014

Ans : (d) According to Gibb's phase rule

$F = C - P + 2$ Where P = Phase

$0 = 3 - p + 2$ C = No. of component

$P \leq 5$ F = 0

277. If atoms are assumed to a hard spheres, then the value of highest APF (Atomic packing factor) in metals will be:

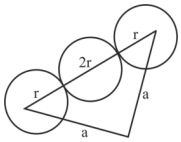
(a) 0.65 (b) 0.84

(c) 0.74 (d) 0.98

AAI Jr. Executive 29.11.2018

Ans. (c) : Atomic packing factor

$$= \frac{\text{number of atom per unit cell} \times \text{Volume of each atom}}{\text{Volume of unit}}$$



$$APF = \frac{N_{avg} \times \frac{4}{3}\pi r^3}{a^3}$$

Number of atom for FCC

$$n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$$

$(r + r + 2r)^2 = a^2 + a^2$

$(4r)^2 = 2a^2$

$4r = a\sqrt{2}$

$$r = \frac{a\sqrt{2}}{4}$$

$$APF = \frac{4 \times \frac{4}{3}\pi\left(\frac{a\sqrt{2}}{4}\right)^3}{a^3} = 0.74$$

278. The crystal structure of diamond is less close-packed than that of Cu, because the C-C bonds are

(a) Shorter than the Cu-Cu bonds

(b) Directional in nature

(c) Non-directional in nature

(d) Only partially covalent

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : The crystal structure of diamond is less close-packed than of copper because C–C bonds are shorter bonds than Cu-Cu bonds.

279. The stress required to cause slip in a metal single crystal depends upon

(a) Crystal Structure

(b) Temperature

(c) Orientation of the crystal with respect to the applied stress

(d) Cross-sectional area

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : The stress required to cause slip in a metal single crystal depends upon the orientation of the crystal with respect to the applied stress. Such as normal stress, shear stress etc.

280. Closed packed hexagonal space lattice is found in

(a) Alpha-iron, tungsten, chromium and molybdenum

(b) Gamma-iron, aluminium, copper, lead, silver and nickel

(c) Zinc, magnesium, cobalt, cadmium, antimony and bismuth

(d) None of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Zinc, magnesium, cobalt, cadmium, antimony and bismuth.

281. Region of disorder created by movement of dislocations in super lattice is called:

(a) Twin (b) Stacking fault

(c) Anti-phase boundary (d) Orowon loop

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Anti-phase boundary- An antiphase boundary (APB) separates two domains of the same ordered phase. It results from symmetry breaking that occurs during ordering processes, which can start at different locations in a disordered lattice.

Region of disorder created by movement of dislocations in super lattice is called anti-phase boundary.

282. Which of the following is classified as a secondary bond in materials?

(a) Ionic bonding (b) Covalent bonding

(c) Metallic bonding (d) Hydrogen bonding

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : Hydrogen bonding is the stronger form of secondary bonding and is formed from polar molecular bonding.

Hydrogen bonding is classified as two types-

(a) The intramolecular hydrogen bonding

(b) The intermolecular hydrogen bonding

283. In a two component system, if the non-compositional variable is only temperature, the number of degrees of freedom in the case of a single phase field as per Gibbs Phase Rule is?

(a) 0 (b) 1

(c) 2 (d) 3

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : From Gibb's phase rule with only temperature as variable,

$P + F = C + 1$

$1 + F = 2 + 1$

$F = 2$

284. In an iron-carbon diagram, the percentage by weight of carbon at eutectoid composition is

(a) 1.12 (b) 0.76

(c) 0.24 (d) 0.03

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : Eutectoid Reaction-

$$\gamma \underset{0-76\%C}{\overset{727^\circ C}{\rightleftharpoons}} \alpha + Fe_3C$$

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285. Of the various microstructures that may be produced for a given steel alloy, which among the following is the hardest form

(a) Martensite (b) Pearlite
(c) Bainite (d) Spheroidite

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : ● Martensite microstructure is the hardest form among various microstructure that produced for given steel alloy.

Reason- The strength and hardness is due to elastic strain within the martensite, which is a result of too many carbon atoms being in the spaces between the iron atoms in the martensite.

286. Which of the following order of crystal structure will match with metals Iron - Nickel - Titanium in that order at room temperature?

(a) BCC - HCP - FCC (b) FCC - BCC - BCP
(c) HCP - FCC- BCC (d) BCC -FCC - HCP

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : Iron → BCC
Nickel → FCC
Titanium → HCP

287. The reaction in which a liquid phase transform into two different solid phases is called

(a) Eutectoid reaction
(b) Peritectic reaction
(c) Eutectic reaction
(d) Peritectoid reaction

ISRO Scientist/Engineer 17.12.2017
TSPSC AEE 2015, TNPSC AE 2019

Ans. (c) : (1) Eutectoid Reaction–
Solid 1 → Solid 2 + Solid 3
(2) Peritectic Reaction–
Liquid + Solid 1 → Solid 2
(3) Eutectic Reaction–
Liquid → Solid 1 + Solid 2
(4) Peritectoid Reaction–
Solid 1 + Solid 2 → Solid 3

288. The crystal structure of cementite is

(a) FCC (b) BCC
(c) Tetragonal (d) Orthorhombic

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : Cementite is a very hard and brittle inter-metallic compound and has a significant influence on the properties of steels. The crystal structure of cementite is orthorhombic.
Cementite Fe_3C
FCC → Austenite
BCC → Ferrite

289. Which of the following order of crystal structure will match with metals Iron – Copper – Zinc in that order?

(a) BCC – HCP – FCC (b) FCC – BCC – HCP
(c) HCP – FCC – BCC (d) BCC – FCC – HCP

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : BCC (Body Centered Cubic)– BCC elements are generally hard and brittle BCC elements are tungsten (W), Chromium (Cr), Vanadium (V), Molybdenum (Mo), Iron [Fe except in (910 – 1400°C)], Titanium (Ti)

FCC(Face Centered Cubic)– "FCC element are generally strong and ductile."
FCC elements are Iron Fe [temperature rang of 910 – 1400°C], Aluminium (Al), Copper, Nickel, Gold, Silver, Platinum etc.
HCP (Hexagonal Closed Packed Structure)– HCP elements are generally less ductile.
HCP elements are titanium, zinc, zirconium, Mg, Cobalt etc.

290. Usually Materials with the following crystal structure fail in ductile mode

(a) FCC (b) BCC
(c) HCP (d) None of these

ISRO Scientist/Engineer 12.05.2013

Ans. (a) : **FCC** - (Face Centered Cubic)
FCC is strong in Brittle materials.
BCC - (Body Centered Cubic)
BCC is strong in ductile materials
HCP - (Hexagonal Cubic Packing)
HCP is weak in ductile materials.

291. Which of the following regions of the electromagnetic spectrum would be used to determine the structure of crystalline solids?

(a) Microwave (b) Infrared
(c) X-ray (d) Visible

ISRO Scientist/Engineer 2011

Ans. (c) : Crystalline structure is determined by x-ray diffraction method.

292. A plane intersects the co-ordinate axes at x = 2/3, y = 1/3, z = 1/2. What is the Miller index of this plane?

(a) (9 3 2) (b) (4 3 2)
(c) (3 6 4) (d) (4 2 3)

ISRO Scientist/Engineer 2009
ESE 2015

Ans. (c) : Plane Intersect the co-ordinates axes

$$x = \frac{2}{3}, y = \frac{1}{3}, z = \frac{1}{2}$$

Miller indices are inverse of the intersection of plane and co-ordinate axis, therefore

$$\frac{3}{2} : \frac{3}{1} : \frac{2}{1} \quad ...(1)$$

Multiply eq. (1) by 2
3 : 6 : 4
So, the Miller index is (3 6 4).

293. The defect responsible for the phenomenon of slip, by which most metals deform plastically, is known as

(a) Fracture (b) Twinning
(c) Dislocation (d) Strain hardening

ISRO Scientist/Engineer 2008
HPPSC Poly. Lect. 05.07.2021

Ans. (c) : The defect responsible for the phenomenon of slip, by which most metals deform plastically, is known as dislocation.

294. When FCC iron and BCC iron coexist in equilibrium the degrees of freedom are :

(a) 1 (b) 2
(c) 3 (d) 0

Mizoram PSC (Insp. of Factories) 2019 Paper-III

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Ans. (a) $P + F = C + 2$

$F = C - P + 2$

$= 1 - 2 + 2$, $\boxed{F = 1}$

295. Sharing of electrons between neighbouring atoms results in :

(a) metallic bond (b) ionic bond

(c) covalent bond (d) none of these

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : Ionic Bond– Develop between ion of opposite type and these bond are quite stable and strong.

e.g. → NaCl, CaO, MgO etc.

Covalent Bond– Sharing of electrons between neighbour atoms results in covalent bond.

Covalent bonds develop between two atoms of different or similar type when they share electrons in a localized manner.

e.g. → CH_4, CO_2 etc.

296. The atomic number of an atom is equal to :

(a) atomic weight (b) atomic mass

(c) number of protons (d) mass number

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : The atomic number of an atom is equal to the number of protons.

The atomic mass of an atom is equal to summation of mass of protons and mass of nuetrons.

297. The material property which depends only on the basic crystal structure is :

(a) Fatigue strength (b) Work hardening

(c) Fracture strength (d) Elastic constant

Mizoram PSC (Insp. of Factories) 2019 Paper-III
VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2010

Ans. (c) : Fracture strength is a material property which depends on the basis crystal structure. Fracture strength depends on the strength of the material.

Elastic constant depends not only on material parameters but also on the experimental geometry.

298. Match List-I (Crystal Structure) with List-II (Example) and select the correct answer using the codes given below the lists :

List-I (Crystal Structure)	List-II (Example)
A. Simple Cubic	1. Zinc
B. Body-centered Cubic	2. Copper
C. Face-centered Cubic	3. Alpha iron at room temperature
D. Hexagonal Close Packed	4. Manganese

Code :

	A	B	C	D
(a)	4	3	1	2
(b)	4	3	2	1
(c)	3	4	2	1
(d)	3	4	1	2

Mizoram PSC (Insp. of Factories) 2019 Paper-III
ESE 2003

Ans. (b)

Simple cubic (SC)	→	Po, Mn
Body-centered cubic (BCC)	⇒	Fe (except in 910^0 – 1400^0C), W, Cr, V, Mo etc.
Face centered cubic (FCC)	⇒	Fe (in 910–1400^0C), Cu, Ni, Al, Au, Ag, Pt. etc.
Hexagonal close packed (HCP)	⇒	Ti, Zn, Zr, Mg, Be, Cd, Co

299. Which one of the following is the correct ascending order of packing density for the given crystal structures of metals?

(a) Simple cubic – Face centred cubic – Body centred cubic

(b) Body centred cubic – Simple cubic – Face centred cubic

(c) Simple cubic – Body centred cubic – Face centred cubic

(d) Body centred cubic – Face centred cubic – Simple cubic

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c)

Crystal Structure	SC	BCC	FCC	HCP
Atomic packing factor	0.52	0.68	0.74	0.74

300. A display of a 3-dimensional model that shows all of its edges is known as

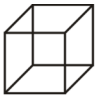
(a) 2D model (b) 3-D model

(c) Wire frame (d) Solid model

APPSC AEE Screening Test 2016

Ans. (c) : A display of a 3-dimensional model that shows all of its edges is known as wire frame.

e.g.–



3-dimensional wire frame model of cube

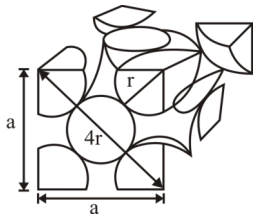
301. Volume of an FCC unit cell in terms of the atomic radius R is:

(a) $8R\sqrt{2}$ (b) $16R^3\sqrt{2}$

(c) $16R\sqrt{2}$ (d) $8R^3\sqrt{2}$

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
ESE 18.07.2021

Ans. (b) :



Volume of FCC unit cell in terms of the atomic radius R.

The atoms touch one another across the face diagonal of the length which is 4R

Since the unit cell is cube its volume is a^3 where 'a' cell edge length

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From the right triangle on the face.

$$a^2 + a^2 = (4R)^2 \Rightarrow a = 2R\sqrt{2}$$

$\therefore$ The FCC unit cell volume may be derived from.

$$V_c = a^3 = \left(2R\sqrt{2}\right)^3 = 16R^3\sqrt{2}$$

302. Choose the correct order of the microstructures of steel in ascending values of their hardness.

(a) Martensite, pearlite, bainite, sorbite
(b) Bainite, martensite, sorbite, pearlite
(c) Pearlite, sorbite, martensite, bainite
(d) Pearlite, sorbite, bainite, martensite

Assam PSC AE (PHED), 18.10.2020

Ans. (d) : Correct order of microstructure of steel in ascending values of their hardness is pearlite, sorbite, bainite and martensite.

303. Which of the following statements is true?

(a) Below 910°C, pure iron exists in FCC crystal structure.
(b) Between 1537°C and 1400°C, iron exists as pearlite
(c) In the iron-carbon equilibrium diagram the eutectic point is at 6.67% of C.
(d) None of these

Karnataka PSC AE, 10.09.2017

Ans. (d) :
- Above 910°C pure iron exist in FCC crystal structure.
- Below 725°C iron exist as pearlite ($\alpha + Fe_3C$)
- In Iron-carbon Equilibrium diagram Eutectic point is at 4.3% of carbon.

304. The Curie point for most ferrous magnetic materials is about

(a) 390 °C (b) 540 °C
(c) 760 °C (d) 880 °C

ESE 2019

Ans. (c) : If a ferromagnetic substance is heated and its temperature exceeds a specified temperature T_C, this substance becomes paramagnetic. During cooling, the substance becomes ferromagnetic again when reaching this temperature. The temperature T_C is called the curie temperature and is different for each ferromagnetic substance. Most of ferromagnetic substance have a relatively high curie temperature for nickel curie temperature is about 360°C and iron 770°C.

305. Relative amounts of phases in a region can be deduced using:

(a) Phase rule (b) Lever rule
(c) Either (A) or (B) (d) None of these

OPSC AEE 2019 Paper-I

Ans : (b) : Relative amounts of phase in a region can be deduced using Lever rule.

306. The [110] direction in a cubic unit cell is parallel to the following :

(a) Face diagonal of unit cell
(b) Edge of the cube
(c) Body diagonal of the cube
(d) None of the above

UKPSC AE 2012, 2007 Paper-I

Ans. (a) : Face diagonal of unit cell

307. Vectorial sum of the Burgers vector of dislocations meeting at a nodal point is

(a) zero (b) 1
(c) –1 (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (a) : The vectorial sum of Berger's vectors of dislocations meeting at a nodal point is must be zero. A dislocation line can not end abruptly within the crystal, it can close on it self as a loop or ends either at a node or at the surface.

308. Zn and Mg have the following crystal structure:

(a) B.C.C. (b) F.C.C.
(c) diamond cubic (d) H.C.P.

UKPSC AE 2007 Paper -I

Ans. (d) : Zn and Mg, H.C.P. crystal structure.

309. Closed packed planes are formed in

(a) simple cubic crystals
(b) body centered cubic crystals
(c) diamond cubic crystals
(d) face centered cubic crystals

UKPSC AE 2007 Paper -I

Ans. (c) : Closed packed planes are formed in diamond cubic crystals.

310. A given steel test specimen is studied under metallurgical microscope (magnification used is 100X). In that different phases are observed one of them is Fe_3C. The observed phase Fe_3C is also known as:

(a) Ferrite (b) Austenite
(c) Cementite (d) Martensite

UJVNL AE 2016, ESE 1997

Ans : (c) phase Fe_3C is known as cementite.
(i) It is also called iron carbide
(ii) It contains 6.67% carbon by weight
(iii) Strength 35 MPa
(iv) High compressive strength

311. If steel is cooled in still air, the structure obtained is

(a) pearlite (b) sorbite
(c) troosite (d) acicular

BPSC Poly. Lect. 2016

Ans : (b) If steel is cooled in still air, the structure obtained is sorbite.

Sorbite:-

It is a microstructure consisting of ferrite and finally decide cementite produced on tempering martensite above 450 ^{0}C . The constituents also known as sorbite, pearlite is produced by the decomposition of outside when cooled at a rate faster than that which will produce sorbite and slower than that which will produce troostite.

312. The weld bead between a heavy steel section and a thin section occurs mainly due to the formation of:

(a) bainite (b) spheroidite
(c) martensite (d) none of these

(HPPSC LECT. 2016)

Ans : (a) The weld bead between a heavy steel section and a thin section occurs mainly due to the formation of bainite.

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Bainite :- It is formed by isothermal decomposition of austenite in alloy steels. when the decomposition takes place at a temperature above that at which martensite is formed the resulting bainite is known as upper bainite and when it occurs below that temperature pearlite is formed the bainite formed is called lower bainite. It has a needle like structure which is less marked at high formation temperature.

313. Which stainless steel is used in cryogenic vessels?
(a) Ferritic (b) Austenitic
(c) Martensitic (d) None of these

Gujarat PSC AE 2019

Ans : (b) : Austenitic stainless steel is used in cryogenic vessels.

314. Pure graphite contains
(a) 0% of carbon (b) 2% of carbon
(c) 6.67% of carbon (d) 100% of carbon

BPSC AE 2012 Paper - VI

Ans : (d) : Pure graphite contains 100% of carbon

315. Which of the following does not aid in graphitization of cementite?
(a) Presence of Si (b) Presence of Al
(c) Presence of Ni (d) Low temperature

BPSC AE Mains 2017 Paper - VI

Ans : (d) : Low temperature does not aid in graphitization of cementite.

316. Which of the following phases exhibits highest solid solubility of carbon?
(a) Delta iron
(b) Gamma iron (austenite)
(c) Alpha iron
(d) Ferrite

BPSC AE Mains 2017 Paper - VI

Ans : (b) : Gamma iron (austenite) phases exhibits highest solid solubility of carbon.

317. The number of phases present in equilibrium at eutectic point
(a) 0 (b) 1
(c) 2 (d) 3

TSPSC AEE 2015

Ans. (d) : The number of phases present in equilibrium at eutectic point is 3.

318. When a low carbon steel is heated up to upper critical temperature :
(a) There is no change in grain size
(b) The average grain size is a minimum
(c) The grain size increases very rapidly
(d) The grain size first increases and then decreases very rapidly

OPSC Civil Services Pre. 2011

Ans. (b) : When a low carbon steel is heated upto upper critical temperature the average grain size is a minimum.

319. Which one of the following sets of constituents is expected in equilibrium cooling of hyper eutectoid steel from austenitic state ?
(a) Ferrite and Pearlite
(b) Cementite and Pearlite
(c) Ferrite and Bainite
(d) Cementite and Martensite

OPSC Civil Services Pre. 2011, ESE 1995

Ans. (b) : In equilibrium cooling of hypereutectoid steel from austenitic state is Cementite and Pearlite.

320. The hardness of steel increases if it contains
(a) pearlite (b) ferrite
(c) cementite (d) martensite

JPSC AE 2013, Paper-V

Ans : (c) : Cementite or iron carbide, is very hard, brittle intermettalic compound of iron and carbon, as Fe_3C, contains 6.67% carbon.
- It is the hardest structure.
- Its crystal structure is orthorhombic.
- It has low tensile strength.
- It has high compressive strength.

321. In if the steel at room temperature is magnetic, the presence of which constituent can be ruled out
(a) Austenite (b) Ferrite
(c) Cementite (d) Pearlite

APPSC AEE Mains 2016, Paper-III

Ans. (a) : Austenite

322. If plain carbon steel in the austenitic condition is quenched in water to room temperature, its structure changes from 'austenite' to
(a) Pearlite (b) Martensite
(c) Cementite (d) Ferrite

JWM 2017

Ans. (b) : When plain carbon steel in the austenitic condition is quenched in water to room temperature, its structure change to Martensite.

323. Consider the following statements:
1. Ferrite is the softest constituent's phase of steel
2. Pearlite is the mixture of Bainite and Ledeburite

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

JWM 2017

Ans. (a) : Pearlite is a mixture of ferrite and cementite forming distinct layers or bands in slowly cooled carbon steels.

324. Which is the isothermal reversible reaction in which a solid phase is converted into two or more intimately mixed solids on cooling?
(a) Peritectoid (b) Peritectic
(c) Eutectic (d) Eutectoid

CIL MT 26.03.2017

Ans. (d)
Peritectic reaction : Liquid + solid 1 $\rightarrow$ solid 2
Peritectoid reaction : solid 1 + solid 2 $\rightarrow$ solid 3
Eutectic reaction : Liquid 1 $\rightarrow$ solid 2 + solid 3
Eutectoid reaction : solid 1 $\rightarrow$ solid 2 + solid 3
Monotectic reaction : Liquid 1 $\rightarrow$ liquid 2 + solid

325. Which of the following is/are the characteristic features of eutectic alloys?
1. The possess the lowest melting temperature
2. They undergo phase transformation at the same temperature while heating as well as cooling

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

JWM 2017

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Ans. (c) : The characteristics features of eutectic alloys-
(a) They undergo phase transformation at the same temperature while heating as well as cooling.
(b) They possess the lowest melting temperature.

326. Formation of Ferrite and Cementite from Austenite takes place at:
(a) Temperature = 723ºC, carbon composition = 2%
(b) Temperature = 1130ºC, carbon composition = 4.3%
(c) Temperature = 723ºC, carbon composition = 0.8%
(d) Temperature = 1130ºC, carbon composition = 0.8%

UPRVUNL AE 07.10.2016

Ans. (c) : Formation of Ferrite and Cementite from Austenite takes place at temperature = 723ºC, carbon composition = 0.8%.

327. Consider the following statements about Pearlite
1. It is a Eutectoid alloy of ferrite and cementite
2. It is a Eutectoid alloy of ferrite and carbide
3. The ratio of ferrite to cementite is 87:13
4. The ratio of ferrite to carbide is 86:14
5. The temperature at which the alloy is formed is 726ºC
Out of these statements which are true
(a) 1, 3 and 5 (b) 2, 4 and 5
(c) 1 and 5 (d) 1 and 3
(e) 2 and 4

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Pearlite is a two phased, layered structure composed of alternating layers of ferrite (87.5 wt%) and cementite (12.5 wt%) that occurs in some steels and cast iron. During slow cooling of an iron-carbon alloy, pearlite forms by a eutectoid reaction as austenite cools below 723ºC.

328. _____ is NOT a type of malleable cast iron.
(a) Ferritic malleable cast iron
(b) Pearlitic malleable cast iron
(c) Pearlitic-ferritic malleable cast iron
(d) Black heat malleable cast iron
(e) Grey malleable cast iron

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Grey malleable cast iron is not a type of malleable cast iron.

329. In the iron-iron carbide equilibrium diagram, which of the following phases is NOT present?
(a) Ferrite (b) Austenite
(c) δ-ferrite (d) Cementite
(e) Eutectite

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Eutectite phase is not present in the iron-iron carbide equilibrium diagram.

330. Which of the following is NOT a transformation product of Austenite?
(a) Pearlite (b) Bainite
(c) Lath martensite (d) Plate martensite
(e) Graphite

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Graphite is not a transformation product of austenite.

331. The percentage of pearlite in a slowly cooled melt of 0.5% of carbon steel is
(a) 48.5% (b) 52.5%
(c) 58.5% (d) 62.5%

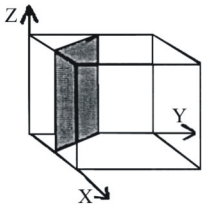
ESE 2020

Ans. (d) : Lever Rule (at Room temperature)

$$m_{pearlite} = \frac{C_o - C_\alpha}{C_p - C_\alpha} = \frac{0.5 - 0.008}{0.8 - 0.008} = 0.6212$$

Percentage of pearlite at 0.5% C is 62.12%

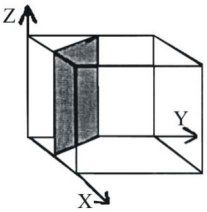
332. The Miller indices for the plane in the figure below are :



(a) [120] (b) [210]
(c) [220] (d) [002]

BHEL ET 2019

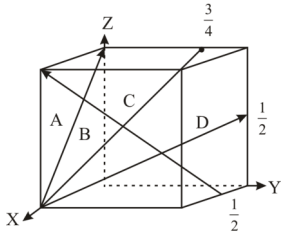
Ans. (c) :



Miller Indices –

Axis -	x	y	z
Intercepts	$\frac{1}{2}$	$\frac{1}{2}$	∞
Miller Index/Reciprocal	2	2	0

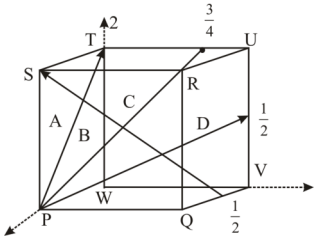
333. In the figure shown below, Miller indices [021] have the direction of :



(a) B (b) A
(c) D (d) C

BHEL ET 2019

Ans. (c) : Miller indices
[021]



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for planet 'TUVW'

Axis	X	Y	Z
intercepts	∞	$\frac{1}{2}$	1
Reciprocals	$\frac{1}{\infty}$	$\frac{1}{\frac{1}{2}}$	$\frac{1}{1}$
Rationalization	0	2	1
Miller Indices	0	2	1

334. The average numbers of atoms per unit cell in simple cubic structures of solid are _____.
(a) 1 (b) 2
(c) 4 (d) 6
(e) 8

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) :

	BCC	FCC	HCP	SC
Average No. of Atoms (N_{avg})	2	4	6	1

335. Which of the following is not a type of crystal imperfections?
(a) Lattice vibrations (b) Point defects
(c) Line defects (d) Planar defects
(e) Slip

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Slip is not a type of crystal imperfections.

336. A process in which the atoms in a part of the crystal subjected to stress rearrange themselves, such that the orientation of the part changes in such a way that the distorted part becomes a mirror image of the other part, is called:
(a) Slip
(b) Twinning
(c) Geometrical softening
(d) Geometrical hardening
(e) Realignment

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : A process in which the atoms in a part of a crystal subjected to stress rearrange themselves, such that the orientation of the part changes in such a way that the distorted part becomes a mirror image of the other part, is called twinning.

337. Crystal structure of Copper is
(a) Face Centred Cubic Structure
(b) Body Centred Cubic Structure
(c) Hexagonal Closed Packed Structure
(d) Simple Cubic Crystal Structure

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : FCC elements generally strong and ductile in nature.
Example- Al, Cu, Ni, Au, Ag, Pt.

338. Binary phase diagrams are maps that represent the relationships between _____of phases at equilibrium
(a) Temperature and the compositions
(b) Compositions and quantities
(c) Temperature and quantities
(d) Temperature and the compositions and quantities

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) Binary phase diagrams are maps that represent the relationship between temperature and the compositions and quantities of phases at equilibrium, which influence the microstructure of an alloy, this may involve the transition from one phase to another or appearance or disappearance of a phase.

339. In metallurgical and materials engineering system, the Gibb's Phase Rule is given by : [pressure is maintained at one atmosphere] [where F = Number of degrees of freedom, c = Number of components, P = Number of phase which can coexist at equilibrium]
(a) F = C + 1 – P (b) F = C + 1 + P
(c) F = C – 1 + P (d) F = C – 1 – P

SJVN ET 2019

Ans. (a) : Gibb's Phase rule describes the possible number of degree of freedom in a closed system at equilibrium in term of number of separate phases and number of chemical constituents in the system.
F = Number of degree of freedom
C= Number of chemical component
P = Number of Phase
F = C – P + 1

340. The Crystal structure of Chromium (Cr) at room temperature is
(a) Body-Centered Cubic (BCC)
(b) Face-Centered Cubic (FCC)
(c) Hexagonal Close-Packed (HCP)
(d) Octagonal Close-Packed (OCG)
(e) Simple Cubic (SC)

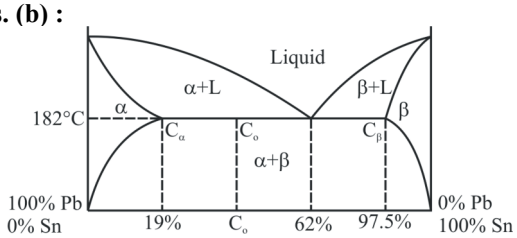
CGPSC AE 26.04.2015 Shift-I

Ans. (a) : At room temperature chromium has a BCC crystal structure with a basis of one Cr atom.

341. In the Pb-Sn system, the fraction of total α phase is 3 times the fraction of β phase at eutectic temperature of 182°C, Pb with 19% Sn dissolved in it, Sn with 2.5% Pb dissolved in it, and liquid is in equilibrium. The alloy composition of tin (Sn) and lead (Pb) are nearly
(a) 28.6% and 71.4% (b) 38.6% and 61.4%
(c) 48.6% and 51.4% (d) 58.6% and 41.4%

ESE 2019

Ans. (b) :



Pb – Sn System–

$C_\alpha = 19\%$ Sn

$C_\beta = 2.5\%$ Pb $= 97.5\%$ Sn

$m_\alpha = 3\, m_\beta$

$$\left[\frac{C_\beta - C_o}{C_\beta - C_\alpha}\right] = 3\left[\frac{C_o - C_\alpha}{C_\beta - C_\alpha}\right]$$

$$\left[\frac{97.5 - C_o}{97.5 - 19}\right] = 3\left[\frac{C_o - 19}{97.5 - 19}\right]$$

$97.5 - C_o = 3C_o - 57$

$C_o = 38.6\%$, Pb = 100 – 38.6 = 61.4%

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342. The structure of sodium chloride is considered as
(a) A body-centered crystal
(b) A simple cubic crystal
(c) Two interpenetrating FCC sub lattices of Cl^- ions and Na^+ ions
(d) A cubic crystal with Na^+ and Cl^- alternatively at the cubic corners

ESE 2019

Ans. (c) : The structure of sodium chloride is face centered cubic (FCC) lattice.

343. A metal has lattice parameter of 2.9 Å, density of of 7.87 g/cc, atomic weight of 55.85, and Avogadro's number is 6.0238×10^{23}. The number of atoms per unit cell will be nearly.
(a) 1 (b) 2
(c) 8 (d) 16

ESE 2019

Ans. (b) : Given,

$a = 2.9\ Å = 2.9 \times 10^{-10}\ m = 2.9 \times 10^{-8}\ cm$

$\rho = 7.87\ gm/cc$

$A = 55.85$

$N_A = 6.023 \times 10^{23}$

$$\text{Density } \rho = \frac{nA}{N_A \times (\text{Volume of unit cell})}$$

$$\rho = \frac{n \times A}{N_A \times a^3}$$

$$7.87 = \frac{n \times 55.85}{6.023 \times 10^{23} \times (2.9 \times 10^{-8})^3}$$

$n = 2.069$

344. Statement I: The phase of a substance is characterized by its distinct molecular arrangement which is homogeneous throughout and is separated from the others by easily identifiable boundary surfaces.
Statement II: Phase change is not characterized on molecular structure and/or behavior of the different phases.

ESE 2018

Ans. (b)
- A Phase is identified as having a distinct molecular arrangement that is homogeneous throughout and separated from the other by easily identifiable boundary surface.
- Phase change is not concerned with the molecular structure and behaviour of different materials.

345. Which of the following statements are correct?
1. Steel and cast iron are multi phase alloys.
2. Ferrite is a single phase interstitial solid solution of carbon in iron.
3. Wrought iron is a highly refined iron with a small amount of slag which gives resistance to progressive corrosion
4. Stellite contains large amounts of metals like cobalt and tungsten resulting in high hardness.

Select the correct answer using the code given below.
(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only
(c) 1, 3 and 4 only (d) 2 and 4 only

ESE 2018

Ans. (a)
(i) Steel and cast iron are multiphase alloys of α-Ferrite and cementite.
(ii) Ferrite is a single phase interstitial solid solution of carbon in iron.
(iii) Wrought iron is a pure form of Fe with small amount of slag ($CaSiO_3$) that gives resistance to progressive corrosion.
(iv) Stellite is a good cutting tool material that have large quantity of cobalt and tungsten.

346. Atoms are arranged in face centred cubic (FCC) structure as
(a) All eight corners of the cube
(b) All eight corners of the cube and one of the body center
(c) All eight corners of the cube and one at the center of each face
(d) All eight corners of the cube and one at the body center with one at the center of each face

HPPSC AE 2018

Ans. (c) : Atoms are arrange in face centred cubic structure as all eight corners of the cube and one at the centre of each face.

347. The edge length of an FCC unit cell of atomic radius R is given by:
(a) $2\sqrt{2}R$ (b) 2R
(c) $\sqrt{2}R$ (d) 3R
(e) $3\sqrt{2}R$

CGPSC AE 25.02.2018

Ans. (a) : Edge length of various crystal structure-

BCC	FCC	HCP	SC
$a = \frac{4R}{\sqrt{3}}$	$\frac{4R}{\sqrt{2}} = 2\sqrt{2}R$	2R	2R

348. What is the coordination number for simple cubic structure?
(a) 4 (b) 8
(c) 12 (d) 6

CIL MT 26.03.2017

Ans. (d) :

Crystal structure	Coordination No.
Simple cubic	6
BCC	8
FCC, HCP	12

349. Which of the following are zero dimensional defects?
1. Vacancy
2. Interstitial defect
3. Substitutional defect

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

JWM 2017

Ans. (d) : Defects in metal-
(1) Point defect or zero dimensional defect
(a) Vacancy defect
(b) Interstitial defect
(c) Substitutional defect
(d) Frenkle defect
(e) Schottky defect
(2) Surface defect
(a) Grain boundary defect

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(b) Twin boundary defect
(c) Tilt boundary defect
(d) Staking faults

(3) Line defect
(a) Edge dislocation defect
(b) Screw dislocation defect

350. Plastic deformation by twinning mechanism occurs in metals due to the movement of atoms where:
1. Atoms move through distances proportional to their (original) distance from the twinning plane
2. Atoms move in integral steps of full atomic distances from their adjacent planes
3. Atoms jumps from one plane to another

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

JWM 2017

Ans. (b) : At the point, where atoms move in integral steps of full atomic distance from their adjacent planes.

351. $\frac{\alpha\sqrt{3}}{4}$ is the atomic radius of
(a) BCC lattice (b) FCC lattice
(c) HCP lattice (d) Simple cube

TNPSC AE 2018

Ans. (a)

Crystal structure	α to r relation
SC	$r = \frac{\alpha}{2}$
BCC	$r = \frac{\alpha\sqrt{3}}{4}$
FCC	$r = \frac{\alpha\sqrt{2}}{4}$
HCP	$r = \alpha/2$

where, α = lattic parameters
r = atomic radius

352. In which of the process line defects were not formed
(a) Solidification of metals
(b) Recrystalisation of metals
(c) Deformation of metals
(d) Melting of metals

TNPSC AE 2017

Ans. (d) : In melting of metals process line defects were not formed.

353. Which type of materials are used as bridges between human tissues and metals?
(a) Polymeric biomaterials
(b) Ceramic biomaterials
(c) Metallic biomaterials
(d) Polymarised plastic materials

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Metallic biomaterials are used as bridges between human tissues and metals.

354. The number of electrons found in the outer most orbit of semi-conductor materials is,
(a) 2 (b) 4
(c) 6 (d) 8

TNPSC AE 2019

Ans. (b) : The number of electrons found in the outer most orbit of semi-conductor materials is 4.

355. Crystal structure of metals is studied by
(a) metallographic techniques
(b) X-Ray techniques
(c) ultrasound method
(d) electron microscopy

TNPSC AE 2019

Ans. (b) : Crystal structure of metals is studied by X-ray techniques.

356. A solid + a liquid result in a liquid upon heating during _____ reaction.
(a) Eutectic (b) Peritectic
(c) Monotectic (d) Syntectic

OPSC AEE 2019 Paper-I

Ans : (c) : Invariant reactions-
- Monotectic reaction : $Liquid_1 \rightarrow Liquid_2 + Solid$
- Eutectic reaction : $Liquid \rightarrow Solid_1 + Solid_2$
- Eutectoid reaction : $Solid_1 \rightarrow Solid_2 + Solid_3$
- Peritectic reaction : $Liquid + Solid_1 \rightarrow Solid_2$
- Peritectoid reaction : $Solid_1 + Solid_2 \rightarrow Solid_3$

357. Eutectoid product in Fe-C system is called :
(a) Pearlite (b) Bainite
(c) Ledeburite (d) Spheroidite

OPSC AEE 2019 Paper-I

Ans. (a) : 1. Eutectic Reaction- It takes place at 1147°C, 4.35%C.

$$L \rightleftharpoons (\gamma + Fe_3C) \rightarrow \text{Ledeburite}$$

$$L \rightleftharpoons S_2 + S_3$$

2.Peritectic Reaction- It takes place at 1492°C, 0.18% C

$$\delta + L \rightleftharpoons \gamma$$

3.Eutectoid Reaction- It takes place at 727°C, 0.8% C

$$\gamma \rightleftharpoons (\alpha + Fe_3C) \rightarrow \text{Pearlite}$$

$$S_2 \rightleftharpoons S_4 + S_3$$

358. The Burger's vector lies parallel to the dislocation line along the axis of a line of atoms in the same plane in :
(a) Screw dislocation (b) Edge dislocation
(c) Cracks (d) Vacancies

HPPSC W.S. Poly. 2016

Ans : (a) Screw dislocation:- When a part of Crystal displaced angularly over the remaining about some edge, the dislocation is called screw dislocation.
(i) Burger's vector is parallel to the screw dislocation line, whose direction is perpendicular.
(ii) Screw dislocation creates shear stress and strain in the crystal.

359. The locations of atoms and their particular arrangement in a given crystal are described by means of :
(a) Potential energy (b) Space lattice
(c) Intermolecular bond (d) Diffusion

HPPSC W.S. Poly. 2016

Ans : (b) The locations of atoms and their particular arrangement in a given crystal are described by means of Space lattice.
Space lattice:- A space lattice is defined as an infinite array of points in three dimensional space in which each points is identically spaced and located with respect to the other.

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360. Which one of the following parameter is significant to ascertain chemical equilibrium of a system ?
(a) Clapeyron of equation
(b) Maxwell relation
(c) Gibb's function
(d) Helmholtz function

(KPSC AE. 2015)

Ans : (c) Gibb's function is signification to ascertain Chemical equilibrium of a system

Gibb's Phase rule : -

F= C - P + 2

F = Degree of freedom of the system

C = No of Components of the system

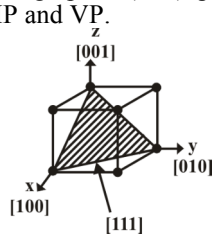
P = No of phase in equilibrium in the system.

- The degree of freedom decreases as the no of phases increases
- A two Component System Can't have more than 4 phase in equilibrium.

361. (111) plane of a cube is:
(a) Vertical
(b) Horizontal
(c) Inclined to HP
(d) Inclined to both HP and VP

OPSC AEE 2015 Paper-I

Ans : (d) Crystallographic (111) plane of a cube is inclined to both HP and VP.



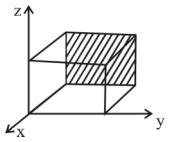
362. The atomic bond found in diamond is:
(a) Ionic (b) Covalent
(c) Metallic bond (d) None of these

OPSC AEE 2015 Paper-I

Ans : (b) The atomic bond found in diamond is covalent bond.

Covalent Bond : It is also called homopolar bond. This bond result by mutual sharing of valence electrons of identical atoms. The electronic structure of an atom is relatively stable. It has eight electrons in its outer valence shell, which sometimes an atom acquires by shearing electrons with an adjacent similar atom. Diamond, methane chlorine are typical examples.

363. The set of miller indices of the plane shown in the given figure is:



(a) $(\bar{1}00)$ (b) (100)
(c) (101) (d) (110)

UJVNL AE 2016, ESE 1999

Ans : (a) Intercept on Y and Z axis is ∞. So taking reciprocal, it will be (0,0) on both z and y -axis . Intercept on $\bar{x}$ axis is on the negative side of X-axis $(\bar{1}00)$

364. Schottky imperfection is a
(a) Point imperfection
(b) Line imperfection
(c) Surface imperfection
(d) Volume imperfection

UPPSC AE 12.04.2016 Paper-I

Ans. (a) : Point defect or zero dimensional defect.
(a) Vacancy defect
(b) Interstitial defect
(c) Substitutional defect
(d) Frenkel defect
(e) Schottky defect

365. Match the List- I with the List- II and select the correct answer using the codes given below:

List–I (Crystal Structure)	List–II (Packing Efficiency)
A. Simple cubic	1. 0.34
B. Diamond cubic	2. 0.74
C. Body centred cubic	3. 0.52
D. Face centred cubic	4. 0.68

	A	B	C	D
(a)	4	3	1	2
(b)	3	1	4	2
(c)	1	2	4	3
(d)	3	2	1	4

UPPSC AE 12.04.2016 Paper-I

Ans : (b)

Crystal Structure	Packing Efficiency
Simple cubic	0.52
Body centred cubic	0.68
Face centred cubic	0.74
Diamond cubic	0.34

366. Which of the following statements is not true for austenitic stainless steels?
(a) They are hardened and strengthened by cold working
(b) They are most corrosion resistant amongst stainless steels
(c) Austenitic phase is extended to room temperature
(d) They are magnetic in nature

UKPSC AE-2013, Paper-I

Ans. (d) : For austenitic stainless steels –

- They are not magnetic in nature
- They are hardened and strengthened by cold working
- They are most corrosion resistant amongst stainless steels
- Austenitic phase is extended to room temperature

367. Austenite decomposes into ferrite and cementite at a temperature of:-
(a) 727ºC (b) 1148ºC
(c) 1495ºC (d) 1539ºC

UKPSC AE-2013, Paper-I

Ans. (a) : Austenite decomposes into ferrite and cementite at a temperature of 727ºC.

368. The eutectic percentage of carbon in iron is
(a) 0.025% (b) 0.15%
(c) 2.03% (d) 4.33%

BPSC AE Mains 2017 Paper - VI

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Ans. (d) : Eutectic reaction
At 4.30%C and 1147°C

$$L(4.30\%C) \leftrightarrow \gamma(2.14\%C) + Fe_3C$$

In eutectic reaction, the liquid solidifies as a phase mixture of austenite (containing 2.14%C) and cementite. This phase mixture is known as ledeburite.

Where, γ = Austenite
Fe_3C = Cementite

369. Relationship between atomic radius 'R' and unit cell length 'a' for BCC crystal structure is:-

(a) $a = \frac{4R}{\sqrt{3}}$ (b) $a = 2R\sqrt{2}$

(c) $a = \frac{2R}{\sqrt{3}}$ (d) $a = 3R\sqrt{2}$

UKPSC AE-2013, Paper-I

Ans. (a) : Relationship between atomic radius 'R' and unit cell length 'a' for BCC crystal structure is $a = \frac{4R}{\sqrt{3}}$.

370. Which statement is not true in case of martensite?

(a) Crystal structure is BCC
(b) Transformation does not involve diffusion
(c) Grains are plate like or needle like in appearance
(d) It is a non-equilibrium phase

UKPSC AE-2013, Paper-I

Ans. (a) : Crystal structure is BCC is not true in case of martensite. It has body centered tetragonal structure.

371. Iron is 'Face Centered Cubic (FCC) at which one of the following temperatures ?

(a) Room temperature (b) 1400 °C
(c) 910 °C (d) None of the above

UKPSC AE 2012 Paper-I

Ans. (c) : Face centered cubic temperature at 910°C.

372. Gibbs phase rule for binary phase diagram is

(a) F + P = C + 2 (b) F + P = C + 3
(c) F + P = C + 1 (d) F + P = C

TSPSC Managers, 2015

Ans. (c) : In Binary phase diagram (2 component systems) pressure is constant and therefore Gibbs phase rule becomes.

$$P + F = C + 1$$

Where, P = Phase
F = No. of degree of freedom (DOF)
C = No. of component

373. The predominant phase in the iron-carbon alloy system for a composition with 99% Fe at room temperature is which of the following?

(a) austenite (b) cementite
(c) delta (d) ferrite

GPSC ARTO Pre 30.12.2018

Ans. (d) : The predominant phase in the iron –carbon alloy system for a composition with 99% Fe at room temperature is ferrite.

374. The softest crystalline form in ferrous metal is:

(a) Austenite (b) Pearlite
(c) Bainite (d) Martensite

Punjab PSC SDE 12.02.2017

Ans. (b) : Pearlite- Pearlite consists of alternate layers of ferrite and cementite and it is layered in structure. When thin and closely packed layer, it is fine pearlite and when thick and widely spread layer, it is coarse pearlite is eutectoid mixture at 723°C containing 0.8% C (Fe-C phase diagram).

375. Which of the following metals can be easily drawn into wire?

(a) tin (b) copper
(c) lead (d) zinc

TNPSC AE 2018

Ans. (b) : Copper can be easily drawn into wire.

376. Annealing twins are typically found in metals that have the _____ crystal structure.

(a) BCC (b) HCP
(c) FCC (d) Unpredictable

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Annealing twins, most prevalent in FCC metals. e.g. aluminium, copper, brass & iron etc.
Annealing twins are formed as a consequence of growth accidents during the recrystallisation of deformed cubic close packed metals.

377. Strengthening of single phase metals can be achieved by :

(a) Grain size reduction
(b) Solid solution alloying
(c) Strain hardening
(d) All of these

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Different ways to hinder dislocation motion /strengthening mechanism.

(1) In single phase material.
- Grain size reduction
- Solid solution strengthening
- Strain hardening.

(2) In multiphase material.
- Precipitation strengthening
- Dispersion strengthening
- Martensite strengthening.

378. The softest and hardest phases in plain carbon steel are

(a) ferrite and cementite respectively
(b) austenite and martensite respectively
(c) pearlite and cementite respectively
(d) ferrite and bainite respectively

RPSC IOF, 2020

Ans. (a) : The layers of cementite in the pearlite microstructure are transformed into spheroid resulting in the softest and most ductile form of steel.
Ferrite is the softest phase and cementite is the hardest phase in plain carbon steel.

379. The boundary between (alpha) and (alpha + beta) regions is called

(a) liquidus (b) solidus
(c) solvus (d) None of the above

Assam PSC AE (IWT) 14.03.2021

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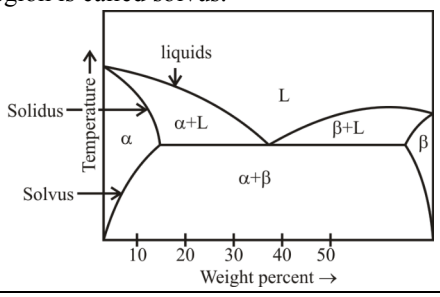
Ans. (c) : The boundary between (alpha) and (alpha + Beta) region is called solvus.



380. The hardness of martensite in a steel is a function of

(a) cooling rate (b) nickel content
(c) carbon content (d) nose location

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : The carbon content defines the maximum quenched hardness of the stainless steel. Heating the steel to high temperatures which are then followed by a rapid cooling process results in hardening. Hardness of a steel at 99.9% martensite as a function of carbon content.

381. For full annealing of hypereutectoid steel requires temperature 30 °C to 50 °C:

(a) above the lower critical temperature
(b) below the lower critical temperature
(c) below the upper critical temperature
(d) above the upper critical temperature

CIL MT 27.02.2020

Ans. (a) : The object of full annealing is to soften the metal, refine the grain structure.
This process consist of heating the steel 30°C to 50°C above the lower critical temperature for hypereutectoid.

382. The spherodising process is carried out at:

(a) 550 degree C (b) 680 degree C
(c) 980 degree C (d) 250 degree C

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Spherodising process is carried out at 680°C (1200°F).
It is conducted at temperatures that are slightly below the eutectoid temperature followed by the slow cooling process.

383. Heating the medium carbon steel to above recrystallization temperature and rapidly cooling by quenching causes.

(a) hardening due to formation of troosite
(b) annealing of steel and relieving of stresses
(c) normalization of steel and forming of fine grain structure
(d) hardening due to formation of martensite

TSGENCO AE 14.11.2015

Ans. (d) : Heating the medium carbon steel to above recrystallization temperature and rapidly cooling by quenching causes hardening due to formation of martensite.

Austenite (f) $\xrightarrow{\text{Transforms}}$ Martensite $\rightarrow$ (RC 65) (under dynamic fast cooling).

384. During the analysis of microstructure of pearlite, it is found that it is composed of A% ferrite and B% cementite. The value of A and B are _____ respectively.

(a) 88 and 12 (b) 50 and 50
(c) 60 and 40 (d) 12 and 88

AAI Jr. Executive 29.11.2018

Ans. (a) : Pearlite structure, it is composed of 88% ferrite and 12% cementite.

385. A mixture of chemical compounds or elements that have a single chemical composition and solidifies at the lowest temperature than any other composition made up of the same ingredients.

(a) Phase balanced system
(b) Eutectic system
(c) Binary thermally balanced system
(d) intrinsic system

ISRO Scientist/Engineer 24.05.2014

Ans. (b) : A eutectic system is a mixture of chemical compounds or elements that have a single chemical composition that solidifies at lower temperature than any other composition made up of the same ingredients. This composition is known as the eutectic composition and the temperature which it solidifies is known as the eutectic temperature.

386. The essential ingredient of any hardened steel is

(a) Austenite (b) Pearlite
(c) Martensite (d) Cementite

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : The essential ingredient of any hardened steel is martensite.

387. Fe_3C is known as .....

(a) Cementite (b) Ferrite
(c) Austenite (d) None

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : Cementite (Fe_3C) – It has orthorhombic crystal structure, having 6.67% C by weight. It is extremely hard and brittle in nature.

388. In which of the following phases of steel cementite is in particle form?

(a) Martensite (b) Ferrite
(c) Pearlite (d) Bainite

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) : In Bainite phases of steel cementite is in particle form.
Bainite – It is the product of isothermal decomposition of austenite and it cannot be produced by continuous cooling Bainite is aggregate of ferrite and carbide. Also it is tougher

389. On heating, if one solid phase splits into two solid phase, the reaction is :

(a) eutectoid (b) peritectoid
(c) peritectic (d) eutectic

Mizoram PSC (Insp. of Factories) 2019 Paper-III

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Ans. (a)

- **Eutectic reaction**

$$\text{Liquid} \underset{\text{Heating}}{\overset{\text{Cooling}}{\rightleftharpoons}} \text{Solid}_1 + \text{Solid}_2$$

- **Peritectic reaction**

$$\text{Liquid} + \text{Solid} \underset{\text{Heating}}{\overset{\text{Cooling}}{\rightleftharpoons}} \text{New Solid}$$

- **Eutectoid reaction**

$$\text{Solid} \underset{\text{Heating}}{\overset{\text{Cooling}}{\rightleftharpoons}} \text{Solid}_1 + \text{Solid}_2$$

390. The microstructure composition of pearlite for a Fe_3C diagram consists of :

(a) Carbon dissolved in alpha iron having a body cantered cubic structure
(b) Carbon dissolved in gama iron having a face cantered cubic structure
(c) A mixture of body-cantered alpha iron and face-entered gamma iron
(d) Carbon dissolved in body-centered alpha iron and an Fe, Fe_3C

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) : The microstructure composition of pearlite for a Fe_3C diagram consists of carbon dissolved in body centered alpha iron and an Fe, Fe_3C.

391. Which one of the following materials, deformation of crystals was not by twinning?

(a) Zinc (b) Tin
(c) Iron (d) Aluminium

TNPSC AE 2017

Ans. (d) : Aluminium

392. The temperature at which component losses or gains magnetic properties is

(a) Curie temperature (b) Upper critical point
(c) Lower critical point (d) Eutectic temperature

TSPSC AEE 2015

Ans : (a) The temperature at which component losses or gains magnetic properties is curie temperature.

Material	Curie temperature
Cobalt	1121°C
Iron	770°C
Nickel	358°C

393. The hardness of various structures in decreasing order during heat treatment of steel is:

(a) fine pearlite, coarse pearlite, martensite, spherodite
(b) martensite, fine pearlite, coarse pearlite, spherodite
(c) fine pearlite, martensite, spherodite, coarse pearlite
(d) coarse pearlite, fine pearlite, martensite, spherodite

HPPSC LECT, 2016

Ans : (b)
Martensite>fine pearlite>Coarse pearlite>spherodite.
Martensite : - It is a metastable phase of steel formed by transformation of austenite below a temperature 240°C (MS). It is an interstitial supersaturated solid state of carbon in α iron which is super saturated with carbon. It forms as a result of shear type transformation with virtually no diffusion. Martensite normally is product of quenching. It posses an accicular or needle like structure.

Martensite

394. The resultant microstructure after normalizing should be

(a) martensitic (b) pearlitic
(c) bainitic (d) ferritic

BPSC AE Mains 2017 Paper - VI

Ans : (b) : The resultant microstructure after normalizing should be pearlitic.

395. A steel containing 0.85% carbon is known as

(a) eutectoid steel
(b) hypo-eutectoid steel
(c) hyper-eutectoid steel
(d) none of these

TNPSC AE, 2008

Ans. (c) : (0.008% – 0.83%) → Hypo-eutectoid
(0.83% – 2.1%) → Hyper-eutectoid

396. The structure of hypereutectoid steels comprise

(a) Phases of ferrite and cementite
(b) Cementite only
(c) Phases of cementite and pearlite
(d) Pearlite only

TSPSC AEE 2017

Ans. (c) : hypereutectoid steels contains 0.8%-2.0% wt% carbon below 1340ºF, 727ºC the remaining austenite transforms to pearlite. Hence at room temperature the microstructure consists of proeutectoid cementite and pearlite.

397. Which of the following display properties similar to that of steel?

1. Grey cast iron
2. Black-hearth cast iron
3. Pig iron
4. White-hearth cast iron

(a) 1 and 3 only (b) 3 and 4 only
(c) 1 and 2 only (d) 2 and 4 only

UPRVUNL AE 21.08.2016

Ans. : (d) Black-hearth cast iron and White-hearth cast iron display properties similar to that of steel.

398. Mechanical deformation by twinning occurs in metals that have :

(a) bcc structure at low temperature and at high rate of loading
(b) bcc structure at low temperature and at low rate of loading
(c) hcp structure at high temperature and at low rate of loading
(d) hcp structure at high temperature and at high rate of loading

DRDO Scientists 2009

Ans. (a) : bcc structure at low temperature and at high rate of loading

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399. Cold rolled carbon sheet steel contains maximum carbon of :

(a) 0.25% (b) 0.5%
(c) 0.6% (d) 0.4%

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : Cold rolled carbon sheet steel contains maximum carbon of 0.25%

400. Miller indices denotes :

(a) plane (b) direction
(c) number of atoms (d) family of direction

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : Miller indices area a symbolic vector representation for the orientation of an atomic plane in a crystal lattice and are defined as the reciprocals of the fractional intercepts which the plane makes with the crystallographic axes.

401. The specific surface area of solids is :

(a) surface area per unit volume
(b) surface area per unit weight
(c) surface weight per unit area
(d) surface area per unit mass

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Specific surface area is a property of solid defined as the total surface area of a material per unit mass. It is a physical valve that can be used to determine the type and properties of a material.

402. What is the Frenkel defect in the lattice crystal?

(a) Defect in which positive and negative ions are missing
(b) Atoms missing in the given lattice
(c) Defect in which interstitial position is occupied by extra atom in the crystal without disorganising the parent atom
(d) Defect in which interstitial position is occupied by missing atoms

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : The Frenkel defect (also known as the Frenkel pair/disorder) is a defect in the lattice crystal where an atom or ion occupies a normally vacant site other than its own lattice site vacant. Frenkel defect are mostly exhibited in ionic solids where the smaller ion is dislocated.

Example– AgBr, ZnS AgCl, AgI the defect occurs mainly due to the smaller size Zn^{2+} and Ag^{+} ions.

403. The hardness of steel depends upon amount of

(a) pearline (b) ferrite
(c) cementite (d) martensite

GPSC ARTO 01.05.2016

Ans. (c) : The maximum attainable hardness of steel solely depends on the amount of carbon it contains. It is also affected by the distribution of carbon present in the iron. As various shapes of carbon have different hardness properties.

Cementite is a form of carbon.

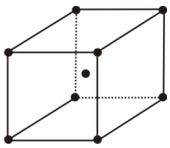
If carbon was not in the option, then cementite will be the answer but out of cementite and carbon, the carbon is the main reason for the hardness of steel.

404. In a unit-cell of a body centred cubic space lattice, the total number of atoms is

(a) 6 (b) 9
(c) 14 (d) 17

GPSC ARTO 01.05.2016
WBPSC AE 2003
UKPSC AE 2007 Paper -I
APPSC Poly. Lect. Automobile Engg. 2020
GPSC ARTO 01.05.2016

Ans. (b)



The total numbers of atom as unit cell in a Body centred cubic space lattice s equal to T.N.A = 8 Corner atoms + 1 body centred atom.

T.N.A. = 9

405. Wrought iron contains carbon up to

(a) 1.5% (b) 0.15 %
(c) 1.0 % (d) 2.0 %

GPSC Lect. (Poly.) 07.12.2014

Ans. (b) : Wrought Iron contains
Carbon up to – 0.15 % C
Steel contains – upto 2%C
Cast Iron contains 2–6.62 %C

406. In crystal structure of metal, the imperfection is called.

(a) Slip (b) Impurity
(c) Dislocation (d) Air bubble

GPSC Lect. (Poly.) 07.12.2014
TSPSC AEE 2017
GPSC Lect. (Poly.) 17.12.2014
UKPSC AE 2007 Paper -I

Ans. (c) : Crystal defect, imperfection in the regular geometrical arrangement of the atoms in a crystalline solid. These imperfection results from deformation of solid, rapid cooling from high temperature (or) high energy radiation striking the solid imperfection may be–

- Point imperfection
- Line imperfection
- Surface imperfection

407. In iron carbide diagram, pearlite is

(a) Eutectoid mixture of austenite and ferrite
(b) Eutectoid mixture of cementite and ferrite
(c) Eutectoid mixture of austenite and cementite
(d) Eutectoid mixture of austenite and ferrite

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : In iron carbide diagram, pearlite is eutectoid mixture of cementite and ferrite.

408. Gamma iron exists at temperatures in the range

(a) Room temperature and lower critical temperature
(b) Between 500°C and 850°C
(c) Between 900°C and 1400°C
(d) Between 1400°C and 1600°C

Vizag Steel MT (Re-Exam) 24.11.2013

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Ans. (c) : Temperature range from 1400°C to 910°C it contains FCC structure which is also known as γ-Austenite.

409. Microstructure of annealed hypereutectoid plain carbon steel consists of

(a) Pearlite with grain boundary cementite
(b) Full pearlite
(c) Ferrite and pearlite
(d) Tempered martensite

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Microstructure of annealed hypereutectoid plain carbon steel consists of pearlite with grain boundary cementite.

410. Stacking fault is a :

(a) Point defect
(b) Line defect
(c) Surface and grain boundary defect
(d) None of these

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : A stacking fault is a planar defect or surface defect that can occur in crystalline material. Crystalline materials from repeating patterns of layers of atoms. Errors can occur in the sequence at these layers and are known as stacking faults.

411. When 1.0% plain carbon steel is slowly cooled from the molten state to 740ºC, the resulting structure will contain

(a) Austenite and ferrite
(b) Austenite and cementite
(c) Ferrite and cementite
(d) Pearlite and cementite

GPSC Executive Engineer 23.12.2018

Ans. (b) : From iron carbon diagram we can observe that the temperature 740ºC and 1% carbon lies in hypoeutectic region which contains mixture of Austenite and cementite.

412. Increase in carbon content in plain carbon steel raises its

(a) Tensile strength and malleability
(b) Tensile strength and hardness
(c) Ductility and ultimate tensile strength
(d) Ductility and melting temperature

GPSC Executive Engineer 23.12.2018

Ans. (b) : Increase in carbon content in plain carbon steel raises its tensile strength and hardness.

413. Which is the magnetic allotrope of iron?

(a) α iron (b) β iron
(c) γ iron (d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) : Magnetic allotrope of iron α iron.

414. What structure is Quartz an example of ?

(a) Chain structure
(b) Double tetrahedral structure
(c) Sheet structure
(d) Framework structure

VIZAG STEEL MT 2011

Ans. (d) : Quartz an example of framework structure.

415. In each type of crystal structure a certain fundamental grouping of atoms is repeated indefinitely in all three dimensions. What is this fundamental grouping called as?

(a) A unit molecule (b) A bond
(c) A unit cell (d) An atomic space

RPSC AE 2018
VIZAG STEEL MT 2011

Ans. (c) : crystal structure a certain fundamental grouping of atoms is repeated indefinitely in all three dimensions. What is this fundamental grouping called as a unit cell.

416. A space lattice found in α - iron is called :

(a) body centered cubic space lattice
(b) face centered cubic space lattice.
(c) close packed hexagonal space lattice.
(d) none of these

Kerala PSC IOF 19.04.2016

Ans. (a) A space lattice found in α - iron is called body centered cubit space lattice.

417. Percentage reduction in the area of cast iron specimen during tensile test would be of the order of

(a) negligible (b) more than 50%
(c) 10 to 25% (d) 25 to 50%

GPSC AE 26.07.2015

Ans. (a) : negligible

418. Chromium as an alloying element in alloy steel is used principally to

(a) Improve harden ability
(b) Improve mechanical properties at low temperature
(c) Improve mechanical properties at elevated temperature
(d) Improve the corrosion and oxidation resistance

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Chromium as an alloying element in alloy steel is used principally to Improve the corrosion and oxidation resistance.

419. The compositions of some of the alloy steels are as under

1. 18W4Cr1V **2. 12$M_0$1W4Cr1V**
3. 5$M_0$6W4Cr2V **4. 18W8Cr1V**

The composition of commonly used high speed steels would include

(a) 1 and 2 (b) 2 and 3
(c) 1 and 4 (d) 1 and 3

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Composition of alloy steels

W-Series	M_0 Series
18 -W	5 - M_0
4 - Cr	6 - W
1 - V	4 - Cr
	2 - V

420. Atomic radius of Face Centred Cubic (FCC) crystal is
a = lattice parameter

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(a) $\frac{a\sqrt{2}}{4}$ (b) $\frac{a\sqrt{3}}{2}$

(c) $\frac{a\sqrt{3}}{4}$ (d) $\frac{a\sqrt{2}}{3}$

UPPSC AE 13.12.2020, Paper-I

Ans. (a) :

Structure	Atomic radius
BCC	$\frac{\sqrt{3}a}{4} = r$
FCC	$\frac{\sqrt{2}a}{4} = r$

421. Which of the following phase of steel is NOT present in iron-carbon phase diagram?

(a) Ferrite (b) Cementite

(c) Austenite (d) Martensite

UPPSC AE 13.12.2020, Paper-I, ESE 2013

Ans. (d) : Martensite phase of steel is not present in iron carbon phase diagram.

3. Mechanical Testing

422. The "Jominy test" is used to find

(a) Young's modulus

(b) hardenability

(c) yield strength

(d) thermal conductivity

GPSC Asst. director of transport 05.03.2017
GPSC ARTO Pre 30.12.2018
GPSC EE Pre, 28.01.2017
Oil India Senior Officer 23.12.2020
APPSC IOF 2009
ESE 2019

Ans. (b) : The "Jominy test" is used to find hardenability.

423. Compressive test performed on cast iron will have fracture occurring:-

(a) Along an oblique plane

(b) Along the axis of load

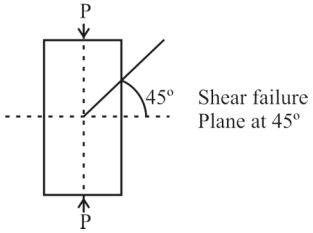
(c) Perpendicular to the axis of load

(d) None of the above

HPSC Civil Services Pre 2014
Nagaland CTSE 2016, 2017 Ist Paper
UPPSC AE 12.04.2016 Paper- I
UKPSC AE-2013, Paper-I
HPPSC Poly. Lect. 05.07.2021

Ans. (a) : In Compression test the fracture in cast iron Specimen would occur along the oblique plane.

Failure in Compression for Brittle material



424. In tensile test of mild steel, necking will start:-

(a) At lower yield stress

(b) At upper yield stress

(c) At ultimate tensile stress

(d) Just before fracture

Assam PSC AE (IWT) 14.03.2021
UKPSC AE-2013, Paper-I

Ans. (c) : In tensile test of mild steel, necking will start at the ultimate tensile stress.

425. When mild steel is subjected to tensile loading, the fracture will be of

(a) Star Shape

(b) Flat type

(c) Cup-and cone shape

(d) Granular shape

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
HPPSC Poly. Lect. 05.07.2021
J&K PSC Civil Services Pre, 2010

Ans. (c) : For ductile – Cup and cone shape
For brittle – granular shape

426. During tensile-testing of a specimen using a Universal Testing machine, the parameter actually measured include

(a) True stress and true strain

(b) poisson's ratio and Young's modulus

(c) engineering stress and engineering strain

(d) load and elongation

Kerala PSC Poly Lect. 2017
TNPSC AE, 2008
APPSC AE 04.12.2012
APPSC AEE 2012
UPPSC AE 13.12.2020, Paper-I
ESE 1996

Ans. (d) : It is a load controlled machine and works on the principle of elongation/deformation of material on the application of load. Universal Testing machine, the parameter actually measured load and elongation in tensile test.

427. Brinell tester use hardness steel ball of size.

(a) 1 mm (b) 5 mm

(c) 10 mm (d) 15 mm

RPSC IOF 2020
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Brinell tester use a hardness steel ball of size 10 mm.

428. Indenter used in Brinell Hardness test is

(a) Square base pyramid diamond

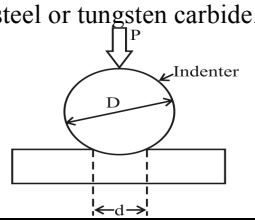
(b) Rectangular based pyramid diamond

(c) Sphere made of steel or tungsten carbide

(d) Spero conical diamond

GPSC Asst. Prof. Prod 21.08.2016
HPPSC AE 2018

Ans. (c) : Indenter used in Brinell hardness test is sphere made of steel or tungsten carbide.



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D – Ball diameter
d – Impression diameter
P – Load
HB – Brinell result

$$HB = \frac{2P}{\pi D\left[D - \sqrt{\left(D^2 - d^2\right)}\right]}$$

429. The percentage reduction in area of a cast iron specimen during tensile test would be of the order of

(a) more than 70% (b) 30 - 50%
(c) 10 - 25% (d) negligible

Rajasthan AE (Nagar Nigam) 2016 Shift-III
Gujarat PSC AE 2019

Ans : (d) : The percentage reduction in area of a cast iron specimen during test would be of the order of negligible because cast iron is a brittle materials.

430. For applied load P kg, diameter of ball D mm and diameter of indentation d mm, the Brinell hardness number is given by

(a) $B_{HN} = \dfrac{P}{\frac{\pi}{2}(D - \sqrt{D^2 - d^2})}$

(b) $B_{HN} = \dfrac{D}{\frac{\pi}{2}(D - \sqrt{D^2 - d^2})}$

(c) $B_{HN} = \dfrac{P.D}{\frac{1}{2}(D - \sqrt{D^2 - d^2})}$

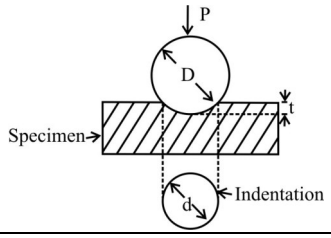
(d) $B_{HN} = \dfrac{P}{\left\{\frac{\pi D}{2}\right\}\left[D - \sqrt{D^2 - d^2}\right]}$

BPSC Poly. Lect. 2016
TNPSC AE 2018
ISRO Scientist/Engineer 2009

Ans : (d)

Brinell hardness Number = $\dfrac{P}{\left\{\frac{\pi D}{2}\right\}\left[D - \sqrt{D^2 - d^2}\right]}$

In Brinell hardness test hardened steel ball is used as indenter.

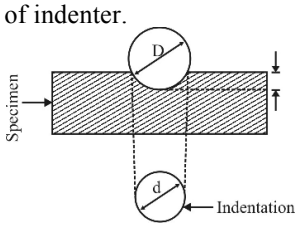


431. A measure of Rockwell hardness is the _____

(a) Depth of penetration of indenter
(b) Surface area of indentation
(c) Height of rebound
(d) Projected area of indentation

Mizoram PSC (Insp. of factories) 2019 paper III
UPRVUNL AE 21.08.2016

Ans. : (a) A measure of Rockwell hardness is the depth of penetration of indenter.



Brinell hardness number = $\dfrac{P}{\frac{\pi D}{2}[D - \sqrt{D^2 - d^2}]}$

432. Cup and cone fracture is observed in the fracture of

(a) brittle material
(b) ductile material
(c) advanced ceramic material
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : A cup and cone fracture is a type of failure observed in ductile materials and plastics that are subjected to a uniaxial force. It is essentially the separation of a body into two separate pieces due to the application of excessive tensile stress.

433. The Rockwell test for hardness employs __ as an indenter.

(a) steel pyramid (b) diamond cone
(c) diamond pyramid (d) steel ball

UPRVUNL AE 05.07.2021

Ans. (b) : The Rockwell test for hardness employs diamond cone as an indenter.

434. The empty portion of cupola above the preheating zone is called as _____.

(a) Tuyere (b) Well
(c) Fettling zone (d) stack

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : The empty portion of cupola above the preheating zone is called as stack.

435. The charpy test is conducted to measure

(a) creep strength
(b) toughness
(c) fatigue strength
(d) elastic strength of a material

HPPSC Poly. Lect. 05.07.2021

Ans. (b) : Charpy test is conducted to measure toughness.

436. Brinell number of a metal or an alloy is a measure of its

(a) hardness (b) tensile strength
(c) malleability (d) toughness

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : Brinell number of a metal or an alloy is a measure of its hardness.

437. In Brinell hardness test, a _____ ball is used for a standard time usually for_____

(a) Steel, 10 seconds
(b) Tungsten Carbide, 10 seconds
(c) Steel ball, 30 seconds
(d) Tungsten Carbide, 30 seconds

PPSC Asstt. Municipal Engineer 15.06.2021

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Ans. (d) : In Brinell hardness test, a tungsten carbide ball is used for a standard time usually for 30 seconds.

438. Which one of the following mechanical properties is provided by the Notched bar test?
(a) impact strength (b) hardness
(c) resilience (d) endurance limit

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : Notched bar test–impact strength.

439. Consider the following statements for ductile fracture:
1. The material undergoes substantial plastic deformation with high energy absorption before fracture.
2. Presence of cracks on the surface of material initiates this type of failure.
3. Fracture occurs due to necking. Which of the above statements is/are correct?
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

ESE 18.07.2021

Ans. (d) : Ductile fracture- In ductile fracture the material undergoes substantial plastic deformation with high energy absorption before fracture and so the toughness value of ductile materials is high. Fracture occurs in these materials due to necking.

440. The ultimate tensile strength of low Carbon Steel by working at high strain rate will
(a) increase
(b) decrease
(c) remain constant
(d) first increase, then decrease

UKPSC AE 2012 Paper-I

Ans. (a) : increase

441. How many necks appears during tensile test of specimen prepared according to ASTM standard.
(a) One (b) Two
(c) Three (d) Unpredictable

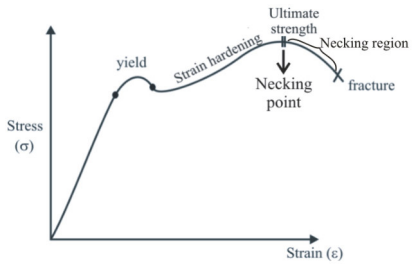
TSPSC Managers, 2015

Ans. (a) : According to ASTM standard, one necks appears during the tensile test of specimen.

442. When does the phenomenon of necking become prominent in a simple tension test?
(a) After elastic limit.
(b) After yield point
(c) After proportional limit
(d) From around the ultimate stress

RPSC AEN Pre-2013

Ans. (d) : Necking point become prominent in the tensile test around the ultimate tensile strength.



443. Toughness of a material is measured by the
(a) Vicker's hardness test
(b) Scratch test
(c) Universal testing machine
(d) Notched bar test

RPSC IOF, 2020

Ans. (d) : The charpy impact test also known as the charpy V-notch test is a standardized high strain-rate test which determines the amount of energy absorbed by material during fracture absorbed energy is a measure of the materials notch toughness.

444. Fluorescent dye method of testing is used for detecting the_____ of an object
(a) surface crack
(b) interior defects
(c) structural characteristics
(d) None of these

RPSC IOF, 2020

Ans. (a) : Fluorescent dye penetrate inspection is a non destructive testing method that is used to detect surface cracks, seams or discontinuities in parts. These surface defects show up during the inspection process as indications under black light.

445. Principle of dye penetrant test is based on
(a) magnetic domain
(b) absorption of X-ray
(c) capillary action
(d) polarized sound waves in liquid

RPSC IOF, 2020

Ans. (c) : The principle of liquid penetrant testing is that the liquid penetrant is drawn into the surface-breaking crack by capillary action and excess surface penetrant then removed.

446. In which NDT test piezoelectric transducer converts the electric energy to acoustic energy
(a) Radiography
(b) Magnetic particle test
(c) Ultrasonic
(d) Eddy current

RPSC IOF, 2020

Ans. (c) : Ultrasonic is generated with a transducer. A piezoelectric element in the transducer converts electrical energy into mechanical vibrations and vice versa. The transducer is capable of both transmitting and receiving sound energy.

447. Which of the following types of rays is used in radiography for the inspection of welded joints?
(a) X-rays (b) Infrared rays
(c) Ultraviolet rays (d) Visible rays

RPSC IOF, 2020

Ans. (a) : X-rays is used in radiography for the inspection of welded joints. This method of weld testing makes use of X- rays, produced, by an X-ray tube or X-rays, produced by a radioactive isotope.

448. In case of low or medium strength carbon steel, the rough estimate of ultimate tensile strength (S_{ut} in MPa) can be obtained from Brinell Hardness number (HB) as :

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(a) $S_{ut} \approx (3.43 \text{ to } 3.47) * HB$

(b) $S_{ut} \approx (1.5 \text{ to } 2.0) * HB$

(c) $S_{ut} \approx (0.5 \text{ to } 0.9) * HB$

(d) $S_{ut} \approx (30 \text{ to } 37) * HB$

APPSC Poly Lect. 13.03.2020

Ans. (a) : Ultimate tensile strength (S_{ut} in MPa) can be obtained from Brinell Hardness Number (HB) is –

$$S_{ut} \approx (3.43 \text{ to } 3.47) \times HB$$

For low or medium strength carbon steel.

449. The behavior of the materials under dynamic loading is assessed by:

(a) Brinell Test (b) Torsion test

(c) Impact test (d) Universal Test

NLC GET 17.11.2020, Shift-II

Ans. (c) : The behavior of the material under dynamic loading is assessed by Impact test determines the amount of energy absorbed by a material during fracture. It is to determine whether the material is brittle or ductile in nature.

450. UTS is always …… breaking strength of mild steel.

(a) Equal to (b) Less than

(c) More than (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Ultimate tensile strength (UTS) is always more than breaking strength of mild steel.

451. Which one of the phenomena is not used to measure hardness?

(a) indentation (b) wear

(c) scratch (d) fracture

VIZAG MT, 14.12.2020

Ans. (d) : Phenomenon used to measure hardness–

(1) Indentation

(2) Wear

(3) Scratch

Fracture is related to toughness.

452. Which is not the method to measure hardness?

(a) Brinell test (b) Rockwell test

(c) Vickers test (d) Cupping test

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Cupping test is used to measure ductility or formability of sheet metal and strips Brinell test, Rockwell test and Vickers test are method to measure hardness.

453. The type of test, in which the capillary action principle is used, is

(a) Probe test (b) Bend liquid test

(c) Dye penetrant test (d) None of the above

WBPSC AE, 2017

Ans. (c) : The type of test, in which the capillary action principle is used is dye penetrant test.

454. Eddy current test is used to detect

(a) cracks (b) hardness

(c) conductivity (d) All of the above

WBPSC AE, 2017

Ans. (a) : Eddy current testing can be used for a wide range of applications. It is often applied for surface crack detection and material sorting. Material sorting is used to ensure that the proper materials are in use and to verify component materials or assembly features (such as the orientation or position of a subcomponent in an assembly).

455. Erichsen cupping test is known as

(a) Creep test (b) Torsion test

(c) Fatigue test (d) Formability test

WBPSC AE, 2017

Ans. (d) : The Erichsen cupping test is a standard method used to asses the formability, ductility and adhesion of a coatings which induces similar strains to those experienced during forming.

456. Ultrasonic test is used to detect and locate____ in weldments

(a) external defects (b) internal defects

(c) roughness (d) none of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Ultrasonic is a kind of Non-Destructive testing technique based on ultrasonic waves propagation in objects to find internal defects.

457. Match List I with List II and select the correct answer using the codes given below the lists.

List-I (Property)	List-II (Testing Machine)
A. Tensile strength	1. Rotating Bending Machine
B. Impact strength	2. Three-point loading machine
C. Bending strength	3. Universal Testing machine
D. Fatigue strength	4. Izod Testing Machine

(a) 4, 3, 2, 1 (b) 3, 2, 1, 4

(c) 2, 1, 4, 3 (d) 3, 4, 2, 1

GPSC ARTO Pre 30.12.2018

Ans. (d) :

Tensile strength – Universal Testing Machine

Impact strength – Izod Testing Machine

Bending strength – Three –Point loading Machine

Fatigue strength – Rotating Bending Machine

458. Match List-I with List-II and select the correct answer using the codes given below the list :

List-I (Material properties)	List-II (Test to determine material properties)
A. Ductility	1. Impact test
B. Toughness	2. Fatigue test
C. Endurance limit	3. Tension test
D. Resistance to penetration	4. Hardness test

Code :

	A	B	C	D
(a)	3	2	1	4
(b)	4	2	1	3
(c)	3	1	2	4
(d)	4	1	2	3

Punjab PSC (Lect.) 06.08.2017

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Ans. (c)

A	B	C	D
3	1	2	4

(A) Ductility → Tension test
(B) Toughness → Impact test
(C) Endurance limit → Fatigue test
(d) Resistance to Penetration – Hardness

459. The Charpy impact test is carried out on a specimen of 55mm × 10 mm × 10 mm in size and has a 2mm deep notch at its centre with _____ angle.

(a) 60 degree (b) 55 degree
(c) 50 degree (d) 45 degree

Punjab PSC SDE 12.02.2017

Ans. (d)

460. Vicker's Hardness Test has an indenter of shape

(a) flat cuboidal (b) spherical
(c) conical (d) pyramidal

OPSC Civil Services Pre 2006

Ans. (d) : Brinell- carbide ball
Rockwell Hardness- steel ball or diamond cone
Knoop test - Rhombus shape
Vickers- Square- shaped diamond pyramid

461. For fatigue design of machine members, the relevant material property to be considered is

(a) yield strength (b) endurance strength
(c) hardness (d) impact strength

OPSC Civil Services Pre 2006

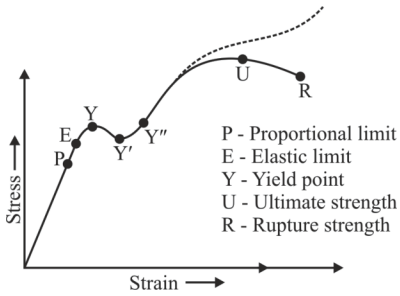
Ans. (b) : Fatigue test define endurance strength.

462. A point at which the extension of steel material is faster than the load increment is:

(a) Breaking point
(b) Yield point
(c) Plastic point
(d) Ultimate stress point

CIL MT 27.02.2020

Ans. (b)



When the load is further increased, plastic deformation occurs.
At yield point the extension of steel material faster than the load increment.

463. In case of Izod test, the specimen is kept as:

(a) Simply supported beam
(b) Overhanging beam
(c) Cantilever beam
(d) Fixed ended beam

CIL MT 27.02.2020

Ans. (c) : In case of Izod test, specimen kept vertical on the anvil notch is facing the striking hammer. One end is fixed at anvil other end free. Hammer strike at free end as like cantilever beam.

464. .....test is used to determine the scratch hardness of a thin materials.

(a) Mohr's scale of hardness
(b) Shore scleroscope hardness
(c) file hardness
(d) None of the above

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Mohr's scale of hardness test is used to determine the scratch hardness of a thin material.

465. Rockwell reading is measure of the penetration caused by the

(a) minor load only
(b) major load only
(c) both major and minor loads
(d) none of the above

WBPSC AE 2008

Ans. (b) : Rockwell reading is measure of the penetration caused by the major load only.

466. In Vicker's hardness test, a square base pyramid diamond indentor having.....between the opposite faces is used.

(a) 90º (b) 105º
(c) 152º (d) 136º

HPPSC AE (PWD) 10.09.2016

Ans. (d) : The Vickers hardness test method consists of indenting the test material with a diamond indenter in the form of a right pyramid with a square base and an angle of 136º between the opposite faces.

467. Match List-I (Mechanical property) with List-II (Mechanical test) and select the correct answer using the code given below the lists:

List-I (Mechanical property)	List-II (Mechanical test)
(A) Percentage elongation	1. Mohr test
(B) Hardness	2. Tensile test
(C) Toughness	3. Torsion test
(D) Modulus of rigidity	4. Impact test

(a) A -2, B-3, C-4, D-1 (b) A-2, B-1, C-4, D-3
(c) A-4, B-3, C-2, D-1 (d) A-4, B-1, C-2, D-3

UPRVUNL AE 21.08.2016

Ans. : (b)

Mechnical property	Mechanical test
Percentage elongation	Tensile test
Hardness	Mohr test
Toughness	Impact test
Modulus of rigidity	Torsion test

468. Diamond pyramid indenter is used for

(a) Brinell Hardness Test
(b) Vickers Hardness Test
(c) Rockwell B Hardness Test
(d) Rockwell C Hardness Test

GPSC Lect. 23.10.2016

Ans. (b) : Diamond pyramid indenter is used for Vickers hardness testing and resulting indentation in the workpiece. Vickers hardness tester uses a square based diamond pyramid indenter.

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469. The hardness testing method which does not involve resistance to plastic deformation is

(a) Shore scleroscope (b) Brinell
(c) Rockwell (d) Vickers

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : Shore scleroscope measures hardness in terms of the elasticity of the material. The harder the material, the harder the rebound.

470. Which of the following hardness test uses diamond cone type indenter?

(a) Brinell (b) Vickers
(c) Knoop (d) Rockwell

ISRO Scientist/Engineer 12.01.2020

Ans. (d) :

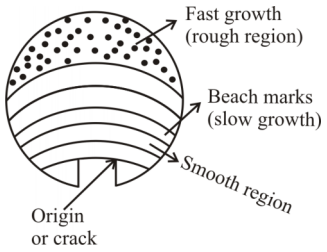
Hardness test	Type of Indenter
Brinell's Hardness test	Steel ball indenter
Vicker hardness test	Square base diamond pyramid indenter.
Rockwell hardness test	"Brale" indenter (Conical black diamond)

471. Beach or clamshell marks on a failed metallic surface are typical of

(a) Ductile fracture (b) Brittle fracture
(c) Creep failure (d) Fatigue failure

ISRO Scientist/Engineer 12.01.2020

Ans. (d) :



Beach or clamshell marks on a failed metallic surface are typical type of fatigue failure.

472. The main objective of 'shot peening' is to improve which property of metal parts

(a) Surface finish (b) Ductility
(c) Fatigue strength (d) None of the above

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : Shot peening process used to improve fatigue strength.

473. Match the items in Column I and Column II :

Column-I	Column-II
P. Charpy test	1. Fluidity
Q. Knoop test	2. Micro hardness
R. Spiral test	3. Formability
S. Cupping test	4. Toughness
	5. Permeability

(a) P - 4, Q - 5, R - 3, S - 2
(b) P - 3, Q - 5, R - 1, S - 4
(c) P - 2, Q - 4, R - 3, S - 5
(d) P - 4, Q - 2, R - 1, S - 3

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) Charpy test → Toughness
Knoop test → Micro hardness
Spiral test → Fluidity
Cupping → Formability

474. Which of the following scale is not used to measure the hardness of very hard materials?

(a) Rockwell 'C' scale
(b) Brinell hardness scale
(c) Rockwell 'A' scale
(d) Rockwell 'B' scale

UPRVUNL AE 07.10.2016

Ans. (d) : Rockwell 'B' scale is not used to measure the hardness of very hard materials.

475. Brittle fracture takes place without any appreciable deformation and by rapid crack propagation. The direction of crack propagation is very nearly:

1. Perpendicular to the direction of the applied tensile stress
2. Parallel to the direction of the applied tensile stress
3. At an angle 45° to the direction of the applied tensile stress

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

JWM 2017

Ans. (a) : In Brittle fracture, the direction of crank propagation is perpendicular to the direction of applied tensile stress.
Brittle failure is due to the principle stress.

476. Which of the following is non-destructive test

(a) tensile test (b) charpy test
(c) cupping test (d) radiography test

TNPSC AE 2018

Ans. (d)

Non destructive test	Destructive test
Used for find out defect of materials	Used for find out the properties of the materials
Load is not applied on the material	Load is applied on the material
No load application, so no chance for material damage	Due to load application, material gets damaged.
Not expensive	Expensive
Example- Dye penetrant test, ultrasonic, radiography x-ray test, visual test etc.	**Example-** Tensile, compression, hardness, charpy test, cupping test etc.

477. The most inexpensive non-destructive method of material testing is

(a) Dye penetrante testing
(b) Ultrasonic testing
(c) X-ray testing
(d) Visual testing

TNPSC AE 2018

Ans. (a) : Dye Penetrante Test– Dye penetrante test also called liquid penetrante inspection or penetrante testing is widely applied and low-cost inspection method used to check surface braking defects in all non-porous materials.

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478. Which of the following tests can be used to detect surface cracks in the welding of non-magnetic alloys?
(a) Gamma ray test (b) X-ray test
(c) Flourescent test (d) Magnaflux test

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Flourescent test can be used to detect surface crack in the welding of non-magnetic alloys.

479. The Non-Destructive Inspection (NDI) technique employed during inspection for castings of tubes and pipes to check the overall strength of a casting in resistance to bursting under hydraulic pressure is
(a) Radiographic inspection
(b) Magnetic particle inspection
(c) Fluorescent penetrant
(d) Pressure testing

ESE 2020

Ans. (d) : Pressure testing

480. The relationship between the gauge length (L) and the cross sectional area (A) of the test specimen in a tensile testing is given by
(a) $L = 5.65\sqrt{A}$ (b) $A = 5.65\sqrt{L}$
(c) $L = 8.65\sqrt{A}$ (d) $A = 8.65\sqrt{L}$

HPPSC AE 2018

Ans. (a) : The relationship between the gauge length (L) and the cross sectional area (A) of the test specimen in a tensile testing is given by $L = 5.65\sqrt{A}$

481. The indentation on a steel sample has been taken using 10 mm tungsten carbide ball at 500 kgf load. If the average diameter of the indentation is 2.5 mm, the BHN will be nearly
(a) 90 (b) 100
(c) 110 (d) 120

ESE 2019

Ans. (b) : Given,
D = 10 mm
d = 2.5 mm
P = 500 kgf
Brinell hardness number

$$BHN = \frac{2P}{\pi D\left[D - \sqrt{D^2 - d^2}\right]}$$

$$= \frac{2 \times 500}{\pi \times 10\left[10 - \sqrt{10^2 - 2.5^2}\right]}$$

$$= 100.24$$

482. A solid rod of circular cross-section made of brittle material, when subjected to torsion, fails along a plane at 45° to the axis of the rod. Consider the following statements as pertaining there to:
1. Distortion energy is maximum on this 45° plane
2. Shear stress is maximum on this 45° plane
3. Normal stress is maximum on this 45° plane

Which of the above is/are correct?
(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

ESE 2018

Ans. (c) : Brittle material shaft under pure torsion fail along a plane at 45° to the axis of the shaft because they are weak in tension and maximum tensile stress plane will be at angle of 45° to the cross-section of shaft or long axis of shaft.

483. Which one of the following pairs of tests has been developed to evaluate the fracture resistance of engineering materials, subjected to dynamic loads or impacts?
(a) Tension impacts and bending impacts
(b) Tensile test and Brinell hardness test
(c) Vickers hardness test and Tensile test
(d) Scleroscope test and file test

ESE 2018

Ans. (a) : To evaluate toughness or fracture resistance of a material which are subjected to a rapidly applied load or impact, basically two types of tests are applied.
(i) Bending impact
(ii) Tension impact

484. Vicker's Pyramid Number (VPN) is equal to :
(a) $\frac{2P\sin\theta}{d^2}$ (b) $\frac{P\sin\theta}{d^2}$
(c) $\frac{\frac{P}{2}\sin\theta}{d^2}$ (d) $\frac{P\sin\theta}{2d^2}$

where P = Load in kg, θ = angle between opposite faces of diamond pyramid, d = diameter of the impression.

OPSC Civil Services Pre. 2011

Ans. (c) : Vicker's pyramid Number (VPN) is equal to $= \frac{\frac{P}{2}\sin\theta}{d^2}$
■ Angle between opposite faces is 136^0.

485. The method of testing hardness by Brinell test is based on the principle of
(a) Indentation (b) Fracture
(c) Scratching (d) Rebound

Gujarat PSC AE 2019

Ans : (a) : Brinell hardness number hardened steel ball is used as indentor.
Brinell hardness number

$$= \frac{2P}{\pi D\left[D - \sqrt{D^2 - d^2}\right]}$$

486. In a tensile test on mild steel specimen, the breaking stress as compared to ultimate tensile stress is
(a) more
(b) less
(c) same
(d) more/less depending on composition

Gujarat PSC AE 2019

Ans : (b) : In a tensile test on mild steel specimen the breaking stress as compared to ultimate tensile stress is less.

487. Which of the following tests is also called Micro hardness Test?
(a) Brinell test (b) Rockwell test
(c) Knoop test (d) Vickers test

UPPSC AE 12.04.2016 Paper-I

Ans : (c) Micro hardness Test is knoop test.

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488. The ductility of a material _____with the increase in percentage reduction in area of a specimen under tensile test.

(a) increases (b) decreases
(c) remains same (d) None of these

Sikkim PSC (Under Secretary), 2017

Ans. (a) : The ductility of a material increases with increase in percentage reduction in area of specimen under tensile test.

489. Spheroidal or nodular graphite iron is designed as SG 500/7. Here '500' and '7' stand for

(a) Proof stress in N/mm^2 and elongation in %
(b) Tensile strength in N/mm^2 and impact strength in N-m
(c) Tensile strength in N/mm^2 and elongation in %
(d) Tensile strength in $kg.mm^2$ and elongation in %

GPSC EE Pre, 28.01.2017

Ans. (c) : Material iron grade - SG 500/7
Elongation min% - 7
Tensile strength (N/mm^2)- 500

490. In a tensile test, neck appears in the material

(a) At the centre of specimen
(b) In the gauge length only
(c) Anywhere in parallel length
(d) Anywhere between fixed points

TSPSC AEE 2015

Ans. (c) : In a tensile test, neck appears in the material anywhere in parallel length.

491. The notch angle of Izod impact test specimen is

(a) 10º (b) 20º
(c) 30º (d) 45º

UKPSC AE 2007 Paper -I

Ans. (d) : Notch angle of Izod impact test specimen is 45º.

492. Age hardness is related to

(a) cast iron (b) duralumin
(c) stainless steel (d) brass

GPSC ARTO 01.05.2016

Ans. (b) : Age Hardening– Age hardening, also known as precipitation hardening, is a type of heat treatment that is used to impart strength to metal and their alloys. Malleable metals and alloys of nickel, magnesium and titanium are suitable for age hardening process.
Ex. Duralumin

493. Which one of the following is NOT used for measurement of hardness?

(a) Brinnel test (b) Rockwell test
(c) Vicker's test (d) Die penetrant test

UPSC JWM Adv. No-16/2009

Ans. (d) : Die penetrant test is not used for measurement of hardness.

494. Surface endurance limit of gear material is dependent upon its

(a) elastic strength
(b) yield strength
(c) brinell hardness number
(d) toughness

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : The fatigue or endurance limit is the stress level below which an infinite number of loading cycles can be applied to a material without fatigue failure. Similarly surface endurance limit of a gear material is dependent upon the hardness of the surface or on brinell hardness number.

495. Doubling the frequency of current in an induction hardening process

(a) doubles the depth of hardening
(b) halves the depth of hardening
(c) increases the depth of hardening by a factor of $\sqrt{2}$
(d) reduces the depth of hardening by a factor of $\sqrt{2}$

DRDO Scientists 2008

Ans. (d) : $x = 5000\sqrt{\frac{\rho}{\mu f}} \Rightarrow x \propto \frac{1}{\sqrt{f}}$

where : x = depth of hardening
ρ = electrical resistivity
f = frequency
μ = magnetic permeability

496. Which one of the following properties cannot be evaluated by static tension test?

(a) Shear strength
(b) Modulus of elasticity
(c) Ductility
(d) Poisson's ratio

UPSC JWM Adv. No-16/2009

Ans. (a) : Shear strength cannot be evaluated by static tension test.

497. In radiographic test, type of defect not detectable by X-Ray is

(a) Delamination in cladded sheet
(b) Porosity in castings
(c) Tungsten inclusion in TIG weld
(d) Under cut in metal arc welding

ISRO Scientist/Engineer 2011

Ans. (d) : In radiographic test, type of defect not detectable by X-Ray is under cut in metal arc welding.

498. In order to measure/detect materials by non-destructive testing the method generally used is

(a) Acoustic emission
(b) Infrared radiometer
(c) Liquid crystallography
(d) Thermochemic point

ISRO Scientist/Engineer 2010

Ans. (a) : NDT (Non-Destructive Testing)
Types of NDT

(1) Acoustic Emission testing
(2) Electromagnetic testing
(3) Laser testing
(4) Leak testing
(5) Magnetic flux leakage.
(6) Liquid Penetrant testing
(7) Magnetic Partical testing
(8) Radiography

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(9) Thermographic testing
(10) Infrared testing
(11) Eddy current testing
(12) X-ray testing
(13) Ultra sonic testing
(14) Visual Testing

499. The hardness of a lathe bed material should be measured by :
(a) Rockwell tester
(b) Brinell Hardness tester
(c) Vickers Hardness tester
(d) None

Kerala PSC Poly. Lect. 09.01.2014

Ans. (b) :

4. Heat Treatment Process

500. Machine tool guide ways are generally hardened by :
(a) Flame hardening
(b) Vacuum hardening
(c) Induction hardening
(d) Martempering

TRB Asstt. Prof., 2012
J&K PSC Civil Services Pre, 2010
Nagaland PSC (CTSE) 2018 Paper-I
UPPSC AE 13.12.2020 Paper-I
TSPSC AEE 2015
UPSC JWM Adv. No-16/2009
ESE - 1997, ESE - 1996

Ans. (a) : Machine tool guide ways are generally hardened by flame hardening. Flame hardening is a common surface hardening method. Metal surfaces, such as steel, are heated with high-temperature flame and then quenched. The result of flame hardening is hard surface that is more resistant against wear and corrosion.

501. Quenching is not necessary when hardening is done by :
(a) Case carburizing
(b) Flame hardening
(c) Nitriding
(d) Any of the above processes

GPSC Engineer, Class-II Pre-19.01.2020
GPSC ARTO Pre 30.12.2018
GPSC ARTO Pre 30.11.2018
PSPCL AE, 2012
WBPSC AE, 2008

Ans. (c) : Case hardening or surface hardening is the process of hardening the surface of a metal deeper underneath to remain soft, thus forming a thin layer of harder metal (called the case) at the surface.
Flame hardening is a heat treatment process where fuel gas flames are directly impinged hardened which is then subjected to quenching. It results in a hard surface layer of martensite over a softer interior care.
Nitriding – It is a heat treatment process that diffuses nitrogen into the surface of a metal to create a case - hardened surface. These processes are most commonly used on high carbon, low-alloy steels they are also used on medium and high carbon steels, titanium, aluminium and molybdenum.

502. The process in which carbon and nitrogen both are absorbed by the metal surface to get it hardened is known as :
(a) Carburising (b) Cyaniding
(c) Flame hardening (d) Induction hardening.

RPSC ACF & FRO, 26.02.2021
MECON MT 2020
NLC GET 17.11.2020, Shift-II

Ans. (b) : The process in which carbon and nitrogen both are absorbed by the metal surface to get it hardened is known as cyaniding. In this, workpiece is immersed in molten salt bath containing NaCN which is heated to about 820°C to 860°C. This is usually followed by water quenching.

503. The lower critical temperature-
(a) decreases as the carbon content in steel increases
(b) increases as the carbon content in steel increases
(c) is same for all steels
(d) depends upon the rate of heating.

RPSC ACF & FRO, 26.02.2021
GPSC Engg. Class-2 Pre 19.01.2020

Ans. (c) : The lower critical temperature is same for all steels. Lower critical temperature for steel's austenite to pearlite transformation is 723°C, below this temperature, austenite does not exist.

504. Nitriding process is used to increase surface hardness of
(a) cast iron (b) alloy steel
(c) high carbon steel (d) none of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016
CGPSC AE 16.10.2016

Ans. (b) : Nitriding process is used for alloy steel, titanium and molybdenum.

505. To reduce brittleness of the hardened steel and thus to increase ductility and to make steel sufficiently tough to resist shock and fatigue, which heat treatment process is used
(a) Nitriding (b) Tempering
(c) Normalizing (d) Hardening

Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
APPSC IOF 2009

Ans. (b) : Tempering : It is a process of improving the characteristic of metal especially steel by heating to a high temperature (below melting point) and then cooling it in air. This process reduces brittleness of hardened steel and make steel tough to resist shock and fatigue.

506. Heat treatment given to hardened steel to improve its ductility is :
(a) Annealing (b) Normalizing
(c) Tempering (d) Hardening

APPSC IOF, 2009, APGENCO AE. 2017
KPSC AE. 2015, TNPSC AE 2017
OPSC AEE 2015 Paper-I
TSPSC AEE 2015

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Ans. (c) : Tempering is done after hardening of steel to improve its ductility and toughness.
Tempering is a process of heating the metal to a temperature below lower critical temperature.

507. Recrystallization temperature for pure metals is (T_m is absolute melting temperature) :
(a) 0.1 T_m (b) 0.3 T_m
(c) 0.5 T_m (d) 0.7 T_m

MPPSC State Forest Service Exam, 2014
GPSC DEE Class-2 (GWSSB) 04.07.2021

Ans. (b) : Generally for pure metals recrystallization temperature is in the range of approx. 0.3 to 0.4 T_m and for alloys it is 0.5 T_m.
(T_m is the melting point).

508. Annealing of white cast iron produces
(a) grey cast iron (b) nodular iron
(c) malleable iron (d) wrought iron

APGENCO AE, 2017
APPSC Poly. Lect. Automobile Engg. 2020

Ans. (c) : Malleable Iron is obtained by heat treatment of white cast Iron in which Leadeburite is converted into matrix of ferrite or pearlite.

509. The process of reheating the martensitic steel to reduce its brittleness without any significant loss in its hardness is :
(a) normalizing (b) annealing
(c) quenching (d) tempering

GPSC ARTO Pre 30.12.2018
GATE - 2006
ISRO Scientist/Engineer (RAC), 10.03.2019
GPSC Executive Engineer 23.12.2018

Ans. (d) : Tempering is a process of annealing martensite at low temperature (below 727°C). Tempering heat treatment gives desired combination of hardness, ductility, toughness strength and structured stability.

510. The main purpose of spheroidising treatment is to improve
(a) Hardness of low carbon steel
(b) Machinability of low carbon steel
(c) Machinability of high carbon steel
(d) Hardness of high carbon steel

GPSC ARTO Pre 30.12.2018
VIZAG Steel MT 10.06.2012
APPSC Poly. Lect. Automobile Engg. 2020

Ans. (c) : Spheroidising is a form of heat treatment for iron-based alloys, commonly carbon steels, in order to convert them into ductile and machinable alloys. The main purpose of spheroidising treatment is to improve machinability of high carbon steels.

511. During heat treatment of steel, the hardness of various structures in increasing order is :
(a) martensite, fine pearlite, coarse pearlite, spheroidite
(b) fine pearlite, coarse pearlite, spheroidite, martensite
(c) martensite, coarse pearlite, fine pearlite, spheroidite
(d) spheroidite, coarse pearlite, fine pearlite, martensite

GPSC ARTO Pre 30.12.2018
GATE 2003
VIZAG Steel MT 10.06.2012
VIZAG Steel MT 18.08.2013

Ans. (d) : The hardness of various structures in increasing order is spheroidite, coarse pearlite, fine pearlite, martensite.

512. Which medium is used for fastest cooling during quenching of steel?
(a) Air (b) Oil
(c) Water (d) Brine (salt water)

BPSC AE Mains 2017 Paper - VI
UKPSC AE-2013, Paper-I
ESE - 2009

Ans. (d) : Brine medium is used for fastest cooling during quenching of steel.

513. Case hardening is the only method suitable for hardening:
(a) high alloy steel (b) high carbon steel
(c) high speed steel (d) low carbon steel

Kerala PSC AE (Irrigation) 05.04.2017
APPSC Poly. Lect. Automobile Engg. 2020
GPSC Lect. Poly. 07.12.2014

Ans. (b) : Both carbon and alloy steels are suitable for case-hardening.
Typically mild steels are used, with low carbon contains usually less than 0.3% of carbon.

514. The aim of the tempering heat treatment is to reduce ____ while substantially reducing
(a) fatigue strength, ductility
(b) hardness; brittleness
(c) harness; ductility
(d) fatigue strength; hardness

UPRVUNL AE 05.07.2021

Ans. (b) : Heat treatment involve the use of heating or chieling, normly to extreme temperatures to achieve a desired result such as hardening or softening of a material. Heat treatment technique include annealing case hardening, precipitation strengthening, temperature and quenching.

515. In austempering process of heat treatment, the austenite changes in to
(a) martensite (b) troosite
(c) sorbite (d) bainite

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Oil India Senior Officer 23.12.2020
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Austempering– The steel is heated above the upper critical temperature then it is quickly quenched a little above Martensite temperature so as to just miss the nose of TTT curve and then maintained there are at the same temperature for prolonged period so that complete austenite is transformed into bainite and then quenched it to room temperature.

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516. Austempering is employed to obtain

(a) 100% martensite structure
(b) 100% bainitic structure
(c) 50% martenisite and 50% bainitic structure
(d) 100% pearlite structure

RPSC IOF, 2020
KPSC AE 2015
TSPSC AEE 2015
ESE 2005

Ans. (b) : Austempering is a heat treatment process for medium - to - high carbon ferrous metals which produces a metallurgical structure called bainite. It is used to increase strength, toughness and reduce distortion and to obtain 100% bainitic structure.

517. T.T.T. diagram indicates time and temperature transformation of

(a) Cementite (b) Pearlite
(c) Ferrite (d) Austenite
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
Mizoram PSC (Insp. of Factories) 2019 Paper-III
GPSC DEE Class-2 (GWSSB) 04.07.2021

Ans. (d) : T.T.T. diagram is a practical diagram. It graphically describes the cooling rate required for the transformation of austenite to pearlite, bainite or martensite, TTT diagram also gives the temperature at which such transformation takes place.

518. Work hardening strengthens an alloy by

(a) Removing Internal defects in the crystal structure
(b) Increasing the dislocation density
(c) Decreasing the grain size of the alloy
(d) Increasing the lattice resistance to dislocation motion

ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Work hardening strengthens an alloy by increasing the dislocation density.

519. Hardening of steel does not increase its

(a) residual internal stresses
(b) corrosion resistance
(c) tensile strength
(d) hardness

RPSC IOF, 2020

Ans. (b) : Hardening of steel does not increase its corrosion resistance.
Hardening is the process known as cold working or cold forming processes in cold forming metal is formed at high speed and high pressure using tool steel or carbide dies the cold working of the metal increases the hardness, yield strength and tensile strength.

520. The effects of strain hardening be removed by ____ heat treatment.

(a) Tempering (b) Annealing
(c) Normalizing (d) Spheroidising

RPSC ACF & FRO, 26.02.2021

Ans. (b) : The effect of strain hardening may be removed by annealing heat treatment.
Annealing consists of three steps for relieving stresses after strain hardening.
(i) Recovery, (ii) Recrystalization, (iii) Grain growth

521. Fastest method at case hardening a samples is:

(a) Carburizing (b) Nitriding
(c) Induction hardening (d) Cyaniding

TNPSC AE 2018
TSPSC Managers, 2015

Ans. (c) : Induction hardening are processes in which the surface of the steel is heated very rapidly to high temperatures (by direct application of an oxy-gas flame, or by induction heating) then cooled rapidly, generally using water, this creates a 'case' of martensite on the surface.

522. Austempering produces

(a) Hard surface and tough core
(b) Hard core and tough surface
(c) Hard surface and hard core
(d) Tough surface and tough core

TSPSC Managers, 2015

Ans. (a) : Austempering produces a structure that is stronger and tougher than comparable structures produced with convectional heat treatments. Austempering is a heat treating process for medium to high carbon ferrous metals which produces a metallurgical structure called bainite.

523. Objective of Tempering is

(a) improve ductility
(b) reduce ductility
(c) improve brittleness
(d) improve internal stresses

APPSC Poly. Lect. 2013
WBPSC AE 2008

Ans. (a) : Purpose of tempering is-
(i) It reduces brittleness and hardness of steel.
(ii) It increases ductility.
(iii) It relieves internal stresses.
(iv) It improve toughness of steel.
Tempering–Martensite is very hard and brittle material. It can not use directly in engineering application, so tempering is performed to reduce hardness and brittleness of material and introduce toughness.

524. Spheroidizing is extensively employ

(a) low and medium carbon steel
(b) medium and high carbon steel
(c) low carbon steel
(d) None of the above

APPSC Poly. Lect. 2013

Ans. (b) : Spheroidizing is extensively employs to medium carbon steel by heating between 680ºC and lower critical temperature holding it at this temperature & then cooling it slowly to about 600ºC. Cementite is converted into granular form by spheroidizing process to increase machinability.

525. Gas carburizing can be done with

(a) CO_2 (b) CO
(c) NO (d) O_2

APPSC Poly. Lect. 2013

Ans. (b) : Gas carburizing is carried out in a gaseous atmosphere containing CO and CH_4

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526. Which of the following is not the purpose of annealing?

(a) Increasing hardness
(b) Increasing ductility
(c) Improving machinability
(d) Refining grain size.

RPSC IOF, 2020
RPSC AEN Pre-2013

Ans. (a) : Annealing is the process involving heating above recrystallization temperature holding at this temperature for some time and then slow cooling. The various purposes of annealing heat treatment are.

- Internal stress relieving.
- restore ductility and toughness
- refine grain size.
- elements chemical non-uniformity and gaseous contents in steel,
- It softness the steel.
- It enhance and improve the machinability of steel.
- It increases the ductility of steel.
- It enhance the toughness of steel.

527. Which is false statement about normalizing. Normalising is done to

(a) Relieve internal stresses
(b) Improve grain properties
(c) Refine grain structure
(d) Induce stresses

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (d) : Normalizing is done to achieve –

- To refine the grain structure and to eliminate coarse grain structure.
- To improve certain mechanical and electrical properties.
- To remove strain caused by cold working processes like hammering, Rolling, bending.
- Improve toughness

528. In normalising operation of a steel casting, the cooling is done in

(a) Oil bath (b) Water
(c) Air (d) Furnace

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : Normalising operation of steel is carried out by heating approximately 100ºF above upper critical temperature followed by cooling in air to room temperature.

829. Which of the following case hardening processes, result in a change in the composition in a steel component?

1. Carburizing
2. Cyaniding
3. Nitrding
4. Flame hardening

(a) 2, 3 and 4 only (b) 1, 3 and 4 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE - 2012

Ans. (c) : 1, 2 and 3 only

530. Case hardening of low and medium carbon steel is done by

(a) Spheroidising (b) Cyaniding
(c) Normalising (d) Tempering

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) : Cyaniding is a process used for case-hardening, of low and medium carbon steels, it is fast and efficient. The part is heated to 871ºC - 954ºC in bath of sodium cyanide & then is quenched and rinsed in water or oil to remove any residual cyanide.

531. Which one of the following heat-treatment operation is carried out to modify the properties of steel hardened by quenching?

(a) Normalizing (b) Annealing
(c) Induction heating (d) Tempering

RPSC AE GWD, 2011

Ans. (d) : Tempering- It consists of reheating the quenched metal below the critical temperature followed by desired rate of cooling to restore some of the ductility and reduce the brittleness. Increased toughness is obtained at the expense of high strength.
Tempering is carried out to modify the properties of steel hardened by quenching.

532. During annealing process, the temperature at which the new grains are formed in the metal is called

(a) Eutectic temperature
(b) Lower critical temperature
(c) Recrystallisation temperature
(d) Upper critical temperature

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Annealing – It is the process to obtain softness, improve machinability, increase ductility and toughness, relieve internal stresses, reduce or eliminate structural inhomogenity, refine grain size, and to prepare steel for subsequent heat treatment.
So in this process the temperature at which new grains are formed in the metal is called recrystallization process.

533. "The formation of a new set of strain free grains within a previously cold worked material" is known as :

(a) recrystallization (b) resilience
(c) ductility (d) passivity

APPSC Poly Lect. 13.03.2020

Ans. (a) : "The formation of a new set of strain free grains within a previously cold worked material" is known as Recrystallization.

534. Heat treatment of steel is done primarily to change the

(a) Physical properties
(b) Chemical composition
(c) Corrosion properties
(d) Mechanical properties

TSPSC AEE 2017

Ans. (d) : Heat treatment on most type of the steel is done for improving mechanical properties, such as tensile and yield strength.

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535. Which one of the processes is used for refining grains of metal after it has been distorted by cold working:

(a) Annealing (b) Softening
(c) Normalizing (d) Tempering

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : Annealing is a heat treatment process used mostly to increase the ductility and reduce the hardness of a material. It refine grains of metal after it has been distorted by cold working.

536. Heating the steel to just above its upper critical temperature for the required period of time and cooling in still air in called :

(a) Spheroidizing (b) Tempering
(c) Normalizing (d) Hardening

GPSC Asst. Prof (Prod.) 21.08.2016
CGPSC AE 15.01.2021
GPSC Executive Engineer 23.12.2018

Ans. (c) : Normalizing- Heating the steel to just above its upper critical temperature for the required period of time and cooling in still air.
Tempering- The process reduces the hardness a little but toughens the steel, significantly reduces the brittleness of steel so that it does not break easily.

537. The purpose of tempering is to :

(a) decrease toughness
(b) relieve residual stresses
(c) decrease ductility
(d) increase hardness

NLC GET 17.11.2020, Shift-II

Ans. (b) : Tempering in metallurgy, process of improving the characteristics of a metal, especially steel, by heating it to a high temperature, through below the melting point, then cooling it, usually in air. The process has the effect of toughening by leaving brittleness and reducing internal stresses and relieves residual stresses.

538. Nitriding of steel shaft improves:

(a) its fatigue strength
(b) its surface Finish
(c) its machinability
(d) its torsional stiffness
(e) both its fatigue strength and torsional stiffness

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : Nitriding can be used for hardening long parts such as the steel shaft because it delivers uniform diffusion of nitrogen along the entire port due to increase in hardness, the steel shaft is capable to with stand fatigue load.

539. Which of the following is intermetallic compound ?

(a) Ferrite (b) Austenite
(c) Pearlite (d) Cementite

J&K PSC Civil Services Pre, 2010

Ans. (d) : Fe_3C (cementite) is intermetallic compound of iron and carbon while α(ferrite) and γ (austenite) is a solid solution of carbon in iron. Solid solution are usually strong and ductile while intermetallic compound are usually hard and brittle.

540. The following treatment is used to avoid cracking of hardened steel :

(a) martempering (b) normalizing
(c) austempering (d) annealing

APPSC IOF, 2009

Ans. (d) : Annealing is a heat treatment process used mostly to increase the ductility and reduce the hardness of a material. This results in reduction of dislocations in crystal & such as crack etc.

541. The coarse grain size in heat treatment is due to.

(a) slow cooling (b) fast cooling
(c) quenching (d) oil cooling

MPPSC State Forest Service Exam, 2014

Ans. (a) : The coarse grain size in heat treatment is due to slow cooling because metal cohesiveness get increased with their near by molecules due to slow cooling.

542. The main objects of annealing are

(a) To refine grain size and to relieve internal stresses
(b) To alter electrical and magnetic properties
(c) To remove gases trapped in the metal during initial casting
(d) All of the above

Karnataka PSC AE, 10.09.2017

Ans. (d) : Annealing is done in furnace to obtain-
- Grained refine structure and internal stress relieve.
- To alter electrical and magnetic properties.
- To remove trapped gas in metal during initial casting.

543. Increased cooling rate in the heat treatment of mild steel results into :

(a) Coarse pearlite
(b) Fine pearlite
(c) Heat treatment does not affect the pearlite morphology
(d) Mixture of fine and coarse pearlite

GPSC ARTO Pre 30.12.2018

Ans. (b) : Increased cooling rate in the heat treatement of mild steel results into fine pearlite.

544. Mechanical properties of the metal improve in hot working due to :

(a) Recovery of grains
(b) Recrystallisation
(c) Grain growth
(d) Refinement of grain size

APGENCO AE 2012

Ans. (d) : Improvement in mechanical properties of metal in its hot working is due to refinement of grain size.

545. Austempering heat treatment process is practiced to improve:

(a) hardness (b) ductility
(c) brittleness (d) toughness

CIL MT 27.02.2020
Kerala PSC Poly. Lect. 09.06.2014

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Ans. (d) : Austempering the process of tempering at which steel is heated above critical temperature at about 875°C then suddenly cooled by quenching it in salt bath temperature about 250°C to 525°C after complete transformation the steel is cooled in air.
In this process good impact strength or toughness is obtained the degree of cracking is reduced.

546. Which of the following processes uses heat treatment to alters the physical and sometimes chemical properties of a material to increase its ductility and to make it more workable?
(a) Smelting (b) Annealing
(c) Forging (d) All the above

KPCL AE 2016

Ans. (b) : Annealing is a heat treatment process that change the physical and sometimes also the chemical properties of a materials to increase ductility and reduce the hardness to make it more workable. Annealing process requires the material above its recrystallization temperature for a set amount of time before cooling.

547. Which of the following is the heat treatment process in which iron or steel absorbs carbon liberated when the metal is heated in the presence of a carbon bearing material, in order to make the metal harder?
(a) Carbonization (b) Carbonation
(c) Carburizing (d) Carburation

KPCL AE 2016

Ans. (c) : Carburizing or carburization is a heat treatment process in which iron or steel absorbs carbon liberated when the metal is heated in the presence of a carbon bearing material push as charcoal or carbon monoxide, with the intent of making the metal harder.

548. In hot working process :
(a) Grain structure of the metal is refined
(b) Porosity of the metal is largely eliminated
(c) mechanical properties are improved
(d) All of the above

MPSC Poly. Lect. 2014

Ans. (d) : In hot working process.
(i) Grain structure of metal is refine
(ii) Porosity of metal is largely eliminated.
(iii) Mechanical properties are improved.

549. Which of the following hardening processes is not generally used for steels?
(a) Nitriding
(b) Cyaniding
(c) Age hardening
(d) Induction hardening

WBPSC AE 2008

Ans. (c) : Age Hardening–It is commonly used to process aluminum alloys and other non-ferrous metals for commercial use. The example are aluminum–Copper, Copper–beryllium, copper–tin, magnesium–aluminum and some non ferrous alloys.

550. Heating to beyond critical temperature and maintaining this temperature for a definite period to allow complete transformation into austenite there after cooling slowly in the furnace is known as:
(a) Normalizing (b) Stress relieving
(c) Spheroidising (d) Full Annealing

MPSC Dy. Engineer 2014

Ans. (d) : Full annealing is a process of heating beyond critical temperature and maintaining this temperature for a definite period to allow complete transformation into austenite there after cooling slowly in the furnace.
According to TTT-diagram slow cooling results in coarse pearlite.

551. Heating medium carbon steel by means of an alternating magnetic field to a temperature within or above the transformation range followed by quenching is known as:
(a) Case hardening (b) Carbonitriding
(c) Induction hardening (d) Carburizing

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Induction hardening is a type of surface hardening in which a metal parts are induction heated and then quenched. The quenched metal undergoes a martensitic. Transformation increasing the hardness and brittleness of the part. Induction hardening is used to selectively harden areas of a part or assembly without affecting the properties of the parts as a whole.

552. Which of the following heat treatment process is not a method of case-hardening
(a) nitriding
(b) quenching
(c) cyaniding
(d) induction hardening

TSGENCO AE 14.11.2015

Ans. (b) : Surface (case) hardening –
Low carbon steel : • Carburizing
• Nitriding
• Cyaniding
High carbon steel : • Flame hardening
• Induction hardening

553. Creep becomes important at 0.4 T_m, T_m being the absolute melting temperature of the metal. For Al, 0.4 T_m:
(a) 373 K (b) 473 K
(c) 573 K (d) 673 K

HPPSC Asstt. Prof. 2014

Ans. (a) : Aluminium melting temperature
$T_m = 660^oC$
$= 660 + 273$
$T_m = 933$ K
Creep = $0.4\ T_m = 0.4 \times 933$
$= 373.2$ K

554. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I (Heat treatment)	List-II (Effect on the properties)
A. Annealing	1. Redefined grain structure
B. Nitriding	2. Improves the hardness of the whole mass

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C. Martempering	3. Increases surface hardness
D. Normalising	4. Improves ductility

Code :

	A	B	C	D
(a)	4	3	2	1
(b)	1	3	4	2
(c)	4	2	1	3
(d)	2	1	3	4

(e) None of the above

UK PSC AE 2012 Paper- I
CGPSC Poly. Lect. 22.05.2016
ESE-1995

Ans. (a)

Annealing – Improves ductility
Nitriding – Increases surface hardness
Martempering – Improves the hardness of the whole mass
Normalising – Redefined grain structure

555. What is the yield stress for a polycrystalline alloy, when the grain size is ASTM 8 (grain diameter = 0.002 mm)?

Assume : Yield stress for crystal having no grain boundaries = $\delta_1 = 80\ MNm^{-2}$

Hall pitch constant = $K = 0.63\ MNm^{-3/2}$

(a) 2143 MNm^{-2} (b) 214.3 MNm^{-2}
(c) 21.43 MNm^{-2} (d) 2.143 MNm^{-2}

MPSC Lect. (Prod.) 2018

Ans. (*) : Hall pitch equation

$$\sigma_y = \sigma_0 + \frac{k}{\sqrt{d}}$$

$\sigma_0 = 80$ MPa
$k = 0.63\ MNm^{-3/2}$
$d = 0.002\ mm = 2\times10^{-6}m$

$$\delta_y = 80 + \frac{0.63}{\sqrt{2\times10^{-6}}} = 525.47\,MPa$$

σ_y = Yield stress of a polycrystalline material (MPa)
σ_0 = Yield stress of the material at an ∞ grain size (MPa)
k = hall pitch constant
d = Average grain size

556. Heating or cooling in a cycle designed to produce granular forms of carbide within the microstructure is known as

(a) Quenching (b) Normalizing
(c) Process Annealing (d) Spheroidizing

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : Spheroidizing process in which carbide in the granular form is produced in the structure of steel. The operation consists of heating steel upto a temperature slightly above the lower critical temperature (730^0C to 770^0C). It is held at this temperature for some time and then cooled slowly to temperature of 600^0C.

557. Temperature required for full annealing in hyper-eutectoid steel is

(a) 50^0C above upper critical temperature
(b) 50^0C above lower critical temperature
(c) 50^0C below lower critical temperature
(d) 50^0C below upper critical temperature

UPRVUNL AE 21.08.2016

Ans. : (b) 50^0C above lower critical temperature

558. If a plain carbon steel is rapidly cooled to the room temperature by quenching it in water, its structure is changed from

(a) austenite to pearlite
(b) austenite to martensite
(c) austenite to cementite
(d) austenite to ferrite

GPSC Lect. 23.10.2016

Ans. (b) : If a plain carbon steel is rapidly cooled to the room temperature by quenching it in water, its structure is changed from austenite to martensite form iron phase change graph, it become hard, due to quenching.

559. Pickling treatment is cleaning the casting with

(a) Soda ash
(b) Dilute acid
(c) Compressed air and sand particles
(d) Iron shots

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Pickling is a metal surface treatment used to remove impurities, such as stains, inorganic contaminants, rust or scale from ferrous metals.
A solution called pickle liquor, which usually contains acid is used to remove the surface impurities.

560. Hardness will be highest after which of the following heat treatment process?

(a) Furnace cooling (b) Air cooling
(c) Oil quenching (d) Water quenching

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : Hardness will be highest after water quenching. The rate of cooling decides the hardness of the alloy and this is different for different fluids used because of the difference in the thermal conductivities the specific heats and the heats of vaporization of these medium.

Quenching medium	**– Cooling rate compared to water**
Water	– 1
Circulated Air	– 0.032
Still Air	– 0.0152

561. Which of the following treatment does not improve hardness?

(a) Quenching (b) Cyaniding
(c) Normalizing (d) Annealing

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : Quenching–It is hardening process. It is also increases surface hardening.
Cyaniding– It is increases hardness.
Normalizing– Hardness increases.
Annealing– It is relieving stress process.
Note: Annealing process does not improve hardness, instead it is used to increase ductility and reduce hardness and strength.

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562. What is the common surface hardening treatment in steel?

(a) Carburizing (b) Tempering
(c) Quenching (d) None of the above

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : Carburizing–Carburizing is the common surface hardening treatment in steel which is used for flow carbon steel.

563. A material is dimensionally stable at room temperature if its glass transition temperature (T) is

(a) Below room temperature
(b) Just above room temperature
(c) Equal to room temperature
(d) Well above room temperature

ISRO Scientist/Engineer 12.05.2013

Ans. (d) : A material is dimensionally stable at room temperature if its glass transition temperature (T) is well above room temperature.

564. Which of the following thermocouple is capable of measuring highest temperature?

(a) Chromel - alumel (b) Platinum - rhodium
(c) Iridium - rhodium (d) Iron - constantan

ISRO Scientist/Engineer 2011

Ans. (c) :

Chromel - alumel	1100-1300ºC
Platinum - rhodium	1650-1700ºC
Iridium - rhodium	2000-2100ºC
Iron - constantan	850-1100°C

565. Materials after cold working are subjected to following process to relieve stresses

(a) Hot working (b) Tempering
(c) Normalizing (d) Annealing

ISRO Scientist/Engineer 2010

Ans. (d) : Before cold working, the lattice of the material exhibits a regular, nearly defect-free pattern (almost no dislocations). The defect-free lattice is created or restored by annealing.

566. Anodising is

(a) A process of coating of zinc by hot dipping
(b) A zinc diffusion process
(c) A process used for making thin phosphate coatings on steel to act as a base or primer for enamels and paints
(d) An oxidizing process used for aluminium and magnesium articles

ISRO Scientist/Engineer 2009

Ans. (d) : Anodising is an oxidizing process used for aluminium and magnesium articles.

567. "Troosite" is obtained when

(a) Quenching steel during transformation
(b) A fully hardened steel is finally drawn at about 677ºC
(c) Steel is rapidly quenched in oil
(d) When alloy steels are rapidly quenched in water

ISRO Scientist/Engineer 2008

Ans. (a) : "Troosite" is obtained when quenching steel during transformation.

568. Induction hardening is the process of

(a) Hardening the core
(b) Uniform hardening
(c) Selective hardening
(d) Hardening surface for wear resistance

ISRO Scientist/Engineer 2008
TNPSC AE 2018
Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Induction hardening is the process of hardening surface of work piece to obtain hard and wear resistance surface.

569. Which one is not the purpose of heat treatment of steels?

(a) Changing the composition of steel on the surface
(b) Changing the percentage of carbon and Si in the bulk.
(c) Increasing or decreasing the grain size
(d) Removing undesirable residual stresses

ISRO Scientist/Engineer 2008

Ans. (b) : Changing the percentage of carbon and silicon in the bulk is not the purpose of heat treatment of steels.

570. Which is false statement about annealing. Annealing is done to

(a) Relieve stresses
(b) Harden steel slightly
(c) Improve machining characteristic
(d) Soften material

ISRO Scientist/Engineer 2007
ISRO Scientist/Engineer 2012

Ans. (b) : Annealing is not done to harden steel slightly. It use for relieve stresses, improve machining characteristics and soften material.

571. The heat treatment of carbon steel, consisting of austenitization, followed by slow cooling in air is called as :

(a) Nitriding (b) Tempering
(c) Quenching (d) Annealing

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (d) : The heat treatment of carbon steel, consisting of austenitization, followed by slow cooling in air is called as annealing.

572. The complete phase recrystallization and fine grain structure is obtained in casting, forging and rolled parts by :

(a) Recrystallization annealing
(b) Normalizing
(c) Spheroidizing
(d) Austenising

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : The complete phase recrystallization and fine grain structure is obtained in casting, forging and rolled parts by normalizing.

573. Annealing heat treatment process is

(a) refines grain structure
(b) improve the hardness
(c) improves ductility
(d) increases surface hardness

TSPSC AEE 2015

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Ans : (c) Annealing heat treatment process is improve ductility.

Annealing

* Quenching medium	:	Furnace
* Full annealing	:	Ductility increase and toughness increase
* Process annealing	:	Stress relieving.
* Spheroidise annealing	:	Machinability increases
* Diffusion annealing	:	To homogenize the chemistry of material.

574. In the induction hardening process, it is high

(a) current (b) voltage
(c) frequency (d) temperature

TSPSC AEE 2015

Ans : (c) In the induction hardening process it is high frequency (2000 cycles/sec).

Induction Hardening:- The work is placed in heating coils called inductor the coil consist of several turns of water cooled Cu tubing. Alternate current is passed through the inductor and thus alternate magnetic field sets up. The field induces eddy currents on the surface layers and heat is generated the current density is not uniform throughout the cross-section of the workpiece. It is more along the surface, layer of the workpiece Approximate the 90% of the heat is generated in the work layer of thickness (x) represented by

$$x = 5000\sqrt{\frac{\rho}{\mu f}}$$

575. Annealing heat treatment is a

(a) Slow cooling process
(b) very slow cooling process
(c) very rapid cooling process
(d) non-cooling process

TSPSC AEE 2015

Ans : (b) Annealing heat treatment is a very slow cooling process.

Annealing:- Annealing is a general term applied to several softening processes. Annealing may be defined as a softening process in which the metal is heated to austenite phase, hold at that temperature for sometime to allow for internal changes and then cooled very slowly through the transformation range when the internal structure of the metal gets stablized.

576. A Zinc diffusion process is called

(a) Galvanizing (b) Parkerizing
(c) Sherardizing (d) Anodizing

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Galvanizing- Galvanization is the process in which steel requires a zinc coating to be applied to the steel in a molten hot bath. It also requires the steel to go through a series of other baths to ensure that the zinc will completely adhere to iron in the steel without any other foreign bodies getting in the way.

Sherardizing- Sherardizing is also called vapour galvanizing or dry galvanizing. The process of sherardizing involves heating the steel upto a very high temperature it is then put into a rotating drum which also contains zinc dust and possibly some other filler such as sand. The steel is then rolled in the drum full of zinc at a high temperature where the zinc will then evaporate and diffuse, bonding to the steel and filling in very small crevices.

577. Which of the following is not a heat treatment process?

(a) Tempering (b) Nitriding
(c) Honing (d) Quenching

UJVNL AE 2016

Ans : (c) Honing is not a heat treatment process where Honing is surface finishing process. In honing operation tool is called hone.

578. The diffusion coefficient of carbon through iron during a hardening process

(a) decreases with temperature
(b) remains constant as temperature increases
(c) remains constant as pressure increases
(d) increases with temperature

TNPSC AE 2018

Ans. (d) : The diffusion coefficient of carbon through iron during a hardening process increases with temperature.

579. Flame and induction hardening process is used for

(a) A big job
(b) Small portion of a job
(c) Deeper hardness of a job
(d) Inner hardness of a job

HPPSC AE 2018

Ans. (b) : Induction hardening–Induction hardening is a process used for the surface hardening of steel and other alloys components. The parts to be heat treated are placed inside a water cooled copper coil and then heated above their transformation temperature by applying an alternating current to the coil. The alternating current in the coil induce an alternating magnetic field within the work piece, which if made from steel, caused the outer surface of the part to heat a temperature above the transformation range. Parts are held at that temperature until the appropriate depth of hardening has been achieved, and then quenched in oil, or another media, depending upon the steel type and hardness desired.

Flame Hardening–Flame hardening is similar to induction hardening, in that it is a surface hardening process.

Heat is applied to the part being hardened, using an oxy-acetylene flame on the surface of the steel being hardened and heating the surface above the upper critical temperature before quenching the steel in a spray of water.

580. Precipitation hardening is a strengthening method for steels and alloys, which is also called as:

(a) Age hardening
(b) Strain hardening
(c) Solid solution hardening
(d) Quenching
(e) Dispersion hardening

(CGPSC Polytechnic Lecturer 2017)
TSGENCO AE 14.11.2015

Ans. (a) : Precipitation Hardening or Age Hardening–Precipitation hardening is commonly used to process aluminum alloys and other non-ferrous metals for commercial use. The example are aluminum–Copper, Copper–beryllium, copper–tin, magnesium–aluminum and some ferrous alloys.

The Process is called precipitation hardening because the small particles of the new phase are termed "Precipitates"

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581. Pick the odd statement out with regard to objectives of heat treatments of steels.

(a) To enhance hardness, wear resistance and cutting ability of steel.
(b) To re-soften the steel after cold working.
(c) To reduce or eliminate internal residual stresses.
(d) To decease or increase the grain size of steels.
(e) To destablise the steel so that it changes in dimension with time.

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Objectives of Heat Treatment–
- Increase hardness, wear and abrasion resistance and cutting ability of steels.
- Re-softening the hardened steel.
- To adjust mechanical, physical or chemical properties like, hardness, tensile strength, ductility, electrical and magnetic properties, microstructure or corrosion resistance.
- Eliminate internal residual stress.
- Refine grain size.
- Eliminate gases like hydrogen which embrittle steel.

582. Box annealing is also called as:

(a) Black annealing
(b) Isothermal annealing
(c) Bright annealing
(d) Spheroidise annealing
(e) Sub critical annealing

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Box-Annealing–A process of annealing a ferrous alloy in a suitable closed metal container, with or without backing material, in order to minimize oxidation. The charger is usually heated slowly to a temperature below the transformation range but sometimes above or within it, and is then cooled slowly. This process is also called "close annealing" or "Pot annealing" or black annealing.

583. The liquid carbonitriding process of case hardening is also called as:

(a) Case carburising (b) Cyaniding
(c) Nitriding (d) Flame hardening
(e) Induction hardening

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Cyaniding–Cyaniding is a process of hardening the surface of steel components through the addition of nitrogen and carbon. It is done by immersing the work piece in a bath of molten sodium Cyanide and sodium carbonate. This process is conducted at a temperature of 950°C.

584. Nitrided steel is used for gears where

(a) High hardness is desired on teeth
(b) Low speed operation is desired
(c) Variable speed operation is desired
(d) Variable load operation is desired

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Nitriding has been known for its effectiveness in resisting shear stresses due to friction and prevention of the tooth flank sub-surface fatigue failure, that can occur in highly loaded core-hardened gears teeth is desired.

585. Consider the following statements:

1. The quenching of steel results in an increase in wear resistance, strength and hardness.
2. By the process of case hardening, hard wearing resistant surface is produced on mild steel. This is an effective method for low carbon steels because they cannot be hardened by the process of quenching.
3. When a metal is mixed with small atoms of non metallic element in such a manner that invading atoms occupy interstitial positions in the metal lattice, and interstitial alloy results.

What of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : (i) Quenching of steel causes increase in hardness, wear resistance and strength.
(ii) Case hardening process is applicable for low carbon steels because they are not heat treated by hardening process and depth of hardenability is very low.
(iii) When a metal is mixed with small size atom of non-metallic elements they occupy interstitial positions of metal and form interstitial alloy.
Hence all the statements are correct.

586. On completion of heat treatment, the austenite structure would be retained if

(a) The rate of cooling is greater than the critical cooling rate.
(b) The rate of cooling is less than the critical cooling rate
(c) The initiating temperature of martensite formation is above the room temperature
(d) The finishing temperature of martensite formation is below the room temperature.

GATE 1997, ESE 2017

Ans. (d) : When the finishing temperature of martensite is below the room temperature then we get the retained austenite.
The martensite start and martensite finish temperature are related and these temperatures are controlled by various alloying element including carbon and chromium.
Note- The answer given by UPSC is option (c).

587. Which of the steel is best candidate for heat treatment?

(a) Low and medium carbon steel
(b) Low carbon steel
(c) High carbon steel
(d) Low and high carbon steel

UPRVUNL AE 07.10.2016

Ans. (c) : High carbon steel is best candidate for heat treatment.

588. What causes transformation of deformed martensite into austenite phase?

(a) Heating (b) Cooling
(c) Both (a) and (b) (d) Quenching

RPSC AE 2018

Ans. (a) : Heating causes transformation of deformed martensite into austenite phase.

589. Which one of the following heat treatment process improves machinability?

(a) Annealing (b) Process annealing
(c) Normalizing (d) Spheroidizing

JWM 2017
TNPSC AE 2018

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Ans. (d) : Process annealing is a stress relieving process. It is used in wire industry.
- Spheroidise annealing is a heat treatment process which increases the machinability.
- Normalizing process used to produced hard surface and tough core.

590. Hardening by carburizing is limited to
(a) 0.05 mm (b) 0.1 mm
(c) 2 mm (d) 5 mm

TNPSC AE 2017

Ans. (c) : Hardening by carburizing is limited to 2 mm.

591. The slowest cooling rate is achieved when steel is quenched in
(a) Fused salt (b) Air
(c) Brine (d) Mixture of water

TNPSC AE 2017

Ans. (b) : The slowest cooling rate is achieved when steel is quenched in air.

592. In which of the following methods, surface of a steel component becomes hard due to phase transformation austenite to Martensite?
(a) Carbonitriding (b) Flame hardening
(c) Nitriding (d) lonisation

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : In flame hardening method, surface of a steel component becomes hard due to phase transformation austenite to martensite.

593. The following hardening process is generally free from scale formation.
(a) flame hardening (b) cyaniding
(c) nitriding (d) induction hardening

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : Induction hardening process is generally free from scale formation.

594. A steel is heated at about 875 ºC where the structure consists of entirely austenite. It is then cooled suddenly at a temperature of about 250 ºC-520 ºC. This process of heat treatment is known as
(a) normalising (b) annealing
(c) austempering (d) martempering

JPSC AE 2013, Paper-V

Ans : (c) : Austempering- The austempering is misnomer because it is not a tempering process, It is also known as isothermal quenching. In this process, the steel is heated above the upper critical temperature, at about 875 ºC where the structure consists entirely of austenite. It is then suddenly cooled by quenching it in a salt bath or lead bath maintained at a temperature of about 250 ºC to 525 ºC.

595. Surface hardening is principally employed for
(a) low-carbon steel
(b) medium-carbon steel
(c) high-carbon steel
(d) cast iron

BPSC AE 2012 Paper - VI

Ans : (a) : Surface hardening is mainly employed for the application where high surface hardness, here resistance are of main concern and impact strength is not a much important parameter. It is mainly employed in the low and medium duty applications.

596. Cyaniding is the process of :
(a) Dipping steel in cyanide bath
(b) Reacting steel surface with cyanide salts
(c) Adding carbon and nitrogen by heat treatment of steel to increase its surface hardness
(d) Making corrosion resistance steel

OPSC AEE 2019 Paper-I

Ans : (c) : Cyaniding is a case hardening method which increases the hardness and wear-resistance of the outer surface of the steel.
- Steel is heated in molten cyanide at about 850ºC followed by quenching.
- Carbon and nitrogen are absorbed by steel.

597. Which is false statement about normalizing?
(a) Normalizing is done to refine grain structure
(b) Normalizing is done to reduce segregation in casting
(c) Normalizing is done to improve mechanical properties
(d) Normalizing is done to induce stresses

OPSC AEE 2019 Paper-I

Ans : (d) : The normalizing is done for the following purpose-
(i) To refine the grain structure of the steel to improve machinability tensile strength and structure of weld.
(ii) To remove strains caused by cold working processes.
(iii) To remove dislocations caused in the internal structure of the steel due to hot working.
(iv) To improve certain mechanical and electrical properties.

The process of normalizing consists of heating the steel 30ºC–50ºC above its upper critical temperature for hypo-eutectoid steels or ACM line for hyper-eutectoid steels. It is held at this temperature for about 15 minutes and then allowed to cool down in still air. The process of normalizing is frequently applied to casting and forging etc.

598. By what process, maximum hardness is obtained for a steel part?
(a) Carburizing (b) Nitriding
(c) Cyaniding (d) Annealing

APPSC Poly Lect. Automobile Engg., 2020
PSPCL AE. 2012
Gujarat PSC AE 2019

Ans : (b) : Nitriding is a heat treatment process that diffuses nitrogen into the surface of a metal to create a case hardened surface. These processes are most commonly used on low carbon alloy steels.

599. Age hardening is generally applicable to
(a) cast iron
(b) medium carbon steel
(c) high alloy steel
(d) alloys of aluminum, magnesium, nickel, etc.

BPSC Poly. Lect. 2016

Ans : (d) It was first observed by A Wilmin 1906 that where Commercial Aluminum alloy with Cu was quenched from a relatively high temperature, it showed increase in hardness on standing subsequently at room temperature. This process is called precipitation hardening or age hardening. A similar phenomenon occurs in various ferrous and non ferrous alloys.

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600. Annealing is used to make the steel components
(a) hard (b) soft
(c) brittle (d) none of these
(HPPSC LECT. 2016)

Ans : (b) Annealing:-
Quenching medium : Furnace
Full annealing → Ductility increase
Process annealing → stress relieving
Spheroidise annealing →Machinability increases
Diffusion annealing → To homogenize the chemistry of material

601. If there are bad effects on strain hardening on a cold formed parts, the part must be:-
(a) Annealed (b) Tampered
(c) Hardened (d) Normalised
UKPSC AE-2013, Paper-I

Ans. (a) : If there are bad effects on strain hardening on a cold formed parts, the part must be annealed.

602. Heat treatment is done to
(a) change grain size and soften the metal
(b) improve electrical and magnetic properties
(c) relieve internal stresses
(d) all of the above
UKPSC AE 2007 Paper-I

Ans. (d) : All of the above

603. In nitriding _____ is used.
(a) ammonia (b) slat bath
(c) methane (d) water bath
NLCIL GET 17.11.2020, Shift-II

Ans. (a) : In gas nitriding uses ammonia or ammonia hydrogen mixture to enhance the nitrogen activity. Ammonia easily dissociates into gaseous nitrogen and hydrogen according to the chemical equilibrium.

604. Precipitation hardening is applicable for
(a) Pure aluminium
(b) Low carbon steel
(c) Non-metal
(d) Aluminium - Copper alloy
ISRO Scientist/Engineer 2011

Ans. (d) : Precipitation hardening is also called age hardening or particle hardening. In a heat treatment technique used to increase the yield strength of aluminium, magnesium, nickel, titanium and some steels and stainless steel.

605. The process of introducing carbon and nitrogen into a solid ferrous alloy is known as:
(a) Carbonitriding (b) Nitriding
(c) Carburizing (d) Cyaniding
HPPSC W.S. Poly. 2016

Ans : (a) Carbonitriding–The process of Introducing carbon and nitrogen into a solid ferrous alloy is known as carbonitriding.
Cyaniding–It is the process of introducing carbon & nitrogen into the surfaces of steel by heating the surface to a suitable temperature and holding it in contact with molten cyanide.

606. The advantage of nitriding process is
(a) It does not produce brittle case
(b) It gives good fatigue resistance
(c) All types of steel can be satisfactorily treated
(d) It take less cycle time
GPSC ARTO 01.05.2016

Ans. (b) : Advantage of Nitriding process–
- High fatigue life
- Nitriding produce great surface hardness of materials.
- Nitriding process produce material have greater resistance to corrosion and wear.
- Nitriding process required less time.
- Nitriding process is clean process because this process not produce carbon.

607. Maximum hardenability of any steel depends on
(a) The grain size
(b) The chemical composition
(c) The carbon content
(d) The iron content
GPSC Lect. (Poly.) 07.12.2014

Ans. (c) : Maximum hardenability of any steel depends on the carbon content.

608. The approximate height of a blast furnace is
(a) 10 m (b) 20 m
(c) 30 m (d) 40 m
Nagaland PSC (CTSE) 2018, Paper-I

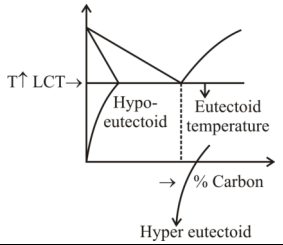
Ans. (c) : A blast furnace is a special type of furnace for smelting iron from ore. Blast furnace are very large. They can be up to 30 meter (100ft) tall & to 15m (30 to 49ft) in diameter.

609. Specify the sequence correctly
(a) grain growth, recrystallisation, stress relief
(b) stress relief, grain growth, recrystallisation
(c) stress relief, recrystallisation, grain growth
(d) grain growth, stress relief, recrystallisation
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : As we know, the sequence after cold working of metals are subjected to annealing to relieve internal stresses are induced during cold working & that annealing carried put by subsequent re-crystallization & grain growth.

610. In process annealing, the hypo-eutectoid steel is
(a) heated from 30°C to 50°C above the upper critical temperature and then cooled in still air
(b) heated from 30°C to 50°C above the upper critical temperature and then cooled suddenly in a suitable cooling medium
(c) heated from 30°C to 50°C above the upper critical temperature and then cooled slowly in the surface
(d) heated below or close to the lower critical temperature and then cooled slowly
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : As we know, in process annealing, the hypo-eutectoid steel is heated below or close to the lower critical temperature and then cooled slowly.



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611. The process extensively used for making bolts and nuts is

(a) hot piercing (b) extrusion
(c) cold peening (d) cold heading

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : The process extensively used for making bolts & nuts is cold heading. It is a method of forming metal in progressive steps to net shaped at room temperature.

612. The type of case hardening using 50% charcoal, $BaCO_3$, $CaCO_3$ at temperature close to 950 degrees is _____ carburising.

(a) liquid and gas (b) gas
(c) liquid (d) pack

AAI Jr. Executive 26.03.2021

Ans. (d) : Pack carburising is a type of case hardening using 50% charcoal, $BaCO_3$, $CaCO_3$ at temperature close to 950°.

613. Pick up the wrong statement. Annealing results in:

(a) refining grain structure
(b) relieving internal stresses
(c) improving wear resistance
(d) improving machinatbility

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : Annealing is a heat treatment process used mostly to increase the ductility and reduce the hardness of a material.
This change in hardness and ductility is a result of the reduction of dislocations in the crystal structure of the material being annealed.
Increase ductility so increase machinability refining grain structure and reliving thermal stresses.

614. Which of the following are heat treatment process?

(a) Normalizing
(b) Annealing
(c) hardening and quenching
(d) All of the above

GPSC AE 26.07.2015

Ans. (d) : • Normalizing
• Annealing
• hardening and quenching

615. The gas obtained from smelting operation in which air is forced through layers of coke and iron ore is known as _____.

(a) Coke Oven illuminating gas
(b) Blast furnace gas
(c) Hydro-Carbonyl gas
(d) Sewer gas

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : The gas obtained from smelting operation in which air is forced through layers of coke and iron ore is known as blast furnace gas.

616. Copper sheets are manufactured by

(a) drawing (b) hammering
(c) extruding (d) rolling

HPPSC Poly. Lect. 05.07.2021

Ans. (d)
- Copper sheets are manufactured by rolling.
- Roll forming is most common method of sheet metal production.

617. Which of the following refractory material is recommended for steel furnaces containing CaO slag?

(a) Alumina (b) Silika
(c) Magnesia (d) Fireclay

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : Magnesia

618. The lower critical point for all steels is

(a) 600ºC (b) 700ºC
(c) 723ºC (d) 913ºC

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (c) : The lower critical temperature of all steel is the same i.e. 723°C it is the temperature of pearlite to austenite transformation.

619. Consider the following quenching media :

1. Oil 2. Water
3. Water + NaOH 4. Brine

The correct sequence of these media in order of increasing hardness of steel undergoing heat treatment is

(a) 1 < 3 < 2 < 4 (b) 2 < 1 < 3 < 4
(c) 1 < 2 < 3 < 4 (d) 4 < 3 < 2 < 1

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) :

620. As per the Indian Standard Designation of general steels, which of the following is mild steel?

(a) C20 (b) C35
(c) C45 (d) C75
(e) C100

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : As per the Indian standard designation of general steels C20 is mild steel.
C-20 is an unalloyed low carbon mild steel grade supplied as a hot rolled or bright drawn finish bar. As a low carbon steel grade it provides low strength with good machinability and is suitable for welding.

621. In tempering process the steel is heated to at

(a) Below the lower critical point
(b) Higher than the lower critical point
(c) In between the lower and upper critical point
(d) Higher than the upper critical point

PTCUL AE 25.06.2017

Ans. (a) : Tempering in metallurgy, process of improving the characteristics of metal expecially steel by heating it to a high temperature, through below the melting point, then cooling it, usually in air the process has the effect of toughing by lessening brittleness and reducing internal stresses.

622. The temperature required for full annealing in hypo-eutectoid steel is:

(a) 30ºC to 50ºC above the upper critical temperature to the austinitic condition
(b) 30ºC to 50ºC below upper critical temperature
(c) 30ºC to 50ºC above lower critical temperature

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(d) 30ºC to 50ºC below lower critical temperature
(e) equal to lower critical temperature

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Full annealing is the process of slowly raising the temperature about 50ºC above the Austenitic temperature line A_3 or line ACM in the case of Hypereutectoid steel (steel with < 0.77% carbon) and 50ºC into the Austenite-Cementite region in case of Hypereutectoid steel (steel with > 0.77% corbon).

623. The TTT diagram shows the times required for isothermal transition from
(a) austenite to pearlite
(b) austenite to ferrite
(c) ferrite to pearlite
(d) martensite to pearlite

ESE 18.07.2021

Ans. (a) : Time Temperature Transformation (TTT) diagram is used to indicate time and temperature transformation of austenitic stainless steel to pearlite at eutectoid temperature. This transformation of austenite is time dependent.

624. The process used for relieving the internal stresses previously set up in the metal and for increasing the malleability of steel is
(a) Normalising (b) Full annealing
(c) Process annealing (d) Spheroidising

TANGEDCO AE 2015

Ans. (c) : Process annealing is used to relieve stress and increase malleability of low carbon steel.
Spheroidizing-Increases machinability for medium and high carbon steel.
Full annealing–Ductility and toughness increases.

625. A steel specimen is heated to 780 °C and is then cooled at the slowest possible rate in the furnace. The property imparted to the specimen by this process is
(a) toughness (b) hardness
(c) softness (d) tempering

ESE 2018

Ans. (c) : Toughness depends on grain size and impurity concentration. As grain size decreases its strength increases. This is one of the few instances in which strength and toughness can be increased at the same time. But when material is cooled very slowly its grain size increases thus its toughness decreases and ductility increases hence softness also increases.

626. The Time-Temperature-Transformation (TTT) curves are also called as:
(a) T-curves (b) M-curves
(c) N-curves (d) S-curves
(e) Q-curves

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : TTT-Curve– Time - Temperature - Transformation curve also known as isothermal transformation diagram. This curve also termed as S-curve or Ben curve. This curve measure the rate of transformation at a constant temperature i.e it shows time relationships for the phases during isothermal transformation. Information regarding the time to start the transformation and the time required to complete the transformation can be obtained from set of TTT diagrams.

627. Retained austenite will be produced during quenching of materials
(a) Hypoeutectic steels
(b) Eutectoid steels
(c) Hypereutectoid steels
(d) Cast irons

TSPSC AEE 2015

Ans. (c) : Retained austenite will be produced during quenching of materials hypereutectoid steels.

628. Which one of the following was not used for understanding the mechanics of the heat treatment?
(a) TTT diagrams (b) CCT diagrams
(c) Hardenability curves (d) Phase diagrams

TNPSC AE 2017

Ans. (d) : Phase diagrams are graphical representations of the phase present in particular alloy being held at a particular temperature. Phase diagrams can be used to predict the phase changes that have occurred in an alloy that has been exposed to a particular heat treatment process.

629. The temperature at which new stress-free grains are formed in the metal is called,
(a) upper critical temperature
(b) melting point
(c) recrystallization temperature
(d) eutectic temperature

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : The temperature at which new stress-free grains are formed in the metal is called, recrystallization temperature.

630. Which is false statement about annealing?
(a) Annealing is done to relieve stresses
(b) Annealing is done to harden steel slightly
(c) Annealing is done to improve machine characteristics
(d) Annealing is done to soften material

OPSC AEE 2019 Paper-I

Ans : (b) : Annealing reduces the hardness of the steel completely and improve toughness, it is complete softening process and increase ductility.

631. The abbreviation T.T.T. Diagrams stand for :
(a) Tensile Temperature Time Diagrams
(b) Time Temperature Transformations diagrams
(c) Temperature Time Testing Diagrams
(d) Time Transformation Testing Diagrams

OPSC AEE 2019 Paper-I

Ans : (b) : Time, Temperature, Transformations (TTT) diagrams indicates transformation of austenite phase.

632. Cold working of steel is defined as working
(a) below its recrystallisation temperature
(b) above is recrystallisation temperature
(c) at its recrystallisation temperature
(d) at zero Kelvin

GPSC Executive Engineer 23.12.2018

Ans. (a) : Cold working of steel is defined as working below its recrystallisation temperature.

633. A circular is obtained when steel is
(a) Quenched in oil
(b) Cooled in still air
(c) Slowly cooled in furnace
(d) Quenched in water

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) : A circular is obtained when steel is quenched in water.

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634. Cementite consist of

(a) 13% carbon and 87% ferrite
(b) 13% cementite and 87% ferrite
(c) 13% ferrite and 87% cementite
(d) 6.67% carbon and 93.33% iron

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : As we know, cementite (or iron carbide) is a compound of iron & carbon, more precisely an intermediate transition metal carbide with formula Fe_3C. It consists of 6.67% carbon & 93.33% of iron.

635. Hardness of steel greatly improves with

(a) annealing (b) cyaniding
(c) normalizing (d) tempering

GPSC Executive Engineer 23.12.2018, GATE 2003

Ans. (b) : hardness of steel greatly improves with cyaniding.

5. Engineering Materials

636. 18/8 stainless steel consists of :

(a) 18% Nickel and 8% Chromium
(b) 18% Chromium and 8% Nickel
(c) 18% Nickel and 18% Chromium
(d) 8% Nickel and 8% Chromium

ISRO Scientist /Engineer (RAC) 10.03.2019
GPSC ASST. Prof. 28.08.2016
OPSC Civil Services Pre. 2011
UKPSC AE-2013, Paper-I
Mizoram PSC (Insp. of Factories) 2019 Paper-III
Jharkhand Urja Vikas Nigam Ltd. AE 2017
Nagaland PSC (CTSE) 2018 Paper-I
Kerala PSC Poly. Lect. 19.01.2014

Ans. (b) : Stainless steel contains 18% chromium and 8% Nickel. It is austenitic stainless steel.

637. 18-4-1 HSS is having 18%

(a) Vanadium (b) Chromium
(c) Tungsten (d) Carbon

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Punjab PSC SDE 12.02.2017
APPSC Poly. Lect. 2013
ISRO Scientist/Engineer 2012
APPSC IOF 2009
WBPSC AE 2008
WBPSC AE 2003
HPPSC Workshop Suptd. 08.07.2021

Ans. (c) : 18-4-1 HSS is having.
18% → Tungsten
4% → Chromium
1% → Vanadium.

Uses of HSS– Generally, high speed steels are used for machining at high speeds. With their excellent wear resistance, high temperature properties and excellent hardness, HSS used in drills, tapes, milling cutters tool bits, gear cutters, saw blades and rolls etc.

638. Corrosion resistance of stainless steel is due to:

(a) Chromium (b) Carbon
(c) Vanadium (d) Copper

RPSC Lect. (Tech. Edu. Dept.)13.03.2021 Paper-I
TSPSC AE 2015, 2018, ESE 2017
RPSC IOF 2020, OPSC AEE 2015 Paper-I
UJVNL AE 2016
HPPSC W.S. Poly. 2016

Ans : (a) Stainless steel: Stainless steel are also called corrosion resistance steel. Their major alloying element is high amount of chromium. Chromium reacts with oxygen to from a stronger layer of chromium oxide on the surface of the metals which acts as a barrier to retard further oxidation, rust or corrosion and staining. On the basis of their structure, the stainless steel are classified as follows:
(i) Austenitic stainless steel
(ii) Ferritic stainless steel
(iii) Martensitic stainless steel

639. Monel metal is a

(a) nickel-zinc alloy (b) copper-zinc alloy
(c) nickel-copper alloy (d) copper-tin alloy

UPRVUNL AE 05.07.2021
BPSC AE 2012 Paper - VI
WBPSC AE, 2017
ESE 2004
Nagaland PSC (CTSE) 2018, Paper-I
Gujarat PSC AE 2019

Ans : (c) : Monel metal- It is an important alloy of nickel and copper. It contains 68% nickel, 29% copper and 3% other constituents. It resembles nickel in appearance and is strong, ductile tough. It is superior to brass and bronze in corrosion resisting properties.

640. Brass is an alloy of:

(a) Copper and zinc (b) Copper and tin
(c) Copper, tin and zinc (d) None of the above

UP Jal Nigam AE 2016
RPSC AE 2018
SJVN ET 2019
APPSC IOF, 2009
CGPSC ARTO & TSI 07.01.2017
APPSC AEE 2012
UPRVUNL AE 05.07.2021
Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : Brass is an alloy of copper and zinc.

641. Corrosion resistance of steel is increased by addition of :

(a) Sulphur, phosphorus, lead
(b) Chromium and nickel
(c) Vanadium, aluminum
(d) Tungsten, molybdenum, vanadium, chromium

APPSC AE Poly Lect. 2013
ISRO Scientist/Engineer 2009
JPSC AE 10.04.2021 Paper-II
GWSSB AAE 27.12.2015
OPSC AEE 2019 Paper-I

Ans : (b) : Nickel improve tensile strength, raises elastic limit, imparts hardness, toughness and increases corrosion resistance.
Chromium increases strength, hardness and corrosion resistance.

642. Gun metal used in journal bearings, contains :

(a) 88% Cu, 10% Sn, 2% Zn
(b) 80% Cu, 10% Sn, 10% Zn
(c) 85% Cu, 5% Mg, 10% Al
(d) 85% Cu, 5% Sn, 10% Pb

RPSC ACF & FRO, 26.02.2021
ISRO Scientist/Engineer (RAC) 07.05.2017

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TSPSC AEE 2015
GPSC ARTO 01.05.2016
ESE - 2004

Ans. (a) : Gun metal, also known as red brass is an alloy of copper (Cu), tin (Sn) and zinc (Zn) 88% Cu, 10% Sn & 2% Zn.

Application- Guns, hydraulic castings, valves gears, statues etc.

643. Which of the element of element pairs is added to steel in small amounts to increase machinability?
(a) Nickel
(b) Sulphur and phosphorus
(c) Silicon
(d) Manganese and copper

TRB Asstt. Prof., 2012
Punjab PSC (Lect.) 06.08.2017
GPSC AE 26.07.2015

Ans. (b) : Sulphur is added up to 0.33% in free cutting steels to increase machinability.
Similarly, phosphorus is added up to 0.12% for free cutting steel.

644. White cast iron contains carbon in the form of
(a) Free carbon (b) Graphite
(c) White carbon (d) Cementite

Rajasthan Nagar Nigam AE 2016, Shift-II
APPSC AEE 2012

Ans. (d) : In white cast iron, the white colour is due to the fact that the carbon is in the form of cementite (known as carbide) which is hardest constituent of iron. This cementite is caused by quick cooling of molten iron.
Composition
Carbon = 2% to 2.3%
Silicon = 0.85% to 1.2%
Manganese = 0.1 % to 0.47%
Phosphorus = 0.05% to 0.27%
Sulphur = 0.12% to 0.35%
and the remaining iron.

645. ……. are strong and ductile materials.
(a) Polymer fibers (b) Porcelain
(c) Metals (d) Charcoal

APGENCO AE 2012
HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Metal are strong and ductile material metals are good conductor of electricity and heat.

646. Toughness of steel is increased by adding:-
(a) Nickel (b) Sulphur
(c) Chromium (d) Tungsten

ESE 2020
UKPSC AE-2013, Paper-I
Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Toughness of steel is increased by adding nickel.

647. Gun metal is a type of _____.
(a) Aluminum bronze (b) Tin bronze
(c) Beryllium bronze (d) Silicon bronze
(e) Copper and nickel alloy

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Gun Metal– Gun metal, also known as red brass, is a type of bronze – an alloy of copper, tin and zinc. Proportions vary but 88% copper, 8-10% tin, and 2-4% zinc is an approximation originally used chiefly for making guns.

Use–
- Intricate, pressure-tight castings like pipe fitting, valves and pumps are made from gun metal.
- Bearing, where speeds and loads are moderate and manufactured from gun metal and used in machinery.

648. Titanium in the steel improves the
(a) hardness
(b) strength
(c) resistance to corrosion of steel
(d) strength and resistance to corrosion of steel

Gujarat PSC AE 2019

Ans : (d) : Titanium in the steel improves the strength and resistance to corrosion of steel.

649. Points of arrest for iron correspond to :
(a) Stages at which allotropic forms changes
(b) Stages at which further heating does not increase temperature for some time
(c) Stages at which properties do not change with increase in temperature
(d) There is nothing like points of arrest

OPSC AEE 2019 Paper-I

Ans : (a) : Points of arrest for iron correspond to stages at which allotropic forms change.

650. Steel containing 0.8 to 1.5 percent carbon, is known as
(a) mild steel
(b) dead-mild steel
(c) medium-carbon steel
(d) high-carbon steel

JPSC AE 2013, Paper-V

Ans : (d) : Steel containing 0.8 to 1.5 percent carbon is known as high carbon steel.

651. Addition of vanadium to steel results in the improvement of
(a) heat treatability by quenching
(b) hardenability
(c) fatigue strength
(d) resistance to oxidation at elevated temperature

BPSC AE 2012 Paper - VI
ESE 2000

Ans : (b) : Addition of vanadium to steel results in improvement of hardenability.

652. The limitations of plain carbon steels include
(a) poor hardenability
(b) major loss of hardness on tempering
(c) low corrosion resistance
(d) All of the above

BPSC AE Mains 2017 Paper - VI

Ans : (d) : The limitations of plain carbon steels includes-
- poor hardenability
- major loss of hardness on tempering.
- Low corrosion resistance.

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653. Connecting rod is usually made of :

(a) Aluminium (b) Low carbon steel
(c) Medium carbon steel (d) High carbon steel

GMB AAE 25.06.2017
OPSC AEE 2019 Paper-I

Ans : (c) : Connecting rod is usually made of medium carbon steel.

654. Railway rails are normally made of

(a) mild steel (b) nickel alloy steel
(c) high carbon steel (d) tungsten steel

TNPSC AE 2019
OPSC AEE 2019 Paper-I
GPSC GWSSB DEE 07.07.2016

Ans. (c) : Railway rails are normally made of high carbon steel.

655. An alloy produced by adding 1% of tin to Muntz metal is called as

(a) a brass (b) Admiralty brass
(c) Naval brass (d) Naval brass

ESE 2019
APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : Alpha (α)–beta (β) brass (Muntz metal), also called duplex brass, is 35 – 45% zinc and is suited for hot working.

- An alloy produced by adding 1% of tin to Muntz metal is called as Naval brass.
  It contains 60% Cu, 39% Zn and 1% Sn
- α-brass containing 75% Cu and 25% Zn. Prince's metal or prince Rupert's metal is a type of α-brass.
- Leaded brass is an alpha–beta brass with an addition of lead. It has excellent machinability.
- Brass contain 29% Zn, 70% Cu and 1% Sn which inhibits dezincification in the environment.

656. Addition of magnesium to cast iron increases its

(a) hardness
(b) corrosion resistance
(c) creep strength
(d) ductility and strength in tension.

RPSC ACF & FRO 26.02.2021
ESE 2018
Nagaland PSC (CTSE) 2018, Paper-I
Kerala PSC poly lect. 2017
UKPSC AE 2013 Paper I

Ans. (d) : By addition of magnesium in cast iron, the shape of graphite flakes can be modified into nodules (spheres), which increases its ductility, strength and fluidity.

657. Mild steel is a

(a) low carbon steel (b) medium carbon steel
(c) high carbon steel (d) high speed steel

TSPSC AEE 2015
APPSC AEE 2012
GWSSB AEE 27.12.2015

Ans : (a)
(i) Dead mild steel:- up to 0.15% carbon
(ii) Low carbon steel or mild steel:- 0.15% to 0.45% carbon
(iii) Medium carbon steel:- 0.45% to 0.8% carbon
(iv) High carbon steel:- 0.8% to 1.5% carbon.

658. Babbit metal is an alloy of

(a) Sn and Cu (b) Sn, Cu, Sb and Pb
(c) Sn, Cu and Pb (d) Sn, Cu and Sb

Vadodara Muncipal Corp. DEE, 2018
UKPSC AE 2012, 2007 Paper-I
UPPSC AE 12.04.2016 Paper-I
PPSC Asstt. Municipal Engg. 15.06.2021
ESE - 2011

Ans : (b) Babbit metal is an alloy of Sn, Cu, Sb and Pb.
Babbit metal:- The term white metal covers both tin-base and lead-base bearing alloys. These are usually referred to as babbits. Being alloys of several metals, they have a non- homogeneous microstructure. This imparts to it antifrictional properties.

659. Which one of the following is closest to the purest form of iron?

(a) Cast iron (b) Wrought iron
(c) Grey cast iron (d) Mild steel

RPSC Lect (tech edu. dept.) 13.03.2021 Paper I
UKPSC AE-2013, Paper-I

Ans. (b) : Wrought iron is closest to the purest form of iron.

660. Which one of the following is the ferrous material ?

(a) Zinc (b) Iron
(c) Silicon Carbide (d) Copper

UKPSC AE 2012 Paper-I
VIZAG Steel MT 2011

Ans. (b) : Ferrous material is iron.

661. Babbit materials are used for

(a) Gears (b) Bearings
(c) Bolts (d) Clutch liners

JPSC AE 10.04.2021 Paper II
UKPSC AE 2012 Paper-I

Ans. (b) : Babbits also called Babbitt metal or bearing metal is an alloy used for bearing surface.
Babbitt metal is used as the lining for bearing shall of most iron, steel and bronze. The Babbitt lining prevents friction and wear that is common when a lubricant is not able to prevent bearing is moving parts from welding together. Therefore it extends life of bearing.
Composition- 88% Sn, 8% Sb & 2% Cu.

662. Bronze contains

(a) 70% Cu and 30% Zn
(b) 90% Cu and 10% Zn
(c) 75% Cu and 25% Zn
(d) None of the above

APPSC Poly. Lect. Automobile Engg. 2020
UKPSC AE 2012 Paper-I
JPSC AE 2013, Paper-V
GPSC Lect. (Auto) 16.10.2016
UPRVUNL AE 05.07.2021

Ans. (d) : Bronze is an alloy of copper and tin.

663. The percentage of carbon in grey cast iron is the range of-

(a) 0.25 to 0.75% (b) 1.25 to 1.75%
(c) 3 to 4% (d) 8 to 10%

RPSC AE 2018
Nagaland PSC CTSE 2017, Paper-I
HPPSC Asstt. Prof. 18.11.2016
VIZAG STEEL MT. 10.06.2012

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VIZAG STEEL MT. 18.08.2013
Nagaland PSC (CTSE) Paper-I 2018
GATE 2004

Ans. (c) : The percentage of carbon in grey cast iron is the range of 3 to 4%.

664. A material whose molecular properties allow considerable reversible extension is referred to as

(a) A polymer (b) A Plastic
(c) A elastomer (d) An isomer

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : An Elastomers are amorphous polymers maintained above their glass transition temperature, so that considerable molecular reconfirmation, without breaking covalent bonds, is feasible. They are polymer with viscoelasticity and with weak intermolecular forces, generally low Young's Modulus and high failure strain compared with other materials.

665. Consider the following statements relation to composite materials.

A. A composite material is a structural material created synthetically or artificially by combining two or more materials having dissimilar characteristics.

B. The constituents are combined at macroscopic level and are not soluble in each other.

C. One constituent is called as matrix phase and the other is called as reinforcing phase.

D. Reinforcing phase is embedded in the matrix phase

Which of the above statements is/are correct?

Select the code for the correct answer from the options given below

(a) A only (b) A, B and C only
(c) A and C only (d) None of the above

Karnataka PSC AE, 10.09.2017

Ans. (d) : • All options are correct.

666. Thermoplastics are :

(a) Non-linear polymers in which rise in temperature increases plasticity
(b) Linear polymers in which rise in temperature increases plasticity
(c) Linear polymers in which temperature has no effect on plasticity.
(d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (b) : Thermoplastics are linear polymers in which plasticity increases as temperature rises. They can be reshaped by heating and various parts are produced using various polymer processing technique.

667. High speed steel tool material contains carbon

(a) 0.6 – 1.0 % (b) 2 – 4 %
(c) 4 – 6 % (d) 6 – 10 %

TNPSC AE, 2008

Ans. (a) : High Speed Steel (HSS) tool material contains carbon (0.6 – 1.0 %).

668. Cupronickle is an alloy of :

(a) 30 percent Nickel and 70 percent Copper, an is extensively used in condenser tubes.
(b) 10 percent Chromium, 20 percent Nickel and the rest Copper, and is recommended for boiler tubes.
(c) 30 percent Copper and the rest Nickel, and is recommended for gas turbine blades.
(d) 70 percent Nickel and 30 percent Copper, and is extensively used in condenser tubes.

RPSC ACF & FRO, 26.02.2021

Ans. (a) : Cupronickle is an alloy of copper that contains nickel and strengthening elements, such as iron and manganese. The copper content typically varies from 60 to 90 percent. An alloy containing 30 percent nickel and 70 percent copper is widely used for condenser tubes in steam power plants.

669. The electric process of steel making is especially adopted to :

(a) Alloy and carbon tool steel
(b) Magnet steel
(c) High speed tool steel
(d) All of these

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Most of the steel is produced by the electric arc method, which uses high current electric arc to melt steel scrap and convert it into liquid steel of specified chemical composition and temperature.
This process is adopted to alloy and carbon tool steel, magnet steel and High speed tool steel.

670. Stainless steel contains following elements:

(a) Cr, Fe, Sn, Ni (b) Cr, Sn, Cu, Ni
(c) Cr, Ni, Fe, C (d) Fe, Pb, Sn, C

TSPSC Managers, 2015
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Stainless steel contains carbon, manganese, phosphorus, sulphur, silicon, chromium, nickel, nitrogen, iron etc.
The term stainless steel is used to describe a family of about 200 alloys of steel with remarkable heat and corrosion resistance properties.

671. The binding material for cemented carbide is

(a) Iron (b) Chromium
(c) Nickel (d) Cobalt

TSPSC Managers, 2015

Ans. (d) : Cobalt is used as a binding agent in cemented carbide tool, which increases the grain size transverse rupture strength increases and hot hardness decreases.

672. What are the constituent of phosphor bronze?

(a) Nickel, Copper and Cobalt
(b) Copper, Zinc and Manganese
(c) Copper, Tin and Phosphorous
(d) Aluminium, Copper and Nickel

Kerala PSC Poly. Lect., 2017

Ans. (c) : Copper, Tin and Phosphorous
Phosphor bronze is a member of family of copper alloys. It is composed of copper that is alloyed with 0.5-11% of tin and 0.01-0.35% phosphorus.

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673. Which of the following alloys of aluminium is often used for aeroplane automobile parts?

(a) Duralumin (b) Magnalium
(c) RR - 53 (d) Aluminium bronze

TRB Poly. Lect., 2012

Ans. (a) : Duralumin, strong, hard and lightweight alloy of aluminium, widely used in aircraft construction.

674. Carbon content of maraging steel is

(a) nickel, cobalt and 0.05% carbon.
(b) 10% carbon
(c) 4.3% carbon
(d) 1% carbon

APPSC Poly. Lect. 2013

Ans. (a) : Maraging steel contains an extremely low amount of Carbon (0.03% maximum) and large amount of Nickel (17-19%) together with lower amount of Cobalt (8-12%) molybdenum (3-5%), Titanium (0.2-1.8%) and aluminium (0.1-0.15%)

675. Strength of the epoxides depends on

(a) C - O - C ring (b) mixture
(c) molecules (d) None of the above

APPSC Poly. Lect. 2013

Ans. (a) : Strength of the epoxides depends on C-O-C ring opening reactions. Formation of an epoxide occurs in the presence of a peroxide reactant such as MCBA or DEI.

676. Ball bearings are generally made of

(a) cast iron (b) malleable cast iron
(c) carbon steel (d) carbon chrome steel

VIZAG STEEL MT 2015
TNPSC AE 2017

Ans. (d) : Ball bearings are generally made of carbon chrome steel

677. Which of the following material can bear sudden and excessive shocks better?

(a) Cast iron (b) Pig iron
(c) White iron (d) Wrought iron

RPSC ADE 2016

Ans. (d) : Wrought iron can bear sudden and excessive shocks better.
Pig iron is the metallic ore that contains 90% of iron. Pig iron is used for making steel and pure iron units.
White cast iron is hard and brittle their casting cannot be machined.
Cast iron, an alloy of iron that contains 2 to 4 percent carbon, along with varying amounts of silicon and manganese and traces of impurities such as sulphur and phosphorus.

678. What is the theoretical limit of carbon content in iron-carbon alloys upto which they can be called as steel?

(a) 6% (b) 3%
(c) 1% (d) 2%

RPSC AEN Pre-2013

Ans. (d) : Carbon content in iron-carbon alloys upto 2%.

679. Which of the following is a thermosetting plastic?

(a) Polyethylene (b) Bakelite
(c) Polyvinyl chloride (d) Nylon

RPSC AEN Pre-2013

Ans. (b) : A thermosetting plastic is a polymer that irreversibly becomes rigid when heated. Such a material is also known as a thermoset or thermosetting polymer. Initially, the polymer is a liquid or soft solid.
Many common plastics are thermosets. They include
- Vulcanized rubber.
- Polyester resin
- Polyurethane
- Melamine
- Bakelite
- Silicon resin
- epoxy resin

680. What is the corrosion that occurs in certain materials where the loader surfaces are in contact and have relative motion called?

(a) Fretting corrosion (b) Stress corrosion
(c) Hertzian corrosion (d) Galvanic corrosion

RPSC AEN Pre-2013

Ans. (a) : The corrosion that occurs in certain materials where the loading surfaces are in contact and have relative motion called fretting corrosion.

681. Which of the following is not a constituent of babbits?

(a) Tin (b) Antimony
(c) Nickel (d) Copper

RPSC AEN Pre-2013

Ans. (c) : Babbit metal is content of tin or lead base alloy. Its used as bearing material for axles and crankshafts.
Babbit material composition–
89.3% Tin
7.1% Antimony
3.6% Copper

682. The percentage of carbon in pig iron varies from

(a) 4 to 4.5% (b) 2.5 to 2.9%
(c) 1.1 to 2.5 % (d) 0.1 to 1.0 %

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (a) : Pig iron has very high carbon content - %C = 3.5% to 4.5% With silica and other constituents, which makes it very brittle and not useful directly as a material except for limited applications. It is very impure form of iron.

683. The metal suitable for bearings subjected to heavy loads is

(a) Silicon bronze (b) White metal
(c) Monel metal (d) Phosphor bronze

RPSC IOF, 2020

Ans. (b) : White metals are any of several light colored alloy used as a base for plated silverware, ornaments or novelties, as well as any of several lead-based or in based alloy. used for things like bearings jewellary, miniature figures, fusible plugs and many other applications.

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684. Which alloying element does not improve the machinability of steel?

(a) Sulphur (b) Silicon
(c) Phosphorous (d) Lead

RPSC IOF, 2020

Ans. (b) : Silicon is the alloying element does not improve the machinability of steel.

685. Which of the following is not strengthening mechanism for multiphase structural steel?

(a) precipitation strengthening
(b) dispersion strengthening
(c) solid-solution strengthening
(d) none of these

RPSC IOF, 2020

Ans. (c) : Grain size reduction, solid solution strengthening and strain hardening are all strengthening mechanism which increase the hardness of the material. Grain size increment isn't a strengthening mechanism.

686. Addition of Tungsten in steel

(a) improve strength at elevated temperature
(b) improve corrosion resistance
(c) decrease hardness
(d) improve coarse grain

RPSC IOF, 2020

Ans. (a) : Tungsten is a good carbide former that prevents grain growth, enhances toughness and increases red hardness and high temperature strength. Tungsten is used in hot forming tool steels and high speed steels.

687. Corrosion of metals involves

(a) Physical reactions (b) Chemical reactions
(c) Both (d) None

RPSC IOF, 2020

Ans. (b) : Corrosion of the metals happen due to the oxidation on the external surface of a metal and oxidation is itself a chemical reaction and that's why the corrosion of metal mainly involves chemical reactions.

688. Which of the following iron is the purest one?

(a) Pig iron (b) Mild Steel
(c) Wrought iron (d) Cast iron

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : As we know the carbon percent wrought iron is just 0.12 to 0.25% of carbon. So, it is a purest form of iron.

Material		Carbon content
Wrought iron	–	Does not exceed 0.15%
Mild steel	–	Upto 0.25%
Cast Iron	–	2 to 4%
Pig Iron	–	3.5 to 4.5%

689. If T_m is absolute melting temperature, then the recrystallization temperature for pure metals in terms of T_m is normally

(a) 0.7-0.8 T_m (b) 0.8-0.9 T_m
(c) 0.6-0.7 T_m (d) 0.3-0.4 T_m

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : Recrystallization mostly depends on the melting points and the impurities present in the metals. Generally for pure metals recrystallization temperature is in the range of 0.3 to 0.4 T_m, for alloys it is 0.5 to 0.6 T_m

690. Cast iron possessing which one of the following metallographic structures is best suited for damping capacity in engineering applications?

(a) Excess cementite
(b) Carbon in temper form
(c) Silicon carbide in flake structure
(d) Spheroidal form of graphite

ESE 2014
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : Cast iron possess Silicon carbide in flake structure which is best suited for damping capacity in engineering applications.

691. Ball bearing are made from:

(a) Low carbon steel
(b) Medium carbon steel
(c) High speed steel
(d) Chrome nickel steel

RPSC ADE 2016
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Ball bearing are made from chrome nickel steel.

692. Which of the following, is produced form blast furnace?

(a) Cast iron (b) Wrought iron
(c) Mild Steel (d) Pig iron

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : Blast furnace produces pig iron from iron ore/scrap by the reducing action of carbon at a high temperature in the presence of a fluxing agents such as limestone.

693. Which of the following is NOT the correct reason for using cast iron for components like machine bed, housings of gear boxes, etc.?

(a) It has higher compressive strength
(b) It has high tensile strength
(c) It can be given any complex shape by casting
(d) It has excellent ability to damp vibrations and has good wear resistance.

RPSC AE (GWD) 06.05.2016

Ans. (b) : For using cast iron for components like machine bed, housing of gear boxes etc.

- It has higher compressive strength
- It can be given any complex shape by casting
- It has excellent ability to damp vibration and has good wear resistance.

694. Which one of the following is WRONG statement?

(a) Nickel and chromium in steel helps in providing high strength at elevated temperature
(b) Nickel and chromium in steel helps in decreasing harden ability
(c) Nickel and chromium in steel helps in providing corrosion resistance
(d) Nickel and chromium in steel helps in raising the elastic limit

Rajasthan Nagar Nigam AE 2016, Shift-II

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Ans. (b) : The combination of toughening effect of nickel and the hardening effect of chromium produces a steel of high tensile strength with great resistance to shock.
Nickel– Refines grain structure increase strength and toughness increases resistance to corrosion but no effect on ductility.
Chromium– Increase wear resistance and heat resistance.

695. Depth of hardness of the steel is increased by addition of
(a) Calcium (b) tungsten
(c) Nickel (d) Chromium

Rajasthan Nagar Nigam AE 2016, Shift-I
Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Depth of hardness of the steel is increased by addition of chromium.
In Steels-
Vanadium – Increases endurance strength.
Molybedenum – Improves creep strength.
Silicon – temperature oxidation.
Tungsten – Imparts hot hardness to steel.
Nickel – Toughness
Chromium – Enhancing the hardness

696. Cemented carbide tools are unsuitable for cutting
(a) Aluminium (b) Steel
(c) Cast iron (d) Brass

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Cemented carbide tools are unsuitable for cutting steel.

- **Cemented Carbide Tools**
  Capable of cutting speeds 3 to 4 times high-speed steel tool bits.
- Low toughness but high hardness and excellent red-hardness.
- Consist of tungsten carbide sintered in cobalt matrix.
- Straight tungsten used to machine cast iron.

697. Solar cells are made of
(a) silica (b) steel
(c) sulphur (d) antimony

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Solar cells are made of silica silicon atoms connected to one another to form a crystal lattice. This lattice provides an organized structure that makes conversion of light into electricity more efficient.

698. High speed steel tools retain their hardness up to a temperature of
(a) 250 °C (b) 350 °C
(c) 500 °C (d) 900 °C

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : High speed steel tools retain their hardness up to a temperature of 900°C.

699. Which of the following element results in the presence of free graphite in Cast Iron:
(a) Phosphorus (b) Sulphur
(c) Manganese (d) Silicon

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Silicon is responsible for the presence of free graphite in cast Iron.

700. High carbon steels contain.
(a) 2.5 to 4.5 percent Carbon
(b) 0.5 to 0.7 percent Carbon
(c) 0.7 to 1.5 percent Carbon
(d) 0.2 to 0.5 percent Carbon

ISRO Scientist/Engineer 2010
NLC GET 17.11.2020, Shift-II

Ans. (c) : High Carbon Steel (HCS) of course, has the highest ratio of carbon to iron it consists of more than 0.60% carbon. thereby changing its physical properties also known as carbon tool steel, it has around 0.7 to 1.5% of carbon
(1) White cast iron has 2.5 to 4.5% of carbon
(2) Medium carbon steel and high speed steel has 0.35 to 0.7% of carbon.
(3) Mild steel has 0.15 to 0.31 carbon.

701. Tungsten content in High speed steel cutting tool material is about :
(a) 16% (b) 4%
(c) 0.1% (d) 1%

J & K PSC Screening, 2006

Ans. (a) : High speed steel → 18W – 4Cr – 1V
18% W (~ 16%)
4% Cr
1% V

702. Which of the following is the hardest cutting tool material next to diamond?
(a) Cemented carbides (b) Ceramics
(c) Silicon (d) Cubic boron nitride

NPCIL ET, 2015

Ans. (d) : CBN (Cubic Boron Nitride) : CBN is the hardest material known presently next to diamond. CBN is artificial abrasive and its make in laboratory at high pressure and temperature.

703. PVC sheets are produced by-
(a) Melt Spinning (b) Injection Moulding
(c) Calendaring (d) Rotomoulding

NPCIL ET, 2015

Ans. (c) : PVC sheets are produced by calendaring process. Polyvinyl chloride is formed in flat sheets in a variety of thickness and colours. As flat sheets, PVC is often expanded to create voids in the interior of the material, providing additional thickness without additional weight and minimal extra cost. Sheets are cut by saw and rotary cutting equipment.

704. Which of the following is a smart material ?
(a) Ni-Cd Alloy
(b) Ni-Ti shape memory alloys
(c) Steels
(d) C-C composites

NPCIL ET, 2015

Ans. (b) : Ni-Ti shape memory alloys is a smart material. Smart materials exhibit properties such as ferroelectricity, pyroelectricity, piezoelectricity and magnetization.

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705. Alloys are-

(a) Homogenous (b) Non-homogeneous
(c) Orthotropic (d) Anisotropic

NPCIL ET, 2015

Ans. (a) : Alloys are homogenous. Alloy, metallic substance composed of two or more elements, as either a compound or a solution.

706. Teflon is a-

(a) Thermoset fluoro carbon polymer
(b) Thermo plastic fluoro carbon polymer
(c) Inorganic compound of fluorine and carbon
(d) Laminated phenolic material

NPCIL ET, 2015

Ans. (b) :

707. What is the Zinc – Copper proportion in brasses used for cartridges and drawn tubes?

(a) 10 : 90 (b) 15 : 85
(c) 30 : 70 (d) 50 : 50
(e) 85 : 15

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : The composition of brass, generally 66% copper and 34% of zinc, makes it a favorable substitute for copper based jewellery as it exhibits greater resistance to corrosion.

708. Carbon content in steel affects following property

(a) Hardness (b) Machinability
(c) Tensile strength (d) All of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Carbon content in steel affects.
1. Hardness
2. Tensile strength
3. Machinability

709. ..... resins repeatedly softened by heat and hardened by cooling.

(a) Thermosetting (b) Thermoplastic
(c) Both (a) and (b) (d) None of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : Thermoplastic – Thermoplastics softens when heated and becomes hard when cooled. The process can be repeated many times.
Exp- PVC, Polypropylene, polyethylene, polystyrene, PTFE or Teflon, acrylic etc.
Thermosetting – Thermosetting plastic once shaped cannot be recycled. Upon heating they get burned but does not melt. It is because of cross linking between adjacent molecular chains
Exp. Epoxy, Polyester, Phenol formaldehyde (Bakelite etc.)

710. Delta Metal is an alloy of -

(a) Iron, Nickel and Copper
(b) Copper, Zinc and Iron
(c) Iron, lead and Tin
(d) Copper, Zinc and Antimony

GMB AAE 25.06.2017

Ans. (b) : A brass alloy consisting of copper (55%), zinc (41-43%) and iron (1-3%), with the balance consisting of various other metals.
The proportions used make the material harder and suitable for valves and bearing.

711. Compared to metals, ceramics exhibit better :

(a) electrical conductivity
(b) oxidation resistance
(c) ductility
(d) all the above

APPSC IOF, 2009

Ans. (b) : Compared to metals, ceramics exhibit better oxidation resistance.

712. Which of the following material is known for its high damping capacity ?

(a) Cast iron (b) Stainless steel
(c) Aluminium alloys (d) Titanium

APPSC IOF, 2009

Ans. (a) : Gray cast iron is used for machine tool beds because of its high damping capacity & compressive strength.

713. Mechanite is used in

(a) cutting tool (b) rolling mill
(c) machine tool bed (d) forging mill

WBPSC AE, 2007

Ans. (c) : machine tool bed

714. The percentage of carbon in cast iron varies from

(a) 0.1 to 0.5 (b) 0.5 to 1
(c) 1 to 1.7 (d) 1.7 to 4.5

Sikkim PSC (Under Secretary), 2017

Ans. (d) : The percentage of carbon in cast iron varies from 1.7 to 4.5%.
Usually, it contains around 4% C. it is hard and brittle.

Material	**% of Carbon**
Carbon steel	0.05% - 2.0%
Wrought iron	0.02% - 0.03%
White cast iron	1.75 % - 2.30%
Malleable cast iron	2.2% - 3.60%
Grey cast iron	2.5% - 3.75%
Pig iron	3.00% - 4.00%

715. Structural constituents of equilibrium cooling of plain carbon steel (0.4% C) at room temperature are

(a) Austenite and Pearlite
(b) Aerinite and Pearlite
(c) Cementite and Pearlite
(d) Ferrite and Martensite

Karnataka PSC Lect., 27.05.2017

Ans (a) : Structural constituents of equilibrium cooling of plain carbon steel (0.4% C) at room temperature are austenite and pearlite.

716. Match List I with List II and select the correct answer using codes given below :

List-I (Steel type)	**List-II (Product)**
(a) mild steel	**1. screw driver**
(b) tool steel	**2.commercial beams**
(c) medium carbon steel	**3. crane hooks**
(d) high carbon steel	**4. blanking dies**

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Codes :

	(a)	(b)	(c)	(d)
(a)	1	4	3	2
(b)	2	4	1	3
(c)	1	3	4	2
(d)	2	4	3	1

TRB Asstt. Prof., 2012

Ans. (d) :

717. Elastic modulus of a unidirectional glass/epoxy lamina, with a 60% fibre volume fraction (assuming iso strain condition, E_f = 80 GPa E_m = 3 GPa) is given by :

(a) 24.6 GPa (b) 49.2 GPa
(c) 33.8 GPa (d) 140.8 GPa

KPSC ADF 2015

Ans. (b) : For unidirectional

$E_c = E_f . F_f + E_m F_m$
$= 80 \times 0.60 + 0.40 \times 3$
$= 48 + 1.2$
$= 49.2$ GPa

718. What is the general name for the class of structures made of rolled up carbon lattices?

(a) Nano rods (b) Nano sheets
(c) Nano tubes (d) Fuller rods

KPSC ADF 2015

Ans. (c) : Carbon nanotubes (CNTs) are cylindrical molecules that consist of single layer carbon atoms (graphene).

719. A cast iron designated by FG 300 is

(a) Grey cast iron with carbon content of 3%
(b) Grey cast iron with ultimate compressive strength of 300 N/mm^2
(c) Grey cast iron with tensile yield strength of 300 N/mm^2
(d) Grey cast iron with ultimate tensile strength of 300 N/mm^2

GPSC EE Pre, 28.01.2017

Ans. (d) : A cast iron designated by FG 300 is grey cast iron with ultimate tensile strength of 300 N/mm^2.

720. Fillers are added to plastic to :

(a) Improve flow
(b) Reduce brittleness
(c) Facilitate possibility
(d) Reduce cost

PSPCL AE, 2012

Ans. (d) : Reduce cost – Fillers are added to a polymer formulation to reduce the costs and improve the properties fillers can be either solid, liquid or gas, they occupy space and replace the expressive resin with less expensive compounds without modifying other characteristics.

721. Addition of which of the following improves machining of copper?

(a) Sulphur (b) Vanadium
(c) Tin (d) Zinc

PSPCL AE, 2012

Ans. (b) : Vanadium

722. What is the approximate value of Young's modulus of elasticity of cast iron :

(a) 100 – 105 GPa (b) 200 – 210 GPa
(c) 240 – 250 GPa (d) None of these

PSPCL AE, 2012

Ans. (a) :

Metal	Young's modulus
Cast iron	100 to 108 GPa
Brass	102 GPa
Aluminium	70 GPa
Plastic	25 GPa

723. The depth of penetration of hardened steel ball in specimen is 0.140 mm. What is HRB of specimen :

(a) 70 (b) 60
(c) 50 (d) None of these

PSPCL AE, 2012

Ans. (b) :

$$HRB = 130 - \frac{\text{Depth of indentation in mm}}{0.002}$$

$$HRC = 100 - \frac{\text{Depth of indentation in mm}}{0.002}$$

$$HRB = 130 - \frac{0.140}{0.002} \quad \text{(Depth of indentation= 0.140 mm)}$$

$HRB = 130 - 70$
$= 60$

724. What is the percentage of nickel in Nimonic 90 alloy (a creep resistant alloy) –

(a) 25 (b) 58
(c) 76 (d) None of these

PSPCL AE, 2012

Ans. (b) : Nimonic 90
Ni – 54% min, Cr – 18-21%, Co – 15-21%, Ti – 2-3%, Al – 1-2%

725. Which of the following cast iron is resistant to attack by sea water?

(a) Low chromium cast iron
(b) Low nickel cast iron
(c) Low sulphur cast iron
(d) Low phosphorous cast iron

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (*) : High nickel cast iron offer resistance to attack by sea water.
Note- No answer is given by commission.

726. Alloy containing 65-70% of nickel is called as :

(a) Monel metal (b) White cast iron
(c) Stainless steel (d) Brass

GPSC ARTO Pre 30.12.2018
UPRVUNL AE 05.07.2021

Ans. (a) : Monel metal composition.
65 – 70% Nickel and 31.5% copper with small amounts of iron, Manganese, carbon and silicon.

Stainless steel → Steel containing atleast 10.5% chromium, less than 1.2% carbon and other alloying elements.

Brass → 60.66% Cu, 36.58% Zn, 1.02% Tin and 1.74% Iron.

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727. The carbon content in wrought iron is
(a) less than 0.01 percent
(b) between 0.01 to 1.0 percent
(c) between 1.0 and 3.0 percent
(d) between 3.0 and 5.0 percent

Punjab PSC (Lect.) 06.08.2017

Ans. (a,b) : wrought iron has iron as much as 99% wrought iron has purest form of iron. Percentage of carbon purest to 0.02%.

728. Which one of the following is a copper free alloy?
(a) German silver (b) Muntz metal
(c) White metal (d) Gun metal

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : A white metal alloy may include antimony, tin, lead cadmium, bismuth, and zinc (some of which are quite toxic). Not all of these metals are found in all white metal alloys metals are mixed to achieve a desired goal or need. As an example, a base metal for jewellery needs to be castable, polishable, have good flow characteristics have the ability to cast fine detail without an excessive amount of porosity and cast at between 230^0 and 300^0C.

729. Which one of the following is a n-type semiconductor?
(a) Phosphorus in Silicon (b) Boron in Silicon
(c) Aluminum in Silicon (d) Indium in Silicon

Punjab PSC (Lect.) 06.08.2017

Ans. (a) : N–Type Semiconductor–It is an extrinsic semiconductor which is obtained by doping the impurity pentavalent impurity atoms such as antimony, phosphorous, arsenic etc. To the pure germanium or silicon semiconductor.

730. Which of the following tools are harder and more wear resistant than tungsten carbide but are weaker in tension?
(a) low carbon steel tools
(b) high carbon steel tools
(c) HSS tools
(d) Ceramic tools

OPSC Civil Services Pre 2006

Ans. (d) : Ceramic tools are weaker in tension but harder and more resistance to wear than tungsten carbide.

731. 18-4-1 high speed steel has
(a) 1% chromium (b) 4% tungsten
(c) 18% vanadium (d) 0.7% carbon

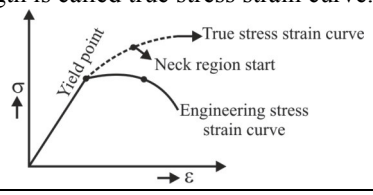
OPSC Civil Services Pre 2006

Ans. (d) : A common type of high speed steel contains 18% tungsten, 4% chromium, 1% vanadium and only 0.5 – 0.8% carbon.

732. For a Mild Steel, the Engineering stress-strain curve and True stress-strain curve are looks similar up to:
(a) Ultimate strength point
(b) Proportional limit
(c) Yield point
(d) Elastic limit

CIL MT 27.02.2020

Ans. (c)
- The curve based on the original cross-section and gauge length is called engineering stress strain curve.
- The curve based on instantaneous cross-section area and length is called true stress strain curve.



733. Manganese is added to Carbon steels:
(a) Decrease the elastic deformation
(b) Raise the yield point
(c) Make the steel ductile and good bending qualities
(d) Make the steel tougher and harder

CIL MT 27.02.2020

Ans. (d) : Manganese is a hard metal and is very brittle. It is hard to melt but easily oxidized.
Manganese make steel hard and brittle due to its property.

734. The carbon content in tool steels is:
(a) 0.3 to 0.8% (b) 1.5 to 3.6%
(c) 0.1 to 0.4% (d) 0.8 to 1.5%

CIL MT 27.02.2020

Ans. (d) : The % carbon content in –
Tool steel – 0.8 to 1.5%
Mild steel – up to 0.25%
High carbon steel – 0.61 to 1.50%.

735. The tool made of cemented carbide wear out faster at:
(a) fast speed (b) slow speed
(c) medium speed (d) very fast speed

CIL MT 27.02.2020

Ans. (b) : The tool made of cemented carbide wear out faster at slow speed because cemented carbide tool made by powder metallurgy technique and cobalt binder used and its temperature is high.
In slow speed the contact between surfaces are more so it is obvious that at slow speed it will wear more.
$\rightarrow$ Speed decreases
Friction between tool & w/p increases.
So wear also increase.

736. Stellite preserve hardness up to a temperature of:
(a) 1100 °C (b) 500 °C
(c) 350 °C (d) 900 °C

CIL MT 27.02.2020

Ans. (a) : Stellite alloys are composed of various amount of iron, carbon, aluminium, nickel, cobalt, chromium, manganese, silicon, sulphur, titanium, phosphorus, molybdenum stellite preserves hardness upto a temperature of 1100 °C.

737. The connecting rod of an IC engine is made from :
(a) cast iron (b) cast steel
(c) forged steel (d) stainless steel

BPSC Asstt. Prof. 29.11.2015

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Ans. (c) : The connecting rod of on IC engine is made from forged steel.

738. Blade of power saw is made from :
(a) lead alloys (b) aluminium alloys
(c) high-speed steel (d) cast iron

BPSC Asstt. Prof. 29.11.2015
APPSC AEE 2012

Ans. (c) : High speed steels
Properties- maintain high hardness upto temperature about of 550°C and hence can be used for cutting metals high speeds.
- High wear resistance and cutting ability.
- Classification depending upon process alloying elements.
- T-type HSS [18 : 4 : 1 : steel/Tungsten steels] composition : 0.7%C, 18%W, 4% Cr, 1% V
- M-type HSS [Molybdenum steels] composition : 0.85% C, 6%W, 5% Mo, 4% Cr, 2% V.

739. The upper critical temperature for steel
(a) is constant
(b) depends upon the rate of heating
(c) varies with carbon-content of the steel
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : The upper critical temperature for steel varies with carbon content of the steel

740. Carbon content of common HSS material ranges from :
(a) 0.2% to 0.4% (b) 0.4% to 0.6%
(c) 0.6% to 0.8% (d) 0.8% to 1.0%

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Carbon content of common HSS material ranges from 0.6% to 0.8%.

741. Cast iron is
(a) ductile (b) malleable
(c) tough (d) brittle

WBPSC AE 2003

Ans. (d) : Cast Iron– The cast-iron is obtained by remelting of pig iron in cupola furnace carbon contain 1.7% to 4.2%. It has low cost, good casting characteristic wear resistance and excellent machinability and good damping characteristics.

742. Which of the following materials is brittle ?
(a) Gray cast iron (b) Steel
(c) Aluminium (d) Gold

APPSC IOF, 2009

Ans. (a) : Gray cast iron is a Brittle material.

743. The property desirable for transformer core material is :
(a) high hysterisis (b) low hysterisis
(c) low conductivity (d) none of the above

APPSC IOF, 2009

Ans. (b) : The desirable properties of core material are as follows :
(1) Low hysteresis loss
(2) High permeability

744. As per I.S. specification, C 65 means plain carbon steel containing carbon
(a) 0.40 to 0.90% (b) 0.50 to 0.80%
(c) 0.60 to 0.70 (d) 0.30 to 1.0%

WBPSC AE 2003

Ans. (c) : Carbon steel contains 0.60 to 0.70 (0.65 on average carbon)

745. ..... is an example for thermosetting resins.
(a) PVC
(b) ABS
(c) Methacrylate
(d) Phenol-formaldehyde

HPPSC AE (PWD) 10.09.2016

Ans. (d) : The most frequently used thermosetting resins are polyester.
Epoxy
Phenolic
Vinyl ester
Silicon
Polyamide and polyamide-lmide.

746. 'Gliding Metal' covers a range from.....
(a) 5% to 15% Zn (balance Cu)
(b) 15% to 35% Cu (balance Zn)
(c) 2% to 4% Al (balance Zn)
(d) 1% to 2.5% Cu (balance Ph)

HPPSC AE (PWD) 10.09.2016

Ans. (a) : Gliding Metals are a form of brass with a much higher copper content than zinc content.
Range from 95% copper and 5 to 8% zinc.

747. Babbit material used in ball bearing consists of
(a) Nickel, magnesium and lead
(b) Zinc, magnesium and tin
(c) Tin, lead and copper
(d) Nickel, lead and copper
(e) None of the above

CGPSC AE 16.10.2016

Ans. (e) : The babbit metal or bearing metal is any of several alloys used for bearing surface in a plain bearing. It contains, 93% of Tin, 5% of Antimony & 2% of Copper.
Lead or Tin Based Alloys (Babbit Materials)
Typical compositions of

A lead based alloy	A tin based alloy
(1) Lead (Pb) 75%	(1) Tin (Sn) 88%
(2) Antimony (Sb) 15%	(2) Antimony (Sb) 8%
(3) Tin (Sn) 10%	(3) Copper (Cu) 4%

748. An electrician solder consists of _____ lead and ____ tin.
(a) 72 percent ; 28 percent
(b) 58 percent ; 42 percent
(c) 37 percent ; 63 percent
(d) 60 percent ; 40 percent

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : An electrician solder consists of 58% lead and 42% tin.

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749. Select from the following ascending order of cutting tool materials hardness

(a) Ceramics - High carbon steel - High speed steel - Diamond
(b) Ceramics - High speed steel - High carbon steel - Diamond
(c) High speed steel - High carbon steel - Ceramics - Diamond
(d) High carbon steel - High speed steel - Ceramics - Diamond

TSGENCO AE 14.11.2015

Ans. (d) : Ascending order of cutting tool material hardness.
High carbon steel - High speed steel - ceramics-diamond.

750. The straight grades of cemented carbide cutting tool materials contain :

(a) Tungsten carbide only
(b) Tungsten carbide and titanium carbide
(c) Tungsten carbide and cobalt
(d) Tngsten carbide and coal carbide
(e) None of the above

Mizoram PSC (Insp. of Factories) 2019 Paper-III
CGPSC Poly. Lect. 22.05.2016

Ans. (c) : Straight grades, sometimes referred to as unalloyed grades are nominally pure WC–Co composites. The grade of cemented carbides in this group contain WC and Co as the main elements, although small additions or trace levels of other elements are often added to optimize properties. They have the widest range of strength and toughness of all the cemented carbide types and this is in combination with excellent wear resistance.

751. Sheets of ---------- can be produced by calendering process.

(a) PVC (b) PVDC
(c) Rubber (d) ABS

MPSC Lect. (Prod.) 2018

Ans. (a) : Sheets of PVC can be produced by calendering process.

752. Duralumin alloy contains copper and aluminium in the ratio of :

(a) 10 : 88 (b) 4 : 94
(c) 8 : 90 (d) 12 : 86

UPRVUNL AE 21.08.2016

Ans. : (b) Duralumin composition –
3.5 – 4.5% cu + 0.4 – 0.7% Mn and Mg + Aluminium
Application – Forging, stamping, tubes, rivets, sheet

753. Largest forbidden energy gap is exhibited by :

(a) Extrinsic semiconductor
(b) Intrinsic Semiconductor
(c) Conductor
(d) Insulator

UPRVUNL AE 2014

Ans. : (d) Insulator does not have free electron because of it, largest forbidden energy gap is exhibited by it.

754. Dielectrics are :

(a) Insulating materials
(b) Semiconducting materials
(c) Magnetic materials
(d) Ferroelectric materials

UPRVUNL AE 2014

Ans. : (a) Dielectrics are insulating material or a very poor conductor of electric current. When dielectrics are placed in an electric field, practically no current flow in them because unlike metal they have no loosely bound, or free electron that may drift through the material. Instead, electric polarization occurs.

755. The element which has ferrite stabilizing effect is:

(a) Nickel (b) Manganese
(c) Copper (d) Chromium

UPRVUNL AE 2014

Ans : (d) The ferrite stabilizers are chromium
Ferrite stabilizing- It is the balance between the two types of alloying elements that control the quantity of ferrite in the weld metal.

756. How many electrons are there in valence shell of all semiconductor elements?

(a) 8 (b) 2
(c) 4 (d) 0

UPRVUNL AE 2014

Ans : (c) All seminconductor has 4 electron in their valence cell that well known pure semiconductors.

$_{32}$Ge	= K	L	M	N	
	2	8	18	④	Remaining electron
$_{14}$Si	= K	L	M		
	2	8	④		Remaining electron

757. Which of the following material has the lowest resistivity?

(a) Constantan (b) Silver
(c) Manganin (d) Nichrome

UPRVUNL AE 2014

Ans : (b) Silver has lowest resistivity because it is good conductor of electricity. It means there is no restriction while flowing in current in it because of this, it has lowest resistivity.

758. What type of materials are ceramics?

(a) Insulators (b) Conductors
(c) Semiconductors (d) Superconductors

UPRVUNL AE 2014

Ans : (a) Ceramic is an insulator material. It is a mixture of silica and clay. Silica and clay do not have free electron.

759. Holes are the majority charge carriers in :

(a) P-type semiconductors
(b) N-type semiconductors
(c) Insulators
(d) Superconductors

UPRVUNL AE 2014

Ans : (a) Holes are the majority charge carriers in P-type semiconductor while electrons are the majority charge carriers in N-type semiconductor to make the n-type semiconductor, pentavalent impurities like phosphorus arsenic are added and to make the P-type semiconductor a trivalent imparity is added to a pure semiconductor in a small amount and as a result , a large amount of holes are created in it.

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760. How many valence electrons are possessed by Germanium?

(a) 0 (b) 1
(c) 4 (d) 8

UPRVUNL AE 2014

Ans : (c) Germanium (Ge) has 4 valence electron Ge has atomic number 32.

32 Ge = K L M N
2 8 18 ④

The remaining 4 electron are in the outer incomplete shell. which in known as valance electron.

761. For characterizing flexibility and short cycle times the commonly used term is :

(a) Customer-driven
(b) Proactiveness
(c) Agility
(d) Customer-readiness

UPRVUNL AE 2014

Ans : (c) For characterizing flexibility and short cycle times the commonly used term is agility.
A very fast response to sudden market changes and emerging threats by intensive customer interaction.

762. The process is deemed to be out of control _____causes are present.

(a) root (b) special
(c) random (d) common

UPRVUNL AE 2014

Ans : (b) Special cause variation is present when the control chart of a process measure shows either plotted points outside the control limits or a non-random pattern of variation.
When a control chart shows special cause variation a process measure is said to be out of control or unstable.
The process is deemed to be out of control special cause are present.

763. When we view processes as interconnected components of a system, we tend to avoid :

(a) root causes (b) capacity utilization
(c) process variance (d) sub optimization

UPRVUNL AE 2014

Ans : (d) When we view processes as interconnected components of system, we tend to avoid sub-optimization.

764. Vanadium in HSS tool is added to increase

(a) Hardenability
(b) Corrosion resistance
(c) Wear resistance
(d) Cutting ability at very high temperature

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : High speed steel (HSS)- Alloying composition

W/Mo	Cr	V	Fe	C
18%	4%	1%	76%	1%

W : Increase the hot hardness temperature.
Cr : Increase the corrosion resistance and strength.
V : Increase the wear resistance.

765. Coke to metal ratio in the charge of a cold blast cupola operating with indigenous high ash coke is of the order of

(a) 1:5 (b) 5:1
(c) 1 :10 (d) 10:1

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Coke to metal ratio in the charge of a cold blast cupola operating with indigenous high ash coke is of the order of 1:12 to 1:8.
So, 1:10 is correct.

766. An alloy steel which is work hardenable and which is used to make the blades of bulldozers, bucket wheel excavators and other earth moving equipment contain iron, carbon and

(a) Silicon (b) Magnesium
(c) Manganese (d) Chromium

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Maganese and carbon, iron is an alloy steel which is work hardenable and which is used to make the blades of bulldozers, bucket wheel excavators and other earth moving equipment.

767. Smelting is the process of

(a) Expelling moisture, carbon dioxide, sulphur and arsenic from the iron ore by heating in shallow kilns
(b) Removing the impurities like clay, sand etc. from the iron ore by washing with water
(c) Reducing the ore with carbon in the presence of a flux
(d) All of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Smelting is the process of reducing ore with carbon in the presence of a flux.

768. Which of the following element is added to high speed tool steels as a scavenger to remove slag impurities during melting and also for increasing cutting efficiency of tools.

(a) Chromium (b) Vanadium
(c) Molybdenum (d) Manganese

ISRO Scientist/Engineer 12.01.2020

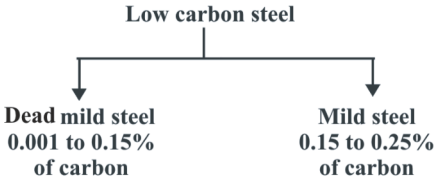
Ans. (b) : Vanadium is added to high speed tool steels as a scavenger to remove slag impurities during melting and also for increasing cutting efficiency of tools.

769. Carbon content of mild steel can be

(a) 0.51% (b) 0.85%
(c) 0.15% (d) 1.5%

ISRO Scientist/Engineer (RAC) 07.05.2017
UKPSC AE 2007 Paper -I

Ans. (c) : Carbon content of low alloys are,



Medium carbon steel (0.25% to 0.75% of carbon)
High carbon steel (0.75% to 1.5% of carbon)

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770. Which of the following carbon steels is most weldable

(a) 1.0% carbon steel (b) 0.3% carbon steel
(c) 0.5% carbon steel (d) 0.15% carbon steel

ISRO Scientist/Engineer 22.04.2018

Ans. (d) : Low carbon steel (0.15% carbon) is most weldable among them because they can welded with low preheat and no post-treatment is required.

771. A small percentage of boron is added to steel in order to

(a) increase endurance strength
(b) reduce machinability
(c) increase water resistance
(d) increase hardenability

ISRO Scientist/Engineer 22.04.2018

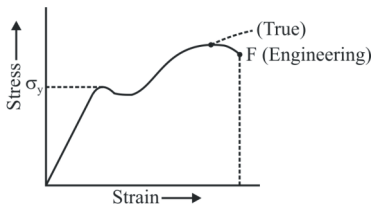
Ans. (d) : Boron gives hardness to steel. So, a small percentage of boron is added to steel in order to increase hardenability.

772. Which of the following is true for ductile materials?

(a) Engineering stress-strain curve cannot have negative slope
(b) Most applicable failure theory is maximum principal stress theory
(c) Ultimate strain is the strain at ultimate stress
(d) Strain hardening is represented by a negative slope in engineering stress strain curve

ISRO Scientist/Engineer 17.12.2017

Ans. (c) :



Stress-strain diagram for ductile material

(1) Engineering stress-strain curve can have negative slope hence statement (a) is wrong.
(2) Most applicable failure theory is maximum principal stress theory for brittle materials not for ductile material hence statement (b) is also wrong.
(3) From the above figure it is clear that ultimate strain is the strain at ultimate stress.
(4) Strain hardening is represented by a constant slope is engineering stress-strain curve.

Hence statement (d) is also wrong.

773. White iron structure consists of

(a) Pearlite
(b) Cementite
(c) Ferrite
(d) Pearlite and Cementite

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : The structure of white cast iron consist of pearlite and ledeburite, a eutectic mixture of pearlite (converted from austenite) and cementite.

774. Element which makes steel stainless

(a) Mg (b) Ni
(c) C (d) Cr

ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Stainless steel remains stainless or does not rust because of the interaction between its alloying elements and the environment. Stainless steel contains iron (Fe), chromium (Cr), manganese (Mn), Silicon (Si), Carbon (C) and in many cases, significant amounts of nickel (Ni) and molybdenum (Mo).

These elements react with oxygen from water and air to form a very thin, stable film that consists of such corrosion products as metal oxides and hydroxides. Chromium plays a dominant role in reacting with oxygen to form this corrosion product film.

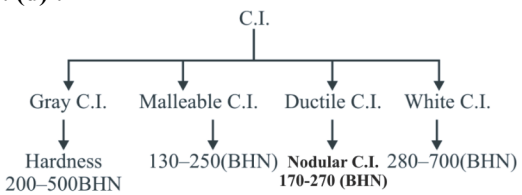
In fact, all stainless steels by definition contain at least 10 percent chromium.

775. The variety of cast iron which has maximum hardness is

(a) Gray iron (b) Malleable iron
(c) Ductile iron (d) White iron

ISRO Scientist/Engineer 07.05.2017

Ans. (d) :



White cast iron has maximum hardness among various varieties of cast iron. The white cast iron structure is very hard, wear-resistant and brittle because of the presence of large amount of iron carbide instead of graphite. It is obtained either by cooling gray cost iron and rapidly or by a adjusting the composition by keeping the carbon and silicon content low. This type of cast iron is called white cast iron because of the white crystalline appearance of the fracture surface.

776. Below the ductile to brittle transition temperature, weld joints show

(a) High ductility (b) High toughness
(c) Low toughness (d) Ductile fracture

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Low toughness

777. Which one of the following steels contains least percentage of carbon?

(a) Mild steel
(b) Stainless steel 304L
(c) Stainless steel 316L
(d) IF steel

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : IF steel—Interstitial Free Steels (IF Steel) are ultra low carbon steel.

Steel is an alloy of iron and upto 2 percent carbon. Carbon is small enough to fit into the interstices of primarily iron matrix making it an "interstitial element" in steel. If the steel alloy has an ultra low carbon level, most of these gaps will not be interstitial free steel.

778. Knife line or hairline cracking is observed in

(a) Ferritic stainless steel
(b) Martensitic stainless steel

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(c) Austenitic stainless steel
(d) Stabilized stainless steel

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : Knife line (or) hairline cracking is obtained in stabilised stainless steel.

779. Thermosetting plastics have in general
(a) molecular chains that slip past one
(b) be a ceramic since all ceramics are glass and glass is a kind of ceramic
(c) have a monolithic crystal structure
(d) have no long range crystalline lattice structure

ISRO Scientist/Engineer 11.10.2015

Ans. (d)
- Thermosetting plastic having high melting point.
- Thermosetting plastic can not re-cycle.
- Thermosetting plastic have no long rang crystalline lattice structure.
- Thermosetting plastic having cross link chain crystalline lattice structure no long chain.

780. German silver contains
(a) 12.5% silver (b) 5% silver
(c) 1% silver (d) None of the above

ISRO Scientist/Engineer 11.10.2015

Ans. (d) : German sliver varies in composition, the percentage of the three element ranging approximately as follows :
Copper, from 50% to 61.6%
Zinc, from 19% to 17.2%
Nickel from 30% to 21.1%
The proportions are always specified in commercial alloys.
Hence there is no silver.

781. Diamond cutting tools are not recommended for machining of ferrous metals due to
(a) high tool hardness
(b) chemical affinity of tool material with iron
(c) Poor tool toughness
(d) High Thermal conductivity of work material

ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Diamond have high thermal conductivity. So, diamond cutting tools are not recommended for machining with ferrous metal due to chemical affinity of tool material.

782. Tungsten in High Speed Steel provides
(a) Hot hardness (b) Toughness
(c) Wear resistance (d) Sharp cutting edge

ISRO Scientist/Engineer 2011

Ans. (a) : Tungsten is a good carbide former that prevents grain growth, enhances toughness and increases red hardness and high temperature strength. Tungsten is used in hot forming tool steels and high-speed steels.

783. Match the lists I and II using code given below

A. Car dash board	**1. Polyvinylchloride (PVC)**
B. Aircraft windows	**2. TEFLON**
C. Conduit pipes	**3. Polyacrylonitrile**
D. Bearings and gears	**4. Polymethylmethacrylate**

	A	B	C	D
(a)	1	2	3	4
(b)	3	4	1	2
(c)	4	3	2	1
(d)	2	3	4	1

ISRO Scientist/Engineer 2011

Ans. (b) :

784. A material is classified as brittle, if the elongation in a gauge length of 59mm, is less than
(a) 1% (b) 3%
(c) 4% (d) 5%

ISRO Scientist/Engineer 2010

Ans. (d) : Typically brittle materials have a fracture strain less than 0.05 ($\varepsilon_f < 0.05$) or percent elongation less than five percent and ductile materials have a fracture strain greater than or equal to 0.05 ($\varepsilon_f \geq 0.05$)

785. The main alloying elements in high speed steel in order of increasing proportion are
(a) Vanadium, Chromium, tungsten
(b) Tungsten, titanium, vanadium
(c) Chromium, titanium, vanadium
(d) Tungsten, chromium, titanium

ISRO Scientist/Engineer 2010

Ans. (a) : In HSS
Tungsten - 18%
Chromium - 4%
Vanadium - 1%
Carbon - 0.7%
and Iron - Rest

786. In nodular iron, graphite is in the form of
(a) Spheroids (b) Nodular carbide
(c) Flakes (d) Cementite

ISRO Scientist/Engineer 2006
ISRO Scientist/Engineer 2009

Ans. (a) : In nodular iron, graphite is in the form of spheroids.

787. Addition of copper to aluminium results in
(a) Precipitation hardening system
(b) Corrosion resistance
(c) Grain refinement
(d) High machinability

ISRO Scientist/Engineer 2009

Ans. (a) : Addition of copper to aluminium results in precipitation hardening system.

788. A cold chisel is made of
(a) Mild steel (b) Cast iron
(c) H.S.S. (d) High carbon

ISRO Scientist/Engineer 2008

Ans. (d) : A cold chisel is made of high carbon.

789. Which of the following has maximum hardness-
(a) Austenite (b) Pearlite
(c) Troostite (d) Martensite

ISRO Scientist/Engineer 2006

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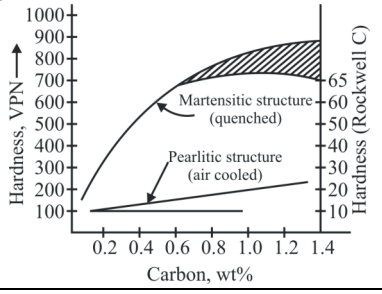
Ans. (d) : The highest hardness of martensite is 700 (BHN) brinell whereas hardness of a pearlite steel is 400 (BHN) brinell.



790. Composite materials are :
(a) made mainly to improve temperature resistance
(b) used for improved optical properties
(c) made with strong fibres embedded in weaker and softer matrix to obtain strength better than strength of matrix
(d) made with strong fibres embedded in weaker and softer matrix to obtain strength better than strength of both matrix and filler

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) : Composite materials are made with strong fibres embedded in weaker and softer matrix to obtain strength better than strength of both matrix and filler.

791. Ceramic materials are :
(a) inorganic compounds of metallic and non-metallic elements
(b) basically crystalline oxides or metals
(c) good conductors of electricity
(d) none of these

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : Ceramics material are inorganic compounds of metallic and non-metallic elements.

792. Which one are inorganic materials :
(a) biological materials
(b) minerals and ceramics
(c) plastics
(d) wood

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : Inorganic materials are minerals and ceramics.

793. Using cutting material which can sustain at high temperature :
(a) cermet
(b) high carbon steel alloy
(c) composite of two metals
(d) none of these

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : A cermet is a composite material composed of ceramic and metal. A cermet is ideally designed to have the optimal properties of both a ceramic, such as high temperature resistance and hardness and those of a metal, such as the ability to undergo plastic deformation.

794. Non ferrous alloys and stainless steels are heat treated by a process called
(a) Annealing
(b) Quenching
(c) Tempering
(d) Precipitation hardening

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d) : Non ferrous alloys and stainless steels are heat treated by a process called precipitation hardening.

795. The thermoplastics among the following is
(a) Acrylics (b) Phenolics
(c) Polyester (d) Polyamides

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : Acrylic is a transparent thermoplastic homopolymer known more commonly by the trade name "plexiglass".

796. Which of the following thermocouples has the lowest measuring range?
(a) Iron - Constantan
(b) Chromel -Alumel
(c) Copper - Constantan
(d) Chromel - Constantan

TNPSC AE 2014

Ans. (c) : Copper - Constantan thermocouple has the lowest measuring range.

797. Bumper and other collision absorbing materials is made up of
(a) Light alloys of Brass
(b) Light alloys of Copper
(c) Light alloys of Aluminium
(d) Wood blocks

TNPSC AE 2017

Ans. (c) : Bumper and other collision absorbing materials is made up of Light alloys of Aluminium.

798. Wear resistance of steel is increased by adding,
(a) nickel (b) chromium
(c) sulphur (d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Wear resistance of steel is increased by adding chromium.

799. Hardness of steel is increased by adding,
(a) nickel (b) chromium
(c) sulphur (d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Hardness of steel is increased by adding, molybdenum.(some other also responsible for inc. Hardness like- chromium, tungsten

800. In free cutting steels, important alloying element is–
(a) nickel (b) chromium
(c) sulphur (d) tungsten

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : In free cutting steels, important alloying element is sulphur.

801. Bronze is an alloy of copper and :
(a) aluminium (b) nickel
(c) iron (d) tin

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Bronze is an alloy of Copper and Tin.

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802. Medium carbon steels have carbon content in the range of _____ %.
(a) 0.30-0.60 (b) 0.08-0.30
(c) 0.60-1.80 (d) 0.30-1.80
(e) 0.60-2.00
(CGPSC Polytechnic Lecturer 2017)
Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Medium carbon steels have carbon content in the range of 0.30 to 0.60%.

803. Theoretically, steels are the alloys of iron and carbon, in which carbon contents vary between:
(a) 0.008 – 2.1% (b) 0.008 – 2.05%
(c) 0.0008 – 2.1% (d) 0.0008 – 2.05%
(e) 0.0008 – 1.8%
(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Steel	**Carbon Percent (C%)**
Mild steel (Low Carbon)	0.008 to 0.25% (Approx.)
Medium-carbon steel	Approximately 0.3 to 0.6%
High-carbon steel	Approximately 0.6 to 1.5%
Ultra-high carbon steel	Approximately 1.5 to 2.1%

804. Neat oils can be preferred while machining of :
(a) Gray cast iron (b) Copper alloys
(c) Aluminium alloys (d) Alloy steels
TRB Polytechnic Lecturer 2017

Ans. (d) : Neat oils can be preferred while machining of alloy steels.

805. Mild steel contains–
(a) less than 0.3% carbon
(b) 0.3 to 0.5% carbon
(c) 0.5 to 1.4% carbon
(d) 3 to 4% carbon
Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : % C in mild or low carbon steel varies from 0.05% to 0.30%.

806. Steels used for welded assemblies are–
(a) medium carbon steel (b) mild steel
(c) high carbon steel (d) alloy steel
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Steels used for welded assemblies are mild steel.

807. Steels used for automobile bodies and hoods are-
(a) medium carbon steel (b) high carbon steel
(c) alloy steel (d) mild steel
Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Steels used for automobiles bodies and hoods are mild steel.

808. Steels used for helical springs are–
(a) medium carbon steel (b) mild steel
(c) high carbon steel (d) alloy steel
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Steels used for helical spring are high carbon steel.

809. Material used for machine tool beds is–
(a) cast iron (b) mild steel
(c) high carbon steel (d) alloy steel
RPSC AE 2018
UKPSC AE 2012 Paper I
Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Material used for machine tool beds is cast iron because it has high damping capacity and absorbs vibrations

810. Material used for Bearing bushes is–
(a) Phosphor bronze (b) gunmetal
(c) Babbitt (d) any one of the above
Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Material used for bearing bushes is, phosphor bronze, gun metal, Babbitt.

811. Material used for self-lubricated bearing is–
(a) Acetal
(b) Polyurethane
(c) Polytetrafluroethylene (Teflon)
(d) any one of the above
Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Any one of the above

812. Which of the following materials has maximum strength–
(a) grey cast iron (b) plain carbon steel
(c) alloy steel (d) aluminium alloy
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Alloy steel

813. Plain carbon steels are designated by,
(a) carbon content (b) tensile strength
(c) composition of alloying element
(d) none of the above
Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Plain carbon steels are designated by carbon content.

814. Plain carbon steel designated by 40C8 means,
(a) plain carbon steel with ultimate tensile strength of 400 N/mm^2 and 0.8% carbon
(b) plain carbon steel with 0.35 to 0.45% carbon and 0.7 to 0.9% manganese
(c) plain carbon steel with 0.8% carbon and 4% manganese
(d) plain carbon steel with 40% carbon and 8% manganese
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Plain carbon steel designated by 40C8 means, plain carbon steel with 0.35 to 0.45% carbon and 0.7 to 0.9% manganese.

815. Consider the percentages of the following metals
1. Co (40 to 50%) 2. Cr (34% to 39%)
3. W (14 to 19%) 4. C (2%)
Identify the metals that make up the cast alloy Satellite.
(a) 2, 3 and 4 (b) 1, 2 and 4
(c) 1, 2, 3 and 4 (d) 1, 3 and 4
(e) 1 and 4
CGPSC AE 26.04.2015 Shift-I

Ans. (c) : 1, 2, 3 and 4

816. The distinct characteristic of Invar is
(a) It is magnetic
(b) it has low coefficient of thermal expansion
(c) It has high tensile strength
(d) It is non corrosive
ESE 2019

Ans. (b) : Invar is a nickel-iron alloy notable for its uniquely low coefficient of thermal expansion.
It contains 64% iron and 36% nickel.

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817. Consider the following statements regarding solid solution of metals:

1. The solubility of metallic solids is primarily limited by size factor.
2. A metal with high valence can dissolve large amount of metal of lower valence.
3. A metal with same lattice crystal structure can form a series of solid solutions
4. The limit of solid solubility is indicated by a phase boundary called Liquidus.

Which of the above statements are correct?

(a) 1 and 4 (b) 2 and 4
(c) 1 and 3 (d) 1 and

ESE 2018

Ans. (c) :
(i) The solubility of metallic solid is limited by relative size factor i.e. difference between the atomic radius of atoms must not be greater than 15%.
(ii) A metal with lower valence can dissolve large amount of metal having higher valence.
(iii) When crystal structure of solute and solvent is same then it can form a series of solid solution or if results into extensive solubility.
(iv) The limit of solid solubility is indicated by a phase boundary called solvus.

818. Consider the following statements:

1. HSS tools wear very rapidly, whereas in cemented carbide tools, even though hardness is retained, crater wear can occur due to solid-state diffusion.
2. Cutting tools made of Super-HSS, also known as cobalt-based HSS, are made by adding 2% to 15% of cobalt which increases the cutting efficiency at heavier cuts by increasing the hot hardness and wear resistance.
3. Tools failure due to excessive stress can be minimized by providing small or negative rake angles on brittle tool materials, protecting tool tip by providing large side-cutting edge angles, and honing a narrow chamfer along the cutting edge.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : (i) When HSS tool is used at higher cutting temperature, then due to plastic deformation and crater wear, it wears very rapidly. Crater wear occurs at some distance away from tool tip on tool face. It is mainly due to high temperature where solid state diffusion can cause rapid wear.
(ii) Super HSS is a molybdenum series high-speed steel alloy with an additional 8% or 10% cobalt.
(iii) By reducing rake angle the strength of tool increases hence by providing small or negative rake angles the tool failure can be avoided.

819. Statement (I): The ideal material for shafts transmitting power is CI.
Statement (II): CI resists compression well.

ESE 2017

Ans. (d) : Cast iron (CI) is brittle in nature, so its torsional strength is less hence can not used as shaft.
So, statement-I is false.
Cast iron (CI) is strong in compression but weak under tension.

820. The correct order of increasing resistivity among the following materials is

(a) nickel, doped silicon, sodium silicate, pure silica
(b) doped silicon, nickel, pure silica, sodium silicate
(c) nickel, pure silica, sodium silicate, doped silicon.
(d) sodium silicate, nickel, pure silica, doped silicon.

ESE 2017

Ans. (a) : Increasing order of resistivity of metal
Nickel (10^{-7}) < Doped silicon (10^{-4}) < Sodium Silicate (10^{2}) < Pure silica (10^{4})

821. The maximum percentage of the carbon content in steels is

(a) 1.2% (b) 2%
(c) 4% (d) 6.67%

GPSC ASST. Prof. (prod.) 21. 08. 2016
HPPSC AE 2018

Ans. (b) : Mild [low carbon] steel – Approximately 0.05% to 0.25% carbon
Medium carbon steel – Approximately 0.30% to 0.60% carbon content
High carbon steel – Approximately 0.60% to 0.95% carbon content
Very high carbon steel – Approximately 0.95% to 2.1% carbon content
Cast Iron – Approximately 2.1% to 4% carbon

822. If a steel is designated as C8O, which of the following is correct for it?

(a) C ≡ tool steel
80 ≡ 80% Iron
(b) C ≡ plain carbon steel
80 ≡ 0.8% carbon
(c) C ≡ tool steel
80 ≡ 0.8% carbon
(d) C ≡ plain carbon steel
80 ≡ 80% Iron

UPRVUNL AE 07.10.2016

Ans. (b) : If a steel is designated as C8O means
C ≡ plain carbon steel
80 ≡ 0.8% carbon

823. An addition of manganese to aluminium results in the improvement of :

(a) Strength characteristics
(b) Corrosion resistance
(c) Hardernability
(d) Toughness system

JWM 2017

Ans. (a) : Strength characteristics of aluminium is improved by adding manganese in it.

824. Which of the following ceramic materials are applications?

(a) Alumina and zirconium oxides
(b) Boron titanate and silicon carbide
(c) Barium titanate and lead-zirconate-titanate
(d) Porcelain and fused silica glass

JWM 2017

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Ans. (c) : • Barium titanate is mostly used in thermistor
• Lead zirconate titanate is used to make ultrasound transducers and sensors.

825. Among the following materials, the most suitable material for which standing shock and vibration without danger of cracking is
(a) chilled cast iron (b) gray cast iron
(c) malleable cast iron (d) white cast iron

TNPSC AE 2018

Ans. (c) : Malleable cast iron have the most suitable material for with standing shock and vibration without danger of cracking.

826. The principal mechanism by which non-ferrous alloys are heat treated is called:
(a) Flame hardening
(b) Induction hardening
(c) Age hardening
(d) Precipitation hardening
(e) Case hardening

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : Precipitation Hardening or Age Hardening–Precipitation hardening is commonly used to process aluminum alloys and other non-ferrous metals for commercial use. The example are Aluminum–Copper, Copper–Beryllium, Copper–Tin, Magnesium–Aluminum and some ferrous alloys.
The Process is called precipitation hardening because the small particles of the new phase are termed "Precipitates"
Mechanism of Hardening–
During plastic deformation:
- Zones or Precipitates act as obstacles to dislocation motion.
- Stress must be increased to 'Push' the dislocation through the distribution of precipitates.
- Consequently the alloy becomes harder and stronger.

827. Cast irons contain a higher carbon content, normally between 3.0 and 4.5 wt% C, and other alloying elements, notably _____
(a) Silicon (b) Phosphorous
(c) Magnesium (d) Aluminium

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Cast-iron is obtained by remelting pig iron with coke and lime stone in a furnace (cupola). It is primarily alloying of iron & carbon. The carbon content in cast iron varies from 1.7 to 4.5%. The cast iron contain some small amount of impurities such as silicon (4%), Sulphur magnesium and phosphorous (1%).

828. While machining cast iron generally cutting fluid is not required because
(a) Cast iron jobs revolves at low speeds
(b) Cast iron surface is generally rough
(c) Cast iron contains graphite which does not allow temperatures to rise
(d) Cast iron contains carbon which cools the job

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : In machining of cast iron, no cutting fluid is required because, soluble oils are used for machining metals of high machinability. Cast iron contains graphite which work as a lubricant which does not allow the temperatures to rise.

829. A cast iron designated by BM350 is,
(a) blackheart malleable cast iron with carbon content of 3.5%
(b) blackheart malleable cast iron with ultimate tensile strength of 350N/mm^2
(c) blackheart malleable cast iron with ultimate compressive strength of 350 N/mm^2
(d) Blackheart malleable cast iron with tensile yield strength of 350 N/mm^2

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : A cast iron designated by BM350 is blackheart malleable cast iron with ultimate tensile strength of 350 N/mm^2.

830. The pouring temperature for grey cast iron is
(a) 1000 ^oC (b) 1250 ^oC
(c) 1400 ^oC (d) 1550 ^oC

SJVN ET 2013

Ans. (c) : The pouring temperature of grey cast iron is 1400^oC

831. In cast iron components, shrinkage cavities are formed due to–
(a) cored holes at the junction of walls
(b) very small fillet radius
(c) concentration of metal at the junction of walls
(d) very thin wall thickness

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : In cast iron components, shrinkage cavities are formed due to concentration of metal at the junction of walls.

832. Piston compression rings are made of
(a) Cast iron (b) Bronze
(c) Aluminium (d) White metal

RPSC Vice Principal ITI 2018

Ans. (a) : Piston compression rings are made of cast iron.

833. An important property of malleable cast iron in comparison to grey cast iron is having high
(a) Compressive strength (b) Yield strength
(c) Hardness (d) Surface finish

TNPSC AE 2019

Ans. (b) : An important property of malleable cast iron in comparison to grey cast iron is having high Yield strength.

834. Which of the following alloys possess highest specific strength of all structural materials?
(a) Magnesium alloys (b) Titanium alloys
(c) Chromium alloys (d) Magnetic steel alloys

TNPSC AE 2019

Ans. (b) : Titanium alloys are metals that contain a mixture of titanium and other chemical elements. Such alloys have very high tensile strength and toughness (even at extreme temperature). They are light in weight, have extraordinary corrosion resistance and the ability to withstand extreme temperature.

835. Duralumin alloy contains aluminum and copper in the ratio of

	% Al	% Cu
(a)	94%	4%
(b)	90%	8%
(c)	88%	10%
(d)	86%	12%

TNPSC AE 2019

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Ans. (a) : Duralumin is an alloy made up of 94% aluminium, 4% copper, 1% magnesium and 0.5% to 1% manganese. It is a very hard alloy.
Duralumin, strong, hard, lightweight alloy of aluminium widely used in aircraft construction.

836. A cast iron designated by FG300 is,
(a) grey cast iron with carbon content of 3%
(b) grey cast iron with ultimate tensile strength of 300 N/mm^2
(c) grey cast iron with ultimate compressive strength of 300 N/mm^2
(d) grey cast iron with tensile yield strength of 300 N/mm^2

HPPSC ASST. Prof. 20.11.2017
RPSC Vice Principal ITI 2018
Nagaland PSC CTSE 2017, Paper-I
GPSC EE Pre 28.01.2017

Ans. (b) : Grey cast iron with ultimate tensile strength of 300 N/mm^2.

837. The presence of hydrogen in steel causes :
(a) Reduced neutron absorption cross-section
(b) Improved weld ability
(c) Embrittlement
(d) Corrosion resistance

OPSC AEE 2019 Paper-I

Ans : (c) : The presence of hydrogen in steel causes embrittlement.

838. Wt.% of carbon in mild steels :
(a) < 0.008 (b) 0.008-0.3
(c) 03-0.8 (d) 0.8-2.11

OPSC AEE 2019 Paper-I

Ans : (b) : Mild steel is a low carbon steel.
(i) Dead mild steel : Upto 0.15% carbon
(ii) Low carbon steel or mild steel : 0.15% to 0.45% carbon
(iii) Medium carbon steel : 0.45% to 0.8% carbon
(iv) High carbon steel : 0.8% to 1.5% carbon

839. High carbon steel contains carbon
(a) Between 0.05 and 0.6%
(b) Between 0.6 and 1.3%
(c) Between 1.3 and 1.7%
(d) More than 1.7%

Gujarat PSC AE 2019

Ans : (b) : High carbon steel contains carbon 0.6 and 1.3%.

840. The machinability of steel is improved by adding
(a) Nickel
(b) Chromium
(c) Nickel and Chromium
(d) Sulphur, lead and phosphorus

AAI Jr. Executive 26.03.2021
Gujarat PSC AE 2019

Ans : (d) : The machinabiltiy of steel is improved by adding sulphur, lead and phosphorous.

841. Which of the following material is used for manufacturing of extrusion nozzles?
(a) Grey cast iron (b) Malleable cast iron
(c) White cast iron (d) Nodular cast iron

UPPSC AE 12.04.2016 Paper-I

Ans : (c) White cast iron material is used for manufacturing of extrusion nozzle because it is very hard and brittle.
White cast iron:- It is a particular variety of cast iron having 1.75 to 2.3% carbon. The white colour is due to the fact that the carbon is in the form of carbide, which is the hardest constituent of iron. The white cast iron has a high tensile strength and a low compressive strength.

842. The melting point of mild steel is about
(a) 1250°C (b) 1500°C
(c) 850°C (d) 1750°C

KPSC AE 2015
Kerala PSC Poly. Lect. 09.01.2014

Ans : (b) The melting point of mild steel is about 1500°C

Metal	Melting point
Aluminum	659°C
Copper	1081°C
Gold	1064°C
Silver	960°C
Tin	232°C
Zinc	419°C

843. Major constituents of stellite are
(a) Nickel, copper and zinc
(b) Zinc, lead and tin
(c) Cobalt, vanadium and nickel
(d) Cobalt, chromium and tungsten

(KPSC AE 2015)

Ans : (d) Major constituents of satellite are cobalt, chromium and tungsten.

844. Which the percentage increase of carbon in steel
(a) brittleness of steel decreases
(b) Strength of steel decreases
(c) Ductility of steel decreases
(d) Hardness of steel decreases

(KPSC AE 2015)

Ans : (c)

Alloying elements	Effect on steel
Carbon	Ductility of steel decrease
Vanadium	Increase endurance strength
Molybdenum	Improves creep properties
Chromium	Increase hardness.

845. In low carbon steels, the presence of small quantities of sulphur improves
(a) Formability (b) Machinability
(c) Weldability (d) Hardenability

CGPSC AE 15.01.2021
Maharashtra PSC AE Pre 2019
KPSC AE 2015

Ans : (b) In low carbon steels, the presence of small quantities of sulphur improves machinability.

Low carbon steel Alloy metal	Properties on low carbon steel
Sulphur	Machinability
Manganese	Yield point increase
Silicon	Prevent them from becoming porous OR make the steel tougher and harder

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846. Which of the following is used for bearing liner?

(a) Bronze (b) Babbit metal
(c) Gun metal (d) brass

Nagaland PSC (CTSE) 2018, Paper-I
(KPSC AE 2015)

Ans : (b) Babbit metal is used for bearing liner.
Babbit metal:- The tin base and lead base babbits are widely used as a bearing materials because they satisfy most requirements for general application. The babbits are recommended where the maximum bearing pressure is not over 7 to 14 N/mm^2.

Babbit metal

Tin base babbit metal	**Lead base babbit metal**
Tin = 90%	Lead - 84%
Copper = 4.5%	Tin - 6%
Antimony = 5%	Antimony = 9.5%
Lead = 0.5 %	Copper = 0.5%

847. ..... is used as an alloying element to enhance the endurance strength of steel materials.

(a) Tungsten. (b) Molybdenum
(c) Nickel (d) Vanadium

GWSSB DEE 07.07.2016
HPPSC LECT. 2016

Ans : (d) Vanadium is used as an alloying element to enhance the endurance strength of steel materials.

Alloy element	**Effect**
Nickel	Hardenability
Chromium	hardness and corrosion resistance
Vanadium	Increase yield strength
Tungsten	Hot Hardness

848. The first product in the process of converting iron ore into useful metal from a blast furnace is known as :

(a) Cast iron (b) Wrought iron
(c) Pig iron (d) Steel

HPPSC W.S. Poly. 2016

Ans : (c) The First product in the processes of converting iron ore into useful metal from a blast furnace is known as pig iron.
Pig iron:- It is the crude form of iron and is used as a raw materials for the production of various other ferrous metals, such as cast iron, wrought iron and steel. The pig iron is obtained by smelting iron ores in a blast furnace. The haematite is widely used for the production of pig iron since pyrite contains only 30 to 40% iron, therefore it is not used for manufacturing pig iron.

849. When carbon in the cast iron is mostly in free state, the cast iron is known as :

(a) Molted cast iron (b) White cast iron
(c) Grey cast iron (d) Black cast iron

HPPSC W.S. Poly. 2016

Ans : (c) When carbon in the cast iron is mostly in free state, the cost iron is known as grey cast iron.
Grey cast iron:- It contains 3 to 3.5 percent carbon. The gray colour is due to the fact that carbon is present in the form of free graphite. It is widely used for machine tool bodies, flywheels. Pipes and pipe fittings etc.

850. The percentage of carbon in cast iron usually varies between :

(a) 0.1 to 0.2% (b) 0.5 to 1.0%
(c) 1.5 to 2.5 % (d) 2.5 to 3.5 %

(KPSC AE. 2015)
JPSC AE 2013, Paper-V

Ans : (d) The Cast iron is obtained by re-melting pig iron with coke and lime stone in a furnace known as Cupola. It is primarily an alloy of iron and carbon. The Carbon Contents in Cast iron varies from 1.7% to 4.5%. It is also Contains small amount of silicon. Manganese, Phosphorous and sulphur.

851. Bronze used for church bells contain:

(a) No copper (b) 4 to 8% of tin
(c) 1 to 4% of tin (d) 15 to 25% of tin

OPSC AEE 2015 Paper-I

Ans : (d) Bell metal : Bell metal is a hard alloy used for making bells and related instruments, such as cymbals. It is a form of Bronze, usually in approximately a 4:1 ratio to tin (e.g 78% copper, 22% tin by weight)

852. Among the following characteristics, one is NOT the characteristic of austenitic stainless steel :

(a) Toughness (b) Corrosion resistance
(c) Brittleness (d) Resilience

OPSC AEE 2015 Paper-I

Ans : (c) Brittleness is not the characteristic of austenitic stainless steel.
Austenitic stainless steel: 15-20% Cr, 7-10% Ni
* High ductility (18/8 stainless steel) are non magnetic
* Titanium and niobium added to stabilise the carbon.

853. Which of the following has least percentage of carbon?

(a) Malleable cast iron (b) Pig iron
(c) Stainless steel (d) Wrought iron

OPSC AEE 2015 Paper-I

Ans : (d) Wrought iron : It is the purest iron which contains 99.5% iron but may contain upto 99.9%. The carbon content is about 0.02% It is a tough, malleable and ductile material. It cannot stand sudden and excessive shocks. It can be easily forged or welded.

854. White metal is an alloy of :

(a) Lead + Tin (b) Lead + Bismuth
(c) Lead + Zinc (d) Lead + Aluminum

OPSC AEE 2015 Paper-I
Kerala PSC Poly. Lect. 19.01.2014

Ans : (a) The white metals are any of several light–coloured alloys used as a base for plated silverware, ornaments or novelties, as well as any of several lead–based or tin–based alloys used for things like bearings, jewellery, miniature figures, fusible plugs, some medals and metal type.
A white metal alloy may include antimony, tin, lead, cadmium, bismuth, and zinc (some of which are quite toxic). Not all of these metals are found in all white metal alloys. Metals are mixed to achieve a desired goal or need. As an example,. base metal for jewellery needs to be castable, polishable, have good flow characteristics, have the ability to cast fine detail without an excessive amount of porosity and cast at between 230^0C and 300^0C (450^0F and 575^0F).

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855. Which of the following generally made of High carbon steel?

(a) Hammers (b) Angle iron
(c) Solid drawn tubes (d) Boiler plates

OPSC AEE 2015 Paper-I

Ans : (a) High Carbon Steel : High carbon steels has a carbon content of 0.7% to 1.5%. it has a maximum strength of 1400 N/mm^2 when carbon is 0.8% thereafter its hardness increases but ductility decreases with increase of carbon. It is hard, less ductile and is almost always fully heat–treated before being used. Its properties vary with the carbon content and method of heat treatment, but in general the lower the carbon content the tougher the steel is, and the higher the carbon content the harder and less shock–resistance the steel. High carbon steel with carbon content of 1% to 1.5% is called tool steel.

Use : It is used for hand tools, press tools, machine parts, files reamers, metal cutting saws, hammer, twist drills, Mandrels etc.

856. Y- alloy is an alloy of

(a) Nickel (b) Aluminum
(c) Copper (d) Tin

TSPSC AEE 2015

Ans : (b) Some widely used aluminum alloys are
(i) Duralumin (ii) Y – alloy
(iii) Maghalium (iv) Hindalium.

Y - alloy composition:-
Copper = 4 %
Nickel = 2%
Magnesium = 1.5%
Silicon = 0.5 %
Iron = 0.5 %
Aluminum = Remainder (92.5%)

857. The raw material required for making pig iron ore

(a) coal, limestone, ore
(b) ore, coke, wrought iron
(c) coke, limestone, ore
(d) coal, coke ore

TSPSC AEE 2015

Ans : (c) The raw material required for making pig iron are coke, limestone, ore.

858. Yellow brass also known as

(a) Cartridge brass (b) Gun metal
(c) Naval brass (d) Muntz metal

TSPSC AEE 2015

Ans : (d) Muntz-metal OR yellow metal:- It is best known α + β (60:40) alloys hot working brass. In the cast state the microstructure is of a widmanstatten type, with α at the grain boundaries and along the crystal planes of the β. constituent. Hot working refines the structure giving a more uniform distribution of α in a matrix of β.

Composition:-
Copper = 60%
Zinc = 40%

859. The bourdon tubes are generally made of

(a) Mild steel
(b) Bronze or Nickel steel
(c) Tin
(d) Copper

TSPSC AEE 2015

Ans : (b) The Bourden tubes are generally made of Bronze or nickel steel.

860. Iron carbon alloy with less than 2% carbon is called

(a) cast iron (b) pig iron
(c) steel (d) wrought iron

TSPSC AEE 2015

Ans : (c) Iron carbon alloy with less then 2% carbon is called steel.

Type of Steel:-

Steel Type	**% of carbon**
Dead mild steel	upto-0.15%
Low carbon or mild steel	0.15-0.30%
Medium Carbon Steel	0.3-0.6%
High Carbon Steel	0.6-1.5%
Tool Steel	0.9-1.5%

861. While machining cast iron which of its elements turns hands black

(a) Iron (b) Sulphur
(c) Graphite (d) Nickel

UJVNL AE 2016

Ans : (c) Cast Iron:- The cast iron is obtained by re-melting pig iron with coke and limestone in a furnace known as cupola. It is primarily an alloy of iron and carbon. the carbon contents in cast iron varies from 1.7 per cent to 4.5 percent. It also contains small amounts of silicon, manganese, phosphorous and sulphur. the carbon in a cast iron is present in either of the following two forms:-
1. Free carbon or graphite
2. Combined carbon or cementite

862. Which of the following characteristics is possessed by Nickel?

(a) Paramagnetic (b) Ferromagnetic
(c) Non- magnetic (d) Die electric

UJVNL AE 2016

Ans : (b) Ferromagnetic characteristics are possessed some metal—Fe, Ni and CO.
Ferromagnetism:- Iron, nickel, cobalt and some of the rare earths (gadolinium, dysprosium) exhibit a unique magnetic behavior which is called ferromagnetism because iron (ferrum in Latin) is the most common and most dramatic example. Sararium and neodymium in alloy with cobalt have been used to fabricate very strong rare-earth magnets.

863. Which of the following elements determine maximum attainable hardness in the steel?

(1) Chromium (2) Manganese
(3) Carbon (4) Molybdenum

Select the correct answer using codes given below.

(a) 1 only (b) 1 and 2
(c) 3 only (d) 2 and 4

UPPSC AE 12.04.2016 Paper-I

Ans : (c) Carbon elements determine maximum attainable hardness in the steel. Effects of alloying carbon elements in steel:-
(i) Increase hardness
(ii) Increase Tensile strength
(iii) Lower melting point
(iv) Reduces ductility

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864. Important property requirements for tool materials employed for high speed machining are:-
(a) Impact strength, melting point and hardness
(b) Hot hardness, wear resistance and toughness
(c) Melting point, toughness and shear strength
(d) Shear strength, wear resistance and impact strength

UKPSC AE-2013, Paper-I

Ans. (b) : Important property requirements for tool materials employed for high speed machining are hot hardness, wear resistance and toughness

865. Match the items in List – 1 to that of the List – 2 and choose the correct alternative.

List – 1	List – 2
A. Alnico V	1. Metallic Magnet
B. Ferrexodur	2. Ceramic Magnet
C. Nickel Oxide	3. Anti ferromagnetic
D. Ferrites	4. Compounds containing trivalent iron
	5. Ferrimagnetic
	6. Soft magnetic

Alternatives :

	A	B	C	D
(a)	1	2	3	4
(b)	6	2	3	4
(c)	4	6	1	2
(d)	2	1	6	1

UKPSC AE 2012 Paper-I

Ans. (a) : A-1, B-2, C-3, D-4

866. Griffith theory of failure is suitable for
(a) Mild Steel (b) Low Carbon Steel
(c) Alloy Steel (d) Glass

NLCIL 17.11.2020 Shift II
UKPSC AE 2012 Paper-I

Ans. (d) : Griffith theory of failure is suitable for glass.

867. Mild Steel is an example of
(a) Substitution solid solution
(b) Interstitial solid solution
(c) Inter metallic compound
(d) None of the above

UKPSC AE 2012 Paper-I

Ans. (b) : Mild Steel is an example of interstitial solid solution.

868. The processes, used to make the steel magnetically softer, are
(a) Annealing and Decarburization
(b) Decarburization and Quenching
(c) Annealing, Grain growth and Decarburization
(d) Grain growth and Quenching

UKPSC AE 2012 Paper-I

Ans. (c) : Annealing, Grain growth and Decarburization

869. German silver is an alloy of
(a) Silver and Tin
(b) Silver and Gold
(c) Nickel and Copper
(d) Nickel, Copper and Zinc

JPSC AE 10.04. 2021 Paper II
UKPSC AE 2012 Paper-I

Ans. (d) : Nickel, Copper and Zinc

870. Very high strength in Aluminium alloys is obtained by
(a) precipitation hardening
(b) solid solution hardening
(c) cold working
(d) annealing

UKPSC AE 2007 Paper -I

871. Which of the following metals has the lowest specific gravity?
(a) Monal metal (b) Copper
(c) Magnesium (d) Bronze

UKPSC AE 2007 Paper -I

Ans. (c) : Magnesium

872. Nickel is mostly found in
(a) Russia (b) Canada
(c) Russia and Canada (d) Russia and India

UKPSC AE 2007 Paper -I

Ans. (c) : Russia and Canada

873. Increase in ferrite phase in steel leads to increase in:-
(a) Strength (b) Hardness
(c) Ductility (d) Brittleness

UKPSC AE-2013, Paper-I
UPPSC AE 12.04.2016 Paper-I
ESE 2005

Ans. (a) : Increase in ferrite phase in steel leads to increase in strength.

874. Which one of the following has the highest value of specific stiffness?
(a) Steel (b) Aluminium
(c) Fibre glass (d) Carbon fibre composite

UKPSC AE-2013, Paper-I

Ans. (d) : Carbon fibre composite has the highest value of specific stiffness.

875. Which statement is wrong about diamagnetic materials ?
(a) Their susceptibility is positive.
(b) Their permeability is less than one.
(c) Super-conductors are diamagnetic.
(d) They repel the external magnetic flux.

UKPSC AE 2012 Paper-I

Ans. (a) : Their susceptibility is positive.

876. Super conductivity is that state of a material at which it electrical resistance
(a) becomes zero.
(b) becomes infinite.
(c) starts showing a change.
(d) stops being affected by temperature change.

UKPSC AE 2012 Paper-I

Ans. (a) : becomes zero.

877. Assertion (A) : Soft magnets are the obvious choice for arc or high frequency applications.
Reason (R) : They must be magnetised and demagnetized many times per second.
Code :
(a) Both (A) and (R) are true, but (R) does not explain (A) correctly.
(b) Both (A) and (R) are true, and (R) explains (A) correctly.
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

UKPSC AE 2012 Paper-I

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Ans. (b) : Both (A) and (R) are true, and (R) explains (A) correctly.

878. Dielectric strength can be reduced by
(a) removing cracks
(b) absence of imperfections
(c) absence of flaws
(d) impurities, cracks and pores

UKPSC AE 2012 Paper-I

Ans. (d) : impurities, cracks and pores

879. Assertion (A) : Metallic Magnets cannot be used in high frequency circuits.
Reason (R) : The low resistivity of metallic magnets permits heating from induced currents.
Code :
(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true, but (R) does not explain (A) correctly.
(d) Both (A) and (R) are true and (R) explains (A) correctly.

UKPSC AE 2012 Paper-I

Ans. (d) : Both (A) and (R) are true and (R) explains (A) correctly.

880. Choose the correct statement from the following :
(a) Ceramics compounds involve simple coordination than their corresponding components.
(b) Ceramics compounds are more ductile.
(c) Ceramics compounds are more stable with respect to thermal and chemical environments than their components.
(d) Ceramics compounds have less resistance to slip.

UKPSC AE 2012 Paper-I

Ans. (c) : Ceramics compounds are more stable with respect to thermal and chemical environments than their components.

881. Which of the following is made of Ceramic materials?
(a) Heating element (b) Spark plug
(c) Pyrometer (d) Furnace linings

UKPSC AE 2007 Paper -I

Ans. (d) : Furnace linings

882. A material in superconducting state is
(a) paramagnetic (b) diamagnetic
(c) ferromagnetic (d) anti ferromagnetic

UKPSC AE 2007 Paper -I

Ans. (b) : Diamagnetic

883. The constituents which most ceramic material contain is
(a) Calcium (b) Nitrate
(c) Silicate (d) None of the above

UKPSC AE 2007 Paper -I

Ans. (c) : Silicate

884. Energy of a photon is
(a) hv (b) h.λ
(c) $\frac{1}{hv}$ (d) v. λ

UKPSC AE 2007 Paper -I

Ans. (a) : hv

885. The electrical conductivity of semi-conductor is of the order of
(a) 10^{-3} mho cm^{-1} (b) 10^{-6} mho cm^{-1}
(c) 10^{-8} mho cm^{-1} (d) 10^{3} mho cm^{-1}

UKPSC AE 2007 Paper -I

Ans. (a) : 10^{-3} mho cm^{-1}

886. In a dielectric, the power loss is proportional to
(a) w (b) w^2
(c) $\frac{1}{w}$ (d) $\frac{1}{w^2}$

UKPSC AE 2007 Paper -I

Ans. (a) : w

887. Dielectric materials are used primarily for
(a) insulation (b) charge storage
(c) reducing electric loss (d) none of these

UKPSC AE 2007 Paper -I

Ans. (b) : Charge storage

888. By doping, electrical conductivity of a semi-conductor
(a) decreases (b) increases
(c) remains unaffected (d) none of the above

UKPSC AE 2007 Paper -I

Ans. (b) : Increases

889. Ferrites are sub-group of
(a) ferromagnetic material
(b) ferrigmagnetic material
(c) diamagnetic material
(d) paramagnetic material

UKPSC AE 2007 Paper -I

Ans. (b) : Ferrigmagnetic material

890. Plywood is an example of which type of composite structure–
(a) Laminar (b) Sandwich
(c) Honeycomb (d) Snow skis

Nagaland PSC CTSE 2016 Paper-I
Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Plywood composite structure comes in sandwich form structure.

891. Thermoplastic polymers are
(1) Formed by addition polymerization
(2) Formed by condensation polymerization.
(3) Softened on heating and hardened on cooling for any number of times.
(4) Moulded by heating and cooling
Of these statement, select the correct answer from the options given below
(a) (ii) and (iii) are true (b) (ii) and (iv) are true
(c) (i) and (iv) are true (d) (i) and (iii) are true

UPPSC AE 12.04.2016 Paper-I

Ans : (d) Thermoplastic polymers are
(i) Formed by addition polymerization
(ii) Soft on heating and hardened on cooling for any number of times.

892. Carbon-carbon composites are extensively used at :
(a) A very low temperature applications
(b) Room temperatures
(c) Temperatures around 3000^0C
(d) Around 300^0C

OPSC AEE 2015 Paper-I

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Ans : (c)
Composite Materials :
* These are combination of two or more materials that has properties different than its constituents.
* It has almost an unlimited potential for higher strength, stiffness and high temperature capabilities over conventional materials.
* Base material which is present in large amount is called matrix the other material is referred as reinforcing material.

893. Statement (I) : The greater the chemical affinity of two metals, the more restricted is their solid solubility and greater is the tendency of formation of compound.
Statement (II) : Wider the separation of elements in the periodic table, greater is their chemical affinity.
(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).
(c) Statement (I) is true, but Statement (II) is false.
(d) Statement (I) is false, but Statement (II) is true.

ESE 2020

Ans. (a) : Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).

894. Which of the following statements is true?
(a) Ceramic materials have low melting point
(b) Porcelain is used as insulating material in spark plugs
(c) Graphite is viscoelastic in nature
(d) Compacting iron oxide powder ceramic tools are prepared

Gujarat PSC AE 2019

Ans : (b) : Porcelain is used as insulating material in spark plugs.

895. When magnetic field is removed, which type of materials maintain magnetic properties?
(a) Ferromagnetic materials
(b) Diamagnetic materials
(c) Paramagnetic materials
(d) Piezo magnetic materials

TNPSC AE 2019

Ans. (a) : Ferromagnetic materials- Ferromagnetic materials have a large positive susceptibility to an magnetic filed. They exhibit a strong attraction to magnetic fields and are able to retain their magnetic properties after the external field has been removed.
Example- Iron, Cobalt, Nickel etc.
Paramagnetic material- Paramagnetic substance are those substance which are attracted by a magnet.
Example- Aluminium, Chromium etc.
Diamagnetic materials- Diamagnetic substance are those substance which are feebly replaced by a magnet.
Example- Antimony, Bismuth, Gold, Copper, Silver, Quartz, Water . Air, Hydrogen etc,

896. Phenol and formaldehyde and polymerized to produce
(a) Bakelite (b) Polyester
(c) PVC (d) Polyethylene

UPRUVNL AE 2011
TNPSC AE 2018

Ans. (a) To produce Bakelite phenol and formaldehyde and polymerized is used.

897. Following is unique to polymeric materials
(a) Elasticity (b) Viscoelasticity
(c) Plasticity (d) None

Gujarat PSC AE 2019

Ans : (b) : A polymer is a large molecule, or macro-molecule composed of million of repeated linked units, each a relatively light and simple molecule.
The most basic property of a polymer's the identity of its constituent monomers. A second set of properties, known as microstructure, essentially describes the arrangement of these monomers within the polymer at the scale of a single chain.

898. Which among the following rubbers are used for automobile tyres?
(a) Only Polyurethane rubbers
(b) Only Butadiene rubbers
(c) Both Polyurethane and Butadiene are used
(d) None of these

Gujarat PSC AE 2019

Ans : (c) : Both Polyurethane and butadiene are used for automobile types.

899. Mechanical properties of fiber-reinforced composites depend on
(a) Properties of constituents
(b) Interface strength
(c) Fiber length, orientation and volume fraction
(d) All of the above

Gujarat PSC AE 2019

Ans : (d) : Mechanical properties of fiber - reinforced composites depend on-
• Properties of constituents.
• Interface strength.
• Fiber length, orientation and volume fraction.

900. A fatigue crack in a sound and smooth specimens takes :
(a) longer time in initiation than propagation
(b) longer time in propagation than initiation
(c) equal time in initiation and propagation
(d) no time in propagation

(HPPSC AE 2014)

Ans : (a) A fatigue crack in a sound and smooth specimen takes longer time in initiation than propagation.

901. Usually stronger constituent of a composite is :
(a) Matrix
(b) Reinforcement
(c) Both are of equal strength
(d) Can't define

OPSC AEE 2019 Paper-I

Ans : (b) : Usually stronger constituent of a composite is reinforcement.

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902. Most commercial glasses consist of :

(a) Lime (b) Soda
(c) Silica (d) All of these

OPSC AEE 2019 Paper-I

Ans : (d) : Most commercial glasses consists of Lime, soda and silica.

903. Longitudinal strength of fibre reinforced composite in mainly influenced by :

(a) Fibre strength
(b) Fibre orientation
(c) Fibre volume fraction
(d) Fibre length

OPSC AEE 2019 Paper-I
Gujarat PSC AE 2019

Ans : (a) : Longitudinal strength of fibre reinforced composite is mainly influenced by fiber strength.

904. Quartz is a

(a) ferro electric material
(b) Ferro magnetic material
(c) piezoelectric material
(d) dimagnetic material

TNPSC AE 2019
APPSC AEE 2012

Ans. (c) : Piezoelectric effect- Piezoelectric effect is the ability certain materials to generate an electric change in response to applied mechanical stress.

Piezoelectric materials-

• Quartz, • Rochelle salt, • Berilinite, • Lead titanale etc.

905. A unidirectional fibre-epoxy composite contains 65% by volume fibre and 35% epoxy resin. If the relative density of the fibre is 1.48 and of the resin is 1.2, the percentage weight of fibre will be nearly

(a) 70% (b) 75%
(c) 80% (d) 85%

ESE 2020

Ans. (a) : $\text{Density} = \frac{\text{mass}}{\text{volume}}$

Volume fraction of fibre (V_f) = 0.65

Volume fraction of epoxy (V_e) = 0.35

$$\frac{V_f}{V_e} = \frac{\frac{m_f}{\rho_f}}{\frac{m_e}{\rho_e}} \Rightarrow \frac{0.65}{0.35} = \frac{\frac{m_f}{1.48}}{\frac{m_e}{1.2}}$$

$$\frac{m_f}{m_e} = 2.29$$

$$m_f + m_e = 100$$

$$m_f + \frac{m_f}{2.29} = 100$$

$$m_f = 69.6$$

$$\approx 70$$

906. Which of the following are the advantages of Nano-composite materials?

1. Decreased thermal expansion coefficients
2. Higher residual stress
3. Reduced gas permeability
4. Increased solvent resistance

(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 1, 2 and 4 only (d) 2, 3 and 4 only

ESE 2020

Ans. (b) : 1, 3 and 4 only.

907. Filler is used in plastics to

(a) completely fill up the voids created during manufacturing
(b) improve plasticity, strength and toughness
(c) provide colour, strength, impact resistance and reduce cost
(d) to accelerate the condensation and polymerisation

TNPSC AE 2018

Ans. (c) : Filler is used in plastics to provide colour, strength, impact resistance and reduce cost.

908. Which of the following is not an example of plastic

(a) polyethylene, polystyrene,
(b) poly (ethylene terephthalate),
(c) and poly (vinyl chloride)
(d) Kovar

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : Kovar is not plastic material, Kovar is actually an iron nickel - Cobalt alloy.

909. Which of the following is not a ceramic

(a) Alumina (b) Magnesia
(c) Spinel (d) Super Invar

Nagaland PSC CTSE 2017, 2016 Paper-I

Ans. (d) : Here, super invar is not a ceramic material. It is an alloy that is made up of about 32% Ni, 5% Cobalt, Fe & other materials like. Copper, aluminium and manganese.

910. Thermosetting plastic is one,

(a) which softens when heated and hardens upon cooling
(b) which can be molded and remolded repeated
(c) which once having cured by chemical reaction, does not soften or melt upon subsequent heating
(d) which has linear polymer chain

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Thermosetting plastic is one, which once having cured by chemical reaction, does not soften or melt upon subsequent heating.

911. Thermoplastic is one,

(a) which softens when heated and hardens upon cooling
(b) which can be moulded and remoulded
(c) which has linear polymer chain
(d) which has all three characteristics

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Thermoplastic is one, which has all three characteristics.

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912. Fibers used for fiber-reinforced-plastics are made of–

(a) steel wires (b) hemp
(c) glass and carbon (d) asbestos

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Fibers used for fiber-reinforced-plastics are made of glass and carbon.

913. Synthetic rubber is used for–

(a) V belt (b) gasket
(c) seals (d) all of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Synthetic rubber is used for have V-belt, gasket, seals.

914. Which of the following statement is incorrect about thermoplastic polymer?

(a) Polyvinyl chloride (PVC) is an example of thermoplastic polymer
(b) They cannot be repeatedly soften under heating and harden on cooling
(c) They can be reshaped and recycled
(d) These are linear or branched linear polymers

UPRVUNL AE 07.10.2016

Ans. (b) : Thermoplastic polymer they cannot be repeatedly soften under heating and harden on cooling is incorrect statement.

915. Which among the following is the characteristics of polymers?

(a) High tensile strength
(b) High coefficient of friction
(c) Low density
(d) Higher viscosity

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) Low density is the characteristics of polymers.

916. Thermosetting plastics are those materials which

(a) are formed to shape under heat and pressure and results in a permanently hard product
(b) do not become hard with application of heat and pressure
(c) are flexible and can withstand considerable wear under suitable conditions
(d) are used as a friction lining for clutches and brakes

JPSC AE 2013, Paper-V

Ans : (a) : Thermosetting Plastics are those which are formed into shape under heat and pressure and results in permanently hard product. The heat first softens the material, but as additional heat and pressure is applied, it becomes hard by a chemical change known as phenol formaldehyde (Bakelite), Phenol-furfural (Durite), urea-formaldehyde etc.

917. The chemical name of bakelite is

(a) polytetrafluoroethylene
(b) phenol formaldehyde
(c) phenol acetaldehyde
(d) urea formaldehyde

BPSC AE 2012 Paper - VI
GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans : (b) : The chemical name of bakelite is phenol formaldehyde.

918. Which of the following is a thermoplastic?

(a) Bakelite (b) Epoxy resins (araldite)
(c) Polypropylene (d) Polyurethanes

BPSC AE Mains 2017 Paper - VI

Ans : (c) : Some of the common thermoplastics are-
- Cellulose nitrate (celluloid)
- Polythene
- Polyvinyl acetate
- Polyvinyl chloride (P.V.C.)
- Polypropylene

919. Which of the following are weldable plastic?

(a) Thermosets alone
(b) Thermoplastics alone
(c) Both thermosets and thermoplastics
(d) Neither thermosets and thermoplastics

BPSC AE 2012 Paper - VI

Ans : (b) : The thermoplastic materials do not become hard with the application of heat and pressure and no chemical change occurs. They remain soft at elevated temperatures until they are hardened by cooling. These can be remelted repeatedly by successive application of heat. Some of the common thermoplastics are cellulose nitrate, polyethylene, polyvinyl acetate, polyvinyl chloride (P.V.C.).

920. The process of growing large molecules from small molecules is known as :

(a) Polymerization (b) Polymorphism
(c) Hysteresis (d) Allotropy

HPPSC W.S. Poly. 2016

Ans : (a) The process of growing large molecules from small molecules is known as polymerization.

921. These polymers cannot be recycled:

(a) Elastomers (b) Thermoplasts
(c) Thermosets (d) All polymers

(HPPSC LECT. 2016)

Ans : (c) Thermosets Polymers Cannot be recycled.
Thermosetting Plastics:- They are formed from the intermediate products which under the influence of heat pressure etc. undergo chemical changes of condensation and Polymerisation to form a rigid final shape which is unaffected by heat or solvents.

922. Which one of the following is a basic refractory material?

(a) Dolomite (b) Quartz
(c) Sand (d) Silicon carbide

UPPSC AE 12.04.2016 Paper-I

Ans : (a)
Acidic refectories :- Ziricon, fire clay and silica
Basic refectories :- Dolomite and magnetic
Neutral refectories :- Alumina and chromites.

923. Example of large particle composites is

(a) cemented carbide
(b) natural composites
(c) particle reinforced composite
(d) None of the above

APPSC Poly. Lect. 2013

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Ans. (c) : Large-particle composite is concrete, which is composed of cement (the matrix), sand and graval (the particulates)

924. Melting point of iron is -

(a) 1539°C (b) 1601°C
(c) 1489°C (d) 1712°C

Mizoram PSC (Insp. of factories) 2019 Paper - III
MPSC lect (Prod.) 2018
Rajasthan AE Nagar Nigam 2016 shift-3
GMB AAE 25.06.2017
WBPSC AE 2007
Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : Melting point of iron 1539°C
Boiling point of iron 2862°C

925. Plain carbon steels are designated in the AISI code system by which of the following?

(a) 01XX (b) 10XX
(c) 11XX (d) 12XX

GPSC ARTO Pre 30.12.2018

Ans. (b) : Plain carbon steels are designated in the AISI code.

10	XX
Type of Material Selected	Amount of Carbon present in steel

926. Which of the following is not a common alloying ingredient in steel?

(a) carbon (b) chromium
(c) nickel (d) molybdenum

GPSC ARTO Pre 30.12.2018

Ans. (a) : Carbon is not a common alloying ingredient in steel.

927. Which of the following has the lowest melting point?

(a) aluminum (b) lead
(c) magnesium (d) tin

GPSC ARTO Pre 30.12.2018

Ans. (d) :

Material		Melting Point
Aluminium	-	660.3°C
Magnesium	-	650°C
Lead	-	327.5°C
Tin	-	231.9°C

928. Alloy steel as compared to carbon steel is more:

(a) Tough (b) Strong
(c) Fatigue resistance (d) All of the above

APGENCO AE 2012

Ans. (d) : Alloy steel have a high percentage of other elements apart from iron and carbon. Commonly alloysteel has comparatively high strength, low, weld ability, high melting points, high ductility and high corrosion resistance.

Carbon steel- Carbon steel is also divided into high carbon, steel, medium, carbon steel and low carbon steel. However, unlike alloy steel, this is the main distinction between carbon steel types.

929. High oxygen levels in duplex stainless steel results into :

(a) Reduction of toughness
(b) Increase in weld corrosion
(c) Both of above
(d) None of above

Punjab PSC SDE 12.02.2017

Ans. (c) : High oxygen levels in duplex stainless steel results into.
- Reduction of toughness.
- Increases in weld corrosion.

930. The maximum amount of carbon that can be alloyed with iron is :

(a) 1.4% (b) 3.4%
(c) 3.67% (d) 6.67%

Punjab PSC SDE 12.02.2017

Ans. (d) : The maximum amount of carbon that can be alloyed with iron is 6.67%.

931. The American designation for mild steel En 3A is:

(a) AISI 1020 (b) AISI 1060
(c) AISI 4142 (d) AISI 5140

Punjab PSC SDE 12.02.2017

Ans. (a) : The American designation for mild steel En 3A is AISI 1020.

932. In grey cast iron, carbon is present in the form of

(a) Cementite (b) Free carbon
(c) Flakes (d) Spheroids
(e) None of the above

CGPSC Poly. Lect. 22.05.2016
APPSC AEE 2012

Ans. (c) : Iron-carbon alloys containing flakes of graphite embedded in steel matrix, which show a grey blackish coloured fracture due to graphite the free form of carbon are called gray cast iron.
The strength of grey cost iron depends on the strength of steel matrix and the size and character of graphite flakes in it.

933. Which spring steel is widely used for aircraft engine valves

(a) Oil tempered wire (0.6 C to 0.7 C)
(b) Chrome vanadium
(c) Hard drawn wire (0.6 C to 0.7 C)
(d) Phosphor bronze wire

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : Chrome vanadium spring steel is widely used for aircraft engine valves.

934. Conductivities of semi conductors range from

(a) 10^{-9} to 10^{4} $ohm^{-1}cm^{-1}$
(b) 10^{-8} to 10^{3} $ohm^{-1}cm^{-1}$
(c) 10^{-7} to 10^{4} $ohm^{-1}cm^{-1}$
(d) 10^{-9} to 10^{3} $ohm^{-1}cm^{-1}$

TNPSC AE 2017

Ans. (a) : Conductivities of semi conductors range from 10^{-9} to 10^{4} $ohm^{-1}cm^{-1}$.

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935. Which of the following has maximum malleability?

(a) Lead (b) Brass
(c) Wrought iron (d) Aluminum

KPSC AE 2015

Ans : (a) Lead has maximum malleability in given option.
Malleability:- It is a special case of ductility which permit material to be rolled or hammered into thin sheets. A malleable material should be plastic but it is not essential to be so strong.

936. A piece of steel was found to contain 0.65% C, 4% Cr, 18% W and 1% V. Which kind of steel is it?

(a) High alloy steel (b) Low alloy steel
(c) Carbon steel (d) High speed steel
(e) Stainless steel

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) : The 18-4-1 means high steel, having 18% of tungsten, 4% of Cr and 1% of vanadium, used as cutting tool.

937. "A very weak form of magnetism that is non-permanent, persist only when an external field is applied and having a negative value of magnetic susceptibility" is known as :

(a) dielectric strength (b) diamagnetism
(c) ferromagnetism (d) ferrimagnetism

APPSC Poly Lect. 13.03.2020

Ans. (b) : Diamagnetism is "A very weak form of magnetism that is non-permanents persist only when external field is applied and having a negative value of magnetic susceptibility.

Magnet	**Magnetic Susceptibility (χ)**
Ferromagnetism →	$\chi >> 1$
Paramagnetism →	$\chi > 1$
Diamagnetism →	$\chi < 1$

938. An oxidizing process used for aluminium and magnesium articles is called

(a) Galvanizing (b) Anodizing
(c) Parkerizing (d) Sheradizing

ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Anodizing is an electrolytic passivation process used to increase the thickness of the neutral oxide layer on the surface of metal parts. The process is called anodizing because the part to be treated forms the anode electrode of an electrical circuit. Anodic films are most commonly applied to protect aluminium and magnesium alloys/articles.

939. One of the characteristics of Polymer is

(a) high Temperature Stability
(b) High Mechanical Strength
(c) High Elongation
(d) Low Hardness

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : One of the characteristics of Polymer is high elongation

940. The property of corrosion resistance of chromium Stainless Steel is due to

(a) Predominant nature of Cr present in stainless steel
(b) The formation of thin film of oxygen and moisture absorbed from the atmosphere
(c) The formation of a thin film of Cr_2O_3 on the surface of steel
(d) The inherent property of Cr to resist corrosion

ISRO Scientist/Engineer 2012

Ans. (c) : The property of corrosion resistance of chromium Stainless Steel is due to the formation of a thin film of Cr_2O_3 on the surface of steel

941. The iron ore mostly used for the production of pig iron is ..... .

(a) Magnetite (b) Hematite
(c) Limonite (d) Pyrite

Mizoram PSC (Insp. of Factories) 2019 Paper-III

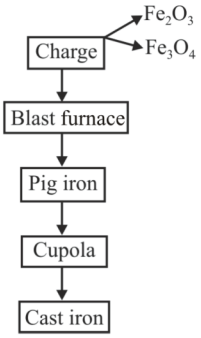
Ans. (b) : The iron ore mostly used for the production of pig iron is Hematite because it contains more % of iron than others.

942. The Cupola is used to manufacture .....

(a) Pig iron (b) Cast iron
(c) Steel (d) None

GPSC ARTO Pre 30.12.2018
WBPSC AE 2003
Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : Cupola furnace – By melting pig iron, large quantities of cast iron can be produced.
Melting Ratio / Change Ratio
Metal : Fuel
4:1 → 12:1
Cupola → 10:1



943. In case of ferromagnetic materials, the spin moments associated with two sets of atoms are aligned

(a) parallel to each other
(b) anti-parallel to each other
(c) anti-parallel but unequal magnitude
(d) randomly

TNPSC AE 2018

Ans. (a) : In case of ferromagnetic materials, the spin moments associated with two sets of atoms are aligned parallel to each other.

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944. Perm alloy is

(a) a non-ferrrous alloy used in aircraft industry
(b) a polymer
(c) a nickel and iron alloy having high permeability
(d) a kind of stainless steel

TNPSC AE 2018

Ans. (c) : Perm alloy is a nickel and iron alloy having high permeability.

945. Which of the following is a copper free alloy?

(a) Brass (b) Phosphor bronze
(c) Invar (d) Muntz metal

TNPSC AE 2017
GPSC AE 26.07.2015

Ans. (c) : Invar - Invar is a nickel steel alloy notable for its uniquely low coefficient of thermal expansion.

946. Match the List I alloys with List II applications and select the correct answer using the codes given below.

	List I	List II
(A)	Chromel	1. Journal bearing
(B)	Babbit alloy	2. Milling cutter
(C)	Nimonic alloy	3. Thermo couple wire
(D)	High speed steels	4. Gas turbine blade

	(A)	(B)	(C)	(D)
(a)	3	1	4	2
(b)	3	4	1	2
(c)	2	4	1	3
(d)	2	1	4	3

TNPSC AE 2017
GPSC ARTO 01.05.2016

Ans. (a) :

	List I		List II
(A)	Chromel	→	Thermo couple wire
(B)	Babbit alloy	→	Journal bearing
(C)	Nimonic alloy	→	Gas turbine blade
(D)	High speed steels	→	Milling cutter

947. According to Indian standard specifications, SG 400/15 means

(a) Spheroidal graphite cast iron with B.H.N 400 and minimum tensile strength 15 MPa
(b) Spheroidal graphite cast iron with minimum tensile strength 400 MPa and 15% elongation
(c) Spheroidal graphite cast iron with minimum compressive strength 400 MPa and 15% reduction in area
(d) Spherioidal graphite caston with maximum tensile strength 400 MPa with 15% elongation.

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : According to Indian standard specifications, SG 400/15 means Spheroidal graphite cast iron with minimum tensile strength 400 MPa and 15% elongation.

948. Elastic deformation in polymers is due to

(a) Slippage of molecular chains
(b) Straightening of molecular chains
(c) Slight adjust of molecular chains
(d) Covalent bonds
(e) Stretching of molecular chains

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Elastic deformation—Bending and stretching of covalent bonds and slight adjustments of secondary Vander Waals forces.

949. Silicon steel is widely used in

(a) cutting tools (b) dies and punches
(c) electrical industry (d) chemical industry

ISRO Scientist/Engineer 11.10.2015

Ans. (c) : It is also called electrical steel or transformer steel. It is widely used in electrical system because of low hysterisis loss and high magnetic permeability. Silicon steel is widely used in making transformer cores.

950. Shock resisting steel should have higher?

(a) stiffness (b) hardness
(c) toughness (d) shear strength

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : Shock resisting steel offers good ductility and toughness at high hardness levels.

951. Fillers like asbestos are used in plastics to :

(a) Enhance the strength and hardness
(b) Improve heat and corrosion resistance
(c) Reduce the production cost
(d) All of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : Fillers like asbestos are used in plastics to reduce the production cost.

952. Which of the following elements does not impart hardness to steel?

(a) copper (b) chromium
(c) nickel (d) silicon

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (a) : Copper is a soft, malleable and ductile metal with very high thermal and electrical conductivity and it is not impart hardness to steel.

953. Carbon steel is

(a) made by adding carbon in steel
(b) refined from cast iron
(c) an alloy of iron and carbon with variable quantities of phosphorous and sulphur
(d) extremely brittle

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (c) : Carbon steel is an alloy of iron and carbon with variable quantities of phosphorous and sulphur.
Most common type of steel that contains about 0.1 to 0.3%, carbon.
In general, increase in the amount of carbon reduces ductility but increases tensile strength and the ability to harden through tempering. As an industry wide practice, steel that does not contain any spur thread or standard amount of one or more alloying elements to be classified as alloy steel is called carbon steel.

954. Which one of the following tool material exhibits highest cutting speed?

(a) High speed steel (b) Ceramic
(c) Carbon steel (d) Coated carbide

UPSC JWM Adv. No-16/2009

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Ans. (b) : Ceramic is a tool material which exhibits highest cutting speed. In the ceramic tool material Negative rake angle is provide.

955. Consider the following materials :

1. Carbon
2. Mica
3. Bakelite
4. Fibre glass

Which of these materials are good conductors of heat, but bad conductors of electricity?

(a) 1 only (b) 2 only
(c) 2 and 3 (d) 3 and 4

UPSC JWM Adv. No-16/2009

Ans. (a) : Carbon

956. The material commonly used for crane hooks is

(a) cast iron (b) wrought iron
(c) mild steel (d) aluminium

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : The material which is suitable for crane is wrought iron, because it has good toughness strength as compared to other materials.

957. The steel widely used for motor car crankshafts is

(a) nickel steel (b) chrome steel
(c) nickel chrome steel (d) high speed steel

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : The steel widely used for motor car crankshaft is chrome steel because it has high wear & corossion resistance properties & with stand high temperature and have high tensile strength.

958. Silicon in cast iron

(a) makes the iron soft and easily machinable
(b) increases hardness and brittleness
(c) makes the iron white and hard
(d) aids fusibility and fluidity

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : As we know, a high percentage of silicon, forces carbon out of solution forming graphite & the production of grey cast iron. It makes iron soft & easily machinable sure to free carbon.

959. The addition of magnesium to cast iron increases its

(a) hardness
(b) ductility and strength in tension
(c) corrosion resistance
(d) creep strength

Nagaland PSC (CTSE) 2018 Paper-I, ESE 2018

Ans. (b) : Magnesium is a powerful inoculant in cast iron, when added in concentrations exceeding 0.04%, it produces a spheroidal form of graphite that improves the iron's strength & ductility.

960. According to Indian standard specifications, a plain carbon steel designated by 40C8 means that the carbon content is

(a) 0.04% (b) 0.35 to 0.45%
(c) 0.4 to 0.6% (d) 0.6 to 0.8%

Nagaland PSC (CTSE) 2018 Paper-I

Ans. (b) : Carbon steel → 40C8

Avg $C = \frac{40}{100}$% carbon range → 0.35 to 0.45%

8 → Denotes 0.8% of manganese

40 → C% × 100 → Avg of C%

C → Carbon

8 → Mn% × 10

961. Muntz metal (Yellow brass) contains

(a) 70% copper and 30% zinc
(b) 60% copper and 40% zinc
(c) 59% copper 40% zinc 1% tin
(d) 60.45% copper 35.2% zinc and 5.35% nickel

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : Muntz metal also called Yellow Metal, variety of the alloy brass consisting of 60% of copper & 40% of zinc used to make machine parts that require resistance to corrosion.

962. Monel metal contains

(a) 65% nickel, 15% chromium and 20% iron
(b) 68% nickel, 29% copper and 3% other constituents
(c) 80% nickel, 20% chromium
(d) 80% nickel, 14% chromium and 6% iron

Nagaland PSC (CTSE) 2018, Paper-I
Kerala PSC Poly. Lect. 09.01.2014

Ans. (b) : Monel metal is of a group of nickel, copper alloys. It containing about 66% nickel & 31.5% copper & very small amount of iron, Mn, C & Si.

963. The cold chisels are made from

(a) cast iron (b) mild steel
(c) high speed steel (d) cast tool steel

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Cold chisel are made from carbon-tool steel, which is usually octagonal in cross-section. Tool steel is used as it can be hardened to form a hard and tough cutting edge. To make a cold chisel heat the end of the bar of tool steel until it is bright red.

964. The binding material used in cemented carbide tools is

(a) tungsten (b) chromium
(c) silicon (d) cobalt

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : The material that makes up the tools insert grains of tungsten carbide, along with particles of other materials, cemented together using the metal cobalt as a binder.

965. Thermosetting plastic among the following is

(a) Nylon (b) Teflon
(c) PVC (d) Bakelite

Vizag Steel MT (Re-Exam) 24.11.2013

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Ans. (d) : Thermoplastic–These are the polymers that can be easily softened repeatedly when heated and hardened when cooled with little change in their properties.
Example–Polystyrene, polyethylene, Teflon, polyvinylchloride etc.
Thermosetting Plastics–These are the polymers that undergo permanent change on heating. On heating they undergo extensive cross-linking in mould and become hard and infusible, therefore, they can not be reduced.
Example–Bakelite, Terrylene, melamine etc.

966. The percentage of carbon in Grey cast iron is in the range of
(a) 0.25 to 0.75 percent
(b) 1.25 to 1.75 percent
(c) 3 to 4 percent
(d) 8 to 10 percent

VIZAG STEEL MT 10.06.2012
VIZAG STEEL MT 18.08.2013
GATE 2004

Ans. (c) : The percentage of carbon in gray cast is 3 to 4 percent.

967. The most widely used reinforcement in modern day FRP tennis racket is
(a) Aluminum (b) Carbon fibre
(c) Magnesium (d) Glass fibre

GPSC Executive Engineer 23.12.2018

Ans. (b) : Carbon fibre

968. Muntz metal contains copper and zinc in the ratio :
(a) 1 : 1 (b) 2 : 3
(c) 3 : 2 (d) 1 : 4

Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Muntz metal – 3 : 2
It contains 60% Cu, 39% Zn and 1% Sn.

969. In brass, copper is alloyed with:
(a) tin (b) lead
(c) zinc (d) nickel
(e) duralumin

CGPSC ARTO & TSI 07.01.2017

Ans. (c) :
- Brass = Copper + Zinc $\Rightarrow$ Cu+Zn
- Bronze = Copper + Tin $\Rightarrow$ Cu+Sn
- Gun metal is an mixture of copper with zinc and tin in proportion of 88%, 2-4% and 8-10% respectively.

970. Monel metal has chemical composition primarily by A = 70 percent and B = 30 percent approximately. A and B are _____ and _____, respectively.
(a) Cu and Ni (b) Cu and Sn
(c) Ni and Cu (d) Sn and Cu

AAI Jr. Executive 26.03.2021

Ans. (c) : Monel metal composition,
70% Nickel (Ni)
30% Copper (Cu)

971. Which of the following properties of tin increases when it is added in lead?
(a) Strength
(b) Hardness
(c) Strength and hardness
(d) Strength and ductility

VIZAG STEEL MT 2011

Ans. (c) : Strength and hardness

972. What does the polymeric materials composed of carbon compounds known as?
(a) Organics (b) Ceramic
(c) Ferrous (d) Non-ferrous

VIZAG STEEL MT 2011

Ans. (b) : Ceramic

973. Property of cast iron is :
(a) good wear resistance
(b) good casting characteristic
(c) good machinability
(d) all of these

Kerala PSC IOF 19.04.2016

Ans. (d)
- Wear resistance
- Casting characteristic
- Machinability

974. Phosphor bronze has :
(a) high resistance to corrosion
(b) good wearing qualities and high elasticity
(c) valuable cold working property
(d) all the above

Kerala PSC IOF 19.04.2016

Ans. (b) Phosphor bronze has good wearing qualities and high elasticity.

975. The effect of alloying zinc to copper is to:
(a) improve welding characteristics
(b) improve hardness
(c) impart free machining properties
(d) none of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Zinc has very minor effect on the thermal conductivity of copper, followed by increasing effects from nickel, tin, manganese, silicon and serious effects from phosphorus. Phosphorous is often used to deoxidizing copper, which can increase the hardness and strength but severally affect the conductivity.
So, correct option is none of these.

976. Permalloy is a:
(a) nonferrous alloy
(b) combination of nickel and iron alloy
(c) alloy of tin, lead and cadmium
(d) kind of stainless steel

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (b) : Permalloy is a nickel-iron magnetic alloy, with about 80% nickel and 20% iron content.

977. Which of the following materials cannot be forged?
(a) wrought iron (b) cast iron
(c) mild steel (d) high speed steel

Kerala PSC AE (Irrigation) 05.04.2017

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Ans. (b) : If the carbon content is over 2%, the metal is called cast iron, because if has a relatively low melting point and is easily cast. It is quite brittle, however, and cannot be forged so therefore not used for blacksmithing.
Material can be forged–Carbon alloy and stainless steels, very hard tool steels, Aluminium, titanium, brass and copper and high temperature alloy can be forged.

978. Simple cast iron with no major alloying additions
(a) is the product of blast furnace
(b) has a carbon content in the ranges of 0.2 to 2.0 wt%
(c) has a carbon content in the range of 2.0 to 5.0 wt%
(d) always contains martensite

APPSC AEE 2012

Ans. (c) : Simple cast iron with no major alloying additions has a carbon content in the range of 2.0% to 6.7% (wt%).

979. Martensite in steel is
(a) Hard and body centered cubic
(b) Soft and body centered cubic
(c) Hard and body centered tetragonal
(d) Soft and body centered tetragonal

APPSC AEE 2012

Ans. (c) : Martensite in steel is – Hard and body centered tetragonal

980. Corrosion resistance of steel is increased by adding
(a) Zinc
(b) Tungsten and Molybdenum
(c) Chromium and Nickel
(d) Vanadium and Sulphur

APPSC AEE 2012

Ans. (c) : Corrosion resistance of steel is increased by adding chromium and Nickel.

981. Manganese in steel increases its
(a) tensile strength (b) hardness
(c) ductility (d) malleability

APPSC AEE 2012

Ans. (a) : Manganese increases hardenability and tensile strength of steel, but to a lesser extent than carbon. It is also able to decrease the critical cooling rate during hardening thus increasing the steel hardenability mach more efficient than any other alloying elements.

982. Presence of Sulphur makes steel brittle. Its effect can be reduced by adding
(a) Copper (b) Silicon
(c) Vanadium (d) Mangnese

APPSC AEE 2012

Ans. (d) : Presence of sulphur makes steel brittle. Its effect can reduced by adding mangnese.

983. Low carbon steel can be hardened by
(a) heating and quenching in oil
(b) heating and quenching in water
(c) carburizing and cyaniding
(d) Any of the above

APPSC AEE 2012

Ans. (c) : Carburizing & cyaniding – It is also referred to as case hardening, is a heat treatment process that produced a surface which is resistant to wear, while maintaining toughness and strength of core. This treatment is applied to low carbon steel parts after machining, as well as high alloy steel bearing

984. Incoloy principally contains nickel, chromium and _____.
(a) molybdenum (b) carbon
(c) iron (d) magnesium

UPRVUNL AE 05.07.2021

Ans. (c) : Incoloy is an alloy of nickel, iron, chromium contains less than 50% nickel.

985. Residual stresses are called ____ stresses.
(a) load (b) locked-in
(c) surface (d) top in

UPRVUNL AE 05.07.2021

Ans. (b) : Residual stress are called looked in stresses. Residual stresses result in significant plastic deformation, leading to warping and distortion of an object.

986. Galvanised iron sheets have a coating of _____
(a) aluminium (b) tin
(c) zinc (d) copper

UPRVUNL AE 05.07.2021

Ans. (c) : Galvanization is the process of applying a protective zinc coating to steel or iron in order to prevent it from rusting.

987. The minimum corrosion allowance for pressure vessel is
(a) 0.5 mm (b) 1.5 mm
(c) 1.0 mm (d) 2.0 mm

GPSC AE 26.07.2015

Ans. (b) : Generally, a minimum corrosion allowance of 1.5 mm shall be provided for carbon steel material unless otherwise specify.

988. The class- C insulation can withstand
(a) 180ºC (b) 150ºC
(c) 200ºC (d) 105ºC

GPSC AE 26.07.2015

Ans. (a) : Class-C insulation withstand a temperature of up to 180°C or higher, it is typically made of 100% inorganic material as explained above electrical insulation is classified with it's maximum allowable temperature.

989. If in polyethylene, all the hydrogen atoms are replaced by fluorine atoms, we shall get _____.
(a) Polystyrene
(b) Polyurethane
(c) Polytetrafluoroethylene
(d) Polyamides

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : If in polyethylene, all the hydrogen atoms are replaced by fluorine atoms, we shall get polytetrafluoroethylene.

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990. Resinoid bond wheels are made from bakelite and other resinous material and its is denoted by letter _____.

(a) R (b) C
(c) A (d) B

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Resinoid bond wheels are made from bakelite and other resinous material and its is denoted by letter B.

991. _____ increases fusibility and fluidity in cast iron but induces brittleness.

(a) Phosphorus (b) Aluminum
(c) Zinc (d) Silver

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : Phosphorus increases fusibility and fluidity in cast iron but induces brittleness.

992. Which of the following are the advantages of polymer composite materials?

1. Higher Specific Strength
2. High Specific Modulus
3. Higher Corrosion Resistance
4. Higher Residual Stresses

(a) 1, 2, 3 (b) 1, 2, 4
(c) 1, 3, 4 (d) 1, 2, 3, 4

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : The advantages of polymer composite materials are -
1. Higher Specific Strength
2. High Specific Modulus
3. Higher Corrosion Resistance
4. Higher Residual Stresses

993. 'Killed Steels' are those steels ..... .

(a) which are destroyed by burning
(b) which after their destruction are recycled to produce fresh steel
(c) which are deoxidised in the ladle with silicon and aluminium
(d) in which carbon is completely burnt

GWSSB DEE 07.07.2016

Ans. (c) : which are deoxidised in the ladle with silicon and aluminium

994. Which of the following materials has a non-linear stress-strain curve?

(a) copper (b) rubber
(c) low carbon steel (d) aluminium

HPPSC Poly. Lect. 05.07.2021

Ans. (b) : rubber

995. AISI 1060 steel contains :

(a) 0.106% by wt of carbon
(b) 10.6% by wt of carbon
(c) 1.06% by wt of carbon
(d) 0.6% by wt of carbon

DRDO Scientists 2009

Ans. (d) : AISI 1060 carbon steel composition

Iron	→	98.35-98.85%
Managanese	→	0.60-0.90%
Carbon	→	0.55-0.66%
Sulphur	→	≤ 0.05%

996. Graphene is

(a) New material made from carbon nanotubes
(b) One-atom thick sheet of graphite
(c) A thin film made from fullerence
(d) A graphical representation of C60 molecule

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : One-atom thick sheet of graphite

997. Which of the following insulting material has the least affinity to moisture?

(a) Cotton (b) Paper
(c) Asbestos (d) Mica

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : insulting material has the least affinity to moisture– Asbestos.

998. Which one of the following features improves the fatigue strength of a metallic material?

(a) Increasing the temperature
(b) Scratching the surface over stressing
(c) Over stressing
(d) Under stressing

APPSC AEE 2012

Ans. (d) : Fatigue strength of metallic materials is improved by under stressing.

999. Cermets are

(a) metals for high temperature use with ceramic like properties
(b) ceramics with metallic strength and luster
(c) coated tool materials
(d) metal-ceramics composites

APPSC AEE 2012
APPSC AE 04.12.2012

Ans. (d) : Cermets- is a composite material composed of ceramic and metal materials. A ceramic is ideally designed to have the optimal properties as both a ceramic such as high temperature resistance and hardness.

1001. α-stabilizers are

(a) Si, Al, Cr (b) Mg, Cu, Co
(c) Al and Mn (d) Ni, Mn, Cu

TANGEDCO AE 2015

Ans. (a) : Alpha and beta phase are used in Titanium processing. HCP structure (α phase) and BCC structure (β Phase).
α stabilizers – Al, Si, Ga, Zr, C, O, N.
β stabilizers – V, Mo, Nb, Ta, Cr

1002. The mass fraction of α ferrite and cementite in pearlite is

(a) 0.77 and 0.33 respectively
(b) 0.7 and 0.3 respectively
(c) 0.02 and 0.98 respectively
(d) 0.888 and 0.112 respectively

TANGEDCO AE 2015

Ans. (d) : Mass fraction of α-ferrite and cementite in pearlite is 0.888 and 0.112 respectively since C_c = 6.67%, C_x = 0.22% and CD = 0.77%.

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1003. Magnetic permeability is maximum for ____ material.

(a) Diamagnetic (b) Paramagnetic
(c) Ferromagnetic (d) None of the above

TANGEDCO AE 2015

Ans. (c) : Magnetic permeability is the measure of the degree to which magnetic field can penetrate through material. It is scalar quantity and maximum for ferromagnetic material.

1004. Composite materials generally have

(a) High Elastic Modulus
(b) High Specific Strength
(c) Both 1 and 2
(d) Low Stiffness

TANGEDCO AE 2015

Ans. (c) : Composite materials are generally formed to increase elastic modulus and specific strength.

1005. Nodular cast iron is produced by adding _____ to the molten cast iron.

(a) Copper (b) Chromium
(c) Nickel (d) Magnesium

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Nodular cast iron is produced by adding magnesium to the molten cast iron, Mn = 0.28% to 0.49%.

1006. Mircrometer measuring faces are tipped with _____ to prevent rapid wear.

(a) Chromium (b) Tungsten carbide
(c) Molybdenum (d) Copper

Oil India Senior Officer 23.12.2020

Ans. (b) : Mircrometer measuring faces are tipped with tungsten carbide to prevent rapid wear.

1007. Which one of the following is most ductile?

(a) Copper (b) Silver
(c) Platinum (d) Iron

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : Gold and platinum are the earth's most ductile metals, but gold has a significantly greater ductility than platinum.

1008. The process of reducing ore with carbon in the presence of flux is called as

(a) Hardening (b) Smelting
(c) Quenching (d) Shaping

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : The process of reducing ore with carbon in the presence of flux is called as smelting.

1009. When a Ge crystal doped with phosphorus atom it becomes

(a) P-type semiconductor
(b) N-type semiconductor
(c) Photosensitive
(d) An insulator

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : When a Ge crystals are doped with phosphorus it becomes N-type semiconductor.
If it is doped with trivalent impurity like boron then it becomes a P-type semiconductor.

1010. An engineer's hammer is made of :

(a) cast iron (b) bell metal
(c) high carbon steel (d) gun metal

GWSSB AAE, 27.12.2015

Ans. (c) : Engineer's hammer is made of high carbon steel.

1011. When _____ is added to, its tensile strength of steel can be increased by adding

(a) Chromium (b) Manganese
(c) Magnesium (d) Tungsten

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : Chromium–Increase resistance to corrosion, abrasion, wear and oxidation.
Manganese–Increase hardinability, tensile straight, act as deoxidizer.
Tungsten–Increase wear resistance and imparts red hardness.

1012. Which one of the following alloying ingredients increases the harden ability and forms carbides for wear resistance?

(a) Chromium (b) Molybdenum
(c) Nickel (d) Manganese

ESE 18.07.2021

Ans. (b) : Carbide forming tendency in Chromium is greater than Manganese (Mn) but lesser than tungsten (W). Hardenability increases moderately.
Manganese has consider formation tendency greater than iron (Fe) in steel but lesser than chromium. Manganese increases handenabilty moderately.
Nickel leads to formation graphite that it has negative effect on carbide formation but it stabilized the Austenite.
Molybdenum leads to formation of carbide more than chromium and increases hardenability strongly.

1013. The dielectric constant of rubber varies between

(a) 0.5 and 1.0 (b) 1.0 and 1.5
(c) 1.5 and 2.0 (d) 2.5 and 5.0

ESE 18.07.2021

Ans. (d) : Dielectric Constant :
Rubber - 2 to 4
Nylon - 3.4 to 22.4
Air - 1 to 1
Backelite - 5 to 22
Glass - 3.8 to 14.5

1014. Consider the following statements regarding polymeric materials:

1. **A plastic material that requires heat to make it formable (plastic) and upon cooling, retains its shape is known as thermosetting plastic.**
2. **The chemical reaction in which high molecular mass molecules are formed from two or more monomers is called chain polymerization.**
3. **A polymer chain consisting of two or more types of monomeric units is called copolymer.**

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Which of the above statements is/are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 2 only (d) 3 only

ESE 18.07.2021

Ans. (b) :

- Thermoplastic materials are those which soften on the application of heat with or without pressure, but then require cooling to set them to shape.
- Thermosetting materials are those plastic which require heat and pressure to mold them into shape. They can not be resoftened as once they have set and hardened. Hence statement 1 is wrong.

1015. As per mechanical properties, which one of the following microconstituents is soft and ductile?

(a) Bainite
(b) Martensite
(c) Spheroidite
(d) Tempered martensite

ESE 18.07.2021

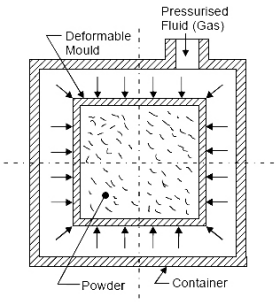
Ans. (c) : Decreasing order of Ductility-
Spheroidite > Coarse pearlite > fine pearlite > bainite > tempered martensite > martensite.
Decreasing order of hardness-
Tempered martensite > bainite > fine pearlite > coarse pearlite > spheroidite.

1016. The simultaneous compaction and shaping of a ceramic powder (and binder) by pressure applied uniformly in all directions is known as

(a) Glaze pressing (b) Porcelain pressing
(c) Slip pressing (d) Isostatic pressing

ESE 18.07.2021

Ans. (d) : The simultaneous compaction and shaping of a ceramic powder (and binder) by pressure applied uniformly in all directions is known as Isostatic pressing.



1017. Match the following :

(List-I) (Alloying element in Steel)	(List-II) (Effect)
A. Copper	1. Free Machinning
B. Phosphorus	2. Restricts grain growth
C. Lead	3. Raises yield point
D. Aluminimum	4. Reduces strength to impact

(a) A-3, B-4, C-1, D-2 (b) A-1, B-2, C-3, D-4
(c) A-4, B-1, C-2, D-3 (d) A-2, B-3, C-4, D-1

UPRVUNL AE 21.08.2016

Ans. : (a)

Alloying element in steel	Effect
Copper	Raises yield point
Phosphorus	Reduces strength to impact
Lead	Free machining
Aluminium	Restricts grain growth

1018. Beryllium is used chiefly as an alloy addition to copper for producing

(a) precipitation - hardenable alloy
(b) corrosion resistance alloy
(c) high - strength alloy
(d) non - magnetic and non-sparking alloy

TNPSC AE 2018

Ans. (d) : Beryllium is used chiefly as an alloy addition to copper for producing non - magnetic and non-sparking alloy.

1019. Low carbon steel:

(a) has lowest tensile strength relative to high carbon steel
(b) has carbon percentage upto 1.5%
(c) is ideal steel for heat treatment
(d) is harder than high carbon steel

UPRVUNL AE 07.10.2016

Ans. (a) : Low carbon steel—Low carbon steel contains approximately 0.05 to 0.30% carbon making it malleable and ductile. Mild steel has a relatively low tensile strength relative to high carbon steel, but it is cheap and easy to form, surface hardness can be increased through carburizing.

1020. The effect of cobalt in a steel alloy is

(a) to increase red hardness and sustain hardness during tempering.
(b) to increase softness and weld ability
(c) to promote deoxidization and nitriding.
(d) to promote grain growth and machinability

Karnataka PSC AE, 10.09.2017

Ans. (a) : Cobalt is added in steel alloys to increase its strength and hardness and red hardness of HSS.

1021. Crankshaft and gears are made from

(a) Grey cast iron (b) White cast iron
(c) Nodular cast iron (d) None of these

Gujarat PSC AE 2019

Ans : (c): Crankshafts are made from forged steel or cast iron. Crankshafts for high volume, low load production vehicles are generally constructed from Nodular cast iron.

1022. Chilled cast iron is:

(a) Soft on surface
(b) Machine freely
(c) High Resistance to wear
(d) None of these

RPSC ADE 2016

Ans. (c) : High resistance to wear chilled cast iron is high resistance to wear.

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06.

PRODUCTION TECHNOLOGY

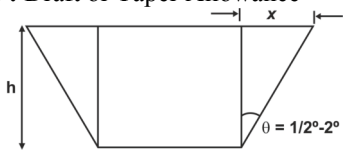
1. Metal Casting Processes

1. Draft on pattern for casting is provided for

(a) Shrinkage allowance
(b) Identification
(c) Taper to facilitate removal from mould
(d) Machining allowance

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
JPSC AE PRE 2019, MECON MT 2019
Nagaland PSC CTSE 2017, Paper-I
Kerala PSC Poly. Lect., 2017
Kerala PSC IOF 19.04.2016
GWSSB DEE 07.07.2016, GWSSB AAE, 27.12.2015
MPPSC State Forest Service Exam, 2014
J&K PSC Civil Services Pre, 2013
ISRO Scientist/Engineer 2012

Ans. (c) : Draft or Taper Allowance



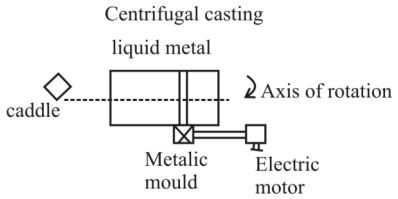
For easy removal of the pattern from the mould for the vertical surface of the pattern draft or taper allowance is provided.
This will be express in terms of taper angle and this value will depends upon vertical height of the pattern.

2. Which one of the following moulding processes does not require use of core?

(a) Sand moulding (b) Shell moulding
(c) Centrifugal casting (d) Plaster moulding

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Assam PSC AE (IWT) 14.03.2021
UJVNL AE 2016, UPPSC AE 12.04.2016 Paper-I
GPSC AE 26.07.2015, Assam PSC CCE Pre 2015
TSPSC Managers, 2015, HPPSC AE 2014
GPSC Lect. (Poly.) 07.12.2014
JPSC AE 2013 Paper-V

Ans. (c) :



Centrifugal casting is a type of casting method in which a liquid metal is poured in a rotating metallic mould. Due to centrifugal force metal is force to outer periphery. The impurities, which is lower density is collected at the centre. At the outer periphery due to high heat transfer heat loss which produces a fine particle. Which have high density and good surface finish. This method does not require core.

3. Chaplets are placed between mould and core in order to:

(a) reduce directional solidification
(b) help local alloying of molten metal
(c) prevent core movement due to buoyancy
(d) help easy removal of core from costing
(e) help improve the purity of metal

TNPSC AE 2018
CGPSC Asstt. Workshop Supt., 17.07.2016
RPSC LECTURER 16.01.2016
UPPSC AE 12.04.2016 Paper-I
GWSSB DEE 07.07.2016
Kerala PSC IOF 19.04.2016, KPSC AE. 2015
Kerala PSC AE 06.08.2015, GATE 2003, 1992
MPPSC State Forest Service Exam, 2014
BPSC AE 2012 Paper-VI, DRDO Scientists 2008

Ans. (c) : Chaplets are placed between mould and core in order to support the core and to prevent core movement due to buoyancy of liquid metal.

4. In Investment casting, the pattern is made of

(a) Wax (b) Plastics
(c) Plaster of Paris (d) Ceramics

Haryana PSC AE (PHED) 05.09.2020, Paper-II
GPSC Executive Engineer 23.12.2018
TSPSC AEE 2017
GPSC Asst. (Prof.) 21.08.2016
GPSC Asstt. Prof. (Prod.) 21.08.2016
GWSSB DEE 07.07.2016
Kerala PSC Asstt. Prof. 27.09.2016
GWSSB AAE, 27.12.2015
Kerala PSC AE 06.08.2015

Ans. (a) : Investment casting- This is the process where the mould is prepared around on expendable pattern.

- The pattern is dipped into a slurry made by suspending fine ceramic materials in a liquid such as sodium silicate. The excess liquid is allowed to drain off from the pattern. Thus a small sheet is formed around the work pattern.

Application of investment casting:-

- Gas turbine blades
- Jet engine parts
- Medical implants (Titanium alloys)
- Dentures (Stainless steel)
- Gold ornaments etc.

5. A_____ normally provides a steeper temperature gradient so that directional solidification as required in a casting can be obtained

(a) chill (b) runner
(c) riser (d) gating

GPSC DEE, Class-2 (GWSSB) 04.07.2021
UPRVUNL AE 05.07.2021
Haryana PSC AE (PHED) 05.09.2020, Paper-II
HPPSC AE 2018, GPSC Asstt.Prof. (Prod) 21.08.2016
UPRVUNL AE 07.10.2016
GPSC AE 26.07.2015, HPPSC Asstt. Prof. 2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
APPSC AEE 2012, TRB Poly. Lect., 2012

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Ans. (a) : Chills are metallic objects placed in mould to increase directional solidification. They have high thermal conductivity.

6. The solidification time in castings is proportional to

(a) $\left\{\frac{\text{volume}}{\text{surface area}}\right\}^2$ (b) $\left\{\frac{\text{volume}}{\text{surface area}}\right\}^4$

(c) $\left\{\frac{\text{volume}}{\text{surface area}}\right\}^3$ (d) $\left\{\frac{\text{volume}}{\text{surface area}}\right\}^1$

HPCL Engineer 12.08.2021 Shift-I
NLCIL GET 17.11.2020, Shift-II
VIZAG STEEL MT 2015
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TRB Poly. Lect., 2012

Ans. (a) : Solidification time of a molten metal in a castings is proportional to, (where, V=Vol. of metal A= its surface area)

Solidification Time = $K\left(\frac{V}{A}\right)^2$ (Chvorinov's rule)

Where K = Solidification factor

7. Hot tears are formed in the castings because of

(a) the uniform cooling rates and high strength of metal at higher temperature
(b) the differential cooling rates and low strength of metal at higher temperature
(c) the misalignment of caves in mould
(d) none of the above

APPSC Poly Lect. Automobile Engg., 2020
TNPSC AE 2018
Jharkhand Urja Vikas Nigam Ltd. AE 2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
APPSC AEE Screening Test 2016
OPSC AEE 2015 Paper-I

Ans. (b) : When metal is heated its strength decreases and residual stress in the material cause the casting to fail as it cools. This is called hot tears or hot cracking kind of casting defect.

8. Core prints are used for

(a) Strengthen core
(b) Form seat to support and hold the core in place
(c) Fabricate core
(d) None of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE 2018
UJVNL AE 2016
HPPSC AE 2014
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Core prints are used for making form seat to support and hold the core in place.

9. Cores are used

(a) In symmetrical casting only
(b) To get smooth surface on casting
(c) To get cavity or recess in a casting
(d) To strengthen mould

MECON MT 2019
WBPSC AE, 2017
APPSC AEE Screening Test 2016
Kerala PSC IOF 19.04.2016
GWSSB AAE, 27.12.2015

Ans. (c) : Cores are used to form internal cavities in the casting and improve mould surface and used to form a part of a green sand mould.
Core is a used in casting and moulding processes to produce internal cavities. The core is normally a disposable item that is destroyed to get it out of the piece.

10. Misrun is a casting defect which occurs due to

(a) Very high pouring temperature of metal
(b) Absorption of gases by liquid metal
(c) Insufficient fluidity of molten metal
(d) Improper alignment of mould flasks

HPPSC Workshop Suptd. 08.07.02021
HPPSC Asstt. Prof. 18.11.2016
ISRO Scientist/Engineer 03.07.2016
Kerala PSC Asstt. Prof. 27.09.2016
VIZAG STEEL MT 10.06.2012
ESE 1996, GATE 2004

Ans. (c) :

Due to lack of fluidity of molten liquid metal if the molten liquid metal will solidify before reaching the extreme end of mould cavity will form defect known as misrun.

11. Green sand is a mixture of

(a) 30% sand and 70% clay
(b) 50% sand and 50% clay
(c) 70% sand and 30% clay
(d) 90% sand and 10% clay

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Mizoram PSC (Insp. of Factories) 2019 Paper-III
TSPSC AEE 2015, OPSC AEE 2015 Paper-I

Ans. (c) : • Green sand is a Mixture of silica sand with 18 to 30% clay and 2 to 6% water
• Composition of green sand Silica- 70-80%, Clay- 18-30%, Water- 2 to 6%, Additives- 1 to 6%
• Coal dust is mixed in green sand to prevent defects casting.

12. When two streams of molten metal do not fuse with each other during casting of an automobiles part, the defect is known as :

(a) Pore (b) Scab
(c) Cold shut (d) Dry shut

PPSC Asstt. Municipal Engineer 15.06.2021
HPPSC Lect. (Auto) 23.04.2016, GATE 2009
PSPCL AE, 2012, VIZAG STEEL MT 10.06.2012

Ans. (c) : Cold shut occurs when two relatively cold streams of molten metal from different gates meet and do not fuse together properly during casting process.

13. In.....casting expandable pattern is used.

(a) die (b) squeeze
(c) investment (d) continuous

(HPPSC LECT. 2016)
Nagaland PSC CTSE 2016 Paper-I
UJVNL AE 2016, (KPSC AE. 2015), GATE 2006

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Ans : (c) In investment casting expandable pattern is used.

Investment Casting :- In this mould is prepared by an expandable pattern wax. It is used for making Jewellary surgical equipments, blade for gas turbine.

It is also known as lost wax process or precision casting. The casting produced by this method are within very close tolerance (± 0.05 mm) and do not require subsequent machining.

14. Swell is a casting defect which :

(a) results in a mismatching of the top and bottom part of a casting
(b) is due to enlargement of the mould cavity by metal pressure causing partial or overall enlargement of the casting
(c) occurs near the ingates at rough lumps on the surface of the casting
(d) is due to a thin projection of metal not intended as part of the casting

VIZAG MT, 14.12.2020
Vadodara Muncipal Corp. DEE, 2018
HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : Swell is a casting defect which is due to enlargement of mould cavity by metal pressure causing partial or overall enlargement of the casting.

15. The pattern allowance which takes care of contraction of a casting is known as :

(a) shrinkage allowance
(b) machining allowance
(c) draft
(d) none of the above

APGCL AM, 2021
GPSC Lect. (Auto) 16.10.2016
GWSSB AAE, 27.12.2015

Ans. (a) : The pattern allowance which takes care of contraction of a casting is known as shrinkage allowance.

16. A casting defect which occurs due to improper venting of sand is known as

(a) cold shut (b) blow hole
(c) shift (d) swell

JPSC AE 10.04.2021, Paper-II
Assam PSC CCE Pre 2015
TSPSC AEE 2015

Ans. (b)

- **Blow hole–** It is an excessively smooth depression on the outer surface of a casting. Blow holes are casting defects there are caused due to excessive gas content in the molten metal. When permeability of sand is low then these gases are not able to escape out & generates blow holes casting defect in the casting product.
- **Swell–** It is an enlargement of the mould cavity by molten pressure resulting in localised or general enlargement of the casting.
- **Shift–** It results in a mismatching of the top and bottom parts of a casting, usually at the parting line.
- **Cold Shut–** Cold shuts occur when two relatively cold streams of molten metal from different gates meet and do not fuse together properly during the casting process. A misrun occurs when the molten metal freezes before it reaches all parts of the mould cavity which leaves a completely unfilled part of the mould.

17. Centrifugally cast automotive parts have:

(a) Rough grain structure with low density
(b) Fine grain structure with high density
(c) Very large grain structure with low density
(d) Very large grain structure with high density

GPSC DEE, Class-2 (GWSSB) 04.07.2021
HPPSC Lect. (Auto) 23.04.2016
TANGEDCO AE 2015
TRB Poly. Lect., 2012
GATE 1993

Ans. (b) : Centrifugal casting is used for axis symmetric object & hollow parts. In which liquid molten metal is poured in a rotating metallic mould. Due to centrifugal force liquid metal is forced to outward and impurities are collected at centre. Due to which at the contact of liquid metal and metallic mould surface heat transfer is maximum which produce fine grain structure with high density.

18. A mould has a downsprue whose length is 20 cm and the cross sectional area at the base of the downsprue is 1cm^2. The downsprue feeds a horizontal runner leading into the mould cavity of volume 1000 cm^3. The time required to fill the mould cavity will be

(a) 4.05 s (b) 5.05 s
(c) 6.05 s (d) 7.25 s

HPPSC Asstt. Prof. 18.11.2016
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2005

Ans. (b) : Height of sprue, h = 20 cm = 0.2m
Area of downsprue (ingate) = 1 cm^2
$= 1 \times 10^{-4}$ m^2
Volume of casting = 1000 cm^3 (Mold cavity) = 10^{-3} m^3
Velocity of sprue base

$$V = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 0.2}$$

V = 1.98 m/sec.
Volume of casting = Area × Velocity × time
$10^{-3} = 10^{-4} \times 1.98t$

$$t = \frac{10^{-3} \times 10^{4}}{1.98}$$

$$\boxed{t = 5.05 \text{ sec.}}$$

19. While cooling a cubical casting of side 40 mm undergoes 3%, 4% and 5% volume shrinkage during liquid state, phase transition and solid state, respectively. The volume of the metal compensated from the riser is

(a) 2% (b) 7%
(c) 8% (d) 9%

BHEL ET 2019
BEL PE 2015
VIZAG STEEL MT 10.06.2012
GATE 2008

Ans. (b) : The riser can compensate for volume shrinkage only in the liquid stage and transition stage and not in the solid state.
Hence volume of metal that needs to be compensated from the riser = 3 + 4 = 7%

20. Which of the following materials requires the largest shrinkage allowance, while making a pattern for casting?

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(a) Cast iron (b) Brass
(c) Aluminum (d) Plain carbon steel

OPSC AEE 2015 Paper-I
(KPSC AE 2015), UKPSC AE 2012 Paper-I
VIZAG STEEL MT 10.06.2012, GATE 1999

Ans : (d)

Pattern Materials	Shrinkage allowance (mm/meter)
Cast iron	7-10.5
White cast iron	21
Malleable CI	15
Steel	20
Brass	14
Aluminium	18
Aluminium alloy	13–16
Bronze	10.5–20

21. What is the function of 'skim-bob' provided in the gating system of moulds?
(a) To trap heavier and lighter impurities.
(b) To promote directional solidification.
(c) To control the flow of molten metal
(d) To smoothen the flow of molten metal.

UPRVUNL AE 05.07.2021
GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC AEN Pre-2013

Ans. (a) : Light impurities in the molten metal are prevented from reaching the mould cavity by skim-bob. A small hollow spherical bulge above the top runner surface.

22. Directional solidification in castings can be improved by
(a) chills and chaplets
(b) chaplets and padding
(c) chills and padding
(d) chills, chaplets and padding

Nagaland PSC (CTSE) 2018, Paper-I
UJVNL AE 2016
TRB Asstt. Prof., 2012
ESE 1995

Ans. (c) : Directional solidification in castings can be improved by chills and padding.
Chaplets is used to provide support to castings.

23. During Un-pressurized gating system the meaning of 1:3:3 is:
(a) Area of bottom of sprue, runner cross-section and total ingate opening
(b) Runner Cross-section, total in gate opening and area of bottom of sprue
(c) Area of bottom of sprue, total in gate opening and runner cross-section
(d) Total in gate opening, runner cross-section and area of bottom of sprue.

HPPSC ADF 05.03.2019
Punjab PSC SDE 12.02.2017
UPRVUNL AE 21.08.2016

Ans. (a) :
Gating Ratio for non-pressurised gating system
Sprue area : Runner area : Ingate area
1 : 3 : 3
Gating Ratio for pressurized gating system
Sprue area : Runner area : Ingate area
1 : 2 : 1

24. .....are used to support cores inside the mould cavity to take care of its own weight and overcome the metallostatic forces.
(a) Sprue (b) Chill
(c) Chaplet (d) Striker

HPPSC Asstt. Prof. 29.10.2016
HPPSC AE (PWD) 10.09.2016
TRB Poly. Lect., 2012

Ans. (c) : Chaplet are used to support cores inside the mould cavity to take care of its own weight and over come the metallostatic force. If the core is very long or only one edge can protrude from the mould, then chaplets are often made of the same metal as the casting.

25. _____is most suitable process for casting of turbine blades made of high temperature and high strength alloy:
(a) Investment casting (b) Slush casting
(c) Centrifugal casting (d) Die casting

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
Nagaland PSC (CTSE) 2018, Paper-I
GPSC Executive Engineer 23.12.2018
CGPSC ARTO & TSI 07.01.2017
PSPCL AE, 2012

Ans. (a) : Investment casting or lost wax casting or precision casting- This is the process where the mould is prepared around an expandable pattern.
Advantages-
1. Complex shapes which are difficult to produced can be produced by investment casting.
2. Very fine details and thin section can be produced.
3. Casting produced by this process not required any machining.
4. Controlled mechanical, properties can be obtained
Applications- Product made by this process are vanes and blades for gas turbines, stainless steel valve bodies, impeller for turbines.

26. Sweep patterns are used for moulding parts having the shape of
(a) rectangular (b) complex shape
(c) square (d) symmetrical

UPRVUNL AE 05.07.2021
MPSC Lect. (Prod.) 2018, TSPSC AEE 2015

Ans. (d) : Sweep patterns are used for moulding parts having the shape of symmetrical.

27. Riser is designed so as to
(a) Freeze after the casting freezes
(b) Freeze before the casting freezes
(c) Freeze at the same time as the casting
(d) Minimize the time of pouring

GPSC DEE, Class-2 (GWSSB) 04.07.2021
ISRO Scientist/Engineer 07.05.2017
Nagaland PSC CTSE 2016 Paper-I, GATE 1987
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : Riser– It is a reservoir of molten metal provided in the casting so that hot metal can flow back into the mould cavity when there is a reduction in volume of metal due to solidification.

28. A casting process in which molten metal is poured in a mould and allowed to solidify while the mould is revolving is known as:
(a) Die casting (b) Investment casting
(c) Centrifugal casting (d) Continuous casting

SJVN ET 2019, APPSC AEE MAINS 2016, PAPER-III

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Ans. (c) : Centrifugal casting a casting process in which molten metal is poured in a mould and allowed to solidify while the mould is revolving is called centrifugal process. The casting produced under this centrifugal force is called centrifugal casting. This process is especially employed for casting articles of symmetrical shape. The ferrous and nonferrous metal can be casted by this process have dense and fine grained structure.

29. Which of the following is not a part of gating system?

(a) Pouring basin (b) Sprue
(c) Gates and riser (d) Vents

Nagaland PSC CTSE 2017, Paper-I
Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : Vents is not a part of gating system.

30. Ornaments are cast by

(a) Die casting (b) Continuous casting
(c) Pressed casting (d) Centrifugal casting

SJVN ET 2013
J&K PSC Civil Services Pre, 2013

Ans. (c) : Ornaments are cast by pressed and slush casting.

31. Pouring basin is provided in sand moulds

(a) to reduce the turbulence in the liquid metal
(b) to maintain high temperature of liquid metal
(c) to increase fluidity
(d) to improve properties of sand

Nagaland PSC CTSE 2017, Paper-I
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Pouring basin is provided in sand moulds to reduce the turbulence in the liquid metal.

32. Cylinder blocks of internal combustion engine are made by

(a) Forging (b) Extrusion
(c) Casting (d) Drilling
(e) Broaching

Nagaland PSC CTSE 2017, Paper-I
CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Manufacturing of engine blocks are mainly done using sand casting, although die casting also used it is more cost effective as the die wear out easily due to the high temperature of the molten metal. The casted engine block is then machined to get the surface finish and coolant passages.

33. Investment casting is used for

(a) Mass production
(b) Shapes that are made with difficulty using complex patterns in sand casting
(c) Stainless steel parts
(d) Shapes which are complex, intricate and cannot be cast by any other method

JWM 2017
ISRO Scientist/Engineer 2010

Ans. (d) : Investment casting is used for-

- Complex shapes which are difficult to produce by any other method are possible since the pattern is withdrawn by melting it.
- Very fine details and thin section can be produced by this process.
- Very close tolerance and better surface finish can be produced.

34. According to Chvorinov, following is the correct formula for solidification time (T_s), where k = Mould casting constant, V = volume of casting, SA = surface area.

(a) $T_s = V\left(\frac{SA}{k}\right)^2$ (b) $T_s = k\left(\frac{SA}{V}\right)^2$

(c) $T_s = k\left(\frac{V}{SA}\right)^2$ (d) $T_s = V\left(\frac{k}{SA}\right)^2$

CIL MT 26.03.2017
BPSC Poly. Lect. 2016
ESE 2011

Ans. (c) : According to Chvorinov's rule

$$\text{Solidification time} = k\left(\frac{V}{SA}\right)^2$$

Where V = volume of casting
S.A. = surface area of casting

35. Which of the following processes is suitable for making hollow parts with thin sections?

(a) Thermoforming
(b) Blow moulding
(c) Injection moulding
(d) Reaction injection moulding
(e) Extrusion

HPPSC AE 2018
CGPSC Polytechnic Lecturer 2017

Ans. (b) : Blow Moulding–Blow moulding is a specific manufacturing process by which hollow plastic parts are formed and can be joined together. It is also used for forming glass bottles or other hollow shapes.
The blow molding process begins with melting down the plastic and forming it into a parison or, in the case of injection and injection stretch blow moulding, perform. The parison is a tube-like piece of plastic with a hole in one end through which compressed air can pass.
The parison is then clamped into a mold and air is blown into it.

36. Molten metal is transported down into the mould by the _____.

(a) Gate (b) Sprue
(c) Runner (d) Riser

Assam Engg. College AP/Lect. 18.01.2021
TNPSC AE 2017
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) :
(1) Gate–The actual entry point (end of the runner) through which molten metal enters the mould cavity.
(2) Sprue–The passage through which the molten metal from the pouring basin, reaches the mould cavity.
(3) Runner–The horizontal channel which connects the sprue to the gates.
(4) Riser–It is a reservoir of molten metal provided in the casting.

37. Which one of the following is the permanent mould casting process?

(a) Investment casting process
(b) Full mould process
(c) Vacuum casting process
(d) Die casting process

ISRO Scientist/Engineer 17.12.2017
HPPSC Asstt. Prof. 2014

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Ans. (d) : Permanent mould casting—Permanent mould casting is a process for producing a large number of castings using a single reversible mould. The casting process simply involves pouring molten metal into a mould where it cools and solidifies. The mould is then opened the casting removed and the mould is reused.

38. The part of a gating system which regulates the rate of pouring of molten metal is

(a) Pouring basin (b) Runner
(c) Choke (d) Ingate

GPSC EE Pre, 28.01.2017
GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : The part of a gating system which regulates the rates of pouring of molten metal is choke.

39. A spherical drop of molten iron of radius 2 mm solidifies in 10 s. A similar drop of radius 4 mm will solidify in:

(a) 40 s (b) 20 s (c) 10 s (d) 30 s

GPSC DEE, Class-2 (GWSSB) 04.07.2021
AAI Jr. Executive 29.11.2018
UPRVUNL AE 21.08.2016

Ans. (a) : $t = k\left(\frac{V}{A}\right)^2$

k = Solidification factor
V = Volume of casting
A = Surface area of casting

$$t \propto \frac{1}{A}$$

$$\frac{t_2}{t_1} = \frac{d_2^2}{d_1^2}$$

$$\frac{t_2}{10} = \frac{4^2}{2^2}$$

$$\frac{t_2}{10} = \frac{16}{4} \qquad \boxed{t_2 = 40\,\text{sec}}$$

40. In carbon dioxide moulding process, the binder used is:

(a) Sodium silicate
(b) Sodium bentonite
(c) Calcium bentonite
(d) Phenol formaldehyde

VIZAG Steel MT 24.01.2021
Shift-I, CIL MT 27.02.2020

Ans. (a) : In CO_2 moulding for making cores and occasionally for moulds is to use sodium silicate as a binder. Mould is prepare with a mixture of sodium silicate and sand and then treated with CO_2.
Sodium silicate gives better dimensional accuracy.

41. Cast iron pipes are manufactured by;

(a) Sand casting method
(b) Centrifugal casting method
(c) Die casting method
(d) Lost was method

CIL MT 27.02.2020
ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (b) : The casting produced under the centrifugal force is called centrifugal casting. The casting produced by this process have dense and fine grained structure.
→ This casting employed only for symmetrical shape. Such as cast iron pipe, thin walled cylinders.

42. Two cubical castings of the same metal and sizes of 2 cm side and 4 cm side are moulded in green sand. If the smaller casting solidifies in 2 mins, the expected time of solidification of larger casting will be:

(a) 16 mins (b) 8 mins
(c) 2 mins (d) 4 mins

CIL MT 27.02.2020
GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Cubical casting of same metal
Side of first casting $A_1 = 2$cm
Side of first casting $A_2 = 4$cm
Solidification time of 1st casting $t_1 = 2$ min

$$t \propto \left(\frac{V}{A}\right)^2$$

$$\frac{t_2}{t_1} = \frac{\left(\frac{V_2}{A_2}\right)^2}{\left(\frac{V_1}{A_1}\right)^2}$$

$$t_2 = t_1\left[\frac{V_2 \times A_1}{A_2 \times V_1}\right]^2$$

$$t_2 = 2\left[\frac{4^3 \times 2^2}{4^2 \times 2^3}\right]^2$$

$t_2 = 8$ minutes

43. Petrol engine carburetors are manufactured by:

(a) Die casting (b) Centrifugal casting
(c) Shell casting (d) Sand casting

CIL MT 27.02.2020
GMB AAE 25.06.2017
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Carburetor usually manufactured by pressure die casting, followed by machining carburetor have many fine passages, for idle mixture.

44. The moulding process employed for thermoplastic materials is

(a) Injection and extrusion method
(b) Compression and transfer molding method
(c) Die casting method
(d) Continuous casting method

Punjab PSC (Lect.) 06.08.2017
GWSSB AAE, 27.12.2015

Ans. (a) : The moulding process employed for thermoplastic materials is injection and extrusion method.

45. The ratio of surface area of volume for a unit volume to riser is minimum in case of

(a) Cylindrical riser (b) Spherical riser
(c) Hemispherical riser (d) Cuboids riser

GPSC EE Pre, 28.01.2017
ESE 2012

Ans. (b) : Because area of riser minimum then sprue area minimum.

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46. Which one of the following materials will require the largest size of riser for the same size of casting?

(a) Aluminium (b) Cast iron
(c) Steel (d) Copper

GPSC DEE, Class-2 (GWSSB) 04.07.2021
RPSC AE 2018
Nagaland PSC (CTSE) 2018, Paper-I
GWSSB DEE 07.07.2016
TRB Asstt. Prof., 2012

Ans. (a) : Aluminium will require the largest size of riser for the same size of casting.

- Liquid shrinkage maximum for aluminium → 65%.
- Liquid and solidification shrinkage is max for aluminium which required more volume of riser.
- Solid shrinkage is maximum for brass which required large size of pattern.
- Total shrinkage maximum for steel.

47. The property of sand due to which it evolves a great amount of steam and other gases is called:

(a) Permeability (b) Cohesiveness
(c) Adhesiveness (d) None

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
HPPSC Workshop Suptd. 08.07.02021
Mizoram PSC (Insp. of Factories) 2019 Paper-III
RPSC Lecturer 16.01.2016

Ans. (a) : Permeability – The gas evolving capability of moulding sand is known as permeability. This will expressed by permeability number (P_n).

$$P_n = \frac{VH}{PAT} = \frac{3007.2}{T}$$

$V \Rightarrow$ Volume of air passing through the specimen.
$H \Rightarrow$ Height of standard specimen.
$P \Rightarrow$ Pressure of the air passing through the specimen.
$A \Rightarrow$ Cross-sectional area of cylindrical specimen.
$T \Rightarrow$ Time required to pass through the specimen.

48. Metal casting process that is characterized by forcing molten metal under high pressure into a mold cavity is called as

(a) Slush casting
(b) Die casting
(c) Permanent mould casting
(d) Centrifugal casting

Rajasthan AE (Nagar Nigam) 2016 Shift-3
Assam PSC CCE Pre 2015, ESE 2005

Ans. (b) : Die casting involves the preparation of components by injecting molten metal at high pressure into metallic die so it is also called pressure die casting. Die casting and permanent mould casting both uses reusable metallic mould but die casting operate at high pressure compared to permanent mould casting.
In die casting any narrow sections, complex shapes and fine surface details can easily be produced.
Note-Die casting machines are two types–
(i) Hot Chamber Die Casting–It is generally used for low melting point metal & its alloy (such as zinc, tin and lead) with attached melting point.
(ii) Cold Chamber Die Casting–It is generally used for high melting point metal and its alloy with separate melting point.

49. Match the following manufacturing processes with the manufactured products:

(a) Investment casting	**(1) Turbine rotors**
(b) Die casting	**(2) Turbine blades**
(c) Centrifuged casting	**(3) Connecting rods**
(d) Drop forging	**(4) Gl pipes**
(e) Extrusion	**(5) Cl Pipes**
(f) Shell moulding	**(6) Carburetor body**

(a) 1, 6, 5, 3, 2, 4 (b) 2, 6, 5, 3, 4, 1
(c) 6, 1, 5, 3, 2, 4 (d) 1, 6, 3, 5, 4, 2

OPSC AEE 2015 Paper-I, GATE-1994

Ans : (b)

Manufactured Process	Manufactured Products
(i) Investment casting	(i) Turbine blades.
(ii) Die casting	(ii) Carburetor body
(iii) Centrifuged casting	(iii) CI pipes
(iv) Drop forging	(iv) Connecting rods.
(v) Extrusion	(v) GI pipes.
(vi) Shell moulding	(vi) Turbine rotors

50. The process of making hollow castings of desired thickness by permanent mould without the use of cores is known as :

(a) die casting (b) slush casting
(c) pressed casting (d) centrifugal casting

PPSC Asstt. Municipal Engineer 15.06.2021
HPPSC W.S. Poly. 2016
TSPSC AEE 2015, ESE 2012

Ans : (b) The process of making hollow casting of desired thickness by permanent mould without the use of cores is known as slush casting.
Slush Casting:-This method is a special application of permanent mould casting in which hollow casting are produced without the use of cores. Molten metal is poured into the metallic mould and allowed to solidify upto the required thickness. The mould is then turned over so that the remaining liquid metal falls out and casting of desired thickness can be obtained. Normally small thickness casting of lead zinc and law melting alloy are obtained by slush casting method.

51. Swab is :

(a) a welding defect (b) a gear cutter
(c) a tool used in foundry (d) a forging die

RPSC AE 2016, RPSC ACF-2011

Ans. (c) : Swab– It is a hemp fiber brush used for moistening the edges of sand mould, which are in contact with the pattern surface before withdrawing the pattern. It is also used for coating the liquid blacking on the mould faces in dry sand.

52. In centrifugal casting, the impurities are

(a) Uniformly distributed
(b) Forced towards the outer surface
(c) Trapped near the mean radius of the casting
(d) Collected at the centre of the casting

UJVNL AE 2016, GPSC AE 26.07.2015

Ans : (d) In centrifugal casting, the impurities are collected at the centre of the casting.
Centrifugal casting:- In centrifugal casting no core is needed to form the hole in the middle. Centrifugal casting finds its best use in mass production, The use in machinery and equipment for centrifugal casting can be justified only when a large quantity of identical casting are required.

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53. Match plate pattern is used in :

(a) green sand moulding (b) bench moulding
(c) pit moulding (d) machine moulding

WBPSC AE, 2017
Kerala PSC Poly. Lect., 2017

Ans. (d) : Match plate pattern is used in machine moulding. Match plate pattern is used for small casting with higher accuracy.

54. Match List I correctly with List II and select your answer using the codes given below the lists:

List I	List II
a) Aluminium brake shoe	1. Deep drawing
b) Plastic water bottle	2. Blow moulding
c) Stainless steel cups	3. Sand casting
d) Soft drink can (aluminium)	4. Centrifugal casting
	5. Impact extrusion
	6. Upset forging

Codes:

	a	b	c	d
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	2	5	1
(d)	3	6	5	4

MPSC Poly. Lect. 2014, TRB Poly. Lect., 2012

Ans. (c) :

List-I	List-II
1) Aluminum brake shoe	Sand casting
2) Plastic water bottle	Blow moulding
3) Stainless steel cups	Impact extrusion
4) soft drink can (aluminum)	Deep drawing

55. Surface finish of casting depends upon

(a) Mould dressing (b) Pattern finish
(c) Sand compactness (d) All of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Surface finish of casting depends upon, mould dressing for a good mould dressing you will get sharp edges, pattern surface finish also affect the surface of mould, and sand compactness provide smooth mould and good surface finish.

56. The negative allowance on the pattern is :

(a) Draft allowance
(b) Machining allowance
(c) Distortion allowance
(d) Rapping allowance

UPRVUNL AE 05.07.2021
NLC GET 17.11.2020, Shift-II

Ans. (d) : In rapping or shake allowance, the pattern is taken out of the mould and it is rapped or shaken, so, as to free it from the adjoining sand. Due to this there may be a little increase in the size of the mould cavity.

Taper or draft allowance :- It is a taper which is given to all the vertical walls of the pattern for easy and clean withdrawal of the pattern from the sand without damaging the mould cavity.

Finishing or machining allowance- This allowance is provided on the pattern if the casting is to be machined. This allowance is given in addition to shrinkage allowance.

Distortion allowance– This allowance is provided on patterns used for castings of such design in which the contraction is not uniform throughout.

57. Blow holes in sand casting are caused by :

(a) lower strength of the solidified metal
(b) higher moisture in the moulding sand
(c) faulty moulding flask
(d) poor casting design

CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (b) : When higher moisture is present in the moulding sand. As temperature increases, voids formed. Then liquid metal absorbs gases by atmosphere during heating results in blow holes.

Blow holes is a gas defect in sand casting also called as pin holes.

58. Core sand mixture contains the following as binder:

(a) Turpentine oil (b) Linseed oil
(c) Cement (d) Thermosetting plastic

(CGPSC Polytechnic Lecturer 2017)
GPSC Asstt. Prof. 28.08.2016
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Core sand is highly rich silica sand mixed with oil binders such as core oil which composed of linseed oil, resin, light mineral oil and other bind materials. Linseed oil is used as binder.

59. The addition of coal dust to the green moulding sand is to improve

(a) Permeability (b) Surface finish
(c) Mouldability (d) Green strength

ISRO Scientist/Engineer 11.10.2015
OPSC AEE 2015 Paper-I
MPPSC State Forest Service Exam, 2014

Ans. (b) : Addition of coal dust to the moulding sand is to improve sand porosity and good compaction that gives result to improve surface finish and increases dry strength reduces expansion, etc.

60. A casting defect which results in general enlargement of a casting is known as

(a) Shift (b) Sand wash
(c) Swell (d) Scab

Gujarat PSC AE 2019
OPSC Civil Services Pre. 2011

Ans : (c) : Swell- A casting defect which results in general enlargement of a casting is known as swell.

61. Toys and ornaments of non-ferrous alloys are made by

(a) die casting
(b) centrifugal casting
(c) permanent mold casting
(d) Slush casting

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GPSC Executive Engineer 23.12.2018
APPSC AEE 2012

Ans. (d) : Toys and ornaments of non-ferrous alloys are made by slush casting.

62. A sprue hole is

(a) an eccentric hole
(b) a blind hole
(c) an opening is a mould into which molten metal is poured
(d) a thorough hole drilling through two mating parts

Nagaland PSC CTSE 2016 Paper-I
RPSC AE 2016, RPSC ACF-2011

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Ans : (c) A sprue hole is an opening in a mould into which molten metal is poured.
The sprue cross-section may be circular, square or rectangular. The size of the sprue usually varies from 10 mm square for small casting to about 20mm square for heavy casting. The size of the sprue should be so enough that it is kept full during the entire pouring operation, and the metal does not enter the mould cavity with high velocity causing splattering and turbulence.

63. The property of moulding sand which does not provide any resistance to contraction of the solidified casting is called?
(a) Collapsibility (b) Refractoriness
(c) Adhesiveness (d) Permeability

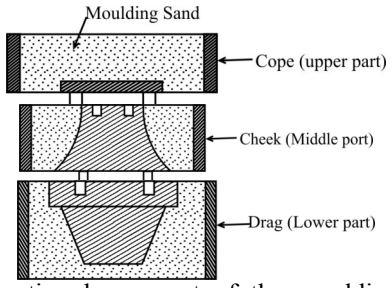
PPSC Asstt. Municipal Engineer 15.06.2021
UPRVUNL AE 07.10.2016

Ans. (a) : Collapsibility which is the property of moulding sand which does not provide any resistance of contraction of the solidified casting.
Adhesiveness—Adhesiveness is the property of sand due to which the sand particles sticks to the sides of the moulding box.
Adhesiveness of sand enables the proper lifting of cope along with the sand.
Refractoriness—Refractoriness is the property of sand to withstand high temperature of molten metal without fusion or soften.

64. In sand casting lower part of the moulding flask is called:
(a) cope (b) riser
(c) drag (d) none of these

WBPSC AE, 2017
(HPPSC LECT. 2016)

Ans : (c)

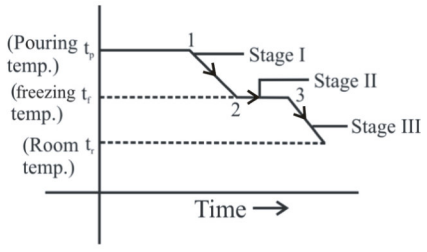


In sand casting lower part of the moulding flask is called drag

65. Shrinkage allowance is added to the pattern dimensions to take care of
(a) liquid shrinkage (b) pattern shrinkage
(c) solid shrinkage (d) none of the above

HPPSC AE 2018
TRB Polytechnic Lecturer 2017
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) :



1-2 → liquid shrinkage
2-3 → solidification shrinkage
3-4 → Solid shrinkage
State III is solid
Shrinkage is compensated by shrinkage allowance or oversize pattern.

66. In casting process, which of the following cross section is preferred for runner
(a) square (b) rectangular
(c) trapezoidal (d) circular

RPSC ADE 2016
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : The circular cross section is the most recommended one as it has the least flow resistivity (least perimeter) among the same cross section size. However, due to easier ejection /removal semicircle and trapezoidal shapes are also used. So, most preferred section is trapezoidal.

67. Collapsibility of green sand-mould can be increased by adding
(a) Water (b) Clay
(c) Wood flour (d) Synthetic resin

TSPSC Managers, 2015
ISRO Scientist/Engineer 2012

Ans. (c) : Collapsibility of green sand mould can be increased by addition of wood flour saw dust. And also additives of wood flour and saw dust increases permeability.

68. Match list-I (Process) with -II (Products or raw material) and Select the correct Answer using the codes given below the list:

	List-I		**List-II**
(a)	Die casting	1.	Phenol formaldehyde
(b)	Shell moulding	2.	C.I. pipes
(c)	CO_2 moulding	3.	Non-ferrous alloys
(d)	Centrifugal casting	4.	Sodium silicate

Codes:

	A	**B**	**C**	**D**
(a)	3	1	4	2
(b)	1	3	4	2
(c)	1	3	2	4
(d)	3	1	2	4

APGENCO AE, 2017
UJVNL AE 2016, ESE 2000

Ans : (a) List-I (Products)	**List-II (Process)**
None - ferrous alloys	Die casting
Phenol formaldehyde	Shell moulding
Sodium silicate	CO_2 Moulding
C.I. pipes.	Centrifugal castting

69. Gears are casted by
(a) sand mould casting
(b) slush casting
(c) permanent mould casting
(d) centrifugal casting

Nagaland PSC (CTSE) 2018, Paper-I
VIZAG STEEL MT 2011

Ans. (c) : Permanent mould casting is a process for producing a large number of casting using a single reusable mould. The casting process is used for producing gears, simply involves pouring molten metal into a mould where its cools & solidifies & mould can be used.

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70. In metal casting, gating ratio is defined as the ratio of _____.

(a) sprue area: total runner area: total gate area
(b) total gate area: sprue area: total runner area
(c) total gate area: total runner area: sprue area
(d) total runner area: total gate area: sprue area

VIZAG MT, 14.12.2020
CIL MT 26.03.2017

Ans. (a) : Gating ratio → Sprue : Runner : In gate
For non pressurized gating system : 1 : 4 : 4, 1 : 2 : 4
For pressurized gating system : 2 : 2 : 1, 3 : 2 : 1

71. A casting of size 100 mm × 100 mm × 50 mm is required. Assume volume shrinkage of casting is 2.6%. If the height of the riser is 80 mm and riser volume desired is 4 times the shrinkage in casting, what is the appropriate riser diameter in mm?

(a) 14.38 (b) 20.34
(c) 28.76 (d) 57.52

RPSC IOF, 2020
GATE 2003

Ans. (d) : Size of casting = 100 × 100 × 50
$= 500000\ mm^3$

$$\text{Volume of the riser} = 4\times\frac{2.6}{100}\times100\times100\times50$$

$$\frac{\pi}{4}d^2h = 52000mm^3 \quad (\because h = 80\ mm)$$

$$d = 28.77\ mm$$

$(t_s)_r = (t_s)_c$

Time taken to solidification of riser = Time taken to solidification of castings

$$\frac{\text{Volume of the riser}}{\text{Surface area of riser}} \geq \frac{\text{Volume of casting}}{\text{Surface area of casting}}$$

$$\frac{\frac{\pi}{4}d^2\times h}{2\frac{\pi}{4}d^2+\pi dh} \geq \frac{100\times100\times50}{2(100\times100+100\times50+100\times50)}$$

Substituting the value of d and h

$6.08 \geq 12.5$

So, diameter of riser should not be 28.77 mm
Diameter of riser greater than 28.77 mm
So, option (d) d = 57.52 is correct.

72. The process of making hollow casting from permanent mould by close fitting core is called

(a) Die casting (b) Slush casting
(c) pressed casting (d) Centrifugal casting

GPSC Engineer, Class-II Pre-19.01.2020
J & K PSC Screening, 2006

Ans. (c) : The process of making hollow casting from permanent mould by close fitting core is called pressed casting. Die casting, slush casting and centrifugal casting does not require core.

73. Organic binder used in moulding sand is

(a) Dextrin (b) Bentonite
(c) Kaolonite (d) Ball clay

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : An organic binder is an organic ingredient used to bind together two or more other materials in mixture. e.g. coal dust and bentonite etc.

74. The fluidity of molten metal of cast alloys (without any addition of fluxes) increases with increase in

(a) Viscosity
(b) Surface tension
(c) Freezing range
(d) Degree of super heat

PPSC Asstt. Municipal Engineer 15.06.2021
GATE 2019

Ans. (d) : When the pouring temperature of molten metal increases, the viscosity decreases with increasing fluidity.
Hence, fluidity of molten metal increases with increase in degree of superheat.
Hence, the correct option is (d).

75. A cube shaped casting solidifies in 10 minutes. Time taken by the same material cube 27 times heavier than the original one in minutes will be:

(a) 10 (b) 90
(c) 75 (d) 60

AAI Jr. Executive 26.03.2021

Ans. (b) : $t_1 = 10$ minutes
New cube is 27 times heavier than original one,

$m_2 = 27m_1$

$\rho V_2 = 27\rho V_1$ (Same material)

$a_2^3 = 27a_1^3$

$$\boxed{a_2 = 3a_1}$$

$$t_1 = \left(\frac{V}{A}\right)^2 = \left(\frac{a_1^3}{6a_1^2}\right)^2 = \left(\frac{a_1}{6}\right)^2 = 10\,\text{minutes}$$

$$t_2 = \left(\frac{V}{A}\right)^2 = \left(\frac{a_2^3}{6a_2^2}\right)^2 = \left(\frac{a_1}{6}\right)^2$$

$$t_2 = \left(\frac{a_2}{6}\right)^2 = \left(\frac{3a_1}{6}\right)^2 = 9\left(\frac{a_1}{6}\right)^2$$

$= 9 \times 10$
$= 90$ minutes

76. In a non-pressurized gating system, a metal alloy is poured at a volumetric flow rate of $8 \times 10^{-4}\ m^3/s$ through a sprue with a base cross-sectional area 400 mm^2. Neglecting any frictional losses, what is the height of sprue in mm?

(a) 200 (b) 175
(c) 225 (d) 275

AAI Jr. Executive 26.03.2021

Ans. (a) : Given, flow rate = $8 \times 10^{-4}\ m^3/sec$
Area of base of sprue = 400 mm^2

$Q = A \times v$
$8 \times 10^{-4} = 400 \times 10^{-6} \times v$
$v = 2\ m/s$
$v = \sqrt{2gh}$
$v^2 = 2gh$
$(2)^2 = 2 \times 10 \times h$

$$h = \frac{2}{10}m$$

Height of sprue, h = 200 mm

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77. As per casting cooling equation, the solidification time T_s of casting and its volume V have a relation represented as:

(a) $T_S \propto V^2$ (b) $T_S \propto V^{0.5}$

(c) $T_S \propto V^{-2}$ (d) $T_S \propto V$

AAI Jr. Executive 26.03.2021

Ans. (a) : As per casting cooling equation, the solidification time T_s of casting and its volume V have relation.

$$T_s = K\left(\frac{V}{A}\right)^2$$

$$\Rightarrow \quad T_s \propto (V)^2$$

78. Which of the following is NOT a preferred material for the making of patterns?

(a) Metal (b) Plastic

(c) Wood (d) Ceramic

UPRVUNL AE 05.07.2021

Ans. (d) : Ceramic is NOT a preferred material for the making of patterns. Whereas metal, wood and plastics are preferred material for the making of patterns.

79. _____ is an organic binder used in core making during the casting process.

(a) Bentonite (b) Kaolinite

(c) Fire clay (d) Cereal binder

UPRVUNL AE 05.07.2021

Ans. (d) : Cereal binder is an organic binder used in core making during the casting process.

80. _____ is a replica of the final product and used to prepare mould cavity

(a) sprue (b) pattern

(c) core (d) riser

UPRVUNL AE 05.07.2021

Ans. (b) : Pattern is a replica of the final product and used to prepare mould cavity.

81. Which of the following is a permanent mould casting process?

(a) Centrifugal casting (b) investment casting

(c) vacuum casting (d) shell moulding

UPRVUNL AE 05.07.2021

Ans. (a) : Centrifugal casting is a permanent mould casting process.

82. _____ sand used in casting contains up to 50 percent clay.

(a) lean (b) loam

(c) weak (d) silica

UPRVUNL AE 05.07.2021

Ans. (b) : Loam sand used in casting contains up to 50 percent clay.

83. Which of the following components feed(s) molten metal to casting set-up?

(a) ladle (b) dowel pins

(c) riser (d) core

UPRVUNL AE 05.07.2021

Ans. (a) : Ladle is used to feed molten metal into mould cavity.

84. Whenever a hole, recess, undercut or internal cavity is required in a casting, a _____, which is usually made up of a refractory material like sand is use.

(a) vent (b) core

(c) cope (d) pattern

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : Whenever a hole, recess, undercut or internal cavity is required in a casting, a core, which is usually made up of a refractory material like sand is use.

85. Disk shaped components are cast by

(a) true centrifugal casting

(b) semi centrifugal casting

(c) centrifuging

(d) die casting

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Semi-centrifugal casting is used to produce larger part that are symmetrical along the vertical axes. Such as gear blank and flywheel.

86. During solidification of a pure metal, the grains in the casting near the mould wall are

(a) coarse and randomly oriented.

(b) fine and randomly oriented.

(c) fine and ordered.

(d) coarse and ordered.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : During solidification of a pure metal, the grains in the casting near the mould wall are fine and randomly oriented because the rate of solidification is high at the surface of mould during solidification.

87. In directional solidification of a casting, the solidification continues progressively from:

(a) thinnest section to thickest section

(b) blind riser wall surface

(c) core wall surface

(d) None

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Directional solidification is that the casting should be cool and solidify progressively from thin sections to heavy sections with constant metal feed into the heavy sections.

88. One of the limitations of Green-sand casting is:

(a) it is inappropriate for the common metals like cast iron, steel, aluminium

(b) usually there is a limit of size of 100 cm length and weight of more than 10 kg

(c) tolerances and surface finish are poorer than other casting processes

(d) the casting is relatively expensive than other casting methods.

CGPSC AE 15.01.2021

Ans. (c) : One of the limitations of Green sand casting is tolerances and surface finish are poorer than other casting processes.

89. In sand casting, the following allowance is added to all linear dimensions of the casting?

(a) Shake allowance

(b) Draft allowance

(c) Machining allowance

(d) Shrinkage allowance

VIZAG Steel MT 24.01.2021

Ans. (d) : Shrinkage allowance is always to be added to all linear dimensions as the material has tendency to shrink during solidification so to compensate this solidification shrinkage pattern is made larger.

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90. _____ used for producing non-symmetrical small castings having intricate details.

(a) Slush casting (b) Squeeze casting
(c) Die casting (d) Centrifuging casting

VIZAG Steel MT 24.01.2021

Ans. (d) : Centrifuging Casting–This casting process is generally used for producing non-symmetrical small castings having intricate details. A number of such small jobs are joined together by means of a common radial runner with a central sprue on a table which is possible in vertical direction of mould rotation.

91. Distorsion allowance is not provided in the following shape of casting.

(a) U (b) T
(c) C (d) O

JPSC AE 10.04.2021, Paper-II

Ans. (d) : Distortion allowance – When casting of very thin surfaces like, V shape, U shape, T shape & C shape etc. It will distort or damage during pattern removal or during casting. To avoid this problem, a camber is provided on pattern to compensate distortion is known as distortion allowance.

Distortion allowance is not provided in O shape casting.

92. During the freezing of a metal, the possible casting structure is

(a) columnar structure
(b) dendritic structure
(c) equi-axed grains structure
(d) party columnar and partly equi-axed

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : A structure consisting of more or less regular column, usually six sided but sometimes with eight cone. The column are often fracture transvely with a cup joint, showing concave surface above.

93. If only one product is to be made we should adopt

(a) Die casting
(b) Continuous casting
(c) Machine moulding
(d) Sand casting by hand

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Sand casting- Sand casting also known as sand moulded casting is a metal casting process characterized by using sand as the mould material.
Sand castings are produced in specialized factories called foundaries.

94. In gated patterns

(a) No gates are made in the moulding sand
(b) No runners are made in the moulding sand
(c) Gates and runners are built in the pattern itself
(d) Only gates are built in to the pattern

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : In gated patterns gates and runners are built in the pattern itself.

95. In small castings, which of the following allowances can be ignored.

(a) Shrinkage Allowance
(b) Rapping Allowance
(c) Draft Allowance
(d) Machining Allowance

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : Rapping or Shaking Allowance–When the pattern is shaken for easy withdrawal, the mould cavity, hence the casting is slightly increased in size. In order to compensate for this increase, the pattern should be initially made slightly smaller.
For small and medium castings, this allowance can be ignored.

96. Which of the following is NOT the smith forging operation?

(a) Upsetting (b) Bending
(c) Setting down (d) Casting

APPSC Poly Lect. 13.03.2020

Ans. (d) : Forging is the operation where the metal is heated and then a force is applied to manipulate the metal in such a way that the required final shape is obtained. Forging is generally a hot-working operation though cold forging is used sometimes.
Types of Forging–
(i) Smith Forging
(ii) Drop Forging
(a) Fullering impression (b) Edging impression
(c) Bending impression (d) Blocking impression
(e) Finishing impression (f) Trimming
(iii) Press Forging
(iv) Machine Forging
"Casting" is not the smith forging operation.

97. The machine used in a foundry for mixing large quantity of sand is known as :

(a) feeder (b) batch muller
(c) screener (d) incinerator

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The machine used in a foundry for mixing large quantity of sand is known as batch muller.

98. A pattern marker's shrinkage rule considers

(a) All pattern allowances
(b) Only shrinkage allowance
(c) All materials to be cast
(d) All materials of the pattern

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : A pattern marker's shrinkage rule considers only shrinkage allowance.

99. Metal patterns are used for

(a) Small castings
(b) Large castings
(c) Complicated castings
(d) Large scale production of castings

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) : Metal patterns are used for large scale production of castings.

100. Which of the following casting process is known as expandable pattern casting process?

(a) Sand casting (b) Lost foam casting
(c) Shell mould casting (d) Die casting

Oil India Senior Officer 23.12.2020

Ans. (b) : Investment Casting–The process full mould casting also called as cavity less casting or investment casting or disposable pattern casting is variant of sand casting in the sand that the process utilizes an expandable pattern.

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101. Which of the following is not a casting defect?

(a) hot tear (b) blow hole
(c) scab (d) decarburisation

RPSC IOF, 2020

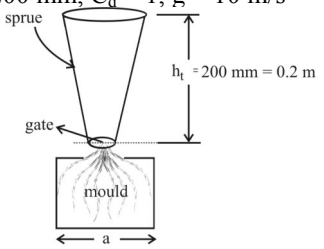
Ans. (d) : Hot tear scab air inclusion and pin hole porosity are known as gas defect in castings.
Blow hole and open blows are spherical flattened or elongated cavities present inside the casting or on the surface of the casting.
So, decarburisation is not casting defects.

102. In a sand casting process, a sprue of height 200 mm and 10 mm base diameter leads to a runner which fills a cubical mould cavity of 200 mm side. Time required to fill the cavity will be: [discharge coefficient = 1, g = 10 m/s^2, $\pi = \frac{22}{7}$]

(a) $\frac{560}{11}$ sec (b) $\frac{460}{11}$ sec
(c) $\frac{360}{11}$ sec (d) $\frac{260}{11}$ sec

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given, $h_t = 200$ mm, $d_1 = 10$ mm
$a = 200$ mm, $C_d = 1$, $g = 10$ m/s^2



$$\text{Time to fill (t)} = \frac{V_m}{A_g C_d \sqrt{2gh_t}} = \frac{a^3}{\frac{\pi}{4}d^2 \times \sqrt{2gh_t}}$$

$$\frac{a^3 \times 4}{\frac{22}{7} \times d^2 \times \sqrt{2gh_t}} = \frac{(200)^3 \times 4 \times 7}{22 \times (10)^2 \times \sqrt{2 \times 10 \times 1000 \times 200}}$$

$$t = \frac{8 \times 10^6 \times 28}{22 \times 100 \times 2 \times 10^3}$$

$$t = \frac{560}{11} \text{ seconds}$$

103. Consider the following statements for different patterns.
A. Polystyrene is a disposable pattern
B. Wood is a non-disposable pattern
C. Rubber is a disposable pattern
D. Metal is a non-disposable pattern
Select the correct option from below.

(a) Only A is correct
(b) Both A and B are correct
(c) Only C is correct
(d) A, B and D are correct

APPSC Poly Lect. 13.03.2020

Ans. (d) : Polystyrene is a disposable pattern
Wood is a non-disposable pattern
Metal is a non-disposable pattern

104. A sphere-shaped casting solidifies in 10 min. What will be the solidification time in minutes for another sphere of the same material, which is 8 times heavier than the original casting?

(a) 80 (b) 10
(c) 40 (d) 20

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given data :
Solidification time $(t_1) = 10$ minutes
$t_2 = ?$
For solidification time (t) $\propto \left(\frac{V}{A}\right)^2$
Where;
V = volume of casting
A = Surface area

For a sphere; $\frac{V}{A} = \frac{\frac{4}{3}\pi R^3}{4\pi R^2} = \frac{R}{3}$

If casting is 8-times heavier then –

$$V_2 = 8V_1$$

$$\frac{4}{3}\pi R_2^3 = 8 \times \frac{4}{3}\pi R_1^3$$

$$\Rightarrow \boxed{R_2 = 2R_1}$$

$$\frac{t_1}{t_2} = \frac{\left(\frac{V_1}{A_1}\right)^2}{\left(\frac{V_2}{A_2}\right)^2} = \frac{\left[\left(\frac{R_1}{3}\right)^2\right]_1}{\left[\left(\frac{R_2}{3}\right)^2\right]_2} = \frac{\left[\left(\frac{R_1}{3}\right)^2\right]_1}{\left[\left(\frac{2R_1}{3}\right)^2\right]_2} = \frac{\frac{R^2}{9}}{\frac{4R^2}{9}} = \frac{1}{4}$$

Given, $t_1 = 10$ min

$$\frac{10}{t_2} = \frac{1}{4}$$

$$t_2 = 40 \text{ min}$$

105. A spherical drop of molten of radius 5 mm was found to solidify in 12 s. In how much time will a similar drop of radius 10 mm solidify?

(a) 6 s (b) 12 s
(c) 24 s (d) 48 s

APPSC Poly Lect. 13.03.2020

Ans. (d) : Given data;
Radius $(R_1) = 5$ mm, $t_1 = 12$ sec.
$R_2 = 10$ mm; $t_2 = ?$

$\because$ Solidification time (t) $\propto \left(\frac{V}{A}\right)^2 = \left(\frac{R}{3}\right)^2$

[for spherical droplet]

$$\frac{t_1}{t_2} = \left(\frac{R_1/3}{R_2/3}\right)^2 = \left(\frac{5}{10}\right)^2 = \frac{1}{4}$$

$$\Rightarrow \frac{t_1}{t_2} = \frac{1}{4} \Rightarrow \boxed{t_2 = 48 \text{ sec}}$$

106. Large size symmetrical moulds and cores are shaped by:

(a) Shell pattern (b) Loose piece pattern
(c) Sweep pattern (d) Skeleton pattern

NLC GET 17.11.2020, Shift-II

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Ans. (c) : Sweep pattern– It is used to sweep the complete casting by means of a plane sweep. These are used for generating large shapes which are symmetrical or prismatic in nature such as a bell shaped or cylindrical. This type of pattern is particularly suitable for very large castings such as bells.
A sweep pattern is preferred for producing large casting of circular sections and symmetrical shapes.
Loose Piece Pattern– Loose piece are used in order to enable their easy withdraw from the mould. These pieces form an integral part of pattern during moulding
Skeleton pattern– A Skeleton pattern is the skeleton of a desired shape.
⇒ The skeleton is made from wooden strips and is thus a wooden work.
⇒ Skeleton patterns are employed for producing a few large castings.
⇒ A Skeleton pattern is very economical because it involves less material costs.

107. The foundry crucible is made with:
(a) Graphite (b) Steel
(c) Cast iron (d) Lead

CIL MT 27.02.2020

Ans. (a) : Foundry crucibles are designed to withstand the highest temperatures encountered in the metal casting works and it is made with graphite.

108. Which pattern is suitable for circular work such as wheel rims and gears?
(a) Boxed up pattern (b) Shell pattern
(c) Split pattern (d) Segmental pattern

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Segmental Pattern–Pattern of this type is generally used for circular castings. For example wheel rim, gear blank etc. Such as turns are sections of a pattern so arranged as to form a complete mould by being moved to form each section of the mould.

109. In casting, the height to diameter ratio of riser is in the range of:
(a) 5 to 6.5 (b) 3 to 5.5
(c) 1 to 1.5 (d) 6 to 8.5

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : In casting, the height to diameter ratio of riser is in the range of 1 to 1.5.

110. Shell moulding can be used for
(a) producing milling cutters
(b) making gold ornaments
(c) producing heavy and thick walled casting
(d) producing thin casting

ESE 2006

Ans. (d) : Shell moulding can be used for producing thin casting.

111. A zone of cupola furnace which starts from above the melting zone and extends upto bottom of charging door is known as:
(a) Stack zone (b) Well
(c) Preheating zone (d) Combustion zone

SJVN ET 2019

Ans. (c) : A zone of cupola furnace which starts from above the melting zone and extends upto bottom of charging door is known as preheating zone.

112. Shell patterns are often used for
(a) Pipe work (b) Bends
(c) Drainage fittings (d) All of the above

Gujarat PSC AE 2019

Ans : (d) : Shell patterns are often used for
■ Pipe work
■ Bends
■ Drainage fittings

113. Cohesiveness of sand depends on
(a) Grain size and shape of the sand particles
(b) Bonding materials
(c) Moisture content
(d) All of the above

Gujarat PSC AE 2019

Ans : (d) : Cohesiveness- It is the property of sand due to which the sand grain stick to together during ramming. It is defined as the strength of the moulding sand. Cohesiveness of sand depend on-
■ Grain size and shape of the sand particles.
■ Bonding materials
■ Moisture content

114. According to Chvorinov's rule, the solidification time of casting is proportional to (Volume/surface area)^n. The value of the exponent 'n for metal is equal to :
(a) 2 (b) 3
(c) 2.5 (d) 1.5

BHEL ET 2019

Ans. (a) : Given -

$$\text{Solidification Time} = \left(\frac{V}{A}\right)^n \quad ...(1)$$

According to chvorinov's rule -

$$\text{solidification time } t = \left(\frac{V}{A}\right)^2 \quad ...(2)$$

comparing equation (1) and (2)

$$\left(\frac{V}{A}\right)^2 = \left(\frac{V}{A}\right)^n$$

so, $\boxed{n = 2}$

115. The pattern which is made in two or more pieces are called-
(a) Follow board pattern
(b) Split pattern
(c) Gated pattern
(d) Sweep pattern

MECON MT 2019

Ans. (b) : A pattern for a casting made in two or more piece and doweled together permit separate removal.

116. Which casting process has no size and shape limits?
(a) Sand casting
(b) Plaster mould casting
(c) Investment casting
(d) Die casting

MECON MT 2019

Ans. (a) : Sand casting, also known as sand mold casting is a metal casting process characterized by using sand at the mold material.
• Sand casting are product in specialized factories called foundries.

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117. In which type of casting is the mould completely free of metal as it is spin about its vertical axis and riser and cores may be employed?

(a) True centrifugal casting
(b) Semi-centrifugal casting
(c) Slush casting
(d) Investment casting

MECON MT 2019

Ans. (b) : Semi centrifugal casting is used to produce larger parts that are symmetrical along the vertical axis, such as gear blanks and flywheel.
• The mould used is arranged horizontally and thin rotation.

118. Which of the following is used for removing loose sand from inside the mould cavity?

(a) Lifter (b) Swab
(c) Bellow (d) Trowel

MECON MT 2019

Ans. (a) : A lifter is a finishing tool used for repairing the mould and finishing sand.
• Lifter is also used for removing loose sand from mould.

119. Spherical of molten metal of radius 4 was found out to solidify in 10 sec. A similar drop of radius 8 mm would solidify in:

(a) 4 sec (b) 40 sec
(c) 20 sec (d) 10 sec

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (b) : Given,
Let, Radius of spherical drop A(r_A) = 4 mm
Solidification time (t_A) = 10 sec.
Radius of spherical drop B(r_B) = 8 mm
Solidification time (t_B) = ?
∴ According to chvorinov's equation

Solidification time $(t_S) = K\left(\frac{V}{SA}\right)^2$

Where K – Solidification factor
V – Volume of casting
SA – Surface area of casting

For spherical drop A

$$(t_s)_A = K\left[\frac{V_A}{(SA)_A}\right]^2 = K\left[\frac{r_A}{3}\right]^2 = K\left[\frac{4}{3}\right]^2$$

$$K = (t_s)_A \times \frac{9}{16} = \frac{10\times 9}{16} = 5.625\,\text{sec/mm}^2$$

$$\therefore\quad (t_s)_B = K\left[\frac{(V)_B}{(SA)_B}\right]^2 = K\left[\frac{r_B}{3}\right]^2 = 5.625\left[\frac{8}{3}\right]^2 = 40\,\text{sec.}$$

So A similar drop of radius 8 mm would solidify in 40 sec.

120. The main advantages of using CO_2 moulding is that

(a) much hardened mould is obtained
(b) gases can be made to escape more easily
(c) gases formed react with CO_2 to form a colloidal solution
(d) carbon % in the molten metal can be increased

TNPSC AE 2019

Ans. (a) : The main advantages of using CO_2 moulding is that much hardened mould is obtained.

121. The ratio of surface energy term to volume energy term in the nucleation energy equation at critical condition during the solidification process is :

(a) -1/2 (b) -3/2
(c) -1 (d) -2/3

BHEL ET 2019

Ans. (b) : Surface energy = $\gamma \times 4\pi r^2$
Volume energy = $\Delta G \times (4/3)\pi r^3$

Critical radius $r^* = \left(\frac{-2\gamma}{\Delta G}\right)$

Ratio of surface energy to volume energy –

$$\frac{\gamma \times 4\pi r^2}{\Delta G \times \frac{4}{3}\pi r^3}$$

$$\Rightarrow \quad \frac{3\gamma}{r\Delta G}$$

At critical radius:

$$\frac{3\gamma\Delta G}{-2\gamma\Delta G} \Rightarrow -\frac{3}{2}$$

122. In a casting process, a mould of dimension 60 cm × 30 cm × 14 cm is to be filled by liquid the top pouring. The liquid metal height above the top surface of the mould is 14 cm and the area of the gate is 6 cm^2. The time taken to fill the mould will be :

(a) 32 s (b) 12 s
(c) 25.34 s (d) 28.28 s

BHEL ET 2019

Ans. (c) : Given - Volume of mould = 60 × 30 × 14 cm^3
Liquid metal height h_t = 14 cm
A_g = 6 cm^2

$$\text{time filling } (t_f) = \frac{A_m h_m}{A_g V_g}$$

$$= \frac{V_m}{A_g \times \sqrt{2gh_t}}$$

$$= \frac{60\times 30\times 14}{6\times\sqrt{2\times 9.81\times 14}}$$

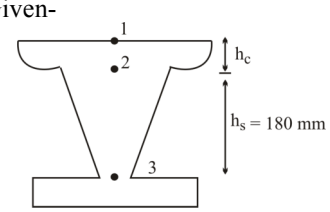
$$= 25.34 \text{ sec.}$$

123. In a gating system design, a down sprue of 180 mm length has a diameter of 20 mm at its top end. The liquid metal n pouring cup is maintained up to 60 mm height. The diameter, in mm. of the down sprue at its lower end to avoid aspiration will be :

(a) 11.11 (b) 14.14
(c) 25 (d) 17.32

BHEL ET 2019

Ans. (b) : Given-



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$h_s = 180$ mm
$D_2 = 20$ mm
$h_c = 60$ mm
Applying continuity equation (2) to (3) section -

$$Q = A_2 V_2 = A_3 V_3$$

$$\frac{A_2}{A_3} = \frac{V_3}{V_2} = \frac{\sqrt{2gh_t}}{\sqrt{2gh_c}}$$

$$\left(\frac{A_2}{A_3}\right) = \sqrt{\frac{h_t}{h_c}}$$

$$\frac{\frac{\pi}{4}\times(D_2)^2}{\frac{\pi}{4}(D_3)^2} = \sqrt{\frac{240}{60}} = 2$$

$$\left(\frac{D_2}{D_3}\right)^2 = 2$$

$$\frac{(20)^2}{(D_3)^2} = 2$$

$$\frac{400}{2} = (D_3)^2$$

$$D_3 = \sqrt{200}$$

$$\boxed{D_3 = 14.14 \text{ mm}}$$

124. The diameter of an optimum cylindrical riser (height = diameter) attached to the side of a steel plate casting having dimensions of 25 cm × 12.5 cm × 5 cm, by assuming that volumetric shrinkage of solidification is 3% for steel and that the volume of the riser is 3 times that by the shrinkage consideration alone, will be :

(a) 8.32 cm (b) 7.26 cm
(c) 3.24 cm (d) 5.64 cm

BHEL ET 2019

Ans. (d) : Cylindrical riser

$$\boxed{\text{height} = \text{diameter}}$$

$$\boxed{h = d}$$

Volume of casting = 25 cm × 12.5 cm × 5cm
= 1562.5 cm^3

Volumetric shrinkage = $v = \frac{\pi}{4}d^2h$

$$\boxed{h = d}$$

Assuming volumetric shrinkage = 3 × 1562.5 × 0.03
= 140.625 cm^3

Volume of riser when shrinkage consideration alone

$$v = \frac{\pi}{4}d^3 = 140.625 = d = 5.636 \text{ cm}$$

or $\boxed{h = d = 5.64 \text{ cm}}$

125. Riddle is basically used to:

(a) Mixing and tempering the moulding sand
(b) Remove foreign particles from the sand
(c) Shape and smoothen the mould surface
(d) Drawn pattern from mould

HPPSC ADF 05.03.2019

Ans. (b) : Riddle is basically used to cleaning the moulding sand and remove foreign particles from the sand.

126. A cubic casting of 100 mm side undergoes solidification shrinkage of 5% and phase transformation shrinkage of 4%. The net side of cube after solidification would be, in mm:

(a) 96.26 (b) 95.43
(c) 96.97 (d) 91.00

HPPSC ADF 05.03.2019

Ans. (c) : Side of cube, a = 100 mm
Solidification shrinkage = 5%
Transformation shrinkage = 4%
After considering allowance, size of cube is reduced
$V = (100)^3 \times 0.95 \times 0.96$
$V = 912000$
Net side of cube after solidification
$a^3 = 912000$
$a = 96.976$ mm

127. The draft or taper allowance on casting is generally

(a) 1 to 2 mm/m (b) 2 to 5 mm/m
(c) 5 to 10 mm/m (d) 10 to 15 mm/m

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : The draft or taper allowance is provided on the pattern for easily removal from mould, it is generally in the range of 1mm/m to 2mm/m.

128. Swab is used for

(a) smoothing and cleaning out depressions in the mould
(b) cleaning and moulding sand
(c) moistening the sand around the edge before removing the pattern
(d) reinforcement of sand in the top part of the moulding sand

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : Swab is a temperature fiber brush used for moistening the edges of sand mould, which are in contact with the pattern surface, before withdrawing the pattern. It is also used for coating the liquid blacking on the mould faces in dry sand moulds.

129. The purpose of a gate is to

(a) deliver molten metal into the mould cavity
(b) act as a reservoir for the molten metal
(c) feed the molten metal to the casting in order to compensate for the shrinkage
(d) deliver molten metal from pouring basin to gate

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : The purpose of a gate is to delivers molten metal into mould cavity, a proper method of gating system is that it leads the pure molten metal to flow through a ladle to casting cavity which ensures proper & smooth fitting of the cavity.

130. For feeding the liquid molten metal into the mould, converging passage is used

(a) to increase the rate of feeding
(b) to reduce the wastage of cast metal
(c) to reduce air aspiration
(d) to easily break off the protruding portion of casting

GPSC Executive Engineer 23.12.2018

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Ans. (c) : To avoid air aspiration effect for feeding the liquid molten metal into the mould, converging passage is used.

131. In a green sand moulding process, uniform ramming leads to
(a) less chance of porosity
(b) uniform flow of molten metal into mold cavity
(c) greater dimensional stability of the casting
(d) less sand expansion type of casting defect

TNPSC AE 2018, GATE 1992

Ans. (c) : In a green sand moulding process, uniform ramming leads to greater dimensional stability of the casting.

132. "A reservoir of molten metal provided in the casting so that hot metal can flow back into the mould cavity due to the reduction in volume of metal due to solidification" is known as:
(a) Spure (b) Runner
(c) Riser (d) Core
(e) Chills

CGPSC AE 25.02.2018

Ans. (c) : Riser- A reservoir of molten metal provided in the casting so that hot metal can flow back into the mould cavity due to the reduction in volume of metal due to solidification.
Riser compensate two type contraction—
1. Liquidus
2. Solidification.

133. The ability of casting material to fill the mould cavity is described by .....
(a) cohesiveness (b) reactiveness
(c) fluidity (d) permeability

RPSC AE 2018

Ans. (c) : The ability of casting material to fill the mould cavity is described by fluidity.
Fluidity—Fluidity an ability of metals and alloys to flow through the getting system filling the cavity of the casting mould and conforming its shape.
Factor determining fluidity
Solidification interval—The wider solidification interval of an alloy, the lower fluidity of the alloy.
Viscosity—Viscosity of the alloy, the lower viscosity of an alloy, the better its fluidity.

134. A cylindrical blind riser with diameter 'd' and height 'h' is placed on the top of mold cavity of a closed type sand mold. If the riser is of constant volume, then the rate of solidification in the riser is the least when the ratio h/d is
(a) 1 : 2 (b) 2 : 1
(c) 1 : 4 (d) 4 : 1

TNPSC AE 2018

Ans. (a) : Optimum condition for top blind riser- solidification time (t_s) ↑ and surface are A ↓

$$A = \frac{\pi}{4}d^2 + \pi dh$$

$$V = \frac{\pi}{4}d^2 \times h$$

$$A = \frac{\pi}{4}d^2 + \pi d.\frac{4V}{\pi d^2}$$

$$\frac{dA}{d(d)} = 0$$

$$2\frac{\pi}{4}d + (-1)\frac{4V}{d^2} = 0$$

$$\frac{2\pi}{4}d = \frac{4V}{d^2}$$

$$V = \frac{\pi}{8}d^3 = \frac{\pi}{4}d^2 \times h$$

$$\boxed{\frac{h}{d} = \frac{1}{2}}$$

135. An aluminium cube of 12 cm side has to be cast along a cylindrical riser of height equal to its diameter. The riser is not insulated on any surface. If the volume shrinkage of aluminium during solidification is 6 percent, calculate shrinkage volume on solidification
(a) 72 cm^3 (b) 103.68 cm^3
(c) 1728 cm^3 (d) 1624.32 cm^3

TNPSC AE 2018

Ans. (b) : Data given as,
a = 12 cm
Volume shrinkage of aluminium during solidification = 6% × V
$V = a^3 = (12)^3 = 1728\ cm^2$
Then, shrinkage volume on solidification

$$= 1728 \times \frac{6}{100} = 103.68\ cm^3$$

136. Surfaces to be left unfinished are to be painted
(a) red (b) blue
(c) black (d) yellow

TNPSC AE 2018

Ans. (c) : Those surfaces which are to be left unfinished, painted by black colour.

137. The process in which expendable pattern of wax, polystyrene plastic or frozen mercury is used for making the casting mould is called.
(a) Plaster mould casting (b) Die-casting
(c) Slush casting (d) Investment casting

MPSC Lect. (Prod.) 2018

Ans. (d) : The process in which expendable pattern of wax, polystyrene plastic or frozen mercury is used for making the casting mould is called investment casting.

138. Three dimensional solid construction, often called as
(a) Solid modeling
(b) Primitive instancing
(c) Constructive solid geometry
(d) All above

MPSC Lect. (Prod.) 2018

Ans. (a) : Three dimensional solid construction, often called as solid modeling.

139. In casting of a metal, the surface area is kept constant but volume is doubled. The solidification time will:
(a) increase to 2 times (b) decrease by 4 times
(c) decrease by 2 times (d) increase to 4 times

AAI Jr. Executive 29.11.2018
DRDO Scientists 2008

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Ans. (d) : $t \propto \left(\frac{V}{A}\right)^2$

$t_1 = V^2$

$t_2 = 4V^2$

$\frac{t_1}{t_2} = \frac{V^2}{4V^2}$

$\boxed{t_2 = 4t_1}$

140. Decreasing the clay content in the moulding sand increases the

(a) Permeability (b) Refractoriness
(c) Green strength (d) Flowability

HPPSC AE 2018

Ans. (a) : Decreasing the clay content in the moulding sand increase the permeability. The addition of clay in sand increase the optimum moisture content.
The addition of clay reduces the hydraulic conductivity of sand. The effect is very prominent upto 10% clay content and beyond that reduction in permeability is gradually decrease.

141. Refractoriness of the moulding sand can be increased by

(a) Increasing the grain size
(b) Decreasing the grain size
(c) Increasing the water content
(d) Increasing the clay content

HPPSC AE 2018

Ans. (a) : Refractoriness –Refractoriness is the ability of the moulding material to withstand the high temperatures of the molten metal to be poured so that it does not get fused with the metal. The refractoriness of the silica sand is highest. Refractoriness is measured by the sinter point of the sand.
Refractoriness of the moulding sand can be increased by increasing the grain size.

142. Which of the following generally has square cross section

(a) Ingot (b) Bloom
(c) Slab (d) Billet

HPPSC AE 2018

Ans. (d) : Billet has square cross-section.
Ingot–Ingot is a brick of cast iron poured off a foundry or drained out of a bloomery. Size is dependent on the size of the foundry and the intended used for the ingot.
Bloom–A bloom is an intermediate product, rectangular is shape with a cross sectional area typically large than 36 sq.in.
Slab–A slab is an intermediate shape between an Ingot and a plate with a width at least twice its thickness.
Billet–A billet can be round or square with cross-sectional area.

143. In sand-casting, when it is not possible to get very fine details on the surface then some modifications are done in a pattern termed as:

(a) Pattern allowance
(b) Shrinkage allowance
(c) Machine allowance
(d) Core prints
(e) Elimination of details

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : In sand-casting, when it is not possible to get very fine details on the surface then some modification are done is known as Elimination of details.
Pattern allowance, shrinkage allowance and machine allowance are provided on the pattern if the casting is to be machined whereas core print is an open space provided in the mould for locating, positioning and supporting core.

144. The ability of the moulding material to withstand the high temperatures of molten metal so that it does not cause fusion is known as:

(a) Refractoriness (b) Green strength
(c) Dry strength (d) Hot strength
(e) Permeability

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Refractoriness is the ability of the moulding material to withstand the high temperature of molten metal so that it does not cause fusion. Moulding sands with poor refractoriness may burn when the molten metal is poured into the mould cavity. Usually sand moulds should be able to withstand upto 1650°C.

145. _____ is not the slag trap method used in sand molding.

(a) Bottom gate (b) Pouring basin
(c) Strainer core (d) Runner extension
(e) Whirl gate

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Bottom gate is not the slag trap method used in sand molding.

146. Die cast parts are used when–

(a) material of the parts has low melting point
(b) parts have small size
(c) all three objectives are desired
(d) parts are made on large scale

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Die cast parts are used when all three objectives are desired.

147. In die casting process

(a) Any metal can be cast
(b) Any size of casting can be prepared
(c) The cost of dies is generally insignificant
(d) Very high production rates are possible

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : In die casting process a high production rates are possible.

148. The main advantage of investment casting is–

(a) Raw materials required are cheap
(b) No special tooling is required
(c) There is no sophisticated technology involved
(d) It can be conveniently adopted for heavy castings

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Investment casting is suitable for both small cast part & heavy or large casting.

149. The molten metal is not introduced directly into the mould cavity as it will cause:

(a) Erosion of the mould cavity
(b) Corrosion of the mould cavity
(c) Crack in the bottom surface
(d) Breakage of the mould

TRB Polytechnic Lecturer 2017

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Ans. (a) : The molten metal is not introduced directly into the mould cavity as it will cause erosion of the mould cavity.

150. Which one does not form a part of the gating system for a casting:
(a) Pouring basin (b) Sprue
(c) Choke (d) Pattern

TRB Polytechnic Lecturer 2017

Ans. (d) : Pouring basin, sprue, and choke are the parts of gating system for a casting whereas pattern is not a part of gating system.

151. Fine grains size during the solidification of a metal is achieved by:
(a) Lower nucleation rate
(b) Higher nucleation rate with lower growth rate
(c) Higher nucleation rate with higher growth rate
(d) Larger growth rate

TRB Polytechnic Lecturer 2017

Ans. (b) : Fine grains size during the solidification of a metal is achieved by higher nucleation rate with lower growth rate.

152. Statement (I): Thicker sections of casting take longer to solidify than thinner sections.
Statement (II): Thicker sections of casting carry residual stresses.

ESE 2017

Ans. (b) : Thicker sections take longer time for solidification because they have higher (V/A). Thicker sections have more residual stresses due to higher differential cooling rate. Thicker sections of casting will take longer duration because, amount of molten metal in cavity is high, heat to be removed from the molten metal is high and hence time taken is longer.
As the solidified metal is shrinking, it induces residual stresses in the thicker sections. Hence both the statements are correct but II is not the correct explanation of I.

153. Statement (I): Sand with grains of uniform round shape is preferred for preparing moulds.
Statement (II): If grains are large and regular in shape, the air-venting property of the mould prepared with them would be better.

ESE 2017

Ans. (d) : Uniform round shape grains do not provide good strength. Generally good sand is one in which most of the sand is refined in three consecutive sieves. Round grains have more void space. So air venting capability is more. As grain size is large, the inter space will be high hence porosity property is high.
Hence statement-I is false and statement-II is correct.

154. For heavy and large size casting, which gating system is used?
(a) Step gate (b) Top gate
(c) Bottom gate (d) parting line gate
(e) laminar gate

CGPSC ARTO & TSI 07.01.2017

Ans. (a) : Step gate :-They used for heavy and large castings. The molten metal enters mold cavity through a number of in-gates, which are arranged in vertical steps. The size of in-gates is normally increased from top to Bottom. This ensures the gradual filling of the mold without mold erosion and produce sound casting.

155. The linear shrinkage of brass is approximately
(a) 10 mm/meter (b) 20 mm/meter
(c) 15 mm/meter (d) 13 mm/meter

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : The linear shrinkage of brass is approximately
15 mm/meter or about 1.4%.

156. The surface at which the two halves of the sand mould meet when cope and drag are filled together is called
(a) Middle Surface (b) Casting surface
(c) Parting Surface (d) None of the above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : The surface at which the two halves of the sand mould meet when cope and drag are filled together is called parting surface.

157. A cubic casting of 50mm side undergoes volumetric solidification shrinkage and volumetric solid contraction of 4% and 6% respectively. No. riser is used. Assume uniform cooling in all directions. The side of the cube after solidification and contraction is
(a) 48.32mm (b) 49.90mm
(c) 49.94mm (d) 49.96mm

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : Volumetric solidification shrinkage and volumetric solid contraction cause decrease in dimension volume of cube = $50^3 = 125000\ mm^3$
After considering both allowances
$V_n = 125000 \times (1 - 0.04) \times (1 - 0.06)$
$= 112800\ mm^3$
Side of cube $= \sqrt[3]{112800} = 48.32\,mm$

158. A cube shaped casting solidifies in 5 minutes. The solidification time in minutes for a cube of the same material, which is 8 times heavier than the original casting will be
(a) 10 (b) 20
(c) 24 (d) 40

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Given,
$t \propto (V/A)^2$ and $m_2 = 8\ m_1$
$\rho V_2 = 8\rho V_1$
$\Rightarrow \quad V_2 = 8V_1$
$a_2 = 2a_1 \Rightarrow t_2 = 4t_1$
$t_2 = 4 \times 5 = 20$ min

159. The hot tearing in a metal casting is due to
(a) High fluidity
(b) High melt temperature
(c) Wide range of solidification temperature
(d) Low coefficient of thermal expansion

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : The hot tearing in a metal casting is due to wide range of solidification temperature.

160. Process in which lowest tolerances in cast products of ferrous and non ferrous metals can be achieved is
(a) Die casting
(b) Shell casting
(c) Investment casting
(d) Sand casting

ISRO Scientist/Engineer 17.12.2017

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Ans. (c) : Investment casting uses a piece of ceramic mould. The mould is prepared by surrounding the ceramic material over the wax or plastic pattern. Once the ceramic material solidifies, the wax replica is melted and drained out from the mould and the metal is poured into the mould cavity.
A wide variety of materials such as both ferrous and non-ferrous can be used in investment casting. Any metal that can be melted in standard induction furnace or vacuum furnace can be considered for this case.

161. In continuous casting process, the mould is generally made of
(a) Cast iron (b) Bronze
(c) Copper (d) Tungsten carbide

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Continuous casting process, the mould is generally made of copper or graphite.

162. Consider the following statements with regard to the feeding system for a casting:
1. The feeder should be thermally adequate
2. The feeder should be volumetrically adequate
3. The feeder should have adequate feeding range.

Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

JWM 2017

Ans. (d) : In feeding system for casting, the feeder should be thermally adequate, volumetrically adequate and have adequate feeding range.

163. In any casting process, shrinkage allowance on the pattern is provided in order to compensate for shrinkage when
(a) The temperature of the liquid metal drops from the pouring to its freezing temperature
(b) The metal changes from the liquid to the solid state at its freezing temperature
(c) The temperature of solid phase drops from it freezing temperature to room temperature
(d) The temperature of metal drops from its pouring temperature to room temperature

JWM 2017

Ans. (c) : Solid shrinkage are compensated by giving shrinkage allowance is oversize pattern.
Solid Shrinkage- When the temperature of solid phase drops from its freezing temperature to room temperature.

164. Cold chamber die casting is suitable for which of the material listed below?
(a) Brass (b) Tin
(c) Zinc (d) Lead

CIL MT 26.03.2017

Ans. (a) : Cold chamber die casting : In this the furnace is separated from casting unit. It is used to produce high melting point material casting. Force required to move the liquid metal into the die will be high when compare to hot chamber die casting. **Brass,** Copper Aluminium and Magnesium is commonly cold die casting metals.

165. Sand and oxide layers adhering to the casting are removed by which of the following processes?
(a) Heating (b) Shot blasting
(c) Gas cutting (d) Air cooling

CIL MT 26.03.2017

Ans. (b) : Shot blasting is a post-production finishing service that is processed through the rapid impacting of a part's surface with a defined amount of abrasive material. It helps in removing sand and oxide layer.

166. Which of the following are provided in mould so as to increase the heat extraction capacity of the sand mould?
(a) Chaplets (b) Core
(c) Chills (d) Cope

CIL MT 26.03.2017

Ans. (c) : Chaplet: Chaplets are used to support cores inside the mould cavity.
Chill: Chills are metallic objects, which are placed in the mould to increase the heat extraction capacity of castings.
Core: Used for making hollow cavities in castings.
Cope: Upper part of mould is cope and lower part is drag. Sometime moulding box contains three parts at that time middle part is known as cheek.

167. Which of the following is the best suited for production of hollow pipes?
(a) Centrifugal casting
(b) Investment casting
(c) Continuous casting
(d) Hot chamber die casting

CIL MT 26.03.2017

Ans. (a) : Centrifugal casting is used for making bigger size hollow symmetrical pipes. For producing a hollow part, the axis of rotation is placed at the center of the desired casting. The speed of rotation is maintained as high as to produce a centripetal acceleration of the order of 60g to 70g. It should be noted casting of hollow parts need no core in this process.

168. Which force is acting on the core when the metal is poured into mould cavity?
(a) Gravity (b) Inertia
(c) Buoyancy (d) Drag

CIL MT 26.03.2017

Ans. (c) : As the density of core (made of sand) is less than the density of metal being poured in the cavity. So there will be an upward buoyancy force on the core. To overcome this force the core print are used.

169. Which of the following materials is not used for pattern making?
(a) Mica (b) Wax
(c) Wood (d) Plaster of Paris

TSPSC AEE 2017

Ans. (a) : Wax, POP and wood are used for making pattern, mica is not used for pattern, used for making mould.

170. Pitch obtained as a by product of coke is used in moulding sand for improving its
(a) Permeability (b) Hot strength
(c) Green strength (d) Dry strength

TSPSC AEE 2017

Ans. (b) : Pitch obtained as a byproduct of coke is used in moulding sand for improving its hot strength and casting finish.

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171. Which of the following helps to remove steam and gasses formed as result of molten metal contact with moulding sand.
(a) Riser (b) Vents
(c) Gating system (d) Core prints
TSPSC AEE 2017

Ans. (b) : Vents (Small holes) helps to remove steam and gases formed as result of molten metal contact with moulding.

172. The solidification of alloys starts as soon as the metal temperature falls below the
(a) Freezing range
(b) Solidus
(c) Liquidus
(d) Average of Melting temperatures of the constituent metals
TSPSC AEE 2017

Ans. (c) : The solidification of alloys starts as soon as the metal temperature falls below the liquidus.

173. The minimum thickness limit for sand casting of grey cast iron is
(a) 0.5 mm (b) 1 mm
(c) 5 mm (d) 3 mm
TSPSC AEE 2017

Ans. (d) : The minimum thickness limit for sand casting of grey cast iron is 3 mm.

174. The tool used for lifting the pattern from the Mould is called
(a) Draw spike (b) Slick
(c) Lifter (d) Trowel
GMB AAE 25.06.2017

Ans. (a) : Draw spike are used to lift the pattern from the sand mould. Two types of draw spike.
(i) Draw pin is inserted into a wooden pattern.
(ii) Draw plate fastened to the large wooden plate tapped hole of metal pattern.

175. Loam moulds are used for
(a) large castings
(b) very small castings
(c) symmetrical castings
(d) centrifuging
WBPSC AE, 2017

Ans. (a) : Loam moulds are employed chiefly in the making of large casting for which it would be expensive to use full pattern and ordinary flask equipments.

176. A gagger is tool used for
(a) cleaning welds.
(b) cleanings.
(c) supporting hanging part of cope
(d) cleaning materials before welding.
WBPSC AE, 2017

Ans. (c) : A gagger tool is used for reinforcing the moulding sand in the cope part of the moulding box.

177. The furnace used for casting in a foundry shops is known as
(a) Blast furnacey
(b) Electric induction furnace
(c) Cupola furnacey
(d) muffle furnace
WBPSC AE, 2017

Ans. (b) : Electric induction furnace commonly used in foundries because are high quality and simple to operate.

178. The machining allowance provided on patterns depends upon
(a) type of casting metal
(b) size and shape of casting
(c) method of casting used
(d) all of these
Sikkim PSC (Under Secretary), 2017

Ans. (d) : The machining allowance provided on pattern depends upon-
1. Type of casting metal
2. Size and shape of casting
3. Method of casting
4. Shrinkage allowance
5. Draft allowance
6. Distortion allowance
7. Shake allowance

179. The centrifugal casting method, is used for casting articles of
(a) symmetrical shape about vertical axis
(b) symmetrical shape about horizontal axis
(c) irregular shape
(d) non-ferrous metal only
Sikkim PSC (Under Secretary), 2017, ESE 2005

Ans. (b) : The centrifugal casting is normally carried out in permanent mould which is rotated during the solidification of casting for producing hollow parts, axis of rotation placed at centre of the desired casting. The centrifugal action less nonmetallic inclusion near centre. Here no core is used for casting of hollow parts.
Centrifugal casting are of three types-
(1) True centrifugal casting
(2) Semi centrifugal casting
(3) Centrifuging
True centrifugal casting- This is normally used for hollow pipes, tubes, hollow bushes.
Semi centrifugal casting- Semi centrifugal casting is used for jobs which or more complicated than those possible in true centrifugal casting.
Centrifugal- In order to obtain metal pressure during solidification, when casting shapes or not axis symmetrical, the centrifuging process is used.

180. The defects as blow holes, cracks, shrinkage cavities and slag inclusions in metal casting can be inspected by the following method.
(a) Radiography test
(b) Visual inspection
(c) Damping test
(d) Magnetic particle test
Karnataka PSC Lect., 27.05.2017

Ans (a) : X-ray testing is the best nondestructive inspection method for detecting internal defect of casting. In this method a casting is exposes to radiation from an X-ray tube. The casting absorb part of radiation and the remaining part of the radiation exposes the radiography film.

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181. The solidification time for a metal casting 't', depends on the density of metal 'ρ', temperature of liquid metal θ_1 and temperature of mould surface θ_2 in such a way that

(a) $t \propto \frac{\rho}{(\theta_1 - \theta_2)}$ (b) $t \propto \frac{\rho}{(\theta_2 - \theta_1)}$

(c) $t \propto \frac{(\theta_1 - \theta_2)}{\rho}$ (d) $t \propto \frac{(\theta_2 - \theta_1)}{\rho}$

Karnataka PSC AE, 10.09.2017

Ans. (a) : $t \propto \rho$

$$t \propto \frac{1}{\theta_1 - \theta_2}$$

$$\therefore t \propto \left(\frac{\rho}{\theta_1 - \theta_2}\right)$$

Where,

θ_1 = temp. of liquid metal

θ_2 = temp. of mould surface

ρ = density of metal.

182. Rat tail defect in a casting is due to

(a) mould expansion
(b) metal expansion
(c) metal mould expansion
(d) none of the above

Punjab PSC (Lect.) 06.08.2017

Ans. (a)

- Rat tails and buckles are caused by the expansion of a thin outer layer of moulding sand on the surface of the mould cavity due to metal heat.
- A rat tail is caused by depression of a part of the mould under compression which appears as an irregular line on the surface of the casting.
- The mould must provide for proper expansion instead of forming compressed layers to avoid this defect.

183. Shell moulding process requires pattern made of

(a) wood (b) sand
(c) plastic (d) metal

Punjab PSC (Lect.) 06.08.2017

Ans. (d) : Shell Moulding– It is process in which the sand mixed with a thermosetting resin is allowed to come into contact with a heated metallic pattern plate, so that a thin and strong shell of mould is formed around the pattern.

- The synthetic resins used in shell moulding are essentially thermosetting resins which get hardened irreversibly by heat. Resin used are phenol formaldehyde resins.

184. Which of the following rule is used in designing of riser of metal casting mould?

(a) Chvorinov's rule (b) Lever rule
(c) Phase rule (d) Bernoulli's rule

Punjab PSC (Lect.) 06.08.2017

Ans. (a) : Chvorinov's rule–The solidification time is a function of the volume of a casting and its surface area.

$$\text{Solidification time} = K\left(\frac{V}{A}\right)^2$$

K = Solidification factor

185. When pure metals and some eutectic alloys solidify, the interface between the liquid and solid metals is a well defined smooth surface. As the solidification advances towards the center the grains become larger, the thickness of grains 'x' at time 't' is given by

(a) $K_1\sqrt{t}$ (b) $K_1/\sqrt{t}$
(c) $K_1\sqrt{2t}$ (d) K_1/t

where K_1 is a constant, which depends upon the metal poured and the mould material.

Punjab PSC (Lect.) 06.08.2017

Ans. (a) $x = K_1\sqrt{t} + K_2$

186. Zinc mould in permanent casting is preheated in temperature range of :

(a) 150°–200° (b) 250°–275°
(c) 225°–300° (d) 300°–330°

Punjab PSC SDE 12.02.2017

Ans. (a) : Zinc mould in permanent casting is preheated in temperature range of 150° - 200°.

187. The most effective way to reduce the temperature drop in a heat generating solid is to :

(a) reduce the linear dimension
(b) reduce the thermal conductivity
(c) reduce the convection coefficient on the surface
(d) reduce the heat generation rate.

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : The most effective way to reduce the temperature drop in a heat generating solid is to reduce the linear dimension.

188. A trap placed in a horizontal gate to prevent heavier and lighter impurities from entering the mould is

(a) Splash core (b) Pouring basin
(c) Skim bob (d) Strainer

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (c) : Skin bob : It is a trap placed in horizontal gate to prevent heavier and lighter impurities from entering the mould.

189. Plastic parts are generally made by

(a) Investment casting (b) Injection moulding
(c) Shell moulding (d) Continuous casting

UJVNL AE 2016

Ans : (b) Plastics parts are generally made by injection moulding.

Injection moulding:-

It makes use of heat softening characteristics of thermoplastics materials. These materials soften when heated and reharden when cooled. No chemical change takes place when the material is heated or cooled, the change being entirely physical.

190. Moulding sand should posses which property

(a) Collapsibility (b) Flow ability
(c) Cohesion (d) All of the above

UJVNL AE 2016

Ans : (d) Property of moulding sands.:-

i) Permeability
ii) Cohesivness
iii) Adhesivness
iv) Plasticity
v. Refractoriness
vi. Chemical resistivity
vii. Binding Property
viii. Flowability

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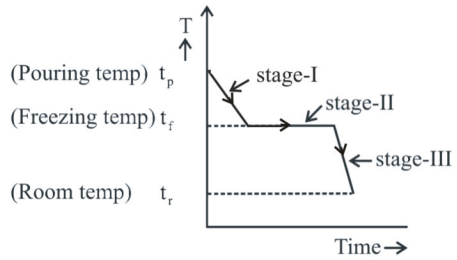
191. A pattern is made slightly larger in dimension to give:
(a) Rapping allowance (b) Shrinkage allowance
(c) Draft allowance (d) None of the above

UJVNL AE 2016

Ans : (b) A pattern is made slillghtly lorger in dimension to give shrinkage allowance
shrinkage allowance:-



1-2 :- Liquid shrinkage
2-3 :- Solidification shrinkage
3-4 :- Solid shrinkage
liquid shrinkage > Solid shrinkage > Phase transfer motion shrinkage.

192. Which of the following is a casting defect?
(a) Blow hole
(b) Slag inclusion
(c) Pour short
(d) All above are casting defects

UJVNL AE 2016

Ans : (d) Casting defects:-

i) Blow hole and open blow	ii) Pin hole porosity
iii) Cuts and washes	iv) Metal penetration
v) Fusion	vi) Runout
vii) Misruns and Cold shuts	viii) Slag inclusion
ix) Hot tear	x) Hot spot

193. Gate in a mould connects
(a) Sprue base with mould cavity
(b) Riser with mould cavity
(c) Pouring basin with mould cavity
(d) Pouring basin with runner

UJVNL AE 2016

Ans : (a) Gate in a mould Connects sprue base with mould cavity. The actual entry point through which molten metal enters the mould cavity.

194. Pipes of large length and diameters are made by :
(a) Extrusion
(b) Slush casting
(c) Pressed casting
(d) Semi-centrifugal casting

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (d) : Pipes of large length and diameters are made by semi centrifugal casting.
Semi-centrifugal casting is used to produce larger parts that are symmetrical along the vertical axis, such as gear blanks and flywheels. The mold used is arranged horizontally and then rotated. While the process generates sufficient level of centrifugal force to cast outer walls, it needs a core positioned in the center of the mold to cast the hollow section.

195. For small and intricate casting the sand grain should be ..... .
(a) fine (b) coarse
(c) medium (d) rounded

GWSSB DEE 07.07.2016

Ans. (a) :

196. Match List-I (Material to be cast) with List-II (Shrinkage allowance in mm/m) and select the correct answer using the codes given below the list:

List-I (Material to be used)	List-II (Shrinkage allowance in mm/m)
A. Steel	**1. 20**
B. Zinc	**2. 24**
C. Brass	**3. 7-10**
D. Grey cast iron	**4. 15**

(a) A-3, B-2, C-1, D-4 (b) A-1, B-2, C-4, D-3
(c) A-3, B-4, C-1, D-2 (d) A-1, B-2, C-3, D-4

UPRVUNL AE 21.08.2016

Ans. : (b)

Materials	Shrinkage allowance (mm/m)
Steel	20
Zinc	24
Brass	15
Grey cast iron	7–10

197. Due to the shake allowance:
(a) The fillets are provided on pattern
(b) The pattern is made slightly bigger than required size
(c) The tapper is provided on vertical faces of pattern
(d) The pattern is made slightly smaller than required size

UPRVUNL AE 07.10.2016

Ans. (d) : Due to the shake allowance the pattern is made slightly smaller than required size.

198. "Bell" shapes are generally made by
(a) Single piece pattern
(b) Gated pattern
(c) Cope and drag pattern
(d) Sweep pattern

RPSC LECTURER 16.01.2016

Ans. (d) : Sweep pattern is used to "Bell" shapes or 3-D symmetrical object casting.

199. Shrinkage allowance may be expressed in:
(a) mm/kg.f (b) mm/meter
(c) mm/kg (d) radian/sec

UPRVUNL AE 07.10.2016

Ans. (b) : Shrinkage allowance may be expressed in mm/meter.

200. For Gating system of casting, choke are (A) may be written as (where, m = mass of casting, ρ = density of molten metal, H = effective metal height, C = efficiency factor, t = pouring time):

(a) $A = \frac{m}{\rho t C\sqrt{2gH}}$ (b) $A = \frac{\rho m}{tC\sqrt{2gH}}$
(c) $A = \frac{m\sqrt{2gH}}{\rho tC}$ (d) $A = \frac{tm}{\rho C\sqrt{2gH}}$

UPRVUNL AE 07.10.2016

Ans. (a) : Choke area,

$$A = \frac{m}{\rho t C\sqrt{2gH}}$$

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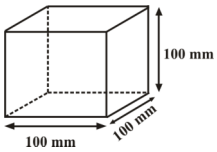
201. Which of the following is not the casting defect?
(a) Hot tears (b) Stop-off
(c) Run out (d) Blow holes
UPRVUNL AE 07.10.2016

Ans. (b) : Stop-off is not the casting defect.
Hot-tears—Hot tears, also known as hot cracking, are failures in the casting that occur as the casting cools.
Run-out—A run-out is caused when the molten metal leaks out of the mould caused by faulty mould making or faulty flask.
Blow holes—When gases entrapped on the surface of the casting due to solidifying metal, a rounded or oval cavity is formed called as blow holes.

202. If a cube of 100 mm dimension is to be made of Aluminium (shrinkage allowance = 10 mm/m). What will be the dimension of wooden pattern? Assume only shrinkage allowance.
(a) 101mm × 101mm × 101mm
(b) 100mm × 99mm × 98mm
(c) 99mm × 99mm × 99mm
(d) 102mm × 101mm × 100mm
UPRVUNL AE 07.10.2016

Ans. (a) : Shrinkage allowance = 10 mm/m
= 1 mm/100 mm



Then dimension of wooden pattern
= (100 + 1) × (100 + 1) × (100 × 1)
= 101 × 101 × 101 mm^3

203. A casting of size 100 × 100 × 100 mm^3 solidifies in 20 minutes. Find out the solidification time for casting of size 100 × 100 × 50 mm^3 under same condition.
(a) 16.3 minutes (b) 14.3 minutes
(c) 12.3 minutes (d) 11.3 minutes
RPSC LECTURER 16.01.2016

Ans. (d) : Casting size = 100 × 100 × 100 mm^3
Solidifies time = 20 minutes
Solidification time = t_2
For size = 100 × 100 × 50 mm^3
Solidification time $t_s = K\left[\frac{V}{A}\right]^2$
where
K - Solidification area
A - surface area
For same condition

$$t_{s1} = K\left[\frac{100\times100\times100}{6\times100^2}\right]^2$$

$$t_{s1} = K\times\left[\frac{100}{6}\right]^2 \quad ...(1)$$

$$K = \frac{20}{\left(\frac{100}{6}\right)^2} = 0.072\,\text{mint/mm}$$

$$t_{s2} = K\left[\frac{100\times100\times50}{2\times100\times100+2\times100\times50+2\times100\times50}\right]^2$$

$$t_{s2} = 0.072\left[\frac{500000}{40000}\right]^2$$

$$t_{s2} = 0.072\times156.25$$

$$t_{s2} = 11.25 \simeq 11.3\,\text{minutes}$$

204. In shell moulding, the dimensional accuracy that can be expected is
(a) 0.001 to 0.002 mm/mm
(b) 0.003 to 0.005 mm/mm
(c) 0.1 mm to 0.3 mm/mm
(d) 0.3 mm to 0.5 mm/mm
Nagaland PSC CTSE 2016 Paper-I
SJVN ET 2013

Ans. (b) : Shell moulding is a process that uses a risk covered sand to form the mold. It has better dimensional accuracy of the order of 0.002 to 0.003 mm. Shell mould requires less floor space for casting.

205. The disadvantage of die casting process is
(a) High production rates are not possible
(b) Surface finish is poor
(c) Heavy sections can not be easily cast
(d) Cost is less
Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Disadvantage of die-casting process are :-
1) All metal & alloy cannot be cast.
2) Cost of machines, dies and other equipment is high
3) Heavy casting cannot be cast.

206. The firing of a cupola is started
(a) 2½ to 3 hours before the first metal is tapped
(b) 6 to 8 hours before the first metal is tapped
(c) 8 to 12 hours before the first metal is tapped
(d) 12 to 18 hours before the first metal is tapped
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : The firing of a cupola is stared 2½ to 3 hours before the first metal is tapped for pouring.

207. True centrifugal casting is used to
(a) Cast objects symmetrical about an axis
(b) Get accurate castings
(c) Get dynamically balanced castings
(d) Get statistically balanced castings
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : True centrifugal casting is generally used for the making of hollow pipes and tubes, which are axi-symmetric with a concentric hole.

208. The fuel used in pit furnace is generally
(a) Wood (b) Charcoal
(c) Oil (d) Coal
Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : The fuel used in pit furnace is generally coal.

209. Riser is generally located
(a) near the deepest section of the mould
(b) near the parting line
(c) on the upper most section of the casting
(d) Opposite to the pouring end
Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Riser is generally located on the upper most section of the casting.

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210. In die casting the moulds are made of

(a) metal (b) ceramic
(c) sand (d) glass

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : In die casting, the mould is made by metallic material. Metallic mould is used in die-casting which result for the extraction of heat from the molten metal will not be easy. So, water cooling channels is used near the castings. This made the proper solidification of the castings.

211. Tuyeres are provided in cupola furnace for

(a) air blasting
(b) collecting slag
(c) collecting liquid metal
(d) filling in the charge

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : The tuyeres are the components in cupola furnace are responsible for the supply of air for combustion in cupola with the help of blower.

212. Shift is a casting defect which

(a) Results in a mismatching of the top and bottom parts of a casting
(b) Results in general enlargement of a casting
(c) Occurs near the ingrates as rough lumps on the surface of a casting
(d) Occurs as sand patches on the upper surface of a casting

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : Shift is a casting defect which results in a mismatching of the top and bottom parts of a casting.

213. Dye penetration inspection is generally used to locate

(a) Temporary defects
(b) Surface breaking defects
(c) Artificial defects
(d) Core defects

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : Dye penetration inspection is generally used to locate Surface breaking defects.

214. Refractoriness in the moulding sand is due to the presence of

(a) clay (b) silica
(c) additives and binders (d) dust

BPSC Poly. Lect. 2016

Ans : (b) Refractoriness in the moulding sand is due to the presence of silica.
It is the ability of the silica sand to withstand high heat without breaking down or fusing. Sand with poor refractoriness may burn at high temperature.

215. Hardness of green sand mould increases with :

(a) Increase in permeability.
(b) Increase in moisture content beyond 4 percent
(c) Increase in both moisture content and permeability
(d) Decrease in permeability.

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Hardness of green sand mould increases with decrease in permeability.
Permeability- It is the property of moulding sand which allow to escape the gases from moulds. If gas is trapped by liquid metal. Otherwise it solidify and form gas defects hence permeability will decrease when hardness of moulds increases.

Permeability No. $\left(p_n = \frac{VH}{pAT}\right)$

p = Difference of pressure of air passing through specimen (g/cm^2)
A = $(\pi/4d^2)$ [Cross-sectional area of cylindrical specimen (cm^2)]
T = Time taken by air to escape (minute)
V = Volume of air passing through specimen (cm^3)
H = Height of cylindrical specimen (cm)

216. .....force is present inside the mould cavity because of the head with which the metal is entering the mould cavity during sand casting.

(a) Metallostatic (b) Centripetal
(c) Coriolis (d) Shear

HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Metallostatic force is present inside the mould cavity because of the head with which the metal is entering the mould cavity during sand casting.
Chaplets are used to support the cores inside the mould cavity to take core of its own weight and overcome the metallostatic force.
Pressure developed within a volume of molten metal. In mechanical penetration, the metal penetrates into the green sand mould because of the metallostatic pressure of molten metal.

217. Casting shapes which are axi-symmetrical in shapes are manufactured using .....casting.

(a) Centrifuging (b) Centrifugal
(c) Sheave (d) None of these

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : Advantages of centrifugal casting-
(1) There is no gating element used.
(2) Casting yield is 100%.
(3) Accuracy and surface finish is vey high.
(4) It is generally, used for hollow-axi-symmetrical cylindrical shape of objects only.
Application of centrifugal casting-
(1) Hollow cylindrical pipes
(2) Propeller shafts.
Centrifuging casting is used for unsymmetrical shapes.

218. Due to chilling of the castings the casting defect occurs is known as :

(a) Swell (b) Drop
(c) Buckles (d) Hot spots

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : Due to chilling of the castings the casting defect occurs is known as hot spots.
Hot spots defect occur when an are on the casting cools move rapidly than the surrounding materials. Hot spot are areas on the casting which is harder than the surrounding area.
It is also called as hard spot.

219. .....type of pattern can be used when the contour of the part is such that withdrawing the pattern from the mould is not possible due to obstructing part.

(a) Follow board (b) Skeleton

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(c) Loose piece (d) Sweep

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Loose piece– When a one piece solid pattern has projections or back drafts which lie above or below the parting plane, it is impossible to with draw it from the mould with such patterns, the projections are made with the help of loose pieces.

220. Goose-neck injector is used in..... process.

(a) precision investment casting
(b) shell moulding
(c) sand casting
(d) hot chamber die casting

HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : Goose neck injector is used in hot chamber die casting.

221. .....generally refers to the stress required to rupture the specimen under compressive loading.

(a) Refractoriness (b) Green strength
(c) Shear stress (d) Rigidity

HPPSC AE (PWD) 10.09.2016

Ans. (b) : The strength of the sand in green or moist termed as green strength. The green sand particle have the ability to clinging to each other to impart sufficient strength to the mould.

222. "Nishiyama Process" a process comes under:

(a) Metal cutting
(b) Foundry
(c) Non-conventional machining
(d) Plastic fabrication

HPPSC AE (PWD) 10.09.2016

Ans. (b) : To use Granulated Blast Furnace (GBF) slag as mould material for either full or partial replacement of existing silica sand in foundry Industry Nishiyama process was adopted for evaluating the same. A series of sand tests were carried out on sand and slag individually and also combinations of these two. Three types of mould were made with sand slag individually and combination of these two.

223. The compression failure of the skin of the mould cavity because of the excessive heat in the molten metal causes.....

(a) Run out (b) Drop
(c) Rat tail (d) Swell

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Rat tail and buckle caused by compression failure of the skin of the mould cavity because of excessive heat in the molten metal. A buckle is more severe failure of the sand surface under compression.

224. .....pattern is adopted for those castings where there are some portions which are structurally weak and if not supported properly are likely to break under the force of ramming.

(a) Skeleton (b) Sweep
(c) Loose Piece (d) Follow Board

HPPSC AE (PWD) 10.09.2016

Ans. (d) : Follow board pattern is a pattern that consists of a simple wooden board which is used for several reasons the wooden board is used as a base in follow board pattern for molding process. This patterns is used in processes where casting structures are weak and they may break after the application of force.

225. Which one of the following processes consists of central sprue to feed metal into the cavities through a number of radial gates?

(a) Centrifuging
(b) true centrifugal casting
(c) Precision casting
(d) Semi-centrifugal casting

UPRVUNL AE 21.08.2016

Ans. : (a) Centrifuging processes consists of central sprue to feed metal into the cavities through a number of radial gates.

226. In which of the following is mould rotated rapidly about the central axis as the metal is poured into it?

(a) Vacuum sealed moulding process
(b) Investment casting
(c) Centrifugal casting
(d) Evaporative pattern casting

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : This is the process where the mould is rotated rapidly about the central axis as the metal is poured into it. Because of the centrifugal force, a continuous pressure will be acting on the metal as it solidify. The slag oxides and other inclusions being lighter get separated from the metal and segregates toward centre.
Three main type of centrifugal casting are–
(i) True centrifugal casting
(ii) Semi centrifugal casting
(iii) Centrifuging

227. The pattern used for the production of large quantity is

(a) Skeleton (b) Match plate
(c) Split (d) Single plate

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : These are generally used for small casting with higher dimensional accuracy and large production the gating system is already made and attached to the match plate. These patterns are used for machine moudling. They are expensive since they increase productivity.
Skeleton– It is used to prepare shells and drums by loam sand (loam sand = 50% clay + 50% silica used)
Split– It is being used when complexity of part is more and the withdraw of pattern is difficult.
Single plate– It is used for very simple casting. Atleast one surface must be flate surface.

228. Residual stresses in casting occur due to:

(a) Insufficient gate size
(b) Impurities in molten metal
(c) Differential rate of solidification of thin and thick sections of casting
(d) Sand particles mixing into molten metal while pouring

GPSC Asstt. Prof. 28.08.2016

Ans. (c) : Residual stresses are developed during the solidification process due to the temperature gradients between different parts of casting or due to the mechanical constraints imposed by the mould during shrinkage of the cast.

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229. In cold chamber method of die casting:
(a) Low melting point metals can be cast
(b) High melting point metals can be cast
(c) Die is kept hot by electrical heating
(d) Die is kept cold by circulating water

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : Cold chamber die casting is a preferred for non-ferrous materials have high melting points and is used by most of the top parts manufactures in the industry. Typically, this will include metal alloys of aluminium, brass and copper.
Applications-carburetors, crank case, valve body, Toilet fixtures.

230. The minimum thickness of a complicated shape of casting made of steel as compared to cast iron preferably is:
(a) Less (b) Same
(c) More (d) All of these

GPSC Asstt. Prof. 28.08.2016

Ans. (c) : It totally depends on shrinkage allowance. Steel has more (0.0200 mm/mm) shrinkage allowance than cast iron (0.0160 mm/mm), So the thickness is more in steel as compared to cast iron.

231. Two cubical castings of the same metal and sizes of 3 m and 6 m sides are moulded in green sand. If the smaller casting solidifies in 16 mins, what is the expected time of solidification of the larger casting?
(a) $16\sqrt{8}$ min (b) 32 min
(c) 48 min (d) 52 min
(e) 64 min

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : As we know,

$$\text{Solidification time} = \left[\frac{V}{A}\right]^2$$

$$\frac{T_1}{T_2} = \frac{\left[\frac{V}{A}\right]_1^2}{\left[\frac{V}{A}\right]_2^2} = \frac{\left[\frac{3^3}{6\times 3^2}\right]^2}{\left[\frac{6^3}{6\times 6^2}\right]^2} = \frac{\left(\frac{1}{2}\right)^2}{(1)^2}$$

$$\frac{T_1}{T_2} = \frac{1}{4} \Rightarrow T_2 = 4\times 16$$

$T_2 = 64$ min

232. With a solidification factor of 0.826×10^5 s/m^2, what is the solidification time for a spherical casting of 600 mm diameter?
(a) 8.26 sec (b) 82.6 sec
(c) 826 sec (d) 8260 sec
(e) 826 minutes

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (c) : As we know,
Solidification time = Solidification factor × (modulus)^2

$$= 0.826\times 10^5 \times \left(\frac{0.600}{6}\right)^2$$

= 826 sec

233. When there is no room temperature change, the total shrinkage allowance on a pattern is independent of:
(a) Pouring temperature of liquid metal
(b) Freezing temperature of liquid metal
(c) Component size
(d) Coefficient of thermal contraction of solidification metal
(e) Freezing temperature of liquid metal and component size

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : The total shrinkage allowance on a pattern, is dependent on freezing temperature of the liquid metal, component size, and coefficient of thermal contraction of solidification metal, and independent of pouring temperature of liquid metal.

234. Hot chamber die casting is not suited for:
(a) Zinc and its alloys
(b) Aluminium and its alloys
(c) Lead and its alloys
(d) Tin and its alloys
(e) Zinc and its alloys as well as Tin and its alloys

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : Hot chamber die casting is not suited for aluminium and its alloys.

235. Permeability of a moulding sand is increased with
(a) Increase in the moisture content
(b) Coarse grain size
(c) Increasing the compacting pressure
(d) Fine grain size

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Permeability, $Pn = \frac{VH}{PAT}$
V = Volume of air passing through specimen
H = Height of standard specimen
P = Pressure of air passing through the specimen
A = Cross-section of area of cylinder
T = Time
Gas evolving capacity of moulding sand can be enhanced by increasing coarse grain size

236. Fetling of casting is done to
(a) produce uniformly cooled casting
(b) remove extra metals from casting
(c) smoothen surface
(d) All of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Fetling- A process of removing excess material from casting formed by the divided lines by the casting process.

237. Thickness of slush casting process is given by $t = 0.577\tau^{0.5} + 0.413$. After pouring metal if we immediately invert the mould then what is the thickness of casting?
(a) 0 (b) 0.413
(c) 0.99 (d) 1.567

RPSC ADE 2016

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Ans. (b) : Slush casting is a traditional method of permanent mould casting process, where the molten metal is not allowed to completely solidify in the mould. When the desired thickness is obtained the remaining molten metal is poured out. Mainly hollow products are cast by this process.
Example- bowls, candle stick, lamps statues, jewelleries hollow handle.
t (thickness) = $0.577\tau^{0.5} + 0.413$
where τ is time
for $\tau = 0$
$t = 0.413$

238. In a gravity feed system in casting, the cross-section area of horizontal runner
(a) Remains same
(b) Keeps on increasing
(c) Keeps on decreasing
(d) Depends upon material

TSPSC Managers, 2015

Ans. (b) : In a gravity feed system in casting, the cross section area of horizontal furnace keep on increasing. The distinction between gravity and low pressure casting is that the force of gravity is used to fill the mould in gravity casting. Molten metal is poured into a vertical opening (sprue) and flows through the bottom of the sprue and horizontal along a single runner into the casting cavity.

239. Identify the odd one out in casting process
(a) chills (b) padding
(c) inoculation (d) loam sand

TSPSC AEE 2015

Ans. (d) : Loam Sand– Loam sand is mixture of sand and clay with water to a thin plastic paste.
• Loam sand posses high clay as much as 30-35% and 18% water.
• Pattern are not used for loam moulding and shapes is given to mould be sweeps.

240. Chills are not used in the following application
(a) directional solidifications
(b) smaller casting for eutectic alloy
(c) change in crossections
(d) complex geometry

TSPSC AEE 2015

Ans. (b) : Chills are used in the application of directional solidifications, change in cross section and complex geometry not for smaller casting for eutectic alloy.

241. Loam sand comprises of
(a) 40% of clay and 10% moisture
(b) 50% of clay and 10% moisture
(c) 80% of clay and 20% moisture
(d) 50% of clay and 18% moisture

(KPSC AE 2015)

Ans : (d) Loam sand contains upto 50% clay and dries hard. It also contains fire clay. It must be sufficiently adhesive to hold on to the vertical surface of the rough structure of the mould. Chopped stray and manure are commonly used to assist in binding. The moisture content is from 18 to 20%

242. Which of the following materials can be used for making pattern?
(a) Stainless steel (b) Lead alloys
(c) Sand (d) All of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Lead Alloys – It is applied where cheap and corrosion resistance material is required. An alloy contains 8 to 10% Pb are used as bearing metals.
Properties of Lead- Resistance to corrosion and low melting point.
• It is Poisonous.
• Heavy weight, high density, softness, malleability.
• High coefficient of expansion and low electrical conductivity.

243. Chaplets are made of :
(a) graphite
(b) plastic
(c) same metal as the cast metal
(d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Chaplets are used to support core inside the mould cavity to take care of buoyancy force due to molten metal as that of casting.

244. Read the following statements with respect to casting of parts and select to correct answer.
(a) Shrinkage allowance for a casting are independent of the casting material
(b) Distortion allowance is provided to those castings which exhibit uniform cooling rates in their sections.
(c) Shaking or rapping allowance is a negative allowance
(d) Draft allowance is provided only those walls of casting which are in the direction of disengagement of cope and drag
(a) Only statement (a) and (b) are true
(b) Only statement (b) and (c) are true
(c) Only statement (c) and (d) are true
(d) Only statement (a) and (d) are true

TSGENCO AE 14.11.2015

Ans. (c) : Shaking or rapping allowance is a negative allowance.
Draft allowance is provided to only those wall of the casting which are in the direction of disengagement of cop and drag.

245. A riser compensates for the shrinkage that happens in the casting process during –
(a) only molten stage
(b) only solidification stage
(c) solid stage
(d) both molten and solidification stage

TSGENCO AE 14.11.2015

Ans. (d) : A riser compensates for the shrinkage that happen in the casting process during both liquid matter and solidification stages whereas pattern compensates for solid shrinkage stage.

246. Slush casting process is used to produce
(a) hollow castings (b) intricate castings
(c) large size castings (d) thin walled castings

ISRO Scientist/Engineer 11.10.2015

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Ans. (a) : Slush Casting– (Without using core)

■ To produce thin casting and hollow thin casting without using the core, this technique can be used.

■ This is known as partial solidification, generally used for non-ferrous low melting point materials.

Applications–

Hollow thin castings
Hollow statues
Toys
Decorative items
Lamp shades
Thin ornaments etc.

247. Two castings of the same metal have the same surface area. One casting is in the form of a sphere and the other is a cube. What is the ratio of the solidification time for the sphere to that of the cube?

(a) 3/4 (b) $6/\pi$
(c) $5/4\pi$ (d) $3\pi/8$

ISRO Scientist/Engineer 11.10.2015

Ans. (b) : We know that,

Solidification time

$$t_c = k \times \left(\frac{V}{A}\right)^2$$

k = Solidification characteristic constant

For cube–

Volume = a^3

Surface area = $6a^2$

$$t_c = k \times \left(\frac{a^3}{6a^2}\right)^2 = \frac{k \times a^2}{36}$$

For sphere–

$$\text{Volume} = \frac{4}{3}\pi r^3$$

Surface area = $4\pi r^2$

$$t_s = k \times \left[\frac{\frac{4\pi r^3}{3}}{4\pi r^2}\right]^2$$

$$t_s = k \times \left[\frac{r}{3}\right]^2 = \frac{kr^2}{9}$$

Surface area is same–

$$6a^2 = 4\pi r^2$$

$$\frac{a^2}{r^2} = \frac{4\pi}{6}$$

$$\frac{t_c}{t_s} = \frac{\frac{k \times a^2}{36}}{\frac{k \times r^2}{9}} = \frac{a^2}{r^2} \times \frac{9}{36}$$

$$= \frac{4\pi}{6} \times \frac{9}{36} = \frac{\pi}{6}$$

$$\therefore \quad \frac{t_{sphere}}{t_{cube}} = \frac{6}{\pi}$$

248. Consider the following statements

1. Permanent Moulding produces a sound dense casting with superior mechanical properties.
2. The castings produced are quite uniform in shape have a higher degree of dimensional accuracy than castings produced in sand
3. The permanent mold process is also capable of producing a consistent quality of finish on castings
4. The cost of tooling is usually higher than for sand castings
5. The process is generally limited to the production of small castings of simple exterior design, although complex castings such as aluminium engine blocks and heads are now common place.

Out of the above which are the characteristics of permanent mold casting?

(a) 1, 2, 3 and 4 (b) 3, 4 and 5
(c) 1, 4 and 5 (d) 1, 2, 3, 4 and 5
(e) 2, 3, 4 and 5

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Characteristics of permanent mould casting

(i) Castings produced by this method have very close grain texture and hence porosity is decreased by half to that of sand casting.
(ii) Castings produced by this method are superior in hardness and mechanical properties.
(iii) The castings are free from sand with good surface finish.
(iv) The cost of tooling is usually higher than for sand casting.
(v) The process is capable of producing more complex shapes and design that sand casting.

249. Consider the following statements

1. To minimize turbulence to avoid trapping gasses into the mould
2. To get enough metal into the mould cavity before the metal starts to solidify
3. To avoid shrinkage
4. Establish the best possible temperature gradient in the solidifying casting so that the shrinkage if occurs must be in the gating system not in the required cast part.
5. Does not Incorporate a system for trapping the non-metallic inclusions

From the above options choose the statements that best describe the goals for a gating system

(a) 2, 4, 5 (b) 1, 3, 4
(c) 1, 2, 3, 4 (d) 1, 3, 4, 5
(e) 2, 3, 4

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : The assembly of channels which facilitates the molten metal to enter into the mould cavity is called the gating system.

The goals for the gating are:

(i) To minimise turbulence to avoid trapping gases into the mould.
(ii) To get enough metal into the mould cavity before the metal starts to solidify.
(iii) To avoid shrinkage.
(iv) Establish the best possible temperature gradient in the solidifying casting so that the shrinkage if occurs must be in the gating system not in the required cast part.
(v) Incorporates a system for trapping the non-metallic inclusion.

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250. Gray cast iron blocks 200 × 100 × 10 mm are to be cast in sand moulds. Shrinkage allowance for pattern making is 1%. The ratio of the volume of pattern to that of the casting will be

(a) 0.97 (b) 0.99
(c) 1.01 (d) 1.03
(e) 0.96

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : $V_{casting} = 200 \times 100 \times 10$

$$V_{pattern} = \left(200 + 200 \times \frac{1}{100}\right) \times \left(100 + 100 \times \frac{1}{100}\right) \times \left(10 + 10 \times \frac{1}{100}\right)$$

$$= 202 \times 101 \times 10.1$$

Then, $$\frac{V_{pattern}}{V_{casting}} = \frac{202 \times 101 \times 10.1}{200 \times 100 \times 10} = 1.03$$

251. A model of casting which is constructed in such a way it can be used for forming an impression in damp sand is defined as

(a) molding (b) pattern
(c) damper (d) prototype

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : A model of casting, constructed to use for forming a mould in damp sand is called as pattern. The first step in casting process is to prepare a good model for casting. The model for casting is called as pattern.

252. In sand casting, draft allowance is added to

(a) All linear dimension
(b) Only the interior dimensions
(c) Only the exterior dimensions
(d) Only to the dimensions that are perpendicular to the parting plane

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : A draft allowance is used for those dimension which are perpendicular to the parting line.

253. Hot-chamber die-casting process is used for

(a) Aluminium (b) Zinc
(c) Cast iron (d) Steel

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : Hot chamber die-casting machine are primarily used for zinc alloys, copper, lead and other low melting alloys.

254. Resistance offered by the surface of a green sand mould to penetration by a loaded plunger is?

(a) Compactibility (b) Mouldability index
(c) Mould Hardness (d) Permeability

MPSC Dy. Engineer 2014

Ans. (c) : The resistance offered by the surface of a green sand mould to penetration by a loaded plunger is mould hardness.

255. Which of the following is not a function of flux that is added during casting of cast iron?

(a) absorbs impurities
(b) replenishes material loss
(c) protects casting from oxidation
(d) forms slag

Haryana PSC Civil Services Pre, 2014

Ans. (c)

256. Misrun is the casting defect due to

(a) Moulding and core making materials
(b) Mould and core making technique
(c) Casting process
(d) Impurities in molten metal

MPPSC State Forest Service Exam, 2014

Ans. (b) : Misrun occurs when the molten metal freezes before it reaches all parts of the mould cavity which leaves a completely unfilled part of the mould it occur due to wrong technique of moulds core making.

257. It is the best for shallow flasks and suited for relatively small work.

(a) Hand Ramming (b) Jolt ramming
(c) Squeezing. (d) Sand slinger

MPSC Dy. Engineer 2014

Ans. (c) : Squeezing moulding machines produce greatest sand density at the top of the flask and softest near the parting line of pattern. These machines are suitable for shallow moulds and for relatively small work.

258. Casting defect refers to rough lumps of metal at some point of the casting and a corresponding area of rough holes or depression at other points.

(a) Swelling of the casting on the top side
(b) Variations in wall thickness
(c) Core blow
(d) Sand wash

Mizoram PSC (Insp. of Factories) 2019 Paper-III
MPSC Lect. (Prod.) 2018
TANGEDCO AE 2015
MPSC Dy. Engineer 2014

Ans. (d) : Sand wash is a casting defect which occurs near the ingates or rough lumps on the surface of a casting.

Casting Defects–It is an unwanted irregularities that appear in the casting during metal casting process. Casting defects can be categorised into 5 types:

- Gas porosity – Blowholes, open holes, pin holes.
- Shrinkage defects – Shrinkage cavity
- Mould material defects – Cut and washes, swell, drops, metal penetration, rat tail.
- Pouring metal defects – Cold shut, misrun, slag inclusion.
- Metallurgical defects – Hot tears, hot spots.

259. The longest cylindrical job can be produced by:

(a) sand casting
(b) continuous casting
(c) investment casting
(d) powder metallurgy

HPPSC Asstt. Prof. 2014

Ans. (b) : The longest cylindrical job can be produced by continuous casting.

260. Moulding is carried out in moulding boxes called ___.

(a) Cores (b) Pots
(c) Flasks (d) Cavities

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : Moulding is carried out in moulding boxes called flasks.

261. Which of the following metal expands during solidification of casting?

(a) Aluminium (b) Copper
(c) Gray cast iron (d) Zinc

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

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Ans. (c) : Gray cast iron expands upon solidification as graphite has a relatively high specific volume and its precipitates as graphite flakes during solidification, it cause a net expansion of the metal.

262. Which casting method is limited to non ferrous metals?
(a) Sand casting
(b) Shell mould casting
(c) Plaster mould casting
(d) centrifugal casting

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : Non-ferrous alloy do not contain iron in appreciable amount. Common material are aluminium, brass, copper and titanium.
Non-ferrous casting can be made using sand casting process, investment process or metal mold process including low pressure permanent mold process and high pressure die casting.

263. Slag inclusion in casting is a :
(a) surface defect (b) internal defect
(c) superficial defect (d) none of the above

J&K PSC Civil Services Pre, 2013

Ans. (a) : • Slag inclusion is caused when molten metal containing slag particle is poured into the mold cavity and solidify, remove slag particle from the molten metal before pouring into mould cavity.
• You can remove slag by melting the metal with a flux in a vacuum or in an inert atmosphere.

264. The most suitable material for die casting is :
(a) Steel (b) Cast Iron
(c) Nickel (d) Copper

J&K PSC Civil Services Pre, 2013

Ans. (d) : The most suitable material for die casting is copper.

265. Sprue in casting refers to :
(a) gate (b) runner
(c) riser (d) vertical passage

J&K PSC Civil Services Pre, 2013

Ans. (d) : Sprue in casting refers to vertical passage to flow molten metal.

266. The sand is packed on pit moulds with
(a) Manually (b) Squeezers
(c) Jolt Machine (d) Sand Slingers

SJVN ET 2013

Ans. (d) : The sand is packed on pit moulds with sand slingers.

267. Which is not a casting defect?
(a) Misruns (b) Cope and Drag
(c) Lifts and shifts (d) sponginess

SJVN ET 2013

Ans. (b) : Casting defects-
• Blow holes
• Air inclusions
• Pinhole porosity
• Cuts and washes
• Fusion
• Drop

268. Wood for pattern is considered dry when moisture content is
(a) 0% (b) 5%
(c) Less than 15% (d) Less then 25%

SJVN ET 2013

Ans. (c) : Wood for pattern is considered dry when moisture content is less than 15%

269. Tumbling is a process of :
(a) Improving the fatigue limit
(b) Imparting residual stress to surface
(c) Cleaning the surface of small parts
(d) Improving the creep limit

GPSC AE 26.07.2015
TNPSC AE (Industries) 09.06.2013

Ans. (c) : Tumbling is a process of cleaning the surface of small parts.

270. Match List I (Components) correctly with List II (Processes) and select your answer using the codes given below the lists.

List I (Components)	List II (Processes)
(a) Car body (metal)	1. Machining
(b) Clutch lining	2. Casting
(c) Gears	3. Sheet metal processing
(d) Engine block	4. Powder metallurgy

Codes:

	a	b	c	d
(a)	3	4	2	1
(b)	4	3	1	2
(c)	4	3	2	1
(d)	3	4	1	2

TRB Poly. Lect., 2012

Ans. (d) :

(Component)		(Processes)
Car body	-	Sheet-metal processing
Clutch lining	-	Powder metallurgy
Gear	-	Machining
Engine block	-	Casting

271. Zinc alloys are preferentially cast by
(a) investment casting
(b) die casting
(c) centrifugal casting
(d) shell-mould casting

BPSC AE 2012 Paper - VI

Ans : (b) : Die casting- The die casting (also known as pressure die casting) may be defined as that casting which uses the permanent mould (called die) and the molten metal is introduced into it by means of pressure. Even though lead zinc and its alloys also can be casted but they are not used commonly in engineering hence aluminum and its alloys are the most commonly used materials in the die casting.

272. Match List I correctly with List II and select your using the codes given below.

List I (Moulding tools)	List II (Operations)
(A) Rammers	1. Corner repair
(B) Draw spike	2. Flattening and smoothing sand
(C) Trowel	3. Tamping the sand
(D) Slick	4. Lifting the patterns out of mould

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Codes

	a	b	c	d
(a)	3	2	4	1
(b)	3	4	2	1
(c)	4	3	1	2
(d)	4	3	2	1

TNPSC ACF 2012

Ans. (b) : Rammers – Tamping the sand
Draw spike – Lifting the pattern out of mould
Trowel – flattening and smooth sand
Slicks – Corner repair

273. The methods of casting for producing ornamental pieces are
(a) pressed and slush casting
(b) slush and gravity casting
(c) gravity and semi-permanent mould casting
(d) semi-permanent mould and pressed casting

TNPSC ACF 2012

Ans. (a) : The best process for producing ornamental process are pressed and slush casting.

274. Which of the following materials has more shrinkage allowance?
(a) Cast iron (b) Brass
(c) Lead (d) Aluminium alloy

TNPSC ACF 2012

Ans. (c) : In the given option lead has more shrinkage allowance.
- Total shrinkage is maximum for steel material.
- Solid shrinkage is maximum for brass.
- All is heavy more liquid and solidification shrinkage.

275. Which of the following engineering materials is the most suitable candidate for hot chamber die casting?
(a) low carbon steel (b) titanium
(c) copper (d) tin

VIZAG STEEL MT 10.06.2012, GATE 2007

Ans. (d) : Hot chamber die casting is used for low melting point material like, lead, tin and zinc.

276. Which of the following is not a casting process?
(a) Extrusion
(b) Semi-centrifuge method
(c) Shell moulding
(d) Slush process

APPSC AEE 2012

Ans. (a) : Semi– Centrifuge method, shell moulding and slush process are casting process but extrusion is the forming process.

277. The addition of iron oxide to the foundry sand improves the
(a) Bonding (b) Green strength
(c) Hot strength (d) Permeability

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : The addition of iron oxide to the foundry sand improves the hot strength, because it minimize expansion & gas related defects, soften when heated, to allow for sand grains to expand without cracking the core or mould surface.

278. Which of the following qualifies as precision casting process?
(a) Ingot casting (b) Investment casting
(c) Sand casting (d) Shell molding

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Investment casting- It is an industrial process based on lost-wax casting it have excellent surface finish, high dimensional accuracy. Extremely intricate part are castable. Smaller part can be easily casted with these method.

279. Which of the following are advantages of die casting over sand casting?
(a) Higher melting temperature metals
(b) Higher production rates
(c) Larger parts can be cast
(d) Mold cannot be reused

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Compared to sand casting, die casting produces parts that have thinner walls & smoother surface, have closer dimensional limits, much more economical, production is faster, finishing costs are lower. So, mass production is possible in die casting.

280. Fluidity in casting (CI) operation is greatly influenced by
(a) Melting temperature of molten metal
(b) Pouring temperature of molten metal
(c) Finish of the mould
(d) Carbon content of molten metal

ISRO Scientist/Engineer 2011

Ans. (b) : Fluidity in casting (CI) operation is greatly influenced by pouring temperature of molten metal.
• The ability of a metal to flow and fill a mold is known as fluidity.
• Pouring temperature is the most important controlling factor of fluidity.

281. The property of sand due to which the sand grains stick together is called :
(a) Collapsibility (b) Permeability
(c) Cohesiveness (d) Adhesiveness

OPSC Civil Services Pre. 2011

Ans. (c) : The property of sand due to which the sand grains stick together is called cohesiveness.

282. The purpose of gate is used to :
(a) Deliver molten metal into the mould cavity
(b) Act as a reservoir for the molten metal
(c) Feed the molten metal to the casting in order to compensate for the shrinkage
(d) Deliver molten metal from the pouring basin to gate

OPSC Civil Services Pre. 2011

Ans. (a) :
■ Gate is provided to feed mould cavity to feed casting at desired flow rate.
■ Airvents may be provided for escape of gases.
■ Riser and shrinkage allowance is provided to compensate the shrinkage.

283. Assertion (A) : In die-casting method, small thickness can be filled with liquid metal.
Reason (R) : The air in die cavity trapped inside the casting causes problems.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A

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(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : As we know, in die casting method, small thickness can be filled with liquid metal. But if air trapped can cause problems (defects) so, to avoid defects care should be taken.

284. A sphere, a cube and a disc all of the same material, quality and volume are heated to 900 K and left in air. Which of these will have the lowest rate of cooling
(a) Cube
(b) Disc
(c) Sphere
(d) All will have the same rate of cooling

ISRO Scientist/Engineer 2010

Ans. (c) : Cooling time given by

$$t \propto \left(\frac{V}{SA}\right)^2$$

As volume is same for all

$$t \propto \frac{1}{(SA)^2}$$

It is known that the sphere has the least surface area, hence without doing any calculation we can infar that cooling time will be more in a sphere.
Note:-Disc is basically a cylinder with negligible height.

285. Consider the following materials :
1. Wax 2. Wood
3. Plastic

Which of these materials can be used as pattern in investment casting process ?
(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-50/2010

Ans. (d) : As we know, investment casting used to design very device, casting (job), in which the pattern is made up of wax and plastic, when hot metal poured they melt & metal take that area.

286. The INCORRECT guide line for the good gating system is :
(a) The riser volume should be sufficient for compensating the shrinkage in the casting
(b) less pouring time contributes turbulent flow in the mold
(c) long pouring requires higher pouring temperature
(d) the molten material in the riser should solidify in the beginning

DRDO Scientists 2009

Ans. (d) :

287. The casting defect which is not caused by the high pouring temperature of melt is :
(a) cuts (b) metal penetration
(c) fusion (d) rat tails

DRDO Scientists 2009

Ans. (a) :

288. When the molten metal is poured in a mold, it may pass through the following items in sequence:
(a) sprue, runner, gate, top riser
(b) runner, sprue, gate, top riser
(c) top riser, runner, sprue, gate
(d) gate, sprue, top riser, runner

DRDO Scientists 2008

Ans. (a) :

289. Shrinkage allowance is made by
(a) Adding to external and internal dimensions
(b) Subtracting from external and internal dimensions
(c) Subtracting from external dimension and adding to internal dimensions
(d) Adding to external dimensions and subtracting from internal dimensions

ISRO Scientist/Engineer 2007

Ans. (d) : Shrinkage allowance is made by adding to external dimensions and subtracting from internal dimensions.

290. Large castings are gradually subjected to
(a) tempering (b) annealing
(c) normalizing (d) tumbling

OPSC Civil Services Pre 2006

Ans. (c) : Large castings are gradually subjected to normalizing because in normalizing casting will be cooled in open air whereas in annealing, it will be cooled in furnace which is inconvenient.

291. An instrument to study the behaviour of moulding sand at high temperature :
(a) Dilatometer (b) Pointometer
(c) Cinometer (d) None of the above

HPPSC Asstt. Prof. 20.11.2017

Ans. (a) : Dilatometer is used to study the behaviour of moulding sand at high temperature.

292. The strength of the sand is high if grains are
(a) round and small
(b) sharp and irregular in shape
(c) large and regular in shape
(d) regular in shape but varying in size

TNPSC ACF 2012

Ans. (b) : The strength of the sand is high if grains are sharp and irregular in shape.

293. Plastic bottles are manufactured using the process of
(a) Blow moulding (b) Injection moulding
(c) Atomizing (d) Die casting

GPSC Asstt. Director of Transport 05.03.2017
Haryana PSC Civil Services Pre, 2014
UPSC JWM Advt. No.-52/2010

Ans. (a) : Blow moulding is a manufacturing process for forming and joining together hollow plastic parts. It is also used for forming glass bottles or other hollow shapes.

294. Machining allowance in case of die casting is
(a) Large
(b) Same as in other methods of casting
(c) Small
(d) Generally not provided

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : In die casting, machining allowances is not provided. Die casting is a moulding process in which the molten is injected under high pressure and velocity into a split mould die.

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2. Metal Joining Processes

295. In thermit welding, heat is generated from :
(a) chemical reaction involving finely divided aluminium and iron oxide
(b) combustion of gases
(c) electric arc
(d) none of the above

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Vadodara Muncipal Corp. DEE, 2018
GPSC Asstt. Prof. (Prod.) 21.08.2016
TSGENCO AE 14.11.2015, TSPSC AEE 2015
Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) : In thermit welding heat is generated from chemical reaction involving finely divided aluminium and iron oxide.
$8Al + 3\ Fe_3O_4 \rightarrow 4Al_2O_3 + 9Fe + Heat$ (Exothermic)
Ratio of Al and Fe (Iron oxide Fe_3O_4) according to atomic weight is 1 : 3.

296. The range of voltage (open circuit) in arc welding is generally
(a) 20–40V (b) 40–90V
(c) 100–150V (d) 110–220V

HPCL Engineer 12.08.2021 Shift-I
WBPSC AE, 2017
GWSSB DEE 07.07.2016
Nagaland PSC CTSE 2016 Paper-I
GPSC AE 26.07.2015
GWSSB AAE, 27.12.2015

Ans. (b) : An open circuit voltage in case of D.C. welding is 60 to 80V. While a closed circuit voltage is 15 to 25V.

297. Thermit welding is a form of :
(a) Resistance welding (b) Gas welding
(c) Fusion welding (d) Arc welding

JPSC AE 10.04.2021, Paper-II
GWSSB DEE 07.07.2016
ISRO Scientist/Engineer (RAC) 29.11.2015
GWSSB AAE, 27.12.2015, SJVN ET 2013

Ans. (c) : Thermit welding is a fusion welding process that makes use of the intense heat produced when a mixture containing iron oxide and powdered aluminium is ignited. The reaction is based on the high affinity of aluminium is ignited. It reduces iron oxide to thermit steel and slag, like.
$8\ Al + 3\ Fe_3O_4 \rightarrow 9\ Fe + 4\ Al_2O_3$ (Slag) + Heat (Exothermic)

298. Welding spatter is
(a) Welding defect
(b) Flux
(c) Electrode Coating
(d) Welding technique

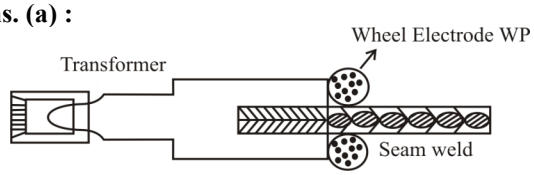
GMB AAE 25.06.2017
HPPSC W.S. Poly. 2016
GPSC AE 26.07.2015
Kerala PSC Poly. Lect. 09.01.2014
ISRO Scientist/Engineer 2012

Ans. (a) : Welding spatter–Welding spatter is a defect in weld.
Spatter refers to particles of metal scattered around the vicinity of the weld along its length. It occurs due to–
(1) Excessive current
(2) Too high arc voltage
(3) Arc blow making arc uncontrollable.

299. A continuous spot welding process is known as
(a) Seam welding (b) Projection welding
(c) Thermit welding (d) Percussion welding

Assam PSC AE (IWT) 14.03.2021
Nagaland PSC (CTSE) 2018 Paper-I
GPSC Asstt. Prof. (Prod.) 21.08.2016
GWSSB DEE 07.07.2016
Kerala PSC IOF 19.04.2016

Ans. (a) :

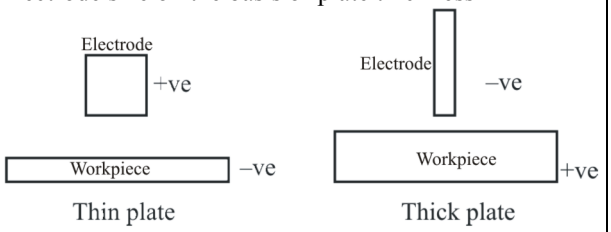


Continuous spot welding process for leak proof joints is seam welding.

300. Two plates of the similar metal having equal thickness are to be butt welded with electric arc. When the plate thickness changes, welding is achieved by
(a) Changing the electrode coating
(b) Adjusting the duration of current and voltage
(c) Adjusting the current and voltage
(d) Changing the electrode size

Rajasthan AE (Nagar Nigam) 2016 Shift-3
GPSC Asstt. Prof. 28.08.2016
BPSC Asstt. Prof. 29.11.2015
GPSC AE 26.07.2015
VIZAG Steel MT 10.06.2012

Ans. (d) : When the plate thickness changes, welding is achieved by changing the electrode size. Electrode size depends on plate thickness in electric arc welding.
Electrode size on the basis of plate thickness –



• The amount of current flow is decided by the electrode size.

301. The commonly used flux in brazing process is
(a) NaCl (b) Lead
(c) Slag (d) Borax

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
TSPSC Manager (Engg.) HMWSSB 12.11.2020
GMB AAE 25.06.2017
CGPSC AE 16.10.2016
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d) : Commonly used fluxes are ammonium chloride or resin acids in soldering or HCL and $ZnCl_2$, due to high temperature in brazing borax is used as flux in Brazing.

302. The most commonly used flame in gas welding is
(a) neutral (b) oxidizing
(c) carbrizing (d) All of the above

Assam PSC AE (IWT) 14.03.2021
JPSC AE 10.04.2021, Paper-II
GPSC Asstt. Prof. 28.08.2016
GPSC Lect. 23.10.2016
GWSSB AAE, 27.12.2015

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Ans. (a) :
(1) Neutral Flame–Neutral flame is most commonly used flame in gas welding.
Neutral flame is obtained by supplying equal volumes of oxygen and acetylene. The most of the oxy-acetylene welding is done with neutral flame.
i.e., mild steel, cast iron, low carbon steel and medium carbon steel can be welded by this flame.

- $\frac{O_2}{C_2H_2} = 1:1$

(2) Oxidising Flame–The oxidising flame is obtained when there is an excess of oxygen. It is used for welding brass and bronze.
i.e., copper, zinc and brass can be welded by this flame.

- $\frac{O_2}{C_2H_2} = 1.15 \text{ to } 1.5$

(3) Reducing or Carburising Flame–The reducing flame is obtained when there is an excess of acetylene.
i.e., nickel alloy, high carbon steel can be welded by this flame.

- $\frac{O_2}{C_2H_2} = 0.85 \text{ to } 0.95$

303. Two 1 mm thick steel sheets are to be spot welded at a current of 5000 A. Assuming effective resistance to be 200 μ ohm and current flow time of 0.2 second, heat generated during the process will be
(a) 0 Joule (b) 500 Joule
(c) 1000 Joule (d) 0.2 Joule

GPSC Executive Engg. 23.12.2018
Rajasthan AE (Nagar Nigam) 2016 Shift-3
VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012

Ans. (c) : Given, I = 5000 A, R = 200 μ ohm
Heat generated (H) = I^2 Rt
$= (5000)^2 \times 200 \times 10^{-6} \times 0.2$
= 1000 Joule

304. In oxyacetylene gas welding process, the temperature of the innermost cone of the flame is around
(a) 2200^0C (b) 3200^0C
(c) 4200^0C (d) 5200^0C

Haryana PSC AE (PHED) 05.09.2020, Paper-II
MECON MT 2019
OPSC AEE 2015 Paper-I
VIZAG Steel MT 10.06.2012

Ans. (b) : Temperature in oxyacetylene gas welding approx 3200^0C.

305. In resistance welding
(a) voltage and current are high
(b) voltage is high and current is low
(c) voltage and current are low
(d) voltage low and current is high

UPRVUNL AE 07.10.2016
APPSC AEE MAINS 2016, PAPER-III
TSGENCO AE 14.11.2015
Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : Resistance welding has low voltage and high current.

306. Principal materials used in soldering are
(a) Tin and lead (b) Copper and tin
(c) Zinc and copper (d) Copper and lead

Gujarat PSC AE 2019
Nagaland CTSE 2016, 2017 Ist Paper
APGENCO AE, 2017
APPSC Poly. Lect. 13.03.2017
WBPSC AE 2017, RPSC AE 2016
GPSC Asstt. Prof. (Prod.) 21.08.2016
RPSC ACF-2011

Ans : (a) : Soldering is a method of joining similar or dissimilar metals by means of a filler metal whose liquids temperature is below 450^oC. Solder is an alloy of tin and lead used to create electrical joints.

307. Following gases are used in tungsten inert gas welding.
(a) CO_2 and H_2 (b) Argon and neon
(c) Argon and Helium (d) Helium and neon

PPSC Asstt. Municipal Engineer 15.06.2021
TNPSC AE 2019
Gujarat PSC AE 2019
Vadodara Muncipal Corp. DEE, 2018
ISRO Scientist/Engineer (RAC) 29.11.2015
Kerala PSC Poly. Lect. 09.01.2014
UKPSC AE 2012 Paper-I
ISRO Scientist/Engineer 2009

Ans. (c) : Helium and Argon gases used in tungsten inert gas welding.

308. Which of the following is a solid state joining process?
(a) gas tungsten arc welding
(b) resistance spot welding
(c) friction welding
(d) submerged are welding

RPSC 2016, HPPSC LECT. 2016
ISRO Scientist/Engg. 11.10.2015
VIZAG Steel MT 10.06.2012

Ans : (c) Cold welding, friction welding, diffusion welding, forge welding are examples of solid state welding. Cold welding includes pressure welding, ultra sonic welding and explosive welding.

309. Pre-heating before welding is done to
(a) make the steel softer
(b) burn away oil, grease, etc. from the plate surface
(c) prevent cold cracks
(d) prevent plate distortion

GPSC Executive Engineer 23.12.2018
ISRO Scientist/Engineer 11.10.2015
GPSC AE 26.07.2015, GATE 1996

Ans. (c) : Assumptions of Welding–
- By virtue of metal being melted at the interface in a welded joint it's necessary that the interface are very clean. If the interface are with any oil, dirt or grease left, then these would interface with proper using of the metal and thus weaken the joint.
- Heavier oxides films may be removed by acid picking wire brushing or emery.
- Pre-heating material before welding to prevent cracks, scratches by cold i.e. cold cracks, cold scratches etc.

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310. The ratio of aluminium and iron oxide used in thermit welding is

(a) 1 : 3 (b) 3 : 1
(c) 1 : 2 (d) 2 : 1

GPSC Engg. Class-II Pre 19.01.2020
TSPSC AEE 2015, HPPSC AE 2018, 2014

Ans. (a) : The ratio of aluminium and iron oxide used in thermit welding is 1 : 3.

Thermit welding–Thermit welding is welding process that employs molten metal to permanently join the conductors.

The process employs an exothermic reaction of a thermit composition to heat the metal, and require no external source of heat or current. The chemical reaction that produce the heat is an aluminothermic reaction between aluminium powder and metal oxide [1:3]

$$Fe_2O_3 + 2Al \longrightarrow 2Fe + Al_2O_3$$

Application–Cast iron] stainless steel, brass, bronze. It is especially useful for joining dissimilar metals.

311. In which type of welding a pool of molten metal is used

(a) Electro slag (b) Submerged arc
(c) MIG (d) TIG

Vizag Steel (MT) 2017
Kerala PSC Poly. Lect. 09.01.2014
ISRO Scientist/Engg. 2012

Ans. (a) : In Electro slag welding a pool of molten metal is used.

312. Grey cast iron is usually welded by

(a) Gas welding (b) Resistance welding
(c) Arc welding (d) TIG welding

Vizag Steel (MT) 2017
Kerala PSC AE (Irrigation) 05.04.2017
Nagaland PSC CTSE 2016 Paper-I
Kerala PSC Poly. Lect. 09.01.2014
ESE 2009, J & K PSC Screening, 2006

Ans. (a) : Grey cast iron is usually welded by gas welding or oxy-acetylene welding having neutral flame.

313. In D.C. reverse polarity are welding:

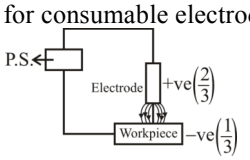
(a) The W.P. is made positive and the electrode is made negative
(b) The W.P. is made negative and the electrode is made positive
(c) Both the W.P. and the electrode are made positive
(d) Both the W.P. and the electrode are made negative

JPSC AE 10.04.2021, Paper-II
Karnataka PSC Lect., 27.05.2017
RPSC VPITI 14.02.2016

Ans. (b) : For thinner materials, where less heat is required in the weld zone the polarity could be reversed by making the work piece as negative. This is termed reversed polarity or (DCRP) direct current reverse polarity.

In this the penetration is small

It is used mainly for consumable electrode



DC-reverse polarity

Also direct current straight polarity can be used for thicker material because penetration in this case is high and is mainly used in case of non consumable electrode.

314. In arc welding process the current value is decided by:

(a) Thickness of plate
(b) Length at weld portion
(c) Voltage across arc
(d) Speed at travel

VIZAG Steel MT 2017
TSPSC Managers, 2015
Kerala PSC Poly. Lect. 09.01.2014

Ans. (a) : Welding current depends on the size of the electrode i.e. core diameter. Electrode size is governed by the thickness of the metal and joint preparation and the electrode type by welding position.

• If the current supplied is in excess then it will cause spatter. If current supplied is less than the required then it will cause less penetration.

The approximate average welding current for structural steel electrodes is 35d (where d is electrode diameter in mm).

315. If the proportional of oxygen is less than the proportion of acetylene in oxy acetylene gas welding, the flame produced is _____ .

(a) Plasma arc (b) Carburizing flame
(c) Oxidizing flame (d) Neutral flame

APPSC Poly Lect. 13.03.2020
Mizoram PSC (Insp. of Factories) 2019 Paper-III
CIL MT 26.03.2017
RPSC LECTURER 16.01.2016

Ans. (b) : • A carburizing flame (reducing flame) is one in which the supply of acetylene is in excess as compared to the oxygen $\left(\text{i.e.}\frac{O_2}{C_2H_2} = \frac{0.95}{1}\right)$.

• If the supply of acetylene is less than the supply of oxygen than the flame is known as oxidizing flame.

• If both acetylene and oxygen are in the ratio of 1 : 1 then flame is called neutral flame.

316. Which of the following welding processes uses consumable electrodes?

(a) TIG welding (b) MIG welding
(c) Thermit welding (d) Gas welding

Assam PSC AE (IWT) 14.03.2021
MECON MT 2019
HPPSC AE 2018
Kerala PSC IOF 19.04.2016
RPSC LECTURERE 16.01.2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
Mizoram PSC AE/SDO 2014 Paper-III
JPSC AE 2013

Ans. (b) : The Metal Inert Gas (MIG) or Gas Metal Arc Welding (GMAW) process uses a consumable electrodes which is usually in the form of copper-coated coiled wire. Argon is used to shield the weld and direct current with electrode being positive to generate more heat for melting.

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317. Resistance spot welding is performed on two plates of 1.5 mm thickness with 6mm diameter electrode, using 15000 A current for time duration of 0.25 second. Assuming the interface resistance to be 0.0001 Ω, the heat generated to form the weld is :

(a) 5625 W-sec (b) 8437 W-sec
(c) 22500 W-sec (d) 33750 W-sec

Vadodara Muncipal Corp. DEE, 2018
ISRO Scientist/Engineer 11.10.2015
MPSC Poly. Lect. 2014, GATE 2001

Ans. (a) Given, Thickness of plate (t) = 1.5mm
diameter of electrode (d) = 6mm
$I = 15000A$ $t = 0.25$ sec. $R = 0.0001\Omega$
Heat is generated to form weld = I^2Rt
$= (15000)^2 \times 0.0001 \times 0.25 = 5625$ W-sec

318. The electrodes used in spot welding have a tip of

(a) stainless steel (b) aluminium
(c) copper (d) brass

Mizoram PSC (Insp. of Factories) 2019 Paper-III
Sikkim PSC (Under Secretary), 2017
HPPSC AE (PWD) 10.09.2016

Ans. (c) : The electrodes are one of the most important factors in the resistance welding process but often the most abused. It is important to consider the electrode material, shape, size, tip profile and cooling.
The electrodes used in spot welding have a tip of copper.

319. TIG welding is preferred for

(a) Mild steel (b) Aluminium
(c) Silver (d) All options are correct

PPSC Asstt. Municipal Engineer 15.06.2021
MPPSC State Forest Service Exam, 2014
GPSC Poly. Lect. 07.12.2014

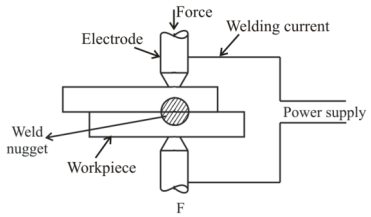
Ans. (b) : TIG (GTAW) welding is a stronger weld, when you use TIG welding for thinner metals, it makes for a much stronger weld. TIG welding creates a stronger bond another reason why it is preferred for aluminium & strainless steel.
Welding of Al, Mg and its alloys is automatic aerospace and chemical industry.

320. For resistance spot welding of 1.5 mm thick steel sheets, the current required is of the order :

(a) 10 Amp (b) 100 Amp
(c) 1000 Amp (d) 10000 Amp

TNPSC AE 2018 , TSPSC AEE 2017
HPPSC Asstt. Prof. 18.11.2016
GPSC AE 26.07.2015, GATE 1990

Ans. (d) : Resistance Spot Welding–In spot welding a pair of sheet is held between two copper electrode. Indentation is produced over the sheet by applying the pressure. When high current nearly 110,000 Amp passed through the conductor maximum resistance will be there at interface between the two sheets due to presence of air gap. As a result of that material in both the sheets liquefied and a spot (nugget) will form.



321. In oxy-acetylene gas welding the volume of oxygen required per unit volume of acetylene :

(a) 1 (b) 1.5
(c) 2 (d) 2.5

TNPSC AE 2017, Kerala PSC AE 06.08.2015
J&K PSC Civil Services Pre, 2010

Ans. (d) : • $C_2H_2 + \frac{5}{2}O_2 \rightarrow 2CO_2 + H_2O$

Here, 1 mole C_2H_2 requires 2.5 moles of O_2
1 mole O_2 from cylinder
1.5 mole O_2 from atm
} Total 2.5 mole O_2

Neutral flames –
Outer flame (Blue colour) – 1275°C
$O_2 : C_2H_2 = 1$
Inner flame (inner cone) (white bright) – 3200°C
Oxidizing flames (reducing flames)
– 3300°C (inner cone)
$O_2 : C_2H_2 = 1.15 – 1.5$
Carburizing flames – 2900°C (inner core)
$O_2 : C_2H_2 = 0.85 – 0.95$
Application-
Neutral flame – Mild steel, low carbon steel, Al alloys, cr etc welded.
Oxydizing flame – Bronze or fusing brases.
Carburising flame- High carbon steel.

322. The welding set up is said to have straight polarity, when

(a) work is connected to the positive terminal and the electrode holder to the negative terminal
(b) work is connected to the negative terminal and the electrode holder to the positive terminal
(c) work is connected to the positive terminal and the electrode holder earthed
(d) work is connected to the negative terminal and the electrode holder earthed

TNPSC AE 2018
GPSC Lect. 23.10.2016, CGPSC AE 16.10.2016

Ans. (a) : Straight polarity- This occurs when the electrode is made negative and the plates are positive. As a result the electron flows from the electrode tip to base plate.
1. Used for high thick and high melting point material.
2. Depth of penetration is more.
3. Weld deposition rate is low.
Reverse polarity- This occurs when the electrode is made positive and the plates are negative. The electrons flows in the reverse direction from the base plates to the electrode tip.
1. Used for high thin and low melting point material.
2. Depth of penetration is less.
3. Weld deposition rate is maximum.

323. In which of the following welding process flux in the form of granules are used?

(a) Spot welding (b) Submerged arc welding
(c) TIG welding (d) MIG welding

PPSC Asstt. Municipal Engg. 15.06.2021
Haryana PSC AE (PHED) 05.09.2020, Paper-II
Kerala PSC AE (Irrigation) 05.04.2017
APPSC AEE 2012, ESE 2006

Ans. (b) : Submerged arc welding

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324. Consider the following processes
1. Arc welding
2. Resistance welding
3. Thermit welding
4. Gas welding
The correct sequence of these processes in increasing order of their welding temperatures is
(a) 1, 2, 3, 4 (b) 1, 3, 4, 2
(c) 4, 1, 3, 2 (d) 4, 3, 1, 2

APGENCO AE, 2017
UPRVUNL AE 21.08.2016
ESE 2000

Ans. : (*)

Types of welding	Temperature
Arc welding	$3600\text{-}15000^0C$
Resistance welding	$750\text{-}1850^0C$
Thermit welding	$1200\text{-}2400^0C$
Gas welding	$2300\text{-}3300^0C$

The correct sequence is 2 < 3 < 4 < 1.

325. The welding process used extensively in joining mild steel shanks to high speed drills is
(a) Upset butt welding (b) Projection welding
(c) Flash butt welding (d) None of these

APPSC Poly Lect. 13.03.2020
MPSC Lect. (Prod.) 2018

Ans. (c) : The welding process used extensively in joining mild steel shanks to high speed drills is flash butt welding.

326. Electron beam welding can be carried out in :
(a) A pressurized inert gas chamber
(b) A shielding gas environment
(c) Open air
(d) Vaccum

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Oil India Ltd. Sr. Engg. (Production) 30.11.2019
ISRO Scientist/Engineer (RAC) 07.05.2017
TANGEDCO AE 2015, GPSC AE 26.07.2015
Kerala PSC Poly. Lect. 09.01.2014
UPSC JWM Advt. No.-52/2010

Ans. (d) : Electron beam welding carried out in Vacuum to avoid Scattering of electron beam.

327. Defect in welding where a long and continuous visual separation line between the base of metal and HAZ is:
(a) Lamellar tearing (b) Hot cracking
(c) Undercut (d) Porosity

HPPSC AE (PWD) 10.09.2016
HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Lamellar tearing- Lamellar tearing is a defect in welding where a long and continuous visual separation line between the base of metal and HAZ is there.

328. In solid-state welding, the contamination layers between the surfaces to be welded are removed by:
(a) Alcohol (b) Plastic deformation
(c) Water jet (d) Sand blasting
(e) Mechanical press

Assam Engg. College AP/Lect. 18.01.2021
CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : As we know, in solid state welding, the contamination layers between the surfaces to be welded are removed by plastic deformation or by used of chipping hammer.

329. In electric resistance welding, voltage (V) required for heating is in the range
(a) 1 to 5 V (b) 6 to 10 V
(c) 11 to 15 V (d) 16 to 20 V

MPPSC State Forest Service Exam, 2014
OPSC Civil Services Pre. 2011

Ans. (b) : In electric resistance welding, voltage (v) required for heating is in the range of 6 to 10 V.

330. Which two of the following joining processes are autogenous?
(i) Diffusion welding
(ii) Electroslag welding
(iii) Tungsten inert gas welding
(iv) Friction welding
(a) (ii) and (iii) (b) (i) and (iv)
(c) (ii) and (iv) (d) (i) and (iii)

Assam Engg. College AP/Lect. 18.01.2021
CGPSC Poly. Lect. 22.05.2016

Ans. (b) : Autogenous weldings are those welding in which no filler material is used.
Diffusion welding and friction welding are autogenous weldings.

331. The major difficulty during welding of aluminium is due to its_____
(a) high tendency of oxidation
(b) high thermal conductivity
(c) low melting point
(d) low density

Assam Engg. College AP/Lect. 18.01.2021
CGPSC Poly. Lect. 22.05.2016, TSPSC AEE 2015

Ans. (a) : Aluminium is very difficult to weld because it forms a layer of Al_2O_3 on its surface.
Major difficulty during welding of aluminium is due to its high tendency of oxidation to avoid this TIG welding is used for aluminium.

332. Arc blow is encountered in
(a) AC welding machines
(b) DC welding machines
(c) Thermit welding
(d) None of the above

HPPSC AE 2018
GPSC Asstt. Prof. (Prod.) 21.08.2016
Haryana PSC Civil Services Pre, 2014

Ans. (b) : Arc blow is more common in DC welding.
While welding near the edges, since most of the magnetic lines will be concentrated in the material, arc will be defected towards the workpiece. Arc blow result in spatter.

333. The Brazing process is carried out in the temperature range
(a) 60°C – 150°C (b) 180°C – 250°C
(c) 350°C – 500°C (d) 600°C – 750°C

GMB AAE 25.06.2017, KPSC AE 2015

Ans. (c) : Brazing is a joining process when metal are bounded together using a filter metal with a melting temperature in the range of 350°C – 500°C but actual range is 700°C – 900°C in soldering process the temperature range 180°C – 250°C.

334. Pinch effect in welding is result of :
(a) Expansion at gases in arc
(b) Electromagnetic forces
(c) Electric forces

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(d) Surface tension molten metal

TSPSC Managers, 2015
NPCIL ET, 2015, ESE 2001

Ans. (b) : In a metal inert gas welding-
• droplet transfer – at low current – only gravity force.
• spray transfer → at high current → gravity + electromagnetic force (pinch effect).

335. In resistance seam welding, the electrode is in the form of a

(a) cylinder (b) flat plate
(c) coil of wire (d) circular disc

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
GATE 2011

Ans. (d) : In resistance seam welding, to workpieces or provided between to copper electrode which or in the form of rollers or wheels. Welding speed is very high. It can be used mass production. There is a possibility of indentation between electrode and workpiece.

336. What is the type of shielding gas for GTAW (Gas Tungsten Arc Welding) used for carbon steels only called?

(a) Oxygen (b) Argon
(c) Nitrogen (d) Hydrogen

Kerala PSC AE 05.04.2017
Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Tungsten inert gas welding also known as gas tungsten Arc welding (GTAW), is an arc welding process that uses a non-consumable tungsten electrode to produce the weld.
TIG welding utilizing power supply to generate an electric arc between the tungsten and the workpiece, using the resultant heat to create weld.
The weld area is generally protected from atmospheric contamination by use of an inert shielding or cover gas (argon or helium)

337. Submerged arc welding is

(a) a process which uses a mixture of iron oxide and granular aluminium
(b) accomplished by maintaining a hot molten metal pool between plates
(c) a process in which arc is maintained under a blanket of flux
(d) All of the above

Assam PSC AE (IWT) 14.03.2021
ISRO Scientist/Engineer 22.04.2018

Ans. (c) : Submerged arc welding is a process in which arc is maintained under a blanket of flux. Arc is produced between a bare metal electrode and the workpiece. The submerged arc welding is mostly done on low carbon and alloy steels but it may be used on many of the non-ferrous metals.

338. The number of zones of heat generation in spot welding is

(a) two (b) three
(c) five (d) seven

Assam PSC CCE Pre 2015
Kerala PSC Poly. Lect. 09.01.2014

Ans. (c) : Spot Welding– It is used for welding lap joints, joining components made from plate material having 0.025 mm to 1.25 mm in thickness. The plates to be joined together are placed between the two electrode tips of copper or copper alloy. It may be noted that
(i) The electrode tip diameter (d) should be equal to $\sqrt{\tau}$, where t is the thickness of plate to be welded.
(ii) The distance between the nearest edge of plate and centre of weld should be at least 1.5d.
(iii) The spacing between two spot welds should not be less than 3d.

339. A fusion welding process in which the weld is effected by pouring superheated liquid around the parts to be united, is known as :

(a) Diffusion welding (b) percussion welding
(c) slag welding (d) thermit welding

NLC GET 17.11.2020, Shift-II
WBPSC AE, 2017

Ans. (d) : Thermit welding is the only welding process employing an exothermal chemical reaction for the purpose of developing a high temperature. It is a Fusion process in which weld is effected by pouring superheated liquids thermit steel around the parts to be united.
Diffusion welding- It is a solid state welding technique used in metal working capable of joining similar and dissimilar metals. It operates on the principal of solid state diffusion where in the atoms of two solid metallic surface intersperse themselves over time.
Percussion welding- Percussion welding is similar to Flash welding except are power by a rapid discharge of stored electrical energy. The duration is only 1 to 10 millisecond, heat is intense and highly concentrated. Small weld metal is produced little or no upsetting and low HAZ.
Slag welding- Welding slag is a form of clay or vitreous material produced as a byproduct of some arc welding process most specially shielded metal arc welding process, submerged arc welding and flux-cored arc welding.

340. Two pipes of inner diameter 100 mm and outer diameter 110 mm each joined by flash butt welding using 30 V power supply. At the interface, 1 mm of material melts from each pipe which has a resistance of 42.4Ω. If the unit melt energy is 64.4 MJm^{-3}, then time required for welding in seconds is

(a) 1 (b) 5
(c) 10 (d) 20

VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012

Ans. (c) : Energy required for melting/volume
$E = 64.4\ MJ/m^3$
$V = 30$ volts
$R = 42.4\Omega$
$d_i = 100$ m
$d_o = 110$ mm

$$\text{Melting energy} = \frac{\pi}{4}\left(d_o^2 - d_i^2\right)\times 64.4\times 10^6$$
$$= \frac{\pi}{4}\times\left(110^2 - 100^2\right)\times 64.4\times 10^6$$
$$= 212.32 \text{ Joules}$$

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$$H = I^2RT = \frac{R^2T}{R} \qquad (\because V = IR)$$

$$T = \frac{H}{V^2} \times R = \frac{212.32 \times 42.4}{30^2}$$

$$\boxed{T = 10 \text{ Second}}$$

341. In resistance welding, heat is generated due to the resistance between-

(a) Electrode and work piece
(b) Asperities between touching plates
(c) Two dissimilar metals being in contact
(d) Inter-atomic forces

NPCIL ET, 2015, ESE 2003, GATE 2003

Ans. (b) : In resistance welding, heat is generated due to the resistance between asperities between touching plates because of air resistance is very high.

342. Brasses and bronze are welded using

(a) Neutral Flame
(b) Carburizing Flame
(c) Reducing Flame
(d) Oxidizing Flame

GPSC Asstt. Director of Transport 05.03.2017
Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : Brasses and bronze are welded using oxidizing flame.

343. The welding process used to join ends of two pipes of uniform cross section is

(a) spot welding (b) flash butt welding
(c) upset butt welding (d) seam welding

Mizoram PSC (Insp. of Factories) 2019 Paper-III
ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (c) : Here option (b) and (c) are quite close, but option (c) will be more appropriate as the actual welding/joining happens during upset. In flash state arc is generated while the actual joining happens during upset.

344. Projection Welding is a

(a) Continuous Spot Welding Process
(b) Multi-spot welding process
(c) Are Welding Process
(d) Process used for joining round bars

Kerala PSC Asstt. Prof. 27.09.2016
ISRO Scientist/Engineer 12.05.2013

Ans. (b) : Projection Welding is a multi-spot welding process. Projection electrodes are capable of carrying more current than spot welding electrodes and can, therefore weld much thicker materials. During the projection welding process two flat electrodes cover the face of a fastener.

345. A flux is used in brazing to :

(a) Improve appearance of finished bond
(b) Provide filler material for brazing
(c) Prevent oxidation and remove rust
(d) Help heat flow for stronger bond.

UPRVUNL AE 05.07.2021
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : The flux commonly used in Brazing is Borax. A flux is required in all brazing operations. The purpose of the flux is to remove oxides from the base material and to prevent oxidation during the heating process, thus promoting the free flow of the brazing filler metal.

346. Fabrication weld ability test is used to determine:

(a) mechanical properties required for satisfactory performance of welded joint
(b) susceptibility of welded joint for cracking
(c) suitability for joint design
(d) appropriate machining process

RPSC ACF & FRO, 26.02.2021
Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : Fabrication weld ability test is used to determine susceptibility of welded joint for cracking.

Fabrication weldability test is applied by steel fabricators to access the cold cracking susceptibility of their steels.

Objective of the test is cold cracking susceptibility of butt welds (HAZ–WM Combination)

347. The material used for coating the electrode is called :

(a) protective layer (b) slag
(c) deoxidizer (d) flux

(HPPSC AE 2014)
Mizoram PSC AE/SDO 2014 Paper-III

Ans : (d) The material used for coating the electrode is called flux.

Fluxes:- In welding of certain metals, the fusion of the weld does not take place very readily. Particularly when the oxides of the base metal have a higher melting point than the metal it self. Thus these oxides remain on the surface and become entrapped in the solidifying metal instead of flowing from the welding zone. This oxide can be removed from the weld location by use of certain fluxes which react chemically with the oxides of most metals and form fusible slag at welding temperature.

348. In submerged arc welding, an arc is produced between:

(a) Carbon electrode and the work-piece
(b) Metal electrode and the work-piece
(c) Bare metal electrode and the work-piece
(d) Two tungsten electrodes and the work

GPSC Engg. Class-II Pre 19.01.2020
OPSC AEE 2015 Paper-I

Ans : (c) Submerged arc welding, is a common arc welding process that involves the formation of an arc between a continuously fed electrode and the workpiece gas shield and a slag which protects the weld zone.

349. Which of the following is the fusion welding process?

(a) Forge welding
(b) Gas welding
(c) Resistance welding
(d) Thermit welding with pressure

SJVN ET 2019
Kerala PSC Poly. Lect. 09.01.2014

Ans. (b) : Gas welding the fusion welding process.

350. In an arc welding process, the voltage and current are 25 Volts and 300 amperes respectively. The arc heat transfer efficiency is 0.85 and welding speed is 8 mm/s the net heat input (in J/mm) will be:

(a) 796.8 (b) 684.2
(c) 620.7 (d) 721.6

AAI Jr. Executive 26.03.2021, SJVN ET 2019
HPPSC ADF 05.03.2019, GATE 2006

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Ans. (a) : Given, Voltage (V) = 25 volt
current (I) = 300 amperes
Heat transfer efficiency (η_h) = 0.85
Welding speed (v) = 8 mm/s
Heat supplied,
$$H = \frac{VI}{v} = \frac{25 \times 300}{8}$$
= 937.5 J/mm
Actual heat input,
$H_a = \eta_h \times H_s$
$H_a = 0.85 \times 937.5$
= 796.875 J/mm

351. The colour of oxygen cylinder in oxyacetylene welding is
(a) black (b) maroon
(c) blue (d) green

Nagaland PSC (CTSE) 2018, 2016 Paper-I

Ans. (a) : Oxygen cylinder is black and acetylene is maroon, color in oxyacetylene welding.

352. In brazing the usual melting temperature of the filler rod is :
(a) Above 1500 ºC
(b) Above 450 ºC
(c) Below 450 ºC
(d) Above melting temp. of parent metal

OPSC AEE 2019 Paper-I
APPSC AEE MAINS 2016, PAPER-III

Ans : (b) : Brazing- Brazing is the coalescence of a joint with the help of a filler metal whose liquids temperature is above 427ºC and below the solids temperature of the base metal. Filler metal in drawn into the joint by capillary action. Flux usually used is borax. Due to the many advantages over welding, brazing is very widely used in industries. Brazing can joint almost all metals. Dissimilar metals can be brazed. For example cast iron to stainless steel.

353. Stud and projection welding belong to the following category of welding
(a) Gas welding (b) Arc welding
(c) Resistance welding (d) Pressure welding

Vizag Steel (MT) 2017
UPRVNL AE 07.10.2016

Ans. (c) : Type of Resistance welding
i. Seam welding
ii. Spot welding
iii. Projection welding
iv. Upset welding
v. Flash welding.

354. The welding of stainless is generally difficult because of the following reason
(a) Rust formation takes place
(b) High melting temperature of stainless steel
(c) Formation of oxide film
(d) Fear of cracking

Vizag Steel (MT) 2017
HPPSC W.S. Poly. 2016

Ans. (c) : The welding of stainless steel is generally difficult because formation of oxide film.

355. Which of the following resistance weldings is used for making leakproof joint?
(a) Spot welding
(b) Projection welding
(c) Seam welding
(d) Percussion welding

Haryana PSC AE (PHED) 05.09.2020, Paper-II
BPSC AE 2012 Paper - VI

Ans : (c) : Seam welding- Seam welding is a fusion welding. When the spot welds on two over lapping piece of metal are spaced the process is known as roll spot welding. It the spot welds are sufficiently made close, then the process is called seam welding.
Seam welding is used to make leak proof joint. Example container, drum, leak proof fuel tanks in automobile etc. This process is best adopted for metal thickness ranging from 0.025 mm to 3 mm

356. Ratio of oxygen to acetylene in neutral flame is:
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 3 : 4

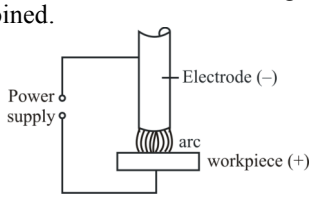
AAI Jr. Executive 26.03.2021, RPSC AE 2018
Kerala PSC Poly. Lect. 09.01.2014, HPPSC AE 2014

Ans. (a) : The ratio of oxygen to acetylene in neutral flame is 1 : 1.

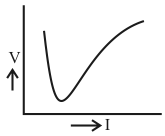
357. In arc welding, the electric arc is produced between the work and the electrode by
(a) voltage (b) flow of current
(c) contact resistance (d) all of these

GPSC Engg. Class-II Pre 19.01.2020
Sikkim PSC (Under Secretary), 2017

Ans. (c) : In arc welding, the electric arc is produced between the work and the electrode by contact resistance. Principle of arc welding is the production of heat to on struck between the welding electrode and metal to be joined.



An arc is generated between two conductors of electricity cathode and anode, when they are touched to establish the flow of current and then separated by small distance.



$V = A + Bl$
V = Voltage l = Arc length A, B = Constant

358. An example of welding in plastic state is
(a) Arc welding (b) Gas welding
(c) Forge welding (d) All of the above

GPSC Engg. Class-II Pre 19.01.2020
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : Forge welding- Joining two pieces of metal by heating to high temperature then hammering together, creating enough pressure to cause plastic deformation.

359. Residual stress in welding causes
(a) enhanced dimensional stability.
(b) welding cracking.
(c) reduction in creep strength.
(d) both (b) and (c)

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) :

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360. The acetylene cylinder is filled with a material saturated with

(a) Calcium carbide (b) Acetylene
(c) Calcium oxide (d) Acetone

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) : The acetylene cylinder is filled with a material saturated with acetone.

361. Which of the following welding methods provides the highest heat flux (W/mm^2)?

(a) Oxy-acetylene gas welding
(b) Tungsten inert gas welding
(c) Plasma arc welding
(d) Laser beam welding

PPSC Asstt. Municipal Engineer 15.06.2021
GATE 2019

Ans. (d) : In laser beam welding heat flux is highest upto 1 MW/mm^2. Decreasing order of heat flux –
Laser beam welding > Plasma arc welding > Tungsten inert gas welding > Oxy-acetylene gas welding.

362. Brazing is a metal joining process whereby

(a) a filler metal is heated above melting point and distributed between two or more close fitting parts by capillary action.
(b) a filler metal is heated below melting point and distributed between two or more close fitting parts by capillary action.
(c) a filler metal is heated at melting point and distributed between two or more close fitting parts by capillary action.
(d) None of the above

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : A filler metal is heated above melting point and distributed between two or more close fitting parts by capillary action

363. Which one of the following is a joining process that may employ acetylene, natural gas, butane in combination with oxygen to supply the heat required to melt the filler rod and diffuse it into the surface of the base metal?

(a) Furnace brazing (b) Torch brazing
(c) Induction brazing (d) Dip brazing

ESE 18.07.2021

Ans. (b) : Torch Brazing is a joining process that may employ acetylene, natural gas, butane or propane in combination with air or oxygen to supply the heat required to melt the filler rod and diffuse it into the surface of the base metal. This technique is not extensively used for continuous mass production. Flux used is in the form of paste or powder.

364. Volt ampere characteristics is given by $I^2 = 600(V-60)$. Arc characteristic is given as $I = 20(V-60)$. Here I is in Amperes, V is in volts. The power of stable arc is _____ kW.

(a) 3.463 (b) 2.55
(c) 6.75 (d) 5.46

AAI Jr. Executive 26.03.2021

Ans. (*) : Given, $I_t^2 = 600(V-60)$ [Parabolic relation]

$I_a = 20(V-60)$

Stable arc generation condition,

$I_a = I_t$

$I_t^2 = I_a^2$

$\Rightarrow \quad 600(V-60) = [20(V-60)]^2$

$\Rightarrow \quad 600(V-60) = 400(V-60)^2$

$\Rightarrow \quad V = 61.5$ Volts

Now, $I = 20(V-60)$

$= 20(61.5-60) = 30$ A

Power of stable arc,

$P = VI = 61.5 \times 30 = 1.845$ kW

365. A joint between two components lying approximately in the same plane is called ____

(a) corner joint (b) T-joint
(c) lap joint (d) butt joint

UPRVUNL AE 05.07.2021

Ans. (d) : A joint between two components lying approximately in the same plane is called butt joint.

366. _____ defect can never occur in a gas tungsten arc welding process

(a) crack (b) porosity
(c) tungsten inclusion (d) slag inclusion

UPRVUNL AE 05.07.2021

Ans. (c) : Tungsten inclusion defect can never occur in a gas tungsten arc welding process.

367. Which of the following is NOT the purpose of flux in brazing ?

(a) Promoting wetting properties
(b) Chemically cleaning the metal
(c) Increasing the weight of brazed joint
(d) Shielding from the atmosphere

UPRVUNL AE 05.07.2021

Ans. (c) : The purpose of flux in brazing–
- Promoting wetting properties
- Chemically cleaning the metal
- Shielding from the atmosphere

368. In manual metal arc welding, flux is _____

(a) Applied as a wire roll
(b) Not applied
(c) Bonded to electrode
(d) Applied as a loose power

UPRVUNL AE 05.07.2021

Ans. (c) : In manual metal arc welding, flux is bonded to electrode.

369. Which of the following is NOT a phenomenon that occurs in welding arc?

(a) Thermionic emission of electrons
(b) Emission of UV rays
(c) Avalanche motion of electrons
(d) Formation of the inner cone of gas flame

UPRVUNL AE 05.07.2021

Ans. (d) : Formation of the inner cone of gas flame

370. _____ in a weld is provided to compensate for the flaws in the weld

(a) Backing strip (b) reinforcement
(c) j-groove (d) V-groove

UPRVUNL AE 05.07.2021

Ans. (b) : Reinforcement in a weld is provided to compensate for the flaws in the weld.

371. Magnetic arc blow is also known as _____

(a) arc seamer (b) arc stopper
(c) arc wander (d) arc striker

HPPSC Workshop Suptd. 08.07.02021

Ans.(c) : Magnetic arc blow is also known as arc wander.

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372. "Spelter" is _____.
(a) combination of 62% lead and 38% tin
(b) combination of 50% lead and 50% tin
(c) non-ferrous filler material
(d) combination of 62% lead and 38% zinc

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Spelter is a zinc – lead alloy that ages to resemble bronze.
$\Rightarrow$ 62% lead, 38% zinc.

373. _____type of oxy-acetylene flame tend to reduce its strength and ductility of weld metal and also make it more brittle.
(a) Neutral (b) Carburising
(c) Oxydising (d) All of these

RPSC ACF & FRO, 26.02.2021

Ans. (b) : Carburizing flame tends to reduce its strength and ductility of weld metal and also make it more brittle.

374. During a TIG welding process, the arc current and arc voltage were 50 A and 60V, respectively, when in the welding speed was 150 mm/min. In another process, the TIG welding is carried out at a welding speed of 120 mm/min at the same arc voltage and heat input to the material so that weld quality remains the same. The welding current (in A) for this process is
(a) 44.72 (b) 40
(c) 55.90 (d) 62.25

Assam Engg. College AP/Lect. 18.01.2021

Ans. (b) : Given,
$I_1 = 50$ A $u_2 = 120$ mm/min
$V_1 = 60$ V $V_2 = V_1 = 60$
$u_1 = 150$ mm/min. $I_2 = ?$
Heat Input I = Heat input in II

$$\frac{50\times 60}{150} = \frac{60\times I}{120}$$

I = 40 amp

375. During the arc welding:
(a) 2/3 of the heat is developed near the positive pole and 1/3 is developed near the negative pole
(b) 3/4 of the heat is developed near the positive pole and 1/4 is developed near the negative pole
(c) 1/3 of the heat is developed near the positive pole and 2/3 is developed near the negative pole
(d) 1/4 of the heat is developed near the positive pole and 3/4 is developed near the negative pole

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : Arc welding is a fusion welding process used to join metals. An electric arc from an AC or DC power supply creates an intense heat of around 6500°F which melts the metal at the join between two work pieces. Once cooled the molten metals solidify to form a metallurgical bond.
During this welding 2/3 of the heat is developed near the positive pole and 1/3 is developed near the negative pole.

376. Voltage in the case of resistance welding process is,
(a) 15 to 20 V (b) 5 to 15 V
(c) 10 to 15 V (d) 0.5 to 10 V

Oil India Senior Officer 23.12.2020

Ans. (c) : Voltage in the case of resistance welding process is, 10 to 15 V.

377. In brazing, the solder fills the joint by,
(a) Capillary action (b) Fusion
(c) Diffusion (d) Surface tension

Oil India Senior Officer 23.12.2020

Ans. (a) : In brazing, the solder fills the joint by, capillary action.

378. In an arc welding process, two weld coupons were made using two different welding processes. For the first coupon, the voltage, current and welding speed used are 15V, 300 A and 30 mm/min respectively. Whereas the second coupon is welded with 60 kV, 200 mA and 25 mm/s. If the heat transfer efficiency for welding the first coupon is half of that of second coupon, the ratio of heat input per unit length is?
(a) 50:2 (b) 75:8
(c) 5:8 (d) 5:2

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : Given, $V_1 = 15$V $I_1 = 300$A
$V_2 = 60$ kV $= 60\times10^3$ $I_2 = 200$mA $= 200\times10^{-3}$
$v_1 = 30$ mm/min $= \frac{1}{2}$ mm/sec.
$v_2 = 25$ mm/sec $\eta_t \Rightarrow$ Heat transfer efficiency.

$$\eta_{t_1} = \frac{\eta_{t_2}}{2}$$

Heat supplied per unit length,

$$H_s = \eta_t \times \frac{VI}{v} \text{ J/sec}$$

$$\frac{H_{s_1}}{H_{s_2}} = \frac{\frac{V_1I_1}{v_1}\eta_{t_1}}{\frac{V_2I_2}{v_2}\eta_{t_2}} = \frac{\frac{15\times300}{1/2}\times\frac{\eta_{t_2}}{2}}{\frac{60\times10^3\times200\times10^{-3}}{25}\eta_{t_2}}$$

$$= \frac{150}{16} = \frac{75}{8}$$

$H_{s_1} : H_{s_2} = 75 : 8$

379. In Gas tungsten arc welding (GTAW) which of the following polarity is generally used for getting higher penetration
(a) Direct current straight polarity (DCSP)
(b) Direct current reverse polarity (DCRP)
(c) Alternating current high frequency (ACHF)
(d) All of the above

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : In GTAW process, non-consumable electrode is used, depth of penetration is more if heat generation is more on the workpiece.
In direct current straight polarity (DCSP) heat generation is more on the workpiece hence high penetration $2/3^{rd}$ of heat transfer to workpiece and $1/3^{rd}$ heat transfer to electrode.

380. When a 12V lead acid battery is fully charged, its open circuit voltage is :
(a) 2 V (b) 2.5 V
(c) 10 V (d) 12.6 V

APPSC Poly Lect. Automobile Engg., 2020

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Ans. (d) : 12V battery will measure at about 12.9V when its fully charged and about 11.4 volts when it is fully discharged. That's a total of 1.5 volts that represents the full range of charge on a 12 volt battery.

381. Straight polarity in arc welding is obtained with
(a) Alternating current electrode with electrode being positive
(b) Direct current electrode with electrode being positive
(c) Direct current electrode with electrode being negative
(d) Alternating current electrode with electrode being negative

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : Direct current straight polarity occurs when electrode is made negative and base plates are made positive. Thus electrons flow from electrode tip to base plates.

382. In which of the following welding process flux is fed separately?
(a) Electric arc welding
(b) Plasma arc welding
(c) Tungsten inert gas arc welding
(d) Submerged arc welding

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Submerged arc welding uses a continuous bare wire electrode and a separate flux added over the joint separately in the form of granular or powder.

383. In GTAW, the material coated on pure tungsten electrode to enhance its current capacity is:
(a) Thorium (b) Radium
(c) Titanium (d) Manganese

CIL MT 27.02.2020

Ans. (a) : Thorium is added to the tungsten because it increase the current carrying capacity of the electrode and it reduces contamination of the weld. In addition, it is easier to start the arc and the later is more stable.

384. For non-consumable electrode in arc welding
(a) Straight polarity is preferred
(b) Reverse polarity is preferred
(c) Neutral polarity is preferred
(d) Polarity is no matter

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : For non-consumable electrode in arc welding straight polarity is preferred because in straight polarity electrode is negative and workpiece is positive due to which $1/3^{rd}$ of heat generation is on electrode and $2/3^{rd}$ of heat generated on workpiece.

385. In Oxy-Acetylene welding reducing flame is mainly used for welding
(a) Aluminium
(b) Cast Iron
(c) Zinc, Lead and Tin
(d) Nickel, Monel and Stellite

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Application of Oxyacetylene Welding–
Neutral Flame–Low carbon steel, Mild steel, Cast iron, Aluminium alloys.
Oxidizing Flame–Brass, Bronze, Copper, Zn.
Carburizing Flame–Nickel alloys, High carbon steel, etc.

386. In which type of welding molten metal is poured for joining the metals?
(a) Gas welding (b) MIG welding
(c) Thermit welding (d) Arc welding

RPSC IOF, 2020

Ans. (c) : Thermit welding- In this molten metal is poured for joining the metals.
- Thermit is a mixture of Aluminium powder and Iron oxide.
- Small percentage of manganese is added as alloying to increase strength of joint.
- It used for repair of railway track.

387. The quantum of heat generated in resistance welding depends upon
(a) welding current alone
(b) resistance of current conducting path alone
(c) area of welding electrode alone
(d) None of these

RPSC IOF, 2020

Ans. (d) : The quantity of heat required to melt a given volume of metal depends on
(1) The heat to rise the temperature of the solid metal to its melting point.
(2) The melting point of the metal.
(3) The heat to transform the metal from solid to liquid phase at the melting point. So, answer is none of these.

388. Weld decay mostly found in
(a) Stainless steel (b) Mild steel
(c) Cast iron (d) Wrought iron

RPSC IOF, 2020

Ans. (a) : Weld decay is a corrosion process that mainly occurs as a result of sensitization in the heat affected zones of metal during welding operations.
This process mostly occurs in stainless steels or certain nickel-based alloys. It is form of intergrannular corrosion.

389. Percussion welding is a type of :
(a) gas welding (b) arc welding
(c) resistance welding (d) solid state welding

APPSC Poly Lect. 13.03.2020

Ans. (c) : Percussion welding (PEW) is type of resistance welding that joints dissimilar metals together.
Types of resistance welding–
- Spot welding
- Percussion welding
- Seam welding
- Projection welding
- Flash butt welding
- Stud welding

390. Which of the following is NOT an Arc welding?
(a) MIG
(b) TIG
(c) Atomic hydrogen welding
(d) Projection welding

APPSC Poly Lect. 13.03.2020

Ans. (d) : Projection welding is a resistance welding process in which the workpieces are joined by the heat generated due to the resistance of the workpieces to the flow of electric current through them.

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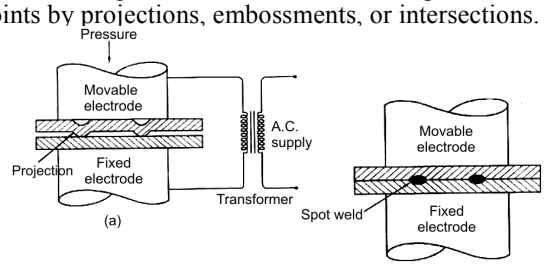
The resulting welds are localized at predetermined points by projections, embossments, or intersections.



Projection welding is not an arc-welding.

391. Which one of the following symbols is used to indicate square butt weld?
(a) V (b) ϒ
(c) || (d) Y

APPSC Poly Lect. 13.03.2020

Ans. (c) : "||" symbol is used to indicate square butt weld.
Following symbols is used to indicate weld joint.

No.	Designation	Illustration	Symbol
1.	Butt weld between plates with raised edges (the raised edges being melted down completely)		⋀
2.	Square butt weld		‖
3.	Single-V butt weld		V
4.	Single-bevel butt weld		⟋
5.	Single-V butt weld with broad root face		Y
6.	Single-bevel butt weld with broad root face		⟋

392. In the..... process, the high frequency vibratory energy is used to weld the part.
(a) thermit welding (b) SAW
(c) GTAW (d) ultrasonic welding

APPSC Poly Lect. 13.03.2020

Ans. (d) : Ultrasonic Welding (USW) is a solid state welding process that produces coalescence through the simultaneous application of localised high frequency vibratory energy. Ultrasonic metal welding can be applied for both similar or dissimilar metal work pieces.
- The frequency of vibration most commonly used is 20 kHz.
- No change in metallurgical structure of weldment.
- It is used in semiconductor, microcircuits and fine wires.

393. Brazing is done with silver alloys with melting range of :
(a) 400°C to 650°C (b) 600°C to 850°C
(c) 500°C to 650°C (d) 300°C to 450°C

NLC GET 17.11.2020, Shift-II

Ans. (b) : Brazing is the joining of metals through the use of heat and a filler metal whose melting temperature is above 450°C but below the melting point of the metals being joined.
Brazing is done with silver alloys with melting range of 600°C to 850°C.

394. Which type(s) is arc welding is/are suitable for joining non-ferrous metals?
(a) DC arc welding
(b) AC arc welding
(c) Both DC as well as AC arc welding
(d) None of the above

Assam PSC AE (PHED), 18.10.2020

Ans. (a) : D.C. arc welding is suitable for joining-non-ferrous metals. Because it creates a more stable arc easier welding and less spatter. D.C. polarity is used in most welding applications. It produces a smoother welding output compared to A.C.

395. The welding process which can be used for welding of plastics is
(a) Electron Beam Welding
(b) Resistance Welding
(c) Ultrasonic Welding
(d) Arc welding

Haryana PSC AE (PHED) 05.09.2020, Paper-II
Gujarat PSC AE 2019

Ans. (c) : Ultrasonic welding is commonly used for plastics and metals and especially for joining dissimilar materials.

396. Which one of the following is a solid state welding process?
(a) Tungsten inert Gas welding
(b) Diffusion welding
(c) Spot welding
(d) Submerged Arc Welding

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Solid state welding is a welding process, in which two work piece are joined under a pressure providing an intimate contact between them and at a temperature essentially below the melting point of the parent material. Bonding of the material is a result of diffusion of their interface atoms.

397. In case of arc welding of steel with a potential of 20 V and current of 200 A, the travel speed is 5 mm/s and the cross-sectional area of the joint is 20 mm^2. The heat required for melting steel may be taken as 10 J/mm^3 and heat transfer efficiency as 0.85. The melting efficiency will be nearly
(a) 18% (b) 29%
(c) 36% (d) 42%

ESE 2019

Ans. (b) : Given,
Voltage (V) = 20 Volt
Current (I) = 200 A
Welding speed (v) = 5 mm/s
Area (A) = 20 mm^2
Heat required (H_m) = 10 J/mm^3
Heat transfer efficiency (η) = 0.85

$$\text{Melting efficiency } (\eta_m) = \frac{H_m}{H} = \frac{H_m}{\frac{VI}{Av}\times\eta}$$

$$= \frac{10}{\frac{20\times200}{20\times5}\times0.85}$$

$$= 0.29 = 29\%$$

398. The metal powder used in thermit welding of steel is
(a) Tungsten (b) Copper
(c) Lead (d) Aluminium

TNPSC AE 2019
UPPSC AE 12.04.2016 Paper-I

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Ans. (d) : Thermit welding- Thermit welding is welding process utilizing heat generated by exothermic chemical reaction between the components of the thermit (a mixture of a metal oxide and aluminium powder). The molten metal, produced by the reaction, acts as a filler material joining the work pieces after solidification.

399. In a fusion welding process,
(a) only heat is used
(b) only pressure is used
(c) combination of heat and pressure is used
(d) none of these

Gujarat PSC AE 2019

Ans : (a) : Fusion welding- It is defined as melting together and joining metals by means of heat. It uses heat to melt the base metals and may add a filler metal. The thermal energy required for these operations is usually supplied by chemical or electrical means. Filler metals may or may not be used.

400. The arc in the arc welding is created by
(a) Current (b) Voltage
(c) Frequency (d) Contact resistance

Gujarat PSC AE 2019

Ans : (d) : The arc in arc welding is created by contact resistance.

401. The symbol ◺ refers to :
(a) single fillet weld (b) square but weld
(c) seam weld (d) single v-butt weld

SJVN ET 2019

Ans. (a) : The symbol used in welding
Single fillet weld – ◺
Square butt weld – ||
Seam weld– ⊖
Spot weld – ○

402. For soldering of brass and copper, the flux used is
(a) Zinc chloride
(b) Hydrochloride acid
(c) Cupric acid
(d) Ammonium chloride

Gujarat PSC AE 2019
HPPSC Lect. (Auto) 23.04.2016

Ans : (a) : For soldering of brass and copper, the flux used is zinc chloride.

403. A carburizing flame is obtained by supplying :
(a) Equal volumes of oxygen and acetylene
(b) More volume of oxygen than acetylene
(c) More volume of acetylene than oxygen
(d) None of these

OPSC AEE 2019 Paper-I

Ans : (c) : Carburizing flame- The carburizing flame is obtained when there is an excess of acetylene, It is used for welding of molten metal a certain alloys steels, many of non-ferrous hard surfacing materials such as satellite.

404. In laser welding the laser material used is :
(a) Molybdenum (b) Ruby crystal
(c) Titanium (d) Lithium

OPSC AEE 2019 Paper-I

Ans : (b) : Laser welding is a welding process which produces coalescence of materials with the heat obtained from the application of a concentrate coherent light beam impinging upon the surfaces to be joined. The focused laser beam has the highest energy concentration or heat flux of any known source of energy.

405. After fusion welding, the nature of residual stress in weldment area is :
(a) Compressive (b) Tensile
(c) Shear (d) No stress

OPSC AEE 2019 Paper-I

Ans : (a) : After fusion welding, the nature of residual stress is weldment area is compressive.

406. In TIG welding, the electrode is made of :
(a) Graphite
(b) Tungsten
(c) Same composition that the weld material
(d) Copper

OPSC AEE 2019 Paper-I

Ans : (b) :
Tungsten inert gas (TIG)-
This process is also known as gas tungsten arc welding (GTAW). This is similar to MIG welding. But non-consumable electrode of tungsten is used. For shielding, inert gas such as argon is used.
In the tungsten electrode 1 to 2% thorium and zirconium are added improve electron emission, arc stability, arc striking and current carrying capacity etc.

407. What happens to metal deposition if welding operation is performed with reverse polarity?
(a) Increases
(b) Decreases
(c) Remain same
(d) First decreases then increases

OPSC AEE 2019 Paper-I

Ans : (a) : If welding operation is performed with reverse polarity metal deposition increases.

408. The welding torch angle in forehand gas welding technique is:
(a) 60º (b) 50º
(c) 60º-70º (d) 90º

OPSC AEE 2019 Paper-I

Ans : (c) :
Leftward or forehand gas welding- In this method, the welding torch is held torch is held in the operator's right hand, the tip pointing towards the left and the weld is made from right to left. The torch makes an angle 60º-70º with the plate and the welding rod makes on angle of 30º-40º. The plates above 6 mm thickness are not economical to weld with this method.
Rightward or backhand gas welding- In this method, the welding torch is held in the right hand the filler rod in the left-hand. The welding begins at the left hand end of the joint and proceeds towards the right. The torch makes an angle 40º-50º with the plate and the welding rod makes an angle of 30º-40º. This method is better and economical for plates over 6 mm thickness.

409. A DC welding power source has a linear voltage-current (V-I) characteristic with an open circuit voltage of 80 V and a short circuit current of 300 A. For maximum arc power, the current (in Amperes) should be set at _____.
(a) 1200 A (b) 150 A
(c) 1500 A (d) 120 A V

BHEL ET 2019

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Ans. (b) : Given - $V_{OC} = 80$ V

$I_{SC} = 300$ A

Linear voltage (V.I)-

$$\frac{V}{V_{OC}} + \frac{I}{I_{SC}} = 1$$

$$\frac{V}{80} + \frac{I}{300} = 1$$

$$\frac{V}{80} = 1 - \frac{I}{300}$$

$$\frac{V}{80} = \frac{300 - I}{300}$$

$$\frac{V}{4} = \frac{300 - I}{15}$$

$$15V = 1200 - 4I$$

$$V = \frac{1200 - 4I}{15}$$

Power $P = VI$

$$= \left(\frac{1200 - 4I}{15}\right) I = \frac{1200\,I - 4I^2}{15}$$

$$= \frac{1200}{15} I - \frac{4I^2}{15}$$

$$P = 80I - \frac{4I^2}{15}$$

For maximum Arc power –

$$\frac{dP}{dI} = 80 - \frac{4}{15}(2I)$$

$$80 - \frac{4 \times 2}{15} I = 0$$

$$\frac{8}{15} I = 80$$

$$I = \frac{80 \times 15}{8}$$

$$\boxed{I = 150A}$$

410. Copper is welded by:

(a) Neutral flame (b) Oxidising flame
(c) Carburising flame (d) None of the above

HPPSC ADF 05.03.2019

Ans. (b) : In oxidizing flame copper, zinc and brass.

411. Which of the following comes under the arc welding process?

(a) Tungsten Inert Gas Welding
(b) Oxy-Hydrogen Welding
(c) Projection Welding
(d) Electron beam Welding

MECON MT 2019

Ans. (a) : Arc welding is one of many fusion welding process used to join metals. A power source generates an electric arc between a consumable or non-consumable electrode and base metal. In tungsten inert gas welding (TIG) is used to non-consumable electrode.

412. Which of the following weldings works on the same principal as electro-slag welding with the addition of some of the principles of submerged arc welding?

(a) Electron-Beam welding
(b) Electro-gas welding
(c) Ultrasonic welding
(d) Plasma arc welding

MECON MT 2019

Ans. (b) : Electro gas welding is an arc welding process that uses an arc between a continuous filler metal electrode and the weld pool, employing vertical position welding with backing to confine the molten weld metal.

413. Forge welding is best suited for-

(a) stainless steel (b) High carbon steel
(c) Wrought iron (d) Cast iron

MECON MT 2019

Ans. (c) :

414. In which of the following welding is the material around the joint melted in both the parts to be joined?

(a) Fusion welding
(b) Percussion welding
(c) Flash welding
(d) Resistance spot welding

MECON MT 2019

Ans. (a) :

415. Neutral flame cannot be used for welding :

(a) Cast iron
(b) Mild steel
(c) Copper and aluminium
(d) none of these

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (d) Application of the Flames
Neutral flame– Mild steel, low carbon steels, aluminium alloys, alloy steels, stainless steel, cast iron.
Oxidising flame– Copper, Zinc, Brass and Bronze, Grey cast iron (Not used in steels)
Carburising flame (Reducing flame)– High Carbon steels, Nickel base alloys and Cast iron (sometimes)

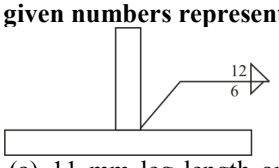
416. A process of joining pipes usually in the form of overlapped joints by making a filler metal flow into the gap between them by capillary action is known as:

(a) Brazing (b) Braze welding
(c) Soldering (d) Rolling

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (c) : Soldering is a method of joining similar or dissimilar metals by means of a filler metal whose liquids temperature is below 450ºC
Soldering is used mainly to provide continuity and leak proof joints.
In soldering filler metal alloys of tin (sn) and lead (pb).
In this process of joining metal pipes usually in the form of overlapped joint by making a filler metal flow into the gap between them by capillary action.

417. In the below figure shown for a fillet weld, the given numbers represent :



(a) 11 mm leg length on arrow side; 6 mm leg length on other side
(b) 6 mm leg length on arrow side; 12 mm leg length on other side

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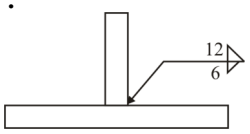
(c) 6 mm leg length on arrow side; 12 mm leg length arrow on side
(d) 6 mm leg length on other side; 12 mm leg length on arrow side

BHEL ET 2019

Ans. (c) :



• '6 mm' represents leg length on arrow side, 12 mm represents leg length arrow on side.

418. In a welding process the welding parameters used are : welding current = 250 A; welding voltage = 25V and welding traverse speed = 6 mm/s. Find the welding power.
(a) 6.55 kW (b) 65.5 kW
(c) 62.5 kW (d) 6.25 kW

BHEL ET 2019

Ans. (d) : Given - Welding current = 250 A
Welding voltage = 25V
Welding power (P) = VI
= 25 × 250
= 6250 W
= 6.250 kW

419. In a oxy-acetylene welding, the ratio of oxygen to acetylene in reducing flame is
(a) 0.5 : 1 (b) 0. 9 : 1
(c) 1 : 1.2 (d) 1 : 1.5

HPPSC AE 2018

Ans. (b) : In a oxy-acetylene welding, the ratio of oxygen to acetylene in reducing flame (carburizing) is 0.9 : 1.

420. Polarity change is required in
(a) AC welding
(b) DC welding
(c) Both AC and DC welding
(d) Neither AC nor DC welding

HPPSC AE 2018

Ans. (b) : Arc welding power sources can supply either AC or DC or both forms of current. In case of DC Polarity, Current flows only in one direction, whereas, in case of AC, Current flow direction reverses in every cycle so polarity change is required in DC welding.

421. For spot welding of 1 mm thick sheet with a current flow time of 0.2 s, the heat generated is 1000J. If the effective resistance is 200μ, the current required is
(a) 4000 A (b) 5000 A
(c) 6000 A (d) 7000 A

ESE 2018

Ans. (b) : $H = I^2 Rt$

$$I = \sqrt{\frac{H}{Rt}} = \sqrt{\frac{1000}{200\times10^{-6}\times0.2}}$$

$I = 5000$ A

422. The temperature produced by oxy-hydrogen flame is about
(a) 1800°C (b) 2100°C
(c) 2400°C (d) 3200°C

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : The temperature of oxyacetylene flame is about 3200°C and that of oxy-hydrogen flame is approximately 2400°C. Through the combustion of pure oxygen and acetylene the oxyacetylene process generates a high temperature flame above 3000°C.

423. In back-hand welding, the angle between the welding torch and the work is kept as
(a) 30°-40° (b) 40°-50°
(c) 50°-60° (d) 60°-70°

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : Backhand welding is a welding technique in which the workers welds the objects from left to right. It is also known as pull welding. It involves applying the torch before the rod itself. The torch and rod angles are 45° degree.

424. Which of the following welding process results in the smallest heat affected zone
(a) Shielded metal arc welding
(b) Gas welding
(c) Laser beam welding
(d) Thermit welding

GPSC Executive Engineer 23.12.2018
GATE 2011, ESE 2006

Ans. (c) : Laser beam welding- Utilizes a high power laser beam as the source of heat, to produce a fusion weld. Because the beam can be focused into a very small area, it has high energy density and therefore deep penetrating capability.

425. Determine the heat required for melting in case of arc welding of steel with a potential of 22V and current of 230A. The cross sectional area of the joint is 25mm^2 and the travel speed is 6 mm/sec. Heat required to melt the steel may be taken as 10 J/mm^3.
(a) 1365 J/s (b) 1295 J/s
(c) 1500 J/s (d) 250 J/s

MPSC Lect. (Prod.) 2018

Ans. (c) Given,
$H_m = 10 J/mm^3$ $A_b = 25\ mm^2$ $V = 6$ mm/sec
Heat required for melting = $H_m \times A_b \times V$
$= 10 \times 25 \times 6 = 1500$ J/sec

426. The primary cause for deflection of arc during arc welding is .....
(a) hydrostatic field (b) magnetic field
(c) explosive field (d) wind velocity field

RPSC AE 2018

Ans. (b) : The primary cause for deflection of arc during arc welding is magnetic field called arc blow.
Arc blow is basically the deflection of electric arc from its axis of electrodes. Arc blow is mainly encounted during DC arc welding due to interaction between different electromagnetic fields in and around the welding arc.

427. Which of the followings is NOT the "resistance welding process"?
(a) Percussion (b) Spot
(c) Ultrasonic (d) Projection
(e) Butt

CGPSC ARTO & TSI 07.01.2017

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Ans. (c) : Types of Resistance Welding
1. Resistance spot welding
2. Resistance seam welding
3. Resistance projection welding
4. Resistance butt welding
5. Flash butt wiling

428. Preheating is essential in welding of:
(a) High speed steel (b) Stainless steel
(c) Cast iron (d) German silver

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : Preheating is specially important welding highly restrained weld joints, thick materials. (The rule of thumb on thickness (The rule of thumb on thickness and when to preheat varies by material type). Base material that tend to be more brittle such as cast iron and when welding dissimilar materials.

429. The vacuum maintained in the electron beam welding is of the order:
(a) 1 torr (b) 5 torr
(c) 10^{-1} torr (d) 10^{-5} torr

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : Electron beam welding also shows high strength properties, maintaining up to 95% of strength of the base materials. As the technique is performed in a vacuum environment there are no impurities left by the process vacuum maintained in electron beam welding is of the order of 10^{-5} torr.

430. A cylindrical section having no joints is called as:
(a) seamless (b) perfect
(c) pointless (d) ideal

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (a) : A cylindrical section having no joints is known as a jointless section or seamless section.

431. Weld spatter defect in welding is generally the result of
(a) using too high welding current
(b) using too low welding current
(c) low voltage
(d) high voltage

Karnataka PSC Lect., 27.05.2017
Nagaland PSC CTSE 2016 Paper-I

Ans (a) : Weld spatter defect in welding is generally the result of
(i) Excessive current
(ii) Excessive voltage
(iii) Arc blow making arc uncontrollable

432. A welding process in which it uses a non consumable electrode and a separate filler metal with an inert shielding gas is known as
(a) Gas welding
(b) Electric are welding
(c) T.I.G. welding
(d) M.I.G. welding

Karnataka PSC Lect., 27.05.2017

Ans (c) : A welding process in which it uses a non consumable electrode and a separate filler metal with an inert shielding gas is known as T.I.G. welding. Tungsten Inert Gas (TIG) welding also known as Gas Tungsten Arc Welding (GTAW) is an arc welding process that produces. The weld with non-consumable tungsten electrode.

433. Two stainless steel foils of 0.1 mm thickness are to be joined. Which of the following processes would be best suited
(a) Gas welding (b) TIG welding
(c) MIG welding (d) Plasma arc welding

Vizag Steel (MT) 2017

Ans. (d) : Plasma arc welding– Plasma is a partially ionized gas and is produced by the passage of gas through an electric field which separates into electrons and positive ions plasma arc welding a direct current (DC) power source having open circuit voltage of about 70 Volt is used. Currents may vary from 50 to 350 Amperes.

434. Following equipment is used for arc welding a material by carbon electrode
(a) A.C. welding set
(b) Rectifier
(c) Motor generator
(d) D.C. welding set with straight polarity

Vizag Steel (MT) 2017

Ans. (d) : D.C. welding set with straight polarity is used for arc welding a material by carbon electrode.

435. In MIG welding, helium or argon is used in order to
(a) Provide cooling effect
(b) Act as flux
(c) Act as shielding medium
(d) Facilitate welding process

Vizag Steel (MT) 2017

Ans. (c) : In MIG welding helium or argon is used in order to Act as shielding medium.

436. For welding metals less than 5 mm thick, following method of gas welding would give best results
(a) Forehand (b) Backhand
(c) Straight hand (d) Inclined hand

Vizag Steel (MT) 2017

Ans. (a) : For welding metals less than 5mm thick, fore hand method of gas welding would give best results.

437. The purpose of applying flux coating on AC arc welding electrode includes
(a) maintaining continuity of arc
(b) protection against corrosion
(c) removal of impurities
(d) All of the above

BPSC AE Mains 2017 Paper - VI

Ans : (d) : The purpose of applying flux coating on AC arc welding electrode is for maintaining continuity of arc, a protection against corrosion and removal of impurities.

438. Hard-zone cracking in low alloy steel due to welding is the result of an absorption of
(a) N_2 (b) O_2
(c) H_2 (d) C

TNPSC AE 2017

Ans. (c) : Hard-zone cracking in low alloy steel due to welding is the result of an absorption of H_2.

439. Bead welds are generally used for
(a) thin sheets (b) circular rods
(c) thick plates (d) hollow rods

TSPSC AEE 2017

Ans. (a) : Bead welds are generally used for thin plates and sheets.

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440. When fragile components are to be joined without any significant addition of weight, the recommended method of joining is
(a) Soldering
(b) Brazing
(c) Adhesive bonding
(d) Mechanical fastening

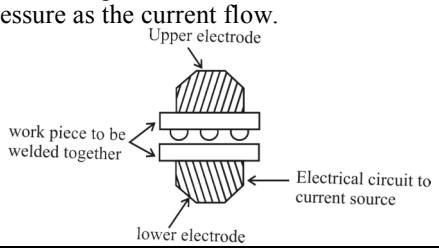
TSPSC AEE 2017

Ans. (c) : Adhesive bonding is the joining technique used in the manufacture and repair of a wide range of products along with welding and soldering, Adhesive bonding is one of the basic joining processes.

441. The type of welding in which the pieces to be welded are arranged between two flat electrodes which exert pressure as the current flows, is known as
(a) Projection welding (b) Friction welding
(c) Explosive welding (d) Thermit welding

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : Projection welding in which the pieces to be welded are arranged between two flat electrodes which exert pressure as the current flow.



442. Which one is the strongest among the following brazing joints?
(a) Inclined (b) Lap
(c) Butt (d) V-butt

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Lap joint is the strongest among the following brazing.

443. Edge preparation for welding is not required upto a thickness of
(a) 15 mm (b) 25 mm
(c) 30 mm (d) 4 mm

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Edge preparation for welding is not required upto a thickness of 4 mm, because till 4 mm the welding can be done without edge preparation.

444. In resistance welding the voltage supplied is:
(a) 100 V (b) 1 V
(c) 500 V (d) 1000 V

CIL MT 26.03.2017

Ans. (b) : In resistance welding very high-current (up to 100,000 A) and very low-voltage (0.5 to 10 V) is used.

445. Which of the following welding method is not a solid state welding process?
(a) Friction welding
(b) Forge welding
(c) Resistance spot welding
(d) Ultra-sonic welding

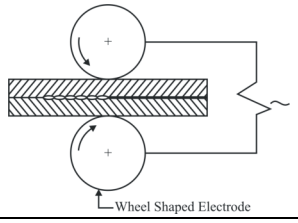
CIL MT 26.03.2017

Ans. (c) : Solid state welding is a group of welding processes which produces coalescence at temperature essentially below the melting point of the base materials being joined, without the addition of brazing filler metal. Pressure may or may not be used. Some examples of solid state welding are (friction welding, ultrasonic welding, forge welding etc). Resistance spot welding is not a solid state welding.

446. _____ does not require a filler metal.
(a) Gas metal arc welding
(b) Submerged arc welding
(c) Oxy-acetylene welding
(d) Oxy-hydrogen welding
(e) Resistance seam welding

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Gas metal arc welding, submerged arc welding, oxy-acetylene welding and oxy-hydrogen welding require a filler metal where as resistance seam welding does not require a filler metal.
Seam Welding–Seam welding uses a wheel – shaped electrode to make either a series of overlapping spot welds to form a continuously welded and leak tight seam or a number of spot welds spaced apart roll-spot welding.



447. The heat generated in the resistance welding can be expressed as:
(where H-total heat generated, I-current, R-resistance of the joint, t-time and k-constant to account heat loss)
(a) $H = kI^2Rt$ (b) $H = kI^3Rt$
(c) $H = kIR^2t$ (d) $H = kIR^3t$
(e) $H = kI^2R^3t$

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : We know the electrical power (P)

$P = VI \quad V \rightarrow$ Voltage

$= RI^2 \quad R = \frac{V}{I}$

But total heat generated for time t

$H = RI^2t$

Some heat also loss and we take constant k to account heat loss

$H = kRI^2t$

448. In gas welding, the maximum flame temperature occurs at–
(a) The tip of the flame
(b) The inner cone
(c) The outer cone
(d) Next to the inner cone

Nagaland PSC CTSE 2016,2017, Paper-I

Ans. (b) : The maximum flame temp occurs at the inner cone about 3200^0 C

449. The electrode used for GTAW with yellow colour is of the type of
(a) Pure tungsten (W)
(b) 1% Thorium tungsten (W)
(c) 2% Thorium tungsten (W)
(d) Zirconium tungsten (W)

Karnataka PSC AE, 10.09.2017

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Ans. (b) : Non-Consumable electrode of tungsten are used with alloying elements like Thorium and Beryllium 1%. Thorium gives yellow colour.

450. In L-D steel making, the nature of final slag can be described as
(a) Oxidizing
(b) Basic
(c) Oxidizing and basic
(d) Reducing and basic

APGENCO AE, 2017

Ans. (c) : Linz- Donawitz steel making is a method of primary steel making in which carbon rich molten pig iron is made into steel. Blowing through molten pig iron lowers the carbon content of the alloy and changes into low carbon steel.

451. Which of the following statement is CORRECT?
(a) No flux is used in gas welding of mild steel
(b) Borax is the commonly used flux coating on welding electrodes
(c) Laser beam welding uses a vacuum chamber instead of a shielding method
(d) Alternating current can be used in Gas Tungsten are welding

APGENCO AE, 2017

Ans. (a, b&d) :

452. Which of the following welding processes use consumable electrode?
(a) GTAW
(b) Resistance spot welding
(c) Friction welding
(d) SAW

TSPSC AEE 2017

Ans. (d) : Submerged arc welding is a common arc welding process that involves the formation of an arc between a continuously fed electrodes and the work piece. SAW (Submerged Arc Welding) is used to consumable electrode.

453. What would be general selection of number of runs (R), size of electrode (E) in mm and current (C) in ampere during fillet welding of 6mm thick plate?
(a) R = 1, E = 4, C = 160
(b) R = 3, E = 5, C = 200
(c) R = 1, E = 5, C = 200
(d) R = 1, E = 4, C = 320

Punjab PSC SDE 12.02.2017

Ans. (a&c) : R = 1, E = 4, C = 160
R = 1, E = 5, C = 200

454. As per the Indian (BIS) system of the coding of electrodes, 2nd digit indicates____.
(a) Position of welding
(b) Percentage of elongation & impact value
(c) Tensile strength
(d) Current, Polarity and OCV

Punjab PSC SDE 12.02.2017

Ans. (b) : As per the Indian (BIS) system of the coding of electrodes, 2nd digit indicates percentage of elongation and impact value.

455. While Brazing of Stainless steel, Nickel- and Cobalt base alloys which type of filler metal is used?
(a) Aluminium-silicon
(b) Copper-phosphorus
(c) Gold
(d) Nickel-silver

Punjab PSC SDE 12.02.2017

Ans. (d) : Brazing– Welding gives higher joint strength but the problem is phase transformation and oxide formation.
• In brazing the base metal is not melted but the joint is obtained by means of filler metal.
• It is a process in which a filler metal is placed between surfaces to be joined. Temperature is raised enough to melt filler metal.
• Molten metal fills fitting space by capillary action. Filler metal used for Brazing melt about 450°C (copper-zince, Cu-Ag, Al-Si). The use of flux essential in brazing, in order to prevent oxidation. It is suitable for carbide drill bits and carbide inserts on steel shanks.

456. The advantage of the welding process is
(a) It relieves the joint from residual stresses
(b) It helps in checking of distortion of work piece
(c) Large number of metals and alloys, both similar and/or dissimilar can be joined
(d) Heat produced during the welding does not produce metallurgical changes

ESE 2012

Ans. (c) : The advantage of the welding process is large number of metals and alloys, both similar and/or dissimilar can be joined.

457. Arc blow is more common is ..... .
(a) dc
(b) dc welding with straight polarity
(c) ac welding with bare electrodes
(d) dc welding with bare electrodes

GWSSB DEE 07.07.2016
TSPSC Managers, 2015, ESE 2001

Ans. (d) : Deflection of electric arc due to deflection of electromagnetic flux lines at the beginning and end of the workpiece is known as arc blow. The problem of arc blow is more serious in DC bare electrodes.
In AC welding due to change in polarity, the magnetic field is not constant on workpiece.

458. In atomic hydrogen welding, the arc is struck between the :
(a) tungsten electrode and work piece
(b) copper electrode and work piece
(c) two tungsten electrodes
(d) between two copper electrodes

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : In atomic hydrogen welding, the are s struck between the two tungsten electrodes.
In atomic hydrogen welding.

$$H_2 \rightarrow H + H - Ah_1$$
$$H + H \rightarrow H_2 + Oh_2$$

Hydrogen acts as a heating agent, and hydrogen gas will also acts as a shielding gas.

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459. Distortions in welding causes due to:
(a) Impurities (b) Slag
(c) Shrinkage (d) Cavitations
HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Distortions in welding causes due to shrinkage when a weld is laid it is molten metal and therefore hot, as it cools it shrinks, this shrinkage impose stress on the weld and the parent material new the weld. Distortion is also created by localized heating of the base metals which causes local expansion and contractions during the welding operation.

460. In Oxy-hydrogen welding the flame temperature is around :
(a) 874°C (b) 1074°C
(c) 2870°C (d) 4852°C
HPPSC Asstt. Prof. 29.10.2016

Ans. (c) :

Gas flames	$\frac{O_2}{C_2H_2}$ value	Temperature range
Neutral flames	1	3200°C
Oxidizing flames	1.15 to 1.5	3300°C
Carburizing flames	.85 to 0.95	2850-2900°C

461. While welding of Nickel using GMAW process, the shielding gas commonly used is:
(a) Oxygen + carbon dioxide
(b) Carbon dioxide
(c) Oxygen
(d) Argon
HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : Argon, helium, CO_2 and oxygen are the most can common shielding gases used in GMAW. Each gas has benefits and drawbacks in any given application.
While welding of Nickel using GMAW process, the shielding go gas commonly used is 'Argon'.

462. Consider the following statements:
The magnitude of residual stresses in welding depends upon
1. Support and clamping of components
2. Design of weldment
3. Amount of metal melted/deposited
4. Welding process used
Which of the statements given above are correct?
(a) 2 and 3 (b) 1 and 3
(c) 1, 2 and 4 (d) 1, 2 and 3
UPRVUNL AE 21.08.2016

Ans. : (c) The magnitude of residual stresses in welding depends upon
(i) Support and clamping of components
(ii) Design of weldment
(iii) Welding process used

463. Hot cracks occur in the weld and fusion zone as the metal solidifies. Which of the following are the causes for hot cracks?
1. High carbon or alloy content of the base metal.
2. Presence of sulphur and phosphorous for hot cracks.
3. Joint restraint
4. Moisture in the joint or electrode
Select the correct answer using the code given below
(a) 1, 2 and 3 (b) 3 and 4
(c) 1, 2, 3 and 4 (d) 1, 2 and 4
UPRVUNL AE 21.08.2016

Ans. : (a) Hot cracks occur in the weld and fusion zone as the metal solidifies, following causes for hot cracks-
(i) High carbon or alloy content of the base metal.
(ii) Presence of sulphur and phosphorous for hot cracks.
(iii) Joint restaint.

464. For ultrasonic welding the thickness of metal is usually restricted to
(a) 3 mm (b) 6 mm
(c) 10 mm (d) 15 mm
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Ultrasonic welding is restricted to the lap joint welding of thin sheets foil and wires and the attaching of thin sheet to heavier structural members. The maximum thickness of weld is about 2.5 mm for aluminium and 1 mm for harder materials.

465. In a gas welding using oxygen and acetylene cylinders the pressure of gas is more in
(a) Oxygen cylinder
(b) Acetylene cylinders
(c) Equal in both cylinders
(d) Gas welding does not use oxygen cylinders
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : In gas welding, there are two gauges on the body of the regulator, one showing the pressure in the cylinder, while the other, supplied pressure at the torch, the pressure in Oxygen at the welding torch is between 70 and 280 kN/m^2 and for acetylene it is between 7 and 103 kN/m^2.

466. Plains and butt welds may be used on materials upto approximately
(a) 25 mm thick (b) 40 mm thick
(c) 50 mm thick (d) 75 mm thick
Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Plain and butt welds may be used on materials upto approximately 25 mm thickness.

467. Two 3 mm thick MS sheets are to be welded. Electrodes of 16, 12, 10 Nos. are available which one would you choose.
(a) 16 (b) 12
(c) 10 (d) Any of the above
Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Two 3 mm thick mild steel sheet are to be welded. The electrodes 10 Nos. is required for welding the sheet.

468. In a linear arc welding process, the heat input per unit length is inversely proportional to :
(a) Welding current (b) Welding voltage
(c) Welding speed (d) None of the above
ISRO Scientist/Engineer 22.04.2018
UP Jal Nigam AE 2016

Ans. (c) : In a arc welding process, the heat input is given as

$$H = I^2 RT \quad ...(i)$$

$$H = \frac{VI}{S} \quad ...(ii)$$

Where,

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I = welding current
V = Voltage
R = Resistance
T = time of welding cutting
S = welding speed

From above equation we see that heat input is directly proportional to welding current and voltage.

Duty cycle is the time out of 10 minutes, while which power source is working. Thus duty cycle is related to time of welding.

Also, from equation (ii) it is clear that as welding speed (s) increases, heat input (H) decreases.

469. In shielded metal are welding the shielding gas is provided by

(a) cylinder connected to the electrode
(b) electrode coating material
(c) electrode core material
(d) work piece material

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : In shielded metal arc welding, shielding is provided by decomposition of the electrode covering (coating). The main function of the shielding is to protect the arc and the hot metal from the chemical reaction with constituents of the atmosphere.

470. Carburizing flame is formed in oxyacetylene gas welding when

(a) quantity of oxygen is increased
(b) quantity of acetylene is increased
(c) both are at same level
(d) it does not depend on the quantities of oxygen or acetylene

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : Carburizing or reducing flame are used to weld or braze easily oxidized alloys such as aluminium. As oxygen is increased the fuel is turned down and carburizing feather pulls off and disappears.

471. The range of temperature of soldering process is

(a) 10°C to 50°C (b) 50°C to 100°C
(c) 180°C to 250°C (d) 250°C to 400°C

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Alloy of lead & tin (Pb–Sn) mixture is used for soldering in electric work. The temperature in soldering at which metal melts at 188°C (370°F) to 250°C (482°F).

472. A non consumable electrode is used in

(a) Gas welding (b) DC Arc welding
(c) AC Arc welding (d) Argon arc welding

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : A non consumable electrode is used in Argon arc welding.

473. The welding process by Metal Inert-Gas (MIG) welding is

(a) slower than the welding process by Tungsten inert-Gas (TIG) welding
(b) faster than the welding process by Tungsten inert-Gas (TIG) welding
(c) at same speed as the welding process by Tungsten inert-Gas (TIG) welding
(d) at unpredictable speed

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : The welding process by Metal Inert-Gas (MIG) welding is faster than the welding process by Tungsten inert-Gas (TIG) welding.

474. In welding process, how is penetration increased?

(a) By increasing current and decreasing speed
(b) By decreasing both current and speed
(c) By increasing both current and speed
(d) By increasing both arc voltage and speed
(e) By increasing current and decreasing voltage

CGPSC Asstt. Workshop Supt., 17.07.2016
GATE 1990

Ans. (a) : The welding variable that has the greatest effect on the degree of weld penetration, is current, as welding current increases, weld penetration increases and as welding current decreases, weld penetration decreases.

475. In case of arc welding with reverse polarity

(a) Electrode is kept as positive pole
(b) Electrode is kept as negative pole
(c) Electrode is kept as neutral pole
(d) Electrode is kept as alternating changing pole

UJVNL AE 2016

Ans : (a)

476. TIG welding:

(a) Uses no inert gas for shielding
(b) Is a resistance welding process
(c) Uses electrode of high current carrying capacity
(d) Uses consumable electrode

UPRVUNL AE 07.10.2016

Ans. (c) : TIG welding—Tungsten inert gas welding, also known as gas tungsten arc welding (GTAW), is an arc welding process that uses a non-consumable tungsten electrode to produce the weld.

TIG welding utilizing power supply to generate an electric arc between the tungsten and the workpiece, using the resultant heat create weld.

The weld area is generally protected from atmospheric contamination by use of an inert shielding or cover gas (argon or helium).

477. Which of the following is not true for brazing?

(a) It causes the metallurgical damage by phase transformation
(b) Gap between the parent material is filled by filler material
(c) Gap between the parent material is filled by capillary action
(d) Filler material must have low melting point

UPRVUNL AE 07.10.2016

Ans. (a) : It causes the metallurgical damage by phase transformation is not true for brazing.

478. Porosity in welding:

(a) is created due to presence of H_2 in the weld pool
(b) is created due to lamellar tearing
(c) is created due to cold cracking
(d) causes increase in strength of joint

UPRVUNL AE 07.10.2016

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Ans. (a) : Porosity—Porosity is the presence of cavities in the weld metal caused by the freezing in of gas released from the weld pool as it solidifies.
Cause—Porosity is caused by the absorption of nitrogen, oxygen and hydrogen in the molten weld pool which is then released on solidification to become trapped in the weld metal.
Prevention—
Air entrainment—
- Seal any air leak
- avoid weld pool turbulence
- reduce excessively high gas flow

Hydrogen—
- dry the electrode and flux
- clean and decrease the workpiece surface.

Surface coatings—
- Clean the joint edges immediately before welding
- Check that the weldable primer is below the recommended maximum thickness.

479. The flame which contains excess oxygen that theoretical required:
(a) is acetylene flame
(b) is neutral flame
(c) is oxidizing flame
(d) is carburizing flame

UPRVUNL AE 07.10.2016

Ans. (c) : Oxidizing flame contains access oxygen than theoretical required.

480. Which of the following welding method uses a pool of molten metal?
(a) carbon arc welding
(b) submerged arc welding
(c) TIG arc welding
(d) MIG arc welding

GPSC Lect. 23.10.2016

Ans. (b) : Submerged arc welding, process that involves the formation of an electric arc between a continuously fed electrode and the workpiece. A blanket of powdered flux generates a protective gas shield and slag (No item metal pool) which protects the weld zone.

481. Spot welding is an example of
(a) Gas welding
(b) Arc welding
(c) Resistance welding
(d) Tungsten inert gas welding

GPSC Lect. 23.10.2016

Ans. (c) : Spot welding is a form of resistance welding which is a method of welding two or more metal sheets together without using any filler materials.

482. A resistance spot-welding operation is performed on two pieces of 1.5 mm thick sheet steel using 12000 amps current for a duration of 0.20 second. The electrodes are 6 mm in diameter at the contacting surfaces. Resistance is assumed to be 0.0001 ohms and the resulting weld nugget is 6 mm in diameter and 2.5 mm thick. The unit melting energy for the metal is 12 J/mm^3. What portion of heat generated was used to form the weld?
(a) 29.4% (b) 70.6%
(c) 58.8% (d) 41.2%

ISRO Scientist/Engineer 03.07.2016

Ans. (a) : I = 12000 A, t = 0.2 sec
R = 0.0001 Ω (ohms)
d = 6 mm, h = 2.5 mm, E = 12 J/mm^3
Heat supplied = I^2Rt
$= (12000)^2 \times 0.0001 \times 0.2$
= 2880 J
Heat required to melt–

$$= \frac{\text{Energy}}{\text{Volume}} \times \text{Volume}$$

$$= 12 \times \frac{\pi}{4} d^2 \times h = 12 \times \frac{\pi}{4} (6)^2 \times 2.5 = 848.23$$

$$\text{Portion of Heat Generated } (\eta_{melting}) = \frac{\text{Heat required}}{\text{Heat supplied}} \times 100$$

$$= \frac{848.23}{2880} \times 100 = 29.45\%$$

483. A soldering iron is used mainly to
(a) Heat the metal (b) Melt the solder
(c) Heat the flux (d) None

APPSC AEE Screening Test 2016

Ans. (b)
Soldering
- Formation of metal to metal joint using solder.
- The joint is made by alloy formation of bare metal and solder.
- In soldering, there are four key elements Iron, solder, flux and component.
- Flux cored solder wire is used for hand soldering.

484. During MIG welding, the metal is transferred in the form of
(a) continuous flow of molten metal
(b) a fine spray of metal
(c) electron beam
(d) solution

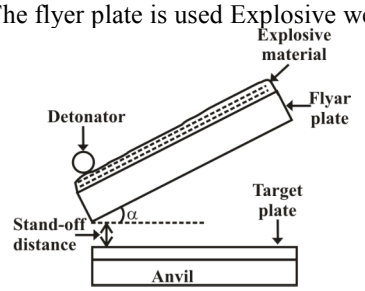
BPSC Poly. Lect. 2016
ESE 2007

Ans : (b) During MIG welding, the metal is transferred in the form of a fine spray of metal.

485. The 'flyer plate' is used ..... welding.
(a) Ultrasonic (b) Explosive
(c) Electron beam (d) Laser beam

(HPPSC LECT. 2016)

Ans : (b) The flyer plate is used Explosive welding.



α = initial stand-off angle.
Explosive welding:- In this process, the metals to be joined are placed at an angle to each other. The energy derived from an Explosion forces the plates together at high velocity causing surface ripples in the metal. As the force is dissipated the ripples lock or weld the two metals.

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486. Inadequate penetration in case of welding operation will lead to

(a) Crack formation (b) Corrosion
(c) Diffusion (d) Undercutting

HPPSC W.S. Poly. 2016

Ans : (a) Inadequate penetration in case of welding operation will lead to crack formation.

487. In.....welding, the joint to be welded is filled with a stick electrode is kept in place by means of a glass fibre tape or copper retaining bar.

(a) Stud (b) Plasma arc
(c) Atomic hydrogen (d) Fire cracker

HPPSC AE (PWD) 10.09.2016

Ans. (d) : In fire cracker welding, the joint to be welded is filled with a stick electrode is kept in place by means of a glass fibre tape or copper retaining bar.

488. In.....resistance welding the heat required for melting is obtained by means of an arc rather than simple resistance heating.

(a) Upset (b) Flash
(c) Seam (d) Projection

HPPSC AE (PWD) 10.09.2016

Ans. (b) : In flash resistance welding the heat required for melting is obtained by means of an arc rather than simple resistance heating.

489. In an arc welding.....is the depression in the weld metal pool at the point where the arc strikes base metal plate.

(a) Root (b) Puddle
(c) Crater (d) Flank

HPPSC AE (PWD) 10.09.2016

Ans. (c) : A crater is a tiny explosion which will send droplets of molten metal in energy direction. A crater is the unfilled end of a weld and is considered a weld defect that causes stopping the arc too soon and not allowing weld end to fill completely.

490. As per IS : 815-1966 for a coated electrode with designation "E 3 2 5 411 P", the designation "411" refers to:

(a) Type of covering
(b) Strength and temperature related information
(c) Welding position
(d) Extrusion related

HPPSC AE (PWD) 10.09.2016

Ans. (b) : 411–Strength and temperature related information.

491. "Transferred arc' process comes in.....

(a) USM (b) LBM
(c) PAM (d) EDM

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Plasma Arc Machining (PAM) is a metal cutting process where metals are cut with plasma arc produced by passing electric current through it.

492. The predominant problem faced with DC arc is the....., the deflection of the arc by means of the magnetic field setup due to the flow of the welding current.

(a) arc pit (b) spatter
(c) arc blow (d) pinch spot

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Magnetic arc blow, also known as arc wander is the arc deflection caused by distortion of the magnetic field produced by an electric arc current. When welding ferromagnetic steels lack of fusion imperfection can be caused through uncontrolled deflection of the arc usually termed arc blow is the usually unwanted deflection of the arc during arc welding. There are two types of arc blow commonly known in the electric welding industry magnetic and thermal.

493. With increase in torch angle, flame density :

(a) Increases
(b) Decreases
(c) Remain constant
(d) First increases then decreases

RPSC ADE 2016

Ans. (a) : The angle made by the axis of a torch with an axis of the work piece is called a torch angle it varies from 0^0 to 90^0.
With increases of torch angle, the area of weld bead to which the flame is exposing is reducing hence density is increasing.

494. Cycle time of resistance welding consists of

(a) Squeeze time, weld time, hold time & on time
(b) On time, weld time , hold time & off time
(c) On time, weld time & off time
(d) Squeeze time, weld time, hold time & off time

RPSC ADE 2016

Ans. (d) : Cycle time of resistance welding consist of squeeze time, weld time, hold time and off time.
Squeeze time – It is time interval between the initial application of the electrode force on the work and the first application of current.
Weld time – Time during which welding current is applied to the metal sheet.
Hold time – Time, after the welding when the electrodes are still applied to the sheet to chill the weld. Hold time is necessary to allow the weld nugget to solidify before releasing the weld parts.

495. Which of the following rays is absent during welding?

(a) Visible light rays (b) Infrared ray
(c) Gamma rays (d) Ultra violet rays

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Gamma rays is absent during welding.
Gamma rays are the most energetic form of electromagnetic radiation, with a very short wavelength of less than one-tenth of a manometer. Gamma radiation is the product of radioactive atoms that undergo radioactive decay.

496. In DC welding, the straight polarity (electrode negative), results in

(a) deep penetration
(b) lower deposition rate
(c) less heating workpiece
(d) smaller weld pool

GPSC Asstt. Prof. (Prod.) 21.08.2016
HPPSC Asstt. Prof. 18.11.2016

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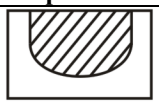
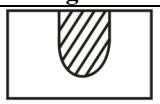
Ans. (a, b) :

		DC Electrode positive	DC Electrode negative
(1)	Penetration	 Shallow	 Deep
(2)	Heat generation	$2/3^{rd}$ at electrode	$1/3^{rd}$ at electrode
		$1/3^{rd}$ at workpiece	$2/3^{rd}$ at workpiece
(3)	Metal deposition rate	High	Low
(4)	Polarity	Reverse polarity (W.P)	Straight polarity (W.P)

497. The flame type normally used to weld mild steel in oxy-acetylene welding is

(a) Carburizing flame (b) Neutral flame
(c) Oxidizing flame (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : It is a general purpose flame used for welding of mild steel, low carbon steel, medium carbon steel, aluminium alloy, cast iron etc.
In natural flame, the ratio of volume (O_2) to C_2H_2 is 1:1.

498. The neutral flame is created by supplying the flame with the amount of oxygen which is_____

(a) precisely enough for reduction but no oxidation occurs
(b) precisely enough for equal amounts of oxidation and reduction occurs
(c) precisely enough for oxidation but no reduction occurs
(d) precisely enough for burning, and neither oxidation nor reduction occurs

KPCL AE 2016

Ans. (d) : The neutral flame is created by supplying the flame with the amount of oxygen which is precisely enough for burning and neither oxidation nor reduction occurs.

499. In which of the welding technique, the weld pool is protected by inert gas?

(a) SMAW (b) TIG welding
(c) SAW (d) Resistance weeding

TSPSC Managers, 2015

Ans. (b) : In TIG (Tungsten Inert Gas) welding, weld pool is protected by Helium and Argon inert gases. The weld area and electrode are protected from oxidation or other atmospheric contamination.

500. Neutral flame is used to weld

(a) steel (b) cast iron
(c) copper (d) All of the above

Assam PSC CCE Pre 2015

Ans. (d)

- **Neutral flame–** The neutral flame is obtained by supplying equal volume of oxygen and acetylene. It has the following two sharply defined zones :
  (i) An inner luminous cone (3200^0C)
  (ii) An outer cone or envelope of bluish colour (1250^0C)

The most of the oxy-acetylene welding (e.g. welding of steel, Cast iron, Copper, Aluminium etc.) is done with the neutral flame.

- **Oxidizing flame–** The oxidizing flame is obtained when there is an excess of oxygen. It is used for welding brass and bronze.
- **Reducing or carburizing flame–** The reducing flame is obtained when there is an excess of acetylene. It is used for welding molten metal, a certain alloy steels, many of non-ferrous, hard surfacing materials such as satellite.

501. Which one of the following welding process uses non-consumable electrodes

(a) MIG (b) Manual arc welding
(c) SAW (d) TIG

GPSC Engg. Class-II Pre 19.01.2020
Gujarat PSC AE 2019
GPSC Asstt. Prof. 28.08.2016
GWSSB DEE 07.07.2016
Kerala PSC Asstt. Prof. 27.09.2016
GPSC Asstt. Prof. (Prod.) 21.08.2016
UP Jal Nigam AE 2016
GWSSB AAE, 27.12.2015
Kerala PSC Poly. Lect. 09.01.2014
GPSC Poly. Lect. 07.12.2014
RPSC AEN Pre-2013, GATE 2011

Ans. (d) : In tungsten inert gas welding electrode of tungsten is non-consumable.

Consumable Electrodes-

- Shielded Metal Arc Welding (SMAW)
- Gas Metal Arc Welding (GMAW)
- Flux-cored arc Welding (FCAW)
- Sub Merged Arc Welding (SAW)
- Electro Slag Welding (ESW)
- Electro-Gas Welding (EGW)
- Gas Turgsten Arc Welding (GTAW)

Non- Consumable Electrodes-

- Gas Tungsten Arc Welding (GTAW) or Tungsten Inert Gas welding (TIG)
- Atomic Hydrogen Welding (AHW)
- Carbon Arc Welding (CAW).

502. Seam welding is ----- process

(a) multi spot welding
(b) arc welding
(c) thermit welding
(d) continuous spot welding

GWSSB AAE, 27.12.2015

Ans. (d) : Seam welding is continuous spot welding

503. In a butt welding process using arc welding, the arc power is found to be 2.5 KVA. The process is used to weld two steel plates each of 3 mm thickness as shown in the figure. Determine maximum possible welding speed. It is assumed that the metal transfer is a short circuit type and the arc is on for 85% of the total time. Take $\alpha_{steel} = 1.2 \times 10^{-5}$, m^2/s, $k_{steel} = 43.6$ W/m°C, melting point of steel = 1530°C and ambient temperature = 30°C

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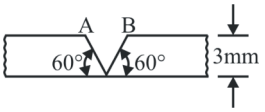


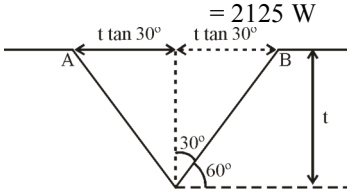
(a) 0.035 m/s (b) 0.0146 m/s
(c) 0.02 m/s (d) 0.0156 m/s

ISRO Scientist/Engineer 11.10.2015

Ans. (b) : We know that:
$\Rightarrow$ the rate of heat input is,
$Q = C.VI$ Where, C = constant, I = current
V = voltage
and
then rate of heat input in term of velocity is-

$$= 8\, k\theta_m\, t \left(\frac{1}{5}+\frac{VW}{4\alpha}\right)$$

Where,
k = conductivity , t = thickness of plate
θ_m = temperature difference, w = width of plate
V = velocity
Given,
C = 0.85 = 85% , t = 3 mm
$VI = 2.5 \times 10^3$, $\alpha_{steel} = 1.2 \times 10^{-5}\ m^2/s$
$k_{steel} = 43.6\ W/m^oC$ $\theta_m = 1530 - 30 = 1500\ ^oC$
therefore, the rate of heat input $= 0.85 \times 2.5 \times 10^3$
$= 2125\ W$



Width of plate
$W = AB = 2t \tan 30^o$

$$= 2\times3\times\frac{10^{-3}}{\sqrt{3}} = 2\sqrt{3}\times10^{-3}\,m$$

Rate of heat input $Q = 8k\theta_m t\left(\frac{1}{5}+\frac{VW}{4\alpha}\right)$

$$2125 = 8\times43.6\times1500\times3\times10^{-3}\times\left[\frac{1}{5}+\frac{2\sqrt{3}\times10^{-3}\times V}{4\times1.2\times10^{-5}}\right]$$

$$2125 = 1569.6\times\left[\frac{1}{5}+72.17V\right]$$

$1.35 = 0.2 + 72.17\ V$
$1.15 = 72.17\ V$
$V = 0.0159\ m/s$

504. Which of the following resistance welding processes essentially uses wheels as electrodes
(a) spot welding (b) projection welding
(c) seam welding (d) flash butt welding

TSGENCO AE 14.11.2015

Ans. (c) : Seam welding is a resistance welding process essentially uses wheel (rollers) as electrodes. This is used for light pressure and leak proof fuel tanks in automobiles, sheet welded tubes, drums and a small container.

505. Composition of most commonly used soldering alloy is
(a) Tin 60% and Lead 40%
(b) Tin 50% and Lead 50%
(c) Tin 40% and Lead 60%
(d) Tin 35% and Lead 65%

TSGENCO AE 14.11.2015

Ans. (a) : Composition of most commonly used soldering alloy is Tin 60% and Lead 40%.

506. Match List-I with List-II and select the correct answer using the code given below the lists :

	List I		List II
A.	Atomic hydrogen welding	1.	Two pieces are brought together and power supply is switched on
B.	Plasma-arc welding	2.	Nugget is formed at the interface of two plates
C.	Spot welding	3.	Gas is ionized
D.	Flash welding	4.	Inert gas shielded arc welding

Code:

	A	B	C	D		A	B	C	D
(a)	4	3	2	1	(b)	1	3	2	4
(c)	4	2	3	1	(d)	1	2	3	4

UPSC JWM Advt. No.-50/2010

Ans. (a) :
- Atomic hydrogen welding is done, in which inert gas shielded arc welding is used.
- Plasma arc welding, in which gas is get ionized produce high temperature.
- Flash welding is a type of resistance welding in two pieces are brought together & power supply is switched on.
- Spot welding is formed a Nugget at the interface of two plates.

507. The current in resistance welding can be regulated by
(a) varying input supply
(b) changing primary turns of transformer
(c) changing secondary turns of transformer
(d) None of the above

TSPSC AEE 2015

Ans : (b) The current in resistance welding can be regulated by changing primary turns of transformer.
Electric Resistance Welding:-
It is a type of pressure welding. It is used for joining pieces of sheet metal or wire. the welding heat is obtained at the location of the desired weld by the electrical resistance through the metal pieces to a relatively short duration, low voltage (from 6 to 10 volts only) high ampere (varying from 60 to 4000 amperes) electric current.

508. In arc welding, the temperature of heat produced by the electric arc is the order of
(a) 3000°C to 4000°C (b) 4000°C to 4000°C
(c) 5000 °C to 6000°C (d) 6000 °C to 7000°C

TSPSC AEE 2015

Ans : (d) In arc welding the temperature of heat produced by the electric arc, is the order of 6000°C- 7000°C.

Welding process	Temperature
arc welding	6000-7000°C
Metal arc welding	2400°C-2700°C
oxy-acetylene flame	3200°C
oxy-hydrogen flame	2400°C

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509. For Arc welding
(a) Alternating current with high frequency is used.
(b) Alternating current with low frequency is used
(c) Direct current is used.
(d) All the above

TSPSC AEE 2015

Ans : (d) For arc welding:-
(i) A.C with high frequency is used
(ii) A.C with low frequency is used
(iii) Direct current is used.

510. The acetylene cylinder is usually painted with
(a) black colour (b) white colour
(c) marron colour (d) yellow colour

TSPSC AEE 2015

Ans : (c) (i) The Acetylene cylinder is usually painted with maroon colour
(ii) The oxygen cylinder is usually painted with black colour
(iii) Acetylene gas is stored in cylinder in liquid form.

511. Which statement is correct in coated electrode arc welding process
(a) Both rod and flux melts simultaneously
(b) Coating melts first then rod material
(c) Rod melts first then coating material
(d) Rod only melts and material coating does not melt

TSPSC AEE 2015

Ans. (c) : In coated electrode arc welding process, rod melts first then coating material.

512. The temperature of oxidized flame in gas welding is
(a) 2900^oC (b) 3200^oC
(c) 3480^oC (d) 3840^oC

TSPSC AEE 2015

Ans. (c) : 3480^0C

513. Filler metal is used in
(a) seam welding (b) spot welding
(c) projection welding (d) gas welding

Assam PSC CCE Pre 2015

Ans. (d) : Gas welding– It is a type of fusion welding, in which the heat for welding is obtained by the combustion of a fuel gas. The most widely used gas combination for producing a hot flame for welding metals is oxygen and acetylene (C_2H_2). The approximate flame temperature produced by oxy-acetylene flame is 3200^0C.

514. Two 8 mm thick plates are placed 5 mm apart and welded by a butt joint. Welding is carried out at 20 V and speed of 5 mm/sec. Heat transfer efficiency is 0.8. If the heat required to melt steel is 10 J/mm^3 and melting efficiency is 0.625, the weld current in ampere will be -
(a) 100 (b) 200
(c) 300 (d) 400

NPCIL ET, 2015

Ans. (b) : Given, t = 8 mm
gap = 5 mm, height = x
V = 20 volts, velocity = 5 mm/sec.
η_{HT} = 0.8, Welding rate (H) = 10 J/mm, $\eta_{melting}$ = 0.625, I = ?
Volume of weld pool = 8 × 5
Heat required = Volume of weld pool × Welding rate
= Area × Velocity × Welding rate
= 8 × 5 × 5 × 10
Heat supplied = $\eta_{HT} \times VI$

$$\eta_{melting} = \frac{\text{Heat required}}{\text{Heat supplied}} = \frac{8\times5\times5\times10}{\eta_{HT}\times VI}$$

$$0.625 = \frac{8\times5\times5\times10}{0.8\times20\times I}$$

I = 200 A

515. Selection of incorrect parameters such as travel speed and arc voltage cause -
(a) Undercut
(b) Porosity
(c) Cracks
(d) Incomplete penetration

NPCIL ET, 2015

Ans. (d) : Incomplete penetration, lack of input in the weld area. It is due to little heat and this can be corrected by–
(1) Increasing the wire-feed speed and reducing the stick out distance.
(2) Reducing the speed of travelled.
(3) Using proper welding techniques

516. In direct arc furnace a flux containing lime silica sand and flour spar about 20% by weight is charged because of :
(a) Forming a basic oxidizing slag which can remove phosphorous from the molten metal
(b) It helps in rapid melting of the metal
(c) This enables easy initial arcing, which helps in melting the metal charge
(d) Is to have very good control over metal temperature and to a large extent on metal composition

KPSC ADF 2015

Ans. (a) : In direct arc furnace a flux containing lime silica sand and flour spar about 20% by weight is charged because of forming a basic oxidizing slag which can remove phosphorous from the molten metal.

517. In gas welding process (oxy-acetylene) acetylene is prepared by dropping water into calcium carbide or dropping calcium carbide into water. From the following select a correct relation between water and calcium carbide :
(a) $CaC_2 + 2H_2 \rightarrow C_2H_2 + Ca(OH)_2$
(b) $CaC_2 + 2H_3O \rightarrow C_2H_2 + Ca(OH)_2$
(c) $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$
(d) $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_4$

KPSC ADF 2015

Ans. (c) :

518. Spot welding, projection welding and seam welding belong to the category of :
(a) Arc welding (b) Thermit welding
(c) Forge welding (d) Resistance welding

Kerala PSC Poly. Lect. 09.01.2014

Ans. (d) : Spot welding, projection welding and seam welding belong to the category of resistance welding.

519. The temperature developed during a thermit welding process is in the order of :
(a) 1500^0C (b) 2000^0C
(c) 2500^0C (d) 3500^0C

Kerala PSC Poly. Lect. 09.01.2014

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Ans. (c) : The temperature developed during a thermit welding process is in the order of 2500^0C.

520. High speed electron beam of electron beam welding is focused on the weld spot using :
(a) Vacuum lens (b) Inert gas lens
(c) Optical lens (d) Magnetic lens

Kerala PSC Poly. Lect. 09.01.2014
GATE 2003, ESE 2003

Ans. (d) : High speed electron beam of electron beam welding is focused on the weld spot using magnetic lens.

521. Which of the following welding process does not require any consumable?
(a) Gas welding
(b) Electroslag welding
(c) Friction stir welding
(d) Thermit welding

HPPSC Asstt. Prof. 2014

Ans. (c) : Friction stir welding is a solid state welding which does not require any consumable material.

522. The following gas can be used for gas welding:
(a) Hydrogen (b) LPG
(c) Acetylene (d) All of these

HPPSC Asstt. Prof. 2014

Ans. (d) : All of gases like Hydrogen, LPG and Acetylene can be used for welding purpose.

523. In welding, the electrodes are coated with clay like material to :
(a) Control the rate at which the electrode melts
(b) Provide a shiny and clean finish over the weld
(c) Create notches in the surfaces of the joining parts
(d) None of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (a) : In welding the electrodes are coated with clay like materials to control the rate at which electrode melts.

524. Arc welding process that non consumable electrode is
(a) Shielded metal arc welding
(b) Submerged arc welding
(c) Electro slag welding
(d) Atomic hydrogen welding

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d) : Atomic hydrogen welding

525. An example for inorganic adhesive is
(a) Epoxy (b) Acrylic
(c) Sodium silicate (d) Phenolic

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Sodium silicate

526. Weldability is poor for
(a) Low carbon steels
(b) medium carbon steels
(c) High carbon steels
(d) Niobium

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) :

527. Consider the following statements
1. The process is extremely versatile over a wide range of thickness and all welding positions for both ferrous and non-ferrous metals
2. Low quality welds are produced with the problem of slag removal.
3. Process is costly and less portable than manual metal arc welding.
4. High deposition rate
What are the characteristics of gas metal arc welding?
(a) 1, 2 and 3 (b) 1, 2, 3 and 4
(c) 1, 3 and 4 (d) 2, 3 and 4
(e) 3 and 4

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Characteristic of gas metal arc welding:
(i) The ability to join a wide range of material types and thickness.
(ii) Welding can be done in all positions
(iii) No slag removal required.
(iv) High weld quality
(v) Process is costly and less portable than manual metal arc welding.
(vi) Less distortion of work piece and high deposition rate.

528. Consider the following defects
1. cracks 2. porosity
3. solid inclusion 4. Cold shuts
Which of these defects are found during welding?
(a) 1, 3 and 4 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) 1 and 3
(e) 1, 2 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (e) : External welding defects are
(i) Crater
(ii) Cracks
(iii) Spatter and surface porosity
(iv) Incomplete filled groove
(v) Distortion
Internal welding defects are
(i) Blowholes and internal porosity
(ii) Cracks
(iii) Inclusion
(iv) Lack of fusion
(v) Incomplete fusion.

529. In which of the following processes does electrode get consumed?
(a) TIG welding (b) resistance welding
(c) thermit welding (d) arc welding

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : Arc welding is a method of permanently joining two or more metal parts. It consist of a combination of different welding process where in coalescence is produced by heating with an electric arc and with or without the use of filler metal depending upon the base plate thickness. They can use either consumable or non consumable electrodes. The welding is usually protected by some type shielding gas.

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530. Filler metal is not required in the following type of welding process:
(a) Oxy-Oxy acetylene welding
(b) Arc welding
(c) Resistance welding
(d) Oxy-oxy-hydrogen welding

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (c) : Filler metal is not required in the resistance welding.

531. Consumable electrode for manual metal arc welding
(a) tungsten (b) steel
(c) copper (d) cadmium copper

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (b) : Consumable electrode for manual metal arc welding of steel.

532. In gas welding of mild steel using an oxyacetylene flame, the total amount of acetylene consumed was 10 litre. The oxygen consumption from the cylinder is
(a) 5 litre (b) 10 litre
(c) 15 litre (d) 20 litre

CSE Pre 1994

Ans. (b) : In gas welding of mild steel using an oxyacetylene flame, the total amount of acetylene consumed was 10 litre. The oxygen consumption from the cylinder is also 10 litre.

533. In oxidizing flame the inner core attains a temperature of _____ ^{0}(C)
(a) 2100 (b) 2800
(c) 3150 (d) 3500

Haryana PSC Civil Services Pre, 2014

Ans. (d)

Flame temperature		Welding
2800^0C	–	Carburizing
3100^0C	–	Neutral
3500^0C	–	Oxidizing

534. During welding when the weld reduces the cross-section thickness of the base metal it is called _____
(a) lack of penetration (b) spattering
(c) undercutting (d) blow hole

Haryana PSC Civil Services Pre, 2014

Ans. (c) During welding when the weld reduces the cross-section thickness of the base metal it is called undercutting.

535. In arc welding, the temperature of arc is in the range of
(a) 1000°C to 2000°C (b) 2000°C to 4000°C
(c) 4000°C to 6000°C (d) 6000°C to 7000°C

MPPSC State Forest Service Exam, 2014

Ans. (c) : The temperature of the arc is about 4000°C to 6000°C. The melting temperature of iron about 1500°C. Consequently the base material & electrode are heated to a high temperature and fuse together.

536. In MIG welding
(a) No flux is required
(b) Welding speed is high
(c) Non consumable electrode is used
(d) Even unclean surface can be welded

MPPSC State Forest Service Exam, 2014

Ans. (d) : MIG welding is an arc welding process in which a continuous solid wire electrode is fed through a welding gun and into the weld pool joining the two base materials together. A shielding gas also sent through the welding gun & protects the weld pool from contamination. So a uncleaned surfaces can be welded.

537. Arc length in arc welding should be equal to
(a) Half the diameter of electrode rod
(b) Rod diameter
(c) Twice the rod diameter
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (b) : The stable arc required high quality welding can be with an arc length equal to 0.6 to 0.8 of the electrode diameter arc length in arc welding should be equal to rod diameter.

538. Which gas is used for welding steel by GMAW or MIG Process?
(a) CO_2
(b) Argon-Helium Mixtures
(c) Argon-Oxygen Mixtures
(d) Pure Argon Gas

MPSC Dy. Engineer 2014

Ans. (a) : MIG or GMAW Process– It is an arc welding process that uses a continuous solid wire element rod heated and fed into the weld pool from a welding gun.
Shielding Gas- It performs a number of important functions.
(1) Forms the arc plasma.
(2) Stabilizes the arc roots on the material surface.
(3) Ensures smooth transfer of molten droplets from the wire to the weld pool.
General purpose shielding gases for MIG are mixtures of Argon, Oxygen and CO_2 and special gas mixtures may contains helium.
For steel–
(1) CO_2
(2) Argon + 2 to 5% Oxygen
(3) Argon + 5 to 25% CO_2
Non- Ferrous.
(1) Argon.
(2) Argon /helium

539. The metal particles expelled during fusion welding that do not form a part of the weld is known as:
(a) Spatter (b) Size of weld
(c) Weld crack (d) Weld tab

MPSC Dy. Engineer 2014

Ans. (a) : The metal particles expelled during fusion welding that do not form a part of the weld is known as spatter.
Weld spatter due to excessive amount of heat input and lack of welding speed.

540. The principle of applying heat and pressure is widely used in
(a) spot welding (b) seam welding
(c) projection welding (d) all of the above

APPSC Poly. Lect. 2013

Ans. (d) : The principle of applying heat and pressure is widely used in, resistance welding is a type of hot pressure welding, spot, seam & projection welding are some examples.

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541. Pick the wrong statement about AC and DC power sources for arc welding:
(a) The problem of magnetic blow is greatly reduced with AC power source.
(b) Arc stability is higher with DC power source.
(c) The DC power source provides high efficiency.
(d) AC power sources are considerably less expensive.

RPSC AEN Pre-2013

Ans. (c) : The D.C. power source provide high efficiency is the wrong statement.

542. In gas welding, which gases are not used:
(a) $O_2 + C_2H_2$ (b) $N_2 + H_2$
(c) $O_2 + H_2$ (d) $CO + H_2$

SJVN ET 2013

Ans. (b) : In gas welding (N_2+H_2) gases are not used.

543. Which of the carbon steel has maximum weldability:
(a) 0.15% Carbon steel (b) 0.30% Carbon steel
(c) 0.50% Carbon steel (d) 0.75% Carbon steel

SJVN ET 2013

Ans. (a) : 0.15% carbon steel has maximum weldability.

544. The amount of current necessary in resistance welding is of the order of:
(a) 1 - 2 KVA/cm^2 (b) 2.5 - 4.0 KVA/cm^2
(c) 4.5 - 6.2 KVA/cm^2 (d) 6.5 - 7.9 KVA/cm^2

SJVN ET 2013

Ans. (c) : The amount of current necessary in resistance welding is of the order of 4.5 – 6.2 KVA/cm^2.

545. Under cuts in weldments are caused due to :
(a) Low welding current
(b) Excessive welding current
(c) Wrong selection of weld rods
(d) Wrong flux

SJVN ET 2013

Ans. (b) : Under cuts in weldments are caused due to excessive welding current.

546. Magnesium sheets can be easily welded by using :
(a) tungsten inert gas welding
(b) metal inert gas welding
(c) electro slag welding
(d) none of these

PSPCL AE, 2012

Ans. (a) : Magnesium sheets can be easily welded by using TIG welding to prevent oxidation.

547. In resistance welding, which is a correct statement :
(1) heat energy is supplied and mechanical pressure is applied in making
(2) if zinc coating is present in electrode it will erode and pit the surface, resulting in decrease in weld quality
(a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

PSPCL AE, 2012

Ans. (c) : Zinc coating in electrode causes erosion and pitting the Furnace thus decreases the weld quality.

Resistance welding is the joining process of metal by applying pressure and passing current for a length of time through the metal area which is to be jointed.

548. Poor fusion in a welded joint is due to:
(a) High welding speed
(b) Dirty metal surface
(c) Improper current
(d) Lack of flux

APGENCO AE 2012

Ans. (c) : Poor fusion in a welded joint is due to improper current. Incomplete fusion causes –
– low heat input
–surface contamination
– electrode angle is incorrect
– the electrode diameter is incorrect for the material thickness
– Travel speed is too fast.

549. Carburising flame is used to weld metals like
(a) Stainless steel (b) Low carbon steel
(c) Copper and Brass (d) High carbon steel

APPSC AEE 2012

Ans. (d) : Carburizing flame– It is used in low alloy steel steel, non-ferrous metals that do not tend to absorb carbon, High carbon steel. this flame is used for welding aluminium non-ferrous metals like monel metal, nickel etc.

550. In arc welding, penetration is minimum for
(a) DCSP (b) DCRP
(c) AC (d) DCEN

ISRO Scientist/Engineer 2011

Ans. (b) : In arc welding:
Penetration is minimum for D.C. reverse polarity and DCRP is used to thin plate because in case of reverse polarity the electrons are transferred from the plate to the electrode and hence more heat (2/3 part of heat) generation takes place to electrode.

551. Match list-I (welding effect) with list-II (causes) and select the correct answer using the codes given below the lists :

List-I (Welding effect)	**List-II (Causes)**
A. Spatter	**1. Damp electrodes**
B. Distortion	**2. Arc blow**
C. Slag inclusion	**3. Improper cleaning in multipass Welding**
D. Porosity	**4. Poor joint selection**

	A	**B**	**C**	**D**
(a)	2	4	3	1
(b)	4	2	1	3
(c)	2	4	1	3
(d)	4	2	3	1

ISRO Scientist/Engineer 2011

Ans. (a) : Welding design and defect:

Welding effect	**Causes**
A. Spatter	1. Arc blow
B. Distortion	2. Poor joint selection
C. Slag inclusion	3. Improper cleaning in multipass Welding
D. Porosity	4. Excessive H_2, O_2, N_2 in the welding atmosphere or Damp electrodes

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552. **Assertion (A) :** Bead is the metal added during single pass of welding.
Reason (R) : Bead material is same as base metal.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : Bead is the metal added during single pass of welding form a joint between the material. In most of the cases, uses, bead material is same as base metal due to good strength.

553. **Assertion (A) :** Spatter is one of the welding defects.
Reason (R) : In submerged arc welding process, there is no spatter of molten metal.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : Spatter is a type of welding defects, in submerged arc welding, thick layer of flux. Completely lowers the molten metal thus preventing spatter & sparks on the sides of workpiece.

554. **Match List I with List II and select the correct answer using the code given below the List :**

	List I (Basic form of weld)		**List II (weld symbol)**
A.	Fillet	1.	
B.	Square butt	2.	
C.	Spot weld	3.	
D.	Seam weld	4.	

Code:

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	2	3	1
(c)	1	3	2	4	(d)	4	3	2	1

UPSC JWM Advt. No.-50/2010

Ans. (a) : As we know,
→ Shows the symbol for fillet weld.
→ Used to show the square weld.
→ Used to show the spot weld.
→ Used to show the seam weld.

555. **Which is not correct statement about the function of flux in brazing**
(a) To avoid thermal distortion and cracking
(b) To dissolve surface oxide coatings which have formed prior to brazing
(c) To prevent oxides from forming during the brazing operation on both the base metal and the brazing material
(d) To facilitate the wetting process by reducing the viscosity of the melt

ISRO Scientist/Engineer 2010

Ans. (a) : The function of flux in brazing is
- To dissolve surface oxide coatings which have formed prior to brazing
- To prevent oxides from forming during the brazing operation on both the base metal and the brazing material
- To facilitate the wetting process by reducing the viscosity of the melt

So, to avoid thermal distortion and cracking is not the correct statement.

556. **Consider the following statement :**
1. They give off inert gases.
2. They stabilize the arc.
3. They control the viscosity of the slag.
Which of the above statements are correct with respect to coating on electrodes?
(a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 3 only

UPSC JWM Adv. No-16/2009

Ans. (a) : With respect to coating on electrodes are –
(1) They give off inert gasses.
(2) They stabilize the arc.
(3) They control the viscosity of the slag.

557. **Consider the following statements :**
1. The electrode is non consumable
2. Both DC and AC power supplies can be used
3. Fluxes are required for thinner sections.
Which of the above statements is/are correct for TIG welding?
(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 3 only

UPSC JWM Adv. No-16/2009

Ans. (b) : For TIG welding –
- The electrode is non consumable.
- Both DC and AC power supplies can be used depending on workpiece material.

558. **To reduce metallurgical damage, it is desirable to melt material with :**
(a) Minimum energy input with high heat density
(b) maximum energy input with low heat density
(c) minimum energy input with low heat density
(d) maximum energy input with high heat density

DRDO Scientists 2009

Ans. (b) :

559. **Which of the following is a correct statement about spot welding, seam welding, friction welding and roll bonding?**
(a) All are fusion welding processes
(b) All are solid-state welding processes
(c) Seam welding is a fusion welding process and other three are solid-state welding processes

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(d) Roll bonding is a fusion welding process and other three are solid-state welding processes

DRDO Scientists 2008

Ans. (b) :

560. The temperature employed in brazing are

(a) below the melting point of the metals to be joined
(b) above the melting point of the metals to be joined
(c) below the melting point of the filter metal
(d) same as those employed in soldering

DRDO Scientists 2008

Ans. (a) : The temperature employed in brazing are below the melting point of the metals to be joined.

561. When two plates are butt welded, residual stresses get generated in the welded structure. The residual stresses are

(a) tensile at the butt joint and compressive away from the joint
(b) compressive at the butt joint and tensile away from the joint
(c) tensile everywhere
(d) compressive in one plate and tensile in other plate

DRDO Scientists 2008

Ans. (a) :

562. Suggest the welding process often used in joining rails.

(a) thermal welding
(b) submerged arc welding
(c) thermit welding
(d) percussion welding

J & K PSC Screening, 2006

Ans. (c) : Thermit welding- In this molten metal is poured for joining the metals.

- Thermit is a mixture of Aluminium powder and Iron oxide.
- Small percentage of manganese is added as alloying to increase strength of joint.
- It used for repair of railway track.

563. Thermit welding differs from other methods of welding in the following way :

(a) It does not use heating
(b) It employs exothermic chemical reaction for developing high temperature
(c) It takes less time
(d) It does not require electrodes

J & K PSC Screening, 2006

Ans. (b) : Thermit welding differs from other methods of welding because it employes exothermic chemical reaction for developing high temperature.
Thermit welding is a mixture of aluminium powder and iron oxide the ratio of about 1 : 3 by weight.

564. In resistance welding, voltage used for heating is

(a) 1 V (b) 10 V
(c) 100 V (d) 1000 V

OPSC Civil Services Pre 2006

Ans. (b) : In welding transformer in resistance welding lowers the line voltage to 1-15 volts and increase the current of 1 to 200 kilo amperes.

565. Which colour of flame represent highest temperature?

(a) blue (b) bright red
(c) dark yellow (d) white

OPSC Civil Services Pre 2006

Ans. (a) : The hottest fires are from oxyacetylene torch (about 3000ºC) that combine oxygen and gas to create pin point blue flames. The inner core of the candle flame is light blue with a temperature of around 1800K (1500ºC)

566. Too high welding current in arc welding would result in

(a) Excessive spatter, under cutting along edges, irregular deposits, wasted electrodes
(b) Excessive piling up of weld metal, poor penetration, wasted electrodes
(c) Too small bead, weak weld and wasted electrodes
(d) Excessive piling up of weld metal, overlapping without penetration of edges, wasted electrodes

ISRO Scientist/Engineer 2006

Ans. (a) : Very high welding current in arc welding would result in excessive spatter, under cutting along edges, irregular deposits, wasted electrodes.

567. The melting point of silver alloys used in brazing is in the range of:

(a) 200 to 350 degree C
(b) 600 to 850 degree C
(c) 100 to 250 degree C
(d) 400 to 550 degree C

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Silver brazing, frequently called hard soldering or silver soldering is a low-temperature brazing process with rods having melting points ranging from 600 to 850 degree centigrade. This is considerably lower than that of the copper alloy brazing filler metals.

568. The bonding of a rubber sheet with a metal is done by

(a) Welding
(b) High frequency die electric heating
(c) Induction welding
(d) Adhesive bonding

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : The bonding of a rubber sheet with a metal is done by Adhesive bonding.

569. Which of the following operating is known as 'sweating'?

(a) Tinning two surfaces to be joined, applying flux and heating
(b) Heating two surfaces to be joined, to the high temperature in presence of flux
(c) Heating two surfaces at low temperature
(d) Joining of two surfaces without use of flux

BPSC Poly. Lect. 2016

Ans : (a) Tinning two surfaces to be joined applying flux and heating is known as sweating.

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3. Metal Cutting Processes

570. In a single point turning operation with a cemented carbide and steel combination having a Taylor exponent of 0.25, if the cutting speed is halved, then the tool life will become

(a) half (b) two times
(c) eight times (d) sixteen times

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
UPPSC AE 13.12.2020, Paper-I
MECON MT 2019,
Nagaland PSC (CTSE) 2018, Paper-I,
GATE 2016, ISRO 2013
ISRO Scientist/Engineer 12.05.2013
UKPSC AE 2013 Paper-I, APPSC AEE 2012
UKPSC AE 2007 Paper -I, ESE 1999, CSE Pre 1995

Ans. (d) : As we know, $VT^n = C$, $n = 0.25$

Given, $V_1 = 2V_2 \Rightarrow \frac{V_1}{V_2} = 2$

So, $V_1T_1^n = V_2T_2^n$

$$\frac{T_2}{T_1} = \left[\frac{V_1}{V_2}\right]^{1/n} = \left[\frac{V_1}{V_2}\right]^{\frac{1}{0.25}} = \left[\frac{V_1}{V_2}\right]^4$$

$$\frac{T_2}{T_1} = [2]^4 = 16$$

$T_2 = 16\ T_1$

571. Taylor's tool life equation (with usual notation) is:

(a) $V^nT = C$ (b) $V^n/T = C$
(c) $V/T^n = C$ (d) $VT^n = C$

NLC GET 17.11.2020, Shift-II
TNPSC AE 2018, GPSC ARTO 01.05.2016
HPPSC W.S. POLY. 2016, KPSC AE 2015
Kerala PSC AE 06.08.2015, UPRVUNL AE 2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TRB Poly. Lect., 2012, WBPSC AE, 2007
UKPSC AE 2007 Paper -I

Ans. (d) : Taylor's tool life equation.

$VT^n = C$

where V = cutting speed (m/min.)
T = time (min)
n = exponent of tool life (depends on tool material)
C = machining constant (depends on tool and work material and cutting condition)

value of exponent (*n*)

n = 0.08 to 0.2 For HSS tool
n = 0.1 to 0.15 For cast alloys
n = 0.2 to 0.4 For carbide tool
n = 0.5 to 0.7 For ceramic tool

572. The number of element on tool signature of ASA system are:

(a) 5 (b) 6
(c) 7 (d) 8

CGPSC AE 25.02.2018, CIL MT 26.03.2017
Nagaland CTSE 2016, 2017 Ist Paper
UPRVUNL AE 07.10.2016
MPPSC State Forest Service Exam, 2014
UKPSC AE 2007 Paper -I, J & K PSC Screening, 2006

Ans. (c) : The number of element on tool signature of ASA system are 7.

Example-

10º, 12º, 8º, 6º, 15º, 20º, 3

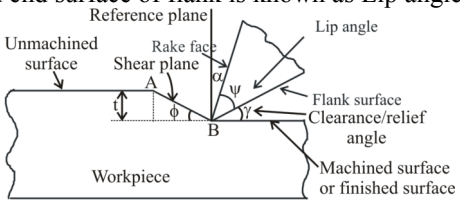
1. Back rake angle (α_b) - 10º
2. Side rake angle (α) - 12º
3. End relief angle (γ_e) - 8º
4. Side relief angle (γ_s) - 6º
5. End edge angle (C_e) - 15º
6. Side cutting edge angle (C_s) - 20º
7. Nose radius (R) - 3 inch

573. The angle between the tool face and the ground end surface of flank is known as

(a) lip angle (b) rake angle
(c) clearance angle (d) nose angle

RPSC ACF & FRO, 26.02.2021
Nagaland PSC (CTSE) 2018, Paper-I
SJVN ET 2013, VIZAG STEEL MT 2011
TNPSC AE, 2008, ESE 1995

Ans. (a) : The angle between the tool face and the ground end surface of flank is known as Lip angle.



The lip angle of a single point tool is usually 60º to 80º.

574. The approximate value of the tool life exponent 'n' of ceramic tool is

(a) 0.03 to 0.08 (b) 0.08 to 0.20
(c) 0.20 to 0.40 (d) 0.40 to 0.60

HPCL Engineer 12.08.2021 Shift-I
RPSC IOF, 2020, CIL MT 27.02.2020
Nagaland PSC (CTSE) 2018, Paper-I
AAI Jr. Executive 29.11.2018
ESE 2008, CSE Pre 1999

Ans. (d) :

Tool Life Exponent (n)		Tool
0.08 to 0.2	–	HSS
0.1 to 0.15	–	Cast alloy
0.2 to 0.4	–	Carbides
0.4 to 0.6	–	Ceramics

575. According to the Ernst and Merchant theory, the relation between the shear angle (□), friction angle (β) and rake angle (α) in single point cutting tool in turning is as follows:

(a) $2□ + \beta + \alpha = 90°$ (b) $2□ + \beta - \alpha = 90°$
(c) $□ + 2\beta - \alpha = 45°$ (d) $2□ + \beta - \alpha = 45°$

CIL MT 26.03.2017
BPSC Poly. Lect. 2016, BPSC AE 2012
UKPSC AE 2007, 2012 Paper -I

Ans. (b) : From the Merchant's analysis

$$2\phi + \beta - \alpha = \frac{\pi}{2} = 90^o$$

Where, ϕ = shear angle
β = friction angle
α = cutting rake angle

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576. Which one of the following materials will have highest cutting speed?

(a) Cast iron (b) Tool steel
(c) Brass (d) Aluminium

Nagaland PSC CTSE 2017, 2016 Paper-I
J&K PSC Civil Services Pre-2013
TRB Poly. Lect. 2012

Ans. (d) : Ductile material will have the highest cutting speed (Aluminium).

577. An orthogonal cutting operation is being carried out under the following conditions : Cutting Speed = 2 m/sec, Depth of cut = 0.5 mm, Chip thickness = 0.6 mm. What is the chip velocity ?

(a) 2 m/sec (b) 2.4 m/sec
(c) 1 m/sec (d) 1.66 m/sec

RPSC ACF & FRO, 26.02.2021
UPPSC AE 13.12.2020, Paper-I
TNPSC AE (Industries) 09.06.2013
UKPSC AE 2012 Paper-I, ESE 2003

Ans. (d) : Given data
Cutting speed = 2 m/sec.
Depth of cut = 0.5 mm
Chip of thickness = 0.6 mm

$$r = \frac{\text{Uncut chip thickness}}{\text{Chip thickness}} = \frac{t}{t_c} = \frac{0.5}{0.6}$$

$\therefore$ velocity of chip $(V_c) = V \times r$

$$= 2 \times \frac{0.5}{0.6} = 1.66 \text{ m/sec.}$$

578. Thrust force will increase with the increase in :

(a) side cutting edge angle
(b) tool nose radius
(c) rake angle
(d) end cutting edge angle

Mizoram PSC (Insp. of Factories) 2019 Paper-III
HPPSC Asstt. Prof. 18.09.2017
APPSC AEE 2012, CSE Pre 1995

Ans. (a) : Side cutting-edge angle (C_s) – It is the angle between the side cutting edge and the side of the tool shank. Angle is measured in a plane parallel to the base. Side cutting edge angle increases, thrust force will also increases.

579. In cutting tools, crater wear develop at :

(a) The principal flank
(b) The tool nose radius
(c) The side shank
(d) The rake surface

Oil India Senior Officer 23.12.2020
HPPSC Lect. (Auto) 23.04.2016
(HPPSC LECT. 2016)
MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (d) : In cutting tools, crater wear develops at the rake surface of tool because at this point tool tip temperature is maximum due to continuous flow of chip.

580. Using Taylor's equation for tool life $VT^{0.5} = C$, what is the percentage increase in tool life when the cutting speed is halved?

(a) 50% (b) 75%
(c) 200% (d) 300%

APPSC Poly Lect. 13.03.2020
Haryana PSC AE (PHED) 05.09.2020, Paper-II
CGPSC Polytechnic Lecturer 2017, CSE Pre 2002

Ans. (d) : Given,
Taylor's equation Life; $VT^{0.5} = C \Rightarrow V_1T_1^{0.5} = V_2T_2^{0.5}$

Given, $V_2 = \frac{V_1}{2}$

$$\Rightarrow V_1T_1^{0.5} = \frac{V_1}{2}(T_2)^{0.5}$$

$$\Rightarrow \frac{T_1}{T_2} = \frac{1}{4} \Rightarrow T_2 = 4T_1$$

$\therefore$ % increases in tool life =

$$\frac{T_2 - T_1}{T_1} \times 100 = \left(\frac{4T_1 - T_1}{T_1}\right) \times 100$$

$$= \frac{3T_1}{T_2} \times 100 = 300\%$$

581. Tool life is most affected by machine

(a) cutting speed
(b) tool geometry
(c) feed and depth
(d) microstructure of material being cut

Assam PSC AE (IWT) 14.03.2021
Assam PSC AE (PHED), 18.10.2020
GWSSB DEE 07.07.2016
UKPSC AE-2013, Paper-I

Ans. (a) : Factors Influencing Tool Life–
1. Cutting speed
2. Physical properties of work piece
3. Area of cut
4. Ratio of feed and depth of cut (f/d)
5. Shape and angles of tools
6. Effect of lubricant

582. In orthogonal cutting of metals the cutting edge is:

(a) perpendicular to the shear plane
(b) perpendicular to the work piece
(c) perpendicular to the direction of tool travel
(d) none of the above

Nagaland PSC CTSE 2017, 2016 Paper-I
TRB Asstt. Prof., 2012, TNPSC AE, 2008
CSE Pre-2008, J & K PSC Screening, 2006

Ans. (c) : In orthogonal cutting of metals cutting edge is perpendicular to tool travel.

583. While machining which of the following improves surface finish :

(a) Increased depth of cut
(b) Increased cutting speed
(c) Increased feed rate
(d) Formation of built up edge

Kerala PSC IOF 19.04.2016
RPSC Lect. (Tech. Edu. Dept.) 2011
J & K PSC Screening, 2006

Ans. (b) : Method of improve surface finish –
(1) Increased cutting speed
(2) Decrease depth of cut
(3) decrease feed rate.

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584. In a tool wear study experiment, it is found that doubling cutting speed reduces the tool life to 1/8 the of the original. The Taylor's tool life index is

(a) 1/2 (b) 1/3
(c) 1/4 (d) 1/8

TSPSC AEE 2017, WBPSC AE, 2017
APGENCO AE, 2017, MPPSC AE 2016
UPPSC AE 12.04.2016 Paper-I
NPCIL ET, 2015, GATE 2004, ESE 2000

Ans. (b) : $V_2 = 2V_1$

$T_2 = \frac{T_1}{8}$, $n = ?$

$V_1T_1^n = V_2T_2^n$

$V_1 \times T_1^n = 2V_1 \times \left(\frac{T_1}{8}\right)^n$

$V_1 \times T_1^n = 2V_1T_1^n \times \left(\frac{1}{8}\right)^n$

$\left(\frac{1}{2}\right)^1 = \left(\frac{1}{2}\right)^{3n}$

$3n = 1$

$n = 1/3$

585. The angle which is found between the face of the cutting tool and the normal to the machined surface at the cutting edge is called :

(a) rake angle (b) relief angle
(c) clearance angle (d) cutting angle

ISRO Scientist/Engineer (RAC) 22.04.2018
Kerala PSC AE 06.08.2015
(KPSC AE. 2015), UKPSC AE 2007 Paper -I

Ans. (a) : Rake angle is between the face of cutting tool and the normal to the machined surface at the cutting edge.

Positive rake –
- Reduce shear.
- low power consumption
- low cutting force
- less tendency of BUE
- low cutting temperature
- Continuous chip

Negative rake –
- Strong tool tip
- longer tool life
- suitable for old machines
- case of high feed
- High productivity

Zero rake –
- Easy to design
- Easy to manufacture
- Easy to re-sharpen
- Easy to analysis.

586. Angle on which strength of the tool depends is:

(a) lip angle (b) rake angle
(c) cutting angle (d) clearance angle

Nagaland PSC (CTSE) 2018, Paper-I
Kerala PSC Poly. Lect., 2017
RPSC VPITI 14.02.2016
HPPSC W.S. Poly. 2016
OPSC Civil Services Pre. 2011

Ans. (a) : Angle on which strength of the tool depends is lip angle.
Note-Official answer given as option (b).

587. Chip breaker is :

(a) an obstruction attached to face of the tool
(b) a hammer used to break the chips
(c) a coolant forced under pressure on the chip
(d) none of these

VIZAG STEEL MT 2015, 2011
J & K PSC Screening, 2006

Ans.(a) : Chip Breakers– Chip breaker is a piece of metal clamped to the rake surface of the tool which bends the chip and breaks it. Chips can also be broken by changing the tool geometry there by controlling the chip flow.
Chip breakers are used in cutting tools to –
(1) increase tool life
(2) provides smooth path for long chips
(3) break the chips into short segments.

588. According to Lee and Shaffer theory, the shear angle ϕ is given by relation (α = rake angle, β = friction angle)

(a) $\phi = \frac{\pi}{2} + \frac{\beta}{2} + \frac{\alpha}{2}$ (b) $\phi = \frac{\pi}{4} - \beta + \alpha$
(c) $\phi = \frac{\pi}{2} - \frac{\beta}{2} + \frac{\alpha}{2}$ (d) $\frac{\phi}{2} = \frac{\pi}{2} + \alpha + \beta$

Punjab PSC SDE 12.02.2017
BPSC Poly. Lect. 2016
VIZAG STEEL MT 2015

Ans. (b) : $\phi = \frac{\pi}{4} - \beta + \alpha$

Merchant theory–

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \frac{\beta}{2}$$

Stabler's theory–

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \beta$$

Lee and Shaffer theory–

$$\phi = \frac{\pi}{4} - \beta + \alpha$$

589. If a cutting tool is designated as 0^0 -10^0-6^0 - 6^0-8^0-75^0-1 mm, what is the side cutting edge angle of the tool?

(a) 0^0 (b) 10^0
(c) 6^0 (d) 75^0

WBPSC AE, 2017
Punjab PSC (Lect.) 06.08.2017
VIZAG STEEL MT 2015

Ans. (d) : Back rake angle -0^o
Side rake angle- 10^o
End relief angle- 6^o
Side relief angle-6^o
End cutting edge angle-8^o
Side cutting edge angle-75^o
Nose radius - 1 mm

590. The velocity of tool along the tool face is known as-

(a) Shear velocity (b) Chip velocity
(c) Cutting velocity (d) Mean velocity

MECON MT 2019
ISRO Scientist/Engineer 11.10.2015
VIZAG STEEL MT 2011

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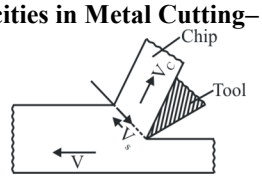
Ans. (b) : Velocities in Metal Cutting–



"The velocity of the chip (V_C) relative to tool is called 'chip velocity'
i.e. velocity of tool along the tool face.

591. Which of the following is a single point cutting tool?
(a) milling cutter (b) parting off tool
(c) hacksaw blade (d) grinding wheel
Mizoram PSC (Insp. of Factories) 2019 Paper-III
RPSC AE 2016, RPSC ACF-2011, ESE 2006

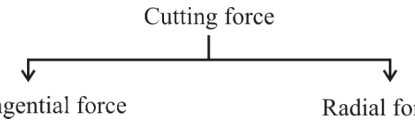
Ans. (b) : Parting off tool is a single point cutting tool while all other is a multipoint cutting tool. e.g., milling, hacksaw, grinding wheel.

592. In an orthogonal cutting, depth of cut is halved and feed rate is doubled. If the chip thickness ratio is unaffected with the changed cutting conditions, the actual chip thickness will be:
(a) Doubled (b) Halved
(c) Unchanged (d) Quadrupled
OPSC AEE 2019 Paper-I
Nagaland PSC (CTSE) 2018, Paper-I
APPSC AEE 2012, CSE Pre 1995

Ans : (a) : Chip thickness ratio = depth of cut/chip thickness
So, as given depth of cut = 1/2,
= 1/2/chip thickness
= 2 × chip thickness
So, double chip thickness

593. The primary tool force is used in calculating the total power consumption in machining is
(a) radial force (b) tangential force
(c) axial force (d) frictional force
WBPSC AE, 2017, MPPSC AE 2016
UJVNL AE 2016, KPSC AE 2015
TSPSC Managers, 2015, ESE 2001, 1995

Ans : (b)



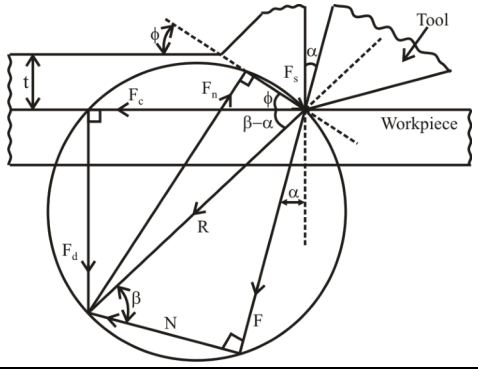
* Tangential force is a cutting operation largest force.
* If radial force increase then chattering start.
* Tangential force is used in calculating the total power consumption in machining.

594. Crater wear occurs due to which one of the following phenomenon?
(a) Adhesion (b) Diffusion
(c) Oxidation (d) All of the above
MPPSC AE 08.11.2015
TSPSC AEE 2015, CSE Pre. 2008
TNPSC AE, 2008, UKPSC AE 2007 Paper -I

Ans. (b) : Crater wear occurs due to diffusion only. Crater is form on the face of the cutting edge by the action of chip particles following over it because of very high temperature. It is mainly due to diffusion.

595. The diameter of the Merchant's Circle diagram represents the magnitude of the
(a) tangential force (b) Frictional force
(c) normal force (d) resultant force
Haryana PSC AE (PHED) 05.09.2020, Paper-II
WBPSC AE, 2017

Ans. (d) : Diameter of merchant's circle diagram represents the resultant force.



596. In an orthogonal machining process, the feed is 0.5 mm/rev and principal cutting edge angle is 90^0. The cutting speed used is 2 m/s . If the chip thickness is found to be 0.75 mm, the chip velocity is
(a) 3 m/s (b) 2.5 m/s
(c) 2 m/s (d) 1.33 m/s
Haryana PSC AE (PHED) 05.09.2020, Paper-II
AAI Jr. Executive 29.11.2018

Ans. (d) : Given,
Feed (f) = 0.5 mm/rev
Cutting speed (v) = 2 m/sec
Chip thickness (t_c) = 0.75 mm
We know that,
For orthogonal cutting,
Thickness feed t = f = 0.5 mm

$$\therefore \text{ Chip thickness ratio } (r_c) = \frac{t}{t_c} = \frac{0.5}{0.75}$$

$$(r_c) = 0.666$$

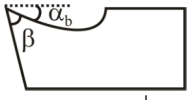
$$\text{And also } (r_c) = \frac{v_c}{v} = \frac{\text{Chip velocity}}{\text{Cutting velocity}}$$

$$0.666 = \frac{v_c}{2}$$

$$\boxed{v_c = 1.332\,m/s}$$

597. On increasing the value of rake angle, the strength of the tool _____
(a) increase (b) decrease
(c) remains constant (d) is unpredictable
APPSC AEE MAINS 2016, PAPER-III
Haryana PSC Civil Services Pre, 2014

Ans. (b) : As Rake angle increases, lip angle decreases means tool become sharp and stress concentration at tip increases which leads to reduction in strength of tool.



Rake angle (α_b)↑→lip angle (β)↓→strength of tool↓

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598. Angle between portion of side flank immediately below major (side) cutting edge and a line perpendicular to the base of the tool is called _____ angle.

(a) end relief (b) side rake
(c) side relief (d) side clearance

MECON MT 2019
Haryana PSC Civil Services Pre, 2014

Ans. (c) : Side Relief Angle – It is the angle between the side flank and the line passing through the tip perpendicular to the base and the angle is measured in a plane perpendicular to the tool axis.
This angle varies in the range of 5^0 to 16^0.

599. Tool life is generally better when

(a) grain size of the metal is large
(b) grain size of the metal is small
(c) hard constituents are present in the microstructure of the tool material
(d) none of the above

Mizoram PSC (Insp. of Factories) 2019 Paper-III
Sikkim PSC (Under Secretary), 2017

Ans. (a) : Tool life generally better when grain size of metal is large because when grain size is large the material will be soft so, that machining can be done smoothly with low wear and tear so tool life is greater when grain size material is large.

600. Continuous chips are formed during machining of

(a) Cast iron (b) Mild steel
(c) Aluminium (d) None of these

Kerala PSC IOF 19.04.2016
MPPSC State Forest Service Exam, 2014

Ans. (b) : Continuous chips are formed during machining of ductile materials.
Mild steel is the appropriate answer.
Discontinuous chips are formed during machining of cast iron.

601. If α is the rake angle of cutting, ϕ is the shear angle and V is the cutting velocity, the velocity of chip sliding along the shear plane is

(a) $\frac{v\cos\alpha}{\cos(\phi-\alpha)}$ (b) $\frac{v\sin\alpha}{\cos(\phi-\alpha)}$
(c) $\frac{v\cos\alpha}{\sin(\phi-\alpha)}$ (d) $\frac{v\sin\alpha}{\sin(\phi-\alpha)}$

APGENCO AE, 2017
CSE Pre 2002, ESE 2001

Ans. (a) :

$$V_C = \frac{V\cos\alpha}{\cos(\phi-\alpha)}$$

$$\frac{V}{\cos(\phi-\alpha)} = \frac{V_c}{\sin\phi} = \frac{V_s}{\cos\alpha}$$

$$\frac{V_s}{\cos\alpha} = \frac{V}{\cos(\phi-\alpha)}$$

$$V_s = \frac{V\cos\alpha}{\cos(\phi-\alpha)}$$

V = cutting velocity
V_C = chip flow velocity
V_S = shear velocity
ϕ = shear angle
α = Rake angle
V_s = sliding velocity

602. In a single point cutting tool, the angle between the face of the tool and a line parallel to base of the tool and measured in a plane (perpendicular) through the side cutting edge is known as :

(a) Side cutting edge angle
(b) End cutting edge angle
(c) Side relief angle
(d) Back rake angle

Punjab PSC SDE 12.02.2017
GPSC ARTO 01.05.2016, ESE 2003

Ans. (d) : Back rake angle– In a single point cutting tool, the angle between the face of the tool and a line parallel to base of the tool and measured in a plane (perpendicular) through the side cutting edge is known as back rake angle.
Side cutting Edge angle- It is the angle which prevent interference material. This angle effects tool life and surface finish.
End cutting Edge Angle-The end cutting edge angle is the amount that the end cutting edge slopes away from the nose of the tool so that it will clear the finished surface.
Side relief Angle – It is the angle between the surface of the flank immediately below the point and a plane at right angles to the centre line of the point of tool.

603. In orthogonal turning of a low carbon steel bar of diameter 150 mm with uncoated carbide tool, the cutting velocity is 90 m/min. The feed is 0.24 mm/ rev and the depth of cut is 2 mm. The chip thickness obtained is 0.48 mm. If the orthogonal rake angle is zero and cutting edge angle is 90º, the shear angle is degree is:

(a) 20.56 (b) 26.56
(c) 30.56 (d) 36.56

HPPSC ADF 05.03.2019
VIZAG STEEL MT 10.06.2012, GATE 2007

Ans. (b) : Diameter of bar D = 150 mm
Cutting velocity V = 90 m/min
Feed (f) = 0.24 mm/rev
depth of cut (d) = 2mm
chip thickness t_c = 0.48 mm
orthogonal rake angle $\alpha = 0°$

$$\Rightarrow \text{Chip thickness ratio (r)} = \frac{t}{t_c}$$

$$r = \frac{0.24}{0.48} = 0.5$$

$\Rightarrow$Shear angle (ϕ)

$$\tan(\phi) = \frac{r\cos\alpha}{1-r\sin\alpha} = \frac{0.5\cos(0^{0})}{1-0.5\sin(0^{0})} = 0.5$$

$$\phi = \tan^{-1}(0.5)$$

$$\boxed{\phi = 26.56}$$

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604. The negative rake is usually provided on
(a) high carbon steel tools
(b) high speed steel tools
(c) cemented carbide tools
(d) all of these

GPSC Lect. 23.10.2016
GPSC Asstt. Prof. 28.08.2016
MPPSC State Forest Service Exam, 2014

Ans. (c) : Cutting tool with negative rake angle is stronger and used to cut high strength material.
Cemented carbide tools are normally given negative rake angle. It can be used in high speed cutting operation.

605. When the cutting edge of the tool is dull, then during machining
(a) Discontinuous chips are formed
(b) Continuous chips are formed
(c) Continuous chips with built-up edge are formed
(d) No chips are formed.

Oil India Senior Officer 23.12.2020
GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : When cutting edge of the tool is dull, then during machining, continuous chips with built up edge are formed.

606. The basis of slip line field theory in metal cutting is
(a) Merchant theory
(b) Lee and Shaffer theory
(c) Ernst-Merchant theory
(d) None of these

HPPSC ADF 05.03.2019
ISRO Scientist/Engineer 07.05.2017

Ans. (b) : Slip line field theory in metal cutting– "Lee and Shaffer theory".

607. In metal cutting operation shear angle is defined as the angle
(a) Made by the plane of shear with the direction of tool travel
(b) Made by the shear plane with the tool axis
(c) Made by the shear plane with the central plane of work piece
(d) None of the above

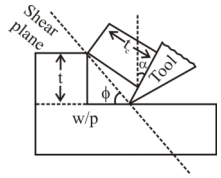
Nagaland PSC CTSE 2017, 2016 Paper-I

Ans. (a) : In metal cutting operations shear angle is defined as the angle made by the plane of shear with the direction of tool travel. The shear angle may be determined by the following relation: $\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$
ϕ = shear angle
r = cutting ratio or chip thickness ratio
α = Rake angle.

608. Which of the following relation is correct for orthogonal cutting? [where, r = chip thickness ratio, α = rake angle, ϕ = shear angle]
(a) $\tan\phi = \frac{\cos\alpha}{1 - r\cos\alpha}$ (b) $\tan\phi = \frac{r\sin\alpha}{1 - r\cos\alpha}$
(c) $\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$ (d) $\tan\phi = \frac{r\sin\alpha}{1 - r\sin\alpha}$
(e) $\tan\phi = \frac{r}{1 - r\sin\alpha}$

CGPSC AE 25.02.2018
GPSC Asstt. Prof. (Prod.) 21.08.2016
UKPSC AE 2007 Paper -I

Ans. (c) : $\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$



$r = \frac{t}{t_c}$ = chip thickness ratio
t = uncut chip thickness
t_c = after cut chip thickness
α = back rake angle

609. The cemented carbide tools are generally poor in
(a) Compression (b) Tension
(c) Shear (d) Torsion

Nagaland PSC CTSE 2017, 2016 Paper-I
MPPSC State Forest Service Exam, 2014

Ans. (c) : cemented carbide tools are generally poor in shear. It is a hard brittle material used extensively as cutting tools material.

610. In oblique cutting of metals, the cutting edge of the tool is :
(a) Perpendicular to the workpiece
(b) Perpendicular to the direction of tool travel
(c) Parallel to the direction of tool travel
(d) Inclined at an angle less than 90^0 to the direction of tool travel

HPSC W.S. POLY. 2016
OPSC Civil Services Pre. 2011

Ans. (d) : Inclined at an angle less than 90^0 to the direction of tool travel.

611. Which of the following statements is correct for orthogonal cutting system?
(a) The cutting edge of the tool is perpendicular to the direction of tool travel.
(b) The cutting edge clears the width of the work piece on either ends.
(c) The chip flows over the tool face and the direction of the chip flow velocity is normal to the cutting edge
(d) All of the above

Kerala PSC IOF 19.04.2016
JPSC AE 2013, Paper-V

Ans : (d) : Orthogonal cutting system
- The cutting edge of the tool is perpendicular to the direction of tool travel.
- The cutting edge clears the width of the work piece on either ends.
- The chip flows over the tool face and the direction of the chip flow velocity is normal to the cutting edge.

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612. The tool life is affected by :

(a) Depth of cut (b) Cutting speed
(c) Feed (d) All of these

OPSC Civil Services Pre. 2011
UKPSC AE 2012, 2007 Paper-I

Ans. (d) : The tool life is affected by,
(a) Depth of cut
(b) Cutting speed
(c) Feed

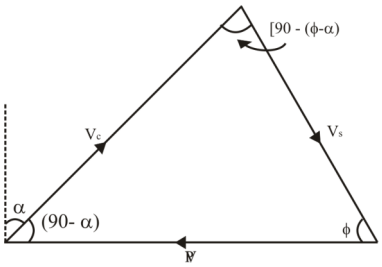
613. The chip thickness ratio 'r' is given by where ϕ = shear plane angle, and α = rake angle

(a) $\frac{\cos\phi}{\sin(\phi-\alpha)}$ (b) $\frac{\sin(\phi-\alpha)}{\cos\phi}$

(c) $\frac{\cos(\phi-\alpha)}{\sin\alpha}$ (d) $\frac{\sin\phi}{\cos(\phi-\alpha)}$

UPPSC AE 12.04.2016 Paper-I, CSE Pre.2000

Ans : (d)



From this figure we get,

$$\frac{V}{\cos(\phi-\alpha)}=\frac{V_c}{\sin\phi}=\frac{V_s}{\cos\alpha}$$

we know that chip thickness ratio (r) is given as

$$r=\frac{t}{t_c}=\frac{l_c}{l}=\frac{V_c}{V}=\frac{\sin\phi}{\cos(\phi-\alpha)}$$

$$\boxed{r=\frac{\sin\phi}{\cos(\phi-\alpha)}}$$

614. Crater wear occurs mainly on the

(a) nose part, front relief face and side relief face of the cutting tool
(b) cutting edge only
(c) front face only
(d) face of the cutting tool at a short distance from the cutting edge only

RPSC Vice Principal ITI 2018
Nagaland PSC (CTSE) 2018, Paper-I
HPPSC Asstt. Prof. 18.09.2017
UPPSC AE 12.04.2016 Paper-I

Ans. (d) :
Crater Wear - occur mainly on the face of the cutting tool at a short distance from the cutting edge only.
Flank Wear – Occurs mainly on the nose part, front relief face and side relief face of the cutting tool.

615. In an orthogonal cutting operation, the chip thickness and the uncut thickness are equal 0.45mm each. If the tool rake angle is 0°, the shear plane angle is :-

(a) 18º (b) 30º
(c) 45º (d) 60º

GPSC ARTO 01.05.2016
UKPSC AE-2013, Paper-I, GATE 1998

Ans. (c) : Given as,
$t = 0.45$ mm
$t_c = 0.45$ mm
$\alpha = 0^o$
then

$$r=\frac{t}{t_c}=\frac{0.45}{0.45}=1$$

We know that

$$\tan\phi=\frac{r\cos\alpha}{1-r\sin\alpha}$$

$$\tan\phi=\frac{1\times\cos 0^o}{1-1\times\sin 0^o}$$

$$\tan\phi=1$$

$$\phi=\tan^{-1}(1)$$

$$\boxed{\phi=45^o}$$

616. Friction at the tool-chip interface can be reduced by

(a) decreasing the rake angle
(b) increasing the depth of cut
(c) decreasing the cutting speed
(d) increasing the cutting speed

GPSC Executive Engineer 23.12.2018
VIZAG STEEL MT 10.06.2012, GATE 2009

Ans. (d) : As the speed increases the time of surface contact between tool and chip reduces and so, the friction will also reduces.

617. The heat generated in metal cutting can conveniently be determined by:

(a) Installing thermocouple on the job
(b) Installing thermocouple on the tool
(c) Calorimetric set up
(d) Using radiation pyrometer

HPPSC Asstt. Prof. 18.09.2017, CSE Pre 2003

Ans. (c) : The heat generated in metal cutting can conveniently be determined by calorimetric set up.

618. Discontinuous chips during machining forms when :

(a) Work is brittle, speed is low
(b) Work is brittle, speed is high
(c) Work is ductile, speed is low
(d) Work is ductile, speed is high

UPRVUNL AE 05.07.2021
TSPSC Managers, 2015

Ans. (a) : Discontinuous chips during machining forms when work is brittle, speed is low.

619. The lip angle of a single point tool is usually

(a) 60º to 80º (b) 40º to 60º
(c) 20º to 40º (d) None of these

APPSC Poly Lect. Automobile Engg., 2020
RPSC VPITI 14.02.2016

Ans. (a) : Lip angle/wedge angle/knife angle (ψ) is the angle between Rake face and flank of the tool.
Lip angle of single point cutting tool is 60º to 80º
Rake angle 10°-15°
Side cutting edge angle 15°-30°

620. Cast iron during machining produces

(a) Discontinuous chips
(b) Continuous chips

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(c) None of these
(d) Continuous chips with built up edge

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
Haryana PSC AE (PHED) 05.09.2020, Paper-II
WBPSC AE, 2017, RPSC VPITI 14.02.2016

Ans. (a) : Cast iron is a brittle material, discontinuous chip forms. Deformed material gets fractured very easily and thus, the chip produced is in the form of discontinuous segment.
It this deformed material instead of flowing continuously gets ruptured periodically.

621. Chip ratio is defined as ratio of
(a) Chip thickness to uncut thickness
(b) Uncut thickness to chip thickness
(c) Length of uncut material to length of chip
(d) None of these

RPSC IOF, 2020
HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : Chip ratio is defined as ratio of uncut thickness to chip thickness

Chip thickness ratio (r) $= \frac{t_0}{t_c} = \frac{\sin\phi}{\cos(\phi - \alpha)}$

Where, ϕ = Shear plane angle
α = Back rake angle

622. Merchant theory relationship between shear angle ϕ, friction angle β and the cutting rake angle α and the machining constant C for the work material is:
(a) $2\beta + \alpha - \phi = C$ (b) $2\alpha + \beta - \phi = C$
(c) $2\phi + \alpha - \beta = C$ (d) $2\phi + \beta - \alpha = C$

GPSC ARTO Pre 30.12.2018
GPSC Asstt. Prof. 28.08.2016, ESE 2010

Ans. (d) : Merchant Circle Theory– It was confirmed by the observation of the curvature of the chips. The increased curvature correspond to a decrease in shear stress, decrease in shear area, thus an increase in cutting force.

Shear angle (ϕ), $\phi = 45 + \frac{\alpha}{2} - \frac{\beta}{2}$

$$\boxed{2\phi + \beta - \alpha = C}$$

623. Tool signature is:
(a) Giving orthogonal views of the tool
(b) The plan of tool
(c) The complete specification of tool
(d) There is nothing like tool signature

ESE 2018, GPSC Asstt. Prof. 28.08.2016

Ans. (c) : Tool signature specifies the tool, so it is also called tool signature, Orthogonal Rake System (ORS) clearly explained in short with inclination angle, clearance angle, rake angle, approach angle.
Note-Option (a) is given by commission.

624. A carbide tool (having *n* = 0.25) with mild steel work piece was found to give life of 1 hour and 21 minutes while cutting at 60 rpm. The value of C in Taylor's equation $VT^n = C$ would be equal to :
(a) 180 (b) 150
(c) 100 (d) 200

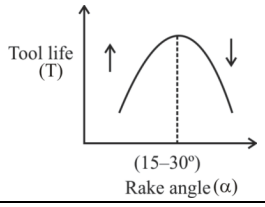
CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (a) : Given,
n = 0.25
Tool life, T = 1 hour 21 minutes
T = 81 minutes
Taylor's equation, $VT^n = C$
$(60)(81)^{0.25} = C$
$\boxed{C = 180}$

625. With increase in rake angle, tool life -
(a) Increases
(b) Decreases
(c) First increases then decreases
(d) Remains constant

NPCIL ET, 2015, DRDO Scientists 2009

Ans. (c) :



626. In metal cutting operation, the maximum heat (i.e., 80-85%) is generated in
(a) the shear zone
(b) the chip-tool interface zone
(c) the tool-work interface zone
(d) None of the above

Assam PSC AE (PHED), 18.10.2020
APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : In metal cutting operation the maximum temperature occurs in shear zone.
Shear zone is the contact zone between the chip and the tool. In the entire heat generated 80% of the total heat is carried away by the chip about 10% is transferred to the tool and remaining 10% is retained by the workpiece.

627. Force and angles acting during metal machining are related by _____diagram.
(a) Mohr's Circle (b) Merchant Circle
(c) Maxwell's Circle (d) None of the above

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : Force and angles acting during metal machining are related by Merchant circle diagram.

628. The values of principal cutting edge angle ϕ of the turning tool whose geometry is designated as :10°, 10°, 8°, 6°,15°,30°, 0 (inch).
(a) 6° (b) 15°
(c) 30° (d) 60°

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) : Side cutting edge angle (C_s) = 30°
Principal cutting edge angle = $90 - C_s$
$(\lambda) = 90 - 30$
$= 60°$

629. While orthogonal turning at a cutting speed of 100 m/min, the tool life is 10 minutes, while the tool life increases to 40 minutes if the cutting speed is reduced to 50 m/ min. The value of index (n) in the Taylor's tool life equation is
(a) 2 (b) 0.5
(c) 5 (d) 0.2

PPSC Asstt. Municipal Engineer 15.06.2021

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Ans. (b) : $V_1T_1^n = V_2T_2^n$

$100 \times 10^n = 50 \times 40^n$

$\frac{100}{50} = \left(\frac{40}{10}\right)^n$

$2^1 = (2)^{2n}$

$2n = 1$

$\boxed{n = \frac{1}{2} = 0.5}$

630. In Taylor's tool life equation $VT^n = C$, the constants n and C depend upon

1. Work piece material

2. Tool material

3. Coolant

(a) 1,2 and 3 (b) 1 and 2 only

(c) 2 and 3 only (d) 1 and 3 only

PPSC Asstt. Municipal Engineer 15.06.2021

ESE 2012

Ans. (a) :

631. In metal cutting operation, the approximate ration of heat distributed among chip, tool and work, in that order is

(a) 80 : 10 : 10 (b) 33 : 33 : 33

(c) 20 :60 : 10 (d) 10 : 10 : 80

PPSC Asstt. Municipal Engineer 15.06.2021

ESE 1998

Ans. (a) :

632. The chip thickness along shear plane is

(a) equal to uncut thickness.

(b) equal to cut chip thickness.

(c) greater than uncut thickness and cut chip thickness.

(d) not related to chip thickness before or after cut.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : The chip thickness along shear plane is greater than uncut thickness and cut chip thickness.

633. With reference to single point cutting tool, the angle between the end cutting edge and a line perpendicular to the shank of tool is called

(a) End cutting edge angle.

(b) Side cutting edge angle.

(c) End relief angel.

(d) side relief angel.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : End cutting edge angle

634. Tool life in turning will decrease by

(a) depth of cut. (b) feed.

(c) cutting velocity. (d) tool rake angle.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) : Tool life in turning will decrease by cutting velocity.

635. Consider the following statements regarding desirable properties of cutting fluid:

1. It should get oxidised when left in air.
2. It should react with the materials of machine tool parts.
3. It should wet the surface of cutting tool and work piece.

Which of the above statements is/are correct?

(a) 2 only (b) 3 only

(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (b) :

Cutting fluids are used for decreasing power requirement and increasing heat dissipation. The desirable properties of cutting find are

- It should not get oxidised when left in air
- It should not react with the material of machine tools parts
- It should easily available
- It should have low viscosity
- It should wet the surface of cutting tool and workpiece for better heat dissipation

636. In metal cutting terminology a layer of compressed metal from the material being cut, which adheres to and piles up on the face of the cutting tool edge during a machining operation, is called as :

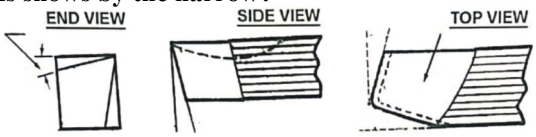
(a) The chip-tool interface (b) The rupture

(c) A built-up edge (d) Shear angle

CGPSC AE 15.01.2021

Ans. (c) : In metal cutting terminology, a layer of compressed metal from the material being cut which adheres to and piles on the face of the cutting tool edge during a machining operation is called as a built-up edge.

637. In the figure given below, what angle or radius is shows by the narrow?



(a) Side cutting angle (b) Side relief angle

(c) End relief angle (d) Side rake angle

CGPSC AE 15.01.2021

Ans. (d) : Side Rake Angle

638. The velocity relations in orthogonal cutting are:

(a) $V_c = \frac{V\cos\alpha}{\cos(\phi-\alpha)}, V_s = \frac{V\sin\phi}{\cos(\phi-\alpha)}$

(b) $V_c = \frac{V\cos\phi}{\cos(\phi-\alpha)}, V_s = \frac{V\sin\alpha}{\cos(\phi-\alpha)}$

(c) $V_c = \frac{V\sin\alpha}{\cos(\phi-\alpha)}, V_s = \frac{V\cos\phi}{\cos(\phi-\alpha)}$

(d) $V_c = \frac{V\sin\phi}{\cos(\phi-\alpha)}, V_s = \frac{V\cos\alpha}{\cos(\phi-\alpha)}$

where V_c = velocity with which chip moves over the rake face of the cutting tool, V_s = velocity with which metal of the work piece shears along the shear plane, V = velocity of tool relative to the work piece, ϕ = shear angle and α = tool rake angle.

CGPSC AE 15.01.2021

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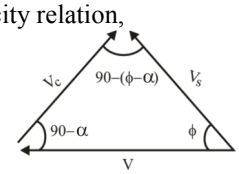
Ans. (d) : Velocity relation,



$$\frac{V}{\sin[90-(\phi-\alpha)]}=\frac{V_C}{\sin\phi}=\frac{V_S}{\cos\alpha}$$

$$\frac{V}{\cos(\phi-\alpha)]}=\frac{V_C}{\sin\phi}=\frac{V_S}{\cos\alpha}$$

$$\boxed{V_C=\frac{V\sin\phi}{\cos(\phi-\alpha)}} \quad , \quad \boxed{V_S=\frac{V\cos\alpha}{\cos(\phi-\alpha)}}$$

639. In a single point cutting tool, the point where the side cutting edge and end cutting edge intersects is called _____.

(a) neck (b) nose
(c) heel (d) face

VIZAG Steel MT 24.01.2021

Ans. (b) : In a single point cutting tool the point where the side cutting edge and end cutting edge intersect is called nose.

640. Better surface finish is obtained with a large rake angle because

(a) the area of shear plane decreases resulting in the decrease in shear force and cutting force
(b) the tool becomes thinner and the cutting force is reduced
(c) less heat is accumulated in the cutting zone
(d) the friction between the chip and tool is less

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) : For better surface finish large rake angle is used because as rake angle increase shear angle also increase which reduces area of shear plane due to which cutting force and shear force decreases.

641. The main cutting force acting on a tool during the turning (orthogonal cutting) operation of a metal is 400 N. The turning was performed using 2 mm depth of cut and 0.1 mm/rev feed rate. The specific cutting pressure (in N/mm^2) is _____

(a) 1000 (b) 2000
(c) 20000 (d) 4000

Assam Engg. College AP/Lect. 18.01.2021
GATE 2014

Ans. (b) : 2000
Cutting force (F_c) = 400 N
depth of cut (d) = 2 mm
feed rate (f) = 0.1 mm/rev.

$$\text{Specific cutting pressure} = \frac{F_c}{wt_1}=\frac{F_c}{d\times f}=\frac{400}{2\times 0.1}$$

$$=2000\frac{N}{mm^2}$$

642. The tool life equation is given by $VT^{0.5}$ = const. If tool life is reduced to one-fourth of its initial life, the percentage increase in cutting velocity is-

(a) 100% (b) 200%
(c) 125% (d) 80%

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Given,
$VT^{0.5}$ = constant
$V_1T_1^{0.5}$ = constant
$V_2T_2^{0.5}$ = constant
given as $T_2 = T_1/4$

$$V_1T_1^{0.5}=V_2\left(\frac{T_1}{4}\right)^{0.5}$$

$$V_1=V_2\times\left(\frac{1}{4}\right)^{0.5}$$

$$\frac{V_2}{V_1}=2$$

$$\text{Percentage increase in cutting velocity}=\frac{V_2-V_1}{V_1}\times 100$$

$$=\frac{2V_1-V_1}{V_1}\times 100 \ =100\%$$

643. The orientation of cutting edge in cutting line for an ASA system of cutting is termed as .....

(a) cutting edge nose radius
(b) cutting edge inclination angle
(c) cutting edge relief angle
(d) cutting edge rake angle

UPRVUNL AE 05.07.2021

Ans. (b) : The orientation of cutting edge in cutting line for an ASA system of cutting is termed as cutting edge inclination angle.

644. _____ affects the position of the cutting edge with respect to the workpiece and alters dimensional accuracy of machined parts

(a) Thermal wear (b) Flank wear
(c) Notch wear (d) Crater wear

UPRVUNL AE 05.07.2021

Ans. (b) : Flank wear affects the position of the cutting edge with respect to the workpiece and alters dimensional accuracy of machined parts.

645. Which of the following processes employees a single-point cutting tools

(a) Hobbing (b) Shaping
(c) Reaming (d) Broaching

UPRVUNL AE 05.07.2021

Ans. (b) : Shaping

646. Which type of wear occurs locally in an area where the main cutting edge contacts

(a) thermal wear (b) Notch wear
(c) Crater wear (d) Flank wear

UPRVUNL AE 05.07.2021

Ans. (b) : Notch wear

647. which of the following is NOT true about the tool material used in cutting an machining?

(a) it shall possess a high coefficient of frictions.
(b) it shall possess high hot hardness.
(c) it shall possess a high thermal conductivity
(d) It shall possess a high wear resistance

UPRVUNL AE 05.07.2021

Ans. (a) : It shall possess a high coefficient of frictions

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648. _____ rake angles are generally used for machining of ductile and tough materials.

(a) zero (b) negative
(c) positive (d) exponential

UPRVUNL AE 05.07.2021

Ans. (c) : Positive rake angles are generally used for machining of ductile and tough materials.

649. Crater wear starts at some distance from the tool tip because.

(a) cutting fluid cannot penetrate that region.
(b) stress on rake face is maximum at the region.
(c) tool strength is minimum at that region.
(d) tool temperature is maximum at that region.

RPSC ACF & FRO, 26.02.2021
Mizoram PSC (Insp. of Factories) 2019 Paper-III
APGENCO AE, 2017
CSE Pre-2007, ESE 2002, 2000, GATE 1989

Ans. (d) : Crater wear is mainly due to the diffusion between the chip and the face of the tool, a short distance from the cutting edge. The crater is formed on the surface of the tool by action of chip particles flowing over it because of very high temperature.

650. Feed rate for a cutting tool having nose radius of 1.8 mm and theoretical roughness of R_n = 5μm is:

(a) 0.36 mm/rev (b) 0.187 mm/rev
(c) 0.036 mm/rev (d) 0.0187 mm/rev

RPSC ACF & FRO, 26.02.2021

Ans. (*) : Given,
Nose radius (r) = 1.8 mm
Roughness $R_n = 5\mu m = 5 \times 10^{-3}$ mm

$$R_n = \frac{f^2}{8r}$$

$$f^2 = 8r \times R_n$$

$$f = \sqrt{8 \times 1.8 \times 5 \times 10^{-3}}$$

$$\boxed{f = 0.268 mm/rev}$$

651. The cutting speed for maximum profit rate should be chosen as

(a) Below the speed for minimum cost
(b) In between the speeds for minimum cost and maximum production rate
(c) Equal to the speed for minimum cost
(d) Higher than the speed for maximum production rate

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
ESE 2000

Ans. (b) : The determination of the maximum profit cutting speed by application of a fundamental economic principle that maximum profit occurs at the production rate, where marginal revenue equals the marginal costs. Pre effect of demand function, feed and lost and time parameters on determination of maximum profit cutting speed are analysed.
So, the maximum profit rate is between the speeds for minimum cost and maximum production rate.

652. In a cutting tool, abrasion and adhesion are primarily responsible for

(a) crater wear
(b) plastic deformation
(c) flank wear
(d) mechanical breakage

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : Wear on the flank face of the tool is called flank wear. It is most important wear that appears on the flank surface parallel to the cutting edge. It is most commonly results from abrasive/adhesive wear of the cutting edge against the machined surface. It is generally result from high temperatures, which affect tool and work material properties.

653. For a single-point cutting tool, the continuous chip with builtup edge is formed due to

(a) stronger adhesion between chips and tool face
(b) low rake angle
(c) large uncut thickness
(d) All of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : In single point cutting of metals a built up edge is an accumulation of material against the rake face, that seizes to the tool tip stronger adhesion between chip and tool face and it is due to low rake angle and large uncut thickness.

654. A good machinability rating would indicate

(a) Long tool life, high power requirement and less machining time
(b) Long tool life, low power requirement and a good surface finish
(c) Short tool life and a good surface finish
(d) Long tool life, high power requirement and a good surface finish

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Machinability index designates the degree of difficulty or easy with which a particular material can be machined. A good machinability rating or index represents long tool life, low power requirement and a good surface finish.

655. Wear ratio of tungsten carbide work is:

(a) 0.5 (b) 1
(c) 2 (d) 3

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) :

656. During turning a metallic rod at a given condition, the tool life was found to increase from 25 min to 50 min, when cutting velocity V_c alone was reduced from 141.4 m/min to 100 m/min. How much will be the life of that tool if machined at 353.5 m/min?

(a) 20 min (b) 15 min
(c) 10 min (d) 4 min

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (d) : $T_1 = 25$ mm, $V_1 = 141.4$ m/min
$T_2 = 50$ min, $V_2 = 100$ m/min
$T_3 = ?$, $V_3 = 353.5$ m/min
From Taylor's tool life equation, n → exponent

$$V_1 T_1^n = V_2 T_2^n$$

$$141.4 \times 25^n = 100 \times 50^n$$

$$1.414 = 2^n$$

$$\boxed{n = 0.5}$$

$$V_2 T_2^n = V_3 T_3^n$$

$$100 \times 50^{0.5} = 353.5 \times T_3^{0.5}$$

$$T_3 = 4 \text{ min}$$

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657. Shaping grey cast iron will produce
(a) Continuous chip with Built Up Edge (BUE)
(b) Continuous chip without Built Up Edge (BUE)
(c) Discontinuous chip of irregular size and shape
(d) Discontinuous chip of regular size and shape

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (c) : Condition for Forming Discontinuous Chip–
(a) of irregular size and shape –
→ Work material 'brittle' (such as grey cast iron).
(b) of regular size and shape –
→ Work material ductile but hard and work hardenable.
- Feed → Large
- Depth of cut → Large
- Tool rake → Negative

658. For same tool-work materials and speed, feed and depth of cut, the average cutting temperature will decrease
(a) with the increase in principal cutting edge angle (ϕ)
(b) with the decrease in principal cutting edge angle (ϕ)
(c) with the increase in auxiliary cutting edge angle (ϕ_1)
(d) with the decrease in auxiliary cutting edge angle (ϕ_1)

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : Principal cutting edge = 90° – side cutting edge angle.
- At its increased value is will have more of its length in which for a given depth of cut.
- At its increased value is produce thinner and wider chip that will distribute the cutting heat (increase tool life).

659. Extreme Pressure Additive (EPA) is mixed with cutting fluid for improving its power of
(a) cooling
(b) lubrication
(c) cleaning of cutting zone
(d) protection of machined surface

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : Extreme Pressure Additive (EPA)–Where high pressure and rubbing action are encountered, hydrodynamic lubrication cannot be maintained, so Extreme Pressure (EP) additive must be added to the chemical components such as boron, phosphorus, sulphur, chlorine or combination of these.
The components are activated by the higher temperature resulting from extreme pressure. As the temperature derivatives such as iron chloride or iron sulphide and forms a solid protective coating.
So, Extreme Pressure Additive (EPA) is mixed with cutting fluid for improving its power of 'lubrication'.

660. If a tool shape is specified as 8-14-6-6-6-15-4, then 15 refers to:
(a) side relief angle
(b) side rake angle
(c) end relief angle
(d) side cutting edge angle

Karnataka PSC AE (WRD) 31.07.2021
NLCIL GET 17.11.2020, Shift-II

Ans. (d) : The seven elements that comprise the signature of a single-point cutting tool are always started in the following order–
Back rake angle
Side rake angle
End relief angle
Side relief angle
End cutting edge angle
Side cutting edge angle
Nose radius
Thus a tool with shaped specified as 8-14-6-6-6-15-4 has 8° back rake angle, 14° side rake angle, 6° end relief angle and 6° side relief angle, 6° end cutting edge angle and 15° side cutting edge angle and 4 mm nose radius.

661. The back rake angle of a cutting tool
(a) determines the profile of tool
(b) prevents rubbing
(c) decides the type of cutting action
(d) controls the chip formation

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : The back rake angle of cutting tool controls the chip formation.

662. Chips in metal cutting are produced due to
(a) Fracture process
(b) Peeling process
(c) Crack formation and Propagation
(d) Shearing and Plastic flow process

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Chip in metal cutting are produced due to shearing and plastic flow process.

663. Nose radius on single point cutting tool is chiefly given for
(a) Increasing tool life
(b) Increasing surface finish
(c) Decreasing cutting force
(d) Decreasing frequency of breakdown

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : Nose radius provide good surface finish and longer tool life.
- A sharp point on the end of tool is highly stressed under heat and pressure, wear out faster and leaves a groove in the path of cut.
- As nose radius is increased there is an improvement in surface finish and allowed cutting speed but too large nose radius will induce chatter.

664. During turning of a metallic rod at a given condition, the tool life was found to decrease from 100 min to 25 min. When cutting speed was increased from 50 m/min. to 100 m/min. How much will be life of tool if machined at 80 m/min.?
(a) 38.06 min. (b) 39.06 min.
(c) 40.06 min. (d) 41.06 min.

ISRO Scientist/Engineer 12.01.2020

Ans. (b) : $T_1 = 100$ min $V_1 = 50$m/min
$T_2 = 25$ min $V_2 = 100$ m/min
$T_3 = ?$ $V_3 = 80$ m/min
Taylor's tool life equation,
$V_1T_1^n = V_2T_2^n$
$50\times100^n = 100\times25^n$

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$$\left(\frac{100}{25}\right)^n = \frac{100}{50}$$

$4^n = 2 \Rightarrow n = 0.5$

$V_1T_1^n = V_3T_3^n$

$50\times100^{0.5} = 80 \times T_3^{0.5}$

$50 \times 10 = 80\times T_3^{0.5}$

$$T_3^{0.5} = \frac{25}{4}$$

$$T_3 = \left(\frac{25}{4}\right)^{1/0.5} = \left(\frac{25}{4}\right)^2 = \frac{625}{16}$$

$T_3 = 39.0625$ min.

665. The secondary shear zone during metal machining lies between :

(a) metal chip and cutting tool
(b) cutting tool and metal workpiece
(c) metal workpiece and metal chip
(d) cutting tool and metal dust

APPSC Poly Lect. 13.03.2020

Ans. (a) The secondary shear zone during metal machining lies between metal chip and cutting tool, this is because the thermal conductivity of the tool is less than the chip.

666. In the metal cutting operation, what will be the shear angle when a cutting tool has positive rake $\alpha = 20^0$. The friction angle is 40^0. (Use Merchant's shear angle theory)

(a) 35^0 (b) 30^0
(c) 90^0 (d) 40^0

APPSC Poly Lect. 13.03.2020

Ans. (a) : According to Merchant's shear angle theory–

$2\phi + \beta - \alpha = 90^0$

Where, ϕ = shear angle = ?

β = friction angle = 40^0

α = Rake angle = 20^0

$\therefore \quad 2\phi + 40^0 - 20^0 = 90^0$

$2\phi = 70^0$

$\phi = 35^0$

667. Shear angle depends upon _____.

(a) friction between chip and tool interface only
(b) tool rake angle only
(c) tool rake angle, friction between chip and tool interface only
(d) tool rake angle, friction between chip and tool interface, built up edge formation

VIZAG MT, 14.12.2020

Ans. (d) : Shear angle depends upon–
(1) Tool rake angle
(2) Friction between chip and tool interface
(3) Built up edge formation

668. In machining, the built-up edge develops at the chip tool interface mainly due to

(a) cohesion (b) diffusion
(c) abrasion (d) adhesion

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Adhesion of chip at the chip-tool interface is the reason of built-up edge development.

669. Side rake angle of a single point cutting tool is the angle:

(a) between the surface of the flank below the point and a line drawn from the point perpendicular to the base
(b) by which the face of the tool is inclined sideways
(c) between the surface of the flank below the point and a plane at right angles to the centre line of the point of the tool
(d) by which the face of the tool is inclined towards back

CIL MT 27.02.2020

Ans. (b) : The angle by which the face of the tool is inclined towards side is called side rake angle it is provided on tool to control chip flow.

670. During orthogonal machining with cutting tool having a 10° rake angle, the chip thickness is measured to be 0.4 mm. uncut chip thickness being 0.15 then shear strain is_____

(a) 2.71 (b) 4.71
(c) 5.51 (d) 3.78

Oil India Limited Sr. Engg. (Production) 30.11.2019

Ans. (a) : Given,
Rake angle, $\alpha = 10^o$
Chip thickness, $t_c = 0.4$ mm
Uncut chip thickness, $t_o = 0.15$ mm
Shear strain (ϕ) = ?
$\therefore$ We know that

$$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha} = \frac{(t_0/t_c)\cos\alpha}{1-(t_0/t_c)\sin\alpha}$$

$$= \frac{(.15/.4)\cos10^o}{1-(.15/.4)\sin10^o} = 0.395$$

$\phi = 21.55$

$\therefore$ We know that,

Shear strain (γ) = $\cot\phi + \tan(\phi - \alpha)$
= $\cot(21.55) + \tan(11.55^o)$
= $2.53 + 0.20$
$\gamma = 2.73$

671. In an experiment for the evaluation of a cutting tool life using the Taylor's tool life equation, $VT^n = C$, The value of n = 0.5. The tool life has a life of 180 minutes at a cutting speed of 18 m/min. If the tool life is reduced to 45 minutes, then the cutting speed in m/min will be:

(a) 36 (b) 32
(c) 40 (d) 16

BHEL ET 2019

Ans. (a) : n = 0.5
T_1 = 180 min
V_1 = 18 m/min
T_2 = 45 min
V_2 = ?
Applying Taylor's tool life equation

$\boxed{VT^n = C}$

$V_1 T_1^n = V_2 T_2^n$

$18 \times (180)^{0.5} = V_2 \times (45)^{0.5}$

$241.4953 = V_2 \times 6.7082$

$\boxed{V_2 = 36\ m/min}$

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672. Which of the following is not the characteristics of cutting tool material?
(a) it should have high toughness
(b) it should have high resistance to softening at higher temperature
(c) Is should have higher hardness than that of the work material
(d) Coefficient of friction between chip and tool should be very high

SJVN ET 2019

Ans. (d) : For cutting tool material, co-efficient of friction between chip and tool should be very low. Option (d) is not characteristic.

673. Which of the following is correct for Earnest-Merchant Theory? [where ϕ = Shear angle, α = Rake angle, β = Friction angle]

(a) $\phi = \frac{\pi}{2} - \frac{1}{2}(\alpha - \beta)$ (b) $\phi = \frac{\pi}{4} + \frac{1}{2}(\alpha - \beta)$

(c) $\phi = \frac{\pi}{2} + \frac{1}{2}(\alpha - \beta)$ (d) $\phi = \frac{\pi}{4} - \frac{1}{2}(\alpha - \beta)$

SJVN ET 2019
UKPSC AE 2007 Paper -I

Ans. (b) : Earnest merchant theory of metal cutting state that shear occurs along a thin plane which have minimum energy. They give relation among shear angle (ϕ), rake angle (α) and friction angle (β)

$$\phi = \frac{\pi}{4} - \frac{1}{2}(\beta - \alpha) \text{ or } \phi = \frac{\pi}{4} + \frac{1}{2}(\alpha - \beta)$$

674. Data of an orthogonal cutting is:
Chip thickness ratio = 0.4
Rake angle = 20^o,
What will be the shear plane angle?
[given, cos 20^o = 0.94, sin 20^o = 0.34]
(a) $\cos^{-1}(0.711)$ (b) $\tan^{-1}(0.435)$
(c) $\tan^{-1}(0.711)$ (d) $\tan^{-1}(0.211)$

SJVN ET 2019

Ans. (b) : Chip thickness ratio (r) = $\frac{\sin\phi}{\cos(\phi - \alpha)}$

ϕ = Shear angle
α = Rake angle = 20°

$$0.4 = \frac{\sin\phi}{\cos\phi.\cos 20 + \sin\phi.\sin 20}$$

$0.4 \times 0.94 \cos\phi + 0.4 \times \sin\phi \times 0.34 = \sin\phi$

$$\cot\phi = \frac{1 - 0.4 \times 0.34}{0.4 \times 0.94} = 2.29$$

$\tan\phi = 0.435$

$\phi = \tan^{-1}(0.435)$

675. The orthogonal shear angle is equal to the normal shear angle in :
(a) Orthogonal cutting (b) Oblique cutting
(c) Simple cutting (d) Uniform cutting

MECON MT 2019

Ans. (b) : Oblique cutting

676. Chip equivalent is increased by :
(a) an increases in side-cutting edge angle of tool
(b) an increase in nose radius and side cutting edge angle of tool
(c) increasing the plant area of cut
(d) increasing the depth of cut

Mizoram PSC (Insp. of Factories) 2019 Paper-III
ESE 1996

Ans. (b) : Chip equivalent is increased by an increase in nose radius and side cutting edge angle of tool.

677. For cutting of brass with single-point cutting tool on a lathe, tool should have :
(a) negative rake angle
(b) positive rake angle
(c) zero rake angle
(d) zero side relief angle

NLC GET 17.11.2020, Shift-II
Mizoram PSC (Insp. of Factories) 2019 Paper-III
UPPSC AE 12.04.2016 Paper-I
MPPSC State Forest Service Exam, 2014
ESE 2001

Ans. (c)
- Brass is turned with zero rake angle.
- Cast iron uses zero rake angle.

678. Single point thread cutting tool should ideally have :
(a) zero rake (b) positive rake
(c) negative rake (d) normal rake

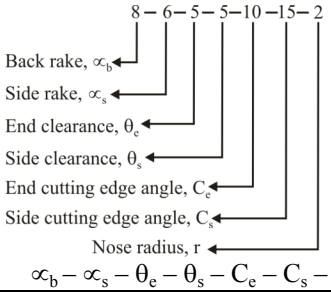
Mizoram PSC (Insp. of Factories) 2019 Paper-III
Sikkim PSC (Under Secretary), 2017
ESE 1995

Ans. (a) : Single point thread cutting tool should ideally have zero rake.

679. In ASA System, if the tool nomenclature is 8-6-5-5-10-15-2-mm, then the side rake angle will be :
(a) 5^0 (b) 6^0
(c) 8^0 (d) 10^0

Mizoram PSC (Insp. of Factories) 2019 Paper-III
Nagaland PSC (CTSE) 2018, Paper-I
TSPSC AEE 2017, UJVNL AE 2016

Ans. (b)



$\alpha_b - \alpha_s - \theta_e - \theta_s - C_e - C_s - r -$

680. A built-up-edge is formed while machining :
(a) ductile materials at high speed
(b) ductile materials at low speed
(c) brittle materials at high speed
(d) brittle materials at low speed

Mizoram PSC (Insp. of Factories) 2019 Paper-III
RPSC AE 2018, TANGEDCO AE 2015
TRB Asstt. Prof., 2012, GATE 2002

Ans. (b)
Continuous chips with built up edge–

Material type	→	Ductile
Rake angle	→	Low
Depth of cut	→	High
Cutting speed	→	Low
Feed	→	High

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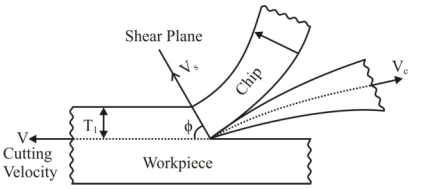
681. In orthogonal cutting, shear angle is the angle between :

(a) shear plane and the cutting velocity
(b) shear plane and the rake plane
(c) shear plane and the vertical direction
(d) shear plane and the direction of elongation of crystals in the chip

Mizoram PSC (Insp. of Factories) 2019 Paper-III
CSE Pre-2003

Ans. (a)



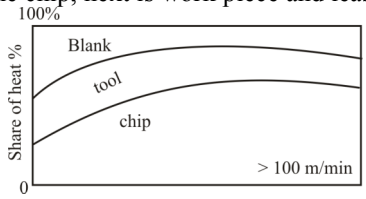
□ → Shear angle which is angle between shear plane and the cutting velocity.

682. As the cutting speed increases :

(a) more heat is transmitted to the workpiece and less heat is transmitted to the tool
(b) more heat is carried away by the chip and less heat is transmitted to the tool
(c) more heat is transmitted to both the chip and the tool
(d) more heat is transmitted to both the workpiece and the tool

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : As the cutting speed increases, the power consumption increases and heat generation increases out of the heat generated maximum heat is carried away by the chip, next is work piece and least is tool.



Approximate of heat amongest chip, tool and blank.

683. Flank wear occurs mainly on which of the following?

(a) Nose part and top face
(b) Cutting edge only
(c) Nose part, front relief face, and side relief face of the cutting tool
(d) Face of the cutting tool at short distance from the cutting edge

Mizoram PSC (Insp. of Factories) 2019 Paper-III
UKPSC AE 2007 Paper -I, ESE 2016, 2007

Ans. (c) : Flank wear mainly occur on nose part, front relief face and side relief face of the cutting tool.

684. Which condition is non favourable for continuous chips?:

(a) Low feed and depth of cut
(b) Brittle work piece
(c) High speed
(d) High back rake angle

BHEL ET 2019

Ans. (b) : Brittle work piece is non favourable condition for continuous chips.

685. In an orthogonal cutting operation the tool has a rake angle = 30°, chip thickness before the cut = 0.3 mm and the cut yields a deformed chip thickness = 0.6mm. Calculate the chip thickness ratio.

(a) 1/2 (b) 1/3
(c) 1/4 (d) 1/6

MECON MT 2019

Ans. (a) : Chip thickness ratio-

$$r = \frac{t_1}{t_2}$$

$$r = \frac{0.3}{0.6} = \frac{1}{2}$$

Where,

t_1 = Chip thickness before machining
t_2 = Chip thickness after machining

686. A single point cutting tool has -

(a) Rake angle (b) Cutting edge angle
(c) Lip angle (d) All of these

MECON MT 2019

Ans. (d) :

687. Side rake angle is:

(a) Positive if its slope is towards cutting edge
(b) Positive if its slope is away from cutting edge
(c) Cannot be zero
(d) Always negative

HPPSC ADF 05.03.2019

Ans. (b) : Positive if its slope is away from cutting edge

688. For cutting brass, the hacksaw blade should have ..... teeth per 25mm.

(a) 14 (b) 24
(c) 32 (d) 40

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : As we know, brass is a soft material it can easily cut by hacksaw, the blade should have, 14 teeth per 25 mm.

689. In metal cutting, use of low feeds and high cutting speeds is desired when the objective is

(a) high metal removal
(b) dry machining
(c) use of soft cutting tool
(d) surface finish

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : As we know, due to low cutting speed & large depth of cut produces rough surface, for smooth/fine surface finish we increases the speed & reduces the depth of cut get good surface finish.

690. Crater wear is predominant in

(a) carbon tool steels
(b) tungsten carbide tools
(c) high speed steel tools
(d) ceramic tools

Nagaland PSC (CTSE) 2018, Paper-I, ESE 1995

Ans. (a) : Crater wear is predominant in tungsten carbide tools because tungsten is good heat dissipation, so diffusion will be less and have good chemical affinity. So crater wear of tool is less.

691. A better machinable metal is one which gives

(a) lower chip-tool contact area and larger shear angle

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(b) higher chip tool contact area and smaller shear angle
(c) lower chip tool contact area and smaller shear angle
(d) higher chip-tool contact area and larger shear angle

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : A better machinable metal is on which gives lower chip tool contact area & larger shear angle, because the cutting forces decrease due to low areas & amount of chip removal increases as shear angle increases.

692. Which of the following statement is wrong?
(a) the diamond is the hardest tool material and can run at cutting speeds about 50 times that of high speed steel tool
(b) the ceramic tools can be used at cutting speeds 40 times that of high speed steel tools
(c) the cemented carbide tools can be used at cutting speeds 10 times that of high speed steel tools
(d) the ceramic tools can withstand temperature upto 600°C only

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : In the following all the statement is true but the ceramic tools can with stand temperature up to 1350°C due to covalent bond posses high melting point, high hardness & high chemical resistance.

693. Cutting tools are provided with large positive rake angle mainly for
(a) Reducing the magnitude of the cutting force
(b) Increasing the strength of the cutting edge
(c) Avoiding rubbing action with the finished surface
(d) Better heat dissipation

GPSC Executive Engineer 23.12.2018, GATE 1997

Ans. (a) : By providing large positive rake angle, the chance of discontinuous chip is less and friction will be reduced.

694. The useful tool life of HSS tool machining mild steel at 18 m/min is 3 hours. Calculate the tool life when the tool operates at 24 m/min. (Use Taylor's tool life equation $VT^n = C$, where n = 0.125)
(a) 18 min (b) 30 min
(c) 16 min (d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (a)
$V_1 = 18$ m/min, $T_1 = 3$ hours $= 3 \times 60 = 180$ min
$V_2 = 24$ m/min, $T_2 = ?$, $n = 0.125$
$VT^n = C$
$V_1T_1^n = V_2T_2^n$
$18 \times (180)^{0.125} = 24 \times (T_2)^{0.125}$
$T_2 = 18.02$ min
$T_2 \approx 18$ min

695. Using Lee and Shaffer relation, calculate shear angle if normal rake angle of the tool = 10^0, coefficient of friction (μ) = 0.6.
(a) 14^0 (b) 18^0
(c) 24^0 (d) None of the above

MPSC Lect. (Prod.) 2018

Ans. (c) : Lee and shaffer relation,

$\phi + \beta - \alpha = \frac{\pi}{4}$ $\alpha = 10^0$, $\mu = 0.6$

$\tan \beta = \mu$

$\beta = \tan^{-1}(\mu) = \tan^{-1}(0.6) = 30.963$

$\phi + \beta - \alpha = \frac{\pi}{4}$

$\phi + 30.963 - 10 = 45^0$

$\boxed{\phi = 24.03^0}$ $\boxed{\phi = 24^0}$

696. During orthogonal machining of a bar, using zero rake angle cutting tools, minimum strain in work piece is desired. To achieve this, the shear plane angle should be:
(a) 60° (b) 0°
(c) 45° (d) 30°

AAI Jr. Executive 29.11.2018

Ans. (c) : Shear strain,

$\gamma = \cot\beta + \tan(\beta - \alpha)$

α = Rake angle

$\alpha = 0$

γ = Shear strain

$\gamma = 2$

β = Shear angle

$\gamma = \cot\beta + \tan\beta$

$$\frac{d\gamma}{d\beta} = \frac{d}{dx}\left(\frac{2}{\sin 2\beta}\right)$$

$$\frac{d\gamma}{d\beta} = \frac{0 - 2.2 \cdot \cos 2\beta}{(\sin 2\beta)^2} = \frac{4\cos 2\beta}{(\sin 2\beta)^2}$$

$$\frac{d^2\gamma}{d\beta^2} = \frac{(\sin 2\beta)^2(-8\sin 2\beta) - 4(\cos 2\beta)(4\cos 2\beta \sin 2\beta)}{(\sin 2\beta)^4}$$

$$\frac{d^2\gamma}{d\beta^2} = \frac{8(\sin 2\beta)^2 - 16(\cos 2\beta)^2}{(\sin 2\beta)^3}$$

$$\frac{d\gamma}{d\beta} = 0$$

$$\frac{-4\cos 2\beta}{(\sin 2\beta)^2} = 0$$

$\cos 2\beta = 0$

$\cos 2\beta = \cos 90$

$\boxed{\beta = 45°}$

697. In a machining operation cutting speed is reduced by 50%. Assuming n = 0.5, C = 300 in Taylor's equation the increase in tool life is
(a) 2 (b) 4
(c) 8 (d) 16

ISRO Scientist/Engineer 22.04.2018

Ans. (b) : $V_1 = V, V_2 = \frac{V}{2}, n = 0.5, C = 300$

Taylor's equation

$VT^n = C$

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$$V(T_1)^{0.5} = \frac{V}{2}(T_2)^{0.5}$$

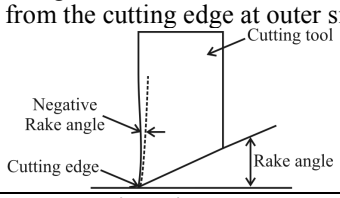
$$\left(\frac{T_2}{T_1}\right)^{\frac{1}{2}} = 2$$

$T_2 = 4T_1$

698. Negative rake angle is used in
(a) High carbon steel tool
(b) Ceramic tool
(c) Super steel
(d) Cubic boron nitride tool

HPPSC AE 2018

Ans. (b, d) : Negative Rake Angle–A tool has a negative rake angle when the face of the cutting tool slopes away from the cutting edge at outer side.



699. The Merchant circle is used to
(a) Improve the quality of a product
(b) Find the principal stress and principal strain
(c) Fix the price of a product in the market
(d) Establish the shear angle relationship in machining

HPPSC AE 2018

Ans. (d) : The Merchant circle is used to establish the shear angle relationship in machining.

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \frac{\beta}{2}$$

$$\phi = 45^\circ + \frac{\alpha}{2} - \frac{\beta}{2}$$

Merchant's theory gives higher shear plane angle means lower shear force due to smaller shear plane.

700. While turning a 60 mm diameter bar, it was observed that the tangential cutting force was 3000 N and the feed force was 1200 N. If the tool rake angle is 32°, then the coefficient of friction is nearly (may take sin32° = 0.53, cos32° = 0.85 and tan32° = 0.62)
(a) 1.37 (b) 1.46
(c) 1.57 (d) 1.68

ESE 2018

Ans. (a) : Given,
$F_c = 3000N$, $F_T = 1200$ N
$\alpha = 32°$
We know that

$$\mu = \tan\beta = \frac{F}{N} = \frac{F_c \sin\alpha + F_T \cos\alpha}{F_c \cos\alpha - F_T \sin\alpha}$$

$$= \frac{(3000 \times \sin 32°) + (1200 \times \cos 32°)}{(3000 \times \cos 32°) - (1200 \times \sin 32°)}$$

$\mu = 1.363$

701. In an orthogonal cutting operation, the length of the cut is 76 mm, length of the chip measured is 61 mm and depth of cut is 0.2 mm. What is the thickness of the chip?
(a) 0.01 mm (b) 0.8 mm
(c) 0.2 mm (d) 0.25 mm

TNPSC AE 2018

Ans. (d) : Data given as -
$f = t = 0.2$ mm
$\ell = 76$ mm
$\ell_c = 61$ mm
We know that chip thickness ratio

$$r = \frac{t}{t_c} = \frac{\ell_c}{\ell}$$

$$t_c = \frac{76 \times 0.2}{61}$$

$t_c = 0.24918 \simeq 0.25$ mm

702. If n = 0.5 and C = 300 for the cutting speed and the tool life relation, when cutting speed is reduced by 30%, the tool life is increased by :
(a) 95% (b) 104%
(c) 78% (d) 50%

GPSC ARTO Pre 30.12.2018

Ans. (b) : Given, $n = 0.5$
$C = 300$
$V_1 T_1^n = C$
$V_1 T_1^{0.5} = 300$
$V_2 T_2^{0.5} = 300$
$V_1 T_1^{0.5} = 0.7 V_1 T_2^{0.5} \quad (\because V_2 = 0.7V_1)$

$$\frac{T_2}{T_1} = \left(\frac{1}{0.7}\right)^2 = 2.0408$$

$$\frac{T_2 - T_1}{T_1} = 2.0408 - 1 = 1.0408$$

Tool life is increased, $\frac{T_2 - T_1}{T_1} \times 100 = 104.08\%$

703. In a machining test, a cutting speed of 100 m/min indicated the tool life as 16 min and a cutting speed of 200 m/min indicated the tool life as 4 min. The values of n and C are :
(a) 0.5 and 200 (b) 0.25 and 200
(c) 0.5 and 400 (d) 0.25 and 400

GPSC ARTO Pre 30.12.2018

Ans. (c) : Given,
$V_1 = 100$ m/min. , $T_1 = 16$ min
$V_2 = 200$ m/min. , $T_2 = 4$ min
$V_1 T_1^n = V_2 T_2^n$
$100(16)^n = 200(4)^n$
$(4)^n = 2$
$(2)^{2n} = 2^1$
$2n = 1$
$\boxed{n = 0.5}$
$V_1 T_1^n = C \Rightarrow 100(16)^{0.5}$
$\Rightarrow 100 \times 4 = 400$
$\boxed{C = 400}$

704. Tool geometry of a single point cutting tool is specified by the following elements :
1. Back rake angle (α_b)
2. Side rake angle (α_s)
3. End cutting edge angle (ϕ)
4. Side cutting edge angle (ψ)

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5. Side relief angle (c_s)
6. End relief angle (c_b)
7. Nose radius (r)

The correct sequence of these tool elements used for correctly specifying tool geometry is :

(a) 1, 2, 3, 6, 5, 4, 7 (b) 1, 2, 6, 5, 3, 4, 7
(c) 1, 2, 5, 6, 3, 4, 7 (d) 1, 2, 6, 3, 5, 4, 7

GPSC ARTO Pre 30.12.2018

Ans. (b) : Tool geometry of single point cutting tool is specified by :

$$\alpha_b - \alpha_s - c_b - c_s - \phi - \psi - r$$
$$1 - 2 - 6 - 5 - 3 - 4 - 7$$

705. If the cutting speed is increased, then the built-up edge:

(a) Becomes longer
(b) May or may not form
(c) Becomes smaller and finally does not form at all
(d) Has nothing to do with speed

HPPSC Asstt. Prof. 18.09.2017

Ans. (c) : If the cutting speed is increased, then the built-up edge becomes smaller and finally does not form at all.

706. Pick up the correct statement:

(a) The surface roughness increases with the decrease in coefficient of friction between tool and the material
(b) At slow speed roughness decreases as the positive rake angle increases
(c) The surface roughness decreases with the decreases of nose radius
(d) The surface finish is independent of cutting speed, feed and depth of cut

PTCUL AE 25.06.2017

Ans. (b) :

707. In metal cutting Amonton's Laws of Friction are not obeyed. This could be attributed to

(a) Higher strain rate associated with metal cutting
(b) High sliding velocity of the chip over the tool face
(c) Higher order normal stresses causing sticking zone on the tool face
(d) Reduction in hardness of tool material at temperatures prevailing at the tool-chip interface

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : In metal cutting Amonton's laws of friction are not obeyed because higher order normal stresses causing sticking on the tool face.

708. A single-point cutting tool with 12° rake angle is used to machine a steel work-piece. The depth of cut, i.e. uncut chip thickness is 0.81 mm. The chip thickness under orthogonal machining condition is 1.8mm. The shear angle is approximately

(a) 22° (b) 26°
(c) 56° (d) 76°

GPSC Asstt. Director of Transport 05.03.2017
VIZAG STEEL MT 10.06.2012, GATE 2011

Ans. (b) : Relation between shear angle (ϕ), chip thickness ratio (r) and rake angle (α) is given by

$$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$$

Where, $r = \frac{0.81}{1.8} = 0.45$

$$\tan\phi = \frac{0.45 \times \cos 12°}{1 - (0.45 \times \sin 12)}$$

$$\phi = 26°$$

709. A steel bar 200 mm in diameter is turned at a feed of 0.25 mm/rev with a depth of cut of 4 mm. The rotational speed of the workpiece is 160 rpm. The material removal rate in mm^3/s is

(a) 160 (b) 167.6
(c) 1600 (d) 1675.5

GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (d) : Material removal rate (MRR) in mm^3/s

$$MRR = f.d \times V, \quad V = \frac{\pi DN}{60}$$

$$MRR = 0.25 \times 4 \times \frac{\pi \times 200 \times 160}{60}$$

$$= 1675.5 \text{ mm}^3/\text{s}$$

710. Cutting tool is much harder than the workpiece. Yet the tool wears out during the tool-work interaction, because

(a) Extra hardness is imparted to the workpiece due to coolant used
(b) Oxide layers on the workpiece surface impart extra hardness to it
(c) Extra hardness is imparted to the workpiece due to severe rate of strain
(d) Vibration is induced in the machine tool.

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Metal cutting takes place in plastic deformation zone, where work hardening takes place as chip get strained. So, tool wears out during the tool work interaction.

711. During orthogonal cutting, an increase in cutting speed causes

(a) An increase in longitudinal cutting force
(b) An increase in radial cutting force
(c) An increase in tangential cutting force
(d) Cutting forces to remain unaffected

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : During orthogonal cutting, an increase in cutting speed causes, No effect on cutting forces, but decrease in radial cutting force, and cutting force does not depend upon cutting speed depends only on feed, depth of cut etc.

712. In the metal cutting process, the high cutting speed and large angle of the tool will result in the formation of

(a) continuous chips
(b) discontinuous chips
(c) continuous chips with built up edge
(d) chips of irregular shapes

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (a) : In the metal cutting process, the high cutting speed and large angle of the tool will result in the formation of continuous chips.

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713. The specific metal cutting energy is expressed as

(a) $\frac{\tau\cos(\beta-\alpha)}{\sin\phi\cos(\phi+\beta-\alpha)}$ (b) $\frac{\tau\sin(\beta-\alpha)}{\sin\phi\cos(\phi-\beta+\alpha)}$

(c) $\frac{\tau\cos(\alpha-\beta)}{\sin\phi\cos(\phi-\beta+\alpha)}$ (d) $\frac{\tau\sin(\alpha-\beta)}{\sin\phi\cos(\phi+\beta-\alpha)}$

where 'α' is rake angle, 'β' is friction angle, 'ϕ' is shear angle and 'τ' shear stress

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Cutting force

$$F_C = \frac{\tau_s\, bt_1\cos(\beta-\alpha)}{\sin\phi\cos(\phi+\beta-\alpha)}$$

Where τ_s = Ultimate shear strength of workpiece material

b = width of workpiece

t_1 = Uncut chip thickness

Specific cutting energy (U_C)

$$= \frac{\text{Cutting Energy}}{\text{MRR}}$$

$$= \frac{F_C\cdot V_C}{bt_1 V_C} = \frac{F_C}{bt_1} = \frac{\tau_s\cos(\beta-\alpha)}{\sin\phi\cos(\phi+\beta-\alpha)}$$

714. The relationship of the shear angle $\phi = \frac{\pi}{4} - \beta + \frac{\alpha}{2}$ is as per ('α' is rake angle and 'β' is friction angle)

(a) Ernst-Merchant theory
(b) Merchant's theory
(c) Stabler theory
(d) Lee and Shaffer theory

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Ernst-Merchant theory

$$2\phi + \beta - \alpha = \frac{\pi}{2}$$

Merchants theory,

$2\phi + \beta - \alpha$ = Constant

Stabler theory,

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \beta$$

Lee and Shaffer theory,

$$\phi + \beta - \alpha = \frac{\pi}{4}$$

715. The optimum cutting speed for maximum production is ('n' is tool constant and 'Tc' is time required to change the tool)

(a) $\frac{k}{\left[\left\{\frac{1}{n}+1\right\}T_c\right]^{\frac{n}{2}}}$ (b) $\frac{k}{\left[\left\{\frac{1}{n}-1\right\}T_c\right]^{n}}$

(c) $\frac{k}{\left[\left\{1-\frac{1}{n}\right\}T_c\right]^{2n}}$ (d) $\frac{k}{\left[\left\{\frac{1}{n}-1\right\}\frac{T_c}{2}\right]^{n}}$

ISRO Scientist/Engineer 07.05.2017

Ans. (b) : Various formulae for find out optimum cutting speed according to given parameters.

$$V_{opt.} = \frac{k}{\left[\left(\frac{1}{n}-1\right)T_c\right]^n}$$

(used if only T_c given)

$$V_{opt} = \frac{k}{\left[\left\{\frac{1}{n}-1\right\}\left(T_c + \frac{C_t}{C_m}\right)\right]^n}$$

(used if T_c, C_t and C_m given)

$$V_{opt} = \frac{k}{\left[\left\{\frac{1}{n}-1\right\}\left(\frac{C_t}{C_m}\right)\right]^n}$$

(used if C_t and C_m given)

where,

C_t = tooling cast (Rs/service)

C_m = Labour cost + overhead cost (Rs/min)

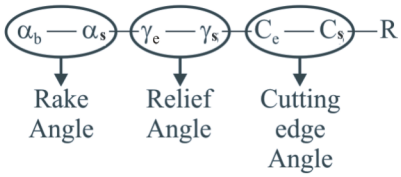
716. The side relief angle in a cutting tool signature of 10, 20, 7, 6, 8, 15, 2 is

(a) 10 (b) 7
(c) 6 (d) 8

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Tool designation (ANSI) or ASA

OR Tool signature–



[∵ b = Back, s = side]

Rake, Relief, cutting edge

Tool Signature–

10	20	7	6	8	15	2
α_b	α_s	γ_e	γ_s	C_e	C_s	R

717. In a single point cutting tool signature ($\alpha_b - \alpha_s - \varphi_e - \varphi_s - C_e - C_s - r$), the last term r stands for nose radius in:

(a) Micrometre (b) Millimetre
(c) Centimetre (d) Decametre
(e) Metre

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : $\alpha_b - \alpha_s - \varphi_e - \varphi_s - C_e - C_s - r$

Tool signature in American standards Association (ASA)

α_b – Back rake angle

α_s – Side rake angle

φ_e – End clearance angle

φ_s – Side clearance angle

C_e – End cutting edge angle

C_s – Side cutting or approach angle

r – Nose radius [in inch]

In ASA system, nose radius is measured in inch unit, whereas, in orthogonal Rake system (ORS) it is measured in mm.

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718. The strength of a single point cutting tool depends upon:

1. The clearance angle
2. The rake angle
3. The lip angle

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

JWM 2017

Ans. (d) : Strength of single point cutting tool depends on rake angle but rake angle also depends on lip and clearance angle.

719. In turning, chip thickness ratio will be _____.

(a) Equal to zero (b) Greater than two
(c) Greater than one (d) Less than one

CIL MT 26.03.2017

Ans. (d) :

$$\text{Chip thickness ratio (r)} = \frac{\text{uncut chip thickness}}{\text{chip thickness}}$$

For turning operation, r is less than 1

720. The heat generated in metal cutting is dissipated in difference proportions into environment, tool, chip, workpiece. The correct order of this proportion in decreasing magnitude is:

(a) took, work, chip, environment
(b) work, tool, environment, chip
(c) work, tool, chip, environment
(d) chip, tool, work, environment

TRB Polytechnic Lecturer 2017
GATE 1994

Ans. (d) : The heat generated in metal cutting is dissipated maximum with chip and minimum to environment.
The correct order of this proportion in decreasing magnitude is
Chip, tool, work, environment

721. In machine tools, chatter is due to–

(a) free vibrations
(b) Random vibrations
(c) Forced vibrations
(d) Self-excited vibrations

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : In machine tools, chatter is due to self-excited vibrations.

722. Cutting power consumption in turning can be significantly reduced by

(a) increasing rake angle of the tool
(b) increasing the cutting angle of the tool
(c) widening the nose radius of the tool
(d) increasing the clearance angle

TNPSC AE 2017, GATE 1994

Ans. (a) : Cutting power consumption in turning can be significantly reduced by increasing rake angle of the tool
We know that

$$\phi = \frac{\pi}{4} + \frac{\alpha}{2} - \frac{\beta}{2}$$

If $\alpha\uparrow \rightarrow \phi\uparrow \rightarrow$ Area of shear plane (A) $\downarrow \rightarrow F_C\downarrow \rightarrow P_C\downarrow$

723. Chisels used for metal cutting are–

(a) Hardened
(b) Annealed, hardened
(c) Annealed
(d) Hardened and tempered

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Chisels used for metal cutting are hardened, all over.

724. Minimum shear strain in orthogonal turning with a cutting tool of zero rake angle is

(a) 0.0 (b) 0.5
(c) 1.0 (d) 2.0

TSPSC AEE 2017
VIZAG STEEL MT 10.06.2012, GATE 2009

Ans. (d) : Shear Strain, $\gamma = \frac{\cos\alpha}{\sin\phi.\cos(\phi-\alpha)}$

$\alpha = 0$, ϕ = Shear Angle

$$\gamma = \frac{1}{\sin\phi\cdot\cos\phi}$$

For γ to be minimum $\sin\phi\cdot\cos\phi$ should be maximum

$$\frac{\partial}{\partial\phi}(\sin\phi\cdot\cos\phi) = -\sin\phi\cdot\sin\phi + \cos\phi\cdot\cos\phi = 0$$

$$\boxed{\tan^2\phi = 1}$$

$$\phi = 45°$$

$$\gamma = \frac{1}{\sin\phi\cdot\cos\phi} = \frac{1}{\sin 45°\cdot\cos 45°}$$

$$\boxed{\gamma = 2}$$

725. In machining, Merchant circle diagram deals with

(a) cutting temperature (b) cutting forces
(c) tool life (d) chip conditions

WBPSC AE, 2017
HPPSC Asstt. Prof. 29.10.2016
HPPSC Lect. (Auto) 23.04.2016
UKPSC AE 2007 Paper -I

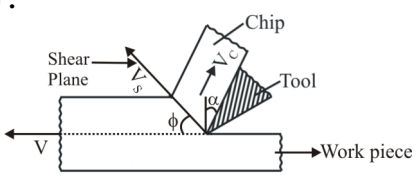
Ans. (b) : Merchant's circle diagram is used for the analysis of cutting forces acting during orthogonal cutting of work piece.

726. In theory of metal cutting, the shear plane angle is the angle between

(a) shear plane and job surface.
(b) shear plane and vertical plane.
(c) shear plane and horizontal plane.
(d) shear plane and tool surface.

WBPSC AE, 2017
J&K PSC Civil Services Pre, 2013

Ans. (a) :



$V \rightarrow$ Cutting velocity
$V_s \rightarrow$ Shear plane velocity
$\phi \rightarrow$ Shear plane angle
$\alpha \rightarrow$ Side rake angle

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727. Tool material should be hard so that it can resist

(a) Adhesion wear (b) Abrasion wear
(c) Diffusion wear (d) All of the above

WBPSC AE, 2017

Ans. (d) :

728. Which one does not fit with the rest in the following?

(a) Hot chisel (b) Hot punch
(c) Reamer (d) Flatter

WBPSC AE, 2017

Ans. (d) : Flatter is used basically for mallable material. We apply force to convert sheets into a mannered way (plane way) on other hand hot chisel, hot punch and reamer is used for cutting of work piece.

729. In an orthogonal machining experiment, cutting force is found to be 1000 N and thrust force is 500 N. What is the co-efficient of friction at the tool-chip interface if rake angle is zero?

(a) 1/2 (b) 2
(c) $1/\sqrt{2}$ (d) 1/8

APGENCO AE, 2017

Ans. (a) : $F_c = 1000$ N
$F_T = 500$ N
$\alpha = 0$ (Rake angle)

$$\tan\beta = \frac{F_T}{F_C} = \frac{500}{1000} = \frac{1}{2} \quad [\beta = \text{friction angle}]$$

$$\tan\beta = \frac{1}{2}$$

$$\mu = \tan\beta = \frac{1}{2} \quad [\mu = \text{coefficient of friction}]$$

730. Which of the following stresses are important for chip formation?

1. Tensile stress
2. Shear stress
3. Compressive stress
4. Bending stress

(a) 2 and 3 are true (b) 3 and 4 are true
(c) 4 and 1 are true (d) 1 and 2 are true

APGENCO AE, 2017

Ans. (a) : Shear stress and compressive stress are important for chip formation.

731. Depth of cut in turning operation is the total reduction in the

(a) diameter of the work
(b) radius of the work
(c) radius of the work in one pass of the tool
(d) diameter of the work in one pass of the tool

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : Depth of cut in turning operation is the total reduction in the radius of the work in one pass of the tool.

732. If the tool designation, under ASA system is $8 - 14 - 6 - 6 - 15 - \frac{1}{8}$, it means that back rake is

(a) 14 (b) 8
(c) 15 (d) 6

GPSC ARTO 01.05.2016

Ans. (b) : Tool Designation under ASA system-

$\alpha_b - \alpha_s - Q_e - Q_s - C_e - C_s - r$
- r → Nose radius (mm)
- C_s → Side cutting edge angle
- C_e → End cutting edge angle
- Q_s → Side relief angle
- Q_e → End relief angle
- α_s → Side rake angle
- α_b → Back rake angle

So,

8 - 14 - 6 - 6 - 6 - 15 - 1/8
→ Back rake angle

733. Orthogonal cutting refers to cutting with :

(a) Least chip strain
(b) Minimum radial force of cutting
(c) Zero degree inclination angle
(d) Negative rake

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) : Orthogonal cutting has been coined to cover the case where the cutting tool generate a plane surface parallel to an original plane surface of the material being cut and set with its cutting edges perpendicular to the direction of relative motion of tool and workpiece.
Orthogonal cutting refers to zero degree inclination angle.
Used in–grooving, parting, slotting, pipe cutting etc.

734. During machining on a Lathe excess metal is removed in the form of chips as in the case of turning. Which of the following conditions are required for continuous ribbon like chip to be formed in turning?

(i) At a higher cutting speed
(ii) At a lower cutting speed
(iii) A brittle material
(iv) A ductile material

Select the correct Answer amongst the following:

(a) i and iii (b) i and iv
(c) ii and iii (d) ii and iv

UJVNL AE 2016

Ans : (b) During machining on a lathe, excess metal is removed in the form of chips as in the case of turning. following conditions are required for continious ribbon like chip to be formed in turning
(i) At a high Cutting speed
(ii) A ductile material.

735. Negative rake angle tool is recommended to machine

(a) Ductlie material at high speed
(b) Brittle material at high speed
(c) Ductile material at low speed
(d) Very hard and ductile material at high speed

UJVNL AE 2016

Ans : (b) Negative rake angle tool is recommended to machine Brittle material at high speed.
Ductile material rake angle kept:- 10°-15°
Brittle material rake angle kept:- Negative or minimum
HSS tool for Aluminum rake angle:- 35°
C.C tool for Aluminum rake angle:- 15°

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736. Which of the following statements is incorrect about the continuous chip?

(a) It is formed while machining ductile materials at high cutting speeds.
(b) It is formed when feed and depth of cut are low.
(c) It results in good surface finish.
(d) None of the above

UPPSC AE 12.04.2016 Paper-I

Ans : (d)

Continuous chip	Dis- continuous chip	Chips with buit-up edge
Ductile work	Brittle work	Ductile work
High Speed	Low Speed	Low Speed
Low feed	High Feed	High feed
Low depth of cut	High depth of cut	High depth of cut
High back rake angle	Low back rake angle	Low back rake angle

737. In a machining operation chip thickness ratio is 0.3 and tool back rake angle is 10^o. The value of shear strain is

(a) 0.86 (b) 2.24
(c) 3.10 (d) 3.34

UPPSC AE 12.04.2016 Paper-I

Ans : (d) shear strain $= \gamma = \cot\phi + \tan(\phi - \alpha)$

$$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$$

$$\tan\phi = \frac{0.3\cos 10^o}{1 - 0.3\sin 10^o} = 0.3116$$

$$\phi = 17.31°$$

$$\gamma = \cot\phi + \tan(\phi - \alpha)$$

$$\gamma = 3.208 + 0.1282$$

$$\gamma = 3.34$$

$$\boxed{\text{ShearStrain}(\gamma) = 3.34}$$

738. Which is the main reason for poor surface finish

(a) Heavy depth of cut (b) High cutting speed
(c) High feed (d) Low side rake angle

UPPSC AE 12.04.2016 Paper-I

Ans : (a) High feed is the main reason for poor surface finish. Effect of various machining variables on surface finish.

Increasing of following machining variables	Leads to improved /deteriorated surface finishing
Cutting speed	Improves
Feed rate	Deteriorates
Depth of cut	Deteriorates
True rake angle	Improves

739. Excessive heat generated during metal cutting is due to

(a) Built up edge formed on the cutting tool
(b) Correctly grounded tool
(c) Low friction between tool and workpiece
(d) Cutting tool of good surface finish

RPSC Lecturer 16.01.2016

Ans. (a) : Excessive heat generated during metal cutting is due to built up edge formed on the cutting tool.

740. Cutting speed for some materials are given below. Which one of the below could be cutting speed for brass

(a) 20 m/min (b) 30 m/min
(c) 40 m/min (d) 80 m/min

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : Cutting speed of brass, Aluminium and bronze is around 60-90 m/min.

741. Spindle speed in a lathe for turning a 40 mm diameter rod at a cutting speed of 30 m/min would be

(a) 100 rpm (b) 206 rpm
(c) 238 rpm (d) 468 rpm

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : D = 40 mm,
V = 30 m/min = 30×1000 mm/min
N = ?

$$V = \frac{\pi DN}{1000},$$

$$N = \frac{30\times 1000}{(3.14\times 40)} = 238.85 \text{ rpm}$$

742. For cutting mild steel the cutting angle a chisel should be

(a) 55 degree (b) 60 degree
(c) 70 degree (d) 75 degree

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : For cutting of different material, cutting angle of a chisel should be,
Mild steel – 55°
Cast iron – 60°
Aluminium – 35°

743. A 250 mm diameter is turned at a cutting speed of 20 m/min feed 0.5 mm per revolution of spindle at a spindle speed of 25 R.P.M. Now if the diameter of work piece is double, and spindle speed is also double, the time taken to turn the job will be

(a) Four times the original time
(b) Double the original time
(c) No change in turning time
(d) One fourth the original time

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : D_1 = 250 mm, V_1 = 20 m/min, F = 0.5 mm/rev., N_1 = 25 rpm, Now if diameter is double, D_2 = 500 mm, & speed is double (V_2) = 40 m/min, then, N_2 =?

$$V_2 = \frac{\pi D_2 N_2}{1000} \Rightarrow \frac{1000\times 40}{3.14\times 500} = N$$

N = 25.47 rpm. [No change in turning time]

744. A work piece of 100 mm diameter is to be machined at a cutting speed of 100 meters per minute. The rpm of the work piece should be

(a) 100 (b) 159
(c) 200 (d) 318

Nagaland PSC CTSE 2016 Paper-I

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Ans. (d) : D = 100 mm, V = 100 m/min, N = ?

$$V = \frac{\pi DN}{1000} \text{ (m/min)}$$

$$N = \frac{100 \times 1000}{\pi \times 100}$$

$$N = 318.47 \text{ rpm}$$

745. In metal cutting use of low feed and high cutting speed is desired when the objective is

(a) High metal removal rate
(b) Dry Machining
(c) Use of soft cutting tool
(d) Surface finish

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : For high surface finish in metal cutting use of low and high cutting speed.

746. The relation for cutting tool life is given as $VT^n = C$, where V is cutting speed, T is corresponding life, n and c are constants depending on cutting conditions. The numerical value of n for roughing cut as compared to that for light cuts in mild steel would be

(a) More
(b) Less
(c) Same
(d) Does not depend on type of cut

BPSC Poly. Lect. 2016

Ans : (a) The numerical value of n for roughing cut compared to that for light cuts in mild steel would be more.

$$VT^n = C$$

n= the index or exponent which depends upon machine tool and workpice characteristics.

Generally the Value of n Varies from $\left(\frac{1}{5} \text{to} \frac{1}{10}\right)$ for all tool materials.

747. The value of side rake angle of the turning tool having tool signature :
0°, 10°, 8°, 6°, 20°, 60°, 0(mm) will be:

(a) 60° (b) 20°
(c) 10° (d) 8°

(HPPSC LECT. 2016)

Ans : (c) Tool signature:-
(i) Back rake angle - 0°
(ii) Side rake angle - 10°
(iii) End relif angle - 8°
(iv) Side relif angle - 6°
(v) End cutting edge angle - 20°
(vi) Side cutting edge angle - 60°
(vii) Nose radius 0 mm

748. Oblique cutting system is also known as:

(a) One-dimensional cutting system
(b) Two-dimensional cutting system
(c) Three-dimensional cutting system
(d) None of the above

(HPPSC LECT. 2016)

Ans : (c) Oblique Cutting system is also known as three-dimensional cutting system .
Three dimensional cutting:-The simplest three dimensional cutting tool is a straight cutting edge that is inclined to the velocity vector. The inclination 'i' of the tool represent the point of departure from the orthogonal cutting case and it significantly alters the chip flow and hence the performance of a tool.

749. The rake angle of a cutting tool is 10°, shear angle 35° and cutting velocity 25 m/min. What is the chip velocity along tool face?

(a) 1.9 m/min (b) 3.9 m/min
(c) 7.9 m/min (d) 15.8 m/min

MPPSC AE 2016

Ans : (d) $\alpha = 10^o$

$\phi = 35^o$

$V = 25 \text{m/min}$

$$V_c = \frac{V \sin \phi}{\cos(\phi - \alpha)}$$

$$V_c = \frac{25 \sin 35^o}{\cos(35^o - 10^o)}$$

$$V_c = 15.8 \text{m/min}.$$

750. The crater wear of a cutting tool is due to

(a) Chemical action of the coolant
(b) Excessive heat generated during cutting
(c) Rubbing of tool against workplace
(d) Abrasive action of the chip

MPPSC AE 2016

Ans : (d) i. The crater wear of a cutting tool is due to Abrasive action of the chip
ii. Flank wear occurs mainly on the nose part, front relief and side relief face of the cutting tool.

751. Small nose radius

(a) Increases tool life
(b) Produces chipping and decreases tool life
(c) Results in excessive stress concentration
(d) Decreases tool life

Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (c) : Small nose radius results in excessive stress concentration. But too large nose radius will induced chatter vibration.
If nose radius increases, cutting force and cutting power also increases.

752. With increasing cutting velocity, the total time for machining a component

(a) First decreases and then increases
(b) Reduces to half
(c) Remains unaffected
(d) Gets doubled

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : With increase cutting velocity initially machining time decrease but due to frequent total wearing machining time increase.

753. Find the incorrect statement

(a) Brittle materials produce discontinuous chips
(b) Notches cannot be seen on continuous chips
(c) Ductile materials produce continuous chips
(d) Ductile materials in presence of fine feed and high cutting speed produce chips with built up edges

Rajasthan Nagar Nigam AE 2016, Shift-II

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Ans. (d) : The condition that normally induce the formation of BUE are low cutting speed high feed and low rake angle.

754. Choose the wrong statement in respect of cutting tool

(a) It should be able to retain hardness at cutting temperature
(b) It should have low coefficient of frictions
(c) It should have low thermal conductivity
(d) It should be able to resist shock

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (c) : Cutting Tool Materials

(i) **Red or Hot Hardness–** It is the ability of tool material to with-stand high temperature without losing its cutting edge. The red hardness of tool material can be increased by adding of molybdenum, Tungsten etc.
All these makes carbides.
(ii) **Wear Resistance–** Cutting tool material should have ability to resist wear due to adhesion and abrasion.
(iii) **Toughness–** The tool material should have enough toughness to withstand impact loads and vibrations during cutting.
(iv) **Lower friction–** Between tool and chip interface.
(v) Low coefficient of thermal expansion.
(vi) Good thermal conductivity.
(vii) High specific heat.

755. The tools made of cemented carbide wear out faster at

(a) Fast speed (b) Very fast speed
(c) Medium speed (d) Slow speed

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (d) : The tools made of cemented carbide wear out faster at slow speed.

756. Ductile materials produce:

(a) Discontinuous chips
(b) Continuous chips
(c) In-homogenous chips
(d) Continuous chips with built up edge

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : Continuous chips are formed by the continuous plastic deformation of metal without fracture in front of the cutting edge of the tool and deformed by the smooth flow of the chip up the tool face. So, ductile material produces continuous chips.

757. A cutting tool has a nose radius of 2 mm. What is the feed rate for a theoretical surface roughness of 40 microns?

(a) 0.2 mm/rev (b) 0.4 mm/rev
(c) 0.5 mm/rev (d) 0.6 mm/rev
(e) 0.8 mm/rev

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : As we know,
$f = ?$
Nose radius = 2 mm, = R_n
$R_t = 40$ microns

$$R_t = \frac{f^2}{8R_n} \Rightarrow f^2 = 8 \times 2 \times 0.040$$

$$\boxed{f = 0.8\,mm/rev}$$

758. In orthogonal machining operation, the chip thickness and the uncut chip thickness are equal to 0.6 mm. If the tool rake angle is 0°, what is the shear plane angle?

(a) 18° (b) 25°
(c) 30° (d) 45°
(e) 60°

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) : Given, $\alpha = 0°$, $t = t_c = 0.6$

$$r = \frac{t}{t_c} = \frac{0.6}{0.6} = 1$$

$$\tan\phi = \frac{r\cos\alpha}{1 - r\sin\alpha}$$

$$\tan\phi = \frac{1\cos 0°}{1 - 1 \times \sin 0°} = 1$$

$\tan\phi = 1$
Shear plane angle, $\phi = 45°$

759. To increase the metal removal rate by keeping same surface finish, the nose radius of the tool is increased 4 times. The maximum metal removal rate:

(a) remains unchanged (b) becomes double
(c) becomes 4 times (d) becomes 8 times
(e) becomes 16 times

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : As we know, $(R_a)_{max} = \frac{f^2}{8r}$

So, $\frac{r_2}{r_1} = \left(\frac{f_2}{f_1}\right)^2 \Rightarrow \frac{f_2}{f_1} = \sqrt{\frac{4r_1}{r_1}}$

So, become = 2 times

760. As per American Standards Association (ASA) system, the tool singular is given by $\alpha_b - \alpha_s - \theta_e - \theta_s - C_e - C_s - R$ where 'θ_s'_____

(a) Side rake angle
(b) Side relief angle
(c) Side cutting edge angle
(d) none of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : In ASA system of tool signature θ_s is side relief angle.

761. For any given underformed chip thickness a smaller shear angle results in ____ chips

(a) soft
(b) heavily strain hardened
(c) undeformed
(d) none of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Shear angle is of fundamental importance in chip formation. Smaller the shear angle larger the strain, the machining forces and the power requirements.

762. The failure of a cutting tool could be due to

(a) Adhesive wear (b) Abrasive wear
(c) Diffusion wear (d) All of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016
CGPSC AE 16.10.2016

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Ans. (d) : Abrasive Wear– Dislodging of material from the other surface by abrasive action.
Adhesion Wear– High temperature and pressure between chip and tool, metallic bounding takes place at the point of content and fracture occurs due to yielding.
Diffusion Wear– Movement of atoms in metallic crystal from a region of high concentration to low concentration.

763. In a Taylor's differential piece-rate system .....followed.
(a) 8 piece rates are
(b) 1 piece rate is
(c) 2 piece rates are
(d) infinite piece rates are

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : In a Taylor's differential piece-rate system 2-piece rates are followed.

764. The cutting fluid
(a) Cools the tool and work piece
(b) Reduces friction
(c) Improves surface finish
(d) Does all the above
(e) Does none of these

CGPSC AE 16.10.2016

Ans. (d) : Cutting fluid are the fluids which are generally applied while the machining operation is takes place, it helps in cooling both tool & workpiece, reduce friction & improves surface finish.

765. Economics of machining is concerned with
(a) Minimization of machining cost
(b) Maximizing the production rate
(c) Maximizing the profit
(d) All of the above
(e) None of these

CGPSC AE 16.10.2016

Ans. (d) : Economics of machining is concerned with the product mix & the production rate at each manufacturing stages which gives the greatest return on the overhead investment and the running cost including both raw material cost & machine operating cost.

766. In an orthogonal cutting, the cutting force and thrust force were observed to the 400 N and 200 N respectively. If the rake angle of tool is zero, the coefficient of friction in chip-tool interface will be
(a) 2 (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$
(e) None of these

CGPSC AE 16.10.2016

Ans. (b) : We know that,
$F = F_c\sin\alpha + F_t\cos\alpha$
$N = F_c\cos\alpha - F_t\sin\alpha$
F_c & F_t is cutting & thrust force.
Coefficient of friction in chip tool interface,
$$\mu = \frac{F}{N}$$
$$\alpha = 0,\ \mu = \frac{F_t}{F_c} = \frac{200}{400} = \frac{1}{2}$$

767. Which of the following forces are measured directly by strain gauges or force dynamometers during metal cutting?
1. Frictional resistance of the tool against the chip flow acting along the tool face.
2. Vertical force which helps in holding the tool in position.
3. Horizontal cutting force exerted by the tool on the work piece
4. Force exerted by the tool on the chip acting normally to the tool face.
(a) 1 and 3 (b) 1 and 4
(c) 3 and 4 (d) 2 and 4

UPRVUNL AE 21.08.2016

Ans. : (c) 3 and 4

768. In machining process the ratio of thickness of chip to the uncut chip thickness is known as
(a) chip thickness ratio
(b) chip compression factor
(c) cutting ratio
(d) none of these

GPSC Lect. 23.10.2016

Ans. (b) : Chip Thickness Ratio or Cutting Ratio (r)
$$= \frac{\text{Uncut chip thickness before machining}}{\text{Chip thickness after machining}}$$
Chip Reduction Ratio or Coefficient of Chip Contraction (k) or chip compression factor
$$= \frac{\text{Chip thickness after machining}}{\text{Uncut chip thickness before machining}}$$

769. In machining process, the cutting force can be expressed as.....if ϕ is the shear plane angle and α is the rake angle.
(a) $F_C = F_S\cos\phi + F_N\sin\phi$
(b) $F_C = F_S\cos\phi - F_N\sin\phi$
(c) $F_C = F\sin\alpha - N\cos\alpha$
(d) $F_C = F\cos\alpha - N\sin\alpha$

GPSC Lect. 23.10.2016

Ans. (a) : Cutting force (F_c)
$F_c = F_s\cos\phi + F_N\sin\phi$

770. In a cutting operation the cutting speed was reduced by 20%. Assuming n = 0.5 and C = 350 in Taylor's equation, the increase in tool life is
(a) 46% (b) 48%
(c) 59% (d) 56%

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : $V_2 = 0.8\ V_1$
$n = 0.05,\ C = 350$
Taylor's equation $V_1T_1^n = V_2T_2^n$
$$V_1T_1^{0.5} = 0.8V_1T_2^{0.5}$$
$$T_2 = \left(\frac{10}{8}\right)^2 T_1 = 1.56\ T_1$$
% Increase in tool life
$$= \frac{T_2 - T_1}{T_1} = \frac{1.56T_1 - T_1}{T_1} = 56\%$$

771. Which of the following is incorrect for 'continuous chips' formation?
(a) It generates very high frictional heat

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(b) It requires low rake angle of cutting tool
(c) It requires ductile material of work piece
(d) It is difficult to handle

UPRVUNL AE 07.10.2016

Ans. (b) : Continuous chips—Continuous chips are formed by the continuous plastic deformation of metal without fracture in front of the cutting edge of the tool and is formed by the smooth flow the chip up the tool face. Mild steel and copper are considered to be most desirable material for obtaining continuous chips.

- It generates very high frictional heat.
- It requires ductile material of work piece.
- It is difficult to handle.

772. If in a machining operation, cutting speed is reduced to 80% then increase in tool life will be: (In Taylor equation, n = 0.5)

(a) 56% (b) 66%
(c) 46% (d) 70%

UPRVUNL AE 07.10.2016

Ans. (a) : Ist

Cutting speed $V_1 = V$

Tool life $T_1 = T$

IInd

Cutting speed = 0.8V

$T_2 = ?$

n = 0.5

We know that Taylor equation

$$V_1T_1^n = V_2T_2^n = \text{constant}$$

$$V \times (T_1)^{\frac{1}{2}} = (0.8) \times (V) \times (T_2)^{\frac{1}{2}}$$

$$(T_2)^{\frac{1}{2}} = \frac{1}{0.8} \times T_1^{\frac{1}{2}}$$

$$\frac{T_2}{T_1} = \left(\frac{5}{4}\right)^2 = \frac{25}{16}$$

$T_2 = 25x$, $T_1 = 16x$

Then increase in tool life will be

$$= \frac{T_2 - T_1}{T_1} \times 100$$

$$= \frac{25x - 16x}{16x} \times 100 = \frac{9}{16} \times 100 = 56.25\%$$

$$\simeq 56\%$$

773. When 50 mm rod is turning at a spindle speed of 200 rpm the cutting speed would be

(a) 10-12 m/min (b) 12-15 m/min
(c) 15-20 m/min (d) 30-35 m/min

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : D = 50 mm, N = 200 rpm, V = ? m/min

$$V = \frac{\pi DN}{1000} = \frac{\pi \times 50 \times 200}{1000} = 31.4 \text{ m/min}$$

774. Tool life equation for tools under consideration are as follows-
HSS : $VT^{0.2} = 150$, Carbide : $VT^{0.3} = 250$.
where V is cutting speed in m/min and T is the tool life in min. The breakeven cutting speed above which the carbide tool will be beneficial is :

(a) 54 m/min (b) 150 m/min
(c) 194 m/min (d) 250 m/min

RPSC ADE 2016

Ans. (a) : For HSS $T = \frac{150^{1/0.2}}{V^{1/0.2}}$

for carbide tool $T = \frac{250^{1/0.3}}{V^{1/0.3}}$

At break even $T_{HSS} = T_{carbide}$

$$\frac{150^5}{V^5} = \frac{250^{10/3}}{V^{10/3}}$$

V= 54.01 m/min

775. The factor responsible for the formation of continuous chips with built up edge is

(a) High cutting speed and large rake angle
(b) High cutting speed and small rake angle
(c) Low cutting speed and large rake angle
(d) Low cutting speed and small rake angle

MECON MT 2019
RPSC VPITI 14.02.2016

Ans. (d) : The condition that normally induced the formation of BUE are low cutting speed, high feed and low rake angle. One of the prerequisites for the formation of BUE is the work hardenability of the work piece material. Higher the work hardenability, rougher is the machined surface produced.

776. Chip flow over rake face is :

(a) Sliding model
(b) Sticking model
(c) Sliding and sticking model
(d) Sticking and sliding model

TSPSC Managers, 2015

Ans. (d) : Chip flow on rake surface means chip produced due to shearing action during machining over rake surface and 70-80% of total cutting temperature comes from heat produced by rubbing of chip on rake surface.
So, chip flow over rake surface is sticking and sliding model.

777. Identify the wrong statement. If side cutting edge angle of a single point cutting tool increases.

(a) Tool life increases
(b) Chips becomes thinner
(c) Chips becomes wider
(d) Temperature over rake face increases

TSPSC Managers, 2015

Ans. (d) : If side cutting edge angle (C_s) of a single point cutting tool increases.

- tool life increase.
- chips becomes thinner.
- chips becomes wider.

w = width

d = depth

$$w = \frac{d}{\cos C_s}$$

$C_s = \uparrow$

$\cos C_s = \downarrow$

So, $w = \uparrow$ (wider)

Temperature over rake face does not increases. So, option (d) is answer.

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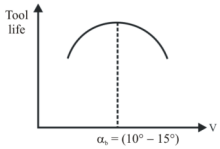
778. Identify the wrong statement related to back rake angle in single point cutting tool. Increase in back rake angle.

(a) Lip angle increases
(b) Tool life initially increases
(c) Contact length on rake face will decrease
(d) Cutting forces will be relatively low

TSPSC Managers, 2015

Ans. (a) : Related to back rake angle (α_b) in single point cutting tool. Increasing in back rake angle (α_b).

- tool life initially increases



- contact length on rake face will decreases.
- cutting force will be relatively low.

Lip angle does not increase due to back rake angle so, option (a) is the answer.

779. Tool life of 10 hrs is obtained when cutting with single point tool at 63 m/min. If Taylor's constant C = 257.35, tool life on doubling the velocity will be

(a) 5 hrs (b) 25.7 min
(c) 38.3 min (d) No change

ISRO Scientist/Engineer 11.10.2015

Ans. (b) : Given, $T_1 = 10$ hrs
$= 10 \times 60$
$= 600$ min
$V_1 = 63$ m/min

Using Taylor's tool life equation

$$\boxed{VT^n = C}$$

$$\therefore \quad V_1 T_1^n = C$$

$$63 \times (10 \times 60)^n = 257.35$$

$$600^n = \frac{257.35}{63}$$

$$600^n = 4.035$$

Taking both side natural logarithmic

$\ln 600^n = \ln 4.035$

$n \ln 600 = \ln 4.035$

$$n = \frac{\ln 4.035}{\ln 600}$$

$n = 0.22$

When velocity is doubled

$V_2 = 2V_1 = 2 \times 63 = 126$ m/min

$$126 \times T_2^{0.22} = 257.35$$

$$T_2^{0.22} = \frac{257.35}{126}$$

$$T_2 = \left[\frac{257.35}{126}\right]^{\frac{1}{0.22}}$$

$= 25.69$ min ≈ 25.7 min

780. During orthogonal machining with a HSS tool, the rake angle is 5°, un-deformed chip thickness is 0.25 mm and width of cut is 4 mm. Assuming shear strength of work material to be 350 N/mm^2 and coefficient of friction as 0.5, determine cutting and thrust force.

(a) 1029.4 N, 406.8 N (b) 1000 N, 500 N
(c) 1110.5 N, 425.8 N (d) 1025.4 N, 410.8 N

ISRO Scientist/Engineer 11.10.2015

Ans. (a) : Shear stress $= (\tau_s) = \frac{F_s}{A_s}$

Area of shear plane $\boxed{A_s = \frac{b}{\sin\phi} \times t_c}$

$$\boxed{F_s = \tau_s \times \frac{bt_c}{\sin\phi}}$$

Friction angle

$$\tan\beta = \mu$$
$$\beta = \tan^{-1}(\mu)$$
$$= \tan^{-1}(0.5)$$
$$\beta = 26.57$$

Apply merchant theory or analysis (single plane theory)

$$\phi = 45° + \frac{\alpha}{2} - \frac{\beta}{2}$$

$$\phi = \frac{90 + \alpha - \beta}{2}$$

$$\phi = \frac{90 + 5 - 26.57}{2} = 34.22°$$

$$\therefore \quad F_s = \tau_s \times \frac{bt_c}{\sin\phi}$$

$$= \frac{350 \times 4 \times 0.25}{\sin 34.22}$$

$F_s = 622.37$ N

Cutting force $(F_c) = R\cos(\beta - \alpha)$

Shear force $(F_s) = R\cos(\phi + \beta - \alpha)$

$$F_c = \frac{F_s \cos(\beta - \alpha)}{\cos(\phi + \beta - \alpha)}$$

$$= \frac{622.37\cos(26.57 - 5)}{\cos(34.22 + 26.57 - 5)}$$

$= 1029.45$ N

Thrust 'force' $(F_T) = F_c \tan(\beta - \alpha)$

$= 1029.45 \tan(26.57 - 5)$

$= 406.96$ N

781. Discontinuous chips occur in case of:

(a) Ductile materials (b) Plastics
(c) Composites (d) Brittle materials

OPSC AEE 2015 Paper-I
UKPSC AE 2007 Paper -I

Ans : (d)

Continuous chip	Dis- continuous chip	Chips with buit-up edge
Ductile workpiece	Brittle workpiece	Ductile workpiece
High Speed	Low Speed	Low Speed
Low feed	High Feed	High feed
Low depth of cut	High depth of cut	High depth of cut
High back rake angle	Low back rake angle	Low back rake angle

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782. For proper cutting, cutting materials does not require .....

(a) Higher hardness (b) Hot hardness
(c) High wear resistance (d) High friction

OPSC AEE 2015 Paper-I

Ans : (d) For proper cutting materials does not requires high friction.
The cutting speed depends upon the work piece and took materials.

783. Continuous chips during machining forms when

(a) speed is high, feed is low
(b) speed is high, feed is high
(c) speed is low, feed is low
(d) speed is low, feed is high

TSPSC AEE 2015

Ans. (a) : Continuous chips during machining forms when speed is high, feed is low.

784. The most important requisite of a cutting tool material is

(a) Carbon percentage
(b) Percentage of alloying elements
(c) Red (hot) hardness
(d) Easy fabrication

GPSC AE 26.07.2015

Ans. (c) : The most important requisite of a cutting tool material is red (hot) hardness.

785. A dynamometer is a device used for the measurement of

(a) chip thickness ratio
(b) forces during metal cutting
(c) wear of the cutting tool
(d) deflection of the cutting tool

APPSC Poly. Lect. 13.03.2020
Oil India Ltd. Sr. Engg. (Production) 30.11.2019
CGPSC AE 25.02.2018
WBPSC AE, 2017, TANGEDCO AE 2015

Ans. (b) : Dynamometer is a device used for measuring force, torque and rpm.

786. For a certain tool it was that the relationship between cutting speed V and tool life T was as follows :

$$V = \frac{1540}{2.73 + T} + C$$

If for a cutting speed of 80 m/min the tool life is 19.27 minutes the cutting speed corresponding to tool life of 6.27 minutes should be :

(a) 283.7 m/min (b) 217.3 m/min
(c) 144 m/min (d) 154 m/min

VIZAG STEEL MT 2015

Ans. (*) : given,

V_1=80 m/min
T_1= 19.27 minutes
V_2= ?
T_2= 6.27 minutes

$$V_1 = \frac{1540}{2.73 + T} + C$$

$$80 = \frac{1540}{2.73 + 19.27} + C$$

$$C = 10$$

$$V_2 = \frac{1540}{2.73 + 6.27} + 10$$

$$= 171.11 + 10 = 181.11 \text{ m/min}$$

None of the given answer correct.

787. Relief angle on HSS tools usually varies from :

(a) 3^o to 10^o (b) 11^o to 15^o
(c) 16^o to 150^o (d) 22^o to 27^o

VIZAG STEEL MT 2015

Ans. (a) :

- Relief angle on HSS tools usually varies from 3^0 to 10^0.
- It minimize rubbing contact between machined surface.
- It helps to eliminate tool breakage and increase tool life.
- Small angle for machining hard material.
- It prevents side flank of tool from rubbing against the work.

788. In an orthogonal machining operation: Uncut thickness = 0.5 mm Cutting speed = 20 m/min Width of cut = 5 mm Chip thickness = 0.7 mm Thrust force = 200 N Cutting force = 1200 N Rake angle = 15° Assume Merchant's theory. The coefficient of friction at the tool-chip interface is:

(a) 0.23 (b) 0.46
(c) 0.85 (d) 0.95

NPCIL ET, 2015

Ans. (b) : In orthogonal machining-
Uncut thickness (t) = 0.5 mm
Cutting speed (v) = 20 m/min
Width (w) = 5 mm, chip thickness (t_c) = 0.7 mm
Thrust force (F_t) = 200N, F_c (cutting force) = 1200 N
Rake angle (α) = 15º
We know that-
According Merchant Cycle Theory,

$F = F_c \sin\alpha + F_t \cos\alpha$
$= 1200 \sin 15^o + 200 \cos 15^o$
$= 310.582 + 193.185$
$F = 503.767$ N
$N = F_c \cos\alpha + F_t \sin\alpha$
$= 1200 \cos 15^o + 200 \sin 15^o$
$= 1159.110 + 51.763$
$N = 1210.873$ N

Coefficient of friction (μ) = F/N

$$= \frac{503.767}{1210.873}$$

$(\mu) = 0.416$

789. The following data relates to an orthogonal turning process: Chip thickness 0.62; Feed 0.2mm/rev; Rake angle 15°. Calculate cutting ratio.

(a) 0.322 (b) 0.4
(c) 0.35 (d) 0.3

NPCIL ET, 2015

Ans. (a) : Orthogonal cutting process $\lambda = 90^o$
Chip thickness (t_c) = 0.62 mm
feed = 0.2 mm/rev
We know that,

$t = f \sin\lambda$

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$t = f \sin 90^\circ = t = f$

$t = 0.2$ mm

Chip thickness ratio/cutting ratio (r) $= \frac{t}{t_c}$

$(r) = \frac{0.2}{0.62}$

Cutting ratio (r) = 0.322

790. In metal cutting at moderate speed, the maximum heat is carried by :

(a) work
(b) tool
(c) chip
(d) Almost equally by all of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Generated in metal cutting is as follows :

$(Heat)_{chip} > (Heat)_{workchip} > (Heat)_{tool}$

So, the most of the heat goes into the moving chip.

- Largest% of heat generated goes to the chip

60% to 70% → chip
10% to 20% → work chip
10% to 20% → Tool

791. In the signature of cutting tools, nose radius is indicated :

(a) in the beginning
(b) in the end
(c) somewhere in the middle
(d) Nose radius not included in tool signature

BPSC Asstt. Prof. 29.11.2015
(KPSC AE 2015)

Ans. (b) : American Standard Association (ASA)–

$\alpha_b - \alpha_s - \gamma_e - \gamma_s - C_e - C_s - R$

α_b – Back Rake angle
α_s – Side Rake Angle
γ_e – End Relief Angle
γ_s – Side Relief Angle
C_e – End cutting Edge Angle
C_s – Side cutting edge Angle
R – Nose Radius

- **Orthogonal Rake System (ORS)–**

$i - \alpha - \gamma_s - \gamma_e - C_e - \lambda - R$

i – Inclination angle
α – Side rake angle
γ_s – Side relief angle
γ_e – End relief angle
C_e – Side cutting edge angle
λ – Principal cutting edge angle
R – Nose Radius.

792. Rake angle may be :

(a) positive (b) negative
(c) zero (d) All of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (d) : Rake angle is the angle of inclination of rake surface from reference plane i.e., normal to horizontal machined surface.

- It allows chip flow direction.
- Rake angle may be positive, negative and zero.
- Zero rake angle is used in cast iron machining due to toughness of cast iron.
- In thread cutting, zero degree rake angles are used.

793. In case of built-up edge in a machining process, which of the following statement is true –

(a) It is an edge provided on the cutting tool by the tool manufacture
(b) It consists of layers of material from the workpiece that are gradually deposited on the tool
(c) A thick built-up edge is desirable and improves cutting efficiency
(d) A thick built-up edge improves the surface finish of the material surface

TSGENCO AE 14.11.2015

Ans. (b) : In case of built-up-edge in a machining process, it consists of layers of material from the workpiece that are gradually deposited on the tool. The built up edge effectively changes tool geometry and rake steepness. It also reduces the contact area between the chip and the cutting tool. The formation of BUE have negative effects on the quality of the workpiece.

794. Which one of the following is true in case of tool life

(a) It is directly proportional to the cutting speed
(b) It is inversely proportional to the cutting speed
(c) Does not depend on the cutting speed
(d) No equation is available that helps roughly calculate the cutting speed

TSGENCO AE 14.11.2015

Ans. (b) : According to taylors tool life equation –

$VT^n = C$

$T^n = \frac{C}{V}$

$T^n \propto \frac{1}{V}$

Where, T = tool life
V = cutting speed

795. Which tool geometry of a single point cutting tool affects surface finish?

(a) Rake angles (b) Relief angles
(c) Cutting edge angles (d) Nose radius

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d) : Nose radius

796. In a typical metal cutting operation, using a cutting tool of positive rake $\gamma = 20^\circ$, it was observed that the shear angle was 30°. The friction angle is

(a) 45° (b) 35°
(c) 25° (d) 40°
(e) 50°

CGPSC AE 26.04.2015 Shift-I

Ans. (e) : Relation between rake, shear and friction angle

$2\phi + \beta - \alpha = \frac{\pi}{2}$

$(2 \times 30) + \beta - 20 = 90$

$\beta = 50^\circ$

797. The lip angle used in a drill for drilling the hard materials

(a) 128° (b) 136°
(c) 90° (d) 60°

TNPSC AE 2014

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Ans. (b) : Lip angle (α) used in a drilling
α for wood → $60^0 - 90^0$,
α for soft and ductile material → 118^0
α for cast iron → 116^0
α for high strength material - carbide → $135^0 - 150^0$.

798. A cutting fluid address two main problems
1. **heat generation at the shear zone and friction zone**
2. **friction at the tool-chip and tool-work interfaces**
3. **reduce the cutting work**
4. **ease of cutting**

Out of above which are correct?
(a) 1 (b) 1 and 2
(c) 1, 3 and 4 (d) 1, 2 and 4
(e) 1 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : A cutting fluid is any liquid or gas that is applied directly to the machining operation to improve cutting performance. Cutting fluids address two main problems
(i) Heat generation at the shear zone and friction zone.
(ii) Friction at the tool-chip and tool work interfaces.

799. Tool life in turning will decrease by maximum extent if the following is doubled
(a) depth of cut (b) feed
(c) cutting velocity (d) tool rake angle
(e) 1 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : By Taylor's tool life equation
$VT^n = \text{constant}$
$V_1T_1^n = V_2T_2^n$
at $V_2 = 2V_1$ and $n = 1$
$V_1T_1 = 2V_1T_2$
$T_2 = \frac{T_1}{2}$
Hence tool life decrease to half as doubled the cutting velocity.

800. In turning, the ratio of the optimum cutting speed for minimum cost and optimum cutting speed for maximum rate of production is always
(a) Equal to 1
(b) In the range of 0.6 to 1
(c) In the range of 0.1 to 0.6
(d) Greater than 1
(e) Equal to 0.4

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : In the range of 0.6 to 1.

801. Serrated chips are observed in machining
(a) Aluminium (b) Titanium
(c) Cast iron (d) Tungsten

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Serrated chips are observed in machining titanium alloy at medium speed.

802. Which of the following conditions does not favour discontinuous chip?
(a) brittle material
(b) large chip thickness
(c) low cutting speed
(d) large rake angle

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : Discontinuous chips tends to be formed when one or more or the following condition exit–
- Brittle material, such as cast iron and bronze
- Large chip thickness
- Low cutting speed
- Small rake angle

803. The cutting speed is minimum when machining _____ with a high-speed steel tool
(a) Cast iron (b) Mild steel
(c) Brass (d) Aluminium

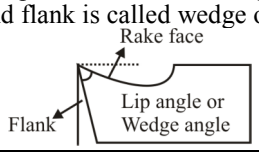
GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : The cutting speed is minimum when machining cast iron with a high-speed steel tool because cast iron is brittle material.

804. The portion of the cutting part enclosed between the face and the flank is called _____
(a) wedge (b) shank
(c) base (d) rake face

Haryana PSC Civil Services Pre, 2014

Ans. (a) : The portion of the cutting part enclosed between face and flank is called wedge or lip.



805. Only two perpendicular components of cutting force act on the tool in case of _____ cutting
(a) oblique (b) orthogonal
(c) 3D (d) inclined

Haryana PSC Civil Services Pre, 2014

Ans. (b) : In case of orthogonal cutting two perpendicular component of force are acting which are cutting force (F_c) and thrust force (F_T).

806. A single point tool has :
(a) rake angle (b) cutting angle
(c) lip angle (d) All of these

MPSC Poly. Lect. 2014

Ans. (d) : All of these

807. Which of the following is performed with a cutting tool moving at a cutting speed "V" in the direction of primary motion?
(a) Cutting
(b) Griding
(c) Non-traditional machining process, utilizing electrical, chemical and optimal sources of energy
(d) Casting

UPRVUNL AE 2014

Ans : (a) Cutting work performed when cutting tool, in the direction of cutting velocity.
$VT^n = C$

808. For a specific cutting energy, if "F" is power required for cutting "Z" is the metal removal rate, then which of the following expressions can be used to calculate the energy per unit volume "U"?
(a) U = F/Z
(b) U = FZ
(c) U = Z/F
(d) Cannot be determined, more data required

UPRVUNL AE 2014

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Ans : (a) Specific cutting Energy $(p) = \frac{F_c}{wt_1}$

Energy Per unit volume $(U) = \frac{F}{Z}$

When Z = metal Removed rate

809. Poor surface finish results due to :

(a) heavy depth of cut (b) low cutting speed
(c) high cutting speed (d) coarse feed

J&K PSC Civil Services Pre, 2013

Ans. (a) : Poor surface finish results due to heavy depth of cut.

810. Identify the correct statements:
In the tool life formula $VT^n = C$, n is
I. 0.1 to 0.5 for high speed steel tools.
II. 0.2 to 0.4 for tungsten carbide tools.
III. 0.4 to 0.6 for ceramic tools.
IV. 1 to 1.5 for alloy tools.
of these statements, which are correct?

(a) I and II (b) II and III
(c) I, II and III (d) All of the above

APPSC Poly. Lect. 2013

Ans. (b) : In the tool life formula, $VT^n = C$
where, V = Cutting velocity
T = Tool life
n = Taylor exponent based on the tool material
C = Constant based on tool and work
n is different for different materials
n = 0.08 to 0.2 for HSS tool
n = 0.1 to 0.15 for cast Alloys
n = 0.2 to 0.4 for Carbide tool
n = 0.5 to 0.7 for ceramic tools

811. According to modified merchant theory

(a) $\beta = \frac{\pi}{4} - \frac{v}{2} + \frac{\alpha}{2}$ (b) $\beta = \frac{\pi}{4} + \frac{v}{2} - \frac{\alpha}{2}$

(c) $\beta = \frac{\pi}{4} - \frac{v}{2} - \frac{\alpha}{2}$ (d) $\beta = \frac{\pi}{4} + \frac{v}{2} + \frac{\alpha}{2}$

where β- shear angle
v - friction angle
α - rake angle

APPSC Poly. Lect. 2013

Ans. (a) : According to modified merchant theory are-

$\beta = \frac{\pi}{4} - \frac{v}{2} + \frac{\alpha}{2}$ $\rightarrow$ β = Shear Angle

α = rake angle
v = friction angle

$\Rightarrow$ According to Lee and Shaffer

$\beta = \frac{\pi}{4} - v + \alpha$

$\Rightarrow$ According to Stabler

$\beta = \frac{\pi}{4} - v + \frac{\alpha}{2}$

812. In a tool wear test with high speed steel cutting tool, the following data were recorded:

Tool life	Cutting speed
30 min	25 m/min
2 min	70 m/min

The Taylor's equation is

(a) $30 \times 25^{2.98} = 70 \times 2^{2.98}$
(b) $25 \times 30^{2.98} = 70 \times 2^{2.98}$
(c) $25 \times 30^{2.98} = 2 \times 70^{2.98}$
(d) $30 \times 25^{2.98} = 2 \times 70^{2.98}$

APPSC Poly. Lect. 2013

Ans. (b) :

Tool life	Cutting speed
30 min	25 m/min
2 min	70 m/min

Tool wear test with HSS, Cutting tool-
n(HSS) = 0.08 to 0.2
$VT^n = C$
$\rightarrow$ $25 \times 30^n = 70 \times 2^n$

813. A larger rake angle will result in:

(a) Higher shear plane angle and lower cutting force.
(b) Lower shear plane angle and lower cutting force.
(c) Higher shear plane angle and higher cutting force.
(d) Lower shear plane angle and higher cutting force.

RPSC AEN Pre-2013

Ans. (a) : A larger rake angle will result
- High shear plane angle (ϕ) and lower cutting force (F_c).
- Lower velocity, lower power.
- Reduce shear strain.
- Less tendency of BUE (Built up edge)
- Low cutting temperature and continuous chip.

814. As per laws of adhesive wear the volume of the wear material is proportional to…

(a) the yield stress
(b) the distance of travel
(c) the load
(d) the distance of travel and load

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : As per laws of adhesive wear the volume of the wear material is proportional to the distance of travel and the load.

815. Continuous chips with built-up edge are formed during the machining of

(a) brittle materials (b) ductile metals
(c) hard metals (d) soft metals

JPSC AE 2013, Paper-V

Ans : (b) : Continuous chips with built-up edge are formed during the machining of ductile materials.

816. The quality of machined surface depends on:-

(a) The material of the workpiece
(b) Rigidity of machine work-tool system
(c) Cutting conditions
(d) All of the above

UKPSC AE-2013, Paper-I

Ans. (d) : All of the above.

817. A cylinder of 25 mm diameter and 100 mm length is turned with a tool, for which the relation $VT^{0.25} = 55$ is applicable. The cutting velocity is 22 m/min. For a tool feed of 0.046 mm/rev, the number of tool regrinds required to produce 425 cylinders is

(a) 12 (b) 22
(c) 43 (d) 85

Vizag Steel MT (Re-Exam) 24.11.2013

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Ans. (d) : Given,

$d = 25$ mm, $L_e = 100$ mm, $v = 22$ m/min,

$f = 0.046$ mm/rev, No. of cylinder = 425

$$v = \frac{\pi DN}{1000} = \frac{\pi \times 25 \times N}{1000}$$

$\therefore$ $N = 280$ rpm

Machining time for turning is given by–

$$T_m = \frac{L_e}{f \times N}$$

$$T_m = \frac{100}{0.046 \times 280} = 7.76 \text{ min}$$

$$22 \times T^{0.25} = 55$$

$$\boxed{T = 39.0625 \text{ min}}$$

No. of tool regrind

$$= \frac{\text{Machining time of one cylinder} \times \text{No. of cylinder}}{\text{Tool life}}$$

$$\text{No. of tool regrind} = \frac{7.76 \times 425}{39.0625} = \frac{3298}{39.0625} = 84.56 \approx 85$$

818. Which part of the cutting tool is prone to crater wear ?

(a) Base (b) Shank

(c) Flank (d) Face

TNPSC AE (Industries) 09.06.2013

Ans. (d) :

819. The process of shearing in a cutting process takes place

(a) at tool-chip interface

(b) along work surface behind cutting edge

(c) along work surface behind cutting edge

(d) along shear plane inclined to tool chip interface and work surface

TNPSC ACF 2012

Ans. (d) : The process of shearing in a cutting process takes place are along shear plane inclined to tool chip interference and work surface.

820. Materials that produce continuous chip are

(a) cast iron

(b) brass castings

(c) low carbon steel

(d) ductile materials with coarse feed and small rake angle

TNPSC ACF 2012

Ans. (c) : For continuous chip are-

Work material – Ductile (low carbon steel)

Cutting velocity – High

Feed – low

Rake angle – Positive and high

Cutting fluid – Both cooling and lubrication.

821. An ideal tool material is one which

(a) has very long life

(b) has same life at all cutting speeds

(c) produces no heat during cutting

(d) produces smooth surface finish

TNPSC ACF 2012

Ans. (b) : An ideal tool material is one which has same life at all cutting speeds.

822. What is approximate percentage change in the life of a tool (t) with zero rake angle used in orthogonal cutting when its clearance angle (α) is changed from 10° to 7°?

(a) 30% increase (b) 30% decrease

(c) 70% increase (d) 70% decrease

VIZAG STEEL MT 10.06.2012, GATE 1999

Ans. (b) : Wear rate is proportional to tangle i.e., wear rate $\alpha \tan \gamma$.

Where, γ = clearance angle.

- When clearance angle decreased from 10° to 7°, then

$$\text{Percentage Change in tool life} = \frac{\tan 10^0 - \tan 7^0}{\tan 10^0} \times 100$$

$$= 30.37\% \text{ decrease}$$

$$\boxed{\text{Percentage Change in tool life} \cong 30\% \text{ Decrease}}$$

823. The magnitude of cutting power consumption depends mainly on the cutting tool's

(a) Rake angle (b) Cutting angle

(c) Clearance angle (d) Lip angle

APPSC AEE 2012

Ans. (a) : Positive rake- helps reduce cutting force and thus cutting power. the farm of machined chips depends mainly upon cutting power consumption.

824. In metal cutting at speed above 20 m/min, maximum heat is carried away by

(a) Work piece (b) Tool

(c) Chip (d) Equally by all

APPSC AEE 2012

Ans. (c) : In metal cutting at speed above 20m/min, maximum heat is carried away by the chip.

825. Tool life testing on a lathe under dry cutting condition gave 'n' and 'c' of Taylor's tool life equation as 0.12 m/min and 130 m/min respectively. When a coolant was used, 'C' decreased by 10%. The percentage or increase in tool life with the use of coolant at a cutting speed of 90 m/min is

(a) 21.29% (b) 128.29%

(c) 121.29% (d) 26.29%

TRB Asstt. Prof., 2012

Ans. (*) : Given, $n = 0.12$, $C = 130$

$v_1 = 90$ m/min

$$vT_1^n = C$$

$$90(T_1)^{0.12} = 130$$

$T_1 = 21.42$ minutes

When coolant was used, 'C' decreased by 10%

$C_{new} = 0.9 \times 130 = 117$

$$vT_2^n = C_{new}$$

$$90(T_2)^{0.12} = 117$$

$T_2 = 8.902$ minutes

$$\% \text{ change} = \frac{T_2 - T_1}{T_1} \times 100$$

$$= \frac{8.902 - 21.42}{21.42} \times 100 = -58.44\%$$

826. A good machinability rating would indicate

(a) long tool life, high power requirement and less machining time.

(b) long tool life, low power requirement and a good surface finish.

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(c) short tool life and a good surface finish.
(d) long tool life, high power requirement and a good surface finish.

UKPSC AE 2012 Paper-I

Ans. (b) : long tool life, low power requirement and a good surface finish.

827. Continuous chips will be formed when machining speed is
(a) low (b) medium
(c) high (d) independent of speed

UKPSC AE 2012 Paper-I

Ans. (c) : high

828. A cutting tool is turning a work piece of 40 mm diameter, revolving at 300 rpm. If tool life is 120 min, find the value of constant C as per the Taylor's tool life equation, Assuming n = 1/7.
(a) 85 (b) 80
(c) 70 (d) 75

UKPSC AE 2012 Paper-I

Ans. (d) : Given,
D = 40 mm
N = 300 rpm
T = 120 minutes
$VT^n = C$

$$\left(\frac{\pi DN}{1000}\right)(120)^{1/7} = C$$

$$\frac{\pi(40)(300)}{1000}(120)^{1/7} = C$$

$C = 74.70$

$\boxed{C = 75}$

829. Which of the following should be more to reduce wear of a tool ?
(a) Weight (b) Density
(c) Hardness (d) (b) & (c) both

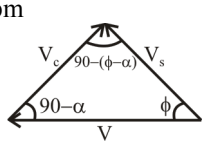
UKPSC AE 2012 Paper-I

Ans. (c) : Hardness

830. The rake angle of a cutting tool is 15°, the shear angle is 45° and the cutting velocity is 35 mpm. What is the velocity of chip along the tool face ?
(a) 28.5 mpm (b) 27.3 mpm
(c) 25.3 mpm (d) 23.5 mpm

UKPSC AE 2012 Paper-I

Ans. (a) : Given,
$\alpha = 15°$
$\phi = 45°$
v = 35 mpm



$$\frac{V}{\cos(\phi-\alpha)} = \frac{V_c}{\sin\phi}$$

$$V_c = \frac{V\sin\phi}{\cos(\phi-\alpha)}$$

$$= \frac{35\times\sin 45}{\cos(45-15)}$$

$\boxed{V_c = 28.5\,\text{mpm}}$

831. What effect does the low cutting speed have on the carbide tool/workpiece ?
(a) Reduces tool life
(b) Increases tool life
(c) Have no effect on tool life
(d) Spoils the work piece

VIZAG STEEL MT 2011

Ans. (a) : For carbide tool/workpiece low cutting speed reduces tool life.

832. In metal machining, the zone where the heat is generated due to friction between the moving chip and the tool face is called
(a) Friction zone
(b) Work-tool contact zone
(c) Shear zone
(d) Tool contact zone

VIZAG STEEL MT 2011

Ans. (a) : In metal machining, the zone where the heat is generated due to friction between the moving chip and the tool face is called friction zone.

833. Which of the following terms best describes cemented carbide?
(a) Ceramic (b) Cermet
(c) Composite (d) Metal

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Cemented carbide belong to a class of hard, wear resistance, refractory materials in which the hard carbide particle are bound together, or cemented by a ductile metal binder. Cermet refers to a composite of ceramic material with a metallic binder.

834. A cutting tool having tool signature as 10, 9, 6, 6, 8, 8, 2 will have side rake angle
(a) 10° (b) 9°
(c) 8° (d) 2°

ISRO Scientist/Engineer 2011

Ans. (b) : Tool signature or Tool designation (ANSI) or (ASA)

$$\alpha_b - \alpha_s - \gamma_e - \gamma_s - C_e - C_s - R$$

$10 - 9 - 6 - 6 - 8 - 8 - 2$

Side Rake Angle = 9

835. A 50mm diameter steel rod was turned at 284 rpm and tool failure occurred in 10 minutes. The speed was changed to 232 rpm and the tool failed in 60 minutes. Assuming straight line relationship between cutting speed and tool life, the value of Taylorian Exponent is
(a) 0.21 (b) 0.13
(c) 0.11 (d) 0.23

ISRO Scientist/Engineer 2011

Ans. (c) : By Taylor's Equation
$T_1 = 10$, $N_1 = 284$ rpm
$T_2 = 50$, $N_2 = 232$ rpm

$$V_1T_1^n = V_2T_2^n$$

$$\pi DN_1(10)^n = \pi DN_2(60)^n$$

$n = 0.11287 \approx 0.11$

836. The cutting fluid mostly used for machining steel is :
(a) Water (b) Soluble Oil
(c) Dry (d) Heavy Oils

OPSC Civil Services Pre. 2011

Ans. (b) : The cutting fluid mostly used for machining steel is soluble oil.

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837. The following is true about the built-up edge.
(a) It forms at high speeds
(b) It forms due to low friction
(c) It improves tool life
(d) If improves surface finish

UPSC JWM Advt. No.-52/2010

Ans. (c) : Built-up edge improves tool life.

838. The tool life increases
(a) With decrease in nose radius
(b) With decrease in hot hardness
(c) Increase in cutting velocity
(d) By adding lead to steel

UPSC JWM Advt. No.-52/2010

Ans. (d) : The tool life increases by adding lead to steel Because lead acts as an internal Lubricant, reducing friction, permits higher speeds as its Lubricating action, due to which it prolong or increases tool life.

839. A steel bar of 50 mm diameter was turned at 284 r.p.m. and the tool failure occurred in 10 minutes. The speed was changed to 232 r.p.m. and the tool failed in 60 minutes. What cutting speed should be used to obtain a 30 minutes tool life?
(a) 252 r.p.m. (b) 174 r.p.m.
(c) 228 r.p.m. (d) 302 r.p.m.

APPSC IOF, 2009

Ans. (a) : Given,

$N_1 = 284$ rpm, $T_1 = 10$ min
$N_2 = 232$ rpm, $T_2 = 60$ min
$N_3 = ?$ $T = 30$

Using Taylor's tool life equation,

$VT^n = C$

$284 \times (10)^n = 232 \times (60)^n$

$$\frac{71}{58} = (6)^n$$

$$\ln\left(\frac{71}{58}\right) = n\ln(6) \qquad \boxed{n = 0.11}$$

$$\Rightarrow \quad N_3 \times (30)^{0.11} = 232 \times (60)^{0.11}$$

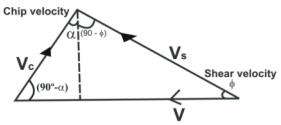
$$\boxed{N_3 = 250.38\,\text{rpm}}$$

Nearest option is (a).

840. The rake angle of a cutting tool is 15º, shear angle 45º and cutting velocity 35 m/min. What is the velocity of chip along the tool face?
(a) 25.3 m/min (b) 27.3 m/min
(c) 28.5 m/min (d) 23.5 m/min

ISRO Scientist/Engineer 2009

Ans. (c) :



Shear angle $\phi = 45°$
Rake Angle $\alpha = 15°$
V = Cutting velocity = 35 m/min

$$\frac{V}{\sin(90-\phi+\alpha)} = \frac{V_c}{\sin\phi}$$

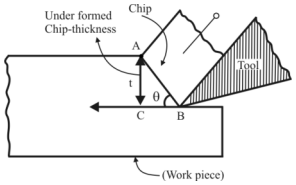
$$V_c = \frac{35}{\sqrt{2}\left(\frac{\sqrt{3}}{2}\right)}$$

$= 28.5$ m/min

841. If the underformed chip thickness is 2mm and shear angle equal to 30°, then what is the length of the shear plane?
(a) 1 mm (b) 2 mm (c) 4 mm (d) 8 mm

UPSC JWM Adv. No-16/2009

Ans. (c) :



Given,

t = Undefomed chip thickness = 2mm
ϕ = shear angle
$= 30°$

let, AB = length of shear plane.

$\because \quad \frac{AC}{AB} = \sin\phi$

$$\Rightarrow \quad AB = \frac{AC}{\sin\phi} = \frac{t}{\sin\phi}$$

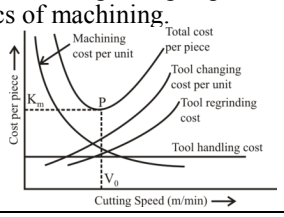
$$\Rightarrow \quad AB = \frac{2mm}{\sin 30°} = 4mm$$

$\Rightarrow$ Length of shear plane (AB) = 4mm

842. Which one of the following costs remain constant in an economics of machining?
(a) Tool cost per piece
(b) Tool changing cost per piece
(c) Tool handling cost per piece
(d) Machining cost per piece

UPSC JWM Adv. No-16/2009

Ans. (c) : Tool handling cost per piece remain constant in an economics of machining.



843. The cutting tool that performs orthogonal cutting is :
(a) twist drill (b) broach
(c) helical milling cutter
(d) single point cutting tool for turning

DRDO Scientists 2009

Ans. (d) : The cutting tool that performs orthogonal cutting is single point cutting tool for turning.

844. By which one of the following the coefficient of friction between chip and tool can be reduced?
(a) By reducing the effective rake angle
(b) By reducing the width of the tool
(c) By reducing the depth of cut
(d) By reducing the clearance angle

UPSC JWM Adv. No-16/2009

Ans. (a) : The coefficient of friction between chip and tool can be reduced by reducing the effective rake angle.
We know,

$\mu = \tan\beta$

Varying α will affect β as per merchant theory and hence μ will also be affected.

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845. A cutting tool having tool signature as 10, 10, 10, 6, 6, 8, 8, 2 will have rake angle as

(a) 2 (b) 6
(c) 8 (d) 10

WBPSC AE 2008

Ans. (d) : According to ASA nomenclature

$\alpha_b - \alpha_s - \gamma_c - \gamma_s - C_e - C_s - R$

where

α_b = back R_a/c_e angle
α_s = side R_a/c_e angle
γ_c = end Relief angle
γ_s = side Relief angle
c_e = end cutting edge angle
c_s = side cutting edge angle
R = Nose Radius

846. On which of the following parameters in the design of cutting tool based?

(a) Bending stress (b) Tensile strength
(c) Rigidity (d) Compactness

WBPSC AE 2008

Ans. (c) : Rigidity

847. On which factors does the machinability depend?

(a) Life of tool
(b) Cutting forces
(c) Work piece profile
(d) Physical and mechanical properties, compression of work piece material and microstructure

WBPSC AE 2008
ISRO Scientist/Engineer 2007

Ans. (d) : machinability depends on
(i) Physical and mechanical properties.
(ii) Compression of work material
(iii) Microstructure

848. Regarding tool life, which the following statements is correct?

(a) It is measured by the volume of material removed between tool resharpening
(b) It is the time for which the tool is in contact with the work piece
(c) It is the time between tool sharpening
(d) All of the above

WBPSC AE 2008

Ans. (d) : All of the above

849. Why negative rake angles are used?

(a) To give better finish
(b) To decrease the temperature at the tool tip
(c) To increase the strength of the cutting tool point
(d) All of the above

WBPSC AE 2008

Ans. (c) : Negative rake angles are used to increase the strength of the cutting tool point.

850. What is the correct sequence of the following parameters in order of their maximum to minimum influence on tool life?

1. Feed rate
2. Depth of cut
3. Cutting speed

Select the correct answer using the code given below

(a) 1, 2, 3 (b) 3, 2, 1
(c) 2, 3, 1 (d) 3, 1, 2

ISRO Scientist/Engineer 2008

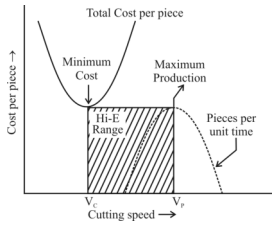
Ans. (d) : Cutting speed > Feed rate > Depth of cut

851. For a single-pass turning process for which the tool life decreases with increase in cutting speed, the optimum cutting speed for maximum production rate is

(a) always less than the optimum cutting speed for minimum cost per piece
(b) always more than the optimum cutting speed for minimum cost per piece
(c) always equal to the optimum cutting speed for minimum cost per piece
(d) less than the optimum cutting speed for minimum cost per piece, if the tool cost is very high

DRDO Scientists 2008

Ans. (b) :



High efficiency range of cutting speed

852. In orthogonal machining process, the shear angle decreases if the

(a) rake angle decreases
(b) rake angle increases
(c) friction at the tool-chip interface decreases
(d) friction at the tool-work interface decreases

DRDO Scientists 2008

Ans. (b) : In orthogonal machining process, the shear angle decreases if the rake angle increases.

853. The following is not true for the tool life equation:

(a) The smaller the value of 'n' steeper the slope of log V-log T line.
(b) For ideal tool material n = 1
(c) The larger the value of C smaller is the tool life
(d) The range of variation of n for HSS is 0.08 – 0.20

UKPSC AE 2007 Paper -I

Ans. (c) : The larger the value of C smaller is the tool life

854. Chip velocity in machining of metals is equal to

(a) rV (b) rV_s
(c) $\frac{V}{r}$ (d) $\frac{V_s}{r}$

where V = cutting of velocity
V_s = velocity along the shear plane
r = chip thickness ratio

UKPSC AE 2007 Paper -I

Ans. (a) : rV

855. If the cutting speed is increased, what will happen to the built-up edge?

(a) It becomes longer
(b) It becomes smaller and may vanish later
(c) It does not form

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(d) Has no relation to speed

UKPSC AE 2007 Paper -I

Ans. (b) : It becomes smaller and may vanish later.

856. The metal removal during machining is by:

(a) cutting (b) tearing
(c) shearing (d) plastic distortion

J & K PSC Screening, 2006

Ans. (d) : The metal removal during machining is by plastic distortion.

857. In Taylor's tool life equation, $VT^n = C$, index 'n' depends upon

(a) material of work-piece
(b) condition of machine
(c) material of tool
(d) coolant used

OPSC Civil Services Pre 2006

Ans. (c) : Index 'n' in Taylor's tool life equation is depends on tool material.

858. Merchant's theory of metal cutting is based on the assumption that

(a) The shear stress is maximum at the shear plane and shear will take place in a direction in which the energy required for shearing is maximum
(b) The cutting energy depends on microstructures and coefficient of friction
(c) During cutting operation electrons are disintegrated causing generation of heat and new surfaces which are softer than original
(d) Shear occurs in a single plane and there exists a shear plane which separates the chip and work-piece

OPSC Civil Services Pre 2006

Ans. (d) : Merchant Theory Assumption–
(i) That expenditure is minimum in process i.e shear will take place in a direction in which energy required for shearing is minimum.
(ii) That shear stress is maximum at the shear plane and it remains constant.
(iii) Shear surface is a plane extending upwards from the cutting edge.
(iv) The tool is perfectly sharp and there is no contact along the clearance face.
(v) Width of the tool is greater than that of the work.

859. Consider the following statements

1. Very high production rate
2. Lower dimensional and form accuracy and surface finish of the product
3. Roughing and finishing in single stroke of the same cutter
4. Needs only two motion for cutting, so design, construction, operation and control are simpler

Which of these are correct about Broaching

(a) 1, 2 and 4 (b) 1, 2 and 3
(c) 1 and 2 (d) 1 and 3
(e) 1, 2, 3 and 4

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Advantages and limitations of broaching
Major advantages
(i) Very high production rate (much higher than milling, planning, boring etc.)
(ii) High dimensional and form accuracy and surface finish of the product.
(iii) Roughing and finishing in single stroke of the same cutter.
(iv) Needs only one motion (cutting), so design, construction, operation and control are simpler.
(v) Extremely suitable and economic for mass production.

860. If the cutting speed is reduced by 40%, when the Taylor's tool life exponent n = 0.5, then the ratio of new tool life to original tool life is ____.

(a) 2.5 (b) 6.25
(c) 5 (d) 4

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (*) : Given,
$V_1 = V$

$$V_2 = \frac{60}{100}V = 0.60V$$

Taylor's tool life exponent, n = 0.2
According taylor's tool life –

$$T1\ V_1^n = T_2 V_2^n$$

$$\frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^n$$

$$\frac{T_1}{T_2} = \left(\frac{0.60V}{V}\right)^{0.5}$$

$$\frac{T_1}{T_2} = 0.744$$

861. In an orthogonal turning operation with zero rake angle tool, the shear plane angle in radian for minimum shear strain is:

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

AAI Jr. Executive 26.03.2021

Ans. (d) : Shear strain,

$$\boxed{\gamma = \cot\phi + \tan(\phi - \alpha)}$$

For min shear strain with zero rake angle.

Shear angle, $\phi = \frac{\pi}{4}$

$$\boxed{\gamma_{min} \geq 2}$$

862. The Taylor's tool life exponent n is 0.4 and tool changing time is 2 minutes. For maximum production rate the tool life in minutes is:

(a) 1 (b) 6
(c) 3 (d) 2

AAI Jr. Executive 26.03.2021

Ans. (c) : Tool life exponent = 0.4
T_c = 2 minutes
For max production,

$$\text{Tool life } (T_o) = T_c\left(\frac{1-n}{n}\right)$$

$$= 2\left(\frac{1-0.4}{0.4}\right) = 3 \text{ minutes}$$

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863. A cutting tool has signature as per ASA, 6° – 11° – 10°– 7° – 10° – 30° – 0.5mm. The difference in side cutting edge angle and end cutting edge angle is:

(a) 19° (b) 20° (c) 23° (d) 24°

AAI Jr. Executive 26.03.2021

Ans. (b) : 6° – 11° – 10°– 7° – 10° – 30° – 0.5

α_b = 6° (Back rake angle)

α_s = 11° (Side rake angle)

γ_e = 10° (End relief angle)

γ_s = 7° (Side relief angle)

C_e = 10° (End cutting edge angle)

C_s = 30° (Side cutting edge angle)

R = 0.5 (Nose radius)

The difference in SCEA and ECEA is 30° – 10° = 20°

864. _____ system for cutting is a tool-in-use type system for machining with a single point cutting tool.

(a) ASA (b) Orthogonal

(c) Coinciding (d) parallel

UPRVUNL AE 05.07.2021

Ans. (b) :

865. _____ are used in cutting operations for breaking of continuous chips formed during machining

(a) Band saws (b) chip breakers

(c) chip smashers (d) Hacksaws

UPRVUNL AE 05.07.2021

Ans. (b) : Chip breakers are used in cutting operations for breaking of continuous chips formed during machining.

866. For a machining system, the cutting speed is 20m/min, Taylor exponent is 1 and the Taylors coefficient is 2000. the tool life in seconds is.

(a) 6000 (b) 1980

(c) 100 (d) 2020

UPRVUNL AE 05.07.2021

Ans. (a) : Given, V = 20 m/min., n = 1, C = 2000

$VT^n = C$

$20 \times T^1 = 2000$

$T = \frac{2000}{20} = 100\,\text{min}$

T = 100 × 60 = 6000 sec.

867. Which of the following is NOT a factor responsible for continuous chips?

(a) small rake angles

(b) Sharp cutting edges

(c) small depth of cuts

(d) High cutting speeds

UPRVUNL AE 05.07.2021

Ans. (a) : Factors responsible for continuous chips-

- Sharp cutting edges
- Small depth of cuts
- High cutting speeds

868. The primary shear zone during metal machining lies between the :

(a) cutting tool & metal dust

(b) cutting tool & metal work piece

(c) metal chip & cutting tool

(d) metal work piece & metal chip

BHEL ET 2019

Ans. (d) :

- Primary shear zone during metal cutting machining lies between metal workpiece and metal cutting chip.
- Secondary shear zone during metal cutting lies between cutting tool and chip. Secondary shear zone exerted due to friction between cutting tool and cutting chip.

869. The angle between side cutting edge and side of the tool shank is known as :

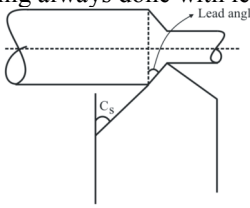
(a) Lead angle (b) Rake angle

(c) Nose angle (d) Clearance angle

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : Lead Angle- A common name for the side cutting edge angle. It lies between side cutting edge and side of the tool shank.

Actual machining always done with lead angle.



870. A built up edge is formed while machining :

(a) Ductile material at high speed

(b) Ductile material at low speed

(c) Brittle material at high speed

(d) Brittle material at low speed

OPSC AEE 2019 Paper-I

Ans : (b) : Chips with built-up edge-

- Ductile materials
- Low speed
- High feed
- High depth of cut

871. Carbide tipped tools usually have

(a) negative rake angle (b) positive rake angle

(c) any rake angle (d) no rake angle

GPSC Executive Engineer 23.12.2018

Ans. (a) : Carbide tipped tools usually have negative rake angle.

872. The relationship of tool life and cutting speed is expressed as $VT^n = C$. Which of the following is not true for this relation?

(a) V=cutting speed (m/min), T=temperature (oC)

(b) V = cutting speed (m/min), T= tool life (min)

(c) Tool life decreases at high cutting speeds

(d) 'C' is numerically equal to the cutting speed that gives tool life of 1 min

(e) It is known as Taylor tool life equation

CGPSC AE 25.02.2018

Ans. (a) : Taylor tool life equation–

$VT^n = C$

where,

V – cutting speed (m/min)

T – Tool life (min)

C – Numerically equal to the cutting speed that gives tool life of 1 min.

n – Exponent depend on tool material

n = 0.08 - 0.2 for HSS

n = 0.1 - 0.15 for Cast Alloys

n = 0.2 - 0.41 for carbide tool

n = 0.5 - 0.7 for ceramic tool

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873. Why does crater wear start at some distance from the tool tip?

(a) Tool strength is minimum at that region
(b) Cutting fluid cannot penetrate that region
(c) Tool temperature is maximum in that region
(d) Stress on rake face is maximum at that region

RPSC AE 2018

Ans. (c) : Crater wear start at some distance from the tool tip because of tool temperature is maximum in that region.

Crater wear—

- Crater wear occur on the rake face.
- At very high speed crater wear predominates
- For crater wear, temperature is main culprit and tool defuse into chip material and tool temperature is maximum at some distance from the tool tip.

874. Tool-life criterion normally used is:

(a) Flank wear
(b) Crater wear
(c) Crater wear and flank wear
(d) Flank wear and nose radius

TRB Polytechnic Lecturer 2017

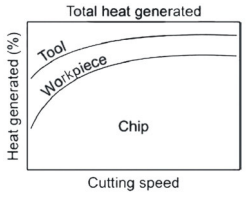
Ans. (b) : Crater wear is used normally for tool life criterion.

875. In turning operation, the typical ratio of heat generated in chip, tool and work piece is in the order of:

(a) 70 : 20 : 10 (b) 10 : 70 : 20
(c) 10 : 20 : 70 (d) 70 : 10 : 20

CIL MT 26.03.2017

Ans. (d) :



b

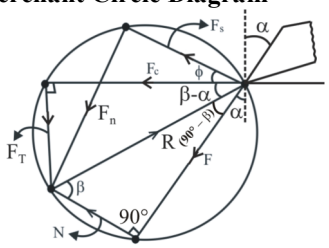
It is clear from the figure that at high speed, maximum heat is carried away by the chip, then work piece and least is carried by tool.
So the ratio 70: 10: 20 will be correct.
Note-CIL gives option (a) as correct answer.

876. Graphical method to find forces acting on single point cutting tool is known as :

(a) Merchant circle diagram
(b) Euler circle diagram
(c) None of three mentioned here.
(d) Newton circle diagram.

GWSSB AAE, 27.12.2015

Ans. (a) : Merchant Circle Diagram–



F_c = cutting force
F_n = Normal force on shear plane
R = Resultant force
F_T = thrust force
F = friction force
N = Normal friction force on tool chip interface
ϕ = shear angle
β = friction angle
α = Rake angle.

877. Which tool reference system is generally easy to use for re-sharpening the cutting tool using grinding?

(a) ORS (b) NRS
(c) ASA (d) WRS

HPPSC Asstt. Prof. 2014

Ans. (b) : NRS system is used for re-sharpening the cutting tool using grinding.

878. In ASA system, if the tool nomenclature is 8-6-5-5-10-15-2 mm, then the side rake angle will be :

(a) 5° (b) 6°
(c) 8° (d) 10°

HPPSC Asstt. Prof. 2014

Ans. (b) : Back rake angle– 8°
Side rake angle- 6°
End relief angle- 5°
Side relief angle- 5°
End cutting edge angle- 10°
Side cutting edge angle- 15°
Nose radius- 2 mm.

879. If the value of rake angle is a large positive value, then :

(a) material is not deformed
(b) material is forced back to itself
(c) material is deformed less in the chip
(d) material is deformed more in the chip

UPRVUNL AE 2014

Ans : (c) The value of rake angle is a large positive value then material is deformed less in the chip.
At higher positive rake angle, a tool material deforms less in the chip as the tool losses strength.

880. Which of the following is not the assumption in Merchant's theory :-

(a) Tool is perfectly sharp
(b) Shear is occurring on a plane
(c) Uncut chip thickness is constant
(d) A continuous chip with built up edge (BUE) is produced

UKPSC AE-2013, Paper-I

Ans. (d) : A continuous chip with built up edge (BUE) is produced, not a assumption in merchant's theory.

881. Which of the following are the reasons for reduction of tool life in a machining operation?

1. Temperature rise of cutting edge.
2. Chipping of tool edge due to mechanical impact.
3. Gradual wear at tool point.
4. Increase in feed of cut at constant cutting force

Select the answer from the following :

(a) 1, 2 & 4 (b) 1, 2 & 3
(c) 1, 3 & 4 (d) 1, 2, 3 & 4

UKPSC AE 2012 Paper-I

Ans. (b) : 1, 2 & 3

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882. In machining processes, the percentage of heat generated in shear action is carried away by the chips to the extent of:-

(a) 10% (b) 25%
(c) 50% (d) 90%

UPPSC AE 13.12.2020 Paper-I
UKPSC AE-2013, Paper-I

Ans. (d) : 90%

883. The maximum height of surface roughness H_{max} produced by a single point produced by a single point turning tool having feed, *f*mm/rev and nose radius, R mm is

(a) $\frac{f^2}{8R}$ (b) $\frac{f}{8R}$
(c) $\frac{f^2}{2R}$ (d) $\frac{f}{2R}$

PPSC Asstt. Municipal Engineer 15.06.2021
UKPSC AE 2007 Paper -I

Ans. (a) : $\frac{f^2}{8R}$

884. Which angle prevents the sudden engagement of the entire depth of cut when the tool enters the work material?

(a) Side cutting edge angle
(b) Side rake angle
(c) End cutting edge angle
(d) Back rake angle

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Side cutting edge angle-
Side cutting edge angle (SEGA) prevents the sudden engagement of the entire depth of cut when the tool enters the work material. As a result it affects the resulting tool life and surface finish. It can vary from 0º to 90º. When the SCEA is increased the entry of the tool is smooth to start the cut since the depth of cut would be gradually increasing until the entire cutting edge is in contact with the workpiece.

885. The carbide tools operating at very low cutting speeds (below 30 m/min)

(a) reduces tool life
(b) increases tool life
(c) have no effect on tool life
(d) spoils the workpiece

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : The carbide tool operating at very low cutting speed below 30 m/min reduces tool life.

886. A cutting tool has a nose radius of 1.8 mm. The feed rate for a theoretical surface roughness of R_t = 5 micron is

(a) 0.360 mm/rev (b) 0.268 mm/rev
(c) 0.036 mm/rev (d) 0.0187 mm/rev

RPSC IOF, 2020

Ans. (b) : Nose radius = 1.8 mm
Feed rate = ?
R_t = 5 microns = 0.005 mm

$$0.005 = \frac{F^2}{8R_t}$$

$$F = \sqrt{0.005 \times 8 \times 1.8}$$

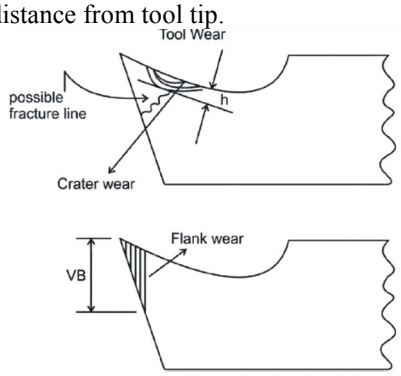
F = 0.268 mm/rev

887. In a turning tool, crater wear occur on _____.

(a) Base (b) Flank face
(c) Rake face (d) Shank

CIL MT 26.03.2017

Ans. (c) : Crater wear occur on the rake face. For crater wear temperature is the main culprit and tool diffuse into the chip material and tool temperature is maximum at some distance from the tool tip. So crater wear starts at some distance from tool tip.



888. The recommended average cutting speed in meters per minute for HSS tool for cutting mild steel is

(a) 15 (b) 30 (c) 60 (d) 90

TNPSC AE 2014

Ans. (b) :

Material	speed (m/min)
Diamond	200 - 2000
CBN	30 - 310
Ceramic	150 - 650
Carbides	30 - 150
High speed steel	10 - 60

889. The metal in machining operation is removed by :

(a) Tearing chips
(b) Distortion of metal
(c) Shearing the metal across a zone
(d) Cutting the metal across a zone

(HPPSC AE 2014)

Ans : (c) The metal in machining operation is removed by shearing the metal across a zone.
To remove metal by machining the tool first penetrate the surface once the tool has penetrated the surface, the chip formed should readily break. The ideal method of breaking the chip is to provide a built in breaker. It is wrong to assume that because a metal is soft it is easy to machine. In steel a completely spheroidized structure is considered to be easiest to machine.

890. For achieving a specific surface finish in single point turning the most important factor to be controlled is

(a) Depth of cut (b) Cutting speed
(c) Feed (d) Tool rake angle

ISRO Scientist/Engineer 2008

Ans. (c) : specific of surface finish

$$H = \frac{f^2}{8R}$$

where f = feed
R = Nose radius

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4. Machining and Machine Tool Operations

891. Milling cutter is mounted on the part of a milling machine called

(a) Dividing head (b) Spindle
(c) Bracket (d) Arbor

UJVNL AE 2016
Rajasthan AE (Nagar Nigam) 2016 Shift-3
RPSC VPITI 14.02.2016, (KPSC AE 2015)
TSPSC Managers, 2015, MPSC Poly. Lect. 2014
TNPSC AE 2013, WBPSC AE 2008

Ans : (d) Milling cutter is mounted on the part of a milling machine called arbor.
Arbor:- The arbor is an accurately machined shaft for holding and driving the arbor type cutter. It is tapered at one end to fit the spindle nose and has two slots to fit the nose keys for locating and driving it.

892. The time taken to drill a hole through a 25 mm thick plate with the drill rotating at 300 rpm and moving at a feed rate of 0.25 mm/rev is:

(a) 10 sec (b) 20 sec
(c) 60 sec (d) 100 sec

RPSC ADE 2016, GPSC AE 26.07.2015
(HPPSC AE 2014), OPSC Civil Services Pre 2006

Ans. (b) : $T_m = \frac{T + A + O + X}{fN} \times$ Number of hole

T = Thickness of plate
A = Approach
O = Over travel

$X = \frac{D}{2\tan(\theta/2)}$

θ = lip/point angle.

Given, T= 25 mm (Number of hole = 1)
f = 0.25m/rev
N = 300 rpm

Neglect approach and over travel –
Here lip angle is not give, hence time taken to drill a hole

$T_m = \frac{T}{fN} = \frac{25}{0.25 \times 300} = \frac{1}{3}\text{min} = \frac{60}{3}\text{sec} = 20\,\text{sec}$

893. Internal gear cutting operation can be performed by :

(a) Milling
(b) Shaping with rack cutter
(c) Shaping with pinion cutter
(d) Hobbing

MECON MT 2019
HPPSC Asstt. Prof. 18.11.2016
APPSC AEE MAINS 2016, PAPER-III
Sikkim PSC (Under Secretary), 2017
JPSC AE 2013, Paper-V
CSE Pre-1996, ESE 1996

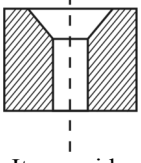
Ans. (c) : Internal gear cutting operation can be performed by shaping with pinion cutter.
In the case of 'rotating pinion type cutter', such an indexing is not required therefore, this type is more productive and so common.

894. The operation of making a cone shaped enlargement of the end of a hole is known as :

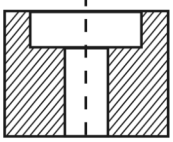
(a) counter sinking (b) counter boring
(c) trepanning (d) spot facing

GPSC Lect. 23.10.2016
HPPSC W.S. Poly. 2016
GWSSB AAE, 27.12.2015
MPSC Poly. Lect. 2014
TNPSC AE (Industries) 09.06.2013

Ans. (a) : Counter sinking – Counter sinking makes a leveled section at the end of drilled hole to provide a proper seat for a flat headed screw or rivet. The most common angles are 60°, 80° an 90°.

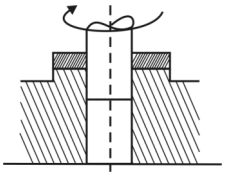


Counter Boring – It provides an enlargement of cylindrical hole with a flat bottom so that a bolt head, or a nut will have a smooth seating surface.



Counter bouring

Trepanning – Trepanning is a technique used for drilling larger hole diameters where machine power is limited as it is not power consuming as conventional drilling. This method is for through hole applications.
Spot Facing– It is the operation of smoothing and squaring the surface around and at the end of a hole so as to provide a smooth seat for a nut or for the head of a cap screw.



895. Gang Milling is:

(a) A process of cutting gears
(b) A milling process for generating hexagonal surfaces
(c) A process in which two or more cutters are used
(d) A milling operation combined with turning

GPSC Asstt. Prof. 28.08.2016
Nagaland PSC CTSE 2016 Paper-I
Nagaland PSC CTSE 2017, Paper-I, WBPSC AE, 2017
UPSC JWM Advt. No.-52/2010, ESE 2006

Ans. (c) : Gang milling refers to the use of two or more milling cutters mounted on the same arbor (that is ganged) in a horizontal milling setup.

896. The point angle of a standard ground drill is:

(a) 100º (b) 90º
(c) 118º (d) 130º

GPSC Asstt. Prof. 28.08.2016
GPSC Asstt. Prof. (Prod.) 21.08.2016
Punjab PSC SDE 12.02.2017
TRB Poly. Lect., 2012, APPSC AEE 2012

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Ans. (c) : Point angle is located on the head of the twist drill. The smaller the point angle, the easier the centering in the materials. For standard drill the point angle is 118º.

897. The size of a lathe is expressed by:
(a) diameter of chuck
(b) maximum speed of chuck
(c) swing of lathe
(d) height of centers from ground

WBPSC AE, 2017
HPPSC Asstt. Prof. 20.11.2017
RPSC AE 2016, RPSC ACF-2011

Ans. (c) : The size of an engine lathe is determined by the largest piece of stock that can be machined. Before machining a work piece, the following measurement must be considered. The diameter of the work that will swing over the bed and the length between lathe centers.

898. Power is transmitted by lead screw to carriage through
(a) gear system
(b) pulley drive
(c) rack and pinion arrangement
(d) half nut

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
WBPSC AE, 2017, RPSC AE 2016, RPSC ACF-2011

Ans : (d) Power is transmitted by lead screw to carriage through half nut. The apron is fitted to the front position of the saddle facing the operator. It consists of a hand wheel for saddle movement, pinion to engage with the rack for saddle movement, a lever to engage the automatic feed for the saddle automatic feed clutches, split nut (half nut) and lead screw.

899. The size of a shaper is given by
(a) stroke length (b) motor power
(c) mass of machine (d) rate size

Sikkim PSC (Under Secretary), 2017
WBPSC AE, 2017
Nagaland PSC CTSE 2016 Paper-I
TNPSC AE 2013, CSE Pre-1995

Ans. (a) : • This size of a shaper is given by the maximum stroke length of the ram in mm.
• The size of shaper is determined by maximum length of cut or stroke it can make.
• The length of stroke is always principal dimension, but a number of other details are also required to specify a shaper fully.

900. The spindle speeds of machine tools are usually designed to follow ——— progression.
(a) logarithmic (b) arithmetical
(c) harmonical (d) geometrical

HPPSC Workshop Suptd. 08.07.02021
GPSC Asstt. Director of Transport 05.03.2017
HPPSC LECT. 2016, WBPSC AE 2008

Ans. (d) : $N_{max} = N_{min}\, r^{n-1}$

$$r = \left(\frac{N_{max}}{N_{min}}\right)^{\frac{1}{n-1}}$$

901. In centre less grinding the regulating wheel is inclined at an angle
(a) 1–8° (b) 9–15°
(c) 15–18° (d) 12–24°

Nagaland PSC (CTSE) 2018, 2016 Paper-I
VIZAG STEEL MT 2015
TNPSC AE (Industries) 09.06.2013

Ans. (a) : In center less grinding, the regulating wheel is inclined at (1°–8°), creating a feed angle relative to the grinding wheel, this allow to perform the dual purpose of rotating the work piece & driving the work piece, across the face of the grinding wheel.

902. The usual ratio of forward and return stroke in shaper is–
(a) 2 : 1 (b) 1 : 2
(c) 2 : 3 (d) 3 : 2

Nagaland PSC CTSE 2017, Paper-I
J&K PSC Civil Services Pre, 2013
TNPSC AE (Industries) 09.06.2013

Ans. (d) : The usual ratio of forward & return stroke in shaper is 3:2

903. The main purpose of a boring operation, as compared to drilling, is to:
(a) Drill a hole
(b) Finish the drilled hole
(c) Correct the hole
(d) Enlarge the existing hole

GPSC Asstt. Prof. 28.08.2016
UPSC JWM Advt. No.-52/2010
WBPSC AE 2003

Ans. (d) : Boring is a process that involves the use of a single point cutting tool or boring head to enlarge an existing hole in a work-piece.

904. Which method is used for turning internal tapers only?
(a) compound rest (b) tailstock off set
(c) taper attachment (d) reamer

Nagaland PSC CTSE 2017, 2016 Paper-I
RPSC ACF-2011

Ans. (c) : Tapper attachment is used for turning internal taper only. They set the angle on the attachment the cross slide is move producing the taper as the saddle moves along the bed.

905. A grinding wheel of 150 mm diameter is rotating at 3000 rpm. The grinding speed is
(a) 7.5 π m/s (b) 15π m/s
(c) 45 π m/s (d) 450 π m/s

ISRO Scientist/Engineer (RAC) 22.04.2018
TNPSC AE 2017, APPSC AEE 2012

Ans. (a) :
Data given
D = 150 mm
N = 3000 rpm
we know that

$$V = \frac{\pi DN}{60 \times 1000}$$

$$V = \frac{\pi \times 150 \times 3000}{1000 \times 60}$$

$$V = 7.5\pi \text{ m/s}$$

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906. Quality screw threads are produced by
(a) thread milling
(b) thread chasing
(c) thread cutting with single point tool
(d) thread casting

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2003

Ans. (a) : Quality screw threads are produced by thread milling.

907. A lead-screw with half nuts in a lathe, free to rotate in both directions has
(a) V-threads (b) whit worth threads
(c) buttress threads (d) acme threads

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2002

Ans. (d) : Acme threads– The Acme thread form a 29° thread angle with a thread height half of the pitch the apex (or crest) and valley (or root) are flat.

908. Knurling is an operation :
(a) of cutting smooth collars
(b) of under cutting
(c) of generally roughing the surface for hand grip
(d) done prior to screw cutting

TSGENCO AE 14.11.2015
RPSC AE 2016, RPSC ACF-2011

Ans. (c) : The knurling is a process of embossing (Impressing) a diamond shaped or straight-line pattern into the surface of workpiece. knurling is essentially a roughening of the surface and is done to provide a better gripping surface.

909. Which of the following is fastest method of cutting gears?
(a) Milling (b) Gear shaping
(c) Gear hobbing (d) Gear burnishing

TSPSC Managers, 2015
APPSC Poly. Lect. 2013, TNPSC AE 2013

Ans. (c) : Hobbing is fastest gear cutting method. It is special types of milling machine. The teeth or splines of the gear are progressively cut into the material (a flat cylindrical piece of metal) by a series of cuts made by a cutting tool called a hob.

910. Surface grinding is done to produce
(a) Flat surface only
(b) Internal cylinder holes
(c) Flat and Tapered surface
(d) Tapered surface only

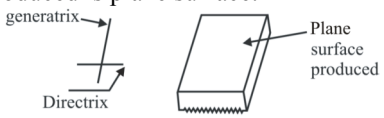
MECON MT 2019
Rajasthan AE (Nagar Nigam) 2016 Shift-3

Ans. (a) : Surface grinding is done to produce flat surface and smooth finish.
It is widely used in abrasive machining process in which a spinning wheel covered in rough particles cut the chips of metallic or nonmetallic substance from a workpiece to make flat face.

911. In a machining operation, if the generatrix and directix both are straight lines, the surface obtained
(a) cylindrical (b) helical
(c) plane (d) surface of revolution
(e) None of the above

Assam Engg. College AP/Lect. 18.01.2021
HPPSC Asstt. Prof. 20.11.2017
CGPSC Poly. Lect. 22.05.2016

Ans. (c) : Generatrix – It is a point line or surface that when moved along a given path, generates a new surface.
Directrix– The path followed by the generatrix is called the directrix different surface can be generated depending upon the generatrix and directrix.
If generatrix is a line and directrix is also a line then surface produced is plane surface.



912. For producing more accurate holes, the sequence of operations to be followed is :
(a) centering, drilling, boring, reaming
(b) centering, boring, drilling, reaming
(c) drilling, centering, boring, reaming
(d) drilling, reaming, boring, centering

BPSC AE Mains 2017 Paper - VI
Kerala PSC AE 06.08.2015
(KPSC AE. 2015)

Ans. (a) : For more accurate holes, the sequence of operations to be centering, drilling, boring, reaming.
Centering– Centering is the process of making tiny hole in the job in a work holding device chuck, drive plate and face plate on centering of lathe machine. It is required to turn any job concentric to lathe centers for cylindrical machining.
Drilling- It is a cutting process that uses a drill bit to cut a hole of circular cross-section in solid material.
Boring- Boring is a cutting operation that uses a single point cutting tool or a boring head to produce conical or cylindrical surface by enlarging an existing opening in a work piece.
Reaming- Reamers are designed to enlarge the size of a previously formed hole by a small amount but with the high degree of accuracy to leave smooth sides. The process of enlarging hole is called reaming.

913. It is required to cut threads of 2 mm pitch on lathe. The lead screw has a pitch of 6 mm. If the spindle speed is 60 r.p.m, then the speed of lead screw will be:
(a) 10 r.p.m. (b) 20 r.p.m.
(c) 120 r.p.m. (d) 180 r.p.m.

Nagaland PSC (CTSE) 2018 Paper-I
HPPSC Asstt. Prof. 2014
ISRO Scientist/Engineer 2009

Ans. (b) : Speed of lead screw $= \frac{60}{6} \times 2$
$= 20$ rpm

914. In centre less grinding the work piece advances by
(a) Push given by operator
(b) Hydraulic force
(c) Force exerted by grinding wheel
(d) Remains stationary

WBPSC AE, 2017
Nagaland PSC CTSE 2017, 2016 Paper-I

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Ans. (c) : In centre less grinding the work piece advances by the force exerted by grinding wheel. Grinding wheel are stationary wheel, which rotate on a fixed axis, at high speed.

915. Which of the following is a single point cutting tool?

(a) Hacksaw blade (b) Milling cutter
(c) Grinding wheel (d) Parting tool

GWSSB AAE, 27.12.2015
Assam PSC CCE Pre 2015
UPRVUNL AE 2014

Ans : (d) Single point cutting tools contain only one main cutting edge in the cutter body.
Multi point cutting tools contain more than one (even up to hundreds) cutting edges in the cutter body.
- Parting tool is a single point cutting tool
- Hacksaw, milling cutter, grinding wheel are the multipoint cutting tool.

916. A milling cutter having 8 teeth is rotating at 150 RPM. If the feed per tooth is 0.1, the table speed in mm per minute is

(a) 120 (b) 187
(c) 125 (d) 70

ISRO Scientist/Engineer 11.10.2015
MPSC Poly. Lect. 2014

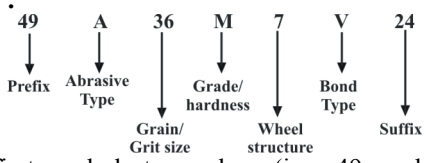
Ans. (a) : Given, No. of teeth (z) = 8, Speed (N) = 150 RPM, Feed per tooth (f_t) = 0.1 mm
Table speed
= feed rate = $f_t \times z \times N$
= $0.1 \times 8 \times 150$
= 120 mm/min

917. A grinding wheel is specified as 49 A 36 M 7 V 24. The number 36 stands for

(a) Structure (b) Grade
(c) Grit size (d) Bond

APPSC Poly. Lect. 2013
ISRO Scientist/Engineer 2009

Ans. (c) :



Here, first and last number (i.e. 49 and 24) is manufacturer's identification number.

918. The cutting speed of the tool in turning operation is:

(a) Directly proportion to diameter of the workpiece
(b) Inversely proportional to the workpiece
(c) Inversely proportional to the square of the workpiece
(d) Directly proportional to the square of the diameter of the workpiece

CIL MT 26.03.2017
BPSC AE 2012 Paper - VI

Ans. (a) :
Cutting speed = $r \times \omega = \pi DN$ mm/min
Where D is in mm and N is in RPM.
So cutting speed is directly proportional to diameter of the workpiece.

919. The gear is manufactured on generating principle in

(a) milling (b) hobbing
(c) forming (d) broaching

(KPSC AE. 2015)
TNPSC AE 2013

Ans. (b) : The gear is manufactured on generating principle in hobbing.

920. The _____ operation is usually used to obtained smooth and close toleranced holes.

(a) Drilling (b) Tapping
(c) Boring (d) Countersinking
(e) Reaming

TSPSC Manager (Engg.) HMWSSB 12.11.2020
HPPSC AE 2018
(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Reaming–Reaming has been defined as a machining process that uses a multi-edged fluted cutting tool to smooth, enlarge or accurately size an existing hole. Reaming is performed using the same types of machines as drilling.
Tapping–Tapping is the process of cutting a thread inside a hole so that a cap screw or bolt can be threaded into the hole. Also, it is used to make thread on nuts. Tapping can be done on the lathe by power feed or by hand.
Countersinking–Countersinking is a process that creates a V-shaped edge near the surface of the hole. It is often used to deburr a drilled or tapped hole, or to allow the head of a counter sunk-head screw to sit flush or below a surface.

921. The binding material used in cemented carbide tools is

(a) Nickel (b) Cobalt
(c) Chromium (d) Carbon

Nagaland CTSE 2016, 2017 Ist Paper
UKPSC AE 2012 Paper -I

Ans. (b) : The binding material used in cemented carbide is cobalt. Cobalt acts as a binder & also reduces the brittleness of the carbides tool. In cemented carbide composition are 82% tungsten carbide, 10% titanium carbide and 8% cobalt.

922. A straight teeth slab milling cutter of 100 mm diameter and 10 teeth rotating at 200 rpm is used to remove a layer of 3 mm thickness from a steel bar. If the table feed is 600 mm/minute, the feed per tooth in this operation will be

(a) 0.2 mm (b) 0.3 mm
(c) 0.5 mm (d) 0.6 mm
(e) None of these

CGPSC AE 16.10.2016
GPSC AE 26.07.2015

Ans. (b) : Table feed = f_t NZ
$600 = f_t \times 200 \times 10$
$f_t = \frac{3}{10} = 0.3\,\text{mm}$

923. Operation of producing large holes by removing metal along the circumference of a hole is known as

(a) spot facing (b) counter sinking
(c) lapping (d) trepanning

OPSC AEE 2019 Paper-I
WBPSC AE 2003

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Ans. (d) : Operation of producing large holes by removing metal along the circumference of a hole is known as Trepanning.

924. A mild steel rod having 50 mm diameter and 500 mm length is to be turned on a lathe. Determine the machining lime to reduce the rod to 45 mm in one pass when cutting speed is 30 m/min and a feed of 0.7 mm/rev is used :

(a) 1.74 min (b) 2.74 min
(c) 3.74 min (d) 4.74 min

Kerala PSC AE 06.08.2015 (KPSC AE. 2015)

Ans. (c) : given,

$d_p = 50$ mm (diameter) $d_f = 45$ mm (to be reduce)
$L = 500$ mm (length) $V = 30$ m/min.
feed = 0.7 mm/rev.

$$V = \frac{\pi dN}{1000} m/min.$$

$$N = \frac{V \times 1000}{\pi d}$$

$$= \frac{30 \times 1000}{\pi \times 50mm}$$

$$= 190.98 = 191 rpm$$

$$\because \left[time = \frac{L}{f.N} \right] min$$

$$= \frac{500}{0.7 \times 191} = 3.739 = 3.74 min$$

925. Consider the following cutting tool materials used for metal-cutting operation at high speed

1. High-speed steel
2. Ceramics
3. Tungsten carbide
4. Cemented titanium carbide

(a) 3, 1, 4, 2 (b) 1, 3, 2, 4
(c) 3, 1, 2, 4 (d) 1, 3, 4, 2

UPRVUNL AE 21.08.2016
CGPSC AE 16.10.2016

Ans. : (d) The correct sequence in increasing order of the range of cutting speed for optimum use of the these materials
High speed steel < Tungsten carbide < Cemented titanium cabide < Ceramics.

926. Which of the machine tools can be used for boring?

1. Vertical milling machine
2. Horizonal milling machine
3. Lathe
4. Drilling machine

(a) 2 and 4 (b) 1, 2, 3
(c) 1, 2, 3, 4 (d) 1, 3, 4

UPRVUNL AE 21.08.2016
ISRO Scientist/Engg. 2010

Ans. : (d)
Following machine tools can be used for boring
(i) Vertical milling machine
(ii) Lathe
(iii) Drilling machine

927. In which operation on a work piece on lathe the spindle speed will be least?

(a) plain turning (b) taper turning
(c) finishing (d) thread cutting

RPSC AE 2016, TNPSC AE 2013
RPSC ACF-2011, WBPSC AE 2003

Ans. (d) : Thread cutting operation a work piece on lathe the spindle speed will be least.

928. A mandrel is :

(a) a slightly tapered hardened steel shaft that supports works which cannot be otherwise.
(b) a tapered gauge used for inspection of tapered holes
(c) auxiliary chuck used on lathe for holding small loads
(d) is used in lathe work to hold castings

RPSC AE 2016, RPSC ACF-2011

Ans. (b) : A mandrel is a tapered gauge used for inspection of tapered holes.

929. Tumbler gears in lathe are used to

(a) reduce the spindle speed
(b) cut gears
(c) give desired direction of movement to the lathe carriage
(d) drill a workpiece

MECON MT 2019
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : Tumbler gear mechanism is used to change the direction of lead screw sand feed rod in lathe machines. By engaging tumbler gear, the carriage can be moved automatically from tailstock end to headstock.

930. The angle between the lathe centres is

(a) 30º (b) 45º
(c) 60º (d) 90º

TNCSC AE 2020, MECON MT 2020

Ans. (c) : The angle between the lathe centres is usually have an included 60º, but in heavy machining situations an angle of 75º is used.

931. Drilling can be termed as :

(a) Oblique cutting (b) Simple cutting
(c) Orthogonal cutting (d) Uniform cutting

Assam PSC AE (PHED), 18.10.2020
Sikkim PSC (Under Secretary), 2017
Kerala PSC Poly. Lect., 2017
VIZAG Steel MT 2011

Ans. (a) : Drilling can be termed as oblique cutting.

932. Which of the following machines does not require quick return mechanism

(a) slotter (b) planer
(c) shaper (d) broaching

TNPSC AE 2013
J&K PSC Screening 2006

Ans. (d) : Broaching does not require quick return mechanism.

933. The best process for mass production of good quality gears is

(a) hobbing (b) gear shaping
(c) casting (d) milling

Sikkim PSC (Under Secretary), 2017
APPSC Poly. Lect. 2013

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Ans. (a) : Hobbing is the best process for mass production of good quality gear and Cutting splines or cutting sprockets on a hobbing machine.
Due to continuous indexing, gear hobbing is highly accurate. High rate of production process because of multi point cutting tool.

934. If N is number of strokes per minute, L is stroke length in mm, C is cutting ratio (cutting time/ total time) the cutting speed of shaper in m/min is given by
(a) NL/1000C (b) NLC/100
(c) 1000L/NC (d) 1000LC/N
Nagaland PSC CTSE 2016 Paper-I
VIZAG STEEL MT 2015

Ans. (a) : Cutting speed of shaper

$$V = \frac{NL(1+m)}{1000}$$

$$\left[\because \text{Cutting ratio}(m) = \frac{t_{return}}{t_{cutting}}\right]$$

$$C = \frac{t_{cutting}}{t_{total}} = \frac{1}{\frac{t_{total}}{t_{cutting}}} = \frac{1}{\frac{t_{cutting}+t_{return}}{t_{cutting}}} = \frac{1}{1+\frac{t_{return}}{t_{cutting}}} = \frac{1}{1+m}$$

$$V = \frac{NL}{1000C}$$

935. A hacksaw blade cuts on the
(a) forward stroke
(b) return stroke
(c) both forward and return strokes
(d) None of the above
TNPSC AE 2017
Assam PSC CCE Pre 2015

Ans. (a) : A hacksaw blade cuts on the forward stroke.

936. A taper pin of 90 mm has a taper length of 50 mm. The larger diameter of taper is 95mm and the smaller diameter is 85mm. The taper in mm/meter and in degree and the tail stock set over will be respectively.
(a) 20 mm/meter, 5.71" and 9 mm
(b) 200 mm/meter, 57.1" and 90 mm
(c) 20 mm/meter, 57.1" and 90 mm
(d) 200 mm/meter, 5.71" and 9 mm
CGPSC AE 15.01.2021
CGPSC AE 16.10.2016

Ans. (d) : Given, $L_w = 90$ mm
$D_i = 95$mm
$D_f = 85$ mm

(i) Taper length $L_t = 50$ mm

$$\tan(\alpha/2) = \frac{(D_i - D_f)}{2L_t} = \frac{(95-85)}{2\times 50}$$

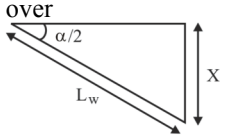
$$\tan(\alpha/2) = 0.1 \Rightarrow \boxed{\alpha/2 = 5.71^\circ}$$

(taper angle)

(ii) Taper in mm/meter $= \frac{(D_i - D_f)}{L_t(\text{in metres})}$

$$= \frac{(95-85)}{50\times 10^{-3}} = \frac{10\times 1000}{50} = 200\,\text{mm/meter}$$

(iii) Tail stock set over



$$\sin\left(\frac{\alpha}{2}\right) = \frac{X}{L_w}$$

$$X = \sin\left(\frac{\alpha}{2}\right)\times L_w$$

$$\Rightarrow 0.099\times 90 = 8.955\,\text{mm} = 9\text{mm}$$

937. A twist drill is specified by
(a) the size of hole it can drill
(b) a number of specifying hole size
(c) a letter of alphabet
(d) Any of the above
Assam PSC AE (PHED), 18.10.2020
WBPSC AE, 2017

Ans. (d) : Twist drill is specified by its shank material and diameter.
(1) The size of hole it can drill
(2) A number of specifying hole size
(3) A letter of alphabet

938. The operation(s) performed on a shaper is/are
(a) machining horizontal surface
(b) machining vertical surface
(c) machining angular surface
(d) All of the above
Assam PSC AE (PHED), 18.10.2020, TSPSC AEE 2015

Ans. (d) : The following are the different shaper machine operations.
(1) Angular surface
(2) Irregular surface
(3) Vertical surface
(4) Horizontal surface
(5) Cutting slots grooves and keyways
(6) Machining splines and cutting gears

939. Single point thread cutting tool should ideally have :
(a) Zero rake (b) Positive rake
(c) Negative rake (d) None of these
HPPSC Asstt. Prof. 2014,
J&K PSC Civil Services Pre, 2010

Ans. (a) : Single point thread/gear cutting tool should ideally have zero rake angle.

940. What is the meaning of '46' and 'L' in a grinding wheel carrying the marking 250 × 25 × 32 WA46L4V17?
(a) Grain size and Structure
(b) Grade and Structure
(c) Grain size and Grade
(d) Structure and Bond
Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : $\underbrace{250\times 25\times 32}_{\text{W}}\boxed{\text{A46LUV}}17$

Special information-
A – type of abrasive (Aluminium Oxide Al_2O_3)
46- Grain size (Medium)
L- Grade (Medium)
U - Structure (Close)
V- Bond type (Vitrified)

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941. For the simple indexing, which formula holds true

(a) $\frac{30}{n}$ (b) $\frac{40}{n}$

(c) $\frac{50}{n}$ (d) $\frac{60}{n}$

HPPSC Workshop Suptd. 08.07.02021
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Simple indexing- In this case different index plates with varying number of holes are used to increase the range indexing. $T = 40/n$
T = No. of turn
N = no of holes

942. If 'L' is the length of the stroke in shaping, 'B' is the width of the workpiece including allowance, 'f' is the feed per cycle in mm, 'V_1' is the cutting speed in m/min and 'K' is the ratio of idle stroke to cutting stroke time, then machining time can be calculated by

(a) $\frac{LB(1+K)}{f \times V_1}$ (b) $\frac{LB}{f \times V_1}$

(c) $\frac{KLB}{f \times v_1}$ (d) $\frac{KLB}{f}$

Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : $\frac{LB(1+K)}{f \times V_1}$

943. Guide ways of lathe beds are hardened by which of the following methods?

(a) Carburizing (b) Cyaniding
(c) Nitriding (d) Flame hardening

Karnataka PSC AE, 10.09.2017
GPSC AE 26.07.2015

Ans. (d) : Flame Hardening– It is used to get most wear and corrosion resistant surface. Guide ways of lathe are hardened by this method.

944. Low helix angle drills are preferred for drilling holes in :

(a) Plastic (b) Copper
(c) Cast steel (d) Carbon steel

UPPSC AE 13.12.2020 Paper-I
PSPCL AE, 2012

Ans. (a) : Low helix angle drills are preferred for drilling holes in plastic.

945. The process of improving the cutting action of the grinding wheel is called.

(a) truing (b) dressing
(c) facing (d) clearing

Nagaland PSC (CTSE) 2018, Paper-I
MPSC Poly. Lect. 2014, APPSC AEE 2012

Ans. (b) : The process of improving the cutting action of the grinding wheel is called dressing and process of maintaining circularity or shape of grinding wheel is called truing.

946. While machining a brass casting on a lathe, the type of chuck used is–

(a) Collect chuck (b) Magnetic chuck
(c) Three jaw chuck (d) Four jaw chuck

Nagaland PSC CTSE 2017, 2016 Paper-I

Ans. (d) : Four jaw chuck used in machining a brass casting on a lathe.

947. Standard taper generally used on milling machine spindles is

(a) Morse taper
(b) Chapman taper
(c) Brown and sharp taper
(d) Jamo and Reed taper

Nagaland PSC CTSE 2016 Paper-I
SJVN ET 2013

Ans. (c) : Standard milling machine taper is used on most machines of recent manufacture & the taper which used is brown and sharp taper.

948. Hobbing process is not suitable for cutting following type of gear

(a) Spur (b) Helical
(c) worm (d) Bevel

Vizag Steel (MT) 2017
GPSC AE 26.07.2015

Ans. (d) :
Hobbing process– Hobbing process is carried out on gear hobbing machine using hob as tool.
–Hob is a cylinder on the surface of which a continuous thread has been cut having the cross section of involute gear teeth.
So, hobbing process is not suitable for cutting bevel type of gear.

949. It is desired to perform the operations like drilling, reaming, counterboring etc. on a work-piece. Which of the following machines will be used?

(a) Sensitive drilling machine
(b) Radial drilling machine
(c) Gang drilling machine
(d) Multiple spindle drilling machine

TSPSC AEE 2015
JPSC AE 2013, Paper-V

Ans : (c) : When a number of single spindle drilling machine columns are placed side by side on a common base and have common worktable, the machine is known as the gang drilling machine.
It is desired to perform the operations like drilling, reaming, counter boring, etc. on a work-piece.

950. Which of the following processes results in the best accuracy of the hole made?

(a) Reaming (b) Tapping
(c) Broaching (d) Boring

GPSC AE 26.07.2015
BPSC AE 2012 Paper-VI, CSE Pre-2010

Ans : (a) : Reaming- It is an operation of slightly enlarging a machined hole to proper size with a smooth finish.

951. Which of the following operations is also known as internal turning?

(a) Milling (b) Facing
(c) Tapping (d) Boring

Haryana PSC AE (PHED) 05.09.2020, Paper-II
BPSC AE 2012 Paper - VI

Ans : (d) : Boring is also known as internal turning.

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952. Which of the following is a chipless machining process?

(a) Metal spinning (b) Knurling
(c) Hobbing (d) Lapping

Gujarat PSC AE 2019
HPPSC Workshop Supt. 08.07.2015

Ans : (a) : Metal spinning also known as spin forming or spinning or metal turning most commonly, is a metal working process by which a disc or tube of metal is rotated at high speed and formed into an axially symmetric part. Spinning can be performed by hand or by a CNC lathe.

953. Cutting tool material 18 - 4 - 1 HSS has which one of the following compositions?

(a) 18% W, 4% Cr, 1% V
(b) 18% Cr, 4% W, 1% V
(c) 18% W, 4% Ni, 1% V
(d) 18% Cr, 4% Ni, 1% V

RPSC AE 2018, UJVNL AE 2016

Ans. (a) : High speed steel is a subset of tool steels, commonly used as cutting tool material.
18 - 4 - 1 (HSS)
18% - Tungsten (W)
4% - Chromium (Cr)
1% - Vanadium (V)
High speed steel properties and applications—The well balanced composition produce both excellent toughness and cutting power and makes this economical molybdenum high speed steel a standard grade for all types of roughness and finishing tools, as well as for all tools demanding a high degree of toughness.

954. Which of the following method cannot be used for manufacturing internal gears?

(a) Casting (b) Die casting
(c) Broaching (d) Hobbing

Nagaland PSC CTSE 2017, 2016 Paper-I

Ans. (d) : Hobbing method here not used for manufacturing internal gear

955. The hardness with which the bond holds the cutting points or abrasives in place defined by the term :

(a) structure (b) grit size
(c) grain size (d) grade

(KPSC AE. 2015), Kerala PSC AE 06.08.2015

Ans : (d) The bond holds the cutting points or abrasive in place defined by the grade.
Grade of a grinding wheel: The grade of a wheel indicates the strength of the grains, and the holding power of the bond. It is usually referred to as hardness of the wheel. The thickness of the bonding layer holding abrasive Controls the grade of the grinding wheel. Hard wheel is used for precision grinding.

Very Soft	-	A to G
Soft	-	H to K
Medium	-	L to O
hard	-	P to S
Very hard	-	T to Z

956. Which method is used for turning external tapers only?

(a) compound rest (b) tailstock offset
(c) taper attachment (d) reamer

RPSC AE 2016
RPSC VPITI 14.02.2016

Ans : (b) Tailstock offset is used for turning external tapers only.

957. Stamping is the process of

(a) printing letters and numbers on metal sheets
(b) producing flutes or corrugations
(c) producing seamless tubes
(d) shaping thin sheets by pressing them against a form

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : Stamping is the process of printing letters and numbers on metal sheets.

958. The process of putting a rod though several dies of decreasing diameter is known as

(a) wire drawing (b) trimming
(c) rolling (d) stretch forming

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : The process of putting a rod though several dies of decreasing diameter is known as wire drawing.

959. Which of the following machine tools provide higher rate of production compared to others?

(a) Special purpose machine tools
(b) Production machine tools
(c) General purpose machine tools
(d) Single purpose machine tools

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Single purpose machine tools provide higher rate of production compared to others. Single purpose machine tools are designed and produced with various levels of automation i.e. from workplaces with manual setting of components up to fully automated machines with automatic feeding of semi-finished products and storage of the final product.

960. _____ generally has a thickness of more than 50 mm.

(a) sheet (b) foil
(c) slab (d) palte

UPRVUNL AE 05.07.2021

Ans. (c) : Slab generally has a thickness of more than 50 mm.

961. _____ system for cutting is a tool-in-hand type system for sharpening and grinding with a single-point cutting tool.

(a) ASA (b) Parallel
(c) Orthogonal (d) Coinciding

UPRVUNL AE 05.07.2021

Ans. (a) : ASA system for cutting is a tool-in-hand type system for sharpening and grinding with a single-point cutting tool.

962. Milling two sides of a piece of work by employing two side-milling cutters at the same time is _____milling.

(a) Form (b) Plane
(c) Straddle (d) End

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : Milling two sides of a piece of work by employing two side-milling cutters at the same time is straddle milling.

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963. **_____shears are provided with two straight blades of adequate length made of die steel, in which longer blade is fixed to the ram while the other machine bed.**

(a) Sigma (b) Carotene
(c) Delta (d) Guillotine

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Guillotine shears are provided with two straight blades of adequate length made of die steel, in which longer blade is fixed to the ram while the other machine bed.

964. **Among the various drilling machines, which is the most versatile machine compared to other drilling machines in the options given below?**

(a) Radial drilling machine
(b) Sensitive drilling machine
(c) Portable drilling machine
(d) Bench drilling machine

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : Radial drilling machine is the most versatile machine compared to other drilling machines.

965. **A _____ fitted through the spindle, screws in the the arbor and holds its firmly in the spindle.**

(a) draw-in-bar (b) chuck
(c) live centre (d) lug

HPPSC Workshop Suptd. 08.07.02021

Ans. (a) : A draw-in-bar fitted through the spindle, screws in the the arbor and holds its firmly in the spindle.

966. **In milling operation, the minimum table traverse required is _____where L is the length of job and D is the milling cutter diameter.**

(a) L – D (b) L × D
(c) L + D (d) L / D

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : In milling operation, the minimum table traverse required is L + D where L is the length of job and D is the milling cutter diameter.

967. **Swing diameter over bed is a technical specification of _____**

(a) Planer (b) Lathe
(c) Slotter (d) Shaper

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : Swing diameter over bed is a technical specification of lathe.

968. **_____ is that portion of the reamer which stands between two flutes.**

(a) Tang (b) Land
(c) Margin (d) Chamfer

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : Land is that portion of the reamer which stands between two flutes.

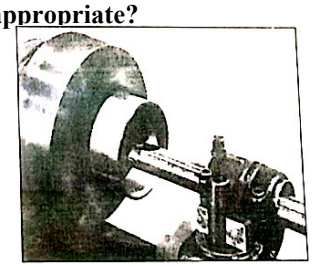
969. **_____ permit a variety of milling cutter types and sizes to be mounted to the milling machine spindle.**

(a) Clapper box (b) Adapter
(c) Indexer (d) Snap tool

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : Adapter permits a variety of milling cutter types and sizes to be mounted to the milling machine spindle.

970. **In the figure given below, which process is the most appropriate?**



(a) Boring (b) Facing
(c) Parting (d) Milling

CGPSC AE 15.01.2021

Ans. (a) : Boring

971. **A steel work piece is to be milled. Material removal rate is 25 cm^3 /min, depth of cut and width of cut are 5 mm and 100 mm respectively. The rate of feed (table feed) is _____.**

(a) 50 cm/min (b) 75 mm/min
(c) 50 mm/min (d) 100 mm/min

VIZAG Steel MT 24.01.2021
TNPSC AE (Industries) 09.06.2013

Ans. (c) : Given,
MRR = 25 cm^3/min
Depth of cut (d) = 5 mm
Width of cut (w) = 100 mm
We know that,
MRR (material removal rate)
MRR = wdf_r (f_r = rate of feed or table feed)
$5 \times 100 \times f_r = 25$

$$f_r = \frac{25}{5\times100} = \frac{25\,cm^3/min}{500\,mm}$$

$$= \frac{25\times10^3}{500}\,mm/min = 50\ mm/min$$

972. **Which one of the following milling operations is used to produce gears?**

(a) Form milling (b) Slot milling
(c) Slab milling (d) Face milling

VIZAG Steel MT 24.01.2021

Ans. (a) : In form milling, the cutter called a form cutter travels axially along the length of the gear tooth at the appropriate depth to produce the gear tooth.

973. **Drilled holes may be reamed for improving hole _____.**

(a) dimensional tolerances and surface finish
(b) dimensional tolerances and tool life
(c) size but reduce surface finish
(d) dimensional tolerances at the expense of surface finish

VIZAG Steel MT 24.01.2021

Ans. (a) : Drilled hole may be reamed for improving hole dimensional tolerance and surface finish.

974. **Which of the following indexing method is also known as rapid indexing ?**

(a) Direct indexing method
(b) Angular indexing method
(c) Compound indexing method
(d) Differential indexing method

APGCL AM, 2021

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Ans. (a) : The Direct indexing is also called rapid indexing the rapid index plate located on the nose of the spindle. This plate usually has 24,.30 and 36 equally spaced holes and only the numbers which will devils evenly into 24, 30 and 36 can be indexed.

975. Material used for lathe bed is
(a) Cast iron (b) Cast steel
(c) Mild steel (d) wrought iron

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : Lathe bed is heavy rigid structure which is having high damping capacity for vibrations generated by machines during machining, so that cast iron have these desired properties.

976. A flat surface can be produced by a lathe machine, if the cutting tool moves
(a) parallel to the axis of rotation of workpiece
(b) perpendicular to the axis of rotation of workpiece
(c) at an angle of 45°
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : A lathe machine basically produces two types of surface flat and cylindrical surface.
Flat surface can be obtained on a lathe machine, if cutting tool moves perpendicular to the axis of rotation.
Cylindrical surface can be obtained if cutting tool moves parallel to the axis of rotation.

977. The process used to finish workpieces with high surface quality, accuracy of shape and dimension is
(a) milling (b) turning
(c) grinding (d) brazing

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : Grinding is the process used to finish workpiece with high surface quality, accuracy of shape and dimension.

978. A file removes the metal during
(a) return stroke
(b) forward stroke
(c) both forward and return strokes
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : A file removes the metal during forward stroke.

979. Which of the following is the example of multipoint cutting tool?
(a) Milling cutter
(b) Broaching tool
(c) Both milling cutter and broaching tool
(d) None of the above

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : Milling cutter and broaching tools use multipoint cutting tool to perform their operations. In milling, material is removed as the work is fed against a rotating multipoint cutter. Broaching also remove material but high accuracy.

980. In a planer work, the length of the stroke is 240mm and double stroke per minute is 25. The ratio between return time and cutting time is 1:2. Determine the cutting speed.
(a) 7 minute (b) 18 minutes
(c) 9 minutes (d) 6 minutes

TNCSC AE 2020

Ans. (c) : Given–
length of cutting stroke (L) = 240 mm
n = No. of double strokes per minute = 25
m = ratio return time to cutting time = $\frac{1}{2}$

$$\text{Cutting speed (v)} = \frac{nL(1+m)}{1000}$$

$$= \frac{25\times 240\times 2}{1000}$$

$$= \frac{3000\times 3}{1000} = 9\,\text{m/min}$$

$$\text{Time taken by cutting stroke (min)} = \frac{\text{length of stroke (L), mm}}{\text{cutting speed (v), m/min}}$$

$$= \frac{240}{9000} = 0.0266\,\text{min}$$

981. Burnishing process can give _____ surface finish on a component.
(a) Lowest (b) Highest
(c) Bi-directional (d) Unidirectional

GPSC Engineer, Class-II Pre-19.01.2020
TNPSC AE (Industries) 09.06.2013

Ans. (b) : Burnishing process can give highest surface finish on a component.

982. The job reciprocates in which type of machine
(a) shaping machine (b) planning machine
(c) slotting machine (d) drilling machine

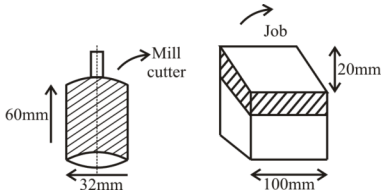
ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (b) : Shaper Machine–Feed motion is given to the workpiece/job and cutting motion given to the tool.
Planner Machine–Feed motion is given to the tool and cutting (reciprocating) motion given to the workpiece/job.

983. Estimate the machining time that will be required to finish a vertical flat surface of length 100 mm and depth 20 mm by an 8 teeth HSS (High speed steel) end mill cutter of 32 mm diameter and 60 mm length in a milling machine. Assume cutting velocity (V_c) = 30 m/min and feed (s_0) = 0.12 mm/tooth. Take approach and run over both 2 mm
(a) 0.4 minutes (b) 0.6 minutes
(c) 0.8 minutes (d) 1.0 minute

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) :



z = 8 teeth, D = 32 mm
f_t = 0.12 mm/teeth
v_c = 30 m/min
$v_c = \pi DN$

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$$N = \frac{v_c}{\pi D} = \frac{30 \times 10^3}{\pi \times 32} = 298.41$$

$N \approx 300$ rpm

Approach (A) = Over run (0) = 2 mm, L = 100 mm

$$\text{Machining time } (t_m) = \frac{L + A + 0}{f_t \times z \times N} = \frac{100 + 2 + 2}{0.12 \times 8 \times 300}$$

$t_m \approx 0.4$ mm

984. The manufacturer has a marking on a grinding wheel as 'A 27 K 7 V'. The code 'K' represents the :

(a) structure (b) bond
(c) grade (d) abrasive particle size

APPSC Poly Lect. 13.03.2020

Ans. (c) : Grinding wheel specification : "A 27 K 7 V"
Code "K" represent grade.

Code		Represents
A	→	Abrasive
27	→	Grain size
K	→	Grade
7	→	Structure
V	→	Bond

985. The _____ process is used to obtain high surface finish.

(a) drilling (b) trepanning
(c) reaming (d) Tapping

APPSC Poly Lect. 13.03.2020

Ans. (c) : Reaming is a multi-tooth cutter, which rotates and moves linearly into an already existing hole. The previous operation could be drilling or preferably boring. Reaming will provide smooth surface (high surface finish) as well as close tolerance on the diameter of the hole.

986. Loading of grinding wheels occurs when

(a) part of grinding surface gives way
(b) hollow cavities are formed on the surface of grinding wheel
(c) abrasive grains becomes clogged with particles of metal being ground
(d) long cracks develop on grinding wheel from center to outer diameter

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Loading of grinding wheels occurs when abrasive grains becomes clogged with particles of metal being ground.

987. Geneva wheel is used for

(a) Quick swining (b) Quick return
(c) Indexing (d) None of the above

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c) : Geneva wheel / Geneva stop, one of the most commmony used devices for producing intermittent rotary motion, characterized by alternate periods of motion and rest with no. reversal in direction. It is also used for indexing (i.e. rotating a shaft through a prescribed angle).

988. In which of the following grindings must the work piece be roted about its own axis between the centres as it passes lengthwise across the face of a revolving grinding wheel?

(a) Internal cylindrical grinding
(b) External cylindrical grinding
(c) Surface grinding
(d) Form grinding

MECON MT 2020

Ans. (b) : In external cylindrical grinding work piece be roted about its own axis between the centres as it passes lengthwise across the face of a revolving grinding wheel.

989. In which part, is a spindle mounted on the centre lathe?

(a) Headstock (b) Tailstock
(c) Carriage (d) Tool-post

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : Head stock is a spindle mounted on the centre lathe.

990. Why are shaper machines being replaced in shops?

(a) Due to advances in casting and other manufacturing processes, surface finishing on components is achieved itself and therefore shaper machines become irrelevant.
(b) The shaper machine is relatively a slow machine tool with very low metal removal rate. Moreover, versatile milling machine can do the work of shaper machine faster.
(c) The shaper machine is a reciprocating machine tool and creates a lot of vibration and noise whereas, a milling machine can do the work of a shaper machine silently.
(d) Shaper machines are not able to remove the metal from the surface of the workpiece and consume high amounts of energy compared to planner machines.

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The shaper machine is relatively a slow machine tool with very low metal removal rate. Moreover versatile milling machine can do the work of shaper machine faster.

991. What type of cutting tool is used in a slotting machine and which part is used to hold it?

(a) Single point cutting tool is held by ram
(b) Multi-point cutting tool is held by ram
(c) Single point cutting tool is held by carriage
(d) Single point cutting tool is held by arbor

APPSC Poly Lect. Automobile Engg., 2020

Ans. (a) : Slotting machine is a reciprocating machine tool in which the ram holding the tool reciprocates in a vertical axis and the cutting action of the tool is only during the downward stroke feed motion. Single point cutting tool is used held by the ram

992. What is the basic difference between shaper machine and planner machine?

(a) No difference as planner machine can also remove metal with same finishing.
(b) In shaper, the cutting tool reciprocates during the cutting motion, while in planner, the workpiece reciprocates
(c) In shaper, the cutting tool reciprocates during the cutting motion, while in planner, the workpiece rotates
(d) In shaper, the cutting tool rotates during the cutting motion, while in planner, the workpiece reciprocates

APPSC Poly Lect. Automobile Engg., 2020

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Ans. (b)

Shaper	Planer
(I) In shaper the cutting tool reciprocates during the cutting motion	(I) In planer work piece reciprocates
(II) Shaper machine is used for small working	(II) Planer is used for large working
(III) Cutting stroke is slower than interactive stroke	(III) Cutting stroke remains slower
(IV) Stroke length is small	(IV) Stroke length is bigger
(V) Cutting tool move up and down in horizontal direction as make a cutting process.	(V) Cutting tool remains motion less.

993. What is name of the machining operation which is used to make a fresh hole in solid material?

(a) Boring operation (b) Drilling operation
(c) Reaming operation (d) Knurling operation

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : Drilling is a cutting process that uses a drill to cut a hole of circular cross-section in solid materials. The drill bit is usually a rotary cutting tool often multi point.

994. Which machine and what type of operation is used to enlarge the existing drilled hole?

(a) Planner machine is used for reaming operation
(b) Boring machine is used for drilling operation
(c) Milling machine is used for drilling operation
(d) Boring machine is used for boring operation

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Boring machine device for producing smooth and accurate holes in a workpiece by enlarging existing hole with a bore which may bear a single cutting tip of steel, cemented carbide or diamond or may be a small grinding wheel.

995. The process of obtaining final finishing operation done with loose abrasive is known as :

(a) planning (b) grinding
(c) honing (d) lapping

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Lapping : Lapping is a machining in which two surfaces are rubbed together with an abrasive between them by hand movement or using a machine.

Honning–

- Honning is the operation of producing very smooth hole by using reciprocating and rotating tool.
- Mainly used for finishing IC engine cylinder.

996. What will be the effect on surface finish on a workpiece at maximum depth of cut during an operation?

(a) Surface finish and rate of material removal rate increases but cutting tool gets heated
(b) Surface finish and cutting tool cooling rate decreases, but material removal rate increases
(c) Surface finish and cutting tool cooling rate increases, but material removal rate decreases
(d) Surface finish and rate of material removal rate increases but cutting tool temperature remains constant

APPSC Poly Lect. Automobile Engg., 2020

Ans. (b) : The effect on surface finish on a workpiece at maximum depth of cut during an operation surface finish and cutting tool cooling rate decreases but material removal rate increases.

997. What is a machine tool?

(a) It is a single point cutting tool used in sophisticated CNC machines which can be operated from remote locations.
(b) It is a type of advanced abrasive multipoint cutting tool and which gives highest quality of surface machining.
(c) It is a tool which holding the cutting tool and able to remove metal from workpiece in order to give required size, configuration and surface finish.
(d) It is type of turret lathe machine which holding the multiple cutting tools and able to perform all the basic operations metal from workpiece in order to give required size, configuration and surface finish.

APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Machine Tool–It is a tool which while holding the cutting tool and able to remove metal from workpiece in order to give required size configuration and surface finish.

998. In a centre lathe, which component is used to provide motion to the cutting tool in direction perpendicular direction to the axis of rotation of the workpiece?

(a) Tool post (b) Lead screw
(c) Carriage (d) Cross-slide

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : The cross slide allows for tool travel 90 degree to the bed of the lathe. The cross slide makes upto the axis of the machine. The x axis of the machine controls the diameter of your work compound rest.

999. In machining operation if path of generatrix and directrix are circular and straight line respectively, the surface obtained will be

(a) Cylindrical
(b) Helical
(c) Plain
(d) Surface of revolution

UPPSC AE 13.12.2020, Paper-I

Ans. (a) : In machining operation if path of generatrix and directrix are circular and straight line respectively, the surface obtained will be cylindrical.

1000. The type of reamer used for reaming operation in a blind hole is:

(a) Right-hand spiral fluted reamer
(b) Left-hand spiral fluted reamer
(c) Straight fluted reamer
(d) No reaming possible

CIL MT 27.02.2020

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Ans. (a) : Right hand spiral reamers should be selected for blind holes. A right hand spiral also helps bridge interruptions such as cross-holes or keyway. They perform very well in highly ductile materials.

1001. In lathe operation, the depth of the cut required to reduce the diameter of a bar from100 mm to 90 mm is:

(a) 40 mm (b) 5 mm
(c) 10 mm (d) 20 mm

NLCIL GET 17.11.2020, Shift-II

Ans. (b) : In lathe operation, the depth of the cut

$$= \frac{d_1 - d_2}{2}$$

$$= \frac{100 - 90}{2}$$

Depth of cut = 5 mm

1002. The time taken to face a work piece of 80 mm diameter for the spindle speed of 90 rpm and cross feed of 0.3 mm/rev will be

(a) 4.12 min (b) 3.24 min
(c) 2.36 min (d) 1.48 min

ESE 2020

Ans. (d) : Given,
d = 80 mm
N = 90 rpm
f = 0.3 mm /rev
L = d/2

$$T_m = \frac{L}{fN}$$

$$L = \frac{80}{2} = 40 \text{ mm}$$

$$T_m = \frac{40}{90 \times 0.3}$$

T_m= 1.48 min

1003. A feed f for the lathe operation is

(a) $\frac{N}{L \times T_m}$ mm/rev (b) $\frac{L}{N \times T_m}$ mm/rev

(c) $\frac{T_m}{N \times L}$ mm/rev (d) $\frac{T_m \times L}{N}$ mm/rev

Where : T_m = Machining time t in min.
N = Speed in rpm
L = Length of cut in mm

ESE 2020

Ans. (b) : Machining time, $T_m = \frac{L}{fN}$

Where, L = length of cut
f = feed
N = rpm

$$\boxed{f = \frac{L}{N \times T_m}}$$

1004. The main advantage of the radial drilling machine is that

(a) It is very compatible and handy for machining
(b) It is accurate, economical, portable and least time consuming while machining
(c) Heavy workpieces can be machined in any position without moving them
(d) Small workpieces can be machined and it can be used for mass production as well

ESE 2020

Ans. (c) : Heavy workpieces can be machined in any position without moving them.

1005. Facing operation _____ of the work piece.

(a) increases the length.
(b) decreases the diameter.
(c) decreases the length.
(d) increases the diameter.

NLCIL GET 17.11.2020, Shift-II

Ans. (c) : Facing is the process of removing metal from the end of work piece to produce a flat surface.
Facing operation decreases the length of the work piece.

1006. Circular saw blades are specified by their diameter, number of teeth and

(a) gauge (b) maximum rpm
(c) arbor-hole (d) number of gullets

RPSC IOF, 2020

Ans. (c) : Circular saw blades are specified by their diameter, number of teeth and arbor- hole.
The size of an arbor hole on a circular saw blade will vary depending on the blades outside diameter.

1007. What is the unit of wear coefficient (k)?

(a) mm^3/sec (b) mm^3/N-m
(c) mm^3/N (d) dimensionless

RPSC IOF, 2020

Ans. (b) : Wear factor W = k × F × V × T
Where,
k = wear factor (mm^3)
F = Force (N)
V = Velocity (m/se)
T = Time (sec)
Unit of wear coefficient (k) is mm^3/N-m

1008. The time required in drilling a 8 mm diameter hole through a 24 mm thick mild steel plate with a drill bit running at 360 rpm and a feed of 0.5 mm per revolution is :
(neglect the approach and over travel of drill)

(a) 8 s (b) 16 s
(c) 4 s (d) 0.33 s

APPSC Poly Lect. 13.03.2020

Ans. (a) : Given data,
Hole diameter (d) = 8 mm
Thickness or Hole length (L) = 24 mm
Speed (N) = 360 rpm, feed (f) = 0.5 mm/rev
Time required to drilling $(T) = \frac{L}{fN}$ (if neglect the approach and our travel of drill)

$$= \frac{24\,\text{mm}}{\frac{0.5\,\text{mm}}{\text{rev}} \times \frac{360}{60}\frac{\text{rev}}{\text{sec}}}$$

$$= \frac{24 \times 60}{0.5 \times 360}\text{sec} = \frac{4}{0.5}\text{sec}$$

$$T = \frac{40}{5} = 8\,\text{sec}.$$

$$\therefore \Rightarrow \left[\text{Time}(T) = 8\,\text{sec}\right]$$

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1009. Which of the following manufacturing processes provides better dimensional control and surface finish?
(a) machining (b) casting
(c) cold and hot forming (d) welding

VIZAG MT, 14.12.2020

Ans. (a) : Machining process provides better dimensional control and surface finish.

1010. Which one of the following milling operations is used to produce key ways?
(a) Slitting (b) Slotting
(c) Straddle Milling (d) Form Milling

VIZAG MT, 14.12.2020

Ans. (b) : Slotting process is used to produce key ways.

1011. The grinding wheel which is recommended for the grinding of ceramic materials is
(a) aluminium oxide (b) siliconcarbide
(c) diamond (d) garnet

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : For grinding of material such as ceramic diamond abrasives are preferred.

1012. Severe grinding operations can cause _____.
(a) compressive residual stresses
(b) tensile residual stresses
(c) angular distortion
(d) zero residual stress

VIZAG MT, 14.12.2020

Ans. (a) : Compressive residual stress.

1013. Lathe machine cannot carry out:
(a) facing (b) planning
(c) turning (d) drilling

BHEL ET 2019

Ans. (b) : Lathe machine operation–
- Facing
- Planning
- Contour turning
- Chamfering
- From turning
- Threading
- Drilling
- Planning

1014. The manufacture has marking on a grinding wheel as 'A 36 L 5 V'. The code 'V' represents the :
(a) grade (b) structure
(c) abrasive particle size (d) bond

BHEL ET 2019

Ans. (d) : Grinding wheel designation
A 36 L 5 V
A - Abrasine (Alphabet)
36 - grain size (Numerical)
L - Grade / Hardness (Alphabet)
5 - Structure (Numerical)
V - Bond (Alphabet)

1015. A copper bar of diameter 200 mm is turned with a feed rate of 0.25 mm/rev with depth of cut of 4 mm. Spindle speed is 160 rpm. The material removal rate (MRR) in mm^3/s is:
(a) 150 (b) 167.55
(c) 1500 (d) 1675.5

BHEL ET 2019

Ans. (d) : diameter d = 200 mm
feed rate = 0.25 mm/rev
depth of cut = 4 mm
spindle speed = 160 rpm
$v = \pi DN$ mm/min
$v = \pi \times 200 \times 160$
= 100530.9649 mm/min
Material removal rate (MRR) = fdv mm/min
$= \frac{0.25 \times 4 \times 100530.9649}{60}$
= 1675.516 mm/sec.

1016. In a shaping process, the cutting velocity is provided to
(a) neither work piece nor cutting tool
(b) work piece and cutting tool
(c) cutting tool
(d) work piece

BHEL ET 2019

Ans. (c) : Shaping process - Shaping is defined as a process for removing metal surfaces in horizontal, vertical and angular planes by the use of a single point cutting tool held on the ram and reciprocates the tool.
Operations of shaper– The shaper is a machine tool used primarily for making slots, grooves and keyways.

1017. The obtuse angle, included between the chisel edge and the lip as viewed from the end of a drill, is called-
(a) Chisel edge angle
(b) Helix or rake angle
(c) Point angle
(d) Lip clearance angle

MECON MT 2019

Ans. (a) : Chisel edge angle

1018. In which milling operation are the entry angle zero and the exit angle non-zero?
(a) Up-milling (b) Down-milling
(c) Slot-milling (d) Face milling

MECON MT 2019

Ans. (a) : Up-milling

1019. In a shaper, the length of stroke is increased by-
(a) Increasing the length of the arm
(b) Increasing the centre distance of bull gear and crank pin
(c) Decreasing the centre distance of bull gear and crank pin
(d) Decreasing the length of the slot in the slotted lever

MECON MT 2019

Ans. (b) : Increasing the centre distance of bull gear and crank pin

1020. The facing is an operation of -
(a) Embossing a diamond shaped pattern on the surface of a workpiece
(b) Reducing the diameter of a workpiece over a very narrow surface
(c) Machining the ends of a workpiece to produce a flat surface square with the axis
(d) Bevelling the extreme end of a workpiece

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Ans. (c) : The facing is an operation of Machining the ends of a workpiece to produce a flat surface square with the axis.

1021. Which of the following is mounted directly on the lathe bed on the same side as a tail stock in the centre lathe?
(a) Carriage (b) Headstock
(c) Turret saddle (d) Legs

MECON MT 2019

Ans. (c) : Turret saddle is mounted directly on the lathe bed on the same side as a tail stock in the centre lathe.

1022. Which of the following machines is used for milling both internal, external short threads and surfaces?
(a) Drum type milling machine
(b) Bed type milling machine
(c) Planer type milling machine
(d) Planetary milling machine

MECON MT 2019

Ans. (d) : Planetary milling machine is used for milling both internal, external short threads and surfaces.

1023. The thickness of the layer of material removed in one pass of the work piece under the cutter is known as-
(a) Depth of cut (b) Feed
(c) Cutting speed (d) Machining time

MECON MT 2019

Ans. (a) : Depth of cut is the thickness of the layer of material removed in one pass of the work piece under the cutter.

1024. The saw milling is an operation of-
(a) Reproduction of an outline of a template on a workpiece
(b) Producting groove around the periphery of a cylindrical or conical workpiece
(c) Producing narrow slots or grooves on a workpiece
(d) Machining several surfaces of a workpieces simultaneously

MECON MT 2019

Ans. (c) : The saw milling is an operation of producing narrow slots or grooves on a workpiece.

1025. The operation of finishing a predrilled hole is known as :
(a) Boring (b) Reaming
(c) Counter boring (d) Spot facing

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : The operation of finishing a predrilled hole is known as reaming.
• It is the process of exacting the hole and finishing it. Expected accuracy of finish is ± 0.005 mm.
Finally, reaming is a cutting process that involves the use of a rotary cutting tool to create smooth interior walls in an existing hole in a workpiece.
• The primary purpose of reaming is simply to create smooth walls in an existing hole.
Manufacturing companies perform reaming using a milling machine or drill press.

1026. Abrasive grain size of 10 on grinding wheel refers to :
(a) Fine grain wheel
(b) Coarse grain wheel
(c) Medium grain wheel
(d) Very fine grain wheel

SJVN ET 2019

Ans. (b) : Abrasive grain size of 10 on grinding wheel refers to coarse grain wheel.

1027. Which of the following is an artificial abrasive used in grinding wheel?
(a) Sandstone (b) Corundum
(c) Silicon carbide (d) Emery

GPSC Engineer, Class-II Pre-19.01.2020
SJVN ET 2019

Ans. (c) : Silicon carbide is an artificial abrasive used in grinding wheel.

1028. What is the purpose of inscribing the concentric circles on the face of the chuck in lathe
(a) To improve the appearance
(b) To facilitate lubrication in the chuck
(c) To minimize the weight of the chuck
(d) To facilitate quick centering of work pieces

TNPSC AE 2019

Ans. (d) : To facilitate quick centering of work pieces

1029. A steel shaft of 25 mm diameter is turned at a cutting speed of 50 m/min. Find the RPM of the shaft.
(a) 637 (b) 159
(c) 6280 (d) 1570

TNPSC AE 2019

Ans. (a) : Data given-
D = 25 mm, V = 50 m/min
We know that

$$V = \frac{\pi DN}{1000}$$

$$50 = \frac{3.14 \times 25 \times N}{1000}$$

$$N = 636.94 \simeq 637 (RPM)$$

1030. Process of shaping thin metal sheets by processing them against a form tool is called
(a) Upsetting (b) Spinning
(c) Reaming (d) Deep drawing

TNPSC AE 2019

Ans. (b) : Metal spinning, also known as spin forming or spinning or metal turning most commonly, is a metal working process by which a disc or tube of metal is rotated at high speed and formed into an axially symmetric part. Spinning can be performed by hand or by a CNC Lathe.

1031. Dressing is
(a) the process of changing the shape of the grinding wheel
(b) the process of removing the metal loading and breaking away the glazed surface of the grinding wheel
(c) the process of disposing the grinding wheel as it no longer useful
(d) the process of adding excess abrasive grains on the grinding wheel

TNPSC AE 2019

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Ans. (b) : Dressing is the process of removing the metal loading and breaking away the glazed surface of the grinding wheel.

1032. Abrasive are not used in :
(a) Buffering process
(b) Burnishing process
(c) Polishing process
(d) Super finishing process

OPSC AEE 2019 Paper-I

Ans : (b) : In burnishing process, the jig-jag edges of material gets converted into straight and smooth edges, which is not required any of abrasive. So, abrasives are not used in burnishing process.

1033. A milling cutter is having 8 teeth rotating at 100 rpm. Workpiece feed is set at 40 mm/min. So feed per tooth is :
(a) 5 mm (b) 0.05 mm
(c) 0.4 mm (d) 0.2 mm

OPSC AEE 2019 Paper-I

Ans : (b) :
Number of teeth (Z) = 8
Speed (N) = 150 rpm
Workpiece feed f = 40 mm/min
Feed per tooth f_t = ?
Workpiece feed (f) = $f_t ZN$
$40 = f_t \times 8 \times 100$
$f_t = 0.05$ mm

1034. In HSS, tungsten can be substitute by :
(a) Chromium (b) Nickel
(c) Molybdenum (d) Cobalt

OPSC AEE 2019 Paper-I

Ans : (c): In HSS, tungsten can be substituted by molybdenum.

1035. Cutting speeds for Turning with HSS on cast iron is about
(a) 20 m/min (b) 30 m/min
(c) 35 m/min (d) 15 m/min

Gujarat PSC AE 2019

Ans : (a) : Cutting speeds for turning with HSS on cast iron is about 20 m/min.

1036. As per ISO, the grading for carbide tool for cast iron and non-ferrous steel is expressed as:
(a) K_{01} to K_{40} (b) P_{01} to P_{60}
(c) M_{10} to M_{30} (d) None of these

HPPSC ADF 05.03.2019

Ans. (a) : K_{01} to K_{40}

1037. A clamping device for holding work in the lathe is known as -
(a) Apron (b) Bed
(c) Carriage (d) Chuck

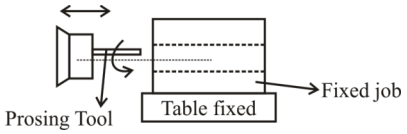
MECON MT 2019

Ans. (d) : A clamping device for holding work in the lathe is known as chuck.

1038. In which of the following machine, the work remains stationary and the tool is rotated?
(a) Vertical boring machine
(b) Horizontal boring machine
(c) Precision boring machine
(d) Jig boring machine

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : As we know, the horizontal boring machine, in which the work is supporting on a table which is constant/fixed and the tool turns into a horizontal axis for rotates.



1039. The abrasive recommended for grinding materials of high tensile strength is
(a) silicon carbide (b) aluminium oxide
(c) sand stone (d) diamond

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : Aluminium oxide abrasive is recommended for grinding of materials which are high strength & tensile strength and are hard enough.

1040. The operation of machining several surfaces of a work piece simultaneously is called
(a) profile milling (b) gang milling
(c) saw milling (d) helical milling

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : Gang Milling–Simultaneous machining of a number of flat horizontal and vertical surfaces of a workpiece by using a combination of more than two tools/cutter mounted on a common arbor.

1041. In gear hobbing
(a) only hob rotates
(b) only gear blank rotates
(c) both hob and gear blank rotates
(d) neither hob nor gear blank rotates

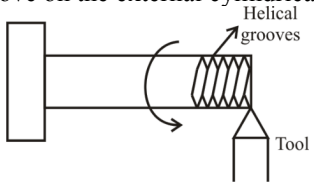
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : Hobbing is a machining process for gear, cutting, cutting splines etc. The teeth or splines of the gear are progressively cut into material by a series of cuts made by a cutting tool hob. In gear hobbing both hob & blank rotates.

1042. Threading is an operation of
(a) smoothing and squaring the surface around a hole
(b) sizing and finishing a small diameter hole
(c) producing a hole by removing metal along the circumference of a hollow cutting tool
(d) cutting helical grooves on the external cylindrical surface

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Threading is a process of producing/cutting of helical groove on the external cylindrical surface.



1043. The relief or clearance at the cutting edge of a flat drill varies from
(a) 3° to 8° (b) 20° to 30°
(c) 60° to 90° (d) 90° to 120°

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : The relief or clearance at the cutting edge of a flat drill varies from 3 to 8°.

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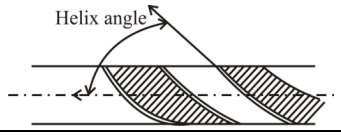
1044. The usual value of the helix angle of a drill is
(a) 20° (b) 30°
(c) 45° (d) 60°

Nagaland PSC (CTSE) 2018, Paper-I

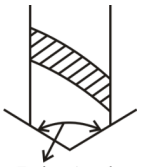
Ans. (b) : The usual or most common helix angle of a drill is about 30°C. It is the angle between the leading edge of the land & axis of the drill.



1045. The point angle of a drill, for drilling stainless steel, is
(a) 90° (b) 118°
(c) 135° (d) 150°

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : The point angle of drill for drilling stainless steel is 135°. Point angle increase as the hardness of the material increase, it decrease or less for after material most common is 118°.



1046. In a planer
(a) tool is stationary and work reciprocate
(b) work is stationary and tool reciprocates
(c) tool moves over stationary work
(d) tool moves over reciprocating work

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : A planer is a type of metal working machine tool uses linear relative motion between the workpiece & a single point cutting tool to cut the workpiece.
In which tool is at stationary & work is moving with table. (reciprocates)

1047. The method of grinding used to produce internal cylindrical holes and tapers, is
(a) internal cylindrical grinding
(b) form grinding
(c) external cylindrical grinding
(d) surface grinding

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : The internal cylindrical grinding method is used to produce or grind the internal cylindrical holes & tapers, which produce higher surface finish.

1048. A fine grained grinding wheel is used to grind
(a) hard and brittle materials
(b) soft and ductile materials
(c) hard and ductile materials
(d) soft and brittle materials

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : A fine grained grinding wheel is used to grind hard and brittle materials, because fine grained grinders have good adhesive property with stand high forces.

1049. The hardness of a grinding wheel is specified by
(a) Brinell hardness number
(b) Rockwell hardness number
(c) Vickers pyramid number
(d) Letter of alphabet

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Specification of grinding wheel.

51	C	45	O	13	R
↓	↓	↓	↓	↓	↓
Manufacter's symbol	Abrassive type C → Carbide A → Al_2O_3	Grain size	Grade [Soft, medium & hard] [Hardness]	Structure	Bond type

1050. An exact centering can be done on a lathe by
(a) Four jaw chuck (b) Three jaw chuck
(c) Dog clutch (d) Face plate

HPPSC AE 2018

Ans. (a) : An exact centering can be done on a lathe by four jaw chuck.
Type of Chuck in Lathe machine are as follows:
- Four jaws independent chuck.
- Three jaws universal chuck.
- Combination chuck.
- Magnetic chuck.
- Collet chuck.
- Air or hydraulic operated chuck.

1051. Knurling operation is performed on
(a) Shaper machine (b) Milling machine
(c) Grinding machine (d) Lathe machine

HPPSC AE 2018

Ans. (d) : Knurling–Knurling is a process of impressing a diamond shaped or straight line pattern into the surface of workpiece by using specially hardend metal wheels to improve its appearance and provide a better gripping surface. Knurling operation is performed on lathe machine.

1052. To produce involute profile of the gear on milling machine, which one is required
(a) Differential indexing head
(b) Angle milling cutter
(c) Slab milling cutter
(d) Profile milling cutter

HPPSC AE 2018

Ans. (d) : To produce involute profile of the gear on milling machine profile milling cutter are used.

1053. The surface finish of a cylindrical hole is maintained by
(a) Honing (b) Lapping
(c) Internal cylindrical grinding (d) Boring

HPPSC AE 2018

Ans. (a) : Honing process–Honing is a finishing process, in which a tool called hone carries out a combined rotary and reciprocating motion while the work piece does not perform any working motion. The Honning process is carried out by mechanically rubbing the honing stone (cutting tool) against the workpiece surface (normally internal) along the controlled path. The surface finish of cylindrical hole is maintained by honing.

1054. In shaping process
(a) Tool reciprocates and feed is given to job
(b) Job reciprocates and feed is given to job
(c) Both tool and job reciprocate relative to each other
(d) Both tool and job remain static

HPPSC AE 2018
Haryana PSC Civil Services Pre, 2014

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Ans. (a) : In shaping process, The shaper holds the single point cutting tool in ram and workpiece is fixed in the table and feed is given to job.
During the forward stroke, the ram is holding the tool is reciprocating over the workpiece to cut into the required shape. During the return stroke, no metal is cutting in the shaper machine, the rotary motion of the drive is converted into reciprocating motion of ram holding the tool.

1055. The operation of cutting a work piece after it has been machined to the desired shape and size
(a) tapping (b) undercutting
(c) parting-off (d) counter boring

TNPSC AE 2018

Ans. (c) : After parting-off, a work piece has been machined to the desired shape and size.

1056. Lapping is-
(a) the operation of sizing and finishing a small diameter hole by removing a very small amount of material
(b) the operation of making a cone-shaped enlargement of the end of a hole
(c) the operation of smoothing and squaring the surface around a hole
(d) the operation of enlarging the end of a hold cylindrically

TNPSC AE 2018

Ans. (a) : Lapping is the operation of sizing and finishing a small diameter hole by removing a very small amount of material.

1057. Cutting tool is much harder than the work piece. Yet the tool wears out during the tool-work interaction, because
(a) extra hardness is imparted to the work piece due to coolant used
(b) oxide layers formed on the work piece surface impart extra hardness to it
(c) extra hardness is imparted to the work piece due to severe rate of strain
(d) vibration is induced in the machine tool

TNPSC AE 2018

Ans. (c) : Cutting tool is much harder than the work piece. Yet the tool wears out during the tool-work interaction, because extra hardness is imparted to the work piece due to severe rate of strain.

1058. A block of length 200 mm is machined by a slab milling cutter 34 mm in diameter. The depth of cut and table feed are set at 2 mm and 18 mm/minute, respectively. Considering the approach and the over travel of the cutter to be same, the minimum estimated machining time (minutes) per pass is
(a) 12 (b) 10
(c) 11 (d) 15

RPSC Vice Principal ITI 2018

Ans. (a) : Given that
Length (L) = 200 mm
Diameter (D) = 34 mm
Depth of cut (t) = 2 mm
Table feed (F_m) = 18 mm/min
Approach and over travelled A = B = Same
Machine time (T_m) = ?

$$T_m = \frac{L + A + B}{f\,N}$$

where, L = length
A = Approach
B = Over travelled

$$\text{Approach (A)} = \sqrt{d(D-d)} = \sqrt{2(34-2)}$$

A = B = 8 mm
Then machine time

$$T_m = \frac{L + A + B}{f\,N} = \frac{200 + 8 + 8}{18 \times 1} = \frac{216}{18}$$

T_m = 12 min

1059. In cold working
(a) strength, hardness and ductility increase
(b) strength and hardness increase, but ductility decreases
(c) strength, hardness and ductility decrease
(d) strength and hardness decrease, but ductility increases

GPSC ARTO Pre 30.12.2018

Ans. (b) : In cold working, due to strain hardening process strength and hardness increases but ductility decreases.

1060. A hole is drilled in an aluminium alloy block with a 15 mm drill at a feed of 0.1 mm/rev. The spindle rotational speed is 500 RPM. The MRR in mm^3/min will be:
(a) 7000 (b) 9800
(c) 7500 (d) 8840

AAI Jr. Executive 29.11.2018

Ans. (d) : d = 15 mm
f = 0.1 mm/rev
N = 500 rpm

$$MRR = \frac{\pi}{4} d^2 \times f \times N \text{ mm}^3/\text{min}$$

$$= \frac{\pi}{4} \times 15 \times 15 \times 0.1 \times 500$$

$$= 8835.729 \text{ mm}^3/\text{min}$$

1061. Select the operation that does NOT require a jig.
(a) Drilling (b) Tapping
(c) Turning (d) Reaming

AAI Jr. Executive 29.11.2018

Ans. (c) : Turning does not require jig.

1062. Which of the following is not the characteristics of cutting tool material?
(a) High hot hardness
(b) High wear resistance
(c) High toughness
(d) High coefficient of friction at tool chip interface
(e) High thermal conductivity

CGPSC AE 25.02.2018

Ans. (d) : Characteristics of cutting tool material-
1. High hot hardness
2. High wear resistance
3. High toughness
4. High thermal conductivity
5. Low coefficient of friction at tool chip interface.

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1063. Cutting fluid used in machining operation should not:
(a) have high heat absorbing capacity
(b) have low flash point
(c) have good lubricating property
(d) wash-off the chips
(e) be non-toxic

CGPSC AE 25.02.2018

Ans. (b) : Common properties of cutting fluid-
1. High heat absorbing capacity
2. High flash point
3. Non corrosive to tool and work material.
4. It should be harmless to operate and bearing
5. It has good lubrication properties to reduce frictional force and to reduce the power consumption.
6. Wash-off the chips easily.
7. Also be non-toxic.

1064. Feed in a lathe is expressed in :
(a) mm (b) mm per revolution
(c) mm per degree (d) rpm

Vadodara Muncipal Corp. DEE, 2018

Ans. (b) : Feed is expressed as mm per revolution.

1065. Find the time required for drilling a 18 mm hole in a work piece having thickness 50 mm. Assume cutting speed 12 metres/minute and feed 0.2 mm/revolution. Neglect the length of approach.
(a) 1.14 minutes (b) 1.16 minutes
(c) 1.18 minutes (d) 1.20 minutes

MPSC Lect. (Prod.) 2018

Ans. (c)

$D = 18$ mm, $T = 50$ mm

$V = 12$ m/min, $f = 0.2$mm/rev.

$V = \pi DN$

$$N = \frac{V}{\pi D} = \frac{12}{\pi \times 0.018} = 212.206\,\text{rpm}$$

$$\text{Drilling time} = \frac{T}{fN}$$

$$= \frac{50}{0.2 \times 212.6} = 1.178\,\text{minutes}$$

1066. The angle at which the compound rest is set for turning a taper job is:
(a) $\tan^{-1}\frac{D+d}{l}$ (b) $\tan^{-1}\frac{D+d}{2l}$
(c) $\tan^{-1}\frac{D-d}{l}$ (d) $\tan^{-1}\frac{D-d}{2l}$

PTCUL AE 25.06.2017

Ans. (d) :

1067. In upmilling, the cutter rotates:
(a) Against the stationary work piece
(b) In the same direction of travel of the work piece
(c) Against the same direction of travel of the work piece
(d) None of these

PTCUL AE 25.06.2017

Ans. (c) : In upmilling, the cutter rotates against the same direction of travel of the work piece.

1068. The unwanted vibrational motion between tool and work is known as :
(a) Deflection (b) Amplitude
(c) Chatter (d) friction

Punjab PSC SDE 12.02.2017

Ans. (c) : The unwanted vibrational motion between tool and work is known as chatter.

1069. The grinding of long, slender shafts or bars is usually done by:
(a) infeed grinding
(b) through feed grinding
(c) end feed grinding
(d) any one of these

HPPSC Asstt. Prof. 18.09.2017

Ans. (b) : The grinding of long, slender shafts or bars in usually done by through feed grinding.

1070. Z-Z method of cutting fluid application is done in
(a) Turning (b) milling
(c) Grinding (d) Broaching

WBPSC AE, 2017

Ans. (c) : Z-Z method of cutting fluid application in grinding. Application of cutting fluid at high pressure through the hole in the tool.

1071. The chamfering is an essential operation after
(a) knurling (b) rough turning
(c) boring (d) thread cutting

Sikkim PSC (Under Secretary), 2017

Ans. (d) : The chamfering is an essential operation thread cutting because after chamfering bolt can be easily assembled into the bolt.
Chamfering provide cone shape at the end of the bolt.
Chamfering can be done by after thread cutting only because nut screwed easily.

1072. In a lathe operation, the machining time is calculated by the relation
(a) $T = \frac{\text{Length of cut}}{\text{RPM}}$
(b) $T = \frac{\text{Length of cut} \times \text{Feed}}{\text{RPM}}$
(c) $T = \frac{\text{Length of cut}}{\text{Feed} \times \text{RPM}}$
(d) $T = \frac{\text{RPM}}{\text{Length of cut} \times \text{Feed}}$

Karnataka PSC Lect., 27.05.2017

Ans (c) : Machining time $T = \frac{\text{Length of cut}}{\text{Feed} \times \text{RPM}}$

1073. A gear has 14 teeth. Which of the following indexing movements can be used?
(a) 3 complete turns on a 42 hole circle
(b) 2 complete turns and 6 holes in an 18 hole circle
(c) 2 complete turns and 18 holes in a 21 hole circle
(d) 2 complete turns and 36 holes in a 49 hole circle

Karnataka PSC Lect., 27.05.2017

Ans (c) : 2 Complete turns and 18 holes in a 21 hole circle.

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1074. The function of taper turning process is to

(a) reduce the diameter of a workpiece along its length
(b) Reduce the diameter by removing the material about an axis offset from the axis of a workpiece.
(c) Remove the material from the end surface of a workpiece
(d) All of the above

Karnataka PSC Lect., 27.05.2017

Ans (b) : Taper Turning- It is an operation of producing on external conical surface on a work piece. A small taper may be produced with the help of a forming tool or chamfering tool. But the larger tapers are produced by swiveling the compound rest at the required angle or by offsetting the tail stock or by taper turning attachment.

1075. A steel punch can be stressed to a maximum compressive stress of 800 MN/m^2. The least diameter of the hole which can be punched through a steel plate 14 mm thick with its ultimate shear strength being 300 MN/m^2 is

(a) 21 mm (b) 31 mm
(c) 10 mm (d) 15 mm

Karnataka PSC Lect., 27.05.2017

Ans (a) : $\sigma_c = 800\ MN/m^2$

$t = 14\ mm$

$\tau_u = 300\ MN/m^2$

$d = ?$

$F_{comp.} = \sigma_{comp.} A$

$= 800 \times \frac{\pi}{4} d^2$

$F_c = \pi d \tau_u t$

$= 800 \times \frac{\pi}{4} d^2 = \pi d \times 300 \times 14$

$d = 21\ mm$

1076. The rake angle in a drill

(a) increases from centre to periphery
(b) decreases from centre to periphery
(c) remains constant
(d) is irrelevant to the drilling operation

TNPSC AE 2017

Ans. (a) : The rake angle in a drill varies from minimum near the dead center and maximum at the outermost periphery. At the outermost periphery the rake angle (α) become equal to helix angle (Ψ)

$\therefore \boxed{\alpha_{max} = \psi}, \boxed{\alpha \le \psi}$

1077. Gear hobbing process is faster than milling because

(a) Indexing time is less
(b) Hob rotates faster
(c) Work rotates faster
(d) Several teeth cut a time

Vizag Steel (MT) 2017

Ans. (d) : Gear hobbing process is faster than milling because several teeth cut a time.

1078. Gear shaper can be used to cut following types of gear

(a) Internal (b) External
(c) Accurate (d) All of the above

Vizag Steel (MT) 2017

Ans. (d) : Gear shaping- is similar to the rack type cutting process, excepting that, the linear type rack cutter is replaced by a circular cutter.
–This process is suitable for cutting various type of gear like spur, herringbone, sprockets, internal gears and cluster gears etc.

1079. The correct sequence of operations for making alloy steel gears is

(a) turning, hobbing, heat treatment, shaving
(b) heat treatment, turning, hobbing, shaving
(c) shaving, turning, hobbing, heat treatment
(d) turning, hobbing, shaving, heat treatment

BPSC AE Mains 2017 Paper - VI

Ans : (d) : The correct sequence of operations for making alloy steel gears is
- Turning, hobbing, shaving, heat treatment.

1080. The correct sequence in increasing hardness of tool materials is

(a) HSS, tungsten carbide, cubic boron nitride, diamond
(b) HSS, cubic boron nitride, tungsten carbide, diamond
(c) cubic boron nitride, HSS, tungsten carbide, diamond,
(d) tungsten carbide, cubic boron nitride, HSS, diamond

BPSC AE Mains 2017 Paper - VI

Ans : (a) : HSS, tungsten carbide, cubic boron nitride, diamond

1081. Which of the following is not the function of a cutting fluid?

(a) Removal of heat
(b) Lubrication
(c) Protection against corrosion
(d) Abrasion of workpiece

BPSC AE Mains 2017 Paper - VI

Ans : (d) : Function of cutting fluid:
- Cooling of the job and the tool.
- Lubrication at the chip-tool interface.
- Cleaning the machining zone by washing away the chip particles and debris.
- Protection of the nascent finished surface.

1082. Choose the correct order of tool materials arranged, according to the decreasing order of their hot hardness.

(a) Ceramics, Cermets, Tungsten Carbide, HSS
(b) Cermets, Ceramics, Tungsten Carbide, HSS
(c) Tungsten Carbide, Ceramics, Cermets, HSS
(d) Cermets, Tungsten Carbide, Ceramics, HSS

CIL MT 26.03.2017

Ans. (a) : The correct order of decreasing hot hardness is Ceramics, Cermets, Tungsten Carbide and HSS.

1083. HSS has the property to–

(a) withstand high temperature
(b) provide lubrication by itself
(c) become hard during machining
(d) reduce impact of the tool

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : HSS has the property to reduce impact of the tool.

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1084. A work piece of 100 mm diameter is to be machined at a cutting speed of 1000 meters per minute. The rpm of the work piece should be–

(a) 1000 (b) 1590
(c) 2000 (d) 3183

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Given,
D = 100 mm, V = 1000 m/min, N=?

$$N = \frac{1000 \times V}{\pi \times D} = \frac{1000 \times 1000}{\pi \times 100} = 3183.00$$

1085. Consider just only the following parameters:

1. Grinding wheel diameter
2. Regulating wheel diameter
3. Speed of grinding wheel
4. Speed of regulating wheel

Which of the above parameters will influence the axial feed rate in centre-less grinding?

(a) 2 and 4 (b) 2 and 3
(c) 1 and 3 (d) 1 and 4

ESE 2017

Ans. (a) : Feed rate in centre-less grinding is given by

$f = \pi DN \sin\alpha$
$f \propto D$
$f \propto N$

Where D = diameter of regulating wheel
N = Speed of regulating wheel

1086. The process for producing smooth surfaces by plastically moving away the raised micro-irregularities on the surface and pressing it into the micro-cavities is known as

(a) Buffing (b) Burnishing
(c) Super finishing (d) Polishing

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Burnishing process for producing smooth surfaces by plastically moving away the raised micro-irregularities on the surface and pressing it into the micro-cavities.

1087. In a drilling operation–

(a) Torque is equal to the axial force.
(b) Torque is more than the axial force
(c) Torque is less than the axial force
(d) Torque is half the axial force

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : In a drilling operation, torque is half of the axial force.

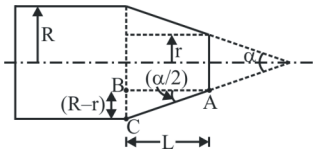
1088. For taper turning, a work piece having r- as radius at the small end, and R- as the radius at the big end with L- as the length of the taper will have a taper angle α given by:

(a) $\tan\alpha = \left(\frac{R-r}{L}\right)$ (b) $\tan 2\alpha = \left(\frac{R-r}{L}\right)$
(c) $\tan\alpha = \left(\frac{R-r}{2L}\right)$ (d) $\tan 2\alpha = \left(\frac{R-r}{2L}\right)$
(e) $\tan\alpha = 2\left(\frac{R-r}{L}\right)$

(CGPSC Polytechnic Lecturer 2017)

Ans. (*) : ΔBAC

$$\tan\left(\frac{\alpha}{2}\right) = \frac{BC}{BA}$$



$$\tan\frac{\alpha}{2} = \frac{(R-r)}{L}$$

$$\alpha = 2\tan^{-1}\left[\frac{R-r}{L}\right]$$

Note– Option (a) is given by CGPSC as answer.

1089. Which of the following lathes is suitable for a small engineering workshop involved in repair work?

(a) Centre lathe
(b) Tool-room lathe
(c) Special purpose lathe
(d) Hollow spindle lathe
(e) Capstan lathe

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Centre-Lathe Machine–A centre lathe is a machine tool that carries out many machine shop tasks. It can cut and smooth the face of a work piece. It can make holes to an exact size and position. It can also make external or internal threaded parts on a work piece. This ability to do different tasks makes it an important tool. In most tasks a lathe use is only limited by the size of the work piece.

The five most common tasks on a centre lathe that you will be perform are:

- Facing
- Turning
- Drilling
- Boring
- Threading
- Knurling
- Chamfering

1090. In a shaper, let N be the rotational speed of the bull gear and L be the length of the stroke, then the time T for completing one stroke is given by:

(a) $T = \frac{2L}{N}$ (b) $T = \frac{L}{2N}$
(c) $T = \frac{L}{N}$ (d) $T = 2\left(\frac{L}{N}\right)$
(e) $T = \frac{1}{2}\left(\frac{L}{N}\right)$

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : Time for completing one stroke

$$T = \frac{L}{N}$$

1091. Suggest a suitable machine from the following options for cutting T- slots on a very long and heavy apparatus table:

(a) Slotting machine
(b) Planer machine
(c) Shaper machine
(d) Vertical lathe
(e) Horizontal milling machine

(CGPSC Polytechnic Lecturer 2017)

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Ans. (b) : Planer Machine–A planer is a type of metal working machine tool that uses linear relative motion between the workpiece and a single-point cutting tool to cut the workpiece. A planer is similar to a shaper, but larger, and with workpiece moving, whereas in a shaper the cutting tool moves. The workpiece is securely fixed on a table called platen, and it reciprocates horizontally against a single edged cutting tool. The surface machined may be horizontal, vertical or at an angle.

Operations of Planer Machine–

- Planning flat horizontal, vertical and curved surface.
- Planning at an angle and machining dovetails.
- Planning slots, T-slots and grooves.

1092. In simple indexing method, the rotation of the index crank = $6\frac{2}{3}$ turns cannot be obtained by:

(a) Rotating indexing crank by 6 full rotation +10 holes in a 15- hole circle
(b) Rotating indexing crank by 6 full rotation +12 holes in an 18- hole circle
(c) Rotating indexing crank by 6 full rotation +14 holes in a 21- hole circle
(d) Rotating indexing crank by 6 full rotation +18 holes in a 27- hole circle
(e) Rotating indexing crank by 6 full rotation +21 holes in a 31- hole circle

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : In simple indexing method, the rotation of the index crank $= 6\frac{2}{3}$ turns cannot be obtained by 31 hole circle plate because 31 can not be divided by 3 completely

	Fully Rotation of Crank	Holes	Hole Circle
$6\frac{2}{3}\times\frac{5}{5}=6\frac{10}{15}$	6	10	15
$6\frac{2}{3}\times\frac{6}{6}=6\frac{12}{18}$	6	12	18
$6\frac{2}{3}\times\frac{7}{7}=6\frac{14}{21}$	6	14	21
$6\frac{2}{3}\times\frac{9}{9}=6\frac{18}{27}$	6	18	27

1093. A 16 mm hole is to be drilled through a mild steel plate of 25 mm thickness. The over-travel of the drill is 5 mm. What will be the drilling time if the feed rate is 150 mm/min?

(a) 0.2 min (b) 0.02 min
(c) 2.0 min (d) 2.2 min
(e) 2.22 min

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Data given
t = 25 mm
Over-travel (x) = 5 mm
f × N = 150 mm/min
D = 16 mm
Then drilling time T = ?

$$T = \frac{L_e}{f\times N} = \frac{(t+x)}{f\times N} = \frac{(25+5)}{150}$$

$$\boxed{T = 0.2\,\text{min}}$$

1094. _____ is not an abrasive used in grinding wheels.

(a) Aluminum oxide (b) Silicon carbide
(c) Cubic boron nitride (d) Sodium silicate
(e) Diamond

(CGPSC Polytechnic Lecturer 2017)

Ans. (d) : Aluminum oxide, silicon carbide, CBN and Diamond are used to an abrasive in grinding wheel whereas sodium silicate is not used as abrasive in grinding wheel.

1095. The maximum peripheral speed a grinding wheel is limited by

(a) The work material
(b) The diameter of the wheel
(c) The drive limitations
(d) The kind of bond used in the wheel

Nagaland CTSE 2016, 2017 Ist Paper

Ans. (d) : The maximum peripheral speed of a grinding wheel is limited by the kind of bond used in the wheel.

1096. Which machine can be used in for grinding the ball bearing outer races

(a) Cylindrical grinding machine
(b) Centre less grinding machine
(c) Magnetic grinding machine
(d) Surface grinding machine

Nagaland CTSE 2016, 2017 Ist Paper

Ans. (b) : Center less grinding machines is used for grinding the ball bearing outer races. Center less grinding may be external or internal.

1097. In machined components,

(a) fiber lines are arranged in a predetermined way
(b) fiber lines of rolled stock are broken
(c) there are no fiber lines
(d) fiber lines are scattered

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : In machined component fiber lines of rolled stock are broken.

1098. The finishing operation on a C40 steel component of length 120mm is done at 60 m/minute cutting speed. If the spindle speed and feed are 440 rpm and 0.1 mm/rev, respectively while the over travel of the tool beyond the length of the job is 2mm, then machining time for one pass in minutes is

(a) 1.77 (b) 2.17
(c) 2.77 (d) 3.77

Punjab PSC (Lect.) 06.08.2017

Ans. (c) : Given as, L = 120 mm
over travel = 2 mm
f = 0.1 mm / rev
N = 440 rpm

$$T = \frac{(\text{Length of over travel + component length})}{fN}$$

$$= \frac{(120+2)}{0.1\times 440} = \frac{61}{22} = 2.77\ \text{min}$$

1099. Which is the property of a tool that prevents plastic deformation at its cutting edge during machining process?

(a) High value of fracture toughness
(b) High value of hardness under heating

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(c) Chemical stability
(d) Adhesion resistance

JWM 2017

Ans. (b) : Hot hardness is the property of a tool that prevents plastic deformation at its cutting edge during machining process.

1100. The difference between (i) a horizontal borer and (ii) a lathe is
(a) In (i) the spindle is horizontal; in (ii) it is vertical
(b) In (i) the tool rotates; in (ii) the object rotates
(c) In (i) the object rotates; in (ii) tool rotates
(d) (i) produces a plain surface; (ii) produces a cylindrical surface

JWM 2017

Ans. (b) : In a horizontal borer, the tool rotates where as in a lathe, the object rotates.

1101. The chip thickness during slab milling process is directly proportional to
(a) Cutter's rotational speed
(b) Angle of contact of the tooth
(c) Number of teeth of the cutter
(d) Feed rate of work piece

JWM 2017

Ans. (d) : The chip thickness during slab milling process is directly proportional to feed rate of work piece.

1102. Consider the following statements with regard to lapping process:
1. Any type of flat material can be lapped to any shape
2. As the parts are not clamped during lapping no heat is generated so there is no warping
3. No burrs are produced

Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

JWM 2017

Ans. (b) : Lapping process-
• Use of loose abrasive between lap and the work piece.
• Any type of flat material can be lapped to any shape.
• No burrs are produced.

1103. HSS tool is used to machine a 20 mm diameter steel shaft, at a spindle speed of 1000 revolutions per minute. What is the cutting speed?
(a) π m/min (b) 20 π m/min
(c) π mm/min (d) 20 π mm/min

CIL MT 26.03.2017

Ans. (b) : Cutting speed = $\pi DN = \pi \times 20 \times 1000 = 20000\ \pi$ mm/min = 20 π m/min

1104. In which of the following operations performed on lathe machine, chips does not occur?
(a) Knurling (b) Boring
(c) Reaming (d) Threading cutting

CIL MT 26.03.2017

Ans. (a) : Knurling: Knurling is a manufacturing process whereby a visually-attractive diamond-shaped (crisscross) pattern is cut or rolled into metal. This pattern allows human hands or fingers to get a better grip on the knurled object than would be provided by the originally-smooth metal surface.
Boring: Boring always involves the enlarging of an existing hole, which may have been made by a drill or may be the result of a core in a casting.
Reaming: Reaming removes a small amount of material from the surface of holes. It is done for two purposes: to bring holes to a more exact size and to improve the finish of an existing hole.
Threading: It is the process of making internal or external threads on the workpiece.
Out of above four processes except knurling all other involves the chips. So Knurling is right option.

1105. A workpiece 200 mm long is to have 3° included angle taper turned on to it by offset centre method. The offset required is
(a) 200 sin 3° (b) 100 sin 3°
(c) $200\sin 1\frac{1^\circ}{2}$ (d) $100\sin 1\frac{1^\circ}{2}$

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Given, Long = 200 mm

$\theta = 3^\circ$

$\frac{\theta}{2}$ = Taper turning angle.

$$\text{Offset} = 200 \cdot \sin\left(\frac{\theta}{2}\right)$$

$$= 200\sin\frac{3^\circ}{2} = 200\sin 1\frac{1^\circ}{2}$$

1106. A drilling machine has spindle speeds from 70 to 800 in eight steps. The fourth speed from the lowest will be nearly
(a) 175 rpm (b) 236 rpm
(c) 320 rpm (d) 435 rpm

GPSC Asstt. Director of Transport 05.03.2017

Ans. (*) : Speeds are in geometric progression –
70, 70r, 70r^2, 70r^3, 70r^4, 70r^5, 70r^6, 70r^7 .....
8th speed = 70r^7

$800 = 70r^7$

$$\frac{800}{70} = r^7$$

$$r = \left(\frac{800}{70}\right)^{1/7} \qquad r = 1.416$$

Hence fourth speed = $70r^3 = 70 \times 1.416^3$

$$\boxed{N_4 = 198.84\,\text{rpm}}$$

1107. In a lathe gear box if the first four consecutive speeds are 19, 30, 47 and 74 rpm. The next speed would be
(a) 93 rpm (b) 104 rpm
(c) 111 rpm (d) 116 rpm

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : We know,

$$\frac{N_1}{N_2} = \frac{N_2}{N_3} = \frac{N_3}{N_4} = \frac{N_4}{N_5}$$

Given, $N_1 = 19$, $N_2 = 30$, $N_3 = 47$, $N_4 = 74$

So, $$\frac{N_3}{N_4} = \frac{N_4}{N_5} \Rightarrow N_5 = \frac{74 \times 74}{47} = 116.5$$

$$\boxed{N_5 \approx 116}$$

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1108. In a rack and pinion arrangement, the pinion has 30 teeth of 10mm pitch and rotates at 45 rpm. The rack is fixed to the table of a planning machine. The speed of the table will be

(a) 54 m/min (b) 27 m/min
(c) 13.5 m/min (d) 6.5 m/min

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Given,
Number of teeth, Z = 30
Pitch, p = 10 mm
Speed, N = 45 rpm
Speed of table,

$$v = \frac{Z \times p \times N}{1000}$$

$$= \frac{30 \times 10 \times 45}{1000}$$

$$\boxed{v = 13.5\,m/min}$$

1109. A dense structure of a grinding wheel is used for

(a) Hard materials (b) Brittle materials
(c) Finishing cuts (d) All of these

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : A dense structure of a grinding wheel is used for hard, brittle materials and finishing cuts.

1110. What is swing over carriage?

(a) The maximum diameter of workpiece that can be rotated over the bed ways
(b) The minimum diameter of workpiece that can be rotated over the bed ways
(c) The maximum diameter of workpiece that can be rotated over lathe saddle
(d) The minimum diameter of workpiece that can rotated over lathe saddle

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Swing over carriage is the maximum diameter of workpiece that can be rotated over lathe saddle.

1111. Mandrels are used to hold

(a) Drill Bits
(b) Hollow Work Pieces
(c) Cutting Tools
(d) Face Plate

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Mandrels is a gently tapered cylinder against which material can be forged or shaped. It is used to hold hollow work pieces.

1112. Gear hobbing produces more accurate gears than milling, because in hobbling

(a) There is continuous indexing operation
(b) Both hob and work piece are rotating
(c) Pressure angle is larger than in milling
(d) A special multi-tooth cutter is used

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (a) : Gear hobbling produces more accurate gears than milling, because in hobbling there is continuous indexing operation.

1113. The type of tool used in milling machine and broaching machine is

(a) single point cutting tool
(b) two point cutting tool
(c) three point cutting tool
(d) multi-point cutting tool

ISRO Scientist/Engineer (RAC) 07.05.2017

Ans. (d) : In milling and broaching machine, multi-point cutting tools are used.
In milling machine a multi point cutter is used where as cutting tool of broaching machine called broach also have number of teeth to remove material from work piece.

1114. Thumb rule between feed and nose radius in rough turning is

(a) f = 0.3r (b) f = 0.5r
(c) f = 0.7r (d) f = 0.9r

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : A simple rule of thumb connecting feed and nose radius is for rough turning
f(mm/rev) = 0.5 × Nose radius

1115. In a grinding wheel specification of 'wa36k8vbe', "wa" indicates

(a) Grain size (b) Type of grade
(c) Type of abrasive (d) Type of structure

ISRO Scientist/Engineer 07.05.2017

Ans. (c) : Grinding wheel specification of 'wa36k8vbe'

wa	–	36	–	K	–	8	–	vbe
↓		↓		↓		↓		↓
Abrasive type		Grain size		Type of grade		Type of structure		Bond materials

1116. Thrust force will increase with the increase in -

(a) Side cutting edge Angle
(b) Tool Nose radius
(c) Rake angle
(d) End cutting edge Angle

GMB AAE 25.06.2017

Ans. (a) : Thrust force will increase with the increase in side cutting edge angle.
• The angle between the side cutting edge and the side of the tool shank is called side cutting edge angle. It is often referred to as the lead angle.
• The angle between the end cutting edge and a line perpendicular to the shank of the tool shank is called end cutting edge angle.

1117. The design of cutting tools is based on :

(a) Tensile strength
(b) Rigidity
(c) Bending stress
(d) Compressive strength

GMB AAE 25.06.2017

Ans. (b) : The design of the cutting tools is based on rigidity.

1118. In a 48-A-100-H-8-C-BE grinding wheel, the grit size is represented by

(a) 48 (b) 100
(c) H (d) 8

WBPSC AE, 2017

Ans. (b) : A = Abrasive
100 = Grain size
H = Grade
8 = Structure
C = Bond

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1119. In an all geared headstock, a few consecutive spindle speeds are, 70, 98, 137, 192....rpm. The next speed is

(a) 269 rpm (b) 220 rpm
(c) 247 rpm (d) 230 rpm

WBPSC AE, 2017

Ans. (a) : $r = \frac{N_2}{N_1} = \frac{98}{70} = 1.4$

$N_n = N_1(r)^{n-1}$

$N_5 = N_1(r)^{5-1}$

$N_5 = 70(1.4)^4$

$N_5 = 268.91$ rpm

1120. For indexing 9 division, the hole circle to be chosen is

(a) 15 hole circle (b) 18 hole circle
(c) 19 hole circle (d) 49 hole circle

WBPSC AE, 2017

Ans. (b) : For direct indexing

$\frac{24}{N}$ N = No. of division

$\frac{24}{9}$

$2\frac{6}{9} = 2\frac{12}{18}$

For indexing 9 division, 18 holes circle to be chosen.

1121. Taper shank of twist drills are provided with

(a) Morse taper
(b) Brown and Sharp taper
(c) Metric taper
(d) Jarno taper

WBPSC AE, 2017

Ans. (b) : Brawn and sharp taper also referred as to as Aircraft taper reamers are used to hold cutting tools or holders in the spindle of machines such as a drill process, milling machine or machining centre.

Jarno Taper– This taper is used by a number of manufacture for taper pins, sockets and shanks used on machine tools.

Morse Taper–Morse taper is used on spindle nose of the lathe headstock.

1122. Automatic turning of internal taper can be done by

(a) swiveling the compound slide.
(b) offsetting the tailstock.
(c) using taper turning attachment.
(d) swiveling the tool post.

WBPSC AE, 2017

Ans. (c) : Automatic turning of internal taper can be done by using taper turning attachment.

1123. Which of the following operations can be performed with milling cutters?

(a) cutting key ways on shafts
(b) cutting external screw threads
(c) cutting teeth of spur gears
(d) all of these

Sikkim PSC (Under Secretary), 2017

Ans. (d) : Milling cutters are cutting tool is typically used in milling machine or machining centers to perform milling operation.

Milling cutters come in several shapes and many

1. Cutting key ways on shaft
2. Cutting external screw threads
3. Cutting tooth of spur gear
4. Cutter location is the topic of where to locate the cutter in order to achieve desired contour of the work piece.
5. High helix angle are typically best for soft metals and low helix angle hard or tough metal.

1124. In 'Helical plain milling cutter', the helix angle of the teeth ranges from

(a) 15^o to 30^o (b) 45^o to 60^o
(c) 75^o to 90^o (d) 105^o to 120^o

Karnataka PSC Lect., 27.05.2017

Ans (b) : In helical plain milling cutter the helix angle of the teeth varies from 45° to 60°. The helical or spiral milling is an operation of producing grooves around the periphery of a cylindrical or concial workpiece.

1125. Internal or external threads of different pitches can be produced by

(a) Pantograph Milling Machine
(b) Profiling Machine
(c) Planetary Milling machine
(d) Plano-Miller

Karnataka PSC Lect., 27.05.2017

Ans (c) : Internal or external threads of different pitches can be produced by planetary milling machine.

1126. Which geometry of form is judged by the difference in diameters at the middle and end cross sections when middle diameter is less compared to end cross section?

(a) Taperness (b) Lobedness
(c) Bowness (d) Barrelness

Punjab PSC SDE 12.02.2017

Ans. (c) : Bowness

1127. What would be machining time for cutting one pass on 118mm long work piece with 2mm depth of cut, feed of 0.1 mm/rev and 400 rpm?

(a) 3 min (b) 2.95 min
(c) 3.05 min (d) 6 min

Punjab PSC SDE 12.02.2017

Ans. (b) : Given,

L = 118 mm
d = 2 mm
f = 0.1 mm/rev
N = 400 rpm

Machining time for cutting one pass –

$t_m = \frac{L}{fN} = \frac{118}{0.1 \times 400}$

$\boxed{t_m = 2.95 \text{min}}$

1128. A block of length 200 mm is machined by a slab milling cutter 34 mm in diameter. The depth of cut and table feed are set at 2mm and 18 mm/minute, respectively. Considering the approach and the over travel of the cutter to be same, the minimum estimated machining time (minutes) per pass is :

(a) 12 (b) 10
(c) 11 (d) 15

HPPSC Asstt. Prof. 20.11.2017

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Ans. (a) : Given, L = 200 mm, D = 34 mm, d = 2mm, f_n = 18mm/min.

Necessary approach $(L_n) = \sqrt{(D-d).d}$

$L_n = \sqrt{(34-2)\times 2} = 8\text{mm}$

OT (Over Travel) = L_n = 8mm (Given)

Minimum time require to machining $= \dfrac{L + L_n + OT}{f_n}$

$= \dfrac{200+8+8}{18} = \dfrac{216}{18} = 12\,\text{min}.$

1129. Spring back in a sheet metal bending operation is the result of

(a) Elastic recovery of the metal
(b) Over bending
(c) Over straining
(d) Yield strength of the metal

TSPSC AEE 2017

Ans. (a) : Spring back is the geometric change made to a part at the end of the forming process, when the part has released from the forces, of the forming tool.
It is elastic recovery of the metal.

1130. In orthogonal turning of medium carbon steel, the specific machining energy is 2.0 J/mm^3 the cutting velocity, feed and depth of the cut are 120 m/min, 0.2 mm/rev and 2 mm respectively. The main cutting force in N is

(a) 40 (b) 80
(c) 400 (d) 800

TSPSC AEE 2017

Ans. (d) : Given
f= 0.2 mm/rev
E= 2.0 J/mm^3
d= 2 mm

specific energy consumption= $\dfrac{\text{Cutting power}}{\text{MRR}}$

$E = \dfrac{F_c \times v}{1000 \times fd \times v}$

$2.0 = \dfrac{F_c}{1000 \times 2 \times 0.2}$

$\boxed{F_c = 800\text{N}}$

1131. A hole is to be punched in a 15 mm thick plate having an ultimate shear strength of 3 kN/mm^2. If the allowable crushing stress in the punch is 6 kN/mm^2, the diameter of the hole for no failure of punch

(a) 120 mm (b) 60 mm
(c) 30 mm (d) 15 mm

GPSC ARTO 01.05.2016

Ans. (c) : Given, thickness of plate (t) = 15 mm
Ultimate shear strength (τ_u) = 3 kN/mm^2
Allowable crushing stress (σ_c) = 6 kN/mm^2

So, $\sigma_c \times \dfrac{\pi}{4} d^2 = \tau_u \times \pi dt$

$\Rightarrow d = \dfrac{\tau_u \times 4 \times t}{\sigma_c} = \dfrac{3 \times 4 \times 15}{6}$

$\boxed{d = 30\text{ mm}}$

1132. A symmetrical cup of circular cross section with diameter 40 mm an height 60 mm with a corner radius of 2 mm is to be obtained in C20 steel of 0.6 mm thickness. The diameter of blank necessary for this is approximately

(a) 78 mm (b) 90 mm
(c) 106 mm (d) 125 mm

GPSC ARTO 01.05.2016

Ans. (c) : Given, Diameter (d) = 40 mm
height(h) = 60 mm
Corner radius (r) = 2mm, thickness(t) = 0.6mm]
Blank diameter (D) = ?
Where, $15r < d \le 20r$

So, Formula $\boxed{D = \sqrt{d^2 + 4dh - 0.5r}}$

$D = \sqrt{40^2 + 4 \times 40 \times 60 - 0.5 \times 2} = 105.83$

$\boxed{D \cong 106\text{ mm}}$

1133. Flank wear occurs mainly on ..... .

(a) nose part and top face.
(b) nose part, front relief face and side relief face
(c) cutting edges
(d) face of the cutting tool at a short distance from the cutting edge

GWSSB DEE 07.07.2016

Ans. (b) : Flank wear occurs mainly on nose part, front relief face and side relief face.

1134. Which one of the following lathe machine operation tool point is placed exactly on work centre line?

(a) Knurling (b) Facing
(c) Turning (d) Boring

Rajasthan Nagar Nigam AE 2016, Shift-I

Ans. (b) : Facing–Facing is a turning operation in which the work piece is machined to its center. It involves moving the cutting tool perpendicular to the workpiece's axis of rotation.
Turning–Turning is a form of machining, a material removal process, which is used to create rotational parts by cutting away unwanted material. Turning is also commonly used as a secondary process to add or refine features on parts that were manufactured using a different process.
Knurling– It is an operation of providing knurled (diamond shaped pattern) surface on the workpiece.
Boring– It is an operation of enlarging of a hole already made in a workpiece.

1135. When the tool of centre lathe is fed in radial direction normal to axis of rotation.

(a) it produces cylindrical surface
(b) it produces flat surfaces
(c) it produces tapered surfaces
(d) none of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (b) : When the tool of centre lathe is fed in radial direction normal to axis of rotation. It produces flat surfaces.

1136. The pistons are usually given coating such as tin plating in order to

(a) Reduce weight
(b) Reduce friction

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(c) Reduce possibility of scoring
(d) Increase lubrication effect

APPSC AEE MAINS 2016, PAPER-III

Ans. (c) : To reduce possibility of scoring, coating of tin plating is done on piston.

1137. The productivity of honing operation is
(a) less than the productivity of lapping operation
(b) more than the productivity of lapping operation
(c) equal to the productivity of lapping operation for the same work piece
(d) unpredictable

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : The productivity of honing operation is more than the productivity of lapping operation.

1138. Tungsten carbide tipped drill bits are used for
(a) slow drilling
(b) smooth surface
(c) drilling without coolant
(d) drilling on glass

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : Tungsten carbide tipped drill bits are used for drilling on glass.

1139. Which of the following is not a fine finishing operation?
(a) Tumbling (b) Grinding
(c) Honing (d) Lapping

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : Tumbling is not a fine finishing operation process it is only smoothen & polishing the small object in large quantity.

1140. High speed steel contains carbon
(a) 0.15 to 0.3% (b) 0.6 to 1.0%
(c) 4 to 6% (d) 6 to 10%

UPPSC AE 12.04.2016 Paper-I

Ans : (b) High Speed Steel (HSS)
* Speed range 30–15 m/min
* Temperature–650^{0C}
* Toughness of HSS is highest among all cutting tool materials.
* The rake angle required for machining aluminum by carbides tool is 15^0

1141. A plate of 600 × 900 mm is to be machined on a shaper. Cutting speed is 6 m/min, return time to cutting time ratio is 1 : 4 and the feed is 2 mm/stroke. Clearance at each end is 75 mm. The time required in minutes on shaper to complete one cut would be :
(a) 45 (b) 60
(c) 80 (d) 90

HPPSC W.S. Poly. 2016

Ans : (b) $V = \frac{6000}{60}\text{mm/sec}$

lets length = 900 mm
width = 600 mm
$\because$ feed = 2mm/stroke

$\therefore$ total number of stroke $= \frac{\text{width}}{\text{feed}} = \frac{600}{2} = 300$

required time for one stroke = cutting time + return time

$$= \left(\frac{900+150}{100}\right) + \frac{1}{4}\left(\frac{900+150}{100}\right)$$

(10.5 + 2.625) sec = 13.125 sec
= 300× 13.125 sec

$\therefore$ total time required = number of stroke × time return stroke
= 300 × 13.125 sec = 3937.5 sec or 65.625 minutes
Ans is near about 60 minutes.

1142. In which milling operation, the cutting force tends to lift the work piece?
(a) Climb (b) Down
(c) Conventional (d) Face

HPPSC W.S. Poly. 2016

Ans : (c) Conventional milling (Up milling) and climb milling (down milling) are two distinct ways to cut material.
The difference between these two techniques is the relationship of the rotation of the cutter to the direction of feed. In conventional milling, the cutter rotates against the direction of the feed so the cutting force tends to lift the workpiece.

1143. The.....in progressive dies are used in order to bring the stock into the correct position for the succeeding blanking or piercing operation.
(a) Plunger (b) Pilot
(c) Stack pin (d) Piston

HPPSC AE (PWD) 10.09.2016

Ans. (b) : Inside of a progressive stamping die, we can do piercing trimming forming and coining to name just a few of the possible operations. The pilots in progressive dies are used as a locating tool. When a punch pierces a hole in one station the press cycles the die and the material is moved to the next station. The pilot punch enters the hole and positions the material in an exacting location before the press cycles the next stroke.

1144. The role of lead screw in lathe machine is to aid in
(a) knurling (b) spinning
(c) thread cutting (d) surface finishing

GPSC Lect. 23.10.2016

Ans. (c) : The lead screw transmits power from the head stock to carriage for screw thread cutting operations are used for both the feed and screw cutting power transmission.

1145. When the surface of the grinding wheel develops a smooth and shining surface it is said to be
(a) loading (b) glazing
(c) truing (d) none of these

HPPSC W.S. Poly. 2016

Ans : (b) When the surface of grinding wheel developed a smooth and shining surface it is said to be glazing.
Causes of wheel's glazing:-
(i) To hard wheel.
(ii) To fine grain.
(iii) To fast speed of wheel.
(iv) To slow speed of work.
(v) Wheel loaded with chip.

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1146. A 50 mm diameter disc is to be punched out from a carbon steel sheet 1.0 mm thick. The diameter of the punch should be

(a) 42.925 mm (b) 50.00 mm
(c) 50.075 mm (d) None of these are correct

MPPSC AE 2016

Ans : (d) Clearance of low carbon steel = 3% of thickness.

C = 0.03 × 1
C = 0.03 mm.

for punching operation
Punch size = blank size -2C = 50 - 2× 0.03
punch size = 50 - 0.06
punch size = 49.94 mm

1147. Which of the following is true for grinding wheel?

(a) Hard grinding wheel is used to grind hard steel
(b) 'A' denotes the hardest abrasive wheel
(c) Grade of grinding wheel is related to its hardness
(d) Soft grinding wheel is used to grind the soft steel

UPRVUNL AE 07.10.2016

Ans. (c) : Hard grinding wheel is used to grind soft steel.
Grade of grinding wheel is related to its hardness.
Soft grinding wheel is used to grind the hard steel.
From A (soft) to Z (hard), determines how tightly the bond holds the abrasive. A to H for softer structure, I to P for moderately hard structure and Q to Z for hard structure.

1148. A slot is to be milled by a side and face milling cutter with 12 teeth and 100 mm diameter. Feed rate is 0.2 mm/tooth and cutting speed is 100 rpm. Table feed will be:

(a) 240 mm/min (b) 600 mm/min
(c) 260 mm/min (d) 300 mm/min

UPRVUNL AE 07.10.2016

Ans. (a) : We know that, table feed of milling machine

$f_m = f_t \times z \times N$

f_t = 0.2 mm/tooth
z = 12, D = 100 mm
N = 100 rpm
$f_m = 0.2 \times 12 \times 100$
f_m = 240 mm/min

1149. Which of the following is not a part of carriage in lathe mechanism?

(a) Tail stock (b) Saddle
(c) Cross slide (d) Tool post

UPRVUNL AE 07.10.2016

Ans. (a) : Tool post, saddle, cross slide these are parts of carriage in lathe mechanism whereas tail stock is another important part of lathe machine.

1150. What will be step ratio of headstock spindle for a lathe machine of following data?
Maximum speed = 2100 rev/min
Minimum speed = 70 rev/min
No. of speeds = 21

(a) $(30)^{1/22}$ (b) $(30)^{1/19}$
(c) $(30)^{1/21}$ (d) $(30)^{1/20}$

UPRVUNL AE 07.10.2016

Ans. (d) : Maximum speed = 2100 rev/min
Minimum speed = 70 rev/min
No. of speeds = 21
then step ratio (r) = ?
We know that

$$\frac{N_{max}}{N_{min}} = (r)^{n-1}$$

$$(r)^{21-1} = \left(\frac{2100}{70}\right) = 30$$

$$(r) = (30)^{\frac{1}{20}}$$

1151. Universal rolling mill contains:

(a) No rolls
(b) Vertical rolls
(c) Horizontal rolls
(d) Both Horizontal and vertical rolls

UPRVUNL AE 07.10.2016

Ans. (d) : Universal rolling mill—The universal rolling mill consists of two vertical rolls and two horizontal rolls. The vertical rolls are ideal and are arranged between the bearing chocks of the horizontal rolls in the vertical plane.
Universal rolling mills are used for producing blooms from ingots and for rolling wide - flange H-section beams.

1152. Finishing of existing hole is known as:

(a) Boring (b) Counter boring
(c) Counter sinking (d) Reaming

UPRVUNL AE 07.10.2016
Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : Reaming process—Reaming is a cutting operation with a multi-edge cutting tool which is constantly in action. Reaming is applied to finish drilled holes accurately to size and with a good surface finish.
Boring—Boring, also called internal turning, is used to increase the inside diameter of hole. The original hole is made with drill, or it may be a cored hole in a casting.
Counter boring—Counter boring is the operation of enlarging one end of an existing hole concentric with the original hole with square bottom. It is done to accommodate the heads of bolts, studs and pins.
Counter sinking—A countersink is a core-shaped hole that is cut into PCB to allow the flat head of a socket cap screw to fit flush with the surface of the board.

1153. In most high speed milling cutters positive radial rake angle is

(a) 10-15 degree (b) 15-25 degree
(c) 25-35 degree (d) less than 50 degree

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : For most of high speed milling cutters, positive radial rake angles of 10° to 15° are used. These angles are satisfactory for most materials and represent a compromise between good shearing or cutting ability and strength.

1154. In order to lower the cutting forces and also to decrease the cutting wear, alloying element generally added to steel for free machining is

(a) Manganese (b) Lead
(c) Chromium (d) Aluminium

Nagaland PSC CTSE 2016 Paper-I

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Ans. (b) : In, order to lower the cutting forces and also to decrease the cutting wear, alloying element (Lead) added to steel for free machining and act as a solid lubricants.

1155. In a machine operation on free cutting mild steel the cutting speed is mentioned as 70 m/min. What should be material of cutting tool

(a) Carbon steel (b) High speed steel
(c) Carbide (d) Diamond

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : For, cutting mild steel, in machining operation, the cutting tool of high speed steel material is used for cutting, at speed 70 m/min

1156. On a triple thread screw

(a) Lead = Pitch (b) Lead = 3 Pitch
(c) Lead = 1/3 Pitch (d) Lead = 9 Pitch

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : On a triple thread screw, lead is three times the pitch.

Lead = 3 × pitch

1157. Which of the following is gear finishing process?

(a) Gear lapping (b) Gear shaving
(c) Gear grinding (d) All of the above

Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : For gear finishing process :- gear lapping, gear shaving, gear grinding all are used for fine finishing process.

1158. Quick return mechanism is associated with

(a) Lathe (b) Milling
(c) Shaping (d) None

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Shaping

1159. When the dia of work piece changes from one end to the other, it is called

(a) Straight turning (b) Taper turning
(c) Knurling (d) facing

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : When the diameter of work-piece changes from one-end to the other, it is called taper turning.

1160. Boring operation is used to

(a) drill a hole
(b) correct the hole
(c) enlarge the existing hole
(d) finish a drilled hole

Nagaland PSC CTSE 2016 Paper-I

Ans. (c) : Boring operation is used after the drilling take place it help to enlarge the existing hole by means of a single point cutting tool.

1161. For machining, which one of the following gang milling operations is employed?

(a) Thread
(b) Bores
(c) Grooves
(d) Step on prismatic parts
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (d) : Gang Milling– Simultaneous machining of a number of flat horizontal and vertical surface of a workpiece using combination of more than two cutters mounted on a common arbor.

1162. Consider the following statements associated with the lathe accessories–

1. Collect are used for turning disc–shaped job.
2. Steady rest is used for supporting a in long job in between head stock and tail stock
3. Mandrel is used for turning cylindrical job of these statements

(a) 1 alone is correct (b) 2 and 3 are correct
(c) 1 and 2 are correct (d) 2 alone is correct

UPRVUNL AE 21.08.2016

Ans. : (b) (i) Steady rest is used for supporting a long job in between head stock and tail stock
(ii) Mandrel is used for turning cylindrical job.

1163. Conisider the following statements

The set-up for internal centre less grinding consists of a regulating wheel a pressure roll and a support roll. Between which the tubular work piece is supported with the grinding wheel within the tube wherein

1. the direction of rotation of work piece and grinding wheel are same
2. the grinding wheel work piece and regulating wheel centers must lie on one line
3. the directions of rotation of grinding wheel and regulating wheel are same
4. the directions of rotation of pressure roll support roll and regulating wheel are same

Which of the statements are correct?

(a) 1,2 and 3 (b) 3 and 4
(c) 1, 3 and 4 (d) 2 and 3

UPRVUNL AE 21.08.2016

Ans. : (*) 1, 2 and 4 are correct.

1164. In deciding the clearance between punch and die in press work in shearing the following rule is helpful

(a) Punch size controls hole size and die size controls blank size
(b) Die size controls hole size and punch size controls blank
(c) Punch size controls both hole size and blank size
(d) die size controls both size and blank size

UPRVUNL AE 21.08.2016

Ans. : (a) Punch size controls hole size die size controls blank size.

1165. A milling cutter of 70 mm diameter with 12 teeth operating at a cutting speed of 22m/min and a feed of 0.05 mm/tooth. The feed per minute is:

(a) 110 mm/min (b) 6 mm/min
(c) 60 mm/min (d) 35 mm/min

UPRVUNL AE 21.08.2016

Ans. : (c) D = 70 mm

V = 22 m/min

$$\therefore V = \frac{2\pi N \times r}{60}$$

$$22 \times 1000 = \frac{3 \times \pi \times N \times r}{60}$$

$$\frac{22 \times 1000}{60} = \frac{2 \times \pi \times N \times \frac{70}{2}}{60}$$

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$$\therefore N = \frac{22 \times 1000}{70 \times \pi} = 100 \text{RPM}$$

Feed = 0.05 m/tooth

Feed in One revolution = 0.6 mm

In 100 Revolution the feed = 0.6×100mm/min

= 60 mm/min

1166. A hole of 30 mm diameter is to be produced by reaming. The minimum diameter permissible is 30.00 mm while the maximum diameter permissible is 30.05 mm In this regard consider the following statements about the reamer size.

1. The maximum diameter of the reamer must be less than 30.05 mm
2. The maximum diameter of the reamer can be more than 30.05 mm
3. The minimum diameter of the reamer cannot be less than 30 mm
4. The minimum diameter of the reamer can be less than 50 mm

Of these statements.

(a) 2 and 4 are correct (b) 1 and 3 are correct
(c) 1 and 4 are correct (d) 2 and 3 are correct

UPRVUNL AE 21.08.2016

Ans. : (b)

1167. Down milling is also called

(a) conventional milling
(b) climb milling
(c) end milling
(d) face milling

GPSC Lect. 23.10.2016

Ans. (b) : Down milling is also called as climb milling. Up milling is also called conventional milling.

1168. In down milling, the thickness of chip is

(a) minimum at the beginning of the cut and maximum at the end of the cut
(b) maximum at the beginning of the cut and minimum at the end of the cut
(c) uniform through the cut
(d) none of these

GPSC Lect. 23.10.2016

Ans. (b) : In down milling the chip thickness is maximum at the beginning of the cut and minimum at the end of cut. The cutting force varies from maximum to zero.

1169. Gear finishing operation is called

(a) shaping (b) milling
(c) hobbing (d) burnishing

GPSC Lect. 23.10.2016

Ans. (d) : Gear burnishing is used to remove or reduce the burrs along with improving the surface finish. The action of burnishing dies on tooth profile allows the improvement in finish without altering the tooth profile.

1170. Which one of the following abrasive materials, has highest hardness?

(a) Aluminum oxide (b) Cubic boron nitride
(c) Silicon carbide (d) None of these

GPSC Lect. 23.10.2016

Ans. (b) : CBN tool material is produced by mixing the main component CBN, which has a hardness second next to diamond, with a special ceramic or metal binder.

1171. In shaper machine the ratio of return time to cutting time is

(a) equal to one (b) less than one
(c) greater than one (d) zero

GPSC Lect. 23.10.2016

Ans. (b) : In shaper return time is less than cutting time, due quick return mechanism. So, the ratio of return time to cutting time is less than one.

1172. A 10 kW drilling machine is used to drill a bore in a small aluminium block of mass 8 kg. How much is the rise in temperature of the block in 2.5 minutes, assuming 50% of power is used up in heating the block? (specific heat of aluminum : 0.91 kJ/(g°C))

(a) 50°C (b) 206°C
(c) 103°C (d) 227°C

ISRO Scientist/Engineer 03.07.2016

Ans. (c) : 10 kW drilling machine is used to drill bore 50% of power i.e. 5 kW used to heat block.

$$\therefore \quad \text{Power} = \frac{\text{Heat Supplied}}{\text{Time}}$$

Heat supplied = $5 \times 10^3 \times 2.5 \times 60$

$m C_p \Delta T = 75 \times 10^4$

$8 \times 0.91 \times 10^3 \times \Delta T = 75 \times 10^4$

$\Delta T = 103.02°C$

1173. A grinding wheel gets glazed due to:

(a) Wear of abrasive grains
(b) Wear of bond
(c) Breaking of abrasives
(d) Cracks in the wheel

GPSC Asstt. Prof. 28.08.2016

Ans. (a) : A glazed grinding wheel is the result of high abrasive wear causing the grains to become dull. A loaded grinding wheel is a result of chips clogging the grains on the grinding wheel due to the grinding of soft materials.

1174. A hot chisel is one which is:

(a) Used when red hot
(b) Used to cut red hot material
(c) Used to cut cold materials
(d) Non existing tool

GPSC Asstt. Prof. 28.08.2016

Ans. (b) : A hot chisel is used to cut metal that has been heated in a forge to soften the metal.

1175. …… is an example for machine tool.

(a) Single point cutting tool
(b) Multipoint cutting tool
(c) Milling cutter
(d) Shaper

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Shaper is machine tool.

A machine tool is a machine for handling or machining metal or other rigid materials, usually by cutting boring, grinding, shearing.

1176. The column of Milling machine is made of :

(a) Aluminium (b) Cast Iron
(c) High speed steel (d) Any metal

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : The column of milling machine is made of cast Iron. Cast iron have high damping capacity.

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1177. Last element in tool signature of single point cutting tool indicates value of :
(a) side clearance angle (b) front rack angle
(c) back rake angle (d) nose radius

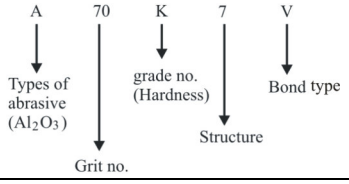
GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Tool Signature- Back rake angle – side rake angle – End relief angle – side relief angle– End cutting edge angle – side cutting edge angle – nose radius (mm)

1178. Specification of grinding wheel is A 70 K 7 V. In this specification. A describes.
(a) grade of the wheel
(b) bond material
(c) abrasive type : Al_2O_3
(d) none of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (c) :



1179. An end mill having 4 teeth and rotating at 250 rpm. If the feed per tooth is 0.1 mm, what is the table feed?
(a) 100 mm/min (b) 10 mm/min
(c) 250 mm/min (d) 25 mm/min

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : Feed per teeth = 0.1 mm/teeth
4 teeth and N = 250 rpm.
$\therefore$ Table feed = 250 × 0.1 mm × 4 mm/min
= 100 mm/min

1180. Guide ways used in machine tools with sliding friction is known as
(a) Antifriction ways (b) Friction free ways
(c) Slide ways (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : Slide ways
There are kind of guide ways used in machine tools with sliding friction.

1181. Which type of slide way profile combination is preferred for beds of general purpose lathes
(a) Open 'V' + Open 'V'
(b) Closed flat + Open 'V'
(c) Closed flat + Closed 'V'
(d) Open 'V' + Closed 'V'

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : Closed flat and closed V type of slide way profile combination is preferred for beds of general purpose lathes.

1182. The process where there is no reduction of material, only a modification in its geometry, is called_____
(a) Lancing (b) Machining
(c) Notching (d) None of these

KPCL AE 2016

Ans. (a) : The process where there is no reduction of metal, only a modification in its geometry is called lancing. Lancing consists of cutting the steel metal through a small length and bending this small cut portion downwards.

1183. Example for machine tool is:
(a) single point tool (b) grinding wheel
(c) milling machine (d) drill bit

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Example for machine tool is milling machine.

1184. ATC is commonly used in :
(a) CNC ECM
(b) CNC machining centre
(c) Conventional shaper
(d) NC PAM

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : ATC is commonly used in CNC machining centre.

1185. The front hanging portion of the carriage of a lathe is known as:
(a) Lead screw (b) Saddle
(c) Tool post (d) Apron

HPPSC AE (PWD) 10.09.2016

Ans. (d) : Apron is a key component of carriage, a part of lathe that is clamped to the saddle. It is designed to hold gears levers and clutch for power transmission.

1186. Compound indexing is used in milling machine for.....operation.
(a) turning (b) knurling
(c) gear cutting (d) none of these

HPPSC AE (PWD) 10.09.2016

Ans. (c) : Compound indexing- Compound indexing is used in milling machine for gear cutting operation. The word compound indexing is an indicative of compound movements of indexing crank and then plate along with crank. In this case indexing plate is normally held stationary by a lock pin first we rotate the indexing crank through required number of holes in a selected hole circle then crank is fixed through pin. It is followed by another movement by designing the rear lock pin. The indexing plate along with indexing crank is rotated in forward or backward direction.

1187. The true sliding strain for a low carbon steel bar which is doubled in length by forging is :
(a) 0.307 (b) 0.5
(c) 0.693 (d) 1.0

HPPSC Asstt. Prof. 18.11.2016

Ans. (c) : Engineering strain, $\in = \frac{2\ell - \ell}{\ell} = 1$

True strain, $\in_T = \ell n(1 + \in)$

$\in_T = \ell n2 = 0.693$

1188. One of the index plates of a milling machine dividing head has the following hole circles : 15; 16; 17; 18; 19; 20. A gear wheel of 34 teeth has to be milled by simple indexing method. To machine each tooth, the index crank has to rotated through
(a) 17 holes in the 20 hole circle
(b) 18 holes in the 20 hole circle
(c) 1 revolution and 3 holes in 17 hole circle
(d) 1 revolution and 2 holes in 18 hole circle
(e) None of these

CGPSC AE 16.10.2016

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Ans. (c) : As we know that,

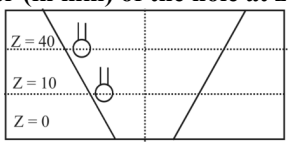
$$\frac{N_1}{N_2}=\frac{N_2}{N_3}$$

So, $\frac{40}{N}=1\frac{6}{34}$

So 1 revolution & 3 holes on 17 holes circles.

$\therefore \quad (17 \times 2 = 34)$

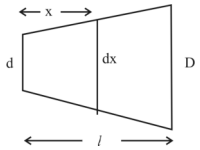
1189. A taper hole is inspected using a CMM, with a probe of 2 mm diameter. At a height, Z = 10 mm from the bottom, 5 points are touched and diameter of circle (not compensated for probe size) is obtained as 20 mm. Similarly, a 40 mm diameter at height z = 40 mm, the smaller diameter (in mm) of the hole at z = 0 is



(a) 13.334 (b) 15.334
(c) 15.442 (d) 15.542

RPSC ADE 2016

Ans. (a) : For tapper rod, diameter varies linearly given by at a distance x from smaller end



$$dx = d + \frac{D-d}{l} \times x$$

here, D = 40, l = 40

dx = 20, x = 10

d = ?

$$20 = d + \frac{40-d}{40} \times 10$$

$$d = \frac{40}{3} = 13.333$$

1190. Which of the following statement is correct regarding grinding of high carbon steel?

(a) Grinding at high speed results in the reduction of chip thickness and cutting forces per grit.
(b) The grinding wheel has to be open structure
(c) Aluminum oxide wheels are employed
(d) All of these

RPSC VPITI 14.02.2016

Ans. (d) : Grinding of high carbon steel

1. Grinding of high speed results in the reduction of chip thickness and cutting forces per grit.
2. The grinding wheel has to be open structure
3. Aluminium oxide wheels are employed.

All these is correct in Grinding of high carbon steel.

1191. A medium carbon steel workpiece is turned on a lathe at 50 m/min cutting speed, 0.8 mm/rev feed and 1.5 mm depth of cut, what is the rate of metal removal:

(a) $100mm^3/min$ (b) $20000\ mm^3/min$
(c) $60000\ mm^3/min$ (d) Data insufficient

RPSC VPITI 14.02.2016

Ans. (c) : Given- cutting speed (v) = 50 m/min
$= 50\times10^3$ mm/min

Feed (f) = 0.8 mm/rev

Depth of cut (d) = 1.5 mm

Metal removal rate (MRR) = v × f × d

MRR = $50 \times 10^3 \times 0.8 \times 1.5$

MRR = $60000\ mm^3/min$

1192. Which taper is found on a spindle nose of the lathe head stock?

(a) Brown and sharp taper
(b) Morse taper
(c) Jarno taper
(d) Pin taper

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (b) : Morse taper is used on spindle nose of the lathe headstock.

Brawn and sharp taper also referred as to as Aircraft taper reamers are used to hold cutting tools or holders in the spindle of machines such as a drill process, milling machine or machining centre.

Jarno Taper– This taper is used by a number of manufacture for taper pins, sockets and shanks used on machine tools.

Pin Taper– It is used mostly to lock two shafts together.

1193. The ability of a material to resist softening at high temperature is known as

(a) Hot tempering (b) Fatigue
(c) Hot hardeness (d) Specific heat

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (c) : Hot hardness or red hardness corresponds to hardness of a material at high temperatures as the temperatures of material increases, hardness decreases and at some point a drastic change in hardness occurs.

1194. The soft grade grinding wheels are denoted by the letters

(a) A to H (b) A to P
(c) I to P (d) Q to Z

Rajasthan Nagar Nigam AE 2016, Shift-II

Ans. (a) : Soft grinding wheel are generally used for hard materials and hard wheels are used for soft materials.

Soft grade	Medium grade	Hard grade
ABCDEFGH	IJKLMNOPQ	RSTUVWXYZ

1195. The gears manufactured on generating principle is:

(a) hobbing (b) milling
(c) broaching (d) shaping

Kerala PSC AE 06.08.2015

Ans. (a) : Gear Hobbing- It is a maching processes in which gear tooth are progressively generated by a series of cuts with a helical cutting tool. All motion in hobbing are rotary and the hob and gear blank rotate continuously as in two gear meshing until all teeth are cut.

1196. Quick returning mechanism is used in :

(a) milling machine (b) broaching machine
(c) slotting machine (d) lathe machine

Kerala PSC AE 06.08.2015
KPSC AE 2015

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Ans. (c) : Quick Return Mechanism – It is used in slotting machine, Hydraulic drive in QRM, the circular motion is converted into reciprocating motion so that the slider moves forward and backwards. In the forward direction, the cutting process occurs while in return there is no such cutting.

1197. Among the conventional machining processes maximum specific energy is consumed in
(a) Turning (b) Chilling
(c) Grinding (d) Planning

GPSC AE 26.07.2015

Ans. (c) : Grinding consumes maximum specific energy as compare to turning, planning etc.

1198. Gear cutting on a milling machine using an involute profile cutter is a
(a) Gear forming process
(b) Gear generating process
(c) Shaping process
(d) Highly accurate gear producing process

GPSC AE 26.07.2015

Ans. (a) : Gear cutting on a milling machine using an involute profile cutter is a gear forming process.

1199. The hardness of a grinding wheel is determined by the
(a) hardness of abrasive grains
(b) ability of the bond to retain abrasives
(c) hardness of the bond
(d) ability of grinding wheel to penetrate the work piece

TANGEDCO AE 2015

Ans. (b) : Hardness of a grinding wheel is determined by the ability of the bond to retain abrasives.

Sl - Ⓒ - 30 - Ⓜ - Ⓢ - Ⓥ - 20
1 2 3 4 5

1-Type of abrasive
2. Grain size
3. Hardness
4. Structure
5. Type of bond

1200. Honing operation is used for high finish:
(a) For a plane surface
(b) Inside a cylindrical surface
(c) Outside cylindrical surface
(d) Not a surface finish operation

OPSC AEE 2015 Paper-I

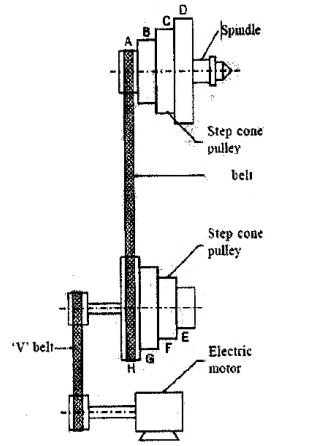
Ans : (b) Honing operation is used for high finish Inside a cylindrical surface.

Honing : Honing is an abrasive process for removing stock from metallic and non–metallic surfaces. It is used to correct local irregularities such as ovality, waviness at axis or non–parallelism of cylindrical features and to develop a particular texture. Honing is the application of bonded abrasive stones to a surface for the purpose of limited stock removal and attainment of a surface finish. The stock removal and attainment of a surface finish. The stock removal is very low (0.5 mm for prime honing and 0.01 mm for secondary or mirror honing).

1201. A lathe with four steps on the cone pulley and with back gears will have
(a) four direct speeds
(b) four direct and four indirect speeds
(c) four indirect speeds
(d) eight indirect speeds

TSPSC AEE 2015

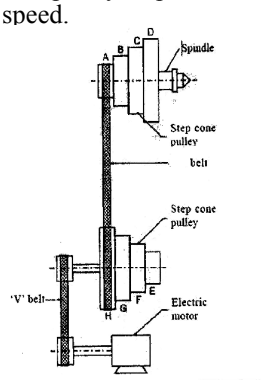
Ans : (b) A lathe with four steps on the cone pulley and with back gears will have four direct and four indirect speed.



1202. A step cone pulley is provided on a lathe to
(a) reverse the direction of spindle rotation
(b) change the spindle speed
(c) drive the lead screw
(d) give desired direction of movement of lathe carriage

(KPSC AE 2015)

Ans : (b) A step cone pulley is provided on a lathe to change the spindle speed.



1203. When the tool moves parallel to the lathe axis, it is called:
(a) Cross feed
(b) Sellar's taper
(c) Chapman's taper
(d) Brown and Sharpe taper

OPSC AEE 2015 Paper-I

Ans : (b) When the tool moves parallel to the lathe axis, it is called Sellar's taper.

1204. In order to grind soft material:
(a) Fine grained grinding wheel is used
(b) Medium grained grinding wheel is used
(c) Coarse grained grinding wheel is used
(d) None of these

OPSC AEE 2015 Paper-I

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Ans : (c) The hardness of a grinding wheel is specified by a letter of alphabet. The alphabets A to H are soft grades. I to P are medium grades and Q to Z are hard grades. The soft grade grinding wheels are used for grinding hard materials and hard grade grinding wheels for softer materials. A soft grade grinding wheel is one in which the abrasive grains can be easily dislodged and in hard grade grinding wheel, the abrasive grains are held more securely.

1205. A mandrel is used to hold:

(a) An eccentric work (b) A heavy work
(c) A thin work (d) None of these

OPSC AEE 2015 Paper-I

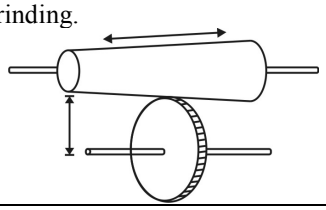
Ans : (a) A mandrel is used to hold an eccentric work.
Mandrel : It is a hardened piece of round bar with centers and flats at each end. It is used for holding and obtaining a hollow piece of work that has been previously drilled or bored. It is held between two centers and should be true with accurate centre holes for machining outer surface of the work piece. It is made of high carbon steel to avoid distortion and wear. It is used for holding the bored jobs (gear blank, Pulleys or tubes) on a lathe for the purpose of turning outside surface of the jobs.

1206. The method of grinding used to produce a straight of tapered surface on a work-piece is

(a) internal cylindrical grinding
(b) form grinding
(c) external cylindrical grinding
(d) surface grinding

TSPSC AEE 2015

Ans : (c) The method of grinding used to produced a straight or tapered surface on a work-piece is external cylindrical grinding.

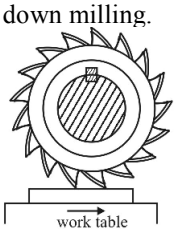


1207. The process of removing metal by a cutter which is rotating in the same direction of travel of work-piece is called

(a) up milling (b) down milling
(c) face milling (d) end milling

TSPSC AEE 2015

Ans : (b) The process of removing metal by a cutter which is rotating in the same direction of travel of work-piece is called down milling.



Climb milling or down milling

1208. A broach has the following teeth:

(a) Rough teeth (b) Semi-finishing teeth
(c) Finishing teeth (d) All of the above

TSPSC AEE 2015

Ans : (d) A broach has:-
(i) Rough teeth
(ii) Semi- finishing teeth
(iii) Finishing teeth.
Broaching is a method of removing metal by a tool that has successively higher cutting edges in a fixed path. Each tooth removes a predetermined amount of material.

1209. The broaching operation in which either the work or tool moves across the other is known as

(a) pull broaching (b) push broaching
(c) surface broaching (d) continuous broaching

TSPSC AEE 2015

Ans : (c) The broaching operation in which either the work or tool moves across the other is known as surface broaching

1210. The surface roughness value for drilling process is

(a) 0.4 to 3.2 microns (b) 1.6 to 20 microns
(c) 0.063 to 5 microns (d) 0.8 to 6.3 microns

TSPSC AEE 2015

Ans : (b) The surface roughness value for drilling process is 1.6 to 20 microns.

1211. Which of the following is not a part of Capstan lathe?

(a) chuck (b) tailstock
(c) spindle (d) tool post

Kerala PSC AE 06.08.2015
KPSC AE 2015

Ans. (b) : Capstan lathe part- Bed, legs, feed Rod, Head stock, chuck Guide ways, saddle, Cross-slide, Tool-post saddle, Ram, Hexagonal tool post or turret head and hand wheel.
So tail stock is not the part of capstan lathe.

1212. Which bond gives the softest bond?

(a) silicate bond (b) Shellac bond
(c) vitrified bond (d) all of equal strength

ISRO Scientist/Engineer 11.10.2015

Ans. (a) : Vitrified Bond (V)– It is mode of clay, field spar (rack forming mineral).
This is the strongest bond of all and not affected by water, oil, acid. It is suitable for high stock (material) removal, even at dry condition.
Resinoid Bond (R)–It is a synthetic thermosetting resign (phenolic resign). Which becomes hard after heating it occupies next place to vitrified bond and widely used for heavy duty grinding and have ability withstand shock loads.
Silicate Bond (S)–It is made of sodium silicate and used for operations that generate less heat.
Shellac Bond (E)–It is also carbonic bond and has considerable strength but it is not suitable for heavy duty grinding it is limited to grinding wheel used in fine finishing.
E.g. Fine rolls in rolling mill.
Rubber Bond (R)–It is made of synthetic or natural rubber and may used is in thin parting wheels its also used for making regulating wheels in center less grinding.

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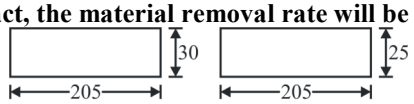
1213. Fifty flat pieces of 1 mm thick and initial dimensions as shown in the figure (a) to be milled in a single cut to the final dimensions shown in figure (b) using end milling. The cutter of diameter 25 mm has 10 teeth and rotates at 100 rpm. Horizontal feed of the table is 10 mm/min. Assuming single tooth in contact, the material removal rate will be



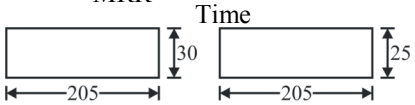
(a) (b)

(a) 35.7 mm^3/s (b) 37.7 mm^3/s
(c) 41.7 mm^3/s (d) 47.7 mm^3/s

ISRO Scientist/Engineer 11.10.2015

Ans. (c) : Given, t = 1 mm, d_{cutter} = 25 mm, no. of teeth = 10 teeth, n = 100 rpm, f = 10 mm/min, MRR = ?

$$MRR = \frac{Volume}{Time}$$



(a) (b)

$$\text{Cutting velocity } V = \frac{\pi DN}{60} m/s$$

$$= \frac{\pi \times 25 \times 100}{60}$$

= 130.8997 mm/s

Thickness = 1 mm

Width of work piece remove = 5 mm

i.e. According to 'figure'

= 30 – 25 = 5 mm

Table horizontal feed = 10 mm/min

Volume of metal removed from one work-piece

$= w \times t \times \ell$

$= 5 \times 1 \times 205\ mm^3$

$= 1025\ mm^3$

Time taken to travel 205 mm

$\Rightarrow \quad \frac{205}{10} min = 20.5\ min$

$\therefore$ Material removal rate from one work piece

$= \frac{1025}{20.5 \times 60} = 0.833\ mm^3/s$

Total MRR from 50 work pieces

$= 50 \times 0.833$

$= 41.67\ mm^3/s$

1214. Which of the following abrasive will be used for grinding tool steel and high speed steel

(a) Diamond (b) SiC
(c) Al_2O_3 (d) Boron Carbide

TSPSC AEE 2015

Ans. (c) : Al_2O_3 abrasive will be used for grinding tool steel and high speed steel.

1215. When the space between two consecutive abrasive is less in griding upon machining ductile materials, how chips weld these abrasives and after some time wheel is rubbing over work without cutting. This phenomenon is called

(a) Glazing (b) Loading
(c) Dressing (d) Trueing

TSPSC AEE 2015

Ans. (b) : In grinding process this phenomenon is called "wheel loading" which can be defined as the state of a grinding wheel when particles of a work-piece material either adhere to the grits or become embedded in the spaces between abrasive grains on grinding wheels.

1216. Spinning operation is carried out on a

(a) Lathe machine (b) Milling machine
(c) Hydraulic press (d) Mechanical press

TSPSC AEE 2015

Ans. (a) : Spinning operation is carried out on a lathe machine.

1217. Point angle in drills are kept low for machining ductile materials to take advantage of

(a) Thinner chips
(b) Thicker chips
(c) Low head developed
(d) Lesser cutting forces

TSPSC AEE 2015

Ans. (a) : For thinner chips, point angle in drills are kept low for machining of ductile materials.

1218. Very accurate screw threads can be manufactured by

(a) milling (b) grinding
(c) rolling (d) chasing

TSPSC AEE 2015

Ans. (b) : With the help of grinding, very accurate screw threads can be manufactured.

1219. When machining parameters are kept constant, tool life will be less in

(a) down milling (b) up milling
(c) face milling (d) end milling

TSPSC AEE 2015

Ans. (b) : In up-milling tool life will be less.

1220. Which of the following methods of taper turning is limited for short tapered lengths?

(a) Taper turning by offsetting the tailstock
(b) Taper turning by swiveling the compound tool test
(c) Taper turning by taper turning attachment
(d) None of these

KPSC ADF 2015

Ans. (b) : Taper turning by swiveling the compound tool rest methods of taper turning is limited for short tapered lengths.

1221. What is Lobbing?

(a) It is an operation in grinding process
(b) It is a process of centreless internal surface grinding
(c) It is a process of centreless external surface grinding
(d) It is a defect in external centreless grinding

KPSC ADF 2015

Ans. (d) : Lobbing is a defect in external centre less grinding.
Lobbing may cause significant workpiece roundness error in centerless grinding. A critical issue for any shop that produces or uses centerless grinding product.

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1222. Average cutting speed in shaping/planer :
(a) $V = NL (1 - H_m)^2/1000$ m/min
(b) $V = N + L (H_m)/1000$ m/min
(c) $V = N - L (1 + H_m)^2/1000$ m/min
(d) $V = NL (1 + H_m)/1000$ m/min

KPSC ADF 2015

Ans. (d) : Average cutting speed in shaping/planer

$$V = \frac{NL(1+H_m)}{1000}$$

Where, V = Cutting speed (m/min)
N = The number of double stroke of the ram per minute or RPM of the bull gear.
L = Length of cutting stroke in 'mm'
H_M = The ratio between return time to cutting time.

1223. A gear is to have 14 teeth. What is the indexing movement required?
(a) 2 complete turns and 18 holes in 21 hole circle
(b) both (a) and (c)
(c) 2 complete turns and 42 holes in 49 circles
(d) 2 complete turns and 18 holes in 49 circles

KPSC ADF 2015

Ans. (a) : 2 completer turns and 18 holes in 21 hole circle in 14 teeth gear indexing movement required.

1224. Chasing dial is an element of :
(a) capstan lathe (b) centre lathe
(c) milling machine (d) drilling machine

BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Chasing dial is an element of centre lathe. The purpose of chasing dial on lathes is to achieve picking up the thread accurately at the beginning of each cut.

1225. Select the correct sequence of manufacturing processes in the ascending order of accuracy
(a) Reaming - Honing - Boring - Drilling
(b) Drilling - Boring - Honing - Reaming
(c) Drilling - Reaming - Boring - Honing
(d) Drilling - Honing - Boring - Reaming

TSGENCO AE 14.11.2015

Ans. (c) : Ascending order –
Drilling – Reaming – Boring – Honing.

1226. In grinding wheel, when the binding agent is strong, so upon grinding hard materials, slowly all the abrasives wear out and grinding wheel is rubbing over work without cutting. This phenomenon is called:
(a) Glazing (b) Loading
(c) Dressing (d) Truing

TSPSC Managers, 2015

Ans. (a) : Glazing- In grinding wheel, when the binding agent is strong, so upon grinding wheel hard materials, slowly all the abrasive wear out and grinding wheel is rubbing over work without a machining (cutting). This phenomenon is called glazing.

1227. Larger clearance angles are used in drills to machine ductile materials because ductile materials have tendency for :
(a) Elastic recovery (b) Larger elongation
(c) Tough (d) Malleable

TSPSC Managers, 2015

Ans. (a) : larger clearance angles are used in drills to machine ductile material because ductile material have tendency for elastic recovery.

1228. Rake angle in twist drill varies from the centre of the drill towards periphery according to :
(a) Increases and then decreases
(b) Decreased and then increases
(c) Increases continuously
(d) Decreases continuously

TSPSC Managers, 2015

Ans. (c) : The rake angle varies from minimum at the DEAD centre to the maximum at the outer periphery. Rake angle at the outer periphery is equal to helix angle.

1229. Diamond pin location is used in fixture because
(a) It does not wear out
(b) Adjust centre to centre distance between two holes
(c) Easy to clamp
(d) Easy to manufacture

TSPSC Managers, 2015

Ans. (b) : Diamond pin- Diamond pin produce more lateral clearance. We should not use two diamond pin because clearance is more. Therefore, we use round and diamond pin.
Diamond pin location is used in fixture because adjust centre to centre distance between two holes.

1230. Coolant used while turning cast iron is :
(a) lard oil (b) soluble oil
(c) kerosene (d) none of these

VIZAG STEEL MT 2015

Ans. (d) : For turning cast iron, the coolant used is a dry air.

1231. Mother of all machine tool is :
(a) Milling Machine (b) Lathe
(c) Drilling (d) Shaper

GWSSB AAE, 27.12.2015

Ans. (b) : Lathe machine is known as mother of all machine because approximately all process can become on lathe using attachments.

1232. A steady rest is the lathe unit that is used for
(a) holding very long and heavy job.
(b) External taper turning
(c) Performing the knurling operation
(d) supporting the bed

GWSSB AAE, 27.12.2015

Ans. (a) : Steady rest are used in lathe to support very long circular work piece fixed between the centres.

1233. Which of the following is not a part of shaper.
(a) table (b) knee
(c) cross slide (d) ram

GWSSB AAE, 27.12.2015

Ans. (b) : Knee is not part of shaper

1234. Consider the following statements about desirability of High velocity in grinding because of
1. To reduce force per grit
2. To increase overall grinding force
3. To offset effect of high negative rake angle
4. To decrease overall grinding force

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(a) 1 and 3 (b) 2 and 3
(c) 3 and 4 (d) 1, 3 and 4
(e) 1, 2 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Desirability of high velocity in grinding because, it is desired to offset the adverse effect of very high negative rake angle of the working grit, to reduce the force per grit as well as the overall grinding force.

1235. In Shearing operations on sheet metal, the operation in which the slug is the part and the rest is scrap is known as :
(a) Blanking (b) Punching
(c) Slitting (d) Bending.

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : In shearing operations on sheet metal the operation in which the slug is the part and rest is scrap is known as punching operations.
In punching operations clearance is provided on die.

1236. The cutting edges of a standard twist drill are called
(a) Flutes (b) Lips
(c) Wedges (d) Flanks

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : The cutting edge of the standard twist drill are called flutes.
flutes are the spiral grooves which run to the length of the drill. The flutes helps to
- To form the cutting edge
- To curl the chip and allow these to comes out.
- The coolant to flow to the cutting edge.

1237. Which one of the following is the cause of tool failure?
(a) excessive angle (b) relief angle
(c) edge angle (d) nose angle

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : Cause of tool failure
(i) Excessive temperature
(ii) Excessive stress
(iii) Flank wear

1238. The tool life under oblique cutting in comparison of the orthogonal cutting is
(a) same (b) less
(c) more (d) none of these

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (c) : Tool life in oblique cutting is more than tool life in oblique cutting.

1239. In grinding wheel a dense structure is required for:
(a) tough material (b) ductile material
(c) heavy cuts (d) finishing cut

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (d) : A dense structure of a grinding wheel is used for–
- Hard material
- Brittle material
- Finishing cuts

1240. A left hand tool on a lathe cut most efficiently when it travels
(a) From right so left end of the lathe bed
(b) Across the bed
(c) With the help of compound slide
(d) From left to right end of the Lathe bed

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : A left hand tool on a lathe cut most efficiently when it travels form-
Left to right end of the lathe bed.

1241. The depth of cut in drilling a blind hole is _____ the drill diameter
(a) One half (b) Double
(c) Equal to (d) One fourth

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : The depth of cut in drilling a blind hole is half the drilling diameter, because these are two cutting edge of drill bits.

1242. The lathe spindles are usually made below and provided with
(a) Internal taper
(b) External taper
(c) Internal and external taper
(d) No taper

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : The lathe spindles are usually made below and provided with internal taper.
Lathe spindles are used for m/k cutting of workpieace

1243. A straight teeth slab milling cutter of 100 mm diameter and 10 teeth rotating at 200 r.p.m. ie. used to remove a layer of 3 mm thickness from a steel bar. If the table feed is 400 mm/minute, the feed per tooth in this operation will be:
(a) 0.2 mm (b) 0.4 mm
(c) 0.5 mm (d) 0.6 mm

HPPSC Asstt. Prof. 2014

Ans. (a) : Given, Teeth (t) = 10
RPM (N) = 200
Table feed = 400 mm/minute
Feed per tooth = (f_t) = ?
Table feed = f_t t N
$400 = f_t \times 10 \times 200$

$$f_t = \frac{400}{10 \times 200}$$

$f_t = 0.2$ mm

1244. During drawing operation in sheet metal formation, the stresses tend to cause _____ in the sheet metal component.
(a) Bubbles (b) Wrinkles
(c) Pits (d) Notches

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : During drawing operation in sheet metal formation the stresses tend to cause Wrinkles in the sheet metal wrinkling occurs in the flange when compressive stresses in the circumference direction reach a critical point of instability. It can occurs in region where the workpiece is unsupported or when the blanking force is insufficient. It can be prevented by increasing the blank holder force and by using a draw feed.

1245. In which of the following milling operation the surface finish is better
(a) climb (b) up
(c) drop (d) face

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Ans. (a) : Climb milling is generally the best way to machine parts today since it reduces the load from the cutting edge, leaves a better surface finish and improves tool life. During conventional milling, the cutter tends to dig into the work piece and may cause the part to be cut out of tolerance.

1246. The process of removing metal by feeding the work past a rotating multipoint cutter is known as

(a) broaching (b) sawing
(c) milling (d) grinding

TNPSC AE 2014

Ans. (c) : The process of removing metal by feeding the work past a rotating multipoint cutter is known as milling.

1247. The thickness of the chip is minimum at the beginning of the cut and maximum at the end of the cut is case of

(a) climb milling (b) up milling
(c) down milling (d) face milling

TNPSC AE 2014

Ans. (b) : The thickness of the chip is minimum at the beginning of the cut and maximum at the end of the cut is case of up milling. Up milling also know as conventional milling.

1248. The best machine tool to cut an internal spline in steel is

(a) milling machine (b) slotting machine
(c) lathe (d) grinding machine

TNPSC AE 2014

Ans. (b) : The slotting machine can be used to cut slots, splines keyways for both internal and external job such as machining internal and external gears.

1249. The cutting speed for reaming aluminium and its alloys is

(a) 47-70 m/min (b) 15-20 m/min
(c) 10-12 m/min (d) 5-20 m/min

TNPSC AE 2014

Ans. (a) : The cutting speed for reaming aluminium and its alloys is 47-70 m/min

1250. Depth of cut for roughing operation normally varies from

(a) 5 mm to 10 mm (b) 1 mm to 5 mm
(c) 0.2 mm to 1 mm (d) 0.01 mm to 0.2 mm

TNPSC AE 2014

Ans. (b) : Depth of cut for roughing operation normally varies from 1 mm to 5 mm.

1251. One of the disadvantages of a hydraulic shaper compared to the mechanical shaper is

(a) stopping point of the cutting stroke can vary depending upon the resistance offered to cutting
(b) less strokes per minute
(c) power available varies during the cutting stroke
(d) cutting speed remains constant throughout the cutting stroke

TNPSC AE 2014

Ans. (a) : One of the disadvantages of a hydraulic shaper compared to the mechanical shaper is stopping point of the cutting stroke can vary depending upon the resistance offered to cutting.

1252. Consider the units in following operations

1. turning-mm/rev 2. shaping-mm/stroke
3. drilling-mm/rev 4. milling-mm/min

Out of the above which are correct

(a) 1, 2 and 3 (b) 1, 2 and 4
(c) 1, 3 and 4 (d) 1, 2, 3 and 4
(e) 1 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : 1, 2, 3 and 4.

1253. Consider the following machines

1. Shaping machine
2. Planing machine
3. Slotting machine

Internal keyway in gears can be cut by which of the above

(a) 1 and 2 (b) 2 and 3
(c) 3 (d) 1
(e) 2

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Internal keyways usually require a slotted busing or horn to fit the hole, with the keyway broach pulled through the horn, guided by the slot.

1254. Characteristic features of a universal cylindrical grinder not possessed by plain cylindrical grinder are:

1. Swiveling wheel head
2. Swiveling wheel head slide
3. Swiveling head stock
4. Rotation of grinding wheel

(a) 1, 3 and 4 (b) 1, 2 and 3
(c) 1, 2 and 4 (d) 1, 2, 3 and 4
(e) 2 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (b) : Characteristic features of a universal cylindrical grinder not possessed by plain cylindrical grinder are

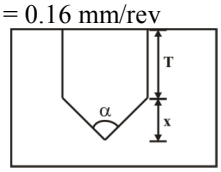
(i) Swiveling wheel head
(ii) Swiveling wheel head slide
(iii) Swiveling wheel stock.

1255. Determine the time that will be required to drill a blind hole of diameter 25 mm and depth 40 mm in a mild steel solid block by a HSS drill of 118° cone angle. The value of V_c = 25 m/min and S_o = 0.16 mm/rev.

(a) 1.0 minutes (b) 2.0 minutes
(c) 1.5 minutes (d) 2.5 minutes
(e) 3 minutes

CGPSC AE 26.04.2015 Shift-I

Ans. (a) : Given, diameter (d) = 25 mm
depth (T) = 40 mm
cone angle (α) = 118°
velocity (v_c) = 25 m/min = 25000 mm/min
feed (S_o) = 0.16 mm/rev



$v_c = \pi dN$
$25000 = 3.14 \times 25 \times N$
$N = 318.47$ rpm

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$$x = \frac{d}{2\tan\left(\frac{\theta}{2}\right)} = \frac{25}{2\tan(59^\circ)} = 7.51\,\text{mm}$$

$$\text{Time } (T_m) = \frac{T + x}{s_o N} \times (\text{No. of holes})$$

$$= \frac{40 + 7.51}{0.16 \times 318.47} = 0.93\,\text{min} \simeq 1\,\text{min}$$

1256. Feed rate in milling operation is equal to :
(a) RPM
(b) RPM × No. of teeth
(c) RPM × Feed per tooth × No. of teeth
(d) None of the above

(HPPSC AE 2014)

Ans : (c) Feed rate in milling operation is equal to RPM Feed per tooth × no. of teeth

(i) RPM of cutter = $\frac{\text{cutter speed}}{\pi D}$

(ii) Feed/min = feed/teeth × no. of teeth × RPM

(iii) Time for milling = $\frac{\text{length}}{\text{feed / mt}}$

(iv) feed rate = RPM × feed per tooth × No of teeth.

1257. Machinability depends on :
(a) microstructure, physical and mechanical properties and composition of workpiece material
(b) cutting forces
(c) type of chip
(d) tool life

(HPPSC AE 2014)

Ans : (a) Factor affecting the machinability
(i) The type of work - piece
(ii) Type of tool material
(iii) Size and shape of tool
(iv) Size, shape and velocity of cut
(v) Type and quality of machine used
(v) co efficient of friction between chip and tool
(vi) Shearing strength of work-piece material

1258. Operation of making ridge of uniform cross-section in the form of a helix formed on the inside or outside of a cylinder is known as :
(a) Spinning (b) Knurling
(c) Threading (d) Trepanning

MPSC Dy. Engineer 2014

Ans. (c) : Threading– Threading operation of making ridge of uniform cross sections in the form of a helix formed on the inside or outside of a cylinder is known as threading.
Knurling- Generating patterns of straight, angled or crossed lines is rolled into the material.
Spinning- It is a method of forming rotationally symmetrical sheet metal parts.
Trepanning- It is a manufacturing process used to produce holes or circular grooves by using one or more cutters and revolving them around a center.

1259. Inclination angle of a turning tool is measured on its:
(a) reference plane (b) cutting plane
(c) orthogonal plane (d) normal plane

HPPSC Asstt. Prof. 2014

Ans. (b) : Inclination angle of a turning tool is measured on its cutting plane.

1260. Which abrasive is generally used for superabrasive grinding?
(a) SiC (b) Al_2O_3
(c) CBN (d) None of these

HPPSC Asstt. Prof. 2014

Ans. (c) : CBN (Cubic Boron Nitride) is used for superabrasive grinding.

1261. Consider the following statements
For precision machining of non-ferrous alloys, diamond is preferred because it has:
(1) low coefficient of thermal expansion
(2) high wear resistance
(3) high compression strength
(4) low fracture toughness
Which of these statements are correct?
(a) (1) and (2) (b) (1) and (4)
(c) (2) and (3) (d) (3) and (4)

HPPSC Asstt. Prof. 2014

Ans. (a) : For precision machining diamond is preferred because it has low coefficient of thermal expansion and high wear resistances.

1262. The parts in contact with the job for doing different operations of generation of surfaces of products are known as :
(a) Products (b) Tools
(c) Chips (d) Cutter

UPRVUNL AE 2014

Ans : (b) Tools part in contact with the job for doing different operations of generation of surface.

1263. Consider the following statements regarding cutting tools and identify the one which is not true.
(a) Nose radius in cutting tool reduces tool life and surface finish.
(b) Shank is that portion of the tool which does not ground to form cutting edges.
(c) Positive back rake angle of a single point tool is the slope downward the shank.
(d) Higher depth of cut increases rate of metal removal.

UPRVUNL AE 2014

Ans : (a) Nose Radius in cutting tool reduces tool life and surface finish.

1264. During metal cutting process, temperature at job-tool contact point is measured by :
(a) Wattmeter (b) Dynamometer
(c) Pyrometer (d) Hydrometer

UPRVUNL AE 2014

Ans : (c) During metal cutting process, temperature at job-tool contact point is measured by pyrometer.
Pyrometer- It is a remote sensing thermometer. It measures the work place and the cutting tool.

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1265. Perforation in Aluminum plate occurs when it is connected with the Mild steel plate and exposed to the same environment. This phenomenon is explained by

(a) Galvanic corrosion
(b) cold welding corrosion
(c) crevice corrosion
(d) high temperature corrosion

ISRO Scientist/Engineer 24.05.2014

Ans. (a) : Two dissimilar metal (i.e. Aluminum plate and mild steel) are in contact with each other from a potential galvanic cells. The atmosphere containing oxygen and moisture can act as the electrolyte.

1266. Which of the following is not a specification of lathe machine tool?

(a) chuck size
(b) swing over diameter
(c) distance between centres
(d) bed length

Haryana PSC Civil Services Pre, 2014

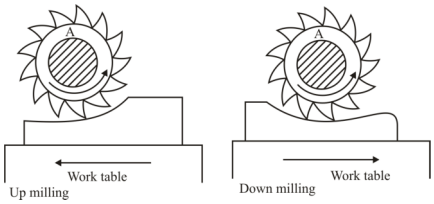
Ans. (a) : Chuck size is not a specification of lathe machine tool. Lathe can be specified by swing over diameter, distance between centers and bed length.

1267. In _____ milling process, feed direction and direction of rotation of cutter are in _____ direction.

(a) up, opposite (b) up, same
(c) down, opposite (d) down, reverse

Haryana PSC Civil Services Pre, 2014

Ans. (a) : In Up milling operation, cutter and workplace movements are in opposite direction but in down milling both move in same direction.



1268. A 20 mm diameter through hole is to be drilled in a 30 mm thick plate using a double fluted, 120° lip angle drill. The drill tip is at a distance of 3 mm from the plate surface when cutting started and an over travel of 2 mm is recommended. If the drill rotates at 500 rev/min and the feed per tooth is 0.1 mm, the machining time of operation (in sec) will be

(a) 48.92 (b) 19.80
(c) 21.00 (d) 24.46

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : For drilling a through-hole of diameter D and length L_w, the machining time T_c is evaluated from

$$T_c = \frac{L_c}{s_o N}$$

Where, s_o = Feed (mm/rev)
N = rpm of tool (rev/min)
$L_c = L_w + A + O + C$
A = Approach, O = over-travel and C = compulsory approach

$C = \frac{D}{2}\cot\rho$ where ρ = half of drill point angle.

Note: If the angle is not given then neglect compulsory approach

Calculation:

Given,

D = 20 mm, L_w = 30 mm, $2\rho = 120° \Rightarrow \rho = 60°$, A = 3 mm, O = 2 mm, N = 500 rev/min and s_o = 0.1 mm/tooth.

$$C = \frac{D}{2}\cot\rho$$

$$C = \frac{20}{2}\cot 60° \Rightarrow 5.773 \text{ mm}$$

$L_c = L_w + A + O + C \Rightarrow (30 + 3 + 2 + 5.773)$ mm = 40.773 mm.

$$T_c = \frac{L_c}{s_o N}$$

$$T_c = \frac{40.773}{0.1 \times 500} \Rightarrow 48.92 \text{ sec}$$

1269. A brass pin has a length of 500 mm and 40 mm diameter. The pin is to be reduced to 38.8 mm in one pas. The cutting speed is 60 m/min feed is 0.8 mm/min and rpm is 478. The machining time will be

(a) 1.31 min (b) 2.62 min
(c) 3.93 min (d) 5.24 min

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Given,
Cutting speed = 478 rpm, f = 0.8 mm/rev
L = 500 mm, D = 40 mm, V = 60 m/min

$$\text{Time for machining} = \frac{L}{fN} = \frac{500}{0.8 \times 478} = 1.31 \text{ min}$$

1270. A slot is to be milled centrally on a block with a dimension of 40 ± 0.05 mm. A milling cutter of 200 mm width is located with reference to the side of the block within ± 0.02 mm the maximum offset in mm between the centrelines of the slot and the block is

(a) ± 0.070 (b) ± 0.070
(c) ± 0.020 (d) ± 0.045

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : Since the reference is taken from the side.
Therefore only tolerance on the tool position will affect the product.
Hence, the maximum offset between the centre lines of the slot and the block tolerance on the side = ± 0.020.

1271. Process of removing surface of oxides and scale from metals by acid solution is :

(a) Buffing (b) Polishing
(c) Galvanizing (d) Pickling

TNPSC AE (Industries) 09.06.2013

Ans. (d) : Process of removing surface of oxides and scale from metals by acid solution is pickling.

1272. The operation of dividing the periphery workpiece into any number of equal parts is called :

(a) Milling (b) Indexing
(c) Gear cutting (d) Hobbing

TNPSC AE (Industries) 09.06.2013

Ans. (b) : The operation of dividing the periphery workpiece into any number of equal parts is called indexing.

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1273. Milling the workpiece simultaneously on both sides is :
(a) Climb milling (b) Gang milling
(c) Square milling (d) Straddle milling

TNPSC AE (Industries) 09.06.2013

Ans. (d) : Milling the workpiece simultaneously on both sides is straddle milling.

1274. The machine tool which is used as alternative to shaper when the workpiece is too large or too heavy is called :
(a) Slotter (b) Planner
(c) Shaper (d) Milling

TNPSC AE (Industries) 09.06.2013

Ans. (b) : The machine tool which is used as alternative to shaper when the workpiece is too large or too heavy is called planner.

1275. The drilling machine which has several spindles is called :
(a) Horizontal drilling (b) Mass drilling
(c) Vertical of drilling (d) Gang drilling

TNPSC AE (Industries) 09.06.2013

Ans. (d) : The drilling machine which has several spindles is called gang drilling.

1276. Abrasive used for grinding ceramics and tungsten carbide is :
(a) Diamond (b) Alumina
(c) Silicon carbide (d) Boron carbide

J&K PSC Civil Services Pre, 2013

Ans. (a) : Hard materials are used to make cutting tool can be nearly as hard as conventional abrasions, the job of sharpening then falls to a special class of abrasive known as super abrasives like diamond and CBN (Cubic Boron Nitride).

1277. A steel workpiece is to be milled. Metal removal rate is 40 cm^3/min. Depth of cut is 10 mm and width of cut is 200 mm. Find the Table feed?
(a) 20 mm/min (b) 30 mm/min
(c) 40 mm/min (d) 50 mm/min

TNPSC AE 2013

Ans. (a) : MRR $= d \times f \times b$
$40 \times 10^3 = 10 \times f \times 200$
Table feed, f = 20 mm/min.

1278. 119 divisions can be indexed by using
(a) differential indexing (b) simple indexing
(c) double indexing (d) direct indexing

TNPSC AE 2013

Ans. (a) : 119 divisions can be indexed by using differential indexing.

1279. In radial drilling machine, large and odd shaped workpieces are hold by using
(a) V-block
(b) Machine
(c) Strap clamps and T-bolts
(d) Three jaw chuck

TNPSC AE 2013

Ans. (c) : In radial drilling machine, large and odd shaped workpieces are hold by using strap clamps and T-bolts.

1280. When the grinding wheels loose their geometry, the original shape is restored by
(a) burnising (b) truing
(c) buffing (d) honing

TNPSC AE 2013

Ans. (b) : The purpose of truing is to bring every point of the grinding surface concentric with the machine spindle and to introduce a form into a wheel.

1281. Gear shaving in the operation pertaining to
(a) Gear cutting
(b) Gear tooth cutting
(c) Gear finishing
(d) Gear tooth correction

TNPSC AE 2013

Ans. (c) : Gear shaving in the operation pertaining to gear finishng.

1282. The name Carborundum refers to
(a) Silicon Carbide (b) Silicon Oxide
(c) Silicon Nitrate (d) Silicon Acid

TNPSC AE 2013

Ans. (a) : The name Carborundum refers to Silicon Carbide.

1283. Which of the following is not a natural abrasive?
(a) Garnet (b) Emery
(c) Carborundum (d) Diamond

TNPSC AE 2013

Ans. (c) : Carborundum is not a natural abrasive where as Gamet, Emery and Diamond are a natural abrasive.

1284. Buffing is the operation of
(a) Cleaning the casting
(b) Producing smooth surface
(c) Depositing metal by spraying
(d) Broaching in reverse direction

TNPSC AE 2013

Ans. (b) : Polishing and buffing are finishing processes for smoothing a workpiece's surface using an abrasive and a work wheel or a leather strop. Technically polishing refers to processes that use an abrasive that is glued to the work the work wheel, while buffing uses a loose abrasive applied to the work wheel.

1285. The progression ratio of the gear box is
(a) $\frac{N_{min}}{N_{max}} = \phi^n$ (b) $\frac{N_{min}}{N_{max}} = \phi^{n-1}$
(c) $\frac{N_{max}}{N_{min}} = \phi^{n-1}$ (d) $\frac{N_{max}}{N_{min}} = \phi$

TNPSC AE 2013

Ans. (c) : The progression ratio of the gear box is given as $\frac{N_{max}}{N_{min}} = \phi^{n-1}$
where -
n = number of division
ϕ = Ratio between two consecutive spindle speed.

$$N_{max} = \frac{V_{max} \times 1000}{\pi \times D_{min}}$$

$$N_{min} = \frac{V_{min} \times 1000}{\pi \times D_{max}}$$

$$N_2 = \phi \times N_1$$

$N_3 = \phi \times N_2$ ..... so an,

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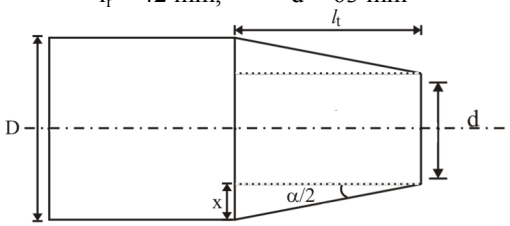
1286. A taper job of length 90 mm, has a taper length of 42 mm. The larger diameter of taper is 73 mm and the smaller diameter is 63 mm. Determine the tail-stock setting over.

(a) 10.51 mm (b) 10.61 mm
(c) 10.71 mm (d) 10.81 mm

TNPSC AE 2013

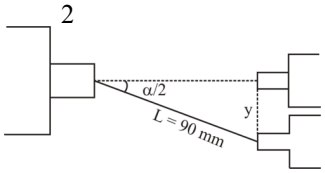
Ans. (c) : Given, L = 90 mm, D = 73 mm
l_r = 42 mm, d = 63 mm



$$\tan\frac{\alpha}{2}=\frac{x}{l_t}$$

$$x=\frac{D-d}{2}$$

$$=\frac{73-63}{2}$$



x = 5 mm

$$\tan\frac{\alpha}{2}=\frac{5}{42}$$

$$\boxed{\alpha=13.578^{o}}$$

Where α is tapper angle then, tail stock setting (y) is given as

$$y=L\times\sin\frac{\alpha}{2}$$

$$y=90\times0.1182$$

$$\boxed{y=10.639\text{ mm}}$$

1287. In lathe, which work holding device is used if the job is complex and irregular shape, which is inconvenient (or) even impossible to clamp/hold in chucks.

(a) Collets (b) Mandrels
(c) Lathe dogs (d) Face plate

TNPSC AE 2013

Ans. (d) : Face plate is used as a work holding device in lathe when the job is complex and irregular shape.

1288. Any number of equal divisions on the periphery of a circle can be obtained on a milling machine by

(a) direct indexing
(b) simple indexing
(c) compound indexing
(d) differential indexing

JPSC AE 2013, Paper-V

Ans : (b) : simple indexing

1289. In centreless grinding work piece, centre will be

(a) above the line joining the two-wheel centres
(b) below the line joining the two-wheel centres
(c) on the line joining the two-wheel centres
(d) at the intersection of the line joining the wheel centres with the plane of the workpiece

JPSC AE 2013, Paper-V

Ans : (a) : Centreless grinding- The centreless grinding is the process of grinding the diameter of a workpiece not mounted on centres or otherwise held. The centreless grinding may be external or internal. In centreless grinding workpiece, centre will be above the line joining the two-wheel centres.

1390. The correct sequence of tool materials in increasing order of their ability to retain their hot hardness is

(a) carbide, ceramic, cermet, borazon
(b) ceramic, carbide, borazon, cermet
(c) cermet, carbide, ceramic, borazon
(d) borazon, ceramic, carbide, cermet

JPSC AE 2013, Paper-V

Ans : (c) : The correct sequence of tool materials in increasing order is-
cermet, carbide, ceramic, borazon

1391. Binding material used in cemented carbide tools is :

(a) Nickel (b) Cobalt
(c) Chromium (d) Silicon

J&K PSC Civil Services Pre, 2013

Ans. (b) : Cobalt is used as binding material in cemented carbide tool.

1392. The number of strokes required to complete the shaping operation is

(a) $\frac{w}{f}$ (b) $\frac{f}{w}$
(c) $\frac{1}{fw}$ (d) $\frac{fw}{1}$

APPSC Poly. Lect. 2013

Ans. (a) : The number of strokes per min. (N) required to complete shaping operation is,

(i) $N=\frac{w}{ft}$ stroke/min

(ii) Cutting speed (v) $=\frac{NL(1+m)}{1000}$

(iii) MRR = fdNL(1 + m) mm^3/min

(iv) No. of Strokes $=\frac{w}{f}$

Where, t = time of machining
w = width
d = depth of cut
f = feed
m = ratio of return stroke time to cutting stroke time.
c = clearance
$L = L_j + 2c$
2c = Sum of clearance from both side of job

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1293. The index movement to mill hexagonal bolt direct indexing, if the rapid index plant 24 slots, is

(a) 2 (b) 4
(c) 8 (d) 16

APPSC Poly. Lect. 2013

Ans. (b) : Rapid indexing is used when a large number of duplicate parts are to be milled.
To find the no. of holes to move the index plate, divide 24 by the no. of divisions required.
Number of holes to move = 24/N
where, N = required no. of divisions
→ Direct indexing for a hexagonal head bolt : because of hexagonal head has six faces.

$$\text{No. of holes to move} = \frac{24}{6} = 4 \text{ holes}$$

The index plate has to move by 4 holes for the next face to the machine.

1294. Which is the correct answer for specification of Broaching machine?
I. Weight of the machine
II. Speed and feed range
III. Power rating
IV. Type of motion

(a) I, II and III (b) II and III
(c) I, II, III and IV (d) I alone

APPSC Poly. Lect. 2013

Ans. (c) : Specification of broaching machine-
(1) Power of the motor and its speed
(2) Rated pulling force
(3) Cutting stroke speed
(4) No. of speeds
(5) Weight of the machine
(6) Linear motion

1295. Which is the composition of Corborundum?

(a) 75 to 90% crystalline Al_2O_3 + iron oxide
(b) 20 to 30% crystalline Al_2O_3 + iron oxide
(c) 100% Al_2O_3
(d) 100% iron oxide

APPSC Poly. Lect. 2013

Ans. (a) : The composition of Corborundum consist of 75 to 90% crystalline Al_2O_3 + iron oxide.

1296. Which surface is not produced by shaper?

(a) Curved surface (b) Inclined surface
(c) Horizontal surface (d) Vertical surface

APPSC Poly. Lect. 2013

Ans. (a) : Shaper can produce flat surface in all horizontal vertical and angular planes. Curved surface is not produced by Shaper.

1297. Which of the following statements is not correct about climb or down milling?

(a) It gives poorer surface finish than up milling.
(b) The work is fed in the same direction as the cutter rotation.
(c) The chip thickness is maximum at the beginning of the cut.
(d) Power required is less.

RPSC AEN Pre-2013

Ans. (a) : About climb milling/down milling
- Its gives good surface finish than up milling.
- The work is fed in the same direction as the cutter rotation.
- The chip thickness is maximum at the beginning of cut and chip thickness is minimum at the end cut.

1298. Clapper block (or box) is used in which machine tool?

(a) Lathe (b) Shaper
(c) Milling (d) Turret lathe

RPSC AEN Pre-2013

Ans. (b) :
- Clapper block (or box) is used in shaper machine tool.
- The clapper box houses the clapper block which is connected to it by means of hinged pin.
- The tool post is mounted upon the clapper block to prevent interference between the tools and workpiece on the return stroke.
- On the forward cutting stroke the clapper block fits securely to the clapper box to make a rigid tool on the work.
- It prevents the tool cutting edge from dragging and wear.
- The work surface is also prevented from any damage due to dragging.
- Clapper box of shaper is used to lift the tool during return stroke.

1299. In taper turning method by swiveling the compound slide, the feed motion is given by:

(a) moving the carriage.
(b) moving the cross slide.
(c) engaging the half nut.
(d) moving the compound slide.

RPSC AEN Pre-2013

Ans. (d) : In tapper turning method by swiveling the compound slide, the feed motion is given by moving the compound slide.
In compound slide method one part of compound rest is swiveled (at half the included angle of the taper). We can easily set this compound slide.

1300. Which of the following is not a constituent of high speed steel tool material?

(a) Tungsten (b) Molybdenum
(c) Titanium (d) Chromium

RPSC AEN Pre-2013

Ans. (c) : High speed steel series
(1) T-series → 18W – 4 Cr – 1V
(2) M-series → 12Mo – 5 Cr – 2V
W = Tungsten
Cr = Chromium
Mo = Molybdenum
V = Vanadium

1301. It is required to cut screw threads of pitch 2 mm in a lathe. The lead screw a pitch of 6 mm. If the spindle speed is 60 rpm, then the speed of the screw will be

(a) 10 rpm (b) 20 rpm
(c) 120 rpm (d) 180 rpm

TRB Asstt. Prof., 2012

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Ans. (b) : Speed of screw

$$N = \frac{60 \times 2}{6} = 20 \text{ rpm}$$

1302. Match List I (Milling process) correctly with List II (Results) and select your answer using the codes given below :

List I (Milling process)	List II (Results)
(A) Face milling	1. surface parallel to the axis of cutter is produced
(B) Peripheral milling	2. surface perpendicular to the axis of cutter is produced sand
(C) End milling	3. surface parallel and perpendicular to the axis of cutter are produced
(D) Slick	4. Lifting the patterns out of mould

Codes

	a	b	c
(a)	1	2	3
(b)	1	3	2
(c)	2	1	3
(d)	2	3	1

TNPSC ACF 2012

Ans. (c) : Face Milling– Surface perpendicular to the axis of cutter is produced.
Peripheral Milling– Surface parallel to the axis of cutter is produces.
End Milling– Surface parallel and propounding cutter to the axis of cutter are produced.

1303. A 600 mm × 30 mm flat surface of a plate is to be finish machined on a shaper. The plate has been fixed with the 600 mm side along the tool travel direction. If the tool over-travel at each end of the plate is 20 mm, average cutting speed is 8 m/min, feed rate is 0.3 mm/stroke and the ratio of the return time to cutting time of the tool is 1:2, the time required for machining will be

(a) 8 min (b) 12 min
(c) 16 min (d) 20 min

VIZAG STEEL MT 10.06.2012
GATE 2005

Ans. (b) :
Length travelled in forward stroke = 640 mm
Number of strokes = 100
Time for cutting = 8 min
Return time = 4 min
Total time = 4 + 8
Total time = 12 min

1304. Stroke of a shaping machine is 250 mm. It makes 30 double strokes per minute. Overall average speed of operation is

(a) 3.75 m/min (b) 5.0 m/min
(c) 7.5 m/min (d) 15.0 m/min

APPSC AEE 2012
CSE Pre-1994

Ans. (c) : Data given,
Stroke of shaping machine = 250 mm
No of Double stroke per minute = 30
Solution–
Maximum speed = 0.25 × 30 × 2 = 15 m/min
Average cutting speed = $V_{max}/2 = 15/2 = 7.5$ m/min

1305. In lathe, the carriage and tail stock are guided on

(a) Different guideways
(b) Same guideways
(c) Any one of the two
(d) None of the above

APPSC AEE 2012

Ans. (b) : In lath, the carriage and fail stock arc guided on same guide ways.

1306. Cutting tool used on lathe, shaper and planer is

(a) single point (b) multi-point
(c) two-point (d) three-point

TRB Poly. Lect., 2012

Ans. (a) : Cutting tool used on lathe, shaper and planer is single point, A Shaper is a type of machining tool that uses linear relative motion between the work-piece & a single point cutting tool to machining linear tool path. A planar is a type of metal working machining tool that uses linear relative motion between the works pieces.

1307. The material that has poor machinability is

(a) aluminium (b) copper
(c) white cast iron (d) gray cast iron

TRB Poly. Lect., 2012

Ans. (c) : White cast iron is harder & so exhibits less machinability It contain less amount of silicon which help for better machinability.

1308. Machinability

(a) tends to increase with increase in hardness
(b) tends to decreases with increase in hardness
(c) remains unaffected with hardness
(d) is not related to any of these

TRB Poly. Lect., 2012

Ans. (b) : Machinability is the ease with which a metal can be cut (machined) permitting the removal of the material with a satisfactory finish at low cost. Material with good machinability (free machining materials) requires little power to cut, can be cut quickly with good surface finish and do not wear tool.
Hence with increase of hardness of material machinability of material decrease.

1309. Reaming is the operation of

(a) enlarging the end of a hole cylindrically
(b) cone shaped enlargement at the end of hole
(c) smoothing and squaring the surface around a hole
(d) sizing and finishing a hole

TRB Asstt. Prof., 2012

Ans. (d) : Reaming is a finishing operation of high-precision holes performed with a multi-edge tool high surface finish, superb hole quality and close dimensional tolerance are achieved at high penetration rates and small depths of cut.

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1310. The number of slots in a 25 mm castel nut is

(a) 2 (b) 4
(c) 6 (d) 8

PSPCL AE, 2012

Ans. (c) : Castel nut also called crown nut is a nut with slots cut into one end visually they look like hexagonal nut and are caused along side a bolt or a thread rod that feature a pre drilled holt where cotton pin can be used to secure the fixing and prevent movement these are used as lock nut and are ideal for used in application where vibrating may undo a fixing. Generally nut has six faces and have one slots on each face.

1311. Machine tool frame should have :

(a) High rigidity to weight ratio
(b) Graphite in the form of nodules
(c) Low hardness
(d) High work hardness

PSPCL AE, 2012

Ans. (a) : High rigidity to weight.

1312. A mild steel block of width 40mm is being milled using a straight slab cutter 70mm diameter with 30 teeth. If the cutter rotates at 40 rpm, and depth of cut is 2mm, what is the value of maximum uncut chip thickness when the table feed is 20 mm/min?

(a) 0.00263 mm (b) 0.00363 mm
(c) 0.00463 mm (d) 0.00563 mm

ISRO Scientist/Engineer 2012

Ans. (d) : Given,
Diameter of straight slab cutter D = 70 mm
N = 40 rpm
depth of cut d = 2mm
table feed f = 20mm/min, Z = 30

$$t_{max} = \frac{2f}{NZ}\sqrt{\frac{d}{D}}$$

$$t_{max} = \frac{2\times 20}{40\times 30}\sqrt{\frac{2}{70}}$$

$$t_{max} = 5.634\times 10^{-3} mm = 0.00563$$

1313. The most common type of drill is

(a) Step drill (b) Twist drill
(c) Spot drill (d) Spade drill

BPSC AE 2012 Paper - VI

Ans : (b) : It is an end cutting tool with two, three or four cutting lips. It has a cylindrical body on which the groove are cuts. These grooves are called flutes while drilling, the drill is held by a shank.

1314. Which of the following grinding processes is used for high production run?

(a) Surface grinding
(b) Cylindrical grinding
(c) Centreless grinding
(d) Internal grinding

BPSC AE 2012 Paper - VI

Ans : (a) : Surface grinding is used to produce that surface in a horizontal position.

1315. What is the material commonly used for machine tool bodies is?

(a) Mild steel (b) Aluminium
(c) Brass (d) Cast iron

VIZAG STEEL MT 2011

Ans. (d) : Machine tool bodies are commonly made up cast iron because it has excellent damping properties.

1316. What is lapping used for?

(a) To draw hole
(b) To smoothen the edges of the workpiece
(c) To reduce the chip thickness
(d) To obtain truly flat and smooth surfaces

VIZAG STEEL MT 2011

Ans. (d) : Lapping is used to obtain truly flat and smooth surfaces.

1317. With a punch for which the maximum crushing stress is 4 times the maximum shearing stress of the plate, the biggest hole that can be punched in the plate would be of diameter equal to

(a) $\frac{1}{4}$ × Thickness of plate
(b) $\frac{1}{2}$ × Thickness of plate
(c) Plate thickness
(d) 2 × Plate thickness

ISRO Scientist/Engineer 2011
OPSC Civil Services Pre 2006

Ans. (c) : Crushing stress

$$\sigma_c = \frac{4P}{\pi d^2}$$

Shear stress

$$\tau = \frac{4P}{\pi dt}$$

(Where d is diameter of hole and t is thickness of plate)

$$\sigma_c = \tau$$

$$\frac{4P}{\pi d^2} = \frac{4P}{\pi dt}$$

d = t

1318. At what rpm should a lathe be run to give a cutting speed of 20 m/min when turning a rod of diameter 40 mm?

(a) 100 rpm (b) 160 rpm
(c) 200 rpm (d) 250 rpm

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : $\text{RPM} = \frac{\text{Speed}\times 1000}{\pi\times \text{Diameter of Rod(D)mm}}$

$$= \frac{V\times 1000}{\pi D}$$

Given- D = 40 mm
Speed = 20 m/min

$$= \frac{20\times 1000}{3.14\times 40} = 159.8\text{rpm}$$

RPM ≈ 160 rpm

1319. The coolant generally used while machining with carbide tools is

(a) Kerosene (b) Water
(c) Soluble oil (d) Graphite

RPSC Lect. (Tech. Edu. Dept.) 2011

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Ans. (c) : For high speed machining with carbide tools, it is very important to provide adequate water-based fluids but, if you are using high-speed steel tappings than heavier oil-based coolant is recommended (Soluble oil)
Note- No answer is given by commission.

1320. Among the following, which material is having highest machine-ability?
(a) Cast iron (b) Mild steel
(c) High carbon steel (d) Aluminium

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Machinability is the ease with which a metal can be cut permitting the removal of material with a satisfactory finish at low lost, It require little power to cut quickly, easily obtain good finish and not wear tool, So in this Aluminium is having all these properties as compared to others.

1321. In a turning operation, the change in radius of the work part is equal to which one of the following?
(a) 1 × depth of cut (b) 2 × depth of cut
(c) 1 × feed (d) 2 × feed

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : When a piece (workpiece) of radius having 'R' is turned, the depth of cut is given toward the centre of work piece which directly reduce its radius & gives new radius, which is equal to depth of cut subtracted from old radius. So, change in radius = 1 × depth of cut.

1322. Which of the following abrasive materials is most appropriate for grinding hardened tool steel?
(a) Aluminium oxide (b) Cubic boron nitride
(c) Diamond (d) Silicon carbide

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : CBN is produced in approximately, is the same way as synthetic diamond is an abrasive that is used primarily for grinding hardened high carbide tool steel & high speed steel. Al_2O_3 is commonly used for grinding steel.

1323. Which of the following abrasive materials is most appropriate for grinding steel and cast iron?
(a) Aluminium oxide (b) Cubic boron nitride
(c) Diamond (d) Silicon carbide

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : For grinding cast Iron, non-ferrous metals & Non metallic material, silicon carbide abrasive is used, which is hard, brittle material, generally required with a fine grit size & a softer grade.

1324. A milling machine has the following two index plates supplied along with the indexing head:
Plate I: 15, 16, 17, 18, 19, 20, hole circles
Plate 2 : 21, 23, 27, 29, 31, 33, hole circles
It is proposed to mill a spur gear of 28 teeth using simple indexing method. Which one of the following combinations of index plate and number of revolutions is correct?
(a) Plate I : 1 revolution and 9 holes in 18 hole circles
(b) Plate 2 : 1 revolution and 9 holes in 21 hole circles
(c) Plate 2 : 1 revolution and 9 holes in 33 hole circles
(d) Plate I : 1 revolution and 9 holes in 15 hole circles

ISRO Scientist/Engineer 2011

Ans. (b) : The rotation of the index crankn (for simple indexing)

$$= \frac{40}{28} = 1\frac{3}{7} \text{ turns}$$

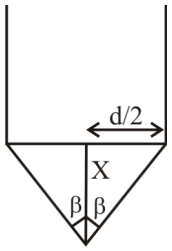
i.e. means $= 1\frac{3\times 3}{7\times 3} = 1\frac{9}{21}$ turns

One full rotation + 9 hole in 21 hole circle, so plate-2 is used.

1325. The depth of cut in drilling is measured at right angles to the axis of the drill and is numerically equal to :
(a) d/4 (b) d/2
(c) d (d) 2d
Where d is diameter of drill.

UPSC JWM Advt. No.-52/2010

Ans. (b) : Solution –



$$\tan\beta = \frac{d/2}{X}$$

$$\text{Depth of cut} = \frac{\text{Diameter of the drill}}{2\tan\beta}$$

Given; d= diameter of the drill

$$\therefore \text{Depth of cut} = \frac{d}{2\tan 45^\circ} \Rightarrow \begin{cases}\tan\beta = \tan 45^\circ \\ \tan\beta = 1\end{cases}$$

$$= \frac{d}{2}$$

1326. Which one of the following statements is correct for shaper?
(a) It is used for machining large flat surfaces
(b) Cutting takes place by reciprocating the work
(c) Only one tool is used
(d) Indexed feed is given to the tool

UPSC JWM Advt. No.-52/2010

Ans. (c) : Explanation- In the shaper, only one tool is used.
Shaper Machine–
(1) It is used for short length of flat surface because motion of, slotted level behave as cantilever beam so higher the length, high will be the moment on the fixed end of slotted lever (arm). So shaper uses for shorter length of workpiece.
(2) Cutting takes place by reciprocating tool.
(3) Only one tool is used
(4) Indexed feed is given to the workpiece.

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1327. Match List I with List II and select the correct answer using the code given below the Lists :

List I (Machining)	List II (Tolerance in mm)
A Turning	1. 0.1 to 0.2
B. Drilling	2. 0.05 to 0.1
C. Grinding	3. 0.02 to 0.035
D. Reaming	4. 0.007 to 0.01

Code :

	A	B	C	D
(a)	3	1	4	2
(b)	2	1	4	3
(c)	3	4	1	2
(d)	2	4	1	3

UPSC JWM Advt. No.-52/2010

Ans. (b) :

(Machining)	(Tolerance in mm)
• Turning	–(2)→ 0.05 to 0.1
• Drilling	–(1)→ 0.1 to 0.2
• Grinding	–(4)→ 0.007 to 0.01
• Reaming	–(3)→ 0.02 to 0.035

1328. Consider the following operations regarding boring machines :

1. Counter boring
2. Counter sinking
3. Trepanning

Which of the above operations is/are correct ?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 only

UPSC JWM Advt. No.-50/2010

Ans. (a) : Counter Boring– Enlarging from the end of circular hole to provide seat for head of bolt.
Counter Sinking–It is a process that creates v-shaped edge near the surface of a hole.
Trepanning–It is a technique used for drilling larger hole diameters where machine power is limited as it is not as power consuming as conventional drilling where the entire hole is converted into chips.

1329. Assertion (A) : Extrusion speed depends on work material.
Reason (R) : High extrusion speed causes cracks in the material.
Codes :

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

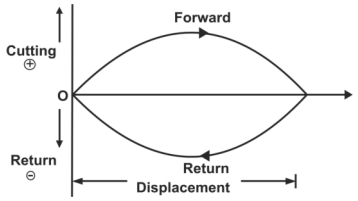
Ans. (b) : As we know extrusion is material forming process, the speed of extrusion is depends on the material properties. Such as plastic, metals etc. The speed may increases or decreases, cause changes in then properties of the materials.

1330. The cutting speed of the tool in a mechanical shaper is

(a) Maximum at the beginning of the cutting stroke
(b) Maximum at the end of the cutting stroke
(c) Maximum at the middle of the cutting stroke
(d) Minimum at the middle of the cutting stroke

ISRO Scientist/Engineer 2010

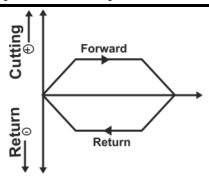
Ans. (c) : <u>Mechanical shaper</u>
In mechanical shaper, speed is maximum at the mid of stroke.



"Velocity diagram in a crank shaper"

<u>Hydraulic shaper</u>

→ In hydraulic shaper, speed remain constant throughout the stroke.
→ Hydraulic system is having due to oil tank.



1331. The approximate variation of the tool life exponent 'n' of cemented carbide tool is :

(a) 0.03 to 0.08 (b) 0.08 to 0.20
(c) 0.2 to 0.48 (d) 0.48 to 0.70

J&K PSC Civil Services Pre, 2010

Ans. (c) : Value of exponent 'n'
n = 0.08 to 0.2 for HSS tool
n = 0.1 to 0.15 for cast alloy
n = 0.2 to 0.4 for carbide tool
n = 0.5 to 0.8 for ceramic tool

1332. What is the correct sequence of the following parameters in order of their maximum to minimum influence on tool life ?

(i) Feed rate (ii) Depth of cut
(iii) Cutting speed
(a) (iii) , (i), (ii) (b) (i), (ii), (iii)
(c) (iii), (ii), (i) (d) (ii), (iii), (i)

J&K PSC Civil Services Pre, 2010

Ans. (a) : Maximum to minimum influence on tool life
Cutting speed > Feed rate > Depth of cut
(III) (I) (II)

1333. While cutting cast iron by a single point cutting tool the average cutting speed is :

(a) 11 m/min (b) 22 m/min
(c) 33 m/min (d) 55 m/min

APPSC IOF, 2009

Ans. (b) : For machining a cast iron workpiece by a high speed steel tool, the average cutting speed is 22 m/min and for a mild steel is 30 m/min.

1334. Machinability can be calculated and predicted by the following :

(a) tensile strength (b) brittle hardness
(c) shear angle (d) all of the above

APPSC IOF, 2009

Ans. (d) : Machinability can be calculated and predicted by tensile strength, brittle hardness & shear angle.

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1335. Find the time required in turning a bar of length L mm and diameter D mm, with a cutting speed V m/min and a feed rate of f mm/rev :

(a) $\frac{\pi DL}{1000\,fN}$ (b) $\frac{2\pi DL}{1000\,Nf}$

(c) $\frac{\pi DL}{1000\,fV}$ (d) $\frac{2\pi DL}{1000\,fV}$

APPSC IOF, 2009

Ans. (a) : $\frac{\pi DL}{1000\,fN}$ = Feed rate or linear speed of tool along workpiece length in mm/min.

1336. In a simple micrometer with screw pitch 0.5 mm and division on thimble 50, the reading corresponding to 5 divisions on barrel and 12 divisions on thimble is

(a) 2.620 mm (b) 2.512 mm

(c) 5.120 mm (d) 5.012 mm

ISRO Scientist/Engineer 2009

Ans. (a) : The reading corresponding to 5 division on barrel and 12 divisions on thimble is

$$0.5\times5+\frac{0.5}{50}\times12=2.62 \text{ mm}$$

1337. Which one of the following statement is correct for a centre lathe?

(a) Compound rest is mounted on the top of the saddle

(b) Three jaw chuck is used for holding irregular jobs.

(c) A mandrel is used for holding bored parts.

(d) Tail stock centre is called live centre.

UPSC JWM Adv. No-16/2009

Ans. (c) : A mandrel is used for holding bored parts in the centre lathe i.e., for outer surface machining.

1338. Consider the following statements with respect to automatic lathe :

1. Head stock can be moved forward and backward.

2. The radial movement of tool slides is controlled by cam.

3. Bar stock is held in a collet chuck.

Which of the above statements is/are correct?

(a) 1, 2 and 3 (b) 1 and 2

(c) 2 and 3 (d) 1 only

UPSC JWM Adv. No-16/2009

Ans. (c) : With respect to automatic lathe–

- The radial movement of tool slides is controlled by cam.
- Bar stock is held in a collet check.

1339. Which of the following is the characteristic of a capstan lathe?

(a) Capstan is mounted directly on the saddle

(b) High rigidity

(c) For feeding the tools entire saddle unit is moved

(d) It suits for relatively small sized bars

UPSC JWM Adv. No-16/2009

Ans. (d) : Capstan lathe has less rigidity, vibrations occur, hence suitable for lighter and smaller jobs (up to 60 mm) and precision work.
i.e., It suits for relatively small sized bars.

1340. Which one of the following statement is correct for down milling?

(a) In this a better surface finish is obtained

(b) This is used for machining castings

(c) In this chip thickness increases gradually

(d) In this the cutter rotates in a direction opposite to the work feed.

UPSC JWM Adv. No-16/2009

Ans. (a) : In the down milling operation a better surface finish is obtained. The metal is removed by the cutter which is rotated in the same direction of travel of the workpiece.

1341. Consider the following statements :

1. Stronger teeth

2. More space for chip flow

3. More life of the cutter

Which of the above advantages is/are correct offered by a coarse milling cutter?

(a) 1, 2 and 3 (b) 1 and 2

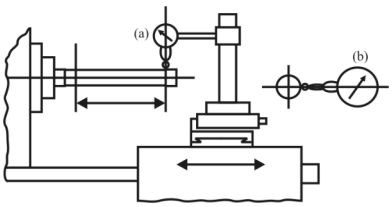
(c) 1 and 3 (d) 2 only

UPSC JWM Adv. No-16/2009

Ans. (a) : Advantages of a coarse milling cutter are –

- Stronger teeth
- More space for chip flow
- More life of the cutter

1342.



Which one of the following can be determined by an above shown machine tool testing carried out in a centre lath?

(a) The center distance between the head-stock and tailstock.

(b) The center distance between head-stock spindle axis and carriage.

(c) Parallellism between the saddle slide way and the headstock spindle axis.

(d) Parallelism between tailstock spindle axis and the headstock spindle axis.

UPSC JWM Adv. No-16/2009

Ans. (d) :

1343. A plate of 25 mm thickness is to be drilled at the feed rate of 0.025 mm/rev within 20 seconds. Neglecting breakthrough distance, the drilling speed would be :

(a) 1000 rpm (b) 1500 rpm

(c) 200 rpm (d) 3000 rpm

DRDO Scientists 2009

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Ans. (d) : $T = \frac{L}{fN}$

$$N = \frac{L}{fT} = \frac{25}{0.025 \times 20} \text{ rps} = 50 \text{rps}$$

$\therefore$ N = 3000 rpm

1344. The precision grinding of round and flat parts with loose, dust type abrasives is known as
(a) lapping (b) honing
(c) polishing (d) buffing

TNPSC AE, 2008

Ans. (a) : Lapping is a machining operation in which two surfaces are rubbed together with an abrasive between them by way of a machine.
The first type of lapping involves rubbing a brittle material such as glass against a surface such as iron or glass itself with an abrasive such as aluminum oxide, jeweler's rough, optician's rough, emery, silicon carbide, diamond etc.

1345. How a portable drilling machine specified?
(a) By spindle speeds and feed
(b) By maximum diameter it can hold
(c) By maximum thickness of job it can hold
(d) By maximum travel of spindle

WBPSC AE 2008

Ans. (b) : A portable drilling machine is specified by maximum diameter that it can hold.

1346. On which of the following machines gear cannot be cut?
(a) Hobbing machine (b) Lathe
(c) Milling machine (d) Shaper

WBPSC AE 2008

Ans. (d) : Shaper

1347. Given that cutting speed = 9m/min, the return time to cutting time ratio is = 1:2 the feed rate = 0.3 mm/stroke, the clearance at each of cut = 25 mm and that the plate is fixed 700 mm side along the direction of tool travel, the time required for finishing one flat surface of a plate of size 700 × 300 mm in a shaper, will be
(a) 10 min (b) 12.5 min
(c) 15 min (d) 20 min

ISRO Scientist/Engineer 2008

Ans. (*) : Given,
v = 9 m/min

$$\lambda = \frac{t_r}{t_f} = \frac{1}{2} = 0.5$$

Feed rate, f = 0.3 mm/stroke
L = 300 mm

$$N = \frac{v \times 1000}{L(1+\lambda)}$$

$$N = \frac{9 \times 1000}{300(1+0.5)}$$

N = 20 strokes/min.

$$t_m = \frac{B + 25 + 25}{fN}$$

$$= \frac{700 + 50}{0.3 \times 20}$$

Machining time, t_m = 125 min

1348. For drilling operation, cylindrical job should always be clamped on a
(a) Vice (b) Socket
(c) V-block (d) Clamp

TNPSC AE, 2008

Ans. (c) : For drilling operation, cylindrical job should always be clamped on a V-block.

1349. For drilling brass, a drill with
(a) high helix angle is required
(b) low helix is required
(c) any helix angle is required
(d) zero helix angle is required

TNPSC AE, 2008

Ans. (b) : Low helix is required for medium workpiece brass, bronze etc.

1350. Time taken to drill a hole through a 25 mm thick plate at 300 r.p.m. at a feed rate of 0.25 mm/rev. will be
(a) 10 sec (b) 20 sec
(c) 25 sec (d) 40 sec

TNPSC AE, 2008

Ans. (b) : Given, L = plate thickness = 25 mm.
Speed (N) = 300 rpm,
Feed (f) =0.25 mm/rev

$$\text{Time (t)} = \frac{L}{fN} = \frac{25\,\text{mm}}{0.25\ \text{mm/rev} \times \frac{300}{60}\ \text{rps}}$$

$$= \frac{25 \times 100 \times 60}{25 \times 300}$$

time (t) = 20 second

1351. The job length for a shaper is
(a) unlimited
(b) equal to that for planer
(c) limited to smaller size
(d) more than that for planer

TNPSC AE, 2008

Ans. (c) : The job length for a shaper is limited to smaller size. Because shaper is used for the purpose of machining smaller jobs. It is mainly used for producing flat and plan surfaces.

1352. Gear shaper can be used to cut which of the following types of gear ?
(a) Internal (b) External
(c) Non-conventional (d) All of these

TNPSC AE, 2008

Ans. (d) : Gear shaper can be used to cut internal, external and non-conventional types of gear.

1353. Which is incorrect?
A hob cutter
(a) rotates about its axis during cutting
(b) moves axially after gear blank has made one rotation
(c) moves into workpiece during feed
(d) Moves parallel to the axis of workpiece during cutting

TNPSC AE, 2008

Ans. (d) : About to hob cutter-
- Rotates about its axis during cutting.
- Moves axially after gear blank has made one rotation.
- Moves into workpiece during feed.

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1354. A grinding wheel is designated as A 60L5V. In this designation the number 60 represents

(a) type of abrasive (b) type of bond
(c) grain size (d) type of wheel

WBPSC AE, 2007

Ans. (c)				
A	**60**	**L**	**5**	**V**
Abrasive	Grain size	Hardness or Grade	Structure	Bond

1355. A low abrading process which uses bonded abrasive sticks for removing stock from metallic and non-metallic surfaces is called

(a) lapping (b) grinding
(c) honing (d) machining

WBPSC AE, 2007

Ans. (c) : A low abrading process which uses bonded abrasive sticks for removing stock from metallic and non-metallic surfaces is called honing.

1356. Normally the surface generated by turning in a lathe is

(a) cylindrical (b) plane
(c) hypoid (d) circular

WBPSC AE, 2007

Ans. (a) : Normally the surface generated by turning in a lathe is cylindrical.

1357. Dividing head is an attachment which is used in a

(a) lathe (b) drilling machine
(c) milling machine (d) rolling mill

WBPSC AE, 2007

Ans. (c) : Dividing head is an attachment used is milling machine.

1358. Spinning operation is carried out on

(a) Hydraulic press (b) Mechanical press
(c) Lathe (d) Milling machine

ISRO Scientist/Engineer 2007

Ans. (c) : Spinning operation is carried out on lathe.

1359. With high speed steel tools, the maximum safe operating temperature is of the order of

(a) 200ºC (b) 540ºC
(c) 760ºC (d) 870ºC

UKPSC AE 2007 Paper -I

Ans. (b) : 540ºC

1360. External screw threads can be mass produced fastest by :

(a) Milling (b) Chasing
(c) Casting (d) Rolling

J & K PSC Screening, 2006

Ans. (d) : External screw threads can be mass produced fastest by - Rolling

1361. In most high speed milling cutters, positive radial rake angle is :

(a) 2 – 6° (b) 7 – 10°
(c) 10 – 15° (d) 15 – 20°

J & K PSC Screening, 2006

Ans. (c) : In most high speed milling cutters positive rake angle is 10 – 15°.

1362. Broaching is used for machining

(a) internal and external surfaces
(b) rounding to irregular shaped holes
(c) teeth of a gear
(d) All of the above

J & K PSC Screening, 2006

Ans. (d) : Broaching is used for machining internal as well as external surface, rounding and irregular shaped holes, teeth of gears, splines, keyways and flat surfaces.

1363. Which of the following processes would produce strongest components?

(a) Hot rolling (b) Extrusion
(c) Cold rolling (d) Forging

ISRO Scientist/Engineer 2006

Ans. (c) : Hot Rolling– Hot rolling is a mill process which involves rolling the steel at high temperature which is above the steel's recrystallization.

Cold Rolling– Cold rolling is a metal working process in which metal is deformed by passing it through rollers at a temperature below its recrystallization temperature.

Cold Rolling increases the yield strength and hardness of metal by introducing defects into the metal's crystal structure. These defects prevent further slip and can reduce the grain size of the metal, resulting in Hall-Petch hardening.

Since strength and hardness of a cold working component is always more than that of hot working due to strain hardening. Therefore cold working would be a better option as cold is specifically mentioned whereas in forging it could be hot or cold.

If the option were forging and rolling only, in that case forging would have been better option.

1364. Grinding cannot correct irregularities but honing can correct because it uses

(a) small contact area at slow speed
(b) small contact area at high speed
(c) large contact area at high speed
(d) large contact area at slow speed

WBPSC AE 2003

Ans. (b) : Grinding cannot connect irregularities but honing can correct because it uses small contact area at high speed. Grinding refers to removal of large amounts of material from the surface of the stroke. Honing is the smoothing of the stone, not as aggressive as grinding and removal of very little material from stone.

1365. The cutting force is maximum when the tooth begins its cut and reduces to minimum when the tooth leaves the work in

(a) up milling (b) down milling
(c) end milling (d) face milling

WBPSC AE 2003

Ans. (b) : UP milling (conventional milling) The process of removing metal by cutter which is rotate against the direction of travel work piece is called up milling.

Cutting force varies from zero to maximum.

Down milling (Climb milling) - In this case cutter rotate in same direction of travel of the work piece. Cutting force varies from maximum to zero.

1366. Broaching process, in which the work moves past the stationary tool, is known as

(a) push broaching
(b) pull broaching
(c) surface broaching
(d) continuous broaching

WBPSC AE 2003

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Ans. (d) : The difference between push, pull, surface and continuous broaching are as follows–
(i) Pull Broaching–Work piece is clamped to the broaching machine in a stationary position and the broach is pulled through the work. Mostly used for internal broaching.
(ii) Push Broaching–Work piece is held in stationary position and the broach is pushed through the work. Most used for sizing and finishing hole cavities and keyways.
(iii) Surface Broaching–Anyone of two is kept moving and other kept stationary.
Most used for surface finishing.
(iv) Continuous Broaching–Broach is held stationary and work piece is moved continuously.
Generally used for broaching a large number of similar work piece at a time.

1367. Operation of beveling the extreme end of a work piece is known as
(a) chamfering (b) grooving
(c) facing (d) knurling

WBPSC AE 2003

Ans. (a) : Operation of beveling (reduce to sloping edge) the extreme end of a work piece is done by chamfering. It is an essential operation after thread cutting.

1368. In high speed cutting tools, the cobalt matrix
(a) provides cushion for absorbing the shock as well as required toughness
(b) surrounds the carbides when heated below the melting point of cobalt
(c) is used as binding agent between tungsten and polymer
(d) molecules are exposed to ultraviolet rays to get high strength

Karnataka PSC Lect., 27.05.2017

Ans (a) : In high speed cutting tools, the cobalt matrix provides cushion for absorbing the shock as well as required toughness.

1369. The laws of adhesive wear, commonly referred to as Archard's law can be expressed as_____ (If 'Q' is the total volume of wear debris produced, 'W' is the normal load, 'L' is the sliding distance, 'H' is the hardness of the softest contacting surfaces being worn away, and 'k' is a non-dimensional wear coefficient dependent on the materials in contact and their exact degree of cleanliness)
(a) $Q = kWL/H$ (b) $Q = kW/LH$
(c) $Q = kHW/L$ (d) $Q = k/LWH$

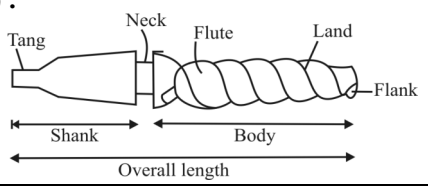
ISRO Scientist/Engineer 12.01.2020

Ans. (a) : The Archard wear equation is a simply model used to describe sliding wear and is based on the theory of 'asperity contact' $Q = kWL/H$

1370. The portion of the twist drill which is used to grip it in the sleeve is known as:
(a) Point (b) Shank
(c) Flute (d) Dead center
(e) Land

CGPSC ARTO & TSI 07.01.2017

Ans. (b) :



1371. A drill of diameter 20 mm makes a hole on a steel part at a cutting speed of 22 m/mm. Find the spindle speed.
(a) 200 rpm (b) 220 rpm
(c) 250 rpm (d) 330 rpm
(e) 350 rpm

CGPSC ARTO & TSI 07.01.2017

Ans. (e) : Given-
$D = 20$ mm
$V = 22$ m/min
$V = \pi DN$
$N = \dfrac{22}{\pi D}$
$N = \dfrac{22}{\pi \times 20 \times 10^{-3}}$
$N = 350$ rpm

1372. Grinding of hard materials requires
(a) Fine grit size and hard grades
(b) Fine grit size and soft grades.
(c) Coarse grit size and hard grades.
(d) Coarse grit size and soft grades.

UPPSC AE 12.04.2016 Paper-I

Ans : (b) Fine grit size and soft grades.

1373. The lathe bed is made of
(a) Mild steel (b) Alloy steel
(c) Pig iron (d) Chilled cast Iron

TSPSC AEE 2015

Ans : (d) The lathe bed is made of chilled cast iron.
Chilled cast iron:- It is a white cast iron produced by quick cooling of molten iron. The quick cooling is generally called chilling and the cost iron so produced is called chilled cost iron.

1374. The Indian Standard Marking system for a grinding wheel is read as WA46K5V17. The alphabet 'K' denotes
(a) bond type (b) grain size
(c) abrasive (d) grade

TNPSC AE 2018

Ans. (d) : WA → Abrasive material
46 → Grit size
K → Grade
5 → Structure
V → Bond and (vitrified)

1375. In which of the following operations, motion of job is rotary and motion of cutting tool is forward?
(a) turning (b) Planning
(c) Milling (d) Drilling

Assam PSC AE (IWT) 14.03.2021

Ans. (a) : Turning is done when motion of job is rotary and motion of cutting tool is forward translating. In turning operation metal is removed from the outer diameter of the surface.

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1376. A twist drill is specified by its shank, material and _____

(a) Diameter (b) Lip angle
(c) Flute size (d) Length of body

GMB AAE 25.06.2017

Ans. (a) : A twist drill is specified by its shank, material and diameter.

1377. Highly accurate gears are manufactured by

(a) Gear milling (b) Gear shaping
(c) Gear rolling (d) Gear turning

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : Gear of accurate size can be made by gear hobbing or gear shaping process.

1378. The grinding wheels are produced by mixing the appropriate grain size of the abrasive with the required bond and pressed into the shape. The standard marking system of a specified grinding wheel is

51 A 36 L 5 V 23

For which characteristic of the grinding wheel the letter 'L' stands for

(a) type of bond
(b) type of grain
(c) type of grade
(d) manufacturer's reference

Punjab PSC (Lect.) 06.08.2017

Ans. (c)

(1)	(2)	(3)	(4)	(5)
51	A	36	L	5 V 23

A – Abrasive type
36 – Grain size
L → Grade / hardness
15 → Structure
V → Bond type

1379. ---------- is regarded as property by which the metal can be removed easily by grinding wheel with less wear and good surface finish.

(a) Sensitivity (b) Grindability
(c) Finishability (d) Grinding ratio

MPSC Lect. (Prod.) 2018

Ans. (b) : Grindability is regarded as property by which the metal can be removed easily by grinding wheels with less wear and good surface finish.

1380. Internal and external threads can be produced on tapered surfaces conveniently by

(a) Universal milling machine
(b) Plano miller
(c) Planetary milling machine
(d) Lathe

ISRO Scientist/Engineer 2010

Ans. (a) : Universal milling machine can produce internal and external threads on tapered surfaces. However lathe machine is the universal machine which helps in producing threads on flat surface conveniently.

1381. Two cutters are mounted on the arbor so that two faces are machined simultaneously in

(a) Gang milling (b) Straddle milling
(c) Pendulum milling (d) Profile milling

ISRO Scientist/Engineer 2010

Ans. (b) : Two cutters are mounted on the arbor so that two faces are machined simultaneously in straddle milling. In Gang milling, three or more cutters are mounted on arbor.

1382. Enlarging an existing circular hole with rotating single point tool is called

(a) Boring (b) Drilling
(c) Reaming (d) Internal turning

ISRO Scientist/Engineer 2008

Ans. (a) : Enlarging an existing circular hole with rotating single point tool is called boring.

1383. Which of the following processes is used when a small batch of gears are needed to be produced without much accuracy?

(a) Broaching (b) Gear forming
(c) Gear shaping (d) Gear hobbing
(e) Gear planning

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Gear forming process is used when a small batch of gears are needed to be produced without much accuracy.

1384. In following operations are to performed on a job, which would be performed first

(a) Drilling (b) Boring
(c) Counter sinking (d) Tapping

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : In following, operations, are to performed on a job,
(1) Drilling – make a specific diameter of hole.
(2) Boring – to enlarge of this existing hole.
(3) Tapping – used to five finishing of the internal surface of a hole.
(4) facing – Smoothing and squaring of the surface around a hole.

1385. In broaching operation the accuracy that can be obtained is of the order of

(a) 0.1 micron (b) 1 micron
(c) 10 micron (d) 100 micron

Nagaland PSC CTSE 2016 Paper-I

Ans. (b) : Broaching is a machining process it is used when precision machining is required especially for odd shapes. The accuracy that can be obtained is of the order of 0.5 μm –1 μm .

1386. Thread rolling is restricted to

(a) Ferrous Materials (b) Ductile Materials
(c) Hard Materials (d) None of these

SJVN ET 2013
Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (b) : Thread rolling is restricted to ductile materials.
Thread rolling is a cold forging process that can be performed on any ductile material.

1387. A drill considered as a cutting tool having zero rake, is known as a:

(a) Flat drill
(b) Straight fluted drill
(c) Parallel shank twist drill
(d) Tapered shank twist drill

SJVN ET 2013, VIZAG Steel MT 2011

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Ans. (b) : A drill considered as a cutting tool having zero rake is known as straight fluted drill.

1388. Which of the following is not the part of a shaper?
(a) Ram (b) Table
(c) Cross slide (d) Clapper box

(KPSC AE 2015)

Ans : (c) Cross slide is not the part of a shaper.
Main part of shaper machine:-
(i) Shaper head
(ii) Column
(iii) Base
(iv) Cross - rail
(v) Saddle and table.
(vi) Ram
(vii) Table

1389. Which of the following is not a surface finishing operation:
(a) Lapping (b) Annealing
(c) Polishing (d) Grinding

UJVNL AE 2016

Ans : (b) Annealing is not a surface finishing operation, Annealing is a heat treatment process.
Annealing:- It is one of the most important process of heat treatment of steel. Following are four types of annealing
(i) Full Annealing
(ii) Process Annealing
(iii) Spheroidise Annealing
(iv) Diffusion Annealing

1390. In which of the following turning processes tool does not travel parallel to the spindle axis?
(a) Thread cutting
(b) Internal boring
(c) Taper turning by offsetting tailstock
(d) Taper turning by using attachment to guide lathe spindle

TNPSC ACF 2012

Ans. (d) : In the process of taper turning by using attachment to guide. lathe spindle does not travel parallel to the spindle axis.

1391. Consider the following statements :
I. Silicate bonded abrasive wheels more efficient than vitrified bonded wheels.
II. Silicate bonded wheels being weaker allow the abrasive grains to fall off thus exposing fresh grains to work piece.
Of these
(a) Both I and II are correct and II is the reason for I
(b) Both I and II are correct, but II is not the reason for I
(c) I is correct, but II is not correct
(d) I is not correct, but II is correct

TNPSC ACF 2012

Ans. (a) : Silicate Board
This bonds are made by mixing abrasive particles with silicate and soda or water glass it is moulded to required shape allowed to dried up and taken out of mould. These wheel exhibits water proofing properly so those can be used with coolant.
Vitrified Bond
This bond consist of mixture of clay and water clay and abrasives are thoroughly mixed with water is moulded a uniform mixture. The mixture is moulded to shape of grinding wheel and dried up to take in out from mould. This wheel process a good strength and porosity to allow high stock removal with cual cutting.

1392. Lapping, honing etc are which of the following types of machining processes?
(a) High speed abrasive
(b) Low speed abrasive
(c) Medium speed abrasive
(d) Large material removal

TNPSC ACF 2012

Ans. (b) : Honing is a low velocity abrasive process lapping is a final abrasive finishing operation that produces extreme dimensional accuracy corrects minor imperfection of shape, refines surface finish and produce close fit between mating surface.

1393. The average cutting speed for turning mild steel with a high speed steel tool is
(a) 15 to 19 m/min (b) 25 to 31 m/min
(c) 60 to 90 m/min (d) 90 to 120 m/min

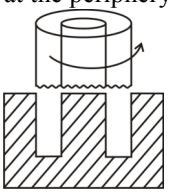
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : The average cutting speed for turning mild steel with a high speed steel tool is 25 to 31 m/min.

1394. Trepanning is an operation of
(a) making a cone shaped enlargement of the end of a hole
(b) smoothing and squaring the surface around a hole
(c) sizing and finishing a small diameter hole
(d) producing a hole by removing metal along the circumference of a hollow cutting tool

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Trepanning is a technique used for drilling larger hole diameters when machine power is limited as it is not as power consuming as conventional drilling. The trepanning tool does not machine the whole diameter only a ring at the periphery.



1395. In grinding irregular, curved, tapered, convex and concave surface, the grinder used is
(a) cylindrical grinder
(b) internal grinder
(c) surface grinder
(d) tool and cutter grinder

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : Surface grinder is used to grinding irregular, curved, tapered, convex and concave surface. It is used to produce smooth finish on flat surfaces.

1395. Glazing in grinding wheels takes place when the

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(a) wheel is too hard or wheel revolves at a very high speed
(b) wheel is too soft or wheel revolves at a very slow speed
(c) wheel is too hard or wheel revolves at a very slow speed
(d) wheel is too soft or wheel revolves at a very high speed

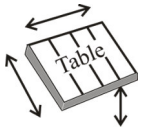
Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : When a cutting edge of a grinding wheels takes a glass-like appearance due to wear of abrasive grains then it is called Glazing of the grinding wheel. The Glazing in the Grinding wheels takes place when the wheel is to hard or its revolves at a very high speed.

1396. In which of the following milling machine, the table can be titled in a vertical plane by providing a swivel arrangement at the knee?
(a) universal milling machine
(b) plain milling machine
(c) omniversal milling machine
(d) hand milling machine

Nagaland PSC (CTSE) 2018, Paper-I

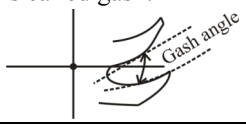
Ans. (a) : Universal milling machine in which the table can be tilted in a vertical plane by providing a swivel arrangement at the knee. They are based machine tools that use spinning cutting tools driven by milling spindles to remove materials.



1397. In a plain milling cutter the chip space between the back of one tooth and the face of the next tooth is called
(a) face (b) fillet
(c) gash (d) land

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : As we know, that in a plain milling cutter, the chip space between the back of one tooth and the face of the next tooth is called gash.



1398. Gear burnishing is a process for
(a) surface finishing
(b) under cut gears
(c) cycloidal gears
(d) removing residual stresses from teeth roots

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (a) : Gear brushing is done when gear surface bad surface, get distorted & any builted edge formed.
So it get brushed, to get a better surface finish.

1399. Which of the following statement is wrong as regard to down milling?
(a) it can not be used on old machine due to backlash between the feed screw of the table and the nut
(b) The chips are disposed off easily and do not interfers with the cutting
(c) the surface milled appears to be slightly wavy
(d) the coolant can be poured directly at the cutting zone where the cutting force is maximum.

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (c) : As we know in down milling the cutting tool is feed with the direction of rotation. It produced a flat surface not a way surface.

1400. The broaching operation in which the work moves past the stationary tool is called
(a) pull broaching
(b) push broaching
(c) surface broaching
(d) continuous broaching

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Continuous broaching operation in which the work moves past the stationary tool. The continuity of operation can be maintained by keeping the broach stationary & moving workpiece through the tool to perform cutting.

1401. The process of changing the shape of grinding wheel as it becomes worn due to breaking away of the abrasive and bond, is called:
(a) truing (b) dressing
(c) facing (d) clearing

HPPSC Asstt. Prof. 18.09.2017

Ans. (a) : The process of changing the shape of grinding wheel as it becomes worn due to breaking away of the abrasive and bond, is called truing.

1402. Which of the following processes cannot be applied for machining of internal gears?
(a) Hobbing (b) Shaping
(c) Milling (d) Broaching

BPSC AE Mains 2017 Paper - VI

Ans : (a) : Internal gears cannot be made by hobbing or rack cutter shaper and gang milling. They can be made by pinion cutter shapes. This is because the hob and a rack do not have enough space to move on internal surfaces for cutting.

1403. Typical range of surface roughness (Ra, μm) values 0.05 to 0.4 can be obtained by___ process
(a) Roller burnishing (b) Grinding
(c) Polishing (d) Lapping

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) : Lapping is a super finishing process which has 0.05 to 0.4 (Ra, μm) surface roughness achievable.

1404. An abrasive process that uses loose abrasive grains is known as_____
(a) Honing (b) Lapping
(c) Grinding (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

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Ans. (b) : Lapping uses loose abrasive grains for super finishing process.

1405. Which one of the following is the cutter with a curved tooth outline of the same shape as the profile of the workpiece?

(a) Plain milling cutter
(b) Face milling cutter
(c) End milling cutter
(d) Profile milling cutter

ESE 18.07.2021

Ans. (d) : Profile milling-It is a common milling operation. Round inserts and concepts with radius are milling cutters used for roughing and semi-roughing while ball nose end mills are milling cutters used for finishing and super-finishing.

1406. Some high speed steels have cobalt (Co) added to them in amounts ranging from 2% to 15%, since this element improves the

(a) cutting efficiency, especially at high temperature
(b) depth hardening ability of the HSS tool
(c) red hardness of the HSS tool
(d) grain structures of the HSS tool

RPSC IOF, 2020

Ans. (c) : Cobalt (Co) Increases strength and hardness and permits higher quenching temperature and increases the red hardness of high speed steel. It also intensifies the individual effects of other major elements in more complex steel.

1407. A milling cutter having 8 teeth is rotating at 125 rpm. If the feed per tooth is 0.075 mm, the speed (in mm per minute) is:

(a) 75 (b) 100
(c) 125 (d) 150
(e) 175

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : As we know,
$F = f_t ZN$
$= 0.075 \times 8 \times 125$
$= 75$ mm/min

1408. For cutting double start screw threads of pitch 0.75 mm on a lathe, the thread cutting tool should have a feed rate of:

(a) 0.375 mm/rev
(b) 0.75 mm/rev
(c) 1 mm/rev
(d) 1.2 mm/rev
(e) 1.5 mm/rev

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (e) : As we know,
Pitch = 0.75 mm
Start = Double start
Feed rate = 2×0.75
$= 1.5$ mm/rev.

1409. A twist drill is a :

(a) side cutting tool
(b) front cutting tool
(c) end cutting tool
(d) none of these

J&K PSC Civil Services Pre, 2013

Ans. (c) : As cutting in twist drill is made up of end cutting tool.

1410. The chip is cut off at thinnest place and then chip thickness increases along chip length in :

(a) up milling
(b) down milling
(c) end milling
(d) climb milling

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) : In up milling operation chip thickness varied from minimum to maximum i.e. chip is cut off at thinnest place and then chip thickness increased along the length of chip.

1411. In a drilling operation, the tool life was found to decrease from 20 min to 5 min due to increase in drill speed from 200 rpm to 400 rpm. What will be the tool life of that drill under same condition if the drill speed is 300 rpm

(a) 7.9 min (b) 8.9 min
(c) 6.4 min (d) 9.8 min

MPPSC AE 08.11.2015

Ans. (b) : Applying Taylor's tool-life equation-
where; T = Time in minutes
V = Velocity in rpm.

VT^n = Constant (K)

$\therefore V_1T_1^n = V_2T_2^n \Rightarrow 200\times(20)^n = 400\times(5)^n$

$\Rightarrow \left(\frac{20}{5}\right)^n = \frac{400}{200} \Rightarrow (4)^n = 2 \Rightarrow (2)^{2n} = (2)^1$

$\Rightarrow 2n = 1 \Rightarrow n = ½$

& $K = V_1T_1^n = V_2T_2^n$

$\Rightarrow K = 200(20)^{½} = 894.43$

$\because$ V = speed of the drill = 300 rpm

$\therefore$ T = Time in minutes = ?

$\therefore 300\times(T)^{½} = 894.43$

$T^{½} = 2.9814$

$\Rightarrow$ T = 8.888 minutes

$\Rightarrow T \approx 8.9$ minutes

1412. Margin wear in drill is due to

(a) abrasion (b) vibration
(c) thermal softening (d) diffusion

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Margin and crater wear is due to excessive heat and flank wear is due to abrasion.

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5. Metal Forming Processes

1413. In blanking process, clearance is provided:
(a) Both on punch and die
(b) Neither on die nor on punch
(c) On the die
(d) On the punch

GPSC DEE Class-2 (GWSSB) 04.07.2021
RPSC IOF, 2020, GATE 2002, CSE Pre 1995, 2005
Oil India Limited Sr. Engineer (Production) 30.11.2019
CIL MT 26.03.2017, GPSC Asstt. Prof. 28.08.2016
UPPSC AE 12.04.2016 Paper-I
UJVNL AE 2016, SJVN ET 2013
TRB Asstt. Prof., 2012, APPSC AEE 2012

Ans. (d) : Blanking is a shearing operation in which punch removes a portion of material from the stock which is a sheet metal strip of the necessary thickness and width. The removed portion is called a blank.
In blanking die size is same as the component size so clearance provided on the punch and shear on die.
In piercing, the punch size same as the actual hole size to be obtained so clearance provided on the die and shear provided on punch.

1414. Steel balls are manufactured by:
(a) Machining (b) Cold heading
(c) Casting (d) Spinning

Nagaland PSC CTSE 2017, Paper-I
GPSC Asstt. Prof. 28.08.2016
UJVNL AE 2016, RPSC AE 2016
TSPSC Managers, 2015
KPSC AE 2015, RPSC ACF-2011

Ans. (b) : Cold heading process produces ball blanks by forming the steel between two hemispherical cavities.

1415. In metals subjected to cold working, strain hardening effect is due to :
(a) slip mechanism
(b) twinning mechanism
(c) dislocation mechanism
(d) fracture mechanism

GPSC DEE Class-2 (GWSSB) 04.07.2021
CGPSC AE 15.01.2021
Mizoram PSC (Insp. of Factories) 2019 Paper-III
RPSC ADE 2016, GWSSB DEE 07.07.2016
TRB Asstt. Prof., 2012, ESE 1997

Ans. (c) : In metals subjected to cold working strain hardening effect is due to dislocation mechanism.
Dislocation Mechanism-
- Produces plastic deformation in crystals
- Proceeds by incremental, step by step braking & remarking of molecular bonds.
- Without dislocation motion, plastic deformation does not occur.

1416. In cold working of metals, the temperature of material is at :
(a) Room temperature
(b) Higher then the room temperature
(c) Lower than the room temperature
(d) Below recrystallisation temperature

JPSC AE 10.04.2021, Paper-II
HPPSC Poly. Lect. 05.07.2021
OPSC AEE 2019 Paper-I, CSE Pre 2005, GATE 2003
Mizoram PSC (Insp. of Factories) 2019 Paper-III
Nagaland PSC CTSE 2017, Paper-I,
GPSC Lect. 23.10.2016

Ans : (d) : Cold working- Working temperature for cold working process will be less than 0.6 times melting temperature. ($T_w < 0.6\ T_m$)
- Stress required for deformation is high.
- It produces better surface finish.
- Porosity is not eliminated.
- It produces better dimensional accuracy.

1417. Hot working of metal in forming process is___.
(a) Working below melting temperature
(b) Working below recrystallization temperature
(c) Working above recrystallization temperature
(d) Working at ambient temperature

Mizoram PSC (Insp. of Factories) 2019 Paper-III
TNPSC AE 2017,
Nagaland PSC CTSE 2017, Paper-I
GPSC Asstt. Prof. (Prod.) 21.08.2016
GWSSB DEE 07.07.2016, TRB Asstt. Prof., 2012

Ans. (c) : Hot working process is carried above recrystallization temperature with or without actual heating. Even brittle materials can be hot worked due to no ductility limit.

1418. Collapsible tooth paste tubes are manufactured by
(a) Direct extrusion (b) Piercing
(c) Impact extrusion (d) Indirect extrusion

GPSC DEE Class-2 (GWSSB) 04.07.2021
RPSC IOF, 2020
Gujarat PSC AE 2019, TNPSC AE 2019
OPSC AEE 2019 Paper-I, RPSC AE 2018
TSPSC AEE 2017, TSPSC AEE 2015
KPSC AE 2015, GPSC Poly. Lect. 07.12.2014
ESE 2008, 2003, GATE 2004, 1989

Ans. (c) : Impact extrusion is commonly used to make collapsible tubes such as toothpaste tubes, cans usually using soft materials such as aluminum, lead, tin usually a small shot of solid materials is placed in the die and is impacted by ram, which causes cold flow in the materials.

1419. In the rolling process, roll separating force can be decreased by
(a) reducing the roll diameter
(b) increasing the roll diameter
(c) providing back-up rolls
(d) increasing the friction between the rolls and the metal

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
APGENCO AE, 2017, CIL MT 26.03.2017
GWSSB DEE 07.07.2016
GATE 2010, ESE 2000

Ans. (a) : In rolling process, when a metal sheet passes through a roller, it touches or is in contact with roller, as we decrease the contact area by decreasing the diameter of roller, so, we need less roll separating forces.
Roll separating force,

$F = \sigma_0 L_P \times b$

L_P = Projected Length

$L_P = \sqrt{R\Delta h}$

$$\boxed{F = \sigma_0 \times \sqrt{R\Delta h} \times b}$$

$F \propto \sqrt{R}$

1420. To reduce the shearing force during blanking operation of large components on small capacity punch press, shear is provided on the_____.

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(a) Punch (b) Die
(c) Die and Punch (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016
GWSSB DEE 07.07.2016
BPSC Asstt. Prof. 29.11.2015
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Shear is provided on the die during blanking operation of large components for reducing maximum force. Providing shear reduces only max force not the total work done in shearing the component.

$F = \frac{F_{max.} \times pt}{S}$

p- Penetration of punch as a fraction
t- Thickness
S- Shear on the die

1421. Which of the following operation can produce a seamless tube ?
(a) Rolling (b) Blanking
(c) Piercing (d) Embossing

HPPSC AE 2018, TNPSC AE 2018
RPSC Lect. (Tech. Edu. Dept.) 2011
J&K PSC Civil Services Pre, 2010

Ans. (c) : Piercing – This production is employed for the production of seamless tube. It consists of passing hot rolled billet at 1100^0C between two conical rollers over a mandrel, one pass is applicable upto 15 m diameter. For large diameter second pass is necessary.

1422. The shear strength of a sheet metal is 300MPa. The blanking force required to produce a blank of 100mm diameter from a 1.5 mm thick sheet is close to
(a) 45 kN (b) 70 kN
(c) 141 kN (d) 3500 kN

GPSC Asstt. Director of Transport 05.03.2017
UP Jal Nigam AE 2016
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012

Ans. (c) : Given, $\tau_s = 300$ MPa $= 300 \times 10^6$ N/m^2
Blank diameter (d_b) = 100 mm = 0.1 m
Thickness of sheet (t_s) = 1.5 mm = 0.0015 m
For, blanking force [F] = $\tau_s \times A_s$

$A_s = \pi d_b t_s$

$F = 300 \times \frac{22}{7} \times 0.0015 \times 0.1 \times 10^6$

F = 141.3 kN

1423. A moving mandrel used in :
(a) Wire drawing (b) Tube drawing
(c) Metal cutting (d) Forging

OPSC AEE 2019 Paper-I,
Assam PSC CCE Pre 2015
HPPSC Asstt. Prof. 2014,
GATE 2014, 1994, ESE 1993

Ans : (b) : Moving mandrel is used in seamless tubes drawing operation.

1424. In drop forging, forging is done by dropping-
(a) the work piece at high velocity
(b) the hammer at high velocity
(c) the die with hammer at high velocity
(d) a weight on hammer to produce the requisite impact

HPPSC Poly. Lect. 05.07.2021, RPSC 2016
ISRO Scientist/Engg. 2010, ESE 1994

Ans : (c) In closed die forging die with hammer is dropped on to the component with high velocity, where as in open die forging only hammer is dropped on the work.

1425. Wrinkling is a common defect found in
(a) Bend components
(b) Deep drawn components
(c) Embossed components
(d) Blanked components

Gujarat PSC AE 2019
Kerala PSC AE (Irrigation) 05.04.2017
HPPSC Asstt. Prof. 18.11.2016, GATE 1991

Ans : (b) : Wrinkling is a common defect found in deep drawn components.

1426. Connecting rod is generally manufactured by:
(a) Casting (b) Drop forging
(c) Machining (d) Upset forging

OPSC AEE 2015 Paper-I
APPSC Poly. Lect. Automobile Engg. 2020
Haryana PSC Civil Services Pre, 2014

Ans : (b) Connecting rod is generally manufactured by Drop forging.
Drop forging : Repeated hammering and closed dies are used in drop forging to obtain required shape. The forging consists of two halves, lower half of die is fixed to the anvil of machine and upper half is in the ram the stock is placed in lower part and hammering is done to fill the cavity in die. To ensure complete filling extra material is provided and hence extra metal comes cut circumferentially to form flash which is trimmed. Two type of hammer is used (i) gravity drop hammer (ii) direct powered drop hammer.

1427. The cutting force in punching and blanking operations mainly depends on :
(a) The modulus of elasticity of material
(b) The shear strength of material
(c) The bulk modulus of material
(d) The yield strength of material

HPPSC Asstt. Prof. 18.11.2016
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2001

Ans. (b) : The cutting force in punching and blanking operations mainly depends on the shear strength of material.
Cutting force in punching and blanking operations–

$\boxed{F_{max} = P \times t \times \tau_u}$

P = Perimeter,
τ_u – Ultimate shear stress

1428. In deep drawing of sheets the limiting draw depends on
(a) percentage of elongation of sheet metal
(b) yield strength of sheet metal
(c) types of press used
(d) thickness of sheet metal

HPPSC Poly. Lect. 05.07.2021, CIL MT 27.02.2020
TRB Poly. Lect., 2012, GATE 1994

Ans. (d) : In deep drawing of sheets, the limiting draw ratio depends on the thickness of sheet metal.

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1429. Match List I with List II and select the correct answer using the codes given below the lists:

List I	List II
Metal forming process	**Type of stress**
A. Coining	**i. Tensile**
B. Wire drawing	**ii. Shear**
C. Blanking	**iii. Tensile and compressive**
D. Deep drawing	**iv. Compressive**

Code:

	A	B	C	D
(a)	iv	i	ii	iii
(b)	iv	i	iii	ii
(c)	i	ii	iv	iii
(d)	i	iii	ii	iv

RPSC 2016, APPSC Poly. Lect. 2013, GATE 2012

Ans. (a) :

Metal forming process	Type of stress
(1) Coining	Compressive
(2) Wire drawing	Tensile
(3) Blanking	Shear
(4) Deep drawing	Tensile and Compressive

1430. What is the process in which a flat shape is cut out from a sheet, the cut out portion being the desire part, called?

(a) Blanking (b) Piercing
(c) Notching (d) Perforating

HPPSC Poly. Lect. 05.07.2021
RPSC AEN Pre-2013

Ans. (a) : The process in which a flat shape is cut out from a sheet, the cut out portion being the desired part, called blanking.

1431. In order to get uniform thickness of the plate by rolling process, one provides:

(a) camber on the rolls
(b) offset on the rolls
(c) hardening of the rolls
(d) none of these

HPPSC Poly. Lect. 05.07.2021
HPPSC Asstt. Prof. 2014, ESE 1993

Ans. (a) : Camber on the rolls are provided is get uniform thickness of the plate by rolling process.

1432. A number of cold rolling passes are required in a two high rolling mill to reduce the thickness of a plate from 50mm to 25mm. The roll diameter is 700 mm and the coefficient of friction at roll work interface is 0.1. It is required that the draft in each pass must be the same. Assuming no front and back tension, the minimum number of passes required will be :

(a) 3 (b) 4
(c) 5 (d) 7

Vadodara Muncipal Corp. DEE, 2018
MPSC Poly. Lect. 2014

Ans. (d) : Given, $t_i = 50$ mm

$t_f = 25$ mm

$R = \frac{700}{2} = 350\text{mm}$

$\mu = 0.1$

Total draft $(\Delta h) = t_i - t_f$

$= 50 - 25$

$\Delta h = 25$ mm

max. possible draft in one pass $(\Delta R)_{max} = \mu^2 R$

$(\Delta R)_{max} = 0.1^2 \times 350$

$(\Delta R)_{max} = 3.5$

no. of pass required $= \frac{\Delta h}{(\Delta R)_{max}} = \frac{25}{3.5} = 7.14$

1433. A shell of 100 mm diameter and 100 mm height with the corner radius of 0.4 mm is to be produced by cup drawing. The required blank diameter is :

(a) 118 mm (b) 161 mm
(c) 224 mm (d) 312 mm

GPSC ARTO Pre 30.12.2018
AAI Jr. Executive 29.11.2018

Ans. (c) : Given,

$d = 100$ mm

$h = 100$ mm

$r = 0.4$ mm

Blank diameter, $D = \sqrt{d^2 + 4dh}$

$= \sqrt{(100)^2 + 4(100)(100)}$

$= 223.40 = 224\,\text{mm}$

1434. Cold working of metal increases

(a) tensile strength (b) yield strength
(c) hardness (d) All of the above

Assam PSC CCE Pre 2015
MPPSC State Forest Service Exam, 2014

Ans. (d) : Cold working – The working of metals below their recrystallization temperature is known as cold working.

Most of the cold working processes are performed at room temperature. The cold working distorts the grain structure and does not provide and appreciable reduction in size. It requires much higher pressures than hot working. The extent to which a metal can be cold worked depends upon its ductility. Cold working increases yield strength and hardness of steel but lowers its ductility. The increase in hardness due to cold working is called work hardening.

Cold working produce the following effects :

(i) The stresses are setup in the metal which remain in the metal, unless they are removed by subsequent heat treatment.
(ii) A distortion of the grain structure is created.
(iii) The strength and hardness of the metal are increased with a corresponding loss in ductility.
(iv) The recrystalline temperature for steel is increased.
(v) The surface finish is improved.
(vi) The close dimensional tolerance can be maintained.

1435. Dies for wire drawing are made of

(a) cast iron (b) wrought iron
(c) mild steel (d) carbides

GPSC Executive Engg. 23.12.2018
Assam PSC CCE Pre 2015

Ans. (d) : Dies for wire drawing are made of carbides.

1436. When steel undergoes a cold working process it becomes progressively

(a) softer (b) harder
(c) ductile (d) malleable

TRB Asstt. Prof., 2012
UPSC JWM Advt. No.-52/2010

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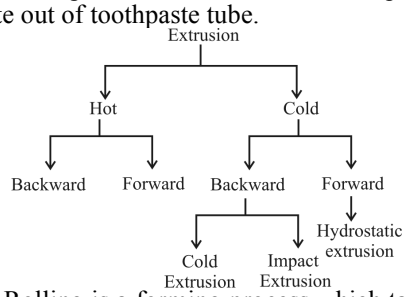
Ans. (b) : Work hardening is that which occurs in metal working process that intentionally induce plastic deformation to exact a shape change.
These process are known as cold working or cold forming process.
The cold working of the metal increase the hardness yield strength and tensile strength.

1437. Which of the following methods can be used for manufacturing 2 metre long seamless metallic tubes?
(a) drawing (b) extrusion
(c) rolling (d) extrusion and rolling

HPPSC Poly. Lect. 05.07.2021
Nagaland PSC (CTSE) 2018 Paper-I
Sikkim PSC (Under Secretary), 2017

Ans. (d) : Extrusion- Extrusion is the process of confining the metal is closed cavity and then allow it to flow only one opening so that metal takes shape of opening. The operation is identical to the squeezing of toothpaste out of toothpaste tube.



Rolling- Rolling is a forming process which takes place between cylindrical rolls where cross-section area of bar or plate is reduced with help of frictional forces regulated between power drive rolls.
The metal is thinned and gets elongated due to shear and compressive stress.
This process has advantage of high productivity and low cost rolling is generally hot working process.

1438. In drawing operation the metal flows mainly due to.
(a) Ductility (b) Work hardening
(c) Plasticity (d) Shearing

MPPSC State Forest Service Exam, 2014
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : In drawing operations the metal flows mainly due to its plasticity. Because in drawing the metal deformed permanently.

1439. A hole of diameter (d) is to be punched through a sheet metal of thickness (t). How much force is required to punch the hole if the ultimate shear stress of the sheet metal is (τ)?
(a) $dt\tau$ (b) $\frac{\pi}{4}d^2t\tau$
(c) $\pi dt\tau$ (d) $\frac{\pi}{4}d^2\tau$

CIL MT 26.03.2017
UPSC JWM Adv. No.-16/2009

Ans. (c) : Punching force $F = Lt\tau$
Where L is length of periphery
t = thickness
τ = ultimate shear strength
For circular section $L = \pi d$
$\therefore F = \pi dt\tau$

1440. Spring back during the sheet metal operation is caused because of the :
(a) release of stored energy during elastic and plastic deformation
(b) release of stored energy during plastic deformation
(c) release of stored energy during elastic deformation
(d) excess energy that was utilized during the forming process

UPPSC AE 13.12.2020 Paper-I
TRB Polytechnic Lecturer 2017

Ans. (c) : Spring back during the sheet metal operation is caused because of the release of stored energy during elastic deformation.

1441. Which type of gears can be manufactured by extrusion process?
(a) Bevel gears (b) Herringbone
(c) Worm gears (d) Spur gears

Nagaland PSC CTSE 2017, Paper-I
Nagaland PTSC CTSE 2017 Paper-I

Ans. (b) : Herringbone gears (which also called double helical gear) can be manufactured by extrusion process. In helical gear, teeth are straight, but inclined to the axis of rotation.

1442. "A process through which the cross-section of a metal piece is increased with a corresponding reduction its length" is known as :
(a) Punching (b) Bending
(c) Upsetting (d) Drawing out

SJVN ET 2019
TSPSC AEE 2015

Ans. (c) : Upsetting- "A process through which the cross-sectional of a metal piece is increased with a corresponding reduction in its length" is known as upsetting.

1443. Trimming is the process associated with
(a) Forging (b) Press work
(c) Polishing of metals (d) Electroplating

HPPSC Poly. Lect. 05.07.2021
(KPSC AE 2015)

Ans : (a) Trimming is the process associated with forging.
Trimming:- In operations such as drop forging and die casting, a small amount of extra metal gets spread out near the parting plane, which is termed flash. The flash is to be trimmed before the casting or forging is to be used.

1444. In rolling process, the state of stress of the material undergoing deformation is:-
(a) Pure compression
(b) Pure shear
(c) Compression and shear
(d) Tension and shear

GPSC Asstt. Director of Transport 05.03.2017
UKPSC AE-2013, Paper-I
GATE 2013

Ans. (c) : In rolling process, the state of stress of the material undergoing deformation is compression and shear.

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1445. Process in which cross-sectional area of bars, rods or tubes in the desired area is reduced by repeated blows is known as

(a) swaging (b) piercing
(c) cold peening (d) extrusion

HPPSC Poly. Lect. 05.07.2021
UKPSC AE 2007 Paper -I

Ans. (a) : Swaging is a forging process in which the dimension of an item are altered using dies into which the items is forced.
Swaging is usually a cold working process, but also may be hot worked the term swage may apply to the process or to a die or tool used in that process.

1446. The process of removing the burrs or flash from a forged component in drop forging is called -

(a) Swaging (b) Perforating
(c) Trimming (d) Fettling

RPSC AE 2018
Kerala PSC Poly. Lect. 09.01.2014, ESE 2005

Ans. (c) : Trimming—Trimming is the process of removing the burrs or flash from a forged component in drop forging.
Swaging—Swaging is a forging process in which the dimension of an item are altered using dies into which the item is forced.
Perforating—A perforation is a small hole in thin material. There is usually more than one perforation in organized fashion, where all of the holes collectively are called a perforation. The process of creating perforations is called perforating.

1447. Drawing out is a forging operation used to

(a) reduce the cross section of a part
(b) increase the cross section of a part
(c) finish the part
(d) flatten the part

Kerala PSC IOF 19.04.2016
Mizoram PSC AE/SDO 2014 Paper-III

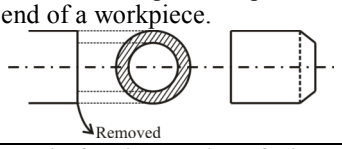
Ans. (a) : Drawing–This is the operation in which metal gets elongated with a reduction in the cross section area.

1448. The chamfering is an operation of

(a) bevelling the extreme end of a workpiece
(b) embossing a diamond shaped pattern on the surface of a workpiece
(c) reducing the diameter of a workpiece over a very narrow surface
(d) enlarging the end of a hole cylindrically

Nagaland PSC (CTSE) 2018, Paper-I
VIZAG Steel MT 2011

Ans. (a) : The chamfering is an operation of beveling the extreme end of a workpiece.



1449. In open-die forging a disc of diameter 200 mm and height 60 mm is compressed without any barreling effect. The final diameter of the disc is 400 mm. The true strain is

(a) 1.986 (b) 1.686
(c) 1.386 (d) 0.602

VIZAG STEEL MT 18.08.2013, 10.06.2012
GATE 2007

Ans. (c) : $\frac{\pi}{4}D_1^2h_1 = \frac{\pi}{4}D_2^2h_2$

$(200)^2 \times 60 = 400^2 \times h_2$

$h_2 = 15$ mm

Change in length = 60 – 15
= 45 mm

True strain = $\ln\left(\frac{h_1}{h_2}\right) = \ln\left(\frac{60}{15}\right)$

True strain = 1.386

1450. The extrusion process (es) used to make aluminium collapsible tube is/are :
1. Tube extrusion
2. Forward extrusion
3. Impact extrusion.
Select the correct answer using the codes given below :

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 3 only

RPSC ACF & FRO, 26.02.2021
HPPSC Poly. Lect. 05.07.2021

Ans. (d) : Impact extrusion is commonly used to make collapsible tubes such as toothpaste tubes, by using soft materials such as aluminium, lead, tin. Usually a small shot of solid material is placed in the die and is impacted by a ram which causes cold flow in the material.

1451. Hot rolling of mild steel is carried out at :

(a) Below recrystallization temp
(b) Between 100º to 150º C
(c) Above recrystallization temp
(d) At 100º C

OPSC AEE 2019 Paper-I,
UJVNL AE 2016, GATE 2002

Ans : (c) : Hot rolling/Hot working of metal is done when the metal is worked above the recrystallisation temperature of metal. Hot rolling refines grain structure. Hot rolling is often used to convert ingots into smaller sections known as billets. Hot rolling is used for producing plate, sheet and strip, but can also be used to produce rods, bars and various structural shapes by making successive passes through rollers which are grooved to produce the desired shape.

1452. Metal extrusion process is generally used for producing

(a) Uniform solid sections
(b) Uniform hollow sections
(c) Uniform solid and hollow sections
(d) Varying solid and hollow sections

GPSC DEE Class-2 (GWSSB) 04.07.2021, ESE 1994
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Metal extrusion is a metal forming manufacturing process in which cylindrical billet inside a closed cavity is forced to flow through a die of desired cross-section. These fixed cross-section profile extruded parts are called extrudates and pushed out using either a mechanical or hydraulic press.

1453. A 4 mm thick sheet is rolled with 300 mm diameter rolls to reduce thickness without any change in its width. The friction coefficient at the work-roll interface is 0.1. The minimum possible thickness of the sheet that can be produced in a single pass is:

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(a) 1 mm (b) 2.5 mm
(c) 1.5 mm (d) 3.7 mm

NPCIL ET, 2015, VIZAG Steel MT 10.06.2012, GATE 2006

Ans. (b) : Given, Initial thickness of sheet
$t_i = 4 \text{ mm} = 4 \times 10^{-3} \text{ m}$
Diameter of roll, (D) = 300 mm = 0.3m
Friction coefficient at work-roll interface, (μ) = 0.1
Let, t_f = final thickness of sheet after rolling for single pass without slipping minimum possible thickness

$\Delta h = t_i - t_f = \mu^2 R$
$4\times10^{-3} - t_f = (0.1)^2 \times 0.15$
$t_f = 4\times10^{-3} - (0.1)^2 \times 0.15$
$= 0.004 - 0.0015 = 0.0025$
$t_f = 2.5$ mm

1454. Which one of the following manufacturing processes requires the provision of 'gutters'?
(a) Closed die forging (b) Centrifugal casting
(c) Investment casting (d) Impact extrusion

CGPSC Asstt. Workshop Supt., 17.07.2016
NPCIL ET, 2015, ESE 1993, GATE 1994

Ans. (a) : In closed die forging process metal is kept in between a pair of dies and a gutter is provided in the lower die. In impression die forging process metal is kept in between a pair of dies and no gutter is provided in the lower die.

1455. In hot working process :
(a) grain structure of the metal is refined
(b) porosity of metal is largely eliminated
(c) mechanical properties such as ductility, toughness, elongation, etc, are improved
(d) all of the above

Assam PSC AE (PHED), 18.10.2020
Vadodara Muncipal Corp. DEE, 2018

Ans. (d) : In hot working process– Grain structure of metal is refined.
- Porosity of metal is eliminated
- Mechanical properties such as ductility, toughness elongation etc are improved.

1456. A tooth paste tube can be produced by:
(a) Solid forward extrusion
(b) Hollow forward extrusion
(c) Hollow backward extrusion
(d) Solid backward extrusion

GPSC Asstt. Prof. 28.08.2016
MPPSC State Forest Service Exam, 2014, ESE 2014

Ans. (c) : Backward impact extrusion is a method of forming hollow metal parts by striking a metal billet which is confined within a cylindrical or square die.
Example- toothpaste tube etc.

1457. Mismatch defect in forging takes place due to:
(a) Incorrectly aligned dies
(b) Weak striking force
(c) Improper heating and cooling of forging
(d) Wrong material composition

HPPSC ADF 05.03.2019

Ans. (a) : Mismatch defect in forging takes place due to incorrectly aligned dies.

1458. Plastic deformation of the metals below the recrystallisation temperature is known as:
(a) Joining process
(b) Cold working process
(c) Hot working process
(d) Warm working process

Oil India Ltd. Sr. Engg. (Production) 30.11.2019

Ans. (b) : Those processes, working or plastic deformation above the recrystallisation temperature, are termed as hot working processes where as those below are termed as cold-working process
Advantage of cold working
(i) Cold working increase the strength and hardness of the material due to strain hardening.
(ii) Better dimensional accuracy and cold tolerances can be achieve
(iii) No oxide formation

1459. Directional properties can be imparted to the metal in
(a) Cold working
(b) Hot working
(c) Warm working
(d) Both cold and hot working

TSPSC AEE 2017

Ans. (a) : The directional properties can be imported to the metal in cold working.

1460. For obtaining a cup of diameter 20 mm and height 20mm by drawing the size of the round blank should be approximately.
(a) 35 mm (b) 40 mm
(c) 45 mm (d) 50 mm

RPSC ACF & FRO, 26.02.2021

Ans. (c) : Given,
d = 20 mm
h = 20 mm

$\text{Blank dia (D)} = \sqrt{d^2 + 4dh}$
$= \sqrt{20^2 + 4\times20\times20}$
$\boxed{D = 44.721\text{mm}} = 45\text{mm}$

1461. A cup without flanges of 12 cm height and 3 cm diameter is to be made from a sheet metal of 2 mm thickness. The number of deductions necessary will be :
(a) one (b) two
(c) three (d) four

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Blank size $D = \sqrt{d^2 + 4dh}$
$= \sqrt{3^2 + 4\times(3)\times12}$
$D = 12.369\,\text{cm}$

Standard reductions
(i) 50%
(ii) 30%
(iii) 25%
(iv) 16%

Reductions
$D_1 = 12.369 \times 0.5 = 6.1845$
$D_2 = 6.1845 \times 0.7 = 4.329$
$D_3 = 4.329 \times 0.75 = 3.245$
$D_4 = 3.246 \times 0.84 = 2.727$

Hence to reach 2mm thickness we have four deductions required.

1462. The process of hot extrusion is used to produce.
(a) Curtain rods made of aluminium

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(b) Stainless steel tubes used in furniture.
(c) Steel pipes/or domestic water supply
(d) Large size pipes used in city water mains.

RPSC ACF & FRO, 26.02.2021

Ans. (a) : Hot extrusion process is used to produce curtain rods made of aluminium.
Lead, copper, aluminium, magnesium and alloys of these metals are commonly extruded. steels, stainless steels are difficult to extrude. Tubes used in furniture are produced by sheet metal operations.

1463. Reduction in flat rolling is defined as :
(a) The ratio of original work thickness to final work thickness.
(b) The ratio of final work thickness to original work thickness
(c) The ratio of draft to original work thickness
(d) Difference in original work thickness and final work thickness

RPSC ACF & FRO, 26.02.2021

Ans. (d) : Reduction in flat rolling is defined as difference in original work thickness and final work thickness.

Draft or Reduction $(\Delta h) = h_i - h_f$

1464. In cold working material gets stronger due to:
(a) Wear mechanism
(b) Slip system
(c) Twin mechanism
(d) Dislocation multiplication

TSPSC Managers, 2015

Ans. (d) : Cold working cause the crystal grains and inclusions to dislocation of grains the flow of the metal. Its cause of work hardening which makes the metal harder, stiffer and stronger, but less plastic and may cause cracks of the piece.

1465. Maximum reduction can be given to any material in tube drawing using
(a) Tube sinking
(b) Drawing with floating mandrel
(c) Drawing with fixed mandrel
(d) Drawing with moving mandrel

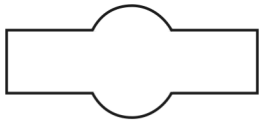
TSPSC Managers, 2015

Ans. (d) : maximum reduction can be given to any material in tube drawing using drawing with moving mandrel.

1466. The process in which during forging material moves towards centre.
(a) Edging (b) Fullering
(c) Hubbing (d) Open die forging

TSPSC Managers, 2015

Ans. (a) : Edging is the process of concentrating material using a concave shaped open die in which forging material moves towards centre.



edging

1467. In wire drawing operation the bright shining surface on wire is obtained if :
(a) Not using a lubricant
(b) Using thin film lubricant
(c) Using solid lubricant
(d) Good surface finish dies

TSPSC Managers, 2015, ESE 1996

Ans. (b) : In wire drawing operation the bright shining surface on wire is obtained if using thin film lubricant.

1468. Extrusion force does not depend upon :
(a) Extrusion ratio
(b) Type of extrusion process
(c) Material of die
(d) Working temperature

TSPSC Managers, 2015, CSE Pre 1997

Ans. (c) : Extrusion force depends upon extrusion ratio, types of extrusion process and working temperature.

$$\text{Extrusion stress } (\sigma_E) = \frac{P}{A_0}$$

$$= \sigma_0 \ln\left(\frac{A_0}{A_f}\right) = \sigma_0 \ln(r)$$

1469. Circular billet forging is a case of :
(a) Plane stress
(b) Plane strain
(c) Unidirectional stress
(d) Unidirectional strain

TSPSC Managers, 2015

Ans. (a) : Circular (rounded) billet forging is a case of plane stress.

1470. Work hardening exponent of any material can be calculated by calculating
(a) Maximum true stress
(b) Maximum true strain
(c) Maximum engineering stress
(d) Maximum engineering strain

TSPSC Managers, 2015

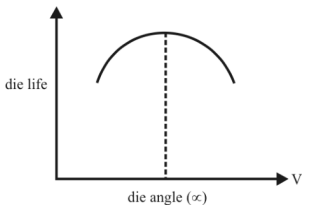
Ans. (b) : work hardening exponent of any material can be calculated by maximum true strain. True strain (ε_T), the rate of instantaneous increasing in the instantaneous gauge length.

$\varepsilon_T = \ln(1 + \varepsilon)$

1471. Increase in semi die angle in metal forming, the die life.
(a) Increases
(b) Decreases
(c) Initially increases then decreases
(d) Initially decreases then increases

TSPSC Managers, 2015

Ans. (c) :



If increasing in semi die angle in metal forming, the die life initially increasing and then decreasing. For large die angle, more turbulence in the metal flow is caused during reduction, increasing the ram force required.

1472. Dead metal zones form in extrusion process because.
(a) Higher extrusion pressure

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(b) Slip lines do not appear in certain regions
(c) Lack of smooth flow of material
(d) Improper lubrication

TSPSC Managers, 2015

Ans. (b) : Dead metal flow zone form in extrusion process because slip lines do not appear in certain regions. Dead metal zone, the corner of the material stays stationary. Bright finishes result due to material flowing past die angle.

1473. Maximum reduction can be given to any material in what kind of rolling mill :
(a) Planetary mill (b) Sendzmier mill
(c) 3 high mill (d) 4 high mill

TSPSC Managers, 2015

Ans. (a) : Maximum reduction can be given to any material in planetary mill. Planetary mills are ideally suited for fine grinding of hard, medium hard, soft, brittle, tough and moist materials. The communication of the material to be grouped takes place primarily through the high energy impact of grinding ball in rotating grinding bowls.

1474. The carbon content of forging dies is
(a) 0.1 to 0.2% (b) 0.55 to 0.65%
(c) 0.8 to 1% (d) 1.1 to 1.4%

APPSC Poly. Lect. 2013

Ans. (b) : Carbon content in drop dies forging is around 0.6 to 0.7.

1475. The force required to punch a circular blank of 60 mm diameter in a plate of 5 mm thickness is (The ultimate shear stress of the plate is 350 N/mm^2)
(a) 329.91 kN (b) 658.9 kN
(c) 128 kN (d) 497 kN

APPSC Poly. Lect. 2013

Ans. (a) : Given, d = 60 mm, t = 5 mm

$\tau_{ult.} = 350\ N/mm^2$

$$\tau_{max} = \frac{F}{A}$$

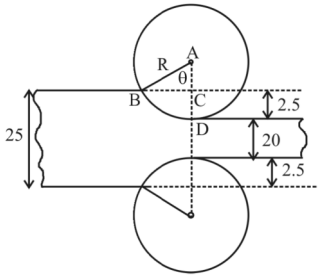
$$350 = \frac{F}{\pi.d.t}$$

Force(F) = 350 × π × 60 × 5 = 329.9 kN

1476. In rolling process, sheet of 25 mm thickness is rolled to 20 mm thickness. Roll is of diameter 600 mm and it rotates at 100 rpm. The rod strip contact length will be
(a) 5 mm (b) 39 mm
(c) 78 mm (d) 120 mm

RPSC ADE 2016, Gate 2004

Ans. (b) :



R = 300 mm
Δh = 25 -20 = 5

$$\cos\theta\ \frac{AC}{AB} = \frac{300-2.5}{300}$$

$\theta = 7.402^0$

contact length
= BD ≃ BC
Arc = radius × angle

$$= 300 \times 7.402 \times \frac{\pi}{180}$$

= 38.75 mm

Can also calculate BC and BD $= \sqrt{300^2 - 297.5^2}$
$= 38.64 \approx 39$ mm

Or
Projected length of arc of contact
$= \sqrt{R\Delta h} = \sqrt{300 \times 5} = 38.72 \approx 39$ mm

1477. Calculate the smallest punch diameter that can be designed for piercing sheet metal strip with the following data: [Crushing strength of the punch material is 1500 MPa. Thickness of the sheet is 2mm, factor of safety is 3, and shear strength of the sheet material is 500 MPa]
(a) 2 mm (b) 6 mm
(c) 8 mm (d) 24 mm

RPSC ADE 2016

Ans. (c) : Punching force = πdτ = π × d × 2 × 500

$$\text{Crushing strength} = \frac{1500}{3} \times \frac{\pi d^2}{4}$$

Crushing strength = Punching Force

$$\frac{1500}{3} \times \frac{\pi d^2}{4} = \pi d \times 1000$$

d = 8 mm

1478. Which of the following is a false statement about work hardening?
(a) It increases hardness.
(b) It increases yield strength
(c) It increases ductility
(d) It involves plastic deformations at temperature lower than the recrystallization temperature.

RPSC AEN Pre-2013

Ans. (c) : In cold working process, when the force is applied on the object, then due to resistance of the deformation, strength, hardness and ductility reduces of that object known as work hardening.

1479. Which of the following is a feature of forging process?
(a) Cost of tooling is less.
(b) Mechanical properties of parts are improved.
(c) Even intricate shapes with cavities can be forged.
(d) The product has very good surface finish.

RPSC AEN Pre-2013

Ans. (b) : Forging reduces the grain size of metal which increases the strength and toughness.
The forgings have high strength and ductility and offer great resistance to impact and fatigue loads due to extra working during the process and the opportunity of aligning the grain flow. Forging improves the structures of the metal and hence its mechanical properties.

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1480. Drop forging is used to make parts in
(a) open-impression dies that involves only upsetting operation
(b) closed-impression dies that involves only upsetting operation
(c) closed-impression dies that involves only drawing out operation
(d) open-impression dies that involves only drawing but operation

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : Drop forging, process of shaping metal and increasing its strength. In most forging, an upper die is forced against a heated workpeice positioned on a stationary lower die.
If upper die or hammer is dropped the process is know as drop forging.

1481. A disadvantage of cold working of metals is
(a) maximum deformation that can be given is limited
(b) dimensional accuracy obtained is less
(c) surface finish obtained is poor
(d) None of these

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : Cold working is the plastic deformation is carried out in a temperature region and over a time interval such that strain hardening is not relieved is called cold working. So, material in harder, calling for greater forces, become less ductile, Limiting of deformation etc.

1482. Mass production on metallic cans are usually done by
(a) embossing (b) coining
(c) spinning (d) drawing

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Drawing process is forming of a flat metal sheet into a hollow or concave shape like a cup, is formed by stretching a metal in punch and die.

1483. An important product manufactured by rolling is
(a) I-section (b) Tubes
(c) Meal Rolls (d) Rollers

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (a) : Some important product manufactured by rolling are as rods, rails (I-section), plates, sheets & strips, structural bars. Rolling is a process in which metal stock is passed through one or more pairs of rolls to reduce the thickness to make the thickness uniform of desired shape.

1484. A drawing that furnishes all the dimensions, limits and special finishing process _____ such as heat treatment, honing, lapping surface finish, etc._____ to guide the craftsman on the shop floor in producing the component is called :
(a) production drawing
(b) machine drawing
(c) assembly drawing
(d) design assembly drawing

APPSC Poly Lect. 13.03.2020

Ans. (a) : Production drawings (sometimes called as working drawing are complete sets of drawings that the manufacturing and assembly of products. They are widely used as orthographic views of machine parts and their assembly.
Elements of Production Drawings–
- Size and shape of component
- Format of drawing sheet
- Process sheet
- Projection method
- Limits, fits, and tolerances of size, form, and position
- Production method
- Indication of surface roughness and other heat treatments
- Material specification and shape such as castings, forgings, plates, rounds, etc.

1485. If 'L' is the length of contact between strip and roll, 'W' is the width of the strip, and σ_{avg} is the average true stress of the strip in the roll gap, the roll force can be determined as
(a) σ_{avg} L.W (b) σ_{avg} L/W
(c) σ_{avg} /L.W (d) $1/\sigma_{avg}$ L.W

TSPSC AEE 2017

Ans. (a) : Roll force = σ_{avg} L.W

1486. The name of the forging operation in which a heated metal bar is fed into a pair of cylindrical rolls that have the shape of the required component cut into their surfaces is
(a) Auto forging (b) Roll forging
(c) Swaging (d) Cylindrical forging

TSPSC AEE 2017

Ans. (b) : Roll forging operation in which a heated metal bar is fed into a pair of cylindrical Rolls that have the shape of the required component cut into their surface.

1487. By drawing operation, the range of reduction in cross section per pass is generally
(a) 5% to 10% (b) 5% to 20%
(c) 15% to 45% (d) 40% to 75%

TSPSC AEE 2017

Ans. (c) : By drawing operation the range of reduction increase section per pass is generally of 15% to 45%.

1488. Desirable properties of metal so that it can undergo metal forming processes are:
(a) Hardness and Toughness
(b) Ductility and creep
(c) Brittleness and hardness
(d) Ductility and yield strength

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Metal forming is the metal working process of fashioning metal parts and objects through mechanical deformation. For metal forming the desired properties are ductility and yield strength.

1489. Large size bolt heads are made by:
(a) Swaging (b) Roll forging
(c) Upset forging (d) Tumbling

GPSC Asstt. Prof. 28.08.2016

Ans. (c) : Upset Forging– This manufacturing process is used extensively in the production of fasteners, to form bolt heads, screw heads etc.

1490. Edging impression is an operation in drop-forging, and is done to :
(a) gather the exact amount of material required at each cross-section

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(b) decrease the material cross-section and increase the length
(c) acquire the shape very near to the final shape
(d) increase the material cross-section and decrease the length

CGPSC AE 15.01.2021

Ans. (a) : Edging impression is an operation in drop forging and is done to gather the exact amount of material required at each cross-section.

1491. Burr height _____ with increasing clearance and ductility of the sheet metal.
(a) increases
(b) decreases
(c) not influenced
(d) first increases and then decrease

VIZAG Steel MT 24.01.2021

Ans. (a) : Burr height increases with increasing clearance and ductility of the sheet metal. Burr is most commonly created by machining operations, such as grinding, drilling, milling, engraving or turning. It may be present in the form of fine wire on the edge of a freshly sharpened tool or as a raised portion of a surface.

1492. In closed die forging, the amount of material initially taken is very carefully controlled, so that :
(a) flash acts as cushioning for the dies
(b) flash is formed
(c) it is suitable for batch production
(d) no flash is formed

VIZAG Steel MT 24.01.2021

Ans. (d) : In closed die forging the amount of material initially taken is very carefully controlled, so that no flash is formed.

1493. Hot tube extrusion process is used in the manufacture of :
(a) Stainless steel tubes (b) Large size rods
(c) Aluminium rods (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : • The temperature range for Hot extrusion of aluminium is 430°-480°C.
• Used to produce curtain rods of aluminium.
• Design of die is a problem.
• Either direct or indirect method is used.
Note-Official answer given by commission is (a).

1494. Blanking and piercing operations can be performed simultaneously in:
(a) simple die (b) progressive die
(c) compound die (d) combination die

J & K PSC Screening, 2006

Ans. (c) : Blanking and piercing operation can be performed simultaneously in compound die.

1495. Flash and gutter are provided in drop forging dies at the following stages:
(a) Blocking (b) Performing (Edgering)
(c) Finishing (d) Fullering

NPCIL ET, 2015

Ans. (c) : Flash and gutter are provided in drop forging dies at the finishing stages. In closed die forging process metal is kept in between a pair of dies and a gutter is provided in the lower die.

1496. The maximum possible draft in cold rolling of sheet increases with the-
(a) Increase in coefficient of friction
(b) Decrease in coefficient of friction
(c) Decrease in roll radius
(d) Increase in roll velocity

NPCIL ET, 2015, GATE 2011

Ans. (a) : Maximum possible draft (Δh)
$\uparrow(\Delta h)_{max} = \uparrow\mu^2 R$
μ = Coefficient of friction between roll workpiece-interface
R = Radius of roll

1497. In rolled and forged product, limiting thickness of the plate is inversely proportional to _____.
(a) flow stress of workpiece
(b) coefficient of friction
(c) elastic modulus of rolls
(d) roll radius

VIZAG MT, 14.12.2020

Ans. (c) : In rolled and forged product, limiting thickness of plate in inversely proportional to elastic modulus of rolls.
$(\Delta h)_{max} = \mu^2 R$
$\Delta h \propto R$
$\Delta h \propto \mu^2$
If elastic modulus of rolls are high then limiting thickness decreases.

1498. In drawing, spinning and bending operations, _____.
(a) tensile stresses are involved
(b) shear stresses are involved
(c) compressive stresses are involved
(d) both tensile and compressive stresses are involved

VIZAG MT, 14.12.2020

Ans. (d) : In drawing, spinning and bending operations, both tensile and compressive stresses are involved.

1499. A 6 mm thick sheet is rolled with 200 mm diameter rolls to reduce thickness without any change in its width. The friction coefficient at work roll interface is 0.2. What is the minimum possible thickness of the sheet that can be produced in a single pass?
(a) 1 mm (b) 2 mm
(c) 3 mm (d) 4 mm
(e) 5 mm

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (b) : Given,
t_1 = initial thickness = 6 mm
t_2 = final thickness
As we know,

$$\mu = \sqrt{\frac{t_1 - t_2}{R}} = \sqrt{\frac{0.006 - t_2}{0.100}}$$

$$0.2 = \sqrt{\frac{0.006 - t_2}{0.100}}$$

t_2 = 2 mm

1500. Cold working produces which of the following effects?
(i) Stresses are set up in the material.
(ii) Grain structure get distorted.

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(iii) Strength and hardness of the metal are decreased.
(iv) Surface finish is reduced.
(a) Only (i) and (ii) (b) Only (i) and (iv)
(c) Only (ii) and (iii) (d) Only (iii) and (iv)
(e) Only (i), (ii) and (iii)

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (a) : Cold working produces, different defects such as, stresses are set up in the material, grain structure get distored, & surface have get distored.

1501. The force requirement in a blanking operation of low carbon steel sheet is 5.0 kN. The thickness of the sheet is 't' and diameter of the blanked part is 'd'. For the same work material, if the diameter of the blanked part is increased to 1.5 d and thickness is reduced to 0.4 t, the new blanking force in kN is :
(a) 3.0 (b) 4.5
(c) 5.0 (d) 8.0

Vadodara Muncipal Corp. DEE, 2018, GATE 2007

Ans. (a) : $F_1 = 5kN$ $F_2 = ?$
thickness = 0.4 t
dia of blank = 1.5 d

$$\tau_s = \frac{F}{\pi Dt}$$

$$\frac{F_1}{\pi D_1 t_1} = \frac{F_2}{\pi D_2 t_2}$$

$$\frac{5}{\pi \times d \times t} = \frac{F_2}{\pi \times 1.5d \times 0.4t}$$

$$\boxed{F_2 = 3kN}$$

1502. It is desired to make a product having T-Shaped cross-section from a rectangular aluminium block. Which one of the following processes is expected to provide the highest strength of the product?
(a) Metal Forming (b) Casting
(c) Welding (d) Machining

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) : In metal forming operation strength of products is high as compared to casting, welding and machining.

1503. The cold forming process in which a hardened tool is pressed against a workpiece (when there is relative motion between the tool and the workpiece) to produce a roughened surface with a regular pattern is
(a) Chamfering (b) Knurling
(c) Roll forming (d) Strip rolling

Assam Engg. College AP/Lect. 18.01.2021

Ans. (b) : Knurling is a manufacturing process, typically conduct on a lathe whereby a pattern of straight, angled or crossed lines is rolled into material.

1504. …….. is one of the forming process.
(a) punching (b) blanking
(c) rolling (d) none of the above

GPSC Lect. (Auto) 16.10.2016

Ans. (c) : Punching–It is forming process that uses a punch press to force a tool, called a punch, through the workpiece to create a hole via shearing. The punch often pases through the work into a die. A scrap slug from the hole is deposited into the die in the process.
Rolling–In metal working rolling is a metal forming process in which metal stock is passed through one or more pairs of rolls to reduce the thickness to make the thickness uniform a desired mechanical property.
Blanking and Piercing–Blanking and piercing are shearing process in which a punch and die are used to produce parts from coil or sheet stock, blanking product the outside feature of the component while piercing produce internal holes on shaper.

1505. Pipes for bicycle frames are made of
(a) cold rolled steel (b) hot rolled steel
(c) forged steel (d) cast iron

GMB AAE 25.06.2017

Ans. (a) : Pipes for bicycle frames are made of cold rolled steel.

1506. For punching a hole of 20mm diameter in a plate 20mm thick, if the shear stress of plate is 30kg/mm^2, the force required will be
(a) 10000 kg (b) 18850 kg
(c) 28850 kg (d) 37714 kg

WBPSC AE, 2017

Ans. (d) : Given, d = 20 mm, t = 20 mm,
$\tau = 30$ kg/mm^2
We know that,
$F = \pi dt\tau$
$= \pi \times 20 \times 20 \times 30 = 37687$
≈ 37714 kg

1507. The operation 'coining' is very similar to
(a) Upset forging (b) Swaging
(c) Open die forging (d) Flashless forging

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (d) : Coining is a cold working process is similar in the other respects to forging, which takes place at elevated temperature, it uses a great deal of force to plastically deform a workpiece, so that if conforms to a die. The coining process is preferred when there is high tonnage.

1508. Metallic toothpaste tubes can be manufactured by
(a) Impact extrusion (b) Tube spinning
(c) Deep drawing (d) Rolling

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Impact extrusion are used for manufacturing metallic toothpaste.

1509. Blanking is an expression used in
(a) sheet metal working (b) casting
(c) drawing (d) rolling

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (a) : Blanking is sheet metal operation in which punched out is our workpiece and rest is scrap. Shear failure takes place in the sheet during operation.

1510. Materials true stress-strain relationship is expressed by $\sigma = k\varepsilon^n$, where 'k' and 'n' are called _____ and____ respectively
(a) strength coefficient and yield strain
(b) yield strain and work hardening exponent
(c) strength coefficient and work hardening exponent.
(d) none of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

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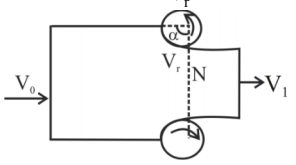
Ans. (c) : $\sigma = k\varepsilon^n$ where,
k = strength coefficient
n = work hardening exponent
ε = extrusion strength

1511. In sheet metal operations involving blanking and punching, the____ controls the blank size and the_____ controls the hole size.
(a) shear and punch (b) die and punch
(c) die and shear (d) none of the above
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : The hole size is controlled by die and punch in sheet metal operation involving blanking and punching respectively.

1512. During a two-high plate rolling operation, plate velocity at the entrance is V_0 and roller constant surface velocity is V_r. The plate velocity at exit is V_1. The backward slip is calculated by
(a) $\frac{V_1 - V_r}{V_r}$ (b) $\frac{V_1 - V_0}{V_r}$
(c) $\frac{V_r - V_o}{V_r}$ (d) $\frac{V_r - V_1}{V_r}$
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : Backward slip $= \frac{V_r - V_0}{V_r}$



Where
V_r = Roller velocity
V_o = Input velocity
V_1 = Output velocity

1513. Swaging is basically a
(a) Drawing Operation
(b) Forging Operation
(c) Piercing Operation
(d) Case hardening Operation
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Swaging is forging operation used for reducing or shaping the cross-section of rods of tubes by means of repeated impacts or blows.

1514. The mechanical properties normally get improved during the cold working process due to____
(a) Refinement of grains (b) Strain hardening
(c) Grain growth (d) None of the above
GPSC Asstt. Prof. (Prod.) 21.08.2016

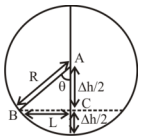
Ans. (b) : During cold working process the strength and hardness of the material is increased due to strain hardening no formation of oxide on the surface improves surface finish.

1515. The fracture produced by any metallurgical weakness along the centerline of the slab during rolling is called
(a) Zipper cracks (b) Wavy edge
(c) Edge cracks (d) Alligatoring
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) : Due to weakness at the center. Sheet bifurcates into two parts known as alligatoring or crocodile crack.

1516. A two high plate rolling, using rollers of radius 'R', produces a reduction in plate thickness. If this thickness reduction is 'Δh' what would be the projected length of the arc of contact (L)
(a) $L = \Delta h\sqrt{R}$ (b) $L = R\sqrt{\Delta h}$
(c) $L = \sqrt{R\Delta h}$ (d) $L = (R\Delta h)^2$
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) :



θ = Bite angle
In ΔABC
$R^2 = L^2 + (R - \Delta h/2)^2$
$R^2 = L^2 + R^2 + \frac{\Delta h^2}{4} - R\Delta h$
$L^2 = R\Delta h \quad \therefore \quad L = \sqrt{R\Delta h}$

1517. A blank holder is used in deep drawing to
(a) Guide the punch through the die
(b) Exactly locate the blank on the die
(c) Avoid wrinkles
(d) None of the above
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : Blank holder or drawn pad or binder is a thick tool steel ring, either flat or contoured, surrounds the punch and controls the inward flow of materials and avoid wrinkles formation.

1518. The maximum diameter of the hole that can be punched from a plate of maximum shear stress 1/4th of its maximum crushing stress of punch is equal to (where t = thickness of the plate)
(a) t (b) 2t
(c) 4t (d) 8t
Haryana PSC Civil Services Pre, 2014

Ans. (a) :
Maximum shear stress $(\tau) = \frac{P}{\pi Dt}$
Maximum crushing stress $(\sigma_c) = \frac{P}{\frac{\pi}{4}D^2}$
given $\tau = \frac{1}{4}\sigma_c$
$\frac{P}{\pi Dt} = \frac{1}{4}\frac{P}{\frac{\pi}{4}D^2}$
$\boxed{D = t}$
Option (a) is the correct answer but Haryana PSC gives (c) as a right answer.

1519. Operation of bending a partially cut hole on one side is called _____
(a) nibbling (b) slitting
(c) lancing (d) spiral
Haryana PSC Civil Services Pre, 2014

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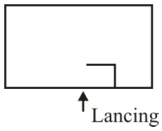
Ans. (c) : Bending a partial cut on a sheet is known as lancing operation.



1520. In a single pass rolling process using 410 mm diameter steel rollers, a strip of width 140 mm and thickness undergoes 10% reduction of thickness. The angle of bite in radians is

(a) 0.006 (b) 0.031
(c) 0.062 (d) 0.6

APGCL AM, 2021, GATE 2012

Ans. (c) :
Given–
$h_o = 8$ mm, $D = 410$ mm
$\Delta h = 10\%$ of $(h_o) = 0.8$ mm
we know that
$\Delta h = D(1-\cos\alpha)$
$0.8 = 410(1-\cos\alpha)$
$\alpha = 3.58$

$$= 3.58\alpha\frac{\pi}{180}\text{rad}$$
$$= 0.062 \text{ rad}$$

1521. Mechanical properties of the metal improve in hot working due to

(a) Recovery grains
(b) Recrystallization
(c) Grain growth
(d) Refinement of grain size

MPPSC State Forest Service Exam, 2014

Ans. (d) : Hot working is used for altering the form of iron & steel without fracture & use of excessive force the advantage of hot working is that it does not produce strain because it refine the grain size above its recrystallization temperature.

1522. Upsetting or cold heading is a

(a) Rolling process (b) Extruding process
(c) Bending process (d) Forging process

MPPSC State Forest Service Exam, 2014

Ans. (d) : Cold heading or Upsetting forging is a process by which stock is upset at its end in order to increase the cross section of the material in this area.
This metal forging process may be hot, but is often a cold working operation.

1523. Cold or hot rolling does not produce

(a) A hollow circular section
(b) A T-section
(c) An I-section
(d) A channel section

MPPSC State Forest Service Exam, 2014

Ans. (a) : Technically cold rolled applies only to sheets that undergo compression between rollers.
e.g T-section, I-section, Channel section etc.
Hollow design are not produced by this.

1524. The process of pulling a wire or bar through a die opening is called :

(a) Rolling process
(b) Open die forging process
(c) Drawing process
(d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Drawing – The process of pulling a wire or bar through a die opening is called drawing process.
Rolling- It is the metal forming process in which metal stock is passed through one or more pairs of rolls to reduce the thickness.
Open die forging- It is the process of deforming a piece of metal between multiple dies that do not completely enclose the material. The metal is altered as the dies hammer or stamp the material through a series of movements untill the desired shape is achieved.

1525. Periodic circumferential cracks on the surface is known as :

(a) Bamboo defect (b) Pipe defect
(c) Fir tree defect (d) Centre crack

JPSC AE 10.04.2021, Paper-II

Ans. (a) : Bamboo defects are periodic surface cracks that develop due to the extruded product stacking to the die land.

1526. Specific roll pressure 'P_{sp}' not depend upon

(a) The projected length of arc of contact 'L_P'
(b) Rolling load 'P'
(c) Draft 'Δh'
(d) No-slip angle 'β'

Karnataka PSC AE, 10.09.2017

Ans. (d) :

$$P_{sp} = \frac{P}{bL_p}$$
$$L_p = \sqrt{R\Delta h}$$

P_{sp} (specific roll pressure) is independent of no-slip angle β.

1527. The average reduction in deep drawing process is about

(a) 20% (b) 30%
(c) 50% (d) 80%

TRB Asstt. Prof., 2012

Ans. (c) : The average reduction in deep drawing process is about 50%

1528. Which one of the following is the main advantage of forging?

(a) good surface finish
(b) low tooling cost
(c) close tolerance
(d) improved physical property

TRB Asstt. rof., 2012, ESE 1996

Ans. (d) : Advantages of Forging-
$\rightarrow$ As in forging operation, refinement of grains takes place which improved the physical property of the material.

1529. Strain ageing is usually associated with the yield point is a phenomena in which :

(a) The ductility is increased on heating at a relatively low temperature after cold working
(b) The strength of the metal is increased and the ductility is decreased on heating at a relatively low temperature after cold working
(c) The strength of the metal and ductility will be reduced on heating at a relatively low temperature after cold working

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(d) None of the above

KPSC ADF 2015

Ans. (b) : Strain aging is behavior associated with the yield-point phenomenon in which the strength of the metal is increased and the ductility is decreased on heating at a relatively low temperature after cold-working.

1530. For most of the extrusion process the extrusion pressure is expressed as :

(a) $P = (k + A_o)\, A_f / A_o \ln$

(b) $P = k\, A_o / A_f A_o \ln$

(c) $P = k\, A_o / A_f \ln$

(d) $P = k\, A_o \ln A_o / A_f$

KPSC ADF 2015

Ans. (d) :

Extrusion pressure $(P) = kA_o \ln \left(\frac{A_o}{A_f}\right)$

k = extrusion constant

1532. The purpose of approach angle in drawing dies is :

(a) To provide better movement to the metal

(b) To avoid frictional effect on the tool

(c) Provision for the easy removal of the product

(d) The place where actual reduction in diameter occurs

KPSC ADF 2015

Ans. (b) : The purpose of approach angle in drawing dies is to avoid frictional effect on the tool.

1532. Which of the following statements are CORRECT about forward extrusion process

1. The ram and the extruded product travel in the same direction

2. The ram and extruded product travel in opposite direction

3. The speed of travel of the extruded product is same as that of the ram

4. The speed of travel of the extruded product is greater than that of the ram

(a) 1 and 3 (b) 2 and 3

(c) 1 and 4 (d) 2 and 4

APGENCO AE, 2017, ESE 2000

Ans. (c) : Direct extrusion, also known as forward extrusion, is the most common extrusion process.

It work by placing the billet in a heavy walled container. The billet is pressed through the die by ram or screw.

There is a reusable dummy block between the ram and the billet to keep them separated.

1533. Hot working of metals refers to which one of the following temperature regions relative to the melting point (Tm) of the given metal on an absolute temperature scale?

(a) Room temperature (b) 0.2 Tm

(c) 0.4 Tm (d) 0.6 Tm

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : The lower limit of the hot working temperature is determined by its recrystallization temperature. As a guideline, the lower limit of the hot working temperature of a material is 60% of its melting temperature.

1534. Which one of the following metal forming processes is not a high energy rate forming process?

(a) Electro-mechanical forming

(b) Roll-forming

(c) Explosive forming

(d) Electro-hydraulic forming

GPSC ARTO Pre 30.12.2018, ESE 2007

Ans. (b) : High Energy Rate Forming Processes are –

(i) Electro-mechanical forming

(ii) Explosive forming

(iii) Electro –hydraulic forming

(iv) Magnetic forming

Whereas Roll forming is not High Energy Rate forming process.

1535. Magnetic forming is an example of :

(a) Cold forming

(b) Hot forming

(c) High energy rate forming

(d) Roll forming

GPSC ARTO Pre 30.12.2018, ESE 2005

Ans. (c) : Magnetic forming is an example of High Energy Rate Forming.

1536. A metal disc of 20 mm diameter is to be punched from a sheet of 2 mm thickness. The punch and the die clearance is 3%. The required punch diameter is :

(a) 19.88 mm (b) 19.94 mm

(c) 20.06 mm (d) 20.12 mm

GPSC ARTO Pre 30.12.2018, GATE 2003

Ans. (a) : This is a blanking operation

Clearance (C) = 3% of thickness

$C = 0.03 \times 2 = 0.06$

Punch diameter = Die size – 2C

$= 20 - 2\,(0.06)$

$= 19.88$ mm

1537. Chemical embossing process is basically

(a) an etching process

(b) electroplating process

(c) galvanizing process

(d) engraving process

OPSC Civil Services Pre 2006

Ans. (d) : Embossing and debossing are the processes of creating either raised or recessed relief images and design in paper and other material. An embossed pattern is raised against the background, while the debossed pattern is sunken into the surface of the material.

1538. Which of the following operations involves straightening a curved sheet metal?

(a) Notching (b) Planishing

(c) Straightening (d) Deforming

KPCL AE 2016

Ans. (b) : Task of rectifying a bended sheet (curved sheet) known as plainshing. It is a metal working that includes completing the surface by at long test molding and smoothing sheet metal.

1539. In rolling of metal, one of the principal breakdown pass sequence is:

(a) Trapezoidal series

(b) Diamond - square series

(c) Blasting series

(d) Geometric series

HPPSC Asstt. Prof. 29.10.2016

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Ans. (b) : There are three types of breakdown passes required in proper sequence to produced desired shape nearer to the roll product-
(a) Box-pass series.
(2) Oval-square series.
(3) Diamond - square series.

1540. The internal ruptures caused due to improper cooling of the large forging is known as :
(a) Flakes (b) Stacks
(c) Burn outs (d) Cold shut
HPPSC Asstt. Prof. 29.10.2016

Ans. (a) : Flakes– Flakes are internal fissures seen in large forging. Hydrogen picked up during melting and casting segregates at internal voids and discontinuities and produces these defects during forging.

1541. Rotary swaging is the operation where the two dies which are free to move .....are held in a spindle which rotates continuously.
(a) eccentrically (b) vertically
(c) elliptically (d) radically
HPPSC Asstt. Prof. 29.10.2016

Ans. (d) : Rotary swing is the operating where the two dies which are free to move radically are held in a spindle which rotates continuously.

1542. .....stop is one type of stop used n sheet metal as die stop.
(a) Slab (b) Latch
(c) Hook (d) Barrel
HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : Latch stop is one type of stop used in sheet metal as die stop.

1543. It is desired to punch a hole of 20 mm diameter in a plate of 20 mm thick. If the shear stress of mild steel is 30 kg/mm^2, the force necessary for punching would be approximately in the range of :
(a) 15-20 tonnes (b) 20 – 25 tonnes
(c) 25 – 35 tonnes (d) 35 – 40 tonnes
MPSC Poly. Lect. 2014

Ans. (d) : Given, D = 20 mm, t = 20 mm,
$\tau = 30$ kg/mm^2
Punch force ($F_{p)}$) = $\pi Dt\tau$

$$= \pi \times 20 \times 20 \times 30 \left[\text{mm} \times \text{mm} \times \frac{\text{kg}}{\text{mm}^2}\right]$$

= 37699 kg
= 37.69 tonne

1544. 'Hardie's is a tool used in smithying and forging belongs to the tool category of.....
(a) hammers (b) tongs
(c) fullers (d) chisels
HPPSC AE (PWD) 10.09.2016

Ans. (d) : The term hardie is used to describe the tools used in the square hole in the anvil referred to as the hardie hole black smithying is a skill in which various articles are forged from wrought iron or steel. The black smith uses various tools to create the items that is made they are include the blacksmith hardies, chisels and punches.

1545. The sheet metal operation in which metal is removed in small increments is known as:
(a) Trimming (b) Shaving
(c) Nibbling (d) Notching
HPPSC AE (PWD) 10.09.2016

Ans. (c) : Nibbling is the removing the metal in small increment, when a specific counter is to be cut in sheet metal a small punch is used to punch repeatedly along the necessary contour generating the required profile.

1546. Which of the following characteristics is not true for spinning?
(a) It is comparable to drawing for making cylindrical shaped parts
(b) No thinning takes place anywhere on the blank
(c) It is suitable for large scale production
(d) Large parts are much more easily made by this process than drawing.
UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation- No thinning takes place anywhere on the blank is not true for spinning characteristics.

1547. Match List I with List II and select the correct answer using the code given below the Lists :

List I	List II
A Rolling	1. Solid and hollow sections
B. Extrusion	2. Shallow containers
C. Forging	3. Plate, sheet
D. Spinning	4. Simple shapes

Code :

	A	B	C	D
(a)	3	1	4	2
(b)	2	1	4	3
(c)	3	4	1	2
(d)	2	4	1	3

UPSC JWM Advt. No.-52/2010

Ans. (a) : Explanation –
(A) Rolling —(3)→ Plate, Sheet
(B) Extrusion —(1)→ Solid and hollow sections
(C) Forging —(4)→ Simple Shapes
(D) Spinning —(2)→ Shallow Containers.

1548. Which of the following materials is NOT commonly hot rolled?
(a) Aluminium (b) Titanium
(c) Magnesium (d) Copper
NLCIL GET 17.11.2020, Shift-II

Ans. (b) : Titanium alloys are resistant to corrosion, chemicals and heat and can withstand temperature of more than 600 degree.
Titanium is not commonly hot rolled.

1549. Which of the following statements is not a sheet metal working operation
(a) perforating (b) notching
(c) slitting (d) upsetting
TSGENCO AE 14.11.2015

Ans. (d) : Perforating, notching and slitting is a sheet metal working operation while upsetting is upset forging process. The diameter of material by compressing its length.

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1550. Which one is not the advantage of cold working?
(a) A high quality surface finish
(b) The mechanical properties may be varied
(c) Close dimensional tolerances
(d) A loss of ductility

HPPSC Asstt. Prof. 2014

Ans. (d) : In cold working strength increases because of work hardening in expires of a loss of ductility.

1551. The rolling of metal foils is usually carried out on a:
(a) two high rolling mill
(b) three high rolling mill
(c) four high rolling mill
(d) cluster high rolling mill

HPPSC Asstt. Prof. 2014

Ans. (d) : The rolling of a metal foils is usually carried out on a cluster high rolling mill.

1552. Crocodile crack occurs in which type of process?
(a) Welding (b) Forging
(c) Casting (d) Metal cutting
(e) None of these

CGPSC AE 16.10.2016

Ans. (e) : It is a rolling defect. Crocodile cracking is also called aligator cracking & may have erroneously fatigue cracks, is a common type of distress in asphalt.

1553. Assertion (A) : Electric arc furnace can be used for acid and basic method of steel making.
Reason (R) : Impurities are eliminated extensively in acid are process.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (b) : Electric arc furnace can be used for acids and basic method of steel making. In acid process, deoxidation can take place in furnace leaving a reasonable time for inclusion to rise into the slag & to be removed before casting but in basic process, deoxidation is rarely carried out. So, in electric furnace, impurities are eliminated by acidic process.

1554. Assertion (A) : In magnetic pulse-forming method, magnetic field produced by eddy currents is used to create force between coil and workpiece.
Reason (R) : It is necessary for the workpiece material to have magnetic properties.
Codes :
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true and R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

Ans. (c) : EM forming is a high velocity, cold forming process for electrically conductive metals most commonly used for copper & aluminium. It does not required magnetic properties.

1555. The strain hardening exponent n of stainless steel SS 304 with distinct yield and UTS values undergoing plastic deformation is
(a) $n < 0$ (b) $n = 0$
(c) $0 < n < 1$ (d) $n = 1$
(e) None of the above

CGPSC Poly. Lect. 22.05.2016, GATE 2015

Ans. (c) : Formula, $\sigma = k\varepsilon^n$,
where,
σ = represents the applied stress on the material
ε = is the strain, k is the strength coefficient.
The value of the strain hardening exponent lies between 0 and 1.
- If n = 0 means that a material is a perfectly plastic
- n = 1 means perfectly elastic solid for stainless steel SS 304 the value of n lies between 0.10 and 0.50.

1556. What will be the blanking pressure to produce a washer 5 cm outside diameter and 2.4 cm diameter hole from a material 4 mm thick having a shear strength of 360 N/mm^2?
(a) 220 kN (b) 390 kN
(c) 312 kN (d) 226 kN

MPSC Lect. (Prod.) 2018

Ans. (d) Given,
$D_1 = 5$ cm
$D_2 = 2.4$ cm $t = 4$ mm $\tau = 360$ N/mm^2
for blanking operation,
Blanking pressure (in kN) $= \pi D_1 \times t \times \tau$
$= \pi \times 0.05 \times 0.004 \times 360 \times 10^6$
$= 226$ kN

1557. Triple triangle finish with N1 as a roughness grade number has the roughness value Ra in μ m
(a) 0.0025 (b) 0.05
(c) 0.1 (d) None of these

MPSC Lect. (Prod.) 2018

Ans. (d)

Roughness Grade Number	Roughness Value Ra(μm)
N_1	0.025
N_2	0.05
N_3	0.1
N_4	0.2
N_5	0.4
N_6	0.8

So roughness grade number N_1 has roughness value R_a is 0.025 μm.

1558. Consider the following operations involved in forging a hexagonal bolt from a round bar stock whose diameter is equal to the bolt diameter
1. Swaging **2. Flattening**
3. Upsetting **4. Cambering**
The correct sequence of these operations is
(a) 2, 1, 3, 4 (b) 3, 2, 1, 4
(c) 2, 3, 4, 1 (d) 1, 2, 3, 4

UPRVUNL AE 21.08.2016, ESE 1999

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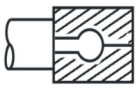
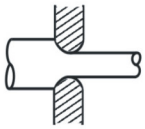
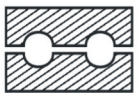
Ans. : (b) The correct sequence of the operation is
Upsetting
Flattening
Swaging
Cambering
Note– This Question is deleted by UPRVUNL

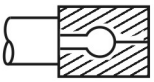
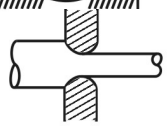
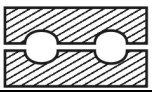
1559. Match List-I (Forging Operation) with List-II (View of the forging Operation) and select the correct answer using the codes given below the list:

List-I (Forging Operation)		List-II (View of the forging operation)
A. Drawing	1.	
B. Edging	2.	
C. Swaging	3.	
D. Fullering	4.	

(a) A-3, B-1, C-2, D-4 (b) A-4, B-3, C-2, D-1
(c) A-3, B-2, C-1, D-4 (d) A-4, B-1, C-2, D-3

UPRVUNL AE 21.08.2016
CSE Pre 2003

Ans. : (a)

View of the forging operation	Forging operation
	Edging
	Swaging
	Drawing
	Fullering

1560. Rolling, Forging etc, are examples of metal ____operations.
(a) Grinding (b) Cutting
(c) Forming (d) Drilling

UPRVUNL AE 2014

Ans : (c) Rolling, forging etc, are the metal forming operation.
Forming operations- These are those type of operation in which without changing in volume, we change the shape of the object, there is no chip formed in metal forming.

1561. During a single pass rolling process, the thickness of metallic sheet is reduced from 18 mm to 12 mm, Roll diameter is 500 mm. Angle of bite in degrees is:
(a) 5.24 (b) 4.79
(c) 8.83 (d) 6.68

HPPSC ADF 05.03.2019

Ans. (c) : We know,

$$\tan\theta = \sqrt{\frac{h_1 - h_2}{\left(\frac{d}{2}\right)}}$$

$$= \sqrt{\frac{18-12}{\left(\frac{500}{2}\right)}}$$

$$= \sqrt{\frac{6}{250}} = \sqrt{0.024}$$

$$\theta = \tan^{-1}(0.1549) = 8.83^\circ$$

1562. For extrusion process the metal
(a) Needs well lubrication
(b) Needs no lubrication
(c) Needs lubrication for steel products only
(d) Needs lubrication for non-ferrous metals only

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : Extrusion may be defined as process by which block of metal (placed in a container) is reduced in cross-section by forcing it to flow through a die under high pressure. For extrusion process the metal needs lubrication.

1563. Diamond tipped tools are not used for turning ferrous materials because
(a) Diamond has poor thermal conductivity
(b) Carbon has good solubility in iron
(c) Diamond has poor toughness
(d) All of above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Diamond tools (tipped tools) are super hard material, used for cutting glasses and other hard material. It is not use for iron, because of a chemical interaction between diamond and iron, and diamond have poor toughness.

1564. The amount of reduction in a single pass which can be achieved in wire drawing of a non-strain hardening material is limited. The limit depends on
(a) The yield strength of the material
(b) The die angle
(c) The coefficient of friction between the die and the material
(d) Ductility of the material measured by percentage reduction in area in a tension test

GPSC Asstt. Director of Transport 05.03.2017

Ans. (a) : In wire drawing, the process and minimum diameter of wire produced is depend upon the elasticity of the material. So, to achieved in wire drawing of a non-strain hardening material is limited by its yield strength of material, because till that point. Material behave as elastic material.

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1565. The state of stress during deep drawing forming operation of a cup is

(a) In the flange of blank, uni-axial compression and in wall of cup, bi-axial tension and compression

(b) In the flange of the blank, uni-axial tension and in wall of cup, bi-axial tension and compression

(c) In the flange of the blank, bi-axial tension and compression and in the wall of cup, simple uni-axial tension

(d) Both flange and wall of the cup will have bi-axial compression and tension

ISRO Scientist/Engineer 12.01.2020

Ans. (c) : Deep Drawing–When cup height is more than half the diameter is termed as deep drawing. During deep drawing, in the flange of blank, there is biaxial tension and compression stresses and in the wall of cup uniaxial tensile stress occurs.

1566. In a single pass rolling operation, the thickness of a 100 mm wide plate is reduced from 20 mm to 15 mm. The roller radius is 125 mm and rotational speed is 8 rpm. The average flow stress for the plate material is 400 MPa. The power required for rolling operation in kW is?

(a) 12.5 π/3 (b) 10 π

(c) 40 π/3 (d) 20 π/3

ISRO Scientist/Engineer 12.01.2020

Ans. (d) : Given, w = 100 mm, h_1 = 20 mm, h_2 = 15 mm

R = 125 mm, N = 8 rpm, $\bar{\sigma}_0$ = 400 MPa

Projected length, $L_p = \sqrt{R.\Delta h} = \sqrt{125 \times 5} = 25\text{mm}$

Force, $F_{avg} = \bar{\sigma}_0 \times L_p \times w$

$= 400 \times 25 \times 100 = 1000$ kN

Arm length, $x = 0.5\sqrt{\Delta h.R} = 0.5 L_p$

$= 0.5 \times 25 = 12.5$ mm

Torque = $F_{avg} \times x = 1000 \times 12.5 = 12500$ N-m

Power required for rolling operation by both roller

$$P = 2T\omega \Rightarrow P = 2 \times T \times \frac{2\pi N}{60}$$

$$= \frac{2 \times 12500 \times 2 \times \pi \times 8}{60} = \frac{20\pi}{3} \times 10^3 \text{ W}$$

$$P = \frac{20\pi}{3} \text{ kW}$$

1567. Tumbling is the process of-

(a) increasing fatigue limit

(b) imparting luster to surface

(c) cleaning the surface of small parts

(d) improving the creep limit

ISRO Scientist/Engineer (RAC) 29.11.2015

Ans : (c) Tumble finishing also known as tumbling or rumbling is a technique for smoothing and polishing a rough surface on relatively small parts.
In the field of metal working a similar process called barreling or barrel finishing.

1568. Coining is the operation of-

(a) cold forging (b) hot forging

(c) cold extrusion (d) piercing

ISRO Scientist/Engineer (RAC) 29.11.2015
SJVN ET 2013

Ans : (a) Coining is a form of precision stamping in which a workpiece is subjected to a sufficiently high stress to induce plastic flow of the material.

1569. The main objective of 'shot peening' is to improve which property of metal parts

(a) fatigue strength (b) ductility

(c) surface finish (d) none of these

ISRO Scientist/Engineer 22.04.2018

Ans. (a) : Shot peening is a cold working process used to finish metal parts to prevent failure by improving fatigue strength.

1570. In a metal sheet of 3 mm thickness, a hole of 7mm diameter needs to be punched. The yield strength in tension of the sheet material is 120 MPa and its ultimate shear strength is 100 MPa. The force required to punch the hole is

(a) 7.92 kN (b) 11.6 kN

(c) 6.6 kN (d) 13.9 kN

ISRO Scientist/Engineer 22.04.2018

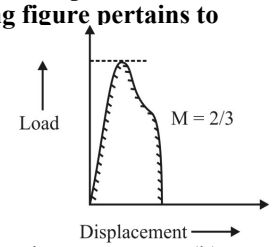
Ans. (c) : Given, t = 3mm, d = 7mm, τ = 100 MPa
σ_y = 120 MPa

Punching force, $F = \pi D t \times \tau_s$

$= 3.14 \times 7 \times 3 \times 100$

$\boxed{F = 6.594\ kN}$

1571. Load Vs Displacement curve shown in the following figure pertains to



(a) Extrusion (b) Bending

(c) Blanking (d) Coining

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : This given curve is for blanking operation as it is done by shearing.

1572. In metal forming, ring compression test is carried out to find out

(a) Compressive strength

(b) Interface friction

(c) Upset ratio

(d) None of these

ISRO Scientist/Engineer 07.05.2017

Ans. (b) :

- For investigating the influence of friction in hot metal forming and for the measurement of friction factor, a ring compression test is carried out.
- In this test, a flat ring is taken and as its height is reduced, its outside diameter increases.
- If there were no friction between the dies and workpiece, both the inner and outer diameters of the ring would expand.
- However, for large friction, the internal diameter of the ring is reduced with increasing deformation.
- In metal forming, ring compression test is carried out to find out interface friction.
- We can determine the magnitude of the friction coefficient μ, at the die (or) workpiece interface.

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1573. A drawing which shows the product with its components in their correct physical relationship is called

(a) Part drawing
(b) Layout drawing
(c) Assembly drawing
(d) Fabrication drawing

ISRO Scientist/Engineer 11.10.2015

Ans. (c) : Machine and mechanism consist of numerous parts and a drawing which shows the complete product with all its components in their correct physical relationship is known as an assembly drawing.

1574. Which of the following processes induce more stress in the metal?

(a) Hot rolling (b) Forging
(c) Swaging (d) Turning

ISRO Scientist/Engineer 12.05.2013

Ans. (c) : Forging

In a forging mechanical properties and reliability of the materials increases due to improve in crystal structure.
Forging reduces the grain size of metal, which increases strength and toughness.
Fatigue and creep strength increases.

Hot rolling

Hot rolling done above the recrystallization temperature.
Hot rolling is an effect way to reduce grain size in metal for Improved strength and ductility.

Swaging or Fullering

Reducing cross-section and making it longer i.e. more stress is obtained.

1575. A 5 mm diameter aluminium alloy test bar is subjected to a load of 500 N. If the diameter of the bar at this load is 4mm, the true strain is

(a) 0.56 (b) 0.22
(c) 0.25 (d) 0.45

ISRO Scientist/Engineer 12.05.2013

Ans. (d) : $\in_t = ln\left(\frac{A_o}{A_f}\right) = 2ln\left(\frac{d_o}{d_t}\right)$

$= 2ln\left(\frac{5}{4}\right) = 0.446$

1576. Stretch forming is a process in which

(a) All deformations occur in the direction of stretch
(b) All forces are applied in the direction of stretch
(c) Advantage is taken of plastic state indicated due to stretch
(d) No dies are used

ISRO Scientist/Engineer 2012

Ans. (c) : Stretch forming is a process in which advantage is taken of plastic state indicated due to stretch.

1577. The initial blank diameter required to form a cylindrical cup of outside diameter 'd' and total height'h' having a corner radius 'r' is obtained using the formula

(a) $D_o = \sqrt{d^2 + 4dh} - 0.5r$
(b) $D_o = d + 2h + 2r$
(c) $D_o = d^2 + 2h^2 + 2r$
(d) $D_o = \sqrt{d^2 + 4dh - 0.5r}$

ISRO Scientist/Engineer 2011

Ans. (a) Blank size,

$D = \sqrt{d^2 + 4dh}$ when $d > 20r$

$D = \sqrt{d^2 + 4dh} - 0.5r$ when $15r \le d \le 20r$

$D = \sqrt{(d-2r)^2 + 4d(h-r) + 2\pi r(d-0.7r)}$ when $d < 10r$

1578. The force required to punch a 25 mm hole in a mild steel plate 10 mm thick, when ultimate shear stress of the plate is 500 N/mm^2 will be nearly

(a) 78 kN (b) 393 kN
(c) 98 kN (d) 158 kN

ISRO Scientist/Engineer 2009

Ans. (b) : Given, Hole size = 25 mm
Thick = 10 mm
Ultimate shear stress = 500 N/mm^2
$F = (\pi dt) \times \tau$
$= 3.14 \times 25 \times 10 \times 500$
$= 392, 500$ N $= 392.5$ kN

1579. Ring rolling is used

(a) To decrease the thickness and increase diameter
(b) To increase the thickness of a ring
(c) For producing a seamless tube
(d) For producing large cylinder

ISRO Scientist/Engineer 2009

Ans. (a) : Ring Rolling is used to decrease the thickness and increase diameter.

1580. In the metal forming process, the stresses encountered are

(a) Greater than yield strength but less than ultimate strength
(b) Less than yield strength of the material
(c) Greater than the ultimate strength of the material
(d) Less than the elastic limit

ISRO Scientist/Engineer 2009

Ans. (a) : In the metal forming process, the stresses encountered are greater than yield strength but less than ultimate strength.

1581. Which one of the following is FALSE about Boronizing process?

(a) it is pack cementation process
(b) it can be done only on high carbon steels
(c) it is used to obtain extremely high wear resistant surfaces on steels
(d) it leads to high distortion

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Boronizing (Boriding) is a chemical vapour deposition (CVD) pack cementite process whereby boron atoms are diffused into steel and other substrates of iron, nickel and cobalt base alloys.
Boronizing provides maximum resistance to abrasion, erosion and wear with negligible growth to the original part. It increases component wear life. Ensuring a very hard, wear-resistance surface, our erosion protection system uses a boron gas diffusion coating or boronized process.

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1582. The maximum value of reduction in rolling process is

(a) $\frac{D}{2}\left[1-\frac{\cos\alpha}{2}\right]$ (b) $\frac{D}{2}[1-\cos\alpha]$

(c) $D(1-\sin\alpha)$ (d) $\frac{D}{2}[1-\sin\alpha]$

Where 'D' is roll diameter and 'α' is the angle of bite

ISRO Scientist/Engineer 17.12.2017

Ans. (*) : Total reduction or draft taken in rolling

$\Delta h = h_0 - h_f = 2(R - R\cos\alpha)$

$\Delta h = 2R(1 - \cos\alpha)$

$\Delta h = D(1 - \cos\alpha)$

Note: This question is excluded for evaluation by the ISRO.

1583. Which one of the following is the process to refine the grains of metal after it has been distorted by hammering or cold working?

(a) Annealing (b) Softening

(c) Re-crystallizing (d) Normalizing

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : Re-crystallization is a process by which deformed grains are replaced by a new set of defects free grains that Nucleate and grow until the original grains have been entirely consumed.

1584. The blank diameter used in thread rolling will be :

(a) equal to minor diameter of the thread

(b) equal to pitch diameter of the thread

(c) a little large than the minor diameter of the thread

(d) a little larger than the pitch diameter of the thread

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : The blank diameter used in thread rolling will be a little larger than the minor diameter of the thread.

1585. If the rate of cooling of a liquid metal is rapid, the temperature of freezing/crystallization will:

(a) decrease (b) increase

(c) remain constant (d) none of these

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (a) : If the rate of cooling of a liquid metal is rapid, the temperature of freezing/crystallization will decrease.

1586. Mannesmann process is a

(a) Casting process

(b) Rolling process

(c) Forging process

(d) Machining operation

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Roll piercing is also called Mannesmann process.

1587. Which of the following is not a significant problem in sheet forming?

(a) Bulging (b) Spring back

(c) Buckling (d) Wrinkling

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Buckling is not a significant problem in sheet forming.

1588. A piercing operation in which the entire contour is not cut is known as

(a) Burnishing (b) Broaching

(c) Notching (d) Lancing

APPSC AEE Screening Test 2016

Ans. (d) : Lancing– A special form of piercing in which the entire contour is not cut, the blanked material remaining as a tab.

1589. The process of raising a design or form above the surface of a component by means of pressing or squeezing action is called :

(a) Drawing (b) Embossing

(c) Spinning (d) Blanking

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : Embossing– It is the operation of forming surface details of figures letters or designs on sheet metal or on the surface of a component by means of pressing or squeezing action between two metal dies.

Coining- It is the process where impressions on both the faces can be different.

Drawing (Shallow & Deep)- It is the operation of sheet metal forming to make cup-shaped, box-shaped or other complex –curved, hollow –shaped parts.

Spinning- It is also an metal forming process in which an axially symmetric part is gradually shaped over a rotating mandrel or form block using a rounded tool or roller.

1590. A punch is used for making holes in steel plates with thickness 8 mm. If the punch diameter is 20 mm and force required for creating a hole is 110 kN, the average shear stress in the plate will be nearly

(a) 139 MPa (b) 219 MPa

(c) 336 MPa (d) 416 MPa

ESE 2020

Ans. (b) : Given, Thickness (t) = 8 mm

Diameter (d) = 20 mm

Force (F) = 110 kN

$F = \tau\pi dt$

$$\tau = \frac{F}{\pi dt} = \frac{110\times10^3}{\pi\times 20\times 8}$$

$= 218.83$ MPa

1591. Plug rolling is used to

(a) produce seamless tube

(b) reduce wall thickness and increase diameter of tubes

(c) produce collapsible tubes

(d) increase wall thickness of tubes

(KPSC AE 2015)

Ans : (b) Plug rolling is used to reduce wall thickness and increase diameter of tubes.

1592. It is required to punch a hole of diameter 10 mm on a sheet of thickness 3. The shear strength of the work material is 420 MPa. The required punch force is

(a) 19.87 kN (b) 39.56 kN

(c) 98.9 kN (d) 359.6 kN

TNPSC AE 2018, GATE 2006

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Ans. (b) : Data given-
d = 10 mm, t = 3 mm
τ = 420 MPa
We know that
Punching force F
$= \pi dt \times \tau_s = 3.14 \times 10 \times 3 \times 420$
= 39.564 kN

1593. Rotary swaging is the operation
(a) employed to expand a turbular or cylindrical
(b) part to reduce the cross-section area of rods and tubes
(c) in which the edges of sheet are turned over to provide stiffness and smooth edge
(d) causes a steadily applied pressure on work piece instead of impact force

TNPSC AE 2018

Ans. (b) : For reduce the cross-sectional area and increase length of the part, rotary swaging operation is used.

1594. In the metal forming processes, the stresses encountered are, less than the:
1. Yield strength of the material
2. Fracture strength of the material and greater than yield strength.

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

JWM 2017

Ans. (b) : In metal forming processes, the stresses produced are less than the fracture strength of the material and greater than yield strength.

1595. A 2 mm thick metal sheet is to be bent at an angle of 1.6 radian with the radius of 150 mm. Distance of neutral axis from inside surface is 0.5 × sheet thickness. Find out bend allowance.
(a) 321.6 mm (b) 241.6 mm
(c) 301.6 mm (d) 281.6 mm

UPRVUNL AE 07.10.2016

Ans. (b) : Bend allowance
$L_b = \alpha (R + kt)$
where
k = constant or stretch factor
$L_b = 1.6 (150 + 0.5 \times 2)$
$= 1.6 \times 151$
$L_b = 241.6$ mm

1596. Which of the following is correct for cold working?
(a) Poor dimensional accuracy
(b) Material loss is very high
(c) Poor surface finish is obtained
(d) It increases strength and hardness better than hot working

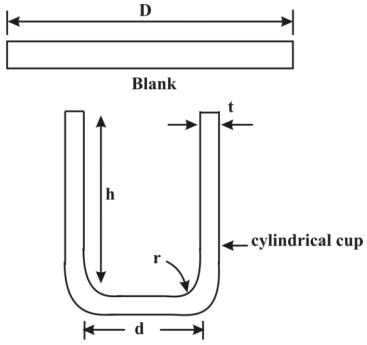
UPRVUNL AE 07.10.2016

Ans. (d) : In cold working, it increase strength and hardness better than hot working.

1597. A cylindrical cup of circular cross section of 50 mm diameter and 60 mm height with corner radius 2.5 mm is to be made of 0.5 mm thick sheet of steel. Its blank diameter (approximately) will be:
(a) 180 mm (b) 120 mm
(c) 140 mm (d) 160 mm

UPRVUNL AE 07.10.2016

Ans. (b) : Data given,
d = 50 mm
h = 60 mm
r = 2.5 mm
t = 0.5 mm
D = ?



$$D = \sqrt{d^2 + 4dh} - 0.5r$$
$$= \sqrt{(50)^2 + 4 \times 50 \times 60} - 0.5 \times 2.5$$
$$= \sqrt{2500 + 12000} - 1.25$$
$$= \sqrt{14500} - 1.25 = 119.16 \approx 120 \text{ mm}$$

1598. In the sheet metal operations shear stress is induced in
(a) Blanking (b) Punching
(c) Trimming (d) All of the above

HPPSC AE 2018

Ans. (d) : In the sheet metal operations shear stress is induced in Blanking, Punching and Trimming.

1599. The force required to punch a circular hole of 30 mm diameter in a plate of thickness 2mm, when ultimate shear stress for the plate material is 300 N/mm^2.
(a) $\pi(18 \times 10^2)$N (b) $\pi(18 \times 10^3)$N
(c) $\pi(18 \times 10^4)$N (d) $\pi(18 \times 10^5)$N
(e) $\pi(18 \times 10^6)$N

(CGPSC Polytechnic Lecturer 2017)

Ans. (b) : Data given,
Hole diameter (d) = 30 mm
Plate thickness (t) = 2 mm
Ultimate shear stress = 300 N/mm^2
Force required to punch a hole is given as
$F = \pi dt \times \tau = \pi \times 30 \times 2 \times 300$
$F = 18\pi \times 10^3$ N

1600. Sheet metal drawing operation is used to make:
(a) Wire (b) Cup shaped parts
(c) Tubes (d) Rods

TRB Polytechnic Lecturer 2017

Ans. (b) : Sheet metal drawing operation is used to make cup shaped parts.

1601. The angular clearance provided on dies is usually
(a) 1/2° to 1° (b) 3° to 5°
(c) 5° to 7° (d) 7° to 9°

Nagaland PTSC CTSE 2017 Paper-I

Ans. (a) : The angular clearance on dies varies from ½° to 1° normal value of angular clearance is 1.5° per sides.

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1602. Extrusion is the process of
(a) Producing holes by arc
(b) Marking cup shaped articles using the force of explosion
(c) Pushing the heated billet of metal through the orifice
(d) Metal sharping presses

Nagaland CTSE 2016, 2017 Paper-I

Ans. (c) : Extrusion is the process of pushing the heated billet of metal through the orifice, which is shaped to provide the desired form of the finished part.

1603. Better surface will be obtained in one of the following process
(a) Direct extrusion (b) Indirect extrusion
(c) Hydraulic extrusion (d) Impact extrusion

Nagaland PTSC CTSE 2017 Paper-I

Ans. (c) : Hydraulic extrusion a high pressure liquid medium is used for the transmission of the force to the billet/blank. The pressurized fluid act as a lubricant and because of this the extruded product has a good surface finish & dimensional accuracy.

1604. With increase in thickness of plate in rolling bite angle
(a) increases
(b) decreases
(c) remains constant
(d) no relation between bite angle and plate thickness

Nagaland PTSC CTSE 2017 Paper-I

Ans. (a) :

1605. The process used to improve fatigue resistance of metal by setting up compressive stresses in its surface is
(a) Hot piercing (b) Extrusion
(c) Cold peening (d) Cold heading

SJVN ET 2013

Ans. (c) : Cold peening- The process to improve fatigue resistance of metal by setting up compressive stress in its surface is known as cold peening.

1606. Metals like lead and tin are hot worked at temperature around
(a) 500 - 600 ^{o}C (b) 200 - 30 ^{o}C
(c) 100 ^{o}C (d) Room Temperature

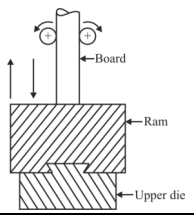
SJVN ET 2013

Ans. (d) : Same metal such as lead and tin, have a low recrystallization temperature and can be hot worked even at room temperature, but most commercial metals require some heating.

1607. In hammer forging, the hammer does not consist of:
(a) Falling weight
(b) Anvil
(c) Die
(d) Lifting mechanism for the ram

TRB Polytechnic Lecturer 2017

Ans. (b) : The hammer does not consist of anvil in hammer forging.



1608. Valve bodies of internal combustion engine are made by–
(a) Forging (b) Rolling
(c) Casting (d) Turning from bar stock

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Valve bodies of internal combustion engine are made by Rolling process.

1609. In forged components,
(a) fiber lines are arranged in a pre-determined way
(b) fiber lines of rolled stock are broken
(c) there are no fiber lines
(d) fiber lines are scattered

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : In forged components, fiber lines are arranged in a predetermined way.

1610. Cold working-
(a) increases toughness and ductility
(b) reduces residual stresses
(c) increases hardness and strength
(d) produces favorable pattern of fiber lines

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : Cold working increases hardness and strength.

1611. Hot working–
(a) increases toughness and ductility
(b) increases surface finish
(c) increases hardness and strength
(d) produces accurate dimensions for the parts

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Hot working increases toughness and ductility.

1612. A prismatic bar with rectangular cross-section of 20 mm × 40 mm and length of 2.8 m is subjected to an axial tensile force of 70 kN. The elongation of bar is 1.2 mm. What is the tensile stress and strain of the bar respectively?
(a) 8.75 MPa, 0.000428
(b) 87.5 MPa, 0.000428
(c) 8.75 MPa, 0.00428
(d) 87.5 MPa, 0.00428

SJVN ET 2019

Ans. (b) :

$$\text{Tensile stress} = \frac{\text{Force}}{\text{Area}}$$

$$= \frac{70\times10^{3}}{20\times40\times10^{-6}}$$

$$= 87.5\times10^{6}\ \text{N}$$

$$= 87.5\ \text{MPa}$$

$$\text{Tensile strain } \in = \frac{\delta l}{l} = \frac{1.2}{2.8\times1000}$$

$$= 0.000428$$

1613. Consider the following operations
1. Coining and sizing 2. Forging
3. Impregnation 4. Sintering
Out of the above which are the secondary and finishing operations in powder metallurgy?
(a) 2 and 3 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1 and 2

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(e) 1, 2 and 3

CGPSC AE 26.04.2015 Shift-I

Ans. (e) : A number of secondary and finishing operations can be applied after sintering, some of them are:

(i) Sizing : Cold pressing to improve dimensional accuracy
(ii) Coining : Cold pressing to press details into surface.
(iii) Impregnation : Oils fills the pores of the part.
(iv) Infiltration : Pores are filled with a molten metal.
(v) Heat treating, planting, painting.

1614. Statement (I) : In ordinary drop-forging with flash, die cavity is properly filled with metal.
Statement (II) : Forging load increases by increasing flash thickness.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

JWM 2017

Ans. (c) : In drop forging, forging load is greatly influenced by the flash thickness and width. The load can be decreased by increasing the flash thickness.

1615. What is the force required for 90° bending of St50 steel of 2 mm thickness in a V-die, if the die opening is taken as 8 times the thickness and the length of the bent part is 1 m, ultimate tensile strength is 500 MPa and K = 1.33?

(a) 166.25 kN (b) 155.45 kN
(c) 154.65 kN (d) 143.85 kN

ESE 2019

Ans. (a) : Given,

$t = 2$ mm
$w = 8t$
$L = 1$ m $= 1000$ mm
$\sigma_u = 500$ MPa
$k = 1.33$

$$\text{Force required for bending} = \frac{kLt^2\sigma_u}{w}$$

$$= \frac{1.33 \times 1000 \times 2^2 \times 500}{8 \times 2} = 166.25 \text{ kN}$$

1616. The maximum possible draft in rolling, which is the difference between initial and final thicknesses of the sheet metal, depends on

(a) rolling force
(b) roll radius
(c) roll width
(d) yield shear stress of the material

ESE 2018

Ans. (b) : Maximum possible reduction in rolling

$(\Delta H)_{max} = \mu^2 R$
$(\Delta H)_{max} \propto R$

1617. Metal Forming process used for making shallow and also for straightening parts is

(a) Peen forming
(b) Magnetic Pulse forming
(c) Stamping
(d) Stretch forming

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : Peen forming– It is also called as shot peening. Peen forming is a die less process performed at room temperature. During the process, the surface of the workpiece is impacted by pressure from small, found steel shot. The impact pressure of the peening shot causes local plastic deformation that manifested itself as a residual compressive stress.

1618. Which of the following process belongs to forging operation

(a) Fullering (b) Welding
(c) Drawing (d) Piercing

TSPSC AEE 2015

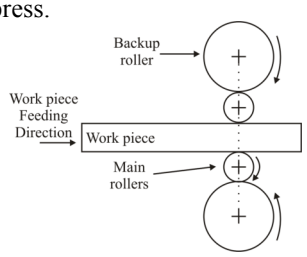
Ans. (a) Fulluring more generally refers to any forging process creating a sharp transition in cross-dimensional area with care, some types of fullring can be achieved using only hammer and the edge of the anvil.

1619. Deep drawing process can be performed on

(a) single action press (b) double action press
(c) triple action press (d) screw press

TSPSC AEE 2015

Ans. (c) : Deep drawing process can be performed on triple action press.



For high rolling mill

1620. The maximum reduction that can be given to any material in wire drawing (in most idealistic condition is)

(a) 53% (b) 63%
(c) 73% (d) 83%

TSPSC AEE 2015, ESE 2014

Ans. (b) : Ideal deformation of a perfectly plastic material.

$$\sigma_d = y l n\left(\frac{A_o}{A_f}\right)$$

at yield limit

$\sigma_d = y$

then,

$$1 = \ln\left(\frac{A_o}{A_f}\right)$$

$$\frac{A_o}{A_f} = e$$

Maximum reduction per pass-

$$\frac{A_o - A_f}{A_o} = 1 - \frac{1}{e}$$

$$= 1 - \frac{1}{2.718} = 0.632$$

$= 63.2\%$

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1621. A solid cylinder of diameter 100 mm and height 50 mm is forged between two frictionless flat dies to a height of 25 mm. The percentage change in diameter is

(a) 0 (b) 2.07
(c) 20.7 (d) 41.4

TNPSC AE 2019, GATE 2012

Ans. (d) : $[\text{Volume of cylinder}]_{\text{initial}} = [\text{Volume of cylinder}]_{\text{final}}$

$$\frac{\pi}{4}D^2H = \frac{\pi}{4}d^2h$$

$$(100)^2 \times 50 = d^2 \times 25$$

$$d = 141.4$$

$$\% \text{ change in diameter} = \frac{d-D}{D} \times 100$$

$$= \frac{141.4-100}{100} \times 100 = 41.4\%$$

1622. Structural sections such as rails, angles, I-beams are made by :

(a) Hot rolling (b) Hot drawing
(c) Hot piercing (d) Hot extrusion

OPSC Civil Services Pre. 2011

Ans. (a) : Structural sections such as rails, angles I-beams are made by Hot rolling.

1623. Coining resembles

(a) upset forging (b) flashless forging
(c) open-die forging (d) swaging

BPSC AE 2012 Paper - VI

Ans : (b) : Flashless forging- It is true closed die forging in which metal deformed in a die cavity permits virtually no excess metal to escape. During this process the perform is totally enclosed in the die cavity so that no flash is formed. In actual practice, a thin fin for ring of flash may form in the clearance between the upper punch and die.

1624. If coefficient of friction μ in a rolling process is 0.5 and radius of roller is 1,000 mm, what is the maximum reduction or draft possible?

(a) 500 mm (b) 250 mm
(c) 750 mm (d) 1,000 mm

OPSC AEE 2019 Paper-I

Ans : (b) : Maximum reduction or draft possible
$h = \mu^2 R$
$h = 0.5^2 \times 1000 = 250$ mm

1625. Forging hammers are called :

(a) Stroke restricted machines
(b) Energy restricted machines
(c) Load restricted machines
(d) Stress restricted machines

OPSC AEE 2019 Paper-I

Ans : (b) : Forging hammers are called energy restricted machines.

1626. In punching operation clearance is provided in:

(a) Punch (b) Die
(c) Blank holder (d) Blank

OPSC AEE 2019 Paper-I

Ans : (b) :
Punch size = blank size
Die size = blank size + 2 clearance

1627. Statement (I) : A cold bend part has no spring back because deformation is plastic.
Statement (II) : In cold bending, parts are normally over-bent slightly.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I).
(c) Statement (I) is true but Statement (II) is false.
(d) Statement (I) is false but Statement (II) is true.

JWM 2017

Ans. (d) : In cold bending, parts are normally over-bent slightly.

1628. In rolling operation, wavy edges appears due to the property of work material

(a) Anisotropy (b) Ductility
(c) Toughness (d) Malleability

TSPSC AEE 2015

Ans. (a) : Formation of wavy edge on top of the cup, called earing, happens due to anisotropy of the material.

1629. Identify the wrong statement related to yield criteria in metal forming

(a) stresses along hydrostatic component do not cause any plastic deformation
(b) Deviation component causes deformation
(c) Plastic deformation appears only when resultant stresses reach non-miser cylinder
(d) Tresca is plane strain case

TSPSC AEE 2015

Ans. (d) : Tresca is plane strain case is wrong statement.

1630. In the wall region of drawn cup in a deep drawing operation, the type of stresses are

(a) Uniaxial stretching
(b) tension-compression
(c) pure compression
(d) tension, compression & shear

TSPSC AEE 2015

Ans. (a) : In the wall region of drawn cup in a deep drawing operation, the type of stresses are uniaxial stretching.

1631. Thickness in the drawn cup by deep drawing process keeps on increasing in the wall region because of

(a) compressive normal stress
(b) compressive hoop stress
(c) shear stress in flange region
(d) biaxial pure compression

TSPSC AEE 2015

Ans. (b) : Thickness in the drawn cup by deep drawing process keeps on increasing in the wall region because of compressive hoop stress.

1632. Maximum strength of the product can be achieved when it is produced by

(a) open die cold working
(b) close die worm working
(c) impression die hot working
(d) open die hot working

TSPSC AEE 2015

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Ans. (a) : An open die cold working maximum strength of the product can be achieve.

1633. To prolong the life of shaper toots, after they are ground they should be undergone through the following operation

(a) Sand blasting (b) Shot peening
(c) Lapping (d) Hardening

UPPSC AE 12.04.2016 Paper-I

Ans : (d) Hardening

1634. A 30 ton press means

(a) gross weight of the press is 30 tons
(b) weight of die is 30 tons
(c) pressure exerted by slide is 30 tons
(d) flywheel of the press weighs 30 tons

RPSC AE 2016
WBPSC AE 2017

Ans : (c) A 30 ton press means pressure exerted by slide is 30 tons.

Power presses:- Power presses are used for producing large quantities of articles quickly, accurately and economically from the cold working of mild steel and other ductile materials.

1635. Bending operation should be performed

(a) At 30° to the grain direction
(b) Parallel to the grain direction
(c) There is no such criteria
(d) At right angle to the grain direction

(KPSC AE 2015)

Ans : (d) Bending operation should be performed at right angle to the grain direction

Bending operation:- Bending is the cold working process involving plastic deformation in which the total surface area remains constant. Bending of pipe, tube, rolled or extruded shapes involves practices and theories of the simple supports beam, the cantilever beam and many of the problem of forming and deep drawing

1636. Thread rolling is somewhat like :

(a) cold extrusion (b) cold machining
(c) cold rolling (d) cold forging

(HPPSC AE 2014)

Ans : (c) Cold rolling:- It is generally employed for bars of all shapes, rods, sheets and strips, in order to provide a smooth and bright surface finish. It is also used to finish the hot rolled components to close tolerance and improve their toughness and hardness.

1637. Spanners are manufactured by :

(a) dry casting (b) forging
(c) sheet rolling (d) cup blanking

(HPPSC LECT. 2016)

Ans : (b) Spanners are manufactured by forging.

Forging:- Forging can be defined as the controlled plastic deformation of metals at elevated temperature into a predetermined size or shape using compressive force exerted through some type of die by a hammer, a press or an upsetting machine.

1638. The fullers are used for

(a) finishing flat surfaces
(b) punching a hole
(c) necking down operation
(d) finishing a punched hole

TSPSC AEE 2015

Ans : (c) The fullers are used for necking down operation.

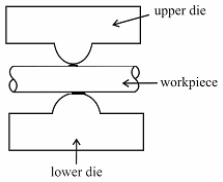
Fullering:- It is the term applied to the first operation is an impression- die forging process. It is actually same as setting down and consists of reducing cross-section of the workpiece or lengthening a preparation of the stock in preparation for subsequent operation.

1639. In the forging operation, fullering is done to:-

(a) Draw out the material
(b) Bend the material
(c) Upset the material
(d) Extrude the material

UJVNL AE 2016, ESE 2001

Ans : (a) In the forging operation. fullering is done to draw out the material.



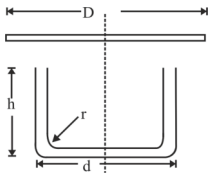
Fullering is an operation metal is displaced form the centre and produces dumbell shape.

1640. The relationship between blank diameter D and cup diameter d during deep drawing process is given as

(a) $D = \sqrt{d^2 + 4dh}$ (b) $D = \sqrt{d^2 + 2dh}$
(c) $D = \sqrt{d^2 + \frac{dh}{2}}$ (d) $D = \sqrt{d^2 + dh}$

UPPSC AE 12.04.2016 Paper-I

Ans : (a)



$D = \sqrt{d^2 + 4hd}$ when $\frac{d}{r} \geq 20$

$D = \sqrt{d^2 + 4hd - 0.5r}$ when $15 \leq \frac{d}{r} < 20$

$D = \sqrt{d^2 + 4hd - r}$ when $10 \leq \frac{d}{r} < 15$

$D = \sqrt{(d-2r)^2 + 4d(h-r) + 2\pi r(d-0.7r)}$ when $\frac{d}{r} < 10$

d = outer diameter to shell mm
D = Blank diameter, mm
h = height of the shell, mm
r = corner radius of punch, mm

1641. The force F required to cut a sheet metal is given by

(a) $F = \tau_s p / t$ (b) $F = \tau_s p / t^2$
(c) $F = \tau_s pt$ (d) $F = \tau_s / pt$

Where τ = shear strength of sheet metal
p = perimeter of the cut
t = thickness of the sheet metal

UPPSC AE 12.04.2016 Paper-I

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Ans : (c) $F = \tau_s pt$

Force required to cut a sheet metal where

τ_s = Shear strength of sheet metal.

p = Perimeter of the cut.

t = Thickness of the sheet metal.

Punching force: - It is the force required to be exerted by the punch in order to shear out the blank from the stock.

$$F = \tau_s . p . t$$

Sometime the tensile strength may replace the shear strength in the above expression because shear is not the only active force the press has to overcome.

1642. The temperature at which new stress free grains are formed in the metal is called:-

(a) Critical temperature
(b) Eutectic temperature
(c) Recrystallization temperature
(d) Yield temperature

UKPSC AE-2013, Paper-I

Ans. (c) : The temperature at which new stress free grains are formed in the metal is called recrystallization temperature

1643. In High velocity forming, the forming speed is greater than:-

(a) 3 m/s (b) 5 m/s
(c) 8 m/s (d) 15 m/s

UKPSC AE-2013, Paper-I

Ans. (b) : High velocity forming (HVF)- HVF or high velocity forming refers to a set of techniques which are used for metal forming. These techniques could include methods such as explosive forming, electromagnetic forming etc. All these techniques involve imparting a high kinetic energy to the work piece accelerating it to a highly velocity, before it is made to hit the appropriate die or made to undergo the process of plastic the formation.

In conventional forming conditions, inertia is neglected, as the velocity of forming is typically less than 5 m/s, while typical high velocity forming operations are carried out at work-piece velocities of about 100 m/s.

1644. The following is not the characteristics of explosive forming :

(a) Low capital cost of the set up.
(b) Very large components can be formed.
(c) Only a simple die is required.
(d) The tooling material is very expensive.

UKPSC AE 2012, 2007 Paper-I

Ans. (d) : The tooling material is very expensive.

1645. Explosive forming is not used for the following:

(a) Making very small complex parts.
(b) For large parts typical of aerospace industry.
(c) Both (a) & (b) above are correct.
(d) None of the above is correct.

UKPSC AE 2012, 2007 Paper-I

Ans. (a) : Making very small complex parts.

1646. The hot forging of steel specimen is carried out at a temperature of

(a) 400ºC (b) 600ºC
(c) 500ºC (d) 950ºC

UKPSC AE 2007 Paper -I

Ans. (d) : 950ºC

1647. In deep drawing the drawing force (neglecting the friction between the job and the die wall) is approximately equal to

(a) $2\pi r_p t$ (b) $3\pi r_p t$
(c) $4\pi r_p t$ (d) $6\pi r_p t$

where r_p = radius of the punch
t = sheet thickness
σ_z = stress in the direction of drawing

UKPSC AE 2007 Paper -I

Ans. (a) : $2\pi r_p t$

1648. The deformation rates as high as following are used in high velocity forming of metals:

(a) 2 m/s (b) 25 m/s
(c) 250 m/s (d) 500 m/s

UKPSC AE 2007 Paper -I

Ans. (c) : 250 m/s

1649. The force required to punch a 10mm diameter hole in a mild steel plate 10mm thick, if the shear strength of mild steel is 360MPa is

(a) 9πkN (b) 36πkN
(c) 18πkN (d) 2.25kN

(APPSC AE 2012)

Ans : (b) hole dia = 10mm = 10×10^{-3}m
thickness of plate = 10mm = 10×10^{-3} m
Shear strength of mild steel = 360 MPa.
Force required to punch = $\pi dt \times \tau$

$$F = \pi \times 10 \times 10^{-3} \times 10 \times 10^{-3} \times 360 \times 10^{6} \text{ N}$$

$$\boxed{F = 36\pi \text{ kN.}}$$

1650. A hole of 20 mm diameter is punched in a plate of 10 mm thick. The shear stress induced in the plate in 0.4 kN/mm^2 Then, the force exerted by the punch is

(a) $500\ \pi$ kN (b) 80π kN
(c) $800\ \pi$kN (d) $8000\ \pi$kN

TSPSC AEE 2015

Ans : (b) hole diameter (d) = 20 mm
plate thickness (t) = 10 mm
shear stress (τ) = 0.4 kN/mm^2
Force exerted by the punch. (F) = $\pi dt \times \tau$
$F = \pi \times 20 \times 10 \times 0.4 \times 10^3$
$F = 80\ \pi$ kN

1651. A metal rod has diameter 8 mm and true stress-strain relationship is given by $\sigma = 900\varepsilon^{0.6}$ MPa is under action of tensile loading. The minimum diameter in mm that can be achieved is :

(a) 8.54 (b) 4.76
(c) 5.92 (d) 3.87

BHEL ET 2019

Ans. (c) : Given diameter = 8 mm
From stress-strain relationship

$$\sigma = 900 \in^{0.6} \text{ MPa}$$

at UTS $\varepsilon_T = n = 0.6$

$$\varepsilon_T = 0.6 = \text{In}\left(\frac{A_o}{A_f}\right)$$

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$$= \ln\left(\frac{\frac{\pi}{4}d_0^2}{\frac{\pi}{4}d_f^2}\right) = 0.6$$

$$\ln\left(\frac{8^2}{d_f^2}\right) = 0.6$$

$$\frac{8^2}{d_f^2} = e^{0.6} = 1.822$$

$$\boxed{d_f = 5.92 \text{ mm}}$$

1652. The process of decreasing the cross-section of a bar and increasing its length is called

(a) drawing-down (b) upsetting
(c) spinning (d) peening

Assam PSC CCE Pre 2015

Ans. (a) : Drawing down– The process of decreasing the cross-section of a bar and increasing its length is called drawing down.
Upsetting–The process of increasing the cross-section of a bar at the expense of its length, is called upsetting.

1653. Mass production of seamless tubes is by the process of

(a) Rolling (b) Spinning
(c) Welding (d) Extrusion

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Extrusion is used for mass production of seamless tubes.
Extrusion having good mechanical property.

1654. In a rolling operation, the thickness of a rectangular piece is reduced from t_1 to t_2. The radius of both the rolls are equal to R. The angle of bite (θ) will be given by :

(a) $\theta = \cos^{-1}\left[1 - \frac{t_1 - t_2}{2R}\right]$

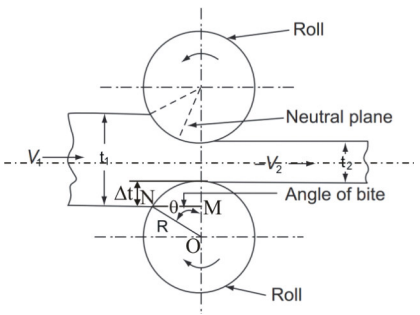
(b) $\theta = \cos^{-1}\left[\frac{t_1 - t_2}{2R}\right]$

(c) $\theta = \cos^{-1}\left[1 - \frac{2R}{t_1 - t_2}\right]$

(d) $\theta = \cos^{-1}\left[\frac{2R}{2R}\right]$

APPSC Poly Lect. 13.03.2020

Ans. (a) : Let, θ = angle of bite



In ΔOMN

$$\cos\theta = \frac{OM}{ON} = \frac{R - \Delta t}{R}$$

But $\Delta t = \frac{t_1 - t_2}{2}$

So, $\cos\theta = \frac{R - \left(\frac{t_1 - t_2}{2}\right)}{R} = 1 - \left(\frac{t_1 - t_2}{2R}\right)$

$$\theta = \cos^{-1}\left[1 - \left(\frac{t_1 - t_2}{2R}\right)\right]$$

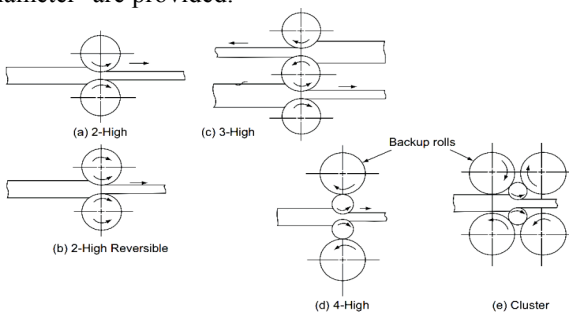
$$\Rightarrow \theta = \cos^{-1}\left[1 - \frac{(t_1 - t_2)}{2R}\right]$$

1655. "Two working rolls of small diameter and four or more back up rolls of large diameter" are provided on :

(a) three-high rolling mill
(b) two high rolling mill
(c) four-high rolling mill
(d) cluster mill

APPSC Poly Lect. 13.03.2020

Ans. (d) : In the cluster mill "Two working rolls of small diameter and four or more backup rolls of large diameter" are provided.



1656. "A process of pushing the heated billet or slug of metal through an orifice provided into a die" is known as :

(a) extrusion (b) deep drawing
(c) piercing (d) rolling

APPSC Poly Lect. 13.03.2020

Ans. (a) : Extrusion is a process of pushing the heated billet or slug of metal through an orifice provided into a die.

1657. A process of making cup-shaped parts from sheet metal blanks with the help of punch is known as :

(a) deep drawing (b) extrusion
(c) piercing (d) upsetting

APPSC Poly Lect. 13.03.2020

Ans. (a) : A process of making cup-shaped parts from the sheet metal blanks with the help of punch is known as deep drawing.

1658. "Hems and seams" are related to :

(a) sheet metal (b) casting
(c) alloying (d) heat treatment

APPSC Poly Lect. 13.03.2020

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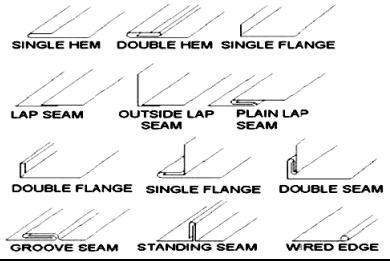
Ans. (a) : "Hems and seams" are two similar metal working processes which is related to sheet metal. Hemming is the process in which the edge is rolled flush to itself , while a seam joins the edges of two materials.



1659. A rolling mill is used to reduce the thickness of a plate from 50 mm to 25 mm. The roll diameter is 700 mm and the coefficient of friction at the roll work interface is 0.1. It is required that the draft in each pass must be the same. Assuming no front and back tensions, the minimum number of passes required in the rolling are :

(a) 16 (b) 8
(c) 4 (d) 12

APPSC Poly Lect. 13.03.2020

Ans. (b) : Given data : $h_0 = 50$ mm, $h_f = 25$ mm
Roll diameter (D) = 700 mm,
Coefficient of friction (μ) = 0.1
Let; n = Minimum number of passes required in the rolling.

$$n = \frac{(\Delta h)_{required}}{(\Delta h)_{maximum}}$$

$$\therefore (\Delta h)_{required} = h_0 - h_f = 50 - 25 = 25 \text{ mm}$$

$$\text{and; } (\Delta h)_{maximum} = \mu^2 R = (0.1)^2 \times \frac{700}{2} = 3.5 \text{ mm}$$

$$\therefore \text{ Number of pass (n)} = \frac{25}{3.5} \approx 7.143$$

$$\therefore \quad \boxed{n = 8 \text{ passes}}$$

1660. During two high strip rolling using roller radius of 'R' and thickness reduction is 'Δh', the coefficient of friction between strip and roller is 'μ', the following equation holds true for the maximum draft:

(a) $\Delta h = \mu R$ (b) $\Delta h = \mu R^2$
(c) $\Delta h = (\mu R)^2$ (d) $\Delta h = \mu^2 R$

Punjab PSC SDE 12.02.2017

Ans. (d) : $\Delta h = \mu^2 R$
μ = Coefficient of friction
R = Radius of rolls

- $\cos\theta = 1 - \frac{\Delta h}{2R}$
- $\theta \rightarrow$ bite angle
- self entry condition ; $\mu \geq \tan\theta$
- Velocity of roll :

$V_r = \pi DN/60$
D = diameter of roll
N = revolution in rpm

1661. The rolling defect seen at the centre due to the tensile stresses set-up in the rolled part because of bending of rolls under the rolling pressure is known as :

(a) Alligatoring (b) Zipper Cracks
(c) Edge Cracks (d) None of the above

Punjab PSC SDE 12.02.2017

Ans. (b) : The rolling defect seen at centre the due to the tensile stresses set-up in the rolled part because of bending of rolls under the rolling pressure is known as zipper cracks.

1662. The principal breakdown pass sequences in rolling are:

(a) box pass series, diamond-square series, oval-square series
(b) round pass series, hexagon-square series, oval-round series
(c) triangle pass series, pentagon-round series, tube-rod series
(d) None of the above

HPPSC AE (PWD) 10.09.2016

Ans. (a) : The principal breakdown pass sequences in rolling are: box pass series, diamond square series, Oval-square series.

1663. Cold heading is a process as

(a) Chipless machining
(b) Explosive forming
(c) High energy ratio forming
(d) Electro machining process

APPSC AEE Screening Test 2016

Ans. (a) : Cold heading (or cold forming) is a method of forming metal in progressive steps into net shaped or near net shaped parts. The cold heading process creates a stronger part, with smooth continuous surfaces while enhancing the grain flow of the material. Cold heading is a chipless (without cutting chip) process.

1664. In cold working

(a) strength, hardness and ductility increase
(b) strength and hardness increase, but ductility decreases
(c) strength, hardness and ductility decrease
(d) strength and hardness decrease, but ductility increases

GPSC ARTO Pre 30.12.2018

Ans. (b) : In cold working, due to strain hardening process strength and hardness increases but ductility decreases.

1665. The forging force in impression-die forging is

(a) Proportional to the projected area of the forging
(b) Proportional to the surface area of the forging
(c) Inversely proportional to the projected area of the forging
(d) Inversely proportional to the surface area of the forging

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (a) : The forging force in impression-die forging is proportional to the projected area of the forging.

1666. Automobile bodies can be manufactured by employing

(a) deep drawing (b) tube drawing
(c) wire drawing (d) rod wiring

Mizoram PSC AE/SDO 2014 Paper-III

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Ans. (a) : Deep drawing is one of the most widely used processes in sheet metal. Apart from its use in many other sectors, if is applied in the automotive industry for the manufacturing of car body parts.

1667. A forming operation which is carried on cylindrical rolls is termed as

(a) casting (b) drawing
(c) rolling (d) grinding

Mizoram PSC AE/SDO 2014 Paper-III

Ans. (c) : A forming operation which is carried on cylindrical rolls is termed as rolling.

1668. Consider the following statements :

1. It is a single pass process.
2. Brittle material cannot be extruded
3. As compared to rolling more complex parts can be produced by this method.

Which of the above statements is/are correct with respect to extrusion process?

(a) 1 only (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

UPSC JWM Adv. No-16/2009

Ans. (d) : Extrusion Process–
- It is a single pass process
- As compared to rolling more complex parts can be produced by extrusion process.
- We use hydrostatic extrusion for brittle material.

1669. Cold shut is a forging defect caused by which of the following reason?

(a) Improper cleaning of the stock
(b) Improper design of die
(c) Misalignment of the two die halves
(d) Improper cooling of the large forging

RPSC Vice Principal ITI 2018
UPSC JWM Adv. No-16/2009

Ans. (b) : Causes of cold shut defect in forging–
(1) Improper design of forging die.
(2) Due to sharp corner.
(3) Excessive chilling in forged product.

1670. Cold or hot rolling does not produce

(a) a hollow circular section
(b) a T section
(c) an I section
(d) a channel section

TNPSC ACF 2012

Ans. (a) : Cold or hot rolling produces– T section I section a channel section Discs, Door and shelving, stamping guard rails for streets and highway.

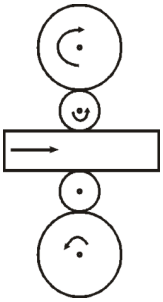
1671. Correlate the following statements :

I. In a four high rolling mill the rollers in contact with metal are larger than the outer driving rollers
II. Smaller rolls concentrate the pressure over smaller areas
Off these

(a) I and II are correct of II is the reason for I
(b) I and II are independently correct
(c) I is correct, but II is not correct
(d) I is not correct, but II is correct

TNPSC ACF 2012

Ans. (d) :



4 high rolling mill

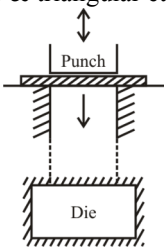
(a) So in four high rolling mill the roller contact with metal are smallest that the outer driving rollers. So first statement is incorrect.
(b) Smaller rolls concentrate the pressure over smaller area this statement is true.

1672. Piercing is an operation of cutting

(a) a cylindrical hole in a sheet of metal by the punch and the die
(b) a hole (other than cylindrical) in a sheet of metal by the punch and the die
(c) a flat sheet to the desired shape
(d) a number of holes evenly spaced I a regular pattern on a sheet metal

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : Piercing is an operation of cutting a hole (other than cylindrical) in a sheet of metal by the punch & the die, eg. square & triangular etc.



1673. The chamfering is an essential operation after

(a) knurling (b) rough turning
(c) boring (d) thread cutting

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Chamfering is an operation of beveling the extreme end of a workpiece, it is essential in thread cutting operation so that a nut can be inserted or assembled easily on bolt.

1674. A 1.5 mm thick sheet is subjected to unequal biaxial stretching and the true strains in the directions of stretching are 0.05 and 0.09 the final thickness of the sheet in mm is

(a) 1.473 (b) 1.304
(c) 1.362 (d) 289

Vizag Steel MT (Re-Exam) 24.11.2013, GATE 2000

Ans. (b) : Method-I
Stretching–The true strain in a stretching operation is given by,

$$\in_x + \in_y + \in_z = 0$$

Where, $\in_x$, $\in_y$ and $\in_z$ represent true strain in x, y and z respectively.
True strain is given by,

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$$\epsilon_T = \ln\left(\frac{\ell_f}{\ell_i}\right) = \ln\left(\frac{A_i}{A_f}\right)$$

Given, $\epsilon_x = 0.05$, $\epsilon_y = 0.09$, $\ell_i = 1.5$ mm

$$\epsilon_x + \epsilon_y + \epsilon_z = 0$$
$$0.05 + 0.09 + \epsilon_z = 0$$
$$\epsilon_z = -0.14$$
$$\epsilon_T = \ln\left(\frac{\ell_f}{\ell_i}\right)$$
$$-0.14 = \ln\left(\frac{\ell_f}{\ell_i}\right)$$
$$e^{-0.14} = \frac{\ell_f}{\ell_i}$$
$$\boxed{\ell_f = 1.304\,\text{mm}}$$

Method-II

$$(\epsilon_T)_{Total} = (\epsilon_T)_1 + (\epsilon_T)_2 = \ln\left(\frac{\ell_f}{\ell_i}\right)$$
$$= 0.09 + 0.05 = 0.14$$
$$\ell_i\omega_i t_i = \ell_f\omega_f t_f$$
$$\boxed{\omega_i = \omega_f}$$
$$\ell_i t_i = \ell_f t_f$$
$$\frac{\ell_f}{\ell_i} = \frac{t_i}{t_f}$$
$$(\epsilon_{Total}) = 0.14 = \ln\left(\frac{t_i}{t_f}\right) = \ln\left(\frac{1.5}{t_f}\right)$$
$$\frac{1.5}{t_2} = 1.15027$$
$$\boxed{t_2 = 1.304}$$

But official answer is given option (a).

1675. Identify the stress state in the flange portion of a partially drawn cylindrical cup when deep drawing without a blank holder

(a) Tensile in all three directions
(b) No stress in the flange at all, because there is no blank holder
(c) Tensile stress in one direction and compressive in the other direction
(d) Compressive in two directions and tensile in the third direction

Vizag Steel MT (Re-Exam) 24.11.2013, GATE 1999

Ans. (b) : Stress state in the flange portion of a partially drawn cylindrical cup when deep drawing without a blank holder. No stress in the flange at all, because there is no blank holder.

1676. Which of the following is an advantage of forging?

(a) good surface finish
(b) improved physical property
(c) low tooling cost
(d) close tolerance

GPSC Executive Engineer 23.12.2018

Ans. (b) : Forging is shaping of metal by squeezing them two or more dies to obtain the desired shape. Forging operation given best mechanical properties. Forging can be done either at room temperature at elevated temperature. It the forging operation is carried out below re-crystallisation temperature it is known as cold forging. It at temperature above are crystallization is called not forging.

1677. In a rolling process, sheet of 30 mm thickness is rolled to 21 mm thickness. Roll is of Diameter 800 mm and it rotates at 100 rpm. The strip contact length will be

(a) 60 mm (b) 63 mm
(c) 80 mm (d) 90 mm

GPSC Executive Engineer 23.12.2018

Ans. (a) : Given, $t_i = 30$ mm, $R = \frac{800}{2} = 400\,\text{mm}$

$t_f = 21$ mm

$\Delta h = t_i - t_f = 30 - 21 = 9$ mm

Strip contact length $(L_c) = \sqrt{R\Delta h}$

$= \sqrt{400 \times 9} = \sqrt{3600}$

$\boxed{L_c = 60\,\text{mm}}$

1678. Which one of the following is not an application of forging?

(a) Rail sections
(b) Chisels
(c) Brake pedal of an automobile
(d) Steel balls of ball bearing

Kerala PSC Poly. Lect. 09.01.2014

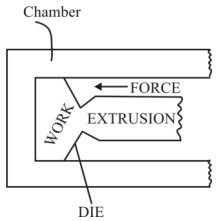
Ans. (d) : Steel balls of ball bearing is not an application of forging.

1679. In backward extrusion process, which type of pressure is applied on a billet of ductile metal to force it through the die orifice?

(a) Tensile
(b) Compressive
(c) Shear
(d) Tensile and Shear
(e) Tensile, Compressive and Shear

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Indirector backward extrusion is a type of metal extrusion process where in the billet (work piece) is kept inside the chamber. Which is completely closed on one side. In this type of extrusion the billet and the container move simultaneously where as the lie remains stationery.



1680. In drawing operation of large diameter wires, the range of reduction in cross-sectional area per pass in generally from:

(a) 5% to 10% (b) 10% to 20%
(c) 15% to 45% (d) 50% to 65%
(e) 70% to 85%

CGPSC ARTO & TSI 07.01.2017

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Ans. (c) : In drawing, reductions in the cross-sectional area per pass range upto about 45%.
Usually the smaller the intial cross-section the smaller the reduction per pass.
Fine wires usually are drawn at 15% to 25% reduction per pass and larger sizes at 20% to 45% per pass.

1681. Centre burst is a common defect in extruded parts. This defect is also known as:
(a) Arrow and bow
(b) Arrow-head fracture
(c) Fish tailing
(d) Snake tail
(e) Hollow sphere

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Central bursts, or chevrons are the internal arrows shaped defects occasionally encountered in cold extrusions or drawn wire. In cold extrusion the defect is usually observed in the final, light step of a multistep extrusion.

1682. A bloom is a solid mass of metal having square or rectangular cross-section with thickness:
(a) equal to 50 mm
(b) ranging between 50 and 75 mm
(c) ranging between 75 and 100 mm
(d) less than 150 mm
(e) equal to or greater than 150 mm

CGPSC ARTO & TSI 07.01.2017

Ans. (e) : Bloom :-It has rectangular/square cross-section. The cross-section area of bloom is always greater than 36 in (230 cm^2)
Bloom are cased as rolling material in the manufacturing process of rails, seamless, piper etc.
Generally a bloom is solid mass of metal having square or rectangular cross-section with thickness equal to or greater than 150 mm.

1683. A solid aluminium cylinder is forged between two frictionless flat dies increasing diameter twice. The percentage change in length from original one will be:
(a) 50 percent (b) 25 percent
(c) –75 percent (d) 75 percent

AAI Jr. Executive 26.03.2021

Ans. (c) : Given,
$D_i = D$
$D_f = 2D$
Vol. of cylinder after forging = Vol. of cylinder before forging

$$\frac{\pi}{4}D_f^2 L_f = \frac{\pi}{4}D_i^2 L_i$$

$$\frac{\pi}{4}(2D)^2 L_f = \frac{\pi}{4}D^2 L_i$$

$$4L_f = L_i$$

$$\frac{L_f}{L_i} - 1 = \frac{1}{4} - 1$$

$$\left(\frac{L_f - L_i}{L_f}\right) = -\frac{3}{4}$$

Percentage change in length, $\frac{L_f - L_i}{L_f} \times 100 = -75\%$

1684. The thickness of copper sheet is reduced from a thickness of 16 cm to 10 cm in a single pass by a roller of diameter 400 cm. What is the bite angle in degrees for this process?
(a) 5.82° (b) 7.62°
(c) 9.82° (d) 0.982°

AAI Jr. Executive 26.03.2021

Ans. (c) : Given,
Initial thickness, $h_i = 16$ cm
Final thickness, $h_f = 10$ cm
Roll diameter, D = 400 cm
Draft, $\boxed{h_i - h_f = D(1 - \cos\alpha)}$
Where, α is bite angle
$16\text{-}10 = 400(1 - \cos\alpha)$
$\cos\alpha = 0.985$
Bite angle, $\alpha = 9.82°$

1685. Which of the following rolling mills is required to reduce a slab directly to strip in one pass?
(a) continuous rolling mill (b) two high mill
(c) planetary mill (d) none of the above

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (c) : Plantary mill–Consist of a pair of heavy backing rolls surrounded by a large number of planetary rolls. The overall reduction is the summation of a series of small reductions by each pair of rolls.
Therefore, the planetary mill can hot reduces a slab directly to strip in one pass through the mill.
Two high mill, Pullover–The stock is returned to the entrance for further reduction.
Two high mill, reverse–The work can be passed back and forth through the rolls by reversing their direction of rotation.
Continuous rolling–The strip will be moving at different velocities at each stage in the mill.

1686. Symmetrical hollow parts of circular cross-section are made by hot:
(a) forging (b) drawing
(c) extrusion (d) spinning

Kerala PSC AE (Irrigation) 05.04.2017

Ans. (d) : The parts of circular cross section which are symmetrical about the axis of rotation are made by hot spinning.

1687. Which of the following processes is used to produce only axisymmetric cup shaped articles from sheet metal?
(a) Shallow drawing (b) Nibbling
(c) Spinning (d) Deep drawing
(e) Bending

(CGPSC Polytechnic Lecturer 2017)

Ans. (c) : Spinning– Spinning is a sheet metal forming process in which a metal blank is pressed over a rotating chuck or form mandral with the help pressing tool to obtain axisymmetric hollow shell.
Nibbling–The nibbling process cuts a contour by producing a series of overlapping slits or notches. This allows for complex shapes to be formed in sheet metal upto 6 mm thick using simple tool.
Shallow Drawing–Shallow drawing is used to describe the process where the depth of draw is less than the smallest dimension of the opening; otherwise, it is considered deep drawing.

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1688. The rolling process is carried out on rolling machines also termed as

(a) rolling jackets (b) rolling structures
(c) rolling platforms (d) rolling mills

UPRVUNL AE 05.07.2021

Ans. (d) : The rolling process is carried out on rolling machines also termed as rolling mills.

1689. Which of the following is NOT a stage of rolling

(a) Finishing stage (b) Leader stage
(c) Breakdown stage (d) Back-up stage

UPRVUNL AE 05.07.2021

Ans. (d) : Back-up stage is not a stage of rolling.

1690. The first product obtained after the breakdown of the ingots is called _____

(a) foil (b) bloom
(c) slab (d) billet

UPRVUNL AE 05.07.2021

Ans. (b) : The first product obtained after the breakdown of the ingots is called bloom.

1691. Which of the following statement is INCORRECT?

(a) Oxide layers are formed during hot forming
(b) Lubrication is crucial for cold forming
(c) the surface finish is poor in hot working as compared to cold forming
(d) Recrystallization takes place in cold forming

UPRVUNL AE 05.07.2021

Ans. (d) :

1692. Which of the following is NOT a general method of forming

(a) Hot (b) Warm
(c) cold (d) cryogenic

UPRVUNL AE 05.07.2021

Ans. (d) : Cryogenic is not a general method of forming.

1693. In the _____ extrusion process, the material remains stationary and the die moves.

(a) direct (b) hooker
(c) indirect (d) impact

UPRVUNL AE 05.07.2021

Ans. (c) : In the indirect extrusion process, the material remains stationary and the die moves.

1694. In case of hot rolling when maximum reduction is cross-section per pass is aimed at, it may be necessary to artificially increase the value of μ by _____rolls.

(a) milling (b) ragging
(c) pressing (d) swaging

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : In case of hot rolling when maximum reduction is cross-section per pass is aimed at, it may be necessary to artificially increase the value of μ by ragging rolls.

1695. _____is an example for mechanical working of sheet metals.

(a) Welding (b) Bending
(c) Casting (d) Milling

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : Bending is an example for mechanical working of sheet metals.

1696. For successful extrusion, the metal should be

(a) ductile (b) malleable
(c) plastic (d) tough

HPPSC Poly. Lect. 05.07.2021

Ans. (c) : For successful extrusion, the metal should be plastic.

1697. Which process involves increasing of the cross-sectional area by pressing or hammering in a direction parallel to the original ingot axis?

(a) swaging (b) setting down
(c) upsetting (d) peening

HPPSC Poly. Lect. 05.07.2021

Ans. (c) : Upsetting involves increasing of the cross-sectional area by pressing or hammering in a direction parallel to the original ingot axis.

1698. Which one is the wrong statement about forging?

(a) are uniform in density as well as in dimensions
(b) have high ductility and strength
(c) offer great resistance to fatigue and impact loads
(d) are less reliable

HPPSC Poly. Lect. 05.07.2021

Ans. (d) :

1699. A sheet of metal is being cut along its length in a straight line. Such a cutting operation is called

(a) crimping (b) parting
(c) slitting (d) lancing

HPPSC Poly. Lect. 05.07.2021

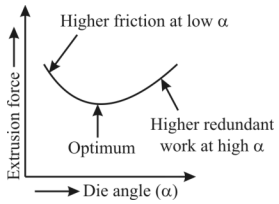
Ans. (c) : A sheet of metal is being cut along its length in a straight line. Such a cutting operation is called slitting.

1700. In the metal extrusion process, when the die angle increases, the required extrusion force :

(a) increases
(b) decreases
(c) decreases and then increases
(d) increases and then decreases

DRDO Scientists 2009

Ans. (c) :



From figure it is clear that in the metal extrusion process, when the die angle increases, the required extrusion force decreases and then increases.

1701. Spring back factor of the sheet metal depends on :

(a) both bend radius and thickness
(b) bend radius only
(c) thickness only
(d) surface area

DRDO Scientists 2009

Ans. (c) : Spring back factor of the sheet metal depends on thickness only.

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1702. Mechanical property that decides the cutting force in the blanking operation is :
(a) compressive strength
(b) shear strength
(c) tensile strength
(d) modulus of elasticity

DRDO Scientists 2009

Ans. (b) : Mechanical property that decides the cutting force in the blanking operation is shear strength.

1703. Mass production of cooking utensils is done by
(a) metal spinning (b) deep drawing
(c) coining (d) embossing

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Mass production of cooking utensils is done by deep drawing.

1704. If the extrusion ratio is 20, the percentage reduction in the cross-sectional area of the billet after the extrusion will be
(a) 98% (b) 95%
(c) 20% (d) 5%

DRDO Scientists 2008

Ans. (b) : A_0 = Area of billet
A_f = Area of part to be extracted

$$\text{Extrusion ratio} = \frac{A_0}{A_f} = 20$$

$$\% \text{ reduction in C/S area} = \left(1 - \frac{A_f}{A_0}\right) \times 100$$

$$= \left(1 - \frac{1}{20}\right) \times 100$$

$$= 95\%$$

1705. Recrystallization temperature depends on:
(a) Grain size (b) type of metal
(c) Annealing time (d) All of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) : Recrystallization temperature depends on–
- Grain size
- Type of metal
- Annealing time

1706. _____ drawing should be drawn prior to ____ drawing.
(a) Assembly, Part (b) Part, Assembly
(c) Part, Working (d) Assembly, Working

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) : Assembly drawing should be drawn prior to part drawing.

1707. Match the following products with the suitable manufacturing process :

Product		Manufacturing Process
P.	Toothpate tube	1. Centrifugal casting
Q.	Metallic pipes	2. Blow moulding
R.	Plastic bottles	3. Rolling
S.	Threaded bolts	4. Impact extrusion

	P	Q	R	S
(a)	4	3	1	2
(b)	2	1	3	4
(c)	4	1	2	3
(d)	1	3	4	2

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (c) :

1708. For cold rolling of sheets and strips, the value of bite angle varies from
(a) 2- 10 degrees (b) 15- 20 degrees
(c) 20-30 degrees (d) 30- 40 degrees

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) : For cold rolling of sheets and strips, the value of bite angle varies from 2-10 degrees.

1709. The increase in hardness due to cold working is called :
(a) Work-hardening (b) Flame hardening
(c) Induction hardening (d) Age-hardening

MECON MT 2020

Ans. (a) : The increase in hardness due to cold working is called work-hardening.

1710. Upset forging causes by
(a) Steadily applied pressure instead of impact force
(b) Force the end of a heated bar into a desired shape
(c) Forging method to reduce the diameter of a bar
(d) None of above

MPSC Lect. (Prod.) 2018

Ans. (b) : Upset forging causes by force the end of a heated bar into a desired shape.

1711. When a steel undergoes a cold-worked process, it becomes progressively
(a) softer (b) harder
(c) ductile (d) malleable

Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Cold working causes the crystals grains and inclusions to distort the flow of the metal, which may cause work hardening and an isotropic material properties work hardening makes the metal harder stiffer and stronger, but less plastic and may cause crack of the workpiece.

1712. Forging operation in which the thickness of bar is reduced by successive forging steps at specific intervals is called _____.
(a) Barreling (b) Drawing in
(c) Cogging (d) Upsetting

Oil India Senior Officer 23.12.2020

Ans. (c) : $\eta_{mech} = \dfrac{\text{Power available at the turbine shaft}}{\text{Power developed by the runner}}$

1713. During which one of the forming operations, tensile stresses are induced?
(a) Stretch forming (b) Ironing
(c) Sizing (d) Coining

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Stretching forming is stamping process where by means of pressure, tensile stress and a locking mechanism, sheet metal is changed into a curved body.

1714. The forming operation used in making raised figures on sheets with its corresponding relief on the other side is _____.
(a) blanking (b) embossing
(c) coining (d) stretch forming

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : Embossing– Embossing is the operation through which raised figures are mode on metal sheet with its corresponding relief on other side. Embossing involves drawing and bending. Die set consists of die and punch with desired contours.

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1715. In a blanking operation to produce steel washer, the maximum punch load used is 2×10^5 N. The thickness of the plate is 4 mm and the penetration is 25%. The work done during this shearing operation is

(a) 200 J (b) 400 J
(c) 600 J (d) 800 J

APPSC AEE 2012, CSE 1994

Ans. (a) : Given,
f = 2×105 N, thickness of plate = 4 mm
penetration (k) = 25% = 0.25
Work done (w) = f.k.t = $2\times10^5 \times0.25\times4\times10^{-3}$
W = 200 Joule

1716. On a cluster roll, there are ____ working rolls.

(a) four (b) eight
(c) two (d) six

UPRVUNL AE 05.07.2021

Ans. (c) :

1717. Find the blanking force required to punch 10 mm diameter holes in a steel sheet of 3 mm thickness. Given shear strength of material = 400 MPa, penetration = 40% and shear provided on the punch = 2 mm.

(a) 22.6 kN (b) 37.7 kN
(c) 61.6 kN (d) 94.3 kN

UPPSC AE 13.12.2020, Paper-I, GATE 2004

Ans. (a) : Given,
Shear strength (τ) = 400 N/mm^2
Sheet thickness (t) = 3 mm
Shear on punch (t_1) = 2 mm
Dia of hole (d) = 10 mm
Penetration (p) = 0.4
Blanking force when shear applied on the punch

$$= \pi dt\tau\left(\frac{pt}{t_1}\right) = \pi\times10\times3\times400\left(\frac{0.4\times3}{2}\right)$$

= 22.6 kN

1718. In sheet metal working, shear is provided on punches and dies so that

(a) good cut edge is obtained
(b) cut blanks are straight
(c) press load is reduced
(d) warping of sheet is minimized

HPPSC Poly. Lect. 05.07.2021, ESE 1994

Ans. (c) : In sheet metal working, shear is provided on punches and dies so that press load is reduced.

1719. In piercing and blanking operation, the clearance between punch and die solely depends on

(a) thickness of sheet metal
(b) capacity and type of press
(c) diameter of the hole required
(d) number of pieces to be made

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : In piercing and blanking operation, the clearance between punch and die solely depends on thickness of sheet metal.

1720. A 50 mm diameter disc is to be punched out from a carbon steel sheet 1.0 mm thick. The diameter of the punch should be

(a) 49.925 mm (b) 50.00 mm
(c) 50.075 mm (d) None of the above

PPSC Asstt. Municipal Engineer 15.06.2021
GATE 1996

Ans. (b) : Clearance–Die opening must be larger than punch and known as 'clearance'.
Punching–
Punch = size of hole
Die = punch size + 2 clearance
Remember–In punching, punch is correct size.
Blanking–
Die = size of product
Punch = Die size – 2 clearance
Remember–In blanking, die size will be correct.
Note-In punching clearance is provided on die
In blanking clearance is provided on punch.
So, this is punching operation and as we know in punching, punch is correct size.
Hence, The diameter of the punch should be 50.00 mm.

1721. With a punch for which the maximum crushing stress is 4 times the maximum shearing stress of the plate, the biggest hole that can be punched in the plate would be of diameter equal to

(a) $\frac{1}{4}\times$ Thickness of plate
(b) $\frac{1}{2}\times$ Thickness of plate
(c) Plate thickness
(d) 2 × plate thickness

ISRO Scientist/Engineer 2008

Ans. (c) : Crushing stress
=4 × shearing stress

$$\sigma = 4\tau$$

$$\frac{F}{\frac{\pi}{4}D^2} = 4\times\frac{F}{\pi Dt}$$

$$\Rightarrow \quad \frac{4}{D} = \frac{4}{t}$$

or D = t

1722. Which one of the following processes is most commonly used for the forging of bolt heads of hexagonal shape?

(a) Open die upset forging
(b) Closed die drop forging
(c) Closed die press forging
(d) Open die progressive forging

RPSC ACF & FRO, 26.02.2021
UPSC JWM Adv. No-16/2009, ESE 1998

Ans. (a) : Open die upset is most commonly used for the forging of bolt heads of hexagonal shape.

1723. Typical hot working temperature of aluminium alloy is

(a) 200 to 300ºC (b) 350 to 485ºC
(c) 500 to 700ºC (d) 700 to 1000ºC

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) :

1724. Which one of the following lubricants is used in forward hot extrusion of steel?

(a) Molten glass (b) Soap solution
(c) Copper sulphate (d) Vegetable oil

ESE 18.07.2021, HPPSC AE 2018

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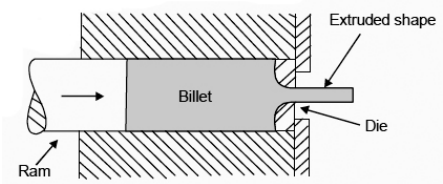
Ans. (a) : Forward Hot extrusion-
Hot extrusion involves prior heating of the billet billet temperature, lubrication is critical hot extrusion for certain metal's (eg. steels) and special lubricants have been developed that are effective under harsh conditions in hot extrusion. For hot extrusion molten glass is an excellent lubricants with steels, stainless steels and high temperatures metals and alloys.

1725. Which one of the following statements is not correct for forward or direct extrusion process?
(a) High friction forces must be overcome.
(b) High extrusion forces are required but mechanically simple and uncomplicated.
(c) Low scrap or material waste only 5 – 6% of billet weight.
(d) Simple, but the material must slide along the chamber wall.

ESE 18.07.2021

Ans. (c) :
- A ram drives the entire billet to and through a stationary die and must provide additional power to overcome the frictional resistance between the surface of the moving billet and the confining chamber, Hence material must slide along the chamber wall.
- Huge amount of frictional energy are developed due to which more amount of force are required for extraction.



1726. Consider the following statements regarding defects in forgings:
1. Flakes are internal breaks or ruptures occurring in some grades of alloy steel.
2. Die shift is caused by misalignment between the top and bottom forging dies.
3. Fins and rags are small projections or loose metal driven into the surface of the forging.

Which of the above statements are correct?
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (d) : The correct statements are-
- Flakes are basically internal ruptures caused by the improper cooling of the large forging.
- Die shift is caused by the misalignment of the two die halves, making the two halves of the forging to be of improper shape.
- Fins and rags are small projections or loose metal driven into the surface of the forging.

1727. Which of the following does not from a part of metal working process:
(a) Shearing (b) Extrusion
(c) Stretch forming (d) Welding

TRB Polytechnic Lecturer 2017

Ans. (d) : Shearing, extrusion and strength forming are form a part of metal working process whereas welding is the fabrication method to join job.

1728. The frictional shear stress (τ) in metal forming is expressed as
(a) $\frac{m\bar{\sigma}}{\sqrt{3}}$ (b) $\frac{\bar{\sigma}}{\sqrt{3m}}$
(c) $\frac{\bar{\sigma}}{m\sqrt{3}}$ (d) $\frac{\sqrt{3}}{m}\bar{\sigma}$

Where 'm' is the friction shear factor and '$\bar{\sigma}$' is the flow stress

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : The relationship between the flow stress in uniaxial tension and shear flow stress.

$$\tau = \frac{m}{\sqrt{3}}\bar{\sigma}$$

1729. Choose the wrong statement.
(a) In a forged part a change of direction is associated with break in granular continuity
(b) In a hot rolled part temperature in vicinity of recrystallization temperature resultant in grain refinement
(c) in a cold rolled part the grains become elongated and not refined
(d) in am extruded part the mechanical properties improve because of strain

TNPSC ACF 2012

Ans. (a) : In forging process because of manipulative ability of forging process, it is possible to closely control the grain flow in specific direction. Such that the best mechanical properties can be obtained based on specific application.

1730. Match List-I (Products) wit List-II (Suitable processes) and select the correct answer using the codes given below the list:

List-I	List-II
A. Collapsible	**1. Forging**
B. Machine tool beds	**2. Casting**
C. Connecting rods	**3. Extrusion**
D. Pressure vessel	**4. Welding**

(a) A-4, B-1, C-3, D-2 (b) A-4, B-2, C-3, D-1
(c) A-3, B-2, C-1, D-4 (d) A-3, B-1, C-4, D-2

UPRVUNL AE 21.08.2016, CSE Pre 2001

Ans. : (c)

Product	**Suitable process**
Collapsible	Extrusion
Machine tool beds	Casting
Connecting rods	Forging
Pressure vessel	Welding

1731. Spring back in metal forming depends on
(a) Modulus of Elasticity
(b) Load Applied
(c) Strain Rate
(d) None of these

ISRO Scientist/Engineer 12.05.2013

Ans. (a) : Spring back in metal forming depends on modulus of elasticity.

1732. Hot working operation is carried out at
(a) Re-crystallization temperature
(b) Near plastic stage temperature
(c) Below re-crystallization

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(d) Above re-crystallization

TSPSC Manager (Engg.) HMWSSB 12.11.2020
ISRO Scientist/Engineer 2012

Ans. (d) : Hot working operation is carried out above the recrystallization temperature.

The porosity of the metal is largely eliminated.

The grain structure of the metal is refined.

Mechanical properties such as toughness percentage elongation, percentage reduction in area and resistance to shock and vibration are improved due to the refinement of grains.

1733. Recrystallization temperature is one at which

(a) crystals first start forming from molten metal when cooled

(b) new spherical crystals first begin to form from the old deformed ones when that strained metal is heated

(c) the allotropic form changes

(d) crystals grow bigger in size

ESE 2018, GWSSB DEE 07.07.2016

Ans. (b) : If a deformed polycrystalline metal is subsequently heated to a high enough temperature, new crystal nucleate and grow to consume and replace the original structure.

The temperature at which this occurs is known as recrystallization temperature.

1734. Which one of the following is the advantage of hot working?

(a) No strain hardening

(b) No oxide formation

(c) Good surface finish

(d) Good accuracy

UPSC JWM Adv. No-16/2009

Ans. (a) : Advantages of Hot Working–

- Significant plastic deformation.
- Significant change in work piece shape.
- Lower forces are required.
- Forming of materials with low ductility.
- No strain hardening.

Disadvantage of Hot Working–

- Shorter tool life
- Poor surface finish
- Lower dimensional accuracy
- Higher production cost

1735. Fatigue strength may be improved by

(a) hot working (b) cold working

(c) innealing (d) sand blasting

TNPSC ACF 2012

Ans. (b) : Fatigue strength can be increased by

(1) Cold working

(2) Shot peeing

(3) Grinding and lopping surface

(4) Using gradual changes of section.

1736. Which is not true in the context of effect of cold working on metals?

(a) increase of strength and hardness of metal

(b) loss of ductility

(c) **refinement in grain structure**

(d) surface finish is improved and close tolerances maintained

HPPSC Poly. Lect. 05.07.2021

Ans. (b) :
- The cold worked processes less ductility.
- Strain hardening occurs.
- Metal surfaces must be clean and scale free before cold working.
- Imparted directional properties may be detrimental.

1737. The recrystallisation temperature is the temperature at which new strain-free grains are formed from the earlier deformed once. For steel this temperature is close to

(a) 800°C (b) 350°C

(c) 560° (d) 1025°C

HPPSC Poly. Lect. 05.07.2021

Ans. (a) : The recrystallization temperature of steel to earlier deformed to 725°C.

1738. Cold working of metals results in

(a) increased strength and hardness but reduced dislocation density.

(b) increased strength and hardness as well as increased dislocation density.

(c) decreased strength and hardness as well as reduced dislocation density.

(d) decreased strength and hardness but increased dislocation density.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) :

1739. In sheet metal work, the cutting force on the tool can be reduced by

(a) grinding the cutting edges sharp

(b) increasing the hardness of die

(c) increasing the hardness of tool

(d) providing shear angle on the tool

HPPSC Poly. Lect. 05.07.2021

Ans. (d) : In sheet metal work, the cutting force on the tool can be reduced by providing shear angle on the tool.

1740. Cutting and forming operations can be done in a single operation on :

(a) Simple die (b) Compound die

(c) Combination die (d) None of these

(HPPSC AE 2014)

Ans : (c) Cutting and forming operation can be done in a single operation on combination die.

Compound die:-In a compound die two or more cutting operation are performed at one station of the press in every stroke of the ram.

Combination die:-In a combination die, two or more operation are performed simultaneously at one station in the single stroke of the ram. It differs from compound die in that the cutting operation is combined with a bending or forming operation in combination die.

Progressive die:-In a progressive die, two or more operation are performed simultaneously, but at different stations.

1741. The forging operation that is carried out in closed dies and in which the flow of metal occurs only at the top layers of the material but not throughout the material is called:

(a) Auto forging (b) Coining

(c) Incremental forging (d) Roll forging

(e) Swaging

CGPSC ARTO & TSI 07.01.2017

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Ans. (b) : Coining :- It is a closed die forging process in which pressures is applied on the surface of the forging in order to obtain closer tolerances, smoothes surface and eliminate draft.
Roll forging:- It is a process for simul tenuously reducing the cross-sectional area and changing the shape of heated bares, bittet or plates by passing them between two driven rolls that rotates in opposite directions and have one or more matching impression in each roll.

1742. In a four high rolling mill, there are four rolls out of which
(a) one is working roll and the remaining three are backing rolls
(b) Two are working rolls and the remaining two are backing rolls
(c) three are working rolls and the fourth one is backing roll
(d) all are working rolls

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : In a four high rolling mill, there are four rolls out of which two are working rolls and the remaining two are backing rolls.

1743. In sheet metal blanking operation, the blank size equals
(a) Die opening size
(b) Punch size
(c) Die opening size plus clearance between the punch and die
(d) Punch size plus clearance between the punch and die

TSPSC AEE 2017

Ans. (a) : In blanking operations the die size is made equal to the blank size and clearance in provided only on the punch.

1744. If D is the blank diameter and 'd' is the punch diameter, the drawing ratio (DR) of a cylindrical shaped part is given by
(a) (D – d)/d (b) d/D
(c) (D – d)/D (d) D/d

TSPSC AEE 2017

Ans. (d) :
Drawing ratio (DR) of a cylindrical shaped part $= \frac{D}{d}$ for cylindrical shaped.

1745. .....is a mechanical deformation technique of reducing or shaping the cross-section of rods or tubes by means of repeated impacts or blows.
(a) Rolling (b) Drawing
(c) Stamping (d) Swaging

HPPSC AE (PWD) 10.09.2016

Ans. (d) : Swaging is a forging process that involves the use of a compressive forces to deform and manipulate the shape of a work piece by a die. During this forging process a worker forces the workpiece through a die. As the workpiece is pushed through the die it takes the die's shape.

1746. Which of the following is not a drawing process
(a) Tube drawing (b) Lancing
(c) Spinning (d) Embossing

Nagaland PTSC CTSE 2017 Paper-I

Ans. (b) : Lancing is not a drawing operation here, lancing is a piercing operation in which the work-piece is sheared and bent with one strike of the die. A key part of this process is that there is not reduction of material, only a modification in its geometry. Lancing operations used to make tabs, vents and louvers.

1747. One of the advantages given below of the forged parts is not correct, find it :
(a) High production rate
(b) Higher forging head
(c) Less die water
(d) Lower forging temperature

TRB Polytechnic Lecturer 2017

Ans. (c) : High die wear occure in the forged parts.

1748. For sheet metal cutting operations, the clearance between the punch and the die should be
(a) 1 to 2 mm
(b) 2 to 5 mm
(c) 1 to 2% of punch diameters
(d) 2 to 10% sheet thickness

TSPSC AEE 2017

Ans. (d) : For sheet metal cutting operations, the clearance between the punch and the die should be 2 to 10% of sheet thickness, to get better edge finish.

1749. Which of the following processing techniques does not produce continuous product?
(a) Extrusion (b) Pultrusion
(c) Calendaring (d) Injection molding

UPRVUNL AE 07.10.2016

Ans. (d) : Pultrusion process—Pultrusion is a continuous moulding process whereby reinforcing fibers are saturated with a liquid polymer resin and then carefully formed and pulled through a heated die to form a part. Pultrusion results in straight constant cross-section parts of virtually any shippable length.
Calendaring process—A calendar is a series of hard pressure rollers used to finish or smooth a sheet of material such as paper, textiles or plastics. Calendar rolls are also used to form some type of plastic films.
Injection molding processing technique does not produce continuous product.

6. Unconventional Manufacturing Methods

1750. In EDM, the material of the tool is
(a) Diamond (b) High Speed Steel
(c) Copper (d) Tungsten Carbide

Mizoram PSC (Insp. of Factories 2019 Paper-III)
GPSC ARTO PRE 30.12.2018, HPPSC AE 2018
GPSC Asstt. Prof. Prod. 21.08.2016, TSPSC AEE 2015
GWSSB AAE 27.12.2015, TNPSC AE 2013
SJVN ET 2013, GATE 1999
UKPSC AE 2007, 2012 Paper -I
VIZAG STEEL MT 10.06.2012

Ans. (c) : The following are the different electrode materials which are used commonly in the industry.
- Graphite
- Electrolytic oxygen free copper
- Brass
- Tellurium copper - 99% Cu + 0.5% tellurium.

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1751. In ultrasonic machining process, the material removal rate will be higher for materials with :

(a) higher toughness
(b) higher ductility
(c) lower toughness
(d) higher fracture strain

CGPSC AE 15.01.2021
UPPSC AE 13.12.2020 Paper-I
Vadodara Muncipal Corp. DEE, 2018
ESE 2018, GATE 1993, 2010
TNPSC AE 2017, MPPSC AE 08.11.2015
TRB Asstt. Prof. 2012, UKPSC AE 2007 Paper-I

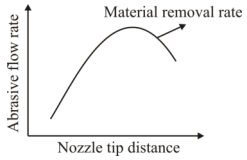
Ans. (c) : In USM (Ultrasonic machining) process, the material removal rate will be higher for materials with lower toughness.

1752. In abrasive jet machining, as the distance between the nozzle tip and the work surface increases, the material removal rate:

(a) Increases continuously
(b) Decreases continuously
(c) Increases, becomes stable and then decreases
(d) None of the above

UP Jal Nigam AE 2016, MPPSC AE 2016
TSPSC Managers, 2015, TNPSC AE 2014, GATE 2012

Ans. (c) : The figure shows the abrasive jet machining for the distance between the workpiece surface and the nozzle tip and flow rate of abrasive.



We can see that the material removal rate is increase due to increase in distance of jet then becomes stable and suddenly starts decreasing due to decrease in the velocity of jet.

1753. The machining process which needs vacuum for its operation is

(a) Electron beam machining
(b) Electrical discharge machining
(c) Electro chemical machining
(d) Plasma machining

GPSC DEE Class-2 (GWSSB) 04.07.2021
UPRVUNL AE 05.07.2021
Oil India Ltd. Sr. Engg. Production 30.11.2019
GPSC Asstt. Director of Transport 05.03.2017
CIL MT 26.03.2017, GPSC AE 26.07.2015
TNPSC AE 2013, GATE 2016, ESE 1993
TRB Poly Lect. 2012

Ans. (a) : Vacuums must be used to avoid contamination and minimize electron collisions with air molecules. EBM is best suitable for small parts. The interaction of the electron beam with the work piece produces hazardous x-rays, and only highly trained personnel should use EBM equipment.

1754. The machining process in which the workpiece is dissolved into an electrolyte solution is known as

(a) electro-chemical machining
(b) electro-discharge machining
(c) laser machining
(d) electron beam machining

OPSC AEE 2019 Paper-I, TNPSC AE 2019
GPSC Asstt. Prof. (Prod.) 21.08.2016
APPSC Poly Tech. Lect. 2013,
ISRO Scientist/Engg. 2009
OPSC Civil Services Pre 2006

Ans. (a) : Electrochemical machining.
A shaped tool or electrode is used as cathode and workpiece as anode. Workpiece is dissolved into an electrolyte solution by anodic dissolution. Electrolyte is pumped at high pressure and low voltage direct current is used.

1755. Which of the following non-conventional machining methods does not cause tool wear?

(a) Anode mechanical machining
(b) Ultrasonic machining
(c) Electro-discharge machining
(d) Electro-Chemical Machining

Mizoram PSC (Insp. of Factories) 2019 Paper-III
UPPSC AE 12.04.2016 Paper-I
MPPSC AE 2016, NPCIL ET 2015, ESE 1997

Ans : (d) Electro-chemical machining methods does not cause tool wear**.**
Electrochemical Machining (ECM):- This process is an extension of already known process of electroplating with modification in a reverse direction. A shaped tool or electrode is used in the process which forms cathode and workpiece forms anode. A small gap is maintained between the tool and workpiece and an electrolyte is pumped through it. Low voltage direct current is used which in the presence of electrolyte enables a control removal of metal form workpiece by anodic dissolution. The tool is provided with a constant feed motion towards workpiece. the electrolyte is pumped at a high pressure through the small gap between tool and work piece.

1756. During ultrasonic machining the metal removal is achieved by :

(a) high frequency eddy currents
(b) high frequency sound waves
(c) hammering action of abrasive particles
(d) rubbing action between tool and work piece

RPSC IOF 2020
Mizoram PSC (Insp. of Factories 2019 Paper-III)
(KPSC AE. 2015), ESE 2006
Kerala PSC AE 06.08.2015, CSE Pre 1996

Ans : (c) During ultrasonic machining the metal removal is achieved by hammering action of abrasive particles.
Ultrasonic machining:-Ultrasonic waves are sound waves which propagate at frequency of 18 to 30 lc/sec and are thus beyond the human ear to respond, no heat is generate in case of ultrasonic machining and it provides better working performance. This process is best suited for machining circular and non-circular holes in hard to machine and brittle metals whether Conducting or non-Conducting Ultrasonic waves can be generated by piezoelectric or magnetostrictive effects.

1757. Following electrolyte is used in electro-chemical machining process:

(a) kerosene (b) transformer oil
(c) brine solution (d) water

HPPSC WS Poly. 2016, GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015, UKPSC AE 2007 Paper -I

Ans. (c) : Brine solution

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1758. Which of the following processes has very high material removal rate efficiency?
(a) Electron beam machining
(b) Electrochemical machining
(c) Electro discharge machining
(d) Plasma arc machining

GPSC ARTO Pre 30.12.2018
GPSC Asstt. Director of Transport 05.03.2017
UJVNL AE 2016, ESE 2012

Ans. (a) : Descending order of metal removal rate efficiency –
LBM > EBM > USM > ECM > EDM > PAM > AJM

1759. In electro chemical machining the material removal is due to
(a) Corrosion (b) Erosion
(c) Fusion (d) Ion displacement

GPSC DEE Class-2 (GWSSB) 04.07.2021
TSPSC Manager (Engg.) HMWSSB 12.11.2020
TSPSC Manager (Engg.) HMWSSB 12.11.2020
Mizoram PSC (Insp. of Factories) 2019 Paper-II
MPPSC AE 08.11.2015, UKPSC AE 2013 Paper-I
VIZAG STEEL MT 18.08.2013, 10.06.2012, GATE 2001

Ans. (d) : In electro chemical machining (ECM) the removal of material is due to Ion displacement.
Electro Chemical Machining– This process, in fact is an extension of an already known process of electroplating with some modifications, but reverse in direction.

Machining process	**Mechanism of material removed**
• ECM	- Ion displacement
• Chemical machining	- Corrosive reaction
• EDM	- Fusion and vaporization
• Ultrasonic machining	- Erosion
• LBM	- Vaporization and ablation
• PAM	- Melting and Partial vaporization
• AJM	- Chipping and erosion

1760. Holes in nylon button are made by:
(a) EDM (b) CHM
(c) USM (d) LBM

GPSC DEE, Class-2 (GWSSB) 04.07.2021
HPPSC ADF 05.03.2019
TNPSC AE (Industries) 09.06.2013

Ans. (d) : Holes in nylon button are made by LBM.

1761. The mechanism of metal removal in EDM process is
(a) Melting and evaporation
(b) Melting and corrosion
(c) Erosion and cavitation
(d) Cavitation and evaporation

PPSC Asstt. Municipal Engg. 15.06.2021
MECON MT 2019, TSPSC AEE 2017
HPPSC Asstt. Prof. 18.11.2016, GATE 2004

Ans. (a) : The metal removal in EDM is done by melting and evaporation of the material at high temperature.

1762. Match List I with List II and select the correct answer using the codes given below the lists:

List I **Machining process**	**List II** **Associated medium**
A. USM	**i. Kerosene**
B. EDM	**ii. Abrasive slurry**
C. ECM	**iii. Vacuum**
D. EBM	**iv. Salt solution**

Code:

	A	**B**	**C**	**D**
(a)	iv	iii	ii	i
(b)	i	ii	iii	iv
(c)	ii	i	iv	iii
(d)	ii	iii	iv	i

APPSC Poly. Lect. 2013
ISRO Scientist/Engg. 2009, ESE 1998

Ans. (c) :

Machining Process	**Associated medium**
(1) USM	Abrasive slurry
(2) EDM	Kerosene
(3) ECM	Salt Solution
(4) EBM	Vacuum

1763. Electron beam machining process is suitable for the following type of material-
(a) low melting point and high thermal conductivity
(b) low melting and low thermal conductivity
(c) high melting point and low thermal conductivity
(d) high melting point and high thermal conductivity

ISRO Scientist/Engineer (RAC) 29.11.2015
TNPSC AE (Industries) 09.06.2013

Ans : (c) Electron beam machining process- Electron beam machining (EBM) is a thermal non-traditional machining process that uses a beam of high-energy electrons focused on the workpiece to melt and vapourise metal. This micro machining process is performed in a vacuum chamber [under vacuum pressure of order 10^{-5}] for (mm/Hg)] and suitable for high melting point and low thermal conductivity material.

1764. Metal removal in Abrasive Jet Machining (AJM) is due to:
(a) Erosion (b) Machining
(c) Melting (d) Grinding

Oil India Limited Sr. Engineer (Drilling) 30.11.2019
TNPSC AE 2013

Ans. (a) : In abrasive jet machining material is removed by the erosive action of high velocity stream of fine particles imparting on the work surface. High pressure air or gas is supplied to the mixing chamber containing the fine abrasive particle. These high pressure abrasive particles and gas mixture are passed through the small nozzle and the erosion caused by their impact at high speed removes the metal from workpiece.

1765. Ultrasonic machining is best suited for
(a) Non-ferrous metal
(b) Amorphous solids
(c) Hard and Brittle materials
(d) Dielectrics

Gujarat PSC AE 2019, TNPSC AE 2018
TSPSC AEE 2015, Kerala PSC Asstt. Prof. 27.09.2016
GWSSB AAE 27.12.2015, (HPPSC AE 2014)
UKPSC AE 2013 Paper–I, APPSC AEE 2012

Ans : (c) Ultrasonic machining- Ultrasonic machining (USM) are suitable methods for making through holes in a glass plate.
In ultrasonic machining, a vibrating formed tool is employed which vibrates at frequencies 20 to 30 kHz with amplitude between 0.01 and 0.06 mm and is pressed lightly on to the work surface while an abrasive

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slurry (such as boron carbide mixed with water) is made to flow under pressure through the gap between the tool face and the work surface to be machined. The tool has the same shape as that of the cavity to machined.

1766. For unconventional machining process, following is the correct sequence of decreasing order of Material Removal Rate (MRR)
(a) ECM > EDM > USM
(b) ECM > USM > EDM
(c) USM > EDM > ECM
(d) USM > ECM > EDM

TSPSC AEE 2015
TSP Managers 2015

Ans. (a) : ECM > EDM > USM

1767. Which of the following non-traditional machining process has highest material removal rate?
(a) Chemical machining
(b) Laser beam machining
(c) Electro discharge machining
(d) Electro chemical machining

Oil India Senior Officer 23.12.2020

Ans. (d) :

1768. In EDM, the tool and work piece are separated by
(a) Positive ions (b) Negative ions
(c) Metal conductor (d) Dielectric fluid

Gujarat PSC AE 2019
UKPSC AE 2013 Paper–I, APGENCO AE 2012

Ans : (d) : Electric discharge machining (EDM) is an advanced machining process which are primarily used for metals difficult to machine with the conventional machining process. Only electrically conducting materials are machined by this process. The EDM process is best suited for making intricates cavities and contours which would be difficult to produce with conventional machining cutting.
The material removed takes place in EDM through a rapid series of electric discharges. The fine pieces of material removed for the workpiece gets flushed away by the continuous flowing dielectric fluid.

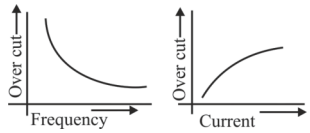
1769. In Electro Discharge machining better surface finish is obtained at
(a) Low frequency and low discharge current
(b) Low frequency and high discharge current
(c) High frequency and low discharge current
(d) High frequency and high discharge current.

CIL MT 27.02.2020
UPPSC AE 12.04.2016 Paper-I
DRDO Scientist 2009

Ans : (c) It Electro Discharge Machining better surface finish is obtained at high frequency and low discharge current.
The Surface produced by this process consist of microscopic craters and the quantity of the machined surface mainly depends on the energy per pulse. If the energy content per pulse is high then the depth of crater will increase causing a poor surface finish and vice-versa. The surface roughness is inversely proportional to the frequency of cutting which is equal to-

$$\frac{1}{(\text{Sparking time} + \text{Charging time})}$$

Sparking time being very small compared to charging it could be said that frequency of cutting = $\frac{1}{\text{Sparking time}}$



1770. Which one of the following processes is the reverse of electroplating process?
(a) EDM (b) ECM
(c) PAM (d) LBM

TNPSC AE 2014, 2018
Kerala LBS Center for Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Electrochemical machining (ECM) process is the reverse of electroplating process.

1771. Which of the following is used as a dielectric medium in EDM?
(a) Salt solution
(b) Silicon carbide solution
(c) AI_2O_3 solution
(d) Kerosene

CIL MT 26.03.2017, UKPSC AE 2012 Paper-I

Ans. (d) : In EDM a fluid is used to act as a dielectric and to help carry away debris. Quite often kerosene based oil is used as dielectric in EDM. The dielectric fluid is circulated through the tool at a pressure of 0.35 N/m^2 or less to free it from eroded metal particles, it is circulated through a filter.

1772. Which of the following statement is wrong about ultra-sonic machining
(a) It is best suited for hard and brittle material
(b) It cuts material at very slow speed
(c) It produces good surface finish
(d) It removes large amount of material

HPPSC Asstt. Prof. 18.09.2017
GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : Wrong Statement-
It removes large amount of material.
In ultrasonic machining due to hammering action abrasive particles material removes which is very less.

1773. Dielectric fluid is used in
(a) electro-chemical machining
(b) ultrasonic machining
(c) electro-discharge machining
(d) laser machining

HPPSC Workshop Suptd. 08.07.2021
TSPSC Manager (Engg.) HMVSSB 12.11.2020
GPSC ARTO 01.05.2016
APPSC AEE 2012, TNPSC AE 2008

Ans. (c) : Dielectric fluids are needed in the gap between tool electrode and workpiece to initiate controlled electrode discharges for achieving removal of material during EDM operation. These dielectric fluids possess high dielectric strength to remain electrically nonconductive.

1774. Which one of the following is most important parameter for EDM ?
(a) Thermal capacity (b) Hardness
(c) Strength (d) Geometry

UKPSC AE 2012, 2007 Paper-I

Ans. (a) : Thermal capacity

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1775. For machining ceramics, glass and plastics which method is not applicable?
(a) AJM (b) LBM
(c) EDM (d) USM
UPPSC AE 13.12.2020 Paper-I
TNPSC AE, 2008

Ans. (c) : AJM (Abrasive Jet Machining), LBM (Laser Beam Machining) and USM (Ultrasonic Machining) used for machining of ceramics, glass and plastics.

1776. The ruby rod used in Laser Beam Machining is made up of
(a) crystalline aluminum oxide or sapphire
(b) copper oxide
(c) zinc oxide
(d) none of these
TNPSC AE, 2008

Ans. (a) : The ruby rod used in Laser Beam Machining is made up of crystalline aluminum oxide or sapphire.

1777. A hole at 1mm is to be drilled in glass. It could best be done by:
(a) Laser drilling (b) Plasma arc drilling
(c) USM (d) EDM process
TSPSC Managers, 2015

Ans. (c) : A hole at 1mm is to be drilled in glass. It could best done by USM (ultrasonic machining).

1778. A big advantage of electro-chemical machining over electro-discharge machining is that
(a) it can cut harder material
(b) it is more accurate
(c) it consumes less power
(d) tool wear is negligible
APPSC Poly. Lect. 2013

Ans. (d) : In ECM (electro-chemical machining) the tool wear is negligible as compare to EDM (electro-discharge machining).

1779. In abrasive jet machining process, the abrasive particles should be
(a) perfectly round
(b) made of diamond power
(c) around 1 mm in size
(d) irregular shape
APPSC Poly. Lect. 2013

Ans. (d) : In abrasive jet machining process, the abrasive particle should have a sharp and irregular shape.

1780. Which one is not the type of EDM?
(a) RAM type EDM (b) Orbital type EDM
(c) Wire cut EDM (d) Straight EDM
APPSC Poly. Lect. 2013

Ans. (b) : Orbital is not the type of EDM, EDM having three types-
(i) Wire cut EDM
(ii) Ram type EDM (also called sinker or cavity EDM) or volume EDM
(iii) Fast load drilling EDM produced (fast, accurate, very small and deep holes)

1781. The surface finish obtainable in ultrasonic machining is of the order of :
(a) 0.2 to 0.5 μm CLA
(b) 2 to 5 μm CLA
(c) 20 to 50 μm CLA
(d) 200 to 500 μm CLA
VIZAG MT 2015

Ans. (a) : The surface finish obtainable in ultrasonic machining is of the order 0.2 to 0.5 μm CLA. In ultrasonic machining, a tool of the desired shape vibrates at an ultrasonic frequency (19-25 kHz) with an amplitude of around 15-20 μm over the workpiece.
The surface finish achieved during the process varies from 0.25 μm to 0.75 μm CLA and it is mainly governed by abrasive grain size.

1782. Which of the following methods uses combination of electrical and chemical energy for machining?
(a) Ultrasonic machining
(b) Abrasive jet machining
(c) Electrochemical machining
(d) Electron beam machining
RPSC IOF, 2020

Ans. (c) : Electrochemical machining uses a combination of electrical and chemical energy for machining.
- Complete the electrical circuit between the tool and workpiece and allow large current pass through it.
- To carry away the heat and waste product of reactions.
- Allow electrochemical reactions at a faster rate.

1783. Non-consumable electrode is employed in
(a) Normal arc welding
(b) Submersed arc welding
(c) MIG Welding
(d) TIG Welding
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) : As we know, In Non-consumable electrodes. They are not consumed during welding, with a separate welding rod is used as filler material, e.g. TIG or GTAW, Atomic hydrogen welding.

1784. The following process is suitable for machining brittle material such as glass :
(a) Electric Discharge Machining (EDM)
(b) Electrochemical Machining (ECM)
(c) Ultrasonic Machining (USM)
(d) Chemical Machining (CHM)
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : USM is used for machining hard and brittle material, which are poor conductors of electricity and thus cannot be processed by ECM or EDM.

1785. Limitation of the electro chemical machining (ECM) process is
(a) Use of corrosive media as electrolytes made it difficult to handle
(b) Poor surface finish
(c) Poor accuracy of the work piece dimensions because of the large tool wear
(d) There will be thermal damage to the work piece
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) ECM has many limitations such as. Electrolyte may cause corrosion of the equipment large production floor is required, only electrically conductive materials may be machined etc.

1786. In abrasive jet machining process, if Q = flow rate of abrasives and d = mean diameter of the abrasives, then MRR is proportional to
(a) Q/d^2 (b) Qd
(c) Qd^2 (d) Qd^3
TSPSC AEE 2017

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Ans. (d) : In abrasive jet machining

$$MRR \propto Qd^3\sqrt{v^3}$$

1787. A process which cuts metal by discharging electric current in a capacitor bank across a thin gap between the tool (cathode) and the work piece (anode), is known as :

(a) Electron beam machining
(b) Electrical discharge machining
(c) Water-jet machining
(d) Laser beam machining

CGPSC AE 15.01.2021

Ans. (b) : A process which cuts metal by discharging electric current in a capacitor bank across a thin gap between the tool (cathode) and the work piece (anode) is known as Electric discharge machining.

Electric Discharge Machining.

- High Voltage, low current.
- Mechanism : Fusion & Vapourization.
- Electrode material : Graphite, Cu, Brass, Steel, Tungsten.
- Kerosene and other mineral oil can be used as dielectric.

1788. In a wire-cut EDM process, a successful cut is made when:

(a) wire and workpiece are conductive materials
(b) wire is conductive and workpiece is non conductive material
(c) wire and workpiece are non conductive materials
(d) wire is non conductive and workpiece is conductive material

VIZAG Steel MT 24.01.2021

Ans. (a) : In wire cut EDM process, a thin metallic wire is fed on to the conducting workpiece, which is submerged in a tank of dielectric fluid such as de-ionized water. Wire is fed in the programmed path and material is cut from the workpiece accordingly.

1789. Which one of the following is an additive manufacturing process?

(a) Electric discharge machining
(b) Electro-magnetic forming
(c) Selective Laser Sintering
(d) Water jet cutting

VIZAG Steel MT 24.01.2021

Ans. (c) : Selective Laser Sintering (SLS), is an additive manufacturing technology that uses a high power laser to sinter small particles of polymer powder into a solid structure based on a 3D model.

1790. …….. is not related to non-conventional machining.

(a) EDM (b) LBM
(c) PAM (d) Machining centre

HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : The aim of the Non-traditional machining is to provide efficient cost effective solution for the manufacture of traditionally difficult to machine material. These process are generally used when traditional methods are not technically or economically feasible.

Non-conventional machining is used mainly for-
(i) Machining of very hard or tough materials
(ii) Machining of very complex slopes.
(iii) High surface finish and accuracy requirement.
(iv) Low cutting forces requirements.

Non- traditional machining processes are-
(i) EDM
(ii) LBM
(iii) PAM etc.

Hence machining centre is not non-traditional machining.

Machining centres are multi-functional machines that allows with great flexibility to carry out an elevated number of mechanical operations, such as milling, drilling, boring etc.

1791. Etchants are used in …… process.

(a) USM (b) EDM
(c) AJM (d) ECM

HPPSC Workshop Supt. 08.07.2021
HPPSC Lect. (Auto) 23.04.2016

Ans. (d) : Etchants are used in electro chemical machining.

Etching is non-traditionally the process of using strong acid or mordant to cut into the unprotected parts of a metal surface to create a design in the metal.

1792. In ultrasonic machining, the material is removed by _____.

(a) thermal dissolution
(b) electrochemical oxidation
(c) anodic dissolution
(d) abrasive action

VIZAG MT, 14.12.2020

Ans. (d) : Process ⟶ Material removed by
(1) USM ⟶ Abrasive action
(2) EDM ⟶ Melting and vaporization
(3) ECM ⟶ Anodic dissolution

1793. Which one of the following statements is TRUE for ultrasonic machining (USM) process?

(a) USM often uses a slurry comprising abrasive particles and water.
(b) USM does not employ magnetostrictive transducer.
(c) USM is an excellent process for machining ductile materials.
(d) In USM, the tool vibrates at subsonic frequency.

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) : In ultrasonic machining tool is made up of ductile material and there is slurry of abrasive and where in between tool and workpiece. This is suitable for machining of brittle materials like glasses and ceramics.

1794. A resistance-capacitance relaxation circuit is used in an electrical discharge machining process. The discharge voltage is 100 V. At a spark cycle time of 25 μs the average power input required is 1 kW. The capacitance (in μF) in the circuit is_____

(a) 10.0 (b) 5.0
(c) 7.5 (d) 2.5

Assam Engg. College AP/Lect. 18.01.2021

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Ans. (b) : Time (t) = 25μs

V = 100 volt

Power (P) = 1 kW = 1000 watt

$$\text{Power (P)} = \frac{\text{Work}}{\text{time}} = \frac{\frac{1}{2}CV^2}{t}$$

$$C = \frac{2t \times P}{V^2}$$

$$= \frac{2 \times 25 \times 10^{-6} \times 1000}{(100)^2}$$

C = 5 μF

1795. The average frequency of spark in EDM is :
(a) 1000
(b) 2000
(c) 5000
(d) 10000 sparks per second

GMB AAE 25.06.2017

Ans. (d) : In electrical discharge machining (EDM). Sparks occur at a frequency on the order of 10,000 sparks per second. In order to yield a good surface finish a randomized spark location is desired, which results in discharge craters evenly distributed on the work piece.

1796. Which of the following is a non-traditional machining method ?
(a) milling
(b) drilling
(c) grinding
(d) ultrasonic machining

J&K PSC Civil Services Pre, 2010

Ans. (d) : Traditional machining process such as turning, drilling and milling.
Non - traditional machining process such as Abrasive Jet Machining (AJM), Ultrasonic Machining (USM), water jet machining (WJM), electrochemical machining, electro-discharge machining etc. Non-traditional machining process also called advanced manufacturing process.

1797. The slurry used in USM process is most commonly_____.
(a) Oil based (b) Alkali based
(c) Water based (d) Ethylene based

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : USM uses abrasive like Boron Carbide, Silicon Carbide, Al_2O_3 in a fluid medium to form a slurry water.

1798. Abrasive jet machining is not suitable for cutting soft materials because abrasive particles tend to_____.
(a) become embedded (b) get deflected
(c) break (d) none of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : Abrasive jet machining is not suitable for cutting soft materials because abrasive particles get embodied on the workpiece.

1799. In ultrasonic machining (USM) the frequency of oscillation is _____.
(a) 10-15 k Hz (b) 5-10 k Hz
(c) Above 15 k Hz (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : USM uses high wave frequency for machining process ranging above 15 kHz with abrasive particles line. BC, SiC and Al_2O_3.

1800. In USM the range of frequency used is
(a) 2-6 kHz (b) 18-30 kHz
(c) 7-9 kHz (d) 10-25 kHz

WBPSC AE, 2007

Ans. (b) : USM uses high wave frequency for machining process ranging above 15 kHz with abrasive particles line. BC, SiC and Al_2O_3. So suitable answer is (b).

1801. In EDM, tool is made of ..... .
(a) stainless steel (b) diamond
(c) tungsten carbide (d) brass

GWSSB DEE 07.07.2016

Ans. (d) :

1802. In EDM, the tool is made up of:
(a) Cast iron (b) Copper
(c) Brass (d) All of the above

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) :

1803. In ultrasonic waves, the frequencies for nondestructive testing of materials are in the range of
(a) 0.5 MHz to 10 MHz (b) 10 MHz to 20 MHz
(c) 20 MHz to 30 MHz (d) 30 MHz to 40 MHz

ESE 2019

Ans. (a) : Generally sound waves range is from 0.5 MHz to 15 MHz but in answer option (a) is close to above.

1804. The vibration frequency used for the tool in the Ultrasonic machining is of the order of-
(a) 10,000 oscillations per second
(b) 20,000 oscillations per second
(c) 35,000 oscillations per second
(d) 45,000 oscillations per second

RPSC AE 2018
UKPSC AE 2012, 2007 Paper-I

Ans. (b) : The vibration frequency used for the tool in the Ultrasonic machining is of the order of 20,000 oscillations per second.

1805. The taper of hole produced by EDM is because of
(a) In accuracy of electrodes size
(b) Large flow of current between electrode bottom and workpiece surface
(c) Side spark between the tool and machined surface
(d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : The taper of hole produced by EDM is because of Side spark between the tool and machined surface.

1806. A re-cast is produced by ____process.
(a) USM (b) EDM
(c) ECM (d) AJM

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : In EDM material removal takes places due to erosion caused by rapid discharge between the tool and work. Positive terminal erodes out faster so workpiece is made anode.

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1807. While removing material from (atomic weight = 56, valency = 2 and density = 7.8 g/cc by electrochemical machining, a metal removal rate of 2 cc/min in desired. The current (in A) required for achieving material removal rate is

(a) 896.07 A (b) 14.96 A
(c) 448.03 A (d) 53764.29 A

APGCL AM, 2021

Ans. (a) : Given:
Atomic weight = 56
Valency = 2

Then chemical equivalent = $\frac{\text{Atomic weight}}{\text{valency}} = \frac{56}{2} = 28$

Density = ρ = 7.8g/cc

MRR = 2 cc/min = $\frac{2}{60}$ cc/sec

We know in ECM $\rightarrow$ MRR = $\frac{eI}{96500 \times \rho}$

$\Rightarrow \frac{2}{60} = \frac{28 \times I}{96500 \times 7.8}$

$\boxed{I = 896.07A}$

1808. With the increase in the abrasive slurry concentration, the MRR in USM will :

(a) first increase and then decrease
(b) first decrease and then increase
(c) first increase and then remain constant
(d) first decrease and then remain constant

APPSC IOF, 2009

Ans. (a) : At the beginning with an increase in the static force MRR increases for various abrasive slurry, and after some time due to increase in tool wear, that causes a decrease in amplitude of vibration, decreases MRR.

1809. In an ECM process for machining iron it is desired to obtain a metal removal rate of 1 cm^3/min. At. wt. of iron = 56 gm, valency = 2, density of iron = 7.8 gm/cm^3, F = 1609 amp-min. The amount of current required in the process is:

(a) 228 Amp (b) 276 Amp
(c) 336 Amp (d) 448 Amp

APPSC IOF, 2009

Ans. (d) : We know, MRR = $\frac{eI}{F\rho}$, e = $\frac{56}{2} = 28$

$1 = \frac{28 \times I}{1609 \times 7.8}$

I = 448 Amp.

1810. In EDM the energy available for the reverse spark is so small because of the resistance that there is no reverse spark, then the frequency of discharge oscillation f_d is given by :

(a) $\frac{1}{2\pi\sqrt{(1/LC)}}$ (b) $\frac{1}{2\pi\sqrt{(LC)}}$
(c) $\frac{1}{3\pi\sqrt{(1/LC)}}$ (d) $\frac{1}{3\pi\sqrt{(LC)}}$

APPSC IOF, 2009

Ans. (b) : In EDM the energy available for the reverse spark is so small because of the resistance that there is no reverse spark, then the frequency of discharge oscillation f_d is given by $\frac{1}{2\pi\sqrt{(LC)}}$.

1811. For machining a complex contour on tungsten carbide work piece, which process will be used:

(a) ECM (b) EDM
(c) USM (d) LBM

APPSC IOF, 2009

Ans. (b) : EDM is used for machining a complex contour on tungsten carbide workpiece.

1812. The process economy is very low for one of the following non-traditional machining process :

(a) USM (b) AJM
(c) ECM (d) EDM

APPSC IOF, 2009

Ans. (a) : USM has low MRR for hardened material and high tool wear and are used for Non-metal cutting.

1813. In plasma arc machining the input values of voltage and amperage is :

(a) 100-200 V, 100-300 A
(b) 10-50 V, 100-300 A
(c) 100-200 V, 50-100 A
(d) 10-50 V, 50-100 A

APPSC IOF, 2009

Ans. (b) : In plasma arc machining the input values of voltage and amperage is 10-50V and current 50 to 350 Amps.

1814. The current density used in ECM is of the order of :

(a) 10000 A/cm^2 (b) 1000 A/cm^2
(c) 200 A/cm^2 (d) 10 A/cm^2

APPSC IOF, 2009

Ans. (c) : In electro Chemical Machining (ECM), current densities are typically of 20 – 200A/cm^2.

1815. Which machining process is developed on the principle of Faraday's and Ohm's laws?

(a) EDM (b) ECM
(c) LBM (d) AJM

AAI Jr. Executive 26.03.2021
Assam Engg. College AP/Lect. 18.01.2021
Karnataka PSC Lect., 27.05.2017
CGPSC Poly. Lect. 22.05.2016
DRDO Scientist 2008

Ans (b) : This process, in fact is an extension of an already known process of electroplating with some modifications, but in a reverse direction. A shaped tool or electrode is used in the process, which forms cathode. The workpiece form anode. A small gap is maintained between the tool and workpiece and an electrolyte is pumped through it. The principle of ECM process is based on Faraday's laws of electrolysis.

1816. In electro chemical machining,

(a) The gap between tool and workpiece is 0.2 mm while tool is connected to cathode and electrolyte flows at a velocity of 50 m/s
(b) When tool is connected to cathode the electrolyte is static while gap between workpiece and tool is 0.25 mm.

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(c) The gap between tool and workpiece is 0.25 mm while tool is connected to anode, the electrolytic pressure is maintained at 20 bar
(d) The gap between tool and work piece is 0.25 mm while tool is connected to anode, the electrolyte is maintained at atmospheric pressure

Karnataka PSC Lect., 27.05.2017

Ans (a) : In electro chemical machining the gap between tool and workpiece in 0.2 mm while tool is connected to cathode and electrolyte flows at a velocity of 50 m/s

1817. Match List-I (materials) with List-II (machining process)

List-I (Material)		List-II (Machining process)	
A.	Conducting material	**I.**	ECM
B.	Ruby rod	**II.**	EDM
C.	Electrolyte	**III.**	USM
D.	Abrasive slurry	**IV.**	LBM

Select the code for the correct answer from the options given below :

	A	B	C	D
(a)	IV	II	I	III
(b)	IV	II	III	I
(c)	II	IV	III	I
(d)	II	IV	I	III

Karnataka PSC AE, 10.09.2017, ESE 2003

Ans. (d) :
- EDM requires conducting material.
- LBM have Ruby rod
- ECM have electrolyte for completing circuit.
- USM have abrasive slurry for machining.

1818. Match the processes according to power consumed :

Process		Order of power consumed	
A.	**Ultrasonic machining**	**1.**	**2700 kW/ cm^3/ min**
B.	**Electro chemical machining**	**2.**	**450 kW/cm^3/ min**
C.	**Electrical discharge machining**	**3.**	**9 kW/cm^3/ min**
D.	**Electron beam machining**	**4.**	**7 kW/cm^3/ min**
E.	**Laser beam machining**	**5.**	**2 kW/cm^3/ min**

	A	B	C	D	E
(a)	4	3	1	5	2
(b)	3	4	5	2	1
(c)	5	2	4	1	3
(d)	1	5	2	3	4

KPSC ADF 2015

Ans. (d) :

1819. The cutting tool used in the spark erosion machining process is called
(a) Arc (b) Capacitor
(c) Electrode (d) Dielectric

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : Spark erosion machining is typically used when producing intricate pieces, by removing material from a piece with current discharge between two electrode that are separated by a dielectric liquid, which is subjected to voltage. So, electrode is used as tool here.

1820. What is the process of removal of material by the impingement of particles at high velocity on component surfaces called?
(a) Corrosive wear (b) Erosive wear
(c) Fretting wear (d) Adhesive wear

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : In Erosive wear mechanism material is removed by the impingement of particles at high velocity on workpiece surface.

1821. Arrange the processes in the increasing order of their maximum material removal rate
(1) Electrochemical Machining (ECM)
(2) Ultrasonic Machining (USM)
(3) Electron Beam Machining (EBM)
(3) Laser Beam Machining (LBM) and
(5) Electric Discharge Machining (EDM)
(a) USM, LBM, EBM, EDM, ECM
(b) EBM, LBM, USM, ECM, EDM
(c) LBM, EBM, USM, ECM, EDM
(d) LBM, EBM, USM, EDM, ECM

GPSC ARTO Pre 30.12.2018
HPPSC Asstt. Prof. 18.11.2016, GATE 2006

Ans. (d) : Increasing order of maximum material removal rate –
LBM < EBM < USM < EDM < ECM < PAM

1822. Of the following processes, which one is noted for the highest material removal rates?
(a) Electric discharge machining
(b) Electrochemical machining
(c) Laser beam machining
(d) Plasma arc cutting

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Plasma arc cutting having highest temperature as compared to all these so it melts large amount of material, and its material removal rate is highest.
Decreasing order of MRR of various unconventional machining process –
PAM > ECM > EDM > USM > AJM > EBM > LBM

1823. Kerosene is required while machining
(a) aluminum (b) magnesium alloy
(c) brass (d) low carbon steel

OPSC Civil Services Pre 2006

Ans. (a) : Paraffin or Kerosene can be used for machining material such as Aluminium.

1824. In chemical machining, the etch factor is expressed as:
(a) tool wear/workpiece wear
(b) Workpiece wear / tool wear
(c) Undercut/depth of cut
(d) Depth of cut/Undercut

CIL MT 27.02.2020, GATE 2017

Ans. (c) : Etch factor is the ratio between the difference in the widths of the widest and narrowest parts of the feature after etching is completed and the thickness of metal etched.

$$\text{Etch factor} = \frac{\frac{P-S}{2}}{T} \times 100\%$$

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1825. In EDM process, the workpiece is connected to:

(a) positive (b) negative
(c) earth (d) None of the above

BPSC Asstt. Prof. 29.11.2015

Ans. (a) : In EDM a potential difference is applied between is tool and workpiece. Both the tool and the work material are to be conductors of electricity. Generally the tool is connected to the negative terminal of the generator and the work piece is connected to positive terminal.

1826. In electro-chemical machining, the gap between the tool and work is kept as :

(a) 0.1 mm (b) 0.25 mm
(c) 0.4 mm (d) 0.75 mm

Nagaland PSC (CTSE) 2018 Paper-I
GMB AAE 25.06.2017
OPSC Civil Services Pre. 2011

Ans. (d) : In electro chemical machining the gap between the tool and work is kept as 0.75 mm.

1527. In Electro-Discharge machining, the work piece is connected to

(a) Cathode (b) Anode
(c) Earth (d) Electrolyte

GPSC DEE, Class-2 (GWSSB) 04.07.2021
TNPSC AE 2017, UKPSC AE 2012 Paper–I

Ans. (b) : In Electro-Discharge machining, the work piece is connected to Anode whereas EDM tool is connected to cathode.

1828. Laser is produced by

(a) graphite (b) diamond
(c) ruby (d) emerald

HPPSC Workshop Suptd. 08.07.2021
BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Laser is the term used for the phenomenon of Amplification of light by stimulated emission of radiation. The setup consist of a stimulating light source (like xenon flash lamp) and a lasser rod.

The light radiated from the flash lamp is focused on to the laser rod (Laser tube) from where it is reflected on a accelerated in the path. This light is emitted in the form of a slightly divergent beam.

1829. .....non-conventional machining process in which "Xenon flash light" is some times used.

(a) PAM (b) AJM
(c) LBM (d) USM

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : Laser is the term used for the phenomenon of Amplification of light by stimulated emission of radiation. The setup consist of a stimulating light source (like xenon flash lamp) and a lasser rod.
The light radiated from the flash lamp is focused on to the laser rod (Laser tube) from where it is reflected on a accelerated in the path. This light is emitted in the form of a slightly divergent beam.

1830. AJM is best suited for machining ..... .

(a) nano-ductile (b) brittle materials
(c) plastics (d) fibres

HPPSC Asstt. Prof. 29.10.2016

Ans. (b) : AJM is best suited for machining brittle materials. No heat generation in AJM

1831. Match the following components with the appropriate machining process :

Component	Process
A. Square hole in a high strength alloy	(i) Milling
B. Square hole in a ceramic component	(ii) Drilling
C. Blind holes in a die	(iii) ECM
D. Turbine blade profile on a high strength alloy	(iv) Jig boring
	(v) EDM
	(vi) USM

	A	B	C	D
(a)	(iii)	(iv)	(v)	(vi)
(b)	(v)	(vi)	(iv)	(iii)
(c)	(i)	(ii)	(v)	(vi)
(d)	(iv)	(iii)	(ii)	(vi)

MPSC Poly. Lect. 2014, GATE 1992

Ans. (b) : Square hole in a high strength alloy – EDM.
Square hole in a ceramic component – USM
Blind holes in a die. – Jig boring.
Turbine blade profile on a high strength alloy = ECM.

1832. The metal is removed by using abrasive slurry between the tool and the work in the process known as

(a) ultrasonic machining (USM)
(b) electro chemical machining (ECM)
(c) electro discharge machining (EDM)
(d) plasma are machining (PAM)

Nagaland PSC (CTSE) 2018, Paper-I
WBPSC AE 2003

Ans. (a) : (USM) Ultrasonic Machining Process- It is the process of removing material used to erode holes and cavities in the hard or brittle work piece by using shaped tools, high –frequency mechanical motion and an abrasive slurry.

1833. Thermo electric energy is used in

(a) ion beam machining (IBM)
(b) laser beam machining (LBM)
(c) electro beam machining (EBM)
(d) electro discharge machining (EDM)

WBPSC AE 2003

Ans. (d) :

Energy	Source	Process
Thermal	Electric spark	EDM
	High speed electron	EBM
	Power full radiation	LBM
	Ionized substance	IBM
	High temperature plasma	PAM

1834. The type of energy used in electrical discharge wire cutting is :

(a) Mechanical energy
(b) Chemical energy
(c) Electrical energy
(d) Thermal energy

TRB Polytechnic Lecturer 2017

Ans. (c, d) : Electrical energy and Thermal energy

1835. In EDM which factor is essential?

(a) Workpiece should be conductor of heat
(b) Workpiece should be magnetic
(c) Workpiece should be conductor of electricity
(d) Workpiece should be non-magnetic

HPPSC Asstt. Prof. 2014

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Ans. (c) : In EDM workpiece must be electrically conductive.

1836. In chemical machining, if selective machining is desired, the portions required to be left unmachined are covered with a resistant material, called a ----------. Which can be stripped away after machining.

(a) Resist (b) Abrasive
(c) Felt (d) Composite

MPSC Lect. (Prod.) 2018

Ans. (a) : Areas from where the material is not to be removed are protected by an etching resistant material, known as 'maskant' or 'resist'.

1837. Which of the following is an advantage of 'Laser beam machining'?

(a) Unskilled operators are needed
(b) High production rate
(c) Low capital investment required
(d) The workpiece is not subjected to large mechanical forces

MPSC Lect. (Prod.) 2018

Ans. (d) : The workpiece is not subjected to large mechanical forces is main advantage of laser beam machining.

1838. What is the value of dielectric constant of vacuum?

(a) Infinity (b) Unity
(c) Zero (d) Less than unity

UPRVUNL AE 2014

Ans : (b)
The value of dielectric constant in vacuum is 1.

$$(E_T) = \frac{E}{Eo}$$

Where E = Permittivity of the substance
Eo = Permittivity of vacuum.

1839. Match the different non-traditional machining processes given in Group-A with their respective basic mechanism of metal removal in Group-B and select the correct answer from the options given below the lists :

Group-A	Group-B
A. USM	(i) Vaporization by thermo-electric method
B. ECM	(ii) Ion-displacement by electrochemical method
C. EDM	(iii) Erosion by mechanical abrasion
D. LBM	(iv) When heated, melts and vaporizes instantly by thermoelectric method

(a) A-ii, B-iii, C-i, D-iv
(b) A-I, B-iii, C-ii, D-iv
(c) A-iv, B-iii, C-ii, D-i
(d) A-iii, B-ii, C-I, D-iv

UPRVUNL AE 2014

Ans : (d) USM –Erosion by mechanical abrasion
ECM – Ion-displacement by electrochemical method
EDM –Vaporization by thermo-electric method
LBM –When heated, melts and vaporizes instantly by thermoelectric method

1840. The process utilizing mainly thermal energy for removing material is

(a) Ultrasonic Machining
(b) Electrochemical Machining
(c) Abrasive Jet Machining
(d) Laser Beam Machining

UPRVUNL AE 05.07.2021, GATE 2014
GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : The machining by laser beam is achieved through thermo-optic interaction between the beam and work material.

1841. During the electrochemical machining (ECM) of iron (atomic weight = 56, valency = 2) at current of 1000 A with 90% current efficiency, the material removal rate was observed to be 0.26 gm/s. If Titanium (atomic weight = 48, valency = 3) is machined by the ECM process at the current of 2000 A with 90% current efficiency, the expected material removal rate in gm/s will be

(a) 0.11 (b) 0.23
(c) 0.30 (d) 0.52

GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (c) : Material Removal Rate (MRR) $= \frac{AI \times \eta}{\rho ZF}$

Given, A = 56, I = 1000 A, η = 90%
MRR = 0.26, Z = 2, F = 96500

$$0.26 = \frac{56 \times 1000 \times 0.9}{\rho \times 2 \times 96500}$$

$\rho = 0.9956$

Now expected MRR $= \frac{48 \times 2000 \times 0.9}{0.9956 \times 3 \times 96500} = 0.30$ gm/s

1842. The Laser Beam Machining process can be carried out, when the media for energy transfer between tool and workpiece is

(a) Air
(b) Liquid
(c) Vacuum
(d) Any of the above medium

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : The Laser Beam Machining process can be carried out, when the media for energy transfer between tool and workpiece is any of the above medium.

1843. When the metal is removed by erosion caused by rapidly recurring spark discharges between the tool and work, the process is known as

(a) Electro chemical machining
(b) Ultra-sonic machining
(c) Electro-discharge machining
(d) Electro-static machining

ISRO Scientist/Engineer (RAC) 22.04.2018
TSPSC AEE 2015

Ans. (c) : In electro-discharge machining the metal is removed by erosion caused by rapidly recurring spark discharges between the tool and work.

1844. The metal removal rate in abrasive jet machining process is

(a) high (b) very high
(c) low (d) very low

ISRO Scientist/Engineer 07.05.2017

Search on Telegram @apna_pdf

Ans. (c) : AJM (Abrasive Jet Machining)–

- The metal removal rate is low.
- Stray cutting could not be avoided (low accuracy of ± 0.1 mm)
- The tapering effect may occur especially when drilling in metals.
- The abrasive may jet impeded in the work surface.
- Suitable dust-collecting systems should be provided.
- Soft materials can't be machined by the process.
- Silica dust may be a health hazard.
- Ordinary shop air should be filtered to remove moisture and oil.

Machining	Metal removal rate (cm^3/sec)
Plasma Jet	1.5
ECM	1
EDM	0.1
USM	0.005
EBM	0.001
AJM	0.0001
LBM	0.0001

1845. In ultrasonic machining process, the material removal rate would

(a) increase
(b) decrease
(c) increase and then decrease
(d) decrease and then increase

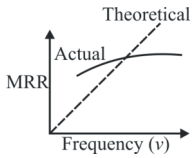
ISRO Scientist/Engineer 11.10.2015

Ans. (a) : USM (Ultra Sonic Machining)–The basic USM process involves a tool made of ductile and tough material vibrating with a very high frequency and a continuous flow of an abrasive slurry in the small gap between the tool and the work surface.

Assumptions are–

- The rate of work material removal is proportional to the volume of work material per impact.
- The rate of work material removal is proportional to the number of particles making impact per cycle.
- The rate of work material removal ϕ is proportional to the frequency.
- All impacts are identical.
- All abrasive grains are identical and spherical in shape.

Frequency– The material removal rate increase linearly with frequency.



- When the amplitude of vibration is increased, the MRR is expected to increase.

1846. A titanium sheet of 5.0 mm thickness is cut by wire-cut EDM process using a wire of 1.0 mm diameter. A uniform spark gap of 0.5mm on both sides of the wire is maintained during cutting operation. If the feed rate of the wire into the sheet is 20 mm/min, the material removal rate (in (mm^3/min) will be

(a) 150 (b) 200
(c) 300 (d) 400

ISRO Scientist/Engineer 12.05.2013, GATE 2009

Ans. (b) : $MRR = V_f hb$

V_f (feed rate of wire) = 20 mm/min
h (titanium sheet thickness) = 5 mm
b (kerf) = $d_w + 2S$
d_w = wire diameter
S = gap
$b = 1 + 2 \times 0.5$
b = 2mm
$\therefore$ $MRR = 20 \times 5 \times 2$
$= 200\ mm^3/min$

1847. Laser Beam machining is suitable for :

(a) Metal Cutting (b) Heat Treatment
(c) Welding (d) All of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (d) : Laser beam can be used for welding, cladding, marking, surface treatment, drilling and cutting among other manufacturing processes. It is used in the automobile, shipbuilding, aerospace etc.

1848. _____setup consists of the tank in which the dielectric fluid is filled.

(a) Electrochemical machining
(b) Ultrasonic machining
(c) Electro discharge machining
(d) Laser machining

MECON MT 2019

Ans. (c) : Electro discharge machining setup consists of the tank in which the dielectric fluid is filled.

1849. Which one of the following statements is correct in respect of unconventional machining processes?

(a) The cutting tool is in direct contact with the job
(b) The tool material needs to be harder than the job material
(c) The tool is never in contact with the job
(d) There has to be a relative motion between the tool and the job

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : Unconventional machining process– These are known as unconventional method because compared to conventional method there is no contact between tool and workpiece specific power consumption is very large, Metal Removal Rate (MRR) is low and is used for machining very hard materials for precision work.

1850. In Electrochemical machining (ECM), rate of material removal is:

(a) Inversely proportional to workpiece hardness
(b) Directly proportional to workpiece hardness
(c) Independent of the workpiece hardness
(d) Directly proportional to square of the hardness

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
TNPSC AE 2008

Ans. (c) : In Electro chemical machining (ECM) material removal takes place due to atomic dissociation of work material which is governed by Faraday's laws.

$$MRR = \frac{\dot{M}}{\rho} = \frac{M}{t\rho} = \frac{IA}{F\rho V}; MRR = f(I, A, \rho, V)$$

Where,

$\dot{M}$ = It A/FV = mass of material dissolve rate

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I = Current, A = Atomic weight, V= Valency
F = Faraday constant (96500C)
ρ = Density of material
So, $(MRR)_{ECM}$ is independent of the workpiece harness.

1851. Which of the following processes has the highest metal removal rate?
(a) Ultrasonic machining
(b) Abrasive machining
(c) Chemical machining
(d) Electron beam machining

Gujarat PSC AE 2019

Ans : (c) : Material removal rate (MRR) is highest in case of electrochemical machining (ECM) and minimum in laser beam machining (LBM). The correct sequence in increasing order is as follows-

ECM > EDM > USM > EBM > LBM

1852. In electro chemical machining, the metal is removed by
(a) dissolution (b) evaporation
(c) sputtering (d) shearing

Karnataka PSC Lect., 27.05.2017
TNPSC AE 2013

Ans. (a) : ECM– ECM is a non-traditional machining process belonging to electrochemical category. ECM is opposite of electrochemical or galvanic coating or deposition process. In ECM, material removal takes place due to atomic dissolution of work material.

1853. The function of servo mechanism in EDM process is to:
(a) Clean the dielectric fluid
(b) Provide dielectric fluid of required flow rate
(c) Provide power to generate the spark
(d) Maintain a small gap between tool and work piece

UPRVUNL AE 07.10.2016

Ans. (d) : The function of servo mechanism in EDM process is to maintain a small gap between tool and work piece.

1854. Which of the following is incorrect for plasma arc machining?
(a) Temperature generated approximately upto 1000ºC
(b) Power supply with approximate range of current 100 - 600 amp
(c) Power supply with approximate range of voltage 50 - 200 V
(d) All electrically conducting materials can be machined

UPRVUNL AE 07.10.2016

Ans. (a) : In plasma arc machining temperature generated approximately upto 11000ºC to 28000ºC.

1855. Which of the following is incorrect for Ultrasonic machining?
(a) Tool is made of ductile material
(b) Very low material removal rate
(c) Abrasive are not used
(d) Tool vibrates with very high frequency

UPRVUNL AE 07.10.2016

Ans. (c) : Ultrasonic machining—Ultrasonic machining is a non-traditional process, in which abrasive contained in a slurry are driven against the work by a tool oscillating at a low amplitude (25-100 microns) and high frequency (15-30 kHz). In ultrasonic machining tool is made of ductile material and tool vibrates with very high frequency to remove very low material. (low material removal rate)

1856. In ECM (Electro Chemical Machining):
(a) As the cutting forces are very high, thin section of work piece cannot be machined
(b) Tool material is conductor of electricity
(c) Surface finish is bad
(d) Tool wear is high

UPRVUNL AE 07.10.2016

Ans. (b) : Electro Chemical Machining (ECM)—In ECM process in which electrochemical process is used to remove materials from the workpiece. In the process, workpiece is taken as anode and tool is taken as cathode. The two electrodes workpiece and tool is immersed in an electrolyte (such as NaCl). When the voltage is applied across the two electrodes the material removal from the workpiece starts. The workpiece and tool is placed very close to each other without touching. In ECM the material removal takes place at atomic level so it produces a mirror finish surface.

1857. In electrochemical machining, which of following material tools is NOT used?
(a) Aluminum (b) Copper
(c) Brass (d) Stainless steel
(e) Diamond

(CGPSC Polytechnic Lecturer 2017)

Ans. (e) : Electrochemical Machining (ECM)–ECM is a machining process in which electrochemical process is used to remove materials from the workpiece. In this process, workpiece is taken as anode and tool is taken as cathode. The two electrodes workpiece and tool is immersed in an electrolyte. When the voltage is applied across the two electrodes, the material removal from the workpiece starts. The tool & workpiece is placed very close to each other without touching.
In ECM, most common tool materials are brass, copper and bronze, but stainless steel, titanium, sintered copper/tungsten, aluminium and graphite are also used. Tool life is longer as compared with EDM and USM, since there is no appreciable tool wear.

1858. The following four unconventional machining processes are available in a shop floor. The most appropriate one to drill a hole of square cross section of 6 mm × 6 mm and 25 mm deep:
(a) Is Electro discharge Machining
(b) Is Plasma Arc Machining
(c) Is Laser Beam Machining
(d) None of the above

UP Jal Nigam AE 2016, GATE 2014

Ans. (a) : Since the depth of hole is mentioned i.e., hole is blind so electric discharge machining is the best for any type of die sinking, cavity sinking, blind complex cavities etc.

1859. In Electric discharge machining, temperature produced by the spark between tool and work piece while machining will be in the order of _____.
(a) 10^oC (b) 10000^oC
(c) 100^oC (d) 1000^oC

CIL MT 26.03.2017

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Ans. (b) : Electro Discharge Machining (EDM) is an electro-thermal non-traditional machining process, where electrical energy is used to generate electrical spark and material removal mainly occurs due to thermal energy of the spark. The kinetic energy of the electrons and ions on impact with the surface of the job and tool respectively would be converted into thermal energy or heat flux. Such intense localized heat flux leads to extreme instantaneous confined rise in temperature which would be in excess of 10,000°C. Such localized extreme rise in temperature leads to material removal.

1860. Which of the following is not the unconventional machining process?
(a) EDM (b) ECM
(c) USM (d) LBM
(e) Turning

CGPSC AE 25.02.2018

Ans. (e) : EDM, ECM, USM, LBM, AJM, WJM, EBM etc. are unconventional machining process whereas turning, milling, facing etc are conventional machinery processes.

1861. Tool vibrates with very high frequency in:
(a) EDM (b) ECM
(c) USM (d) LBM
(e) AJM

CGPSC AE 25.02.2018

Ans. (c) : Tool vibrates with very high frequency in ultrasonic machining.

1862. The wear ratio for tungsten carbide work in EDM is
(a) 0.5 (b) 1.0
(c) 2.0 (d) 3.0

TNPSC AE 2014

Ans. (a) : The wear ratio for tungsten carbide work in EDM is 0.5.

1863. EDM stands for
(a) Energy Discharge Method
(b) Electro-Discharge Machining
(c) Energy Direct Machining
(d) Efficient Direct Method

TNPSC AE 2014

Ans. (b) : EDM → Electro-discharge machining

1864. In ECM process, the MRR is given by
(a) $\frac{F.I}{\rho.A.V}$ (b) $\frac{I.At}{F.\rho.A.V}$
(c) $\frac{F.\rho.A.V}{I.At}$ (d) $\frac{F.I.At}{\rho.A.V}$

TNPSC AE 2014

Ans. (b) : In electro-chemical process, the MRR is given by $MRR = \frac{I\ A\ t}{F.\ \rho.\ A.\ V.}$

1865. In ECG process, a continuous steam of non-corrosive salt solution is passed through work and tool. This solution acts as
(a) an electrolyte
(b) a coolant
(c) an electrolyte and coolant
(d) dielectric medium

TNPSC AE 2014

Ans. (c) : In ECG process, a continuous steam of non-corrosive salt solution is passed through work and tool. This solution acts as an electrolyte and coolant.

1866. The cutting rate in mm/min for LBM process is equal to
(a) $\frac{C\times P\times t}{E\times A}$ (b) $\frac{P\times t}{C\times E\times A}$
(c) $\frac{E\times A\times t}{C\times P}$ (d) $\frac{C\times P}{E\times A\times t}$

TNPSC AE 2014

Ans. (d) : Cutting rate in mm/min for LBM process is given as $= \frac{C\times P}{E\times A\times t}$

1867. High rate of oscillation of tools in ultrasonic machining is produced by
(a) D.C. motor (b) A.C. motor
(c) Pneumatic generator (d) Oscillator

Nagaland PTSC CTSE 2017 Paper-I

Ans. (d) : An electric oscillator in the control unit produces a high rate (frequency) oscillation, usually between 18 to 40 kHz. The transducer converts the oscillation current to a mechanical Vibration.

1868. For the same capacity which of the following process has the lowest capital cost?
(a) Electro chemical machining
(b) Abrasive jet machining
(c) Electron jet machining
(d) Chemical machining

Nagaland CTSE 2016, 2017 Ist Paper

Ans. (b) : Abrasive jet machining in which material removal through the used of Abrasive (which are cheaper) which can cut the sensitive materials without damage, and the initial cost is low, with same capacity obtained as compare to other machining process like, electro-chemical machining.

1869. For the same capacity which of the following process has the highest capital cost?
(a) Electro chemical machining
(b) Plasma arc machining
(c) Electro discharge machining
(d) Laser beam machining

Nagaland PTSC CTSE 2017 Paper-I

Ans. (a) : For the same capacity achieved, but the highest capital cost requires in electro-chemical machining process, in which material removal take-place through chemical reaction of ion dissolution.

1870. In abrasive jet machining the achievable tolerances are nearly
(a) 0.1 μm (b) 1 μm
(c) 50 μm (d) 250 μm

Nagaland PTSC CTSE 2017 Paper-I

Ans. (c) : In abrasive jet machining the achievable tolerance are nearly 50 μm .

1871. Reproduction of sharp corners in the limitation of :
(a) ECM (b) EDM
(c) Laser (d) Plasma

SJVN ET 2013

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Ans. (b) : Limitation of EDM process-
- Because of low material removal rates the process is economical only for very hard and difficult materials to machine.
- The EDM process is not suitable for non-conductors.
- Rapid electrode wear makes the process more costly.
- The surface produced by EDM generally have a matt type appearance, requiring further polishing to attain a glossy finish.

1872. EDM process is suited for the machining of–
(a) conducting materials
(b) non-conducting materials
(c) both (1) and (2)
(d) non-conducting but magnetic

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Irrespective of its hardness strength any material which is an electrical conductor, can be machined.
– Fine holes can be easily drilled.
– Quicker process even harder materials can be machines a much faster rate than conventional machining.

1873. Which of the following is correct of MRR variation in EDM process using RC relaxation circuit?
(a) MRR decreases with increase in capacitance of circuit
(b) MRR decreases with increase in resistance of circuit
(c) MRR decreases with increase in mean current in circuit
(d) MRR remain constant with increase in resistance of circuit

SJVN ET 2019

Ans. (b) : MRR variation in EDM process using RC relaxation circuit decreases with increase in resistance of circuit.

1874. In an ECM with a pure iron work piece, MRR of 5 cm^2/min is required. What will be the required current? [gram atomic weight of iron = 56 g, valency = 2, density of iron = 7.8 g/cm^2 and F = 96500 coulomb]
(a) 224 amp (b) 2.24 amp
(c) 2240 amp (d) 22.4 amp

SJVN ET 2019

Ans. (c) : Given,

$MRR = 5\ cm^3/min = \frac{5}{60}\ cm^3/sec$

$A = 56\ g$
$Z = 2$
$P = 7.8\ g/cm^3$
$F = 96500$ coulamb

$$MRR = \frac{AI}{\rho FZ}$$

$$\frac{5}{60} = \frac{56 \times I}{7.8 \times 96500 \times 2}$$

$$I = 2240.17\ amp$$

1875. In laser beam machining process, the usable range of wavelength of laser beam is:
(a) 400 - 600 µm (b) 0.4 - 0.6 µm
(c) 0.001 - 0.01 µm (d) 600 - 1000 µm

SJVN ET 2019

Ans. (b) : Laser beam machining (LBM)-
- The rays of laser are perfectly parallel and they can be focused to a very small diameter.
- It is a costly method and is used only when it is not feasible to machine with other process.
- It is used to drill micro holes cutting very narrow slot in very hard material like diamond.
- Used for making wire drawing dies. Heat affected zone in this process is very less.

1876. Maximum MRR of EDM process is approximately given by:
(a) 5000 mm^3/min (b) 5×10^5 mm^3/min
(c) 50 mm^3/min (d) 0.5 mm^3/min

SJVN ET 2019

Ans. (a) : 5000 mm^3/min

1877. In Electochemical machining, the gap maintained between tool and work-piece is the order of
(a) 0.05 mm (b) 0.1 mm
(c) 0.5 mm (d) 1 mm

TSPSC AEE 2015

Ans : (c) In electochemical machining the gap maintained between tool and work-piece is the order of 0.5 mm.
Electrochemical Machining:-This process is an extension of already known process of electroplating with modification in a reverse direction. A shaped tool or electrode is used in the process which forms cathode and workpiece forms anode.

1878. Consider following about Beam Machining
1. Low initial cost
2. Not very efficient process
3. Presence of Heat affected zone
4. Micro-holes cannot be drilled in difficult to machine materials
5. Not suitable for heat sensitive materials
Of above, which are properties of LBM
(a) 1, 2, 4 and 5 (b) 3, 4 and 5
(c) 2, 3 and 5 (d) 1, 3 and 4
(e) 1, 2 and 5

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Laser Beam Machining Limitations
(i) High initial capital cost.
(ii) High maintenance cost
(iii) Not very efficient process
(iv) Presence of heat affected zone specially in gas assist CO_2 laser cutting.
(v) Thermal process not suitable for heat sensitive materials like aluminium glass fiber laminate.
(vi) Micro-holes can be drilled in difficult to machine materials.

1879. Consider the following statement concerning mechanism of material removal in Electron Beam Machining
1. Mechanical erosion due to impact of high of energy electrons

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2. **Chemical etching by the high energy electron**
3. **Sputtering due to high energy electrons**
4. **Melting and vaporisation due to thermal effect of impingement of high energy electron**

Of the above, which are correct

(a) 1 (b) 1 and 3
(c) 2 and 3 (d) 4
(e) 3 and 4

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Melting and vaporisation due to thermal effect of impingement of high energy electron.

1880. Ablation mechanism is used in which of the following machining process?

(a) Chemical machining
(b) Ultrasonic machining
(c) Laser beam machining
(d) Electron beam machining

Nagaland PTSC CTSE 2017 Paper-I

Ans. (c) : Ablation mechanism is used in laser beam machining process in which metal can be machined with ablation and evaporation.

1881. Which of the following laser source can give the highest power density?

(a) Plasma Arc (b) Electron beam
(c) Laser continuous (d) laser pulsed

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : Laser pulsed can give the highest power density.

1882. An iron container 10cm × 10cm at its base is filled to a height of 20 cm with a corrosive liquid. A current is produced as a result of an electrolytic cell, and after four weeks, the container has decreased in weight by 70 g. If n = 2, F = 96500 C and M = 55.84 g/mole, the current will be

(a) 0.05 A (b) 0.10 A
(c) 0.20 A (d) 0.40 A

ESE 2019

Ans. (b) : Given,

Material removal = 70 g

$$\text{Time} = 4 \times 7 \times 24 \times 3600 = 2419200 \text{ sec}$$

$$\text{MRR (g/s)} = \frac{70}{2419200} \text{ g/s}$$

$$n = Z = 2$$

$$F = 96500 \text{ C}$$

$$M = A = 55.84 \text{ g/mole}$$

$$\text{MRR} = \frac{AI}{ZF}$$

$$I = \frac{\text{MRR} \times ZF}{A}$$

$$= \frac{\left(\frac{70}{2419200}\right) \times 2 \times 96500}{55.84} = 0.10 \text{ A}$$

1883. In which process the material is removed due to the action of abrasive grains?

(a) Electro-Chemical Grinding (ECG)
(b) Ultrasonic Machining (USM)
(c) Laser Beam Machining (LBM)
(d) Electrical Discharge machining (EDM)

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : In ultrasonic machining process the material is removed due to the action of abrasive grains.

1884. While machining copper alloys by Electro chemical Machining the electrolyte used is

(a) Common salt (b) Sodium sulphate
(c) Sodium Chlorate (d) Sodium Nitrate

APPSC AEE MAINS 2016, PAPER-III

Ans. (b) : Sodium sulphate

1885. In ultrasonic machining, Material Removal Rate with respect for grain size increase in

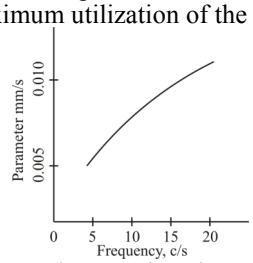
(a) increases
(b) decreases
(c) increases and decreases
(d) decreases and increases

TSPSC AEE 2015, GATE 1992

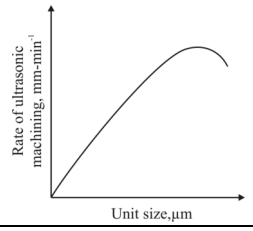
Ans. (c) : Effect of various parameter on MRR in ultrasonic machining–

1. Effect of amplitude on MRR- Increase in amplitude of vibration increase MRR.

2. Effect of frequency on MRR- Frequency used for maching process must be resonant frequency to obtain the greatest amplitude at the tool tip and thus achieve the maximum utilization of the acoustic system.



3. Effect of abrasive grain size- An increase in abrasive grain size results in higher MRR but is achieved when abrasive grain size is comparable with amplitude of vibration of the tool. Hardness of the abrasives and method of introducing the slurry has also effect on MRR.



1886. Keeping all other parameters unchanged, the tool wear in electrical discharge machining would be less if the tool material has

(a) High thermal conductivity and high specific heat
(b) High thermal conductivity and low specific heat
(c) Low thermal conductivity and low specific heat
(d) Low thermal conductivity and high specific heat

RPSC Vice Principal ITI 2018, GATE 2010

Ans. (a) : High thermal conductivity and high specific heat.

1887. In electro discharge machining dielectric is used to :

(a) Help in the movement of the sparks

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(b) Control the spark discharges
(c) Act as coolant
(d) All of these

OPSC Civil Services Pre. 2011

Ans. (d) : In electro discharge machining dielectric is used to,
- Help in the movements of the sparks.
- Control the spark discharges
- Act as coolant.

1888. Which of the following machining processes involves cutting of a hard workpiece by a soft material?
(a) Turning
(b) Grinding
(c) Electro-discharge machining
(d) Honing

BPSC AE Mains 2017 Paper - VI

Ans : (c) : Electro-discharge machining

1889. Ductile materials can be machined in Ultrasonic machining process.
(a) True (b) False
(c) Partly true (d) Partly false

OPSC AEE 2019 Paper-I

Ans : (b) : An ultrasonic machining is primarily used for machining hard and brittle material such as glass, carbide, ceramic and harden steel most of time these materials are not conductive except hard steel. The material of the tool is wear slowly during machining because of softness.

1890. Non-conductive material can be machined by :
(a) LBM (b) EDM
(c) ECM (d) All of these

OPSC AEE 2019 Paper-I

Ans : (a) : Non- conductive material can be machined by LBM.

1891. The selection of non-traditional machining process for a particular application depends on
(a) Physical parameters of the process
(b) Shapes to be machined
(c) Process capability
(d) All of the above

Gujarat PSC AE 2019

Ans : (d) : The selection of non-traditional machining process for a particular application depends on-
- Physical parameters of the process.
- Shape to be machined.
- Process capability.

1892. In electro-discharge machining process, in order to remove maximum metal and have minimum wear on the tool :
(a) the tool is made cathode and work piece as anode
(b) the tool is made anode and the work piece as cathode
(c) the tool and work piece should be of different metals
(d) none of the above

HPPSC W.S. Poly. 2016

Ans : (a) In electro - discharge machining process, in order to remove maximum metal and have minimum wear on the tool, the tool is made cathode and work piece as Anode.
The Spark frequency is normally in the range 200-50000 Hz, the spark gap being of the order of 0.025-0.05mm. An metal removal rate up to 300mm^3/min can be obtained with this process. The efficiency and the accuracy of performance have been found to improve when a forced circulation of the dielectric fluid is provided. The most commonly used dielectric fluid is kerosene. The tool is generally made of brass or copper only.

1993. As tool and work are not in contact in EDM process
(a) no relative motion occurs between them
(b) no wear of tool occurs
(c) no power is consumed during metal cutting
(d) no force between tool and work occurs

MPPSC AE 2016

Ans : (d) As tool and work are not in contact in EDM because no force between tool and work occurs.

1894. Consider the following work piece materials:
(i) Carbides (ii) Glass
(iii) Copper and (iv) Ceramics
Which of the above materials are best suited for ultrasonic Machining.
(a) (ii) Only (b) (ii) and (iii)
(c) (i), (ii) and (iv) (d) (ii), (iii) and (iv)

UPPSC AE 12.04.2016 Paper-I

Ans : (c)
Ultrasonic Machining (USM):- Ultrasonic means those vibrational waves having a frequency above the normal hearing range (e.g. 20K-30K Hz). In this method slurry of the abrasive grains are hammered on to the work surface by a vibrating tool normal to work surface removing the workpiece material in the form of extremely small chips. Tools is feed gradually is used to increase the amplitude to the level needed for machining.
Application :-
(i) Normally used for making dies
(ii) Machining hard carbides, glasses and precious stones
(iii) Also used for dental application.

1895. Metal in electro-chemical machining process is removed by:-
(a) Migration of ions towards the tool
(b) Ionization and shearing
(c) Chemical action and abrasion
(d) Chemical etching

UKPSC AE-2013, Paper-I

Ans. (a) : Metal in electro-chemical machining process is removed by migration of ions towards the tool.

1896. Which of the following is not a limitation for ECM process:-
(a) Very expensive
(b) Sharp corners are difficult to produce
(c) Surface finish is not good
(d) Use of corrosive media as electrolyte makes it difficult to handle

UKPSC AE-2013, Paper-I

Ans. (c) : Surface finish is not good

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1897. The rate of work material removal in USM operation is proportional to the:-
(a) Volume of work material removed per impact
(b) Number of particles making impact per cycle
(c) Frequency of vibration
(d) All of the above

UKPSC AE-2013, Paper-I

Ans. (d) : All of the above.

1898. The most suitable manufacturing process for machining a turbine blade made of nimonic alloy is:-
(a) Milling and lapping
(b) Electric discharge machining
(c) Ultrasonic machining
(d) Electro-chemical machining

UKPSC AE-2013, Paper-I

Ans. (d) : The most suitable manufacturing process for machining a turbine blade made of nimonic alloy is electro-chemical machining.

1899. The electrolyte used in ECM process is:-
(a) Transformer oil
(b) White spirit
(c) Aqueous solution of common salt
(d) None of the above

UKPSC AE-2013, Paper-I

Ans. (c) : The electrolyte used in ECM process is aqueous solution of common salt.

1900. A hole of 1 mm is to be drilled in glass. It could be best done by
(a) Laser drilling
(b) Plasma drilling
(c) Ultrasonic drilling
(d) Electron beam drilling

UKPSC AE 2012 Paper-I

Ans. (c) : Ultrasonic drilling

1901. The following is not true for ECM :
(a) It can machine highly complicated shapes in a single pass.
(b) Tool life is very high.
(c) Machinability of the work material is independent of its physical and mechanical properties.
(d) Kerosene is use as electrolyte.

UKPSC AE 2012 Paper-I

Ans. (d) : Kerosene is use as electrolyte.

1902. Which is the false statement about electro discharge machining ?
(a) It can machine very hard material.
(b) Very good surface finish is obtained.
(c) Section to be machined should be thick.
(d) Metal removal rate is very slow.

UKPSC AE 2012 Paper-I

Ans. (c) : Section to be machined should be thick.

1903. In EDM, metal removal rate is proportional to
(a) Frequency of charging
(b) Energy delivered in each spark
(c) Both (a) and (b)
(d) None of the above

UKPSC AE 2012 Paper-I

Ans. (c) : Both (a) and (b)

1904. Choose the alternative, which explains the correct relationship between the given statements, (A) & (R) from the code given below :
Assertion (A) : In ECM, the shape of the cavity is the mirror image of the tool, but unlike EDM, the tool wear in ECM is a cathode.
Reason (R) : The tool in ECM is a Cathode.
Code :
(a) Both (A) & (R) are true. (R) is the correct explanation of (A).
(b) Both (A) & (R) are true. (R) is not the correct explanation of (A).
(c) (A) is false, but (R) is true.
(d) (A) is true, but (R) is false.

UKPSC AE 2012 Paper-I

Ans. (c) : (A) is false, but (R) is true.

1905. The mechanism of metal removal in Electron Beam Machining is
(a) melting and vapourization
(b) shear
(c) ion-displacement
(d) chemical action

UKPSC AE 2007 Paper -I

Ans. (a) : Melting and vapourization

1906. The tool material used in USM is
(a) Mild Steel (b) High Speed Steel
(c) Carbides (d) Ceramics

UKPSC AE 2007 Paper -I

Ans. (a) : Mild Steel

1907. The current used during EDM is
(a) AC (b) pulsed AC
(c) DC (d) pulsed DC

UKPSC AE 2007 Paper -I

Ans. (d) : Pulsed DC

1908. For efficient cutting of material with Laser Beam Machining, the material must be
(a) good conductor of electricity
(b) bad conductor of heat
(c) good conductor of heat
(d) good conductor of heat and electricity

UKPSC AE 2007 Paper -I

Ans. (b) : Bad conductor of heat

1909. Which of the following process has least power consumption in kW/cm^2/min?
(a) Plasma arc machining
(b) Laser beam machining
(c) Ultrasonic machining
(d) Electro discharge machining

Nagaland PTSC CTSE 2017 Paper-I

Ans. (a) : Plasma arc machining process has least power consumption in kW/cm^2/min. In this machining the arc is formed between an electrode and the work-piece.

1910. In electro-chemical milling, the gap between tool and work kept is of the order of
(a) No gap (b) 0.25 mm
(c) 0.75 mm (d) 1.25 mm

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : In electro-chemical milling, the gap between tool and work kept is of the order of 0.25 mm.

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1911. In electro discharge machining (EDM), if the thermal conductivity of tool is high and the specific heat of work piece is low, then the tool wear rate and material removal rate are expected to be respectively.

(a) high and high (b) low and low
(c) high and low (d) low and high

VIZAG STEEL MT 10.06.2012, GATE 2007

Ans. (d) : Material removal rate $\propto$ Thermal conductivity of tool.

$$\text{Material removal rate} \propto \frac{1}{\text{Thermal capacity of workpiece}}.$$

1912. A copper rod of diameter 100 mm is turned on lathe machine at a feed of 0.50 mm/rev with depth of cut of 2 mm. The rotational speed of the rod is 320 RPM. MRR (material removal rate) in mm^3/s is:

(a) 1675.5 (b) 167.55
(c) 16.755 (d) 16755

AAI Jr. Executive 26.03.2021

Ans. (a) : Given,
(D) dia = 100 mm
Feed, (f) = 0.50 mm/rev
Depth of cut (d) = 2 mm
Speed, N = 320 rpm
Metal removal rate (MRR) = fdv

$$MRR = fd\left(\frac{\pi DN}{60}\right)$$

$$= 0.5 \times 2\left(\frac{\pi \times 100 \times 320}{60}\right)$$

$= 1674.66\ mm^3/sec$
$\simeq 1675.5\ mm^3/sec$

1913. During calculation of material removal rate in electro-discharge machining, supply voltage was used 60 V in place of the actual supply voltage 40 V. Condition for maximum power delivery to the discharge circuit is satisfied. The ratio of actual to calculated material removal rate will be

(a) $\frac{3}{2}$ (b) $\frac{4}{9}$
(c) $\frac{9}{4}$ (d) $\frac{2}{3}$

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Actual supply voltage (V_1) = 40 V
Calculated supply voltage (V_0) = 60 V

$$\text{Power (P)} = VI = \frac{V^2}{R}$$

$$MRR \propto \text{Power supply}$$

$$\propto \frac{V^2}{R}$$

$$\frac{MRR_1}{MRR_2} = \frac{V_1^2}{V_2^2}$$

$$= \frac{(40)^2}{(60)^2} = \frac{4}{9}$$

1914. In ultrasonic machining, the property which contributes to the better performance is :

(a) electrical conductivity
(b) magnetic permeability
(c) thermal conductivity
(d) hardness

DRDO Scientists 2009

Ans. (d) :

1915. The method of machining with help of ionized stream of gas is known as:

(a) Plasma arc machining
(b) Laser beam machining
(c) Electro discharge machining
(d) None of these

PTCUL AE 25.06.2017

Ans. (a) :

1916. Which of the following is not generally machined by Ultra Sonic Machining (USM)?

(a) Copper (b) Glass
(c) Silicon (d) Germanium

ISRO Scientist/Engineer (RAC) 12.01.2020

Ans. (a) : USM is generally uses for brittle materials i.e. glass, ceramics, refractory materials.
USM is also used in dental operations to produce holes on human teeth.
USM does not used for ductile material i.e. copper, aluminium etc.
Tool in USM is generally made of steel.

1917. In an ultrasonic machining (USM) process, the material removal rate (MRR) is plotted as a function of the feed force of the USM tool. With increasing feed force, the MRR exhibits the following behaviour :

(a) Increases linearly
(b) Decreases linearly
(c) does not change
(d) First increases and then decreases

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) :

1918. Abrasive water jet machining uses water jets at _____.

(a) sub-sonic speeds (b) ordinary speed
(c) sonic speeds (d) super sonic speeds

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Abrasive Water jet machining (AWJW) uses a mixture of water and on abrasive substance to cut hard materials such as metal or granite. Abrasive water jet machining uses water jet at supersonic speeds.

1919. Non traditional machining processes are introduced to overcome the limits of _____ properties of the workpiece to be machined

(a) chemical (b) atomic
(c) electrical (d) mechanical

UPRVUNL AE 05.07.2021

Ans. (a) :

1920. Which of the following is non traditional machining process that uses mechanical action for removing material?

(a) Ultrasonic machining
(b) Electrical discharge machining
(c) Electron beam machining
(d) laser beam machining

UPRVUNL AE 05.07.2021

Ans. (a) :

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1921. In chemical milling process, the chemical reagent of steel work piece is:
(a) Nitric acid
(b) Sodium sulphate
(c) Caustic soda
(d) Sodium chloride

GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) :

1922. Find the wrong answer from the following n case of ultrasonic machining
(a) tapering of slots and holes occurs
(b) high viscosity liquid medium increases material removal rate
(c) surface roughness value ranging 0.2 – 1.6 μm Ra
(d) production rates very low

GPSC ARTO 01.05.2016

Ans. (b) : Slurry composition and property in ultrasonic machining– Since the viscosity of slurry dampens the oscillation of grits, the material removal rate decreases as the viscosity increases.

1923. In ultra-sonic machining, tool is made of
(a) tungsten carbide (b) brass or copper
(c) diamond (d) stainless steel

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (b) : In ultra-sonic machining, tool is made of brass or copper.

1924. Which of the following process is used for preparing parts having large curved surfaces and thin sections?
(a) hot machining
(b) ultra sonic machining
(c) ECM process
(d) chemical milling

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Chemical milling or industrial etching is the subtractive manufacturing process of using baths of temperature regulated etching chemical to remove material to create an object with the desired shape. It produce large curved surfaces & thin sections as compared to others.

1925. To drill holes of size 0.025 mm, which is the most appropriate process?
(a) Plasma drilling
(b) CNC drilling
(c) LASER drilling
(d) Electro discharge drilling

ISRO Scientist/Engineer 24.05.2014

Ans. (c) : To drill holes of size 0.025 mm, the most appropriate process is LASER drilling which can drill small holes with high accuracy.

1926. Which material removed process is also known as advanced material removal process?
(a) Bulk-machining process
(b) Primitive casting process
(c) Traditional
(d) Non-traditional

BHEL ET 2019

Ans. (d) : Non-traditional.

1927. Which of the following is/are used as low wearing tool material(s) in electric discharge machining?
(a) Copper and brass
(b) Aluminium and graphite
(c) Silver tungsten and copper tungsten
(d) Cast iron

Mizoram PSC (Insp. of Factories) 2019 Paper-III

Ans. (c) : Silver tungsten and copper tungsten are used as low wearing tool materials in electric discharge machining. Because they have low wear resistance and these tools are not suitable for prolonged machining.

1928. Deep hole drilling of small diameter (say 200 μm) is done with EDM by selecting the tool material as
(a) Copper wire
(b) Tungsten wire
(c) Brass wire
(d) Tungsten carbide wire

MPPSC AE 08.11.2015

Ans. (c) : Brass wire is selected for the tool material to deep hole drilling of small diameter (say 200 μm) with EDM (Electric Discharge Machining).

1929. Inter electrode gap in electro-chemical grinding is controlled by controlling the:-
(a) Pressure of electrolyte flow
(b) Applied static load
(c) Size of abrasives in the wheel
(d) Texture of the workpiece

UKPSC AE-2013, Paper-I

Ans. (c) : Inter electrode gap in electro-chemical grinding is controlled by controlling the size of abrasives in the wheel.

1930. The abrasive slurry used in ultra-sonic machining contains fine particles of :
(a) Aluminum Oxide
(b) Boron Carbide
(c) Silicon Carbide
(d) All of these

OPSC Civil Services Pre. 2011

Ans. (d) : The abrasive slurry used in ultra-sonic machining contains fine particles of Aluminum Oxide, Boron Carbide, Silicon Carbide.
- Aluminum oxide
- Boron carbide
- Silicon carbide

1931. Which one of the following process conditions leads to higher MRR in ECM?
(a) Higher current, larger atomic weight
(b) Higher valency, lower current
(c) Lower atomic weight, lower valency
(d) Higher valency, lower atomic weight

TSPSC AEE 2017, GATE 2007

Ans. (a) : $Q = \frac{AI}{\rho ZF}$

I = current (A)
Q = Metal removal rate (cm^3/sec)
ρ = density (g/cm^3)
Z = valency of cation
F = 96500 Coulomb
A = atomic weight

→ higher current
→ Larger atomic weight increases MRR

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7. Numerical Control and Computer Numerical Control Systems

1932. During the execution of a CNC part program block N020G02X45, 0Y25.0R5.0 the type of tool motion will be

(a) Circular interpolation-clockwise
(b) Circular interpolation-counter clockwise
(c) Liner interpolation
(d) rapid feed

AAI Jr. Executive 26.03.2021
GPSC ARTO Pre 30.12.2018
RPSC Vice Principal ITI 2018
GPSC Asstt. Prof. (Prod.) 21.08.2016
VIZAG Steel MT 18.08.2013
ISRO Scientist/Engineer 12.05.2013
VIZAG Steel MT 10.06.2012
GATE 2004

Ans. (a) : N020G02X45, 0Y25.0R5.0
G02- used for clockwise rotation to tool in radial direction.
R5.0 - Radius for rotation

1933. NC contouring is an example of -

(a) Continuous path positioning
(b) Point to point positioning
(c) Absolute positioning
(d) Incremental positioning

GPSC DEE Class-2 (GWSSB) 09.07.2021
TNPSC AE 2019
HPPSC ADF 05.03.2019
APPSC AEE MAINS 2016, PAPER-III
GPSC Asstt. Prof. 28.08.2016
NPCIL ET, 2015
VIZAG Steel MT (Re Exam) 24.11.2013
OPSC Civil Services Pre 2006
GATE 2006

Ans. (a)
- NC machine is a continuous path positioning takes in the NC contouring.
- The points that are present a predetermined such that the path or the track that is occupied or followed has to go through those points that are predetermined and the method of creating search path is contouring.
- Hence as the name suggest NC contouring comes under the path positioning due to the predetermined position of the points.

1934. Feed drives in CNC milling machines are provided by:

(a) stepper motor (b) induction motor
(c) servo motor (d) synchronous motors

AAI Jr. Executive 29.11.2018
TNPSC AE 2017
GWSSB DEE 07.07.2016
GWSSB AAE, 27.12.2015
J&K PSC Civil Services Pre, 2010
APPSC IOF, 2009

Ans. (c) : Servo Motor- Feed drives in CNC milling machine are provided by servo motor.
CNC = Computer Numerical Control
NC = Numerical Control.

1935. The interpolator in a CNC machine controls

(a) Spindle Speed (b) Coolant flow
(c) Feed rate (d) Tool change

GPSC DEE Class-2 (GWSSB) 04.07.2021
UPPSC AE 13.12.2020 Paper-I
RPSC VPITI 14.02.2016
UP Jal Nigam AE 2016
CGPSC Poly. Lect. 22.05.2016
APPSC AEE MAINS 2016, PAPER-III
GATE 2015, 2007

Ans. (c) : The interpolator is either an electronic hardware device for a NC system or a software program for a CNC system.
An interpolator provide two functions.
(i) It calculates individual axis velocity
(ii) drive the tool along the programmed path at the given feed rate.

1936. G codes are used for creating CNC programs to control CNC machines. G17 refers to _____.

(a) selection of XZ plane
(b) input in mm
(c) selection of XY plane
(d) selection of YZ plane

VIZAG Steel MT 24.01.2021
Punjab PSC SDE 12.02.2017
KPSC ADF 2015

Ans. (c) : G codes are used for creating CNC programs to control (CNC) machines. G17 refers to selection of XY plane.

1937. Which motor is NOT used in axis or spindle drives of CNC machine tools?

(a) induction motor (b) DC servo motor
(c) Stepper motor (d) linear servo motor

GPSC DEE, Class-2 (GWSSB) 04.07.2021
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2007

Ans. (a) : Induction motor– An induction motor or a synchronous motor s an AC electric motor in which the electric current in the rotor acceded to produce torque is obtained by electromagnetic induction from the magnetic field of the stator winding.

1938. In a CNC program block, N002 G02 G91 X40 Z40 ....., G02 AND G91 refer to

(a) circular interpolation in counterclockwise direction and incremental dimension
(b) circular interpolation counterclockwise direction and absolute dimension
(c) circular interpolation in clockwise direction and incremental dimension
(d) circular interpolation in clockwise direction and absolute dimension

VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012, GATE 2010

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Ans. (c)
G02 → Circular interpolation in clockwise.
G03 → Circular interpolation anti-clockwise
G01 → Linear interpolation
G91 → Incremental dimension.

1939. The APT (Automatically Programmed Tools) language is used with
(a) Drafting systems
(b) NC machines
(c) Programmable controllers
(d) Large automation systems

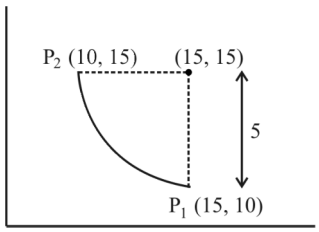
TNPSC AE 2018, BPSC Poly. Lect. 2016
TRB Poly. Lect., 2012

Ans. (b) : (Automated Programmed Tool) APT is a high level computer programming language, most commonly used to generate instructions for Numerically controlled m/c tools,

1940. In a 2-D CAD package, clockwise circular arc of radius 5, specified from P_1(15,10) to P_2(10,15) will have its centre at :
(a) (10,10) (b) (15,10)
(c) (15,15) (d) (10,15)

AAI Jr. Executive 26.03.2021, VIZAG Steel MT 2017
RPSC ADE 2016, GATE 2004

Ans. (c) : Given,
P_1 (15,10)
P_2 (10,15)



From the figure
Radius = 5
Centre of arc having radius 5 = [15, (10+5)]
= [15,15]

1941. In a point-to-point control type of NC system
(a) control of position and velocity of tool is essential
(b) control of only position of the tool is essential
(c) control of only velocity of the tool is sufficient
(d) neither position nor velocity need be controlled

TNPSC AE (Industries) 09.06.2013
TRB Poly. Lect., 2012

Ans. (b) : In point-to-point (PTP) is also sometime called a positioning system. In PTP, the objective of the m/c tool control system is to more the cutting tool to a predefined location.

1942. CIM is considered as natural evaluation by:
(a) Integration of CAM, FMS & SCM
(b) Integration of CAD & CAM
(c) Integration of TQM, CAM & SCM
(d) Integration of CAD & FMS

TRB Asstt. Prof., 2012, RPSC AE GWD, 2011

Ans. (b) : CIM (Computer Integrated Manufacturing) is considered a natural evolution of the technology of CAD/CAM which by itself evolved by the integration of CAD and CAM.

1943. 'YZ' plane is selected as active plane, during CNC programming by
(a) G17 (b) G18
(c) G19 (d) G04

Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : G17 – XY plane selection
G18 – ZX plane selection
G19 – YZ plane selection

1944. G00 code is used in :
(a) Robots (b) AGV
(c) CNC machines (d) PTER

HPPSC Lect. (Auto) 23.04.2016
WBPSC AE, 2007

Ans. (c) G00 code is used in CNC machine
G00 – Rapid movements
G01 – linear interpolation
G02 – Circular interpolation in CW
G03 – Circular interpolation in ACW
G04 – Dwell

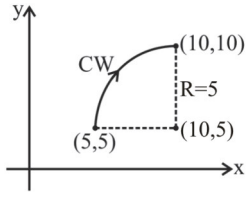
1945. The tool of an NC machine has to move along a circular arc from (5, 5) to (10, 10) while performing an operation. The center of the arc is at (10, 5). Which one of the following NC tool path commands performs the above mentioned operations?
(a) N010 G02 X10 Y10 X5 Y5 R5
(b) N010 G03 X10 Y10 X5 Y5 R5
(c) N010 G01 X5 Y5 X10 Y10 R5
(d) N010 G02 X5 Y5 X10 Y10 R5

HPPSC ADF 05.03.2019
Vizag Steel MT (Re-Exam) 24.11.2013
GATE 2005

Ans. (d) : When cutter is else where except start point, then general tool path commands is–

$$N010\left\{\begin{matrix}G02\\ \text{or}\\ G03\end{matrix}\right\}\underbrace{X_Y_}_{\text{Start}}\underbrace{X_Y_}_{\text{End}}\underbrace{R_}_{\text{Radius}}$$

GO2 : CW circular interpolation
GO3 : CCW circular interpolation
So commands will be



N010 GO2 X5 Y5 X10 Y10 R5

1946. In CNC programming G-code used for circular interpolation (counter clockwise) is
(a) G 03 (b) G 02
(c) G 01 (d) G 00

GPSC ARTO 01.05.2016
HPPSC Asstt. Prof. 29.10.2016

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Ans. (a) : G-Code is a software programming language used to control a CNC machine. A "G" is followed by a number, which is a command to change geometry where.

G00 → Rapid movement

G01 → Linear Interpolation

G02 → Circular Interpolation clockwise

G03 → Circular interpolation counter clockwise.

1947. In a CNC machine tool, encoder is used to sense and control :

(a) Spindle speed (b) Spindle position

(c) Table position (d) All of these

HPPSC W.S. Poly. 2016, UJVNL AE 2016

Ans : (c) In a CNC machine tool, encoder is used to sense and control Table position.

1948. The core product model required for any CAD/CAM system :

(a) Data model (b) Solid model

(c) Prototype (d) Geometric model

Kerala PSC AE 06.08.2015, (KPSC AE. 2015)

Ans. (d) : The core product model required for any CAD/CAM system is Geometrical model.

⇒ CAD is concerned with only the creation of a design. This design may later be fed into computer aided machines to create manufactured product through the case of a CAM program. CAM tools differ from CAD tools. For CAD only a computer and CAD software are required for a technician or artist to create a design.

1949. A block information in NC machine program means

(a) one row on tape

(b) a word comprising several tapes

(c) one complete instructions

(d) completion of job

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : A block information in NC machine program means one complete instructions.

1950. In NC part programming spindle speed of 730 rpm will be coded by the magic-3 rules as

(a) S673 (b) S730

(c) S630 (d) S037

GPSC DEE, Class-2 (GWSSB) 04.07.2021

GATE 1989

Ans. (b) : S730

1951. In CNC feed drive, a stepper motor with step angle of 1.8 degree drives a lead screw with pitch of 2 mm the basic length unit BLU for this drive is

(a) 10 microns (b) 20 microns

(c) 40 microns (d) 100 microns

GPSC DEE, Class-2 (GWSSB) 04.07.2021

GATE 2008

Ans. (a) : Step angle = 1.8°/pulse

Pitch = 2 mm

No. of pulses in 360° $= \frac{360}{1.8} = 200$ pulse

Distance travelled in one pulse $= \frac{2}{200}$ mm

$= 0.01$ mm

$= 10$ microns

1952. Feed motion can be provided with stepper motors in CNC

(a) lathe (b) drilling machine

(c) milling machine (d) grinding machine

GPSC DEE, Class-2 (GWSSB) 04.07.2021

GATE PI 1991

Ans. (b) : Feed motion can be provided with stepper motors in CNC drilling machine.

1953. For generating Coons surface we require

(a) a set of grid points on the require

(b) a set of grid control points

(c) four bounding curves defining the surface

(d) two boundry curve points

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : For generating Coons surface we require four bounding curves defining the surface.

1954. In a numerical control programming, the M code used for automatic clamping of fixtures is

(a) M3 (b) M10

(c) M6 (d) M5

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) :

M10 - Clamps on

M11 - Clamps off

M03 - Spindle on clockwise

M06 - Tool change

M05 - Spindle stop

1955. Most popular NC programming language is

(a) AUTOMAP (b) APT

(c) ADAPT (d) COBOL

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (b) APT (Automatically Programmed Tool) Programming Language is a language for defining tool path on CNC machines. It is most commonly used programming language.

1956. In a CNC milling operation, the tool has to machine the circular arc from point (20, 20) to (10, 10) at sequence number 5 of the CNC part program. If the center of the arc is at (20, 10) and the machine has incremental mode of defining position coordinates, the correct tool path command is

(a) N 05 G90 G01 X-10 Y-10 R10

(b) N 05 G91 G03 X-10 Y-10 R10

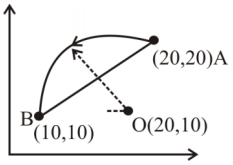
(c) N 05 G90 G03 X-20 Y-20 R10

(d) N 05 G91 G02 X-20 Y-20 R10

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

GATE 2015

Ans. (b) :



So, Incremental system

N05 G91 G03 X-10 Y- 10 R 10

G91- Incremental

G03- Counter clock wise circular arc.

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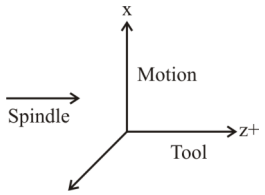
1957. While specifying the axes of CNC machine tool (as per the ISO standards), the spindle axis is considered as

(a) X-axis (b) Y-axis
(c) Z-axis (d) A-axis

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) :



ISO standard
Spindle axis is Z-axis

1958. What is the correct order of the development of the following?

(1) Numerical control
(2) Computer numerical control
(3) Direct numerical control
(4) Distributed numerical control

(a) 4, 3, 2 and 1 (b) 1, 3, 2 & 4
(c) 1, 2, 3 & 4 (d) 4, 2, 3 & 1

VIZAG Steel MT 24.01.2021

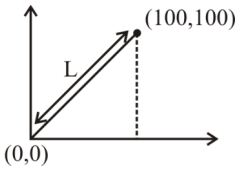
Ans. (c) : Correct order of development–
- Numerical Control (NC)
- Computer Numerical Control (CNC)
- Direct Numerical Control (DNC)
- Distributed Numerical Control (DNC)

1959. A CNC vertical milling machine cut a straight slot of 10 mm width and 2 mm depth by a cutter of 10 mm diameter between points (0, 0) and (100, 100) on the XY plane (dimensions in mm). The feed rate used for milling is 50 mm/min. What is the milling time for the slot (in seconds)?

(a) 180 (b) 170
(c) 120 (d) 240

Assam Engg. College AP/Lect. 18.01.2021
GATE 2012

Ans. (b) :



Milling time

$$T_m = \frac{\text{Length travelled}}{\text{Feed rate}}$$

$$= \frac{\sqrt{100^2 + 100^2}}{50}$$

$$= \frac{141.42}{50}$$

$$= 2.82 \text{ min}$$

$$= 2.82 \times 60$$

$$T_m = 169.70 \approx 170 \text{ sec.}$$

1960. Match the following part programming codes with their respective functions:

Part Programming Codes		Functions	
P.	G01	I.	Spindle stop
Q.	G03	II.	Spindle rotation, clockwise
R.	M03	III.	Circular interpolation, anticlockwise
S.	M05	IV.	Linear interpolation

(a) P-II, Q-I, R-IV, S-III
(b) P-IV, Q-III, R-II, S-I
(c) P-IV, Q-II, R-III, S-I
(d) P-III, Q-IV, R-II, S-I

Assam Engg. College AP/Lect. 18.01.2021

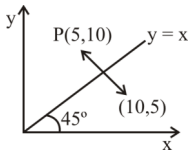
Ans. (b) : P- IV, Q- III, R- II, S-I
G01- Linear interpolation
G03- Circular interpolation in anticlockwise direction
M03- Spindle rotation in clockwise direction
M05- Spindle stop

1961. In a CAD package, mirror image of a 2D point P(5, 10) is to be obtained about a line which passes through the origin and makes an angle of 45° counter-clockwise with the X-axis. The coordinates of the transformed point will be____.

(a) (7.5, 5) (b) (10, –5)
(c) (7.5, –5) (d) (10, 5)

Assam Engg. College AP/Lect. 18.01.2021
GPSC Asstt. Director of Transport 05.03.2017
GATE 2013

Ans. (d) : (10, 5)



Points (10, 5) will be the mirror image of points P (5, 10) about line (y = x) at angle 45° and passes through origin.

1962. Which CAD display configurability allows work without any host support ?

(a) One with no secondary storage capacity
(b) Medium intelligence
(c) High intelligence
(d) Low intelligence

APGCL AM, 2021

Ans. (c) High intelligence CAD display configurability allows work without any host support.

1963. The following is basically a material flow simplification technique

(a) A simple and visual method of cell design
(b) Family formation by classification and codification
(c) Cell formation using Production Flow Analysis
(d) Both (B) and (C)

APGCL AM, 2021

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Ans. (c) : Material flow simplification technique is cell formation using Production flow Analysis

1964. Before automating solely in search of cost savings, which one of the following is NOT a valid consideration

(a) What support activities does the technology need in order to function effectively ?
(b) Is it worth getting rid of the human potential along with its cost ?
(c) Can the technology perform the task better (in a broader sense) of safer than a human ?
(d) Which is the most appropriate technology?

APGCL AM, 2021

Ans. (d)

1965. Statement (I) : The stepper motor is a device that produces rotation through equal angles, the so-called steps, for each digital pulse supplied to its input.
Statement (II) : Stepper motors can be used to give controlled rotational steps but cannot give continuous rotation, as a result their applications are limited to step angles only.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).
(c) Statement (I) is true, but Statement (II) is false.
(d) Statement (I) is false, but Statement (II) is true.

ESE 2020

Ans. (c) : Statement (I) is true, but Statement (II) is false.

1966. Hard automation is also called

(a) Selective automation
(b) Total automation
(c) Group technology
(d) Fixed position automation

ESE 2020

Ans. (d) : Hard automation is also called fixed position automation.

1967. The method of CNC programming which enables the programmer to describe part geometry using variables is

(a) Computer assisted part programming
(b) Computer aided drafting programming
(c) Conversational programming
(d) Parametric programming

ESE 2020

Ans. (d) : Parametric programming

1968. In computer aided part programming by Automatically Program Tool (APT), "COOLANT/ ON" is a

(a) Geometry Statement
(b) Motion Statement
(c) Post Processor Statement
(d) Set up Statement

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : COOLANT/ON–Coolant fluid to be turned on.
The above automatically program tool is post processor statement.

1969. Which G-code is used for Threading-Constant lead

(a) G-31 (b) G-33
(c) G-32 (d) G-35

TNCSC AE 2020

Ans. (c) : G-32 code for constant lead threading on CNC lathe machine.
• G-3– Skip function (used for probes and tool length measurement system)

1970. A drill bit is positioned in absolute system at A (2.5, 1.2) and it has to move to point B (12.5, 24.2). In CNC program, the coordinates of B in an incremental system will be :

(a) (12.5, 24.2) (b) (24.2, 1.2)
(c) (10.0, 23.0) (d) (23.0, 10.0)

APPSC Poly Lect. 13.03.2020

Ans. (c) : Given data
Absolute positioned = A (2.5, 1.2) [In CNC program]
Move to point = B (12.5, 24.2)
∴ The coordinates of B in an incremental system will be–
$= (12.5 - 2.5, \quad 24.2 - 1.2)$
$= (10.0, \ 23.0)$

1971. In a CNC program block
N10 G02 G91 X52 Z25 .....,
G02 represents :

(a) linear interpolation
(b) clockwise circular interpolation
(c) incremental command
(d) absolute command

APPSC Poly Lect. 13.03.2020

Ans. (b) : CNC – Program –
N10 G02 G91 X52 Z25 .....
Where;
G02 represents, clockwise circular interpolation.
(1) G01 Linear interpolation
(2) G02 Circular CW
(3) G03 Circular ACW
(4) G90 Absolute positioning system
(5) G91 Incremental command

1972. G codes and M codes are the instructions to various CNC machine tools. G 01 and M 05 refer to ____.

(a) linear interpolation, spindle stop
(b) circular interpolation, spindle on
(c) linear interpolation, spindle on
(d) circular interpolation, spindle stop

VIZAG MT, 14.12.2020

Ans. (a) : G 01 – Linear interpolation
M 05 – Spindle stop

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1973. In a computer Numerical controlled turning center, the backlash error during longitudinal feed motion is accomplished by

(a) Pulley and belt
(b) Recirculating ballscrew and nut
(c) gears
(d) worm wheels

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (b) : In a computer Numerical controlled turning center, the backlash error during longitudinal feed motion is accomplished by recirculating ballscrew and nut.

1974. Pick the analytic surface entity from the list:

(a) Ruled surface (b) Gordon surface
(c) Spline surface (d) Coons patches

CIL MT 27.02.2020

Ans. (a) : Analytic Surface–Surface entities which are defined by the analytic equation are known as analytic surface.

Types of Analytic Surfaces–

(i) Plane surface, (ii) Ruled surface, (iii) Tabulated surface, (iv) Surface of revolution

Synthetic Surfaces–The surface entities are defined by the set of data point are known as synthetic surfaces.

Types of Synthetic Surfaces–

(i) Bezier surface, (ii) B-spline surface, (iii) Coons surface, (iv) Fillet surface, (v) Gordon surface

1975. CNC machines allow the machine operator to set the zero point at any position on the machine table. This feature is called:

(a) Floating zero (b) Fixed zero
(c) Machine zero (d) Fixed origin

HPCL Engg. 12.08.2021 Shift-I
CIL MT 27.02.2020
UPPSC AE 13.12.2020 Paper-I

Ans. (a) : In CNC lathe, it is a point where tool change is carried out. The part origin can be set at any point inside the machine. Establishing the part origin is also known as zero shift, floating zero, or datum and work shift.

1976. In CAD, a solid model is generated by using 10 numbers of primitives. The total number of Boolean operations required to generate the solid model would be

(a) 7 (b) 9
(c) 3 (d) 4

Gujarat PSC AE 2019

Ans : (b) : In CAD, a solid model is generated by using 10 numbers of primitives. The total number of Boolean operations required to generate the solid model would be 9.

1977. The completeness and unambiguity of a solid model are attributed to fact that:
Database stores both ..... and ..... information.

(a) Geometry and mathematical
(b) Geometry and topology
(c) Pictorial and size
(d) Design and data

Gujarat PSC AE 2019

Ans : (b) : The completeness and unambiguity of a solid model are attributed to fact that databases stores both geometry and topology information.

1978. Design of tools is a part of CAD or CAM?

(a) CAD (b) CAM
(c) Both (d) None

Gujarat PSC AE 2019

Ans : (b) : Design of tools is a part of (CAM) computer-aided manufacturing.

1979. Mass property calculations depend upon

(a) Geometry only
(b) Topology only
(c) Geometry and topology
(d) None

Gujarat PSC AE 2019

Ans : (c) : Mass property calculation depends upon geometry and topology.

1980. What type of format the IGES system uses?

(a) Binary (b) Non-binary
(c) Program format (d) None of these

Gujarat PSC AE 2019

Ans : (a) : Binary.

1981. Scaling factors are always positive. Negative scale factors produce .....

(a) Reflection (b) Crap
(c) Zoom (d) Protrusion

Gujarat PSC AE 2019

Ans : (a) : Scaling factors are always positive. Negative scale factors produce reflection.

1982. In NC motion control system, the system which is capable of continuous and simultaneous control of two or more axis refers to

(a) Continuous path system
(b) Point to point path system
(c) Discrete system
(d) None of the above

Gujarat PSC AE 2019

Ans : (a) : Continuous path system- In continuous path system, cutting tool usually remains in contact with workpiece as it travels from one programmed point to next continuous path controlled cause the tool to maintain continuous contact with the part as the tool cuts a contour shape.

1983. How many translators are required for 7 CAD/CAM systems using (a) direct and (b) indirect translation?

(a) Direct: 42 and Indirect: 14
(b) Direct: 22 and Indirect: 14
(c) Direct: 12 and Indirect: 24
(d) None of the above

Gujarat PSC AE 2019

Ans : (a) : Direct : 42 and indirect : 14

1984. If the z-axis and x-axis of CNC lathe are provided with straight line controls, it is possible to carry out

(a) Turning and facing only
(b) Turning, facing and taper turning
(c) Turning, facing and thread cutting
(d) Turning, facing and drilling

GPSC DEE, Class-2 (GWSSB) 04.07.2021
Gujarat PSC AE 2019
GPSC Executive Engineer 23.12.2018

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Ans : (b) : If the z-axis and x-axis of CNC lathe are provided with straight line controls, it is possible to carry out turning, facing and taper turning.

1985. Which one of the following systems is consisting of processing stations, material handling and storage, computer control system and human labour?
(a) Portable manufacturing system
(b) Focused integrated system
(c) Flexible manufacturing system
(d) Automated integrated system

ESE 2019

Ans. (c) : Flexible manufacturing system (FMS) consist of a group of NC machines connected together by an automated material handling system and operating under computer control.
The basic components of FMS are machine tools and the related equipment, material handling equipment, computer control system and human labour.

1986. CNC machines are also called:
(a) Soft-wired
(b) Hard-wired
(c) Hard as well as soft wired
(d) None of these

HPPSC ADF 05.03.2019

Ans. (a) : CNC machines are also called soft-wired.

1987. Which of the following option best describes a Computer Aided Manufacturing (CAM) technology?
(a) Scheduling (b) Drafting
(c) Documenting (d) Numerical control

Oil India Ltd. Sr. Engg. (Drilling) 30.11.2019

Ans. (d) : Computer-Aided Manufacturing (CAM)– It is use of computer to program, direct and control production equipment in the fabrication of manufactured items. Specially computer, hardware, software, database and telecommunication, technologies are used together with numerical control machines expert system, vision system, LASER, industrial robots.

1988. The function of graphic software is :
(a) To design and manufacture application
(b) To control the computers work flow organize and process the data
(c) To modify the system for specific application
(d) To provide users various functions of geometric modeling and construction.

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (d) : Graphic Software–Graphic software is used to create, edit and manage 2D images.
Common graphics software are photoshop, illustrator, paint shop pro, adobe lightroom etc.
This provides users various function of geometric modeling and construction.

1989. An example of the delimiter in a FORTRAN program is
(a) Semi colon (b) Double colon
(c) Single colon (d) Comma

JPSC AE PRE 2019

Ans. (d) : Comma

1990. Which one is a valid variable declaration in FORTRAN?
(a) Real : : Celcius (b) Real Celcius
(c) Celcius Real (d) Real : : Celcius

JPSC AE PRE 2019

Ans. (a) : Real : : Celcius

1991. A flow chart that outlines the main segments of any program:
(a) Micro (b) Queue
(c) Macro (d) Union

JPSC AE PRE 2019

Ans. (c) : A flow chart that outlines the main segments of any program is macro.

1992. "SCALE" command in Autocad is used to:
(a) To create multiple copies of the object
(b) Measure the dimension of the object
(c) Enlarge or reduce the object proportionally
(d) To rotate the object or a part of it around a point

SJVN ET 2019

Ans. (c) : "Scale" command in Autocad is used to enlarge or reduce the object proportionally.

1993. The command in AutoCAD which is used to bevel the edge of object is known as:
(a) offset (b) chamfer
(c) fillet (d) stretch

SJVN ET 2019

Ans. (b) : chamfer

1994. Out of the following which type of coding system is used in NC programming?
(a) Indian coding system
(b) Binary coding system
(c) Discrete coding system
(d) None of the above

Gujarat PSC AE 2019

Ans : (b) : NC machining is the form of programmable automation in which the process is controlled by the number letters and symbols, here automation of medium and small volume production (batch type) is done by some controls under the instructions of a program. It requires more time for the execution of the job in comparison with special purpose machine tool.

1995. In the G code of NC part programming, G21 refers to
(a) Input values specified in millimeters
(b) Return to reference point
(c) Thread cutting in turning
(d) Dwell for a specified time

Gujarat PSC AE 2019

Ans : (a) : G21- Input values specified in milimeters.

1996. In NC part programming, M code specifies
(a) End of block function
(b) Miscellaneous or auxiliary functions
(c) Tool selection function
(d) Preparatory function

Gujarat PSC AE 2019

Ans : (b) : M codes specifies miscellaneous or auxiliary functions.

1997. In automatically programmed tool (APT), motion commands are used to specify the
(a) Tool path (b) Dimensions
(c) Tolerance value (d) None of the above

Gujarat PSC AE 2019

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Ans : (a) : In automatically programmed tool (APT), motion command are used to specify the tool path.

1998. Which of the following techniques enables the designer to mould and shape, rather than construct an object using a series of lines?

(a) solid modeling
(b) surface modeling
(c) wireframe modeling
(d) FEM

Vadodara Muncipal Corp. DEE, 2018

Ans. (a) : Solid modeling– Solid modeling enables the designs to mould and shape, rather than construct an object using a series of lines.

1999. In a point to point control NC machine, the slides are positioned by an integrally mounted stepper motor drive. If the specification of the motor is 1° pulse, and the pitch of the lead screw is 3.6 mm, what is the expected positioning accuracy?

(a) 1 μm (b) 10 μm
(c) 50 μm (d) 100 μm

Vadodara Muncipal Corp. DEE, 2018
GATE 1997

Ans. (b) : 360° turn gives movement equal to pitch

i.e. $360^{\circ} = 3.6$ mm

$$1^{\circ} = \frac{3.6}{360}\text{mm}$$

$$\text{Position accuracy} = \frac{3.6}{360}\text{mm}$$

$$= 10\ \mu\text{m}$$

2000. A display of a volume model that shows all of its edges is called :

(a) 3-D Model (b) Surface model
(c) Wire frame (d) Solid model

Vadodara Muncipal Corp. DEE, 2018

Ans. (c) : A display of a volume model that shows all of its edges is called a wire frame model.

2001. Which among the NC operations given below are continuous path operations?

Arc welding (AW)
Milling (M)
Drilling (D)
Punching of sheet metal (P)
Laser cutting of sheet metal (LC)
Spot welding (SW)

(a) AW, LC and M (b) AW, D, LC and M
(c) D, LC, P and SW (d) D, LC and SW

GPSC ARTO Pre 30.12.2018
MPPSC AE 08.11.2015, GATE 2005

Ans. (a) : NC continuous path operations – Arc Welding (AW), Laser Cutting of sheet metal (LC), Milling (M).

NC non- continuous path operation –
Drilling (D)
Punching (P)
Spot Welding (SW)

2002. Match the following :

P. M05	**1.** Absolute coordinate system
Q. G01	**2.** Dwell
R. G04	**3.** Spindle stop
S. G90	**4.** Linear interpolation

(a) P-2, Q-3, R-4, S-1 (b) P-3, Q-4, R-1, S-2
(c) P-3, Q-4, R-2, S-1 (d) P-4, Q-3, R-2, S-1

GPSC ARTO Pre 30.12.2018, GATE 2009

Ans. (c) :

Code	Function
M05	Spindle stop
G01	Linear Interpolation
G04	Dwell
G90	Absolute Coordinate System

2003. Which of the following statement is correct about generative CAPP system?

(a) No standard manufacturing plans are predefined or stored
(b) System based on principles of 'Group Technology'
(c) System are based on part classification and coding
(d) Route sheet is stored in computer files

MPSC Lect. (Prod.) 2018

Ans. (c) : CAPP system are based on part classification and coding.

2004. Which one of the following is not the division of Flexible Manufacturing System (FMS) Class?

(a) Flexible Manufacturing Module (FMM)
(b) Flexible Fabrication - Machining - Assembly System (FFMAS)
(c) Flexible Manufacturing Group (FMG)
(d) Flexible Manufacturing Technology (FMT)

TNPSC AE 2018

Ans. (d) : Flexible manufacturing technology (FMT) is not the division of FMS class.

2005. CAD/CAM technology was initiated in the

(a) Die industry (b) Nuclear industry
(c) Aerospace industry (d) Medical industry

TNPSC AE 2018

Ans. (c) : CAD/CAM technology was initiated in the Aerospace industry.

2006. In NC part programming F-code indicates

(a) the type of motion or action is to be carried out
(b) the spindle speed at which the spindle rotates
(c) the type of tool
(d) the rate at which the spindle moves along a programming axis

TNPSC AE 2018

Ans. (d) : In NC part programming F-code indicates the rate at which the spindle moves along a programming axis.

2007. The roles of rover operator in CIM system are

(a) the supervision of other Human Resources in the system
(b) making the tools ready for production
(c) setting up the fixtures, pallets and tools for the system
(d) reacting to unscheduled machine shops, identifying broken tools and tool adjustments

TNPSC AE 2018

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Ans. (d) : The roles of rover operator in CIM system are reacting to unscheduled machine shops, identifying broken tools and tool adjustments.

2008. The translation distances dx, dy is called as
(a) translation vector
(b) shift vector
(c) both (a) and (b)
(d) neither translation or shift but δ

TNPSC AE 2018

Ans. (c) : The translation distance dx, dy is called translation vector and shift vector.

2009. MACRO subroutine is defined by the format
(a) MACRO = Parameter
(b) Symbol = MACRO / Parameter
(c) MACRO / Parameter
(d) MACRO

TNPSC AE 2018

Ans. (b) : MACRO subroutine is defined by the format Symbol = MACRO / Parameter.

2010. Choose the wrong statement:
(a) CNC computers control only one machine while DNC computer control many machines using local networking
(b) CNC computers is an integrated part of the machine whereas DNC computer is located at a distance from the machine
(c) DNC computers are having higher processing power than CNC computers
(d) DNC software does not take care of management of information flow to a group of machines

TNPSC AE 2018

Ans. (d) : DNC software does not take care of management of information flow to a group of machines is a wrong statement.

2011. In Retrieval CAPP system.
(a) No standard manufacturing plans are stored
(b) A standard process plan is stored in computer files for each part code number
(c) Engineering drawing specifications are translated into compute interpretable data
(d) Manufacturing plans are prepared

TNPSC AE 2018

Ans. (b) : In Retrieval CAPP system A standard process plan is stored in computer files for each part code number.

2012. Most FEM software use
(a) displacement method (b) force method
(c) skyline method (d) stress method

TNPSC AE 2018

Ans. (a) : Most FEM software use displacement method.

2013. In which one of the following machining manual part programming is done?
(a) CNC machining (b) NC machining
(c) DNC machining (d) FMS machining

GPSC Executive Engineer 23.12.2018

Ans. (b) : In NC machining manual part programming is done.

2014. The most common interpolation methods in continuous path NC machining are
(a) linear and circular
(b) linear and parabolic
(c) circular and parabolic
(d) circular and elliptic

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GPSC Executive Engineer 23.12.2018

Ans. (a) : Linear and circular is the most common interpolation methods in continuous path NC machining.

2015. CAE and CAM are linked through
(a) A common database and communication system
(b) NC type programming and automated design
(c) Assembly automation and tool production
(d) Parts production and testing

TNPSC AE 2017
BPSC Poly. Lect. 2016

Ans. (a) : CAE and CAM are linked through a common database and communication system.

2016. Calligraphic is
(a) coloured image (b) coloured drawing
(c) line drawing (d) dot matrix

TNPSC AE 2017

Ans. (c) : Calligraphic is line drawing.

2017. A technique for displaying applications where complex 3-D geometric are required for the exterior shell of a product is called
(a) 2-D modeling
(b) Solid modeling
(c) 3-D modeling
(d) Surface modeling

TNPSC AE 2017

Ans. (d) : A technique for displaying applications where complex 3-D geometric are required for the exterior shell of a product is called Surface modeling.

2018. The difference between CAD and CAM is that CAD software is directed at product design while (CAM software is)
(a) concerned with production and control of tool design
(b) Concerned with management programs
(c) Specifically for PC board design
(d) designed for communications

TNPSC AE 2017

Ans. (a) : CAM software is concerned with production and control of tool design.

2019. The widely employed computer architecture for CAD/CAM application
(a) Mainframe-based system
(b) Minicomputer-based system
(c) Microcomputer-based system
(d) Workstation-based system

Vizag Steel (MT) 2017

Ans. (d) : The widely employed computer architecture for CAD/CAM application is workstation based system.

2020. A line with tapering width can be easily created by using the.....tool.
(a) Line (b) Polyline
(c) Eclipse (d) Circle

Vizag Steel (MT) 2017

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Ans. (b) : A line with tapering width can be easily created by using the polyline tool.

2021. When setting up a mechanical drawing in AutoCAD the drafter should set units to.....

(a) Metric (b) Decimal
(c) Fractional (d) Architectural

Vizag Steel (MT) 2017

Ans. (b) : Decimal

2022. In computer aided drafting practice, an arc is defined by

(a) Two end points only
(b) Center and radius
(c) Radius and one center point
(d) Two end points and center

AAI Jr. Executive 26.03.2021
Vizag Steel (MT) 2017, GATE 2000

Ans. (d) : In computer aided drafting practice, an arc is defined by
– Two points and center.
– One end point and angle of arc and center.
– Two end points angle of arc and radius.

2023. CNC machines are best suited, when

(a) production volumes are very high and there is no product variety
(b) holes are to be bored at precise locations
(c) very small production volumes and product varieties are involved, while accuracy requirements are not very stringent
(d) a moderate production volume is combined with a moderate product variety and contoured shapes are to be cut

BPSC AE Mains 2017 Paper - VI

Ans : (a) : CNC machines are best suited, when production volumes are very high and there is no product variety.

2024. Consider the following statements with reference to NC machines:

1. Both closed-loop and open-loop systems are used.
2. Papers, tapes, floppy tapes and cassettes are used for data storage.
3. Digitizers may be used as interactive input devices.
4. Post-processor is an item of hardware.

Which of the above statements are correct?

(a) 1, 2 and 4 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 3

ESE 2017

Ans. (d) : Post processor is used in CNC machines only but no post processor is used in NC machines.
So, statement-4 is wrong.
A post processor is a unique "driver" specific to a CNC machine, robot or mechanism.

2025. Consider the following benefits of CIM:

1. Less direct labour
2. Less scrap and rework
3. Higher machine use

Which of the above are correct?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

ESE 2017

Ans. (a)
(i) With computer integrated manufacturing (CIM), it is possible to make the manless manufacturing therefore less direct labour.
(ii) Due to correct machining and consistency in manufacturing no rework is required and scrap is minimised.
(iii) Maximum utilisation machines is possible. Hence all the statement are correct.

2026. Extended Binary - coded decimal interchange code uses

(a) 8 - bit code (b) 16 - bit code
(c) 32 - bit code (d) 7 - bit code

TNPSC AE 2017

Ans. (a) : Extended Binary - coded decimal interchange code uses 8 - bit code.

2027. Which of the following are the principal functions of a CNC machine?

1. Machine tool control
2. In process compensation
3. Improved programming and operating features

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

JWM 2017

Ans. (d) : Function of CNC machine-
• Machine tool control precisely
• Speed up production
• Consistent machine each and every time
• Improved programming and operating features.

2028. In which machine tool is the point-to-point Numerical controlled system applied?

(a) Machining centre (b) Drilling machine
(c) Turning centre (d) Grinding machine
(e) Shaping machine

CGPSC ARTO & TSI 07.01.2017

Ans. (b) : Numerical control (NC) is a relatively new form of automation which is generally applied in metal cutting, machine tools, although faminer drafting, assembling, wire drawing, welding and inspection machines are also making use of numerical control nowadays.
The latest development in the numerical control machine tool is the versatile machining centre. This is a single machine capable of doing a number of operation such as milling, boring, drilling, reaming and tapping by automatic tool changer.

2029. In CNC part program, what does M05 stand for?

(a) End of program
(b) End of tope and Coolant ON
(c) Tape rewind
(d) Spindle CW
(e) Spindle OFF

CGPSC ARTO & TSI 07.01.2017

Ans. (e) : In CNC part program.
M02 - End of program
M03 - Spindle start- forward or clockwise rotation
M04 - Spindle start reverse or counter clockwise rotation
M05 - Spindle stop
M06 - Tool Change

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2030. A Bezier surface is used for

(a) Machine tool industry
(b) Automobile industry
(c) Utensils
(d) Aerospace industry

TSPSC AEE 2017

Ans. (b) : Bezier surface are a species of mathematical spline used in computer graphics, CAD and FEM is used in automobile industry.

2031. In APT language, the cutter motion in incremental coordinate mode is addressed as :

(a) GO TO/…. (b) GO/TO ….
(c) GO DL TA/…. (d) GO FWD/….

Kerala PSC Poly. Lect., 2017
ESE 1995

Ans. (c) : GO DL TA/….
Note-Official answer given by commission is (b).

2032. Expansion of CIMS is

(a) Computer Integrated Manufacturing system
(b) Computer Integral Machining system
(c) Control Information Manufacturing simulation
(d) Computer Integrated Machining simulation

TSPSC AEE 2017

Ans. (a) : CIMS → Computer Integrated Manufacturing System.

2033. Point P (X, Y) located in XY plane is to be rotated in counter clockwise direction to new point P_1 (X_1, Y_1) by an angle ϕ. Which transformation matrix is to be used?

(a) $\begin{bmatrix} -\cos\phi & \sin\phi \\ \sin\phi & -\cos\phi \end{bmatrix}$ (b) $\begin{bmatrix} -\cos\phi & -\sin\phi \\ -\sin\phi & -\cos\phi \end{bmatrix}$

(c) $\begin{bmatrix} \cos\phi & -\sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix}$ (d) $\begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$

Punjab PSC SDE 12.02.2017

Ans. (d) : $\begin{bmatrix} -\cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$

2034. What is the purpose of the flush ( ) function?

(a) Flushes all streams and specified streams
(b) Flushes only specified stream
(c) Flushes input/output buffer
(d) Flushes file buffer

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : The flush ( ) method in python file handling clears the internal buffer of the file.

2035. Consider the following statements in the context of a CNC lathe machine:

1. The X axis controls the cross motion of the cutting tool
2. The Z axis controls the travel of the carriage towards or away from the headstock
3. The Y axis controls the motion of the job
4. The X-Y-Z axes are used for rotational movement of the table

Which of the above statements are correct?

(a) 1 and 3 (b) 1 and 2
(c) 3 and 4 (d) 2 and 3

JWM 2017

Ans. (b) : For lathes, the x-axis is the cross-slide motion at a right angle to the spindle axis.
Simple CNC lathes are two-axis machines and have no y-axis.
The axis of motion that is parallel to the spindle axis is always the z-axis.

2036. In measures of capacity, the system efficiency is expressed as ratio of–

(a) Actual output to the system capacity
(b) Actual output to Design capacity
(c) System capacity to Design capacity
(d) Design capacity to system capacity

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : In measure capacity, the system efficiency ratio is expressed as the ratio of actual output to the system capacity.

2037. SIMD represents an organization that.....

(a) Refers to a computer system capable of processing several programs at the same time.
(b) Represents organization of single computer containing a control unit, processor unit and a memory unit.
(c) Includes many processing units under the supervision of a common control unit
(d) None of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : SIMD represent an organization that computer of single computer represents organization of single computer containing a control unit, processor unit and a memory unit.

2038. Oval shaped symbol is used in flow charts to show the–

(a) decision box (b) statement box
(c) error box (d) start and end

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : Oval shaped symbol is used in flow chart to represent the start and end.

2039. In C programs, a compiler is used to convert the given program into–

(a) Assembly language (b) Machine language
(c) Decimal language (d) None of these

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : In C-language the complier is used to convert a given program in machine language.

2040. Language used to solve complex mathematical computations is–

(a) FORTAN (b) COBOL
(c) Pascal (d) Ada

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : FORTAN language is used for complex calculation of Mathematical computations.

2041. The command M02 is used in CNC programming for–

(a) Program Stop (b) End of Program
(c) Spindle ON (d) Optional Stop

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : The command M02 is used to end the program in CNC machine.

2042. Consider the following components :

(1) A dedicated computer
(2) Bulk memory
(3) Telecommunication lines

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Which of these components are required for a DNC system?

(a) (1), (2) and (3) (b) (2) and (3)
(c) (1) and (2) (d) (1) and (3)

TRB Polytechnic Lecturer 2017

Ans. (a) : A dedicated computer, bulk memory and also telecommunication lines are required for DNC system.

2043. Machining of complex shapes on CNC machines requires

(a) Simultaneous control of x, y and z axes
(b) Simultaneous control of x and y axes
(c) Independent control of x, y and z axes
(d) Independent control of x, y and z axes

APPSC AEE MAINS 2016, PAPER-III

Ans. (a) : Machining of complex shapes on CNC machines requires simultaneous control of x, y and z axes.

2044. A simulation model uses the mathematical expressions and logical relationships of the

(a) Estimated inferences
(b) Performance measures
(c) Computer model
(d) Real system

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : A simulation model uses the mathematical expressions and logical relationships of the real system.

2045. Which of the following is not a CAPP approach

(a) Variant approach
(b) Generative approach
(c) Hybrid approach
(d) Expert system approach

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : Expert system approach in not a CAPP approach.

2046. In relation to CNC machines which of the following statement is not correct

(a) machine utilization is very high
(b) ideal for mass production
(c) jigs and fixtures are mostly eliminated
(d) easy handling of sudden change in demand

GPSC ARTO 01.05.2016

Ans. (b) : According to CNC machines, following statement correct.
(i) Machine utilization is very high
(ii) Easy handling of sudden change in demand.
(iii) Jigs and Fixtures an mostly eliminated.

2047. Which is not a graphics standard

(a) NURBS (b) GKS
(c) CORE (d) PHIGS

GPSC ARTO 01.05.2016

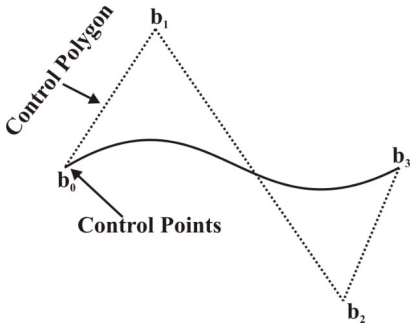
Ans. (a) : Graphics standards- CORE, GKS, PHIGS
Enveloping property– NURBS.

2048. A cubic Bezier curve has "n" controlling points where n is equal to

(a) 2 (b) 3
(c) 4 (d) 5

GPSC ARTO 01.05.2016

Ans. (c) :



Cubic Bezier Curve–

- Cubic bezier curve is a bezier curve with degree of freedom is 3.
- The total no. of Control point n = 4. Here, b_0, b_1, b_2, b_3 are control points.
- Degree of this curve is 3 so it is a cubic bezier curve.

Parametric equation $P(t) = \sum_{i=0}^{n} Bi\ J_{ni}(t)$

2049. Finite element method can be applied to

(a) stress analysis
(b) seepage analysis
(c) magnetic-flux studies
(d) All of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) : Finite element analysis is used for solving numerically different equations arising in engineering and mathematical modeling like stress analysis, seepage analysis, magnetic flux studies etc.

2050. Which elements is considered as three-dimensional, in FEM

(a) Triangular (b) Quadrilateral
(c) Tetrahedral (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) : Typically, a 3D solid element can be tetrahedron or hexahedron in shape with either flat or curved surface. Each node of the element will have three translational degrees of freedom. The element can thus deform in all three directional in space.

2051. In which one of the following machining manual part programming is done?

(a) NC machining (b) CNC machining
(c) DNC machining (d) FMS machining

APPSC AEE Screening Test 2016

Ans. (a) In NC machining manual part programming is done.
Part programming consists of a sequence of instructions to be performed on the NC/CNC machine.

2052. Which of the step of "operator controlled machine tool" and "NC machine tool" is similar?

(a) Component drawing (b) Control unit
(c) Recording on the tape (d) Programming

UPRVUNL AE 07.10.2016

Ans. (a) : Component drawing step of "operator controlled machine tool" and "NC machine tool" is similar.

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2053. Which of the following is incorrect for NC machine tool?
(a) Instruction undergoes electronic processing and outcome is the pulsed command
(b) Instruction for manufacturing the components are given verbally to the instructive and he executes the given instruction
(c) Control unit sends signals to drive unit of machine tool and to magnetic box
(d) Instruction for manufacturing the components are written in coded language and read by tape recorder

UPRVUNL AE 07.10.2016

Ans. (b) : Instruction for manufacturing the components are given verbally to the instructive and he executes the given instruction is incorrect for NC machine tool.

2054. Which of the following is not the actuation system of NC machine tool?
(a) Contour system
(b) Pneumatic system
(c) Electromechanical system
(d) Hydraulic system

UPRVUNL AE 07.10.2016

Ans. (a) : Contour system is not the actuation system of NC machine tool.

2055. Which of the following is not a true statement for CNC system?
(a) Operation of CNC machine is very easy relative to NC machine
(b) Programme to manufacture a component can be easily called in CNC relative to NC machine
(c) Reprogramming in CNC is more difficult than NC system
(d) CNC machine has self-diagnostic features

UPRVUNL AE 07.10.2016

Ans. (c) : Reprogramming in CNC is more difficult than NC system is not true statement for CNC system.

2056. Adaptive control feature is used:
(a) For automation
(b) Only to give better surface finish to the components
(c) Only to reduce power consumed by machine
(d) For inventory control

UPRVUNL AE 07.10.2016

Ans. (a) : Adaptive control feature is used for automation.
Adaptive control—Improvement in CNC machine tools depend on the refinement of adaptive control, which is the automatic monitoring and adjustment of machining conditions in response to variation in operation performance with a manually controlled machine tool, the operator watches for changes in machining performance and makes the necessary mechanical adjustment. An essential element of NC and CNC machining, adaptive control is needed to protect the tool, the workpiece, and the machine from damage caused by malfunctions or by unexpected changes in machine behaviour. Adaptive control is also a significant factor in developing unmanned machining techniques.

2057. Which of the following is incorrect?
(a) CNC machine can diagnose program
(b) CNC machine has higher flexibility in comparison to NC machine
(c) Data reading error in CNC machine is low in comparison to NC machine
(d) Conversion of units in CNC machine is not possible

UPRVUNL AE 07.10.2016

Ans. (d) : Conversion of units in CNC machine is not possible is incorrect.

2058. General description of CAD does not consist of:
(a) Implementation (b) Synthesis
(c) Presentation (d) Optimization

MPPSC AE 2016

Ans : (a) General description of CAD does not consist of Implementation

2059. Volume of work produced in FMS environment is determined from :
(a) Number of machine used in the FMS
(b) Kind of material handing equipment used in FMS
(c) Kind of layout used in FMS
(d) All are correct

MPPSC AE 2016

Ans : (d) Volume of work produced in FMS environment is determined from
i. Number of machine used in the FMS
ii. Kind of material handling equipment used in FMS.
iii. Kind of layout used in FMS.

2060. Machining time in NC and CNC machine tools is_____ in comparison to conventional machine tool:
(a) More (b) Less
(c) Unpredictable (d) Equal

MPPSC AE 2016

Ans : (b) Machining time in NC and CNC machine tools is less Comparison to conventional machine tool.

2061. What is the purpose of satellite computers in Distributed Numerical Control machines?
(a) To act as stand by systems
(b) To share the processing of large size NC
(c) To serve a group of NC machines
(d) To network with another DNC setup

MPPSC AE 2016

Ans : (c) To serve a group of NC machines

2062. In which machining system, the highest level of automation is found?
(a) CNC machine tools
(b) Automatic transfer machines
(c) Machine tools with electro hydraulic positioning and control
(d) DNC machining system

MPPSC AE 2016, ESE 2009

Ans : (c) The highest level of automation is Machine tools with electro hydraulic positioning and Control.

2063. In a CIM industry ..... is also used as transportation means for work parts and tools.
(a) AGV (b) NC machine tool
(c) RCC device (d) None of these

(HPPSC LECT. 2016)

Ans : (a) In a CIM industry AGV is also used as transportation means for work parts and tools.

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2064. Transfer machines can be defined as :
(a) Material Processing machines
(b) Material handling machines
(c) Material Processing and Material handling machines
(d) Components feeders for automatic assembly

MPPSC AE 2016

Ans : (c) Transfer machines can be define as materials processing and materials handling machines.

2065. Punched tape is used in ?
(a) NC machine (b) CNC machine
(c) NC and CNC both (d) DNC machine

MPPSC AE 2016

Ans : (a) Punch tape is used in NC machine.

2066. Flexible manufacturing allows for
(a) automated design
(b) factory management
(c) tool design and tool production
(d) quick and inexpensive product changes

TNPSC AE 2017, BPSC Poly. Lect. 2016
APPSC AEE MAINS 2016, PAPER-III

Ans : (d) Flexible manufacturing allows for quick and inexpensive product changes.

2067. The limitations of a pneumatically operated automatic control system is
(a) Complexity of components
(b) Costly maintenance
(c) Suffers from the transmission lags and slow response
(d) None of the above

Nagaland PSC CTSE 2016 Paper-I

Ans. (a) : The limitation of a pneumatically operated automatic control system is more difficult or complexity because of the compressed air is used in system, to transmit and control energy.

2068. Two and half axis control in NC machines means
(a) Control of three axis simultaneously
(b) Control of two axis and spindle speed simultaneously
(c) Control of two axis simultaneously and one axis independently
(d) Control of two axis simultaneously and one axis is stationary

RPSC VPITI 14.02.2016

Ans. (c) : Two and half dimension means that you are cutting a specific part of multiple features which is varying depth.
Controls of two axis (x, y) simultaneously and one axis independently.

2069. NC machine tool is operated by:
(a) Feed back system
(b) Output-input modules
(c) A series of coded instructions
(d) Digitizing

GPSC Asstt. Prof. 28.08.2016
APPSC Poly. Lect. 2013

Ans. (c) : NC is a form of programmable automation in which the mechanical actions of a machine tool or other equipment are controlled by a program containing coded alpha numerical data.

2070. During CNC milling machine programming, tool radius compensation for the tool on right of the programmed path is specified by
(a) G41 (b) G42
(c) G40 (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : G and M codes are used for machining and commonding action in CNC milling programming. G42 is for right radius tool compensation G41 for left and G40 for cancellation.

2071. Which type of motors are used as feed drives of CNC machine tools?
(a) AC servo motors (b) DC servo motors
(c) Stepper motors (d) All of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) : All of the above.

2072. Coolant ON/OFF can be executed in a CNC machine using:
(a) G-code (b) M-code
(c) N-code (d) C-code

HPPSC AE (PWD) 10.09.2016

Ans. (b) : Coolant ON/OFF can be executed in CNC machine using by the M codes.

Mode	Description
M07	Coolant ON (Mist)
M08	Coolant on (flood)
M09	Coolant off
M19	Spindle orientation

2073. If you use the absolute coordinate system to create a line from a starting point of 0.0 with 8 units on the X axis and 5 units on the Y axis. What do you enter for the second point?
(a) 8,5 (b) 5,0
(c) 0,8 (d) 5,8

UPRVUNL AE 21.08.2016

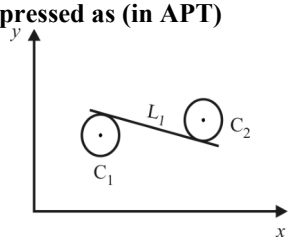
Ans. : (a) If use the absolute co-ordinate system to create a line from a starting point of (0.0) with 8 units on the X axis and 5 units on the Y axis. then 8.5 enter for the second point.

2074. Rapid plane in NC machine programming is :
(a) Imaginary plane just above the work piece.
(b) Imaginary plane just below the work piece.
(c) Plane on which work is kept
(d) Plane at tool change.

TSPSC Managers, 2015

Ans. (a) : Rapid plane in NC machine programming is imaginary plane just above the work piece.

2075. L_1 is expressed as (in APT)



(a) L_1=LINE/LFT TANTO C_1, RGT TANTO C_2
(b) L_1=LINE/RGT TANTOC_1, LFT TANTO C_2
(c) L_1=LINE/LFT TANTOC_1, LFT TANTO C_2
(d) L_1=LINE/RGT TANTO C_1, RGT TANTO C_2

TSPSC Managers, 2015

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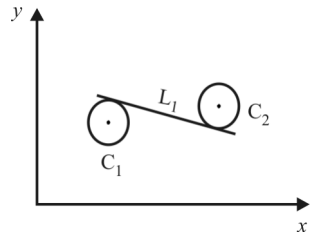
Ans. (a) :



L_1=LINE/LFT TANTO C_1, RGT TANTO C_2

2076. In CNC milling 'z' axis represents :
(a) Tool axis
(b) Work axis
(c) Table axis
(d) Both work and tool axis

TSPSC Managers, 2015

Ans. (a) : In CNC milling 'z' axes represents tool axis.

2077. CIM II represents the
(a) second generation of development in manufacturing
(b) third generation of development in manufacturing
(c) fourth generation of development in manufacturing
(d) fifth generation of development in manufacturing

VIZAG STEEL MT 2015

Ans. (d) : Fifth generation of development is manufacturing.

2078. CIM Wheel was developed by
(a) CASA/SME in 1975
(b) ASTI/ MRP in 1875
(c) MRSA/ SEM in 1965
(d) CIMA/ASM in 1985

VIZAG STEEL MT 2015

Ans. (a) : CASA/ SME in 1975

2079. The Software system used for translating neutral instruction from the CAM system into the specific instructions required by the NC machine is.
(a) Post-processor (b) CIM software
(c) CAD/CAM software (d) operating system

VIZAG STEEL MT 2015

Ans. (a) : Post - processor

2080. For increasing the productivity, CNC system can be interface with
(a) CAD/CAM (b) DNC
(c) FMS (d) All of these

VIZAG STEEL MT 2015

Ans. (d) : For increasing the productivity, CNC system can be interface with
- CAD/CAM
- DNC
- FMS

2081. A part programme for any arbitrary object is given as follows.

N001	**G71**	**G91 M03 S500 EOB**
N002	**G00**	**X 10.00 Y10.00 EOB**
N003	**G00**	**Z-10.00 EOB**
N004	**G84**	**Z- 25.00 F60 EOB**
N005	**G80**	**EOM**
N006	**M02**	**EOB**

The above programming format will be used for canned cycle for
(a) drilling (b) tapping
(c) boring (d) grooving

VIZAG STEEL MT 2015

Ans. (b,c) : Tapping, Boring

2082. Which type of motor is used in axial or spindle drives of CNC machine tools?
(a) Induction motors
(b) DC servomotors
(c) Stepper motors
(d) Linear servo motors

NPCIL ET, 2015

Ans. (c) : Stepper motors are used in axial or spindle drives of CNC machine tools.

2083. The code "G04" is used for-
(a) Absolute co-ordinate system
(b) Dwell
(c) Spindle stop
(d) Linear interpolation

NPCIL ET, 2015

Ans. (b) : G04- Dwell- Takes an address for dwell period (may be X, U or P)
The dwell period is specified in the controllers parameter, typically in milliseconds.

2084. CNC machines are generally used for -
(a) Mass production
(b) Batch Production
(c) Continuous Flow Production
(d) Single Piece production

NPCIL ET, 2015

Ans. (a) : CNC machines are generally used for mass production. CNC (computer numerical control) taking digitized data, a computer and CAM program is used to control, automated and monitor the movements of a machine. The machine can be a milling machine, lathe, router, welder, grinder, laser or water jet cutter, sheet metal stamping machine, robot or many other types of machines.

2085. In CNC programming code G means
(a) Machine Code (b) Preparatory Code
(c) Auxillary code (d) Standard code

GWSSB AAE, 27.12.2015

Ans. (b) : G– Preparatory code

2086. What is the meaning of the following NC code statement : N 20 G 91 X 20 Y 10
(a) Statement number 20 move the tool incrementally by X = 20 , Y = 10 and Z = 0
(b) Statement number 20 move the tool to X = 20, Y = 10 and Z = 0 position
(c) Statement number 20 drill a hole with dia = 20 and depth = 10
(d) Statement number 20 drill a step hole with dia 20 and 10

TSGENCO AE 14.11.2015

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Ans. (a) : NC code statement = N20 G91 X20 Y10
Number 20 move the tool G91 incrementally positioning by
X = 20, Y = 10 and Z = 0

2087. Which one of the following is most suitably applicable for a five axis CNC machine
(a) it has three rotational and two translation axis
(b) it has three transitional and two rotational axis
(c) is a CNC machine where machining can be done 5/3 times faster than a 3 axis machine
(d) is useful only for complex geometry parts and normal 3D parts cannot be machined on it.

TSGENCO AE 14.11.2015

Ans. (b) : 5- Axis CNC machining offers three linear axes and two rotational axes that work simultaneously to achieve complex surface machining.
5-Axis machine helps you work on light tolerance to achieve complex geometries.

2088. The essential physical components of FMS is
(a) potentially independent NC machine tools
(b) a conveyance network to more parts and sometimes tools between machines and fixturing stations
(c) an overall control network that coordinates the machine tools, the part-moving elements and the workpieces
(d) all options are correct

MPPSC AE 08.11.2015

Ans. (d) The essential physical components of FMS is-
- Potentially independent NC machine tools.
- A conveyance network to more parts and some times tools between machines and fixturing stations.
- An overall control network that coordinates the machine tools, the part moving elements and the work pieces.
- FMS stands for flexible manufacturing system.

2089. CAD/CAM is inter-relationship between
(a) Marketing and design
(b) Manufacturing and marketing
(c) Engineering and marketing
(d) Engineering and manufacturing

MPPSC AE 08.11.2015

Ans. (d) : CAD/CAM is the inter-relationship between Engineering and Manufacturing.
CAD (Computer aided design)
CAM (Computer aided manufacturing)

2090. What is computer numerical control (CNC)?
(a) CNC is a self contained NC system for a single machine tool using a computer controlled by a part program to perform basic NC functions.
(b) CNC is a self contained system for a single operated manual controlled machine
(c) CNC is a self contained NC system for a multi machine tool operated by a conventional technique.
(d) None of these is correct

MPPSC AE 08.11.2015

Ans. (a) : Computer Numerical Control (CNC): CNC is a self contained NC system for a single machine tool using a computer controlled by a part program to perform basic NC functions.

2091. In FMS the tools are identified by means of :
(a) colour code (b) bar code
(c) PLC (d) digital code

(KPSC AE. 2015)

Ans : (b) In FMS the tools are Identified by means of bar code.

2092. The term ATC in CNC machine tool means
(a) Automatic tool control
(b) Automatic tool changer
(c) Automatic table contour
(d) automatic tool coding

TSPSC AEE 2015

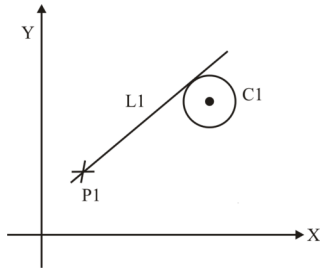
Ans. (b) : The term ATC in CNC machine tool means automatic tool changer.

2093. M06 represents the following instruction in CNC programming.
(a) Program stop (b) tool change
(c) coolant on (d) spindle on

TSPSC AEE 2015

Ans. (b) : M06 represents the tool change instruction in CNC programming.

2094.



As shown figure 'L1' is expressed in APT as
(a) L1 = LINE / TANTO C1
(b) L1 = LINE / INTOF C1
(c) L1 = LINE / P1, LFT TANTO C1
(d) L1 = LINE / P1, RGT TANTO C1

TSPSC AEE 2015

Ans. (c) : L1 = LINE / P1, LFT TANTO C1

2095. Automated Manufacturing device is a component of :
(a) FMS (b) CAD/CAM
(c) CAPA (d) ASRS

UPRVUNL AE 2014

Ans : (a) Automated manufacturing is a component of flexible manufacturing system (FMS).
An Automated Manufacturing System is an interconnected system of material processing stations capable of material processing a wide variety of part types simultaneously under computer control such system exhibits flexibility in part routing, part processing, part handling, and tool changing.

2096. What is the name given to a system using an automated work cell controlled by electronic signals from a common centralized computer facility?

Search on Telegram @apna_pdf

(a) Adaptive control system
(b) Robotics
(c) Flexible manufacturing system (FMS)
(d) Automatic guided vehicle (AGV) system

UPRVUNL AE 2014

Ans : (c) Flexible manufacturing system which use electronic signal to control automated work cell.

2097. Currently, the need for a single neutral format for Computer Aided Design programming, is filled mainly by :
(a) STL file (b) STEP file
(c) TIFF file (d) JEPEF file

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Neutral file exchange uses an intermediary neutral file format to translate data between CAD systems. This method starts from a preprocessor embedded in the original CAD system, which generates the neutral file from the originating CAD format.

Some neutral formats are defined by standards organizations such as IGES and STEP while others are properietery but still widely used and regarded as quasi industry standards.

IGES- Initial Graphics Exchange Specifications.

STEP – (ISO 10303 – Standard for Exchange of Product model data).

2098. _____Manufacturing System is a system based on multi-operation machine tools, incorporating integrated computer control with associated support function and material handling
(a) Flow (b) Fixed
(c) Flexible (d) None of these

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (c) : Flexible manufacturing systems (FMS) is a system base on multi operation machine tools, incorporating integrated computer control with associated support function and material handling.

2099. What difference does the 5th generation computer have form other generation computers?
(a) Technological advancement
(b) Scientific code
(c) Object oriented programming
(d) Need of the today

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Technological advancement difference the 5th generation computer have form other generation computers.

2100. Use of CIM technology offers a layout architecture where computation takes place
(a) Using on line information
(b) Using virtual memory
(c) On line information and multi-processing
(d) Numerically

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (c) : Computer Integrated Manufacturing (CIM)–CIM presupposes that measurements can be made online, which is not possible for all manufacturing processes then it uses multi-stage processing of online information.

2101. In a closed loop control NC machine, tool system is the design of control loop that decides
(a) Position error, accuracy and repeatability
(b) Cost of the machine tool, maintenance and operation
(c) Programming of a part, load and torque
(d) Cost of machining

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (a) : Closed Loop Control System–
• In a closed loop control system the actual output from the system i.e. the actual displacement of the machine slide, is compared with input signal.
The closed loop control system is characterised by the presence of feedback device in the system closed loop control system decides position, accuracy and repeatability.

2102. The numerical control system which is applicable to a drilling machine is :
(a) Point to point system
(b) Continuous path system
(c) Straight cut system
(d) Zig Zag machining system

TNPSC AE (Industries) 09.06.2013

Ans. (a) : The numerical control system which is applicable to a drilling machine is point to point system.

2103. Which functional code indicates the parabolic interpolation?
(a) G04 (b) G03
(c) G06 (d) G08

APPSC Poly. Lect. 2013

Ans. (c) : The method by which control system, calculates the intermediate point and speed of the motor, is known as interpolation.
This interpolation is of three types-
(i) Linear-G01 (ii) Circular-G02 (iii) Parabolic-G06

2104. Goal Programming can involve:
(a) One objective; one constraint
(b) One objectives; multi constraints
(c) Multi objectives; multi constraints
(d) Multi objectives; one constraint

SJVN ET 2013

Ans. (c) : Goal programming can involve multi objectives ; multi constraints.

2105. Machining centre is a
(a) NC machine tool
(b) Transfer machine tool
(c) Group of automatic machine tools
(d) Next logical step beyond NC machine

SJVN ET 2013
ESE 2001

Ans. (c) : Machining centre is a group of automatic machine tools.
• It includes an automatic tool changer and a table that clamps the workpiece in place against the tool.

2106. Two major categories of software are application software and
(a) System software (b) Unix software
(c) CAD software (d) High-level languages

TRB Poly. Lect., 2012

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Ans. (a) : Two major categories of software are application software and system software.

2107. Number control
(a) applies only to milling machines
(b) is a method for producing exact number of parts per hour
(c) is a method for controlling by means of a set of instructions
(d) all of these

TRB Poly. Lect., 2012

Ans. (c) Numerical control is a method for controlling by means of a set of instructions

2108. Which one of the following items best describes the CAM technology?
(a) Documentation
(b) Numerical control
(c) Drafting
(d) Geometric modeling

TRB Asstt. Prof., 2012

Ans. (b) : Numerical control

2109. In manual programming and tape preparation for an NC drilling machine, the spindle speed was coded as S684. The spindle in rpm will be
(a) 68.4 (b) 684
(c) 840 (d) 6840

TNPSC ACF 2012
ESE 1996

Ans. (b) :
S 6 8 4 → rpm
↳ Spindle
The spindle in rpm well be 684.

2110. Which of the following pairs are correctly matched ?
I. CNC machine – Post processor
II. Machining centre – Tool magazine
III. DNC – FMS
(a) I, II and III (b) I and II
(c) I and III (d) II and III

TNPSC ACF 2012

Ans. (a) : CNC Machine – Post Processor
Machining centre – Tool magazine
DNC – FMS

2111. Which of the following are rules the programming NC machine tools in APT language?
I. Only capital letters are used
II. A period is placed at the end statement
III. Insertion of space does not affect the APT word
Now select the correct answer using the codes given below
(a) I and II (b) II and III
(c) I and III (d) I alone

TNPSC ACF 2012

Ans. (c) : The following rule of programming NC machine tools in APT language-
(1) The cycle time is total time taken by all the machining operation clear.
(2) Act the machine tools must be. Automatic clear.
(3) All the machine tools must be conventional and general purpose type clear.
(4) Only capital letter type clear.
(5) Insertion of space does not affect the APT word.

2112. In a computer application where a direct dialogue is carried on between the computer and the user it is referred to as
(a) Interfacing
(b) Decision application
(c) Video conferencing
(d) Real time application

APPSC AEE 2012

Ans. (a) : Interfacing– In computing, a interface is a shares boundary across which two or more separate components of a computer system exchange information. the exchange can be between software, computer hardware, peripheral devices, humans, and Combination of these.

2113. FMS system is best suited to :
(a) Mass production system
(b) High variety production system
(c) Job-shop production system
(d) Mid variety and mid volume production system

RPSC AE GWD, 2011
DRDO Scientists 2009

Ans. (d) : FMS (Flexible manufacturing system) is best suited for mid variety and mid volume production system.

2114. Lean manufacturing focuses on :
(a) Reduction of inventory
(b) Elimination of non-value added activities (waste)
(c) Grouping of machine for product family
(d) Use of CNC machines for production

RPSC AE GWD, 2011

Ans. (b) : Lean manufacturing focuses on Elimination of wastes (non-value added activities).

2115. CNC system stands for :
(a) Complex Numerical Control
(b) Computer Numerical Control
(c) Conventional Numerical Control
(d) Continuous Numerical Control

RPSC AE GWD, 2011

Ans. (b) : CNC – Computer Numeric Control.
DNC– Distributed or Direct Numeric Control.
NC – Numeric Control.

2116. In CAM, "Part programming" refers to
(a) Generation of cutter location data
(b) On-line Inspection
(c) Machine Selection
(d) Tool Selection

ISRO Scientist/Engineer 2011

Ans. (a) : In CAM, "Part programming" refers to generation of cutter location data.

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2117. Consider the following components regarding numerical control system :

1. Programme of instructions
2. Machine control unit
3. Processing equipment

Which of these are correct ?

(a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only

UPSC JWM Advt. No.-50/2010

Ans. (a) : The components of the N-C machine systems are–
- Programme of instructions
- MCU (Machine Control Unit)
- Processing equipment

2118. Consider the following advantages of DNC systems :

1. Time-sharing
2. Greater computational capability
3. Remote computer location

Which of the above is/are correct ?

(a) 1 and 2 only (b) 2 and 3 only
(c) 2 only (d) 1, 2 and 3

UPSC JWM Advt. No.-50/2010

Ans. (d) : Direct numerical control systems have many advantages such as:
- Time sharing
- Greater computational capability
- Remote computer location etc.

2119. Interpolator is enclosed in :

(a) Data processing unit
(b) Control loop unit
(c) Machine tool
(d) None of the above

APPSC IOF, 2009

Ans. (b) : Control loop unit

2120. Which of the following pairs are correctly matched?

(i) CNC machine—post processor
(ii) Machining center—tool magazine
(iii) DNC-FMS

(a) (i) (ii) and (iii) (b) (i) and (ii)
(c) (i) and (iii) (d) (ii) and (iii)

APPSC IOF, 2009

Ans. (a) : All three are correctly matched–
(i) CNC machine – Post processor
(ii) Machining center – Tool magazine
(iii) DNC – FMS

2121. The following has integrated tool changing unit and a component indexing device :

(a) NC system (b) Turning center
(c) CNC system (d) Machining center

MPPSC AE 2016
APPSC IOF, 2009

Ans. (d) : Machining center has integrated tool changing unit and a component indexing device.

2122. In the context of computer-aided design (CAD). STEP means

(a) standard for the Exchange of Product Model Data
(b) Solid Three-dimensional Engineering Product
(c) Solid Technology and Engineering Product
(d) Standard Three-dimensional Engineering Package

DRDO Scientists 2008

Ans. (a) : STEP = Standard for The Exchange of Product Model Data.

2123. A stepping motor has 300 step angles and its output shaft is directly connected to the idea screw having a pitch of 5 mm. The work table of a positioning system is driven by the lead screw. The required pulse rate to move the work table at 200 mm/min is :

(a) 100 Hz (b) 200 Hz
(c) 100 Hz (d) 40 Hz

DRDO Scientists 2009

Ans. (b) : Pulse rate = $\frac{200}{60}\times\frac{1}{5}\times 300 = 200$ Hz

2124. The tool of NC machine has to cut circular arc from strating point (10, 16) to end of the arc (18, 24) having centre of the arc (18, 16). Which N.C. program statement would cut the required arc?

(a) G03 X 18.0 Y 24.0 | 0 J 8.0
(b) G02 X 18.0 Y 24.0 | 8.0 J0
(c) G03 X 18.0 Y 24.0 | 8.0 J 8.0
(d) G02 X 18.0 Y 24.0 | 8.0 J 8.0

Punjab PSC SDE 12.02.2017

Ans. (b) : G02 X 18.0 Y 24.0 | 8.0 J0

2125. Which of the following methods is/are of projecting the pictorial views?

(a) Axonometric projection
(b) Oblique projection
(c) Perspective projection
(d) All of these

MECON MT 2019

Ans. (d) : Perspective projection is of projecting the pictorial view.

2126. CIM technology if implemented integrates various production function if the factory has

(a) line layout (b) cell layout
(c) functional layout (d) mixed type

VIZAG STEEL MT 18.08.2013

Ans. (d) : CIM technology if implemented integrates various production function if the factory has mixed type.

2127. While producing a few parts with an accuracy of 0.01 mm, it would be preferable to select

(a) an automated machine tool
(b) NC machine tool
(c) transfer machine
(d) general purpose machine

VIZAG STEEL MT 18.08.2013

Search on Telegram @apna_pdf

Ans. (b) : For production in batch size with high accuracy as 0.01 mm, it would be preferable NC machine tools, which executes the operations. It may consist of worktable, cutting tools jigs and fixtures, motors for driving spindle and coolant and lubricating system.

2128. A 100 mm long work piece is to be machined on an NC lathe at 500 rpm and feed rate of 0.1 mm/rev. A stepper motor having a step of 1.8° is used to provide the feed motion through a lead screw having a pitch of 5 mm. Pulses required to machine the work piece:

(a) 4000 (b) 3000
(c) 3500 (d) 4500

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (a) :

2129. A control system that measures certain output process variables and uses these to control speed and feed is called:

(a) adaptive control (b) in-process control
(c) operator control (d) feed-back control

CIL MT 27.02.2020

Ans. (a) : A control system that measures certain output process variables and uses these to control speed and feed is called adaptive control.

2130. Which of the following devices produces incremental motion through equal pulses?

(a) AC servo motor (b) DC servo motor
(c) Stepper motor (d) Series motor

UPPSC AE 13.12.2020, Paper-I
ESE 2018

Ans. (c) : In a stepper motor, there are salient poles on rotor as well as stator. The motion of the rotor can be controlled by energizing the stator windings in a sequence, by using equal pulses.

2131. Scaling is to be applied to point P (X, Y) by scaling factor Sx, Sy. What would be scaling matrix?

(a) $\begin{bmatrix} Sx & Sy \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ Sx & Sy \end{bmatrix}$

(c) $\begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix}$ (d) $\begin{bmatrix} 0 & Sx \\ Sy & 0 \end{bmatrix}$

Punjab PSC SDE 12.02.2017

Ans. (c) : $\begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$

2132. Reflection of point P (X, Y) to P1 (-X, -Y) required transformation matrix as :

(a) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

Punjab PSC SDE 12.02.2017

Ans. (c) : $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

2133. The magnetic storage chip used to provide non-volatile direct access storage of data and that have no moving parts are known as

(a) Magnetic core memory
(b) Magnetic tape memory
(c) Magnetic disk memory
(d) Magnetic bubble memory

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : The magnetic storage chip used to provide non-volatile direct access storage of data and that have no moving parts are known as magnetic bubble memory.

2134. Which of the following have low failure rate?

(a) Mechanical devices
(b) Electronic devices
(c) Electro-mechanical devices
(d) None of above

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Electronic devices have low failure rate as compared to all these.

2135. Easily reloctable language is

(a) Machine language
(b) Assembly language
(c) High level language
(d) Medium level language

GPSC Asstt. Director of Transport 05.03.2017

Ans. (b) : Easily reloctable language is assembly language.

2136. Among the following, identify the one which is not a component of a control system.

(a) A standard
(b) A flow of authority to enforce the standards
(c) A means of measuring accomplishment
(d) A comparison of actual results with the standard

UPRVUNL AE 2014

Ans : (b) A flow of authority to enforce the standards is not a component of a control system.

2137. Generally, the three components of control systems are checked to determine the responsibility for control. If any of these criteria is not met, then the process is the responsibility of :

(a) Management
(b) Quality control team
(c) The process owner
(d) The supervisor

UPRVUNL AE 2014

Ans : (a) Generally for checking the control system, there are three criteria responsible. If any of these criteria got failed then this failure responsibility is given to Manager.

2138. $(1111\ 0111\ 1101\ 1011)_2 = (\quad)_{16}$

(a) D734 (b) ABC1
(c) F7DB (d) D7BF

APPSC IOF, 2009

Ans. (c) : $1111 \rightarrow F$
$0111 \rightarrow 7$
$1101 \rightarrow D$
$1011 \rightarrow B$

Search on Telegram @apna_pdf

2139. ASCII is how many bit code ?

(a) 7 (b) 8
(c) 6 (d) 9

APPSC IOF, 2009

Ans. (b) : Computer manufacturers agreed to use code called the ASCII. ASCII is an 8-bit code.

2140. 1011 equivalent Gray Code is :

(a) 1011 (b) 1101
(c) 1001 (d) 1110

APPSC IOF, 2009

Ans. (d) :

Decimal Number	4 - bit Binary Code	4 - bit Gray Code
11	1011	1110

2141. The ability of a positioning system to return to a previously defined location is measured by its:

(a) accuracy (b) sensitivity
(c) control resolution (d) repeatability

DRDO Scientists 2009

Ans. (d) : The ability of a positioning system to return to a previously defined location is measured by its repeatability.

2142. Determine the correctness or otherwise of the following Assertion (A) and the Reason (R) :

Assertion (A) : The generative system of computer-aided process planning is more complex than the variant system.

Reason (R) : The generative system has the capability of creating a new plan instead of using and modifying the existing plan. This system has to use some type of expert system.

(a) Both (A) and (R) are true and (R) is the correct reason for (A)
(b) Both (A) and (R) are true but (R) is not the correct reason for (A)
(c) Both (A) and (R) are false
(d) (A) is true but (R) is false

DRDO Scientists 2008

Ans. (a) :

2143. With numerical control equipment, which one of the following is not true?

(a) Single-piece machining is possible and feasthic because of the natures of the control system
(b) Fixture cost is considerably lower for numerical control machining than for conventional machining
(c) The initial cost of N.C. machine is low
(d) Programming and tape writing take much less time than building jig and fixtures and locating gases

TNPSC AE, 2008

Ans. (c) : The initial cost of N.C. machine is low because its costly setup.

8. Jigs and Fixtures

2144. 3-2-1 method of location in a jig or fixture would collectively restrict the work piece in 'n' degrees of freedom where the value of 'n' is

(a) 6 (b) 8
(c) 9 (d) 1

GPSC ARTO Pre 30.12.2018
MPPSC AE 2016
MPPSC AE 08.11.2015
ISRO Scientist/Engineer 11.10.2015
TRB Asstt. Prof., 2012

Ans. (c) : 3-2-1 method of location in a jig or fixture would collectively restrict the work piece in 'n' degrees of freedom where the value of 'n' is 9.

2145. A device which holds and locates a work piece during an inspection or for a manufacturing operation, is known as :

(a) Fixture (b) Jig
(c) Lathe (d) Templates

HPPSC W.S. Poly. 2016
UPRVUNL AE 2014
Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans : (a) A device which holds and locates a work picce during an inspection or for a manufacturing operation is know as fixture.
Fixture:- This may be described as a structure for locating, holding and supporting a component or workpiece securely in a definite position for a specific operation but it does not guide the cutting tool. The cutting tools are set in position by machine adjustment or by trial and error method.

2146. Diamond pin location is used in a fixture because

(a) It does not wear out
(b) It takes care of any variation in centre distance between two holes
(c) It is easy to clamp the part on diamond pins
(d) It is easy to manufacture

RPSC IOF, 2020
UKPSC AE 2007 Paper-I
ESE 1999

Ans. (b) : Diamond pin location is used in a fixture because it takes care of any variation in centre distance between two holes.

2147. A diamond pin is used in conjunction with a round locator for radial location in a jig because

(a) Diamond is harder material and hence has long life
(b) Reduces jamming possibility if two round locators are used
(c) Diamond pin is less expensive to make
(d) None of these

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I
RPSC Lect. (Tech. Edu. Dept.) 2011

Search on Telegram @apna_pdf

Ans. (d) : A diamond pin is used with a round locator for radial location in a jig because, it is easy to clamp the part of diamond pins.
A diamond locating pin is used in jigs and fixtures because it helps in assembly with tolerance on centre distance.

2148. In the 3-2-1 principle of fixture design 3 refers to the number of
(a) Clamps required
(b) Locators on the primary datum face
(c) Degrees of freedom of the work place
(d) Operations carried out on the primary datum face

TSPSC AEE 2017, WBPSC AE, 2007
UJVNL AE 2016
UPPSC AE 12.04.2016 Paper-I

Ans : (b) In the 3-2-1 principle of fixture 3 refers to number of locating position. According to principle, 3 pins are used in primary datum, 2 pins used on secondary datum, which is perpendicular to 1st and 1 in tertiary datum which arrests 9 DOF.
3-2-1 principle of location:-3-2-1 principle of location states that to locate a piece fully, it has to be placed and held against three points in a base plane, two points in a vertical plane, and one point in a plane Square with the first two. It is important that the above planes be square with each other and the points should be spaced as for apart as possible.

2149. A device which holds and locates a work piece during an inspection or for a manufacturing operation, but does not guide the tool, is called
(a) Jig (b) Fixture
(c) Saddle (d) Carriage

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
UPRVUNL AE 2014

Ans. (b) : A device which holds and locates a work piece during an inspection or for a manufacturing operation, but does not guide the tool, is called fixture.

2150. Which of the following is not true in case of jigs and fixtures ?
(a) Consistency in dimension
(b) Fast production speed is not possible
(c) Auto-location control
(d) None of the above

UKPSC AE 2012, 2007 Paper-I

Ans. (b) : Fast production speed is not possible

2151. A diamond-locating pin is used in jigs and fixture, because
(a) diamond is very hard and wear-resistant
(b) it occupies very little space
(c) it helps in assembly with tolerance on central distance
(d) it has a long life

Nagaland PSC (CTSE) 2018, Paper-I
JPSC AE 2013, Paper-V
BPSC AE 2012 Paper - VI

Ans : (c) : Diamond pins are relieved on two sides to allow for variations in the centre to centre distance of the holes. They locate at right angles to the axis between pin centres and should be placed in the largest hole.

2152. Which of the following operation does NOT use a jig?
(a) Tapping (b) Reaming
(c) Turning (d) Drilling

UPPSC AE 13.12.2020, Paper-I
GPSC Executive Engineer 23.12.2018
UKPSC AE-2013, 2012 Paper-I

Ans. (c) : Applications of jigs are :
(i) Drilling, (ii) Reaming, (iii) Tapping
Applications of fixtures are :
(i) Milling, (ii) Grinding, (iii) Planning, (iv) Turning, (v) Shaping.

2153. The jig bush most commonly used in jigs is
(a) Liner bush (b) Headless drill bush
(c) Headed drill bush (d) Shaped drill bush

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (c) : A jig bush is also know as drill bush, is a tool used in metal cutting to guide by jigs to cutting tools, most used for drill tool. Most common are headed drill bush.

2154. _____Jig is a type of drill jig.
(a) Root (b) Stem
(c) Leaf (d) Sand

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) :

2155. Which one of the following is work holding device used in lathes?
(a) Chuk (Chuks) (b) Tool post
(c) Cross slide (d) Compound rest

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Chuck–
It's the most important device for holding the work piece, particularly of short length and large diameter or irregular shape which can not be conveniently mounted between centers. It can be attached to the lathe by screwing on the spindle nose.

2156. Which of the following can hold the work, locate the work and guide the drill at the desired position
(a) drill bush (b) drill fixture
(c) metal locator (d) drill jig

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : Drill jig hold the work locate the work and guide the drill at the desired position.

2157. Jigs cannot be used in
(a) drilling (b) reaming
(c) milling (d) tapping

Haryana PSC AE (PHED) 05.09.2020, Paper-II

Ans. (c)igs cannot be used in milling.

2158. When a cylindrical is located in Vee block, how many number of degree of freedom will be restricted:
(a) 0 (b) 2
(c) 7 (d) 12

Oil India Ltd. Sr. Engg. (Production) 30.11.2019
HPPSC Lect. (Auto) 23.04.2016

Ans. (c) : Number of degree of freedom will be restricted is 7.

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2159. V-blocks (Vee locators) are used for clamping as well as locating when faces are inclined upto
(a) 30 degree (b) 12 degree
(c) 9 degree (d) 3 degree
GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : V-blocks are used for clamping as well as locating when faces are inclined upto 3° degree.

2160. Complex irregular three dimensional surfaces can be located using the following method
(a) Sighting (b) Nesting
(c) Integral locators (d) Buttons
ISRO Scientist/Engineer 17.12.2017

Ans. (b) : Nesting locators position a part by enclosing it is a depression or recess of the same shape as the part. Complex irregular 3-D surfaces can be located using nesting locators. It is the most accurate locating device for profile locator.

2161. The type of locator used to prevent jamming during locating of a work piece is
(a) concentric locator (b) integral locator
(c) equilateral locator (d) spherical locator
ISRO Scientist/Engineer 17.12.2017

Ans. (d) : Length of the locator should be short as much as possible to prevent how a workpiece is positioned on a spherical locator, the diameter (d) of the locator is always the same.
This feature makes the spherical locator impossible to jam or blind in the locating hole.
Concentric locators, for the most part, locate a workpiece from a central axis. This axis may or may not be in the centre of the workpiece.

2162. The principle most commonly followed for locating work pieces in a fixture is:
(a) 2 – 3 – 1 (b) 1 – 2 – 3
(c) 3 – 2 – 1 (d) 1 – 3 – 2
CIL MT 26.03.2017

Ans. (c) : 3-2-1 is the principle to locate the work piece firmly so that the required operations like drilling, reaming etc. can be done. According to the principle 3 pins are used in primary datum, 2 pins used on secondary datum which is perpendicular to 1st and 1 in tertiary datum which arrest 9 degrees of freedom.

2163. Which of the following locating device is used to locate cylindrical jobs?
(a) Drill jigs (b) V-blocks
(c) Angle plates (d) Metal pins
CIL MT 26.03.2017

Ans. (b) : V(vee) Locators are used mainly for round work. So for locating cylindrical jobs, V – block locators are used

2164. _____ is NOT a clamping device used in jigs and fixtures.
(a) Diamond pin (b) Slotted strap
(c) Swing latch (d) Quick acting nut
(e) C-clamp
(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : Diamond pin is not a clamping device used in jigs and fixtures.
Jig and Fixture–Fixtures, being used in machine shop, are strong and rigid mechanical device which enable easy, quick and consistently accurate locating, supporting and clamping, blanks against cutting tools and result faster and accurate machining with consistent quality, functional ability and interchangeability. Jig is a fixture with an additional feature of tool guidance.

2165. The simplest and low-cost clamp used in Jigs and Fixtures is:
(a) Strap clamp (b) Can clamp
(c) Toggle clamp (d) Equiliser
TRB Polytechnic Lecturer 2017

Ans. (a) : The simplest and low-cost clamp used in Jigs and Fixtures is strap clamp.

2166. Which of the following is not a work holding device?
(a) V-block (b) Chuck
(c) Steady rest (d) None of the above
UJVNL AE 2016

Ans : (d) None of the above
Chuck:- It is the most important device for holding the workpieces, particularly of short length and large diameter or a irregular shape which can't be conveniently mounted between centres.
Steady rest:- The steady rest is used when a long piece is machined or drilled at its end by holding the job in a chuck. It avoids the undue deflection of the job at the other end.

2167. Drill jigs are used to hold the work piece when ..... .
(a) drilling and reaming only
(b) counter boring and countersinking only
(c) spot facing and tapping only
(d) All three mentioned here
GWSSB DEE 07.07.2016

Ans. (d) : Drill jigs are used to hold the work piece when –
- Drilling and reaming only
- Counter boring and countersinking only
- Spot facing and tapping only

2168. The selection of non-traditional machining process for a particular application depends on
(a) Physical parameters of the process
(b) Shapes to be machined
(c) Process capability
(d) All of the above
Nagaland PSC CTSE 2016 Paper-I

Ans. (d) : The selection of non-traditional machining process for a particular application depends on Various factors.
(1) Physical parameter of the process (Surface quality high).
(2) Shape to be machined (Complexity shape)
(3) Process capability.

2169. If the diameter of the hole is subjected to considerable variation, then for location in jigs and fixtures, the pressure type of locator used is :
(a) Conical locator (b) Cylindrical locator
(c) Diamond pin locator (d) Vee locator
(e) None of the above
CGPSC Poly. Lect. 22.05.2016

Ans. (a) : If the diameter of the hole is subjected to considerable variation, then for location in jigs and fixtures, the pressure type of locator used is conical locator.

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2170. Consider the following statement regarding fixtures employed for holding work piece during machining (i) The location is based on 3-2-1 principle, (ii) The number refers to pins employed in three mutually perpendicular planes to arrest all the degree of freedom (iii) Fixtures also provides the tool guidance

(a) (i), (ii) and (iii) are true
(b) (ii) and (iii) are true
(c) (i) and (iii) are true
(d) (i) and (ii) are true

RPSC VPITI 14.02.2016

Ans. (d) : Fixtures employed for holding work piece during machining-
(i) The location is based on 3-2-1 principle
(ii) The number refers to pins employed in three mutually perpendicular planes to arrest all the degree of freedom. These are correct statements.
Jigs are used to guide the tool.

2171. Locator duplication not only cost____but also could cause_____.

(a) less and accuracies
(b) more and inaccuracies
(c) less and inaccuracies
(d) more and accuracies

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Locator duplication costs more but also causes more in accuracies.

2172. To accurately locate a part in a jig or fixture, movements must be restricted. This is done with ____ and _____.

(a) bushes and locators
(b) locators and clamps
(c) clamps and bushes
(d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Locators provide a positive stop for the workpiece. Clamps are only intended to hold the workpiece against the locators.

2173. When holes are to be drilled in several faces in a small workpiece_____ is used

(a) Template jig (b) Turnover jig
(c) Pot jig (d) Box jig

APGCL AM, 2021
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (d) : Box jig- A jig in which work is rigidly held so that it can be drilled or machined from different angles at angle setting depending on which face of the jig in turned towards the tool.

Box type jig:
⇒ The box jig can permits drilling of holes in three sides of the work piece and saves time and increase production.
⇒ Used the part that must be machined on more than one side.

Template jig:
⇒ It is normally used for accuracy rather than speed.
⇒ Used for locating hole pattern on large workspace.
⇒ Self - clamping

Plate type jig:
⇒ Modified form of the template jig.
⇒ They have built in clamps to hold the work

Open type jig:
⇒ Used for simple operations
⇒ Work is done only on one side of the part

2174. How a jig differs from a fixture in its true form?

(a) As a locator
(b) As a holder
(c) As a supporter
(d) With all the above guides the tool or cutter

KPSC ADF 2015

Ans. (d)ig acts like as a locator, holder and provide the support with the guide the tool or cutter.

2175. Jigs are not used in one of the followings:

(a) Location
(b) Clamping of the component
(c) Safety
(d) None of these

OPSC AEE 2015 Paper-I

Ans : (c) Jig : A jig is defined as a device which hold and locates a work–piece and guides and controls one more cutting tools. The Jigs are used for holding and guiding the tool in drilling, reaming or tapping operations.

2176. Renewable drill bushes are a type of _____ elements of jig design.

(a) Clamping (b) Locating
(c) Tool Setting (d) Tool Guiding

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (d) : Renewable drill bushes are a type of Tool guiding elements of jig design.
Jig – locate as well as guide the tool
Fixture – clamping and locating the tool.

2177. _____is the dimensional and positional relationship between the work-piece and the cutting tool

(a) Jigs (b) Clamping
(c) Fixing (d) Locating

UPRVUNL AE 2014

Ans : (d) Locating is the dimensional and positional relationship between the work place and the cutting tool.

2178. Consider the following statements regarding Jigs and Fixtures and identify the one which is not true.

(a) Elements of Jigs are body, clamping and locating devices and also tool guide
(b) Elements of fixture are body, clamping and locating devices
(c) Fixtures are usually massive and heavier in construction and bolted rigidly on machine table
(d) Jig is a device which can hold and locate the medium and heavier sized work pieces

UPRVUNL AE 2014

Ans : (d) Jigs is a device which can hold and locate the medium and heavier sized work piece.

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2179. Use of jigs and fixtures leads to:-

(a) High operational cost
(b) High maintenance cost
(c) High Initial cost
(d) High manufacturing cost

UKPSC AE-2013, Paper-I

Ans. (c) : Use of jigs and fixtures leads to high Initial cost.

2180. The purpose of jigs and fixture is to:

(a) Increase machining accuracy
(b) Facilitate interchangeability
(c) Decrease cost on quality control
(d) All of these

SJVN ET 2013

Ans. (d) : The purpose of jigs and fixture is to-
- Increase machine accuracy.
- Facilitate interchangeability.
- Decrease cost on quality control.

2181. A jig is defined as a device which

(a) holds and locates a work-piece and guides and controls one or more cutting tools
(b) holds and locates a work piece during an inspection or for a manufacturing operation
(c) is used to check the accuracy of a work piece
(d) All of the above

PPSC Asstt. Municipal Engineer 15.06.2021
JPSC AE 2013, Paper-V

Ans : (a) : A jig is defined as a device which holds and locates a work piece and guides and controls one or more cutting tools. The jigs are used for holding and guiding the tool in drilling, reaming or tapping operations.

2182. The floating position of the holding fixture in a rotary transfer device is used to

(a) improve the accuracy of location
(b) reduce the tendency to over index
(c) reduce the cycle time
(d) improve upon the acceleration and deceleration characteristics

BPSC AE 2012 Paper - VI

Ans : (a) : The floating position of the holding fixture in a rotary transfer device is used to improve the accuracy of location. The rotary transfer device moves the work piece in a circular path, which usually has a maximum diameter of about 3 meters.

2183. The jigs and fixture are used for

(a) reducing the cost of manufacture
(b) facilitating interchangeability, increasing, productivity and accuracy
(c) accuracy and removal of chips
(d) increasing productivity and allowing rest time to operator

TNPSC ACF 2012

Ans. (b) : The jigs and fixture are used for facilitating inter change ability, increasing productivity and accuracy. Jigs are commonly used in drilling, boring, reaming and tapping, while fixtures are adopted for milling, slotting, shape turning and planning.

2184. Bench vice is an example of

(a) Jig (b) Fixture
(c) Locator (d) Clamping device

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (b) : Bench vice is an object with two parallel jaws in which one is fixed and other is movable. It just hold the workpiece for any operation hence it can be said as a fixture.

2185. The following fixture is used to guide tools for machining contours which the machine cannot follow normally :

(a) Indexing fixture (b) Plate fixture
(c) Profile fixture (d) None of the above

APPSC IOF, 2009

Ans. (c) : Profile fixture is used to guide tools for machining contours which the machine cannot follow normally.

2186. The following is not true for Jigs:

(a) These are used on drilling operation
(b) These guide the tools
(c) These hold the components
(d) Increased machining accuracy

UKPSC AE 2007 Paper -I

Ans. (c) : These hold the components

2187. In order to locate a rectangular block in a jig or fixture accurately the following number of movements must be restricted:

(a) 3 (b) 6
(c) 9 (d) 12

UKPSC AE 2007 Paper -I

Ans. (d) : 12

2188. Match the following

Components used in jigs and fixtures		Function	
i.	Jack pin	1.	To guide the drill bit during machining
ii.	V-locator	2.	For accommodating the variation in the distance of a hole from a plane surface
iii.	Bushes	3.	To locate the circular or semicircular objects in a jig a fixture
iv.	Diamond pin	4.	To locate workpiece whose dimensions are subject to variation

(a) i-3, ii-4, iii-1, iv-2 (b) i-3, ii-4, iii-2, iv-1
(c) i-4, ii-3, iii-1, iv-2 (d) i-4, ii-3, iii-2, iv-1

APGENCO AE, 2017

Ans. (c) :

2189. In 3-2-1 principle of jig design, 3 hemispherical pins are provided in base. Indentify the wrong statement in this regard.

(a) It ensures machining in one plane
(b) It arrests 5 degrees of freedom
(c) It positions the tool
(d) For heavier work pieces we use 3 hemispherical plugs.

TSPSC Managers, 2015

Ans. (c) : The 3-2-1 principle of location (six point location principle) is used to constrain the movement of workpiece along the three axes xx, yy and zz. This is achieved by providing six locating points 3 pins in base plate, 2 pins in vertical plane and 1 pin in plane which is perpendicular to first two planes.

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2190. Which is a device for holding and locating component for specific operation and also guide tools

(a) clamp
(b) fixture
(c) None of three mentioned here.
(d) Jig

GWSSB AAE, 27.12.2015

Ans. (d)ig- It is a work holding device that holds, supports and locates the workpiece and guides the cutting tool for a specific operation.
Fixtures–It is a work holding device that holds, supports & locate the workpiece for a specific operation but does not guide the cutting tool.

9. Powder Metallurgy

2191. In powder metallurgy, the process of heating the cold pressed metal powder is known as:

(a) Sintering (b) Granulation
(c) Deposition (d) Precipitation

CGPSC AE 15.01.2021
GPSC Asstt. Prof. 28.08.2016

Ans. (a) : The sintering process consists of heating the compared green mould in a furnace to a temperature below the highest melting point from the constituents.
• Sintering is applied to achieve all possible final strength and hardness required for the finished product.

2192. The following operations are involved in producing powder metallurgy parts.

L. Sintering
M. Pressing
N. Atomising
O. Blending

The correct sequence of operations is :

(a) M, N, L, O (b) M, N, O, L
(c) N, O, L, M (d) N, O, M, L

Punjab PSC (Lect.) 06.08.2017
TSGENCO AE 14.11.2015

Ans. (d) : The correct sequence of operation is
- Atomising
- Blending
- Pressing
- Sintering

2193. Powders of two or more pure metals are mixed in a ball mill. The process is known as

(a) Electrolytic deposition
(b) Reduction
(c) Communication
(d) Mechanical Alloying

JPSC AE 10.04.2021, Paper-II

Ans. (d) : Mechanical alloying is a solid state processing technique allowing production of machniscopically homogeneous materials starting from various powder mixtures. This process is generally carried out in high energy ball mills and involves repeated welding, fracturing and rewelding of powder particles.

2194. The type of bonding in ceramic is

(a) Purely covalent
(b) Predominantly covalent with small ionic character.
(c) Predominantly ionic with small covalent character
(d) Purely ionic

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) : The type of bonding in ceramic is predominantly ionic with small covalent character.

2195. In powder metallurgy process the sequence of operations is given by

(a) atomization, mixing, sintering, pressing.
(b) atomization, sintering, mixing, pressing.
(c) atomization, pressing, mixing, sintering.
(d) atomization, mixing, pressing ,sintering.

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d)

2196. Spherical metal powders are produced by:

(a) Atomization (b) Reduction
(c) Oxidation (d) Electrolytic process

CIL MT 27.02.2020

Ans. (a) : Atomization– Atomization breaks molten metal into small droplets by rapidly freezing before the droplet come into contact with each other or with a solid surface.
The particle shape is determined largely by the rate of solidification and varies from spherical, if low heat capacity gas is employed.
To highly irregular shape if water jet is used.

2197. A powder metallurgy part in comparison with cast part has :

(a) Low in strength, ductility and density
(b) Low strength, high ductility and density
(c) High in strength, ductility and density
(d) High strength, lower density and ductility

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (*) : Powder metallurgy is the process of blending fine powdered materials, pressing them into a desired shape or form (compacting) and then heating the compressed material in a controlled atmosphere to bond the material (sintering and presintering).
The final component usually does not require further machining process and they have low in strength, ductility and density.
Note-Official answer is (a).

2198. Which of the following is not the secondary processes of powder metallurgy?

(a) Heat treatment (b) Impregnation
(c) Infiltration (d) Sintering

SJVN ET 2019

Ans. (d) : Primary process in powder metallurgy, blending, compaction, sintering.
Secondary process in powder metallurgy - coining, forging, heat treating impregnation infiltration, plating.

2199. During sintering densification is not due to :

(a) Atomic diffusion (b) Surface diffusion
(c) Bulk diffusion (d) Grain growth

OPSC AEE 2019 Paper-I

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Ans : (d) : Sintering is a heat treatment applied to a powder compact in order to impart strength and integrity. The temperature used for sintering is below the melting point of the major constituent of the powder metallurgy material.

2200. Sintering in powder metallurgy:
(a) Strengthens the component
(b) Increases electrical conductivity
(c) Increases density and ductility
(d) All of the above

HPPSC ADF 05.03.2019

Ans. (d) :

2201. "Sintering" is referred to as:
(a) mixing of different powders to obtain requisite properties
(b) compression of loose powder
(c) heating of the green compact
(d) cleaning of powder by sieving
(e) cleaning of powder by water

CGPSC AE 25.02.2018

Ans. (c) : Heating of the green compact called "Sintering" process.

2202. For joining of powder metallurgy components the best suited process is
(a) Oxy acetylene welding
(b) Arc welding
(c) Electric resistance welding
(d) Thermit welding

TSPSC AEE 2017

Ans. (a) : For joining of powder metallurgy components the best suited process is oxy acetylene welding.

2203. Heating the metal powder to a temperature above its recrystallisation temperature but below its melting point is called
(a) Blending (b) Compacting
(c) Sintering (d) Pulverization

Karnataka PSC AE, 10.09.2017

Ans. (c) : Sintering is heating of metal powder above its recrystallisation temperature but below its melting point.

2204. What is the correct sequence of operations in powder metallurgy?
(a) Compacting, Sintering, Blending, Production of metal powder
(b) Production of metal powder, Compacting, Sintering, Blending
(c) Production of metal powder, Blending, Compacting, Sintering
(d) Production of metal powder, Blending, Sintering, Compacting

CIL MT 26.03.2017

Ans. (c) : Powder metallurgy is the name given to the process by which fine powdered materials are blended, pressed into a desired shape (compacted), and then heated (sintered) in a controlled atmosphere to bond the contacting surfaces of the particles and establish the desired properties.

2205. Consider the following objectives of sintering in powder metallurgy:
1. To achieve good bonding of powder particles
2. To produce a dense and compact structure
3. To achieve high strength
3. To produce parts free of any oxide.

Which of the above objectives are correct?
(a) 1, 2 and 3 only (b) 1, 2 and 4 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

JWM 2017

Ans. (d) : Objectives of sintering in powder metallurgy-
• To achieve good bonding of powder particles
• To produce a dense and compact structure
• To achieve high strength
• To produce parts free of any oxide.
• To reduce brittleness and porosity.

2206. In powder metallurgy, the weight of the powder used for compaction is _____ the weight of the green powder compact.
(a) equal to (b) greater than
(c) less than (d) greater than or equal
(e) less than or equal

(CGPSC Polytechnic Lecturer 2017)

Ans. (a) : In powder metallurgy, the weight of the powder used for compaction is equal to the weight of the green power compact.

2207. To improve the self lubricating capacity of a powder metallurgy part the following finishing operation is used:
(a) Repressing (b) Sizing
(c) Infilliation (d) Impregnation

MPSC Lect. (Prod.) 2018
TRB Polytechnic Lecturer 2017

Ans. (d) : To improve the self lubricating capacity of a powder metallurgy part the following finishing operation is used impregnation.

2208. In powder metallurgy, the metal powder particle size is reduced during automization by:
(a) Decreasing gas velocity
(b) Increasing gas velocity
(c) Decreasing gas pressure
(d) Decreasing metal volume

TRB Polytechnic Lecturer 2017

Ans. (b) : By increasing gas velocity, the metal powder particle size is reduced during automization.

2209. Sintering is done:
(a) To achieve the atomization of molten metal
(b) Just after blending of metal powder and additives
(c) To cool the compact in water
(d) To increase the strength of compact

UPRVUNL AE 07.10.2016

Ans. (d) : Sintering—Sintering is a heat treatment applied to a power compact in order to impact strength and integrity. The temperature used for sintering is below the melting point of the major constituent of power metallurgy material.
After compaction, neighbouring powder particles are held together by cold welds, which gives the compact sufficient "green strength" to be handled. At sintering temperature, diffusion process cause necks to form and grow at these contact points.

2210. During sintering of a powder metal compact, the following process takes place:
(a) some of the pores grow
(b) powder Particles do not melt but a bond is formed between them
(c) all the pores reduce in size and bond occurs due to melting
(d) Powder particles fuse and join together

(HPPSC LECT. 2016)

Ans : (b) During sintering of powder metal compact, powder particles do not melt but a bond is formed between them.
Sintering:- Heating of powder compact in a furnace to below the melting point of at least one of the major constituents under a controlled atmosphere is called sintering. In the sintering furnaces, the components are gradually heated and soaked at the required temperature which depends on the type of material. The sintering temperature and time vary with the compressive load used, the type of powder and strength requirement of the finished parts.

2211. 'Compacting process' used in powder metallurgy is done:
(a) To achieve atomization of molten metal
(b) After secondary finishing of sintered object
(c) After blending of metal powder and additives
(d) After sintering of blended metal powder and additives

UPRVUNL AE 07.10.2016

Ans. (c) : Powder compaction—Powder compaction is the process of compacting metal powder in a die through the application of high pressure. Typically the tools are held in the vertical orientation with the punch tool forming the bottom of the cavity. The powder is than compacted into a shape and the ejected from the die cavity. This process used in power metallurgy is done after blending of metal powder and additives.

2212. In powder metallurgy, the operation carried out to improve the bearing property of a bush is called
(a) Infiltration (b) Impregnation
(c) Plating (d) Heat treatment
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (b) : In powder metallurgy, the operation carried out to improve the bearing property of a bush is called impregnation.

2213. The factors influencing shrinkage during sintering include (i) particle size, (ii) pressure used, (iii) sintering temperature, (iv) pressing time
(a) (i), (ii) and (iv) (b) (ii) and (iv)
(c) (i) and (iv) (d) (i), (ii) and (iii), (iv)

TSPSC AEE 2015

Ans : (d) Factor influencing shrinkage during sintering
(i) Particle size
(ii) Pressure used
(iii) Sintering temperature
(iv) Pressing time.

2214. The tiny balls for ball point pens are made by
(a) Rolling (b) Casting
(c) Powder metallurgy (d) Shot peening

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) :

2215. The correct sequence of the given processes in manufacturing by powder metallurgy is:
(a) blending, compacting, sintering and sizing
(b) blending, compacting, sizing and sintering
(c) compacting, sizing, blending and sintering
(d) compacting, blending, sizing and sintering

HPPSC Asstt. Prof. 2014

Ans. (a) : Powder metallurgy operation sequence,
Blending → Compacting → Sintering → Sizing

2216. Following is a process used to form powder metal to shape
(a) Sintering (b) Explosive Compacting
(c) Isostatic Molding (d) All of these

ISRO Scientist/Engineer 12.05.2013

Ans. (d) : All sintering, explosive compacting and isostatic molding process used to form powder metal to shape.

2217. Powder metallurgy techniques are used in the production of
(a) High carbon steel tool
(b) HSS tools
(c) Tungsten carbide tools
(d) Twist drills

APPSC AEE MAINS 2016, PAPER-III
SJVN ET 2013

Ans. (c) : Powder metallurgy techniques are used in the production of tungsten carbide tools.

2218. Which of the following processes for the new ceramic materials accomplishes shaping and sintering simultaneously?
(a) Doctor-blade process (b) Freeze drying
(c) Hot pressing (d) Injection molding

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (c) : Hot pressing is similar to dry pressing, except it is carried out at elevated temperatures so, sintering of the product is accomplished simultaneously with pressing and eliminates the need for a separate sintering step.

2219. The processes are commonly used for producing powder
(a) Atomization (b) Reduction process
(c) Electrolytic process (d) All of the above

RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : There are four main processes used in powder production.
(1) Solid state reduction (2) Atomization
(3) Electrolysis (4) Chemical Removal

2220. Consider the following methods :
1. Mechanical pulverization
2. Atomization
3. Sintering
4. Electrolytic deposition

Which of these methods is/are used for production of metal powders ?
(a) 1 only (b) 2 and 3
(c) 3 and 4 (d) 1, 2 and 4

UPSC JWM Advt. No.-50/2010

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Ans. (d) : As we know, for the production of metal powders, we use different techniques such as, mechanical pulverization, atomization and electrolytic deposition, depends upon size and use of metal powders.

2221. The process of infiltration in sintered product is to improve

(a) porosity (b) dimensional
(c) surface finish (d) coherent property

Ans : (a) The process of infiltration in sintered product is to improve porosity.
Sintering:- Heating of powder compact in a furnace to below the melting point of at least one of the major constituents under a controlled atmosphere is called sintering.

2222. In order to have interference fit, it is essential that the lower limit of the shaft should be

(a) greater than the upper limit of the hole
(b) lesser than the upper limit of the hole
(c) greater than the lower limit of the hole
(d) lesser than the lower limit of the hole

GPSC Executive Engineer 23.12.2018

Ans. (a) : In order to have interference fit, it is essential that the lower limit of the shaft should be greater than the upper limit of the hole.

10. Metrology

2223. A feeler gauge is used to check

(a) radius
(b) surface
(c) roughness
(d) thickness of clearance

Gujarat PSC AE 2019
GPSC Executive Engg. 23.12.2018
WBPSC AE, 2017
GPSC Asstt. Prof. 28.08.2016
RPSC AE 2016, KPCL AE 2016
GWSSB AAE, 27.12.2015
Kerala PSC AE 06.08.2015
VIZAG STEEL MT 2015
KPSC AE 2015, RPSC ACF-2011

Ans. (d) : Feeler gauges are used to measure gap width between two parallel force e.g. finding clearance between two parallel surfaces.

2224. Auto collimator is used to check

(a) roughness (b) flatness
(c) angles (d) automobile balance

GPSC Executive Engg. 23.12.2018,
MPPSC AE 2016
GPSC ARTO 01.05.2016
BPSC Asstt. Prof. 29.11.2015
TSPSC AEE 2015
GPSC AE 26.07.2015
GPSC Poly. Lect. 07.12.2014
TNPSC ACF 2012, TRB Poly. Lect., 2012
GATE 1998

Ans. (c) : Auto collimator is an optical instrument for non- contact measurement of small angles on small angular tilt of a reflecting surface used to align components and measure deflections in optical or mechanical system A visual auto collimator can measures angles as small as 0.5 arc second, while an electronic auto collimator can be up to l00 times more accurate.

2225. A Ring gauge is used to measure.

(a) Only internal threads
(b) Outside Diameter
(c) Outside diameter and roundness
(d) Only external threads

GPSC DEE Class-2 (GWSSB) 14.07.2021
MPSC Lect. (Prod.) 2018
TNPSC AE 2017, KPCL AE 2016
GWSSB DEE 07.07.2016
MPSC Dy. Engineer 2014
VIZAG Steel MT 18.08.2013
VIZAG Steel MT 10.06.2012, GATE 2006

Ans. (c) : A ring gauge is used to measure outside diameter and roundness of shaft.
While plug gauge is used to measure inner diameter of hole.

2226. The standard tolerance unit 'i' equal to :

(a) $i = 0.45\sqrt{D} + 0.001D$
(b) $i = 0.45\sqrt[3]{D} + 0.001D$
(c) $i = 0.45\sqrt{D} + 0.01D$
(d) $i = 0.45\sqrt[3]{D} + 0.01D$

TNPSC AE 2019
Nagaland PSC (CTSE) 2018 Paper-I
Punjab PSC SDE 12.02.2017
ISRO Scientist/Engg. 11.10.2015
TNPSC AE 2014, HPPSC Asstt. Prof. 2014

Ans. (b) : $i = 0.45\sqrt[3]{D} + 0.001D$
Standard Tolerance– There are 18 grades of standard tolerances IT01, IT0, IT1, IT2,…..IT16 which are obtained in terms of standard tolerance limit "i" in microns.

2227. Profile of a gear tooth can be checked by :

(a) Sine bar
(b) Bench micrometer
(c) Optical pyrometer
(d) Optical projector

ISRO Scientist/Engg. 22.04.2018
RPSC AE 2016, UPRVUNL AE 2014
UKPSC AE 2012 Paper-I, RPSC ACF-2011

Ans : (d) Profile of gear tooth are checked by optical projector.
Sine Bar- Sine bar is used to measure angle.
Optical Pyrometer- It is use for measuring high temperature.

2228. Precision is defined as

(a) the degree of repetitiveness of the measuring process
(b) lesser number of controllable errors
(c) lesser number of random errors
(d) none of these

TNPSC AE 2018
GPSC Asstt. Prof. (Prod.) 21.08.2016
GPSC Lect. 23.10.2016

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Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014
TRB Poly. Lect., 2012, RPSC ACF-2011

Ans. (a) : Ability of an instrument to reproduce a certain reading with a given accuracy is called precision. Precise data have small dispersion.

2229. Plug gauges are used to

(a) measure outside diameter of workpieces
(b) measure the diameter of holes of workpieces
(c) check the diameter of holes in workpieces
(d) check the outside diameter of workpieces

GMW AAE 25.06.2017, UKPSC AE 2012 Paper-I
TRB Asstt. Prof., 2012, RPSC ACF-2011
ISRO Scientist/Engineer 2008

Ans. (c) : They are used for checking holes that whether the hole dimensions are with in specified tolerance.

2230. Clinometer is used to measure :

(a) the internal diameter
(b) the thickness
(c) the external diameter
(d) the angular faces :

HPPSC Workshop Suptd. 08.07.2021
NLC GET 17.11.2020, Shift-II
Punjab PSC SDE 12.02.2017
TNPSC AE 2013, TRB Poly. Lect., 2012
TNPSC AE, 2008

Ans. (d) : Clinometer-A clinometer is a special case of the application of the spirit level. A clinometer is used for the measurement of the included angles of two adjacent faces of a component with help of parallel and surface plate.

(i) caliper is used to measure internal diameter and external diameter of the object.

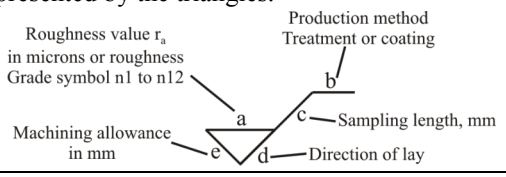
(ii) Thick gauges are measuring instruments that can be used to determine the thickness or gauges of a material.

2231. The surface roughness on a drawing is represented by :

(a) zig-zag lines (b) squares
(c) triangles (d) circles

GWSSB AAE, 27.12.2015, APGENCO AE 2012
TRB Asstt. Prof., 2012, RPSC ACF-2011
ISRO Scientist/Engineer 2010

Ans. (c) : Triangles- Surface Roughness on a drawing is represented by the triangles.



2232. The M and E system in metrology are related with measurement of

(a) gear (b) screw thread
(c) flatness (d) surface finish

GPSC DEE Class-2 (GWSSB) 04.07.2021
GMB AAE 25.06.2017, GPSC AE 26.07.2015
TNPSC ACF 2012

Ans. (d) : M and E system related to surface finish.

M system After plotting the characteristic of any surface a horizontal line is drawn by joining two points. The line is shifts up and draw in such a way that 50% area is above the line and 50% area below the line.

E system- A sphere of 25 mm diameter is rolled over the surface and the locus of its centre is being traced out called envelope. This envelope is shifted in down word direction fill the area above line is equal to area below the line. This is called mean envelope and the system of datum is called E system.

2233. A fit on a hole- shaft system is specified as H7-s6. The type of fit is

(a) Clearance fit (b) Running (Sliding) fit
(c) Transition fit (d) Interference fit

GPSC Asstt. Director of Transport 05.03.2017
GPSC Asstt. Workshop Suptd. 17.07.2016
APPSC AEE MAINS 2016, PAPER-III
NPCIL ET, 2015, GATE 1996

Ans. (d) : Interference fit, is a form of fastening between two light fitting mating parts that produces a joint which is held together by friction after the parts are pushed together.

H7/p6 → Interference fit [P to z]
H7/h6 → Clearance fit [A to h]
H7/k6 → Transition fit [I to o]

2234. Tolerances are specified

(a) To obtain desired fits
(b) Because it is not possible to manufacture a size exactly
(c) To obtain high accuracy
(d) To have proper allowance

Vizag Steel (MT) 2017, RPSC 2016, RPSC AE 2016

Ans. (b) : Because it is not possible to manufacture a size exactly.

2235. LVDT is used for measuring:

(a) displacement (b) roughness
(c) pressure (d) speed

HPPSC LECT. 2016, BPSC Poly. Lect. 2016
(UJVNL-AE-2016)

Ans : (a) LVDT is used for measuring displacement

LVDT:- Linear Variable Displacement Transducer. A very basic transducer which is always useful in the field of instrumentation.

Principle of LVDT:-LVDT works under the principle of mutual induction, and the displacement which is a non-electrical energy is converted into an electrical energy. And the way how the energy is getting converted is described in working of LVDT in a detailed manner.

2236. Autocollimator is used for

(a) Parallelism measurement
(b) Straightness measurement
(c) Flatness measurement
(d) Angular measurement

GPSC ARTO 01.05.2016
GPSC Lect. (Poly.) 07.12.2014
TNPSC ACF 2012

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Ans. (d) : Autocollimator- An optical instrument for non-contact measurement of small angles or small angular tilts of reflecting surface used to align components and measure deflections in optical or mechanical system.
An autocollimators works by projecting an image onto a target mirror, and measuring the deflection a scale either visually or by means an electronic defector.

2237. Basic hole means
(a) upper deviation is zero
(b) lower deviation is zero
(c) Both (A) and (B)
(d) None of the above

Nagaland CTSE 2017 Paper-I8
ISRO Scientist/Engineer 11.10.2015
BPSC Asstt. Prof. 29.11.2015

Ans. (b) : Algebraic difference between the minimum limit of size and the corresponding basic size. It is a positive quantity when the minimum limit of size is greater than the basic size and a negative quantity when the minimum limit of size is less than the basic size.

2238. Accuracy of measuring instrument is :
(a) The closeness with which a measurement can be read directly from a measuring instrument
(b) A measure of how close the reading is to the true size
(c) The difference between measured and actual value
(d) The capability to indicate the same reading again and again for a given measured

Punjab PSC SDE 12.02.2017, RPSC AE 2016
TNPSC AE (Industries) 09.06.2013
RPSC ACF-2011

Ans. (b) : The ability of the instrument to measure the accurate value is known as accuracy. In other words, the closeness of the measured value to a standard or true value. Accuracy is obtained by talking small reading.

2239. The path difference between one bright and nearest dark band in an interferometer is
(a) one wavelength (b) two wavelength
(c) half of wavelength (d) none of the above

HPPSC ADF 05.03.2019
GPSC Asstt. Prof. (Prod.) 21.08.2016
MPPSC AE 08.11.2015

Ans. (c) : Interferometry uses electromagnetic waves for superimposing and causing interference of bright and dark fringes. Path difference between one bright and nearest dark band in an interferometer is $\lambda/2$.

2240. Bevel protector is used for -
(a) Angular measurements
(b) Thickness measurements
(c) Flatness measurements
(d) Height measurements

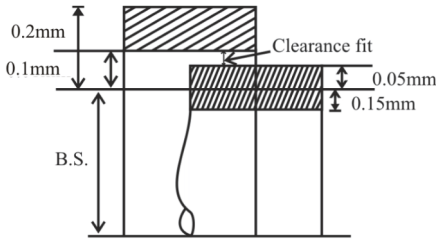
GMB AAE 25.06.2017
RPSC ACF-2011, TNPSC AE, 2008

Ans. (a) : The Bevel protector is used to establish and test angle stovery close tolerances. It reads to S arc minutes (s' and 112°) and can measure angle from 0° to 360°. The Bevel Protector consists of a beam, a graduated dial and a blade which is connected to a swivel plate (with vernier scale) by thumb not and clamp.

2241. A shaft (diameter $= 20^{+0.05}_{-0.15}$ mm) and a hole (diameter $= 20^{+0.20}_{+0.10}$ mm) when assembled would yield.
(a) transition fit (b) interference fit
(c) clearance fit (d) none of these

GPSC DEE Class-2 (GWSSB) 04.07.2021
Vadodara Muncipal Corp. DEE, 2018
TRB Poly. Lect., 2012

Ans. (c) : Clearance fit.



Maximum limit of shaft is less than lower limit of hole this is a clearance fit.

2242. Which of the following is not the angle measuring device?
(a) Angle plate (b) Sine bar
(c) Bevel protractor (d) Angle gauge

Vadodara Muncipal Corp. DEE, 2018
WBPSC AE, 2017
GPSC Asstt. Prof. 28.08.2016

Ans. (a) : Angle plate is a work holding device. It is not used for angle measurement.

2243. Clearance in a fit is the difference between _____.
(a) maximum hole size and minimum shaft size
(b) maximum hole size and maximum shaft size
(c) minimum hole size and minimum shaft size
(d) minimum hole size and maximum shaft size

PPSC Asstt. Municipal Engg. 15.06.2021
VIZAG MT, 14.12.2020
GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : Clearance fit = Minimum hole size – Maximum shaft size

2244. Sine bar is used to measure :
(a) the diameter (b) the angle
(c) the distance (d) the depth of hole

GPSC DEE Class-2 (GWSSB) 04.07.2021
NLC GET 17.11.2020, Shift-II
VIZAG STEEL MT 2015

Ans. (b) : A sine bar is used in conjuction with slip gauge block for precise angular measurement. A sine bar is used either to measure an angle very accurately or Face locate any work to a given angle offering a high degree of accuracy in measuring angles for milling, grinding and inspection applications.

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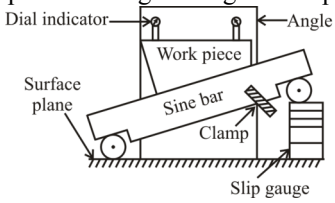
2245. A sine bar is specified by

(a) its total length
(b) the centre distance between the two rollers
(c) the size of the rollers
(d) the weight of sine bar

TSPSC AEE 2015, ISRO Scientist/Engineer 2011
TNPSC AE, 2008

Ans. (b) : A sine bar is specified by the centre distance between the two rollers. A sine bar measuring instrument precision angle along with slip gauge.



2246. Wear allowance is provided on :

(a) go gauge
(b) no go gauge
(c) both go and no go gauges
(d) when both are combined in one gauge

Vizag Steel (MT) 2017
MPPSC AE 08.11.2015, RPSC ACF-2011

Ans. (a) : Wear allowance is applied to the nominal gauge diameter before gauge tolerance is applied. The wear allowance must be kept as small as possible. Wear allowance is usually as 5% of work tolerance. This wear allowance is generally applied to only "Go gauge".

2247. Two slip gauges in precision measurement are joined by :

(a) sliding (b) adhesion
(c) wringing (d) slipping

TNPSC AE 2018, GWSSB DEE 07.07.2016
RPSC ACF-2011

Ans. (c) : Two slip gauges in precision measurement are joined by wringing. Wringing is the act of joining the slip gauges together while building upto sizes.

2248. The interchangeability can be achieved by

(a) Bonus plan
(b) Better process planning
(c) Standardization
(d) Better product planning

RPSC VPITI 14.02.2016, UJVNL AE 2016
ISRO Scientist/Engg. 2008

Ans. (c) : Interchangeability is the ability to select components for assembly at random and fit them together within proper tolerances.
Interchangeability can be achieved by standardization.

2249. Optical flats are made of

(a) Quartz (b) Glass
(c) Steel (d) Plastic

Gujarat PSC AE 2019, TNPSC ACF 2012

Ans : (a) : Optical flats are made of Quartz.

2250. Which of the following is an interference fit?

(a) Push fit (b) Running fit
(c) Sliding fit (d) Shrink fit

OPSC AEE 2019 Paper-I
Nagaland CTSE 2017, 2016 Ist Paper

Ans : (d) : Shrink fit is an interference fit, running and sliding fits are clearance fit while push fit is a transition fit.

2251. Which one of the following measuring instruments is supposed to be most accurate

(a) micrometer (b) vernier caliper
(c) vernier dial gauge (d) optical projector

RPSC AE 2016
Nagaland PSC CTSE 2016 Paper-I

Ans : (d) Optical projector most accurate measuring instrument.

2252. Which of the following gives an idea about the ability of the equipment to detect small variation is the input signal (quantity being measured)?

(a) readability (b) accuracy
(c) sensitivity (d) precision

TNPSC AE 2018, RPSC AE 2016, RPSC ACF 2011

Ans : (c) Sensitivity:- Is the ratio of the magnitude of response (output signal) to the magnitude of quantity being measured (input signal.)
Sensitivity is represented by the slope of calibration curve, and it has wide range of units and these depend upon the instrument being investigated.

2253. Plug gauge is used to measure

(a) Taper bores (b) Cylindrical bores
(c) Spherical holes (d) None of the above

TNCSC AE 2020
ISRO Scientist/Engineer 11.10.2015

Ans. (b) :

Gauge	For Measuring
Snap gauge	External dimensions
Plug gauge	Internal dimensions
Taper plug gauge	Taper hole
Ring gauge	External Diameter
Gap gauge	Gaps and grooves
Radius gauge	Gauging radius
Thread pitch gauge	External thread
Filler gauge	Clearance or gap between components
Fillet gauge	Radius of an object
Cylindrical plug gauge	Inside diameter of straight hole

2254. A fit is specified as 25 H8/e8. The tolerance value for a nominal diameter of 25 mm in IT8 is 33 micron and fundamental deviation for the shaft is –40 micron. The maximum clearance of the fit micron is

(a) –7 (b) 7
(c) 73 (d) 106

GPSC AE 26.07.2015
Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) : Given,
Basic size = 25 mm and the fit is 25 H8/e8
Tolerance = 33 micron
Fundamental deviation = –40 micron
Where, H in the hole tolerance represents
Hole basis system

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For the hole, we have, lower limit = 25 mm
Upper limit = Lower limit + Tolerance
Upper limit = 25 + 0.033 = 25.033 m
For the shaft
We have, upper limit = (basic size) – (fundamental deviation)
Upper limit = 25 – 0.04 = 24.96 mm
Lower limit = Upper limit – Tolerance
Lower limit = 24.96 – 0.033 = 24.927 mm
Here, the maximum clearance = (Upper limit of hole) – (Lower limit of the shaft)
Maximum clearance = 25.033 – 24.927 = 0.106 mm
Maximum clearance = 106 microns

2255. The dimensional limits on a shaft of 25h7 are
(a) 25.000, 25.021 mm
(b) 25.000, 24.979 mm
(c) 25.000, 25.007 mm
(d) 25.000, 24.993 mm

GPSC ARTD Pre-30.12.2018
TSPSC AEE 2017
VIZAG STEEL MT 18.08.2013
VIZAG STEEL MT 10.06.2012
GATE 2003

Ans. (b) : Taking step size as geometric mean diameter
i.e D = 25 mm
$i = 0.45(25)^{1/3} + 0.001 \times 25$
$i = 1.3408 \times 10^{-3}$ mm
$IT7 = 16i = 16 \times 1.3408 \times 10^{-3} = 0.0214528$ mm
UL = 25 mm
LL = D – Tolerance zone
= 25 – 0.0214528 = 24.979 mm

2256. In _____ the tolerance zones of the hole and shaft always overlap.
(a) clearance fit (b) interference fit
(c) combined fit (d) transition fit

UPRVUNL AE 05.07.2021
HPPSC Workshop Suptd. 08.07.2021

Ans. (d) : In transition fit the tolerance zones of the hole and shaft always overlap.

2257. What terms is used to designate the direction of the predominant surface pattern produced by machining operation?
(a) roughness (b) lay
(c) waviness (d) cutoff

GPSC DEE, Class-2 (GWSSB) 04.07.2021
GPSC Engg. Class-2 Pre 19.01.2020

Ans. (b) :

2258. An optical gauge works on the principle of
(a) Reflection of light rays
(b) Polarization of light rays
(c) Interferences of light rays
(d) Dispersion of light rays

Gujarat PSC AE 2019
Punjab PSC (Lect.) 06.08.2017

Ans. (c) : An optical gauge works on the principle of interference of light rays.

2259. In screw thread measurement if P is the pitch of the screw & θ is the included angle then best wire size is calculated by :
(a) P / (2 sin θ/2) (b) P / (2 tan θ/2)
(c) P / (2 cot θ/2) (d) P / (2 cos θ/2)

TNPSC AE 2018
Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016
TNPSC AE (Industries) 09.06.2013

Ans. (d) : Best wire diameter $(d_w) = \frac{P}{2}\sec\left(\frac{\theta}{2}\right)$
where, P = pitch (mm)
θ = Thread angle

2260. Match the measuring instruments with the appropriate applications. :

Measuring instruments	**Applications.**
(a) Talysurf	(i) T-slots
(b) Telescopic gauge	(ii) Flatness
(c) Transfer calipers	(iii) Internal diameter
(d) Autocollimator	(iv) Roughness

	A	B	C	D
(a)	(i)	(ii)	(iii)	(iv)
(b)	(iv)	(iii)	(i)	(ii)
(c)	(iv)	(iii)	(ii)	(i)
(d)	(iii)	(iv)	(i)	(ii)

TNPSC AE 2017
MPSC Poly. Lect. 2014

Ans. (b) :

Measuring Instruments	**Applications.**
Talysurf	Roughness
Telescopic gauge	Internal diameter
Transfer calipers	T-slots
Auto collimators	Flatness.

2261. Which one of the following statements is TRUE?
(a) The 'GO' gage controls the upper limit of a hole
(b) The 'NO' gage controls the lower limit of a shaft
(c) The 'GO' gage controls the lower limit of a hole
(d) The 'NO GO' gage controls the lower limit of a hole
(e) None of the above

Assam Engg. College AP/Lect. 18.01.2021
CGPSC Poly. Lect. 22.05.2016

Ans. (c) : Go gauge limit is applied that lower limit of hole and upper limit of shaft while NO GO gauge is applied to minimum material condition that is upper limit of a hole and lower limit of a shaft.

2262. The least count of a metric Vernier caliper having 25 divisions on Vernier scale, matching with 24 divisions of main scale (1 main scale divisions = 0.5mm) is
(a) 0.005 mm (b) 0.01 mm
(c) 0.02mm (d) 0.005 mm

RPSC ACF-2011
ISRO Scientist/Engineer 2008

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Ans. (c) :

$$\text{Least Count} = \frac{\text{length of smallest main scale}}{\text{Number of divisions on the vernier scale}}$$

$$= \frac{0.5}{25} = 0.02mm$$

2263. Which of the following errors are inevitable in the measuring system and it would be vainful exercise to avoid them
(a) Systematic errors
(b) Random errors
(c) Calibration errors
(d) Environmental errors
TNPSC AE 2017, ISRO Scientist/Engineer 2007

Ans. (b) :

2264. _____ can be used to measure the flatness of a machine bed.
(a) Height gauge
(b) Auto collimator
(c) Tool makers microscope
(d) Vernier calipers
GPSC DEE Class-2 (GWSSB) 04.07.2021
Oil India Ltd. Sr. Engg. (Production) 30.11.2019
GPSC ARTO Pre 30.12.2018
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (b) : Auto collimator is an optical instrument which is used for measure the small angular differences, changes or defection plane surface inspection etc.
For small angular measurement, auto collimator provides a very sensitive and accurate approach.
Other applications are-
1. Checking of an internal right angle
2. Comparative measuring using master gauge.
3. Straightness and flatness of the surface.
4. For machine tool adjustment setting.
5. Assessment of sureness and parallelism of component.

2265. A dead weight tester is used for:
(a) Calibrating pressure measuring instruments
(b) Producing high pressure
(c) Measuring weight
(d) Measuring resistive force
KPSC AE 2015, SJVN ET 2013

Ans. (a) : A dead weight tester is used for calibrating pressure measuring instruments.

2266. According to Indian Standard, 50 H8-g7 means
(a) upper limit is (50+8) mm and lower limit (50–7) mm
(b) designation of tolerance with basic size 50 mm
(c) designation of fit of two parts with basic size 50 mm
(d) None of above
RPSC Vice Principal ITI 2018
Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : 50 H8-g7
■ Tolerance grade for hole is 8.
■ Tolerance grade for shaft is 7.
■ Designation of fit of two parts with size 50 mm.

2267. Gratings are used in connection with
(a) Flatness measurement
(b) Roundness measurement
(c) Surface texture
(d) Linear displacement
TNPSC AE 2017, ISRO Scientist/Engineer 2007

Ans. (d) : Gratings are used in connection with linear displacement.

2268. Pertho condenser method is used as a
(a) direct method to measure surface roughness
(b) indirect method to measure surface roughness
(c) direct method to measure surface texture
(d) indirect method to measure surface texture
TNPSC AE 2019
Kerala PSC Asstt. Prof. 27.09.2016

Ans. (a) : Pertho condenser method is used as a direct method to measure surface roughness.

2269. In limits and fits system, basic shaft system is one whose
(a) Lower deviation is zero
(b) Upper deviation is zero
(c) Minimum clearance is zero
(d) Maximum clearance is zero
Vizag Steel (MT) 2017
Nagaland PSC CTSE 2017, Paper-I
TSPSC Manager 2015
RPSC ACF 2011, RPSC AE (GWD) 2011

Ans. (b) : In limits and fits system, basic shaft system is one whose upper deviation is zero.

2270. V-block is used in workshop to check :
(a) roundness of a cylindrical job
(b) surface roughness
(c) dimensions of oval job
(d) type on a job
GMB AAE 25.06.2017, RPSC ACF-2011

Ans. (a) : V-Blocks are generally used for cylindrical workpiece to check their roundness and for making centres. V-Blocks are like precision metalworking jigs which are used to hold workpieces.

2271. The term "allowance" in limits and fits is usually referred to :
(a) minimum clearance between shaft and hole
(b) maximum clearance between shaft and hole
(c) difference of tolerances of hole and shaft
(d) difference between maximum size and minimum size of hole
GPSC ARTO Pre 30.12.2018, RPSC ACF-2011

Ans. (a) : Allowance in limits and fits refers to minimum clearance between shaft and hole.

2272. While designing limit gauges, manufacturing tolerance and wear allowance to be provided are _____% of work tolerance, respectively.
(a) 10% and 5% (b) 5% and 10%
(c) 5% and 5% (d) 10% and 10%
Punjab PSC SDE 12.02.2017
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : Manufacturing tolerance and wear allowance are 10% and 5% respectively of work tolerance while designing limit gauges.

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2273. In a hole basis system

(a) The nominal size and the limits of the shaft are maintained constant and the hole limits are varied.

(b) The nominal size and the limits of both the shaft and hole are made constant

(c) The nominal size and the limits of the hole are maintained constant and the shaft limits are varied

(d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

TRB Poly. Lect., 2012

Ans. (c) : Hole basis system

The nominal size and the limits of the hole are constant while limits of shaft are varied.

2274. Gear teeth vernier is used to measure

(a) Circular pitch (b) Depth of tooth

(c) Tooth thickness (d) Pitch line thickness

Nagaland CTSE 2017 Paper-I

GPSC Asstt. Prof. 21.08.2016

Nagaland CTSE 2016 Paper-I

RPSC Lect. (Tech. Edu. Dept.) 2011

TNPSC AE 2008

Ans. (d) : Gear teeth Vernier calliper is used to measure the theoretical value of gear tooth thickness at the pitch line using the distance from the top of a tooth to the chord. It can also be used for measuring hobs, form & thread tools etc.

2275. A metric thread of pitch 2 mm and thread angle 60° is inspected for its pitch diameter using the 3-wire method. The indicated diameter of the wire will be nearly

(a) 0.85 mm (b) 1.05 mm

(c) 1.15 mm (d) 2.05 mm

GPSC Asstt. Director of Transport 05.03.2017

ESE 2017

Ans. (c) : Wire diameter $(d) = \dfrac{P}{2\cos\left(\dfrac{\theta}{2}\right)}$

$= \dfrac{2}{2\cos\left(\dfrac{60}{2}\right)} = 1.15 \text{ mm}$

2276. Consider the following statements regarding limits and fits:

1. Actual size is the standard size of the part and is the same both for the hole and its shaft.
2. Basic size is the dimension as measured on the manufactured part.
3. Deviation is the algebraic difference between a size and the corresponding basic size.

Which of the above statements is/are correct?

(a) 3 only (b) 1 and 3 only

(c) 2 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (a) :

- Actual size is the actual measured dimension of the part not the standard size of the part and if is not the same fg both hole and shaft. Hence wrong.
- Basic size is the size of a part to which all limits of variation are applied. Basic dimension is theoretical dimension. Hence wrong.
- Deviation is the algebric difference between a size (actual max etc) and the corresponding basic size. Hence correct.

2277. Parallel misalignment is present when

(a) two shafts are parallel to each other but are not in the same plane.

(b) two shafts are parallel to each other and are in the same plane.

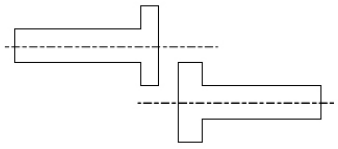
(c) the shafts are not parallel to each other.

(d) the shafts are aligned with each other.

ESE 18.07.2021

Ans. (b) :

Parallel misalignment is present when the centre lines of both shafts are parallel to each other and are in same plane.



2278. The major limitation with displacement or proximity probes is

(a) Size (b) Time

(c) Accuracy (d) Cost

ESE 18.07.2021

Ans. (a) : The major limitation with displacement or proximity probes is size.

2279. Appropriate instrument to check the flatness of slip gauges

(a) dial indicator

(b) pneumatic comparator

(c) optical comparator

(d) tool makers microscope

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) :

2280. The fit on a hole-shaft pair system is specified as H7-H6, t he type of fit is

(a) Clearance fit (b) sliding fit

(c) push fit (d) force fit

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) :

2281. A threaded nut of M16,ISO metric type having 2 mm pitch with a pitch diameter of 14.701 mm is to be checked for its pitch diameter using two or three number of balls or rollers.

(a) rollers of 2 mm dia

(b) rollers of 1.55 mm dia

(c) balls of 2 mm dia

(d) balls of 1.55 mm dia

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

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2282. For angle measurement in metrology , the following pair can be used in conjunction with each other

(a) sine bar and vernier calipers
(b) bevel protractor and slip gauges
(c) slip gauges and sine bar
(d) bevel protractor and sine bar

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (c) :

2283. Abbe's principal of alignment is used to be followed in

(a) vernier calipers
(b) depth vernier
(c) internal caliper micrometer
(d) height vernier

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) : Abbe's principal of alignment is used to be followed in depth vernier.

2284. The geometric tolerance that does not need a datum for its specification is

(a) concentricity (b) runout
(c) perpendicularity (d) flatness

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) : The geometric tolerance that does not need a datum for its specification is flatness.

2285. Which one of the instruments is comparator?

(a) tool makers microscope
(b) GO/NO gauge
(c) optical interferometer
(d) dial guage

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (d) :

2286. External taper can be accurately measured wit h the help of

(a) slip guages and sine bar
(b) GO/NO guage
(c) optical interferometer
(d) dial guage

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (a) : External taper can be accurately measured wit h the help of slip guages and sine bar.

2287. In a unilateral system of tolerance, the tolerance is allowed on

(a) one side of the actual size
(b) one side of the nominal size
(c) both sides of the actual size
(d) both sizes of the nominal size

GPSC DEE, Class-2 (GWSSB) 04.07.2021

Ans. (b) :

2288. A sine bar is generally used with

(a) Slip gauges. (b) Dial gauge.
(c) Surface plate. (d) All of the above

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (d) : A sine bar is generally used for measure inclination of surface some bar used with
(i) Slip gauges
(ii) Dial gauge
(iii) Surface plate

2289. Which of the following is the tester in which the gear being inspected is made to mesh with a standard gear and a dial indicator is used to capture the radial errors?

(a) Parkinson Gear Tester
(b) Macmillan Gear Tester
(c) Jhonson Gear Tester
(d) Pitch Checking Instrument

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (a) : Parkinson gear tester in which the gear being inspected is made to mesh with a standard gear and a dial indicating is used to capture the radial errors.

2290. Group of geometric tolerance applied to individual features are called _____.

(a) run outs
(b) orientation tolerances
(c) positional tolerances
(d) form tolerances

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (d) : Group of geometric tolerances applied to individual feature are called form tolerance (form derivation).
Geometrical tolerances are used to convey in a brief and precise manner complete geometrical requirements on engineering drawings.

2291. The maximum angle that can be set using a sine bar (without errors) is _____ degrees.

(a) 15 (b) 60
(c) 45 (d) 30

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (c) : The sine bars become impractical and inaccurate as the angle exceeds 45º because the bar is physically clumsy to hold in that position and even a slight error of the sine bar can cause large angular errors after this angle.

2292. _____refers to relatively finely spaced surface irregularities of small wavelength on the part which results from the inherent action.

(a) Create (b) Waviness
(c) Roughness (d) Concentricity

HPPSC Workshop Suptd. 08.07.02021

Ans. (c) : Roughness refers to relatively finely spaced surface irregularities of small wavelength on the part which results from the inherent action.

2293. In a hole-shaft assembly, when the largest permissible diameter of the shaft is smaller than the smallest diameter of the hole, _____ can be achieved.

(a) interference fit (b) clearance fit
(c) transition fit (d) combined fit

UPRVUNL AE 05.07.2021

Ans. (b)

2294. In a hole basis system, size of hole is always kept _____

(a) zero (b) constant
(c) variable (d) negative

UPRVUNL AE 05.07.2021

Ans. (b) : In a hole basis system, size of hole is always kept constant.

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2295. Which of the following is NOT applicable for interchangeability?
(a) Easies the assembly of mating parts
(b) Enhances production rates
(c) Limits mass production
(d) Reduces assembling costs

UPRVUNL AE 05.07.2021

Ans. (c) :

2296. _____ is a type of geometric tolerance applied to limit the direction of a feature in relation to other features in an assembly.
(a) Unilateral tolerance
(b) form tolerance
(c) positional tolerance
(d) orientation tolerance

UPRVUNL AE 05.07.2021

Ans. (d)

2297. Alignment of dowel pins is type of ____
(a) combined fit (b) transition fit
(c) clearance fit (d) interference fit

UPRVUNL AE 05.07.2021

Ans. (d) : Alignment of dowel pins is type of interference fit.

2298. In a limit system, the following limits are specified for hole and shaft. Calculate the tolerance on the shaft.
Hole $= 30^{+0.02}_{-0.00}$ Shaft $= 30^{-0.02}_{-0.05}$
(a) 0.02 (b) –0.03
(c) 0.07 (d) 0.03

UPRVUNL AE 05.07.2021

Ans. (d) : Tolerance on the shaft = 0.05 – 0.02 = 0.03

2299. _____ zone is the zone between the maximum and minimum size.
(a) Dynamic (b) Allowance
(c) Deviation (d) Tolerance

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) : Tolerance zone is the zone between the maximum and minimum size.

2300. The _____is the ratio of the explained variation to the total variation
(a) coefficient of assessment
(b) coefficient of determination
(c) coefficient of attenuation
(d) coefficient of conversion

HPPSC Workshop Suptd. 08.07.02021

Ans. (b) : The coefficient of determination is the ratio of the explained variation to the total variation

2301. One of the main elements that make a microscope a measuring instrument is the _____
(a) Crank (b) Ratina
(c) Spigot (d) Recticle

HPPSC Workshop Suptd. 08.07.02021

Ans. (d) :

2302. Identify the wrong statement pertaining to hot working of metals.
(a) rapid and economical process
(b) close tolerances can be maintained
(c) metal becomes plastic and causes growth of grains
(d) power required to finish the part from ingot is less

HPPSC Poly. Lect. 05.07.2021

Ans. (b) :

2303. If parts in a sample are measured and the means of the sample measurement is outside the tolerance limits :
(a) The process is out of control and the cause should be established
(b) The process is in control but not capable of producing within the established control limits
(c) The process is within the established control limits with only natural causes of variation.
(d) All of the above are true

CGPSC AE 15.01.2021

Ans. (a) : If parts in a sample are measured and the means of the sample measurement is outside the tolerance limits the process is out of control and cause should be established.

2304. Compute the limit dimensions for a clearance fit on the hole basis system, for a basic size of 40 mm diameter, with a minimum clearance of 0.05 mm; with the tolerance on the hole being 0.021 and the tolerance on the shaft being 0.15 mm.
(a) 0.171 mm (b) 0.200 mm
(c) 0.121 mm (d) 0.221 mm

VIZAG Steel MT 24.01.2021

Ans. (d) : For a clearance fit on the hole basis system
= Minimum clearance + tolerance on the hole (0.021) mm + shaft of tolerance (0.15) mm
= 0.05 + 0.021 + 0.15
= 0.221 mm

2305. The gauge useful for checking the component dimension which lies within the permissible limits is _____.
(a) plug gauge (b) limit gauge
(c) snap gauge (d) go and no-go gauge

VIZAG Steel MT 24.01.2021

Ans. (b) : The gauge useful for checking the component dimension which lies within the permissible limit is limit gauge.

2306. Gear-tooth thickness and its profile are measured using _____ and _____ respectively.
(a) mircometer, dial indicator
(b) vernier calliper, mircometer
(c) micrometer, gear-tooth calliper
(d) gear-tooth calliper, micrometer

VIZAG Steel MT 24.01.2021

Ans. (d) : Gear-tooth thickness and its profile are measured using gear tooth calliper and micrometer respectively.

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2307. Match the following:

P.	Feeler gauge	I.	Radius of an object
Q.	Fillet gauge	II.	Diameter within limits by comparison
R.	Snap gauge	II I.	Clearance or gap between components
S.	Cylindrical plug gauge	I V.	Inside diameter of straight hole

(a) P-III, Q-II, R-I, S-IV
(b) P-III, Q-I, R-II, S-IV
(c) P-IV, Q-II, R-I, S-III
(d) P-IV, Q-I, R-II, S-III

Assam Engg. College AP/Lect. 18.01.2021

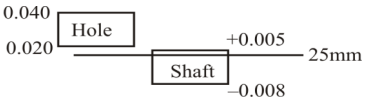
Ans. (b) : Feeler gauge- Clearance or gap between components
Fillet gauge- Radius of an object
Snap gauge- Diameter within limits by comparison
Cylindrical plug gauge- Inside diameter of straight hole

2308. Holes of diameter $25.0^{+0.040}_{+0.020}$ mm are assembled interchangeably with the pins of diameter $25.0^{+0.005}_{-0.008}$ mm. The minimum clearance in the assembly will be_____

(a) 0.015 mm (b) 0.048 mm
(c) 0.005 mm (d) 0.008 mm

Assam Engg. College AP/Lect. 18.01.2021

Ans. (a) : 0.015 mm



Minimum clearance = (25 + 0.020) – (25 + 0.005)
= 0.015 mm

2309. Which of the following is true for the calibration of instruments ?

(a) In casting and fabrication shops the measurements made are of more precise nature as compared to those made in machine shop and tool room
(b) Periodical calibration is made for optical measuring instruments
(c) There is no need of periodic calibration in universal microscope
(d) Preventive maintenance is not necessary for optical measuring instruments

APGCL AM, 2021

Ans. (b) : Calibration of instruments to provides consistency in reading and reduces errors, thus validating the measurements universally.

2310. Which comparator utilizes the principle of a button spinning on a loop string

(a) Sigma comparator
(b) Abramson comparator
(c) Optical comparator
(d) Eden-Rolt comparator

APGCL AM, 2021

Ans. (b) : Abramson comparator–The basic element of this type of comparator is a light pointer mode of glass fixed to a thin twisted metal spring.
In johanson microkator, instead of gear or rack and pinion it uses a twisted strip to magnify the small linear movement of the plunger into indication so it can also be called as twisted strip comparator it is also known as Abramson comparator.

Sigma comparator :

⇒ It uses a partially wrapped band wrapped about a driving drum to turn the pointer needle. The assembly provides a frictionless movement with a resistant pressure provided by the springs.

Eden-Rolt Reed:
It uses a pointer attached to the end of two reeds. One reed is pushed by a plunger, while the other is fixed.

2311. V block has an angle of _____ degrees.

(a) 45 (b) 140
(c) 90 (d) 60

AAI Jr. Executive 26.03.2021

Ans. (d) : V-block has an angle of 60°.

2312. How many grades of tolerances are specified according to BIS?

(a) 18 (b) 20
(c) 16 (d) 24

GMB AAE 25.06.2017
Nagaland PSC CTSE 2017, Paper-I
(KPSC AE. 2015)
VIZAG STEEL MT 18.08.2013
UPSC JWM Advt. No.-52/2010

Ans. (a) : Total 18 grades of tolerances are specified according to BIS.

2313. The magnitude of the permissible variation of a dimension or other measured value or control criteria from the specified value is termed as:

(a) limit (b) lower-upper limit
(c) tolerance (d) fit

AAI Jr. Executive 26.03.2021

Ans. (c) : The magnitude of the permissible variation of a dimension of other measured value or control criteria from the specified value is termed as tolerance.

2314. A shaft having limits of size as $20^{+0.050}_{+0.035}$ and a hole with limits of size as $20^{+0.000}_{+0.025}$ will have

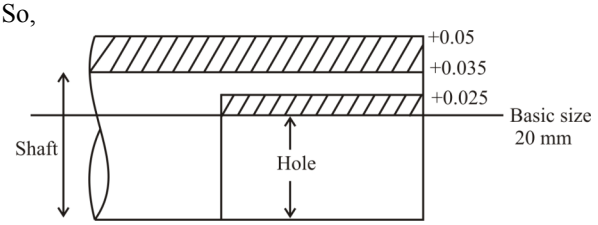
(a) Interference fit
(b) Clearance fit
(c) Transition fit
(d) Cannot be determined

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (a) : Hole with limit size = $20^{+0.000}_{+0.025}$ mm

Shaft with limit size = $20^{+0.050}_{+0.035}$

So,



It shows interference fits

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2315. Which of the following is an interference fit?
(a) Push fit (b) Running fit
(c) Sliding fit (d) Shrink fit
RPSC Lect. (Tech. Edu. Dept.) 13.03.2021, Paper-I

Ans. (d) Shrink fits is a process where a size change after assembly creates an interference fit. By heating and cooling one component prior to assembly and push fit are transition fit.

2316. The accuracy of micrometres, callipers, dial indicators can be checked by a
(a) feeler gauge (b) slip gauge
(c) ring gauge (d) plug gauge
Assam PSC AE (IWT) 14.03.2021

Ans. (b) : Slip gauges are a system for producing precision lengths the individual gauge block is a metal or ceramic block that has been precision ground and lapped to a specific thickness.

2317. If ball bearing is to be fitted on a shaft, the fit between the inner race of bearing and the concerned shaft will be _____ fit.
(a) Clearance (b) Interference
(c) Transition (d) None of the above
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : If ball bearing is to be fitted on a shaft, the fit between the inner race of bearing and the concerned shaft will be Interference fit.

2318. In case of which fit, the tolerances are most tight?
(a) Clearance (b) Interference
(c) Transition (d) Can not say
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (d) :

2319. _____ is useful for angular measurement
(a) Angular gauge (b) Interferometry
(c) Both (a) and (b) (d) None of the above
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) :

2320. Snap gauge is used to check
(a) Inner dia of component
(b) Outer dia of component
(c) Both (a) and (b)
(d) None of the above
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (b) : Snap gauge is used to check outer dia of component.

2321. The algebraic difference between the minimum limit and the basic size is called
(a) Actual deviation
(b) Upper deviation
(c) Lower deviation
(d) Fundamental deviation
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (c) :

2322. _____ tolerance is related to alignment of components
(a) Roughness (b) Perpendicularity
(c) Straightness (d) Runout
GPSC Engineer, Class-II Pre-19.01.2020

Ans. (a) : Roughness tolerance is related to alignment of components.

2323. Ceramic slip gauges are better than steel slip gauges due to its :
(a) wringing capability
(b) resistance to impact
(c) higher thermal expansion
(d) resistance to wear
APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : Ceramic slip gauges are more resistant to wear as compared to steel slip gauges. Natural stability and durability due to its low friction co-efficient make it more resistant.

2324. Temperature compensation can be achieved in strain gauge circuits using :
(a) a potentiometer
(b) high input voltages
(c) a pair of strain gauges
(d) a quarter bridge
APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Temperature compensation aims to reduce the effect of temperature fluctuations on component that are sensitive to changes of temperature. NTC thermistar are often used for temperature compensation of oscillator frequency of a quartz crystal oscillator. Temperature compensation can be achieved in strain gauge circuit using a pair strain gauge.

2325. For effective measurement, the errors should be minimised in transducers by
(a) Power compensation
(b) Voltage compensation
(c) Resistance compensation
(d) Temperature compensation
APPSC Poly Lect. Automobile Engg., 2020

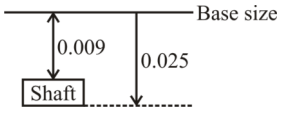
Ans. (d) : For effective measurement the errors should be minimised in transducers by temperature compensation.

2326. The diameters of holes are checked for aggregable using a :
(a) ring gauge (b) slip gauge
(c) plug gauge (d) screw pitch gauge
APPSC Poly Lect. Automobile Engg., 2020

Ans. (c) : Plug gauge– In same cases called pin gauge are metrology tools whose purpose is to gauge the inside diameters of holes that have been drilled or machined into a manufactured part component or assembly.

2327. A shaft has dimension φ 35 (– 0.009 to – 0.025). The respective values of fundamental deviation and tolerance are
(a) – 0.025, ± 0.008 (b) – 0.025, 0.016
(c) – 0.009, ± 0.008 (d) – 0.009, 0.016
UPPSC AE 13.12.2020, Paper-I

Ans. (d) :



Fundamental deviation = – 0.009
Tolerance = 0.025 – 0.009 = 0.016

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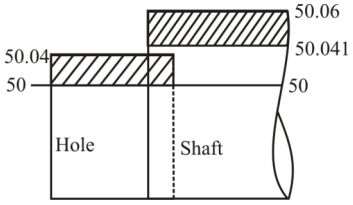
2328. Which of the following represents the type of fit for a hole and shaft pair? Given that hole $= 50^{+0.04}_{+0.00}$ mm and shaft $= 50^{+0.060}_{+0.041}$ mm

(a) Clearance fit (b) Loose fit
(c) Transition fit (d) Interference fit

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : When the tolerance zone of shaft crosses tolerance zone of hole. Such type of fit is called interference fit.



Upper limit of hole = 50 + 0.04
= 50.04 mm
Lower limit of shaft = 50 + 0.041
= 50.041
Since upper limit of hole is less than lower limit of shaft.
So it is a interference fit.

2329. A comparator for its working depends on

(a) Accurately calibrated scale
(b) Comparison with standard such as slip gauges
(c) Optical device
(d) Limit gauges

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : A comparator works on relative measurements. It gives only dimensional differences in relation to a basic dimension. So a comparator compares the unknown dimensions of a part with some standard such as slip gauges etc.

2330. Least count of the Vernier instrument is equal to:

(a) value of the smallest division on the main scale minus value of the smallest division on the Vernier scale
(b) value of the smallest division on the Vernier scale minus value of the smallest division on the main scale
(c) value of the smallest division on the Vernier scale
(d) value of the smallest division on the main scale

NLCIL GET 17.11.2020, Shift-II

Ans. (a) : Least count of Vernier instrument is equal to value of the smallest division on the main scale minus value of the smallest division on Vernier scale.

2331. _____ is a dial type adjustable snap gauge used for external measurement.

(a) Dial gauge (b) Vernier gauge
(c) Screw gauge (d) Pass meter

NLCIL GET 17.11.2020, Shift-II

Ans. (d) : Pass meter is a dial type adjustable snap gauge used for external measurement.

2332. The strain gauge should have high value of gauge factor

(a) to reduce hysterisis effects
(b) to increase sensitivity
(c) to reduce or eliminate the effect of variation in ambient temperature
(d) to give linear relation between applied strain and resistance change

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : The strain gauge should have high value of gauge factor to increase sensitivity.

2333. A set of readings of a measurement has a wide range, it indicates

(a) High precision (b) High accuracy
(c) Low precision (d) Low accuracy

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : A set of readings of a measurement has a wide range it indicates low precision.

2334. In a shaft basic system, the upper deviation of the size of shaft is

(a) One (b) Less than one
(c) More than one (d) Zero

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (d) : In a shaft basic system the upper deviation of the size of shaft is zero.

2335. Which of the following statements is true?

(a) the longer the sine bar, the better the accuracy
(b) the shorter the sine bar, the better the accuracy
(c) accuracy of sine bar does not depend on ambient temperature
(d) a sine bar cannot measure unknown angles

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (a) : The longer the sine bar, the better the accuracy.

2336. Double reflection of light using a pair of mirrors is a unique feature of

(a) an autocollimator
(b) a clinometer
(c) a Zeiss ultra optimeter
(d) an optical projector

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (c) : Double reflection of light using a pair of mirrors is a unique feature of a zeiss ultra optimeter.

2337. Limit gauges are used to

(a) measure flatness of a component
(b) measure exact size of a component
(c) check if the component dimension lies within permissible limits
(d) measure surface roughness of the component

TSPSC Manager (Engg.) HMWSSB 12.11.2020
Oil India Senior Officer 23.12.2020

Ans. (c) : Limit gauges are used to check if the component dimension lies within permissible limits.

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2338. The main use of tool maker's microscope is in measuring

(a) phase shift of monocromatic light
(b) shape, size and angle of small components
(c) biological degradation of small machine components
(d) contours of large machine parts

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : The main use of tool markers microscope is in measuring shape, size and angle of small components.

2339. Surface texture depends to a large extent on

(a) material composition
(b) type of manufacturing operation
(c) skill of the operator
(d) accuracy of measurement

TSPSC Manager (Engg.) HMWSSB 12.11.2020

Ans. (b) : Surface texture depends on a large extent on type of manufacturing operation.

2340. Which of the following is preferred for selecting a combination of slip gauges?

(a) selective halving
(b) writing out all combinations and selecting the closest one
(c) average
(d) determining the fewest number of blocks

TSPSC Manager (Engg.) HMWSSB 12.11.2020

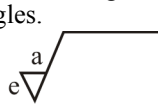
Ans. (d) : Combination of slip gauges determining the fewest number of blocks.

2341. What does 'a' and 'e' indicate in the surface texture symbol shown below

(a) Roughness value, machining allowance
(b) Production method, roughness value
(c) Machining allowance, sampling length
(d) Heat treatment method, sample length

ISRO Scientist/Engineer 12.01.2020

Ans. (a) : The surface roughness on a drawing is represented by triangles.



'a' represents roughness value
'e' represents machining allowance

2342. The amount by which the actual size of a shaft is less than the actual size of mating hole in an assembly is called -

(a) Clearance (b) Interference
(c) Allowance (d) Tolerance

MECON MT 2020

Ans. (a) : The amount by which the actual size of a shaft is less than the actual size of mating hole in an assembly is called clearance.

2343. In dimensioning, ϕ12, DEEP 20 indicates :

(a) 12 mm diameter of hole & 20 mm depth of drilled hole
(b) 20 mm diameter of hold & 12 mm depth of drilled hole
(c) 12 mm diameter of hole & 20 mm fillet on drilled hole
(d) 20 mm diameter of hole & 12 mm fillet on drilled hole

APPSC Poly Lect. 13.03.2020

Ans. (a) : ϕ 12, DEEP 20 – dimensioning indicates
⇒ 12 mm diameter of hole – ϕ 12
⇒ 20 mm depth of drilled hole – DEEP 20

2344. The divisions on the main scale of the vernier caliper are 0.5 mm apart. The vernier has 100 divisions equal in length to 98 main scale divisions. The least count of the instrument is :

(a) 0.04 (b) 0.02
(c) 0.03 (d) 0.01

NLC GET 17.11.2020, Shift-II

Ans. (d) : Least count of vernier scale (LC) = value of divisions on main scale – value of division on vernier scale.

Value of division on main scale = 0.5 mm

and 98 division on main scale = 100 division on vernier scale

One division of vernier scale $= \frac{98}{100}$ division on main scale

vernier scale $= 0.5 \times \frac{98}{100} = 0.49$

Now, (LC) least count = 1MSD – 1 VSD

$L.C = 0.5 - 0.49$

$\boxed{L.C = 0.01mm}$

2345. The workshop-grade slip gauges which are used to setting machine tools, positioning milling cutter etc., is _____.

(a) grade 2 (b) grade 00
(c) grade 0 (d) grade 1

VIZAG MT, 14.12.2020

Ans. (a) : The workshop-grade slip gauges which are used to setting machine tools, positioning milling cutter etc, is grade 2.

(1) 0 Grade → Use for inspection
(2) 00 Grade → Use as a standard in the standard room only
(3) Calibration Grade → Has actual size for calibration
(4) 2 Grade → Work shop grade for machine tools etc.
(5) 1 Grade → For more precise work such as with sin bar

2346. External taper can be accurately measured with the help of

(a) Dividing head
(b) Sine bar and slip gauge
(c) Clinometers
(d) Combination set

Gujarat PSC AE 2019

Ans : (b) : External taper can be accurately measured with the help of sine bar and slip gauge.

2347. Sensitivity and range of measuring instrument have

(a) Direct relation (b) Linear relation
(c) Inverse relationship (d) No relationship

Gujarat PSC AE 2019

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Ans : (c) : Sensitivity and range of measuring instruments have inverse relationship.

2348. Minimum limit of the hole is taken as the ..... in a hole basis system:

(a) Maximum size (b) Actual size
(c) Basic size (d) Nominal size

Gujarat PSC AE 2019

Ans : (c) : Minimum limit of the hole is taken as the basic size in a hole basic system.

2349. In a unilateral system of tolerance, the tolerance is allowed on

(a) one side of the actual size
(b) one side of the nominal size
(c) both sides of the actual size
(d) both sides of the nominal size

Gujarat PSC AE 2019

Ans : (b) : In a unilateral system of tolerance, the tolerance is allowed on one side of the nominal size.

2350. Piezoelectric material functions on the principle of

(a) Newton's Second law of motion
(b) Conservation of energy
(c) Thomson's Effect
(d) None of the above

Gujarat PSC AE 2019

Ans : (a) : Piezoelectric material function on the principle of Newton's second law of motion.

2351. A graph is drawn to a vertical magnification of 10000 and horizontal magnification of 100, and the areas above and below the datum line are as follows:

Above	150mm^2	80mm^2	170mm^2	40mm^2
Below	80mm^2	60mm^2	150mm^2	120mm^2

The average roughness Ra for sampling length of 0.8 mm will be

(a) 1.14 μm (b) 1.10 μm
(c) 1.06 μm (d) 1.02 μm

ESE 2019

Ans. (c) : Given,

L = 0.8 mm

Horizontal magnification (HM) = 100

Vertical magnification (VM) = 10,000

$\sum A = 150 + 80 + 170 + 40 + 80 + 60 + 150 + 120$

$= 850\ mm^2$

Average Roughness (R_a)

$$= \frac{\Sigma A}{L} \times \frac{1000}{VM} \times \frac{1}{HM}$$

$$= \frac{850}{0.8} \times \frac{1000}{10,000} \times \frac{1}{100}$$

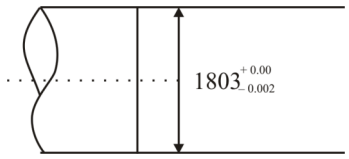
$= 1.06\ \mu m$

2352. Expressing a dimension as $18.3^{+0.00}_{-0.02}$ mm in the case of

(a) unilateral tolerance (b) Bilateral tolerance
(c) limiting tolerance (d) Trilateral tolerance

TNPSC AE 2019

Ans. (a) : Bilateral tolerance- Variation is permitted in both positive and negative directions from the nominal dimension.



Example- X^{+a}_{-b}

Unilateral tolerance- Variation from the specific dimension is permitted in only one direction either positive or negative, but not both.

Example- $X^{+0}_{-b}, \chi^{-a}_{+0}$

2353. The angles of angle gauges in the degree series are

(a) 1º, 3º, 9º, 25º, 42º (b) 1º, 3º, 9º, 27º, 81º
(c) 1º, 3º, 9º, 27º, 41º (d) 1º, 5º, 10º, 25º, 45º

TNPSC AE 2019

Ans. (c) : The angles of angle gauges in the degrees series are

$1^o, 3^o, 9^o, 27^o, 41^o$

2354. Dummy strain gauges used for

(a) Comparison of temperature changes
(b) Increasing the sensitivity of bridge in which they are included
(c) Compensating for differential expansion
(d) Calibration of strain gauges

TNPSC AE 2019

Ans. (a) : Dummy strain gauges used for comparison of temperature changes.

2355. The methods for reducing the errors of approximation infinite element solution are

(a) p - version and h - version
(b) a - version and d - version
(c) y - version and k - version
(d) m - version and n - version

TNPSC AE 2019

Ans. (a) : The methods for reducing the errors of approximation infinite element solution are p - version and h - version.

2356. A pitch is the rotation about

(a) Vertical axis (b) Horizontal axis
(c) Own axis (d) Z-axis

TNPSC AE 2019

Ans. (b) : A pitch is the rotation about horizontal axis.

2357. Pick out the wrong statement about Taylor's principle of gauging

(a) Go gauges should be full form gauges
(b) Go gauges should check all the related dimensions simultaneously
(c) It is sufficient to use Go gauges on the width and length of the component
(d) Not go gauges should check only one dimension at a time

TNPSC AE 2019

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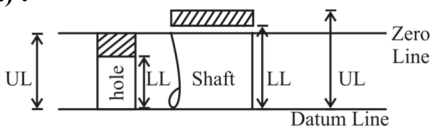
Ans. (c) : Taylor's Principle- It states that GO gauge should check all related dimensions. Simultaneously NOGO gauge should check only one dimension at a time. It state that GO gauge should check all related dimension.

2358. In order to have interference fit, it is essential that the lower limit of the shaft should be:

(a) Greater than the upper limit of the hole
(b) Lesser than the upper limit of the hole
(c) Greater than the lower limit of the hole
(d) Lesser than the lower limit of the hole

OPSC AEE 2019 Paper-I

Ans : (a) :



In order to have interface fit, minimum interface must be negative.

$\therefore$ min (I) = UL of hole – LL of shaft

0 > UL of hole – LL of shaft

LL of shaft > UL of hole

2359. Which of the following is not a limit gauge?

(a) Go and Not Go gauges
(b) Thread gauge
(c) Taper gauge
(d) Voltmeter

HPPSC ADF 05.03.2019

Ans. (d) : Voltmeter

2360. Which of the following is a line standard of measurement ?

(a) Measuring tape (b) Slip gauge
(c) Micrometer (d) End bars

MECON MT 2019

Ans. (a) : Measuring tape

2361. The Hole $80.00^{+0.30}_{+0.00}$ & shaft $80.00^{+0.021}_{+0.002}$ when assembled will result in:

(a) Clearance fit (b) Drive fit
(c) Interference fit (d) Transition fit

Oil India Ltd. Sr. Engg (Production) 30.11.2019

Ans. (d) : Given,

Dimension of Hole $80.00^{+0.30}_{+0.00}$

Dimension of shaft = $80.00^{+0.021}_{+0.002}$

According to given dimensions both have unilateral tolerance.

$\because$ Maximum clearance = HLH – LLS = 80.30 – 80.002 = + 0.298

Minimum clearance = LLH – HLS = 80.00 – 80.021 = – 0.021

Maximum clearance is positive but minimum clearance is negative so it will be a transition fit.

Note- Its can be divide into three categories based on actual limits of hole & shaft.

1. Clearance fit
2. Interference fit
3. Transition fit

Transition fit- In this case the diameter of the largest permissible hole is greater than the diameter of the smallest shaft and the diameter of smallest hole is smaller than the diameter of the largest shaft.

2362. 'Not Go' limit refers to :

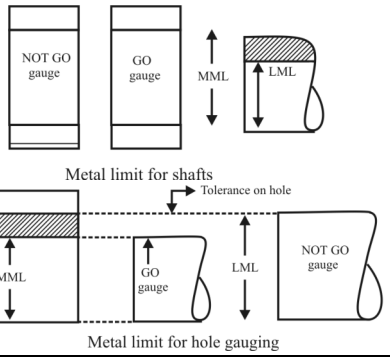
(a) The lower limit of a shaft and upper limit of a hole
(b) The lower limit of a hole and upper limit of a shaft
(c) Hole whose lower deviation is zero
(d) Maximum material condition.

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (a) : The gauge required to check the dimensions of the components correspond to two sizes conforming to the maximum and minimum limits of the components. They are called Go gauges and NOGO or NOT Go gauges, which correspond respectively to the MML and LML of the component

MML (Maximum Metal Limit)– Lower limit of a hole and higher limit of the shaft

LML (Least Metal Limit or Minimum Metal Limit)– Higher limit of a hole and lower limit of the shaft.



2363. Which of the following is not the grade of slip gauge specified by IS 2984 – 1966?

(a) Grade 0 (b) Grade III
(c) Grade I (d) Grade II

Oil India Limited Sr. Engineer (Drilling) 30.11.2019

Ans. (b) : Slip gauges- Also known as Johnson gauges or gauges blocks. They have high degree of surface finish and accuracy. They are rectangular blocks of steel having cross-section of 30mm × 10 mm and are most commonly used end standards in engineering practices.

Indian standard on slip gauge (IS: 2984 - 1966)

Slip gauges are graded according to their accuracy as grade 0, Grade I and Grade II.

Note– Grade II → Used in workshop during the actual production of components, tools and gauges.

Grade I → Higher accuracy and used in Inspection department.

Grade 0 → Used in laboratory and standard room.

2364. A mortise gauge is a

(a) striking tool (b) planning tool
(c) boring tool (d) marking tool

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : A mortise gauge is a wood working tool used by a carpenter to scribe mortise and fenon joints on wood prior to cutting. It is a marking tool.

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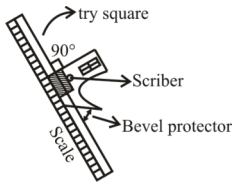
2365. The instrument which has all the features of try-square, bevel protractor, rule and scriber is

(a) outside micrometer
(b) inside micrometer
(c) depth gauge micrometer
(d) combination set

Nagaland PSC (CTSE) 2018, Paper-I

Ans. (d) : Combination set is the instrument as shown is figure, has all features of try square, bevel protector rule and scriber.



2366. In a bilateral system of tolerance, the tolerance is allowed on

(a) one side of the actual size
(b) one side of the nominal size
(c) both sides of the actual size
(d) both sides of the nominal size

Nagaland PSC (CTSE) 2018, 2017 Paper-I

Ans. (d) : In a bilateral system of tolerance, the tolerance is allowed on both sides of the nominal size. It is concerned with equal variation, in both the direction, either plus or mines.

$20^{+0.00}_{-0.005} \rightarrow$ Unilateral

$20^{+0.05}_{-0.005} \rightarrow$ Bilateral

2367. Which one of the following instruments is a comparator?

(a) Tool Maker's Microscope
(b) GO/NO GO gauge
(c) optical Interferometer
(d) dial gauge

GPSC Executive Engineer 23.12.2018

Ans. (d) : Dial gauge.

2368. The reflector combined with auto collimator can be used for checking

(a) parallelism (b) circularity
(c) surface finish (d) alignment

ISRO Scientist/Engineer 22.04.2018

Ans. (a&d) : Auto-Collimator is an instrument used to measure small angular inclinations. It is also used to check straightness, flatness and alignment. Autocollimator is a tube for projecting parallel rays of light. It consists simply of a light source and a collimating lens (converging lens).

2369. For a strain gauge (gauge factor = 2.1 and resistance = 50 Ω), subjected to a maximum strain of 0.001, the maximum change in resistance is

(a) 0.084 Ω (b) 0.105 Ω
(c) 0.135 Ω (d) 0.156 Ω

ESE 2018

Ans. (b) : Given,
Gauge factor (G) = 2.1
Resistance (R) = 50Ω
Strain ($\in$) = 0.001

$$\text{Gauge factor (G)} = \frac{\Delta R / R}{\in}$$

$$\Delta R = G \in \times R = 2.1 \times 0.001 \times 50 = 0.105\ \Omega$$

2370. The general way of describing cylindrictiy of a component is by the

(a) minimum - zone method
(b) maximum - zone method
(c) limited zone method
(d) cylinder zone method

TNPSC AE 2018

Ans. (a) : The general way of describing cylindrictiy of a component is by the minimum - zone method.

2371. The digitized frame of the image in a machine vision system is referred as

(a) ADC (b) Frame buffer
(c) Vision buffer (d) DAC

TNPSC AE 2018

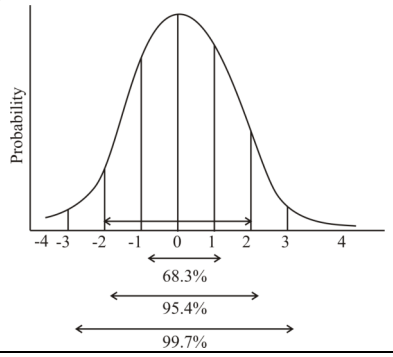
Ans. (b) : The digitized frame of the image in a machine vision system is referred as Frame buffer.

2372. Approximately what area is covered under the normal distribution curve between ± 3 standard deviation

(a) 80% (b) 88.00%
(c) 99.73% (d) 68.00%

TNPSC AE 2018

Ans. (c) : Approximately 99.73% area is covered under the normal distribution curve between ±3 standard deviation.



2373. An example of an attribute gauge is a

(a) plug gauge (b) micrometer
(c) slip gauge (d) an angle gauge

TNPSC AE 2018

Ans. (a) : Attribute gauging- Attribute gauges include Go/ No Go, functional and bench gauging system. These gauges measure part characteristics and either accept or reject the part based on accuracy.

2374. The type of gauge which has gauging sections combined on one end is

(a) combination gauge (b) progressive gauge
(c) limit gauge (d) fixed gauge

TNPSC AE 2018

Ans. (b) : Progressive gauge has gauging sections combined on one end.

2375. Which one of the following is the most accurate measuring device/instrument?

(a) steel rule (b) digital micrometer
(c) vernier caliper (d) laser micrometer

TNPSC AE 2018

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Ans. (d) : Laser micrometer is the most accurate measuring device.

2376. According to Taylor's principle, No Go gauge checks

(a) only one feature at a time
(b) only important dimensions at a time
(c) all the dimensions at a time
(d) only the related dimensions at a time

TNPSC AE 2018

Ans. (a) : According to Taylor's principle, No Go gauge checks only one feature at a time.

2377. Variable Head flow meters can be used for measurement of flow of

(a) liquids only
(b) liquids and gases
(c) slurries only
(d) liquids, gases and slurries

TNPSC AE 2018

Ans. (d) : Variable Head flow meters can be used for measurement of flow of liquids, gases and slurries

2378. Standardization will

(a) Ensure interchangeability in local market
(b) Ensure competitiveness in local market
(c) Ensure perfectness
(d) Ensure interchangeability and competitiveness in world market place

MPSC Lect. (Prod.) 2018

Ans. (d) : Standardization will ensure interchangeability and competitiveness in world market place

2379. Which one of the following instruments is widely used to check and calibrate geometric features of machine tools during their assembly?

(a) Ultrasonic probe
(b) Coordinate Measuring Machine (CMM)
(c) Laser interferometer
(d) Vernier calipers

GPSC ARTO Pre 30.12.2018

Ans. (c) : Laser interferometer is widely used to check and caliberate geometric features of machine tools during their assembly.

2380. The total amount a dimension may vary and is the difference between the maximum and minimum limits is called

(a) Tolerance (b) Limits
(c) Fit (d) Offset

MPSC Lect. (Prod.) 2018

Ans. (a) : The total amount a dimension may vary and the difference between the maximum and minimum limits is called 'tolerance'.

2381. Which of the following gives actual measurement of any specific dimension?

(a) Inspection of variables
(b) Inspection of attributes
(c) Both (a) and (b)
(d) None of the above

MPSC Lect. (Prod.) 2018

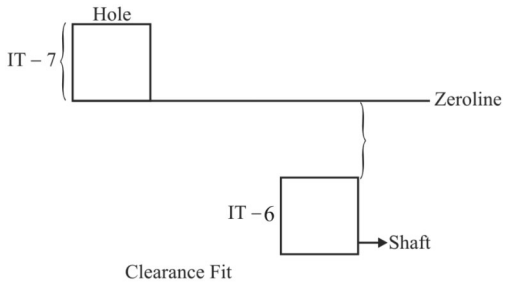
Ans. (a) : Inspection of variables gives actual measurement of any specific dimension.

2382. 20 H_7g_6 is an example of:

(a) clearance fit (b) transition fit
(c) interference fit (d) push fit

AAI Jr. Executive 29.11.2018
DRDO Scientists 2008

Ans. (a) : Suppose a hole and shaft assembly is designated by 20 H_7g_6
20 indicates basic size, H is the hole with width of tolerance zone IT – 7. g is type of shaft with IT – 6 is the tolerance zone.



2383. The tolerance value for grade IT7 is equal to:

(a) 14i (b) 7i
(c) 16i (d) 25i

Gujarat PSC AE 2019
AAI Jr. Executive 29.11.2018

Ans. (c) :

IT 6	10 i
IT 7	16 i
IT 8	25 i
IT 9	40 i

2384. A hole is of dimension $\phi 60^{(+0.015/+0.000)}$ and the corresponding shaft is of dimension $\phi 60^{(+0.011/+0.001)}$. When they are assembled they will form a

(a) interference fit (b) clearance fit
(c) transition fit (d) none of these

ISRO Scientist/Engineer 22.04.2018

Ans. (c) : Upper size of hole

$\phi 60^{(+0.015/+0.000)} = 60 + 0.015 = 60.015$ mm

Upper size of shaft

$\phi 60^{(+0.011/+0.001)} = 60 + 0.011 = 60.011$ mm

Upper size of shaft is more than lower size of hole (60mm) but less than upper size of hole.
So the assembled parts make transition fit.

2385. Least count of vernier caliper is

(a) Ratio of main scale to vernier scale
(b) Ratio of vernier scale to main scale
(c) Difference between one division of main scale and one division of vernier scale
(d) Sum of main scale to the vernier scale

HPPSC AE 2018

Ans. (c) : Least count of a vernier caliper is difference between one division of main scale and one division of vernier scale.

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2386. The amount by which diameter of hole is larger than the diameter of shaft is known as:
(a) Interference (b) Clearance
(c) Tolerance (d) Limit
(e) Margin

CGPSC AE 25.02.2018

Ans. (b) : The amount by which diameter of hole is larger than the diameter of shaft is known as clearance.

2387. Which of the following can be used to scribe lines parallel to the edges of a part
(a) Vernier calipers (b) Screw gauge
(c) Divider (d) Hermaphrodite

Gujarat PSC AE 2019
Vizag Steel (MT) 2017

Ans. (d) : Hermaphrodite calliper– It is also known as odd leg calliper consisting of one divider and one calliper leg.
–It is used for layout work like scribing lines parallel to the edge of the work and for finding the centre of a cylindrical work.

2388. A comparator for its working depends on
(a) Accurately calibrated scale
(b) Comparison with standard such as slip gauges
(c) Accurate micrometer gauge
(d) Optical devices

Vizag Steel (MT) 2017

Ans. (b) : Comparison with standard such as slip gauges.
Comparator– is an instrument used for comparing the dimensions of a component with a standard of length.
Purpose of a comparator is to detect and display the small differences between the unknown linear dimensions and standard length.

2389. The term "Allowance" in limits and fits is usually referred to
(a) Minimum clearance between shaft and hole
(b) Maximum clearance between shaft and hole
(c) Difference of tolerance of hole and shaft
(d) Difference between maximum size & minimum size of the hole

Vizag Steel (MT) 2017
PTCUL AE 25.06.2017

Ans. (a) : Allowance is the amount of designed (intentional) deviation between two mating dimensions in a fit.

2390. In interferomatric method, the path differences between one bright band and the next is varied by
(a) Half wave length
(b) Two half wave length
(c) One quarter wave length
(d) Two wavelengths

Vizag Steel (MT) 2017

Ans. (b) : Two half wave length

2391. The method of fractional coincidences in interferometric techniques is used for
(a) Measurement of end gauges
(b) Flatness of surface
(c) Linear displacement measurement
(d) Convexity/concavity of surfaces

Vizag Steel (MT) 2017

Ans. (a) : The method of fractional coincidences in interferometric techniques is used for measurement of end gauges.

2392. Grating are used in connection with:
(a) Angular measurement
(b) Surface flatness measurement
(c) Linear displacement measurement
(d) None of these

PTCUL AE 25.06.2017

Ans. (c) : Grating are used in connection with linear displacement measurement.

2393. Which of the following device can not be used to measure pressure:
(a) Strain gauge (b) LVDT
(c) Pirani gauge (d) Pyrometer

PTCUL AE 25.06.2017

Ans. (d) :

2394. Wickman gauge employed to inspect:
(a) Screw threads (b) Tapers
(c) Holes (d) None of these

PTCUL AE 25.06.2017

Ans. (a) :

2395. The tolerance of grade 6 is obtained by–
(a) die casting
(b) turning on capstan and turret lathes
(c) grinding
(d) sand casting

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : The tolerance of grade 6 is obtained by grinding.

2396. In clearance fit,
(a) tolerance zones of hole and shaft overlap
(b) tolerance zone of hole is completely below that of shaft
(c) tolerance zone of hole is entirely above that of shaft
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (c) : In clearance fit, tolerance zone of hole is entirely above that of shaft.

2397. In interference fit,
(a) tolerance zones of hole and shaft overlap
(b) tolerance zone of hole is completely below that of shaft
(c) tolerance zone of hole is entirely above that of shaft
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : In interference fit, tolerance zone of hole is completely below that of shaft.

2398. In transition fit,
(a) tolerance zones of hole and shaft overlap
(b) tolerance zone of hole is completely below that of shaft
(c) tolerance zone of hole is entirely above that of shaft
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

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Ans. (a) : In transition fit, tolerance zones of hole and shaft overlap.

2399. The tolerance of grade 8 is obtained by–
(a) die casting
(b) turning on capstan and turret lathes
(c) grinding
(d) sand casting

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : The tolerance of grade 8 is obtained by turning on capstan and turret lathes.

2400. The tolerance of grade 16 is obtained by-
(a) die casting
(b) turning on capstan and turret lathes
(c) grinding
(d) sand casting

Nagaland PSC CTSE 2017, Paper-I

Ans. (d) : The tolerance of grade 16 is obtained by sand casting.

2401. Speedometer is a–
(a) display giving quantitative measurements
(b) display giving state of affair
(c) display indicating predetermined settings
(d) none of above

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Display giving quantitative measurements

2402. Moving scale or moving dial is used for-
(a) display giving quantitative measurements
(b) display giving state of affair
(c) display indicating predetermined settings
(d) none of above

Nagaland PSC CTSE 2017, Paper-I

Ans. (a) : Display giving quantitative measurements

2403. When large force is required to operate, the type of control used is–
(a) knobs and switches
(b) levers and wheels
(c) push buttons
(d) none of the above

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : Levers and wheels

2404. Which micrometer is used for measuring the span between the teeth of a gear?
(a) Blade micrometer
(b) Screw thread micrometer
(c) Disc micrometer
(d) Dial micrometer

CIL MT 26.03.2017

Ans. (c) : Disc micrometer is used for measuring the span between the teeth of a gear.

2405. Statement (I): Pyrometer is used to measure diffuse solar radiation by blocking the direct radiation with a shadow band.
Statement (II): Pyrheliometer is used to measure diffuse radiation.

ESE 2017

Ans. (c) : The pyrometer is a broadband instrument that measures the direct component of solar radiation at normal incidence. It does not measure diffuse radiation. The diffuse radiation is measured by pyrometer.

2406. Statement (I): In sugarcane crushing rollers, the fit between the cast roll and the forged steel shaft is of interference type.
Statement (II): This helps in removing the roll from the shaft whenever not needed.

ESE 2017

Ans. (a) : In sugarcane crusher an interference fit exist between shaft and mounted roll.

2407. Which of the following methods is not concerned with the surface finish measurement?
(a) Spectrophotometry method
(b) Ultrasonic method
(c) Field emission method
(d) Critical angle of attack method

TNPSC AE 2017

Ans. (a) : Spectrophotometry method is not concerned with the surface finish measurement.

2408. A least accurate measuring device was
(a) Air gauge
(b) Micrometer screw gauge
(c) Steel rule
(d) Optical projector

TNPSC AE 2017

Ans. (c) : A least accurate measuring device was steel rule.

2409. The resolution of electrostatic plotter is expressed in terms of
(a) number of lines per unit area
(b) number of dots per inch
(c) ratio of darkened area to gross area
(d) number of lines per inch

TNPSC AE 2017

Ans. (b) : The resolution of electrostatic plotter is expressed in terms of number of dots per inch.

2410. For general use the measuring tip of a comparator should be
(a) Flat (b) Spherical
(c) Conical (d) Grooved

TNPSC AE 2017

Ans. (b) : For general use the measuring tip of a comparator should be Spherical.

2411. The gauges which are only used for checking the size and condition of other gauge are called
(a) Plug gauge (b) Master gauge
(c) Limit gauge (d) Inspection gauge

TNPSC AE 2017

Ans. (b) : These gauge are used in laboratories and test houses to calibrate industrial gauges.

2412. A regulator problem is where the close loop system must
(a) try to follow a series of set point changes
(b) remove any disturbance acting on the system
(c) respond very quickly
(d) respond very slowly

TNPSC AE 2017

Ans. (b) : A regulator problem is where the close loop system must remove any disturbance acting on the system.

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2413. A Push (Location) fit is used for

(a) Gears fixed on a shaft by means of a key
(b) Shaft rotating at a slow speed in a journal bearing
(c) Shaft rotating at a high speed in a journal bearing
(d) Swinging parts such as the fit between gudgeon pin and the connecting rod of an automobile engine

GPSC Asstt. Director of Transport 05.03.2017

Ans. (d) : A push fit fitting is a compression type of fitting with an easy installation process and a water tight seal. It seals the gaps between piston and other parts.

2414. Micrometer measurements are more accurate than those by Vernier Caliper, primarily because:

(a) The micrometer used flat parallel anvils
(b) Component dimension lies in the axis of measurement
(c) Micrometer does not use a linear sliding scale
(d) Micrometer uses a calibrated drum

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : Micrometer measurements are more accurate than those by vernier caliper, primarily because micrometer does not use a linear sliding scale and having smaller least count.

2415. What does N, P and L mean in N.P.L. Gauge interferometer?

(a) Nikon pulsed laser
(b) Nuclear plasma laboratory
(c) National Physics Laboratory
(d) Nuclear physics laboratory

GPSC Asstt. Director of Transport 05.03.2017

Ans. (c) : NPL gauge interferometer, NPL stands form National Physics Laboratory.

2416. In tolerancing of a part, unless otherwise specified, individual features of size must

(a) not violate size limits of MMC at any cross section
(b) not violate an envelope of perfect form at LMC
(c) not violate an envelope of perfect form at MMC
(d) None of these

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Maximum material condition (MMC) ensure further machining for correction if nothing is specified.

2417. Identify the geometric characteristic that describe the feature geometry and the inter-relationship of part features

(a) form tolerance (b) form
(c) orientation tolerance (d) dimension

ISRO Scientist/Engineer 17.12.2017

Ans. (b) : There are four geometric characteristics that describe feature geometry and the inter-relationship of part features. These are
- Size
- Form
- Orientation
- Location

Consequently, there are four types of tolerances that are possible for each feature. These are
- Size tolerances
- Form tolerances
- Orientation tolerances
- Location tolerances

2418. Smallest thickness which can be measured by a slip gauge is

(a) 1.001 mm (b) 0.01 mm
(c) 0.001 mm (d) none of these

ISRO Scientist/Engineer 17.12.2017

Ans. (a) : Smallest thickness which can be measured by a slip gauge is 1.001 mm.

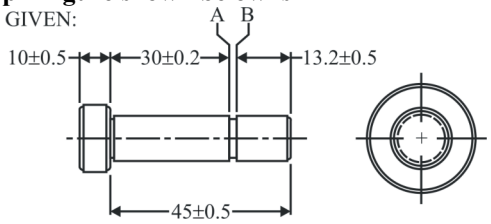
2419. Tolerances generally followed for tooling design are

(a) 5 – 10% of work piece tolerance
(b) 11 – 30% of work piece tolerance
(c) 31 – 50% of work piece tolerance
(d) none of these

ISRO Scientist/Engineer 17.12.2017

Ans. (c) : Tolerances generally followed for tooling design are 31 – 50% of work piece tolerance.

2420. The missing distance between A and B in the pin figure shown below is



(a) 1.8 (b) 1.8 ± 1.1
(c) 0.6 (d) 1.8 ± 1.2

ISRO Scientist/Engineer 07.05.2017

Ans. (d) : Basic size of AB
$= 45 - (30 + 13.2) = 1.8$
Tolerance = Cumulative sum
$= 0.2 + 0.5 + 0.5 = 1.2$
So, the missing distance between A and B
$= 1.8 \pm 1.2$

2421. The following limits are specified in a limit system, to give a clearance fit between a hole and a shaft:

Hole = $25^{+0.02}_{0-.01}$ mm and shaft = $25^{-0.004}_{-0.02}$ mm

Determine tolerance on shaft.

(a) 0.012 mm (b) 0.016 mm
(c) 0.018 mm (d) 0.014 mm

CIL MT 26.03.2017

Ans. (b) : Upper limit of shaft = 25 - 0.004 = 24.996
Lower limit of shaft = 25 - 0.02 = 24.98
Tolerance = Upper limit of shaft – Lower limit of shaft = 24.996 – 24.98 = 0.016

2422. Tolerance for a hold and shaft assembly having a nominal size of 40 mm are as follows:

Hole = $40^{+0.06}_{+0.02}$ mm **and shaft** $= 40^{-0.06}_{-0.08}$ mm

Determine MML of hole

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(a) 39.94 mm (b) 40.06 mm
(c) 40.02 mm (d) 39.92 mm

CIL MT 26.03.2017

Ans. (c) : Maximum material Limit (MML) in case of hole is the condition when hold is at lower limit.

So MML for hole

$40^{+0.06}_{+0.02} = 40 + 0.02 = 40.02\text{mm}$

2423. In unilateral system for tolerances,

(a) tolerances are given on both positive and negative sides of basic size
(b) one tolerance is zero and other tolerance is given on any one side of basic size
(c) one tolerance is zero and other tolerance is given only on higher side of basic size
(d) one tolerance is zero and other tolerance is given only on lower side of basic size

Nagaland PSC CTSE 2017, Paper-I

Ans. (b) : In unilateral system for tolerances, one tolerance is zero and other tolerance is given only on higher side of basic size.

2424. The size by which it is referred to as a matter of convenience:

(a) Actual size (b) Basic size
(c) True size (d) Nominal size

TRB Polytechnic Lecturer 2017

Ans. (d) : Nominal size is referred to as a matter of convenience.

2425. Four pats with the following dimensions are to be assembled in random. What should be tolerance on the assembled length?

Part 1 = $25.32^{+0.02}_{-0.01}$ mm
Part 2 = 18.91 ± 0.03 m
Part 3 = $62.17^{+0.05}_{-0.07}$ mm
Part 4 = 46.25 ± 0.04 mm

(a) $152.65^{+0.15}_{-0.14}$ (b) 152.65
(c) $152.65^{+0.14}_{-0.15}$ (d) $145.52^{+0.02}_{-0.07}$

TRB Polytechnic Lecturer 2017

Ans. (c) : Tolerance an the assembled length

$= [25.32 + 18.91 + 62.17 + 46.25]^{+[0.02+0.03+0.05+0.04]}_{-[0.01+0.03+0.07+0.04]}$

$= [152.65]^{+0.14}_{-0.15}$

2426. The hole $40^{+0.20}_{+0.00}$ and shaft $40^{+0.01}_{-0.01}$

(a) Clearance fit (b) Interference fit
(c) Transition fit (d) Drive fit

TSPSC AEE 2017

Ans. (c) :

2427. Flatness of slip gauges is checked with

(a) Interferometer
(b) Optical flat
(c) Electronic comparator
(d) Linear measuring machine

TSPSC AEE 2017

Ans. (a) : Interferometer and optical flat is used to measure flatness but interferometer is more accurate than the optical flat.

2428. A hole of diameter $25^{+0.01}$ mm is to be inspected by using GO/NO GO gauges. The size of the GO plug gauge should be

(a) 25.00 mm (b) 25.01 mm
(c) 25.02 mm (d) 24.99 mm

TSPSC AEE 2017

Ans. (b) : High limit of hole = 25.01
Low limit of hole = 25.00mm
Work tolerance = 0.01 mm
Gauge tolerance = 10% of work tolerance
= 0.001 mm
Dimension of go gage = 25 + 0.001
= 25.001 mm
Without gauge tolerance
go gauge = 25.01 mm

2429. Strain gauge rosettes are used when the direction of the

(a) Principal stress is unknown
(b) Hoop stress is known
(c) Principal stress is known
(d) Hoop stress is unknown

TSPSC AEE 2017

Ans. (a) : Generally, if the direction of principal stress is uncertain in structure stress measurement, a triaxial rosette gauge is used and measured strain values are calculated, to find the direction of principal stress.

2430. When an elastic member is used in conjunction with a strain gauge, it is called a

(a) Proving ring
(b) Absorption dynamometer
(c) Prony brake
(d) Load cell

TSPSC AEE 2017

Ans. (d) : A strain gauge elastic member combination arrangement used for the measurement of load or weight is called load cell. It utilizes an elastic member as primary transducer and strain gauge as a secondary transducer.

2431. Circular scale of the micrometer is marked on-

(a) Anvil (b) Barrel
(c) Thimble (d) Spindle

GMB AAE 25.06.2017

Ans. (c) : • Circular scale of the micrometer is marked on thimble
• Millimeter Scale in a micrometer is marked on Barrel.

2432. Johansson Mikrokator is a type of

(a) Mechanical Optical Comparator
(b) Mechanical Comparator
(c) Optical Comparator
(d) Electrical Comparator

Karnataka PSC Lect., 27.05.2017

Ans (b)ohansson Mikrokator is a type of mechanical comparator.

2433. A 30 mm hole/shaft assembly results in minimum and maximum clearance of 0.03 mm and 0.3 mm respectively. The hole has a unilateral tolerance with zero fundamental deviation. If the tolerance of the shaft is 0.08 mm, the maximum size of the hole is

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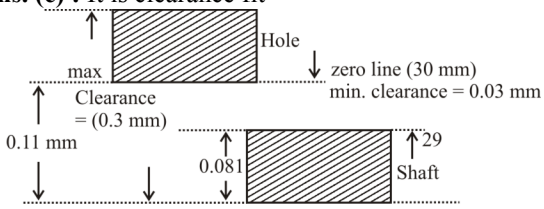
(a) 30.11 mm (b) 30.16 mm
(c) 30.19 mm (d) 30.27 mm

Karnataka PSC AE, 10.09.2017

Ans. (c) : It is clearance fit



Max. clearance = Min. clearance + tolerance of shaft + tolerance of hole.

0.30 mm = 0.03 mm + 0.08 mm + x

$\therefore$ x = 0.19 mm

$\therefore$ Upper deviation of hole = 0.19 mm

Max. size of hole = Basic size + UD

= 30 mm + 0.19 mm

= 30.19 mm

2434. Two full turn of the circular scale of a micrometer cover a distance of 1 mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that this micrometer has a zero error of minus 0.03 mm (i.e. –0.03 mm) while measuring the diameter of a thin wire, engineer notes the main scale reading of 4 mm and the number of circular scale in line with the main scale as 35. The diameter of the wire is-

(a) 4.35 mm (b) 4.32 mm
(c) 4.38 mm (d) 4.03 mm

GPSC EE Pre, 28.01.2017

Ans. (c) : Main scale reading (MSR) = 4 mm

No. of circular divisions = 35

Since 100 divisions = 1 mm

So, 35 division = 0.35 mm

Since zero error is = – 0.03 mm

It will add up in the measurement

4 + 0.35 + 0.03

= 4.38 mm

2435. Which one of the following combination results in a press fit?

(a) G7-h6 (b) F7-n6
(c) H7-h6 (d) F7-j6

APGENCO AE, 2017

Ans. (b) : F7-h6

2436. The intentional difference between the maximum material limits of mating parts is known as :

(a) Tolerance (b) Allowance
(c) Deviation (d) Fit

Punjab PSC SDE 12.02.2017

Ans. (b) : Allowance- It is defined as the difference between maximum material limits of mating parts.

Tolerance- It is equal to difference of the two limit size of the part.

Tolerance = upper limit (UL) – Lower limit (LL)

Deviation- It is the difference between the size and the correspoinding basic size.

Fit- The fit between two mating parts is the relationship existing between them with respect to the amount of interference which is present when they are assembled together. For example a shaft running in a bush can move in relation to it and so forms a moving joint whereas a pulley mounted fixed joint.

2437. The tolerance of grade 6 is obtained by :

(a) Die casting
(b) Grinding
(c) Turning on capstan and turret lathes
(d) Sand casting

HPPSC Asstt. Prof. 20.11.2017

Ans. (b) : The tolerances of grade 6 is obtained by grinding. As IT grade increases the accuracy decreases.

2438. Plain ring gauges are used for checking external diameter of parts for

(a) Cylindrical objects (b) Spherical objects
(c) Conical objects (d) Rectangular objects

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : Plain ring gauge ideal for inspecting outside diameter of cylindrical manufacturing parts with GO/NO GO gauge.

• GO ring gauge are built with minus tolerance while NO GO ring gage plus tolerance.

2439. In a fit system there are two ways of representing a system, hole basis and shaft basis. Which of the following statements hold TRUE for either of them?

(a) In hole basis system, dimension of shaft is considered to be datum
(b) Dimension of shaft cannot be easily controlled by standard manufacturing processes
(c) In shaft basis system, dimension of hole is considered to be datum
(d) In hole basis system, dimension of hole is considered to be datum

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : In hole basis system the size of hole remains constants and it is the diameter of the shaft that is varied to determine the fit.

In a shaft basis system, the size of shaft remain constant and the hole diameter is varied to determine the fit.

2440. For the hole of basic size 60 mm with upper deviation of 0.046 mm, lower deviation of 0 mm and for the shaft of basic size 60 mm with upper deviation of -0.010 mm, lower deviation of -0.029 mm, state the correct type of fit which is suitable for the mating hole and shaft.

(a) Forced fit (b) ransition fit
(c) Interference fit (d) Clearance fit

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : Hole lower size – shaft upper size

= (60–0) – (60 – 0.029)

= + 0.029

If deviation is positive then fit is clearance fit.

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2441. Which of the following instruments uses mechanical-cum-optical means for magnification?

(a) Johansson Mikrokator
(b) Talysurf Surface Roughness tester
(c) Profilometer
(d) Tomlinson Surface meter

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : Tomlinson surface meter uses mechanical-cum-optical means for magnification.
The vertical movement of the probe caused by the surface irregularities makes the horizontal lapped cylinder to roll This rolling of lapped cylinder causes the movement of the arm.

2442. Cross staff is essentially an equipment used for

(a) setting out right angles
(b) measuring horizontal distance only
(c) measuring vertical angles of 20 degrees only
(d) taking oblique offsets

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (a) : The cross-staff is an instrument used to measure angles and altitudes consisting a trigonometrically graduated staff and one or more perpendicular vanes moving over it.

2443. Choose the correct surface roughness symbol for the value of 0.25 to 1.6 (R(a). (Symbols and notations carry their usual meaning)

(a) ∇ (b) $\nabla\nabla$
(c) $\nabla\nabla\nabla\nabla$ (d) $\nabla\nabla\nabla$

Jharkhand Urja Vikas Nigam Ltd. AE 2017

Ans. (d) : Surface roughness symbol for the value of 0.25 to 1.6 (R(a) arithmetic average is $\nabla\nabla\nabla$.

2444. A gauge block is checked for it's taperness using interferometer and the wavelength of the light is 0.4 'μm'. The number of fringes obtain are ten and five when the job is fitted by 180°. The taperness is :

(a) 0.4 μm (b) 0.5 μm
(c) 0.8 μm (d) 0.6 μm

Kerala PSC Poly. Lect., 2017

Ans. (b) : Taperness along the length can be considered as error in its parallelism which is given by

$$= \frac{n_2 - n_1}{4} \times \lambda$$

$$= \frac{10 - 5}{4} \times 0.4$$

$$= 0.5 \ \mu m$$

2445. The most accurate surface roughness measuring instrument is

(a) Perthometer
(b) Light Section Microscope
(c) Tomlinson Surface Tester
(d) Profile projector

Kerala PSC Poly. Lect., 2017

Ans. (a) : The most accurate surface roughness measuring instrument is perthometer.

2446. _____ is a means by which the tool designer ensures that the part will fit into the tool only in its correct position.

(a) Clamping (b) Foolproofing
(c) Locating (d) None of the above

GPSC Asstt. Prof. (Prod.) 21.08.2016

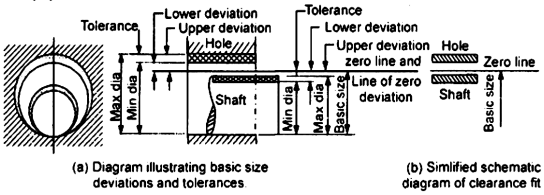
Ans. (b) : Foolproofing
It is a method which ensures the correct loading of components. It ensures that the component will fit into work holder only in its correct position.

2447. The algebraic difference between maximum limit and minimum limit of size is called :

(a) Clearance (b) Allowance
(c) Tolerance (d) Maximum deviation

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (c) :



(a) Diagram illustrating basic size deviations and tolerances. (b) Simlified schematic diagram of clearance fit

Allowance–It is a minimum clearance or maximum interference between mating parts.
Deviation–The algebraic difference between a size (actual, maximum etc.) and corresponding basic size.
Tolerance–It is the zone bounded by the two limits of size of the parts and defined by its magnitude that is tolerance and by its position in relation to the zero line.
Tolerance = Maximum limit – Minimum Limit.
Clearance–It is the positive difference between the hole size and shaft size.

2448. The tool employed to measure the shaft run out is the :

(a) Feeler gauge (b) Micrometer
(c) Dial gauge (d) Caliper

Kerala PSC Poly. Lect. (Auto) 05.05.2016

Ans. (c)

2449. The algebraic difference between the maximum limit and basic size is called :

(a) upper deviation (b) lower deviation
(c) actual deviation (d) mean deviation

Kerala PSC IOF 19.04.2016

Ans. (a) The algebraic difference between the maximum limit and basic size is called upper deviation.

2450. Which one of the following tolerances set on inner diameter and out diameter respectively of headed jig bush for press fit is correct?

(a) G7h6 (b) H7h6
(c) F7h6 (d) F7j6

GWSSB DEE 07.07.2016, UPRVUNL AE 21.08.2016

Ans. (b) :

2451. Which one of following steps would lead to interchangeability?

(a) Quality control (b) Operator training
(c) Process planning (d) Product design

GWSSB DEE 07.07.2016

Ans. (a) :

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2452. GO-NO GO gauges are used for inspection of:
(a) variables
(b) attributes
(c) both variables and attributes
(d) none of these

(HPPSC LECT. 2016)

Ans : (b) Go - No Go gauges are used for inspection of attributes.
* Go gauges are designed at maximum material limit condition and NO GO gauges are design at minimum material limit condition.
* Go gauges are designed to check shape as well size. And has to be in the full form.

2453. Vernier caliper gauge improves the :
(a) measuring sensitivity (b) accuracy
(c) repeatability (d) resolution

HPPSC W.S. Poly. 2016

Ans : (a) Vernier caliper gauge improves the measuring sensitivity.
Vernier caliper is used to measure external as well as internal diameter, diameters of shaft thickness of parts, depth of slots and holes, to an accuracy of 0.02 mm.

2454. The ratchet mechanism in a micrometer screw gauge serves to :
(a) Check wear out
(b) ensure a uniform measuring force
(c) eliminate play
(d) use it as a snap gauge

HPPSC W.S. Poly. 2016

Ans : (b) The ratchet mechanism in a micrometer screw gauge serves to ensure a uniform measuring force.

2455. All the working surfaces and the cylindrical surfaces of the rollers of sine bar have a surface finish of the order of :
(a) 0.2 micron (b) 0.5 micron
(c) 1 micron (d) 5 micron

HPPSC W.S. Poly. 2016

Ans : (a) All the working surfaces and the cylindrical surfaces of the rollers of sine bars have a surface finish of the order of 0.2 micron
Sine bar:- It is used either to measure angles more precisely than a bevel protractor or for locating any work to given angle within very close limits. It is generally used with slip gauge.

2456. The difference between the upper limit and lower limit of a dimension is known as :
(a) Basic size (b) Nominal size
(c) Tolerance (d) Actual size

HPPSC W.S. Poly. 2016

Ans : (c) Tolerance:- It is the difference between the upper limit and lower limit of a dimension. The tolerance may be unilateral or bilateral
Unilateral System of tolerance:- $20^{+0.004}_{-0.000}$
Bilateral System of tolerance:- $20^{+0.002}_{-0.002}$

2457. Resistance wire strain gauge works on the principle
(a) that resistance changes in proportion to strain on material
(b) the resistance of wire changes with load
(c) that conductivity is directly proportional to load on member
(d) due to elongation length increases and diameter reduces thereby resistance changes

RPSC AE 2016

Ans : (d) Resistance wire strain gauge works on the principle due to elongation length increases and diameter reduce by resistance changes.

2458. Thickness of light gauge sheet steel can be best checked with a
(a) finely divided steel scale
(b) depth gauge
(c) micrometer
(d) thickness measuring machine fitted with dial gauge

RPSC AE 2016

Ans : (c) Thickness of light gauge sheet steel can be best checked with a micrometer

2459. Which of the following is used as GO & NO GO gauge in measurement?
(a) Slip gauge (b) Snap gauge
(c) Angle gauge (d) Sprit level

UJVNL AE 2016

Ans : (b) Snap gauge used as Go and No Go gauge in measurement.
Go - No Go gauges are used for inspection of attributes.
* Go gauges are designed at maximum material limit condition and NO GO gauges are design at minimum material limit condition.
* Go gauges are designed to check shape as well size. And has to be in the full form.

2460. Flatness of a surface can be measured by
(a) Profile projector
(b) Slip gauges
(c) Coordinate measuring machine
(d) Talysurf

UJVNL AE 2016

Ans : (c) Flatness of a surface can be measured by co-ordinate measuring machine. Co-ordinate measuring machine is device which measured the all dimension, roughness, flatness, groove angle etc.

2461. One micron is equal to
(a) 0.001 mm (b) 0.001 mm
(c) 0.001 A° (d) None of the above

UJVNL AE 2016

Ans : (b) 1 micron = 0.001 mm

2462. The ratio of standard deviation and square root of number of observation is called
(a) RMS value (b) Variance
(c) Optimum value (d) Standard error

UJVNL AE 2016

Ans : (d) Standard error is the ratio of standard deviation and square root of number of observation.

2463. Which instrument is used to measure the inclination of a plane surface precisely?
(a) Snap gauge (b) Sine bar
(c) Angle plate (d) 1 and 2 both

UJVNL AE 2016

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Ans : (b) Sine bar is used to measure the inclination of a plane surface precisely.
Sine bar used either to measure angles more precisely than a bevel protractor or for location any work to give angle within very close limits.

2464. ..... micrometer consists of two micrometer heads installed in parallel in one frame and cab adjusted to go and not go dimensions of the gauge.
(a) Disc type (b) Limit
(c) Point (d) Screw thread
HPPSC AE (PWD) 10.09.2016

Ans. (b) : Limit micrometers can be used as GO/NO-GO gauge by setting the upper and lower limits for measurements. The measuring faces of these limit micrometers are carbide tipped.

2465. The optical instrument used for the measurement of small angular differences, changes or deflection, plane surface inspection etc. is.....
(a) Auto-collimator
(b) Optical transfer calipers
(c) Johansson Mikrokator
(d) Sine bar
HPPSC AE (PWD) 10.09.2016

Ans. (a) : Auto-collimator is an optical instrument used for the measurement of small angular differences. For small angular measurements it provides a very sensitive and accurate approach. It is essentially an infinity telescope and a collimator combined into one instrument.

2466. The stylus probe instrument used for surface finish is:
(a) Clinometer
(b) Profilometer
(c) Differential comparator
(d) None of the above
HPPSC AE (PWD) 10.09.2016

Ans. (b) : A Profilometer is a measuring instrument used to measure a surface profile in order to quantify its roughness critical dimensions as step curvature, flatness are computed from the surface topography.

2467. In representing the surface texture, the following symbols are used to represents
II, ⊥, X, M, C, R
(a) Flaws (b) Surface texture
(c) Lag (d) Waviness
(e) None of these
CGPSC AE 16.10.2016

Ans. (e) : In surface texture, symbols are represented as,
= → Parallel to plane of projection of view in which symbol is used.
⊥ → Perpendicular to plane
X → Crossed in two oblique
M → Multi directional
C → Approx circular relative
R → Approx radial relative to centre surface

2468. CLA value for Honing process is :
(a) 0.05- 1.0 (b) 0.05- 3.0
(c) 0.025- 0.1 (d) 6
UPRVUNL AE 21.08.2016

Ans. : (a) CLA value for honing process is 0.05- 1.0

2469. Match List-I with List-II and select the correct answer using the codes given below the list:

List-I (Measuring device)	**List-II (Parameter measured)**
A. Optical flat	1. Measurement of very small displacements
B. Laser scan micrometer	2. Measurment of gear pitch
C. Diffraction grafting	3. On-line measurment of moving parts
D. Auto collinatiors	4. Smaller angular deviation on long flat surfaces
	5. Surface texture using interferometry Interferometry

(a) A-5, B-3, C-1, D-4 (b) A-3, B-5, C-4, D-1
(c) A-5, B-4, C-1, D-2 (d) A-3, B-5, C-1, D-2
UPRVUNL AE 21.08.2016

Ans. : (a)

Meausuring device	**Parameter measured**
Optical flat	Surface texutre using interferometry
Laser scan micrometer moving	On-line measurement of part
Diffraction grafting	Measurement of very small displacement
Auto collinatiors	Smaller angular deviation on long flat surface

2470. The diameter of the hole at the gauge point while for making connection for various devices for pressure measurement should be about
(a) 3 mm to 6 mm (b) 1 mm to 3 mm
(c) 6 mm to 9 mm (d) 9mm to 12 mm
APPSC AE Subordinate Service Civil/Mech. 2016

Ans. (a) : The diameter of the hole at the gauge point while for making connection for various devices for pressure measurement should be about 3mm to 6 mm.

2471. Maximum Material Condition for a shaft is
(a) higher limit of size (b) lower limit of size
(c) basic size (d) none of these
GPSC Lect. 23.10.2016

Ans. (a) : Maximum material condition, is the maximum limit of an external feature eg. a shaft manufactured to its high limits would contain maximum amount of material, for shaft is higher limit of size.

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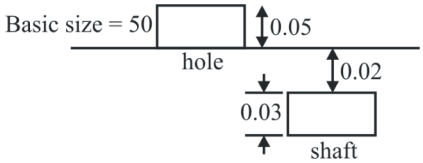
2472. A hole is specified as $\phi\ 50^{(+0.050/-0.000)}$ mm. The mating shaft has a clearance fit with minimum clearance of 0.02 mm. The tolerance on the shaft is 0.03 mm. Maximum clearance between hole and shaft is

(a) 0.100 mm (b) 0.030 mm
(c) 0.080 mm (d) 0.070 mm

ISRO Scientist/Engineer 03.07.2016

Ans. (a) :



Maximum clearance between hole and shaft
= max. hole size – min shaft size
= (50 + 0.05) – (50 – 0.02 – 0.03)
= 0.1 mm

2773. A wire has a mass (0.3 ± 0.003) g, radius (0.5 ± 0.005) mm and length (6 ± 0.06) cm. The maximum percentage error in the measurement of density is

(a) 1 (b) 2
(c) 3 (d) 4

ISRO Scientist/Engineer 03.07.2016

Ans. (d) : Mass = (0.3 ± 0.003) g
Radius = (0.5 ± 0.005) mm
Length = (6 ± 0.06) cm

So, $d_{min} = 2(0.5 - 0.005) = 0.99$ mm
$d_{max} = 2(0.5 + 0.005) = 1.01$ mm
$L_{min} = 6 - 0.06 = 5.94$ cm
$L_{max} = 6 + 0.06 = 6.06$ cm

Density $\rho_{max} = \dfrac{m_{max}}{V_{min}}$

$$= \frac{0.3 + 0.003}{\frac{\pi}{4}(d_{min})^2 \times L_{min}}$$

$$= \frac{4 \times 0.3003}{\pi \times (0.99)^2 \times 5.94}$$

$= 0.06627$ g/cm^3

$$\rho_{min} = \frac{m_{min}}{V_{max}} = \frac{0.3 - 0.003}{\frac{\pi}{4}(1.01)^2 \times 6.06}$$

$= 0.06117$ g/cm^3

$$\rho_{actual} = \frac{m}{V} = \frac{0.3}{\frac{\pi}{4}(1)^2 \times 6} = 0.06366 \text{ g/cm}^3$$

$$\% \text{ error} = \frac{0.06366 - 0.06117}{0.06366}$$

$= 0.039 = 3.9\%$

2774. The case with which observations can be made accurately is referred to as

(a) Readability (b) Sensibility
(c) Accuracy (d) Precision
(e) None of the above

CGPSC Poly. Lect. 22.05.2016

Ans. (a) : The case with which observations can be made accurately is referred to as readability.

2475. The accuracy of a limit gauge can be checked by

(a) Sine bar (b) Micrometer
(c) Slip gauge (d) Vernier calipers

APPSC AEE Screening Test 2016

Ans. (c) : Slip gauge can measures a finer tolerance limit.
Slip gauges are very precisely manufactured for measuring, setting up of sizes and also to calibrate precision measuring instruments or inspection gauges. They can be used to measure tolerances in the range of 0.001 to 0.005 mm very accurately. These are used as standard unit length. Therefore, slip gauges are used for checking the accuracy of the limit gauge.

2476. In hole basis system, basic size is equal to

(a) maximum hole size
(b) Lower deviation is zero
(c) upper deviation is zero
(d) minimum hole size

APPSC AEE MAINS 2016, PAPER-III

Ans. (d) : In hole basis system [Lower deviation of hole is zero], basic size is equal to minimum hole size.
In shaft basis system [Upper deviation of shaft is zero] basic size is equal to maximum size of shaft.

2477. In India, the work for standardization, of production has been mainly done by

(a) IPC (b) SI
(c) ISI (d) UPSC

BPSC Poly. Lect. 2016

Ans : (c) In India, the work for Standardization of production has been mainly done by ISI.

2478. Consider the following statements:

(i) Mechanical comparators are used for higher accuracy.
(ii) Optical comparators use both optical and mechanical means to get magnification.
(iii) Pneumatic comparators are used for very high magnification.
(iv) Dial indicator is the most widely used mechanical comparator.

(a) (ii), (iii) and (iv) are true
(b) (iii) are true
(c) (i), (ii) and (iii) are true
(d) (i) and (ii) are true

UPPSC AE 12.04.2016 Paper-I

Ans : (a)
(i) Optical comparator used both optical and mechanical means to get magnification.
(ii) Pneumatic comparators are used for very high magnification
(iii) Dial indicator is the most widely used mechanical comparator.

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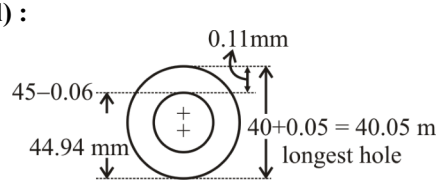
2479. A bush was turned after mounting the same on a mandrel. The mandrel diameter, in millimeters, is $45^{+0.000}_{-0.06}$ and bore diameter of bush is $45^{+0.05}_{-0.00}$. The maximum eccentricity of the bush, in mm, will be:

(a) 0.01 (b) 0.1
(c) 0.5 (d) 0.055
(e) 0.11

CGPSC Asstt. Workshop Supt., 17.07.2016

Ans. (d) :



Diametrical clearance = 0.11 mm

$$\text{Eccentricity} = \frac{\text{D.C.}}{2} = \frac{0.11}{2} = 0.055\,\text{mm}$$

2480. Resolution of a measuring instruments is :

(a) Retardation of response
(b) Difference between true value and input signal
(c) Range of the accuracy which can be in tolerance limit
(d) Smallest change in the measure and that can be measured.

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : Resolution of a measuring instruments is the smallest change in the measure and that can be measured.

2481. What is the purpose of tachometer?

(a) Distance measurement
(b) Angular velocity measurement
(c) Linear velocity measurement
(d) Stresses due to temperature measurement

GPSC Lect. (Auto) 16.10.2016
SJVN ET 2013

Ans. (b) : Tachometer is used for angular velocity measurement.

2482. The conformity of the output to the true value of the measurement is called

(a) precision (b) sensitivity
(c) accuracy (d) none of the above

GPSC Lect. (Auto) 16.10.2016
GPSC Lect. 23.10.2016

Ans. (c) : Accuracy – Conformity of the output to the true value.
Precision – It represents degree of repetitiveness
Sensitivity – Change of output which can be measured when small change happens in input.

2483. Metric micrometers have precision of :

(a) 0.1 mm (b) 0.02 mm
(c) 0.05 mm (d) 0.01 mm

GPSC Lect. (Auto) 16.10.2016

Ans. (d) : 0.01 mm

2484. As per I.S.O., roughness symbol of '∇∇' represents 'Ra' values (in micron) in the range of

(a) 0.025 to 0.1 (b) 0.2 to 0.8
(c) 1.6 to 6.3 (d) 12.5 to 25

GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (c) :

Roughness (μm)	Roughness Grade number	Roughness symbol
6.3	N9	∇∇
3.2	N8	
1.6	N7	

2485. Which of the following is a comparator?

(a) Dial indicator (b) Vernier caliper
(c) Micrometer (d) None of the above

Gujarat PSC AE 2019
GPSC Asstt. Prof. (Prod.) 21.08.2016

Ans. (a) : Dial indicator is a kind of comparator used to measure small distance and angles accurately, used for calibration of machine.

2486. Go gauge is made for _____ limit to inspect hole size .

(a) Maximum material
(b) Minimum material
(c) Intermediate material
(d) Optimum material

RPSC ADE 2016

Ans. (a) : Hole size is measured by plug gauge. It should be used in GO-NOGO combination. The purpose of the Go gauge is to determine whether the hole has met the low limit of the specification. Go gauge measure maximum material limit for plug gauge.

2487. Errors generally distributed in accordance with the Gaussian distribution are :

(a) Controllable errors (b) Calibration errors
(c) Avoidable errors (d) Random errors

RPSC ADE 2016

Ans. (d) : With the Gaussian distribution or normal distribution the errors are generally distributed in accordance to random error.

2488. A discrepancy is :

(a) The difference between a measurement and true value of the quantity measured
(b) The difference between true value of the quantity and error
(c) The difference between measured value and actual value
(d) The difference between measured values of the same quantity

RPSC ADE 2016

Ans. (d) : The difference between measured values of the same quantity.
A discrepancy is the difference between measured value of the same quantity.

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2489. A piezoelectric crystal can be used to measure:
(a) Temperature (b) Velocity
(c) Acceleration (d) Flow

RPSC ADE 2016

Ans. (c) : A piezoelectric crystal can be used to measure acceleration.

2490. A hole size $30.00^{+0.03}_{-0}$ mm. The corresponding shaft is of size $30.00^{+0.08}_{+0.04}$ mm. The resulting assembly has
(a) Interference fit (b) Transition fit
(c) Clearance fit (d) None of these

RPSC VPITI 14.02.2016

Ans. (a) : Upper limit of hole-Lower limit of shaft
$= 0.03 - (+0.04)$
$= -0.01$
Lower limit of hole- Upper limit of shaft
$= -0 - (+0.08)$
$= -0.08$
This value in negative so fit is interference fit.
or
As we can see clearly upper limit of hole is less than lower limit of shaft so this implies interference fit.

2491. Which one of the following instruments is used for conducting alignment tests?
(a) Strain gauge (b) Dial gauge
(c) Dynamometer (d) Tachometer

RPSC VPITI 14.02.2016

Ans. (b) : To test this the feeler of the dial gauge rests on the face of the locating spindle shoulder and dial gauge holder is clamped to the bed. The locating cylinder is then rotated and the change in reading noted down.

2492. Interferometry is used to measure
(a) Straightness (b) Flatness
(c) Roundness (d) Angularity

RPSC VPITI 14.02.2016

Ans. (b) : Interferometry is used to measure flatness, small displacement and surface irregularities.

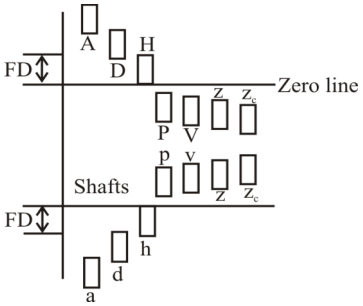
Parallelism error $= \frac{(n_2 - n_1)}{4} \times \lambda$

Where, n = no of fringes
λ = Wavelength

2493. What type of Fit is indicated by H_7/p_6 ?
(a) easy running Fit (b) Clearance Fit
(c) Transition Fit (d) Interference Fit

GPSC Asstt. Prof. 28.08.2016

Ans. (d) : Limits and fits

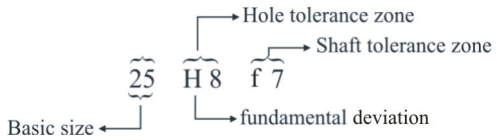


So, H_7/p_6 indicated, interference fits.

2494. The 'fit' made of a 25 mm H-hole with tolerance grade IT8 and a 25 mm f-shaft with tolerance grade IT7 is represented by:
(a) 25 O7-f8 (b) 25 F7-g8
(c) 25 O8-h7 (d) 25 H8-f7

HPPSC Lect. (Auto) 23.04.2016

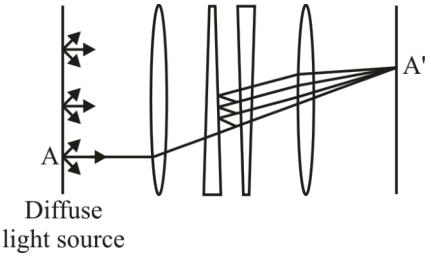
Ans. (d) :



2495. Fabry-Petrot Interferometer consists of two optical flats coated with high efficiency…. film on the two facing surfaces.
(a) Dark
(b) Semi-transparent
(c) Opaque and black
(d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : Fabry-Petrot Interferometer (FPI)–
In optics, a FPI or etalon is an optical cavity mode from two parallel reflecting surfaces (i.e. thin mirrors). Optical waves can pass through the optical cavity only when they are in resonance with it.
FPI are widely used in telecommunication, lasers and Spectro scopy to control and measure the wavelength of the light.



The heart of FPI is a pair of partially reflective glass optical flats spaced micrometer to centimeter aparts, with the reflective surfaces facing each other. The flats in an interferometer are often made in a wedge shape to prevent the rear surfaces from producing interference fringes the rear surfaces often also have an anti-reflective contains.

2496. Microkater are used :
(a) As comparator
(b) As slip micro gauge
(c) For force measurement
(d) As hardness tester

HPPSC Lect. (Auto) 23.04.2016

Ans. (a) : Microkater are used as comparator A Johansson microkator is a mechanical comparator used to obtain mechanical magnification of the difference in length as compared to a standard.

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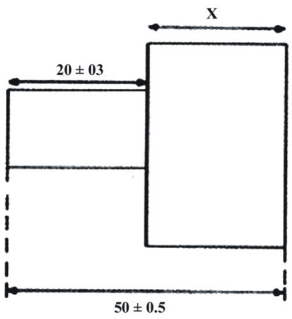
2497.



the tolerance of 'x' in the given figure.

(a) 30 ± 0.2 (b) 30 ± 0.5
(c) 30 ± 0.3 (d) 30 ± 0.8

TSPSC AEE 2015

Ans. (d) : For equal bilateral tolerances, the tolerance of the gap.

$$t_x = \sum_{all} t = 0.3 + 0.5 = 0.8$$

$x = 30 \pm 0.8$ mm

2498. The term allowance in limits and fits is usually referred by

(a) Minimum clearance between shaft and hole
(b) Maximum clearance between shaft and hole
(c) Difference of tolerance at shaft and hole
(d) Difference between maximum and minimum size of hole

TSPSC AEE 2015

Ans. (a) : The term allowance in limits and fits is usually referred by minimum clearance between shaft and hole.

2499. Change of resistance with intensity of light takes place in

(a) Photo-voltaic cell (b) Photo-emissive cell
(c) Photo-conductive cell (d) None of these

(KPSC AE 2015)

Ans : (c) Change of resistance with intensity of light takes place in photo conductive cell.

2500. The diameter of a round bar with close tolerance is quickly measured by

(a) External micro meter (b) Vernier caliper
(c) Slip gauge (d) snap gauge

(KPSC AE 2015)

Ans : (d) The diameter of a round bar with close to clearance is quickly measured by snap gauge.
(ii) Snap gauge is used to check the external dimension.

2501. Which of the following thermocouples can measure temperature in a comparatively higher range?

(a) Chromel -alumel (b) Iron-constantan
(c) Iridum-rhodium (d) Platinum-rhodium

(KPSC AE 2015)

Ans : (c) Iridium- rhodium thermocouples measure temperature in a comparatively higher range.

2502. The difference between the maximum material limits of mating parts is known as

(a) Deviation (b) Clearance
(c) Allowance (d) Tolerance

(KPSC AE 2015)

Ans : (c) The difference between the maximum material limit of mating parts is known as allowance.
Allowance:- It is the difference between the basic dimensions of the mating parts. The allowance may be positive or negative when the shaft size is less then hole size, then the allowance is positive and when the shaft size is greater than hole size, then the allowance is negative.

2503. All the thread characteristics can be measured precisely with

(a) Thread gauge
(b) Screw pitch gauge
(c) Micrometer with anvil
(d) Tool room microscope

(KPSC AE 2015)

Ans : (d) All the thread characteristic measured precisely with tool room microscope.

2504. Mechanical strain gauges can measure

(a) static and dynamic strains
(b) dynamic strains only
(c) static strains only
(d) static and quasi-static strains

(KPSC AE 2015)

Ans : (d) Mechanical strain gauge measure static and quasi-static strains.
Strain measurement:-
There is a wide variety of strain measuring techniques such as photo- elasticity, interferometry, holography and strain gauges.
Operation of resistance strain gauges is based on the principle that electrical resistance of a conductor changes when the resistance element is strained by an external force. the resistance change is measured and correlated to strain or the physical effect causing strain. for a material strained within elastic limits.

2505. Acute angle attachment is available in

(a) Taper plug gauge (b) Sine bar
(c) Angle gauges (d) Bevel protractor

TSPSC AEE 2015

Ans : (d) Acute angle attachment is available in bevel protractor.
Bevel protractor:- It is also called vernier bevel protractor. It is used for measuring and testing angles, within the limit of 5 minuts ($^1/_{12}$ of a degree)

2506. Auto collimator uses

(a) white light (b) monochromatic light
(c) mercury light (d) sodium vapour light

TSPSC AEE 2015

Ans : (a) Auto collimator uses white light.

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2507. Roughness grade symbol "two triangles" corresponds to the roughness grade number

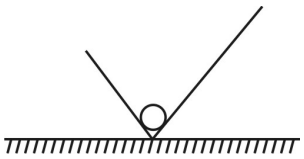
(a) three (b) four
(c) five (d) eight

TSPSC AEE 2015

Ans : (d) Roughness grade symbol "two triangle" corresponds to the roughness grade number eight. is less surface finishing.

Roughness symbol	Finish
~	Very rough
∇	Coarse
∇∇	Smooth
∇∇∇	fine
∇∇∇∇	super fine.

2508. State the meaning of following machining symbol.



(a) the surface is machined
(b) removal of material by machining is required
(c) removal of material is not permitted
(d) obtained by any production method

APSC Poly. Lect. 13.03.2020
TSPSC AEE 2015

Ans : (c) Removal of material is not permitted.

2509. According to Indian Standard specification 50 H7/g6 means that the

(a) actual size is 50 mm.
(b) basic size is 50 mm.
(c) difference between actual size and basic size is 50 mm.
(d) max. size is 50 mm.

TSPSC AEE 2015

Ans : (b) According to Indian standard specification 50 H7/g6 means that the basic size in 50 mm.

2510. As per Indian standard, the slip gauges according to accuracy are graded into

(a) 2 categories (b) 4 categories
(c) 5 categories (d) 3 categories

GPSC AE 26.07.2015

Ans. (d) :

2511. The accuracy of micrometer, callipers, dial indicators can be checked by :

(a) Feeler gauge (b) Slip gauge
(c) Ring gauge (d) Plug gauge

KPSC ADF 2015

Ans. (b) : Slip gauges are the blocks made up of metal and are made with a high degree of precision. Slip gauges are available in all sizes. Slip gauge are used for dial indicators, etc.

2512. A hole is of dimension shaft $\phi 9^{+0.015}_{+0}\,mm$. The corresponding shaft is of dimension $\phi 9^{+0.010}_{+0.001}\,mm$. The resulting assembly has :

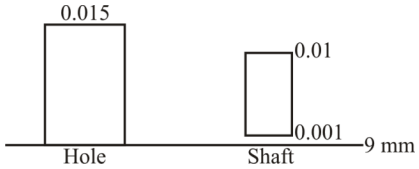
(a) loose running fit (b) close running fit
(c) transition fit. (d) interference fit.

BPSC Asstt. Prof. 29.11.2015

Ans. (c) : Given,

Hole size = $\phi 9^{+0.015}_{+0.000}\,mm$

Shaft size = $\phi 9^{+0.010}_{+0.001}\,mm$



Maximum clearance = maximum hole size – minimum shaft size
= Upper limit of hole – lower limit of shaft
= 9.015 – 9.001
= 0.0014 mm

Maximum (I) = minimum hole size – maximum shaft size
= 9 – 9.010
= –0.01 mm

Since maximum clearance is positive and maximum interference is negative. Hence the assembly is transition fit.

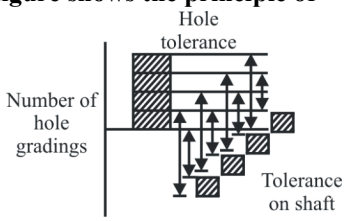
2513. In the tolerance specification 25D6, the letter 'D' represents

(a) Grade of tolerance (b) Upper deviation
(c) Lower deviation (d) Type of fit

GPSC DEE, Class-2 (GWSSB) 04.07.2021
MPPSC AE 08.11.2015

Ans. (c) : In the tolerance specification 25 D6, the letter 'D' represents lower deviation. 6 represents grade of tolerance.

2514. The figure shows the principle of



(a) traceability (b) interchangeability
(c) matched fits (d) selective assembly

ISRO Scientist/Engineer 11.10.2015

Ans. (d) : Selective Assembly is a cost-effective approach for reducing the overall variation and thus improving the quality of an assembled product. In this process, components of a mating pair are measured and grouped into several classes (bins) as they are manufactured. So, the figure shows the principle of selective assembly.

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2515. Circular shapes appear in this fashion when viewed at an angle other than 90 degrees

(a) Circular (b) Elliptical
(c) Lengthened (d) Angular

ISRO Scientist/Engineer 11.10.2015

Ans. (b) : Circular shapes appear in elliptical fashion when viewed at an angle other than 90 degrees.

2516. Consider the following Instruments

1. Profilometer
2. The Tomlinson surface meter
3. The Taylor Hobson Talysurf
4. Profilograph

Out of the above opinion which instruments are used to measure surface finish

(a) 1 and 2 (b) 1, 3 and 4
(c) 1 and 3 (d) 1, 2, 3 and 4
(e) 1, 2 and 4

CGPSC AE 26.04.2015 Shift-I

Ans. (d) : Method of measuring surface roughness
(i) Profilometer
(ii) Tomlinson surface tester
(iii) The Taylor-Hobson Talysurf
(iv) Profilograph
(v) Light cross-section method

2517. Ring and gap gauges are used to check the

(a) hole (b) key
(c) shaft (d) spline
(e) angle

CGPSC AE 26.04.2015 Shift-I

Ans. (c) : Ring gauge, is a cylindrical ring of a thermally stable material, often steel, whose inside diameter is finished to gauge tolerance and is used for checking the external diameter of a cylindrical object.

2518. Tolerance of gauges can be checked by

(a) optical flat (b) sine bar
(c) auto collimator (d) spirit level

TSPSC AEE 2015

Ans. (a) : Tolerance of gauges can be checked by optical flat.

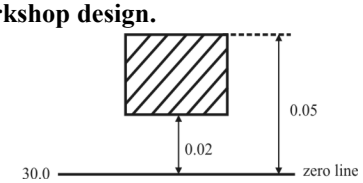
2519. According to Taylor's principle "No Go" gauge checks

(a) Only one feature at a time
(b) Only important dimensions at a time
(c) All the dimensions at a time
(d) Only related dimensions at a time

TSPSC Managers, 2015

Ans. (a) : According to Taylor's principle "No Go" gauge checks only one feature at a time.

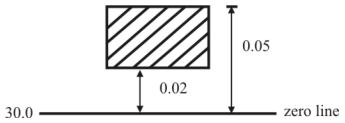
2520. Tolerance of hole are shown in figure. Maximum size of Go gauge according to workshop design.



(a) 30.02 (b) 30.023
(c) 30.017 (d) 30.053

TSPSC Managers, 2015

Ans. (b)



Lower limit of hole = 30.02
Maximum size of GO gauge
= Lower limit + 10% of Tolerance
= 30.02 + 10% of 0.03
= 30.023

2521. A dimension expressed as 20 ± 0.03 is the case of :

(a) bilateral tolerance
(b) unilateral tolerance
(c) accuracy limiting tolerance
(d) any of these

VIZAG STEEL MT 2015

Ans. (a) : $20^{\pm 0.03}$ – Bilateral tolerance
$20^{+0.00}_{+0.03}$ – Unilateral tolerance

2522. Which of the following is not the part of products planning competitive strategy?

(a) Technology leadership
(b) Cost leadership
(c) Customer focus
(d) Endurance

VIZAG STEEL MT 2015

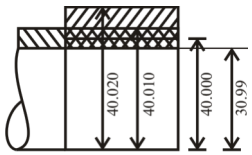
Ans. (d) : Endurance

2523. Hole = $40^{+0.020}_{+0.000}$, Shaft = $40^{+0.010}_{-0.010}$. This assembly will result in-

(a) Clearance fit (b) Interference fit
(c) Transition fit (d) Drive fit

NPCIL ET, 2015

Ans. (c) : Hole = $40^{+0.020}_{+0.000}$ Shaft = $40^{+0.010}_{-0.010}$



Transition fit

2524. In three wire inspection of external thread of M16*2, the measurement was found to be 16.455 mm. The effective pitch diameter of thread is-

(a) 14.723 (b) 16.705
(c) 17.122 (d) 17.455

NPCIL ET, 2015

Ans. (a) : P = 2

$$\text{Best wire size} = \frac{P}{2}\sec 30°$$

$$dw = \frac{2}{2} \times \frac{2}{\sqrt{3}}$$

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dw = 1.155 mm

m = 16.445 mm

$$\text{Effective diameter} = m - \left[dw + \frac{P}{2}\tan\alpha \right]$$

$$= 16.445 - (1.155 + 1 \times \tan 30)$$

$$= 14.723 \text{ mm}$$

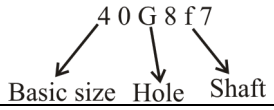
2525. Assembly is designated by 40G8f7. The position of tolerance of hole is indicated by-

(a) Letter G (b) Letter f

(c) Number 7 (d) Number 8

NPCIL ET, 2015

Ans. (d) :



2526. Which of the following instruments has minimum least count?

(a) Vernier Calliper (b) Micro meter

(c) Dial indicator (d) Vernier micrometer

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : Least count- The least count of an instrument is the smallest measurement that can be taken accurately with it.

Vernier calipers	–	0.02 mm
Micrometer	–	0.01 mm
Dial indicator	–	0.01 – 0.001
Vernier micrometer	–	0.005

2527. Load cell is essentially a

(a) Strain gauge (b) Photovoltaic cell

(c) Thermistor (d) Pressure gauge

GPSC Lect. (Poly.) 07.12.2014

Ans. (a) : Load cell – A load cell is a force transducer if converts a force such as tension compression pressure or torque into a electrical signal that can be measured and standarised. As the force applied to the load cell increases, the electrical signal changes proportionally. The most common type of load cell used are

Strain gauge

Pneumatic

Hydraulic

2528. The least count of the Vernier level protractor is

(a) 10 min (b) 1°

(c) 5° (d) 5 min

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : The least count of the Vernier level protractor is 5 minutes.

2529. _____ of an instrument is the ability of an instrument to respond to small changes of the stimulus.

(a) Resolution (b) Precision

(c) Discrimination (d) Sensitivity

UPRVUNL AE 2014

Ans : (c) Discrimination– It is the ability of the instrument to respond to small changes of the stimulus. It is defined as the largest change in a stimulus that produces not detectable change in response of the measuring instrument.

Resolution– It is defined as the smallest measurable input change. Thus, the smallest increment in input which can be detected with certainty by an instrument is its resolution or discrimination.

Precision– The most repeatable value (or) the reproducible value out of the set of records is known as precision. It is the measure of consistency. It gives the closeness of individual measured value to the average of all measured values.

Sensitivity– The ratio of the change of instrument indication to the change of quantity being measured is termed as sensitivity.

2530. _____ is the property of a measuring instrument, whereby its metrological properties remain constant over time.

(a) Resolution (b) Precision

(c) Discrimination (d) Stability

UPRVUNL AE 2014

Ans : (d) Stability is the property of a measuring instrument, where by its metrological properties remain constant overtime.

2531. H_7g_7 is

(a) clearance fit (b) interference fit

(c) shrinkage fit (d) transition fit

TNPSC AE 2014

Ans. (a) : H_7g_7 is a clearance fit

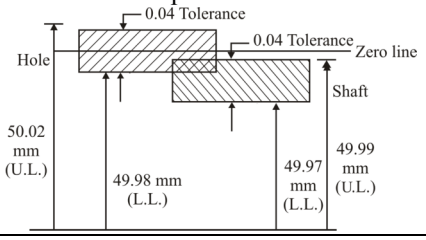
2532. Hole dimension is 50.00 ± 0.02 mm and Shaft dimension is $50.00^{-0.01}_{-0.03}$ mm. The fit is

(a) clearance (b) interference

(c) transition (d) shrinkage

TNPSC AE 2014

Ans. (c) : Transition Fit → Transition fit, which may provide either a clearance or on interference depending upon the actual value of individual tolerance of the mating parts. In this case, the tolerance zones of the hole and the shaft overlap.



2533. θ, Taper angle of dead centre measured in a sine bar is

where h – height of slip gauges

L – length of sine bar

(a) $\theta = \sin^{-1} h / L$ (b) $\theta = \sin^{-2} h / L$

(c) $\theta = \sin^{-1} L / h$ (d) $\theta = \sin^{-1} h / 2L$

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Ans. (A) : $\theta = \sin^{-1}\left[\frac{h}{L}\right]$

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2534. The cross-sectional area of slip gauges of above 10 mm is m mm^2

(a) 30 × 5 (b) 30 × 9
(c) 35 × 9 (d) 40 × 10

TNPSC AE 2014

Ans. (c) : 35 × 9

2535. Composite error of gear is measured by

(a) base tangent comparator
(b) Double Vernier
(c) Gear tooth caliper
(d) Parkinson rolling gear tester

TNPSC AE 2014

Ans. (d) : Composite error of gear is measured by Parkinson rolling gear tester.

2536. The following is the standard press fit for easy dismantling of ferrous and non-ferrous parts assembly

(a) H_7g_7 (b) H_7h_6
(c) H_7n_6 (d) H_7p_5

TNPSC AE 2014

Ans. (d) : H_7p_5

2537. H7/g6 is a

(a) Transition fit (b) Sliding fit
(c) Interference fit (d) Force fit

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Sliding fit

2538. The type of gauge used to measure holes is

(a) Snap gauges (b) Plug gauges
(c) Ring gauges (d) Strain gauges

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (b) : Plug gauges

2539. The symbol used to represent flatness is

(a) – (b) ⊥
(c) // (d) ▱

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (d) : ▱

2540. The instrument used to accurately measure small angular deviations on a flat surface is

(a) Straight edge (b) Dial indicator
(c) Auto collimator (d) Optical flat

Kerala LBS Centre For Sci. & Tech. Asstt. Prof. 2014

Ans. (c) : Auto collimator

2541. The following surface roughness parameter is expressed in percentage

(a) R_a (b) R_p
(c) R_q (d) t_p

TNPSC AE 2014

Ans. (d) : R_a – Arithmetical mean deviation
R_p – Highest peak
R_q – Root mean square roughness
t_p – the difference in the profile bearing ratio [Expressed as a percentage] between the upper and lower t_p % value.

2542. RMS value of surface finish, R_q is obtained from the following equation:

(a) $R_q = \frac{1}{L}\int_0^L |y|dx$ (b) $R_q = \left[\frac{1}{L}\int_0^L |y|dx\right]^{1/2}$

(c) $R_q = \frac{1}{L}\int_0^L y^2dx$ (d) $R_q = \left[\frac{1}{L}\int_0^L y^2dx\right]^{1/2}$

HPPSC Asstt. Prof. 2014

Ans. (d) : RMS (Root means square) (R_q)

$$R_q = \sqrt{\frac{\int_0^L y^2dx}{L}}$$

$$R_q = \left[\int_0^L \frac{y^2dx}{L}\right]^{1/2} = \left[\frac{1}{L}\int_0^L y^2dx\right]^{1/2}$$

2543. The thickness of oil film at the surface of slip gauges is of the order of :

(a) 0.005 micron (b) 0.1 micron
(c) 1 micron (d) 10 micron

UPRVUNL AE 2014

Ans : (a) The thickness of oil film at the surface of slip gauges is of the order of 0.005 micron.

2544. Which of the following is not a typical example of interference fit?

(a) Shrunk on couplings (b) Iron tyres
(c) Cross head slides (d) Crank pins

UPRVUNL AE 2014

Ans. : (c) An interference fit is a fit where there will always be overlap in the joint between the specified mating shaft/hole even at minimum material condition values allowed by shaft and maximum value allowed by the hole tolerance value.

2545. Dimension of shaft is $100^{+0.01}_{-0.00}$ and corresponding Hole diameter is $100^{+0.02}_{-0.01}$. The maximum material condition of the fit is

(a) Shaft diameter: 100.01 and hole diameter: 99.99
(b) Shaft diameter: 100.00 and hole diameter: 99.99
(c) Shaft diameter: 100.01 and hole diameter: 100.02
(d) Shaft diameter: 100.005 and hole diameter: 100.015

ISRO Scientist/Engineer 24.05.2014

Ans. (a) :

Shaft, — $100^{+0.01}_{-0.00}$

In hole — $100^{+0.02}_{-0.01}$

"Maximum material condition refers to a feature of size that contains the greatest amount of material, but yet remains within its tolerance zone.

i.e. Largest shaft diameter.

Smallest hole size.

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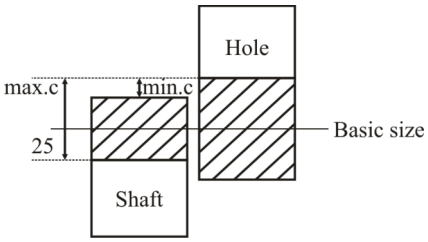
Largest or upper limit of shaft	= 100 + 0.01
	= 100.01
Smallest or lower Limit of hole	= 100 - 0.01
	= 99.99

2546. In a interchangeable assembly, shafts of size $25.00^{(+0.040/-0.0100)}$ mm mate with holes of size $25.000^{(+0.020/-0.000)}$ mm. The maximum possible clearance in the assembly will be

(a) 10 μm (b) 20 μm
(c) 30 μm (d) 60 μm

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (d) :



Max. clearance (c) = UL of hole – LL of shaft
Min. clearance = LL of hole – UL of shaft
UL of hole = 25.020 mm
LL of shaft = 24.96 mm
Maximum clearance = 25.020 – 24.96 = 0.060 mm
= 60 μm

2547. GO and NO-GO plug gauges are to be designed for a hole of $20.000^{-0.050/-0.010}$ mm. Gauge tolerances can be taken as 10% of the hole tolerance. Following ISO system of gauge design, sizes of GO and NO-GO gauge will be respectively

(a) 20.010 mm and 20.050 mm
(b) 20.014 mm and 20.046 mm
(c) 20.006 mm ad 20.054 mm
(d) 20.014 mm and 20.054 mm

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : UL of hole = 20 + 0.050 = 20.050 mm
LL of hole = 20 + 0.010 = 20.010 mm
Tolerance of hole = UL – LL = 20.050 – 20.010 = 0.040
Wear allowance = 10% of tolerance = 0.1 × 0.04 = 0.004
Size of Go gauge = LL of hole + wear allowance
Size of Go gauge = 20.010 + 0.004 = 20.014 mm
Size of No Go gauge = UL of hole + wear allowance
Size of No Go gauge = 20.050 – 0.004 = 20.046 mm

2548. A thin coat of _____ is applied on slip gauges to wring properly :

(a) Water (b) Silicone
(c) Air speed (d) Density

TNPSC AE (Industries) 09.06.2013

Ans. (b) : A thin coat of silicone is applied on slip gauges to wring properly.

2549. A Vernier scale consists of 25 divisions on 12 mm spacing and the main scale has 24 divisions on 12 mm spacing. What is the leat count ?

(a) 0.01 mm (b) 0.02 mm
(c) 0.03 mm (d) 0.04 mm

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

2550. Ionization gauges are used to measure :

(a) Higher pressures (b) Lower pressures
(c) Sound (d) Stress

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

2551. Millimeter scale in a micrometer is marked on

(a) anvil (b) barrel
(c) thimble (d) spindle

TNPSC AE 2013

Ans. (b) : Total reading = MSR + (HSR × LC)
MSR – Main scale reading
HSR – Head scale reading
LC – Least count
L.C. = 0.01 mm

2552. Eden-Rolt comparator is a popular instrument for calibration of

(a) slip gauges (b) vernier
(c) micrometer (d) sine bar

TNPSC AE 2013

Ans. (a) : Eden-Rolt comparator is a popular instrument for calibration of slip gauges.

2553. Expressing a dimension as 32.5/32.3 is case of

(a) Unilateral tolerance (b) Bilateral tolerance
(c) Limiting dimensions (d) All of above

SJVN ET 2013

Ans. (c) : Expressing a dimension as 32.5/32.3 is the case of limiting dimensions.

2554. The flow error in case of normal nozzles is of the order of :

(a) 0.1% (b) 0.2%
(c) 0.5% (d) 1%

SJVN ET 2013

Ans. (d) : The flow error in case of normal nozzles is of the order of 1%.

2555. Sine bars inherently become increasingly impractical and inaccurate as the angle exceeds

(a) 15° (b) 30°
(c) 60° (d) 45°

TNPSC AE 2013

Ans. (d) : Sine bars are more accurate for angles between 15° to 45°, below and above to this limit, angles measured are not much more accurate.

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If a sine bar is to be accurate then the following properties must exist:

1. The distance between centres (L) must be precisely known.
2. The axes of rollers must be parallel to each other.
3. The upper surface of the sine bar must be flat and parallel.
4. The rollers must be of identical diameters and round to within a close tolerance.

2556. The maximum amount by which the result differs from the true value is called

(a) accuracy (b) correction
(c) precision (d) discrepancy

TNPSC AE 2013

Ans. (a) : The maximum amount by which the result differs from the true value is called accuracy.

2557. Which one of the following is not a direct surface roughness measuring instrument?

(a) Tomlinson Surface meter
(b) Taylor-Hobson Talysurf
(c) Wallace Surface Dynamometer
(d) Profilometer

TNPSC AE 2013

Ans. (c) : Wallace Surface Dynamometer is not a direct surface roughness measuring instrument**.**

2558. Which of the following is not a basic form of slip gauges?

(a) Rectangular
(b) Square with centre hole
(c) Circular with centre hole
(d) Square without centre hole

TNPSC AE 2013

Ans. (c) : Circular with centre hole is not a basic form of slip gauges.

2559. The error committed by a person in the measurement are

(a) Gross Errors
(b) Random Errors
(c) Instrumental Errors
(d) Environmental Errors

TNPSC AE 2013

Ans. (a) : The error committed by a person in the measurement are gross errors.

2560. The behavior between the time that the input value changes and the time that the value given by the transducer settles down to the steady state value is known as

(a) Static Characteristics
(b) Design Characteristics
(c) Steady State Characteristics
(d) Dynamic Characteristics

TNPSC AE 2013

Ans. (d) : The behavior between the time that the input value changes and the time that the value given by the transducer settles down to the steady state value is known as dynamic characteristics.

2561. The dead zone in a pyrometer is 0.125% of the span, the instrument is calibrated from 1000ºC to 1800ºC. What temperature change must occur before it can be detected?

(a) 2.5ºC (b) 1.25ºC
(c) 3ºC (d) 4ºC

TNPSC AE 2013

Ans. (b) : Span $= 1800 - 800$
$= 1000^{o}C$

$$\text{Dead zone} = \frac{0.125}{100} \times 1000$$
$$= 1.25^{o}C$$

2562. Limit imposed by the firm itself of its environment on the decision maker are called

(a) parameters (b) constraints
(c) constructs (d) perimeters

VIZAG STEEL MT 18.08.2013

Ans. (b) : Constraints

2563. Optical flats are used in conjunction with

(a) angular measurement
(b) surface flats
(c) surface parallelism
(d) interferometrc measurement

VIZAG STEEL MT 18.08.2013

Ans. (b) : Optical flat– An optical flats is a precisely palished flat surface, used as a reference against which the flatness of an unknown surface may be compared. The flatness of an optical flat is measured in traction of a reference wavelength, 632.8 mm.

2564. The ratio of "Theoretical minimum number of parts" to "Actual number of part" is called

(a) part count ratio (b) assembly efficiency
(c) independent axon (d) zero defect

VIZAG STEEL MT 18.08.2013

Ans. (a) : $\text{Parts count ratio} = \frac{\text{Theoretical minimum no. of parts}}{\text{Actual number of parts}}$

2565. Taylor's principle is concerned with

(a) Pneumatic comparators
(b) Interferometric measurements
(c) Gauging measurements
(d) Angular measurement

SJVN ET 2013

Ans. (c) : Taylor's principle is concerned with gauging measurements.

2566. Standardization deals with the characteristics of product that include:-

(a) Its dimensions
(b) Method of testing the product
(c) Composition and properties of its material
(d) All of the above

UKPSC AE-2013, Paper-I

Ans. (d) : All of the above

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2567. Pneumatic comparators work on following theory:-

(a) Newton's theory (b) Bernouli's theory
(c) Pascal's theory (d) Legendre's theory

UKPSC AE-2013, Paper-I

Ans. (b) Pneumatic comparators wok on Bernouli's theory.

2568. When the dimension is expressed as $20^{+0.035}_{-0.025}$ mm then the tolerance is

(a) 0.035 mm (b) 0.025 mm
(c) 0.01 mm (d) 0.06 mm

APPSC AEE 2012

Ans. (d) : Tolerance = Upper limit – Lower limit

Dimension expressed $20^{+0.035}_{-0.025}$

Upper limit = 20 + 0.035 = 20.035

Lower limit = 20 – 0.025 = 19975

$\therefore$ Tolerance = 20.035 – 19.975 = 0.06

2569. To measure _____ strain, strain rosettes are used.

(a) Linear (b) Shear
(c) Volumetric (d) Any of the above

APPSC AEE 2012

Ans. (a) : Strain rosettes are generally used for longitudinal strain or linear strain.

2570. The number of slip gauge in a set is

(a) 87 (b) 48
(c) 103 (d) all of these

TNPSC ACF 2012

Ans. (d) :

2571. The thickness of oil film at the surface slip gauge is of the order of

(a) 0.005 micron (b) 0.1 micron
(c) 1 micron (d) 10 micron

TNPSC ACF 2012

Ans. (a) : The thickness of oil film at the surface slip gauge is of the order of 0.005 minutes.

2572. The maximum number of faces in precision polygon is

(a) 6 (b) 12
(c) 20 (d) 72

TNPSC ACF 2012

Ans. (d) : The maximum number of facts in precision polygon is72.

2573. Millimetre in scale in a micrometer is marked on

(a) barrel (b) thimble
(c) spindle (d) anvil

TNPSC ACF 2012

Ans. (a) : Millimeter is scalene a micrometer is marted on barrel.

2574. The micrometer thimble is graduated with 50 divisions and 2 revolutions of the thimble are required to move 1 mm, the degree of precision obtained is

(a) 1 mm (b) 0.01 mm
(c) 0.001 mm (d) none

TNPSC ACF 2012

Ans. (b) :

2575. Proving ring are used to measure

(a) displacement (b) force
(c) velocity (d) acceleration

TNPSC ACF 2012

Ans. (b) : Proving ring are used to measure force.

2576. The best size wire for 150 metre thread for measuring pitch diameters of screw thread in terms of P is

(a) 0.5 P (b) 0.75 P
(c) 0.5773 P (d) 0.6 P

TNPSC ACF 2012

Ans. (c) : for best wire size $d_P = \frac{P}{2}\sec\theta$

for metric thread $\theta = \frac{60}{2} = 30$

$d_P = \frac{P}{2} \times \sec 30$

$= \frac{P}{2} \times \frac{2}{\sqrt{3}} = \frac{P}{\sqrt{3}} = 0.5733\ P$

2577. Which of the following errors are regularly repetitive in nature?

(a) Systematic error (b) Random error
(c) Controllable error (d) Avoidable error

TNPSC ACF 2012

Ans. (a) : Systematic error are that are not determined by by change but are introduced by an inaccuracy inherent to the system.

Random Error- They are unpredictable and can't be replicated by repeating the experiment again.

Controllable Errors – These errors are controllable in there magnitude as well as sense which can be determined and reduced. Controllable error either have a constant value or changing value according to definite law.

2578. A sprit-level is required to have 2 mm of bubble movement of 2 inch inclination. The radius of the tube is

(a) 2 m (b) 80 m
(c) 120 m (d) 200 m

TNPSC ACF 2012

Ans. (d) :

2579. The basic unit in angular measurement is

(a) degree (b) minute
(c) second (d) right angle

TNPSC ACF 2012

Ans. (d) : Basic unit in angular measurement is right angle.

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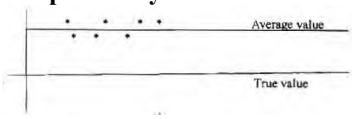
2580. The figure shows the dimensions obtained on a component by a certain instrument-



(a) precise but not accurate
(b) accurate but not precise
(c) both accurate and precise
(d) neither precise nor accurate

TNPSC ACF 2012

Ans. (a) : In given diagram shows precise but not accurate.

2581. Surface gauge is used for

(a) leveling the surface plate
(b) checking the surface finish
(c) laying out the work accurately
(d) checking the outer diameter

TNPSC ACF 2012

Ans. (c) : Surface gauge is a dimensionless measuring tool made of steel, generally used to transfer measurement to work place by scrubbing oretchery of line. It is also used to indicate the accuracy or parallelism of surfaces.

2582. Work is usually required to be held in a vertical position for laying out. For this purpose it is clamped to a/an

(a) surface plate (b) angle plate
(c) V block (d) machine bed

TNPSC ACF 2012

Ans. (b) : Work is usually required to be held in vertical position for laying out for this purpose it is clamped to an angle brute.

2583. According to accuracy, slip gauges are classed under ……. number of accuracy classes.

(a) two (b) three
(c) five (d) seven

TNPSC ACF 2012

Ans. (c) : According to accuracy slip gauge are closed under free number of accuracy classes.

2584. Which one of the following methods can be used for forecasting the sales potential of a new product?

(a) Time series analysis
(b) Jury of executive opinion method
(c) Sales force composite method
(d) Direct survey method

TNPSC ACF 2012

Ans. (d) : Survey Method is one of the most common and direct methods of forecasting demand in the short term. This method it encompasses the future purchase plane of consumer and their intentions.

2585. A hole is specified as $40^{0.050}_{0.000}$ mm. The tolerance on the shaft is 0.04 mm. The maximum clearance in mm between the hole and the shaft is

(a) 0.04 (b) 0.05
(c) 0.10 (d) 0.11

VIZAG STEEL MT 10.06.2012

Ans. (c) : Minimum Clearance = Minimum hole size – max. shaft size.
∴ Max shaft size = 40.00 – 0.01 =39.99 mm
Tolerance on shaft = 0.04 mm
Min. shaft size = 39.99 – 0.04 = 39.95
Max. Clearance = max. hole size – min. shaft size
Max. Clearance = 40.05 - 39.95 = 0.10

2586. Go gauges are meant to check

(a) one feature (b) multiple features
(c) wear (d) none of these

TRB Poly. Lect., 2012

Ans. (b) : Go gauge is designed to check maximum metal condition i.e. MML (Maximum Material Limit) & minimum clearance or maximum interference.

2587. Angle dekkor is a type of

(a) clinometer (b) auto-collimator
(c) sine bar (d) interferometer

TRB Poly. Lect., 2012

Ans. (b) : Angle dekkor is a type of Auto-collimator, There is an illuminated Scale in the focal plane of the collimating lens It is capable of measuring small variation in angular setting i.e, determining angular tilt.

2588. Comparators are used to

(a) measure dimensions
(b) measure dimensional accuracy
(c) zero errors
(d) composite errors

TRB Poly. Lect., 2012

Ans. (b) : Comparator is another type of linear measuring method, which is quick and more convenient for checking large number of identical dimensions, It can not measure absolute dimensions but can only compare two highly reliable dimension.

2589. Interference occurs when light rays are

(a) obtained from two different sources and their travel path lengths differ by odd number of half wavelengths
(b) obtained from a single source and their travel path lengths differ by odd number of half wavelengths
(c) obtained from a single source and their travel path lengths differ by even number of half wavelengths
(d) obtained from two different sources and travel path lengths differ by even number of half wavelengths .

TRB Poly. Lect., 2012

Ans. (b) : Interference occurs when light rays are obtained from a single source and their travel path length differ by odd number of half wave lengths.

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2590. Which of the following is not connected with surface finish?

(a) R_a (b) R_z

(c) R_{max} (d) R_n

TRB Poly. Lect., 2012

Ans. (d) : R_n is not conducted with surface finish here

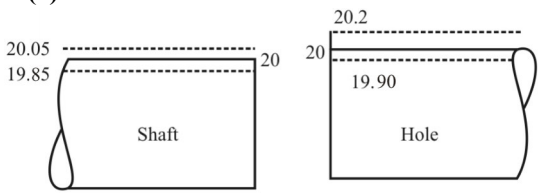
2591. A shaft of diameter $20^{+0.05}_{-0.15}$ and a hold of diameter $20^{+0.20}_{-0.10}$, when assembled would yield

(a) transition fit (b) interference fit

(c) clearance fit (d) none of these

TRB Asstt. Prof., 2012

Ans. (a) :



Clearance Fit – Lower limit hole is larger than upper limit of shaft.

Interference Fit- lower limit of shaft is larger than the upper limit of hole.

Transition Fit- In this type of fit upper limit of one of the component lies between upper and lower limit at the other component.

2592. Match the list I with list II and select the correct answer from the codes given in the lists below

List I (Measuring instruments)	List II (Applications)
A. Talysurf	1. T slots
B. Telescopic gauge	2. Flatness
C. Transfer Calipers	3.Internal Diameter
D. Autocollimator	4. Roughness

Codes :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	1	2
(c)	4	3	2	1
(d)	3	4	1	2

ISRO Scientist/Engineer 2012

Ans. (b)

2593. A shaft has a dimension $\phi 35^{-0.009/-0.025}$, the respective values of fundamental deviation and tolerance are

(a) -0.025, 0.008 (b) -0.0025, 0.016

(c) -0.009, 0.008 (d) -0.009, 0.016

AAI Jr. Executive 26.03.2021

ISRO Scientist/Engineer 2012

Ans. (d) : Shaft-

Dimension 35 - 0.009/-0.025

Basic size of shaft = 35

Upper Deviation (UD) = - 0.025 mm

Lower Deviation (LD) = - 0 .009 mm

(Fundamental Deviation) = -0.009

Tolerance = UL - LL

(BS + UD) - (BS - LD)

= -0.025 - (-0.009)

= - 0.016

2594. A comparator for its working depends on

(a) comparison with standard such as slip gauges

(b) accurately caliberated scale

(c) optical device

(d) limit gauge

UKPSC AE 2012 Paper-I

Ans. (a) : comparison with standard such as slip gauges

2595. Which of the following instruments is used to measure smoothness of a metallic surface ?

(a) Talysurf

(b) Coordinate Measuring Machine

(c) Profile Projector

(d) None of the above

UKPSC AE 2012 Paper-I

Ans. (a) : Talysurf

2596. Which one of the following measuring instruments is supposed to be most accurate

(a) micrometer (b) vernier caliper

(c) vernier dial gauge (d) optical projector

RPSC ACF-2011

Ans. (d) : Optical projector measuring instruments is supposed to be most accurate.

2597. Resistance wire strain gauge works on the principal :

(a) that resistance changes in proportion to strain on material

(b) the resistance of wire changes with load

(c) that conductivity is directly proportional to load on member

(d) due to elongation length increases and diameter reduces, thereby resistance changes

RPSC ACF-2011

Ans. (d) : When force is applied to any metallic wire its length increases due to the strain. The more is the applied force, more is the strain and more is the increase in length of the wire. L_1 is the initial length of the wire and L_2 is the final length after application of the force, the strain is given as :

$$\varepsilon = \frac{L_2 - L_1}{L_1}$$

2598. Tolerances are specified

(a) to obtain desired fits

(b) because it is not possible to manufacture a size exactly

(c) to obtain high accuracy

(d) to have proper allowance

RPSC ACF-2011

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Ans. (b) : Tolerance is specified on both the sides of the basic dimension. This usually have + and – tolerance is equal amount. Tolerances are provided to the parts because variations in the material properties introduce error and production machines themselves have inherent inaccuracies. An other reason to introduce tolerance is that it is not possible to make perfect setting by operator so some tolerance are provided.

2599. Which of the following is the most important characteristic of a measuring instrument?
(a) precision (b) accuracy
(c) repeatability (d) sensitivity

TNPSC ACF 2012
RPSC ACF-2011

Ans. (c) : Accuracy – It represent to true value.
Precision- It represents the degree of repetition.
Repeatability It is defined as the closeness of the agreement between independent test result obtained with same method on the same test material.

2600. Thickness of light gauge sheet steel can be best checked with a :
(a) finely divided steel scale
(b) depth gauge
(c) micrometer
(d) thickness measuring machine fitted with dial gauge

RPSC ACF-2011

Ans. (c) : Thickness of light gauge sheet steel can be best checked with a "micrometer".

2601. The term "traceability' in engineering metrology is concerned with :
(a) measuring machines
(b) standards
(c) pneumatic comparators
(d) optical instruments

RPSC ACF-2011

Ans. (b) : The term "traceability' in engineering metrology is concerned with standards.

2602. The diameter of finished turned shaft can best be checked with a
(a) combination set
(b) slip gauge
(c) height gauge
(d) micrometer screw gauge

RPSC ACF-2011

Ans. (d) : The diameter of finished turned shaft can best be checked with a micrometer screw gauge.

2603. It is desirable to handle the slip gauges with a cloth or chamois leather in order to :
(a) avoid injury to hands
(b) protect the surfaces of slip gauges
(c) insulate them from the heat of the hand
(d) ensure that the varnish applied on gauges does not come out

RPSC ACF-2011

Ans. (c) : Its desirable to handle the slip gauges with a cloth or chamois leather in order to insulate them from the heat of the hand.

2604. Polygons in metrology are concerned with :
(a) method of circular dividing
(b) testing of parallelism
(c) testing of circularity
(d) interferometer measurements

RPSC ACF-2011

Ans. (a) : Any precision polygon is usually calibrated by two autocollimeters before using, in order to know the errors. For this method, precision polygon is mounted on a table and this table is a rotating table.

2605. Universal surface gauge is used for :
(a) checking straightness
(b) checking flatness
(c) checking parallelism
(d) layout work and inspection

RPSC ACF-2011

Ans. (d) : Universal surface gauge has the following additional features when compared with fixed type surface gauge. Universal surface gauge can also be used an cylinder surface.

2606. Profilometer is an instrument used to measure
(a) gear involute (b) thread profile
(c) surface roughness (d) surface flatness

RPSC ACF-2011

Ans. (c) : A profilometer is instrument used to measure a surface's profile in order to quantity its roughness. Critical dimensions as step, curvature flatness are compared from the surface topography.

2607. Drilled holes and honed holes could be designated by which of the following grades?
(a) H_5, H_{11} (b) H_6, H_{10}
(c) H_8, H_6 (d) H_{10}, H_5

GPSC Lect. (Poly.) 07.12.2014
RPSC Lect. (Tech. Edu. Dept.) 2011

Ans. (d) : Drilled holes and honed holes could be designated by grades H_{10}, H_5
Note- No answer is given by commission.

2608. CLA value and RMS values are used for measurement of
(a) Metal hardness
(b) Sharpness of tool edge
(c) Surface dimensions
(d) Surface roughness

ISRO Scientist/Engineer 2011

Ans. (d) : CLA value and RMS values are used for measurement of surface roughness.

2609. A shaft and hole pair is designated as 50H7d8. This assembly constitutes
(a) Interference fit (b) Transition fit
(c) Clearance fit (d) None of the above

ISRO Scientist/Engineer 2011

Ans. (c) : 50H7d8 belongs to clearance fit.

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2610. What are dial gauges or dial test indicators used for?

(a) Measuring known angles or to find any work to a given angle.
(b) Checking flatness of the surfaces and parallelism of the bars and rods.
(c) Checking the linear displacement of the tool.
(d) Checking the thickness of the chip.

VIZAG STEEL MT 2011

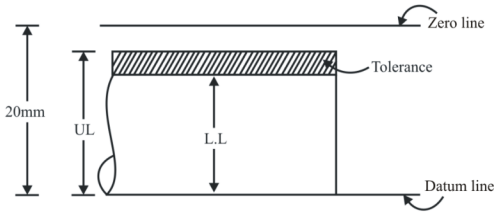
Ans. (c) :

2611. The diameter of a solid shaft is given as $20^{-0.02}_{-0.05} mm$, the value of tolerance over diameter is:

(a) –0.07 (b) 0.03
(c) –0.03 (d) 0.07

UPSC JWM Advt. No.-52/2010

Ans. (b) : Explanation- Solid shaft Dia $\rightarrow 20^{-0.02}_{-0.05} mm$



Upper limit of shaft = 20.00 – 0.02 = 19.98 mm (UL)
Lower limit of shaft = 20.00 – 0.05 = 19.95 mm (LL)
Tolerance = UL – LL
= 19.98 – 19.95 = 0.03 mm

2612. Consider the following

1. Length
2. Angle
3. Position
4. Profile
5. Surface finish

Which one of the following combinations engineering metrology quantifies ?

(a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 1, 2, 3 and 5 only (d) 1, 2, 3, 4 and 5

UPSC JWM Advt. No.-52/2010

Ans. (d) : 1, 2, 3, 4 and 5

2613. A systematic error in judgment is a

(a) Conservative error
(b) Compensating error
(c) Bias error
(d) Random error

UPSC JWM Advt. No.-52/2010

Ans. (c) : Bias error

2614. The vernier reading should not be taken at its face value before an actual check has been taken for

(a) Zero error
(b) Its calibration
(c) Flatness of measuring jaws
(d) Temperature equalization

ISRO Scientist/Engineer 2010

Ans. (a) : The Vernier reading should not be taken at its face value before an actual check has been taken for zero error, because if we take face value before an actual check, then there may be probability of error, which will give wrong readings in the measuring of our experiments.

2615. A vibro meter

(a) Indicates vibration amplitude
(b) Indicates vibration frequency
(c) Indicates nature of vibrations
(d) only indicates second and above harmonics

ISRO Scientist/Engineer 2010

Ans. (a) : <u>Vibro meter</u>
Indicates vibration amplitude "To and fro periodic motions about their equilibrium position."

2616. Optical square is

(a) Engineer's square having stock and blade set at 90º
(b) A constant deviation prism having the angle of deviation between the incident ray and reflected ray, equal to 90º
(c) A constant deviation prism having the angle of deviation between the incident ray and reflected ray, equal to 45 º
(d) Used to produce interference fringes

GPSC Executive Engg. 23.12.2018
ISRO Scientist/Engineer 2010

Ans. (c) : Optical square is a constant deviation prism having the angle of deviation between the incident ray and reflected ray, equal to 45º.

2617. A master gauge is

(a) A new gauge
(b) An international reference standard
(c) A standard gauge for checking accuracy of gauges used on shop floors
(d) A gauge used by experienced technicians

ISRO Scientist/Engineer 2010

Ans. (c) : A master gauge is a standard gauge for checking accuracy of gauges used on shop floors.

2618. Dimension of the hole is $50^{+0.02}_{-0.00}$ and shaft is $50^{+0.02}_{+0.00}$ mm. The minimum clearance is

(a) 0.02mm (b) 0.00mm
(c) -0.02mm (d) 0.01mm

ISRO Scientist/Engineer 2010

Ans. (c) : Minimum clearance = minimum size of hole - maximum size of shaft
= 50.00 - 50.02 = - 0.02 mm

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2619. Expressing a dimension as $25.3^{\pm 0.05}$ mm is the case of

(a) Unilateral tolerance (b) Bilateral tolerance
(c) Limiting dimensions (d) All of the above

ISRO Scientist/Engineer 2010

Ans. (b) : Unilateral Tolerance

A case may occur when one of the limits **coincides** with the basic size.

e.g. $\phi_0^{+0.20}, 25_{-0.10}^{0}$

Bilateral Limits

A case occur when the maximum limit is above and the minimum limit is below the basic size.

e.g. $\phi^{\pm 0.04}$ OR $25.3^{\pm 0.05}$

Basic size = 25.3 mm
Upper limit = 25.30 + 0.05 = 25.35 mm
Lower limit = 25.30 - 0.05 = 25.25 mm
Tolerance = 25.35 - 25.25 = 00.10 mm

2620. In comparison to an open-loop system a closed loop system is

(a) More stable (b) More accurate
(c) More complex (d) All of the above

ISRO Scientist/Engineer 2010

Ans. (d) : In comparison to an open-loop system a closed loop system is more stable, more accurate and more complex.

2621. Which one of the following instruments is used for conducting alignment tests?

(a) Strain gauge (b) Dial gauge
(c) Dynamometer (d) Tachometer

UPSC JWM Adv. No-16/2009

Ans. (b) : The various equipment used for performing alignment tests are.

- Test mandrels
- Straight edges and squares
- Spirit level
- Auto - collimator
- Dial gauges
- Waviness meter

2622. Which one of the following models would be most sensitive to large random fluctuations in the data?

(a) Simple exponential smoothing with alpha $\alpha = 0.1$
(b) Unweighted moving average of 2 periods
(c) Weighted moving average of 10 periods
(d) Time –series regression

UPSC JWM Adv. No-16/2009

Ans. (d) : Time-series regression models is most sensitive to large random fluctuations in the data.

2623. The method of least squares provide a unique line which 'best fits' the past data points. It is the best fit because it minimizes the :

(a) Maximum error
(b) Sum of the errors
(c) Sum of the squares of the error
(d) Slope of the least –squares line.

UPSC JWM Adv. No-16/2009

Ans. (c) : Sum of the squares of the error.

2624. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I (Therblig Symbol)	List-II (Description)
(A)	1. Prolonged grasp of the object
(B)	2. Locate an article
(C)	3. Releasing the objects at the desired place
(D)	4. Taking hold of the bject
	5. Putting parts together.

Code :

	A	B	C	D
(a)	3	1	4	2
(b)	2	4	5	3
(c)	3	4	1	2
(d)	1	5	4	3

UPSC JWM Adv. No-16/2009

Ans. (b) :

Therblig Symbol		Description
(A)	→	Locate an article
(B)	→	Taking hold of the object
(C)	→	Putting part together
(D)	→	Releasing the objects at the desired place.

2625. Consider the following statements :

1. Possibility of placing repeat orders
2. Reoder point
3. Nature of demand
4. Lead time

On which of these, the economic lot size is dependent?

(a) 1 only (b) 1 and 3
(c) 2 and 3 (d) 3 and 4

UPSC JWM Adv. No-16/2009

Ans. (c) : The economic lot size is dependent on

- Reorder point
- Nature of demand

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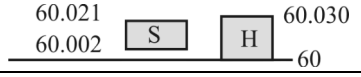
2626. The type of fit between the hole size $60^{+0.030}_{+0.000}$ and the shaft of size $60^{+0.021}_{+0.002}$ is :

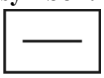
(a) selective fit (b) clearance fit
(c) transition fit (d) interference fit

DRDO Scientists 2009

Ans. (c) : Transition fit



2627. The geometric characteristic representing the symbol :



(a) flatness
(b) symmetry
(c) straightness
(d) high value of skewness

DRDO Scientists 2009

Ans. (c) : straightness

2628. The criterion for a good bearing surface is :

(a) negative value of skewness
(b) positive value of skewness
(c) zero skewness
(d) high value of skewness

DRDO Scientists 2009

Ans. (a) : Negative value of skew is the criterion for a good bearing surface.

2629. An irregularity which occurs on a surface at one place of a machined job or at relatively infrequent or widely varying intervals is known as what?

(a) Roughness (b) Waviness
(c) Flaw (d) Lay

UPSC JWM Adv. No-16/2009

Ans. (c) : Flaw is an irregularity which occurs on a surface at one place of a machined job or at relatively infrequent or widely varying intervals.

2630. Match List I correctly with List II and select your answer using the codes given below :

	List I		List II
(a)	Sensitivity	1.	Ability to reproduce
(b)	Accuracy	2.	Change in output per change in input
(c)	Precision	3.	Change in input for a change in output from a non-zero point
(d)	Resolution	4.	Deviation from theoretical value

	a	b	c	d
(a)	1	2	3	4
(b)	2	4	1	3
(c)	3	4	1	2
(d)	4	1	2	3

TNPSC AE, 2008

Ans. (c) : Sensitivity – Change in input for a change in output from a non-zero point
Accuracy – Deviation from theoretical value
Precision – Ability to reproduce
Resolution – Change in output per change in input

2631. The accessory of slip gauges is

(a) scribing and centre points
(b) measuring jaws
(c) holder
(d) base and (a), (b) and (c)

TNPSC AE, 2008

Ans. (d) : Slip gauge accessories are measuring jaws, holder, base and scribing and centre points.
Field of application of slip gauges is extended by using these accessories.

2632. Scribing block is used to

(a) hold the round bars during marking
(b) check the trueness of flat surfaces
(c) locate the centres of round bars
(d) check the surface roughness

TNPSC AE, 2008

Ans. (b) : Scribing block is used to check the trueness of flat surfaces and scribing block is a gauge consisting of a scriber mounted an on adjustable stand.

2633. Method concerned with surface finish measurement is

(a) ultrasonic method
(b) field emission method
(c) critical angle of attack method
(d) all of these

TNPSC AE, 2008

Ans. (d) : Method concerned with surface finish measurement are ultrasonic method, field emission method and critical angle of attack method.

2634. Element of the indicating device carrying the scale is called

(a) dial
(b) transducer
(c) housing
(d) index

TNPSC AE, 2008

Ans. (a) : Element of the indicating device carrying the scale is called dial.

2635. Sensitivity and range of measuring instrument have

(a) direct relationship
(b) linear relationship
(c) inverse relationship
(d) none of these

TNPSC AE, 2008

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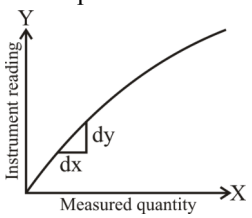
Ans. (c) : Sensitivity and range of measuring instrument have inverse relationship.



2636. Systematic errors are

(a) regular and repetitive
(b) randomly distributed
(c) distributed on either side of mean
(d) unpredictable

TNPSC AE, 2008

Ans. (a) : Systematic errors are usually caused by measuring instruments that are incorrectly calibrated or are used incorrectly. However, they can creep into your experiment from many sources, including consistent and repeatable.

2637. Basic shaft and basic hole are those whose upper deviations and lower deviations respectively are

(a) +ve, -ve (b) -ve, +ve
(c) Zero, Zero (d) None of the above

ISRO Scientist/Engineer 2008

Ans. (c) : Basic shaft and basic hole are those whose upper deviations and lower deviations respectively are zero, zero.

2638. Standards to be used for reference purposes in laboratories and workshops are termed as

(a) Primary standards
(b) Secondary standards
(c) Tertiary standards
(d) Working standards

ISRO Scientist/Engineer 2008

Ans. (c) : Standards to be used for reference purposes in laboratories and workshops are termed as tertiary standards.

2639. The following is true for "GO" and "NO GO" gauges:

(a) Plug gauge can measure the dimension of a hole
(b) Wear allowance is provided on "NO GO" end
(c) "GO" end is smaller than "NO GO" end
(d) Gauge tolerance is 10% of manufacturing tolerance

UKPSC AE 2007 Paper -I

Ans. (d) : Gauge tolerance is 10% of manufacturing tolerance.

2640. Micro-geometrical deviations of surface roughness are a series of repeated deviations of a wave with a ratio of pitch to height approximately equal to

(a) 50 (b) 100
(c) 500 (d) 1000

UKPSC AE 2007 Paper -I

Ans. (a) : 50

2641. The following is not a characteristic of optical comparator:

(a) High magnification
(b) Very few moving linkages
(c) Cheap
(d) Optical lever is weightless

UKPSC AE 2007 Paper -I

Ans. (c) : Cheap

2642. Limit Gauge

(a) checks whether the part is made within the specified tolerance
(b) measures the value of the upper specification limit
(c) measures the value of the lower specification limit
(d) measures both upper and lower specification limit

UKPSC AE 2007 Paper -I

Ans. (a) : Checks whether the part is made within the specified tolerance.

2643. If a quantity Q is dependent on three other quantities q_1, q_2 and q_3 related such that $Q = K \times (q_1)^{n1} \times (q_0)^{n2} \times (q_3)^{n3}$ then overall error $\frac{\delta Q}{Q} =$

(a) $n_1\left(\frac{\delta q_1}{q_1}\right)+n_2\left(\frac{\delta q_2}{q_2}\right)+n_3\left(\frac{\delta q_3}{q_3}\right)$

(b) $\frac{1}{n_1}\frac{\delta q_1}{q_1}+\frac{1}{n_2}\frac{\delta q_2}{q_2}+\frac{1}{n_3}\frac{\delta q_3}{q_3}$

(c) $\frac{\delta q_1}{q_1}+\frac{\delta q_2}{q_2}+\frac{\delta q_3}{q_3}$

(d) $\left(\frac{\delta q_1}{q_1}\right)^{n_1}+\left(\frac{\delta q_2}{q_2}\right)^{n_2}+\left(\frac{\delta q_3}{q_3}\right)^{n_3}$

ISRO Scientist/Engineer 2006

Ans. (a) : $Q = K\times(q_1)^{n1}\times(q_2)^{n2}\times(q_3)^{n3}$

Taking log both side,

$\ln Q = \ln k + n_1 \ln q_1 + n_2 \ln q_2 + n_3 \ln q_3$

[Differentiate both side]

$$\boxed{\frac{\delta Q}{Q}=n_1\left(\frac{\delta q_1}{q_1}\right)+n_2\left(\frac{\delta q_2}{q_2}\right)+n_3\left(\frac{\delta q_3}{q_3}\right)}$$

$\ln k = 0$, because 'k' is constant

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2644. Which of the following errors are generally distributed in accordance with the Gaussian distribution-

(a) Controllable errors (b) Calibration errors
(c) Avoidable errors (d) Random errors

ISRO Scientist/Engineer 2006

Ans. (d) : Random errors are those which are accidental and whose magnitude and sign cannot be predicted from knowledge of measuring system and conditions. Random errors have possitive and negative values and their magnitudes are generally distrubuted in accordance with Gaussian distribution of measurement.

2645. A threaded nut M16, ISO metric type, having 2 mm with a pitch diameter of 14.701 mm is to be checked for its pitch diameter using two or three numbers of balls or rollers of the following series

(a) Rollers of 2 mm ϕ
(b) Rollers of 1.155 mm ϕ
(c) Balls of 2 mm ϕ
(d) Balls of 1.155 mm ϕ

Vizag Steel MT (Re-Exam) 24.11.2013

Ans. (b) : In the given problem, isometric thread m 16 is given.
In case of threads always rollers (wires) are used.
Pitch (P) = 2 mm
Pitch diameter = 14.701 mm
In case of metric thread $2\theta = 60°$
$\theta = 30°$
For a best size wire,

$$d = \frac{P}{2}\sec\theta = \frac{2}{2}\sec 30° = 1.155\,mm$$

2646. The best size wire for measuring the effective diameter diameter of thread is of diameter

(a) P/2 sec θ (b) P/2 cos θ
(c) P sec θ (d) P sec θ

TNPSC ACF 2012

Ans. (a) : Best wire size is $d = \frac{P}{2}\sec\theta$
for iso metric thread $\theta = 60°$

2647. Effective Diameter of screw thread can be measured using

(a) Bench micrometer
(b) Two- wire methods
(c) Floating carriage micrometer
(d) Vernier caliper

PPSC Asstt. Municipal Engineer 15.06.2021

Ans. (b) : Effective Diameter of screw thread can be measured using two- wire methods.

2648. While measuring the major diameter of an external thread, a 35.5 mm diameter plain plug gauge is used as standard. The micrometer readings over the plug gauge and thread are 9.376 mm and 11.876 mm. The major diameter of the external thread is _____.

(a) 38 mm (b) 39 mm
(c) 40 mm (d) 37 mm

VIZAG MT, 14.12.2020

Ans. (a) : The major diameter $D = S \pm (R_1 - R_2)$
S = Size of setting gauge
R_1 = Micrometer reading over setting gauge
R_2 = Micrometer reading over thread
$D = S \pm (R_1 - R_2)$
$D = 35.5 + (11.876 - 9.376)$
$\boxed{D = 38\,mm}$

2649. Taylor's Principle states that Go Gauge should always be designed, so that it will cover the _____metal condition of as many dimensions as possible in the same limit gauge.

(a) Mean (b) Maximum
(c) Minimum (d) Median

MPSC HOD (Govt. Poly. Colleges) 04.10.2014

Ans. (b) : Taylor's principle states that Go Gauge should always be designed so that it will cover the maximum metal condition of as many dimensions as possible in the same limit gauge.

2650. The peak to valley surface roughness in a machining operation is 32 μm. Assuming that the surface roughness profile is triangular, the center line average surface roughness is

(a) 4 μm (b) 8 μm
(c) 16 μm (d) 32 μm

DRDO Scientists 2008

Ans. (b) : Peak to valley surface roughness
$(H_{max}) = 32\ \mu m$
Average surface roughness

$$= \frac{H_{max}}{4} = \frac{32}{4} = 8\mu m$$

2651. Tap cannot be used to cut

(a) BSW threads (b) Metric threads
(c) Square threads (d) Unified threads

GPSC AE 26.07.2015

Ans. (c) :

2652. Cylindrical parts are held on planer by

(a) V-blocks and arrestors
(b) Angle plates
(c) V-block, T-bolts and clamps
(d) T-bolt and clamps

ISRO Scientist/Engineer 2010

Ans. (c) : The standard clamping device used for holding a workpiece on a planer are
(i) Heavy duty vices
(ii) Clamps and T-bolts
(iii) Adjustable block
(iv) Angle plates and step blocks
(v) Planer Jack
(vi) V-Blocks

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07.

MECHATRONICS AND ROBOTICS

1. Given the joint variables, q_i, determination of end effector's position and orientation is known as :

(a) Inverse kinematics
(b) Forward kinematics
(c) Backward kinematics
(d) None of the above

JPSC AE 10.04.2021, Paper-II
HPPSC Workshop Suptd. 08.07.02021
HPPSC Asstt. Prof. 29.10.2016

Ans. (b) :
Forward kinematics or Direct kinematics–
Given Information– The value of the joint variables.
Required Information- The position and orientation of the end effector.
Inverse kinematic-
Given data – The position and the orientation of their end effectors
Required data – The values of the joint variables.

2. End effector of a Robot can have …… motion.

(a) M-P-Y (b) P-P-Z
(c) R-G-B (d) R-P-Y

HPPSC Lect. (Auto) 23.04.2016
HPPSC AE (PWD) 10.09.2016

Ans. (d) : Main End effector motion are.
- Rolling (R)
- Pitching (P)
- Yawing (Y)

In Robotics, An end effector is the devices at the end of a robotic arm, designed to interact with the environment. End effector may consists of a gripper or a tool.

3. The following measurement are carried out by internal state sensors of the end effector

(a) Position
(b) Position and Velocity
(c) Velocity and Acceleration
(d) Position, Velocity and Acceleration

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Internal sensors measure the robot's internal state. They are used to measure position, velocity and acceleration of robot joint or end effectors.

4. Which of the following provides anti-clockwise and clockwise rotation about the vertical axis perpendicular to the arm?

(a) Shoulder swivel (b) Arm sweep
(c) Wrist bend (d) Elbow extension

UPPSC AE 13.12.2020, Paper-I

Ans. (b) : Arm sweep provides anticlockwise and clockwise rotation about the vertical axis perpendicular to arm while elbow extension provides radial movement to the manipulator arm.

5. PLC operates on following signals

(a) Digital (b) Impulse
(c) Analog (d) Frequency

UPPSC AE 13.12.2020, Paper-I

Ans. (a) : PLC stands for programmable logic controller . The PLC of a computer operates with digital instruction in binary code of 0 and 1.

6. A technology for application of mechanical, electronics and computer based systems to control and operate the systems is called

(a) PLC
(b) Sequential controller
(c) Microprocessor based systems
(d) Automation

UPPSC AE 13.12.2020, Paper-I

Ans. (d) : Automation is the application of technology, programs, robotics or processes to achieve outcomes with minimal human input.

7. Hall sensor is used to measure the following

(a) Position of shaft
(b) Angular velocity
(c) Strength of magnetic field
(d) All the above

UPPSC AE 13.12.2020, Paper-I

Ans. (c) : Hall effect is applied in magnetic field and Hall sensor is used to measure strength of magnetic field.
The output of Hall sensor is directly proportional to the magnetic field strength.

8. Industrial Robots used in automobile industry are made of series of links and …….

(a) Bearings (b) Joints
(c) valves (d) None of these

HPPSC Lect. (Auto) 23.04.2016

Ans. (b) : Industrial Robots used in automobile industry are made of series of links and joints.

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9. The structure of Robot

(a) defines its range of motion
(b) determines its potential uses
(c) defines the workspace needed for the robot
(d) all of the above

Vadodara Muncipal Corp. DEE, 2018

Ans. (d) : The structure of robot defines its range of motion
- determines its ptotential uses.
- defines work space needed for the robot.

10. An LVDT has

(a) One primary coil and two secondary coils
(b) Two primary coils and one secondary coil
(c) One primary coil and one secondary coil
(d) Two primary coils and two secondary coils

Karnataka PSC Lect., 27.05.2017

Ans (c) : An LVDT has one primary coil and one secondary coil.

11. Which of the following statements is/are true for LVDT?

(a) It is a mutual capacitive transducer.
(b) Presence of hysteresis gives high repeatability
(c) It can measure displacement and pressure
(d) All of the above

Karnataka PSC Lect., 27.05.2017

Ans (c) : **LVDT–**The linear variable differential transformer (also called linear variable displacement transformer) is a type of electrical transformer used for measuring linear displacement (position). A counter part to this device that is used for measuring rotary displacement is called a rotary variable differential transformer (RVDT). The LVDT has three solenoidal coils placed end to end around a tube.

12. Robots which have rotational movement in both horizontal and vertical planes with single linear movement are

(a) Cartesian or rectangular configuration
(b) Cylindrical configuration
(c) Polar configuration
(d) Articulated configuration

Karnataka PSC Lect., 27.05.2017

Ans (c) : **Polar configuration-** This configuration has a telescopic arm which pivots about a horizontal axis and also rotates about a vertical axis, owing to the mechanical and/or actuator connection limitations.

13. How many storage locations are available when a memory device has twelve address lines?

(a) 4096 (b) 512
(c) 144 (d) 2048

MPPSC State Forest Service Exam, 2014

Ans. (a) : If a memory device has 'n' address lines, then it can access 2^n locations in memory,

So $n = 12$

$2^{12} = 4096$

14. Which of the following approach is followed by Object Oriented Language during the execution of a program?

(a) bottom up approach
(b) Top down approach
(c) Random approach
(d) None of these

MPPSC State Forest Service Exam, 2014

Ans. (a) : If we use functional approach it is top down, and when we follow object oriented approach then it is bottom up approach. Because we use main function in C++ at anywhere but complier starts to compile the program in main function so it is called as bottom & approach.

15. Which Memory looses its content when power is lost

(a) Nonvolatile (b) Static
(c) Volatile (d) Random

MPPSC State Forest Service Exam, 2014

Ans. (c) : As we know, the volatile memory looses its content when power is lost.

16. Net Strain Gauge is

(a) An active device and converts mechanical displacement into change of resistance.
(b) A passive device and converts mechanical displacement into change of resistance.
(c) A passive device and converts electrical displacement into change of resistance.
(d) An active device and converts electrical displacement into change of resistance.

Karnataka PSC AE, 10.09.2017

Ans. (b) : A passive device which converts mechanical displacement into change of resistance is Net strain gauge.

17. If 'n' number of bits are denoted for a robotic joint and 'R' is the range of movement, then control resolution (CR) is

(a) $CR = \frac{R}{2^n}$

(b) $CR = \frac{R+n}{2}$

(c) $CR = \frac{nR}{2}$

(d) $CR = \frac{2n}{R}$

Karnataka PSC AE, 10.09.2017

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Ans. (a) :

$$\left[CR = \frac{R}{2^n}\right]$$

Where,

CR = Control resolution

R = Range of movement

n = No. of bits for a robotic joint.

18. In the barcode symbol 'Code 39', digit 3 indicates

(a) Wide to narrow ratio

(b) Number of elements in the character

(c) Number of wide elements in the character

(d) Number of characters in the code

Karnataka PSC AE, 10.09.2017

Ans. (c) : Barcode is used for representing data indicates, machine readable from. In (code 39) 3 represents no. of wide elements in the character.

19. Which Logic circuit would you use for addressing memory?

(a) Full adder

(b) Multiplexer

(c) Decoder

(d) Direct memory access circuit

GPSC ARTO Pre 30.12.2018

Ans. (c) : Address decoders are fundamental building blocks for systems that uses buses.

Decoder is used for addressing memory.

20. A Flip-Flop circuit can be used for

(a) Counting (b) Scaling

(c) Rectification (d) Demodulation

GPSC ARTO Pre 30.12.2018

Ans. (a) : Flip flops are essential in data storage.

They are electronic circuits with two stable states used to store binary data. They are used for counting of pulses and synchronizing variably timed input signals to some reference timing signal a flip flop can be level triggered or edge triggered.

21. The communication between central system and the outside environment is done by :

(a) Control system (b) Logic system

(c) Memory system (d) I/O subsystem

GPSC ARTO Pre 30.12.2018

Ans. (d) : The communication between central system and outside environment is done by I/O subsystem.

22. In a articulated joint robot the number of degrees of freedom associated with arm and body motion will be :

(a) 1 (b) 2

(c) 3 (d) 4

HPPSC Asstt. Prof. 29.10.2016

Ans. (c) : In a articulated joint robot the number of degrees of freedom associated with arm and body motion will be 3.

23. Which of the following best describes a pick-and –place manipulator?

(a) mainly used for large parts handling

(b) capable of point to point operation

(c) most flexible of all robots

(d) an intelligent robot

MPSC Poly. Lect. 2014

Ans. (b) : Pick-and-place manipulator are capable of point to point operation.

24. The axis of movement of a robot may include

(a) elbow rotation

(b) wrist rotation

(c) X-Y coordinate motion

(d) all options are correct

MPPSC AE 2016

MPPSC AE 08.11.2015

Ans. (d) : The axis of movement of a robot may include-

- Elbow rotation (yaw)
- Wrist rotation (roll)
- X-Y coordinate rotation

25. Automatic loading and unloading of materials can be accomplished by means of

(a) power push or pull device

(b) power roller, belt and chain

(c) automatic couple and uncouple

(d) all options are correct

MPPSC AE 08.11.2015

Ans. (d) : Automatic loading and unloading of a materials can be accomplished by means of-

- Power push or pull device
- Power roller, belt and chain
- Automatic couple and uncouple.

26. Assertion (A) : Thin wires should be used in thermocouples for measuring transient temperatures.

Reason (R) : The value of time constant for thin wire is low which leads to better response.

Codes :

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true and R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

UPSC JWM Advt. No.-50/2010

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Ans. (a) : As we know,

Time constant of thermocouples $= \frac{\rho V C_p}{hA}$ for circular

$\therefore \quad \frac{V}{A} = \frac{D}{6}$

$= \frac{\rho D C_p}{6h}$

Time constant $\propto D \rightarrow$ diameter of wire.

Thin wire should be used in thermocouples for measuring transient temperatures, because for thin wire time constant is low, leads to better response.

27. ———— are used to move the joint of robots.

(a) Transducers (b) Actuators

(c) Controllers (d) Vacuum

MPSC Lect. (Prod.) 2018

Ans. (b) : Actuators are used to move the joint of robots.

28. Statement (I) : Microprocessors which have memory and various input/output arrangements, all on the same chip, are called microcontrollers.

Statement (II) : The microcontroller is the integration of a microprocessor with RAM, ROM, EPROM, EEPROM and I/O interfaces, and other peripherals such as timers, on a single chip.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).

(b) Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).

(c) Statement (I) is true, but Statement (II) is false.

(d) Statement (I) is false, but Statement (II) is true.

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Ans. (a) : Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).

29. Statement (I) : Capacitive proximity sensor can only be used for the detection of metal objects and is best with ferrous metals.

Statement (II) : One form of capacitive proximity sensor consists of a single capacitor plate probe with the other plate being formed by the object, which has to be metallic and earthed.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).

(b) Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).

(c) Statement (I) is true, but Statement (II) is false.

(d) Statement (I) is false, but Statement (II) is true.

ESE 2020

Ans. (d) : Statement (I) is false, but Statement (II) is true

30. Revolving joint of the robot is referred to as

(a) L joint (b) O joint

(c) T joint (d) V joint

ESE 2020

Ans. (d) : V joint

31. Which of the following factors are to be considered while selecting a microcontroller?

1. Memory requirements

2. Processing speed required

3. Number of input/output pins

(a) 1 and 2 only (b) 1 and 3 only

(c) 2 and 3 only (d) 1, 2 and 3

ESE 2020

Ans. (d) : Following factors are to be considered while selecting a micro controller-

(i) Memory requirements

(ii) Processing speed required

(iii) Number of input/output pins

(iv) Interfaces required

(v) The number of interrupts required

32. Which of the following statements regarding interface circuit are correct?

1. Electrical buffering is needed when the peripheral operates at a different voltage or current to that on the microprocessor bus system or there are different ground references.

2. Timing control is needed when the data transfer rates of the peripheral and the microprocessor are different.

3. Changing the number of lines is needed when the codes used by the peripherals differ from those used by the microprocessor.

(a) 1 and 2 only (b) 1 and 3 only

(c) 2 and 3 only (d) 1, 2 and 3

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Ans. (a) :

1. Electrical buffering is needed when the peripheral operates at a different voltage or current to that on the microprocessor bus system or there are different ground references.
2. Timing control is needed when the data transfer rates of the peripheral and the microprocessor are different.

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33. Alternative paths provided by vertical paths from the main rung of a ladder diagram, that is, paths in parallel, represent

(a) Logical AND operations
(b) Logical OR operations
(c) Logical NOT operations
(d) Logical NOR operations

ESE 2020

Ans. (b) : Logical OR operations

34. The resolution of an encoder with 10 tracks will be nearly

(a) 0.15° (b) 0.25°
(c) 0.35° (d) 0.45°

ESE 2020

Ans. (c) :

$$\text{Resolution} = \frac{360}{2^n}$$

$$= \frac{360}{2^{10}} = 0.35^\circ$$

35. The selection of the right controller for the application depends on

1. The degree of control required by the application.
2. The individual characteristics of the plants.
3. The desirable performance level including required response, steady-state deviation and stability.
Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

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Ans. (d) : 1) The degree of control required by the application.
2) The individual characteristics of the plant.
3) The desirable performance level including response, steady-state deviation and stability.

36. Consider a system described by
x = Ax Bu
y = Cx Du
The system is completely output controllable if and only if

(a) The matrix
$[CB \vdots CBA \vdots CB^2A \vdots \cdots \vdots CB^{n-1}A \vdots D]$
is of rank n
(b) The matrix
$[CB \vdots CAB \vdots CA^2B \vdots \cdots \vdots CA^{n-1}B \vdots D]$
is of rank m
(c) The matrix
$[BC \vdots BAC \vdots BA^2C \vdots \cdots \vdots BA^{n-1}C \vdots D]$
is of rank m
(d) The matrix
$[BC \vdots ABC \vdots CA^2B \vdots \cdots \vdots CB^{n-1}A \vdots D]$
is of rank n

where:
x = State vector (n-vector)
u = Control vector (r-vector)
y = Output vector (m-vector)
A = n × n matrix
B = n × r matrix
C = m × n matrix
D = m × r matrix

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Ans. (b) : The matrix
$[CB \vdots CAB \vdots CA^2B \vdots \cdots \vdots CA^{n-1}B \vdots D]$
is of rank m

37. Which one of the following symbols is used as the notation for designating arm and body of a robot with jointed arm configuration?

(a) TRL (b) TLL, LTL, LVL
(c) LLL (d) TRR, VVR

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Ans. (d) :

Robot configuration (Arm & Body)	**Symbol**
Jointed arm configuration →	TRR, VVR
Polar configuration →	TRL
Cylindrical configuration →	TLL, LTL, LVL
Cartesian configuration →	LLL

38. A compliant motion control of robots can be understood by the problem of controlling of

(a) Position and velocity of joints
(b) Position and acceleration of the end-effector
(c) Manipulator motion and its force interactions with the environment
(d) Joint velocities of given end-effector velocity

ESE 2020

Ans. (c) : Manipulator motion and its force interactions with the environment.

39. For the vector v = 25 i + 10 j + 20 k, perform a translation by a distance of 8 in the x-direction, 5 in the y-direction and 0 in the z-direction. The translated vector Hv will be

(a) $\begin{bmatrix} 1 \\ 20 \\ 33 \\ 15 \end{bmatrix}$ (b) $\begin{bmatrix} 33 \\ 15 \\ 20 \\ 1 \end{bmatrix}$

(c) $\begin{bmatrix} 15 \\ 33 \\ 1 \\ 20 \end{bmatrix}$ (d) $\begin{bmatrix} 1 \\ 15 \\ 20 \\ 33 \end{bmatrix}$

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Ans. (b) : Given vector, $v = 25\hat{i} + 10\hat{j} + 20\hat{k}$
Translation in x direction = 8
Translation in y-direction = 5

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Translation in z-direction = 0

$$\text{Translation vector} = \begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 25 \\ 10 \\ 20 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 33 \\ 15 \\ 20 \\ 1 \end{bmatrix}$$

40. Statement (I) : SCARA configuration provides substantial rigidity for the robot in the vertical direction, but compliance in the horizontal plane.

Statement (II) : A special version of the cartesian coordinate robot is the SCARA, which has a very high lift capacity as it is designed for high rigidity.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).
(c) Statement (I) is true, but Statement (II) is false.
(d) Statement (I) is false, but Statement (II) is true.

ESE 2020

Ans. (c) : Statement (I) is true, but Statement (II) is false.

41. Which of the following is one of the basic units of memory controller in micro-controller?

(a) Microcode engine
(b) Master program counter
(c) Program status word
(d) Slave program counter

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Ans. (d) : The register Arithmetic-logic unit (RALU) communicates with all memory, except the register file and peripheral SFRs, through the memory controller. The memory controller contains the prefetch queue, the slave program counter (slave PC), address and data registers, and the bus controller.

42. In ladder logic programming, an alternative in place of using same internal relay contact for every rung is to use

(a) battery-backed relay
(b) dummy relay
(c) one-shot operation
(d) master control relay

ESE 2019

Ans. (d) : Master control relay is used in ladder logic for whole block of outputs which can be simultaneously turned on/off.

43. Consider the following statements:

(1) The term 'attenuation' is used to describe the process of removing a certain band of frequencies from a signal and permitting others to be transmitted.
(2) The Wheatstone bridge can be used to convert a voltage change to an electrical resistance change.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 and 2

ESE 2019

Ans. (d) : Attenuation– Attenuation is a general term that refers to any reduction in the strength of a signal of particular frequencies but not of removing signal of some frequencies. So statement-1 is wrong.

Wheatstone Bridge–It can be used to convert a resistance change to a voltage change. So statement-2 is also wrong.

44. At time t, the excitation voltage to a re-solver is 24 V. The shaft angle is 90°. The output signals from the re-solver V_{S1} and V_{S2} will be

(a) 12 V and 0 V (b) 24 V and 0 V
(c) 12 V and 12 V (d) 24 V and 12 V

ESE 2019

Ans. (b) : Given,

$V_1 = 24$ V, $\alpha = 90°$

Output signals of re-solver are

$V_{s_1} = V_1 \sin\alpha = 24 \sin 90 = 24$ V

$V_{s_2} = V_1 \cos\alpha = 24 \cos 90 = 0$ V

45. An actuator having a stem movement at full travel of 30 mm is mounted with a control valve having an equal percentage plug and with minimum flow rate of 2 m^3/s and maximum flow rate of 24 m^3/s. When the stem movement is 10 mm, the flow rate will be

(a) 3.4 m^3/s (b) 3.8 m^3/s
(c) 4.2 m^3/s (d) 4.6 m^3/s

ESE 2019

Ans. (d) : Given,

Maximum flow rate (Q_m) = 24 m^3/s

$$\text{Valve range ability (R)} = \frac{\text{Maximum flow}}{\text{Minimum flow}} = \frac{24}{2} = 12$$

Maximum valve travel (T) = 30 mm

x = 10 mm

$$Q = Q_n R^{\left(\frac{x}{T}-1\right)}$$

$$= 24(12)^{\left(\frac{10}{30}-1\right)} = 4.57$$

≈ 4.6 m^3/s

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46. For the control signal to change at a rate proportional to the error signal, the robotic controller must employ

(a) integral control
(b) proportional-plus-integral control
(c) proportional-plus-derivative control
(d) proportional-plus-integral-plus-derivative control

ESE 2019

Ans. (a) : Given that the control signal to change at a rate proportional to the error signal it means

$$\frac{d}{dt}[\Delta I(t)] \propto E(t)$$

$\Delta I(t) = k\int E(t)dt$

$\Delta I(t)$ = change of control signal (controller output)

$E(t)$ = Error signal

As the change of control signal is proportional to the integral of error, it represents "integral controller".

47. What is the minimum number of degrees of freedom that a robot needs to have in order to locate its end effectors at an arbitrary point with an arbitrary orientation in space?

(a) 3 (b) 4
(c) 5 (d) 6

ESE 2019
GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : For industrial robot the degree of freedom is six. which is more flexible and versatile.

48. Using a robot with 1 degree of freedom and having 1 sliding joint with a full range of 1 m, if the robot's control memory has a 12-bit storage capacity, the control resolution for the axis of motion will be

(a) 0.236 mm (b) 0.244 mm
(c) 0.252 mm (d) 0.260 mm

ESE 2019

Ans. (b) : Given,

Total range of sliding joint = 1m = 1000 mm

Control memory = 12 bit, n = 12

Number of increments = $2^{12} = 4096$

$$\text{Control resolution} = \frac{\text{Stroke length}}{\text{Number of increment}}$$

$$= \frac{\text{Stroke length}}{2^n}$$

$$= \frac{1000}{4096} = 0.244 \text{ mm}$$

49. Assume that the joint mechanisms at serial link manipulators are frictionless. The joint torque τ required to bear an arbitrary end point force F is

(a) $J^{-1}F$ (b) JF
(c) J^TF (d) $J^{-1}F^T$

ESE 2019

Ans. (c) : Joint torque $\tau = J^TF$

J^T = Transpose of Jacobian matrix

F = End-point force vector

50. Rotate the vector v = 5i + 3j + 8k by an angle of 90° about the x-axis. The rotated vector (Hv) would be

(a) $\begin{bmatrix}1\\3\\-8\\5\end{bmatrix}$ (b) $\begin{bmatrix}-8\\5\\1\\3\end{bmatrix}$

(c) $\begin{bmatrix}3\\-8\\5\\1\end{bmatrix}$ (d) $\begin{bmatrix}5\\-8\\3\\1\end{bmatrix}$

ESE 2019

Ans. (d) : Rotational transformation is given by

$$\text{Rot}(x, 90) = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & \cos\theta & -\sin\theta & 0\\ 0 & \sin\theta & \cos\theta & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & \cos 90 & -\sin 90 & 0\\ 0 & \sin 90 & \cos 90 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & -1 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

$V = 5i + 3j + 8k$

Rotated vector = Rot (x, θ) of V

$$= \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & -1 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}\begin{bmatrix}5\\3\\8\\1\end{bmatrix}$$

$$= \begin{bmatrix}5\\-8\\3\\1\end{bmatrix}$$

51. Statement (I): The function of arithmetic logic unit (ALU) in microprocessor is to perform data manipulation.

Statement (II): The status register is where data for an input to the arithmetic and logic unit is temporarily stored.

ESE 2019

Ans. (c) : The ALU is one of the most important components in microprocessor and is typically the part of the processor that is designed first. Once the ALU is designed the rest of the microprocessor is implemented to feed operands and control codes to the ALU.

Status register (flag register) consist of flag bits indicates status of ALU after data manipulation.

So statement-1 is true while statement-2 is false.

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52. **Statement (I):** To use a sensor, we generally need to add signal conditioning circuitry, such as circuits which amplify and convert from analog to digital, to get the sensor signal in the right form, take account of any nonlinearities, and calibrate it.

Statement (II): A smart sensor is integrated with the required buffering and conditioning circuitry in a single element and provides functions beyond that of just a sensor.

ESE 2019

Ans. (b) : To use a sensor, we generally need to add signal conditioning circuitry, such as circuits which amplify and convert from analog to digital, to get the sensor signal in the right form, take account of any nonlinearities and calibrate it. Additionally, we need to take account of drift i.e., a gradual change in the properties of a sensor over time. Some sensors have all these elements taken care of in single package; they are called smart sensors.
Hence both the statements are true and second is the correct explanation of first.

53. **Statement (I):** The count-up overflow (OV) bit is 1 when the up-counter increments above the maximum positive value.

Statement (II): The count-down underflow (UN) bit is 1 when the counter decrements below the minimum negative value.

ESE 2019

Ans. (c) : The count-down underflow (UN) bit is 1 when the counter decrements below the maximum negative value.

54. **Statement (I):** The multiplexer is essentially an electronic switching device which enables each of the inputs to be sampled in turn.

Statement (II): A multiplexer is a circuit that is able to have inputs of data from a number of sources and then, by selecting an input channel, gives an output from just one of them.

ESE 2019

Ans. (a) : The multiplexer is a combinational logic circuit designed to switch one of several input lines to a single common output line.

55. **Statement (I):** The term 'encoder' is used for a device that provides an analog output as a result of angular or linear displacement.

Statement (II): An increment encoder detects changes in angular or linear displacement from some datum position where as an absolute encoder gives the actual angular or linear position.

ESE 2019

Ans. (d) : An encoder is used for providing a digital output as a result of angular or linear displacement.
- An increment encoder detects changes in angular or linear displacement from some datum position.
- An absolute encoder gives the actual angular or linear position.

56. **Statement (I):** Process control valves are used to control the rate of fluid flow and are used where, perhaps, the rate of flow of a liquid into a tank has to be controlled.

Statement (II): A common form of pneumatic actuator used with process control valves is the diaphragm actuator.

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Ans. (b) : Process control valves are used to control the rate of fluid flow and are used where, perhaps, the rate of flow of a liquid into a tank has to be controlled. The basis of such valves is an actuator being used to move a plug into the flow pipe and so alter the cross-section of the pipe through which the fluid can flow.
A common form of pneumatic actuator used with process control valve is the diaphragm actuator. Essentially it consists of a diaphragm with the input pressure signal from the controller on one side and atmospheric pressure on the other, this difference in pressure being termed the gauge pressure.

57. **The degrees of freedom of a SCARA robot are**

(a) six (b) five
(c) four (d) three

ESE 2018
UPPSC AE 13.12.2020, Paper-I

Ans. (c) : SCARA robot has 4 DOF. Its full form is "Selective Compliance Assembly Robot Arm" It is similar in construction to the jointer-arm robot, except the shoulder and elbow rotational axes are vertical.

58. **Consider the following statements regarding electromechanical devices**

1. A potentiometer has an input of rotation and an output of a potential difference.
2. An electric motor has an input of a potential difference and an output of rotation of a shaft.
3. A generator has an input of rotation of a shaft and an output of a potential difference.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (d) : (i) A potentiometer has an input of displacement (either linear or angular) and an output of potential difference.
(ii) For electric motor, the input is potential difference and output is rotation of shaft.
(iii) For electric generator, the input is rotation of shaft and output is potential difference.

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59. The indirect operation of solenoid valve in pneumatic circuit is designed to reduce

1. valve size towards lowering the cost
2. coil size and electrical power consumption
3. response time

Which of the above is/are relevant to the context?

(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

ESE 2018

Ans. (b) : A solenoid valve is an electromechanical actuated valve to control the flow of liquids and gases. The indirect operation of solenoid valve in pneumatic circuit is designed to reduce coil size and electrical power consumption.

60. Consider the following statements:

1. Robots take permissible actions only.
2. All actions that are obligatory for robots are actually performed by them subject to ties and conflicts among available actions.
3. All permissible actions can be proved by the robot to be permissible and it can be explained in ordinary English.

Which of the above statements are correct?

(a) 1 and 3 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (b) :
(i) Robots only take permissible actions.
(ii) Robots perform all obligatory actions relevant to them, subject to ties and conflicts among available actions.
(iii) All robot actions cannot be explained by ordinary English for e.g. in lead through programming mode, robot joints are moved with a teach pendant.

61. Consider the following statements relating to the term 'Robot Repeatability':

1. It is a statistical term associated with accuracy in the action.
2. It is a measure of the ability of the robot to position the tool tip in the same place repeatedly.
3. It does not describe the error with respect to absolute coordinates.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2018

Ans. (c) : Accuracy is how closely a robot can reach a commanded position. When the absolute position of the robot is measured and compared to the commanded position the error is a measure of accuracy. Repeatability is how well the robot will return to a programmed position. This is not the same as accuracy. Accuracy and repeatability are different measures.
Note : The answer given by UPSC is option (d)

62. Consider the following statements regarding homogeneous coordinate transformation matrix :

1. A homogeneous transformation matrix can be considered to consist of four sub-matrices.
2. The upper left 3 × 3 sub-matrix represents the position vector.
3. The upper right 3 × 1 sub-matrix represents the rotation matrix.
4. The lower left 1 × 3 sub-matrix represents perspective transformation.

Which of the above statements are correct?

(a) 1 and 3 (b) 1 and 4
(c) 2 and 3 (d) 2 and 4

ESE 2018

Ans. (b) : Statement 2 : Upper left 3 × 3 sub-matrix represent rotation matrix.
Statement 3 : Upper right 3 × 1 sub-matrix represent position.
So statement 2 and statement 3 both are wrong.

63. With reference to a microprocessor, RISC stands for

(a) Redefined Instruction Set Computer
(b) Reduced Instruction Set Computer
(c) Restructured Instruction Set Computer
(d) Regional Instruction Set Computer

UPPSC AE 13.12.2020, Paper-I

ESE 2018

Ans. (b) : RISC stands for reduced instruction set computer.

64. An OR logic control in pneumatic systems is possible with the help of

(a) sequence valve (b) shuttle value
(c) dual pressure valve (d) delay valve

ESE 2018

Ans. (b) : Shuttle valve allows fluid flow from one of two sources, like of logic.
The basic structure of a shuttle valve is like a tube with three opening.

65. Which one of the following is not an Addressing Mode in 8085?

(a) Immediate (b) Indirect
(c) Register (d) Segment

ESE 2018

Ans. (d) : The way of specifying data to be operated by an instruction is called addressing mode.
In 8085 microprocessor there are 5 types of addressing modes–
(i) Immediate addressing

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(ii) Register addressing
(iii) Direct addressing mode
(iv) Register indirect addressing mode
(v) Implied/Implicit addressing mode

66. Consider the following statements regarding Programming Logic Controller (PLC):

1. It was developed to replace the microprocessor.
2. Wiring between device and relay contracts are done in its program.
3. Its I/O interface section connects it to external field devices.
4. It requires extensive wiring in the application.

(a) 1 and 3 (b) 1 and 4
(c) 2 and 3 (d) 2 and 4

ESE 2018

Ans. (c) : Option 1 and 4 are wrong because
Option 1 : PLC not replace microprocessor as it is a part of PLC.
Option 4 : Not require extensive wiring by PLC.

67. If B is the magnetic flux density at right angles to a plate, I is the current flow through the plate, t is the plate thickness and K_H is Hall coefficient, the resultant transverse potential difference V for Hall sensor is given by

(a) $K_H \frac{Bt}{I}$ (b) $K_H \frac{t}{BI}$
(c) $K_H \frac{BI}{t}$ (d) $K_H \frac{I}{Bt}$

ESE 2018

Ans. (c) : A hall effect sensor is a device that is used to measure the magnitude of a magnetic field. Its output voltage is directly proportional to the magnetic field strength through it.
Hall effect voltage is given by

$$V_H = K_H \frac{BI}{t}$$

68. Which one of the following statements is correct?

(a) Microprocessor is more suitable for general purpose and micro controller is more suitable for special purpose and custom built application
(b) Microprocessor and micro-controller are suitable for general purpose application
(c) Microprocessor and micro-controller and suitable for special purpose application
(d) Microprocessor and microcontroller are suitable for special purpose and custom-built application

ESE 2018

Ans. (a) : Microprocessors has off-chip peripherals, suitable for general purpose applications and microcontrollers has on-chip peripherals with fixed quantity to suit special purpose.

69. Which one of the following is correct with respect to microcontrollers?

(a) Integration of a microprocessor with I/O interfaces and memory and other peripherals in a single IC chip
(b) A single very large scale integrated (VLSI) chip that contains programmable electronic components that perform control functions.
(c) Digital circuits for data handling and computation.
(d) The primary computation and system control operations.

ESE 2017

Ans. (a) Microcontroller is integration of all microprocessor, input and memory other peripheral in a single chip.

70. Which one of the following statements is correct?
Seismic transducer working in the displacement mode should have

(a) weak springs and heavy mass
(b) stiff springs and light mass
(c) weak springs and light mass
(d) stiff springs and heavy mass

ESE 2017

Ans. (a) : In displacement mode ratio $r = \frac{\omega}{\omega_n} > 2$

$\omega > \omega_n$

Natural frequency is low $= \sqrt{\frac{k}{m}}$

So, k is low
$\Rightarrow$ weak spring and m is high hence heavy mass.

71. A stepper motor is to be used to drive the linear axis of a certain Mechatronics system. The motor output shaft is connected to a screw thread with a 30 mm pitch. Linear resolution of 0.5 mm is stipulated. What is the needed step angle?

(a) 9° (b) 8°
(c) 7° (d) 6°

ESE 2017

Ans. (d) : Step angle $= \frac{\text{Linear Resolution}}{\text{Pitch}} \times 360°$

$= \frac{0.5}{30} \times 360°$

$= 6°$

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72. Consider the following statements regarding a stepper motor:

1. The rotation angle of the motor is proportional to the input pulse.
2. The motor has full torque at standstill.
3. Speed and electric control signal of the motor vary mutually linearly.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) : Advantages of stepper motor are

- The rotation angle of the motor is proportional to the input pulse.
- The motor has full torque at standstill.
- Its rotational speed varies directly proportional to the frequency of the pulse.
- Excellent response to starting, stopping and reversing.

73. Consider the following statements with reference to SCARA Robot:

1. It has four degrees of freedom.
2. It has only one forward kinematic solution.
3. It has two inverse kinematic solutions.

Which of the above statements are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) : 1. SCARA Robol has 4 degree of freedom
2. Forward solution is only one
3. Inverse kinematic solution are two (+ or –)

74. The number of degrees of freedom in a 3D Robot of TRL : R type configuration is

(a) 4 (b) 3
(c) 2 (d) 1

ESE 2017

Ans. (a) :

Robot	DOF
Planar	3
PUMA Industrial	6
T3	6
SCARA	4
TRL : R type	4

75. Which of the following are the basic differences between vibration signature and noise signature?

1. Vibration signature is essentially in the frequency range zero to 100 cps whereas noise signature is in the range 20 cps to 3000 cps.
2. Vibration signature has well-defined peaks whereas the noise signal is smeared.
3. The intensities of noise signatures are much less than that of vibration signatures.
4. Detection of vibration signature calls for a microphone whereas that of noise can do with a pickup.

Select the correct answer using the code given below.

(a) 1 and 4 (b) 2 and 3
(c) 1 and 2 (d) 3 and 4

ESE 2017

Ans. (b) : Vibration signature range might go beyond 100 cps and may be even more than 20,000 cps
Vibration signature → Acceleration (pick-up)
Noise → Microphone

76. Consider the following features relating to Robot kinematics with reference to SCARA Robot:

1. Shoulder and elbow rotational axes are vertical.
2. The Robot could perform insertion tasks along the vertical direction.
3. Its general configuration is analogous to that of a human arm.

Which of the above features are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 2017

Ans. (d) : Its full form is 'selective compliance Assembly Robot Arm'.
It is similar in construction to the jointed arm robot, except the shoulder and elbow rotational axes are vertical. It means that the arm is very rigid in the vertical direction, but compliant in the horizontal direction. Its general configuration is analogous to that of a human arm.

77. In robotics, precision of movement is a complex issue and it is described as three attributes namely spatial resolution, repeatability and

(a) soundness (b) accuracy
(c) speed (d) sensation

TNPSC AE 2017

Ans. (b) : In robotics, precision of movement is a complex issue and it is described as three attributes namely spatial resolution, repeatability and accuracy.

78. A Robot is basically a

(a) machining device
(b) inspection device
(c) material handler
(d) machine tool

TNPSC AE 2017

Ans. (c) : A Robot is basically a material handler.

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78. .....sensor is an example for proximity sensor used in Robots.

(a) Micro switch (b) Ultrasonic
(c) Touch and tactile (d) None of these

(HPPSC LECT. 2016)

Ans : (b) Ultrasonic sensor is an example for proximity sensor used in Robots.

80. Which Robot configuration represent a human arm?

(a) Mechanical configuration
(b) Polar configuration
(c) Cylindrical configuration
(d) Jointed arm configuration

TNPSC AE 2018

Ans. (d) : Jointed arm configuration represent a human arm in robot configuration.

81. Which of the following is not a robot programming method?

(a) Manual programming method
(b) Walk through programming method
(c) Teach pendant method
(d) Numerical programming method

TNPSC AE 2018

Ans. (d) : Numerical programming method is not a robot programming method.

82. Consider the following statements with respect to Laws of Robotics:

1. A robot may not injure a human being or, through inaction, allow a human being to come to harm
2. A robot must obey orders to it by human beings, except where such orders would conflict with the above first law
3. A robot may or may not obey any order given to it by human beings

Which of the above statements are correct?

(a) 1, 2 and 3 (b) 1 and 3 only
(c) 1 and 2 only (d) 2 and 3 only

JWM 2017
ESE 2017

Ans. (c) : Law of Robotics-

1. A robot may not injured a human being or through inaction, allow a human being to come to harm.
2. A robot must obey the orders give it by human beings except where such orders would conflict with the first law.
3. A robot must protect its own existence as long as such protection does not conflict with the first or second laws.

83. The most accurate robot is

(a) Cartesian robot (b) Polar robot
(c) Spherical robot (d) Articulated

TSPSC AEE 2015

Ans. (a) : The most accurate robot is Cartesian robot.

84. What type of motion must be produced in servo mechanism actuator?

(a) Angular or Reciprocating
(b) Linear or rotary
(c) Translatory and Angular
(d) Linear and Angular

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : Linear or rotatory motion must be produced in servo mechanism actuator.

85. Consider the following statements regarding modulation:

1. The modulation is essential in Communication systems, where a weak signal is transmitted by the use of a carrier signal.
2. When the frequency of the high frequency signal is varied in accordance with the intensity of the low-frequency weak signal, the modulation is said to be frequency modulation.
3. The process of recovering original baseband signal from the modulated signal is called phase modulation.

Which of the above statements are correct?

(a) 1 and 3 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (b) : The correct statements are-

- The modulation is essential in Communication systems, where a weak signal is transmitted by the use of a carrier signal.
- When the frequency of the high frequency signal is varied in accordance with the intensity of the low-frequency weak signal, the modulation is said to be frequency modulation.

86. Consider the situation where a microprocessor gives an output of an 8-bit word. This is fed through an 8-bit digital-to-analog converter to a control valve. The control valve requires 6.0 V to be fully open. If the fully open state is indicated by 11111111, then what is the output to the valve for a change of 1 bit?

(a) 0.033 V (b) 0.053 V
(c) 0.043 V (d) 0.023 V

ESE 18.07.2021

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Ans. (d) :

The full scale output voltage of 6V will be divided into 2^8 intervals. A change of 1 bit is thus a change in the output voltage of

$$= \frac{6}{2^8} = 0.023\text{V}$$

87. Which one of the following is not an application of Hall effect sensor?

(a) Magnetic switch for electric transducer
(b) Measurement of current
(c) Measurement of acceleration
(d) Measurement of power

ESE 18.07.2021

Ans. (c) :

Application of Hall effect sensor-

1. Hall sensors are used for proximity switching, positioning, speed detection and current sensing application.
2. They are used in brushless DC motors to detect the position of the permanent magnet
3. It is used as a magnetic switch for electric transducer.
4. It is used for the measurement of the position, displacement and proximity.
5. It is used for measurement of current.
6. It is used for measurement of power.

88. Consider the following statements regarding mechatronics systems:

1. The anti-lock brakes on a car are a simple example of a real time computing system.
2. The completion of an operation after its deadline is considered useless in soft real time system.
3. The hard real time system tolerates lateness and may respond with decreased service quality.

Which of the above statements is/are correct?

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

ESE 18.07.2021

Ans. (a) : Real time mechatronics system-

Real time control and computing is the study of hardware and software systems which are subject to a "real time constraint".

A system is said to be a real time system if there is total correctness of an operation. This depends upon not only the operation's logical correctness, but also the time in which it is performed and used. The antilock brakes on a car are a simple example of a real-time computing system and the real time constraint in the system is the short time in which the brakes must be released to prevent the wheel from locking.

Two types of real time system-

1. **Hard real time system**: In this completion of an operation after its deadline is considered useless.
2. **Soft real time system**: It tolerates such lateness and may respond with decreased service quality.

89. Which one of the following materials has least piezoelectric charge sensitivity?

(a) Quartz (b) Barium Titan ate
(c) PZT (d) PVDF

ESE 18.07.2021

Ans. (a) :

Piezoelectric charge sensitivity of following materials:

Quartz - 2.2 pC/N

Barium Titanate - 130 pC/N

Lead zirconate Titanate (PZT) - 265 pC/N

PVDF - 3.10 pC/N

Hence, Quartz has least piezoelectric charge sensitivity.

90. The ideal hydraulic rotary actuator provides shaft torque, T, which is

(a) equal to displaced volume measured.
(b) inversely proportional to the displaced volume measured.
(c) proportional to the differential pressure.
(d) inversely proportional to the differential pressure.

ESE 18.07.2021

Ans. (c) : The ideal hydraulic rotary actuator provides shaft torque, T, which is proportional to the differential pressure, ΔP , across the servo valve.

$$T = kD\Delta P$$

Where,

k = Proportionality constant relating torque and differential pressure.

D = displaced volume measured in mm^3

Hence option (c) will be correct answer.

91. An ammeter requires a change of 3 A in its coil to produce a change in deflection of the pointer by 12 mm. What is the stat ic sensitivity?

(a) 36 mm/A (b) 9 mm/A
(c) 4 mm/A (d) 15 mm/A

ESE 18.07.2021

Ans. (c) :

$$\text{Sensitivity} = \frac{\text{Change in output}}{\text{Change in input}}$$

$$= \frac{12}{3} = 4\text{mm/A}$$

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92. The settling time for a unit step response of a second-order system is

(a) proportional to the natural frequency

(b) inversely proportional to the natural frequency

(c) equal to the damping ratio

(d) proportional to the damping ratio

ESE 18.07.2021

Ans. (b) :

In second order system, the setting time is used as a measure of the time taken for the oscillations to die away.

$$t_s \propto \frac{1}{\xi\omega_n}$$

Here ω_n is natural frequency

ξ is damping ratio

93. A typical wrist mechanism with three rotational joints would be indicated by

(a) TRL (b) TRT

(c) LLL (d) TRR

ESE 18.07.2021

Ans. (d) : A typical wrist mechanism with three rotational joints would be indicated by TRR.

94. The Analog-to-Digital conversion process involves:

1. Quantizing

2. Sampling

3. Encoding

What is the correct sequence?

(a) 2-1-3 (b) 2-3-1

(c) 3-1-2 (d) 3-2-1

ESE 18.07.2021

Ans. (a) : The correct sequence of Analog-to-Digital conversion process is-

Sampling → Quantizing → Encoding.

95. In the first order system, the response is given by (x is constant input and y is the output of the system)

(a) $y = x\,(e^{-t/T} - 1)$ (b) $y = x\,(1 - e^{-t/T})$

(c) $y = x - e^{-t/T}$ (d) $y = x\,(1 - x\,e^{-t/T})$

GPSC ARTO 01.05.2016

Ans. (b) :

$$\boxed{y = x\left(1 - e^{t/T}\right)}$$

96. Which of the following power source is used for the robot manipulator

(a) Hydraulic (b) Pneumatic

(c) Electric (d) All of the above

GPSC Lect. (Poly.) 07.12.2014

Ans. (d) : power source is used for the robot manipulator is

- Hydraulic
- Pneumatic
- Electric

97. Transducer can give different output from the same value of quantity being measured according to whether that value has been reached by a continuously increasing or decreasing change. The effect is known as :

(a) Static Error

(b) Dynamic Error

(c) Hysteresis Error

(d) Repeatability Error

TNPSC AE (Industries) 09.06.2013

Ans. (c) :

98. When a beam of charged particles passes through a magnetic field, forces act on the particles and the beam is deflected from its straight line path. This is known as :

(a) Hall effect (b) Piliter effect

(c) Archimedes (d) Faraday's law

TNPSC AE (Industries) 09.06.2013

Ans. (a) :

99. In pneumatic comparator, which one of the following is not correct ?

(a) Internal dimensions com be readily measured not only with respect to tolerance but also geometric form

(b) No physical contact is made during measurement

(c) Eliminates error due to dirt and foreign matter

(d) Single gauging head is required for different dimensions

TNPSC AE (Industries) 09.06.2013

Ans. (d) :

100. Consider a resistance thermometer with three essential elements namely A, B and C. If 'A' takes the temperature signal and transforms it into resistance signal, element 'B' transforms the resistance signal in to current signal, element 'C' transforms the current signal into display of a movement at a pointer across a scale, which of these is a sensor, signal processor and data presentation ?

(a) B – Sensor, A – Signal processor, C – Data presentation

(b) A – sensor, B – signal processor, C – Data Presentation

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(c) C – sensor, A – signal processor, B – Data Presentation
(d) B – Sensor, C – Signal processor, A – Data presentation

TNPSC AE (Industries) 09.06.2013

Ans. (b) :

101. The management Information Systems structure with only one main computer system is called
(a) distributed MIS (b) centralized MIS
(c) decentralized MIS (d) Hierarchical MIS

VIZAG Steel MT 24.01.2021, Shift-I

Ans. (b) : The management Information System (MIS) structure with only main computer system is called centralized (MIS). Its easier to get a complete view of data. Manage update and back-up data.

102. Word Robot was coined by a Czech novelist from a Czech word :
(a) Rabota (b) Rebota
(c) Ribota (d) Robota

JPSC AE 10.04.2021, Paper-II

Ans. (d) : "Robota" is the root word of "robot" which is derived from slavic language. The word "robots means slave. It was first introduced by a Czech writer Karol capek in his play "Rossum's Universal Robots".

103. The PUMA 560, a Programmable Universal Machine for assembly robot has the 6 degree of freedom from :
(a) 3 revolute joints and 3 prismatic joints
(b) 2 revolute joints and 4 prismatic joints
(c) 4 revolute joints and 2 prismatic joints
(d) 6 revolute joints

JPSC AE 10.04.2021, Paper-II

Ans. (d) : The PUMA 560, a programmable universal machine for Assembly robot has the 6 degree of freedom from 6 revolute joints.

104. What is the name for space inside which a robotic manipulator operates ?
(a) Environment (b) Spatial base
(c) Work envelope (d) Exclusion zone

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Work envelope is the space inside which a robotic manipulator operates.

105. The Robotic manipulator designed with Cartesian co-ordinate system has :
(a) three linear movements
(b) three rotational movements
(c) two linear and one rotational movements
(d) two rotational and one linear movement.

JPSC AE 10.04.2021, Paper-II

Ans. (a) : The Robotic manipulator designed with Cartesian co-ordinate system has three linear movements in x, y and z directions.
Cartesian Co-ordinate Robots – Three linear movements
Spherical Co-ordinate Robots – Two rotational and one linear movement.
Cylindrical robots – Two linear and one rotational movements.
Jointed arm – Three rotational movements.

106. Which part of the Robot provides motion to the manipulator and end effectors?
(a) Controller (b) Sensor
(c) Actuator (d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : Actuators – Part of Robot which provides motion to the manipulator and end effectors.
Sensor – Robotic sensors are used to estimate a robot's condition and environment
Controller- It is the brain of the robot that control all the movements of the manipulator and the end effectors.

107. θ_i, in a serial robotic manipulator according to Denavit-Hartenberg notation is :
(a) Rotation about X axis
(b) Rotation about Y axis
(c) Rotation about Z axis
(d) None of the above

JPSC AE 10.04.2021, Paper-II

Ans. (c) : θ_i, in a serial robotic manipulator according to Denavit –Hartenberg notation is rotation about Z axis.

108. The number of movable joints in the base, arm and end effector of the robot determines
(a) degrees of freedom (b) payload capacity
(c) operational limits (d) flexibility

JPSC AE 10.04.2021, Paper-II

Ans. (a) : The number of moveable joints in the base, arm and end effector of the robot determines degree of freedom.
Flexibility- Ability to perform several different kinds of application.
Payload- It is the weight that robot can lift. It includes the weight of the end arm tooling and the weight of product being picked.
Operational limits – The workspace of robots is limited by singularities and joint limits.

109. A device, which does the general population associate most with automation is ..... .
(a) robots
(b) flexible manufacturing
(c) computer graphics workstation
(d) numerical control machine

GWSSB DEE 07.07.2016

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Ans. (a) : A device, which does the general population associate most with automation is robots.

110. Why the Bezier curve is smoother than the cubic spline curve ?

(a) Because of higher order derivatives
(b) Because of control points
(c) Because the curve passes through all the data points
(d) All of the above

TRB Polytechnic Lecturer 2017

Ans. (a) : Bezier curve is smoother than the cubic spline curve because of higher order derivatives.

111. The alternator produces electricity in its

(a) rotor field coil
(b) stator windings
(c) regulator
(d) armature commutator

TNPSC AE 2017

Ans. (b) : The alternator produces electricity in its stator windings.

112. Yaw velocity can be measured using

(a) Proximity sensor (b) Speed sensor
(c) Gyro sensor (d) Torque sensor

TNPSC AE 2017

Ans. (c) : Yaw velocity can be measured by measuring the ground velocity at two geometrically separated points on the body, or by a gyroscope, or it can be synthesized from accelerometers and the lie. It is the primary measure of how drivers sense a car's turning visually.

113. In control system terminology PID control stands for

(a) Proportional Integrated Decimal
(b) Proportional Intelligent Definite
(c) Proportional Integral Derivative
(d) Principal Intelligent Derivative

TNPSC AE 2014

Ans. (c) : Proportional Integral Derivative– A PID controller is an instrument used in industrial control applications to regulate temperature, flow pressure, speed and other process variable . PID controllers use a control loop feedback mechanism to control process variables and are the most accurate and stable controller.

114. In an electromagnetic flow meter the induced voltage is proportional to

(a) Flow rate
(b) Square root of flow rate
(c) Square of flow rate
(d) Logarithm of flow rate

TNPSC AE 2014

Ans. (a) : In an electromagnetic flow meter the induced voltage is proportional to flow rate.

115. Statement (I) : The size of a memory unit is specified in terms of the number of storage locations available; 1 K is 2^{10} = 1024 locations and thus a 4 K memory has 4096 locations. Statement (II) : Erasable and programmable ROM (EPROM) is a form of memory unit used for ROMs that can be programmed and their contents altered.

(a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I).
(b) Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).
(c) Statement (I) is true, but Statement (II) is false.
(d) Statement (I) is false, but Statement (II) is true.

ESE 2020

Ans. (b) : Both Statement (I) and Statement (II) are individually true, but Statement (II) is not the correct explanation of Statement (I).

116. Which of the following features is/are relevant to variable reluctance stepper motors?

1. Smaller rotor mass; more responsive
2. Step size is small
3. More sluggish

(a) 1 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

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Ans. (a) : Smaller rotor mass; more responsive.

117. Consider the situation where a microprocessor gives an output of an 8-bit word. This is fed through an 8-bit digital-to-analogue converter to a control valve. The control valve requires 6.0 V being fully open. If the fully open state is indicated by 11111111, the output to the valve for a change of 1-bit will be

(a) 0.061 V (b) 0.042 V
(c) 0.023 V (d) 0.014 V

ESE 2020

Ans. (c) : $\frac{6.0}{2^8} = 0.023\,\text{V}$

118. Which of following register of the processor is connected to memory Bus?

(a) PC (b) MAR
(c) RAM (d) IR

JPSC AE PRE 2019

Ans. (b) : The Memory Address Register (MAR) is the CPU register that either stores the memory address from

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which data will be fetched to the CPU or the address to which data will be sent and stored. In other words, MAR holds the memory location of data that need to be accessed.

119. A temperature control system is known as:

(a) Process control system
(b) Servomechanism
(c) Cascade control system
(d) None of these

PTCUL AE 25.06.2017

Ans. (a) : A temperature controlled system is a type of control system that automatically control the temperature of an object an area . We commonly use temperature control systems in air conditioners, refrigerators, geysers etc.
Where the temperature is automatically adjusted as per the input setting. In order to implement a temperature control system. We need a temperature sensor, a controller and a cooling system.

120. A photovoltaic cell converts :

(a) Electromagnetic radiation directly into electricity
(b) Solar radiation into thermal energy
(c) Thermal energy into electricity
(d) All the above

Kerala PSC Asstt. Prof. 27.09.2016

Ans. (a) : Photovoltaic cell is an electrical device that converts the energy of light directly into electricity by the photovoltaic effect.

121. Components of a control system may include ..... .

(a) amplifier only
(b) sensors only
(c) actuators only
(d) All three mentioned here

GWSSB DEE 07.07.2016

Ans. (d) : Components of a control system may include–
- Amplifier only
- Sensors only
- Actuators only

122. A flow-measuring device element which is inherently linear and requires no signal characterisation is a/an :

(a) pitot tube (b) Venturi
(c) orifice plate (d) Turbine

APPSC Poly Lect. Automobile Engg., 2020

Ans. (d) : A flow-measuring device element which is inherently linear and requires no signal characterisation is a turbine.

123. What are the components present in basic instrumentation systems?

(i) sensor
(ii) actuator
(iii) signal processor

(a) (i), (ii) and (iii) (b) (ii) and (iii)
(c) (i) and (ii) (d) (i) and (ii)

GPSC Lect. (Auto) 16.10.2016

Ans. (a) : Components present in basic instrumentation systems –
1. Sensor
2. Actuator
3. Signal processor.

124. Program Counter (PC) contains :

(a) address of next instruction which is to be executed
(b) address of present instruction which has to be executed
(c) address of relocatable instruction
(d) none of the above

APPSC IOF, 2009

Ans. (a) : The program counter contains the address of next instruction to be executed.

125. MOV A, B statement comes under the ..... mode statement.

(a) immediate addressing
(b) indexed addressing mode
(c) direct addressing
(d) relative addressing mode

APPSC IOF, 2009

Ans. (b) : MOV A, B statement comes under the indexed addressing mode statement.

126. Which of the following storage device stores data electronically ?

(a) RAM (b) CD-ROM
(c) Floppy Drive (d) DA tape

APPSC IOF, 2009

Ans. (b) : CD-ROM is a storage device stores data electronically.

127. For traction applications, the type of motor best suited is

(a) Induction motor
(b) Synchronous motor
(c) DC shunt motor
(d) DC series motor

ISRO Scientist/Engineer 2008

Ans. (d) : DC series motor have high initial torque so for traction applications, this type motor best suited.

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08.

RENEWABLE SOURCES OF ENERGY

1. Cosmic rays, such as gamma rays, are a source of

(a) soil pollution
(b) noise pollution
(c) thermal pollution
(d) radiation pollution

Assam PSC AE (IWT) 14.03.2021

Ans. (d) : Cosmic rays are high energy protons and atomic nucli that move through space at nearly the speed of light. They originated from the sun, from outside of the solar system in our own galaxy and from distant galaxies cosmic rays such as gamma rays are a source of radiation pollution.

2. Wind mill called as 'Darriens' mill is :

(a) Flexible boom mill
(b) Vertical axis mill
(c) Horizontal axis single blade mill
(d) Horizontal axis bicycle wheel mill

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (b) : Darriens mill is a type of vertical axis wind turbine used to generate electricity from wind energy.

3. The device used to generate power, utilizing energy of wind, is

(a) air motor (b) aerogenerator
(c) windmill (d) wind generator

Assam PSC AE (IWT) 14.03.2021

Ans. (c) : A wind mill or wind energy converter is a device that converts the winds kinetic energy into electrical energy wind turbines are manufactured in a wide range of size with either horizontal or vertical axis.

4. A wind turbine extracts maximum power from wind, when the downstream wind speed reduces to :

(a) one-third that of upstream wind
(b) half that of upstream wind
(c) two third that of upstream wind
(d) zero

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (a) : A wind turbine extracts maximum power from wind; when the downstream speeds of wind reduces to one third of upstream wind.

5. Which of the following turbine is normally used in a tidal range plant ?

(a) Pelton turbine
(b) Kaplan turbine with variable pitch blades
(c) Kaplan turbine with fixed pitch blades.
(d) rancis turbine.

RPSC Lect. (Tech. Edu. Dept.) 13.03.2021

Ans. (b) : The Kaplan turbine is a propeller type reaction turbine that is usually immersed completely in the fluid. Kaplan turbine is beneficial in that as it is able to operate in lower pressure situations where pelton or Francis cannot.

6. Which of the following power plant is least polluting and causing heat environmental concern?

(a) Hydro electric plant (b) solar power plant
(c) Nuclear plant (d) gas power plant

ISRO Scientist/Engineer (RAC) 07.05.2017

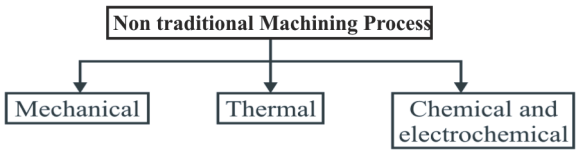
Ans. (b) : Solar power plant is least polluting and causing least environment concern.

7. The source of energy in ultrasonic machining process is

(a) Mechanical (b) Electrical
(c) Electric spark (d) None of these

ISRO Scientist/Engineer 07.05.2017

Ans. (a) :



Mechanical	Thermal	Chemical and electrochemical
USM	EDM	
AJM	EBM	CHM
AWJM	LMB	PCM
	IBM	ECM
	PBM	

8. With respect to the wind turbine blades, TSR means

(a) Tip Swift Ratio (b) Tip Sharp Ratio
(c) Tip Speed Ratio (d) Tip Swing Ratio

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Ans. (c) : TSR = Tip Speed ratio

$$= \frac{\text{blade tip speed}}{\text{wind speed}}$$

9. For a wind turbine 10 m long running at 20 rpm in 12.9 kmph wind, the TSR will be nearly

(a) 3.6 (b) 5.8

(c) 7.6 (d) 9.8

ESE 2020

Ans. (b) : $\text{TSR} = \frac{\text{Tip speed}(R_\omega)}{\text{Wind velocity}(V)}$

R= 10 m

$\omega = \frac{2\times\pi\times 20}{60}$ rad/sec

$V = 12.9\times\frac{5}{18}$ m/sec

$\text{TSR} = \frac{10\times 2\times\pi\times 20}{12.9\times\frac{5}{18}}$

TSR = 5.84

10. The available wind power in wind turbines increases with

(a) Decrease in rotor diameter

(b) Increase in rotor diameter

(c) Decrease in wind velocity

(d) Decrease in air velocity

RPSC LECTURER 16.01.2016

Ans. (b) : The available wind power in wind turbines increases with increase in rotor diameter.

11. Which of the following is not a renewable source of energy

(a) Solar energy (b) Wind energy

(c) Natural gas (d) Tidal energy

HPPSC AE 2018

Ans. (c) : Renewable energy is energy that is collected from renewable resources, which are naturally replenished on a human timescale, such as sunlight, wind, rain, tides, waves and geothermal heat, agricultural residue, firewood and animal dung. The non-renewable sources are the fossil fuels such as coal, crude oil and natural gas.

12. In various solar energy storage systems, pumped hydro-electric storage system falls under which one of the following categories?

(a) Thermal energy storage

(b) Electrical energy storage

(c) Mechanical energy storage

(d) Electromagnetic energy storage

ESE 18.07.2021

Ans. (c) : Mechanical energy stored in various forms and the storage methods are classified on the basis of the form in which it is stored.

1. Mechanical storage
(a) Pumped storage
(b) Compressed air storage
(c) Flywheel storage
2. Chemical energy storage
(a) Batteries storage
(b) Hydrogen storage
(c) Reversible chemical reaction storage
3. Electromagnetic energy storage
4. Electrostatic energy storage
5. Thermal energy storage
6. Biological storage

13. Greenhouse gases are those that absorb and emit infrared radiation. Examples include

(a) nitrogen

(b) ozone

(c) argon

(d) carbon dioxide

Assam PSC AE (IWT) 14.03.2021

Ans. (b&d) : A greenhouse gas is a gas which absorbs infrared radiation in thermal range and causes global warming. The primary greenhouse gases in earth's atmosphere are water vapour, carbon dioxide methane, nitrous oxide and ozone.

Note-(b) given by commission.

14. What is the standard value of solar constant adopted by World Radiation Centre?

(a) 1192 W/m^2

(b) 1084 W/m^2

(c) 1927 W/m^2

(d) 1367 W/m^2

ESE 18.07.2021

Ans. (d) : Solar constant is the total radiation energy received from sun per unit time per unit area on a theoretical surface perpendicular to the sun rays and at earth's mean distance from the sun.

Solar constant value adopted by world Radiation centre is 1367 W/m^2.

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15. What is the tip speed ratio of savonius wind turbine rotor?

(a) 1 (b) 3

(c) 5 (d) 7

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Ans. (a) : Tip-speed ratio λ is defined as

$$\lambda = \frac{\text{speed of the rotor blade}}{\text{speed of incoming air}} = \frac{R\omega}{u_0}$$

Rotor Type	Tip speed ratio
1. Propeller (1-3 Blades)	6 – 20
2. Sailwing	4
3. Chalk multiblade	3 – 4
4. American multiblade	1
5. Dutch type	2 – 3
6. Savoanious	1
7. Darrieus	5 – 6
8. Musgrove and Evan	3 - 4

16. What is the solidity of American multiblade wind turbine rotor?

(a) 0.4 (b) 0.7

(c) 0.9 (d) 1

ESE 18.07.2021

Ans. (b) :

Rotor Type	Solidity
Savonious rotor	1
American multiblade	0.7

Solidity is defined as the ratio of the projected area of the rotor blades on the rotor plane to the swept areas of the rotor.

17. The energy density of Bio-ethanol is

(a) 8.3 MJ/kg (b) 14.6 MJ/kg

(c) 26.9 MJ/kg (d) 34.7 MJ/kg

ESE 18.07.2021

Ans. (c) : The energy density of Bio-ethanol is 26.9 MJ/kg.

18. The percentage of hydrogen in producer gas is

(a) 34% (b) 27%

(c) 18% (d) 8%

ESE 18.07.2021

Ans. (d) :

19. In single basin, double effect scheme, power is generated

(a) during filling. (b) during emptying.

(c) on ebb only. (d) on both flood and ebb.

ESE 18.07.2021

Ans. (d) : In single basin, double effect scheme, power is generated on both flood and ebb. Two- way (reversible) hydraulic turbines are used. In single effect scheme, power is generated either during filling or emptying the basin.

20. The operating temperature of alkaline fuel cell is

(a) 39°C (b) 90°C

(c) 127°C (d) 192°C

ESE 18.07.2021

Ans. (b) : The operating temperature of alkaline fuel cell is 90°C

Cell type	Operating temperature
Phosphoric Acid fuel cell	150 – 200°C
Alkaline fuel cell	90°C
Solid polymer fuel cell	40 – 60 °C
Direct methanol fuel cell	50 – 120 °C
Molten Carbonate fuel cell	600 – 700 °C
Solid oxide fuell cell	600 –1000 °C

21. The ideal emf produced by polymer electrolyte membrane fuel cell at 25°C is

(a) 3.57 V (b) 2.94 V

(c) 1.23 V (d) 0.73 V

ESE 18.07.2021

Ans. (c) : Polymer electrolyte membrane fuel cell (PEMFC) or solid polymer fuel cell (SPFC) operates at 40 – 60°C. The ideal emf produced is 1.23 V at 25°C

22. Which one of the following fuel cells has highest efficiency?

(a) PAFC (b) MCFC

(c) PEMFC (d) AFC

ESE 18.07.2021

Ans. (d) : Characteristics of various fuel cells

Fuel Cell	Operating temp.	Fuel	Efficiency
1. PEMFC	40-60°C	H_2	48-58%
2. AFC	90°C	H_2	64%
3. PAFC	150-200°C	H_2	42%
4. MCFC	600-700°C	H_2 & CO	50%
5. SOFC	600-1000°C	H_2 & CO	60-65%

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23. How many kilograms of steam per day is produced by 15 m diameter community solar cooker developed by Centre for Scientific Research, Auroville (Puducherry)?

(a) 100 kg

(b) 300 kg

(c) 600 kg

(d) 1000 kg

ESE 18.07.2021

Ans. (c) : A 15 m diameter community solar cooker has been developed at the Centre for Scientific Research (CSR), Auroville. Around 600 kg of steam per day could be generated from this bowl.

24. In a solar passive space heating system, the south-facing thick wall is called

(a) Vent wall

(b) Trombe wall

(c) Damper wall

(d) Ventilation wall

ESE 18.07.2021

Ans. (b) : The south facing thick wall in a solar passive heating system, called Trombe wall is made of concrete, adobe, stone or composites of brick blocks and sand, designed for thermal storage.

25. Which one of the following turbines is used in underwater power stations?

(a) Pelton turbine

(b) Deriaz turbine

(c) Tubular turbine

(d) Turgo-impulse turbine

UKPSC AE 2007 Paper -II

Ans. (c) : Tubular turbine

26. The most common material used in making solar cells is…

(a) silver

(b) silicon

(c) Aluminium

(d) Iron

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (b) : The most common material used in making solar cells is silicon.

27. Energy available when temperature of water at the surface of ocean is different from that at deeper levels is called.

(a) Ocean thermal energy

(b) Tidal energy

(c) Sea water energy

(d) None of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (a) : Energy available when temperature of water at the surface of ocean is different from that at deeper levels is called ocean thermal energy.

28. Which instrument is used to measure solar radiation?

(a) Pyrometer only

(b) Thermocouple only

(c) Thermometer only

(d) Pyranometer only

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Solar radiation is measured by pyranometer.

29. Maximum power that can be developed by a windmill is given by… where A-Area swept by wind mill rotor blades in m^2, ρ-Density of air in kg/m^3, m-mass flow rate of air in kg/s, V-Velocity of wind in m/s.

(a) $\frac{1}{2}mV^2$ only

(b) $AV^3/2$ only

(c) mV only

(d) none of the above

MPSC Dy. Engg. (GWD) 19.05.2013

Ans. (d) : Maximum Power developed by wind mill.

$P = C_P \rho A U^3$

P = Power

ρ = density of air

A = Rotor swept area

U = wind speed.

C_P = Power coefficient

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